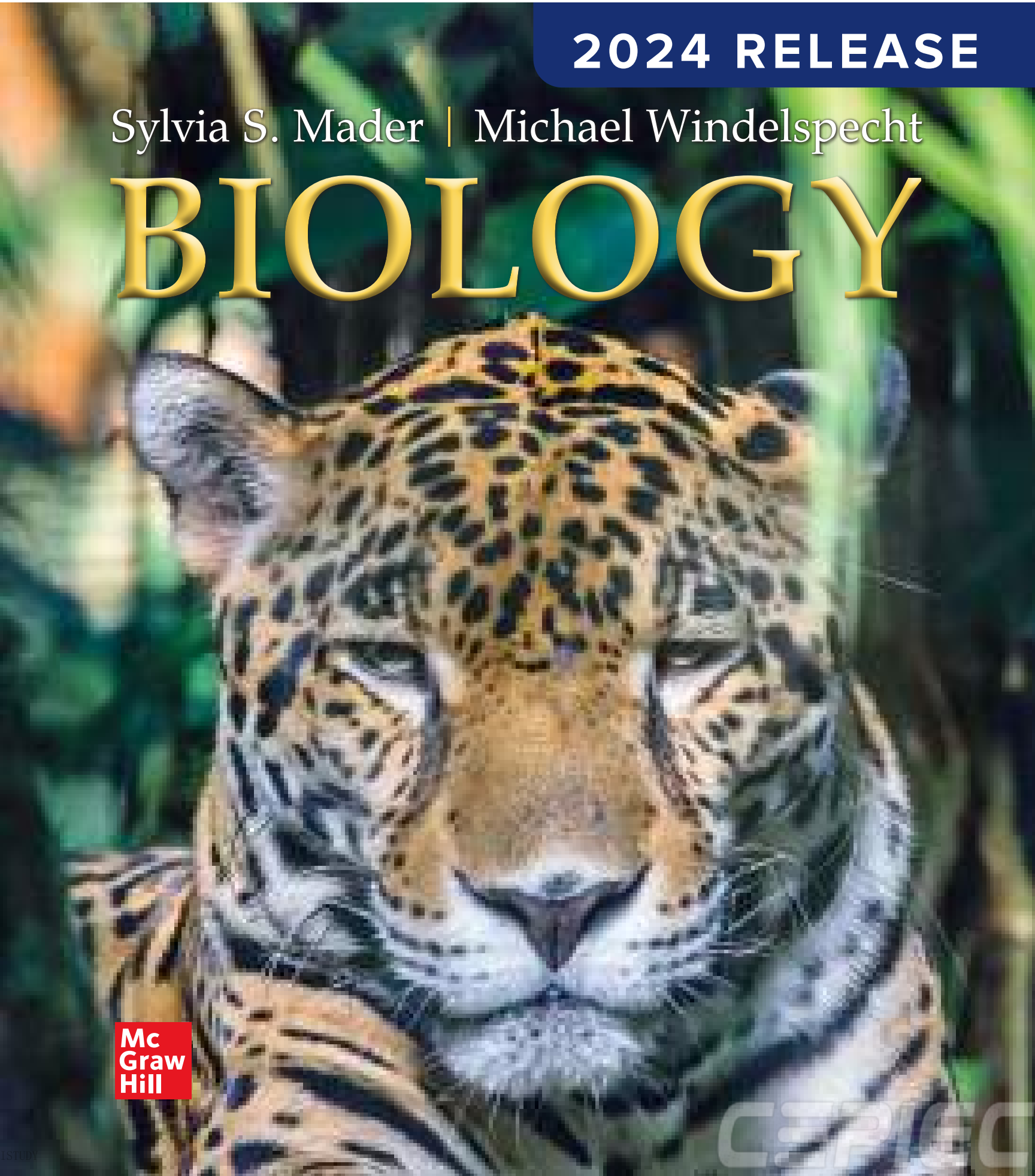
This International Student Edition is for use outside of the U.S.



2024 RELEASE

Sylvia S. Mader | Michael Windelspecht

BIOLOGY

Mc Graw Hill

BIOLOGY

ISTUDY

2024 Release

BIOLOGY

Sylvia S. Mader

Michael Windelspecht

With contributions by

Jason Carlson
St. Cloud Technical and Community College

David Cox
Lincoln Land Community College

Gretel Guest
Durham Technical Community College

Mc Graw Hill

ISTUDY

BIOLOGY

Published by McGraw Hill LLC, 1325 Avenue of the Americas, New York, NY 10019. Copyright ©2024 by McGraw Hill LLC. All rights reserved. Printed in the United States of America. No part of this publication may be reproduced or distributed in any form or by any means, or stored in a database or retrieval system, without the prior written consent of McGraw Hill LLC, including, but not limited to, in any network or other electronic storage or transmission, or broadcast for distance learning.

Some ancillaries, including electronic and print components, may not be available to customers outside the United States.

This book is printed on acid-free paper.

1 2 3 4 5 6 7 8 9 LWI 29 28 27 26 25 24

ISBN 978-1-266-94002-6
MHID 1-266-94002-2

Cover Image: ©*Eric Weber*

All credits appearing on page or at the end of the book are considered to be an extension of the copyright page.

The Internet addresses listed in the text were accurate at the time of publication. The inclusion of a website does not indicate an endorsement by the authors or McGraw Hill LLC, and McGraw Hill LLC does not guarantee the accuracy of the information presented at these sites.

mheducation.com/highered

ISTUDY

Brief Contents

1 Biology: The Study of Life 1

UNIT 1 **The Cell 18**

2 Basic Chemistry 19
3 The Chemistry of Organic Molecules 36
4 Cell Structure and Function 56
5 Membrane Structure and Function 80
6 Metabolism: Energy and Enzymes 98
7 Photosynthesis 111
8 Cellular Respiration 126

UNIT 2 **Genetic Basis of Life 142**

9 The Cell Cycle and Cellular Reproduction 143
10 Meiosis and Sexual Reproduction 162
11 Mendelian Patterns of Inheritance 181
12 Molecular Biology of the Gene 202
13 Regulation of Gene Expression 221
14 Biotechnology and Genomics 237

UNIT 3 **Evolution 258**

15 Darwin and Evolution 259
16 How Populations Evolve 276
17 Speciation and Macroevolution 293
18 The Origin and History of Life 312
19 Organizing Life 333

UNIT 4 **Microbial Evolution 348**

20 Viruses, Bacteria, and Archaea 349
21 Protist Evolution and Diversity 370
22 Fungi Evolution and Diversity 390

UNIT 5 **Plant Evolution and Biology 406**

23 Plant Evolution and Diversity 407
24 Flowering Plants: Structure and Organization 429
25 Flowering Plants: Nutrition and Transport 449
26 Flowering Plants: Control of Growth Responses 469
27 Flowering Plants: Reproduction 488

UNIT 6 **Animal Evolution and Diversity 504**

28 Invertebrate Evolution 505
29 Vertebrate Evolution 536
30 Human Evolution 556

UNIT 7 **Comparative Animal Biology 572**

31 Animal Organization and Homeostasis 573
32 Circulation and Cardiovascular Systems 589
33 The Lymphatic and Immune Systems 609
34 Digestive Systems and Nutrition 629
35 Respiratory Systems 646
36 Body Fluid Regulation and Excretory Systems 663
37 Neurons and Nervous Systems 676
38 Sense Organs 698
39 Locomotion and Support Systems 716
40 Hormones and Endocrine Systems 732
41 Reproductive Systems 751
42 Animal Development and Aging 773
43 Animal Behavior 794

UNIT 8 **Ecology 812**

44 Population Ecology 813
45 Community and Ecosystem Ecology 835
46 Major Ecosystems of the Biosphere 862
47 Conservation of Biodiversity 883

About the Authors

Sylvia S. Mader

Sylvia S. Mader Sylvia S. Mader has authored several nationally recognized biology texts published by McGraw Hill. Educated at Bryn Mawr College, Harvard University, Tufts University, and Nova Southeastern University, she holds degrees in both biology and education. Over the years, she has taught at University of Massachusetts, Lowell; Massachusetts Bay Community College; Suffolk University; and Nathan Mayhew Seminars. Her ability to reach out to science-shy students led to the writing of her first text, *Inquiry into Life,* which is now in its 17th edition. Highly acclaimed for her crisp and entertaining writing style, her books have become models for others who write in the field of biology.

Inspire EdVentures LLC

Michael Windelspecht As an educator, Dr. Windelspecht has taught introductory biology, genetics, and human genetics in the online, traditional, and hybrid environments at community colleges, comprehensive universities, and military institutions. For over a decade, he served as the Introductory Biology Coordinator at Appalachian State University, where he directed a program that enrolled over 4,500 students annually.

He received degrees from Michigan State University (B.S., zoology–genetics) and the University of South Florida (Ph.D., evolutionary genetics) and has published papers in areas as diverse as science education, water quality, and the evolution of insecticide resistance. His current interests are in the analysis of data from digital learning platforms for the development of personalized microlearning assets and next-generation publication platforms. He is currently a member of the National Association of Science Writers and several science education associations. He has served as the keynote speaker on the development of multimedia resources for online and hybrid science classrooms. In 2015, he won the DevLearn HyperDrive competition for a strategy to integrate student data into the textbook revision process.

His current passion is the development of innovative educational environments to reach people in our modern age. This involves not only working with his close friends to establish a network of trail cameras in central Belize to enhance jaguar conservation efforts, but also developing an innovative network of virtual experiences to promote conservation programs to a diverse audience.

As an author and editor, Dr. Windelspecht has over 30 reference textbooks and multiple print and online lab manuals to his credit. He has founded several science communication companies, including Ricochet Creative Productions, which actively develops and assesses new technologies for the science classroom, and Inspire EdVentures, which provides experiential learning opportunities online and in Belize. You can learn more about Dr. Windelspecht by visiting his website at www.windelspectrum.com.

Contributors

Courtesy of Jason Carlson

Jason Carlson is a Biology Instructor at St. Cloud Technical and Community College in Minnesota, where he teaches introductory biology, microbiology, nutrition, and human biology. Before entering higher education, he was a middle and high school science teacher with an education from the University of Idaho, Bemidji State University, and St. Cloud State University. In the classroom, he supports a student-driven applied curriculum with relevant and hands-on research and investigation.

Courtesy of Dave Cox

Dave Cox serves as Professor of Biology at Lincoln Land Community College, in Springfield, Illinois. He was educated at Illinois College and Western Illinois University. As an educator, Professor Cox teaches introductory biology for nonmajors in the traditional classroom format, as well as in a hybrid format. He also teaches biology for majors, and marine biology and biological field studies as study-abroad courses in Belize. He is the co-owner of Howler Publications, a company that specializes in scientific study abroad courses. Professor Cox has served as a contributor to multiple McGraw Hill titles over the past 15 years. He also develops educational resources for the ecotourism industry in Belize.

©Gretel Guest

Gretel Guest is a Professor of Biology at Durham Technical Community College, in Durham, North Carolina. She has been teaching nonmajors and majors biology, microbiology, and genetics for more than 20 years. Dr. Guest was educated in the field of botany at the University of Florida, and received her Ph.D. in plant sciences from the University of Georgia. Dr. Guest was a contributor to *Essentials of Biology 4e,* a contributor to the digital content in *Stern's Introductory Plant Biology 13e,* and she co-authored *The Biology Book* published by DK Books. Gretel Guest is also a Spanish medical interpreter.

ISTUDY

Preface

Goals of the 2024 Release

The mission of Dr. Sylvia Mader's *Biology* text has always been to give students an understanding of biological concepts and a working knowledge of the scientific process. However, like the world around us, the process of teaching science is changing rapidly. Increasingly, instructors are being asked to engage their students by making content more relevant, while still providing students with a firm foundation in those core principles on which biology is founded.

As we have all learned over the past few years, the challenges facing society—from climate change to the COVID-19 pandemic—are demanding that we as a society not only become more scientifically literate, but also understand and appreciate the importance of the process of scientific thought and the roles of scientists in our culture.

As educators, the authors of this text understand the needs of our colleagues in developing curricula that increasingly focus on relevancy and delivery in the online environment. McGraw Hill Education has long been an innovator in the development of digital resources, and the *Biology* text, and its authors, are at the forefront of the integration of these technologies into the science classroom.

In this release, the authors focused on the following areas:

- Making the content more relevant to the current generation of students by updating chapter openers and featured readings to focus on issues and topics important to the discussions that students are hearing in the world around them.
- Integrating relevancy modules to supplement the format of a traditional textbook and provide another avenue for students to engage with the content.
- Redesigning the artwork to ensure it transitions to the digital world of mobile devices.
- Developing a new boxed feature, Diversity in Science, to emphasize the contributions of unrecognized scientists to our understanding of biology.
- Transitioning to the evergreen model of content delivery that involves the ability to provide continuous updates over the life of the project.

The Themes of This Text

The Vision and Change document (2009) clearly identifies the need to integrate core concepts throughout the curriculum. We recognize that scientific literacy is not based upon the memorization of a series of facts. Instead, learning is based on establishing associations and links between what, at first glance, appear to be diverse topics.

The main themes we have chosen to emphasize include biological systems, evolution, and the nature of science. These themes are integrated into all aspects of the textbook, from the unit learning outcomes to the theme-based feature readings in the text. At the start of each chapter, "Following the Themes" introduces the relationship of the chapter's content to each of the themes. At the end of each chapter, "Connecting the Concepts" not only reminds the student of the relationships between chapter content and the three core themes, but also acts as a prelude to topics in the next few chapters of the text. In essence, the themes act as the threads that unite the concepts throughout the text, enabling the student to see relationships from the molecular to ecosystem levels of biology.

Diversity, Equity, and Inclusion in the 2024 Release

In preparation for this release, the authors worked with a diversity, equity, and inclusion reviewer to identify areas in the narrative and visual presentation where content could be made more sensitive to, and representative of, the broad spectrum of learners taking biology courses today. A new feature, Diversity in Science, features scientists from diverse backgrounds who have contributed to our understanding of our world. In addition, continual improvements have been made in the accessibility of the artwork and media that support the text.

Throughout this release of *Biology,* the authors have worked to make the text as gender-neutral as possible to reflect the changing attitudes in society on gender identification. The terms *male* and *female* are used where needed to reflect the biological sex of the organism.

Virtual Labs and Lab Simulations

While the biological sciences are hands-on disciplines, instructors are now often being asked to deliver some of their lab content online, as full online replacements, supplements to prepare for in-person labs, or make-up labs.

These simulations help each student learn the practical and conceptual skills needed, then check for understanding and provide feedback. With adaptive pre-lab and post-lab assessment available, instructors can customize each assignment.

From the instructor's perspective, these simulations may be used in the lecture environment to help students visualize complex scientific processes, such as DNA technology or Gram staining, while at the same time providing a valuable connection between the lecture and lab environments.

McGraw Hill

Relevancy

The use of real-world examples to demonstrate the importance of biology in the lives of students is widely recognized as an effective teaching strategy for the introductory biology classroom. Students want to learn about the topics they are interested in. The development of relevancy-based resources is a major focus for the authors of the Mader series of texts. Some examples of how we have increased the relevancy content of this edition are explained in the following paragraphs.

Connecting the Concepts Graphics

To help students better understand the relevancy of the opening essay to the contents of the chapter, the authors developed the Connecting the Concepts images at the end of the chapter. These assets explore the relationship between the chapter opener and the three major themes of the text.

Relevancy Modules

A series of relevancy modules, available as a supplementary eBook to the existing text within Connect, have been designed to accompany the *Essentials of Biology* content. These modules demonstrate the connections between biological content and topics that are of interest to society as a whole. Each module consists of an introduction to the topic, an overview of basic relevant scientific concepts, and then a closer look at the application of these concepts back to the topic. An infographic at the end of each module may be easily used in the lecture environment to initiate discussion of the topic. Discussion and assessment questions, specific to the modules, are available at the end of the module and for automatic assessment in the Connect platform.

A new "Thinking Scientifically" feature in each module uses a fake blog post about the module topic, similar to what students may encounter in their social media feeds, to increase science literacy. Assessments, available in the module or assignable in Connect, encourage the reader to analyze the author's credentials, consider potential conflicts of interest, and look for unscientific thinking and data misinterpretation. The reader is then challenged to determine what conclusions are appropriate.

These modules are available as a supplementary eBook to the existing text within Connect, and may be assigned by the instructor for use in a variety of ways in the classroom.

Slavica/E+/Getty Images

BioNOW Videos

The BioNOW series of videos, narrated and produced by educator Jason Carlson, provide a relevant, applied approach that allows your students to feel they can actually do and learn biology themselves. While tying directly to the content of your course, the videos help students relate their daily lives to the biology you teach and then connect what they learn back to their lives.

Each video provides an engaging and entertaining story about applying the science of biology to a real situation or problem. Attention is taken to use tools and techniques that the average person would have access to, so your students see the science as something they could do and understand.

McGraw Hill

RicochetScience Website

The RicochetScience.com website, managed by Michael Windelspecht, provides updates on news and stories that are interesting to science and nonscience majors alike. The PopScience articles on this site provide an excellent focus for classroom discussions on topics that are currently being debated in society. The site also features videos and tutorial animations to assist the students in recognizing the relevancy of what they are learning in the classroom.

Ricochet Science™

Mc Graw Hill connect®

A complete course platform

Connect enables you to build deeper connections with your students through cohesive digital content and tools, creating engaging learning experiences. We are committed to providing you with the right resources and tools to support all your students along their personal learning journeys.

65%
Less Time Grading

Laptop: Getty Images; Woman/dog: George Doyle/Getty Images

Every learner is unique

In Connect, instructors can assign an adaptive reading experience with SmartBook® 2.0. Rooted in advanced learning science principles, SmartBook 2.0 delivers each student a personalized experience, focusing students on their learning gaps, ensuring that the time they spend studying is time well spent. **mheducation.com/highered/connect/smartbook**

Study anytime, anywhere

Encourage your students to download the free ReadAnywhere® app so they can access their online eBook, SmartBook® 2.0, or Adaptive Learning Assignments when it's convenient, even when they're offline. And since the app automatically syncs with their Connect account, all of their work is available every time they open it. Find out more at **mheducation.com/readanywhere**

"I really liked this app—it made it easy to study when you don't have your textbook in front of you."

Jordan Cunningham, a student at *Eastern Washington University*

Effective tools for efficient studying

Connect is designed to help students be more productive with simple, flexible, intuitive tools that maximize study time and meet students' individual learning needs. Get learning that works for everyone with Connect.

ISTUDY

Education for all

McGraw Hill works directly with Accessibility Services departments and faculty to meet the learning needs of all students. Please contact your Accessibility Services Office, and ask them to email **accessibility@mheducation.com**, or visit **mheducation.com/about/accessibility** for more information.

Affordable solutions, added value

Make technology work for you with LMS integration for single sign-on access, mobile access to the digital textbook, and reports to quickly show you how each of your students is doing. And with our Inclusive Access program, you can provide all these tools at the lowest available market price to your students. Ask your McGraw Hill representative for more information.

Solutions for your challenges

A product isn't a solution. Real solutions are affordable, reliable, and come with training and ongoing support when you need it and how you want it. Visit **supportateverystep.com** for videos and resources both you and your students can use throughout the term.

Updated and relevant content

Our new Evergreen delivery model provides the most current and relevant content for your course, hassle-free. Content, tools, and technology updates are delivered directly to your existing McGraw Hill Connect® course. Engage students and freshen up assignments with up-to-date coverage of select topics and assessments, all without having to switch editions or build a new course.

ISTUDY

Acknowledgments

Dr. Sylvia Mader is one of the icons of science education. Her dedication to her students, coupled with her clear, concise writing style, has benefited the education of thousands of students over the past four decades. As an educator, it is an honor to continue her legacy and to bring her message to this newest generation of students.

As always, I had the privilege to work with a phenomenal group of people on this edition. I would especially like to thank you, the numerous instructors who have shared emails with me or have invited me into your classrooms, both physically and virtually, to discuss your needs as instructors and the needs of your students. You are all dedicated and talented teachers, and your energy and devotion to quality teaching is what drives a textbook revision.

Many dedicated and talented individuals assisted in the development of this 2024 release of *Biology.* A special thanks to my coauthors, Jason Carlson, David Cox, and Gretel Guest, for their hard work on this edition. I am very grateful for the help of so many professionals at McGraw Hill who were involved in bringing this book to fruition. Therefore, I would like to thank the following:

- My product developer, Anne Winch, for her incredible ability to manage all aspects of this project simultaneously.
- Portfolio director Michelle Vogler and portfolio manager Lora Neyens, for their guidance and for reminding me why what we do is important.
- My marketing manager, Erin Martin, for placing me in contact with great instructors, on campus and virtually, throughout this process.
- My content project manager, Diane Nowaczyk, and program manager, Kelly Hart, for guiding this project throughout the publication process.
- Content licensing specialist, Lori Hancock, and photo researcher, David Tietz, for the photos within this text. Biology is a visual science, and their contributions are evident on every page.
- Michael McGee and Sharon O'Donnell, who acted as my copyeditor and proofreader, respectively, for this edition.
- Jane Peden, for her behind-the-scenes work that keeps us all functioning.

As both an educator and an author, communicating the importance of science represents one of my greatest passions. Our modern society is based largely on advances in science and technology over the past few decades. As I present in this text, there are many challenges facing humans, and an understanding of how science can help analyze, and offer solutions to, these problems is critical to our species' health and survival. The only solution to these problems is an increase in scientific literacy, and more importantly, a greater appreciation for the roles of scientists in society. It is my hope that this text helps with that process. I would like to give special thanks to Junior Buddy, the jaguar on the front cover of this text, and the staff of the Belize Zoo. I have had the opportunity to meet you over multiple visits to your beautiful country, and your dedication to educating the world on the importance of rain forests and the creatures that live in them is inspirational. I can only hope to carry on the legacy of Junior Buddy in this edition.

I also want to acknowledge my family and friends for all of their support. My family team of Sandy, Devin, and Kayla have always been my motivation and encouragement. And for my dearest friends, who have never waivered in their support, thank you for believing.

Michael Windelspecht
Blowing Rock, NC

ISTUDY

Readings

THEME Evolution

Human Adaptation to Life at High Elevations 7
Metagenomics 254
The Anatomy of Speciation 295
Possible Extraterrestrial Components of RNA 316
Flu Viruses: Seasonal Versus Pandemic Flu 352
Carboniferous Forests 416
The Evolutionary History of Maize 422
Survival Mechanisms of Plants 432
Plants Can "Smell" and "Talk" to Each Other 478
Plants and Their Pollinators 493
Evolution of the Animal Body Plan 508
The Complexity of Our Evolutionary Past 567
Sexual Selection in Male Bowerbirds 809
Interactions and Coevolution 845
The Sixth Mass Extinction Event 891

THEME Biological Systems

Isotopes of a Meltdown 24
The Impact of Acid Deposition 33
Tropical Rain Forest Destruction and Climate Change 120
Zika Virus 357
African Sleeping Sickness 374
Deadly Fungi 398
The Concept of Water Potential 460
Would You Eat Insects? 530
UV Rays: Too Much Exposure or Too Little? 583
AIDS and Opportunistic Infections 622
Dietary Guidelines for Human Health 641
Urinalysis 672
Drug Use and Substance Abuse Disorders 684
Preventing Transmission of STDs 767
Preventing and Testing for Birth Defects 788
Sustainability in the Lobster Industry 825
Destruction of the Amazon Rain Forest 872

THEME Nature of Science

Microscopy Today 59
Carbon Monoxide Can Lead to Enzyme Inhibition and Even Death 107
Fermentation and Food Production 132
The Importance of the G_1 Checkpoint 146
Reproductive and Therapeutic Cloning 153
Meiosis and the Parthenogenic Lizards 167
Hemophilia and the Royal Families of Europe 199
Exercise and Gene Expression 229
CRISPR Gene Editing in Humans 244
Inbreeding in Populations 283
Genetic Basis of Beak Shape in Darwin's Finches 304
DNA Barcoding of Life 337
Sushi, Science, and Beautiful Skin 382
Fungi as a Biological Control 402
The *Cannabis* Plant and Its Products 441
Plants Can Be Used for Cleaning and Discovery of Minerals 456
Why So Many Scientists Work with *Arabidopsis* 480
Vertebrates and Human Medicine 545
A Genomic Comparison of *Homo sapiens* and Chimpanzees 560
Recent Findings About Preventing Cardiovascular Disease 601
Medicinal Leeches 606
Adult Vaccinations 624
The Rise and Fall of Artificial Trans Fatty Acids 642
SARS-CoV-2 and the Respiratory System 658
Are E-cigs Safe? 660
Understanding Parkinson's Disease 689
Artificial Retinas Come into Focus 707
The Accidental Discovery of Botox 726
Identifying Insulin as a Chemical Messenger 746
Preimplantation Genetic Diagnosis 765
Jaguar Behavior 802
Marine Protected Areas and Ocean Biodiversity 838
Global Climate Change 858
Wildlife Conservation and DNA 867

▶ Diversity in Science

The Legacy of Marie Curie 23
The Immortal Henrietta Lacks 155
Nettie Stevens and the Discovery of the Sex Chromosomes 165
CRISPR Pioneers 242
The Contributions of John Edmonstone to Science 265

ISTUDY

Contents

Preface vii

1 Biology: The Study of Life 1
- 1.1 The Characteristics of Life 2
- 1.2 Evolution and the Classification of Life 6
- 1.3 The Process of Science 10
- 1.4 Science and the Challenges Facing Society 13

UNIT 1 The Cell 18

2 Basic Chemistry 19
- 2.1 Chemical Elements 20
- 2.2 Molecules and Compounds 25
- 2.3 Chemistry of Water 27
- 2.4 Acids and Bases 31

3 The Chemistry of Organic Molecules 36
- 3.1 Organic Molecules 37
- 3.2 Carbohydrates 39
- 3.3 Lipids 43
- 3.4 Proteins 46
- 3.5 Nucleic Acids 50

4 Cell Structure and Function 56
- 4.1 Cellular Level of Organization 57
- 4.2 Prokaryotic Cells 61
- 4.3 Introduction to Eukaryotic Cells 63
- 4.4 The Nucleus and Ribosomes 66
- 4.5 The Endomembrane System 68
- 4.6 Microbodies and Vacuoles 71
- 4.7 The Energy-Related Organelles 72
- 4.8 The Cytoskeleton 74

5 Membrane Structure and Function 80
- 5.1 Plasma Membrane Structure and Function 81
- 5.2 Passive Transport Across a Membrane 85
- 5.3 Active Transport Across a Membrane 88
- 5.4 Modification of Cell Surfaces 91
- 5.5 Cell-to-Cell Communication 94

6 Metabolism: Energy and Enzymes 98
- 6.1 Cells and the Flow of Energy 99
- 6.2 Metabolic Reactions and Energy Transformations 101
- 6.3 Metabolic Pathways and Enzymes 103
- 6.4 Oxidation-Reduction Reactions and Metabolism 107

7 Photosynthesis 111
- 7.1 Photosynthetic Organisms 112
- 7.2 The Process of Photosynthesis 113
- 7.3 Plants Convert Solar Energy 116
- 7.4 Plants Fix Carbon Dioxide 119
- 7.5 Other Types of Photosynthesis 122

8 Cellular Respiration 126
- 8.1 Overview of Cellular Respiration 127
- 8.2 Outside the Mitochondria: Glycolysis 129
- 8.3 Outside the Mitochondria: Fermentation 131
- 8.4 Inside the Mitochondria 133
- 8.5 Metabolism 137

UNIT 2 Genetic Basis of Life 142

9 The Cell Cycle and Cellular Reproduction 143
- 9.1 The Cell Cycle 144
- 9.2 The Eukaryotic Chromosome 147
- 9.3 Mitosis and Cytokinesis 148
- 9.4 The Cell Cycle and Cancer 154
- 9.5 Prokaryotic Cell Division 158

10 Meiosis and Sexual Reproduction 162
- 10.1 Overview of Meiosis 163
- 10.2 Genetic Variation 166
- 10.3 The Phases of Meiosis 168
- 10.4 Meiosis Compared to Mitosis 170
- 10.5 The Cycle of Life 172
- 10.6 Changes in Chromosome Number and Structure 174

11 Mendelian Patterns of Inheritance 181
- 11.1 Gregor Mendel 182
- 11.2 Mendel's Laws 183
- 11.3 Mendelian Patterns of Inheritance and Human Disease 189
- 11.4 Beyond Mendelian Inheritance 192

12 Molecular Biology of the Gene 202
- 12.1 The Genetic Material 203
- 12.2 Replication of DNA 208
- 12.3 Gene Expression: RNA and the Genetic Code 210
- 12.4 Gene Expression: Transcription 212
- 12.5 Gene Expression: Translation 214

13 Regulation of Gene Expression 221
- 13.1 Prokaryotic Regulation 222
- 13.2 Eukaryotic Regulation 225
- 13.3 Gene Mutations 232

14 Biotechnology and Genomics 237
- 14.1 Biotechnology 238
- 14.2 Biotechnology Products 245
- 14.3 Gene Therapy 249
- 14.4 Genomics 250

ISTUDY

UNIT 3 Evolution 258

15 Darwin and Evolution 259
15.1 The History of Evolutionary Thought 260
15.2 Darwin's Theory of Natural Selection 263
15.3 Evidence for Evolution 269

16 How Populations Evolve 276
16.1 Genes, Populations, and Evolution 277
16.2 Natural Selection 284
16.3 Maintenance of Diversity 288

17 Speciation and Macroevolution 293
17.1 How New Species Evolve 294
17.2 Modes of Speciation 300
17.3 Principles of Macroevolution 307

18 The Origin and History of Life 312
18.1 The Origin of Life 313
18.2 The History of Life 318
18.3 Geological Factors that Influence Evolution 328

19 Organizing Life 333
19.1 Systematic Biology 334
19.2 The Three-Domain System 338
19.3 Phylogeny 340

UNIT 4 Microbial Evolution 348

20 Viruses, Bacteria, and Archaea 349
20.1 Viruses, Viroids, and Prions 350
20.2 Emerging Infectious Diseases and SARS-CoV-2 356
20.3 Prokaryotes 359
20.4 Bacteria 361
20.5 Archaea 365

21 Protist Evolution and Diversity 370
21.1 Overview of Protists 371
21.2 Supergroup Excavata 371
21.3 Supergroup Chromalveolata 375
21.4 Supergroup Rhizaria 380
21.5 Supergroup Archaeplastida 381
21.6 Supergroup Amoebozoa 385
21.7 Supergroup Opisthokonta 386

22 Fungi Evolution and Diversity 390
22.1 Evolution and Characteristics of Fungi 391
22.2 Diversity of Fungi 393
22.3 Symbiotic Relationships of Fungi 400

UNIT 5 Plant Evolution and Biology 406

23 Plant Evolution and Diversity 407
23.1 Ancestry and Features of Land Plants 408
23.2 Evolution of Bryophytes: Colonization of Land 411
23.3 Evolution of Lycophytes: Vascular Tissue 413
23.4 Evolution of Pteridophytes: Megaphylls 415
23.5 Evolution of Seed Plants: Full Adaptation to Land 418

24 Flowering Plants: Structure and Organization 429
24.1 Cells and Tissues of Flowering Plants 430
24.2 Organs of Flowering Plants 434
24.3 Organization and Diversity of Roots 436
24.4 Organization and Diversity of Stems 439
24.5 Organization and Diversity of Leaves 444

25 Flowering Plants: Nutrition and Transport 449
25.1 Plant Nutrition and Soil 450
25.2 Water and Mineral Uptake 454
25.3 Transport Mechanisms in Plants 458

26 Flowering Plants: Control of Growth Responses 469
26.1 Plant Hormones 470
26.2 Plant Growth and Movement Responses 477
26.3 Plant Responses to Phytochrome 483

27 Flowering Plants: Reproduction 488
27.1 Sexual Reproductive Strategies 489
27.2 Seed Development 494
27.3 Fruit Types and Seed Dispersal 496
27.4 Asexual Reproductive Strategies 500

UNIT 6 Animal Evolution and Diversity 504

28 Invertebrate Evolution 505
28.1 Evolution of Animals 506
28.2 The Simplest Invertebrates 512
28.3 Diversity of the Lophotrochozoans 516
28.4 Diversity of the Ecdysozoans 524
28.5 Invertebrate Deuterostomes 532

29 Vertebrate Evolution 536
29.1 The Chordates 537
29.2 The Vertebrates 539
29.3 The Fishes 540
29.4 The Amphibians 543
29.5 The Reptiles 546
29.6 The Mammals 551

30 Human Evolution 556
30.1 Evolution of Primates 557
30.2 Evolution of Humanlike Hominins 561
30.3 Evolution of Early Genus *Homo* 564
30.4 Evolution of Later Genus *Homo* 566

ISTUDY

UNIT 7 Comparative Animal Biology 572

31 Animal Organization and Homeostasis 573
31.1 Types of Tissues 574
31.2 Organs, Organ Systems, and Body Cavities 580
31.3 The Integumentary System 581
31.4 Homeostasis 584

32 Circulation and Cardiovascular Systems 589
32.1 Transport in Invertebrates 590
32.2 Transport in Vertebrates 592
32.3 The Human Cardiovascular System 594
32.4 Blood 602

33 The Lymphatic and Immune Systems 609
33.1 Evolution of Immune Systems 610
33.2 The Lymphatic System 611
33.3 Innate Immune Defenses 613
33.4 Adaptive Immune Defenses 616
33.5 Immune System Disorders and Hypersensitivity Reactions 625

34 Digestive Systems and Nutrition 629
34.1 Digestive Tracts 630
34.2 The Human Digestive System 633
34.3 Digestive Enzymes 638
34.4 Nutrition and Human Health 640

35 Respiratory Systems 646
35.1 Gas-Exchange Surfaces 647
35.2 Breathing and Transport of Gases 652
35.3 Respiration and Human Health 656

36 Body Fluid Regulation and Excretory Systems 663
36.1 Animal Excretory Systems 664
36.2 The Human Urinary System 667

37 Neurons and Nervous Systems 676
37.1 Evolution of the Nervous System 677
37.2 Nervous Tissue 680
37.3 The Central Nervous System 686
37.4 The Peripheral Nervous System 691

38 Sense Organs 698
38.1 Sensory Receptors 699
38.2 Chemical Senses 700
38.3 Sense of Vision 702
38.4 Senses of Hearing and Equilibrium 708
38.5 Somatic Senses 713

39 Locomotion and Support Systems 716
39.1 Diversity of Skeletons 717
39.2 The Human Skeletal System 718
39.3 The Muscular System 724

40 Hormones and Endocrine Systems 732
40.1 Animal Hormones 733
40.2 Hypothalamus and Pituitary Gland 737
40.3 Other Endocrine Glands and Hormones 739

41 Reproductive Systems 751
41.1 How Animals Reproduce 752
41.2 Human Male Reproductive System 754
41.3 Human Female Reproductive System 758
41.4 Control of Human Reproduction 762
41.5 Sexually Transmitted Diseases 766

42 Animal Development and Aging 773
42.1 Early Developmental Stages 774
42.2 Developmental Processes 777
42.3 Human Embryonic and Fetal Development 782
42.4 The Aging Process 789

43 Animal Behavior 794
43.1 Inheritance Influences Behavior 795
43.2 The Environment Influences Behavior 797
43.3 Animal Communication 801
43.4 Behavioral Ecology 805

UNIT 8 Ecology 812

44 Population Ecology 813
44.1 Scope of Ecology 814
44.2 Demographics of Populations 815
44.3 Population Growth Models 818
44.4 Regulation of Population Size 821
44.5 Life History Patterns 823
44.6 Human Population Growth 826
44.7 A Sustainable Future 828

45 Community and Ecosystem Ecology 835
45.1 Ecology of Communities 836
45.2 Community Development 844
45.3 Dynamics of an Ecosystem 847
45.4 Ecological Consequences of Climate Change 856

46 Major Ecosystems of the Biosphere 862
46.1 Climate and the Biosphere 863
46.2 Terrestrial Ecosystems 865
46.3 Aquatic Ecosystems 875

47 Conservation of Biodiversity 883
47.1 Conservation Biology and Biodiversity 884
47.2 Value of Biodiversity 886
47.3 Causes of Extinction 889
47.4 Conservation Techniques 894

Appendix
Periodic Table of Elements & The Metric System A

Glossary G

Index I

ISTUDY

(marine biologist): Images & Stories/Alamy Stock Photo; (natural river): ESB Professional/Shutterstock; (CT Brain): Gorodenkoff/Shutterstock; (microscope): Budimir Jevtic/Shutterstock

1

Biology: The Study of Life

Chapter Outline

1.1 The Characteristics of Life

1.2 Evolution and the Classification of Life

1.3 The Process of Science

1.4 Science and the Challenges Facing Society

Diversity in Science

Our planet is home to a staggering diversity of life. Not counting bacteria, it is estimated that over 8.7 million species—including ours, *Homo sapiens*—inhabit the globe. Some estimates place that number even higher. Furthermore, life can be found everywhere, from the deepest trenches in the oceans to the tops of the highest mountains. Biology is the area of scientific study that focuses on understanding all aspects of living organisms. To further our understanding of what it means to be alive, biologists explore life from the molecular level found in our genes all the way up to the large-scale interactions of multiple species in the environments around us.

Equally as important as the biodiversity of life on our planet is the diversity of those who study biology. It is because of our diverse experiences that scientists can ask a broad spectrum of questions, develop hypotheses, and design experiments or models to explain natural phenomena. Therefore, in order for the scientific community to address the challenges facing human society, from climate change to emerging diseases, we need a diverse population of individuals, with unique experiences and viewpoints, to contribute their ideas and opinions.

As we will see throughout this text, there are many ways to study our world, and our diversity is one of our greatest strengths in developing insights into understanding our world and in generating solutions for the problems it faces.

As you read through this chapter, think about the following questions:

1. What are some of the many ways a scientist can study biology?
2. Why would diversity in the scientific community play an important role in addressing how science can address the needs of society?

Following *the* Themes

CHAPTER 1 – BIOLOGY: THE STUDY OF LIFE

Evolution	Evolution is the unifying theme of biology and explains the history and diversity of life on this planet.
Nature of Science	The study of science involves the processes of making observations, forming hypotheses, and conducting experiments in an attempt to understand the principles of life.
Biological Systems	From communities of organisms to individual cells, all life is connected, from the smallest atom to the largest ecosystem.

1.1 The Characteristics of Life

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among the levels of biological organization.
2. Identify the basic characteristics of life.

Life. Everywhere we look, from the deepest trenches of the oceans to the geysers of Yellowstone, we find that planet Earth is teeming with life. Without life, our planet would be nothing but a barren rock hurtling through space. The variety of life on Earth is staggering. Recent estimates suggest there are around 8.7 million species on the planet, but this number does not include bacteria, which historically have been difficult to identify. The variety of living organisms ranges in size from bacteria, much too small to be seen by the naked eye, all the way up to 300-foot-tall giant sequoia trees or 40-ton humpback whales. Humans are just one of those species.

The science of **biology,** the subject of this textbook, is the study of life. As we will see, just as life is diverse, so is the study of biology. Biologists may study life from a variety of different perspectives.

Figure 1.1 illustrates the major groups of living organisms. Bacteria are widely distributed, microscopic organisms with a very simple structure. A *Paramecium* is an example of a microscopic protist, which are larger in size and more complex than bacteria. The other organisms in Figure 1.1 are easily seen with the naked eye. They can be distinguished by how they get their food. A morel is a fungus that digests its food externally. A sunflower is a photosynthetic plant that makes its own food, and a humpback whale is an aquatic animal that ingests its food.

Although life is tremendously diverse, it may be defined by several basic characteristics shared by all organisms. Like nonliving things, living organisms are composed of chemical elements. Also, organisms obey the same laws of chemistry and physics that govern everything in the universe. The characteristics of life, however, provide insight into the unique nature of life, and help to distinguish living organisms from nonliving things.

Life Is Organized

Life can be organized in a hierarchy of levels. These levels of biological organization (Fig. 1.2) begin with **atoms,** the basic units of matter. Atoms combine to form small **molecules,** which join to form larger molecules (macromolecules) within a **cell,** the smallest, most basic unit of life.

Some cells are independent organisms, such as single-celled bacteria. Many living organisms are **multicellular,** meaning they contain more than one cell. In multicellular organisms, similar cells may combine to form a **tissue**—for example, the nerve and muscle tissues of animals. Tissues make up **organs,** such as the brain or a leaf. Organs work together to form **organ systems;** for example, the brain works with the spinal cord and a network of nerves to form the nervous system. Organ systems are joined together to form an **organism,** such as an elephant.

The levels of biological organization extend beyond the individual organism. The members of similar organisms within a particular area belong to a **population.** A savanna in Africa may have populations of elephants and trees, for example. All of the populations of similar organisms that are capable of interbreeding are collectively called a **species,** such as the humans, zebras, and trees in Figure 1.2. They form a **community.** The community of populations interacts with the physical environment (water, land, climate) to form an **ecosystem.** Collectively, all the Earth's ecosystems make up the **biosphere.**

You should recognize from Figure 1.2 that each level of biological organization builds upon the previous level and is more complex. Moving up the hierarchy, each level acquires new, *emergent properties,* or new, unique characteristics, that are determined by the interactions between the individual parts. For example, when cells are broken down into bits of membrane and liquids, these parts themselves cannot carry out all the basic characteristics of life. However, all the levels of biological organization are interconnected and function as biological systems. For example, a change in the concentration of carbon dioxide (a small molecule) in the atmosphere may negatively influence the operation of organs, organisms, and entire ecosystems. As we will see, life is interconnected at a variety of levels.

Bacteria

Paramecium

Morel

Sunflower

Whale

Figure 1.1 The diversity of life. Biology is the scientific study of life. The photos shown are a sample of the many diverse forms of life found on planet Earth. (bacteria): Eye of Science/Science Source; (*Paramecium*): Michael Abbey/Science Source; (morel): Carol Wolfe; (sunflower): MedioImages/PunchStock/Getty Images; (whale in Alaska): Image Source/Getty Images

ISTUDY

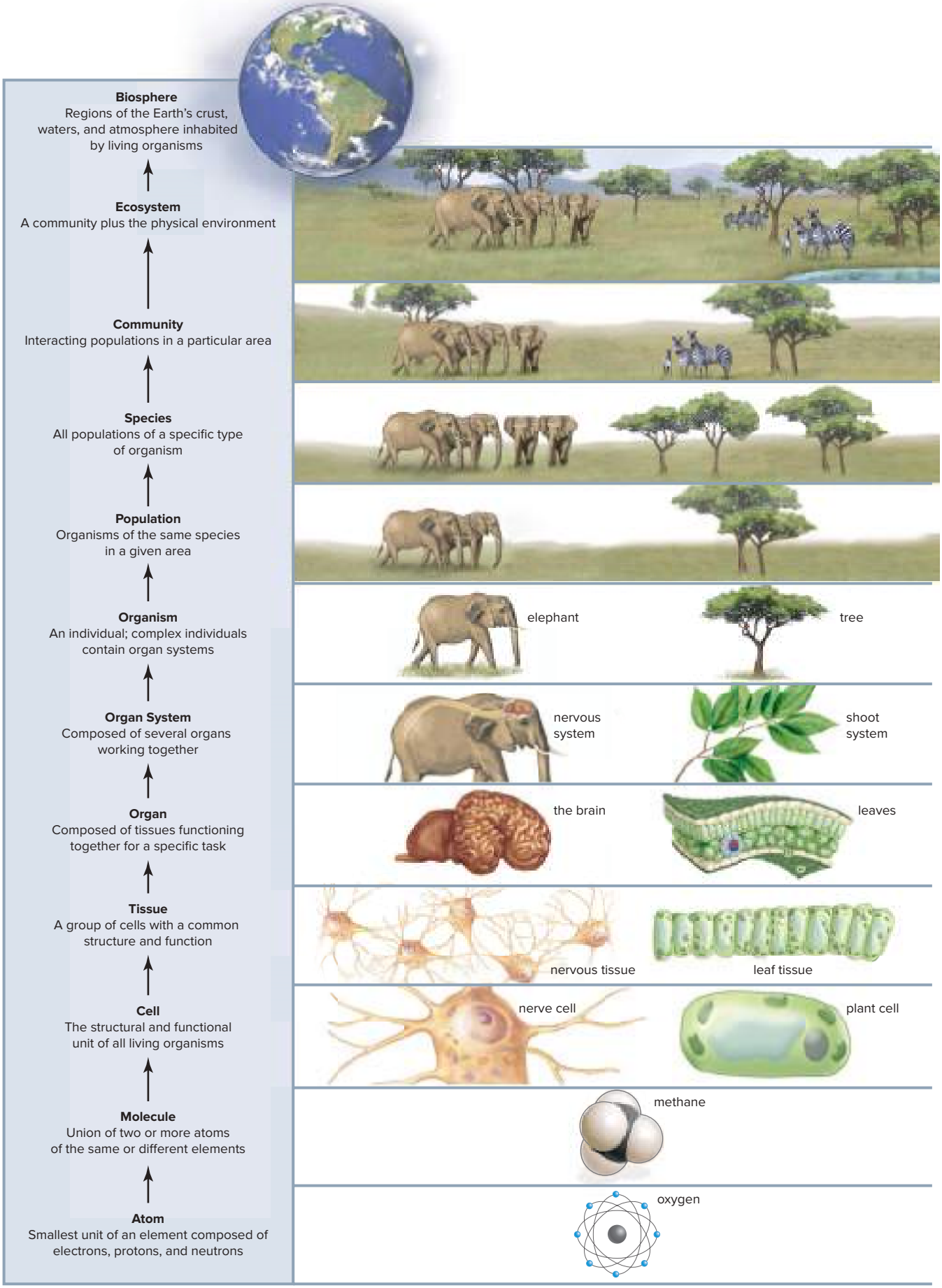


Figure 1.2 Levels of biological organization. All life is connected by levels of biological organization. These extend from atoms to the biosphere.

ISTUDY

Life Requires Materials and Energy

Living organisms cannot maintain their organization or carry on life's activities without an outside source of nutrients and energy (Fig. 1.3). Food provides nutrients, which are used as building blocks or for energy. **Energy** is the capacity to do work, and it takes work to maintain the organization of the cell and the organism. When cells use nutrient molecules to make their parts and products, they carry out a sequence of chemical reactions. The term **metabolism** (Gk. *meta,* "change") encompasses all the chemical reactions that occur in a cell.

The ultimate source of energy for nearly all life on Earth is the sun. Plants and certain other organisms are able to capture solar energy and carry on **photosynthesis,** a process that transforms solar energy into the chemical energy of organic nutrient molecules. All life on Earth acquires energy by metabolizing nutrient molecules made by photosynthetic organisms. This applies even to plants themselves.

The energy and chemical flow between organisms also defines how an ecosystem functions (Fig. 1.4). Within an ecosystem, chemical cycling and energy flow begin when producers, such as grasses, take in solar energy and inorganic nutrients to produce food (organic nutrients) by photosynthesis. Chemical cycling (aqua arrows in Fig. 1.4) occurs as chemicals move from one population to another in a food chain, until death and decomposition allow inorganic nutrients to be returned to the producers once again. Energy (red arrows), on the other hand, flows from the sun through plants and the other members of the food chain as they feed on one another. The energy gradually dissipates and returns to the atmosphere as heat. Because energy does not cycle, ecosystems could not stay in existence without solar energy and the ability of photosynthetic organisms to absorb it.

Energy flow and nutrient cycling in an ecosystem climate largely determine not only where different ecosystems are found in the biosphere but also which communities are found in the ecosystem. For example, deserts exist in areas of minimal rain, while forests require much rain. The two most biologically diverse ecosystems—tropical rain forests and coral reefs—occur where solar energy is most abundant. One example of an ecosystem in North America is the grasslands, which are inhabited by populations of rabbits, hawks, and various types of grasses, among many others. These populations interact with each other by forming food chains in which one population feeds on another. For example, rabbits feed on grasses, while hawks feed on rabbits and other organisms.

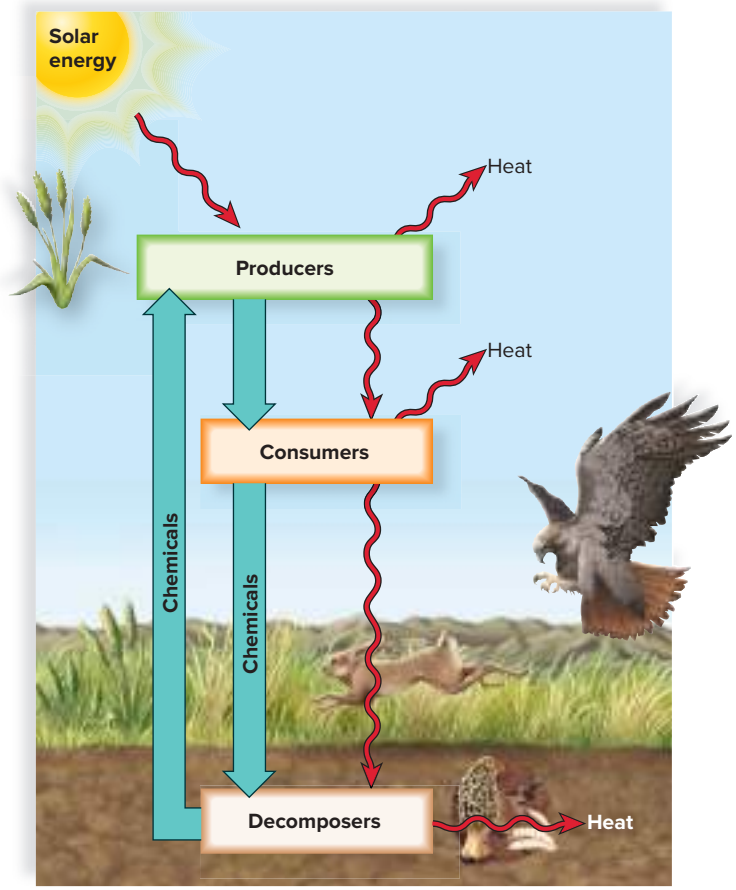


Figure 1.4 Chemical cycling and energy flow in an ecosystem. In an ecosystem, chemical cycling (aqua arrows) and energy flow (red arrows) begin when plants use solar energy and inorganic nutrients to produce their own food. Chemicals and energy are passed from one population to another in a food chain. Eventually, energy dissipates as heat. With the death and decomposition of organisms, chemicals are returned to living plants once more.

Figure 1.3 Acquiring nutrients and energy. All life, including this bear, the fish, and the trees, need to acquire energy. Justinreznick/E+/Getty Images

Living Organisms Maintain Homeostasis

To survive, it is imperative that an organism maintain a state of biological balance, or **homeostasis** (Gk. *homoios,* "like"; *stasis,* "the same"). For life to continue, temperature, moisture level, acidity, and other physiological factors must remain within the tolerance range of the organism. Homeostasis is maintained by systems that monitor internal conditions and make routine and necessary adjustments.

Organisms have intricate feedback and control mechanisms that do not require any conscious activity. These mechanisms may

ISTUDY

be controlled by one or more tissues themselves or by the nervous system. When you are studying and forget to eat lunch, your liver releases stored sugar to keep blood sugar levels within normal limits. Many organisms depend on behavior to regulate their internal environment. In animals, these behaviors are controlled by the nervous system and are usually not consciously controlled. For example, a lizard may raise its internal temperature by basking in the sun, or cool down by moving into the shade.

Living Organisms Respond

Living organisms interact with the environment, as well as with other organisms. Even single-celled organisms can respond to their environment. In some, the beating of microscopic hairs or, in others, the snapping of whiplike tails moves them toward or away from light or chemicals. Multicellular organisms can manage more complex responses. A vulture can detect a carcass a kilometer away and soar toward dinner. A monarch butterfly can sense the approach of fall and begin its flight south, where resources are still abundant.

The ability to respond often results in movement: The leaves of a land plant turn toward the sun, and animals dart toward safety. Appropriate responses help ensure the survival of the organism and allow it to carry on its daily activities. All together, these activities are termed the *behavior* of the organism. Organisms display a variety of behaviors as they maintain homeostasis and search and compete for energy, nutrients, shelter, and mates. Many organisms display complex communication, hunting, and defense behaviors.

Living Organisms Reproduce and Develop

Life comes only from life. All forms of life have the ability to **reproduce,** or make another organism like itself. Bacteria, protists, and other single-celled organisms simply split in two. In most multicellular organisms, the reproductive process begins with the pairing of a sperm from one partner and an egg from the other partner. The union of sperm and egg, followed by many cell divisions, results in an immature stage, which proceeds through stages of **development,** or change, to become an adult.

When living organisms reproduce, their **genes,** or genetic instructions, are passed on to the next generation. Random combinations of sperm and egg, each of which contains a unique collection of genes, ensure that the offspring has new and different characteristics. An embryo develops into a whale, a yellow daffodil, or a human because of the specific set of genes it inherits from its parents. In all organisms, the genes are made of long **DNA (deoxyribonucleic acid)** molecules. DNA provides the blueprint, or instructions, for the organization and metabolism of the particular organism. All cells in a multicellular organism contain the same set of genes, but only certain genes are turned on in each type of specialized cell. You may notice that not all members of a species are exactly the same, and that there are obvious differences among species. These differences are the result of **mutations,** or inheritable changes in the genetic information. Mutation provides an important source of variation in the genetic information. However, not all mutations are bad—the observable differences in eye and hair color are examples of mutations.

Mutations help create a staggering diversity of life, even within a group of otherwise identical organisms. Sometimes, organisms inherit characteristics that allow them to be more suited to their way of life.

Living Organisms Have Adaptations

Adaptations are modifications that make organisms better able to function in a particular environment. For example, penguins are adapted to an aquatic existence in the Antarctic. An extra layer of downy feathers is covered by short, thick feathers, which form a waterproof coat. Layers of blubber also keep the birds warm in cold water. Most birds have forelimbs proportioned for flying, but penguins have stubby, flattened wings suitable for swimming. Their feet and tails serve as rudders in the water, but the flat feet also allow them to walk on land. Penguins also have many behavioral adaptations to living in the Antarctic. They often slide on their bellies across the snow in order to conserve energy when moving quickly (Fig. 1.5). They carry their eggs—one or at most two—on their feet, where the eggs are protected by a pouch of skin. This also allows the birds to huddle together for warmth while standing erect and incubating the eggs.

From penguins to giant sequoia trees, life on Earth is very diverse because, over long periods of time, organisms respond to

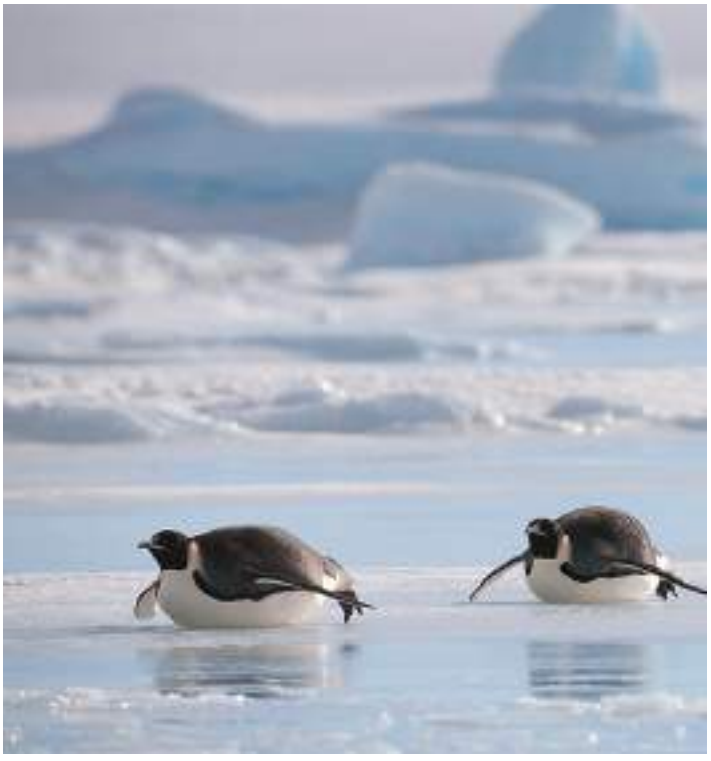


Figure 1.5 Living organisms have adaptations. Penguins have evolved complex behaviors, such as sliding across ice to conserve energy, to adapt to their environment. Paul Souders/Photodisc/Getty Images

ISTUDY

ever-changing environments by developing new adaptations. These adaptations are unintentional, but they provide the framework for evolutionary change. **Evolution** (L. *evolutio,* "an unrolling") includes the way in which populations of organisms change over the course of many generations to become more suited to their environments. All living organisms have the capacity to evolve, and the process of evolution constantly reshapes every species on the planet, potentially providing a way for organisms to persist, despite a changing environment. We will take a closer look at this process in the next section.

Check Your Progress 1.1

1. Distinguish between an ecosystem and a population in the levels of biological organization.
2. List the common characteristics of all living organisms.
3. Explain how adaptations relate to evolutionary change.

1.2 Evolution and the Classification of Life

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the relationship between the process of natural selection and evolutionary change.
2. Distinguish among the three domains of life.

Despite diversity in form, function, and lifestyle, organisms share the same basic characteristics. As mentioned, they are all composed of cells organized in a similar manner. Their genes are composed of DNA, and they carry out the same metabolic reactions to acquire energy and maintain their organization. The unity of life suggests they are descended from a common ancestor—the first cell or cells.

Evolution: The Core Concept of Biology

The phrase "common descent with modification" sums up the process of evolution, because it means that as descent occurs from common ancestors, so do modifications that cause organisms to be adapted to their environment. Through many observations and experiments, Charles Darwin came to the conclusion that **natural selection** is the process that makes modification—that is, adaptation—possible.

Natural Selection

During the process of natural selection, some aspect of the environment selects which traits are more apt to be passed on to the next generation. The selective agent can be an abiotic agent (part of the physical environment, such as altitude), or it can be a biotic agent (part of the living environment, such as a deer). Figure 1.6 shows how the dietary habits of deer might eventually affect the characteristics of the leaves of a particular land plant.

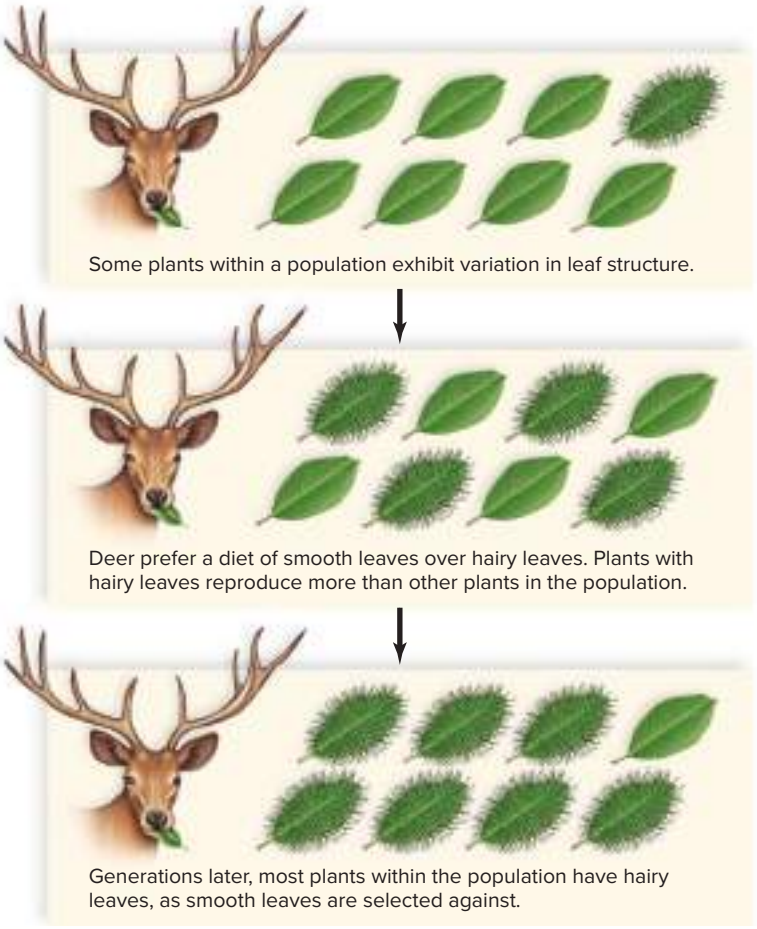


Figure 1.6 Natural selection. Natural selection selects for or against new traits introduced into a population by mutations. Over many generations, selective forces such as competition, predation, and the physical environment alter the makeup of a population, favoring those more suited to the environment and lifestyle.

Mutations fuel natural selection, because mutation introduces variations among the members of a population. In Figure 1.6, a plant species generally produces smooth leaves, but a mutation occurs that causes one plant to have leaves that are covered with small extensions, or "hairs." The plant with hairy leaves has an advantage, because the deer (the selective agent) prefer to eat smooth leaves, not hairy leaves. Therefore, the plant with hairy leaves survives best and produces more seeds than most of its neighbors. As a result, generations later most plants of this species produce hairy leaves.

As with this example, Darwin realized that although all individuals within a population have the potential to reproduce, not all do so with the same success. Prevention of reproduction can be the result of a number of factors, including an inability to capture resources, as when long-necked, but not short-necked, giraffes can reach their food source, or an inability to escape being eaten because long legs, but not short legs, can carry an animal to safety. As is presented in the Evolution feature, "Human Adaptation to Life at High Elevations," humans also exhibit adaptations to their environment.

Whatever the example, it can be seen that organisms with advantageous traits can produce more offspring than those that

ISTUDY

THEME Evolution

Human Adaptation to Life at High Elevations

Humans, like all other organisms, have an evolutionary history. This means not only that we share common ancestors with other animals, but also that, over time, we demonstrate adaptations to changing environmental conditions. One study of populations living in the high-elevation mountains of Tibet (Fig. 1A) demonstrates how the processes of evolution and adaptation influence humans.

Normally, if a person moves to a higher altitude, their body responds by making more hemoglobin, the component of blood that carries oxygen, which thickens the blood. For minor elevation changes, this does not present much of a problem. But for people who live at extreme elevations (some people in the Himalayas can live at elevations of over 13,000 ft, or close to 4,000 m), this can present a number of health problems, including chronic mountain sickness, a disease that affects people who live at high altitudes for extended periods of time. The problem is that, as the amount of hemoglobin increases, the blood thickens and becomes more viscous. This can cause elevated blood pressure, or hypertension, and an increase in the formation of blood clots, both of which have negative physiological effects.

Because high hemoglobin levels would be a detriment to people at high elevations, it makes sense that natural selection would favor individuals who produced less hemoglobin at high elevations. Such is the case with the Tibetans in this study. Researchers have identified an allele of a gene that reduces hemoglobin production at high elevations. Comparisons between Tibetans at both high and low elevations strongly suggest that selection has played a role in the prevalence of the high-elevation allele.

The gene is *EPSA1,* located on chromosome 2 of humans. *EPSA1* produces a transcription factor, which basically regulates which genes are turned on and off in the body, a process called gene expression. The transcription factor produced by *EPSA1* has a number of functions in the body. For example, in addition to controlling the amount of hemoglobin in the blood, this transcription factor also regulates other genes that direct how the body uses oxygen.

When the researchers examined the variations in *EPSA1* in the Tibetan population, they discovered that their version greatly reduces the production of hemoglobin. Therefore, the Tibetan population has lower hemoglobin levels than people living at lower altitudes, allowing these individuals to escape the consequences of thick blood.

How long did it take for the original population to adapt to living at higher elevations? Initially, the comparison of variations in these genes between high-elevation and low-elevation Tibetan populations suggested that the event may have occurred over a 3,000-year period. But researchers were skeptical of those data because they represented a relatively rapid rate of evolutionary change. Additional studies of genetic databases yielded an interesting finding—the *EPSA1* gene in Tibetans was identical to a similar gene found in an ancient group of humans called the Denisovans (see Section 30.4). Scientists now believe that the *EPSA1* gene entered the Tibetan population around 40,000 years ago, either through interbreeding between early Tibetans and Denisovans, or from one of the immediate ancestors of this lost group of early humans.

Figure 1A Humans' adaptations to their environments. Humans have adaptations that allow them to live at high altitudes, such as these individuals in Tibet. Michael Freeman/Photodisc/Getty Images

Questions to Consider

1. What other environments do you think could be studied to look for examples of human adaptation?
2. In addition to hemoglobin levels, do you think that people at high elevations may exhibit other adaptations?

lack them. In this way, living organisms change over time, and these changes are passed on from one generation to the next. Over long periods of time, the introduction of newer, more advantageous traits into a population may drastically reshape a species. Natural selection tends to sculpt a species to fit its environment and lifestyle and can create new species from existing ones. The end result is the diversity of life, which historically has been classified into three large groups: Bacteria, Archaea, and Eukarya (Fig. 1.7).

Organizing Diversity

An evolutionary tree is like a family tree. Just as a family tree shows how a group of people have descended from one couple, an evolutionary tree traces the ancestry of life on Earth to a common ancestor (Fig. 1.7). One couple can have diverse children, and likewise a population can be a common ancestor to several other groups, each adapted to a particular set of environmental conditions. In this way, over time, diverse life-forms have arisen. Evolution may be considered the unifying concept of biology, because it explains so many aspects of it, including how living organisms arose from a single ancestor.

Because life is so diverse, it is helpful to group organisms into categories. **Taxonomy** (Gk. *tasso,* "arrange"; *nomos,* "usage") is the discipline of identifying and grouping organisms according to certain rules. Taxonomy makes sense out of the bewildering variety of life on Earth and is meant to provide valuable insight into evolution. **Systematics** is the study of the evolutionary relationships between organisms. As systematists learn more about living

ISTUDY

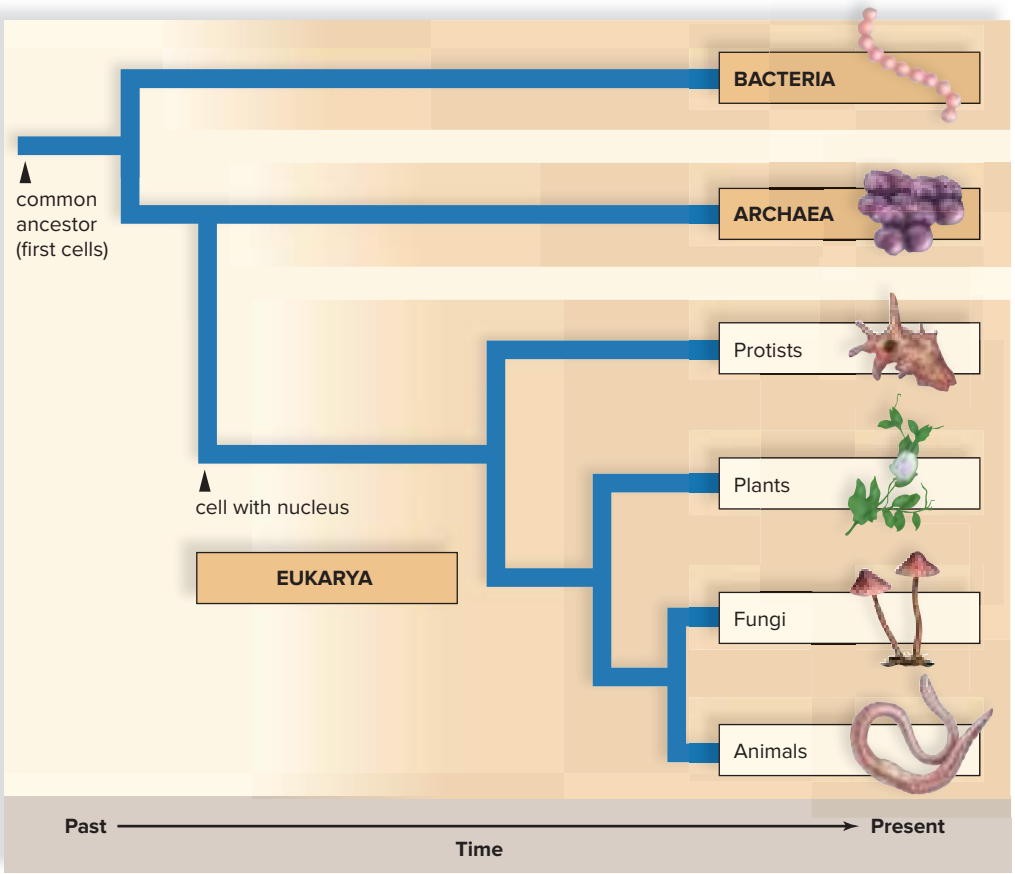


Figure 1.7 Evolutionary tree of life. As existing organisms change over time, they give rise to new species. Evolutionary studies show that all living organisms arose from a common ancestor about 4 billion years ago. Domain Archaea and domain Bacteria include the prokaryotes. Domain Eukarya includes both single-celled and multicellular organisms that possess a membrane-bound nucleus.

organisms, the taxonomy often changes. DNA technology is now widely used by systematists to revise current information and to discover previously unknown relationships between organisms.

Several of the basic classification categories, or *taxa,* going from least inclusive to most inclusive, are **species, genus, family, order, class, phylum, kingdom, supergroup,** and **domain** (Table 1.1). The least inclusive category, species (L. *species,* "model, kind"), is defined as a group of interbreeding individuals. Each successive classification category above species contains more types of organisms than the preceding one. Species placed within one genus share many specific characteristics and are the most closely related, while species placed in the same kingdom share only general characteristics with one another. For example, all species in the genus *Pisum* look pretty much the same—that is, like pea plants—but species in the plant kingdom can be quite varied, as is evident when we compare grasses to trees. Species placed in different domains are the most distantly related.

Table 1.1 Levels of Classification

Category	Human	Corn
Domain	Eukarya	Eukarya
Supergroup*	Opisthokonta	Archaeplastids
Kingdom	Animalia	Plantae
Phylum	Chordata	Anthophyta
Class	Mammalia	Monocotyledones
Order	Primates	Commelinales
Family	Hominidae	Poaceae
Genus	*Homo*	*Zea*
Species**	*H. sapiens*	*Z. mays*

*Supergroups are present only in domain Eukarya.
**To specify an organism, you must use the full binomial name, such as *Homo sapiens.*

Domains

Current biochemical evidence suggests there are three domains: **domain Bacteria, domain Archaea,** and **domain Eukarya.** Figure 1.7 shows the evolutionary relationship of these three domains. Both domain Bacteria and domain Archaea may have evolved from the first common ancestor soon after life began. These two domains contain the **prokaryotes,** which lack the membrane-bound nucleus found in the **eukaryotes** of domain Eukarya. However, archaea organize their DNA differently than bacteria, and their cell walls and membranes are chemically more similar to eukaryotes than to bacteria. So, the conclusion is that eukarya split off from the archaeal line of descent. Some scientists have suggested that biology actually be divided into a two-domain system instead, with one domain constituting the bacteria, and the other domain the archaeans and the eukaryotes, to reflect these relationships. For this text, we will proceed with the three-domain system.

Prokaryotes are structurally simple but metabolically complex. Archaea (Fig. 1.8) can live in aquatic environments that lack oxygen or are too salty, too hot, or too acidic for most other organisms. Perhaps these environments are similar to those of the primitive

Domain Archaea

Sulfolobus, an archaean

- Prokaryotic cells of various shapes
- Adaptations to extreme environments
- Absorb or chemosynthesize food
- Unique chemical characteristics

Figure 1.8 Domain Archaea. Eye of Science/Science Source

ISTUDY

Domain Bacteria

- Prokaryotic cells of various shapes
- Adaptations to all environments
- Absorb, photosynthesize, or chemosynthesize food
- Unique chemical characteristics

Escherichia coli, a bacterium

Figure 1.9 Domain Bacteria. A. Barry Dowsett/Science Source

Earth, and archaea (Gk. *archae,* "ancient") are the least evolved forms of life, as their name implies. Bacteria (Fig. 1.9) are variously adapted to living almost anywhere—in the water, soil, and atmosphere, as well as on our skin and in our mouth and large intestine.

Kingdoms and Supergroups

Historically, the classification of domain Eukarya divided organisms into one of four kingdoms (Fig. 1.10).

Protists (kingdom Protista), which comprise a very diverse group of organisms, range from single-celled forms to a few multicellular ones. Some are photosynthesizers, and some must acquire their food. Common protists include algae, the protozoans, and the water molds.

Figure 1.7 shows that plants, fungi, and animals evolved from protists. **Plants** (kingdom Plantae) are multicellular, photosynthetic organisms. Example plants include azaleas, zinnias, and pines. Among the **fungi** (kingdom Fungi) are the familiar molds and mushrooms that, along with bacteria, help decompose dead organisms. **Animals** (kingdom Animalia) are multicellular organisms that must ingest and process their food. Aardvarks, jellyfish, and zebras are representative animals.

However, the development of improved techniques in analyzing the DNA of organisms (see Section 14.1) suggests that not all of the protists share the same evolutionary lineage, meaning that the evolution of the eukaryotes has occurred along several paths. A new taxonomic group, called a **supergroup,** has been developed to explain these evolutionary relationships. There are currently six supergroups for domain Eukarya (Table 1.2). Over the past several years, changes have been made to the supergroup classification as new research is unveiling the relationships between these organisms. We will explore the structure of the eukaryotic supergroups in more detail in Section 21.1.

Table 1.2 Eukaryotic Supergroups

Supergroup	Sample Organisms
Excavata	Diplomonads, euglenozoans
Chromalveolata	Dinoflagellates, ciliates, diatoms, golden algae, brown algae, and water molds
Rhizaria	Foraminiferans, radiolarians
Archaeplastida	Red algae, green algae, plants
Amoebozoa	Amoeboids, slime molds
Opisthokonta	Fungi, choanoflagellates, animals

Domain Eukarya: Kingdom Protista

- Algae, protozoans, slime molds, and water molds
- Complex single cell (sometimes filaments, colonies, or even multicellular)
- Absorb, photosynthesize, or ingest food

Paramecium, a single-celled protozoan

Domain Eukarya: Kingdom Fungi

- Molds, mushrooms, yeasts, and ringworms
- Mostly multicellular filaments with specialized, complex cells
- Absorb food

Amanita, a mushroom

Domain Eukarya: Kingdom Plantae

- Certain algae, mosses, ferns, conifers, and flowering plants
- Multicellular, usually with specialized tissues, containing complex cells
- Photosynthesize food

Phalaenopsis, orchid, a flowering plant

Domain Eukarya: Kingdom Animalia

- Sponges, worms, insects, fishes, frogs, turtles, birds, and mammals
- Multicellular with specialized tissues containing complex cells
- Ingest food

Vulpes, a red fox

Figure 1.10 Domain Eukarya. Domain Eukarya currently consists of four kingdoms. (*Paramecium*): M. I. Walker/Science Source; (mushroom): Ingram Publishing/Getty Images; (orchid): Emilio Ereza/Pixtal/age fotostock; (fox): Fuse/Getty Images

Scientific Name

Biologists use **binomial nomenclature** to assign each living organism a two-part name called a scientific name. For example, the scientific name for mistletoe is *Phoradendron tomentosum.* The first word is the genus, and the second word is the species designation (also called a *specific epithet*) of each species within a genus. The genus may be

abbreviated (e.g., *P. tomentosum*) and, if the species has not been determined, it may simply be indicated with a generic abbreviation (e.g., *Phoradendron* sp.). Scientific names are universally used by biologists to avoid confusion. Common names tend to overlap and often differ depending on locality and the language of a particular country. But scientific names are based on Latin, a universally used language that not long ago was well-known by most scholars.

Check Your Progress 1.2

1. Explain how natural selection results in new adaptations within a species.
2. List the levels of taxonomic classification from most inclusive to least inclusive.
3. Describe the differences that might be used to distinguish among the various kingdoms of domain Eukarya.

1.3 The Process of Science

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the components of the scientific method.
2. Distinguish between a theory and a hypothesis.
3. Analyze a scientific experiment and identify the hypothesis, experiment, control groups, and conclusions.

The process of science pertains to the study of biology. As you saw in Figure 1.2, the multiple stages of biological organization mean that life can be studied at a variety of levels. Some biological disciplines are cytology, the study of cells; anatomy, the study of structure; physiology, the study of function; botany, the study of plants; zoology, the study of animals; genetics, the study of heredity; and ecology, the study of the interrelationships between organisms and their environment.

Religion, aesthetics, ethics, and science are all ways in which humans seek order in the natural world. The nature of scientific inquiry differs from other ways of knowing and learning, because the scientific process uses the **scientific method,** a standard series of steps used in gaining new knowledge that is widely accepted among scientists. The scientific method (Fig. 1.11) acts as a guideline for scientific studies. Scientists often modify or adapt the process to suit their particular field of study.

Observation

Scientists believe that nature is orderly and measurable—that natural laws, such as the law of gravity, do not change with time—and that a natural event, or *phenomenon,* can be understood more fully through **observation—**a formal way of "seeing what happens."

Scientists use all of their senses in making observations. The behavior of chimpanzees can be observed through visual means, the disposition of a skunk can be observed through olfactory means, and the warning rattles of a rattlesnake provide auditory information of imminent danger. Scientists also extend the ability of their senses by using instruments; for example, the microscope enables us to see objects that could never be seen by the naked eye. Finally, scientists may expand their understanding even further by taking advantage of the knowledge and experiences of other scientists. For instance, they may look up past studies at the library or on the internet, or they may write or speak to others who are researching similar topics.

Hypothesis

After making observations and gathering knowledge about a phenomenon, a scientist uses inductive reasoning to formulate a possible explanation. **Inductive reasoning** occurs whenever a person uses creative thinking to combine isolated facts into a cohesive whole. In some cases, chance alone may help a scientist arrive at an idea.

One famous case pertains to the antibiotic penicillin, which was discovered in 1928. While examining a petri dish of bacteria

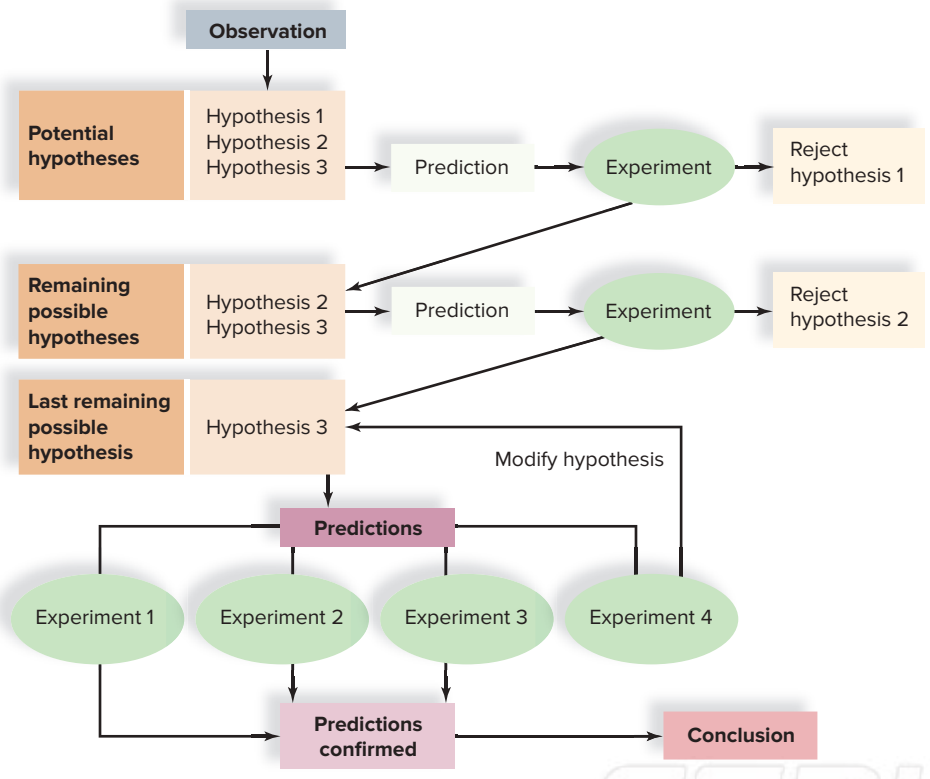


Figure 1.11 Flow diagram for the scientific method. On the basis of new and/or previous observations, a scientist formulates a hypothesis. The hypothesis is used to develop predictions to be tested by further experiments and/or observations, and new data either support or do not support the hypothesis. Following an experiment, a scientist often chooses to retest the same hypothesis or to test a related hypothesis. Conclusions from many different but related experiments may lead to the development of a scientific theory. For example, studies pertaining to development, anatomy, and fossil remains all support the theory of evolution.

ISTUDY

that had become contaminated with the mold *Penicillium,* Alexander Flemming observed an area that was free of bacteria. Flemming, an early expert on antibacterial substances, reasoned that the mold might have been producing an antibacterial compound.

We call such a possible explanation for a natural event a **hypothesis.** A hypothesis is not merely a guess; rather, it is an informed statement that can be tested in a manner suited to the processes of science.

All of a scientist's past experiences, no matter what they might be, have the potential to influence the formation of a hypothesis. But a scientist considers only hypotheses that can be tested. Moral and religious beliefs, while very important in the lives of many people, differ among cultures and through time and may not be scientifically testable.

Predictions and Experiments

Scientists often perform an **experiment,** which is a series of procedures designed to collect data for the purpose of testing a hypothesis. To determine how to test a hypothesis, a scientist uses deductive reasoning. **Deductive reasoning** involves "if, then" logic. In designing the experiment, the scientist may make a **prediction,** or an expected outcome, based on knowledge of the factors in the experiment.

The manner in which a scientist intends to conduct an experiment is called the **experimental design.** A good experimental design ensures that scientists are examining the contribution of a specific variable, called the **experimental variable,** to the observation. The result is termed the **responding variable,** or dependent variable, because it is due to the experimental variable:

Experimental Variable (Independent Variable)	**Responding Variable** (Dependent Variable)
Factor of the experiment being tested	Result or change that occurs due to the experimental variable

To ensure that the results will be meaningful, an experiment contains both test groups and a **control** group. A test group is exposed to the experimental variable, but the control group is not. If the control group and test groups show the same results, the experimenter knows that the hypothesis predicting a difference between them is not supported.

Scientists often use a **model** to test a hypothesis. Model organisms, such as the fruit fly *Drosophila melanogaster* or the mouse *Mus musculus,* are often chosen because they allow the researcher to control aspects of the experiment, such as age and genetic background. Cell biologists may use mice for modeling the effects of a new drug. Like model organisms, model systems allow the scientist to control specific variables and environmental conditions in a way that may not be possible in the natural environment. For example, ecologists may use computer programs to model how human activities will affect the climate of a specific ecosystem. While models provide useful information, they do not always answer the original question completely. For example, medicine that is effective in mice should ideally be tested in humans, and ecological experiments that are conducted using computer simulations need to be verified by field experiments. Biologists, and all other scientists, continuously design and revise their experiments to better understand how different factors may influence their original observation.

Presenting and Analyzing the Data

The **data,** or results, from scientific experiments may be presented in a variety of formats, including tables and graphs. A graph shows the relationship between two quantities. In many graphs, the experimental variable is plotted on the *x*-axis (horizontal), and the result is plotted along the *y*-axis (vertical). Graphs are useful tools to summarize data in a clear and simplified manner. For example, the line graph in Figure 1.12 shows the variation in the concentration of blood cholesterol over a 4-week study. The bars above each data point represent the variation, or standard error, in the results. The title and labels can assist you in reading a graph; therefore, when looking at a graph, first check the two axes to determine what the graph pertains to. By looking at this graph, we know that the blood cholesterol levels were highest during week 2, and we can see to what degree the values varied over the course of the study.

Statistical Data

Most authors who publish research articles use statistics to help them evaluate their experimental data. In statistics, the standard error, or standard deviation, tells us how uncertain a particular value is. Suppose you predict how many hurricanes Florida will have next year by calculating the average number during the past 10 years. If the number of hurricanes per year varies widely, your standard error will be larger than if the number per year is usually about the same. In other words, the standard error tells you how far off the average could be. If the average number of hurricanes is four and the standard error is ±2, then your prediction of four hurricanes is between two and six hurricanes. In Figure 1.12, the standard error is represented by the bars above and below each data point. This provides a visual indication of the statistical analysis of the data.

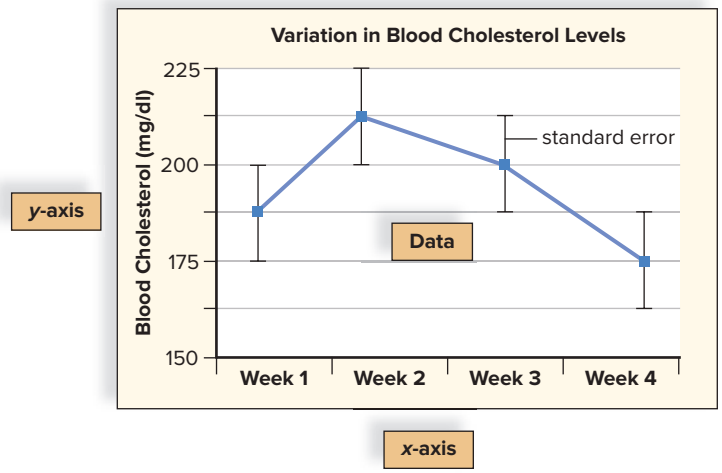


Figure 1.12 Presentation of scientific data. This line graph shows the variation in the concentration of blood cholesterol over a 4-week study. The bars above each data point represent the variation, or standard error, in the results.

Statistical Significance

When scientists conduct an experiment, there is always the possibility that the results are due to chance or to some factor other than the experimental variable. Investigators take into account several factors when they calculate the probability value (p) that their results were due to chance alone. If the probability value is low, researchers describe the results as statistically significant. A probability value of less than 5% (usually written as $p < 0.05$) is acceptable; even so, keep in mind that the lower the p value, the less likely it is that the results are due to chance. Therefore, the lower the p value, the greater the confidence the investigators and you can have in the results. Depending on the type of study, most scientists like to have a p value of < 0.05, but p values of < 0.001 are common in many studies.

Scientific Publications

Scientific studies are customarily published in scientific journals, such as *Nature* and *Science,* so that all aspects of a study are available to the scientific community. Before information is published in scientific journals, it is typically reviewed by experts, who ensure that the research is credible, accurate, unbiased, and well-executed. Another scientist should be able to read about an experiment in a scientific journal, repeat the experiment in a different location, and get the same (or very similar) results. Some articles are rejected for publication by reviewers when they believe there is something questionable about the design of an experiment or the manner in which it was conducted. This process of rejection is important in science because it causes researchers to critically review their hypotheses, predictions, and experimental designs, so that their next attempt will more adequately address their hypothesis. Often, it takes several rounds of revision before research is accepted for publication in a scientific journal.

Scientific magazines, such as *Scientific American* and *New Scientist,* differ from scientific journals in that they report scientific findings to the general public. The information in these articles is usually obtained from articles first published in scientific journals.

Scientific Theory

The ultimate goal of science is to understand the natural world in terms of scientific **theories,** which are concepts that join together well-supported and related hypotheses. In ordinary speech, the word *theory* refers to a speculative idea. In contrast, a scientific theory is supported by a broad range of observations, experiments, and data, often from a variety of disciplines. Some of the basic theories of biology are:

Theory	*Concept*
Cell	All organisms are composed of cells, and new cells come only from preexisting cells.
Homeostasis	The internal environment of an organism stays relatively constant—within a range that is protective of life.
Evolution	All living organisms have a common ancestor, but each is adapted to a particular way of life.

As stated earlier, the theory of evolution is the unifying concept of biology because it pertains to many different aspects of life. For example, the theory of evolution enables scientists to understand the history of life, as well as the anatomy, physiology, and embryological development of organisms. Even behavior can be described through evolution, as we will see in a study discussed later in this chapter.

The theory of evolution has been a fruitful scientific theory, meaning that it has helped scientists generate new hypotheses. Because this theory has been supported by so many observations and experiments for over 100 years, some biologists refer to the **principle** of evolution, a term sometimes used for theories that are generally accepted by an overwhelming number of scientists. The term **law** instead of *principle* is preferred by some. For instance, in a subsequent chapter concerning energy relationships, we will examine the laws of thermodynamics.

An Example of the Scientific Method

We now know that most stomach and intestinal ulcers (open sores) are caused by the bacterium *Helicobacter pylori*. Let's say investigators want to determine which of two antibiotics is best for the treatment of an ulcer. When clinicians do an experiment, they try to vary just the experimental variables—in this case, the medications being tested. A control group is not given the medications, but one or more test groups receive them. If by chance the control group shows the same results as a test group, the investigators immediately know that the results of their study are invalid, because the medications may have had nothing to do with the results. The study depicted in Figure 1.13 shows how investigators may study this hypothesis.

Hypothesis: Newly discovered antibiotic B is a better treatment for ulcers than antibiotic A, which is in current use.

Next, the investigators might decide to use three experimental groups: one control group and two test groups. It is important to reduce the number of possible variables (differences), such as sex, weight, and other illnesses, among the groups. Therefore, the investigators randomly divide a very large group of volunteers equally into the three groups. The hope is that any differences will be distributed evenly among the three groups. This is possible only if the investigators have a large number of volunteers. The three groups are to be treated like this:

- *Control group:* Subjects with ulcers are not treated with either antibiotic.
- *Test group 1:* Subjects with ulcers are treated with antibiotic A.
- *Test group 2:* Subjects with ulcers are treated with antibiotic B.

After the investigators have determined that all volunteers do have ulcers, they will want the subjects to think they are all receiving the same treatment. This is an additional way to protect the results from any influence other than the medication. To achieve this end, the subjects in the control group can receive a placebo, a treatment that appears to be the same as that administered to the other two groups but actually contains no medication. In this study, the use of a placebo would help ensure the same dedication by all subjects to the study.

ISTUDY

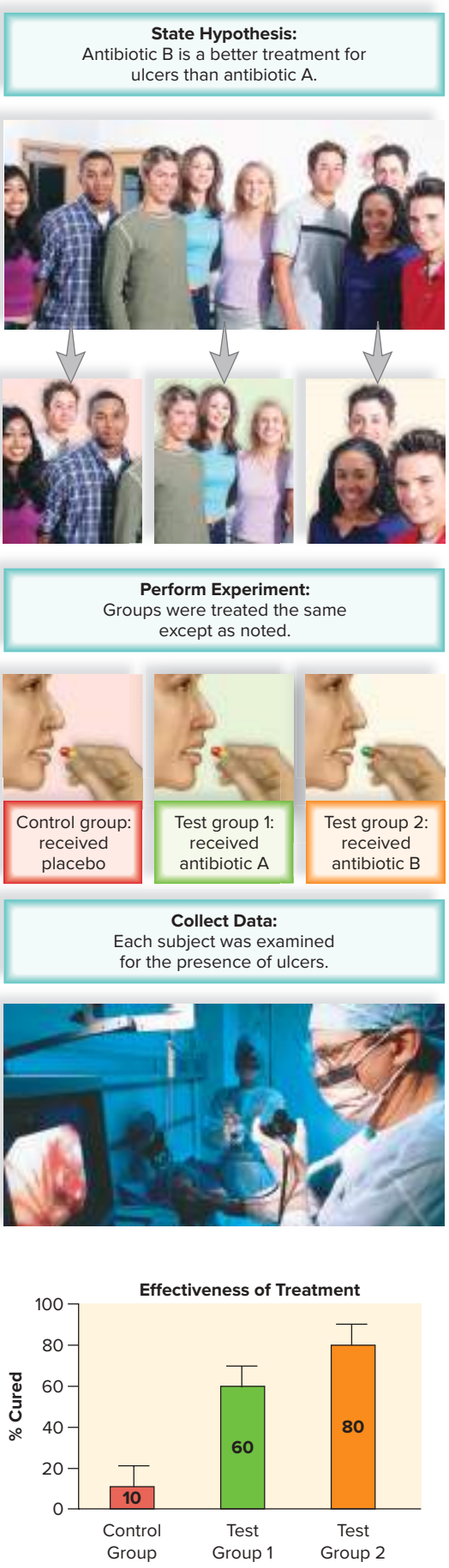


Figure 1.13 Example of a controlled study. In this study, a large number of people were divided into three groups. The control group received a placebo and no medication. One of the test groups received medication A, and the other test group received medication B. The results are depicted in a graph, and it shows that medication B was a more effective treatment than medication A for the treatment of ulcers. (students): Corbis/VCG/image100/Getty Images; (surgery): ALIX/Science Source

Results and Conclusion

After 2 weeks of administering the same amount of medication (or placebo) in the same way, the stomach and intestinal linings of each subject are examined to determine if ulcers are still present. Endoscopy is a procedure that involves inserting an endoscope (a small, flexible tube with a tiny camera on the end) down the throat and into the stomach and the upper part of the small intestine. Then, the doctor can see the lining of these organs and can check for ulcers. Tests performed during an endoscopy can also determine if *Helicobacter pylori* is present.

Because endoscopy is somewhat subjective, it is probably best if the examiner is not aware of which group the subject is in; otherwise, the examiner's prejudice may influence the examination. When neither the patient nor the technician is aware of the specific treatment, it is called a double-blind study.

In this study, the investigators may decide to determine the effectiveness of the medication by the percentage of people who no longer have ulcers. So, if 20 people out of 100 still have ulcers, the medication is 80% effective. The difference in effectiveness is easily read in the graph portion of Figure 1.13.

Conclusion: On the basis of their data, the investigators conclude that their hypothesis has been supported.

Check Your Progress 1.3

1. Identify the role of the experimental variable in an experiment.
2. Distinguish between the roles of the test group and the control group in an experiment.
3. Describe the process by which a scientist may test a hypothesis about an observation.

1.4 Science and the Challenges Facing Society

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between science and technology.
2. Summarize the major challenges facing science and society.

As we have learned in this chapter, science is a systematic way of acquiring knowledge about the natural world. It is a slightly different endeavor than technology. **Technology** is the

application of scientific knowledge to the interests of humans, and scientific investigations are the basis for the majority of our technological advances. As is often the case, a new technology, such as your cell phone or a new drug, is based on years of scientific investigations. However, despite our scientific and technological advances, society faces many challenges. In this section, we will explore a few of those critical challenges that scientists are actively investigating.

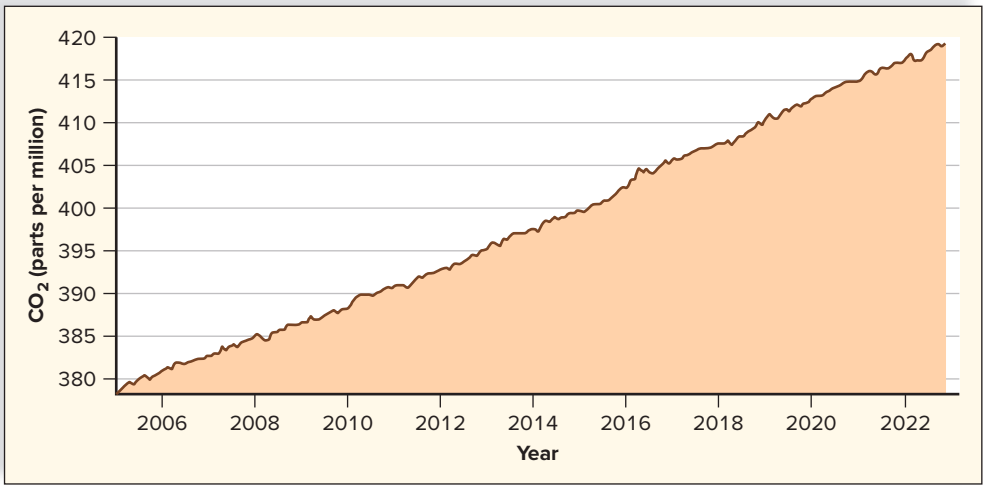


Figure 1.14 Increases in atmospheric carbon dioxide concentrations. The global average carbon dioxide concentration now exceeds 410 ppm and is contributing to climate change and global warming. Source: NOAA, "Global Climate Change: Facts," http://climate.nasa.gov/vital-signs/carbon-dioxide/

Climate Change

Scientists have identified that the single greatest challenge facing science and society, and the greatest threat to both humans and the environment, is climate change. The term **climate change** refers to changes in the normal cycles of the Earth's climate that may be attributed to human activity. Climate change is primarily due to an imbalance in the chemical cycling of the element carbon. Normally, carbon is cycled within an ecosystem. However, due to human activities, more carbon dioxide is being released into the atmosphere than is being removed. In 1850, atmospheric CO_2 was at about 280 parts per million (ppm); today, it is over 420 ppm (Fig. 1.14). This increase is largely due to the burning of fossil fuels and the destruction of forests to make way for farmland and pasture. Today, the amount of carbon dioxide released into the atmosphere is about twice the amount that remains in the atmosphere. It's believed that most of this dissolves in the oceans, which is increasing their acidity. The increased amount of carbon dioxide (and other gases) in the atmosphere is causing a rise in temperature called **global warming.** These gases allow the sun's rays to pass through, but they absorb and radiate heat back to Earth, a phenomenon called the *greenhouse effect.*

There is a consensus among scientists from around the globe that climate change and global warming are causing significant changes in many of the Earth's ecosystems and represent one of the greatest challenges of our time. Throughout this text, we will return to how climate change is affecting ecosystems, reducing biodiversity, and contributing to human disease. We will examine climate change in more detail in Section 45.4.

Biodiversity and Habitat Loss

Biodiversity is the total number and relative abundance of species, the variability of their genes, and the different ecosystems in which they live. The biodiversity of our planet has been estimated to be around 8.7 million species (not counting bacteria), and so far, approximately 2.3 million have been identified and named. **Extinction** is the death of a species or larger classification category. It is estimated that presently we are losing hundreds of species every year due to human activities and that as much as 27% of all species, including most primates, birds, and amphibians, may be in danger of extinction before the end of the century. In many cases, these extinctions are associated with climate change (Fig. 1.15). Many biologists are alarmed about the present rate of extinction and hypothesize it may eventually rival the rates of the five mass extinctions that occurred during our planet's history. The last mass extinction, about 65 million years ago, caused many plant and animal species, including the dinosaurs, to become extinct.

Figure 1.15 Climate change and extinction. The Bramble Cay melomys (*Melomys rubicola*) was native to Australian great barrier reef islands. However, it has not been observed since 2009 and is believed to be the first mammalian species to go extinct as a result of climate change. Polaris/Newscom

ISTUDY

The two most biologically diverse ecosystems—tropical rain forests and coral reefs—are home to many organisms. These ecosystems are also threatened by human activities. The canopy of the tropical rain forest alone supports a variety of organisms, including orchids, insects, and monkeys. Coral reefs, which are found just offshore of the continents and islands near the equator, are built up from calcium carbonate skeletons of sea animals called corals. Reefs provide a habitat for many animals, including jellyfish, sponges, snails, crabs, lobsters, sea turtles, moray eels, and some of the world's most colorful fishes. Like tropical rain forests, coral reefs are severely threatened as the human population increases in size. Some reefs are 50 million years old, yet in just a few decades, human activities have destroyed an estimated 25% of all coral reefs and seriously degraded another 30%. At this rate, nearly three-quarters could be destroyed within the next few decades. Similar statistics are available for tropical rain forests.

The destruction of healthy ecosystems has many unintended effects. For example, we depend on them for food, medicines, and various raw materials. Draining of the natural wetlands of the Mississippi and Ohio Rivers and the construction of levees have worsened flooding problems, making once fertile farmland undesirable. The destruction of South American rain forests by intentional fires has killed many species that may have yielded the next miracle drug and has decreased the availability of many types of lumber. We are only now beginning to realize that we depend on ecosystems even more for the services they provide. Just as chemical cycling occurs within a single ecosystem, so all ecosystems keep chemicals cycling throughout the biosphere. The workings of ecosystems ensure that the environmental conditions of the biosphere are suitable for the continued existence of humans. And several studies show that ecosystems cannot function properly unless they remain biologically diverse. We will explore the concept of biodiversity in greater detail in Chapters 44 through 47.

Emerging Diseases

Over the past decade, avian influenza (H5N1 and H7N9), swine flu (H1N1), severe acute respiratory syndrome (SARS), and Middle East respiratory syndrome (MERS) have been in the news. In 2020, a global pandemic called COVID-19 was caused by a new form of SARS virus named SARS-CoV-2. These are all called **emerging diseases** because they are new to humans.

Reemerging diseases are also a concern. Unlike an emerging disease, a reemerging disease has been known to cause disease in humans for some time, but generally has not been considered a health risk due to a relatively low level of incidence in human populations. However, reemerging diseases can cause problems. An excellent example is the Ebola outbreak in West Africa of 2014–2015. Ebola outbreaks have been known since 1976, but generally have affected only small groups of humans. The 2014–2015 outbreak was a much larger event. Although the exact numbers may never be known, it is estimated that over 28,000 people were infected, with over 11,000 fatalities. Smaller outbreaks occurred in Africa in 2016 and 2018. These outbreaks have the potential to disrupt the societies of several West African nations.

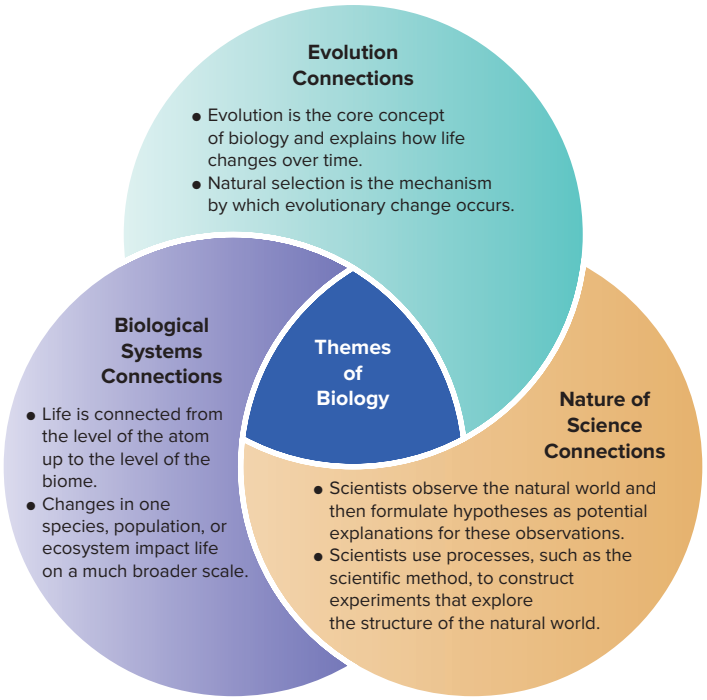
As we have learned from COVID-19, both emerging and r eemerging diseases have the potential to cause health problems for humans across the globe. Scientists investigate not only the causes of these diseases (for example, the viruses) but also their effects on our bodies and the mechanisms by which they are trans-mitted. We will take a closer look at viruses in Section 20.1.

Check Your Progress 1.4

1. Explain how a new technology differs from a scientific discovery.
2. Explain why the conservation of biodiversity is important to human society.
3. Summarize how emerging diseases and climate change have the potential for influencing the entire human population.

ISTUDY

Connecting *the* Concepts



What Did You Learn?

1.1 The Characteristics of Life

Biology is the area of science that studies life. Although living organisms are diverse, they have certain characteristics in common, such as the following:

- *Living organisms are organized.* **Atoms** and **molecules** are the nonliving components of cells. All organisms are composed of cells (the basic unit of life), but they may be single-celled or **multicellular.** Multicellular organisms may contain **tissues, organs,** and **organ systems.** Above the level of the cell, organisms are organized into **populations, species,** and **communities. Ecosystems** and the **biosphere** represent the highest levels of biological organization.
- *Living organisms require materials and energy.* All living organisms need an outside source of materials and **energy. Metabolism** is the term used to summarize these chemical reactions in the cell. Photosynthesis is an example of a metabolic process.
- *Living organisms maintain homeostasis.* **Homeostasis** is the ability to maintain a stable internal environment.
- *Living organisms respond to stimuli.* These stimuli help an organism react to changes in its environment, such as the presence of food.
- *Living organisms reproduce and develop.* Organisms reproduce to pass on their genetic information, included in the **genes** of their **DNA (deoxyribonucleic acid),** to the next generation. **Mutations** introduce variation into the DNA. **Development** is the series of steps that an organism proceeds through to become an adult.
- *Living organisms have adaptations.* **Adaptations** allow an organism to exist in a particular environment. **Evolution** is the accumulation of these changes over multiple generations.

1.2 Evolution and the Classification of Life

Life on Earth is diverse, but the theory of evolution unifies life and describes how all living organisms evolved from a common ancestor. **Natural selection** describes the process by which living organisms are descended from a common ancestor. Mutations occur within a population, creating new traits. The agents of natural selection, present in both biological and physical environments, shape species over time and may create new species from existing ones.

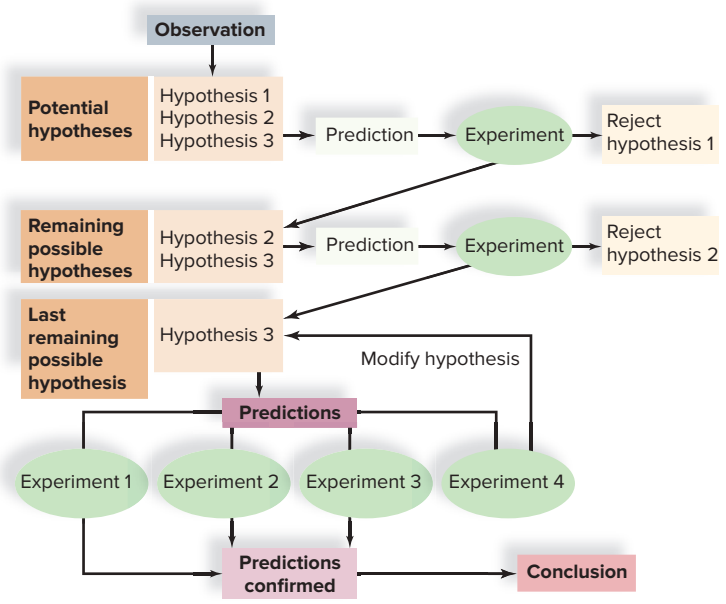
In **taxonomy,** organisms are assigned an italicized **binomial nomenclature** that consists of the genus and the specific epithet. From the least inclusive to the most inclusive category, each **species** belongs to a **genus, family, order, class, phylum, kingdom, supergroup,** and finally, **domain.** Systematics is the study of evolutionary relationships among species.

- The three domains of life are Archaea, Bacteria, and Eukarya. **Domain Archaea** and **domain Bacteria** contain **prokaryotes**, single-celled organisms that are structurally simple but metabolically complex.
- **Domain Eukarya** contains **eukaryotes,** which includes protists, fungi, plants, and animals. **Protists** range from single-celled to multicellular organisms and include the protozoans and most algae. Among the **fungi** are the familiar molds and mushrooms. **Plants** are well-known as the multicellular photosynthesizers of the world, while **animals** are multicellular and ingest their food. Organisms in domain Eukarya are also classified into supergroups to reflect evolutionary relationships.

ISTUDY

1.3 The Process of Science

When studying the natural world, scientists use a process called the **scientific method.**



- **Observations,** along with previous data, are used to formulate a hypothesis. **Inductive reasoning** allows a scientist to combine facts into a **hypothesis.**
- New observations and/or **experiments** are carried out in order to test the hypothesis. **Deductive reasoning** allows for the development of a **prediction** of what may occur as a result of the experiment. A good **experimental design** includes an **experimental variable** and a **control** group. Scientists may use models and **model** organisms in their experimental design.
- The **data** from the experimental and observational results are analyzed, often using statistical methods. The results are frequently presented in tables or graphs for ease of interpretation.
- A conclusion is made as to whether the results support the hypothesis or do not support the hypothesis.
- The results may be submitted to a scientific publication for review by the scientific community.
- Over time, multiple conclusions in a particular area may allow scientists to arrive at a **theory** (or **principle** or **law**), such the cell theory or the theory of evolution. The theory of evolution is a unifying concept of biology.

1.4 Science and the Challenges Facing Society

While science investigates the principles of the natural world, **technology** applies this knowledge to the needs of society. Some challenges that scientists are investigating include:

- The impact of **climate change** and **global warming.**
- The loss of **biodiversity** and habitats such as coral reefs and rain forests. This often results in the **extinction** of species.
- **Emerging diseases,** such as avian influenza, COVID-19, and MERS.

Thinking Critically About Biology

1. You are a scientist working at a pharmaceutical company and have developed a new cancer medication that has the potential for use in humans. Outline a series of experiments, including the use of a model, to test whether the cancer medication works.
2. You want to grow large tomatoes, and you notice that a name-brand fertilizer claims to produce larger plants than a generic brand. How would you test this claim?
3. Scientists are currently exploring the possibility that life may exist on some of the planets and moons of our solar system. Suppose you were a scientist on one of these research teams and were tasked with determining whether a new potential life-form exhibited the characteristics of behavior or adaptation. What would be your hypothesis? What types of experiments would you design?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Characteristics of Life

McGraw Hill

How do you exhibit the general characteristics of life in your daily activities?

Making It Relevant

1. Why is it important for the scientific community to be diverse in order to effectively address the needs facing society due to climate change?
2. How might a diverse scientific community be able to better address the loss of biodiversity at a local level?
3. Emerging diseases often occur in rural areas of the globe, where scientific education is minimal. What are some ways that a diverse, and global, community of scientists can help reduce the risk of these diseases becoming a problem?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 1

The Cell

In this unit, we will begin to explore the basic unit of life, the cell. As we observed in Section 1.1, all life, even a simple single cell, shares some general characteristics. However, before we explore the various functions of the cell, we need to understand the structure and function of a cell. For that, we shall do a quick overview of chemistry. We will also look at the nature of water, which is one of the most important molecules for life as we know it. Once we have covered the chemical foundation of life, we will discuss organic molecules, such as carbohydrates and proteins, which perform the functions of the cell.

Cells, being alive, must acquire energy and materials and maintain an internal environment by homeostasis. The majority of the chapters in this unit will help you develop an understanding of how cells accomplish these goals. We will also explore how the cell's structure relates to its function, either as a single-celled organism or as part of a multicellular tissue, organ, or organism. Later units will discuss the process of cellular reproduction and response to stimuli.

Because the cell forms the foundation for all life, your understanding of these concepts will serve you well as you move into later parts of this text.

Unit Outline

Chapter 2 Basic Chemistry

Chapter 3 The Chemistry of Organic Molecules

Chapter 4 Cell Structure and Function

Chapter 5 Membrane Structure and Function

Chapter 6 Metabolism: Energy and Enzymes

Chapter 7 Photosynthesis

Chapter 8 Cellular Respiration

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Examine how inanimate elements can be combined to produce a living cell.
Nature of Science	Describe how science is used to investigate cellular phenomena.
Biological Systems	Evaluate how cellular components work together in order to function and live.

ISTUDY

2

Basic Chemistry

Paopano/iStock/Getty Images

Life Outside Our Solar System

Ever since the first exoplanet—a planet located outside our solar system—was discovered in 1992, scientists have been expanding their search for Earth-like exoplanets. Over the past few years, a number of new space-based telescopes, such as the Kepler, Hubble, and the recently launched TESS and Webb telescopes, have been delivering a wealth of new planetary discoveries.

Attention has focused on the "habitable zone," regions around a star where the amount of light is believed to allow for liquid water. Recent discoveries have detected a number of interesting candidates. For example, Proxima b is located around our nearest stellar neighbor, and Trappist-1 has seven planets within the habitable zone. To date, over 5,000 exoplanets have been discovered, and telescopes such as Webb are now beginning to study the composition of their atmosphere.

All life as we know it on Earth uses the same elements (carbon, hydrogen, phosphorous, etc.) and survives within specific ranges of both temperature and pH. Understanding the chemical composition of these exoplanets may help scientists determine if conditions elsewhere in the universe are similar to those that exist on Earth.

In this chapter, we will explore the building blocks of all matter—atoms—and the importance of water to life as we know it.

As you read though this chapter, think about the following questions:

1. Why are scientists looking for water on these exoplanets?
2. Why is water considered to be so important to life?

Chapter Outline

2.1 Chemical Elements
2.2 Molecules and Compounds
2.3 Chemistry of Water
2.4 Acids and Bases

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.1 What are the general characteristics shared by all living organisms?

Section 1.3 How does the scientific process help us understand the natural world?

Following *the* Themes

CHAPTER 2 – BASIC CHEMISTRY

Evolution	Chemicals form the basis of living organisms, which in turn evolve by changing their chemistry over time to adapt to changing environments.
Nature of Science	Knowledge of chemicals is used to understand the scientific basis of life.
Biological Systems	Chemical elements are combined into molecular compounds, which are used to build cells, the basic units of life.

ISTUDY

2.1 Chemical Elements

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how protons, neutrons, and electrons relate to atomic structure.
2. Use the periodic table to evaluate relationships between atomic number and mass number.
3. Describe how variations in an atomic nucleus account for its physical properties.
4. Determine how electrons are configured around a nucleus.

Throw a ball, pat your dog, rake leaves, turn a page; everything we touch—from the water we drink to the air we breathe—is composed of matter. **Matter** refers to anything that takes up space and has mass. Although matter has many diverse forms—anything from molten lava to kidney stones—it exists in only four distinct states: solid, liquid, gas, or plasma.

Elements

All matter, both nonliving and living, is composed of basic substances called **elements.** An element is a substance that cannot be broken down into simpler substances by ordinary chemical means. Each element has its own unique properties, such as density, solubility, melting point, and reactivity. It is quite remarkable that, in the known universe, there are only 94 naturally occurring elements (see Appendix A) that serve as the building blocks of matter. Other elements have been artificially constructed by physicists and are not biologically important.

Both the Earth's crust and all organisms are composed of elements, but they differ as to which ones are common. Only 6 elements—carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur—are basic to life and make up about 95% of the body weight of organisms. The properties of these elements are essential to the uniqueness of cells and all living organisms, such as those seen in Figure 2.1. Other elements, such as potassium, calcium, iron, magnesium, and zinc, are also important to life.

Atoms

In the early 1800s, the English scientist John Dalton developed the *atomic theory,* which says that elements consist of tiny particles called **atoms** (Gk. *atomos,* "uncut, indivisible"). An atom is the smallest part of an element that displays the properties of the element. An element and its atoms share the same name. One or two letters create the **atomic symbol** that stands for this name. For example, the symbol H means a hydrogen atom, the symbol Rn stands for radon, and the symbol Na (L. *natrium*) is used for a sodium atom.

Physicists have identified a number of subatomic particles that make up atoms. The three best-known subatomic particles are

Figure 2.1 A comparison of the elements that make up the Earth's crust and living organisms. The graph inset (**a**) shows that the Earth's crust primarily contains the elements silicon (Si), aluminum (Al), and oxygen (O). **b.** Living organisms, such as the trees and the human shown here, primarily contain the elements oxygen (O), nitrogen (N), carbon (C), and hydrogen (H). Biological molecules also often contain the elements sulfur (S) and phosphorus (P). Jordan Siemens/Photodisc/Getty Images

positively charged **protons,** uncharged **neutrons,** and negatively charged **electrons.** Protons and neutrons are located within the nucleus of an atom, and electrons move about the nucleus. Figure 2.2 shows the arrangement of the subatomic particles in a helium atom, which has two electrons. Because the precise location of the electrons is difficult to establish, we often indicate their probable positions using shading (Fig. 2.2*a*). When we are using a model of an atom—for example, to predict a chemical reaction—we indicate the average location of the electrons using **electron shells** (Fig. 2.2*b*).

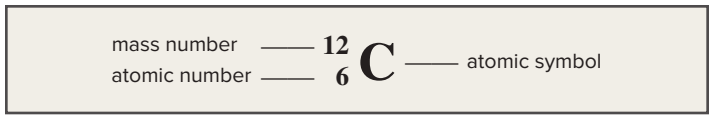
The concept of an atom has changed greatly since Dalton's day. Today's physicists are using high-energy supercolliders, such as the Large Hadron Collider in Europe, to explore the intricate internal structure of the atom.

It is also important to note that the majority of an atom is empty space. If an atom could be drawn the size of a football field, the nucleus would be like a gumball in the center of the field, and the electrons would be tiny specks whirling about in the upper stands. We should also realize that both of the models in Figure 2.2 indicate only where the electrons are expected to be most of the time. In our analogy, the electrons might very well stray outside the stadium at times.

Atomic Number and Mass Number

Atoms have not only an atomic symbol but also an atomic number and a mass number. All the atoms of an element have the same number of protons housed in the nucleus. This is called the **atomic number,** which accounts for the unique properties of this type of atom. Generally, atoms are assumed to be electrically neutral, meaning that the number of electrons is the same as the number of protons in the atom. The atomic number tells you not only the number of protons but also the number of electrons.

Each atom also has its own **mass number,** which is the sum of the protons and neutrons in the nucleus. Protons and neutrons are assigned one atomic mass unit (AMU) each. Electrons are so small that their AMU is considered zero in most calculations (Fig. 2.2*c*). By convention, when an atom stands alone (and not in the periodic table, discussed next), the atomic number is written as a subscript to the lower left of the atomic symbol. The mass number is written as a superscript to the upper left of the atomic symbol:

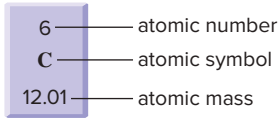


Whereas each atom of an element has the same atomic number, the number of neutrons may vary slightly. **Isotopes** (Gk. *isos,* "equal") are atoms of the same element that differ in the number of neutrons. For example, the element carbon has three naturally occurring isotopes:

$$^{12}_{6}C \qquad ^{13}_{6}C \qquad ^{14}_{6}C$$

It is important to note that the term *mass* is used, not *weight,* because mass is constant, while weight changes according to the gravitational force of a body. The gravitational force of the Earth is greater than that of the moon; therefore, substances weigh less on the moon, even though their mass has not changed.

The term **atomic mass** refers to the average mass for all the isotopes of that atom. Because the majority of carbon is carbon 12, the atomic mass of carbon is closer to 12 than to 13 or 14. To determine the number of neutrons from the atomic mass, subtract the number of protons from the atomic mass and take the closest whole number.



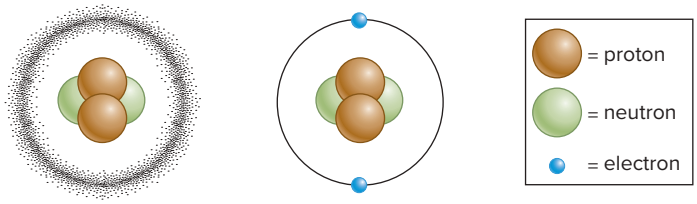
The Periodic Table

Once chemists discovered a number of the elements, they began to realize that even though each element consists of a different atom, certain chemical and physical characteristics recur. The periodic table, developed by the nineteenth-century Russian chemist Dmitri Mendeleev, was constructed as a way to group the elements, and therefore atoms, according to these characteristics.

Figure 2.3 is a portion of the periodic table, which is shown in total in Appendix A. In the periodic table, the horizontal rows are called periods, and the vertical columns are called groups. The atomic number of every atom in a period increases by one if you read from left to right. All the atoms in a group share similar chemical characteristics, namely in the type of chemical bonds they form. For example, the atoms in group VIII are called the noble gases, because they are inert and rarely react with another atom. Helium, neon, argon, and krypton are all examples of noble gases.

Radioactive Isotopes

Some isotopes of an element are unstable, or radioactive. For example, unlike the other two isotopes of carbon, carbon 14 changes over time into nitrogen 14, which is a stable isotope of the element nitrogen. As carbon 14 decays, it releases various types of energy in the form of rays and subatomic particles. The radiation given off by radioactive isotopes can be detected in various ways. The Geiger



a. b.

Subatomic Particles			
Particle	**Electric Charge**	**Atomic Mass Unit (AMU)**	**Location**
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	–1	0	Electron shell

c.

Figure 2.2 General structure of an atom. Atoms contain subatomic particles, which are located as shown. Protons and neutrons are found within the nucleus, and electrons are outside the nucleus. **a.** The shading shows the probable location of the electrons in the helium atom. **b.** The average location of an electron is sometimes represented by an electron shell. **c.** The electric charge and the atomic mass units (AMUs) of the subatomic particles vary as shown.

ISTUDY

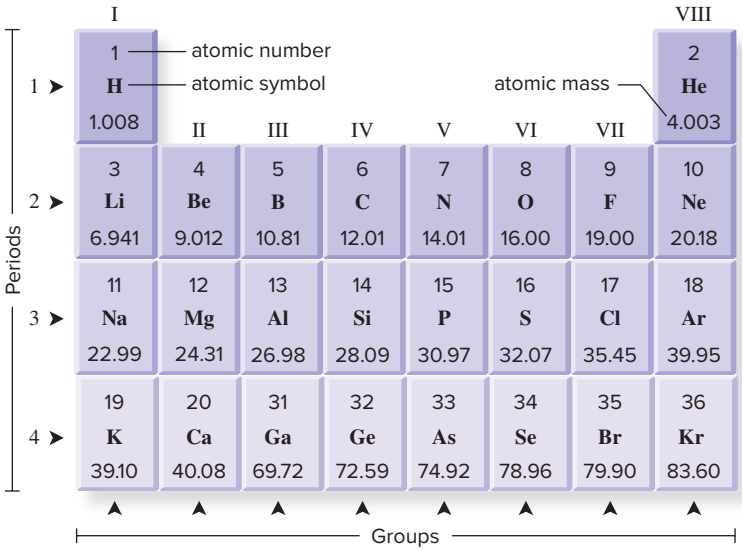


Figure 2.3 A portion of the periodic table. In the periodic table, elements are listed in the order of their atomic numbers but are arranged so each element is placed in a group (vertical column) and period (horizontal row). All the atoms in a particular group have the same number of valence electrons and therefore share common chemical characteristics. Each period shows the number of electron shells for an element. This abbreviated periodic table contains the elements most important in biology. (The complete periodic table is in Appendix A.)

counter is an instrument that is commonly used to detect radiation. In 1896, the French physicist Antoine-Henri Becquerel discovered that a sample of uranium would produce a bright image on a photographic plate even in the dark, and a similar method of detecting radiation is still in use today. As detailed in the feature, "The Legacy of Marie Curie," Marie Curie, who worked with Becquerel, coined the term *radioactivity* and contributed much to its study. Today, biologists use radiation to date objects from our distant past, to create images, and to trace the movement of substances in the body.

Low Levels of Radiation

The chemical behavior of a radioactive isotope is essentially the same as that of the stable isotopes of an element. This means you can put a small amount of radioactive isotope in a sample and it becomes a *tracer* by which to detect molecular changes. Melvin Calvin and his co-workers used carbon 14 to detect all the various reactions that occur during the process of photosynthesis (see Section 7.2).

The importance of chemistry to medicine is nowhere more evident than in the many medical uses of radioactive isotopes. Specific tracers are used in imaging the body's organs and tissues. For example, after a patient drinks a solution containing a minute amount of iodine 131, it becomes concentrated in the thyroid—the only organ to take it up. A subsequent image of the thyroid indicates whether it is healthy in structure and function (Fig. 2.4*a*).

Positron-emission tomography (PET) is a way to determine the comparative activity of tissues. Radioactively labeled glucose, which emits a subatomic particle known as a positron, is injected into the body. The radiation given off is detected by sensors and analyzed by a computer. The result is a color image that shows which tissues have taken up the glucose and are therefore metabolically active. The red areas surrounded by green in Figure 2.4*b* indicate which areas of the brain are most active. PET scans of the brain are used to diagnose a brain tumor, Alzheimer's disease, epilepsy, or whether a stroke has occurred. PET scans, utilizing radioactive thallium, can detect signs of coronary artery disease and low blood flow to the heart.

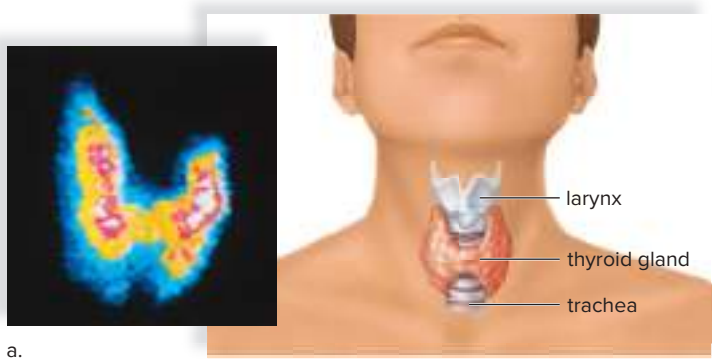
High Levels of Radiation

Radioactive substances in the environment can harm cells, damage DNA, and cause cancer. When Marie Curie was studying radiation, its harmful effects were not known, and she and many of her co-workers developed cancer. The release of radioactive particles following a nuclear power plant accident, as occurred in Japan in 2011 following a tsunami (see the Biological Systems feature, "Isotopes of a Meltdown"), can have far-reaching and long-lasting effects on human health.

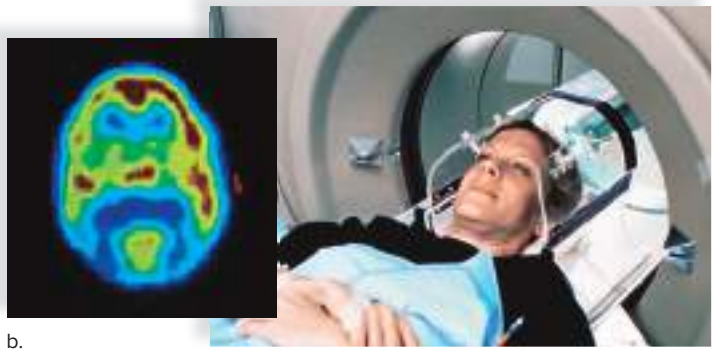
The harmful effects of radiation can be put to good use, however. Radiation from radioactive isotopes has been used for many years to sterilize medical and dental products. Ultraviolet radiation is often used to prevent harmful bacterial contamination of eggs, meats, some shellfish, fruits, and vegetables (Fig. 2.5*a*). Targeted radioisotopes can be introduced into the body, so that the subatomic particles emitted destroy only cancer cells, with little risk to the rest of the body (Fig. 2.5*b*).

Electrons and Energy

Various models may be used to illustrate the structure of a single atom. While the number of neutrons and protons may easily be depicted, because they are located in the nucleus, it is not possible to determine the precise location of any individual electron at any



a.



b.

Figure 2.4 Low levels of radiation. **a.** Medical scan of the thyroid gland (*left*) indicates the presence of a tumor that does not take up radioactive iodine. **b.** A positron-emission tomography (PET) scan reveals which portions of the brain are most active (green and red colors). (photos): (a, scan): Southern Illinois University/Science Source; (b, scan): Dr John Mazziotta ET AL/Science Source; (patient): Courtesy of National Institutes of Health

ISTUDY

▶ Diversity in Science

The Legacy of Marie Curie

Like many people, you may not have a favorable view of the word *radioactivity*. Yet, modern society relies heavily on the concept of radioactivity to provide us with electricity, safe food supplies, and the benefits of modern medicine. Our understanding of biology is also dependent on the use of radioactivity to understand the inner workings of a cell, variations in genes, and how organisms and ecosystems process nutrients and energy.

Our understanding of radioactivity is largely based on the work of Marie Curie, a Polish scientist of the late nineteenth and early twentieth century (Fig. 2A). Yet her story, like the story of so many other female scientists, was also one defined by overcoming barriers to females. Her life and research not only turned science on its head, but proved to those of her time that a woman could be just as brilliant and accomplished—if not more so—than her male peers, during a time when many women were not even permitted to study at universities.

Early in her career, the discovery of radioactivity by French scientist Henri Becquerel inspired Curie and her husband to study the phenomenon of radiation in their own lab. Working in cramped and often poorly ventilated conditions, they discovered that certain minerals containing uranium were, in fact, emitting far more radiation than they should have been. The two believed that this hinted at a new element, which Curie named polonium after her home country of Poland. Not five months later, Curie found an even more radioactive element than polonium, which she called radium.

Curie's research did more than discover new elements; she also proposed a new theory of radiation—that the "rays" emitted by elements such as uranium and polonium were the product of interactions within the atom itself. This challenged the then widely accepted belief that the atom was the smallest particle in physics, thus opening the door to future atomic research.

Curie's research on radiation was immediately recognized as one of the greatest discoveries of its time. But when it was proposed that Curie be awarded the Nobel Prize in Physics in 1903, her name was nearly passed over in favor of her husband's. As men, some members of the committee attributed Curie's work to her husband, because a woman had never been given a Nobel Prize before. In the end—in part due to intensive lobbying on her behalf by her husband—Marie Curie was awarded the Nobel Prize in Physics in 1903, alongside her husband and Henri Becquerel, becoming the first woman in history to win a Nobel Prize.

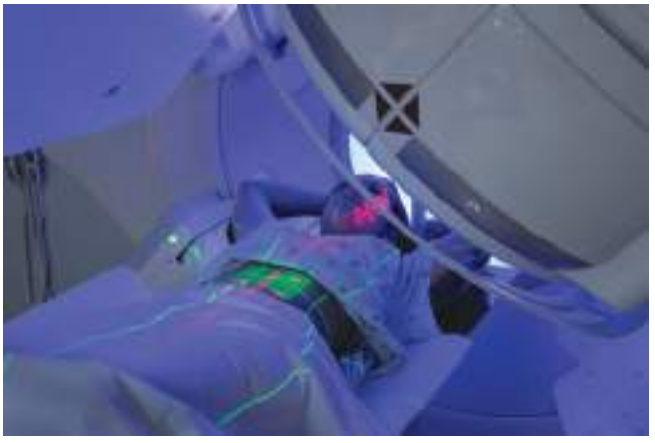
Not 10 years later in 1911, she received a second Nobel, this time in chemistry, and wholly on her own, making her the first person to win two Nobel Prizes, of any gender.

Curie's research into radiation would go on to have several real-world applications that we continue to use today. The use of radiation in medicine would go on to save lives in World War I, when "*petit Curies*"—mobile X-ray carriages—treated Allied soldiers on the Western Front. Today, modern medical radiotherapy used to combat cancer and tumors originates from Curie's research, while her theories that challenged the indivisible nature of the atom would inspire future atomic scientists, giving way to nuclear power, as well as, less benevolently, the atomic bomb.

Figure 2A Marie Curie. Marie Curie was one of the early pioneers in the discovery of radioactivity. Science History Images/Alamy Stock Photo

As we remember Curie's contributions, we must also recognize that her accomplishments were made despite the gender barriers placed before her—barriers still faced by females and underrepresented populations in the sciences over 110 years later.

a.



b.

Figure 2.5 High levels of radiation. **a.** Radiation can also be used to sterilize some foods, such as the eggs shown here, to protect against bacteria. **b.** Physicians use targeted radiation therapy to kill cancer cells.
(a): Phonkrit Ninchak/Shutterstock; (b): Mark Kostich/E+/iStock/Getty Images

ISTUDY

THEME Biological Systems

Isotopes of a Meltdown

On March 11, 2011, a magnitude 8.9 earthquake struck the coast of Japan, producing a massive tsunami, causing the shutdown of 11 of Japan's nuclear power reactors, including reactors 1, 2, and 3 at the Fukushima Daiichi power plant (Fig. 2B). Shortly after the shutdown, the pressure within reactor 1 built up to twice that of normal levels. In an attempt to relieve pressure, steam-containing radiation was vented from the reactors. Over the next several days, the plant continued to have problems and the levels of radiation in the surrounding areas continued to rise. Shortly thereafter, Japanese officials formed a 10-kilometer (km) evacuation zone, which, over the next few days, was increased to 20 km (12.4 miles), resulting in the relocation of close to 390,000 people.

The Fukushima disaster released a number of radioactive isotopes into the atmosphere and the surrounding ocean. These included isotopes of iodine, cesium, and xenon. Each of these isotopes is a by-product of the nuclear fission reactions that occur within a nuclear power plant. As the uranium in these plants breaks down, the uranium atoms release energy (such as gamma radiation) and particles (alpha and beta particles). This is called radioactive decay, and the loss of particles effectively transforms the original uranium atom into isotopes of another element. In many cases, the resulting isotopes are also radioactive, and then undergo additional radioactive decay.

Three of the more common isotopes released by the Fukushima disaster are listed in the following paragraphs. These isotopes differ considerably in their half-lives, in the types of radiation they release, and their potential adverse effects on living organisms, such as humans.

Cesium-134 and Cesium-137

Perhaps the biggest concern following the Fukushima disaster was the release of radioactive cesium. Naturally, cesium (Cs) is a metal with an atomic number of 55 and an atomic mass of approximately 132.9. However, within a nuclear reactor, two isotopes of cesium (^{134}Cs and ^{137}Cs) are formed. Cesium-134 and Cesium-137 have half-lives of 2 years and 30 years, respectively. Both of these isotopes release both beta particles and gamma radiation as they decay. Both beta particles and gamma radiation have the ability to penetrate tissues and damage cells and their DNA. Exposure to beta particles and gamma radiation, either internally or externally, can cause burns and greatly increase the chance of cancer.

Figure 2B The Fukushima Power Plant. The 8.9 magnitude earthquake produced a 15-meter (49.2-ft) tsunami that triggered an explosion of the plant causing radioactive isotopes to be released into the atmosphere and surrounding water. The Yomiuri Shimbun/AP Images

Following the Fukushima disaster, there were significant concerns of ^{137}Cs entering the atmosphere and the Pacific Ocean. Because of the winds at the time, most of the ^{137}Cs was carried back over Japan, forcing evacuations up to 35 kilometers (21.7 miles) inland. Large regions of this evacuation zone still remain uninhabitable due to soil contamination by ^{137}Cs. Small amounts of atmospheric ^{137}Cs were detected on the West Coast of the United States, and some ^{137}Cs has been detected in the waters off the coast of California, but in neither case at levels that are considered hazardous to humans.

Iodine-131

Radioactive iodine, ^{131}I, was also released by the reactors. Iodine is a water-soluble element that has an atomic number of 53 and an atomic mass of approximately 126.9. There are several forms of iodine isotopes, with ^{131}I being formed mainly by nuclear fission reactions. Like ^{137}Cs, ^{131}I emits beta particles, which may damage tissues. It also releases small amounts of gamma radiation.

Radioactive iodine has a half-life of only about 8 days, so its long-term effect on the environment is minimal. In our bodies, iodine is mainly used by the thyroid gland to manufacture the thyroid hormones associated with metabolic functions, including overall metabolic rate. Intense exposure to ^{131}I may cause thyroid problems, including thyroid cancer. Interestingly, isotopes of iodine at low doses are used by the medical profession to diagnose problems with the thyroid gland and treat some forms of thyroid cancer.

Xenon-133

Xenon (Xe) is a noble gas with an atomic number of 54 and an atomic weight of 131.2. One isotope of xenon, ^{133}Xe, has a half-life of only 5 days. Like ^{131}I and ^{137}Cs, ^{133}Xe emits beta particles which may damage tissue, but the short half-life of the isotope limits the chances of severe problems. Further reducing the risk is the fact that, as a noble gas, xenon does not react with other elements, and thus is not easily introduced into the chemical compounds within cells.

Like radioactive iodine, ^{133}Xe is used by the medical profession to diagnose disease. Because it is a gas, it is frequently used in the diagnosis of lung disorders. Studies are now underway to use ^{133}Xe as a form of treatment for certain types of lung cancer, because it easily enters the lungs and the emission of beta particles can help destroy lung cancer cells.

Questions to Consider

1. Describe the differences between ^{137}Cs, ^{131}I, and ^{133}Xe and their non-radioactive isotopes.
2. Explain why ^{137}Cs is more dangerous than ^{131}I or ^{133}Xe.

ISTUDY

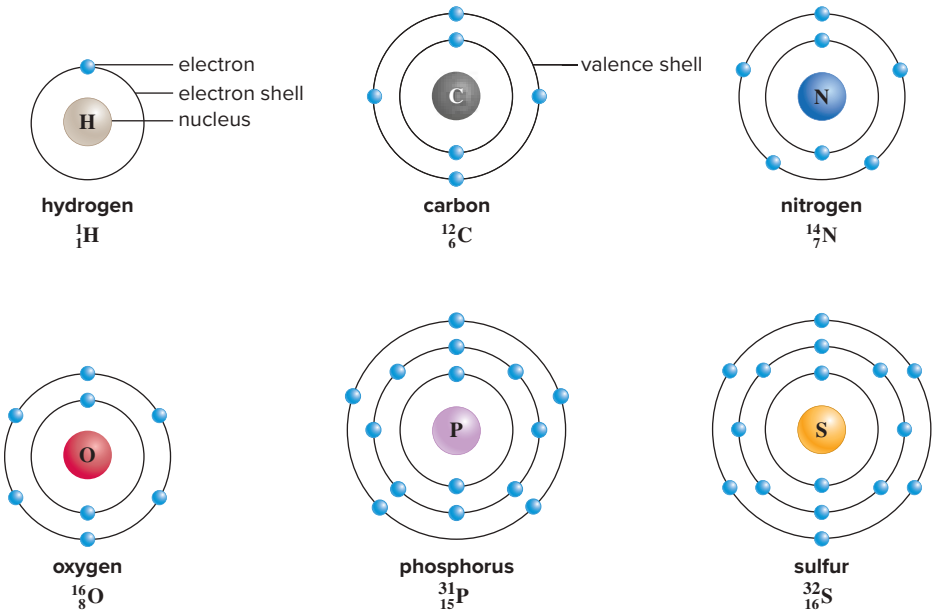


Figure 2.6 Bohr models of atoms. Electrons orbit the nucleus at particular energy levels (electron shells). The first shell contains up to two electrons, and thereafter each shell is most stable when it contains eight electrons. Atoms with an atomic number above 20 may have more electrons in their outer shells. The outermost, or valence, shell helps determine the atom's chemical properties and how many other elements it can interact with.

given moment. One of the more common models is the Bohr model (Fig. 2.6), developed by the physicist Niels Bohr.

In the Bohr model, the electron shells (also called electron orbitals) about the nucleus are used to represent the average energy levels of an electron. Because the negatively charged electrons are attracted to the positively charged nucleus, it takes energy to push them away and keep them in their own shell. The more distant the shell, the more energy it takes. Therefore, it is more accurate to speak of electrons as being at particular energy levels in relation to the nucleus. Electrons may move between energy levels. For example, when we explore the processes of photosynthesis, you will learn that when atoms absorb the energy of the sun, electrons are boosted to a higher energy level. Later, as the electrons return to their original energy level, energy is released and transformed into chemical energy. This chemical energy supports the majority of life on Earth; therefore, our very existence is dependent on the energy of electrons.

Let's take a more detailed look at the Bohr models depicted in Figure 2.6. The first shell is closest to the nucleus and can contain two electrons; the second shell can contain eight electrons. In all atoms, the lower shells are filled with electrons before the next higher level contains any electrons.

The sulfur atom, with an atomic number of 16, has two electrons in the first shell, eight electrons in the second shell, and six electrons in the outer, third shell. Revisit the periodic table (see Fig. 2.3) and note that sulfur is in the third period. In other words, the period tells you how many shells an atom has. Also note that sulfur is in group VI. The group tells you how many electrons an atom has in its outer shell.

Regardless of how many shells an atom has, the outermost shell is called the **valence shell.** The valence shell is important, because it determines many of an atom's chemical properties. If an atom has only one shell, the valence shell is complete when it has two electrons. In atoms with more than one shell, the valence shell is most stable when it has eight electrons. This is called the **octet rule.** Each atom in a group within the periodic table has the same number of electrons in its valence shell. As mentioned previously, all the atoms in group VIII of the periodic table have eight electrons in their valence shell. These elements are also called the noble gases, because they do not ordinarily react.

The electrons in the valence shells play an important role in determining how an element undergoes chemical reactions. Atoms with fewer than eight electrons in the outer shell react with other atoms in such a way that after the reaction each has a stable outer shell. As we will see, the number of electrons in an atom's valence shell determines whether the atom gives up, accepts, or shares electrons to acquire eight electrons in the outer shell.

Check Your Progress 2.1

1. Contrast atomic number and atomic mass.
2. Examine the periods and groups from the periodic table to determine the electron configuration of chlorine.
3. Explain how two isotopes of an element vary with regard to their atomic structure.

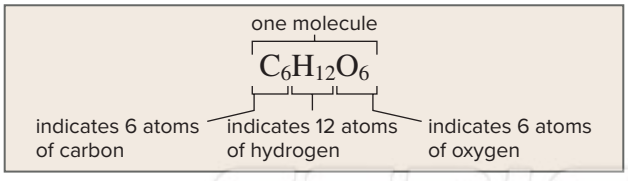
2.2 Molecules and Compounds

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how elements are combined into molecules and compounds.
2. List the different types of bonds that occur between elements.
3. Explain the difference between a polar and a nonpolar covalent bond.

A **molecule** exists when two or more of the same type of atoms bond together; it is the smallest part of a compound that retains its chemical properties. A **compound** is a molecule containing atoms of at least two different elements. In practice, these two terms are used interchangeably, but in biology we usually speak of molecules. Water (H_2O) is a molecule that contains atoms of hydrogen and oxygen. A **formula** tells you the number of each kind of atom in a molecule. For example, the formula for glucose is:



ISTUDY

Electrons possess energy, as do the bonds between atoms. Organisms are directly dependent on chemical-bond energy to maintain their organization. As you may know, organisms routinely break down glucose, the sugar shown above, to obtain energy. When a chemical reaction occurs, as when glucose is broken down, electrons shift in their relationship to one another, and energy is released. Spontaneous reactions, which occur freely, always release energy.

Ionic Bonding

Sodium (Na), with only one electron in its valence shell, tends to be an electron donor (Fig. 2.7*a*). Once it gives up this electron, the second shell, with its stable configuration of eight electrons, becomes its outer shell. Chlorine (Cl), on the other hand, tends to be an electron acceptor. Its valence shell has seven electrons, so if it acquires only one more electron, it has a stable outer shell. When a sodium atom and a chlorine atom come together, an electron is transferred from the sodium atom to the chlorine atom. Now both atoms have eight electrons in their outer shells.

This electron transfer, however, causes a charge imbalance in each atom. After giving up an electron, the sodium atom has one more proton than it has electrons; therefore, it has a net charge of +1 (symbolized by Na^+). After accepting an electron, the chlorine atom has one more electron than it has protons; therefore, it has a net charge of −1 (symbolized by C^-). These charged particles are called **ions.** Sodium (Na^+) and chloride (Cl^-) are not the only biologically important ions. Some, such as potassium (K^+), are formed by the transfer of a single electron to another atom; others, such as calcium (Ca^{2+}) and magnesium (Mg^{2+}), are formed by the transfer of two electrons.

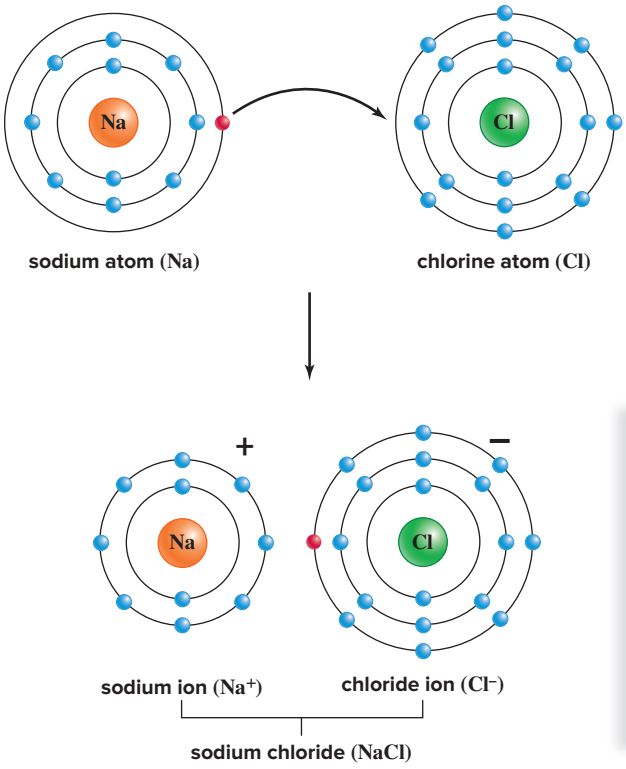
Ionic compounds are held together by an attraction between negatively and positively charged ions, called an **ionic bond.** When sodium reacts with chlorine, an ionic compound called sodium chloride (NaCl) results. Sodium chloride is an example of a salt. It is commonly called table salt, and is used to season food (Fig. 2.7*b*). Salts are solid substances that usually separate and exist as individual ions in water, as discussed in Section 2.3.

Covalent Bonding

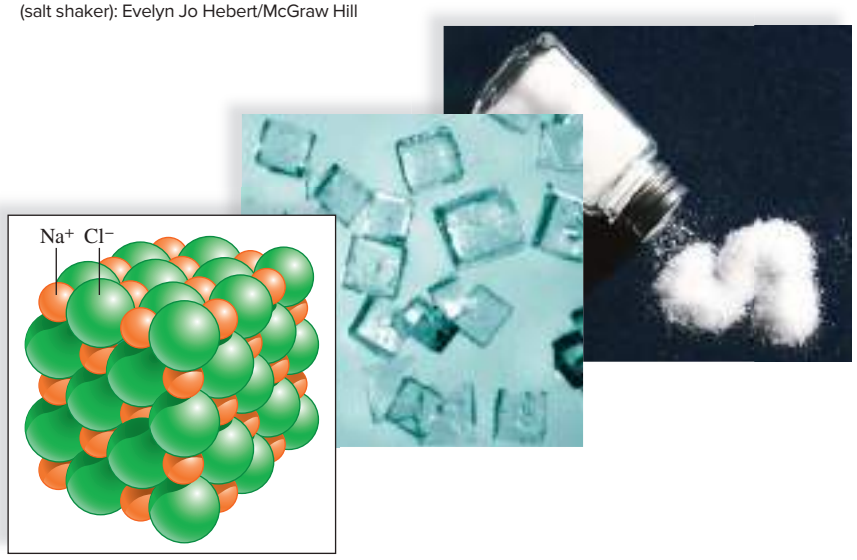
A **covalent bond** results when two atoms share electrons in such a way that each atom has an octet of electrons in the outer shell (or two electrons, in the case of hydrogen). In a hydrogen atom, the outer shell is complete when it contains two electrons. If hydrogen is in the presence of a strong electron acceptor, it gives up its electron to become a hydrogen ion (H^+). But if this is not possible, hydrogen can share with another atom and thereby have a completed outer shell. For example, one hydrogen atom will share with another hydrogen atom. Their two electron shells overlap, and the electrons are shared between them (Fig. 2.8*a*). Because they form a covalent bond and share the electron pair, each atom has a completed outer shell.

A more common way to symbolize that atoms are sharing electrons is to draw a line between the two atoms, as in the structural formula H—H. Just as a handshake requires two hands, one from each person, a covalent bond between two atoms requires two electrons, one from each atom. In a molecular formula, the line is omitted and the molecule is simply written as H_2.

Sometimes, atoms share more than one pair of electrons to complete their octets. A double covalent bond occurs when two atoms share two pairs of electrons (Fig. 2.8*b*). To show that oxygen



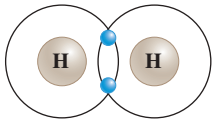
a.



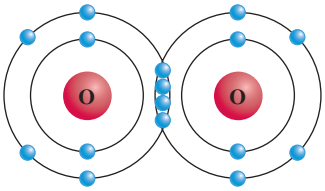
b.

Figure 2.7 Formation of sodium chloride (table salt). **a.** During the formation of sodium chloride, an electron is transferred from the sodium atom (Na^+) to the chlorine atom (Cl^-). At the completion of the reaction, each atom has eight electrons in the outer shell, but each also carries a charge as shown. **b.** In a sodium chloride crystal, ionic bonding between Na^+ and Cl^- causes the atoms to assume a three-dimensional lattice in which each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions. The result is crystals of salt, as in table salt. (salt crystals): Evelyn Jo Johnson/McGraw Hill; (salt shaker): Evelyn Jo Hebert/McGraw Hill

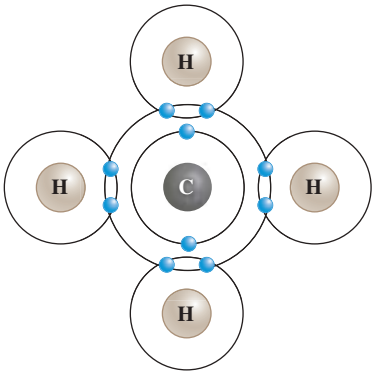
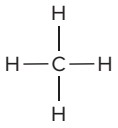
ISTUDY

Electron Model	Structural Formula	Molecular Formula
	H—H	H_2

a. Hydrogen gas

	O═O	O_2

b. Oxygen gas

		CH_4

c. Methane

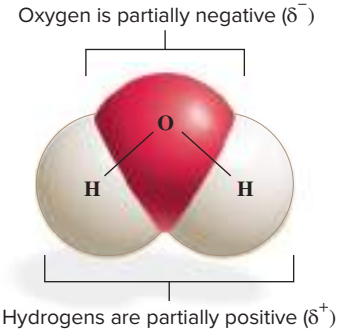
Figure 2.8 Covalently bonded molecules. In a covalent bond, atoms share electrons, allowing each atom to have a completed outer shell. **a.** A molecule of hydrogen (H_2) contains two hydrogen atoms sharing a pair of electrons. This single covalent bond can be represented in any of the three ways shown. **b.** A molecule of oxygen (O_2) contains two oxygen atoms sharing two pairs of electrons. This results in a double covalent bond. **c.** A molecule of methane (CH_4) contains one carbon atom bonded to four hydrogen atoms.

gas (O_2) contains a double bond, the molecule can be written as O═O. It is also possible for atoms to form triple covalent bonds, as in nitrogen gas (N_2), which can be written as N ≡ N. Single covalent bonds between atoms are quite strong, but double and triple bonds are even stronger.

Nonpolar and Polar Covalent Bonds

When the sharing of electrons between two atoms is equal, the covalent bond is said to be a **nonpolar covalent bond.** However, in some cases, one atom is able to attract electrons to a greater degree than the other atom. In this case, we say that the atom that has a greater attraction for a shared pair of electrons has a greater **electronegativity.** When electrons are not shared equally, the covalent bond is a **polar covalent bond.**

The shape of a molecule may also influence whether it is polar or nonpolar. While carbon is larger and has more protons than a hydrogen atom, the symmetrical nature of a methane molecule cancels out any polarities; thus, methane is a nonpolar molecule. This is not so in water, which has this shape:



In water, the oxygen atom is more electronegative than the hydrogen atoms. As a result, water molecules are polar. Moreover, because of its nonsymmetrical shape, the polar bonds cannot cancel each other, and water is a polar molecule. The more electronegative end of the molecule is designated slightly negative (δ^-), and the hydrogens are designated slightly positive (δ^+).

Water is not the only polar molecule in living organisms. For example, the amine group (—NH_2) is polar, and this causes amino acids and nucleic acids to exhibit polarity, as we will see in the next chapter. The polarity of molecules affects how they interact with other molecules.

Check Your Progress 2.2

1. Compare and contrast an ionic bond with a covalent bond.
2. Describe the process by which ions are formed.
3. Explain why methane is nonpolar but water is polar.

2.3 Chemistry of Water

Learning Outcomes

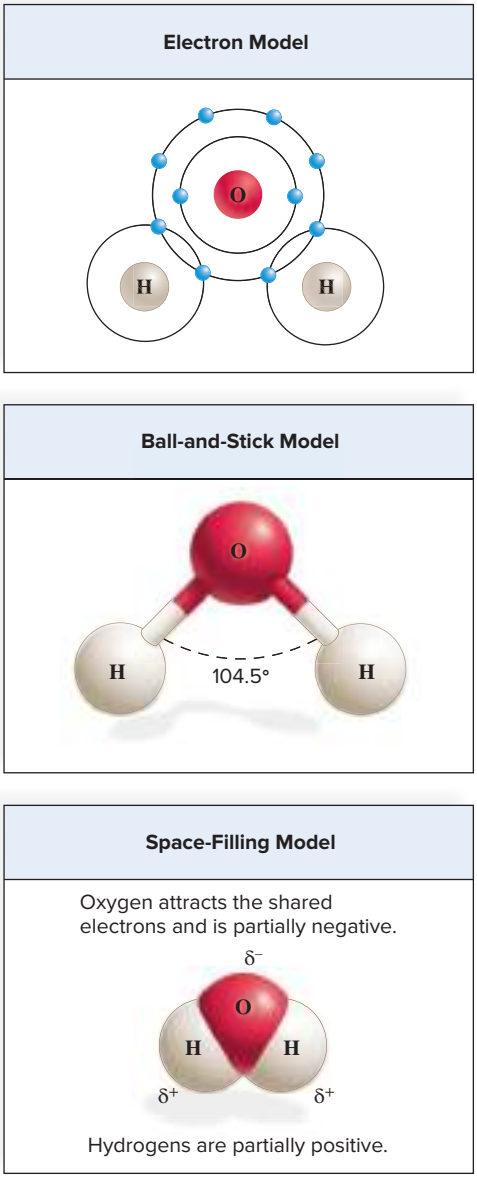
Upon completion of this section, you should be able to

1. Describe how water associates with other molecules in solution.
2. Describe why the properties of water are important to life.
3. Analyze how water's solid, liquid, and vapor states allow life to exist on Earth.

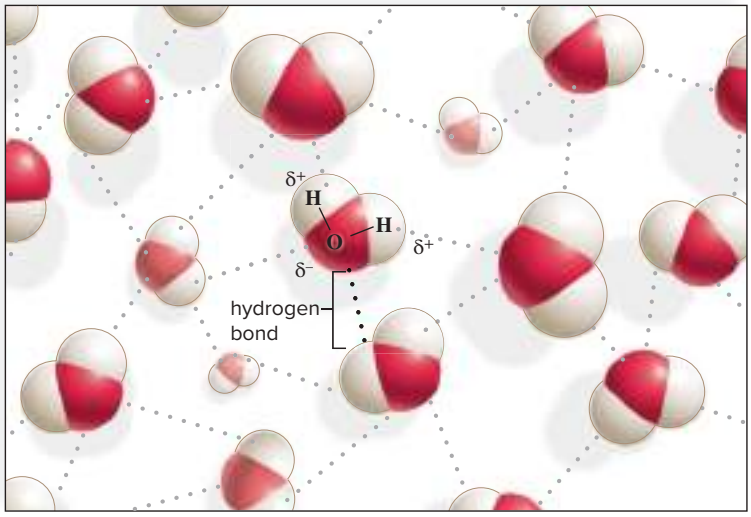
Figure 2.9*a* recaps what we know about the water molecule. The structural formula at the top of the figure shows that when water forms, an oxygen atom is sharing electrons with two hydrogen atoms. The ball-and-stick model in the center shows that the covalent bonds between oxygen and each of the hydrogens are at an angle of 104.5°. Finally, the space-filling model gives us the three-dimensional shape of the molecule and indicates its polarity.

In the study of biology, we often state that structure relates to function. This is true at a variety of organizational levels, including molecules such as water. For example, hormones have specific shapes that allow them to be recognized by the cells in the body.

ISTUDY



a. Water (H_2O)



b. Hydrogen bonding between water molecules

Figure 2.9 Water molecule. **a.** Three models for the structure of water. The electron model does not indicate the shape of the molecule. The ball-and-stick model shows that the two bonds in a water molecule are angled at 104.5°. The space-filling model also shows the V shape of a water molecule. **b.** Hydrogen bonding between water molecules. Each water molecule can hydrogen bond with up to four other molecules, in three dimensions. When in a liquid state, water is constantly forming and breaking hydrogen bonds.

We can stay well only when antibodies recognize the shapes of disease-causing agents, the way a key fits a lock, and are able to remove them.

The shape of a water molecule and its polarity result in the formation of hydrogen bonds. A **hydrogen bond** is caused by the attraction of a slightly positive hydrogen to a slightly negative atom in the vicinity. In carbon dioxide, O═C═O, a slight difference in polarity between carbon and the oxygens is present, but because carbon dioxide is symmetrical, the opposing charges cancel one another and hydrogen bonding does not occur.

Hydrogen Bonding

The dotted lines in Figure 2.9*b* indicate that the hydrogen atoms in one water molecule are attracted to the oxygen atoms in other water molecules. Each of these hydrogen bonds is weaker than an ionic or covalent bond. The dotted lines indicate that hydrogen bonds are more easily broken than the other bonds.

Hydrogen bonding is not unique to water. Other biological molecules, such as DNA, have polar covalent bonds involving an electropositive hydrogen and usually an electronegative oxygen or nitrogen. In these instances, a hydrogen bond can occur within the same molecule or between nearby molecules.

Although a single hydrogen bond is more easily broken than a single covalent bond, multiple hydrogen bonds are collectively quite strong. Hydrogen bonds between cellular molecules help maintain their proper structure and function. For example, hydrogen bonds hold the two strands of DNA together (see Section 3.5). When DNA makes a copy of itself, hydrogen bonds easily break, allowing DNA to unzip. But normally, the hydrogen bonds add stability to the DNA molecule. Similarly, the shape of protein molecules is often maintained by hydrogen bonding between different parts of the same molecule. As we will see, many of the important properties of water are the result of hydrogen bonding.

Properties of Water

The first cell(s) evolved in water, and all living organisms are 70–90% water. Because of hydrogen bonding, water molecules cling together, and this association gives water its unique chemical properties. Without hydrogen bonding between molecules, water would freeze at −100°C and boil at −91°C, making most of the water on Earth steam, and life unlikely. Hydrogen bonding is responsible for water being a liquid at temperatures typically found on the Earth's surface. It freezes at 0°C and boils at 100°C. These and other unique properties of water make it essential to the existence of life as we

ISTUDY

know it. As noted in the chapter opener, the search for life on other planets often begins with the search for water.

Water Has a High Heat Capacity

A **calorie** is the amount of heat energy needed to raise the temperature of 1 g of water 1°C. In comparison, other covalently bonded liquids require input of only about half this amount of energy to rise in temperature 1°C. The many hydrogen bonds that link water molecules together help water absorb heat without the temperature changing significantly. Converting 1 g of the coldest liquid water to ice requires the loss of 80 calories of heat energy (Fig. 2.10*a*). Water holds on to its heat, and its temperature falls more slowly than that of other liquids. This property of water is important not only for aquatic organisms but for all life.

Because the temperature of water rises and falls slowly, organisms are better able to maintain their normal internal temperatures and are protected from rapid changes in temperature.

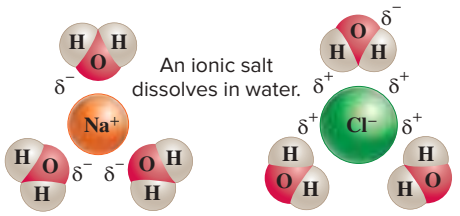
Water Has a High Heat of Evaporation

When water boils, it evaporates, meaning it vaporizes into the environment. Converting 1 g of the hottest water to a gas requires an input of 540 calories of energy. Water has a high heat of evaporation because hydrogen bonds must be broken before water boils.

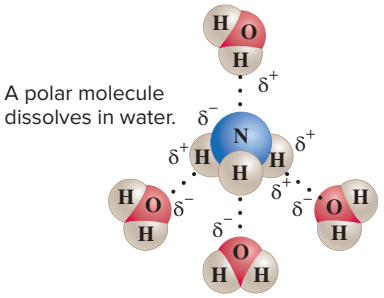
Water's high heat of vaporization gives animals in a hot environment an efficient way to release excess body heat. When an animal sweats, or gets splashed, body heat is used to vaporize water, thus cooling the animal (Fig. 2.10*b*). Because of water's high heat of vaporization and ability to hold on to its heat, temperatures along the coasts are moderate. During the summer, the ocean absorbs and stores solar heat, and during the winter, the ocean releases it slowly. In contrast, the interior regions of continents experience abrupt changes in temperatures.

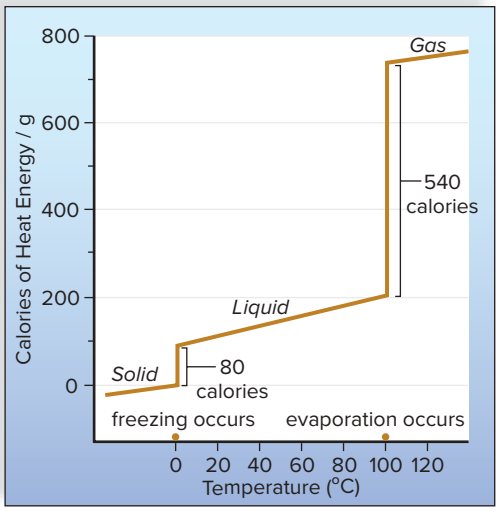
Water Is a Solvent

Due to its polarity, water facilitates chemical reactions, both outside and within living systems. As a solvent, it dissolves a great number of substances, especially those that are also polar. A **solution** contains dissolved substances, which are then called **solutes.** When ionic salts—for example, sodium chloride (NaCl)—are put into water, the negative ends of the water molecules are attracted to the sodium ions, and the positive ends of the water molecules are attracted to the chloride ions. This attraction causes the sodium ions and the chloride ions to separate, or dissociate, in water.



Water is also a solvent for larger polar molecules, such as ammonia (NH_3).





a. Calories lost when 1 g of liquid water freezes, and calories required when 1 g of liquid water evaporates.

b. Bodies of organisms cool when their heat is used to evaporate water.

Figure 2.10 Temperature and water. **a.** Water can be a solid, a liquid, or a gas at naturally occurring environmental temperatures. At room temperature and pressure, water is a liquid. When water freezes and becomes a solid (ice), it gives off heat, and this heat can help keep the environmental temperature higher than expected. On the other hand, when water evaporates, it takes up a large amount of heat as it changes from a liquid to a gas. **b.** This means that splashing water on the body will help keep body temperature within a normal range. (b): Jonathan Hewitt/Alamy Stock Photo

ISTUDY

Water Molecules Are Cohesive and Adhesive

Cohesion refers to the ability of water molecules to cling to each other due to hydrogen bonding. At any moment in time, a water molecule can form hydrogen bonds with at most four other water molecules. Because of cohesion, water exists as a liquid under the conditions of temperature and pressure present at the Earth's surface. The strong cohesion of water molecules is apparent because water flows freely, yet water molecules do not separate from each other.

Adhesion refers to the ability of water molecules to cling to other polar surfaces. This is a result of water's polarity. Multicellular animals often contain internal vessels in which water assists the transport of nutrients and wastes, because the cohesion and adhesion of water allow blood to fill the tubular vessels of the cardiovascular system. For example, the liquid portion of our blood, which transports dissolved and suspended substances throughout the body, is 90% water.

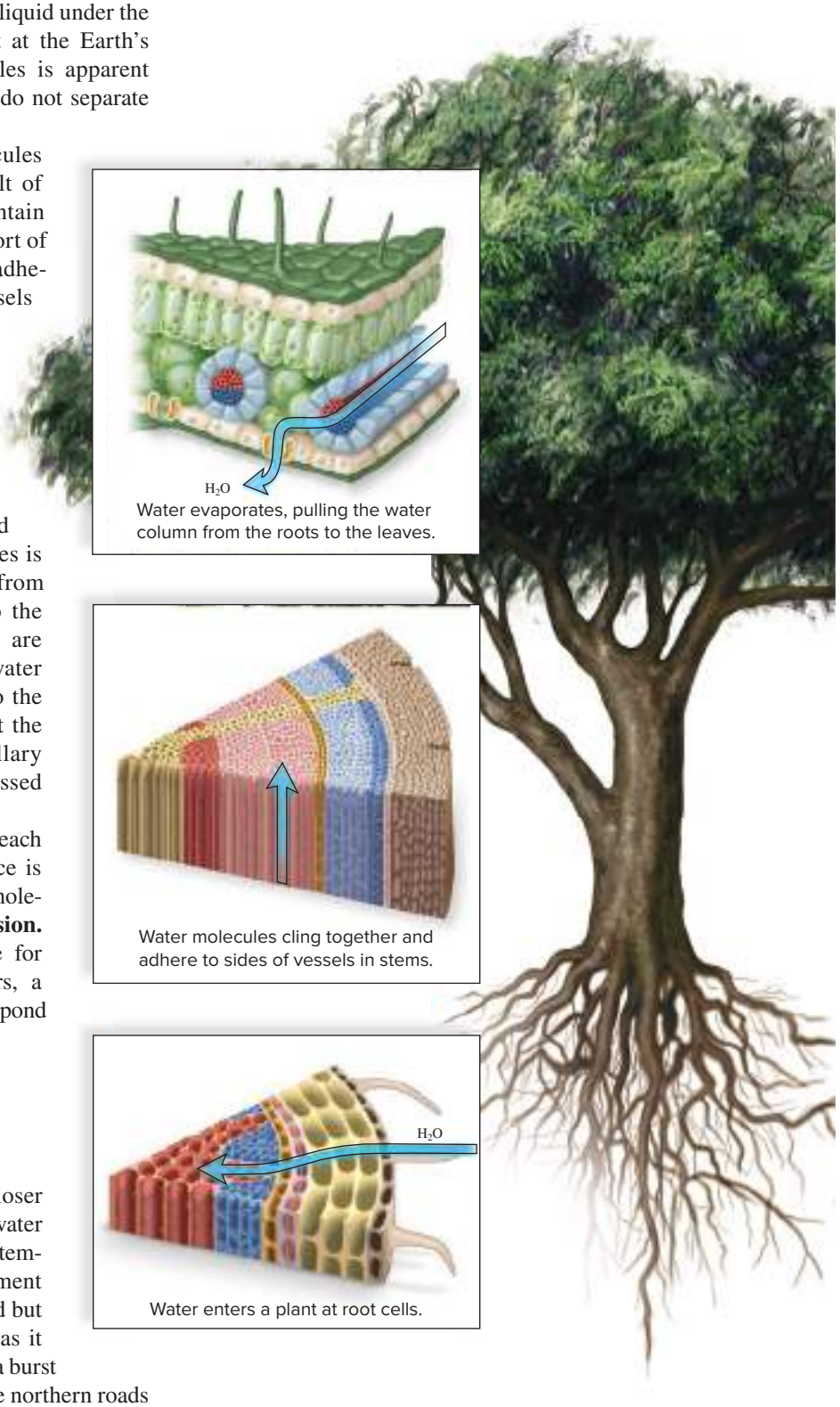
Cohesion and adhesion also contribute to the transport of water in plants. Plants have their roots anchored in the soil, where they absorb water, but the leaves are uplifted and exposed to solar energy. Water evaporating from the leaves is immediately replaced with water molecules from transport vessels that extend from the roots to the leaves (Fig. 2.11). Because water molecules are cohesive, a tension is created that pulls the water column up from the roots. Adhesion of water to the walls of the transport vessels also helps prevent the water column from breaking apart. This capillary action is essential to plant life, as will be discussed in Section 25.3.

Because water molecules are attracted to each other, they cling together where the liquid surface is exposed to air. The stronger the force between molecules in a liquid, the greater the **surface tension.** Water's high surface tension makes it possible for humans to skip rocks on water. Water striders, a common insect, can even walk on the surface of a pond without breaking the surface.

Frozen Water (Ice) Is Less Dense than Liquid Water

As liquid water cools, the molecules come closer together. Water is most dense at 4°C, but the water molecules are still moving about (Fig. 2.12). At temperatures below 4°C, only vibrational movement occurs, and hydrogen bonding becomes more rigid but also more open. This means that water expands as it reaches 0°C and freezes, which is why cans of soda burst when placed in a freezer, or why frost heaves make northern roads bumpy in the winter. It also means that ice is less dense than liquid water, and therefore ice floats on liquid water.

Figure 2.11 Water molecules are cohesive and adhesive.
Cohesion and adhesion play an important role in the movement of water in a plant. When water evaporates from the leaves, the water column is pulled upward due to the cohesion of water molecules to one another and the adhesion of water molecules to the sides of the vessels. This capillary action is critical for plants to function.



ISTUDY

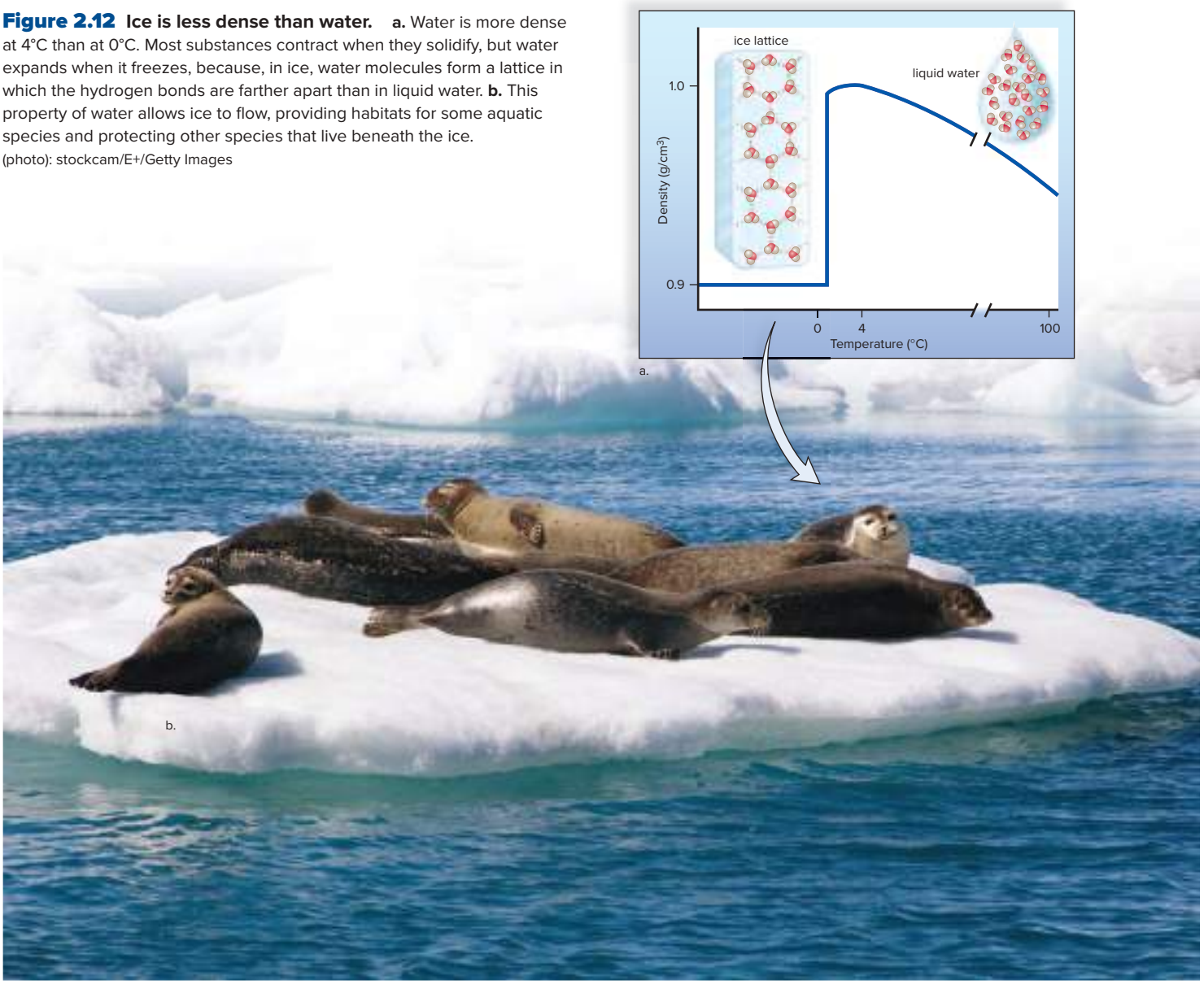


Figure 2.12 Ice is less dense than water. **a.** Water is more dense at 4°C than at 0°C. Most substances contract when they solidify, but water expands when it freezes, because, in ice, water molecules form a lattice in which the hydrogen bonds are farther apart than in liquid water. **b.** This property of water allows ice to flow, providing habitats for some aquatic species and protecting other species that live beneath the ice.
(photo): stockcam/E+/Getty Images

If ice did not float on water, it would sink to the bottom, and ponds, lakes, and perhaps even the ocean would freeze solid, making life impossible in the water and on land. Instead, bodies of water always freeze from the top down. When a body of water freezes on the surface, the ice acts as an insulator to prevent the water below it from freezing. This allows aquatic organisms to survive the winter. As ice melts in the spring, it draws heat from the environment, helping prevent a sudden change in temperature, which might be harmful to life.

Check Your Progress 2.3

1. Explain how water's structure relates to the formation of hydrogen bonds.
2. Explain how hydrogen bonds relate to the properties of water.
3. Explain how spraying water on your body helps cool it off.

2.4 Acids and Bases

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between an acid and a base.
2. Explain the relationship between H^+ or OH^- concentration and pH.
3. Analyze how buffers prevent large pH changes in solutions.

When water ionizes, it releases an equal number of **hydrogen ions (H^+)** (sometimes just called protons because hydrogen consists of only a single proton and electron) and **hydroxide ions (OH^-)**:

$$\underset{\text{water}}{H{-}O{-}H} \rightleftharpoons \underset{\substack{\text{hydrogen}\\\text{ion}}}{H^+} + \underset{\substack{\text{hydroxide}\\\text{ion}}}{OH^-}$$

Only a few water molecules at a time dissociate, and the actual number of H^+ and OH^- is very small (1×10^{-7} moles/liter). A **mole** is a unit of measurement used in chemistry for very small items, such as atoms of an element.

Acidic Solutions (High H^+ Concentrations)

Lemon juice, vinegar, tomatoes, and coffee are all acidic solutions. What do they have in common? **Acids** are substances that dissociate in water, releasing hydrogen ions (H^+). The acidity of a substance depends on how fully it dissociates in water. For example, hydrochloric acid (HCl) is a strong acid that dissociates almost completely in this manner:

$$HCl \longrightarrow H^+ + Cl^-$$

If hydrochloric acid is added to a beaker of water, the number of hydrogen ions (H^+) increases greatly.

Basic Solutions (Low H^+ Concentration)

Milk of magnesia and ammonia are common basic solutions familiar to most people. **Bases** are substances that either take up hydrogen ions (H^+) or release hydroxide ions (OH^-). For example, sodium hydroxide (NaOH) is a strong base that dissociates almost completely in this manner:

$$NaOH \longrightarrow Na^+ + OH^-$$

If sodium hydroxide is added to a beaker of water, the number of hydroxide ions increases, while the number of hydrogen ions decreases.

pH Scale

The **pH scale** is used to indicate the acidity or basicity (alkalinity) of a solution. The pH scale (Fig. 2.13) ranges from 0 to 14. A pH of 7 represents a neutral state in which the hydrogen ion and hydroxide ion concentrations are equal. A pH below 7 is an acidic solution, because the hydrogen ion concentration is greater than the hydroxide concentration. A pH above 7 is basic, because the $[OH^-]$ is greater than the $[H^+]$. Further, as we move down the pH scale from pH 14 to pH 0, each unit is 10 times more acidic than the previous unit. As we move up the scale from 0 to 14, each unit is 10 times more basic than the previous unit. Therefore, pH 5 is 100 times more acidic than pH 7 and 100 times more basic than pH 3. The pH scale was devised to eliminate the use of cumbersome numbers associated with the hydrogen ion concentrations on the left of Figure 2.13.

The Biological Systems feature, "The Impact of Acid Deposition," describes detrimental environmental consequences of low pH rain and snow. In humans, pH needs to be maintained within a narrow range, or there are health consequences. The pH of blood is around 7.4, and blood is buffered in the manner described next to keep the pH within a normal range.

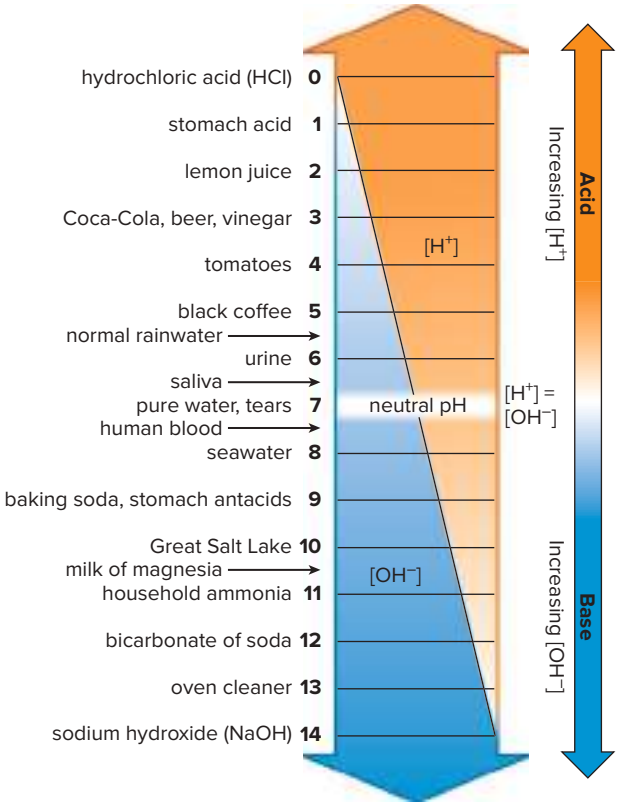


Figure 2.13 The pH scale. The pH scale ranges from 0 to 14, with 0 being the most acidic and 14 being the most basic. pH 7 (neutral pH) has equal amounts of hydrogen ions (H^+) and hydroxide ions (OH^-). An acidic pH has more H^+ than OH^- and a basic pH has more OH^- than H^+.

Buffers and pH

A **buffer** is a chemical or a combination of chemicals that keeps pH within normal limits. Many commercial products, such as shampoos or deodorants, are buffered as an added incentive for us to buy them.

In living organisms, the pH of body fluids has to be maintained within a narrow range, or else molecules don't function correctly and our health suffers. The pH of our blood when we are healthy is always about 7.4—that is, just slightly basic (alkaline). If the blood pH drops to about 7, acidosis results. If the blood pH rises to about 7.8, alkalosis results. Both conditions can be life-threatening, so the blood pH must be kept around 7.4. Normally, pH stability is possible because the body has built-in mechanisms to prevent pH changes. Buffers are one of these important mechanisms.

Buffers help keep the pH within normal limits because they are chemicals or combinations of chemicals that take up excess hydrogen ions (H^+) or hydroxide ions (OH^-). For example,

ISTUDY

THEME Biological Systems

The Impact of Acid Deposition

Acid Deposition

Normally, rainwater has a pH of about 5.6 because the carbon dioxide in the air combines with water to give a weak solution of carbonic acid. Acid deposition includes rain or snow that has a pH of less than 5, as well as dry acidic particles that fall to Earth from the atmosphere.

When fossil fuels such as coal, oil, and gasoline are burned, sulfur dioxide and nitrogen oxides combine with water to produce sulfuric and nitric acids. These pollutants are generally found eastward of where they originated because of wind patterns. The use of very tall smokestacks causes them to be carried even hundreds of miles away. For example, acid rain in southeastern Canada results from the burning of fossil fuels in factories and power plants in the midwestern United States.

Impact on Lakes

Acid rain adversely affects many aspects of biological systems. Aluminum may leach from the soil of lakes, particularly in areas where the soil is thin and lacks limestone (calcium carbonate, or $CaCO_3$) as a buffer. Acid rain may convert mercury in lake bottom sediments to toxic methyl mercury. Methyl mercury accumulates in fish, which wildlife and people eat. Over time, methyl mercury can accumulate in body tissues and cause serious sensory and muscular health problems. Acid rain in Canada and New England has caused hundreds of lakes to be devoid of fish, and in some cases, any life at all.

Impact on Forests

The leaves of plants damaged by acid rain (Fig. 2C) can no longer carry on photosynthesis as before. When plants are under stress, they become susceptible to diseases and pests of all types. Forests on mountaintops receive more rain than those at lower levels; therefore, they are more affected by acid rain. Forests are also damaged when toxic chemicals such as aluminum are leached from the soil. These kill soil fungi that assist roots in acquiring the nutrients trees need. In New England and the southern Appalachians, millions of acres of high-elevation forests have been devastated.

Sulfur dioxide and nitrogen oxides, the main precursors of acid rain, have been steadily decreasing in the United States due to clean air legislation and strict emission limits (Fig. 2D).

Figure 2C Acid rain strips nutrients from the leaves and needles of trees, making it difficult for them to run photosynthesis. Peter Miller/Science Source

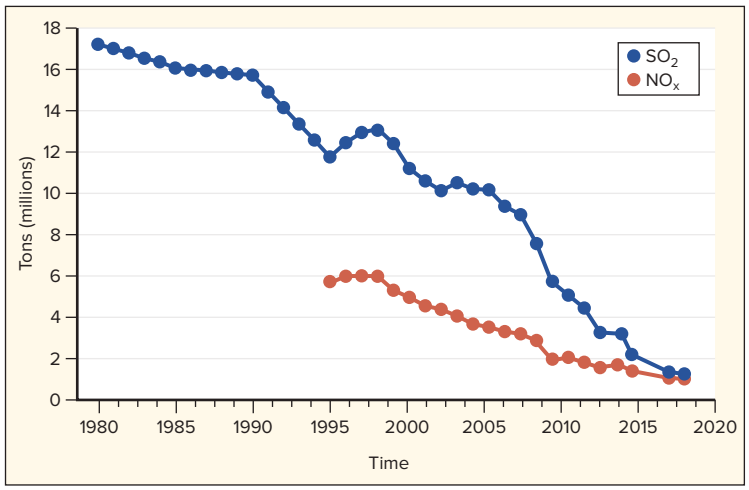


Figure 2D Trends in U.S. acid rain emissions. The burning of fossil fuels in factories, automobiles, and other industrial processes produces chemicals like SO_2 and NO_x that lead to acid deposition and destruction of the environment. Clean air legislation and stricter emission standards over the past several decades have resulted in steady decreases in SO_2 and NO_x, chemicals that lead to acid rain. Source: EPA.gov

Impact on Humans and Structures

Humans may be affected by acid rain. Inhaling dry sulfate and nitrate particles appears to increase the occurrence of respiratory illnesses, such as asthma. Buildings and monuments made of limestone and marble break down when exposed to acid rain. The paint on homes and automobiles is likewise degraded. However, damage to natural systems and human structures due to acid rain is likely to decrease if we continue efforts to reduce chemicals that contribute to acid rain.

Questions to Consider

1. What acid rain trends are evident from the EPA data?
2. Considering that manufacturing is essential to our national interests, how might we modify industrial processing to reduce sulfur dioxide and nitrogen oxide contamination?
3. How might we prevent methyl mercury from entering biological systems and reduce the amount already present?

ISTUDY

carbonic acid (H_2CO_3) is a weak acid that minimally dissociates and then re-forms in the following manner:

$$\underset{\text{carbonic acid}}{H_2CO_3} \underset{\text{re-forms}}{\overset{\text{dissociates}}{\rightleftharpoons}} \underset{\text{bicarbonate ion}}{H^+ + HCO_3^-}$$

Blood always contains a combination of some carbonic acid and some bicarbonate ions. When hydrogen ions (H^+) are added to blood, the following reaction reduces acidity:

$$H^+ + HCO_3^- \longrightarrow H_2CO_3$$

When hydroxide ions (OH^-) are added to blood, this reaction reduces basicity:

$$OH^- + H_2CO_3 \longrightarrow HCO_3^- + H_2O$$

These reactions prevent any significant change in blood pH.

Check Your Progress 2.4

1. Explain the difference in H^+ concentration between an acid and a base.
2. Determine how much more acidic a pH of 2.0 is than a pH of 4.0.
3. Summarize how buffers play an important role in the physiology of living organisms.

Connecting *the* Concepts

What Did You Learn?

2.1 Chemical Elements

Both living organisms and nonliving things are composed of **matter** consisting of **elements.** The most significant elements that are found in living organisms include carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur.

Elements are identified by a unique **atomic symbol,** such as Na for sodium. Elements contain **atoms,** and atoms contain subatomic particles. **Protons** have positive charges, **neutrons** are uncharged, and **electrons** have negative charges. Protons and neutrons in the nucleus determine the **mass number** of an atom. The **atomic number** indicates the number of protons and the number of electrons in electrically neutral atoms.

Isotopes are atoms of a single element that differ in their numbers of neutrons. The **atomic mass** of an element is the average of the naturally occurring isotopes. Radioactive isotopes have many uses, including serving as tracers in biological experiments and medical procedures.

Electrons occupy energy levels (electron shells) at discrete distances from the nucleus. The number of electrons in the **valence shell** determines the reactivity of an atom. The first shell is complete when it is occupied by two electrons. In atoms up through calcium (number 20), every shell beyond the first shell is stable when eight electrons are

ISTUDY

present. The **octet rule** states that atoms react with one another in order to have a stable valence shell. Most atoms do not have filled outer shells, and this causes them to react with one another to satisfy the octet rule. In the process, they form compounds and/or molecules.

2.2 Molecules and Compounds

Compounds and **molecules** are formed when elements associate with each other. A **formula** tells you the number of each kind of element in a compound or molecule. **Ions** are formed when atoms lose or gain one or more electrons to achieve a completed valence shell. The interaction of oppositely charged ions forms **ionic bonds.**

Covalent bonds occur when atoms share one or more pairs of electrons. There are single, double, and triple covalent bonds.

In **nonpolar covalent bonds,** the atoms share the electrons equally. In **polar covalent bonds,** the sharing of electrons is not equal. If the molecule forms polar covalent bonds, the more **electronegative** atom carries a slightly negative charge, and the other atom carries a slightly positive charge.

2.3 Chemistry of Water

Water is an essential molecule for life. The polarity of water molecules forms hydrogen bonds between water molecules. A **hydrogen bond** is a weak attraction between a slightly positive hydrogen atom and a slightly negative oxygen or nitrogen atom within the same or a different molecule. Hydrogen bonds help maintain the structure and function of cellular molecules.

Water's polarity and hydrogen bonding account for its unique properties, which allow life to exist and carry on cellular activities. These properties include the following:

- Water has a high heat capacity. It can absorb heat (measured in **calories**) without a great change in temperature.
- Water has a high heat of evaporation. A large amount of heat is required to cause liquid water to change to a gas.
- Water is a solvent. Water allows the formation of **solutions** with many different **solutes.** Molecules that attract water are **hydrophilic,** whereas molecules that repel water are **hydrophobic.**
- Water is cohesive and adhesive. **Cohesion** allows water molecules to cling together and accounts for the **surface tension** of water. **Adhesion** allows water to cling to surfaces, such as internal transport vessels.
- Frozen water is less dense than liquid water. This allows ice to float on liquid water.

2.4 Acids and Bases

pH is a measure of how acidic or basic a substance is. A small fraction of water molecules dissociate to produce an equal number of **hydrogen ions** and **hydroxide ions.** The term **mole** is used to quantify very small items in chemistry, such as ions. The **pH scale** indicates the concentration of hydrogen ions. Solutions with equal numbers of H^+ and OH^- are termed neutral.

Acids release hydrogen ions in water; these solutions have a pH less than 7. **Bases** either release hydroxide ions or take up hydrogen ions; these solutions have a pH greater than 7. Cells are sensitive to pH changes. Biological systems often contain **buffers** that help keep the pH within a normal range.

Thinking Critically About Biology

1. Natural phenomena often require an explanation. Based on how sodium chloride dissociates in water (see Section 2.3) and Figure 2.12, explain why the oceans don't freeze.
2. Melvin Calvin used radioactive carbon (as a tracer) to discover a series of molecules that form during photosynthesis. Explain why all carbon behaves chemically the same, even when radioactive.
3. On a hot summer day, you decide to dive into a swimming pool. Before you begin your dive, you notice that the surface of the water is smooth and continuous. After the dive, you discover that some water droplets are clinging to your skin and that your skin temperature feels cooler. Explain these observations based on the properties of water.
4. Water has been discovered in the soil of Mars, indicating the possibility that life once existed on the planet. Explain the potential changes that could have occurred to the water on Mars that might prevent life from currently existing on Mars.
5. Antacids are a common over-the-counter remedy for heartburn, a condition caused by an overabundance of H^+ ions in the stomach. Based on what you know about pH, how do the chemicals in antacids work?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Properties of Water

McGraw Hill

What chemical property allows water to have all the unique characteristics shown in the video?

Making It Relevant

1. How do you think the discovery of life on another planet may influence our understanding or appreciation for life on Earth?
2. Would you expect that the chemistry of life on another planet would be similar or different to ours? Why?
3. As scientists continue to explore other planets and moons in our solar system, they are looking for evidence that may indicate life has existed, or currently exists, on those bodies. Other than water, which molecules should scientists look for that might reveal evidence of life?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

3

The Chemistry of Organic Molecules

El Roi/123RF

Chapter Outline

3.1 Organic Molecules
3.2 Carbohydrates
3.3 Lipids
3.4 Proteins
3.5 Nucleic Acids

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 2.1 What is a covalent bond?

Section 2.2 What information about a molecule can be obtained from its structural and molecular formulas?

Section 2.3 What is the difference between hydrophobic and hydrophilic molecules?

Benefits of CBD

Cannabidiol oil, commonly called CBD oil, has recently made its way into a variety of products, such as gummies, additives to your morning coffee, and topical ointments. CBD is an active ingredient that can be derived from the *Cannabis sativa* plant. Hemp is an ideal source of CBD due to the high percentage of CBD molecules and low percentage of THC (tetrahydrocannabinol) molecules it contains. CBD has a molecular formula of $C_{21}H_{30}O_2$, making it an organic molecule that is classified as a lipid.

It is thought that the complex pharmacological mechanisms of CBD are what produce its variety of effects. Managing anxiety, helping individuals fall asleep, and relieving chronic pain are just a few benefits of the CBD molecule. When applied to the skin of animal models as an oil, it helps inhibit inflammation and lower pain associated with arthritis. Due to this scientific evidence, the FDA recently approved the cannabis-derived medicine Epidiolex to help control childhood epilepsy syndromes. More human trials are needed to determine if the CBD molecule can help people manage chronic pain.

In addition to these benefits, there are unfortunately some negative side effects. These range from nausea and fatigue to irritability. It also has the ability to increase the levels of natural blood thinners found in our blood and can elevate the levels and potency of other medications a person may be taking. Because CBD is not regulated by the FDA, the medical community does not yet know the full impact of it on our health or the most effective dose for any particular medical condition.

More research is necessary to fully understand the interaction between CBD and other organic molecules in our bodies before we can feel comfortable using CBD-based products.

In this chapter, you will learn about the structure and function of the major classes of organic molecules, including carbohydrates, lipids, proteins, and nucleic acids.

As you read through this chapter, think about the following questions:

1. What class of biological molecules does CBD belong in?
2. What might a synthetic form of CBD look like?
3. What distinguishes CBD from other biological molecules?

Following *the* Themes

CHAPTER 3 – THE CHEMISTRY OF ORGANIC MOLECULES

Evolution	The diversity of biological life is the result of changes in DNA sequences, one of the many organic molecules found in cells.
Nature of Science	An understanding of a molecule's structure provides insight into its function.
Biological Systems	Organic molecules form the functional basis for all cellular systems.

ISTUDY

3.1 Organic Molecules

Learning Outcomes

Upon completion of this section, you should be able to

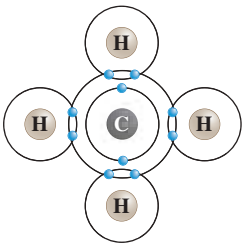
1. Explain how the properties of carbon enable it to produce diverse organic molecules.
2. Explain the relationship between a functional group and the chemical reactivity of an organic molecule.
3. Compare the role of dehydration synthesis and hydrolytic reactions in organic chemistry.

Chemists of the nineteenth century thought that the molecules of cells must contain a vital force, so they divided chemistry into **organic chemistry,** the chemistry of living organisms, and **inorganic chemistry,** the chemistry of nonliving matter. We still use this terminology, even though many types of organic molecules can now be synthesized in the laboratory. Today, we simply use the term *organic* to identify those molecules and compounds that contain both carbon and hydrogen atoms.

Living organisms contain only four classes of organic molecules: carbohydrates, lipids, proteins, and nucleic acids. Collectively, these are called the **biomolecules,** and despite the limited number of types, their functions in a cell are quite diverse. A bacterial cell contains some 5,000 different organic molecules, while a plant or animal cell has twice that number. The diversity of life (Fig. 3.1) is possible because of the diversity of organic molecules. Despite their functional differences, the variety of organic molecules is based on the unique chemical properties of the carbon atom.

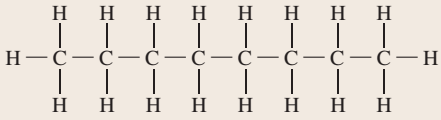
The Carbon Atom

What is there about carbon that makes organic molecules the same but also different? Carbon is quite small, with only a total of six electrons: two electrons in the first shell and four electrons in the outer shell. To acquire four electrons to complete its outer shell, a carbon atom almost always forms covalent bonds. Carbon can form covalent bonds with as many as four other elements.



Methane

Generally, carbon forms those bonds with other atoms of carbon, plus hydrogen, nitrogen, oxygen, phosphorus, and sulfur—the same elements that make up most of the weight of living organisms (see Fig. 2.1). The ability of carbon to share electrons with other carbon atoms plays an important role in establishing the shape, and therefore the function, of the biomolecules. This is because the C—C bond is very stable and allows the formation of long carbon chains. The molecules termed *hydrocarbons,* such as the octane molecule below, are chains of carbon atoms that have additional bonds exclusively with hydrogen atoms.


octane

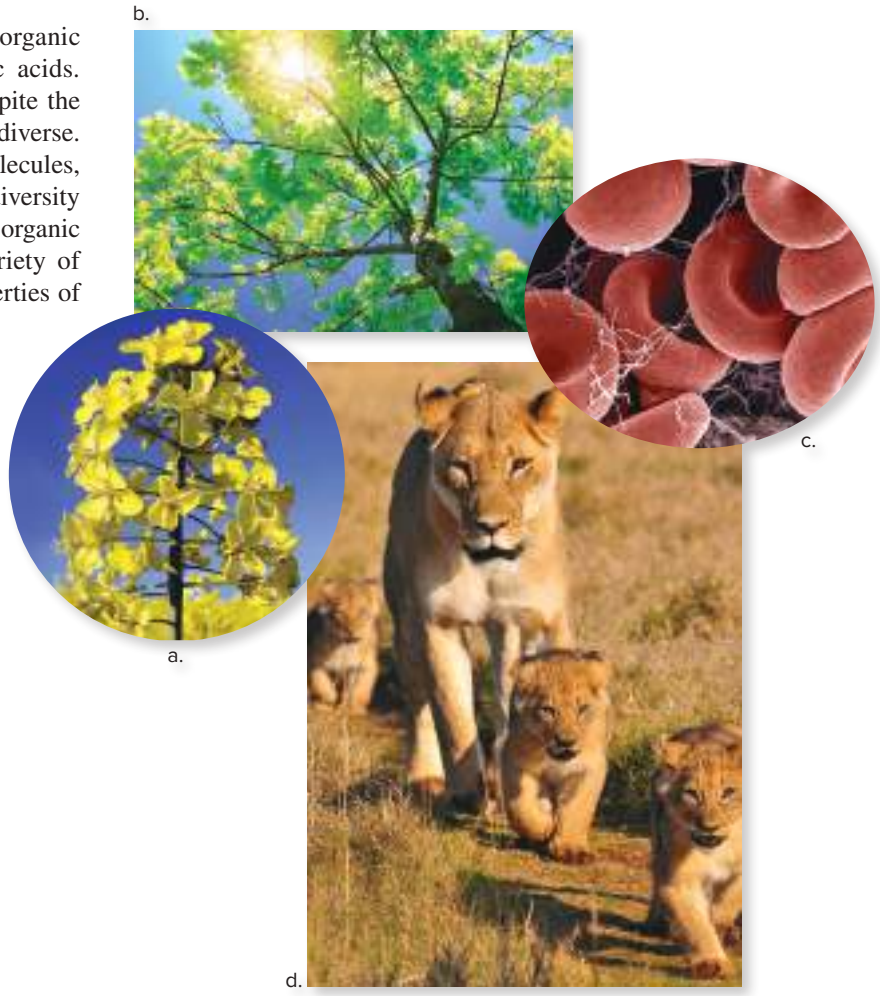
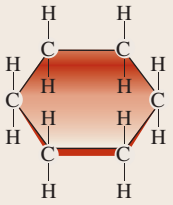


Figure 3.1 Carbon and life. Carbon is the basis of life as we know it. The structure of carbon allows for the formation of (**a**) the lipids that store energy in this canola plant; (**b**) carbohydrates that provide structure for this tree; (**c**) the proteins that form the hemoglobin of red blood cells; and (**d**) the genetic material that the lioness has passed on to her offspring. (a): Peter Austin/E+/Getty Images; (b): Don Hammond/Design Pics; (c): Steve Gschmeissner/Science Photo Library/Getty Images; (d): john michael evan potter/Shutterstock

In addition to long, linear molecules, hydrocarbons may also form a ring structure when placed in a water environment:



cyclohexane

In addition to forming single bonds, carbon can form double bonds with itself and other atoms. Double bonds aren't as flexible as single bonds, and they restrict the movement of bonded atoms. Double bonds affect a molecule's shape and therefore influence its function. The presence of double bonds is one way to distinguish between saturated and unsaturated fats, which are important to heart health. Carbon is also capable of forming triple bonds with itself, as in acetylene, H—C≡C—H, a gas used in industrial welding. Branches may also form at any carbon atom, allowing the formation of long, complex carbon chains. This flexibility makes carbon the ideal building block for biomolecules, and it plays an important role in establishing the diversity of organic molecules we observe in nature.

The Carbon Skeleton and Functional Groups

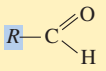
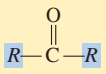
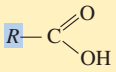
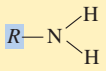
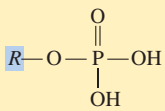
The carbon chain of an organic molecule is called its skeleton, or backbone. Just as a skeleton accounts for your body's shape, so does the carbon skeleton of an organic molecule account for its shape. The diversity of vertebrates, species with a backbone, is a product of the overall shape of the organism and the types of appendages (fins, wings, limbs) they have developed. Likewise, the diversity of organic molecules comes from the attachment of different functional groups to the carbon skeleton.

A **functional group** is a specific combination of bonded atoms that always reacts in the same way, regardless of the carbon skeleton to which it is attached. The majority of the chemical reactivity of a biomolecule can be attributed to its functional groups, rather than to the carbon skeleton to which it is attached.

Typically, the carbon skeleton acts as a framework for the positioning of the functional groups. Table 3.1 lists some of the more common functional groups. The *R* indicates the "remainder" of the molecule. This is the place on the functional group that attaches to the carbon skeleton.

The configuration of the functional groups determines the properties of the biomolecule. For example, the addition of an —OH (hydroxyl group) to a carbon skeleton turns that molecule into an alcohol. When an —OH replaces one of the hydrogens in ethane, a 2-carbon hydrocarbon, it becomes ethanol, a type of alcohol that is familiar because humans can consume it. Whereas ethane, like other hydrocarbons, is hydrophobic (not soluble in water), ethanol is hydrophilic (soluble in water), because the —OH functional group makes the otherwise nonpolar carbon skeleton polar. Because cells are 70–90% water, the ability to interact with and be soluble in water profoundly affects the function of organic molecules in cells.

Table 3.1 Functional Groups

Group	Structure	Compound	Significance
Hydroxyl	R—OH	Alcohol as in ethanol	Polar, forms hydrogen bond Present in sugars, some amino acids
Carbonyl		Aldehyde as in formaldehyde	Polar Present in sugars
		Ketone as in acetone	Polar Present in sugars
Carboxyl (acidic)		Carboxylic acid as in acetic acid	Polar, acidic Present in fatty acids, amino acids
Amino		Amine as in tryptophan	Polar, basic, forms hydrogen bonds Present in amino acids
Sulfhydryl	R—SH	Thiol as in ethanethiol	Forms disulfide bonds Present in some amino acids
Phosphate		Organic phosphate as in phosphorylated molecules	Polar, acidic Present in nucleotides, phospholipids

R = remainder of molecule

Another example is organic molecules that contain carboxyl groups (—COOH). Carboxyl groups are highly polar. In a water environment, they tend to ionize and release hydrogen ions in solution, therefore acting as an acid:

$$—COOH \longrightarrow —COO^- + H^+$$

The attached functional groups determine not only the polarity of an organic molecule but also the types of reactions it can undergo. You will see that alcohols react with carboxyl groups when a fat forms, and that carboxyl groups react with amino groups during protein formation.

Isomers

Isomers (Gk. *isos,* "equal") are organic molecules that have identical molecular formulas but different arrangements of atoms. The two molecules in Figure 3.2 are isomers of one another; they have the same molecular formula but different functional groups. Therefore, we would expect them to have different properties and react

ISTUDY

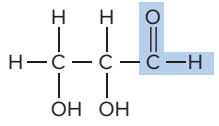
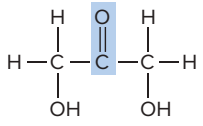
glyceraldehyde	dihydroxyacetone
	

Figure 3.2 Isomers. Isomers have the same molecular formula but different atomic configurations. Both of these compounds have the formula $C_3H_6O_3$. In glyceraldehyde, a colorless crystalline solid, oxygen is double-bonded to an end carbon. In dihydroxyacetone, a white crystalline solid, oxygen is double-bonded to the middle carbon.

differently in chemical reactions. In essence, isomers are variations in the molecular structure of a molecule. Isomers are another example of how the chemistry of carbon leads to variations in the structure of organic molecules.

The Biomolecules of Cells

Many of the biomolecules you are familiar with, such as carbohydrates, lipids, proteins, and nucleic acids, are macromolecules, meaning they contain smaller subunits joined together (Table 3.2). The carbohydrates, proteins, and nucleic acids are referred to as **polymers,** because they are constructed by linking together a large number of the same type of subunit, called a **monomer.** Lipids do not form polymers, because they contain two different types of subunits (glycerol and fatty acids). Polymers may vary considerably in length. Just as a train increases in length when boxcars are hitched together one by one, so a polymer gets longer as monomers bond to one another.

Synthesis and Degradation

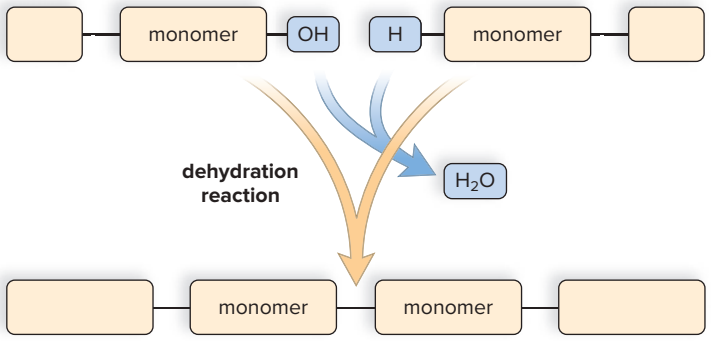
To build, or synthesize, a macromolecule, the cell uses a condensation reaction in which subunits are joined to form a larger structure. This is commonly called a **dehydration reaction,** because the equivalent of a water molecule—that is, an —OH (hydroxyl group) and an —H (hydrogen atom)—is removed as subunits are joined. Therefore, water molecules are formed as biomolecules are synthesized (Fig. 3.3*a*).

To break down biomolecules, a cell uses a degradation type of reaction. During a **hydrolysis reaction** (Gk. *hydro,* "water"; *lyse,* "break"), an —OH group from water attaches to one subunit, and an —H from water attaches to the other subunit. In other words, hydrolytic reactions break down biomolecules by adding water to them (Fig. 3.3*b*).

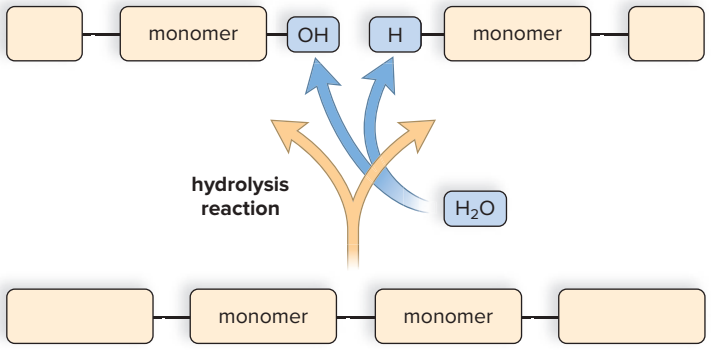
These reactions rarely occur spontaneously. Usually, special molecules called *enzymes* act as catalysts, speeding up the rate of the reaction and allowing it to occur in a functional amount of time. We will take a closer look at enzymes in Section 6.3.

Table 3.2 Biomolecules

Category	Subunits (monomers)	Polymer
Carbohydrates	Monosaccharide	Polysaccharide
Lipids	Glycerol and fatty acids	Does not form polymers
Proteins	Amino acids	Polypeptide
Nucleic acids	Nucleotide	DNA, RNA



a. Synthesis of a biomolecule



b. Degradation of a biomolecule

Figure 3.3 Synthesis and degradation of biomolecules. **a.** In cells, synthesis often occurs when subunits bond during a dehydration reaction (removal of H_2O). **b.** Degradation occurs when the subunits separate during a hydrolysis reaction (the addition of H_2O).

Check Your Progress 3.1

1. Describe the properties of a carbon atom that make it ideally suited to produce varied carbon skeletons.
2. Explain why the substitution of a carboxyl group for a hydroxyl group in a biomolecule would change the molecule's function.
3. Explain why water is needed for the breakdown of a biomolecule.

3.2 Carbohydrates

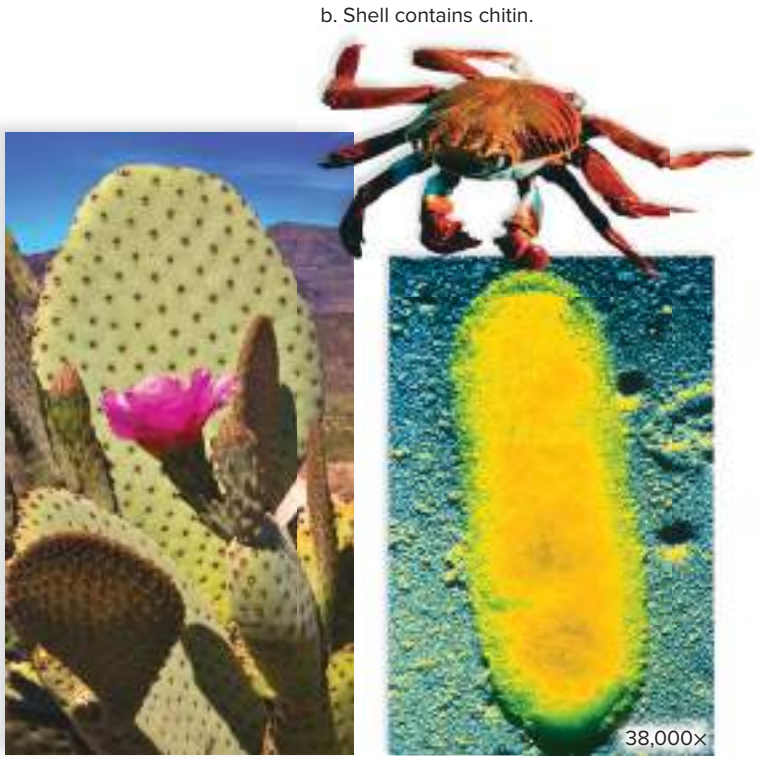
Learning Outcomes

Upon completion of this section, you should be able to

1. Summarize the role of carbohydrates in a cell.
2. Distinguish among the forms of carbohydrates.
3. Compare the energy and structural uses of starch, glycogen, and cellulose.

Carbohydrates are almost universally used as an immediate energy source in living organisms, but in some organisms they also have a structural function (Fig. 3.4). The majority of carbohydrates have a carbon-to-hydrogen-to-oxygen ratio of 1:2:1 (CH_2O). The term *carbohydrate* (literally, carbon-water) includes single sugar molecules

ISTUDY



a. Cell walls contain cellulose. c. Cell walls contain peptidoglycan.

Figure 3.4 Carbohydrates as structural materials. **a.** Plants, such as the cactus shown here, have the carbohydrate cellulose in their cell walls. **b.** The shell of a crab contains a carbohydrate called chitin. **c.** The cell walls of bacteria contain a carbohydrate known as peptidoglycan.
(a): Charles Harker/Shutterstock; (b): Ingram Publishing/Alamy Stock Photo; (c): CNRI/Science Source

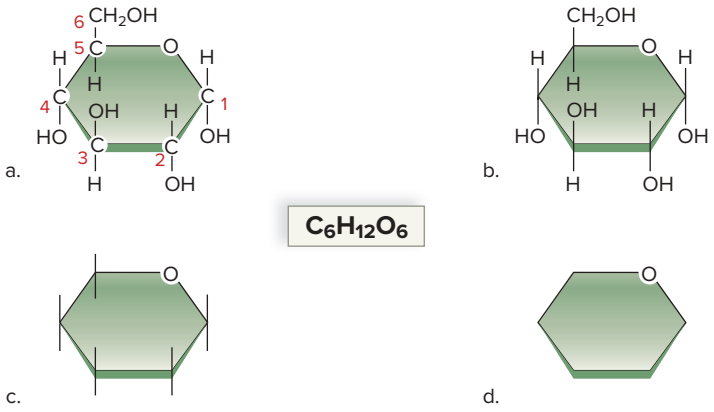


Figure 3.5 Glucose. Glucose is the common form of monosaccharide that provides energy for cells. Each of these structural formulas is glucose. **a.** The carbon skeleton and all attached groups are shown. **b.** The carbon skeleton is omitted. **c.** The carbon skeleton and attached groups are omitted. **d.** Only the ring shape, which includes one oxygen atom, remains.

and chains of sugars. Chain length varies from a few sugars to hundreds of sugars. The monomer subunits, called monosaccharides, are assembled into long polymer chains called polysaccharides.

Monosaccharides

Monosaccharides (Gk. *monos,* "single"; *sacchar,* "sugar") consist of only a single sugar molecule and are commonly called simple sugars. A monosaccharide can have a carbon backbone of three to seven carbons. For example, **pentoses** (Gk. *pent,* "five") are monosaccharides with five carbons, and **hexoses** (Gk. *hex,* "six") are monosaccharides with six carbons. Monosaccharides have a large number of hydroxyl groups, and the presence of this polar functional group makes them soluble in water.

An example of a hexose is **glucose** (Fig. 3.5). Glucose has a molecular formula of $C_6H_{12}O_6$. Despite the fact that glucose has several isomers, such as fructose and galactose, we usually think of $C_6H_{12}O_6$ as glucose. Glucose is critical to biological function and is the major source of cellular fuel for all living organisms. Glucose is the molecule that is broken down and converted into stored chemical energy (ATP) during cellular respiration in nearly all types of organisms.

Ribose and **deoxyribose,** with five carbon atoms, are pentose sugars that are significant because they make up the structural backbone, respectively, in the nucleic acids RNA and DNA. RNA and DNA will be discussed in more detail when we examine nucleic acids later in the chapter.

Disaccharides

A **disaccharide** contains two monosaccharides that have joined during a dehydration reaction. Figure 3.6 shows how the disaccharide maltose (an ingredient used in brewing) is formed when two glucose molecules bond together. Note the position of the bond that results when the —OH groups participating in the reaction

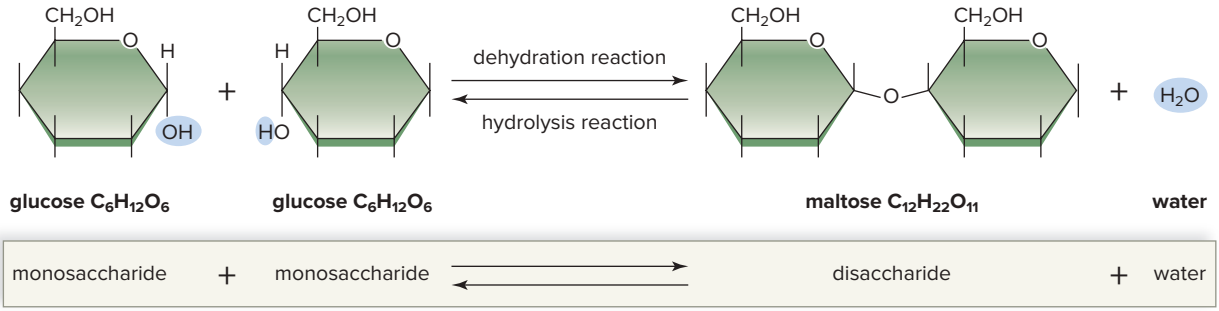


Figure 3.6 Synthesis and degradation of maltose, a disaccharide. Synthesis of maltose occurs following a dehydration reaction when a bond forms between two glucose molecules, and water is removed. Degradation of maltose occurs following a hydrolysis reaction when this bond is broken by the addition of water.

ISTUDY

project below the ring. When the enzymes in our digestive system break this bond, the result is two glucose molecules.

Sucrose is another disaccharide of special interest, because it is sugar we use at home to sweeten our food. Sucrose consists of glucose and fructose monomers. Sucrose is also the form in which sugar is transported in plants. We acquire sucrose from plants such as sugarcane and sugar beets. You may also have heard of lactose, a disaccharide found in milk. Lactose is glucose combined with galactose. Individuals who are *lactose intolerant* lack the enzyme (called lactase) that breaks down lactose. To prevent gastrointestinal discomfort, they can either avoid foods that contain lactose (e.g., dairy products) or take nutritional supplements that contain the enzyme.

Polysaccharides: Energy-Storage Molecules

Polysaccharides are long polymers of monosaccharides. Due to their length, they are sometimes referred to as complex carbohydrates. Some types of polysaccharides function as short-term energy-storage molecules. When an organism requires energy, the polysaccharide is broken down to release sugar molecules. The helical shape of the polysaccharides in Figure 3.7 exposes the sugar linkages to the hydrolytic enzymes that can break them down.

Plants store a large amount of glucose in the form of **starch.** The cells of a potato contain granules, where starch resides during winter until energy is needed for growth in the spring. Notice in Figure 3.7*a*

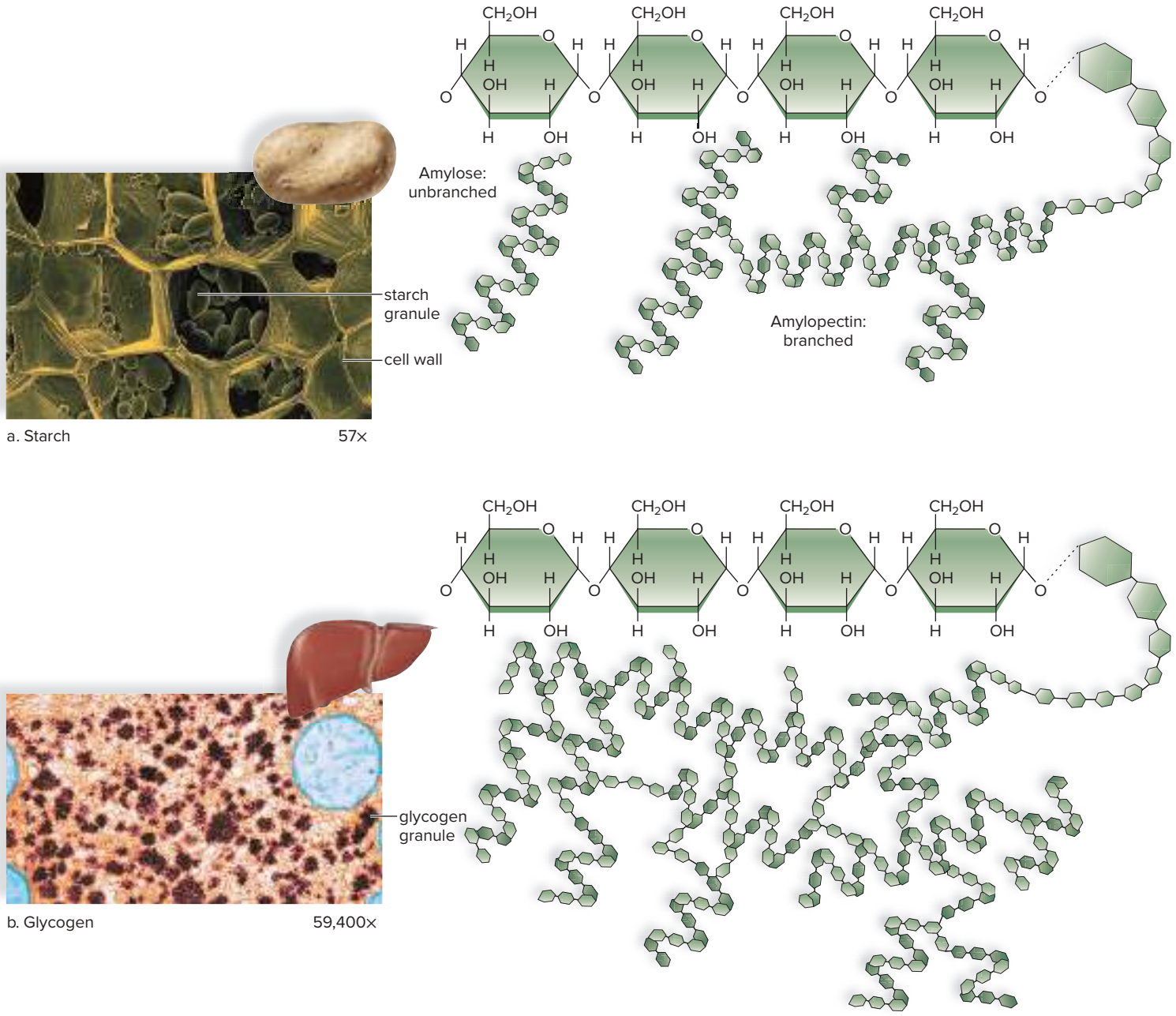


Figure 3.7 Starch and glycogen structure and function. **a.** The electron micrograph shows the location of starch in plant cells. Starch is a chain of glucose molecules that can be branched or unbranched. **b.** The electron micrograph shows glycogen deposits in a portion of a liver cell. Glycogen is a highly branched polymer of glucose molecules. (photos): (a): Dr Jeremy Burgess/Science Source; (b): Don W. Fawcett/Science Source

that starch exists in two forms: One form (amylose) is unbranched and the other (amylopectin) is branched. When a polysaccharide is branched, there is no main carbon chain, because new chains occur at regular intervals, always at the sixth carbon of the monomer.

Animals store a large amount of glucose in the form of **glycogen.** In our bodies and those of other vertebrates, liver cells contain granules, where glycogen is stored until needed. The storage and release of glucose from liver cells are controlled by hormones. After we eat, the release of the hormone insulin from the pancreas promotes the storage of glucose as glycogen. Notice in Figure 3.7*b* that glycogen is even more branched than starch.

Polysaccharides serve as storage molecules because they are not as soluble in water and are much larger than a simple sugar. Therefore, polysaccharides cannot easily pass through the plasma membrane, a sheetlike structure that encloses cells.

Polysaccharides: Structural Molecules

Structural polysaccharides include **cellulose** in plants, **chitin** in animals and fungi, and **peptidoglycan** in bacteria (see Fig. 3.4). In all three, monomers are joined by the type of bond shown for cellulose in Figure 3.8. Glucose is the monomer that makes up cellulose, while in chitin the monomer has an attached amino group. The structure of peptidoglycan is even more complex, with each monomer containing an amino acid chain. In both cases, the addition of a functional group to the glucose monomer changes its chemical properties.

Cellulose is the most abundant carbohydrate and, in fact, the most abundant organic molecule on Earth—over 100 billion tons of cellulose are produced by plants each year. Wood and cotton are both examples of plant products that are composed of cellulose. Because of the structure of the bonds between the glucose molecules, animals are not able to digest cellulose. However, some microorganisms can. The protozoans in the gut of termites enable these insects to digest wood. In cows and other ruminants, microorganisms break down cellulose in a special digestive-tract pouch before the "cud" is returned to the mouth for more chewing and reswallowing. In rabbits, microorganisms digest cellulose in a pouch, where it is packaged into pellets. In order to make use of these nutrient pellets, rabbits have to reswallow them as soon as they pass out at the anus. For other animals, including humans, that have no means of digesting cellulose, cellulose serves as dietary fiber, which maintains regularity of fecal elimination.

Chitin is found in fungal cell walls and in the exoskeletons of crabs and related animals, such as lobsters, scorpions, and insects. Chitin, like cellulose, cannot be digested by animals; however,

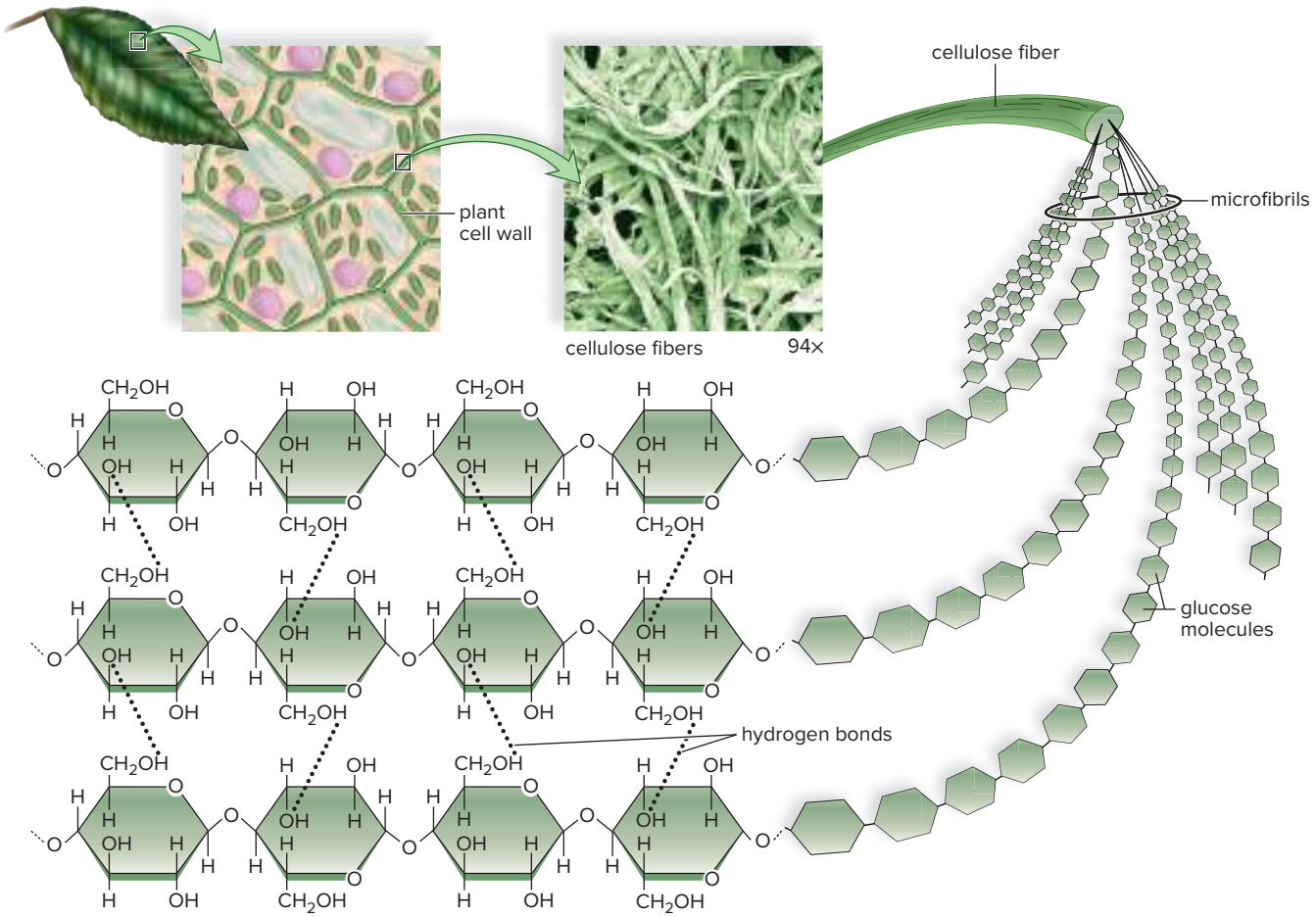


Figure 3.8 Cellulose fibrils. Cellulose fibers criss-cross in plant cell walls for added strength. A cellulose fiber contains several microfibrils, each a polymer of glucose molecules—notice that the linkage bonds differ from those of starch. Every other glucose is flipped, permitting hydrogen bonding and greater strength between the microfibrils. (photo): Scimat/Science Source

ISTUDY

humans have found many other good uses for chitin. Seeds are coated with chitin, and this protects them from attack by soil fungi. Because chitin also has antibacterial and antiviral properties, it is processed and used in medicine as a wound dressing and suture material. Chitin is even useful in the production of cosmetics and various foods.

Check Your Progress 3.2

1. Summarize the general characteristics of carbohydrates and their roles in living organisms.
2. Describe how monosaccharides are combined to form disaccharides.
3. Explain why humans cannot utilize the glucose in cellulose as a nutrient source.

3.3 Lipids

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe why lipids are essential to living organisms.
2. Distinguish between saturated and unsaturated fatty acids.
3. Contrast the structures of fats, phospholipids, and steroids.
4. Compare the functions of phospholipids and steroids in cells.

A variety of organic compounds are classified as **lipids** (Gk. *lipos,* "fat") (Table 3.3). These compounds are insoluble in water due to their hydrocarbon chains. Hydrogens bonded only to carbon are nonpolar and have no tendency to form hydrogen bonds with water molecules. **Fats,** more formally called the triglycerides, are the primary lipid used by animals for both insulation and long-term energy storage. Fat is distributed throughout the body of most animals. In mammals, the majority is found just beneath the skin, where it helps retain body heat. Triglycerides in plants are commonly referred to as **oils.** We are familiar with fats and oils, because we use them in cooking and consume them as food, supplements, and medicine (see the chapter opener). However, increasingly they are also being used as an alternative fuel source, such as biodiesel, for our industrialized societies.

Table 3.3 Lipids

Type	Functions	Human Uses
Fats	Long-term energy storage and insulation in animals	Butter, lard
Oils	Long-term energy storage in plants and their seeds	Cooking oils
Phospholipids	Component of plasma membrane	Food additives
Steroids	Component of plasma membrane (cholesterol), sex hormones	Medicines
Waxes	Protection, prevention of water loss (cuticle of plant surfaces), beeswax, earwax	Candles, polishes

In addition to the triglycerides, phospholipids, steroids, and waxes are important lipids found in living organisms. The next sections will explore the structure and function of these classes of lipids.

Triglycerides: Long-Term Energy Storage

Triglycerides are composed of fatty acids and glycerol subunits. Each **fatty acid** consists of a long hydrocarbon chain with an even number of carbons and a —COOH (carboxyl) group at one end. Most of the fatty acids in cells contain 16 or 18 carbon atoms per molecule, although smaller ones are also found. The fatty acid chains may be either saturated or unsaturated (Fig. 3.9). **Saturated fatty acids** (Fig. 3.9*a*) lack double bonds between the carbon atoms and contain as many hydrogens as they can hold. **Unsaturated fatty acids** (Fig. 3.9*b*) have double bonds in the carbon chain, which reduces the number of bonded hydrogen atoms. In addition, double bonds in unsaturated fatty acids may have chemical groups arranged on the same side (termed *cis configuration*) or on opposite sides (termed *trans configuration*) of the double bond. A **trans fat** is a triglyceride that has at least one bond in a trans configuration. The cis or trans configuration of an unsaturated fatty acid affects its biological activity.

Glycerol is a 3-carbon compound with three —OH groups. The —OH groups are polar, making glycerol soluble in water. When a fat or an oil forms, the —COOH functional groups of three fatty acids react with the —OH groups of glycerol during a dehydration reaction (Fig. 3.10), resulting in a fat molecule and three molecules of water. Fats and oils are degraded during a hydrolysis reaction. Notice that triglycerides have many nonpolar C—H bonds; therefore, they do not mix with water. Even though cooking oils and water are both liquid, they do not mix, even after shaking, because the nonpolar oil and polar water are chemically incompatible.

Triglycerides containing fatty acids with unsaturated bonds melt at a lower temperature than those containing only saturated fatty acids. The reason is that a double bond creates a kink in the fatty acid chain that prevents close packing between the hydrocarbon chains (Fig. 3.10). Butter is a fat that is solid at room temperature and is composed primarily of saturated fatty acids, whereas corn oil, which is a liquid even when placed in the refrigerator, is composed primarily of unsaturated fatty acids. This difference has applications useful to living organisms. For example, the feet of reindeer and penguins contain unsaturated triglycerides, and this helps protect those exposed parts from freezing.

In general, fats, which most often come from animals, are solid at room temperature, whereas oils, which come from plants, are liquid at room temperature. Diets high in animal fat have been associated with cardiovascular disorders, because saturated fats and other molecules can accumulate inside the lining of blood vessels and block blood flow. Health organizations have recommended replacing fat with oils such as olive oil and canola oil in our diet whenever possible.

Nearly all animals use fat rather than glycogen for long-term energy storage. Gram for gram, fat stores more energy than glycogen. The C—H bonds of fatty acids make them a richer source of chemical energy than glycogen, because more bonds with stored energy are present in fatty acids. In contrast, glycogen has many C—OH bonds, which are less energetic bonds. Also, fat droplets do not contain water, because they are nonpolar. Small birds, such as the broad-tailed

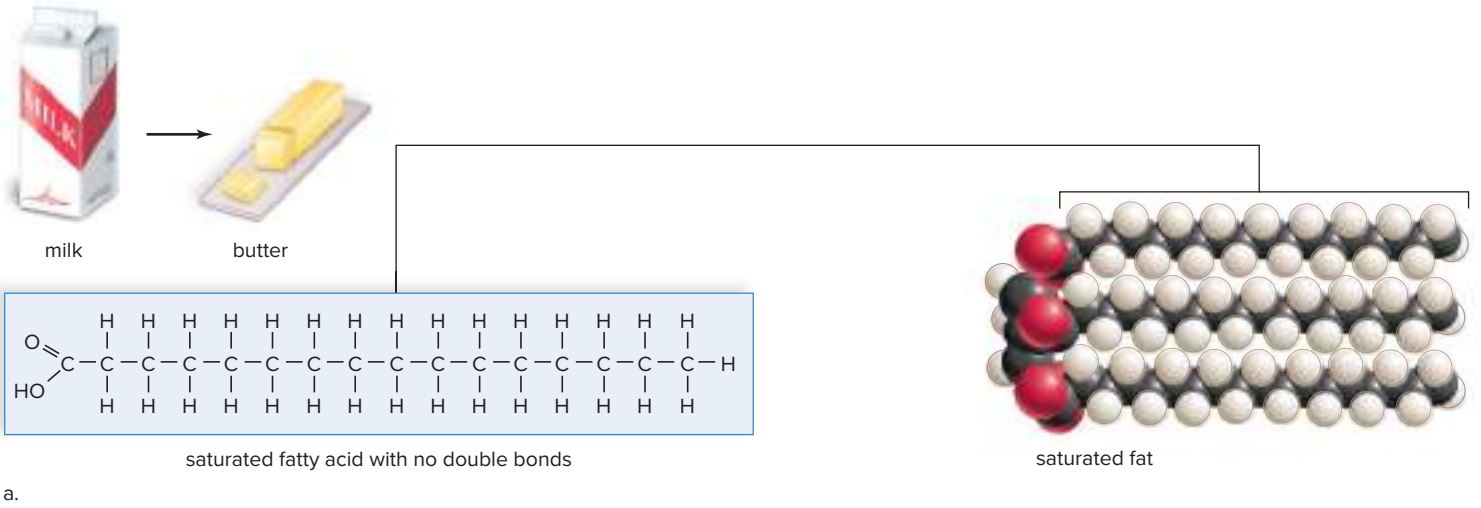


Figure 3.9 Saturated and unsaturated fatty acids. A fatty acid has a carboxyl group attached to a long hydrocarbon chain. **a.** If there are no double bonds between the carbons in the chain, the fatty acid is saturated. **b.** If there are double bonds between some of the carbons (yellow highlights), the fatty acid is unsaturated and a kink occurs in the chain. Trans fats (red highlight) possess a double bond with hydrogen atoms on either side of the carbon chain.

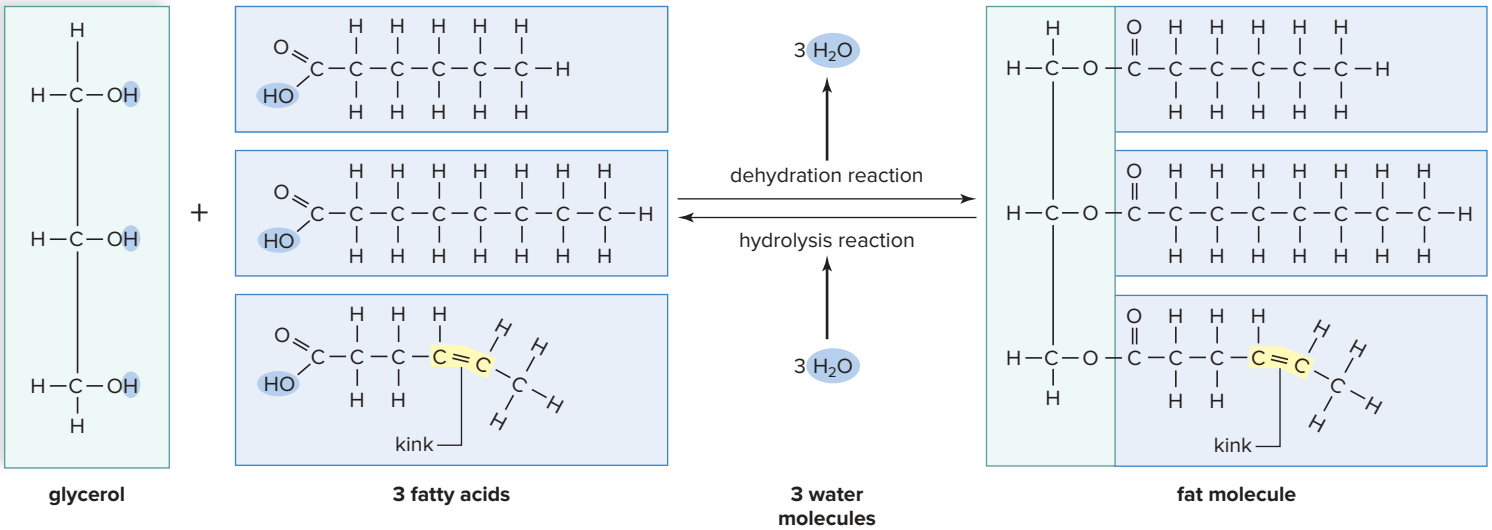


Figure 3.10 Synthesis and degradation of a triglyceride. Following a dehydration reaction, glycerol is bonded to three fatty acid molecules as fat forms and water is given off. Following a hydrolysis reaction, the bonds are broken due to the addition of water.

ISTUDY

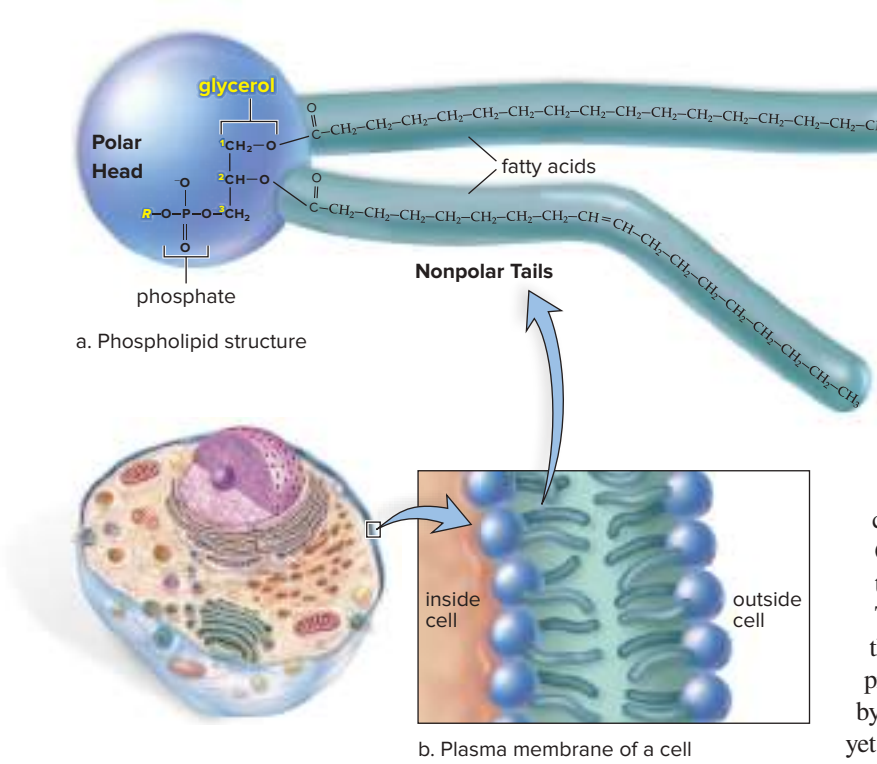


Figure 3.11 Phospholipids form membranes. **a.** Phospholipids are constructed like fats, except that in place of the third fatty acid, they have a polar phosphate group. The hydrophilic (polar) head is soluble in water, whereas the two hydrophobic (nonpolar) tails are not. A tail has a kink wherever there is an unsaturated bond. **b.** Because of their structure, phospholipids form a bilayer that serves as the major component of a cell's plasma membrane. The fluidity of the plasma membrane is affected by kinks in the phospholipids' tails.

hummingbird, store a great deal of fat before they start their long spring and fall migratory flights. About 0.15 g of fat per gram of body weight is accumulated each day. If the same amount of energy were stored as glycogen, a bird would be so heavy it could not fly.

Phospholipids: Membrane Components

Phospholipids are basically triglycerides, except that in place of the third fatty acid attached to glycerol, there is a polar phosphate group (Fig. 3.11*a*). This portion of the molecule becomes the polar head, while the hydrocarbon chains of the fatty acids become the nonpolar tails. Notice in Figure 3.11*a* that a double bond causes a tail to kink.

Phospholipids have hydrophilic heads and hydrophobic tails. When exposed to water, phospholipids tend to arrange themselves so that the polar heads are oriented toward water and the nonpolar fatty acid tails are oriented away from water. In living organisms, which are made mostly of water, phospholipids tend to become a bilayer (double layer). The polar heads prefer to interact with the water, while the nonpolar tails associate together and stay away from the water molecules. Thus, phospholipids arrange themselves like a "sandwich," with the polar heads facing the outside (the bread slices) and the fatty acid tails on the inside (the filling). This phospholipid bilayer is a key component used to keep cells separate from the biological compartments within cells.

The plasma membrane that surrounds cells consists primarily of a phospholipid bilayer (Fig. 3.11*b*). The presence of kinks in the tails causes the plasma membrane to be fluid across a range of temperatures found in nature. A plasma membrane is essential to the structure and function of a cell, and this signifies the importance of phospholipids to living organisms.

Steroids: Four Fused Rings

Steroids are lipids with structures that are entirely different from those of triglycerides and phospholipids. Steroid molecules have skeletons of four fused carbon rings (Fig. 3.12*a*). Each type of steroid differs primarily by the types of functional groups attached to the carbon skeleton.

Cholesterol is an essential component of an animal cell's plasma membrane, where it provides physical stability. Cholesterol is the precursor of several other steroids, such as the sex hormones testosterone and estrogen (Fig. 3.12*b, c*). The male sex hormone, testosterone, is formed primarily in the testes, and the female sex hormone, estrogen, is formed primarily in the ovaries. Testosterone and estrogen differ only by the functional groups attached to the same carbon skeleton, yet each has its own profound effect on the body and sexuality of an animal. Human and plant estrogens are similar in structure, and if estrogen therapy is recommended, some females prefer taking soy products in preference to estrogen from animals.

Cholesterol can also contribute to cardiovascular disorders. The presence of cholesterol encourages the accumulation of fatty material inside the lining of blood vessels, which decreases the size of the opening and thereby can result in high blood pressure. Cholesterol-lowering medications are available.

Waxes

In **waxes,** long-chain fatty acids are connected to carbon chains containing alcohol functional groups. Waxes are solid at normal temperatures, because they have a high melting point. Being hydrophobic, they are also waterproof and resistant to degradation. In many plants, waxes, along with other molecules, form a protective cuticle (covering) that prevents the loss of water from all exposed parts (Fig. 3.13*a*). In many animals, waxes are involved in skin and fur maintenance. In humans, wax is produced by glands in the outer ear canal. Earwax contains cerumen, an organic compound that has both lubricating and antibacterial properties. It also traps dust and dirt, preventing these contaminants from reaching the eardrum. A honeybee produces beeswax in glands on the underside of its abdomen. Beeswax is used to make the six-sided cells of the comb, where honey is stored (Fig. 3.13*b*). Honey contains the sugars fructose and glucose, breakdown products of the sugar sucrose.

Check Your Progress 3.3

1. List the functions of triglycerides, phospholipids, steroids, and waxes.
2. Contrast the structure of a saturated fatty acid with that of an unsaturated fatty acid.
3. Explain why phospholipids form a bilayer in water.

ISTUDY

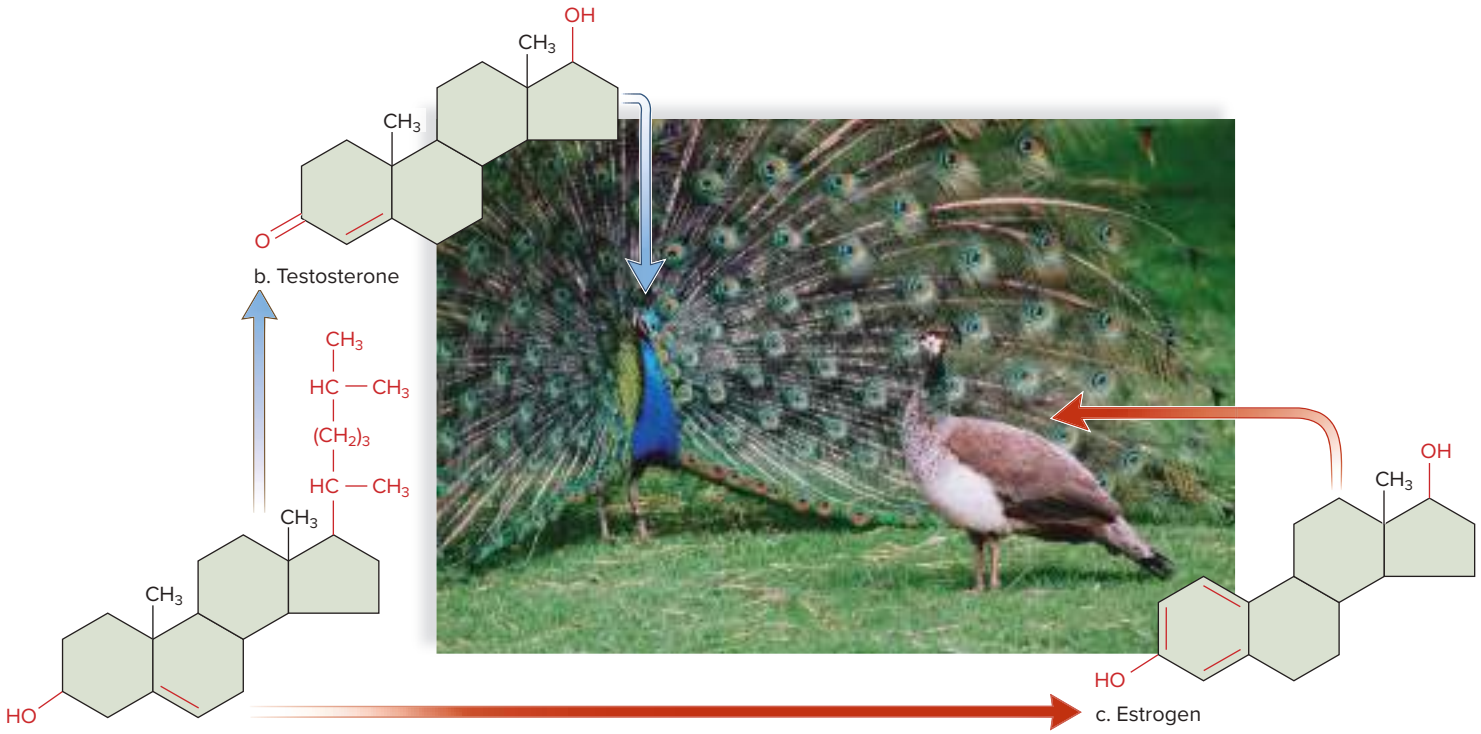


Figure 3.12 Steroid diversity. **a.** Built like cholesterol, (**b**) testosterone and (**c**) estrogen have different effects on the body due to different functional groups attached to the same carbon skeleton. Testosterone is the male sex hormone active in peacocks (*left*), and estrogen is the female sex hormone active in peahens (*right*). These hormones are present in many living creatures. (photo): Ernie Janes/Avalon

a. b.

Figure 3.13 Waxes. Waxes are a type of lipid. **a.** Fruits are protected by a waxy coating, which is visible on these plums. **b.** Bees secrete the wax that allows them to build a comb, where they store honey. (a): Harald Theissen/Imagebroker/Getty Images; (b): Simon Murrell/IT Stock/Alamy Stock Photo

3.4 Proteins

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the functions of proteins in cells.
2. Explain how a polypeptide is constructed from amino acids.
3. Compare the four levels of protein structure.
4. Understand the factors that affect protein structure and function.

Proteins (Gk. *proteios,* "first place") are of primary importance to the structure and function of cells. As much as 50% of the dry weight of most cells consists of proteins. Several hundred thousand proteins have been identified. The following are some of their many functions in animals:

- *Metabolism.* Enzyme proteins bring reactants together, increasing the rate of chemical reactions in cells. They are specific for one particular type of reaction and function best within specific ranges of temperature and pH.

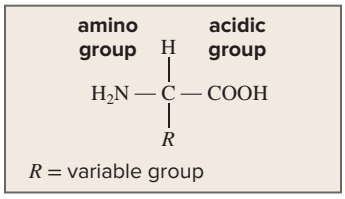
ISTUDY

- *Support.* Some proteins have a structural function. For example, keratin makes up hair and nails, while collagen gives strength to ligaments, tendons, and skin.
- *Transport.* Channel and carrier proteins in the plasma membrane regulate which substances enter and exit cells. Other proteins transport molecules in the blood of animals. For instance, hemoglobin is a complex protein that transports oxygen to tissues and cells.
- *Defense.* Antibodies are proteins of our immune system that combine with foreign substances, called antigens. Antibodies bind and prevent antigens from destroying cells and upsetting homeostasis.
- *Regulation.* Some hormones are protein-based structures that influence cellular behavior. Some serve as intercellular messengers that influence cell metabolism. Examples include insulin, which regulates how much glucose is in the blood and in cells, and growth hormone during childhood and adolescence, which determines the height of an individual.
- *Motion.* The contractile proteins actin and myosin allow parts of cells to move and cause muscles to contract. Muscle contraction allows animals to travel from place to place. All cells contain proteins that move cell components to different internal locations. Without such proteins, cells would not be able to function.

Proteins are such a major part of living organisms that tissues and cells of the body can sometimes be characterized by the proteins they contain or produce. For example, muscle cells contain large amounts of actin and myosin for contraction; red blood cells are filled with hemoglobin for oxygen transport; and support tissues, such as ligaments and tendons, contain the protein collagen, which is composed of tough fibers.

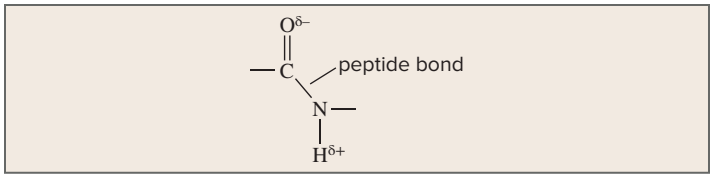
Amino Acids: Protein Monomers

Proteins are polymers constructed from amino acid monomers. The name **amino acid** is used because one of the functional groups in the amino acid is —NH_2 (an amino group) and another is —COOH (an acid group). The third group is called an *R* (variable) group. The structure of an amino acid is as follows:



Note that the central carbon atom in an amino acid bonds to a hydrogen atom and to three other groups of atoms, one of which is the *R* group. Amino acids differ according to their particular *R* group (Fig. 3.14). The *R* groups range in complexity from a single hydrogen atom to complicated ring compounds. Some *R* groups are polar and associate with water, whereas others are nonpolar and do not. Also, the amino acid cysteine has an *R* group that ends with a —SH (sulfide) group, which often covalently connects one chain of amino acids to another by a disulfide bond, —S—S—. Several other amino acids commonly found in cells are shown in Figure 3.14.

Amino acids are linked by dehydration reactions that connect the carboxyl group of one amino acid to the amino group of another amino acid (Fig. 3.15). The resulting covalent bond between two amino acids is called a **peptide bond.** The atoms associated with the peptide bond share the electrons unevenly, because oxygen is more electronegative than nitrogen. Therefore, the hydrogen attached to the nitrogen has a slightly positive charge, whereas the oxygen has a slightly negative charge:



The polarity of the peptide bond means that hydrogen bonding is possible between the —CO of one amino acid and the —NH of another amino acid in a polypeptide. This hydrogen bonding influences the structure, or shape, of a protein.

A **peptide** is two or more amino acids bonded together, and a **polypeptide** is a chain of many amino acids joined by peptide bonds. A protein is a polypeptide that has been folded into a particular shape and has function. Some proteins may consist of more than one polypeptide chain, making it possible for some proteins to have a very large number of amino acids.

The amino acid sequence greatly influences the final three-dimensional shape and function of a protein. Each protein has a sequence of amino acids that is defined by information contained within a gene. This amino acid sequence forms the basis for all levels of protein structure, which directly affect protein function. Proteins that have an abnormal sequence often have a three-dimensional shape that causes them to function improperly. From an evolutionary perspective, we also know that, for a particular protein, the sequences of amino acids are highly similar within a species and are different across species.

Shape of Proteins

Proteins may have up to four levels of structural organization: primary, secondary, tertiary, and quaternary (Fig. 3.16).

Primary Structure

The *primary structure* of a protein is the linear sequence of amino acids. Just as millions of different words can be constructed from just 26 letters in the English alphabet, hundreds of thousands of different polypeptides can be built from just 20 amino acids. To make a new word in a language, all that is usually required is to vary the number and sequence of a few letters. Likewise, changing the sequence of 20 amino acids in a polypeptide can produce a huge array of different proteins. The sequence of the amino acids in the primary structure is determined by the information contained within genes, which are part of the DNA of the cell (see Section 12.3).

ISTUDY

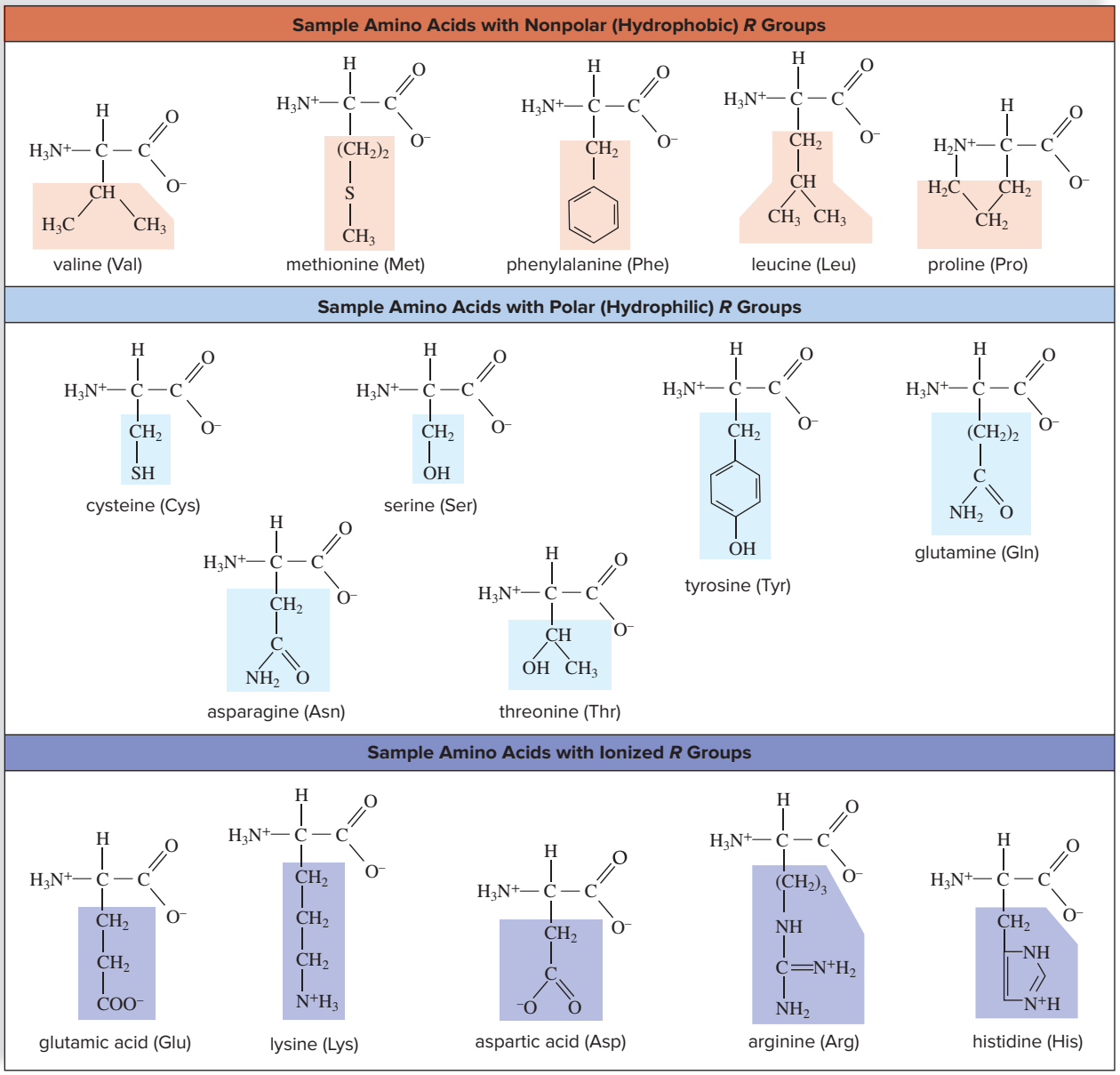


Figure 3.14 Amino acids. There are 20 different kinds of amino acids, some of which are shown here, that are available for the formation of a polypeptide. Amino acids differ by the particular *R* group (shaded area of the molecule) attached to the central carbon. Some *R* groups are nonpolar and hydrophobic (*top*), some are polar and hydrophilic (*center*), and some are ionized and hydrophilic (*bottom*). The amino acids are shown in ionized form.

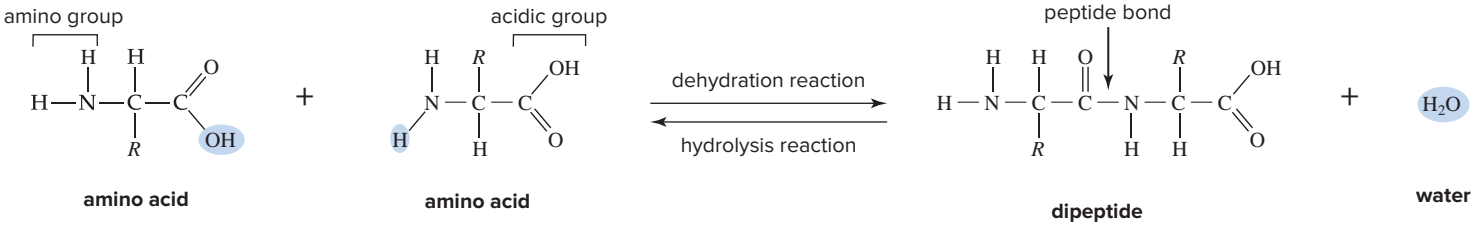


Figure 3.15 Synthesis and degradation of a peptide. Following a dehydration reaction, a peptide bond joins two amino acids and a water molecule is released. Following a hydrolysis reaction, the bond is broken due to the addition of water.

ISTUDY

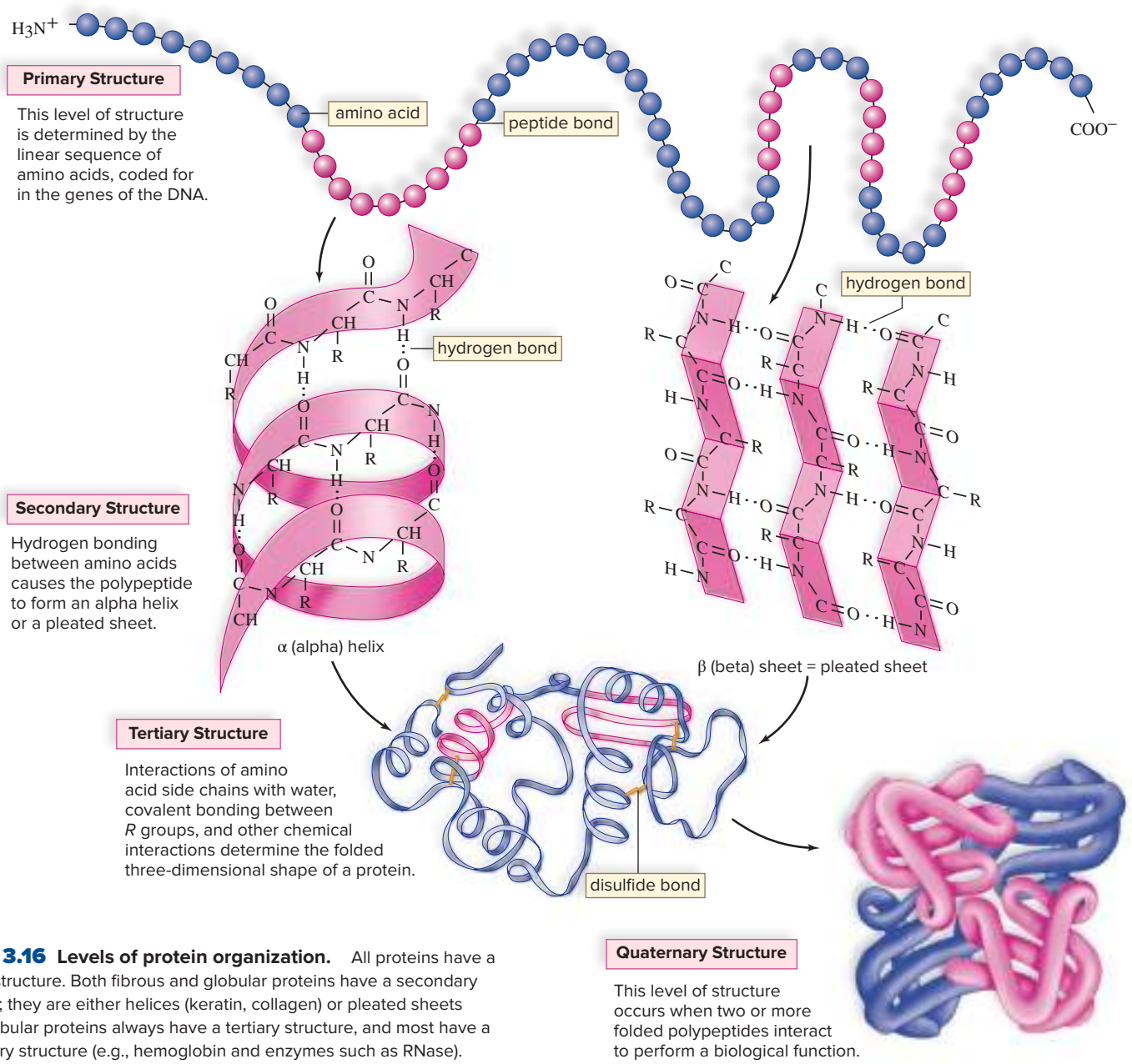


Figure 3.16 Levels of protein organization. All proteins have a primary structure. Both fibrous and globular proteins have a secondary structure; they are either helices (keratin, collagen) or pleated sheets (silk). Globular proteins always have a tertiary structure, and most have a quaternary structure (e.g., hemoglobin and enzymes such as RNase).

Secondary Structure

The *secondary structure* of a protein occurs when the polypeptide coils or folds in a particular way (Fig. 3.16).

Linus Pauling and Robert Corey, who began studying the structure of amino acids in the late 1930s, concluded that a coiling they called an α (alpha) helix, and a pleated sheet they called a β (beta) sheet, were two basic patterns of structure that amino acids assumed within a polypeptide. The names came from the fact that the α helix was the first pattern they discovered, and the β sheet was the second pattern. Each polypeptide can have multiple α helices and β pleated sheets.

The spiral shape of α helices is formed by hydrogen bonding between every fourth amino acid within the polypeptide chain, whereas β sheets are formed when the polypeptide turns back upon itself, allowing hydrogen bonding to occur between extended lengths of the polypeptide. *Fibrous proteins,* which are structural proteins, exist only as helices or pleated sheets that hydrogen bond to each other. Examples are keratin, a protein in hair, and silk, a protein that forms spider webs. Both of these proteins have only a secondary structure (Fig. 3.17).

Tertiary Structure

A *tertiary structure* is the folding that results in the final three-dimensional shape of a polypeptide. *Globular proteins,* which tend to ball up into rounded shapes, have tertiary structure. Most enzymes are globular proteins.

The interaction of hydrophobic amino acids in the polypeptide chain with the surrounding water is a major factor in how proteins fold into, and maintain, their final shape. These nonpolar amino acids tend to group together in the interior of a protein, to be as far away from water as possible. In contrast, the polar hydrophilic and ionic amino acids interact well with water and tend to orient themselves on the protein's surface. These chemical interactions, along with hydrogen bonds, ionic bonds, and covalent bonds between *R* groups,

a. b. c.

Figure 3.17 Fibrous proteins. Fibrous proteins are structural proteins. **a.** Keratin—found, for example, in hair, horns, and hooves—exemplifies fibrous proteins that are helical for most of their length. **b.** A chemical treatment, called a perm, may be used to alter the secondary structure of the keratin proteins. **c.** Silk made by spiders is fibrous proteins that are pleated sheets for most of their length. Hydrogen bonding between parts of the molecule causes the pleated sheet to double back on itself. (photos): (a): vgajic/E+/Getty Images; (b): Image Source/Alamy Stock Photo; (c): francois dion/Moment/Getty Images

all contribute to the tertiary structure of a protein. Strong disulfide linkages (—S—S—) in particular help maintain the tertiary shape.

Enzymes work best at a specific temperature, and each one has an optimal pH, at which the rate of the reaction is highest. At the optimal temperature and pH, the enzyme can maintain its normal shape. A higher temperature and change in pH can disrupt the interactions that maintain the shape of the enzyme. When a protein loses its natural shape, it is said to be **denatured.** An organism can die if too many proteins become denatured, because it can no longer maintain the metabolic processes necessary for life.

Quaternary Structure

Some proteins have a *quaternary structure,* because they consist of more than one polypeptide. Hemoglobin, the protein that transports oxygen in the blood, is a globular protein that consists of four polypeptides. Each polypeptide in hemoglobin has a primary, secondary, and tertiary structure. However, a protein can have only two polypeptides and still have a quaternary structure.

The Importance of Protein Folding

The function of a protein is directly associated with its three-dimensional structure. Therefore, the correct folding of a protein is important. Changes in the instructions in the genes encoding a protein may result in changes in either the structure of the protein or the way it folds. At times, this may be detrimental, as is the case for diseases such as cystic fibrosis. However, sometimes these changes are less damaging. Minor changes in the proteins associated with hair or eye color are examples of variation in protein structure that are not detrimental to an organism.

Cells contain *chaperone proteins,* which help new proteins fold into their normal shape. Initially, researchers thought that chaperone proteins only ensured that proteins folded properly, but now it appears that they might correct any misfolding of a new protein. In any case, without fully functioning chaperone proteins, a cell's proteins may not be functional, because they have misfolded. Several diseases in humans, such as Alzheimer's disease, are associated with misshapen proteins.

Other diseases in humans are due to misfolded proteins, but the cause may be different. For years, investigators have been studying fatal brain diseases, known as transmissible spongiform encephalopathies (TSEs), that have no cure because no infective agent could be found. Mad cow disease is a well-known example of a TSE disease. Now it appears that TSE diseases might be due to misfolded proteins, called **prions,** that cause other proteins of the same type to fold the wrong way, too.

Check Your Progress 3.4

1. List the roles of proteins in living organisms.
2. Describe how two amino acids are combined to form a polypeptide.
3. Summarize the differences among primary, secondary, tertiary, and quaternary structure.
4. Describe the consequences of incorrect protein folding.

3.5 Nucleic Acids

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between a nucleotide and nucleic acid.
2. Compare the structure and function of DNA and RNA nucleic acids.
3. Explain how ATP is able to store energy.

Each cell has a storehouse of information that specifies how a cell should behave, respond to the environment, and divide to make new cells. **Nucleic acids** are composed of nucleotides and have the ability to store information, including the instructions for life, and

ISTUDY

conduct chemical reactions. **DNA (deoxyribonucleic acid)** is one type of nucleic acid that not only stores information about how to copy, or replicate, itself but also specifies the order in which amino acids are to be joined to make a protein.

RNA (ribonucleic acid) is another diverse type of nucleic acid that has multiple uses. Messenger RNA (mRNA) is a temporary copy of a gene in the DNA that specifies what the amino acid sequence will be during the process of protein synthesis. Transfer RNA (tRNA) is also necessary in synthesizing proteins, and it helps translate the sequence of nucleic acids in a gene into the correct sequence of amino acids during protein synthesis. Ribosomal RNA (rRNA) works as an enzyme to form the peptide bonds between amino acids in a polypeptide. A wide range of other RNA molecules also perform important functions within the cell.

Not all nucleotides are made into DNA or RNA polymers. Some nucleotides are directly involved in metabolic functions in cells. For example, some are components of **coenzymes,** nonprotein organic molecules that help regulate enzymatic reactions. **ATP (adenosine triphosphate)** is a nucleotide that stores large amounts of energy needed for cellular reactions and for various other energy-requiring processes in the cell.

Structure of DNA and RNA

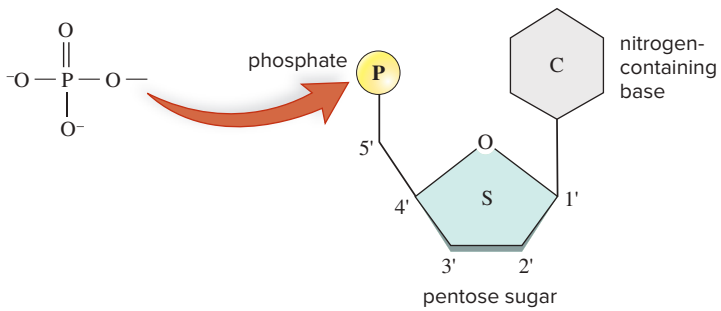
Every **nucleotide** is comprised of three types of molecules: a pentose sugar, a phosphate (phosphoric acid), and a nitrogen-containing base (Fig. 3.18*a*). In DNA, the pentose sugar is deoxyribose, and in RNA the pentose sugar is ribose. A difference in the structure of these 5-carbon sugars accounts for their respective names, because, as you might guess, deoxyribose lacks an oxygen atom found in ribose (Fig. 3.18*b*).

Table 3.4 Comparing DNA and RNA Structure

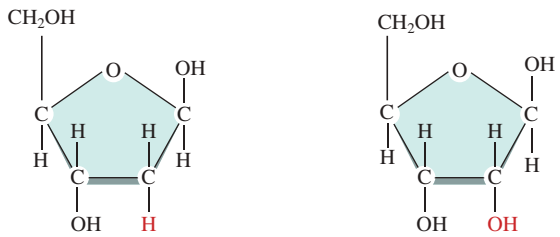
	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	Adenine, guanine, thymine, cytosine	Adenine, guanine, uracil, cytosine
Strands	Double-stranded with base pairing	Usually single-stranded
Helix	Yes	No

Both DNA and RNA contain combinations of four nucleotides (Fig. 3.18*c*), but these differ somewhat between the two nucleic acids (Table 3.4). Nucleotides that have a base with a single ring are called pyrimidines, and nucleotides with a double ring are called purines. In DNA, the pyrimidine bases are cytosine and thymine; in RNA, the pyrimidine bases are cytosine and uracil. Both DNA and RNA contain the purine bases adenine and guanine. These molecules are called bases because their presence raises the pH of a solution. Table 3.4 summarizes the differences between DNA and RNA.

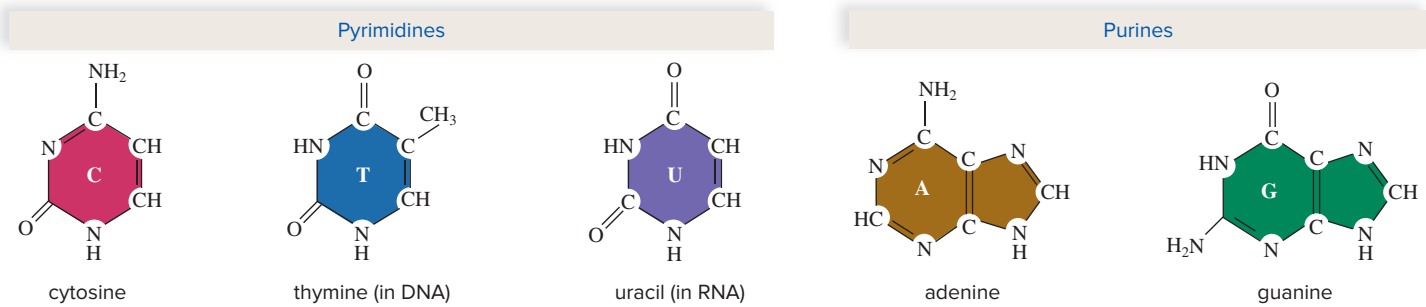
Nucleotides are joined into a DNA or an RNA polymer by a series of dehydration reactions. The resulting polymer is a linear molecule called a strand, in which the backbone is made up of an alternating series of sugar–phosphate–sugar–phosphate molecules. The bases project to one side of the backbone. Nucleotides are



a. Nucleotide structure



b. Deoxyribose versus ribose



c. Pyrimidines versus purines

Figure 3.18 Nucleotides. **a.** A nucleotide consists of a pentose sugar, a phosphate molecule, and a nitrogen-containing base. **b.** DNA contains the sugar deoxyribose, and RNA contains the sugar ribose. **c.** DNA contains the pyrimidines C and T and the purines A and G. RNA contains the pyrimidines C and U and the purines A and G.

ISTUDY

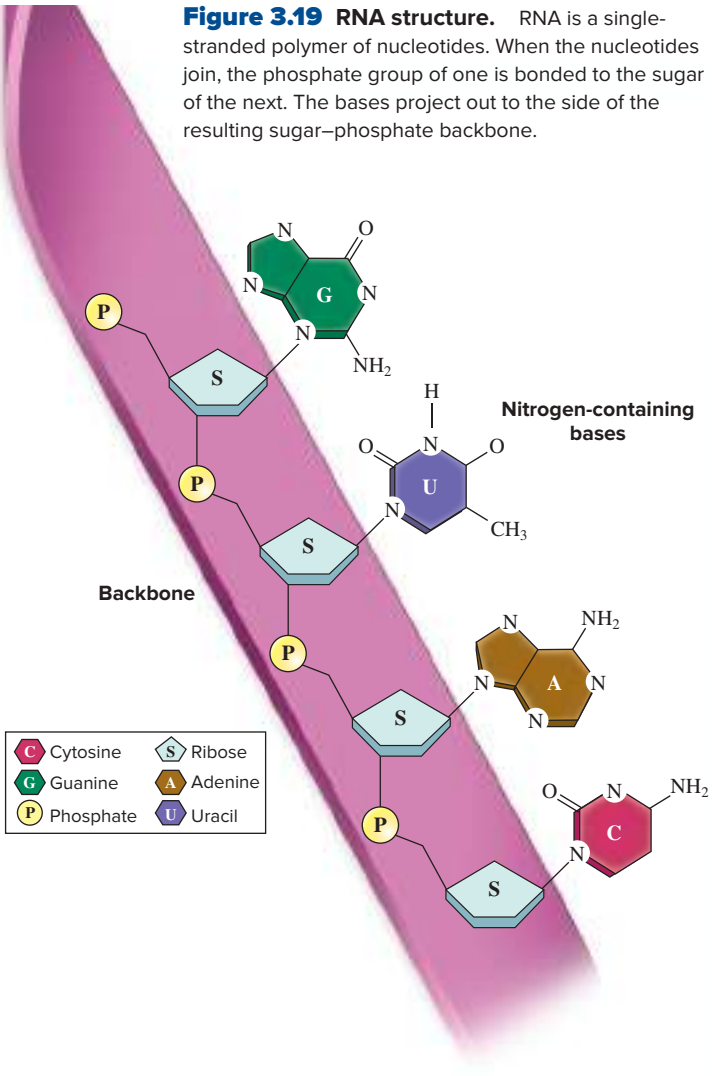


Figure 3.19 RNA structure. RNA is a single-stranded polymer of nucleotides. When the nucleotides join, the phosphate group of one is bonded to the sugar of the next. The bases project out to the side of the resulting sugar–phosphate backbone.

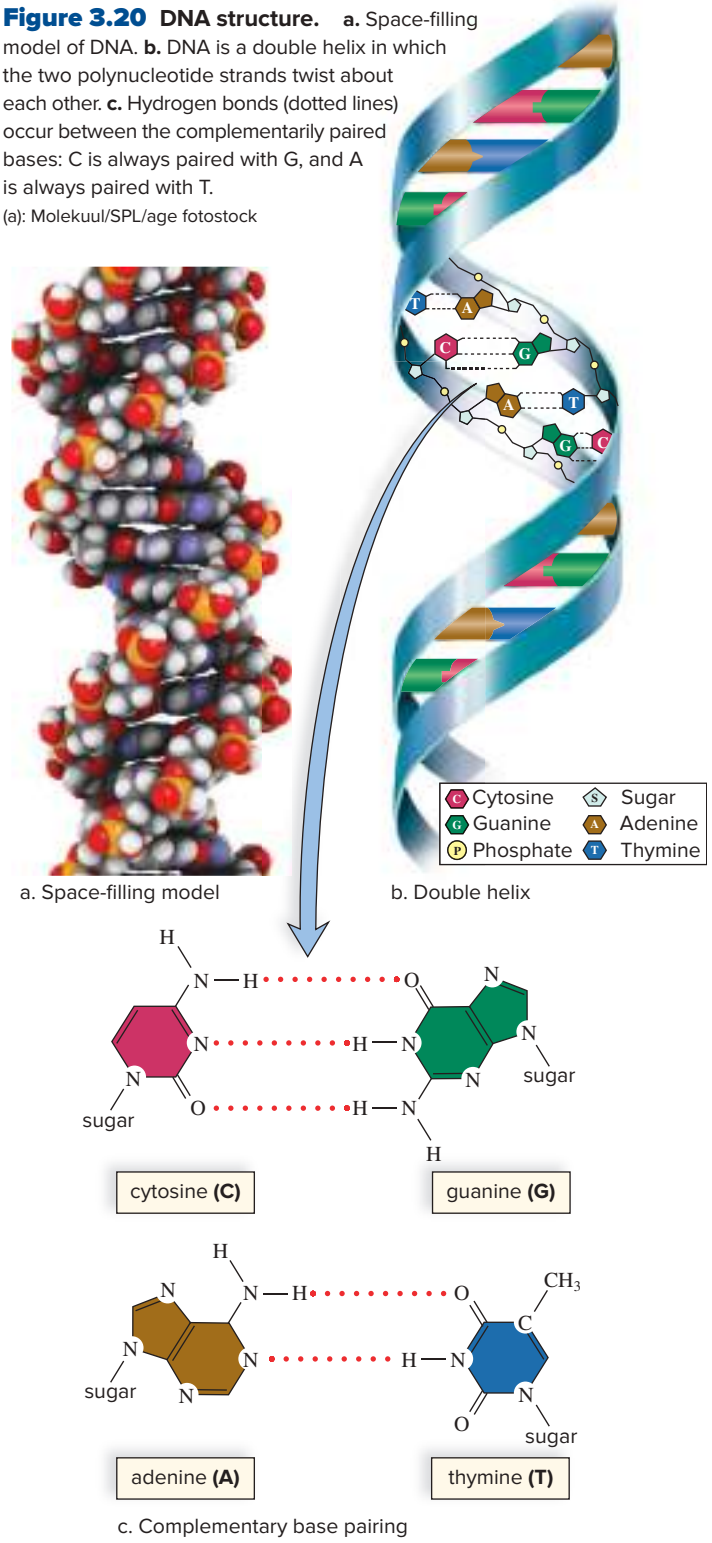


Figure 3.20 DNA structure. **a.** Space-filling model of DNA. **b.** DNA is a double helix in which the two polynucleotide strands twist about each other. **c.** Hydrogen bonds (dotted lines) occur between the complementarily paired bases: C is always paired with G, and A is always paired with T.
(a): Molekuul/SPL/age fotostock

joined in an order specified by the strand they are copied from. DNA is double-stranded, and RNA is single-stranded (Fig. 3.19).

The two strands in double-stranded DNA usually twist around each other to form a double helix (Fig. 3.20*a, b*). The two strands are held together by hydrogen bonds between pyrimidine and purine base pairs. The bases can be in any order within a strand, but between strands, thymine (T) is always paired with adenine (A), and guanine (G) is always paired with cytosine (C). This is called **complementary base pairing.** Therefore, regardless of the order or the quantity of any particular base pair, the number of purine bases (A + G) always equals the number of pyrimidine bases (T + C) (Fig. 3.20*c*). We will take a closer look at the structure of DNA and RNA in Chapter 12.

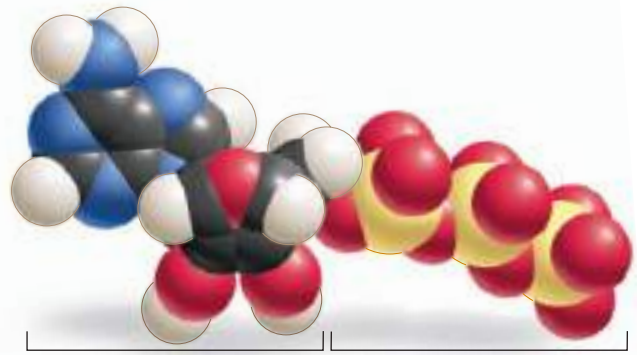
ATP (Adenosine Triphosphate)

ATP is a nucleotide comprised of adenine and ribose (adenosine) and three phosphates (triphosphate). The three phosphate groups are attached together and to ribose, the pentose sugar (Fig. 3.21).

The last two phosphate bonds in ATP are unstable and are easily broken, making it a high-energy molecule. In cells, hydrolysis of the terminal phosphate bond produces the molecule **ADP (adenosine diphosphate),** a phosphate molecule Ⓟ, and lots of energy that is used for cellular work.

The energy released by ATP hydrolysis is used to power many cellular processes, including enzyme reactions, cell communication, and cell division. ATP hydrolysis is chemically favored, because ADP and Ⓟ are more stable than the original ATP molecule. Even

ISTUDY



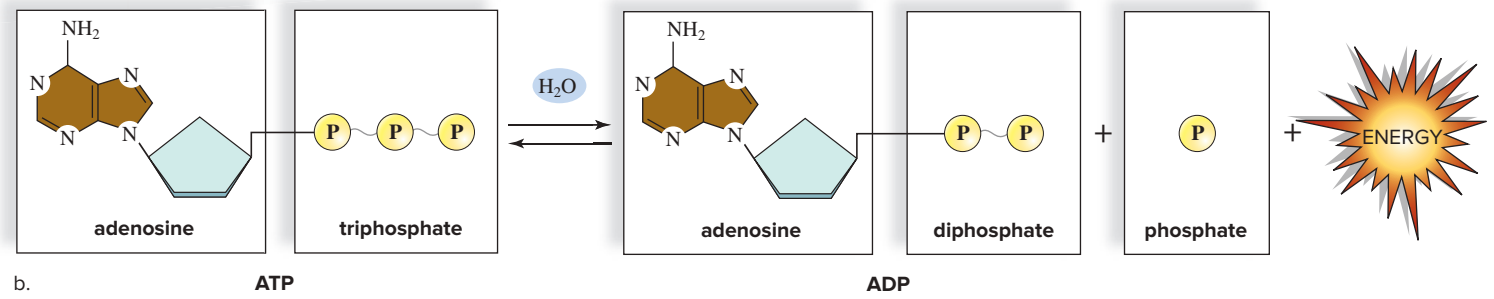


Figure 3.21 ATP. ATP, the universal energy currency of cells, is composed of adenosine and three phosphate groups. **a.** Space-filling model of ATP. **b.** When cells require energy, ATP becomes ADP + Ⓟ, and energy is released.

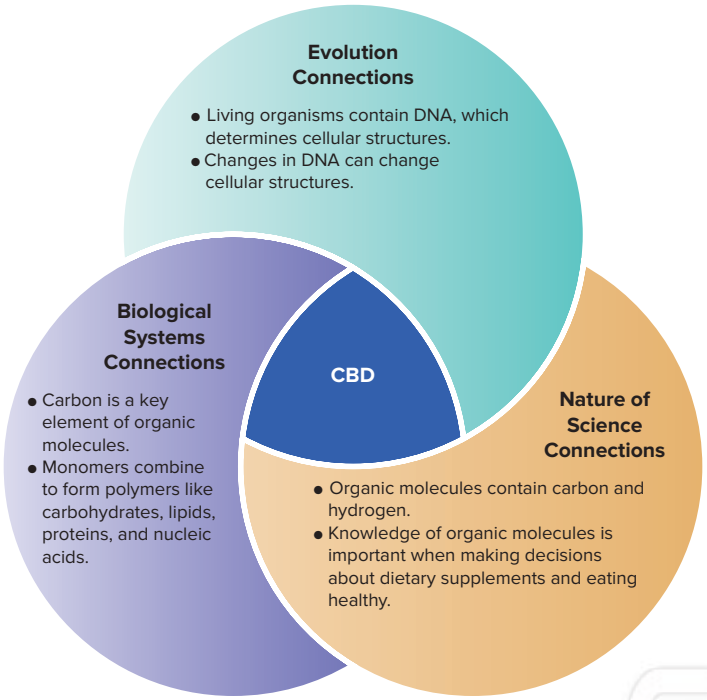
though the third phosphate bond is broken, it is the whole molecule that releases energy.

In many cases, the hydrolysis of the ATP nucleotide is coupled to chemically unfavorable reactions in cells to allow these reactions to proceed. For example, key steps in the synthesis of macromolecules, such as carbohydrates and proteins, are able to proceed because the energy from ATP breakdown is used to pay the energy costs of the chemical reaction. ATP also supplies the energy for muscle contraction and nerve impulse conduction. Just as you spend money when you pay for a product or service, cells "spend" ATP when they need something. That's why ATP is called the energy currency of cells.

Check Your Progress 3.5

1. Examine how a nucleic acid stores information.
2. List the three components of a nucleotide.
3. Evaluate the properties of ATP that make it an ideal carrier of energy.

Connecting *the* Concepts



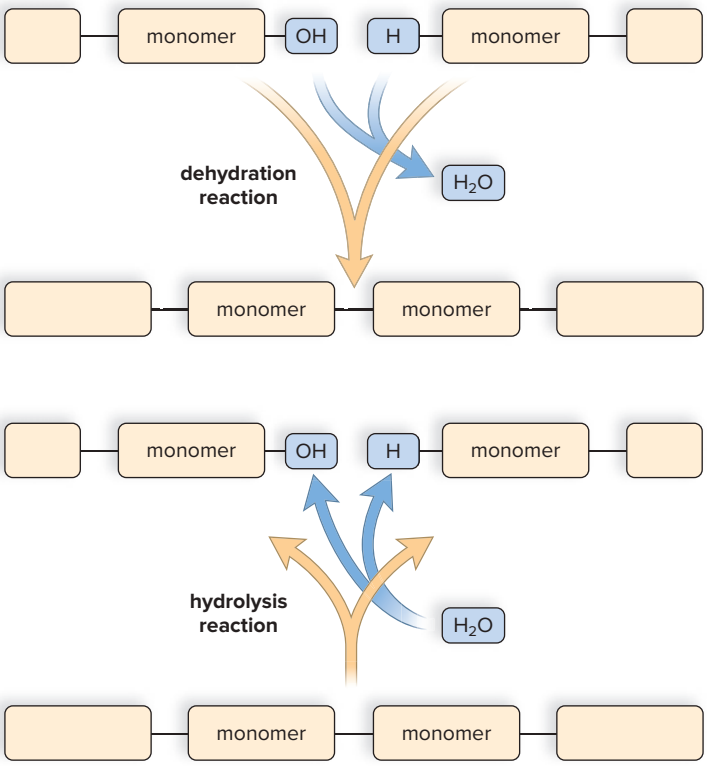
What Did You Learn?

3.1 Organic Molecules

The chemistry of carbon, also called **organic chemistry,** accounts for the diversity of **organic** molecules found in living organisms. **Inorganic chemistry** is the study of nonliving matter. Carbon can bond with as many as four other atoms. It can also bond with itself to form both chains and rings. Differences in the carbon skeleton and attached **functional groups** cause biomolecules to have different chemical properties. Isomers are organic molecules with identical formulas but different structures. Some functional groups are hydrophobic, and others are hydrophilic.

The chemical properties of a molecule determine how it interacts with other molecules and its role in the cell. There are four classes of biomolecules in cells: carbohydrates, lipids, proteins, and nucleic acids.

Polymers are formed by linking monomers. For each bond formed during a **dehydration reaction,** a molecule of water is removed, and for each bond broken during a **hydrolysis reaction,** a molecule of water is added. Enzymes are involved in these processes.



3.2 Carbohydrates

Monosaccharides, disaccharides, and polysaccharides are all **carbohydrates.** Therefore, the term *carbohydrate* includes both the monomers (e.g., glucose) and the polymers (e.g., starch, glycogen, and cellulose). **Monosaccharides** include **glucose** (a **hexose** sugar) and **ribose** and **deoxyribose** (**pentose** sugars). **Dissaccharides** consist of two monosaccharides. **Polysaccharides** may serve as energy-storage molecules, such as **starch** and **glycogen,** or as structural molecules, such as **cellulose, chitin,** and **peptidoglycan.**

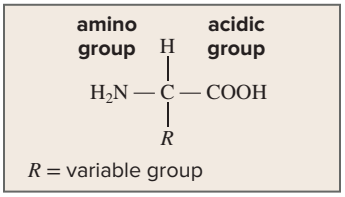
3.3 Lipids

Lipids include a wide variety of compounds that are insoluble in water. The majority of lipids are **triglycerides,** including the **fats** and oils. These are involved in long-term energy storage and contain one **glycerol** and three **fatty acids.** Both glycerol and fatty acids have polar groups, but fats and oils are nonpolar, and this accounts for their insolubility in water. Fats tend to contain **saturated fatty acids,** and oils tend to contain unsaturated fatty acids. Saturated fatty acids do not have carbon–carbon double bonds, but **unsaturated fatty acids** do have double bonds in their hydrocarbon chain. The double bond causes a kink in the molecule that accounts for the liquid nature of oils at room temperature. **Trans fats** are a form of unsaturated fatty acid.

A **phospholipid** replaces one of the fatty acids with a phosphate group. In water, phospholipids form a bilayer, because the head of each molecule is hydrophilic and the tails are hydrophobic. **Steroids** have the same four-ring structure as cholesterol, but each differs by the attached functional groups. **Waxes** are composed of a fatty acid with a long hydrocarbon chain bonded to an alcohol.

3.4 Proteins

Proteins carry out many diverse functions in cells and organisms, including support, metabolism, transport, defense, regulation, and motion. Proteins are polymers of **amino acids.**



A **peptide** consists of two amino acids joined by a **peptide bond.** A **polypeptide** is a long chain of amino acids. There are 20 different amino acids in cells, and they differ only by their *R* groups. Whether or not the *R* groups are hydrophilic or hydrophobic helps determine the structure, and therefore the function, of the protein. A polypeptide has up to four levels of structure: The primary level is the linear sequence of the amino acids, which is determined by the DNA; the secondary level contains α helices and β (pleated) sheets held in place by hydrogen bonding between amino acids along the polypeptide chain; and the tertiary level is the final folded polypeptide, which is held in place by internal bonding and hydrophobic interactions between *R* groups. Proteins that contain more than one polypeptide have a quaternary level of structure as well.

Some proteins serve as enzymes, which regulate and carry out body functions. As with other proteins, the shape of an enzyme is important to its function. Both high temperatures and drastic pH change can cause proteins to **denature,** lose their shape, and decrease their function. Some proteins, called prions, may cause other proteins to fold incorrectly, thus altering their function.

3.5 Nucleic Acids

The **nucleic acids DNA (deoxyribonucleic acid)** and **RNA (ribonucleic acid)** are linear polymers of nucleotides. Each nucleotide has three components: a 5-carbon sugar, a phosphate (phosphoric acid), and a nitrogen-containing base.

DNA, which contains the sugar deoxyribose, phosphate, and nitrogen-containing bases, is the genetic material that stores information for its own replication and specifies the order in which amino acids are sequenced in proteins. DNA uses mRNA to direct protein synthesis. DNA is a double-stranded helix that has complementary base pairing between the strands. This means that A pairs with T, and C pairs with G, through hydrogen bonding. RNA is single-stranded, contains the sugar ribose and phosphate, and has the same bases as DNA except that uracil takes the place of thymine. There are many different types of RNA.

ISTUDY

ATP (adenosine triphosphate) is a nucleotide that, with its unstable phosphate bonds, stores energy to do cellular work. Hydrolysis of ATP to **ADP (adenosine diphosphate)** releases energy needed by the cell to make a product or conduct metabolism. Some nucleic acids also act as **coenzymes** in metabolic reactions.

Thinking Critically About Biology

1. The seeds of temperate plants tend to contain unsaturated fatty acids, whereas the seeds of tropical plants tend to have saturated fatty acids. How would you test your hypothesis? Assuming your hypothesis is supported, give an explanation.
2. Chemical analysis reveals that an abnormal form of an enzyme contains a polar amino acid at the location where the normal form has a nonpolar amino acid. Formulate a testable hypothesis concerning the abnormal enzyme.
3. In order to understand the relationship between enzyme structure and function, researchers often study mutations that swap one amino acid for another. In one enzyme, function is retained if a particular amino acid is replaced by one that has a nonpolar *R* group, but function is lost if the amino acid is replaced by one with a polar *R* group. Why might that be?
4. Triglycerides are complex fats that play a role in long-term energy storage and insulation in animals. Identify the carbohydrate group that animals also use as long-term energy storage and explain how they form it.

Making It Relevant to You

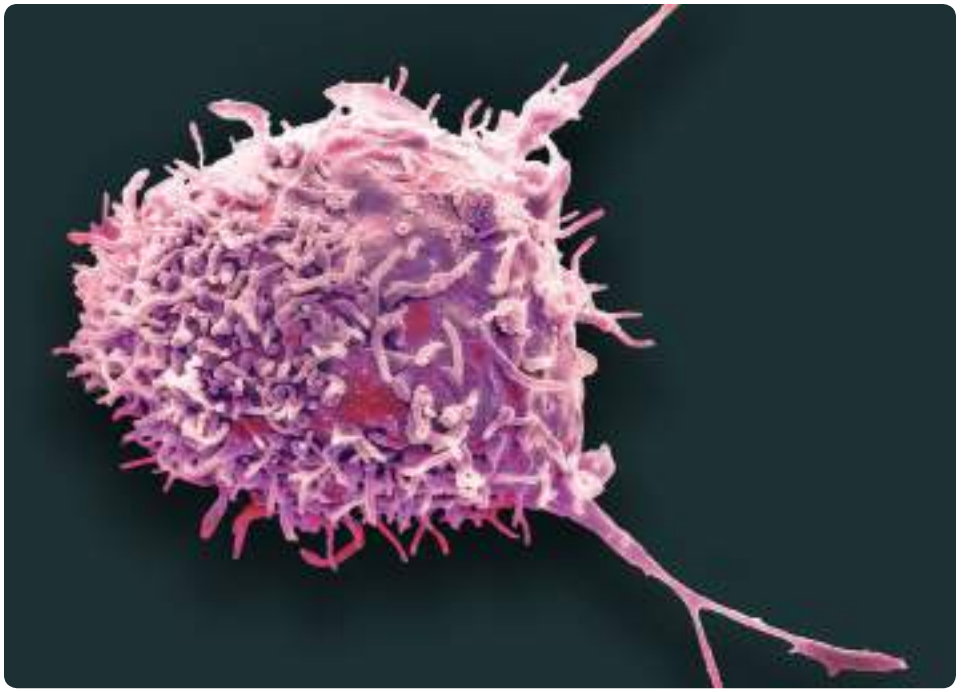
1. In what way could you, or someone you know, benefit from using a CBD-based product?
2. Which biomolecule does CBD need to bind to in order to provide its benefits?
3. What characteristics would a synthetic version of CBD need to have in order to function like the real version?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

4

Cell Structure and Function



Steve Gschmeissner/Science Photo Library/Getty Images

Chapter Outline

4.1 Cellular Level of Organization
4.2 Prokaryotic Cells
4.3 Introduction to Eukaryotic Cells
4.4 The Nucleus and Ribosomes
4.5 The Endomembrane System
4.6 Microbodies and Vacuoles
4.7 The Energy-Related Organelles
4.8 The Cytoskeleton

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Table 3.1 What role do functional groups serve in biological molecules?

Section 3.1 What macromolecules are needed to construct a cell?

Section 3.4 How does the structure of a protein influence its function?

Stem Cells

One of the new frontiers in medicine is the use of human stem cells to regenerate tissue that has been damaged by a disease. Recently, researchers have shown that some forms of a vision disorder called macular degeneration, type 2 diabetes, and even Alzheimer's disease may be treated by using stem cells.

But what is a stem cell? Unlike most cells of our bodies, which are specialized and can divide only a limited number of times, stem cells are unspecialized and have the potential to divide almost indefinitely. As a stem cell divides, it can develop into a wide variety of cell types. This is possible because a stem cell contains all of the genetic information and internal structures to become almost any other type of cell in the body; it just hasn't begun the process of specialization. In other words, a stem cell is a cell that doesn't know what it wants to be when it grows up, but once it is placed in a tissue, such as the eye or pancreas, it starts to assume the functions of the neighboring cells. Often, it can compensate for nearby dysfunctional cells and help correct a disorder or disease.

Cells are the fundamental units of life. Although the diversity of organisms is incredible, the cells of all organisms share many similarities. In this chapter, we will explore the two main evolutionary classes of cells and see how they are similar as well as different. This overview of cell structure will help you understand how biological organisms, such as yourself, function.

As you read through this chapter, think about the following questions:

1. What are the fundamental differences between prokaryotic and eukaryotic cells?
2. What are the roles of the cellular organelles within eukaryotic cells?

Following *the* Themes

CHAPTER 4 – CELL STRUCTURE AND FUNCTION

Evolution	All cells originate from existing cells, creating an unbroken lineage back to cells from early Earth.
Nature of Science	Understanding how cells function allows us to discover ways to treat cell-based diseases.
Biological Systems	Cells play an important role at all levels of biological organization, from tissues to ecosystems.

ISTUDY

4.1 Cellular Level of Organization

Learning Outcomes

Upon completion of this section, you should be able to

1. Understand that cells are the basic units of life.
2. List the basic principles of the cell theory.
3. Recognize how the surface-area-to-volume ratio limits the size of a cell.

Cells are the basic units of life. All of the chemistry and biomolecules we have discussed so far are necessary but insufficient on their own to support life. It is only when these components are brought together and organized into a cell that life is possible.

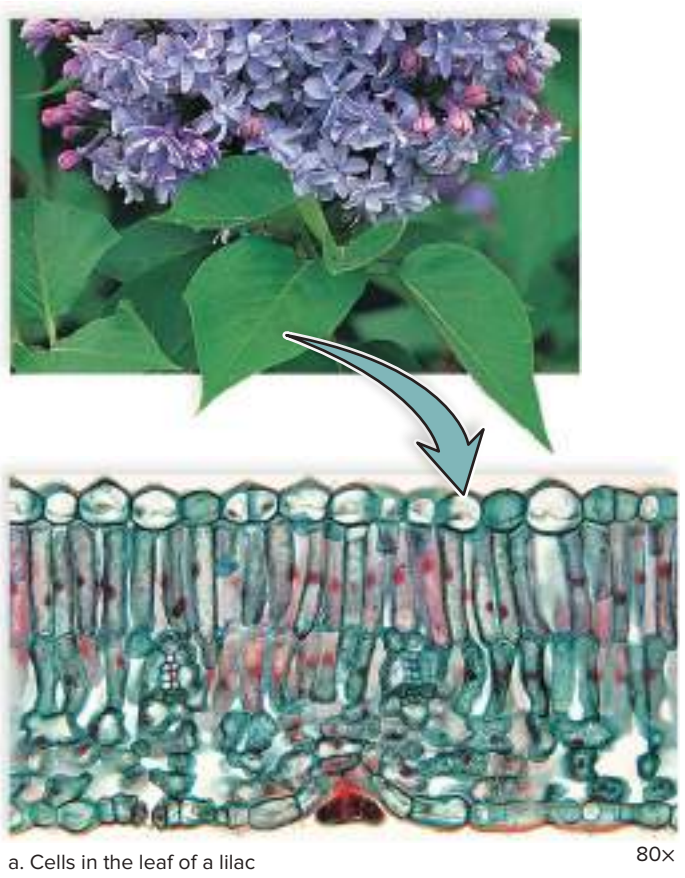
All organisms are made up of cells. When we observe plants, animals, and other organisms, it is important to realize that what we are seeing is a collection of cells that work together in a highly organized, regulated manner and so conduct the business of life. Figure 4.1 shows the connection between whole organisms and their component cells. Although the cellular basis of life is clear to us now, scientists were unaware of this fact as recently as 200 years ago. The link between cells and life became clear to microscopists during the 1830s.

The **cell** is the smallest structure capable of performing all of the functions necessary for life. The collective work of the nineteenth-century scientists Robert Brown, Matthias Schleiden, and Theodor Schwann helped determine that plants and animals are composed of cells. Further work by the German physician Rudolph Virchow showed that cells self-reproduce and that "every cell comes from a preexisting cell." Today, we know that various illnesses of the body, such as diabetes and prostate cancer, are due to cellular malfunction. Countless scientific investigations since that time verify these initial findings. From these results, we can infer that all life on Earth today came from cells in ancient times, and that all cells are related in some way. In reality, a continuity of cells has been present from generation to generation, even back to the very first cell (or cells) in the history of life.

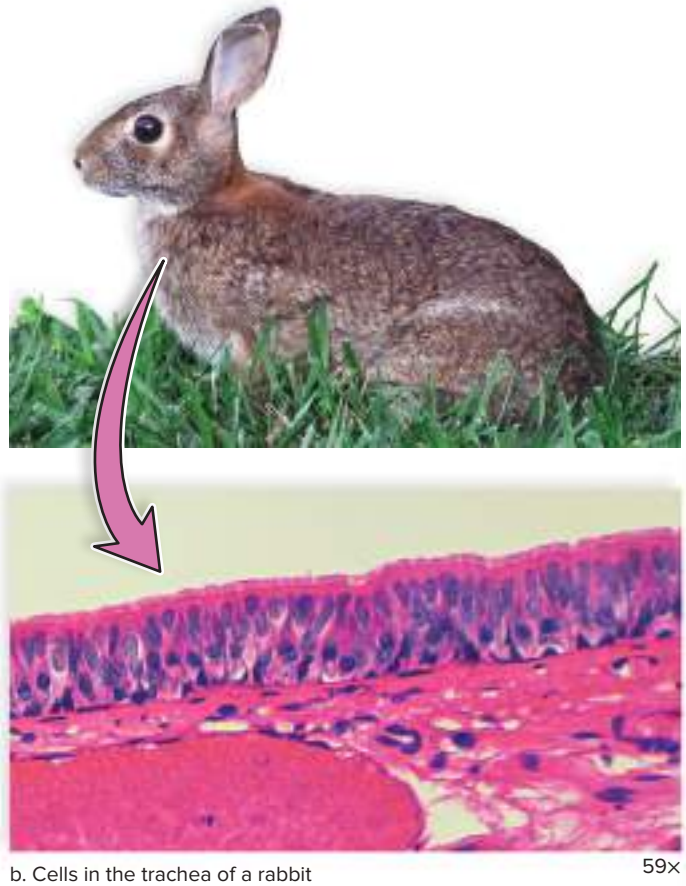
Today, some life-forms exist as single cells, whereas others are complex, interconnected systems of cells. When single-celled organisms reproduce, a single cell divides and becomes two new organisms. When multicellular organisms grow, many cells divide. The presence of many cells allows some to specialize to do particular jobs within the multicellular organism, including the cells that create genetic variation through sexual reproduction.

The work of Schleiden, Schwann, and Virchow helped created the **cell theory.** It states that:

- All organisms are composed of cells.
- Cells are the basic units of structure and function in organisms.
- Cells come only from preexisting cells because cells are self-reproducing.



a. Cells in the leaf of a lilac 80×



b. Cells in the trachea of a rabbit 59×

Figure 4.1 Organisms and cells. While it may be difficult to see without a microscope, all organisms, including plants such as a (**a**) lilac, and animals, such as a (**b**) rabbit, are composed of cells. (a) top: Geoff Bryant/Science Source; (a) bottom: ©Ray F. Evert/University of Wisconsin; (b) top: U.S. Fish & Wildlife Service/Lewis Gorman; (b) bottom: Garry DeLong/Science Source

ISTUDY

Cell Size

Although they range in size, cells are generally quite small. A frog's egg, at about 1 millimeter (mm) in diameter, is large enough to be seen by the human eye. But most cells are far smaller than 1 mm; some are even as small as 1 micrometer (μm)—one-thousandth of a millimeter. Cell inclusions and macromolecules are smaller than a micrometer and are measured in terms of nanometers (nm).

Because of their size, very small biological structures can only be viewed with microscopes, which magnify a visual image. Figure 4.2 shows the visual range of the eye, light microscope, and electron microscope. The discussion of microscopy in the Nature of Science feature, "Microscopy Today," explains why an electron microscope lets us see so much more detail than a light microscope.

Why are cells so small? To answer this question, consider that a cell is a system by itself. As such, it needs a surface area large enough to allow adequate nutrients to enter and have wastes eliminated. Small cells, not large cells, are likely to have an adequate surface area for exchanging wastes for nutrients. As cells increase in size, the surface area becomes inadequate to exchange the materials that the volume of the cell requires.

Figure 4.3 illustrates that dividing a large cube into smaller cubes provides a lot more surface area per volume. This relationship is called the **surface-area-to-volume ratio.** Calculations show that a 1-cm cube has a surface-area-to-volume ratio of 6:1, whereas a 4-cm cube has a surface-area-to-volume ratio of 1.5:1. In general, a higher surface-area-to-volume ratio increases the efficiency of transporting materials into and out of the cell.

A mental image might help you visualize the importance of surface-area-to-volume ratios and why this relationship favors smaller cells. Imagine a small room and a large room filled with people. The small room, which holds 20 people, has only two doors, and the large room, which holds 80 people, has four doors.

	One 4-cm cube	Eight 2-cm cubes	Sixty-four 1-cm cubes
Total surface area (height × width × number of sides × number of cubes)	96 cm^2	192 cm^2	384 cm^2
Total volume (height × width × length × number of cubes)	64 cm^3	64 cm^3	64 cm^3
Surface area/Volume of each cube (surface area ÷ volume)	1.5:1	3:1	6:1

Figure 4.3 Surface-area-to-volume relationships. As cell size decreases from 4 cm^3 to 1 cm^3, the surface-area-to-volume ratio increases.

If a fire occurred in both rooms, it would be faster to get the people out of the smaller room, because it has the more favorable ratio of doors to people. Similarly, a small cell size is more advantageous for exchanging molecules because of its greater surface-area-to-volume ratio.

Check Your Progress 4.1

1. Explain why cells are alive but macromolecules are not.
2. State the components of the cell theory.
3. Explain why a large surface-area-to-volume ratio is needed for the proper functioning of cells.

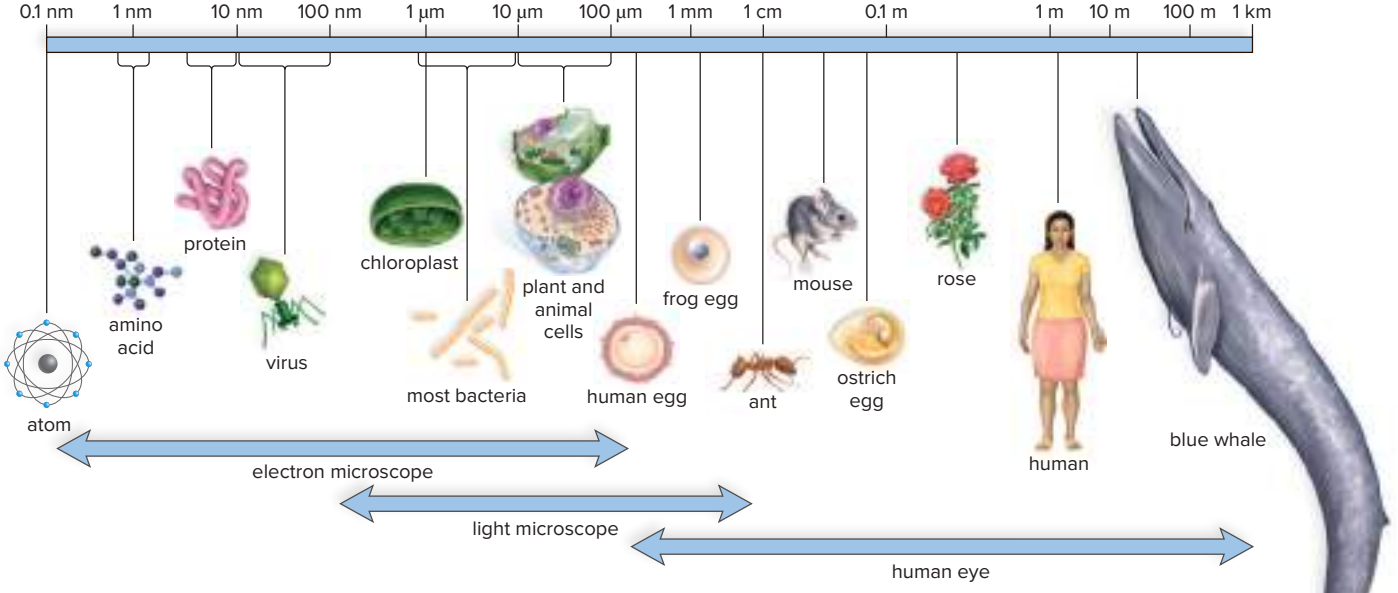


Figure 4.2 The sizes of various objects. The unassisted human eye can usually see macroscopic organisms and a few large cells. Microscopic cells are visible with a light microscope, but do not offer much detail. An electron microscope is necessary to see organelles in detail and to observe viruses and molecules. In the metric system (see Appendix A), each higher unit is ten times greater than the preceding unit. (1 meter = 10^2 cm = 10^3 mm = 10^5 μm = 10^9 nm)

ISTUDY

THEME Nature of Science

Microscopy Today

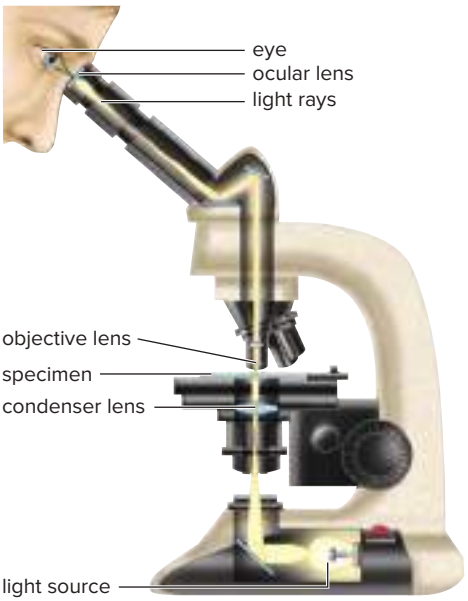
Because cells are the basic units of life, the more we learn about cells, the more we understand life. Cells were not discovered until the seventeenth century, when the microscope was invented. Since that time, various types of microscopes have been developed for studying cells and their components.

Many times when scientists don't have suitable tools to investigate natural phenomena, they invent them. Microscopes have given scientists a deeper look into how life works than is possible with the naked eye. Today, there are many types of microscopes. A *compound light microscope* uses a set of glass lenses to focus light rays passing through a specimen to produce an image that is viewed by the human eye. A *transmission electron microscope* (*TEM*) uses a set of electromagnetic lenses to focus electrons passing through a specimen to produce an image, which is projected onto a fluorescent screen or photographic film. A *scanning electron microscope* (*SEM*) uses a narrow beam of electrons to scan over the surface of a specimen that is coated with a thin metal layer. Secondary electrons given off by the metal are detected and used to produce a three-dimensional image on a television screen. Figure 4A shows these three types of microscopic images.

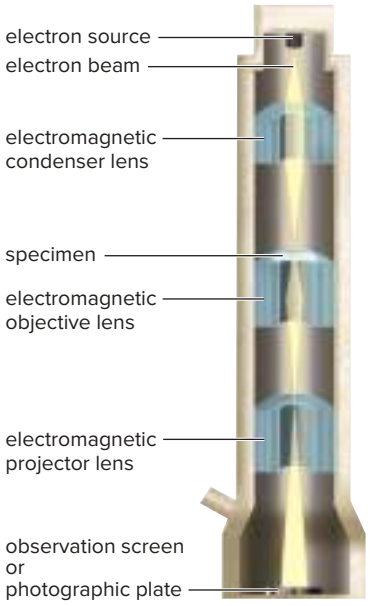
Euglena, light micrograph 200×

Euglena, transmission electron micrograph 12,500×

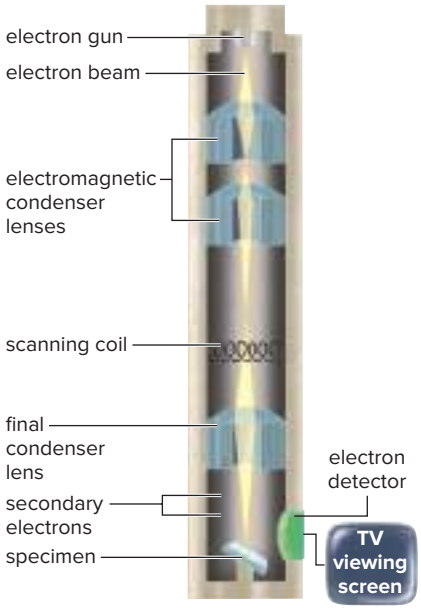
Euglena, scanning electron micrograph 1,520×



a. Compound light microscope



b. Transmission electron microscope



c. Scanning electron microscope

Figure 4A Diagram of microscopes with accompanying micrographs of *Euglena gracilis*. **a.** The compound light microscope and (**b**) the transmission electron microscope provide an internal view of an organism. **c.** The scanning electron microscope provides an external view of an organism.

(light micrograph): Eric V. Grave/Science Source; (TEM): Biophoto Associates/Science Source; (SEM): Andrew Syred/Science Source

(*Continued*)

Magnification, Resolution, and Contrast

Magnification is the ratio between the size of an image and its actual size. Electron microscopes magnify to a greater extent than do compound light microscopes. A light microscope can magnify objects about a thousand times, but an electron microscope can magnify them hundreds of thousands of times. The difference lies in the means of illumination. The path of light rays and electrons moving through space is wavelike, but the wavelength of electrons is much shorter than the wavelength of light. This difference in wavelength accounts for the electron microscope's greater magnifying capability and its greater ability to distinguish between two points (resolving power).

Resolution is the minimum distance between two objects that allows them to be seen as two separate objects. A microscope with poor resolution might enable a student to see only one cellular granule, while the microscope with the better resolution would show two granules next to each other. The greater the resolving power, the greater the detail seen.

If oil is placed between the sample and the objective lens of the compound light microscope, the resolving power is increased, and if ultraviolet light is used instead of visible light, it is also increased. But typically, a light microscope can resolve down to 0.2 µm, while the transmission electron microscope can resolve down to 0.0002 µm. If the resolving power of the average human eye is set at 1.0, then the typical compound light microscope is about 500, and the transmission electron microscope is 500,000 (Fig. 4A*b*).

The ability to make out, or resolve, a particular object can depend on *contrast,* a difference in the shading of an object compared to its background. Higher contrast is often achieved by staining cells with colored dyes (light microscopy) or with electron-dense metals (electron microscopy), which make them easier to see. Optical methods such as phase contrast and differential interference contrast (Fig. 4B) can also be used to improve contrast. Using fluorescently tagged antibodies can also help us visualize subcellular components such as specific proteins (see Fig. 4.19).

Illumination, Viewing, and Recording

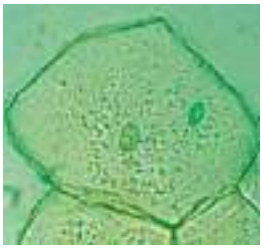
Light rays can be bent (refracted) and brought to focus as they pass through glass lenses, but electrons do not pass through glass. Electrons have a charge that allows them to be brought into focus by electromagnetic lenses. The human eye uses light to see an object but cannot use electrons for the same purpose. Therefore, electrons leaving the specimen in the electron microscope are directed toward a screen or a photographic plate that is sensitive to their presence. Humans can view the image on the screen or photograph.

A major advancement in illumination has been the introduction of *confocal microscopy,* which uses a laser beam scanned across the specimen to focus on a single shallow plane within the cell. The microscopist can "optically section" the specimen by focusing up and down, and a series of optical sections can be combined in a computer to create a three-dimensional image, which can be displayed and rotated on the computer screen.

An image from a microscope may be recorded by placing a television camera where the eye would view the image. The television camera converts the light image into an electronic image, which can be entered into a computer. In *video-enhanced contrast microscopy,* the computer makes the darkest areas of the original image much darker and the lightest areas of the original much lighter. The result is a high-contrast image with deep blacks and bright whites. Even more contrast can be introduced by the computer if shades of gray are replaced by colors.

250×

Bright-field. Light passing through the specimen is brought directly into focus. Usually, the low level of contrast within the specimen interferes with viewing all but its largest components.



225×

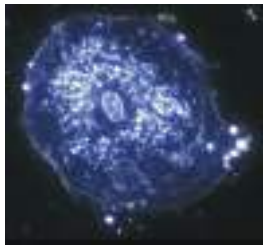
Bright-field (stained). Dyes are used to stain the specimen. Certain components take up the dye more than other components, and therefore contrast is enhanced.

160×

Differential interference contrast. Optical methods are used to enhance density differences within the specimen so that certain regions appear brighter than others. This technique is used to view living cells, chromosomes, and organelle masses.

600×

Phase contrast. Density differences in the specimen cause light rays to come out of "phase." The microscope enhances these phase differences so that some regions of the specimen appear brighter or darker than others. The technique is widely used to observe living cells and organelles.



600×

Dark-field. Light is passed through the specimen at an oblique angle so that the objective lens receives only light diffracted and scattered by the object. This technique is used to view organelles, which appear quite bright against a dark field.

Figure 4B Photomicrographs of cheek cells. Bright-field microscopy is the most common form used with a compound light microscope. Other types of microscopy include differential interference contrast, phase contrast, and dark-field. (bright-field): Ed Reschke; (bright-field stained): Biophoto Associates/Science Source; (differential interference): M. I. Walker/Science Source; (phase contrast): Walter Dawn/Science Source; (dark-field): Stephen Durr

ISTUDY

4.2 Prokaryotic Cells

Learning Outcomes

Upon completion of this section, you should be able to

1. Examine the evolutionary relatedness of prokaryotes, eukaryotes, and archaeans.
2. Describe the fundamental components of a bacterial cell.

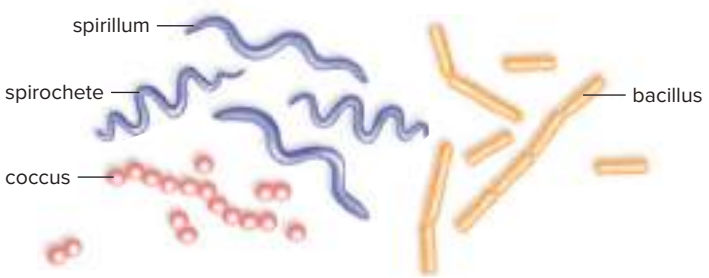
Fundamentally, two different types of cells exist in nature. **Prokaryotic cells** (Gk. *pro,* "before"; *karyon,* "kernel, nucleus") are cells that lack a membrane-bound nucleus. **Eukaryotic cells** (Gk. *eu,* "true") are cells that possess a nucleus. The bacteria and archaeans were once thought to be closely related because of their similar size and shape. Comparisons of DNA and RNA sequences now show archaeans to be biochemically distinct from both bacteria and eukaryotes. These comparisons also suggest that archaeans are more closely related to eukaryotes than bacteria and help define the three domains of life presented in Figure 1.7. Two of the three domains, the Eubacteria and Archaea, are prokaryotic cells, while all eukaryotic cells are assigned to domain Eukarya.

Prokaryotes as a group are one of the most abundant and diverse life-forms on Earth, and they are present in great numbers in the air, water, and soil, as well as living in and on other organisms. Although they are structurally less complicated than eukaryotes, their metabolic capabilities as a group far exceed those of eukaryotes. Prokaryotes are an extremely successful group of organisms whose evolutionary history dates back to the first cells on Earth.

Bacteria are well-known because they cause some serious diseases, such as tuberculosis, anthrax, tetanus, throat infections, and gonorrhea. But many species of bacteria are important to the environment, because they decompose the remains of dead organisms and contribute to ecological cycles. Bacteria also assist humans in still another way: We use them to manufacture all sorts of products, from industrial chemicals to foodstuffs and drugs. For example, today we know how to place human genes in certain cultured bacteria so they can produce human insulin, a necessary hormone for the treatment of diabetes.

The Structure of Prokaryotes

Prokaryotes are quite small; an average size is 1.1–1.5 μm wide and 2.0–6.0 μm long. The majority of the prokaryotes have one of these basic shapes:

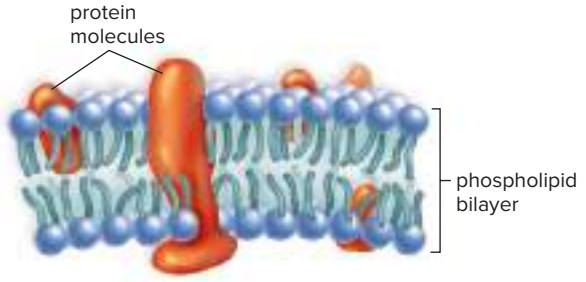


A rod-shaped bacterium is called a **bacillus,** while a spherical-shaped bacterium is a **coccus.** Both of these can occur as pairs or chains, and in addition, cocci can occur as clusters. Some long rods are twisted into spirals, in which case they are **spirilla** if they are rigid or **spirochetes** if they are flexible.

Figure 4.4 shows the generalized structure of a bacterium. "Generalized" means that not all bacteria have all the structures depicted, and some have more than one of each.

Cell Envelope

In bacteria, the **cell envelope** consists of the plasma membrane, the cell wall, and the glycocalyx. The **plasma membrane** is a phospholipid bilayer with embedded proteins:



We will explore the structure of the plasma membrane in greater detail in Section 5.1. The plasma membrane has the important function of regulating the entrance and exit of substances into and out of the cytoplasm within the cell. Regulating the flow of materials into and out of the cytoplasm is necessary in order to maintain the cell's normal composition.

In prokaryotes, the plasma membrane can form internal pouches called *mesosomes.* Mesosomes most likely increase the internal surface area for the attachment of enzymes carrying on metabolic activities.

The **cell wall** maintains the shape of the cell, even if the cytoplasm should happen to take up an abundance of water. The cell wall of a bacterium contains peptidoglycan, a complex molecule containing a unique amino disaccharide and peptide fragments. Because peptidoglycan is unique to the bacteria, several classes of antibiotics target the peptidoglycan component of the cell wall.

The **glycocalyx** is a layer of polysaccharides that lies outside the cell wall in some bacteria. When the layer is well organized and not easily washed off, it is called a **capsule.** A slime layer, on the other hand, is not well organized and is easily removed. The glycocalyx aids against drying out and helps bacteria resist a host's immune system. It also helps bacteria attach to almost any surface.

Cytoplasm

The **cytoplasm** is a semifluid solution composed of water and inorganic and organic molecules encased by a plasma membrane. Among the organic molecules are a variety of enzymes, which speed the many types of chemical reactions involved in metabolism.

While prokaryotes lack a membrane-bound nucleus, their DNA is located in a region of the cytoplasm called the **nucleoid.** Prokaryotes have a single, coiled chromosome, while eukaryotic cells typically have multiple chromosomes. Many prokaryotes also have extrachromosomal pieces of circular DNA called **plasmids.** Plasmids are routinely used in biotechnology laboratories as a vector to transport DNA into a bacterium (see Section 14.1). Procedures such as this are possible because all life on Earth is

ISTUDY

constructed from the same four DNA nucleotides: A, G, C, and T. Biotechnology plays an important role in the production of new medicines and many of the commercial products we use every day.

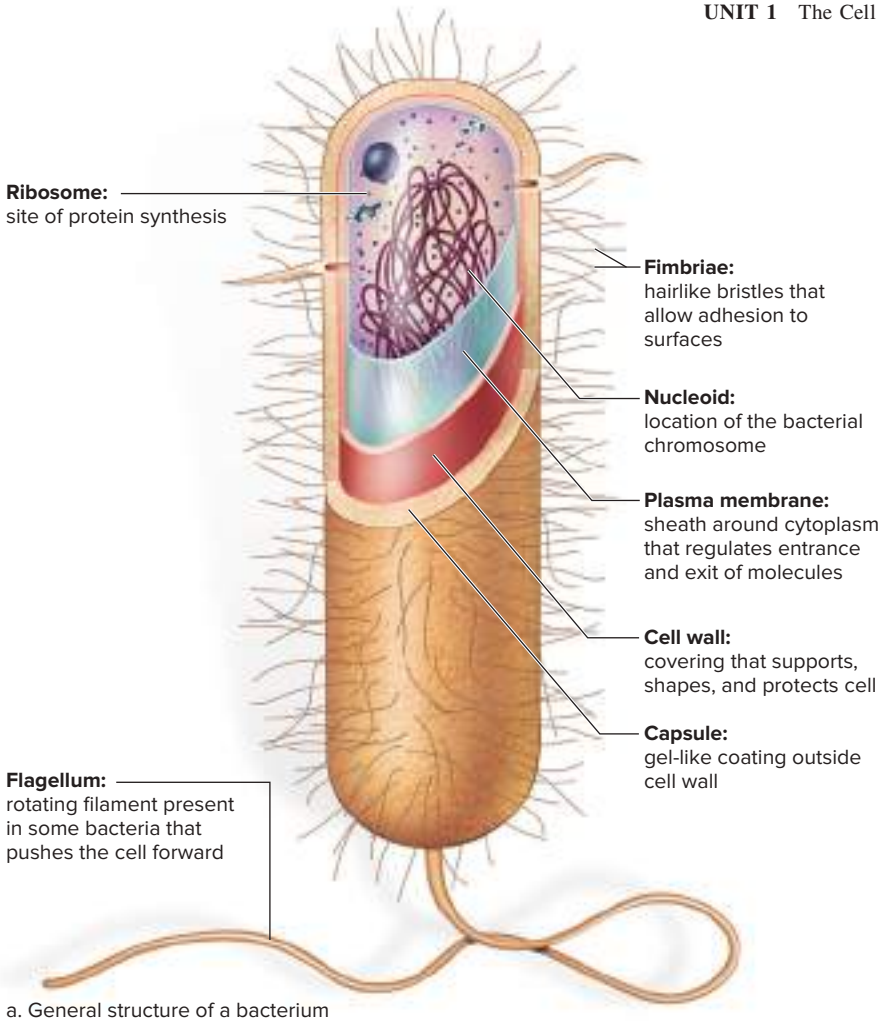
The many proteins encoded by the prokaryotic DNA are synthesized on tiny structures in the cytoplasm called **ribosomes.** A prokaryotic cell contains thousands of ribosomes that are similar in shape and function but are smaller than eukaryotic ribosomes. Like their eukaryotic counterparts, prokaryotic ribosomes still contain RNA and protein in two subunits.

There is a tremendous amount of metabolic diversity in the prokaryotes. Some prokaryotes carry out metabolism in the same manner as animals (by ingesting other organisms), but the cyanobacteria are a form of bacteria that are capable of photosynthesis in the same manner as plants. These organisms live in water, in ditches, on buildings, and on the bark of trees. Their cytoplasm contains extensive internal membranes called **thylakoids** (Gk. *thylakon,* "small sac"), where chlorophyll and other pigments absorb solar energy for the production of carbohydrates. Cyanobacteria are called the blue-green bacteria, because some have a pigment that adds a shade of blue to the cell, in addition to the green color of chlorophyll. The cyanobacteria release oxygen as a by-product of photosynthesis, and ancestral cyanobacteria were some of the earliest photosynthesizers on Earth. Many sources of evidence show that the composition of the early Earth's atmosphere was changed by the addition of oxygen.

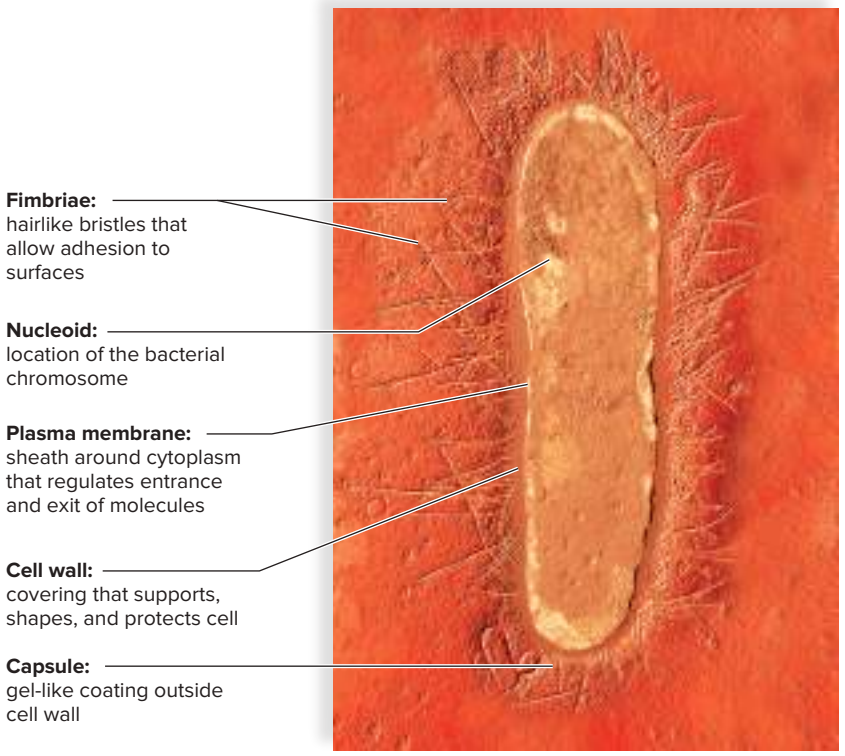
External Structures

The external structures of a prokaryote, namely the flagella, fimbriae, and conjugation pili, are made of protein. Motile prokaryotes can propel themselves in water by the means of appendages called **flagella** (usually 20 nm in diameter and 1–70 nm long). The prokaryotic flagellum is one of the great wonders of nature, and it consists of a filament, a hook, and a basal body. The basal body is a series of rings anchored in the cell wall and membrane. Unlike the flagellum of the eukaryotes, which has a whiplike motion, the flagellum of a prokaryote rotates 360°. Sometimes flagella occur only at the two ends of a cell, and sometimes they are dispersed randomly over the surface. The number and location of flagella can be used to help distinguish different types of prokaryotes.

Fimbriae are small, bristlelike fibers that sprout from the cell surface. They are not involved in locomotion; instead, fimbriae are involved in attaching prokaryotes to a surface. **Conjugation pili** are rigid, tubular structures used by prokaryotes to pass DNA from cell to cell. Prokaryotes reproduce asexually by binary fission, but they can exchange DNA by way of the conjugation pili. They can also take up DNA from the external medium or by way of viruses.



a. General structure of a bacterium



b. Location of these structures on *Escherichia coli* 32,000×

Figure 4.4 Prokaryotic cell structure. **a.** The structure of a general prokaryotic cell, such as a bacterium. **b.** These structures on a *Escherichia coli* bacterium. (photo): Sercomi/Science Source

ISTUDY

Check Your Progress 4.2

1. Explain the major differences between a prokaryotic and eukaryotic cell.
2. Describe the functions of the bacterial cell envelope, cytoplasm, and external structures.

4.3 Introduction to Eukaryotic Cells

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how the endosymbiotic theory explains eukaryotic cell structure.
2. Summarize the functions of the organelles in a eukaryotic cell.
3. Compare and contrast the structure of animal and plant cells.

Eukaryotic cells, like prokaryotic cells, have a plasma membrane that separates the contents of the cell from the environment and that regulates the passage of molecules into and out of the cytoplasm. The plasma membrane is a phospholipid bilayer with embedded proteins. What distinguishes eukaryotic cells from prokaryotic cells is the presence of a nucleus and internal membrane-bound compartments, called **organelles.** Nearly all organelles are surrounded by a membrane with embedded proteins, many of which are enzymes. These enzymes make products specific to that organelle, but their action benefits the whole cell system. Each organelle carries out specialized functions, which together allow the cell to be more efficient and successful. These features would have given the new cell a selective advantage over other cells.

Origin of the Eukaryotic Cell

The fossil record, which is based on the remains of ancient life, suggests that the first cells were prokaryotes. Biochemical data suggest that eukaryotes are more closely related to archaea than bacteria. Evidence suggests that eukaryotic cells evolved from prokaryotic cells in stages. The distinguishing characteristic of the eukaryotic cell, the nucleus, is believed to have evolved due to the invagination of the plasma membrane (Fig. 4.5). The same process also explains the origin of organelles such as the endoplasmic reticulum and the Golgi apparatus.

There is strong scientific evidence that the origin of the energy organelles occurred when larger eukaryotic cells engulfed smaller prokaryotic cells. Observations in the laboratory indicate that an amoeba infected with bacteria can become dependent upon them. Other evidence indicates that mitochondria and chloroplasts are derived from prokaryotes that were taken up by larger cells (Fig. 4.5). In this scenario, mitochondria were originally aerobic heterotrophic bacteria, and chloroplasts were originally cyanobacteria. The eukaryotic host cell would have benefited from an ability to utilize oxygen or synthesize organic food when, by chance, the prokaryote was taken up and not destroyed. After the prokaryote entered the host cell, the two would have begun living together cooperatively. This is known as the **endosymbiotic theory** (*endo,* "in"; *symbiosis,* "living together"). Some of the evidence supporting this includes:

- Mitochondria and chloroplasts are similar to bacteria in size and structure.

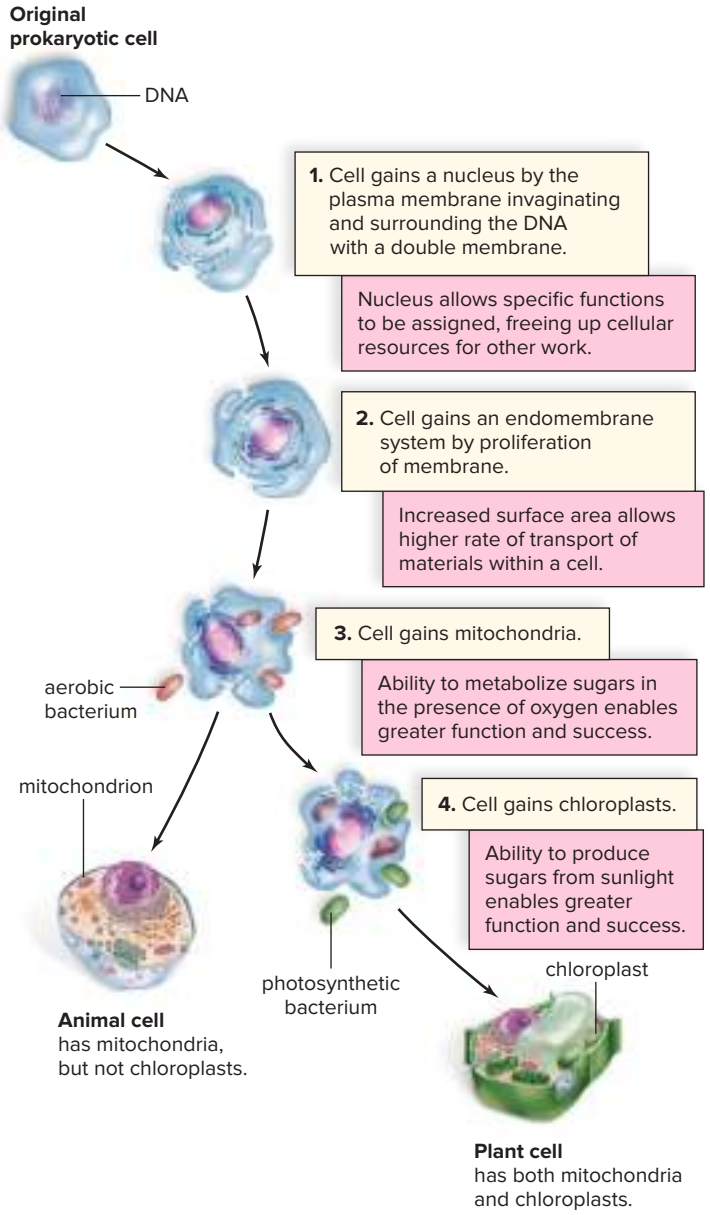
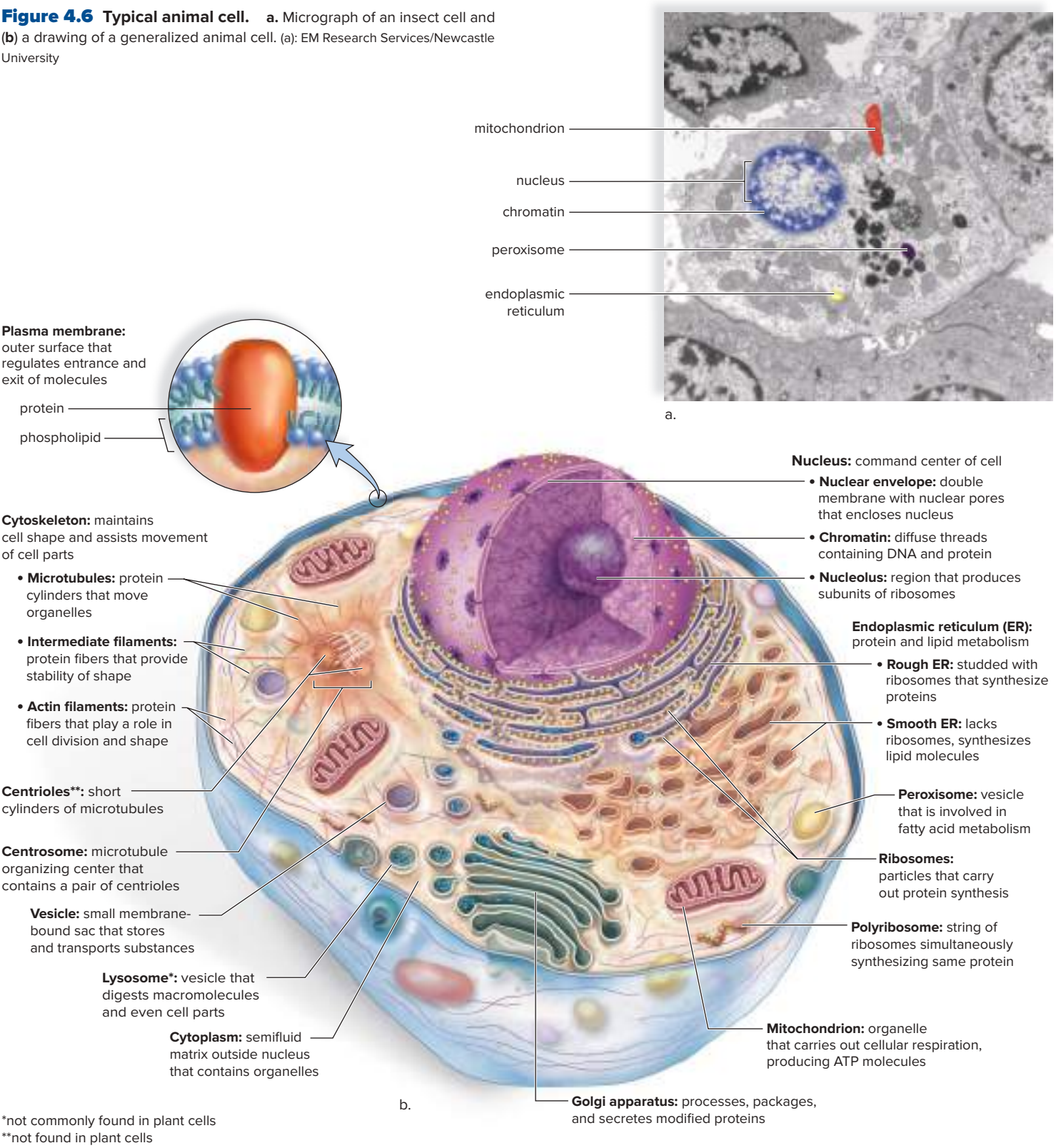


Figure 4.5 Origin of organelles. Invagination of the plasma membrane could have created the nuclear envelope and an endomembrane system that involves several organelles. The endosymbiotic theory states that mitochondria and chloroplasts were independent prokaryotes that took up residence in a eukaryotic cell. Endosymbiosis was a first step toward the origin of the eukaryotic cell during the evolutionary history of life.

- Both organelles are surrounded by a double membrane—the outer membrane may be derived from the engulfing vesicle, and the inner one may be derived from the plasma membrane of the original prokaryote.
- Mitochondria and chloroplasts contain a limited amount of genetic material and divide by splitting. Their DNA (deoxyribonucleic acid) is a circular loop like that of prokaryotes.
- Although most of the proteins within mitochondria and chloroplasts are now produced by the eukaryotic host, they do have their own ribosomes and they do produce some proteins. Their ribosomes resemble those of prokaryotes.
- The RNA (ribonucleic acid) base sequence of the ribosomes in chloroplasts and mitochondria also suggests a prokaryotic origin of these organelles.

ISTUDY

Figure 4.6 Typical animal cell. **a.** Micrograph of an insect cell and (**b**) a drawing of a generalized animal cell. (a): EM Research Services/Newcastle University

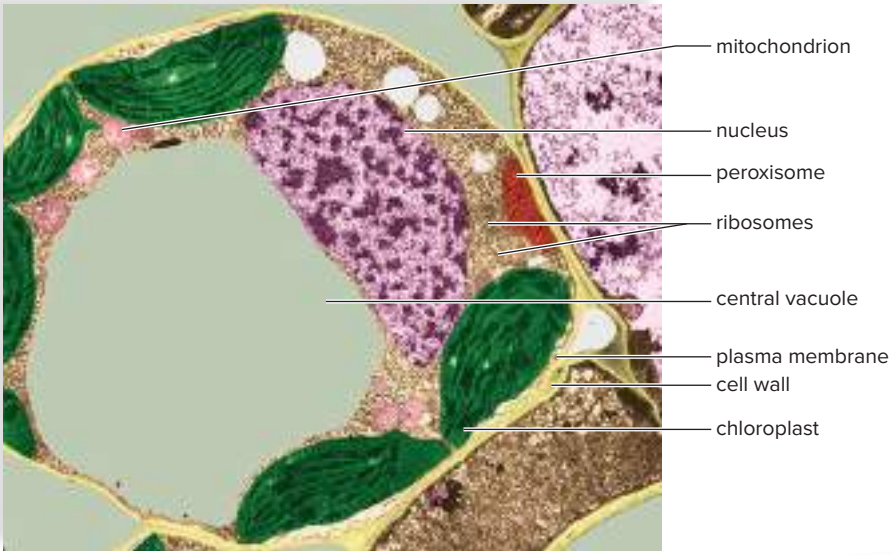


*not commonly found in plant cells
**not found in plant cells

Structure of a Eukaryotic Cell

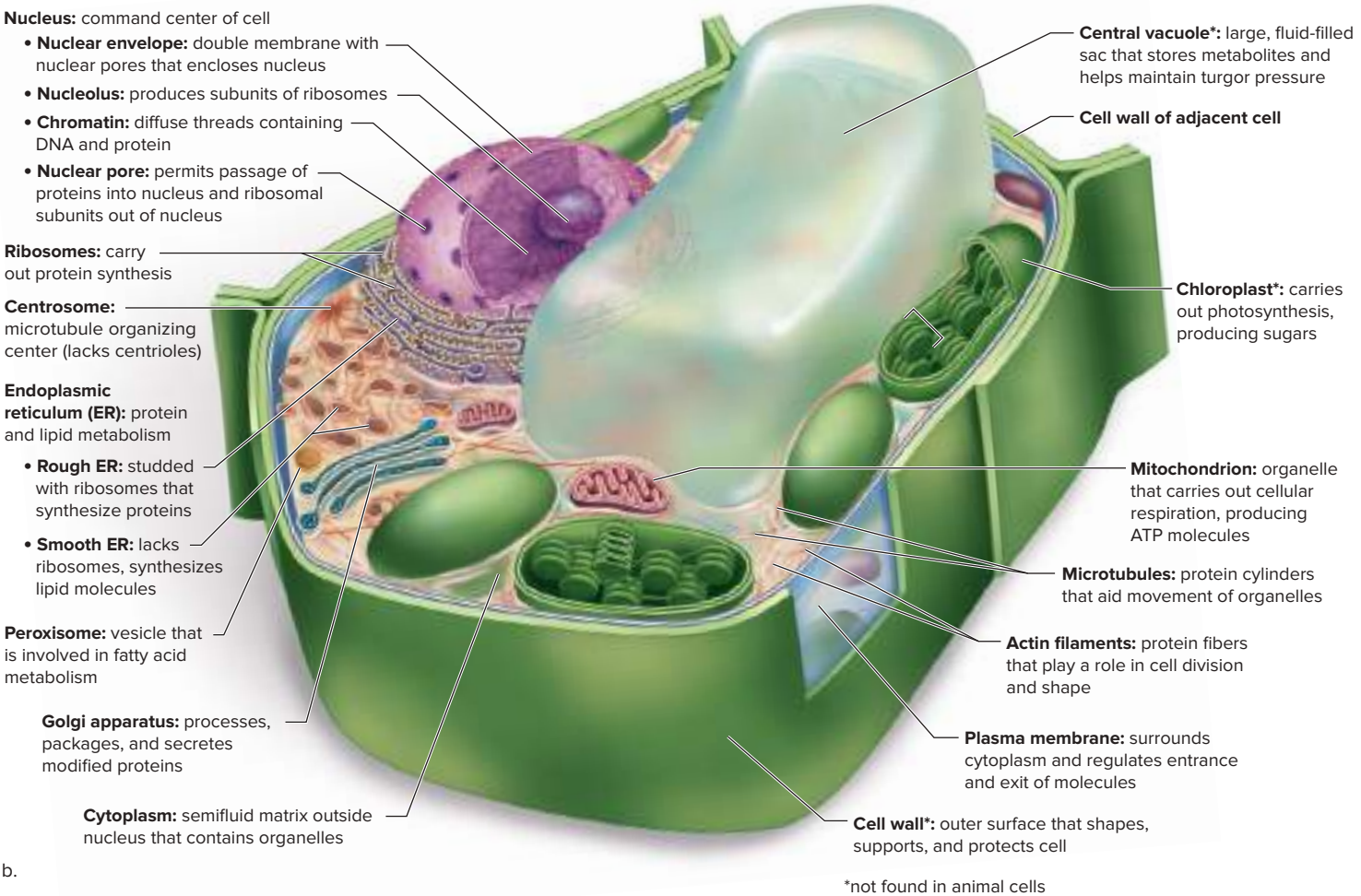
Figures 4.6 and 4.7 show general features of fully evolved animal and plant cells. Specialized cells, as opposed to generalized cells, do not necessarily contain all the structures depicted and may have more or fewer copies of any particular organelle, depending on their particular function. These generalized depictions of plant and animal cells are useful for study purposes. A baseline understanding of cell structure and function will be helpful when you study the function of specialized cells later in this text. Overall, the cell can be seen as a system of interconnected organelles that work together to metabolize, regulate, and conduct life processes. For example, the nucleus is a compartment that houses the genetic material within eukaryotic chromosomes and contains hereditary information. The nucleus communicates with ribosomes in the cytoplasm, and the organelles of the endomembrane system—notably the endoplasmic reticulum and the Golgi apparatus—communicate with one another.

ISTUDY



a. 12,300×

Figure 4.7 Typical plant cell. **a.** False-colored micrograph of a young plant cell and (**b**) a drawing of a generalized plant cell. (a): Biophoto Associates/Science Source



b.

*not found in animal cells

Production of specific molecules takes place inside or on the surface of organelles. As mentioned, enzymes embedded in the organelles' membranes make these molecules. These products are then transported around the cell by transport **vesicles,** membranous sacs that enclose the molecules and keep them separate from the cytoplasm. For example, the endoplasmic reticulum communicates with the Golgi apparatus by means of transport vesicles. Communication with the energy-related organelles—mitochondria and chloroplasts—is less obvious, but it does occur, because they import particular molecules from the cytoplasm.

Vesicles move around by means of an extensive network or lattice of protein fibers called the **cytoskeleton,** which also maintains cell shape and assists with cell movement. The protein fibers serve as tracks for the transport vesicles that are taking molecules from one organelle to another. Organelles are also moved from place to place using this transport system. Think of the cytoskeleton as a three-dimensional road system inside cells used to transport important cargo from place to place. The cytoskeleton is discussed in detail in Section 4.8.

In addition to the plasma membrane, some eukaryotic cells, notably plant cells and those of fungi and many protists, have a

ISTUDY

cell wall. A plant cell wall contains cellulose and, therefore, has a different composition from the bacterial cell wall.

Cells can vary in the proportion of organelles they have, depending on the specialized function of the cell. For example, a liver cell whose function is partly to detoxify drugs and other ingested compounds contains a greater proportion of smooth endoplasmic reticulum, the organelle that accomplishes that task. A nerve cell, whose job is to conduct electrical signals across long distances, contains more plasma membrane relative to other cells. Other cells may specialize so extensively that they completely lose an organelle, like a red blood cell that ejects its nucleus to increase the surface area needed to carry oxygen in the blood.

Check Your Progress 4.3

1. Summarize the benefits of compartmentalization found in cells.
2. Examine why organelles increase cell efficiency and function.
3. Explain the origins of the nucleus, chloroplast, and mitochondria of eukaryotic cells.

4.4 The Nucleus and Ribosomes

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the structure and function of the nucleus.
2. Describe the flow of information from DNA to a protein.
3. Explain the role of ribosomes in protein synthesis.

The **nucleus** is essential to the life and function of a eukaryotic cell. It contains the genetic information (DNA) that is passed on between each generation of cells. The DNA specifies the information that ribosomes use to carry out protein synthesis and thus the metabolic activities of the cell.

The Nucleus

The nucleus, which has a diameter of about 5 µm, is a prominent structure in the eukaryotic cell (Fig. 4.8). It generally appears as an oval structure located near the center of most cells. Some cells, such as skeletal muscle cells, can have more than one nucleus. The interior of the nucleus contains a semifluid matrix called the **nucleoplasm.** The nucleus is the command center of the cell.

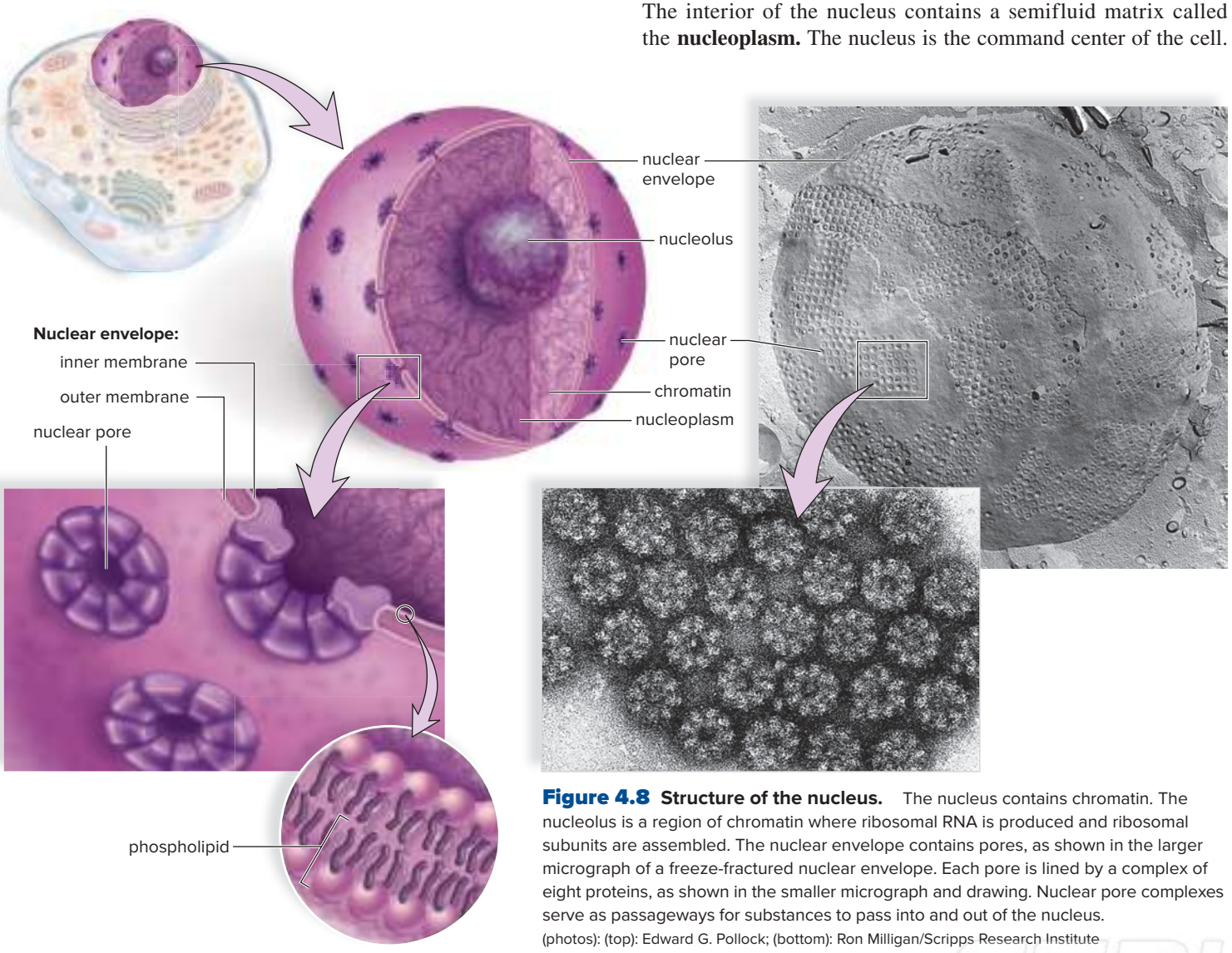


Figure 4.8 Structure of the nucleus. The nucleus contains chromatin. The nucleolus is a region of chromatin where ribosomal RNA is produced and ribosomal subunits are assembled. The nuclear envelope contains pores, as shown in the larger micrograph of a freeze-fractured nuclear envelope. Each pore is lined by a complex of eight proteins, as shown in the smaller micrograph and drawing. Nuclear pore complexes serve as passageways for substances to pass into and out of the nucleus.
(photos): (top): Edward G. Pollock; (bottom): Ron Milligan/Scripps Research Institute

ISTUDY

It contains **chromatin** (Gk. *chroma,* "color"), which is a combination of proteins and nucleic acids. Chromatin has a grainy appearance, but is actually a network of strands that will condense and undergo coiling into rodlike structures called **chromosomes,** just before the cell divides. The chromosomes are the carriers of genetic information. This information is organized on the chromosome as **genes,** the basic units of heredity. All the cells of an individual contain the same number of chromosomes, and the mechanics of nuclear division ensure that each daughter cell receives the normal number of chromosomes, except for the egg and sperm, which usually have half this number.

Three types of ribonucleic acid (RNA) are produced in the nucleus: *ribosomal RNA* (*rRNA*), *messenger RNA* (*mRNA*), and *transfer RNA* (*tRNA*). Ribosomal RNA is produced in the **nucleolus,** a dark region of chromatin where rRNA joins with proteins to form the subunits of ribosomes. Ribosomes are small bodies in the cytoplasm that facilitate protein synthesis. Messenger RNA, a mobile molecule, acts as an intermediary for DNA, a sedentary molecule, which specifies the sequence of amino acids in a protein. Transfer RNA participates in the assembly of amino acids into a polypeptide by recognizing both mRNA and amino acids during protein synthesis.

The nucleus is important to cell structure and function, because it specifies the code to make proteins. Although the nucleus is physically separated from the cytoplasm by a double membrane known as the **nuclear envelope,** it is still able to communicate with the cytoplasm through nuclear pores. **Nuclear pores** are of sufficient size (100 nm) to permit the passage of ribosomal subunits and mRNA out of the nucleus into the cytoplasm, as well as the passage of proteins from the cytoplasm into the nucleus. High-resolution electron micrographs show that nonmembrane components associated with the pores form a nuclear pore complex. Nuclear pore complexes act as gatekeepers to regulate what goes into and out of a nucleus.

Ribosomes

Ribosomes are particles where protein synthesis occurs. A large and small ribosomal subunit, each comprised of a mix of proteins and rRNA, are necessary components of a functional ribosome. In eukaryotes, ribosomes are 20 nm by 30 nm, and in prokaryotes they are slightly smaller. The number of ribosomes in a cell varies depending on its functions; for example, pancreatic cells and those of other glands have many ribosomes because they produce secretions that contain proteins.

In eukaryotic cells, some ribosomes occur freely within the cytoplasm, either singly or in groups called **polyribosomes,** whereas others are attached to the endoplasmic reticulum (ER), a membranous system of flattened saccules (small sacs) and tubules (see Section 4.5). In the nucleus, the information within a gene is copied into mRNA, which is exported through a nuclear pore complex into the cytoplasm. Ribosomes receive the mRNA, which carries a coded message from DNA indicating the correct sequence of amino acids in a particular protein. Proteins synthesized by cytoplasmic ribosomes are used in the cytoplasm, and those synthesized by attached ribosomes end up in the ER.

What causes a ribosome to bind to the endoplasmic reticulum? Binding occurs only if the protein being synthesized by a ribosome begins with a sequence of amino acids called a *signal peptide.* The signal peptide binds a particle (signal recognition particle, SRP), which then binds to a receptor on the ER. Once the protein enters the ER, an enzyme cleaves off the signal peptide, and the protein ends up within the lumen (interior) of the ER, where it folds into its final shape (Fig. 4.9).

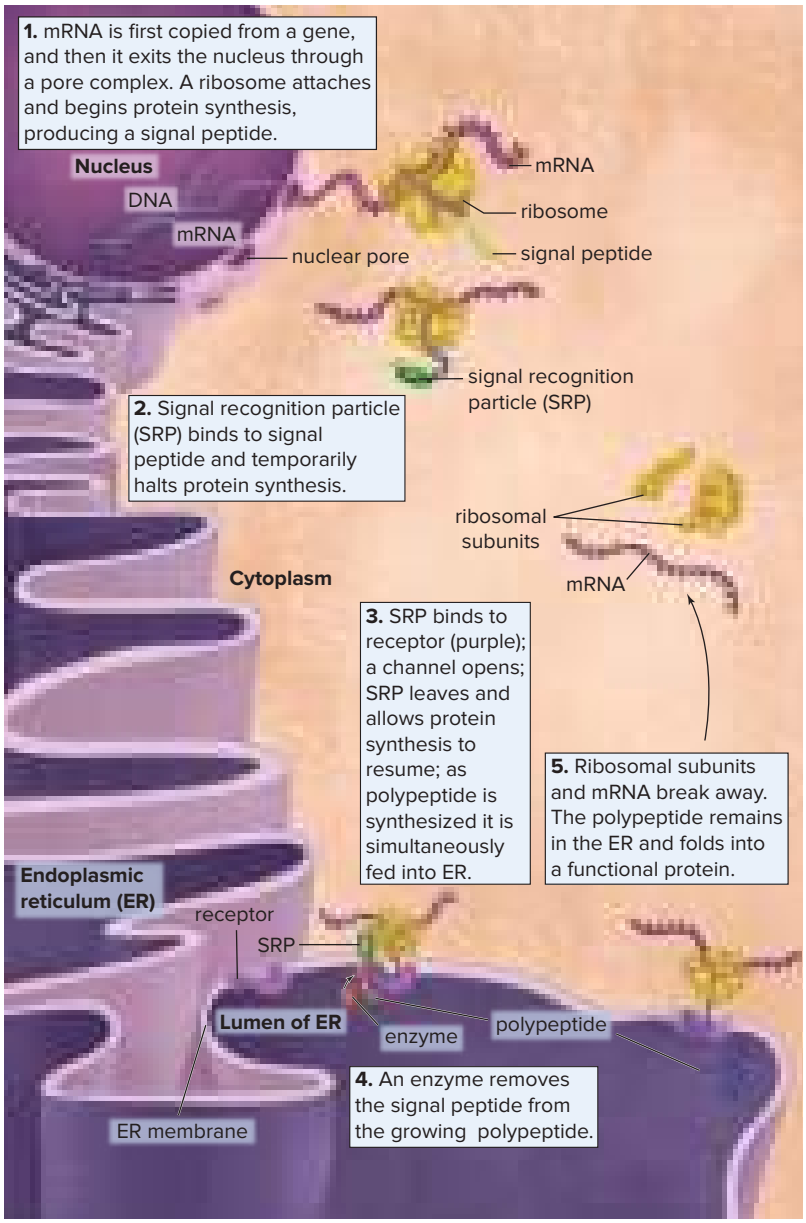


Figure 4.9 Function of ribosomes. Ribosomes are sites of protein synthesis. An mRNA molecule, serving as a temporary copy of a gene from the nucleus, is read by a ribosome in the cytoplasm. Amino acids are connected together by the ribosome in a sequence specified by the mRNA. When a polypeptide is first translated, it begins with a signal peptide; this combines with a signal recognition particle (SRP), which is brought to the rough ER. The SRP leaves, and the polypeptide is made and pushed into the ER lumen. The signal peptide is removed and the polypeptide folds into its final protein shape.

ISTUDY

The process of DNA being transcribed into mRNA, and then being translated into a protein, occurs in all living cells at some point during their life span. Because of its universality, the DNA–mRNA–protein sequence of events is termed the *central dogma of molecular biology* (see Section 12.3).

Check Your Progress 4.4

1. Distinguish between the chromatin and chromosomes within the nucleus.
2. Explain the importance of the nuclear pores.
3. Describe the sequence of events that transfers information from a gene to a functional protein.

4.5 The Endomembrane System

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the importance of the endomembrane system in cellular function.
2. Examine how the ER, Golgi apparatus, and lysosomes differ from one another.
3. Describe how endomembrane vesicles are able to fuse with organelles.

The **endomembrane system** consists of the nuclear envelope, the membranes of the endoplasmic reticulum, the Golgi apparatus, and several types of vesicles. This system compartmentalizes the cell so particular enzymatic reactions are restricted to specific regions and overall cell efficiency is increased. The vesicles transport molecules from one part of the system to another.

Endoplasmic Reticulum

The **endoplasmic reticulum (ER)** (Gk. *endon,* "within"; *plasma,* "something molded"; L. *reticulum,* "net"), consisting of a complicated system of membranous channels and saccules (flattened vesicles), is physically continuous with the nuclear envelope (Fig. 4.10). The ER consists of rough ER and smooth ER, which have different structures and functions.

Rough ER is studded with ribosomes on the exterior side of the membrane that faces the cytoplasm, giving it the capacity to produce proteins. Inside its lumen, the rough ER allows proteins to fold and take on their final three-dimensional shape. The rough ER also contains enzymes that can add carbohydrate (sugar) chains to proteins, forming glycoproteins that are important in many cell functions.

Smooth ER is continuous with the rough ER and does not have attached ribosomes. Certain organs contain cells with an abundance of smooth ER, depending on the organ's function. In some organs, increased smooth ER helps produce more lipids. For example, in the testes, smooth ER produces testosterone, a steroid hormone. In the liver, smooth ER helps detoxify drugs. The smooth ER of the liver increases in quantity when a person consumes alcohol or takes barbiturates on a regular basis. Regardless of functional differences, both rough and smooth ER form vesicles that transport molecules to other parts of the cell, notably the Golgi apparatus.

The Golgi Apparatus

The **Golgi apparatus** is named for Camillo Golgi who discovered its presence in cells in 1898. The Golgi apparatus typically consists of a stack of 3 to 20 slightly curved, flattened saccules whose appearance can be compared to a stack of pancakes (Fig. 4.11). In animal cells, one side of the stack (the *cis,* or inner, face) is

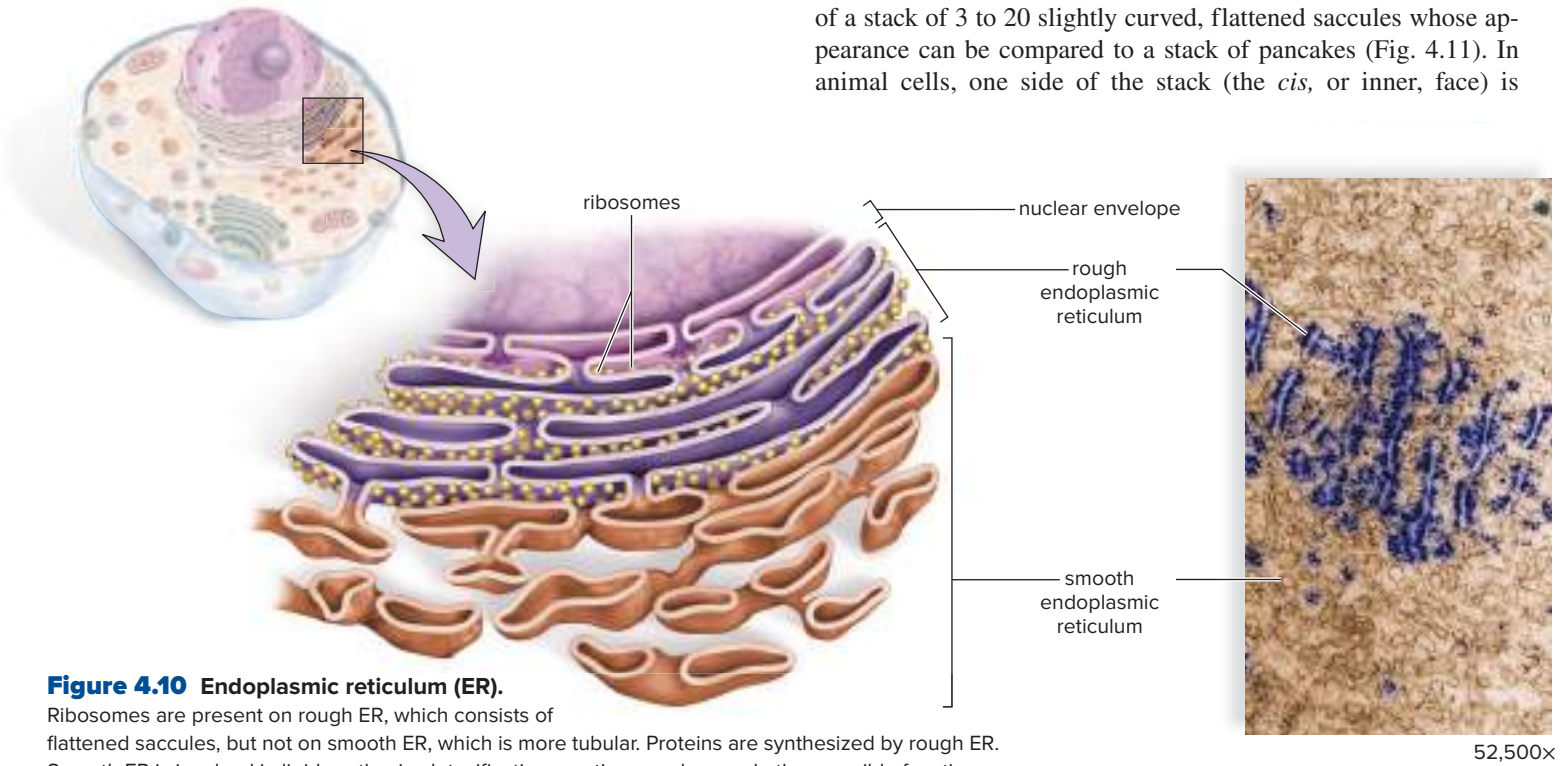


Figure 4.10 Endoplasmic reticulum (ER). Ribosomes are present on rough ER, which consists of flattened saccules, but not on smooth ER, which is more tubular. Proteins are synthesized by rough ER. Smooth ER is involved in lipid synthesis, detoxification reactions, and several other possible functions.
(photo): Martin M. Rotker/Science Source

ISTUDY

directed toward the ER, and the other side of the stack (the *trans,* or outer, face) is directed toward the plasma membrane. Vesicles can frequently be seen at the edges of the saccules.

Protein-filled vesicles that bud from the rough ER, as well as lipid-filled vesicles that bud from the smooth ER, are received by the Golgi apparatus at its inner face. These substances are altered as they move through the saccules. For example, the Golgi apparatus contains enzymes that modify the carbohydrate chains first attached to proteins in the rough ER. It can modify one sugar into another sugar on glycoproteins. In some cases, the modified carbohydrate chain serves as a signal molecule or molecular address label that determines the protein's final destination in the cell.

The Golgi apparatus sorts the modified molecules and packages them into vesicles that depart from the outer face. These vesicles may be transported to various locations within the cell, depending on their molecular address labels. In animal cells, some of these vesicles are lysosomes, which are discussed next. Other vesicles may return to the ER or proceed to the plasma membrane, where they merge and discharge their contents to the outside of the cell by exocytosis.

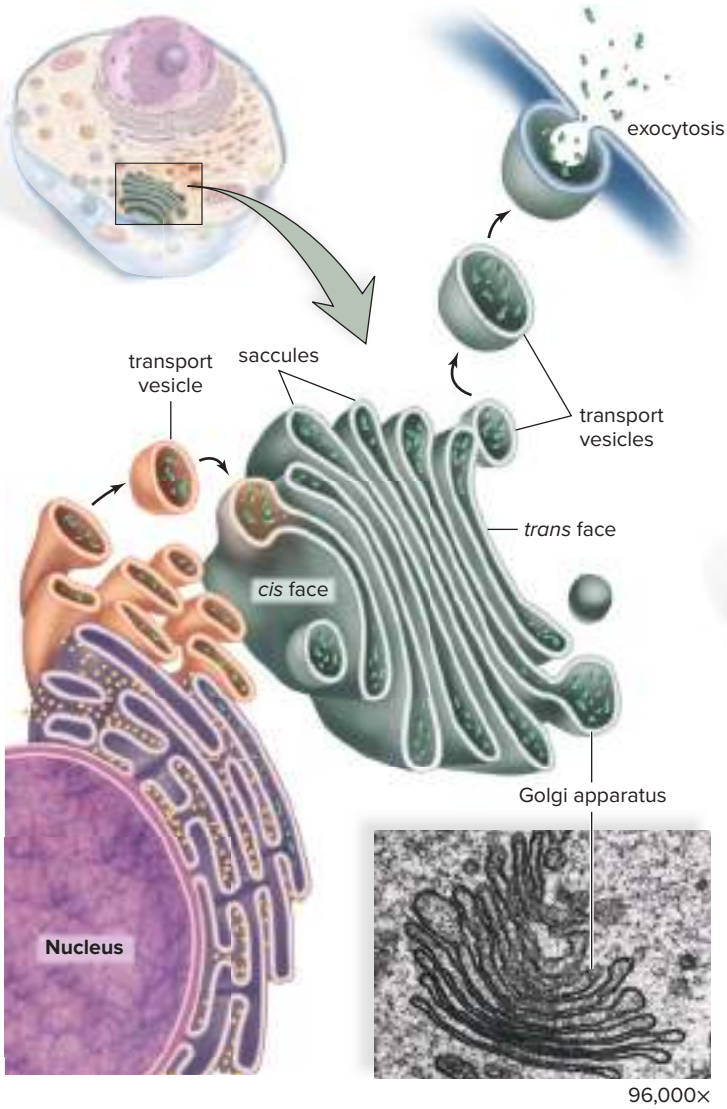


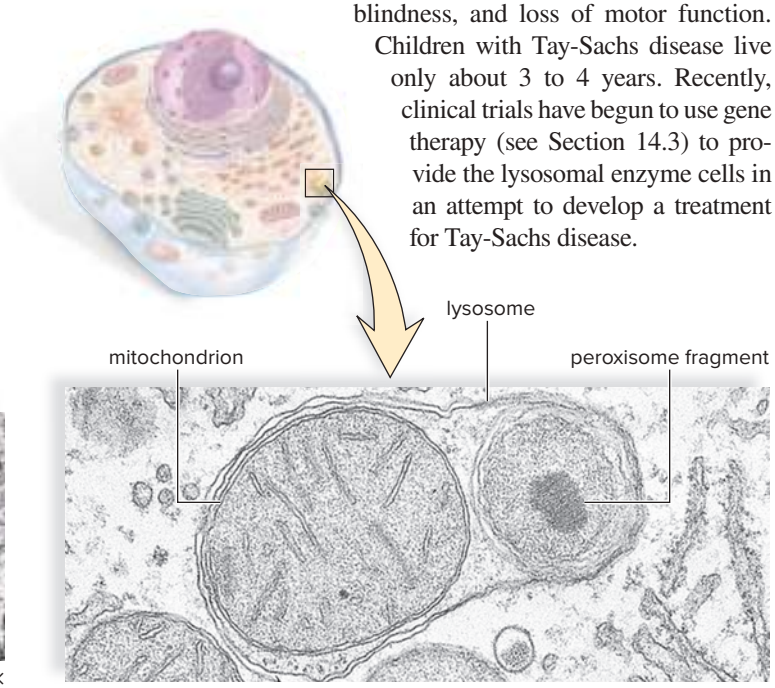
Figure 4.11 Golgi apparatus. The Golgi apparatus is a stack of flattened, curved saccules. It processes proteins and lipids and packages them in transport vesicles that either distribute these molecules to various locations within the cell or secrete them externally. (photo): Biophoto Associates/Science Source

Lysosomes

Lysosomes (Gk. *lysis,* "destruction"; *soma,* "body") are membrane-bound vesicles produced by the Golgi apparatus. They have a very low pH and store powerful hydrolytic-digestive enzymes in an inactive state. Lysosomes act much like your stomach in that they assist in digesting material taken into the cell. They also destroy nonfunctional organelles and portions of cytoplasm (Fig. 4.12) by a process called **autophagy.**

Materials can be taken into a cell by vesicle or vacuole formation at the plasma membrane. When a lysosome fuses with either, the lysosomal enzymes are activated and digest the material into simpler subunits that are exported into the cytoplasm and recycled by other cell processes. White blood cells, specialized to protect the body from foreign entities, are well-known for engulfing pathogens (e.g., disease-causing viruses and bacteria), which are then broken down in lysosomes. White blood cells have a greater proportion of lysosomes than other cells, because their specialized function is the digestion of foreign bodies.

A number of human lysosomal storage diseases are due to a missing lysosomal enzyme. In Tay-Sachs disease, the missing enzyme digests a fatty substance that helps insulate nerve cells and increases their efficiency. The fatty substance accumulates in so many storage bodies that nerve cells die off. Individuals who are affected appear normal at birth but begin to develop neurological problems at 4 to 6 months of age. Eventually, the child suffers cerebral degeneration, slow paralysis, blindness, and loss of motor function. Children with Tay-Sachs disease live only about 3 to 4 years. Recently, clinical trials have begun to use gene therapy (see Section 14.3) to provide the lysosomal enzyme cells in an attempt to develop a treatment for Tay-Sachs disease.



Mitochondrion and a peroxisome in a lysosome

Figure 4.12 Lysosomes. Lysosomes, which bud off the Golgi apparatus in cells, are filled with hydrolytic enzymes that digest molecules and parts of the cell. Here, a lysosome digests a worn mitochondrion and a peroxisome. (photo): Don W. Fawcett/Science Source

ISTUDY

Endomembrane System Summary

You have seen that the endomembrane system is a series of membranous organelles that work together and communicate by means of transport vesicles. The endoplasmic reticulum (ER) and the Golgi apparatus are essentially flattened saccules, and lysosomes are specialized vesicles.

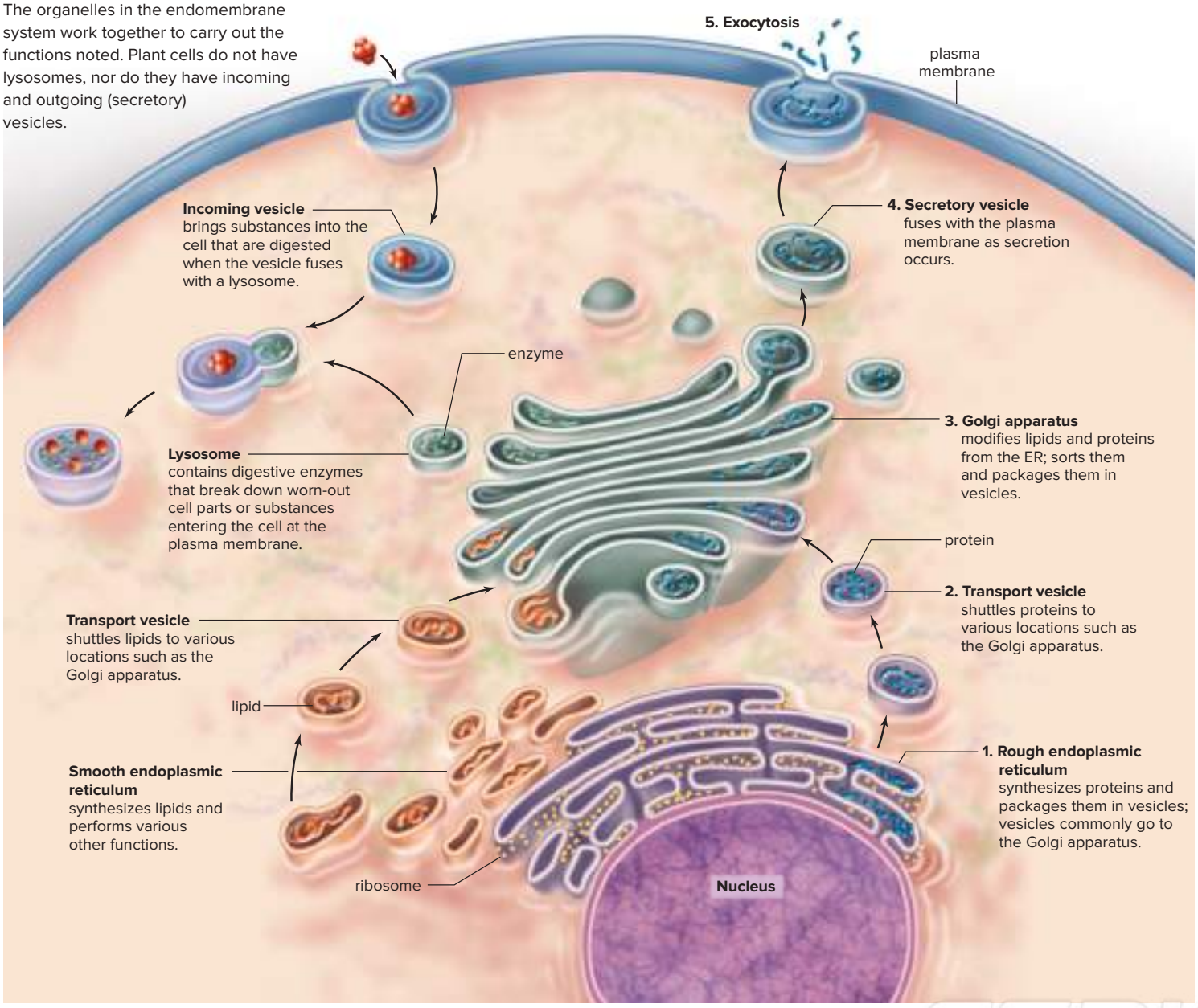
Organelles within the endomembrane system can interact because their membranes readily fuse together, and because membrane-associated proteins enable communication and specialized functions. Figure 4.13 shows how the components of the endomembrane system work together. Products of both rough ER and smooth ER are carried in transport vesicles to the Golgi apparatus, where they are further modified. Using signaling sequences and molecular address labels, the Golgi apparatus sorts these products and packages them into vesicles that transport them to various cellular destinations. Secretory vesicles take the proteins to the plasma membrane, where they exit the cell by exocytosis. For example, secretion into ducts occurs when the mammary glands produce milk or the pancreas produces digestive enzymes.

In animal cells, the Golgi apparatus also produces lysosomes that contain stored hydrolytic enzymes. Lysosomes fuse with incoming vesicles from the plasma membrane and digest macromolecules taken into a cell.

Check Your Progress 4.5

1. Contrast the structure and functions of rough and smooth endoplasmic reticulum.
2. Describe the relationship between the components of the endomembrane system.
3. Examine how cellular function would be affected if the Golgi apparatus ceased to function.

Figure 4.13 Endomembrane system.
The organelles in the endomembrane system work together to carry out the functions noted. Plant cells do not have lysosomes, nor do they have incoming and outgoing (secretory) vesicles.



ISTUDY

4.6 Microbodies and Vacuoles

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the role of peroxisomes and vacuoles in cell function.
2. Contrast peroxisomes and vacuoles with endomembrane organelles.

Eukaryotic cells contain a variety of membrane-bound vesicles, called **microbodies,** that contain specialized enzymes to perform specific metabolic functions. One example is the peroxisome. In addition, cells may contain large storage areas called vacuoles.

Peroxisomes

Peroxisomes are membrane-bound vesicles that contain enzymes used to break down fatty acids. Unlike the enzymes of lysosomes, which are loaded into the vesicle by the Golgi apparatus, the enzymes in peroxisomes are synthesized by free ribosomes and transported into a peroxisome from the cytoplasm. As the enzymes within the peroxisome oxidize fatty acids, they produce hydrogen peroxide (H_2O_2), a toxic molecule. However, peroxisomes also contain an enzyme called catalase that immediately breaks down H_2O_2 to water and oxygen. You can see this reaction when you apply hydrogen peroxide to a wound; the resulting bubbles occur as catalase breaks down the H_2O_2.

Peroxisomes are metabolic assistants to the other organelles. They have varied functions but are especially prevalent in cells that synthesize and break down lipids. In the liver, some peroxisomes produce bile salts from cholesterol, and others break down fats. The disease adrenoleukodystrophy (ALD) is caused when peroxisomes lack a membrane protein needed to import a specific enzyme and/or long-chain fatty acids from the cytoplasm. As a result, long-chain fatty acids accumulate in the brain, causing neurological damage.

Plant cells also have peroxisomes (Fig. 4.14). In germinating seeds, they oxidize fatty acids into molecules that can be converted to sugars needed by the growing plant. In leaves, peroxisomes can carry out a reaction that is opposite to photosynthesis—the reaction uses up oxygen and releases carbon dioxide.

Vacuoles

Like vesicles, **vacuoles** are membranous sacs, but vacuoles are larger than vesicles. The vacuoles of some protists are quite specialized, including contractile vacuoles for ridding the cell of excess water and digestive vacuoles for breaking down nutrients. Vacuoles usually store substances. In general, few animal cells contain vacuoles; however, fat cells contain a very large, lipid-engorged vacuole that takes up nearly two-thirds of the volume of the cell!

Vacuoles are essential to plant function. Plant vacuoles contain not only water, sugars, and salts but also water-soluble pigments and toxic molecules. The pigments are responsible for many of the red, blue, or purple colors of flowers and some leaves. The toxic substances help protect a land plant from herbivorous animals.

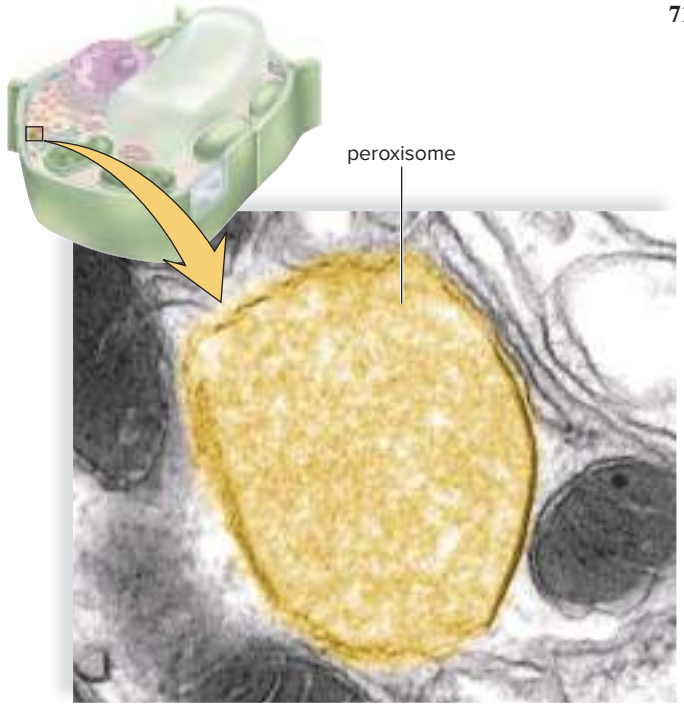


Figure 4.14 Peroxisomes. Peroxisomes contain one or more enzymes that can oxidize various organic substances. (photo): EM Research Services/Newcastle University

Plant Cell Central Vacuole

Typically, plant cells have a large **central vacuole** that may take up to 90% of the volume of the cell. The vacuole is filled with a watery fluid called cell sap that gives added support to the cell (Fig. 4.15). The central vacuole maintains hydrostatic pressure, or turgor pressure, in plant cells, providing structural support. A plant cell can rapidly increase in size by enlarging its vacuole. Eventually, a plant cell also produces more cytoplasm.

The central vacuole functions in the storage of both nutrients and waste products. Metabolic waste products are pumped across the vacuole

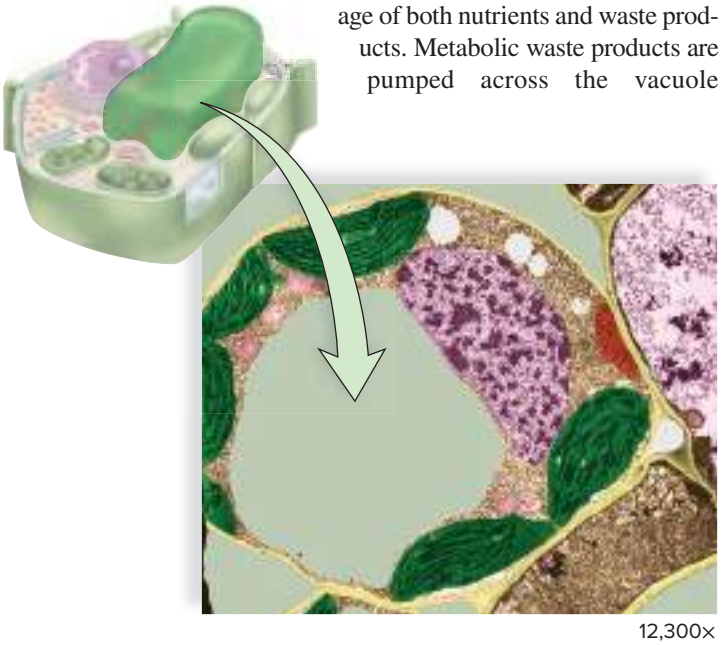


Figure 4.15 Plant cell central vacuole. The large central vacuole of plant cells has numerous functions, from storing molecules to helping the cell increase in size. Biophoto Associates/Science Source

ISTUDY

membrane and stored permanently in the central vacuole. As organelles age and become nonfunctional, they fuse with the vacuole, where digestive enzymes break them down. This is a function analogous to that carried out by lysosomes in animal cells.

Check Your Progress 4.6

1. Compare the structure and functions of a peroxisome with those of a lysosome.
2. Distinguish between where peroxisome and lysosome proteins are produced.

4.7 The Energy-Related Organelles

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between the functions of chloroplasts and mitochondria in a cell.
2. Describe the internal structure of mitochondria and chloroplasts.

Life is possible only because a constant input of energy maintains the structure of cells. Chloroplasts and mitochondria are the two eukaryotic membranous organelles that specialize in converting energy to a form that can be used by the cell. Although animal cells contain only mitochondria, plant cells contain both mitochondria and chloroplasts.

During *photosynthesis,* **chloroplasts** (Gk. *chloros,* "green"; *plastos,* "formed, molded") use solar energy to synthesize carbohydrates, which serve as organic nutrient molecules for plants and all life on Earth. Photosynthesis can be represented by this equation:

solar energy + carbon dioxide + water → carbohydrate + oxygen

Plants, algae, and cyanobacteria are capable of conducting photosynthesis in this manner, but only plants and algae have chloroplasts, because they are eukaryotes.

In *cellular respiration,* **mitochondria** (*sing.*, mitochondrion) break down carbohydrate-derived products to produce ATP (adenosine triphosphate). Cellular respiration can be represented by this equation:

carbohydrate + oxygen → carbon dioxide + water + energy

Here, the word *energy* stands for ATP molecules. When a cell needs energy, ATP supplies it. The energy of ATP is used to drive synthetic reactions, active transport, and all energy-requiring processes in cells. Figure 4.16 provides a summary of the interactions between these two energy-producing organelles.

Chloroplasts

Some algal cells have only one chloroplast, while some plant cells have as many as a hundred. Chloroplasts can be quite large, being twice as wide and as much as five times the length of a mitochondrion.

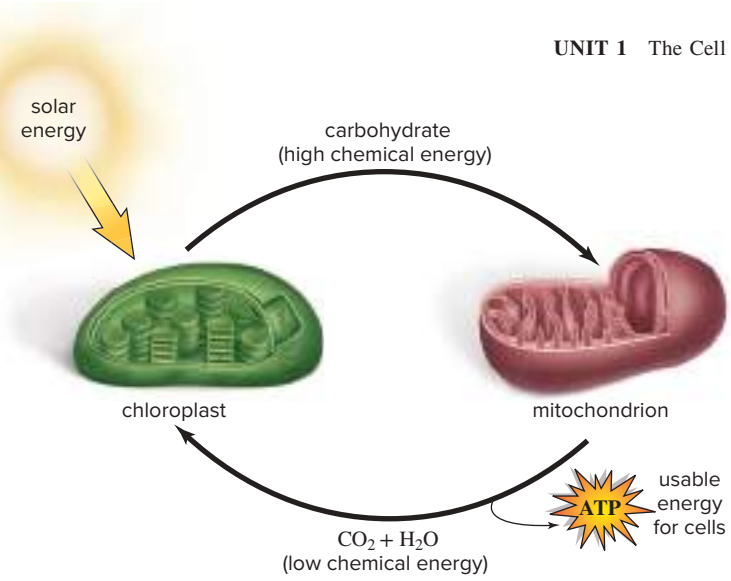


Figure 4.16 Energy-producing organelles. Chloroplasts use sunlight to produce carbohydrates, which in turn are used by the mitochondria. The mitochondria then produce carbon dioxide and water, which is in turn used by the chloroplasts.

Chloroplasts have a three-membrane system (Fig. 4.17). They are surrounded by a double membrane, which includes an outer membrane and an inner membrane. The **stroma** is a semifluid region that is enclosed by a double membrane. Enzymes and **thylakoids,** disklike sacs formed from a third chloroplast membrane are found within the stroma. A stack of thylakoids is a **granum.** The lumens of the thylakoids are believed to form a large, internal compartment called the thylakoid space. Chlorophyll and the other pigments that capture solar energy are located in the thylakoid membrane, and the enzymes that synthesize carbohydrates are located outside the thylakoid in the fluid of the stroma.

The endosymbiotic theory holds that chloroplasts are derived from a photosynthetic bacterium that was engulfed by a eukaryotic cell (see Fig. 4.5). This certainly explains why a chloroplast is surrounded by a double membrane—one membrane is derived from the vesicle that brought the prokaryote into the cell, while the inner membrane is derived from the prokaryote. The endosymbiotic theory is also supported by the finding that chloroplasts have their own prokaryotic-type chromosome and ribosomes, and they produce some of their own enzymes even today.

Other Types of Plastids

A chloroplast is a type of plastid. **Plastids** are plant organelles that are surrounded by a double membrane and have varied functions. *Chromoplasts* contain pigments that result in the yellow, orange, or red color of autumn leaves, fruits, carrots, and some flowers. *Leucoplasts* are generally colorless plastids that synthesize and store starches and oils. A microscopic examination of potato tissue reveals a number of leucoplasts.

Mitochondria

Nearly all eukaryotic cells, and certainly all plant and algal cells in addition to animal cells, contain mitochondria. Even though mitochondria are smaller than chloroplasts, they can usually be seen with a light microscope. The number of mitochondria can vary depending on the metabolic activities and energy needed within a cell. Some cells, such as liver cells, may have as many as 1,000 mitochondria.

ISTUDY

We think of mitochondria as having a shape like that shown in Figure 4.18, but actually they often change shape to be longer and thinner or shorter and broader. Mitochondria can form long, moving chains, or they can remain fixed in one location—typically where energy is most needed. For example, they are packed between the contractile elements of cardiac cells and wrapped around the interior of a sperm's flagellum. In contrast, fat cells contain few mitochondria—they function in fat storage, which does not require energy.

Mitochondria have two membranes, the outer membrane and the inner membrane. The inner membrane is highly convoluted into folds called **cristae** that project into the matrix. These cristae increase the surface area of the inner membrane so much that in a liver cell they account for about one-third the total membrane in the cell. The inner membrane encloses a semifluid **matrix,** which contains mitochondrial DNA and ribosomes. Again, the presence of a double membrane and mitochondrial genes is consistent with the endosymbiotic theory regarding the origin of mitochondria, which was illustrated in Figure 4.5.

Mitochondria are often called the powerhouses of the cell because they produce most of the ATP utilized by the cell. Within the matrix of the mitochondria is a highly concentrated mixture of enzymes that break down carbohydrates and other nutrient molecules. These reactions supply the chemical energy needed for a chain of proteins on the inner membrane to create the conditions that allow ATP synthesis to take place. The entire process, which also involves the cytoplasm, is called *cellular respiration,* because oxygen is used and carbon dioxide is given off, as shown at the beginning of this section.

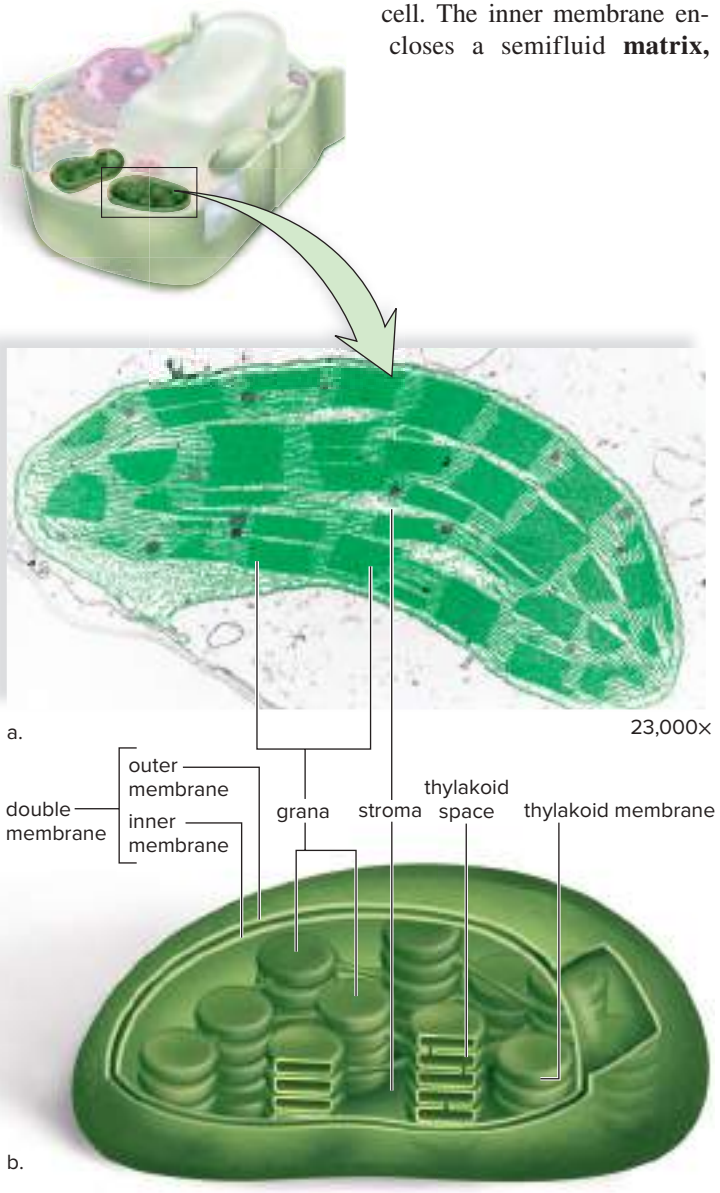


Figure 4.17 Chloroplast structure. Chloroplasts carry out photosynthesis. **a.** Electron micrograph of a longitudinal section of a chloroplast. **b.** Generalized drawing of a chloroplast in which the outer and inner membranes have been cut away to reveal the grana, each of which is a stack of membranous sacs called thylakoids. (a): Omikron/Science Source

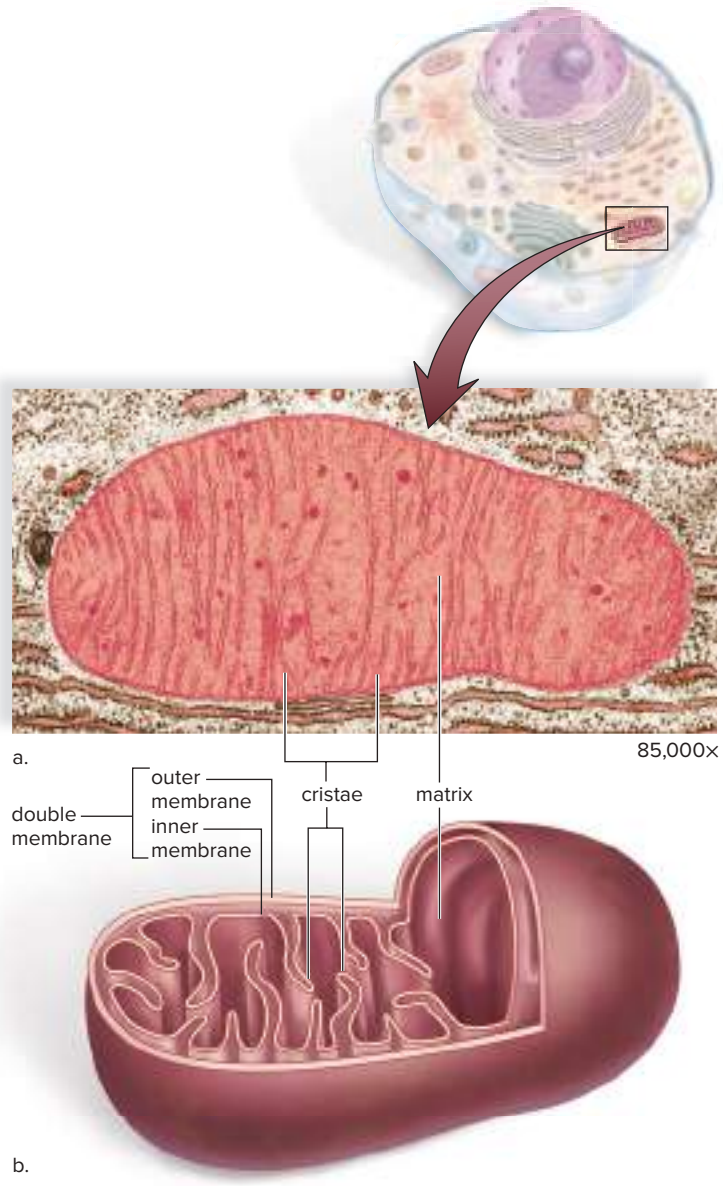


Figure 4.18 Mitochondrion structure. Mitochondria are involved in cellular respiration. **a.** Electron micrograph of a longitudinal section of a mitochondrion. **b.** Generalized drawing in which the outer membrane and portions of the inner membrane have been cut away to reveal the cristae. (a): Keith R. Porter/Science Source

ISTUDY

Mitochondrial Diseases

So far, dozens of different mitochondrial diseases that affect the brain, muscles, kidneys, heart, liver, eyes, ears, or pancreas have been identified. The common factor among these genetic diseases is that the patient's mitochondria are unable to completely metabolize organic molecules to produce ATP. As a result, toxins accumulate inside the mitochondria and the body. The toxins can be free radicals (substances that readily form harmful compounds when they react with other molecules), and these compounds damage mitochondria over time. In the United States, between 1,000 and 4,000 children per year are born with a mitochondrial disease. In addition, it is possible that many diseases of aging are due to malfunctioning mitochondria (see Section 42.4).

Check Your Progress 4.7

1. Summarize the roles of mitochondria and chloroplasts in the cell.
2. Discuss the evidence that chloroplasts and mitochondria are derived from ancient bacteria.
3. Explain why chloroplasts and mitochondria contain complex internal membrane structures.

4.8 The Cytoskeleton

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the structure and function of actin filaments, intermediate filaments, and microtubules.
2. Describe how motor molecules interact with cytoskeletal elements to produce movement.
3. Explain the diverse roles of microtubules within the cell.

Cells are exposed to many physical forces. Cell shape, movement, and internal transport all require structural support, provided by the cytoskeleton. The protein components of the cytoskeleton (Gk. *kytos,* "cell") interconnect and extend from the nucleus to the plasma membrane in eukaryotic cells. Prior to the 1970s, it was believed that the cytoplasm was an unorganized mixture of organic molecules. Then, high-voltage electron microscopes, which can penetrate thicker specimens, showed instead that the cytoplasm is highly organized. The technique of immunofluorescence microscopy identified the makeup of the protein components within the cytoskeletal network (Fig. 4.19).

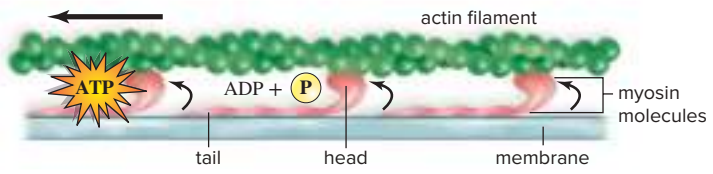
The cytoskeleton contains actin filaments, intermediate filaments, and microtubules, which maintain cell shape and allow the cell and its organelles to move. Therefore, the cytoskeleton is often compared to the bones and muscles of an animal. However, the cytoskeleton is dynamic; it can rearrange its protein components as necessary in response to changes in internal and external environments. A number of different mechanisms appear to regulate this process, including protein phosphatases, which remove phosphates from proteins and bring about assembly, and protein kinases, which phosphorylate proteins and lead to disassembly.

Actin Filaments

Actin filaments (formerly called microfilaments) are long, extremely thin, flexible fibers (about 7 nm in diameter) that occur in bundles or meshlike networks. Each actin filament contains two chains of globular actin monomers twisted about one another in a helical manner.

Actin filaments provide structural support as a dense, complex web just under the plasma membrane, to which they are anchored by special proteins. Sometimes, actin filaments can dynamically rearrange themselves and facilitate cellular movement, such as when an amoeba moves over a surface with pseudopods (L. *pseudo,* "false"; *pod,* "feet"), or when intestinal cell microvilli lengthen and shorten into the gut lumen (the space where ingested food is processed). In plant cells, actin filaments form the tracks along which chloroplasts circulate in a particular direction in a process called cytoplasmic streaming.

Actin filaments move the cell and its organelles by interacting with *motor molecules,* which are proteins that can attach, detach, and reattach farther along an actin filament. The motor molecule myosin uses ATP to pull actin filaments along in this way. Myosin has both a head and a tail. In muscle cells, the tails of several myosin molecules are joined to form a thick filament. In nonmuscle cells, cytoplasmic myosin tails are bound to membranes, but the heads still interact with actin:



During animal cell division, the two new cells form when actin, in conjunction with myosin, pinches off the cells from one another.

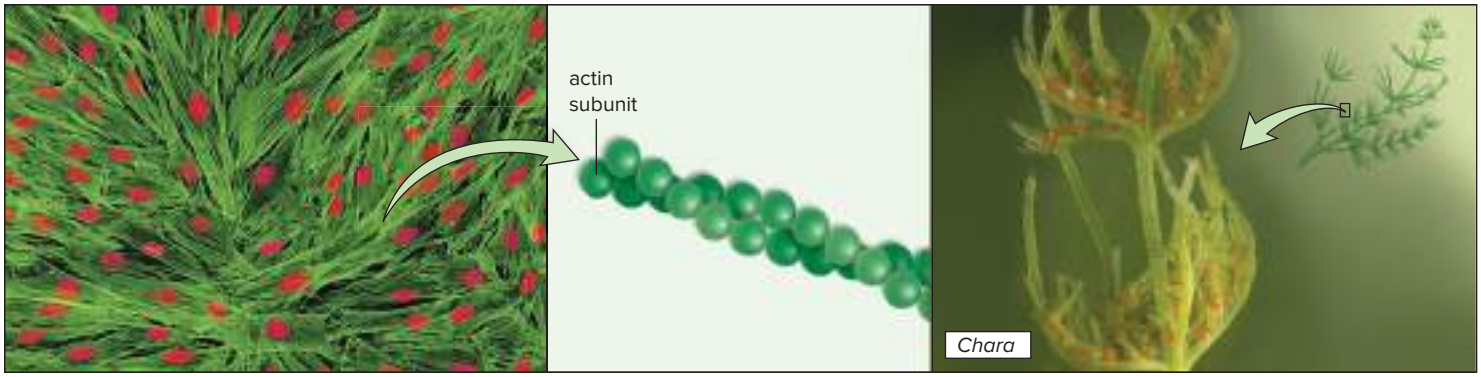
Intermediate Filaments

Intermediate filaments (8–11 nm in diameter) are so named because they are intermediate in size between actin filaments and microtubules. They form a ropelike assembly of fibrous polypeptides, but the specific filament type varies according to the tissue. Some intermediate filaments support the nuclear envelope, whereas others support the plasma membrane and take part in the formation of cell-to-cell junctions. In the skin, intermediate filaments made of the protein keratin give great mechanical strength to skin cells. Like other cytoskeletal components, intermediate filaments are highly dynamic and disassemble when phosphate is added to them by a kinase.

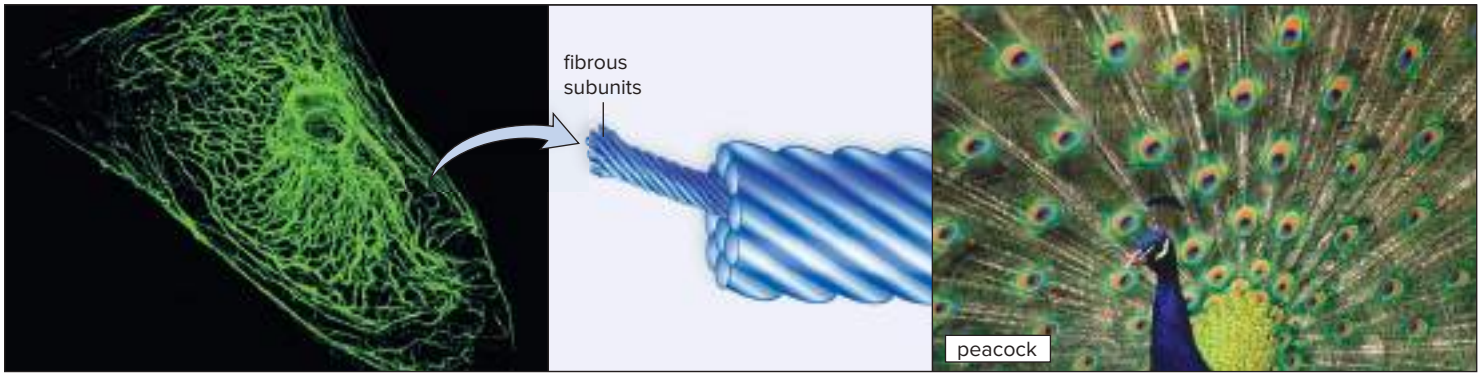
Microtubules

Microtubules (Gk. *mikros,* "small") are small, hollow cylinders about 25 nm in diameter and from 0.2 to 25 μm in length. They are made of a globular protein called tubulin, which is of two types: α and β. Alpha tubulin has a slightly different amino acid sequence than β tubulin. When assembly occurs, α and β tubulin molecules come together as dimers, and the dimers arrange themselves in rows. Microtubules have 13 rows of tubulin dimers, surrounding what appears in electron micrographs to be an empty central core.

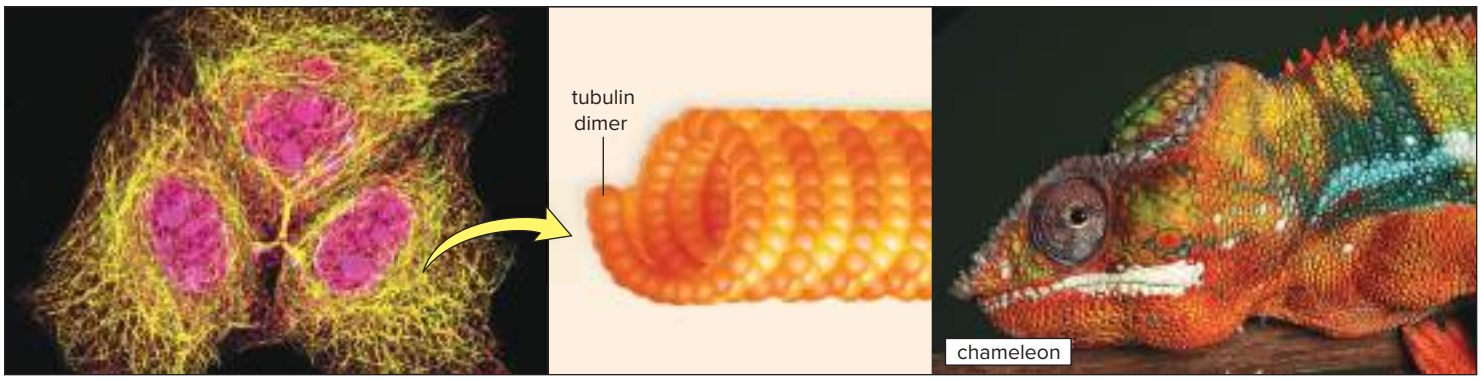
ISTUDY



a. Actin filaments 850×



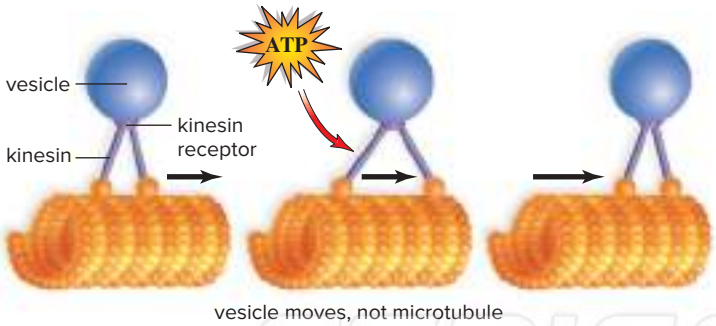
b. Intermediate filaments 1,000×



c. Microtubules

Figure 4.19 The cytoskeleton. The cytoskeleton maintains a cell's shape and allows its parts to move. Three types of protein components make up the cytoskeleton. They can be detected in cells by using labeling and fluorescence microscopy. **a.** *Left to right:* Cells showing a twisted double chain of actin filaments (green fibers). The giant cells of the green alga *Chara* use actin filaments to move organelles within the cell. **b.** *Left to right:* Animal cells showing fibrous, ropelike intermediate filaments (blue fibers). A peacock's colorful feathers are strengthened by intermediate filaments. **c.** *Left to right:* Animal cells showing hollow microtubules made of tubulin dimers (orange fibers). A chameleon's skin cells use microtubules to move pigment granules around so they take on the color of their environment. (actin): Thomas Deerinck, NCMIR/Science Source; (*Chara*): Bob Gibbons/Alamy Stock Photo; (intermediate filaments): Alvin Telser/Science Source; (peacock): Bryan Mullennix/Pixtal/age fotostock; (microtubules): Dr Gopal Murti/Science Source; (chameleon): CathyKeifer/123RF

Microtubule assembly is under the regulatory control of a microtubule-organizing center (MTOC). In most eukaryotic cells, the main MTOC is in the **centrosome** (Gk. *centrum,* "center"), which lies near the nucleus. Microtubules radiate from the centrosome, helping to maintain the shape of the cell and acting as tracks along which organelles can be moved. Whereas the motor molecule myosin is associated with actin filaments, the motor molecules kinesin and dynein are associated with microtubules:



ISTUDY

There are different types of kinesin proteins, each specialized to move one kind of vesicle or cellular organelle. Kinesin moves vesicles or organelles in an opposite direction from dynein. Cytoplasmic dynein is closely related to the molecule dynein found in flagella.

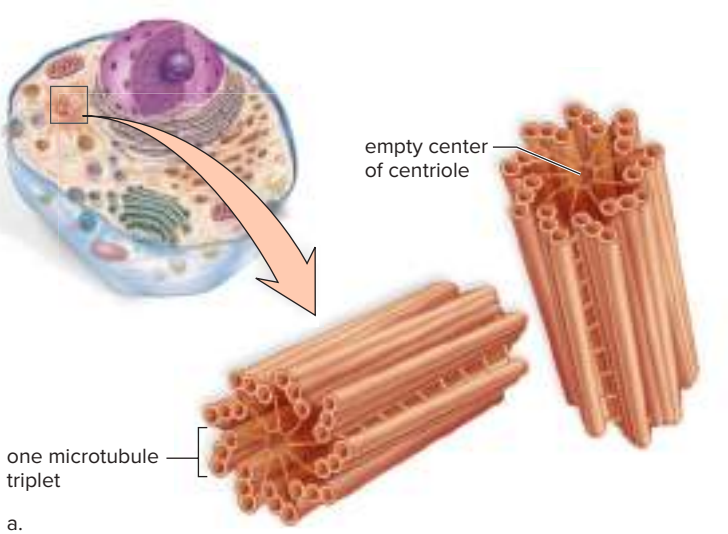
Before a cell divides, microtubules disassemble and then reassemble into a structure called a spindle, which distributes chromosomes in an orderly manner. At the end of cell division, the spindle disassembles, and microtubules reassemble once again into their former array. Plants have evolved various types of poisons that prevent them from being eaten by herbivores. One of these, colchicine, is a plant poison that binds tubulin and blocks the assembly of microtubules.

Centrioles

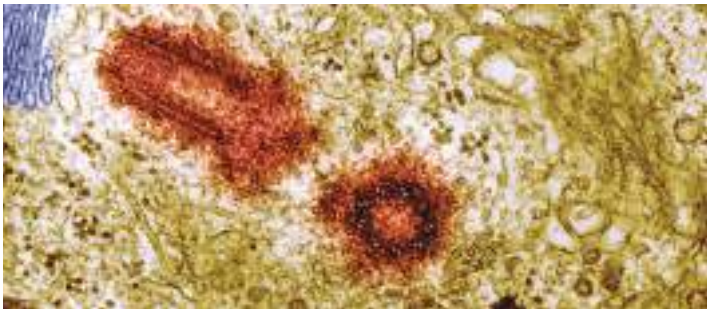
Centrioles are short cylinders with a 9 + 0 pattern of microtubule triplets—nine sets of triplets are arranged in an outer ring, but the center of a centriole does not contain a microtubule. In animal cells and most protists, a centrosome contains two centrioles lying at right angles to each other. A centrosome, as mentioned previously, is the major microtubule-organizing center for the cell. Therefore, it is possible that centrioles are also involved in the process by which microtubules assemble and disassemble.

Before an animal cell divides, the centrioles replicate, and the members of each pair are at right angles to one another (Fig. 4.20). Then, each pair becomes part of a separate centrosome. During cell division, the centrosomes move apart and most likely function to organize the mitotic spindle. In any case, each new cell has its own centrosome and pair of centrioles. Plant and fungal cells have the equivalent of a centrosome, but this structure does not contain centrioles, suggesting that centrioles are not necessary to the assembly of cytoplasmic microtubules.

A *basal body* is a structure that lies at the base of cilia and flagella and may direct the organization of microtubules within these structures. In other words, a basal body may do for a cilium or flagellum what the centrosome does for the cell. In cells with cilia and flagella, centrioles are believed to give rise to basal bodies.



a.



b.

Figure 4.20 Centrioles. **a.** The centrosome of an animal cell contains two centrioles positioned at right angles to each other. **b.** A micrograph of one centrosome containing two centrioles. (b): Don W. Fawcett/Science Source

Cilia and Flagella

Cilia (L. *cilium,* "eyelash, hair") and **flagella** (L. *flagello,* "whip") are hairlike projections that can move either in an undulating fashion, like a whip, or stiffly, like an oar. In free cells, cilia (or flagella) move the cell through liquid. For example, single-celled paramecia are organisms that move by means of cilia, whereas sperm cells move by means of flagella. If the cell is attached to other cells, cilia (or flagella) are capable of moving liquid over the cell. The cells that line our upper respiratory tract have cilia that sweep debris trapped within mucus back up into the throat, where it can be swallowed or expelled. This action helps keep the lungs clean.

In eukaryotic cells, cilia are much shorter than flagella, but they have a similar construction. Both are membrane-bound cylinders enclosing a matrix area. In the matrix are nine pairs of microtubules arranged in a circle around two central microtubules; this is called the 9 + 2 pattern of microtubules (Fig. 4.21). Cilia and flagella move when the pairs of microtubules slide past one another using motor molecules.

As mentioned, each cilium and flagellum has a basal body lying in the cytoplasm at its base. Basal bodies have the same circular arrangement of microtubule triplets as centrioles and are believed to be derived from them. It is possible that basal bodies organize the microtubules within cilia and flagella, but this idea is not supported by the observation that cilia and flagella grow by the addition of tubulin dimers to their tips.

Check Your Progress 4.8

1. Differentiate between the components of the cytoskeleton and how they provide support to the cell.
2. Explain how ATP is used to produce movement in a cell.
3. Describe the role of motor molecules and microtubules in cilia and flagella.

ISTUDY

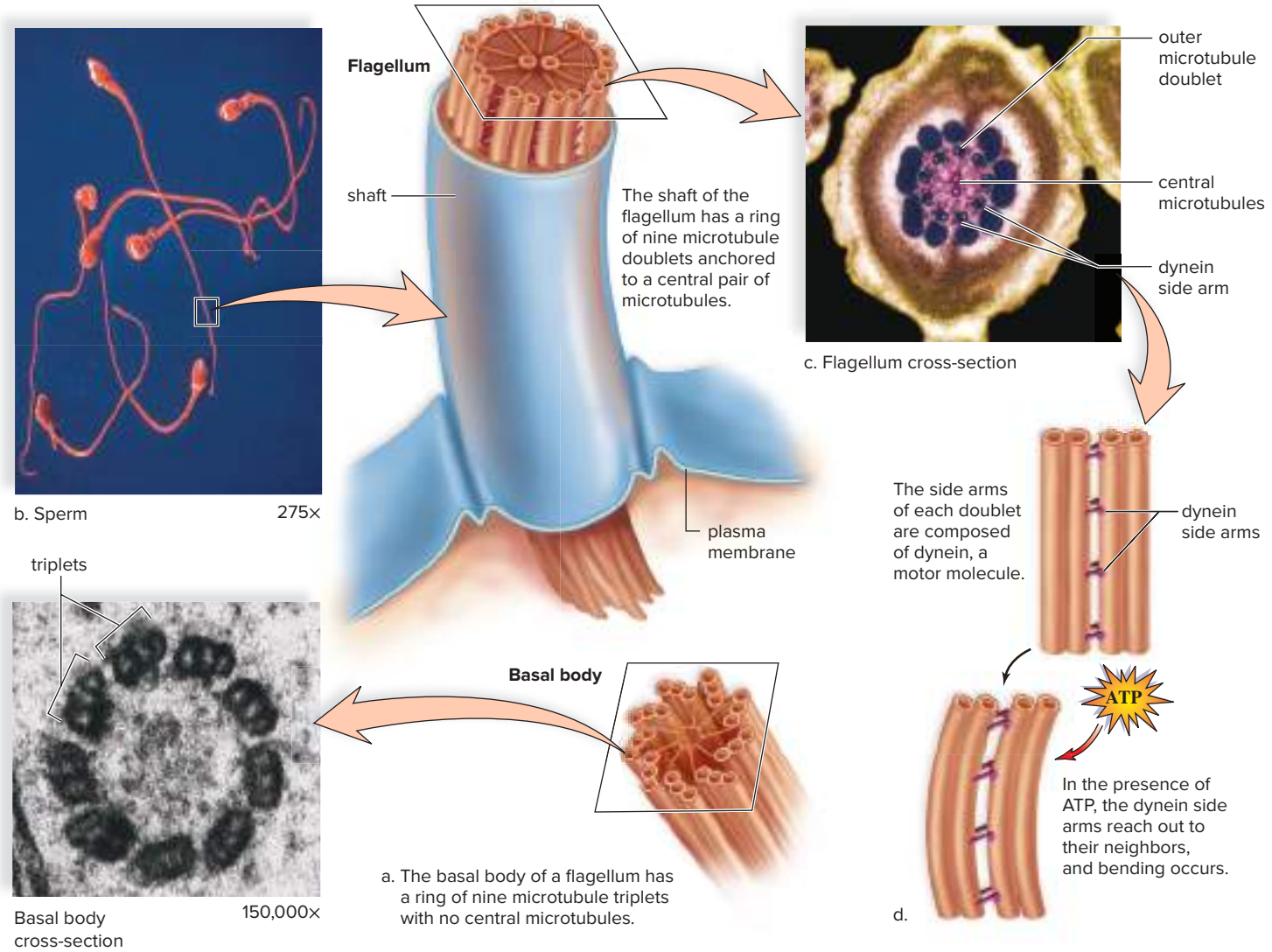
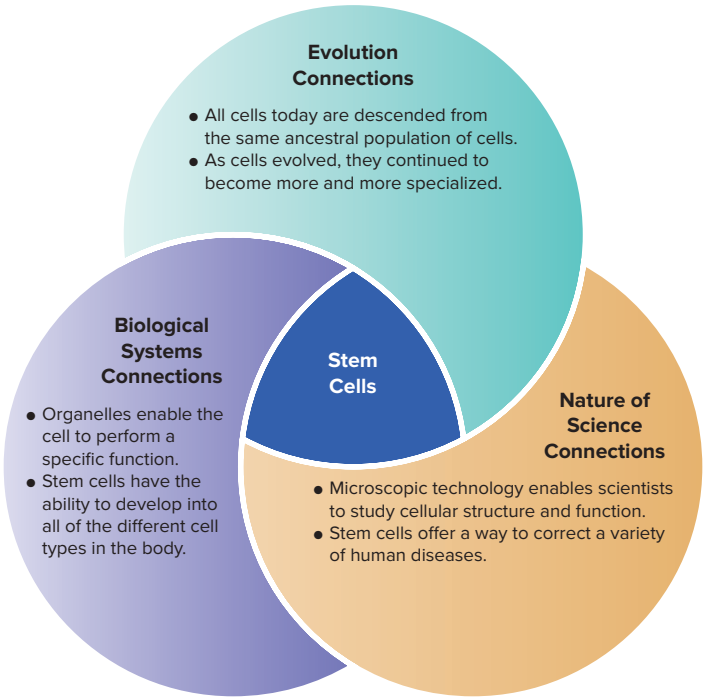


Figure 4.21 Structure of a flagellum. **a.** The basal body of a flagellum has a 9 + 0 pattern of microtubule triplets. Notice the ring of nine triplets, with no central microtubules. **b.** In sperm, the shaft of the flagellum has a 9 + 2 pattern (a ring of nine pairs of microtubules surrounds a central pair of microtubules). **c.** In place of the triplets seen in a basal body, a flagellum's outer pairs have side arms of dynein, a motor molecule. **d.** In the presence of ATP, the dynein side arms reach out and attempt to move along their neighboring pair. Because of the radial spokes connecting the pairs to the central microtubules and motor molecules, bending occurs. (photos): (a): Biophoto Associates/Science Source; (b): David M. Phillips/Science Source; (c): Steve Gschmeissner/Science Source

ISTUDY

Connecting *the* Concepts



What Did You Learn?

4.1 Cellular Level of Organization

The **cell theory** states that all organisms are composed of cells and that all cells come from preexisting cells. **Cells** are very small (measured in micrometers) and must remain small in order to have an adequate **surface-area-to-volume ratio.** The plasma membrane regulates exchange of materials between the cell interior and the external environment.

4.2 Prokaryotic Cells

Prokaryotic cells lack the nucleus of **eukaryotic cells.** Prokaryotes include bacteria and archaeans. A prokaryotic cell may be called a **bacillus, coccus, spirillum,** or **spirochete,** depending on its shape.

- The **cell envelope** of bacteria includes a **plasma membrane,** a **cell wall,** and an outer **glycocalyx,** also called a **capsule.** The **cytoplasm** contains **ribosomes, plasmids,** and a region called the **nucleoid,** where the DNA may be found. The cytoplasm of cyanobacteria also includes thylakoids. The external structures of a bacterium include the **flagella, fimbriae,** and **conjugation pili.**

4.3 Introduction to Eukaryotic Cells

Eukaryotic cells are much larger than prokaryotic cells, and they contain compartmentalized structures called **organelles,** each of which has a specific structure and function (Table 4.1) that increases cell efficiency. The **endosymbiotic theory** helps explain the evolutionary origins of many membrane-bound organelles. Most membranous organelles are in constant communication.

- Eukaryotic cells also contain a variety of transport **vesicles** that move around the cell using the proteins of the **cytoskeleton.**

4.4 The Nucleus and Ribosomes

The **nucleus** of eukaryotic cells is surrounded by a **nuclear envelope** containing **nuclear pores** that regulate transport between the cytoplasm and the **nucleoplasm.** The nucleus contains **chromatin** (proteins and nucleic acids) that is organized into **chromosomes. Genes** on the chromosomes contain specific instructions for traits.

Ribosomes are manufactured in the **nucleolus** of the nucleus. **Ribosomes** are organelles that function in protein synthesis. In order to make a protein, mRNA is copied exactly from the DNA, processed, and exits the nucleus through a nuclear pore. After a ribosome attaches to an mRNA, most of the time this assembly goes to the rough ER to make a protein. Ribosomes may be connected as groups, called **polyribosomes.**

4.5 The Endomembrane System

The **endomembrane system** includes the endoplasmic reticulum (ER), the Golgi apparatus, the lysosomes (in animal cells), and transport vesicles.

- Newly produced proteins made in the **rough ER** are modified before they are packaged in transport vesicles, many of which go to the Golgi apparatus.
- The **smooth ER** has various metabolic functions, depending on the cell type, but it generally makes lipids that are carried by vesicles to different locations, particularly the **Golgi apparatus.** The Golgi apparatus modifies, sorts, and repackages proteins, as well as processes lipids. Some proteins are tagged for transport to different cellular destinations; others are secreted from the cell.
- **Lysosomes** are produced by the Golgi apparatus and contain digestive enzymes that are involved in breaking down macromolecules and **autophagy.**

4.6 Microbodies and Vacuoles

Cells contain numerous vesicles, **microbodies,** and **vacuoles.** Microbodies are vesicles with specific metabolic functions. **Peroxisomes** are microbodies that are involved in the metabolism of long-chain fatty acids. The large **central vacuole** in plant cells functions in storage and in the breakdown of molecules and cell parts.

ISTUDY

Table 4.1 Comparison of Prokaryotic Cells and Eukaryotic Cells

		Eukaryotic Cells	
	Prokaryotic Cells	Animal	Plant
Cell wall	Usually (peptidoglycan)	No	Yes (cellulose)
Plasma membrane	Yes	Yes	Yes
Nucleus	No	Yes	Yes
Nucleolus	No	Yes	Yes
Ribosomes	Yes (smaller)	Yes	Yes
Endoplasmic reticulum	No	Yes	Yes
Golgi apparatus	No	Yes	Yes
Lysosomes	No	Yes	Not usually
Mitochondria	No	Yes	Yes
Chloroplasts	No	No	Yes
Peroxisomes	No	Usually	Usually
Cytoskeleton	No	Yes	Yes
Centrioles	No	Yes	No
9 + 2 cilia or flagella	No	Often	No (in flowering plants); Yes (sperm of bryophytes, ferns, and cycads)

4.7 The Energy-Related Organelles

Cells require a constant input of energy to maintain their structure.

- **Chloroplasts** are **plastids** that capture the energy of the sun and conduct photosynthesis, which produces carbohydrates. The internal structure of chloroplasts includes the **stroma** and **thylakoids.** Thylakoids are stacked as **grana** within the chloroplast.
- Carbohydrate-derived products are broken down in **mitochondria** in the presence of oxygen via cellular respiration, and ATP is produced as a result. The internal structure of a mitochondrion includes the **cristae** and the **matrix.**

4.8 The Cytoskeleton

The cytoskeleton contains actin filaments, intermediate filaments, and microtubules. These maintain cell shape and help transport organelles from place to place within the cell.

- **Actin filaments** interact with motor molecules to allow a range of functions from muscular contraction to cellular division.
- **Intermediate filaments** support the nuclear and plasma membranes and participate in the cell-to-cell junctions that produce tissues.
- **Microtubules** radiate out from the **centrosome** and are present in **centrioles, cilia,** and **flagella.** They serve as an internal transport system along which vesicles and other organelles move.

Thinking Critically About Biology

1. The protists that cause malaria contribute to infections associated with AIDS. Scientists have discovered that an antibiotic that inhibits prokaryotic enzymes will kill the parasite, because it is effective against the plastids in the protist. What can you conclude about the origin of the plastids?
2. In the 1958 movie *The Blob,* a giant, single-celled alien creeps and oozes around, attacking and devouring helpless humans. Why couldn't there be a real single-celled organism as large as the Blob?
3. Calculate the surface-area-to-volume ratio of a 1-mm cube and a 2-mm cube. Which has the smaller ratio?
4. Tay-Sachs disease is due to a faulty gene that does not allow the lysosome to produce the enzymes necessary for normal functioning. Suggest a cellular organelle that could, potentially, be genetically modified with the fewest complications to perform the same function as the lysosome.

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Cell Size

McGraw Hill

Why would cells along your digestive tract have a very high surface-area-to-volume ratio?

Making It Relevant

1. In your opinion, which cellular structure is the most important one in determining the cell's activity?
2. If your cells were to "eject" one cell organelle, which one do you think the cell could still function without?
3. Should scientists use genetic engineering to correct cells that are not functioning correctly?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

5

Membrane Structure and Function

Steve Heap/Shutterstock

Chapter Outline

5.1 Plasma Membrane Structure and Function

5.2 Passive Transport Across a Membrane

5.3 Active Transport Across a Membrane

5.4 Modification of Cell Surfaces

5.5 Cell-to-Cell Communication

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.3 How does the structure of a phospholipid make it an ideal molecule for the plasma membrane?

Section 3.4 How does a protein's shape relate to its function?

Section 4.3 What are the key features of eukaryotic cells?

The Opioid Crisis

Each day, approximately 130 people die from an opioid overdose in the United States. The addiction and misuse of opioids has become a national crisis that impacts the social and economic welfare of our country.

The opioid system within the human body controls pain, reward, and addictive behaviors. Opioid receptors belong to a group of transmembrane proteins. When the receptors are engaged by an opioid like morphine or heroine, the pain pathway is altered and feelings of euphoria and pleasure can result. Three main receptors—mu, delta, and kappa—are responsible for enabling opioids to exert their effect. Mu receptors are molecular switches that trigger the brain reward system. Delta receptors appear to play a role in determining an individual's emotional state, and kappa receptors are associated with pain relief. Morphine is the most widely used opioid to control pain after surgery or for individuals who suffer from chronic pain. Morphine acts on the mu receptor and therefore has the potential for abuse and addiction. A number of drugs may be prescribed to combat opioid abuse. These drugs bind to the opioid receptors and eliminate withdrawal symptoms.

In this chapter, we will explore the structure of the plasma membrane.

As you read through this chapter, think about the following questions:

1. How do transport and channel proteins function in a plasma membrane?
2. What type of transport are these channels performing?

Following *the* Themes

CHAPTER 5 – MEMBRANE STRUCTURE AND FUNCTION

Evolution	The plasma membrane and internal membrane-bound organelles have evolved to support specialized functions for cells.
Nature of Science	Research in membranes and associated proteins leads to new discoveries in medicine and disease treatment.
Biological Systems	Membranes play an important role in cellular communication and response to environmental stimuli.

ISTUDY

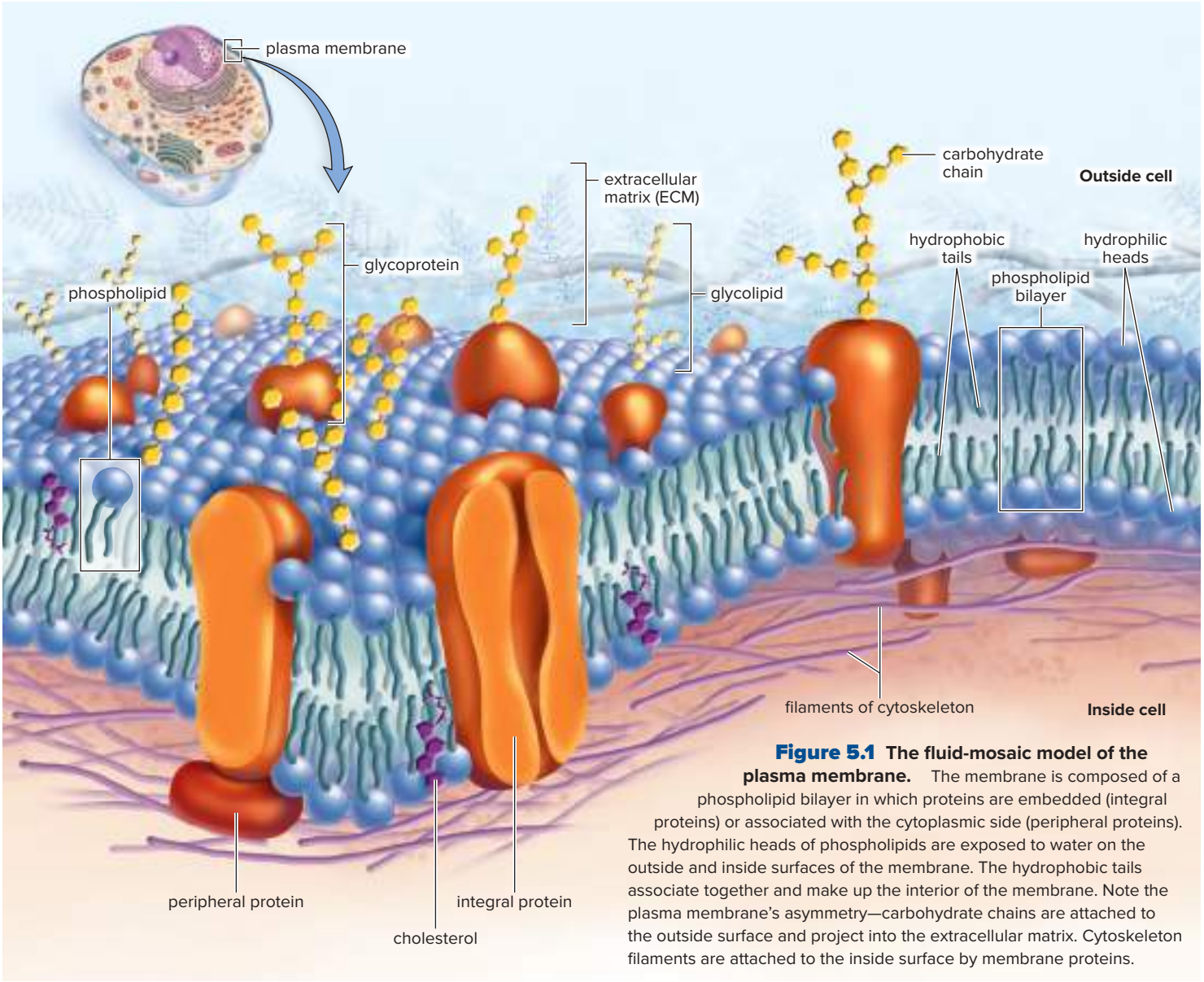


Figure 5.1 The fluid-mosaic model of the plasma membrane. The membrane is composed of a phospholipid bilayer in which proteins are embedded (integral proteins) or associated with the cytoplasmic side (peripheral proteins). The hydrophilic heads of phospholipids are exposed to water on the outside and inside surfaces of the membrane. The hydrophobic tails associate together and make up the interior of the membrane. Note the plasma membrane's asymmetry—carbohydrate chains are attached to the outside surface and project into the extracellular matrix. Cytoskeleton filaments are attached to the inside surface by membrane proteins.

5.1 Plasma Membrane Structure and Function

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between the different structural components of membranes.
2. Describe the nature of the fluid-mosaic model as it relates to membrane structure.
3. Describe the diverse role of proteins in membranes.
4. Explain why the plasma membrane exhibits selective permeability.

The ability to create compartments is a key feature of cells. Membranes, made of a phospholipid bilayer, create separation between the cell and the external environment, as well as compartments within the cell itself. Having separate compartments allows multiple, sometimes incompatible, chemical processes to occur simultaneously. This "division of labor" allows cells to operate more efficiently and respond to changing environmental conditions.

Components of the Plasma Membrane

The structure of a typical animal cell's plasma membrane is depicted in Figure 5.1. In addition to the phospholipid bilayer, membrane components include protein molecules that are either partially or wholly embedded in the bilayer. Cholesterol is another lipid found in the animal plasma membrane; related steroids are found in the plasma membrane of plants. As we will see, cholesterol helps modify the fluidity of the membrane over a range of temperatures.

Phospholipids are considered *amphipathic molecules* because they possess both a hydrophilic (water-loving) region and a hydrophobic (water-fearing) region. The amphipathic nature of phospholipids largely explains why they form a bilayer in water (Fig. 5.2). Because similar substances associate with one another,

ISTUDY

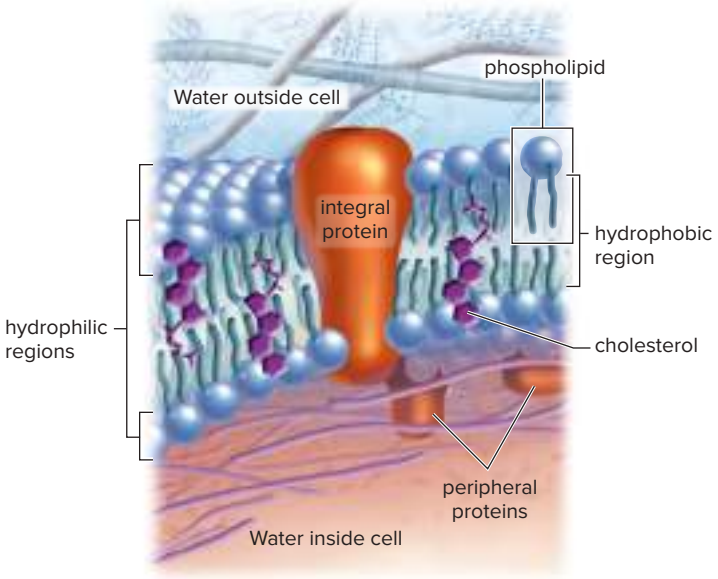


Figure 5.2 Phospholipids in the plasma membrane. The hydrophilic heads of the phospholipids are at the surfaces of the membrane; the hydrophobic tails make up the interior of the membrane. Proteins embedded in the membrane serve a variety of functions.

the hydrophilic polar heads of the phospholipid molecules naturally associate with the polar water molecules found on the outside and inside of the membrane. Likewise, the hydrophobic nonpolar tails associate with each other because they want to "get away" from the polar water.

Cell membranes are highly similar in the types of molecules they contain, which makes them interchangeable and allows them to fuse together fairly easily. What makes one membrane different from another are the types of proteins integrated into the membrane. As shown in Figure 5.1, proteins are scattered throughout the membrane in an irregular pattern, and this pattern can vary from membrane to membrane.

Electron micrographs can be used to study the nature of many membrane proteins. A research method called *freeze-fracture* first freezes, and then splits, the membrane so the upper and lower layers separate. The proteins remain intact and go with one of the layers. Proteins embedded in the plasma membrane are called *integral proteins,* whereas proteins that occur only on the cytoplasmic side of the membrane are called *peripheral proteins*.

Some integral proteins protrude from only one surface of the bilayer, but most span the membrane, with a hydrophobic core region that associates with the nonpolar core of the membrane. Hydrophilic ends of integral proteins protrude from both surfaces of the bilayer, interacting with polar water molecules. Integral proteins can be held in place by attachments to protein fibers of the cytoskeleton (inside) and fibers of the extracellular matrix (outside). Only animal cells have an extracellular matrix (ECM), which contains various protein fibers and very large, complex carbohydrate molecules. The ECM, which is discussed in greater detail in Section 5.4, has a number of functions, from lending external support to the plasma membrane to assisting in communication between cells.

The Fluid-Mosaic Model

Membranes are not rigid but rather flexible structures. They consist of a variety of molecules, including phospholipids, cholesterol, and proteins. The **fluid-mosaic model** is used to describe the interactions of these membrane components.

The lipid content of the membrane is responsible for its fluidity. Cells are flexible because the phospholipid bilayer is fluid. At body temperature, the phospholipid bilayer of the plasma membrane has the consistency of olive oil. The greater the concentration of unsaturated fatty acid residues, the more fluid the bilayer. In each monolayer, the fatty acid tails jostle around, and an entire phospholipid molecule can move sideways at a rate averaging about 2 μm—the length of a prokaryotic cell—per second. Although it is possible for phospholipid molecules to flip-flop from one monolayer to the other, they rarely do so, because this would require the hydrophilic head to move through the hydrophobic center of the membrane. At times, however, special proteins help the phospholipids flip.

The presence of cholesterol molecules prevents the plasma membrane from becoming too fluid at higher temperatures and too solid at lower temperatures. At higher temperatures, cholesterol stiffens the membrane and makes it less fluid than it would otherwise be. At lower temperatures, cholesterol helps prevent the membrane from freezing by not allowing contact between certain phospholipid tails.

A plasma membrane is considered a mosaic because of the presence of many proteins. The number and kinds of proteins can vary in the plasma membrane and in the membranes of the various organelles. The position of these proteins can shift over time, unless they are anchored to another structure, such as the cytoskeleton. Experiments have been conducted in which the proteins were tagged prior to allowing mouse and human cells to fuse. An hour after fusion, the proteins from each cell type were completely mixed, suggesting that at least some proteins are able to move sideways in the membrane.

Scientists once thought that all membrane proteins could freely move sideways within the fluid bilayer. Today, however, we know that membrane proteins are often associated with the ECM, the cytoskeleton, or both. These connections hold a protein in place and partially anchor the otherwise fluid phospholipid bilayer.

It should be noted that the two sides of the membrane are not identical. Carbohydrate chains (see below) are attached only to molecules on the outside surface, and peripheral proteins occur on one surface or the other. Thus, the membrane is said to be asymmetrical.

Glycoproteins and Glycolipids

Phospholipids and proteins that have attached carbohydrate (sugar) chains are called **glycolipids** and **glycoproteins,** respectively. The carbohydrate (sugar) chains on a cell's exterior can be highly diverse. The chains can vary in the number and sequence of sugars, and in whether the chain is branched. Each cell within an individual has its own "fingerprint" because of these chains. For this reason, glycolipids and glycoproteins play an important role in cellular identification. As you probably know, transplanted tissues are often rejected by the recipient. Rejection occurs because the immune

ISTUDY

system is able to detect that the foreign tissue's cells do not have the appropriate carbohydrate chains to be recognized as self. In humans, carbohydrate chains are also the basis for the A, B, and O blood groups.

In animal cells, the carbohydrate chains attached to proteins give the cell a "sugar coat," more properly called a *glycocalyx*. The glycocalyx protects the cell and has various other functions, including cell-to-cell adhesion, reception of signaling molecules, and cell-to-cell recognition.

The Functions of the Proteins

Although the protein components of plasma membranes differ depending on the type of cell and the processes it is undergoing, several types of proteins are likely to be routinely present, such as the following.

Channel proteins. Channel proteins are involved in passing molecules through the membrane. They form a channel that allows a substance to simply move from one side to the other (Fig. 5.3*a*). For example, a channel protein allows hydrogen ions to flow across the inner mitochondrial membrane. Without this movement of hydrogen ions, ATP would never be produced.

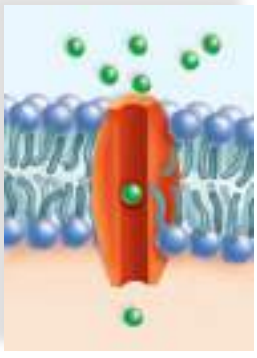
Carrier proteins. Carrier proteins are also involved in passing molecules through the membrane. They receive a substance and change their shape, and this change moves the substance across the membrane (Fig. 5.3*b*). A carrier protein transports sodium and potassium ions across the plasma membrane of a nerve cell. Without this carrier protein, nerve impulse conduction would be impossible.

Cell recognition proteins. Cell recognition proteins are glycoproteins (Fig. 5.3*c*). Among other functions, these proteins help the body recognize when it is being invaded by pathogens, so an immune response can occur. Without this recognition, pathogens would be able to freely invade the body and hinder its function.

Receptor proteins. Receptor proteins have a shape that allows only a specific molecule to bind to it (Fig. 5.3*d*). The binding of this molecule causes the protein to change its shape and thereby bring about a cellular response. The coordination of the body's organs is totally dependent on such signaling molecules. For example, the liver stores glucose after it is signaled to do so by insulin. The opioid receptors discussed in the chapter opener are an example of a receptor protein.

Enzymatic proteins. Some plasma membrane proteins are enzymes that carry out metabolic reactions directly (Fig. 5.3*e*). Without these enzymes, some of which are attached to the various membranes of the cell, a cell would never be able to perform the chemical reactions needed to maintain its metabolism.

Junction proteins. Proteins are involved in forming various types of junctions between animal cells (Fig. 5.3*f*). Signaling molecules that pass through gap junctions allow the cilia of cells that line the respiratory tract to beat in unison.

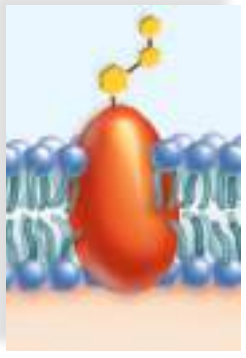


Channel Protein: Allows a particular molecule or ion to cross the plasma membrane freely. Cystic fibrosis, an inherited disorder, is caused by a faulty chloride (Cl^-) channel; a thick mucus collects in airways and in pancreatic and liver ducts.

a.

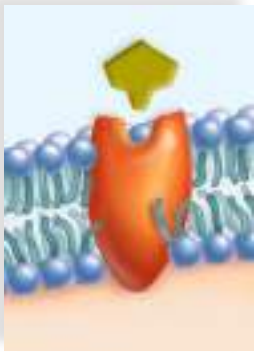
Carrier Protein: Selectively interacts with a specific molecule or ion so that it can cross the plasma membrane. Glucose transporters move glucose in and out of the various cell types in the body. The inability of some persons to use energy for sodium–potassium (Na^+-K^+) transport has been suggested as the cause of their obesity.

b.



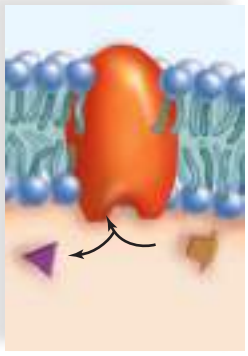
Cell Recognition Protein: The MHC (major histocompatibility complex) glycoproteins are different for each person, so organ transplants are difficult to achieve. Cells with foreign MHC glycoproteins are attacked by white blood cells responsible for immunity.

c.



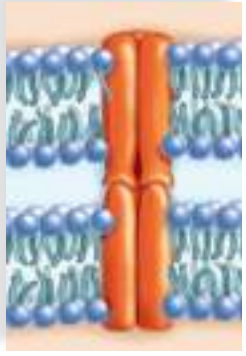
Receptor Protein: Is shaped in such a way that a specific molecule can bind to it. Some forms of dwarfism result not because the body does not produce enough growth hormone, but because their plasma membrane growth hormone receptors are faulty and cannot interact with growth hormone.

d.



Enzymatic Protein: Catalyzes a specific reaction. The membrane protein, adenylate cyclase, is involved in ATP metabolism. Cholera bacteria release a toxin that interferes with the proper functioning of adenylate cyclase; sodium (Na^+) and water leave intestinal cells, and the individual may die from severe diarrhea.

e.



Junction Proteins: Tight junctions join cells so that a tissue can fulfill a function, as when a tissue pinches off the neural tube during development. Without this cooperation between cells, an animal embryo would have no nervous system.

f.

Figure 5.3 Membrane protein diversity. These are some of the functions performed by proteins found in the plasma membrane.

ISTUDY

Table 5.1 Passage of Molecules into and out of the Cell

Name	Direction	Requirement	Examples
Diffusion	Toward lower concentration	Concentration gradient	Lipid-soluble molecules, gases
Facilitated transport	Toward lower concentration	Channels or carrier and concentration gradient	Some sugars, amino acids
Active transport	Toward higher concentration	Carrier plus energy	Sugars, amino acids, ions
Bulk transport	Toward outside or inside	Vesicle utilization	Macromolecules

Permeability of the Plasma Membrane

The plasma membrane regulates the passage of molecules into and out of the cell. This function is critical because the cell must maintain its normal composition under changing environmental conditions. The plasma membrane is essential because it is **selectively permeable,** allowing only certain substances into the cell while keeping others out.

Molecules that can freely cross a membrane generally require no energy to do so. Substances that are hydrophobic and therefore similar to the phospholipid center of the membrane are able to diffuse across membranes at no energy cost. Polar molecules, however, are chemically incompatible with the center of the membrane and so require an expenditure of energy to drive their transport.

Table 5.1 and Figure 5.4 examine which types of molecules can passively cross a membrane (no energy required) and which may require transport by a carrier protein and/or an expenditure of energy. In general, small, noncharged molecules, such as carbon dioxide, oxygen, glycerol, and alcohol, can freely cross the membrane. They are able to slip between the hydrophilic heads of the phospholipids and pass through the hydrophobic tails of the membrane because they are similarly nonpolar.

These molecules follow their **concentration gradient** as they move from an area where their concentration is high to an area where their concentration is low. Consider that a cell is always using oxygen when it carries on cellular respiration. The internal consumption of oxygen results in a low cellular concentration. Because oxygen concentration is higher outside than inside the cell, oxygen tends to move across the membrane into the cell. The concentration of carbon dioxide, on the other hand, is highest inside the cell, because it is produced during cellular respiration. Therefore, carbon dioxide tends to move with its concentration gradient from inside to outside the cell.

Water, a polar molecule, would not be expected to readily cross the primarily nonpolar membrane. However, scientists have discovered that the majority of cells have channel proteins, called **aquaporins,** that allow water to cross a membrane more quickly than expected. Aquaporins also allow cells to equalize water pressure differences between their interior and exterior environments, so that their membranes don't burst from environmental pressure changes.

Ions and polar molecules, such as glucose and amino acids, can slowly cross a membrane. To move as quickly as is necessary, they are often assisted across the plasma membrane by carrier proteins. Each carrier protein recognizes particular shapes of molecules and must combine with an ion, such as sodium (Na^+), or a molecule, such as glucose, before changing its shape and transporting the molecule across the membrane. Therefore, carrier proteins are specific for the substances they transport across the plasma membrane.

Bulk transport is a way that large particles can exit or enter a cell. During exocytosis, fusion of a vesicle with the plasma membrane moves a particle to outside the membrane. During endocytosis, vesicle formation moves a particle to inside the plasma membrane. Vesicle formation is reserved for the movement of macromolecules or even for something larger, such as a virus. As with many other processes, a cell is selective about what enters by endocytosis.

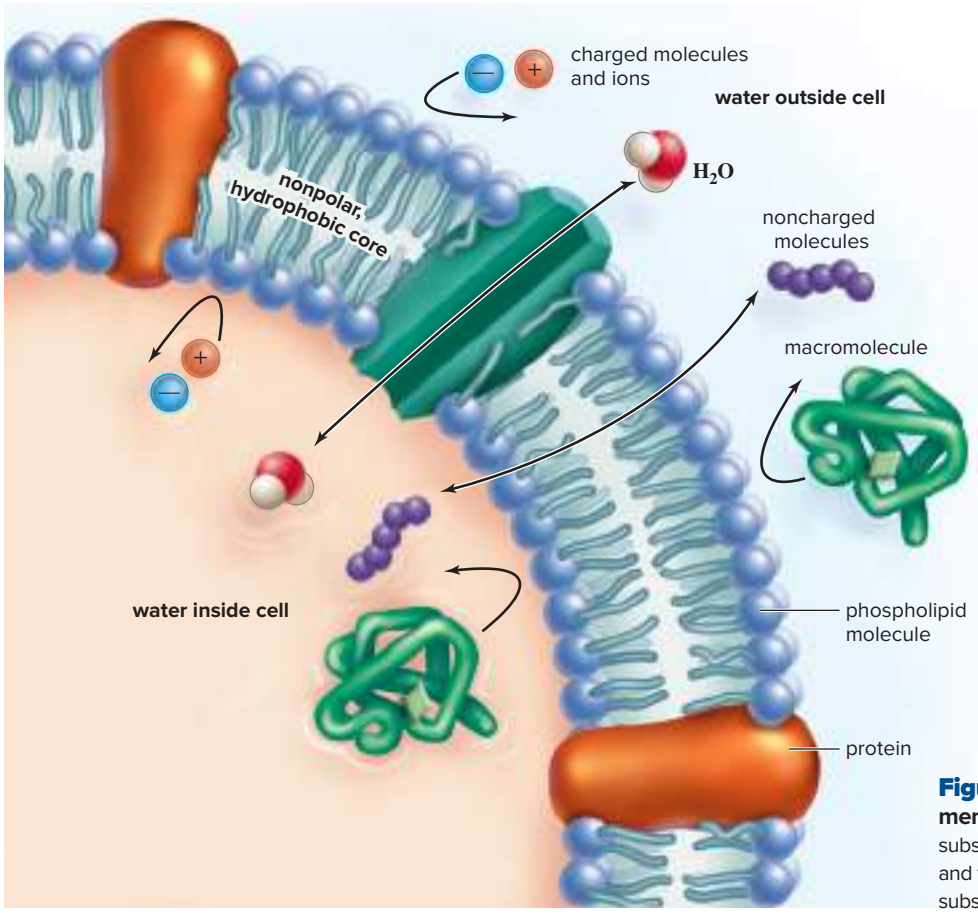


Figure 5.4 How molecules cross the plasma membrane. The curved arrows indicate that these substances cannot passively cross the plasma membrane, and the long, back-and-forth arrows indicate that these substances can diffuse across the plasma membrane.

ISTUDY

Check Your Progress 5.1

1. Explain why phospholipids play such an important role in the structure of the cell membrane.
2. Describe the role of proteins in the fluid-mosaic model.
3. Compare how cells transport polar and nonpolar molecules across a membrane.

5.2 Passive Transport Across a Membrane

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare diffusion and osmosis across a membrane.
2. Describe the role of proteins in the movement of molecules across a membrane.
3. Differentiate among the effects of hypotonic, isotonic, and hypertonic solutions on animal and plant cells.

Diffusion is the movement of molecules from a higher to a lower concentration—that is, down their concentration gradient—until equilibrium is achieved and the molecules are distributed equally. Diffusion is a physical process that results from the random molecular motion that can be observed with any type of molecule. For example, when a crystal of dye is placed in water (Fig. 5.5), the dye and water molecules move in various directions, but their net movement, which is the sum of their motion, is toward the region of lower concentration. Eventually, the dye is dissolved evenly in the water, resulting in equilibrium and a uniformly colored solution.

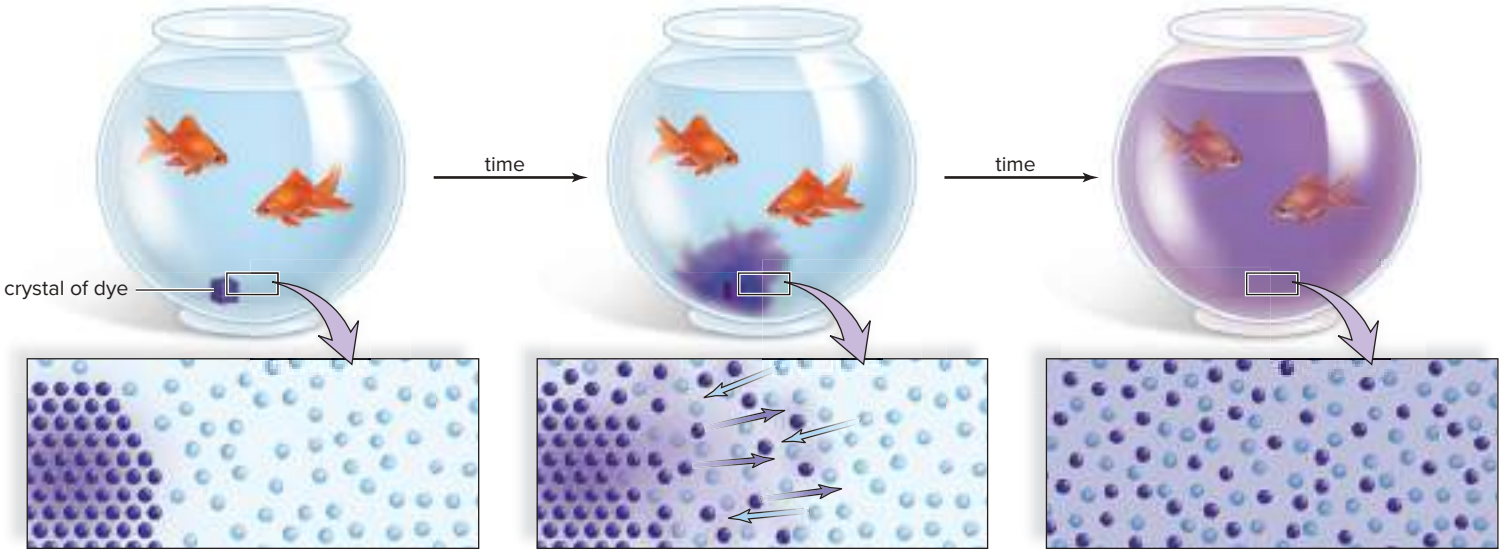
A **solution** contains both a **solute,** usually a solid, and a **solvent,** usually a liquid. In this case, the solute is the dye and the solvent is the water molecules. Once the solute and solvent are evenly distributed, they continue to move about, but there is no net movement of either one in any direction.

The chemical and physical properties of the plasma membrane allow only a few types of molecules to enter and exit a cell simply by diffusion. Gases can freely diffuse through the lipid bilayer because they are small and nonpolar; this is the mechanism by which oxygen enters cells and carbon dioxide exits cells. This is also how oxygen diffuses from the alveoli (air sacs) of the lungs into the blood in the lung capillaries (Fig. 5.6). After inhalation (breathing in), the concentration of oxygen in the alveoli is higher than that in the blood; therefore, oxygen diffuses into the blood along its concentration gradient.

Several factors influence the rate of diffusion, including temperature, pressure, electrical currents, and molecular size. For example, as temperature increases, the rate of diffusion increases. The movement of fishes in the bowl would also speed the rate of diffusion (see Fig. 5.5).

Osmosis

The diffusion of water across a selectively permeable membrane from high to low concentration is called **osmosis.** To illustrate osmosis, a thistle tube is used containing a 10% solute solution. This percentage is based on the number of grams of solute per 100 ml of solvent. In this case, 10 g of sugar would be added to the water to bring the total volume to 100 ml. The thistle tube is then covered at one end by a selectively permeable membrane and placed in a beaker containing a 5% solute solution (Fig. 5.7*a*). The beaker has a higher concentration of water molecules (lower percentage of solute), and the thistle tube has a lower concentration of water molecules (higher percentage of solute). Diffusion always occurs from higher to lower concentration. Therefore, a net movement of water takes place, moving across the membrane from the beaker to the inside of the thistle tube (Fig. 5.7*b*).



a. Crystal of dye is placed in the water. b. Diffusion of water and dye molecules c. Equal distribution of molecules results.

Figure 5.5 The process of diffusion. Diffusion is spontaneous, and no chemical energy is required to bring it about. **a.** When a dye crystal is placed in water, it is concentrated in one area. **b.** The dye dissolves in the water, and over time a net movement of dye molecules from a higher to a lower concentration occurs. There is also a net movement of water molecules from a higher to a lower concentration. **c.** Eventually, the water and the dye molecules are equally distributed throughout the container.

ISTUDY

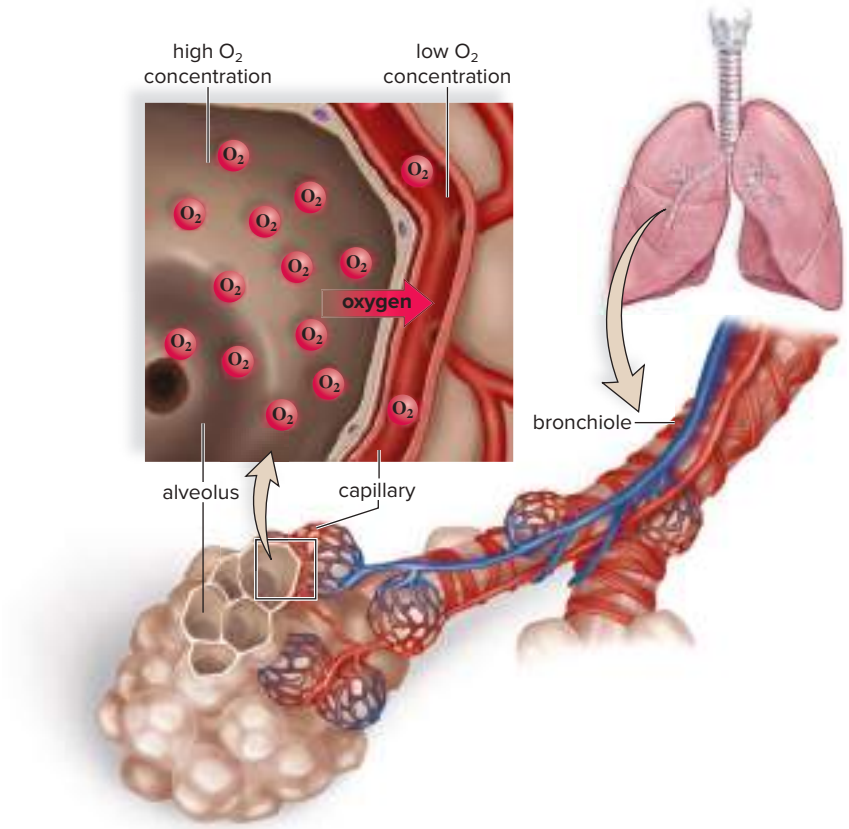


Figure 5.6 Gas exchange in lungs. Oxygen (O_2) diffuses into the capillaries of the lungs because there is a higher concentration of oxygen in the alveoli (air sacs) than in the capillaries.

The solute does not diffuse out of the thistle tube. Why not? Because the membrane is not permeable to the solute. As water enters and the solute does not exit, the level of the solution within the thistle tube rises (Fig. 5.7*c*). In the end, the concentration of solute in the thistle tube is less than 10%. Why? Because there is now less solute per unit volume. And the concentration of solute in the beaker is greater than 5%, because there is now more solute per unit volume.

Water enters the thistle tube due to the osmotic pressure of the solution within the thistle tube until it reaches equilibrium. **Osmotic pressure** is the pressure that develops in a system due to osmosis. In other words, the greater the possible osmotic pressure, the more likely it is that water will diffuse in that direction. Due to osmotic pressure, water is absorbed by the kidneys and taken up by capillaries in the tissues. Osmosis also occurs across the plasma membrane, as we'll see next.

Isotonic Solution

In the laboratory, cells are normally placed in **isotonic solutions.** The prefix *iso* means "the same as," and the term **tonicity** refers to the strength of the solution. In an isotonic solution, the solute concentration and the water concentration both inside and outside the cell are equal (Fig. 5.8), and therefore there is no net gain or loss of water. A 0.9% solution of the salt sodium chloride (NaCl) is known to be isotonic to red blood cells. Therefore, intravenous solutions medically administered usually have this tonicity. Terrestrial animals can usually take in either water or salt as needed to maintain the

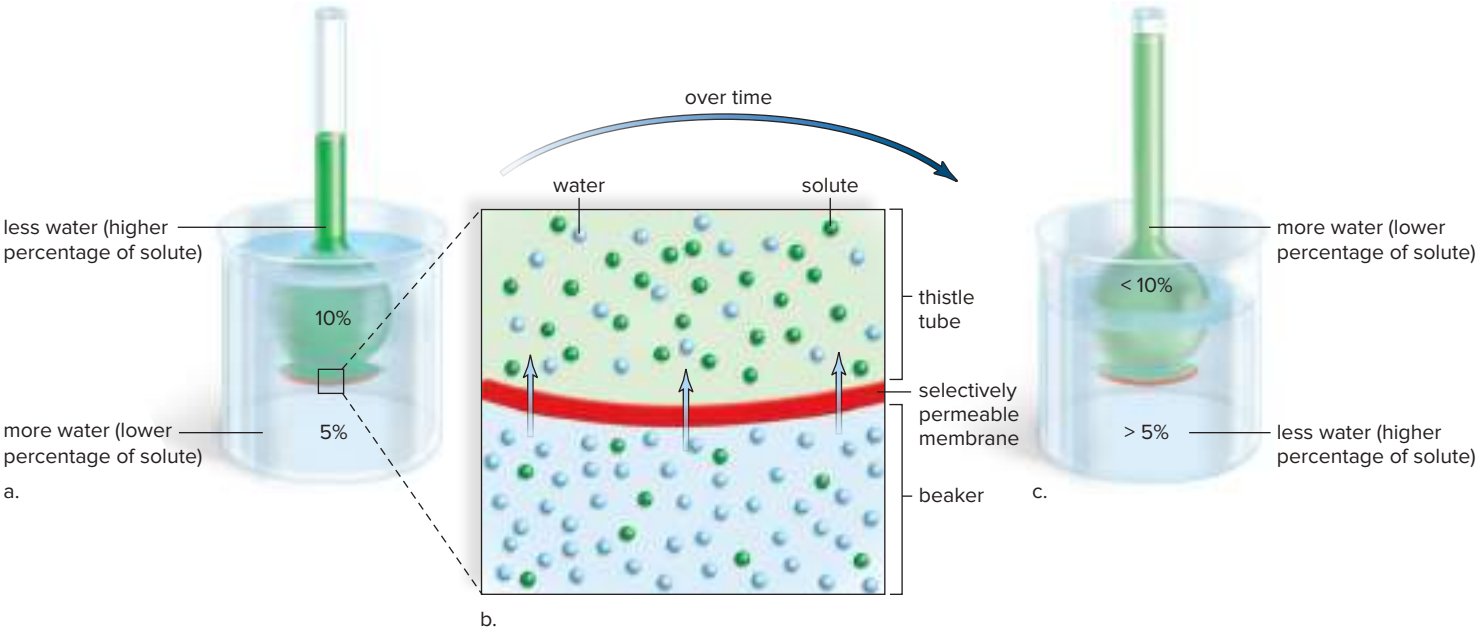


Figure 5.7 Osmosis demonstration. **a.** A thistle tube, covered at the broad end by a selectively permeable membrane, contains a 10% solute solution. The beaker contains a 5% solute solution. **b.** The solute (green circles) is unable to pass through the membrane, but the water (blue circles) passes through in both directions. There is a net movement of water toward the inside of the thistle tube, where there is a lower percentage of water molecules. **c.** Due to the incoming water molecules, the level of the solution rises in the thistle tube. Eventually, the concentration of water across the membrane equalizes.

ISTUDY

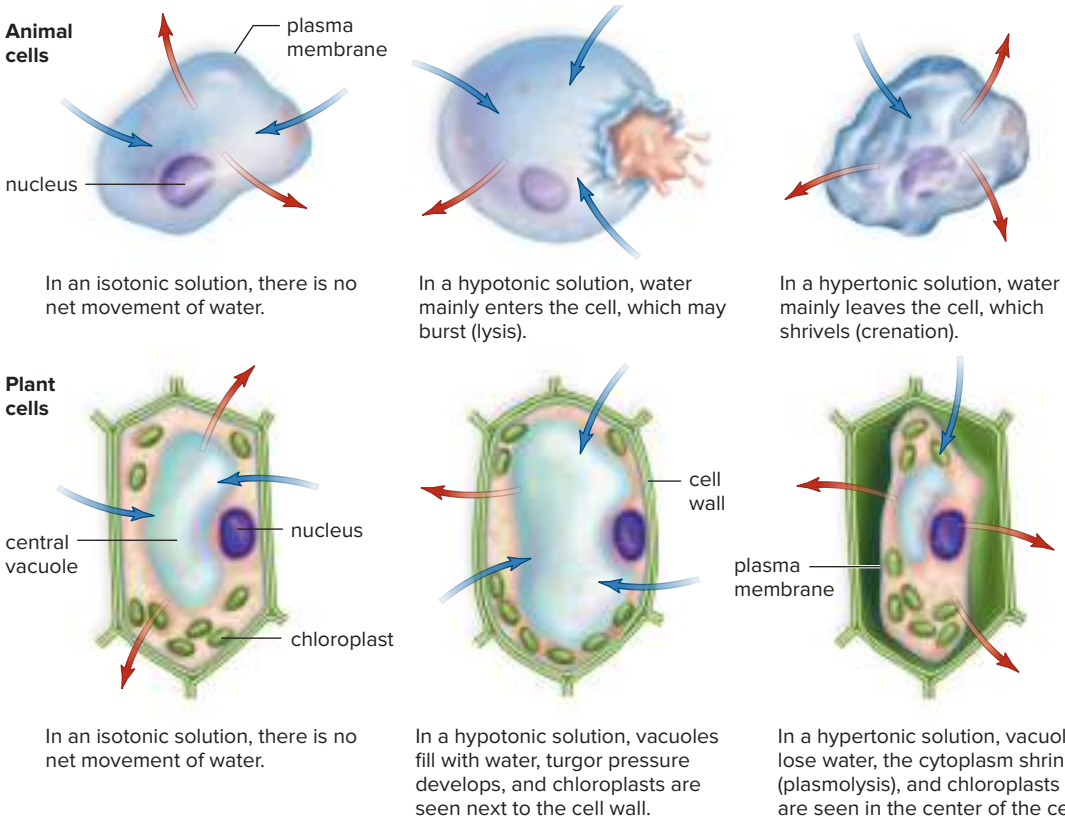


Figure 5.8 Osmosis in animal and plant cells. The arrows indicate the net movement of water molecules. To determine the net movement of water, compare the number of blue arrows (which are taking water molecules into the cell) versus the number of red arrows (which are taking water out of the cell). In an isotonic solution, a cell neither gains nor loses water; in a hypotonic solution, a cell gains water; and in a hypertonic solution, a cell loses water.

tonicity of their internal environment. Many animals living in an estuary, such as oysters, blue crabs, and some fishes, are able to cope with changes in the salinity (salt concentrations) of their environment using specialized kidneys, gills, and other structures.

Hypotonic Solution

Solutions that cause cells to swell, or even to burst, due to an intake of water are said to be **hypotonic solutions.** The prefix *hypo* means "less than" and refers to a solution with a lower concentration of solute (higher concentration of water) than inside the cell. If a cell is placed in a hypotonic solution, water enters the cell, because the lower cellular concentration of water prompts a net movement of water from the outside to the inside of the cell (Fig. 5.8).

Any concentration of a salt solution lower than 0.9% is hypotonic to red blood cells. Animal cells placed in such a solution expand and sometimes burst because of the buildup of pressure. The term *cytolysis* is used to refer to cells that have been disrupted. **Hemolysis** is the term used to describe cytolysis in red blood cells.

The swelling of a plant cell in a hypotonic solution creates **turgor pressure.** When a plant cell is placed in a hypotonic solution, the cytoplasm expands, because the large central vacuole gains water and the plasma membrane pushes against the rigid cell wall. Unlike animal cells that have no cell wall, the plant cell does not burst, because the cell wall does not give way. Turgor pressure in plant cells is extremely important to the maintenance of the plant's erect position. If you forget to water your plants, they wilt due to decreased turgor pressure.

Organisms that live in fresh water have to avoid taking in too much water. Many protozoans, such as paramecia, have contractile vacuoles that rid the body of excess water. Freshwater fishes have well-developed kidneys that excrete a large volume of dilute urine. These fish still have to take in salts through their gills. Even though freshwater fishes are good osmoregulators, they would not be able to survive in either distilled water or a salty marine environment.

Hypertonic Solution

Solutions that cause cells to shrink or shrivel due to loss of water are said to be **hypertonic solutions.** The prefix *hyper* means "more than" and refers to a solution with a higher percentage of solute (lower concentration of water) outside the cell. If a cell is placed in a hypertonic solution, water leaves the cell; the net movement of water is from the inside to the outside of the cell (Fig. 5.8).

Any concentration of a salt solution higher than 0.9% is hypertonic to red blood cells. If animal cells are placed in this solution, they shrink. The term **crenation** refers to red blood cells in this condition. Meats are sometimes preserved by salting them. The bacteria are not killed by the salt but by the lack of water in the meat.

When a plant cell is placed in a hypertonic solution, the plasma membrane pulls away from the cell wall as the large central vacuole loses water. This is an example of **plasmolysis,** a shrinking of the cytoplasm due to osmosis. The dead plants you may see along a salted roadside died because they were exposed to a hypertonic solution during the winter. Also, when salt water invades coastal marshes due to storms and human activities, coastal plants die. Without roots to hold the soil, they wash into the sea, doing away with many acres of valuable wetlands.

Marine animals cope with their hypertonic environment in various ways that prevent them from losing excess water to the environment. Sharks increase or decrease urea in their blood until their blood is isotonic with the environment and, in this way, do not lose too much water. Marine fishes and other types of animals drink no water but excrete salts across their gills. When a marine turtle "cries," it is ridding its body of salt by means of glands near the eye.

ISTUDY

Facilitated Transport

The plasma membrane impedes the passage of all but a few substances. Yet biologically useful molecules are able to rapidly enter and exit the cell either by way of a channel protein or because of carrier proteins in the membrane. These transport proteins are specific; each can transport only a certain type of molecule or ion across the membrane. How carrier proteins function is not completely understood, but after a carrier combines with a molecule, the carrier is believed to undergo a conformational change in shape that moves the molecule across the membrane. Carrier proteins are utilized for both facilitated transport (movement with concentration gradient; requires no energy) and active transport (movement against concentration gradient; requires energy) (see Table 5.1).

Facilitated transport explains how molecules such as glucose and amino acids are rapidly transported across the plasma membrane. Water moves through the membrane by using a channel protein. Glucose and amino acids move through the membrane by combining with specific carrier proteins, which then transport them through the plasma membrane. For example, various sugar molecules of identical size might be present inside or outside the cell, but glucose can cross the membrane hundreds of times faster than the other sugars. This is an example of why the membrane can be called selectively permeable.

A model for facilitated transport (Fig. 5.9) shows that after a carrier has assisted the movement of a molecule to the other side of the membrane, it is free to assist the passage of other solute molecules. Neither diffusion nor facilitated transport requires an expenditure of energy, because the molecules are moving down their concentration gradient.

Check Your Progress 5.2

1. Explain why both osmosis and diffusion are passive processes.
2. Describe how a cell would react to a hypertonic or hypotonic solution.
3. Contrast diffusion with facilitated transport.

5.3 Active Transport Across a Membrane

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how active transport moves substances across a membrane.
2. Compare the energy requirements of passive and active transport.
3. Contrast the bulk transport of large and small substances into a cell.

At times, a cell may need to move additional solutes across a membrane in order to do more work, which requires it to go against the concentration gradient. The process of **active transport** moves molecules against their concentration gradient. Active transport requires energy, usually in the form of ATP. For example, iodine collects in the cells of the thyroid gland; glucose is completely absorbed from the gut by the cells lining the digestive tract; and sodium can be almost completely withdrawn from urine by cells lining the kidney tubules. In each of these instances, molecules move from a lower to a higher concentration, exactly opposite the process of diffusion.

Carrier proteins and an expenditure of energy (ATP) are both needed to transport molecules against their concentration gradient. In this case, the ATP is needed for the carrier to combine with the substance to be transported. Therefore, it is not surprising that cells involved primarily in active transport, such as kidney cells, have a large number of mitochondria near membranes where active transport is occurring.

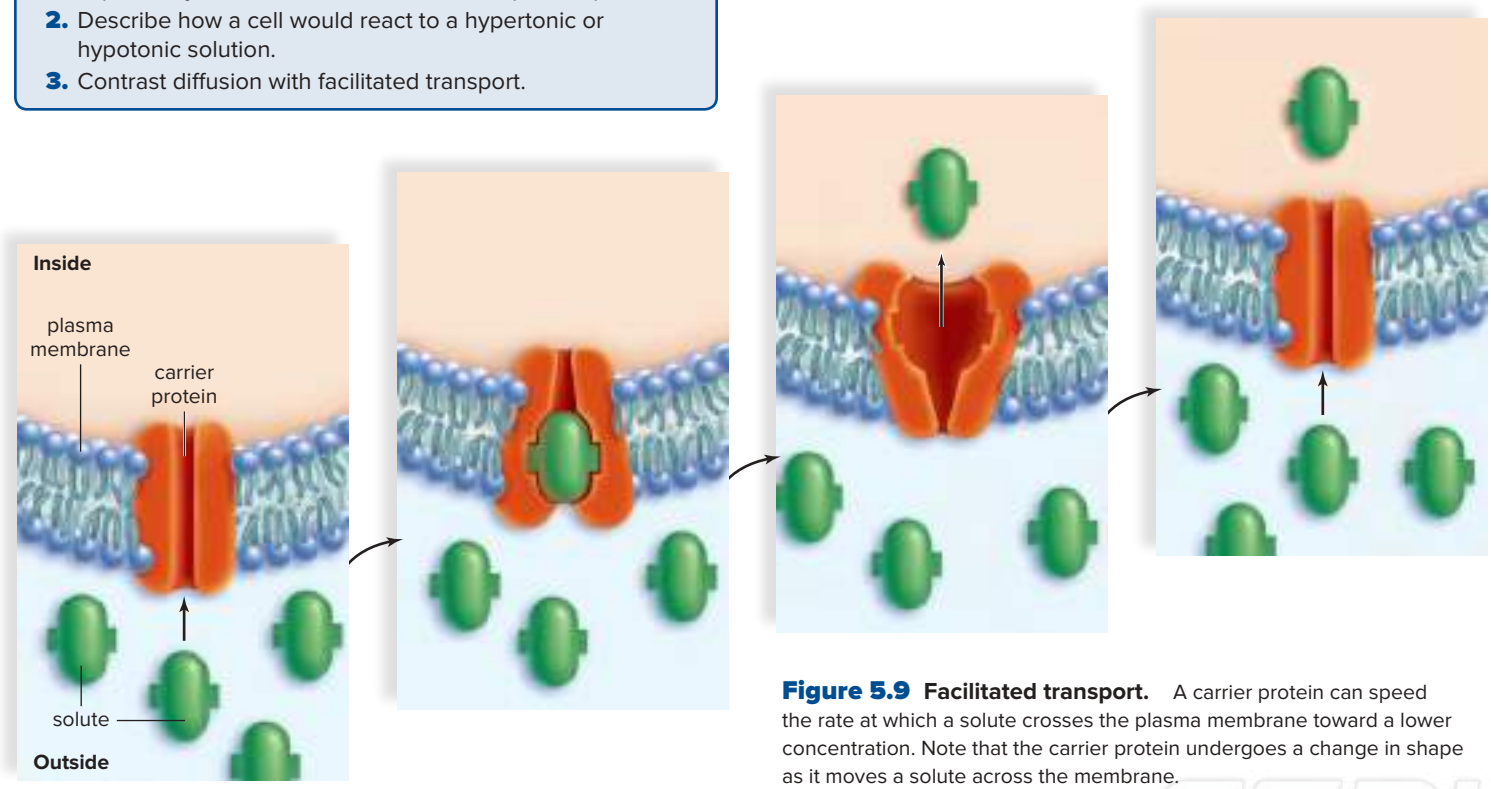


Figure 5.9 Facilitated transport. A carrier protein can speed the rate at which a solute crosses the plasma membrane toward a lower concentration. Note that the carrier protein undergoes a change in shape as it moves a solute across the membrane.

ISTUDY

Proteins involved in active transport often are called pumps because, just as a water pump uses energy to move water against the force of gravity, proteins use energy to move a substance against its concentration gradient. One type of pump that is active in all animal cells, but is especially associated with nerve and muscle cells, moves sodium ions (Na^+) to the outside of the cell and potassium ions (K^+) to the inside of the cell. The transport of sodium and potassium are linked together through the same carrier protein, called a **sodium-potassium pump**.

The sodium-potassium carrier protein has an initial shape that allows it to bind three sodium ions. Phosphate from an ATP molecule is added to the carrier protein, and it changes shape. This shape change moves sodium across the membrane. The new shape is no longer compatible with binding to the sodium, which falls away.

The new shape, however, is compatible with picking up two potassium ions, which bind to their sites. As the phosphate that was added from ATP in an earlier step leaves, the carrier protein assumes its original shape, and the two potassium ions are released inside the cell (Fig. 5.10). The cotransport of three sodium and two potassium creates not only a solute gradient but also an electrical gradient across the plasma membrane.

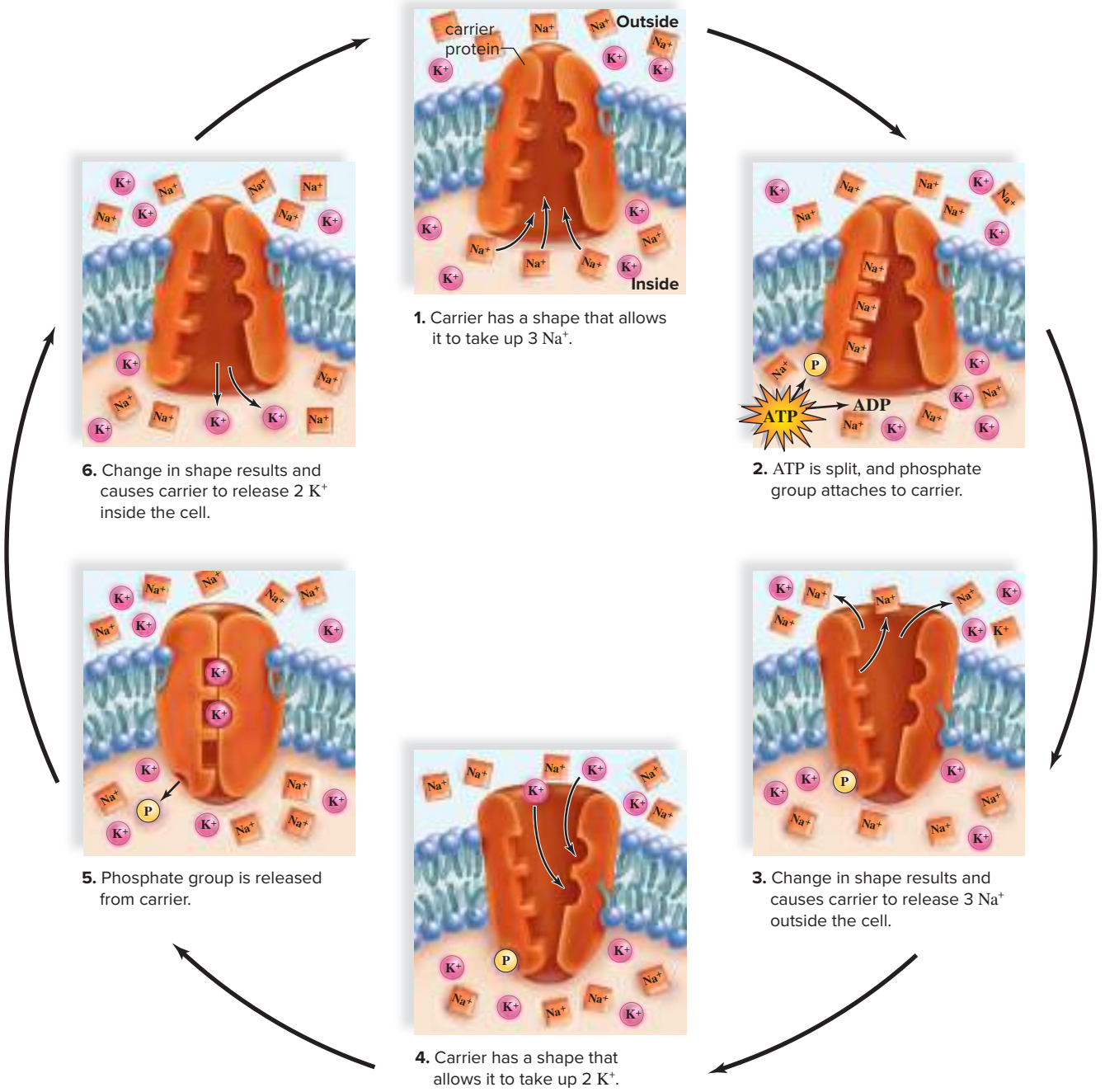


Figure 5.10 The sodium-potassium pump. The same carrier protein transports sodium ions (Na^+) to the outside of the cell and potassium ions (K^+) to the inside of the cell, because it undergoes an ATP-dependent change in shape. Three sodium ions are carried outward for every two potassium ions carried inward. Therefore, the inside of the cell is less positively charged compared to the outside.

ISTUDY

The passage of salt (NaCl) across a plasma membrane is of primary importance to most cells. The chloride ion (Cl^-) usually crosses the plasma membrane because it is attracted by positively charged sodium ions (Na^+). First, sodium ions are pumped across a membrane, and then chloride ions simply diffuse through channels that allow their passage.

As noted in Figure 5.3*a,* the genetic disorder cystic fibrosis results from a faulty chloride channel protein. When chloride is unable to exit a cell, water stays behind. The lack of water outside the cells causes abnormally thick mucus in the bronchial tubes and pancreatic ducts, thus interfering with the function of the lungs and pancreas.

Bulk Transport

How do large molecules such as proteins, polysaccharides, or nucleic acids enter and exit a cell? These molecules are too large to be transported by carrier proteins, so they are instead transported into and out of the cell by vesicles. Membrane vesicles formed around macromolecules require an expenditure of cellular energy, but the cost is worth it, because each vesicle keeps its cargo from mixing with molecules within the cytoplasm that could alter the cell's function. Generally, substances can exit a cell through exocytosis, and enter a cell through endocytosis.

Exocytosis

During **exocytosis,** an intracellular vesicle fuses with the plasma membrane as secretion occurs (Fig. 5.11). Hormones, neurotransmitters, and digestive enzymes are secreted from cells in this manner. The Golgi body often produces the vesicles that carry these cell products to the membrane. During exocytosis, the membrane of the vesicle becomes a part of the plasma membrane, because both are nonpolar. Adding additional vesicle membrane to the plasma membrane can enlarge the cell and is a part of growth in some cells. The proteins released from the vesicle may adhere to the cell surface or become incorporated into an extracellular matrix.

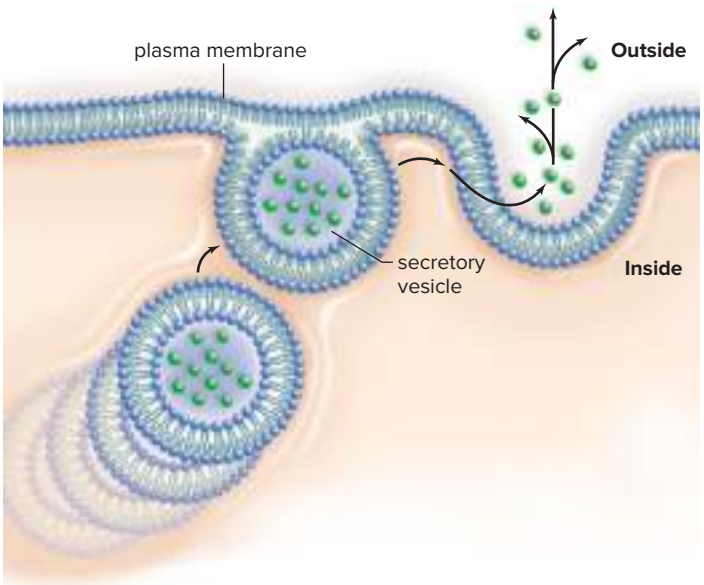


Figure 5.11 Exocytosis. Exocytosis secretes or deposits substances on the outside of the cell.

Cells of particular organs are specialized to produce and export molecules. For example, pancreatic cells produce digestive enzymes or insulin, and anterior pituitary cells produce growth hormone, among other hormones. In these cells, secretory vesicles accumulate near the plasma membrane, and the vesicles release their contents only when the cell is stimulated by a signal received at the plasma membrane. A rise in blood sugar, for example, signals pancreatic cells to release the hormone insulin. This is called regulated secretion, because vesicles fuse with the plasma membrane only when the needs of the body trigger it to do so.

Endocytosis

During **endocytosis,** cells bring substances into the cell by forming vesicles around the material. A portion of the plasma membrane invaginates to envelop the substance, and then the membrane pinches off to form an intracellular vesicle. Endocytosis occurs in one of three ways, as illustrated in Figure 5.12. Phagocytosis transports large substances, such as a virus, and pinocytosis transports small substances, such as solutes, into the cell. Receptor-mediated endocytosis is a specialized form of pinocytosis.

Phagocytosis. When the material taken in by endocytosis is large, such as a food particle or another cell, the process is called **phagocytosis** (Gk. *phagein,* "to eat"). Phagocytosis is common in single-celled organisms, such as amoebas (Fig. 5.12*a*). It also occurs in humans. Certain types of human white blood cells are amoeboid—that is, they are mobile like an amoeba, and they can engulf debris such as worn-out red blood cells or viruses. When an endocytic vesicle fuses with a lysosome, digestion occurs. Later in this text you will see that this process is a necessary and preliminary step toward the development of our immunity to bacterial diseases.

Pinocytosis. **Pinocytosis** (Gk. *pinein,* "to drink") occurs when vesicles form around a liquid or around very small particles (Fig. 5.12*b*). Blood cells, cells that line the kidney tubules or the intestinal wall, and plant root cells all use pinocytosis to ingest substances.

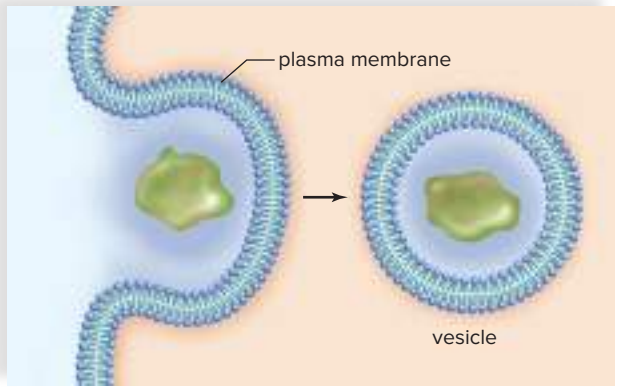
Whereas phagocytosis can be seen with a light microscope, an electron microscope must be used to observe pinocytic vesicles, which are no larger than 0.1–0.2 μm. Still, pinocytosis involves a significant amount of the plasma membrane, because it occurs continuously. Cells do not shrink in size, because the loss of plasma membrane due to pinocytosis is balanced by the occurrence of exocytosis.

Receptor-Mediated Endocytosis. **Receptor-mediated endocytosis** is a specific form of pinocytosis that uses a receptor protein to recognize compatible molecules and take them into the cell. Molecules such as vitamins, peptide hormones, or lipoproteins can bind to specific receptors, found in special locations in the plasma membrane (Fig. 5.12*c, d*). This location is called a coated pit because there is a layer of protein on the cytoplasmic side of the pit. Once formed, the vesicle is uncoated and may fuse with a lysosome. When empty, a used vesicle fuses with the plasma membrane, and the receptors return to their former location.

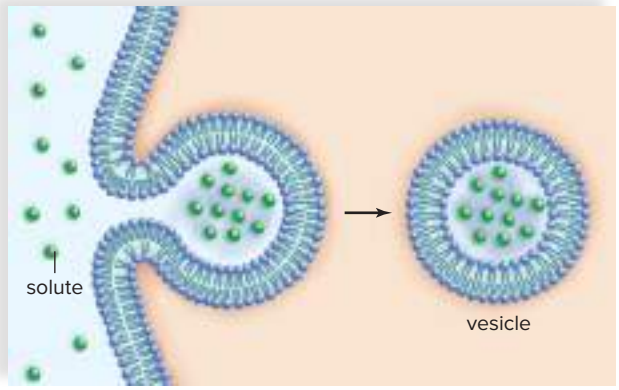
Receptor-mediated endocytosis is selective and much more efficient than ordinary pinocytosis. It is involved in uptake and in the transfer and exchange of substances between cells. Such exchanges take place when substances move from the female parent's blood into fetal blood at the placenta, for example.

The importance of receptor-mediated endocytosis is demonstrated by a genetic disorder called familial hypercholesterolemia.

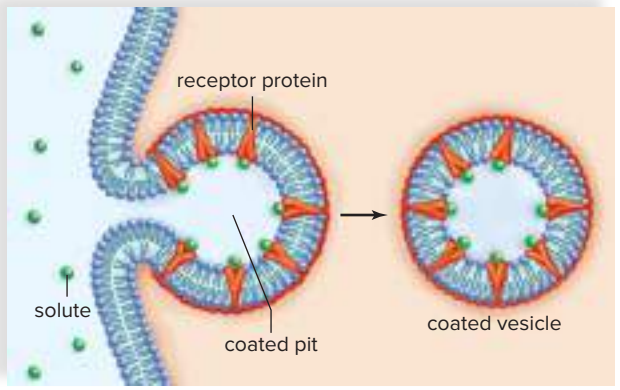
ISTUDY



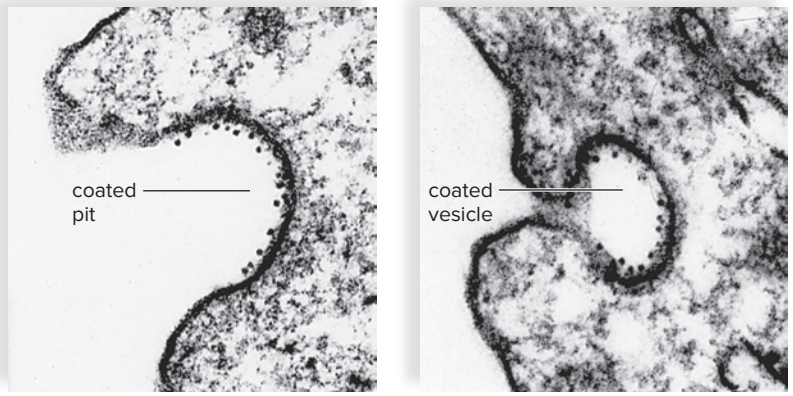
a. Phagocytosis



b. Pinocytosis



c. Receptor-mediated endocytosis



d. Micrographs of receptor-mediated endocytosis

Figure 5.12 Three methods of endocytosis. **a.** Phagocytosis occurs when the substance to be transported into the cell is large. Digestion occurs when the resulting vacuole fuses with a lysosome. **b.** Pinocytosis occurs when a macromolecule, such as a polypeptide, is transported into the cell. **c, d.** Receptor-mediated endocytosis is a form of pinocytosis. Molecules first bind to specific receptor proteins, which migrate to or are already in a coated pit. The coated vesicle that forms contains the molecules and their receptors. (d): Mark Bretscher

Cholesterol is transported in blood by a complex of lipids and proteins called low-density lipoprotein (LDL). Ordinarily, body cells take up LDL when LDL receptors gather in a coated pit. But in some individuals, the LDL receptor is unable to bind properly to the coated pit, and the cells are unable to take up cholesterol. Instead, cholesterol accumulates in the walls of arterial blood vessels, leading to high blood pressure, occluded (blocked) arteries, and heart attacks.

Check Your Progress 5.3

1. Compare facilitated transport with active transport.
2. Explain why active transport requires energy.
3. Summarize why a cell would use bulk transport rather than active transport.

5.4 Modification of Cell Surfaces

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the role of the extracellular matrix in an animal cell.
2. Compare the structure and function of adhesion, tight, and gap junctions in animals.
3. Explain the role of plasmodesmata in plants.

Most cells do not live isolated from other cells. Rather, they live and interact within an external environment that can dramatically influence cell structure and function. This extracellular environment often contains large molecules produced by nearby cells that are secreted from their membranes. In plants, prokaryotes, fungi, and most algae, the extracellular environment is a fairly rigid cell wall, which is consistent with a somewhat sedentary lifestyle. Animals, which tend to be more active, have a more varied extracellular environment, which can change depending on the tissue type.

Cell Surfaces in Animals

Here we examine two types of animal cell surface features: (1) the extracellular matrix outside cells and (2) the junctions between some types of cells. Both of these can connect to the cytoskeleton and contribute to communication between cells, and therefore tissue formation.

Extracellular Matrix

A protective **extracellular matrix (ECM)** is a meshwork of proteins and polysaccharides in close association with the cell that produced them (Fig. 5.13). Collagen and elastin fibers are two well-known structural proteins in the ECM; collagen resists stretching, and elastin gives the ECM resilience. Fibronectin is an

ISTUDY

adhesive protein (colored green in Fig. 5.13) that binds to a protein, called integrin, in the plasma membrane. Integrins are integral membrane proteins that connect to fibronectin externally and to the actin cytoskeleton internally. Through its connections with both the ECM and the cytoskeleton, integrin plays a role in cell signaling, permitting the ECM to influence the activities of the cytoskeleton and, therefore, the shape and activities of the cell.

Amino sugars in the ECM form multiple polysaccharides that attach to a protein and are, therefore, called proteoglycans. Proteoglycans, in turn, attach to a very long, centrally placed polysaccharide. The entire structure, which looks like an enormous bottle brush, resists compression of the extracellular matrix. Proteoglycans assist cell signaling when they regulate the passage of molecules through the ECM to the plasma membrane, where receptors are located. During development, they help bring about differentiation by guiding cell migration along collagen fibers to specific locations. Thus, the ECM has a dynamic role in all aspects of a cell's behavior.

Later on, in the discussion of tissues in Section 31.1, you'll see that the extracellular matrix varies in quantity and consistency. It can be quite flexible, as in loose connective tissue; semiflexible, as in cartilage; or rock solid, as in bone. The extracellular matrix of bone is hard because, in addition to the components mentioned, mineral salts (notably, calcium salts) are deposited outside the cell.

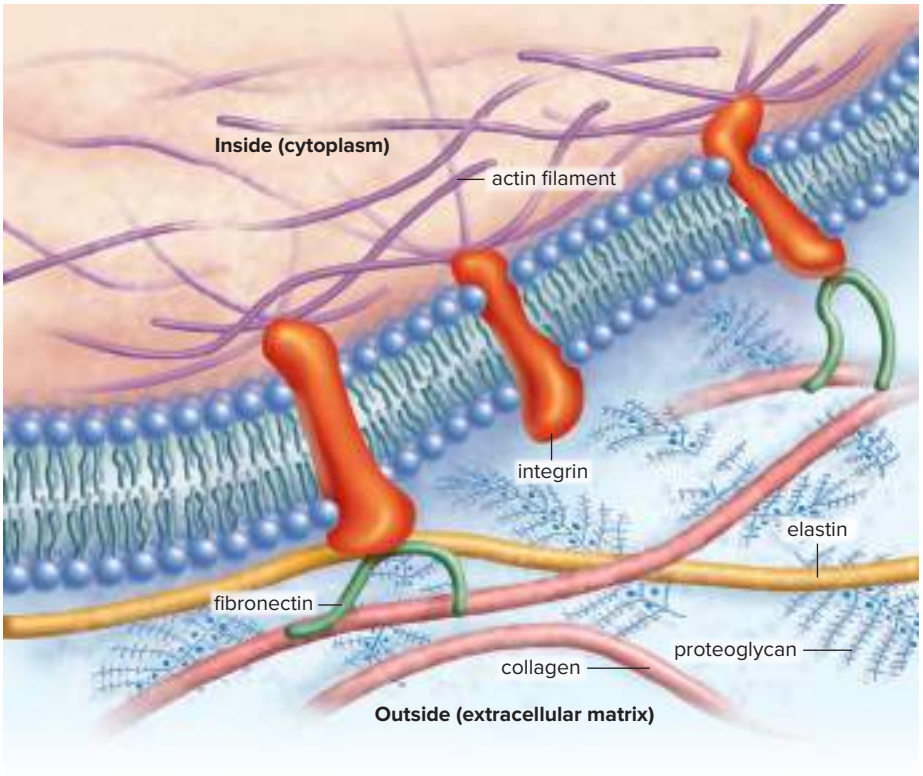


Figure 5.13 Animal cell extracellular matrix. In the extracellular matrix, collagen and elastin have a support function, while fibronectins bind to integrin, and in this way assist communication between the ECM and the cytoskeleton.

The proportion of cells to ECM also varies. In the small intestine, for example, epithelial cells constitute the majority of the tissue, and the ECM is a thin sheet beneath the cells. In bone, the ECM makes up most of the tissue, with comparatively fewer cells.

Junctions Between Cells

Certain tissues of vertebrate animals are known to have junctions between their cells that allow them to behave in a coordinated manner. Three types of junctions are shown in Figure 5.14.

Adhesion junctions mechanically attach adjacent cells (Fig. 5.14*a*). One example of an adhesion junction is the *desmosome.* In desmosomes, internal cytoplasmic plaques firmly attach to the intermediate filament cytoskeleton within each cell and are joined between cells by integral membrane proteins called cadherins. The result is a sturdy but flexible sheet of cells. In some organs—such as the heart, stomach, and bladder, where tissues get stretched—desmosomes hold the cells together. At a *hemidesmosome,* the intermediate filaments of the cytoskeleton are attached to the ECM through integrin proteins. Adhesion junctions are the most common type of intercellular junction between skin cells.

Another type of junction between adjacent cells is the **tight junction,** which brings cells even closer than desmosomes. Tight junction proteins actually connect plasma membranes between adjacent cells together, producing a zipperlike fastening (Fig. 5.14*b*). Tissues that serve as barriers are held together by tight junctions: In the intestine, the digestive juices stay out of the rest of the body, and in the kidneys the urine stays within kidney tubules, because the cells are joined by tight junctions.

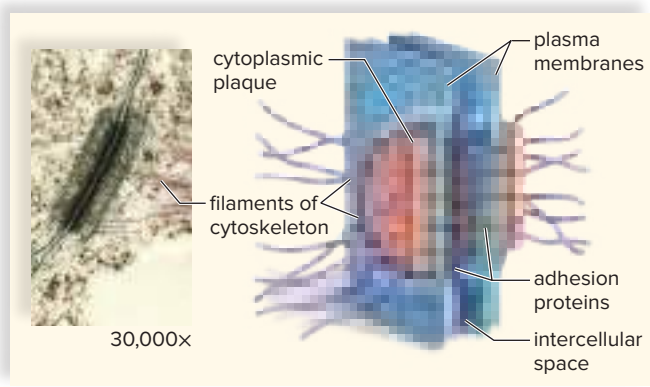
A **gap junction** allows cells to communicate. A gap junction is formed when two identical plasma membrane channels join (Fig. 5.14*c*). The channel of each cell is lined by six plasma membrane proteins. A gap junction lends strength to the cells, but it also allows small molecules and ions to pass between them. Gap junctions are important in heart muscle and smooth muscle, because they permit the flow of ions that is required for the cells to contract as a unit.

Plant Cell Walls

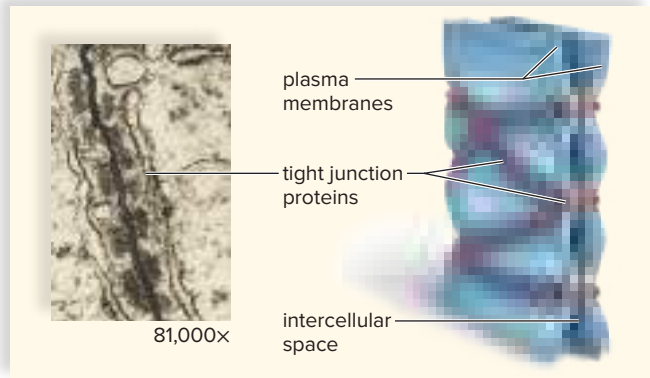
In addition to a plasma membrane, plant cells are surrounded by a porous **cell wall** that varies in thickness, depending on the function of the cell.

All plant cells have a primary cell wall. The primary cell wall contains cellulose fibrils, in which microfibrils are held together by noncellulose substances. Pectins allow the wall to stretch when the cell is growing, and noncellulose polysaccharides harden the wall when the cell is mature. Pectins are especially abundant in the middle lamella, which is a layer of adhesive substances that holds the cells together.

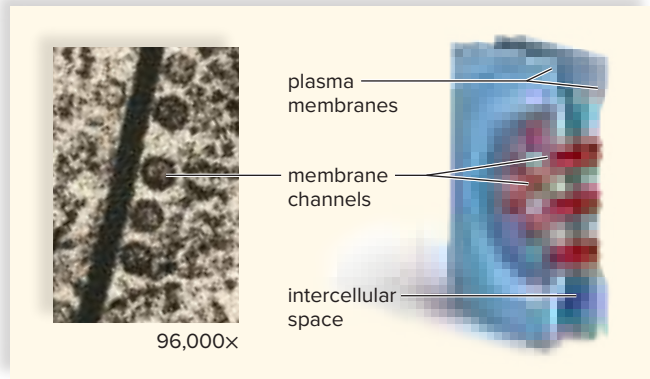
ISTUDY



a. Adhesion junction



b. Tight junction



c. Gap junction

Figure 5.14 Junctions between cells of the intestinal wall. **a.** In adhesion junctions such as a desmosome, adhesive proteins connect two cells. **b.** Tight junctions between cells form an impermeable barrier because their adjacent plasma membranes are joined and don't allow molecules to pass. **c.** Gap junctions allow communication between two cells because adjacent plasma membrane channels are joined. (photos): (a): SPL/Science Source; (b, c): David M. Phillips/Science Source

Some cells in woody plants have a secondary wall that forms inside the primary cell wall. The secondary wall has a greater quantity of cellulose fibrils than the primary wall, and layers of cellulose fibrils are laid down at right angles to one another. Lignin, a substance that adds strength, is a common ingredient of secondary cell walls in woody plants.

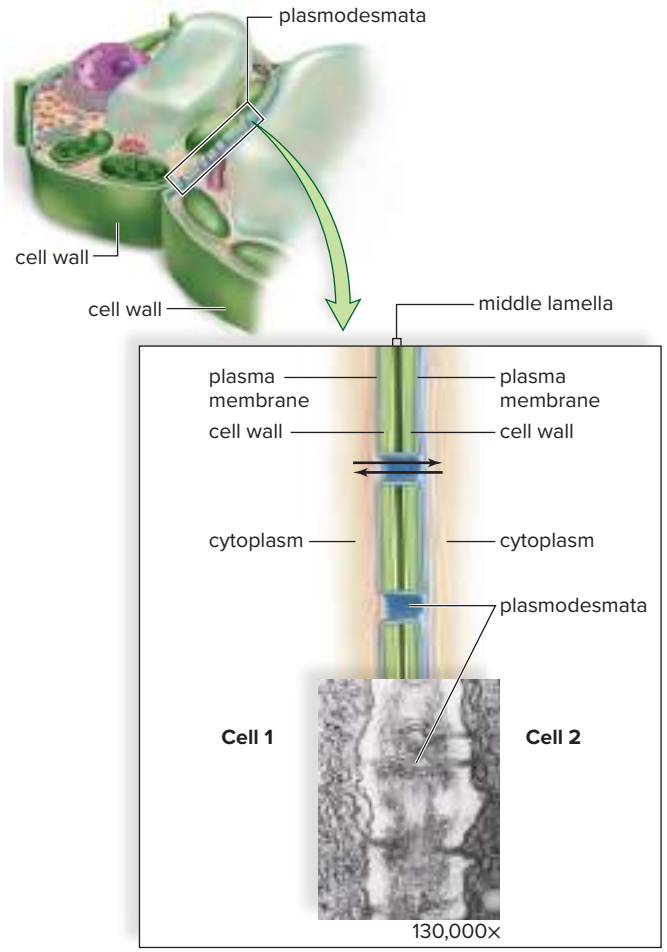


Figure 5.15 Plasmodesmata. Plant cells are joined by membrane-lined channels that contain cytoplasm. Water and small molecules can pass from cell to cell. (photo): Biophoto Associates/Science Source

In a plant, the cytoplasm of living cells is connected by **plasmodesmata** (*sing.*, plasmodesma), numerous narrow, membrane-lined channels that pass through the cell wall (Fig. 5.15). Cytoplasmic strands within these channels allow direct exchange of some materials between adjacent plant cells and eventually connect all the cells within a plant. The plasmodesmata allow only water and small solutes to pass freely from cell to cell. This limitation means that plant cells can maintain their own concentrations of larger substances and differentiate into particular cell types.

Check Your Progress 5.4

1. Describe the composition of the extracellular matrix of an animal cell.
2. Explain why a cell would be connected by a tight junction, rather than a gap junction or an adhesion junction.
3. Explain the role of the plasmodesmata in plant cells.

ISTUDY

5.5 Cell-to-Cell Communication

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the importance of plasma membrane receptors in cell signaling.
2. Explain the purpose of a signal transduction pathway in a cell.
3. Explain the role of second messengers in a cell.

As we learned in Section 1.1, all living organisms are comprised of cells, and these cells are able to sense and respond to specific signals in their environment. Whether those cells are single-celled bacteria, or part of a multicellular organism, the ability of cells to communicate with other cells is an essential part of all biological systems.

For example, in newborn animals, internal signals such as hormones are essential to ensure that specific tissues develop when and how they should. In plants, external signals, such as a change in the amount of light, tell them when it is time to resume growth or to flower. A bacterium that lives in your body responds to signaling molecules when it finds food and escapes immune cells in order to stay alive. Signaling helps the bread mold that grows on stale bread detect an opposite mating strain to begin its sexual life cycle. Internal signaling molecules enable eukaryotic organisms to coordinate their cellular activities, to metabolize, and to better respond in a changing environment.

Cell Signaling

The cells of a multicellular organism "talk" to one another by using signaling molecules, also called **chemical messengers.** The major categories of cell signaling are presented in Table 5.2.

Cells respond to only certain signaling molecules. This is because signaling molecules must bind to a specific receptor protein in the surface of a cell (Fig. 5.16), and only those cells that possess these receptors are able to respond. Each cell has a mix of receptors, which gives it the ability to respond differently to a variety of external and internal stimuli. Each cell is also able to balance the relative strength of incoming signals in order to change its cellular structure or function. If a minimum level of signaling is not met, the cell dies.

Signaling molecules interacting with their receptor is only the beginning of a complex process of communication that tells the cell how to respond. Once a signaling molecule and receptor interact, a cascade of events occurs that increase, decrease, or otherwise change the signal to elicit a cellular response. This process is called a **signal transduction** pathway. As shown in Figure 5.16, the cell response to a transduction pathway can be a change in the shape or movement of a cell, the activation of a particular enzyme, or the activation of a specific gene.

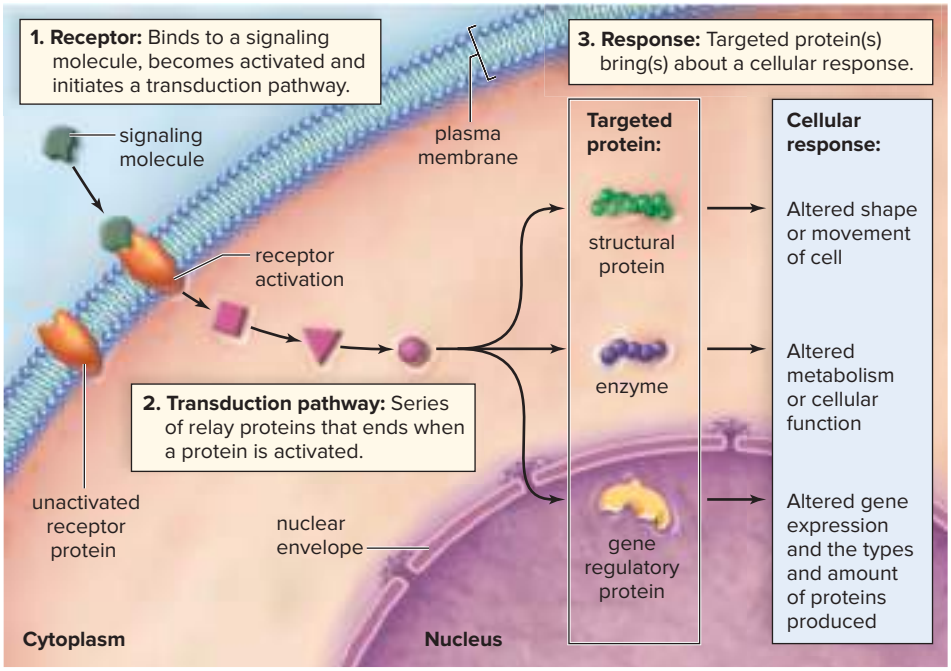


Figure 5.16 Overview of signal transduction pathways. The process of internal cell signaling involves three steps: binding of the signaling molecule, transduction of the signal, and response of the cell depending on what type of protein is targeted.

Table 5.2 Forms of Cell Signaling

Form	Description	Examples
Synaptic	neurotransmitters involved in the transfer of information within the nervous system	epinephrine, serotonin
Endocrine	hormones produced by one organ or tissue of the body to influence the function of another organ or tissue	sex hormones and insulin
Paracrine	molecules released over short distances	blood clotting and tissue repair signals
Direct contact	contact between receptors of neighboring cells or through channels such as gap junctions	contact between cells during early embryonic development
Autocrine	release of chemicals by a cell to influence its own activity	cytokines of the immune system

ISTUDY

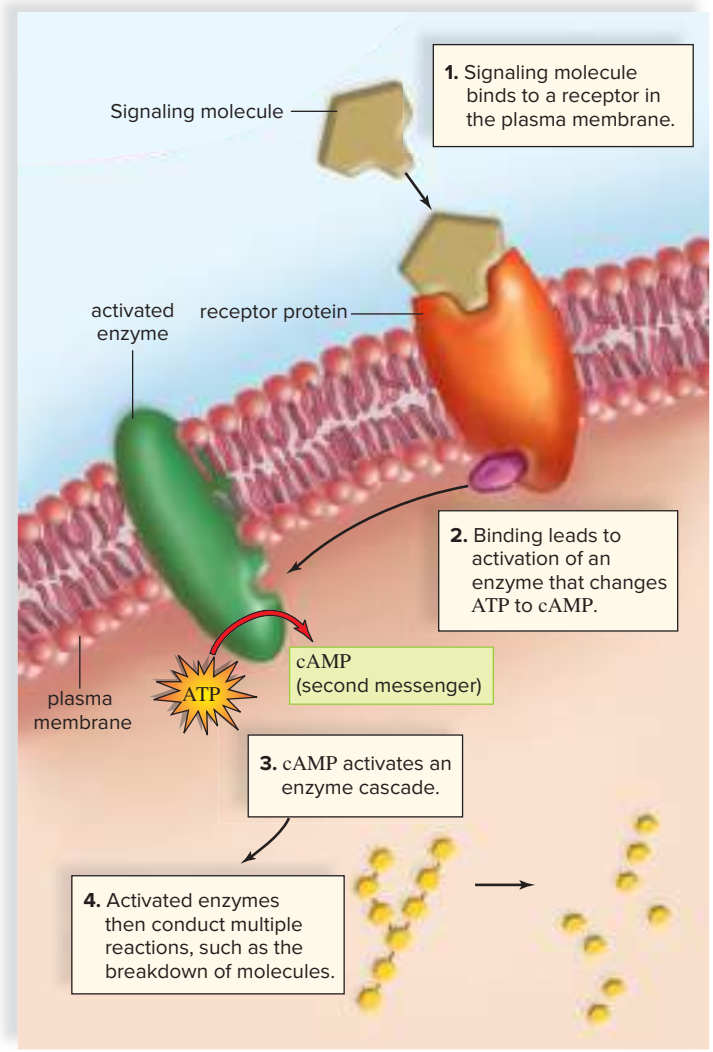


Figure 5.17 cAMP second messenger system. cAMP is used as a second messenger in many signal transduction pathways and is useful in accelerating the response of the cell to the stimuli (signaling molecule).

These changes to protein shape or function are often associated with the activation or deactivation of a molecule (often a protein) by the adding or removing of phosphate groups. When a phosphate has been added to a molecule, it is said to be phosphorylated, and when a phosphate group is removed, it is said to be dephosphorylated. The enzymes that conduct these reactions are called kinases and phosphatases.

In many of these pathways, a protein, called **G protein,** assists in signal transduction. The G protein interacts between the receptor and another membrane-bound protein, called an effector protein, that then initiates the cellular response.

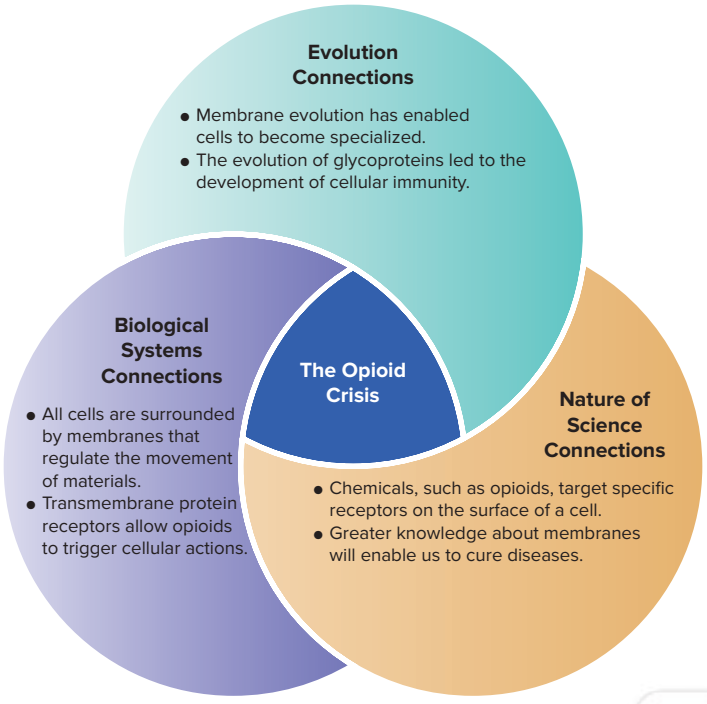
Second Messenger Systems

Signal transduction pathways often function by using a **second messenger.** The two more common second messenger molecules are cAMP (cyclic AMP) and calcium ions (Ca^{2+}). An example of a cAMP second messenger system is shown in Figure 5.17. The role of second messenger systems is to amplify the effect of the signaling molecule and accelerate the response of the cell to the stimuli.

Check Your Progress 5.5

1. Explain why receptors for cell signaling are specific.
2. Describe the stages of a signal transduction pathway.
3. Describe the purpose of a second messenger.

Connecting *the* Concepts

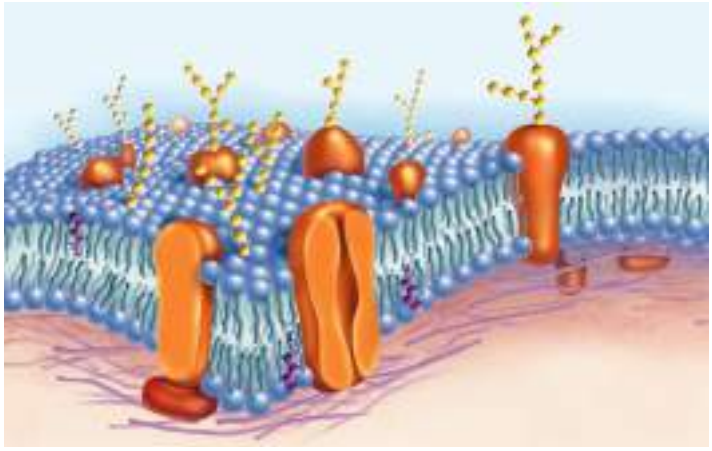


ISTUDY

What Did You Learn?

5.1 Plasma Membrane Structure and Function

The **fluid-mosaic model** describes the interactions of the components of the plasma membrane. In the lipid bilayer, phospholipids are arranged with their hydrophilic (polar) heads adjacent to water, and their hydrophobic (nonpolar) tails buried in the interior. The lipid bilayer has the consistency of oil but acts as a barrier to the entrance and exit of most biological molecules. Membrane **glycolipids** and **glycoproteins** are involved in marking the cell as belonging to a particular individual and tissue.



A variety of proteins may be embedded within the plasma membrane. The hydrophobic portion of an integral protein lies in the lipid bilayer of the plasma membrane, and the hydrophilic portion lies at the surfaces. Some functions include the following.

- **Channel proteins:** Form a channel through the membrane to move molecules. Examples are aquaporins that allow rapid movement of water across a membrane.
- **Carrier proteins:** Interact with a target molecule, causing a change in shape, which allows movement of the molecule through the membrane.
- **Cell recognition proteins:** Glycoproteins involved in cellular identification.
- **Receptor proteins:** Detect specific molecules in the environment and invoke a cellular response.
- **Enzymatic proteins:** Carry out metabolic reactions within the plasma membrane.
- **Junction proteins:** Form connections between cells.

The plasma membrane of a cell is **selectively permeable** and isolates the cell from the external environment. In doing so, the membrane establishes a **concentration gradient** for each type of molecule. These molecules are then moved across the membrane by active, passive, or **bulk transport.**

5.2 Passive Transport Across a Membrane

Diffusion is the passive movement of molecules from a higher to a lower concentration. In a **solution,** the **solute** diffuses within the **solvent** until it is evenly distributed. Some molecules (lipid-soluble compounds, water, and gases) simply diffuse across the membrane. No metabolic energy is required for diffusion to occur.

The diffusion of water across a selectively permeable membrane is called **osmosis.** During osmosis, **osmotic pressure** directs water to move across the membrane into the area of higher solute (less water) content per volume. **Tonicity** is a measure of the relative solute concentration. When cells are in an **isotonic solution,** they neither gain nor lose water. When cells are in a **hypotonic solution,** they gain water. This accounts for the **turgor pressure** of plants and **hemolysis** in red blood cells. When cells are in a **hypertonic solution,** they lose water, which can cause **crenation** and **plasmolysis** (Table 5.3).

Other molecules are transported across the membrane either by a channel protein or by carrier proteins that span the membrane. During facilitated transport, a substance moves down its concentration gradient. No energy is required.

5.3 Active Transport Across a Membrane

During **active transport,** a carrier protein acts as a pump that causes a substance to move against its concentration gradient. One example is the **sodium-potassium pump** that carries Na^+ to the outside of the cell and K^+ to the inside of the cell. Energy in the form of ATP molecules is required for active transport to occur.

Larger substances can enter and exit a cell by exocytosis and endocytosis. **Exocytosis** involves secretion. **Endocytosis** includes **phagocytosis, pinocytosis,** and **receptor-mediated endocytosis.** Receptor-mediated endocytosis makes use of receptor proteins in the plasma membrane. Once a specific solute binds to receptors, a coated pit becomes a coated vesicle. After losing the coat, the vesicle can join with the lysosome or, after discharging the substance, the receptor-containing vesicle can fuse with the plasma membrane.

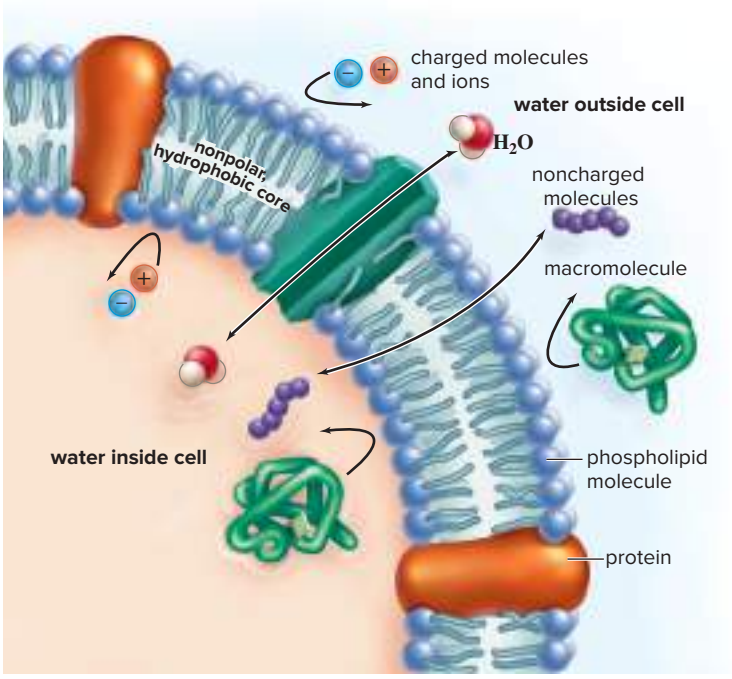


Table 5.3 Effect of Osmosis on a Cell

Tonicity of Solution	Solute Concentration	Net Movement of Water	Effect on Cell
Isotonic	Same as cell	None	None
Hypotonic	Less than cell	Cell gains water	Swells, turgor pressure
Hypertonic	More than cell	Less than cell	Shrinks, plasmolysis

ISTUDY

5.4 Modification of Cell Surfaces

Animal cells have an **extracellular matrix (ECM),** which influences their shape and behavior. The amount and character of the ECM vary by tissue type. Some animal cells have junction proteins that join them to other cells of the same tissue. **Adhesion junctions** and **tight junctions** help hold cells together; **gap junctions** allow the passage of small molecules between cells.

Plant cells have a freely permeable **cell wall,** with cellulose as its main component. Also, plant cells are joined by narrow, membrane-lined channels, called **plasmodesmata,** that span the cell wall and contain strands of cytoplasm that allow materials to pass from one cell to another.

5.5 Cell-to-Cell Communication

All cells, whether they are prokaryotes or eukaryotes, utilize signals to communicate with other cells. These signals are delivered using signaling molecules called **chemical messengers,** which interact with specific receptors within the membrane of the target cell.

Once activated, a **signal transduction** pathway, often with the aid of **G proteins** and **second messengers,** relays the signal to a target protein, which is then often activated by adding or removing phosphates.

Thinking Critically About Biology

1. As mentioned, cystic fibrosis is a genetic disorder caused by a defective membrane transport protein. The defective protein closes chloride channels in membranes, preventing chloride from being exported out of cells. This results in the development of a thick mucus on the outer surfaces of cells. This mucus clogs the ducts that carry digestive enzymes from the pancreas to the small intestine, clogs the airways in the lungs, and promotes lung infections. Why do you think the defective protein results in a thick, sticky mucus outside the cells, instead of a loose, fluid covering?
2. When people bite into a hot pepper, they experience the sensation of burning and pain due to capsaicin binding to a membrane protein. Explain which specific type of membrane protein would be responsible for binding to capsaicin. Also, explain the difference between the cell membranes of individuals who experience greater discomfort when they eat hot peppers compared to individuals who do not experience the discomfort.

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Saltwater Filter

McGraw Hill

Explain why the branches were able to filter out the bacteria but not the salt from the water sample.

Making It Relevant

1. Based on what you have learned, how do your cells interact with opioids? (Refer to Figure 5.3*d*.)
2. Do you think prescription-based opioids should be prescribed to patients suffering from chronic pain or other ailments? What limitations would you impose?
3. Is it possible that individuals who lack or have altered transmembrane receptors will not become addicted to opioids?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

6

Metabolism: Energy and Enzymes

Nick Hawkins/NHPA/Photoshot

Chapter Outline

6.1 Cells and the Flow of Energy

6.2 Metabolic Reactions and Energy Transformations

6.3 Metabolic Pathways and Enzymes

6.4 Oxidation-Reduction Reactions and Metabolism

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.4 What is the relationship between protein structure and function?

Section 3.5 What are the structure and general functions of the nucleic acids?

Section 4.7 What are the roles of the chloroplasts and mitochondria in a cell?

Enzymes and Vampire Bats

Vampire bats were the inspiration for many legends about vampires. Vampire bats do exist, but they are not as frightening as the legends would have you believe. The common vampire bat, *Desmodus rotundus,* is about the size of a human thumb, with a wingspan of only 8 inches. These bats normally feed on the blood of cattle, horses, and pigs.

Vampire bat saliva has an amazing ability to dissolve blood clots, allowing the blood to continue to flow from the wound while the bat feeds. Normally, blood clots due to the action of fibrin, an insoluble protein in the blood plasma. Fibrin is dissolved by the enzyme plasmin, which circulates in the blood in an inactive form called plasminogen. Another enzyme activates plasminogen, and then dissolves the fibrin of the clot. There is an enzyme in the saliva of the vampire bat that is 150 times more potent at activating plasminogen—and thus dissolving clots—than any known drug. It may one day be used to treat victims of stroke, caused when a clot blocks blood supply to the brain.

This chapter describes the general characteristics and functions of enzymes, and how enzymes function in the flow of energy and metabolism.

As you read through this chapter, think about the following questions:

1. What is the role of an enzyme in a metabolic pathway?
2. What environmental factors influence the rate of enzyme activity?

Following *the* Themes

CHAPTER 6 – METABOLISM: ENERGY AND ENZYMES

Evolution	Cells have evolved to metabolize energy in order to support the basic characteristics of life.
Nature of Science	Understanding the principles of metabolism helps us understand how cells and our bodies function.
Biological Systems	Energy flows through all biological systems, creating the ability to do work, although eventually all energy is lost as heat.

ISTUDY

6.1 Cells and the Flow of Energy

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare potential and kinetic energy.
2. Describe the first and second laws of thermodynamics.
3. Examine how the organization and structure of living organisms are related to heat and entropy.

To maintain their structural organization and carry out metabolic activities, cells—and organisms comprised of cells—need a constant supply of energy. **Energy** is defined as the ability to do work or bring about a change. The general characteristics of life, including growth, development, metabolism, and reproduction, all require energy.

Organic nutrients, made by photosynthesizing producers (algae, plants, and some bacteria), directly provide consumers with energy that was originally obtained from sunlight. Considering that producers use light energy to produce organic nutrients, the majority of life on Earth is ultimately dependent on solar energy.

Forms of Energy

Energy occurs in two forms: kinetic and potential energy. **Kinetic energy** is the energy of motion, as when water flows over a waterfall, a ball rolls down a hill, or a moose walks through grass. **Potential energy** is stored energy whose capacity to accomplish work is not being used at the moment. The food we eat has potential energy because the energy stored in chemical bonds can be converted into various types of kinetic energy. Food is a form of **chemical energy** because it is composed of organic molecules, such as carbohydrates, proteins, and fat. When a moose walks, it converts the chemical energy it has eaten into a type of kinetic energy called **mechanical energy** (Fig. 6.1).

Two Laws of Thermodynamics

In nature, energy flows in biological systems. Figure 6.1 illustrates the flow of energy in a terrestrial ecosystem. Plants capture only a small portion of solar energy, and much of it dissipates as **heat.** When plants photosynthesize and then make use of the food they produce, more heat results. Even with this considerable loss of heat, there is enough remaining to sustain a moose and the other organisms in an ecosystem. As organisms metabolize nutrient molecules, all the captured solar energy is eventually dissipated as heat. Therefore, we can see that energy flows through the ecosystem and does not cycle within it.

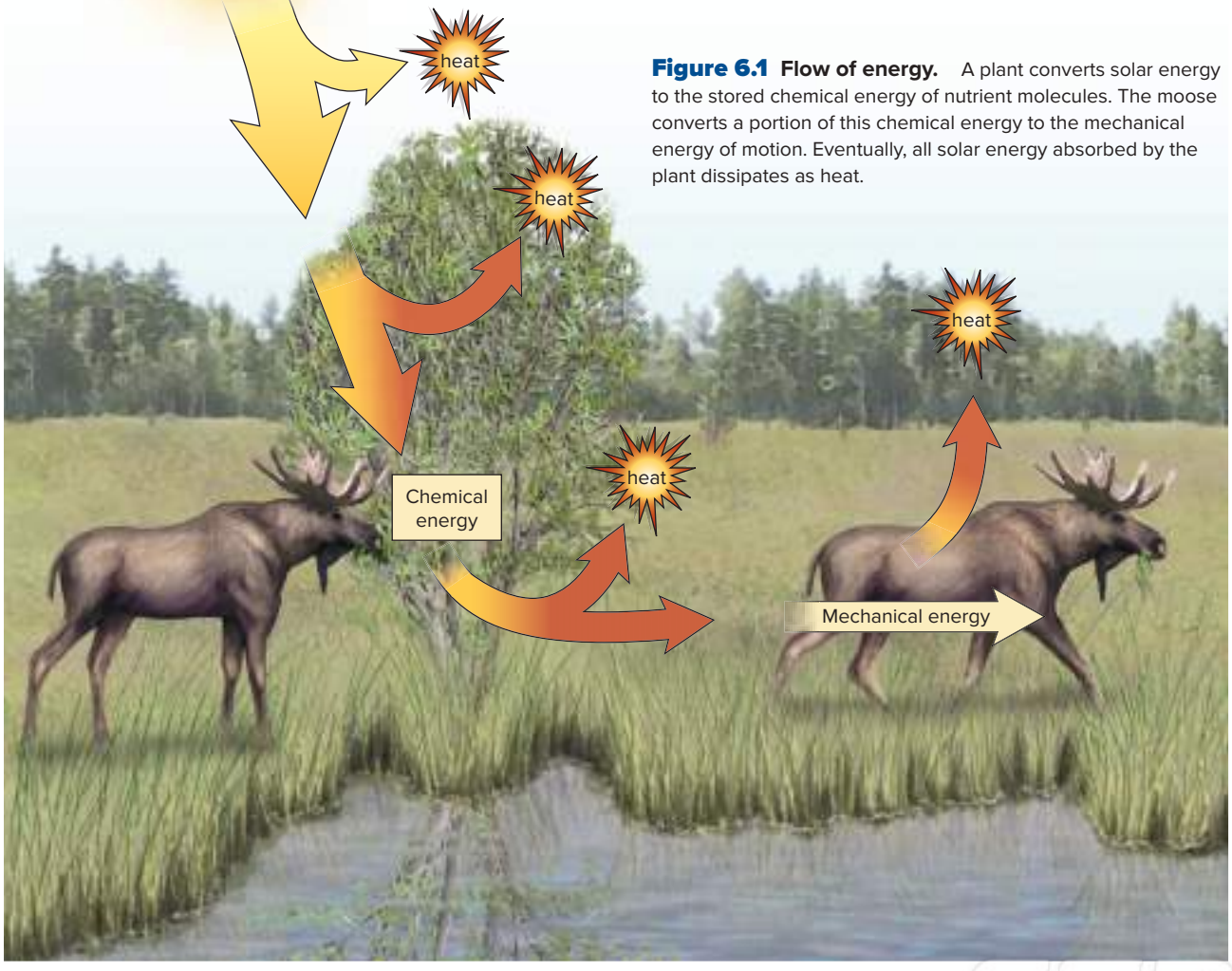
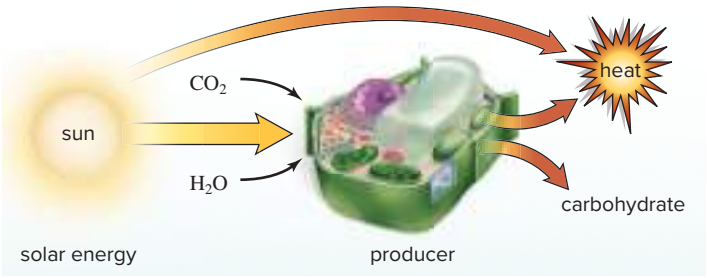


Figure 6.1 Flow of energy. A plant converts solar energy to the stored chemical energy of nutrient molecules. The moose converts a portion of this chemical energy to the mechanical energy of motion. Eventually, all solar energy absorbed by the plant dissipates as heat.

ISTUDY

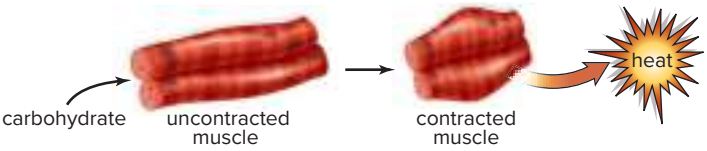
Two **laws of thermodynamics,** formulated by early energy researchers, explain why energy flows through ecosystems and through cells:

> The first law of thermodynamics—the law of conservation of energy—states that energy cannot be created or destroyed, but it can be changed from one form to another.



When leaf cells photosynthesize, they use solar energy, carbon dioxide, and water to form carbohydrate molecules. Carbohydrates are energy-rich molecules, because they have many bonds that store energy. Carbon dioxide and water are energy-poor molecules, because of the relative lack of bonds. Not all of the captured solar energy becomes carbohydrates; some becomes heat.

Obviously, plant cells do not create the energy they use to produce carbohydrate molecules. That energy comes from the sun. Is any energy destroyed? No, because the heat the plant cells give off is also a form of energy. Similarly, as a moose walks, it uses the potential energy stored in carbohydrates to kinetically power its muscles. As its cells use this energy, none is destroyed, but each energy exchange produces some heat, which dissipates into the environment.



The second law of thermodynamics therefore applies to living systems:

> The second law of thermodynamics states that energy cannot be changed from one form to another without a loss of usable energy.

In our example, this law is upheld because some of the solar energy taken in by the plant and some of the chemical energy within the nutrient molecules taken in by the moose become heat. When heat dissipates into the environment, it is no longer usable—that is, it is not available to do work. Each energy transformation moves us closer to a condition where all usable forms of energy become heat that is lost to the environment. Heat that dissipates into the environment cannot be captured and converted to one of the other forms of energy.

As a result of the second law of thermodynamics, no process requiring a conversion of energy is ever 100% efficient. Much of the energy is lost in the form of heat. In automobiles, the internal combustion engine is between 20% and 30% efficient in converting chemical energy stored in gasoline into mechanical energy used to drive the wheels. The majority of energy is lost as dissipated heat. Cells are capable of about 40% efficiency, with the remaining energy being given off to the surrounding environment as heat.

Cells and Entropy

The second law of thermodynamics can be stated another way: Every energy transformation makes the universe less organized, or structured, and more disordered, or chaotic. The term **entropy** (Gk. *entrope,* "a turning inward") is used to indicate the relative amount of disorganization. Because the processes that occur in cells are energy transformations, the second law means that every process that occurs in cells always does so in a way that increases the total entropy of the universe. The second law means that each cellular process makes less energy available to do useful work in the future.

Figure 6.2 shows two processes that occur in cells. The second law of thermodynamics tells us that glucose tends to break apart into carbon dioxide and water over time. Why? Because glucose is more organized and structured, and therefore less stable, than its breakdown products. Also, hydrogen ions on one side of a membrane tend to move to the other side unless they are prevented from

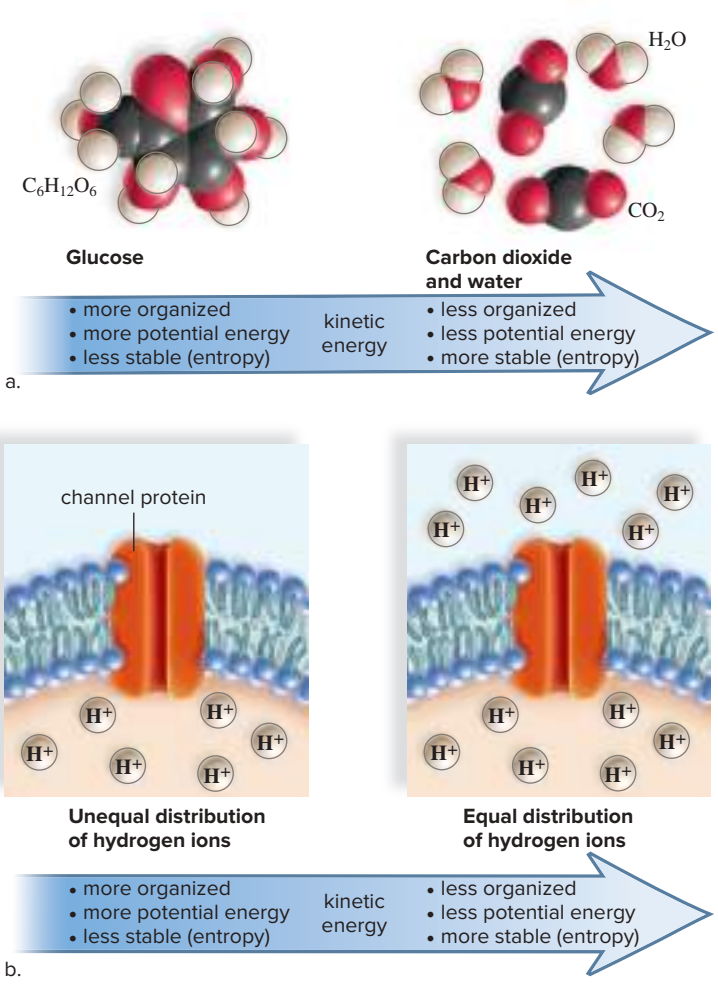


Figure 6.2 Cells and entropy. The second law of thermodynamics tells us that (**a**) glucose, which is more organized, tends to break down into carbon dioxide and water, which are less organized. **b.** Similarly, hydrogen ions (H^+) on one side of a membrane tend to move to the other side, so that the ions are randomly distributed. Both processes result in an increase in entropy.

doing so, because when they are distributed randomly, entropy has increased. As an analogy, you know from experience that a neat room is more organized but less stable than a messy room, which is disorganized but more stable. Energy is required to return a messy room to a more organized, or neat, state.

On the other hand, you know that some cells can make glucose out of carbon dioxide and water, and all cells can actively move ions to one side of the membrane. How do they do it? By the input of energy from an outside source. Photosynthesizing producers use energy from sunlight to create organized structure in biological molecules. Organisms that consume producers are then able to use this potential energy to kinetically drive their own metabolic processes. Thus, the majority of living organisms depend on a constant supply of energy that is originally provided by the sun. The ultimate fate of all solar energy in the biosphere is to become randomized in the universe as heat. A living cell can function because it serves as a temporary repository of order, purchased at the cost of a constant flow of energy.

Check Your Progress 6.1

1. Provide an example of a conversion from potential to kinetic energy.
2. Summarize how the first and second laws of thermodynamics relate to cells.
3. Explain the importance of entropy to a living system.

6.2 Metabolic Reactions and Energy Transformations

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how the ATP cycle involves both endergonic and exergonic reactions.
2. Describe how energy is stored in a molecule of ATP.
3. Examine how cells use ATP to drive energetically unfavorable reactions.

All living organisms maintain their structure and function through chemical reactions. **Metabolism** is the sum of all the chemical reactions that occur in a cell. **Reactants** are substances that participate in a reaction, while **products** are substances that form as a result of a reaction. In the reaction A + B → C + D, A and B are the reactants, while C and D are the products. Whether a reaction occurs spontaneously—that is, without an input of energy—depends on how much energy is left after the reaction. Using the concept of entropy, or disorder, a reaction occurs spontaneously if it increases the entropy of the universe.

In cell biology, which occurs on a small scale, we are less concerned about the entire universe, which is vast. In such specific instances, cell biologists use the concept of free energy instead of entropy. **Free energy** (also called "delta G," or ΔG) is the amount of energy left to do work after a chemical reaction has occurred. The change in free energy after a reaction occurs is determined by subtracting the free energy content of the reactants from that of the products. A negative result (−ΔG) means that the products have less free energy than the reactants causing the reaction to occur spontaneously. In our reaction, if C and D have less free energy than A and B, then the reaction occurs without additional input of energy.

Metabolism includes both spontaneous reactions and energy-requiring reactions. **Exergonic reactions** are spontaneous reactions that release energy, while **endergonic reactions** require an input of energy to occur. In the body, many reactions, such as protein and carbohydrate synthesis, are endergonic. For these nonspontaneous reactions to occur during metabolism, they must be coupled with exergonic reactions, so that a net spontaneous reaction results. Many biological processes use ATP as an energy carrier between exergonic and endergonic reactions.

ATP: Energy for Cells

ATP (adenosine triphosphate) is the common energy currency of cells; when cells require energy, they use ATP. A sedentary oak tree, a flying bat, and a human require vast amounts of ATP. The more active the organism, the greater the demand for ATP. However, cells do not keep a large store of ATP molecules on hand. Instead, they constantly regenerate ATP by using **ADP (adenosine diphosphate)** and inorganic phosphate, Ⓟ. This is called the ATP cycle (Fig. 6.3). This cycle is powered by the breakdown of glucose and other biomolecules during cellular respiration. However, according to the second law of thermodynamics, this process is not very efficient. Only 39% of the free energy stored in the chemical bonds of a glucose molecule is transformed to ATP; the rest is lost as heat.

There are many biological advantages to the use of ATP as an energy carrier in living systems. ATP provides a common and universal energy currency because it can be used in many different types of reactions. Also, when ATP is converted to energy, ADP, and Ⓟ, the amount of energy released is sufficient to efficiently power most biological functions. In addition, ATP breakdown can be coupled to endergonic reactions in such a way that it minimizes energy loss.

Structure of ATP

ATP is a nucleotide composed of the nitrogen-containing base adenine and the 5-carbon sugar ribose (together called adenosine), and three phosphate groups. The three phosphates of ATP repel each other, creating instability and potential energy (Fig. 6.3). ATP is called a "high-energy" molecule, because a phosphate group can be easily removed. Under cellular conditions, the amount of energy released when ATP is hydrolyzed to ADP+ Ⓟ is about 7.3 kcal per **mole.** A mole is a unit of measurement in chemistry that is equal to the molecular weight of a molecule expressed in grams.

Functions of ATP in Cells

In living systems, ATP can be used for the following:

- *Chemical work.* ATP supplies the energy needed to synthesize macromolecules (anabolism) that make up the cell, and therefore the organism.
- *Transport work.* ATP supplies the energy needed to pump substances across the plasma membrane.
- *Mechanical work.* ATP supplies the energy needed to permit muscles to contract, cilia and flagella to beat, chromosomes to move, and so forth. In most cases, ATP is the immediate source of energy for these processes.

ISTUDY

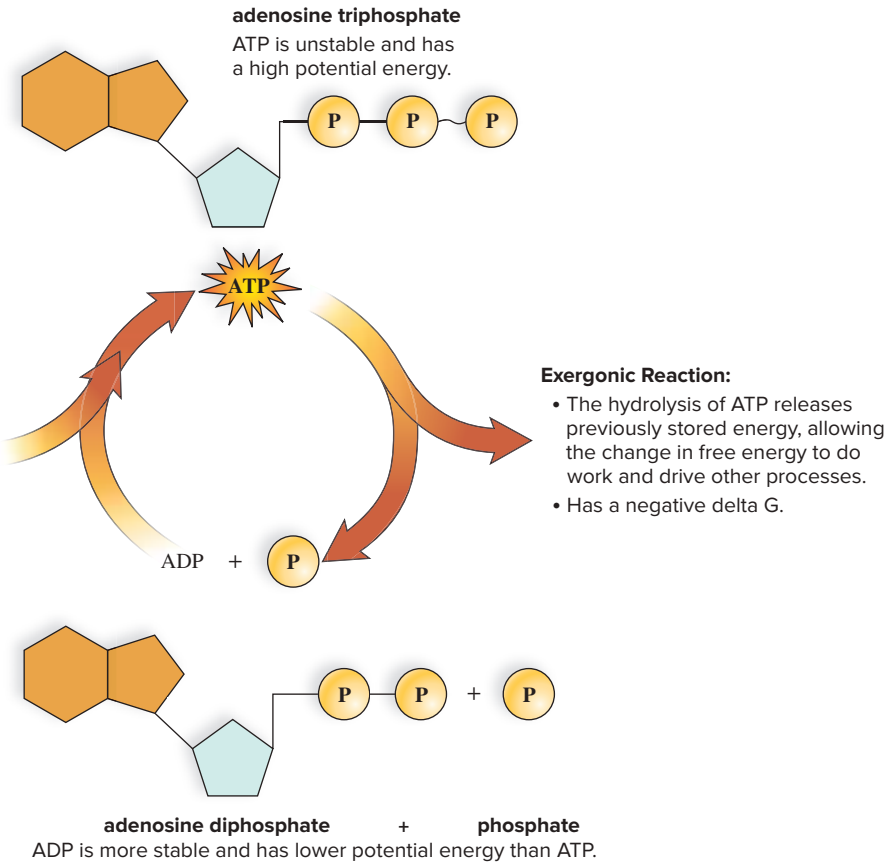
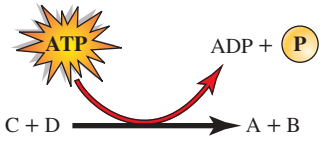


Figure 6.3 The ATP cycle. In cells, ATP carries energy between exergonic reactions and endergonic reactions. When a phosphate group is removed by hydrolysis, ATP releases the appropriate amount of energy for most metabolic reactions.

Coupled Reactions

How can the energy released by ATP hydrolysis be transferred to an endergonic reaction that requires energy and therefore would not ordinarily occur? In other words, how does ATP act as a carrier of chemical energy? How can that energy be transferred efficiently to an energetically unfavorable reaction?

The answer is that ATP breakdown is *coupled* to the energy-requiring reaction, such that both the energetically favorable and unfavorable reactions occur in the same place, at the same time. Usually, the energy-releasing reaction is the hydrolysis of ATP. Because the cleavage of ATP's phosphate groups releases more energy than the amount consumed by the energy-requiring reaction, the net reaction is exergonic, entropy increases, and both reactions proceed. The simplest way to represent a coupled reaction is like this:



This reaction tells you that coupling occurs, but it does not show how coupling is achieved. A cell has two main ways to couple ATP hydrolysis to an energy-requiring reaction: ATP is used to energize a reactant, or ATP is used to change the shape of a reactant. Both can be achieved by transferring a phosphate group to the reactant, so that the product is *phosphorylated.*

For example, when a polar ion moves across the nonpolar plasma membrane of a cell, it requires a carrier protein. In order to make the carrier protein assume a shape conducive to the ion, ATP is hydrolyzed. Then, instead of the last phosphate group floating away, an enzyme attaches it to a carrier protein. The negatively charged phosphate causes the protein to undergo a change in shape that allows it to interact with, and move the ion across, the membrane. Another example of a coupled reaction is when a polypeptide is synthesized at a ribosome. There, an enzyme transfers a phosphate group from ATP to each amino acid in turn, and this transfer supplies the energy needed to overcome the energy cost associated with bonding one amino acid to another.

Through coupled reactions, ATP drives forward the energetically unfavorable processes that must occur to create the high degree of order and structure essential for life. Macromolecules must be made and organized to form cells and tissues; the internal composition of the cell and the organism must be maintained; and movement of cellular organelles and the organism must occur if life is to continue.

Figure 6.4 shows an example of how ATP hydrolysis is coupled to the physiological process, muscle contraction (see Section 39.3). During muscle contraction, myosin filaments pull actin filaments to the center of the cell, and the muscle shortens. First, the myosin head combines with ATP (three connected green triangles) and takes on its resting shape. Next, ATP breaks down to ADP (two connected green triangles) plus Ⓟ (one green triangle). The resulting change in shape allows myosin to attach to actin. Finally, the release of ADP and Ⓟ from the myosin head causes it to change its shape again and pull on the actin filament. The cycle then repeats. During this cycle, chemical energy has been transformed to mechanical energy, and entropy has increased.

Check Your Progress 6.2

1. Explain why ATP is an effective short-term energy storage molecule.
2. Summarize the ATP cycle.
3. Examine how transferring a phosphate from ATP changes a molecule's structure and function.

ISTUDY

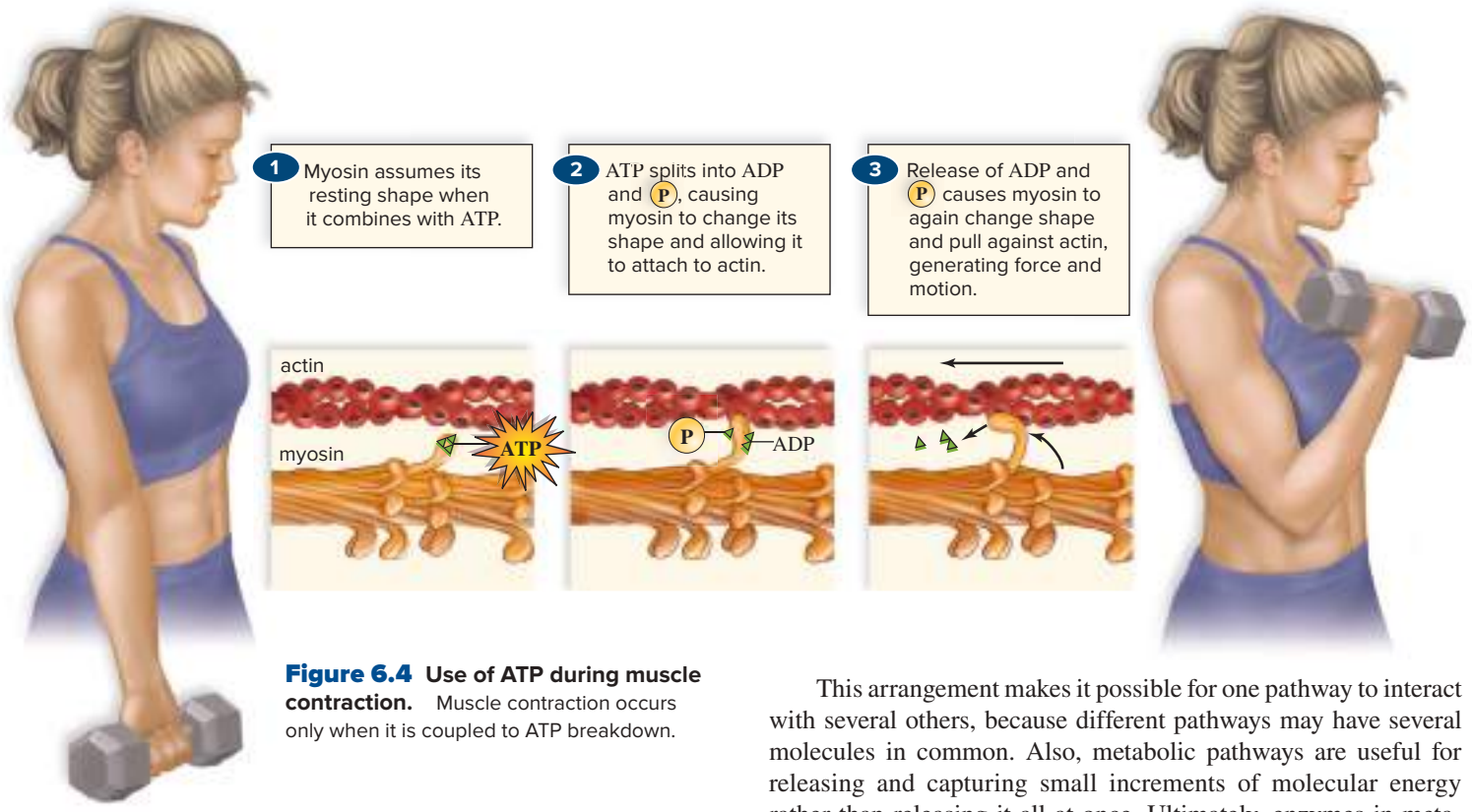


Figure 6.4 Use of ATP during muscle contraction. Muscle contraction occurs only when it is coupled to ATP breakdown.

6.3 Metabolic Pathways and Enzymes

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the purpose of a metabolic pathway and how enzymes help regulate it.
2. Recognize how enzymes influence the activation energy and rates of a chemical reaction.
3. Distinguish between conditions and factors that affect an enzyme's rate of reaction.

The chemical reactions that constitute metabolism would not easily occur without the use of organic catalysts called enzymes. An **enzyme** is a molecule that speeds a chemical reaction without itself being affected by the reaction. While most enzymes are proteins, some are made of RNA. These are called **ribozymes,** and they are involved in the synthesis of RNA and the synthesis of proteins at ribosomes. Enzymes allow reactions to occur under mild conditions, and they regulate metabolism, partly by eliminating nonspecific side reactions.

Chemical reactions do not occur haphazardly in healthy cells. They are usually part of a series of linked reactions called a **metabolic pathway.** Metabolic pathways begin with a particular reactant and end with a final product. Many specific steps can be involved in a metabolic pathway, and each step is a chemical reaction catalyzed by an enzyme. The reactants in an enzymatic reaction are called the **substrates** for that enzyme. The substrates for the first reaction are converted into **products,** and those products then serve as the substrates for the next enzyme-catalyzed reaction. One reaction leads to the next reaction in an organized, highly regulated manner.

This arrangement makes it possible for one pathway to interact with several others, because different pathways may have several molecules in common. Also, metabolic pathways are useful for releasing and capturing small increments of molecular energy rather than releasing it all at once. Ultimately, enzymes in metabolic pathways enable cells to regulate and respond to changing environmental conditions.

The following diagram illustrates a simple metabolic pathway:

$$A \xrightarrow{E_1} B \xrightarrow{E_2} C \xrightarrow{E_3} D$$

In this diagram, A is the substrate for E_1, and B is the product. Now B becomes the substrate for E_2, and C is the product. This process continues until the final product, D, forms. Any one of the molecules (A–D) in this metabolic pathway could also be a reactant in another pathway. Many of the metabolic pathways in living organisms are highly branched, and interactions between metabolic pathways are very common. It is important to note that each step in the metabolic pathway can be regulated because each step requires an enzyme. The specificity of enzymes allows the regulation of metabolism. The presence of particular enzymes helps determine which metabolic pathways are operative. In addition, some substrates can produce more than one type of product, depending on which pathway is open to them. Therefore, which enzyme is present determines which product is produced, as well as determining the direction of metabolism, without several alternative pathways being activated. As we will see, the ability to regulate these pathways gives our cells fine control over how they respond in a changing environment and helps maximize cell efficiency.

Enzyme-Substrate Complex

In most instances, only one small part of the enzyme, called the **active site,** associates directly with the substrate (Fig. 6.5). In the active site, the enzyme and substrate are positioned in such a way that they more easily fit together, seemingly as a key fits a lock. However, an active site differs from a lock and key because it undergoes a slight

ISTUDY

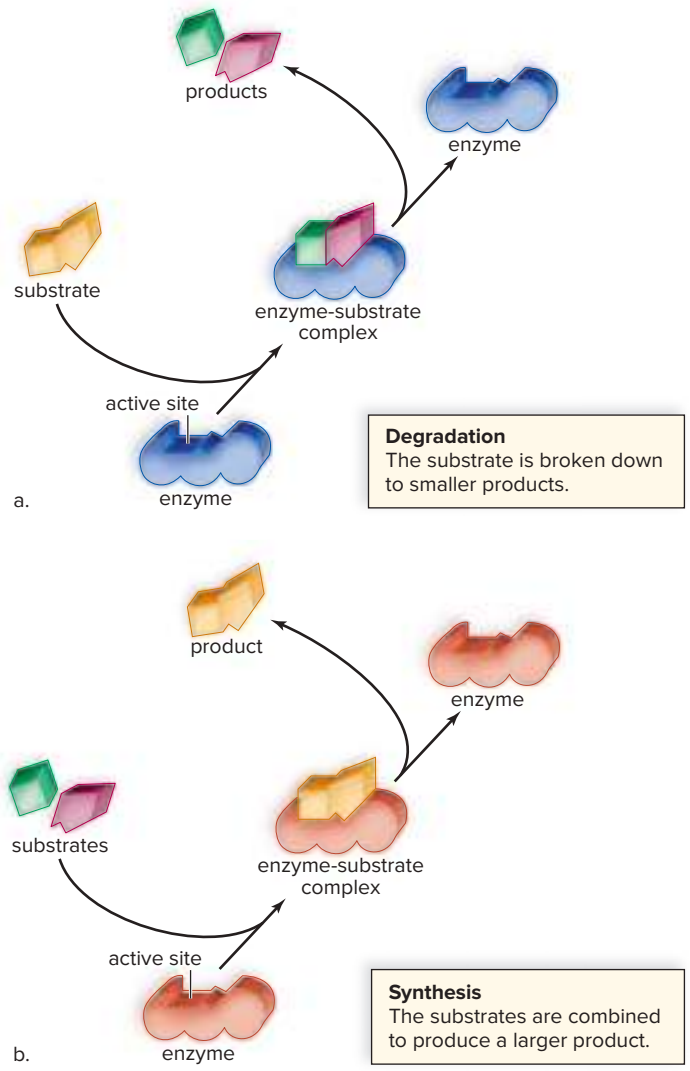


Figure 6.5 Enzymatic actions. Enzymes have an active site where the substrate(s) specifically fits (fit) together, so the reaction will occur. Following the reaction, the product or products are released, and the enzyme is free to act again. Certain enzymes carry out (**a**) degradation, and others carry out (**b**) synthesis.

change in shape to accommodate the substrate(s). This is called the **induced fit model,** because the enzyme is induced to undergo a slight alteration to achieve optimum fit for the substrates.

The change in shape of the active site facilitates the reaction that now occurs. After the reaction has been completed, the product or products are released, and the active site returns to its original state, ready to bind to another substrate molecule. Only a small amount of enzyme is actually needed in a cell, because enzymes are not used up by the reaction; they merely enable it to happen more quickly.

Some enzymes do more than simply form a complex with their substrate(s); they participate in the reaction. Trypsin digests protein by breaking peptide bonds. The active site of trypsin contains three amino acids with *R* groups that actually interact with members of the peptide bond—first to break the bond and then to introduce the components of water. This illustrates that the formation of the enzyme-substrate complex is very important in speeding the reaction. Because enzymes bind only with their substrates, they are sometimes named for their substrates and usually end in the suffix *ase*. For example, lipase is involved in hydrolyzing lipids.

Energy of Activation

Molecules frequently do not react with one another unless they are activated in some way. In the lab, for example, in the absence of an enzyme, molecules may be heated in order to increase the number of effective collisions. The energy that must be added to cause molecules to react with one another is called the **energy of activation (E_a).** Activation energy is essential to keep molecules from spontaneously degrading within the cell. Figure 6.6 shows that an enzyme effectively lowers E_a, thus reducing the energy needed for a chemical reaction to begin. It is important to note that the enzyme has no effect on the energy content of the product; rather, it only influences the rate of the reaction. Reducing the energy of activation increases the rate at which the reaction may occur. For this reason, enzymes are often referred to as catalysts of chemical reactions.

Factors Affecting Enzymatic Rate

Generally, enzymes work quickly, and in some instances they can increase the reaction rate more than 10 million times. The rate of a reaction is the amount of product produced per unit time. This rate depends on how much substrate is available to associate at the active sites of enzymes. Therefore, increasing the amount of substrate and the amount of enzyme can increase the rate of the reaction. Factors like a change in pH or temperature, as well as inhibitors, can alter the shape of the active site causing a change in the shape of the enzyme, called **denaturation.** Denaturation prevents an enzyme from binding to its substrate efficiently and can decrease the rate of a reaction. Thus, enzymes require specific conditions to be met in order to be fully operational. In fact, some enzymes require additional molecules called cofactors, which help speed the rate of the reaction, because they help bind the substrate to the active site, or they participate in the reaction at the active site.

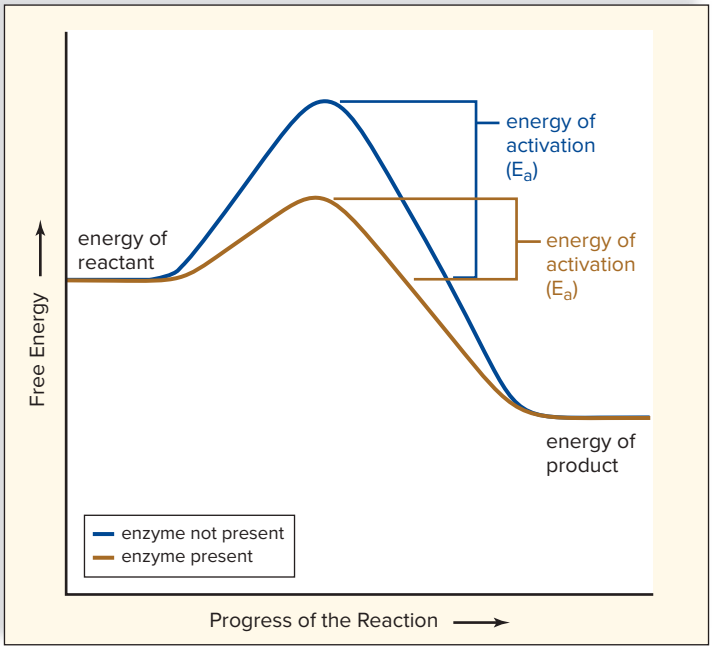


Figure 6.6 Energy of activation (E_a). Enzymes speed the rate of reactions, because they lower the amount of energy required for the reactants to activate. Even spontaneous reactions like this one, in which the energy of the product is less than the energy of the reactant, speed up when an enzyme is present.

ISTUDY

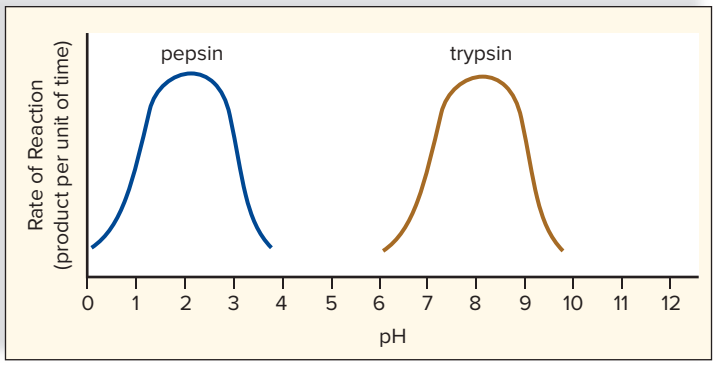


Figure 6.7 The effect of pH on rate of reaction. The optimal pH for pepsin, an enzyme that acts in the stomach, is about 2, while the optimal pH for trypsin, an enzyme that acts in the small intestine, is about 8. Enzyme shape is best maintained at the optimal pH, which allows it to function best and bind with its substrates.

Substrate Concentration

Molecules must collide to react. Generally, enzyme activity increases as substrate concentration increases, because there are more collisions between substrate molecules and the enzyme. As more substrate molecules fill active sites, more product results per unit of time. But when the active sites are filled almost continuously with substrate, the rate of the reaction can no longer increase. Maximum rate has been reached.

Just as the amount of substrate can increase or limit the rate of an enzymatic reaction, so the amount of active enzyme can also increase or limit the rate of an enzymatic reaction. Therefore, sufficient concentrations of substrate and enzymes are necessary to achieve maximum reaction rate.

Optimal pH

Each enzyme also has an optimal pH at which the reaction rate is highest. Figure 6.7 shows the optimal pH for the enzymes pepsin and trypsin. At their respective pH values, each enzyme can maintain its normal structural configuration, which enables optimum function. The globular structure of an enzyme is dependent on interactions, such as hydrogen bonding, between *R* groups. A change in pH can alter the ionization of these side chains, causing the enzyme to denature. Under extreme conditions of pH, the enzyme changes its structure and becomes inactive.

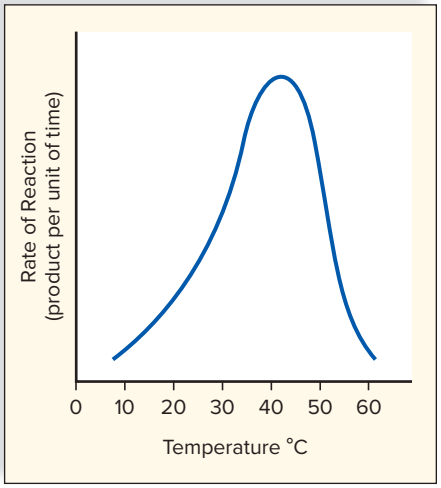
Temperature

Typically, as temperature rises, enzyme activity increases (Fig. 6.8*a*). This occurs because warmer temperatures cause more effective collisions between enzyme and substrate. The body temperature of an animal seems to affect whether it is normally active or inactive (Fig. 6.8*b*, *c*). It has been suggested that mammals are more prevalent today than reptiles because they maintain a warm internal temperature that allows their enzymes to work at a rapid rate.

In the laboratory and in your body, if the temperature rises beyond a certain point, enzyme activity eventually levels out and then declines rapidly because the enzyme is denatured. Exceptions to this generalization do occur. For example, some prokaryotes can live in hot springs because their enzymes do not denature. These organisms are responsible for the brilliant colors of the hot springs. Another exception involves the coat color of animals. Siamese cats have inherited a mutation that causes an enzyme to be active only at cooler body temperatures. The enzyme's activity causes the cooler regions of the body—the face, ears, legs, and tail—to be dark in color (Fig. 6.9). The coat color pattern in several other animals can be explained similarly.

Enzyme Cofactors and Coenzymes

Many enzymes require the presence of an inorganic ion or a nonprotein organic molecule at the active site in order to work properly. These necessary ions or molecules are called **cofactors** (Fig. 6.10). The inorganic ions include metals such as copper, zinc, or iron. The nonprotein organic molecules are called **coenzymes.** These cofactors participate in the reaction and may even accept or contribute



a. Rate of reaction as a function of temperature

b. Body temperature of ectothermic animals often limits rates of reactions.

c. Body temperature of endothermic animals promotes rates of reactions.

Figure 6.8 The effect of temperature on rate of reaction. **a.** Usually, the rate of an enzymatic reaction doubles with every 10°C rise in temperature. This enzymatic reaction is maximum at about 40°C. Then, it decreases until the reaction stops altogether, because the enzyme has become denatured. **b.** The body temperature of ectothermic animals, such as iguanas, which take on the temperature of their environment, often limits rates of reactions. **c.** The body temperature of endothermic animals, such as polar bears, promotes rates of reaction. (b): Philip Coblentz/MediaImages/SuperStock; (c): Walter Huber/moodboard/Glow Images

ISTUDY

Figure 6.9 The effect of temperature on enzymes. Siamese cats have inherited a mutation that causes an enzyme to be active only at cooler body temperatures. Therefore, only certain regions of their bodies are dark in color. axelbueckert/iStock/Getty Images

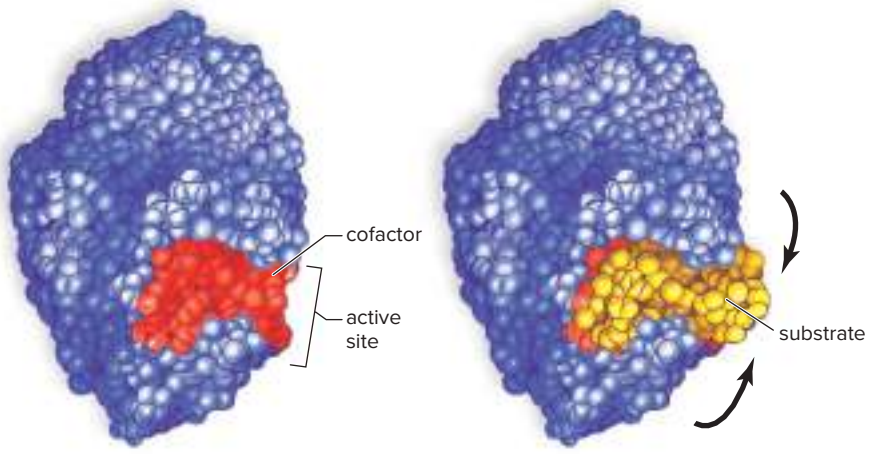


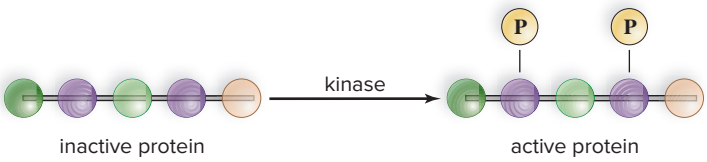
Figure 6.10 Cofactors at active site. **a.** Cofactors, including inorganic ions and organic coenzymes, may participate in the reaction at (**b**) the active site.

atoms to the reactions. Examples of these are NAD^+ (nicotinamide adenine dinucleotide), FAD (flavin adenine dinucleotide), and $NADP^+$ (nicotinamide adenine dinucleotide phosphate), each of which plays a significant role in either cellular respiration or photosynthesis.

Vitamins are often components of coenzymes. **Vitamins** are relatively small, organic molecules that are required in trace amounts in our diet and in the diets of other animals for synthesis of coenzymes. The vitamin becomes part of a coenzyme's molecular structure. For example, the vitamin niacin is part of the coenzyme NAD^+, and riboflavin (B_2) is a component of the coenzyme FAD. If a vitamin is not available, enzymatic activity will decrease, and the result will be a vitamin-deficiency disorder. In humans, a niacin deficiency results in a skin disease called pellagra, and riboflavin deficiency results in cracks at the corners of the mouth.

Enzyme Activation

Not all enzymes are needed by the cell all the time. Genes can be turned on to increase the concentration of an enzyme in a cell or turned off to decrease the concentration. But enzymes can also be present in the cell in an inactive form. Activation of enzymes occurs in many different ways. Some enzymes are covalently modified by the addition or removal of phosphate groups. An enzyme called a *kinase* adds phosphates to proteins, as shown below, and an enzyme called a phosphatase removes them. In some proteins, adding phosphates activates them; in others, removing phosphates activates them. Enzymes can also be activated by cleaving or removing part of the protein, or by associating with another protein or cofactor.



Enzyme Inhibition

Sometimes it is necessary to limit the activity of an enzyme. **Enzyme inhibition** occurs when a molecule (the inhibitor) binds to an enzyme and decreases its activity. In Figure 6.11, F is the end product of a metabolic pathway that can act as an inhibitor. This type of inhibition is beneficial, because once sufficient end product is present, inhibiting further production can conserve raw materials and energy.

Figure 6.11 also illustrates **noncompetitive inhibition,** because the inhibitor (F, the end product) binds to the enzyme E_1 at

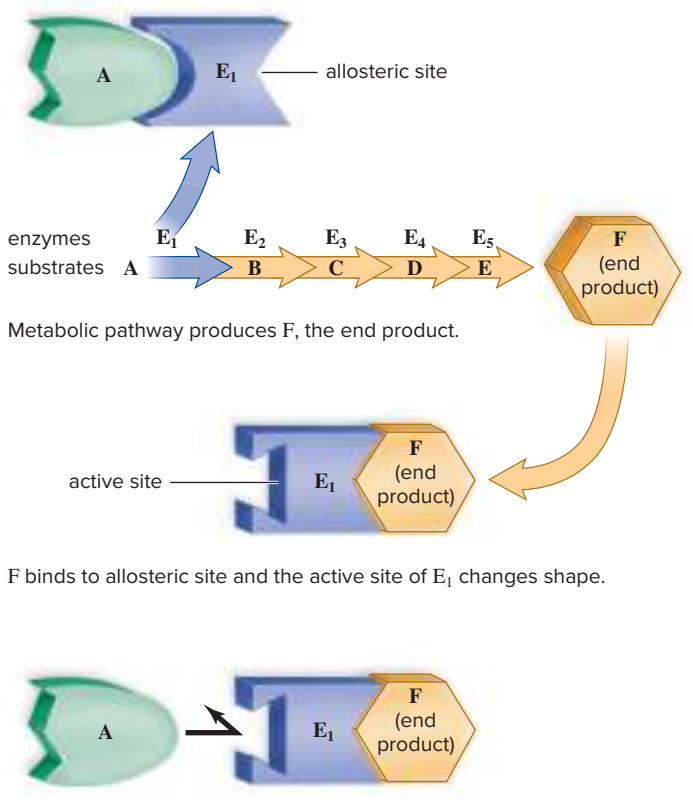


Figure 6.11 Noncompetitive inhibition of an enzyme. In the pathway, A–E are substrates, E_1–E_5 are enzymes, and F is the end product of the pathway that inhibits enzyme E_1. This negative feedback is useful, because it prevents wasteful production of product F when it is not needed.

ISTUDY

THEME Nature of Science

Carbon Monoxide Can Lead to Enzyme Inhibition and Even Death

Enzymes are protein molecules in the cell that function as catalysts in biological reactions. The active site of an enzyme enables it to bind to a specific substrate, increasing the speed of chemical reactions in living organisms. Fortunately, enzymes are not altered during the reaction and can be used over and over to facilitate the same chemical reaction. Changes in temperature and pH can denature an enzyme, altering its shape, which makes it unable to bind to the substrate. This essentially shuts down the biological reaction.

Enzyme inhibition occurs when an enzyme is blocked by a substance that prevents it from binding with their particular substrate. This will also shut down a biological reaction. Competitive inhibition occurs when an inhibitor binds to the enzyme's active site, preventing the substrate from binding with the enzyme. Noncompetitive inhibition occurs when the inhibitor binds to a location other than the active site. This alters the enzyme's shape and prevents it from binding with the substrate.

Cytochrome c oxidase is an enzyme that plays a critical role in the electron transport pathway in the mitochondria of the respiratory system. Inhibition of this enzyme will prevent the respiratory cells from making enough ATP to survive. The protein hemoglobin, which is found in our red blood cells, is responsible for transporting oxygen from the lungs to the cells in our body. If carbon monoxide (CO) is absorbed by the bloodstream, it can inhibit the ability of the cytochrome c oxidase enzyme to bind with its substrate. CO can also bind with the hemoglobin molecule, displacing oxygen in the red blood cells. This inhibition prevents the bloodstream from transporting oxygen, leading to headaches, dizziness, nausea, and even a loss of consciousness. High levels of carbon monoxide can lead to a person's death.

Carbon monoxide is a by-product of fossil fuel combustion. Every fall, when we use our furnaces to heat our homes, we need to be aware of the potential for carbon monoxide poisoning. Because carbon monoxide is colorless and odorless, we must rely on a carbon monoxide detector (Fig. 6A) to alert us if the CO levels in our homes are becoming unsafe.

Maintaining optimal temperature and pH, and avoiding enzyme inhibitors, are critical if we wish to remain healthy.

Questions to Consider

1. Should the landlord, tenant, or furnace company be responsible for ensuring that a home has a working CO detector?
2. If you believe you have CO poisoning, what should you do immediately to prevent further symptoms?

Figure 6A Carbon monoxide (CO) detector. Carbon monoxide can bind to hemoglobin, blocking the attachment of oxygen, which can lead to a person's death. Otto Pleska/123RF

a location other than the active site. The site is called an **allosteric site.** When an inhibitor is at the allosteric site, the active site of the enzyme changes shape, which in turn changes its function.

In Figure 6.11, the enzyme E_1 is inhibited because it is unable to bind to A, its substrate. The inhibition of E_1 means that the metabolic pathway is inhibited and no more end product is produced, until conditions change and more end product is needed.

In contrast to noncompetitive inhibition, **competitive inhibition** occurs when an inhibitor and the substrate compete against each other for the active site of an enzyme. Product forms only when the substrate, not the inhibitor, is at the active site. In this way, the amount of product is regulated.

Normally, enzyme inhibition is reversible, and the enzyme is not damaged by being inhibited. When enzyme inhibition is irreversible, the inhibitor permanently inactivates or destroys an enzyme. The Nature of Science feature, "Carbon Monoxide Can Lead to Enzyme Inhibition and Even Death," discusses how enzyme inhibition can shut down metabolic pathways, resulting in death.

Check Your Progress 6.3

1. Explain how enzymes are involved in metabolic pathways.
2. Describe how an enzyme interacts with a substrate to reduce the energy of activation.
3. List the environmental conditions that may influence enzyme activity.

6.4 Oxidation-Reduction Reactions and Metabolism

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how the reactions for photosynthesis and cellular respiration represent oxidation-reduction reactions.
2. Summarize the relationship between the metabolic reactions of photosynthesis and cellular respiration.

In the next two chapters, you will explore two important metabolic pathways: photosynthesis (see Fig. 7.6) and cellular respiration (see Fig. 8.1). Both of these pathways are based on the use of special enzymes to facilitate the movement of electrons. The movement of these electrons plays a major role in the energy-related reactions associated with these pathways.

Oxidation-Reduction Reactions

When oxygen (O) combines with a metal such as iron or magnesium (Mg), oxygen receives electrons and becomes an ion that is negatively charged. The metal loses electrons and becomes an ion that is positively charged. When magnesium oxide (MgO) forms, it is appropriate to say that magnesium has been oxidized. On the other hand, oxygen has been reduced, because it has gained negative

charges (i.e., electrons). Reactions that involve the gain and loss of electrons are called oxidation-reduction reactions. Sometimes, the terms *oxidation* and *reduction* are applied to other reactions, whether or not oxygen is involved. In a discussion of metabolic reactions, oxidation represents the loss of electrons, and reduction is the gain of electrons. In the reaction Na + Cl →NaCl, sodium has been oxidized (loss of electron), and chlorine has been reduced (gain of electrons). Because oxidation and reduction go hand-in-hand, the entire reaction is called a **redox reaction.** One easy way to remember what is happening in redox reactions is to remember the term OIL RIG:

OIL RIG

Oxidation **I**s **L**oss **R**eduction **I**s **G**ain

The terms *oxidation* and *reduction* also apply to covalent reactions in cells. In this case, however, oxidation is the loss of hydrogen atoms ($e^- + H^+$), and reduction is the gain of hydrogen atoms. Notice that when a molecule loses a hydrogen atom it has lost an electron, and when a molecule gains a hydrogen atom it has gained an electron. This form of oxidation-reduction is exemplified in the overall equations for photosynthesis and cellular respiration.

Chloroplasts and Photosynthesis

The chloroplasts in plants capture solar energy and use it to convert water and carbon dioxide into a carbohydrate. The overall equation for photosynthesis can be written like this:

$$\underset{\text{light energy}}{} + \underset{\text{carbon dioxide}}{6\ CO_2} + \underset{\text{water}}{6\ H_2O} \longrightarrow \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{oxygen}}{6\ O_2}$$

This equation shows that, during photosynthesis, hydrogen atoms are transferred from water to carbon dioxide to form glucose. In this reaction, therefore, carbon dioxide has been reduced and water has been oxidized. It takes energy to reduce carbon dioxide to glucose, and this energy is supplied by solar energy. Chloroplasts are able to capture solar energy and convert it to the chemical energy of ATP, which is used along with hydrogen atoms to reduce carbon dioxide. Oxygen is a by-product that is released (Fig. 6.12).

The reduction of carbon dioxide to form a mole of glucose will store 686 kcal in the chemical bonds of glucose. This is the energy that living organisms utilize to support themselves only because carbohydrates (and other nutrients) can be oxidized in mitochondria.

Mitochondria and Cellular Respiration

Mitochondria, present in both plants and animals, oxidize carbohydrates and use the released energy to build ATP molecules (Fig. 6.12). Cellular respiration therefore consumes oxygen and produces carbon dioxide and water, the very molecules taken up by chloroplasts. The overall equation for cellular respiration is the opposite of the one we used to represent photosynthesis:

$$\underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{oxygen}}{6\ O_2} \longrightarrow \underset{\text{carbon dioxide}}{6\ CO_2} + \underset{\text{water}}{6\ H_2O} + \text{energy}$$

In this reaction, glucose has lost hydrogen atoms (been oxidized), and oxygen has gained hydrogen atoms (been reduced). When oxygen gains electrons, it becomes water. The complete oxidation of a mole of glucose releases 686 kcal of energy, and some of this energy is used to synthesize ATP molecules. If the energy within glucose were released all at once, most of it would

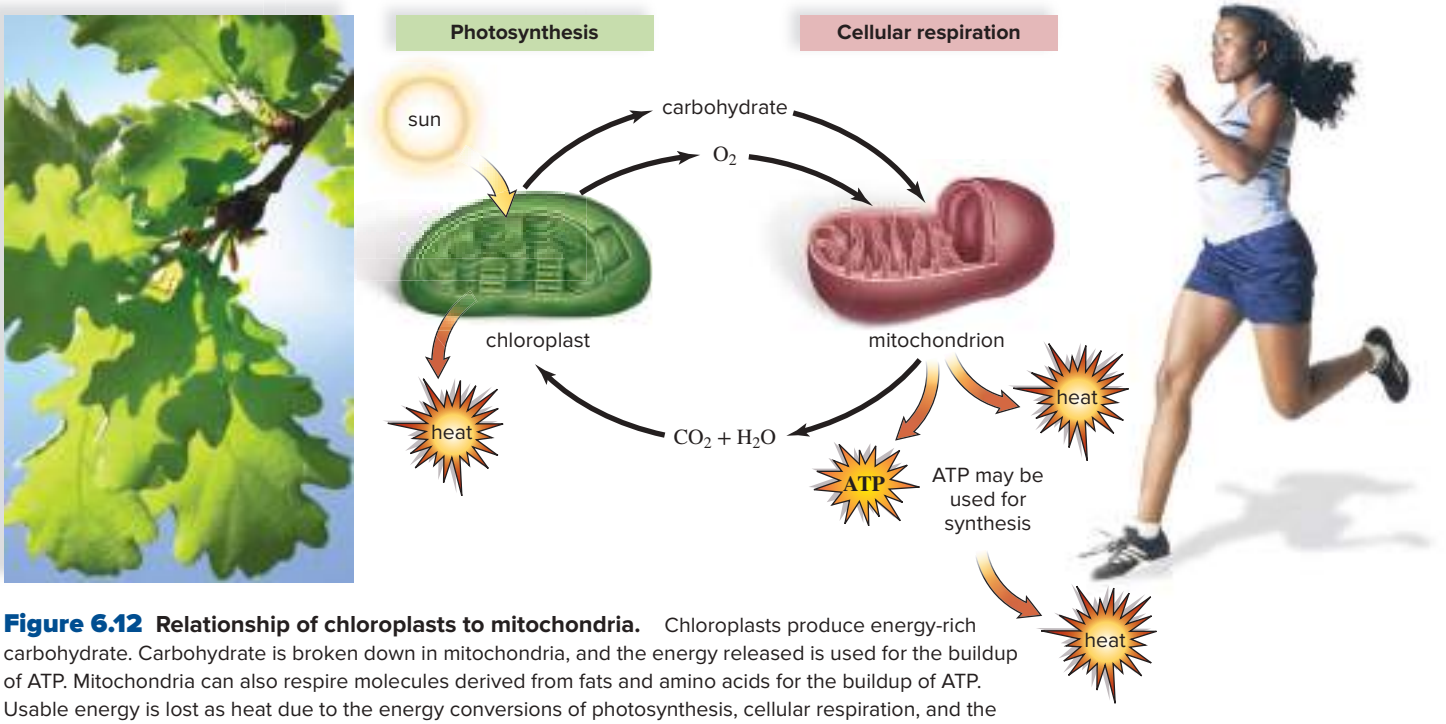


Figure 6.12 Relationship of chloroplasts to mitochondria. Chloroplasts produce energy-rich carbohydrate. Carbohydrate is broken down in mitochondria, and the energy released is used for the buildup of ATP. Mitochondria can also respire molecules derived from fats and amino acids for the buildup of ATP. Usable energy is lost as heat due to the energy conversions of photosynthesis, cellular respiration, and the use of ATP in the body. (leaves): Olliig/Shutterstock; (runner): Alan Bailey/Rubberball Productions/Getty Images

ISTUDY

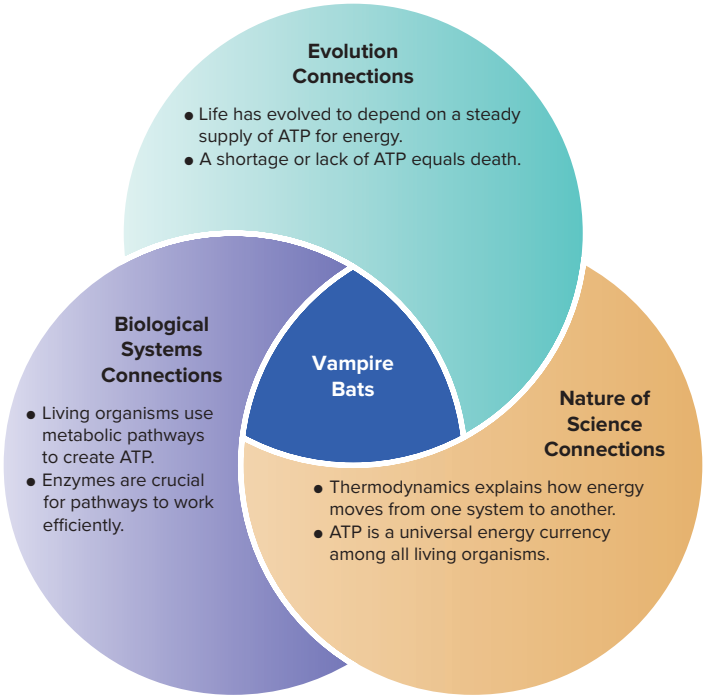
dissipate as heat instead of some of it being used to produce ATP. Instead, cells oxidize glucose step by step. The energy is gradually stored and then converted to that of ATP molecules, which is used in animals in the many ways listed in Figure 6.12.

Figure 6.12 shows us very well that chloroplasts and mitochondria are involved in a cycle. The carbohydrate produced within the chloroplasts becomes a substrate for the cellular respiration reaction that occurs in the mitochondria, while carbon dioxide released by mitochondria becomes a substrate for the photosynthesis reaction that occurs in the chloroplasts. These organelles are involved in a redox cycle, because carbon dioxide is reduced during photosynthesis and carbohydrate is oxidized during cellular respiration. Note that energy does not cycle between the two organelles. Instead, it flows from the sun through each step of photosynthesis and cellular respiration until it eventually is released as unusable heat, while ATP is used by the cell.

Check Your Progress 6.4

1. Compare the role of carbon dioxide in photosynthesis and cellular respiration.
2. Distinguish how energy from electrons is used to establish an electrochemical gradient in chloroplasts and mitochondria.

Connecting *the* Concepts



What Did You Learn?

6.1 Cells and the Flow of Energy

Energy is the ability to do work or bring about a change. Energy may exist in several forms.

- **Kinetic energy** is the energy of motion, and **potential energy** is stored energy.
- **Chemical energy** is a form of potential energy that is stored within the chemical bonds of molecules.

Two **laws of thermodynamics** are basic to understanding energy in biological systems.

- The first law of thermodynamics states that energy cannot be created or destroyed but can only be transferred or transformed.
- The second law of thermodynamics states that one usable form of energy cannot be completely converted into another usable form, such as **heat.** As a result of these laws, we know that the **entropy** of the universe is increasing and that only a flow of energy from the sun maintains the organization of living organisms.

6.2 Metabolic Reactions and Energy Transformations

The term **metabolism** encompasses all the chemical reactions occurring in a cell. Considering individual reactions, only those that result in a negative **free-energy** difference—that is, the **products** have less usable energy than the **reactants**—occur spontaneously. Such reactions, called exergonic reactions, release energy.

An **endergonic** reaction, which requires an input of energy, occurs only when coupled to an exergonic process. The energy released from the exergonic reaction must exceed the net energy cost of the endergonic reaction in order for the overall reaction to proceed. Hydrolysis of ATP, an **exergonic** reaction, is commonly used to drive energetically unfavorable metabolic reactions.

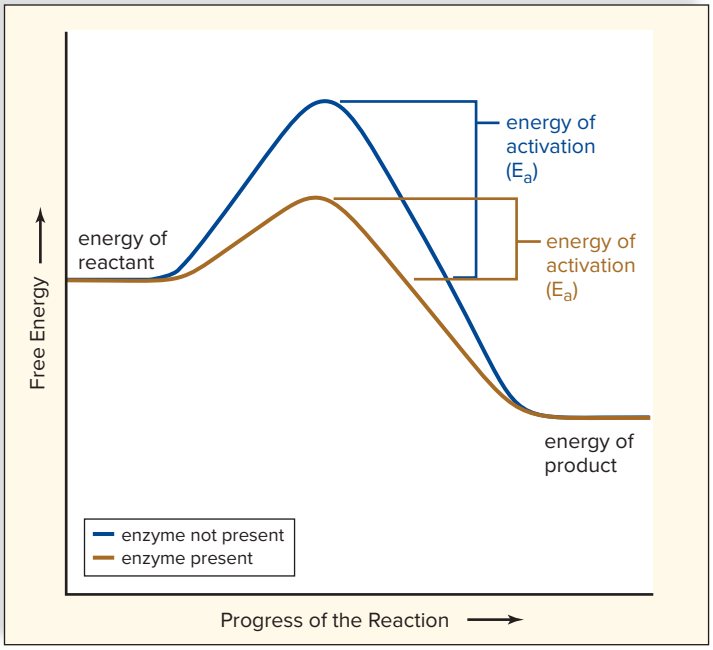
The energy currency of the cell is **ATP (adenosine triphosphate).** As cells use ATP for cellular reactions, **ADP (adenosine diphosphate)** is generated. Metabolic reactions in the cell can break down glucose and other macromolecules to release energy and regenerate ATP.

6.3 Metabolic Pathways and Enzymes

A **metabolic pathway** is a series of reactions that proceeds in an orderly, step-by-step manner. **Enzymes** speed reactions by lowering the **energy**

ISTUDY

of activation when they form a complex with their **substrates.** The **induced fit model** explains how the substrate interacts with the **active site** of the enzyme. The three-dimensional shape of the enzyme is important. When **denaturation** occurs, an enzyme may not be able to interact as efficiently with the substrate. Most enzymes are proteins, but some, called **ribozymes,** may be RNA molecules.



Generally, enzyme activity increases as substrate concentration increases; once all active sites are filled, maximum reaction rate has been achieved. Environmental factors, such as temperature or pH, can affect the shape of an enzyme and therefore its function. Many enzymes need **cofactors** or coenzymes to carry out their reactions. **Vitamins** are often used to build **coenzymes,** such as NAD^+, FAD, and $NADP^+$.

Enzyme inhibition regulates the activity of many metabolic pathways. In addition, **noncompetitive inhibition,** where a molecule binds to an allosteric site on the enzyme, may influence enzyme behavior. In **competitive inhibition,** an inhibitor interferes with the binding of the substrate on the active site of an enzyme.

6.4 Oxidation-Reduction Reactions and Metabolism

The overall equation for photosynthesis is the opposite of that for cellular respiration. Both processes involve oxidation-reduction reactions. **Redox reactions** are a major way in which energy is transformed in cells.

OIL RIG

Oxidation **I**s **L**oss **R**eduction **I**s **G**ain

During photosynthesis, carbon dioxide is reduced to glucose, and water is oxidized. Glucose formation requires energy, and this energy comes from the sun. Chloroplasts capture solar energy and convert it to the chemical energy of ATP molecules, which are used along with hydrogen atoms to reduce carbon dioxide to glucose. During cellular respiration, glucose is oxidized to carbon dioxide, and oxygen is reduced to water. This reaction releases energy, which is used to synthesize ATP molecules in all types of cells. Energy flows through all living organisms. Photosynthesis is a metabolic pathway in chloroplasts that converts solar energy to the chemical energy within carbohydrates, and cellular respiration is a metabolic pathway completed in mitochondria that converts this energy into that of ATP molecules. Eventually, the energy within ATP molecules becomes heat.

Cellular respiration is an aerobic process that requires oxygen and gives off carbon dioxide. Cellular respiration involves oxidation. The air we inhale contains the oxygen, and the food we digest after eating contains the carbohydrate glucose needed for cellular respiration.

Thinking Critically About Biology

1. Enzymes help cells minimize the loss of usable energy that occurs as energy is transformed from one form to another. If enzymes did not have this ability, how might it affect the operation of cells and living organisms in general?
2. Entropy is increased when nutrients break down, so why are enzymatic metabolic pathways required for cellular respiration?
3. Plasminogen is an enzyme that needs to be converted into plasmin so it can dissolve a fibrin blood clot. Vampire bats produce an enzyme that is 150 times more potent at converting plasminogen to plasmin. Explain what information researchers would need to know about the vampire bat enzyme in order to synthesize it in the laboratory.
4. How might a knowledge of the structure of an active site of an enzyme allow you to build a drug to regulate a metabolic pathway?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Energy Part I: Energy Transfers

McGraw Hill

Explain how both laws of thermodynamics apply to the experiments in this video.

Making It Relevant

1. How is the action of fibrin, the protein that causes blood to clot, advantageous to the survival of humans and animals? How is the action of the anti-clotting compound in vampire bat blood advantageous to the bat's survival?
2. What precautions might need to be taken if a person was given the anti-clotting compound from vampire bat saliva?
3. Is it ethical to source drugs from animals if they provide a medical benefit to people?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

7

Photosynthesis

Geert Vanden Wijngaert/Bloomberg/Getty Images

Improving Photosynthesis

Photosynthesis is a very inefficient process. A crop plant like soybean can convert only about 1 percent of the sun's energy into growing and producing food people can eat. Scientists with the Realizing Increased Photosynthetic Efficiency (RIPE) project are trying to improve that number to help feed the world.

Photosynthesis involves over 150 chemical steps and hundreds of different genes in a plant. The RIPE project is altering those genes to create genetically modified plants that can produce more food. For example, RIPE scientists modified three genes that control how a plant adjusts photosynthesis between sunlight and shade conditions. The natural process is slow and causes a leaf to have several minutes of reduced photosynthesis every time it becomes shaded. When they modified the genes to create a faster response, the soybean plants increased their yield by 20% over an entire growing season.

RIPE is especially focused on providing improved crops for sub-Saharan Africa, a region with chronic food shortages. Genetic modifications to improve photosynthesis could increase food without using more water or clearing more land for agriculture. However, many countries are hesitant to allow genetically modified plants.

As you read through this chapter, think about the following questions:

1. How does a plant capture solar energy during photosynthesis?
2. What chemicals are created in photosynthesis that can be used for energy and growth?

Chapter Outline

7.1 Photosynthetic Organisms
7.2 The Process of Photosynthesis
7.3 Plants Convert Solar Energy
7.4 Plants Fix Carbon Dioxide
7.5 Other Types of Photosynthesis

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 6.1 How does energy flow in biological systems?

Section 6.3 What role do enzymes play in regulating metabolic processes?

Section 6.4 How are redox reactions and membranes used to conduct cellular work?

Following *the* Themes

CHAPTER 7 – PHOTOSYNTHESIS

Evolution	Plants have adapted and spread to nearly every environment and climate on Earth.
Nature of Science	Studying biological processes like photosynthesis can lead to innovations that benefit people.
Biological Systems	Ecosystems on Earth depend on plants converting solar energy to chemical energy.

ISTUDY

7.1 Photosynthetic Organisms

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how autotrophs are able to produce their own food.
2. Describe the components of a chloroplast.
3. Compare the roles of oxygen and carbon dioxide in autotrophs and heterotrophs.

Photosynthesis converts solar energy into the chemical energy of a carbohydrate. Photosynthetic organisms, including plants, algae, and cyanobacteria, are called **autotrophs,** because they produce their own food (Fig. 7.1). It has been estimated that all of the world's green organisms together produce between 100 billion and 200 billion metric tons of sugar each year. Imagine enough sugar cubes to re-create the volume of 2 million Empire State Buildings!

No wonder photosynthetic organisms are able to sustain themselves and all other living organisms on Earth. With few exceptions, it is possible to trace any food chain back to plants and algae. In other words, producers, which have the ability to synthesize carbohydrates, feed not only themselves but also consumers, which must take in preformed organic molecules. Collectively, consumers are called **heterotrophs.** Both autotrophs and heterotrophs use organic molecules produced by photosynthesis as building blocks for growth and repair and as a source of chemical energy for cellular work.

Photosynthesizers also produce copious amounts of oxygen gas (O_2) as a by-product. Oxygen, required by organisms for cellular respiration, rises high into the atmosphere where it forms an ozone shield that filters out ultraviolet radiation and makes terrestrial life possible.

Oil and coal provide over 80% of the energy needed to power vehicles, factories, computers, and a multitude of electrically energized appliances. The energy within that oil and coal was originally captured from the sun by plants and algae millions of years ago—thus the name "fossil fuels." Today's trees are also commonly used as fuel. The fermentation of plant materials produces ethanol, which can be used to fuel automobiles directly or as a gasoline additive.

The products of photosynthesis are critical to humankind in a number of other ways. They serve as a source of building materials, fabrics, paper, and pharmaceuticals. Of course, we also appreciate green plants for the simple beauty of an orchid blossom, the scent of a rose, or the majesty of the forests.

Photosynthesis in Flowering Plants

Photosynthesis takes place in the green portions of plants. The leaves of a flowering plant contain mesophyll tissue, in which cells are specialized for photosynthesis (Fig. 7.2). The raw materials for photosynthesis are water and carbon dioxide. The roots of a plant absorb water, which then moves in vascular tissue up the stem to a leaf by way of the leaf veins. Carbon dioxide in the air enters a leaf through small openings called **stomata** (*sing.,* stoma). After entering a leaf, carbon dioxide and water diffuse into **chloroplasts** (Gk. *chloros,* "green"; *plastos,* "formed, molded"), the organelles that carry out photosynthesis.

A double membrane surrounds a chloroplast, and its semifluid interior is called the **stroma** (Gk. *stroma,* "bed, mattress"). A different membrane system within the stroma forms flattened sacs called **thylakoids** (Gk. *thylakos,* "sack"), which in some places are stacked to form **grana** (*sing.,* granum). The space of each thylakoid is thought to be connected to the space of every other thylakoid within a chloroplast, thereby forming an inner compartment within chloroplasts, called the thylakoid space. Overall, chloroplast membranes provide a tremendous surface area for photosynthesis to occur.

The thylakoid membrane contains **chlorophyll** and other pigments that are capable of absorbing the solar energy that drives photosynthesis. The stroma contains an enzyme-rich solution, where carbon dioxide is first attached to an organic compound and then reduced to a carbohydrate.

Humans and other respiring organisms release carbon dioxide into the air. Some of the same carbon dioxide molecules enter a leaf through the stoma and are converted to carbohydrate. Carbohydrate, in the form of glucose, is the chief source of chemical energy for most organisms. Thus, an interdependent relationship exists between organisms that make their own food (autotrophs) and those that consume their food (heterotrophs) (see Fig. 7.3).

Oscillatoria 40× Kelp Sequoias

Figure 7.1 Photosynthetic organisms. Photosynthetic organisms (autotrophs) include cyanobacteria (*left*); algae, such as kelp (*middle*); and plants (*right*). (Oscillatoria): Sinclair Stammers/Science Photo Library/Getty Images; (kelp): Chuck Davis/The Image Bank/Getty Images; (sequoias): Dynamic Graphics Group/Creatas Images/Alamy Stock Photo

ISTUDY

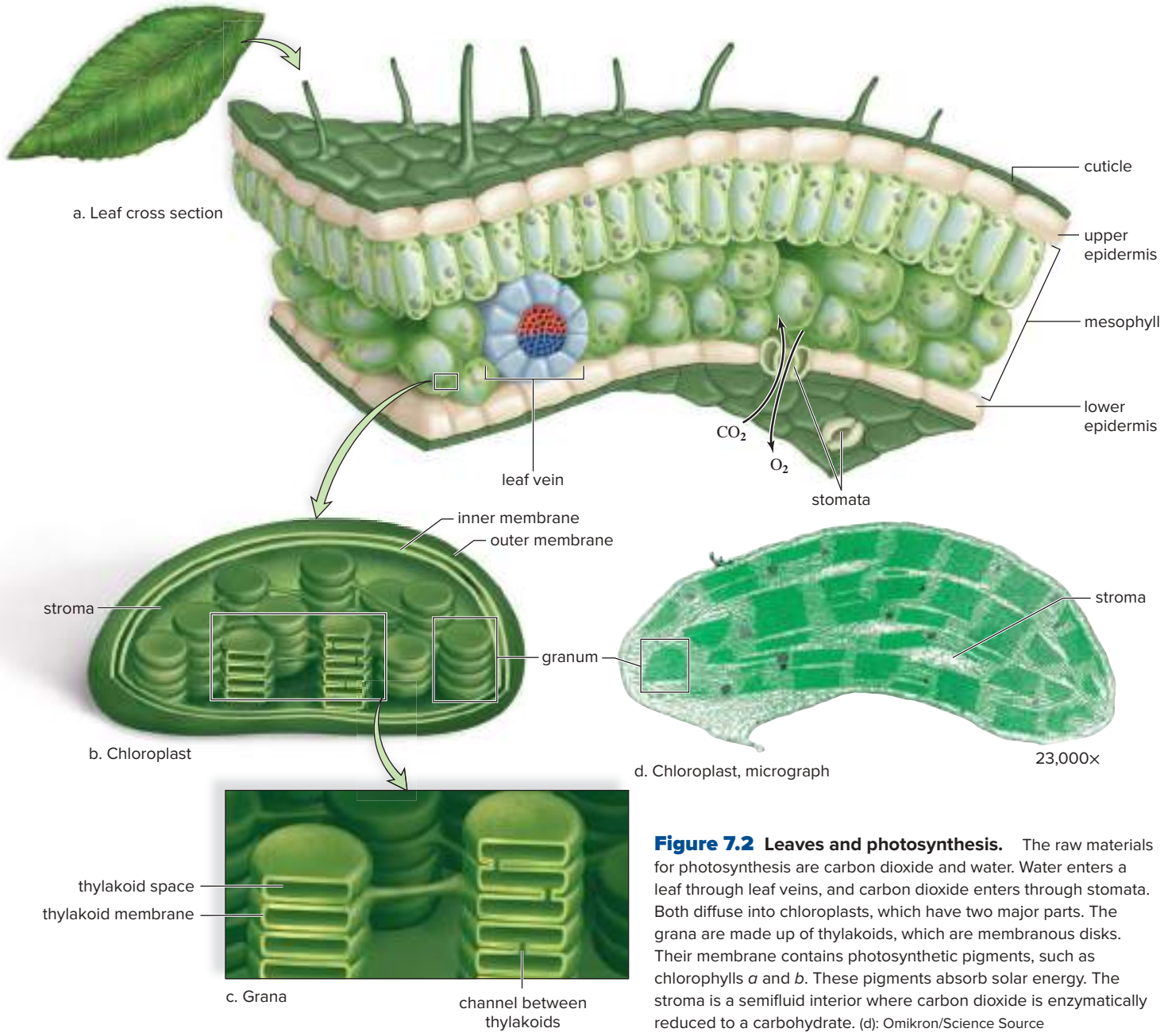


Figure 7.2 Leaves and photosynthesis. The raw materials for photosynthesis are carbon dioxide and water. Water enters a leaf through leaf veins, and carbon dioxide enters through stomata. Both diffuse into chloroplasts, which have two major parts. The grana are made up of thylakoids, which are membranous disks. Their membrane contains photosynthetic pigments, such as chlorophylls *a* and *b*. These pigments absorb solar energy. The stroma is a semifluid interior where carbon dioxide is enzymatically reduced to a carbohydrate. (d): Omikron/Science Source

Check Your Progress 7.1

1. Which process do autotrophs use to produce food?
2. Distinguish the part of a chloroplast that absorbs solar energy from the part that forms a carbohydrate.
3. List the products autotrophs provide heterotrophs through photosynthesis.

7.2 The Process of Photosynthesis

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the overall process of photosynthesis.
2. Compare energy input and output of the light reaction.
3. Compare carbon input and output of the Calvin cycle reaction.

The overall process of photosynthesis can be represented by an equation:

$$6\,CO_2 + 12\,H_2O \xrightarrow{\text{solar energy}} 6(CH_2O) + 6\,H_2O + 6\,O_2$$

In this equation, CH_2O represents carbohydrate. If the carbohydrate were multiplied by 6, it would become $C_6H_{12}O_6$, or glucose.

The overall equation implies that photosynthesis involves oxidation-reduction (redox) and the movement of electrons from one molecule to another. Recall that oxidation is the loss of electrons, and reduction is the gain of electrons. In living organisms, as discussed in Section 6.4, the electrons are very often accompanied by hydrogen ions, so that oxidation is the loss of hydrogen atoms ($H^+ + e^-$) and reduction is the gain of hydrogen atoms. This

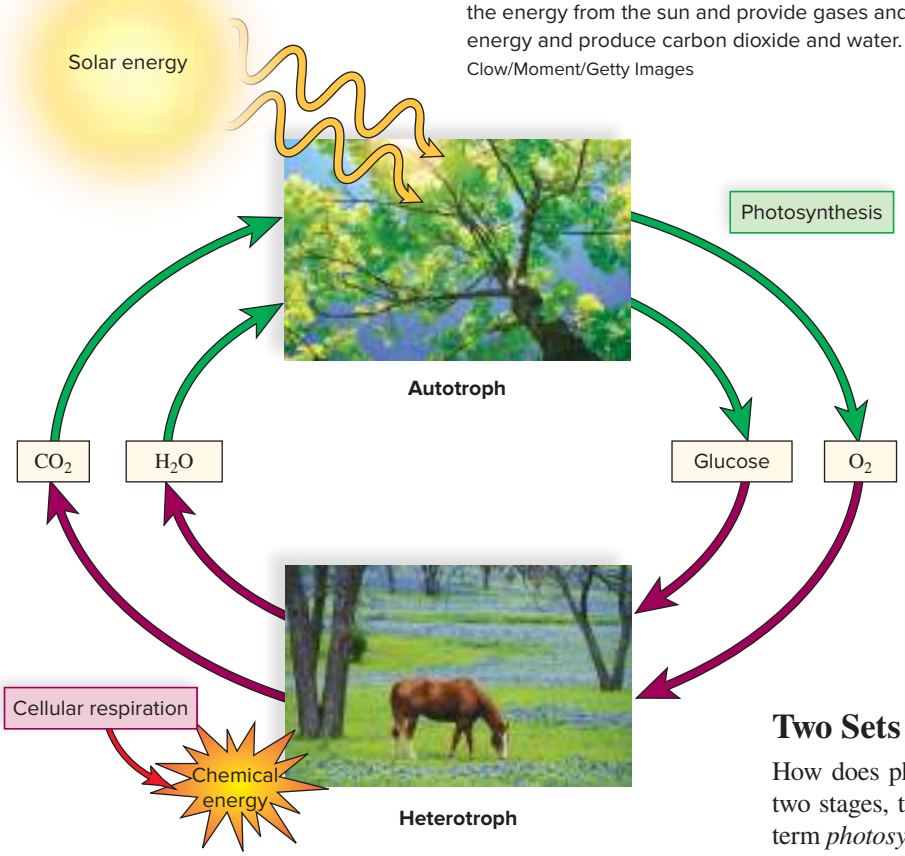
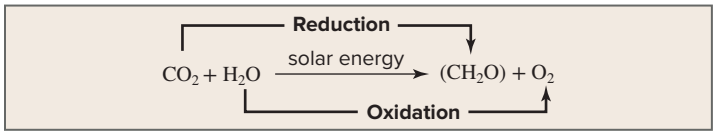


Figure 7.3 Autotrophs and the relationship to heterotrophs. Photosynthetic organisms harness the energy from the sun and provide gases and nutrients for heterotrophs. Heterotrophs generate chemical energy and produce carbon dioxide and water. (photos): (autotroph): Don Hammond/Design Pics; (heterotroph): Jeff R Clow/Moment/Getty Images

simplified rewrite of the equation makes it clear that carbon dioxide has been reduced and water has been oxidized:



It takes hydrogen atoms and a lot of energy to reduce carbon dioxide. From your study of energy and enzymes in Section 6.2, you might expect that solar energy is not used directly during photosynthesis; rather, it is converted to ATP molecules. ATP is the energy currency of cells and, when cells need something, they spend ATP. In this case, solar energy is used to generate the ATP needed to reduce carbon dioxide to a carbohydrate. Of course, this carbohydrate represents the food produced by plants, algae, and cyanobacteria that feeds the biosphere.

$NADP^+$/NADPH

A review of Section 6.4 will also lead you to suspect that the electrons needed to reduce carbon dioxide are carried by a coenzyme. $NADP^+$ is the coenzyme of oxidation-reduction (redox coenzyme) active during photosynthesis. When $NADP^+$ is reduced, it has accepted two electrons and one hydrogen atom, and when NADPH is oxidized, it gives up its electrons:

$$NADP^+ + 2\,e^- + H^+ \rightarrow NADPH$$

What molecule supplies the electrons that reduce $NADP^+$ during photosynthesis? Put a sprig of *Elodea* in a beaker and supply it with light, and you will observe a bubbling (Fig. 7.4). The bubbling occurs because the plant is releasing oxygen as it photosynthesizes.

A famous experiment performed by C. B. van Niel of Stanford University found that the oxygen given off by photosynthesizers comes from water. Van Niel performed two separate experiments. When an isotope of oxygen, ^{18}O, was a part of water, the O_2 given off by the plant contained ^{18}O. When ^{18}O was a part of carbon dioxide supplied to a plant, the O_2 given off by a plant did not contain the ^{18}O. Why not? Because oxygen released by the plant doesn't come from carbon dioxide; it comes from water. This was the first step toward discovering that water splits during photosynthesis. When water splits, oxygen is released and the hydrogen atoms ($H^+ + e^-$) are taken up by $NADP^+$. Later, NADPH reduces carbon dioxide to a carbohydrate.

Two Sets of Reactions

How does photosynthesis occur? The process can be divided into two stages, the light reactions and the Calvin cycle reactions. The term *photosynthesis* comes from the associations between these two stages: The prefix *photo* refers to the light reactions that capture the waves of sunlight needed for the *synthesis* of carbohydrates occurring in the Calvin cycle. The light reactions take place on thylakoids, and the Calvin cycle takes place in the stroma.

Light Reactions

The **light reactions** are so named because they occur only when the sun is out. The green pigment chlorophyll, present in thylakoid membranes, is largely responsible for absorbing the solar energy that drives photosynthesis.

Figure 7.4 Photosynthesis releases oxygen. Bubbling indicates that the aquatic plant *Elodea* releases O_2 gas when it photosynthesizes. Nigel Cattlin/Science Source

ISTUDY

During the light reactions, solar energy energizes electrons, which move down an electron transport chain. As the electrons move down the chain, which consists of proteins in the cell membrane (see Fig. 7.11), energy is released and captured to produce ATP molecules. Energized electrons are also taken up by $NADP^+$, which is reduced and becomes NADPH. This equation can be used to summarize the light reactions, because during the light reactions solar energy is converted to chemical energy:

$$\text{solar energy} \longrightarrow \text{chemical energy (ATP, NADPH)}$$

Calvin Cycle Reactions

The **Calvin cycle reactions** are named for Melvin Calvin, who in 1961 received a Nobel Prize in Chemistry for discovering the enzymatic reactions that reduce carbon dioxide to a carbohydrate in the stroma of chloroplasts (Fig. 7.5). The enzymes that speed the reduction of carbon dioxide during both day and night are located in the semifluid substance of the chloroplast stroma.

During the Calvin cycle reactions, CO_2 is taken up and then reduced to a carbohydrate that can later be converted to glucose. This equation can be used to summarize the Calvin cycle reactions, because during these reactions, the ATP and NADPH formed during the light reactions are used to reduce carbon dioxide:

$$\text{chemical energy (ATP, NADPH)} \longrightarrow \text{chemical energy (carbohydrate)}$$

Summary

Figure 7.6 summarizes our discussion so far and shows that, during the light reactions, (1) solar energy is absorbed, (2) water is split so that oxygen is released, and (3) ATP and NADPH are produced.

Figure 7.5 Melvin Calvin. Melvin Calvin, a chemist, is most noted for his work using a carbon 14 isotope to follow the route that carbon travels through a plant during photosynthesis. Georg Gerster/Science Source

During the Calvin cycle reactions, (1) CO_2 is absorbed and (2) reduced to a carbohydrate (CH_2O) by utilizing ATP and NADPH from the light reactions (bottom set of red arrows). The top set of red arrows takes ADP + Ⓟ and $NADP^+$ back to light reactions, where they become ATP and NADPH once more, so that carbohydrate production can continue.

Check Your Progress 7.2

1. List the reactants and products of photosynthesis.
2. How is energy transformed during the light reactions?

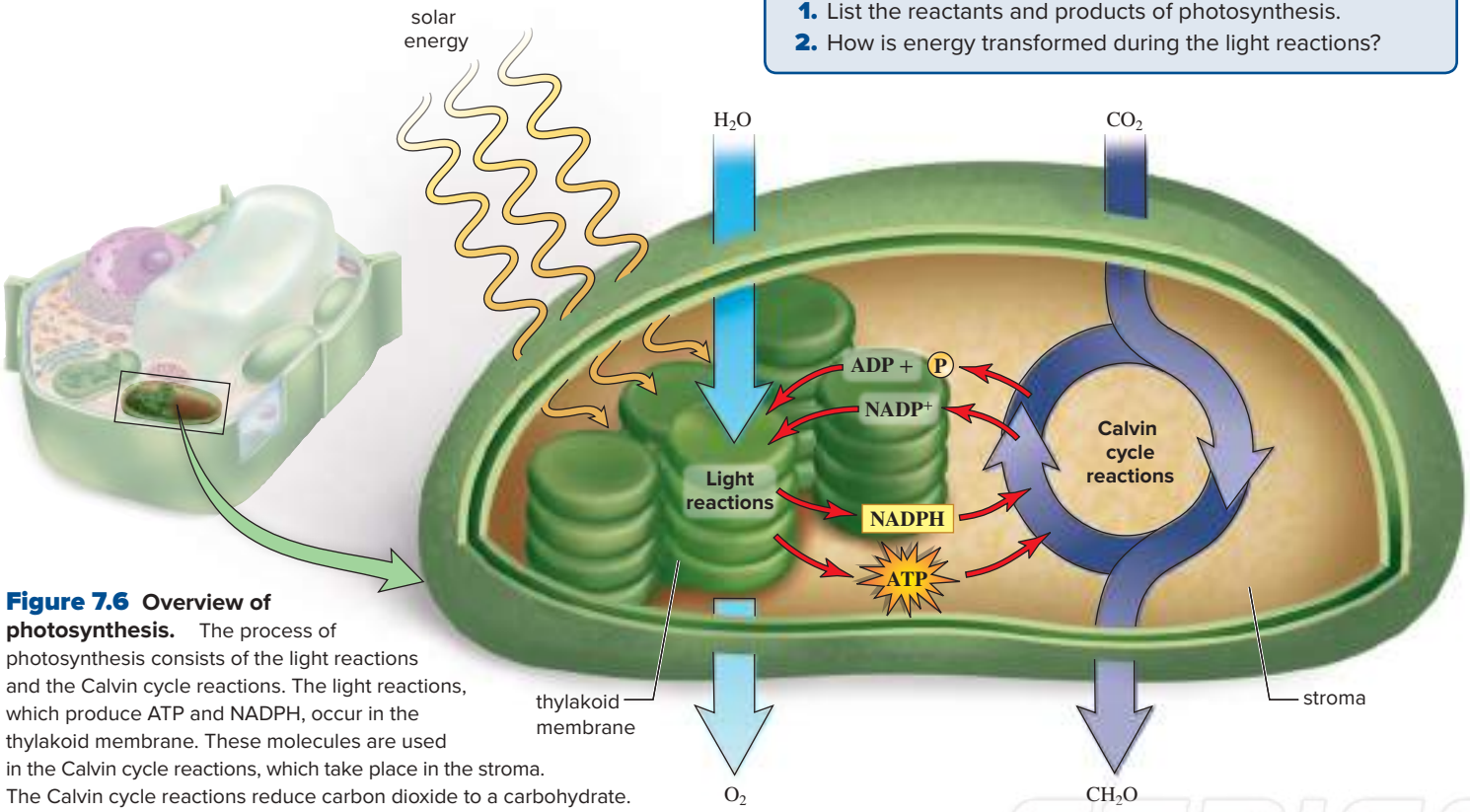


Figure 7.6 Overview of photosynthesis. The process of photosynthesis consists of the light reactions and the Calvin cycle reactions. The light reactions, which produce ATP and NADPH, occur in the thylakoid membrane. These molecules are used in the Calvin cycle reactions, which take place in the stroma. The Calvin cycle reactions reduce carbon dioxide to a carbohydrate.

ISTUDY

7.3 Plants Convert Solar Energy

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the relationship between wavelength and energy in the electromagnetic spectrum.
2. Explain the role of photosynthetic pigments in harnessing solar energy.
3. Examine how ATP and NADPH are produced from redox reactions and membrane gradients.

Solar energy can be described in terms of its wavelength and its energy content. Figure 7.7 shows the types of radiant energy from the shortest wavelength, gamma rays, to the longest, radio waves. Most of the radiation reaching the Earth is within the visible-light range. Higher-energy wavelengths are screened out by the ozone layer in the atmosphere before they reach the Earth's surface, and lower-energy wavelengths are screened out by water vapor and carbon dioxide. Because visible light is the most prevalent in the environment, organisms have evolved to use these wavelengths. For example, human eyes have cone cells that respond to color wavelengths, and plants have pigments that are energized by most of the same wavelengths (Fig. 7.7).

Pigments and Photosystems

Pigment molecules absorb wavelengths of light. Most pigments absorb only some wavelengths; they reflect or transmit the other wavelengths. The pigments in chloroplasts are capable of absorbing various portions of visible light. This is called their **absorption spectrum.**

Photosynthetic organisms differ in the type of chlorophyll they contain. In plants, chlorophyll *a* and chlorophyll *b* play prominent roles in photosynthesis. **Carotenoids** play an accessory role. Both chlorophylls *a* and *b* absorb violet, blue, and red light better than the light of other colors. Because green light is transmitted and reflected by chlorophyll, plant leaves appear green to us. In short, plants are green because they do *not* use the green wavelength! The carotenoids, which are shades of yellow and orange, are able to absorb light in the violet-blue-green range. These pigments become noticeable in the fall when chlorophyll breaks down.

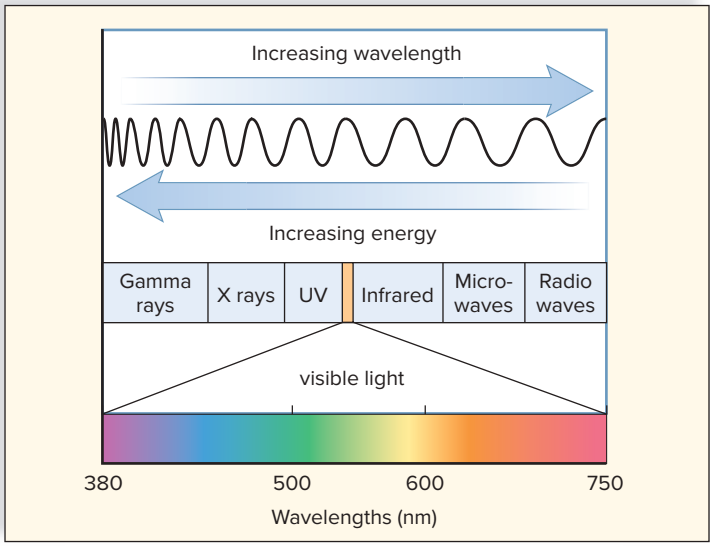
How do you determine the absorption spectrum of pigments? To identify the absorption spectrum of a particular pigment, a purified sample is exposed to different wavelengths of light inside an instrument called a spectrophotometer. A spectrophotometer measures the amount of light that passes through the sample, and from this it is possible to calculate how much was absorbed. The amount of light absorbed at each wavelength is plotted on a graph, and the result is a record of the pigment's absorption spectrum (Fig. 7.8). Notice the low absorbance reading for the green and yellow wavelengths and recall why plants are green.

A **photosystem** consists of a pigment complex (molecules of chlorophyll *a,* chlorophyll *b,* and the carotenoids) and electron acceptor molecules within the thylakoid membrane. The pigment complex serves as an "antenna" for gathering solar energy.

Electron Flow in the Light Reactions

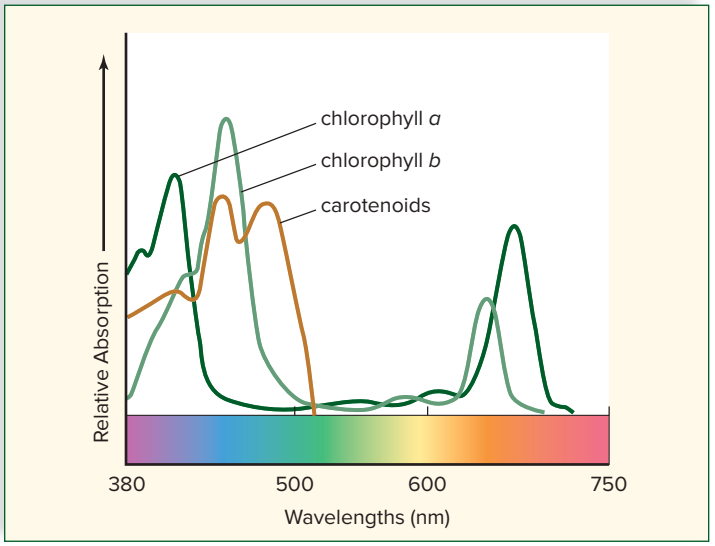
The light reactions utilize two photosystems, called photosystem I (PS I) and photosystem II (PS II). The photosystems are named for the order in which they were discovered, not for the order in which they occur in the thylakoid membrane or participate in the photosynthetic process.

During the light reactions, electrons usually, but not always, follow a **noncyclic pathway** that begins with photosystem II (Fig. 7.9). The pigment complex absorbs solar energy, which is then passed from one pigment to the other until it is concentrated



The electromagnetic spectrum includes visible light.

Figure 7.7 The electromagnetic spectrum. The electromagnetic spectrum extends from the very short wavelengths of gamma rays through the very long wavelengths of radio waves. Visible light, which drives photosynthesis, is expanded to show its component colors. Each color varies in its wavelength and energy content.



Absorption spectrum of photosynthetic pigments.

Figure 7.8 Photosynthetic pigments and photosynthesis. The photosynthetic pigments in chlorophylls *a* and *b* and the carotenoids absorb certain wavelengths within visible light. This is their absorption spectrum. Notice that they do not absorb green light. That is why the leaves of plants appear green to us. You observe the color that is not absorbed by the leaf.

ISTUDY

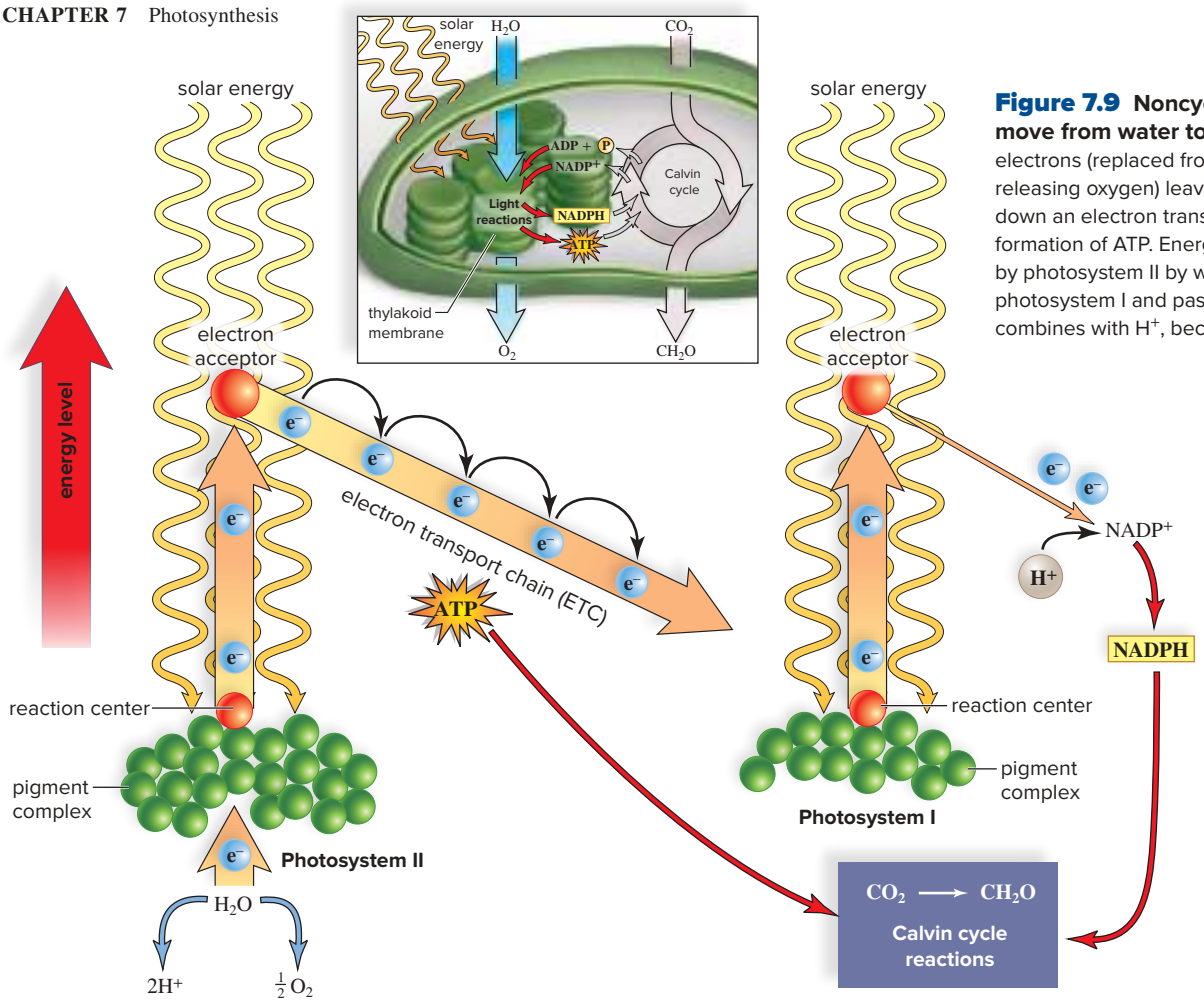


Figure 7.9 Noncyclic pathway: Electrons move from water to $NADP^+$. Energized electrons (replaced from water, which splits, releasing oxygen) leave photosystem II and pass down an electron transport chain, leading to the formation of ATP. Energized electrons (replaced by photosystem II by way of the ETC) leave photosystem I and pass to $NADP^+$, which then combines with H^+, becoming NADPH.

in a particular pair of chlorophyll *a* molecules, called the *reaction center.* Electrons (e^-) in the reaction center become so energized that they escape from the reaction center and move to nearby electron acceptor molecules.

PS II would disintegrate without replacement electrons, and these are removed from water, which splits, releasing oxygen to the atmosphere. Notice that with the loss of electrons, water has been oxidized and that the oxygen released during photosynthesis does come from water. Many organisms, including plants themselves and humans, use this oxygen within their mitochondria to make ATP. The hydrogen ions (H^+) stay in the thylakoid space and contribute to the formation of a hydrogen ion gradient.

An electron acceptor sends energized electrons, received from the reaction center, down an **electron transport chain (ETC),** a series of carriers that pass electrons from one to the other. As the electrons pass from one carrier to the next, energy is captured and stored in the form of a hydrogen ion (H^+) gradient. When these hydrogen ions flow down their electrochemical gradient through ATP synthase complexes, ATP production occurs (see Fig. 7.11). Notice that this ATP is then used by the Calvin cycle reactions in the stroma to reduce carbon dioxide to a carbohydrate.

When the PS I pigment complex absorbs solar energy, energized electrons leave its reaction center and are captured by electron acceptors. Low-energy electrons from the *electron transport chain* adjacent to PS II replace those lost by PS I. The electron acceptors in PS I pass their electrons to $NADP^+$ molecules. Each $NADP^+$ accepts two electrons and an H^+ to become reduced and forms NADPH. This NADPH is then used by the Calvin cycle reactions in the stroma along with ATP in the reduction of carbon dioxide to a carbohydrate.

ATP and NADPH are not made in equal amounts during the light reactions, and more ATP than NADPH is required during the Calvin cycle. Where does this extra ATP come from? Every so often, an electron moving down the noncyclic pathway is rerouted back to an earlier point in the electron transport chain. The cyclic pathway, which occurs in many prokaryotes, and at high oxygen levels in eukaryotes, enables electrons to participate in additional redox reactions, moving more H^+ across the thylakoid membrane and through ATP synthase, ultimately producing more ATP (Fig. 7.10).

Organization of the Thylakoid Membrane

As we have discussed, the following molecular complexes are present in the thylakoid membrane (Fig. 7.11):

- PS II, which consists of a pigment complex and electron acceptor molecules, receives electrons from water as water splits, releasing oxygen.
- The electron transport chain (ETC), consisting of Pq (plastoquinone) and cytochrome complexes, carries electrons from PS II to PS I via redox reactions. Pq also pumps H^+ from the stroma into the thylakoid space.
- PS I, which also consists of a pigment complex and electron acceptor molecules, is adjacent to NADP reductase, which reduces $NADP^+$ to NADPH.
- The **ATP synthase complex,** which has a channel and a protruding ATP synthase, is an enzyme that joins ADP + Ⓟ.

ISTUDY

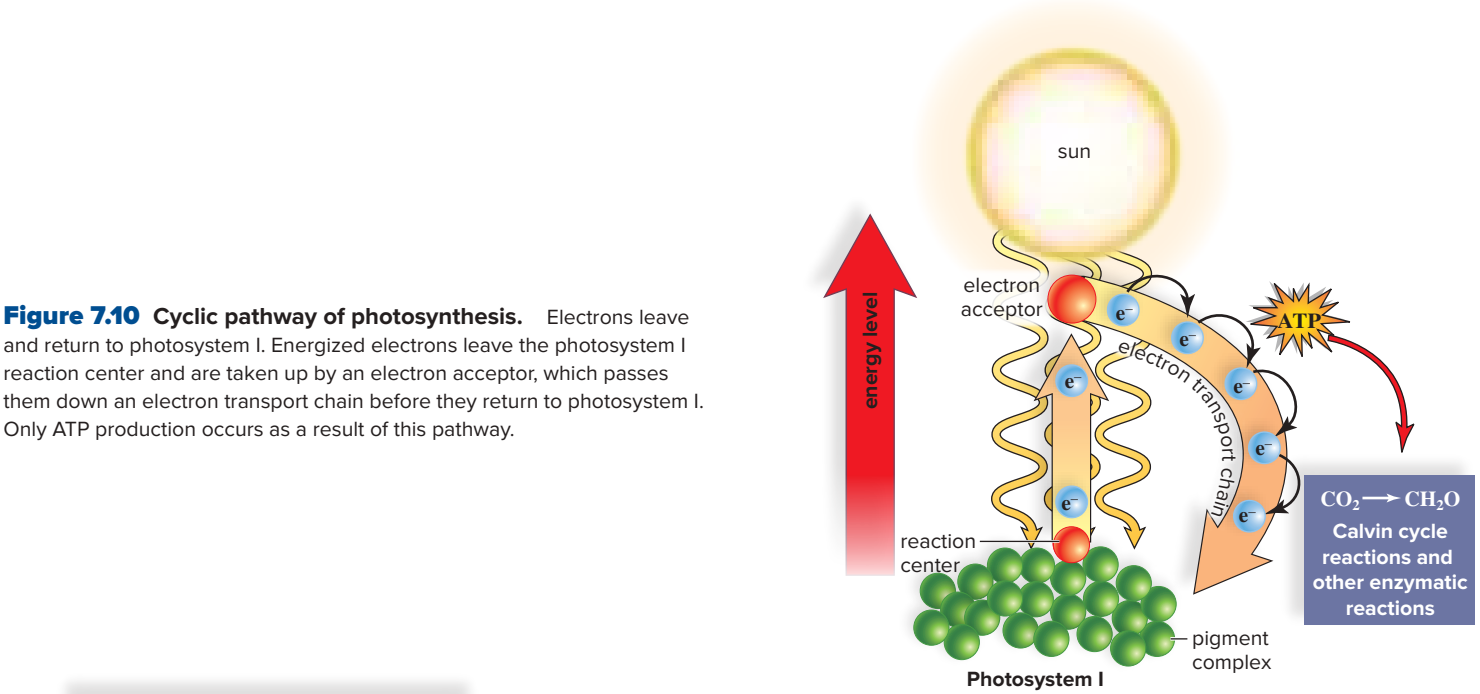


Figure 7.10 Cyclic pathway of photosynthesis. Electrons leave and return to photosystem I. Energized electrons leave the photosystem I reaction center and are taken up by an electron acceptor, which passes them down an electron transport chain before they return to photosystem I. Only ATP production occurs as a result of this pathway.

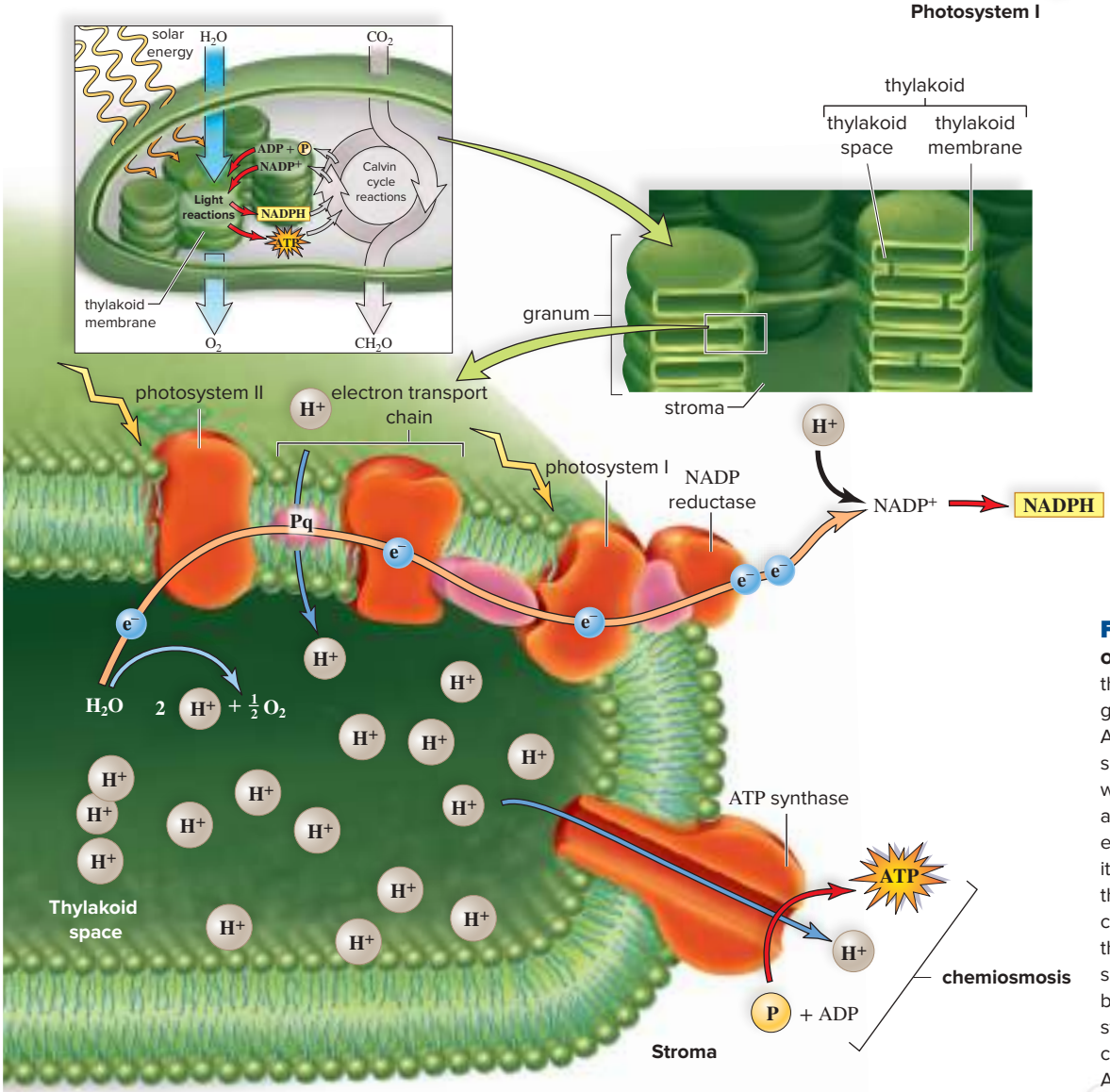


Figure 7.11 Organization of a thylakoid. Each thylakoid membrane within a granum produces NADPH and ATP. Electrons move through sequential molecular complexes within the thylakoid membrane, and the last one passes electrons to $NADP^+$, after which it becomes NADPH. A carrier at the start of the electron transport chain pumps hydrogen ions from the stroma into the thylakoid space. When hydrogen ions flow back out of the space into the stroma through an ATP synthase complex, ATP is produced from ADP + Ⓟ.

ISTUDY

ATP Production

The thylakoid space acts as a reservoir for many hydrogen ions (H^+). First, each time water is oxidized, two H^+ remain in the thylakoid space. Second, as the electrons move from carrier to carrier via redox reactions along the electron transport chain, the electrons give up energy, which is used to pump H^+ from the stroma into the thylakoid space. Therefore, there are more H^+ in the thylakoid space than in the stroma. This difference and the resulting flow of H^+ (often referred to as protons in this context) from high to low concentration provide kinetic energy that allows an ATP synthase complex enzyme to enzymatically produce ATP from ADP + Ⓟ. This method of producing ATP is called **chemiosmosis,** because ATP production is tied to the establishment of an H^+ gradient.

Check Your Progress 7.3

1. Distinguish visible light from the electromagnetic spectrum.
2. Describe the movement of electrons from water to $NADP^+$ in the light reactions.

7.4 Plants Fix Carbon Dioxide

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the three steps of the Calvin cycle and when ATP and/or NADPH is needed.
2. Evaluate the significance of the RuBP carboxylase enzyme to photosynthesis.
3. Explain how glyceraldehyde-3-phosphate (G3P) is used to produce other necessary plant molecules.

During the light reactions, the high-energy molecules ATP and NADPH were produced. The Calvin cycle, another series of chemical reactions, will use these high-energy molecules for an amazing process: carbon dioxide fixation. Carbon dioxide in its gas form is all around us in our atmosphere (see the Biological Systems feature, "Tropical Rain Forest Destruction and Climate Change"). We and other respiring organisms release it as waste during cellular respiration. Unfortunately, CO_2 is unattainable by heterotrophs—we cannot harness or extract CO_2 from the air and then use those carbon atoms to make sugar. Plants and other autotrophs can take the carbon from CO_2 gas and convert (or "fix") it in the bonds of a carbohydrate. The word *fixation* is not limited to photosynthesis. As you will learn in later chapters, some bacteria can undergo fixation by removing nitrogen from the air and fixing it into organic molecules.

The Calvin cycle is a series of reactions that can occur in the dark but that uses the products of the light reactions to reduce carbon dioxide captured from the atmosphere to a carbohydrate. The Calvin cycle is composed of three steps: (1) carbon dioxide fixation, (2) carbon dioxide reduction, and (3) regeneration of RuBP (Fig. 7.12).

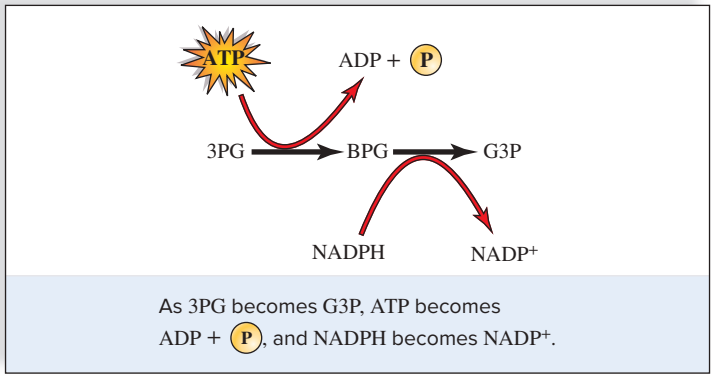
Step 1: Fixation of Carbon Dioxide

Carbon dioxide fixation is the first step of the Calvin cycle. During this reaction, a molecule of carbon dioxide from the atmosphere is attached to RuBP (ribulose-1,5-bisphosphate), a 5-carbon molecule. The result is one 6-carbon molecule, which splits into two 3-carbon molecules.

The enzyme that speeds this reaction, called **RuBP carboxylase,** is a protein that makes up about 20–50% of the protein content of chloroplasts. The reason for its abundance may be that it is unusually slow—it processes only a few molecules of substrate per second compared to thousands per second for a typical enzyme—so there has to be a lot of it to keep the Calvin cycle going.

Step 2: Reduction of Carbon Dioxide

The first 3-carbon molecule in the Calvin cycle is called 3PG (3-phosphoglycerate). Each of two 3PG molecules undergoes reduction to G3P (glyceraldehyde-3-phosphate) in two steps:

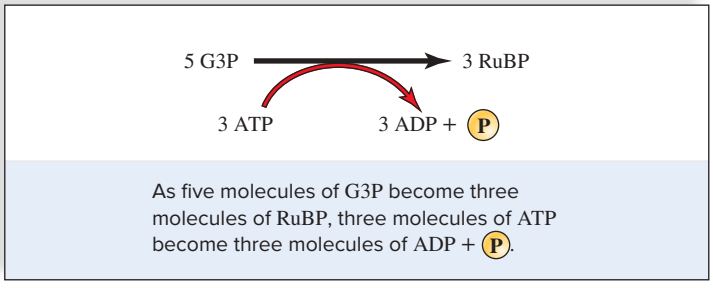


As 3PG becomes G3P, ATP becomes ADP + Ⓟ, and NADPH becomes $NADP^+$.

Energy and electrons are needed for this reduction reaction, and they are supplied by the ATP and NADPH that were made during the light reactions. The difference between 3PG, BPG, and G3P (all with three carbons) is that G3P is reduced, has more electrons, and is now more chemically able to store energy and form larger organic molecules, such as glucose.

Step 3: Regeneration of RuBP

Notice that the Calvin cycle reactions in Figure 7.12 are multiplied by 3 because it takes three turns of the Calvin cycle to allow one G3P to exit. Why? For every three turns of the Calvin cycle, five molecules of G3P are used to re-form three molecules of RuBP, and the cycle continues. Notice that 5 × 3 (carbons in G3P) = 3 × 5 (carbons in RuBP):



As five molecules of G3P become three molecules of RuBP, three molecules of ATP become three molecules of ADP + Ⓟ.

This reaction also uses some of the ATP produced by the light reactions.

ISTUDY

THEME Biological Systems

Tropical Rain Forest Destruction and Climate Change

Leonardo DiCaprio not only is a famous actor but also strives to make global changes through his foundation. One aspect of the Leonardo DiCaprio Foundation is the protection of tropical rain forests. Most people think about saving the fragile species of plants and animals that live in the rain forest, but globally there is a bigger issue at hand.

Climate change is an expected rise in the average global temperature during the twenty-first century due to the introduction of certain gases into the atmosphere. For at least a thousand years prior to 1850, atmospheric carbon dioxide (CO_2) levels remained fairly constant at 0.028%. Since the 1850s, when industrialization began, the amount of CO_2 in the atmosphere has increased to 0.041%, coinciding with increased global temperatures (Fig. 7A).

Role of Carbon Dioxide

In much the same way as the panes of a greenhouse, CO_2 and other gases in our atmosphere trap radiant heat from the sun. Therefore, these gases are called greenhouse gases. Without any greenhouse gases, the Earth's temperature would be about 33°C cooler than it is now. Likewise, increasing the concentration of greenhouse gases makes the Earth warmer and its waters more acidic.

Certainly, the burning of fossil fuels adds CO_2 to the atmosphere. But another factor that contributes to an increase in atmospheric CO_2 is tropical rain forest destruction.

Role of Tropical Rain Forests

Many scientists consider tropical rain forests to be the "lungs" of the Earth. Between 10 and 30 million hectares of rain forest are lost every year to ranching, logging, mining, and other methods of developing the forest for human needs.

Each year, deforestation in tropical rain forests accounts for 10–20% of all CO_2 in the atmosphere. With your body, if you lose lung capacity, you lose body function. Similarly, the consequences of losing forests means greater trouble regarding climate change, because burning a forest adds CO_2 to the atmosphere and removes the trees that would ordinarily absorb CO_2.

The Earth Is a System

Carbon dioxide is removed from the air via photosynthesis, which takes place in forests, oceans, and other terrestrial and marine ecosystems. In fact, photosynthesis produces organic matter, which is estimated to be several hundred times the mass of the people living on Earth. Thus, these environments act as a sink for CO_2, preventing too much CO_2 from accumulating in the atmosphere where it can affect global temperatures and bring about climate change.

Despite their reduction in size from an original 15% to less than 5% of land surface today, tropical rain forests make a substantial contribution to global CO_2 removal. They are a critical element of the Earth's systems and, like any biological system, are essential for normal, healthy function. Tropical rain forests contribute greatly to the uptake of CO_2 and the productivity of photosynthesis, because they are the most efficient of all terrestrial ecosystems.

Tropical rain forests occur near the equator. They can exist wherever temperatures are above 26°C and rainfall is heavy (100–200 cm per year) and regular. Huge trees with buttressed trunks and broad, undivided, dark-green leaves predominate. Nearly all land plants in a tropical rain forest are woody, and woody vines are abundant.

It might be hypothesized that an increased amount of CO_2 in the atmosphere would cause photosynthesis to increase in the remaining portion of the forest. Recent studies, however, are showing that the opposite is true. Too much CO_2 can decrease photosynthesis, because increased temperatures can reduce water and mineral availability. Scientists working with wheat showed a decrease in the production of nitrogen-containing compounds. Another study showed increased herbivory, because plants were unable to produce their defense toxins at higher temperatures.

These and other studies show that, for the Earth, as for any biological system, equilibrium is necessary for health. As a biological system, the Earth is sensitive to environmental change. Our ability to properly balance human activity with the needs of the biosphere requires that we become educated about how the Earth functions.

Questions to Consider

1. How can a rise in temperature affect the production of food crops?
2. How can increased CO_2 levels affect the organisms that live in water?

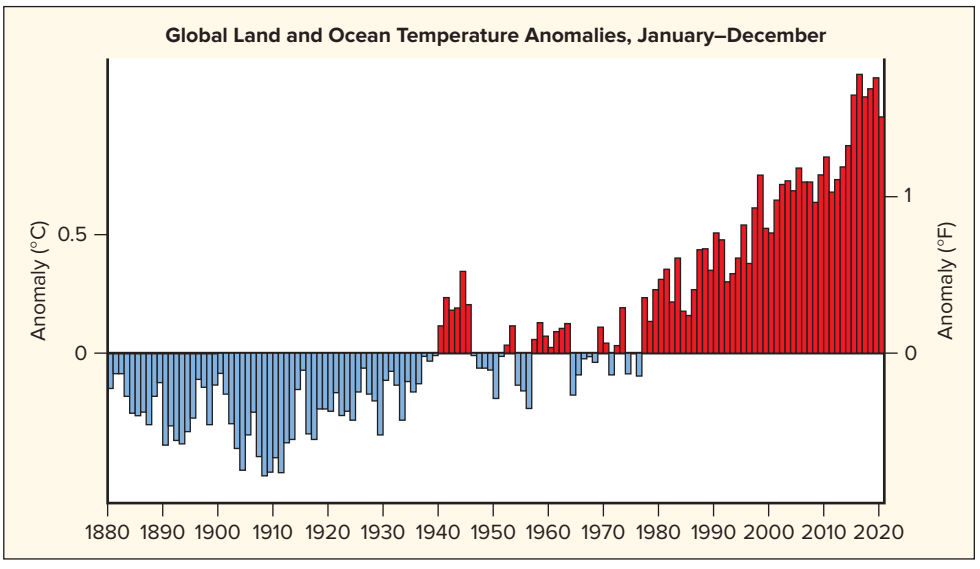


Figure 7A Climate change. Global temperatures have steadily increased above average for the past several decades, coinciding with increased greenhouse gas levels.

Source: NOAA, "Climate at a Glance," 2022, https://www.ncei.noaa.gov/cag/global/time-series/globe/land_ocean/ytd/12/1880-2022

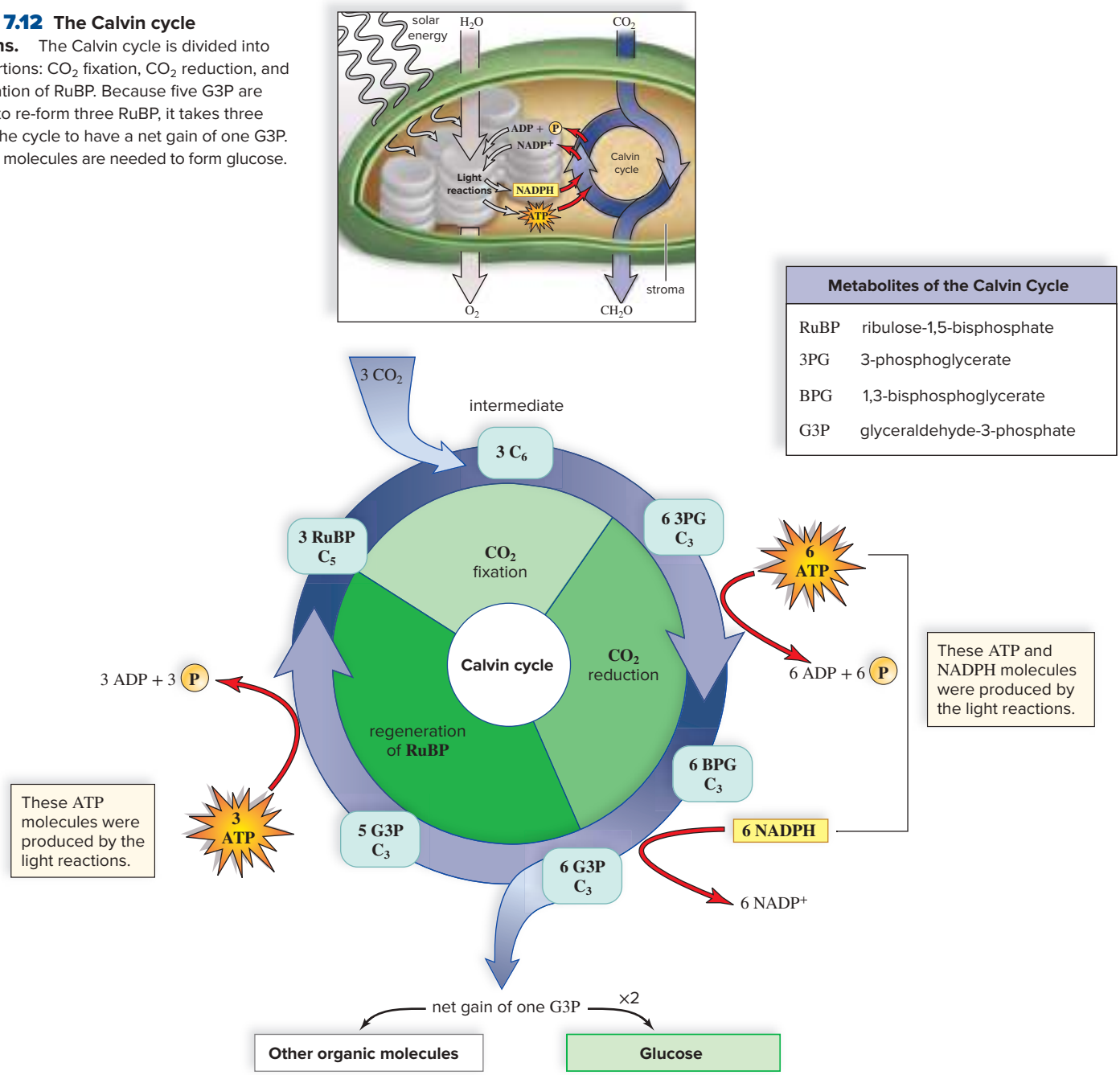


Figure 7.12 The Calvin cycle reactions. The Calvin cycle is divided into three portions: CO_2 fixation, CO_2 reduction, and regeneration of RuBP. Because five G3P are needed to re-form three RuBP, it takes three turns of the cycle to have a net gain of one G3P. Two G3P molecules are needed to form glucose.

The Importance of the Calvin Cycle

G3P is the product of the Calvin cycle that can be converted to other molecules a plant needs. Notice that glucose phosphate is among the organic molecules that result from G3P metabolism (Fig. 7.13). This is of interest to us because glucose is the molecule that plants and animals most often metabolize to produce the ATP molecules they require for their energy needs.

Glucose phosphate can be combined with fructose (and the phosphate removed) to form sucrose, the molecule that plants use to transport carbohydrates from one part of the plant to the other. Glucose phosphate is also the starting point for the synthesis of starch and cellulose. Starch is the storage form of glucose. Some starch is stored in chloroplasts, but most starch is stored in amyloplasts in roots. Cellulose is a structural component of plant cell walls and becomes fiber in our diet, because we are unable to digest it.

A plant can use the hydrocarbon skeleton of G3P to form fatty acids and glycerol, which are combined in plant oils. We are all familiar with corn oil, sunflower oil, and olive oil used in cooking. As mentioned at the beginning of the chapter, researchers are modifying photosynthesis to produce oils that can also be used as fuel. When nitrogen is added to the hydrocarbon skeleton derived from G3P, amino acids are formed, allowing the plant to produce protein.

ISTUDY

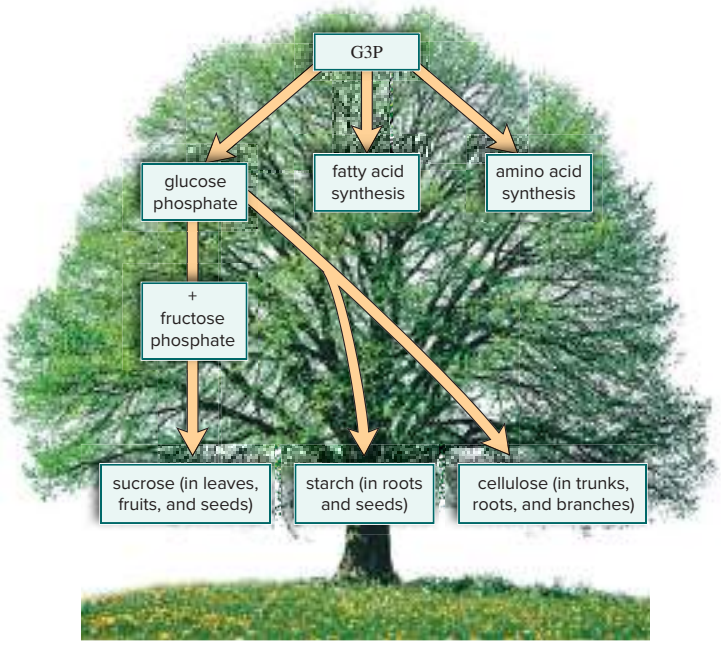


Figure 7.13 Fate of G3P. G3P is the first reactant in a number of plant cell metabolic pathways. From this starting point, different carbohydrates can be produced, such as sucrose, starch, and cellulose. Fatty acid synthesis leads to triglycerides making up plant oils, and production of amino acids allows the plant to make proteins.
(photo): Hermann Eisenbeiss/Science Source

Check Your Progress 7.4

1. Describe the three major steps of the Calvin cycle.
2. Illustrate why it takes three turns of the Calvin cycle to produce one G3P.

7.5 Other Types of Photosynthesis

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the internal location of photosynthesis in C_3 and C_4 plants.
2. Contrast C_3/C_4 modes of photosynthesis with CAM photosynthesis.
3. Explain how different ways of achieving photosynthesis allow plants to adapt to particular environments.

The majority of plants, such as azaleas, maples, and tulips, carry on photosynthesis as described in Section 7.2 and are called **C_3 plants** (Fig. 7.14). C_3 plants use the enzyme RuBP carboxylase to fix CO_2 to RuBP in mesophyll (photosynthetic) cells. The first detected molecule following fixation is the 3-carbon molecule 3PG:

$$\text{RuBP} + CO_2 \xrightarrow{\text{RuBP carboxylase}} 2\ \text{3PG}$$

As shown in Figure 7.2, leaves have small openings called stomata, through which water can leave and carbon dioxide (CO_2) can enter. If the weather is hot and dry, the stomata close, conserving water. (Water loss might cause the plant to wilt and die.) Now the concentration of CO_2 decreases in leaves, while O_2, a by-product of photosynthesis, increases. When O_2 rises in C_3 plants, RuBP carboxylase combines it with RuBP instead of CO_2. The result is one molecule of 3PG and the eventual release of CO_2. This is called **photorespiration,** because in the presence of light (*photo*), oxygen is taken up and CO_2 is released (*respiration*).

An adaptation called C_4 photosynthesis enables some plants to avoid photorespiration.

C_4 Photosynthesis

In a C_3 plant, the mesophyll cells contain well-formed chloroplasts and are arranged in parallel layers (Fig. 7.15). In a C_4 leaf, the bundle sheath cells, as well as the mesophyll cells, contain chloroplasts. Further, the mesophyll cells are arranged concentrically around the bundle sheath cells.

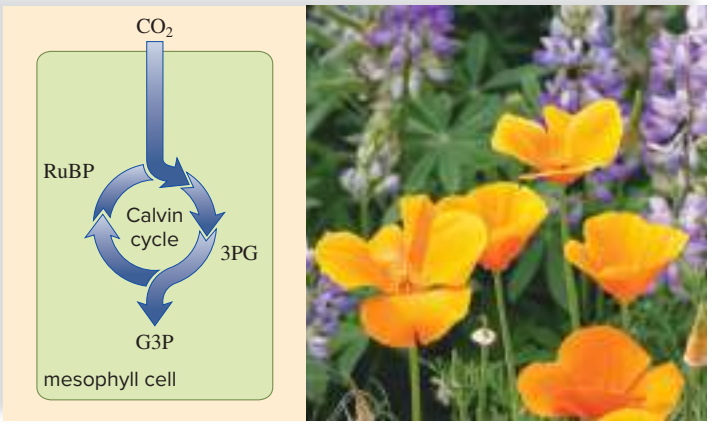


Figure 7.14 Carbon dioxide fixation in C_3 plants. In C_3 plants, like these wildflowers, CO_2 is taken up by the Calvin cycle directly in mesophyll cells. (photo): Don Paulson Photography/Purestock/SuperStock

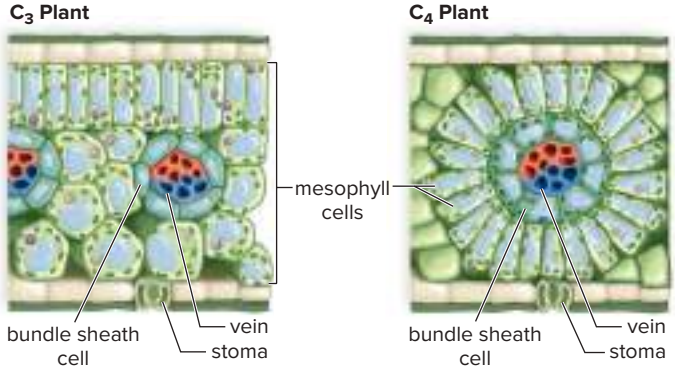


Figure 7.15 Leaf structure of C_3 and C_4 plants. C_3 and C_4 plants differ in their arrangement of photosynthetic mesophyll cells.

ISTUDY

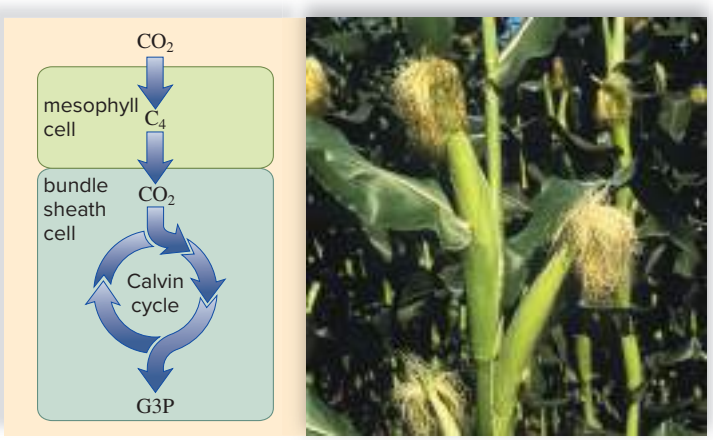
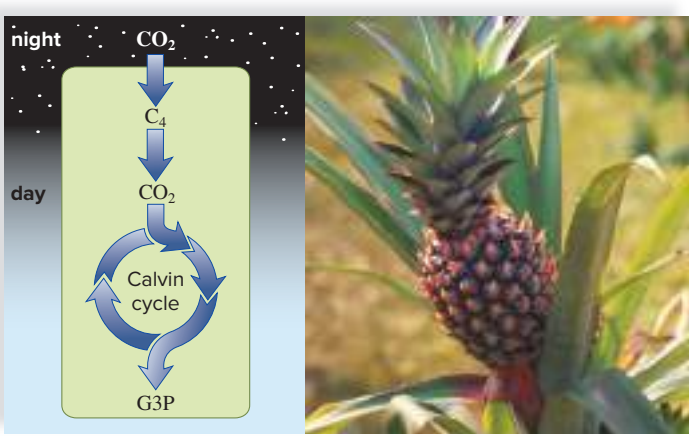


Figure 7.16 Carbon dioxide fixation in C_4 plants. C_4 plants like corn form a C_4 molecule in mesophyll cells prior to releasing CO_2 to the Calvin cycle in bundle sheath cells. (photo): Doug Wilson/USDA



CO_2 fixation in a CAM plant, pineapple, *Ananas comosus*

Figure 7.17 Carbon dioxide fixation in a CAM plant. CAM plants, such as pineapple, fix CO_2 at night, forming a C_4 molecule that is released to the Calvin cycle during the day. (photo): Dinodia/Pixtal/age fotostock

C_4 plants fix CO_2 to PEP (phosphoenolpyruvate, a C_3 molecule) using the enzyme PEP carboxylase (PEPCase). The result is oxaloacetate, a C_4 molecule:

$$\text{PEP} + CO_2 \xrightarrow{\text{PEPCase}} \text{oxaloacetate}$$

In a C_4 plant, CO_2 is taken up in mesophyll cells, and then malate, a reduced form of oxaloacetate, is pumped into the bundle sheath cells (Fig. 7.16). Only here does CO_2 enter the Calvin cycle.

Because it takes energy to pump molecules, you would think that the C_4 pathway would be disadvantageous. Yet in hot, dry climates, the net photosynthetic rate of C_4 plants, such as sugarcane, corn, and Bermuda grass, is about two to three times that of C_3 plants (e.g., wheat, rice, and oats). Why do C_4 plants enjoy such an advantage? The answer is that they can avoid photorespiration, discussed previously. Photorespiration is wasteful, because it is not part of the Calvin cycle. Photorespiration does not occur in C_4 leaves because PEPCase, unlike RuBP carboxylase, does not combine with O_2. Even when stomata are closed, CO_2 is delivered to the Calvin cycle in the bundle sheath cells.

When the weather is moderate, C_3 plants ordinarily have the advantage, but when the weather becomes hot and dry, C_4 plants have the advantage, and we can expect them to predominate. In the early summer, C_3 plants such as Kentucky bluegrass and creeping bent grass predominate in lawns in the cooler parts of the United States, but by midsummer, crabgrass, a C_4 plant, begins to take over.

CAM Photosynthesis

CAM stands for crassulacean-acid metabolism; the Crassulaceae is a family of flowering succulent (water-containing) plants, such as a pineapple plant, that live in warm, dry regions of the world. CAM was first discovered in these plants, but now it is known to be prevalent among other groups of plants as well, such as pineapples.

Whereas a C_4 plant represents partitioning in space—carbon dioxide fixation occurs in mesophyll cells, while the Calvin cycle occurs in bundle sheath cells—CAM is partitioning by the use of time. During the night, CAM plants use PEPCase to fix some CO_2, forming C_4 molecules, which are stored in large vacuoles in mesophyll cells. During the day, C_4 molecules (malate) release CO_2 to the Calvin cycle when NADPH and ATP are available from the light reactions (Fig. 7.17). The primary advantage for this partitioning again has to do with the conservation of water. CAM plants open their stomata only at night; therefore, only at that time does atmospheric CO_2 enter the plant. During the day, the stomata close; this conserves water, but CO_2 cannot enter the plant. Photosynthesis in a CAM plant is minimal, because a limited amount of CO_2 is fixed at night, but it does allow CAM plants to live under stressful conditions.

Photosynthesis and Adaptation to the Environment

The different types of photosynthesis give us an opportunity to consider that organisms are metabolically adapted to their environment. Each method of photosynthesis has its advantages and disadvantages, depending on the climate.

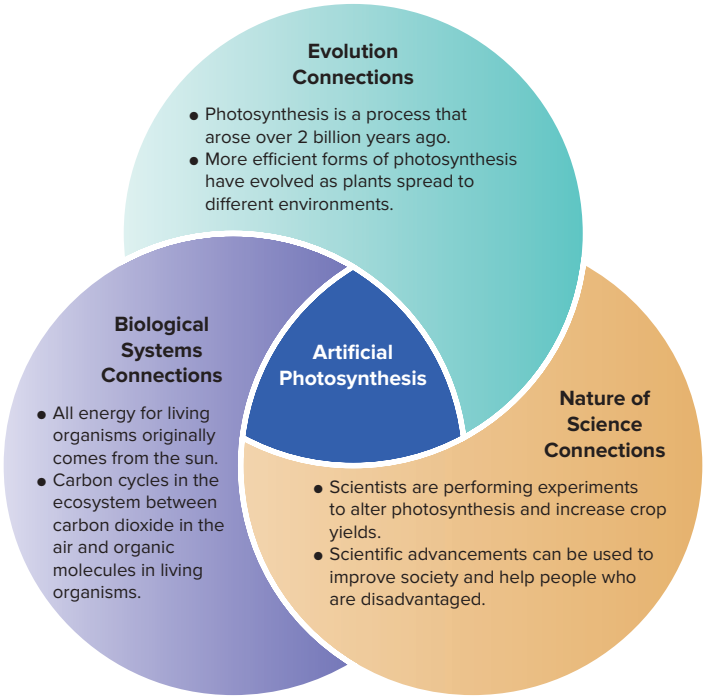
C_4 plants most likely evolved in, and are adapted to, areas of high light intensities, high temperatures, and limited rainfall. C_4 plants, however, are more sensitive to cold, and C_3 plants do better than C_4 plants below 25°C. CAM plants, on the other hand, compete well with either type of plant when the environment is extremely arid. Surprisingly, CAM is quite widespread and has evolved into 23 families of flowering plants, including some lilies and orchids! And it is found among nonflowering plants, including some ferns and cone-bearing trees.

Check Your Progress 7.5

1. List some plants that use a method of photosynthesis other than C_3 photosynthesis.
2. Explain why C_4 photosynthesis is advantageous in hot, dry conditions.

ISTUDY

CONNECTING *THE* CONCEPTS



What Did You Learn?

7.1 Photosynthetic Organisms

Photosynthesis produces carbohydrates and releases oxygen, both of which are used by the majority of living organisms. Cyanobacteria, algae, and plants are **autotrophs** and carry out photosynthesis. **Heterotrophs** consume the products of photosynthesis. In plants, gases enter and exit through **stomata,** and photosynthesis takes place in **chloroplasts.** A chloroplast is enclosed by a double membrane and contains two main components: the semifluid **stroma** and the membranous **grana** made up of **thylakoids.** Thylakoids contain **chlorophyll**—a pigment that captures solar energy.

7.2 The Process of Photosynthesis

The overall equation for photosynthesis shows that it is a redox reaction. Carbon dioxide is reduced, and water is oxidized. During photosynthesis, the **light reactions** take place in the thylakoid membranes, and the **Calvin cycle reactions** take place in the stroma.

7.3 Plants Convert Solar Energy

Photosynthesis uses solar energy in the visible-light range. Specifically, chlorophylls *a* and *b* absorb violet, blue, and red wavelengths best and reflect green light, whereas the **carotenoids** absorb violet-blue-green light and reflect yellow-to-orange light. Specific wavelengths absorbed by pigments determine a pigment's **absorption spectrum.**

The light reactions contain two **photosystems (PSs),** which are pigment complexes that capture solar energy. In the light reactions, a **noncyclic pathway** has an electron flow that begins when solar energy enters PS II and energizes chlorophyll *a* electrons. The oxidation (splitting) of water replaces these electrons in the reaction-center chlorophyll *a* molecules. Oxygen is released to the atmosphere, and hydrogen ions (H^+) remain in the thylakoid space. Electrons are ultimately passed to PS I via an **electron transport chain (ETC),** which pumps hydrogen ions across the thylakoid membrane and results in **chemiosmosis**—a gradient used to make ATP via **ATP synthase complex.** Light-energized electrons from PS I are captured by $NADP^+$, which combines with H^+ from the stroma to become NADPH. The cyclic pathway has an electron flow in the light reactions that pumps additional hydrogen ions and contributes to ATP production.

7.4 Plants Fix Carbon Dioxide

The energy yield of the light reactions is stored in ATP and NADPH. These molecules are used by the Calvin cycle reactions for **carbon dioxide fixation**—a reduction of CO_2 to carbohydrate, namely G3P, which is then converted to all the organic molecules a plant needs.

During the first stage of the Calvin cycle, the enzyme **RuBP carboxylase** fixes CO_2 to RuBP, producing a 6-carbon molecule that immediately breaks down to two C_3 molecules. During the second stage, CO_2 is incorporated into an organic molecule, and then reduced to carbohydrate (CH_2O). This step requires the NADPH and some of the ATP from the light reactions. For every three turns of the Calvin cycle, the net gain is one G3P molecule; the other five G3P molecules are used to re-form three molecules of RuBP, which also requires ATP. It takes two G3P molecules to make one glucose molecule.

7.5 Other Types of Photosynthesis

Plants have adapted ways other than the **C_3 plant** process to photosynthesize in various environments. In C_4 plants, carbon dioxide is first fixed in mesophyll cells via PEPCase, is transported to a different location in bundle sheath cells, and is then released to the Calvin cycle. PEPCase has an advantage over RuBP carboxylase because it doesn't participate in **photorespiration. C_4 plants** avoid the photorespiration complication by separating where carbon fixation occurs from where the Calvin cycle occurs.

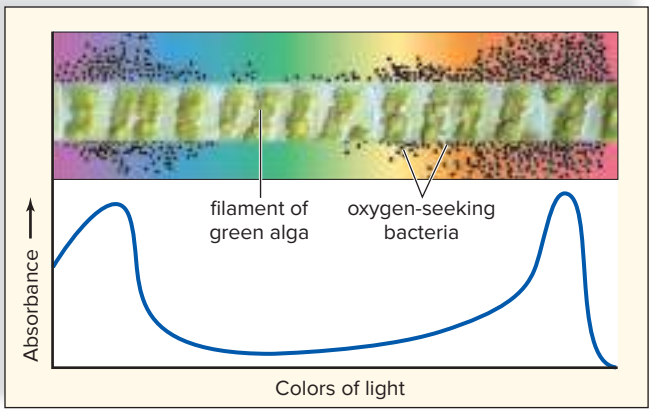
CAM plants, which live in hot, dry environments, cannot leave their stomata open during the day, or they will die from loss of water.

ISTUDY

CAM plants fix carbon only at night, conserving water. Stores of CO_2 are released to the Calvin cycle during the day when photosynthesis is possible. CAM plants avoid drying out by separating when they bring in carbon dioxide from when they release it to the Calvin cycle.

Thinking Critically About Biology

1. In biofuel production, plants are grown in order to create fuels that are eventually burned in vehicles like automobiles. Overall, does the process of biofuel production add carbon dioxide to the atmosphere, remove it, or both?
2. In the image below, the alga is the autotroph and undergoes photosynthesis, whereas the bacteria (small black dots) are the heterotrophs. Explain why the bacteria are clustered at the far ends after the alga was exposed to different color wavelengths of light.



(photo): Stephen Durr

3. A Belgian doctor, Jan Baptista van Helmont (1580–1644), planted a small willow tree in a pot of soil. He weighed the tree and the soil. The tree was watered for 5 years and weighed 74.4 kg more than when he began, and the soil lost 57 g of mass. Explain what accounts for the plant's increase in biomass.

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

Energy Part II: Photosynthesis

McGraw Hill

Why does this experiment only need to produce light in the wavelengths at the two ends of the visible light spectrum?

Making It Relevant

1. Would you have concerns eating food that came from a genetically modified organism, such as soybeans that were modified to have more efficient photosynthesis?
2. Every day, your body requires certain inputs to stay alive. Which of those inputs are provided by photosynthesis and how do they reach you?
3. How is it beneficial to everyone when help is focused toward people who are disadvantaged, such as people in sub-Saharan Africa who chronically face food shortages?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

8

Cellular Respiration

Gregory Shamus/Getty Images

Chapter Outline

8.1 Overview of Cellular Respiration
8.2 Outside the Mitochondria: Glycolysis
8.3 Outside the Mitochondria: Fermentation
8.4 Inside the Mitochondria
8.5 Metabolism

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 6.3 How does an ATP molecule store energy?

Section 6.4 How are high-energy electrons used to make energy for cellular work?

Figure 7.3 Where does the glucose we metabolize come from?

Metabolic Demands of Athletes

During a typical basketball game, such as the 2023 NCAA championship game between UConn and San Diego State, the starting players run an average of 4–5 kilometers (around 3 miles) during a 40-minute game. However, unlike the steady pace maintained by marathoners, soccer players experience periods of intense activity (sprinting), followed by brief periods of rest. This start-and-stop nature of the game means that the athletes' muscles are constantly switching between aerobic and anaerobic metabolism.

During aerobic metabolism, the muscle cells use oxygen to completely break down glucose and produce ATP, the energy "battery" of the cell. This process, which uses carbohydrates and produces carbon dioxide as a waste product, occurs in the cytoplasm and mitochondria of the cell. However, running short fast sprints quickly depletes oxygen levels in the blood and drives the muscles into anaerobic metabolism. Without oxygen, glucose cannot be completely broken down and is changed to lactate, which is responsible for the muscle "burn" we feel after intense exercise. Once oxygen is restored, usually after a brief rest, the body is able to return to aerobic metabolism. Professional athletes are conditioned to reduce the amount of time they are in anaerobic metabolism and to maximize their aerobic metabolism.

In this chapter, we will not only examine these pathways but also explore how nutrients such as fats and proteins can be used to generate ATP for our cellular activities.

As you read through this chapter, think about the following questions:

1. What are the differences between the aerobic and anaerobic pathways?
2. How is the energy of a glucose molecule transformed into energy in ATP?

Following *the* Themes

CHAPTER 8 – CELLULAR RESPIRATION

Evolution	Cellular respiration is the process by which the majority of life on Earth generates ATP for cellular processes.
Nature of Science	An understanding of cellular respiration is useful in exploring the nature of human diseases and nutritional needs.
Biological Systems	The ATP produced at the cellular level is used to power all of the activities of an organism.

ISTUDY

8.1 Overview of Cellular Respiration

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the overall reaction for glucose breakdown and show it is a redox reaction.
2. Examine the role of the NADH and $FADH_2$ redox reactions in cellular respiration.
3. Summarize the phases of cellular respiration.

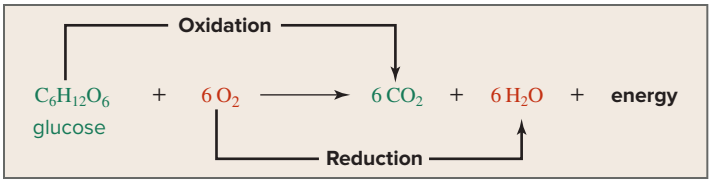


Figure 8.2 The breakdown of glucose. During cellular respiration, glucose is broken down into carbon dioxide and water in order to produce energy. The breakdown requires the removal of electrons from glucose (oxidation), and, for the reaction to proceed, oxygen must be available to accept the electrons (reduction).

Cellular respiration is the process by which cells acquire energy by breaking down organic molecules, such as glucose, to obtain the energy needed to produce ATP molecules for metabolic reactions. Most cellular respiration pathways occur in the mitochondria and require oxygen (O_2). The outputs are ATP, carbon dioxide (CO_2), and water (H_2O) (Fig. 8.1).

Most often, cellular respiration involves the complete breakdown of glucose to carbon dioxide and water (H_2O), as shown in Figure 8.2.

This equation shows that cellular respiration is an oxidation-reduction reaction. Recall that oxidation is the loss of electrons and reduction is the gain of electrons (see Section 6.4); therefore, glucose has been oxidized and O_2 has been reduced. Also remember that a hydrogen atom consists of a hydrogen ion plus an electron ($H^+ + e^-$). Therefore, when hydrogen atoms are removed from glucose, so are electrons. Similarly, when hydrogen atoms are added to oxygen, so are electrons.

You may notice that the reactions in Figure 8.2 are effectively the opposite of the photosynthesis reactions we explored in Section 7.1. Photosynthetic organisms, such as plants, use sunlight to store energy in the form of the chemical bonds within organic molecules. The cellular respiration pathways are used by all organisms, including plants and animals, to release that energy.

Glucose is a high-energy molecule, but its breakdown products, CO_2 and H_2O, are low-energy molecules. Therefore, as the equation shows, energy is released. This is the energy that will be used to produce ATP molecules. The cell carries out cellular respiration in order to build up ATP molecules.

The pathways of cellular respiration allow the energy within a glucose molecule to be released slowly, so that ATP can be produced gradually. Cells would lose a tremendous amount of energy if glucose breakdown occurred all at once—most of the energy would become nonusable heat. The step-by-step breakdown of glucose to CO_2 and H_2O usually produces a maximum yield of 36 to 38 ATP molecules, dependent on the conditions we will explore later. The energy in these ATP molecules is equivalent to about 39% of the energy available in glucose. The remainder is lost as heat.

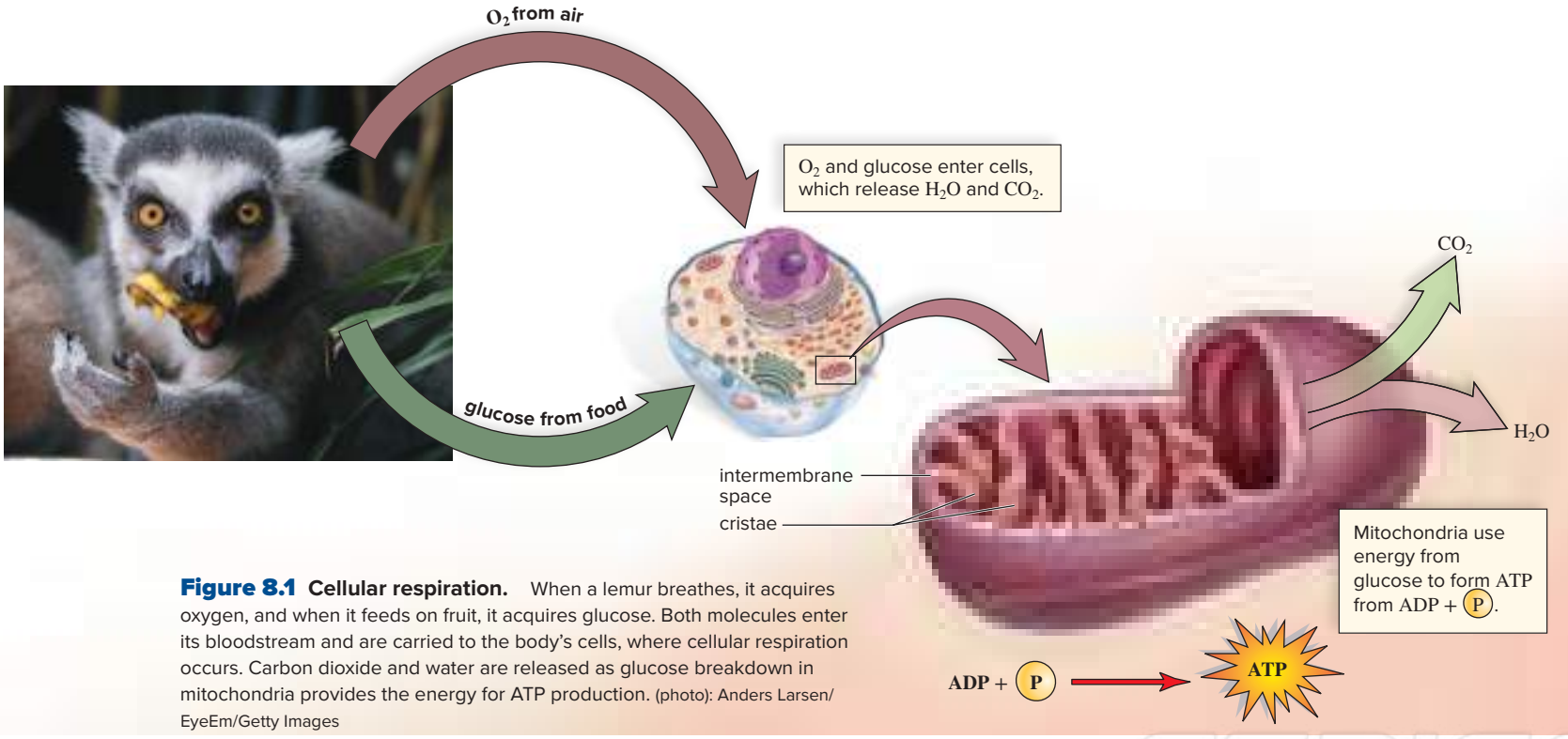


Figure 8.1 Cellular respiration. When a lemur breathes, it acquires oxygen, and when it feeds on fruit, it acquires glucose. Both molecules enter its bloodstream and are carried to the body's cells, where cellular respiration occurs. Carbon dioxide and water are released as glucose breakdown in mitochondria provides the energy for ATP production. (photo): Anders Larsen/EyeEm/Getty Images

NAD^+ and FAD

Cellular respiration involves many individual metabolic reactions, each one catalyzed by its own enzyme. Enzymes of particular significance are those that use **NAD^+ (nicotinamide adenine dinucleotide),** a coenzyme used in oxidation-reduction reactions. When a metabolite is oxidized, NAD^+ accepts two electrons plus a hydrogen ion (H^+), and NADH results. The electrons received by NAD^+ are high-energy electrons usually carried to the electron transport chain:

$$NAD^+ + 2e^- + H^+ \rightarrow NADH$$

NAD^+ can oxidize a metabolite by accepting electrons and can reduce a metabolite by giving up electrons. Only a small amount of NAD^+ needs to be present in a cell, because each NAD^+ molecule is used over and over again. **FAD,** another coenzyme of oxidation-reduction, is sometimes used instead of NAD^+. FAD accepts two electrons and two hydrogen ions (H^+) to become $FADH_2$.

Phases of Cellular Respiration

Cellular respiration involves four phases: glycolysis, the preparatory reaction, the citric acid cycle, and the electron transport chain (Fig. 8.3). Glycolysis takes place outside the mitochondria and does not require the presence of oxygen. Therefore, glycolysis is **anaerobic.** The other phases of cellular respiration take place inside the mitochondria, where oxygen is the final acceptor of electrons. Because they require oxygen, these phases are called **aerobic.**

During these phases, notice where CO_2 and H_2O, the end products of cellular respiration, and ATP, the main outcome of respiration, are produced.

- *Glycolysis* (Gk. *glycos,* "sugar"; *lysis,* "splitting") is the breakdown of glucose (a 6-carbon molecule) to two molecules of pyruvate (two 3-carbon molecules). Oxidation results in NADH and provides enough energy for the net gain of two ATP molecules.
- The *preparatory* (*prep*) *reaction* takes place in the matrix of the mitochondrion. Pyruvate is broken down from a 3-carbon (C_3) to a 2-carbon (C_2) acetyl group, and a 1-carbon CO_2 molecule is released. Because glycolysis ends with two molecules of pyruvate, the prep reaction occurs twice per glucose molecule.
- The *citric acid cycle* also takes place in the matrix of the mitochondrion. Each 2-carbon acetyl group matches up with a 4-carbon molecule, forming two 6-carbon citrate molecules. As citrate bonds are broken and oxidation occurs, NADH and $FADH_2$ are formed, and two CO_2 per citrate are released. The citric acid cycle is able to produce one ATP per turn. Because two acetyl groups enter the cycle per glucose molecule, the cycle turns twice.

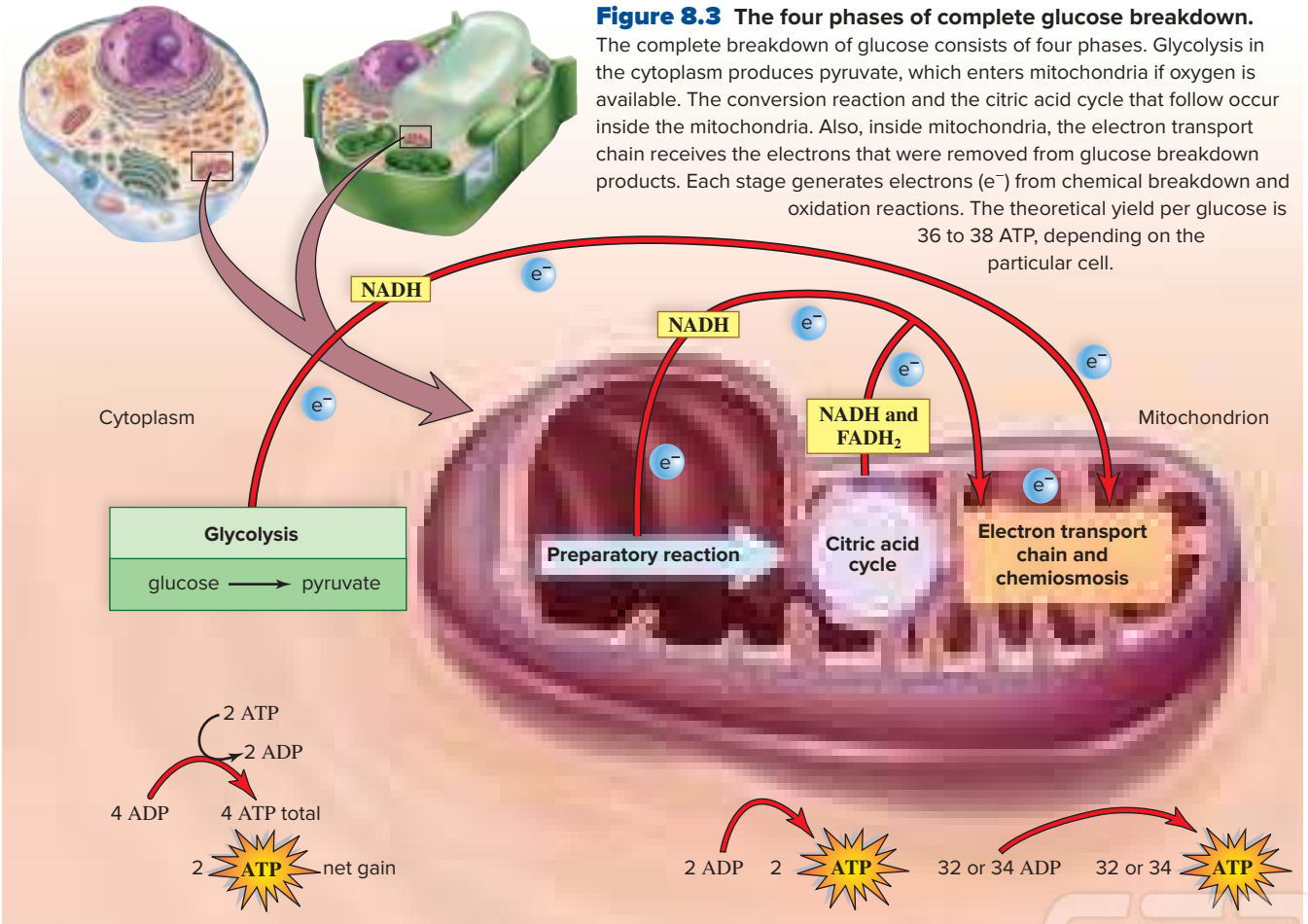


Figure 8.3 The four phases of complete glucose breakdown.
The complete breakdown of glucose consists of four phases. Glycolysis in the cytoplasm produces pyruvate, which enters mitochondria if oxygen is available. The conversion reaction and the citric acid cycle that follow occur inside the mitochondria. Also, inside mitochondria, the electron transport chain receives the electrons that were removed from glucose breakdown products. Each stage generates electrons (e^-) from chemical breakdown and oxidation reactions. The theoretical yield per glucose is 36 to 38 ATP, depending on the particular cell.

ISTUDY

- The *electron transport chain* (*ETC*) is a series of carriers on the cristae of the mitochondria. NADH and $FADH_2$ give up their high-energy electrons to the chain. Energy is released and captured as the electrons move from a higher-energy to a lower-energy state during each redox reaction. Later, this energy is used for the production of between 32 and 34 ATP by chemiosmosis. After oxygen receives electrons at the end of the chain, it combines with hydrogen ions (H^+) and becomes water (H_2O).

Pyruvate, the end product of glycolysis, is a pivotal metabolite; its further treatment depends on whether oxygen is available. If oxygen is available, pyruvate enters a mitochondrion and is broken down completely to CO_2 and H_2O, as shown in the overall cellular respiration equation. If oxygen is not available, pyruvate is further metabolized in the cytoplasm by an anaerobic process called **fermentation.** Fermentation results in a net gain of only two ATP per glucose molecule.

Check Your Progress 8.1

1. Describe how the formula for cellular respiration includes both oxidation and reduction reactions.
2. Explain why NAD^+ and FAD are needed during cellular respiration.
3. Describe the four phases of complete glucose breakdown, including which release CO_2 and which produce H_2O.

8.2 Outside the Mitochondria: Glycolysis

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the role of glycolysis in cellular respiration.
2. List the inputs and outputs of glycolysis.
3. Explain how energy-investment and energy-harvesting steps of glycolysis result in two net ATP.

Glycolysis, which takes place within the cytoplasm outside the mitochondria, is the breakdown of C_6 (6-carbon) glucose to two C_3 (3-carbon) pyruvate molecules. Because glycolysis occurs universally in organisms, it most likely evolved before the citric acid cycle and the electron transport chain. This may be why glycolysis occurs in the cytoplasm and does not require oxygen: There was no free oxygen in Earth's early atmosphere.

Glycolysis is a series of ten reactions, and just as you would expect for a metabolic pathway, each step has its own enzyme. The pathway can be conveniently divided into the energy-investment step and the energy-harvesting steps. During the energy-investment step, ATP is used to "jump-start" glycolysis. During the energy-harvesting steps, four total ATP are made, producing two net ATP overall.

Energy-Investment Step

As glycolysis begins, two ATP are used to activate glucose by adding phosphate. Glucose eventually splits into two C_3 molecules known as G3P, the same molecule produced during photosynthesis. Each G3P has a phosphate group, each of which is acquired from an ATP molecule. From this point on, each C_3 molecule undergoes the same series of reactions.

Energy-Harvesting Steps

Oxidation of G3P now occurs by the removal of electrons accompanied by hydrogen ions. In duplicate reactions, electrons are picked up by coenzyme NAD^+, which becomes

$$2\ NAD^+ + 4\ e^- + 2\ H^+ \rightarrow 2\ NADH$$

When O_2 is available, each NADH molecule carries two high-energy electrons to the electron transport chain and becomes NAD^+ again. In this way, NAD^+ is recycled and used again.

The addition of inorganic phosphate results in a high-energy phosphate group on each C_3 molecule. These phosphate groups are used to directly synthesize two ATP in the later steps of glycolysis. This is called **substrate-level ATP synthesis,** also known as *substrate-level phosphorylation,* because an enzyme passes a high-energy phosphate to ADP, and ATP results (Fig. 8.4). Notice that this is an example of a coupled reaction due to the fact that an energy-releasing reaction is driving forward an energy-requiring reaction on the surface of the enzyme.

Oxidation occurs again, but by the removal of H_2O. Substrate-level ATP synthesis occurs again per each C_3, and two molecules of pyruvate result. Subtracting the two ATP that were used to get started, and the four ATP produced overall, there is a net gain of two ATP from glycolysis (Fig. 8.5).

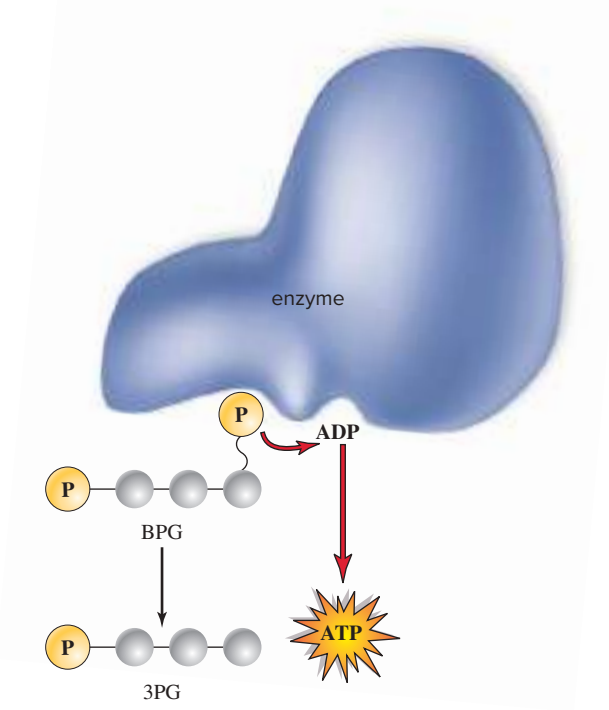


Figure 8.4 Substrate-level ATP synthesis. Substrates participating in the reaction are oriented on the enzyme. A phosphate group is transferred to ADP, producing one ATP molecule. During glycolysis (see Fig. 8.5), BPG is a C_3 substrate (each gray ball is a carbon atom) that gives up a phosphate group to ADP. This reaction occurs twice per glucose molecule.

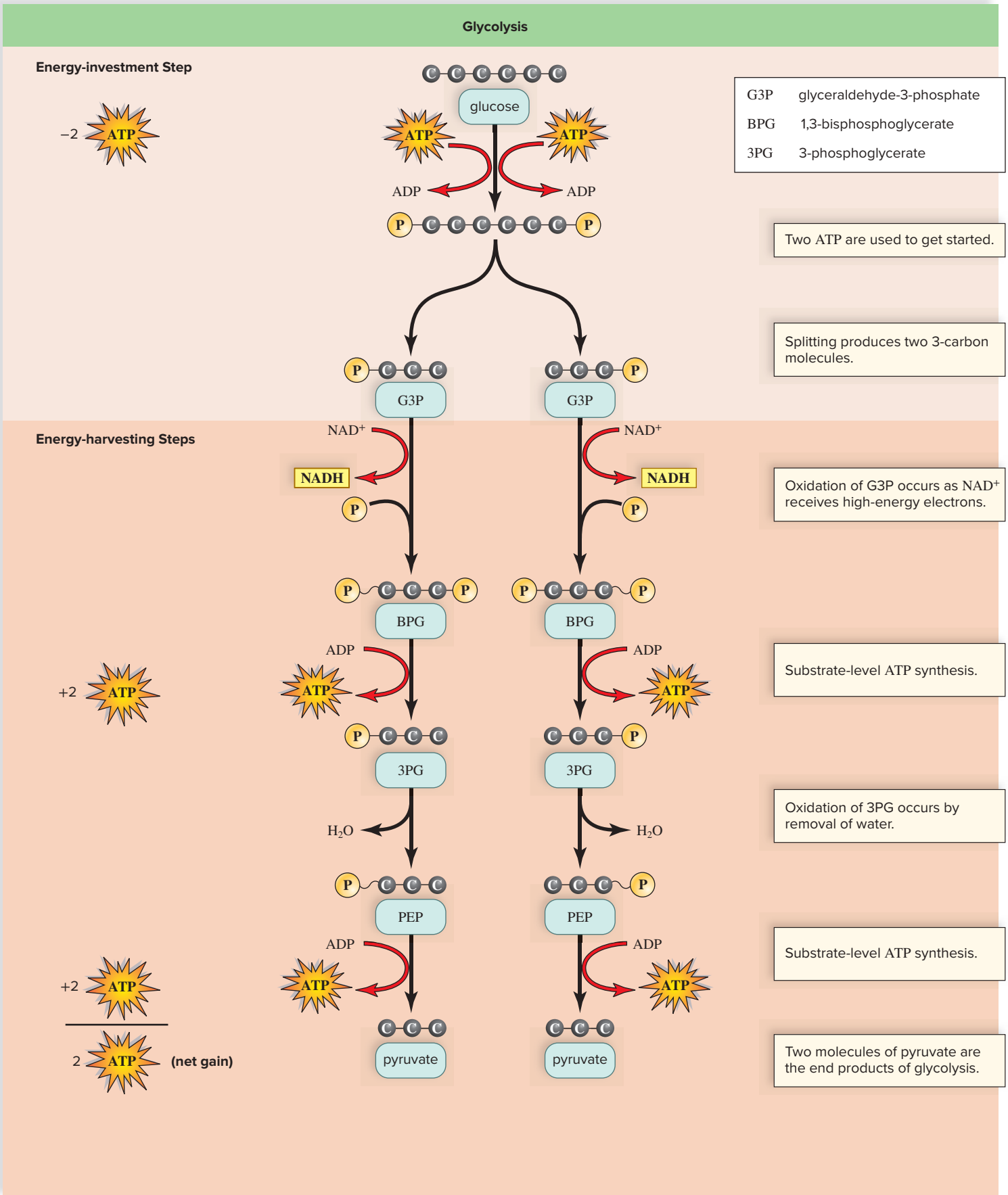
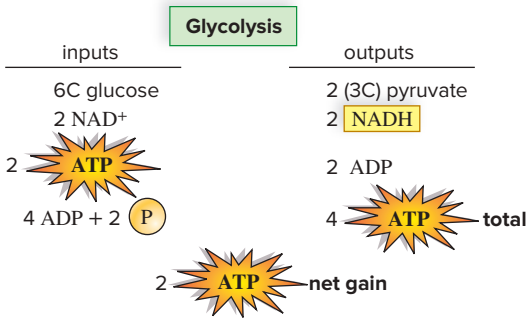


Figure 8.5 Glycolysis. This metabolic pathway begins with C_6 glucose (each gray ball is a carbon atom) and ends with two C_3 pyruvate molecules. Net gain of two ATP molecules can be calculated by subtracting those expended during the energy-investment step from those produced during the energy-harvesting steps. Each of the ten steps is catalyzed by a specialized enzyme.

ISTUDY

Inputs and Outputs of Glycolysis

All together, the inputs and outputs of glycolysis are as follows:



Notice that, so far, we have accounted for only 2 of the 36 to 38 ATP molecules that are theoretically possible when glucose is completely broken down to CO_2 and H_2O. When O_2 is available, the end product of glycolysis—pyruvate—enters the mitochondria, where it is metabolized. If O_2 is not available, fermentation, which is discussed in Section 8.3, occurs.

Check Your Progress 8.2

1. Explain where ATP is used and produced in glycolysis.
2. Explain how ATP is produced from ADP and phosphate during glycolysis.
3. Summarize the location, inputs, and outputs of glycolysis.

8.3 Outside the Mitochondria: Fermentation

Learning Outcomes

Upon completion of this section, you should be able to

1. Summarize the two fermentation pathways.
2. Discuss the conditions under which organisms may switch between cellular respiration and fermentation.
3. Compare the benefits and drawbacks of fermentation.

Complete glucose breakdown requires an input of oxygen to keep the electron transport chain working. So how does the cell produce energy if oxygen is limited? **Fermentation** is an anaerobic process that produces a limited amount of ATP in the absence of oxygen. In animal cells, including human cells, the end product of glycolysis, pyruvate, is reduced by NADH to lactate (Fig. 8.6). Depending on their particular enzymes, bacteria vary as to whether they produce an organic acid, such as lactate, or an alcohol and CO_2. Yeasts are good examples of organisms that generate ethyl alcohol and CO_2 as a result of fermentation.

Why is it beneficial for pyruvate to be reduced when oxygen is not available? Because the cell still needs energy when oxygen is absent. The fermentation reaction regenerates NAD^+, which is required for the first step in the energy-harvesting phase of glycolysis. This NAD^+ is now "free" to return to the earlier reaction (see return arrow in Fig. 8.6) and become reduced once more.

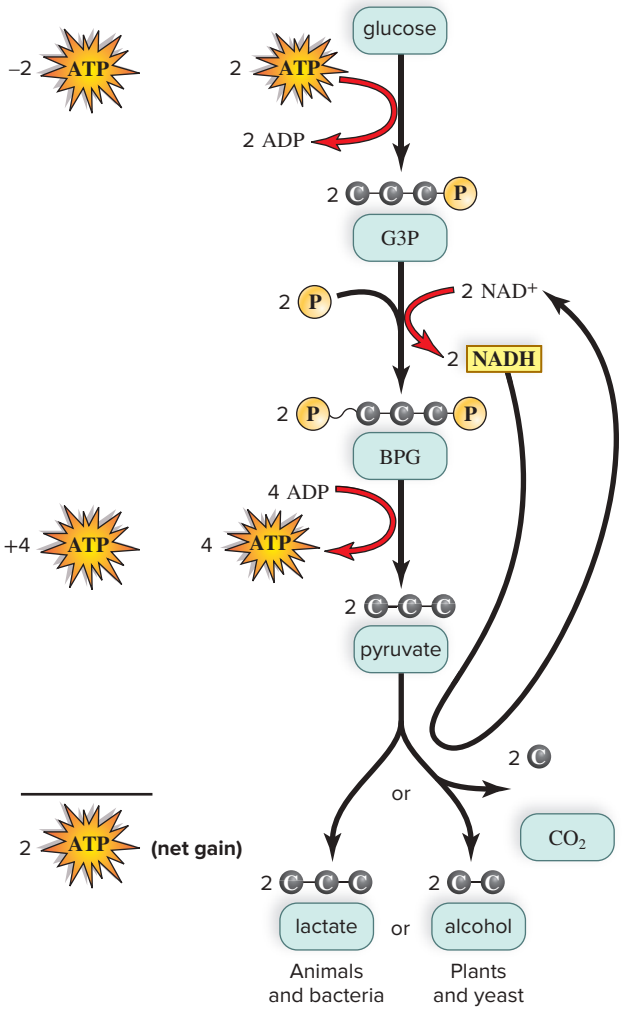


Figure 8.6 Fermentation. Fermentation consists of glycolysis followed by a reduction of pyruvate. This recycles NAD^+ and it returns to the glycolytic pathway to pick up more electrons. As with glycolysis, each step is catalyzed by a specialized enzyme.

Although this process generates much less ATP than when oxygen is present and glucose is fully metabolized into CO_2 and H_2O in the ETC, glycolysis and substrate-level ATP synthesis produce enough energy for the cell to continue working.

Advantages and Disadvantages of Fermentation

As discussed in the Nature of Science feature, "Fermentation and Food Production," people have long used anaerobic bacteria that produce lactate to create cheese, yogurt, and sauerkraut—even before we knew that bacteria were responsible! Other bacteria produce chemicals of industrial importance, including isopropanol, butyric acid, proprionic acid, and acetic acid when they ferment. Yeasts, of course, are used to make breads rise. In addition, alcoholic fermentation is utilized to produce wine, beer, and other alcoholic beverages.

Despite its low yield of only two ATP made by substrate-level ATP synthesis, lactic acid fermentation is essential to certain animals and tissues. Typically, animals use lactic acid fermentation for a rapid burst of energy, such as a cheetah chasing a gazelle. Also, when muscles are working vigorously over a short period of

THEME Nature of Science

Fermentation and Food Production

At the grocery store, you will find such items as bread, yogurt, soy sauce, pickles, and maybe even beer or wine (Fig. 8A). These are just a few of the many foods produced when microorganisms ferment (break down sugar in the absence of oxygen). Foods produced by fermentation last longer, because the fermenting organisms have removed many of the nutrients that would attract other organisms. The products of fermentation can even be dangerous to the very organisms that produced them, as when yeast are killed by the alcohol they produce.

Yeast Fermentation

Baker's yeast, *Saccharomyces cerevisiae,* is added to bread for the purpose of leavening—the dough rises when the yeast give off CO_2. The ethyl alcohol produced by the fermenting yeast evaporates during baking. The many varieties of sourdough breads obtain their leavening from a starter composed of fermenting yeast, along with bacteria from the environment. Depending on the community of microorganisms in the starter, the flavor of the bread may range from sour and tangy, as in San Francisco–style sourdough, to a milder taste, such as that produced by most Amish friendship bread recipes.

Ethyl alcohol in beer and wine is produced when yeast ferment carbohydrates. When yeast ferment fruit carbohydrates, the end result is wine. If they ferment grain, beer results. A few specialized varieties of beer, such as traditional wheat beers, have a distinctive, sour taste, because they are produced with the assistance of lactic acid–producing bacteria, such as those of the genus *Lactobacillus.* Stronger alcoholic drinks (e.g., whiskey and vodka) require distillation to concentrate the alcohol content.

Bacteria that produce acetic acid, including *Acetobacter aceti,* spoil wine. These bacteria convert the alcohol in wine or cider to acetic acid (vinegar). Until the renowned nineteenth-century scientist Louis Pasteur invented the process of pasteurization, acetic acid bacteria commonly caused wine to spoil. Although today we generally associate the process of pasteurization with making milk safe to drink, it was originally developed to reduce bacterial contamination in wine, so that limited acetic acid would be produced. The discovery of pasteurization is another example of how the pursuit of scientific knowledge can positively affect our lives.

Bacterial Fermentation

Yogurt, sour cream, and cheese are produced through the action of various lactic acid bacteria that cause milk to sour. Milk contains lactose, which these bacteria use as a carbohydrate source for fermentation. Yogurt, for example, is made by adding lactic acid bacteria, such as *Streptococcus thermophilus* and *Lactobacillus bulgaricus,* to milk and then incubating it to encourage the bacteria to convert the lactose. During the production of cheese, an enzyme called rennin must also be added to the milk to cause it to coagulate and become solid.

Old-fashioned brine cucumber pickles, sauerkraut, and kimchi are pickled vegetables produced by the action of acid-producing, fermenting bacteria that can survive in high-salt environments. Salt is used to draw liquid out of the vegetables and to aid in their preservation. The bacteria need not be added to the vegetables, because they are already present on the surfaces of the plants.

Soy Sauce Production

Soy sauce is traditionally made by adding a mold, *Aspergillus,* and a combination of yeast and fermenting bacteria to soybeans and wheat. The mold breaks down starch, supplying the fermenting microorganisms with sugar they can use to produce alcohol and organic acids.

As you can see from each of these examples, fermentation is a biologically and economically important process that scientists use for the betterment of our lives.

Questions to Consider

1. How would the production of fermentation products differ from that of other food products?
2. What products of fermentation do you use on a daily basis?

Figure 8A Products from fermentation. Fermentation of different carbohydrates by microorganisms like bacteria and yeast helps produce the products shown. Ricochet Creative Productions LLC/McGraw Hill

ISTUDY

time, lactic acid fermentation provides them with ATP, even though oxygen is temporarily in limited supply.

Efficiency of Fermentation

The two ATP produced per glucose during alcoholic fermentation and lactic acid fermentation are equivalent to 14.6 kcal/mol glucose. Complete glucose breakdown to CO_2 and H_2O represents a possible energy yield of 686 kcal/mol. Therefore, the efficiency of fermentation is only 14.6 kcal/686 kcal × 100, or 2.1% of the total possible for the complete breakdown of glucose. The inputs and outputs of fermentation are shown here:

Fermentation

inputs	outputs
glucose	2 lactate or 2 alcohol and 2 CO_2
2 ADP + 2 P	2 ATP **net gain**

The two ATP produced by fermentation fall far short of the theoretical 36 to 38 ATP molecules that may be produced by cellular respiration. To achieve this number of ATP per glucose molecule, it is necessary to move on to the reactions and pathways that occur with oxygen in the mitochondria.

Check Your Progress 8.3

1. Explain fermentation's role in NAD^+ regeneration.
2. Summarize the two forms of fermentation.
3. List the advantages and disadvantages of fermentation.

8.4 Inside the Mitochondria

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the fate of each carbon during the complete aerobic metabolism of glucose.
2. Contrast substrate-level phosphorylation and chemiosmosis as methods of ATP synthesis.
3. Describe how electron energy from redox reactions is used to create a proton gradient.

The preparatory (prep) reaction, the citric acid cycle, and the electron transport chain, which are needed for the complete breakdown of glucose, take place within the mitochondria. A **mitochondrion** has a double membrane with an intermembrane space (between the outer and inner membrane). Cristae are folds of inner membrane that jut out into the matrix, the innermost compartment, which is filled with a gel-like fluid (Fig. 8.7). Like a chloroplast, a

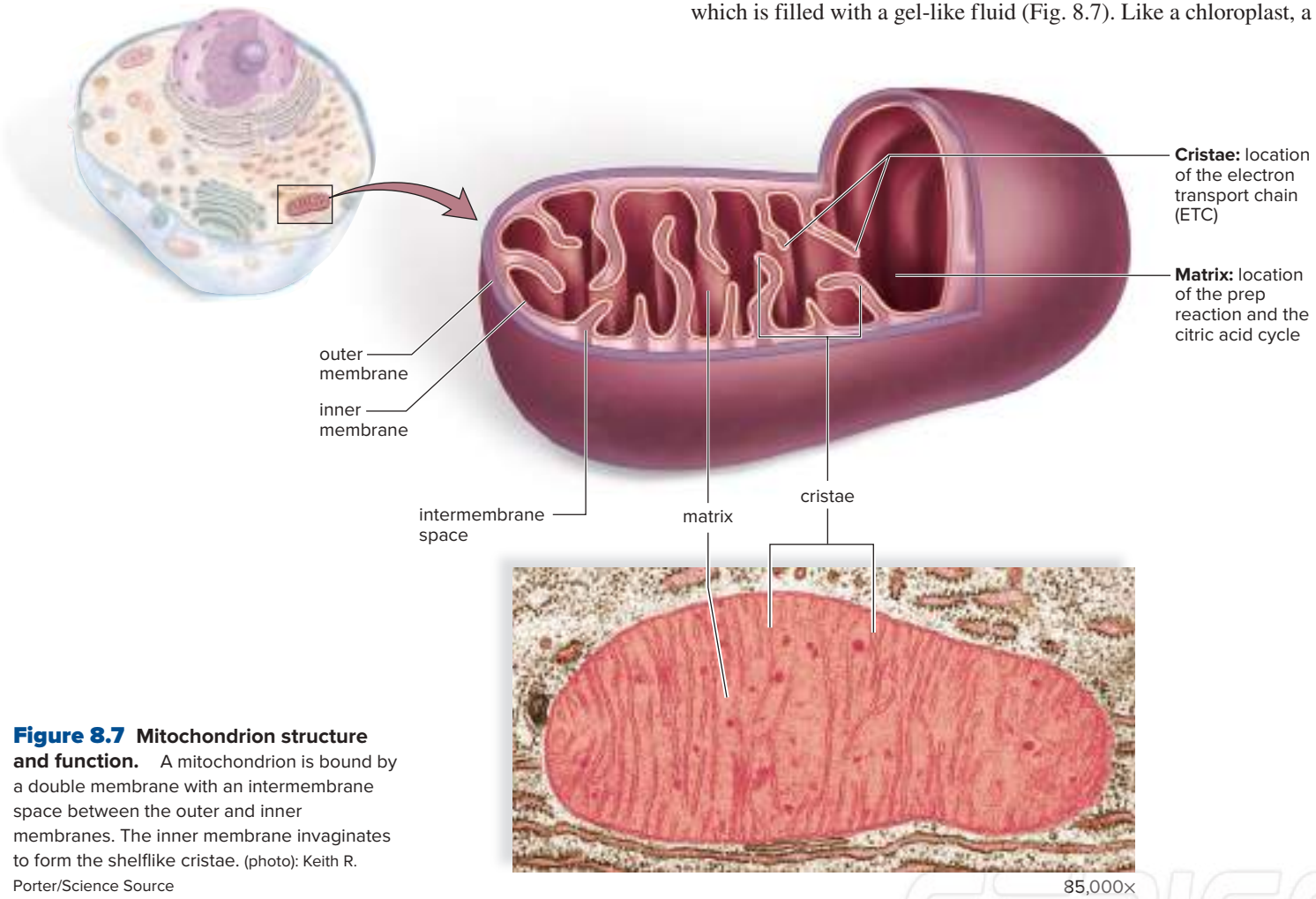


Figure 8.7 Mitochondrion structure and function. A mitochondrion is bound by a double membrane with an intermembrane space between the outer and inner membranes. The inner membrane invaginates to form the shelflike cristae. (photo): Keith R. Porter/Science Source

mitochondrion is highly structured, so we would expect reactions to be located in particular parts of this organelle.

The enzymes that speed the prep reaction and the citric acid cycle are arranged in the matrix, and the electron transport chain is located in the cristae in a very organized manner. Most of the ATP from cellular respiration is produced in mitochondria; therefore, mitochondria are often called the powerhouses of the cell.

The Preparatory Reaction

The **preparatory (prep) reaction** is so called because it converts products from glycolysis into products that enter the citric acid cycle. In this reaction, the C_3 pyruvate is converted to a C_2 acetyl group and CO_2 is given off (Fig. 8.8). This is an oxidation reaction in which electrons are removed from pyruvate by NAD^+ and NADH is formed. One prep reaction occurs per pyruvate, so the prep reaction occurs twice per glucose molecule.

The C_2 acetyl group is combined with a molecule known as CoA. CoA will carry the acetyl group to the citric acid cycle in the mitochondrial matrix. The two NADH carry electrons to the electron transport chain. What about the CO_2? In vertebrates, such as ourselves, CO_2 freely diffuses out of cells into the blood, which transports it to the lungs, where it is exhaled.

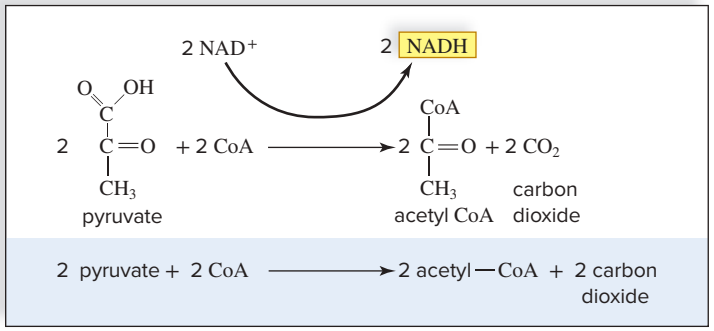


Figure 8.8 Preparatory reaction. The preparatory reaction occurs in the mitochondria. Pyruvate molecules from glycolysis are converted to acetyl—CoA molecules, which enter the citric acid cycle. NADH and CO_2 are generated during the process.

The Citric Acid Cycle

The **citric acid cycle,** also called the Krebs cycle, is a cyclical metabolic pathway located in the matrix of mitochondria (Fig. 8.9). At the start of the citric acid cycle, the (C_2) acetyl group carried by CoA joins with a C_4 molecule, and a C_6 citrate molecule results. During the cycle, oxidation occurs when electrons are accepted by NAD^+ in three instances and by FAD in one instance. Therefore, three NADH and one $FADH_2$ are formed as a result of one turn of the citric acid cycle. Also, the acetyl group received from the prep reaction is oxidized to two CO_2 molecules. Substrate-level ATP synthesis is also an important event of the citric acid cycle. In substrate-level ATP synthesis, you will recall, an enzyme passes a high-energy phosphate to ADP, and ATP results.

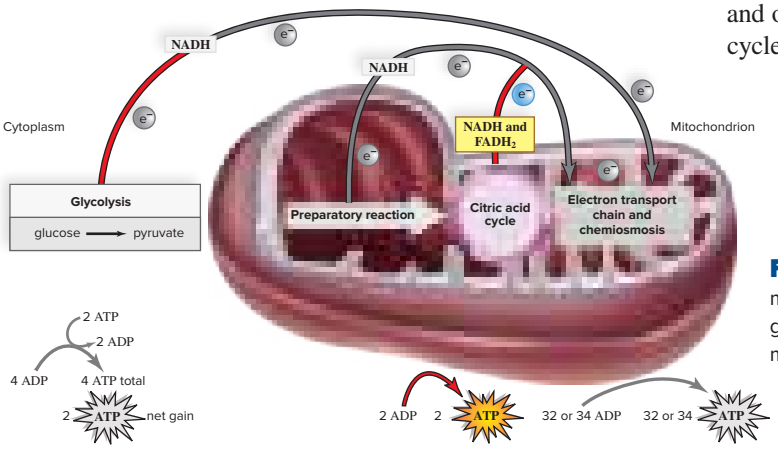
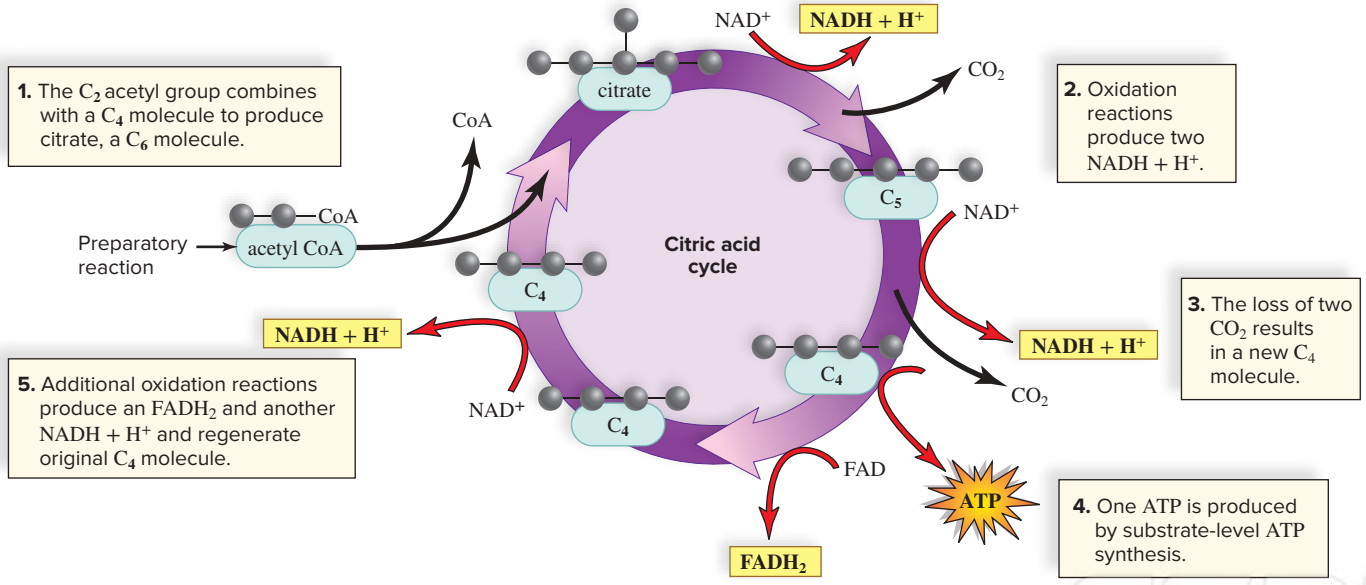


Figure 8.9 Citric acid cycle. The citric acid cycle occurs in the mitochondria and is responsible for the complete reduction of the glucose molecule. The citric acid cycle turns twice per glucose molecule.



ISTUDY

Because the citric acid cycle turns twice for each original glucose molecule, the inputs and outputs of the citric acid cycle per glucose molecule are as follows:

Citric acid cycle

inputs	outputs
2 (2C) acetyl groups	4 CO_2
6 NAD^+	6 NADH
2 FAD	2 $FADH_2$
2 ADP + 2 P	2 ATP

Production of CO_2

The six carbon atoms originally located in a glucose molecule have now become CO_2. The prep reaction produces the first two CO_2, and the citric acid cycle produces the last four CO_2 per glucose molecule. We have already mentioned that this is the CO_2 humans and other animals breathe out.

Thus far, we have broken down glucose to CO_2 and hydrogen atoms. Recall that, as bonds are broken and glucose gets converted to CO_2, energy in the form of high-energy electrons is released. NADH and $FADH_2$ capture those high-energy electrons and carry them to the electron transport chain, as discussed next.

Electron Transport Chain

The **electron transport chain (ETC)** located in the cristae of the mitochondria and the plasma membrane of aerobic prokaryotes, is a series of carriers that pass electrons from one to the other. The high-energy electrons that enter the electron transport chain are carried by NADH and $FADH_2$. Figure 8.10 is arranged to show that high-energy electrons enter the chain and low-energy electrons leave the chain.

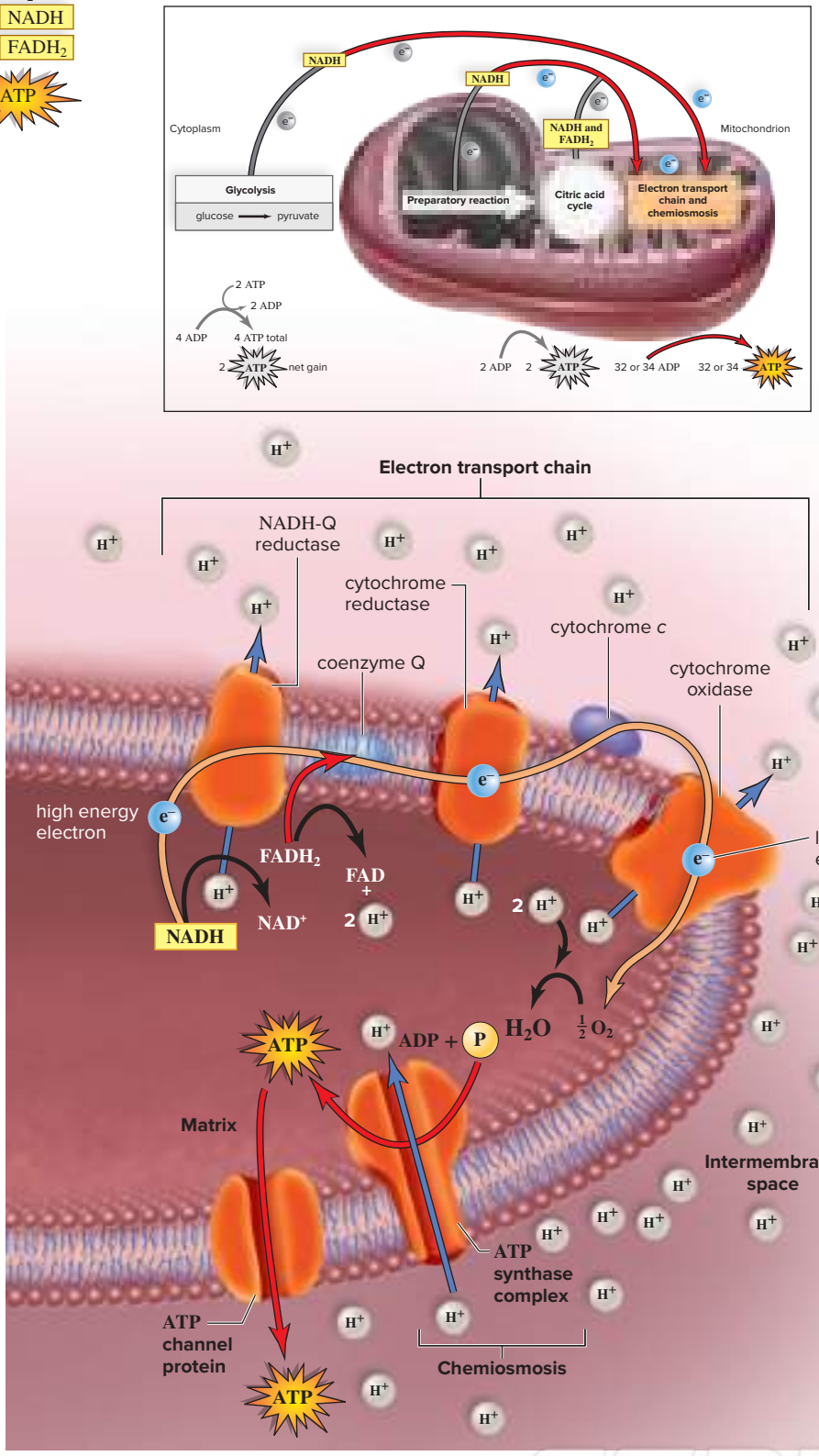


Figure 8.10 Organization and function of the electron transport chain. The electron transport chain is located in the mitochondrial cristae. NADH and $FADH_2$ take electrons to the electron transport chain. As electrons move from one protein complex to the other via redox reactions, energy is used to pump hydrogen ions (H^+) from the matrix into the intermembrane space. As hydrogen ions flow down a concentration gradient from the intermembrane space into the mitochondrial matrix, ATP is synthesized by the enzyme ATP synthase. For every pair of electrons that enters by way of NADH, three ATP result. For every pair of electrons that enters by way of $FADH_2$, two ATP result. Oxygen, the final acceptor of the electrons, becomes a part of water. ATP leaves the matrix by way of a channel protein.

ISTUDY

Members of the Chain

When NADH gives up its electrons, it becomes oxidized to NAD^+, and when $FADH_2$ gives up its electrons, it becomes oxidized to FAD. The next carrier gains the electrons and is reduced. This oxidation-reduction reaction starts the process, and each of the carriers, in turn, becomes reduced and then oxidized as the electrons move down the chain.

Many of the redox carriers are cytochrome molecules. A **cytochrome** is a protein that has a tightly bound heme group with a central atom of iron, the same as hemoglobin does. When the iron accepts electrons, it becomes reduced, and when iron gives them up, it becomes oxidized. As the pair of electrons is passed from carrier to carrier, energy is captured and eventually used to form ATP molecules. A number of poisons, such as cyanide, cause death by binding to and blocking the function of cytochromes.

What is the role of oxygen in cellular respiration and the reason we breathe to take in oxygen? Oxygen is the final acceptor of electrons from the electron transport chain. Oxygen receives the energy-spent electrons from the last of the carriers (i.e., cytochrome oxidase). After receiving electrons, oxygen combines with hydrogen ions, and water forms:

$$\frac{1}{2}\,O_2 + 2\,e^- + 2\,H^+ \longrightarrow H_2O$$

The critical role of oxygen as the final acceptor of electrons during cellular respiration is exemplified by noting that if oxygen is not present, the chain does not function, and no ATP is produced by mitochondria. The limited capacity of the body to form ATP in a way that does not involve the electron transport chain means that death eventually results if oxygen is not available.

Cycling of Carriers

When NADH delivers high-energy electrons to the first carrier of the electron transport chain, enough energy has been captured by the time the electrons are received by O_2 to permit the production of three ATP molecules. When $FADH_2$ delivers high-energy electrons to the electron transport chain, two ATP are produced.

Once NADH has delivered electrons to the electron transport chain and has become NAD^+, it is able to return and pick up more hydrogen atoms. The reuse of coenzymes increases cellular efficiency, because the cell does not have to constantly make new NAD^+; it simply recycles what is already there.

The ETC Pumps Hydrogen Ions. Essentially, the electron transport chain consists of three protein complexes and two carriers. The three protein complexes are the NADH-Q reductase complex, the cytochrome reductase complex, and the cytochrome oxidase complex. The two other carriers that transport electrons between the complexes are coenzyme Q and cytochrome *c* (Fig. 8.10).

We have already seen that the members of the electron transport chain accept electrons, which they pass from one to the other via redox reactions. So what happens to the hydrogen ions (H^+) carried by NADH and $FADH_2$? The complexes of the electron transport chain use the energy released during redox reactions to pump these hydrogen ions from the matrix into the intermembrane space of a mitochondrion.

The vertical arrows in Figure 8.10 show that the protein complexes of the electron transport chain all pump H^+ into the intermembrane space. Energy obtained from electron passage is needed, because H^+ ions are pumped and actively transported against their gradient. This means the few H^+ ions in the matrix will be moved to the intermembrane space, where there are already many H^+ ions. Just as the walls of a dam hold back water, allowing it to collect, so do cristae hold back hydrogen ions. Eventually, a strong electrochemical gradient develops; about ten times as many H^+ are found in the intermembrane space as are present in the matrix.

The ATP Synthase Complex Produces ATP. The ATP synthase complex can be likened to the gates of a dam. When the gates of a hydroelectric dam are opened, water rushes through, and electricity (energy) is produced. Similarly, when H^+ flows down a gradient from the intermembrane space into the matrix, the enzyme ATP synthase synthesizes ATP from ADP + Ⓟ. This process is called **chemiosmosis,** because ATP production is tied to the establishment of an H^+ gradient.

Once formed, ATP moves out of mitochondria and is used to perform cellular work, during which it breaks down to ADP and Ⓟ. These molecules are then returned to mitochondria for recycling. At any given time, the amount of ATP in a human would sustain life for only about a minute; therefore, ATP synthase must constantly produce ATP. It is estimated that mitochondria produce our body weight in ATP every day.

Active Tissues Contain More Mitochondria. Active tissues, such as muscles, require greater amounts of ATP and have more mitochondria than less active cells. When a burst of energy is required, however, muscles still utilize fermentation.

As an example of the relative amounts of ATP, consider that the dark meat of chickens, namely the thigh meat, contains more mitochondria than the white meat of the breast. This suggests that chickens mainly walk or run, rather than fly, about the barnyard.

Energy Yield from Glucose Metabolism

Figure 8.11 calculates the theoretical ATP yield for the complete breakdown of glucose to CO_2 and H_2O during cellular respiration. Notice that the diagram includes the number of ATP produced directly by glycolysis and the citric acid cycle (to the *left*), as well as the number produced as a result of electrons passing down the electron transport chain (to the *right*). A maximum of 32 to 34 ATP molecules may be produced by the electron transport chain.

Substrate-Level ATP Synthesis

Per glucose molecule, there is a net gain of two ATP from glycolysis, which takes place in the cytoplasm. The citric acid cycle, which occurs in the matrix of mitochondria, accounts for two ATP per glucose molecule. This means that a total of four ATP are formed by substrate-level ATP synthesis outside the electron transport chain.

ETC and Chemiosmosis

Most ATP is produced by the electron transport chain and chemiosmosis. Per glucose molecule, ten NADH and two $FADH_2$ take electrons to the electron transport chain. For each NADH formed *inside* the mitochondria by the citric acid cycle, three ATP result, but for each $FADH_2$, only two ATP are produced. Figure 8.10 explains the reason for this difference: $FADH_2$ delivers its electrons to the transport chain after NADH, and therefore these electrons do not participate in as

ISTUDY

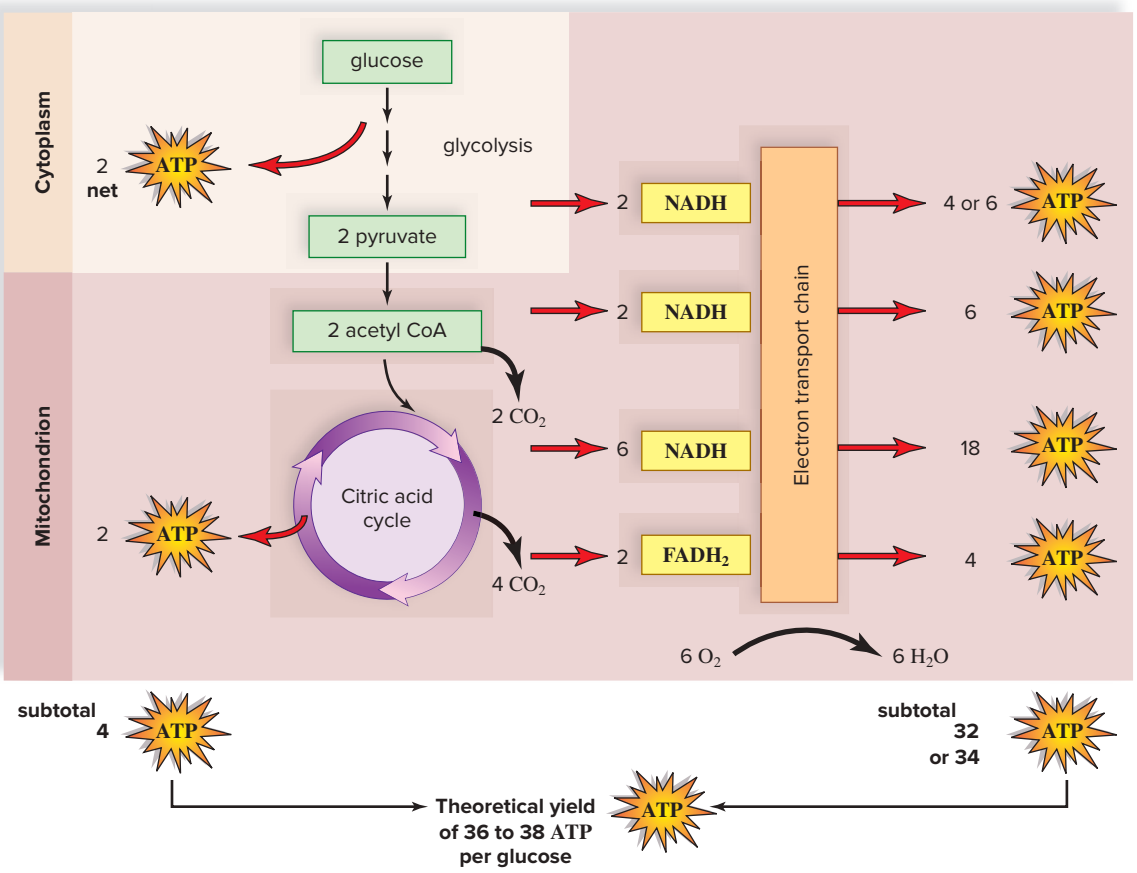


Figure 8.11 Accounting of energy yield per glucose molecule breakdown. Substrate-level ATP synthesis during glycolysis and the citric acid cycle accounts for four ATP. The electron transport chain accounts for 32 or 34 ATP, making the theoretical grand total of ATP between 36 and 38 ATP. Other factors may reduce the efficiency of cellular respiration. For example, cells differ as to the delivery of the electrons from NADH generated outside the mitochondria. If they are delivered by a shuttle mechanism to the start of the electron transport chain, six ATP result; otherwise, four ATP result.

many redox reactions and don't pump as many H^+ as NADH. Therefore, $FADH_2$ cannot account for as much ATP production.

Efficiency of Cellular Respiration

Figure 8.11 provides the theoretical ATP for each stage of cellular respiration. However, we know now that cells rarely ever achieve these theoretical values. Several factors can lower the ATP yield for each molecule of glucose entering the pathway:

- In some cells, NADH cannot cross mitochondrial membranes, but a "shuttle" mechanism allows its electrons to be delivered to the electron transport chain inside the mitochondria. The cost to the cell is one ATP for each NADH that is shuttled to the ETC. This reduces the overall count of ATP produced as a result of glycolysis, in some cells, to four instead of six ATP.
- At times, cells need to expend energy to move ADP molecules and pyruvate into the cell and to establish protein gradients in the mitochondria.

Considerable research is still being done to find out the precise ATP yield per glucose molecule. However, most estimates place the actual yield at around 30 ATP per glucose. Using this number, we can calculate that only 32–39% of the available energy is usually transferred from glucose to ATP. The rest of the energy is lost in the form of heat.

In the next section, we consider how cellular respiration fits into metabolism as a whole.

Check Your Progress 8.4

1. Explain when carbon is converted from glucose into carbon dioxide during cellular respiration.
2. Examine which processes during glucose breakdown produce the most ATP.
3. Compare the function of the mitochondrial inner membrane to a hydroelectric dam.

8.5 Metabolism

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the pathways of carbohydrate, fat, and protein catabolism.
2. Explain how the structure of mitochondria and chloroplasts enables a flow of energy through living organisms.

Key metabolic pathways routinely draw from pools of particular substrates needed to synthesize or degrade larger molecules. Substrates like the end product of glycolysis—pyruvate—exist as a pool that is continuously affected by changes in cellular and environmental conditions (Fig. 8.12). Degradative reactions, termed **catabolism,** that break down molecules must be dynamically

ISTUDY

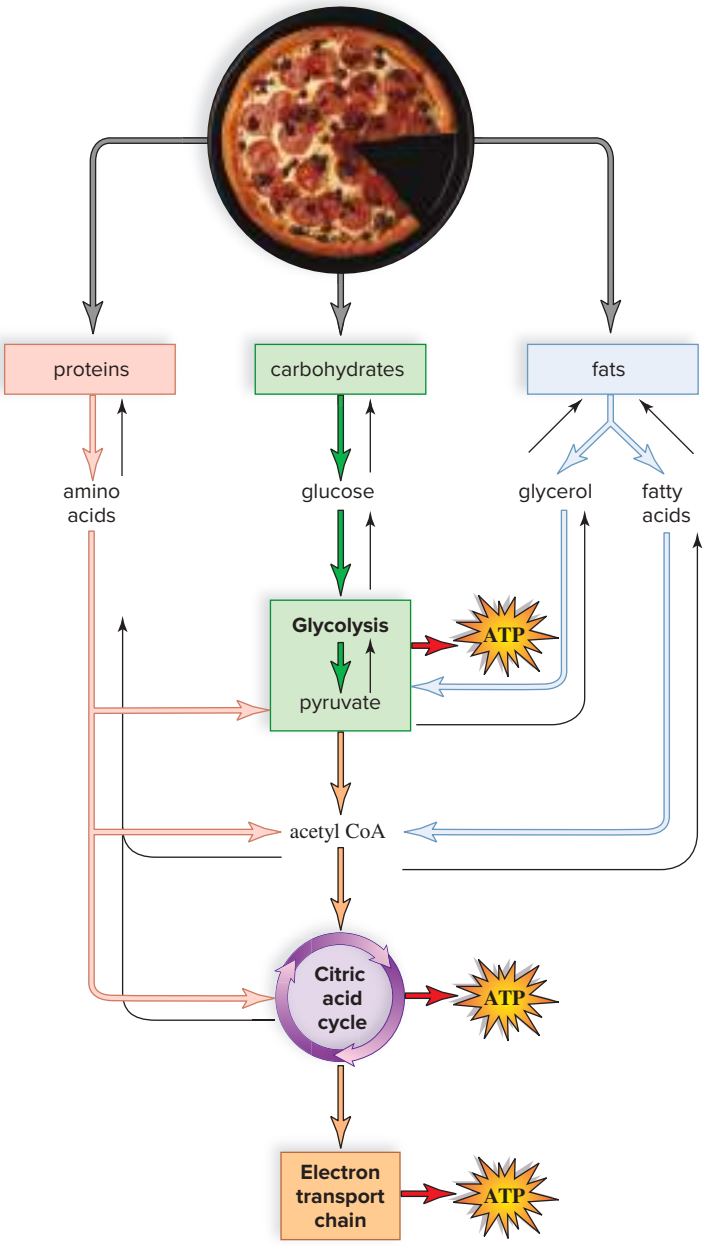


Figure 8.12 The metabolic pool concept. Carbohydrates, fats, and proteins can be used as energy sources, and their monomers (carbohydrates and proteins) or subunits (fats) enter degradative pathways at specific points. Catabolism produces molecules that can also be used for anabolism of other compounds. (photo): C Squared Studios/Photodisc/Getty Images

balanced with constructive reactions, or **anabolism.** For example, catabolic breakdown of fats will occur when insufficient carbohydrate is present; this breakdown adds to the **metabolic pool** of pyruvate. When energy needs to be stored as fat, pyruvate is taken from the pool. This dynamic balance of catabolism and anabolism is essential to optimal cellular function.

Catabolism

We already know that glucose is broken down during cellular respiration. However, other molecules like fats and proteins can also be broken down as necessary. When a fat is used as an energy source, it breaks down to glycerol and three fatty acids. As Figure 8.12 indicates, glycerol can be converted to pyruvate and enter glycolysis. The fatty acids are converted to 2-carbon acetyl CoA that enters the citric acid cycle. An 18-carbon fatty acid results in nine acetyl CoA molecules. Calculation shows that respiration of these can produce a total of 108 ATP molecules. This is why fats are an efficient form of stored energy—the three long fatty acid chains per fat molecule can produce considerable ATP when needed.

Proteins are less frequently used as an energy source, but they are available as necessary. The carbon skeleton of amino acids can enter glycolysis, be converted to acetyl groups, or enter the citric acid cycle at some other juncture. The carbon skeleton is produced in the liver when an amino acid undergoes **deamination,** or the removal of the amino group. The amino group becomes ammonia (NH_3), which enters the urea cycle and becomes part of urea, the primary excretory product of humans. Just where the carbon skeleton begins degradation depends on the length of the *R* group, because this determines the number of carbons left after deamination.

Anabolism

We have already seen that the building of new molecules requires ATP produced during breakdown of molecules. These catabolic reactions also provide the basic components used to build new molecules. For example, excessive carbohydrate intake can result in the formation of fat. Extra G3P from glycolysis can be converted to glycerol, and acetyl groups from glycolysis can be joined to form fatty acids, which in turn are used to synthesize fat. This explains why you gain weight from eating too much candy, ice cream, or cake.

Some substrates of the citric acid cycle can be converted to amino acids through transamination—the transfer of an amino group to an organic acid, forming a different amino acid. Plants are able to synthesize all of the amino acids they need. Animals, however, lack some of the enzymes necessary for synthesis of all amino acids. Adult humans, for example, can synthesize 11 of the common amino acids, but they cannot synthesize the other 9. The amino acids that cannot be synthesized must be supplied by the diet—these are called the essential amino acids. The amino acids that *can* be synthesized are called nonessential. It is quite possible for animals to suffer from protein deficiency if their diets do not contain adequate quantities of all the essential amino acids.

The Energy Organelles Revisited

The equation for photosynthesis in a chloroplast is opposite that of cellular respiration in a mitochondrion:

$$\text{energy} + 6\,CO_2 + 6\,H_2O \underset{\text{cellular respiration}}{\overset{\text{photosynthesis}}{\rightleftharpoons}} C_6H_{12}O_6 + 6\,O_2$$

While you were studying photosynthesis and cellular respiration, you may have noticed a remarkable similarity in the structural organization of chloroplasts and mitochondria. Through evolution, all organisms are related, and the similar organization of these

ISTUDY

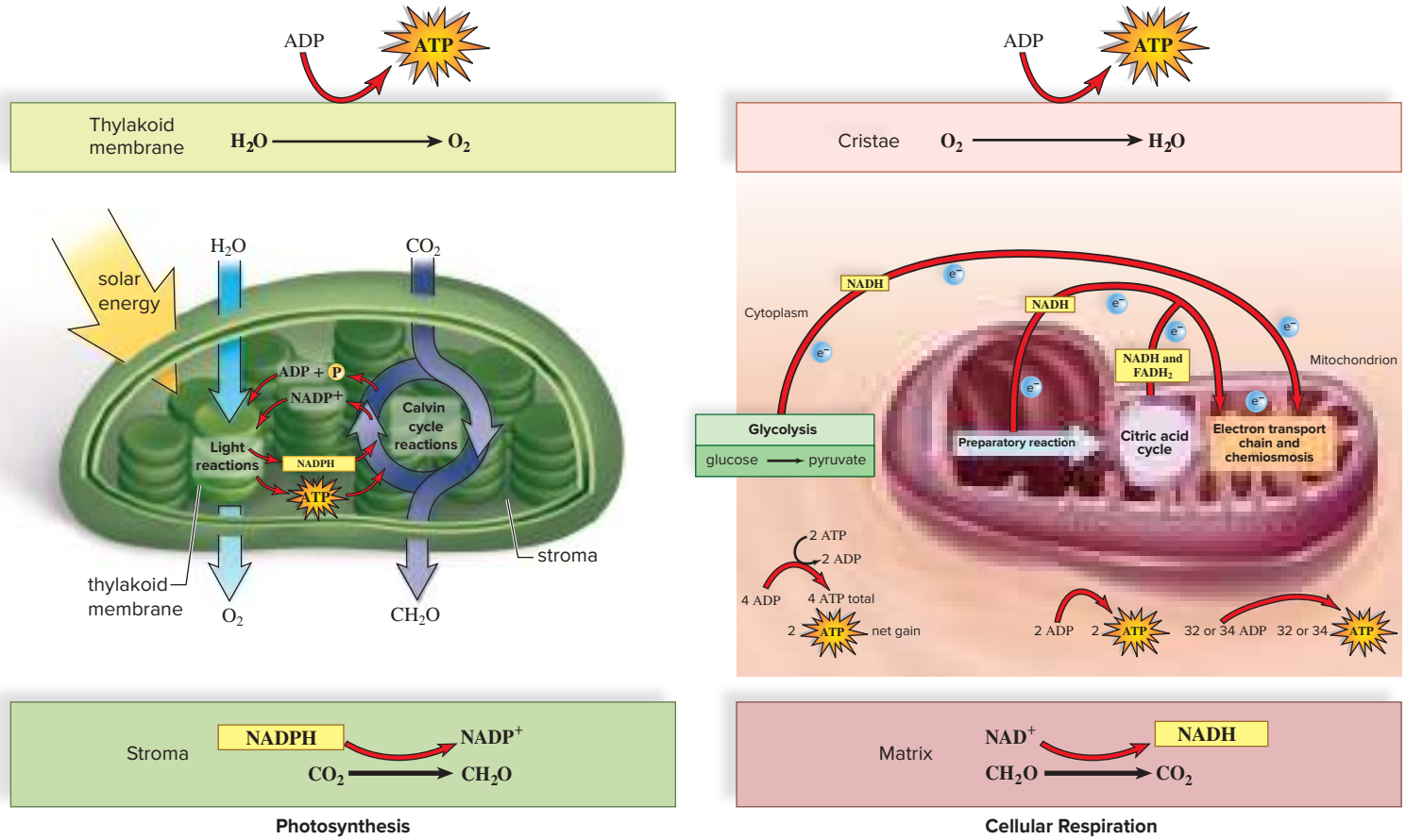


Figure 8.13 Photosynthesis versus cellular respiration. Both photosynthesis and cellular respiration have an electron transport chain located within membranes, where ATP is produced. Both processes have enzyme-catalyzed reactions located within the fluid interior of respective organelles. In photosynthesis, hydrogen atoms are donated by NADPH + H^+ when CO_2 is reduced in the stroma of a chloroplast. During cellular respiration, NADH forms when glucose is oxidized in the cytoplasm and glucose breakdown products are oxidized in the matrix of a mitochondrion.

organelles suggests that they may be related also. The two organelles carry out related but opposite processes (Fig. 8.13):

- *Use of membrane.* In a chloroplast, an inner membrane forms the thylakoids of the grana. In a mitochondrion, an inner membrane forms the convoluted cristae.
- *Electron transport chain (ETC).* An ETC is located on the thylakoid membrane of chloroplasts and the cristae of mitochondria. In chloroplasts, the electrons passed down the ETC have been energized by the sun. In mitochondria, energized electrons have been removed from glucose and glucose products. In both, the ETC establishes an electrochemical gradient of H^+ with subsequent ATP production by chemiosmosis.
- *Enzymes.* In a chloroplast, the stroma contains the enzymes of the Calvin cycle, and in mitochondria, the matrix contains the enzymes of the citric acid cycle. In the Calvin cycle, NADPH and ATP are used to reduce carbon dioxide to a carbohydrate. In the citric acid cycle, the oxidation of glucose products produces NADH and ATP.

Flow of Energy

The ultimate source of energy for producing a carbohydrate in chloroplasts is the sun, whereas the ultimate goal of cellular respiration in a mitochondrion is the conversion of carbohydrate energy into that of ATP molecules. Therefore, energy flows from the sun, through chloroplasts to carbohydrates, and then through mitochondria to ATP molecules.

This flow of energy maintains biological organization at all levels, from molecules to organisms to ultimately the biosphere. In keeping with the energy laws, some energy is lost with each chemical transformation, and eventually the solar energy captured by plants is lost in the form of heat. Therefore, all life depends on a continual input of solar energy.

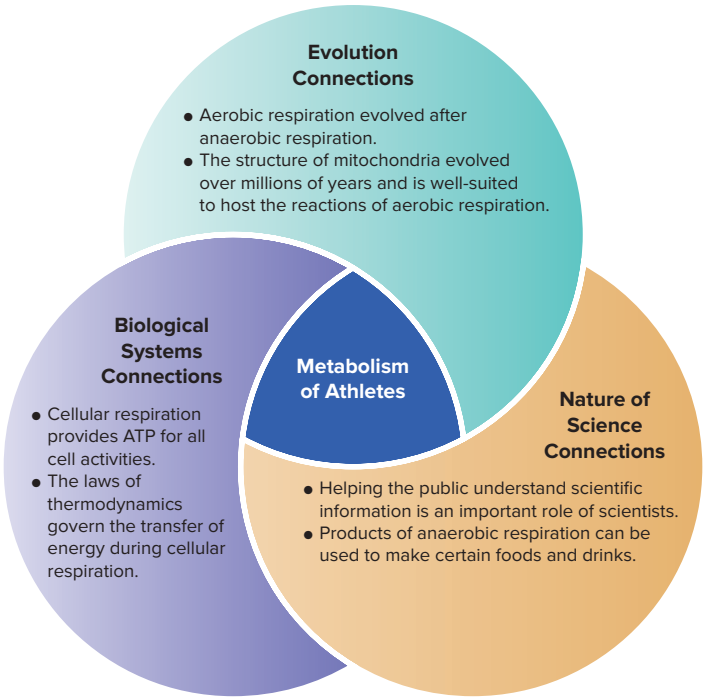
Although energy flows through organisms, chemicals cycle within natural systems. Aerobic organisms utilize the carbohydrate and oxygen produced by chloroplasts to generate energy within the mitochondria to sustain life. Likewise, the carbon dioxide produced by mitochondria returns to chloroplasts to be used in the manufacture of carbohydrates, producing oxygen as a by-product. Therefore, chloroplasts and mitochondria are instrumental not only in allowing a flow of energy through living organisms but also in permitting a cycling of chemicals.

Check Your Progress 8.5

1. Evaluate how catabolism and anabolism are balanced within a cell.
2. Compare the structure and function of chloroplasts and mitochondria.

ISTUDY

Connecting *the* Concepts



What Did You Learn?

8.1 Overview of Cellular Respiration

Cellular respiration is an **aerobic** pathway, during which glucose is completely broken down to CO_2 and H_2O. It consists of four phases: glycolysis, the prep reaction, the citric acid cycle, and the passage of electrons along the electron transport chain. Oxidation of substrates involves the removal of hydrogen atoms ($H^+ + e^-$), usually by redox coenzymes. NAD^+ becomes NADH, and **FAD** becomes $FADH_2$. In the absence of oxygen, the **anaerobic** pathway of glycolysis and **fermentation** produces small amounts of ATP.

8.2 Outside the Mitochondria: Glycolysis

Glycolysis, the breakdown of glucose to two molecules of pyruvate, is a series of enzymatic reactions that occurs in the cytoplasm and is anaerobic. Breakdown releases enough energy to immediately give a net gain of two ATP by **substrate-level ATP synthesis** and the production of two NADH.

8.3 Outside the Mitochondria: Fermentation

Fermentation involves glycolysis followed by the reduction of pyruvate by NADH either to lactate (animals) or to alcohol (yeast) and carbon dioxide (CO_2). The reduction process "frees" NAD^+ so it can accept more hydrogen atoms from glycolysis.

Although fermentation results in only two ATP molecules, it still serves a purpose. Many of the products of fermentation are used in the baking and brewing industries. In vertebrates, it provides a quick burst of ATP energy for short-term, strenuous muscular activity. The accumulation of lactate puts the individual in oxygen debt, because oxygen is needed when lactate is completely metabolized to CO_2 and H_2O.

8.4 Inside the Mitochondria

When oxygen is available, pyruvate from glycolysis enters the **mitochondrion,** where the **prep reaction** takes place.

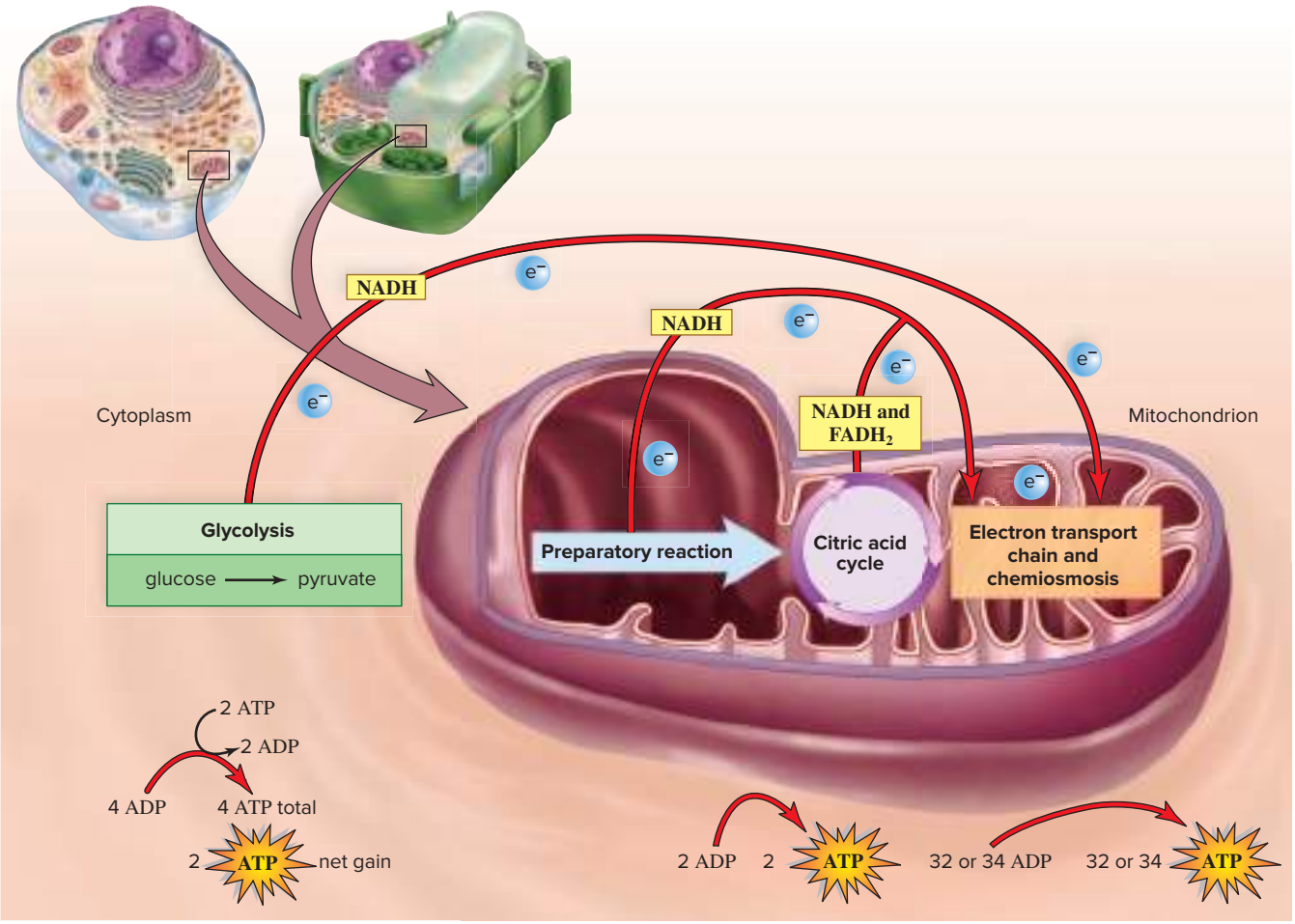
- During this reaction, oxidation occurs as CO_2 is removed from pyruvate. NAD^+ is reduced, and CoA receives the C_2 acetyl group that remains. Because the reaction must take place twice per glucose molecule, two NADH result.
- The acetyl group enters the **citric acid cycle,** a cyclical series of reactions located in the mitochondrial matrix. Complete oxidation follows, as two CO_2 molecules, three NADH molecules, and one $FADH_2$ molecule are formed. The cycle also produces one ATP molecule. The entire cycle must turn twice per glucose molecule.
- The final stage of glucose breakdown involves the **electron transport chain,** located in the cristae of the mitochondria. The electrons received from NADH and $FADH_2$ are passed down a chain of carriers until they are finally received by oxygen, which combines with H^+ to produce water. As the electrons pass down the chain, proteins called **cytochromes** capture energy for use in ATP production.
- The cristae of mitochondria contain complexes of the electron transport chain that not only pass electrons from one to the other but also pump H^+ into the intermembrane space, setting up an electrochemical gradient. When H^+ flows down this gradient through an ATP synthase complex, energy is captured and used to form ATP molecules from ADP and Ⓟ. This is ATP synthesis by **chemiosmosis.**

Theoretically, 36 to 38 ATP are produced by complete glucose breakdown. Four are the result of substrate-level ATP synthesis and the rest are produced as a result of the electron transport chain. For most NADH molecules that donate electrons to the electron transport chain, three ATP molecules are produced. However, in some cells, each NADH formed in the cytoplasm results in only two ATP molecules, because a shuttle, rather than NADH, takes electrons through the mitochondrial membrane. $FADH_2$ results in the formation of only two ATP, because its electrons enter the electron transport chain at a lower energy level.

8.5 Metabolism

Carbohydrate, protein, and fat can be metabolized by entering the degradative pathways at different locations. These pathways also provide metabolites needed for the anabolism of various important substances. Therefore, **catabolism** and **anabolism** both use the same pools of metabolites.

ISTUDY



Similar to the **metabolic pool** concept, photosynthesis and cellular respiration can be compared. For example, both utilize an ETC and chemiosmosis. As a result of the ETC in chloroplasts, water is split, while in mitochondria, water is formed. The enzymatic reactions in chloroplasts reduce CO_2 to a carbohydrate, while the enzymatic reactions in mitochondria oxidize carbohydrate with the release of CO_2.

Thinking Critically About Biology

1. Explain why glycolysis occurs in cells of all living organisms.
2. Different varieties of beer and wine contain different amounts of alcohol produced through fermentation by yeast. Among the reactants required for fermentation, which will determine the amount of alcohol produced?
3. Bacteria do not have mitochondria, yet they contain an electron transport chain. Where might this be located in the bacteria?
4. Cyanide is known to be an inhibitor of the electron transport chain. It functions by inhibiting one of the cytochrome enzymes. How, then, would this cause death to the individual?
5. Some fat-burning compounds accelerate the movement of fatty acids into the cellular respiration pathways. Explain how these compounds may work.

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Energy Part III: Cellular Respiration

McGraw Hill

Explain why more carbon dioxide was exhaled in one breath after exercise compared to one breath at rest.

Making It Relevant

1. From a cellular perspective, what is happening during aerobic exercise and why is this form of exercise considered beneficial?
2. Using Figure 8A as a guide, choose one of your favorite foods and trace how its fats, proteins, and carbohydrates are being processed at the cellular level.
3. During your day, when do you think your metabolism is mostly aerobic? Anaerobic?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 2

Genetic Basis of Life

This unit introduces the basics of cellular reproduction, the Mendelian principles of inheritance, molecular genetics, and biotechnology.

As we learned at the start of this book, all cells come from preexisiting cells. Therefore, cellular reproduction is one of the basic characteristics of life. The central processes of mitosis and meiosis allow organisms to reproduce, grow, and repair damaged tissues. An understanding of cellular reproduction has led to the study of stem cells, an area that is providing treatments for many human diseases.

Mendelian genetics explains the patterns of inheritance that are founded in the process of meiosis. Among the many applications of knowledge of these patterns is the ability to predict the chances of having a child with a specific genetic disorder. An understanding of molecular genetics has led to the development of DNA technologies that have the potential to cure genetic diseases and produce crops to feed an ever-increasing human population.

The field of genetics is making progress in other areas, too. We now understand how cell division is regulated by numerous genes, and how a failure of these regulatory mechanisms may lead to cancer. Many other human diseases are the result of mutations in genes as well. Our understanding of genetics has led to the development of new technologies to sequence, study, and now edit the genetic information of organisms.

Unit Outline

Chapter 9 The Cell Cycle and Cellular Reproduction

Chapter 10 Meiosis and Sexual Reproduction

Chapter 11 Mendelian Patterns of Inheritance

Chapter 12 Molecular Biology of the Gene

Chapter 13 Regulation of Gene Expression

Chapter 14 Biotechnology and Genomics

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Explain how the process of meiosis introduces the variation necessary for evolutionary change.
Nature of Science	Discuss how an understanding of cellular reproduction and molecular biology can be used to treat human disease.
Biological Systems	Understand how the information contained within the DNA is responsible for the physical characteristics of an organism.

ISTUDY

9

The Cell Cycle and Cellular Reproduction

Yuri Arcurs/iStockphoto/Getty Images

Elephants and Cancer

Elephants, like humans, are long-lived organisms. Yet, recent studies have indicated that only about 5% of elephants will develop cancer in their lifetime, compared to up to 25% of humans.

Cancer results from a failure to control the cell cycle, a series of steps that all cells go through prior to initiating cell division. A gene called *p53* is an important component of that control mechanism. This gene belongs to a class of genes called tumor suppressor genes. The proteins encoded by tumor suppressor genes check the DNA for damage. If the cells are damaged, they are not allowed to divide.

Humans have only one pair of *p53* genes. Luckily, you need only a single copy to regulate the cell cycle. But if both genes are not functioning properly, because of inheritance of a defective gene or mutations that inactivate a gene, cells may divide continuously, which is a characteristic of cancer. Interestingly, elephants have 20 pairs of *p53* genes in their genome, which means it is very unlikely their cells will lose all of this gene's functionality.

As you read through this chapter, think about the following questions:

1. What are the roles of the checkpoints in the cell cycle?
2. At what checkpoint do you think *p53* is normally active?

Chapter Outline

9.1 The Cell Cycle
9.2 The Eukaryotic Chromosome
9.3 Mitosis and Cytokinesis
9.4 The Cell Cycle and Cancer
9.5 Prokaryotic Cell Division

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.5 What is the role of DNA in a cell?

Sections 4.2 and 4.3 What are the major differences between prokaryotic and eukaryotic cells?

Section 4.8 What is the role of the cytoskeleton in a eukaryotic cell?

Following *the* Themes

CHAPTER 9 – THE CELL CYCLE AND CELLULAR REPRODUCTION

Evolution	As eukaryotic cells became increasingly complex, a similarly complex series of events evolved to separate their multiple chromosomes, located within a membrane-bound nucleus, into new daughter cells.
Nature of Science	By studying the regulatory mechanisms of the cell cycle, scientists are able to gather a deeper understanding of why cancer occurs.
Biological Systems	Cell division must be regulated for the healthy growth of tissues and organisms.

ISTUDY

9.1 The Cell Cycle

Learning Outcomes

Upon completion of this section, you should be able to

1. List the stages of interphase, and describe the major events that occur during each stage in preparation for cell division.
2. List the checkpoints that regulate the progression of cells through the cell cycle.
3. Explain the mechanisms within the G_1 cell cycle checkpoint that evaluate growth signals, determine nutrient availability, and assess DNA integrity.

The **cell cycle** is an ordered set of stages that takes place between the time a eukaryotic cell divides and the time the resulting daughter cells also divide. When a cell is going to divide, it grows larger, the number of organelles doubles, and the amount of DNA doubles as DNA replication occurs. The two portions of the cell cycle are interphase, which includes a number of stages, and the mitotic stage, when mitosis and cytokinesis occur.

Interphase

Most of the cell cycle is spent in **interphase** (Fig. 9.1). This is the time when a cell performs its usual functions, depending on its location in the body. The amount of time the cell takes for interphase varies widely. Embryonic cells complete the entire cell cycle in just a few hours. For typical adult mammalian cells, interphase lasts for about 20 hours, which is 90% of the cell cycle. In the past, interphase was known as the resting stage. However, today it is known that interphase is very busy and especially with the preparations being made for cellular reproduction and mitosis. Interphase consists of three stages, referred to as G_1, S, and G_2.

G_1 Stage

Cell biologists named the stage before DNA replication G_1, and they named the stage after DNA replication G_2. The G originally stood for "gap," but now that we know how metabolically active the cell is, it is better to think of G as standing for "growth." During G_1, the cell recovers from the previous division. The cell grows in size, increases the number of organelles (such as mitochondria and ribosomes), and accumulates materials that will be used for DNA synthesis. Otherwise, cells are constantly performing their normal daily functions during G_1, including communicating with other cells, secreting substances, and carrying out cellular respiration.

Some cells, such as nerve and muscle cells, typically do not complete the cell cycle and are permanently arrested. These cells exit interphase and enter a stage called G_0. While in the G_0 stage, the cells continue to perform normal, everyday processes, but no preparations are being made for cell division. Cells may not leave the G_0 stage without proper signals from other cells and other parts of the body. Thus, completion of the cell cycle is very tightly controlled.

S Stage

Following G_1, the cell enters the S stage, when DNA synthesis, or replication, occurs. At the beginning of the S stage, each chromosome is composed of one DNA double helix. Following DNA replication, each chromosome is composed of two identical DNA double-helix molecules (Fig. 9.2). Each double helix is called a **chromatid,** and the two identical chromatids are referred to as **sister chromatids.** The sister chromatids remain attached until they are separated during mitosis.

G_2 Stage

Following the S stage, G_2 is the stage from the completion of DNA replication to the onset of mitosis. During this stage, the cell synthesizes the proteins that will assist cell division. For example, it makes the proteins that form microtubules. Microtubules are used during the mitotic stage to form the mitotic spindle that is critical during M stage.

M (Mitotic) Stage

Following interphase, the cell enters the M (for *mitotic*) stage. This cell division stage includes **mitosis** (nuclear division) and **cytokinesis** (division of the cytoplasm). During mitosis, daughter chromosomes are distributed by the **mitotic spindle** to two daughter nuclei. When division of the cytoplasm is complete, two daughter cells are present.

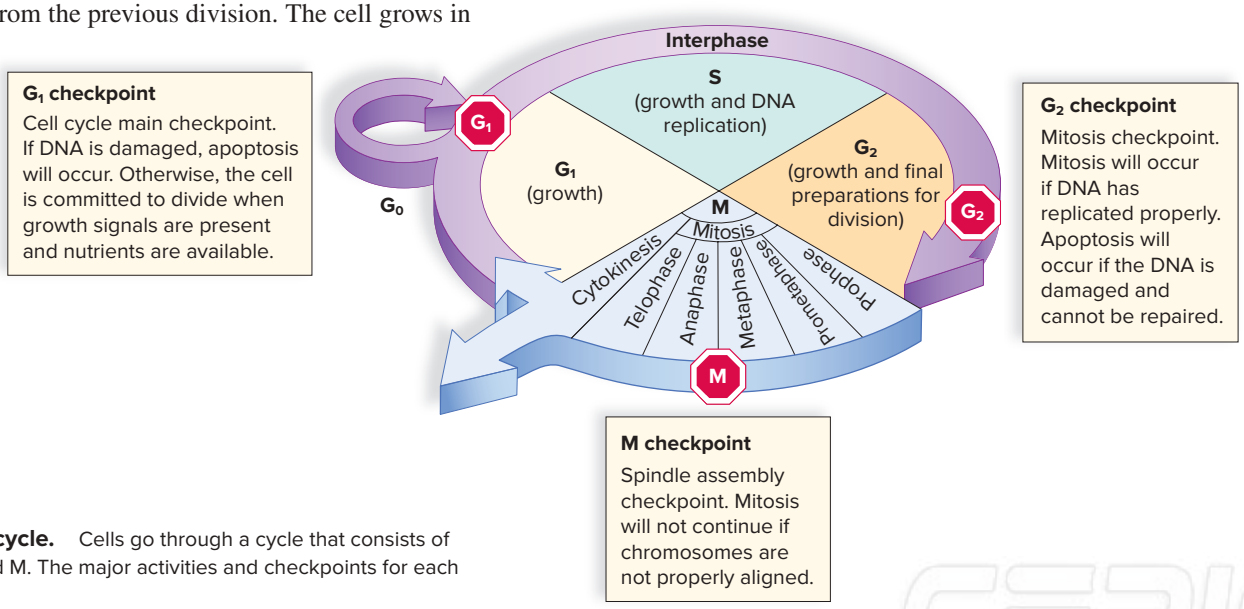


Figure 9.1 The cell cycle. Cells go through a cycle that consists of four stages: G_1, S, G_2, and M. The major activities and checkpoints for each stage are given.

ISTUDY

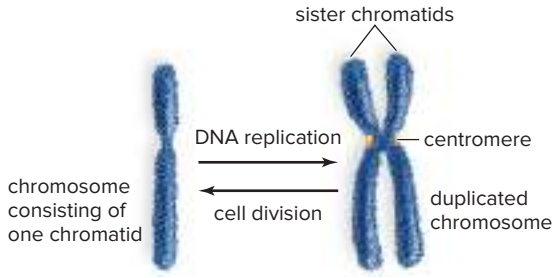


Figure 9.2 Sister chromatids. The process of DNA replication forms sister chromatids, which are separated by the process of mitosis in cell division.

Control of the Cell Cycle

A **signal** is an agent that influences the activities of a cell. **Growth factors** are signaling proteins received at the plasma membrane. Even cells arrested in G_0 will finish the cell cycle if stimulated to do so by growth factors. In general, signals ensure that the cell cycle stages follow one another in the normal sequence.

Cell Cycle Checkpoints

The red stop signs in Figure 9.1 represent three checkpoints at which the cell cycle either stops or continues on, depending on the internal signals received. Researchers have identified a family of internal signaling proteins, called **cyclins,** that increase and decrease as the cell cycle continues. Specific cyclins must be present for the cell to proceed from the G_1 stage to the S stage and from the G_2 stage to the M stage.

As discussed in the Nature of Science feature, "The Importance of the G_1 Checkpoint," the primary checkpoint of the cell cycle is the G_1 checkpoint. In mammalian cells, the signaling protein p53 stops the cycle at the G_1 checkpoint when DNA is damaged. In the name p53, p stands for *p*rotein and 53 represents its molecular weight in kilodaltons. First, p53 attempts to initiate DNA repair, but rising levels of p53 can bring about **apoptosis,** which is programmed cell death (Fig. 9.3). Another protein, called RB, is responsible for interpreting growth signals and nutrient availability signals. RB stands for *r*etino*b*lastoma, a cancer of the retina that occurs when the *RB* gene undergoes a mutation.

The cell cycle may also stop at the G_2 checkpoint if DNA has not finished replicating. This checkpoint prevents the initiation of the M stage before completion of the S stage. If DNA is physically damaged, such as from exposure to solar radiation or X-rays, the G_2 checkpoint also offers the opportunity for DNA to be repaired.

Another cell cycle checkpoint occurs during the mitotic stage. The cycle stops if the chromosomes are not properly attached to the mitotic spindle. Normally, the mitotic spindle ensures that the chromosomes are distributed accurately to the daughter cells.

Apoptosis

Apoptosis is often defined as programmed cell death, because the cell progresses through a typical series of events that bring about its destruction (Fig. 9.3). The cell rounds up, causing it to lose contact with its neighbors. The nucleus fragments, and the plasma membrane develops blisters. Finally, the cell fragments are engulfed by white blood cells and/or neighboring cells.

A remarkable finding of the past few years is that the enzymes that bring about apoptosis, called *caspases,* are always present in the cell. The enzymes are ordinarily held in check by inhibitors, but they can be unleashed by either internal or external signals.

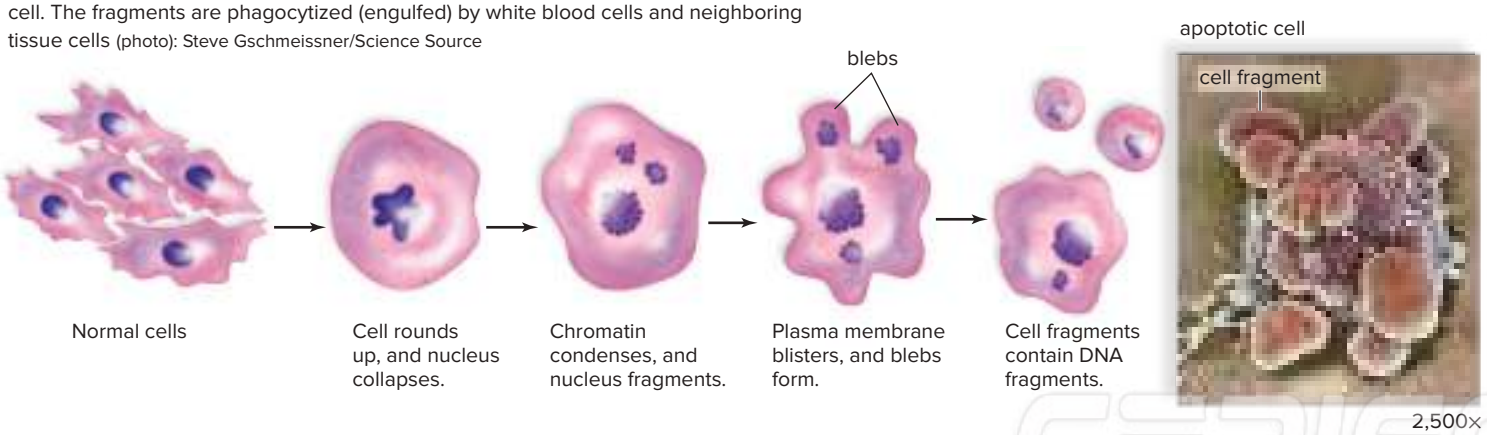
Apoptosis and Cell Division In living systems, opposing events keep the body in balance and maintain homeostasis. Cell division and apoptosis are two opposing processes that keep the number of cells in the body at an appropriate level. Cell division increases and apoptosis decreases the number of **somatic cells.** Both are normal parts of growth and development. An organism begins as a single cell that repeatedly divides to produce many cells, but eventually some cells must die for the organism to take shape. For example, when a tadpole becomes a frog, the tail disappears as apoptosis occurs. In a human embryo, the fingers and toes are at first webbed, but then they are usually freed from one another as a result of apoptosis.

Cell division occurs during your entire life. Even now, your body is producing thousands of new red blood cells, skin cells, and cells that line your respiratory and digestive tracts. Also, if you have a cut, cell division repairs the injury. Apoptosis occurs all the time, too, particularly if an abnormal cell that could become cancerous appears or a cell becomes infected with a virus. Death through apoptosis prevents a tumor from developing and helps limit the spread of viruses.

Check Your Progress 9.1

1. List, in order, the stages of the cell cycle and briefly summarize what is happening at each stage.
2. Explain what conditions might cause a cell to halt the cell cycle and state briefly where in the cycle this would occur.
3. Discuss how apoptosis represents a regulatory event of the cell cycle.

Figure 9.3 Apoptosis is a sequence of events that results in a fragmented cell. The fragments are phagocytized (engulfed) by white blood cells and neighboring tissue cells (photo): Steve Gschmeissner/Science Source



ISTUDY

THEME Nature of Science

The Importance of the G_1 Checkpoint

Cell division is very tightly regulated, so that only certain cells in an adult body are actively dividing. After cell division occurs, cells enter the G_1 stage. Upon completing G_1, they will divide again, but before this happens they have to pass through the G_1 checkpoint.

The G_1 checkpoint ensures that conditions are right for making the commitment to divide by evaluating the meaning of growth signals, determining the availability of nutrients, and assessing the integrity of DNA. Failure to meet any one of these criteria results in a cell's halting the cell cycle and entering G_0 stage, or undergoing apoptosis if the problems are severe.

Evaluating Growth Signals

Multicellular organisms tightly control cell division so it occurs only when needed. Signaling molecules (see Section 5.5), such as hormones, may be sent from nearby cells or distant tissues to encourage or discourage cells from entering the cell cycle. Such signals may cause a cell to enter a G_0 stage, or complete G_1 and enter the S stage. Growth signals that promote cell division cause a cyclin-dependent-kinase (CDK) to add a phosphate group to the RB protein, a major regulator of the G_1 checkpoint.

Ordinarily, a protein called E2F is bound to RB, but when RB is phosphorylated, its shape changes and it releases E2F. Now, E2F binds to DNA, activating certain genes whose products are needed to complete the cell cycle (Fig. 9A). Likewise, growth signals prompt cells that are in G_0 stage to reenter the G_1 stage, complete it, and enter the S stage. If growth signals are sufficient, a cell passes through the G_1 checkpoint and cell division occurs.

Determining Nutrient Availability

Just as experienced hikers ensure that they have sufficient food for their journey, a cell ensures that nutrient levels are adequate before committing to cell division. For example, scientists know that starving cells in culture enter G_0. At that time, phosphate groups are removed from RB (see reverse arrows in Fig. 9A); RB does not release E2F; and the proteins needed to complete the cell cycle are not produced. When nutrients become available, CDKs bring about the phosphorylation of RB, which then releases E2F (see forward arrows in Fig. 9A). After E2F binds to DNA, the proteins needed to complete the cell cycle are produced. Therefore, you can see that cells do not commit to divide until conditions are conducive for them to do so.

Assessing DNA Integrity

For cell division to occur, DNA must be free of errors and damage. The p53 protein is involved in this quality control function. Ordinarily, p53 is broken down because it has no job to do. In response to DNA damage, CDK phosphorylates p53 (Fig. 9B). Now, the molecule is not broken down as usual, and instead its level in the nucleus begins to rise. Phosphorylated p53 binds to DNA; certain genes are activated; and DNA repair proteins are produced. If the DNA damage cannot be repaired, p53 levels continue to rise, and apoptosis is triggered. If the damage is successfully repaired, p53 levels fall, and the cell is allowed to complete G_1 stage—as long as growth signals and nutrients are present, for example.

Actually, many criteria must be met for a cell to commit to cell division, and the failure to meet any one of them may cause the cell cycle to be halted and/or apoptosis to be initiated. The G_1 checkpoint is currently an area of intense research, because understanding it holds the key to possibly curing cancer and to unleashing the power of normal, healthy cells to regenerate tissues, which could be used to cure many other human conditions.

Questions to Consider

1. What is the potential effect of an abnormally high level of a growth hormone on the regulation of the cell cycle?
2. Why might some cancers be associated with a mutation in the gene encoding the p53 protein?

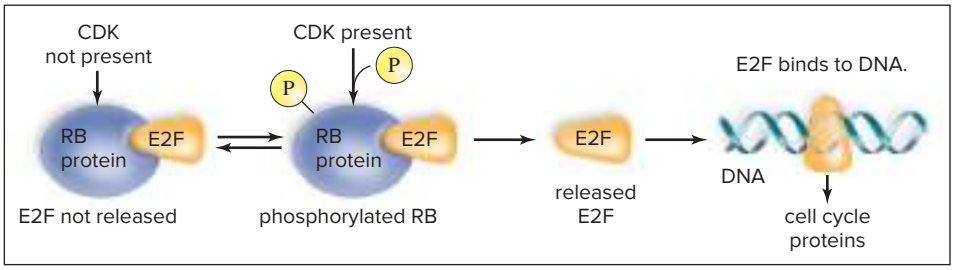


Figure 9A CDK regulation of the G_1 checkpoint. When CDK (cyclin-dependent-kinase) is not present, RB retains E2F. When CDK is present, a phosphorylated RB releases E2F, and after it binds to DNA, the proteins necessary for completing cell division are produced.

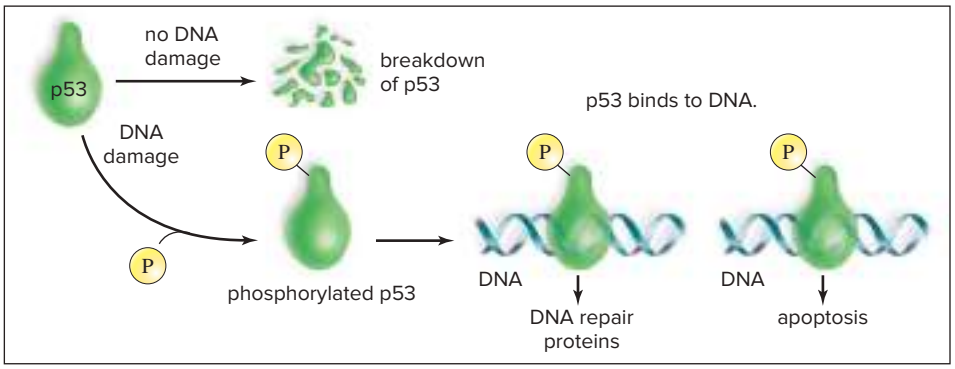


Figure 9B p53 regulation of the G_1 checkpoint. If DNA is damaged, p53 is not broken down; instead, it is involved in producing DNA repair enzymes and in triggering apoptosis when repair is impossible.

ISTUDY

9.2 The Eukaryotic Chromosome

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how DNA becomes sufficiently compacted to fit inside a nucleus.
2. Distinguish between euchromatin and heterochromatin.

Cell biologists and geneticists have been able to construct detailed models of how chromosomes are organized. We will explore the structure of a prokaryotic chromosome in Section 20.3. This chapter will focus on the more complex eukaryotic chromosome.

A eukaryotic chromosome contains a single double-helix DNA molecule, but it is composed of more than 50% protein. Some of these proteins are concerned with DNA and RNA synthesis, but a large majority, termed histones, play primarily a structural role. The five primary types of histone molecules are designated H1, H2A, H2B, H3, and H4 (see Fig. 13.5*b*). Remarkably, the amino acid sequences of H3 and H4 vary little between organisms. For example, the H4 of peas is only two amino acids different from the H4 of cattle. This similarity suggests that few mutations in the histone proteins have occurred during the course of evolution and that the histones, therefore, have essential functions in the cell.

A human cell contains at least 2 m of DNA, yet all of this DNA is packed into a nucleus that is about 6 µm in diameter. The histones are responsible for packaging the DNA so it can fit into such a small space. First, the DNA double helix is wound at intervals around a core of eight histone molecules (two copies each of H2A, H2B, H3, and H4), giving the appearance of a string of beads (Fig. 9.4*a*). Each bead is called a *nucleosome,* and the nucleosomes are said to be joined by "linker" DNA.

a. Nucleosomes ("beads on a string")

b. 30-nm fiber

c. Radial loop domains

d. Heterochromatin

e. Metaphase chromosome

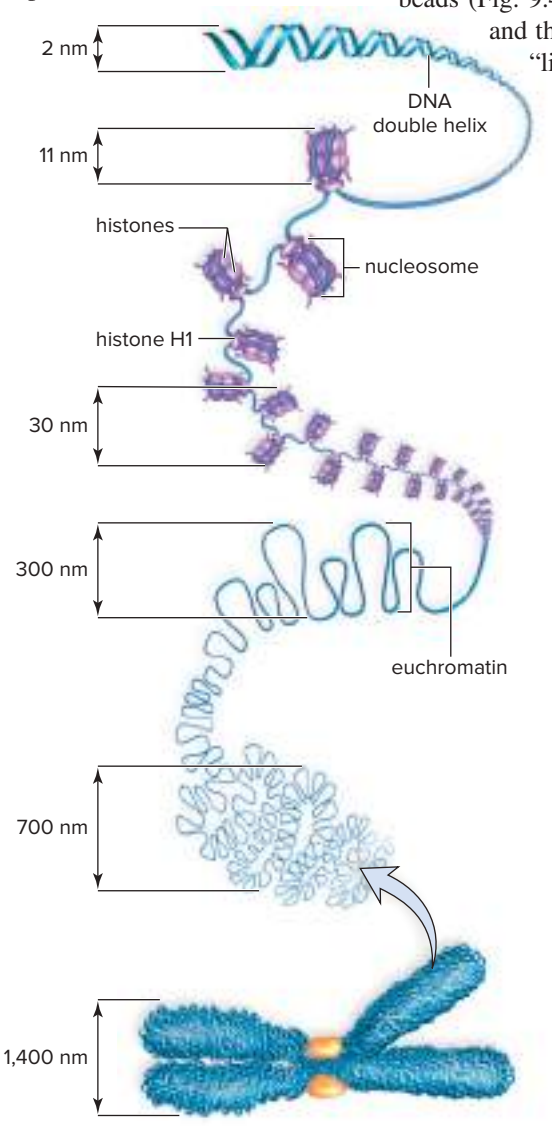


Figure 9.4 Structure of the eukaryotic chromosome. Eukaryotic cells contain nearly 2 m of DNA, yet they must pack it all into a nucleus that is around 6 µm in diameter. Thus, the DNA is compacted by winding it around DNA-binding proteins, called histones, to make nucleosomes. The nucleosomes are further compacted into a zigzag structure, which is then folded upon itself many times to form radial loops, which is the usual compaction state of euchromatin. Heterochromatin is further compacted by scaffold proteins, and further compaction can be achieved prior to mitosis and meiosis. (photos) (a): Don W. Fawcett/Science Source; (b): Courtesy Dr. Jerome Rattner, Cell Biology and Anatomy, University of Calgary; (c): Courtesy of Ulrich Laemmli and J.R. Paulson, Dept. of Molecular Biology, University of Geneva, Switzerland; (d, e): ©Peter Engelhardt, Haartman Institute/University of Helsinki/and NMC/Department of Applied Physics/Aalto University/School of Science and Technology/Helsinki, Finland

ISTUDY

This string is compacted by folding into a zigzag structure, further shortening the DNA strand (Fig. 9.4*b*). Histone H1 appears to mediate this coiling process. The fiber then loops back and forth into radial loops (Fig. 9.4*c*). This loosely coiled **euchromatin** represents the active chromatin containing genes that are being transcribed. The DNA of euchromatin may be accessed by RNA polymerase and other factors that are needed to promote transcription. In fact, recent research seems to indicate that regulating the level of compaction of the DNA is an important method of controlling gene expression in the cell.

Under a microscope, one often observes dark-stained fibers within the nucleus of the cell. These areas within the nucleus represent a more highly compacted form of the chromosome called **heterochromatin** (Fig. 9.4*d*). Most chromosomes exhibit both levels of compaction in a living cell, depending on which portions of the chromosome are being used more frequently. Heterochromatin is considered inactive chromatin, because the genes contained on it are infrequently transcribed, if at all.

Prior to cell division, a protein scaffold helps further condense the chromosome into a form that is characteristic of metaphase chromosomes (Fig. 9.4*e*). As you may recognize, compact chromosomes are easier to move about during cell division than extended chromatin.

Check Your Progress 9.2

1. Summarize the differences between euchromatin and heterochromatin.
2. List the stages of chromosome compacting, starting with a single DNA strand.

9.3 Mitosis and Cytokinesis

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how the cell prepares the chromosomes and centrosomes prior to nuclear division.
2. Summarize the major events that occur during mitosis and cytokinesis.
3. Discuss why human stem cells continuously conduct mitosis.

As mentioned, cell division in eukaryotes involves mitosis, which is nuclear division, and cytokinesis, which is division of the cytoplasm. During mitosis, the sister chromatids are separated and distributed to two daughter cells.

Chromosome Number

As we observed in Section 9.2, the DNA in the chromosomes of eukaryotes is associated with various proteins. When a eukaryotic cell is not undergoing division, the DNA and associated proteins are structured as euchromatin, which has the appearance of a tangled mass of thin threads. Before mitosis begins, chromatin becomes highly coiled and condensed, and it is easy to see the individual chromosomes.

Table 9.1 Diploid Chromosome Numbers of Some Eukaryotes

Type of Organism	Name of Chromosome	Chromosome Number
Fungi	*Saccharomyces cerevisiae* (yeast)	32
Plants	*Pisum sativum* (garden pea)	14
	Solanum tuberosum (potato)	48
	Ophioglossum vulgatum (southern adder's tongue fern)	1,320
Animals	*Drosophila melanogaster* (fruit fly)	8
	Homo sapiens (human)	46
	Carassius auratus (goldfish)	94

When the chromosomes are visible, it is possible to photograph and count them. Each species has a characteristic chromosome number (Table 9.1). This is the full, or **diploid (2n),** number (Gk. *diplos,* "twofold"; *eides,* "like") of chromosomes that is found in all cells of the individual. The diploid number includes two chromosomes of each kind. Most somatic cells of animals are diploid. Half the diploid number, called the **haploid (n)** number (Gk. *haplos,* "simple, single"), contains only one chromosome of each kind. The gametes of animals (egg and sperm) are examples of haploid cells.

Preparations for Mitosis

During interphase, a cell must make preparations for cell division. These arrangements include replicating the chromosomes and duplicating most cellular organelles, including the centrosome, which will organize the spindle apparatus necessary for the movement of chromosomes.

Chromosome Duplication

During mitosis, a 2n nucleus divides to produce daughter nuclei that are also 2n. The dividing cell is called the *parent cell,* and the resulting cells are called the *daughter cells.* Before nuclear division takes place, DNA replicates, duplicating the chromosomes in the parent cell. This occurs during the S stage of interphase. Now each chromosome has two identical double-helical molecules. Each double helix is a *chromatid,* and the two identical chromatids are called *sister chromatids* (Fig. 9.5). Sister chromatids are constricted and attached to each other at a region called the **centromere.** Protein complexes called **kinetochores** develop on either side of the centromere during cell division.

During nuclear division, the two sister chromatids separate at the centromere, and in this way each duplicated chromosome gives rise to two daughter chromosomes. Each daughter chromosome has only one double-helix molecule. The daughter chromosomes are distributed equally to the daughter cells. In this way, each daughter nucleus gets a copy of each chromosome that was in the parent cell.

Division of the Centrosome

The **centrosome** (Gk. *centrum,* "center"; *soma,* "body"), the main microtubule-organizing center of the cell, also divides before

ISTUDY

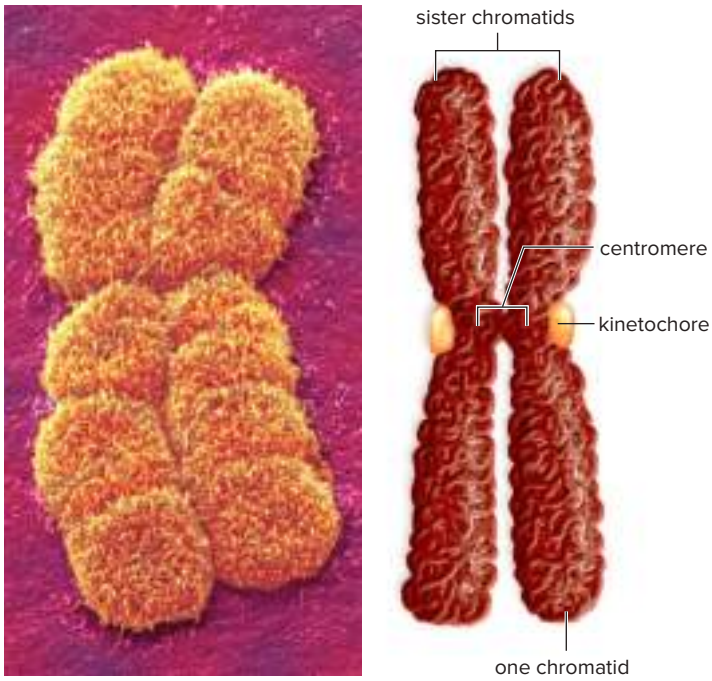


Figure 9.5 Duplicated chromosomes. A duplicated chromosome contains two sister chromatids, each with a copy of the same genes. **a.** Electron micrograph of a highly coiled and condensed chromosome, typical of a nucleus about to divide. **b.** Diagrammatic drawing of a condensed chromosome. The chromatids are held together at a region called the centromere. (a): Andrew Syred/Science Source

mitosis begins. Each centrosome in an animal cell contains a pair of barrel-shaped organelles called **centrioles.** Centrioles are not found in plant cells.

The centrosomes organize the mitotic spindle, which contains many fibers, each of which is composed of a bundle of microtubules. Microtubules are hollow cylinders made up of the protein tubulin. They assemble when tubulin subunits join, and when they disassemble, tubulin subunits become free once more. The microtubules of the cytoskeleton disassemble when spindle fibers begin forming. Most likely, this provides tubulin for the formation of the spindle fibers, or it may allow the cell to change shape as needed for cell division.

Phases of Mitosis

Mitosis is a continuous process that is arbitrarily divided into five phases for convenience of description: prophase, prometaphase, metaphase, anaphase, and telophase (Fig. 9.6).

Prophase

It is apparent during **prophase** that nuclear division is about to occur, because chromatin has condensed and the chromosomes are visible. Recall that DNA replication occurred during interphase, and therefore the *parental chromosomes are already duplicated and composed of two sister chromatids held together at a centromere.* Counting the number of centromeres in diagrammatic drawings gives the number of chromosomes for the cell depicted.

During prophase, the nucleolus disappears and the nuclear envelope fragments. The spindle begins to assemble as the two centrosomes migrate away from one another. In animal cells, an array of microtubules radiates toward the plasma membrane from the centrosomes. These structures are called *asters.* It is thought that asters brace the centrioles during later stages of cell division. Notice that the chromosomes have no particular orientation, because the spindle has not yet formed.

Prometaphase (Late Prophase)

During **prometaphase,** preparations for sister chromatid separation are evident. Kinetochores appear on each side of the centromere, and these attach sister chromatids to the *kinetochore spindle fibers*. These fibers extend from the poles to the chromosomes, which will soon be located at the center of the spindle.

The kinetochore fibers attach the sister chromatids to opposite poles of the spindle, and the chromosomes are pulled first toward one pole and then toward the other before the chromosomes come into alignment. Notice that even though the chromosomes are attached to the spindle fibers in prometaphase, they are still not in alignment.

Metaphase

During **metaphase,** the centromeres of chromosomes are now in alignment on a single plane at the center of the cell. The chromosomes usually appear as a straight line across the middle of the cell when viewed under a light microscope. An imaginary plane that is perpendicular and passes through this circle is called the *metaphase plate.* It indicates the future axis of cell division.

Several nonattached spindle fibers, called *polar spindle fibers,* reach beyond the metaphase plate and overlap. A cell cycle checkpoint, the M checkpoint, delays the start of anaphase until the kinetochores of each chromosome are attached properly to spindle fibers and the chromosomes are properly aligned along the metaphase plate.

Anaphase

At the start of **anaphase,** the two sister chromatids of each duplicated chromosome separate at the centromere, giving rise to two daughter chromosomes. Daughter chromosomes, each with a centromere and single chromatid composed of a single double helix, appear to move toward opposite poles. Actually, the daughter chromosomes are being pulled to the opposite poles as the kinetochore spindle fibers disassemble at the region of the kinetochores.

Even as the daughter chromosomes move toward the spindle poles, the poles themselves are moving farther apart, because the polar spindle fibers are sliding past one another. Microtubule-associated proteins, such as the motor molecules kinesin and dynein, are involved in the sliding process. Anaphase is the shortest phase of mitosis.

Telophase

During **telophase,** the spindle disappears as new nuclear envelopes form around the daughter chromosomes. Each daughter nucleus contains the same number and kinds of chromosomes as the original parent cell. Remnants of the polar spindle fibers are still visible between the two nuclei.

ISTUDY

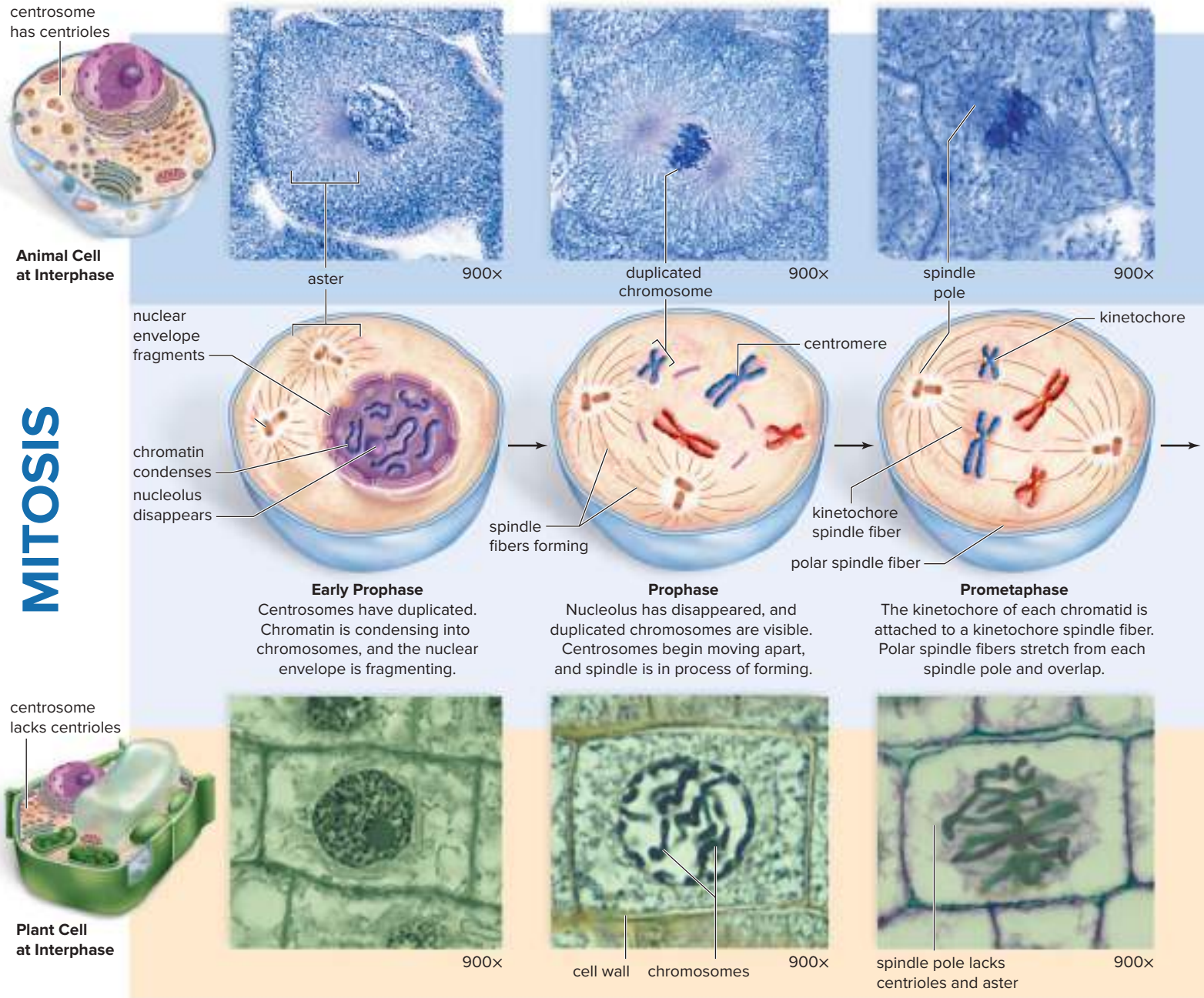


Figure 9.6 Phases of mitosis in animal and plant cells. The blue chromosomes were inherited from one parent, the red from the other parent.
(photos): (animal early prophase, animal prophase, animal metaphase, animal anaphase, animal telophase, plant early prophase, plant prometaphase): Ed Reschke; (animal prometaphase): Michael Abbey/Science Source; (plant prophase, plant metaphase, plant anaphase, plant telophase): Kent Wood/Science Source

The chromosomes become more diffuse chromatin once again, and a nucleolus appears in each daughter nucleus. Division of the cytoplasm requires cytokinesis, which is discussed next.

Cytokinesis in Animal and Plant Cells

As mentioned previously, cytokinesis is division of the cytoplasm. Cytokinesis accompanies mitosis in most cells, but not all. When mitosis occurs but cytokinesis doesn't occur, the result is a multinucleated cell. For example, you will see in Section 27.1 that the embryo sac in flowering plants is multinucleated.

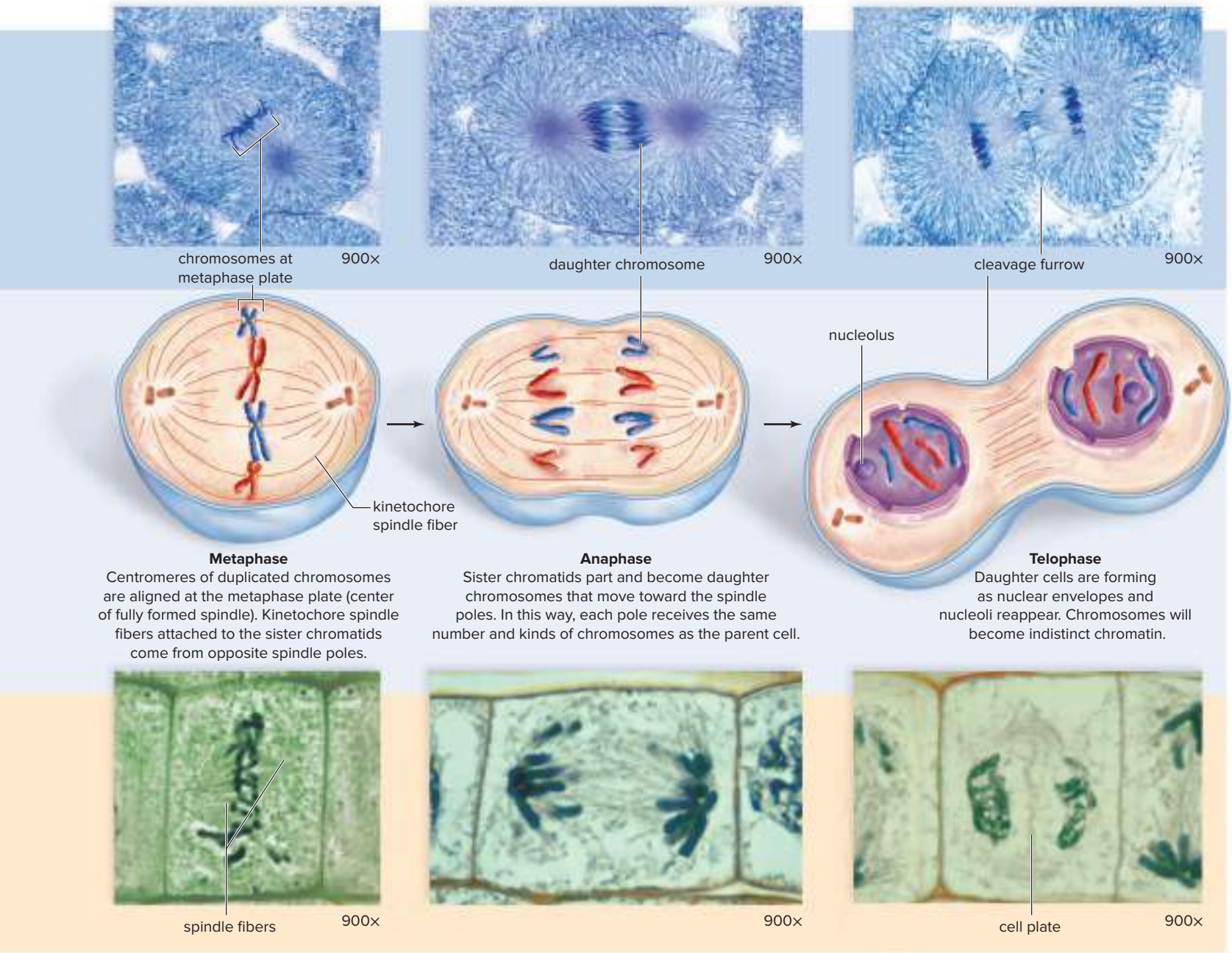
Division of the cytoplasm begins in anaphase, continues in telophase, but does not reach completion until the following interphase begins. By the end of mitosis, each newly forming cell has received a share of the cytoplasmic organelles that duplicated during interphase. Cytokinesis proceeds differently in plant and animal cells because of differences in cell structure.

Cytokinesis in Animal Cells

In animal cells, a **cleavage furrow,** which is an indentation of the membrane between the two daughter nuclei, forms just as anaphase draws to a close. By that time, the newly forming cells have received a share of the cytoplasmic organelles that duplicated during the previous interphase.

The cleavage furrow deepens when a band of actin filaments, called the contractile ring, slowly forms a circular constriction

ISTUDY



between the two daughter cells. The action of the contractile ring can be likened to pulling a drawstring ever tighter about the middle of a balloon. As the drawstring is pulled tight, the balloon constricts in the middle as the material on either side of the constriction gathers in folds. These folds are represented by the longitudinal lines in Figure 9.7.

A narrow bridge between the two cells can be seen during telophase, and then the contractile ring continues to separate the cytoplasm until there are two independent daughter cells (Fig. 9.7).

Cytokinesis in Plant Cells

Cytokinesis in plant cells occurs by a process different from that seen in animal cells (Fig. 9.8). The rigid cell wall that surrounds plant cells does not permit cytokinesis by furrowing. Instead, cytokinesis in plant cells involves the building of new cell walls between the daughter cells.

Cytokinesis is apparent when a small, flattened disk appears between the two daughter plant cells near the site where the metaphase plate once was. In electron micrographs, it is possible that the disk is at right angles to a set of microtubules that radiate outward from the forming nuclei. The Golgi apparatus produces vesicles, which move along the microtubules to the region of the disk. As more vesicles arrive and fuse, a cell plate can be seen. The **cell plate** is simply a newly formed plasma membrane that expands outward until it reaches the old plasma membrane and fuses with this membrane.

The new membrane releases molecules that form the new plant cell walls. These cell walls, known as primary cell walls, are later strengthened by the addition of cellulose fibrils. The space

ISTUDY

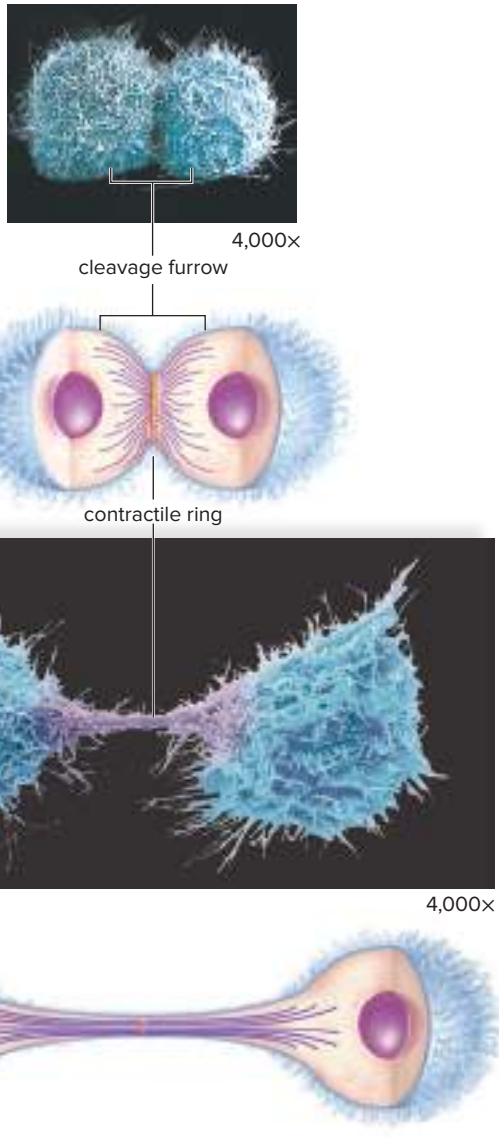


Figure 9.7 Cytokinesis in animal cells. A single cell becomes two cells by a furrowing process. A contractile ring composed of actin filaments gradually gets smaller, and the cleavage furrow pinches the cell into two cells. (photos): (top): National Institutes of Health (NIH)/USHHS; (bottom): Steve Gschmeissner/Brand X Pictures/Science Photo Library/Getty Images

between the daughter cells becomes filled with middle lamella, which cements the primary cell walls together.

The Functions of Mitosis

Mitosis permits growth and repair. In both plants and animals, mitosis is required during development as a single cell develops into an individual. In plants, the individual could be a fern or daisy, while in animals, the individual could be a grasshopper or a human.

In flowering plants, meristematic tissue retains the ability to divide throughout the life of a plant. Meristematic tissue at the shoot tip accounts for an increase in the height of a plant for as long as it lives. Then, too, lateral meristem accounts for the ability of trees to increase their girth each growing season.

In humans and other mammals, mitosis is necessary as a fertilized egg becomes an embryo and as the embryo becomes a fetus. Mitosis also occurs after birth as a child becomes an adult. Throughout life, mitosis allows a cut to heal or a broken bone to mend.

Stem Cells

In Section 9.1, you learned that the cell cycle is tightly controlled, and that most cells of the body at adulthood are permanently arrested in the G_0 stage. However, mitosis is needed to repair injuries, such as a cut or a broken bone. Many mammalian organs contain stem cells (often called adult stem cells) that retain the ability to divide. As one example, red bone marrow stem cells repeatedly divide to produce millions of cells that go on to become various types of blood cells.

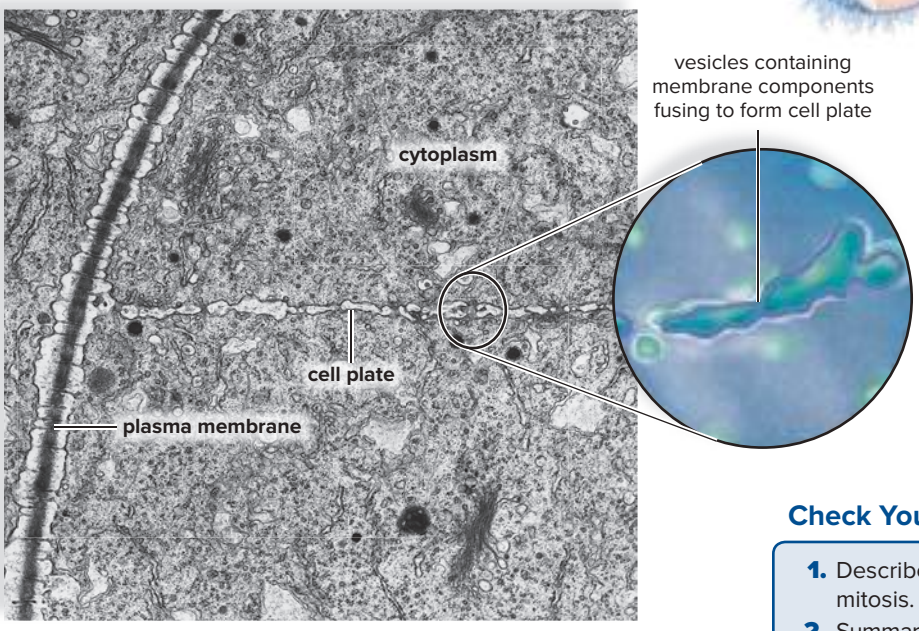


Figure 9.8 Cytokinesis in plant cells. During cytokinesis in a plant cell, a cell plate forms midway between two daughter nuclei and extends to the plasma membrane. (photo): Biophoto Associates/Science Source

Researchers are learning to manipulate the production of various types of tissues from adult stem cells in the laboratory. If successful, these tissues could be used to cure illnesses. As discussed in the Nature of Science feature, "Reproductive and Therapeutic Cloning," **therapeutic cloning,** which is used to produce human tissues, can begin with either adult stem cells or embryonic stem cells. Embryonic stem cells can also be used for **reproductive cloning,** the production of a new individual.

Check Your Progress 9.3

1. Describe the major events that occur during each phase of mitosis.
2. Summarize the differences between cytokinesis in animal and plant cells and explain why the differences are necessary.
3. Discuss the importance of stem cells in the human body.

ISTUDY

THEME Nature of Science

Reproductive and Therapeutic Cloning

Our knowledge of how the cell cycle is controlled has yielded major technological breakthroughs, including reproductive cloning—the ability to clone an adult animal from a normal body cell—and therapeutic cloning, which allows the rapid production of mature cells of a specific type. Both types of cloning are a direct result of recent discoveries about how the cell cycle is controlled.

Reproductive cloning, or the cloning of adult animals, was once thought to be impossible, because investigators found it difficult to have the nucleus of an adult cell "start over" with the cell cycle, even when it was placed in an egg cell that had its own nucleus removed.

In 1997, Dolly the sheep demonstrated that reproductive cloning is indeed possible. The donor cells were starved before the cell's nucleus was placed in an enucleated egg. This caused them to stop dividing and go into a G_0 (resting) stage, and this made the nuclei amenable to cytoplasmic signals for initiation of development (Fig. 9C*a*). This advance has made it possible to clone all sorts of farm animals that have desirable traits, and even to clone rare animals that might otherwise become extinct. Despite the encouraging results, however, there are still obstacles to be overcome, and a ban on the use of federal funds in experiments to clone humans remains firmly in place.

In therapeutic cloning, however, the objective is to produce mature cells of various cell types rather than an individual organism. The purpose of therapeutic cloning is (1) to learn more about how specialization of cells occurs and (2) to provide cells and tissues that could be used to treat human illnesses, such as diabetes, or major injuries like strokes or spinal cord injuries.

There are two possible ways to carry out *therapeutic cloning.* The first way is to use exactly the same procedure as reproductive cloning, except that *embryonic stem cells* (*ESCs*) are separated and each is subjected to a treatment that causes it to develop into a particular type of cell, such as red blood cells, muscle cells, or nerve cells (Fig. 9C*b*). Some have ethical concerns about this type of therapeutic cloning, which is still experimental, because if the embryo were allowed to continue development, it would become an individual.

The second way to carry out therapeutic cloning is to use *adult stem cells.* Stem cells are found in many organs of the adult's body; for example, the bone marrow has stem cells that produce new blood cells. However, adult stem cells are limited in the number of cell types they may become. Nevertheless, scientists are beginning to overcome this obstacle by creating induced pluripotent stem (iPS) cells to mimic embryonic stem cells. Experimentation has recently led to a strategy that allows creation of iPS cells without using an embryo or stem cell. Instead of altering genes or swapping nuclei, scientists simply stressed regular adult cells with trauma, low oxygen, and high acidity. The cells nearly died, but survived by reverting back to an embryonic state. Amazingly, they were then able to divide and differentiate into any new cell type.

Although questions exist about the benefits of iPS cells, these advances demonstrate that scientists are actively investigating methods of overcoming the current limitations and ethical concerns of using embryonic stem cells.

Questions to Consider

1. How might the study of therapeutic cloning benefit scientific studies of reproductive cloning?
2. What types of diseases might not be treatable using therapeutic cloning?

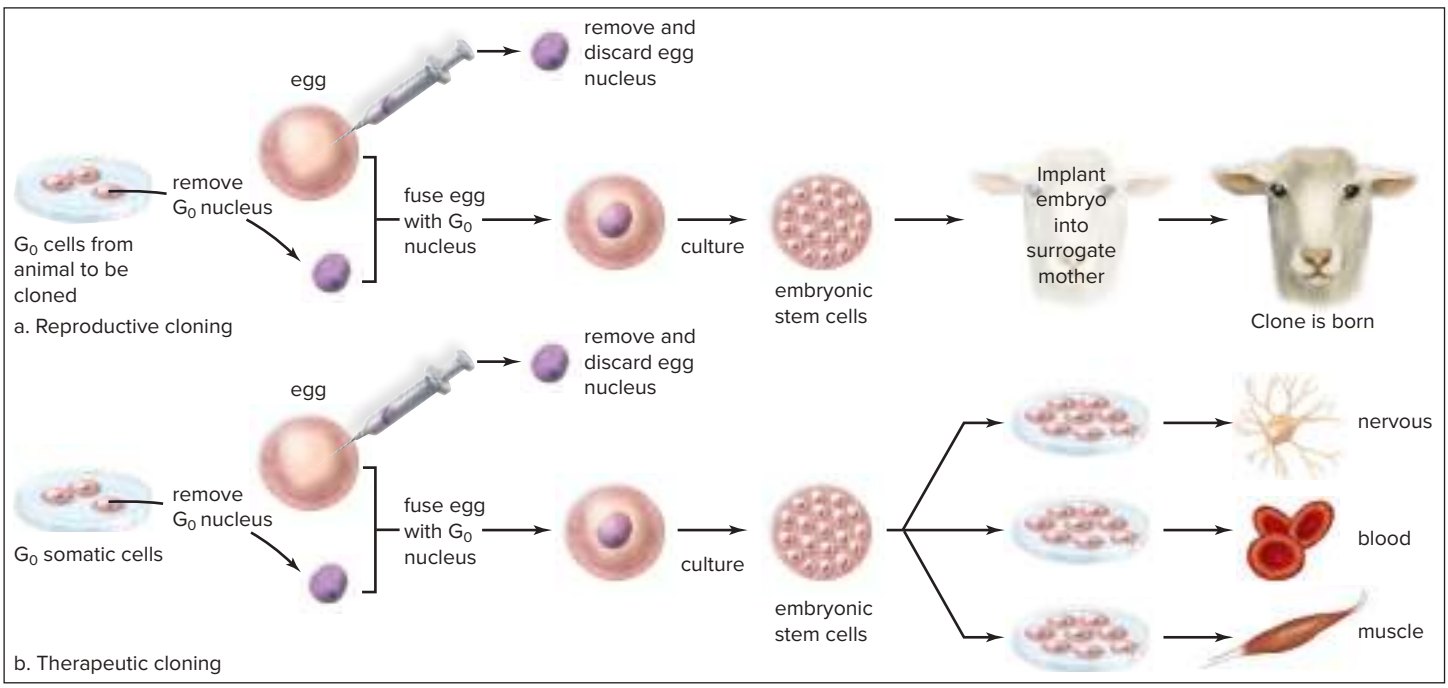


Figure 9C Two types of cloning. **a.** The purpose of reproductive cloning is to produce an individual that is genetically identical to the one that donated a nucleus. The nucleus is placed in an enucleated egg, and, after several mitotic divisions, the embryo is implanted into a surrogate mother for further development. **b.** The purpose of therapeutic cloning is to produce specialized tissue cells. A nucleus is placed in an enucleated egg, and after several mitotic divisions, the embryonic cells (called embryonic stem cells) are separated and treated to become specialized cells.

ISTUDY

9.4 The Cell Cycle and Cancer

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the basic characteristics of cancer cells.
2. Explain the difference between a benign and malignant tumor.
3. Distinguish between the roles of the tumor suppressor genes and proto-oncogenes in the regulation of the cell cycle.

Cancer is a cellular growth disorder that occurs when cells divide uncontrollably. Although causes widely differ, most cancers are the result of accumulating mutations that ultimately cause a loss of control of the cell cycle.

Although cancers vary greatly, they usually follow a common multistep progression (Fig. 9.9). Most cancers begin as an abnormal cell growth that is **benign,** or not cancerous, and usually does not grow larger. However, additional mutations may occur, causing the abnormal cells to fail to respond to inhibiting signals that control the cell cycle. When this occurs, the growth becomes **malignant,** meaning it is cancerous and possesses the ability to spread.

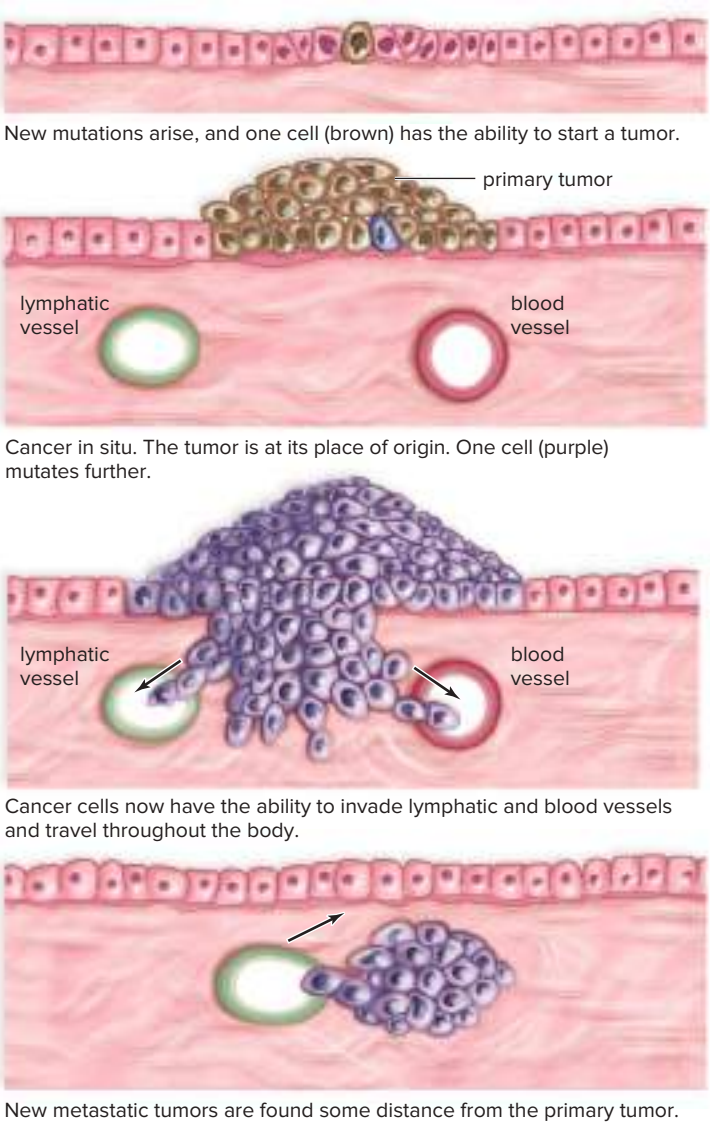
Characteristics of Cancer Cells

The development of cancer is gradual. A mutation in a cell may cause it to become precancerous, but many other regulatory processes within the body prevent it from becoming cancerous. In fact, it may be decades before a cell possesses most or all of the characteristics of a cancer cell (Table 9.2 and Fig. 9.9). Although cancers vary greatly, cells that possess the following characteristics are generally recognized as cancerous:

- *Cancer cells lack differentiation.* Cancer cells are not specialized and do not contribute to the functioning of a tissue. Although cancer cells may still possess many of the characteristics of surrounding normal cells, they usually look distinctly abnormal. Normal cells can enter the cell cycle about 50 times before they are incapable of dividing again. Cancer cells can enter the cell cycle an indefinite number of times and, in this way, seem immortal.
- *Cancer cells have abnormal nuclei.* The nuclei of cancer cells are enlarged and may contain an abnormal number of chromosomes. Extra copies of one or more chromosomes may be present. Often, there are also duplicated portions of some chromosomes present, which causes gene amplification, or extra copies of specific genes. Some chromosomes may also possess deleted portions.
- *Cancer cells do not undergo apoptosis.* Ordinarily, cells with damaged DNA undergo apoptosis, or programmed cell death.

Table 9.2 Cancer Cells Versus Normal Cells

Cancer Cells	Normal Cells
Nondifferentiated cells	Differentiated cells
Abnormal nuclei	Normal nuclei
Do not undergo apoptosis	Undergo apoptosis
No contact inhibition	Contact inhibition
Disorganized, multilayered	One organized layer
Undergo metastasis	Remain in original tissue



New mutations arise, and one cell (brown) has the ability to start a tumor.

Cancer in situ. The tumor is at its place of origin. One cell (purple) mutates further.

Cancer cells now have the ability to invade lymphatic and blood vessels and travel throughout the body.

New metastatic tumors are found some distance from the primary tumor.

Figure 9.9 Progression of cancer. The development of cancer requires a series of mutations, leading first to a localized tumor and then to metastatic tumors. With each successive step toward cancer, the most genetically altered and aggressive cell becomes the dominant type of tumor. The cells take on characteristics of embryonic cells; they are not differentiated; they can divide uncontrollably; and they are able to metastasize and spread to other tissues.

The immune system can also recognize abnormal cells and trigger apoptosis, which normally prevents tumors from developing. Cancer cells fail to undergo apoptosis even though they are abnormal cells.

- *Cancer cells form tumors.* Normal cells anchor themselves to a substratum and/or adhere to their neighbors. They exhibit contact inhibition—in other words, when they come in contact with a neighbor, they stop dividing. Cancer cells have lost all restraint and do not exhibit contact inhibition. The abnormal cancer cells pile on top of one another and grow in multiple layers, forming a **tumor.** During carcinogenesis, the most aggressive cell becomes the dominant cell of the tumor.
- *Cancer cells undergo metastasis and angiogenesis.* Additional mutations may cause a benign tumor, which is usually contained within a capsule and cannot invade adjacent tissue, to become malignant, and spread throughout the body,

ISTUDY

forming new tumors distant from the primary tumor. These cells now produce enzymes that they normally do not express, allowing tumor cells to invade underlying tissues. Then, they travel through the blood and lymph, to start tumors elsewhere in the body. This process is known as **metastasis.**

The Diversity in Science feature, "The Immortal Henrietta Lacks," explores how cancer cells from one individual have helped us understand the general characteristics of cancer cells.

Tumors that are actively growing soon encounter another obstacle—the blood vessels supplying nutrients to the tumor cells become insufficient to support the rapid growth of the tumor. In order to grow further, the cells of the tumor must receive additional nutrition. Thus, the formation of new blood vessels is required to bring nutrients and oxygen to support further growth. Additional mutations occurring in tumor cells allow them to direct the growth of new blood vessels into the

▶ Diversity in Science

The Immortal Henrietta Lacks

The availability of human tissue cells is important for many areas of medical research. Unfortunately, growing human cells has historically been very difficult. While these cells can be coaxed to reproduce using a technique called tissue culture, they typically all die very rapidly, usually after dividing only a few times. However, metastatic cells (a form of cancer cells) can survive and thrive in tissue culture. The techniques used to grow human cells successfully in tissue culture are part of the legacy of Henrietta Lacks (Fig. 9D). The cancer that claimed her life ensured that a part of Henrietta will live forever.

Lacks was a young Black woman and the mother of five children. In February 1951, she experienced unexplained vaginal bleeding and sought help from a Baltimore hospital. Physicians found a quarter-sized tumor on Henrietta's cervix. Samples of the tumor were quickly obtained, then sent to the tissue culture laboratories at Johns Hopkins University. Though radiation treatment was attempted, the tumor ravaged Henrietta's body, appearing on all her major organs within months. After 8 months of suffering, she died in October 1951. She was survived by her husband and five children, three of whom were still in diapers.

Figure 9D Henrietta Lacks. Cancer cells from Henrietta Lacks have revolutionized the study of cancer and cell biology. Photo Researchers/Science History Images/Alamy Stock Photo

Henrietta's tumor cells ended up in the laboratory of George and Margaret Gey at Johns Hopkins. The couple had been trying to culture human cells, with little success, for more than two decades. The cells from Henrietta Lacks's tumor, now termed *HeLa cells* (Fig. 9E), not only lived but multiplied like wildfire. At last, here was a source of human cells that grew rapidly and seemed almost immortal.

Using these cells, the Geys directed their research toward curing polio. Infantile paralysis, or polio, occurred in epidemics. In a small percentage of patients, it damaged the brain and spinal cord, causing paralysis of major muscle groups. If the diaphragm and other respiratory muscles were affected, the disease could be fatal. For the first time, the virus could be grown in human cells—HeLa cells—so that its characteristics could be studied. The results of these studies enabled Dr. Jonas Salk to develop a vaccine for polio. Today, polio is almost unheard of in the Western world.

Research with HeLa cells did not stop with the Geys. For over 50 years, these durable cells have been used worldwide to study many types of viruses, as well as leukemia and other cancers. The cells are human in origin, so they have also been the test subjects to determine the harmful effects caused by drugs or radiation. Effective testing was developed for chromosome abnormalities and hereditary diseases using HeLa. Samples of HeLa cells have even been launched in the space shuttle for experiments involving a zero-gravity environment. HeLa cell colonies can be found around the world. The cells can even be purchased from biological supply companies.

HeLa cells are almost too sturdy. In 1974, researchers discovered that HeLa cells, like bacteria, can be transmitted through the air, on a researcher's glove, or on contaminated glassware. This discovery led to better sterile techniques not only in the laboratory but also in operating rooms where cancerous tumors were removed.

In 2011, Henrietta's story was presented in a book, *The Immortal Life of Henrietta Lacks,* by Rebecca Sloot. The book has received multiple awards for its ability to portray the importance of Henrietta's cells to science.

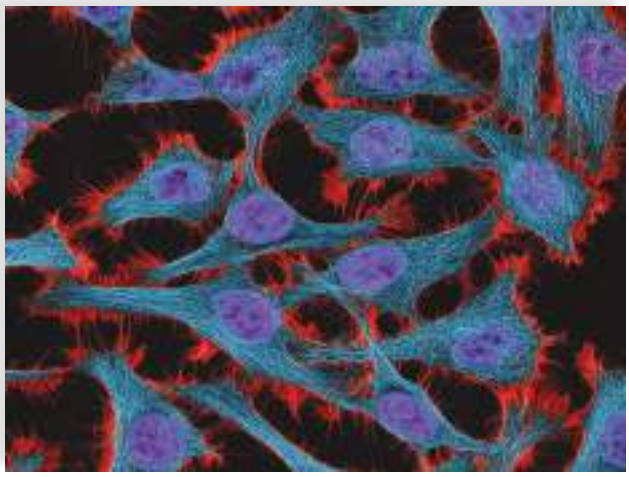


Figure 9E SEM of HeLa cells. The term *HeLa* is derived from the letters of Henrietta Lacks's name. National Institutes of Health (NIH)

tumor in a process called **angiogenesis.** Some modes of cancer treatment are aimed at preventing angiogenesis from occurring.

Origin of Cancer

Normal growth and maintenance of body tissues depend on a balance between signals that promote and inhibit cell division. When this balance is upset, conditions such as cancer may occur. Thus, cancer is usually caused by mutations affecting genes that directly or indirectly affect this balance, such as those shown in Figure 9.10. The following two types of genes are usually affected:

- **Proto-oncogenes** code for proteins that promote the cell cycle and prevent apoptosis. They are often likened to the gas pedal of a car, because they cause the cell cycle to speed up.
- **Tumor suppressor genes** code for proteins that inhibit the cell cycle and promote apoptosis. They are often likened to the brakes of a car, because they cause the cell cycle to go more slowly or even stop.

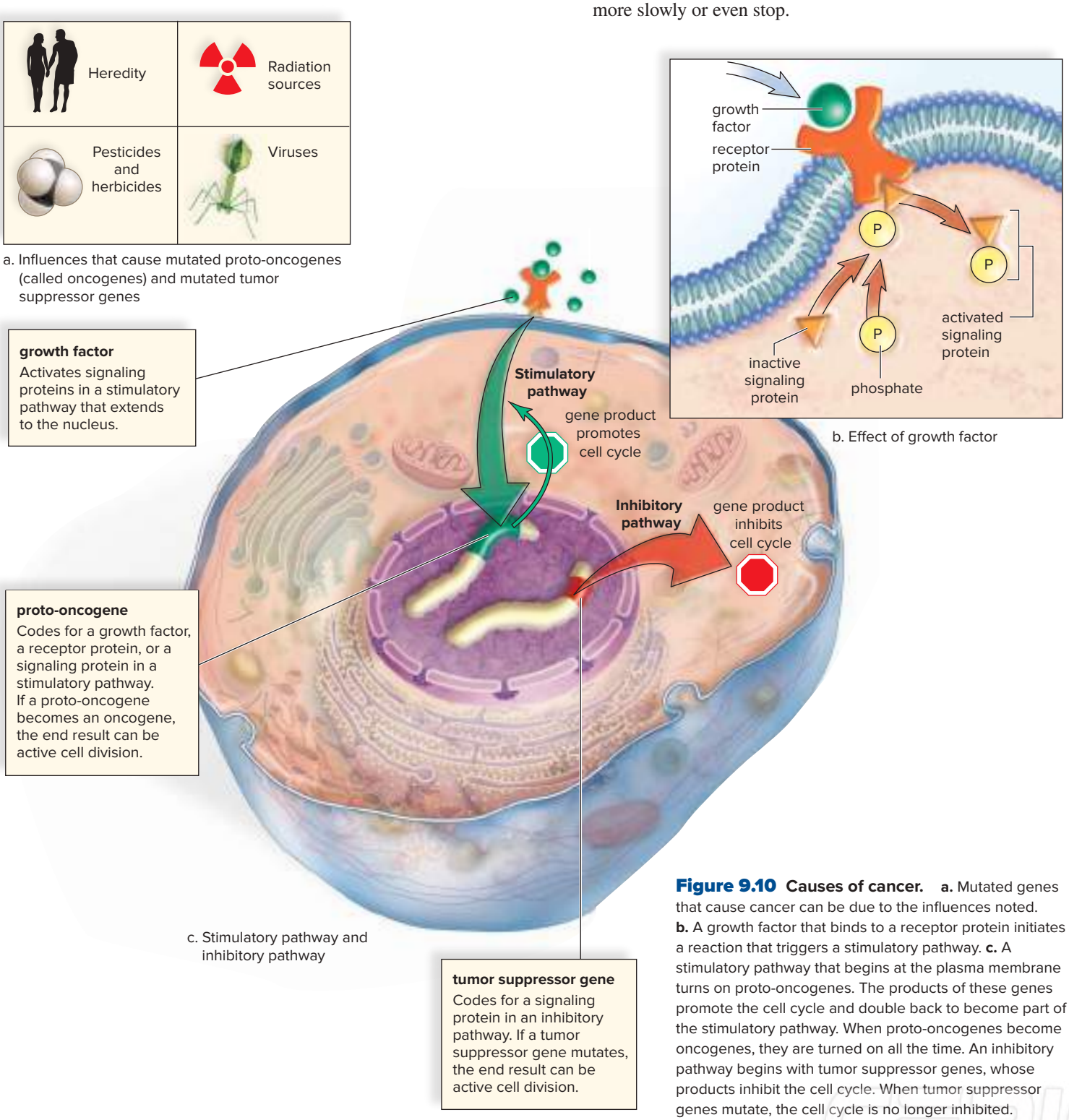


Figure 9.10 Causes of cancer. **a.** Mutated genes that cause cancer can be due to the influences noted. **b.** A growth factor that binds to a receptor protein initiates a reaction that triggers a stimulatory pathway. **c.** A stimulatory pathway that begins at the plasma membrane turns on proto-oncogenes. The products of these genes promote the cell cycle and double back to become part of the stimulatory pathway. When proto-oncogenes become oncogenes, they are turned on all the time. An inhibitory pathway begins with tumor suppressor genes, whose products inhibit the cell cycle. When tumor suppressor genes mutate, the cell cycle is no longer inhibited.

ISTUDY

Proto-oncogenes Become Oncogenes

Proto-oncogenes are normal genes that promote progression through the cell cycle. They are often at the end of a *stimulatory pathway* extending from the plasma membrane to the nucleus. A stimulus, such as an injury, results in the release of a growth factor that binds to a receptor protein in the plasma membrane. This sets in motion a whole series of enzymatic reactions leading to the activation of genes that promote the cell cycle, both directly and indirectly. Proto-oncogenes include the receptors and signal molecules that make up these pathways.

When mutations occur in proto-oncogenes, they become **oncogenes,** or cancer-causing genes as shown in Figure 9.11. Oncogenes are under constant stimulation and keep on promoting the cell cycle regardless of circumstances. For example, an oncogene may code for a faulty receptor in the stimulatory pathway such that the cell cycle is stimulated, even when no growth factor is present! Or an oncogene may either specify an abnormal protein product or produce abnormally high levels of a normal product that stimulate the cell cycle to begin or to go to completion. As a result, uncontrolled cell division may occur.

Researchers have identified approximately 40 oncogenes that can cause increased growth and lead to tumors. The oncogenes most frequently involved in human cancers belong to the *ras* gene family. Mutant forms of the *BRCA1* (*br*east *ca*ncer predisposition gene 1) oncogene are associated with certain hereditary forms of breast and ovarian cancer.

Tumor Suppressor Genes Become Inactive

Tumor suppressor genes, on the other hand, directly or indirectly inhibit the cell cycle and prevent cells from dividing uncontrollably (Fig. 9.12). Some tumor suppressor genes prevent progression of the cell cycle when DNA is damaged. Other tumor suppressor genes may promote apoptosis as a last resort.

A mutation in a tumor suppressor gene is much like brake failure in a car; when the mechanism that slows down and stops cell division does not function, the cell cycle accelerates and does not halt. Researchers have identified about a half-dozen tumor suppressor genes. Among these are the *RB* and *p53* genes that code for the RB and p53 proteins. The Nature of Science feature, "The Importance of the G_1 Checkpoint," in Section 9.1 discusses the function of these proteins in controlling the cell cycle. The *RB* tumor suppressor gene was discovered when the inherited condition retinoblastoma was being studied, but malfunctions of this gene have now been identified in many other cancers as well, including breast, prostate, and bladder cancers. The *p53* gene turns on the expression of other genes that inhibit the cell cycle. The p53 protein can also stimulate apoptosis. It is estimated that over half of human cancers involve an abnormal or deleted *p53* gene.

Other Causes of Cancer

As mentioned previously, cancer develops when the delicate balance between promotion and inhibition of cell division is tilted toward uncontrolled cell division. Other mutations may occur within a cell that affect this balance. For example, while a mutation affecting the cell's DNA repair system will not immediately cause cancer, it leads to a much greater chance of a mutation occurring within a proto-oncogene or tumor suppressor gene. And in some cancer cells, mutation of the telomerase enzyme that regulates the length of **telomeres,** or the ends of chromosomes, causes the telomeres to remain at a constant length. Because cells with shortened telomeres normally stop dividing, keeping the telomeres at a constant length allows the cancer cells to continue dividing over and over again.

Check Your Progress 9.4

1. List the major characteristics of cancer cells that distinguish them from normal cells.
2. Distinguish between a malignant and benign tumor.
3. Compare and contrast the effect on the cell cycle of (a) a mutation in a proto-oncogene and (b) a mutation in a tumor suppressor gene.

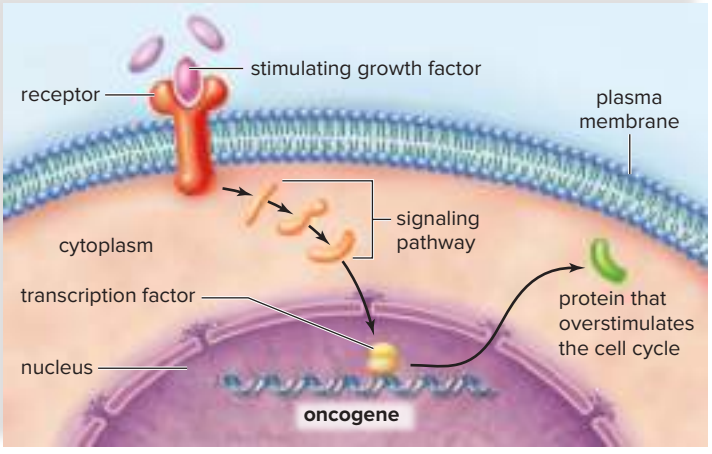


Figure 9.11 Proto-oncogenes mutate to form oncogenes. An oncogene codes for a protein that either directly or indirectly overstimulates the cell cycle.

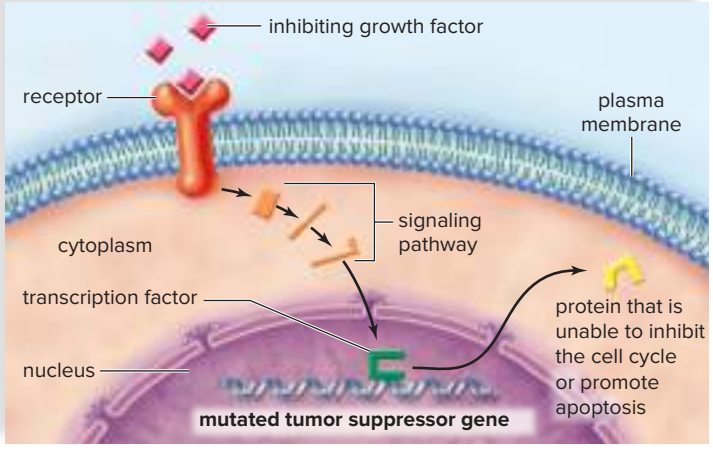


Figure 9.12 Mutations in tumor suppressor genes. A mutated tumor suppressor gene codes for a product that either directly or indirectly fails to inhibit the cell cycle.

ISTUDY

9.5 Prokaryotic Cell Division

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between the structures of a prokaryotic and eukaryotic chromosome.
2. Describe the events that occur during binary fission.

In **asexual reproduction,** cell division may be used to produce two individuals from a single parent. This is common on prokaryotes, and some protistans. During asexual reproduction, the chromosome(s) are duplicated, and each new individual receives a copy of the genetic information. Unless a mutation occurs, the offspring of asexual reproduction are genetically identical to the parent.

The Prokaryotic Chromosome

Prokaryotes (bacteria and archaea) lack a nucleus and other membranous organelles found in eukaryotic cells. Still, they do have a chromosome, which is composed of DNA and a limited number of associated proteins. The single chromosome of prokaryotes contains just a few proteins and is organized differently than eukaryotic chromosomes. A eukaryotic chromosome has many more associated proteins than does a prokaryotic chromosome.

In electron micrographs, the bacterial chromosome appears as an electron-dense, irregularly shaped region called the **nucleoid,** which is not enclosed by a membrane. When stretched out, the chromosome is seen to be a circular loop with a length that is up to about a thousand times the length of the cell. Special enzymes and proteins help coil the chromosome so it will fit within the prokaryotic cell.

Binary Fission

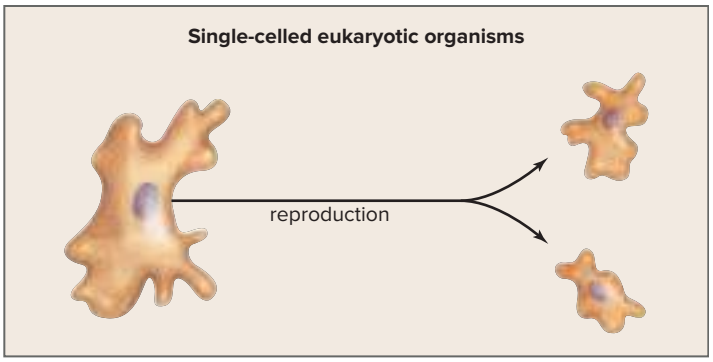
Prokaryotes reproduce asexually by binary fission. The process is termed **binary fission** because division (fission) produces two (binary) daughter cells that are identical to the original parent cell. Before division takes place, the cell enlarges, and after DNA replication occurs, there are two chromosomes. These chromosomes attach to a special plasma membrane site and separate by an elongation of the cell that pulls them apart. During this period, a new plasma membrane and cell wall develop and grow inward to divide the cell. When the cell is approximately twice its original length, the new cell wall and plasma membrane for each cell are complete (Fig. 9.13).

Escherichia coli, which lives in our intestines, has a generation time (the time it takes the cell to divide) of about 20 minutes under favorable conditions. In about 7 hours, a single cell can increase to over 1 million cells! The division rate of other bacteria varies depending on the species and conditions.

Comparing Prokaryotes and Eukaryotes

Both binary fission and mitosis ensure that each daughter cell is genetically identical to the parent cell. The genes are portions of DNA found in the chromosomes.

Prokaryotes (bacteria and archaea), protists (many algae and protozoans), and some fungi (yeasts) are single-celled. Cell division in single-celled organisms produces two new individuals:



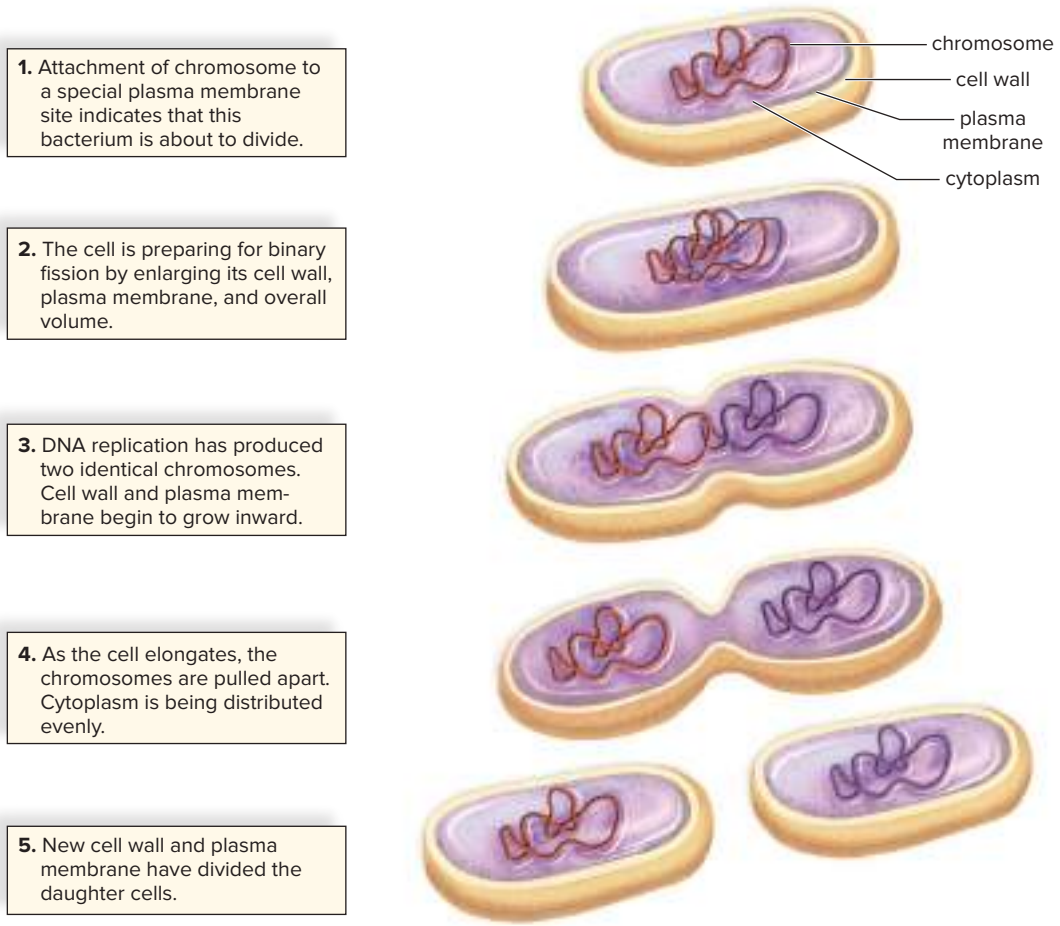


Figure 9.13 Binary fission. First, DNA replicates, and as the cell lengthens, the two chromosomes separate and the cells become divided. The two resulting bacteria are identical. ©Scimat/Science Source

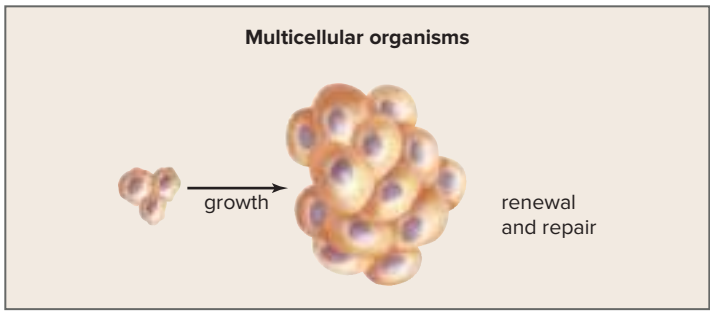
ISTUDY

Table 9.3 Functions of Cell Division

Type of Organism	Cell Division	Function
Prokaryotes		
Bacteria and archaea	Binary fission	Asexual reproduction
Eukaryotes		
Protists and some fungi (yeast)	Mitosis and cytokinesis	Asexual reproduction
Other fungi, plants, and animals	Mitosis and cytokinesis	Development, growth, and repair

This is a form of asexual reproduction because one parent has produced identical offspring (Table 9.3).

In plants, animals, and multicellular fungi (molds and mushrooms), cell division is part of the growth process. It produces the multicellular form we recognize as the mature organism. Cell division is also important in multicellular forms for renewal and repair:



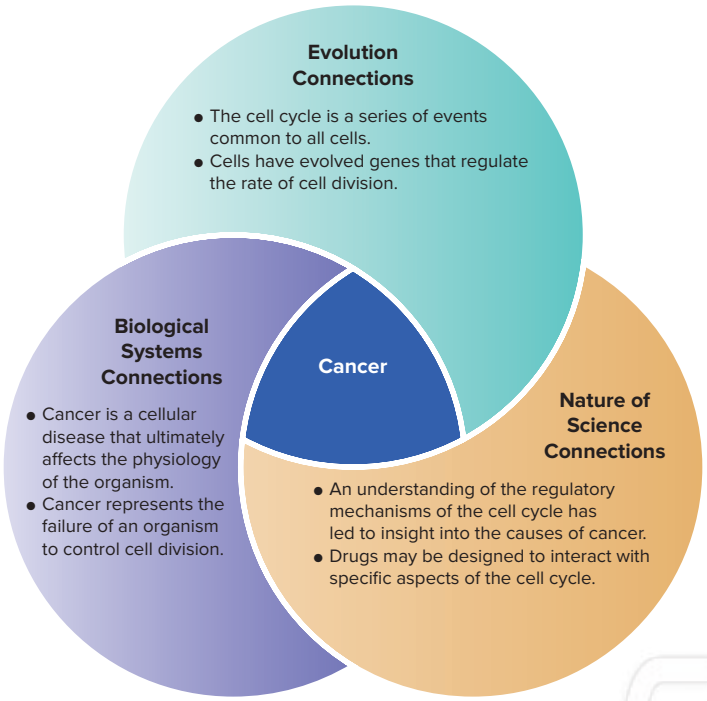
The chromosomes of eukaryotic cells are composed of DNA and many associated proteins. The histone proteins organize a chromosome, allowing it to extend as chromatin during interphase and to coil and condense just prior to mitosis. Each species of multicellular eukaryotes has a characteristic number of chromosomes in the nuclei. As a result of mitosis, each daughter cell receives the same number and kinds of chromosomes as the parent cell. The spindle, which appears during mitosis, is involved in distributing the daughter chromosomes to the daughter nuclei. Cytokinesis, either by the formation of a cell plate (plant cells) or by furrowing (animal cells), is division of the cytoplasm.

In prokaryotes, the single chromosome consists largely of DNA with a few associated proteins. During binary fission, this chromosome duplicates, each daughter cell receives one copy as the parent cell elongates, and a new cell wall and plasma membrane form between the daughter cells. No spindle is involved in binary fission.

Check Your Progress 9.5

1. Explain how binary fission in prokaryotes differs from mitosis and cytokinesis in eukaryotes.
2. Describe the structure of a prokaryotic and a eukaryotic chromosome.

Connecting *the* Concepts



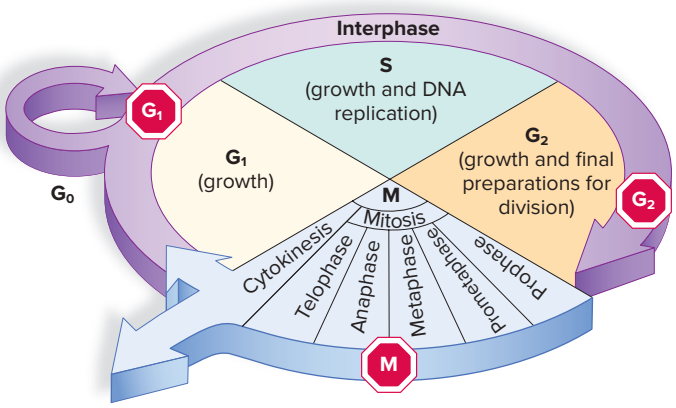
ISTUDY

What Did You Learn?

9.1 The Cell Cycle

The **cell cycle** of a eukaryotic cell includes (1) interphase and (2) a mitotic stage that consists of **mitosis** and **cytokinesis.**

- **Interphase,** in turn, is composed of three stages: G_1 (growth as certain organelles double), S (the synthesis stage, where the **chromatids** are duplicated, forming the **sister chromatids**), and G_2 (growth as the cell prepares to divide). Cells that are no longer dividing are arrested in a G_0 state.
- During the mitotic stage (M), the chromosomes are sorted by the **mitotic spindle** into two daughter cells, resulting in a full complement of chromosomes. During cytokinesis, the cytoplasm is divided between the new daughter cells.
- **Signals,** such as **growth factors** and **cyclin,** are involved in regulating the cell cycle. The cell cycle is regulated by several checkpoints—the G_1 checkpoint; the G_2 checkpoint prior to the M stage; and the M stage checkpoint, or spindle assembly checkpoint, immediately before anaphase. The G_1 checkpoint ensures that conditions are favorable and that the proper signals are present, and it checks the DNA for damage. If the DNA is damaged beyond repair, **apoptosis** may occur. Cell division and apoptosis are two opposing processes that keep the number of **somatic cells** in balance.



9.2 The Eukaryotic Chromosome

Eukaryotic cells contain nearly 2 m of DNA, yet they must pack it all into a nucleus approximately 6 µm in diameter. Thus, the DNA is compacted by winding it around DNA-binding proteins, called histones, to make nucleosomes. The nucleosomes are further compacted into a zigzag structure, which is then folded upon itself many times to form radial loops, which is the usual compaction state of **euchromatin. Heterochromatin** is further compacted by scaffold proteins, and additional compaction occurs prior to mitosis and meiosis.

9.3 Mitosis and Cytokinesis

Interphase represents the portion of the cell cycle between nuclear divisions, and during this time, preparations are made for cell division. These preparations include duplication of most cellular contents, including the centrosome, which organizes the mitotic spindle. The DNA is duplicated during S stage, at which time the chromosomes, which consist of a single chromatid each, are duplicated. The G_2 checkpoint ensures that DNA has replicated properly. This results in a nucleus containing the same number of chromosomes, with each now consisting of two chromatids attached at the **centromere.** During interphase, the chromosomes are not distinct and are collectively called chromatin. Each eukaryotic species has a characteristic number of chromosomes. The total number is called the **diploid** number, and half this number is the **haploid** number.

Among eukaryotes, cell division involves both mitosis (nuclear division) and cytokinesis (division of the cytoplasm). As a result of mitosis, the chromosome number stays constant, because each chromosome is duplicated and gives rise to two daughter chromosomes, which consist of a single chromatid each.

Mitosis: Nuclear division

- **Prophase**—The nucleolus disappears, the nuclear envelope fragments, and the spindle forms between centrosomes. The chromosomes condense and become visible under a light microscope. In animal cells, asters radiate from the **centrioles** within the **centrosomes.** Plant cells generally lack centrioles and, therefore, asters. Even so, the mitotic spindle forms.
- **Prometaphase** (late prophase)—The **kinetochores** of sister chromatids attach to kinetochore spindle fibers extending from opposite poles. The chromosomes move back and forth until they are aligned at the metaphase plate.
- **Metaphase**—The spindle is fully formed, and the duplicated chromosomes are aligned at the metaphase plate. The spindle consists of polar spindle fibers that overlap at the metaphase plate and kinetochore spindle fibers that are attached to chromosomes. The M stage checkpoint, or spindle assembly checkpoint, must be satisfied before progressing to the next phase.
- **Anaphase**—Sister chromatids separate, becoming daughter chromosomes that move toward the poles. The polar spindle fibers slide past one another, and the kinetochore spindle fibers disassemble. Cytokinesis by furrowing begins.
- **Telophase**—Nuclear envelopes re-form, chromosomes begin changing back to chromatin, the nucleoli reappear, and the spindle disappears. Cytokinesis continues and is complete by the end of telophase.

Cytokinesis: Cytoplasmic division

- Cytokinesis in animal cells involves the formation of a **cleavage furrow** that divides the cytoplasm. Cytokinesis in plant cells involves the formation of a **cell plate,** from which the plasma membrane and cell wall are completed.
- In the body, mitosis also produces undifferentiated cells, called stem cells, that are involved in the production of tissues. **Therapeutic cloning** and **reproductive cloning** both utilize stem cells.

9.4 The Cell Cycle and Cancer

The development of **cancer** is primarily due to the mutation of genes involved in control of the cell cycle. Cancer cells lack differentiation, have abnormal nuclei, do not undergo apoptosis, form tumors, and undergo metastasis and angiogenesis. Cancer often follows a progression in which mutations accumulate, gradually causing uncontrolled growth and the development of a **tumor.** Tumors may be either **benign** (not cancerous) or **malignant** (cancerous). Malignant tumors may undergo both **metastasis** and **angiogenesis.**

- **Proto-oncogenes** stimulate the cell cycle after they are turned on by environmental signals, such as growth factors. Oncogenes are mutated proto-oncogenes that stimulate the cell cycle without need of environmental signals. **Tumor suppressor genes** inhibit the cell cycle. Mutated tumor suppressor genes no longer inhibit the cell cycle, allowing unchecked cell division.

ISTUDY

- In some cancer cells, the presence of the telomerase enzyme may stop the **telomeres** from shortening with each cell division, causing the cell to proceed through the cell cycle.

9.5 Prokaryotic Cell Division

Binary fission (in prokaryotes) and mitosis (in single-celled eukaryotic protists and fungi) allow for **asexual reproduction.** Mitosis in multicellular eukaryotes is primarily for the purpose of development, growth, and repair of tissues.

The prokaryotic chromosome has a few proteins and a single, long loop of DNA located in the **nucleoid** region of the cytoplasm. When **binary fission** occurs, the chromosome attaches to the inside of the plasma membrane and replicates. As the cell elongates, the chromosomes are pulled apart. Inward growth of the plasma membrane and formation of new cell wall material divide the cell in two.

Thinking Critically About Biology

1. In a type of skin cancer known as melanoma, skin cells divide uncontrollably. This out-of-control division can be deadly, but a lack of cell division could be harmful as well. Describe why and when skin cells do need to replicate.
2. Survivors of the atomic bombs dropped on Hiroshima and Nagasaki have been the subjects of long-term studies of the effects of ionizing radiation on cancer incidence. The frequencies of different types of cancer in these individuals varied across the decades. In the 1950s, high levels of leukemia and cancers of the lung and thyroid gland were observed. The 1960s and 1970s brought high levels of breast and salivary gland cancers. In the 1980s, rates of colon cancer were especially high. Why do you suppose the rates of different types of cancer varied across time?
3. BPA is a chemical compound that has historically been used in the manufacture of plastic products. However, cells often mistake BPA compounds for hormones that accelerate the cell cycle. Because of this, BPA is associated with an increased risk of certain cancers.
   a. How might BPA interact with the cell cycle and its checkpoints?
   b. Why do you think very small concentrations of BPA might have a large effect on the cell?
4. Why is it advantageous that checkpoints evolved during the cell cycle?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Cell Division

McGraw Hill

Why is it advantageous for cell division to halt in an organism at certain times, such as a plant in the winter?

Making It Relevant

1. Some people mistakenly say that they “inherited cancer” from their parents. Why is this statement not correct, and what would be a better way to rephrase it?
2. Elephants in the chapter opener are resistant to cancer development due to an increase in the number of *p53* genes. Provide a hypothesis on why this may have occurred from an evolutionary perspective. How would you test this hypothesis?
3. Should humans be genetically engineered to contain more *p53* genes? Given what you know about the cell cycle, and the role of checkpoints, what could be some potential problems with this approach?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

10

Meiosis and Sexual Reproduction

kali9/E+/Getty Images

Chapter Outline

10.1 Overview of Meiosis
10.2 Genetic Variation
10.3 The Phases of Meiosis
10.4 Meiosis Compared to Mitosis
10.5 The Cycle of Life
10.6 Changes in Chromosome Number and Structure

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.5 How is genetic information stored in nucleic acids?

Section 4.8 What are microtubules, and how do they interact with chromosomes?

Section 9.3 How are eukaryotic chromosomes organized and replicated prior to cell division?

What Are the Chances of There Being Another You?

What are the chances there will ever be another human like you on this planet? Because of sexual reproduction, and a process called meiosis, two individuals can create offspring that are genetically different from themselves and from each other. In humans, more than 70 trillion different genetic combinations are possible from the mating of just two individuals! In other words, meiosis ensures that you are unique. In animals, meiosis begins the process that produces cells called gametes, which play an important role in sexual reproduction. Sperm are the male gametes, and eggs are the female gametes. The processes by which they are formed are called spermatogenesis and oogenesis, respectively.

While meiosis in the two sexes is very similar, there are some important differences in how spermatogenesis and oogenesis occur. In males, sperm production does not begin until puberty but then continues throughout a male's lifetime. In females, the process of producing eggs has started before the female is even born, and ends around the age of 50, a time called menopause. Another difference concerns the number of gametes that can be produced. In males, sperm production is unlimited, whereas females typically produce only one egg a month. Many more eggs start the process than ever finish it.

As you read through this chapter, think about the following questions:

1. Why is meiosis sometimes called reduction division?
2. What is the role of meiosis in introducing new variation?

Following *the* Themes

CHAPTER 10 – MEIOSIS AND SEXUAL REPRODUCTION

Evolution	The variation introduced during meiosis plays an important role in evolutionary change.
Nature of Science	Changes in chromosome structure and number can have consequences for an individual's physiology.
Biological Systems	The variation introduced by meiosis at a cellular level affects all levels of an organism's physiology.

ISTUDY

10.1 Overview of Meiosis

Learning Outcomes

Upon completion of this section, you should be able to

1. Contrast haploid and diploid chromosome numbers.
2. Explain what is meant by *homologous chromosomes.*
3. Summarize the process by which meiosis reduces the chromosome number.

In sexually reproducing organisms, **meiosis** (Gk. *mio,* "less"; *sis,* "act or process of") is the type of nuclear division that reduces the chromosome number of the cell by 50%. Two terms are important in understanding meiosis. The **haploid (n)** number of chromosomes represents a single set of chromosomes. However, in many eukaryotic organisms, including humans, there are two sets of each chromosome. This is called the **diploid (2n)** number, and reflects the total number of chromosomes in a cell. In humans, we have 23 chromosomes (n) that exist in two sets (2n). Therefore, in humans, meiosis reduces the diploid number of 46 chromosomes to the haploid number of 23 chromosomes.

Gametes, or reproductive cells, usually have the haploid number of chromosomes. In both plants and animals, these are the sperm and egg cells. In **sexual reproduction,** haploid gametes, which are produced during meiosis, subsequently merge into a diploid cell called a **zygote.** In plants and animals, the zygote undergoes development to become an adult organism.

Why undergo meiosis? In sexually reproducing organisms, each parent needs to contribute a single copy of each chromosome so the zygote will be in a diploid state. If each parent contributed two copies, then the number of chromosomes would double with each new generation. Within a few generations, the cells of an animal would be nothing but chromosomes! For example, in humans with a diploid number of 46 chromosomes, in five generations the chromosome number would increase to 1,472 chromosomes (46×2^5). The Belgian cytologist (a biologist that studies cells) Pierre-Joseph van Beneden was one of the first to observe that gametes have a reduced chromosome number. When studying the roundworm *Ascaris,* he noticed that the sperm and egg each contain only two chromosomes, while the zygote and subsequent embryonic cells always have four chromosomes.

Homologous Pairs of Chromosomes

In diploid body cells, the chromosomes occur in pairs. Figure 10.1, a pictorial display of human chromosomes, called a **karyotype,** shows the chromosomes arranged according to pairs. The members of each pair are called homologous chromosomes. **Homologous chromosomes,** *or* **homologues** (Gk. *homologos,* "agreeing, corresponding"), look alike; they have the same length and centromere position. When stained, homologues have a similar banding pattern, because they contain genes for the same traits in the same order in the same locations on both chromosomes in the homologous pair. But while homologous chromosomes have genes for the same traits, such as finger length, the DNA (deoxyribonucleic acid) sequence for the gene on one homologue may code for short fingers and the gene at the same location on the other homologue may code

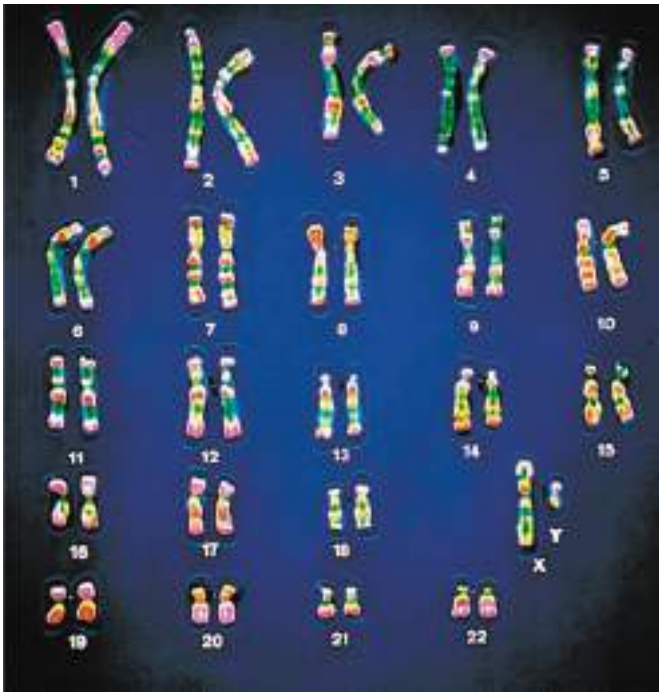


Figure 10.1 Homologous chromosomes. In diploid body cells, the chromosomes occur in pairs called homologous chromosomes. In this karyotype from a human cell, the stained pairs of chromosomes have been numbered 1–22 and XY. Note that chromosome pairs 1–22 are autosomes and the XY pair includes the sex chromosomes and helps determine biological sex. CNRI/Science Source

for long fingers. Alternative forms of a gene (such as for long fingers and short fingers) are called **alleles.** The DNA sequences of alleles are highly similar, but they are different enough to produce alternative physical traits, such as long or short fingers.

To properly produce a haploid number of chromosomes in gametes, you first have to double the amount of DNA. The chromosomes in Figure 10.1 are duplicated as they would be just before nuclear division. Recall that during the S stage of the cell cycle, DNA replicates and the chromosomes become duplicated. The results of the duplication process are depicted in Figure 10.2.

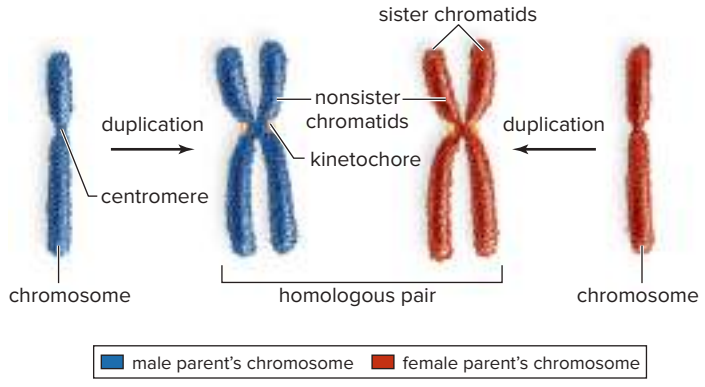


Figure 10.2 Chromatids and DNA replication. Each chromosome in a homologous pair is composed of two chromatids. The sister chromatids contain exactly the same genes; the nonsister chromatids contain genes for the same traits (e.g., type of hair, color of eyes), but the alleles on each gene may be different, such as dark hair versus light hair.

ISTUDY

When duplicated, a chromosome is composed of two identical parts called sister chromatids, each containing one DNA double-helix molecule. The *sister chromatids* are held together at a common region called the centromere.

Why does the zygote have paired chromosomes? One member of a homologous pair was inherited from the male parent, and the other was inherited from the female parent when the haploid sperm and egg fused together. In Figure 10.2 and throughout this chapter, the male parent's chromosome is colored blue, and the female parent's chromosome is colored red. However, this is simply a method of tracking chromosomes in diagrams. Because chromosomes do not have color, geneticists generally use chromosome length and centromere location to identify homologues (see Fig. 10.1). You will see shortly how meiosis reduces the chromosome number. Whereas the zygote and body cells have homologous pairs of chromosomes, the gametes have only one chromosome of each kind—derived from either the male or female parent's homologue.

In humans, 22 pairs are called **autosomes,** and one pair is the sex chromosomes. The **sex chromosomes** are named because they differ between the sexes. Males have the sex chromosomes X and Y, and females have two X chromosomes. The Diversity in Science feature, "Nettie Stevens and the Discovery of the Sex Chromosomes," explores the early research that led to the discovery of the X and Y chromosome.

Meiosis Is Reduction Division

The central purpose of meiosis is to reduce the chromosome number from 2n to n. Meiosis requires two nuclear divisions and produces four haploid daughter cells, each having one of each kind of chromosome. The process begins by replicating the chromosomes, then splitting the matched homologous pairs to go from 2n to n chromosomes during the first division. The second division reduces the amount of DNA in n chromosomes to an amount appropriate for each gamete. Once the DNA has been replicated and chromosomes become a pair, they may exchange genes, creating a genetic mixture different from the parent. The first nuclear division separates each homologous pair, reducing the chromosome number from 2n to n. Even though each daughter cell now has n chromosomes, each chromosome still has a sister chromatid, making a second nuclear division necessary. The end result of meiosis is four gametes with n chromosomes.

Figure 10.3 presents an overview of meiosis, indicating the two nuclear divisions: meiosis I and meiosis II. Prior to meiosis I, DNA replication has occurred; therefore, each chromosome has two sister chromatids. During meiosis I, something new happens that does not occur in mitosis. The homologous chromosomes come together and line up side by side, forming a **synaptonemal complex.** This process is called **synapsis** (Gk. *synaptos,* "united, joined together") and results in a **bivalent** (L. *bis,* "two"; *valens,* "strength")—that is, two homologous chromosomes that stay in close association during the first two phases of meiosis I. Sometimes the term *tetrad* (Gk. *tetra,* "four") is used instead of *bivalent,* because, as you can see, a bivalent contains four chromatids. Chromosomes may recombine or exchange genetic information during this association (see Section 10.2).

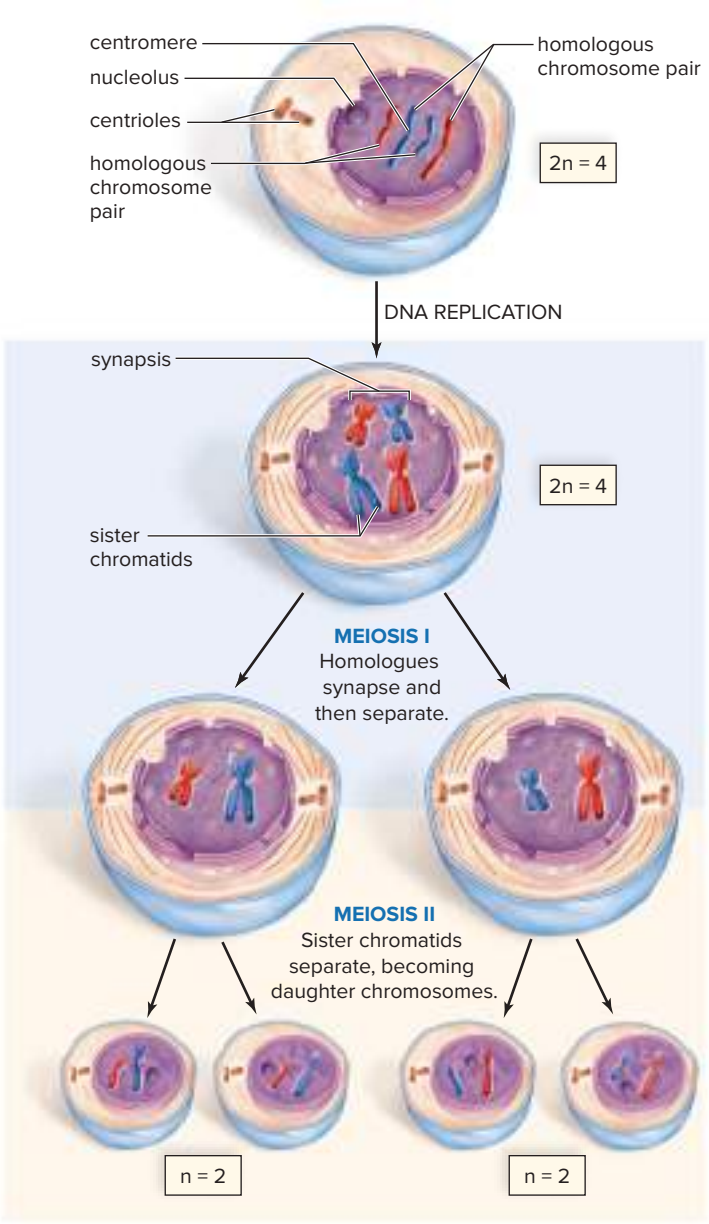


Figure 10.3 Overview of meiosis. Following DNA replication, each chromosome is duplicated and consists of two chromatids. During meiosis I, homologous chromosomes pair and separate. During meiosis II, the sister chromatids of each duplicated chromosome separate. At the completion of meiosis, there are four haploid daughter cells. Each daughter cell has one of each kind of chromosome.

Following synapsis, homologous pairs align at the metaphase plate, and then the members of each pair separate. This separation means that only one duplicated chromosome from each homologous pair reaches a daughter nucleus, reducing the chromosome number from 2n to n. It is important for each daughter nucleus to have a member from each pair of homologous chromosomes, because only in that way can there be a copy of each kind of chromosome in the daughter nuclei. Notice in Figure 10.2 that two possible combinations of chromosomes in the daughter cells are shown: short red with long blue and short blue with long red. Knowing that all daughter cells need to have one short chromosome and one

ISTUDY

▶ Diversity in Science

Nettie Stevens and the Discovery of the Sex Chromosomes

For a female growing up in the aftermath of the American Civil War, a research career in science was next to unheard of. But Nettie Stevens didn't just beat the odds to become one of America's earliest female geneticists; she also helped set the foundation for modern understandings of chromosomes by determining the role of X and Y chromosomes in biological sex.

Largely forgotten in her own life, Stevens has more recently received newfound recognition as an example of the tremendous influence females, such as Rosalind Franklin, have had on the advancement of science—even if they were denied the recognition they deserved in their own lifetimes.

Born in Vermont in 1861, Stevens early on had a great interest in higher education but lacked the financial means to make the jump from high school to college. Instead, she worked as a teacher to make ends meet while studying at Westford Academy in Massachusetts, becoming one of only three females to graduate with a degree in her class. From Westford, Stevens completed degrees at the newly opened Stanford University and Bryn Mawr in Pennsylvania, where she earned a Ph.D. in cellular biology at the age of 39 (Fig. 10A).

Even as Stevens was finishing her degree in 1903 at Bryn Mawr, Gregor Mendel's research on genetics was seeing a renewed burst of interest, with scientists busy diving into research on the factors behind hereditary genetics. At the same time, long-held assumptions, particularly surrounding the conditions that dictated biological sex, remained stubbornly rigid in the scientific community. For centuries, the determining factor of whether a child was biologically male or female was believed based on environmental factors, such as the temperature of the female parent's womb. Even as scientists were uncovering the genetic basis for certain traits, the belief in environmentally determined sex remained stubbornly pervasive in the community.

Stevens's research was one of the first to challenge this assumption. In her research, she looked at the chromosomes of insects—mostly mealworms, beetles, and butterflies—by inspecting thousands of paper-thin cross sections of their gonads under the microscope. After studying and cataloging the individual chromosomes, Stevens found evidence that biological sex was based on the particular assortment of chromosomes—what we now know as XY sex determination.

An Unrecognized Discovery

In 1905, Stevens published her findings, which definitively made the case for biological sex as the makeup of X and Y chromosomes. The discovery effectively made her one of the first scientists in the world to understand how chromosomes may be involved in sex determination.

However, several factors prohibited Stevens from receiving the recognition she deserved. Most prominently was the research of another scientist, Edmund Beecher Wilson, who earlier that year had published his own findings on XY sex determination through his own independent research. While normally two scientists independently coming to the same conclusions within a small window of time would mean they would receive joint recognition, Stevens's discovery was discounted.

Even as Wilson went on to become one of the most famous geneticists of his time, Stevens faded into obscurity, dying not six years later from breast cancer in 1912.

Stevens's Legacy Today

In hindsight, Stevens's findings in the early 1900s were extremely accurate, going further than even Wilson's discovery to make the (ultimately incorrect) claim of chromosomes as being *the* determinate of biological sex, without the influence of any external or environmental factors. Between her and Wilson, the discovery of chromosomal sex determination set the foundations for generations of future geneticists to unlock the mysteries of our chromosomes, especially in how certain traits are passed along the X or Y chromosome.

Today, Stevens's name is being rediscovered, over a hundred years after her death. In 1994, Stevens was added to the National Women's Hall of Fame, and in 2016 on the 155th anniversary of her birth she was recognized with a Google Doodle, which directed internet searches to learn more about her own scientific discoveries. In time, hopefully her name will see its rightful place alongside Wilson's as the co-discoverer of XY sex determination, and she will receive the recognition she deserves.

Figure 10A Nettie Stevens and sex chromosomes. Nettie Stevens conducted some of the early research on sex chromosomes. H.S. Photos/Alamy Stock Photo

ISTUDY

long chromosome, what are the other two possible combinations of chromosomes for these cells?

Notice that DNA replication occurs only once during meiosis. No replication is needed between meiosis I and meiosis II, because the chromosomes are already duplicated. They already have two sister chromatids. During meiosis II, the sister chromatids separate, becoming daughter chromosomes that move to opposite poles. The chromosomes in each of the four daughter cells now contain only one DNA double-helix molecule in the form of a haploid chromosome.

The number of centromeres can be counted to verify that the parent cell has the diploid number of chromosomes. At the end of meiosis I, the chromosome number has been reduced, because there are half as many centromeres present, even though each chromosome still consists of two chromatids each. Each daughter cell that forms has the haploid number of chromosomes. At the end of meiosis II, sister chromatids separate, and each daughter cell that forms still contains the haploid number of chromosomes, each consisting of a single chromatid.

Fate of Daughter Cells

In the plant life cycle (see Section 23.1), the daughter cells become haploid spores that germinate to become a haploid generation. This generation then produces the gametes by mitosis. In the animal life cycle, the daughter cells become the gametes, either sperm or eggs. The body cells of an animal normally contain the diploid number of chromosomes due to the fusion of sperm and egg during fertilization. If meiotic events go wrong, the gametes can contain the wrong number of chromosomes or altered chromosomes. This possibility and its consequences are discussed in Section 10.6.

Check Your Progress 10.1

1. Describe what is meant by *a homologous pair of chromosomes.*
2. Examine how chromosome number changes during meiosis I and meiosis II.
3. Explain the purpose of a bivalent in chromosome pairing.

10.2 Genetic Variation

Learning Outcomes

Upon completion of this section, you should be able to

1. Understand the importance of genetic variation to evolutionary change.
2. Explain how crossing-over contributes to genetic variation.
3. Examine how independent assortment contributes to genetic variation.

We have seen that meiosis provides a way to keep the chromosome number constant generation after generation. Without meiosis, the chromosome number of the next generation would continually increase. The events of meiosis also help ensure that genetic variation occurs with each generation.

Genetic variation is essential for a species to be able to evolve and adapt in a changing environment. Asexually reproducing organisms, such as the prokaryotes, depend primarily on mutations to generate variation among offspring. This is sufficient for their survival, because they produce great numbers of offspring very quickly. Although mutations also occur among sexually reproducing organisms, the reshuffling of genetic material during sexual reproduction ensures that offspring will have a different combination of genes than their parents. Meiosis brings about genetic variation in two key ways: crossing-over and independent assortment of homologous chromosomes.

Genetic Recombination

Crossing-over is an exchange of genetic material between nonsister chromatids of a bivalent during meiosis I. In humans, it is estimated that an average of two to three crossovers occur between the nonsister chromatids during meiosis. At synapsis, homologues line up side by side, and a nucleoprotein lattice appears between them (Fig. 10.4). This lattice holds the bivalent together in such a way that the DNA of the duplicated

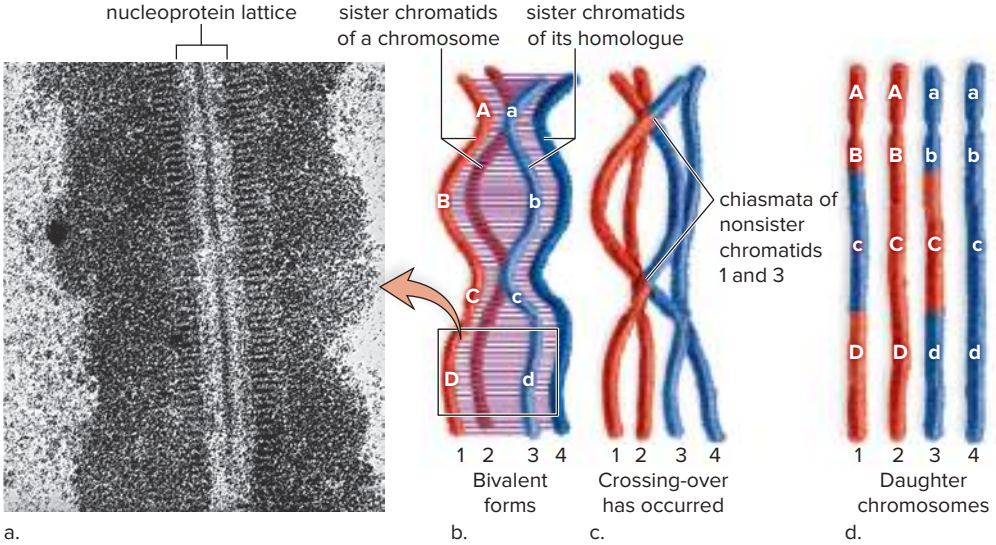


Figure 10.4 Crossing-over during meiosis I. **a.** The homologous chromosomes pair up, and a nucleoprotein lattice develops between them. This is an electron micrograph of the lattice. It "zippers" the members of the bivalent together, so that corresponding genes on paired chromosomes are in alignment. **b.** This visual representation shows only two places where nonsister chromatids 1 and 3 have come in contact. **c.** Chiasmata indicate where crossing-over has occurred. The exchange of color represents the exchange of genetic material. **d.** Following meiosis II, daughter chromosomes have a new combination of genetic material due to crossing-over, which occurred between nonsister chromatids during meiosis I. (a): Courtesy Dr. D. Von Wettstein

ISTUDY

THEME Nature of Science

Meiosis and the Parthenogenic Lizards

The process of crossing-over plays an important role in the generation of genetic variation for sexually reproducing species. For most species that undergo asexual reproduction, fast generation times and mutation allow for the species to introduce enough variation to respond to environmental changes. But what about species, such as the whiptail lizard shown in Figure 10B, that undergo parthenogenesis? Parthenogenesis is the production of new individuals from unfertilized eggs. It is not uncommon in the animal kingdom—many arthropods, lizards, fish, and salamanders are known to be parthenogenic.

Parthenogenesis is a form of asexual reproduction in which only one parent, the female, contributes genetic information to the next generation. Typically, however, these species do not have the short generation times of other asexual organisms, such as bacteria. At least on the surface, parthenogenesis would seem to limit the amount of genetic variation in the species, and thus reduce the ability of the species to respond to changes in its environment. While some species, such as honeybees, avoid this problem by switching between parthenogenesis and sexual reproduction, truly parthenogenic species appear to be at an evolutionary disadvantage.

Researchers from a team at the Howard Hughes Medical Institute discovered that in a parthenogenic species of lizards (whiptail lizards, genus *Aspidoscelis*) there is a variation in the normal process of meiosis. In most cases, crossing-over during meiosis occurs between the nonsister chromatids of homologous chromosomes. However, in the whiptail lizard, crossing-over occurs between the sister chromatids.

Figure 10B In parthenogenic species, such as the whiptail lizard, variations in the process of meiosis allow the species to increase genetic variation with each generation. Suzanne L. Collins/ Science Source

How does this happen? To make this possible, the species doubles the number of chromosomes prior to meiosis—effectively making an additional copy of the genome and forming a pair of homologous chromosomes from a single parent. This doubling allows the reduction division in meiosis to produce diploid (2n) gametes, a requirement for many species that undergo parthenogenesis. Then, the species allows for crossing-over to occur between the sister chromatids themselves. Because there are always slight differences in the sister chromatids (they are never truly identical), small amounts of variation are maintained in the genome, and this is passed on to the next generation. The amount of genetic variation may be small, but this variation in meiosis allows for some level of genetic recombination, thus providing genetic variation to the species.

Questions to Consider

1. Does this process produce the same amount of genetic variation as would occur in normal sexual reproduction?
2. How would you test to determine the amount of genetic variation produced by parthenogenic species?

chromosomes of each homologue pair are aligned. This ensures that the genes contained on the nonsister chromatids are directly aligned. Now crossing-over may occur. As the lattice breaks down, homologues are temporarily held together by *chiasmata* (*sing.*, chiasma), regions where the nonsister chromatids are attached due to DNA strand exchange and crossing-over. After exchange of genetic information between the nonsister chromatids, the homologues separate and are distributed to different daughter cells.

To appreciate the significance of crossing-over, keep in mind that the members of a homologous pair can carry slightly different instructions, or alleles, for the same genetic traits. In the end, due to a swapping of genetic material during crossing-over, the chromatids held together by a centromere are no longer identical. Therefore, when the chromatids separate during meiosis II, some of the daughter cells receive daughter chromosomes with recombined alleles. Due to **genetic recombination,** the offspring have a different set of alleles, and therefore genes, than their parents. This increases the genetic variation of the offspring. See the Nature of Science feature, "Meiosis and the Parthenogenic Lizards," to learn how a special form of crossing-over can increase diversity during asexual reproduction.

ISTUDY

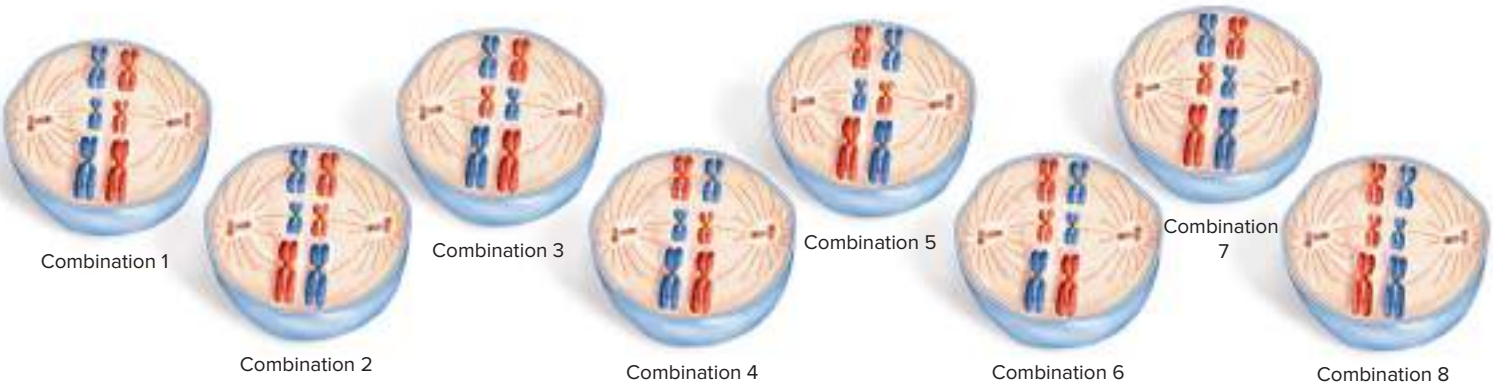


Figure 10.5 Independent assortment. When a parent cell has three pairs of homologous chromosomes, there are 2^3, or 8, possible chromosome alignments at the metaphase plate due to independent assortment. Each possible combination is shown, one in each cell.

Independent Assortment of Homologous Chromosomes

During **independent assortment,** the homologous chromosome pairs separate independently, or randomly. When homologues align at the metaphase plate, the female or male parent's homologue may be oriented toward either pole. Figure 10.5 shows the possible chromosome orientations for a cell that contains only three pairs of homologous chromosomes. Once all possible alignments of independent assortment are considered for these three pairs, the result will be 2^3, or 8, combinations of female and male parents' chromosomes in the resulting gametes from this cell, simply due to independent assortment of homologues.

Significance of Genetic Variation

In humans, who have 23 pairs of chromosomes, the possible chromosomal combinations in the gametes is a staggering 2^{23}, or 8,388,608. The variation that results from meiosis is enhanced by **fertilization,** the union of the male and female gametes. The chromosomes donated by the parents are combined, and in humans this means there are potentially $(2^{23})^2$, or 70,368,744,000,000, different chromosome combinations in the zygote. This number assumes there was no crossing-over between the nonsister chromatids prior to independent assortment. If a single crossing-over occurs, then $(4^{23})^2$, or 4,951,760,200,000,000,000,000,000,000, genetically different zygotes are possible for every couple. Keep in mind that crossing-over can occur several times in each chromosome!

The staggering amount of genetic variation achieved through meiosis is particularly important to the long-term survival of a species, because it increases genetic variation within a population. The process of sexual reproduction brings about genetic recombinations among members of a population.

Asexual reproduction passes on exactly the same combination of chromosomes and genes. Asexual reproduction may be advantageous if the environment remains unchanged. However, if the environment changes, genetic variability among offspring introduced by sexual reproduction may be advantageous. Under the new conditions, some offspring may have a better chance of survival and reproductive success than others in a population. For example, suppose the ambient temperature were to rise due to climate change. This change in the environment could place demands on the physiology of an organism. An animal with less fur, or reduced body fat, could have an advantage over other individuals of its generation.

In a changing environment, sexual reproduction, with its reshuffling of genes due to meiosis and fertilization, might give a few offspring a better chance to survive and reproduce, thereby increasing the possibility of passing on their genes to the next generation.

Check Your Progress 10.2

1. Describe the two main ways in which meiosis contributes to genetic variation.
2. Examine how many combinations of chromosomes are possible in the gametes in a cell with four pairs of homologous chromosomes.
3. Evaluate why meiosis and sexual reproduction are important in responding to the changing environment.

10.3 The Phases of Meiosis

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the phases of meiosis and the major events that occur during each phase.
2. Understand how meiosis reduces the chromosome number from diploid to haploid.

Meiosis consists of two unique, consecutive cell divisions: meiosis I and meiosis II. DNA is replicated in S phase of the cell cycle,

ISTUDY

prior to meiosis I but not meiosis II. Both meiosis I and meiosis II contain a prophase, metaphase, anaphase, and telophase.

Prophase I

It is apparent during prophase I that nuclear division is about to occur, because a spindle forms as the centrosomes migrate away from one another (Fig. 10.6). The nuclear envelope fragments, and the nucleolus disappears.

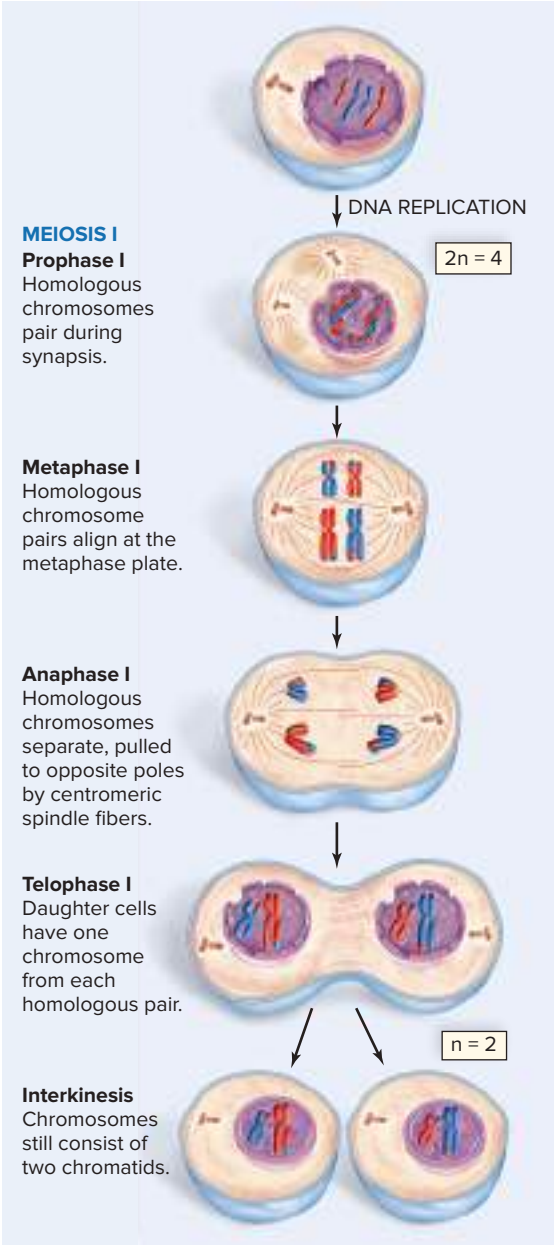


Figure 10.6 Stages of meiosis I. When diploid homologous chromosomes pair during meiosis I, crossing-over occurs, as represented by the exchange of color. Pairs of homologous chromosomes separate during meiosis I, and chromatids separate, becoming haploid daughter chromosomes. Following interkinesis, two cells begin meiosis II (see Fig. 10.7).

The homologous chromosomes, each having replicated during S phase of the cell cycle, consist of two sister chromatids. The homologous chromosomes undergo synapsis to form bivalents. At this time, crossing-over may occur between the nonsister chromatids (see Fig. 10.4). As described earlier, crossing-over increases the genetic diversity of the daughter cells, because after crossing-over, the sister chromatids are no longer identical.

Throughout prophase I, the homologous chromosomes have been condensing, so that by now they have the appearance of compacted metaphase chromosomes.

Metaphase I

During metaphase I, the bivalents held together by chiasmata (see Fig. 10.4) have moved toward the metaphase plate (equator of the spindle). Metaphase I is characterized by a fully formed spindle and alignment of the bivalents at the metaphase plate. As in mitosis, kinetochores are seen, but the two kinetochores of a duplicated chromosome are attached to the same kinetochore spindle fiber.

Bivalents independently align themselves at the metaphase plate of the spindle. Either the female or male parent's homologue of each bivalent may be oriented toward either pole of the cell. The orientation of one bivalent is not dependent on the orientation of the other bivalents. This independent assortment of chromosomes contributes to the genetic variability of the daughter cells, because all possible combinations of chromosomes can occur in the daughter cells.

Anaphase I

During anaphase I, the homologues of each bivalent separate and move to opposite poles, but sister chromatids do not separate. This splitting of the homologous pair reduces the chromosome number from 2n to n. However, each chromosome still has two chromatids (Fig. 10.6).

Telophase I

Completion of telophase I is not necessary during meiosis. That is, the spindle disappears, but new nuclear envelopes need not form before the daughter cells proceed to meiosis II. Also, this phase may or may not be accompanied by cytokinesis, which is separation of the cytoplasm. Notice in Figure 10.6 that the cells have different chromosome combinations than the original parent cell (not all of the combinations are shown in the figure). The cells exiting telophase I are also haploid compared to the diploid parent cell.

Interkinesis

Following telophase, the cells enter interkinesis, a short rest period prior to beginning the second nuclear division: meiosis II. The process of **interkinesis** is similar to interphase between mitotic divisions, except that DNA replication does not occur, because the chromosomes are already duplicated.

ISTUDY

Meiosis II and Gamete Formation

At the beginning of meiosis II, the two daughter cells contain the haploid number of chromosomes, or one chromosome from each homologous pair. Note that these chromosomes still consist of duplicated sister chromatids at this point. During metaphase II, the chromosomes align at the metaphase plate, but they do not align in homologous pairs, as in meiosis I, because only one chromosome of each homologous pair is present (Fig. 10.7). Thus, the alignment of the chromosomes at the metaphase plate is similar to what is observed during mitosis.

During anaphase II, the sister chromatids separate, becoming daughter chromosomes that are not duplicated. These daughter chromosomes move toward the poles. At the end of telophase II and cytokinesis, there are four haploid cells. Because of crossing-over of chromatids during meiosis I, each gamete most likely contains chromosomes with a mixture of the female and male parents' genes.

Following meiosis II in animals, the haploid cells become gametes in animals (see Section 10.5). In plants, they become *spores,* reproductive cells that develop into new multicellular structures without the need to fuse with another reproductive cell. The multicellular structure is the haploid generation, which produces gametes. The resulting zygote develops into a diploid generation. Therefore, plants have both haploid and diploid phases in their life cycle, and plants are said to exhibit an *alternation of generations* (see Section 23.1). In most fungi and algae, the zygote undergoes meiosis, and the daughter cells develop into new individuals. Therefore, the organism is always haploid.

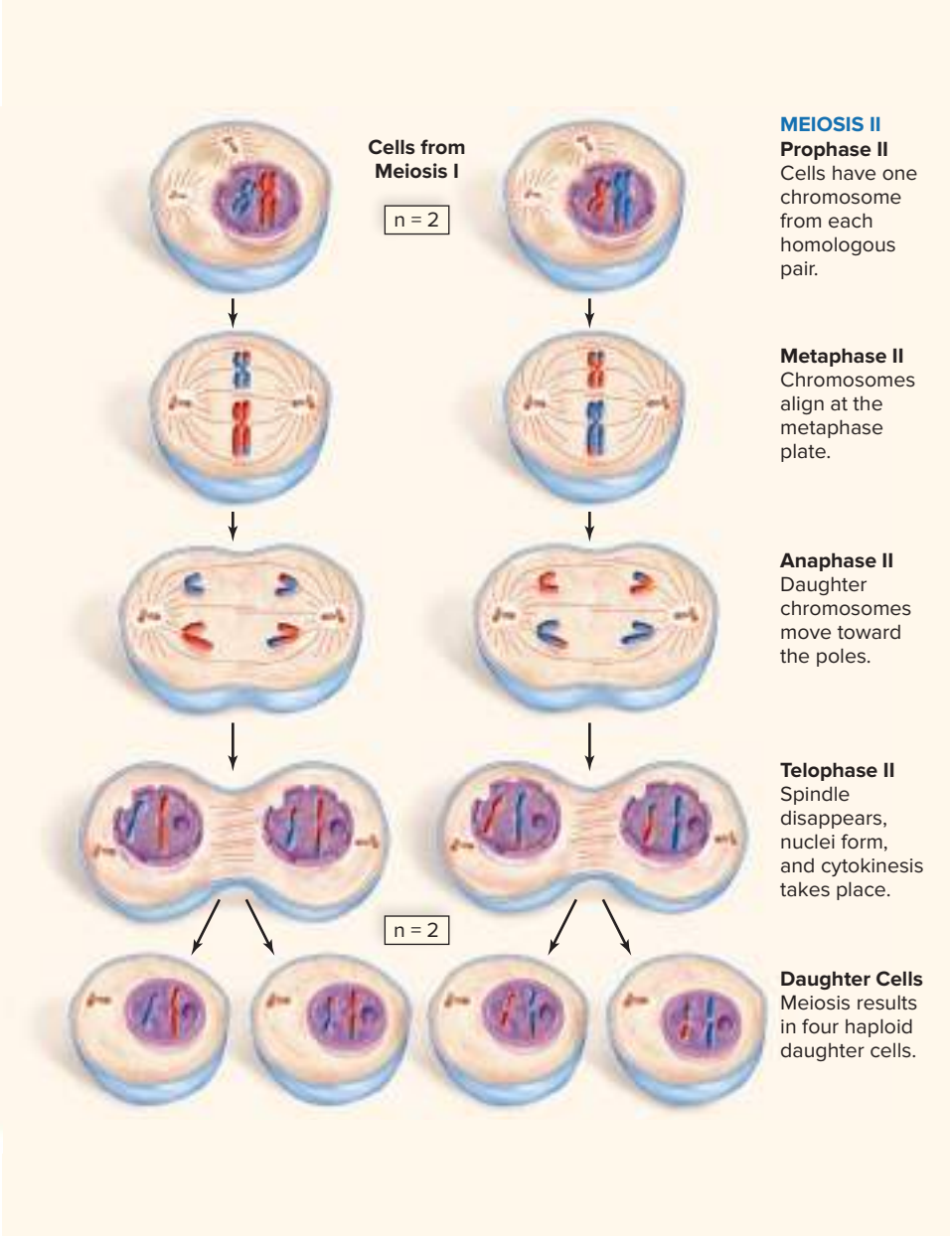


Figure 10.7 Stages of meiosis II. During meiosis II, sister chromatids separate, becoming daughter chromosomes that are distributed to the daughter nuclei. Following meiosis II, there are four haploid daughter cells.

Check Your Progress 10.3

1. Describe the differences between the chromosomal combinations of a cell at metaphase I and metaphase II of meiosis.
2. Explain what would cause daughter cells following meiosis II to contain identical chromosomes or nonidentical chromosomes.
3. Examine what could happen if homologous chromosomes lined up top to bottom instead of side by side during meiosis I.

10.4 Meiosis Compared to Mitosis

Learning Outcomes

Upon completion of this section, you should be able to

1. Contrast changes in chromosome number, genetic variability, and number of daughter cells between meiosis and mitosis.
2. Distinguish the events that occur during prophase I of meiosis that do not occur during prophase of mitosis.
3. Compare chromosome alignment during meiosis I to mitosis.

ISTUDY

There are many similarities between the processes of meiosis and mitosis. In both processes:

- An orderly series of stages, including prophase, prometaphase, metaphase, and telophase, are involved in the sorting and division of the chromosomes.
- The spindle fibers are active in sorting the chromosomes.
- Cytokinesis follows the end of the process to divide the cytoplasm between the daughter cells.

However, the function of mitosis and meiosis in an organism is very different. Mitosis maintains the chromosome number between the cells, whereas meiosis is often referred to as reduction division.

Figure 10.8 compares meiosis and mitosis. Several of the fundamental differences between the two processes include:

- Meiosis requires two nuclear divisions, but mitosis requires only one nuclear division.

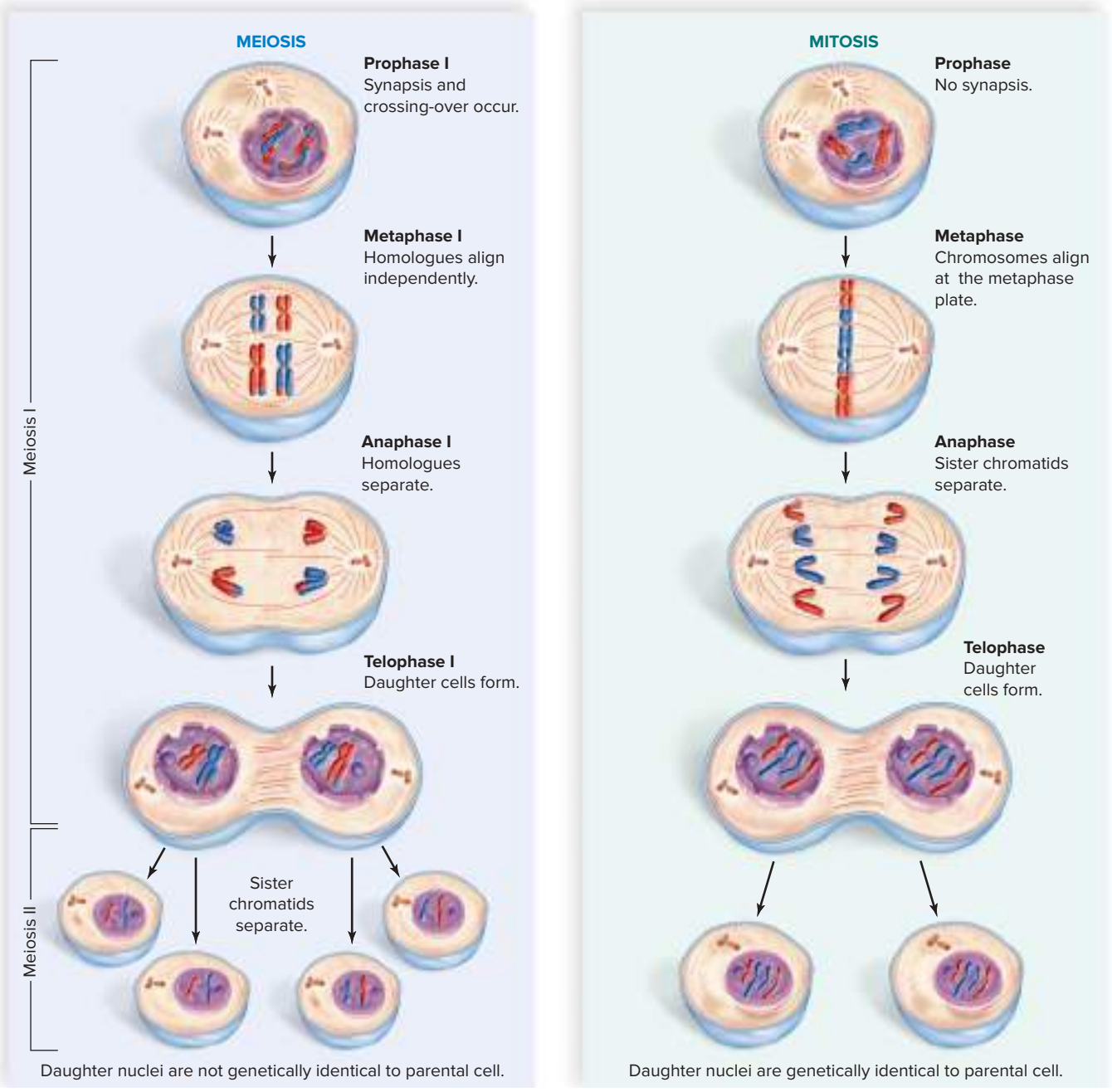


Figure 10.8 Meiosis I compared to mitosis. Why does meiosis produce daughter cells with half the number of chromosomes, whereas mitosis produces daughter cells with the same number of chromosomes as the parent cell? Compare metaphase I of meiosis to metaphase of mitosis. Only in metaphase I of meiosis are the homologous chromosomes paired at the metaphase plate. Members of homologous chromosome pairs separate during anaphase I, and therefore the daughter cells are haploid. The exchange of color between nonsister chromatids represents the crossing-over that occurred during meiosis I. The blue chromosomes were inherited from the male parent, and the red chromosomes were inherited from the female parent.

ISTUDY

- Meiosis produces four daughter nuclei. Following cytokinesis, there are four daughter cells. Mitosis followed by cytokinesis results in two daughter cells.
- Following meiosis, the four daughter cells are haploid and have half the chromosome number of the diploid parent cell. Following mitosis, the daughter cells have the same chromosome number as the parent cell.
- Following meiosis, the daughter cells are genetically identical neither to each other nor to the parent cell. Following mitosis, the daughter cells are genetically identical to each other and to the parent cell.

In addition to the fundamental differences between meiosis and mitosis, two specific differences between the two types of nuclear divisions can be categorized. These differences involve occurrence and process.

Occurrence

Meiosis occurs only at certain times in the life cycle of sexually reproducing organisms. In humans, meiosis occurs only in the reproductive organs and produces the gametes. Mitosis is more common, because it occurs in all tissues during growth and repair.

Process

We now compare the processes of both meiosis I and meiosis II to mitosis.

Meiosis I Compared to Mitosis

Notice that these events distinguish meiosis I from mitosis (Table 10.1):

- During prophase I, bivalents form and crossing-over occurs. These events do not occur during mitosis.
- During metaphase I of meiosis, bivalents independently align at the metaphase plate. The paired chromosomes have a total of four chromatids each. During metaphase in mitosis, individual chromosomes align at the metaphase plate. They each have two chromatids.
- During anaphase I of meiosis, homologues of each bivalent separate, and duplicated chromosomes (with centromeres intact) move to opposite poles. During anaphase of mitosis, sister chromatids separate, becoming daughter chromosomes that move to opposite poles.

Table 10.1 Meiosis I Compared to Mitosis

Meiosis I	Mitosis
Prophase I	**Prophase**
Pairing of homologous chromosomes	No pairing of chromosomes
Metaphase I	**Metaphase**
Bivalents at metaphase plate	Duplicated chromosomes at metaphase plate
Anaphase I	**Anaphase**
Homologues of each bivalent separate, and duplicated chromosomes move to poles	Sister chromatids separate, becoming daughter chromosomes that move to the poles
Telophase I	**Telophase**
Two haploid daughter cells, not identical to the parent cell	Two diploid daughter cells, identical to the parent cell

Table 10.2 Meiosis II Compared to Mitosis

Meiosis II	Mitosis
Prophase II	**Prophase**
No pairing of chromosomes	No pairing of chromosomes
Metaphase II	**Metaphase**
Haploid number of duplicated chromosomes at metaphase plate	Diploid number of duplicated chromosomes at metaphase plate
Anaphase II	**Anaphase**
Sister chromatids separate, becoming daughter chromosomes that move to the poles	Sister chromatids separate, becoming daughter chromosomes that move to the poles
Telophase II	**Telophase**
Four haploid daughter cells, not genetically identical	Two diploid daughter cells, identical to the parent cell

Meiosis II Compared to Mitosis

The events of meiosis II are similar to those of mitosis, except that in meiosis II the nuclei contain the haploid number of chromosomes (Table 10.2). In mitosis, the original number of chromosomes is maintained. Meiosis II produces two daughter cells from each parent cell that completes meiosis I, for a total of four daughter cells. These daughter cells contain the same number of chromosomes as they did at the end of meiosis I.

Check Your Progress 10.4

1. Compare chromosome alignment between metaphase I of meiosis and metaphase of mitosis.
2. Explain how meiosis II is more similar to mitosis than to meiosis I.

10.5 The Cycle of Life

Learning Outcomes

Upon completion of this section, you should be able to

1. Contrast the life cycle of plants with the life cycle of animals.
2. Describe spermatogenesis and oogenesis in humans.

The term **life cycle** refers to all the reproductive events that occur from one generation to the next similar generation. In animals, including humans, the individual is always diploid, and meiosis

ISTUDY

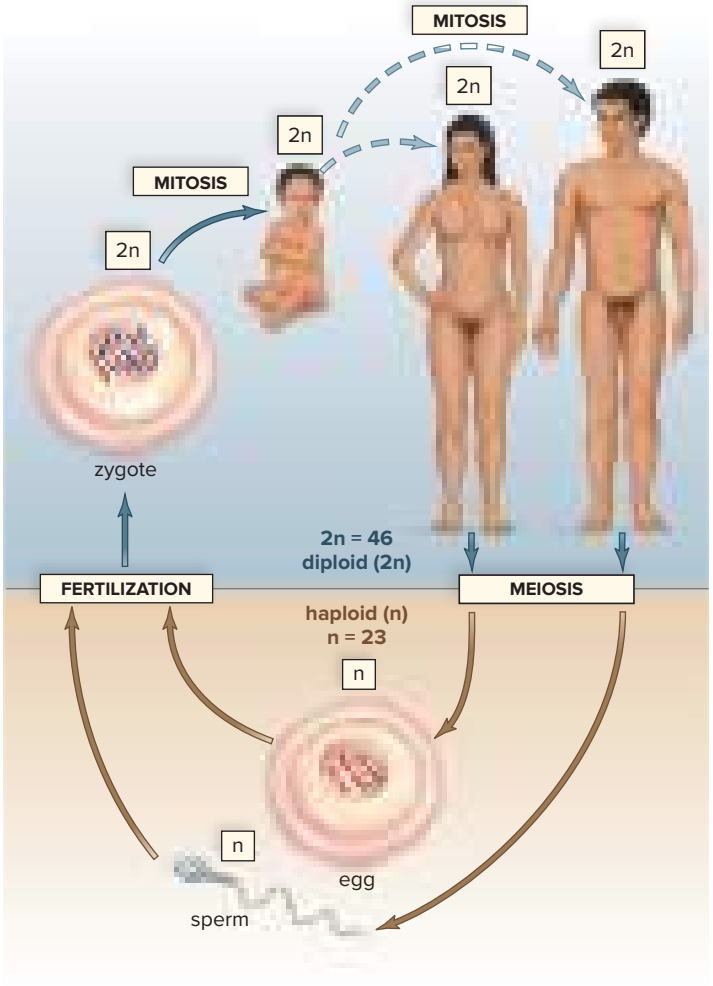


Figure 10.9 Life cycle of humans. Meiosis in males is a part of sperm production, and meiosis in females is a part of egg production. When a haploid sperm fertilizes a haploid egg, the zygote is diploid. The zygote undergoes mitosis as it develops into a newborn child. Mitosis continues throughout life during growth and repair.

produces the gametes, the only haploid phase of the life cycle (Fig. 10.9). In contrast, plants have a haploid phase that alternates with a diploid phase. The haploid generation, known as the **gametophyte,** may be larger or smaller than the diploid generation, called the **sporophyte.**

Mosses growing on bare rocks and forest floors are the haploid generation, and the diploid generation is short-lived. In most fungi and algae, the zygote is the only diploid portion of the life cycle, and it undergoes meiosis. Therefore, the black mold that grows on bread, as well as the green scum that floats on a pond, is haploid.

The majority of plant species, including pine, corn, and sycamore, are usually diploid, and the haploid generation is short-lived. In plants, algae, and fungi, the haploid phase of the life cycle produces gamete nuclei without the need for meiosis, because it has occurred earlier.

Animals are diploid, and meiosis occurs during the production of gametes, called **gametogenesis.** In males, meiosis is a part of **spermatogenesis** (Gk. *sperma,* "seed"), which occurs in the testes and produces sperm. In females, meiosis is a part of **oogenesis** (Gk. *oon,* "egg"), which occurs in the ovaries and produces eggs. A sperm and an egg join at fertilization, restoring the diploid chromosome number. The resulting zygote undergoes mitosis during development of the fetus. After birth, mitosis is involved in the continued growth of the child and the repair of tissues at any time.

Spermatogenesis and Oogenesis in Humans

In human males, spermatogenesis occurs within the testes. In females, oogenesis occurs within the ovaries.

Spermatogenesis

The testes contain stem cells called spermatogonia. These cells keep the testes supplied with primary spermatocytes that undergo spermatogenesis, as described in Figure 10.10. Primary spermatocytes with 46 chromosomes undergo meiosis I to form two secondary spermatocytes, each with 23 duplicated chromosomes. Secondary spermatocytes undergo meiosis II to produce four spermatids with 23 daughter chromosomes. Spermatids then differentiate into viable sperm (spermatozoa). Upon sexual arousal, the sperm enter ducts and exit the penis upon ejaculation.

Oogenesis

The ovaries contain stem cells, called oogonia, that produce many primary **oocytes** with 46 chromosomes during fetal development. They even begin oogenesis, but only a few continue when a female

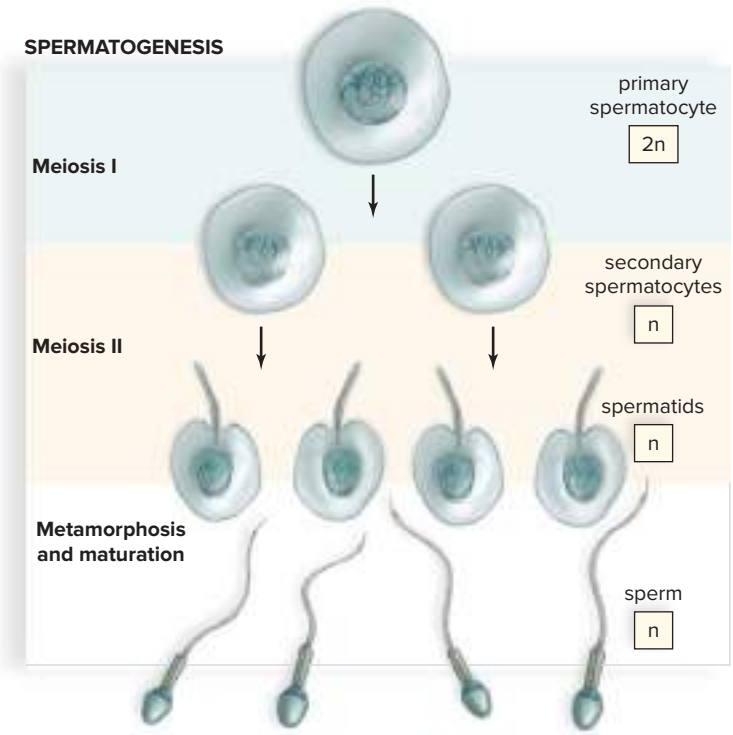


Figure 10.10 Spermatogenesis in mammals. Spermatogenesis produces four viable sperm. In humans, both sperm and egg have 23 chromosomes. Therefore, following fertilization, the zygote has 46 chromosomes.

ISTUDY

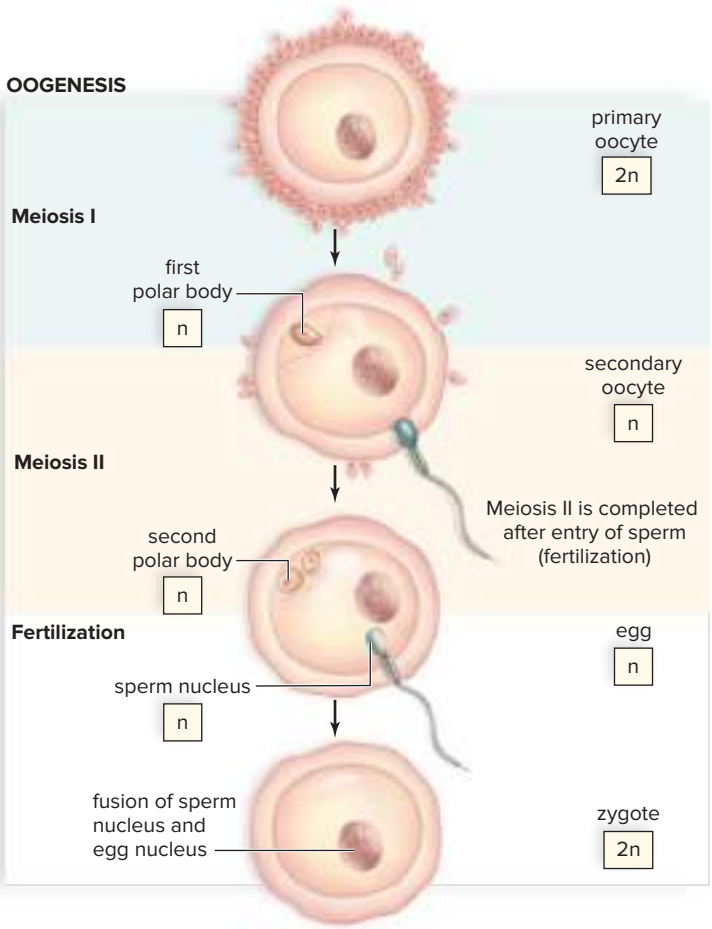


Figure 10.11 Oogenesis in mammals. Oogenesis produces one egg and at least two polar bodies, whereas spermatogenesis produces four viable sperm. In humans, both sperm and egg have 23 chromosomes. Therefore, following fertilization, the zygote has 46 chromosomes.

has become sexually mature. The result of meiosis I is two haploid cells with 23 chromosomes each (Fig. 10.11). One of these cells, termed the **secondary oocyte,** receives almost all the cytoplasm. The other is a **polar body** that may either disintegrate or divide again.

The secondary oocyte begins meiosis II but stops at metaphase II. Then, the secondary oocyte leaves the ovary and enters the uterine tube, where sperm may be present. If no sperm are in the uterine tube, or if a sperm does not enter the secondary oocyte, it eventually disintegrates without completing meiosis. If a sperm does enter the oocyte, some of its contents trigger the completion of meiosis II in the secondary oocyte, and another polar body forms.

At the completion of oogenesis, following entrance of a sperm, there is one egg and two or three polar bodies. The polar bodies are a way to "dispose of" chromosomes while retaining much of the cytoplasm in the egg. Cytoplasmic molecules and organelles are needed by a developing embryo following fertilization. Some zygote components, such as the centrosome, are contributed by the sperm.

The mature egg has 23 chromosomes, but the zygote formed when the sperm and egg nuclei fuse has 46 chromosomes. Therefore, fertilization restores the diploid number of chromosomes. The production of haploid gametes and subsequent fusion of those gametes into a diploid zygote complete a human life cycle.

Check Your Progress 10.5

1. Describe where cells that undergo meiosis are located in humans.
2. Compare the number of gametes produced during oogenesis and spermatogenesis in humans.

10.6 Changes in Chromosome Number and Structure

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between euploidy and aneuploidy.
2. Explain how nondisjunction can cause monosomy and trisomy aneuploidy.
3. Describe human diseases caused by changes in the number of sex chromosomes.
4. Characterize how changes in chromosome structure can lead to human diseases.

We have seen that crossing-over creates variation within a population and is essential for the normal separation of chromosomes during meiosis. Furthermore, the proper separation of homologous chromosomes during meiosis I and the separation of sister chromatids during meiosis II are essential for the maintenance of normal chromosome numbers in living organisms. Although meiosis almost always proceeds normally, a failure of chromosomes to separate, or **nondisjunction,** may occur, resulting in a gain or loss of chromosomes. Errors in crossing-over may result in extra or missing parts of chromosomes.

Aneuploidy

The correct number of chromosomes in a species is known as **euploidy.** A change in the chromosome number resulting from nondisjunction during meiosis is called **aneuploidy.** Aneuploidy is seen in both plants and animals. Monosomy and trisomy are two aneuploid states.

Monosomy ($2n - 1$) occurs when an individual has only one of a particular type of chromosome when there should be two. **Trisomy** ($2n + 1$) occurs when an individual has three of a particular type of chromosome instead of two. Both monosomy and trisomy are the result of nondisjunction during mitosis or meiosis. *Primary nondisjunction* occurs during meiosis I when both members of a homologous pair go into the same daughter cell (Fig. 10.12*a*). *Secondary nondisjunction* occurs during meiosis II when the sister chromatids fail to separate and both daughter chromosomes go into the same gamete (Fig. 10.12*b*).

Notice that when secondary nondisjunction occurs, there are two normal gametes and two aneuploid gametes. In contrast, when primary nondisjunction occurs, no normal gametes are produced. Therefore, primary nondisjunction tends to have more deleterious effects than secondary nondisjunction.

In animals, monosomies and trisomies of nonsex, or autosomal, chromosomes are generally lethal, but a trisomic

ISTUDY

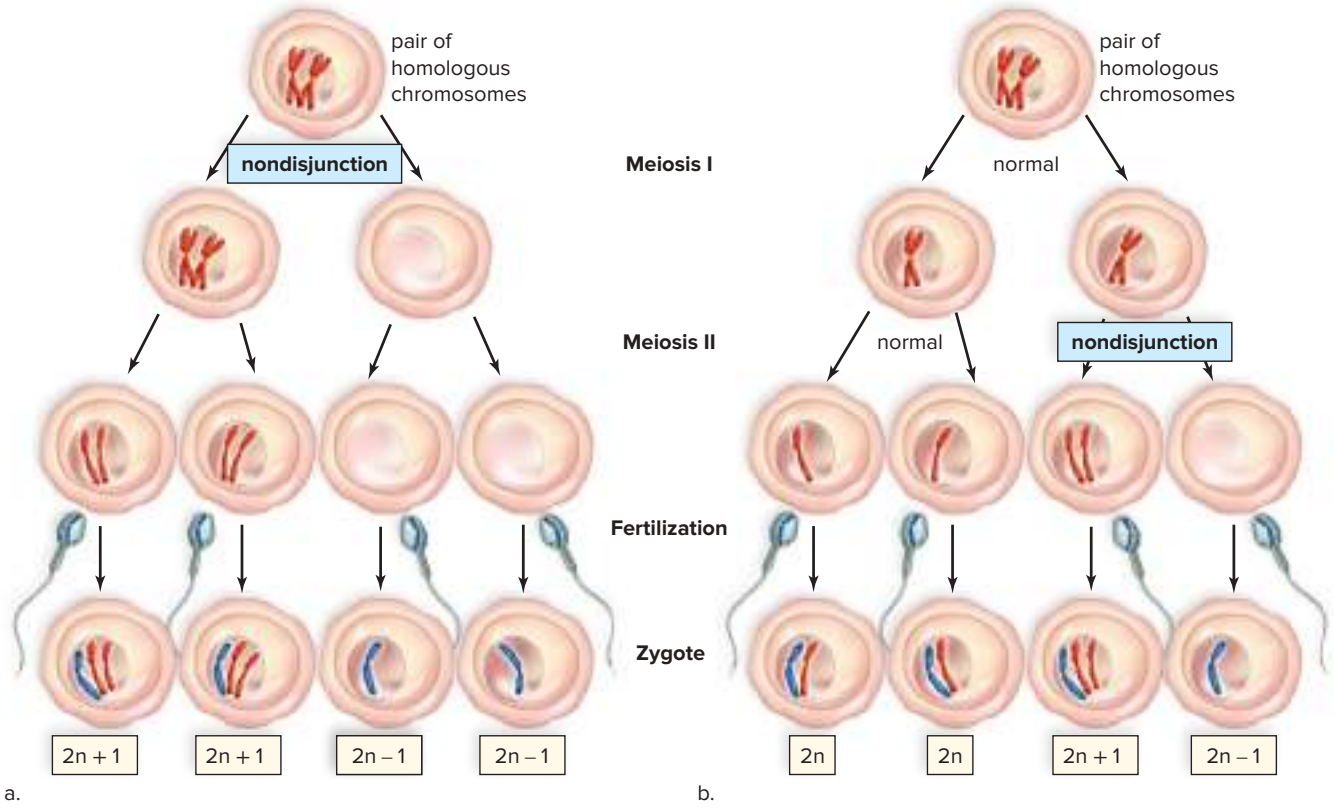


Figure 10.12 Nondisjunction of chromosomes during oogenesis. **a.** Nondisjunction can occur during meiosis I (primary nondisjunction) and results in abnormal eggs that also have one more or one less than the normal number of chromosomes. Fertilization of these abnormal eggs with normal sperm results in a zygote with abnormal chromosome numbers. 2n = diploid number of chromosomes. **b.** Nondisjunction can also occur during meiosis II (secondary nondisjunction) if the sister chromatids separate but the resulting daughter chromosomes go into the same daughter cell. Then, the egg will have one more or one less than the usual number of chromosomes. Fertilization of these abnormal eggs with normal sperm produces a zygote with abnormal chromosome numbers.

individual is more likely to survive than a monosomic one. In humans, only three autosomal trisomic conditions are known to be viable beyond birth: trisomy 13, 18, and 21. Only trisomy 21 is viable beyond early childhood and is characterized by a distinctive set of physical abnormalities and intellectual disabilities. In comparison, sex chromosome aneuploids are better tolerated in animals and have a better chance of producing survivors.

Trisomy 21

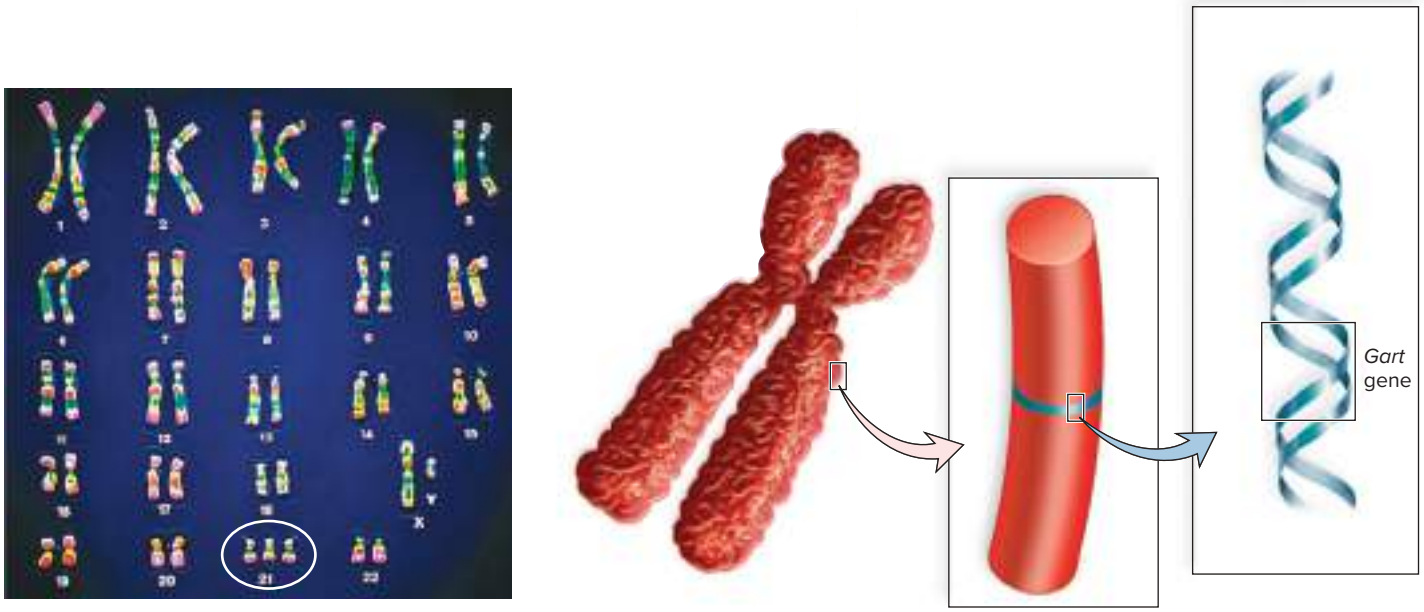
The most common autosomal trisomy seen among humans is trisomy 21, also called Down syndrome. This syndrome is associated with certain characteristics, including the following: a short stature; an eyelid fold; a flattened facial profile and round head; stubby fingers; a wide gap between the first and second toes; a large, fissured tongue; a distinctive palm crease; and an increased rate of heart problems. A degree of intellectual disability might be evident, which may be severe in some individuals. In addition, these individuals have an increased chance of developing Alzheimer's disease later in life. While at one time Down syndrome was correlated with a decrease in life expectancy, individuals with the syndrome may now be expected to live well into their 60s. Despite these potential health challenges, with the help of interventions and support systems, individuals with Down syndrome are able to achieve developmental milestones and live capable and fulfilling lives as adults.

The chances of a woman having a child with Down syndrome increase rapidly with age, starting at about age 40. The reasons for this are still being determined. Although an older woman is more likely to have a child with Down syndrome, most babies with Down syndrome are born to women younger than age 40 because this is the age group having the most babies. Furthermore, studies have shown that in about 20% of the cases studied, the sperm contributed the extra chromosome.

As is the case with all changes in chromosome number, a karyotype may be performed to identify babies with Down syndrome and other aneuploid conditions (Fig. 10.13).

The genes that cause Down syndrome are located on the long arm of chromosome 21 (Fig. 10.13), and extensive investigative work has been directed toward discovering the specific genes responsible for the characteristics of the syndrome. Thus far, investigators have discovered several genes that may account for various conditions seen in persons with Down syndrome. For example, they have located the genes most likely responsible for the increased tendency toward leukemia, cataracts, and an accelerated rate of aging. Researchers have also discovered that an extra copy of the *Gart* gene causes an increased level of purines in the blood, a finding associated with intellectual disability. Research is underway to develop methods of controlling the expression of the *Gart* gene before birth, so as to reduce the level of intellectual disability associated with Down syndrome.

ISTUDY



a. Down syndrome karyotype with an extra chromosome 21

b.

Figure 10.13 Trisomy 21. **a.** Individuals with Down syndrome, or trisomy 21, have an extra chromosome 21. **b.** The karyotype of an individual with Down syndrome shows three copies of chromosome 21. Therefore, the individual has three copies instead of two copies of each gene on chromosome 21. An extra copy of the *Gart* gene, which leads to high levels of purine in the blood, accounts for many of the characteristics of Down syndrome. (a): CNRI/Science Source

Changes in Sex Chromosome Number

An abnormal sex chromosome number is the result of inheriting too many or too few X or Y chromosomes. Nondisjunction during oogenesis or spermatogenesis can result in gametes with an abnormal number of sex chromosomes. However, extra copies of the sex chromosomes are much more easily tolerated in humans than are extra copies of autosomes.

A person with Turner syndrome (XO) is a female, and a person with Klinefelter syndrome (XXY) is a male. However, deletion of the *SRY* gene on the short arm of the Y chromosome results in Swyer syndrome, or an "XY female." Individuals with Swyer syndrome lack a hormone called testis-determining factor, which plays a critical role in the development of male genitals. Furthermore, movement of this gene onto the X chromosome may result in de la Chapelle syndrome, or an "XX male." Men with de la Chapelle syndrome exhibit undersized testes, sterility, and rudimentary breast development. Together, these observations suggest that in humans the presence of the *SRY* gene, not the number of X chromosomes, determines the presence of male sexual characteristics. In its absence, female sexual characteristics develop.

Why are newborns with an abnormal sex chromosome number more likely to survive than those with an abnormal autosome number? Because females have two X chromosomes and males have only one, we might expect females to produce twice the amount of each gene from this chromosome, but both males and females produce roughly the same amount. In reality, both males and females have only one functioning X chromosome. In females, and in males with extra X chromosomes, any additional X chromosomes become an inactive mass called a **Barr body,** named after its discoverer, Murray Barr. X chromosome inactivation provides a natural method for gene dosage compensation of the sex chromosomes (see Section 13.2) and explains why extra sex chromosomes are more easily tolerated than extra autosomes.

Turner Syndrome

From birth, an individual with Turner syndrome has only one sex chromosome, an X. The O signifies the absence of a second sex chromosome (Fig. 10.14). In adult females, Turner syndrome is

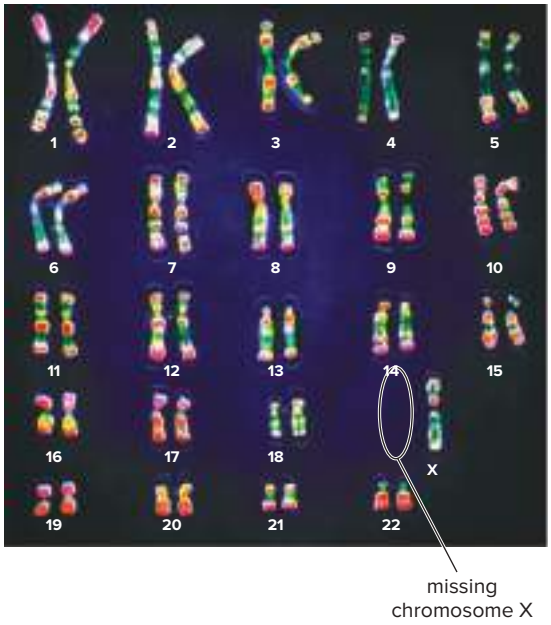


Figure 10.14 Turner syndrome. In Turner syndrome, there is only a single X chromosome. CNRI/Science Source

ISTUDY

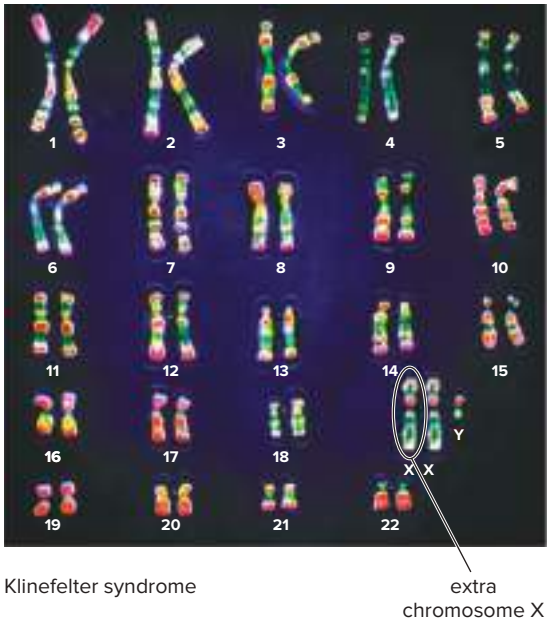


Figure 10.15 Klinefelter syndrome. In Klinefelter syndrome, there are two X and one Y chromosomes. CNRI/Science Source

associated with physical features such as a short stature, a broad chest, and folds of skin on the back of the neck. The ovaries, uterine tubes, and uterus are very small and underdeveloped. Females with Turner syndrome do not undergo puberty or menstruate, and their breasts do not develop. However, some have given birth following in vitro fertilization using donor eggs. They usually do not have an intellectual disability and can lead healthy lives if they receive hormone supplements.

Klinefelter Syndrome

A male with Klinefelter syndrome has two or more X chromosomes in addition to a Y chromosome (Fig. 10.15). The extra X chromosome(s) become(s) a Barr body. The approximate incidence for Klinefelter syndrome is 1 in every 500 to 1,000 males.

In males with Klinefelter syndrome, the testes and prostate gland are underdeveloped and facial hair is lacking. They may exhibit some breast development. Affected individuals may have larger hands and feet, as well as very long arms and legs. They are usually slow to learn but do not have an intellectual disability unless they inherit more than two X chromosomes.

While males with Klinefelter syndrome exhibit no other major health abnormalities, they have an increased risk of some disorders, including breast cancer, osteoporosis, and lupus, which disproportionately affect females. Although men with Klinefelter syndrome typically do not need medical treatment, some have found that testosterone therapy helps increase muscle strength, sex drive, and concentration ability. Testosterone treatment, however, does not reverse the sterility associated with Klinefelter syndrome due to incomplete testicle development.

Poly-X Syndrome

A female with poly-X syndrome has more than two X chromosomes and extra Barr bodies in the nucleus. Females with three X chromosomes have no distinctive phenotype, aside from a tendency to be tall and thin. Although some experience delayed motor and language development, most females with poly-X syndrome do not have intellectual disabilities. Some may have menstrual difficulties, but many menstruate regularly and are fertile. Their children usually have a normal karyotype.

Females with more than three X chromosomes occur rarely. Unlike females with XXX chromosomes, females with XXXX chromosomes are usually tall and tend to have severe intellectual disabilities. Various physical abnormalities are seen, but they may menstruate normally.

Jacobs Syndrome

The condition of Jacobs syndrome (XXY) in males can only result from nondisjunction during spermatogenesis. Males with this syndrome are usually taller than average, have persistent acne, and tend to experience speech and reading problems, but they are fertile and may have children. Despite the extra Y chromosome, there is no difference in behavioral development between males with XYY chromosomes and those with XY chromosomes.

Changes in Chromosome Structure

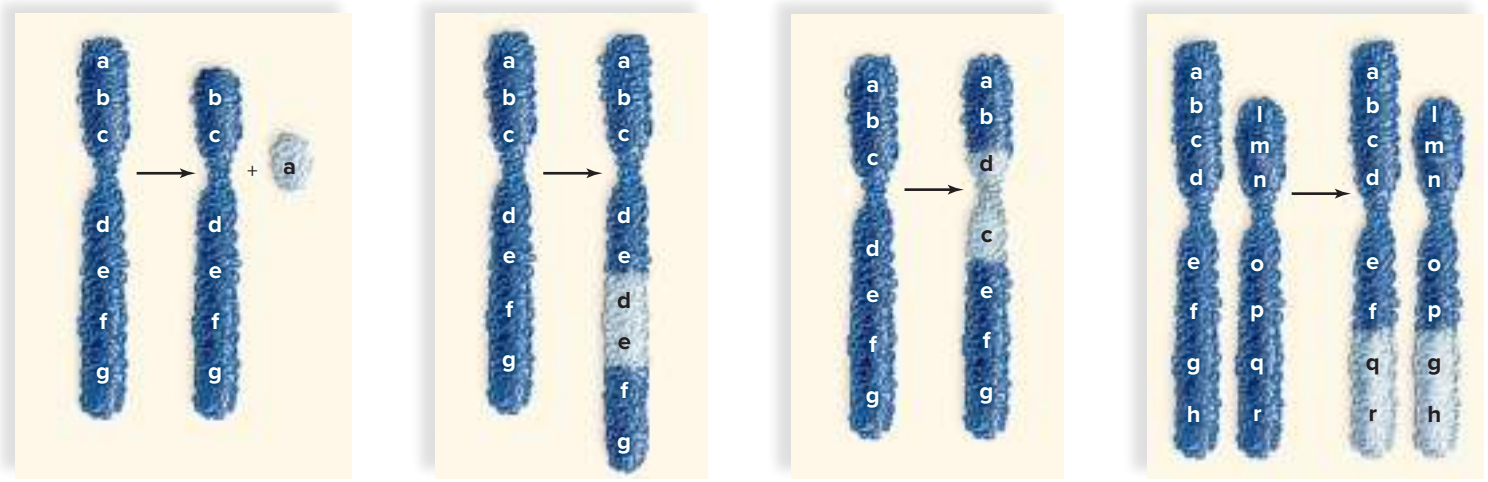
Changes in chromosome structure are another type of chromosomal mutation. Some, but not all, changes in chromosome structure can be detected microscopically. Various agents in the environment, such as radiation, certain organic chemicals, or even viruses, can cause chromosomes to break. Ordinarily, when breaks occur in chromosomes, the two broken ends reunite to give the same sequence of genes. Sometimes, however, the broken ends of one or more chromosomes do not rejoin in the same pattern as before, and the results are various types of chromosomal mutations.

Changes in chromosome structure include deletions, duplications, translocations, and inversions of chromosome segments. A **deletion** occurs when an end of a chromosome breaks off or when two simultaneous breaks lead to the loss of an internal segment (Fig. 10.16*a*). Even when only one member of a pair of chromosomes is affected, a deletion often causes abnormalities.

A **duplication** is the presence of a chromosomal segment more than once in the same chromosome (Fig. 10.16*b*). Duplications may or may not cause visible abnormalities, depending on the size of the duplicated region. An **inversion** has occurred when a segment of a chromosome has been turned around 180° (Fig. 10.16*c*). Most individuals with inversions exhibit no abnormalities, but this reversed sequence of genes can result in duplications or deletions being passed on to their children.

A **translocation** is the movement of a chromosome segment from one chromosome to another, nonhomologous chromosome. The translocation shown in Figure 10.16*d* is *balanced,* meaning there is a reciprocal swap of one piece of the chromosome for the other. Often, there are no visible effects of the swap, but if the individual has children, they receive one copy of the chromosome without the translocation from one parent, and a copy of the chromosome with the translocation from the other parent. The translocation is now said to be *unbalanced,* with extra material from one chromosome and missing material from another chromosome. Embryos with unbalanced translocations usually result in miscarriage, but those individuals who are born often have severe symptoms.

ISTUDY



a. Deletion b. Duplication c. Inversion d. Translocation

Figure 10.16 Types of chromosomal mutations. **a.** Deletion is the loss of a chromosome piece. **b.** Duplication occurs when the same piece is repeated within the chromosome. **c.** Inversion occurs when a piece of chromosome breaks loose and then rejoins in the reversed direction. **d.** Translocation is the exchange of chromosome pieces between nonhomologous pairs.

In some cases, Down syndrome may be caused by an unbalanced translocation between chromosomes 21 and 14. In other words, because a portion of chromosome 21 is now attached to a portion of chromosome 14, the individual has three copies of the genes that bring about Down syndrome when they are present in triplet copy. In these cases, Down syndrome is not caused by nondisjunction during meiosis but is passed on through regular patterns of inheritance (see Chapter 11).

Human Syndromes

Changes in chromosome structure occur in humans and lead to various syndromes, many of which are just now being investigated at the level of the gene. Sometimes changes in chromosome structure can be detected in humans by doing a karyotype. They may also be discovered by studying the inheritance pattern of a disorder in a particular family.

Deletion Syndromes

Distinct physical features associated with this syndrome include a turned-up nose, a wide mouth, a small chin, and larger ears. Children with Williams syndrome often have learning disabilities, but they have been shown to excel in other areas. Common strengths include excellent verbal and musical abilities. In this syndrome, the gene that governs the production of the protein elastin is missing. This affects the health of the cardiovascular system and causes their skin to age prematurely. Many individuals with Williams syndrome are friendly and outgoing in temperament. Other common behavioral and psychological characteristics associated with this syndrome include anxiety, phobias, hyperactivity, and difficulty with attention and concentration, perhaps because of the loss of a gene for a protein normally active in the brain.

Cri du chat ("cat's cry") syndrome is seen when chromosome 5 is missing an end piece. This syndrome is associated with characteristics that include a smaller head, facial abnormalities, and intellectual disabilities. Abnormal development of the glottis and larynx results in the most characteristic symptom—the infant's cry resembles that of a cat.

Translocation Syndromes

A person who has both of the chromosomes involved in a translocation has the normal amount of genetic material and is healthy, unless the chromosome exchange breaks an allele into two pieces. The person who inherits only one of the translocated chromosomes no doubt has only one copy of certain alleles and three copies of certain other alleles. A genetic counselor begins to suspect a translocation has occurred when spontaneous abortions are commonplace and family members have various syndromes.

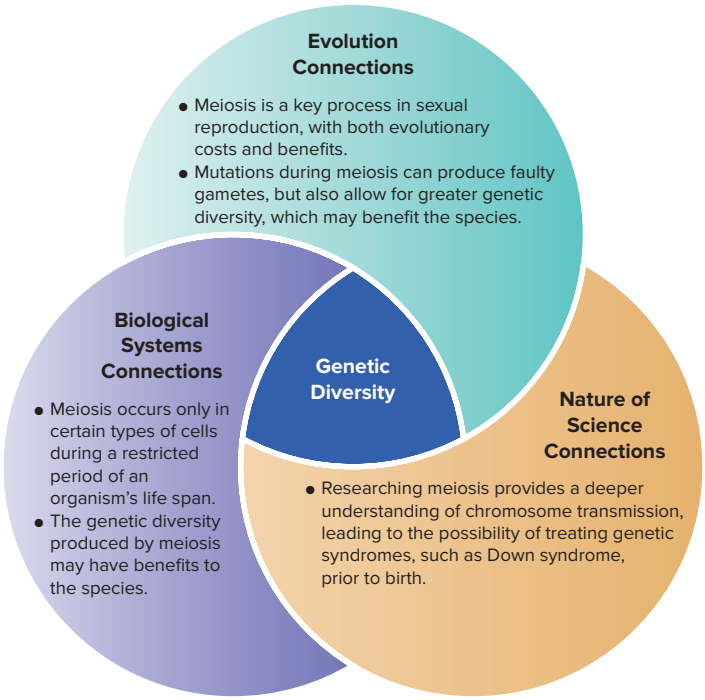
Alagille syndrome occurs due to a translocation between chromosomes 2 and 20. People with this syndrome ordinarily have a deletion on chromosome 20. One consequence of this deletion is a combination of heart defects called *tetralogy of Fallot.* One of these defects is a hole between the chambers of the heart that allows oxygenated and deoxygenated blood to mix, a condition called *cyanosis. Clubbing,* or a widening of the tips of the fingers, may also occur. The symptoms of Alagille syndrome range from mild to severe, so some people may not be aware they have the syndrome.

Check Your Progress 10.6

1. Explain the kinds of changes in chromosome number that can be caused by nondisjunction in meiosis.
2. Examine why sex chromosome aneuploidy is more common than autosome aneuploidy.
3. Compare structural changes between an inversion and a translocation.

ISTUDY

Connecting *the* Concepts



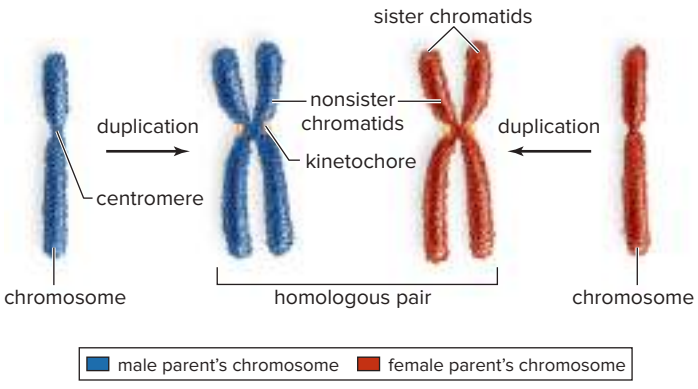
What Did You Learn?

10.1 Overview of Meiosis

Meiosis reduces the chromosome number of a cell from its **diploid** (**2n**) number to its **haploid** (**n**) number. The process ensures that the chromosome number in offspring stays constant generation after generation. In many species, including animals, meiosis is associated with the production of **gametes** for sexual reproduction. Gametes are haploid; on fertilization, they restore the diploid chromosome number in the **zygote.**

The nucleus contains pairs of chromosomes, called **homologous chromosomes** (homologues). **Homologues** contain similar genes, but these genes may have different variations, called **alleles.** Homologous chromosomes may be seen using a picture of the chromosomes, called a **karyotype.** Humans have 23 pairs of homologous chromosomes. One pair are **sex chromosomes** and the other 22 are **autosomes.**

Meiosis requires two cell divisions and results in four daughter cells. Replication of DNA takes place before meiosis begins. During meiosis I, a **synaptonemal complex** forms, allowing the homologues to undergo **synapsis,** resulting in a **bivalent.** The bivalent chromosomes



align independently at the metaphase plate. The daughter cells receive one member of each pair of homologous chromosomes. There is no replication of DNA during interkinesis, the pause between meiosis I and II. During meiosis II, the sister chromatids separate, becoming daughter chromosomes that move to opposite poles, as they do in mitosis. The four daughter cells contain the haploid number of chromosomes and only one of each kind.

10.2 Genetic Variation

Sexual reproduction ensures that the offspring have a different genetic makeup than the parents and increases the ability of a species to survive. Meiosis contributes to genetic variability in two ways: crossing-over and **independent assortment** of the homologous chromosomes. When homologous chromosomes lie side by side during synapsis, **genetic recombination** occurs, and nonsister chromatids may exchange genetic material. Due to crossing-over, the chromatids that separate during meiosis II have a different combination of genes. When the homologous chromosomes align at the metaphase plate during metaphase I, either the chromosome from the female or male parent may be facing either pole. Therefore, there will be all possible combinations of chromosomes in the gametes. Random **fertilization** of an egg by a sperm further increases the level of genetic variation.

10.3 The Phases of Meiosis

Meiosis I, which splits pairs of homologous chromosomes and reduces the chromosome number from 2n to n, is divided into four phases:

- Prophase I—bivalents form, and crossing-over occurs as chromosomes condense; the nuclear envelope fragments.
- Metaphase I—bivalents independently align at the metaphase plate.
- Anaphase I—homologous chromosomes separate, and duplicated chromosomes move to poles.
- Telophase I—nuclei become haploid, having received one duplicated chromosome from each homologous pair. **Interkinesis** follows telophase I.

ISTUDY

Meiosis II, which reduces the amount of DNA by 50% from previously replicated haploid chromosomes, is divided into four phases:

- Prophase II—chromosomes condense, and the nuclear envelope fragments.
- Metaphase II—the haploid number of still duplicated chromosomes align at the metaphase plate.
- Anaphase II—sister chromatids separate, becoming daughter chromosomes that move to the poles.
- Telophase II—four haploid daughter cells are genetically different from the parent cell.

10.4 Meiosis Compared to Mitosis

Meiosis I and mitosis can be compared in this manner:

Meiosis I	Mitosis
Prophase	
Pairing of homologous chromosomes	No pairing of chromosomes
Metaphase	
Bivalents at metaphase plate	Duplicated chromosomes at metaphase plate
Anaphase	
Homologous chromosomes separate and move to poles	Sister chromatids separate, becoming daughter chromosomes that move to the poles
Telophase	
Daughter nuclei have the haploid number of chromosomes	Daughter nuclei have the parent cell chromosome number
Meiosis II	Mitosis
Meiosis II is like mitosis, except that the nuclei are haploid.	

10.5 The Cycle of Life

Meiosis occurs in any **life cycle** that involves sexual reproduction. In the animal life cycle, only the gametes are haploid. In plants, the life cycle alternates between **gametophyte** and **sporophyte** generations. Meiosis produces spores, which develop into a multicellular haploid adult that produces the gametes. In single-celled protists and fungi, the zygote undergoes meiosis, and spores become a haploid adult that gives rise to gametes.

During the life cycle of humans and other animals, meiosis is involved in **gametogenesis,** which includes spermatogenesis and oogenesis. Whereas **spermatogenesis** produces four sperm per meiosis, **oogenesis** produces one egg and two or three nonfunctional **polar bodies.** The polar bodies are inactive cells. When a sperm fertilizes an egg, the zygote has the diploid number of chromosomes. Mitosis, which is involved in growth and repair, also occurs during the life cycle of all animals.

10.6 Changes in Chromosome Number and Structure

Euploidy represents the correct number of chromosomes in each cell of the species. **Nondisjunction** during meiosis I or meiosis II may result in **aneuploidy** (extra or missing copies of chromosomes). A **karyotype** may be used to identify if an aneuploid condition exists. **Monosomy** occurs when an individual has only one of a particular type of chromosome (2n – 1) and is usually lethal; **trisomy** occurs when an individual has three of a particular type of chromosome (2n + 1). Down syndrome is a well-known trisomy in humans, resulting from an extra copy of chromosome 21.

Aneuploidy of the sex chromosomes is tolerated more easily than aneuploidy of the autosomes. Turner syndrome, Klinefelter syndrome, poly-x syndrome, and Jacobs syndrome are examples of sex chromosome aneuploidy. **Barr bodies** help identify that extra X chromosomes exist in a cell.

Abnormalities in crossing-over may result in **deletions, duplications, inversions,** and **translocations** within chromosomes. Many human syndromes, including Williams syndrome, cri du chat syndrome, and Alagille syndrome, result from changes in chromosome structure.

Thinking Critically About Biology

1. Why is the first meiotic division considered to be the reduction division for chromosome number?
2. Although most men with Klinefelter syndrome experience infertility, some are able to father children. It was found that most individuals with Klinefelter syndrome who were fertile exhibit mosaicism, in which some cells are normal (46, XY), but others contain the extra chromosome (47, XXY). How might this mosaicism come about? What effects might result?
3. A male has a balanced translocation between chromosomes 2 and 6. If the male reproduces with a female who does not have a translocation, might the child have the same translocation? Why or why not?
4. In a karyotype, larger chromosomes are given lower numbers, so chromosome 1 is larger than chromosome 21. Trisomy conditions, such as trisomy 21, are usually viable only in the smaller chromosomes. Why do you think this is the case?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Glowing Fish Genetics

McGraw Hill

How did meiosis and genetic recombination play a role in this experiment?

Making It Relevant

1. Why can it accurately be stated that meiosis doesn't benefit an individual, only the species?
2. Why do organisms restrict meiosis to only specialized cells of the body? What would happen if any cell could conduct meiosis?
3. How would an understanding of meiosis benefit assessing potential genetic problems in future offspring?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

11

Mendelian Patterns of Inheritance

Deborah Kolb/Shutterstock

The Genetics of Eye Color

From an early age, we compare our eye color to those of our parents, friends, and people around us. We attempt to establish patterns to explain how we have inherited a certain eye color. However, as with many traits in humans, the real story is a little more complicated.

Eye color is determined by a gene called *OCA2*. Variations in this gene, and the interactions of up to 16 other genes, can influence whether your eyes are brown, green, hazel, or something in between. But what about blue eyes?

Blue eye color is a relatively recent development for humans. It is due to a mutation in a gene called *HERC2,* which controls production of eye-color pigments. In some forms of *HERC2,* a mutation causes the gene to be inactive. You need only one copy of *HERC2* for *OCA2* to produce your eye color, but if both copies of *HERC2* are defective, the result is blue eyes, regardless of which *OCA2* gene you inherited.

This mutation first appeared in northern Europe 6,000–7,000 years ago. Geneticists believe this may have occurred in a single individual, and then it spread across Europe and into other areas of Asia and North Africa. All blue-eyed people are thought to be able to trace their ancestry to this one human.

As you read through this chapter, think about the following questions:

1. At the genetic level, what is meant by the terms *dominant* and *recessive*?
2. What other patterns of inheritance help explain eye color in humans?

Chapter Outline

11.1 Gregor Mendel
11.2 Mendel's Laws
11.3 Mendelian Patterns of Inheritance and Human Disease
11.4 Beyond Mendelian Inheritance

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Chapter 3 What are the roles of proteins and nucleic acids in a cell?

Section 10.3 How do chromosomes segregate during the process of meiosis?

Figure 10.12 What genetic changes are possible in gametes when chromosomes fail to segregate properly?

Following *the* Themes

CHAPTER 11 – MENDELIAN PATTERNS OF INHERITANCE

Evolution	Inheritance of genes within a population is a cornerstone of a species' ability to change over time.
Nature of Science	Gregor Mendel's scientific approach allowed him to establish the basic principles of heredity.
Biological Systems	Inheriting mutated genes can affect many aspects of body function.

ISTUDY

11.1 Gregor Mendel

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how Mendel's scientific approach enabled his genetic experiments to be successful.
2. Contrast blending and the particulate concept of inheritance.

The science of genetics explains the stability of inheritance (why you are human, as are your parents) as well as variations between offspring from one generation to the next (why you have a different combination of traits than your parents). Virtually every culture in history has attempted to explain observed inheritance patterns. An understanding of these patterns has always been important to agriculture, animal science (the science of breeding animals), and medicine.

The Blending Concept of Inheritance

When Gregor Mendel began his work, most plant and animal breeders acknowledged that both sexes contribute equally to a new individual. They thought that parents of contrasting appearance always produced offspring of intermediate appearance. This concept, called the *blending concept of inheritance,* meant that a cross between plants with red flowers and plants with white flowers would yield only plants with pink flowers. When red and white flowers reappeared in future generations, the breeders mistakenly attributed this to instability in the genetic material.

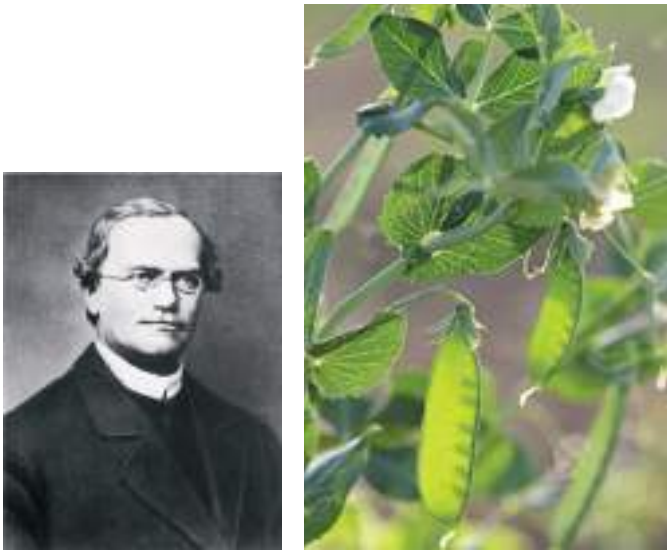
The blending concept of inheritance offered little help to Charles Darwin, the founder of modern evolutionary thought (see Section 15.2). Darwin's theory of natural selection was based on the fact that populations possessed variations that allowed for certain individuals to have a selective advantage. According to the blending concept, over time, variation would decrease as individuals became more alike in their traits.

Mendel's Particulate Theory of Inheritance

Gregor Mendel was an Austrian monk who developed a *particulate theory of inheritance* after performing a series of ingenious experiments in the 1860s (Fig. 11.1*a*). Mendel studied science and mathematics at the University of Vienna, and at the time of his research in genetics, he was a substitute natural science teacher at a local high school.

Mendel was a successful scientist for several reasons. First, he was one of the first scientists to apply mathematics to biology. Most likely, his background in mathematics prompted him to apply statistical methods and the laws of probability to his breeding experiments. He was also a careful, deliberate scientist who followed the scientific method closely and kept very detailed, accurate records. He prepared for his experiments carefully and conducted many preliminary studies with various animals and plants.

Mendel's theory of inheritance is called a particulate theory because it is based on the existence of minute particles, or hereditary units, that we now call *genes*. Inheritance involves the reshuffling of the same genes from generation to generation. The two laws he proposed, the law of segregation and the law of independent assortment,



a. b.

Figure 11.1 Gregor Mendel. **a.** Mendel was a nineteenth-century naturalist who grew and tended the pea plants (**b**) he used for his experiments. His experimental approach allowed him to develop several laws of inheritance. (a): FLHC 52/Alamy Stock Photo; (b): Radius Images/Design Pics/Alamy Stock Photo

which we will explore in Section 11.2, describe the behavior of these particulate units of heredity as they are passed from one generation to the next. While Mendel had no knowledge of DNA or genetic material in general, over time his theories have been well-supported by countless experiments of geneticists and molecular biologists.

Mendel Worked with the Garden Pea

Mendel's preliminary experiments prompted him to choose the garden pea, *Pisum sativum* (Fig. 11.1*b*), as his experimental organism. The garden pea was a good choice for many reasons. The plants were easy to cultivate and had a short generation time. Although peas normally self-pollinate (pollen only goes to the same flower), they could be cross-pollinated by hand by transferring pollen from the anther (male part of a flower) to the stigma (female part of a flower).

Many varieties of peas were available, and Mendel chose 22 of these for his experiments. When these varieties self-pollinated, over generations they became *true-breeding*—meaning that all the offspring were the same and exactly like the parent plants. Unlike his predecessors, Mendel studied the inheritance of relatively simple and discrete traits that were not subjective and were easy to observe, such as seed shape, seed color, and flower color. In his crosses, Mendel observed that the offspring did not possess intermediate characteristics. Instead, they were similar in appearance to one of the parents. As we will see, this disproved the blending concept and supported the notion of a particulate theory of inheritance.

Check Your Progress 11.1

1. Explain the difference between the particulate theory of inheritance and the blending concept.
2. Explain why the garden pea was a good choice for Mendel's experiments.

ISTUDY

11.2 Mendel's Laws

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain Mendel's law of segregation and law of independent assortment.
2. Compare and contrast dominant alleles with recessive alleles and their relation to genotype and phenotype.
3. Use a Punnett square and the law of probability to predict the chances of producing gametes and offspring.

After ensuring that his pea plants were true-breeding—for example, that his tall plants always had tall offspring and his short plants always had short offspring—Mendel was ready to perform cross-pollination experiments (Fig. 11.2) to explore patterns of inheritance. These crosses formed the basis for Mendel's laws of inheritance.

Law of Segregation

For these initial experiments, Mendel chose varieties that differed in only one trait (for example, plant height). If the blending theory of inheritance were correct, a cross of tall and short plants should yield plants with an intermediate appearance of medium height compared to the parents.

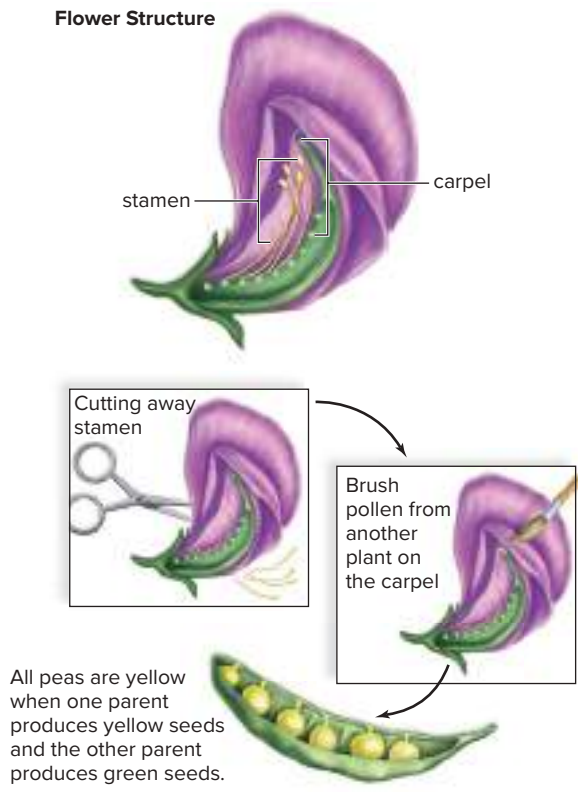


Figure 11.2 Cross-pollination in the garden pea. In the garden pea, the male structures of the plant (stamen) produce pollen (sperm), and the female structures of the plant (carpel) produce eggs. In Mendel's experiments, pollen from one plant was transferred to the female structures of another plant. The open pod shows the seed color trait that resulted from a cross between plants with yellow seeds and plants with green seeds.

Mendel's Experimental Design and Results

Mendel called the original, true-breeding all-tall or all-short parents the *P generation* (for parental). The first generation of offspring were called the F_1, or *filial generation* (L. *filius,* "sons and daughters"). In his experiments, he performed a series of reciprocal crosses, meaning that he first dusted the pollen of tall plants onto the stigmas of short plants, and then he dusted the pollen of short plants onto the stigmas of tall plants.

Figure 11.3 graphically represents Mendel's crosses. These diagrams are commonly referred to as **Punnett squares,** after the twentieth-century geneticist Reginald Punnett, who developed the diagrams to teach genetics to his students. In a Punnett square, all possible types of sperm are lined up vertically, and all possible types of eggs are lined up horizontally (or vice versa), and every possible combination of gametes occurs within the squares. Notice that, in Mendel's crosses, all the F_1 offspring resembled the tall parent.

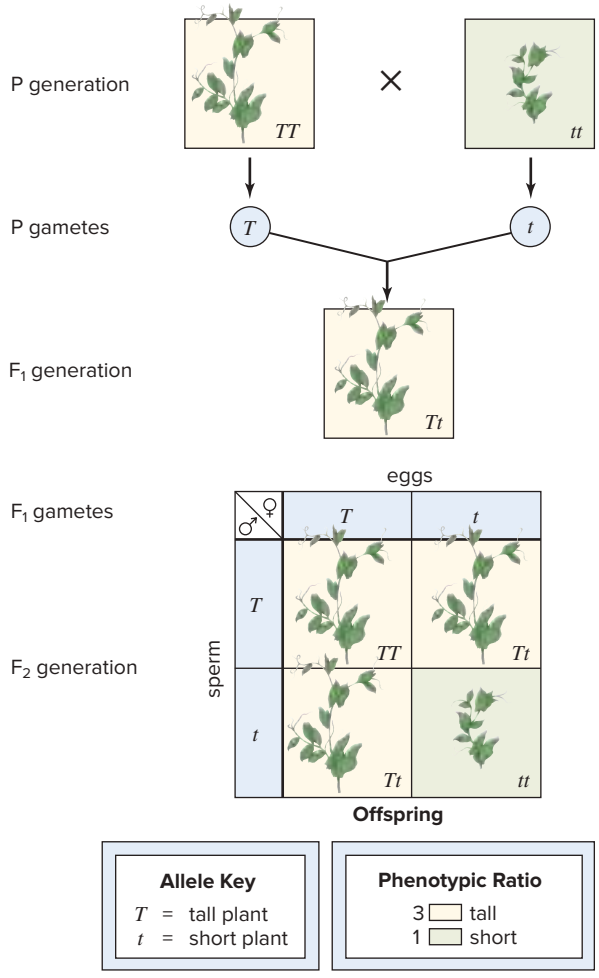


Figure 11.3 Single-trait cross done by Mendel. The P generation pea plants differ in only one trait—length of the stem. The F_1 generation plants are all tall, but the factor for short has not disappeared, because ¼ of the F_2 generation plants are short. The 3:1 ratio allowed Mendel to deduce that individuals have two discrete and separate genetic factors for each trait.

ISTUDY

As you can see, these results were contrary to those predicted by the blending theory of inheritance. Rather than being intermediate, all the F_1 plants were tall and resembled only one parent. Did these results mean that the other characteristic (shortness) had disappeared permanently? Apparently not, because when Mendel allowed the F_1 plants to self-pollinate, three-fourths of the next generation of offspring, or the *F_2 generation,* were tall and one-fourth were short, a 3:1 ratio (Fig. 11.3).

Mendel inferred that the F_1 plants were able to pass on a factor for shortness—it didn't disappear; it just skipped a generation. Because the F_1 plants were tall but clearly still contained the shortness characteristic, Mendel deduced that tallness was dominant to shortness (Fig. 11.3).

Mendel counted many plants in his plant height and other experiments. When he allowed the F_1 pea plants (which were all tall but carried the characteristic for shortness) to self-fertilize and produce offspring, he counted a total of 1,064 plants, of which 787 were tall and 277 were short. This type of experiment is called a **monohybrid cross** (L. *mono,* "single"; *hybrida,* "mixture"), because it is a cross of a single trait (plant height) with organisms that are a hybrid (tall and short characteristics). In fact, in all monohybrid crosses that he performed for the traits shown in Figure 11.4, he found approximately a 3:1 ratio in the F_2 generation. In a monohybrid cross of two heterozygotes, assuming a simple dominant/recessive relationship, the expected phenotypic ratio is 3:1.

Mendel's Conclusion

Mendel's mathematical approach led him to interpret his results differently than previous breeders. He knew that the same ratio was obtained among the F_2 generation time and time again when he did a monohybrid cross involving one of the seven traits he was studying (Fig. 11.4). In each cross, one of the characteristics disappeared in the F_1 generation, only to reappear in one-fourth of the F_2 offspring. Eventually, Mendel arrived at this explanation: A 3:1 ratio among the F_2 offspring was possible if (1) the F_1 parents contained two separate copies of each hereditary factor, one of these being dominant and the other recessive; (2) the factors separated when the gametes were formed, and each gamete carried only one copy of each factor; and (3) random fusion of all possible gametes occurred upon fertilization. Only in this way could the missing

Trait	Characteristics		F_2 Results		
	Dominant	Recessive	Dominant	Recessive	Ratio
Stem length	Tall	Short	787	277	2.84:1
Pod shape	Inflated	Constricted	882	299	2.95:1
Seed shape	Round	Wrinkled	5,474	1,850	2.96:1
Seed color	Yellow	Green	6,022	2,001	3.01:1
Flower position	Axial	Terminal	651	207	3.14:1
Flower color	Purple	White	705	224	3.15:1
Pod color	Green	Yellow	428	152	2.82:1
		Totals:	14,949	5,010	2.98:1

Figure 11.4 Relationship between observed phenotype and F_2 offspring. Mendel was fortunate in choosing the pea plant, because the traits he observed were quite distinct and easily classified. After crossing F_1 hybrids and counting hundreds of F_2 pea plants for each trait, Mendel discovered that each showed a 3:1 ratio.

ISTUDY

trait reoccur in the F_2 generation. Thinking this, Mendel arrived at the first of his laws of inheritance—the law of segregation—which is a cornerstone of his particulate theory of inheritance:

> The **law of segregation** states the following:
> - Each individual has two factors for each trait.
> - The factors segregate (separate) during the formation of the gametes.
> - Each gamete contains only one factor from each pair of factors.
> - Fertilization gives each new individual two factors for each trait.

Mendel's Cross as Viewed by Modern Genetics

We now know that the traits Mendel studied are controlled by single genes. These genes occur on a homologous pair of chromosomes at a particular location, called the gene **locus** (Fig. 11.5). Alternative versions of a gene are called **alleles** (Gk. *allelon,* "reciprocal, parallel"). A **dominant allele** will mask the expression of a **recessive allele** when they are together in the same organism.

The word *dominant* is not meant to imply that the dominant allele is better or stronger than the recessive allele. In both cases, these alleles represent DNA sequences that code for proteins. Often, the dominant allele codes for the protein associated with the normal function of the trait within the cell (such as the production of pigment), while the recessive allele represents a "loss of function," meaning that it codes for a protein that has an altered function or no function within the cell (such as a loss of pigment).

In many cases, the dominant allele is identified by a capital letter, the recessive allele by the same letter but lowercase. Usually, the first letter designating a trait is chosen to identify the allele. Using the plant height example, there is an allele for tallness (*T*) and an allele for shortness (*t*). In Mendel's cross, the original parents (P generation) were true-breeding; therefore, the tall plants had two alleles for tallness (*TT*), and the short plants had two alleles for shortness (*tt*). When an organism has two identical alleles, as these had, we say it is **homozygous** (Gk. *homo,* "same"). Because the parents were homozygous, all gametes produced by the tall plant contained the allele for tallness (*T*), and all gametes produced by the short plant contained an allele for shortness (*t*).

After cross-pollination between different pea plants, all the individuals of the resulting F_1 generation had one allele for tallness and one for shortness (*Tt*). When an organism has two different alleles at a gene locus, we say that it is **heterozygous** (Gk. *hetero,* "different"). Although the plants of the F_1 generation had one of each type of allele, they were all tall. The allele that is expressed in a heterozygous individual is the dominant allele. The allele that is not expressed in a heterozygote is the recessive allele. This explains why shortness, the recessive trait, skipped a generation in Mendel's experiment.

In Mendel's cross (see Fig. 11.3), the F_1 plants produced gametes in which 50% had the dominant allele *T* and 50% had the recessive allele *t.* During the process of fertilization, we assume that all types of sperm (*T* or *t*) have an equal chance to fertilize all types of eggs (*T* or *t*). When this occurs, such a monohybrid cross always produces a 3:1 (dominant-to-recessive) ratio among the offspring. Figure 11.4 gives Mendel's results for several monohybrid crosses, and you can see that the results were always close to 3:1.

As we will see, Mendel's work is compatible with what we know about meiosis (see Section 10.1). Recall that meiosis is the type of cell division that reduces the chromosome number from diploid (2n) to haploid (n). During meiosis I, the members of bivalents (homologous chromosomes, each having sister chromatids) separate. This means that the two alleles for each gene separate from each other during meiosis. Therefore, the process of meiosis gives an explanation for Mendel's law of segregation, as well as why only one allele for each trait is in a gamete.

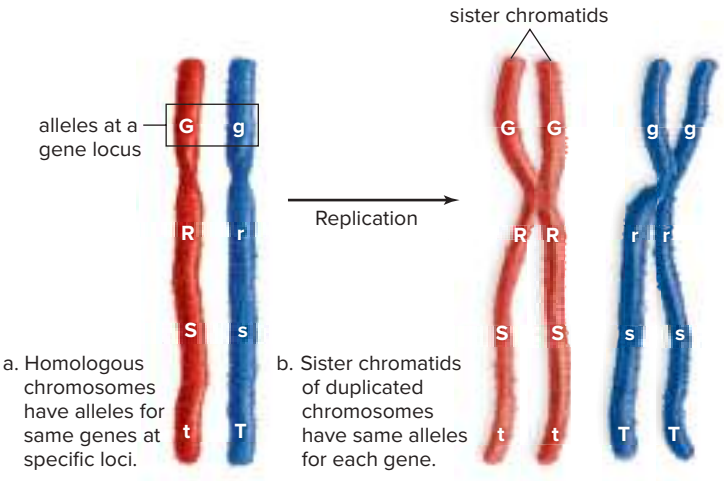


Figure 11.5 Homologous chromosomes. **a.** The letters represent alleles—that is, alternative forms of a gene. Each allelic pair, such as *Gg* or *Tt,* is located on homologous chromosomes at a particular physical location called a gene locus. **b.** Sister chromatids carry the same alleles in the same order. Proteins made from each allele determine the observable traits.

Genotype Versus Phenotype

It is obvious from our discussion that two organisms with different allelic combinations for a trait can have the same outward appearance. For example, pea plants with both the *TT* and *Tt* combinations of alleles are tall. For this reason, it is necessary to distinguish between the alleles present in an organism and the appearance of that organism.

The word **genotype** (Gk. *genos,* "birth, origin") refers to the alleles an individual receives at fertilization. Genotype may be indicated by letters (*TT*) or by short combinations of letters and numbers (*FOXP3*) that represent the DNA sequence for a particular gene. Genotype *TT* is called homozygous dominant, and genotype *tt* is called homozygous recessive. Genotype *Tt* is called heterozygous. These refer to the different ways that alleles can be combined in a cell.

The word **phenotype** (Gk. *phaino,* "appear") refers to the physical appearance of an individual, which is determined by the proteins produced by the corresponding alleles. A homozygous dominant (*TT*) individual and a heterozygous (*Tt*) individual both show the dominant phenotype and are tall, because they make fully

ISTUDY

functional proteins that build the tall trait, while a homozygous recessive individual that shows the recessive phenotype and makes less or nonfunctional protein for that trait is short. Thus, the DNA that makes up the genotype produces the proteins that make up the phenotype.

Mendel's Law of Independent Assortment

Mendel performed a second series of crosses in which true-breeding pea plants differed in two traits. For example, he crossed tall plants having green pods with short plants having yellow pods (Fig. 11.6). The F_1 plants showed both dominant characteristics. As before, Mendel then allowed the F_1 plants to self-pollinate. This F_1 cross is known as a **dihybrid cross** (L. *di,* "two"), because the plants are hybrid in two ways. Mendel recognized that there were two possible results in the F_2 generation of this cross:

- If the dominant factors (*TG*) always segregated into the F_1 gametes together, and the recessive factors (*tg*) always stayed together, then there would be two phenotypes among the F_2 plants—tall plants with green pods and short plants with yellow pods.
- If the four factors segregated into the F_1 gametes independently, then there would be four phenotypes among the F_2 plants—tall plants with green pods, tall plants with yellow pods, short plants with green pods, and short plants with yellow pods.

Figure 11.6 shows that Mendel observed four phenotypes among the F_2 plants, supporting the hypothesis that the alleles segregated independently. This is how Mendel formulated his second law of heredity—the law of independent assortment:

> The **law of independent assortment** states the following:
> - Each pair of factors segregates (assorts) independently of the other pairs.
> - All possible combinations of factors can occur in the gametes.

We know that the process of meiosis explains why the F_1 plants produced every possible type of gamete and, therefore, four phenotypes appeared among the F_2 generation of Mendel's plants. Figure 11.7 shows a parent cell with two homologous pairs of chromosomes, with alleles *Aa* on one pair and *Bb* on the other pair. Following duplication of the chromosomes during interphase, the parent cell undergoes meiosis I. At metaphase I, the homologous pairs line up independently of one another, such that the chromosomes with *A* alleles have an equal chance of lining up with the *B* alleles or the *b* alleles. The subsequent segregation of the homologous pairs during anaphase I reduces the chromosome number from 2n to n. Because *A* alleles can be sorted with *B* or *b,* and so can the *a* allele, it is possible to create gametes with *AB, Ab, aB,* and *ab* allele combinations with equal probability.

The same rule of independent assortment applies for the pea plant example in Figure 11.6. In that case, the possible gametes are the two dominants (such as *TG*), the two recessives (such as *tg*), and the ones that have a dominant and a recessive (such as *Tg* and *tG*). Regardless of whether we are using the A and B chromosome or the T and G chromosome examples, when all possible sperm have an opportunity to fertilize all possible eggs, *the expected phenotypic ratio of a dihybrid cross is always 9:3:3:1.*

It is important to note that the law of independent assortment, and thus the 9:3:3:1 ratio of the dihybrid cross, applies only to alleles of traits on different chromosomes.

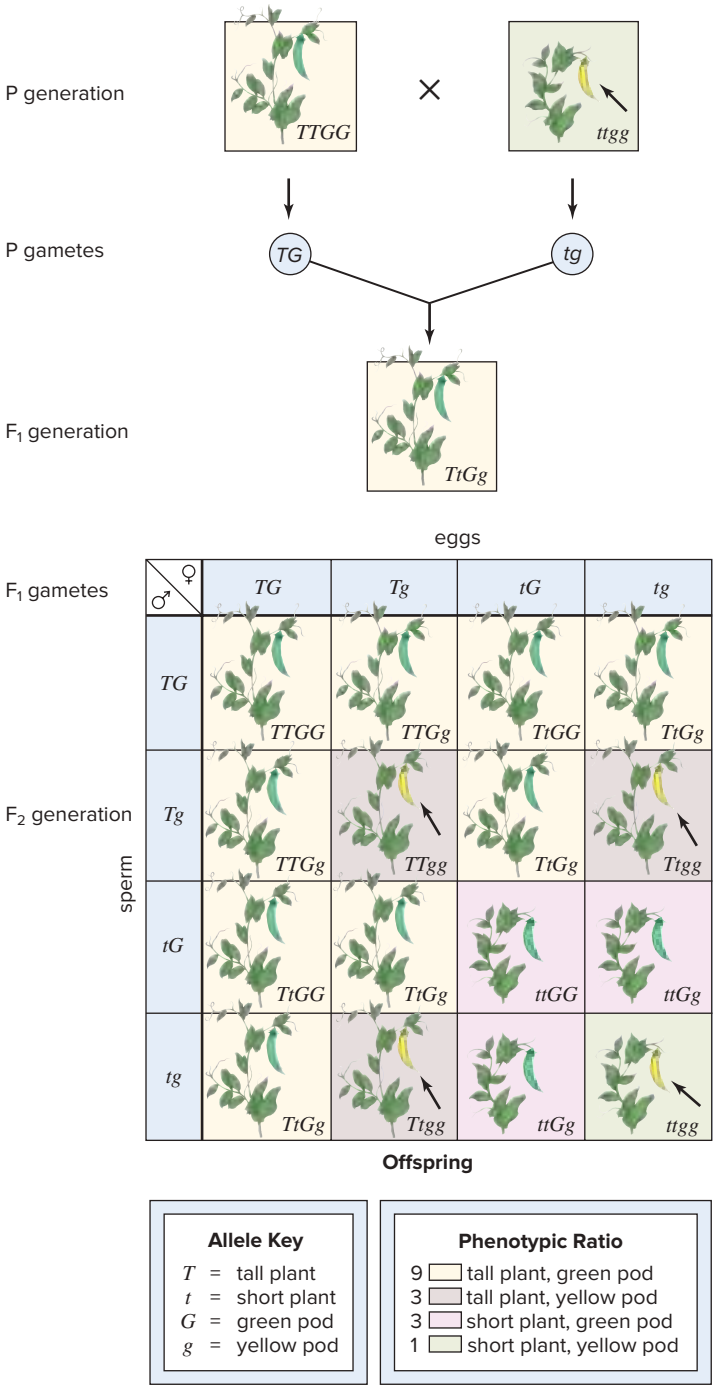


Figure 11.6 Dihybrid cross done by Mendel. P generation plants differ in two traits: length of the stem and color of the pod. The F_1 generation shows only the dominant traits, but all possible phenotypes appear among the F_2 generation, because the F_1 parents are hybrids. The 9:3:3:1 ratio allowed Mendel to deduce that factors segregate into gametes independently of other factors.

ISTUDY

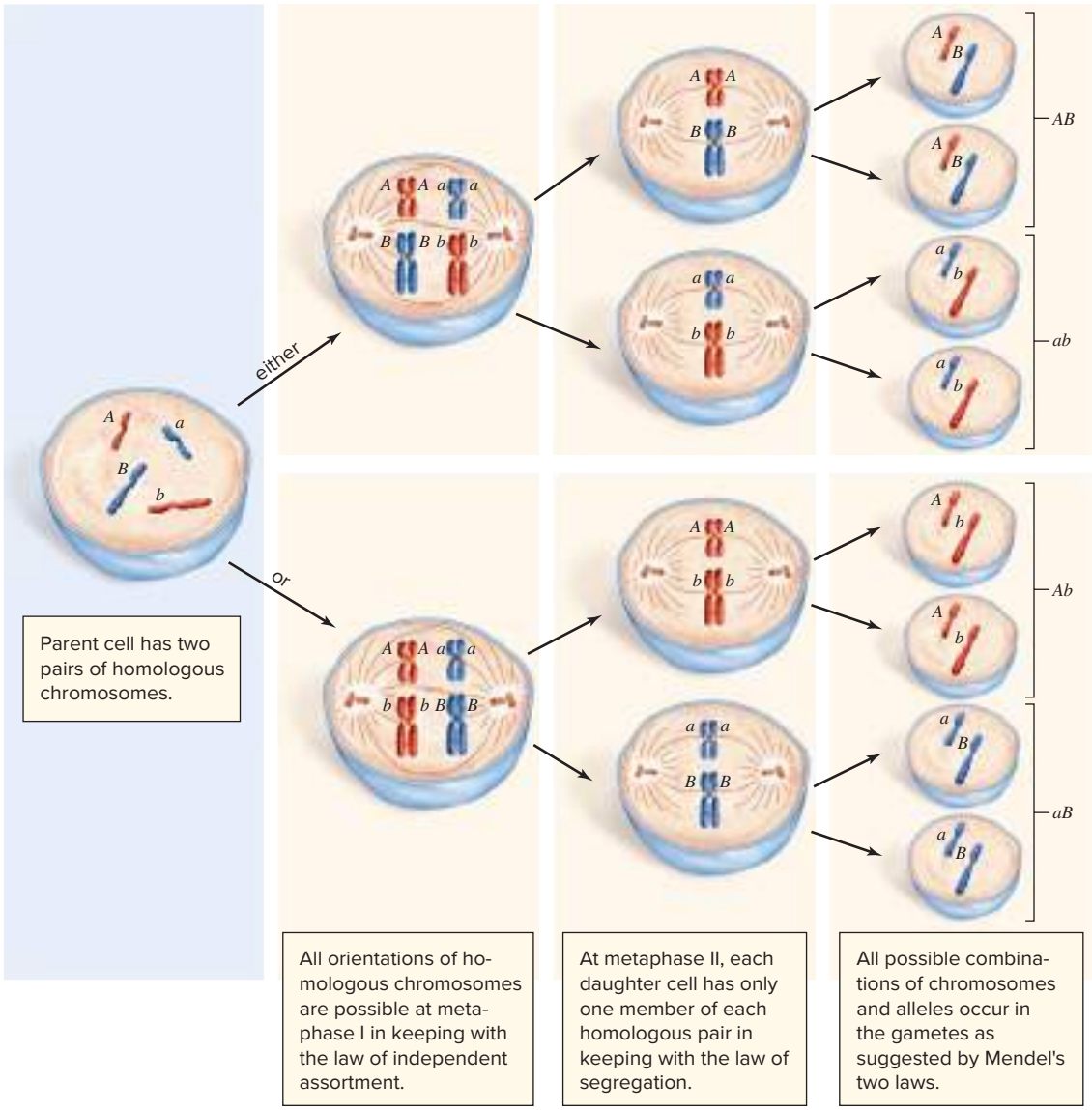


Figure 11.7 Independent assortment and segregation during meiosis. Mendel's laws hold because of the events of meiosis. The homologous pairs of chromosomes line up randomly at the metaphase plate during meiosis I. It doesn't matter which member of a homologous pair faces which spindle pole. In this example, *A* alleles can segregate with *B* or *b* alleles. Likewise, *a* alleles can segregate with *B* or *b* alleles. Therefore, the homologous chromosomes, and alleles they carry, segregate independently during gamete formation. All possible combinations of chromosomes and alleles—that is, *AB, Ab, aB,* and *ab*—occur in the gametes.

Mendel and the Laws of Probability

The Punnett square allows us to easily visualize the possible combinations of gametes in one- and two-trait crosses. This gives us the ability to easily calculate the chances, or the probability, of genotypes and phenotypes among the offspring.

There are not many examples of simple dominant/recessive traits in humans. Scientists have determined that many examples, such as widow's peak and earlobes, are actually under the influence of multiple genes (see Section 11.4). However, one human trait that we can use to demonstrate probability is albinism. Albinism is caused by a recessive mutation that results in a lack of production of melanin, which contributes to pigmentation of skin and hair. Like flipping a coin, an offspring of the cross illustrated in the Punnett square in Figure 11.8 has a 50% (or ½) chance of receiving an *A* for normal pigmentation or an *a* for no pigmentation (albinism) from each parent.

How likely is it that an offspring will inherit a specific set of two alleles, one from each parent? Notice that the chances of each parent contributing an *A* allele is ½ and the chances of contributing an *a* allele is ½. The *product rule* of probability tells us that we have to multiply the chances of independent events to get the answer:

- The chance of *AA* = ½ × ½ = ¼
- The chance of *Aa* = ½ × ½ = ¼
- The chance of *aA* = ½ × ½ = ¼
- The chance of *aa* = ½ × ½ = ¼

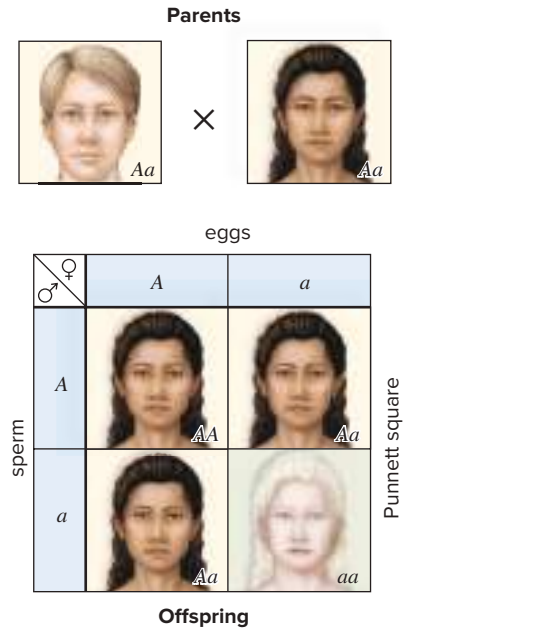


Figure 11.8 Punnett square. A Punnett square can be used to calculate probable results—in this case, a 3:1 phenotypic ratio.

The Punnett square helps us visualize these outcomes, because we can easily see that each of the combinations is ¼ of the total number of squares.

How do we get the phenotypic results? The *sum rule* of probability tells us that when the same event can occur in more than one way, we can add the results. Because 1, 2, and 3 all result in normal pigmentation, we add them up to know that the chance of normal pigmentation is ¾, or 75%. The chance of albinism is ¼, or 25%. The Punnett square doesn't do this for us—we have to add the results ourselves.

The statement "Chance has no memory" is important when considering inheritance across offspring. Every time a couple produces an offspring, the child has the same chances of inheriting the different allele combinations. Thus, for a heterozygous (*Aa*) couple, each child has a 25% chance of having albinism (*aa*). Inheriting a recessive trait becomes especially important when we consider a recessive genetic disorder, such as cystic fibrosis, a debilitating respiratory illness. For a heterozygous couple, there is a 25% chance that their offspring will inherit two recessive alleles and exhibit the disease. And because each offspring is an independent event, it is possible that all their offspring—or none of them—will exhibit cystic fibrosis.

We can use the product rule and the sum rule of probability to predict the results of a dihybrid cross, such as the one shown in Figure 11.6. The Punnett square carries out the multiplication, and we add the results to find that the phenotypic ratio is 9:3:3:1. We expect the same results for every dihybrid cross. Therefore, it is not necessary to do a Punnett square over and over again for either a monohybrid or a dihybrid cross. Instead, we can simply remember the probable results of 3:1 and 9:3:3:1. But we have to remember that the 9 represents the two dominant phenotypes together, the 3s are a dominant phenotype with a hidden recessive, and the 1 stands for the double recessive phenotype.

This tells us the probable phenotypic ratio among the offspring, but not the chances for each possible phenotype. Because the dihybrid Punnett square has 16 squares, the chances are $^{9}/_{16}$ for the two dominants together, $^{3}/_{16}$ for the dominants with each recessive, and $^{1}/_{16}$ for the two recessives together.

Mendel counted the results of many similar crosses to get the probable results, and in the laboratory we, too, have to count the results of many individual crosses to get the probable results for a monohybrid or a dihybrid cross. Why? Consider that each time you toss a coin, you have a 50% chance of getting heads or tails. If you toss the coin only a couple of times, you might very well have heads or tails both times. However, if you toss the coin many times, your results are more likely to approach 50% heads and 50% tails.

Testcrosses

To confirm that the F_1 plants of Mendel's one-trait crosses were, in fact, heterozygous, he crossed his F_1 generation tall pea plants with true-breeding short (homozygous recessive) plants; such a mating is termed a **testcross.** These crosses provided Mendel with further support for his law of segregation.

For the cross in Figure 11.9, Mendel reasoned that half the offspring should be tall and half should be short, producing a 1:1 phenotypic ratio. His results supported the hypothesis that alleles segregate when gametes are formed. In Figure 11.9*a,* the homozygous recessive parent can produce only one type of gamete—*t*—and so the Punnett square has only one column. The use of one column signifies that all the gametes carry a *t. The expected phenotypic ratio for this type of one-trait cross (heterozygous × recessive) is always 1:1.*

One-Trait Testcross

Today, a one-trait testcross is used to determine if an individual with the dominant phenotype is homozygous dominant (e.g., *TT*) or heterozygous (e.g., *Tt*). Because both of these genotypes produce the dominant phenotype, it is not possible to determine the genotype by observation. Figure 11.9*b* shows that if the individual is homozygous dominant, all the offspring will be tall. Each parent has only one type of gamete and, therefore, a Punnett square is not required to determine the results.

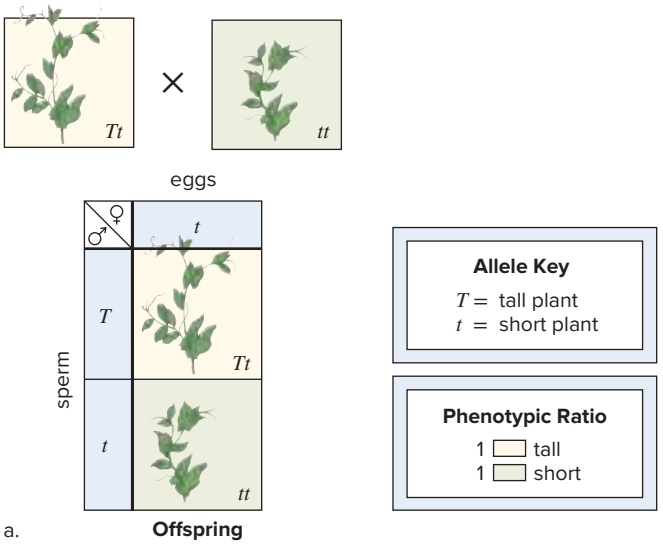
Two-Trait Testcross

When doing a two-trait testcross, an individual with the dominant phenotype is crossed with one having the recessive phenotype. Suppose you are working with fruit flies with the following characteristics:

L = long wings *l* = vestigial (short) wings

G = gray bodies *g* = black bodies

ISTUDY



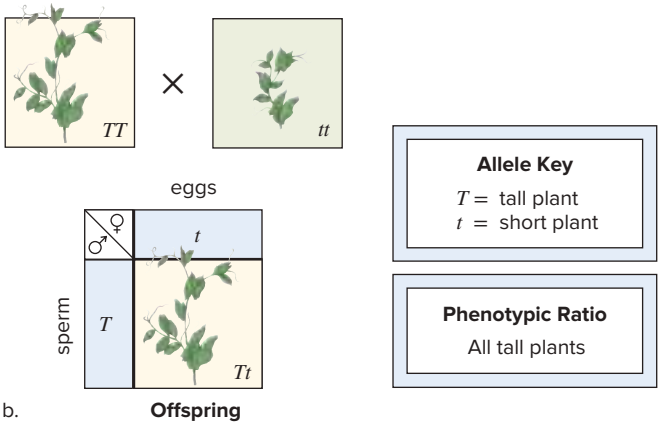
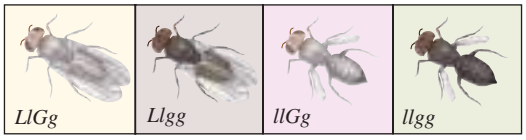


Figure 11.9 One-trait testcrosses. **a.** One-trait testcross when the individual with the dominant phenotype is heterozygous. **b.** One-trait testcross when the individual with the dominant phenotype is homozygous.

You wouldn't know by examination whether the fly on the left was homozygous or heterozygous for wing and body color. To find out the genotype of the test fly, you cross it with the one on the right. You know by examination that this vestigial-winged and black-bodied fly is homozygous recessive for both traits.

If the test fly is homozygous dominant for both traits with the genotype *LLGG,* it will form only one gamete: *LG.* Therefore, all the offspring from the proposed cross will have long wings and a gray body. However, if the test fly is heterozygous for both traits with the genotype *LlGg,* it will form four different types of gametes: *LG, Lg, lG,* and *lg.* Therefore, it can produce four different offspring:



The presence of the offspring with vestigial wings and a black body shows that the test fly is heterozygous for both traits and has the genotype *LlGg*. Otherwise, it could not produce this offspring. In general, the expected phenotypic ratio for this type of two-trait cross (heterozygous for both traits × recessive for both traits) is always 1:1:1:1.

Check Your Progress 11.2

1. Summarize how Mendel's law of independent assortment relates to the process of meiosis.
2. Explain why the *Tt* and *TT* genotypes both have the same phenotype.
3. Calculate the probability of producing an *Aabb* individual from an *AaBb* × *AaBb* cross.

11.3 Mendelian Patterns of Inheritance and Human Disease

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between an autosomal dominant and an autosomal recessive pattern of inheritance.
2. Identify the pattern of inheritance for selected single-gene human disorders.

Many traits and disorders in humans, as well as other organisms, are genetic in origin and follow Mendel's laws. These traits are often controlled by a single pair of alleles on the autosomal chromosomes. An **autosome** is any chromosome other than a sex (X or Y) chromosome. In Section 11.4, we will explore patterns of inheritance associated with the sex chromosomes.

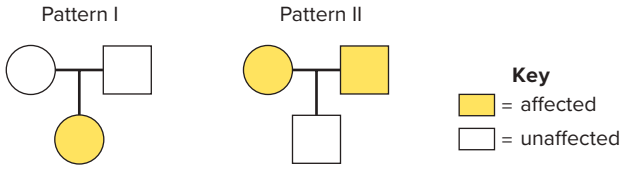
Autosomal Patterns of Inheritance

When a genetic disorder is autosomal dominant, the normal allele (*a*) is recessive, and an individual with the alleles *AA* or *Aa* has the disorder. When a genetic disorder is autosomal recessive, the normal allele (*A*) is dominant, and only individuals with the alleles *aa* have the disorder. A **pedigree** shows the pattern of inheritance for a particular condition; genetic counselors can use a pedigree to determine whether a condition is dominant or recessive.

In a human pedigree, males are designated by squares and females by circles. Shaded circles and squares are the affected individuals. The shaded shapes do not indicate whether the condition is dominant or recessive, only that the individual exhibits the trait. A line between a square and a circle represents a union. In the patterns below, a vertical line leads to a single child. If there are more children, they are lined up horizontally. In pattern I, the child is affected, but neither parent is; this can happen if the condition is recessive and both parents are *Aa.* Notice that the parents are **carriers,** because they do not express the trait but are capable of having offspring with the genetic disorder. In pattern II, the child

ISTUDY

is unaffected, but the parents are affected. This can happen if the condition is dominant and the parents are *Aa*.



Pedigrees may be used to understand how a trait is being inherited across multiple generations to understand its pattern of inheritance. As we will see next, in a pedigree, generations are indicated by Roman numerals on the left side (see Fig. 11.10). It is also common to identify individuals by number (from left to right) in the pedigree. For example, the first individual on the top generation of a pedigree would be given the identification I-1, the second in the same generation I-2, and so forth.

Next, we will take a closer look at autosomal patterns of inheritance for selected human diseases.

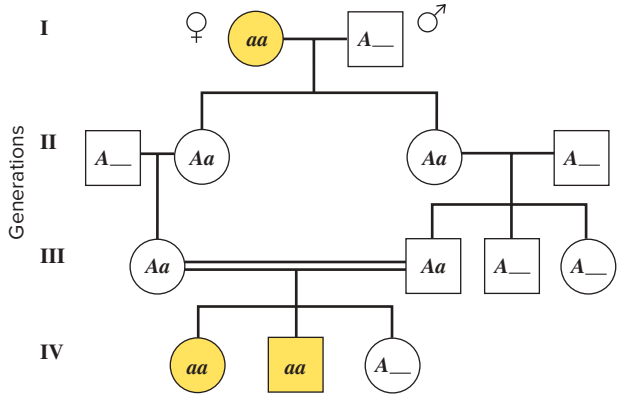
Autosomal Recessive Disorders

Autosomal recessive disorders are typically the result of a loss-of-function mutation in a gene, meaning the gene has lost some (or all) of its ability to produce a gene product, such as a protein. The result is a recessive allele. Since production of the protein from the dominant allele may be enough for the operation of the cell, only individuals who are homozygous recessive for this allele show the phenotype of the disease.

Figure 11.10 provides an example of the inheritance of an autosomal recessive trait in a family. Notice in the third generation that two closely related individuals (III-1 and III-2) have produced three children, two of which have the affected phenotype. In this case, a double line denotes consanguineous reproduction, or inbreeding, which is reproduction between two closely related individuals. This illustrates that inbreeding significantly increases the chances of children inheriting two copies of a potentially harmful recessive allele.

Also notice that some individuals (such as II-1 and II-4) are indicated as *A*__. This means they display the dominant phenotype, but it is not known whether the individual is homozygous dominant (*AA*) or heterozygous (*Aa*).

In humans, a number of autosomal recessive disorders have been identified. In this section, we discuss methemoglobinemia, cystic fibrosis, and phenylketonuria.



Autosomal recessive disorders

- Most children who are affected have parents who are not affected.
- Heterozygotes *(Aa)* have an unaffected phenotype.
- Two parents who are affected will always have children who are affected.
- Close relatives who reproduce are more likely to have children who are affected.

Figure 11.10 Autosomal recessive pedigree. This pedigree and list provide ways to recognize an autosomal recessive disorder. Individuals indicated by an *A*__ have the dominant phenotype (not affected), but may be either *AA* or *Aa*.

Methemoglobinemia

Methemoglobinemia is a relatively harmless disorder that results from an accumulation of methemoglobin in the blood. Hemoglobin, the main oxygen-carrying protein in the blood, is usually converted at a slow rate to an alternate form called methemoglobin. Unlike hemoglobin, which is bright red when carrying oxygen, methemoglobin has a bluish color, similar to that of oxygen-poor blood. Although this process is harmless, individuals with methemoglobinemia are unable to clear the abnormal blue protein from their blood, causing their skin to appear bluish-purple (Fig. 11.11).

Methemoglobinemia was documented for centuries, but its exact cause and genetic link remained mysterious until a persistent and determined physician solved the age-old mystery by doing blood tests and pedigree analysis involving a family known as the "blue Fugates" of Troublesome Creek, Kentucky. Enzyme tests indicated that the blue Fugates lacked the enzyme diaphorase, coded for by a gene on chromosome 22. The enzyme normally converts methemoglobin back to hemoglobin.

Figure 11.11 Methemoglobinemia. The hand on the right appears blue due to chemically induced methemoglobinemia. Courtesy Division of Medical Toxicology, University of Virginia

ISTUDY

The physician treated the disorder in a simple but rather unconventional manner. He injected the Fugates with a dye called methylene blue. This unusual dye can donate electrons to other compounds, successfully converting the excess methemoglobin back into normal hemoglobin. The results were striking but immediate—the patients' skin quickly turned pink after treatment. A pedigree analysis of the Fugates indicated the trait was common in the family because so many members carried the recessive allele.

Cystic Fibrosis

Cystic fibrosis (CF) is the most common lethal genetic disease common among white populations. In the United States, the incidence of CT among whites is about 1 in 3,000 births, while the incidence among Black Americans is approximately 1 in 17,000 births, and among Asian Americans it is 1 in 31,000 newborns. Individuals with CF exhibit a number of characteristic symptoms, the most obvious being extremely salty sweat. In children with CF, the mucus in the bronchial tubes and pancreatic ducts is particularly thick and viscous, interfering with the function of the lungs and pancreas. To ease breathing, the thick mucus in the lungs has to be loosened periodically, but still the lungs frequently become infected. The clogged pancreatic ducts prevent digestive enzymes from reaching the small intestine, and to improve digestion, people with CF take digestive enzymes mixed with applesauce before every meal.

Cystic fibrosis is caused by a defective chloride ion channel that is encoded by the *CFTR* allele on chromosome 7. Research has demonstrated that chloride ions (Cl^-) fail to pass through the defective version of the *CFTR* chloride ion channel, which is located on the plasma membrane (Fig. 11.12). Ordinarily, after chloride ions have passed through the channel to the other side of the membrane, sodium ions (Na^+) and water follow. It is believed that lack of water is the cause of the abnormally thick mucus in the bronchial tubes and pancreatic ducts.

In the past few years, a better understanding of the genetic basis of CF, coupled with new treatments, has raised the average life expectancy for individuals with CF to around 50 years, depending on the severity of the disease. Advances in gene therapy, or the replacement of the faulty allele with a good copy (see Section 14.3), are showing considerable promise as methods of treating CF.

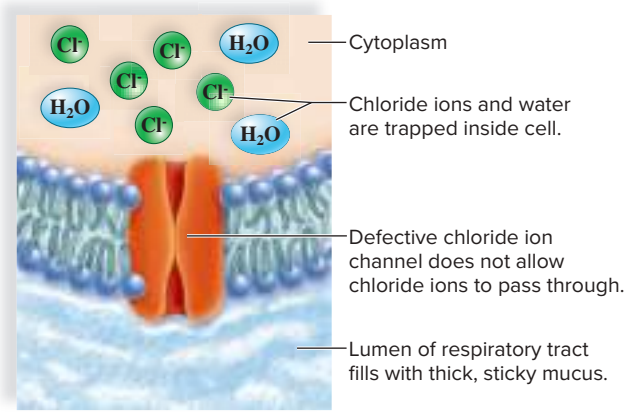


Figure 11.12 Cystic fibrosis. Cystic fibrosis is due to a faulty protein that is supposed to regulate the flow of chloride ions into and out of cells through a channel protein.

Interestingly, the mutated *CFTR* allele is believed to have persisted in the human population as a means of surviving potentially fatal diseases. Individuals who are heterozygous have an increased level of protection against diseases such as cholera. This is called a heterozygote advantage, which we will explore further in Section 16.3.

Phenylketonuria

Phenylketonuria (PKU) is an autosomal recessive metabolic disorder that affects nervous system development. Affected individuals lack the enzyme needed for normal metabolism of the amino acid phenylalanine; therefore, it appears in the urine and the blood. Newborns are routinely tested in the hospital for elevated levels of phenylalanine in the blood. If an elevated level is detected, the newborn will develop normally if placed on a diet low in phenylalanine, which must be continued until the brain is fully developed, around the age of 7, or severe intellectual disabilities will develop. Some doctors recommend that the diet continue for life, but in any case, a female who is pregnant and is affected by phenylketonuria must be on the diet to protect their unborn child.

Autosomal Dominant Disorders

Autosomal dominant disorders (Fig. 11.13) are typically the result of a gain-of-function mutation in a gene, resulting in a dominant allele. Therefore, only a single copy of this allele is needed to show the phenotype.

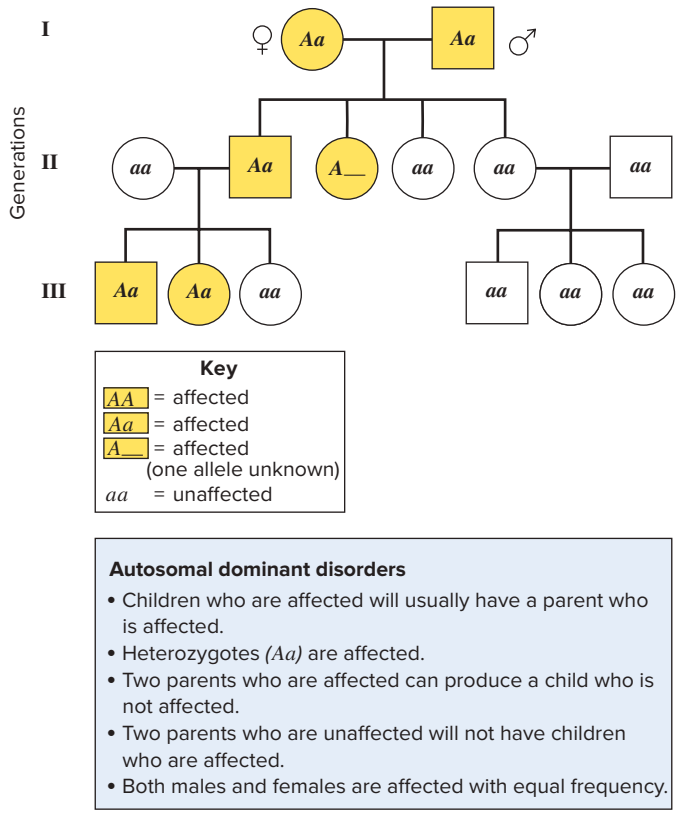


Figure 11.13 Autosomal dominant pedigree. This pedigree and list provide ways to recognize an autosomal dominant disorder. Individuals indicated by an *A*__ have the dominant phenotype (affected), but may be either *AA* or *Aa*.

ISTUDY

A number of autosomal dominant disorders have been identified in humans. Three relatively well-known autosomal dominant disorders are osteogenesis imperfecta, Huntington's disease, and hereditary spherocytosis.

Osteogenesis Imperfecta

Osteogenesis (L. *os,* "bone"; *genesis,* "origin") imperfecta is an autosomal dominant genetic disorder that results in weakened, brittle bones. Although at least nine types of the disorder are known, most are linked to mutations in two genes necessary for the synthesis of type I collagen, one of the most abundant proteins in the human body. Collagen has many roles, including providing strength and rigidity to bone and forming the framework for most of the body's tissues. Osteogenesis imperfecta leads to a defective collagen I that causes the bones to be brittle and weak. Because the mutant collagen can cause structural defects even when combined with normal collagen I, osteogenesis imperfecta is generally considered to be dominant.

Osteogenesis imperfecta, which has an incidence of approximately 1 in 5,000 live births, affects all racial groups similarly and has been documented as far back as 300 years ago. Some historians think the Viking chieftain Ivar Ragnarsson, who was known as Ivar the Boneless and was often carried into battle on a shield, had this condition. In most cases, the diagnosis is made in young children who visit the emergency room frequently due to broken bones. Some children with the disorder have an unusual blue tint in the sclera, the white portion of the eye; reduced skin elasticity; weakened teeth; and occasionally heart valve abnormalities. Currently, the disorder is treatable with a number of drugs that help increase bone mass, but these drugs must be taken long term.

Huntington's Disease

Huntington's disease is a neurological disorder that leads to progressive degeneration of brain cells. Huntington's disease is caused by a mutated copy of the gene for a protein called huntingtin. The defective gene, located on chromosome 4, contains segments of DNA in which the base sequence CAG repeats again and again. This type of structure, called a trinucleotide repeat, causes the huntingtin protein to have too many copies of the amino acid glutamine. The normal version of the huntingtin protein has stretches of between 10 and 25 glutamic acids. If the huntingtin protein has more than 36 glutamic acids, as is coded for in the mutated Huntington's gene, it changes shape and forms clumps inside neurons. These clumps attract other proteins to clump, rendering them inactive. One of these proteins that attaches itself to the clumps, called CBP, helps nerve cells survive.

The onset of symptoms for Huntington's disease is normally not seen until later in life (average age of onset is late thirties to late forties), although it is not unusual for a person with Huntington's disease to develop symptoms as early as their late teens. Huntington's disease has a range of symptoms but is normally characterized by uncontrolled movements, unsteady gait, dementia, and speech impairment. On average, patients live 15 to 20 years after the onset of symptoms, because the disease progresses rapidly. Current effective treatments include medications that slow the progression of the disease. Additionally, dopamine blockers have been found to be very effective in reducing uncontrolled movements and improving eye-hand coordination and steadiness during walking.

Hereditary Spherocytosis

Hereditary spherocytosis is an autosomal dominant genetic blood disorder that results from a defective copy of the *ankyrin-1* gene, found on chromosome 8. The protein encoded by this gene serves as a structural component of red blood cells and is responsible for maintaining their disklike shape. The abnormal spherocytosis protein is unable to perform its usual function, causing the affected person's red blood cells to adopt a spherical rather than disklike shape. As a result, the abnormal cells are fragile and burst easily, especially under osmotic stress. Enlargement of the spleen is also commonly seen in people with the disorder.

With an incidence of approximately 1 in 5,000, hereditary spherocytosis is one of the most common hereditary blood disorders. Roughly one-fourth of these cases result from new mutations and are not inherited from either parent. Hereditary spherocytosis exhibits incomplete penetrance, so not all individuals who inherit the mutant allele will actually show the trait. The cause of incomplete penetrance in these cases and others remains poorly understood.

Check Your Progress 11.3

1. Summarize how to distinguish an autosomal recessive disorder from an autosomal dominant disorder using a pedigree.
2. Construct a pedigree of Ivar Ragnarsson's family tree, assuming that his mother and both her parents did not have the disorder and that Ivar's father's father had osteogenesis imperfecta (mother was unaffected).

11.4 Beyond Mendelian Inheritance

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the inheritance pattern of traits with more than two alleles.
2. Contrast incomplete dominance and incomplete penetrance.
3. Describe the effects of pleiotropy on phenotypic traits.
4. Explain the concept of polygenic and multifactorial traits.
5. Understand how X-linked inheritance differs from autosomal inheritance.

Since the time of Mendel, scientists have discovered that for most traits, the patterns of inheritance do not precisely follow what is predicted by Mendelian genetics. Living organisms are complex biological systems, with many genes and environmental factors interacting to maintain life. However, the principles established by Mendel, including the laws of segregation and independent assortment (see Section 11.2), can be used to help us understand the basis of the complex patterns of inheritance we will explore in this section.

Multiple Alleles and Codominance

When a trait is controlled by **multiple alleles,** the gene exists in several allelic forms within a population. For example, the gene associated with cystic fibrosis (*CFTR*) has over 1,200 known alleles, all of which have the ability to vary the phenotype. We will use the

ISTUDY

human ABO blood type to better understand the influence of multiple alleles on a phenotype. The ABO blood type is controlled by a single gene that is involved with placing antigens (see Section 33.4) on the surface of red blood cells. This gene has three different alleles within the human population that determine blood type.

I^A = A antigen on red blood cells
I^B = B antigen on red blood cells
i = Neither A nor B antigen on red blood cells

Each person receives two of these alleles (one from each parent) to determine the presence or absence of antigens on their red blood cells. The possible phenotypes and genotypes for blood type are as follows:

Phenotype	*Genotype*
A	I^AI^A, I^Ai
B	I^BI^B, I^Bi
AB	I^AI^B
O	ii

Phenotype

The inheritance of the ABO blood group in humans is also an example of **codominance,** because both I^A and I^B are fully expressed in the presence of the other. A person who inherits chromosomes with I^A and I^B alleles will make fully functional A and B protein, and because these alleles are codominant, the resulting mixture of AB protein will give the red blood cell an AB phenotype. On the other hand, both I^A and I^B are dominant over i. Therefore, two genotypes are possible for type A blood, and two genotypes are possible for type B blood.

We can use a Punnett square to confirm that reproduction between a heterozygote with type A blood and a heterozygote with type B blood can result in any one of the four blood types. Such a cross makes it clear that an offspring can have a different blood type than either parent. For this reason, rather than blood type, DNA fingerprinting, also called DNA profiling (see Section 14.1), is used to identify the parents of an individual.

Incomplete Dominance and Incomplete Penetrance

For some genes, the presence of a dominant allele does not completely mask the recessive allele.

Incomplete Dominance

Incomplete dominance is exhibited when a heterozygote has an intermediate phenotype between that of either homozygote. In a cross between a true-breeding, red-flowered four-o'clock plant strain and a true-breeding, white-flowered strain, the offspring have pink flowers. Although this outcome might appear to be an example of the blending theory of inheritance, it is not. The phenotypes have blended, but the individual alleles are not altered. How do we know? When the pink plants self-pollinate, the offspring plants have a phenotypic ratio of 1 red-flowered : 2 pink-flowered : 1 white-flowered. The reappearance of the three phenotypes in this generation makes it clear we are still dealing with a single pair of alleles (Fig. 11.14).

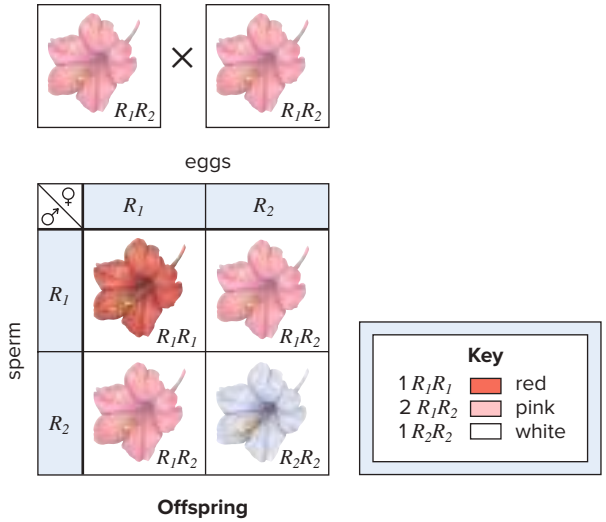


Figure 11.14 Incomplete dominance. When pink four-o'clocks self-pollinate, the results show three phenotypes. This is possible only if the pink parents had an allele for red pigment (R_1) and an allele for no pigment (R_2). Note that alleles involved in incomplete dominance are both given a capital letter.

Incomplete dominance in four-o'clocks actually has more to do with the amount of pigment protein produced in the plant cells. A double dose of pigment results in red flowers; a single dose of pigment results in pink flowers; and a lack of any pigment produces white flowers.

In humans, familial hypercholesterolemia (FH) is an example of incomplete dominance. The gene associated with this condition controls the number number of LDL-cholesterol receptor proteins in the plasma membrane. A person with two mutated alleles lacks LDL-cholesterol receptors. A person with only one mutated allele has half the normal number of receptors, and a person with two normal alleles has the usual number of receptors. People with the full number of receptors do not have familial hypercholesterolemia. When receptors are completely absent, excessive cholesterol is deposited in various places in the body, including under the skin. The presence of excessive cholesterol in the blood causes cardiovascular disease, which may result in death at a young age.

Incomplete Penetrance

A dominant allele may not always lead to the dominant phenotype in a heterozygote, even when the alleles show a true dominant/recessive relationship. The dominant allele in this case does not always determine the phenotype of the individual, so we describe these traits as showing **incomplete penetrance.** In other words, just because a person inherits a dominant allele doesn't mean they will fully express the gene or show the dominant phenotype. Many dominant alleles exhibit varying degrees of penetrance.

The best-known example of incomplete penetrance is polydactyly, the presence of one or more extra digits on the hands, the feet, or both. Polydactyly is inherited in an autosomal dominant manner; however, not all individuals who inherit the dominant allele exhibit the trait. The reasons for this are not clear, but expression of polydactyly may require additional environmental factors or be influenced by other genes, as discussed later in this section.

ISTUDY

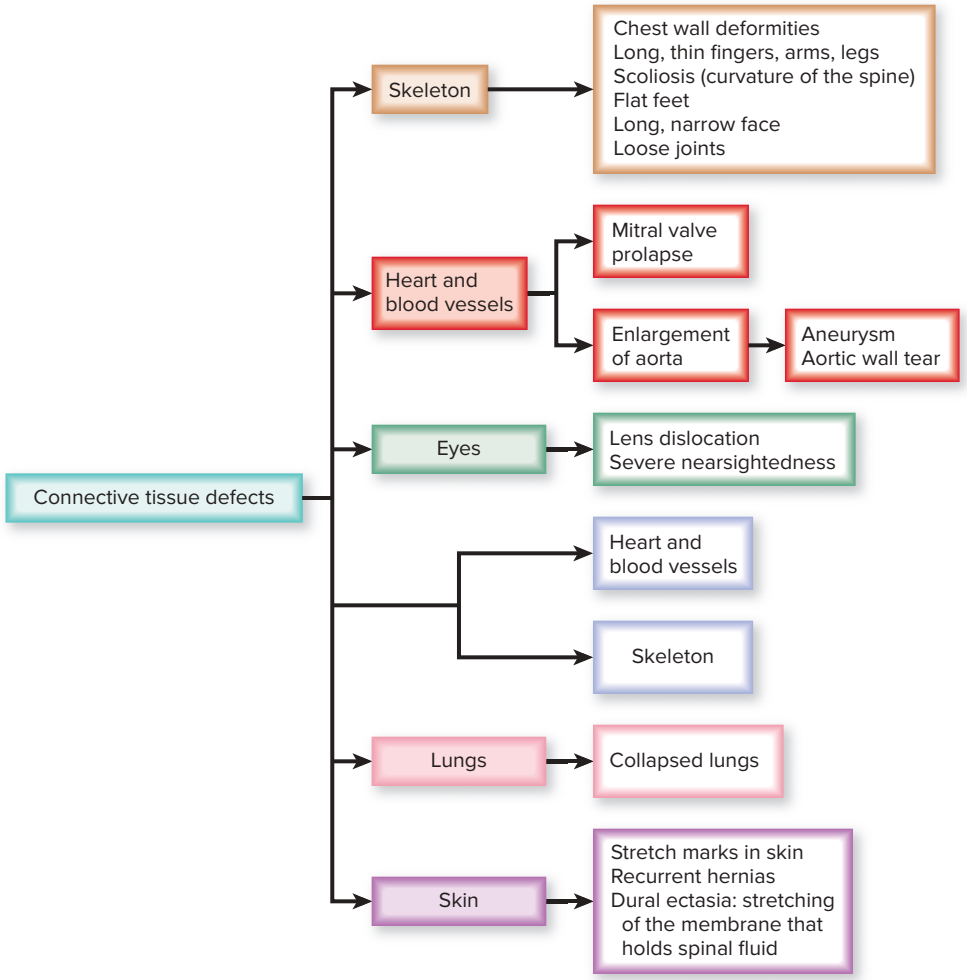


Figure 11.15 Marfan syndrome. Marfan syndrome illustrates the multiple effects a single gene can have. Marfan syndrome is due to any number of connective tissue defects.

Pleiotropic Effects

Pleiotropy occurs when a single mutant gene affects two or more distinct and seemingly unrelated traits. For example, persons with Marfan syndrome have disproportionately long arms, legs, hands, and feet; a weakened aorta; visual impairment; and other characteristics (Fig. 11.15). All of these characteristics are due to the production of abnormal connective tissue.

Marfan syndrome has been linked to a mutated gene (*FBN1*) on chromosome 15 that ordinarily specifies a functional protein called fibrillin. Fibrillin is essential for the formation of elastic fibers in connective tissue. Without the structural support of normal connective tissue, the aorta can burst, particularly if the person is engaged in a strenuous sport, such as volleyball or basketball. Flo Hyman may have been the best American female volleyball player ever, but she fell to the floor and died at the age of 31 because her aorta gave way during a game. Now that coaches are aware of Marfan syndrome, they are on the lookout for it among very tall basketball players.

Many other disorders, including porphyria and sickle-cell disease, are examples of pleiotropic traits. Porphyria is caused by a chemical insufficiency in the production of hemoglobin, the pigment that makes red blood cells red. The symptoms of porphyria are photosensitivity, strong abdominal pain, port-wine-colored urine, and paralysis in the arms and legs. Many members of the British royal family in the late 1700s and early 1800s had this disorder, which can lead to epileptic convulsions, mental changes, such as hallucinations or paranoia, and possibly a coma.

In a person with sickle-cell disease (Hb^SHb^S), the cells are sickle-shaped. The underlying mutation is in a gene that codes for a type of polypeptide chain in hemoglobin. Of 146 amino acids, the gene mutation changes only 1 amino acid, but the result is a less-soluble polypeptide chain that stacks up and causes red blood cells to be sickle-shaped. The abnormally shaped sickle cells slow down blood flow and clog small blood vessels. In addition, sickled red blood cells have a shorter

1,600×, SEM

Sickled red blood cell.
Phototake, Inc./Alamy Stock Photo

ISTUDY

Figure 11.16 Gene interactions and eye color. Individuals with red eyes (*left*) lack the ability to produce eye pigments. The presence of a dominant *OCA2* allele produces the common brown eye color (*center*). Blue eyes (*right*) may occur from either being homozygous recessive for an *OCA2* allele or due to an interaction with a second gene, *HERC2*. (red eye): Mediscan/Alamy Stock Photo; (brown eye): stylephotographs/123RF; (blue eye): lightpoet/123RF

life span than normal red blood cells. Individuals who are affected may exhibit a number of symptoms, including severe anemia, physical weakness, poor circulation, impaired mental function, pain and high fever, rheumatism, paralysis, spleen damage, low resistance to disease, and kidney and heart failure. All of these effects are due to both the tendency of sickled red blood cells to break down and the resulting decreased oxygen-carrying capacity of the blood, which damage the body.

Although sickle-cell disease adversely affects health, from an evolutionary perspective it provides heterozygous individuals with a survival advantage. People who have sickle-cell trait are resistant to the protozoan parasite that causes malaria. The parasite spends part of its life cycle in red blood cells, feeding on hemoglobin, but it cannot complete its life cycle when sickle-shaped cells form and break down earlier than usual. Because of this survival benefit, the sickle-cell allele has been maintained in the human population over evolutionary time (see Section 16.3).

Epistatic Interactions

As we have already noted, genes rarely act alone to produce a phenotype. In many cases, multiple genes may be part of a metabolic pathway consisting of the interaction of multiple enzymes and proteins.

An example of a trait determined by gene interaction is human eye color (Fig. 11.16). Multiple pigments are involved in producing eye color, including melanin. While a number of factors (including the structure of the eye) control minor variations in eye color (such as green or hazel eyes), a gene called *OCA2* is involved in producing the melanin that establishes the brown/blue basis of eye color. The dominant form of the *OCA2* allele produces brown eyes, which is the most common eye color. Individuals who lack *OCA2* completely (as occurs in albinism) have red eyes. Other *OCA2* alleles produce green and hazel eye colors.

However, there is another gene, *HERC2,* that can override the instructions of the *OCA2* gene. If individuals are homozygous for a recessive allele of *HERC2,* they will have blue eyes regardless of the genotype associated with *OCA2*. Geneticists call this type of interaction, where one gene can override another, an **epistatic interaction.**

Polygenic Inheritance

Polygenic inheritance (Gk. *poly,* "many"; L. *genitus,* "producing") occurs when a trait is governed by two or more sets of alleles. Examples include human height, skin color, and the prevalence of diabetes. The individual has a copy of all allelic pairs, possibly located on many different pairs of chromosomes. Each dominant allele has a quantitative effect on the phenotype, and these effects are additive. Therefore, a population is expected to exhibit continuous phenotypic variations, such as a wide variation in human height and weight. In Figure 11.17, a cross between genotypes *AABBCC* and *aabbcc* yields F_1 hybrids with the genotype *AaBbCc.* A range of genotypes and phenotypes results in the F_2 generation, which can be depicted as a bell-shaped curve.

Skin Color

Skin color is the result of pigmentation produced by skin cells called melanocytes, and over 100 different genes influence skin color. It is an example of a polygenic trait that is likely controlled by many pairs of alleles, which results in a range of phenotypes. The vast majority of people have skin colors in the middle range, whereas fewer people have skin colors in the extreme range.

Even so, we will use the simplest model and assume that skin color has only three pairs of alleles (*Aa, Bb,* and *Cc*) and that each capital letter contributes pigment to the skin. When a person with very dark skin reproduces with a person who has very light skin, the children have medium-brown skin. When two people with the genotype *AaBbCc* reproduce with one another, individuals may range in skin color from very dark to very light. The distribution of these phenotypes typically follows a bell-shaped curve, meaning that few people have the extreme phenotypes and most people have the phenotype that lies in the middle. A bell-shaped curve is a common identifying characteristic of a polygenic trait (Fig. 11.18).

However, skin color is also influenced by the sunlight in the environment. Notice again that a range of phenotypes exists for each genotype. For example, individuals who are *AaBbCc* may vary in their skin color, even though they possess the same genotype, and several possible phenotypes fall between the two extremes. The interaction of the environment with polygenic traits is discussed next.

ISTUDY

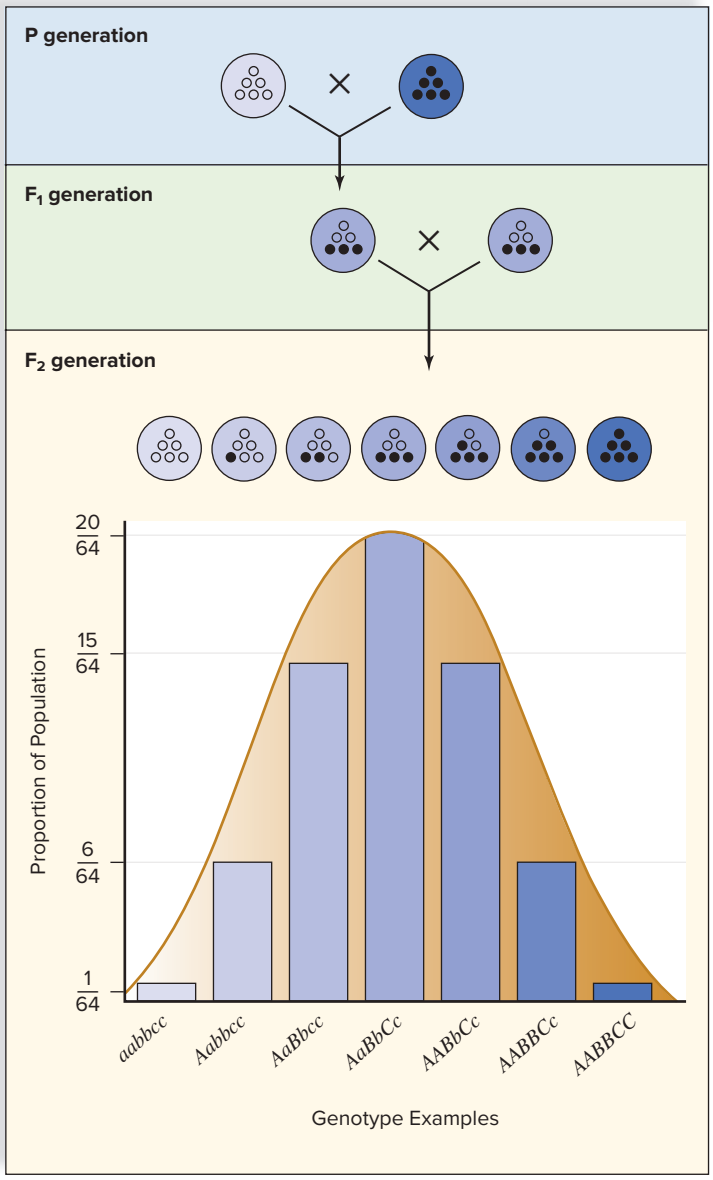


Figure 11.17 Polygenic inheritance. In polygenic inheritance, a number of pairs of genes control the trait. *Top:* Black dots and intensity of blue shading stand for the number of dominant alleles. *Bottom:* Orange shading shows the degree of environmental influences.

Environmental Influences: Multifactorial Traits

Multifactorial traits are those controlled by polygenes subject to environmental influences. Many genetic disorders, such as cleft lip and/or palate, clubfoot, congenital dislocations of the hip, hypertension, diabetes, schizophrenia, and even allergies and cancers, are probably multifactorial, because they are likely due to the combined action of many genes plus environmental influences. The relative importance of genetic and environmental influences on the phenotype can vary, and often it is a challenge to determine how much of the variation in the phenotype may be attributed to each factor. This is especially true in complex polygenic traits for which there may be an additive effect of multiple genes on the phenotype. If each gene has several alleles, and each allele responds slightly differently to environmental factors, then the phenotype can vary considerably.

Multifactorial traits are a challenge for drug manufacturers, since they must determine the response to a new drug based on genetic factors (for example, the ethnic background of the patient) and environmental factors (such as diet). Temperature is an environmental factor that can influence the phenotypes of plants and animals. Primroses have white flowers when grown above 32°C but red flowers when grown at 24°C.

The coats of Himalayan rabbits are darker in color at the ears, nose, paws, and tail. Himalayan rabbits are known to be homozygous for the allele *ch,* which is involved in the production of melanin. Experimental evidence suggests that the enzyme encoded by this gene is active only at a low temperature and that, therefore, black fur occurs only at the extremities, where body heat is lost to the environment. When the animal is placed in a warmer environment, new fur on these body parts is light in color.

Many investigators are trying to determine what percentage of various traits is due to nature (inheritance) and what percentage is due to nurture (the environment). Some studies use identical twins separated since birth, because if identical twins in different environments share the same trait, the trait is most likely inherited. Identical twins are more similar in their intellectual talents, personality traits, and levels of lifelong happiness than are fraternal twins separated at birth. Biologists conclude that all behavioral traits are partly heritable (see Section 43.1), and that genes exert their effects by acting together in complex combinations susceptible to environmental influences.

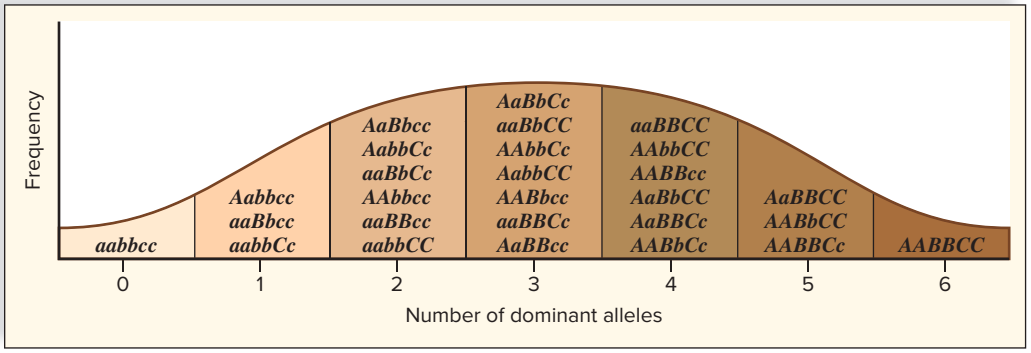


Figure 11.18 The polygenic basis of skin color. Skin color is controlled by many pairs of alleles, which result in a range of phenotypes. The vast majority of people have skin colors in the middle range, whereas fewer people have skin colors in the extreme range.

ISTUDY

X-linked Inheritance

The X and Y chromosomes in mammals determine the sex of the individual. Females are XX, and males are XY. These chromosomes carry genes that control development; in particular, if the Y chromosome contains an *SRY* gene, the embryo becomes a male. The term **X-linked** is used for genes that have nothing to do with biological sex yet are carried on the X chromosome. The Y chromosome does not carry these genes and indeed carries very few genes.

This type of inheritance was discovered in the early 1900s by a group at Columbia University headed by Thomas Hunt Morgan. Morgan performed experiments with fruit flies, *Drosophila melanogaster.* Fruit flies are even better subjects for genetic studies than garden peas. They can be easily and inexpensively raised in simple laboratory glassware; after mating, females lay hundreds of eggs during their lifetimes; and the generation time is short, taking only about 10 days from egg to adult. Fruit flies have a sex chromosome pattern similar to that of humans, and therefore Morgan's experiments with X-linked genes apply directly to humans.

Morgan's Experiment

Morgan took a newly discovered mutant male with white eyes and crossed it with a red-eyed female:

	♀		♂
p	red-eyed	×	white-eyed
F_1	red-eyed		red-eyed

From these results, he knew that red eyes are the dominant characteristic and white eyes are the recessive characteristic. He then crossed the F^1 flies. In the F^2 generation, there was the expected 3 red-eyed : 1 white-eyed ratio, but it struck him as odd that all the white-eyed flies were males:

	♀		♂
$F_1 \times F_1$	red-eyed	×	red-eyed
F_2	red-eyed		1 red-eyed : 1 white-eyed

Obviously, a major difference between the male flies and the female flies was their sex chromosomes. Could it be possible that an allele for eye color was on the Y chromosome but not on the X? This idea could be quickly discarded, because usually females have red eyes, and they have no Y chromosome. Perhaps an allele for eye color was on the X, but not on the Y, chromosome. Figure 11.19 indicates this explanation would match the results obtained in the experiment. These results support the chromosome theory of inheritance by showing that the behavior of a specific allele corresponds exactly with that of a specific chromosome—the X chromosome in *Drosophila.*

Notice that X-linked alleles have a different pattern of inheritance than alleles that are on the autosomes, because the Y chromosome is lacking for these alleles, and the inheritance of a Y chromosome cannot offset the inheritance of an X-linked recessive allele. For the same reason, males always receive an X-linked recessive mutant allele from the female parent—they receive only the Y chromosome from the male parent, and therefore sex-linked recessive traits appear much more frequently in males than in females.

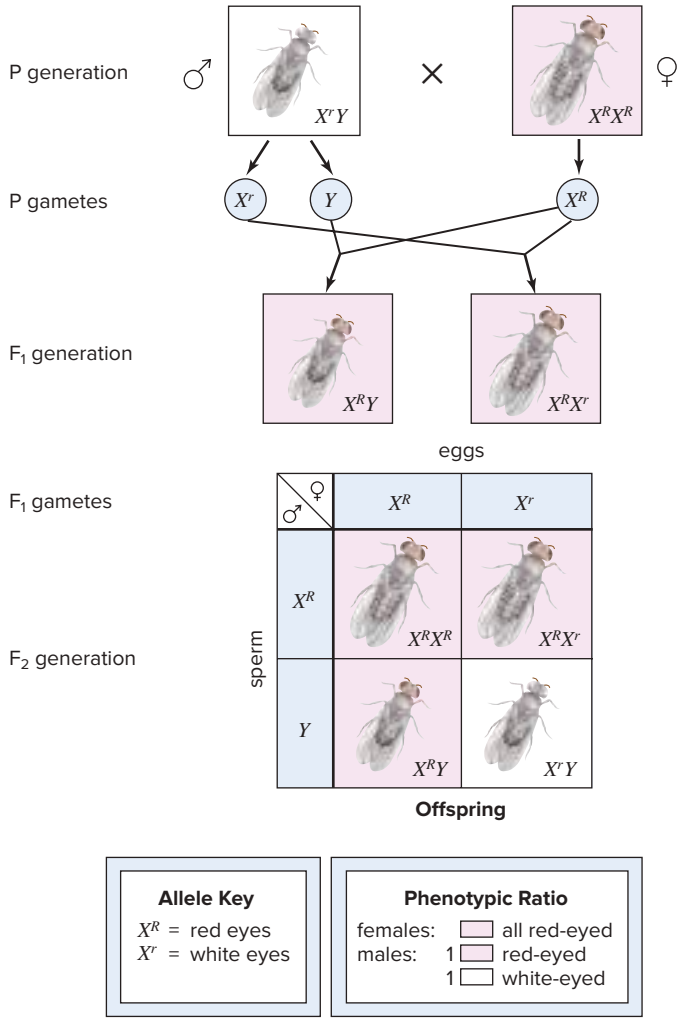


Figure 11.19 X-linked inheritance. Once researchers deduced that the alleles for red/white eye color are on the X chromosome in *Drosophila,* they were able to explain their experimental results. Males with white eyes in the F_2 generation inherit the recessive allele only from the female parent; they receive a Y chromosome lacking the allele for eye color from the male parent.

Solving X-linked Genetics Problems

Recall that when solving autosomal genetics problems, the alleles and genotypes can be represented as follows:

Alleles	*Genotypes*
L = long wings	*LL, Ll*
l = short wings	*ll*

When predicting inheritance of sex-linked traits, however, it is necessary to indicate the sex chromosomes of each individual. As noted in Figure 11.19, however, the alleles for an X-linked gene show an allele attached to the X:

Alleles
X^R = red eyes
X^r = white eyes

ISTUDY

The possible genotypes and phenotypes in both males and females are as follows:

Genotype	*Phenotype*
X^RX^R	red-eyed female
X^RX^r	red-eyed female
X^rX^r	white-eyed female
X^RY	red-eyed male
X^rY	white-eyed male

Notice that there are three possible genotypes for females but only two for males. Females can be heterozygous X^RX^r, in which case they are carriers. Carriers usually do not show a recessive abnormality, but they are capable of passing on a recessive allele for an abnormality. But unlike autosomal traits, males cannot be carriers for X-linked traits; if the dominant allele is on the single X chromosome, they show the dominant phenotype, and if the recessive allele is on the single X chromosome, they show the recessive phenotype. For this reason, males are considered **hemizygous** for X-linked traits, because a male possesses only one allele for the trait and, therefore, expresses whatever allele is present on the X chromosome.

We know that male fruit flies have white eyes when they receive the mutant recessive allele from the female parent. What is the inheritance pattern when females have white eyes? Females can have white eyes only when they receive a recessive allele from both parents.

Human X-linked Disorders

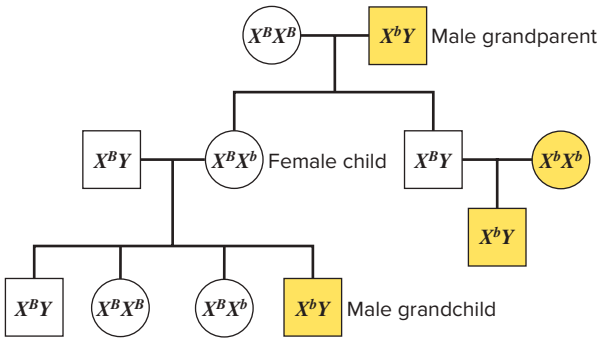
Several X-linked recessive disorders occur in humans, including color blindness, Menkes syndrome, muscular dystrophy, adrenoleukodystrophy, and hemophilia (see the Nature of Science reading "Hemophilia and the Royal Families of Europe").

Color Blindnes In humans, the receptors for color vision in the retina of the eyes are three different classes of cone cells. Only one type of pigment protein is present in each class of cone cell; there are blue-sensitive, red-sensitive, and green-sensitive cone cells. The allele for the blue-sensitive protein is autosomal, but the alleles for the red- and green-sensitive pigments are on the X chromosome. About 8% of white men have red-green color blindness. Most of these individuals see brighter greens as tans, olive greens as browns, and reds as reddish browns. A few cannot tell reds from greens at all. They see only yellows, blues, blacks, whites, and grays.

Pedigrees can also reveal the unusual inheritance pattern seen in sex-linked traits. For example, the pedigree in Figure 11.20 shows the usual pattern of inheritance for color blindness. More males than females have the trait, because recessive alleles on the X chromosome are expressed in males. The disorder often passes from male grandparent to grandchild through a carrier daughter.

Menkes Syndrome Menkes syndrome is caused by a defective allele on the X chromosome. Normally, the gene product controls the movement of the metal copper into and out of cells. The symptoms of Menkes syndrome are due to an accumulation of copper in some parts of the body, and the lack of the metal in other parts.

Symptoms of Menkes syndrome include poor muscle tone, seizures, abnormally low body temperature, skeletal anomalies, and the characteristic brittle, steely hair associated with the disorder. Although the condition is relatively rare, affecting approximately 1 in 100,000, mostly males, the prognosis for people with Menkes syndrome is poor, and most individuals die within the first few years of life. In recent years, some people with Menkes syndrome have been treated with injections of copper directly underneath the skin, but with mixed results, and treatment must begin very early in life to be effective.



Key
X^BX^B = Unaffected female
X^BX^b = Female who is a carrier
X^bX^b = Female who is color blind
X^BY = Unaffected male
X^bY = Male who is color blind

X-linked Recessive Disorders
- More males than females are affected.
- A male child who is affected can have parents who are unaffected.
- For a female to be affected, her male parent must be affected. Her female parent must be a carrier or affected.
- The trait often skips generations from the male grandparent to the male grandchild.
- If a female is affected, all of her male children will be affected.

Figure 11.20 X-linked recessive pedigree. This pedigree for color blindness exemplifies the inheritance pattern of an X-linked recessive disorder. The list gives various ways of recognizing the X-linked recessive pattern of inheritance.

Muscular Dystrophy Muscular dystrophy, as the name implies, is characterized by a wasting away of the muscles. The most common form, Duchenne muscular dystrophy, is X-linked and occurs in about 1 out of every 3,600 male births. Symptoms such as waddling gait, toe walking, frequent falls, and difficulty in rising may appear as soon as the child starts to walk. Muscle weakness intensifies until the individual is unable to walk. Death usually occurs by age 20; therefore, affected males are rarely parents. The recessive allele remains in the population through passage from carrier female to carrier female offspring.

The allele for Duchenne muscular dystrophy has been isolated, and it has been discovered that the absence of a protein called dystrophin causes the disorder. Much investigative work has determined that dystrophin is involved in the release of calcium from the sarcoplasmic reticulum in muscle fibers. The lack of dystrophin causes calcium to leak into the cell, which promotes the action of an enzyme that dissolves muscle fibers. When the body attempts to repair the tissue, fibrous tissue forms, and this cuts off the blood supply, so more and more cells die.

ISTUDY

THEME Nature of Science

Hemophilia and the Royal Families of Europe

About 1 in 10,000 males has hemophilia. There are two common types of hemophili. Hemophilia A is due to the absence or minimal presence of a clotting factor known as factor VIII, and hemophilia B is due to the absence of clotting factor IX. Hemophilia is called the bleeder's disease because the affected person's blood either does not clot or clots very slowly. Although individuals with hemophilia bleed externally after an injury, they also bleed internally, particularly around joints. Hemorrhages can be stopped with transfusions of fresh blood (or plasma) or concentrates of the clotting protein. Also, clotting factors are available as biotechnology products.

The pedigree in Figure 11A shows why hemophilia is often referred to as "the royal disease." Queen Victoria of England, who reigned from 1837 to 1901, was the first of the royals to carry the gene. From her, the disease eventually spread to the Prussian, Spanish, and Russian royal families. In that era, monarchs arranged marriages between their children to consolidate political alliances. This practice allowed the gene for hemophilia to spread throughout the royal families. It is assumed that a spontaneous mutation arose either in Queen Victoria after her conception or in one of the gametes of her parents. However, in the book *Queen Victoria's Gene* by D. M. Potts, the author postulates that Edward Augustus, Duke of Kent, may not have been Queen Victoria's father. Potts suggests that Victoria may have instead been conceived outside the marriage by a male with hemophilia.

Of Queen Victoria's 26 grandchildren, four grandsons had hemophilia and four granddaughters were carriers. Because none of Queen Victoria's brothers and sisters were affected, it seems that the faulty allele she carried arose by mutation, either in Victoria or in one of her parents. Her carrier daughters, Alice and Beatrice, introduced the allele into the ruling houses of Russia, Prussia, and Spain, respectively. Alexi, the last heir to the Russian throne before the Russian Revolution, had hemophilia. There are no individuals with hemophilia in the present British royal family, because Victoria's eldest son, King Edward VII, did not receive the allele.

Questions to Consider

1. How may a pedigree pattern be used to determine if a disease is autosomal dominant or autosomal recessive?
2. Assume that the mutation for hemophilia did not originate with Victoria. What does this tell you about the genotypes of her parents?

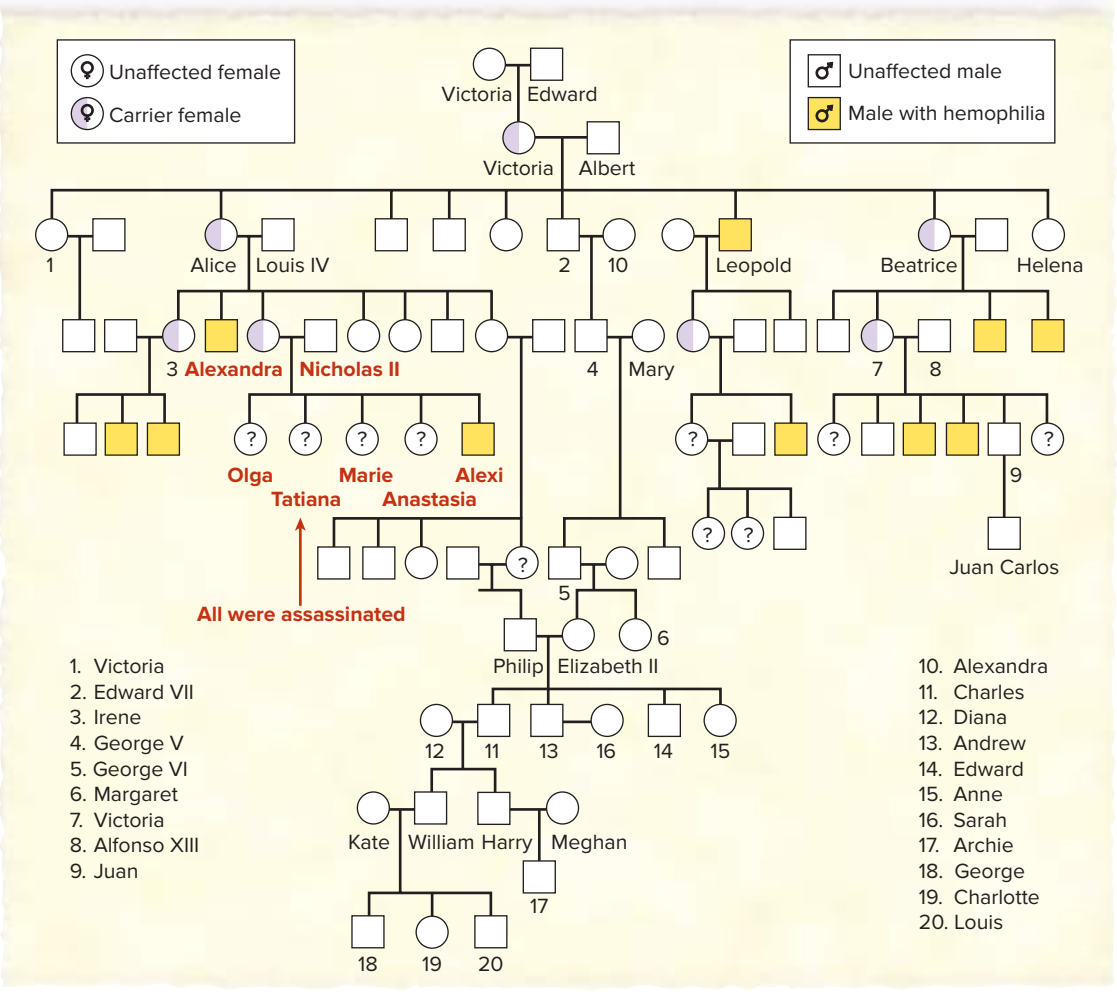


Figure 11A Hemophilia: An X-linked trait. Queen Victoria was a carrier, so each of her sons had a 50% chance of having the disorder, and each of her daughters had a 50% chance of being a carrier.

A test is now available to detect carriers of Duchenne muscular dystrophy. Also, various treatments have been tried. Immature muscle cells can be injected into muscles, and for every 100,000 cells injected, dystrophin production occurs in 30–40% of muscle fibers. The allele for dystrophin has been inserted into thigh muscle cells, and about 1% of these cells then have produced dystrophin.

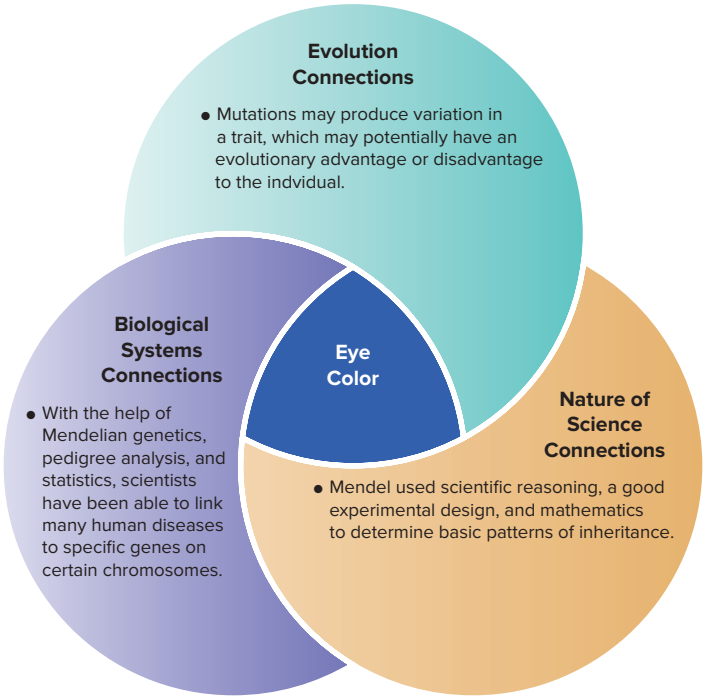
Adrenoleukodystrophy Adrenoleukodystrophy, or ALD, is an X-linked recessive disorder caused by the failure of a carrier protein to move either an enzyme or a very long-chain fatty acid (24–30 carbon atoms) into peroxisomes. As a result, these fatty acids are not broken down, and they accumulate inside the cell; the result is severe nervous system damage.

Children with ALD fail to develop properly after age 5, lose adrenal gland function, exhibit very poor coordination, and show a progressive loss of hearing, speech, and vision. The condition is usually fatal, with no known cure, but the onset and severity of symptoms in children not yet showing symptoms may be mitigated by treatment with a mixture of lipids derived from olive oil. The disease was made well-known by the 1992 movie *Lorenzo's Oil,* detailing a mother and father's determination to devise a treatment for their son with ALD.

Check Your Progress 11.4

1. Summarize why incomplete dominance does not support blending.
2. Summarize how to differentiate between an X-linked trait and an autosomal trait.
3. Explain how a trait may be both polygenic and multifactorial.

Connecting *the* Concepts



What Did You Learn?

11.1 Gregor Mendel

Gregor Mendel used the garden pea as the subject in his genetics studies. In contrast to preceding plant breeders, his study involved nonblending traits of the garden pea. Mendel applied mathematics, followed the scientific method very closely, and kept careful records. His results supported a particulate theory of inheritance, effectively disproving the blending theory of inheritance.

11.2 Mendel's Laws

When Mendel crossed heterozygous plants with other heterozygous plants, called a **monohybrid cross,** he found that the recessive **phenotype** reappeared in about ¼ of the F_2 plants. There was a 3:1 phenotypic ratio. This allowed Mendel to propose his law of segregation, which states that the individual has two factors for each trait (the **genotype**), and the factors segregate with equal probability into the gametes.

- Although Mendel didn't know it, the variations in the traits were due to variations in genes, called **alleles.** Each gene has a specific location, or **locus,** on a chromosome. **Dominant alleles** mask the expression of **recessive alleles.**
- Mendel conducted **dihybrid crosses,** in which the F_1 individuals showed both dominant characteristics, but there were four phenotypes in a 9:3:3:1 ratio among the F_2 offspring. This allowed Mendel to deduce the **law of independent assortment,** which states that the members of one pair of factors separate independently of those of another pair. Therefore, all possible combinations of parental factors can occur in the gametes.
- The laws of probability can be used to calculate the expected phenotypic ratio of a cross. A large number of offspring must be counted in order to observe the expected results, and to ensure that all possible types of sperm have fertilized all possible types of eggs, as is done in a **Punnett square.** The Punnett square uses the product law of probability to arrive at possible genotypes

among the offspring, and then the sum law can be used to arrive at the phenotypic ratio.

- Mendel also crossed the F_1 plants having the dominant phenotype with homozygous recessive plants. The 1:1 results indicated that the recessive factor was present in these F_1 plants (they were heterozygous). Today, we call this a testcross, because it is used to test whether an individual showing the dominant characteristic is **homozygous** dominant or **heterozygous.** The two-trait testcross allows an investigator to test whether an individual showing two dominant characteristics is homozygous dominant for both traits or for one trait only or is heterozygous for both traits.

11.3 Mendelian Patterns of Inheritance and Human Disease

Studies have shown that many human traits and genetic disorders can be explained on the basis of simple Mendelian inheritance. When studying human genetic disorders, biologists often construct **pedigrees** to show the pattern of inheritance of a characteristic within a family. The particular pattern indicates the manner in which a characteristic is inherited, such as **autosomal** dominant or autosomal recessive.

- Autosomal recessive: An individual must possess two copies of the recessive allele to express the trait. Heterozygous individuals may be **carriers** for the trait.
- Autosomal dominant: An individual needs to possess only a single copy of the dominant allele to express the trait. Homozygous dominant and heterozygous individuals both express the trait.

11.4 Beyond Mendelian Inheritance

Other patterns of inheritance have been discovered since Mendel's original contribution.

- For example, some genes have **multiple alleles,** although each organism has only two alleles, as in the inheritance of blood type in humans. Inheritance of blood type also illustrates **codominance.** With **incomplete dominance,** the phenotype of F_1 individuals is intermediate between the parent phenotypes; this does not support the blending theory, because the parent phenotypes reappear in F_2. With incomplete penetrance, some traits that are dominant may not be expressed due to unknown reasons.
- Gene interactions, such as those in metabolic pathways, are examples of **epistatic interactions.** In **pleiotropy,** one gene has multiple effects, as in Marfan syndrome and sickle-cell disease. **Polygenic traits** are controlled by several genes that have an additive effect on the phenotype, resulting in quantitative variations within a population. A bell-shaped curve is seen, because environmental influences bring about many intervening phenotypes, as in the inheritance of height in humans. Skin color and eye color are also examples of **multifactorial traits** (multiple genes plus the environment).
- In *Drosophila,* as in humans, the sex chromosomes determine the sex of the individual, with XX being female and XY being male. Experimental support for the chromosome theory of inheritance came when Morgan and his group were able to determine that the gene for a trait unrelated to sex determination, the white-eyed allele in *Drosophila,* is on the X chromosome.
- Alleles on the X chromosome are called **X-linked** alleles. Therefore, when doing X-linked genetics problems, it is the custom to indicate the sexes by using sex chromosomes and to indicate the alleles by superscripts attached to the X. The Y is blank because it does not carry these genes. Color blindness, Menkes syndrome, adrenoleukodystrophy, and hemophilia are X-linked recessive disorders in humans.

Thinking Critically About Biology

1. In tomatoes, red fruit (*R*) is dominant over yellow fruit (*r*), and tallness (*T*) is dominant over shortness (*t*). A plant that is *RrTT* is crossed with a plant that is *rrTt.* Assuming that the traits are on different chromosomes, what are the chances of an offspring possessing both recessive traits?
2. Assume you are working in a genetics lab with *Drosophila melanogaster,* a model organism for genetic research. You want to determine whether a *Drosophila* characteristic is dominant or recessive. Explain how you would construct such a cross, and the expected outcomes that would indicate dominance.
3. Multiple gene pairs may also interact to produce a single phenotype. In peas, genes *C* and *P* are required for pigment production in flowers. Gene *C* codes for an enzyme that converts a compound into a colorless intermediate product. Gene *P* codes for an enzyme that converts the colorless intermediate product into anthocyanin, a purple pigment. A flower, therefore, will be purple only if it contains at least one dominant allele for each of the two genes (*C*__ *P*__). What phenotypic ratio would you expect in the F_2 generation following a cross between two double heterozygotes (*CcPp*)?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Glowing Fish Genetics

McGraw Hill

What would you conclude if a cross between an orange-colored fish and a yellow-colored fish produced all yellow-colored offspring?

Making It Relevant

1. Variation in a trait may provide an evolutionary advantage or disadvantage for an individual, or sometimes may just be completely neutral, meaning they have no noticeable effect. Ponder some variation of traits in humans that may also be considered neutral.
2. The *OCA2* gene is not only associated with eye color, but also the production of melanin associated with skin pigmentation. Consider why mutations that lighten skin and eye coloration may first develop in northern European countries.
3. At one time, geneticists believed that humans may have as many as 100,000 different genes, but now the number is believed to be closer to 20,000. How do gene interactions, such as *HERC2* and *OCA2* with eye color, help describe how complex organisms, such as humans, can have so few genes?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

12

Molecular Biology of the Gene

(girl): Jodi Matthews/iStock/Getty Images; (flower): Sombat Muycheen/Shutterstock; (DNA): Radius Images/Alamy Stock Photo; (mushroom): IT Stock/age fotostock

Chapter Outline

12.1 The Genetic Material
12.2 Replication of DNA
12.3 Gene Expression: RNA and the Genetic Code
12.4 Gene Expression: Transcription
12.5 Gene Expression: Translation

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.5 What are the components of a nucleotide and the structure of the DNA molecule?

Section 11.2 What is the relationship between an allele and a gene?

Section 11.2 What is the relationship between genotype and phenotype?

Next-Generation DNA

Deoxyribonucleic acid, or DNA, is the genetic material of all life on Earth. Each of the organisms in the photo above, and all of the identified 2.3 million species on the planet, uses a combination of four nucleotides (A, C, G, and T) to code for 20 different amino acids. The combination of these amino acids produces the tremendous variation that gives us the diversity of life.

However, a biotechnology company named Synthorx has expanded this genetic alphabet by creating two new forms of nucleotides, called X and Y. When these nucleotides are incorporated into the DNA of *E. coli* bacteria, the number of potential amino acids that can be coded for increases from 20 to 172. Because of this innovation, it may be possible to develop new drugs (including antibiotics) and vaccines that can help protect us from some diseases. It may also be possible to develop genetically modified organisms that resist pathogens, grow faster, or produce foods with higher nutritional content.

Other companies have been working on using DNA's ability to store information to build the next generation of data storage modules. A single gram of DNA contains 215 million gigabytes of information, or the equivalent of 215,000 large computer hard drives. Because information storage is critical in many industries, including medicine, DNA-based storage systems may soon make conventional hard drives obsolete.

As you read through this chapter, think about the following questions:

1. How is information contained within a molecule of DNA?
2. How does the flow of genetic information from DNA to protein to trait work?

Following *the* Themes

CHAPTER 12 – MOLECULAR BIOLOGY OF THE GENE

Evolution	Changes in the DNA sequence produce different proteins that influence cellular and organismal structure and function.
Nature of Science	Research in gene expression enables scientists to discover innovative and effective ways to treat disease.
Biological Systems	The flow of genetic information in biological systems is highly regulated and responsive to environmental changes.

12.1 The Genetic Material

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the properties a substance must possess in order to serve as the genetic material.
2. Examine how historical researchers demonstrated that DNA is the genetic material.
3. Explain the chemical structure of DNA as defined by the Watson and Crick model.

The middle of the twentieth century was an exciting period of scientific discovery. On one hand, geneticists were busy determining that **DNA (deoxyribonucleic acid)** is the genetic material of all living organisms. On the other hand, biochemists were in a frantic race to describe the structure of DNA. The classic experiments performed during this era set the stage for an explosion in our knowledge of modern molecular biology.

When researchers began their work, they knew that the genetic material must be

- Able to *store information* that pertains to the development, structure, and metabolic activities of the cell or organism.
- Stable, so it *can be replicated* with high accuracy during cell division and be transmitted from generation to generation.
- Able to *undergo rare changes,* called mutations, that provide the genetic variability required for evolution to occur.

In this section, we will explore how DNA is able to fulfill all of these requirements.

Transformation of Bacteria

During the late 1920s, the bacteriologist Frederick Griffith was attempting to develop a vaccine against *Streptococcus pneumoniae* (pneumococcus), which causes pneumonia in mammals. In 1931, he performed a classic experiment with the bacterium. He noticed that when these bacteria are grown on culture plates, some, called S strain bacteria, produce shiny, smooth colonies and others, called R strain bacteria, produce colonies that have a rough appearance. Under the microscope, S strain bacteria have a capsule (mucous coat) that makes them smooth, but R strain bacteria do not.

When Griffith injected mice with the S strain bacteria, the mice died, but when he injected mice with the R strain, the mice did not die (Fig. 12.1). In an effort to determine whether the capsule alone was responsible for the virulence (ability to kill) of the S strain bacteria, he injected mice with heat-killed S strain bacteria. The mice did not die.

Finally, Griffith injected the mice with a mixture of heat-killed S strain and live R strain bacteria. Most unexpectedly, the mice died—and living S strain bacteria were recovered from the bodies! Griffith concluded that some substance necessary for the bacteria to produce a capsule and be virulent must have passed from the dead S strain bacteria to the living R strain bacteria, so that the R strain bacteria were *transformed* (Fig. 12.1*d*). This change in the phenotype of the R strain bacteria must have been due to a change in their genotype. Indeed, couldn't the transforming substance that passed from S strain to R strain be genetic material? Reasoning such as this prompted investigators at the time to begin looking for the transforming substance to determine the chemical nature of the genetic material.

DNA: The Transforming Substance

By the time the next group of investigators, led by Oswald Avery in the 1940s, began their work, it was known that the genes are on the chromosomes and that the chromosomes contain both proteins and nucleic acids. Investigators were having a very heated debate about whether protein or DNA was the genetic material. Many thought the protein component of chromosomes must be the genetic material because proteins contain up to 20 different amino acids that can be sequenced in any particular way. On the other hand, nucleic acids—DNA and RNA—contain only four types of nucleotides as basic building blocks. Some argued that DNA did not have enough variability to be able to store information and be the genetic material.

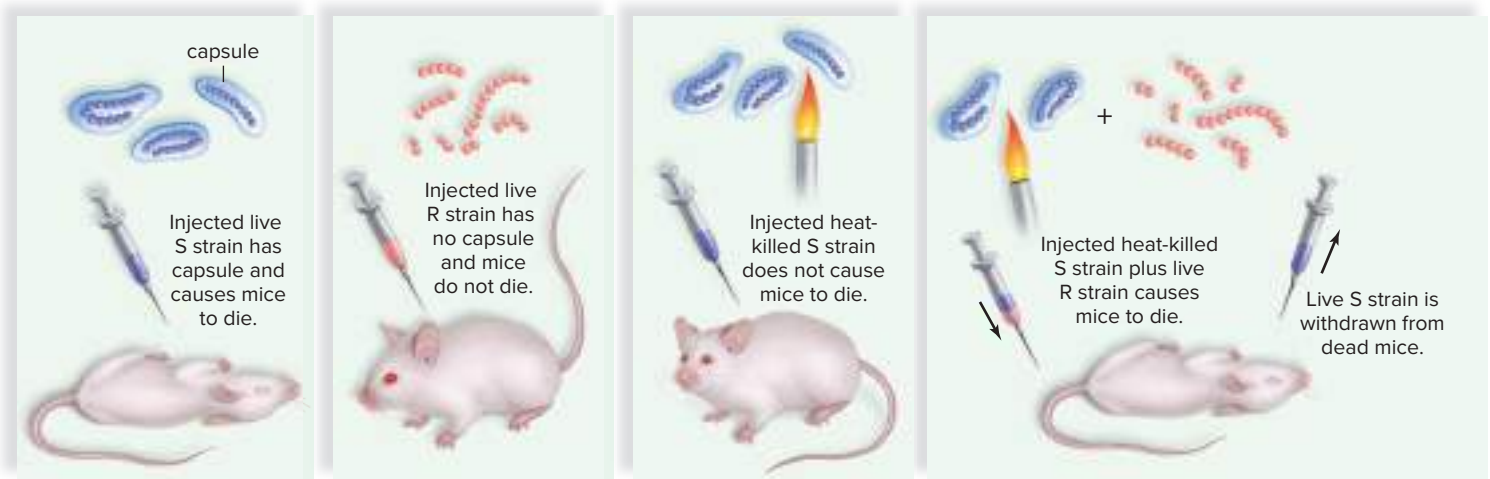
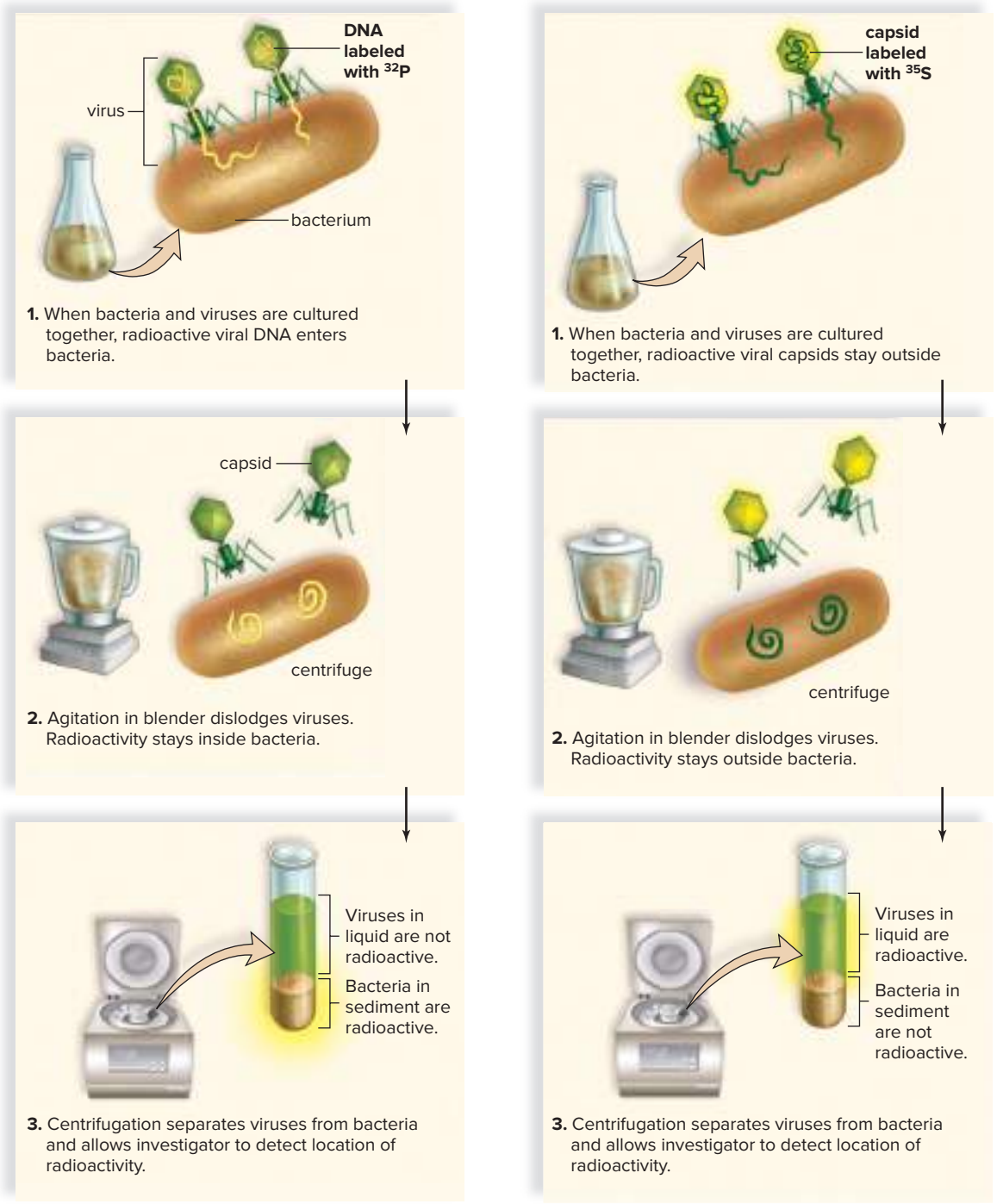


Figure 12.1 Griffith's transformation experiment. **a.** Encapsulated S strain is virulent and kills mice. **b.** Nonencapsulated R strain is not virulent and does not kill mice. **c.** Heat-killed S strain bacteria do not kill mice. **d.** If heat-killed S strain and R strain are both injected into mice, the mice die because the R strain bacteria have been transformed into the virulent S strain.

ISTUDY



a. Viral DNA is labeled (yellow).

b. Viral capsid is labeled (yellow).

Figure 12.2 The Hershey and Chase experiments. In these experiments, the (**a**) genetic material (DNA) and (**b**) protein of a bacteriophage were labeled with radioactive isotopes. The experiments show that the DNA of the bacteriophage, not the protein, enters the host cell and is therefore the genetic material.

In 1944, after 16 years of research, Oswald Avery and his co-investigators, Colin MacLeod and Maclyn McCarty, published a paper demonstrating that the transforming substance that allows *Streptococcus* to produce a capsule and be virulent is DNA. This meant DNA is the genetic material. Here is what they found out:

- DNA from S strain bacteria causes R strain bacteria to be transformed, so they can produce a capsule and be virulent.
- The addition of DNase, an enzyme that digests DNA, prevents transformation from occurring. This supports the hypothesis that DNA is the genetic material.
- The molecular weight of the transforming substance is large. This suggests the possibility of genetic variability.
- The addition of enzymes that degrade proteins has no effect on the transforming substance, nor does RNase, an enzyme that digests RNA. This shows that neither protein nor RNA is the genetic material.

These experiments showed that DNA is the transforming substance and, therefore, the genetic material. Although some scientists remained skeptical, many felt that the evidence for DNA being the genetic material was overwhelming.

A series of experiments by Alfred Hershey and Martha Chase in the early 1950s helped to firmly establish DNA as the genetic material (Fig. 12.2). Hershey and Chase used a virus called a T phage, composed of radioactively labeled DNA and capsid coat

ISTUDY

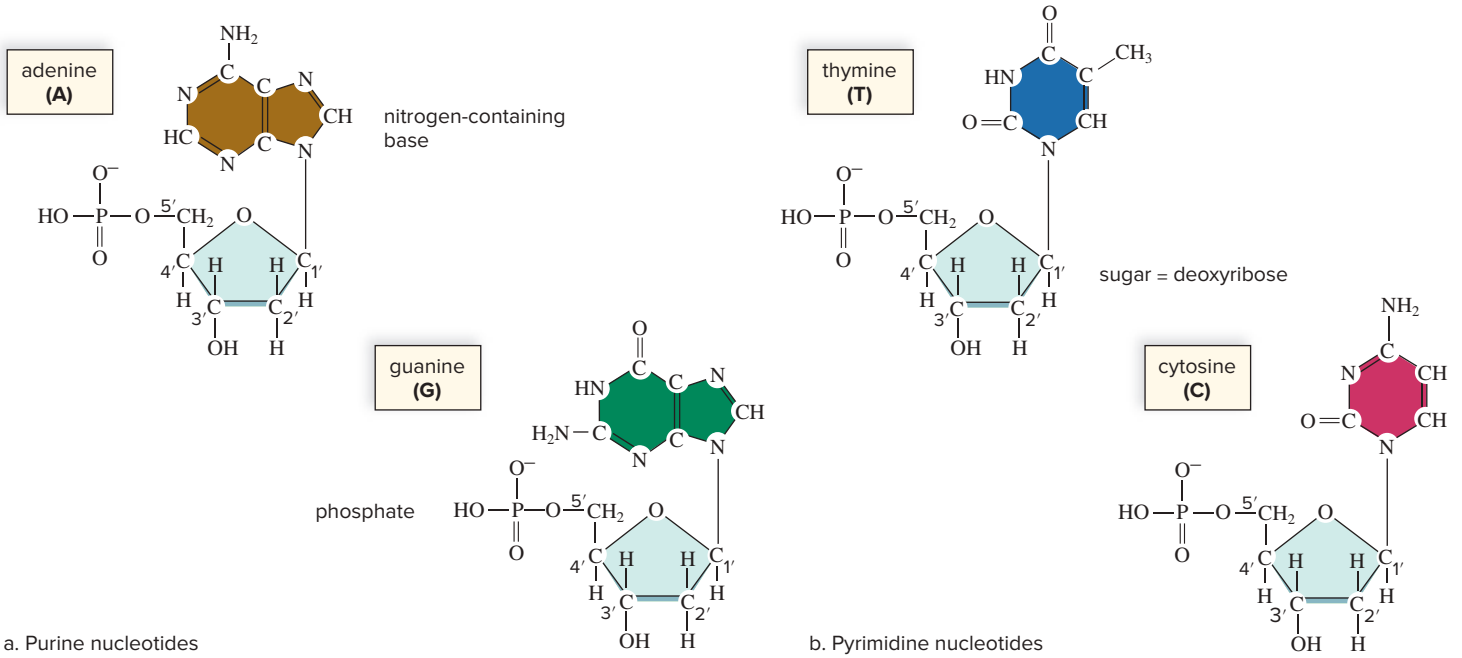


Figure 12.3 Nucleotide composition of DNA. All nucleotides contain phosphate, a 5-carbon sugar, and a nitrogen-containing base. In DNA, the sugar is called deoxyribose, because it lacks an oxygen atom in the 2′ position, compared to ribose. The nitrogen-containing bases are (**a**) the purines adenine and guanine, which have a double ring, and (**b**) the pyrimidines thymine and cytosine, which have a single ring.

proteins, to infect *E. coli* bacteria. They discovered that the radioactive tracers for DNA, but not protein, ended up inside the bacterial cells, causing them to become transformed. Since only the genetic material could have caused this transformation, Hershey and Chase determined that DNA must be the genetic material.

The Structure of DNA

By the early 1950s, DNA was widely accepted as the genetic material of all living organisms. However, the structure of DNA was not known. How can a molecule with only four different nucleotides produce the great diversity of life on Earth?

To understand the structure of DNA, we need to understand how the bases in DNA are composed. Investigators knew that DNA contains four different types of nucleotides: two with *purine* bases, **adenine (A)** and **guanine (G)**, which have a double ring; and two with *pyrimidine* bases, **thymine (T)** and **cytosine (C)**, which have a single ring (Fig. 12.3).

Erwin Chargaff used new chemical techniques developed in the 1940s to analyze in detail the base content of DNA. A sample of Chargaff's data (Table 12.1) shows that while some species—*E. coli* and *Zea mays* (corn), for example—do have approximately 25% of each type of nucleotide, most do not, and that the DNA of various species differs. For example, in humans the A and T percentages are about 31%, but in fruit flies these percentages are about 27%. Therefore, the nucleotide content of DNA is not fixed across species, and DNA does have the *variability* between species required for it to be the genetic material.

Within each species, however, DNA was found to have the *constancy* required of the genetic material—that is, all members of a species have the same base composition. Also, the percentage of A always equals the percentage of T, and the percentage of G equals the percentage of C. It follows that if the percentage of A + T equals 40%, then the percentage of G + C equals 60%. These relationships are called Chargaff's rules.

Table 12.1 DNA Composition in Various Species

DNA Composition in Various Species (%)				
Species	**A**	**T**	**G**	**C**
Homo sapiens (human)	31.0	31.5	19.1	18.4
Drosophila melanogaster (fruit fly)	27.3	27.6	22.5	22.5
Zea mays (corn)	25.6	25.3	24.5	24.6
Neurospora crassa (fungus)	23.0	23.3	27.1	26.6
Escherichia coli (bacterium)	24.6	24.3	25.5	25.6
Bacillus subtilis (bacterium)	28.4	29.0	21.0	21.6

Chargaff's rules:

- The amount of A, T, G, and C in DNA varies from species to species.
- In each species, the amount of A = T and the amount of G = C.

Although only one of four bases is possible at each nucleotide position in DNA, the sheer number of bases and the length of most DNA molecules are more than sufficient to provide for variability. For example, it has been calculated that each human chromosome typically contains about 140 million base pairs. This provides for a staggering number of possible sequences of nucleotides. Because

ISTUDY

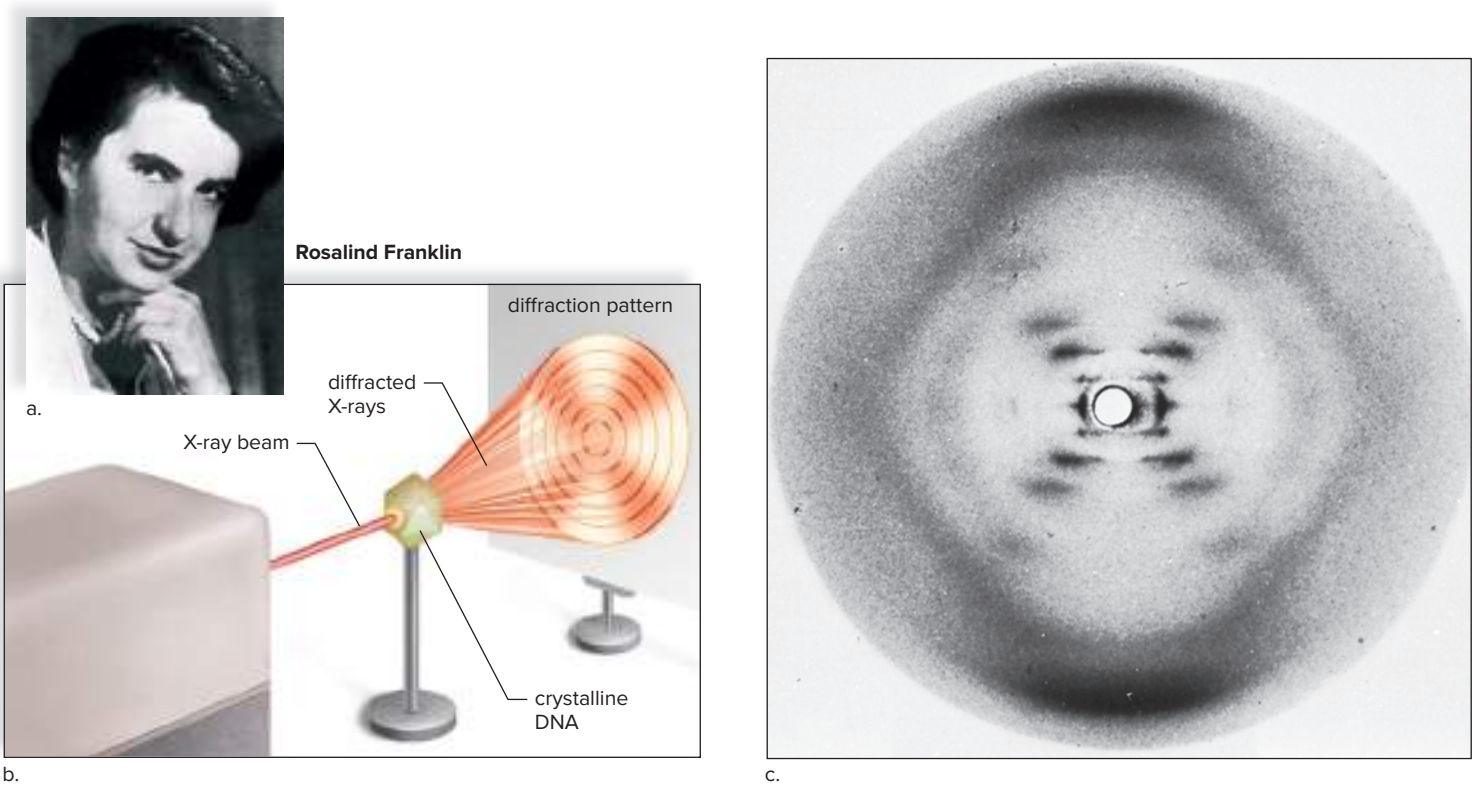


Figure 12.4 X-ray diffraction of DNA. **a.** Rosalind Franklin. **b.** When a crystal is X-rayed, the way in which the beam is diffracted reflects the pattern of the molecules in the crystal. The closer together two repeating structures are in the crystal, the farther from the center the beam is diffracted. **c.** The diffraction pattern of DNA produced by Rosalind Franklin. The crossed (X) pattern in the center told investigators that DNA is a helix, and the dark portions at the top and bottom told them some feature is repeated over and over. Watson and Crick determined this feature was the hydrogen-bonded bases. (a): National Library of Medicine/Science Source; (c): Omikron/Science Source

any of the four possible nucleotides can be present at each nucleotide position, the total number of possible nucleotide sequences is $4^{(140\times106)}$, $4^{140} \times 10^6$ or $4^{140,000,000}$. This amount of variability explains how each species has its own unique base percentages.

X-Ray Diffraction of DNA

Rosalind Franklin (Fig. 12.4*a*), a researcher at King's College in London, studied the structure of DNA using X-rays. She found that if a concentrated, viscous solution of DNA is obtained, it may be separated into fibers. Under the right conditions, the fibers are enough like a crystal (a solid substance whose atoms are arranged in a definite manner) that when X-rayed, an X-ray diffraction pattern results (Fig. 12.4*b*).

The X-ray diffraction pattern of DNA shows that DNA is a double helix. The helical shape is indicated by the crossed (X) pattern in the center of the photograph in Figure 12.4*c*. The dark portions at the top and bottom of the photograph indicate that some portion of the helix is repeated. Maurice H. F. Wilkins, a colleague of Franklin's, showed one of her crystallographic patterns to James Watson, who immediately grasped its significance.

The Watson and Crick Model

In the late 1940s and early 1950s, a number of researchers were actively attempting to develop a model of how DNA, now recognized to be the genetic material, was structured. One of these teams consisted of the American James Watson and the English biophysicist Francis H. C. Crick. Using the information provided by the researchers mentioned earlier in this section, and especially Franklin's X-ray diffraction data and images, they were able to propose a model of DNA in 1953 (Fig. 12.5*a, b*). For their work, they received the Nobel Prize in Physiology or Medicine in 1962.

Based on previous work of other scientists, Watson and Crick knew that DNA is a polymer of nucleotides, but they did not know how the nucleotides were arranged within the molecule. However, by constructing models, they deduced that DNA is a **double helix** with sugar-phosphate backbones on the outside and paired bases on the inside (Fig. 12.5*c*). This arrangement fits the mathematical measurements provided by Franklin's X-ray diffraction data for the spacing between the base pairs (0.34 nm) and for a complete turn of the double helix (3.4 nm).

According to Watson and Crick's model, the two DNA strands of the double helix are *antiparallel,* meaning the sugar-phosphate groups that are chained together to make each strand are oriented in opposite directions. As seen in Figure 12.5*d,* each nucleotide possesses a phosphate group located at the 5′ position of the sugar. Nucleotides are joined together by linking the 5′ phosphate of one nucleotide to a free hydroxyl (–OH) located at the 3′ position on the sugar of the preceding nucleotide, giving the molecule directionality. *Antiparallel* simply means that while one DNA strand runs 5′ to 3′, the other strand runs in a parallel but opposite direction.

ISTUDY

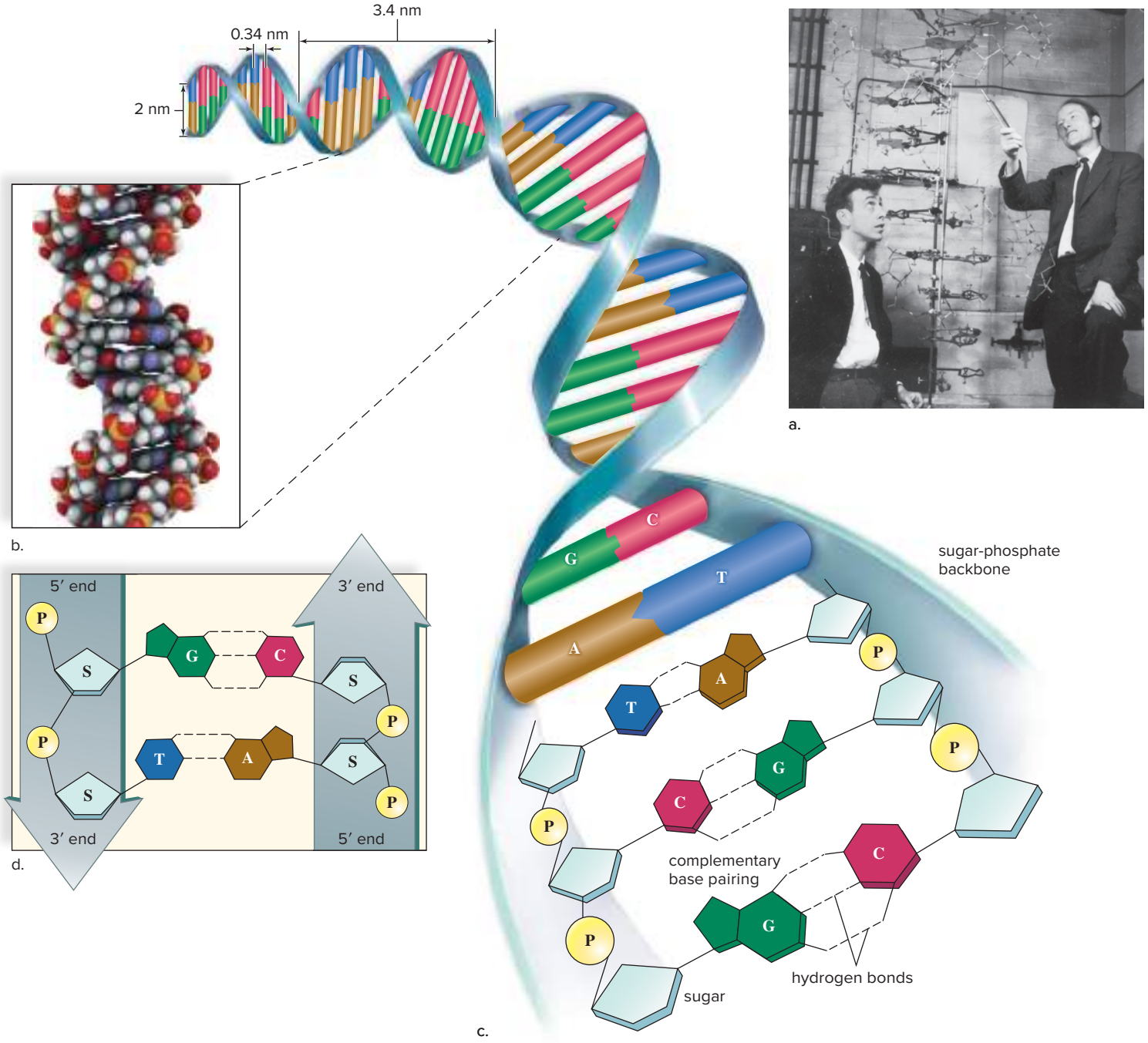


Figure 12.5 Watson and Crick model of DNA structure. **a.** James Watson (*left*) and Francis Crick (*right*) deduced the molecular configuration of DNA. **b.** Space-filling model of DNA. **c.** The double-helix molecules. **d.** The two strands of the molecule are antiparallel. The direction of the strand is said to be 5′ to 3′ when going from top to bottom. The 5′ end of a DNA strand has a phosphate group, while the 3′ end has an –OH group (not shown). (a): A. Barrington Brown/Science Source; (b): Molekuul/SPL/age fotostock

This model also agreed with Chargaff's rules, which state that A = T and G = C. Figure 12.5*d* shows that A is hydrogen-bonded to T, and G is hydrogen-bonded to C. This **complementary base pairing** means that a purine (large, two-ring base) is always bonded to a pyrimidine (smaller, one-ring base). This antiparallel pairing arrangement of the two strands ensures the bases are oriented properly, so they can interact. The consistent spacing between the two strands of the DNA was detected by Franklin's X-ray diffraction pattern, because two pyrimidines together are too narrow, and two purines together are too wide.

The information stored within DNA must always be read in the 5′ to 3′ direction. Thus, a DNA strand is usually replicated in a 5′ to 3′ direction.

Check Your Progress 12.1

1. Explain the major features of DNA structure.
2. Explain the roles of Erwin Chargaff and Rosalind Franklin in elucidating the final structure of DNA.

ISTUDY

12.2 Replication of DNA

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain why the replication of DNA is semiconservative.
2. Describe the enzymes and proteins involved in DNA replication.
3. Contrast DNA replication in eukaryotes and prokaryotes.

When Watson and Crick developed their double-helix model, they commented, "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material." The term **DNA replication** refers to the process of copying a DNA molecule. Following replication, there is usually an exact copy of the parental DNA double helix.

A template is most often a mold used to produce a shape complementary to itself. During DNA replication, each DNA strand of the parental double helix serves as a template for a new strand in a daughter molecule. In the 1950s, Matthew Meselson and Frank Stahl used radioactive isotopes of nitrogen to determine that DNA replication is an example of **semiconservative replication** because each daughter DNA double helix contains an old strand from the parental DNA double helix and a new strand.

Overview of DNA Replication

In Figure 12.6, the backbones of the parental DNA molecule are blue. Following replication, the daughter molecules each have a green backbone (new strand) and a blue backbone (old strand). Because A pairs with T, and G pairs with C, a daughter DNA double helix has the same sequence of bases as the parental DNA double helix had originally.

The process of DNA replication requires three main events: the unwinding of the DNA molecule, complementary base pairing, and the joining of the new DNA strands (Fig. 12.7).

At the molecular level, several enzymes and proteins participate in the synthesis of the new DNA strands (Table 12.2).

Unwinding of the DNA

A **DNA helicase** enzyme unwinds DNA and separates the parental strands. This creates two replication forks that move away from each other. These separated strands now become the template to create two new DNA molecules. DNA is chemically stable as a helix, but not as single strands. **Single-stranded binding proteins** (abbreviated as SSB in Fig. 12.7) attach to newly separated DNA and prevent it from re-forming the helix so replication can occur.

Table 12.2 Enzymes and Proteins Involved in DNA Replication

Protein Name	Function
DNA helicase	Separates double-stranded DNA into single strands
Single-stranded binding protein (SSB)	Binds to single-stranded DNA and prevents it from re-forming a double helix
DNA primase	Synthesizes short RNA primers
DNA polymerase	Synthesizes DNA in the leading and lagging strands; removes RNA primers, filling the gaps with more DNA; and proofreads newly made DNA
DNA ligase	Covalently attaches adjacent Okazaki fragments in the lagging strand

Complementary Base Pairing

DNA replication needs a *primer,* a short strand of RNA, to be put in place before replication can begin. **DNA primase** places short

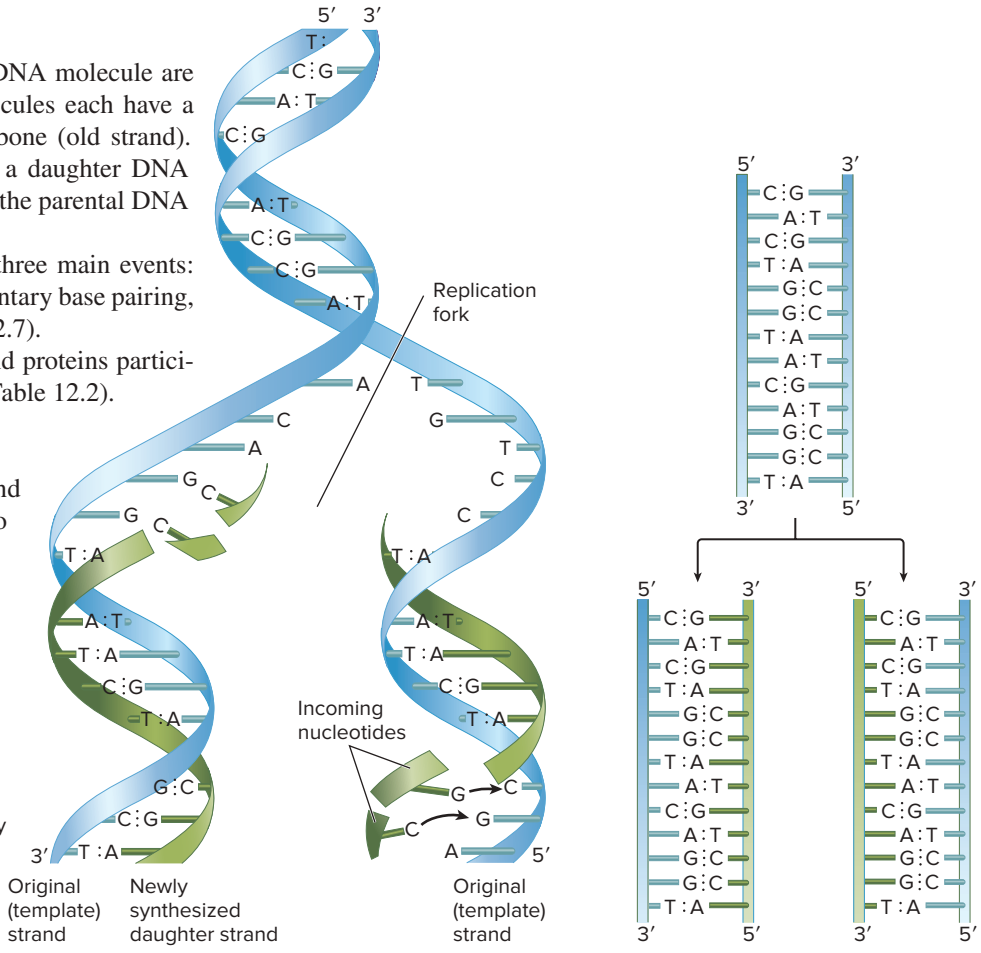


Figure 12.6 A model of semiconservative replication. After the DNA double helix unwinds, each parental strand serves as a template for the formation of the new daughter strands. Complementary free nucleotides hydrogen-bond to a matching base (e.g., A with T; G with C) in each parental strand and are joined to form a complete daughter strand. Two helices, each with a daughter and parental strand, are produced following replication.

ISTUDY

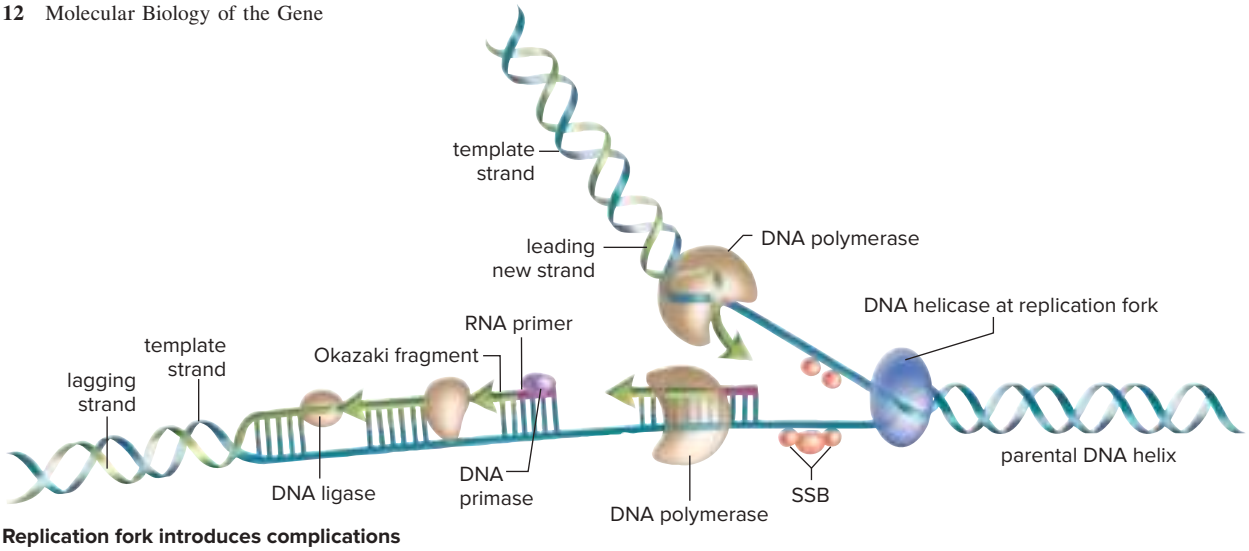


Figure 12.7 Overview of DNA replication. The major enzymes involved in DNA replication. Note that the synthesis of the new DNA molecules occurs in opposite directions due to the orientation of the original DNA strands.

primers on the strands to be replicated. **DNA polymerase** recognizes this RNA target and begins DNA synthesis, allowing new nucleotides to form complementary base pairs with the old strand and connecting the new nucleotides together in a chain. DNA polymerase also proofreads the strands and can correct any mistakes.

The parental strands are antiparallel to each other, and each of the new daughter strands must also be antiparallel to its matching parental strand. This creates a problem. DNA can only be synthesized in a 5′ to 3′ direction (see Fig. 12.5*d*). One strand, the *leading strand,* is exposed so that synthesis in a 5′ to 3′ direction is easier and replication is continuous. The other new strand in the fork must be synthesized in the opposite direction, requiring DNA polymerase to synthesize the new strand in short 5′ to 3′ segments with periodic starts and stops. This strand is called the *lagging strand.* Replication of the lagging strand is therefore made in segments called *Okazaki fragments,* after Japanese scientist Reiji Okazaki, who discovered them.

Joining of the DNA

After both new strands are made, DNA polymerase has yet another role: converting the short RNA sequences, laid down by the primase, into DNA.

Finally, the enzyme **DNA ligase** is the "glue" that binds all the Okazaki fragments together, resulting in the two double-helix molecules that are identical to each other and to the original molecule.

Recall that the DNA is copied during S phase of the cell cycle, before the start of mitosis or meiosis (see Fig. 9.1). Because the goal of these processes is either to create an exact cell copy (mitosis) or to make a gamete for reproduction (meiosis), in either case you have to double the DNA before you can separate it during cell division. DNA replication must occur before a cell can divide. Cancer, which is characterized by rapid, uncontrolled cell division, is sometimes treated with chemotherapeutic drugs that mimic one of the four nucleotides in DNA. When these are mistakenly used by the cancer cells to synthesize DNA, replication stops and the cells die off.

Prokaryotic Versus Eukaryotic Replication

The process of DNA replication is distinctly different in prokaryotic and eukaryotic cells (Fig. 12.8).

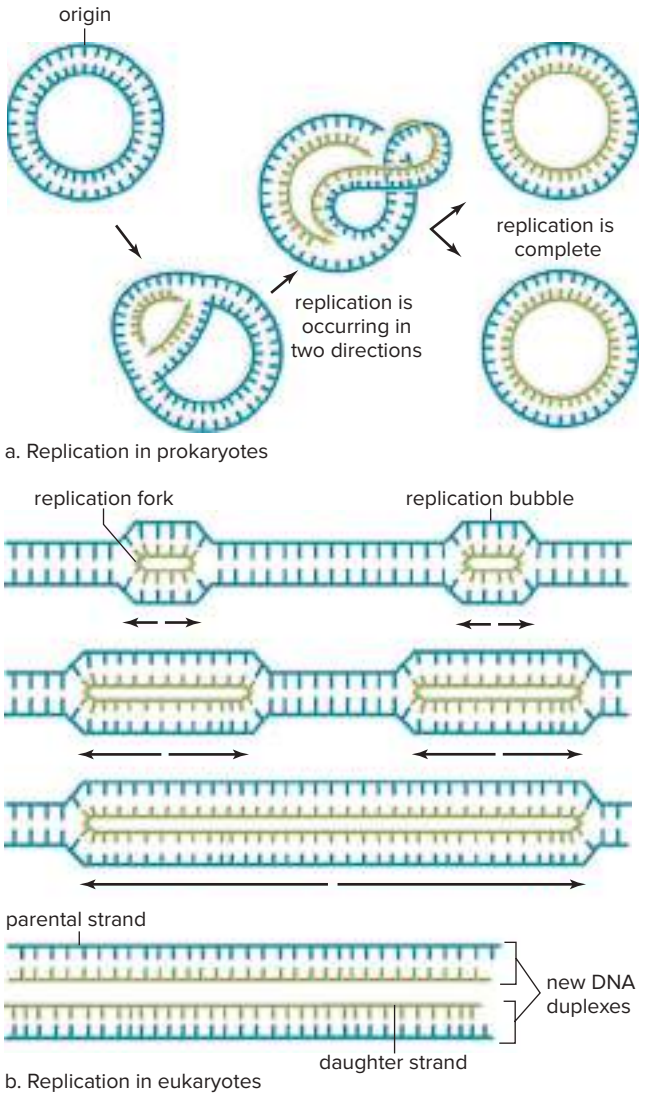


Figure 12.8 Prokaryotic versus eukaryotic replication. **a.** In prokaryotes, replication can occur in two directions at once, because the DNA molecule is circular. **b.** In eukaryotes, replication occurs at numerous replication bubbles, each with two forks. The forks move away from each other until they meet again and the two new daughter helices have been completed.

ISTUDY

Prokaryotic DNA Replication

Bacteria have a single circular loop chromosome, whose DNA must be replicated before the cell divides. In some circular DNA molecules, replication moves around the DNA molecule in one direction only. In others, as shown in Figure 12.8*a,* replication occurs in two directions. The process always occurs in the 5′ to 3′ direction.

It begins at the *origin of replication,* a specific site on the bacterial chromosome. The strands are separated and unwound, and a DNA polymerase enzyme binds to each side of the opening and begins the copying process. When the two DNA polymerases meet at a termination region, replication is halted, and the two copies of the chromosome are separated.

Bacterial cells require about 40 minutes to replicate the complete chromosome. Because bacterial cells are able to divide as often as once every 20 minutes, it is possible for a new round of DNA replication to begin even before the previous round is completed!

Eukaryotic DNA Replication

In eukaryotes, DNA replication begins at numerous origins of replication along the length of the linear chromosome, and replication bubbles spread bidirectionally until they meet. Notice in Figure 12.8*b* that there is a V shape wherever DNA is being replicated. This is called a **replication fork.**

The chromosomes of eukaryotes are long, making replication a more time-consuming process. Eukaryotes replicate their DNA at a slower rate—500 to 5,000 base pairs per minute—but there are many individual origins of replication to accelerate the process. Therefore, eukaryotic cells complete the replication of the diploid amount of DNA (in humans, over 6 billion base pairs) in a matter of hours!

The linear chromosomes of eukaryotes also pose another problem: DNA polymerase is unable to replicate the ends of the chromosomes. The ends of eukaryotic chromosomes are composed of telomeres, which are short DNA sequences that are repeated over and over. Telomeres are not copied by DNA polymerase; rather, they are added by an enzyme called telomerase, which adds the correct number of repeats after the chromosome is replicated. In stem cells, this process preserves the ends of the chromosomes and prevents the loss of DNA after successive rounds of replication. Unregulated telomerase activity can negatively affect cell function, as seen with uncontrolled cell division in cancer cells.

Accuracy of Replication

A DNA polymerase is very accurate and makes a mistake approximately once per 100,000 base pairs at most. This error rate, however, would result in many errors accumulating over the course of several cell divisions. DNA polymerase is also capable of checking for accuracy, or proofreading the daughter strand it is making. It can recognize a mismatched nucleotide and remove it from a daughter strand by reversing direction and removing several nucleotides. Once it has removed the mismatched nucleotide, it changes direction again and resumes making DNA. Overall, the error rate for the bacterial DNA polymerase is only about 1 in 100 million base pairs.

Check Your Progress **12.2**

1. Explain the three major steps in DNA replication.
2. Explain why replication must occur differently on the leading and lagging strands.
3. Compare DNA replication in prokaryotes and eukaryotes.

12.3 Gene Expression: RNA and the Genetic Code

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the function of transcription and translation.
2. Explain how the mRNA nucleotides determine the sequence of amino acids in a polypeptide.

In the early twentieth century, the English physician Sir Archibald Garrod observed that family members often have the same metabolic disorder, and he believed it was likely they all lacked the same functioning enzyme in a metabolic pathway. He introduced the phrase "inborn error of metabolism" to describe this relationship. Garrod's findings were generally overlooked until 1940, when George Beadle and Edward Tatum devised a way to confirm his hypothesis. These investigators performed a series of experiments utilizing a bread mold and found that each of their mutant molds was indeed unable to produce a particular enzyme. This led them to propose that one gene directs the synthesis of one enzyme (or protein).

While today we know that genes are also responsible for specifying not only proteins, but also some forms of RNA molecules involved in regulating **gene expression,** we will follow the path from DNA to protein so we can better understand the general process of gene expression.

RNA Carries the Information

Like DNA, **RNA (ribonucleic acid)** is a polymer composed of nucleotides. The nucleotides in RNA, however, contain the sugar ribose and the bases adenine (A), cytosine (C), guanine (G), and **uracil (U).** In RNA, the base uracil replaces the thymine found in DNA (Table 12.3). Finally, RNA is single-stranded and does not form a double helix in the same manner as DNA (Fig. 12.9).

There are three major classes of RNA that function in protein synthesis:

- **Messenger RNA (mRNA)** takes a message from DNA in the nucleus to the ribosomes in the cytoplasm.

Table 12.3 RNA Structure Compared to DNA Structure

	RNA	DNA
Sugar	Ribose	Deoxyribose
Bases	Adenine, guanine, uracil, cytosine	Adenine, guanine, thymine, cytosine
Strands	Single-stranded	Double-stranded with base pairing
Helix	No	Yes

ISTUDY

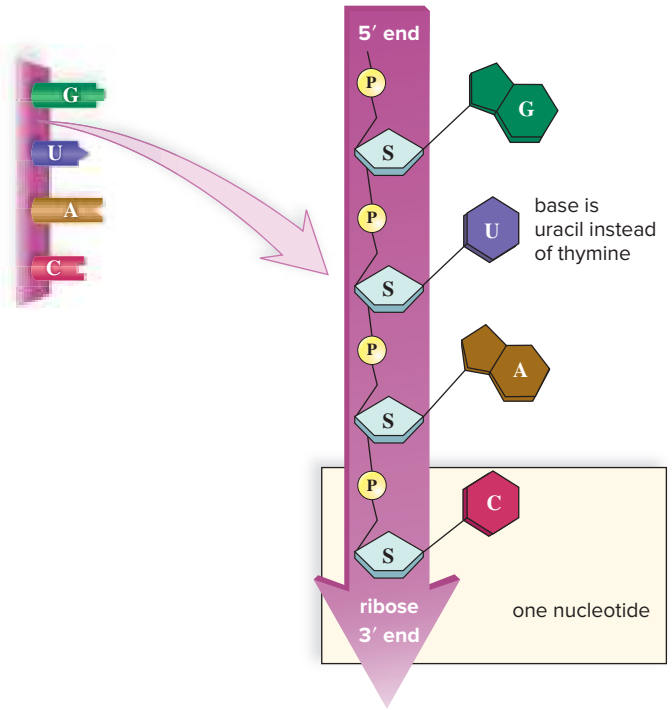


Figure 12.9 Structure of RNA. Like DNA, RNA is a polymer of nucleotides. RNA, however, is single-stranded, the pentose sugar (S) is ribose, and uracil replaces thymine as one of the bases.

- **Transfer RNA (tRNA)** transfers amino acids to the ribosomes.
- **Ribosomal RNA (rRNA)**, along with ribosomal proteins, makes up the ribosomes, where polypeptides are synthesized.

As we will see in Section 13.2, there are other forms of RNA that are involved in regulating the process of gene expression.

Overview of Gene Expression

In the genetic flow of information, two major steps are needed to convert the information stored in DNA into a protein that supports body function (Fig. 12.10). First, the DNA undergoes **transcription** (L. *trans,* "across"; *scriptio,* "a writing"), a process by which an RNA molecule is produced based on a DNA template. DNA is transcribed, or copied base by base, into mRNA, tRNA, and rRNA.

Second, during **translation** (L. *trans,* "across"; *latus,* "carry or bear"), the mRNA transcript is read by a ribosome and converted into the sequence of amino acids in a polypeptide. Like a translator who understands two languages, the cell changes a nucleotide sequence into an amino acid sequence. Together, the flow of information from DNA to RNA to protein to trait is known as the **central dogma** of molecular biology.

The Genetic Code

Now that we know that the DNA sequence within a gene is transcribed into an RNA molecule and, for genes that code for proteins, the mRNA sequence determines the sequence of amino acids in a protein, it becomes necessary to identify the specific **genetic code** for each of the 20 amino acids found in proteins. Although scientists knew that DNA somehow directed protein production, they did not initially know specifically how the code was translated. This discovery was made in the 1960s.

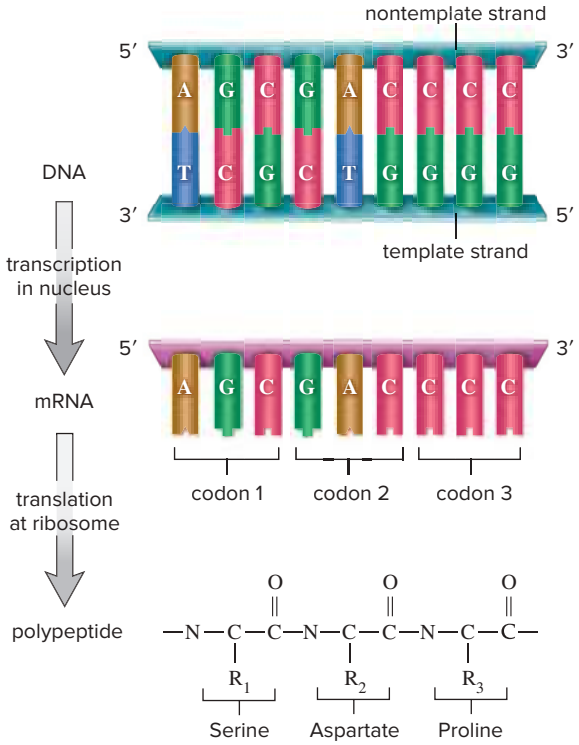


Figure 12.10 The flow of genetic information in a cell. One strand of DNA acts as a template for mRNA synthesis, and the sequence of bases in mRNA determines the sequence of amino acids in a polypeptide.

Logically, the genetic code would have to be at least a triplet of nucleotides. This means that each coding unit, or **codon,** would need to be made up of three nucleotides. The reason is that fewer nucleotides would not provide sufficient variety to encode 20 different amino acids.

In 1961, Marshall Nirenberg and J. Heinrich Matthei performed an experiment that laid the groundwork for cracking the genetic code. First, they found that a cellular enzyme could be used to construct a synthetic RNA (one that does not occur in cells), and then they found that the synthetic RNA polymer could be translated in a test tube that contained the cytoplasmic contents of a cell. Their first synthetic RNA was composed only of uracil, and the protein that resulted was composed only of the amino acid phenylalanine. Therefore, the mRNA codon for phenylalanine was known to be UUU. Later, they were able to translate just three nucleotides at a time. In that way, it was possible to assign an amino acid to each of the mRNA codons (Fig. 12.11).

Like the periodic table and other major works, the genetic code seen in Figure 12.11 is a masterpiece of scientific discovery, because it is a key that unlocks the very basis of biological life. Here are some of its features:

- The genetic code is *degenerate.* This term means that most amino acids have more than one codon; leucine, serine, and arginine have six different codons, for example. The degeneracy (redundancy) of the code helps protect against potentially harmful mutations.
- The genetic code is *unambiguous.* Each triplet codon has only one meaning.
- The code has *start and stop signals.* There is only one start signal, but there are three stop signals.

ISTUDY

First Base	Second Base U	Second Base C	Second Base A	Second Base G	Third Base
U	UUU phenylalanine	UCU serine	UAU tyrosine	UGU cysteine	U
	UUC phenylalanine	UCC serine	UAC tyrosine	UGC cysteine	C
	UUA leucine	UCA serine	UAA *stop*	UGA *stop*	A
	UUG leucine	UCG serine	UAG *stop*	UGG tryptophan	G
C	CUU leucine	CCU proline	CAU histidine	CGU arginine	U
	CUC leucine	CCC proline	CAC histidine	CGC arginine	C
	CUA leucine	CCA proline	CAA glutamine	CGA arginine	A
	CUG leucine	CCG proline	CAG glutamine	CGG arginine	G
A	AUU isoleucine	ACU threonine	AAU asparagine	AGU serine	U
	AUC isoleucine	ACC threonine	AAC asparagine	AGC serine	C
	AUA isoleucine	ACA threonine	AAA lysine	AGA arginine	A
	AUG *(start)* methionine	ACG threonine	AAG lysine	AGG arginine	G
G	GUU valine	GCU alanine	GAU aspartate	GGU glycine	U
	GUC valine	GCC alanine	GAC aspartate	GGC glycine	C
	GUA valine	GCA alanine	GAA glutamate	GGA glycine	A
	GUG valine	GCG alanine	GAG glutamate	GGG glycine	G

Figure 12.11 Messenger RNA codons. In this chart, notice that each of the codons (in boxes) is composed of three letters representing the first base, second base, and third base. For example, find the box where C for the first base and A for the second base intersect. You will see that U, C, A, or G can be the third base. The bases CAU and CAC are codons for histidine; the bases CAA and CAG are codons for glutamine.

With a few exceptions, the genetic code (Fig. 12.11) is universal to all living organisms. In 1979, however, researchers discovered that the genetic code used within the mitochondria, chloroplasts, and some archaebacteria differs slightly from the more familiar genetic code.

The universal nature of the genetic code provides strong evidence that all living organisms share a common evolutionary heritage. Furthermore, because basically the same genetic code is used by all living organisms, it is possible to transfer genes from one organism to another. In Section 14.2, we will explore the implications of this with regard to the development of biotechnology products by many commercial and medicinal companies.

Check Your Progress 12.3

1. Examine the flow of genetic information in a cell.
2. List the three major classes of RNA and provide a function for each.
3. Explain why the term *degenerate* is used to describe the genetic code.

12.4 Gene Expression: Transcription

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among the events of transcription that occur during formation of an mRNA molecule.
2. Describe how eukaryotic mRNA molecules are processed and exported to the cytoplasm.

During *transcription,* a segment of the DNA serves as a template for production of an RNA molecule. Although all types of RNA are produced by transcription, in this section we will focus on mRNA transcription, since this is the type of RNA that eventually leads to the production of a protein.

Overview of Transcription

The sequences of bases in a gene are transcribed into an mRNA molecule based on complementary base pairing: The T base in the DNA pairs with A in the mRNA, G with C, and A with U (note that uracil replaces T in the newly formed mRNA) (Fig. 12.12). When a gene is transcribed, a segment of the DNA helix unwinds and unzips, and complementary RNA nucleotides pair with DNA nucleotides of the strand opposite the gene. This strand is known as the *template strand;* the other strand is the nontemplate, or coding, strand. An **RNA polymerase** moves along the template strand in the 5′ direction. Like DNA polymerase, an RNA polymerase adds a nucleotide only to the 3′ end of the polymer under construction.

It is important to note that either strand of the DNA can be a template strand. In Figure 12.12, RNA polymerase uses one strand as the template, but for another gene, the opposite strand may be the template.

Stages of Transcription

Transcription begins when RNA polymerase attaches to a region of DNA called a promoter (Fig. 12.12). A **promoter** defines the start of transcription, the direction of transcription, and the strand to be transcribed. The binding of RNA polymerase to the promoter is the *initiation* of transcription. The RNA–DNA association is not as stable as the two strands in the DNA helix. Therefore, only the newest portion of an RNA molecule that is associated with RNA polymerase is bound to the DNA, and the rest dangles off to the side.

Elongation of the mRNA molecule occurs as the RNA polymerase reads down the DNA template strand in the 3′ to 5′ direction and continues until RNA polymerase comes to a DNA stop sequence, where *termination* occurs. The stop sequence causes RNA polymerase to stop transcribing the DNA and to release the mRNA molecule, now called an **mRNA transcript.**

It is not necessary for RNA polymerase (Fig. 12.13*a*) to finish making one mRNA transcript before it starts another. As long as they have access to the gene's promoter, many RNA polymerase molecules can be working one after the other to produce mRNA transcripts at the same time (Fig. 12.13*b*). This allows the cell to produce many thousands of copies of the same mRNA molecule, and eventually many copies of the same protein, within a shorter

ISTUDY

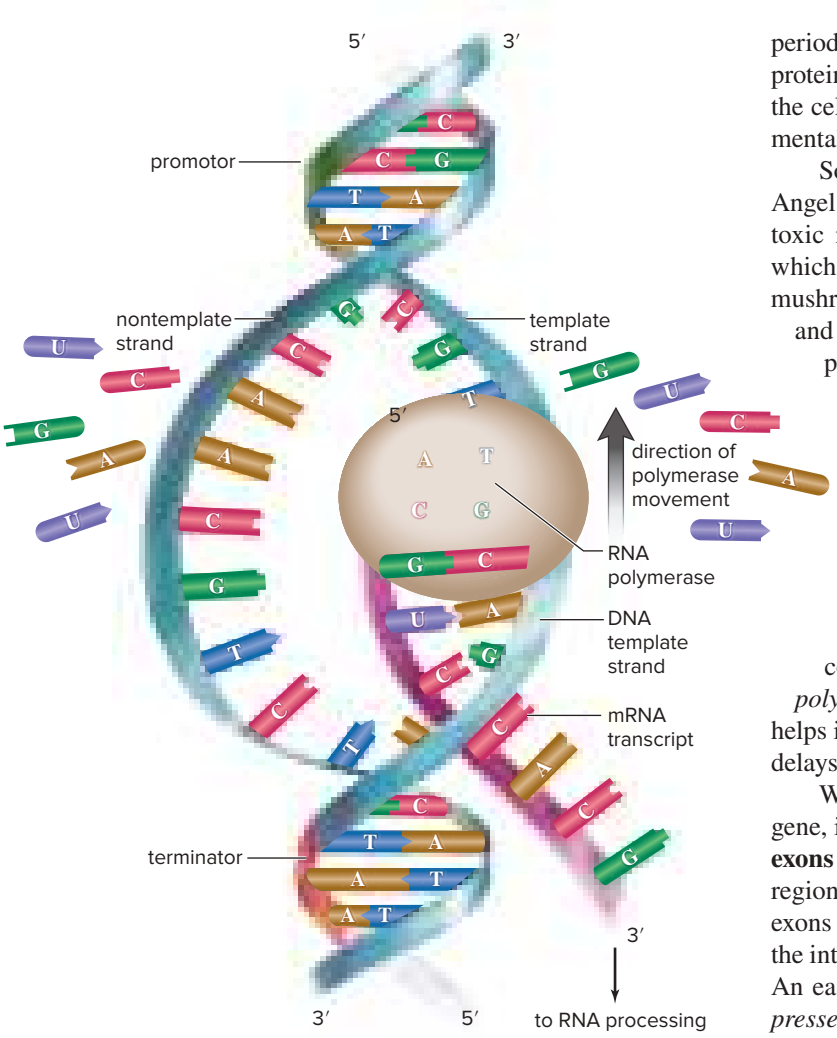


Figure 12.12 Transcription. During transcription, the complementary mRNA is made from a DNA template. At the point of attachment of RNA polymerase, the DNA helix unwinds and unzips, and complementary RNA nucleotides are joined together. After RNA polymerase has passed by, the DNA strands rejoin and the mRNA transcript dangles to the side.

period of time than if a single mRNA copy were used to direct protein synthesis. This ability to rapidly express the gene enables the cell (and the organism) to better respond to changing environmental conditions and have a greater chance at survival.

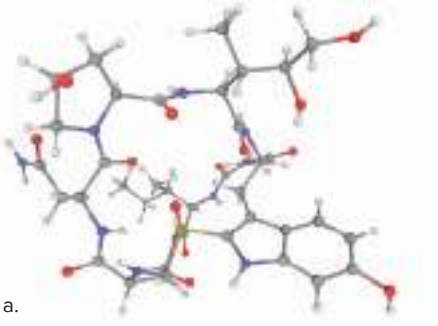
Some species of *Amanita* mushrooms, such as the Destroying Angel and Death Cap (Fig. 12.13*c*), are so named because of their toxic nature. The *Amanita* produces a toxin called α-amanitin, which inhibits the function of RNA polymerase. The *Amanita* mushrooms are responsible for 95% of all mushroom poisonings and can often lead to a quick death due to the destruction of a person's liver.

RNA Molecules Undergo Processing

A newly formed RNA transcript, called a pre-mRNA, is modified or processed before leaving the eukaryotic nucleus. For example, the molecule receives a cap at the 5′ end and a tail at the 3′ end (Fig. 12.14). The *cap* is a modified guanine (G) nucleotide that helps tell a ribosome where to attach when translation begins. The tail consists of a chain of 150–200 adenine (A) nucleotides. This *poly-A tail* facilitates the transport of mRNA out of the nucleus, helps initiate loading of ribosomes and the start of translation, and delays degradation of mRNA by hydrolytic enzymes.

When the mRNA is first made by RNA polymerase from the gene, it is in a rough form. Called pre-mRNA, it contains a mix of **exons** (protein-coding regions) and **introns** (non-protein-coding regions), particularly in multicellular eukaryotes. Because only the exons of the pre-mRNA will be contained in the mature mRNA, the introns, which occur in between the exons, must be spliced out. An easy way to remember this is with the phrase "exons are *expressed* and introns are *in the way.*"

In lower eukaryotes, introns are removed by "self-splicing"—that is, the intron itself has the capability of enzymatically splicing itself out of a pre-mRNA. In higher eukaryotes, the RNA splicing is done by *spliceosomes,* which contain *small nuclear RNAs* (*snRNAs*). By means of complementary base pairing, snRNAs are capable of identifying the introns to be removed. A spliceosome utilizes a **ribozyme** (enzyme made of RNA rather than just protein) to cut and remove the introns. Following splicing of the exons


a.

b. 65,000×

c.

Figure 12.13 RNA polymerase. **a.** The molecular structure of the RNA polymerase. **b.** Numerous RNA transcripts extend from a horizontally oriented gene in an amphibian egg cell. The strands get progressively longer because transcription begins to the left. **c.** The α-amanitin toxin from *Amanita* mushrooms inhibits the activity of the RNA polymerase.
(a): Laguna design/SPL/Alamy Stock Photo; (b): Professor Oscar L. Miller/Science Source; (c): MyLoupe/Universal Images Group/Getty Images

ISTUDY

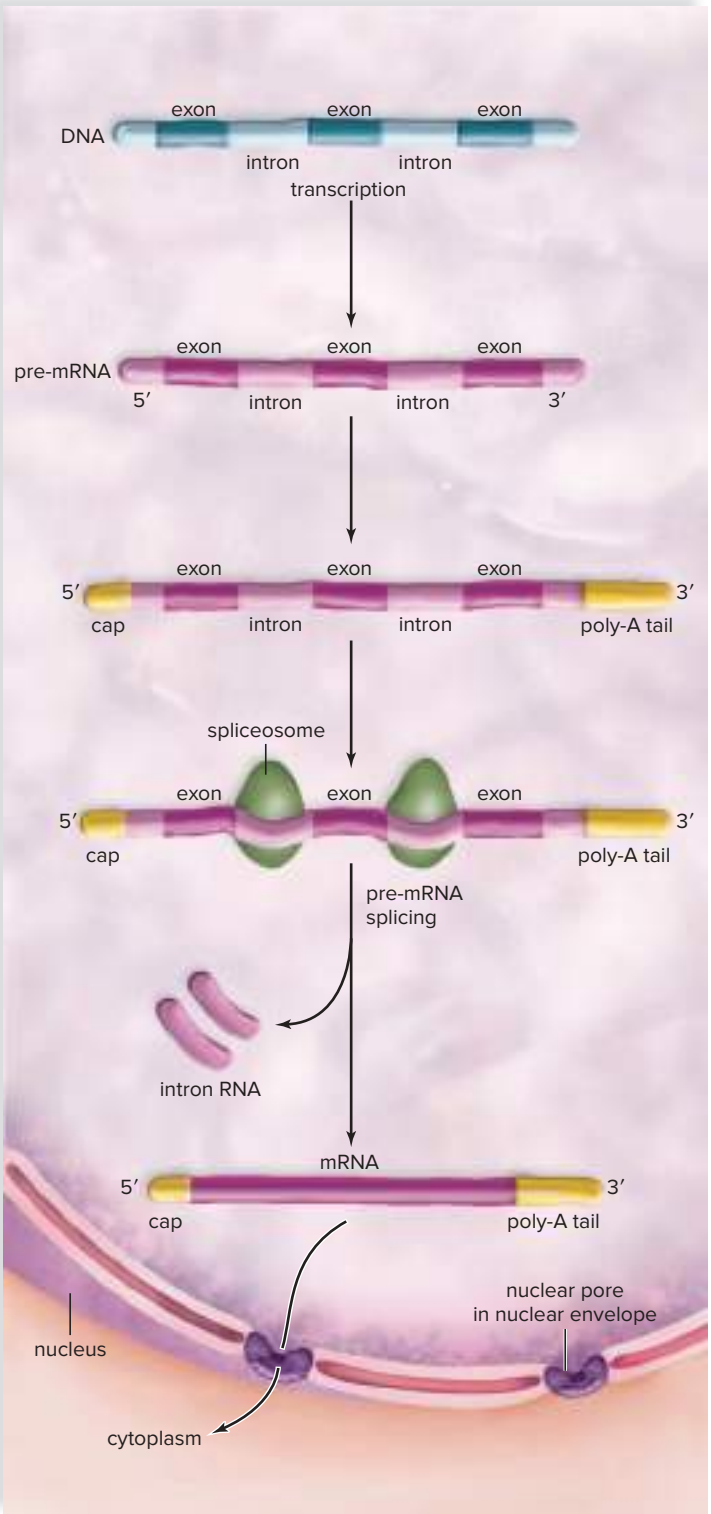


Figure 12.14 Messenger RNA (mRNA) processing in eukaryotes. DNA contains both exons (protein-coding sequences) and introns (non-protein-coding sequences). Both of these are transcribed and are present in pre-mRNA. During processing, a cap and a poly-A tail (a series of adenine nucleotides) are added to the molecule. Also, introns get cut out and the exons get spliced together by complexes called spliceosomes. Once processing is complete, the mRNA molecule is ready to leave the nucleus.

together and the addition of the 5′ cap and 3′ poly-A tail, an mRNA is ready to leave the nucleus and be translated into a protein.

Function of Introns

For many years, scientists thought that introns were simply wasted space within genes. Now, we realize they serve several key functions in the cell. The presence of introns allows a cell to choose which exons will go into a particular mRNA (see Section 13.2). Just because an mRNA has all the exons in its pre-mRNA doesn't mean they will all make it to the final product. For example, if a gene has three exons, then depending on cell need and environmental conditions, it may produce an mRNA with exons 1 and 2 only, or 1 and 3 only, or 1, 2, and 3. This ability is called *alternative mRNA splicing,* and it increases the flexibility and efficiency of the cell. The snRNAs of the spliceosomes that excise the introns play an important role in alternative splicing in eukaryotes.

Some introns give rise to *microRNAs* (*miRNAs*), which are small molecules involved in regulating the translation of mRNAs. These molecules bind with the mRNA through complementary base pairing and, in that way, prevent translation from occurring.

It is also recognized that the presence of introns encourages crossing-over during meiosis, and this permits a phenomenon termed *exon shuffling,* which can play a role in the evolution of new genes.

Check Your Progress **12.4**

1. Explain the role of RNA polymerase.
2. Describe the three major modifications that occur during the processing of an mRNA.
3. Distinguish between the introns and exons of a gene.

12.5 Gene Expression: Translation

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the roles of mRNA, tRNA, and rRNA in translating the genetic code.
2. Examine the stages of translation and the events that occur during each stage.

Translation, which takes place in the cytoplasm of eukaryotic cells, is the second step needed to express a gene into a protein. During translation, the sequence of codons (nucleotide triplets) in the mRNA is read by a ribosome, which connects the sequence of amino acids dictated by the mRNA into a polypeptide. The process is called translation because it requires the conversion of information from a nucleic acid language (DNA and RNA) into an amino acid language (protein).

The Role of Transfer RNA

Transfer RNA (tRNA) molecules transfer amino acids to the ribosomes. A tRNA molecule is a single-stranded nucleic acid that doubles back on itself to create regions where complementary

ISTUDY

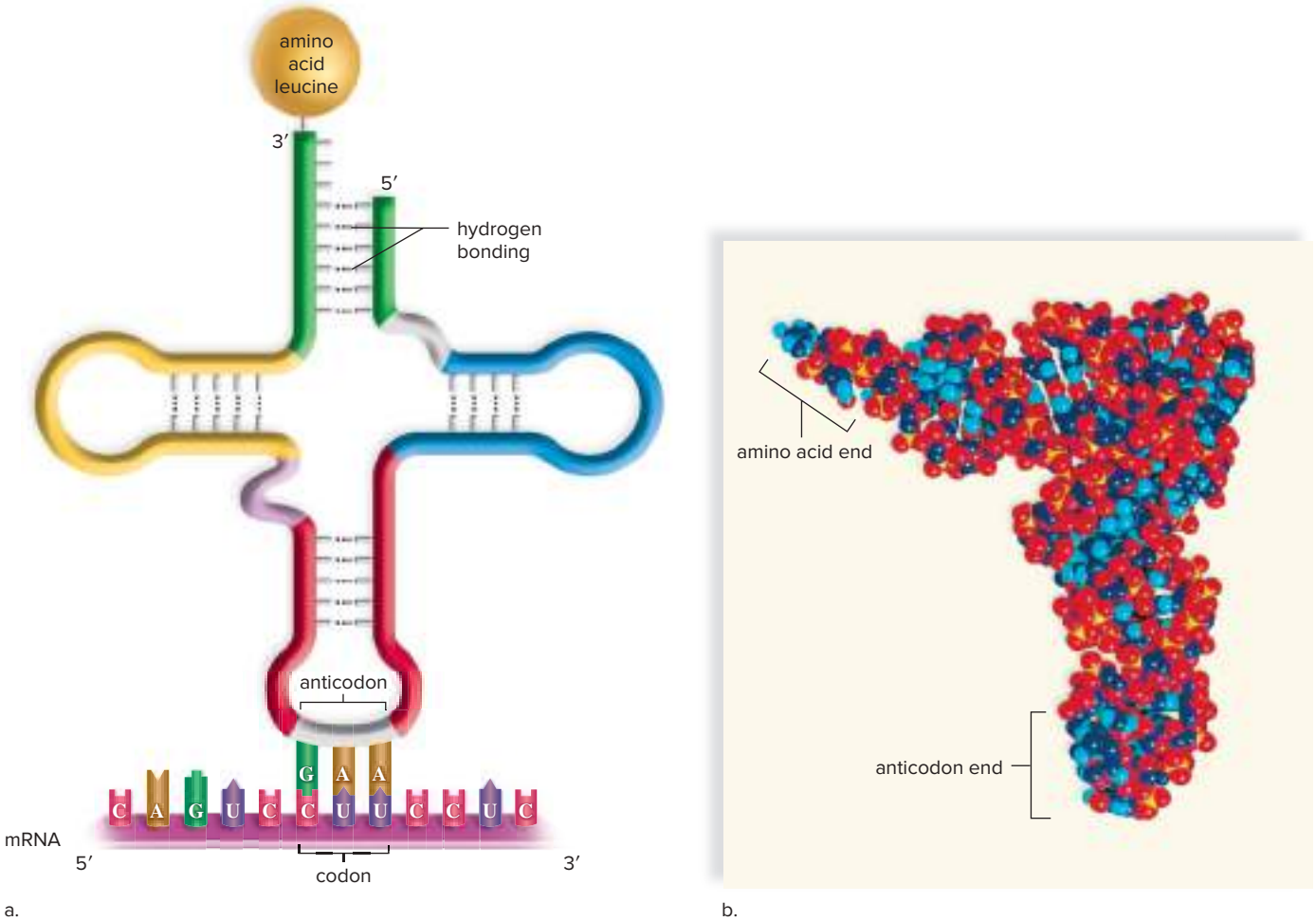


Figure 12.15 Structure of a transfer RNA (tRNA) molecule. **a.** Complementary base pairing indicated by hydrogen bonding occurs between nucleotides within the molecule, and this causes it to form its characteristic loops. The anticodon that base-pairs with a particular messenger RNA (mRNA) codon occurs at one end of the folded molecule; the other two loops help hold the molecule at the ribosome. An appropriate amino acid is attached at the 3′ end of the molecule in the cytoplasm by a tRNA charging enzyme. For this mRNA codon and tRNA anticodon, the specific amino acid is leucine. **b.** Space-filling model of tRNA molecule. (b): Petarg/123RF

bases are hydrogen-bonded to one another. The structure of a tRNA molecule is generally drawn as a flat cloverleaf (Fig. 12.15*a*), but a space-filling model shows the molecule's actual three-dimensional shape (Fig. 12.15*b*).

There is at least one tRNA molecule for each of the 20 amino acids found in proteins. The amino acid binds to the 3′ end. The opposite end of the molecule contains an **anticodon,** a group of three bases that is complementary and antiparallel to a specific mRNA codon. For example, a tRNA that has the anticodon 5′ AAG 3′ binds to the mRNA codon 5′ CUU 3′ and carries the amino acid leucine. In the genetic code, 61 codons specify amino acids; the other 3 serve as stop sequences (see Fig. 12.11).

Approximately 40 different tRNA molecules are found in most cells. There are fewer tRNAs than codons, because some tRNAs can pair with more than one codon. In 1966, Francis Crick observed this phenomenon and called it the *wobble hypothesis.* He stated that the first two positions in a tRNA anticodon pair obey the A–U/G–C configuration rule. However, the third position can be variable. Some tRNA molecules can recognize as many as four separate codons differing only in the third nucleotide. The wobble effect helps ensure that despite changes in DNA base sequences, the resulting sequence of amino acids will produce a correct protein. This is one of the reasons the genetic code is said to be degenerate.

How does the correct amino acid become attached to the correct tRNA molecule? This task is carried out by amino acid–charging enzymes, generically called aminoacyl-tRNA synthetases. Just as a key fits a lock, each enzyme has a recognition site for a particular amino acid to be joined to a specific tRNA. For example, leucine-tRNA synthetase attaches the leucine amino acid to a tRNA with the correct anticodon. This is an energy-requiring process that uses ATP. A tRNA with its amino acid attached is termed a *charged tRNA.* Once the amino acid–tRNA complex is formed, it is added to the large pool of charged tRNAs that exist in the cytoplasm, where it can now be accessed by a ribosome during protein synthesis.

ISTUDY

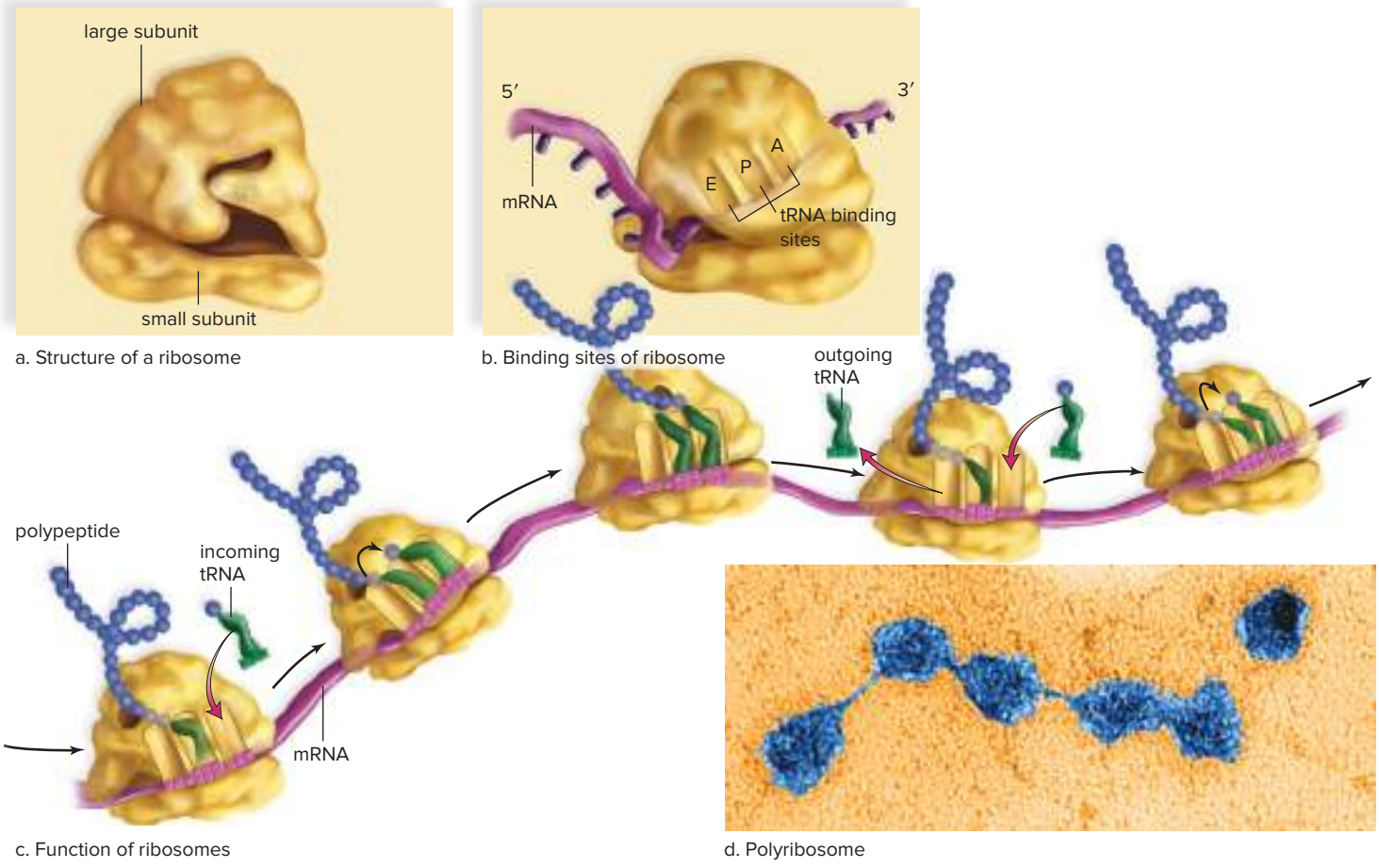


Figure 12.16 Ribosome structure and function. **a.** Side view of a ribosome shows a small subunit and a large subunit. **b.** Frontal view of a ribosome shows its binding sites. mRNA is bound to the small subunit, and the large subunit has three binding sites for tRNAs. **c.** Overview of protein synthesis. The tRNA bearing the growing polypeptide passes the entire chain to the new amino acid carried by the tRNA occupying the A site. The ribosome shifts, and freed of its burden, the "empty" tRNA exits. The new peptide-bearing tRNA moves over one binding site, making the A site accessible once again to a new tRNA. This cycle is repeated until the ribosome reaches the termination codon. **d.** Electron micrograph of a polyribosome, a number of ribosomes all translating the same mRNA molecule. (d): Science Source

The Role of Ribosomal RNA

The ribosome is the site of protein synthesis in a cell. The structure of the ribosome is well-suited to its function in the cell.

Structure of a Ribosome

In eukaryotes, ribosomal RNA (rRNA) is produced from a DNA template in the nucleolus of a nucleus. The rRNA is packaged with a variety of proteins into two ribosomal subunits, one of which is larger than the other. The subunits then move separately through nuclear envelope pores into the cytoplasm, where they join together at the start of translation (Fig. 12.16*a*). Once translation begins, ribosomes can remain in the cytoplasm, or they can become attached to endoplasmic reticulum.

Function of a Ribosome

Both prokaryotic and eukaryotic cells contain thousands of ribosomes per cell, because they play such a significant role in protein synthesis. Ribosomes have a binding site for mRNA and three binding sites for transfer RNA (tRNA) molecules (Fig. 12.16*b*). The tRNA binding sites facilitate complementary base pairing between tRNA anticodons and mRNA codons. The large ribosomal subunit has enzyme activity from rRNA (a ribozyme) that creates the peptide bond between adjacent amino acids. This peptide bond is created many times to produce a polypeptide, which in turn folds into its three-dimensional shape and becomes a protein.

When a ribosome moves down an mRNA molecule, the polypeptide increases by one amino acid at a time (Fig. 12.16*c*). Translation terminates at a stop codon. Once translation is complete, the polypeptide dissociates from the translation complex and folds into its normal shape. Recall from Section 3.4 that a polypeptide twists and bends into a definite shape based on the makeup of its amino acids. This folding process begins as soon as the polypeptide emerges from a ribosome. Chaperone molecules that are often present in the cytoplasm and the ER ensure that protein folding proceeds as it should. For proteins that contain more than one polypeptide, each subunit is folded first, and then subunits join together into a final, functional protein complex.

Like RNA polymerase during transcription, multiple ribosomes often attach and translate the same mRNA at one time. As soon as the initial portion of mRNA has been translated by one ribosome, and the ribosome has begun to move down the mRNA, another ribosome can attach to the mRNA. The entire complex of mRNA and

ISTUDY

multiple ribosomes is called a **polyribosome** (Fig. 12.16*d*), and it greatly increases the efficiency of translation.

Translation Requires Three Steps

During translation, the codons of an mRNA base-pair with the anticodons of tRNA molecules carrying specific amino acids. The order of the codons determines the order of the tRNA molecules at a ribosome and the corresponding sequence of amino acids in a polypeptide. The process of translation must be extremely orderly, so that the amino acids of a polypeptide are sequenced correctly. Even a single amino acid change has the potential to dramatically affect a protein's function, as is the case with individuals who carry the alleles for sickle-cell disease.

Protein synthesis involves three steps: initiation, elongation, and termination. Enzymes are required for each of the three steps to function properly. The first two steps, initiation and elongation, require energy.

Initiation

Initiation is the step that brings all the translation components together. Proteins called initiation factors are required to assemble the small ribosomal subunit, mRNA, initiator tRNA, and the large ribosomal subunit for the start of protein synthesis.

Initiation is shown in Figure 12.17. In prokaryotes, a small ribosomal subunit attaches to the mRNA in the vicinity of the *start codon* (AUG). The first, or initiator, tRNA pairs with this codon.

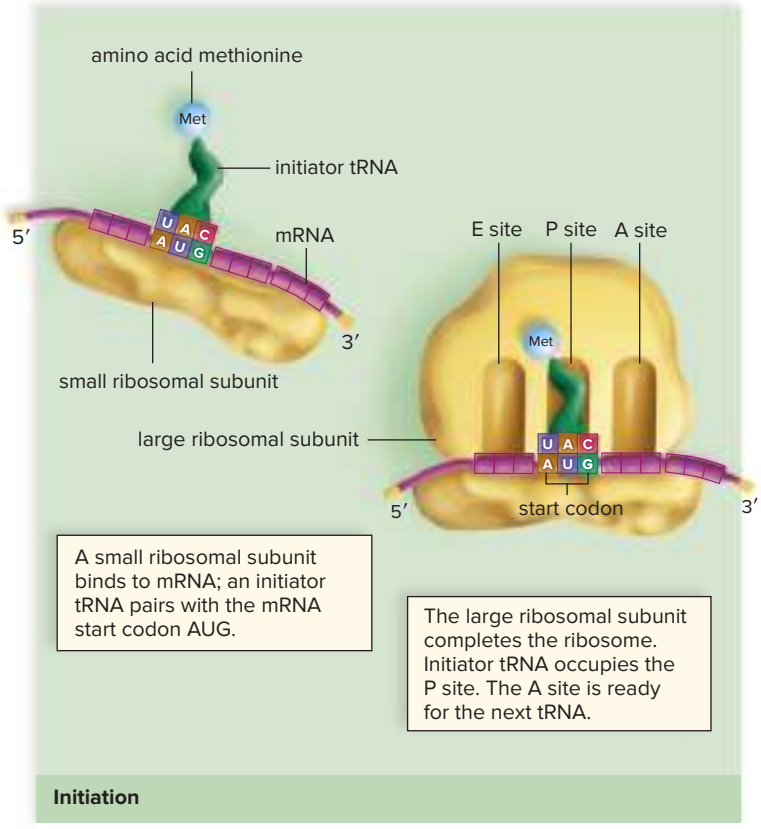


Figure 12.17 Initiation. In prokaryotes, participants in the translation process assemble as shown. The first amino acid is typically a special form of methionine.

Then, a large ribosomal subunit joins to the small subunit (Fig. 12.17). Although similar in many ways, initiation in eukaryotes is much more complex.

As already discussed, a ribosome has three binding sites for tRNAs. One of these is called the E (for "exit") site, second is the P (for "peptide") site, and the third is the A (for "amino acid") site. The initiator tRNA binds to the P site, even though it carries only the amino acid methionine (see Fig. 12.11). The A site is where tRNA carrying the next amino acid enters the ribosome, and the E site is for any tRNAs leaving a ribosome. Following initiation, translation continues with elongation and then termination.

Elongation

Elongation is the stage during protein synthesis when a polypeptide increases in length one amino acid at a time. In addition to the necessary tRNAs, elongation requires elongation factors, which facilitate the binding of tRNA anticodons to mRNA codons within a ribosome.

Elongation is shown in Figure 12.18, where a tRNA with an attached peptide is already at the P site, and a tRNA carrying its appropriate amino acid is just arriving at the A site. Once a ribosome has verified that the incoming tRNA matches the codon and is firmly in place at the A site, the entire growing peptide will be transferred to the amino acid on the tRNA in the A site. A ribozyme, an rRNA-based enzyme that is a part of the large ribosomal subunit, uses energy to transfer the growing peptide and create a new peptide bond. Following peptide bond formation, the peptide is one amino acid longer than it was before. Next, *translocation* occurs: The ribosome moves forward, and the peptide-bearing tRNA is now in the P site of the ribosome. The spent tRNA, now at the E site, exits the ribosome. A new codon is now exposed at the A site and is ready to receive another tRNA.

The complete cycle—complementary base pairing of new tRNA, transfer of peptide chain, and translocation—is repeated at a rapid rate (about 15 times each second in the bacterium *Escherichia coli*).

Eventually, the ribosome reaches a stop codon, and termination occurs, during which the polypeptide is released.

Termination

Termination is the final step in protein synthesis. During termination, as shown in Figure 12.19, the polypeptide and the assembled components that carried out protein synthesis are separated from one another.

Termination of polypeptide synthesis occurs at a *stop codon*—that is, a codon that does not code for an amino acid. Termination requires a protein called a release factor, which can bind to a stop codon and cleave the polypeptide from the last tRNA. After this occurs, the polypeptide is set free and begins to fold and take on its three-dimensional shape. The ribosome dissociates into its two subunits, which are returned to the cytoplasmic pool of large and small subunits, to be used again as necessary.

Overall, proteins do the work of the cell, whether they reside in a membrane within the cell or are free in the cytoplasm. The field of biology called **proteomics** is dedicated to understanding the structure of proteins and how they function in metabolic pathways (see Section 14.4). One of the important goals of proteomics

ISTUDY

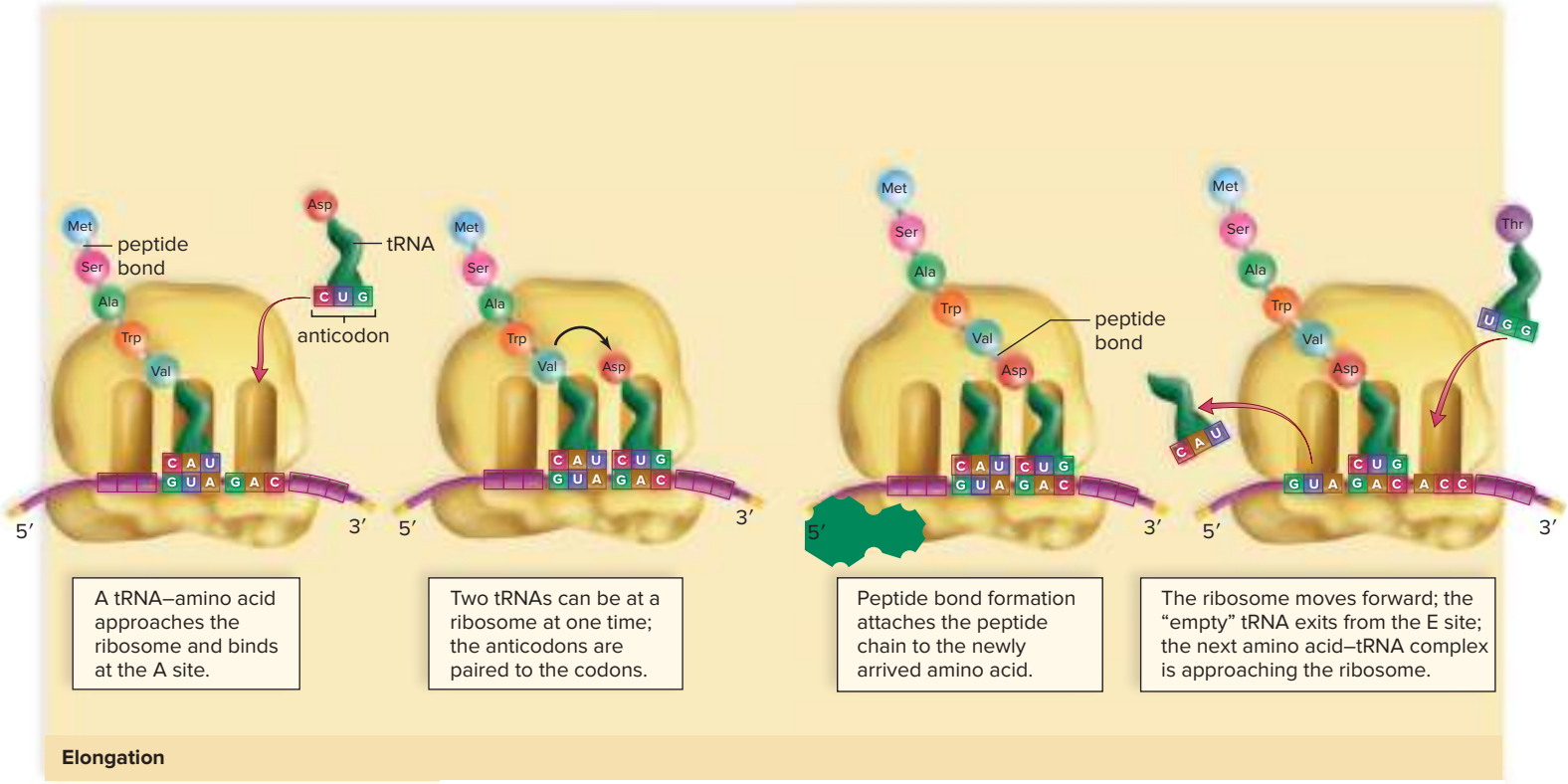


Figure 12.18 Elongation. Note that a polypeptide is already at the P site. During elongation, polypeptide synthesis occurs as amino acids are added one at a time to the growing chain.

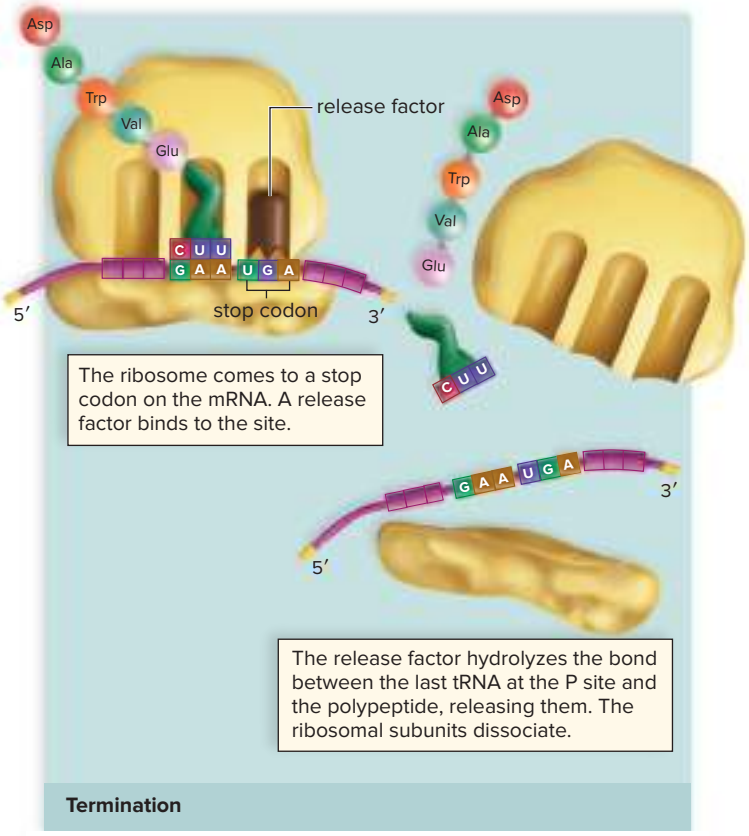


Figure 12.19 Termination. During termination, the finished polypeptide is released, as are the mRNA and the last tRNA.

is to understand how proteins are modified in the endoplasmic reticulum and the Golgi apparatus.

Summary of Gene Expression

A gene has been expressed once its product, a protein (or an RNA), is made and is operating in the cell. For a protein, gene expression requires transcription and translation (Fig. 12.20). Often, modifications to the protein occur after translation so as to prepare the protein for its specific function in the cell (see Section 13.2).

Translation occurs at ribosomes. Some ribosomes (polyribosomes) remain free in the cytoplasm, and some become attached to rough ER. The first few amino acids of a polypeptide act as a signal peptide that indicates where the polypeptide belongs in the cell or if it is to be secreted from the cell. Polypeptides that are to be secreted enter the lumen of the ER by way of a channel and are then folded and further processed by the addition of sugars, phosphates, or lipids. Transport vesicles carry the proteins between organelles and to the plasma membrane as appropriate for that protein.

Check Your Progress 12.5

1. Explain the role of transfer RNA in translation.
2. Describe how the structure of a ribosome contributes to polypeptide synthesis.
3. Examine the events that occur during the three major steps of translation.

ISTUDY

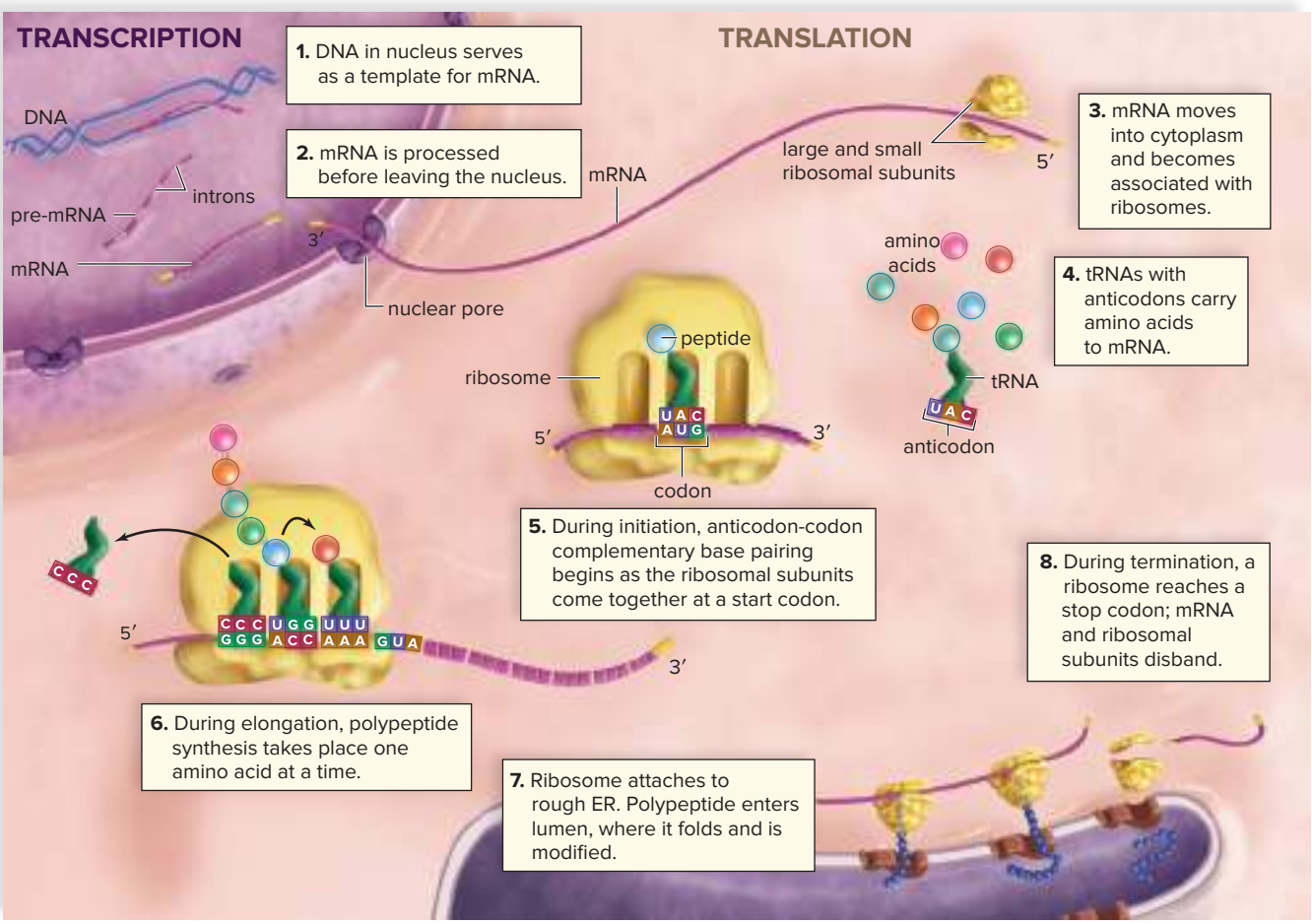
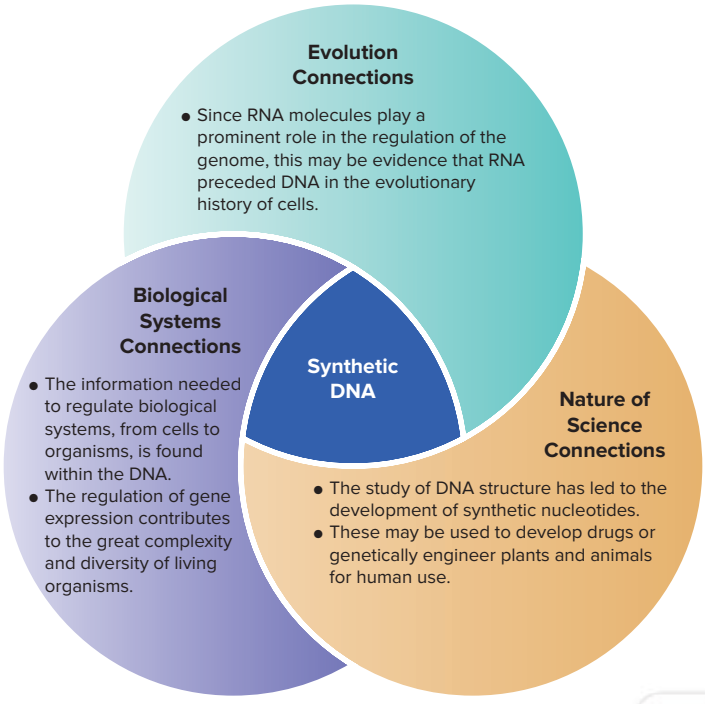


Figure 12.20 Summary of gene expression. In eukaryotes, gene expression involves the processes of transcription, RNA modifications, and translation.

CONNECTING *THE* CONCEPTS



ISTUDY

What Did You Learn?

12.1 The Genetic Material

Early work illustrated that **DNA (deoxyribonucleic acid)** is the hereditary material. Griffith injected strains of pneumococcus into mice and observed that when heat-killed S strain bacteria were injected along with live R strain bacteria, virulent S strain bacteria were recovered from the dead mice. Griffith said that the R strain had been transformed by some substance passing from the dead S strain to the live R strain. Twenty years later, Avery and his colleagues reported that the transforming substance was DNA.

To study the structure of DNA, Chargaff performed a chemical analysis of DNA and found that **adenine (A) = thymine (T)** and **guanine (G) = cytosine (C),** and that the amount of purine equals the amount of pyrimidine. Franklin prepared an X-ray photograph of DNA that showed it is a **double helix,** has repeating structural features, and has certain dimensions. Watson and Crick built a model of DNA in which the sugar-phosphate molecules made up the sides of a twisted ladder and the **complementary base pairs** were the rungs of the ladder.

12.2 Replication of DNA

DNA helicase opens the double-stranded DNA, and **single-stranded binding proteins** ensure the strands stay apart. **DNA primase** helps initiate replication by adding short primers, which are recognized by **DNA polymerase,** and begins replication. **DNA replication** on the leading strand is continuous, unlike synthesis of the lagging strand, which requires making many fragments called Okazaki fragments. Fragments are mended by **DNA ligase,** and the final result is two helices identical to the parental DNA. Replication in prokaryotes typically proceeds in both directions from one point of origin to a termination region until there are two copies of the circular chromosome. Replication in eukaryotes has many points of origin and many bubbles (places where the DNA strands are separating and replication is occurring). Replication occurs at the ends of the bubbles—at **replication forks.**

12.3 Gene Expression: RNA and the Genetic Code

Gene expression is the process by which the information within the DNA is involved in the formation of a functional protein or RNA molecule. The **central dogma** of molecular biology says that the flow of genetic information is from DNA to RNA to protein to traits.

- **RNA (ribonucleic acid)** is a nucleic acid that uses the nucleotide **uracil (U)** instead of thymine (T). **Messenger RNA (mRNA), transfer RNA (tRNA),** and **ribosomal RNA (rRNA)** are all specialized RNAs needed to make proteins. More specifically, (1) DNA is a template for its own replication and for RNA formation during **transcription,** and (2) the sequence of nucleotides in mRNA directs the correct sequence of amino acids of a polypeptide during **translation.**
- The **genetic code** is a series of **codons,** each consisting of three bases. The code is degenerate—that is, more than one codon exists for most amino acids. There are also one start and three stop codons. The genetic code is considered universal, but there are a few exceptions.

12.4 Gene Expression: Transcription

Transcription to produce messenger RNA (mRNA) begins when **RNA polymerase** attaches to the **promoter** of a gene. Elongation occurs until RNA polymerase reaches a stop sequence. An **mRNA transcript** is made, then processed, following transcription. A cap is put onto the 5′ end; a poly-A tail is put onto the 3′ end.

- **Exons** are kept and **introns** are removed in eukaryotes by spliceosomes containing **ribozymes.**
- Small nuclear RNAs (snRNAs) present in spliceosomes help identify the introns to be removed. These snRNAs play a role in alternative mRNA splicing, which allows a single eukaryotic gene to code for different proteins, depending on which segments of the gene serve as introns and which serve as exons.
- Some introns serve as microRNAs (miRNAs), which help regulate the translation of mRNAs. Research is now directed at discovering the many ways small RNAs (sRNAs) influence the production of proteins in a cell.

12.5 Gene Expression: Translation

Translation requires mRNA, transfer RNA (tRNA), and ribosomal RNA (rRNA). Each tRNA has an **anticodon** at one end and an amino acid at the other; amino acid–charging enzymes ensure that the correct amino acid is attached to the correct tRNA. When tRNAs bind with their codon at a ribosome, the amino acids are correctly sequenced in a polypeptide according to the order predetermined by DNA.

In the cytoplasm, many ribosomes move along the same mRNA at a time. Collectively, these are called a **polyribosome.**

Translation requires these steps:

- During **initiation,** mRNA, the first (initiator) tRNA, and the two subunits of a ribosome all come together in the proper orientation at a start codon.
- During **elongation,** as the tRNA anticodons bind to their codons, the growing peptide chain is transferred by peptide bonding to the next amino acid in a polypeptide.
- During **termination** at a stop codon, the polypeptide is cleaved from the last tRNA. The ribosome now dissociates.
- The field of **proteomics** studies how proteins are made, function, and are modified by other organelles.

Thinking Critically About Biology

1. How would you test a hypothesis that a genetic condition, such as cancer, is due to mistakes in transcription and translation?
2. What kind of genes do you think would be included in the category of "housekeeping" genes?
3. DNA replication is fast, virtually error-free, and coordinated with cell division. Discuss which of these three features you think is the most important.
4. A new virus has recently been discovered that infects human blood cells. The virus can be grown in the laboratory using cultured blood cells as host cells. Design an experiment using a radioactive label that would tell you if the virus contains DNA or RNA.

Making It Relevant to You

1. What changes would need to be done to the process of transcription and translation for synthetic DNA to be used to express a unique gene?
2. How might synthetic DNA be used to fight human diseases, such as cancer? How about communicable diseases like the flu?
3. From an evolutionary perspective, what concerns do you have about the development of organisms that include synthetic DNA?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

Rosa Jay/Shutterstock

13

Regulation of Gene Expression

How Do Bats Evade Disease?

Unlike humans and other mammals, bats can carry viruses but never get sick. Bats are the only order of mammals known that can remain healthy while harboring Ebola, influenza, and emerging viruses such as SARS-CoV-2.

When a human becomes infected with these viruses, genes that regulate the inflammatory response are expressed to help fight the infection. In some cases, the inflammatory response goes into overdrive, causing a virus-induced disease and possible death.

Researchers at the Duke-National University of Singapore analyzed the gene expression of key inflammatory genes in mammals. The authors found that bats have a lower expression of these genes. The genetic tolerance to viruses seems to be bat specific. This suggests that bats' immune systems have evolved to ignore these infections. Learning more about immune system gene expression in bats may lead to new ways of treating viral infections in humans.

Gene expression is the information in DNA transcribed and translated into a protein. So, how does a cell know when to undergo gene expression? Gene expression needs to be controlled or regulated, and as you will see in this chapter, many mechanisms do this.

As you read through this chapter, think about the following questions:

1. How does gene regulation differ between prokaryotes and eukaryotes?
2. How might mutations influence the ability of a cell to regulate gene expression?

Chapter Outline

13.1 Prokaryotic Regulation
13.2 Eukaryotic Regulation
13.3 Gene Mutations

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 12.10 What is the central dogma of biology?

Section 12.4 How is an mRNA transcript made?

Section 12.5 What is the role of translation in gene expression?

Following *the* Themes

CHAPTER 13 – REGULATION OF GENE EXPRESSION

Evolution	Whereas prokaryotic gene regulation operates primarily at the level of the gene, eukaryotes have evolved mechanisms to regulate gene expression at multiple levels.
Nature of Science	By understanding how cells regulate gene expression, it is possible to better understand the basis of many human diseases.
Biological Systems	Epigenetics, as well as mutations in the genetic material, may have a profound impact on the function of a cell or the health of an organism.

ISTUDY

13.1 Prokaryotic Regulation

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the structure and function of an operon in prokaryotic gene regulation.
2. Explain how the *trp* and *lac* operons of prokaryotes are regulated.
3. Distinguish between a repressible operon and an inducible operon.

Because their environment is ever changing, bacteria do not always need to express their entire complement of enzymes and proteins. In 1961, French microbiologists François Jacob and Jacques Monod showed that *Escherichia coli* is capable of regulating the expression of its genes. They observed that the genes in a metabolic pathway, called **structural genes,** are grouped on a chromosome and transcribed at the same time. Jacob and Monod, therefore, proposed the **operon** (L. *opera,* "works") model to explain gene regulation in prokaryotes. They were awarded a Nobel Prize in Physiology or Medicine in 1965 for their investigations.

An operon typically includes the following parts:

- **Regulator gene**—Normally located outside the operon, this codes for a DNA-binding protein that acts as a **repressor.** The repressor controls whether the operon is active or not.
- **Promoter**—A short sequence of DNA where RNA polymerase first attaches to begin transcription of the grouped genes. Basically, a promoter signals the start of the operon and the location where transcription begins.
- **Operator**—A short portion of DNA located before the structural genes. If a repressor is attached to the operator, then transcription cannot occur; conversely, if a repressor is not attached, then transcription can occur. In this way, the operator controls transcription of structural genes.
- **Structural genes**—These genes code for the enzymes and proteins involved in the metabolic pathway of the operon. The structural genes are transcribed as a unit.

Next, we will briefly review the findings of Jacob and Monod in their studies of two *E. coli* operons: the *trp* operon and the *lac* operon.

The *trp* Operon

Tryptophan is an essential amino acid synthesized by the enzymes coded for in the *trp* operon. Many investigators, including Jacob and Monod, found that some operons in *E. coli* usually exist in the "on" rather than "off " condition. In the *trp* operon, the regulator codes for a repressor that ordinarily is unable to attach to the operator. Therefore, RNA polymerase can bind to the promoter, and the structural genes of the operon are ordinarily expressed (Fig. 13.1). Their products, five different enzymes, are part of an anabolic pathway for the synthesis of the amino acid tryptophan.

If tryptophan is already present in the medium, the cell does not need these enzymes, and the operon is turned off by the following method. Tryptophan binds to the repressor. A change in shape now allows the repressor to bind to the operator and prevent RNA polymerase from binding to the promoter, and the structural genes are not expressed. The enzymes are said to be repressible (can be turned "off"), and the entire unit is called a *repressible operon*. Tryptophan is called the **corepressor.** Repressible operons are usually involved in anabolic pathways that synthesize a substance needed by the cell.

The *lac* Operon

Bacteria metabolism is remarkably efficient. If proteins or enzymes are needed for metabolism, then the structural genes are expressed. If no metabolism is necessary, then genes are not expressed. For example, if the milk sugar lactose is not present, there is no need to express genes for enzymes involved in lactose catabolism. But when only lactose is present, the cell immediately begins to make the three enzymes needed for lactose metabolism.

The enzymes that break down lactose are encoded by three genes (Fig. 13.2): One gene is for an enzyme called β-galactosidase, which breaks down the disaccharide lactose to glucose and galactose; a second gene codes for a permease that facilitates the entry of lactose into the cell; and a third gene codes for an enzyme called transacetylase, which has an accessory function in lactose metabolism.

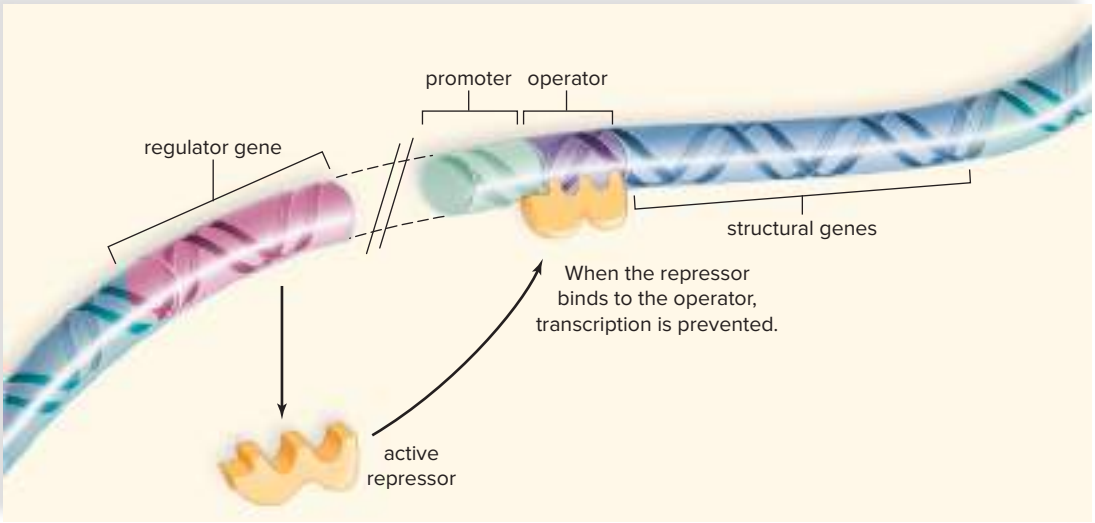
The three structural genes are adjacent to one another on the chromosome and are under the control of a single promoter and a single operator. The regulator gene codes for a *lac* operon repressor that ordinarily binds to the operator and prevents transcription of the three genes. When only lactose (more correctly, allolactose, an isomer formed from lactose) is present, lactose binds to the repressor, and the repressor undergoes a change in shape that prevents it from binding to the operator. Because the repressor is unable to bind to the operator, RNA polymerase is better able to bind to the promoter. After RNA polymerase carries out transcription, the three enzymes of lactose metabolism are synthesized.

Because the presence of lactose brings about expression of genes, it is called an **inducer** of the *lac* operon: The enzymes are said to be inducible enzymes (can be turned "on"), and the entire unit is called an *inducible operon.* Inducible operons are usually found in catabolic pathways that break down a nutrient. Why is that beneficial? Because these enzymes need to be active only when the nutrient is present.

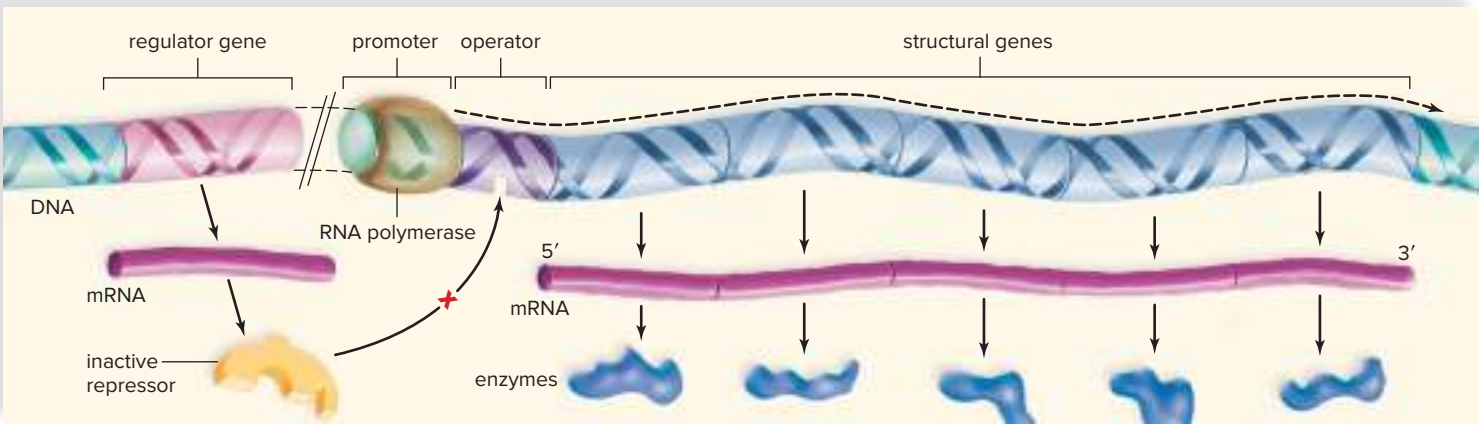
Further Control of the lac *Operon*

If both glucose and lactose are present, then *E. coli* preferentially breaks down glucose. The bacterium has a way to ensure that the lactose operon is fully turned on only when glucose is absent. A molecule called *cyclic AMP* (*cAMP*) accumulates when glucose is absent. Cyclic AMP, which is derived from

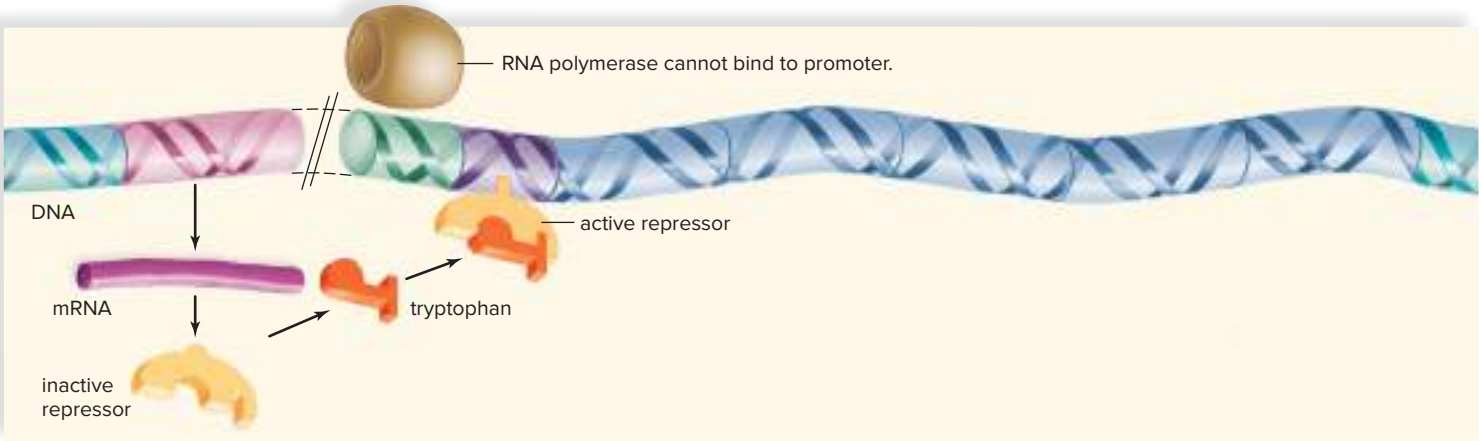
ISTUDY



a. The *trp* operon is regulated by a gene that produces a repressor protein.



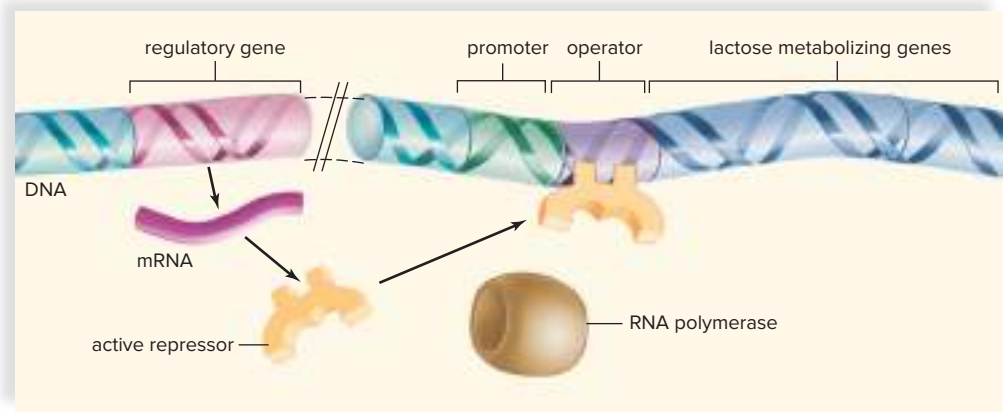
b. Tryptophan absent. Enzymes needed to synthesize tryptophan are produced.



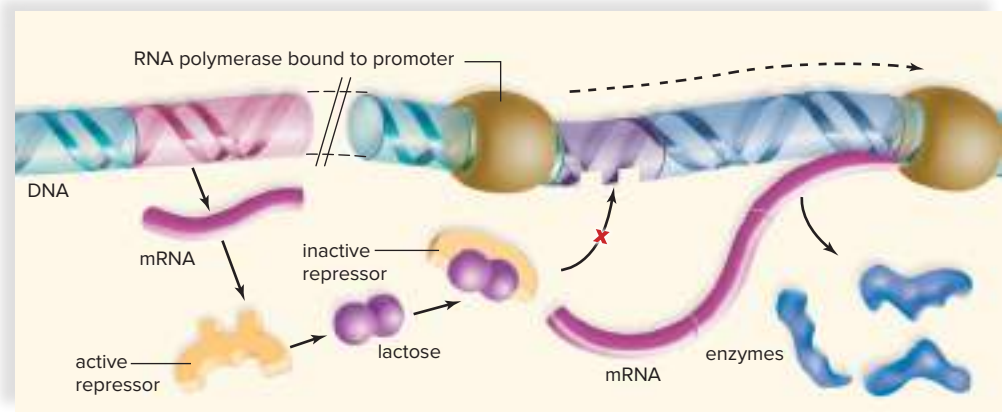
c. Tryptophan present. Presence of tryptophan prevents production of enzymes used to synthesize tryptophan.

Figure 13.1 The *trp* operon. **a.** The regulatory gene codes for a repressor protein that is normally inactive. **b.** When tryptophan is absent, the RNA polymerase attaches to the promoter, and the structural genes are expressed. **c.** When the nutrient tryptophan is present, it binds to the repressor, changing its shape. Now the repressor is active and can bind to the operator. RNA polymerase cannot attach to the promoter, and the structural genes are not expressed.

ISTUDY



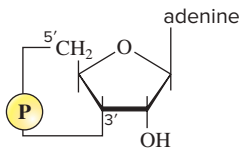
a. Operon when lactose is absent.



b. Operon when lactose is present.

Figure 13.2 The *lac* operon. **a.** The regulator gene codes for a repressor that is normally active. When it binds to the operator, RNA polymerase cannot attach to the promoter, and structural genes are not expressed. **b.** When lactose is present, it binds to the repressor, changing its shape, so it is inactive and cannot bind to the operator. Now RNA polymerase binds to the promoter, and the structural genes are expressed.

ATP, has only one phosphate group, which is attached to ribose at two locations:



cyclic AMP
(cAMP)

Cyclic AMP binds to a molecule called a *catabolite activator protein* (*CAP*), and the complex attaches to a CAP binding site next to the *lac* promoter. When CAP binds to DNA, DNA bends, exposing the promoter to RNA polymerase. RNA polymerase is now better able to bind to the promoter, so that the *lac* operon structural genes are transcribed, leading to their expression (Fig. 13.3).

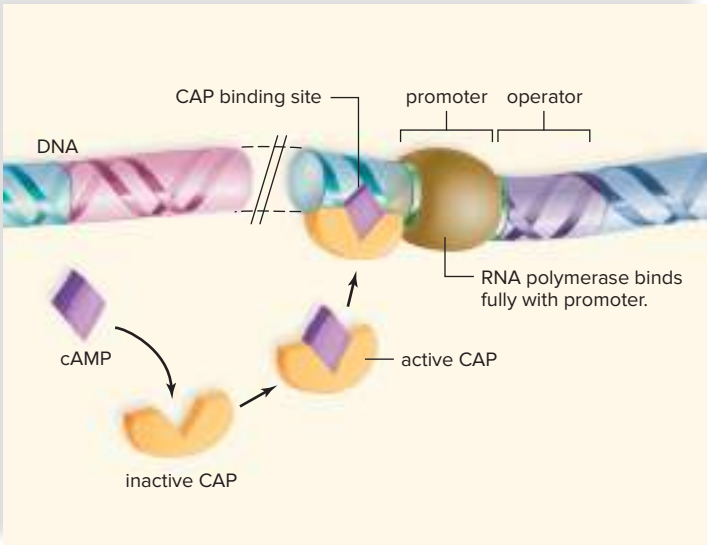
When glucose is present, there is little cAMP in the cell; CAP is inactive, and the lactose operon does not function maximally. CAP affects other operons as well and takes its name for activating the catabolism of various other metabolites when glucose is absent. A cell's ability to encourage the metabolism of lactose and other metabolites when glucose is absent provides a backup system for survival when the preferred energy source, glucose, is absent.

The CAP protein's regulation of the *lac* operon is an example of positive control. Why? When this molecule is active, it promotes the activity of an operon. The use of repressors, on the other hand, is an example of negative control, because when active they shut down an operon. A positive control mechanism allows the cell to fine-tune its response. In the case of the *lac* operon, the operon is only maximally active when glucose is absent and lactose is present. If both glucose and lactose are present, the cell preferentially metabolizes glucose.

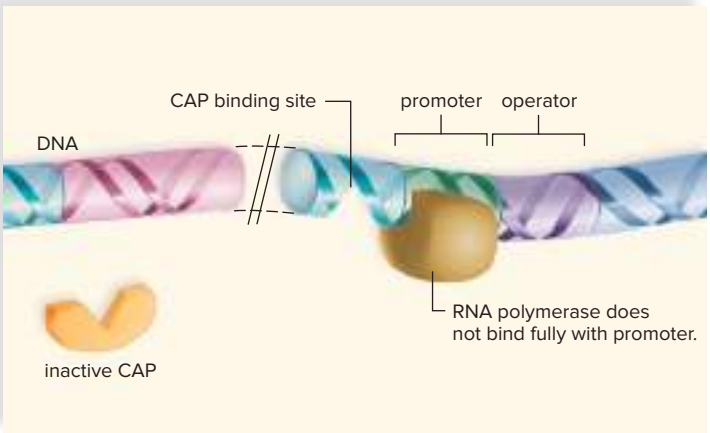
Check Your Progress 13.1

1. Explain the difference between the roles of the promoter and operator of an operon.
2. Summarize how gene expression differs in an inducible operon versus a repressible operon.
3. Describe the difference between positive control and negative control of gene expression.
4. Explain which operon discussed in this section is catabolic and which operon is anabolic.

ISTUDY



a. Lactose present, glucose absent (cAMP level high)



b. Lactose present, glucose present (cAMP level low)

Figure 13.3 Action of CAP. When active CAP binds to its site on DNA, the RNA polymerase is better able to bind to the promoter, so that the structural genes of the *lac* operon are expressed. **a.** CAP becomes active in the presence of cAMP, a molecule that is prevalent when glucose is absent. Therefore, transcription of lactose enzymes increases, and lactose is metabolized. **b.** If glucose is present, CAP is inactive, and RNA polymerase does not completely bind to the promoter. Therefore, transcription of lactose enzymes decreases, and less metabolism of lactose occurs.

13.2 Eukaryotic Regulation

Learning Outcomes

Upon completion of this section, you should be able to

1. List the levels of control of gene expression in eukaryotes.
2. Summarize how chromatin structure may be involved in regulation of gene expression in eukaryotes.
3. Identify the mechanisms of transcriptional, posttranscriptional, and translational control of gene expression.

With a few minor exceptions, each cell of a multicellular eukaryote has a complete complement of genes; the differences in cell types are determined by the different genes that are actively expressed in each cell. For example, in muscle cells, a different set of genes is turned on in the nucleus and a different set of proteins is active in the cytoplasm, compared to nerve or liver cells.

Like prokaryotic cells, a variety of mechanisms regulate gene expression in eukaryotic cells. These mechanisms can be grouped under five primary levels of control; three of them pertain to the nucleus, and two pertain to the cytoplasm (Fig. 13.4). In other words, control of gene activity in eukaryotes extends from transcription to protein activity. The following types of control in eukaryotic cells can modify the amount of the gene product:

- *Chromatin structure:* Chromatin packing is used as a way to keep genes turned off. If genes are not accessible to RNA polymerase, they cannot be transcribed. Chromatin structure is one method of *epigenetic inheritance* (Gk. *epi,* "besides"), the transmission of genetic information outside the coding sequences of a gene.
- *Transcriptional control:* The degree to which a gene is transcribed into mRNA determines the amount of gene product. In the nucleus, transcription factors may promote or repress transcription, the first step in gene expression.
- *Posttranscriptional control:* Posttranscriptional control involves mRNA processing and how fast mRNA leaves the nucleus.
- *Translational control:* Translational control occurs in the cytoplasm and affects when translation begins and how long it continues. Small interfering RNA molecules (siRNA) are known to regulate translation. In addition, any condition that can cause the persistence of the 5′ cap and 3′ poly-A tail can affect the length of translation. Excised introns may also have effects on the life span of mRNA.
- *Posttranslational control:* Posttranslational control, which also takes place in the cytoplasm, occurs after protein synthesis. Only a functional protein is an active gene product.

Let us now explore each of these types of control in greater depth.

Chromatin Structure

The DNA in eukaryotes is always associated with a variety of proteins, and together they make up a stringy material called **chromatin.** One of the more important types of these proteins is the histones. Histones play an important role in the compaction of the DNA (see Fig. 9.4), as well as in eukaryotic gene regulation. Without histones, the DNA would not fit inside the nucleus. Each human cell contains around 2 meters of DNA, yet the nucleus is only 5 to 8 micrometers (μm) in diameter.

The degree to which chromatin is compacted greatly affects the accessibility of the chromatin to the transcriptional machinery of the cell, and thus the expression levels of the genes. Active genes in eukaryotic cells are associated with more loosely packed chromatin called *euchromatin,* while the more tightly packed DNA, called *heterochromatin,* contains mostly inactive genes.

ISTUDY

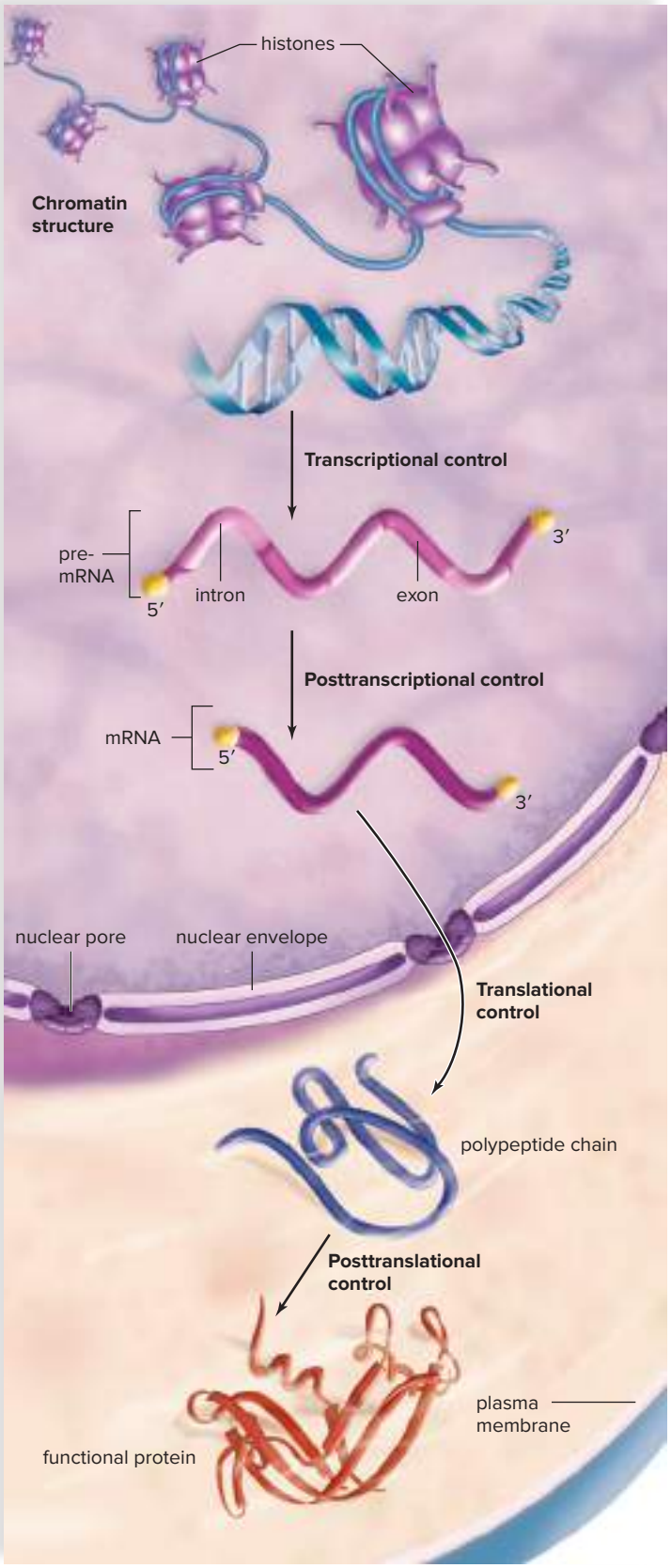


Figure 13.4 Levels of gene expression control in eukaryotic cells. The five levels of control are (1) chromatin structure, (2) transcriptional control, and (3) posttranscriptional control, which occur in the nucleus; (4) translational and (5) posttranslational control, which occur in the cytoplasm.

Under a microscope, the more densly compacted heterochromatin stains darker than euchromatin (Fig. 13.5*a*).

Whether the DNA exists as euchromatin or heterochromatin depends on how tightly the DNA is wrapped around DNA-histone complexes called nucleosomes (see Section 9.2). Histone molecules have *tails,* strings of amino acids that extend beyond the main portion of a nucleosome (Fig. 13.5*b*). In heterochromatin, the histone tails tend to bear methyl groups (—CH_3); in euchromatin, the histone tails tend to be acetylated and have attached acetyl groups (—$COCH_3$).

Histones regulate accessibility to DNA; euchromatin becomes genetically active when histones no longer block access to DNA. When DNA in euchromatin is transcribed, a group of proteins called the *chromatin remodeling complex* pushes aside, or *unpacks,* the histone portion of a nucleosome, so that access to DNA is not blocked and transcription can begin (Fig. 13.5*c*). After unpacking occurs, many decondensed loops radiate from the central axis of the chromosome. These chromosomes have been named lampbrush chromosomes, because their feathery appearance resembles the brushes that were once used to clean kerosene lamps.

In addition to physically moving nucleosomes aside to expose promoters, chromatin remodeling complexes may also affect gene expression by adding acetyl or methyl groups to histone tails.

Heterochromatin Is Not Transcribed

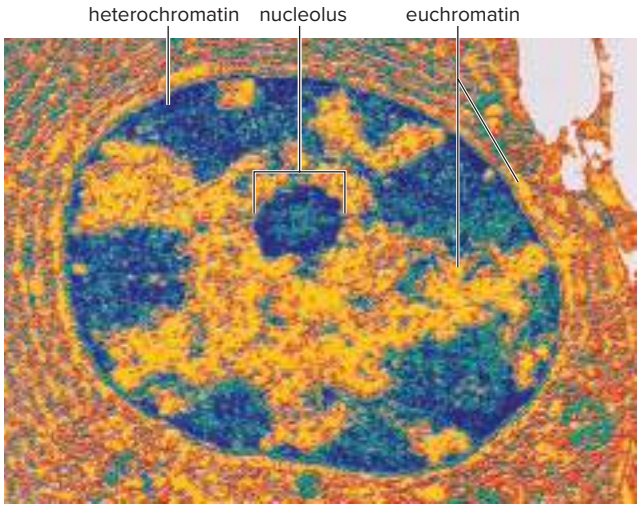
In general, highly condensed heterochromatin is inaccessible to RNA polymerase, and the genes contained within are seldom or never transcribed. A dramatic example of heterochromatin is the **Barr body** in mammalian females. This small, dark-staining mass of condensed chromatin adhering to the inner edge of the nuclear membrane is an inactive X chromosome. To compensate for the fact that female mammals have two X chromosomes (XX), whereas males have only one (XY), one of the X chromosomes in the cells of female embryos undergoes inactivation. The inactive X chromosome does not produce gene products, allowing both males and females to produce the same amount of gene product from a single X chromosome.

How do we know that Barr bodies are inactive X chromosomes that are not producing gene products? In a female who is heterozygous, 50% of the cells have one X chromosome active, and 50% have the other X chromosome active. The body of a female who is heterozygous is therefore a mosaic, with "patches" of genetically different cells. Investigators have discovered that human females who are heterozygous for an X-linked recessive form of ocular albinism have patches of pigmented and nonpigmented cells at the back of the eye.

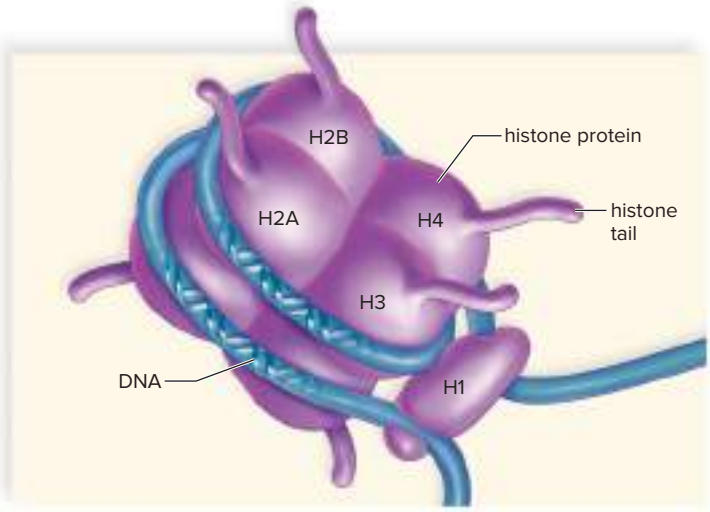
As other examples, females who are heterozygous for X-linked hereditary absence of sweat glands have patches of skin lacking sweat glands. And the female calico cat exhibits a difference in X-inactivation in its cells. In these cats, an allele for black coat color is on one X chromosome, and a corresponding allele for orange coat color is on the other. The patches of black and orange in the coat can be related to which X chromosome is in the Barr bodies of the cells found in the patches (Fig. 13.6).

Epigenetic Inheritance

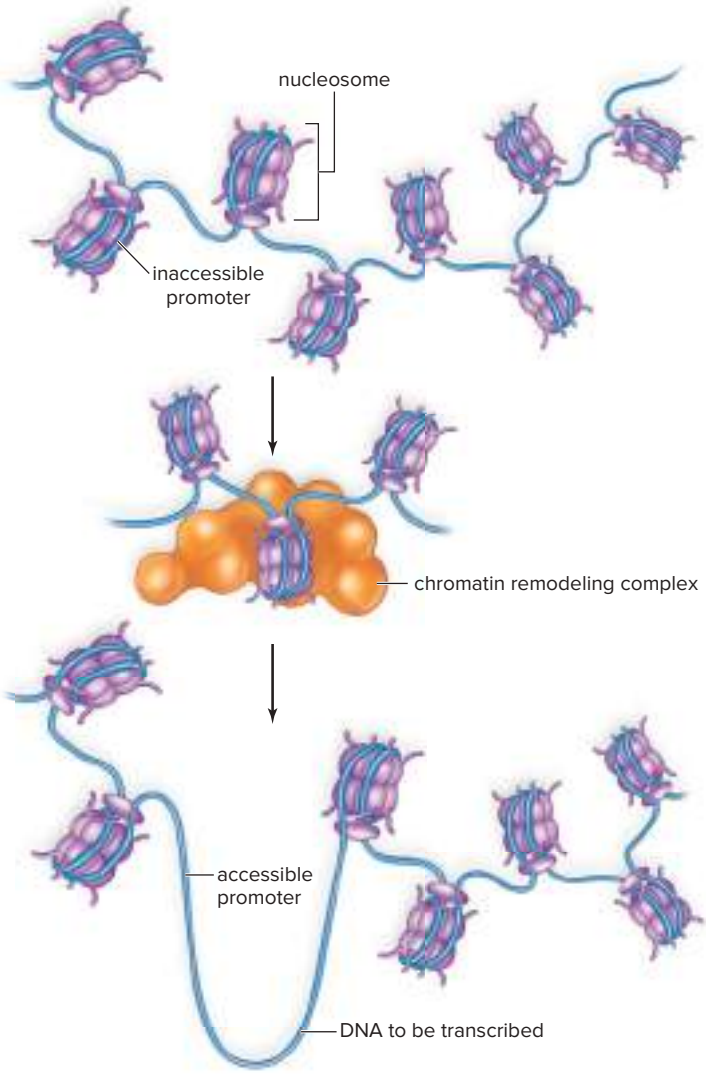
Histone modification is often linked to a phenomenon termed **epigenetic inheritance,** in which variations in the pattern of inheritance are not due to changes in the sequence of the DNA



a. Heterochromatin and euchromatin in a eukaryotic cell

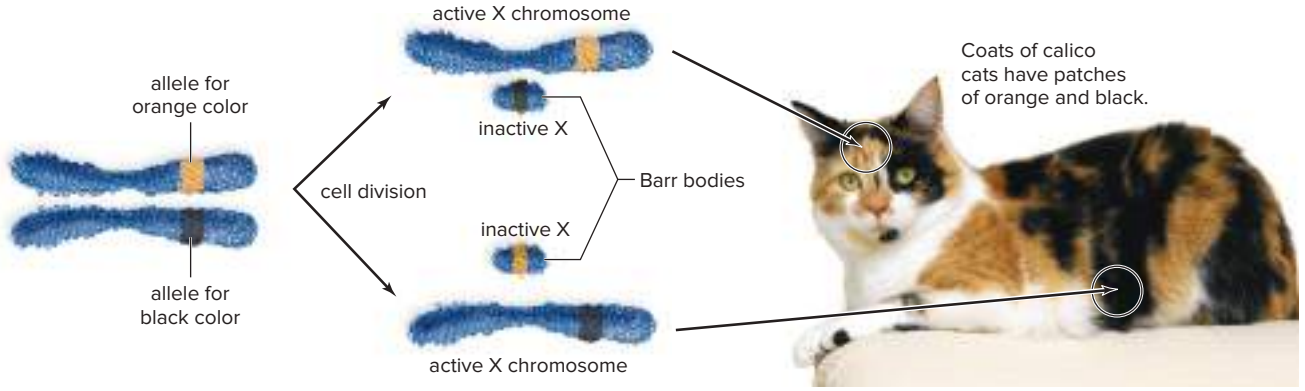


b. A nucleosome



c. DNA unpacking

Figure 13.5 Chromatin structure regulates gene expression. **a.** A eukaryotic nucleus contains highly condensed heterochromatin, and euchromatin, which is not as condensed. **b.** Nucleosomes ordinarily prevent access to DNA, so that transcription cannot take place. If histone tails are acetylated, access can be achieved; if the tails are methylated, access is more difficult. **c.** A chromatin remodeling complex works on euchromatin to make the DNA available, and thus the promoter accessible, for transcription. (a): Alfred Pasieka/Science Source

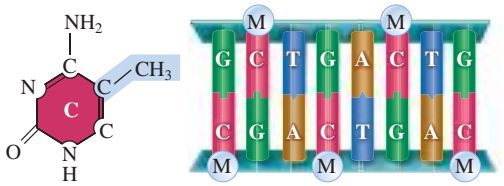


Females have two X chromosomes.

One X chromosome is inactivated in each cell. Which one is by chance.

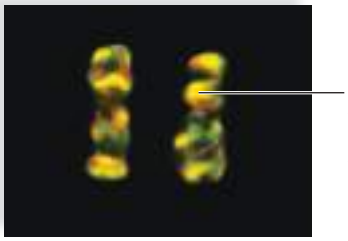
Figure 13.6 X-inactivation in mammalian females. In cats, the alleles for black or orange coat color are carried on the X chromosomes. Random X-inactivation occurs in females. Therefore, in heterozygous females, some of the cells express the allele for black coat color, while other cells express the allele for orange coat color. The white color on calico cats is provided by another gene. (photo): cgbaldauf/E+/Getty Images

ISTUDY



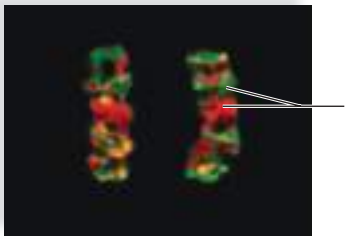
DNA methylation is the addition of a methyl group (M) to the DNA base cytosine (C).

a. Methylation of DNA



Yellow shows where the twins have epigenetic tags in the same place.

3-year-old twins



Red and green show where the twins have epigenetic tags in different places.

50-year-old twins

b. Methylation in twin studies

Figure 13.7 DNA methylation and epigenetic inheritance **a.** DNA is methylated when a methyl group attaches to the cytosine nucleotide. **b.** One of the twin's epigenetic tags are dyed green, and the other twin's tags are dyed red. An overlap in green and red shows up as yellow. The 50-year-old twins have more epigenetic tags in different places than do the 3-year-old twins. (b): National Academy of Sciences, USA

nucleotides, but rather to modifications to the individual nucleotides within the genes. The term is also used broadly to describe inheritance patterns that do not depend on the genes themselves. Epigenetic inheritance explains unusual inheritance patterns and may play an important role in growth, aging, and cancer.

The specific chemical reactions, or epigenetic "tags," can come in different forms but are often associated with DNA methylation, in which a methyl group attaches to the cytosine base of DNA (Fig. 13.7*a*). With a methyl group attached, transcription cannot occur. The methyl group interferes with transcription factors and other proteins in the transcription machinery, thereby silencing or weakening a gene. Over time, the differences in these tags accumulate, making twins increasingly different from each other (Fig 13.7*b*).

For example, during *genomic imprinting,* either the male or female parent's gene (but not both) is methylated during gamete formation. If an inherited allele is highly methylated, the gene is not expressed, even if it is a normal gene in every other respect. For traits that exhibit genomic imprinting, the expression of the gene depends on whether the unmethylated allele was inherited from the female or male parent.

Epigenetics has important implications for medicine. The appearance of tags on genes helps scientists discover the cause of some illnesses that cannot be explained by DNA or genetic mutations alone. Identical twins discordant (different) for autism, psychiatric disorders, and cancer have been shown to have different DNA methylation on certain genes.

In addition, the epigenetic changes are reversible. A study using rats showed that rat pups that are licked and nurtured by their mothers become calm adults. Rat pups that are not nurtured are anxious. Injecting a calm rat with a drug that adds methyl groups creates an anxious rat. Conversely, injecting an anxious rat with a different drug that removes methyl groups creates a calm rat. In drug development, epigenetic medicines could be used to correct or reverse the particular effect of a tag.

Transcriptional Control

Although eukaryotes have various levels of genetic control (see Fig. 13.4), **transcriptional control** remains the most critical of these levels. The first step toward transcription is availability of DNA, which involves chromatin structure. Transcriptional control also involves the participation of transcription factors, activators, and repressors.

Transcription Factors, Activators, and Repressors

Although some operons like those of prokaryotic cells have been found in eukaryotic cells, transcription in eukaryotes is still controlled by DNA-binding proteins. Every cell contains many different types of **transcription factors,** proteins that help regulate transcription by assisting the binding of the RNA polymerase to the promoter. A cell has many different types of transcription factors, and a variety of transcription factors may be active at a single promoter. Thus, the absence of one can prevent transcription from occurring.

Even if all the transcription factors are present, transcription may not begin without the assistance of a DNA-binding protein called a **transcription activator.** These bind to regions of DNA called **enhancers,** which may be located some distance from the promoter. A hairpin loop in the DNA brings the transcription activators attached to the enhancer into contact with the transcription factor complex (Fig. 13.8). Likewise, the binding of repressors within the promoter may prohibit the transcription of certain genes. Most genes are subject to regulation by both activators and repressors.

The promoter structure of eukaryotic genes is often very complex, and a large variety of regulatory proteins may interact with each other and with transcription factors to affect a gene's transcription level. Mediator proteins act as a bridge between transcription factors and transcription activators at the promoter. Now RNA polymerase can begin the transcription process (Fig. 13.8). Such protein-to-protein interactions are a hallmark of eukaryotic gene regulation. Together, these mechanisms can fine-tune a gene's transcription level in response to a large variety of conditions. For example, all the cells in a corn plant contain the gene for the pigment anthocyanin, but where and when anthocyanin is made is transcriptionally controlled. UV light induces anthocyanin production in the leaves where it is controlled by one set of transcription factors. Later, organ development cues anthocyanin production in the kernels controlled by a different set of transcription factors.

ISTUDY

THEME Nature of Science

Exercise and Gene Expression

Studies have shown that humans who stop exercising exhibit anxiety, memory loss, and a decrease in cognitive abilities. Are the changes in mood and behavior the result of differential gene expression?

To test this hypothesis, it is possible to use mice as a model organism since many studies conducted in mice have shown the ability to use this species as a prediction of similar outcomes in humans.

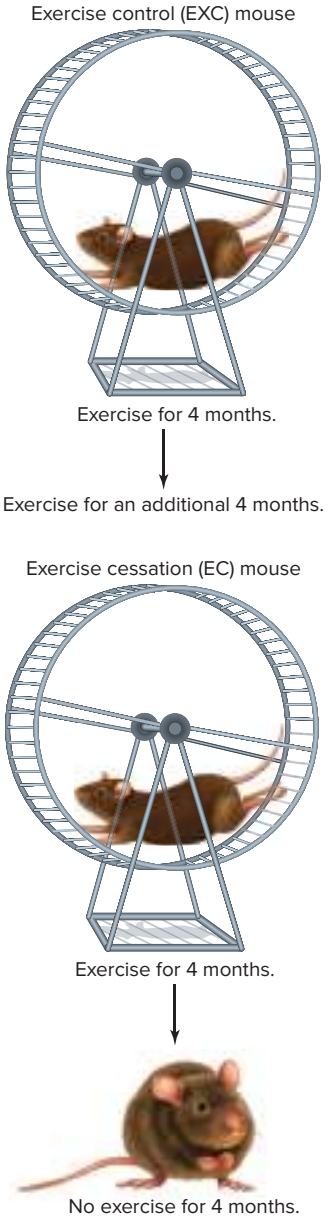


Figure 13A Mice in two different exercise conditions. The exercise control mice (EXCs) were allowed to run on their exercise wheel at any time. After 4 months, the exercise cessation mice (ECs) had their wheel taken away during daylight hours.

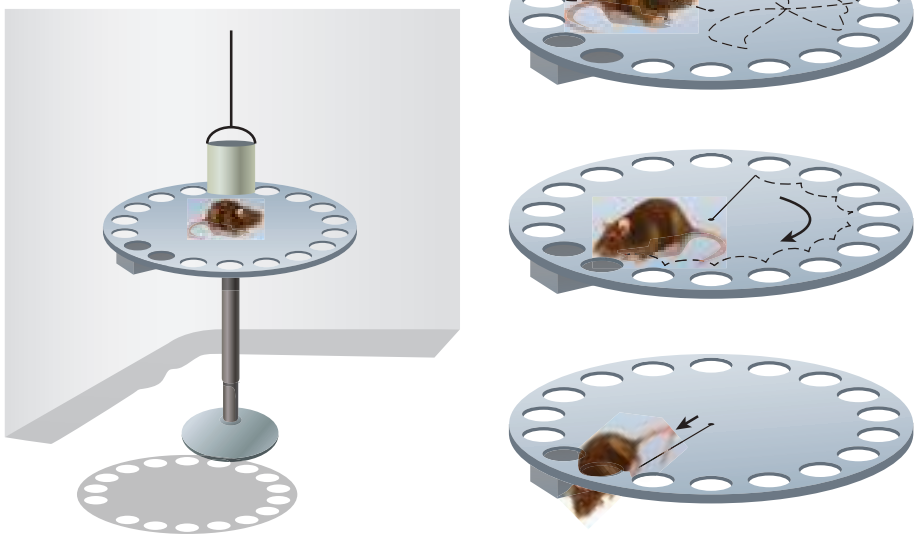


Figure 13B The Barnes maze test. This test is used to determine a mouse's memory and cognitive ability, because at the outset, a mouse takes a long time to reach the escape hole. After several days, the mouse remembers the fastest route to escape.

In one experiment, two random groups of mice were used, one labeled EXC (exercise control) and one labeled EC (exercise cessation). Both sets of mice were allowed to use their exercise wheel at any time in a 24-hour period for 4 months. After 4 months, the wheel was removed during daylight hours for the EC group (Fig. 13A).

Two tests were used to measure memory and cognition. First, the Barnes maze trains a mouse to look through many holes until it reaches the hole that allows it to escape (Fig. 13B). The training takes several days and the mouse should improve over time. The longer it takes a mouse to escape, the weaker their memory and cognition.

Second, dissected brains were tested for gene expression of key genes located in the hippocampus. The hippocampus is the part of the brain responsible for emotion, memory, and the autonomic nervous system (see Figure 37.11). One gene, *Gfap,* tested codes for cytoskeletal proteins in neurons called astrocytes. Astrocytes regulate the transmission of neural impulses in the brain and are linked to cognitive abilities.

Gene expression was tested using molecular techniques that use fluorescence to measure the concentration of the *Gfap* gene product (Fig. 13C).

The data show that EC mice have a decrease in memory and cognitive function and a decrease in gene expression of a gene known to be involved in memory and cognition (Fig. 13C).

Questions to Consider

1. What other controls not mentioned in the reading do you think were implemented for this experiment?
2. Why do you think that exercise improves memory and cognition?

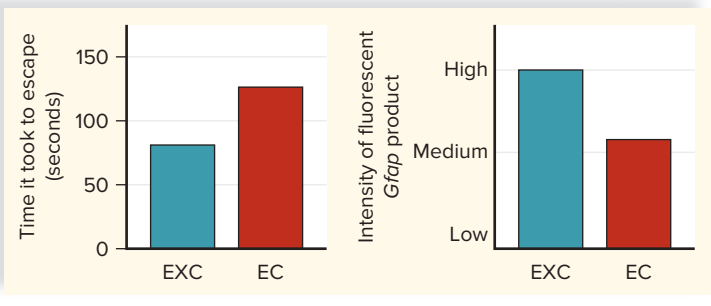


Figure 13C Behavioral and genetic difference between EXC and EC mice. **a.** Mice were timed in the Barnes maze. It took the EC mice longer to complete the maze than the EXC mice. **b.** Gene product *Gfap* was quantified using fluorescent intensity in astrocytes located in the hippocampus. The EXC mice had higher gene expression (because of the higher fluorescence) for *Gfap* than the EC mice.

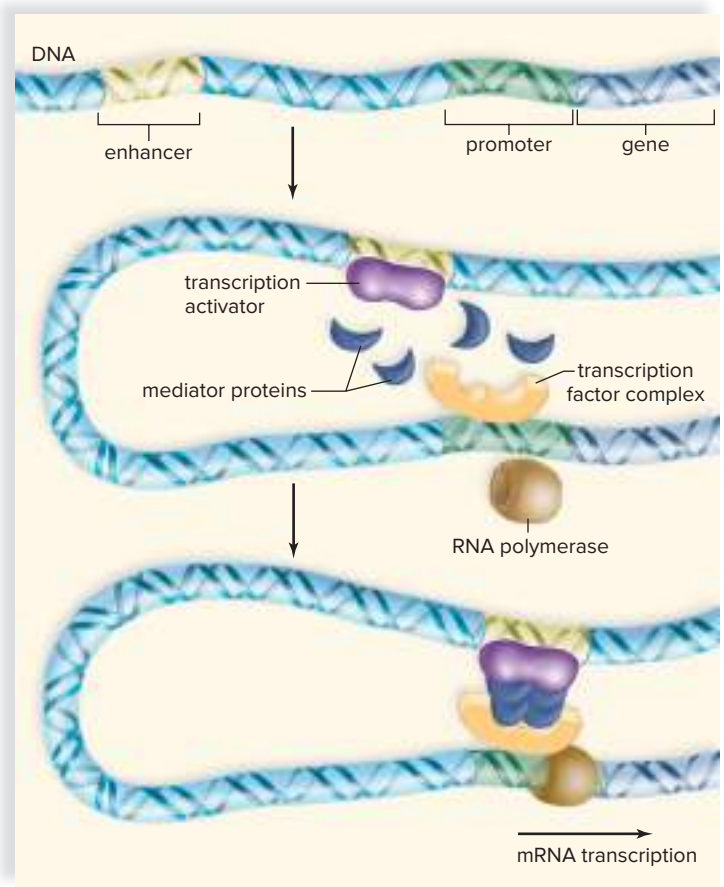
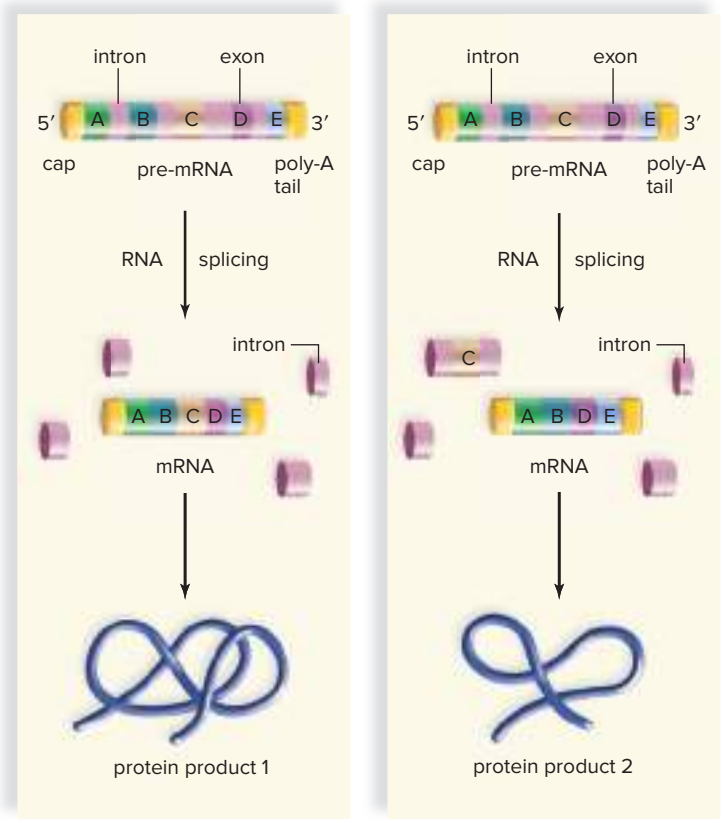


Figure 13.8 Regulation of eukaryotic transcription. Transcription in eukaryotic cells requires that transcription factors bind to the promoter, and transcription activators bind to an enhancer. The enhancer may be far from the promoter, but the DNA loops and mediator proteins act as a bridge joining activators to factors. Only then does transcription begin.



a. b.

Figure 13.9 Alternative processing of pre-mRNA. Because the pre-mRNAs are processed differently in these two cells (**a** and **b**), distinct proteins result. This is a form of posttranscriptional control of gene expression.

Posttranscriptional Control

Posttranscriptional control of gene expression occurs in the nucleus and includes alternative mRNA splicing and controlling the speed with which mRNA leaves the nucleus.

Recall that during pre-mRNA splicing, introns (noncoding regions) are excised, and exons (expressed regions) are joined together to form an mRNA (see Fig. 12.14). When introns are removed from pre-mRNA, differential splicing of exons can occur, and this affects gene expression. For example, an exon that is normally included in an mRNA transcript may be skipped, and it is excised along with the flanking introns (Fig. 13.9). The resulting mature mRNA has an altered sequence, and the protein it encodes is altered. Sometimes introns remain in an mRNA transcript. When this occurs, the protein-coding sequence is also changed.

Examples of alternative pre-mRNA splicing abound. Both the hypothalamus and the thyroid gland produce a protein hormone called calcitonin, but the mRNA that leaves the nucleus is not the same in both types of cells. This results in the thyroid's releasing a slightly different version of calcitonin than does the hypothalamus. Evidence of alternative mRNA splicing is found in other cells, such as those that produce neurotransmitters, muscle regulatory proteins, and antibodies.

Alternative pre-mRNA splicing allows humans and other complex organisms to recombine their genes in novel ways to create the great variety of proteins found in these organisms. Researchers are busy determining how small nuclear RNAs (snRNAs) affect the splicing of pre-mRNA. Alternative mRNA splicing can also result in the inclusion of an intron that brings about destruction of the mRNA before it leaves the nucleus.

Further posttranscriptional control of gene expression is achieved by modifying the speed of transport of mRNA from the nucleus into the cytoplasm. Evidence indicates there is a difference in the length of time it takes various mRNA molecules to pass through a nuclear pore, affecting the amount of gene product realized per unit of time following transcription.

Small RNA (sRNA) Molecules Regulate Gene Expression

For a long time, scientists were faced with a mystery: A cell appeared to contain vastly more DNA than was needed to account for the number of expressed proteins. The DNA that was not expressed as proteins was initially termed "junk" DNA. Recently, however, scientists have begun to understand the role of this DNA in the cell. Although only about 1.5% of the transcribed DNA codes for protein, the remainder is used to form small RNA (sRNA) molecules. We now know that these sRNA molecules represent an

ISTUDY

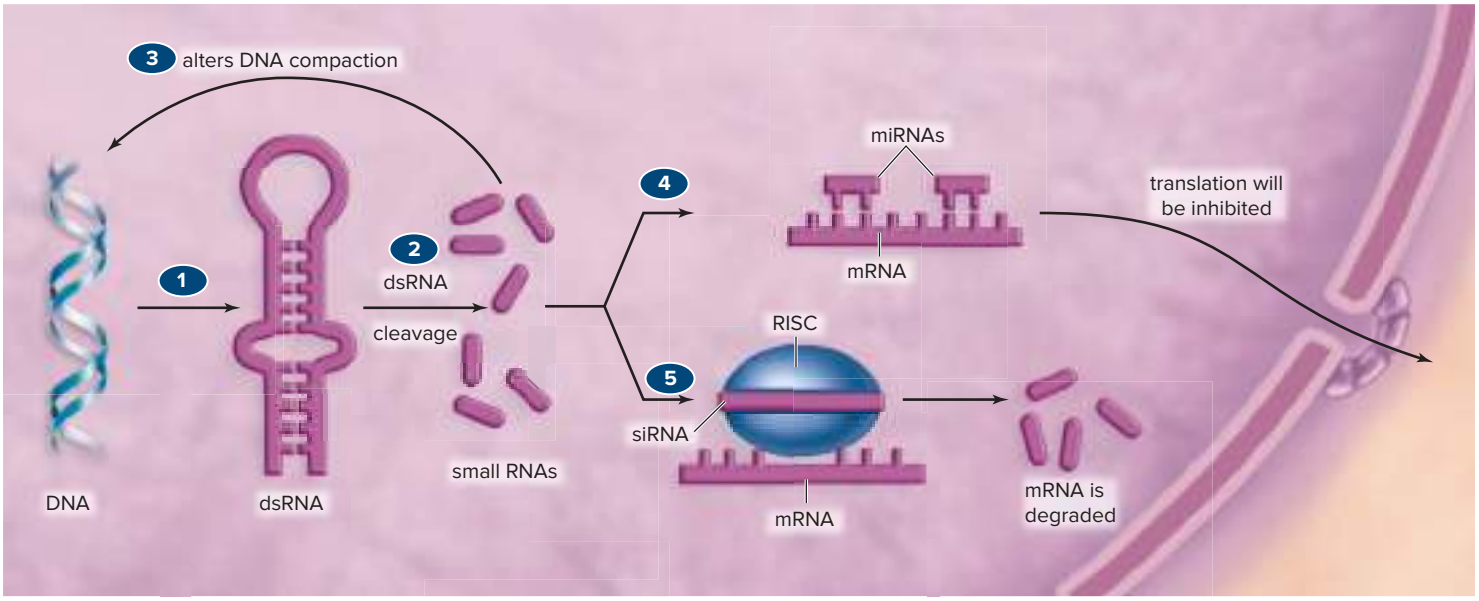


Figure 13.10 Function of small RNA molecules. Transcription of the DNA 1 may lead to looped and double-stranded RNA (dsRNA). The cleavage of the dsRNA 2 produces many small RNA (sRNA) molecules. 3 An sRNA can double back to increase DNA compaction, or it may become an miRNA or siRNA. 4 miRNA reduces translation by binding to complementary mRNA molecules. 5 siRNA forms a complex with RISC, which then degrades any mRNA with a sequence of bases that are complementary to the siRNA.

important form of gene regulation that functions at multiple levels of gene expression.

Let's take a closer look at how these RNA molecules regulate gene expression (Fig. 13.10).

1. The transcribed RNA can form loops as hydrogen bonding occurs between its bases.
2. The double-stranded RNA (dsRNA) is diced up by enzymes in the cell to form sRNA molecules.
3. Some of these sRNA molecules regulate transcription, while others are involved in the regulation of translation. Various ways have been found by which sRNA may regulate gene expression. sRNA molecules have been known to alter the compaction of DNA, so that some genes are inaccessible to the transcription machinery of the cell.
4. Small RNAs are the source of *microRNAs* (*miRNAs*), small snippets of RNA that can bind to and disable the translation of mRNA in the cytoplasm.
5. Small RNAs are also the source of *small-interfering RNAs* (*siRNAs*) that join with an enzyme (an RNA-induced silencing complex, or RISC) to form an active silencing complex. This activated complex targets specific mRNAs in the cell for breakdown, preventing them from being expressed.

By using a combination of miRNA and siRNA molecules, a cell can fine-tune the amount of product being expressed from a gene, much as a dimmer switch on a light regulates the brightness of the room. Because both miRNA and siRNA molecules interfere with the normal gene expression pathways, the process is often referred to as **RNA interference.**

The first scientists to artificially construct miRNA and siRNA molecules to suppress the expression of a specific gene were Andrew Fire and Craig Mello. Following this discovery, medical scientists recognized that it may be possible to use sRNA molecules as therapeutic agents to suppress the expression of disease-causing genes. For their discovery, Fire and Mello received the 2006 Nobel Prize in Physiology or Medicine.

Translational Control

Translational control begins when the processed mRNA molecule reaches the cytoplasm and before there is a protein product. Translational control involves the activity of mRNA for translation at the ribosome.

The presence or absence of the 5′ cap and the length of the poly-A (adenine nucleotide) tail at the 3′ end of a mature mRNA transcript can determine whether translation takes place and how long the mRNA is active. The long life of mRNAs that code for hemoglobin in mammalian red blood cells is attributed to the persistence of their 5′ end caps and their long 3′ poly-A tails. Therefore, any condition that affects the length of the poly-A tail or leads to removal of the cap may trigger the destruction of an mRNA.

Posttranslational Control

Posttranslational control begins once a protein has been synthesized and has become active. Posttranslational control represents the last chance a cell has for influencing gene expression.

If all the proteins produced by a cell during its lifetime remained in the cell, serious problems would arise. Thus, proteins are continually being synthesized and then degraded.

Proteins needed only for a short time can be altered chemically, leaving them nonfunctional. Proteins may not be folded correctly or they may change shape over time, causing them to behave erratically or stick to one another and form aggregates. In fact, a

ISTUDY

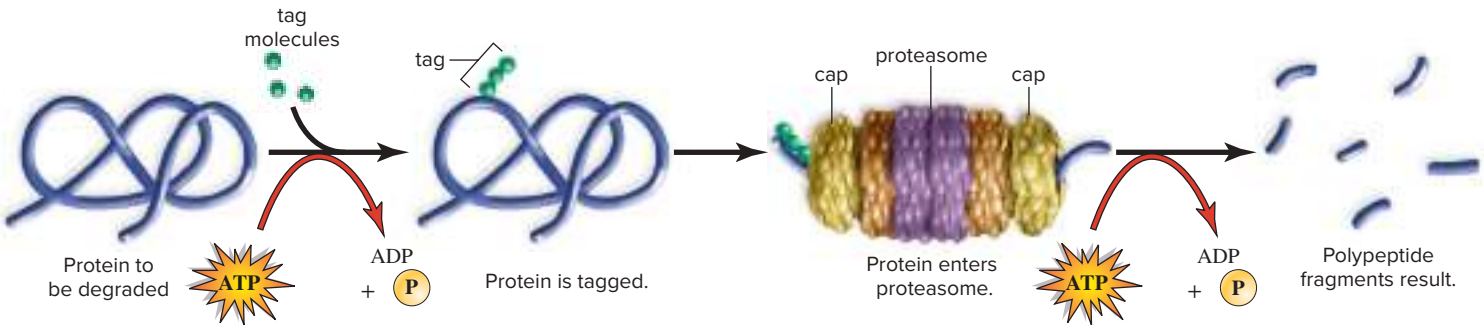


Figure 13.11 Proteasomes and posttranslational control. Proteins to be degraded are first tagged with a signaling molecule. They then enter the proteasome, where they are broken down into polypeptide fragments.

number of neurodegenerative diseases, such as Alzheimer's disease, Parkinson's disease, and mad cow disease, are related to proteins that aggregate, forming plaques in the brain. Thus, in addition to normal turnover of proteins, cells need a way to get rid of old, unused, and incorrectly folded proteins.

Just how long a protein remains active in a cell is usually regulated by the use of **proteases,** enzymes that break down proteins. To protect the cell, proteases are typically confined to the lysosomes or special structures called **proteasomes.** For a protein to enter a proteasome, it has to be tagged with a signaling protein that is recognized by the proteasome cap (Fig. 13.11). When the cap recognizes the tag, it opens and allows the protein to enter the core of the structure, where it is digested to peptide fragments. Notice that proteasomes help regulate gene expression because they help control the amount of protein product in the cytoplasm.

Check Your Progress 13.2

1. List the five levels of genetic control in eukaryotes.
2. Explain how chromatin structure influences gene expression.
3. Discuss how small RNA molecules and proteasomes regulate gene expression.

13.3 Gene Mutations

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between spontaneous and induced mutations.
2. Identify how mutations influence protein structure.
3. Summarize how mutations may cause cancer.

A **gene mutation** is a permanent change in the sequence of bases in DNA. The effect of a DNA base sequence change on protein activity can range from no effect to complete inactivity. Germ-line mutations are those that occur in sex cells and can be passed on to subsequent generations. Somatic mutations occur in body cells and, therefore, may affect only a small number of cells in a tissue. Somatic mutations are not passed on to future generations, but they can lead to the development of cancer.

Causes of Mutations

Some mutations are spontaneous—they happen for no apparent reason—whereas others are induced by environmental influences. In most cases, **spontaneous mutations** arise as a result of abnormalities in normal biological processes. **Induced mutations** may result from exposure to toxic chemicals or radiation, which induce (cause) changes in the base sequence of DNA.

Spontaneous Mutations

Spontaneous mutations can be associated with any number of normal processes. For example, a movable piece of DNA, termed a *transposon,* may jump from one location to another, disrupting one or more genes and leading to an abnormal product (see Section 14.4). On rare occasions, a base in DNA can undergo a chemical change that leads to a mispairing during replication. A subsequent base-pair change may be carried forth in future generations. Spontaneous mutations due to DNA replication errors, however, are rare. DNA polymerase, the enzyme that carries out replication, proofreads the new strand against the old strand and detects any mismatched nucleotides, and each is usually replaced with a correct nucleotide. In the end, only about one mistake occurs for every 1 billion nucleotide pairs replicated.

Induced Mutations

Induced mutations are caused by **mutagens,** environmental factors that can alter the base composition of DNA. Among the best-known mutagens are radiation and organic chemicals. Many mutagens are also **carcinogens** (cancer-causing mutagens).

Chemical mutagens are present in many sources, including some of the food we eat and many industrial chemicals. The mutagenic potential of AF-2, a food additive once widely used in Japan, and of safrole, a flavoring agent once used to flavor root beer, caused them to be banned. Surprisingly, many naturally occurring substances—like aflatoxin, produced in moldy grain and peanuts (and present in peanut butter at an average level of 2 parts per billion), and acrylamide, a natural product found in french fries—are also suspected mutagens.

Tobacco smoke contains a number of organic chemicals that are known carcinogens, and it is estimated that one-third of all cancer deaths can be attributed to smoking. Lung cancer is the most frequent lethal cancer in the United States, and smoking is implicated in the development of cancers of the mouth, larynx, bladder, kidney, and pancreas. The greater the number of cigarettes smoked per day, the earlier smoking starts, and the higher the tar

ISTUDY

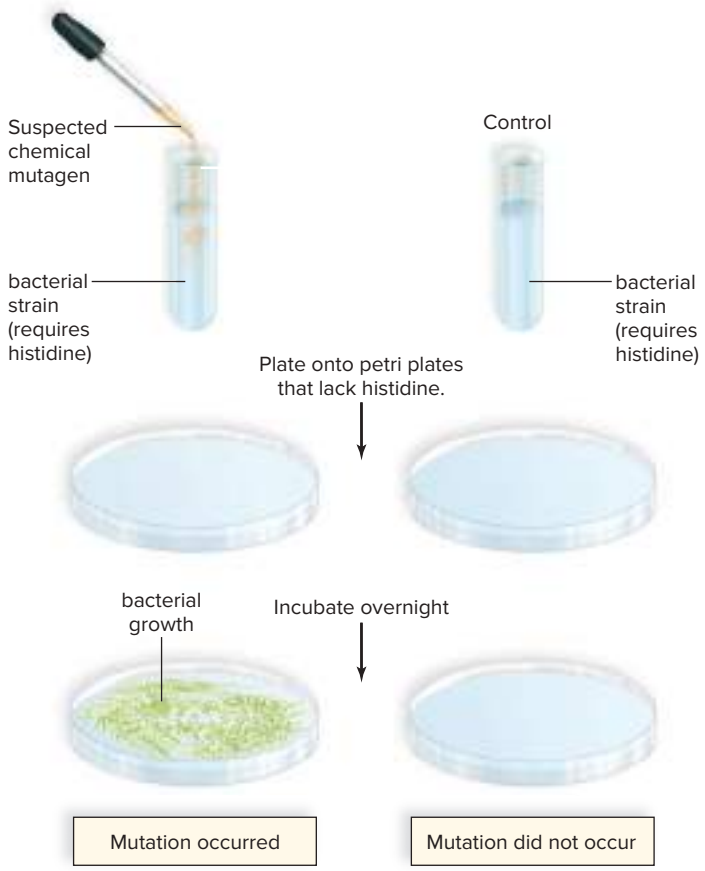


Figure 13.12 The Ames test for mutagenicity. A bacterial strain that requires histidine as a nutrient is exposed to a suspected chemical mutagen, but a control is not exposed. The bacteria are plated on a medium that lacks histidine; only the bacteria exposed to the chemical show growth. A mutation allowed the bacteria to grow; therefore, the chemical can be carcinogenic.

content, the greater is the possibility of these cancers. When smoking is combined with drinking alcohol, the risk of these cancers increases even more.

Scientists use the Ames test for mutagenicity to hypothesize that a chemical can be carcinogenic (Fig. 13.12). In the Ames test, a histidine-requiring strain of bacteria is exposed to a chemical. If the chemical is mutagenic, the bacteria can grow without histidine. A large number of chemicals used in agriculture and industry give a positive Ames test. Examples are ethylene dibromide (EDB), which is added to leaded gasoline (to vaporize lead deposits in the engine and send them out the exhaust), and ziram, which is used to prevent fungal disease on crops. Some drugs, such as isoniazid (used to prevent tuberculosis), are mutagenic according to the Ames test.

Aside from chemicals, certain forms of radiation, such as X-rays and gamma rays, are called ionizing radiation because they create free radicals, ionized atoms with unpaired electrons. Free radicals react with and alter the structure of other molecules, including DNA. Ultraviolet (UV) radiation is easily absorbed by the pyrimidines in DNA. Wherever there are two thymine molecules next to one another, ultraviolet radiation may cause them to bond together, forming *thymine dimers.* A kink results in the DNA. Usually, these dimers are removed by **DNA repair enzymes,** which constantly monitor DNA and fix any irregularities. One enzyme excises a portion of DNA that contains the dimer, another makes a new section by using the other strand as a template, and still another seals the new section in place.

The importance of these repair enzymes is exemplified by individuals with the condition known as xeroderma pigmentosum. They lack some of the repair enzymes, and as a consequence, these individuals have a high incidence of skin cancer because of the large number of mutations that accumulate over time. Also, repair enzymes can fail, as when skin cancer develops because of excessive sunbathing or prolonged exposure to X-rays.

Effect of Mutations on Protein Activity

Point mutations involve a change in a single DNA nucleotide. That change alters transcription and possibly changes the specific amino acid. One type of point mutation is a *base substitution,* resulting in one DNA nucleotide being replaced with another incorrect nucleotide. Notice the base difference in the second row of Figure 13.13*a* and how it changes the resultant amino acid sequence. Sometimes a base substitution has little or no effect on the final protein produced, but in some cases early stop codons can be introduced, or coding for the wrong amino acid can severely alter the protein shape. Such is the case with the genetic disorder sickle-cell disease (Fig. 13.13*b*). In this gene, there is a base substitution that alters the mRNA codon for glutamic acid. Instead, the codon for valine is present, altering the final shape of hemoglobin, the protein that carries oxygen in the blood. The abnormal hemoglobin molecules form semirigid rods, and the red blood cells become sickle-shaped, resulting in decreased blood flow through tiny blood vessels.

Frameshift mutations occur most often when one or more nucleotides are either added or deleted from DNA (Fig. 13.13*a,* bottom two lines). Because all the codons downstream of the mutation are now shifted, the result is a completely new sequence of codons, yielding a nonfunctional protein.

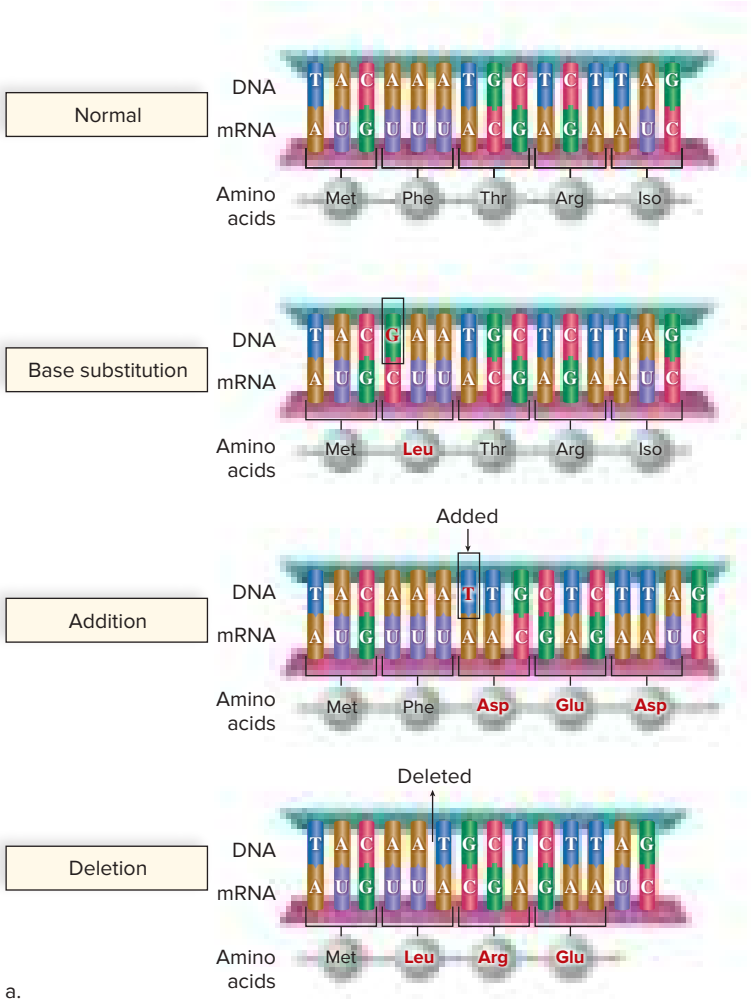
Nonfunctional Proteins

A single nonfunctioning protein can have a dramatic effect on the phenotype, because enzymes are often a part of metabolic pathways. One metabolic pathway in cells is as follows:

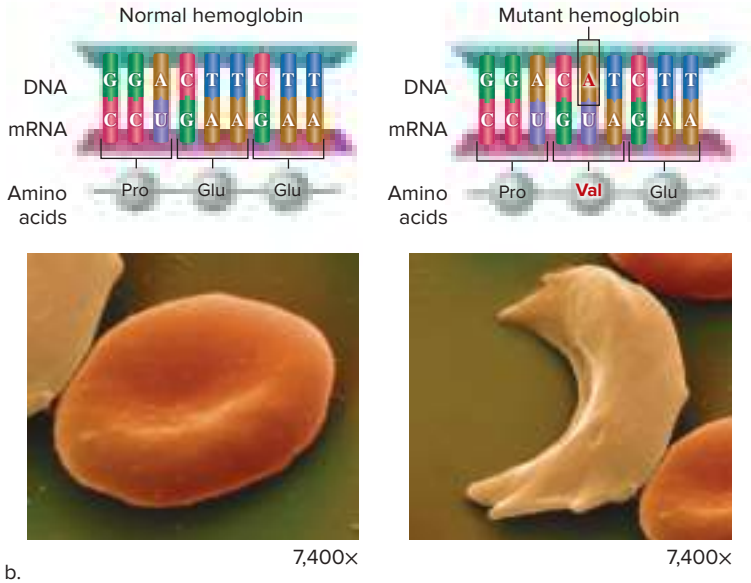
$$\underset{\text{(phenylalanine)}}{\text{A}} \xrightarrow{E_A} \underset{\text{(tyrosine)}}{\text{B}} \xrightarrow{E_B} \underset{\text{(melanin)}}{\text{C}}$$

If a faulty code for enzyme E_A is inherited, a person is unable to convert molecule A to B. Phenylalanine builds up in the system, and the excess causes an intellectual disability and other symptoms of the genetic disorder phenylketonuria (PKU). In the same pathway, if a person inherits a faulty code for enzyme E_B, then B cannot be converted to C, and the individual has albinism.

A rare condition called androgen insensitivity is due to a faulty receptor for androgens, which are male sex hormones, such as testosterone. In a male with this condition, plenty of testosterone is present in the blood, but the cells are unable to respond to it. Female instead of male external genitals form, and female instead of male secondary sex characteristics occur at puberty. The individual, who appears to be biologically female, may be prompted to seek medical advice when menstruation never occurs. The



a.



b.

Figure 13.13 Point mutations. The effect of a point mutation can vary. **a.** Starting at the *top:* Normal sequence of bases results in a normal sequence of amino acids. Next, a base substitution can result in the wrong amino acid. In the final two rows, an addition or deletion can result in a frameshift mutation, altering all the codons downstream of the point mutation. **b.** Due to a base substitution in the hemoglobin gene, the DNA now codes for valine instead of glutamic acid, and the result is that normal red blood cells become sickle-shaped. (b, both): Eye of Science/Science Source

karyotype is that of a male rather than a female, and the individual does not have the internal sexual organs of a female.

Mutations Can Cause Cancer

It is estimated that one in three people will develop cancer at some time in their lives. Of these affected individuals, one-third of the females and one-fourth of the males will die due to cancer. In the United States, the three deadliest forms of cancer are lung cancer, colon and rectal cancer, and breast cancer.

The development of cancer involves a series of accumulating mutations that can be different for each type of cancer. As discussed in Section 9.4, tumor suppressor genes ordinarily act as brakes on cell division, especially when it begins to occur abnormally. Proto-oncogenes stimulate cell division but are usually turned off in fully differentiated, nondividing cells. When proto-oncogenes mutate, they become oncogenes that are active all the time (see Fig. 9.11). Carcinogenesis begins with the loss of tumor suppressor gene activity and/or the gain of oncogene activity. When tumor suppressor genes are inactive and oncogenes are active, cell division occurs uncontrollably, because a cell signaling pathway that reaches from the plasma membrane to the nucleus no longer functions as it should (see Fig. 9.12).

It often happens that tumor suppressor genes and proto-oncogenes code for transcription factors or proteins that control transcription factors. As we have seen, transcription factors are a part of the rich and diverse types of mechanisms that control gene expression in cells. They are of fundamental importance to DNA replication and repair, cell growth and division, control of apoptosis, and cellular differentiation. Therefore, it is not surprising that inherited or acquired defects in transcription factor structure and function contribute to the development of cancer.

For example, the tumor suppressor gene called *p53* is more frequently mutated in human cancers than is any other known gene. It has been found that the p53 protein acts as a transcription factor, and as such it is involved in turning on the expression of genes whose products are cell cycle inhibitors (see Section 9.1). *p53* also promotes apoptosis (programmed cell death) when it is needed. The retinoblastoma protein (RB) controls the activity of a transcription factor for cyclin D and other genes whose products promote entry into the S stage of the cell cycle. When the tumor suppressor gene *p16* mutates, the RB protein is always available, and the result is too much active cyclin D in the cell.

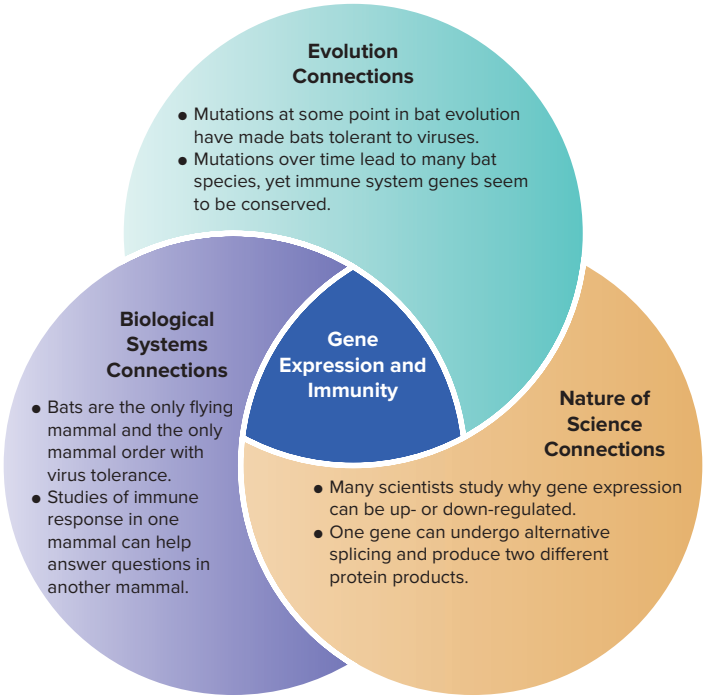
Mutations in many other genes also contribute to the development of cancer. Several proto-oncogenes code for ras proteins, which are needed for cells to grow, to make new DNA, and to not grow out of control. A point mutation is sufficient to turn a normally functioning *ras* proto-oncogene into an oncogene, and abnormal growth results.

Check Your Progress 13.3

1. List some common causes of spontaneous and induced mutations.
2. Explain how a frameshift mutation may disrupt a gene's function.
3. Discuss how a mutation in a tumor suppressor gene and in proto-oncogenes disrupts the cell cycle.

ISTUDY

CONNECTING *THE* CONCEPTS

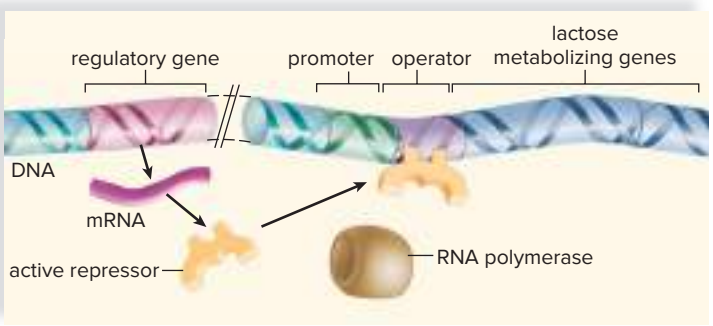


What Did You Learn?

13.1 Prokaryotic Regulation

Prokaryotes often organize genes that are involved in a common process or pathway into **operons** in which the genes are coordinately regulated.

- Gene expression in prokaryotes is usually regulated at the level of transcription. The operon model states that a **regulator gene** codes for a **repressor.** When the repressor binds to the **operator,** RNA polymerase is unable to bind to the **promoter,** and transcription of the **structural genes** of the operon cannot take place. Operons may also be regulated by both activators and repressors.
- The *trp* operon is an example of a repressible operon, because when tryptophan, the **corepressor,** is present, it binds to the repressor. The repressor is then able to bind to the operator, and transcription of structural genes does not take place.
- The *lac* operon is an example of an inducible operon, because when lactose, the **inducer,** is present, it binds to the repressor. The repressor is unable to bind to the operator, and transcription of structural genes takes place if glucose is absent.



- Both the *lac* and *trp* operons exhibit negative control, because a repressor is involved. However, some pathways also provide examples of positive control. The structural genes in the *lac* operon are not maximally expressed unless glucose is absent and lactose is present. At that time, cAMP attaches to a molecule called CAP, and then CAP binds to a site next to the promoter. Now RNA polymerase is better able to bind to the promoter, and transcription occurs.

13.2 Eukaryotic Regulation

The following levels of control of gene expression are possible in eukaryotes: chromatin structure, **transcriptional control,** posttranscriptional control, translational control, and posttranslational control.

- **Chromatin** structure helps regulate transcription. Highly condensed heterochromatin is genetically inactive, as exemplified by **Barr bodies.** Less-condensed euchromatin is genetically active, as exemplified by lampbrush chromosomes in vertebrates.
- Regulatory proteins called **transcription factors,** as well as DNA sequences called **enhancers,** play a role in **transcriptional control** in eukaryotes. Transcription factors bind to the promoter, and **transcription activators** bind to an enhancer. Small RNA molecules, such as microRNAs (miRNAs) and small-interfering RNAs (siRNAs), are involved in **RNA interference** and play a role in gene expression.
- **Posttranscriptional control** is achieved by creating variations in messenger RNA (mRNA) splicing, which may yield multiple mRNA messages from the same gene, and by altering the speed with which a particular mRNA molecule leaves the nucleus.
- **Translational control** affects mRNA translation and the length of time it is translated, primarily by altering the stability of an mRNA.

ISTUDY

- **Posttranslational control** affects whether or not an enzyme is active and how long it is active. **Proteasomes** are specialized structures containing **proteases** that break down protein molecules and participate in the regulation of gene expression.

13.3 Gene Mutations

In molecular terms, a gene is a sequence of DNA nucleotide bases, and a **gene mutation** is a change in this sequence.

- **Spontaneous mutations** occur for no apparent reason, while **induced mutations** are due to environmental **mutagens,** such as radiation and organic chemicals. **Carcinogens** are mutagens that cause cancer. **DNA repair enzymes** monitor and fix irregularities in DNA, notably thymine dimers that come from UV exposure.
- **Point mutations** can have a range of effects, depending on the particular codon change. Sickle-cell disease is an example of a point mutation that greatly changes the activity of the affected gene.
- **Frameshift mutations** result when one or more bases are added or deleted, and the result is usually a nonfunctional protein. Most cases of cystic fibrosis are due to a frameshift mutation. Nonfunctional proteins can affect the phenotype drastically, as in a form of albinism that is due to a single faulty enzyme, and androgen insensitivity, which is due to a faulty receptor for testosterone.
- Cancer is often due to an accumulation of genetic mutations among genes that code for regulatory proteins. The cell cycle occurs inappropriately when proto-oncogenes become oncogenes and tumor suppressor genes are no longer effective. Mutations that affect transcription factors and other regulators of gene expression are frequent causes of cancer.

Thinking Critically About Biology

1. Research has shown that mutations outside of genes may cause disease, such as in some cases of Hirschsprung disease and multiple endocrine neoplasia. Explain how such a mutation might alter the expression of a gene.
2. From an experimental point of view, is it better to use haploid or diploid organisms for mutagen testing? Think about the Ames test when considering your answer.
3. Discuss the times in a person's life when it would be most important to avoid mutagens. Which parts of a person's body should be most protected from mutagens?

Making It Relevant to You

1. Why are bats able to remain healthy when they carry so many viruses?
2. There may have been a time in your life when you and others around you were sick with the same virus (stomach flu, influenza, a common cold, etc.). Sometimes a viral infection is more severe in one person than another. Explain why you think this happens.
3. The scientists studying viral load in bats were from two different medical schools. Why would medical researchers be interested in bat immune systems?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

14

Biotechnology and Genomics

vchal/Shutterstock

CRISPR and Genome Editing

Biotechnology has come a long way since it first began in the early 1950s. Recently, a new form of genome editing called CRISPR is revolutionizing biotechnology. Based on a series of enzymes that protect bacteria from viruses, CRISPR acts as a "molecular scalpel," allowing researchers to target specific sequences of a genome for editing. Early experiments were able to correct defects in mice, and in 2018, researchers in China were able to produce embryos modified by CRISPR. Currently, studies are underway to determine how to modify the pig genome to produce more hearts and organs for human transplants, as well as cures for human diseases such as HIV, Huntington's disease, and high cholesterol. Almost monthly, new applications of this technology are revealed in scientific journals.

These advances present tremendous opportunities to address human disease, but they also give rise to ethical questions. For example, to what extent is it ethical to improve an individual's genome, and should society allow couples to produce "designer babies"?

In this chapter, we will explore the procedures most often used in biotechnology labs, as well as some of the products of biotechnology research.

As you read through this chapter, think about the following questions:

1. What is the process by which a gene from one species is inserted into the genome of another species?
2. What is the difference between a genetically modified organism and a transgenic organism?

Chapter Outline

14.1 Biotechnology
14.2 Biotechnology Products
14.3 Gene Therapy
14.4 Genomics

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 12.1 What is the basic structure of a DNA molecule?

Section 12.2 How is the DNA molecule replicated?

Section 13.2 What is the role of microRNA molecules in a cell?

Following *the* Themes

CHAPTER 14 – BIOTECHNOLOGY AND GENOMICS

Evolution	The comparison of the genomes of humans and model organisms, such as the mouse, is providing insights into the evolution of our species.
Nature of Science	The development of recombinant DNA technology has enabled scientists to produce genetically modified and transgenic organisms that benefit human society.
Biological Systems	Recombinant DNA technology may be used to modify individual cells, as is the case in gene therapy, or produce organisms that assist in the cleanup of polluted ecosystems.

ISTUDY

14.1 Biotechnology

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the steps involved in making a recombinant DNA molecule.
2. Explain the purpose of the polymerase chain reaction (PCR).
3. Identify how PCR may be used to analyze DNA.
4. Describe the process of genome editing.

The term **biotechnology** refers to the use of natural biological systems to create a product or achieve some other end desired by humans. Today, **genetic engineering** allows scientists to modify the genomes of a variety of organisms, from bacteria to plants and animals, to either improve the characteristics of the organism or make biotechnology products. Such modification is possible because decades of research on how DNA and RNA function in cells has allowed for the development of new techniques. These techniques allow scientists to not only clone genes, but also directly edit the genome of an organism.

A **genetically modified organism (GMO)** is one whose genome has been modified in some way, usually by using recombinant DNA technology. A **transgenic organism** is an example of a GMO that has had a gene from another species inserted into its genome. We will take a closer look at both GMOs and transgenics in Section 14.2, but first we need to explore some of the DNA techniques used in biotechnology.

Cloning

In biology, **cloning** is the production of genetically identical copies of DNA, cells, or organisms through some asexual means. When an underground stem or root sends up new shoots, the resulting plants are clones of one another. The members of a bacterial colony on a petri dish are clones because they all came from the division of a single original cell. Human identical twins are also considered clones. Early in embryonic development, the cells separate, and each becomes a complete individual. The Nature of Science feature, "Reproductive and Therapeutic Cloning," in Section 9.3 explores some of the different ways the term *cloning* may be used in biology.

DNA cloning can be done to produce many identical copies of the same gene—that is, for the purpose of **gene cloning.** Scientists clone genes for a number of reasons. They might want to determine the difference in base sequence between a normal gene and a mutated gene. Or they might use the genes to genetically modify organisms in a beneficial way.

Recombinant DNA Technology

Recombinant DNA (rDNA) contains DNA from two or more different sources, such as a human cell and a bacterial cell. One way

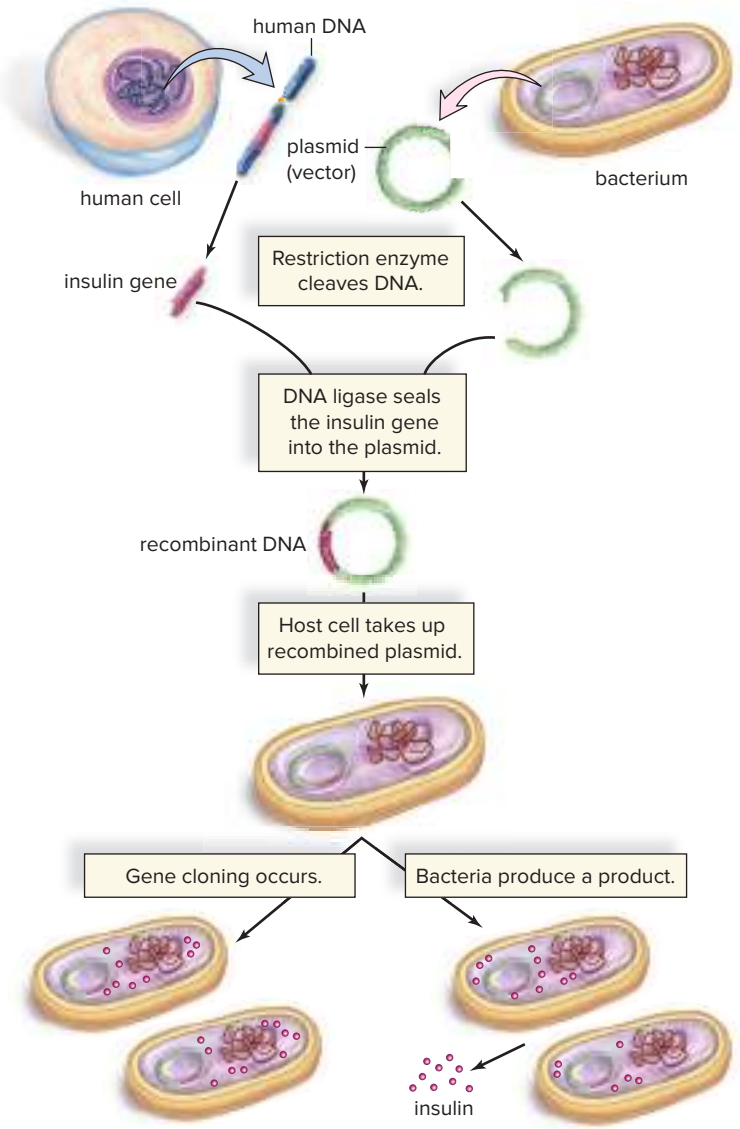


Figure 14.1 Cloning a gene. This figure shows the basic steps in the cloning of a gene. The DNA of interest and plasmid DNA are cleaved by a specific type of restriction enzyme. Then, the DNA, which in this case codes for the human insulin gene, is spliced into a plasmid by the enzyme DNA ligase. Gene cloning is achieved after a bacterium takes up the plasmid. If the gene functions normally, as expected, the product (e.g., insulin) may also be retrieved.

of doing this is shown in Figure 14.1. To make rDNA, a technician needs a **vector** (L. *vehere,* "to carry") by which rDNA will be introduced into a host cell. One common vector is a plasmid. **Plasmids** are small accessory rings of DNA found in bacteria. They were first discovered in the bacterium *Escherichia coli* (*E. coli*). The ring is not part of the main bacterial chromosome. It replicates on its own and can be easily removed from, or introduced into, another bacterial cell.

Two enzymes are needed to introduce foreign DNA into vector DNA: a **restriction enzyme,** which cleaves (cuts) DNA, and an enzyme called **DNA ligase** (L. *ligo,* "bind"), which seals DNA into an opening created by the restriction enzyme. Hundreds of

ISTUDY

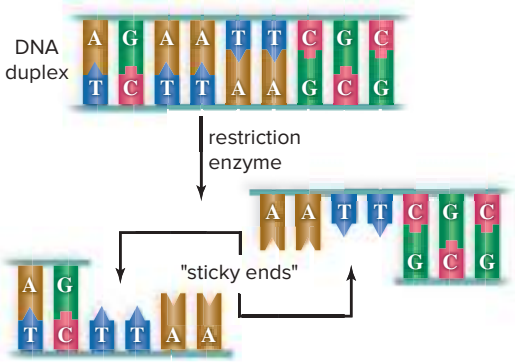


Figure 14.2 Restriction enzymes cut DNA at specific locations. Each restriction enzyme recognizes a specific sequence of nucleotides. After the enzyme cuts the DNA, "sticky ends" may be formed that are useful in the cloning of DNA sequences.

restriction enzymes occur naturally in bacteria, and their role is to cut up any viral DNA that enters the cell. They are called restriction enzymes because they *restrict* the growth of viruses. Scientists take advantage of these enzymes and use them as molecular scissors to cleave any piece of DNA at a specific site.

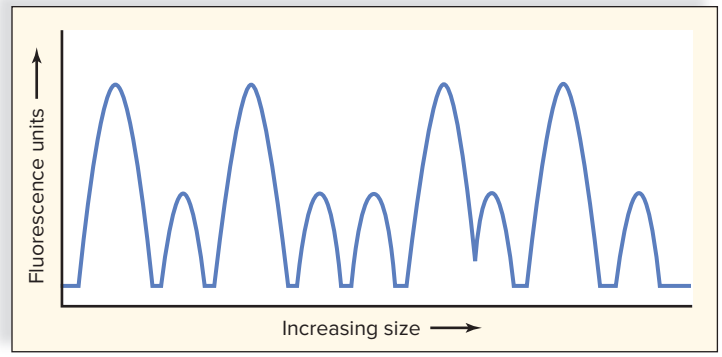
Notice that the restriction enzyme creates a puzzlelike gap in the DNA (Fig. 14.2), into which a piece of foreign DNA can be placed if its ends are complementary to those exposed by the restriction enzyme. The single-stranded, but complementary, ends of the two DNA molecules are called "sticky ends" because they can bind a piece of foreign DNA by complementary base pairing. Sticky ends facilitate the insertion of foreign DNA into vector DNA as long as both are cleaved by the same restriction enzyme.

Next, genetic engineers use the enzyme DNA ligase to seal the foreign piece of DNA into the vector. DNA splicing is now complete; an rDNA molecule has been constructed (see Fig. 14.1). Bacterial cells may take up recombinant plasmids once treated to make their plasma membranes more permeable. Later, as the plasmid replicates, the rDNA is cloned.

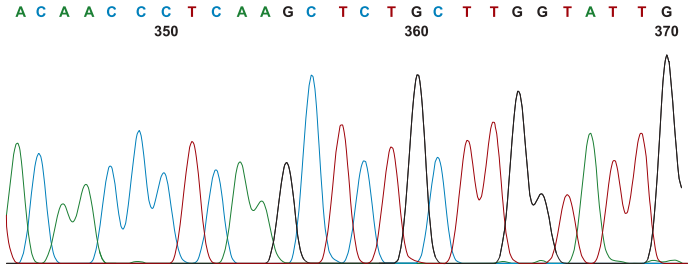
DNA Sequencing

DNA sequencing is a procedure used to determine the order of nucleotides in a segment of DNA, often within a specific gene. DNA sequencing allows researchers to identify specific alleles associated with a disease and thus facilitate the development of medicines or treatments. Information from DNA sequencing also serves as the foundation for the study of forensic biology and even contributes to our understanding of our evolutionary history.

When DNA technology was in its inception in the early 1970s, this technique was performed manually using radioactive tracer elements attached to each of the four nucleotides during DNA replication. These reactions were then loaded into a gel, which in turn was exposed to a piece of photographic film. The results were interpreted from the pattern on the film, which indicated the sequence of nucleotides. Modern-day sequencing involves attaching dyes to the nucleotides and detecting the



a. Automated DNA sequencing



b. An example of a sequenced section of DNA

c. Actual output from a sequencing reaction

Figure 14.3 Automated DNA sequencing. **a.** Modern DNA sequencing techniques use fluorescent dyes that allow researchers to determine the nucleotide sequence of a segment of DNA. **b.** An example of a DNA sequence and (**c**) output on a computer screen.
(b): Alan John Lander Phillips/E+/Getty Images; (c): Nicolas/E+/Getty Images

different dyes via a laser in an automated sequencing machine, which shows the order of nucleotides on a computer screen (Fig. 14.3). To begin sequencing a segment of DNA, many copies of the segment are made, or replicated, using a procedure called the polymerase chain reaction.

The Polymerase Chain Reaction

Another revolution in molecular biology was the development of the **polymerase chain reaction (PCR).** American biochemist

ISTUDY

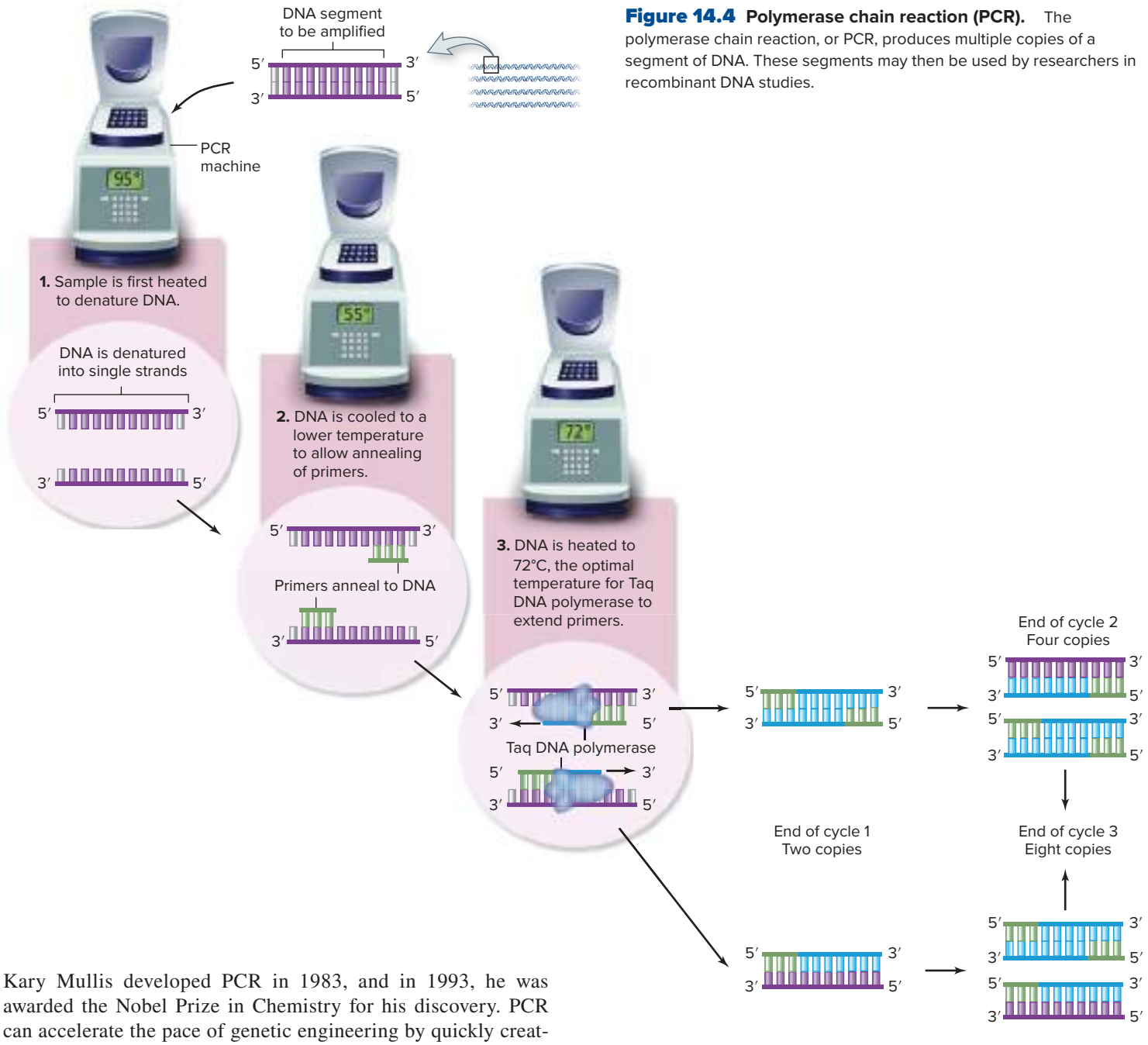


Figure 14.4 Polymerase chain reaction (PCR). The polymerase chain reaction, or PCR, produces multiple copies of a segment of DNA. These segments may then be used by researchers in recombinant DNA studies.

Kary Mullis developed PCR in 1983, and in 1993, he was awarded the Nobel Prize in Chemistry for his discovery. PCR can accelerate the pace of genetic engineering by quickly creating many clones of a piece of DNA without first inserting it into a plasmid. The process mimics DNA replication in the cell (see Section 12.2), except that PCR is very specific—it amplifies (makes copies of) only a targeted DNA sequence. The targeted sequence can be less than one part in a million of the total DNA sample.

PCR requires the use of DNA polymerase, the enzyme that carries out DNA replication, and a supply of nucleotides for the new DNA strands. The DNA polymerase used in the reaction is a heat-stable (thermostable) polymerase that has been extracted from the bacterium *Thermus aquaticus,* which lives in hot springs. The enzyme can withstand the high temperature used to separate double-stranded DNA; therefore, replication does not have to be interrupted by the need to add more enzyme. PCR is a chain reaction because the targeted DNA is repeatedly replicated as long as the process continues. The colors in Figure 14.4 distinguish the old strand from the new DNA strand, but keep in mind that all the newly synthesized strands are identical (clones). Notice that the amount of DNA doubles with each replication cycle.

Analyzing DNA

Analysis of DNA following PCR has improved over the years. At first, the entire genome was treated with restriction enzymes, and because each person has different restriction enzyme sites, this yielded a unique collection of DNA fragments of various sizes. During a process called *gel electrophoresis,* an electrical

ISTUDY

current is used to force DNA through a porous gel material, separating the fragments according to their size. Smaller fragments moved farther through the gel than larger fragments, resulting in a pattern of distinctive bands, called a DNA profile, or **DNA fingerprint.**

DNA amplified by PCR can be analyzed for various purposes. For example, mitochondrial DNA taken from modern living populations was used to decipher the evolutionary history of our species. For identification purposes, DNA taken from a corpse burned beyond recognition can be matched to that on the bristles of the person's toothbrush!

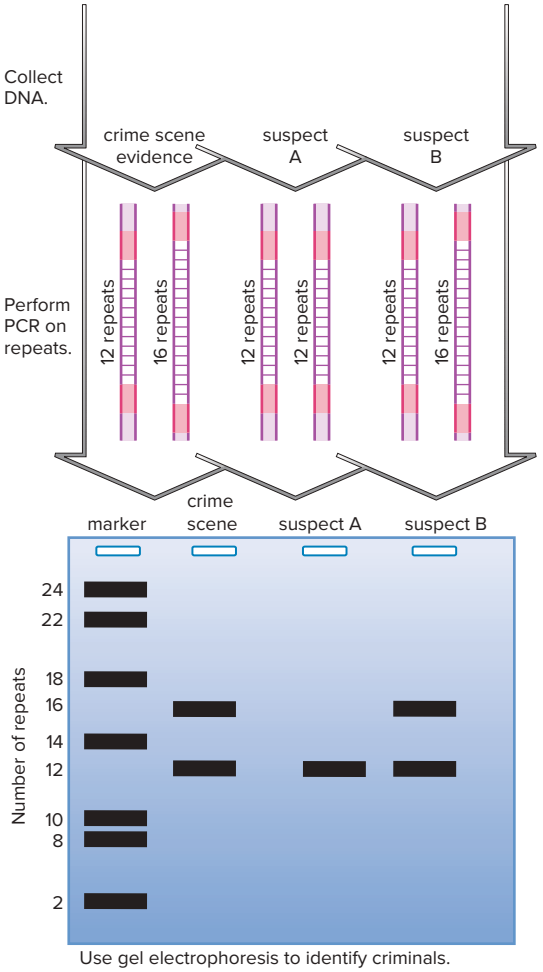
Now, **short tandem repeat (STR) profiling** is the preferred method. STRs are short sequences of DNA bases that recur several times, as in GATAGATAGATA. STR profiling is advantageous because it doesn't require the use of restriction enzymes. Instead, PCR is used to amplify target sequences of DNA, which are fluorescently labeled. The PCR products are placed in an automated DNA sequencer. As the PCR products move through the sequencer, the fluorescent labels are picked up by a laser. A detector then records the length of each DNA fragment. The fragments are different lengths because each person has a specific number of repeats at a particular location on the chromosome (i.e., at each STR locus). That is, the greater the number of STRs at a locus, the longer the DNA fragment amplified by PCR. Individuals who are homozygotes will have a single fragment, and heterozygotes will have two fragments of different lengths (Fig. 14.5). The more STR loci employed, the more confident scientists can be of distinctive results for each person.

Another method of producing DNA fingerprints does away with the need to use gel electrophoresis: The DNA fragments are fluorescently labeled. A laser then excites the fluorescent STRs, and a detector records the amount of emission for each DNA fragment in terms of peaks and valleys. Therefore, the greater the fluorescence, the greater the number of repeats at a location. The printout is the DNA fingerprint, and each person has their own unique printout. Currently, the Federal Bureau of Investigation (FBI) has 13 STR locations on various chromosomes that are routinely used in the identification of individuals in the United States.

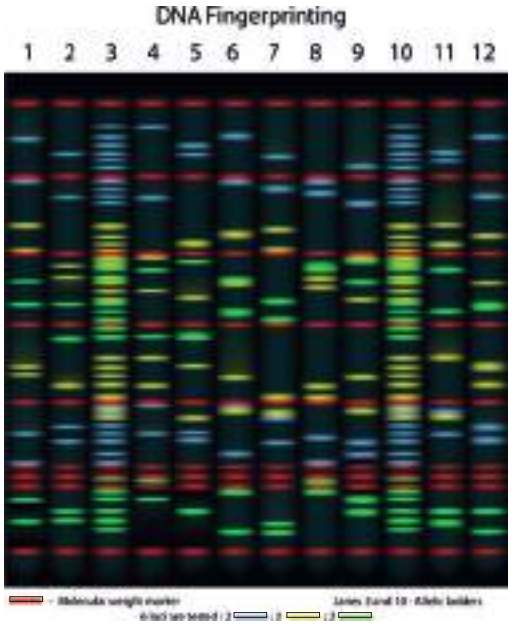
Genome Editing

A relatively new advance in DNA technology is **genome editing,** the targeting of specific sequences in the DNA for removal or replacement. There are several methods by which editing may be done; the most widely used is called CRISPR (*c*lustered *r*egularly *i*nterspaced *s*hort *p*alindromic *r*epeats). The Diversity in Science feature, "CRISPR Pioneers," highlights the two female scientists who developed the process.

CRISPR was first discovered in prokaryotes, where it acts as a form of immune defense against invading viruses. Viruses function by inserting their DNA into host cells, causing those cells to form new viruses (see Section 20.1). The CRISPR system is based on an endonuclease enzyme called Cas9, which is capable of identifying specific sequences of nucleotides in the genomic DNA of the invading virus and breaking both of the DNA strands, thus inactivating the virus.



a. Example of DNA fingerprinting



b. Actual DNA fingerprinting results

Figure 14.5 DNA fingerprinting. **a.** For DNA fingerprinting, PCR is used to generate copies of specific regions of DNA, which are then analyzed for length variations. These variations may be used to identify deceased individuals or suspects in a crime. In this example, the evidence suggests that suspect A is not a suspect. **b.** Actual results from a DNA fingerprinting analysis.
(b): Alila Medical Media Images/Shutterstock

ISTUDY

▶ Diversity in Science

CRISPR Pioneers

Since 1901, the Nobel Prize in Chemistry has been awarded to 186 individuals. Only 7 have been women. Two were added to the list in 2020 when Emmanuelle Charpentier and Jennifer A. Doudna were awarded the prize for their pioneering work on the CRISPR method of genome editing (Fig. 14A).

Nobel Prizes are awarded to people who, according to Alfred Nobel, "have conferred the greatest benefit to humankind." Charpentier and Doudna both studied the CRISPR/Cas9 system as it related to bacterial immunity and began collaborating in 2011 after meeting at a scientific conference. Their experiments first led to important discoveries of how the system worked in bacteria to destroy the DNA of invading viruses. The big breakthrough happened when Charpentier and Doudna were able to simplify the molecules to create a "scalpel" that precisely cuts DNA at a location that could be programmed by the scientists.

The use of CRISPR gene-editing technology has since exploded. In the eight short years between the publication of their work and the Nobel Prize ceremony, other scientists have used the technique to benefit humankind through the development of genetically modified bacteria, plants, and animals. Trials are even underway that deploy the CRISPR scalpel into human patients in order to treat genetic disorders.

In the wake of receiving an award that has historically gone to male scientists, Doudna reflected on its meaning, saying, "I'm proud of my gender. I think it's great, especially for younger women, to see this and to see that women's work can be recognized as much as men's."

Charpentier shared her hope for the next generation: "My wish is that this will provide a positive message to the young girls who would like to follow the path of science, and to show them that women in science can also have an impact through the research they are performing."

Figure 14A Nobel Prize winners. Jennifer A. Doudna (*left*) and Emmanuelle Charpentier (*right*) won the 2020 Nobel Prize in Chemistry for the development of the CRISPR gene-editing method.

(left): Anastasiia Sapon/The New York Times/Redux Pictures; (right): dpa picture alliance/Alamy Stock Photo

Cas9 identifies the specific nucleotides to be cut using a guide RNA molecule that complementary base-pairs to the genomic DNA sequence (Fig. 14.6).

The CRISPR system can be used by researchers to target a specific sequence of nucleotides, in almost any organism, for editing. If the genomic sequence of the target is known, a complementary RNA strand can be used by Cas9 to produce a break in the DNA. This break can be used to inactivate the gene and thus study the role of the gene in the cell, or Cas9 can act as a form of molecular scissors to insert new nucleotides at specific DNA locations.

CRISPR and other genome-editing technologies continue to develop. Scientists are investigating ways of making the processes more efficient, as well as new applications for genome editing in humans and other organisms.

Check Your Progress 14.1

1. Explain the purpose of restriction enzymes in creating an rDNA molecule.
2. Summarize how the polymerase chain reaction (PCR) is used in biotechnology.
3. Explain how DNA fingerprinting distinguishes individuals.
4. Explain why genome editing may be more efficient than recombinant DNA processes.

ISTUDY

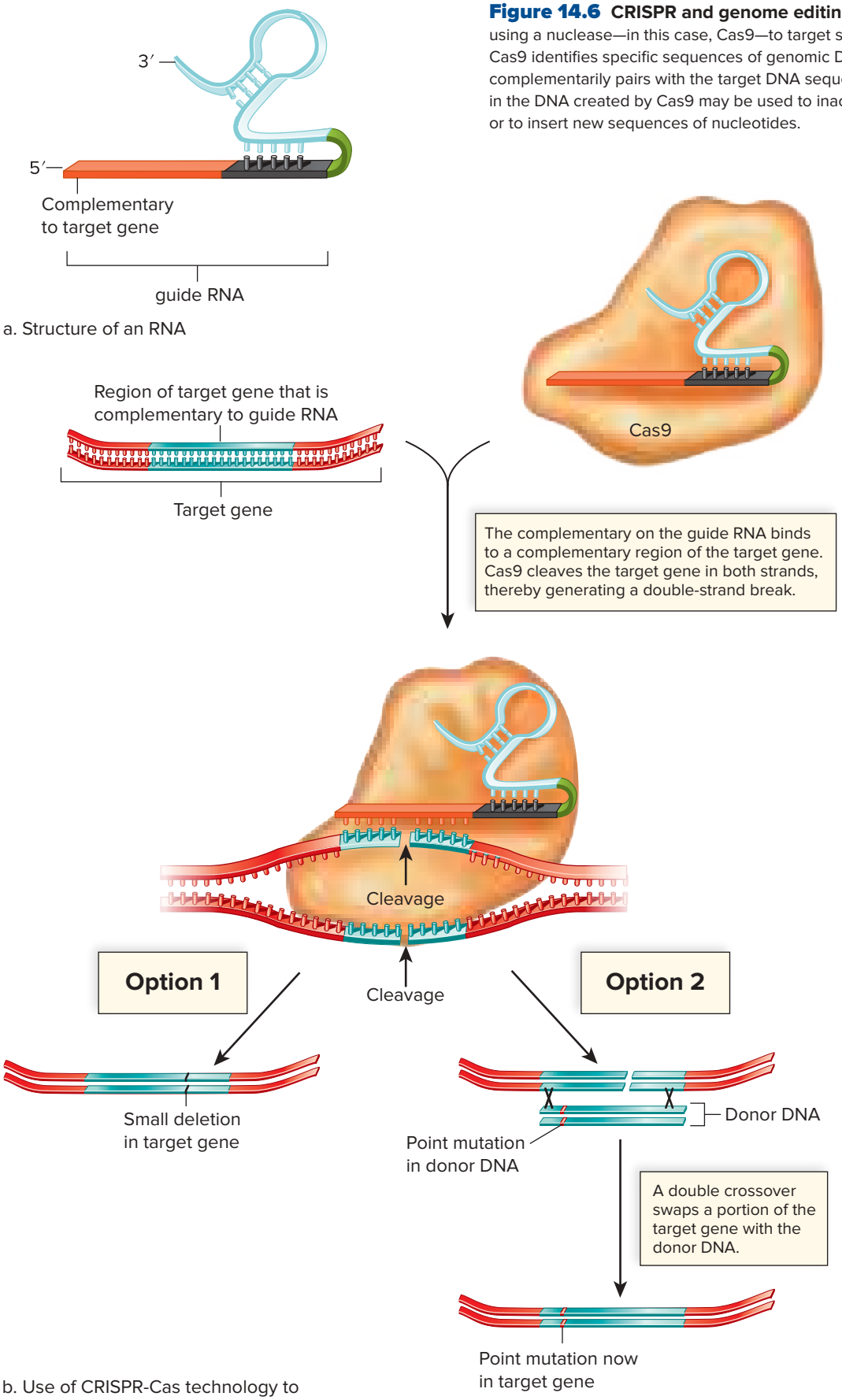


Figure 14.6 CRISPR and genome editing. Genome editing involves using a nuclease—in this case, Cas9—to target specific sequences of DNA. Cas9 identifies specific sequences of genomic DNA using a guide RNA that complementarily pairs with the target DNA sequence. The double-stranded break in the DNA created by Cas9 may be used to inactivate a gene to study its function, or to insert new sequences of nucleotides.

ISTUDY

THEME Nature of Science

CRISPR Gene Editing in Humans

Currently, the vast majority of human-related gene editing is being conducted in animal models. For example, scientists are using mice to study fragile-X syndrome (a chromosomal abnormality), deafness, and neuron diseases such as ALS (amyotrophic lateral sclerosis), to name a few. Ethical and carefully controlled examples of direct human CRISPR gene editing do exist, and scientists are using this technique in clinical trials to treat sickle-cell disease, cancer, and other diseases (Fig. 14B).

Sickle-Cell Disease

Sickle-cell disease affects about 100,000 individuals in the United States and is caused by a single point mutation in a gene that con-tributes to the production of hemoglobin, a protein that carries oxygen in the blood (see Section 32.4). Individuals with sickle-cell disease have two copies of this mutation, resulting in deformed red blood cells that get stuck inside tiny blood vessels and restrict blood flow. When oxygen is not carried around the body normally, individuals can suffer sharp pains and weakness and develop complications that shorten a normal life span.

CRISPR gene editing takes place in blood cells extracted from the bone marrow of a person with sickle-cell disease. The blood cells are commonly harvested from hip bones. Considered an ex vivo gene therapy (see Section 14.3), the blood cells are then edited using CRISPR technology to produce a functioning hemoglobin protein. Afterward, these edited blood cells are reinserted into the individual so the normal protein can be produced, giving affected persons a higher quality of life.

Cancer

A lymphocyte is a type of white blood cell in the human body. A T cell, on the other hand, is a specific type of lymphocyte that is responsible for attacking cancer cells or cells infected with a virus or bacteria (see Section 33.4). T cells have a surface protein called PD-1 (programmed cell death protein 1) that puts the "brakes" on immune system function, preventing it from becoming overactive. This can limit the body's fight against cancer cells.

Patients with certain types of bone or bone marrow cancers are currently in clinical trials at the University of Pennsylvania. Similar to the treatment for individuals with sickle-cell disease, bone marrow cells are extracted from patients with cancer and edited with CRISPR technology ex vivo. The editing involves inserting a gene to make T cells better cancer fighters, and removes the *PD-1* gene so the body's fight against cancer cells is not halted. The Pennsylvania cancer study is just one of many examples where CRISPR gene editing is being used to battle cancer.

Editing Sperm

It may seem that sperm editing should be as controversial as editing an embryo, but editing sperm is considered safer because one cell is being targeted, whereas in an embryo, many cells need to be targeted and some could be missed.

In a New York lab associated with Cornell University, scientists are attempting to use CRISPR technology to edit sperm cells that contain a defective *BRCA2* gene. A mutation in the *BRCA2* gene has been linked to breast, ovarian, and prostate cancer. If these experiments are successful, then future couples may have mutation-free embryos for IVF (in vitro fertilization). As this sperm-editing technology becomes perfected, the fear is it might be used unethically to alter non-disease-producing genes, in order to create "designer" babies.

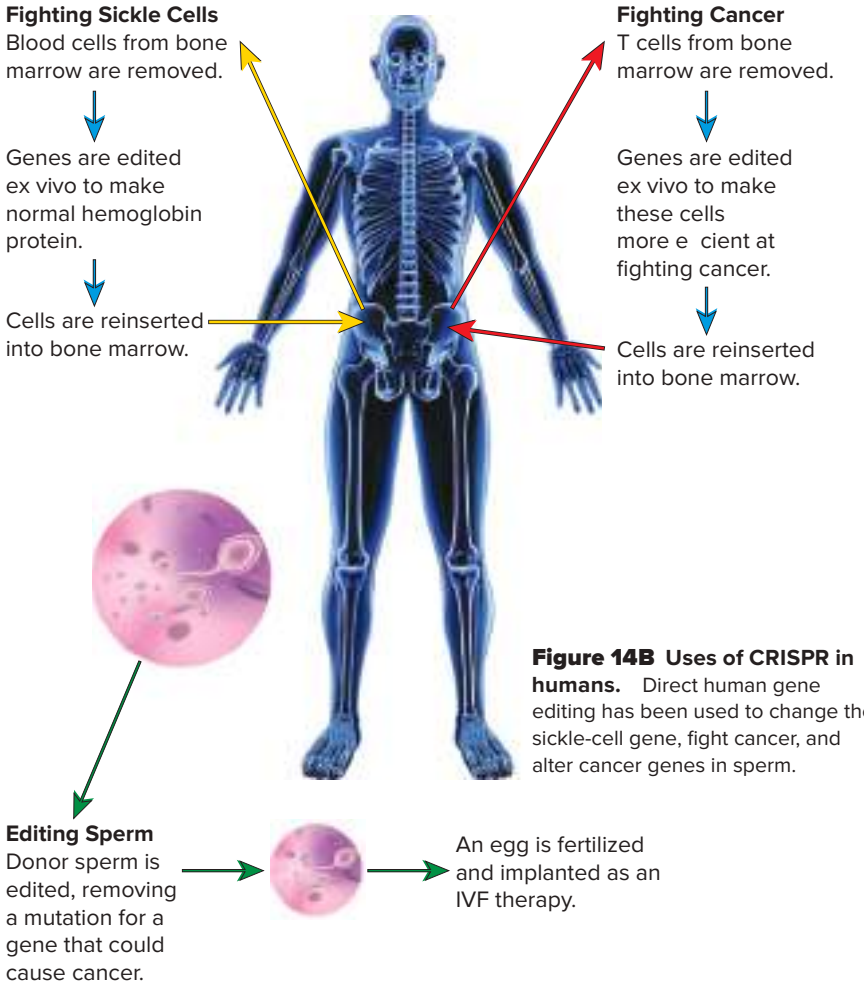


Figure 14B Uses of CRISPR in humans. Direct human gene editing has been used to change the sickle-cell gene, fight cancer, and alter cancer genes in sperm.

ISTUDY

14.2 Biotechnology Products

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the benefits of genetically modified bacteria, plants, and animals to human society.
2. Describe the steps involved in the production of a transgenic animal.

Today, bacteria, plants, and animals may be genetically engineered to produce biotechnology products. Recall that a *genetically modified organism* (*GMO*) is one whose genome has been modified in some way, usually by using recombinant DNA technology. Organisms that have had a foreign gene inserted into their genome are called *transgenic organisms.*

The use of the biotechnology processes outlined in Section 14.1, and specifically the development of the CRISPR genome-editing process, has greatly facilitated the ability of scientists to generate GMOs. Almost every day, new advances are being reported by scientists. In this section, we will take a look at some of the ways bacteria, plants, and animals have been genetically modified to benefit humans.

Genetically Modified Bacteria

Bacteria were the first organisms to be genetically modified, and today recombinant DNA technology is used to produce transgenic bacteria, on an industrial scale, by growing them in huge vats called bioreactors. As the bacteria grow, the product they have been engineered to produce is collected from the medium. Human insulin, food products, pharmaceutical drugs, and even vaccines such as the hepatitis B vaccine can be produced in this manner.

Transgenic bacteria have many other uses as well. Some have been produced to promote the health of plants. For example, bacteria that normally live on plants and encourage the formation of ice crystals have been changed from frost-plus to frost-minus bacteria. As a result, new crops, such as frost-resistant strawberries, have been developed (Fig. 14.7*a*). Also, a bacterium that normally colonizes the roots of corn plants has now been endowed with genes (from another bacterium) that code for an insect toxin. The toxin protects the roots from insects.

Bacteria can be selected for their ability to degrade a particular substance, and this ability can then be enhanced by bioengineering. *Bioremediation* is the process that uses microorganisms or other organisms, such as plants, to detoxify pollutants in the environment. For instance, naturally occurring bacteria that eat oil have been genetically engineered to clean up beaches after oil spills, while others can be used to remove waste oil from ponds (Fig. 14.7b). Bacteria can also remove sulfur from coal before it is burned and help clean up toxic waste dumps. One such strain was given genes that allowed it to clean up levels of toxins that would have killed other strains. Further, these bacteria were given "suicide" genes that caused them to self-destruct when the job had been accomplished.

Organic chemicals are often synthesized by having catalysts act on precursor molecules or by using bacteria to carry out the synthesis. Today, it is possible to go one step further and manipulate the genes that code for these enzymes. For instance, biochemists discovered a strain of bacteria that is especially good at producing phenylalanine, an organic chemical needed to make aspartame, the dipeptide sweetener better known as NutraSweet. They isolated, altered, and formed a vector for the appropriate genes, so that various other bacteria could be genetically engineered to produce phenylalanine.

Figure 14.7 Uses of genetically modified bacteria. **a.** Genetically modified bacteria help protect these strawberries from freezing. **b.** Bacteria capable of decomposing oil have been engineered and used in the cleanup of oil spills, such as in this pond for waste oil in the Amazon.
(green strawberry): Dmytro Skrypnykov/iStock/Getty Images; (bioremediation pond): Morley Read/Alamy Stock Photo

ISTUDY

Genetically Modified Plants

Techniques have been developed to introduce foreign genes into immature plant embryos or into plant cells called *protoplasts* that have had the cell wall removed. The protoplasts are treated with an electric current while they are suspended in a liquid containing foreign DNA. The current creates tiny, self-sealing holes in the plasma membrane, through which the DNA can enter. These treated protoplasts go on to develop into mature plants.

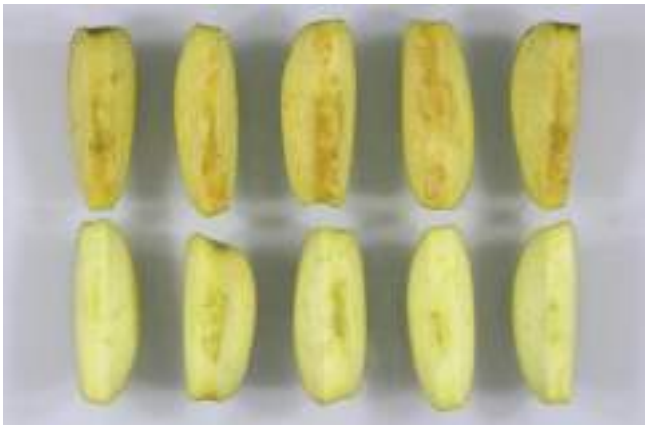
"Bt" plants are so named because they contain a foreign gene taken from a common soil organism, *Bacillus thuringiensis,* which makes a protein that is toxic to many insect pests. About a dozen Bt varieties, including corn, potato, and even a tomato, have now been approved for human consumption (Fig. 14.8*a*). Other species, such as soybean and corn, contain a gene that makes them resistant to a common herbicide, called glyphosate. If crops are resistant to the herbicide and weeds are not, then the herbicide can be used to kill the weeds. When herbicide-resistant plants are planted, weeds are easily controlled, less tillage is needed, and soil erosion is minimized. While there are ecological concerns about generating herbicide-resistant plants that may harm insect populations, the use of these plants remains popular with farmers to increase crop yield.

Foreign genes transferred to strains of cotton, corn, potato, and even bananas have made these plants resistant to pests such as fungi and insects, because their cells now produce a chemical that is toxic to the pest species. Similarly, soybeans have been made resistant to a common herbicide. Some corn and cotton plants are both pest- and herbicide-resistant. A strain of rice called Golden Rice has been engineered to have a higher vitamin A content.

Another focus of genetic engineering in plants has been the development of crops with improved qualities, especially improvements that reduce waste from food spoilage. One of the earliest examples was Golden Rice, which was engineered to have a higher vitamin A content. In another example, by knocking out a gene that causes browning in apples, a company called Okanagan Specialty Fruits produced the Arctic Apple, a genetically modified apple with an increased shelf life (Fig. 14.8*b*). The Innate potato, on the other hand, is a type of genetically modified potato in which a process called RNA interference turns off the expression of genes associated with bruising.

These and other genetically modified crops are now sold commercially, with the goal of increased yields and better nutrient content. Like bacteria, plants are also being engineered to produce human proteins, such as hormones, clotting factors, and antibodies, in their seeds. One type of antibody made by corn can deliver radioisotopes to tumor cells; another, made by soybeans, can be used to treat genital herpes. Currently, tobacco plants are being used to develop a vaccine for a variety of diseases.

a.



b.

Figure 14.8 Example of genetically modified plants. Genetic modifications allow (**a**) corn to resist insects and herbicides that control weeds and (**b**) apples to resist browning.
(a): Zeljko Radojko/iStock/Getty Images; (b): ©Okanagan Specialty Fruits Inc.

Genetically Modified Animals

Techniques have been developed to genetically modify animals by inserting genes into their eggs. It is possible to microinject foreign genes into eggs by hand, but another method uses vortex mixing. The eggs are placed in an agitator with DNA and silicon-carbide needles, and the needles make tiny holes through which the DNA can enter. When these eggs are fertilized, the resulting offspring are transgenic animals. Through this technique, many types of animal eggs have taken up the gene for bovine growth hormone (bGH). The procedure has been used to produce larger fishes, cows, pigs, rabbits, and sheep.

One example is the transgenic salmon. Climate change and overfishing have led to a decline in wild-caught salmon. With demand for salmon high, researchers developed an alternative, a transgenic salmon, called the AquAdvantage salmon.

To produce the AquAdvantage salmon, researchers constructed a transgene composed of the chinook growth hormone gene and the promoter (regulatory region) from ocean pout. This transgene was then microinjected into fertilized Atlantic

ISTUDY

salmon eggs. This experiment was done once in 1989. Of the resulting hatched fish, the first to reach sexual maturity, a female, became the founder fish of the AquAdvantage line in 1992.

The AquAdvantage salmon has the growth hormone gene from the Chinook salmon, for faster growth, and a gene promoter from the Ocean Pout, which keeps the growth hormone gene "on." The gene products are grown on bacterial plasmids, isolated from those plasmids, and then mixed with fertilized Atlantic salmon eggs.

Since then, many generations of AquAdvantage salmon have been produced using conventional breeding methods—by fertilizing conventional Atlantic salmon eggs with milt from AquAdvantage salmon neomales.

The wild Atlantic salmon and the AquAdvantage salmon are the same size when they reach adulthood. The difference is that the transgenic salmon will reach adult size three times faster and consume less food (Fig. 14.9*b*). As an added safety measure to prevent breeding with wild salmon, the transgenic fish are all female and sterile.

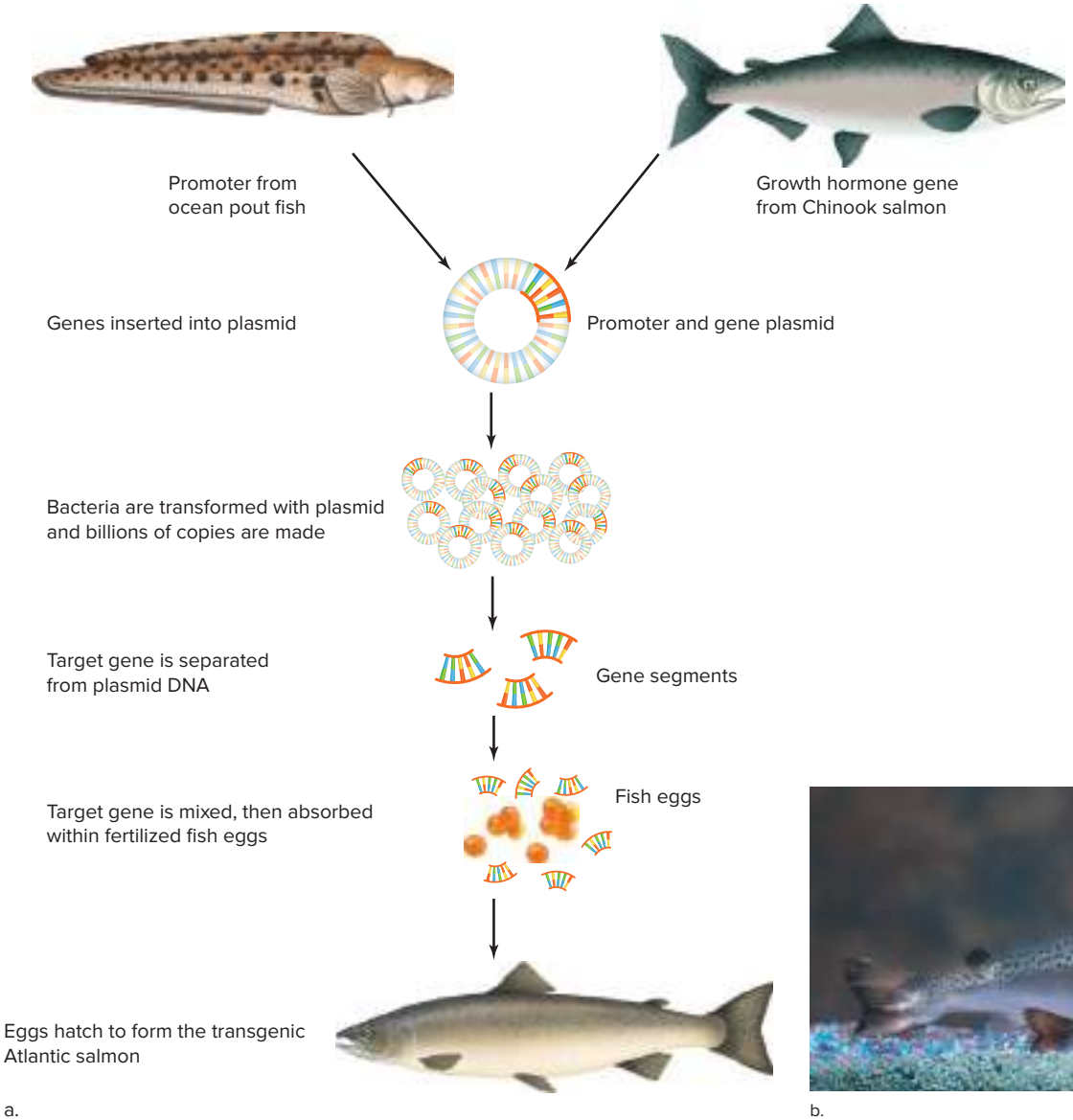


Figure 14.9 Transgenic salmon. **a.** Bioengineered salmon contain gene products from two other fish. **b.** The transgenic Atlantic salmon (background) will reach adult size faster than the non-transgenic salmon (foreground).

(b): ©AquaBounty Technologies, Inc.

ISTUDY

Other examples include the *Aedes aegypti* mosquito, which acts as a vector for several human diseases, notably Zika and dengue fever. A transgenic form of the mosquito being released in Florida contains a genetic "kill switch," which produces proteins that kill the offspring, thus reducing the size of the population.

Transgenic mice are also routinely used in medical research to study human diseases. For example, the allele that causes cystic fibrosis can be cloned and inserted into mouse embryonic stem cells. Occasionally, a mouse embryo homozygous for cystic fibrosis will result. This embryo develops into a mutant mouse that has a phenotype similar to that of a human with cystic fibrosis. New drugs for the treatment of cystic fibrosis can then be tested in such mice. A similar research animal is the OncoMouse, which carries genes for the development of cancer.

Producing and Cloning Transgenic Mammals

Gene pharming is the use of transgenic farm animals to produce pharmaceuticals. Genes that code for therapeutic and diagnostic proteins are incorporated into an animal's DNA, and the proteins appear in the animal's milk or urine. Drugs have been developed using this process that treat cystic fibrosis, cancer, blood diseases, and other disorders. Figure 14.10*a* outlines a procedure for producing transgenic mammals: DNA containing the gene of interest is injected into donor eggs. Following in vitro fertilization, the zygotes are placed in host females, where they develop. After female offspring mature, the product is secreted in their milk.

For many years, researchers believed that adult vertebrate animals could not be cloned, because cloning requires that all the genes of an adult cell be turned on if development is to proceed normally. In 1997, however, Scottish scientists announced they had produced a cloned sheep, which they called Dolly. Since then, calves, goats, pigs, rabbits, and even cats have also been cloned. The techniques can be applied to produce populations of transgenic animals.

As shown in Figure 14.10*b,* after enucleated eggs from a donor are microinjected with 2n nuclei from a single transgenic animal, they are coaxed to begin development in vitro. Development continues in host females until the clones are born. The female clones have the same product in their milk as the original transgenic animal. Now that scientists have a way to clone animals, this procedure will undoubtedly be used routinely to procure biotechnology products. However, animal cloning is a difficult process with a low success rate (usually 1 or 2 viable embryos per 100 attempts). The vast majority of cloning attempts are unsuccessful, resulting in the early death of the clone.

Compared with the production of proteins in bacteria, one advantage of molecular pharming is that certain proteins are more likely to function properly when expressed in mammals. This may be due to specific protein folding or other modifications that occur in mammals but not in bacteria. In addition, certain proteins may be degraded rapidly or folded improperly when expressed in bacteria. Furthermore, the yield of recombinant proteins in milk can be quite large. Each dairy cow, for example, produces about 10,000 liters of milk per year. In some cases, a transgenic cow can produce approximately 1 gram per liter (g/L) of the transgenic protein in its milk.

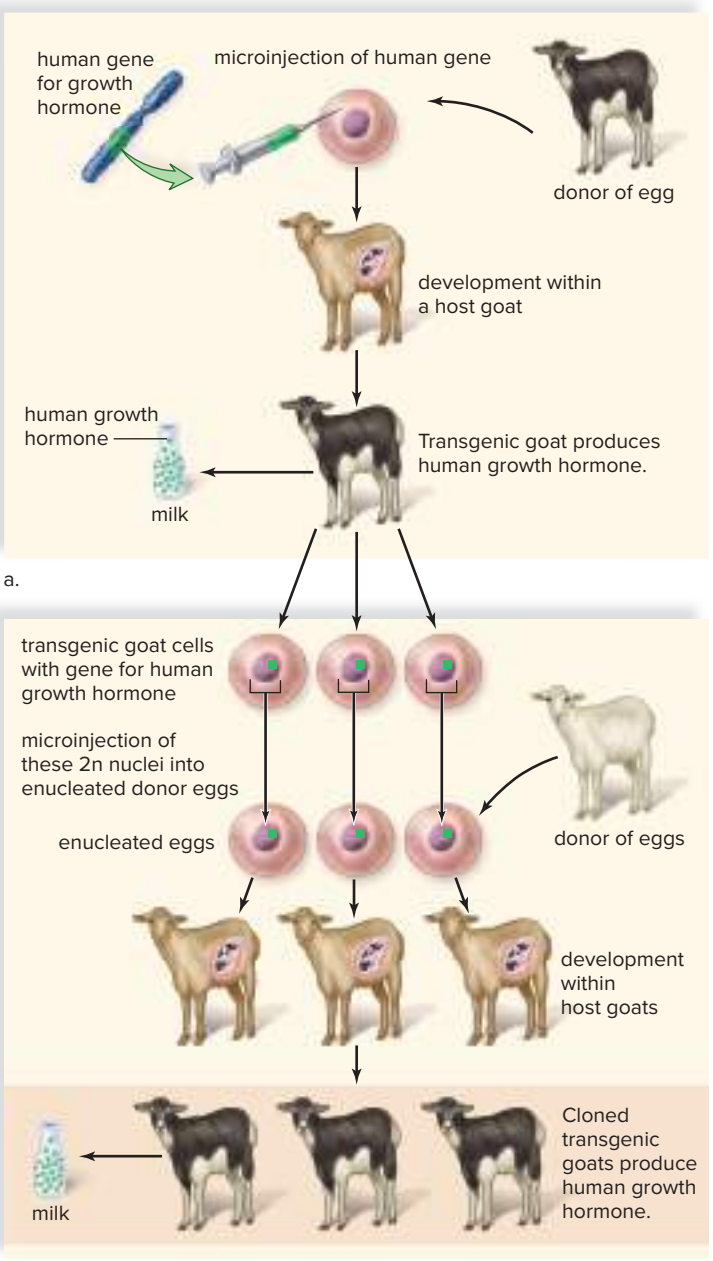


Figure 14.10 Transgenic mammals produce a product. This figure illustrates the basic procedure for generating a transgenic animal. **a.** A bioengineered egg develops in a host to create a transgenic goat, which produces a biotechnology product in its milk. **b.** Nuclei from the transgenic goat are transferred into donor eggs, which develop into cloned transgenic goats.

Check Your Progress 14.2

1. List some of the beneficial applications of transgenic bacteria, animals, and plants.
2. Distinguish between a transgenic animal and a cloned animal.

ISTUDY

14.3 Gene Therapy

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between in vivo and ex vivo gene therapy in humans.
2. List examples of how in vivo and ex vivo gene therapy have been used to treat human disease.
3. Explain how RNA interference represents a form of gene therapy.

The manipulation of an organism's genes can be extended to humans in a process called **gene therapy.** Gene therapy is an accepted therapy for the treatment of a disorder and has been used to cure inborn errors of metabolism, as well as to treat more generalized disorders, such as cardiovascular disease and cancer.

Viruses genetically modified to be safe can be used to transport a normal gene into the body. Sometimes the gene is injected directly into a particular region of the body. In the following sections, we discuss examples of **ex vivo gene therapy,** in which the gene is inserted into cells that have been removed and then returned to the body, and **in vivo gene therapy,** in which the gene is delivered directly into the body.

Ex Vivo Gene Therapy

Children who have SCID (severe combined immunodeficiency) lack the enzyme ADA (adenosine deaminase), which is involved in the maturation of immune cells. Therefore, these children are prone to constant infections and may die unless they receive treatment. To carry out gene therapy, bone marrow stem cells are removed from the bone marrow of the patient and are infected with a virus that carries a normal gene for the enzyme into their DNA (Fig. 14.11). Then, the cells are returned to the patient, where it is hoped they will divide to produce more blood cells with the altered genes.

One of the earliest uses of ex vivo gene therapy was for familial hypercholesterolemia, a condition that develops when liver cells lack a receptor protein for removing cholesterol from the blood. The high levels of blood cholesterol make a person subject to fatal heart attacks at a young age. In this procedure, a small portion of the liver was surgically excised and then infected with a virus containing a normal gene for the receptor before being returned to the individual. Affected persons experienced lowered serum cholesterol levels following this procedure. Scientists are investigating using ex vivo gene therapy to treat other human diseases, including some forms of hemophilia.

In Vivo Gene Therapy

People with cystic fibrosis lack a gene that codes for a transmembrane carrier of the chloride ion. Because of this, these individuals often contract numerous and potentially deadly infections of the respiratory tract. In gene therapy trials, the gene needed to cure cystic fibrosis is sprayed into the person's nose or delivered to the lower respiratory tract by adenoviruses. Another method of delivery is to enclose the gene in a lipid globule called a *liposome.*

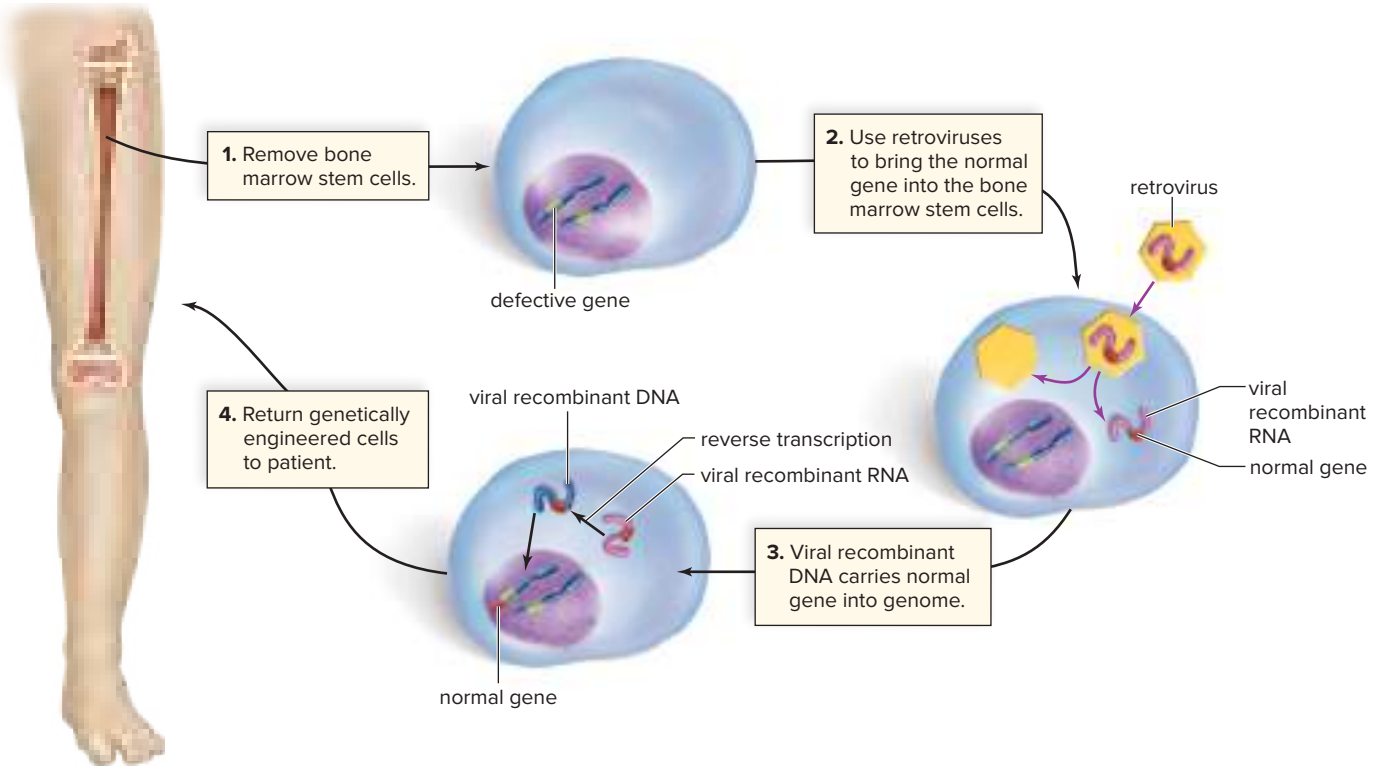


Figure 14.11 Ex vivo gene therapy in humans. Bone marrow stem cells are withdrawn from the body. A virus is used to insert a normal gene into the host genome, and then the cells are returned to the body.

ISTUDY

So far, this treatment has resulted in limited success, but recent advances in the use of lentiviral vectors is promising. Lentiviruses have a long incubation period and have been shown to be effective in infecting lung tissue.

In individuals with cancer, genes are being used to make healthy cells more tolerant of, and tumors more vulnerable to, chemotherapy. The gene *p53* brings about apoptosis, and there is much interest in introducing it into cancer cells that no longer have the gene, and in that way killing them off.

RNA Interference

RNA interference, or RNAi, is a procedure in which small pieces of RNA are used to "silence" the expression of specific alleles. These RNA sequences are designed to be complementary to the mRNA transcribed by a gene of interest. Once the complementary RNA sequences enter the cell, they bind with the target RNA, producing double-stranded RNA molecules. These double-stranded RNA molecules are then broken down by a series of enzymes within the cell. First discovered in worms, RNAi is believed to have evolved in eukaryotic organisms as a protection against certain types of viruses. Research into developing RNAi treatments for a number of human diseases, including cancer and hepatitis, is currently underway.

Check Your Progress 14.3

1. Describe the methods being used to introduce genes into humans for gene therapy.
2. Give an example of ex vivo gene therapy and in vivo gene therapy.
3. Explain the difference between RNA interference and in vivo gene therapy.

14.4 Genomics

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among the sciences of genomics, proteomics, and bioinformatics.
2. Identify the function of repetitive elements, transposons, and unique noncoding DNA sequences in the human genome.
3. Explain how DNA microarrays are used in the study of genomics.

In the preceding century, researchers discovered the structure of DNA, how DNA replicates, and how DNA and RNA are involved in the process of protein synthesis. Genetics in the twenty-first century concerns **genomics,** the study of genomes—our complete genetic makeup and that of other organisms. Knowing the sequence of bases in genomes is the first step, and thereafter we want to understand the function of our genes and their introns, as well as the intergenic sequences. The enormity of the task can be appreciated by realizing there are approximately 3 billion base pairs in the human genome. Many other organisms have a larger number of protein-coding genes but fewer noncoding regions compared to the human genome.

Sequencing the Genome

We now know the order of the base pairs in the human genome. This feat, which has been likened to arriving at the periodic table of the elements in chemistry, was accomplished by the Human Genome Project (HGP), a 13-year effort that involved both university and private laboratories around the world.

In the beginning, investigators developed a laboratory procedure that would allow them to decipher a short sequence of base pairs, and then instruments became available that could carry out sequencing automatically. Over the 13-year span, DNA sequencers were constantly improved, and now modern instruments can automatically analyze up to 2 million base pairs of DNA in a 24-hour period.

Sperm DNA was the material of choice for analysis because it has a much higher ratio of DNA to protein than other types of cells. However, white blood cells from female donors were also used in order to include female-originated samples. The male and female donors were of European, African, American (both North and South), and Asian ancestry.

Many small regions of DNA that vary among individuals, termed polymorphisms, were identified during the HGP. Most of these are *single nucleotide polymorphisms* (*SNPs*); they vary by only one nucleotide. Many SNPs have no effect; others may contribute to enzymatic differences affecting the phenotype. It's possible that certain SNP patterns change an individual's susceptibility to disease and alter their response to medical treatments.

Determining the number of genes in the human genome required a number of techniques, many of which relied on identifying RNAs in cells and then working backward to find the DNA that can pair with each RNA. **Structural genomics**—knowing the sequence of the bases and how many genes we have—is now being followed by functional genomics.

Currently, the number of human genes is just under 20,000. The majority of these genes are expected to code for proteins. However, much of the human genome was formerly described as "junk," because it does not specify the order of amino acids in a polypeptide. However, recall from Section 13.2 that it is possible for RNA molecules to have a regulatory effect on gene expression.

Structure of the Eukaryotic Genome

Historically, genes were defined as discrete units of heredity that corresponded to a locus on a chromosome (see Fig. 11.5). Prokaryotes typically possess a single circular chromosome with genes that are packed together very closely; eukaryotic chromosomes, in contrast, are much more complex: The genes are seemingly randomly distributed along the length of a chromosome and are fragmented into exons, with intervening sequences called introns scattered throughout the length of the gene (Fig. 14.12).

In general, more complex organisms have more complex genes with more and larger introns. In humans, 95% or more of the average protein-coding gene is introns. Once a gene is transcribed, the introns must be removed and the exons joined together to form a functional mRNA transcript (see Fig. 12.14).

ISTUDY

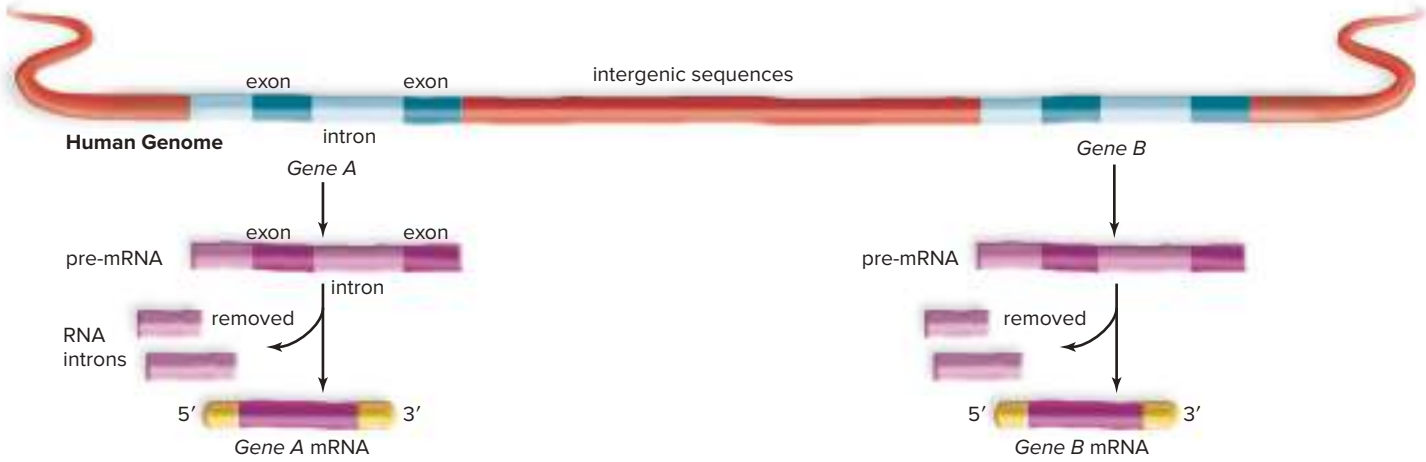
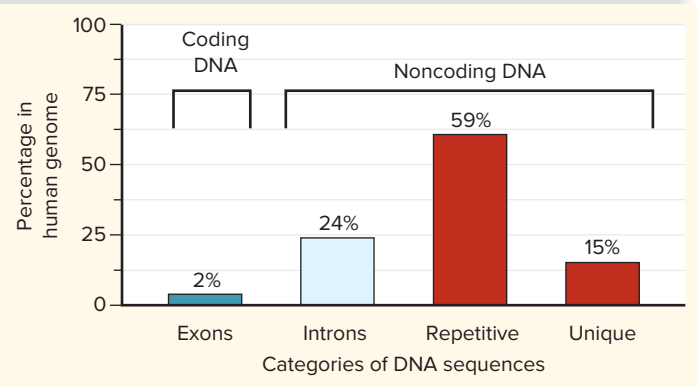


Figure 14.12 Chromosomal DNA. A genome contains protein-coding DNA (exons) and noncoding DNA, including introns (light blue) and other intergenic sequences (red). Only the exons are present in mRNA and specify protein synthesis.

Once regarded as merely intervening sequences, introns are now attracting attention as regulators of gene expression. The presence of introns allows exons to be put together in various sequences, so different mRNAs and proteins can result from a single gene. Introns might also regulate gene expression and help determine which genes are to be expressed and how they are to be spliced. In fact, entire genes have been found embedded within the introns of other genes.

Intergenic Sequences

DNA sequences occur between genes and are referred to as **intergenic sequences** (see Fig. 14.12). In general, as the complexity of an organism increases, so does the proportion of its noncoding DNA sequences. Intergenic sequences are now known to comprise the vast majority of human chromosomes, and protein-coding genes represent only about 1.5–2.0% of our total DNA. The remainder of this DNA, once dismissed as "junk DNA," is now thought to serve many important functions. Several basic types of intergenic sequences are found in the human genome, including (1) repetitive elements, (2) transposons, and (3) unique noncoding DNA. The majority of intergenic sequences belong to this last class.



Repetitive DNA Elements

Repetitive DNA elements occur when a sequence of two or more nucleotides is repeated many times along the length of one or more chromosomes. Repetitive elements are very common—comprising nearly half of the human genome—therefore, many scientists believe that their true significance has yet to be discovered. Although many scientists still dismiss them as having no function, others point out that the centromeres and telomeres of chromosomes are composed of repetitive elements, suggesting that repetitive DNA elements may not be as useless as once thought. For one thing, repetitive DNA of the centromere could possibly help with segregating the chromosomes during cell division.

Repetitive DNA elements include tandem repeats and interspersed repeats. **Tandem repeat** means that the repeated sequences are next to each other on the chromosome. Tandem repeats are often referred to as satellite DNA, because they have a different density than the rest of the DNA within the chromosome. The number and types of tandem repeats may vary significantly from one individual to another, making them invaluable as indicators of heritage. One type of tandem repeat sequence, referred to as *short tandem repeats,* or *STRs,* has become a standard method in forensic science for distinguishing one individual from another and for determining familial relationships (see Section 14.1).

The second type of repetitive DNA element is called an **interspersed repeat,** meaning that the repetitions may be placed intermittently along a single chromosome or across multiple chromosomes. For example, a repetitive DNA element, known as the *Alu* sequence, is interspersed every 5,000 base pairs in human DNA and comprises nearly 5–6% of total human DNA. Because of their common occurrence, interspersed repeats are thought to play a role in the evolution of new genes.

Transposons

Transposons are specific DNA sequences that have the remarkable ability to move within and between chromosomes. Their movement to a new location sometimes alters neighboring genes,

ISTUDY

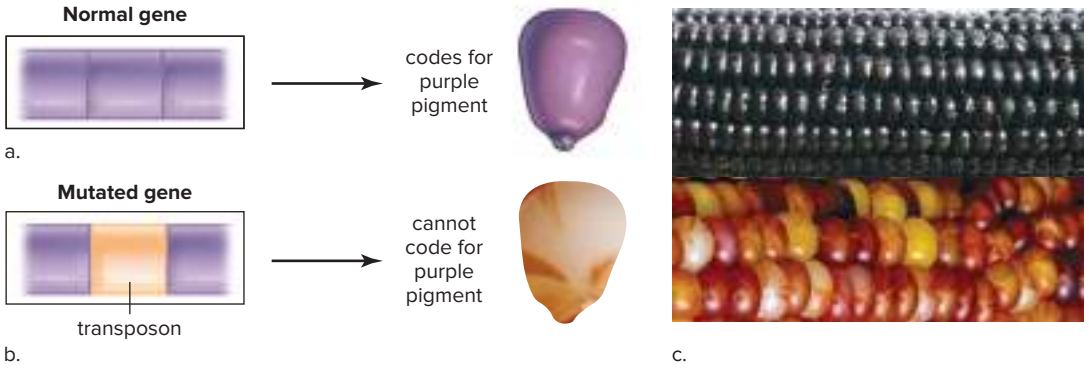


Figure 14.13 Transposons may disrupt gene expression. **a.** A purple-coding gene ordinarily codes for a purple pigment. **b.** A transposon "jumps" into the purple-coding gene. This mutated gene is unable to code for purple pigment, and a white kernel results. **c.** Calico corn displays a variety of colors and patterns due to transposon activity. (c): (top corn): ©Mondae Leigh Baker; (bottom corn): Bob Ross/Shutterstock

particularly decreasing their expression. In other words, a transposon sometimes acts as a regulator gene. The movement of transposons throughout the genome is thought to be a driving force in the evolution of living organisms. The *Alu* repetitive element is an example of a transposon. In fact, many scientists now think that many repetitive DNA elements were originally derived from transposons.

Although Barbara McClintock first described these "movable elements" in corn over 60 years ago, it took time for the scientific community to fully appreciate this revolutionary idea. In fact, their significance was only realized within the past few decades. Transposons, sometimes termed "jumping genes," have now been discovered in bacteria, fruit flies, humans, and many other organisms. McClintock received a Nobel Prize in 1983 for her discovery of transposons and for her pioneering work in genetics (Fig. 14.13).

Unique Noncoding DNA

Genes constitute an estimated 1.5% of the human genome, and repetitive DNA elements make up over 50%; the function of the remaining half, or *unique noncoding DNA,* remains a mystery. Even though this DNA does not appear to contain any protein-coding genes, it has been highly conserved through evolution. In the many millions of years that separate humans from mice, large tracts of this mysterious DNA have remained almost unchanged. But if this DNA has no relevant function, then why has it been so meticulously maintained?

Recently, scientists observed that between 74% and 93% of the genome is transcribed into RNA, including many of these unknown sequences. Thus, what was once thought to be a vast "junk DNA wasteland" may be much more important and may play active roles in the cell. Small-sized RNAs may be able to carry out regulatory functions more easily than proteins at times. Therefore, a previously overlooked RNA signaling network may be what allows humans, for example, to achieve structural complexity far beyond anything seen in the unicellular world. Together, these findings have revealed a much more complex, dynamic genome than was envisioned only a few decades ago.

Revisiting the Definition of a Gene

Perhaps the modern definition of a gene should take the emphasis away from the chromosome and place it on the results of transcription. Previously, molecular genetics considered a gene to be a nucleic acid sequence that codes for the sequence of amino acids in a protein. In contrast to this definition, geneticists have known for some time that all three types of RNA are transcribed from DNA, and that these RNAs are useful products. We also know that protein-coding regions can be interrupted by regions that do not code for a protein but do produce RNAs with various functions. This knowledge expands on the central dogma of genetics and recognizes that a gene product need not be a protein, and a gene need not be on one locus on a chromosome. The DNA sequence that results in a gene product can be split and be present on one or several chromosomes. Also, any DNA sequence can result in one or more products. Furthermore, some prokaryotes have RNA genes. In other words, the genetic material need not be DNA. Again, we can view this as a simple expansion of the central dogma of genetics.

Functional and Comparative Genomics

Since we now know the structure of our genome, the emphasis today is on functional genomics and on comparative genomics. The aim of **functional genomics** is to understand the exact role of the genome in cells or organisms.

To understand the function of genes, it is often useful to study patterns of gene expression. **DNA microarrays,** also known as DNA chips or gene chips, contain microscopic amounts of known DNA sequences fixed onto a small glass slide or silicon chip in known locations (Fig. 14.14). The use of a microarray can tell you what genes are turned on in a specific cell or organism at a particular time and under what environmental circumstances. When mRNA molecules of a cell or an organism bind through complementary base pairing with the various DNA sequences on an array, then that gene is active in the cell.

DNA microarrays are increasingly available that rapidly identify all the mutations in the genome of an individual. This information is called the person's **genetic profile.** The genetic

ISTUDY

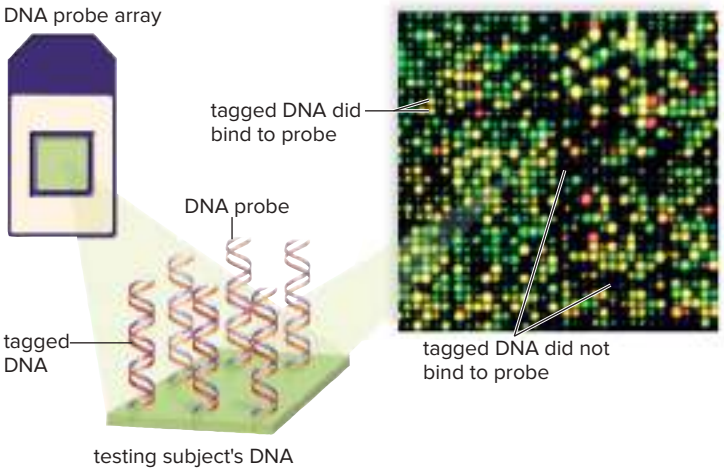


Figure 14.14 Using DNA microarrays to study gene expression. This DNA chip contains rows of DNA sequences for mutations that indicate the presence of particular genetic disorders. If DNA fragments derived from an individual's DNA bind to a sequence representing a mutation on the DNA chip, that sequence fluoresces, and the individual has the mutation.
(photo): Deco/Alamy Stock Photo

profile can indicate if any genetic illnesses are likely and what type of drug therapy for an illness might be most appropriate for that individual.

The aim of **comparative genomics** is to compare the human genome to the genome of other organisms, such as the model organisms listed in Table 14.1. Model organisms are used in genetic analysis because they have many genetic mechanisms and cellular pathways in common with each other and with humans. Functional genomics has also been advanced through the study of these genomes.

Much has been learned by genetically modifying mice; however, other model organisms can also be used. Scientists inserted a human gene associated with early-onset Parkinson's disease into *Drosophila melanogaster,* and the flies showed symptoms similar to those seen in humans with the disorder. This outcome suggested we might be able to use these organisms instead of mice to test therapies for Parkinson's disease.

Comparative genomics also offers a way to study changes in a genome over time, because the model organisms have a shorter generation time than humans. Comparing genomes can also help us understand the evolutionary relationships between organisms. One surprising discovery is that the genomes of all vertebrates are highly similar. Researchers were not surprised to find that the genes of chimpanzees and humans are 98% alike, but they did not expect to find that our sequence is also 85% like that of a mouse. Genomic comparisons will likely reveal evolutionary relationships between organisms never previously considered.

Proteomics

The entire collection of a species' proteins is the **proteome.** At first, it may be surprising to learn that the proteome is larger than the genome until we consider all the many regulatory mechanisms, such as alternative pre-mRNA splicing, that increase the number of possible proteins in an organism.

Proteomics is the study of the structure, function, and interaction of cellular proteins. Specific regulatory mechanisms differ between cells, and these differences account for the specialization of cells. One goal of proteomics is to identify and determine the function of the proteins within a particular cell type. Each cell produces thousands of different proteins, which can vary not only between cells but also within each cell, depending on circumstances. Therefore, the goal of proteomics is an overwhelming endeavor. Microarray technology can assist with this project, as can today's supercomputers.

Table 14.1 Genomes of Model Organisms

Organism	Used to Study	Estimated Genome Size	Estimated Number of Genes
Mouse (*Mus musculus*)	Development, treatment, and biology of human genetic disorders.	2,500 million bases	~20,000
Planarian flatworm (*Schmidtea mediterranea*)	Their regenerative properties and the relationship with brain damage and other neurological diseases.	900 million bases (has a repetitive genome)	~30,000
Fruit Fly (*Drosophila melanogaster*)	Development, neurobiology, and behavior of genetic diseases in animals.	180 million bases	~14,000
Mustard Plant (*Arabidopsis thaliana*)	Genetics and development of flowering plants.	157 million bases	~26,000
Algae (*Chlamydomonas reinhardii*)	Photosynthesis, cell-to-cell recognition, response to stimuli, and motility.	120 million bases	~14,000
Roundworm (*Caenorhabditis elegans*)	Cell death (apoptosis), aging, and diseases such as cancer and diabetes.	100 million bases	~19,000
Yeast (*Saccaromyces cerevisiae*)	The connections between genes and proteins. Function of proteins in eukaryotic cells.	12.1 million bases	~6,300

ISTUDY

THEME Evolution

Metagenomics

In Figure 14C, a microbiologist dips a hand in a pond to extract a muddy substance teeming with life. What microorganisms are in this sample? How do the various microbes interact, and how are they all adapted to living in this environment? These are just a few of many questions that can be answered in the field of metagenomics. Metagenomics studies metagenomes—genetic material obtained directly from environmental samples. A broad sample of collected organisms allows investigators to determine evolutionary interactions in a particular environment by revealing the hidden biodiversity of microscopic life.

Traditionally, if a scientist wanted to know what microbial species were present in a sample, they would have had to isolate one species from another and be able to culture it in a laboratory to obtain enough DNA for sequencing. Only the most abundant species would be isolated and cultured, resulting in a loss of the true biodiversity that actually existed in the sample. Methods have been developed to address this shortcoming.

Shotgun Sequencing

One of the methods employed in metagenomics is the use of shotgun sequencing, a technique also used in the Human Genome Project. Shotgun sequencing got its name from the broad trajectory of buckshot a shotgun blast produces. Shotgun sequencing can be likened to putting ten copies of this textbook in a shredder, and then taking those pieces out and reassembling a complete book. The approach randomly shears DNA, sequences the short fragments, then reassembles these sequences into the correct order in what is called a consensus sequence (Fig. 14D).

When studying microbial biodiversity, the scientist feeds the sequence information into a computer, and bioinformatics software begins the filtering process. Eukaryotic DNA can be identified and removed, leaving the microbial DNA behind for analysis. The biggest challenge scientists face working in metagenomics is the enormous size of the fragmented data. Genes isolated from the microorganisms in the human gut revealed 3.3 million genes, requiring close to 570 gigabases of sequence data. Collecting and analyzing data sets of this size is a difficult computational challenge. Many of these challenges are being met by software developers working in bioinformatics.

Studies Using Metagenomics

Investigators working with the San Diego Zoo were interested in the gut microbes that inhabit various species of mammals. They asked, does each type of mammal have its own community of microbes? The study analyzed the fecal DNA from 34 mammalian species, including humans, and determined which microbes lived in the gut of each species. Most of the fecal samples were obtained from zoo animals with closely monitored diets, as well as humans who kept a strict food diary. The results showed that the *diet* of mammals, not the specific species of that mammal, determines what microorganisms live in the gut. The collection of gut microbes is conserved across mammalian species, depending on what they eat. If a mammal is a herbivore, the set of microbes differs from that of mammals that are omnivores or carnivores.

Metagenomics can also be used as a diagnostic tool to discover causative agents of diseases. Researchers working on a disease affecting boa constrictors suspected that a virus was to blame. DNA from affected and healthy snakes was obtained, and all the snake DNA was filtered out using a computer. The resulting DNA sequences revealed arenaviruses and a virus that was a hybrid of two previously identified viruses. The results not only confirmed that the snake disease was viral but also had larger implications for viral evolution—arenaviruses were only previously known to infect mammals and had never been identified in reptiles. This discovery opened up many questions about host range, evolution, and mechanisms of pathogenesis of viruses.

Questions to Consider

1. Why is metagenomics used versus traditional genomic techniques?
2. Why is shotgun sequencing used in this field of study?
3. Why is metagenomics closely tied to evolutionary biology?

Figure 14C Obtaining an environmental sample. Scientists interested in microbial diversity can collect a sample of mixed species and use metagenomics to analyze which species and genes are present. Bart Coenders/E+/Getty Images

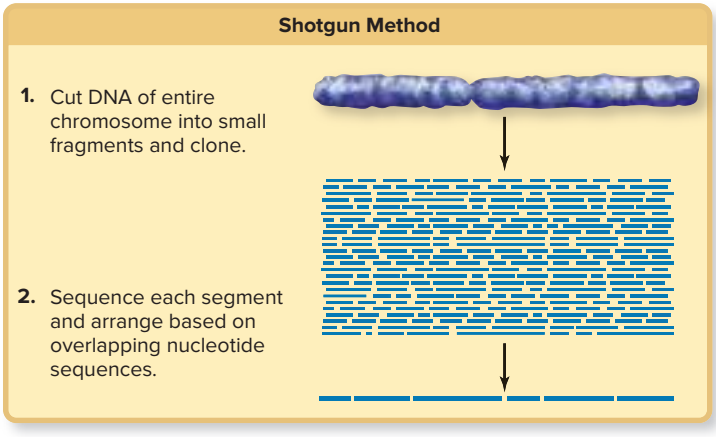


Figure 14D Shotgun sequencing method. The metagenome of an environmental sample is fragmented into small clones and sequenced. Computers assemble and analyze DNA sequences.

ISTUDY

Computer modeling of the three-dimensional shape of cellular proteins is also an important part of proteomics. If the primary structure of a protein is known, it should be possible to predict its final three-dimensional shape, and even the effects of DNA mutations on the protein's shape and function.

The study of protein function is viewed as essential to the discovery of new and better drugs. Also, it may be possible in the future to correlate drug treatment to the particular proteome of the individual to increase its efficiency and decrease side effects.

Bioinformatics

Bioinformatics is the application of computer technologies, specially developed software, and statistical techniques to the study of biological information, particularly databases that contain much genomic and proteomic information. The new, raw data produced by structural genomics and proteomics are stored in databases that are readily available to research scientists. They are called raw data because, as yet, they have little meaning. Functional genomics and proteomics are dependent on computer analysis to find significant patterns in the raw data. For example, BLAST, which stands for *b*asic *l*ocal *a*lignment *s*earch *t*ool, is a computer program that can identify homologous genes among the genomic sequences of model organisms. **Homologous genes** are genes that code for the same proteins, although the base sequences may be slightly different. Finding these differences can help trace the history of evolution among a group of organisms.

Bioinformatics also has various applications in human genetics. For example, researchers found the function of the protein that causes cystic fibrosis by using the computer to search for genes in model organisms that have the same sequence. Because they knew the function of this gene in model organisms, they could deduce the function in humans. This was a necessary step toward possibly developing specific treatments for cystic fibrosis.

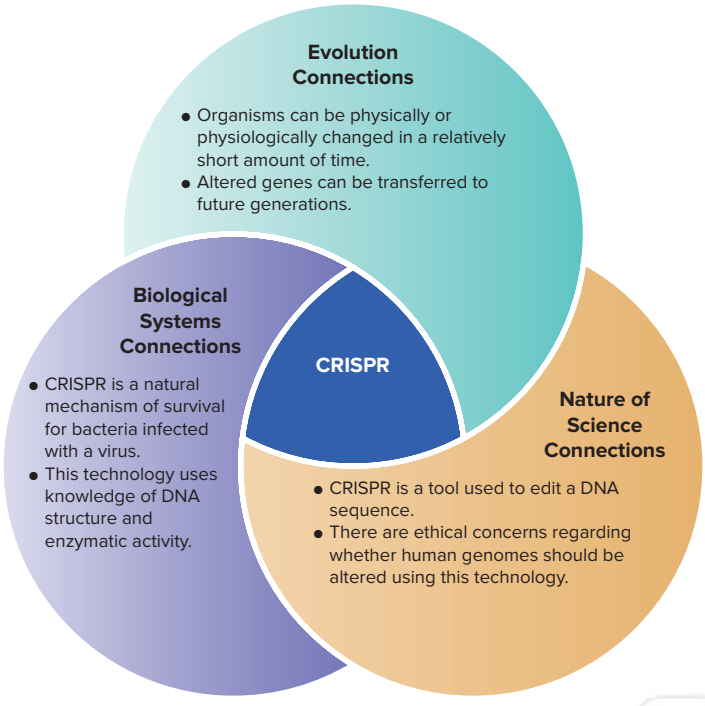
The human genome has 3 billion known base pairs, and without the computer it would be almost impossible to make sense of these data. For example, it is now known that an individual's genome often contains multiple copies of a gene. But individuals may differ as to the number of copies—called *copy number variations.* It seems that the number of copies in a genome can be associated with specific diseases. The computer can help make correlations between genomic differences among large numbers of people and disease.

It is safe to say that, without bioinformatics, progress in determining the function of DNA sequences, comparing our genome to model organisms, knowing how genes and proteins interact in cells, and so forth would be extremely slow. The Evolution feature, "Metagenomics," discusses the use of bioinformatics in the new field of metagenomics.

Check Your Progress 14.4

1. Distinguish between the genome and the proteome of a cell.
2. Summarize the difference between a short tandem repeat and a transposon.
3. Explain how the use of microarrays and bioinformatics aids in the study of genomics and proteomics.

Connecting *the* Concepts



ISTUDY

What Did You Learn?

14.1 Biotechnology

Biotechnology represents the use of a natural system to create a product used by humans. Advances in DNA technology enhanced the ability of **genetic engineering** to produce **genetically modified organisms (GMOs)** and **transgenic organisms.**

- **Cloning** produces genetically identical copies of DNA, cells, or organisms. **Gene cloning** results when a gene is isolated and many copies are produced.
- Two methods are currently available for making copies of DNA: **recombinant DNA (rDNA)** technology and the **polymerase chain reaction (PCR).** Recombinant DNA contains DNA from two different sources. A **restriction enzyme** is used to cleave **plasmid** DNA and to cleave foreign DNA. The resulting "sticky ends" facilitate the insertion of foreign DNA into **vector** DNA. The foreign gene is sealed into the vector DNA by **DNA ligase.** Both bacterial plasmids and viruses can be used as vectors to carry foreign genes into bacterial host cells.
- PCR uses the enzyme DNA polymerase to quickly make multiple copies of a specific piece (target) of DNA. PCR is a chain reaction, because the targeted DNA is replicated over and over again. Analysis of DNA segments, called DNA **gel electrophoresis,** following PCR has all sorts of uses from assisting genomic research to doing **DNA fingerprinting** for the purpose of identifying individuals and confirming paternity. DNA fingerprinting can be accomplished by taking advantage of **short tandem repeat (STR) profiling** present in the genome of all organisms.
- **Genome editing** uses enzyme systems, such as CRISPR, to alter the nucleotide sequence of an organism.

14.2 Biotechnology Products

Genetically modified bacteria, agricultural plants, and farm animals now produce biotechnology products of interest to humans, such as hormones and vaccines. Bacteria usually secrete the product. The seeds of plants and the milk of animals contain the product.

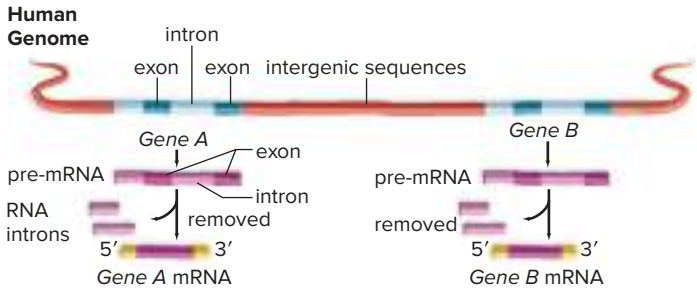
Transgenic bacteria have also been engineered to promote the health of plants, perform bioremediation, extract minerals, and produce chemicals. Transgenic crops, engineered to resist herbicides and pests, are commercially available. Transgenic animals have been given various genes for gene pharming, in particular the one for bovine growth hormone (bGH). Cloning of animals is now possible.

14.3 Gene Therapy

Gene therapy, by either ex vivo or in vivo methods, is used to correct the genotype of humans and to cure various human ills. **Ex vivo gene therapy** has been used to treat diseases such as SCID. **In vivo gene therapy** is being employed in the fight against cancer, cystic fibrosis, and cardiovascular disease. RNA interference may be used to reduce or silence the expression of specific genes.

14.4 Genomics

Researchers in the field of **genomics** now know the sequence of all the base pairs along the length of the human chromosomes. This achievement, known as the Human Genome Project (HGP), helped researchers working in **structural genomics** identify just under 20,000 human genes that code for proteins; the rest of our DNA consists of regions that do not code for a protein and make up the **intergenic sequences.**



- Genes constitute only 1.5% of the human genome. The rest of this DNA is surprisingly more active than once thought. About half of this DNA consists of **repetitive DNA elements,** which may be **tandem repeats** or **interspersed repeats** found throughout several chromosomes. Some of this DNA is made up of mobile DNA sequences called **transposons,** which are a driving evolutionary force within the genome. The role of the remaining portion of the genome is actively being investigated, but it is believed these DNA sequences may play an important role in regulation of gene expression, thus challenging the classical definition of a gene.
- **Functional genomics** aims to understand the function of protein-coding regions and noncoding regions of our genome. To that end, researchers are utilizing tools such as **DNA microarrays.** Microarrays can also be used to create an individual's **genetic profile,** which is becoming helpful in predicting illnesses and reactions to particular medications.
- **Comparative genomics** has revealed that little difference exists between the DNA sequence of our bases and those of many other organisms. Genome comparisons have revolutionized the understanding of evolutionary relationships by revealing previously unknown similarities between organisms.
- **Proteomics** studies genes that code for proteins. A species' entire collection of proteins is the **proteome.** Genes that code for the same proteins even though their DNA is different are **homologous genes,** and computers are instrumental in finding these similar genes. **Bioinformatics** is the use of computers and specialized software programs to process and analyze large amounts of genetic data.

Thinking Critically About Biology

1. Transposons are considered by many researchers to have played a major role in the evolution of life on Earth. Explain how the movement of genetic material within the genome may produce some organisms with a selective advantage.
2. Transgenic viruses that infect humans can be used to help treat genetic disorders. The viruses are genetically modified to contain "normal" human genes that will replace nonfunctional human genes. Using this type of gene therapy, a person is infected with a particular virus, which then delivers the "normal" human gene to cells by infecting them. What are some pros and cons of this viral gene therapy?
3. In a genomic comparison between humans and yeast, what genes would you expect to be similar?

ISTUDY

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video below:

- Glowing Fish Genetics

McGraw Hill

The U.S. Food and Drug Administration (FDA) had to grant approval for the fluorescent fish to be sold in the United States as pets. Some groups were opposed to this approval. What concerns exist regarding genetically modified organisms such as the fluorescent zebrafish?

Making It Relevant

1. Why is CRISPR genome technology referred to as a "molecular scalpel"?
2. How do you feel about genome editing in human embryos? How likely would you be to embrace this technology for a disease affecting a close family member?
3. Specific genes can be knocked out, or inactivated using CRISPR-Cas technology. What research questions could be answered by inactivating genes using this technology?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 3

Evolution

While the term *evolution* is sometimes considered controversial to some, to biologists it is the foundation of their discipline. As we will see, evolution is the change in the gene pool of populations over time. Changes in the gene pool can happen when mutations produce new gene variants, when individuals move into or out of a population, or because of external forces such as natural disasters. If populations change enough over long periods, new populations can branch off from the old. Over time, an evolutionary tree, with many branches, forms.

As with any tree, a branch can be traced back to the root, or the ancestral population, that gave rise to all the others. We can create a family tree of our ancestors as well, back to our great-grandparents and beyond. So, too, can all life trace its ancestry back to the first living cells. A remarkable finding in recent times has been that some of our genes are the same as those of single-celled organisms. This is strong evidence for the shared ancestry of all life on Earth, from single-celled bacteria to multicellular plants and animals.

Darwin proposed that populations change in response to pressures from the environment. Some individuals are better able than others to survive and reproduce in a particular environment. Thus, the environment is the force that determines which individuals contribute offspring to the next generation. Environmental conditions are widely variable, and this variation provides the template by which natural selection shapes the evolution of populations. From carnivorous plants to bats and whales, all life has undergone natural selection in response to the environment. The result is a vast tree of life with many branches, all of which can be traced to a common ancestor of life on Earth.

Unit Outline

Chapter 15 Darwin and Evolution

Chapter 16 How Populations Evolve

Chapter 17 Speciation and Macroevolution

Chapter 18 The Origin and History of Life

Chapter 19 Organizing Life

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Evaluate why Darwin's theory of evolution by natural selection is considered a unifying theory of biology.
Nature of Science	Examine the methods by which scientists obtain evidence in support of evolutionary change.
Biological Systems	Describe how evolution occurs at the molecular, organismal, and population levels of biology.

ISTUDY

15

Darwin and Evolution

©Tyler Keillor, University of Chicago

Evolution of the Tetrapods

In 2004, a fossil "fishapod" was discovered in the Canadian Arctic. *Tiktaalik roseae* is a 375-million-year-old fossil that looks like a cross between an ancient fish and the first four-legged animals, called tetrapods. This unique fossil exhibits a number of transitional features that are both fishlike and tetrapod-like. It had fins, scales, and gills like a fish, but it also had a flexible neck, a flat head, and a wristlike forelimb similar to that of a modern tetrapod. *Tiktaalik* lived in wet, swampy areas, and a wrist may have been advantageous for moving along the bottom of shallow pools and rivers. Transitional fossils, such as *Tiktaalik,* support Darwin's theory that all animals descended from a common ancestor.

The evidence for evolution is not limited to fossils such as *Tiktaalik*—Darwin's theory of evolution is supported by over 150 years of biogeographical, biochemical, developmental, and genetic evidence. Evolution can also be witnessed, in action, over very short periods, such as days, weeks, and years. In this chapter, we take a look at the history of evolutionary thought, beginning with the history of ideas that influenced Darwin as he made his observations before exploring the modern evidence that supports Darwin's theory.

As you read through this chapter, think about the following questions:

1. How do the features of fossil organisms tell us something about an organism's behavior and the environment in which it lived?
2. How does the theory of natural selection explain the evolution of *Tiktaalik*?

Chapter Outline

15.1 The History of Evolutionary Thought
15.2 Darwin's Theory of Natural Selection
15.3 Evidence for Evolution

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.2 What does Darwin mean by evolution as "descent with modification"?

Section 1.3 What is a scientific theory?

Section 12.3 What is the genetic basis of inheritance?

Following *the* Themes

CHAPTER 15 – DARWIN AND EVOLUTION

Evolution	Darwin's theory of natural selection proposes that all life on Earth descends from a common ancestor.
Nature of Science	A scientific theory, such as Darwin's theory of evolution by natural selection, is supported by abundant evidence.
Biological Systems	Evolution by natural selection comes about from interaction between an organism and its environment.

ISTUDY

15.1 The History of Evolutionary Thought

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe Darwin's trip aboard the HMS *Beagle* and some of the observations he made.
2. Identify historical figures and their viewpoints before and during the development of Darwin's theory.

In December 1831, a new chapter in the history of biology had its humble origins. A 22-year-old naturalist, Charles Darwin, set sail on a journey of a lifetime aboard the British naval vessel HMS *Beagle* (Fig. 15.1). Darwin's primary mission on his journey around the world was to serve as the ship's naturalist—to collect and record the geological and biological diversity he saw during the voyage.

As Darwin set sail on the HMS *Beagle,* he was a supporter of the long-held idea that species had remained unchanged since the time of creation. The view of the fixity of species was forged from deep-seated religious beliefs, not by experimentation and observation of the natural world. During the 5-year voyage of the *Beagle,* Darwin's observations challenged his belief that species do not change over time—in fact, his observations of geological formations and species variation led him to propose the mechanism, called natural selection, by which species arise and change. As we will see, natural selection is the core component of **evolution** (L. *evolutio,* "an unrolling"), the genetic change of a species over time. These changes may lead to genetic and phenotypic differences, and potentially, the formation of new species. Evolution is a natural process, and not the result of supernatural forces.

This new view was not readily accepted by Darwin's peers, but it gained gradual credibility as a result of a scientific and intellectual revolution that began in Europe in the late 1800s. Today, more than 150 years since Darwin first published his idea of natural selection, the principle he proposed has been subjected to rigorous scientific tests—so much so that it is now considered one of the unifying theories of biology. Darwin's theory of evolution by natural selection explains both the unity and the diversity of life on Earth, how all living organisms share a common ancestor, and how species adapt to various habitats and ways of life.

Although many have believed that Darwin forged this change in worldview by himself, numerous biologists during the preceding century and some of Darwin's contemporaries had a large influence on Darwin as he developed his theory. The European scientists of the eighteenth and nineteenth centuries were keenly interested in understanding the nature of biological diversity. This was a time of exploration and discovery as the natural history of new lands was being mapped and documented. Shipments of strange plants and animals from newly explored regions were arriving in England to be identified and described by biologists—it was a time of rapid expansion of an understanding of the Earth's biological diversity. In this atmosphere of discovery, Darwin's theory first took root and grew.

Mid-Eighteenth-Century Influences

Many of the beliefs of the eighteenth century, although consistent with Judeo-Christian teachings about special creation, can be traced to the works of the ancient Greek philosophers Plato and Aristotle. Plato said that every species on Earth has a perfect, or "essential," form and species variation is imperfection of this essential form. Aristotle saw that organisms vary in complexity and can be arranged based on their order of increasing complexity.

Georges-Louis Leclerc, better known as Count Buffon, was an eighteenth-century naturalist who worked most of his life writing a 44-volume natural history series that described all known plants and animals. He provided evidence of evolution and proposed various causes, such as environmental influence and the struggle for existence. Buffon's support of evolution seemed to waiver, and often he professed to believe in special creation and the fixity of species.

Taxonomy, the science of classifying organisms, was an important endeavor during the mid-eighteenth century. Chief among eighteenth-century taxonomists was Carolus Linnaeus, who developed the binomial system of nomenclature (a two-part name for a species, such as *Homo sapiens*) and a system of classification for living organisms. Linnaeus, like other taxonomists of his time, believed in the fixity of species, that each species had an "ideal" form. He also believed in the *scala naturae,* a sequential ladder of life where the simplest beings occupy the lowest rungs, and the most complex and spiritual beings—the angels, humans, and then God—occupy the two highest rungs.

Biologists of this time used comparative anatomy, the evaluation of similar structures across a variety of species, to classify organisms into groups. By the late eighteenth century, scientists had discovered fossils and knew they were the remains of plants and animals from the past. Explorers traveled the world and brought back newly discovered extant species (meaning they are still in existence) and fossil organisms to be compared to known living species. At first, scientists believed that each type of fossil had a living descendant in the present time, but eventually some fossils did not seem to match well with known species. In the early nineteenth century, Baron Georges Cuvier was the first to suggest that some species known only from the fossil record had become extinct.

Erasmus Darwin, Charles Darwin's grandfather, was an eighteenth-century physician and a naturalist. His writings on both botany and zoology contained comments and footnotes that suggested the possibility of evolution. He based his conclusions on changes in animals during development, animal breeding by humans, and the presence of vestigial structures (L. *vestigium,* "trace, footprint")—anatomical structures that apparently functioned in an ancestor but have since lost most or all of their function in a descendant (see Section 15.3). Like Buffon, Erasmus Darwin thought species might evolve, but he offered no mechanism by which this change might occur.

Late-Eighteenth/Early-Nineteenth-Century Influences

Baron Georges Cuvier, a distinguished zoologist, used comparative anatomy to develop a system of classifying animals. He also

ISTUDY

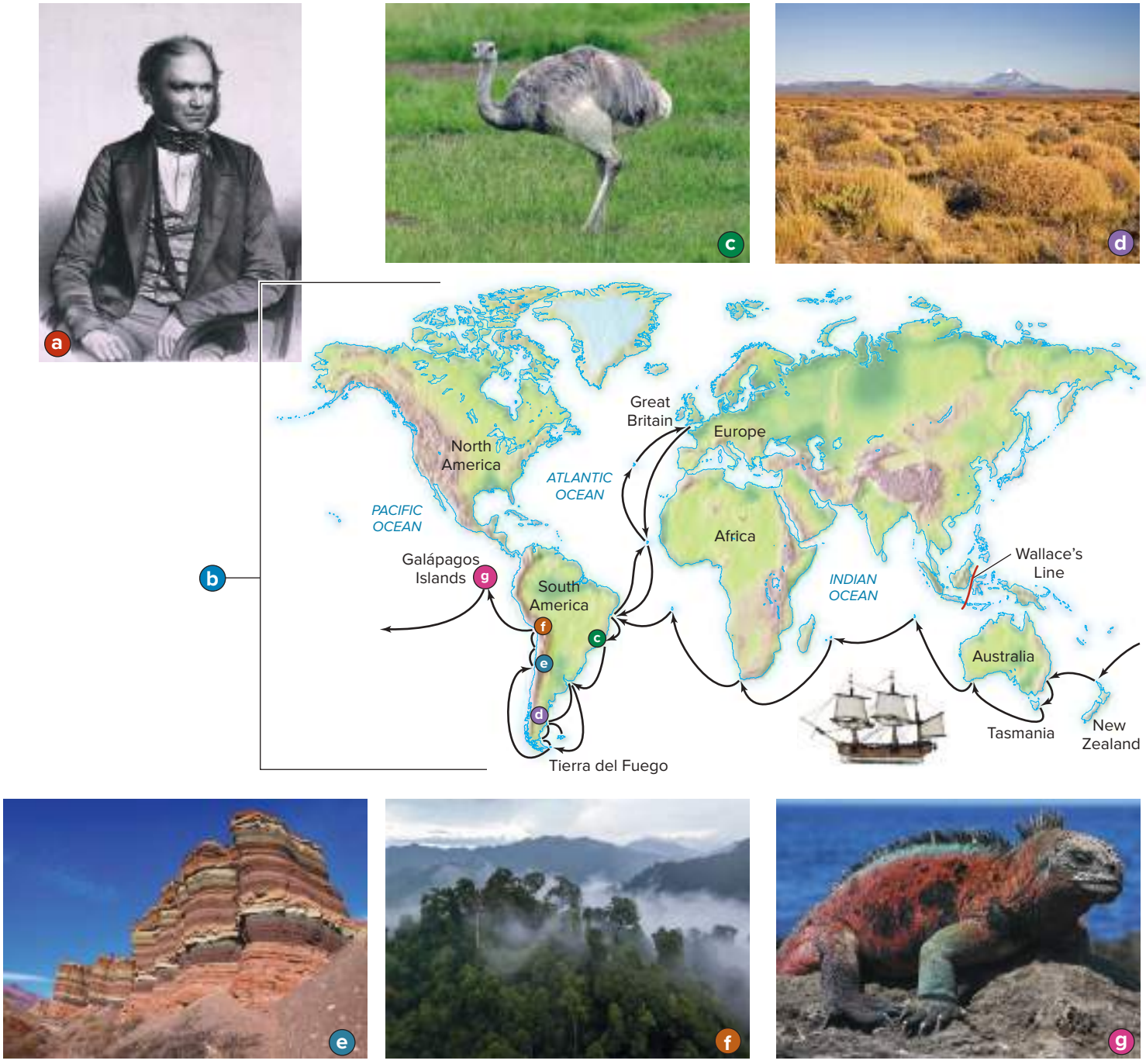


Figure 15.1 Voyage of the HMS *Beagle*. **a.** Young Charles Darwin in 1831 at 22 years old. He did not publish his authoritative book, *On the Origin of Species,* until 1859. **b.** A map of Darwin's journey aboard the HMS *Beagle*. **c.** Along the east coast of South America, he noted a bird called the rhea, which looks like an African ostrich. **d.** In the Patagonian Desert, he observed a rodent that resembles the European rabbit. **e.** In the Andes Mountains, he observed fossils in rock layers. **f.** In the tropical rain forest, he found an abundant diversity of life. **g.** On the Galápagos Islands, he saw marine iguanas with blunt snouts suited for eating algae growing on rocks. (a): Thomas Herbert Maguire/National Library of Medicine; (c): Steve Meese/Shutterstock; (d): Christian Heinrich/imageBROKER/Getty Images; (e): Jeroen Mikkers/Shutterstock; (f): Richard Whitcombe/Shutterstock; (g): Ryan M. Bolton/Shutterstock

founded the science of **paleontology** (Gk. *palaios,* "old"; *ontos,* "having existed"; *logy,* "study of"), the study of fossils, and was quite skilled at using fossil bones to deduce the structure of an animal.

Cuvier was a staunch advocate of the fixity of species and special creation, but his studies revealed that the assembly of fossil varieties changed suddenly between different layers of sediment, or **strata,** within a geographic region. He reconciled his beliefs with his observations by proposing that sudden changes in fossil variation could be explained by a series of local catastrophes, or mass extinctions, followed by repopulation by species from surrounding areas. The result of these catastrophes was a turnover in the assembly of life-forms that occupied a particular region over time. Some of Cuvier's followers suggested there had been

ISTUDY

worldwide catastrophes and God had created new sets of species to repopulate the world. This explanation of the history of life came to be known as **catastrophism.**

In the late eighteenth century, Jean-Baptiste de Lamarck became the first biologist to offer a testable hypothesis that explained how evolution occurs via adaptation to the environment. Lamarck's ideas about descent were entirely different from those of Cuvier. After studying the succession of fossilized life-forms in the Earth's strata, Lamarck proposed that more complex organisms are descended from less complex organisms. He mistakenly concluded, however, that increasing complexity is the result of a natural motivating force—a striving for perfection—that is inherent in all living organisms.

To explain the process of adaptation to the environment, Lamarck proposed the idea of **inheritance of acquired characteristics**—that the environment can produce physical changes in an organism during its lifetime that are inheritable. One example he gave—and for which he is most famous—is that the long neck of giraffes developed over time because their necks grew longer as they stretched to reach food in tall trees, and this longer neck was then passed on to their offspring (Fig. 15.2). His hypothesis of inheritance of acquired characteristics has never been supported by experimentation. The molecular mechanism of inheritance explains why—phenotypic changes acquired during an organism's lifetime do not result in genetic changes that can be passed to subsequent generations.

In the eighteenth century, geologist James Hutton proposed a theory of slow, uniform geological change. Charles Lyell, the foremost geologist of Darwin's time, made Hutton's ideas popular in his book *Principles of Geology,* published in 1830. Hutton explained that the Earth is subject to slow but continuous cycles of rock formation and erosion, not shaped by sudden catastrophes. He proposed that erosion produces dirt and rock debris that is washed into rivers, transported to oceans, and deposited in thick layers, which are converted over time into sedimentary rock. These layers of sedimentary rocks, which often contain fossils, are then uplifted from below sea level to form land during geological upheavals.

Hutton concluded that extreme geological changes can be explained by slow, natural processes, given enough time. Lyell went on to propose the theory of **uniformitarianism,** which states that the natural processes witnessed today are the same processes that occurred in the past. Hutton's general ideas about slow and continual geological change are still accepted today, although modern geologists realize that rates of change have not always been uniform through history. Darwin was not taken by the idea of uniform change, but he was convinced, as was Lyell, that the Earth's massive geological changes are the result of extremely slow processes, and that the Earth, therefore, must be very old.

Thomas Malthus was an economist who studied the factors that influence the growth and decline of human populations. In 1798, Malthus published *An Essay on the Principle of Population,* in which he proposed that the size of human populations is limited only by the quantity of resources, such as food, water, and shelter, available to support it. He related famine, war, and epidemics to the problem of populations overstretching their limited resources.

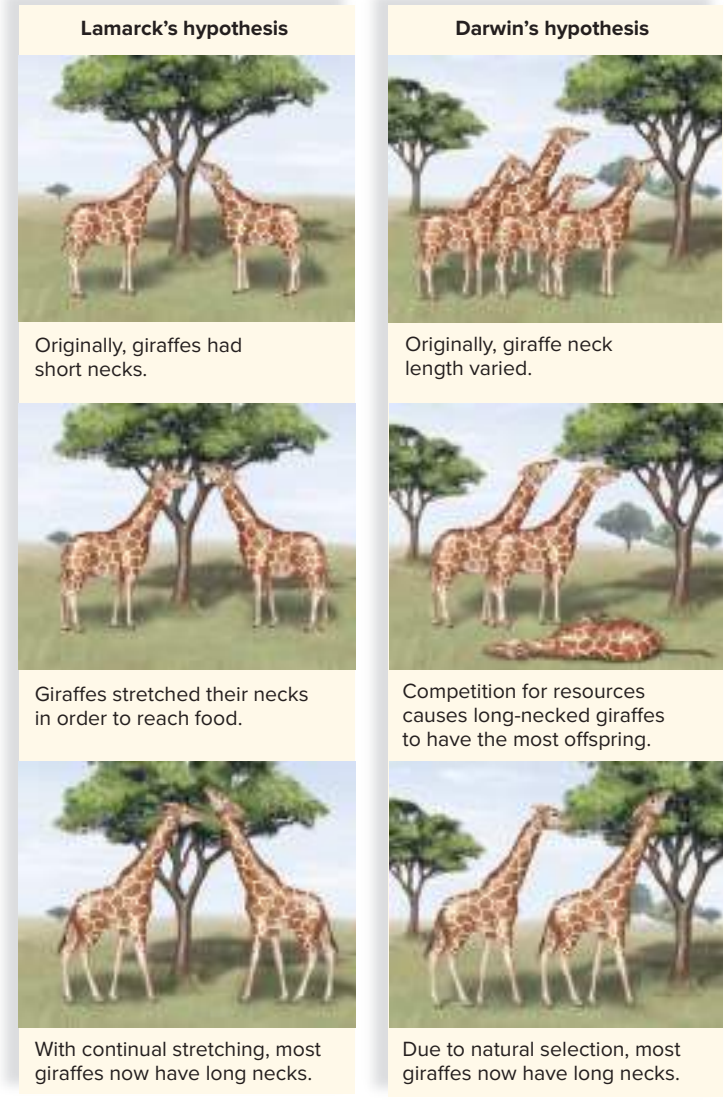


Figure 15.2 A comparison of Lamarck's and Darwin's hypotheses of evolutionary change. **a.** Jean-Baptiste de Lamarck's proposal of the inheritance of acquired characteristics. **b.** Charles Darwin's ideas developed into the theory of natural selection.

Darwin, after reading Malthus's essay in 1838, applied similar principles to animal populations—that is, animals tend to produce more offspring than can survive, and competition for limited resources in the environment is the element that determines survival. Darwin thus used Malthus's principle to formulate his idea of natural selection.

Check Your Progress 15.1

1. Define *catastrophism* and identify who proposed this idea.
2. Evaluate Lamarck's idea of "inheritance of acquired characteristics" as an explanation of biological diversity.
3. Construct a timeline of the history of evolutionary thought. Include major contributors and a brief description of each contribution along the timeline.

ISTUDY

15.2 Darwin's Theory of Natural Selection

Learning Outcomes

Upon completion of this section, you should be able to

1. Summarize the process of evolution by natural selection.
2. List examples of the evidence Darwin gathered from fossils and biogeography that supported his growing idea of shared ancestry.
3. Give examples of how the mechanisms of evolutionary change can be identified and studied.

When Darwin signed on as the naturalist aboard the HMS *Beagle,* he possessed a suitable background for the position. Since childhood, he had been a devoted student of nature and a collector of insects. At age 16, Darwin was sent to medical school to follow in the footsteps of his grandfather and father. However, he did not take to the study of medicine, so his father encouraged him to enroll in the School of Divinity at Christ's College at Cambridge, with the intent of becoming a clergyman.

While at Christ's College, Darwin attended many lectures on biology and geology to satisfy his interest in natural science. During this time, he became the protégé and friend of the botanist John Henslow, from whom he gained skills in the identification and collection of plants. Darwin gained valuable experience in geology in the summer of 1831 by conducting fieldwork with Adam Sedgewick, one of the founders of modern geology. Shortly after Darwin was awarded his BA, Henslow recommended him to serve, without pay, as naturalist aboard the HMS *Beagle,* which was to explore the Southern Hemisphere.

The voyage was to take 2 years—but ended up taking 5 years—and the ship was to traverse the Southern Hemisphere (see Fig. 15.1). Along the way, Darwin encountered species that were very different from those of his native England. As part of his duties as the ship's naturalist, Darwin began to gather evidence that organisms are related through descent with modification from a common ancestor, and that adaptation to various environments results in diversity. Darwin also began contemplating the "mystery of mysteries," the origin of new species.

Observations of Change over Time

On his trip, Darwin observed massive geological changes firsthand. When he explored what is now Argentina, he saw raised beaches for great distances along the coast. Many of the raised beaches had exposed layers of sediment that contained a variety of fossilized shells and bones of extinct mammals. Darwin collected fossil remains of an armadillo-like animal (*Glyptodon*), the size of a small, modern-day car, and a giant ground sloth, *Mylodon darwinii,* the largest of which stood nearly 3 m tall (Fig. 15.3). Darwin also observed marine shells high in the cliffs of the impressive Andes Mountains, which suggested to him that the Earth is very old. Once Darwin accepted that possibility, he began to think that there would have been enough time for descent with modification to occur. Therefore, living forms could be descended from extinct forms known only from the fossil record. It would seem that species were not fixed; instead, they changed over time.

Figure 15.3 Fossils of extinct mammals Darwin found during his exploration of South America. **a.** A giant, armadillo-like glyptodont, *Glyptodon,* is known only by the study of its fossil remains. Darwin came to the conclusion that this extinct animal must be related to living armadillos. The glyptodont weighed 2,000 kg. **b.** Darwin also observed the fossil remains of an extinct giant ground sloth, *Mylodon.*

Biogeographical Observations

Biogeography (Gk. *bios,* "life"; *geo,* "earth"; *grapho,* "writing") is the study of the geographic distribution of organisms throughout the world. The distribution of species and the makeup of species groups in different regions provide hints about past geological events, such as the movement of continents and the formation of volcanic islands, and about ecological change, such as glaciation and river formation.

As Darwin explored the Southern Hemisphere, he compared the animals of South America to those with which he was familiar. He noticed that although the animals in South America were different from those in Europe, similar environments on each continent had similar-looking animals. For example, instead of rabbits, he found the Patagonian cavy in the grasslands of South America. The Patagonian cavy has long legs and ears but the face of a guinea pig, a rodent also native to South America (Fig. 15.4). Did the Patagonian cavy resemble a rabbit because the two types of animals were adapted to the same type of environment? Both animals ate grass, hid in bushes, and moved rapidly using long hind legs.

ISTUDY

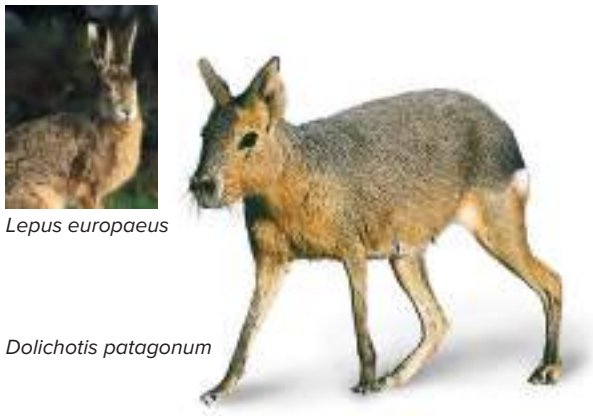


Figure 15.4 The European hare and the Patagonian cavy.
(Lepus): Christian Bauer/Fotosearch/Glow Images; (Dolichotis): Juan Munoz/Science Source

Did the Patagonian cavy have the face of a guinea pig because of having an ancestor in common with guinea pigs?

As he sailed southward along the eastern coast of South America, Darwin saw how similar species replaced one another. For example, the greater rhea (an ostrichlike bird) found in the north was replaced by the lesser rhea in the south. Therefore, Darwin reasoned that related species could be modified according to environmental differences (i.e., northern vs. southern latitudes). When he explored the Galápagos Islands, he found further evidence of this phenomenon.

The Galápagos Islands are a small group of volcanic islands formed 965 km off the western coast of South America. These islands are too far from the mainland for most terrestrial animals and plants to colonize, yet life is present there. The types of plants and animals Darwin found there were slightly different from species he had observed on the mainland, and even more important, they varied from island to island. Where did the animals and plants inhabiting these islands come from? Why were these species different from those on the mainland, and why were different species found on each island?

For example, each of the Galápagos Islands seemed to have its own type of tortoise, and Darwin began to wonder whether this difference was correlated with variation in vegetation among the islands (Fig. 15.5). Long-necked tortoises seemed to inhabit only dry areas where low-growing vegetation was scarce but tall cacti were abundant. In moist regions with relatively abundant ground foliage, short-necked tortoises were found. Had an ancestral tortoise from the mainland of South America given rise to these different types, each adapted to take advantage of food sources in its unique environment?

One of Darwin's most famous observations from the Galápagos Islands was his study of its finches. Darwin almost overlooked the finches because of their nondescript nature compared with many of the other animals in the Galápagos. At the time, Darwin did not recognize that these birds were all finches, because they were very different from the familiar finches from England. However, these birds would eventually play a major role in the formation of his thoughts about geographic barriers and their contribution to the origin of new species. Upon returning to England, Darwin had John Gould identify the birds. Gould identified the birds as "a series of ground finches . . . an entirely new group," and they exhibited significant variation in beak size and shape (see Fig. 15.9).

As outlined in the Diversity in Science feature, "The Contributions of John Edmonstone to Science," Darwin relied on the expertise of others to establish his collections.

Today, many more Galápagos finches have been identified—there are ground-dwelling finches with beaks adapted to eating seeds, tree-dwelling finches with beaks sized according to their insect prey, and a cactus-eating finch with a more pointed beak it uses to punch holes in cactus fruit to extract pulp (see Fig. 15.9). The most unusual of the finches is a woodpecker-type finch. This bird has a sharp beak to chisel through tree bark but lacks the long tongue characteristic of a true woodpecker, which probes for insects. To compensate for this, the bird carries a twig or cactus spine in its beak and uses it to poke into crevices. Once an insect emerges, the finch drops this tool and seizes the insect with its

a. b.

Figure 15.5 Galápagos tortoises. Darwin wondered whether the Galápagos tortoises were descended from a common ancestor. **a.** The tortoises with dome shells and short necks feed at ground level on islands with enough rainfall to support grasses. **b.** Those with shells that flare up in the front have long necks and live on arid islands, where they feed on tall, treelike cacti. (a): Kevin Schafer/Corbis/Getty Images; (b): alblec/iStock/Getty Images

ISTUDY

▶ Diversity in Science

The Contributions of John Edmonstone to Science

Charles Darwin collected hundreds of specimens during his 5-year journey on the HMS *Beagle*. Almost 200 years later, many of those specimens, including the famous finches of the Galápagos Islands, can still be found in museum collections (Fig. 15A). Darwin and his assistants used taxidermy techniques to preserve the specimens so they could be studied after the journey concluded.

A man named John Edmonstone can be credited for teaching Darwin taxidermy. Born sometime in the late 1700s, Edmonstone was a Black enslaved man in what is now Guyana, South America. Edmonstone himself learned about taxidermy from Charles Waterton, a famous naturalist who often visited the plantation where Edmonstone was enslaved.

John Edmonstone gained his freedom after traveling to Scotland with his then-owner in the early 1800s. Living near the University of Edinburgh, he worked as a taxidermist and also taught taxidermy to university students. One such student was Charles Darwin, who came to the university to learn medicine, but changed his mind after enduring boring lectures and witnessing unsettling surgeries that lacked anesthesia.

Darwin then took an interest in natural history, which led to his taxidermy lessons with Edmonstone. Before his first lesson, Darwin wrote to his sister that Edmonstone's instruction "has the recommendation of cheapness, if it has nothing else, as he only charges one guinea, for an hour every day for two months." It would be a few short years before his cheap lessons would be used to preserve specimens that contributed evidence to his theory of evolution.

Figure 15A Darwin's specimens. Mockingbirds collected and preserved by Charles Darwin. Reuters/Alamy Stock Photo

It is also thought that Edmonstone's stories about the natural history of South America helped inspire Darwin's interest in the region. Darwin wrote in his autobiography that Edmonstone "gained his livelihood by stuffing birds, which he did excellently: he gave me lessons for payment, and I used often to sit with him, for he was a very pleasant and intelligent man."

Very little is known about John Edmonstone aside from Darwin's experiences with him. His actual name is not even mentioned by Darwin. Instead, the connection was established by scholars through other records. Despite their influence and contributions, enslaved people like John Edmonstone are regularly omitted from historical accounts.

beak (for more information on the shape of finch beaks, see the Nature of Science feature, "Genetic Basis of Beak Shape in Darwin's Finches," in Section 17.2).

Later, Darwin speculated whether these different species of finches could have descended from a mainland finch species. In other words, he wondered if a finch from South America was the common ancestor of all the types on the Galápagos Islands. Perhaps new species had arisen because the geographic distance between the islands isolated populations of birds long enough for them to evolve independently. And perhaps the present-day species had resulted from accumulated changes occurring within each of these isolated populations.

Natural Selection and Adaptation

Upon returning to England, Darwin began to reflect on the voyage of the HMS *Beagle* and to collect additional evidence in support of his ideas about how organisms adapt to the environment. Darwin concluded early on that species change over time and are not fixed entities crafted by a creator. However, he did not yet have a mechanism to explain how change could happen in existing species and how new species could arise.

By 1842, Darwin had fully developed his idea of natural selection as a mechanism for evolutionary change. In 1858, Alfred Russel Wallace sent an essay to Darwin in which he proposed a similar concept. Eight years of collecting and

ISTUDY

identifying thousands of species new to science in the Malay Archipelago had helped Wallace formulate his views on evolutionary change. The idea of natural selection was first presented to the Linnean Society of London in 1858 as a pair of essays by Darwin and Wallace.

Natural selection is a process based on the following observations:

- Organisms exhibit variation that can be passed from one generation to the next—that is, they have heritable variation.
- Organisms compete for available resources.
- Individuals within a population differ in terms of their reproductive success.
- Organisms become adapted to conditions as the environment changes.

We consider each of these characteristics in detail in the sections that follow.

Organisms Have Heritable Variation

Darwin emphasized that the members of a population vary in their functional, physical, and behavioral characteristics (Fig. 15.6). Before Darwin, variations were viewed as imperfections that should be ignored because they were not important to the description of "fixed" species (see Section 15.1). In contrast, Darwin emphasized that variation is required for the process of natural selection to operate. He suspected that a mechanism of inheritance existed, but he did not have the evidence we have today.

Now we know that genes are the unit of heredity and, along with the environment, determine the phenotype of an organism. Random mutations have been shown to be a source of new genetic variation in a population. Genetic variation can be harmful, helpful, or neutral (have no effect at all) to survival and reproduction. Genetic variation arises by chance and for no particular purpose, and new variation is as likely to be harmful as helpful or neutral to the organism. However, harmful variation is eliminated from the population by natural selection, because individuals with these mutations often do not survive or reproduce. Beneficial or neutral variation can be maintained in a population. Natural selection ignores neutral variation. But a beneficial mutation increases the probability that an individual with this mutation will tend to have greater reproductive success. Biologists have abundant evidence that natural selection operates on heritable variation already present in a population's gene pool, and that this selection process is random—it has no goal of "improvement" in anticipation of future environmental changes.

Organisms Compete for Resources

Darwin applied Malthus's treatise on human population growth to animal populations. He realized that if all offspring born to a population were to survive, insufficient resources would be available to support the growing population. He calculated the reproductive potential of elephants, assuming an average life span of 100 years and a breeding span of 30–90 years. Given these assumptions, a single female probably bears no fewer than six young, and if all these young survive and continue to reproduce at the same rate, then after only 750 years the descendants of a single pair of elephants would number about 19 million! Obviously, no environment has the resources to support an elephant population of this magnitude, and no such elephant population has ever existed. This overproduction potential of a species is often referred to as the geometric ratio of increase.

Figure 15.6 Variation in a population. Variation in populations, such as that seen in these tree snails (*Liguus fasciatus*), is necessary for natural selection to act and result in the adaptation of the species to its environment. James H. Robinson/Science Source

Organisms Differ in Reproductive Success

Some individuals have favorable traits that enable them to better compete for limited resources. The individuals with favorable traits acquire more resources than the individuals with less favorable traits and can devote more energy to reproduction. Darwin called this ability to have more offspring *differential reproductive success.*

Fitness is the reproductive success of an individual relative to other members of a population. The most fit individuals are the ones that capture a larger amount of resources and convert these resources into a larger number of viable offspring. Because organisms vary, as do the conditions of local environments, fitness is

ISTUDY

influenced by different factors for different populations. For example, among western diamondback rattlesnakes (*Crotalus atrox*) living on dark-colored lava flows, the most fit are those that are black in color. But among those living on desert soil, the most fit are those with the typical light coloring with brown blotching. Background matching helps an animal both capture prey and avoid being captured; therefore, it is expected to lead to survival and increased fitness.

Natural selection occurs because certain members of a population happen to have a variation that allows them to survive and reproduce to a greater extent than do other members. For example, a variation in a desert plant that reduces water loss is beneficial; and a mutation in a wild dog that increases its sense of smell helps it find prey.

Organisms Become Adapted

An **adaptation** is any evolved trait that helps an organism be more suited to its environment. Adaptations are especially recognizable when unrelated organisms living in a particular environment display similar characteristics. For example, manatees, penguins, and sea turtles all have flippers, which help them move through the water. In Section 1.1, we saw other ways in which penguins are adapted to their environment. Similarly, a Venus flytrap, a plant that lives in the nitrogen-poor soil of a bog, is able to obtain nitrogen-containing nutrients because it has specialized leaves adapted to catching and digesting flies.

Such adaptations to specific environments result from natural selection. Differential reproduction generation after generation can cause adaptive traits to increase in frequency in each succeeding generation. Evolution includes other processes in addition to natural selection (see Section 16.1), but natural selection is the only process that results in adaptation to the environment.

Artificial Selection

Darwin noted that humans can artificially modify desired traits in plants and animals by selectively breeding certain individuals. For example, the diversity of domestic dogs has resulted from prehistoric humans selectively breeding wolves with particular traits, such as hair length, height, and guarding behavior. This type of human-controlled breeding to increase the frequency of desired traits is called **artificial selection,** also referred to as selective breeding. Artificial selection, like natural selection, is possible only because the original population exhibits a variety of characteristics, allowing humans to select the traits they prefer. For dogs, artificial selection has produced the many breeds of dogs we see today, all of which are descended from the wolf (Fig. 15.7).

As another example, several varieties of vegetables can be traced to a single ancestor. Chinese cabbage, brussels sprouts, and kohlrabi are all derived from a single species, *Brassica oleracea* (Fig. 15.8).

Modern corn, or maize, has a wild ancestor called teosinte. Teosinte looks very different from the corn we grow for food; teosinte has 5–12 kernels in a single row, and a hard, thick outer shell encases each kernel, making it difficult to use as a food source (see the Evolution feature, "The Evolutionary History of Maize," in Section 23.5). Strong evidence from archaeology and genetics supports the hypothesis that prehistoric humans, selecting for softer shells, more kernels, and other desirable traits, produced modern corn. Darwin surmised that if humans could create such a wide variety of organisms by artificial selection, then natural selection

Figure 15.7 Artificial selection. All dogs, *Canis lupus familiaris,* are descended from the gray wolf, *Canis lupus,* which began to be domesticated at least 20,000 years ago. The process of selective breeding by humans has led to extreme phenotypic differences among breeds. (wolf): Nicholas James McCollum/Moment Open/Getty Images; (bulldog): WilleeCole/iStockphoto/Getty Images; (whippet): Daniela Jakob/djakob/123RF

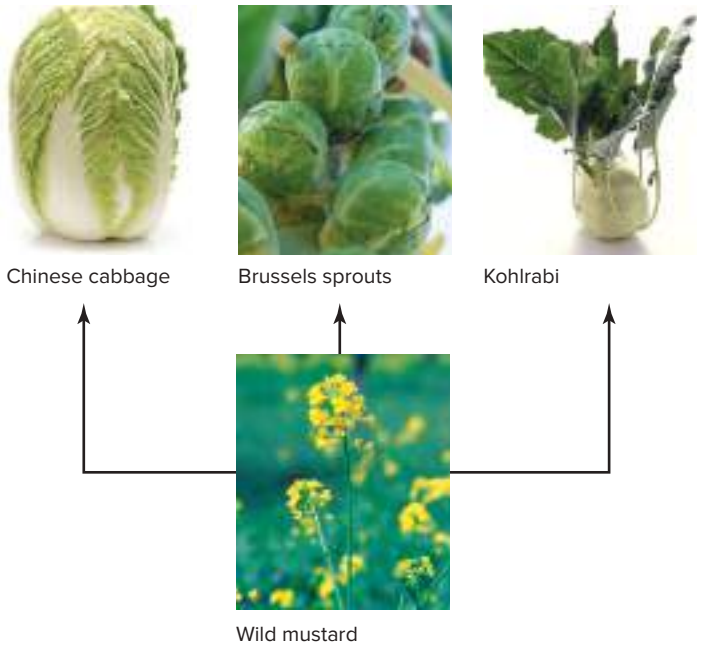


Figure 15.8 Artificial selection of plants. The vegetables Chinese cabbage, brussels sprouts, and kohlrabi are derived from wild mustard, *Brassica oleracea.* Darwin described artificial selection as a model by which to understand natural selection. With natural selection, however, the environment—not human selection—provides the selective force. (cabbage): JIANG HONGYAN/Shutterstock; (brussels sprouts): Anne Murphy/Moment/Getty Images; (kohlrabi): Foodcollection; (mustard): Medioimages/PunchStock

ISTUDY

a. Large, ground-dwelling finch b. Warbler-finch c. Cactus-finch

Figure 15.9 Galápagos finches. Each of the 15 species of finches has a beak adapted to a particular way of life. For example, (**a**) the heavy beak of the large ground-dwelling finch (*Geospiza magnirostris*) is suited to a diet of large seeds; (**b**) the beak of the warbler-finch (*Certhidea olivacea*) is suited to feeding on insects found among ground vegetation or caught in the air; and (**c**) the longer beak, somewhat decurved, and the split tongue of the cactus-finch (*Cactornis scandens*) are suited to extracting the flesh of cactus fruit. (a): Miguel Castro/Science Source; (b): Celia Mannings/Alamy Stock Photo; (c): Michael Stubblefield/Alamy Stock Photo

could also produce diversity, but with the environment, not humans, as the force selecting for particular traits.

Natural Selection in Action

The Galápagos finches have beaks adapted to the food they eat, with different species of finches found on each island (Fig. 15.9). Today, many investigators, including Peter and Rosemary Grant of Princeton University, are documenting natural selection as it occurs on the Galápagos Islands.

In 1973, the Grants began a study of the various finches on Daphne Major, an island near the center of the Galápagos Islands. The weather on this island swings widely back and forth from wet years to dry years. The Grants found that the beak size of the medium ground finch, *Geospiza fortis,* adapted to each weather swing, generation after generation (Fig. 15.10). These finches like to eat small, tender seeds that require a smaller beak. When the weather turns dry, they must eat larger, drier seeds, which are harder to crush. The birds that have a larger beak depth have an advantage during the dry periods, and they have more offspring. Therefore, among the next generation of *G. fortis* birds, the mean, or average, beak size has more depth than the previous generation (for more information on the shape of finch beaks, see the Nature of Science feature, "Genetic Basis of Beak Shape in Darwin's Finches," in Section 17.2). The Grants' research demonstrates that evolutionary change can sometimes be observed within the timeframe of a human life span, rather than over thousands of years.

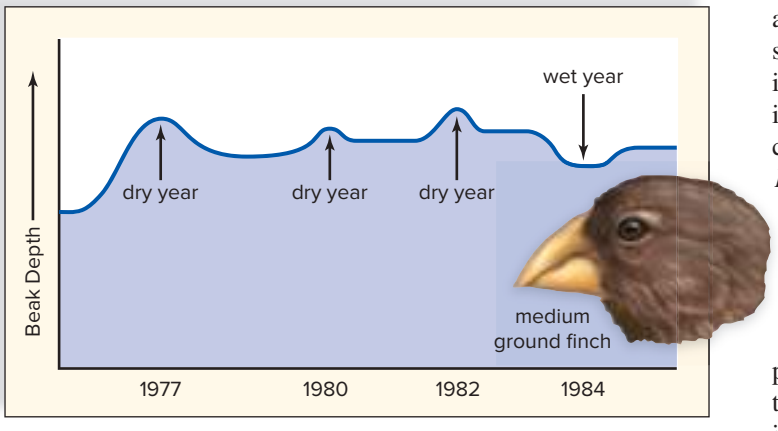
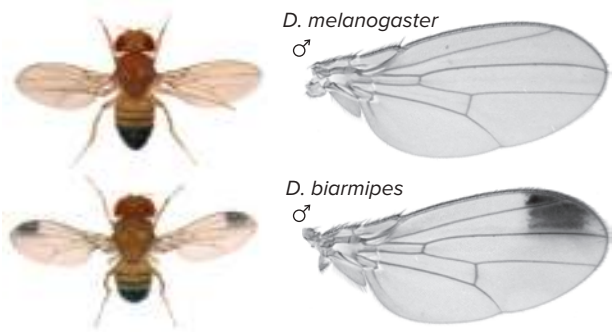


Figure 15.10 Evolution in action. The average beak depth of medium ground finches varies from generation to generation according to the weather. The weather affects the hardness and size of seeds on the islands, and different beak depths are better suited to eating different types of seeds. Average beak features were observed to change many times over a period of a decade. This is one way in which evolution by natural selection has been observable over a short period of time.

Recent advances in biotechnology have produced a set of new and revolutionary tools to document phenotype evolution at the level of the gene. As one example, Sean Carroll (University of Wisconsin, Madison) studies the genes that determine variation in the color patterns on the wings of fruit flies. In one species of fly, *Drosophila biarmipes,* a close relative of the common fruit fly, *D. melanogaster,* males have a black spot on the top, forward edge of the wing (Fig. 15.11). This spot is part of the male fly's courtship dance. Carroll's research shows that the spot on the wing of the male fly has evolved from a few simple mutations that have changed how a wing gene is switched on and off during development. A few simple mutations in the DNA code were enough to produce a change in the wing color pattern of *D. biarmipes.* This study and others like it demonstrate how new traits can evolve as a result of only a few changes in the DNA code that regulate a gene. In the case of *D. biarmipes,* natural selection, in the form of female mate choice, favors the evolution of males with spotted wings.

Industrial melanism is a common example of how natural selection can shape a trait in a population. Prior to the Industrial Revolution in Great Britain, light-colored peppered moths, *Biston betularia,* were more common than dark-colored peppered moths. It was estimated that only 10% of the moth population was dark at this time. With the advent of industry and an increase in pollution, the number of dark-colored moths exceeded 80% of the moth population. After legislation to reduce pollution, a dramatic reversal in the ratio of light-colored moths to dark-colored moths occurred. In 1994, one collecting site recorded a drop in the frequency of dark-colored moths to 19%, from a high of 94% in 1960. (We revisit this example in Section 16.2 for a discussion on the evolution of populations.)

ISTUDY



a.

b.

Figure 15.11 Wing spot in *Drosophila biarmipes*. **a.** Male *D. biarmipes* have a spot on the leading edge of their wings that is not present in their close relative, *D. melanogaster*. **b.** *D. biarmipes* males flash the spot on their wings to attract the attention of females during a courtship dance. (a, b): ©Nicolas Gompel and Benjamin Prud'homme

The rise in bacterial resistance to antibiotics has occurred within the past 30 years or so. Resistance is an expected way of life now, not only in medicine but also in agriculture. New chemotherapeutic and HIV drugs are required because of the resistance of cancer cells and HIV, respectively. Also, pesticides and herbicides have created resistant insects and weeds.

Check Your Progress 15.2

1. List the three categories of observations made by Darwin that support evolution by natural selection.
2. Summarize the components of Darwin's theory of evolution by natural selection.
3. Identify several mechanisms of evolutionary change that can be studied.

15.3 Evidence for Evolution

Learning Outcomes

Upon completion of this section, you should be able to

1. Interpret one example from each area of study—fossil, anatomical, biogeographical, and biochemical—as evidence supporting the descent of all life from a common ancestor.
2. Interpret misconceptions of evolution proposed by critics of evolution.

Many different lines of evidence support the concept that organisms are related through descent from a common ancestor. This is significant because the more varied and abundant the evidence supporting a hypothesis, the more certain it becomes.

Fossil Evidence

Fossils are the remains and traces of past life or any other direct evidence of past life. Traces include trails, footprints, burrows, worm casts, or even preserved droppings. Usually, when an organism dies, the soft parts are either consumed by scavengers or decomposed by bacteria. Occasionally, the organism is buried quickly and in such a way that decomposition is never completed or is completed so slowly that the soft parts leave an imprint of their structure—for example, animals or plants trapped in a landslide or mudflow. Most fossils, however, consist only of hard parts, such as shells, bones, or teeth, because these are usually not consumed or destroyed.

Transitional fossils bear a resemblance to two groups that in the present are classified separately. They often represent the intermediate evolutionary forms of life in transition from one type to another or a common ancestor of these types. Transitional fossils allow us to retrace the evolution of organisms over relatively long periods of time.

In 2004, a team of paleontologists discovered fossilized remains of *Tiktaalik roseae,* nicknamed the "fishapod" because it is the transitional form between fish and four-legged animals, the tetrapods (see the chapter opener). *Tiktaalik* fossils are estimated to be 375 million years old and are from a time when the transition from fish to tetrapods is likely to have occurred. As expected of an intermediate fossil, *Tiktaalik* has a mix of fishlike and tetrapod-like features that illustrate the steps in the evolution of tetrapods from a fishlike ancestor (Fig. 15.12). For example, *Tiktaalik* has a very fishlike set of gills and fins, with the exception of the pectoral, or front, fins, which have the beginnings of wrist bones similar to those of a tetrapod (see the photo in the chapter opener). Unlike a fish, *Tiktaalik* had a flat head, flexible neck, eyes on the top of its head (like a crocodile), and interlocking ribs that suggest it had lungs. These transitional features suggest it had the ability to push itself along the bottom of shallow rivers and see above the surface of the water—features that would come in handy in the river habitat where it lived.

Even in Darwin's day, scientists knew of *Archaeopteryx,* which is an intermediate between dinosaurs and birds. Progressively younger fossils than *Archaeopteryx* have been found: The skeletal remains of *Sinornis* suggest it had wings that could fold against its body like those of modern birds, and its grasping feet had an opposable toe—but it still had a tail. Another fossil, *Confuciusornis,* had the first toothless beak. A third fossil, called *Iberomesornis,* had a breastbone to which powerful flight muscles could attach. Such fossils show how the species of today evolved.

Fossils have been discovered that support the hypothesis that whales had terrestrial ancestors. *Ambulocetus natans* (meaning "the walking whale that swims") was the size of a large sea lion, with broad, webbed feet on its forelimbs and hindlimbs that enabled it to both walk and swim. It also had tiny hooves on its toes and the primitive skull and teeth of early whales (Fig. 15.13).

Modern whales still have a vestigial hindlimb consisting of only a few bones that are very reduced in size. As the ancestors of whales adopted an increasingly aquatic lifestyle, the location of the

ISTUDY

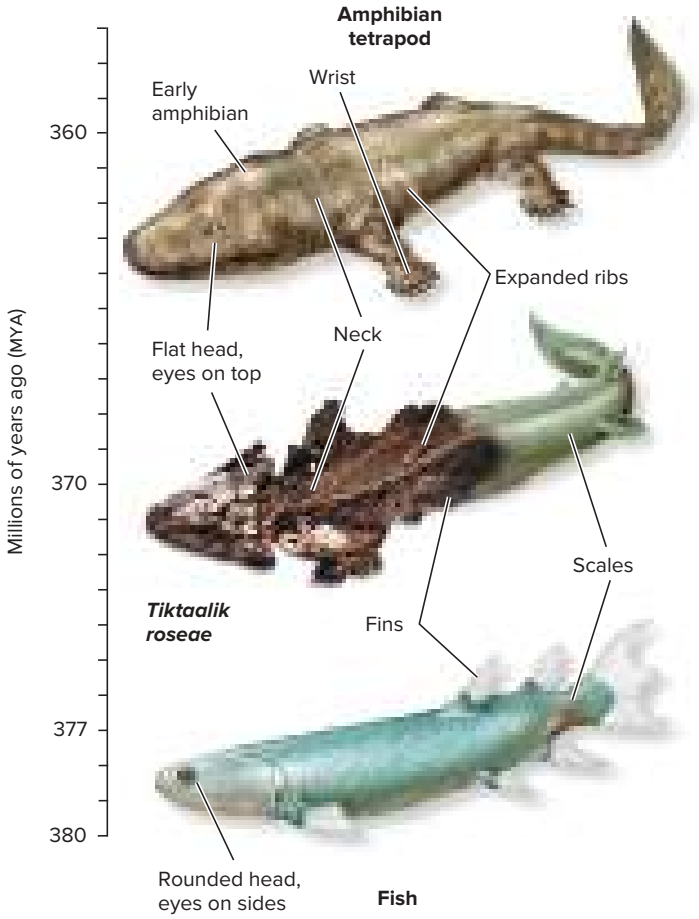


Figure 15.12 Transitional fossils. *Tiktaalik roseae* had a mix of fishlike and tetrapod-like features. Fossils such as *Tiktaalik* provide evidence that the evolution of new groups involves the modification of preexisting features in older groups. The evolutionary transition from one form to another, such as from a fish to a tetrapod, can be gradual, with intermediate forms having a suite of adapted, fully functional features.
(fossil): Corbin17/Alamy Stock Photo

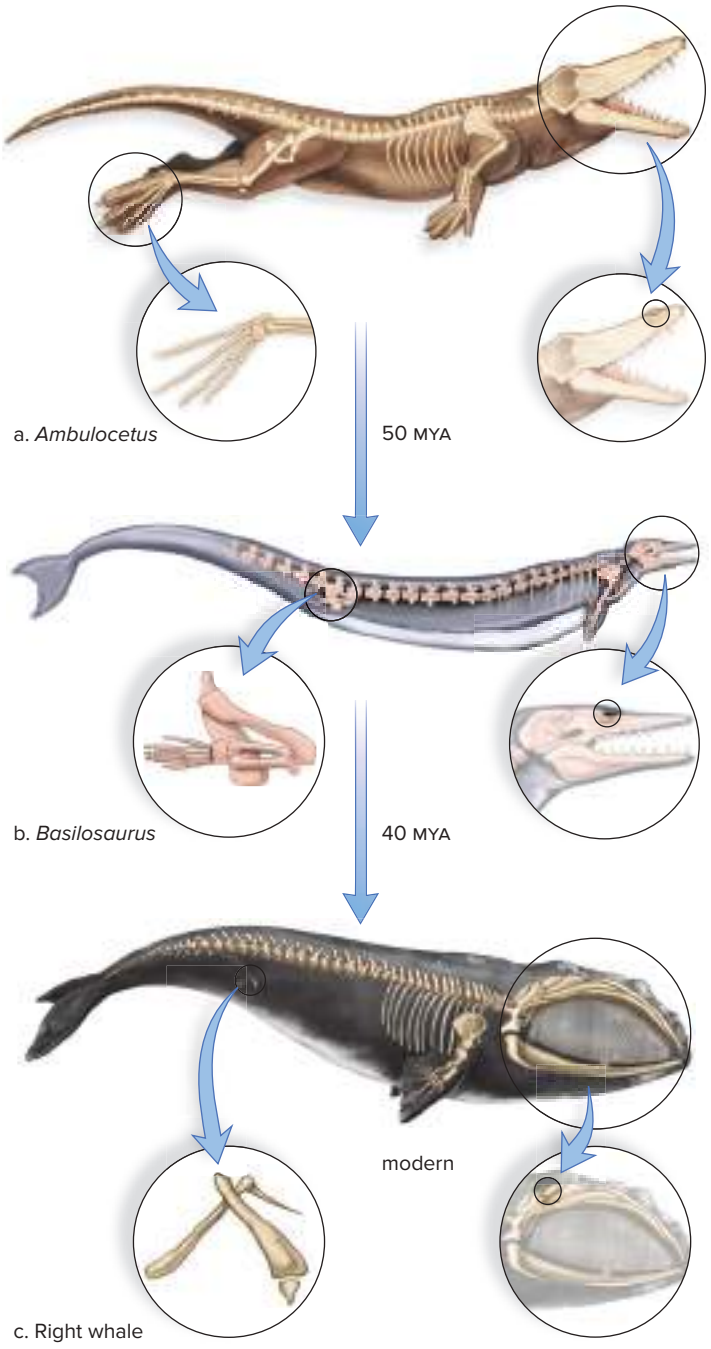


Figure 15.13 Anatomical transitions during the evolution of whales. Transitional fossils, such as *Ambulocetus* and *Basilosaurus,* support the hypothesis that modern whales evolved from terrestrial ancestors that walked on four limbs. These fossils show a gradual reduction in the hindlimb and a movement of the nasal opening from the tip of the nose to the top of the head—both adaptations to living in water.

nasal opening underwent a transition—from the tip of the snout, as in *Ambulocetus;* to midway between the tip of the snout and the skull in *Basilosaurus;* to the very top of the head in modern whales (Fig. 15.13). An older fossil, *Pakicetus,* was primarily terrestrial yet had the dentition of an early whale. A younger fossil, *Rodhocetus,* had reduced hindlimbs that would have been no help for either walking or swimming but may have been used for stabilization during mating.

The origin of mammals is also well documented. The synapsids, an early amniote group, gave rise to the premammals. Slowly, mammal-like organisms acquired features that enabled them to breathe and eat at the same time: a muscular diaphragm and rib cage that helped them breathe efficiently. The earliest true mammals were shrew-sized creatures, which have been unearthed in fossil beds about 200 million years old. (We return to the topic of mammalian evolution in Section 29.6.)

Biogeographical Evidence

We described in Section 15.2 the biogeographical observations that Darwin made during his voyage on the HMS *Beagle.* We noted that in cases where geography separates continents, islands, and seas, we might expect a different mix of plants and animals. For example, during his travels, Darwin observed that South America lacked rabbits, even though the environment was quite suitable for them. He concluded there were no rabbits in South America because rabbits evolved somewhere else and had no means of reaching South America. Instead, a different animal, the Patagonian cavy, occupied the environmental niche that rabbits held elsewhere.

ISTUDY

a. Sugar glider

b. Australian wombat

c. Tasmanian wolf

Figure 15.14 Biogeography. Each type of marsupial in Australia is adapted to a different way of life. **a.** The sugar glider (*Petaurus breviceps*) is a tree-dweller that resembles the placental flying squirrel. **b.** The Australian wombat (*Vombatus*) is nocturnal and lives in burrows. It resembles the placental marmot. **c.** The Tasmanian wolf (*Thylacinus cynocephalus*) is an extinct species that resembled the placental American wolf. (sugar glider): ANT Photo Library/Science Source; (wombat): Marco Tomasini/Shutterstock; (Tasmanian wolf): World History Archive/Alamy Stock Photo

In addition, Darwin noted that the different species of finches on the Galápagos Islands were not found on mainland South America. One reasonable explanation is that an ancestral population of finches from the mainland reached the Galápagos Islands and over time evolved into the different species found on each isolated island.

Much earlier in the history of the Earth, South America, Antarctica, and Australia were originally connected (see Fig. 18.16). Marsupials, mammals in which females have an external body pouch where their young complete development, have evolved from egg-laying mammalian ancestors. Today, marsupials are endemic to South America and Australia. What is now Australia separated and drifted away from the other landmasses. The marsupials then diversified into many different forms suited to various environments (Fig. 15.14).

Marsupials were free to diversify because few, if any, placental mammals were present in Australia. In placental mammals, young complete their development inside the mother's uterus, nourished by the placenta (see Section 29.6). Where placental mammals are abundant, marsupials are not as diverse due to competition. After the formation of the Isthmus of Panama, placental mammals were able to migrate into South America. As a result, marsupial mammals were outcompeted by placental mammals, and the diversity of marsupials in South America declined greatly.

Biogeographical differences, therefore, provided evidence that variability in a single, ancestral population can lead to adaptation to different environments through the forces of natural selection. Competition for resources appears to provide some of the pressure that leads to diversification.

Anatomical Evidence

Darwin was able to show how descent from a common ancestor can explain anatomical similarities among organisms. Vertebrate forelimbs are used for flight (birds and bats), orientation during swimming (whales and seals), running (horses), climbing (arboreal lizards), and swinging from tree branches (monkeys). However, all vertebrate forelimbs contain the same sets of bones organized in similar ways, despite their dissimilar functions (Fig. 15.15). The most plausible explanation for this unity is that this basic forelimb plan was present in a common vertebrate ancestor. This plan was then modified independently in all descendants as each continued along its own evolutionary pathway.

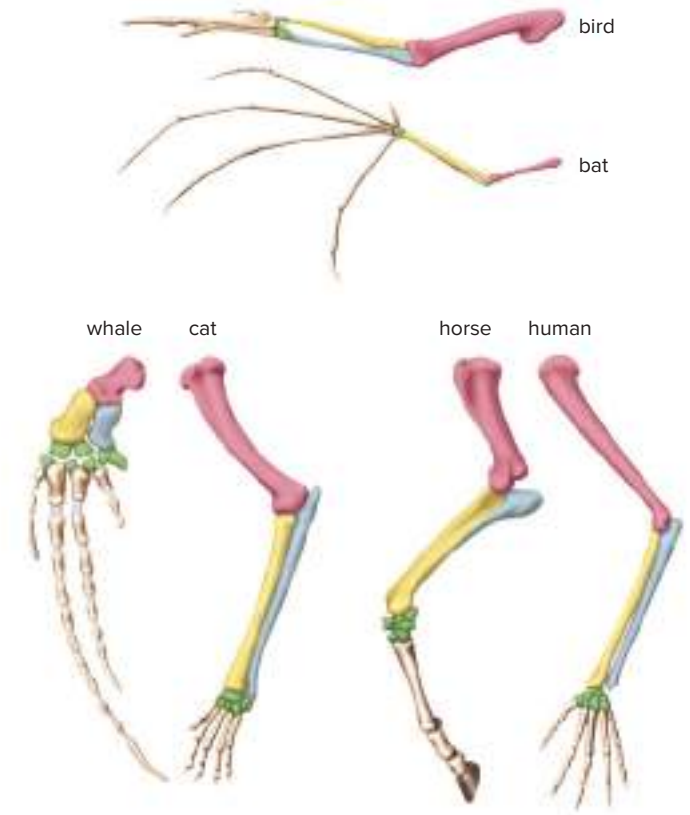


Figure 15.15 Significance of homologous structures. Although the specific details of vertebrate forelimbs are different, the same basic bone structure and position are present (color-coded here). This unity of anatomy is evidence of a common ancestor.

Structures that are anatomically similar because they are inherited from a common ancestor are called **homologous structures.** In contrast, **analogous structures** serve the same function but originated independently in different groups of organisms that do not share a common ancestor. The wings of birds and insects are analogous structures. Thus, homologous, not analogous, structures are evidence for a common ancestry of particular groups of organisms.

As mentioned earlier, *vestigial structures* are anatomical features that are fully developed in one group of organisms but are reduced and may have no function in related groups. Most birds, for example, have well-developed wings used for flight, while some species have greatly reduced wings and do not fly. Similarly, snakes and whales have no use for hindlimbs, yet some species have remnants of a pelvic girdle and hindlimbs. Both the tail bone and wisdom teeth are examples of human structures that have no apparent function in our species today. Vestigial structures occur because organisms inherit their anatomy from their ancestors, and therefore their anatomy carries traces of their evolutionary history.

The homology shared by vertebrates is observable during their embryological development (Fig. 15.16). At some time during development, all vertebrates have a postanal tail and exhibit

ISTUDY

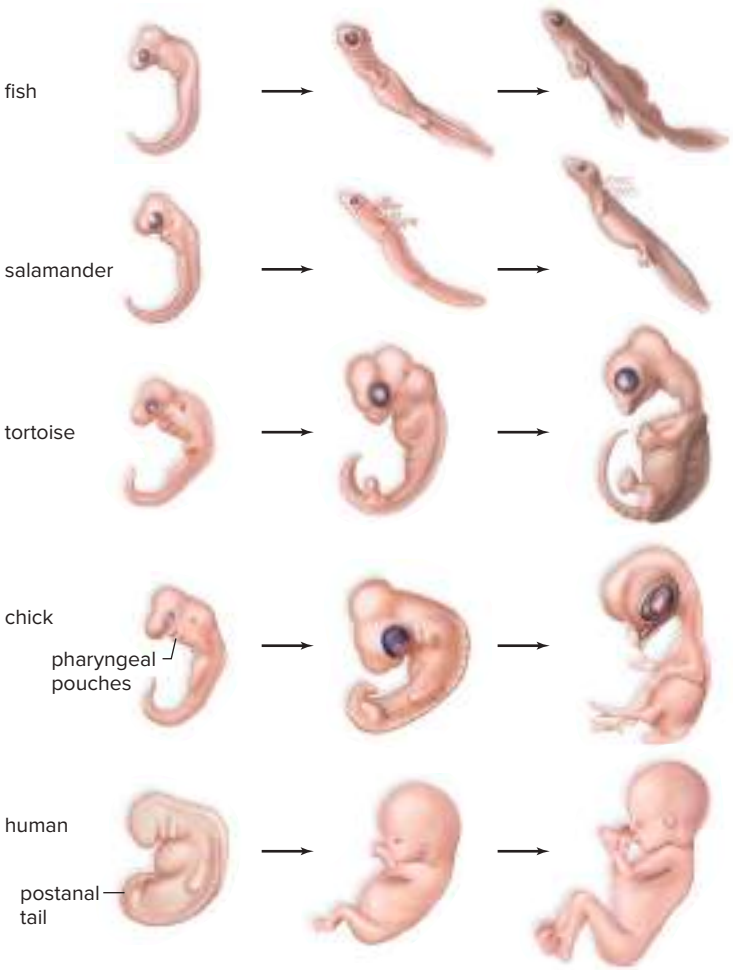


Figure 15.16 Significance of developmental similarities. At these comparable developmental stages, vertebrate embryos have many features in common, which suggests they evolved from a common ancestor. (These embryos are not drawn to scale.)

paired pharyngeal (throat) pouches supported by cartilaginous arches. In fishes and amphibian larvae, these pouches develop into functioning gills. In humans, the first pair of pouches and arches becomes the jawbones, the cavity of the middle ear, and the auditory tube. The second pair of pouches becomes the tonsils and facial muscle and nerves, while the third and fourth pairs become the thymus and parathyroid glands.

Why do structures such as pharyngeal pouches develop in all vertebrate embryos, but then become very different structures with vastly different functions in adults of different groups? New structures or novel functions can originate only through modification of the preexisting structures in one's ancestors. All vertebrates inherited the same developmental pattern from their common ancestor, but each vertebrate group now has a specific set of modifications to this original ancestral pattern.

Biochemical Evidence

All living organisms use the same basic biochemical molecules, including DNA (deoxyribonucleic acid), RNA (ribonucleic acid), and ATP (adenosine triphosphate). We can deduce from this that these molecules were present in the first living cell or cells from which life as we know it today has arisen.

Organisms use a triplet nucleic-acid code in their DNA to encode for 1 of 20 amino acids that will form their proteins. Because the sequences of DNA bases in the genomes of many organisms are now known, clear evidence is available that humans have some genes in common with much simpler organisms, such as prokaryotes. Because the genetic code is universal in living organisms, it is possible to insert a human gene into the genome of a bacterium. The bacterium will then produce the human protein that the gene encodes for.

Also, the sequence of amino acids of some proteins is similar across the tree of life. The sequence of amino acids in the human version of cytochrome *c,* a protein essential to cellular respiration, is remarkably similar to that of yeast (Fig. 15.17). The number of

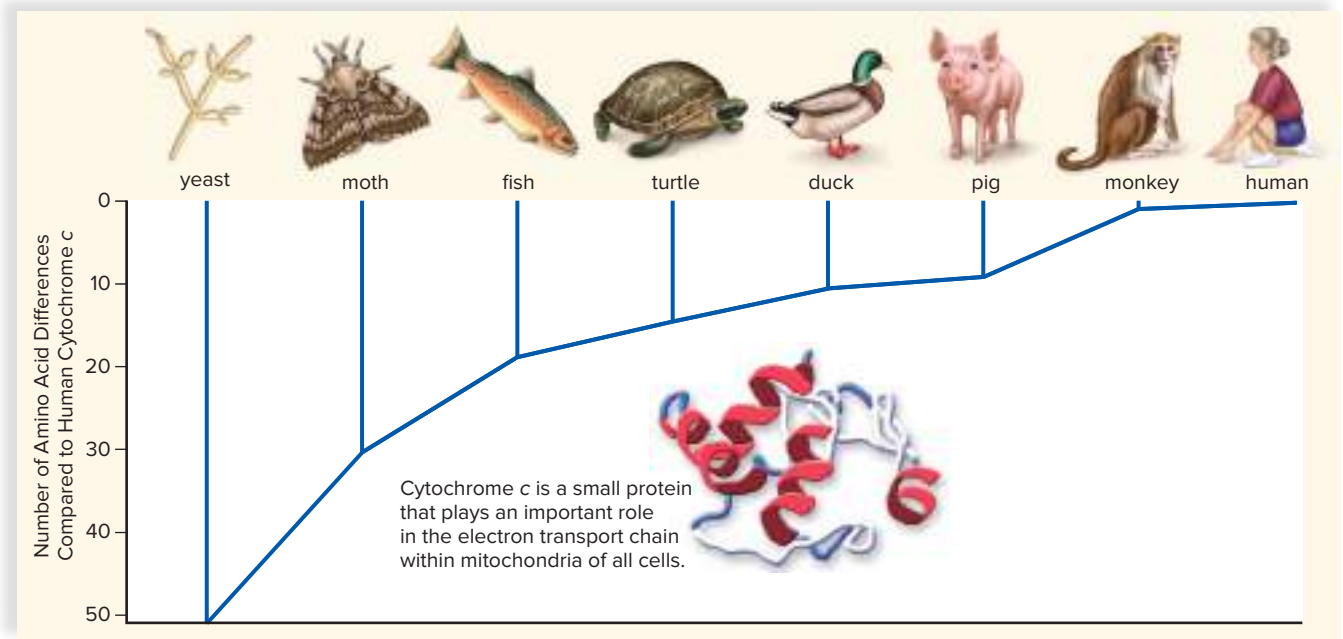


Figure 15.17 Significance of biochemical differences. The branch points in this diagram indicate the number of amino acids that differ between human cytochrome c and the organisms depicted. These biochemical data are consistent with those provided by a study of the fossil record and comparative anatomy.

ISTUDY

differences between the cytochrome *c* amino acid sequence in humans and other organisms increases with the distance in time since they shared a common ancestor—monkey cytochrome *c* differs from that of humans by only 1 amino acid, from that of a duck by 11 amino acids, and from that of yeast by 51 amino acids.

Evidence from Developmental Biology

The study of the evolution of development has discovered that many developmental genes are shared among all animals ranging from worms to humans. It appears that life's vast diversity has come about by a set of regulatory genes that control the activity of other genes involved in development.

For example, *Hox,* or *homeobox,* genes orchestrate the development of the body plan in all animals, from invertebrates (such as sea anemones and fruit flies) to humans (see the Evolution feature, "Evolution of the Animal Body Plan," in Section 28.1). All animals share a *Hox* gene common ancestor, but the number and type of *Hox* genes vary among animal groups. This variation in *Hox* genes is responsible, at least in part, for the wide range of body plans seen in animals. For example, a change in the timing and duration of the expression of *Hox* genes that control the number and type of vertebrae can produce the spinal column of a chicken or the longer spinal column of a snake. Thus, simple changes in how genes are controlled can have profound effects on the phenotype of organisms.

Criticisms of Evolution

Evolution is no longer considered a hypothesis. It is one of the great unifying theories of biology. Evolution is not "just a theory"—in science, the word *theory* is reserved for those concepts that are supported by a large number of observations (see Section 1.2). The theory of evolution has the same status in biology that the theory of heredity has in genetics. However, some people propose mechanisms other than evolution to explain the origin of new species. These alternatives, founded in religious philosophy, cannot be tested, so they are not accepted as scientific evidence.

Many misconceptions about evolution, and the scientific process in general, are commonly used to challenge the legitimacy of the theory of evolution. The following are a few examples of these misconceptions, with a brief scientific explanation after each.

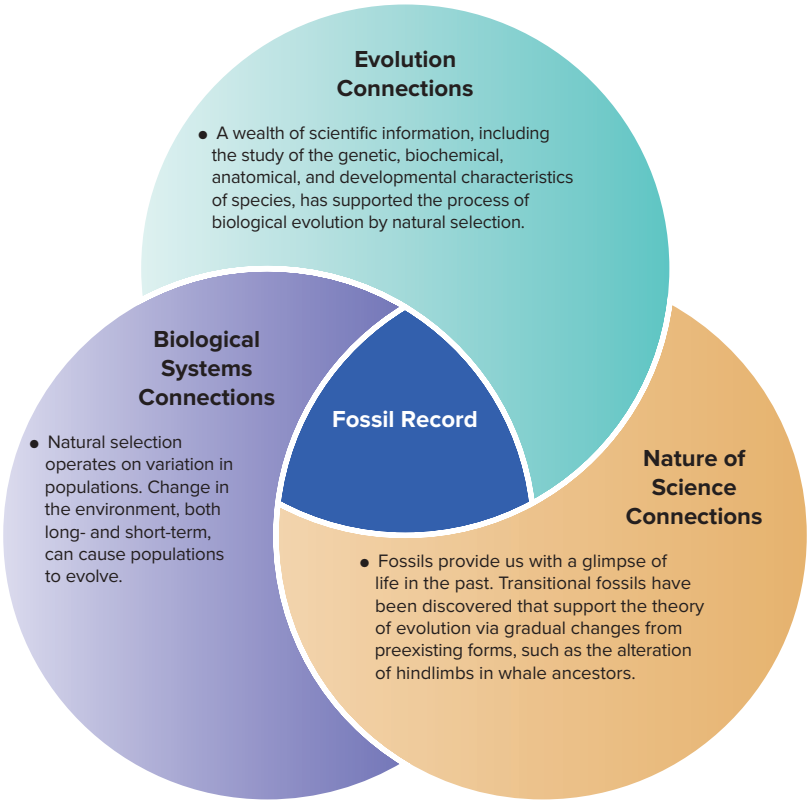
- *Evolution is a theory about how life originated.*
  Evolutionary biologists are concerned with how the diversity of life emerged *following* the origin of life. Certainly, the study of the origin of life is interesting to evolutionary biologists, but this is not the focus of their research.
- *There are no transitional fossils.*
  Biologists do not expect that all transitional forms have been preserved in the fossil record. In fact, scientists *predict* that not all transitional forms will be discovered. The reason is that a series of events must occur before a fossil can be found. First, the organism must have perished in an area that favors the preservation of its skeletal remains. Soft tissue remains are rarely fossilized; thus, many species will not leave fossil remains at all. Second, scientists have to locate and uncover those remains. This is like finding a needle in a haystack!
  Despite this, scientists have unearthed an array of transitional fossils. For example, the fossil record clearly demonstrates transitional forms in the evolution of the whale from its terrestrial ancestor (see Fig. 15.13). Also, a series of intermediate fossils illustrates the transition from fish to tetrapods, such as *Tiktaalik* (see Fig. 15.12).
- *Evolution proposes that life changed as a result of random events; clearly, traits are too complex to have originated "by chance."*
  "Chance" does play a role in evolution, but this is only part of the story. Mutation, the process that *creates* new variation in populations, occurs randomly. However, natural selection, the process that *shapes* variation, is not random. Natural selection can act only on the variation that is present in a population, and it is thus constrained by changes that have occurred in the past. Complex structures, such as the vertebrate eye, did not evolve as a single, functioning unit with all parts intact. Complexity is the result of millions of years of modifications to preexisting traits, each of which provided a useful function at the time. For example, the bacterial flagellum—the long, tail-like propeller of some bacteria—contains a complex, microscopic, rotary motor made from an assembly of many proteins. In a different group of bacteria, a simpler version of this protein assembly does not function as a rotary motor but as a "syringe," called an injectisome, which the bacteria use to "inject" eukaryotic cells with toxins. Scientists hypothesize that the flagellum evolved over time via the addition of proteins to a preexisting, simpler structure, like an injectisome. Both the injectisome and the flagellum, composed of subsets of the same proteins, are totally functional, even though one is less complex than the other.
- *Evolution is not observable or testable; thus, it is not science.*
  Evolution is both observable and testable. Recently, scientists discovered that there are genes that encode more than one type of trait. New variation can arise from small changes to single genes. Several studies show how traits in populations change in response to environmental changes (see Section 15.2). Other branches of science figure out how things work by accumulating evidence from the real world. Particle physicists cannot see the electrons in an atom. Geologists cannot directly observe the past. But like evolutionary biologists, these scientists can learn a lot about the world by gathering evidence from multiple sources. Evolution has 150 years of such supporting evidence from a wide variety of scientific disciplines.

Check Your Progress 15.3

1. Define *transitional fossils* and provide one example.
2. Summarize the differences between homologous and analogous vestigial structures, as well as what each tells us about common ancestry.
3. Explain how biomolecules support the theory of evolution by natural selection.
4. State one misconception about the theory of evolution and explain why it is incorrect.

ISTUDY

CONNECTING *THE* CONCEPTS



What Did You Learn?

15.1 The History of Evolutionary Thought

In general, the pre-Darwinian worldview was different from the post-Darwinian worldview.

- A century before Darwin's trip, most biologists believed in the fixity, or unchanging nature, of species. **Evolution,** the change in species over time, was not widely accepted.
- As explorers traveled the world, questions began to arise. Erasmus Darwin, Charles's grandfather, suggested the possibility of evolution based on the presence of **vestigial structures.**
- Linnaeus, the originator of taxonomy, thought that each species had a place in nature and that classification should describe the fixed features of species.
- Georges Cuvier and Jean-Baptiste de Lamarck, contemporaries of Darwin in the late eighteenth century, differed sharply on evolution. To explain the fossil record of a region, Cuvier, the founder of **paleontology,** proposed that changes in the makeup of fossils in the **strata** of the Earth could be explained by regional catastrophes, or extinctions, followed by repopulation from other regions. Cuvier's idea became known as **catastrophism.** Lamarck was in support of the idea that species can change as they become adapted to their environments. However, he suggested the **inheritance of acquired characteristics** as a mechanism for evolutionary change. We now know that only traits encoded in our genes are heritable.
- Charles Lyell, the foremost geologist of Darwin's time, made popular James Hutton's theory of slow, uniform geological change. Lyell's theory of **uniformitarianism** stated that geological processes today are the same as those in the past. Darwin's observations of geology led him to support Lyell's proposal and that the Earth, therefore, must be very old.
- Thomas Malthus, in his influential work *Essay on the Principle of Population,* proposed that population growth is limited by the availability of resources. Darwin applied this idea to his theory of natural selection.

15.2 Darwin's Theory of Natural Selection

Charles Darwin formulated hypotheses concerning evolution after taking a 5-year voyage as a naturalist aboard the ship HMS *Beagle.* His hypotheses were that descent with modification from a common ancestor does occur and that natural selection results in adaptation to the environment.

- Darwin's study of **biogeography,** including the animals of the Galápagos Islands, led him to conclude that biological diversity arises from adaptation to the environment, which can eventually lead to the formation of new species.
- **Natural selection** is a mechanism of evolutionary change. Members of a population have heritable variations, compete for resources, differ in reproductive success, and adapt to new environmental conditions. **Fitness** is the reproductive success of an organism. **Adaptations** are evolved traits that increase an organism's chances of survival. Darwin observed selection at work as humans used **artificial selection** to increase the frequency of desired traits in various organisms.

ISTUDY

15.3 Evidence for Evolution

The theory that all organisms share a common ancestor is supported by many lines of evidence, including fossils, anatomy, biochemistry, and development.

- The **fossil record** gives us a snapshot of the history of life that allows us to trace the descent of a particular group. **Transitional fossils** bear a resemblance to two groups that are classified separately. These allow us to retrace the evolution of organisms over time.
- Biogeography is the study of the range and distribution of plants and animals in different places throughout the world and how, and when, they came to be distributed as they are today. Therefore, a different mix of plants and animals might be expected in cases where geography separates continents, islands, and seas.
- A comparison of the anatomy and the development of organisms suggests that all life on Earth is closely related. All organisms have certain biochemical molecules and a body plan encoded in genes shared in common, suggesting relatedness. **Homologous structures** are anatomically similar due to inheritance from a common ancestor. **Analogous structures** serve the same function but originated independently in different groups. **Homeobox** genes orchestrate the development of the body plan in all animals.

Thinking Critically About Biology

1. Mutations occur at random and can increase within a population for no particular purpose. Our immune system is capable of detecting and killing certain viruses. Would a virus, such as HIV, that has a frequent rate of mutation, be more or less successful in avoiding the immune system? Explain.
2. A cotton farmer applies a new insecticide against the boll weevil to the crop for several years. At first, the treatment is successful, but then the insecticide becomes ineffective and the boll weevil rebounds. Has evolution occurred? Explain.
3. Both Darwin and Wallace, while observing life on islands, concluded that natural selection is the mechanism for biological evolution. The Hawaiian and nearby islands once had at least 50 species of honeycreeper birds, and they lived nowhere else on Earth. Natural selection occurs everywhere and in all species. What characteristics of islands allow the outcome of natural selection to be so obvious?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video below:

- Quail Evolution

McGraw Hill

Explain how the experiment in this video relates to Darwin's theory of natural selection.

Making It Relevant

1. People often use the word *theory* to refer to a guess about something. In the scientific sense, the word *theory* means a scientific concept that has been supported by observation and experiments, so it is widely accepted by the scientific community as an explanation of natural phenomena. The theory of evolution is referred to as one of the unifying theories of biology, because it offers the best explanation for the diversity of life. Criticisms of Darwin's theory of evolution by natural selection, along with alternative "theories" about the diversity of life, have been presented by various groups. Do you think these alternatives should be presented in a science classroom as an equal to the theory of evolution by natural selection? What criteria should be applied to these theories to make them acceptable?
2. Climate change is causing ecosystems to change at a pace that is more rapid than at any time in recent history. Scientists are concerned that these changes are causing a human-induced mass extinction crisis. How does this relate to the process of evolution? What are some of the potential consequences for humans?
3. Like all species, humans have also evolved over time to adapt to changes in our environment. What do you think some of these changes are, and what evidence would you look for in humans to assess whether an adaptation occurred?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

16

How Populations Evolve

Science Photo Library/Science Source

Chapter Outline

16.1 Genes, Populations, and Evolution
16.2 Natural Selection
16.3 Maintenance of Diversity

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.1 What is a population?

Section 11.2 How is a Punnett square used to estimate genotype frequencies?

Section 11.2 What is an allele?

Antibiotic-Resistant Gonorrhea

Gonorrhea is a sexually transmitted disease (STD) caused by the bacterium *Neisseria gonorrhoeae*. The infection can be found in both males and females, and is one of the most common STDs in the United States, especially among young people 15–24 years of age.

This STD is already resistant to the antibiotics tetracycline, fluoroquinolones, and penicillin, and is currently treated by dosing the patient with two other antibiotics—azithromycin and ceftriaxone. In the last few years, a weaker response to azithromycin by *Neisseria gonorrhoeae* has risen by over 400%. This suggests that azithromycin will be next on the list of antibiotics that will no longer work against gonorrhea.

Use and overuse of antibiotics have resulted in the evolution of resistant bacterial strains. Human activities can accelerate the process of evolution quite rapidly. Some scientists believe these "superbugs" will be a far bigger threat to human health than emerging diseases such as Zika or even COVID-19. The good news is that the study of evolutionary biology, and the processes by which evolution occurs, is helping to combat the superbugs.

As you read through the chapter, think about the following questions:

1. In the case of antibiotic resistance, what type of selection is operating?
2. Discuss strategies that may be used to help reduce the rate at which antibiotic-resistant bacteria evolve.

Following *the* Themes

CHAPTER 16 – HOW POPULATIONS EVOLVE

Evolution	Microevolution, or evolution within populations, is measured as a change in allele frequencies over generations.
Nature of Science	Population geneticists calculate allele and genotype frequencies to characterize how populations evolve.
Biological Systems	Microevolution and the environment are closely linked, such that a phenotype may be adaptive in one environment, but not in another.

ISTUDY

16.1 Genes, Populations, and Evolution

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain how evolution in populations is related to a change in allele frequencies.
2. List the five conditions necessary to maintain Hardy-Weinberg equilibrium.
3. Apply the Hardy-Weinberg principle to estimate equilibrium genotype frequencies.
4. Describe the agents of evolutionary change.

A triathlete who spends months in Denver gradually gets used to being at high altitude. Part of the reason is that the number of oxygen-carrying red blood cells has increased in response to the oxygen-poor environment. Many traits can change temporarily in response to a varying environment. The color change in the fur of an Arctic fox from brown to white in winter, the increased thickness of your dog's fur in cold weather, or the darkening of your skin when exposed to the sun lasts only for a season.

However, these are not evolutionary changes. Changes to traits over an individual's lifetime are not evidence that an individual has evolved, because these traits are not heritable. In order for traits to evolve, they must have the ability to be passed on to subsequent generations. Evolution is about change in a trait within a population over many generations.

A **population** is defined as a group of organisms of a single species living together in the same geographic area. Darwin observed that populations, not individuals, evolve, but he could not explain *how* traits change over time. Now we know that genes interact with the environment to determine traits—the diversity of a population is linked to the genetic diversity of individuals within that population. Because genes and traits are linked, evolution is really about genetic change—or more specifically, *evolution is the change in allele frequencies in a population over time.*

Several mechanisms can cause a population to evolve—to change allele frequencies—over generations. **Microevolution** pertains to evolutionary change within populations. In this chapter, we will use the peppered moth example from Section 15.2 to examine how populations evolve over time.

Microevolution in the Peppered Moth

Population genetics is the field of biology that studies the diversity of populations at the level of the gene. Population geneticists are interested in how genetic diversity in populations changes over generations, as well as in the forces that cause populations to evolve. Population geneticists study microevolution by measuring the diversity of a population in terms of allele and genotype frequencies over generations.

You may recall from Section 10.1 that diploid organisms, such as moths, carry two copies of each chromosome, with one copy of each gene on each chromosome. A single gene can come in many forms, or alleles, that encode variations of a single trait.

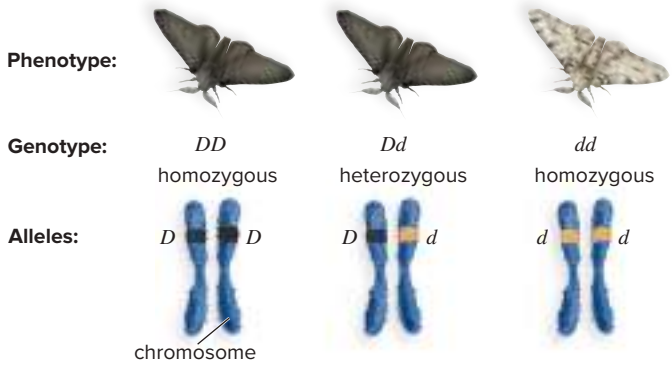


Figure 16.1 The genetic basis of body color in the peppered moth. Light or dark body color in the peppered moth is determined by a gene with two alleles: *D* and *d.* Genotypes *DD* and *Dd* produce dark body color, and *dd* produces light body color. The *D* and *d* alleles are variants of a gene at a particular locus on a chromosome.

In the peppered moth, a single gene for body color has two alleles, *D* (dark color) and *d* (light color), with *D* dominant to *d* (Fig. 16.1).

We know that with only two alleles, there are three possible genotypes (the combinations of alleles in an individual) for the color gene in the peppered moth: *DD* (homozygous dominant), *Dd* (heterozygous), or *dd* (homozygous recessive). *DD* or *Dd* genotypes produce dark moths, and the *dd* genotype produces light moths (Fig. 16.1).

Allele Frequencies

Suppose a population geneticist collected a population of 25 moths, some dark and some light, from a forest outside London (Fig. 16.2). In this population you would expect to find a mixture of *D* and *d* alleles in the **gene pool**—the alleles of all genes in all individuals in a population (Fig. 16.2). The population geneticist ran tests to determine the alleles present in each moth. Of the 50 alleles in the peppered moth gene pool (2 alleles × 25 moths), 10 were *D* and 40 were *d.* Thus, the frequency of the *D* and the *d* alleles would be 10/50, or 0.20, and 40/50, or 0.80 (Fig. 16.2). The **allele frequency,** as illustrated in this example, is the percentage of each allele in a population's gene pool.

Notice that the frequencies of *D* and *d* add up to 1. This relationship is true of the sum of allele frequencies in a population for any gene of any diploid organism. This relationship is described by the expression $p + q = 1$, where p is the frequency of one allele, in this case *D,* and q is the frequency of the other allele, *d* (Fig. 16.2).

For the next three seasons, samples of 25 moths were collected from the same forest, and the allele frequencies were always the same: 0.20 *D* and 0.80 *d.* Because allele frequencies in this population did not change over generations, we could conclude, with regard to color, that this population has not evolved.

Hardy-Weinberg Equilibrium

A population in which allele frequencies do not change over time, such as in the moth population just described, is said to be in *genetic equilibrium,* or **Hardy-Weinberg equilibrium (HWE)**—a stable,

ISTUDY

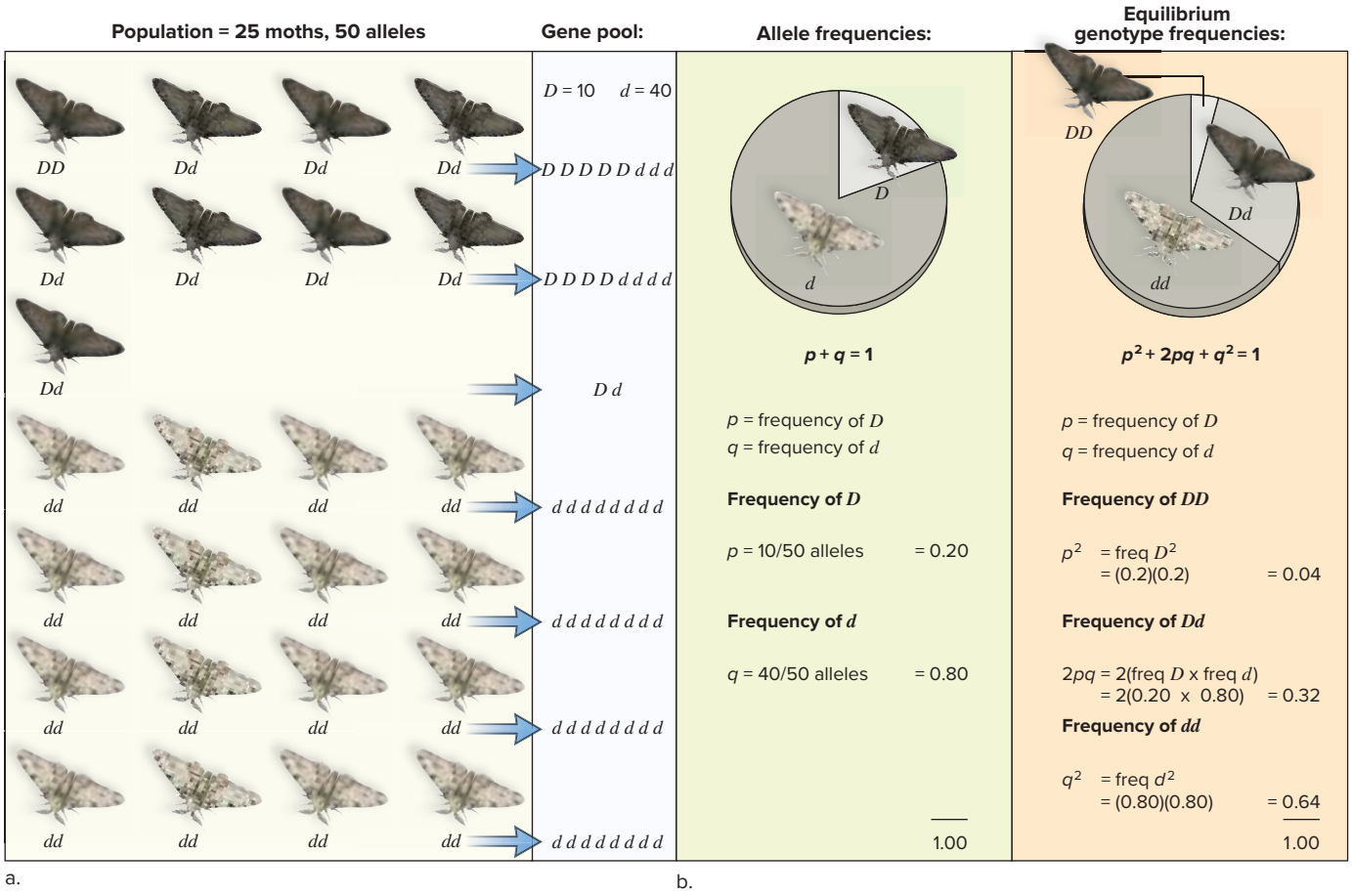


Figure 16.2 How Hardy-Weinberg equilibrium is estimated. **a.** A population of 25 moths contains a gene pool of *D* and *d* alleles. **b.** (*left*) The frequencies of *D* and *d* alleles can be estimated from the gene pool. (*right*) The frequencies of the *D* and *d* alleles should produce, in the next generation, a predictable frequency of genotypes that can be calculated with the Hardy-Weinberg principle.

nonevolving state. Hardy-Weinberg equilibrium is derived from the work of British mathematician Godfrey H. Hardy and German physician Wilhelm Weinberg, who in 1908 developed a mathematical model to estimate genotype frequencies of a population that is in genetic equilibrium. Their mathematical model, called the **Hardy-Weinberg principle,** proposes that the genotype frequencies of a nonevolving population can be described by the expression $p^2 + 2pq + q^2$, again with p and q representing the frequency of alleles *D* and *d*. Recall that in our moth population, $D = p$ and $d = q$, so that D^2 is the frequency of the *DD* genotype, $2Dd$ is the frequency of the *Dd* genotype, and d^2 is the frequency of the *dd* genotype (Fig. 16.2).

A simple Punnett square, first described in Section 11.1, is another way to illustrate how the Hardy-Weinberg principle explains the genotype frequencies of a population. The frequency of the *D* and *d* alleles in the gametes (sperm and egg) in this population would be the same as the allele frequencies, so that 20% of alleles in eggs and sperm will be *D,* and 80% of alleles in eggs and sperm will be *d* (Fig. 16.3). The genotype frequencies from the Punnett square match those predicted by the Hardy-Weinberg principle—that is, the frequency of *DD, Dd,* and *dd* is explained by the expression $D^2 + 2Dd + d^2$ (see Figs. 16.2 and 16.3).

The allele and genotype frequencies of the moth population we are considering follow the Hardy-Weinberg principle only if the population is not evolving. The allele and genotype frequencies

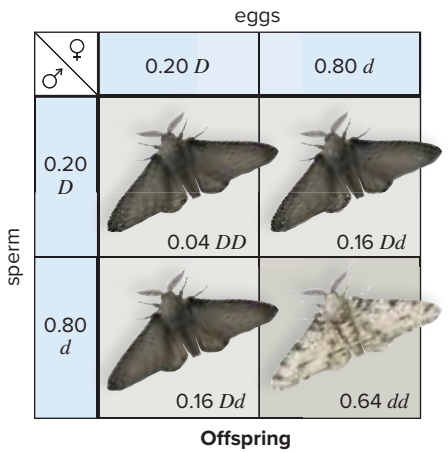


Figure 16.3 Calculation of Hardy-Weinberg equilibrium from gamete frequencies. Notice that the results of the Punnett square match those shown in Figure 16.2.

need to remain constant over time. Thus, the Hardy-Weinberg principle applies only if the following conditions are met:

- *No mutation:* No new alleles can arise by mutation.
- *No migration:* No new members (and their alleles) can join the population, and no existing members can leave the population.
- *Large gene pool:* The population is very large.
- *Random mating:* Individuals select mates at random; mate choice is not biased by genotypes or phenotypes.
- *No selection:* The process of natural selection does not favor one genotype over another.

All of these conditions are required to maintain Hardy-Weinberg equilibrium. Each condition, if not met, can cause allele and/or genotype frequencies to change (Fig. 16.4). For example, individuals migrating into and out of the population would bring alleles into, or remove them from, a population. Natural selection might favor one allele over another, changing allele frequencies, so that some alleles are more or less common than others (Fig. 16.4).

Although possible in theory, Hardy-Weinberg equilibrium is never achieved in wild populations. It is unlikely that all of the five

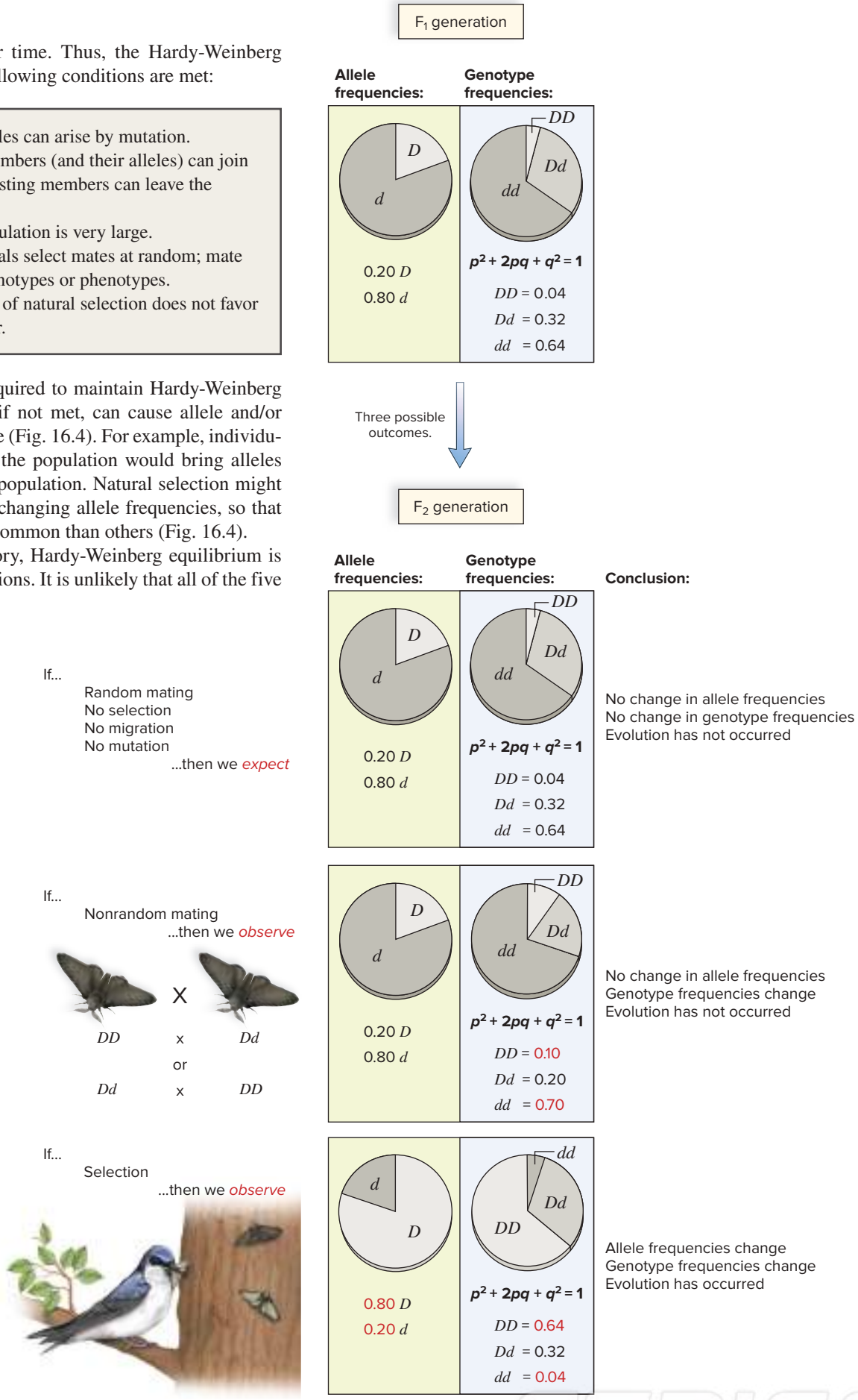


Figure 16.4 Mechanisms of microevolution. Genotype frequencies obey the Hardy-Weinberg principle only if certain conditions are met. Deviations from these conditions change allele or genotype frequencies in a predictable way. Remember that genotype frequencies can change even if allele frequencies do not. Evolution occurs only when allele frequencies change. Changes to genotype frequencies play an important role in evolution, however, because they provide variation on which natural selection can act.

Table 16.1 Hardy-Weinberg Proportions Are Used to Determine If Evolution Has Occurred

HWE Condition	Deviation from Condition	Effect of Deviation on Population	Expected Deviation from HWE	Evolution Occurred?
Random mating	Nonrandom mating	Alleles do not assort randomly	Change in genotype frequencies	No
No selection	Selection	Certain alleles are selected for or against	Change in allele frequencies	Yes
No mutation	Mutation	Addition of new alleles	Change in allele frequencies	Yes
No migration	Immigration or emigration	Individuals carry alleles into, or out of, the population	Change in allele frequencies	Yes
Large population (no genetic drift)	Small population (genetic drift) • Bottleneck effect • Founder effect	Loss of allele diversity; some alleles may disappear	Change in allele frequencies	Yes

required conditions will be met in the real world. The peppered moth population we are using as an example obeys all five of the conditions for genetic equilibrium, but in reality, populations are constantly evolving from one generation to the next.

The Hardy-Weinberg principle does not describe natural populations, but it is an important tool for population geneticists, because the violation of one or more of the five conditions causes the allele and/or genotype frequencies of a population to change in predictable ways. These predictions permit population geneticists to identify the factors that cause microevolution by measuring how allele and genotype frequencies of a population are different from those of a population in Hardy-Weinberg equilibrium (Table 16.1).

Microevolution and Hardy-Weinberg Equilibrium

We can modify the peppered moth scenario to illustrate how the Hardy-Weinberg principle can be used to determine whether, and how, microevolution has occurred. Population genetic theory predicts that Hardy-Weinberg equilibrium can be interrupted by deviation from any of its five conditions. Therefore, we predict that evolutionary change can be caused by mutation, migration, small population size, nonrandom mating, or natural selection.

It is important to remember that evolution is measured as a change in allele frequencies from one generation to another. Thus, we start with allele and genotype frequencies of an F_1 generation, then remeasure the same frequencies in the F_2 generation.

Mutation

A *mutation* is a change to the DNA sequence, which can serve as a source of new genetic variation. Most mutations occur because of errors made to the DNA sequence during DNA replication. The rate of mutations is generally very low, on the order of one mutation per 100,000 cell divisions. Mutations may also occur because of exposure to *mutagens,* chemical or physical agents that cause changes to the DNA code, as described in Section 13.3.

Not all mutations affect the genetic equilibrium of a population. Most mutations that occur during DNA replication and from mutagens are repaired by cellular repair mechanisms. Those mutations that are not repaired can affect the gene pool of a population only if they *are* transmitted from the F_1 to subsequent generations. In other words, the mutations must be carried by the gametes of individuals that successfully reproduce. In addition, mutation is a random process, and the mutation must occur in a gene such that the result is a change in the frequency of the gene's alleles. Thus, it is safe to say that *for any particular gene,* mutation, although possible, is not a major force for evolutionary change, because it generally results in only small changes in the allele frequency of a single allele.

In the peppered moth, a hypothetical example of evolution by mutation could be a mutation that changed a single *D* allele to a *d* allele in gametes of a member of the F_1 generation. This change would alter the frequency of *D* and *d* alleles in the F_2 generation gene pool (Fig. 16.4).

Although inherited mutations are rare, the sum of the effect of mutations is essential to evolution. The reason is that a single organism has many thousands of genes, and each gene can have many different alleles, so that even if the frequency of heritable mutation in a particular allele of a single gene is very low, *over thousands of genes and alleles,* mutation can have a significant impact on the evolution of a population.

For example, the peppered moth is predicted to have around 13,000 genes in its genome. If each gene has only two alleles, then a single moth could carry 26,000 different alleles. In a large population, it is likely a moth would carry at least one new allele that arose because of mutation. These new alleles are sources of new genetic variation, which is essential for evolutionary processes to work. Without new mutations, evolution could not occur, because natural selection must have variation on which to act.

Migration

Gene flow is the movement of alleles between populations. Gene flow occurs when plants or animals migrate, or more specifically their gametes move, between populations. When gene flow brings a new or rare allele into a population, the allele frequency in the next generation changes. Gene flow is a concern of the use of genetically modified plants (see Section 14.2). For example, the movement of a gene that allows a crop plant to be resistant to an

ISTUDY

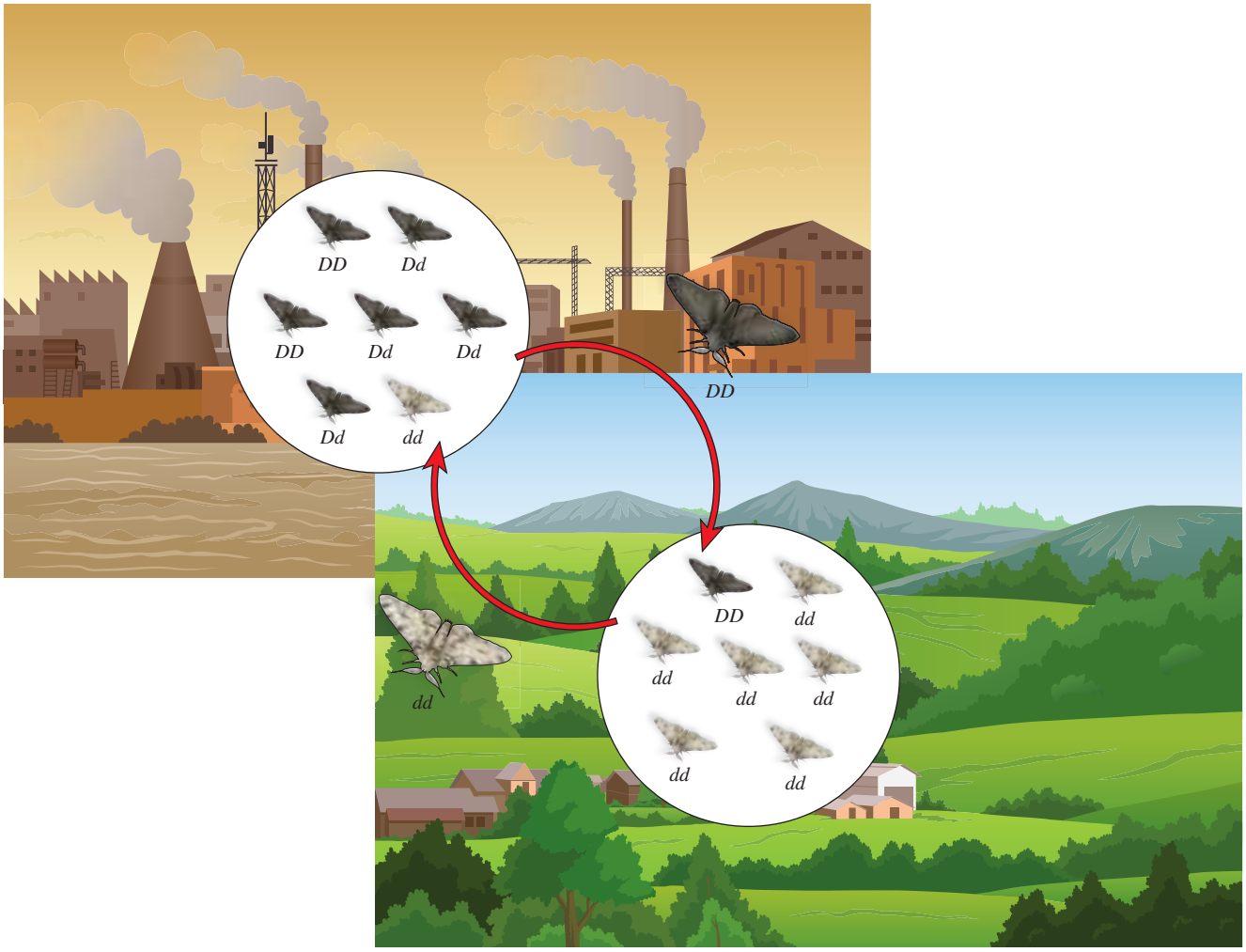


Figure 16.5 Gene flow. Migration of peppered moths with different genotypes would change allele frequencies in different populations.

herbicide, to a neighboring weed species, has the potential for making future weed control more difficult.

In the peppered moth example, the movement of a dark- or light-colored individual into a population would introduce one of the three genotypes: *DD, Dd,* or *dd* (Fig. 16.5). Thus, the numbers of the *D* and *d* alleles would change. In the sample group of 25 moths, the addition of a *Dd* moth would change the number of moths to 26, the total number of alleles to 52, the number of *D* alleles to 11, and the number of *d* alleles to 41. The frequency of *D* would now be 11/52, or 0.21, instead of 0.20; the frequency of *d* would change accordingly, to 0.79. A similar change in allele frequencies would occur if a moth left the population, taking with it its *D* or *d* gametes (see Fig. 16.4).

The amount of gene flow between populations depends on several factors, including the distance between populations, the ability of individuals or their gametes to move between populations, and behavior that determines whether an individual will migrate and mate. When gene flow continuously occurs between populations, the gene pools of each population become more and more similar over time until they appear as if they were a single population. In contrast, if migration between populations does not occur, the gene pools of the populations become more and more different over time. The differences in the genetic makeup of these populations can eventually grow so large they become reproductively isolated—or incapable of interbreeding. Reproductive isolation is the first stage in the formation of new species, which is covered in Section 17.1.

Small Population Size

Genetic drift refers to changes in the allele frequencies of a gene pool due to chance events. Such events remove individuals, and their genes, from a population at random—without regard for genotype or phenotype.

In the equilibrium peppered moth population of 25 individuals, 20% of the alleles are *D,* and 80% are *d.* Suppose that a forest fire kills 5 of the 25 moths, or 20% of the alleles in the gene pool (Fig. 16.6). To simulate the random loss of individuals from the fire, imagine that all of the 25 moths are in a canvas bag. To simulate the fire, 5 moths are removed at random from the bag. The remaining 20 moths in the bag represent the population after the fire. The phenotype, genotype, and allele frequencies are then determined for the 20 moths remaining after the storm. Overall, suppose we find that the loss of the 5 moths has increased the frequency of the *D* allele by 2.5% and decreased the frequency of

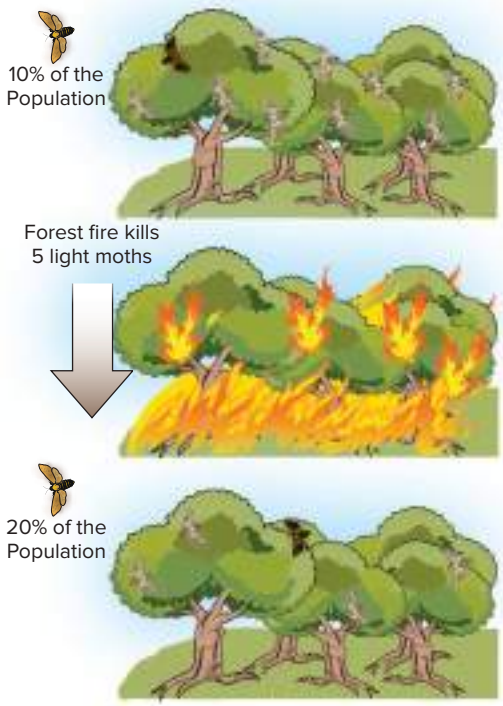


Figure 16.6 Genetic drift. Genetic drift occurs when, by chance, only certain members of a population (in this case, peppered moths) reproduce and pass on their alleles to the next generation. A natural disaster can cause the allele frequencies of the next generation's gene pool to be different from those of the previous generation. Genetic drift can be a powerful force for evolutionary change, especially in small populations.

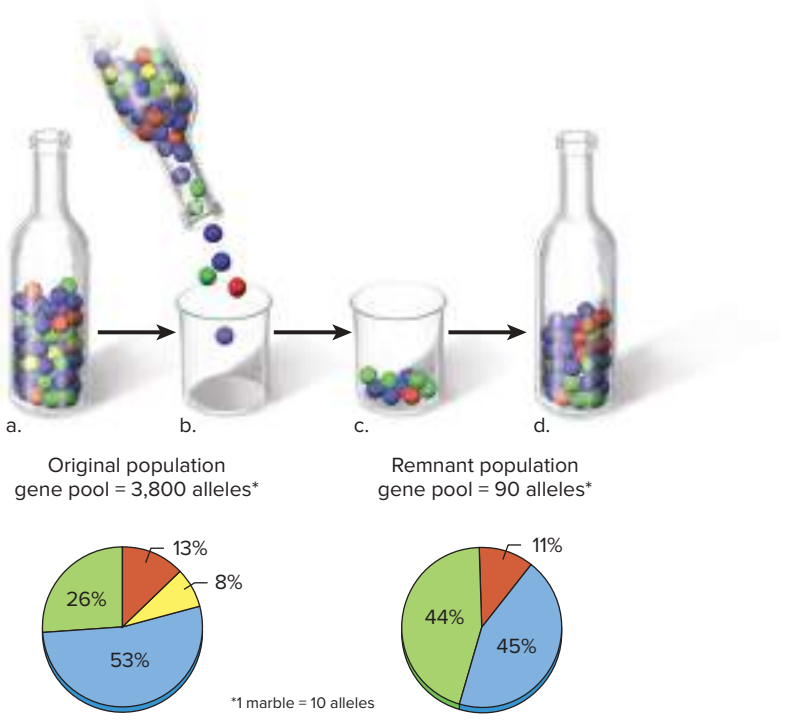


Figure 16.7 Consequences of bottleneck and founder effects. **a.** The gene pool of a large population contains four different alleles, represented by colored marbles in a bottle, each with a different frequency. **b.** A population bottleneck occurs. The marbles, or alleles, that exit the bottle must pass through the narrow neck into the cup. The new gene pool will have a fraction of the alleles from the original population. **c.** The gene pool of the new population has changed from the original. Some alleles are in high frequency, while some are not present. **d.** A founder event is the same as a bottleneck, except that in the founder event the original population still exists.

the *d* allele by 2.5%. The *D* and *d* alleles have changed frequency, and we would conclude that the population has evolved due to genetic drift.

Although genetic drift occurs in populations of all sizes, the gene pool of a smaller population is likely to be more affected by genetic drift. For example, in the population of 25 moths, removing 5 moths reduced the size of the gene pool by 20% (10 alleles from a gene pool of 50). But if the storm had removed 5 moths from a population of 500, or 10 alleles out of a pool of 1,000, only 1% of the gene pool would have been eliminated—with less effect on allele frequency. Thus, the smaller the population, the more genetic drift impacts allele frequencies.

In some cases, a large population can suddenly become very small, such as a *bottleneck,* when natural disasters strike and significantly reduce a population. When this occurs, the effects of genetic drift can be significant. A **bottleneck effect** is a type of genetic drift in which the loss of genetic diversity is due to natural disasters (e.g., hurricane, earthquake, or fire), disease, overhunting, overharvesting, or habitat loss. A **founder effect,** another type of genetic drift, is similar to a bottleneck effect except that genetic variation is lost when a few individuals break away from a large population to found a new population.

The outcome of a founder or a bottleneck effect is a small population with a gene pool made up of a random assortment of alleles from the original population. In some cases, alleles disappear altogether. The gene pool of the small population is often much different from the gene pool of the original population (Fig. 16.7). The greater the reduction in population size, the greater the effects on allele frequencies. Thus, genetic drift is one of the more powerful forces for evolution of small populations.

Another by-product of a very small population is a higher-than-normal occurrence of **inbreeding,** or mating between relatives, because after a few generations only a few, if any, unrelated mates are available. Unlike genetic drift, *inbreeding alone does not affect the frequency of alleles* and thus does not cause a population to evolve. Nevertheless, inbreeding can have a significant impact on the genotypes, and thus the phenotypes, of individuals, sometimes with unfortunate consequences. Of particular concern are rare recessive disorders, which emerge more frequently in inbred populations (see the Nature of Science feature, "Inbreeding in Populations").

Nonrandom Mating

Hardy-Weinberg equilibrium requires individuals in a population to mate randomly. **Nonrandom mating** alone does not cause allele frequencies to change. Nonrandom mating, however, does affect *how the alleles in the gene pool assort into genotypes,* thus

ISTUDY

THEME Nature of Science

Inbreeding in Populations

One of the requirements of a population in Hardy-Weinberg equilibrium is that mates are chosen at random—that is, without preference for a particular trait. Most populations, however, do not meet this requirement, because some traits are more attractive in a mate than others. Humans, for example, select mates based on a set of traits that we find appealing. This type of assortative mating based on trait preference is common in many species. Inbreeding is a unique form of nonrandom, or assortative, mating in which individuals mate with close relatives, such as cousins.

One consequence of inbreeding is an increase in the frequency of homozygous genotypes in a population. For the most part, an increase in homozygotes does not have a large detrimental effect, especially if the population is very large. But when populations are small, inbreeding can have a major impact on the health of a population. Many human diseases are caused by the inheritance of two recessive alleles, such that the disease appears only in persons who are homozygous recessive for the disease-causing allele. In very small populations, the probability that individuals carrying the recessive allele will mate increases, because the mating pool is very small. In this case, inbreeding significantly increases the frequency of homozygous recessive genotypes, and thus those afflicted with a disease.

Human cultures tend to have social rules that discourage inbreeding, but in very small populations, such as those following a bottleneck or founder event, inbreeding is sometimes unavoidable. One example of the effects of inbreeding on human populations is the occurrence of a rare form of non-sex-linked color blindness: achromatopsia (Fig. 16A). People who are achromatic are completely color blind. Normal human color vision is possible because of cells in the back of the eye called cones. Those with achromatopsia do not have any cone cells, because they are homozygous for a rare recessive allele that prevents cone cells from developing. Complete color blindness is so common on Pingelap, a small island in the Pacific Ocean, that it is considered part of everyday life for most families.

Experts propose that the high frequency of achromatopsia appeared on Pingelap following a severe population bottleneck in 1775, when a typhoon killed 90% of the inhabitants. Approximately 20 people survived the typhoon. Four generations after the typhoon struck, achromatopsia began to appear frequently in the population. Figure 16B shows how inbreeding in a population can produce homozygous recessive genotypes. Geneticists explain that the high frequency of this rare genetic disorder on Pingelap is consistent with a large degree of intermarriage among relatives, or inbreeding, following the typhoon.

Mwanenised, a male survivor of the typhoon, was a carrier, or a heterozygote, for the achromatopsia allele. He had ten children, which was a large proportion of the first generation of Pingelapese after the typhoon. Geneticists would predict that, on average, 50% of his children would have been homozygous for the normal allele, and 50% would have been heterozygous carriers of the color blind allele. Thus, none of Mwanenised's children would have been color blind (Fig. 16B). However, intermarriage of successive generations would unavoidably have brought together males and females who were carriers for the achromatopsia allele. Thus, in subsequent generations the homozygous recessive genotype, and complete color blindness, did appear due to inbreeding within a small population (Fig. 16B). On Pingelap, an increase in color blindness was observed by the fourth generation.

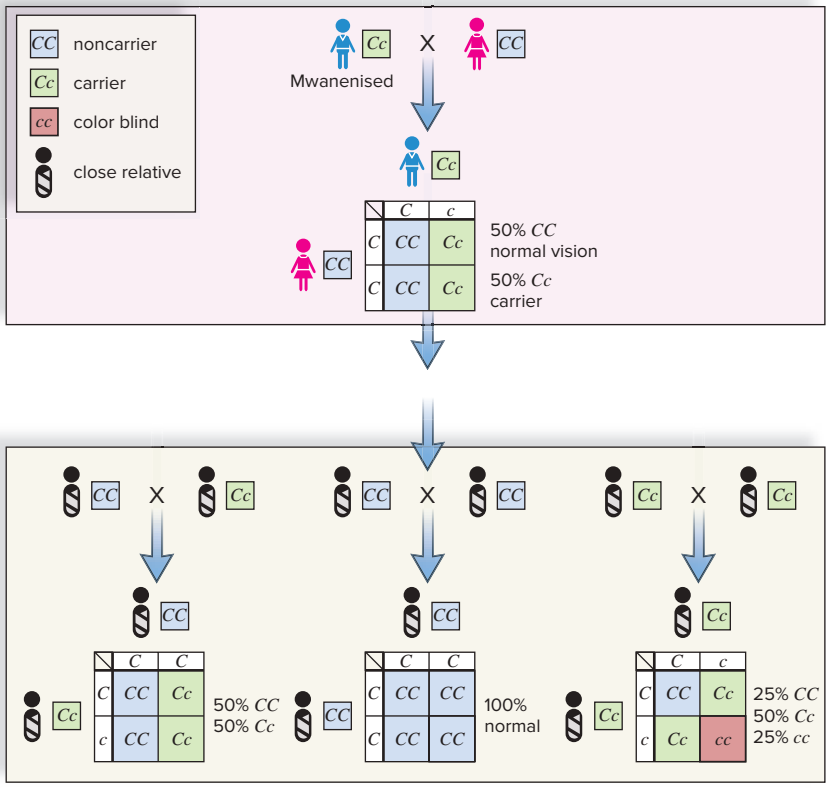
The effect of inbreeding can be long term, especially if a population remains relatively small and isolated. Today, 1 in 12 Pingelapese is afflicted with achromatopsia. On Pingelap, it is 3,000 times more frequent than in the United States, where it occurs in approximately 1/40,000 births.

Figure 16A Achromatopsia. Complete achromatopsia is a recessive genetic disorder that causes complete color blindness. People with complete achromatopsia see no color, only shades of gray. This image shows how a person with this form of color blindness would see the world.

(both): Ingram Publishing/SuperStock

(*Continued*)

ISTUDY



Questions to Consider

1. What might have happened to the color blind allele in the Pingelapese population following the typhoon if Mwanenised had had no children? Only five children?
2. Would you predict that the Pingelapese population is in Hardy-Weinberg equilibrium? How would you measure this?
3. Has the Pingelapese population evolved? Explain. (Hint: Does inbreeding cause a change in allele frequencies?)

Figure 16B Results of inbreeding in a small population. On the island of Pingelap, a high level of incidental inbreeding in a small population has resulted in a higher-than-expected occurrence of complete achromatopsia, a recessive genetic disorder.

affecting the phenotypes in a population. Inbreeding, such as that in the Pingelapese population described in the Nature of Science feature, is an example of nonrandom mating.

In a randomly mating population, the alleles in the gene pool assort at random. When mating is nonrandom, gametes, and thus alleles, assort according to mating behavior. For example, another type of nonrandom mating, called **assortative mating,** occurs when individuals choose a mate with a preferred trait, such as a particular coat color, feather length, or body size. Assortative mating brings together alleles for these traits more often than would happen by chance. The result is that certain genotypes are more frequent than others.

Although nonrandom mating does not, in itself, cause a population to evolve, it can play an important role in the evolution of a population. For example, in the Pingelapese population of the island of Pingelap, nonrandom mating resulted in an increased frequency of color blindness.

Natural Selection

A population in Hardy-Weinberg equilibrium has phenotypes that are equally likely to survive and reproduce. One genotype does not have an advantage over another. But in nature some phenotypes do have a reproductive advantage. Individuals who have an advantageous phenotype often pass on the allele for this trait to their offspring. Over time, selection for this advantageous trait increases the frequency of the alleles associated with it, while other alleles decrease.

Natural selection is the foundation of Darwin's theory of evolution. In the next section, we discuss in detail how natural selection works within populations.

Check Your Progress 16.1

1. List the five conditions necessary for Hardy-Weinberg equilibrium and describe what happens to allele frequencies in a population if these conditions are not met.
2. Estimate the equilibrium genotype frequencies from a population with allele frequencies $p = 0.10$, $q = 0.90$.

16.2 Natural Selection

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare stabilizing, directional, and disruptive selection.
2. Determine the type of natural selection operating on a trait by the change in shape of a phenotype distribution.
3. Explain how sexual selection drives adaptation for increased fitness.

In this chapter, we have been investigating natural selection in a genetic context. For many traits, there may be considerable variation in the phenotype due to multiple alleles for a given gene. Also, traits may be controlled by multiple genes. These are called **polygenic traits.** In both cases, if a graph were constructed for a particular trait in a population, it will often show a wide distribution of variation. When this range of variation is exposed to the environment, natural selection favors the variant that is the most adaptive under the present environmental conditions. Over time, the frequency of that variant in the population changes, and so do the genes associated with that variant.

ISTUDY

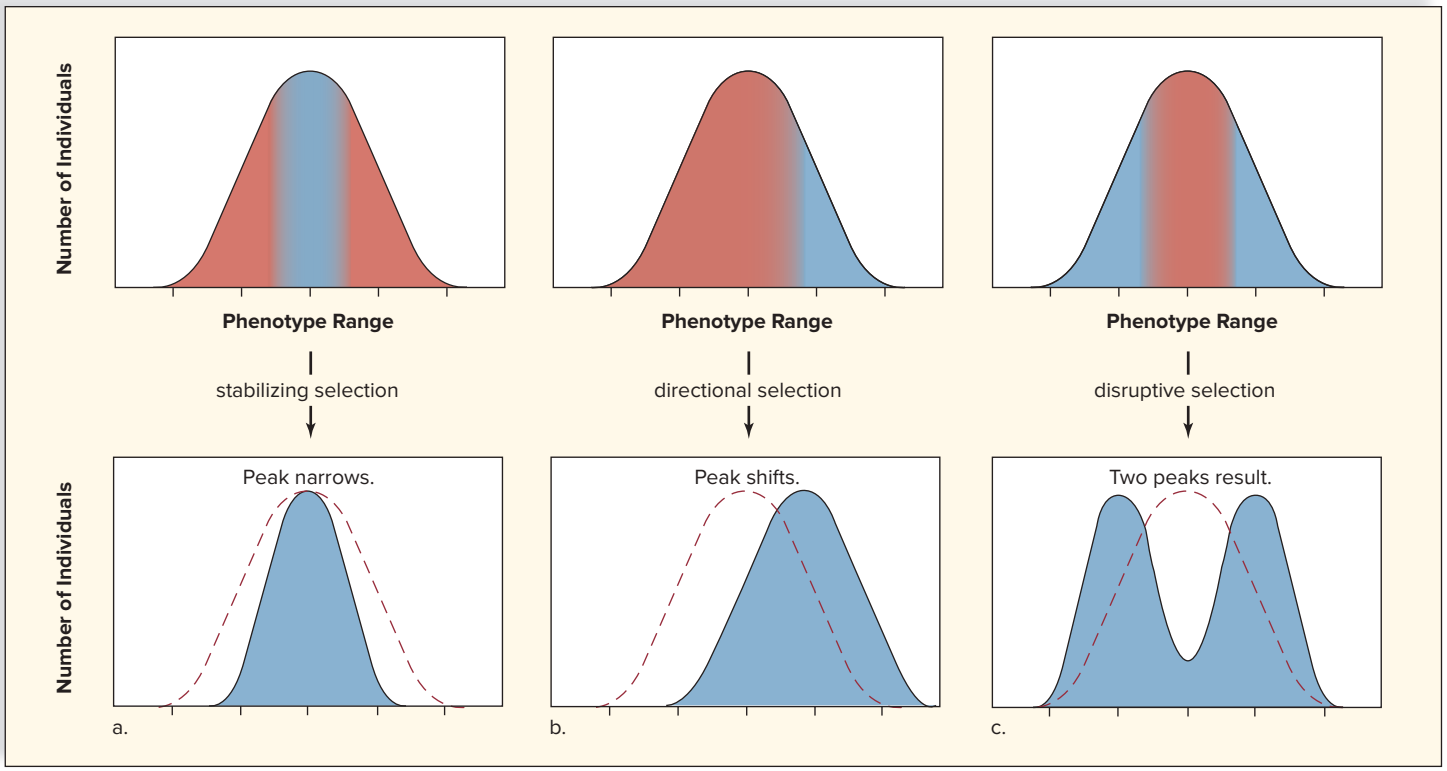


Figure 16.8 Three types of natural selection. Natural selection shifts the average value of a phenotype over time. **a.** During stabilizing selection, the intermediate phenotype increases in frequency; (**b**) during directional selection, an extreme phenotype is favored, which changes the average phenotype value; and (**c**) during disruptive selection, two extreme phenotypes are favored, creating two new average phenotype values, one for each phenotype.

Types of Natural Selection

Investigators have defined three general types of natural selection: stabilizing selection, directional selection, and disruptive selection (Fig. 16.8).

Stabilizing selection occurs when an intermediate phenotype is the most adaptive for the given environmental conditions (Fig. 16.8*a*). With stabilizing selection, extreme phenotypes are selected against, and the intermediate phenotype is favored. One example that is often given is the number of eggs (called the clutch size) produced by wild birds. Clutch size is influenced by genes that determine physiological characteristics, such as the production of yolk, and behavioral characteristics, such as how long the female mates. Birds that produce an intermediate-sized clutch are favored over birds that produce smaller or larger clutches. In the case of a smaller clutch, fewer individuals may make it to the next generation. However, larger clutch sizes are selected against because each individual egg receives fewer nutrients and less parental care.

Human birth weight is another example of stabilizing selection (Fig. 16.9). Over many years, hospital data have shown that human infants born with an intermediate birth weight (3–4 kg) have a better chance of survival than those at either extreme (either much less or much greater than average). When a baby is small, its systems may not be fully functional; when a baby is large, it may have experienced a difficult delivery. Stabilizing selection reduces the variability in birth weight in human populations.

Directional selection occurs when an extreme phenotype is favored, and the distribution curve shifts toward one of the extremes (Fig. 16.8*b*). Over time, directional selection changes the frequency of a phenotype within a population. Such a shift can occur when a population is adapting to a changing environment.

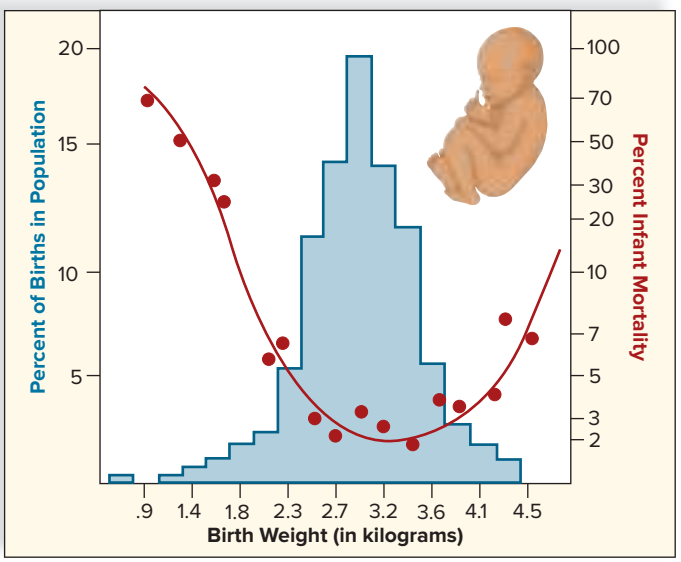


Figure 16.9 Human birth weight. Human birth weight is an example of stabilizing selection. The birth weight (blue) is influenced by the mortality rate (red).

ISTUDY

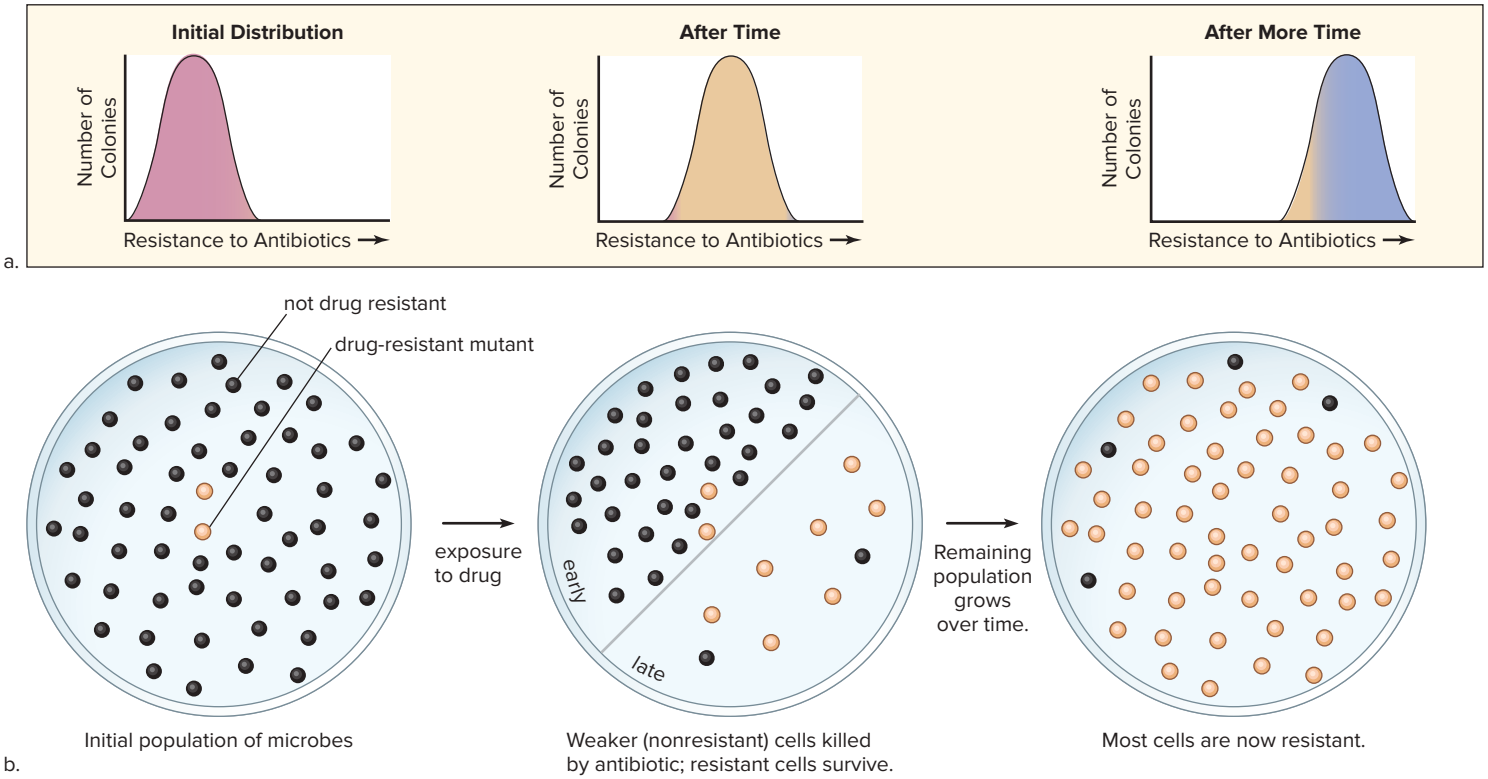


Figure 16.10 Directional selection in a population of bacteria. Directional selection occurs when natural selection favors one extreme phenotype (in this case, antibiotic resistance), resulting in a shift in the distribution curve.

For example, directional selection can be seen in the changes of a bacterial population (Fig. 16.10). Initially, most of the bacteria are susceptible (not resistant) to the antibiotic drug. When the bacteria in the petri dish are exposed to the antibiotic, the nonresistant bacteria die or decline in numbers, mostly before they reproduce. Subsequent generations result in bacteria that are antibiotic resistant and thus have a greater ability to survive and reproduce. This antibiotic resistance allele becomes prevalent in the population and a shift in the distribution curve occurs.

Disruptive selection is found when two or more extreme phenotypes are favored over the intermediate phenotype (see Fig. 16.8*c*). For example, British land snails have a wide habitat range that includes low-vegetation areas (grass fields and hedgerows) and forests. In forested areas, thrushes feed mainly on light-banded snails, and the snails with dark shells become more prevalent. In low-vegetation areas, thrushes feed mainly on snails with dark shells, and light-banded snails become more prevalent. Therefore, these two distinctly different phenotypes are found in the population (Fig. 16.11).

Sexual Selection

Sexual selection refers to adaptive changes in males and females that lead to an increased ability to secure a mate. Sexual selection in males may result in an increased ability to compete with other males for a mate, while females may select a male with the best **fitness** (the ability to produce surviving offspring). In that way, the female increases her own fitness. Sexual selection is often considered a form of natural selection, because it affects fitness.

Female Choice

Females produce few eggs compared to a male's production of sperm, so choosing the best mate becomes important. In a study of satin bowerbirds, two opposing hypotheses regarding female choice were tested.

- *Good genes hypothesis:* Females choose mates on the basis of traits that improve the chance of survival.
- *Runaway hypothesis:* Females choose mates on the basis of traits that improve male appearance. (The term *runaway* pertains to the possibility that, over generations, the trait will become exaggerated in the male until its mating benefit is checked by the trait's unfavorable survival cost.)

As investigators observed the behavior of satin bowerbirds, they discovered that aggressive males were usually chosen as mates by females. It could be that inherited aggressiveness improves the chance of survival, or aggressive males might be good at stealing blue feathers from other males—females prefer blue feathers as bower decorations. Therefore, the data did not clearly support either hypothesis.

Sexual dimorphism—physical differences between males and females of a species with regard to size and other traits—is common in many bird species, meaning that males and females differ in size and other traits. One example is the greater currasow, an endangered species found in the rain forests of Belize (Fig. 16.12). Male currasows have black plumage and a bright yellow-colored knob on their bills. In contrast, females are drab in color and lack the yellow knob. Female choice can explain why male birds are more ornate than females. Consistent with the two hypotheses, it is

ISTUDY

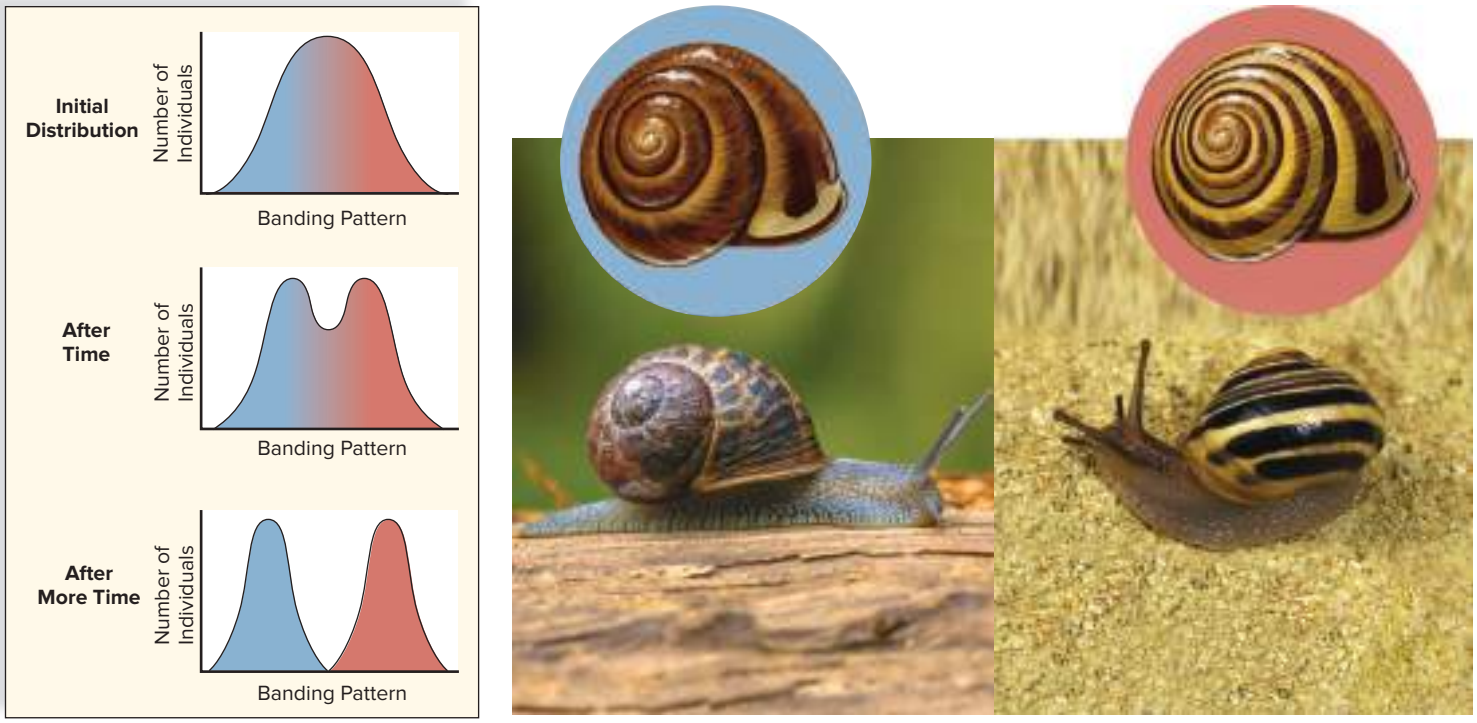


Figure 16.11 Disruptive selection. **a.** Disruptive selection favors two or more extreme phenotypes. **b.** Today, land snails (*Helix aspersa*) comprise mainly two different phenotypes, each adapted to a different habitat. Snails with dark shells are more prevalent in forested areas, whereas light-banded snails have a larger population in areas with low-lying vegetation. (photos): (left): Rudmer Zwerver/Shutterstock; (right): IT Stock Free/Alamy Stock Photo

possible that the yellow knob of the male signifies health and vigor to the female. Or perhaps females' choice of flamboyant males gives their sons an increased chance of being selected by females.

Some investigators have hypothesized that extravagant male features might indicate males that are relatively parasite-free. In barn swallows, females choose males with the longest tails, and investigators have shown that males that are relatively free of parasites have longer tails than those carrying more parasites.

Figure 16.12 Sexual dimorphism. In the great currasow, *Crax rubra,* males have a brilliantly colored knob on their beak that is believed to be a result of sexual selection. The drab females tend to choose males with the most striking examples of this yellow coloring as their mates. Ondrej Prosicky/iStock/Getty Images

Male Competition

Males can father a large number of offspring because they continuously produce sperm in great quantity. We expect males to compete in order to inseminate as many females as possible. **Cost–benefit analyses** have been done to determine whether the *benefit* of access to mating is worth the *cost* of competition among males.

Baboons, a type of Old World monkey, live together in a troop. Males and females have separate **dominance hierarchies,** in which a higher-ranking animal has greater access to resources than does a lower-ranking animal. Dominance is decided by confrontations, resulting in one animal giving way to the other.

Baboons are dimorphic; the males are larger than the females, and they can threaten other members of the troop with their long, sharp canine teeth. One or more males become dominant by frightening the other males. However, the male baboon pays a cost for his dominant position. Being larger means he needs more food, and being willing and able to fight predators means he may get hurt, and so forth. There is often a reproductive benefit to his behavior. Dominant males tend to be the first to monopolize females when they are most fertile. Nevertheless, there may be other ways to father offspring. A male may act as a helper to a female and her offspring; then, the next time she is in estrus, she may mate preferentially with him instead of with a dominant male. Or subordinate males may form a friendship group that opposes a dominant male, making him give up a receptive female.

A **territory** is an area that is defended against competitors. Scientists are able to track an animal in the wild to determine its home range, or territory. **Territoriality** includes the type of defensive behavior needed to defend a territory. Olive baboons travel within a home range, foraging for food each day and sleeping in

ISTUDY

trees at night. Dominant males decide where and when the troop will move. If the troop is threatened, dominant males protect the troop as it retreats and attack intruders when necessary.

Vocalization and displays, rather than outright fighting, may be sufficient to defend a territory (Fig. 16.13). In songbirds, for example, males use singing to announce their willingness to defend a territory. Other males of the species become reluctant to make use of the same area.

Red deer stags (males) on the Scottish island of Rhum compete to be the harem master of a group of hinds (females) that mate only with them. The reproductive group occupies a territory that the harem master defends against other stags. Harem masters first attempt to repel challengers by roaring. If the challenger remains, the two lock antlers and push against one another (Fig. 16.14). If the challenger then withdraws, the master pursues him for a short distance, roaring the whole time. If the challenger wins, he becomes the harem master.

A harem master can father two dozen offspring at most, because he is at the peak of his fighting ability for only a short time. And there is a cost to being a harem master. Stags must be large and powerful in order to fight; therefore, they grow faster and have less body fat. During bad times, they are more likely to die of starvation, and in general, they have shorter lives. Harem master behavior will persist in the population only if its cost (reduction in the potential number of offspring because of a shorter life) is lower than its benefit (increased number of offspring due to harem access).

Check Your Progress 16.2

1. Explain the difference between a population undergoing stabilizing, directional, or disruptive selection.
2. Construct an example of a phenotype distribution curve for a population under directional selection that is increasing in body size.
3. Explain why sexual selection is a form of natural selection.

Figure 16.13 A male olive baboon displaying full threat. In olive baboons, *Papio anubis,* males are larger than females and have enlarged canines. Competition between males establishes a dominance hierarchy for the distribution of resources. Raul Arboleda/AFP/Getty Images

16.3 Maintenance of Diversity

Learning Outcomes

Upon completion of this section, you should be able to

1. List two examples of how diversity is maintained in populations.
2. Describe why heterozygote advantage is a form of stabilizing selection.

Diversity can be maintained in a population through any number of ways. Mutations create new alleles, and sexual reproduction recombines alleles due to meiosis and fertilization. Genetic drift also occurs, particularly in small populations, and the result may be contrary to adaptation to the environment.

a.

b.

Figure 16.14 Competition between male red deer. Male red deer, *Cervus elaphus,* compete for a harem within a particular territory. **a.** Roaring alone may frighten off a challenger, but (**b**) outright fighting may be necessary, and the victor is most likely the stronger of the two animals. (a): Westend61/SuperStock; (b): Jamie Hall/Shutterstock

ISTUDY

Natural Selection

The process of natural selection itself causes imperfect adaptation to the environment. First, it is important to realize that evolution doesn't start from scratch. Just as you can only bake a cake with the ingredients available to you, evolution is constrained by the available diversity. Lightweight titanium bones might benefit birds, but their bones contain calcium and other minerals, the same as other reptiles'. When you mix the ingredients for a cake, you probably follow the same steps taught to you by your elders. Similarly, the processes of development restrict the emergence of novel features. This is why the wing of a bird has the same bones as those of other vertebrate forelimbs.

Imperfections are common because of necessary compromises. The success of humans is attributable to their dexterous hands, but the spine is subject to injury because the vertebrate spine did not originally support the body in an erect position in our ancestors. A feature that evolves has a benefit that outweighs the cost. For example, the benefit of freeing the hands must have outweighed the cost of spinal injuries from assuming an erect posture. We should also consider that sexual selection has a reproductive benefit, but not necessarily an adaptive benefit.

Second, the environment plays a role in maintaining diversity. It's easy to see that disruptive selection in an environment that differs widely can promote polymorphisms within the population (see Fig. 16.11). Then, too, if a population occupies a wide range, as shown in Figure 16.15, it may have several subpopulations designated as subspecies because of recognizable differences.

Figure 16.15 Subspecies help maintain diversity. Each subspecies of rat snake (*Pantherophis obsoleta*) represents a separate population of snakes. Each subspecies has a reservoir of alleles different from another subspecies. Because the populations are adjacent to one another, they may interbreed, and therefore, gene flow may occur among the populations. This interbreeding introduces alleles that may keep each subspecies from fully adapting to its environment. (photos): (*P. obsoleta*): Robert Hamilton/Alamy Stock Photo; (*P. quadrivittata*): Millard H. Sharp/Science Source; (*P. spiloides*): F. Teigler/Blickwinkel/age fotostock; (*P. rossalleni*): Dennis MacDonald/Alamy Stock Photo; (*P. lindheimeri*): Michelle Gilders/age fotostock/SuperStock

ISTUDY

(Subspecies are given a third name in addition to the usual binomial name.) Each subspecies is partially adapted to its own environment and can serve as a reservoir for a different combination of alleles that flow from one group to the next when adjacent subspecies interbreed.

The environment also includes specific selecting agents that help maintain diversity. We have already seen how insectivorous birds can help maintain the frequencies of both light-colored and dark-colored moths, depending on the color of background vegetation. Some predators have a search image that causes them to select the most common phenotype among their prey. This promotes the survival of the rare forms and helps maintain variation. Or a herbivore can oscillate in its preference for food. In Figure 15.10, we observed that the average beak size of the medium ground finch on the Galápagos Islands depended on the available food supply. In times of drought, when only large seeds were available, birds with larger beaks were favored. In this case, we can clearly see that maintenance of variation among a population's members has survival value for the species.

Heterozygote Advantage

Heterozygote advantage occurs when the heterozygote is favored over the two homozygotes. In this way, heterozygote advantage assists the maintenance of genetic, and therefore phenotypic, diversity in future generations.

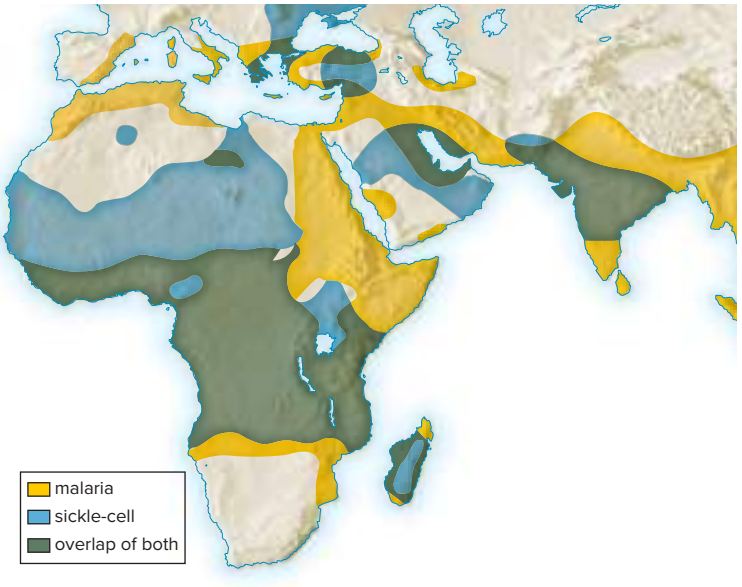
Sickle-Cell Disease

Sickle-cell disease can be a devastating condition. Patients can have severe anemia, physical weakness, poor circulation, impaired mental function, pain and high fever, rheumatism, paralysis, spleen damage, low resistance to disease, and kidney and heart failure. In these individuals, the red blood cells are sickle-shaped and tend to pile up and block flow through tiny capillaries. The condition is due to an abnormal form of hemoglobin (*Hb*), the molecule that carries oxygen in red blood cells. People with sickle-cell disease (Hb^SHb^S) tend to die young and leave few offspring, due to hemorrhaging and organ destruction.

Interestingly, however, geneticists studying the distribution of sickle-cell disease in Africa have found that the recessive allele (Hb^S) has a higher frequency in regions where the disease malaria is also prevalent (blue region in Fig. 16.16). Malaria is caused by a protozoan parasite that lives in and destroys the red blood cells of the normal homozygote (Hb^AHb^A). Individuals with this genotype also have fewer offspring, due to an early death or to debilitation caused by malaria.

Heterozygous individuals (Hb^AHb^S) have an advantage, because they don't die from sickle-cell disease, and they don't die from malaria. The parasite causes any red blood cell it infects to become sickle-shaped. Sickle-shaped red blood cells lose potassium, and this causes the parasite to die. Heterozygote advantage causes the detrimental allele to be maintained in the population. As long as the protozoan that causes malaria is present in the environment, it is advantageous to maintain the recessive allele.

Heterozygote advantage is also an example of stabilizing selection, because the genotype Hb^AHb^S is favored over the two extreme genotypes: Hb^AHb^A and Hb^SHb^S. In the parts of Africa where malaria is common, 1 in 5 individuals is heterozygous (has sickle-cell trait) and survives malaria, while only 1 in 100 is homozygous and dies of sickle-cell disease. In the United States, where malaria is not prevalent, the frequency of the Hb^S allele is declining among African Americans, because the heterozygote has no particular advantage in this country.



Genotype	Phenotype	Result
Hb^AHb^A	Normal	Dies due to malarial infection
Hb^AHb^S	Sickle-cell trait	Lives due to protection from both
Hb^SHb^S	Sickle-cell disease	Dies due to sickle-cell disease

Figure 16.16 Sickle-cell disease. Sickle-cell disease is more prevalent in areas of Africa where malaria is more common.

Cystic Fibrosis

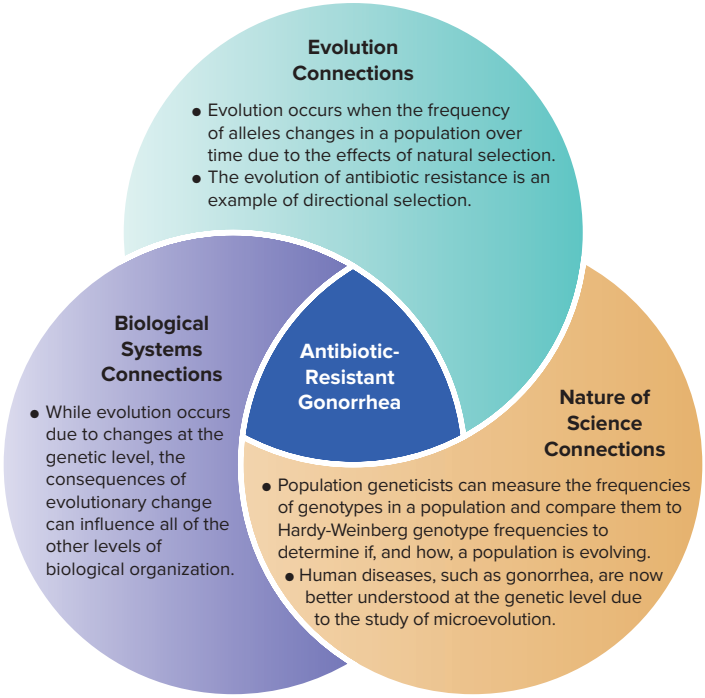
Stabilizing selection is also thought to have influenced the frequency of other alleles. Cystic fibrosis is a debilitating condition that leads to lung infections and digestive difficulties. In this instance, the recessive allele, common among individuals of northwestern European descent, causes the person to have a defective plasma membrane protein. The agent that causes typhoid fever can use the normal version of this protein, but not the defective one, to enter cells. Here again, heterozygote superiority caused the recessive allele to be maintained in the population.

Check Your Progress 16.3

1. Identify ways in which diversity is maintained in a population.
2. Demonstrate how sickle-cell disease is an example of stabilizing selection. Do the same for cystic fibrosis.

ISTUDY

CONNECTING *THE* CONCEPTS



What Did You Learn?

16.1 Genes, Populations, and Evolution

Evolution is the change in the **allele frequency** of a **population** over many generations, not within the individual.

- **Microevolution** pertains to evolutionary change within populations.
- **Population genetics** studies microevolution by measuring the allele frequency over generations. A population in which allele frequencies do not change over time is said to be in **Hardy-Weinberg equilibrium**—a stable, nonevolving state. The Hardy-Weinberg equilibrium is a constancy of **gene-pool** allele frequencies that remains stable from generation to generation if certain conditions are met. The **Hardy-Weinberg principle,** $p^2 + 2pq + q^2$, can measure the genotype frequencies of a nonevolving population. The conditions are no mutations, no gene flow, random mating, no genetic drift, and no selection. Because these conditions are rarely met, a change in allele frequencies is likely.

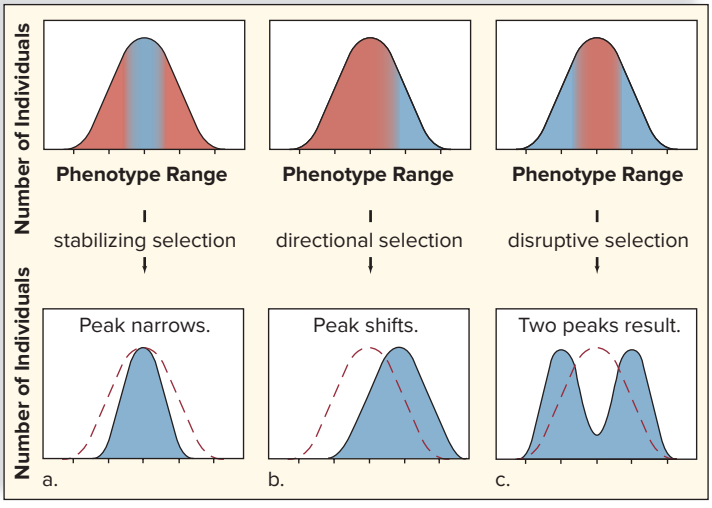
When gene-pool frequencies change, microevolution occurs. Deviations from Hardy-Weinberg equilibrium allow us to detect microevolutionary shifts.

- Mutation occurs when the DNA sequence has changed.
- **Gene flow** is the movement of alleles between populations. If populations are reproductively isolated, then there is a greater chance that gene flow will be restricted.
- **Genetic drift** is when chance events cause allele frequencies to change. Both the **bottleneck effect** and **founder effect** result from the loss of genetic variation within the population. **Inbreeding** can occur as a consequence of genetic drift.
- **Nonrandom mating,** also called **assortative mating,** occurs when individuals are selective about their mates. The manner in which allele frequencies deviate from Hardy-Weinberg equilibrium indicates which of these processes is causing the population to evolve (see Table 16.1).

16.2 Natural Selection

Most of the traits of evolutionary significance are **polygenic,** controlled by many genes. Natural selection favors the most adaptive variant for a given environment. Three types of natural selection occur:

- **stabilizing selection**—the intermediate variation is the most adaptive, as is found in human birth weight;
- **directional selection**—either of the extreme phenotypes is favored, as when antibiotic resistance increases over time; and
- **disruptive selection**—two or more extreme phenotypes are adaptive; the curve forms two peaks, as when British land snails have one of two different banding patterns of shell color.



Sexual selection is about reproductive success, or **fitness.** Males produce many sperm and compete to inseminate females. Females produce few eggs and are selective about their mates.

ISTUDY

- Traits that promote reproductive success, such as **sexual dimorphism,** are shaped by sexual selection. A **cost–benefit analysis** helps a male determine if it is worth competing for mates.
- **Dominance hierarchies** provide dominant males greater reproductive opportunities than lower-ranking males. A **territory** is defended with specific behaviors known as **territoriality.**

Biological differences between the sexes may promote certain mating behaviors because they increase fitness.

16.3 Maintenance of Diversity

Despite constant natural selection, genetic diversity is maintained. Mutations and recombination still occur; gene flow among small populations can introduce new alleles; and natural selection itself sometimes results in variation. In sexually reproducing diploid organisms, the heterozygote acts as a repository for recessive alleles whose frequency is low. In regard to sickle-cell disease, the heterozygote is more fit in areas where malaria occurs. This is known as **heterozygote advantage.**

Thinking Critically About Biology

1. A farmer uses a new pesticide as directed by the manufacturer and loses about 15% of the crop to insects. A farmer in the next state learns of these results, applies three times as much pesticide, and loses only around 3% of the crop to insects. Each farmer follows this pattern for 5 years. At the end of 5 years, the first farmer is still losing about 15% of the crop to insects, but the second farmer is now losing around 40%.
   a. Explain how natural selection may be causing the effect observed at the second farm.
   b. Describe the form of selection that is occurring in the insect population at the second farm.
   c. Which of these insect populations is still in equilibrium? How do you know?
2. You are observing a grouse population in which two feather phenotypes are present in males. One is relatively dark and blends into shadows well, and the other is relatively bright and is more obvious to predators. All of the females are uniformly dark-feathered. Observing the frequency of mating between females and the two types of males, you record the following:

   Matings with dark-feathered males: 13

   Matings with bright-feathered males: 32

   Propose a hypothesis to explain why females apparently prefer bright-feathered males. What selective advantage might there be in choosing a male with alleles that make it more susceptible to predation? What data would help test your hypothesis?
3. If $p^2 = 0.36$, what percentage of the population has the recessive phenotype, assuming a Hardy-Weinberg equilibrium?
4. If 1% of a human population has the recessive phenotype, what percentage has the dominant phenotype, assuming a Hardy-Weinberg equilibrium?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video below:

- Quail Evolution

McGraw Hill

What forms of microevolution are at work in this experiment?

Making It Relevant

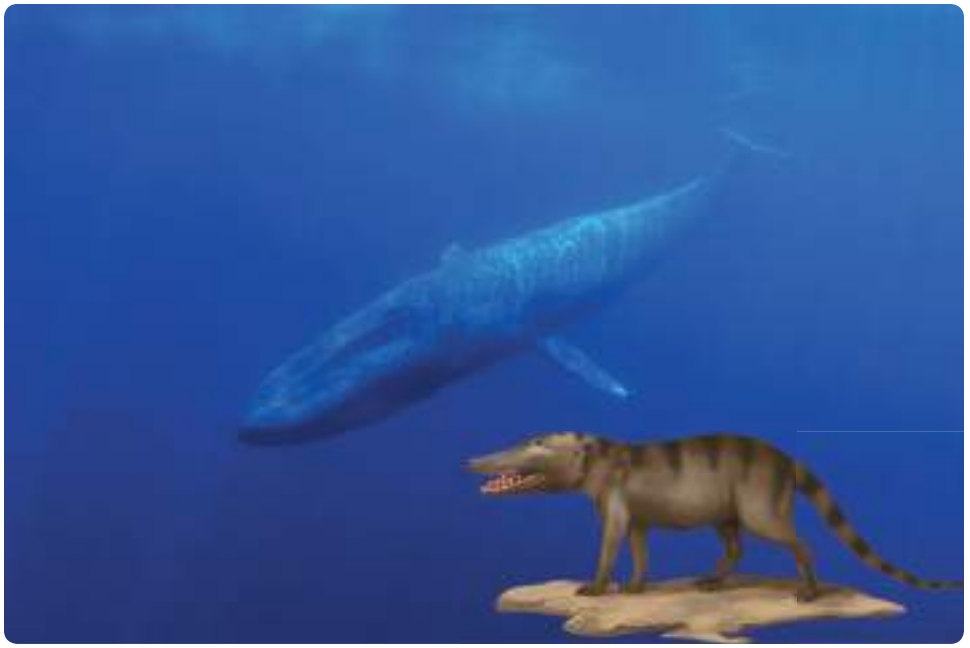
1. How might an understanding of the method by which gonorrhea is evolving resistance to certain antibiotics help researchers develop new forms of antibiotics?
2. How might the use of antibiotics in the food industry (chickens, cows, and pigs) be contributing to the rise in antibiotic-resistant human diseases?
3. Gonorrhea is not the only human disease that is evolving resistance to antibiotics. What other diseases have you heard of in the news that are doing the same, and why has resistance been developing in these diseases??

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

17

Speciation and Macroevolution



(whale): Rich Carey/Shutterstock

Evolution of Whales

Whales are the most majestic creatures of the sea, reaching lengths of almost 100 feet long. Since whales are mammals, their evolutionary history involves ancestors that lived both on land and in the sea.

Scientists have determined that modern whales share a common ancestor with a four-legged animal named *Pakicetus,* a wolf-sized vertebrate that lived in the areas of what is now India and Pakistan around 50 million years ago. *Pakicetus* was not a wolf, but it was definitely a carnivore, and possessed a carnivore-like skull and teeth.

Pakicetus was a terrestrial animal. It lived near lakes and rivers and possessed several physical adaptations—such as where its eyes were located on its head, and its ability to hear underwater—that made it well-adapted to hunt in an aquatic environment.

Over millions of years, the descendants of *Pakicetus* evolved into new species, each with more sophisticated adaptations to the aquatic environment. Legs reduced in size and morphed into fins; in many cases, teeth were replaced with filtering mechanisms; the ability to hear underwater was improved; and other physiological changes allowed these creatures to live permanently in a saltwater environment.

As you read through this chapter, think about the following questions:

1. How do scientists determine whether an organism is a new species?
2. What processes drive the evolution of new species?
3. What can the fossil record tell us about the origin and extinction of species over time?

Chapter Outline

17.1 How New Species Evolve
17.2 Modes of Speciation
17.3 Principles of Macroevolution

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 15.2 What is the role of adaptation in the evolution of a species?

Section 15.3 What is the genetic, fossil, and morphological evidence for evolution?

Section 16.1 What processes drive evolution within populations?

Following *the* Themes

CHAPTER 17 – SPECIATION AND MACROEVOLUTION

Evolution	Macroevolution, or the origin of new species, results from the accumulation of microevolutionary changes over time.
Nature of Science	Evolutionary geneticists research and document the genetic basis of change that can lead to speciation over time.
Biological Systems	New species can arise from specialization within a particular habitat or from adjusting to environmental changes over time.

ISTUDY

17.1 How New Species Evolve

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast microevolution and macroevolution.
2. Identify and compare features of prezygotic and postzygotic reproductive isolation.
3. Explain three ways that species are defined.

In Section 16.1, we examined microevolution, or small-scale evolution within populations. Microevolution is measured as change in allele frequencies in a population over generations. In this chapter, we turn our attention to **macroevolution,** evolution on a large scale. The history of life on Earth is a part of macroevolution. Macroevolution involves **speciation,** or the splitting of one species into two or more species. The same microevolutionary mechanisms that are at play within populations—genetic drift, natural selection, mutation, and migration—are also at play during macroevolution. Thus, microevolution and macroevolution are the result of the same processes, differing only in the scale at which they occur. As discussed in the Evolution feature later in this section, "The Anatomy of Speciation," macroevolution is the result of the accumulation of microevolutionary change that results in the formation of new species.

Species originate, adapt to their environment, and then may become extinct. In fact, much of the biodiversity that has existed on Earth is now extinct. For example, mammals experienced many periods of speciation in the past that resulted in high levels of diversity, but the majority of those species are now extinct. Without the continuous origin and extinction of species, life on Earth would not have the ever-changing history that is found in the fossil record.

Darwin devoted his life to understanding the "mystery of mysteries"—how new species originate. Scientists have learned a lot about this mystery since Darwin's time, including some of the processes that cause species to form. In this chapter, we take a closer look at what constitutes a species and how new species evolve.

What Is a Species?

When you take a walk in a forest, you see a lot of different "types" of plants and animals. If you've had a biology class, you would probably call these different types "species." Although you may not be able to identify each species, you would intuitively recognize many of these organisms as different because of their appearance. Many differences are obvious—for example, clearly an oak tree is a different species from a squirrel. If you look a little closer, you might recognize two different kinds of oak trees, one with large acorns and one with small acorns that has leaves of a slightly different shape. Whether these two oaks are considered different species or variations of the same species depends on which species definition is used.

Scientists define species based on many types of evidence. When presented with two organisms, a **taxonomist,** a scientist who classifies organisms into groups, makes a working hypothesis about whether they are different species based on the evidence, such as their external features. In this manner, each species that is defined is a hypothesis about how the Earth's tree of life is organized.

As with any hypothesis, the addition of new information can result in the redefinition of species. For example, the eighteenth-century scientist Linnaeus (the founder of modern taxonomy) once hypothesized that birds and bats should be together in the same group, because both have wings and fly. We now know that birds and bats are very different organisms on separate branches of the tree of life.

How species are defined is an exciting area of study, because all of the diversity of life on Earth has originated from the evolution of new species. Up until now, we have defined a species as a type of living organism, but in this chapter we characterize species in more depth. First, we examine the major species concepts, or the different ways in which a species can be defined, and then we look at some of the mechanisms by which new species originate.

Morphological Species Concept

Linnaeus identified new species by differences in their appearance, or **morphology.** In the **morphological species concept,** species are distinguished from each other by one or more distinct physical characteristics called **diagnostic traits.** It turns out that Linnaeus was very adept at recognizing species, and many of the morphological species he defined have held up to 200 years of scrutiny.

But the morphological species concept has some disadvantages that Linnaeus could not have predicted. Bacteria and other microorganisms do not have many measurable traits. Also, similarities and differences between organisms can be very subtle and sometimes misleading. Some organisms look so similar that they appear to be the same species. **Cryptic species** are species that look almost identical but are very different in other traits, such as habitat use or courtship behaviors. For example, species of leopard frogs in North America are very difficult to distinguish in appearance, but the males of each species have a unique courtship call (Fig. 17.1).

The morphological species concept is useful for paleontologists as a way to define fossil species based only on traits that are preserved in the fossil record. Unfortunately, because fossils do not provide information about color, the anatomy of soft tissues, or behavioral traits, they are of limited value. However, subtle differences in skeletal features can be used to diagnose the differences between species.

Evolutionary Species Concept

The **evolutionary species concept** relies on identification of certain morphological traits to distinguish one species from another. It was proposed to explain speciation in the fossil record. In addition, the evolutionary species concept, as its name implies, requires that the members of a species share a distinct evolutionary pathway. That is, small, transitional changes in a trait are not used to define new species, because these transitional forms are part of the same evolutionary pathway. However, abrupt changes in traits indicate the evolution of a new species in the fossil record. Consider that the species depicted in

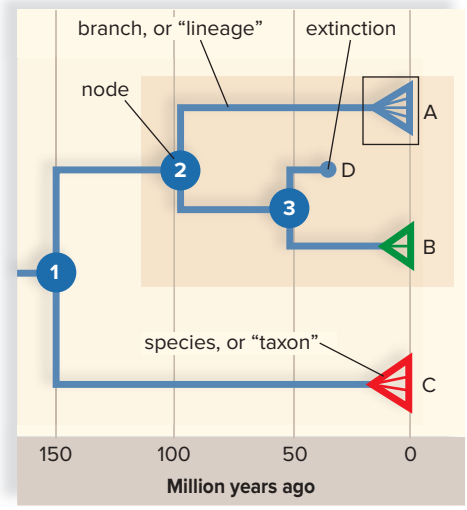
ISTUDY

THEME Evolution

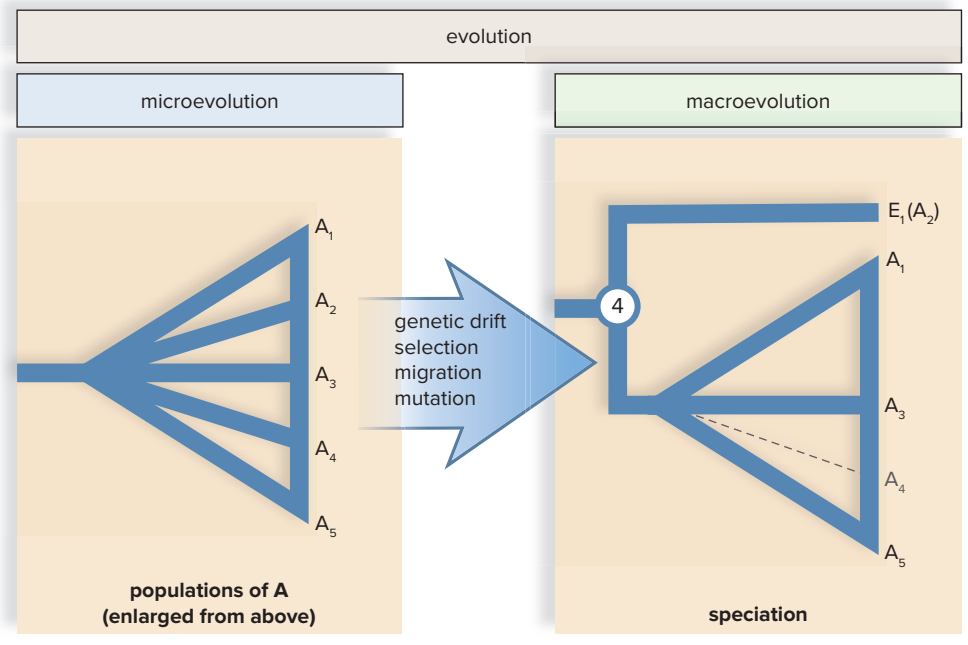
The Anatomy of Speciation

The primary goal of evolutionary biology is to infer the processes of evolution that produce the patterns of diversity seen in nature. More simply put, the evolutionary biologist is interested in explaining biodiversity. At the heart of this endeavor is the phylogeny, the most important tool of the evolutionary biologist, because it represents the history of evolution among organisms. Evolution is considered one of the unifying theories of biology; thus, understanding how to interpret a phylogeny is fundamental to understanding how evolution works. This guide to the anatomy of a phylogeny should help you with the interpretation of a phylogeny.

Macroevolution is about the origin of new species. Darwin's theory proposes that all life on Earth shares a common ancestor. This means that a species originates from evolutionary changes to preexisting species. In this manner, all life on Earth can be envisioned as a large tree of life, with many branches, or species, radiating from it.



a.



b. c.

Figure 17A Anatomy of speciation. A phylogeny, or "family tree," is a hypothesis of evolutionary history of taxa such as species or genera. **a.** A phylogeny has many parts, each of which tells us something about evolutionary relationships among taxa, such as a species or genera. For example, species "A" is more closely related to "B" than to "C," because "A" and "B" share a more recent common ancestor (node 2). **b.** Species "A" is comprised of many populations (A_1–A_5) that are all part of the same branch of the phylogeny. Microevolution occurs at the level of the population. **c.** Microevolution and macroevolution are governed by the same processes—that is, genetic drift, selection, migration, and mutation—but at different scales. Speciation is the result of the accumulation of microevolutionary change in a population over time. Eventually, microevolution can result in the divergence of a population (such as A_2) to form a new species (E_1). Another outcome of evolution is the extinction of populations (A_4) and species (D).

Microevolution is about populations, whereas macroevolution is about the origin of new species (Fig. 17A*b*). However, both are governed by the same processes. Different populations of a single species can accumulate genetic differences as microevolution shapes allele frequencies over time. As more and more genetic differences accumulate, a population may no longer be able to recognize members of another population as potential mates. Under the biological species concept, this would be evidence that the populations had become different species (Fig. 17A*c*). In a phylogeny constructed from DNA nucleotide sequences, for example, these two populations would likely be represented by two different lineages. In this case, both the phylogenetic and biological species concepts would support the origin of a new species.

Phylogenetic Tree Terms

Figure 17A*a* shows the different parts of a phylogeny.

Node: The point at which two branches, or lineages, intersect. The node represents a shared common ancestor for all the species that branch from it. For example, node 1 is the common ancestor of "A," "B," and "C," while node 2 is a common ancestor of "A" and "B."

Root: The point to which all species in the phylogeny can trace their ancestry; the origin of their shared common ancestry.

Extinction: A taxon that is represented in the fossil record, but is now extinct, is represented by a shortened branch correlated with the time at which the extinction occurred.

Monophyletic group (monophyly): A group of species and their common ancestor. For example, all the taxa that share node 2 ("A," "B," and one extinct fossil species) are members of a monophyletic group (light-shaded rectangle), also called a *clade*.

Questions to Consider

1. How could an evolutionary biologist use a phylogeny to find out if two populations have evolved into two different species?
2. Would you expect two different organisms on separate branches of a phylogeny to be able to mate? Why or why not?

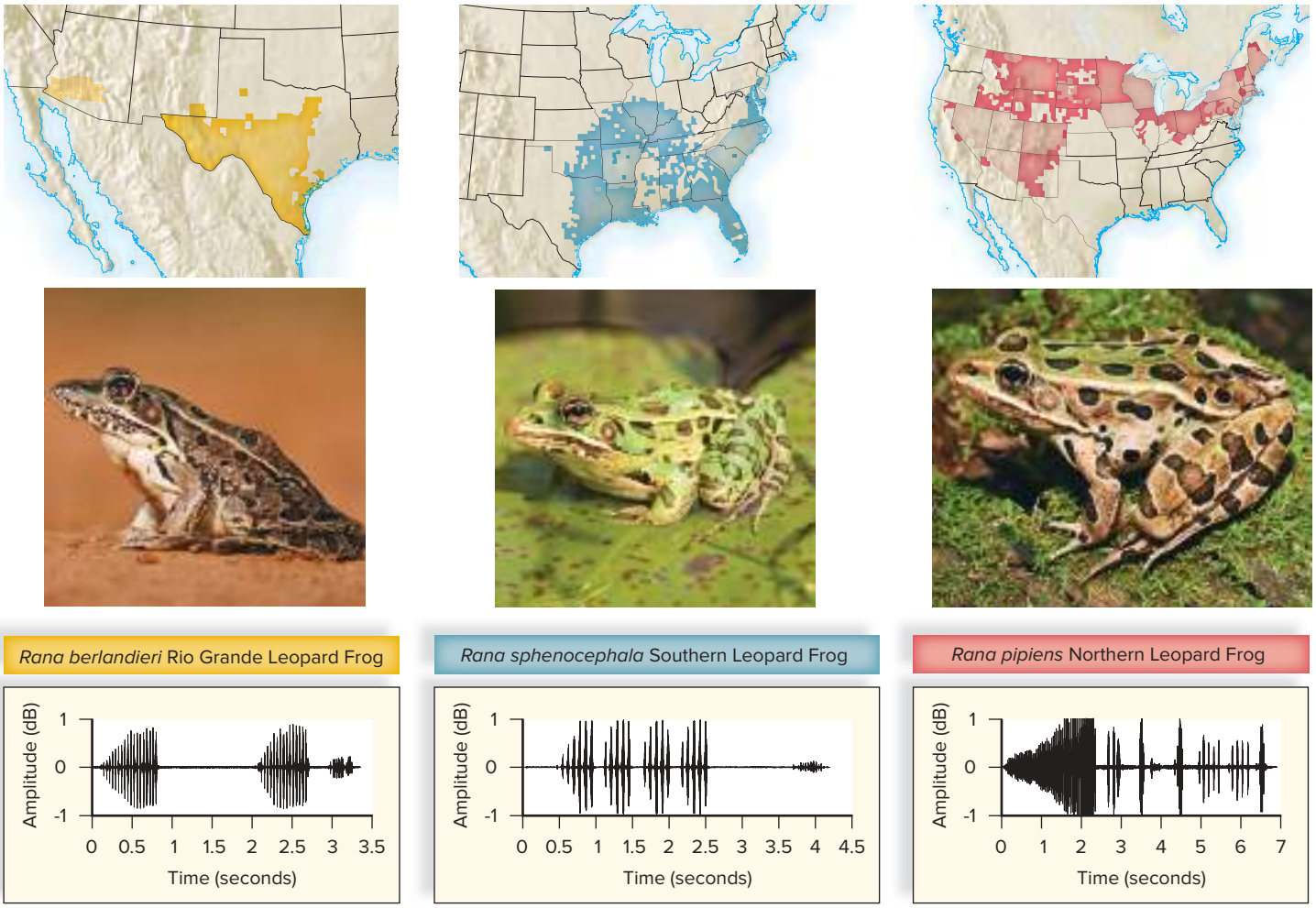


Figure 17.1 Cryptic species of leopard frogs. Leopard frogs are common throughout North America and were once considered to be a single species. Further investigation has revealed that there are at least three species: the Rio Grande, southern, and northern leopard frogs. Although they look similar, they are reproductively isolated, because each has a unique mating call. (photos): (*Rana berlandieri*): Danita Delimont/Alamy Stock Photo; (*Rana sphenocephala*): Robin Chittenden/Alamy Stock Photo; (*Rana pipiens*): Michelle Gilders/Alamy Stock Photo

Figure 17.2 are part of the evolutionary history of *Orcinus orca,* a toothed whale. These species can be recognized individually by differences in diagnostic traits (hindlimbs), but collectively they share an evolutionary pathway distinct from those of other whale species.

Phylogenetic Species Concept

In the **phylogenetic species concept,** an evolutionary "family tree"—or phylogeny—is used to identify species based on a common ancestor—that is, a single ancestor for two or more different groups (see the Evolution feature, "The Anatomy of Speciation"). For you and your cousins, your shared female grandparent is a common ancestor. Similarly, groups of organisms have a common ancestor.

According to the phylogenetic species concept, a species is the smallest set of interbreeding organisms—usually a population—that shares a common ancestor. A phylogeny, a branch that contains all the descendants of a common ancestor, is said to be **monophyletic.** Monophyly is the main criterion for defining species in the phylogenetic species concept.

One advantage of the phylogenetic species concept is that it does not rely only on morphological traits to define a species. The nucleotide sequence of a region of an organism's DNA can be compared to identify the individual A, C, G, or T nucleotide differences that are characteristic of a species. Thus, species of microorganisms and cryptic species can be identified with the phylogenetic species concept, because traits other than morphology can be diagnostic. One example is the giraffe, which has several regional populations distributed around Africa that are distinguishable only by a unique spot shape (Fig. 17.3). Historically, each population was considered as members of a single species: *Giraffa camelopardalis.* A recent phylogeny based on DNA data shows that each regional population represents a monophyletic group, and thus each population should be recognized as an individual species. Based on DNA analysis, there are now four confirmed giraffe species in Africa (Fig. 17.3).

ISTUDY

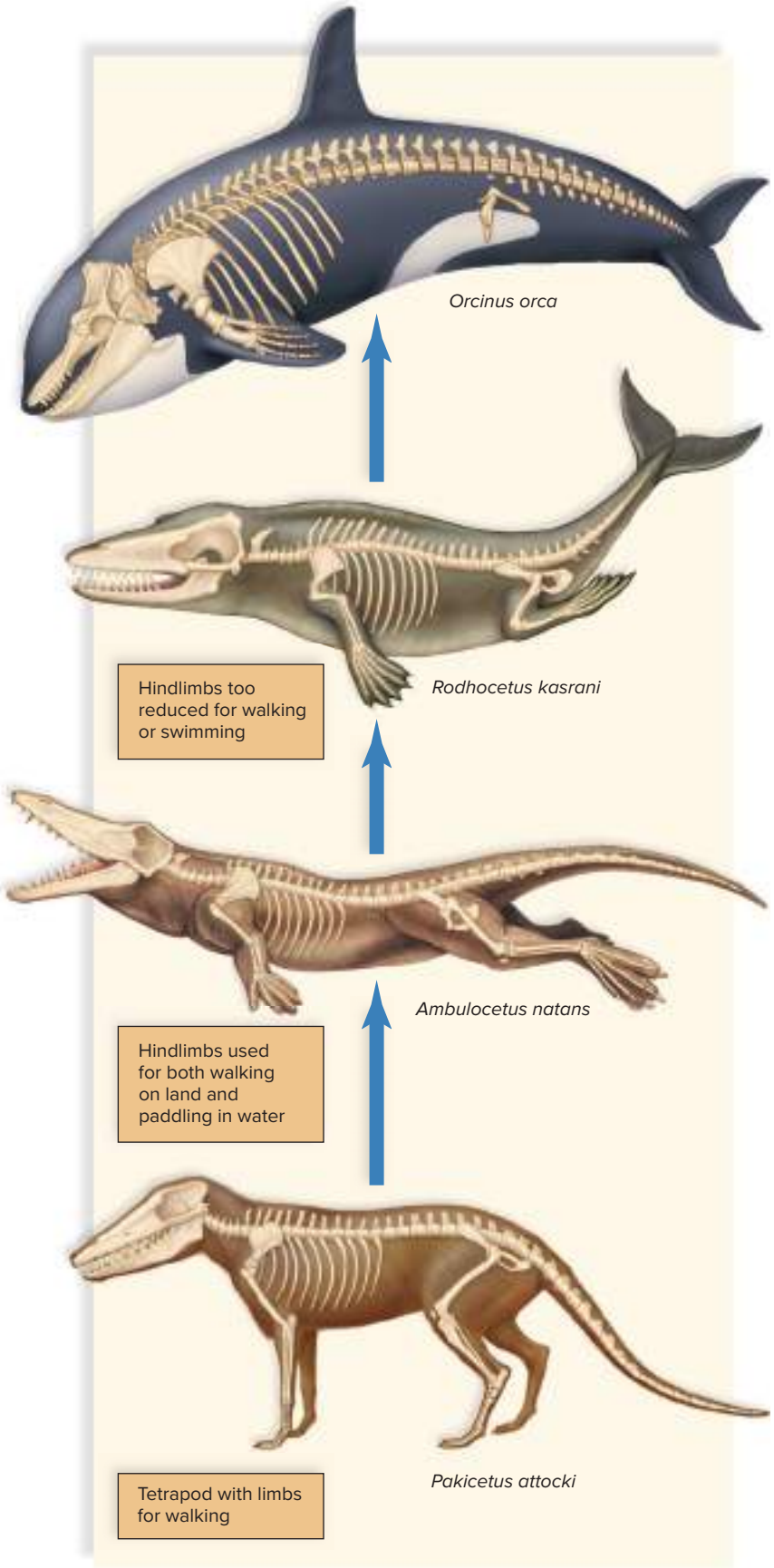


Figure 17.2 Evolutionary species concept. Diagnostic traits can be used to distinguish these species known only from the fossil record. Such traits no doubt would include the anatomy of the limbs.

Biological Species Concept

The **biological species concept** relies primarily on reproductive isolation to identify different species. The most important criterion, according to the biological species concept, is **reproductive isolation**—physiological, behavioral, and genetic processes that inhibit interbreeding. Specifically, if organisms cannot mate and produce offspring in nature, or if their offspring are sterile, they are defined as different species.

Although useful, the biological species concept often cannot be tested in nature, because many potential species do not overlap in their distribution and thus do not have an opportunity to determine whether they are reproductively isolated. Furthermore, the biological species concept cannot be applied to asexually reproducing living organisms or fossils. The benefit of the concept is that, when applicable, it confirms the lack of gene flow—the best indicator that two populations are following independent evolutionary pathways. For example, a group of birds collectively called the flycatchers all look very similar, but they do not reproduce with one another; therefore, they are separate species. Like the leopard frogs (see Fig. 17.1), not only do they live in different habitats but each group has a unique courtship song.

Reproductive Isolating Mechanisms

For two species to remain separate, populations must be reproductively isolated—that is, gene flow must not occur between them. Reproductive barriers that prevent successful reproduction are called isolating mechanisms (Fig. 17.4). Reproductive isolation can occur either before or after fertilization. Reproductive isolation before fertilization is called *prezygotic isolation*; after fertilization: *postzygotic isolation*. A **zygote** is the first cell that results when a sperm fertilizes an egg.

Prezygotic Isolating Mechanisms

Prezygotic isolating mechanisms prevent reproductive attempts or make it unlikely that fertilization will be successful if mating occurs. These isolating mechanisms make it highly unlikely that **hybridization,** or the mating between two species, will occur. Various types of isolating mechanisms can occur between species.

- *Habitat isolation.* When two species occupy different habitats, even within the same geographic range, they are less likely to meet and attempt to reproduce. This is one of the reasons that flycatchers do not mate. In tropical rain forests, many animal species are restricted to a particular level of the forest canopy, and in this way they are isolated from similar species.

ISTUDY

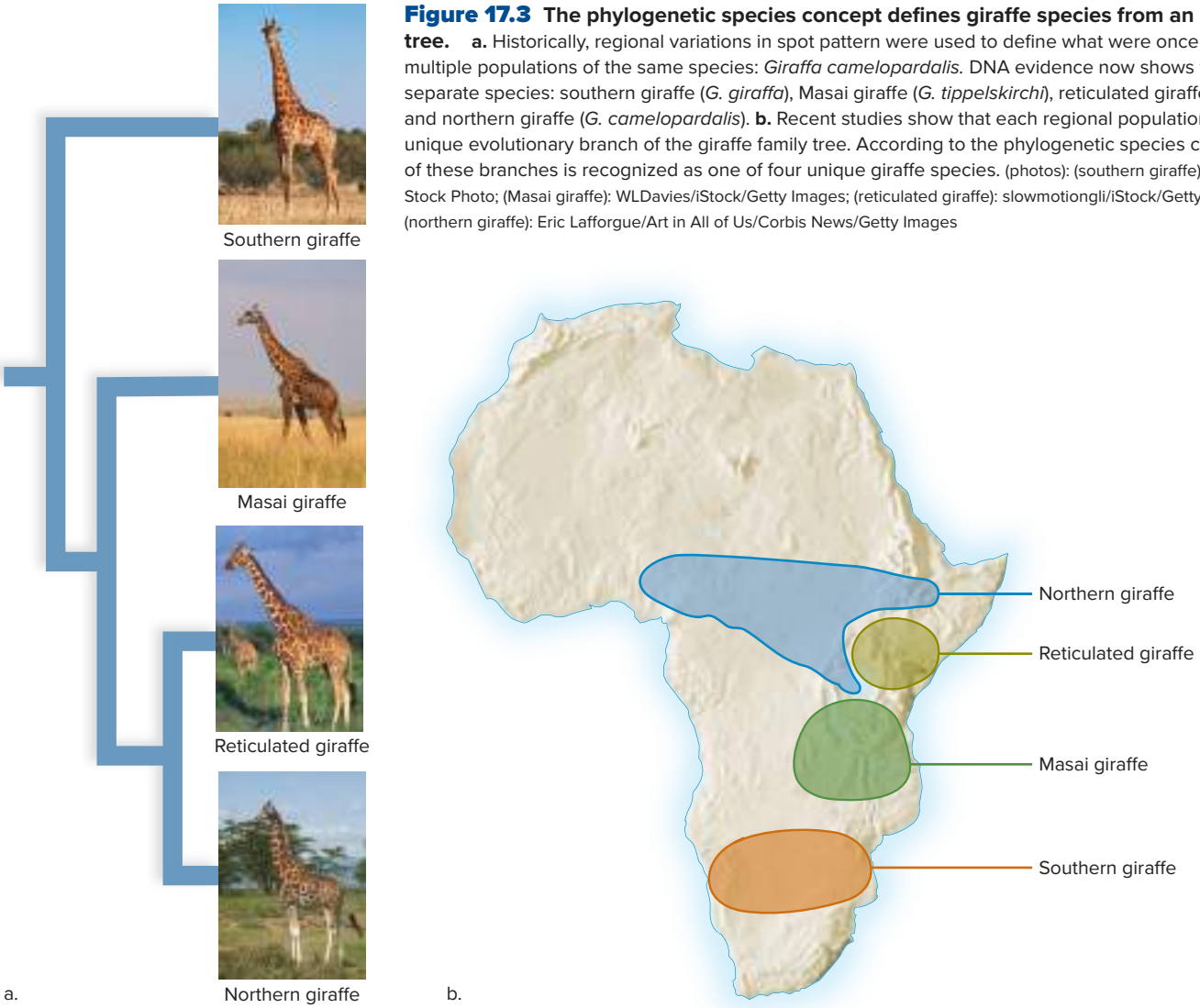
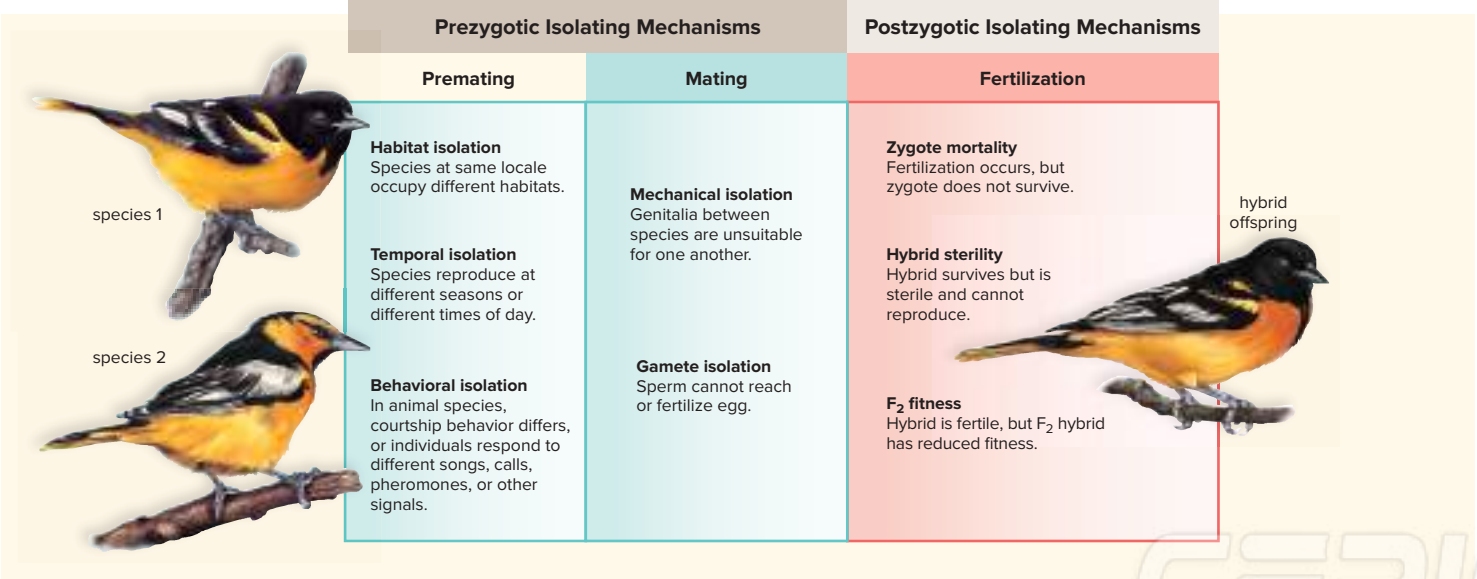


Figure 17.3 The phylogenetic species concept defines giraffe species from an evolutionary tree. **a.** Historically, regional variations in spot pattern were used to define what were once considered multiple populations of the same species: *Giraffa camelopardalis*. DNA evidence now shows there are four separate species: southern giraffe (*G. giraffa*), Masai giraffe (*G. tippelskirchi*), reticulated giraffe (*G. reticulata*), and northern giraffe (*G. camelopardalis*). **b.** Recent studies show that each regional population represents a unique evolutionary branch of the giraffe family tree. According to the phylogenetic species concept, each of these branches is recognized as one of four unique giraffe species. (photos): (southern giraffe): NSP-RF/Alamy Stock Photo; (Masai giraffe): WLDavies/iStock/Getty Images; (reticulated giraffe): slowmotiongli/iStock/Getty Images; (northern giraffe): Eric Lafforgue/Art in All of Us/Corbis News/Getty Images

Figure 17.4 Reproductive barriers. Prezygotic isolating mechanisms prevent mating attempts or a successful outcome, should mating take place. No zygote ever forms. Postzygotic isolating mechanisms prevent the zygote from developing—or should an offspring result, it is not fertile.



ISTUDY

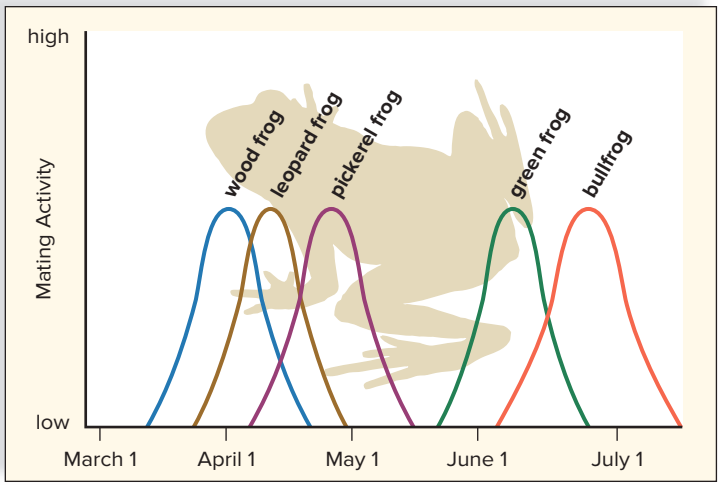


Figure 17.5 Temporal isolation. Five species of frogs of the genus *Rana* are all found at Ithaca, New York. The species remain separate due to breeding peaks at different times of the year, as indicated by this graph.

- *Temporal isolation.* Several related species can live in the same locale, but if each reproduces at a different time of year, they do not attempt to mate. Five species of frogs of the genus *Rana* are all found at Ithaca, New York (Fig. 17.5). The species remain separate because the period of peak mating activity differs, and so do the breeding sites. For example, wood frogs breed in woodland ponds or shallow water, leopard frogs in lowland swamps, and pickerel frogs in streams and ponds on high ground. Having different dispersal times often helps prevent fertilization of the gametes from different species.
- *Behavioral isolation.* Many animal species have courtship patterns that allow males and females to recognize one another. The male blue-footed boobie in Figure 17.6 does a special courtship dance. Male fireflies are recognized by females of their species by the pattern of their flashings. Similarly, female crickets recognize male crickets by their chirping. Many males recognize females of their species by sensing chemical signals called pheromones. For example, female gypsy moths release pheromones that are detected miles away by receptors on the antennae of males.
- *Mechanical isolation.* When animal genitalia or plant floral structures are incompatible, reproduction cannot occur. Inaccessibility of pollen to certain pollinators can prevent cross-fertilization in plants. The genitalia of many insect species are not compatible with those of the members of other species, making mating impossible. For example, male dragonflies have claspers that are suitable for holding only female dragonflies of their own species.
- *Gamete isolation.* Even if the gametes of two different species meet, they may not fuse to become a zygote. In animals, the sperm of one species may not be able to survive in the reproductive tract of another species, or the egg may have receptors only for sperm of its species. In plants, only certain types of pollen grains can germinate, so that sperm successfully reach the egg.

Postzygotic Isolating Mechanisms

Postzygotic isolating mechanisms operate after the formation of a zygote. These mechanisms prevent hybrid offspring from developing and reproducing. If a hybrid is born, it is often infertile (Fig. 17.7) and cannot reproduce. Either way, the genes of the parents are unable to be passed on to successive generations.

Figure 17.6 Prezygotic isolating mechanism. An elaborate courtship display allows the blue-footed boobies of the Galápagos Islands to select a mate. The male lifts up his feet in a ritualized manner that shows off their bright blue color. Henri Leduc/Moment Open/Getty Images

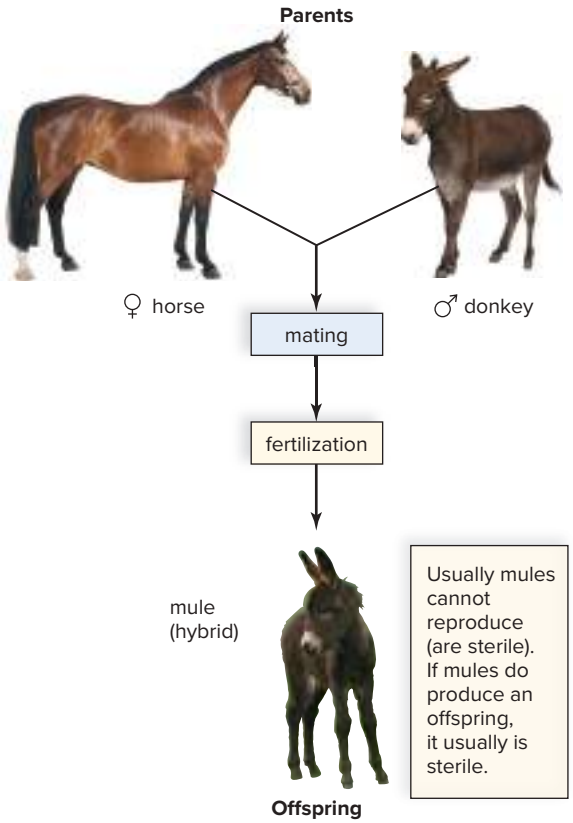


Figure 17.7 Postzygotic isolating mechanism. Mules are infertile. Horse and donkey chromosomes cannot pair to produce gametes. (photos): (horse): Purplequeue/Shutterstock; (donkey): Eric IsselTe/Hemera/360/Getty Images; (mule): Radius Images/Alamy Stock Photo

ISTUDY

- *Hybrid inviability.* A hybrid zygote may die, because it is not viable. A zygote with two different chromosome sets may fail to go through mitosis properly and not develop. The developing embryo may receive incompatible instructions from the maternal and paternal genes, so that it cannot develop properly.
- *Hybrid sterility.* The hybrid zygote may develop into a sterile adult. As is well-known, a cross between a female horse and a male donkey produces a mule. Mules are usually sterile and cannot reproduce (Fig. 17.7). Sterility of hybrids generally results from complications in meiosis that lead to an inability to produce viable gametes. Similarly, a cross between a cabbage and a radish produces offspring that cannot form gametes, most likely because the cabbage chromosomes and the radish chromosomes cannot align during meiosis. On the rare occasion that two hybrids mate and produce offspring, the F_2 hybrids have a reduced fitness.

Check Your Progress 17.1

1. Identify the various factors that can lead to microevolution and macroevolution.
2. List three species concepts and explain the main requirements of each.
3. Explain how frogs that look similar but have different courtship calls can be classified as different species.

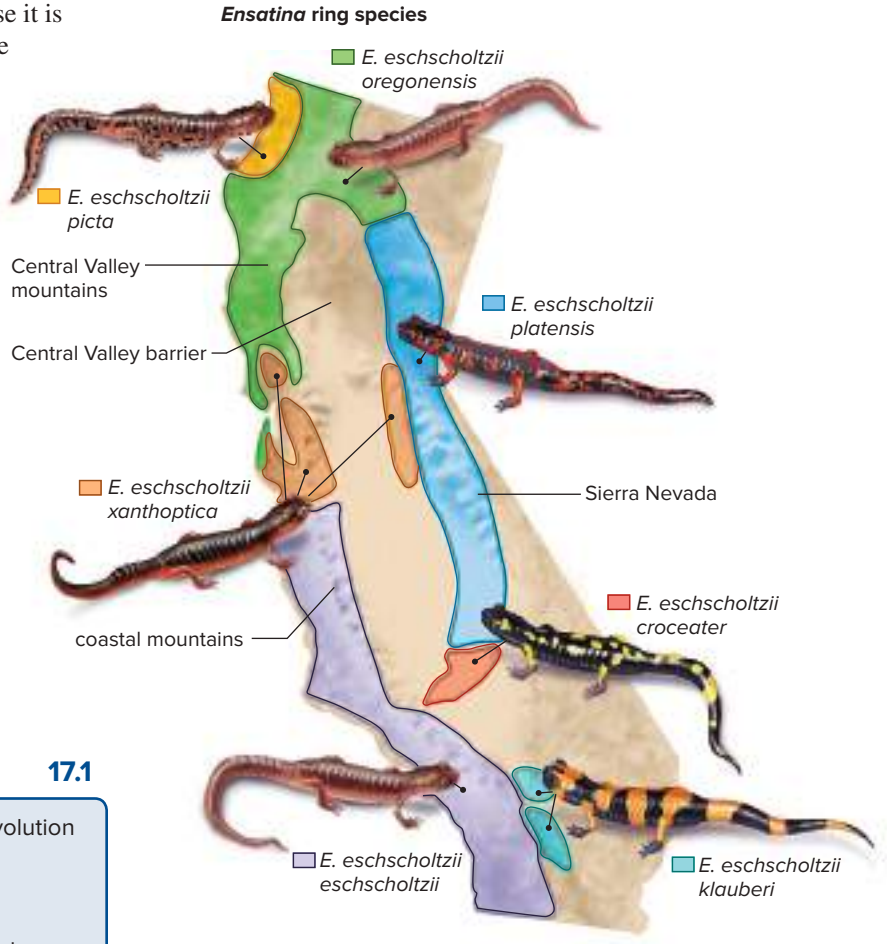


Figure 17.8 Allopatric speciation in progress among *Ensatina* salamanders. The Central Valley of California is reproductively separating a range of populations of *Ensatina eschscholtzii* that are all descended from the same northern ancestral species.

17.2 Modes of Speciation

Learning Outcomes

Upon completion of this section, you should be able to

1. Define two modes of speciation and give examples of each.
2. Identify an example of adaptive radiation.
3. Distinguish between coevolution and convergent evolution.

Speciation is the splitting of one species into two or more species, or the transformation of one species into new species over time. One type of speciation requires populations to be physically isolated from one another, while the other main type does not.

Geographic isolation is helpful, because it allows populations to continue on their own evolutionary path. This can eventually cause the population to be reproductively isolated from other species and from one another. Once reproductive isolation has begun, it can be reinforced by the evolution of more traits that prevent breeding with related species. Geographic isolation can occur repeatedly, so one ancestral species can give rise to several other species.

Allopatric Speciation

In 1942, the evolutionary biologist Ernst Mayr published the book *Systematics and the Origin of Species,* in which he proposed the biological species concept and a process by which speciation could occur. This process, termed **allopatric speciation** (Gk. *allo,* "other"; *patri,* "fatherland"), is the eventual result of populations that have become separated by a geographic or other type of physical barrier. Mayr said that when populations of a species become geographically isolated, microevolutionary processes, such as genetic drift and natural selection, alter the gene pool of each population independently. If the differences between the groups become large enough, reproductive isolation may occur, resulting in the formation of new species.

Examples of Allopatric Speciation

Figure 17.8 features an example of allopatric speciation that has been extensively studied in Southern California. An ancestral population of *Ensatina* salamanders lives in the Pacific Northwest. Members of this ancestral population migrated southward,

ISTUDY

establishing a series of populations. Each population was exposed to unique selective pressures along the coastal mountains and the Sierra Nevada. Due to the presence of the Central Valley of California, gene flow rarely occurs between the eastern populations and the western populations. Genetic differences increased from north to south, resulting in two distinct forms of *Ensatina* salamanders in Southern California that differ dramatically in color and rarely interbreed.

Geographic isolation is even more obvious in other examples. The green iguana of South America is hypothesized to be the common ancestor for both the marine iguana on the Galápagos Islands to the west and the rhinoceros iguana on Hispaniola, an island to the north. Green iguanas are strong swimmers, so by chance a few could have migrated to these islands, where they formed populations that were separate from each other as well as from the parent population in South America. Each population continued on its own evolutionary path as new mutations, genetic drift, and other selection pressures occurred. Eventually, reproductive isolation developed, and the results were three species of iguanas that are reproductively isolated from each other.

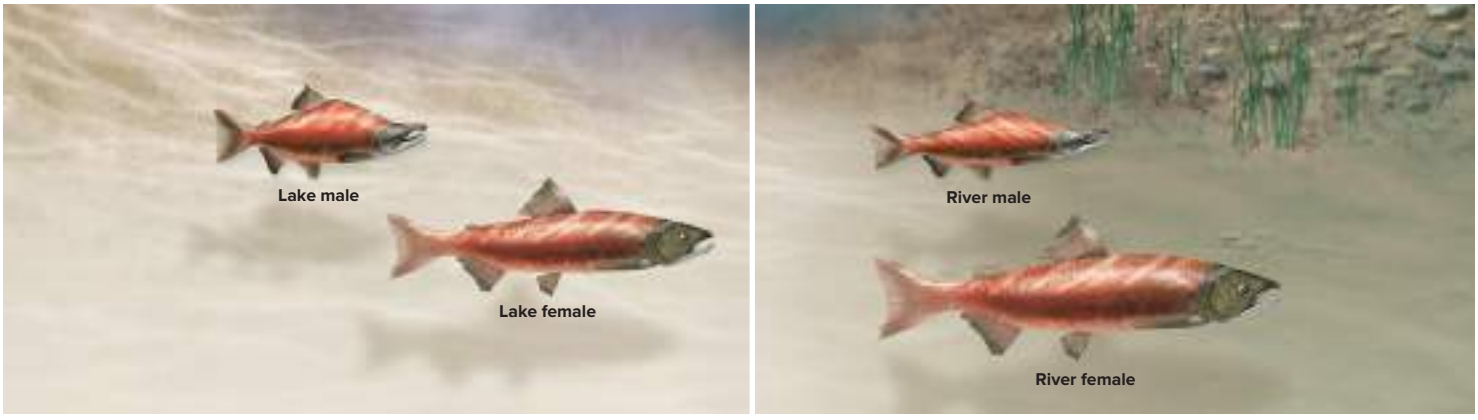
It is interesting to note that the ability of an organism to move about has a large impact on whether allopatric speciation can occur. For example, the oceans of the world are all interconnected, and wide-ranging animals, such as humpback whales, are members of a single species, even though they are seasonally thousands of miles apart. Conversely, the scale of distance and the size of the organism are important, too. Many small organisms, such as parasites, are tightly linked to their hosts. In fact, a species and its parasites can coevolve, because their evolutionary pathways are interdependent.

Another example of allopatric speciation involves sockeye salmon in the state of Washington. In the 1930s and 1940s, hundreds of thousands of sockeye salmon were introduced into Lake Washington. Some colonized an area of the lake near Pleasure Point Beach (Fig. 17.9*a*). Others migrated into the Cedar River (Fig. 17.9*b*). It is possible to tell a Pleasure Point Beach salmon from a Cedar River salmon because of differences in their size and shape due to the demands of reproduction. Males in rivers where the waters are fast-moving tend to be more slender than those at the beach. Sockeye salmon turn sideways into the strong current as part of their mating ritual, and a male with a slender body is better able to perform this maneuver. In contrast, the females in rivers tend to be larger than those at the beach. This larger body helps them dig slightly deeper nests in the gravel beds on the river bottom. Deeper nests are not disturbed by river currents and remain warm enough for eggs to survive and hatch. These differences have resulted in reproductive isolation between these two populations. Not all river salmon remain near the beach their whole lives. In fact, a third of the sockeye males in Pleasure Point Beach grew up in the river population. But the two populations do not interbreed because of the difference in size and shape between the males and females in both populations.

Reinforcement of Reproductive Isolation

As seen in sockeye salmon and other animals, independent evolution of populations can result in reproductive isolation. Another example is seen among *Anolis* lizards, in which males court females by extending a colorful flap of skin, called a dewlap (Fig. 17.10). The dewlap must be seen in order to attract mates. Populations of *Anolis* in a dim forest tend to evolve a light-colored dewlap, while populations in open habitats tend to evolve dark-colored ones. This change in dewlap color causes the populations to be reproductively isolated because females distinguish males of their species by their dewlaps.

As populations become reproductively isolated, postzygotic isolating mechanisms may arise before prezygotic isolating mechanisms. As we have learned, when a horse and a donkey reproduce, the hybrid is not fertile (see Fig. 17.7). Therefore, the process of natural selection tends to favor variations that would prevent the production of hybrids that are unable to reproduce. Indeed, natural selection would favor the continual development of prezygotic



a. Sockeye salmon at Pleasure Point Beach, Lake Washington

b. Sockeye salmon in Cedar River. The river connects with Lake Washington.

Figure 17.9 Allopatric speciation among sockeye salmon. In Lake Washington, salmon that matured (**a**) at Pleasure Point Beach do not reproduce with those that matured (**b**) in the Cedar River. The females from Cedar River are noticeably larger and the males are more slender than those from Pleasure Point Beach, and these shapes help them reproduce in the river.

ISTUDY

Figure 17.10 Reproductive isolation in lizards. The color of the dewlap in *Anolis* lizards acts as a form of reproductive isolation. Dr. Jonathan B. Losos

isolating mechanisms until the two populations were completely reproductively isolated.

The term **reinforcement** is given to the process of natural selection that "reinforces" reproductive isolation. Reinforcement occurs when two populations, formerly of the same species, come back in contact after being isolated. These two species are not able to reproduce when they come in contact, because they no longer recognize each other as mates. An example of reinforcement has been seen in the pied and collared flycatchers of the Czech Republic and Slovakia, where both species occur in close proximity. Only here have the pied flycatchers evolved a different coat color than the collared flycatchers. The difference in color reinforces the choice to mate with their own species.

Sympatric Speciation

Speciation that occurs in the absence of a geographic barrier is termed **sympatric speciation** (Gk. *sym,* "together"; *patri,* "fatherland"). Sympatric speciation is more difficult to observe in nature, because no physical barrier prevents mating between populations, as in allopatric speciation. Some of the best examples of sympatric speciation in nature have involved divergence in diet, microhabitat, or both. In these cases, a new species evolves when a population becomes specialized to live in a different microhabitat.

One example is the midas and arrow cichlid fishes that live in a small lake in Nicaragua. The midas cichlid colonized the lake and occupied its usual rocky, coastal habitat. Over time, a new species of cichlid, the arrow cichlid, evolved from a population of the midas cichlid that adapted to living and feeding in an open water habitat. The partitioning of lake habitats and dietary preferences resulted in a shift in body size, jaw morphology, and tooth size and shape. Now the midas and arrow cichlids are two distinct species.

Sympatric speciation involving **polyploidy** (a chromosome number beyond the diploid [2n] number) is well documented in plants. A polyploid plant can reproduce with itself, but it produces only sterile offspring when mated with 2n individuals, because not all the chromosomes are able to pair during meiosis. Two types of polyploidy are known: autoploidy and alloploidy.

Autoploidy occurs when a diploid plant produces diploid gametes due to nondisjunction during meiosis (see Fig. 10.12). If this diploid gamete fuses with a haploid gamete, a triploid plant results. A triploid (3n) plant is sterile and cannot produce offspring, because the chromosomes cannot pair during meiosis. Humans have found a use for sterile plants, because they produce fruits without seeds. If two diploid gametes fuse, the plant is a tetraploid (4n) and the plant is fertile, as long as it reproduces with another of its own kind. The fruits of polyploid plants are much larger than those of diploid plants. The huge strawberries of today are produced by octaploid (8n) plants.

Alloploidy (Gk. *allo,* "other") occurs when two different but related species of plants hybridize. Hybridization is then followed by a doubling of the chromosomes. For example, the California wildflower *Clarkia concinna* is a diploid plant with 14 chromosomes (seven pairs). A related species, *C. virgata,* is a diploid plant with 10 chromosomes (five pairs). A hybrid of these two species is not fertile, because 7 chromosomes from one plant cannot pair evenly with 5 chromosomes from the other plant (Fig. 17.11).

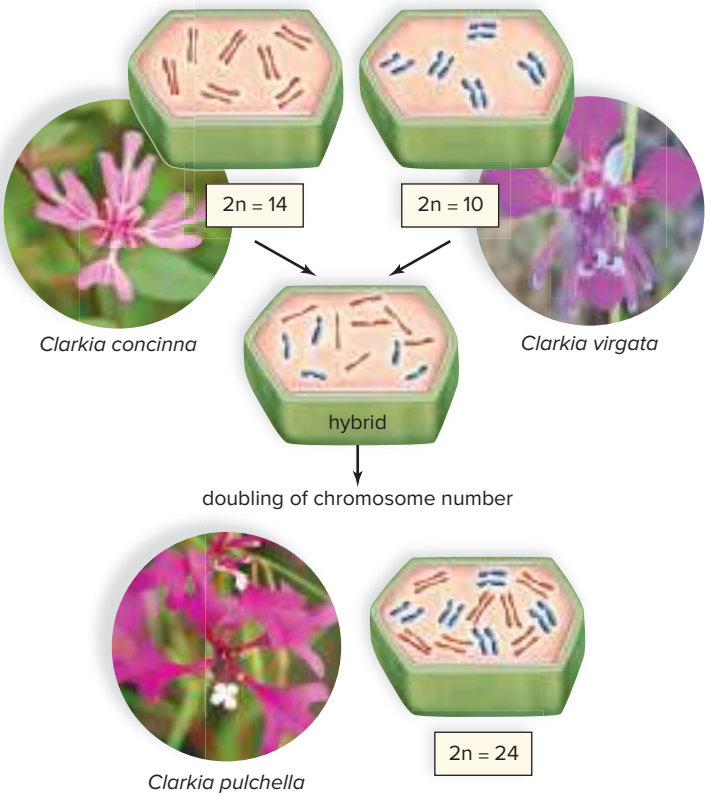


Figure 17.11 Alloploidy produces a new species. Reproduction between two species of *Clarkia* results in a sterile hybrid. Doubling of the chromosome number results in a fertile, third *Clarkia* species that can reproduce with itself only. (photos): (*Clarkia concinna*): Steffen Hauser/botanikfoto/Alamy Stock Photo; (*Clarkia virgata*): ©2016 Christopher Bronny; (*Clarkia pulchella*): age fotostock/Alamy Stock Photo

ISTUDY

However, if the chromosome number doubles in the hybrid, the chromosomes can pair during meiosis, resulting in a fertile plant. The species *C. pulchella* could have arisen this way. Recent molecular data tell us that polyploidy is common in plants and makes a significant contribution to the evolution of new plant species.

Adaptive Radiation

Adaptive radiation is a type of speciation that occurs when a single ancestral species rapidly gives rise to a variety of new species as each adapts to a specific environment. Many instances of adaptive radiation involve sympatric speciation following the removal of a competitor, a predator, or a change in the environment. When competition is reduced, it results in **ecological release.** This is an opportunity for a species to expand its use of resources within habitats that now have less competition. Ecological release provides an opportunity for new species to originate as populations become specialized to newly available microhabitats. Allopatric speciation can also cause a population to undergo adaptive radiation.

Examples of Adaptive Radiation

Darwin proposed that a small population of ancestral finches colonized the Galápagos Islands and their descendants spread out to occupy various niches. Geographic isolation of the various finch populations caused their gene pools to become isolated. Because of natural selection, each population adapted to a particular habitat on its island. In time, the many populations became so genotypically different that now, when by chance they reside on the same island, they do not interbreed and are therefore separate species. During mating, female finches use beak shape to recognize members of the same species, and suitors with the wrong type of beak are rejected.

The 13 species of finches on the Galápagos islands have a range of beak sizes and shapes for feeding on various food sources, including seeds, cactus, flower buds, and insects (Fig. 17.12). Each of the islands in the chain possesses a different variety of foods sources, and the beaks of the finches on the island are adapted to maximize the finch's ability to obtain food.

The genetic mechanisms by which the beak shape changed over time are described in the Nature of Science feature, "Genetic Basis of Beak Shape in Darwin's Finches."

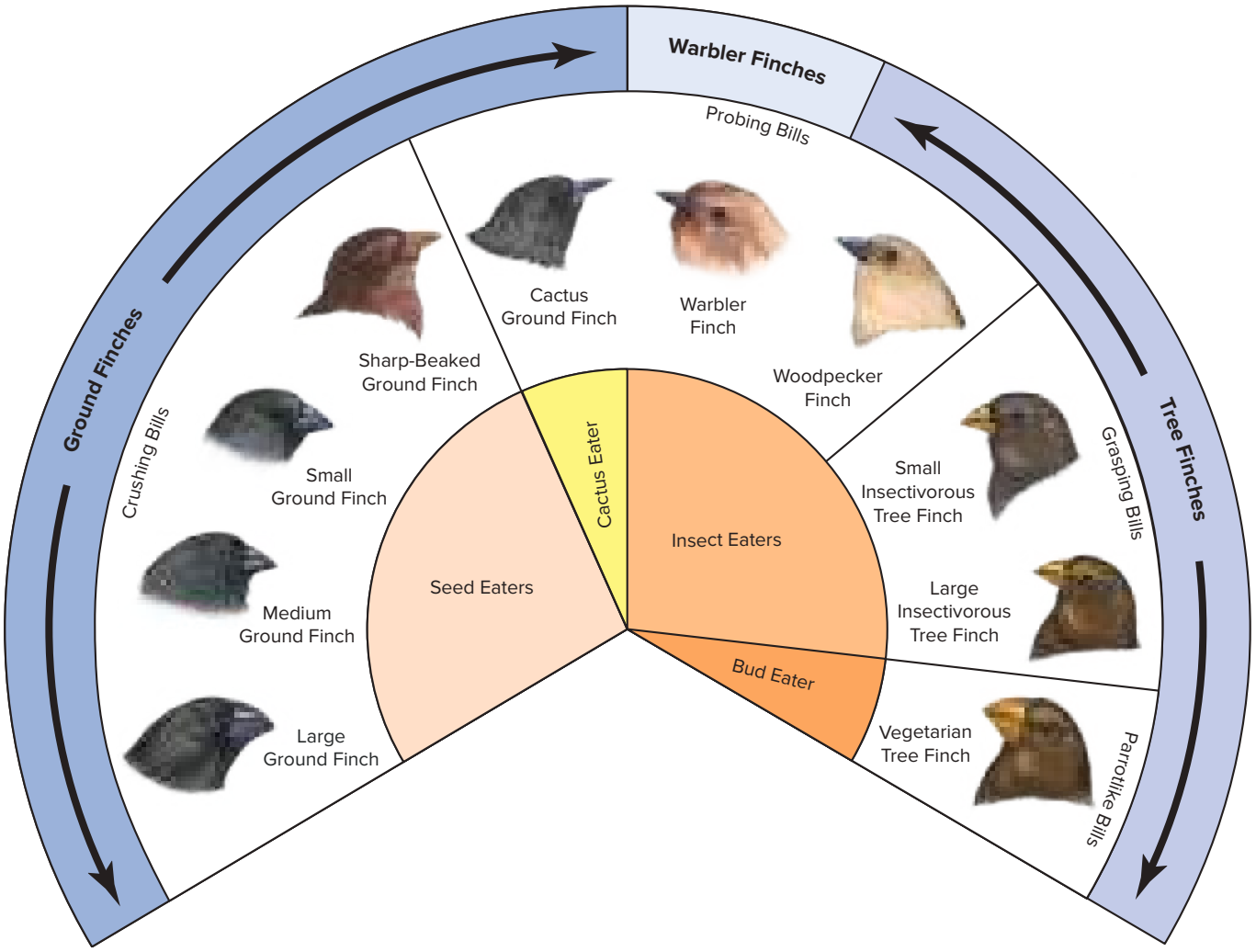


Figure 17.12 Adaptive radiation in Galápagos finches. A single ancestral species of finches colonized the Galápagos Islands and gave rise through adaptive radiation to 13 species of finches, some of which are shown here.

THEME Nature of Science

Genetic Basis of Beak Shape in Darwin's Finches

Darwin's finches are a famous example of how many species originate from a common ancestor. Over time, each species of finch on the Galápagos Islands adapted to a unique way of life, with beak size and shape related to their diets. Ground finches

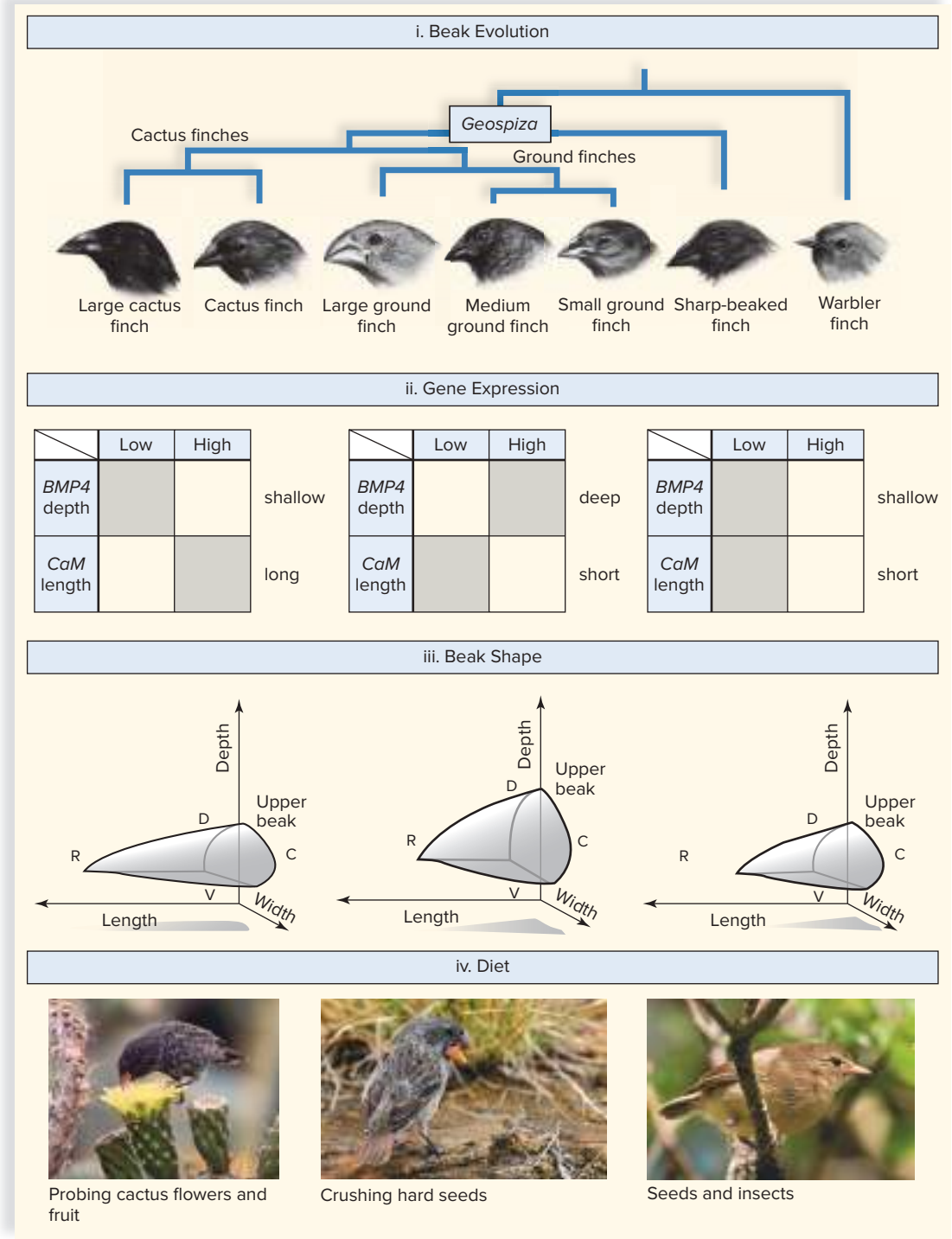


Figure 17B Genetic basis of finch beak size and shape. Bone morphogenetic protein 4 and calmodulin genes regulate the depth and length of the beaks of Darwin's finches. Increases or decreases in gene activity work together to fine-tune beak morphology. (photos): (left): Kevin Schafer/The Image Bank/Getty Images; (center): Alf Jacob Nilsen/Alamy Stock Photo; (right): Ralph Lee Hopkins/National Geographic/Getty Images

have thick, short beaks adept at crushing hard seeds. Cactus finches have long, thin beaks well-suited to probing flowers and the fruit of cacti. The warbler finch feeds on both seeds and insects and has a thin, short beak useful for a mixed diet. Multiple sources of evidence in DNA sequences and morphology support the hypothesis that Darwin's finches are closely related to one another (Fig. 17B).

The differences in beak shape have been recorded by decades of research. Without any additional information, scientists proposed, as did Darwin, that there must be a genetic explanation for the difference in beak shape among species. In 2006, the genes that are responsible for the variation in finch beak shape were discovered. These findings are direct evidence for the mechanism of macroevolution.

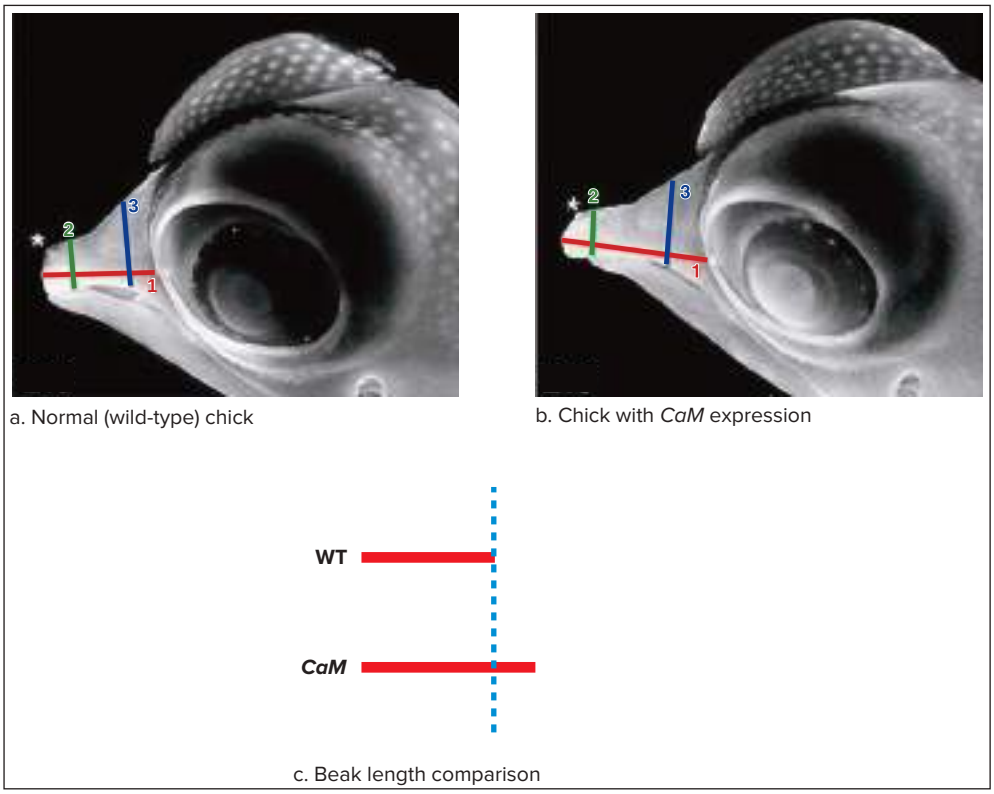
Two genes control beak depth and shape. The gene for bone morphogenetic protein 4 (*BMP4*) determines how deep, or tall, the beak will be. The gene for calmodulin (*CaM*) regulates how long a beak will grow. For example, a high level of *BMP4* creates a deep, wide beak. A high level of *CaM* produces a long beak. In Darwin's finches, a combination of *BMP4* and *CaM* determines overall beak shape (Fig. 17B-ii). The degree of expression of each gene will affect how the beak develops in the embryo.

The cactus finch, for example, has a low level of *BMP4* and a high level of *CaM* expression, which produces a shallow, long beak (Fig. 17B-iii, iv). In contrast, the ground finch has the opposite pattern, with a high level of *BMP4* and a low level of *CaM* expression, producing a short, deep beak (Fig. 17B-iii, iv). One of the most interesting findings of this research is that evolution of beak shape did not require changes to the *BMP4* or the *CaM* gene. An increase or decrease in the expression of these genes during embryo development is enough to change beak shape!

The ability of *BMP4* and *CaM* to affect beak morphology is not limited to Darwin's finches. Variation in beak shape was reproduced in chicken embryos (Fig. 17C). Using molecular tools, finch *BMP4* and *CaM* genes were expressed in the beaks of developing chicken embryos. The expression of *BMP4* caused the chick beaks to deepen, and *CaM* expression caused their beaks to get longer (Fig. 17C*b, c*). Overall, this is strong evidence for the genetic basis of macroevolution.

Questions to Consider

1. How might small, or microevolutionary, changes in *BMP4* and *CaM* in finch populations have resulted in new species of finch?
2. Use Darwin's theory of evolution by natural selection to explain the relationship between finch beak shape and diet on the Galápagos Islands.



a. Normal (wild-type) chick

b. Chick with *CaM* expression

c. Beak length comparison

Figure 17C Expression of *BMP4* and *CaM* in chicks. Chicken embryos were genetically modified to express *CaM* in their beaks during development. **a.** The normal, or wild-type (WT), chick did not have *CaM* expression in the beak during development and the length of the beak was normal. **b.** The *CaM* chick produced an elongated beak. **c.** A side-by-side comparison of the length of WT and *CaM* beaks. Similarly, a separate study of *BMP4* expression in chick embryos produced deeper beaks compared to the WT embryos. (photos): ©Dr. Arkhat Abzhanov, Harvard University, Dept. of OEB

Adaptive radiation has occurred throughout the history of life on Earth when a group of organisms exploited new environments. With the demise of the dinosaurs about 65 million years ago, mammals underwent adaptive radiation as they exploited environments previously occupied by the dinosaurs. Mammals diversified in just 10 million years to include the early representatives of all the mammalian orders, including hoofed mammals (e.g., horses and pigs), aquatic mammals (e.g., whales and seals), primates (e.g., lemurs and monkeys), flying mammals (e.g., bats), and rodents (e.g., mice and squirrels). A changing world presented new environmental habitats and new food sources. Insects fed on flowering plants and, in turn, became food for mammals. Primates lived in trees, where fruits were available. An example of adaptive radiation among plants is the silversword alliance. This group includes plants adapted to both moist and dry environments and even lava fields.

Convergent Evolution

Convergent evolution is said to occur when a biological trait evolves in two unrelated species as a result of exposure to similar environments. Dolphins and tuna do not share a recent common ancestor, yet they both have a dorsal fin for swimming. The ability to swim has evolved in dolphins and tuna independently and, so, has resulted in two different, although similar-looking, solutions to the requirements of moving through water with the greatest amount of efficiency. On both dolphins and fish, the dorsal fin helps keep the animals upright while swimming and prevents them from rolling over. The dolphin's dorsal fin is composed of fibrous connective tissue and blood vessels, whereas the tuna's is composed of bony spines, with skin covering the spines and joining them together.

Traits that evolve convergently in two unrelated lineages because of a response to a similar lifestyle or habitat are said to be analogous—such as the dorsal fin of dolphins and tuna. The opposite of analogous is homologous—traits that are similar because they evolved from a common ancestor (see Fig. 15.15). For example, the wings of butterflies are homologous to the wings of moths, because both are members of the same lineage of insects called the Lepidoptera. All Lepidoptera evolved from a common winged ancestor.

Recent examples of convergent evolution involve adaptive radiations of species in similar, but unconnected, habitats. Lake Malawi and Lake Tanganyika are two African Rift Valley lakes (Fig. 17.13). Within each lake is a set of 200–500 species of cichlid fishes that have adapted to feed on prey in a particular microhabitat of the lake. Each lake has fish that are adapted to feeding on the sandy bottom, as well as species that feed along the rocky shore. Diet specialization has produced a suite of different jaw and tooth shapes and sizes, each adapted to a particular food type. The outcome is an amazing example of convergent evolution. Each lake's assemblage of cichlids has evolved independently of the other, yet if you compare the assemblage in each lake, you find amazing similarities in coloration, body shape, and size and shape of jaws and teeth (Fig. 17.13). Convergent evolution is apparent in the pairing of Lake Malawi and Lake Tanganyika species that have the same features. This is evidence for the evolution of independently derived features that make cichlids adapted to forage in similar habitats.

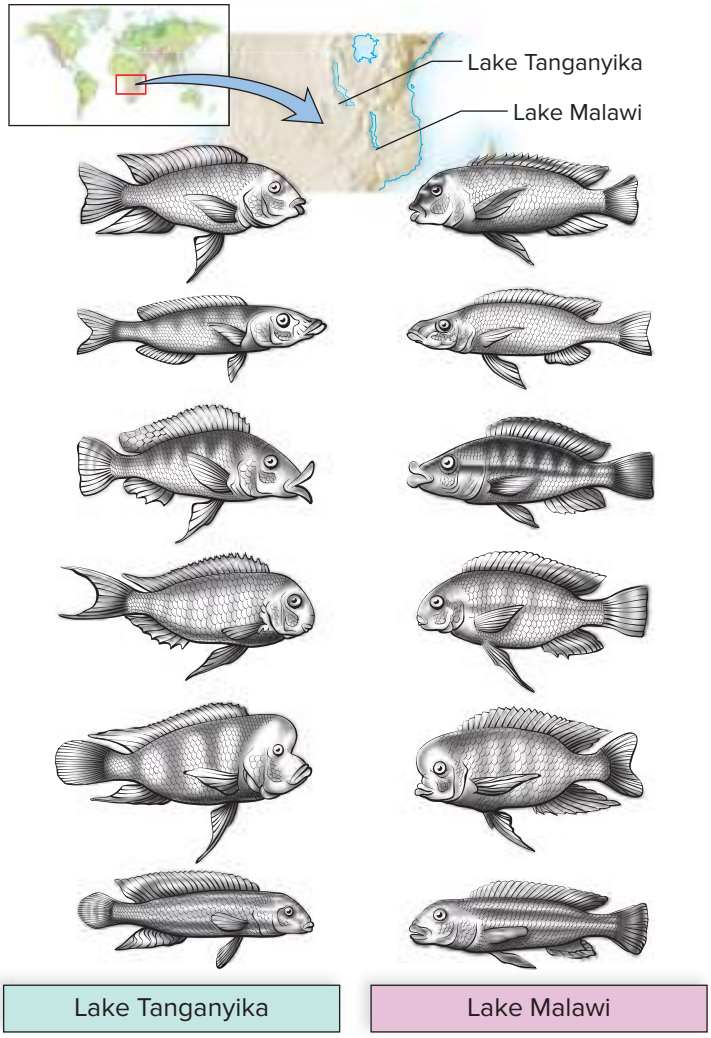


Figure 17.13 Convergent evolution of African lake fish. Cichlids exhibit remarkable evolutionary convergence. In Lake Malawi and Lake Tanganyika, very similar sets of body shapes and sizes have evolved independently of each other, with each type adapted to feed on a different type of food source. Although they appear morphologically similar, all the cichlids from Lake Malawi are more closely related to one another than to any species within Lake Tanganyika.

Check Your Progress 17.2

1. During the last Ice Age, deer mice in Michigan became separated by a large glacial lake and are now two different species. Identify the mode of speciation.
2. List the evidence you would need to show that the five species of big cats—*Panthera leo* (lion), *P. tigris* (tiger), *P. pardus* (leopard), *P. onca* (jaguar), and *P. uncia* (snow leopard)—are an example of an adaptive radiation.
3. Predict the outcome of convergent evolution on the variety of cichlid fish in a newly discovered African Rift Valley lake compared to other lakes with similar microhabitats.

ISTUDY

17.3 Principles of Macroevolution

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between the gradualistic and the punctuated equilibrium models of evolution.
2. Explain how gene expression can influence speciation.
3. Identify how macroevolution is not goal-oriented.

Many evolutionary biologists hypothesize, as Darwin did, that macroevolution occurs gradually. After all, natural selection can only do so much to bring about change in each generation. The gradual evolution of new species is the basis of the *gradualistic model* of evolution. This model proposes that speciation occurs after populations become isolated, with each group continuing slowly on its own evolutionary pathway. The proponents of the gradualistic model often show the history of groups of organisms by drawing the type of diagram shown in Figure 17.14*a*. Note that in this diagram an ancestral species has given rise to two separate species, represented by a slow change in plumage color. The gradualistic model suggests it is difficult to indicate when speciation occurred, because there would be so many transitional links.

After studying the fossil record, some paleontologists tell us that species can appear quite suddenly, and then they remain essentially unchanged during a period of stasis (sameness) until they either undergo extinction or evolve in response to changes in the environment. Based on these findings, they have developed a *punctuated equilibrium model* to explain the fluctuating pace of evolution. This model says that the assembly of species in the fossil record can be explained by periods of equilibrium, or stasis, punctuated (interrupted) by periods of rapid, abrupt speciation, or change. Figure 17.14*b* shows this way of representing the history of evolution over time.

A strong argument can be made that it is not necessary to choose between these two models of evolution, and that both could very well assist us in interpreting the fossil record. In other words, some fossil species may fit one model, and some may fit the other model. In a stable environment, a species may be kept in equilibrium by stabilizing selection for a long period. If the environment changes slowly, a species may be able to adapt gradually. If environmental change is rapid, a new species may arise suddenly before the parent species goes on to extinction. The differences between all possible patterns of evolutionary change are rather subtle, especially when we consider that because geologic time is measured in millions of years, the "sudden" appearance of a new species in the fossil record could actually represent many thousands of years.

Developmental Genes and Macroevolution

Investigators have discovered genes that can bring about radical changes in body shapes and organs. For example, it is now known that the *Pax6* gene is involved in eye formation in all animals, and that homeotic (*Hox*) genes determine the location of repeated structures in all vertebrates (see Section 42.2).

Scientists are working on understanding how evolution could have produced the myriad animals in the history of life. They want to determine how genetic changes brought about such major differences in form. It has been suggested since the time of Darwin that the answer must involve the processes that shape development. In 1917,

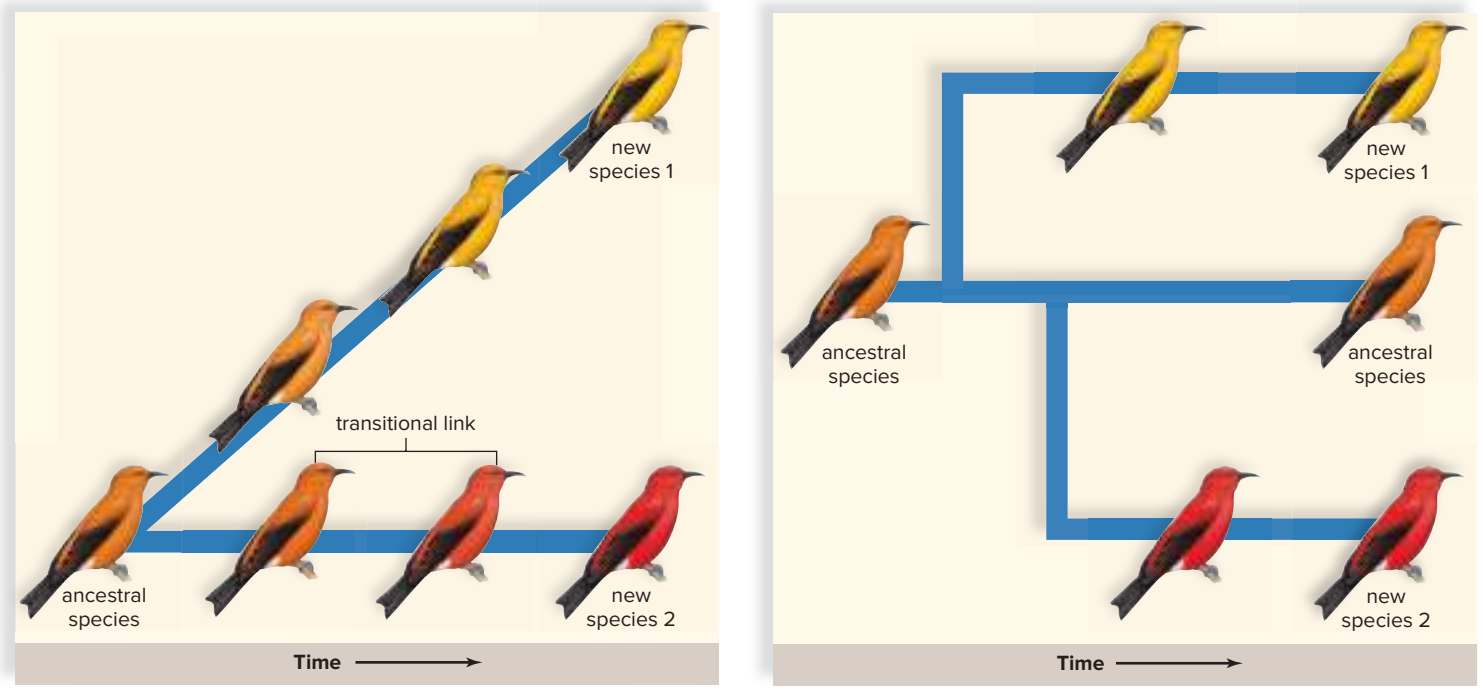


Figure 17.14 Gradualistic and punctuated equilibrium models. **a.** Under the gradualistic model, new species evolve from a series of small changes that occur constantly over time. This process brings about a lot of transitional forms. **b.** Under the punctuated model, new species evolve from a series of abrupt, rapid changes after a period of little or no change. This process would result in different species with few transitional forms.

ISTUDY

D'Arcy Thompson asked us to imagine an ancestor in which all parts are developing at a particular rate. A change in gene expression could stop a developmental process or could continue it beyond its normal time. For instance, if the growth of limb bones were stopped early, the result would be shorter limbs, and if it were extended, the result would be longer limbs compared to those of an ancestor. If the whole period of growth were extended, a larger animal would result, accounting for why some species of horses are so large today.

Using the modern techniques of cloning and manipulating genes, investigators have indeed discovered genes whose differences in expression (the timing and location in the body where proteins they encode are synthesized) can bring about changes in body shapes and organs. This result suggests that these genes must date back to a common ancestor that lived more than 600 MYA, and that despite millions of years of divergent evolution, all animals share the same control switches for development (see Section 28.1).

Development of the Eye

The animal kingdom contains many different types of eyes, which were once thought to require their own sets of genes. Flies, crabs, and other arthropods have compound eyes comprised of hundreds of individual visual units. Humans and other vertebrates have the same camera-type eye with a single lens, as do squids and octopuses. Humans are not closely related to either flies or squids, so it would seem as if all three types of animals evolved "eye" genes separately. Research has shown this is not the case.

In 1994, Walter Gehring and his colleagues at the University of Basel, Switzerland, discovered that a gene called *Pax6* is required for eye formation in all animals (Fig. 17.15).

Mutations in the *Pax6* gene lead to the failure of eye development in both people and mice, and remarkably the mouse *Pax6* gene can cause an eye to develop on the leg of a fruitfly.

Development of Limbs

Wings and arms are very different, but both humans and birds express the *Tbx5* gene in developing limb buds. *Tbx5* encodes a protein that is a transcription factor that turns on the genes needed to make a limb during development. Birds and humans both express *Tbx5* but differ in which genes Tbx5 turns on. Perhaps in an ancestral tetrapod the Tbx5 protein triggered the transcription of only one gene. This could have led to the evolution of limb formation in vertebrates such that changes in the genes regulated by Tbx5 and other transcription factors contributed to the variation in tetrapod limb structure. Therefore, subtle changes in gene control can have profound effects on body shape. This might explain the abundance of variation seen in plant and animal shape and form.

There is also a question of timing. Changing the timing of gene expression, as well as which genes are expressed, can result in a dramatic change in shape.

Development of Overall Shape

Vertebrates have repeating segments, as exemplified by the vertebral column. Changes in the number of segments can lead to changes in overall shape. In general, *Hox* genes control the number and appearance of repeated structures along the main body axes of vertebrates. Shifts in when *Hox* genes are expressed in embryos are responsible for the reason a snake has hundreds of rib-bearing vertebrae and essentially no neck, in contrast to other vertebrates, such as a chick.

Hox genes have been found in all animals. Shifts in the expression of these genes can explain why insects have just six legs but other arthropods, such as crayfish, have ten. In general, the study of *Hox* genes has shown how animal diversity is due to variations in the expression of ancient genes, rather than to wholly new and different genes (see the Evolution feature, "Evolution of the Animal Body Plan," in Section 28.1).

Pelvic-Fin Genes

The three-spined stickleback fish occurs in two forms in North American lakes. In the open waters of a lake, long pelvic spines help protect the stickleback from being eaten by large predators. But on the lake bottom, long pelvic spines are a disadvantage, because dragonfly larvae grab young sticklebacks by their spines and feed on them.

The presence of short spines in bottom-dwelling fish can be traced to a reduction in the development of the pelvic-fin bud in the embryo, and this reduction is due to the altered expression of a gene called *Pitx1*.

Hindlimb reduction has occurred during the evolution of other vertebrates. The hindlimbs became greatly reduced in size as whales and manatees evolved from land-dwelling ancestors into fully aquatic forms (see Fig. 15.13). Similarly, legless lizards have evolved

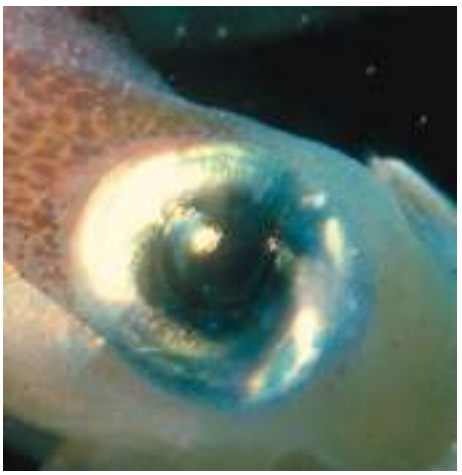


Figure 17.15 *Pax6* gene and eye development. *Pax6* is involved in eye development in a fly (*left*), a human (*center*), and a squid (*right*). (fly eye): Science Source; (human eye): Nick Koudis/Photodisc/Getty Images; (squid eye): Elmer Frederick Fischer/Corbis

ISTUDY

many times. The stickleback study has shown how natural selection can lead to major skeletal changes in a relatively short time.

Human Evolution

The sequencing of genomes has shown that our DNA base sequence is very similar to that of chimpanzees, mice, and indeed all vertebrates. Recent estimates suggest that the human genome has just under 20,000 genes. Based on this knowledge and the work just described, investigators no longer expect to find new genes to account for the evolution of humans. Instead, they predict that differential gene expression, new functions for "old" genes, or both will explain how humans evolved. We discuss the details of human evolution in Section 30.2.

As with all genes, mutations of developmental genes occur by chance, and it is this random process that creates variation. Without variation, evolution cannot occur. Even though mutation is random, natural selection is not a random process. Rather, natural selection acts on the variation that is present, in a way that favors the survival of advantageous traits, in a particular environment at a particular time. This should not be misinterpreted as evidence that evolution is directed or "works" toward an end goal. Evolution is a perpetual process that shapes variation from generation to generation. Where this process leads is unpredictable and depends on a complicated array of external forces. In the following section, we learn that evolution is not directed toward any particular end.

Macroevolution Is Not Goal-Oriented

The evolution of the horse, *Equus,* has been studied since the 1870s, and at first the ancestry of this genus seemed to represent a model for gradual, directed evolution toward the "goal" of the modern horse. Three trends were particularly evident during the evolution of the horse: an increase in overall size, toe reduction, and a change in tooth size and shape.

By now, however, many more fossils have been found, making it easier to tell that the evolutionary history of the horse is complicated by the presence of many lineages that evolved, went extinct, and thus were not on the lineage that led to the modern horse. The evolutionary tree of the horse in Figure 17.16 is an oversimplification, because it is based only on the evidence from the few fossils we have, and there are likely many more fossils yet to be discovered. If the evolution of the horse were directed toward the "goal" of the modern horse, we would expect to see a single branch on this tree with intermediate fossils leading directly from the ancestor to the horse. However, the actual evolutionary tree of *Equus* has many branches, and it will have even more as new fossils are discovered.

Each of the ancestral species of the horse was adapted to its environment. Adaptation occurs only because the members of a population with an advantage are able to have more offspring than other members. Natural selection is opportunistic, not goal-directed.

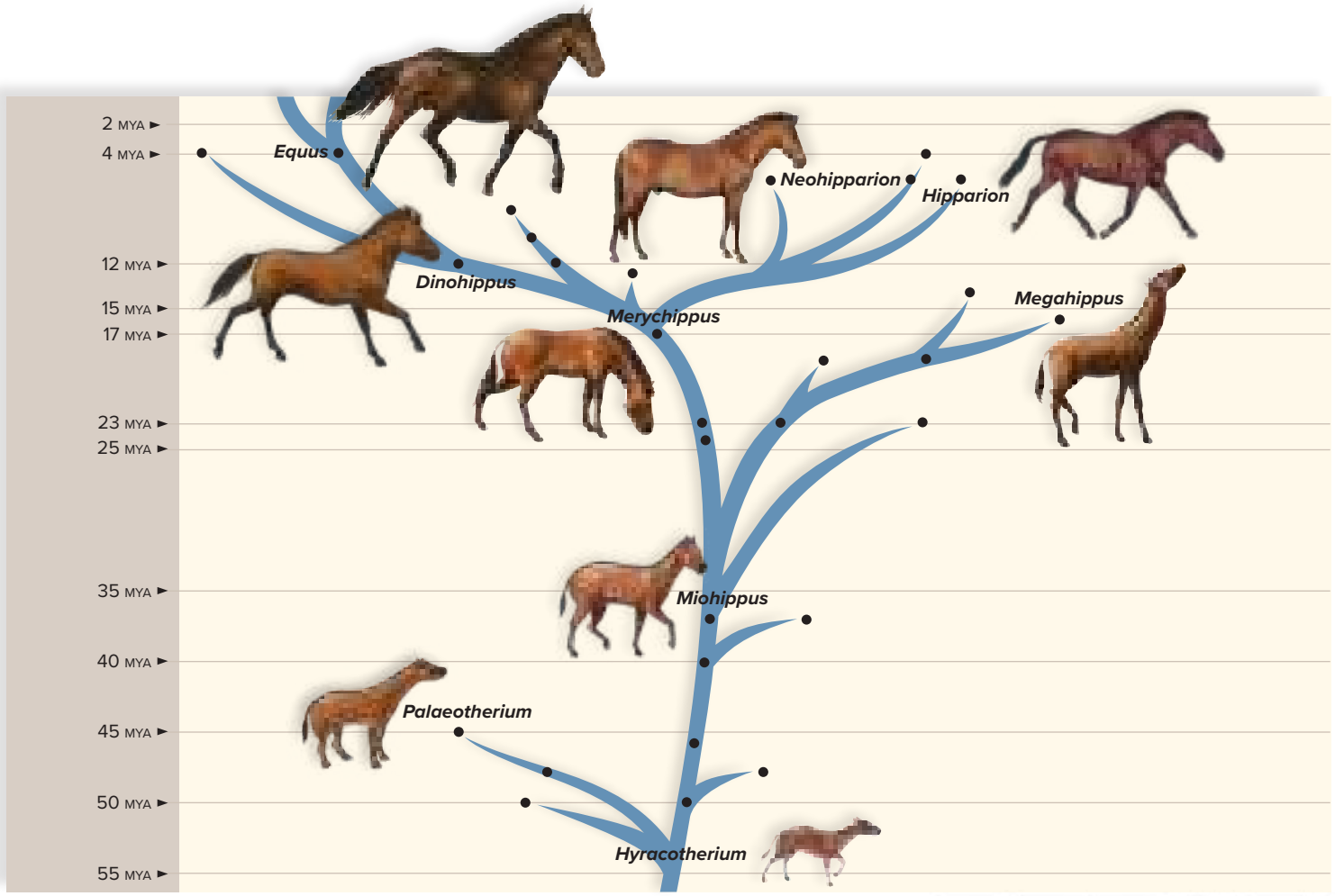


Figure 17.16 Simplified family tree of *Equus*. Every dot on this diagram represents a separate genus in the evolution of the modern horse (*Equus*).

ISTUDY

Fossils named *Hyracotherium* have been designated as the first probable members of the horse family, living about 57 MYA. These animals had a wooded habitat, ate leaves and fruit, and were about the size of a dog. Their short legs and broad feet with several toes would have allowed them to scamper from thicket to thicket to avoid predators. *Hyracotherium* was obviously well-adapted to its environment, because this genus survived for 20 million years.

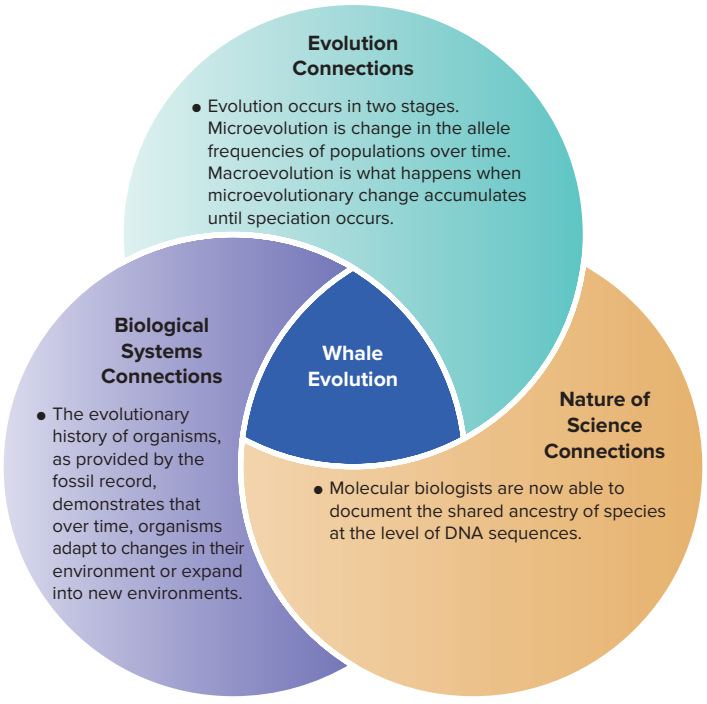
The family tree of *Equus* indicates that speciation, diversification, and extinction are common occurrences during a species' existence. The first adaptive radiation of horses occurred about 35 MYA. The weather was becoming drier, and grasses were evolving. Eating grass requires tougher teeth, and an increase in size and longer legs would have permitted greater speed to escape predators. The second adaptive radiation of horses occurred about 15 MYA and included *Merychippus* as a representative of those groups that were speedy grazers living on the open plain. By 10 MYA, the horse family was quite diversified. Some species were large forest browsers, some were small forest browsers, and others were large plains grazers. Many species had three toes, but some had one strong toe. (The hoof of the modern horse includes only one toe.)

Modern horses evolved about 4 MYA from ancestors with features that were adaptive for living on an open plain, such as large size, long legs, hoofed feet, and strong teeth. The other groups of horses prevalent at the time became extinct, no doubt for complex reasons. Humans have corralled modern horses for various purposes, and this makes it difficult to realize that the traits of a modern horse are adaptive for living in a grassland environment.

Check Your Progress 17.3

1. Explain how the punctuated equilibrium model provides an alternative explanation of the theory of catastrophism proposed by Cuvier (see Section 15.1).
2. Discuss how the study of developmental genes supports the possibility of rapid speciation in the fossil record.
3. Identify the developmental genes that influence macroevolution.

Connecting *the* Concepts



What Did You Learn?

17.1 How New Species Evolve

Large-scale evolutionary changes are known as **macroevolution. Speciation** is a part of macroevolution that involves the formation of new species. **Taxonomists** use one of three species concepts available to define species. Linnaeus, the founder of modern taxonomy, identified species according to **morphology,** or differences in appearance.

- The **morphological species concept** has limitations, because not all organisms have measurable **diagnostic traits. Cryptic species** can look very similar but have other life history or behavioral traits that distinguish them as different species.
- The **evolutionary species concept** requires that each species have its own evolutionary pathway and that each can be recognized by certain morphological traits. The **phylogenetic species concept** uses a family tree to identify a species based on a common ancestor. The branch of the family tree that contains all the descendants of a common ancestor is considered **monophyletic.**
- The **biological species concept** identifies species based on whether two or more populations experience **reproductive isolation** that inhibits inbreeding.

ISTUDY

Prezygotic isolation occurs when reproductive isolation is prior to fertilization, whereas postzygotic isolation occurs after fertilization. A **zygote** is the first cell that forms after fertilization.

- **Prezygotic isolating mechanisms** (habitat, temporal, behavior, mechanical, and gamete isolation) prevent mating from being attempted or prevent fertilization from being successful if mating is attempted. These types of isolating mechanisms prevent **hybridization,** or the mating of two different species.
- **Postzygotic isolating mechanisms** (hybrid inviability, hybrid sterility, and F_2 fitness) prevent hybrid offspring from surviving or reproducing. These mechanisms prevent the genes of the parents from being passed on to the next generation.

17.2 Modes of Speciation

- During **allopatric speciation,** a geographic or physical barrier causes the population to become reproductively isolated. Isolation of populations allows genetic changes to accumulate over time via microevolution. **Reinforcement** occurs when two related populations come back in contact with each other and are unable to reproduce. Various natural selection mechanisms will favor genetic variations that prevent hybridization. Eventually, the ancestral species and the new species no longer breed with one another. As one example, the evolution of a series of salamander subspecies on either side of the Central Valley of California has resulted in two populations of the same species that are unable to successfully reproduce when they come in contact.
- During **sympatric speciation,** a geographic barrier is not required, and speciation is simply a change in genotype that prevents successful reproduction. Sympatric speciation in animals is relatively rare, but it can occur when populations of the same species become specialized on a particular subhabitat and/or food item in the same geographic area. For example, a new species of cichlid, the arrow cichlid, diverged from the midas cichlid in a single lake in Nicaragua because it specialized in feeding in open water. Another example of sympatric speciation is **polyploidy** in plants as a result of nondisjunction during meiosis. **Autoploidy** occurs when a diploid plant produces triploid (3n) offspring. **Alloploidy** is when two different species of plants hybridize, increasing the chromosomal numbers beyond normal for either species.
- **Adaptive radiation** is a type of speciation that occurs when a single ancestral species rapidly gives rise to a radiation of new species as each adapts to a specific environment. Many instances of adaptive radiation involve sympatric speciation following **ecological release.** Ecological release is the reduction of competition that provides opportunity for new species to originate as populations become specialized in newly available microhabitats. Finches on the Galápagos Islands, the honeycreepers of the Hawaiian Islands, and the rise of mammals after the extinction of dinosaurs are examples of adaptive radiation events. Allopatric speciation can also lead to adaptive radiation.
- **Convergent evolution** has occurred when the same biological trait has evolved in two unrelated species as a result of exposure to similar environmental conditions. The wings of birds and bats are examples of convergent evolution. **Analogous** traits are traits found in unrelated lineages that perform a similar function. **Homologous** traits are similar traits found in similar species due to a common ancestor.

17.3 Principles of Macroevolution

Macroevolution is the evolution of new species and higher levels of classification. The fossil record gives us a view of life across millions of years. The hypothesis that species evolve gradually is now being challenged by the hypothesis that speciation can also occur rapidly. In that case, the fossil record could show periods of stasis interrupted by spurts of change—that is, a punctuated equilibrium. Transitional fossils would be expected with gradual change, but not with punctuated equilibrium. It is most likely that species have evolved as the result of a combination of punctuated equilibrium and gradualism. Environmental factors would also have contributed to the process.

Speciation, diversification, and extinction are seen during the evolution of *Equus.* These three processes are commonplace throughout the evolutionary history of a species and illustrate that macroevolution is not goal-directed. The life we see about us represents adaptations to particular environments. Such adaptations have changed in the past and will change in the future.

Thinking Critically About Biology

1. The Hawaiian Islands are located thousands of kilometers from any mainland. Each island arose from the sea bottom and was colonized by plants and animals that drifted in on ocean currents or winds. Each island has a unique environment in which its inhabitants have evolved. Consequently, most of the plant and animal species on the islands do not exist anywhere else in the world. In contrast, on the islands of the Florida Keys, there are no unique or indigenous species. All of the species on those islands also exist on the mainland. Suggest an explanation for these two different patterns of speciation.
2. At a recent school board meeting, a member of the audience stated there is not enough evidence to support how various species have arisen. Due to this lack of evidence, they think that students should be presented with alternative ideas to the theory of evolution. Indicate the scientific lines of evidence employed to support the process of speciation, using the modern horse as an example.
3. Explain the evolutionary mechanisms behind the difference in size between the sockeye salmon of Pleasure Point Beach, Lake Washington, and those of Cedar River, Washington.
4. Over time, enough genetic variations can develop within a population to cause it to undergo speciation. Identify the various mechanisms that will prevent different species from being able to reproduce successfully. Which of these mechanisms is the most influential in keeping species separate?

Making It Relevant to You

1. How do you think the study of development genes in other animals can help benefit the study of human diseases?
2. You are having a discussion in class about evolution and want to use whale evolution as an example of what may have happened in regard to human evolution in the past. What analogies would you make? What evidence would you need to support your claim that humans evolved from species that lived in very different habitats than where they are found today?
3. Given your knowledge of how species evolve, why is it not likely that speciation will occur in the majority of species on the planet as a result of climate change?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

18

The Origin and History of Life

Minakryn Ruslan/iStock/Getty Images

Chapter Outline

18.1 The Origin of Life

18.2 The History of Life

18.3 Geological Factors that Influence Evolution

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.1 What are the basic characteristics that define life?

Section 2.1 What are chemical isotopes?

Section 15.3 How do the fossil and biogeographical records provide evidence for the evolution of new organisms?

The Cretaceous Mass Extinction

The aftermath of an asteroid hitting Earth 66 million years ago was hellish. Computer modeling suggests the impact caused tsunamis, seismic earthquakes, and forest fires nearly worldwide. A new fossil site in North Dakota could contain a real picture of what happened in the very hour after impact. Tanis, as it is called by paleontologists, contains fossils of ancient trees, fish, and animals. Many of the bodies appear broken from the violence of the event, which involved being scooped up and entombed in sediment by massive waves.

Buried with the fossils, and even stuck in the gills of the fish, are tiny spheres of glass called tektites that rained down, molten hot, across the planet. Tektites in gills of the fish indicate they were inhaled as they fell from the sky, and just before the fish were swept up in waves from the shallow sea that existed there 66 million years ago. Amazingly, the site is 1,900 miles from the impact site, yet the tektites rained down swiftly and the wave-causing seismic earthquakes came within 10 minutes.

As with all scientific discoveries, other scientists are peer reviewing this information. Some are skeptical and expect more evidence before concluding the fossils are from the very moment life changed on Earth.

As you read through this chapter, think about the following questions:

1. Do lines of evidence, like fossils, explain the origin of life? Why or why not?
2. What other geological evidence supports the hypothesis of an asteroid hitting Earth 66 million years ago?

Following *the* Themes

CHAPTER 18 – THE ORIGIN AND HISTORY OF LIFE

Evolution	The history of life and origin of species can be summarized by the macroevolutionary changes shown in the fossil record.
Nature of Science	Scientific discoveries are subject to peer review to ensure they are valid and supported by evidence.
Biological Systems	Biogeography, the interface between geology and biology, is used to reconstruct the evolutionary history of life by comparing the geological history of Earth with the distribution of living organisms.

ISTUDY

18.1 The Origin of Life

Learning Outcomes

Upon completion of this section, you should be able to

1. List and describe the four stages of the origin of life.
2. Differentiate between the stages of chemical and biological evolution.
3. Describe a protocell membrane structure and its importance to the evolution of the first living cell.
4. Summarize at least one hypothesis that explains each of the four stages of the origin of life.

At the heart of Darwin's theory of evolution by natural selection is the principle of common ancestry, that all life on Earth can be traced back to a single ancestor. This ancestor, also called the **last universal common ancestor (LUCA),** is common to all organisms that live, and have lived, on Earth since life began (Fig. 18.1). Darwin thought that perhaps the first living organism arose in a "warm little pond with all sorts of ammonia and phosphoric salts, light, heat, electricity, etc." What Darwin unknowingly described were some of the conditions of ancient Earth that gave rise to the first forms of life.

In Section 1.1, we considered the characteristics shared by all living organisms. Living organisms acquire energy through metabolism, or the chemical reactions that occur within cells. Living organisms also respond to and interact with their environment, self-replicate, and are subject to the forces of natural selection that drive adaptation to the environment. The molecules of living organisms, called **biomolecules,** are organic molecules. The first living organisms on Earth would have had all these characteristics, yet early Earth was very different from the Earth we know today, and it consisted mainly of inorganic substances. The challenge is to determine how life started in this inorganic "warm little pond."

Advances and discoveries in chemistry, evolutionary biology, paleontology, microbiology, and other branches of science have helped scientists develop new, and test old, hypotheses about the origin of life. These studies contribute to an ever-growing body of scientific evidence that life originated 3.5–4 billion years ago (BYA) from nonliving matter in a series of four stages:

- *Stage 1. Organic monomers.* Simple organic molecules, called monomers, evolved from inorganic compounds before the existence of the first cells. Amino acids, the basis of proteins, and nucleotides, the building blocks of DNA and RNA, are examples of organic monomers.
- *Stage 2. Organic polymers.* Organic monomers were joined, or polymerized, to form organic polymers, such as DNA, RNA, and proteins.
- *Stage 3. Protocells.* Organic polymers became enclosed in a membrane to form precursors to the first cells, called *protocells* or *protobionts.*
- *Stage 4. Living cells.* Protobionts acquired the ability to self-replicate, as well as other cellular properties.

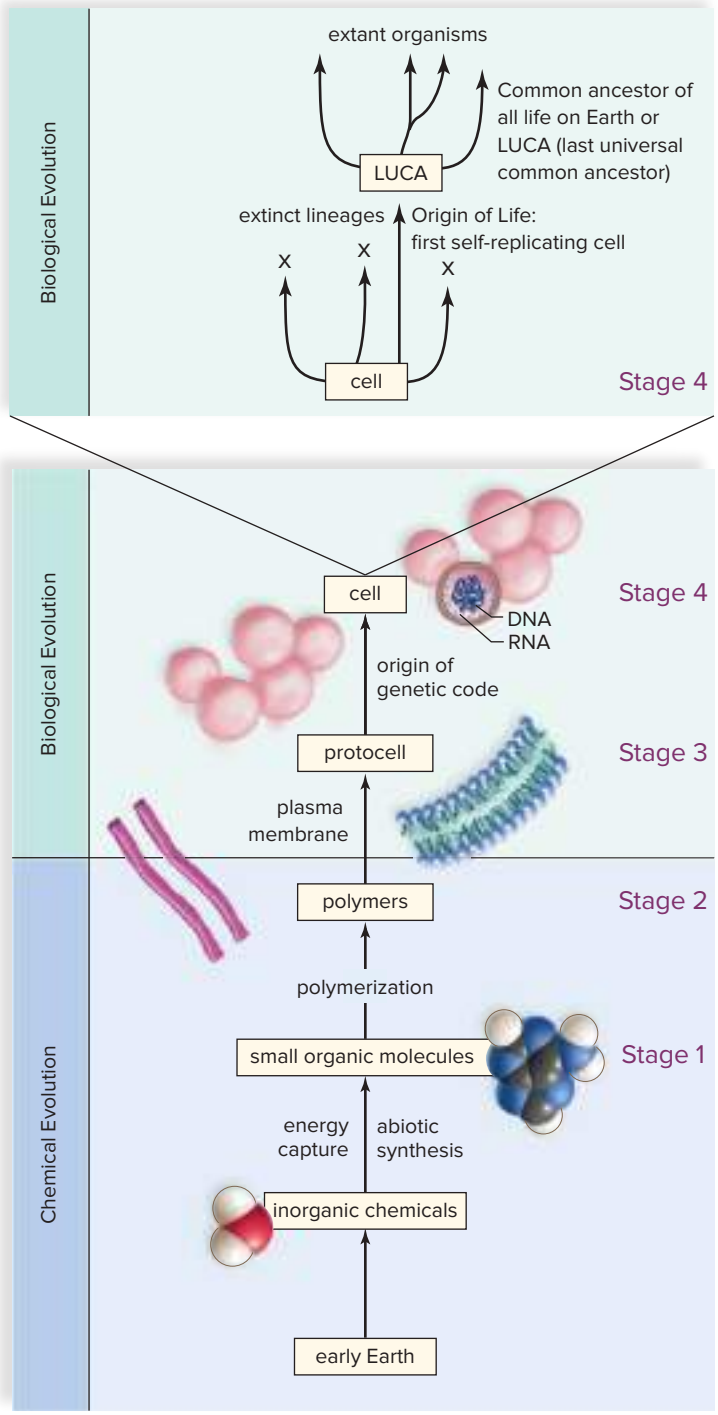


Figure 18.1 Stages of the origin of life. The first organic molecules (*bottom*) originated from chemically altered inorganic molecules present on early Earth (Stage 1). More complex organic macromolecules were synthesized to create polymers (Stage 2), which were then enclosed in a plasma membrane to form the protocells, or protobionts (Stage 3). The protocell underwent biological evolution to produce the first true, self-replicating, living cell (Stage 4). This first living cell underwent continued biological evolution (*top*), with a single surviving lineage, the LUCA, which became the common ancestor of all life on Earth.

ISTUDY

Scientists have performed experiments to test hypotheses at each stage of the origin of life. Stages 1–3 involve the processes of a "chemical evolution," before the origin of life (Fig. 18.1). Stage 4 is when life first evolved through the processes of "biological evolution" (Fig. 18.1). In this chapter, we examine the hypotheses and supporting scientific evidence for each stage of chemical and biological evolution, or **abiogenesis,** the origin of life from nonliving matter.

Stage 1: Evolution of Monomers

In the 1920s, Alexander Oparin, a notable biochemist, and J. B. S. Haldane, a geneticist and evolutionary biologist, independently proposed a hypothesis about the chemical origin of life. They proposed that the first stage in the origin of life was the evolution of simple organic molecules, or monomers (Fig. 18.1), from the inorganic compounds that were present in the Earth's early atmosphere. The Oparin-Haldane hypothesis, sometimes called the **primordial soup hypothesis,** proposes that early Earth had very little oxygen (O_2) but instead was made up of water vapor (H_2O), hydrogen gas (H_2), methane (CH_4), and ammonia (NH_3).

Methane and ammonia are reducing agents, because they readily donate their electrons. In the absence of oxygen, their reducing capability is powerful. **Abiotic synthesis** is the process of chemical evolution, forming organic molecules from inorganic materials. The early Earth's reducing atmosphere could have driven this process. Note that *oxidation* used in this context refers to a chemical redox reaction (see Section 6.4), not to oxygen gas.

The Primordial Soup Hypothesis and the Miller–Urey Experiment

In 1953, Stanley Miller, under the mentorship of Harold Urey, performed an experiment to test Oparin and Haldane's hypothesis of early chemical evolution (Fig. 18.2). The energy sources on early Earth included heat from volcanoes and meteorites, radioactivity from isotopes, powerful electric discharges in lightning, and solar radiation, especially ultraviolet radiation.

For his experiment, Miller placed a mixture of methane (CH_4), ammonia (NH_3), hydrogen (H_2), and water (H_2O) in a closed system, heated the mixture, and circulated it past an electric spark (simulating lightning). After a week's run, Miller discovered that a variety of amino acids and other organic acids had been produced.

The Miller–Urey experiment has been tested and reexamined over the decades since it was first performed. Other investigators have achieved similar results as Miller by using other, less-reducing combinations of gases dissolved in water. In 2008, a group of scientists examined 11 vials of compounds produced from variations of the Miller–Urey experiment and found a greater variety of organic molecules than Miller reported, including all 22 amino acids.

Evidence suggests that nitrogen gas (N_2), not ammonia (NH_3), would have been abundant in the primitive atmosphere. The scarcity of ammonia challenged the Miller–Urey experiment as a valid test of early Earth conditions. However, later experiments showed that ammonia could have been produced when various mixes of iron–nickel sulfides catalyzed the change of N_2 to NH_3. A laboratory test of this hypothesis worked perfectly.

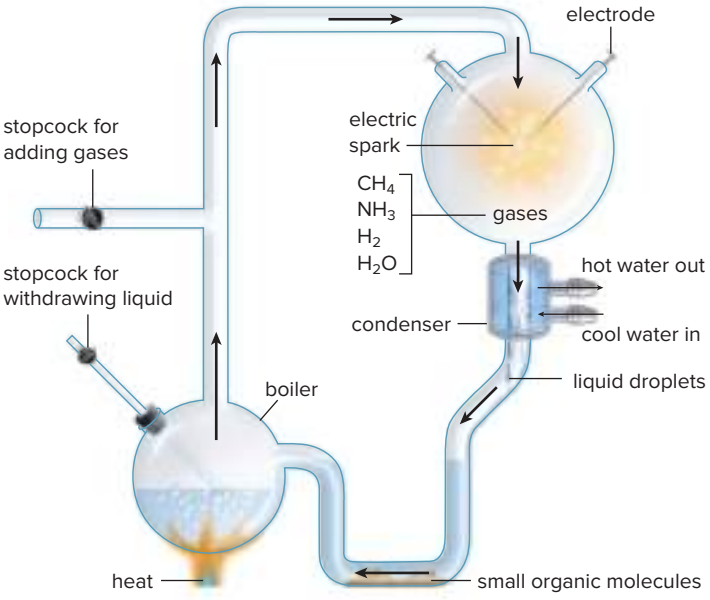


Figure 18.2 Stanley Miller's experiment. Gases that were thought to be present in the early Earth's atmosphere were admitted to the apparatus, circulated past an energy source (electric spark), and cooled to produce a liquid that could be withdrawn. Upon chemical analysis, the liquid was found to contain various small, organic molecules, which could serve as monomers for large, cellular polymers.

Under conditions simulating that of deep-sea thermal vents (Fig. 18.3), 70% of various nitrogen sources were converted to ammonia within 15 minutes. Thus, the early formation of organic monomers could have been supported by the production of ammonia from these vents.

If early atmospheric gases did react with one another to produce small organic compounds, neither oxidation (no free oxygen was present) nor decay (no bacteria existed) would have destroyed these molecules, and rainfall would have washed them into the ocean, where they would have accumulated for hundreds of millions of years. Therefore, the oceans would have been a thick, warm organic soup—much like Darwin's "warm little pond."

The Iron–Sulfur World Hypothesis

The Oparin-Haldane hypothesis was an important contribution to our understanding of the early stages of life's origins, but other hypotheses have also been proposed and tested. In the late 1980s, biochemist Günter Wächtershäuser proposed that thermal vents at the bottom of the Earth's oceans provided all the elements and conditions necessary to synthesize organic monomers. According to his **iron–sulfur world hypothesis,** dissolved gases emitted from thermal vents, such as carbon monoxide (CO), ammonia, and hydrogen sulfide, would pass over iron and nickel sulfide minerals that are present at thermal vents. The iron and nickel sulfide molecules would act as catalysts to drive the chemical evolution from inorganic to organic molecules.

Extraterrestrial Origins Hypothesis

A very different line of thinking involves the comets and meteorites that have pelted Earth throughout history. In recent

ISTUDY

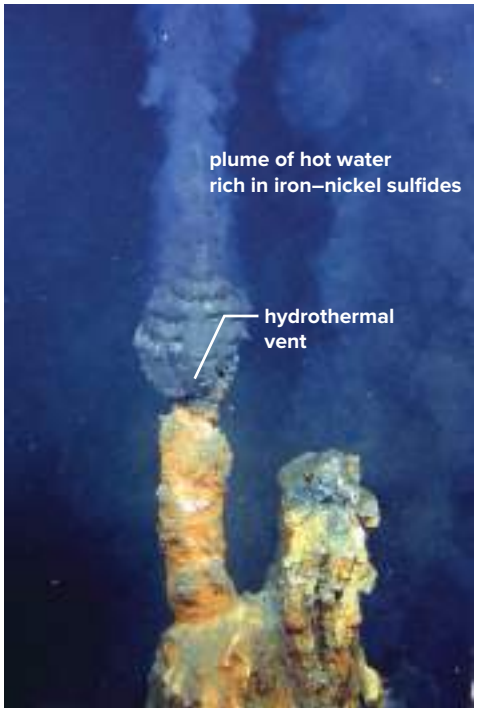


Figure 18.3 Chemical evolution at hydrothermal vents. Minerals that form at deep-sea thermal vents can catalyze the formation of ammonia and even monomers of larger organic molecules that occur in cells. Image courtesy of Submarine Ring of Fire 2006 Exploration, NOAA Vents Program

years, scientists have confirmed the presence of organic molecules in some meteorites. Many scientists feel these organic molecules could have seeded the chemical origin of life on early Earth. Others even hypothesize that bacterium-like cells evolved first on another planet and then were carried to Earth, an idea known as *panspermia.* A meteorite from Mars, labeled ALH84001, landed on Earth some 13,000 years ago. When examined, experts found tiny rods similar in shape to fossilized bacteria. The fact that Mars possessed liquid water in its past, and that life may have evolved there and then was carried to Earth, is one hypothesis being examined. Another event that lends evidence to panspermia is the discovery of glycerine and phosphorus found in the Churyumov-Gerasimenko comet in 2015. Glycerine is important because it is found in fat and is necessary for metabolic processes. The discovery of phosphorus is significant because it is an important element in so many biological molecules such as ATP, DNA, RNA, and phospholipids.

Stage 2: Evolution of Polymers

Within a cell's cytoplasm, organic monomers join to form polymers in the presence of enzymes—such as the synthesis of protein polymers from amino acids by ribosomes. Enzymes themselves are proteins, but proteins were not present on early Earth. The challenge is to determine how the first organic polymers formed if enzymes were not present.

The Iron–Sulfur World Hypothesis

The iron–sulfur world hypothesis of Günter Wächtershaüser and the research of Claudia Huber have shown that organic molecules will react and amino acids will form peptides in the presence of iron–nickel sulfides under conditions found at thermal vents. The inorganic iron–nickel sulfides have a charged surface that attracts amino acids and binds them together to make proteins.

The Protein-First Hypothesis

Sidney Fox has shown that amino acids polymerize abiotically when exposed to dry heat. He suggests that once amino acids were present in the oceans, they could have collected in shallow puddles along the rocky shore line. Then, the heat of the sun could have caused them to form **proteinoids,** small polypeptides that have some catalytic properties.

The formation of proteinoids has been simulated in the laboratory. When placed in water, proteinoids form **microspheres** (Gk. *mikros,* "small, little"; *sphaera,* "ball"), structures composed only of protein that have many properties of a cell. It's possible that even newly formed polypeptides had enzymatic properties. Some may have been more enzymatically active than others, perhaps giving them a selective advantage. If a certain level of enzyme activity provided an advantage over others, this would have set the stage for natural selection to shape the evolution of these first organic polymers. Those that evolved to be part of the first cell or cells would have had a selective advantage over those that did not become part of a cell.

Fox's **protein-first hypothesis** assumes that protein enzymes arose prior to the first DNA molecule. Thus, the genes that encode proteins followed the evolution of the first polypeptides.

The RNA-First Hypothesis

The **RNA-first hypothesis** suggests that only the macromolecule RNA was needed to progress toward formation of the first cell or cells. Thomas Cech and Sidney Altman shared a Nobel Prize in 1989 for their discovery that RNA can be both a substrate and an enzyme. Some viruses today have RNA genes; therefore, the first genes could have been RNA. It would seem, then, that RNA could have carried out the processes of life commonly associated today with DNA and proteins. Those who support this hypothesis say it was an "RNA world" some 4 BYA. The Evolution feature, "Possible Extraterrestrial Components of RNA," explores how this may have been possible.

Stage 3: Evolution of Protocells

Before the first true cell arose, a **protocell (protobiont)** would have emerged—a structure that is characterized by having an outer membrane. After all, life requires chemical reactions to take place within a boundary, protecting them from disruption of conditions.

The Plasma Membrane

The plasma membrane of a cell provides a boundary between the inside of the cell and its outside world. This membrane is critical to the proper regulation and maintenance of cellular activities, and therefore the evolution of a membrane was a critical step in the origin of life.

ISTUDY

THEME Evolution

Possible Extraterrestrial Components of RNA

All living organisms on Earth, including viruses, have genetic material made up of nucleic acids such as DNA and RNA. RNA is thought to have been one of the first biological molecules to appear on Earth and is thought to be more primitive than DNA. Scientists have wondered about the origin of RNA and have hypothesized how the components of RNA (see Section 3.5)—the phosphate group, ribose sugar, and nitrogenous bases—came about.

One theory is that meteors, asteroids, or comets that contained the basic building blocks of RNA carried them to Earth. These components were then transferred to (i.e., *seeded*) early Earth. In support of this theory, scientists have already found several amino acids and nitrogenous bases in meteorites. Recently, two other key components—the phosphate group and ribose sugar—have been created in lab simulations, lending evidence to the hypothesis.

Ribose has not yet been detected in material from space. However, in a recent simulation of an ice comet (basically a dirty snowball with dust particles), French research teams successfully produced ribose in a lab setting. For the simulation, researchers produced an artificial comet by mixing water (H_2O), methanol (CH_3OH), and ammonia (NH_3) in a high-vacuum chamber at 200°C. To produce the dust particles coated in ice, the chemical mixture was irradiated with UV, to mimic the molecular clouds where these particles naturally form. The mixture was then warmed to simulate proximity to the sun. The composition was analyzed, and ribose and other sugars were indeed detected.

The next challenge for astronomers was how the phosphate group may have come about. It is known that, early on our planet, fiery meteorites continually struck the Earth. Recent work has demonstrated that meteorites (Fig. 18A) may have carried an extraterrestrial mineral called schreibersite—$(Fe,Ni)_3P$—that, when in contact with water on Earth, could have corroded and provided the necessary chemical spark leading to the origin of phosphate in biological molecules. This chemical reaction is postulated to have been phosphorylation (see Section 6.2), providing the phosphate necessary to make early nucleic acids. The mechanism of phosphorylation is still unknown and actively being investigated with further lab simulations of early Earth.

Although the formation of ribose and phosphate from real comets and meteors has yet to be confirmed, these findings lend further support to the idea that comets and meteors could have contributed the organic molecules that led to the origin of life on Earth.

Figure 18A A meteor with schreibersite. Meteors, such as those discovered in the Sikhote-Alin mountains in Russia, contains the rare schreibersite mineral. This mineral is postulated to be a possible source of phosphate found in nucleic acids. Matteo Chinellato/Alamy Stock Photo

Questions to Consider

1. Why is the study of the origin of RNA important?
2. What are the benefits and drawbacks of simulating the formation of molecules in a laboratory setting?

The modern plasma membrane is made up of phospholipids assembled in a bilayer (Fig. 18.4). The first plasma membranes were likely made up of fatty acids, which are smaller than phospholipids but, like phospholipids, have a hydrophobic "tail" and a hydrophilic "head" (Fig. 18.4). Fatty acids are one of the organic polymers that could have formed from chemical reactions at deep-sea thermal vents early in the history of life.

In water, fatty acids assemble into small spheres called **micelles.** A micelle is a single layer of fatty acids organized with their heads pointing out and their tails pointing toward the center of the sphere. Under appropriate conditions, micelles can merge to form **vesicles.** Vesicles are larger than micelles and are surrounded by a bilayer (two layers) of fatty acids (Fig. 18.5), similar to the *phospholipid bilayer* of modern plasma membranes. An important feature of a vesicle lipid bilayer is that the individual fatty acids can flip between the two layers, which helps move select molecules, such as amino acids, from the outside to the inside of the vesicle.

The first protocell would likely have been a vesicle with this type of fatty acid bilayer membrane. Interestingly, if lipids are made available to protein microspheres, lipids tend to become associated with microspheres, producing a lipid–protein membrane. Lipid–protein microspheres share some interesting properties with modern cells: They resemble bacteria, they have an electrical potential difference, and they divide and perhaps are subject to selection.

In the 1920s, Alexander Oparin demonstrated that under appropriate conditions of temperature, ionic composition, and pH, concentrated mixtures of macromolecules tend to give rise to complex units called *coacervate droplets.* Coacervate droplets have a tendency to absorb and incorporate various substances from the surrounding solution. Eventually, a semipermeable-type boundary, also a trait of the modern plasma membrane, may form about the droplet.

In the early 1960s, biophysicist Alec Bangham of the Animal Physiology Institute in Cambridge, England, discovered that when he extracted lipids from egg yolks and placed them in water, the lipids naturally organized themselves into double-layered bubbles roughly the size of a cell known as **liposomes** (Gk. *lipos,* "fat"; *soma,* "body").

Later, biophysicist David Deamer, of the University of California, and Bangham realized that liposomes might have provided life's first membranous boundary. Perhaps liposomes with a phospholipid membrane engulfed early molecules that had enzymatic, even

ISTUDY

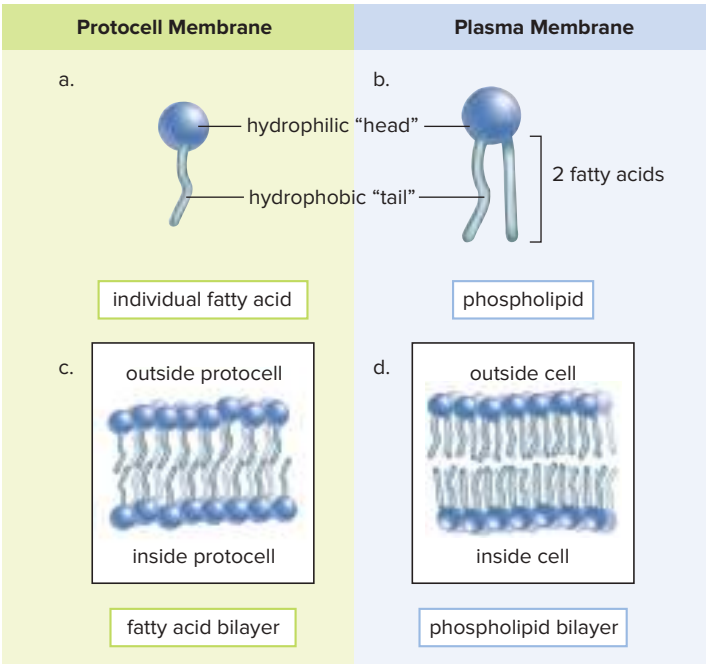


Figure 18.4 A comparison of protocell and modern plasma membranes. The first lipid membrane was likely made of a single layer of fatty acids. These first protocell membranes and the modern plasma membrane have features in common. **a.** Individual fatty acids have a single fatty acid chain with a hydrophilic head and a hydrophobic tail. **b.** Phospholipids of modern plasma membranes are made of two hydrophobic fatty acid chains (the "tails") attached to a hydrophilic head. **c.** Protocell membranes were likely made up of a bilayer of fatty acids, with hydrophilic heads pointing outward and hydrophobic tails pointing inward. **d.** Modern cells are organized in a similar fashion. Both bilayers create a semipermeable barrier between the inside and outside of a cell.

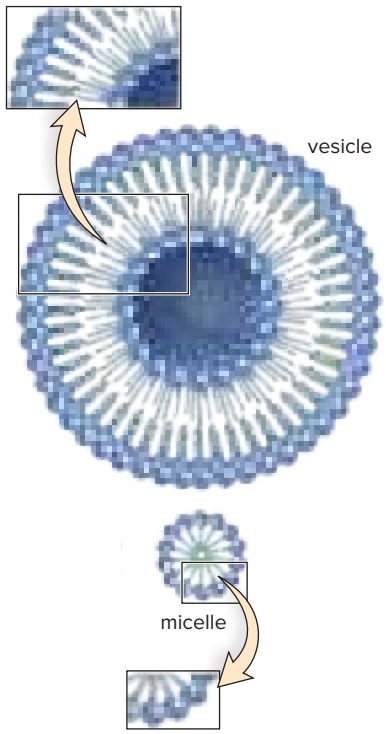


Figure 18.5 Structure and growth of the first plasma membrane. The first plasma membrane was likely made of a fatty acid bilayer, similar to that seen in vesicles. Protocells, the ancestor of modern cells, are thought to have had this type of membrane. The structure of a vesicle reveals the fatty acid bilayer. Micelles are spherical droplets formed by a single layer of fatty acids, and they are much smaller than vesicles.

replicative, abilities. The liposomes would have protected the molecules from their surroundings and concentrated them, so that they could react (and evolve) quickly and efficiently. These investigators called this the **membrane-first hypothesis,** meaning that the first cell had to have a plasma membrane before any of its other parts. Perhaps the first membrane formed in this manner.

Nutrition

A protocell would have had to acquire nutrition in order to grow. One hypothesis suggests that protocells were heterotrophic (Gk. *hetero,* "different"; *trophe,* "food") organisms that consumed preformed organic molecules. If protocells evolved at hydrothermal vents, they may have carried out chemosynthesis—the synthesis of organic molecules from inorganic molecules and nutrients. Chemoautotrophic bacteria obtain energy by oxidizing inorganic compounds, such as hydrogen sulfide (H_2S), a molecule that is abundant at thermal vents. When thermal vents in the deep sea were first discovered in the 1970s, investigators were surprised to discover complex vent ecosystems supported by chemosynthesis instead of photosynthesis.

Glycolysis is a critical metabolic pathway that transforms high-energy chemical bonds into energy for a cell to do work. Glycolysis is the first stage of cellular respiration (see Section 8.2), and it occurs outside of the mitochondria. ATP (adenosine triphosphate) is the most important energy-carrying molecule in living organisms. In the early stages of the origin of life, ATP would have been available to protocells in a preformed state. Over time, the ATP would have been transformed to ADP as it was used up. Natural selection would have favored the evolution of ATP/ADP recycling as a means to provide a renewable energy supply to the first cells. Modern cells have a way to produce ATP from ADP and inorganic phosphate via substrate-level phosphorylation (see Fig. 8.4) during glycolysis.

The greatest amount of ATP is synthesized by oxidative phosphorylation—in particular, via the electron transport chain (see Section 8.2). Because all life on Earth uses ATP to fuel cellular metabolism, the evolution of a means of synthesizing ATP must have occurred very early in the history of life. The first bacteria evolved in the oxygen-poor environment of early Earth. Thus, ATP was likely synthesized first by fermentation (see Section 8.3). The evolution of oxidative phosphorylation in eukaryotes provided an advantage, because it greatly increased the amount of ATP synthesized per unit of energy. Mitochondria share a common ancestor with a group of bacteria that synthesize ATP via an electron transport chain. Oxidative phosphorylation is possible in eukaryotes because mitochondria provide an electron-transport-chain ATP factory.

At first, the protocell must have had limited ability to break down organic molecules. Scientists speculate that it took millions of years for glycolysis to evolve completely. Some evidence suggests that microspheres, from which protocells may have evolved, have some catalytic ability. Oparin's coacervates incorporate enzymes if they are available in the medium.

ISTUDY

Stage 4: Evolution of a Self-replication System

Today's cell is able to carry on protein synthesis in order to produce the enzymes that allow DNA to replicate. The central dogma of genetics states that DNA directs protein synthesis and that information flows from DNA to RNA to protein. It is possible that this sequence developed in stages.

According to the RNA-first hypothesis, RNA would have been the first to evolve, and the first true cell would have had RNA genes. These genes would have directed and enzymatically carried out protein synthesis. Today, ribozymes are enzymatic RNA molecules. We also know a number of viruses have RNA genes. These viruses have a protein enzyme, called reverse transcriptase, that uses RNA as a template to form DNA. Perhaps with time, reverse transcription occurred within the protocell, and this is how DNA-encoded genes arose. If so, RNA was responsible for both DNA and protein formation. Once DNA genes developed, protein synthesis would have been carried out in the manner dictated by the central dogma of genetics.

According to the protein-first hypothesis, proteins, or at least polypeptides, were the first of the three (DNA, RNA, and protein) to arise. Only after the protocell developed a plasma membrane and the necessary enzymes did it have the ability to synthesize DNA and RNA from small molecules provided by the ocean. Because a nucleic acid is a complicated molecule, the likelihood that RNA arose *de novo* is minimal. It seems more likely that enzymes were needed to guide the synthesis of nucleotides and then nucleic acids. Similar to the RNA-first hypothesis, once the information was contained within genes then protein synthesis would occur.

Cairns-Smith proposed that polypeptides and RNA evolved simultaneously. Therefore, the first true cell would have contained RNA genes that could have replicated because of the presence of proteins. This eliminates the baffling chicken-and-egg paradox of whether proteins or RNA came first. It means, however, that these two events would have had to happen at the same time.

After DNA formed, the genetic code had to evolve before DNA could store genetic information. The present genetic code is subject to fewer errors than a million other possible codes. Also, the present code is among the best at minimizing the effect of mutations. A single-base change in a present codon is likely to result in the substitution of a chemically similar amino acid, resulting in minimal changes in the final protein. This evidence suggests that the genetic code did undergo a natural selection process before finalizing into today's code.

Check Your Progress 18.1

1. Identify the stages of the origin of life hypothesis.
2. List three different hypotheses that explain the origin of organic molecules from inorganic matter, and identify the key features of these hypotheses.
3. Compare and contrast the features of the protocell membrane with a modern plasma membrane.

18.2 The History of Life

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the processes of relative and absolute dating of fossils.
2. List three sources of evidence that support the endosymbiotic theory of organelle evolution.
3. Discuss when and where the first multicellular organisms evolved.
4. Describe in chronological order the periods of Earth's history, and identify one major biological event that took place in each.

Macroevolution is the origin of new species and other taxonomic groups. The fossil record documents the history of macroevolution over long periods of time.

Fossils Tell a Story

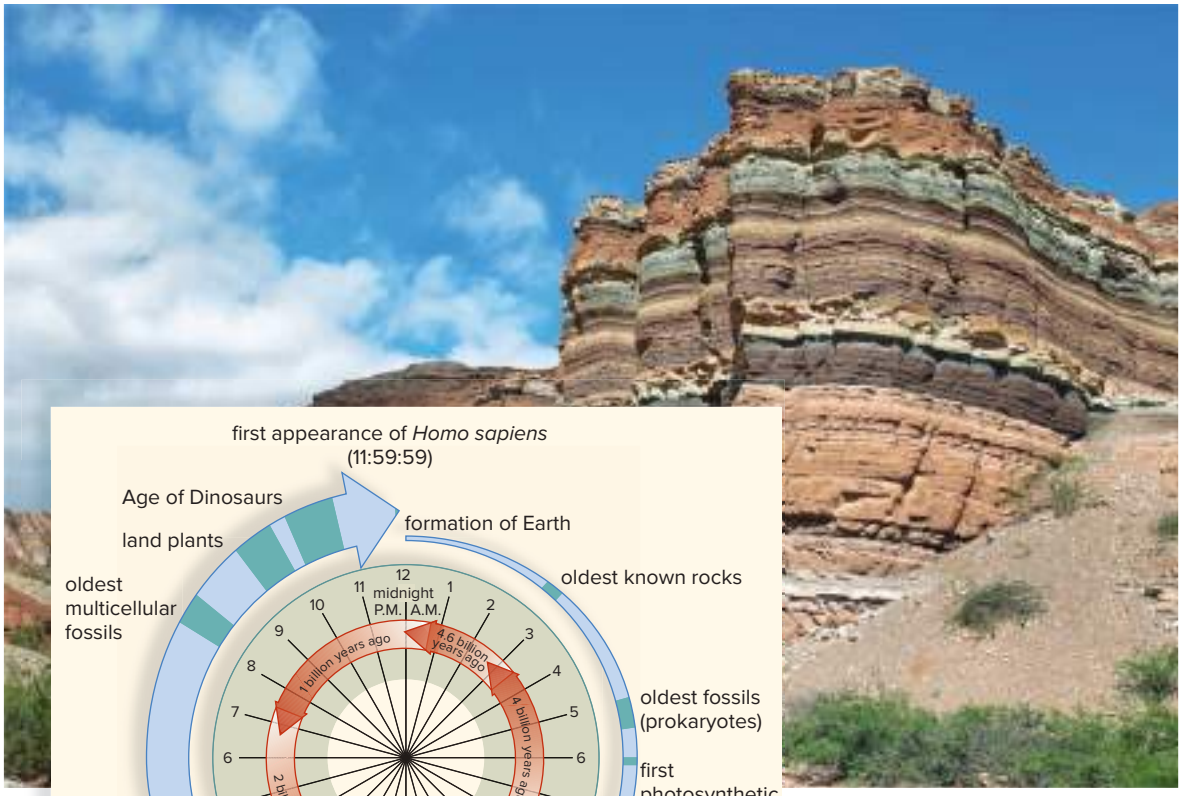
Fossils (L. *fossilis,* "dug up") are the remains and traces of evidence of past life. Recall from Section 15.3 that fossils consist mainly of hard parts, such as shells, bones, or teeth, because these are not destroyed or do not decay easily. However, some fossilized traces can be trails, footprints, or the impressions of soft body parts. **Paleontology** (Gk. *palaios,* "old, ancient"; *logy,* "study of") is the study of the fossil record that results in knowledge about the history of life, ancient climates, and environments.

The great majority of fossils are found embedded in or recently eroded from sedimentary rock. **Sedimentation** (L. *sedimentum,* "a settling") is the gradual settling of particles of eroded and weathered rock and soil, called silt, that are carried by moving water. Silt is deposited gradually, forming layers of particles that vary in size and composition called **sediment.** Sediment becomes a **stratum** (*pl.,* strata), a recognizable layer of sediment in a stratigraphic sequence (Fig. 18.6*a*).

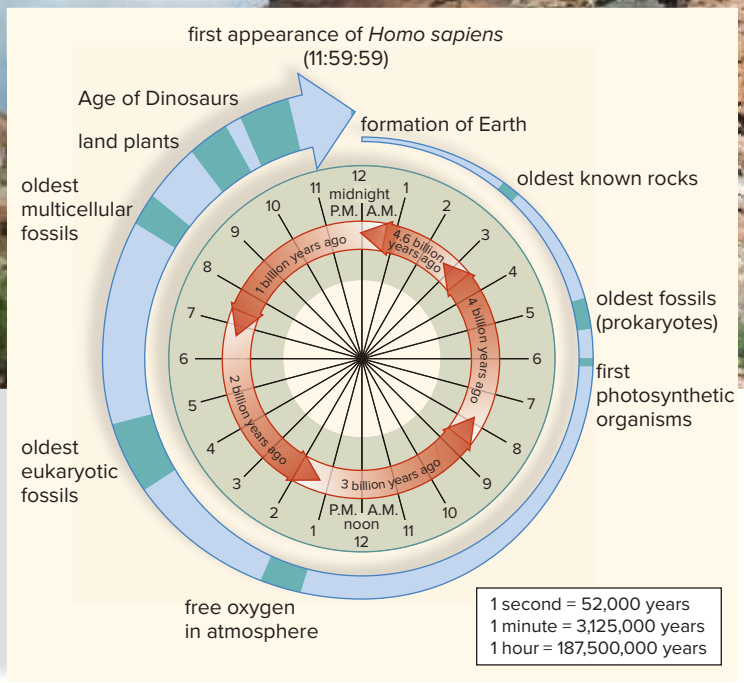
The law of superposition states that a given stratum is considered to be older than the one above it and younger than the one immediately below it. However, the layers of sedimentation can be disturbed by geological forces, and such disturbances can complicate the interpretation of the stratigraphic sequence. Figure 18.6*b* shows the history of Earth as if it had occurred during a 24-hour time span that starts at midnight. The actual years are shown on an inner ring of the diagram. This figure illustrates dramatically that only single-celled organisms were present during most (about 80%) of the history of the Earth.

If the Earth formed at midnight, prokaryotes did not appear until about 5 A.M., eukaryotes are present at approximately 4 P.M., and multicellular forms do not appear until around 8 P.M. Invasion of the land doesn't occur until about 10 P.M., and humans don't appear until 1 second before the end of the day. This timescale has been worked out by studying the fossil record. In addition to sedimentary rock, more recent fossils can be found in tar, ice, bogs, and amber. Shells, bones, leaves, and even footprints can also be found in the fossil record (Fig. 18.7).

ISTUDY



a.



b.

Figure 18.6 The history of life. **a.** Strata, layers of sedimentary rock, are one source of fossils that provide information about the history of life. **b.** The blue ring of this diagram shows the history of life as it would be measured on a 24-hour timescale starting at midnight. (The red ring shows the actual years going back in time to 4.6 BYA.) The fossil record suggests that a very large portion of life's history was devoted to the evolution of unicellular organisms. The first multicellular organisms do not appear in the fossil record until just before 8 P.M., and humans are not on the scene until less than a second before midnight. BYA = billion years ago (a): Marisa Estivill/Shutterstock

Relative Dating of Fossils

In the early nineteenth century, even before the theory of evolution was formulated, geologists sought to correlate the strata worldwide. The problem was that the nature of strata changes across regions. A stratum in England might contain different sediments than one of the same age in Russia. Geologists discovered that each stratum of the same age contained certain **index fossils.** These are fossils that identify deposits made at apparently the same time in different parts of the world. These index fossils are used in **relative dating** methods. For example, a particular species of fossil ammonite (an animal related to the chambered nautilus) has been found over a wide range and for a limited time period. Therefore, all strata around the world that contain this fossil must be of the same age.

Absolute Dating of Fossils

Absolute dating methods rely on radiometric techniques to assign an actual date to a fossil. All radioactive isotopes have a particular half-life, the length of time it takes for half of the radioactive isotope to change into another stable element.

Radiocarbon dating uses the radioactive decay of ^{14}C, a rare carbon isotope. Over time, ^{14}C will stabilize into ^{14}N (see Section 2.1). This process of radioactive decay—that is, when ^{14}C becomes ^{14}N—occurs at a constant rate. Willard Libby won the Nobel Prize in Chemistry for his discovery in 1949 that the half-life of ^{14}C is 5,730 years. This means that half the original ^{14}C in organic matter decays to ^{14}N in this period of time. All living organisms have in them about the same ratio of ^{12}C to ^{14}C. The absolute date of a fossil can be determined by comparing the ^{12}C to ^{14}C ratio of a fossil to that of a living organism. This comparison can be used to calculate the age of the fossil.

Radiocarbon dating is accurate only for fossils up to approximately 100,000 years old. The amount of ^{14}C radioactivity in older fossils is so low that it cannot be used to measure the age of a fossil accurately. ^{14}C is the only radioactive isotope contained within organic matter, but it is possible to use others to date rocks, and from that to infer the age of a fossil contained in the rock. Potassium–argon (K–Ar) dating can be used to date rocks that are 100,000 to 4.5 billion years old, and thus it can be used to indirectly estimate the age of fossils well beyond the limits of radiocarbon dating.

ISTUDY

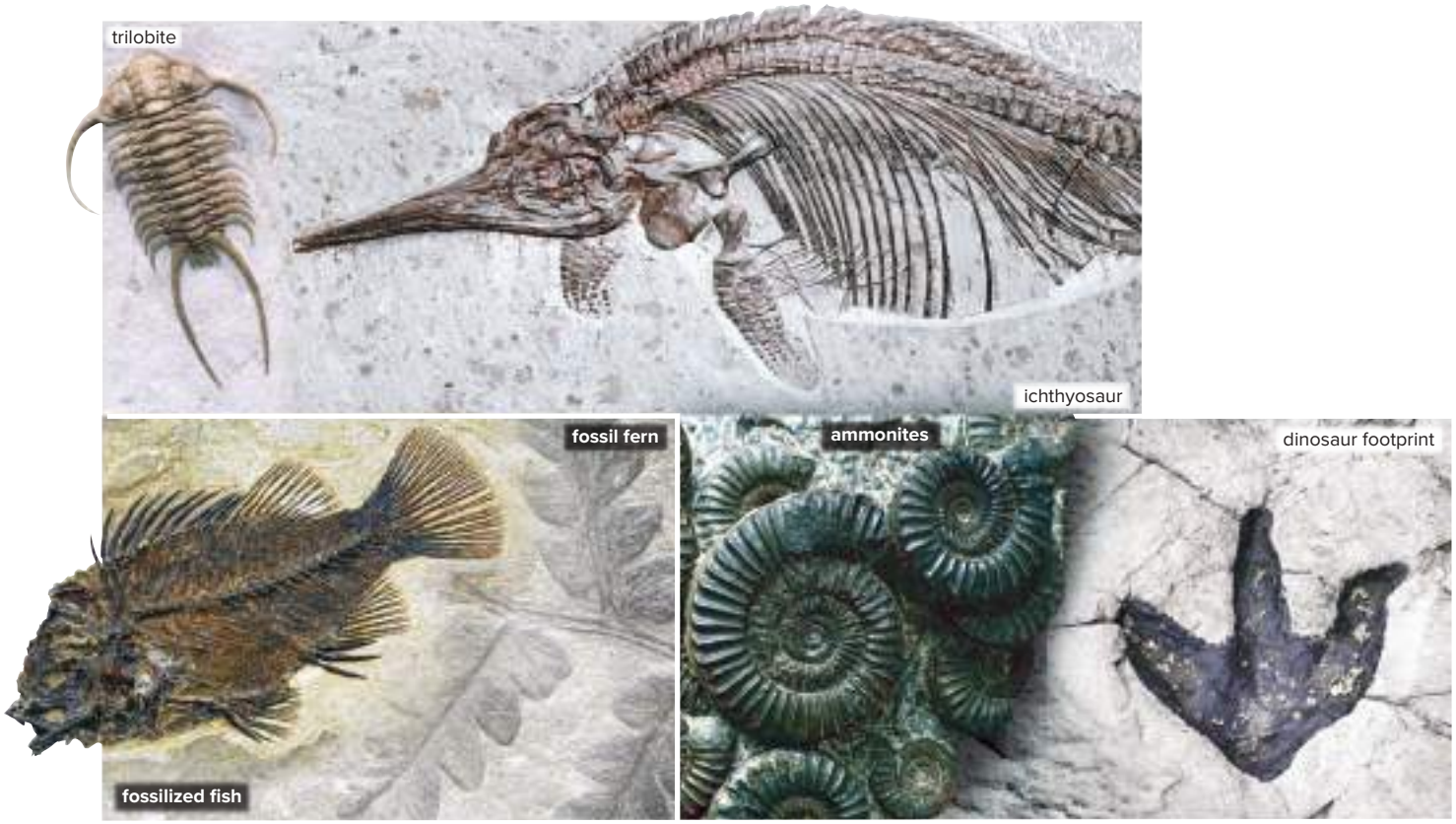


Figure 18.7 Fossils. Fossils are the remains of living organisms from the past. They can be impressions left in rocks, footprints, mineralized bones, shells, or any other evidence of life-forms that lived in the past. (trilobite): Francois Gohier/Science Source; (ichthyosaur): R. Koenig/Blickwinkel/age fotostock; (fish): Alan Morgan; (fern): GEORGE BERNARD/Avalon; (ammonites): Sinclair Stammers/SPL/Science Source; (footprint): fotocelia/iStockphoto/Getty Images

The Geologic Timescale

Geologists have devised the **geologic timescale,** which divides the history of the Earth into eras and then periods and epochs (Table 18.1). The geologic timescale was derived from the accumulation of data from the age of fossils in strata all over the world. Biologists traditionally begin the geologic timescale when life began during Precambrian time (Table 18.1). The Precambrian is a very long period of time, comprising about 87% of the geologic timescale. During this time, life arose and the first cells came into existence.

Evolution of Prokaryotes

The first modern-type cells were probably prokaryotes, which first appeared approximately 3.5 BYA (Table 18.1). Prokaryotes are relatively simple cells—they do not have a nucleus or membrane-bound organelles. Prokaryotes can live in the most inhospitable of environments, such as hot springs, salty lakes, and oxygen-free swamps—all of which may typify habitats on early Earth. The cell wall, plasma membrane, RNA polymerase, and ribosomes of one lineage of prokaryotes, the archaea, are more like those of eukaryotes than those of bacteria (see Section 20.5).

The first identifiable fossils are those of complex prokaryotes. Chemical fingerprints of complex cells are found in sedimentary rocks from southwestern Greenland, dated at 3.8 BYA. The oldest prokaryotic fossils have been discovered in western Australia. These 3.46-billion-year-old microfossils resemble today's cyanobacteria, prokaryotes that carry on photosynthesis in the same manner as plants. At that time, only volcanic rocks jutted above the waves, and there were as yet no continents. Strange-looking boulders, called **stromatolites,** littered beaches and shallow waters (Fig. 18.8*a*). Living stromatolites can still be found along Australia's western coast.

The outer surface of a stromatolite is alive with cyanobacteria. The cyanobacteria in ancient stromatolites added oxygen to the atmosphere (Fig. 18.8*b*). By 2.0 BYA, the presence of oxygen made most environments unsuitable for anaerobic prokaryotes. Photosynthetic cyanobacteria and aerobic bacteria proliferated as new metabolic pathways evolved. Due to the presence of oxygen, the atmosphere became an oxidizing one instead of a reducing one. Oxygen in the upper atmosphere forms ozone (O_3), which filters out the ultraviolet (UV) rays of the sun. Before the formation of the **ozone shield,** the amount of ultraviolet radiation reaching the Earth could have helped create organic molecules, but it would have destroyed any land-dwelling organisms. Once the ozone shield was in place, living organisms were sufficiently protected and able to live on land.

ISTUDY

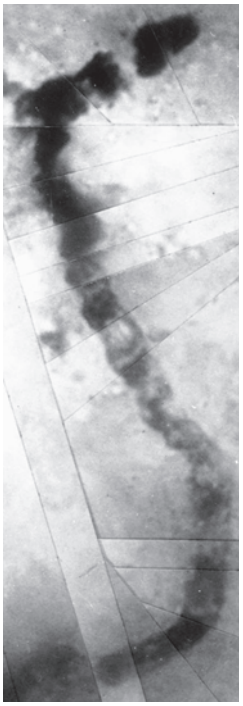
Table 18.1 The Geologic Timescale: Major Divisions of Geological Time and Some Important Evolutionary Events of Each Time Period

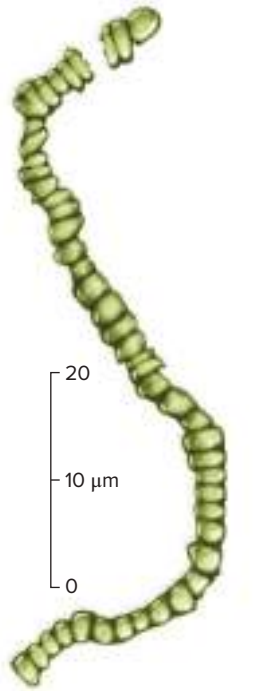
Era	Period	Epoch	Time Frame	Plant Life	Animal Life
Cenozoic	Quaternary	Anthropocene	Current*	Reduction in biodiversity due to human-induced climate change	Reduction in biodiversity due to human-induced climate change
		Significant Extinction Event Underway			
		Holocene	approximately 11,000 years ago	Humans influence plant life.	Age of *Homo sapiens*
	Quaternary	Pleistocene	0.01	Herbaceous plants spread and diversify.	Ice Age mammals and modern humans appear.
	Neogene	Pliocene	2.6 MYA	Herbaceous angiosperms flourish.	First hominids appear.
	Neogene	Miocene	5.3 MYA	Grasslands spread as forests contract.	Apelike mammals and grazing mammals flourish; insects flourish.
	Neogene	Oligocene	23.0 MYA	Many modern families of flowering plants evolve; grasses appear.	Browsing mammals and monkeylike primates appear.
	Paleogene	Eocene	33.9 MYA	Subtropical forests with heavy rainfall thrive.	All modern orders of mammals are represented.
	Paleogene	Paleocene	56.0 MYA	Flowering plants continue to diversify.	Ancestral primates, herbivores, carnivores, and insectivores appear.
		Mass Extinction: 50% of all species, dinosaurs and most reptiles			
Mesozoic	Cretaceous		66.0 MYA	Flowering plants spread; conifers persist.	Placental mammals appear; modern insect groups appear.
	Jurassic		145.5 MYA	Flowering plants appear.	Dinosaurs flourish; birds appear.
		Mass Extinction: 48% of all species, including corals and ferns			
	Triassic		201.3 MYA	Forests of conifers and cycads dominate.	First mammals appear; first dinosaurs appear; corals and mollusks dominate seas.
		Mass Extinction ("The Great Dying"): 83% of all species on land and sea			
Paleozoic	Permian		251.9 MYA	Gymnosperms diversify.	Reptiles diversify; amphibians decline.
	Carboniferous		299.0 MYA	Age of great coal-forming forests occurs; ferns, club mosses, and horsetails flourish.	Amphibians diversify; first reptiles appear; first great radiation of insects occurs.
		Mass Extinction: Over 50% of coastal marine species, corals			
	Devonian		359.2 MYA	First seed plants appear; seedless vascular plants diversify.	First insects and first amphibians appear on land.
	Silurian		416.0 MYA	Seedless vascular plants appear.	Jawed fishes diversify and dominate the seas.
		Mass Extinction: Over 57% of marine species			
	Ordovician		443.7 MYA	Nonvascular land plants appear.	Invertebrates spread and diversify; first jawless and then jawed fishes appear.
	Cambrian		485.4 MYA	Marine algae flourish.	All invertebrate phyla are present; first chordates appear.
			630 MYA		First soft-bodied invertebrates evolve.
			1.0 BYA	Protists diversify.	
			2.1 BYA	First eukaryotic cells evolve.	
			2.7 BYA	O_2 accumulates in atmosphere.	
			3.5 BYA	First prokaryotic cells evolve.	
			4.57 BYA	Earth forms.	

*time frame to still be established
MYA = millions of years ago
BYA = billions of years ago

ISTUDY

a. Stromatolites





b. *Primaevifilum*

Figure 18.8 Prokaryote fossils of the Precambrian. **a.** Stromatolites date back to 3.46 BYA. Living stromatolites are located in shallow waters off the shores of western Australia and in other tropical seas. **b.** The prokaryotic microorganism *Primaevifilum* was found in fossilized stromatolites. (a): Francois Gohier/Science Source; (b, photo): ©Dr. J. William Schopf

Eukaryotic Cells Arise

The eukaryotic cell, which originated somewhere between 1.8 to 2.3 BYA, obtains energy from cellular metabolism in the presence of oxygen. These cells contain a nucleus, as well as other membranous organelles. The eukaryotic cell acquired its organelles gradually. The nucleus may have developed by an invagination of the plasma membrane. The ancestors of the mitochondria in eukaryotic cells were free-living bacteria that synthesized ATP via an electron transport chain, and chloroplasts were free-living photosynthetic prokaryotes. The **endosymbiotic theory** states that a nucleated cell engulfed various prokaryotes, which became the cell's organelles. The nucleated cell and the engulfed bacteria coevolved the ability to synthesize ATP via oxidative phosphorylation (Fig. 18.9). The evidence for the endosymbiotic theory is as follows.

- Present-day mitochondria and chloroplasts have a size that lies within the range of that for bacteria.
- Mitochondria and chloroplasts have their own DNA and make some of their own proteins. (The DNA of the nucleus also codes for some of the mitochondrial proteins.)
- The mitochondria and chloroplasts divide by binary fission, as bacteria do.
- The outer membrane of mitochondria and chloroplasts differ—the outer membrane resembles that of a eukaryotic cell, while the inner membrane resembles that of a bacterial cell.

It has been suggested that flagella (and cilia) also arose by endosymbiosis. First, slender, undulating prokaryotes could have attached themselves to a host cell to take advantage of food leaking from the host's plasma membrane. Eventually, these prokaryotes adhered to the host cell and became the flagella and cilia we know today. The first eukaryotes were single-celled, the same as prokaryotes.

Multicellularity Arises

Fossils of multicellular protists at least as old as 1.4 BYA have been found in Arctic Canada. It's possible that the first multicellular organisms practiced sexual reproduction. Among today's protists we find colonial forms in which some cells are specialized to produce the gametes needed for sexual reproduction.

Separation of germ cells, which produce gametes, from somatic cells may have been an important first step toward the evolution of the Ediacaran invertebrates, which appeared about 630 MYA and died out about 545 MYA. The first fossils of these organisms were found in the Ediacara Hills of Australia. Since then, similar fossils have been discovered on a number of other continents.

Many of the fossils, dated 630–545 MYA, are thought to be of soft-bodied invertebrates (animals without a vertebral column) that most likely lived on mudflats in shallow marine waters. Some may have been mobile, but others were large, immobile, bizarre creatures resembling spoked wheels, corrugated ribbons, and lettuce-like fronds. All were flat and probably had two tissue

ISTUDY

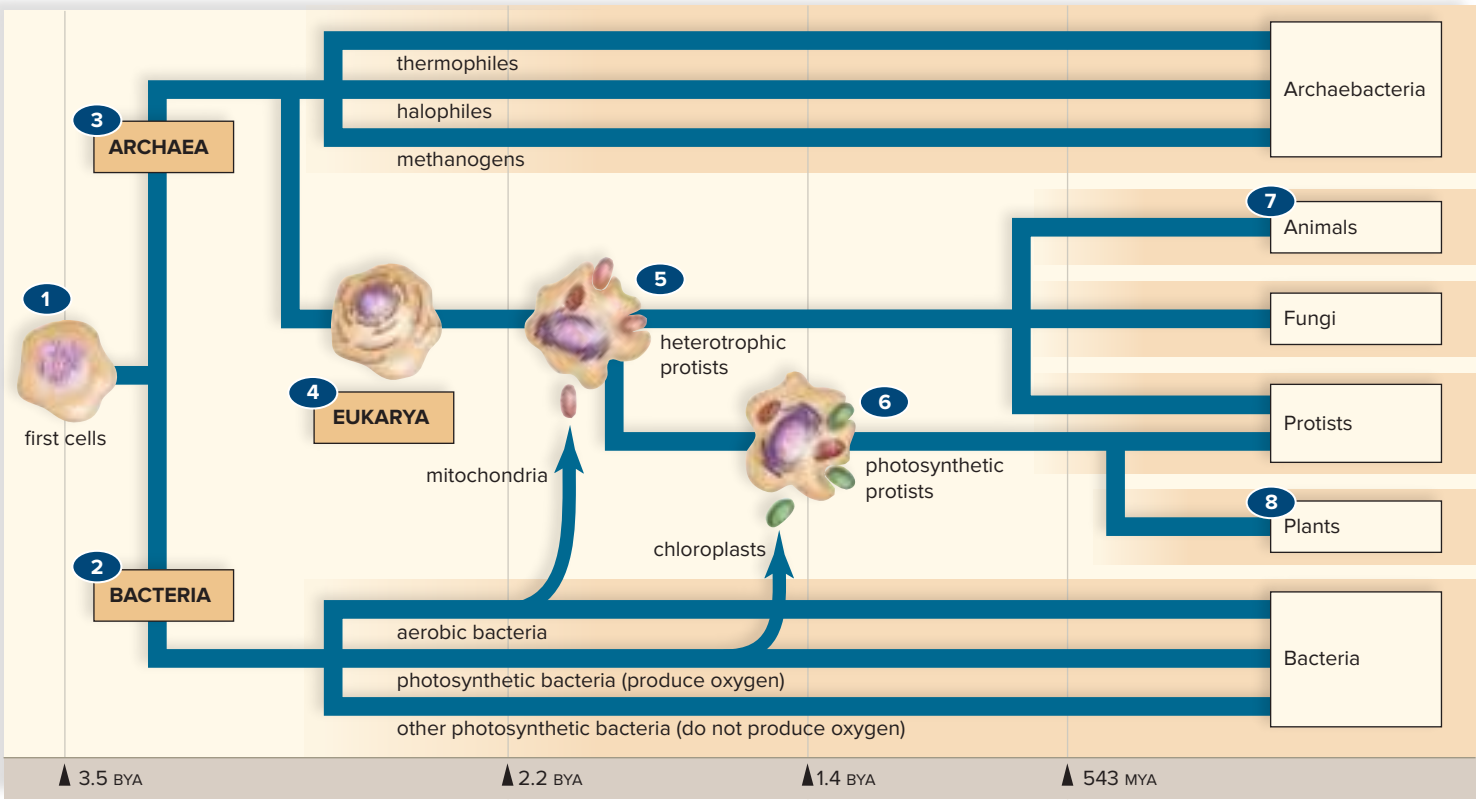


Figure 18.9 The tree of life. During the Precambrian time, 1 the first cell or cells gave rise to 2 bacteria and 3 archaea; 4 the first eukaryotic cell evolved from archaea. 5 Heterotrophic protists arose when eukaryotic cells gained mitochondria by engulfing aerobic bacteria, and 6 photosynthetic protists arose when these cells gained chloroplasts by engulfing photosynthetic bacteria. 7 Animals (and fungi) evolved from heterotrophic protists, and 8 plants evolved from photosynthetic protists. BYA = billion years ago; MYA = million years ago

layers; few had any type of skeleton (Fig. 18.10). They apparently had no mouth; perhaps they absorbed nutrients from the sea or had photosynthetic organisms living on their tissues. With few exceptions, this group disappeared from the fossil record at 545 MYA, but it may have given rise to modern cnidarians and related animals.

The Paleozoic Era

The Paleozoic era lasted about 300 million years. Even though the era was quite short compared to the Precambrian, three major mass extinctions occurred during this era (Table 18.1). An **extinction** is the total disappearance of all the members of a species or higher taxonomic group. **Mass extinctions** are the disappearance of a large number of species or higher taxonomic groups within an interval of just a few million years (see Fig. 18.18).

Cambrian Animals

The seas of the Cambrian period, which began at about 542 MYA, teemed with invertebrate life (Fig. 18.11). Life became so abundant that scientists refer to this period in Earth's history as the Cambrian explosion. All of today's groups of animals can trace their ancestry to this time, and perhaps earlier, according to new molecular clock data. A *molecular clock* is based on the principle that mutations in certain parts of the genome occur at a fixed rate and are not tied to natural selection. Therefore, the number of DNA base-pair differences tells how long two species have been evolving separately.

Animals that lived during the Cambrian possessed protective outer skeletons known as exoskeletons. These hard-body parts were fossilized more readily than Precambrian soft-bodied organisms. Thus, fossils are more abundant during the Cambrian. Cambrian seafloors were dominated by now-extinct trilobites, which had thick, jointed armor covering them from head to tail. Trilobites, a common Cambrian fossil, are classified as arthropods, a major phylum of animals today.

Invasion of Land

Life first began to move out of the ocean and onto land around 500 MYA. The process of adapting to a land-based environment occurred over a long period of time. Different life-forms, such as plants, invertebrates, and vertebrates, had independent histories, and each group colonized the land at different times.

Plants During the Ordovician period, algae, which were abundant in the seas, most likely began to take up residence in bodies of fresh water. Eventually, algae invaded damp areas on land. The first land plants were nonvascular (did not possess water-conducting tissues), similar to the mosses and liverworts that survive today. The lack of water-conducting tissues limited the height of these plants to a few centimeters. Although the Ordovician evidence is scarce, spore fossils from this time support this hypothesis.

ISTUDY

a. b.

Figure 18.10 Ediacaran fossils. The Ediacaran invertebrates lived from about 630 to 545 MYA. They were all flat, soft-bodied invertebrates. **a.** Classified as *Charniodiscus,* this filter-feeder historically was identified as a sea pen, although that classification is being questioned. **b.** Classified as *Dickinsonia,* these fossils are often interpreted to be segmented worms. However, in the opinion of some, they may be cnidarian polyps.
(a): Sinclair Stammers/Science Source; (b): DeAgostini/SuperStock

Figure 18.11 Examples of fossils from the Cambrian period. *Marrella* has been classified as an arthropod. *Olenoides* is a trilobite (arthropod). *Anomalocaris* is another example of an arthropod. (*Marrella*): The Natural History Museum/Alamy Stock Photo; (*Olenoides serratus*): Kevin Schafer/Alamy Stock Photo; (*Anomalocaris canadensis*): Kevin Schaefer/Alamy Stock Photo

Fossils of seedless vascular plants (those having tissue for water and organic nutrient conduction) date back to the Silurian period. They later flourished in the warm swamps of the Carboniferous period. Club mosses, horsetails, and seed ferns were the trees of that time, and they grew to enormous size. A wide variety of smaller ferns and fernlike plants formed an underbrush (Fig. 18.12*a*).

Invertebrates The jointed appendages and exoskeleton of arthropods are adapted to life on land. Various arthropods—spiders, centipedes, mites, and millipedes—all preceded the appearance of insects on land. Insects, also arthropods, enter the fossil record in the Carboniferous period. One fossil dragonfly from the Carboniferous had a wingspan of nearly a meter. The evolution of wings provided advantages that allowed insects to

ISTUDY

a. b. c.

Figure 18.12 Swamp forests of the Carboniferous period. **a.** Vast swamp forests of treelike club mosses and horsetails dominated the land during the Carboniferous period (Table 18.1). The air contained insects with wide wingspans, such as the predecessors of dragonflies. Amphibians diversified greatly in this environment. **b.** Dragonfly fossil from the Carboniferous period. **c.** Modern-day dragonfly. (a): Richard Bizley/Science Source; (b): powerofforever/iStock/Getty Images; (c): Ablestock/Jupiterimages

radiate into the most diverse and abundant group of animals today. Flying provides a way to escape predators, find food, and disperse to new territories.

Vertebrates Vertebrates are animals with a vertebral column. The vertebrate line of descent began in the early Ordovician period with the evolution of jawless fishes. Jawed fishes appeared later in the Silurian period. Fishes are ectothermic (see Section 29.3) aquatic vertebrates that have gills, scales, and fins. The cartilaginous and ray-finned fishes made their appearance in the Devonian period, which is called the Age of Fishes.

At that time, the seas contained giant predatory fishes covered with protective armor made of external bone. Sharks cruised up deep, wide rivers, and smaller, lobe-finned fishes lived at the river's edge in waters too shallow for large predators. Fleshy fins helped the small fishes push aside debris or hold their place in strong currents, and the fins may have allowed these fishes to venture onto land and lay their eggs safely in inland pools. A large amount of data tells us that lobe-finned fishes were ancestral to the amphibians and to modern-day lobe-finned fishes.

Amphibians are thin-skinned vertebrates that are not fully adapted to life on land, mainly because they need water for reproduction. The Carboniferous swamp forests provided the water they needed, and amphibians adaptively radiated into many different sizes and shapes. Some superficially resembled alligators and were covered with protective scales; others were small and snakelike; and a few were larger plant-eaters. The largest measured 6 m from snout to tail. The Carboniferous period is called the Age of Amphibians.

The process that turned the great Carboniferous forests into the coal we use today to fuel our modern society started during the Carboniferous period. The weather turned cold and dry, and this brought an end to the Age of Amphibians. A major mass extinction event occurred at the end of the Permian period, bringing an end to the Paleozoic era and setting the stage for the Mesozoic era.

The Mesozoic Era

Although a severe mass extinction occurred at the end of the Paleozoic era, the evolution of certain types of plants and animals continued into the beginning of the Mesozoic era. Nonflowering seed plants (collectively called gymnosperms) evolved and spread during the Paleozoic, becoming the dominant plant life. Cycads are short and stout, with palmlike leaves that produce large cones. Cycads and related plants were so prevalent during the Triassic and Jurassic periods that these periods are sometimes called the Age of Cycads. Reptiles can be traced back to the Permian period of the Paleozoic era. Unlike amphibians, reptiles can thrive in a dry climate, because they have scaly skin and lay a shelled egg, which hatches on land. Reptiles underwent an adaptive radiation during the Mesozoic era to produce forms that lived in the sea and on the land, as well as forms that could fly. One group of reptiles, the therapsids, had several mammalian skeletal traits.

During the Jurassic period, large, flying reptiles called pterosaurs ruled the air, and giant marine reptiles with paddlelike limbs ate fishes in the sea. But on land, dinosaurs dominated while the evolving mammals remained small and less conspicuous.

ISTUDY

Figure 18.13 Dinosaurs of the late Cretaceous period. *Parasaurolophus walkeri,* although not as large as other dinosaurs, was one of the largest plant-eaters of the late Cretaceous period. The crest atop its head was about 2 m long, and it may have been used as a way to regulate body temperature or as a way to communicate by making booming calls. Also living at that time were the rhinolike dinosaurs represented here by *Triceratops* (*left*), another herbivore. Chase Studio/ Science Source

Although the average size of the dinosaurs was about that of a crow, many giant species developed. The gargantuan *Apatosaurus* and the armored, tractor-sized *Stegosaurus* fed on cycad seeds and conifer trees. The size of a dinosaur such as *Apatosaurus* is hard for us to imagine. It was 4.5 m tall at the hips and 27 m in length, and it weighed about 40 tons. A large body size is beneficial in ectothermic (cold-blooded) animals where the surface-area-to-volume ratio was favorable for retaining heat. Some data also suggest that some dinosaurs were endothermic (warm-blooded).

During the Cretaceous period, great herds of rhinolike dinosaurs, *Triceratops,* roamed the plains, as did *Tyrannosaurus rex. T. rex* was a carnivore, perhaps a part-time scavenger, filling the same ecological role that lions do today. *Parasaurolophus* was a long-crested, duck-billed dinosaur (Fig. 18.13). The long, hollow crest was bigger than the rest of its skull and may have functioned as a thermoregulatory organ, or as a resonating chamber for making booming calls during mating or helping members of a herd locate each other. In comparison to *Apatosaurus, Parasaurolophus* was small. It was less than 3 m tall at the hips and weighed only about 3 tons. Still, it was one of the largest plant-eaters of the late Cretaceous period and fed on pine needles, leaves, and twigs. *Parasaurolophus* was easy prey for large predators; its main defense would have been running away in large herds.

At the end of the Cretaceous period, the dinosaurs became victims of a mass extinction event, which is discussed in Section 18.3.

One group of bipedal dinosaurs, called theropods, includes the Tyrannosaurs and various raptors, such as *Velociraptor.* This group most likely gave rise to the birds, whose fossil record includes the famous *Archaeopteryx.*

Up until 1999, Mesozoic mammalian fossils largely consisted of teeth. This changed when a fossil found in China was dated at 120 MYA and named *Jeholodens.* The animal, identified as a mammal, apparently looked like a long-snouted rat. Surprisingly, *Jeholodens* had sprawling hindlimbs as do reptiles, but its forelimbs were under the belly, as in today's mammals.

Small mammals appeared early in the Mesozoic and were very diverse in the Jurassic and early Cretaceous periods. By the end of the Cretaceous, the common ancestors of each of the modern orders of mammals had evolved. However, by the late Cretaceous, the majority of the Jurassic and early Cretaceous mammalian diversity had gone extinct.

The Cenozoic Era

The Cenozoic era is divided into the Quaternary, Neogene, and Paleogene periods. A recent change to the geologic timescale (see Table 18.1), called the Anthropocene, was introduced in 2020 to reflect the need to identify a new epoch in the geological time frame due to human-induced (anthropogenic) changes to the biosphere, such as biodiversity loss and climate change.

There is some debate as to when this time frame actually began. Some scientists place it at the start of the Agricultural Revolution (8,000–12,000 years ago). Others prefer that it be placed at the beginning of the Industrial Revolution, or with the dropping of the first atomic bomb in the mid-twentieth century.

Mammalian Diversification

At the end of the Mesozoic era, mammals began an adaptive radiation into the many habitats now left vacant by the demise of the dinosaurs. Mammals are endotherms that have hair to help keep body heat from escaping. Their name refers to the presence of mammary glands, which produce milk to feed their young. Mammals were very diverse in the Mesozoic, but by the start of the Paleocene epoch, the majority of Mesozoic mammal diversity had gone extinct.

By the end of the Eocene epoch, all the modern orders of mammals had evolved—representatives of modern mammal lineages were the only mammals left in existence. Mammals underwent adaptive radiation into a number of environments. Several species of mammals, including bats, took to the air. Ancestors of modern whales, dolphins, and manatees began their return to an aquatic lifestyle. On land, herbivorous hoofed mammals populated the

ISTUDY

Figure 18.14 Mammals of the Oligocene epoch. The artist's representation of these mammals and their habitat vegetation is based on fossil remains. Chase Studio/Science Source

Figure 18.15 Woolly mammoths of the Pleistocene epoch. Woolly mammoths were animals that lived along the borders of continental glaciers. Leonello Calvetti/Science Photo Library/Alamy Stock Photo

forests and grasslands and were preyed upon by carnivorous mammals. Many of the types of herbivores and carnivores of the Oligocene epoch are extinct today (Fig. 18.14).

Evolution of Primates

The ancestors of modern primates appeared during the Eocene epoch about 55 MYA. The first primates were small, squirrel-like animals. Flowering plants (collectively called angiosperms) were already diverse and plentiful by the Cenozoic era. Many species of tree-sized angiosperms had evolved by the Eocene. Some primates adapted to living in trees, where protection from predators and food in the form of fruit was plentiful.

Ancestral apes appeared during the Oligocene epoch. These primates were adapted to living in the open grasslands and savannas. Apes diversified during the Miocene and Pliocene epochs and gave rise to the first hominids, a group that includes humans. Many of the skeletal differences between apes and humans relate to the fact that humans walk upright. Exactly what caused humans to adopt bipedalism is still being debated.

The world's climate became progressively colder during the Tertiary period. The Quaternary period begins with the Pleistocene epoch, which is known for multiple ice ages in the Northern Hemisphere. During periods of glaciation, snow and ice covered about one-third of the land surface of the Earth. The Pleistocene epoch was an age of not only humans but also mammalian **megafauna**—giant ground sloths, beavers, wolves, bison, woolly rhinoceroses, mastodons, and mammoths (Fig. 18.15). Humans are still in existence, but the oversized mammals have gone extinct. Mammalian megafauna had died out by the end of the Pleistocene. One hypothesis is that human hunting was at least partially responsible for the extinction of these animals. Climate change is another hypothesis.

Check Your Progress 18.2

1. Determine the age of a fossil that contains 25% of its original ^{14}C.
2. Identify two features of organelles that support the endosymbiotic theory of organelle evolution.
3. List the sequence of events in the Precambrian that led to the evolution of heterotrophic and photosynthetic protists.
4. List the major events in the history of life that occurred during each of the Earth's eras.

ISTUDY

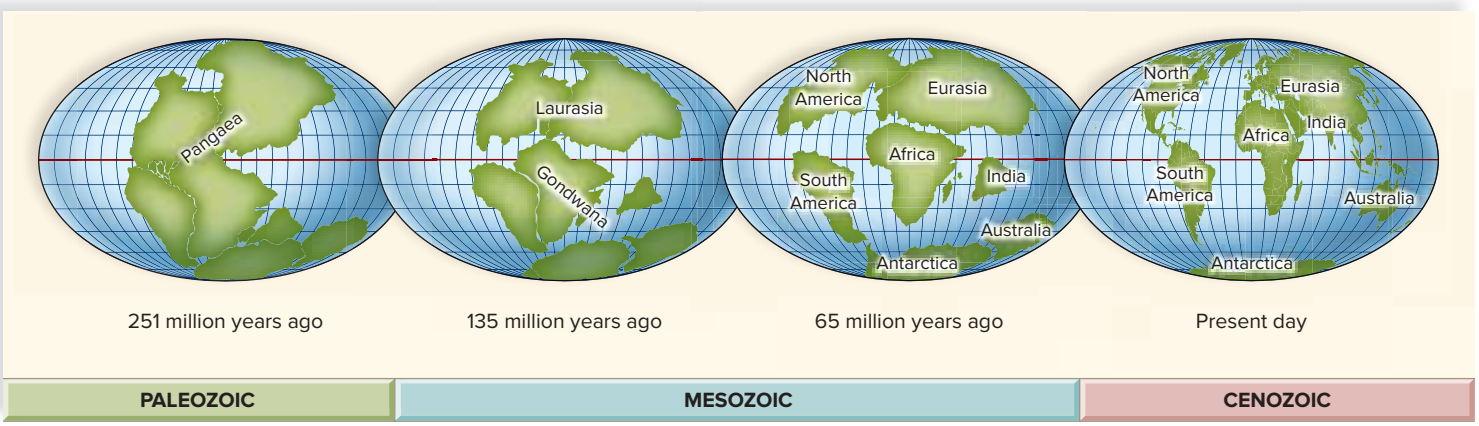


Figure 18.16 Continental drift. About 251 MYA, all the continents were joined into a supercontinent called Pangaea. During the Mesozoic era, the joined continents of Pangaea began moving apart, forming two large continents called Laurasia and Gondwana. Then, all the continents began to separate. Presently, North America and Europe are drifting apart at a rate of about 2 cm per year.

18.3 Geological Factors that Influence Evolution

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify who proposed the theory of continental drift, as well as cite the evidence provided to support this theory.
2. Describe plate tectonics and how it explains the drifting of continents.
3. Interpret biogeographical and geological evidence that supports continental drift.
4. Describe the geological points at which mass extinctions occurred and give one proposed cause for each extinction event.

In the past, it was thought that the Earth's crust was immobile, that the continents had always been in their present positions, and that the ocean floors were only a catch basin for the debris that washed off the land. But in 1920, Alfred Wegener, a German meteorologist, presented data from a number of disciplines to support his hypothesis of continental drift.

Continental Drift

Continental drift was finally confirmed in the 1960s, establishing that the continents are not fixed. Instead, their positions and the positions of the oceans have changed over time (Fig. 18.16). During the Paleozoic era, the continents joined to form one supercontinent, which Wegener called Pangaea (Gk. *pangea,* "all lands"). First, Pangaea divided into two large subcontinents, called Gondwana and Laurasia, and then these also split to form the continents of today. Presently, the continents are still drifting in relation to one another.

Continental drift explains why the coastlines of several continents are roughly mirror images of each other—for example, the outline of the west coast of Africa matches that of the east coast of South America. The same geological structures are also found in many of the areas where the continents touched. For example, based on geological data, a single mountain range runs through South America, Antarctica, and Australia.

Continental drift also explains the unique distribution patterns of several fossils. Fossils of the seed ferns (*Glossopteris*) have been found on all the continents, which supports the theory that all continents were once joined (Fig. 18.17). Similarly, the fossil reptile *Cynognathus* is found in Africa and in South America, and an assemblage of early mammal-like reptiles, the *Lystrosaurus* group, was discovered in Antarctica, far from Africa and Southeast Asia, where it also occurs.

With mammalian fossils, the situation is different: Australia, South America, and Africa share lineages of mammals that were widely distributed across Gondwana. For example, each continent has representatives of the rodent lineage. However, each continent also has its own unique mammal lineages that evolved in isolation after the continents separated.

The mammalian biological diversity of today's world is the result of isolated evolution on separate continents. For example, as mentioned in Section 15.3, marsupials are endemic only to

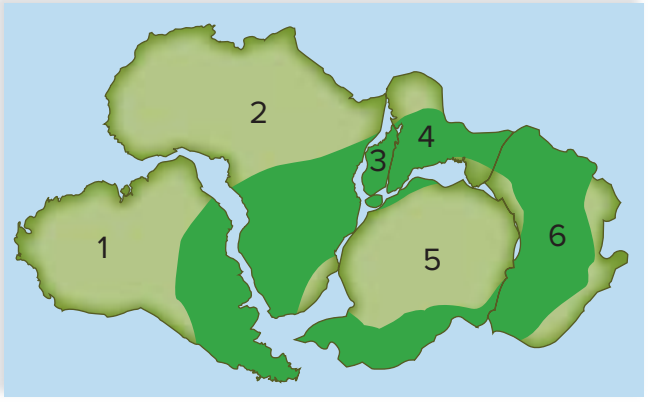


Figure 18.17 Biogeography of *Glossopteris* supports continental drift. Fossils of the fern *Glossopteris* (represented by the dark-green shading) are distributed throughout what was Gondwana.

ISTUDY

South America and Australia, which were connected when the landmasses were a supercontinent (see Fig. 18.16). The fossil record suggests that marsupials were abundant in South America up until about 7 MYA. Marsupials are more prevalent in the Australian biogeographical region than in South America. This is most likely due to marsupials evolving in the Americas and reaching Australia when the southern continents were still joined. Once Australia broke off, marsupials were able to diversify without competition from the placental mammals that evolved elsewhere. Today, the only placental mammals native to Australia are bats and rodents. The rest of the placental mammals were introduced to the continent relatively recently. In contrast, placental mammals are prevalent in the Americas, and much of the marsupial diversity has disappeared on this continent.

Plate Tectonics

The answer to why continents drift has been suggested through a branch of geology known as **plate tectonics** (Gk. *tektos,* "fluid, molten, able to flow"). This idea shows that the Earth's crust is fragmented into slablike plates that float on a lower, hot mantle layer. The continents and the ocean basins are a part of these rigid plates, which move like conveyor belts. At ocean ridges, seafloor spreading occurs as molten mantle rock rises and material is added to the ocean floor. Seafloor spreading causes the continents to move a few centimeters a year, on average. At *subduction zones,* the forward edge of a moving plate sinks into the mantle and is destroyed, forming deep-ocean trenches bordered by volcanoes or volcanic island chains.

The Earth isn't getting bigger or smaller, so the amount of oceanic crust being formed is as much as that being destroyed. When two continents collide, the result is often a mountain range—for example, the Himalayas resulted when India collided with Eurasia. The place where two plates meet and scrape past one another is called a *transform boundary*. The San Andreas fault in Southern California is at a transform boundary, and the movement of the two plates is responsible for the many earthquakes in that region. Earthquakes leave visible evidence of the movement of plates at transform boundaries. In rare cases, the movement of the plates has been witnessed by people as it occurs—roads diverge, the ground uplifts, and cracks emerge during seismic activity. Examples of this were seen in the massive earthquake that took place in Japan in 2011.

Mass Extinctions

At least five mass extinctions have occurred throughout history: at the ends of the Ordovician, Devonian, Permian, Triassic, and Cretaceous periods (Fig. 18.18 and Table 18.1). Mass extinction can be caused by cataclysmic events or by a more gradual process brought on by environmental change.

These ideas were brought to the fore when Walter and Luis Alvarez proposed in 1977 that the Cretaceous extinction, during which the dinosaurs died out, was due to a bolide. A bolide is a large, crater-forming projectile that impacts Earth. The Alvarezes found that Cretaceous clay contains an abnormally high level of iridium, an element that is rare in the Earth's crust but more common in asteroids and meteorites. According to the Alvarezes, the impact of the object that ended the Cretaceous was the equivalent of an explosion 10,000 times that of the world's entire nuclear arsenal. An impact of this magnitude would have blasted into the atmosphere a layer of ash and soot so expansive it would have blocked the sun.

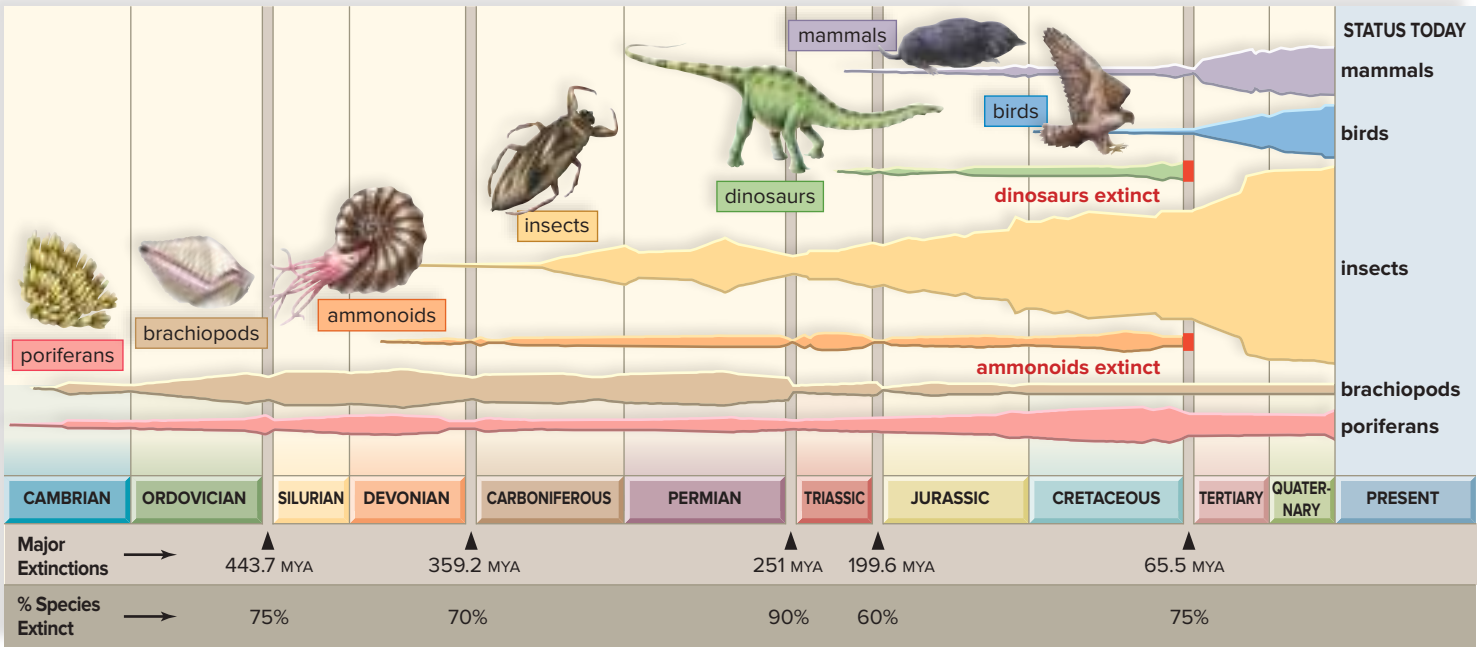


Figure 18.18 Mass extinctions. Five significant mass extinctions and their effects on the abundance of certain forms of marine and terrestrial life. The thickness of the horizontal, colored bars indicates the varying abundance of each life-form considered. MYA = million years ago

A crater that could have been caused by a meteorite impact of this magnitude was found in the Caribbean–Gulf of Mexico region on the Yucatán peninsula. A layer of soot alongside the iridium has dated the impact to the end of the Cretaceous and the beginning of the Tertiary periods. This crater is known as the Chicxulub crater. Certainly, continental drift contributed to the Ordovician extinction. This extinction occurred after Gondwana arrived at the South Pole. Immense glaciers, which drew water from the oceans, chilled even once-tropical land. Marine invertebrates and coral reefs, which were especially hard hit, didn't recover until Gondwana drifted away from the pole and warmth returned.

The mass extinction at the end of the Devonian period saw an end to 70% of marine invertebrates. Helmont Geldsetzer of Canada's Geological Survey notes that iridium has also been found in Devonian rocks in Australia, suggesting that a bolide event was involved. Some scientists believe that this mass extinction could have been due to the movement of Gondwana back to the South Pole.

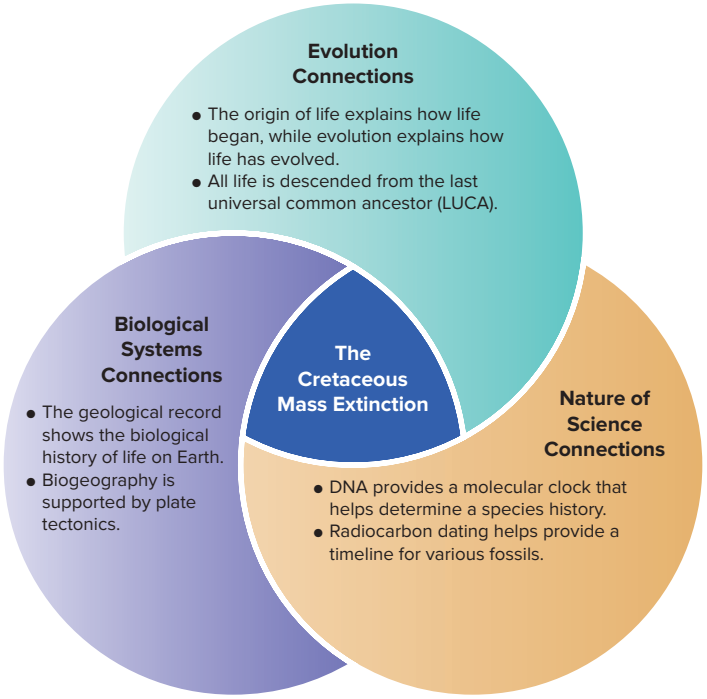
The extinction at the end of the Permian period was quite severe: 90% of species disappeared. The latest hypothesis attributes the Permian extinction to excess carbon dioxide. When Pangaea formed, there were no polar ice caps to initiate ocean currents. The lack of ocean currents caused organic matter to stagnate at the bottom of the ocean. Then, as the continents drifted into a new configuration, ocean circulation switched back on. Then, the extra carbon on the seafloor was swept up to the surface, where it became carbon dioxide, a deadly gas for sea life. The trilobites became extinct, and the crinoids (sea lilies) barely survived. Excess carbon dioxide on land led to climate change, which altered the pattern of vegetation. Areas that were wet and rainy became dry and warm, and vice versa. Burrowing animals that could escape land surface changes seemed to have the best chance of survival.

The extinction at the end of the Triassic period is another that has been attributed to the environmental effects of a meteorite collision with Earth. Central Quebec has a crater half the size of Connecticut that some believe is the impact site. The dinosaurs may have benefited from this event, because this is when the first of the gigantic dinosaurs took charge of the land. A second wave occurred in the Cretaceous period, but it ended in dinosaur extinction, as discussed previously.

Check Your Progress 18.3

1. Explain why continents drift and summarize the geological evidence to support plate tectonics.
2. Defend the theory of continental drift with evidence from the biogeographical distribution of organisms.
3. List the mass extinction events that have occurred on Earth and describe the hypotheses that explain the cause of each.

Connecting *the* Concepts



ISTUDY

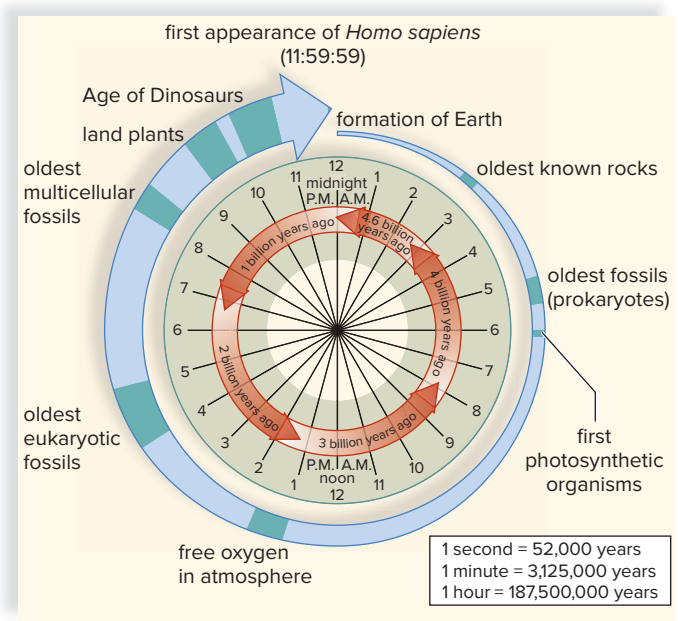
What Did You Learn?

18.1 The Origin of Life

The **last universal common ancestor (LUCA)** is believed to be the ancestor of all life on Earth. **Biomolecules** of living organisms are believed to have formed from nonliving matter, or the process of **abiogenesis.**

- The Oparian-Haldane hypothesis, or **primordial soup hypothesis,** proposes the conditions of early Earth that would have allowed an **abiotic synthesis,** or chemical evolution, of organic molecules from inorganic ones either in the atmosphere or at thermal vents.
- These monomers may have joined together to form polymers at thermal vents according to the **iron–sulfur world hypothesis.** If abiogenesis occurred on land, the sun could have caused **proteinoids** to form, which could have turned into **microspheres** when placed in water.
- The **protein-first hypothesis** assumes that proteins arose prior to DNA. The **RNA-first hypothesis** assumes that RNA evolved first.

The aggregation of polymers inside a fatty acid plasma membrane produced a **protocell (protobiont)** having some enzymatic properties such that it could grow. The basic assemblage of fatty acids into a small sphere is called a **micelle,** which can merge with others to form a **vesicle. Coacervate droplets** are complex units formed from concentrated mixtures of macromolecules under appropriate conditions. A collection of lipids in solution will form bubbles called **liposomes,** leading researchers to formulate the **membrane-first hypothesis.** Biological evolution began with the origin of self-replicating molecules, such as DNA and RNA.



18.2 The History of Life

Fossils provide evidence about the life that has existed on Earth. **Paleontology** is the study of the fossil record.

- **Sedimentation** is the accumulation of particles carried by moving water that leads to the formation of **sediment.** Layers of sediment accumulate to form **strata.**
- **Index fossils** are often found in the stratum and are used as a form of **relative dating. Absolute dating** is accomplished by using **radiometric** techniques, such as **radiocarbon dating,** to assign an actual age to a fossil.
- The accumulation of these data has enabled geologists to establish the **geologic timescale,** which divides the Earth into eras, periods, and epochs.
- The oldest prokaryotic fossils are cyanobacteria, dated about 3.5 BYA, found in strange boulder forms called **stromatolites.** They were the first organisms to add oxygen to the atmosphere, thus forming the early **ozone shield.** The **endosymbiotic theory** states that a nucleated cell engulfed various prokaryotes, forming the first eukaryotic cell about 2.2 BYA. Multicellular animals (the Ediacaran animals) did not evolve until about 630 MYA.
- The Paleozoic era is characterized by the disappearance of all members of a species, or **extinction.** More specifically, there were three **mass extinctions** during this time. A rich animal fossil record starts at the Cambrian period of the Paleozoic era. The evolution of the exoskeleton seems to explain the increased number of fossils at this time. Predation pressures may also have selected for the evolution of a hard external enclosure like an exoskeleton. The fishes were the first vertebrates to diversify and become dominant. Amphibians are descended from lobe-finned fishes.
- Plants invaded the land from water during the Ordovician period. The swamp forests of the Carboniferous period contained seedless vascular plants, insects, and amphibians. This period is sometimes called the Age of Amphibians.
- The Mesozoic era was the Age of Cycads and Reptiles. Mammals and birds evolved from reptilian ancestors. During this era, dinosaurs of enormous size were present. By the end of the Cretaceous period, the dinosaurs were extinct.
- The Cenozoic era is divided into the Tertiary period and the Quaternary period. The Tertiary is associated with the adaptive radiation of mammals and flowering plants that formed vast tropical forests. The Quaternary is associated with the evolution of mammalian **megafauna,** as well as the primates. Grasslands were replacing forests, and this put pressure on primates, which were adapted to living in trees, to respond to an ever-expanding terrestrial habitat. The result may have been the evolution of humans—primates who left the trees.

18.3 Geological Factors that Influence Evolution

The continents are on massive plates that move, carrying the land with them. **Continental drift** helps explain the distribution pattern of today's land organisms. **Plate tectonics** is a branch of geology that helps explain the movement of the Earth's plates, which contributes to continental drift. Five mass extinctions have played a dramatic role in the history of life on Earth. It has been suggested that the extinction at the end of the Cretaceous period was caused by the impact of a large bolide**,** or crater-forming meteorite. Evidence indicates that other extinctions have a similar cause as well. It has also been suggested that tectonic, oceanic, and climatic fluctuations, particularly due to continental drift, can bring about mass extinctions.

ISTUDY

Thinking Critically About Biology

1. Critics of evolution often cite the fact that several hypotheses are being tested to help explain various stages of the origin of life. Explain how having several hypotheses for the evolution of monomers does not negate the fact that the evolution of monomers was the first stage of evolution in the origin of life.
2. You were asked to supply an evolutionary tree of life and decided to use Figure 18.9. How is this tree consistent with evolutionary principles?
3. Discuss how excess carbon dioxide led to a mass extinction in the Permian period.

Making It Relevant to You

1. How did the asteroid impact affect the evolution of mammals?
2. How would life be affected if an asteroid struck Earth today?
3. What can the history of life on Earth tell us about our future?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

19

Organizing Life

Hans Wagemaker/Shutterstock

Classifying Jaguars

Jaguars are the largest of the cats living in the Americas and can weigh up to 148 kg (325 lb) with a body length of 250 cm (98 in.), including the tail. Their current range extends from the southwestern United States, throughout most of Central America, and as far south as Paraguay and northern Argentina. It is estimated there are between 57,000 and 64,000 jaguars across their entire range.

Historically, nine subspecies of jaguars have been recognized. Subspecies classification can be established when a population is geographically isolated from other members of its species and there are significant physical differences. Under the current taxonomy revision, all jaguars have been assigned to a single species, *Panthera onca*. While genetic, physical, and biogeographical analysis of jaguars across their range has not produced enough differences to warrant the continuation of the subspecies classification, four general regional groups have been identified across the jaguar's range: Mexico and Guatemala, southern Central America, north of the Amazon, and south of the Amazon. This information greatly helps scientists better understand jaguar ecology.

As you read through this chapter, think about the following questions:

1. How are the study of macroevolution and systematic biology interrelated?
2. Why is evolutionary history important in the classification of biodiversity?

Chapter Outline

19.1 Systematic Biology
19.2 The Three-Domain System
19.3 Phylogeny

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.4 What is biodiversity?

Section 12.3 Why is DNA called the "genetic code of life"?

Section 17.2 How does adaptive radiation result in the formation of new species?

Following *the* Themes

CHAPTER 19 – ORGANIZING LIFE

Evolution	Systematic biology reconstructs evolutionary history as a tree showing common ancestors and their descendants.
Nature of Science	The tree of life is a phylogeny, or an evolutionary tree, that represents the best hypothesis of the evolutionary history of life on Earth.
Biological Systems	The evolutionary relationship among living organisms, in the form of a phylogeny, serves as the framework for addressing issues in conservation, medicine, and agriculture.

ISTUDY

19.1 Systematic Biology

Learning Outcomes

Upon completion of this section, you should be able to

1. Differentiate among taxonomy, classification, and systematic biology.
2. Reconstruct the levels of the Linnaean classification hierarchy.
3. Identify the genus and species of an organism from its scientific name.

In Section 17.3, we examined macroevolution, evolutionary change that results in the formation of new species. Macroevolution is the source of past and present biodiversity. **Systematics** is the study of biodiversity, which helps us understand the evolutionary relationships between species. Systematics is a quantitative science that uses **traits** of living and fossil organisms to infer the relationships among organisms over time.

The Field of Taxonomy

Taxonomy (Gk. *tasso,* "arrange, classify"; *nomos,* "usage, law") is the branch of systematic biology that identifies, names, and organizes biodiversity into related categories (Fig. 19.1). A **taxon** (*pl.,* taxa) is the general name for a group containing an organism or a group of organisms that exhibits a set of shared traits. **Classification** is the process of naming and assigning organisms or groups of organisms to a taxon. For example, the taxon Vertebrata contains organisms with a bony spinal column. As another example, the taxon Canidae contains the wolf (*Canis lupus*) and its close relative, the domestic dog (*Canis familiaris*).

Taxonomists, scientists who study taxonomy, strive to classify all of the life on Earth. The methods used to classify living organisms have changed throughout history. The ancient Greek philosopher Aristotle was interested in taxonomy, and he sorted organisms into groups, such as horses, birds, and oaks, based on a set of shared traits. Early taxonomists just after Aristotle similarly relied on physical traits to classify organisms. This method proved to be problematic, however; many features of organisms were similar not because they shared a common ancestor but because of convergent evolution (see Section 17.2). For example, animals that have wings could be grouped into a single taxonomic group, but birds, bats, and beetles, all of which have wings, are profoundly different in many other ways.

Today, taxonomists attempt to classify organisms into *natural groups*, groupings of organisms that represent a shared evolutionary history. Natural groups are classified by using a set of traits to construct a **phylogeny,** or evolutionary "family tree," that represents the evolutionary history of taxa. This evolutionary history is then used to classify taxa based on shared ancestry.

Advances in DNA technology allow modern systematic biologists to compare traits other than external features to classify organisms. For example, a phylogeny of animals constructed from DNA sequences clearly shows that beetles, birds, and bats have wings that evolved at different times in the history of life (see Fig. 19.9 for an example of DNA sequence differences). This means that birds, bats, and beetles do not share a common ancestor with wings. Rather, wings originated three times independently, on three different branches of the tree of life, as a result of convergent evolution.

Figure 19.1 Classifying organisms. Taxonomy is the classification of organisms. (Archaea): Steve Gschmeissner/Science Source; (bacteria): Steve Gschmeissner/Science Photo Library/Getty Images; (plant): Mike Liu/honestmike/iStock/Getty Images; (fungi): Worraket/Shutterstock; (sea star): Laura Dinraths/Shutterstock; (mountain lion): IPGGutenbergUKLtd/iStock/Getty Images

Linnaean Taxonomy

The classification hierarchy that taxonomists use today was created by the eighteenth-century scientist Carolus Linnaeus, considered to be the founder of modern taxonomy (see Section 15.1). Linnaeus's system was developed as a way to organize biodiversity. In the mid-eighteenth century, Europeans traveled to distant parts of the world and described, collected, and sent back to Europe examples of plants and animals they had not encountered before. During this time of discovery, Linnaeus developed **binomial nomenclature,**

ISTUDY

a.

b. *Lilium canadense*

c. *Lilium bulbiferum*

Figure 19.2 Carolus Linnaeus. **a.** Linnaeus was the founder of taxonomy and devised the binomial system of naming and classifying organisms. His original name was Karl von Linne, but he later latinized it because of his fascination with scientific names. Linnaeus was particularly interested in classifying plants. **b, c.** Both of these two lilies are species in the same genus, *Lilium.* (a): Pixtal/age fotostock; (b): La_Corivo/iStock/Getty Images; (c): Roberto Facchini/imageBROKER/Getty Images

part of his classification system in which each species receives a unique two-part Latin name (Fig. 19.2).

As an example, *Lilium bulbiferum* and *Lilium canadense* are two different species of lily. The first word, *Lilium,* is the genus (*pl.,* genera), a classification category that can contain many species. The second word, known as the **specific epithet,** refers to one species within that genus. The specific epithet sometimes tells us something descriptive about the organism. Notice that the scientific name is in italics. The species name is designated by the full binomial name—in this case, either *Lilium bulbiferum* or *Lilium canadense*. The specific epithet without the genus gives no clue as to species—just as a house number alone without the street name is useless for finding an address. The genus name can be used alone, however, to refer to a group of related species.

Scientific names are derived in a number of ways. Some scientific names are descriptive—for example, *Acer rubrum* for the red maple (*Acer, "*maple"; *rubrum, "*red"). Other scientific names include geographic descriptions, such as *Alligator mississippiensis* for the *American alligator.* Scientific names can also include eponyms (derived from the name of a person), such as the owl mite *Strigophilus garylarsonii* (named after the cartoonist Gary Larson). Many scientific names are derived from mythical characters, such as *Iris versicolor,* named for Iris, the goddess of the rainbow.

Scientists use Latin, rather than common, names to describe organisms for a variety of reasons. First, a common name varies from country to country because of language differences. Second, even people who speak the same language sometimes use different common names to describe the same organism. For example, bowfin, grindle, choupique, and cypress trout all refer to the same common fish, *Amia calva.* Furthermore, between countries, the same common name is sometimes given to different organisms. A robin (*Erithacus rubecula*) in England is very different from a robin (*Turdus migratorius*) in the United States. Latin, however, is a universal language that not too long ago was well-known by most scholars, many of whom were physicians or clerics. When scientists throughout the world use scientific binomial names, they avoid the confusion caused by common names.

The Classification Hierarchy

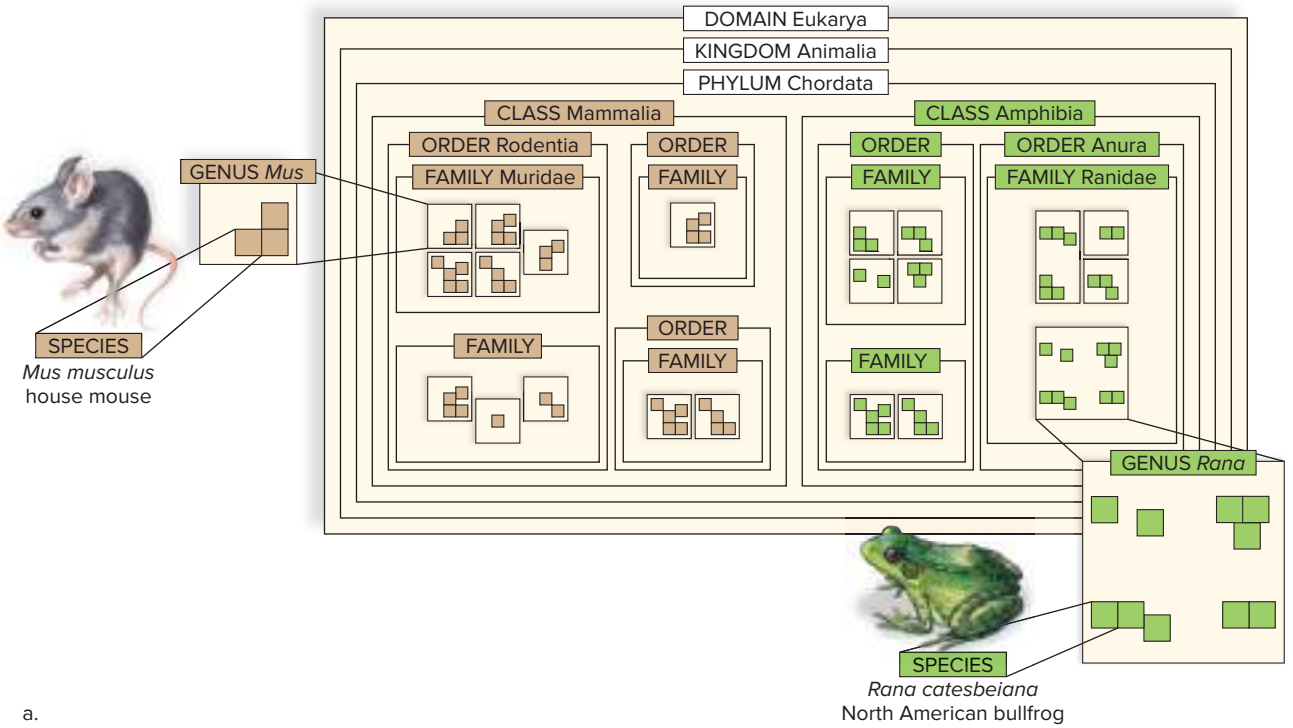
The binomial nomenclature system of Linnaeus is used to classify species. Today, taxonomists use a nested, hierarchical set of categories to classify organisms. Each taxon is given a name and a rank according to which of the following set of major taxonomic groups it belongs. The organisms that fill a particular classification category are distinguishable from other organisms by their shared set of traits. Historically, this included (from most specific to least specific) the classifications of: **species, genus, family, order, class, phylum, kingdom,** and **domain** (Fig. 19.3*a*).

However, science has progressed since the time of Linneaus, and the study of similarities and differences has begun to alter our view of the traditional classification system. In Section 1.2, we introduced the concept of a **supergroup,** which divides the kingdoms previously found in the Eukarya into groups based on genetic similarities. Figure 19.3*b* shows how this compares in the classification of toads and mice. We will explore supergroups more in Section 19.2.

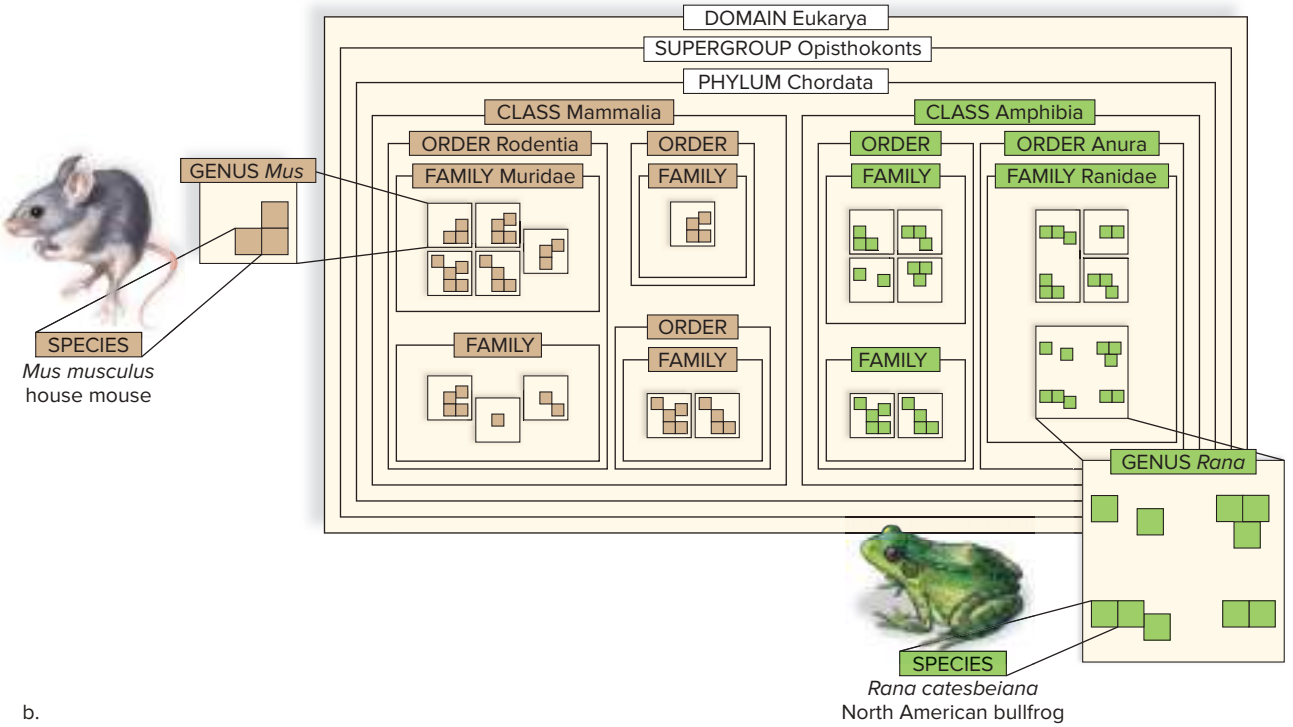
Organisms in the same domain have general traits in common, whereas those in the same species have quite specific traits in common. For example, in Figure 19.3*a*, the kingdom Animalia includes all animals. Within the kingdom Animalia is the phylum Chordata, a taxonomic group that contains only those animals with a spinal cord. Within the phylum Chordata is the class Mammalia, which contains animals that have spinal cords (phylum Chordata) and, among other characteristics, mammary glands. The species is the most exclusive of all the categories, because it contains only a single type of organism. The house mouse, *Mus musculus,* is a single species of mouse in the family Muridae, a family in the order Rodentia (one of several orders in the class Mammalia).

Several species can share the same genus, and many genera can be in the same family, and so on. The number of categories (boxes) in each classification depends on the amount of diversity of a particular

ISTUDY



a.



b.

Figure 19.3 The classification system. Organisms are classified into a series of nested, hierarchical categories. The most nested category is the species. Historically, this classification system (**a**) included species, genus, family, order, class, phylum, kingdom, and domains. **b.** A new classification system replaces kingdoms in the Eukarya with supergroups.

group. For example, the family Muridae (mice, rats, and hamsters) can be divided into 150 to 250 genera (depending on the study) and has up to 1,000 species, whereas the family Ranidae (frogs) is believed to have between 22 to 35 genera and as many as 600 species.

The classification hierarchy is very useful, because it allows scientists to organize the diversity of life, but it is important to remember that the hierarchy was created by scientists and thus does not represent any special relationship among organisms in nature. For example, grouping monkeys and apes into the order Primates tells scientists something about their evolutionary history, but being in the same order does not mean much to the genetics or behavior of monkeys or apes. The classification hierarchy, if

ISTUDY

THEME Nature of Science

DNA Barcoding of Life

Traditionally, taxonomists relied on anatomical traits to tell species apart. Although useful, physical features have several limitations when used as the only means to define a species. The identification of anatomical traits often requires the assistance of taxonomic experts who specialize in a particular group of organisms. In recent times, the number of species has far exceeded the number of available experts. This puts the cataloging of the world's biodiversity on a slow path—much slower than the rate at which our natural areas are disappearing.

The Consortium for the Barcode of Life (CBOL) proposes that any scientist, not just taxonomists, could use a sample of DNA to identify any organism on Earth. Just as a barcode, or UPC code, is used as a unique identifier of products on a store shelf, the CBOL suggests it would be possible to use the base sequence in DNA to develop a barcode (a DNA sequence) for each living organism (Fig. 19A). The order of DNA nucleotides—A, T, C, and G—within a particular gene common to the organisms in each kingdom would fill the role of the store barcode's sequence of numbers.

Speedy DNA sequencing would not only be a boon to efforts to catalog a rapidly disappearing biodiversity, it would also have practical applications. For example, rapid identification of species could help farmers with a pest attacking their crops, doctors having to choose the correct antivenin for people who have experienced snakebites, and customs agents noting invasive species or illegal trade in endangered species. Already, the CBOL has accumulated hundreds of thousands of DNA sequences (the barcode) representing species across the diversity of life.

Interestingly, a pair of New York City high school students, Kate Stoeckle and Louisa Strauss, found a commercial application for the CBOL database by identifying the fish sold in markets and sushi restaurants in Manhattan, New York. They collected 60 fish samples from four restaurants and ten grocery stores in Manhattan, which they sent off to have the DNA segment, the barcode, sequenced and compared to a global library of fish barcodes representing nearly 5,500 fish species. Their results sent a wave of controversy throughout Manhattan and beyond: two of the four restaurants, and six of the ten grocery stores, sold fish that had been mislabeled. Most of the mislabeled fish were being sold as more expensive species. For example, Mozambique tilapia, a commonly farmed fish selling for $1.70 per pound wholesale, was being sold as albacore tuna at $8.50 per pound (Fig. 19B). In one case, they found an endangered fish, the Acadian redfish, being sold as red snapper!

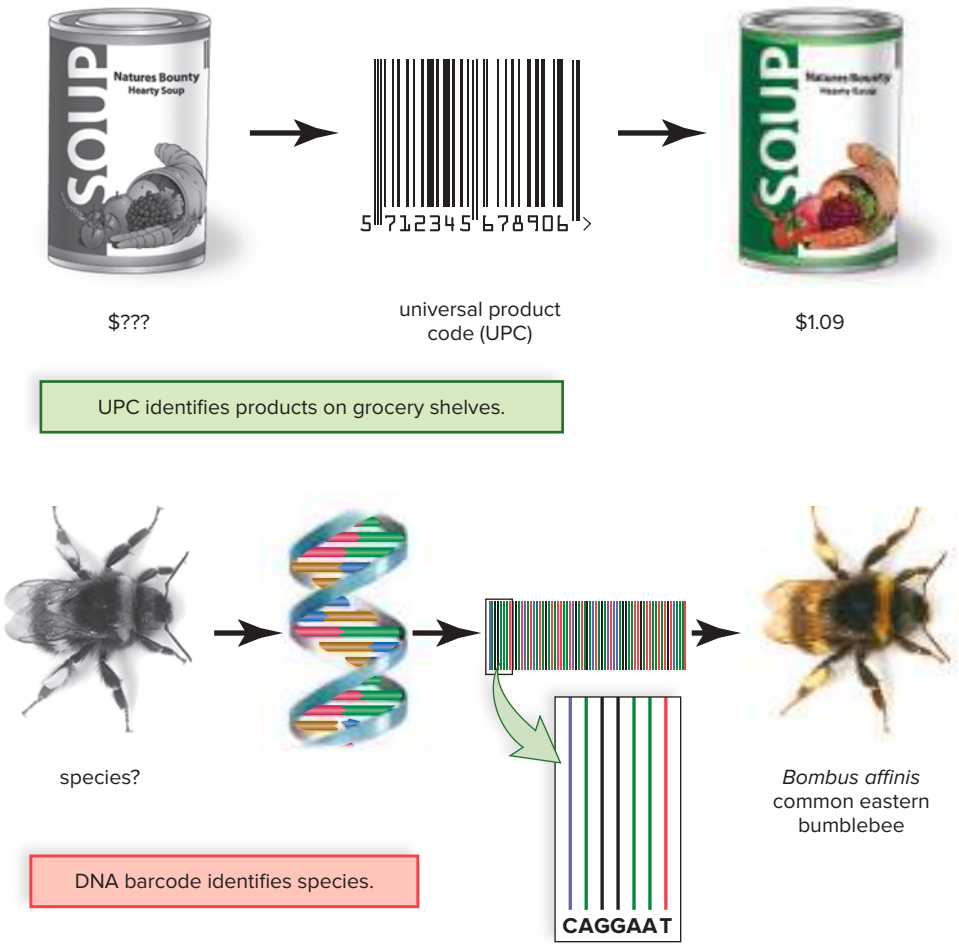


Figure 19A Illustration of the barcoding concept. Much as a barcode (UPC) can identify the type and price of a can of soup, a DNA barcode can be used to identify a species. This is possible because each UPC is unique to a particular product, and each DNA barcode, based on a DNA sequence, is unique to an individual species. (both bumblebees): Stefan Sollfors/Alamy Stock Photo

Questions to Consider

1. How might systematic biologists use DNA barcoding to speed up the classification of biodiversity?
2. Propose additional ways that DNA barcoding could aid in managing modern societal problems, such as global warming, crime, disease, and the conservation of biodiversity.

Figure 19B Mislabeled fish. DNA barcoding identified mislabeled fish in some New York City restaurants. (tuna): MISCELLANEOUSTOCK/Alamy Stock Photo; (tilapia): Ammit/Alamy Stock Photo

ISTUDY

based on natural groups, should be viewed as the best working hypothesis of evolutionary relationships.

As with any scientific hypothesis, the hierarchy, and the placement of organisms within it, is revised with the addition of new information. That is why you can find the same organism classified differently in older textbooks or even among taxonomists. This uncertainty is part of the nature of science and is one of its greatest principles.

Linnaeus set the standard of binomial nomenclature in the mid-1700s, but there were no official rules for classifying organisms until the mid-1800s. During this 100-year period, problems with name confusion arose when scientists in different parts of the world, who did not communicate with each other (remember, there was no internet!), began to give the same Latin names to different species or to develop their own version of Linnaeus's classification system.

The first attempt to make standardized rules of **nomenclature,** the procedure of assigning scientific names to taxonomic groups, occurred in 1842 at a meeting of scientists, among them Charles Darwin. In 1961, after 120 more years of revisions, the International Code of Zoological Nomenclature (ICZN) was officially accepted as the universal guide for naming animals. Today, the International Commission on Zoological Nomenclature is the keeper of the ICZN, and all scientists of the world use its rules to classify animal groups. The International Code of Botanical Nomenclature is responsible for establishing the policies for the naming of plants and certain groups of algae and fungi.

Despite a universal set of rules and over 250 years of taxonomy, only a fraction of the millions of species now living on Earth have been classified. Some taxonomic groups are classified more completely than others; the naming of birds and mammals may be complete, but there are millions of insects and microorganisms that remain to be discovered and classified.

The task of identifying and naming the species of the world is a daunting one. A new fast and efficient way of identifying species based on their DNA is described in the Nature of Science feature, "DNA Barcoding of Life." This method, called DNA "barcoding," compares a short fragment of DNA sequence from an unknown organism to a large database of sequences from known organisms. The similarities and differences between the nucleotide sequence of the unknown organism and the sequences in the database can help determine which taxonomic group the organism likely belongs to or if it is something new to science.

DNA barcoding does not always get the taxonomy correct, however, and for this reason it has been criticized by some taxonomists as being too simplistic. Nevertheless, it is a potentially powerful way to rapidly and inexpensively catalog at least a portion of the world's biodiversity.

Check Your Progress 19.1

1. Describe the relationship among classification, systematic biology, and taxonomy.
2. Use the Linnaean classification system to fully classify the human species *Homo sapiens.*
3. Explain why the grouping together of birds and bats based on having wings does not represent a natural group.

19.2 The Three-Domain System

Learning Outcomes

Upon completion of this section, you should be able to

1. List the three domains of life.
2. Summarize two characteristics that define each of the three domains.

From Aristotle's time to the middle of the twentieth century, biologists recognized only two kingdoms: kingdom Plantae (plants) and kingdom Animalia (animals). Plants were literally organisms that were planted (immobile), whereas animals were animated (moved about). In the 1880s, a German scientist, Ernst Haeckel, proposed adding a third kingdom: the kingdom Protista (protists), which included single-celled microscopic organisms but not multicellular, largely macroscopic ones.

In 1969, R. H. Whittaker expanded the classification system to the **five-kingdom system:** Monera, Protista, Fungi, Plantae, and Animalia. Organisms were placed in these kingdoms based on the type of cell (prokaryotic or eukaryotic), complexity (single-celled or multicellular), and type of nutrition. Kingdom Monera contained all the prokaryotes, which are organisms that lack a membrane-bound nucleus. These single-celled organisms were collectively called the bacteria. The other four kingdoms contain types of eukaryotes, which we describe later.

Defining the Domains

In the late 1970s, Carl Woese and his colleagues at the University of Illinois were studying the relationships among the prokaryotes by comparing the nucleotide sequences of ribosomal RNA (rRNA). Woese found that the rRNA sequences of prokaryotes that lived at high temperatures or produced methane are quite different from that of all the other types of prokaryotes and from all eukaryotes. Therefore, he proposed that there are two groups of prokaryotes (rather than one group, the Monera, in the five-kingdom system). Further, Woese found that these two groups of prokaryotes have rRNA sequences so fundamentally different from each other that they should be assigned to separate domains, the category of classification that is higher than the kingdom category. The two designated domains are **domain Bacteria** and **domain Archaea.** He placed the eukaryotes in a third domain, the **domain Eukarya** (Table 19.1).

The phylogenetic tree shown in Figure 19.4 is based on Woese's rRNA sequencing data. The data suggested that both bacteria and archaea evolved early in the history of life from the last universal common ancestor (LUCA) (see Fig. 18.1). Later, the eukarya diverged from the archaea lineage. This means that the bacteria are members of the oldest lineage of living organisms on Earth, and the eukaryotes, the youngest lineage, are more closely related to the archaea than to the bacteria (Fig. 19.4).

Domain Bacteria

Bacteria are such a diversified group that they can be found in large numbers in nearly every environment on Earth. The

ISTUDY

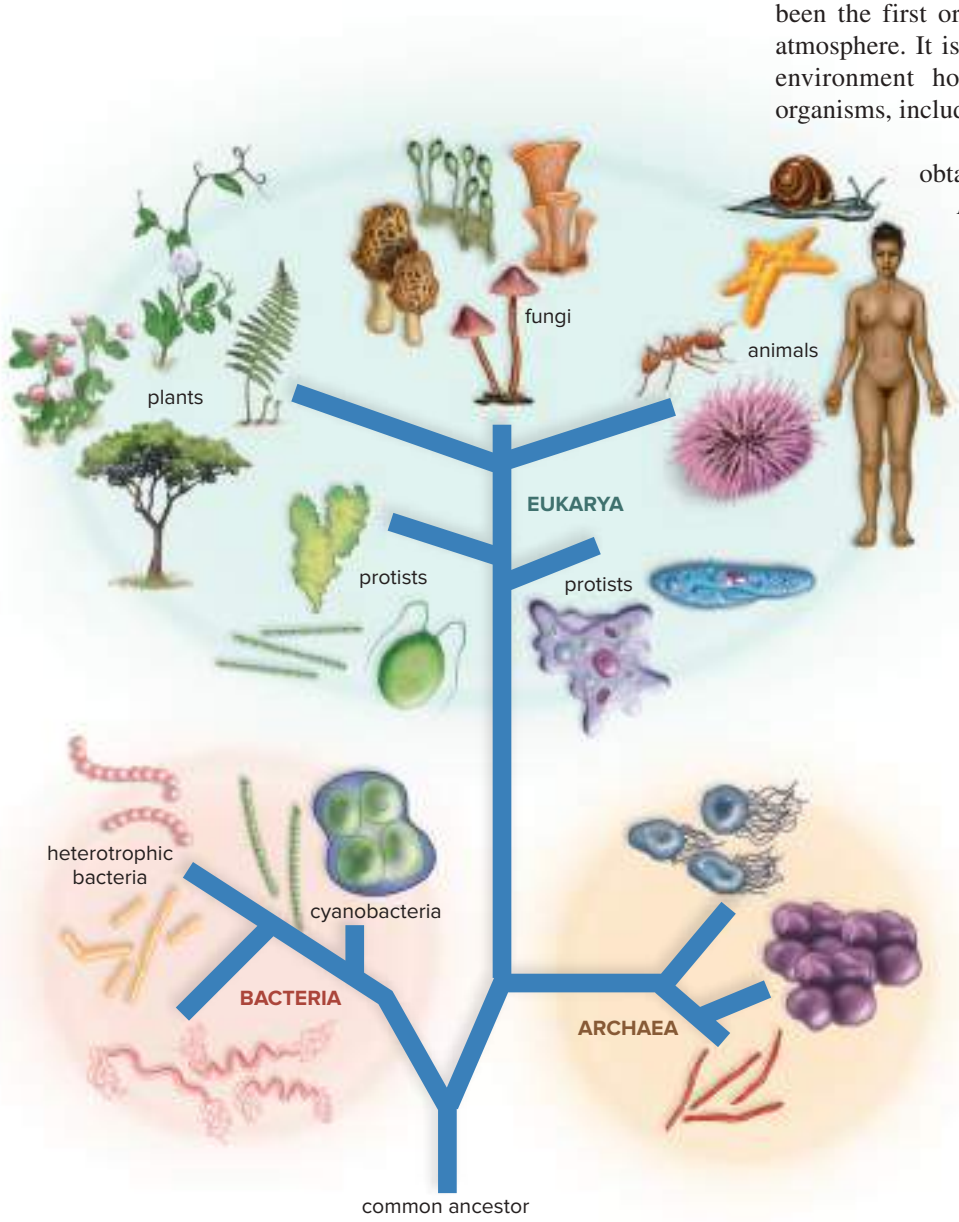
Table 19.1 Major Distinctions Among the Three Domains of Life

	Bacteria	Archaea	Eukarya
Single-celled	Yes	Yes	Some, many multicellular
Membrane lipids	Phospholipids, unbranched	Varied branched lipids	Phospholipids, unbranched
Cell wall	Yes (contains peptidoglycan)	Yes (no peptidoglycan)	Some yes, some no
Nuclear envelope	No	No	Yes
Membrane-bound organelles	No	No	Yes
Ribosomes	Yes	Yes	Yes
Introns	Some	Some	Yes

archaea are structurally similar to bacteria but are placed in a separate domain—domain Archaea—because of biochemical differences. The details of these differences are covered in Sections 20.4 and 20.5.

The cyanobacteria are large, photosynthetic prokaryotes. They carry on photosynthesis in the same manner as plants in that they use solar energy to convert carbon dioxide and water to a carbohydrate, in the process giving off oxygen. The cyanobacteria belong to a very ancient lineage of bacteria, and they may have been the first organisms to contribute oxygen to early Earth's atmosphere. It is possible that they played a role in making the environment hospitable for the evolution of oxygen-using organisms, including animals.

Bacteria have a wide variety of means for obtaining nutrients, but most are heterotrophic. *Escherichia coli,* which lives in the human intestine, is heterotrophic. *Clostridium tetani* (the cause of tetanus), *Bacillus anthracis* (the cause of anthrax), and *Vibrio cholerae* (the cause of cholera) are disease-causing species of bacteria. Heterotrophic bacteria are beneficial in ecosystems, because they break down organic remains. Along with fungi, they keep chemical cycling going, so that plants always have a source of inorganic nutrients.



Domain Archaea

Like bacteria, archaea are prokaryotic single-celled organisms that reproduce asexually. Archaea don't look that different from bacteria under the microscope. The extreme conditions under which many species live has made it difficult to grow them in the laboratory. This may have been the reason that their unique place among the living organisms went unrecognized for a long time.

Figure 19.4 A tree of life showing the three domains. Representatives of each domain are depicted in the shaded areas. The tree shows that domain Archaea and domain Eukarya are more closely related to each other than either is to domain Bacteria.

ISTUDY

Table 19.2 Eukaryotic Supergroups

Supergroup	Sample Organisms
Excavata	Diplomonads, euglenozoans
Chromalveolata	Dinoflagellates, ciliates, diatoms, golden algae, brown algae, and water molds
Rhizaria	Foraminiferans, radiolarians
Archaeplastida	Red algae, green algae, plants
Amoebozoa	Amoeboids, slime molds
Opisthokonta	Fungi, choanoflagellates, animals

Archaea are distinguishable from bacteria by a difference in their rRNA nucleotide sequences and by their unique plasma membrane and cell wall chemistry. The chemical nature of the archaeal cell wall is diverse and is different from that of a bacterial cell. The unique cell wall structure of archaea might help them live in the extreme conditions in which they are found. In addition, the branched nature of diverse lipids in the archaeal plasma membrane is very different from that of the bacterial plasma membrane.

Archaea live in all sorts of environments, but they are known for thriving in extreme environments thought to be similar to those of early Earth. For example, the methanogens live in environments without oxygen, such as swamps, marshes, and the guts of animals where methane is abundant. The halophiles thrive in salty environments, such as the Great Salt Lake in Utah, whereas the thermoacidophiles are both high temperature–loving and acid-loving. These archaea live in extremely hot, acidic environments, such as hot springs and geysers.

Domain Eukarya

Eukaryotes are single-celled or multicellular organisms whose cells contain a membrane-bound nucleus. They also have various organelles, some of which arose through endosymbiosis of other single-celled organisms (see Section 18.2). Sexual reproduction is common in eukaryotes, and various types of life cycles are seen.

Later in this text (see Section 21.1), we will study the individual kingdoms that occur within the domain Eukarya. In the meantime, we can note that the protists are a diverse group of single-celled eukaryotes that are difficult to classify and define. Some protists have filaments and form colonies or multicellular sheets. Even so, protists do not have true tissues. Their nutrition is diverse; some are heterotrophic by ingestion or absorption and some are photosynthetic. Green algae, paramecia, and slime molds are representative protists. There has been considerable debate over the classification of protists, requiring another level of classification, the **supergroup,** which sits below the level of a domain and above that of the kingdom. Presently, the protists are placed in one of six supergroups within the domain Eukarya (Table 19.2).

Fungi are eukaryotes that form spores, lack flagella, and have cell walls containing chitin. They are multicellular, with a few exceptions. Fungi are saprotrophic by absorption—they secrete digestive enzymes and then absorb nutrients from decomposing organic matter. Mushrooms, molds, and yeasts are representative fungi. Despite appearances, molecular data suggest that fungi and animals are more closely related to each other than either is to plants.

Plants are photosynthetic, multicellular organisms that are primarily adapted to a land environment. They share a common ancestor: an aquatic photosynthetic protist. Land plants possess true tissues and have the organ-system level of organization. Examples include cacti, ferns, and cypress trees.

Animals are motile, eukaryotic, multicellular organisms that evolved from a heterotrophic protist. Like land plants, animals have true tissues and the organ-system level of organization. Animals are heterotrophs. Examples include worms, whales, and insects.

Check Your Progress 19.2

1. List the traits that separate archaea from other single-celled organisms.
2. Explain the evidence indicating that fungi are more closely related to animals than to plants.
3. Identify the domain that includes all multicellular organisms.

19.3 Phylogeny

Learning Outcomes

Upon completion of this section, you should be able to

1. Discriminate between ancestral and derived traits.
2. Interpret the evolutionary relationships depicted in a phylogeny.
3. List the types of traits used to construct a phylogeny.

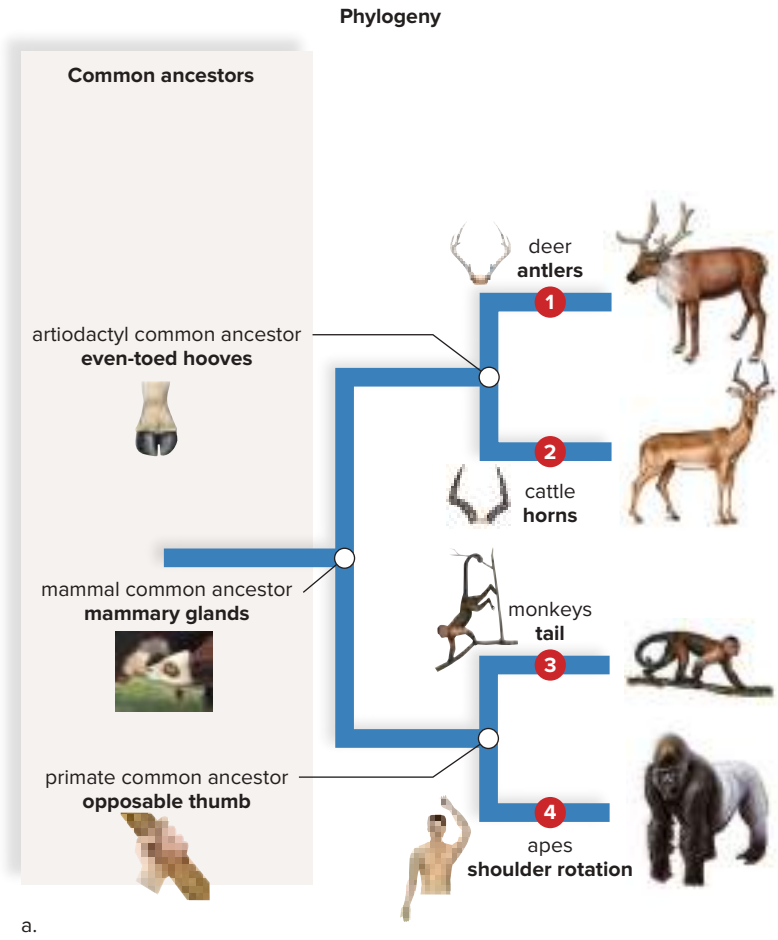
Systematic biologists use characters from the fossil record; comparative anatomy and development; and the sequence, structure, and function of RNA and DNA molecules to construct a phylogeny. Systematic biologists study the evolutionary history of biodiversity, represented by a phylogeny. In essence, systematic biology is the study of the evolutionary history of biodiversity, and a phylogeny is the visual representation of that history.

Interpreting a Phylogeny

Systematic biologists construct a phylogeny from traits that are unique to, and shared by, a taxon and their **common ancestor** (an ancestor to two or more lines of descent). Each branch, or **lineage,** in a phylogeny represents a descendant of a common ancestor. When a new character evolves, a new evolutionary path can begin, or **diverge,** from the old, a new lineage is formed, and a new branch of the phylogeny arises (Fig. 19.5).

Not all traits have equal value when making a phylogeny. **Ancestral traits,** or those found in the common ancestor, are not useful for determining the evolutionary relationships of an ancestor's descendants. For example, deer, cattle, monkeys, and apes, all examples of mammals, share a common ancestor that had mammary glands (Fig. 19.5). Because all mammals have mammary glands, this trait is an ancestral trait and thus is not helpful for

ISTUDY



understanding how deer, cattle, monkeys, and apes are related to each other. In contrast, **derived traits,** or those not found in the common ancestor of a taxonomic group, are the most important traits for clarifying evolutionary relationships. For example, both monkeys and apes have an opposable thumb capable of grasping, a trait not present in the common ancestor of mammals. This shared derived trait places monkeys and apes on a separate lineage of mammals called "primates" (Fig. 19.5).

Whether a trait is derived or ancestral is dependent on whether it is present in the common ancestor, and thus relative to its location within a phylogeny. For example, an opposable thumb is a derived trait for all monkeys and apes when compared to the common ancestor of all mammals. But the opposable thumb, while a derived trait when compared to the mammal common ancestor, is nevertheless an ancestral trait of primates (Fig. 19.5). Similarly, deer and cattle, both artiodactyls, have even-toed hooves, a trait not found in the common ancestor of mammals or in primates. Thus, even-toed hooves is a derived trait when compared to the common ancestor of mammals, but an ancestral trait of artiodactyls (Fig. 19.5). Both even-toed hooves and the opposable thumb suggest that the evolutionary history of artiodactyls and primates became independent as each group diverged from the mammal common ancestor.

Derived traits provide a more detailed phylogeny as they define closer and closer evolutionary relationships. Within the primates, a fully rotating shoulder joint, which allows apes to swing from branch to branch in trees, and the prehensile tail of monkeys are derived traits that define separate ape and monkey branches within the primates. Similarly, horns and antlers are derived traits that divide the cattle and deer into two individual artiodactyl branches (Fig. 19.5).

The hierarchical classification system of Linnaeus defines species as closely related to other species within the same genus. A genus is related to other genera in the same family, and so forth, from order to class to phylum to kingdom to domain (see Fig. 19.3*a*). When we say that two species (or genera, families, etc.) are closely related, we mean that they share a recent common ancestor. For example, all the animals in Figure 19.5 are related, because we can trace their ancestry back to a common ancestor. Taxonomists use the pattern of branching in a phylogeny constructed from an analysis of derived traits to classify taxa into natural groups.

Figure 19.5 The relationship among phylogeny, classification, and traits. For this example, the red-numbered traits are derived traits present in all descendants of a clade: (1) deer have antlers; (2) cattle have horns; (3) monkeys have tails; (4) apes have full shoulder rotation. **a.** The phylogeny is a representation of the evolutionary history of a few members of the class Mammalia. Each common ancestor has a trait that is present in all of its descendants. **b.** As you move toward the tips of the tree, new derived traits provide greater resolution to the classifications. For example, mammals have mammary glands. Artiodactyls have mammary glands and even-toed hooves. Deer have mammary glands, even-toed hooves, and antlers.

ISTUDY

For example, all mammals with even-toed hooves form a single lineage that is assigned to the order Artiodactyla (Fig. 19.5). Furthermore, artiodactyls that have antlers form a lineage within the order Artiodactyla that is classified as the family Cervidae. Antlers are grown only in males during the breeding season, and they can grow quite large and be highly branched. In contrast, artiodactyls that have horns form a lineage within the order Artiodactyla that is classified as the family Bovidae. Unlike antlers, horns are not shed seasonally and are found on both males and females, although they are smaller in females.

Similarly, the order Primates is a lineage of mammals with opposable thumbs. The rotating shoulder of apes and the prehensile tail of monkeys form two independent lineages in the order Primates that are classified as the family Hominidae and family Cebidae, respectively.

Cladistics

When constructing any phylogeny, multiple characteristics from a variety of organisms are compared at the same time (Fig. 19.6). This approach can produce various evolutionary trees, because not all traits are equally useful to the study of evolutionary history. The challenge is determining which phylogeny of the many possible phylogenies is the best hypothesis of evolutionary history. One way to determine the answer is through the use of cladistics.

Cladistics is a method that uses shared, derived traits to develop a hypothesis of evolutionary history. The evolutionary history of derived traits is interpreted into a type of phylogeny constructed with cladistic methods, called a **cladogram.** In a cladogram, a common ancestor and all of its descendant lineages is called a **clade.**

Cladistics applies the principle of **parsimony** (L. *parsimonia,* "frugality, thrift") to a set of traits to construct a cladogram. Parsimony considers the simplest solution to be the "optimal" solution. Thus, the cladogram that represents the simplest evolutionary history—that is, the one that requires the fewest number of evolutionary changes—is considered the best hypothesis based on the traits used to construct the cladogram. As with any hypothesis, a cladogram may change when new traits are discovered and incorporated into the construction of a cladogram. The important point is that our understanding of evolutionary history is a working hypothesis, constantly changing as we learn more about organisms' traits and lifestyles. Thus, cladistics is a hypothesis-based, quantitative science that is subject to rigorous testing.

The first step in developing a cladogram is to construct a table that summarizes the derived traits of the taxa being compared (Fig. 19.6). Derived traits are used to determine shared ancestry among taxa. In cladistics, the **outgroup** is the taxon that is used to determine the ancestral and derived states of characters in the **ingroup,** or the taxa for which the evolutionary relationships are being determined. In Figure 19.6, the outgroup is the lancelet, and the ingroup contains all other vertebrates. Traits present in the ingroup but not in the lancelet (the outgroup) are defined as derived traits. For example, all **chordates,** including the lancelet, have a dorsal or spinal nerve cord, so this is an ancestral trait. Nested within the chordates are clades, each with a uniquely derived trait. Tetrapods are a clade within the chordates that does not include

	Species							
	ingroup							lancelet (outgroup)
Traits	chimpanzee	dog	finch	crocodile	lizard	frog	tuna	
mammary glands	✗	✗						
hair	✗	✗						
gizzard			✗	✗				
epidermal scales			✗	✗	✗			
amniotic egg	✗	✗	✗	✗	✗			
four limbs	✗	✗	✗	✗	✗	✗		
vertebrae	✗	✗	✗	✗	✗	✗	✗	
notochord in embryo	✗	✗	✗	✗	✗	✗	✗	✗

Figure 19.6 Constructing a cladogram: The data. The lancelet is in the outgroup, and all the other species listed are in an ingroup (study group). The species in the ingroup have shared derived traits—derived because a lancelet does not have the trait; shared because certain species in the study group do have them. All the species in the ingroup have vertebrae, all but a fish have four limbs, and so forth. The shared derived traits indicate which species are distantly related and which are closely related. For example, a human is more distantly related to a fish, with which it shares only one trait—namely vertebrae—than to an iguana, with which it shares three traits—vertebrae, four limbs, and an amniotic egg (the amnion layer protects the embryo; see Section 29.5).

fish, because fish have a dorsal nerve cord but do not have four limbs (Fig. 19.7). Likewise, amphibians are tetrapods, but they do not have an amnion, one of several protective membranes that surround a growing embryo, like those found in an amniotic egg. Thus, amniotes are a clade of organisms that possess an amnion, which does not include amphibians.

Once derived and ancestral traits have been identified, the principle of parsimony is applied. The cladogram in Figure 19.7 is considered the best hypothesis, or explanation, of the evolutionary history based on the traits used.

Tracing Phylogeny

Traditionally, systematic biologists relied on morphological data to study evolutionary relationships between taxa. However, morphology can be misleading. For example, recent studies suggest that birds and crocodiles are more closely related to each other than

ISTUDY

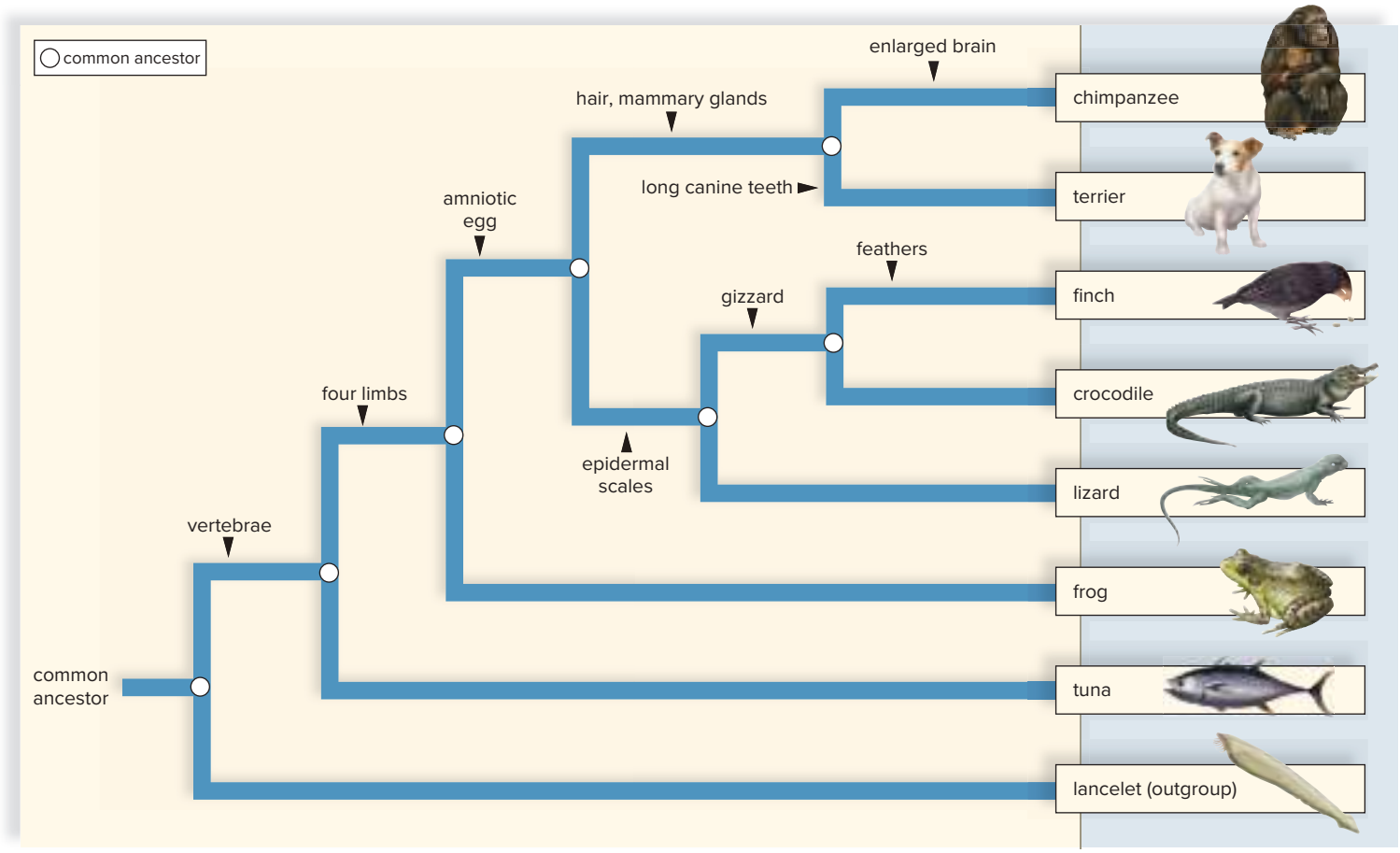


Figure 19.7 Constructing a cladogram: The phylogenetic tree. Based on the data shown in Figure 19.6, this cladogram has six clades. Each clade contains a common ancestor with derived traits that are shared by all members of the clade.

crocodiles are to lizards. The morphological traits that Linnaeus used, namely wings and feathers, led him to classify birds as a different lineage from crocodiles. This makes sense when we consider that Linnaeus had only physical traits at his disposal when classifying organisms—birds do not look much like crocodiles!

Today, we have a wide range of different sources of traits to assist with understanding the evolutionary history of organisms. Armed with these new sources of data, systematic biologists are continually revising the historical classification system to reflect their best understanding of evolutionary history.

Fossil Traits

One of the advantages of fossils is that they can be dated (see Section 18.2), but it is not always possible to tell to which lineage, living or extinct, a fossil is related. At present, paleontologists are discussing whether fossil turtles indicate that turtles are distantly or closely related to crocodiles. On the basis of his interpretation of fossil turtles, Olivier C. Rieppel of the Field Museum of Natural History in Chicago is challenging the conventional interpretation that turtles have traits seen in a common ancestor to all reptiles but are not closely related to crocodiles, which evolved later. His interpretation is being supported by molecular data that show turtles and crocodiles to be closely related (see Section 29.5 for an overview of reptile evolution).

If the fossil record were more complete, fewer controversies might arise about the interpretation of fossils. One reason the fossil record is incomplete is that most fossils are formed from harder body parts, such as bones and teeth. Soft parts are usually eaten or decayed before they have a chance to be buried and preserved. This may be one reason it has been difficult to discover when angiosperms (flowering plants) first evolved. Pollen from the Triassic may help pinpoint the date of origin, whereas a Jurassic fossil recently found may help describe the appearance of the earliest flowering plants (Fig. 19.8). As paleontologists continue to discover new fossils, the fossil record will reveal more traits useful to systematic biologists.

Morphological Traits

Homology (Gk. *homologos,* "agreeing, corresponding") is structural similarity that stems from having a common ancestor. Comparative anatomy, including developmental evidence, provides information regarding homology (see Figs. 15.15 and 15.16).

Homologous structures are similar to each other because of common descent. The forelimbs of vertebrates contain the same bones organized as they were in a common ancestor, despite adaptations to different environments. As Figure 15.15 showed, even though a horse has but a single digit and toe (the hoof), while a bat has four lengthened digits that support its wing, a horse's forelimb and a bat's forelimb contain the same bones.

ISTUDY

Figure 19.8 Ancestral angiosperm. This is an artistic drawing of the fossil *Nanjinganthus dendrostyla* that existed around 174 MYA, during the Jurassic period. It is believed to be the earliest macrofossil of an angiosperm discovered.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Cow	c	c	c	c	g	t	g	g	a	g	g	t	a	c	g	c	t	t	c	a	c	t	c
Pig	c	c	c	c	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	a	c	t	c
Horse	t	c	c	g	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	g	c	c	c
Mouse	c	c	c	c	g	t	g	g	a	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Rat	c	c	c	c	g	t	a	g	a	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Dog	c	c	c	t	g	t	g	g	a	g	g	t	c	c	g	c	t	t	c	a	c	c	c
Guinea Pig	c	c	c	t	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Chimp	c	c	t	g	g	t	g	g	g	g	c	t	a	c	g	c	t	t	c	a	c	c	t
Human	c	c	t	g	g	t	g	g	g	g	g	t	a	c	g	c	t	t	c	a	c	c	t
Orangutan	c	c	c	g	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c
Macaque	a	c	c	g	g	t	g	g	g	g	g	t	g	c	g	c	t	t	c	a	c	c	c

Figure 19.9 DNA sequence alignment. The DNA sequences of a small section of a gene are aligned to determine the evolutionary relationships among some mammals. Each nucleotide is aligned in columns and assigned a number (in this case, 1–23). Just as with physical traits, individual nucleotides provide information about how organisms are related. For example, the sixth nucleotide is a T shared among all mammals. Thus, this T is an ancestral trait of the class Mammalia (see Fig. 19.5). Chimpanzees and humans have in common a T at nucleotides 3 and 23, a trait not found in other mammals. Thus, these two nucleotides are shared derived traits that support a close relationship between chimps and humans. Some nucleotide sequence alignments are thousands of nucleotides long, so computers are used to perform comparisons.

Deciphering homology is sometimes difficult because of convergent evolution. **Convergent evolution** has occurred when distantly related species have a structure that looks the same only because of adaptation to the same type of environment (see Fig. 17.13 for an example of convergent evolution in fish). Similarity due to convergence is termed **analogy.** The wings of an insect and the wings of a bat are analogous.

Analogous structures have the same function in different groups but do not have a common ancestry. Both cacti and spurges are adapted similarly to a hot, dry environment, and both are succulent (thick, fleshy) with spines that originate from modified leaves. However, the details of their flower structure indicate that these plants are not closely related.

The construction of phylogenetic trees is dependent on discovering homologous structures and avoiding the use of analogous structures to uncover ancestry.

Behavioral Traits

Evidence has been found that some dinosaurs cared for their young in a manner similar to crocodilians (including alligators) and birds. These data substantiate the morphological data that dinosaurs, crocodilians, and birds are related through evolution. The mating calls of leopard frogs are another example of a behavioral trait that has been used to decipher evolutionary history. Mating calls support the hypothesis that leopard frogs are an assemblage of multiple species that morphologically look quite similar but are on different evolutionary lineages (see Fig. 17.2).

Molecular Traits

Mutations in the base-pair sequences of DNA accumulate over time. Systematic biologists assume that the more closely species are related, the fewer changes there will be in their DNA base-pair sequences (Fig. 19.9).

A phylogeny of primates based on molecular traits supports a recent common ancestor for humans and chimpanzees (Fig. 19.10). Because DNA codes for amino acid sequences that form proteins, it also follows that, the more closely species are related, the fewer differences there will be in the amino acid sequences within their proteins.

Advances in molecular biology have made it very quick, easy, and inexpensive to collect nucleotide sequences for many different taxa. Technology breakthroughs have made it possible to analyze nucleotide sequences or amino acid sequences quickly and accurately. Archives of DNA sequences from different genes for thousands of organisms are freely available to anyone doing systematic biology research.

Mitochondrial DNA (mtDNA) mutates ten times faster than nuclear DNA. Therefore, when determining the phylogeny of closely related species, investigators often choose to sequence mtDNA instead of nuclear DNA. One such study concerned North American songbirds. Ornithologists believed for a long time that these birds diverged into eastern and western subspecies due to retreating glaciers some 250,000–100,000 years ago. Sequencing of mtDNA allowed investigators to conclude that the two groups of North American songbirds diverged from one another an average of 2.5 million years ago (MYA). Because the old hypothesis based on glaciation is apparently flawed, a new hypothesis is required to explain why eastern and western subspecies arose among these songbirds.

Phylogenetic trees derived from molecular data are routinely used to apply evolutionary theory to all areas of biology. This information has helped human society in areas such as agriculture, medicine, and forensic science. For example, a phylogeny of the HIV types in an individual guides doctors' decisions about antiviral drug treatment. Likewise, phylogenies of insects have assisted agriculturalists in designing effective controls of crop pests.

ISTUDY

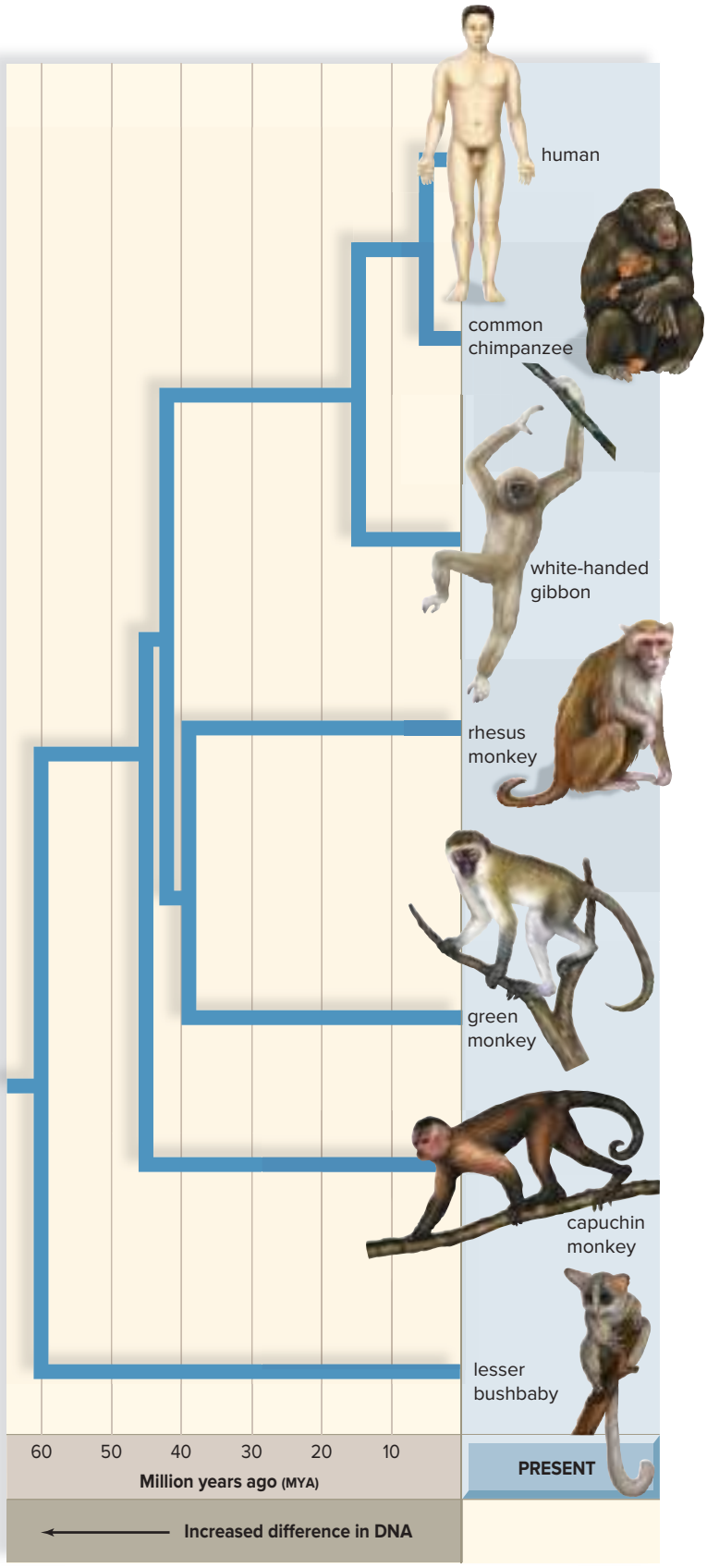


Figure 19.10 A phylogeny determined from molecular data. The relationship of certain primate species based on a study of their genomes. The length of the branches indicates the relative number of nucleotide pair differences that were found between groups. These data, along with knowledge of the fossil record for one divergence, make it possible to suggest a date for the other divergences in the tree.

Protein Comparisons Before amino acid sequencing became routine, immunological techniques were used to roughly judge the similarity of plasma membrane proteins. In one procedure, antibodies are produced by transfusing a rabbit with the cells of one species. Cells of the second species are exposed to these antibodies, and the degree of the reaction is observed. The stronger the reaction, the more similar are the cells from the two species.

Later, it became customary to use amino acid sequencing to determine the number of amino acid differences in a particular protein. Cytochrome *c* is a protein found in all aerobic organisms, so its sequence has been used to examine evolutionary relationships among many organisms. There are three amino acid differences in the cytochrome *c* of chickens and ducks, but between chickens and humans there are 13 amino acid differences. From these data we can conclude that, as expected, chickens and ducks are more closely related than are chickens and humans. In addition to amino acids, differences in individual nucleic acids in DNA and RNA have become a powerful tool for examining evolutionary relationships.

Molecular Clocks Some nucleic acid changes are neutral (not under the influence of natural selection) and thus accumulate at a fairly constant rate. These neutral mutations can be used as a kind of **molecular clock** to construct a timeline of evolutionary history. For example, songbird subspecies have mtDNA with 5.1% nucleic acid differences. Researchers know the average rate at which mtDNA nucleotide changes occur, called the mutation rate, measured in the number of mutations per unit of time. The researchers doing comparative mtDNA sequencing used their data as a molecular clock when they equated a 5.1% nucleic acid difference among songbird subspecies to 2.5 MYA.

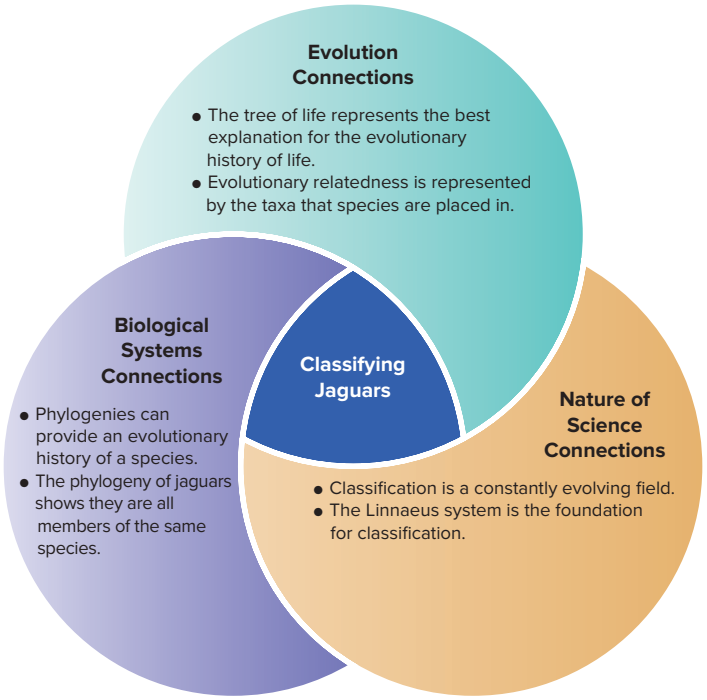
In Figure 19.10, the researchers used their DNA sequence data to suggest how long the different types of primates have been separate. When the fossil record for one divergence is known, it indicates how long it probably takes for each nucleotide pair difference to occur. When the fossil record and molecular clock data agree, researchers have more confidence that the proposed phylogenetic tree is correct.

Check Your Progress 19.3

1. Interpret the ancestral or derived state of traits relative to their position on the phylogeny in Figure 19.5.
2. Compare two phylogenies of the same set of organisms; one requires 10 evolutionary changes, the other 15. Explain which phylogeny would be the best hypothesis for evolutionary history, and why.
3. Recognize the various traits used to construct a phylogeny.

ISTUDY

Connecting *the* Concepts



What Did You Learn?

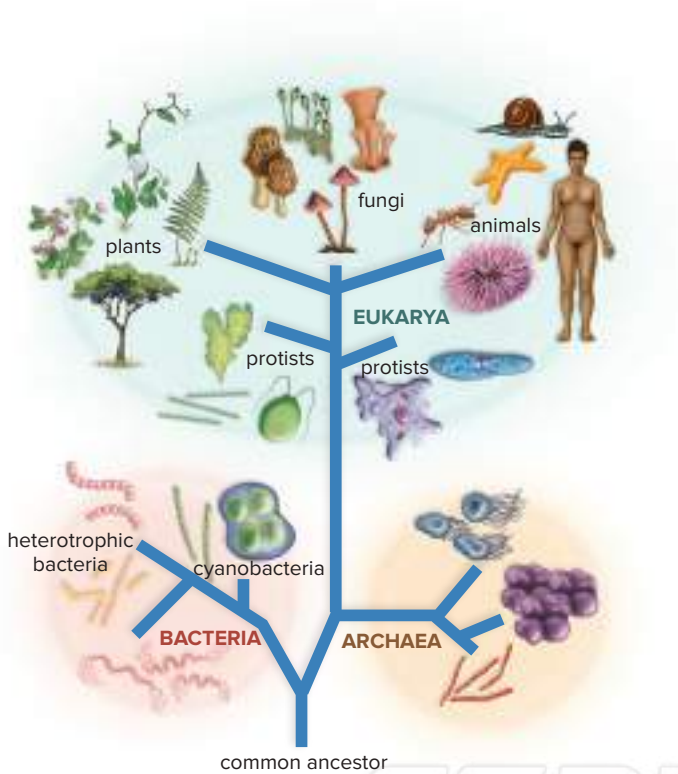
19.1 Systematic Biology

Systematics is dedicated to understanding the evolutionary history of life on Earth. Systematics is a field of science that uses **traits** of living and fossil organisms to determine relationships. **Taxonomy,** a part of systematics, identifies, names, and organizes biodiversity into various **taxa.**

- **Classification** is the process of naming and assigning an organism to a particular taxon. **Taxonomists** strive to classify organisms into natural groups in order to construct a **phylogeny,** or evolutionary "family tree."
- Traditional Linnaean taxonomy uses a **binomial nomenclature** in order to give every species a unique Latin name. The first part is the genus; the second part is the **specific epithet.**
- Taxonomists use a nested, hierarchical set of categories to classify organisms. There are eight main categories of classification: **species, genus, family, order, class, phylum, kingdom,** and **domain.** The categories are hierarchical such that each higher category is more inclusive than the one below it; species in the same kingdom share general charactvers, and species in the same genus share quite specific characters.
- The standardized rules of **nomenclature** are governed by the International Commission on Zoological Nomenclature and the International Code of Botanical Nomenclature.

19.2 The Three-Domain System

Over time, the classification system has been expanded in order to include new information. In 1969, the **five-kingdom system** was introduced. Then, in the late 1970s, the system was expanded to include three domains. Prokaryotes are placed in **domain Bacteria** or **domain Archaea** based on molecular information. All of the protists, fungi, plants, and animals are placed in **domain Eukarya,** and in addition to the eight main categories of classification, eukaryotes are now being classified into **supergroups.**



ISTUDY

19.3 Phylogeny

Phylogenies are constructed from traits that are unique to a taxon and shared by their common ancestor.

- The branches, or **lineages,** on a phylogeny represent different lines of descent that occur when new traits cause lineages to **diverge.**
- **Ancestral traits** are those found in the **common ancestor,** while **derived traits** are unique to a particular taxon and are useful in determining evolutionary relationships.

Cladistics uses shared derived traits to distinguish different groups of species from one another.

- The phylogeny that results from cladistic analysis is called a **cladogram,** with lineages called **clades.**
- The principle of **parsimony** is used to select the tree with the simplest explanation of evolutionary history, given the set of traits used. An **outgroup** is the taxon used to determine the ancestral and derived states of characters in the **ingroup,** or taxa for which the evolutionary relationships are being determined. All **chordates** share the ancestral trait of a spinal nerve cord. The most parsimonious cladogram is the one that requires the fewest number of evolutionary steps. Fossil, morphological, behavioral, and molecular traits help systematic biologists study evolutionary relationships. If the fossil record is complete enough, we can sometimes trace a lineage through time.

Homology helps indicate when species share a common ancestor by using **homologous structures.**

- **Convergent evolution** occurs when distantly related species have evolved similar structures in response to the environment. This convergence is termed **analogy.**
- **Analogous structures** have the same function in different groups but do not share a common ancestor.
- Changes to the genetic code can also be used to construct a timeline of evolutionary history by establishing a **molecular clock.** DNA nucleotide sequence data are commonly used to help determine evolutionary relationships. When multiple lines of evidence agree, it provides more confidence in the phylogenetic tree.

Thinking Critically About Biology

1. Recent DNA evidence suggests to some plant taxonomists that the traditional way of classifying flowering plants is not correct, and that flowering plants need to be completely reclassified. Other botanists disagree, saying it would be chaotic and unwise to disregard the historical classification groups. Argue for and against keeping traditional classification schemes.
2. What data might make you conclude that the eukaryotes should be in more than one domain? What domains would you hypothesize might be required?
3. Because the genomes of chimpanzees and humans are almost identical, and the differences between them are no greater than between any two humans, their classification has been changed. Chimpanzees and humans are placed in the same family and subfamily. They are in different "tribes," which is a rarely used classification category between subfamily and genus. The former classification of chimpanzees and humans placed the two animals in different families. Do you believe chimpanzees should be classified in the same family and subfamily as humans, or do you prefer the classification used formerly? Which way seems prejudicial? Give your reasons for preferring one method over the other.

Making It Relevant to You

1. If jaguar populations were to become isolated, would this be enough to classify them as separate subspecies?
2. Should molecular traits be used to classify humans?
3. Should one line of evidence hold more weight than another when we discuss the classification of species, or should all lines be weighted equally?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 4

Microbial Evolution

Microbes occupy a world unseen by the naked eye. They are the most diverse form of life on the planet, and include viruses, prokaryotic bacteria and archaea, and the eukaryotic protists and fungi. These organisms occur everywhere, from the highest mountain peaks to the deepest ocean trenches, and in every type of environment, even those that are extremely hot and acidic.

Although some microbes cause serious diseases in plants and animals, including humans, we make use of microbes in innumerable ways. For example, bacteria help us accomplish gene cloning and genetic engineering, make foods and antibiotics, and help dispose of sewage and environmental pollutants. The biosphere is totally dependent on the services of microorganisms.

We are aware of how much we rely on land plants, but we may fail to acknowledge that, without microorganisms, land plants could not exist. Decomposing fungi and bacteria make inorganic nutrients available to plants, which they can absorb all the better because their roots are coated with friendly fungi. Photosynthetic bacteria first put oxygen in the atmosphere, and they, along with certain protists, are the producers of food in the oceans.

Microbes represent the earliest form of life on the planet. The prokaryotes were alone on Earth for about 2.5 billion years, and then the single-celled protists gave rise to animals, plants, and fungi. In this unit, we will explore the evolution and biology of microbes.

Unit Outline

Chapter 20 Viruses, Bacteria, and Archaea

Chapter 21 Protist Evolution and Diversity

Chapter 22 Fungi Evolution and Diversity

Unit Learning Outcomes

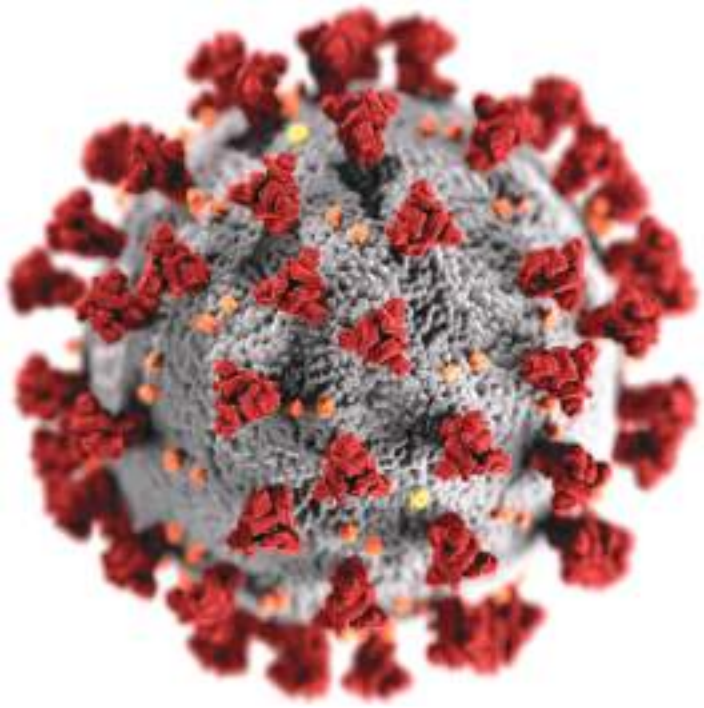
The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Recognize the evolutionary relationships of the microorganisms and their position in the tree of life.
Nature of Science	Identify how microorganisms are used as models in medical, environmental, and genetics research.
Biological Systems	Explain how microorganisms are essential for a healthy ecosystem.

ISTUDY

20

Viruses, Bacteria, and Archaea



Alissa Eckert, MS; Dan Higgins, MAMS/CDC

SARS-CoV-2

In 2020, schools and businesses were shut down, travel was suspended, and the borders of countries were closed, all because of a virus named SARS-CoV-2. The acronym stands for *s*evere *a*cute *r*espiratory *s*yndrome *cor*ona*v*irus *2*. It is likely the virus originated from bats. The virus causes a disease called COVID-19 whose name is a contraction of *cor*ona*v*irus *di*sease 20*19*.

COVID-19 is typically acquired by infected droplets (e.g., due to sneezing, coughing, talking) from someone who has the disease. Droplets enter through the nose, eyes, or mouth. As the virus replicates, the immune response is set in motion. What makes this virus so dangerous is that some people may carry the infection and be asymptomatic, whereas others may have heightened immune responses that require hospitalization.

SARS-CoV-2 belongs to a group of viruses called β-coronavirus, aptly named because the spike glycoproteins protrude from the viral surface and resemble a crown or corona. Understanding the structure and sequence information of a virus allows scientists to develop vaccines and treatments for current pandemics, and to fight those same and similar viruses should they appear in the future.

As you read through this chapter, think about the following questions:

1. How do viruses and prokaryotic bacteria differ from each other and eukaryotes?
2. Although some cause disease, why are microorganisms essential to life?

Chapter Outline

20.1 Viruses, Viroids, and Prions
20.2 Emerging Infectious Diseases and SARS-CoV-2
20.3 Prokaryotes
20.4 Bacteria
20.5 Archaea

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 1.1 What characteristics are necessary for an organism to be considered "living"?

Section 4.2 What is the structure of a prokaryotic cell?

Section 19.2 What characteristics divide microorganisms into the domains Bacteria and Archaea?

Following *the* Themes

CHAPTER 20 – VIRUSES, BACTERIA, AND ARCHAEA

Evolution	All living organisms evolved from the first prokaryotic cells 3.5 billion years ago.
Nature of Science	The study of how microorganisms evolve over time allows researchers to predict disease and to develop vaccines and treatments for it.
Biological Systems	Microorganisms are found in, and are essential to, all environments on Earth.

ISTUDY

20.1 Viruses, Viroids, and Prions

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the basic structures of a virus.
2. Explain the unique characteristics of viruses compared to living cells.
3. Describe the process of viral reproduction.

The term **virus** is associated with a number of plant, animal, and human diseases (Table 20.1). The mere mention of the term brings to mind serious illnesses, such as polio, rabies, and AIDS (acquired immunodeficiency syndrome), as well as formerly common childhood maladies such as measles, chickenpox, and mumps. Viral diseases are of concern to everyone; it is estimated that the average person catches a cold (caused by the influenza virus) several times a year.

Viruses are a biological enigma. They have some characteristics of living organisms, such as a DNA or RNA genome, and the ability to evolve and replicate. However, they can replicate only by using the metabolic machinery of a host cell, they do not have a metabolism, and they do not respond to stimuli. For these reasons, viruses are known as *obligate intracellular parasites* and are either active or inactive, rather than living or nonliving.

Viruses do not fossilize, as other living organisms do, so their history is difficult to study. However, several hypotheses exist regarding their origin and evolution. Hypotheses about the origin of life suggest that proteins or nucleic acids, the two organic molecules present in viruses, were first to evolve. It is possible that viruses arose from these basic polymers at the same time as living cells. Some scientists offer an alternative hypothesis suggesting viruses actually originated after living cells. They propose that viruses were derived from pieces of cell genomes or evolved backward from living cells, or degenerated, into the simplest possible form required for reproduction.

Discovery of Viruses

Our knowledge of viruses began in 1884, when the French chemist Louis Pasteur suggested that something smaller than a bacterium was the cause of rabies. He chose the word *virus* from the Latin word meaning "poison."

In 1892, Dimitri Ivanowsky, a Russian microbiologist, was studying tobacco mosaic disease, which causes damage to the leaves and fruit of tobacco plants. He noticed that even when an infective extract was filtered through a fine-pore porcelain filter that retained bacteria, the extract still caused disease. This substantiated Pasteur's belief, because it meant that the disease-causing agent was smaller than any known bacterium.

In the twentieth century, electron microscopy was born, and viruses were seen for the first time. By the 1950s, virology had become an active field of research; the study of viruses, and now viroids and prions, has contributed much to our understanding of disease, genetics, and the characteristics of living organisms.

Table 20.1 Viral Diseases in Humans

Category	Disease
Sexually transmitted diseases	AIDS (HIV), genital warts, genital herpes, Zika
Childhood diseases	Mumps, measles, chickenpox, German measles
Respiratory diseases	Common cold, influenza, severe acute respiratory syndrome (SARS), COVID-19
Skin diseases	Warts, fever blisters, shingles
Digestive tract diseases	Gastroenteritis, diarrhea
Nervous system diseases	Poliomyelitis, rabies, encephalitis, Zika
Other diseases	Smallpox, Zika, cancer, hepatitis, mononucleosis, yellow fever, dengue fever, conjunctivitis, hepatitis C, Ebola

Classification System of Viruses

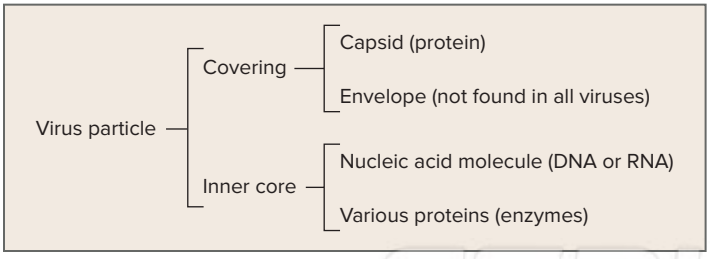
Because viruses mutate rapidly and are not living organisms, they are difficult to classify and name. However, the International Committee on Taxonomy of Viruses (ICTV) has developed a classification system similar to that used for living organisms. Viral classification differs because it includes only the taxonomic levels of order—family, genus, and species—not the higher levels of kingdom, phylum, and class. Over 2,500 species of viruses have been identified, including the species that causes the seasonal flu (influenza A) in the *Influenza A* genus and Orthomyxoviridae family, and SARS-CoV-2 in the β-coronavirus genus and family Coronaviridae.

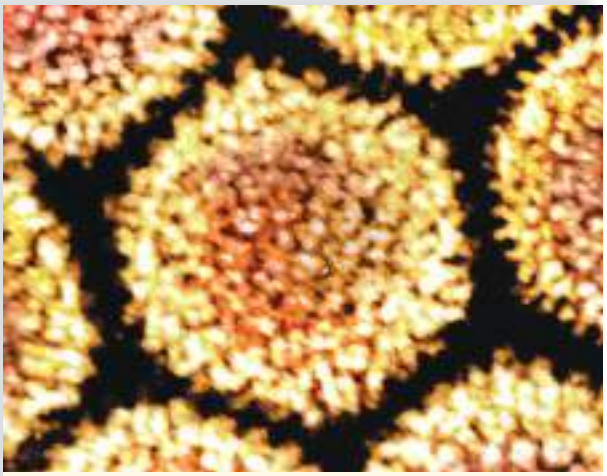
When a new type of virus emerges within a single species of virus, additional classification levels, such as subtype, are required for clear identification. For example, influenza A is classified into subtypes based on the type of two glycoprotein spikes in its envelope: hemagglutinin (H) and neuraminidase (N). Because there are 16 H-type and 9 N-type spikes, many virus subtypes may evolve, as discussed in the Evolution feature, "Flu Viruses: Seasonal Versus Pandemic Flu." Notable examples are the H5N1 "bird flu" and H1N1 "swine flu."

Structure of Viruses

The size of a virus is comparable to that of a large protein macromolecule, approximately 10–400 nm. Viruses are best studied through electron microscopy (Fig. 20.1). Many viruses can be purified and crystallized, and the crystals can be stored just as chemicals are stored. Still, viral crystals become infectious when the viral particles they contain are given the opportunity to invade a host cell.

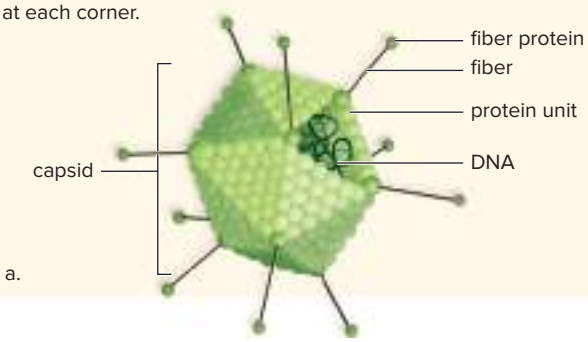
Viruses are categorized by (1) their size and shape; (2) their type of nucleic acid, including whether it is single-stranded or double-stranded; and (3) the presence or absence of an outer envelope. The structure of a virus can be summarized by the following diagram:



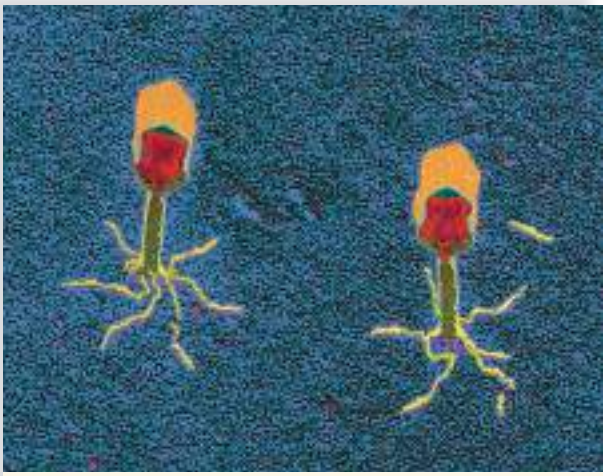


TEM 60,000×

Adenovirus: DNA virus with a polyhedral capsid and a fiber at each corner.

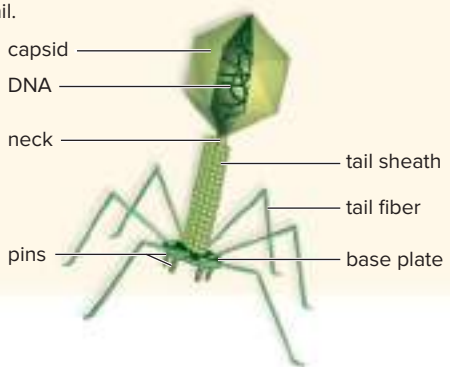


a.

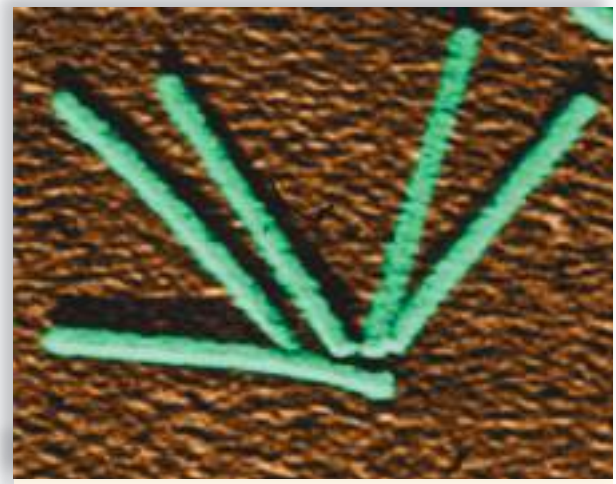


TEM 90,000×

T-even bacteriophage: DNA virus with a polyhedral head and a helical tail.

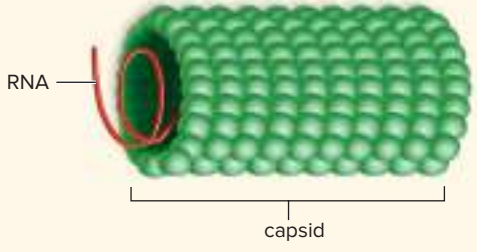


b.

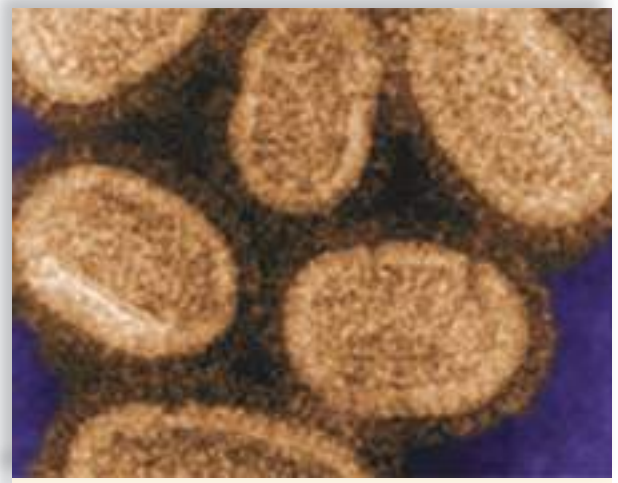


TEM 44,000×

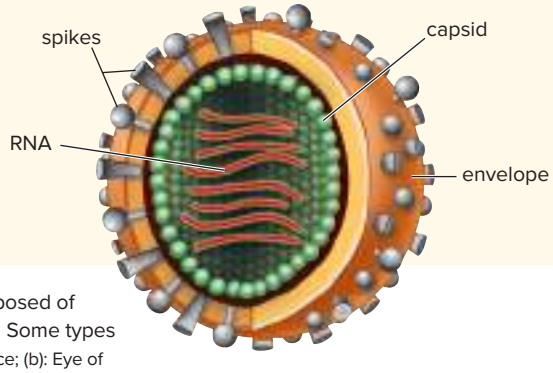
Tobacco mosaic virus: RNA virus with a helical capsid.



c.



Influenza virus: RNA virus with a spherical capsid surrounded by an envelope with spikes.



d.

Figure 20.1 Viruses. Despite their diversity, all viruses have an outer capsid composed of protein subunits and a nucleic acid core—composed of either DNA or RNA, but not both. Some types of viruses also have a membranous envelope. (photos): (a): Biophoto Associates/Science Source; (b): Eye of Science/Science Source; (c): Science Source; (d): Cynthia Goldsmith/Centers for Disease Control and Prevention

ISTUDY

THEME Evolution

Flu Viruses: Seasonal Versus Pandemic Flu

All flu viruses have an H (hemagglutinin) spike and an N (neuraminidase) spike embedded in its plasma membrane (Fig. 20A*a*, *top*). Its H spike allows the virus to bind to its receptor, and its N spike attacks host plasma membranes in a way that allows mature viruses to exit the cell.

At least 16 types of H and 9 types of N spikes exist, allowing different combinations that can infect different hosts. Many of the flu viruses are assigned specific codes based on the type of spike. For example, the H5N1 virus gets its name from its variety of H5 spikes and its variety of N1 spikes.

As is the case with human influenza virus (the seasonal flu), our immune systems can recognize only the particular variety of H spikes and N spikes they have been exposed to in the past by infection or immunization. Each year, the human influenza virus mutates, resulting in a new annual flu vaccination against that flu (Fig. 20A*a*, *bottom*).

In some years, a mixed flu virus arises, one for which there is little or no immunity in the human population. This is when a flu global outbreak (pandemic) may occur (Fig. 20A*b*).

Currently, the H7N9 and H5N1 subtypes of flu virus are of great concern because of their potential to reach pandemic proportions. Both viruses are common in wild birds such as waterfowl, and they can readily infect domestic poultry, such as chickens, which is why they are referred to as "bird flu." More pathogenic strains of these viruses have appeared, with the ability to spread from birds to humans, causing severe illness and death.

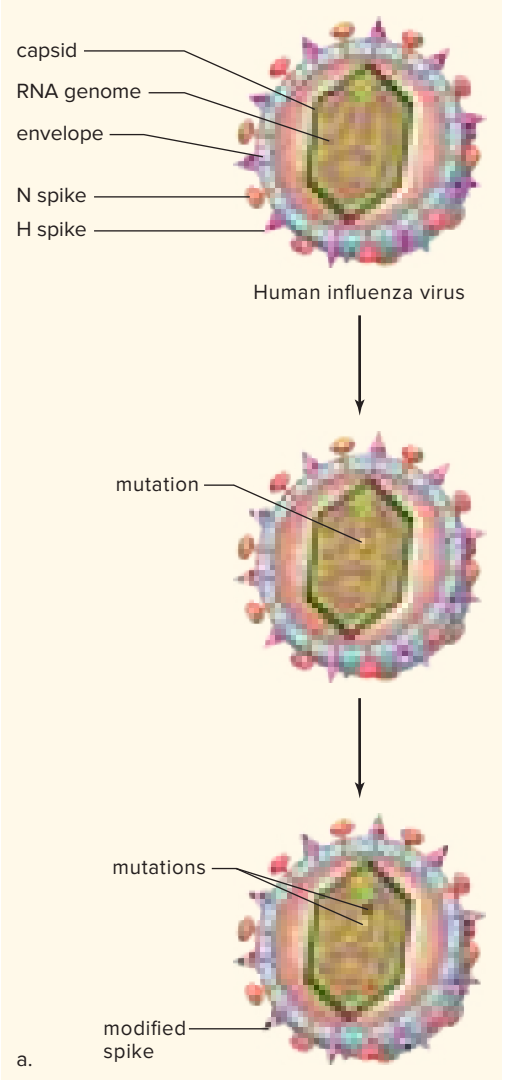
Humans become infected because the viral H spikes can attach to both a bird flu receptor and a human flu receptor. Close contact between domestic poultry and humans is necessary for this to happen, and the virus has rarely been transmitted from one human to another.

At this time, bird flu infects mostly the lungs, or the lower respiratory tract, and transmission by coughing or sneezing is rare. However, a spontaneous mutation in the H spike could enable it to attack the upper respiratory tract, helping the virus spread easily from person to person by coughing and sneezing. Another possibility is that a combining of spikes could occur in a person who is infected with *both* the bird flu and the human flu viruses.

In both cases, the new virus would likely spread rapidly and be lethal. Currently, there are no available vaccines for the H5N1 or H7N9 virus, and approximately half of infected individuals die.

Questions to Consider

1. How may knowing the host-specific nature of a virus prevent a global pandemic?
2. Compare the names of the H5N1 and the H7N9 viruses. What do their names tell you about the differences in their structure?
3. Why do humans need a flu vaccine every year?



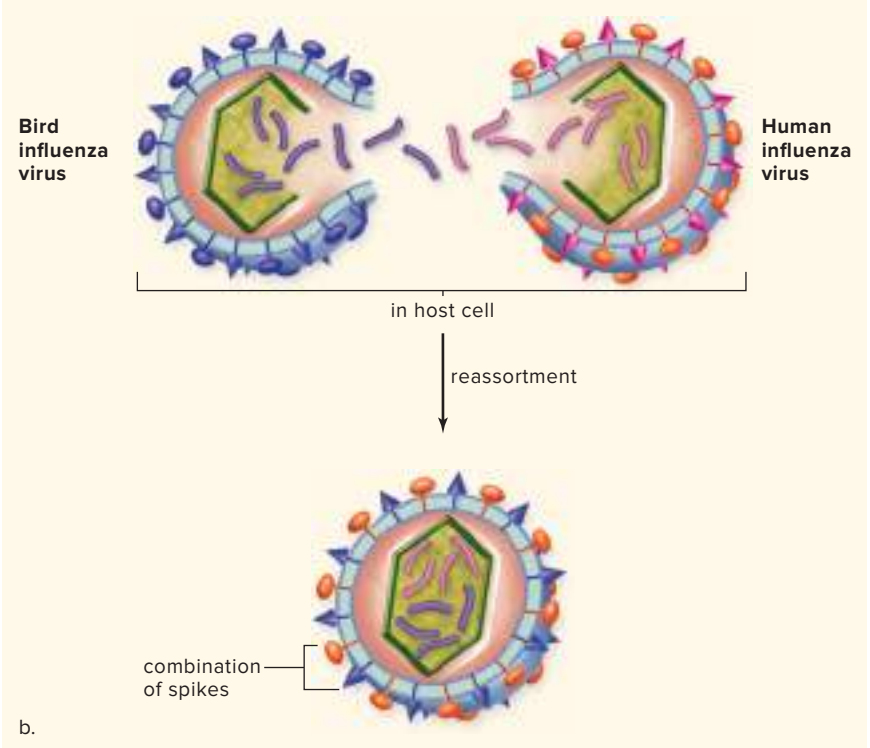


Figure 20A Spikes of a seasonal flu and bird flu virus. a. Genetic mutations in seasonal flu viral spikes change the strain of the flu every year. b. Alternatively, a combination of bird flu spikes and human flu spikes could form a pandemic flu, which is more dangerous to humans.

Viruses vary in shape from threadlike to polyhedral. However, all viruses possess the same basic anatomy: an outer **capsid** composed of protein subunits and an inner core of nucleic acid—either DNA or RNA. A viral genome may have as few as 4, or in the case of the giant viruses, as many as 2,300 genes. In contrast, a human cell contains just under 20,000 genes.

If the viral capsid is the outermost structure on a virus, the virus is said to be naked. Figure 20.1*a, b, c* gives examples of naked viruses. In contrast, enveloped viruses, like influenza virus and coronavirus, are surrounded by an outer membranous envelope actually derived from the host cell's plasma membrane. Figure 20.1*d* is an example of an enveloped virus. Inserted into the envelope are glycoprotein spikes coded for by the viral genome. These molecules vary among viruses and allow each virus to bind to a new host cell. Aside from its genome, a viral particle may also contain various proteins, especially enzymes such as the polymerases, needed to produce viral DNA and/or RNA.

Parasitic Nature of Viruses

As *obligate intracellular parasites,* viruses cannot replicate outside a living cell. Like prokaryotic and eukaryotic cells, viruses have genetic material. Whereas a cell is capable of copying its own genetic material in order to reproduce, a virus cannot duplicate its genetic material or any of its other components on its own. For a virus to reproduce, it must infect a living cell. Once inside a living cell, the virus "hijacks" the cell's protein synthesis machinery to replicate the nucleic acid and other parts of the virus, including the capsid, viral enzymes, and, for some viruses, the envelope.

Cells infected by some viruses are killed or damaged by the replicating virus, causing the symptoms associated with viral infections. For example, cells infected by adenovirus in the respiratory tract are lysed when viral replication is complete, leading to conditions such as bronchitis and pneumonia.

Host Specificity

Viruses infect a variety of cells, but they are *host specific,* meaning that any particular virus is only capable of reproducing within the cells of specific living organisms. The tobacco mosaic virus infects only plants in the tobacco family, and the rabies virus infects only mammals. Some human viruses are even specific to a particular tissue. Human immunodeficiency virus (HIV) enters only certain blood cells, the polio virus reproduces in spinal nerve cells, the hepatitis viruses infect only liver cells, and the SARS-CoV-2 virus infects cells in the lungs.

Host specificity is determined by the structure of molecules in the naked capsid or in the spikes on an enveloped virus. These attach in a lock-and-key manner with a receptor on the host cell's outer surface. A cell that does not have a receptor to match a virus exactly cannot be infected, because the virus will be unable to enter. Many antiviral medications are effective because they interfere with the lock-and-key attachment of viruses to host cells.

Reproduction of Viruses

Viruses are microscopic pirates, commandeering the metabolic machinery of a host cell during their reproduction cycle, which consists of five steps (Fig. 20.2):

1. *Attachment:* A virus binds to a specific host cell based on the host-specific match between virus surface molecules and host cell receptors.
2. *Penetration:* The host cell engulfs the virus or the virus injects its genome into the cytoplasm.
3. *Biosynthesis:* New viral components, including the capsid subunits, spikes, and copies of the genome, are synthesized using the host's ribosomes, enzymes, transfer RNA (tRNA), and energy.
4. *Maturation:* Viral components are assembled into new viruses.
5. *Release:* New viruses exit the host cell through lysis or budding in order to infect new host cells.

This reproductive cycle is most common, but notable exceptions do occur among different types of viruses, including bacteriophages, animal viruses, and retroviruses.

Lytic and Lysogenic Cycles

Bacteriophages (Gk. *bacterion,* "rod"; *phagein,* "to eat") are viruses that parasitize bacteria. They have provided a useful model for scientists to study the reproductive cycle of viruses. Figure 20.2 shows two alternative life cycles of bacteriophages: the lytic cycle and the lysogenic cycle.

In the **lytic cycle,** the five steps of the viral reproduction cycle occur in immediate sequence. The lytic cycle is so named because the bacterial host is lysed at the end of the cycle during release by a virally coded enzyme called lysozyme. In the process, the bacterial cell dies, and several hundred new viral particles are released.

When a virus enters the **lysogenic cycle** instead of the lytic cycle, viral reproduction and release of new viruses does not occur immediately, but reproduction may take place in the future. In the meantime, the infecting phage is *latent*—not actively replicating.

Following attachment and penetration, *integration,* instead of biosynthesis, occurs: Viral DNA becomes incorporated into bacterial DNA with no destruction of host DNA. While latent, the viral DNA is called a *prophage.* The prophage is replicated along with the host DNA, and all subsequent cells, called **lysogenic cells,** carry a copy of the prophage genome. Sometime in the future, certain environmental factors, such as ultraviolet radiation, can induce the prophage to reenter the lytic stage of biosynthesis, followed by maturation and release.

Lysogenic bacterial cells may have distinctive properties due to the prophage genes they carry. The presence of a prophage may cause a bacterial cell to produce a toxin. For example, if the same bacterium that causes strep throat happens to carry a certain prophage, it will cause scarlet fever, so named because the toxin causes a widespread red skin rash as it spreads through the body. Likewise, diphtheria is caused by a bacterium carrying a prophage. The diphtheria toxin damages the lining of the upper respiratory

ISTUDY

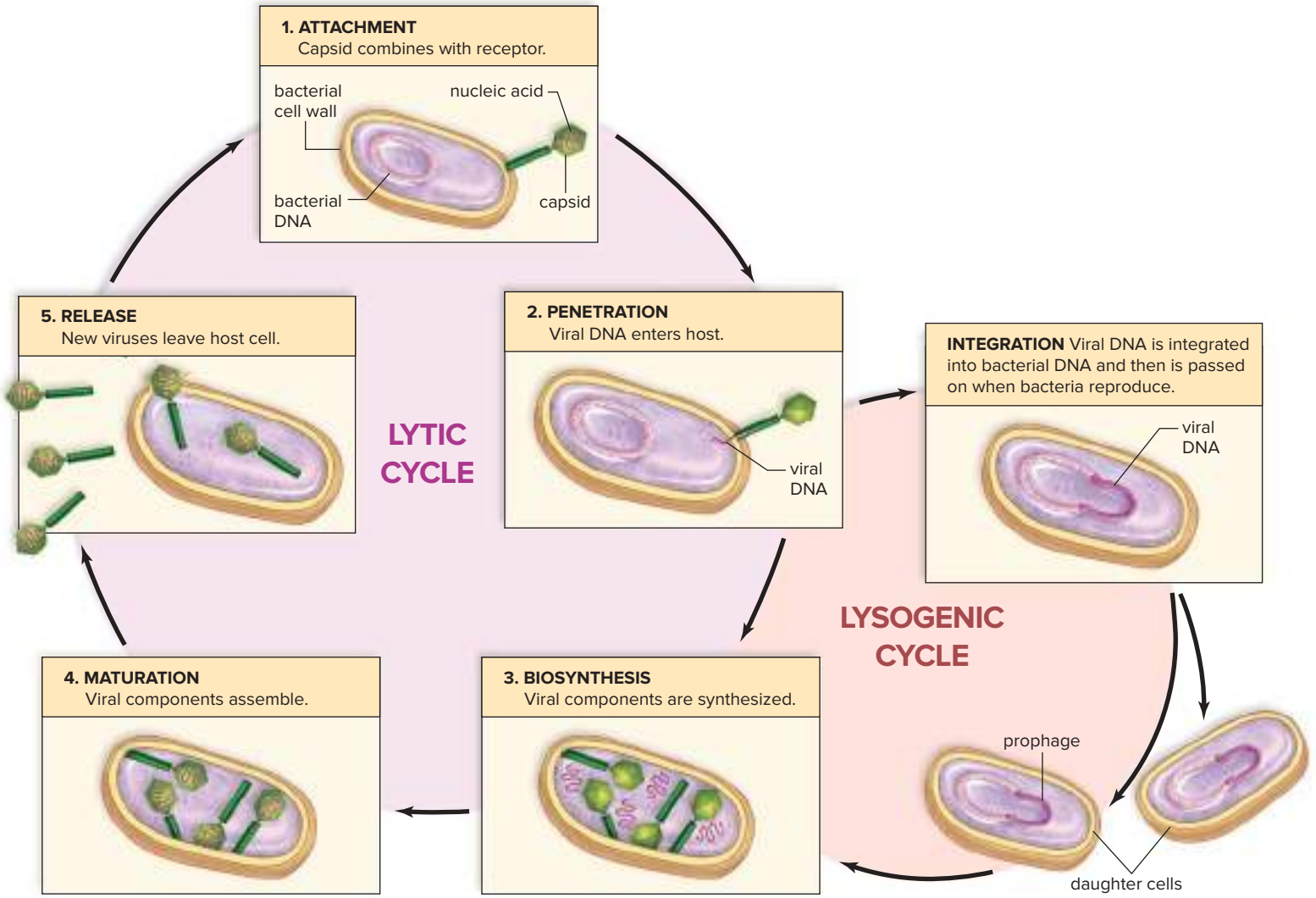


Figure 20.2 Reproduction in viruses. In the reproductive cycle of a virus, viral particles escape when the cell is lysed (broken open). In some cases, the viral DNA is integrated into host DNA.

tract, resulting in the formation of a thick membrane that restricts breathing.

Reproduction of Animal Viruses

Animal viruses reproduce in a manner similar to that of bacteriophages, but various animal viruses have different ways of introducing their genetic material into their host cells. For some enveloped viruses, the process is as simple as attachment and fusion of the spike-studded envelope with the host cell's plasma membrane. Many naked and some enveloped viruses are taken into host cells by endocytosis. Once the virus enters, it is uncoated—that is, the capsid and, if necessary, the envelope are removed. The viral genome, either DNA or RNA, is now free of its covering, and the virus continues the lytic cycle or enters the lysogenic cycle.

Viruses that are highly virulent enter directly into the lytic cycle, causing rapid and severe destruction of host cells. The Ebola virus is highly virulent, and an average of 50% of people die from the disease within 2–21 days of infection. In contrast, HIV enters first into the lysogenic cycle and thus can lie inactive, or latent, for many years before AIDS symptoms emerge.

Viral release is just as variable as penetration for animal viruses. Some mature viruses are released by budding. During budding, the virus picks up its envelope, consisting of lipids, proteins, and carbohydrates, from the host cell. Most enveloped animal viruses acquire their envelope from the plasma membrane of the host cell, but some take envelopes from other membranes, such as the nuclear envelope or Golgi apparatus. Envelope markers, such as the glycoprotein spikes that allow the virus to enter a host cell, are coded for by viral genes. Naked animal viruses are usually released by host cell lysis.

Retroviruses

Retroviruses (L. *retro,* "backward") are animal viruses with an RNA genome that is converted into DNA within the host cell by an enzyme called **reverse transcriptase.** Figure 20.3 illustrates the reproduction of HIV, a type of retrovirus.

Before a retrovirus can integrate into the host's genome, or use the host cell's machinery to transcribe and translate its proteins, it must first convert its RNA to DNA. First, the enzyme reverse transcriptase synthesizes from its RNA genome a single DNA strand, called cDNA because it is a complementary DNA strand to the

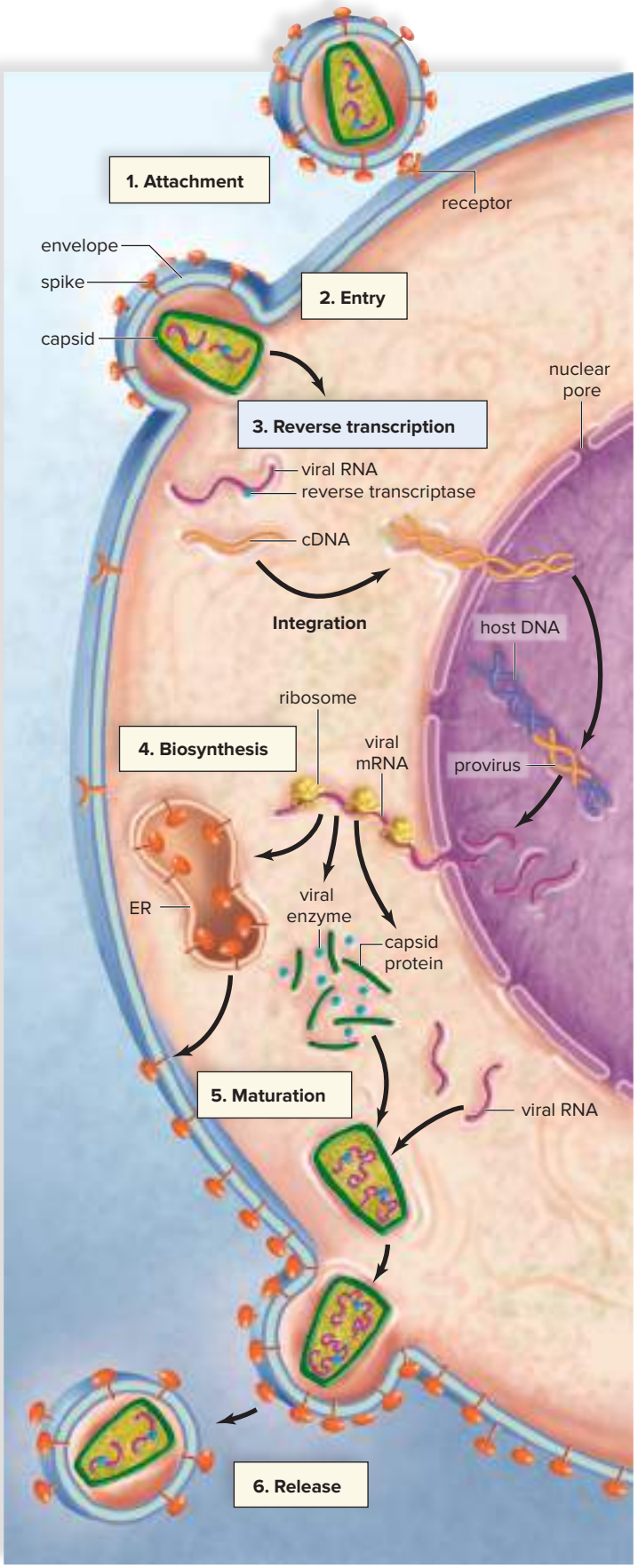


Figure 20.3 Reproduction of the retrovirus HIV. HIV uses reverse transcription to produce a DNA copy (cDNA) of RNA genes; double-stranded DNA integrates into the cell's chromosomes before the virus reproduces and buds from the cell. Reverse transcription is unique to retroviruses.

viral RNA. The single strand of cDNA is used as a template to make a double-stranded DNA.

Using host enzymes, the double-stranded virus DNA is integrated into the host genome. The viral DNA remains in the host genome and is replicated when host DNA is replicated. When and if this DNA is transcribed, new viruses are produced by the steps already cited: biosynthesis, maturation, and release; in the case of HIV, it is released by budding from the host plasma membrane. As mentioned, HIV can remain latent for many years. Without treatment, the median survival time after HIV infection is 9–11 years.

The emergence of AIDS can be delayed by treatment with antiretroviral drugs, which interfere with one or more of the steps of HIV reproduction. For example, one type of antiretroviral drug, AZT, consists of reverse transcriptase inhibitors, which bind to reverse transcriptase and interfere with its function. Another type of drug, Acyclovir, which is also used to treat herpes, inhibits the replication of the HIV viral DNA.

However, HIV is a very rapidly evolving virus, which makes it difficult to find a cure. Viruses such as HIV are often engaged in an "arms race" with an animal's immune system. Its rapid evolution results in many different types of HIV within a patient. Although a new drug or vaccine may work against some, it is very likely that one type within the billions of virus copies will evolve resistance and continue the infection. The rapid evolution of resistance in HIV is why there is currently no cure.

Viroids and Prions

At least a thousand different viruses cause diseases in plants. About a dozen diseases of crops, including potatoes, coconuts, and citrus, have been attributed not to viruses but to **viroids,** which are naked strands of RNA (not covered by a capsid). Like viruses, however, viroids direct the cell to produce more viroids.

A number of fatal brain diseases, known as *transmissible spongiform encephalopathies,* or TSEs, have been attributed to **prions,** a term coined for *proteinaceous infectious* particles. Prions are proteins that normally exist in an animal but have a different conformation, or structure. Like viruses, prions cannot replicate on their own but cause infection by interacting with a normal protein and altering its structure. The process that changes the structure from the normal protein conformation to the prion conformation can be as simple as changing a chemical bond. Once a prion infects tissue, a chain reaction begins that converts normal proteins to prions at an exponential rate.

TSEs are **neurodegenerative diseases,** or those that destroy nerve tissue in the brain. In the brain, prion proteins form clusters that break down normal brain tissue, creating small holes that give the brain a spongy appearance. All TSEs are untreatable and fatal. Mad cow disease, or bovine spongiform encephalopathy (BSE), is a neurodegenerative disease of cattle that is transmissible to humans by eating cattle brains or meat contaminated with prion-infected brain tissue. The discovery of prions began when it was observed that members of a primitive tribe in the highlands of Papua New Guinea died from a disease commonly called kuru (meaning "trembling with fear"). The disease occurred after the individual had participated in the cannibalistic practice of eating a deceased person's brain; the brain was evidently infected with prions.

ISTUDY

Current studies are even linking prions as the causative agent of certain types of dementia and Alzheimer's disease.

Check Your Progress 20.1

1. Describe the structure of a virus.
2. Explain why a virus can only infect specific cells and organisms.
3. Compare the lytic and lysogenic cycles of viral reproduction.
4. Explain, from an evolutionary standpoint, why it is beneficial to a virus if its host lives.

20.2 Emerging Infectious Diseases and SARS-CoV-2

Learning Outcomes

Upon completion of this section, you should be able to

1. Define the term *emerging disease*.
2. Summarize the origins of coronaviruses.
3. Describe the life cycle of the SARS-CoV-2 virus.

An infectious disease may be classified as an **emerging disease** if it is occurring for the first time in human populations, is rapidly increasing in its incidence (or frequency) in humans, or is entering into new geographic regions. Examples of **emerging viruses** are avian influenza (H5N1), swine flu (H1N1), Middle East respiratory syndrome (MERS), Zika virus, and SARS-CoV-2 (Fig. 20.4).

Bird flu (H5N1) remains a constant concern of world-health experts. Wild ducks are resistant to avian influenza viruses that can spread from them to chickens, which increases the likelihood that the disease, another type of influenza, will spread to humans. Another way a virus could emerge is by a change in the mode of transmission—for example, from requiring contact for transmission to being transmitted through the air. Health officials are concerned that this may occur for viruses such as H5N1, thus greatly increasing the number of people who may be exposed to bird flu.

As detailed in the Biological Systems feature, "Zika Virus," the Zika virus was previously found only in West Africa and Southeast Asia, with no outbreaks detected until 2007. However, in 2015, a large outbreak began in Brazil, due to a suitable environment that included high populations of mosquitoes and a susceptible human population.

Other factors can also cause infectious viruses to emerge. Viruses are well-known for their high mutation rates. Some of these mutations affect the structure of the spikes or capsids, so a virus that previously could infect only a particular animal species acquires the ability to infect humans. For example, the diseases AIDS, Ebola fever, and COVID-19 are caused by viruses that at one time infected only primates or bats.

In this section, we will focus on the SARS-CoV-2 emerging virus and the COVID-19 pandemic.

Coronaviruses and COVID-19

SARS-CoV-2, the virus associated with the respiratory disease COVID-19 (see Section 35.3), is an example of a coronavirus. Coronaviruses are named for their distinctive, crownlike spikes emerging from the surface of the virus. They are a relatively common animal virus (zoonotic virus) that may infect a wide variety of animal species, such as bats, camels, birds, livestock, and humans. In some cases, coronaviruses are known to cause respiratory and intestinal diseases, especially in livestock (chickens, pigs, and cattle), and in the case of the SARS/MERS class of viruses, humans.

Coronavirus infections in humans were first recorded in the 1950s, and were typically associated with mild cases of the common cold. A more severe form of human coronavirus, SARS-CoV,

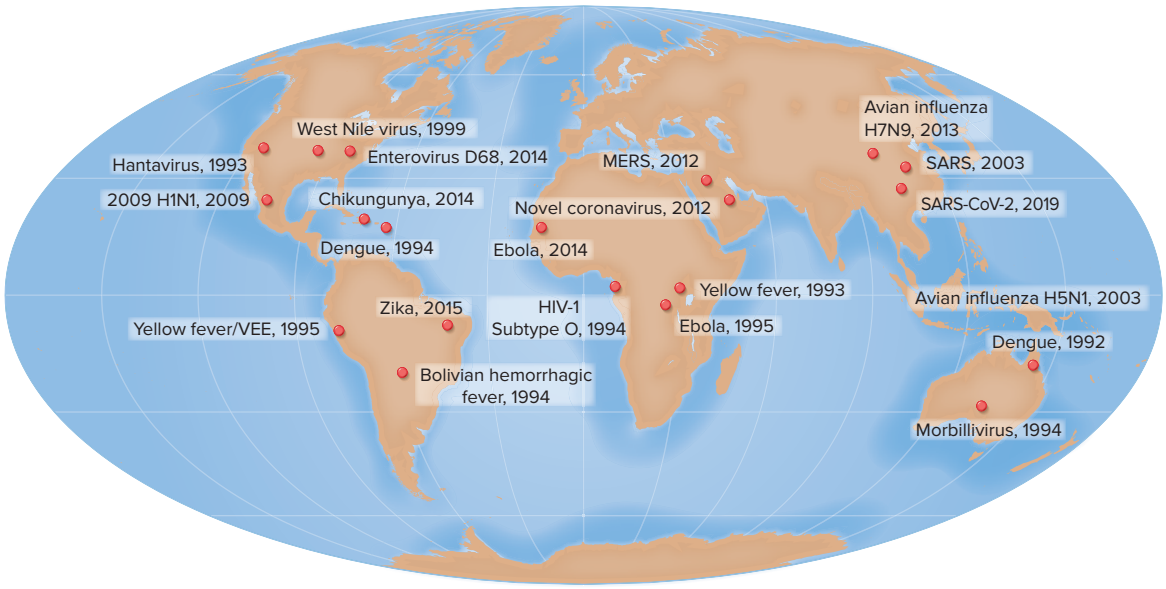


Figure 20.4 Emerging viruses. Emerging viruses, such as those noted here, are new or demonstrate increased prevalence. These disease-causing agents may have acquired new virulence factors, or environmental factors may have encouraged their spread to an increased number of hosts.

ISTUDY

THEME Biological Systems

Zika Virus

An Outbreak

Between 2015 and 2016, Brazil experienced a dramatic increase in the number of babies born with microcephaly (Fig. 20B), or abnormal smallness of the brain. Infants born with this condition can fail to thrive, can have poor coordination and short stature, and may have intellectual, speech, and developmental disabilities.

At the same time, health officials were investigating outbreaks of an illness that caused rash, conjunctivitis, fever, and joint pain. Tests revealed it was the Zika virus, and researchers soon concluded that the infection could pass from a mother to a fetus, causing microcephaly.

Spread of Zika

The Zika virus is spread by certain species of *Aedes* mosquitoes, the same mosquitoes that spread dengue and chikungunya viruses. Once a person is infected, they can pass the virus back to new mosquitoes when bitten, spreading the infection. As mentioned previously, Zika can spread from mother to fetus, but it can also spread through sexual contact with an infected individual.

By tracing mutations in the genetic code of the virus in Brazil, researchers were able to conclude it arrived in Brazil in 2013, likely during a FIFA soccer tournament where a team from Tahiti played. The tournament attracted spectators from Tahiti, where the virus was found, and an infected individual was likely bitten by a mosquito, thus introducing the virus to Brazil.

It has rapidly spread throughout South America, Central America, and the Caribbean—all locations where certain species of *Aedes* mosquitoes are found. Because *Aedes* mosquitoes live in the United States too (Fig. 20C), the virus has the potential to spread, and the CDC, along with scientists, is investigating prevention and control strategies.

Control of Zika

Practical prevention strategies for Zika include:

- Avoiding travel to affected areas
- Using insect repellent
- Wearing long sleeves and pants
- Staying indoors when mosquitoes are active

While prevention is necessary now, some are pursuing a more permanent approach: a mosquito extinction gene. Instead of focusing on the virus itself, the solution may lie in the organism that spreads the disease.

Male *Aedes* mosquitoes can be genetically modified to contain lethal genes they pass to offspring. If they mate with a female, the offspring will not survive. Eventually, the population will decline and possibly disappear. Small-scale trials have reduced populations by over 80% within several months.

While the ethical and biological concerns of this approach are prominent, eradicating mosquitoes could have far-reaching effects because, along with Zika, the *Aedes* mosquitoes are also responsible for dengue fever, chikungunya, and yellow fever, diseases that affect hundreds of thousands of people each year. Aside from *Aedes*, *Anopheles* mosquitoes may be a more important target. They are responsible for spreading malaria, a disease that infects over 200 million people each year, killing over 400,000, mostly children under five.

Figure 20B Microcephaly. Zika infection can spread from a pregnant mother to her infant, causing microcephaly.
Source: Centers for Disease Control and Prevention. "Facts about Microcephaly." Accessed February 11, 2020. https://www.cdc.gov/ncbddd/birthdefects/microcephaly.html.

Questions to Consider

1. How might eradication of mosquitoes affect an ecosystem?
2. What ethical concerns exist surrounding the deliberate extinction of a species?

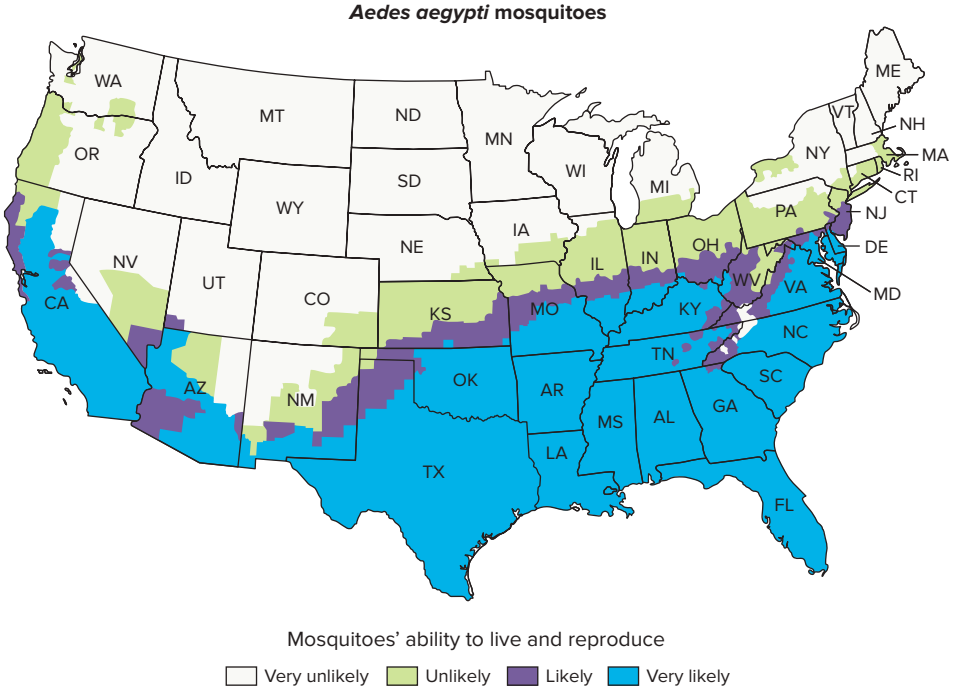


Figure 20C *Aedes* mosquitoes in the United States. *Aedes aegypti* is a common form of mosquito that is likely to carry and spread the Zika virus.
Source: Centers for Disease Control and Prevention. "Estimated potential range of *Aedes aegypti* and *Aedes albopictus* in the United States, 2017." Accessed February 11, 2020. https://www.cdc.gov/zika/vector/range.html.

appeared in China in 2002. Thought to have originated in bats or civet cats and then transmitted to humans, SARS-CoV spread to cause a global outbreak in early 2003. It resulted in over 8,000 known cases and 774 deaths worldwide, with a fatality rate of just under 10%. No human cases of SARS-CoV have been reported since 2004.

Another form of human coronavirus, MERS-CoV, first appeared in Jordan in 2012. Most cases of MERS-CoV occurred in countries on the Arabian Peninsula and were linked to travel from infected areas. The virus is believed to have originated in bats, then spread to camels, and finally was contracted by humans living or working near them. By the end of 2019, there were nearly 2,500 confirmed cases of MERS-CoV, resulting in 858 deaths—a fatality rate of 34.4%.

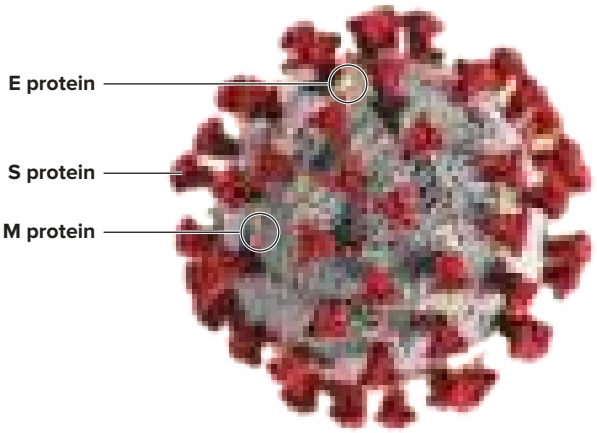


Figure 20.5 The SARS-CoV-2 virus. The surface of a coronavirus showing the different forms of glycoproteins Alissa Eckert, MS; Dan Higgins, MAMS/CDC

While the SARS-CoV-2 virus is a very new emerging virus, researchers have begun to piece together aspects of its life cycle based on their knowledge of other coronaviruses. This life cycle is shown in Figure 20.6.

Because coronaviruses consist of single-stranded RNA, their genetic information can be translated directly into the four structural

SARS-CoV-2 Virus

Coronaviruses are an example of a single-stranded animal RNA virus. They are considerably larger than most RNA viruses, and like many other single-stranded RNA viruses, such as Ebola virus and influenza, they have high mutation rates.

Coronaviruses have an exterior envelope made from the plasma membrane of the host cells. This membrane may be disrupted (or broken down) by the action of soaps and alcohol-based sanitizers. Embedded within this membrane is the "crown," a series of glycoproteins (Fig. 20.5). These include:

- *S protein*. Spike proteins are responsible for the attachment of the virus to specific receptors on its host cell.
- *M protein*. Membrane proteins are involved in the assembly of the virus.
- *E protein.* Envelope proteins are also involved in the assembly of the virus, as well its exit from the host cell.

Inside the coronavirus, the single-stranded RNA genetic material is surrounded by a helical protein structure, called a nucleocapsid (sometimes just referred to as a capsid). The genome of SARS-CoV-2 contains approximately 20 genes, 3 of which code for the S, M, and E proteins, as well as the N protein, the nucleocapsid. In comparison, bacteria, on average, have less than 2,700 genes, and humans have just under 20,000.

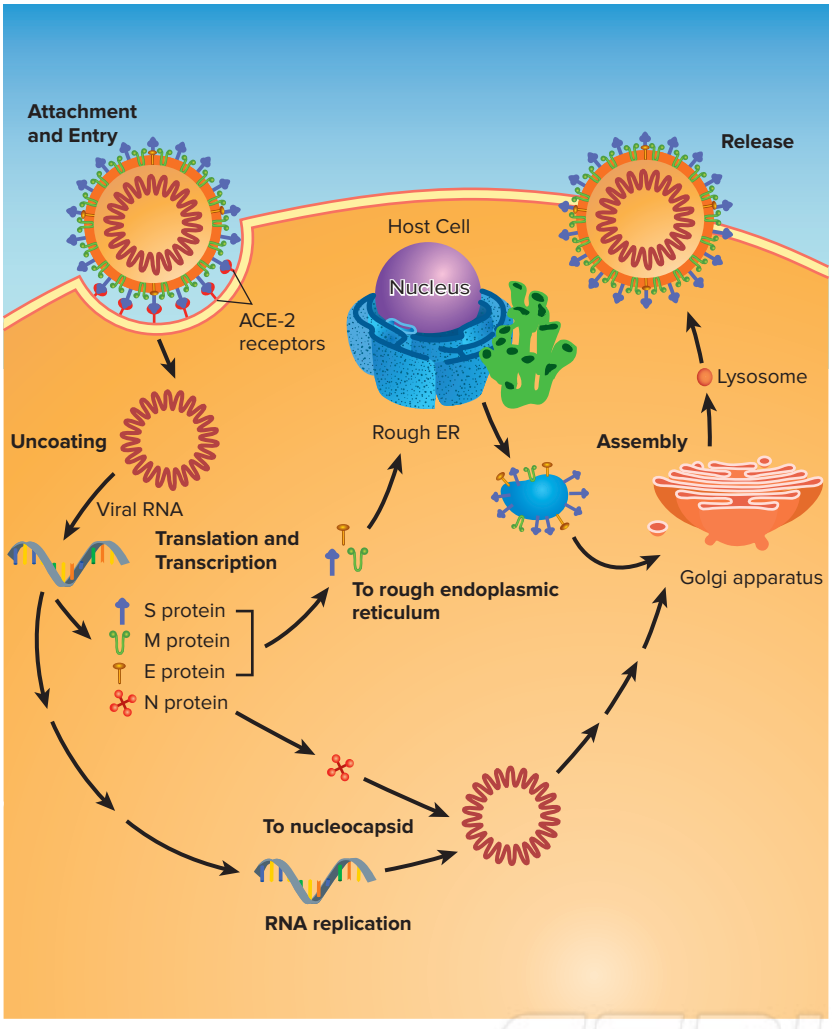


Figure 20.6 Life cycle of SARS-CoV-2. The SARS-CoV-2 virus replicates its proteins and genetic material using the cellular machinery of its host cell.

ISTUDY

proteins (for a review of gene expression, see Section 12.3). With the exception of the N protein, the other structural proteins are sent to the rough ER and Golgi apparatus where they help assemble the new viruses and become part of the envelope surrounding the virus. The N protein is used to assemble the nucleocapsid, which surrounds the new RNA material produced by replication.

We will take a closer look at how SARS-CoV-2 causes COVID-19 in Section 35.3.

Check Your Progress 20.2

1. Distinguish between an emerging disease and a reemerging infectious disease.
2. Explain how emerging viruses arise.
3. Explain the role of the glycoproteins on the surface of the SARS-CoV-2 virus.

20.3 Prokaryotes

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the evolution of prokaryotes.
2. Identify the structural features of prokaryotes.
3. Describe at least four ways in which the cells of prokaryotes differ from eukaryotic cells.

Prokaryotes include bacteria and archaea, which are fully functioning, living, single-celled organisms. Because they are microscopic, the prokaryotes were not discovered until the seventeenth-century Dutch microscopist Antonie van Leeuwenhoek first described them, along with many other microorganisms (see the Nature of Science feature, "Microscopy Today," in Section 4.1).

Leeuwenhoek and others after him believed that the "little animals" he observed could arise spontaneously from inanimate matter. *Spontaneous generation,* the idea that living organisms can emerge from nonliving things, was common at the time. When meat spoiled, for example, it was thought that maggots arose from the meat spontaneously.

For about 200 years, scientists carried out various experiments to determine the origin of microorganisms in laboratory cultures. Finally, in about 1850, Louis Pasteur devised an experiment for the French Academy of Sciences (described in Fig. 20.7). It showed that a previously sterilized broth cannot become cloudy with microorganism growth unless it is exposed directly to the air where bacteria are abundant.

Today, we know that bacteria are plentiful in air, water, and soil and that new bacteria arise from the division of preexisting bacteria—not by spontaneous generation. We also know a lot about the structure of bacteria; their membranes, DNA, and proteins; how and where they live; where they get their nutrition; and how they coordinate replication. First, we will examine the general characteristics of prokaryotes before exploring the specific characteristics of domain Bacteria (see Section 20.4) and domain Archaea (see Section 20.5).

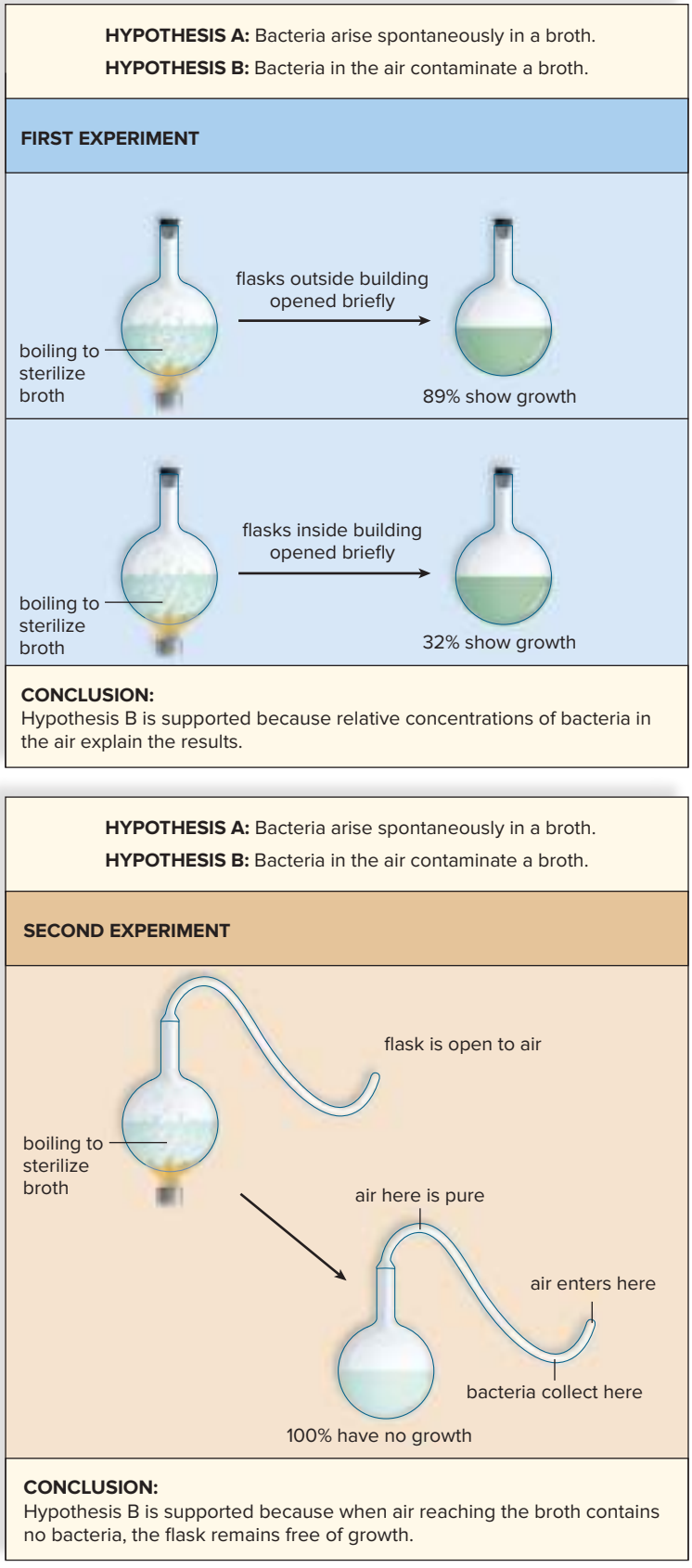


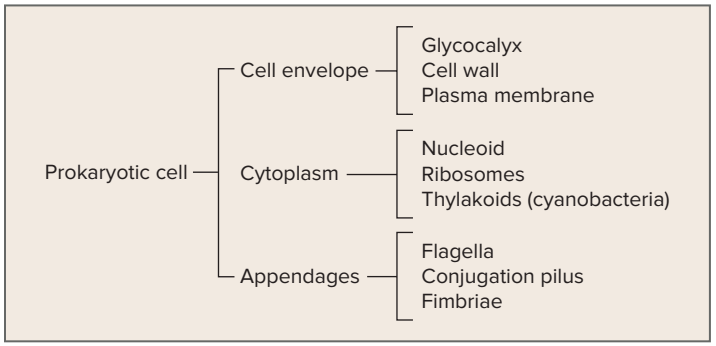
Figure 20.7 Pasteur's experiments. Pasteur disproved the theory of spontaneous generation of microbes by performing these types of experiments.

ISTUDY

Structure of Prokaryotes

Prokaryotes generally range in size from 1 to 10 µm in length and from 0.7 to 1.5 µm in width. To put this into perspective, an average human is about 1.5 m tall, or 1 million times longer than a bacterium (see Fig. 4.2). The term *prokaryote* means "before a nucleus," and these organisms lack a membrane-bound nucleus like that found in eukaryotes. Prokaryotic fossils exist that are over 3.5 billion years old, and the fossil record indicates that the prokaryotes were alone on Earth for about 2.5 billion years. During that time, they became extremely diverse in structure, and especially diverse in metabolic capabilities. Prokaryotes are adapted to living in most environments, because they have a wide variety of ways they can acquire and use energy.

The organization of a typical prokaryotic cell is illustrated in the following diagram:



A prokaryotic cell is surrounded by a plasma membrane and a cell wall situated outside the membrane (Fig. 20.8). The cell wall prevents a prokaryote from bursting or collapsing due to fluctuations in the amount of fluid inside the cell. Yet another layer may exist outside the cell wall, depending on the type of prokaryote.

In many bacteria, the additional cover is a layer of polysaccharides called a *glycocalyx* (Fig. 20.8*a*). A well-organized glycocalyx is called a capsule, whereas a loosely organized one is called a slime layer. Instead of a glycocalyx covering, many bacteria and archaea have a layer comprising protein, or glycoprotein, called an *S-layer.* In parasitic forms of bacteria, these outer coverings help protect the cell from host defenses.

Some prokaryotes move by means of *flagella* (Fig. 20.8*b*). A bacterial flagellum has a filament composed of strands of the protein flagellin wound in a helix. The filament is inserted into a hook anchored by a basal body. The 360-degree rotation of the flagellum causes the cell to spin and move forward. The archaeal flagellum is similar but more slender and apparently lacks a basal body.

Many prokaryotes adhere to surfaces by means of *fimbriae*—short, bristlelike fibers extending from the surface (Fig. 20.8*a*). The fimbriae of the bacterium *Neisseria gonorrhoeae* allow it to attach to host cells and cause gonorrhea, a sexually transmitted disease.

A prokaryotic cell lacks the membranous organelles of a eukaryotic cell; instead, the plasma membrane contains many infolds, where metabolic reactions such as respiration and photosynthesis occur. Although prokaryotes do not have a nucleus, they do have a dense area called a *nucleoid,* where a single chromosome consisting of a circular strand of DNA is found. A typical bacterial chromosome contains a few thousand genes that primarily code for proteins. Depending on the species, a bacterial cell may have one to four identical chromosomes. Many prokaryotes also have accessory rings of DNA called *plasmids,* which contain genes for antibiotic resistance, production of toxins, or degradation

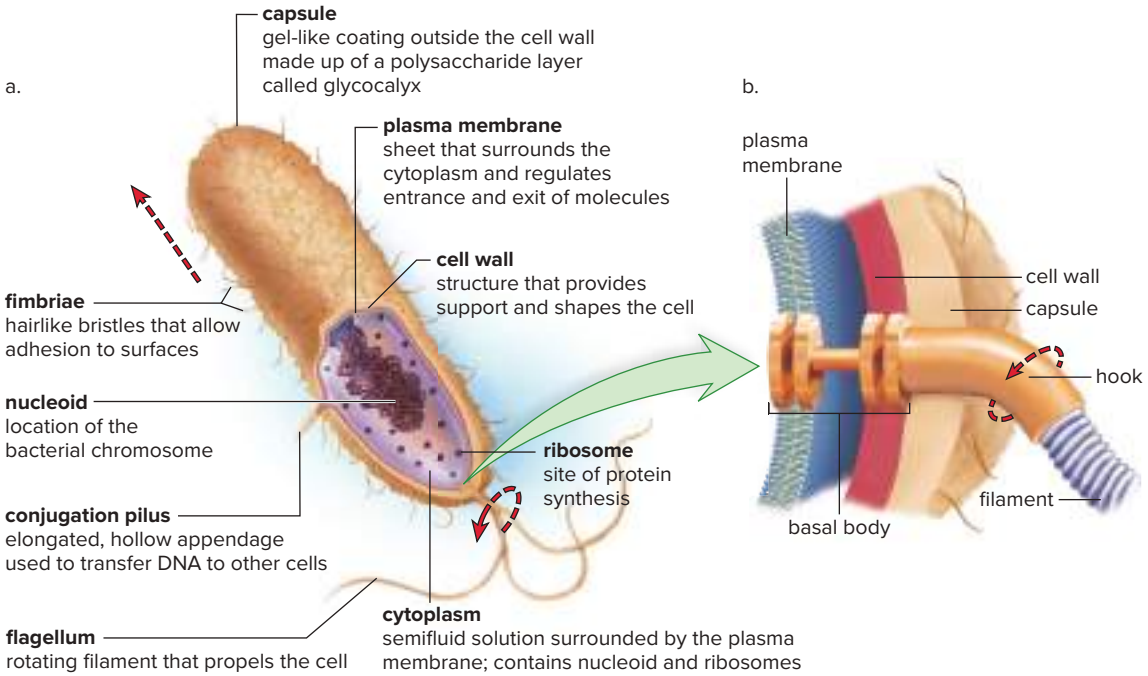


Figure 20.8 Features of prokaryotic cells. **a.** Structural components of a generalized prokaryotic cell. **b.** Each flagellum of a bacterium contains a basal body, a hook, and a filament. The red-dashed arrows indicate that the hook and filament rotate 360°.

ISTUDY

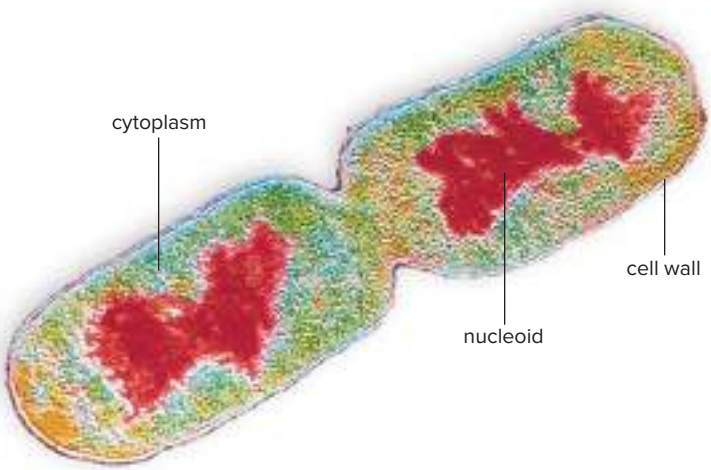


Figure 20.9 Binary fission. When conditions are favorable for growth, prokaryotes divide to reproduce. This is a form of asexual reproduction, because the daughter cells have exactly the same genetic material as the parent cell. CNRI/Science Source

of chemicals. Plasmids can also be extracted and used to carry foreign DNA into host bacteria during genetic engineering processes (see Section 14.1).

Protein synthesis in a prokaryotic cell is carried out by thousands of ribosomes, which are smaller than eukaryotic ribosomes.

Reproduction in Prokaryotes

Mitosis, which requires the formation of a spindle apparatus, does not occur in prokaryotes. Instead, prokaryotes reproduce asexually by means of **binary fission** (Fig. 20.9).

The single circular chromosome replicates, and then the two copies separate as the cell enlarges. Newly formed plasma membrane and cell wall separate the cell into two cells. Prokaryotes have a generation time as short as 12 minutes under favorable conditions. Mutations are generated and passed on to offspring more quickly than in eukaryotes. Also, prokaryotes are haploid, so mutations are immediately subjected to natural selection, which determines any possible adaptive benefit in the particular environment.

In eukaryotes, genetic recombination occurs as a result of sexual reproduction. Sexual reproduction does not occur among prokaryotes, but three means of genetic recombination have been observed in prokaryotes.

- During **conjugation,** two bacteria are temporarily linked together, often by means of a **conjugation pilus** (see Fig. 20.8*a*). While they are linked, the donor cell passes DNA to a recipient cell in the form of a plasmid.
- **Transformation** occurs when a cell picks up free pieces of DNA secreted by live prokaryotes or released by dead prokaryotes.
- During **transduction,** bacteriophages carry portions of DNA from one bacterial cell to another. Viruses have also been found to infect archaeal cells, so transduction may play an important role in gene transfer for both domains of prokaryotes.

Check Your Progress 20.3

1. Explain the connection between Pasteur's experiment and the sterilization of surgical instruments.
2. Describe the difference between a prokaryote nucleoid and a eukaryote nucleus.
3. Define three ways in which prokaryotes can recombine their genetic material without sexual reproduction.

20.4 Bacteria

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the similarities and differences in the cell wall structure of Gram-positive and Gram-negative bacteria.
2. Identify three metabolic types of bacteria and describe how they obtain nutrients from their environments.
3. Describe the unique properties of cyanobacteria.

Bacteria (domain Bacteria) are the more common type of prokaryote. The amount of bacteria on our planet is amazing: Even though we can't see them with the naked eye, the biomass of bacteria on Earth exceeds that of plants and animals combined. They are found in practically every kind of environment on Earth. In this section, we consider the bacteria—their characteristics, metabolism, and lifestyle.

Characteristics of Bacterial Cells

Most bacterial cells are protected by a cell wall composed of the unique molecule **peptidoglycan,** a complex of polysaccharides linked by amino acids. Two types of bacteria have been distinguished based on the structure of their cell wall: Gram-positive bacteria and Gram-negative bacteria.

Gram-positive bacteria have a very thick peptidoglycan cell wall relative to thin-walled *Gram-negative bacteria,* as shown in Figure 20.10. In addition, Gram-negative bacteria are surrounded by a second plasma membrane outside the cell wall, which often blocks antibiotic drugs, making infections difficult to treat.

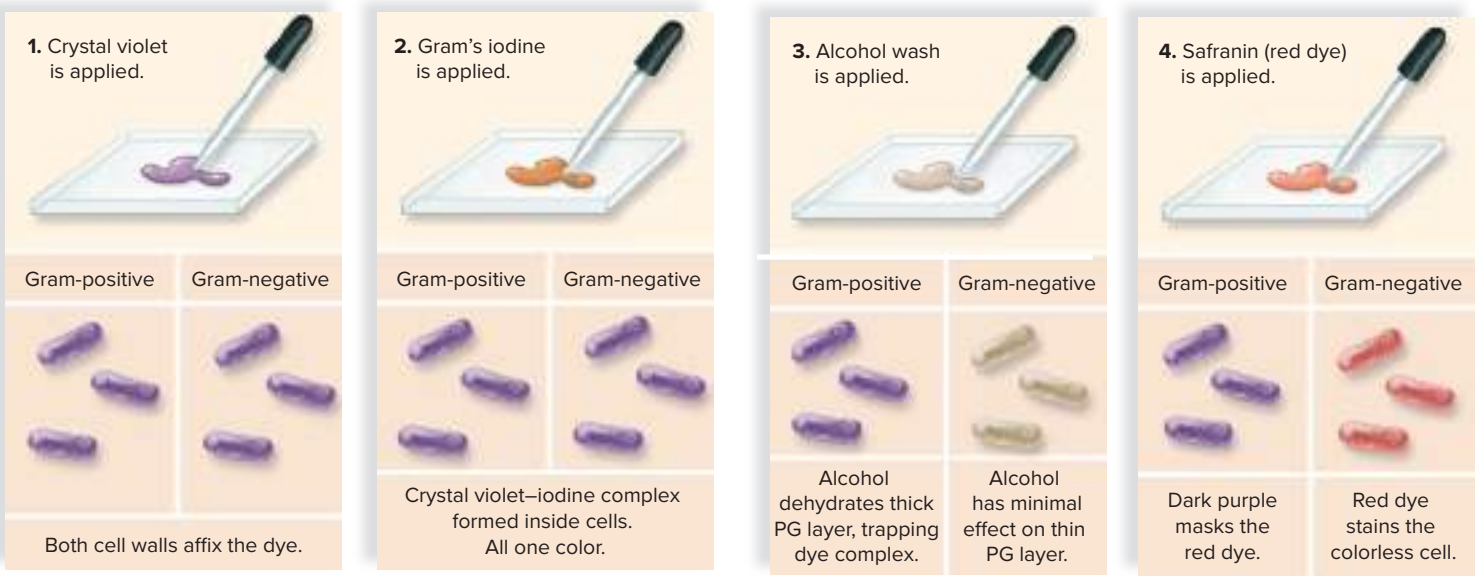
A procedure called the Gram stain, shown in Figure 20.10*a*, lends its name to the types of bacteria, because it distinguishes their cell wall type based on the color bacteria appear after staining. Gram-positive bacteria appear dark purple after the Gram stain process, whereas Gram-negative bacteria appear red, providing a useful first step for identifying unknown bacteria causing an infection.

Bacteria can also be described in terms of their three basic cell shapes (Fig. 20.11):

- Spirilli (*sing.,* spirillum), spiral-shaped or helical-shaped
- Bacilli (*sing.,* bacillus), rod-shaped
- Cocci (*sing.,* coccus), round or spherical

In addition to cell wall type and shape, bacteria can be characterized by their growth arrangement. For example, *staph* arrangement describes clusters of cells (e.g., staphylococcus), *strept*

ISTUDY



a.

b. 1,000×

Figure 20.10 Gram staining. **a.** The thick peptidoglycan (PG) layer encasing Gram-positive bacteria traps crystal violet dye, so the bacteria appear purple after the Gram stain. Because Gram-negative bacteria have much less peptidoglycan (located between the plasma membrane and an outer membrane), they do not retain the crystal violet dye and so exhibit the red counterstain (usually a safranin dye). **b.** A micrograph showing the results of a Gram stain with both Gram-positive and Gram-negative cells.
(photo): (b): CDC/Dr. William A. Clark

arrangement describes chains (e.g., streptobacillus), and *diplo* arrangement describes pairs (e.g., diplococcus).

Describing the characteristics of bacterial cells aids in the identification of species such as *Streptococcus pyogenes,* a Gram-positive streptococcus that causes strep throat, and *Neisseria gonorrhoeae,* a Gram-negative diplococcus responsible for gonorrhea.

Bacterial Metabolism

Bacteria are astoundingly diverse in terms of their metabolic lifestyles. With respect to basic nutrient requirements, bacteria are not much different from other organisms. One difference, however, concerns the need for oxygen. Most bacteria are aerobic and, like animals, require a constant supply of oxygen to carry out cellular respiration. Other bacteria, called **facultative anaerobes,** are able to grow in either the presence or absence of gaseous oxygen. Some bacteria are **obligate anaerobes** and are unable to grow in the presence of free oxygen. A few serious illnesses—such as botulism, gas gangrene, and tetanus—are caused by anaerobic bacteria that infect oxygen-free environments in the human body, such as in the intestine or in deep puncture wounds.

a. Spirillum: *Spirillum volutans* LM 400×
b. Bacilli: *Escherichia coli* SEM 13,300×
c. Cocci: *Streptococcus thermophilus* SEM 6,250×

Figure 20.11 Diversity of bacteria. **a.** Spirillum, a spiral-shaped bacterium. **b.** Bacilli, rod-shaped bacteria. **c.** Cocci, round bacteria. (a): Ed Reschke/Stone/Getty Images; (b): Eye of Science/Science Source; (c): SCIMAT/Science Source

ISTUDY

Autotrophic Bacteria

Bacteria called **photoautotrophs** (Gk. *photos,* "light"; *auto,* "self"; *trophe,* "food") are photosynthetic (for a review of photosynthesis, see Section 7.2). They use solar energy to reduce carbon dioxide to organic compounds. There are two types of photoautotrophic bacteria: those that perform anoxygenic photosynthesis and those that perform oxygenic photosynthesis. Their characteristics are shown next:

Photoautotrophic Bacteria

Anoxygenic Photosynthesis	Oxygenic Photosynthesis
–Does not produce O_2	–Produces O_2
–Photosystem I only	–Photosystems I and II
–Unique type of chlorophyll called bacteriochlorophyll	

Green sulfur bacteria and some purple bacteria carry out anoxygenic photosynthesis. These bacteria usually live in anaerobic (oxygen-poor) conditions, such as the muddy bottom of a marsh. They cannot photosynthesize in the presence of oxygen, and they do not emit oxygen. In contrast, the cyanobacteria (see Fig. 20.14) contain chlorophyll *a* and carry on oxygenic photosynthesis, just as algae and plants do; that is, they reduce carbon dioxide to organic compounds and give off oxygen as a by-product.

Bacteria called **chemoautotrophs** (Gk. *chemo,* "pertaining to chemicals"; *auto,* "self"; *trophe,* "food") carry out chemosynthesis. They oxidize inorganic compounds such as hydrogen gas, hydrogen sulfide, and ammonia to obtain the necessary energy to reduce CO_2 to an organic compound. The nitrifying bacteria oxidize ammonia (NH_3) to nitrites (NO_2^-) and nitrites to nitrates (NO_3^-). Their metabolic abilities keep nitrogen cycling through ecosystems. Other bacteria oxidize sulfur compounds. They live in environments such as deep-sea vents 2.5 km below sea level.

The organic compounds produced by such bacteria and archaea support the growth of a community of organisms found at deep-sea vents (see Section 20.5). This discovery lends support to the suggestion that the first cells originated near these vents.

Heterotrophic Bacteria

Bacteria called **chemoheterotrophs** (Gk. *hetero,* "different") obtain carbon and energy in the form of organic nutrients produced by other living organisms. For example, parasitic bacteria feed on the tissues and fluids of their living host.

In many ecosystems, chemoheterotrophic bacteria called **saprotrophs** serve as *decomposers* that break down organic matter from dead organisms. Probably no natural organic molecule exists that cannot be digested by at least one prokaryotic species, and this plays a critical role in recycling matter and making inorganic molecules available to photosynthesizers.

The metabolic capabilities of chemoheterotrophic bacteria have long been exploited by humans. Bacteria are used commercially to produce chemicals such as ethyl alcohol, acetic acid, butyl alcohol, and acetones. Bacterial action is also involved in the production of butter, cheese, sauerkraut, rubber, silk, coffee, and cocoa. Even antibiotics are produced by some bacteria.

Symbiotic Relationships

Bacteria (and archaea) form symbiotic relationships (Gk. *sym,* "together"; *bios,* "life") in which two different species live together in an intimate way.

- In **mutualism,** both species benefit from the association.
- In **commensalism,** only one species benefits, whereas the other is unaffected.
- In **parasitism,** one species benefits while harming the other.

Mutualistic bacteria live in human intestines, where they release vitamins K and B_{12}, which we can use to help produce blood components. In the stomachs of cows and goats, mutualistic prokaryotes digest cellulose, enabling these animals to feed on grass. Mutualistic bacteria live in the root nodules of soybean, clover, and alfalfa plants, where they reduce atmospheric nitrogen (N_2) to ammonia, a process called nitrogen fixation (Fig. 20.12). Plants are unable to fix atmospheric nitrogen, leaving bacteria their only source for usable nitrogen.

Commensalism often occurs when one population modifies the environment in such a way that a second population benefits. Obligate anaerobes can live in our intestines only because the bacterium *Escherichia coli* uses up the available oxygen.

Parasitic bacteria cause diseases and therefore are called **pathogens;** a few are listed in Table 20.2. In some cases, the growth of microbes themselves does not cause disease; what they release is the pathological portion. When Gram-negative bacteria are killed by an antibiotic, their outer plasma membrane releases a substance called lipopolysaccharide, which acts as a *superantigen* to overstimulate the immune response. The result may be a high fever and a severe drop in blood pressure, leading to shock and possibly death.

Figure 20.12 Nodules of a legume. Some free-living bacteria carry on nitrogen fixation; however, bacteria of the genus *Rhizobium* invade the roots of legumes, with the resultant formation of nodules. Here, the bacteria convert atmospheric nitrogen to an organic nitrogen the plant can use. These are nodules on the roots of a broad bean plant (*Vicia* sp.). Nigel Cattlin/Alamy Stock Photo

ISTUDY

Table 20.2 Bacterial Diseases in Humans

Category	Disease
Sexually transmitted diseases	Syphilis, gonorrhea, chlamydia
Respiratory diseases	Strep throat, scarlet fever, tuberculosis, pneumonia, Legionnaires' disease, whooping cough, inhalation anthrax
Skin diseases	Erysipelas, boils, carbuncles, impetigo, acne, infections of surgical or accidental wounds and burns, leprosy (Hansen's disease)
Digestive tract diseases	Gastroenteritis, food poisoning, dysentery, cholera, peptic ulcers, dental caries
Nervous system diseases	Botulism, tetanus, leprosy, spinal meningitis
Systemic diseases	Plague, typhoid fever, diphtheria
Other diseases	Tularemia, Lyme disease

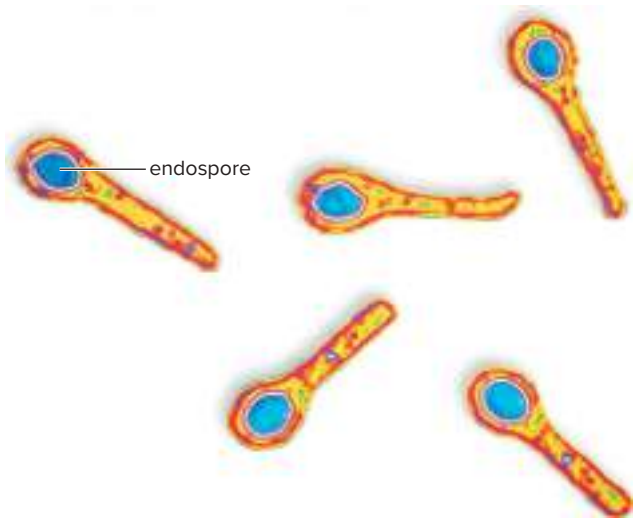


Figure 20.13 The endospore of *Clostridium tetani*. *C. tetani* produces a terminal endospore that causes it to have a drumstick appearance. If endospores gain access to a wound, they germinate and release bacteria that produce a neurotoxin. The patient develops tetanus, a progressive rigidity that can result in death. Immunization can prevent tetanus. Alfred Pasieka/Science Source

When someone steps on a rusty nail, *Clostridium tetani* bacteria can be injected deep into damaged tissue and produce a **toxin** that causes the disease tetanus. The bacteria never leave the site of the wound, but the tetanus toxin they produce does move throughout the body. This toxin prevents the relaxation of muscles. In time, the body contorts, because all the muscles have contracted. Eventually, suffocation occurs.

Fimbriae allow a pathogen to bind to certain cells, and their specificity determines which organs or cells of the body are its host. Like many bacteria that cause dysentery (severe diarrhea), *Shigella dysenteriae* is able to stick to the intestinal wall. In addition, *S. dysenteriae* produces a toxin, called Shiga toxin, that increases the potential for fatality. Also, invasive mechanisms that give a pathogen the ability to move through tissues and into the bloodstream result in a more medically significant disease than if it were localized. Usually, a person can recover from food poisoning caused by *Salmonella.* But some strains of *Salmonella* have virulence factors—including a needle-shaped toxin secretion apparatus—that allow the bacteria to penetrate the lining of the colon and move beyond this organ. Typhoid fever, a life-threatening disease, can then result.

Some of the deadliest pathogens form **endospores** (Gk. *endon,* "within"; *spora,* "seed") when faced with unfavorable environmental conditions. A portion of the cytoplasm and a copy of the chromosome become dehydrated and are then encased by a heavy, protective endospore coat (Fig. 20.13). In some bacteria, the rest of the cell deteriorates, and the endospore is released.

Endospores survive in the harshest of environments—desert heat and dehydration, boiling temperatures, polar ice, and extreme ultraviolet radiation. They also survive for very long periods. When anthrax endospores 1,300 years old germinate, they can still cause a severe infection (usually seen in cattle and sheep). Humans also fear a deadly but uncommon type of food poisoning called botulism, which is caused by the germination of endospores inside cans of food. To germinate, the endospore absorbs water and grows out of the endospore coat. In a few hours' time, it becomes a typical bacterial cell, capable of reproducing once again by binary fission. Endospore formation is not a means of reproduction, but it does allow the survival and dispersal of bacteria to new places.

Antibiotics

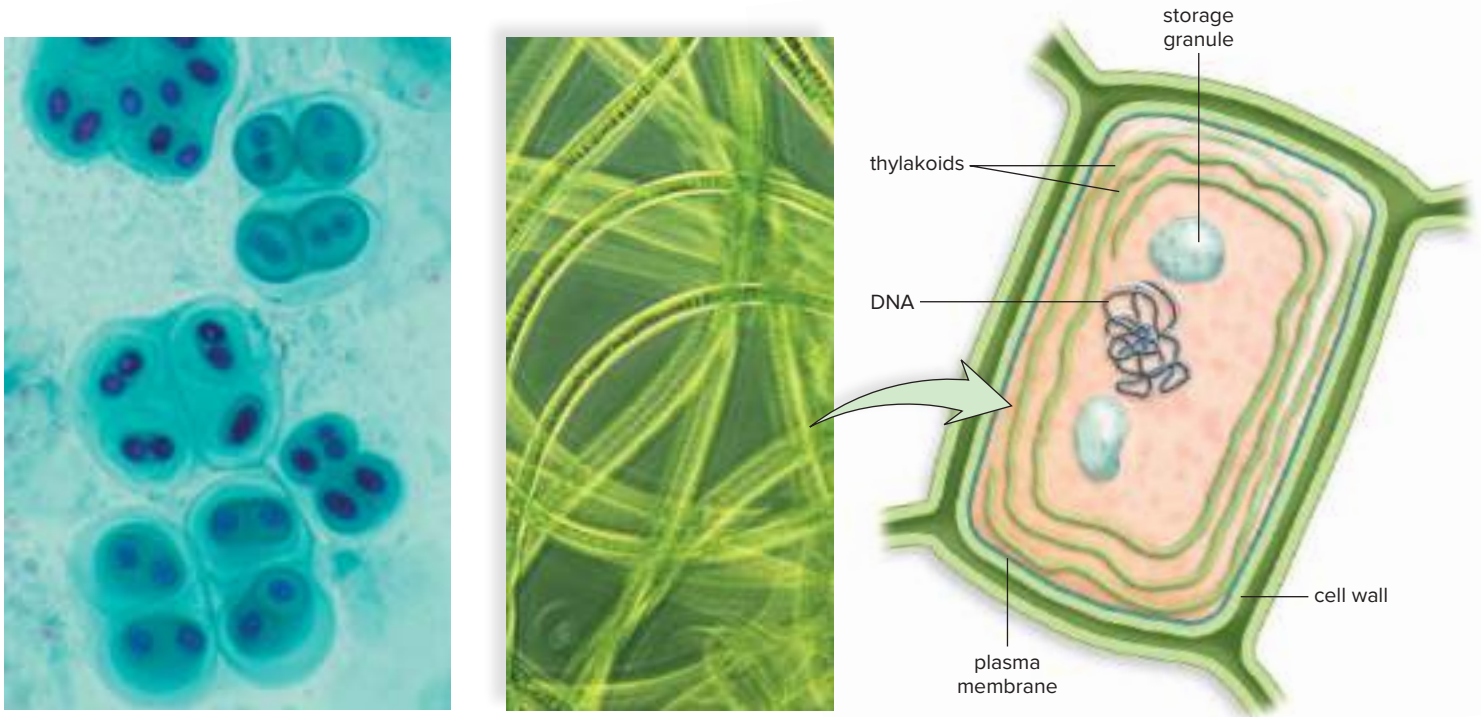
A number of antibiotic compounds are active against bacteria and are widely prescribed. Most antibacterial compounds fall within two classes, those that inhibit protein biosynthesis and those that inhibit cell wall biosynthesis. Two types of antibiotics—erythromycin and tetracyclines—inhibit bacterial protein synthesis by affecting bacterial ribosomes because they function somewhat differently than eukaryotic ribosomes. Antibiotics that inhibit cell wall biosynthesis generally block the formation of peptidoglycan, a compound necessary to maintain the integrity of bacterial cell walls. Penicillin, ampicillin, and fluoroquinolone (e.g., Cipro) inhibit bacterial cell wall biosynthesis without harming animal cells.

Antibiotics are heavily prescribed to treat infection, often when not needed. One outcome of excessive and improper use of antibiotics has been increasing bacterial resistance to antibiotics. Genes conferring resistance to antibiotics can be transferred between infectious bacteria by transformation, conjugation, or transduction. When penicillin was first introduced, less than 3% of *Staphylococcus aureus* strains were resistant to it. Now, because of selective advantage, 90% or more are resistant to penicillin and, increasingly, to methicillin, an antibiotic developed in 1957. A strain of methicillin-resistant *S. aureus* (MRSA) is responsible for many difficult-to-treat infections that have become a threat to human health. MRSA is common in hospitals and nursing facilities, where it causes problems for those with a greater risk of infection, especially those with open wounds and weakened immune systems.

Cyanobacteria

Cyanobacteria are Gram-negative bacteria with a number of unusual traits. They photosynthesize in the same manner as plants and

ISTUDY



a. *Gloeocapsa* LM 250× b. *Oscillatoria* LM 40× c. *Oscillatoria* cell

Figure 20.14 Diversity among cyanobacteria. **a.** In *Gloeocapsa,* single cells are grouped in a common gelatinous sheath. **b.** Filaments of cells occur in *Oscillatoria.* **c.** One cell of *Oscillatoria.* (a): Michael Abbey/Science Source; (b): Sinclair Stammers/Science Photo Library/Getty Images

are believed to be responsible for first introducing oxygen into the primitive atmosphere. Formerly, the cyanobacteria were called blue-green algae and were classified with eukaryotic algae, but now they are classified as prokaryotes. Cyanobacteria can have other pigments that mask the color of chlorophyll, so that they appear red, yellow, brown, or black, rather than only blue-green (Fig. 20.14).

Cyanobacterial cells are rather large, ranging from 1 to 50 μm in width. They can be single-celled, colonial, or filamentous. Cyanobacteria lack any visible means of locomotion, although some glide when in contact with a solid surface and others oscillate (sway back and forth). Some cyanobacteria have a special advantage, because they possess heterocysts, which are thick-walled cells without nuclei, where nitrogen fixation occurs. The ability to photosynthesize and to fix atmospheric nitrogen (N_2) means that their nutritional requirements are minimal. They can serve as food for heterotrophs in ecosystems.

Cyanobacteria are common in fresh and marine waters, in soil, and on moist surfaces, but they are also found in harsh habitats, such as hot springs. They are symbiotic with a number of organisms, including liverworts, ferns, and even at times invertebrates such as corals. In association with fungi, they form **lichens** that can grow on rocks. In a lichen, the cyanobacterium mutualistically provides organic nutrients to the fungus, while the fungus possibly protects and furnishes inorganic nutrients to the cyanobacterium. It is also possible that the fungus is parasitic on the cyanobacterium. Lichens help transform rocks into soil; other forms of life then may follow. It is hypothesized that cyanobacteria were the first colonizers of land during the course of evolution.

Cyanobacteria are ecologically important in still another way. If care is not taken in disposing of industrial, agricultural, and human wastes, phosphates drain into lakes and ponds, resulting in a "bloom" of these organisms. The surface of the water becomes turbid, and light cannot penetrate to lower levels. When a portion of the cyanobacteria die off, the decomposers feeding on them use up the available oxygen, causing fish to die from lack of oxygen.

Check Your Progress 20.4

1. Describe how the peptidoglycan layer is different in Gram-positive and Gram-negative cells.
2. Explain how autotrophic and heterotrophic bacteria differ.
3. Construct a hypothesis about how cyanobacteria may have affected the atmosphere of early Earth.

20.5 Archaea

Learning Outcomes

Upon completion of this section, you should be able to

1. List the biochemical characteristics that distinguish archaea from bacteria and eukaryotes.
2. Describe three different types of archaea and the habitats in which they are found.
3. Explain two ways in which archaea metabolize inorganic compounds in extreme environments.

At one time, **archaea** (domain Archaea) were considered to be a unique group of bacteria. Archaea came to be viewed as a distinct domain of organisms in 1977, when Carl Woese and George Fox discovered that the rRNA of archaea has a different sequence of

bases than the rRNA of bacteria. They chose rRNA because of its involvement in protein synthesis—any changes in rRNA sequence probably occur at a slow, steady pace as evolution occurs.

As discussed in Section 19.2, the tree of life contains three domains: Archaea, Bacteria, and Eukarya. Because archaea and some bacteria are found in extreme environments (hot springs, thermal vents, salt basins), they may have diverged from a common ancestor relatively soon after life began. Then later, the Eukarya diverged from the archaeal line of descent. In other words, the Eukarya are more closely related to the Archaea than to the Bacteria. Archaea and Eukarya share some of the same ribosomal proteins (not found in Bacteria), initiate transcription in the same manner, and have similar types of tRNA. Some scientists have suggested that a two-domain system, instead of a three-domain system, should be in place to reflect the closer relationship between the Archaea and Eukarya.

The Structure of Archaea

Archaea are prokaryotes with biochemical characteristics that distinguish them from both bacteria and eukaryotes. The plasma membranes of archaea contain unusual lipids that allow many of them to function at high temperatures. The lipids of archaea contain glycerol linked to branched-chain hydrocarbons, in contrast to the lipids of bacteria, which contain glycerol linked to fatty acids.

Archaea also evolved diverse cell wall types, which facilitate their survival under extreme conditions. The cell walls of archaea do not contain peptidoglycan, while the cell walls of bacteria do. In some archaea, the cell wall is largely composed of polysaccharides, and in others, the wall is pure protein; a few have no cell wall.

Types of Archaea

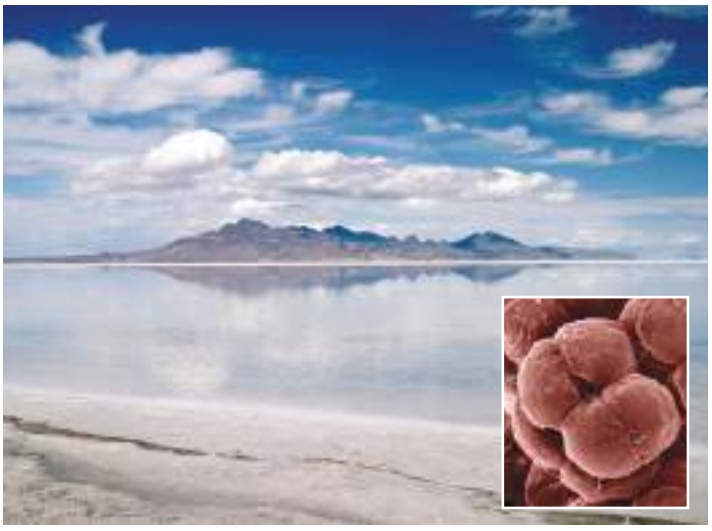
Archaea were originally discovered living in extreme environmental conditions. Three main types of archaea are still distinguished based on their unique habitats: methanogens, halophiles, and thermoacidophiles (Fig. 20.15).

Halophiles

Halophiles require high salt concentrations (usually 12–15%; the ocean, in contrast, is about 3.5%). They have been isolated from highly saline environments in which few organisms are able to survive, such as the Great Salt Lake in Utah, the Dead Sea, solar salt ponds, and hypersaline soils (Fig. 20.15*a*).

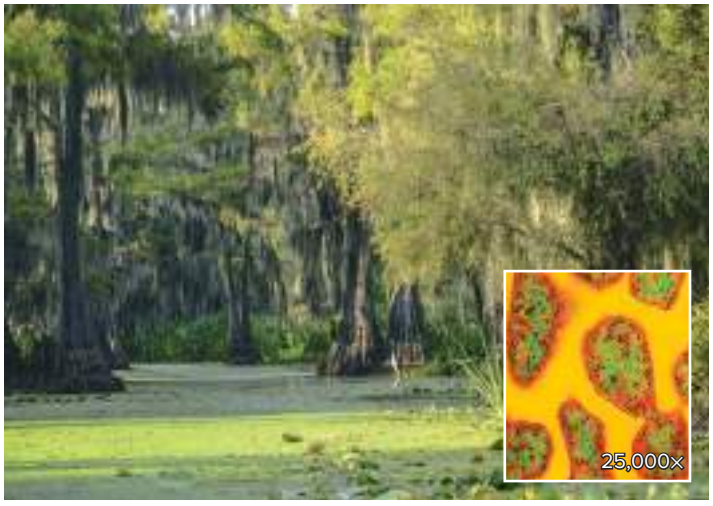
These archaea have evolved a number of mechanisms to survive in environments that are high in salt. To prevent osmotic water loss to the hypertonic environment, halophiles increase solutes such as chloride ions, potassium ions, and organic molecules within the cell, creating an internal environment more isotonic to the outside salt water. This survival ability benefits the halophiles, because they do not have to compete with as many microorganisms as they would encounter in a more moderate environment.

These organisms are aerobic chemoheterotrophs; however, some species can carry out a unique form of photosynthesis if their oxygen supply becomes scarce, as commonly occurs in highly saline conditions. Instead of chlorophyll, these halophiles use a purple pigment called bacteriorhodopsin to capture solar energy for use in ATP synthesis. Interestingly, most halophiles are so adapted to a high-saline environment that they perish if placed in a solution with a low-salt concentration (such as pure water).



a.

b.



c.

Figure 20.15 Extreme habitats. **a.** Halophilic archaea can live in salt lakes. **b.** Thermoacidophilic archaea can live in the hot springs of Yellowstone National Park. **c.** Methanogens live in swamps and in the guts of animals.
(a): (lake): Marco Regalia Sell/Alamy Stock Photo; (halophilic archaea): Eye of Science/Science Source; (b): (geyser): Alfredo Mancia/Flickr/Getty Images; (thermoacidophilic archaea): Eye of Science/Science Source; (c): (swamp): Warren Price Photography/360/iStockphoto/Getty Images; (methanogens): Dr. M. Rohde, GBF/Science Source

ISTUDY

Thermoacidophiles

The second major type of archaea are **thermoacidophiles.** These archaea thrive in extremely hot, acidic environments, such as hot springs, geysers, submarine thermal vents, and volcanoes (Fig. 20.15*b*). They are chemoautotrophic anaerobes that use hydrogen (H_2) as the electron donor, and sulfur (S) or sulfur compounds as terminal electron acceptors, for their electron transport chains. Hydrogen sulfide (H_2S) and protons (H^+) are common products.

Recall that the greater the concentration of protons, the lower (and more acidic) the pH. Thus, it is not surprising that thermoacidophiles grow best at extremely low pH levels, between pH 1 and 2. Due to the unusual lipid composition of their plasma membranes, thermoacidophiles survive best at temperatures above 80°C. Some can even grow at 105°C. (Remember that water boils at 100°C!)

Methanogens

Methanogens (methane makers) are obligate anaerobes found in environments such as swamps, marshes, and the intestinal tracts of animals (Fig. 20.15*c*). Methanogenesis, the ability to form methane (CH_4), is a type of metabolism performed only by some archaea. Methanogens are chemoautotrophs that use hydrogen gas (H_2) to reduce carbon dioxide (CO_2) to methane, and then couple the energy released to ATP production.

Methane, also called biogas, is released into the atmosphere, where it contributes to the greenhouse effect and climate change. About 65% of the methane in our atmosphere is produced by these methanogenic archaea.

Methanogenic archaea may help us anticipate what life may be like on other celestial bodies. Consider, for instance, the unusual microbial community residing in the Lidy Hot Springs of eastern Idaho. The springs, which originate 200 m (660 feet) beneath Earth's surface, are lacking in organic nutrients but are rich in H_2. Scientists have found the springs to be inhabited by vast numbers of microorganisms; over 90% are archaea, and the overwhelming majority are methanogens. The researchers who first investigated the Lidy Hot Springs microbes point out that similar methanogenic communities may someday be found beneath the surfaces of Mars and Europa (one of Jupiter's moons). Because hydrogen is the most abundant element in the universe, it would be readily available for use by methanogens everywhere.

Archaea in Moderate Habitats

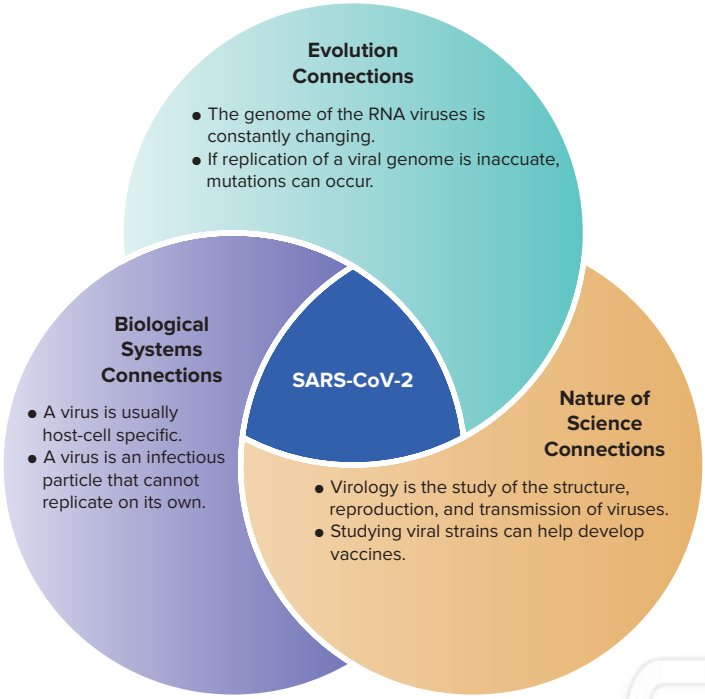
Although archaea are capable of living in extremely stressful conditions, they are found in all moderate environments as well. For example, some archaea have been found living in symbiotic relationships with animals, including sponges and sea cucumbers. Such relationships are sometimes mutualistic or even commensalistic, but there are no parasitic archaea—that is, they are not known to cause infectious diseases.

The roles of archaea in activities such as nutrient cycling are still being explored. For example, a group of nitrifying marine archaea has recently been discovered. Some scientists think these archaea may be major contributors to the supply of nitrite in the oceans. Nitrite can be converted by certain bacteria to nitrate, a form of nitrogen that can be used by plants and other producers to construct amino acids and nucleic acids. Archaea have also been found inhabiting lake sediments, rice paddies, and soil, where they are likely to be involved in nutrient cycling.

Check Your Progress **20.5**

1. Identify the differences between archaea and bacteria.
2. List the three types of archaea distinguished by their unique habitats.
3. Archaea are thought to be closely related to eukaryotes. Explain the evidence that supports this possibility.

CONNECTING *THE* CONCEPTS



ISTUDY

What Did You Learn?

20.1 Viruses, Viroids, and Prions

Viruses are nonliving, obligate intracellular parasites that can only replicate within a living cell. Classification of viruses is difficult, because they are nonliving and mutate rapidly.

- All viruses have at least two parts: an outer **capsid** composed of protein subunits and an inner core of nucleic acid, either DNA or RNA. Some also have an outer membranous envelope that contains glycoprotein spikes used for attachment to host cells.
- Viruses are host specific, meaning they can only attach to and infect the cells of certain organisms. To reproduce, a virus hijacks the host cell, forcing it to make new virus copies during the viral replication cycle.
- The **lytic cycle** of viral replication follows five steps:
  1. *Attachment:* The viral capsid or spikes lock to a host cell receptor.
  2. *Penetration:* The viral genome and certain enzymes enter the cell.
  3. *Biosynthesis:* Host cell enzymes, ribosomes, and ATP are used to create new viral components.
  4. *Maturation:* Viral components are assembled into new viruses.
  5. *Release:* New viruses exit the host cell through lysis or budding.
- Some virus genomes enter the **lysogenic cycle** after penetration and are integrated into the host cell's DNA, creating a **lysogenic cell** that passes copies of the viral genome to its daughter cells. Later (possibly years later), the viral genome will continue the lytic cycle to create new virus particles.
- Replication varies among viruses. **Bacteriophages** are viruses that infect prokaryotes and lyse the host cell during release. Viruses that infect animal cells need to be uncoated to free the genome from the capsid, and either budding or lysis releases the viral particles from the cell. **Retroviruses,** like HIV, use **reverse transcriptase** to produce single-stranded DNA from viral RNA, which is then copied and integrated into the host DNA.
- **Viroids** are naked strands of RNA (not covered by a capsid) that can cause disease in plants. **Prions** are protein molecules that have a misshapen tertiary structure. Prions cause **neurodegenerative diseases,** such as TSEs in humans and mad cow disease in cattle.

20.2 Emerging Infectious Diseases and SARS-CoV-2

Emerging diseases are those occurring for the first time in human populations, or rapidly increasing in their incidence (or frequency) in humans, or entering into new geographic regions. Reemerging diseases are those recurring after a period of decline.

- Examples of **emerging viruses** are SARS, MERS, Zika, and SARS-CoV-2.
- SARS-CoV-2 is a unique form of coronavirus that causes the disease COVID-19. It is an RNA virus that uses surface glycoproteins to infect host cells and reproduce new copies of itself.

20.3 Prokaryotes

Bacteria (domain Bacteria) and archaea (domain Archaea) are **prokaryotes.** Prokaryotes were the first living cells, lack a nucleus, and do not have cytoplasmic organelles.

- Prokaryotes reproduce asexually by **binary fission.** Their chief method for achieving genetic variation is mutation and genetic recombination by means of **conjugation** (using a **conjugation pilus**), **transformation,** and **transduction.**

20.4 Bacteria

Bacteria (domain Bacteria) are the more prevalent type of prokaryote, found in nearly every type of environment on Earth.

- Bacteria possess a cell wall made of **peptidoglycan,** a complex of polysaccharides linked by amino acids. Gram staining highlights the difference between cell wall types. Gram-positive bacteria possess a thick peptidoglycan cell wall, and Gram-negative bacteria have a thinner cell wall surrounded by a second plasma membrane. Bacterial cells occur in three basic shapes: spiral-shaped (spirillum), rod-shaped (bacillus), and round (coccus).
- Prokaryotes differ in their need (and tolerance) for oxygen. There are aerobic prokaryotes requiring oxygen, **facultative anaerobes** capable of living with or without oxygen, and **obligate anaerobes** unable to survive in the presence of oxygen.
- Autotrophic prokaryotes gain energy through photosynthesis **(photoautotrophs)** or oxidation of inorganic compounds **(chemoautotrophs)**. **Chemoheterotrophic** prokaryotes gain organic molecules from living (parasites) or dead organisms **(saprotrophs)**. Saprotrophs are absolutely essential to the cycling of nutrients in ecosystems.
- Many bacteria exist in symbiotic relationships with other organisms, including **lichens,** a symbiotic growth of cyanobacteria and fungi. In **mutualism,** both species benefit, such as nitrogen-fixing bacteria living on the roots of legumes. **Commensal** relationships benefit one organism but have no effect on the other. Some bacteria are **pathogens,** existing in **parasitic,** disease-causing relationships with other living organisms. Production of **endospores, toxins,** and superantigens increases their ability to cause disease.
- **Cyanobacteria** are unique because they photosynthesize as plants do, may grow in colonies or long filaments, and are able to fix their own nitrogen.

20.5 Archaea

Archaea (domain Archaea) are a second type of prokaryote. On the basis of rRNA sequencing, archaea appear to be more closely related to the Eukarya than to the Bacteria.

- Archaea do not have peptidoglycan in their cell walls, whereas bacteria do, and they share more biochemical characteristics with Eukarya than do bacteria.
- Three types of archaea live under harsh conditions, such as anaerobic marshes **(methanogens)**, salty lakes **(halophiles)**, and hot sulfur springs **(thermoacidophiles)**. Archaea are also found in moderate environments.

Thinking Critically About Biology

1. How would the viral characteristic of host specificity allow for viral treatment of cancer?
2. Most viral infections, including the common cold, chickenpox, mononucleosis, and Ebola, are untreatable. Few antiviral drugs exist, and those in use may cause severe side effects in the patient. Why are viral infections so difficult to treat, and why do the drugs cause side effects? Are there steps in viral reproduction that new antiviral medications could target?

ISTUDY

3. Antibiotic medications work by targeting specific structures and functions in bacterial cells. Side effects in the patient are usually minimal, because their eukaryotic cells do not possess the same structures and characteristics as the prokaryotic pathogens. What structures or functions of the prokaryotic cell would serve as good targets for new antibiotics?
4. We know that bacteria and archaea are very diverse in their metabolic capabilities, such as the consumption of organic waste and carbon dioxide and the creation of others such as oxygen, alcohol, methane, nitrates, and antibiotics. Amazingly, these known capabilities come from only the 1–10% of prokaryotes that have been identified. Scientists hypothesize that at least 90% are undiscovered and have potential metabolic capabilities we could harness to benefit human life. If you were a scientist, what metabolic capability of bacteria would you try to discover to solve a human or environmental problem?

Making It Relevant to You

1. What does our experience with SARS-CoV-2 tell us about how we should prepare for the next emerging virus?
2. Why would it take longer to develop treatments and preventive measures against emerging diseases rather than reemerging diseases?
3. The term *herd immunity* refers to many individuals in a population who are immune to an infection, usually by receiving a vaccine. Why would herd immunity be important to individuals who are vulnerable in a population?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

21

Protist Evolution and Diversity

(red tide bloom): Don Paulson Photography/Purestock/Alamy Stock Photo; (plankton): FLPA/Shutterstock

Chapter Outline

21.1 Overview of Protists
21.2 Supergroup Excavata
21.3 Supergroup Chromalveolata
21.4 Supergroup Rhizaria
21.5 Supergroup Archaeplastida
21.6 Supergroup Amoebozoa
21.7 Supergroup Opisthokonta

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 4.3 How does the endosymbiotic theory explain the origin of energy-producing organelles in the eukaryotic cell?

Section 4.4 What is the basic structure of a eukaryotic cell?

Section 6.4 How are eukaryotic organelles involved in the production and flow of energy in a cell?

Harmful Algal Blooms

When heading to a vacation on the Gulf Coast of Florida, you can normally expect to take beautiful walks along the beach and breathe in the fresh, salty air. However, if your beach vacation coincides with a peak year of the Florida red tide, you can expect the stench of rotting fish along the sand and the absence of shore birds and marine mammals.

The "Florida red tide" is a regional name for a harmful algal bloom, or HAB, that occurs in Gulf Coast counties every year, typically in the late summer or early fall. A HAB is a higher-than-normal concentration of algae or cyanobacteria. The organism that primarily causes the Florida red tide is a single-celled dinoflagellate called *Karenia brevis* (inset above). Under a light microscope, *K. brevis* has distinctive reddish-brown photosynthetic pigments that turn the water red during a HAB. An overpopulation of *K. brevis* affects the balance of a marine food chain. In addition, cells produce a toxin called brevetoxin. When cells are broken open by wind and strong tides, they release the toxin into the seawater. Crashing waves may even lift the brevetoxin along with ocean spray into the air, and the toxin becomes aerosolized, affecting beachgoers.

As you read through this chapter, think about the following questions:

1. What explains the amazing diversity of protists?
2. How do microorganisms, such as protists, impact our health and welfare?

Following *the* Themes

CHAPTER 21 – PROTIST EVOLUTION AND DIVERSITY

Evolution	Protists represent the oldest branch of eukaryotes in the tree of life. Endosymbiosis was a key step in the evolution of the first eukaryotes billions of years ago.
Nature of Science	Many protists serve as model organisms critical to many areas of scientific research, including ecosystem ecology, medicine, and epidemiology.
Biological Systems	Protists are at the base of many food chains, serving as the foundation for all life in an ecosystem.

ISTUDY

21.1 Overview of Protists

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the origin of eukaryotic organelles.
2. Assign protists into one of two groups based on mode of nutrition.
3. Understand that protists represent multiple evolutionary lineages.

Historically, the **domain Eukarya** was divided into four kingdoms: Protista, Fungi, Plantae, and Animalia. However, as we learned in Section 1.2, the evolution of the eukaryotes has resulted in the formation of the **supergroup** classification. As we will discover, protists are found in all of the supergroups. Generally, **protists** are single-celled eukaryotes, although some exist as colonies of cells or are multicellular. They are such a diverse group, however, that perhaps the best definition of a protist is any eukaryotic organism that is not a multicellular plant, animal, or fungus.

Protists offer us a glimpse into the past because they are most likely related to the first eukaryotic cell, which is thought to have arisen from a prokaryotic cell around 1.7 billion years ago. According to the **endosymbiotic theory,** mitochondria may have resulted when a nucleated cell engulfed aerobic bacteria, and chloroplasts may have originated when a nucleated cell with mitochondria engulfed cyanobacteria (see Fig. 4.5). Protists also bridge the gap between the first eukaryotic cells and multicellular organisms—the ancestry of the other three types of eukaryotes (fungi, plants, and animals) can be traced to a protist.

Characteristics of Protists

Protists vary in size from microscopic algae and protozoans to kelp that can exceed 200 m in length. Kelp, a brown alga, is multicellular; *Volvox,* a green alga, is colonial; *Spirogyra,* also a green alga, is filamentous. Most protists are single-celled, but despite their small size they have attained a high level of complexity. The amoeboids and ciliates possess unique organelles—their contractile vacuole is an organelle that assists in water regulation.

Protists are sometimes grouped according to how they acquire organic nutrients. The algae are a diverse group of **photoautotrophic** protists that synthesize organic compounds via photosynthesis. Protozoans are a group of **heterotrophic** protists that obtain organic compounds from the environment. Some protozoans, such as *Euglena,* are **mixotrophic,** meaning they are able to combine autotrophic and heterotrophic nutritional modes.

Protists reproduce sexually and asexually. Asexual reproduction by mitosis is the norm in protists. Sexual reproduction generally occurs only when environmental conditions are unfavorable. Protists can form spores or **cysts,** which are dormant phases of the protist life cycle, that can survive until favorable conditions return. Parasitic protists form cysts for the transfer to a new host.

Many protists cause diseases in humans, but many others have significant ecological importance. Aquatic photoautotrophic protists produce oxygen and are the foundation of the food chain in both freshwater and saltwater ecosystems. They are a part of **plankton,** organisms suspended in the water that serve as food for heterotrophic protists and animals. Interestingly, whales, the largest animals in the sea, feed on plankton, one of the smallest.

Table 21.1 Eukaryotic Supergroups

Supergroup	Members
Excavata	Diplomonads, parabasalids, euglenozoans
Chromalveolata	Dinoflagellates, apicomplexans, ciliates, diatoms, golden brown algae, brown algae, water molds
Rhizaria	Cercozoans, foraminiferans, radiolarians
Archaeplastida	Red algae, green algae, land plants
Amoebozoa	Amoeboids, slime molds
Opisthokonta	Fungi, choanoflagellates, animals

Evolution and Diversity of Protists

Protists were once classified together as a single kingdom. Recently, DNA evidence has suggested that protists are not **monophyletic**—that they do not all belong to the same evolutionary lineage. In fact, protists and other eukaryotes, including plants, fungi, and animals, are currently classified into six supergroups. The supergroup level of classification is the taxonomic group that is below domain and above kingdom. Each supergroup represents a separate evolutionary lineage and can be summarized in Table 21.1.

The DNA evidence supports these multiple protist lineages, but the relationships among the lineages are sometimes difficult to decipher. Protist lineages are very long and old, dating back to the origin of the first eukaryotes. As lineages stretch back in time, we can be less and less certain about how they are related to each other, just as the history of humans is less complete the further back in time we look. New research in the evolution of protists has helped clarify some of the evolutionary relationships among eukaryote lineages (Fig. 21.1), but much research still needs to be done.

In the following sections, we examine the various supergroups into which protists and other eukaryotes have been placed based on our current understanding.

Check Your Progress 21.1

1. Explain how mitochondria and chloroplasts originated in eukaryotic cells.
2. Describe how algae and protozoans are nutritionally different from one another.
3. Identify the eukaryote supergroups that (a) include both plants and protists and (b) include fungi, animals, and protists.

21.2 Supergroup Excavata

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the distinguishing characteristics of excavates.
2. Identify pathogenic excavate species.

ISTUDY

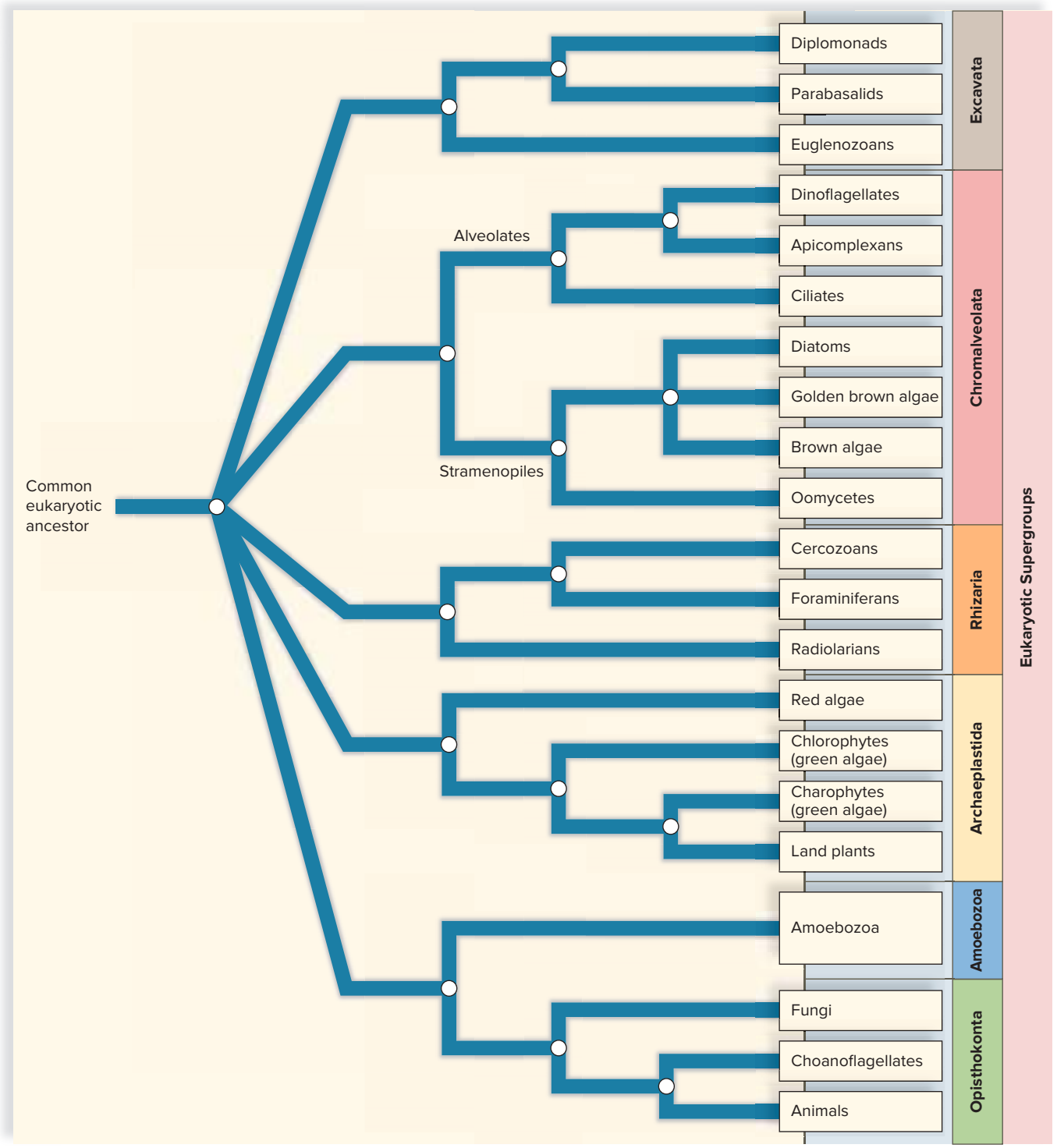


Figure 21.1 Proposed evolutionary relationships among the eukaryotic supergroups. Molecular data are used to determine the relatedness of the supergroups and their constituents. This is a simplified tree that does not include all members of each supergroup.

Excavates (L. *cavus,* "hollow") have atypical or absent mitochondria and distinctive flagella and/or deep (excavated) oral grooves.

Diplomonads and Parabasalids

Diplomonads and parabasalids are single-celled, flagellated excavates that are endosymbionts of animals. They are able to survive in anaerobic, or low-oxygen, environments. These protozoans lack mitochondria; instead, they rely on fermentation for the production of ATP. A variety of forms are found in the guts of termites, where they assist with the breakdown of cellulose.

A **diplomonad** (Gk. *diplo,* "double"; *monas,* "unit") cell has two nuclei and two sets of flagella. The diplomonad *Giardia lamblia* forms cysts, which are transmitted by contaminated water. *Giardia* attaches to the human intestinal wall, causing severe diarrhea (Fig. 21.2). This protozoan lives in the digestive tracts of a variety of other mammals as well. Beavers are known to be a reservoir of *Giardia* infection in the mountains of the western United

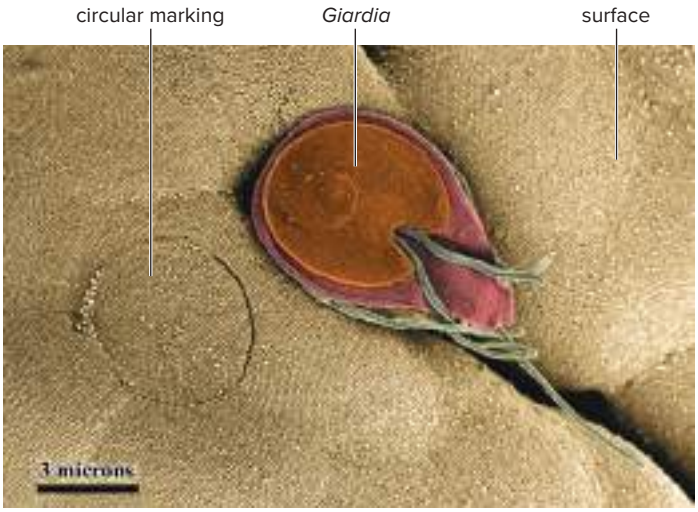


Figure 21.2 ***Giardia lamblia.*** This diplomonad adheres to any surface, including epithelial cells, by means of a sucking disk. Characteristic markings can be seen after the disk detaches. Cultura Creative/Alamy Stock Photo

States, and many cases of infection have been acquired by hikers who filled their canteens at a beaver pond.

Parabasalids have a unique, fibrous connection between the Golgi apparatus and flagella. A common sexually transmitted disease in females, trichomoniasis, is caused by the parabasalid *Trichomonas vaginalis* (Fig. 21.3). The parasite can also infect the male genital tract, but the male may have no symptoms.

Euglenozoans

Euglenozoans diverged from other members of Excavata and do possess a mitochondria. They do however lack a cell wall and instead have strips of proteins encircling the cell. For this reason, euglenozoans have a distinct appearance when they swim, alternating between oval and rounded shapes. The euglenozoans include the free-living euglenids and the parasitic kinetoplastids.

Euglenids are small (10–500 μm), single-celled, freshwater organisms. *Euglena deses* is a common inhabitant of freshwater ditches and ponds. Classifying the thousand or so species of euglenids is problematic, because they are very diverse. Some euglenids are mixotrophic, some are photoautotrophic, and others are heterotrophic. One-third of all genera have chloroplasts; the rest do not. Those that lack chloroplasts ingest or absorb their food.

When present, the chloroplasts are surrounded by three rather than two membranes. Carbohydrates are synthesized in a region of the chloroplast called a pyrenoid. Euglenids produce an unusual type of carbohydrate called paramylon.

Euglenids have two flagella, one much longer than the other (Fig. 21.4). The longer flagellum is called a tinsel flagellum, because it has hairs on it. Near the base of the tinsel flagellum is an eyespot that has a photoreceptor capable of detecting light.

Euglenids are bounded by a flexible *pellicle* composed of protein bands lying side by side. This arrangement allows euglenids to

Figure 21.3 ***Trichomonas vaginalis.*** This organism is an anaerobic, pear-shaped parabasalid with five flagella. Having no mitochondria, *T. vaginalis* obtains energy through phagocytosis and transport of nutrients through its membrane. Kateryna Kon/Shutterstock

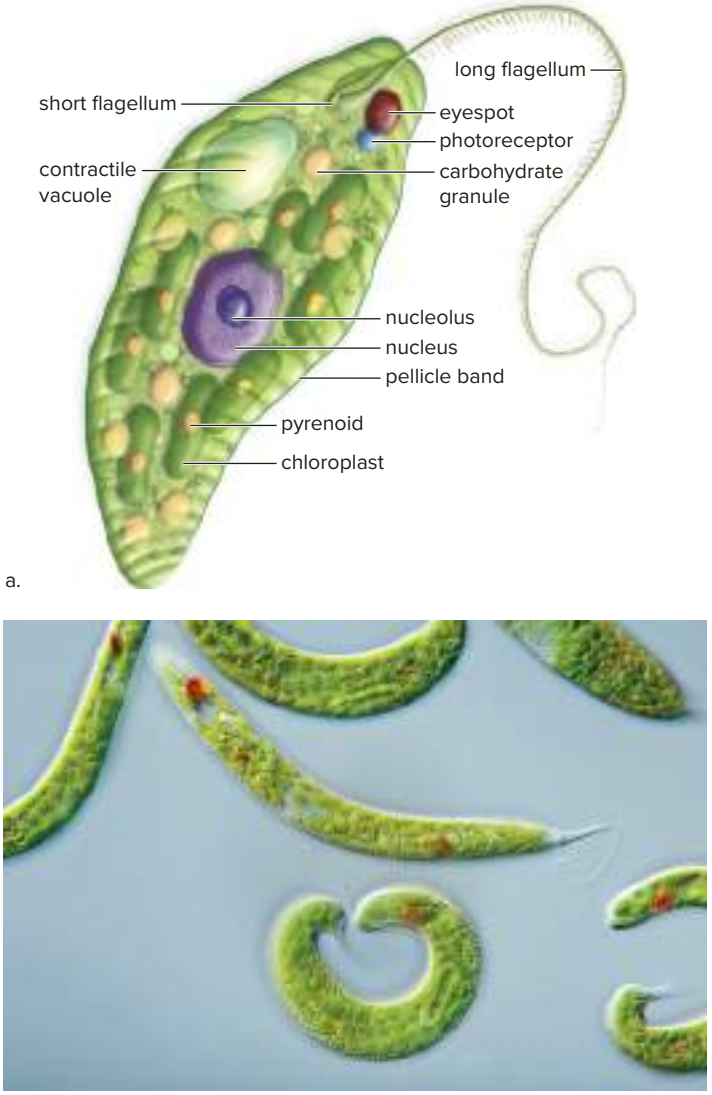


Figure 21.4 ***Euglena.*** **a.** In *Euglena,* a very long flagellum propels the body, which is enveloped by a flexible pellicle. **b.** Micrograph of *Euglena.* (b): Lebendkulturen.de/Shutterstock

ISTUDY

THEME Biological Systems

African Sleeping Sickness

The World Health Organization (WHO) recognizes 17 neglected tropical diseases (NTDs) that affect more than 1 billion people exclusively in the most impoverished communities. African sleeping sickness, also called African human trypanosomiasis, is the deadliest of all of the NTDs. It is caused by a type of parasitic, single-celled protist, *Trypanosoma brucei,* which is transmitted to humans in the bite of the blood-sucking tsetse fly (*Glossina*) (Fig. 21A).

The tsetse fly is the transmission vector of the disease, because it carries the trypanosomes and transmits the protists into the human bloodstream while feeding. The flies first become infected with the parasite when they bite either an infected human or another mammal, such as domestic cattle and pigs, that carry *T. brucei.* Figure 21B illustrates the relationship among humans, the tsetse fly, and domestic livestock.

In the later stages of infection, *T. brucei* attacks the brain, causing behavioral changes and a shift in sleeping patterns. If caught early enough, it can be cured with medication, but because it is most common in very poor nations, medication and treatment are difficult to come by. Without treatment, the disease is fatal.

Recently, an effort to eliminate the tsetse fly from these lands has been successful, so much so that the World Health Organization (WHO) reports that the number of new cases of sleeping sickness has dropped to the lowest levels in 50 years. Currently, there are around a thousand new cases of this disease per year. This decline in the incidences of new cases is a direct result of efforts by public health agencies and the WHO to control tsetse fly populations.

Questions to Consider

1. How does poverty reinforce a high occurrence of African sleeping sickness?
2. Livestock are called "disease reservoirs." Explain why.

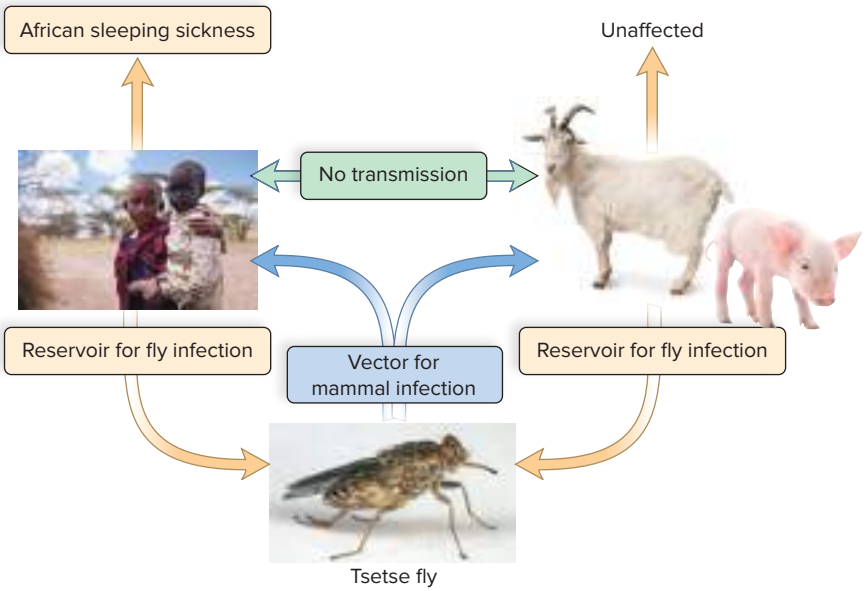


Figure 21A The transmission pattern of *Trypanosoma brucei.* *T. brucei* is a parasitic protist that causes African sleeping sickness. It can be transmitted from livestock or other humans to new human hosts. (children): Zoonar GmbH/Alamy Stock Photo; (tsetse fly): Nigel Cattlin/Alamy Stock Photo; (goat): Eureka/Alamy Stock Photo; (pig): Anatolii Tsekhmister/123RF

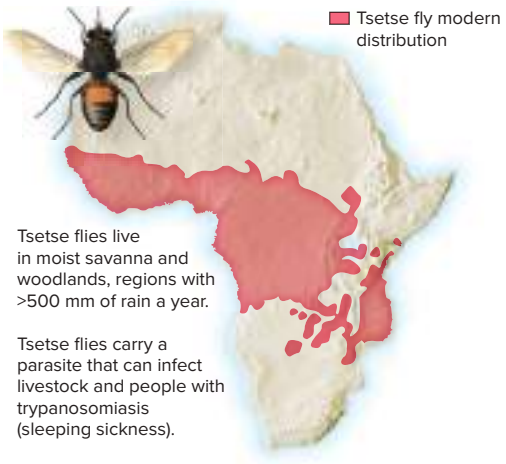


Figure 21B Areas affected by sleeping sickness. Sleeping sickness is found in moist tropical regions of Africa. People who live near wet areas or engage in fishing, raising livestock, and farming are at greater risk.

assume different shapes. A contractile vacuole rids the body of excess water. Euglenids reproduce by longitudinal cell division, and sexual reproduction is not known to occur.

Kinetoplastids are single-celled, flagellated protozoans named for their distinctive *kinetoplasts,* large masses of DNA found in their mitochondrion. Trypanosomes are parasitic kinetoplastids that are passed to humans by insect bites. *Trypanosoma brucei* (Fig. 21.5) is the cause of African sleeping sickness (see the Biological Systems feature, "African Sleeping Sickness"). It is transmitted by an insect vector, the tsetse fly (*Glossina*). The lethargy characteristic of the disease is caused by an inadequate supply of oxygen to the brain. Many thousands of cases are diagnosed each year. Fatalities or permanent brain damage is common.

Trypanosoma cruzi causes Chagas disease in humans in Central and South America. Approximately 8 million people in these regions are infected with this parasite.

Check Your Progress 21.2

1. What are the two distinctive features of the excavates?
2. Match two excavates with the human diseases they cause.

ISTUDY

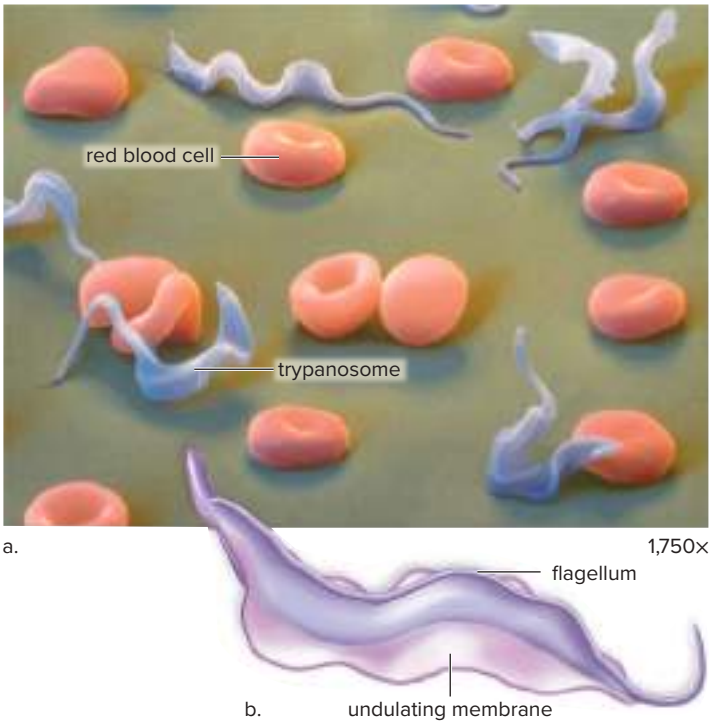


Figure 21.5 ***Trypanosoma brucei.*** **a.** Micrograph of *Trypanosoma brucei,* a causal agent of African sleeping sickness, among red blood cells. **b.** The drawing shows its general structure. (a): Eye of Science/Science Source

21.3 Supergroup Chromalveolata

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the characteristics of the Chromalveolata supergroup.
2. Identify unique species of the Chromalveolata supergroup.
3. Describe the life cycle of *Plasmodium*.

The Chromalveolata supergroup includes two large subgroups: alveolates and stramenopiles.

Alveolates

Alveolates have alveoli (small air sacs) lying just beneath their plasma membranes that are thought to lend support to the cell surface or aid in membrane transport. Alveolates are single-celled organisms.

Dinoflagellates

Dinoflagellates are single-celled, photoautotrophic algae encased by protective cellulose and silicate plates (see Fig. 21.6). Dinoflagellates typically have two flagella: one flagellum acts as a rudder, and the other causes the cell to spin as it moves forward. Dinoflagellates have chlorophylls *a* and *c* and carotenoid accessory pigments that give them a yellow-green to brown color.

There are currently around 1,500 identified species of dinoflagellates. Some species, such as *Noctiluca,* are capable of bioluminescence (producing light). Dinoflagellates are a component of plankton and thus an important source of food for small animals in the ocean. Zooxanthellae are dinoflagellates that form symbiotic relationships with invertebrates such as corals and other protozoans. Zooxanthellae are endosymbionts, living within the bodies of their hosts. Endosymbiotic dinoflagellates lack cellulose plates and flagella. Corals (see Section 28.2), members of the animal kingdom, usually contain large numbers of zooxanthellae, which provide their animal hosts with organic nutrients. In return, the corals provide the zooxanthellae with shelter, nutrients, and protection.

Dinoflagellates are one of the most important groups of primary producers in the marine ecosystem. Under unusually high nutrient conditions, populations of dinoflagellates and other algae can undergo a population explosion, called an algal bloom. During an algal bloom, a single milliliter of water can contain more than 30,000 algae. Some algal blooms caused by dinoflagellates are so large they turn the water brown or red because of the high density of the algae (Fig. 21.7*a*). These colorful algal blooms, called **red tides** (see the chapter opener), are sometimes so extensive they can be seen from space.

The dinoflagellate *Alexandrium catanella* can cause a HAB, because it secretes saxitoxin, a neurotoxin that is responsible for neurotoxic shellfish poisoning. Massive fish kills can occur as a result of saxitoxin produced during a red tide (Fig. 21.7*b*). Humans who consume shellfish that have fed during an *A. catanella* outbreak can die from paralytic shellfish poisoning, which paralyzes the respiratory organs.

Dinoflagellates usually reproduce asexually. Each daughter cell inherits half of the parent's cellulose plates. During sexual reproduction, the daughter cells act as gametes and fuse to form a diploid zygote. The zygote enters a resting stage until signaled to undergo meiosis. The product of meiosis is a single haploid cell, because the other cells disintegrate.

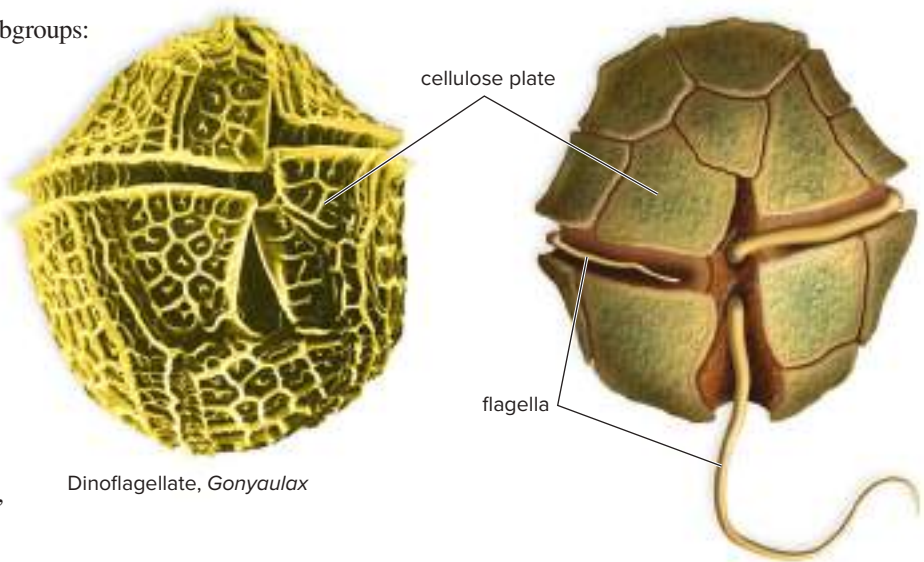


Figure 21.6 **A typical dinoflagellate.** Dinoflagellates, such as *Gonyaulax,* are alveolates with cellulose plates and grooves that have flagella. Biophoto Associates/Science Source

a. b.

Figure 21.7 Dinoflagellate bloom and fish kill. **a.** A dinoflagellate bloom, often called a red tide because of the color of the water, occurring in southeastern Alaska. **b.** Fish kills, such as this one on Padre Island, Texas, can be the result of a dinoflagellate bloom. (a): Julian Nieman/Alamy Stock Photo; (b): Todd Yates/Corpus Christi Caller-Times/AP Images

Apicomplexans

Apicomplexans, also known as *sporozoans,* include about 4,000 species of nonmotile, parasitic, spore-forming protozoans. Apicomplexans have a unique organelle called an apicoplast, which is used to penetrate a host cell. All apicomplexans are parasites of animals, and some can infect multiple hosts.

Plasmodium, an apicomplexan, is responsible for malaria, a disease that affects over 240 million people each year. *Plasmodium* is a parasitic protozoan that infects human red blood cells. The chills and fever of malaria appear when the infected cells burst and release toxic substances into the blood (Fig. 21.8). In humans, malaria is caused by four distinct members of the genus *Plasmodium. Plasmodium vivax,* the cause of one type of malaria, is the most common.

The transmission cycle of *Plasmodium* involves the *Anopheles* mosquito as an intermediate organism, or **vector,** that transmits the disease between the host and other organisms. The life cycle of

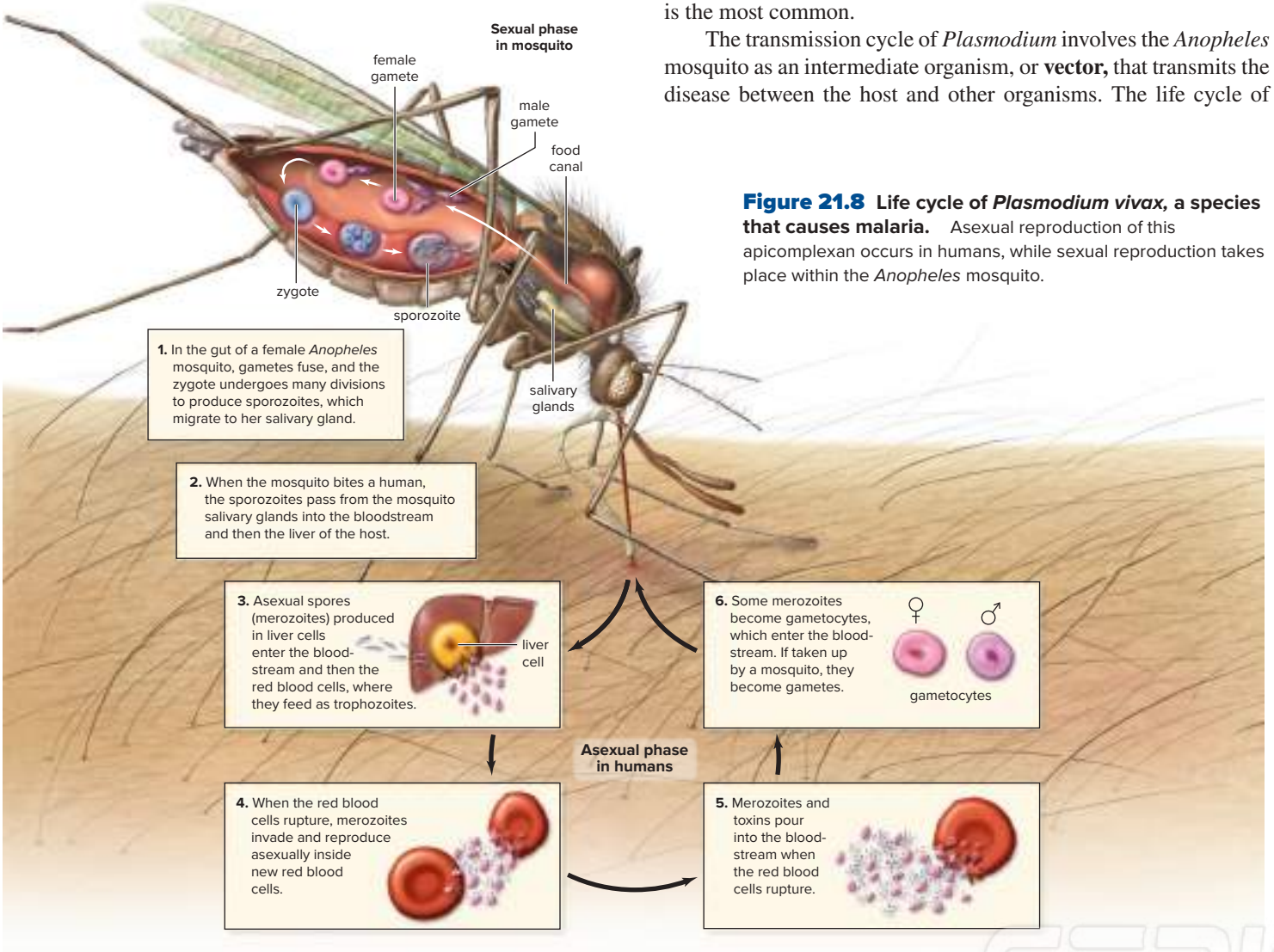


Figure 21.8 Life cycle of *Plasmodium vivax,* a species that causes malaria. Asexual reproduction of this apicomplexan occurs in humans, while sexual reproduction takes place within the *Anopheles* mosquito.

Plasmodium alternates between a sexual and an asexual phase, dependent upon whether reproduction takes place inside the mosquito (sexual) or the human host (asexual) (Fig. 21.8). Female *Anopheles* mosquitoes acquire protein for the production of eggs by biting humans and other animals.

Despite efforts to control malaria, it kills more than 600,000 people each year. Malaria is most common in densely populated, tropical areas of the globe. Because of its prevalence in these areas, almost 4 billion people, or half the world's population, are at risk for the disease.

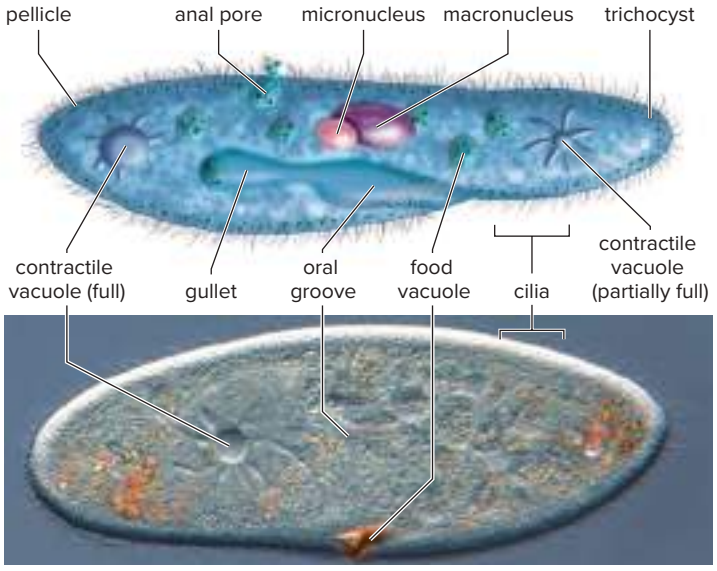
Apicomplexans include other human parasites. Cyclosporiasis is an infection of the intestine caused by the parasitic apicomplexan *Cyclospora cayetanensis.* The cyclosporin parasite is transmitted by feces-contaminated fresh produce and water. Outbreaks of cyclosporiasis in the United States have been attributed to contaminated fresh raspberries, basil, snow peas, and mesclun lettuce. *Toxoplasma gondii,* another apicomplexan, causes toxoplasmosis that is transmitted to humans from the feces of infected cats. In pregnant females, the parasite can infect the fetus and cause birth defects and intellectual disability. In people infected with AIDS, it can infect the brain and cause neurological problems.

Ciliates

Ciliates are single-celled protists that move by means of cilia. They are the most structurally complex and specialized of all protozoa. Members of the genus *Paramecium* are classic examples of ciliates (Fig. 21.9*a*). Hundreds of cilia project through tiny holes in a semirigid outer covering, or pellicle. *Paramecium* beat their cilia in a coordinated and rhythmic manner to "swim" through their environment. Oval capsules that lie just beneath the pellicle contain **trichocysts.** Upon mechanical or chemical stimulation, trichocysts discharge long, barbed threads, which are useful for defense and for capturing prey. Some trichocysts release poisons.

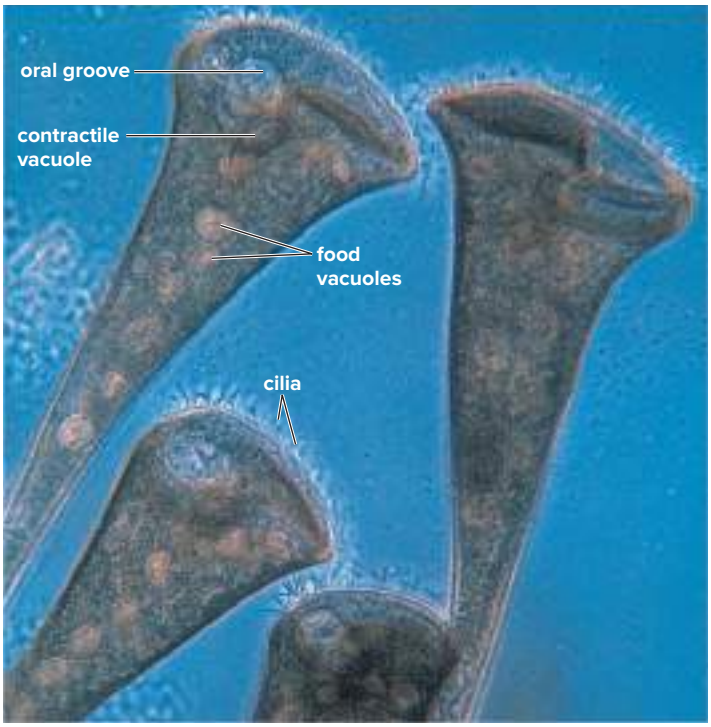
Most ciliates ingest their food. *Paramecium* feed by sweeping food particles down a gullet, below which food vacuoles form. Following digestion, the soluble nutrients are absorbed by the cytoplasm, and the nondigestible residue is eliminated through the anal pore.

The ciliates are a diverse group of protozoans that range in size from 10 to 3,000 μm. The majority of the 4,500 identified species of ciliates are free-living, mobile, and single-celled; however, several parasitic, sessile, and colonial forms exist. *Suctoria* are sessile ciliates that have specialized microtubules for capturing, paralyzing, and ingesting other ciliates. *Stentor* may be the most elaborate ciliate, resembling a giant vase (Fig. 21.9*c*). Cilia at the opening of the



a. *Paramecium*

b. During conjugation two *Paramecia* first unite at oral areas 100×



c. *Stentor* 125×

Figure 21.9 Ciliates. Ciliates are the most complex of the protists. **a.** Structure of *Paramecium,* adjacent to an electron micrograph. Note the oral groove, the gullet, and the anal pore. **b.** A form of sexual reproduction called conjugation occurs periodically. **c.** *Stentor,* a large, vase-shaped, freshwater ciliate. (photos): (a): micro_photo/iStock/Getty Images; (b): Ed Reschke/Stone/Getty Images; (c): Eric Grave/Science Source

"vase" sweep in food particles. They are one of the largest single-celled protozoans, reaching lengths of 2 mm. *Ichthyophthirius* is an ectoparasitic protozoan that causes a common disease in fishes called "ick." If left untreated, it can be fatal.

During asexual reproduction, ciliates divide by transverse binary fission. Ciliates have two types of nuclei: a large macronucleus and one or more small micronuclei. The macronucleus controls the normal metabolism of the cell, whereas the micronuclei participate in reproduction. During sexual reproduction, the micronuclei undergo meiosis. Sexual reproduction involves conjugation (Fig. 21.9*b*), during which two ciliates unite and exchange haploid micronuclei. After conjugation, the macronuclei dissolve and new macronuclei are formed from the fusion of the micronuclei.

Stramenopiles

Stramenopiles include the diatoms, golden brown algae, brown algae, and oomycetes.

Diatoms

A **diatom** is a tiny, single-celled stramenopile with an ornate silica shell. The shell is made up of upper and lower shelves, called valves, that fit together (Fig. 21.10). Diatoms have a carotenoid accessory pigment, which gives them an orange-yellow color. Diatoms make up a significant part of plankton, which serves as a source of oxygen and food for heterotrophs in both freshwater and marine ecosystems.

Diatoms reproduce asexually and sexually. Asexual reproduction occurs by diploid parents undergoing mitosis to produce two diploid daughter cells. Each time a diatom reproduces asexually, the size of the daughter cells decreases until diatoms are about 30% of their original size. At this point, they begin to reproduce sexually. The diploid cell produces gametes by meiosis. Gametes fuse to produce a diploid zygote, which grows and then divides via mitosis to produce new diploid diatoms of normal size.

Figure 21.10 Diatoms. Diatoms are stramenopiles of various colors, but even so their chloroplasts contain a unique golden brown pigment (*fucoxanthin*), in addition to chlorophylls *a* and *c*. The beautiful pattern results from markings on the silica-embedded wall. Dennis Kunkel Microscopy/Science Source

The valves of diatoms are covered with a great variety of striations and markings that form beautiful patterns when observed under the microscope. These are actually depressions or pores through which the organism makes contact with the outside environment. The remains of diatoms, called diatomaceous earth, accumulate on the ocean floor and are mined for use as filtering agents, soundproofing materials, and gentle polishing abrasives, such as those found in silver polish and toothpaste.

Golden Brown Algae

Golden brown algae derive their distinctive color from yellow-brown carotenoid accessory pigments. The cells of these single-celled or colonial protists typically have two flagella with tubular hairs, a characteristic of stramenopiles. Golden brown algae cells may be naked, covered with organic or silica scales, or enclosed in a secreted, cagelike structure called a lorica. Many golden brown algae, such as *Ochromonas* (Fig. 21.11), are mixotrophs, capable of photosynthesis as well as phagocytosis. Like diatoms, the golden brown algae contribute to freshwater and marine phytoplankton.

Brown Algae

Brown algae have chlorophylls *a* and *c* in their chloroplasts and an accessory carotenoid pigment that gives them their characteristic brown color. Food reserves are stored as a carbohydrate called *laminarin.* The brown algae range from small forms with simple filaments to large, multicellular forms that may reach 100 m in length (Fig. 21.12). The vast majority of the 2,000 identified species live in cold ocean waters.

The multicellular brown algae are seaweeds that live along the rocky coasts in the north temperate zone. They are pounded by waves as the tide comes in and are exposed to dry air as the tide goes out. They dry out slowly, however, because their cell walls contain a water-retaining material.

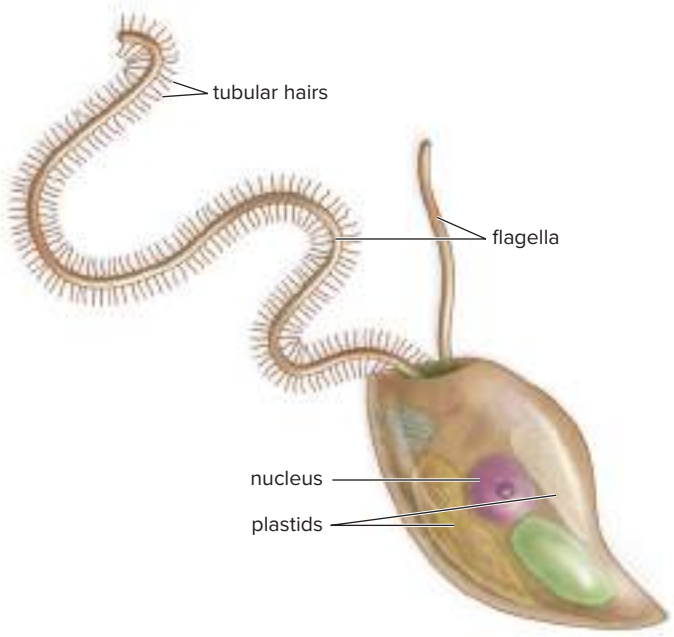


Figure 21.11 *Ochromonas*, a golden brown alga. Golden brown algae have a type of flagella that characterizes stramenopiles. The longer of the two flagella has rows of tubular hairs.

ISTUDY

Figure 21.12 Brown algae. *Laminaria* and *Fucus* are seaweeds known as kelps. They live along rocky coasts of the north temperate zone. The other brown algae featured, *Nereocystis* and *Macrocystis,* form spectacular underwater "forests" at sea.

Laminaria, commonly called kelp, and *Fucus,* known as rockweed, are examples of brown algae that grow along the shoreline. They have a structure, called a holdfast, that allows them to cling to rocks. The giant kelps *Macrocystis* and *Nereocystis* form dense kelp forests in deeper water that provide food and habitat for marine organisms.

Laminaria is unique among the protists in that members of this genus show tissue differentiation—that is, they transport organic nutrients by way of a tissue that resembles phloem in land plants. Most brown algae have an alternation-of-generations life cycle, but some species of *Fucus* have an exclusively sexual life cycle.

Brown algae are harvested for human food and for fertilizer. *Macrocystis* is the source of algin, a pectinlike material that is added to ice cream, sherbet, cream cheese, and other products to give them a stable, smooth consistency.

Oomycetes

Oomycetes ("egg fungi"), also called water molds, are funguslike protists, often seen as furry growths on their food source. In spite of their common name, some water molds live on land and parasitize insects and plants. Over 700 species of water molds have been described. A water mold, *Phytophthora infestans,* was responsible for the 1840s potato famine in Ireland, and another, *Plasmopara viticola,* for the downy mildew of grapes that ravaged the vineyards of France in the 1870s. However, most water molds are saprotrophic and live off dead organic matter. Another well-known water mold is *Saprolegnia,* which is often seen as a cotton-like white mass on dead organisms (Fig. 21.13).

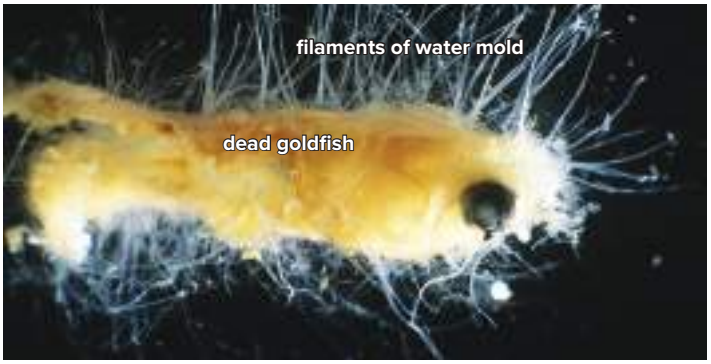


Figure 21.13 Water mold. *Saprolegnia,* a water mold, feeding on a dead goldfish, is not a fungus. Noble Proctor/Science Source

ISTUDY

Water molds used to be grouped with fungi, because they are similar to fungi in many ways. The funguslike water molds have a filamentous body similar to the hyphae of fungi, but their cell walls are composed of cellulose instead of chitin. The life cycle of water molds also differs from that of the fungi.

Water molds are diploid, but they can reproduce both asexually and sexually. During asexual reproduction, water molds produce flagellated, motile, diploid zoospores inside structures called *sporangia* (*zoosporangia*). During sexual reproduction, structures called oogonia, which produce haploid eggs, and antheridia, which produce haploid sperm, form at the tips of filaments. Sperm and eggs come together when the antheridium and oogonium come in contact and sperm is inserted into the oogonium. Eggs and sperm fuse and produce diploid zoospores, which emerge from the oogonium.

Check Your Progress 21.3

1. What characteristics separate alveolates from stramenopiles?
2. Which chromalveolates are useful to humans?
3. Summarize the life cycle of *Plasmodium vivax,* one apicomplexan that causes malaria.

21.4 Supergroup Rhizaria

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify characteristics of the supergroup Rhizaria.
2. Identify unique species of the supergroup Rhizaria.
3. Describe endosymbiosis in the Cercozoa.

Rhizarians (Gk. *rhiza,* "root") consist of groups placed together based on rRNA sequence data. They look very different from each other and do not share many morphological characteristics. Some commonalities are the presence of thin, threadlike **pseudopods,** and many produce a shell that can be fossilized.

Cercozoa

Cercozoans can have pseudopods or flagella, and many exhibit an outer netlike appearance (Fig. 21.14). Some species have sausage-shaped cyanobacteria living in their cells as endosymbionts. This endosymbiosis event took place recently, only 100 million years ago, compared to 1 billion years ago when the endosymbiosis that led to modern-day chloroplasts took place. Cercozoans are therefore important model organisms in the study of evolution of endosymbiosis.

Foraminiferans and Radiolarians

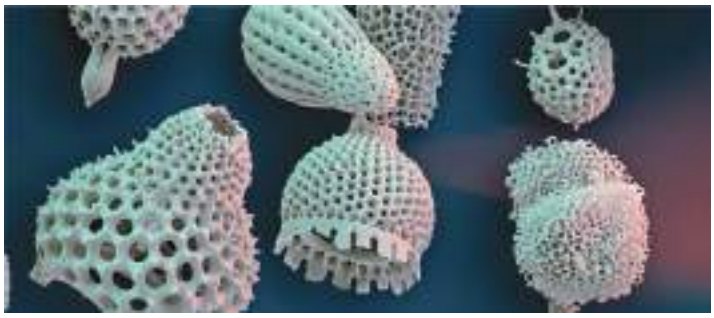
Foraminiferans and **radiolarians** are organisms with fine, threadlike pseudopods. Both have a skeleton called a test.

The tests of foraminiferans and radiolarians are intriguing and beautiful. In foraminiferans, the calcium carbonate test is often multichambered. Pseudopods extend through openings in the test, which covers the plasma membrane (Fig. 21.15*a*). In radiolarians, their glassy silicon test is internal and usually has a radial arrangement of spines (Fig. 21.15*b*). Pseudopods are external to the test.

Figure 21.14 The cercozoan *Pulinella chromatophora*. Cells contain endosymbiotic cyanobacteria. Dr. Oliver Skibbe/Science Source

The tests of dead foraminiferans and radiolarians form a layer of sediment 700–4,000 m deep on the ocean floor. The presence of

a. Foraminiferan, *Globigerina,* and the White Cliffs of Dover, England



b. Radiolarian tests SEM 150×

Figure 21.15 Foraminiferans and radiolarians. **a.** Pseudopods of a live foraminiferan project through holes in the calcium carbonate shell. Fossilized shells were so numerous they became a large part of the White Cliffs of Dover when a geological upheaval occurred. **b.** Skeletal test of a radiolarian. In life, pseudopods extend outward through the openings of the glassy silicon shell. (a): (cliffs): Platslee/Shutterstock; (inset): NHPA/Photoshot/SuperStock; (b): Eye of Science/Science Source

ISTUDY

tests is used as an indicator of oil deposits on land and sea. Their fossils date as far back as the Precambrian and are evidence of the antiquity of protists. Each geological period has a distinctive form of foraminiferan; thus, foraminiferans can be used as index fossils to date sedimentary rock. Millions of years of foraminiferan deposits formed the White Cliffs of Dover along the southern coast of England. Also, the great Egyptian pyramids are built of foraminiferan limestone. One foraminiferan test found in the pyramids is about the size of an old silver dollar—about an inch and a half across! This species, known as *Nummulites,* produced a flattened, coiled test, and its fossils have been found in deposits worldwide, including central-eastern Mississippi.

Check Your Progress 21.4

1. Why are cercozoans, radiolarians, and foraminiferans in the same supergroup?
2. What make cercozoans a good model organism?
3. Why are radiolarians and foraminiferans good geological markers?

21.5 Supergroup Archaeplastida

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the distinguishing characteristics of archaeplastida.
2. Describe the life cycles of archaeplastida.

Archaeplastids (Gk. *archeos,* "ancient"; *plastikos,* "moldable") include land plants and other photosynthetic organisms, such as red and green algae, that have plastids derived from endosymbiotic cyanobacteria (see Fig. 4.5).

Red Algae

Red algae are multicellular seaweeds that possess red and blue **accessory pigments,** which transfer energy from absorbed light to the photopigment chlorophyll during photosynthesis (Fig. 21.16). These algae live in warm seawater, some at depths exceeding 70 m. Their accessory pigments allow them to absorb the wavelengths of light that penetrate into deep water.

Most of the around 7,000 identified species of red algae are much smaller and more delicate than brown algae, but some species can exceed a meter in length. Red algae can be filamentous but most have feathery, flat, or, ribbonlike branches. Coralline red algae have cell walls that contain calcium carbonate, a mineral that contributes to the growth of coral reefs.

Red algae are economically important. As discussed in the Nature of Science feature, "Sushi, Science, and Beautiful Skin," agar is a gelatin-like product made primarily from the algae *Gelidium* and *Gracilaria.* It is used commercially to make capsules for vitamins and drugs, as a material for making dental impressions, and as a base for cosmetics. In the laboratory, agar is a solidifying agent for a bacterial culture medium. When purified, it becomes the gel for electrophoresis, a procedure that separates proteins or nucleotides. Agar is also used in food preparation as an antidrying agent for baked goods and to make jellies and desserts set rapidly.

Carrageenan, extracted from various red algae, is an emulsifying agent for the production of chocolate and cosmetics. *Porphyra,* another red alga, is the basis of a billion-dollar aquaculture industry in Japan. The reddish-black wrappings around sushi rolls consist of processed *Porphyra* blades.

Green Algae

Green algae are protists that contain both chlorophylls *a* and *b.* They inhabit a variety of environments, including oceans, fresh water, snowbanks, the bark of trees, and the backs of turtles. Some of the 8,000 known species of green algae also form symbiotic relationships with plants, animals, and fungi in lichens (see Section 22.3).

Green algae occur in many different forms. The majority are single-celled; however, filamentous and colonial forms exist. Seaweeds are multicellular green algae that resemble lettuce leaves. Despite the name, green algae are not always green; some have additional pigments that give them an orange, red, or rust color.

Biologists propose that land plants are closely related to green algae, because both land plants and green algae have chlorophylls *a* and *b,* a cell wall that contains cellulose, and food reserves made of starch. Molecular data suggest that green algae are subdivided into two groups, the **chlorophytes** and the **charophytes.** Charophytes are thought to be the green algae group most closely related to land plants.

Chlorophytes

Chlamydomonas is a tiny, photoautotrophic chlorophyte that inhabits still, freshwater pools. Its fossil ancestors date back over a billion years. The anatomy of *Chlamydomonas* is best seen in an electron

Figure 21.16 Red algae. Red algae are multicellular seaweeds, represented by this species of *Asparagopsis*. Shutterstock/Antonio Martin

ISTUDY

THEME Nature of Science

Sushi, Science, and Beautiful Skin

Hiding in the world's oceans is a vast resource in the form of algae or seaweeds. Algae can be green, brown, golden brown, and red. Green algae is being developed as a biofuel, and brown again contains various thickening agents. Red algae has many commercial uses, which are described here.

Red Algae in Our Foods

Mainly found in Asian dishes, seaweed has become more mainstream in Western food culture. Large red algae farms in Japan and South Korea have nets "seeded" with red algae that are allowed to grow to optimal size for harvesting. The red algae is then dried and formed into sheets, as done in papermaking. After roasting and seasoning, the sheets of algae may be eaten as snacks (Fig. 21C*a*). Red algae are also used as the seaweed that wraps sushi (Fig. 21C*b*).

Red seaweed is also responsible for the popular food additive carrageenan. Carrageenan is a polysaccharide extracted from red algae and functions as a thickening agent. To demonstrate the importance of carrageenan in foods, look at the two bottles of chocolate milk in Figure 21D. If you add cocoa powder to milk, mix it, and allow it to sit, the cocoa particles will sink, creating sediment at the bottom. Adding carrageenan prevents this unappealing separation by surrounding cocoa particles and keeping them suspended in the bottle. Carrageenan is used for the same reason in many food and bathroom products, including processed meats, ice cream, cheeses, and toothpaste.

a. b.

Figure 21C Red algae in Asian cuisine. **a.** Sheets of dried and seasoned seaweed snacks and (**b**) the seaweed found in sushi comes from red algae. (a): Keith Homan/Shutterstock; (b): StudioPhotoDFlorez/Shutterstock

Red Algae in Laboratory Research

Agar is extracted from the cell walls of certain red algae species and is used to make the jellylike substance found in petri dishes (Fig. 21E), much like how we make gelatin dessert at home. Agar begins as a powder that, when mixed with a hot liquid containing nutrients for bacteria and other organisms, solidifies in the petri dish. In a research lab setting, this is referred to as "solid media." Chemically, agar is made up of a long chain of polysaccharides called agarose, and other unused components. Refined agarose is used to make gels for DNA gel electrophoresis (see Section 14.1).

Red Algae for Skin Care

Red algae also contains carotenoids and fatty acids. The skin care industry claims this combination of ingredients acts as an antioxidant and neutralizes free radicals from the environment that can cause damage to skin.

Questions to Consider

1. How does carrageenan derived from red algae keep products in suspension?
2. Red algae live in the ocean. Why would this organism contain agar in its cell walls?

Figure 21D Chocolate milk without carrageenan (*left*) and with carrageenan (*right*). Ingredion Incorporated

Figure 21E Agar and agarose come from red algae. A petri dish (*left*) and DNA gel (*right*) contain a solid medium derived from red algae. (left): angellodeco/Shutterstock; (right): Andrew Brookes/Cultura/Getty Images

ISTUDY

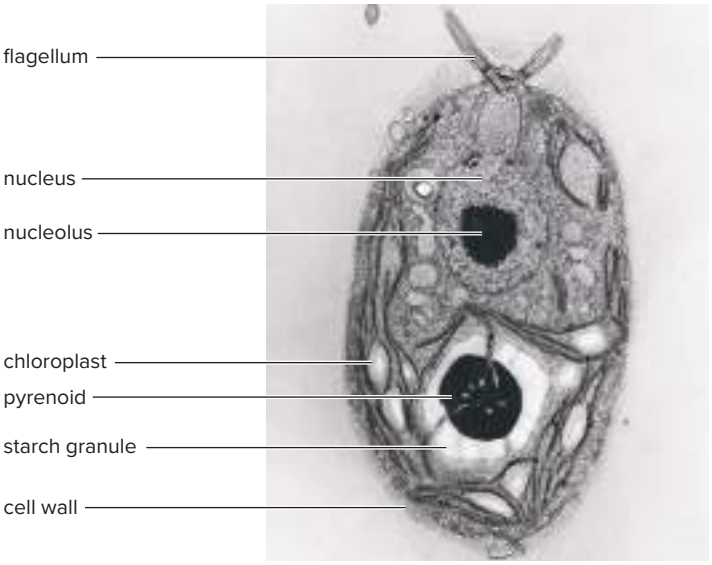


Figure 21.17 **Electron micrograph of *Chlamydomonas*.** *Chlamydomonas* is a microscopic, single-celled chlorophyte. Biophoto Associates/Science Source

micrograph because it is less than 25 µm long (Fig. 21.17). It has a defined cell wall and a single, large, cup-shaped chloroplast that contains a *pyrenoid,* a dense body where starch is synthesized. In many species, a bright red, light-sensitive eyespot helps guide individuals toward light for photosynthesis.

When conditions are favorable—that is, proper nutrients and sunlight are available—*Chlamydomonas* exists as haploid cells. These haploid cells, called *vegetative cells,* have two long, whiplike flagella projecting from the end that operate with a breaststroke-like motion. *Chlamydomonas* vegetative cells often reproduce asexually by mitosis. Each mitosis event produces two haploid daughter cells, and as many as 16 daughter cells can form inside the parent cell wall (Fig. 21.18). Each daughter cell then secretes a cell wall and acquires flagella. The fully formed, functional haploid daughter cells emerge from within the parent by secreting an enzyme that digests the parent cell wall.

When growth conditions are unfavorable, *Chlamydomonas* reproduces sexually. Two haploid vegetative cells of two different mating types come into contact and fuse to form a diploid zygote. A heavy wall forms around the zygote, and it becomes a zygospore, which undergoes a period of dormancy in which it is resistant to unfavorable conditions. When conditions improve, the zygospore emerges from dormancy, undergoes meiosis, and produces four haploid zoospores by meiosis. **Zoospores** are haploid flagellated spores that grow to become adult vegetative cells, thus completing the life cycle.

A number of colonial forms occur among the flagellated chlorophytes. *Volvox* is a well-known colonial green alga. A **colony** is a loose association of independent cells. A *Volvox* colony is a hollow sphere with thousands of cells arranged in a single layer surrounding a watery interior. *Volvox* cells move the colony by coordinating the movement of their flagella. Some *Volvox* cells are specialized for reproduction, and each of these can divide

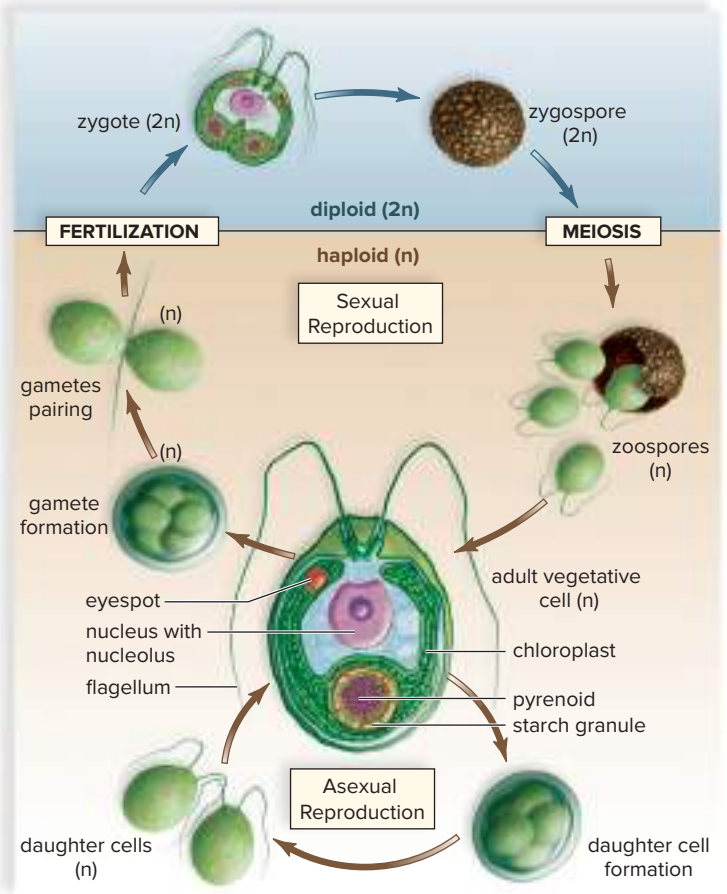


Figure 21.18 **Life cycle of *Chlamydomonas*.** *Chlamydomonas* reproduction is an example of the haploid life cycle common to algae. During asexual reproduction, all structures are haploid; during sexual reproduction, meiosis follows the zygospore stage, which is the only diploid part of the cycle.

asexually to form a new daughter colony (Fig. 21.19). This daughter colony resides for a time within the parent colony, but then escapes by releasing an enzyme that dissolves away a portion of the parent colony.

Ulva is a multicellular chlorophyte that's called sea lettuce because it lives in the sea and has a leafy appearance (Fig. 21.20*a*). The body of *Ulva* is two cells thick and can be as much as a meter long. *Ulva* has an alternation-of-generations life cycle (Fig. 21.20*b*) like that of land plants.

Charophytes

The charophytes are filamentous algae. **Filaments** (L. *filum,* "thread") are end-to-end chains of cells. Charophytes have both branched and unbranched filaments. Charophytes often grow on aquatic flowering plants. Others attach to rocks or similar objects underwater or are suspended in the water column. *Spirogyra* is an example of an unbranched charophyte (Fig. 21.21) found in green masses on the surfaces of ponds and streams. It has ribbonlike, spiraled chloroplasts. *Spirogyra* undergoes sexual reproduction via **conjugation** (L. *conjugalis,* "pertaining to marriage"), a temporary union during which the cells exchange genetic material. Two haploid filaments line up parallel to each other, and the cell contents of one filament move into the cells of the other filament,

ISTUDY

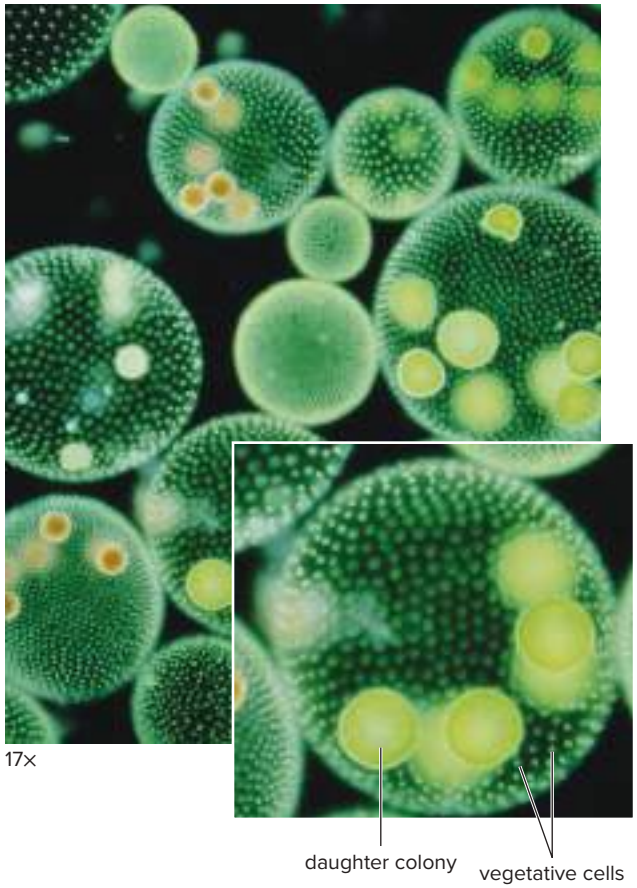


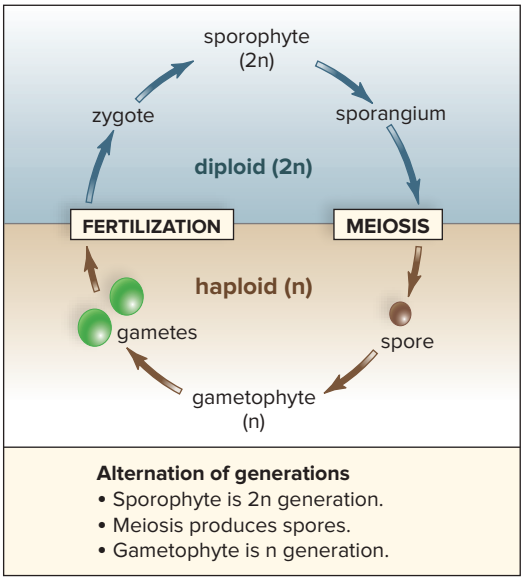
Figure 21.19 *Volvox.* *Volvox* is a colonial chlorophyte. The adult *Volvox* colony often contains daughter colonies, which are asexually produced by special cells. (both): Manfred Kage/Science Source

forming diploid zygospores. Diploid zygospores survive the winter, and in the spring they undergo meiosis to produce new haploid filaments.

Chara (Fig. 21.22) is a charophyte that lives in freshwater lakes and ponds. It is commonly called a stonewort, because it is encrusted with calcium carbonate deposits. The main strand of the alga, which can be over a meter long, is a single-file strand of very long cells anchored by rhizoids, which are colorless, hairlike filaments. Only the cell at the upper end of the main strand produces new cells. Whorls of branches occur at multicellular nodes, regions between the giant cells of the main strand. Each of the branches is also a single-file thread of cells (Fig. 21.22*b*).

Male and female multicellular reproductive structures grow at the nodes, and in some species they occur on separate individuals. The male structure produces flagellated sperm, and the female structure produces a single egg. The gametes fuse to produce a diploid zygote, which is retained until it is enclosed by tough walls.

a. *Ulva lactuca*



b. Alternation-of-generations life cycle

Figure 21.20 *Ulva.* **a.** *Ulva lactuca* is a multicellular chlorophyte known as sea lettuce. A single *Ulva* individual has a flat, leaflike appearance. **b.** Alternation-of-generations life cycle. (a): ©Evelyn Jo Johnson

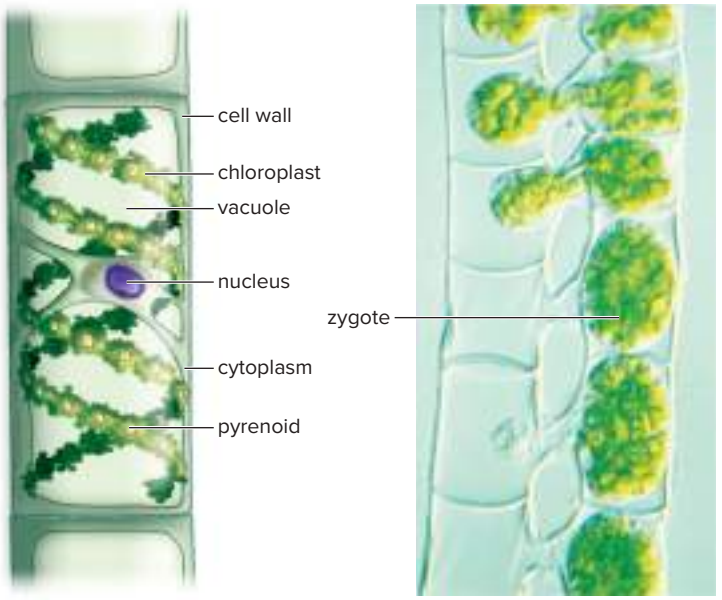
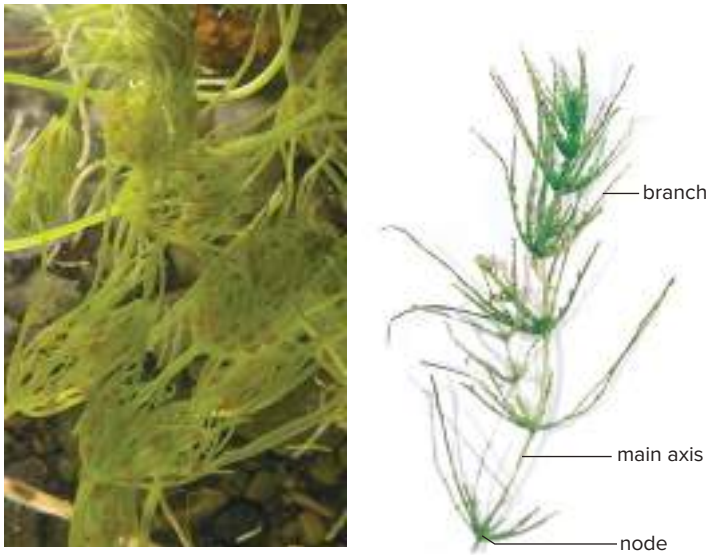


Figure 21.21 *Spirogyra.* **a.** *Spirogyra* is an unbranched charophyte in which each cell has a ribbonlike chloroplast. **b.** During conjugation, the cell contents of one filament enter the cells of another filament. Zygote formation follows. (b): M.I. Walker/Science Source

ISTUDY



a. *Chara*, several individuals b. One individual

Figure 21.22 ***Chara.*** *Chara* is an example of a stonewort, a charophyte that shares a common ancestor with land plants. (a): Bob Gibbons/Alamy Stock Photo; (b): ©Kingsley Stern

DNA sequencing data suggest that among green algae the stoneworts are most closely related to land plants.

Check Your Progress 21.5

1. What are the unique features of red algae?
2. What are the unique features of green algae?

21.6 Supergroup Amoebozoa

Learning Outcomes

Upon completion of this section, you should be able to

1. Define the unique features of amoebozoans.
2. Describe the life cycle of a plasmodial slime mold.

Amoebozoans (Gk. *ameibein,* "to change"; *zoa,* "animal") are protozoans that move by pseudopods. Pseudopods form when an amoebozoan's microfilaments contract and extend as the cytoplasm streams toward a particular direction. Amoebozoans usually live in aquatic environments, where they are often part of the plankton.

Amoeboids

Amoeboids are protists that move and ingest their food with pseudopods. Hundreds of species of amoeboids have been identified. *Amoeba proteus* is a commonly studied freshwater amoeba (Fig. 21.23). Amoeboids feed by phagocytosis, through which they engulf their prey with a pseudopod. Prey organisms may be algae, bacteria, or other protists. Digestion occurs within a *food vacuole.* Freshwater amoeboids, including *Amoeba proteus,* have contractile vacuoles, which excrete excess water from the cytoplasm.

Entamoeba histolytica is a parasitic amoeboid acquired by eating or drinking contaminated food or water. It can live in the human large intestine and cause amoebic dysentery. Another infectious amoeba, *Naegleria fowleri,* is often called the "brain eating amoeba" because it can cause a fatal brain infection. This amoeba lives in warm freshwater and can be acquired by unintentionally inhaling water through the nose while swimming.

Slime Molds

Slime molds are important decomposers that feed on dead plant material. They were once classified as fungi, but unlike fungi, they lack cell walls and have flagellated cells at some point in their life cycle. The vegetative state of the slime molds is mobile and amoeboid. Slime molds produce spores by meiosis; the spores germinate to form gametes. Nearly 1,000 species of slime molds have been identified, including plasmodial and cellular forms.

Usually, **plasmodial slime molds** exist as a *plasmodium,* a diploid, multinucleated, cytoplasmic mass enveloped by a slime sheath. This term should not be confused with the genus *Plasmodium,* which is in the alveolate group. This sluglike slime mass consumes decaying plant material as it creeps along a forest floor or an agricultural field (Fig. 21.24*a*).

When conditions are unfavorable, a plasmodium develops many sporangia. A **sporangium** (Gk. *spora,* "seed") is a reproductive structure that produces spores. An aggregate of sporangia is called a fruiting body. The spores produced by a plasmodial slime mold sporangium can survive until moisture is sufficient for them to germinate. In plasmodial slime molds,

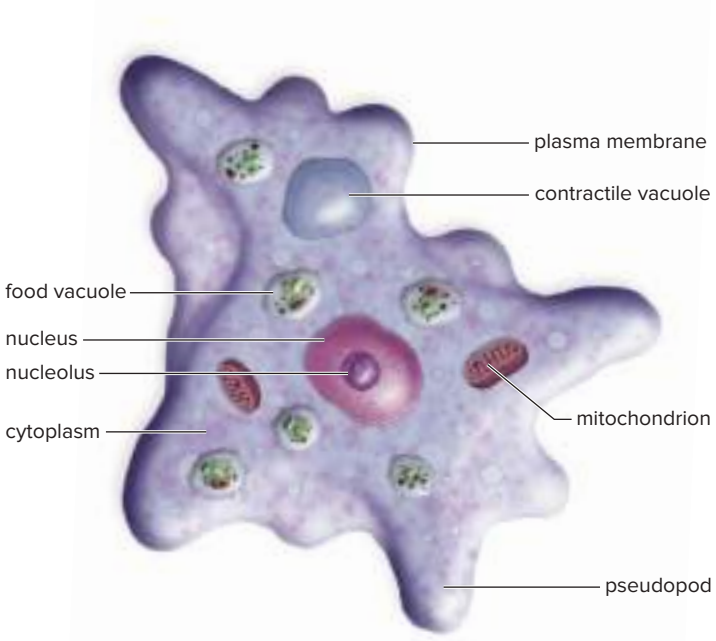
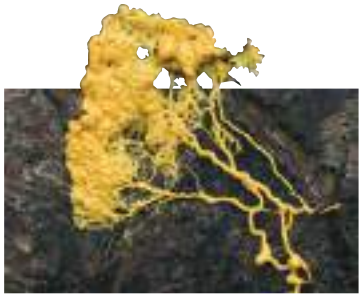


Figure 21.23 ***Amoeba proteus.*** This amoeboid is common in freshwater ponds. Bacteria and other microorganisms are digested in food vacuoles, and contractile vacuoles rid the body of excess water.

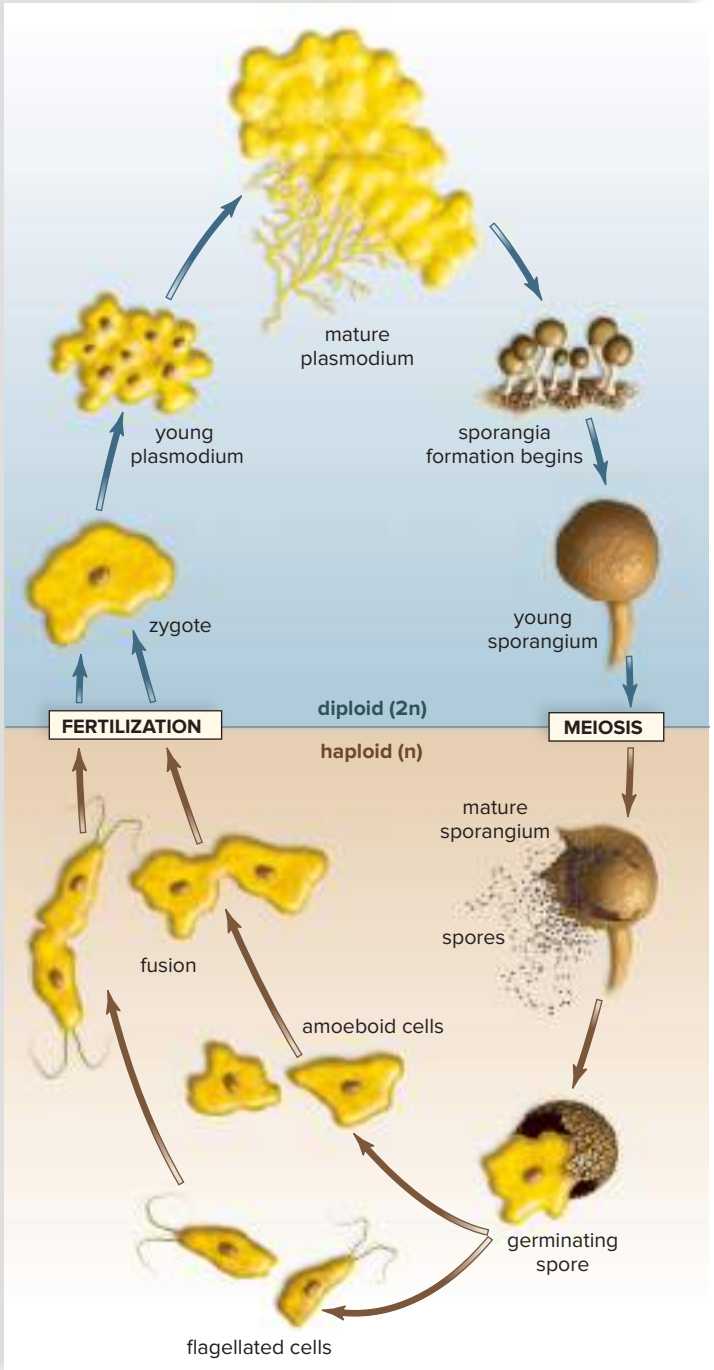
ISTUDY



Plasmodium, *Physarum*

Sporangia, *Hemitrichia*

a. Examples of slime molds



b. Life cycle of a slime mold

Figure 21.24 Plasmodial slime molds. **a.** A few examples of slime molds. **b.** During the life cycle of a slime mold, the diploid adult forms sporangia during sexual reproduction, when conditions are unfavorable to growth. Haploid spores germinate, releasing haploid amoeboid or flagellated cells that fuse. (photos): (left): NHPA/Photoshot/SuperStock; (right): Nature Picture Library/SuperStock

spores release either a haploid flagellated cell or an amoeboid cell. Eventually, two of the haploid cells fuse to form a zygote that feeds and grows, producing a multinucleated plasmodium once again.

Cellular slime molds exist as individual amoeboid cells. They are common in soil, where they feed on bacteria and yeasts. Fewer species of cellular slime molds have been identified compared to plasmodial species.

As unfavorable environmental conditions develop, the slime mold cells release a chemical that causes them to aggregate into a pseudoplasmodium. The pseudoplasmodium stage is temporary and eventually gives rise to a fruiting body, in which sporangia produce spores. When favorable conditions return, the spores germinate, releasing haploid amoeboid cells, and the asexual cycle begins again.

Check Your Progress 21.6

1. How do amoebozoans move?
2. How do plasmodial slime molds and cellular slime molds differ?

21.7 Supergroup Opisthokonta

Learning Outcomes

Upon completion of this section, you should be able to

1. List major groups of organisms found in supergroup Opisthokonta.
2. Describe the structure of choanoflagellates.

The supergroup **Opisthokonta** (Gk. *opisthos,* "behind"; *kontos,* "pole") includes animals, fungi, and several closely related protists. The protists can be multicellular, or single-celled protozoans like the choanoflagellates.

The **choanoflagellates,** including single-celled as well as colonial forms, are filter feeders with cells that bear a striking resemblance to the choanocytes that line the inside of sponges (see Section 28.2).

Each choanoflagellate has a single posterior flagellum surrounded by a collar of slender extensions. Beating of the flagellum creates a water current that flows through the collar, where food particles are taken in by phagocytosis. Colonial

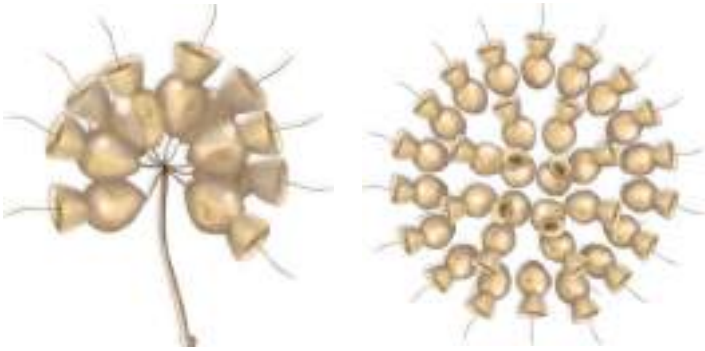
ISTUDY

choanoflagellates such as *Codonosiga* (Fig. 21.25*a*) commonly attach to surfaces with a stalk, but sometimes float freely like *Proterospongia* (Fig. 21.25*b*).

There are over 150 identified species of choanoflagellates distributed worldwide in either freshwater or marine environments, and they can be found at varying ocean depths.

Check Your Progress 21.7

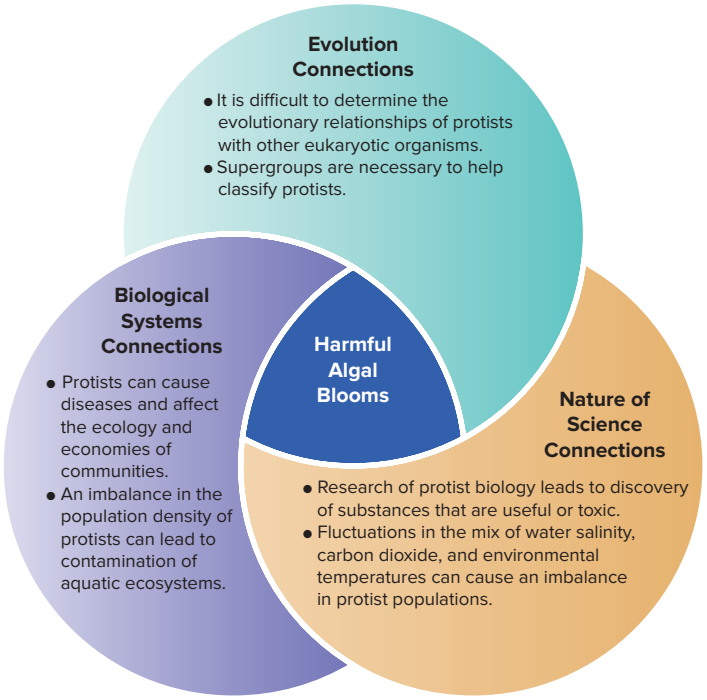
1. Which eukaryotic organisms are classified as opisthokonts?
2. Choanoflagellates are most like the common ancestors of which eukaryotic kingdom?



a. *Codonosiga* b. *Proterospongia*

Figure 21.25 Colonial choanoflagellates. **a.** A *Codonosiga* colony can anchor itself with a slender stalk. **b.** A *Proterospongia* colony is unattached.

Connecting *the* Concepts



What Did You Learn?

21.1 Overview of Protists

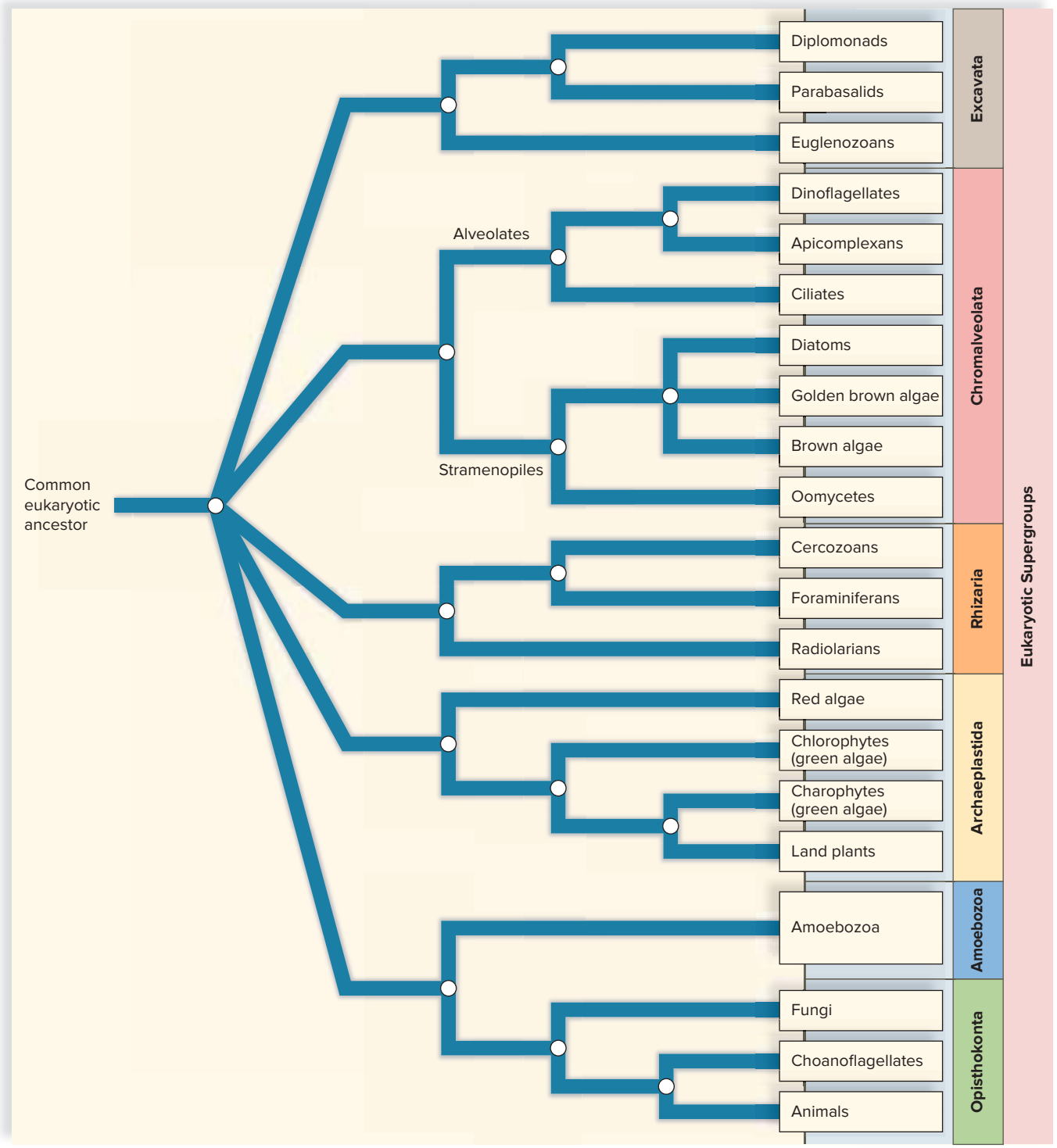
Protists are generally single-celled (sometimes multicellular) organisms in the **domain Eukarya** that acquired mitochondria and chloroplasts by the **endosymbiotic theory.**

- While most protists are single-celled, some may be multicellular or exist in colonies or filaments.
- To obtain nutrients, protists may be **photoautotrophic, heterotrophic,** or **mixotrophic.**
- Reproduction can be sexual, asexual, or both, and employ unique strategies including alternation of generations or conjugation.

Protists are of great ecological importance because they are the main producers in aquatic ecosystems, contributing to the community of **plankton.**

Protists, and other eukaryotes, are classified into six **supergroups,** each representing a separate evolutionary lineage. However, protists are not a **monophyletic** group.

ISTUDY



21.2 Supergroup Excavata

Supergroup **Excavata** includes single-celled, motile protists, such as **diplomonads, parabasalids,** and **euglenozoa.**

- Diplomonads and parabasalids, such as *Giardia lamblia* and *Trichomonas vaginalis,* are common parasites of animal hosts. **Euglenids** are photoautotrophic, flagellated cells with a pellicle instead of a cell wall. Many of the **kinetoplastids** are parasites, including trypanosomes, which cause Chagas disease and African sleeping sickness when transmitted by the bite of an insect vector.

21.3 Supergroup Chromalveolata

Supergroup Chromalveolata consists of **alveolates** and **stramenopiles.**

- Alveolates include **dinoflagellates, apicomlexans,** and **ciliates.** Dinoflagellates can form harmful algal blooms called **red tides.** The apicomplexans are nonmotile parasites that form spores. *Plasmodium,* an apicomplexan that causes malaria, is **vector**-transmitted between disease reservoirs and hosts by *Anopheles* mosquitoes. The ciliates, such as *Paramecium,* move by coordinated movement of their many cilia, capturing prey with specialized barbed threads called **trichocysts.**

ISTUDY

- Stramenopiles include **diatoms, golden brown algae, brown algae,** and **oomycetes** (water molds). Diatoms and golden brown algae have an outer layer of silica. Brown algae have chlorophylls *a* and *c,* plus a brownish carotenoid accessory pigment. Water molds are similar to fungi, except that they produce flagellated, diploid zoospores.

21.4 Supergroup Rhizaria

Supergroup Rhizaria includes **cercozoa, foraminiferans,** and **radiolarians.** Cercozoa can have cyanobacteria living as endosymbionts. Foraminiferans and radiolarians have threadlike **pseudopods** and skeletons called tests. The tests of foraminiferans and radiolarians form a deep layer of sediment on the ocean floor that can be used as index fossils.

21.5 Supergroup Archaeplastida

Supergroup **Archaeplastida** includes land plants and all red and green algae.

- **Red algae** are economically valuable and contain plastids with pigments that give them a reddish or reddish-brown appearance.
- **Green algae** include the **chlorophytes** and **charophytes** that possess chlorophylls, store energy as starch, and have cell walls made of cellulose.
- Chlorophytes include single-celled (*Chlamydomonas*), colonial (*Volvox*), and multicellular (*Ulva*) forms. Chlorophytes form **filaments** and have diverse life cycles, including a haploid life cycle involving **zoospores** common to algae and alternation of generations similar to land plants (*Ulva*).
- Charophytes include *Spirogyra,* as well as stoneworts (*Chara*), and are thought to be the closest living relatives of land plants. Some charophytes, such as *Spirogyra,* reproduce by **conjugation.**

21.6 Supergroup Amoebozoa

Supergroup **Amoebozoa** contains **amoeboids** and **slime molds**—protists that use pseudopods for motility and feeding.

- Amoeboids are heterotrophs that move and feed by phagocytosis. **Plasmodial** and **cellular slime molds** are important decomposers, which produce nonmotile spores during unfavorable conditions using a **sporangium.** They are similar to fungi, except they have an amoeboid stage.

21.7 Supergroup Opisthokonta

Supergroup **Opisthokonta** includes fungi, the animal-like **choanoflagellates,** and animals.

Thinking Critically About Biology

1. Protists, mostly those in the oceans, perform approximately half of the photosynthesis on Earth. How might their large role in the carbon cycle influence climate change?
2. You are trying to develop a new anti-termite chemical that will not harm environmentally beneficial insects. Because termites are adapted to eat only wood, they will starve if they cannot digest this food source. Termites have two symbiotic partners: the protozoan *Trichonympha collaris* and the bacteria it harbors that actually produce the enzyme that digests the wood. Knowing this, how might you prevent termite infestations without targeting the termites directly?
3. Certain protist species cause human diseases that are difficult to treat and are often deadly. How does the evolutionary relatedness of humans and protists explain the difficulty of treating a person infected with a pathogenic protist?
4. Convergent evolution is a process by which unrelated species evolve similar adaptations to their environment. For example, both birds and bats have independently evolved the ability to fly, instead of inheriting the trait from a common ancestor. In protists, an example is parasitism. Certain apicomplexans in the SAR (stramenopiles, alveolates, and rhizarians) supergroup, and some parabasalids in the excavata supergroup, have evolved the ability to parasitize human hosts. What are other examples of traits that have developed through convergent evolution within the protists?

Making It Relevant to You

1. What causes a red tide?
2. How can a red tide directly affect you?
3. Red tides occur every year, but some years are worse than others. What factors do you believe would contribute to a particularly bad year for red tide?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

22

Fungi Evolution and Diversity

NEO6iA M/Shutterstock

Chapter Outline

22.1 Evolution and Characteristics of Fungi
22.2 Diversity of Fungi
22.3 Symbiotic Relationships of Fungi

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 9.3 What is the difference between a haploid and a diploid cell?

Section 10.1 Why is meiosis essential to sexual reproduction?

Section 21.7 What supergroup of protists is most closely related to the fungi?

Fungal Parasites of Animals

Fungi were the first eukaryotes to invade land. They are ancient organisms, with deep, underground networks of mycelium, or "fungal roots," that service the ecosystem by recycling organic debris back into topsoil. The part of the fungi we see, and sometimes eat, is the fruiting body, such as a mushroom, which is the center of reproduction for many species.

The medicinal properties of fungi have been known for centuries—distillates from fungi have antimicrobial and antiviral properties. In many cultures, fungi play a key role in nontraditional medicine—knowledge just now being tapped by pharmaceutical research. For example, many species of fungi are *entomopathogenic*, meaning they specifically attack insects and insect larvae. The *Cordyceps* fungi produces spores that attach onto caterpillars. The spores germinate, and then digest and kill the caterpillar with its fungal hyphae. The spore-producing fruiting body will later emerge from the head. Caterpillars with *Cordyceps* fruiting bodies are an expensive Chinese medicine used to treat respiratory, kidney, and liver disease.

Fungi that attack insects are considered environmentally safe, and there is a growing interest worldwide to use fungi as a biological control against insect pests such as mosquitoes. Products such as antibiotics, antivirals, and biological control, derived from the natural processes of fungi, have the potential to revolutionize human society and our impact on the environment.

As you read through this chapter, think about the following questions:

1. Why is it important to human health to preserve fungi biodiversity?
2. What are the benefits and drawbacks of fungi being pathogenic?
3. How would ecosystems be impacted if fungi were to go extinct?

Following *the* Themes

CHAPTER 22 – FUNGI EVOLUTION AND DIVERSITY

Evolution	Fungi are more closely related to animals than to any other group of eukaryote; both animals and fungi belong in the supergroup Opisthokonta.
Nature of Science	Fungi have properties that make them an important resource for the study of medicine, genetics, and molecular biology, as well as for industrial applications.
Biological Systems	Fungi have huge networks of mycelium beneath the soil that serve as nature's recycling center. Decomposition by fungi is a critical ecosystem service.

ISTUDY

22.1 Evolution and Characteristics of Fungi

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify two traits that are similar between animals and fungi.
2. Distinguish among fungi that are haploid, dikaryotic, or diploid.
3. Define and identify the structural features of fungi.

Fungi, found in the supergroup Opisthokonta, are mostly multicellular eukaryotes that share a common mode of nutrition. Mycologists, scientists who study fungi, expect this number of species to increase to the millions in the future as new species are discovered and molecular biology techniques improve.

Plants are autotrophs and create their own food. Conversely, fungi, like animals, are heterotrophs and consume preformed organic matter. Animals, however, are heterotrophs that ingest food, whereas fungi are **saprotrophs** that absorb food. Their cells send out digestive enzymes into the immediate environment that break down dead and decaying organic matter. The resulting nutrient molecules are then absorbed by the fungus cells.

Evolution of Fungi

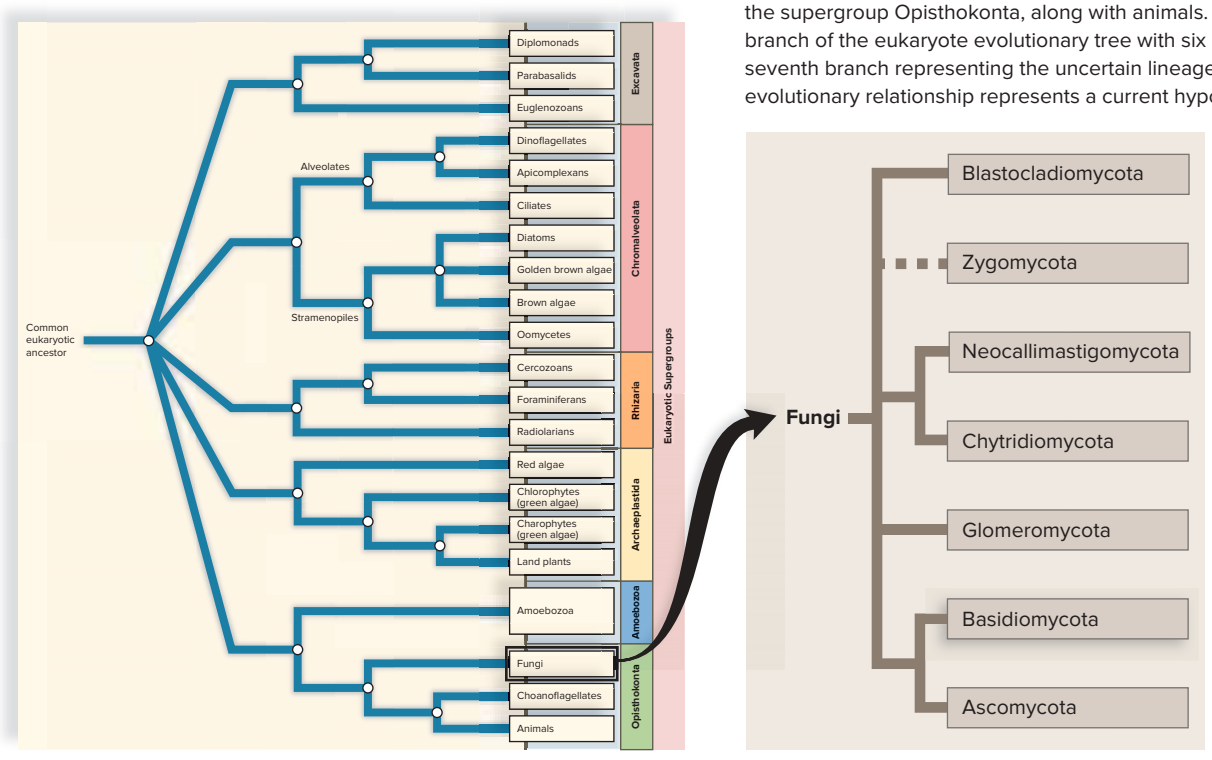
Classification of fungal phyla is ongoing and incomplete. Recent sampling has detected new groups of fungi and determined previously identified groups are not **monophyletic.** Currently, six phyla are recognized, with as many still waiting to be defined. The relationships shown in Figure 22.1 represent a current hypothesis that has been supported by DNA analysis.

Two major hypotheses highlight recent changes in fungal phylogeny. Previously, Neocallimastigomycota and Blastocladiomycota were grouped with chytrids (Chytridiomycota), but are now recognized as distinct phyla. On the other hand, a once familiar phylum called Zygomycota is not recognized in current classification because of increasing evidence it was not monophyletic. Instead, its members, called zygospore fungi, are *paraphyletic,* meaning descendants of a common ancestor may no longer be related. Because of our ever-changing knowledge of these relationships, this chapter will discuss only the major lineages. Our discussion of fungal structure applies mostly to ascomycetes and basidiomycetes.

Protists evolved some 1.5 BYA (billion years ago). Plants, animals, and fungi can all trace their ancestry to protists, but molecular data tell us that animals and fungi shared a more recent common ancestor than animals and plants. Therefore, animals and fungi, both in the supergroup Opisthokonta, are more closely related to each other than either is to plants (Fig. 22.1). The common ancestor of animals and fungi was most likely an aquatic, flagellated, single-celled protist. Multicellular forms evolved sometime after animals and fungi split into two different lineages.

Fungi do not fossilize well, so it is difficult to estimate from the fossil record when they first evolved. The earliest known fossil fungi are dated at 460 MYA (million years ago), but fungi probably evolved a lot earlier. While animals were still swimming in the seas during the Silurian, plants were beginning to live on the land, and

Figure 22.1 Evolutionary relationships among the fungi.
a. A phylogeny of the six eukaryote supergroups. Fungi are members of the supergroup Opisthokonta, along with animals. **b.** A close-up of the fungi branch of the eukaryote evolutionary tree with six recognized phyla and a seventh branch representing the uncertain lineage of the Zygomycota. This evolutionary relationship represents a current hypothesis of fungal evolution.

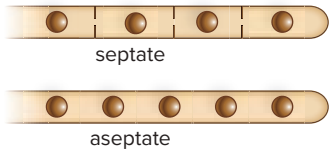


a. b.

ISTUDY

a. Fungal mycelia

b. Individual strands of hyphae 170×



c. Septate and aseptate hyphae

Figure 22.2 Mycelia and hyphae of fungi. **a.** Fungal mycelium made from hyphae, growing as a white mass on strawberries. **b.** Hyphal filaments growing on the surface of a plant. **c.** Hyphae are either septate (have cross-walls) or aseptate (do not have cross-walls.) (a): Jeeranan Thongpan/Shutterstock; (b): Andrew Syred/Science Source

they brought fungi with them. Mycorrhizae are evident in plant fossils, also dated some 460 MYA. Perhaps fungi were instrumental in the colonization of land by plants. Much of the fungal diversity most likely had its origin in an adaptive radiation when organisms began to colonize land.

Structure of Fungi

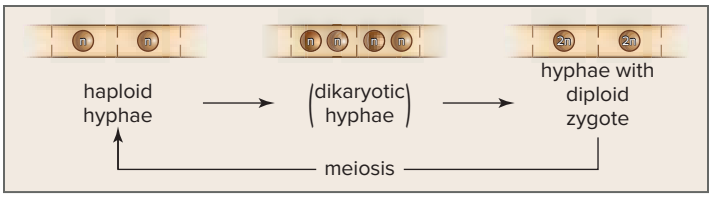
Some fungi, including the yeasts, are single-celled organisms; however, the vast majority of species are multicellular. The body of most fungi is a multicellular structure known as a mycelium (Fig. 22.2*a*). A **mycelium** (Gk. *mycelium,* "fungus filaments") is a network of fungal filaments; these filaments are called **hyphae** (Gk. *hyphe,* "web"). Hyphae give the mycelium quite a large surface-to-volume ratio, and this maximizes the absorption of nutrients into the body of a fungus (Fig. 22.2*b*).

Hyphae grow from their tips, and in some fungi, **septa** (*sing.,* septum), or walls of tissue, are formed behind the growing tip, partitioning the hyphae into individual cells. Fungi that have septa in their hyphae are called **septate** (L. *septum,* "fence, wall"). Pores in the septa allow cytoplasm, and sometimes even organelles, to pass freely from one cell to another. The septa that separate reproductive cells, however, are completely closed in all fungal groups. Aseptate fungi are not divided into cells, and many nuclei are present in the cytoplasm of a single hypha (Fig. 22.2*c*). Some hyphae are capable of penetrating rigid substances, such as plant tissues. When a fungus reproduces, a specific portion of the mycelium becomes a reproductive structure, which is nourished by the rest of the mycelium.

Fungal cells are quite different from plant cells, because they lack chloroplasts and have a cell wall made of chitin, not cellulose. **Chitin,** like cellulose, is a polymer of glucose organized into microfibrils, but each glucose molecule of chitin has a nitrogen-containing amino group attached to it. Chitin is also found in the exoskeleton of arthropods, a major phylum of animals that includes the insects and crustaceans. Unlike plants that store energy as starch, fungi store energy as glycogen, the same molecule that animals use to store energy. Except for the aquatic chytrids, fungi are not motile. Terrestrial fungi lack basal bodies (see Section 4.8) and do not have flagella at any stage in their life cycle. They move toward a food source by hyphae growing toward it. Growing hyphae can cover as much as a kilometer a day!

Reproduction of Fungi

Both sexual and asexual reproduction occur in fungi. Sexual reproduction of terrestrial fungi occurs in these stages:



The relative length of time of each phase varies with the species.

During sexual reproduction, hyphae (or a portion thereof) from two different mating types make contact and fuse. In some species, the nuclei from the two mating types fuse immediately. In other species, the nuclei pair up but do not fuse for days, months, or even years. The nuclei continue to divide in such a way that every cell (in septate hyphae) has at least one of each nucleus. A hypha that contains paired haploid nuclei is said to be n + n, or **dikaryotic** (Gk. *dis,* "two"; *karyon,* "nucleus"). When the nuclei do eventually fuse, the zygote undergoes meiosis prior to spore formation.

How can fungi ensure that the offspring will be dispersed to new locations? As an adaptation to life on land, fungi usually produce nonmotile, windblown spores during both sexual and asexual reproduction. A **spore** is a reproductive cell that develops into a new organism without the need to fuse with another reproductive cell (see Fig. 22.6*b*). A large mushroom may produce billions of spores within a few days. When a spore lands on an appropriate food source, it germinates and begins to grow.

Asexual reproduction usually involves the production of spores by a specialized part of a single mycelium. Alternatively, asexual reproduction can occur by fragmentation—a portion of a mycelium begins a life of its own. Also, single-celled yeasts reproduce asexually by **budding;** a small cell forms and gets pinched off as it grows to full size (see Fig. 22.6*a*).

Check Your Progress 22.1

1. Describe how animals and fungi differ with respect to nutritional mode.
2. Explain how fungal cells differ from plant cells.
3. Describe the function of a fungal spore.

ISTUDY

22.2 Diversity of Fungi

Learning Outcomes

Upon completion of this section, you should be able to

1. List the major lineages of fungi.
2. Summarize the life cycle typical of fungi in each of the six lineages.
3. Identify one benefit and one disadvantage of human and fungi interactions.

In 1969, the plant ecologist R. H. Whittaker classified fungi as a group that was distinct from protists, plants, animals, and the prokaryotic organisms. He based his reasoning on the observation that fungi are the only type of multicellular organism to be saprotrophic. However, fungi are now considered to be the closest multicellular relative of animals. Both fungi and animals are placed in the eukaryote supergroup Opisthokonta, which also includes certain heterotrophic protists (see Fig. 22.1).

As discussed earlier, the phylogenetic relationships among fungi have been the cause of much debate. The understanding of fungal phylogeny is going through rapid and exciting changes, aided by increasing molecular sequence data. In this section, we will describe characteristics of the major fungal lineages listed in Table 22.1: Blastocladiomycota, Zygomycota, Neocallimastigomycota, Chytridiomycota, Glomeromycota, Ascomycota, and Basidiomycota.

As you explore the diversity of fungi in this section, note that phylum names end with "mycota," and when describing members of that phylum, the ending "mycetes" is used. For example "ascomycetes are in the phylum Ascomycota."

Microsporidia Are Parasitic Fungal Relatives

The **microsporidia** are not included in our initial discussion of fungal diversity because it is unclear how they fit in with the kingdom Fungi. Considered a "sister group" to fungi, the microsporidia are single-celled obligate, intracellular, animal parasites, most often seen in insects but also found in vertebrates, such as fish, rabbits, and humans (Fig. 22.3). Biologists once believed that microsporidia were an ancient line of protist, due in part to their lack of mitochondria. However, genome sequencing of the microsporidium *Encephalitozoon cuniculi* revealed genes that are mitochondrial, leading to the hypothesis that this organism once had a mitochondrion, which later became greatly reduced. In addition, microsporidia have the smallest known eukaryotic genome.

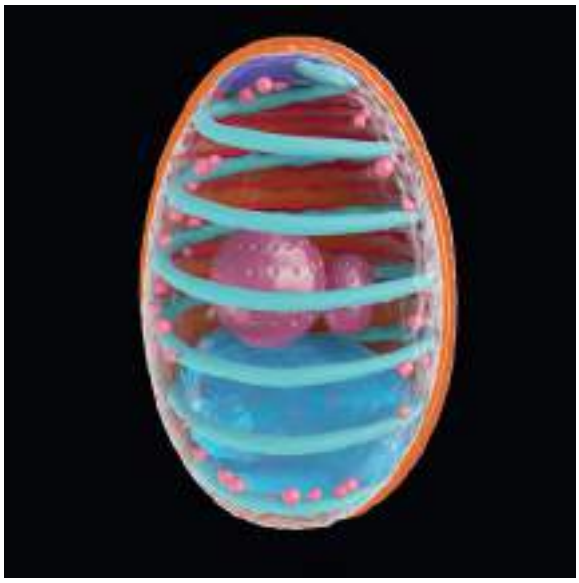


Figure 22.3 Illustration of the infectious spore of microsporidium *Encephalitozoon cuniculi*. The blue spiral is the polar tube, which uncoils from the spore, and then penetrates and deposits its cellular contents into the host cell. SciePro/Shutterstock

E. cuniculi and other microsporidia commonly cause diseases in individuals who are immunocompromised, such as those receiving organ transplants or individuals with AIDS. Microsporidia infect their hosts with spores that contain a polar tube. This tube extrudes the contents of the spore into intestinal or neuronal cells and remains hidden in a vacuole, causing the human host to suffer with diarrhea and neurodegenerative diseases. The parasitic nature of these organisms makes understanding the phylogenetic placement important in identifying effective disease treatments.

Table 22.1 Features of the Fungi Phyla

Fungal Lineage	Characteristics
Blastocladiomycota	Contains zoospores and can exhibit an alternation-of-generations life cycle similar to plants.
Zygomycota	Contain hyphae without septa. Sexual spores are called zygospores. Most commonly associated with species that produce black bread mold.
Neocallimastigomycota	Fungi with no mitochondria. Use anaerobic metabolism to grow in the guts of herbivores. Zoospores found in their life cycle.
Chytridiomycota	Aquatic single-celled fungi with zoospores. Can be animal parasites.
Glomeromycota	Form arbuscular mycorrhizae.
Ascomycota	Asexual reproduction on conidiophores and sexual reproduction in a fruiting body called an ascus.
Basidiomycota	Possesses the most commonly recognized fruiting bodies (i.e., mushrooms, puffballs) with haploid spores called basidiospores.

Fungi with Zoospores

A **zoospore** is an asexual spore with a flagellum. The spore can swim in moist soil or aquatic environments until it lands on a substrate and germinates. The placement of the flagella in their spores suggests a shared ancestry of fungi and choanoflagellates (see Table 22.1). There are three phyla of true fungi with zoospores: Blastocladiomycota, Neocallimastigomycota, and Chytridiomycota.

Blastocladiomycota

Blastocladiomycetes, also known as blastoclads, are found in terrestrial or aquatic environments. All members produce asexual zoospores and some species will also exhibit an alternation-of-generations life cycle similar to plants. As with other fungi, many blastocads decompose cellulose and other dead organic matter, but the most widely studied species are the parasites of plants, invertebrates, and algae.

The blastocad, *Physoderma maydis,* is a serious plant pathogen that causes brown spot and stalk rot on corn (Fig. 22.4*a*).

Spore-producing structures called sporangia remain dormant in the soil and are wind dispersed into the whorled leaves of a corn plant. When that plant gets wet, the sporangia produce zoospores that swim and infect the young tissues of developing plants.

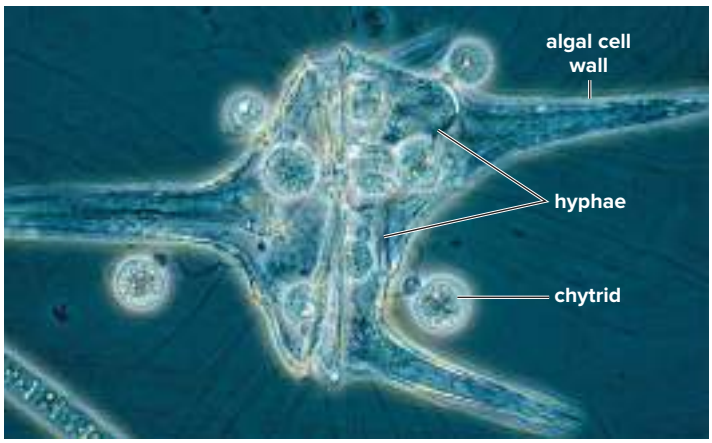
Neocallimastigomycota

Neocallimastigomycetes are fungi that lack mitochondria, are anaerobic, and live exclusively in the guts of ruminants and other herbivores. Ruminants, like sheep and cows, and non-mammalian herbivores like some lizards, depend on these fungi to enzymatically break down cellulose and lignin. Without these fungi, these animals would not be able to obtain sufficient nutrition and calories from their plant-based diets.

The genome of the genus *Neocallimastix* was sequenced and found to have genes that code for digestive enzymes that originally were present in bacteria. The genes were obtained through *horizontal gene transfer,* a method of sharing DNA between unrelated species. Due to the anaerobic growth of these fungi and their abilities to degrade cellulose at relatively high temperatures, they have potential uses in biotechnology for biofuel production.

a.

b.



c.

Figure 22.4 Fungal phyla with zoospores. **a.** The Blastocladiomycete *Physoderma maydis* infection of corn. Purple spots on the midrib of the leaf are the fungal sporangium. **b.** Line drawing of the sporangia and hyphae of *Neocallimastix*. **c.** Chytrids parasitizing a dinoflagellate. (a): Daren Mueller/Iowa State University/Bugwood.org; (c): ©Reproduced with permission of the Freshwater Biological Association on behalf of The Estate of Dr Hilda Canter-Lund. ©The Freshwater Biological Association.

Chytridiomycota

The **chytrids** include the simplest fungi, which may resemble the first fungi to have evolved. Some chytrids are single cells; others form branched aseptate hyphae. Most chytrids reproduce asexually through the production of zoospores within a single cell. The zoospores grow into new chytrids.

Chytrids play a role in the decay and digestion of dead aquatic organisms, but some are parasites of living plants, animals, and protists (Fig. 22.4*c*). The parasitic chytrid *Batrachochytrium dendrobatidis* in the last 40 years has been detected in over 700 amphibian species all around the world, resulting in a staggering loss of biodiversity. The chytrid infects the amphibian's skin, causing a normally permeable layer to thicken, inhibiting oxygen and water intake. The electrolyte imbalances result in cardiac arrest and death.

Zygomycota Produce Zygospores

The phylum Zygomycota is included here with the understanding that this group is not monophyletic, and many of the relationships within this group have yet to be resolved. **Zygospore fungi** live off plant and animal remains in the soil or in bakery goods in the pantry. Some are parasites of soil protists, worms, and insects such as the housefly.

The black bread mold, *Rhizopus stolonifer,* is a common example of this phylum. *Rhizopus* has both a sexual and an asexual phase in its life cycle (Fig. 22.5). The body of this fungus is composed of mostly aseptate hyphae that can specialize to perform various tasks. *Rhizopus* exhibits three kinds of specialized hyphae:

- *stolons*—horizontal hyphae that exist on the surface of the bread.
- *rhizoids*—hyphae that grow into bread, anchor the mycelium, and carry out digestion.
- *sporangiophores*—aerial hyphae that bear sporangia.

A **sporangium** is a capsule that produces haploid spores, called *sporangiospores,* during the asexual phase of reproduction (Fig. 22.5).

Zygospores are diploid spores produced during sexual reproduction. Two mating types of hyphae, termed plus (+) and minus (−), are chemically attracted to one another, and they grow toward each other until they touch (Fig. 22.5). The ends of the touching hyphae swell as nuclei are directed toward, and enter, the tips of the hyphae. Cross-walls then develop a short distance behind the swollen end of each hypha, forming an isolated capsule called a *gametangium* (*pl.,* gametangia). The gametangia of each hypha merge, and the nuclei fuse to form a diploid zygote.

ISTUDY

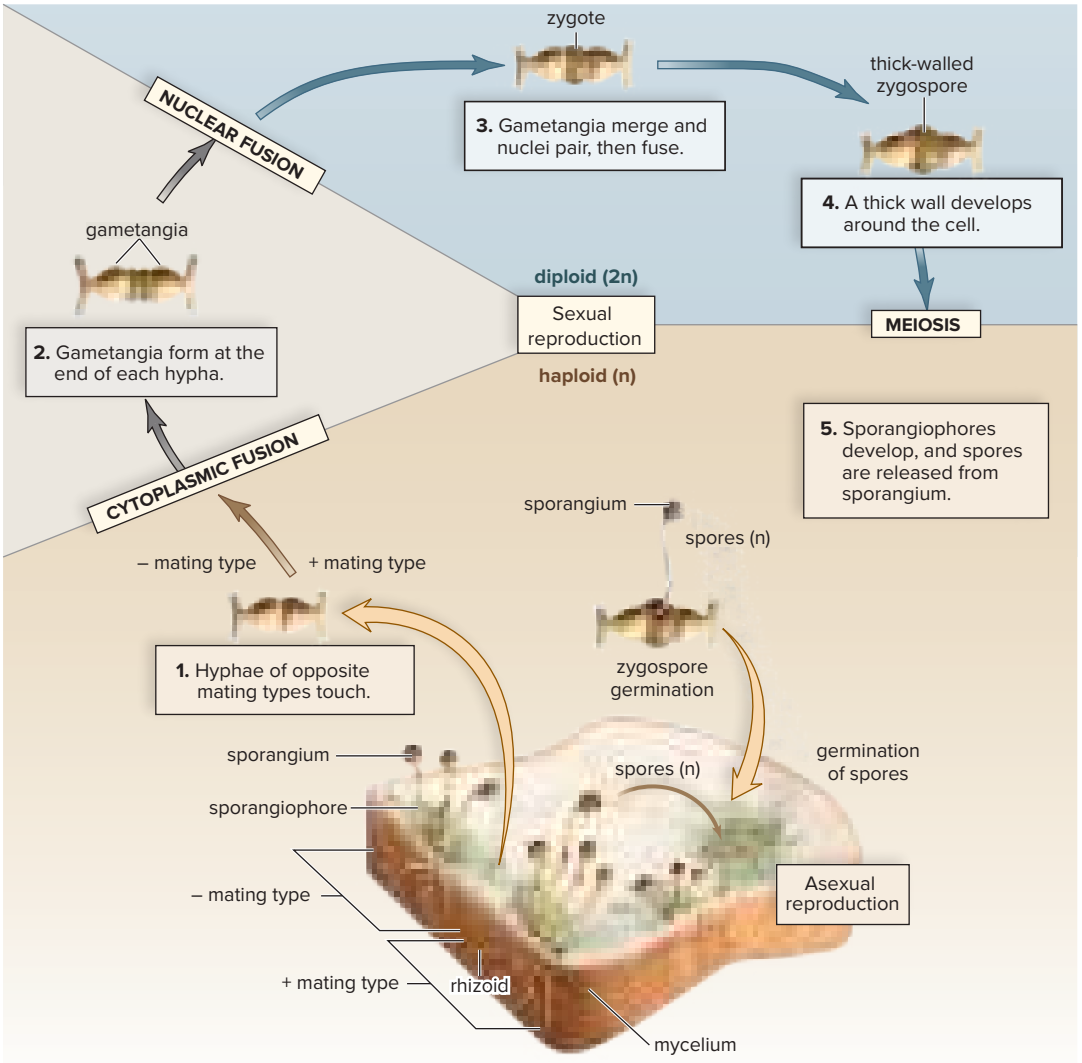


Figure 22.5 **Black bread mold, *Rhizopus stolonifer*.** **1.** At the start of sexual reproduction, two compatible mating types make contact. **2., 3.** First, gametangia fuse, and then the nuclei fuse. **4.** The zygospore is a resting stage that can survive unfavorable growing conditions. **5.** Due to zygotic meiosis, which occurs before or as the sporangiospores are produced, the adult is haploid. Asexual reproduction is the norm.

A thick wall develops around the zygote, which is now called a zygospore. The zygospore undergoes a period of dormancy before new haploid sporangiospores are formed by meiosis. Sporangiophores with sporangia at their tips germinate from the zygospore, and many sporangiospores are released. The spores, dispersed by air currents, give rise to new haploid mycelia, which will continue the sexual phase of the life cycle. Spores from black bread mold have been found in the air above the North Pole, in the tropics, and far out at sea.

Glomeromycota Includes Important Symbiotic Fungi

Glomeromycota, or **AM fungi,** are a relatively small group (230 described species) of fungi. The name AM stands for *arbuscular mycorrhizal* fungi. Arbuscules are branching invaginations that the fungus makes when it invades plant roots (see Fig. 22.14*a*).

Mycorrhizae (see Section 22.3) are a mutualistic association of plants and fungi, and AM fungi are the most common fungi to form symbiotic relationships with plants. They play a critical role in the ability of plants to absorb nutrients with their roots. The majority of plants have a mutually beneficial relationship, or symbiosis, with AM fungi.

Ascomycetes Produce a Fruiting Body Called an Ascocarp

Members of the Ascomycota phylum are called **sac fungi,** and consist of over 64,000 species of fungi that can reproduce asexually or sexually. Multicellular **molds** and single-celled **yeasts** are the most common morphological types. The sac fungi play an essential role in recycling by digesting materials that do not easily decompose, such as cellulose, lignin, and collagen. Some species have even been known to consume jet fuel and wall paint.

Members of this phylum are ecologically important, because some have beneficial relationships with plants, algae, and some animals; others have parasitic relationships with these organisms, too. The ascomycetes have an important economic role. Some species, mainly yeasts, are used to produce foods, while others cause food spoilage and can produce toxins. Fungal infections are of medical importance, as is the production of antibiotics from sac fungi.

ISTUDY

Figure 22.6 Asexual reproduction in sac fungi. **a.** Yeasts, unique among fungi, reproduce by budding, leaving a bud scar behind. **b.** The sac fungi usually reproduce asexually by producing spores called conidia or conidiospores. (a): Steve Gschmeissner/Science Photo Library/Getty Images; (b): Jannicke Wiik-Nielsen/iStock/Getty Images

Reproduction

Asexual reproduction is most common among these fungi. Yeasts usually reproduce by asexual budding, in which a small cell forms and pinches off as it grows to full size (Fig. 22.6*a*). Molds tend to produce spores called conidia, or **conidiospores,** which vary in size and shape. Conidiospores can develop at the tips of aerial hyphae called *conidiophores* (Fig. 22.6*b*). When released, the conidiospores are dispersed by wind.

Sexual reproduction takes place in a **fruiting body.** A fruiting body affords an evolutionary advantage, because it is formed to produce and enhance the release of spores. The name *ascomycota* refers to the **ascus** (*pl.,* asci) (Gk. "bag, sac"), a fingerlike sac that develops during sexual reproduction. In some instances, the asci are surrounded and protected by sterile hyphae within a fruiting body, called an ascocarp. *Ascocarps* can have different shapes; they can be cup-shaped or stalked with saclike pits (Fig. 22.7). Within the ascus, the hyphae are dikaryotic, and two nuclei fuse to create diploid cells that in turn undergo meiosis to become new

Figure 22.7 Sexual reproduction in sac fungi. The sac fungi reproduce sexually by producing asci, in fruiting bodies called ascocarps. **a.** In the ascocarp of cup fungi, dikaryotic hyphae terminate, forming the asci, where meiosis follows nuclear fusion and spore formation takes place. **b.** In morels, the asci are borne on the ridges of pits. (photos): (a): Carol Wolfe; (b, morel ascocarp): Mark Dierker/Bear Dancer Studios; (b, ascospores): Richard Gross/McGraw Hill

ISTUDY

haploid cells called *ascospores.* Ascospores are released and dispersed by the wind after bursting from a swollen ascus.

Food Industry

The yeast *Saccharomyces cerevisiae* is widely used in the baking and brewing industries. Through the process of fermentation (see Section 8.3), yeast can produce ethanol and carbon dioxide gas. Bread rises because of the gas given off, and drinks such as beer, wine, and liquor obtain their alcohol by fermenting yeast. The genus *Aspergillus,* a mold, is used to produce soy sauce and the Japanese food miso by fermentation of soybeans. In addition, the food industry uses *Aspergillus* to produce citric acid and gallic acid. These serve as additives during the manufacture of a wide variety of products, including foods, inks, medicines, dyes, plastics, toothpaste, soap, and chewing gum. The cheese industry uses the mold *Penicillium* to make blue cheese, Roquefort, Camembert, Brie, and Stilton cheeses.

Molds can also be detrimental to the food industry. Molds accelerate spoilage and can release toxins on food. Peanuts and other grains are susceptible to colonization by certain species of *Aspergillus* that produce aflatoxin, which can damage an animal's liver. The U.S. FDA has established guidelines on the amount of aflatoxin allowed in food consumed by humans and livestock.

Importance for Humans

Yeasts can be harmful to humans. *Candida albicans* is a yeast that is normally found on the body. When the balance of *Candida* and other microorganisms is disturbed, the yeast proliferates, causing oral thrush in newborns or vaginal infections in females (Fig. 22.8*a*). Moldlike fungi cause infections of the skin called tineas. Athlete's foot, jock itch, and ringworm are all tineas characterized by itching, peeling skin, or a raised inflammation (Fig. 22.8*b*).

Patients with AIDS or other immunocompromised conditions are particularly susceptible to fungal infections. The strong similarities between fungal and human cells make it difficult to design fungal medications that do not harm patients. Researchers target and exploit any biochemical difference they can discover, such as focusing on the cell wall or fungal-specific steroid production. Fungicides are developed and applied to crops, used on skin to battle tineas, or taken internally to fight systemic fungal infections.

One positive medical benefit is that many sac fungi have antibiotic and antimicrobial properties. Certain members of the genus *Penicillium* can be used to produce the antibiotic penicillin, which is a treatment for bacterial infections; another produces cyclosporine, which controls the immune system to prevent organ rejection after a transplant.

a.

b.

Figure 22.8 Fungal infections. Oral thrush (**a**) and the ringworm tinea (**b**) are fungal infections. (a): Dr M.A. Ansary/Science Source; (b): John Hadfield/Science Source

Many geneticists use ascomycetes as model organisms. With genomic information available, *Aspergillus nidulans* serves as a safe model for more dangerous related species; *Neurospora crassa* was used to develop the "one gene, one enzyme hypothesis," and *Saccharomyces* is a single-celled eukaryote with genes similar to those of other eukaryotic organisms, and they can be easily manipulated in a lab.

Basidiomycetes Produce a Fruiting Body Called a Basidiocarp

The phylum Basidiomycota, or **club fungi,** consists of over 31,000 species. Mushrooms, toadstools, puffballs, shelf fungi, jelly fungi, bird's-nest fungi, and stinkhorns are basidiomycetes. In addition, fungi that cause plant diseases, such as smuts and rusts, are placed in this phylum. Several mushrooms, such as portabella and shiitake mushrooms, are savored as foods by humans. Approximately 75 species of basidiomycetes are considered poisonous. The poisonous "death cap" mushroom is discussed in the Biological Systems feature, "Deadly Fungi."

The Biology of Club Fungi

The body of a basidiomycete is a mycelium composed of septate hyphae. Most members of this phylum are saprotrophs, feeding on dead and decaying organic matter, although several parasitic species obtain nutrition from living hosts.

Although club fungi occasionally produce conidia asexually, they usually reproduce sexually. Their formal name, Basidiomycota, refers to the **basidium** (L. *basidi,* "small pedestal"), a club-shaped structure in which spores called basidiospores develop. Basidia are located within a fruiting body called a *basidiocarp,* which we recognize as a mushroom (Fig. 22.9). Prior to the formation of a basidiocarp, haploid hyphae of opposite mating types meet and fuse, producing a dikaryotic mycelium. The dikaryotic mycelium continues its existence year after year, even for hundreds of years on occasion.

Mushrooms are composed of tightly packed hyphae whose walled-off ends become basidia. In gilled mushrooms, the basidia are located on radiating folds called gills. In shelf fungi and pore mushrooms (Fig. 22.10*a, b*), the basidia terminate in tubes. The extensive surface area of a basidiocarp is lined by basidia, where nuclear fusion, meiosis, and spore production occur. A basidium has four projections, into which cytoplasm and a haploid nucleus enter as the basidiospore forms. Basidiospores are windblown; when they germinate, a new haploid mycelium forms. It is estimated that some large mushrooms can produce up to 40 million spores per hour.

In puffballs, spores are produced inside parchmentlike membranes, and the spores are released through a pore or when the membrane breaks down (Fig. 22.10*c*). In bird's-nest fungi, falling raindrops provide the force that causes the nest's basidiospore-containing "eggs" to fly through the air and land on vegetation (Fig. 22.10*d*). Stinkhorns emit an incredibly disagreeable odor. Flies are attracted by the odor, and when they linger to feed on the sweet jelly, the flies pick up spores, which they later distribute (Fig. 22.10*e*).

Check Your Progress 22.2

1. Compare and contrast the three zoospore-producing phyla.
2. Discuss how spore dispersal is accomplished in different phyla of fungi.
3. Discuss the evolutionary advantage of a fruiting body and how size and complexity of fruiting bodies differ between phyla.

ISTUDY

THEME Biological Systems

Deadly Fungi

It is unwise and potentially fatal for amateurs to collect mushrooms in the wild, because certain mushroom species are poisonous. The red and yellow *Amanitas* are one example. These species are also known as fly agaric, because they were once thought to kill flies (the mushrooms were gathered, crushed, and then sprinkled into milk to attract flies). Its toxins include muscarine and muscaridine, which produce symptoms similar to those of acute alcoholic intoxication. In 1–6 hours, the victim staggers, loses consciousness, and becomes delirious, sometimes suffering from hallucinations, manic conditions, and stupor. Luckily, it also causes vomiting, which rids the system of the poison, so death occurs in less than 1% of cases.

The death cap mushroom, *Amanita phalloides* (Fig. 22A), causes 90% of the fatalities attributed to mushroom poisoning. When this mushroom is eaten, symptoms don't begin until 10–12 hours later. Abdominal pain, vomiting, delirium, and hallucinations are not the real problem; rather, a poison called amanitin interferes with RNA transcription by inhibiting RNA polymerase, and the victim dies from liver and kidney damage.

Some hallucinogenic mushrooms are used in religious ceremonies, particularly among Mexican Indians. *Psilocybe mexicana* contains a chemical called psilocybin, which is a structural analogue of LSD and mescaline. It produces a dreamlike state in which visions of colorful patterns and objects seem to fill up space and dance past in endless succession. Other senses are also sharpened to produce a feeling of "intense reality." Psilocybin is currently being used in clinical trials to help individuals with addiction, severe depression, and anxiety.

The only reliable way to tell a nonpoisonous mushroom from a poisonous one is to be able to correctly identify the species. Poisonous mushrooms cannot be identified with simple tests, such as whether they peel easily, have a bad odor, or blacken a silver coin during cooking. So, only consume mushrooms that have been identified by a bona-fide expert!

Like club fungi, some sac fungi contain chemicals that can be dangerous to people. *Claviceps purpurea,* the ergot fungus, infects rye and replaces the grain with ergot—hard, purple-black bodies consisting of tightly cemented hyphae (Fig. 22B). When ground with the rye and made into bread, the fungus releases toxic alkaloids, which cause the disease ergotism. In humans, vomiting, feelings of intense heat or cold, muscle pain, a yellow face, and lesions on the hands and feet are accompanied by hysteria and hallucinations.

The alkaloids that cause ergotism have medicinal properties. They stimulate smooth muscle and block the sympathetic nervous system, and they can be used to cause uterine contractions and treat migraine headaches. Although the ergot fungus can be cultured in petri dishes, no one has successfully produced ergot in the laboratory. The only way to obtain ergot is to collect it from an infected field of rye.

Ergotism was common in Europe during the Middle Ages. During this period, it was known as St. Anthony's Fire and was responsible for 40,000 deaths in an epidemic in AD 994. We now know that ergot contains lysergic acid, from which LSD is easily synthesized. Based on recorded symptoms, some historians believe that those individuals who claimed to have been "bewitched" in Salem, Massachusetts, during the seventeenth century were actually suffering from ergotism. The ensuing mass hysteria, however, led to the execution of 20 people for the crime of witchcraft.

Questions to Consider

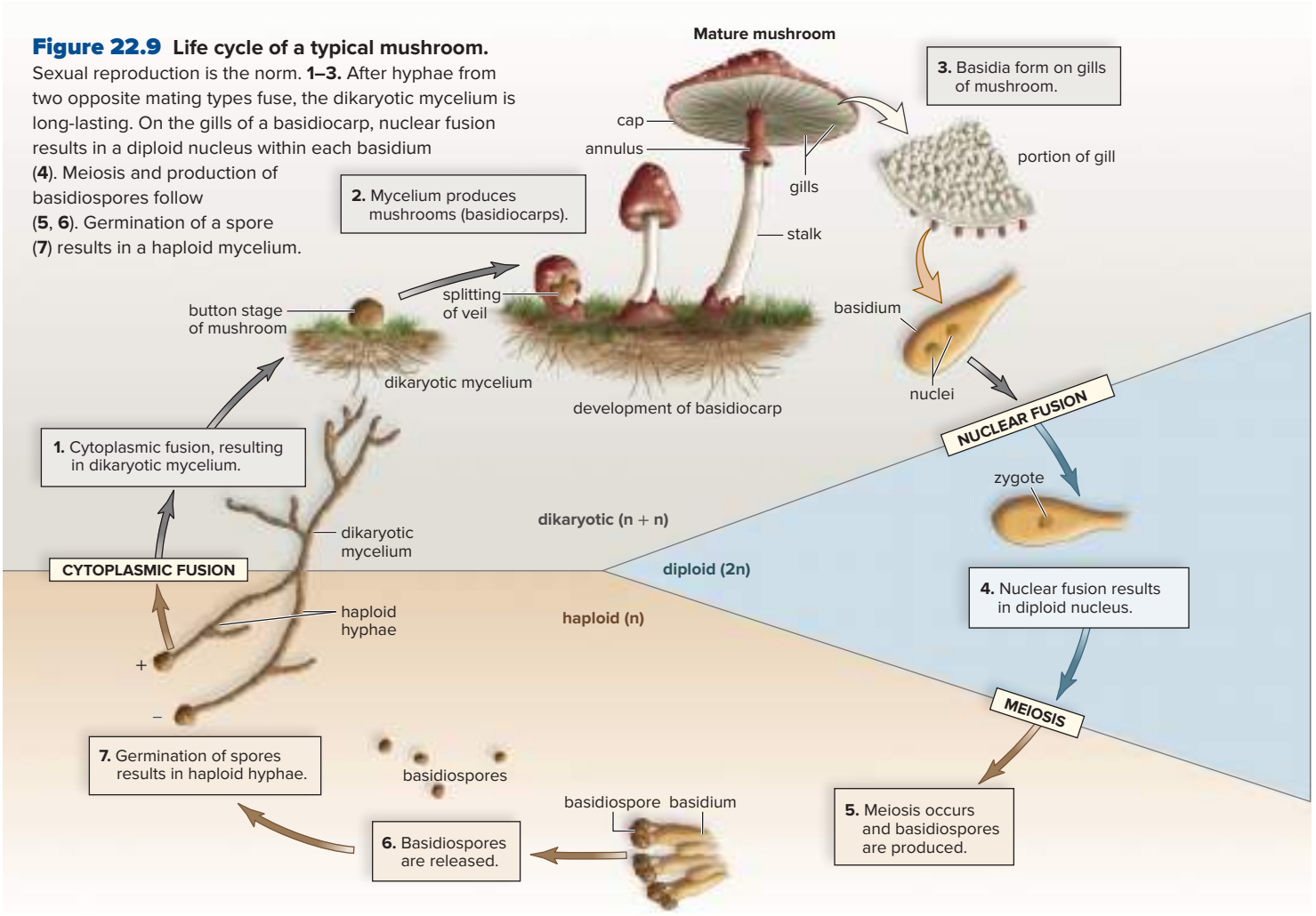
1. How might a sequence of DNA from an unknown species of fungus help identify it?
2. Why are toxins that interfere with RNA polymerase fatal—that is, why is RNA polymerase needed for normal body function?

Figure 22A A poisonous mushroom species, *Amanita phalloides.* De Agostini/Getty Images

Figure 22B Ergot infection of rye, caused by *Claviceps purpurea.* Wildlife GmbH/Alamy Stock Photo

ISTUDY

Figure 22.9 Life cycle of a typical mushroom.
Sexual reproduction is the norm. **1–3.** After hyphae from two opposite mating types fuse, the dikaryotic mycelium is long-lasting. On the gills of a basidiocarp, nuclear fusion results in a diploid nucleus within each basidium (**4**). Meiosis and production of basidiospores follow (**5**, **6**). Germination of a spore (**7**) results in a haploid mycelium.



a. Shelf fungus

b. Pore mushroom, *Boletus*

Figure 22.10 Club fungi. **a.** A shelf fungus. **b.** Fruiting bodies of *Boletus.* This mushroom is not gilled; instead, it has basidia-lined tubes that open on the undersurface of the cap. **c.** In puffballs, the spores develop inside an enclosed fruiting body. **d.** In bird's-nest fungi, spores are contained within white packets that are ejected when hit with a raindrop. **e.** Flies are attracted to the sticky spore mass produced by stinkhorns. (a): Alexander von Duren/imagebroker/Alamy Stock Photo; (b): P. DI PIAZZA/DeAgostini/Getty Images; (c): Dr. Robert Siegel/Stanford University; (d): Ed Reschke/Stone/Getty Images; (e): David Plummer/Alamy Stock Photo

c. Puffball

d. Bird's-nest fungi

e. Stinkhorn

ISTUDY

22.3 Symbiotic Relationships of Fungi

Learning Outcomes

Upon completion of this section, you should be able to

1. Discuss various types of symbiotic relationships that exist among fungi, plants, and animals.
2. Summarize the association that occurs between a fungus and its photosynthesizing partner in lichens.
3. Explain the mutualistic relationship between mycorrhizae and plants.

Symbiosis is a close association between two different species. Sometimes, that association can be **mutualistic** and beneficial to both species. Mutualism will be described in our discussion of lichens and mycorrhizae. A symbiosis can also be **parasitic,** benefiting only one partner and proving harmful to the other. Fungi are able to parasitize plants, animals, and even other fungi.

Fungal Parasitism

Infections in Plants

Fungal parasites of plants exist throughout the Fungi kingdom. The ascomycete, *Ophiostoma,* was responsible for wiping out elm trees in the United States by inducing Dutch elm disease. *Physoderma maydis,* a blastocad, was described earlier as a corn parasite. Smuts and rusts are basidiomycetes that parasitize cereal crops, such as corn, wheat, oats, and rye. They are of great economic importance because of the crop losses they cause every year.

Smuts and rusts don't form basidiocarps, and their spores are small and numerous, resembling soot. Some smuts enter seeds and exist inside the plant, becoming visible only near maturity. Other smuts externally infect plants. In corn smut, the mycelium grows between the corn kernels and secretes substances that cause the development of tumors on the ears of corn (Fig. 22.11*a*).

The life cycle of rusts requires alternate hosts, and one way to keep them in check is to eradicate the alternate host. Wheat rust (Fig. 22.11*b*) is also controlled by producing new and resistant strains of wheat. The process is continuous, because rust can mutate to cause infection once again.

Infections in Animals

Both invertebrates and vertebrates can be parasitized by fungi. Nematodes can be captured by hyphae traps and digested while still alive. Insect exoskeletons can be penetrated by germinating spores that eventually take over their bodies (Fig. 22.12*a*). The use of fungi to decrease mosquito populations is discussed in the Nature of Science feature, "Fungi as a Biological Control."

Vertebrates, such as reptiles, can be attacked by molds, and amphibian populations are declining worldwide because of chytrid infections. Of particular concern is the fungal infection that has decimated bat populations mainly along the east coast of the United States and Canada. An ascomycete called *Pseudogymnoascus destructans* is responsible for a disease called white nose syndrome, because of the characteristic fungi growing on infected bats' muzzles. This infection disturbs hibernating bats by waking them up during the winter months, causing them to look for food and starve to death (Fig. 22.12*b*).

Lichens: Mutualistic or Parasitic?

Lichens are an association between fungi and a photosynthesizing partner, either cyanobacterium or green alga. Until recently, all lichens were thought to include only two organisms, but scientists discovered some actually contained three—a photosynthesizing partner and two fungi. Analyzing DNA from these unique lichens showed both ascomycete and basidiomycete fungi were present along with the green alga.

As one example, a crustose lichen has a body consisting of three layers. The ascomycete fungus forms a thin, tough outer layer called the cortex where the basidiomycete yeast are also embedded (Fig. 22.13*a*). Below the cortex, a second layer is formed by photosynthetic cells enveloped in specialized fungal hyphae that transfer nutrients to the rest of the fungus. A third layer, called the medulla, is formed by loosely packed hyphae. Lichens can reproduce

a. Corn smut, *Ustilago* b. Wheat rust, *Puccinia* 40×

Figure 22.11 Smuts and rusts. **a.** Corn smut. **b.** Micrograph of wheat rust. (a): Marvin Dembinsky/Carol Dembinsky/Dembinsky Photo Associates/Alamy Stock Photo; (b): Patrick Lynch/Science Source

ISTUDY

a.

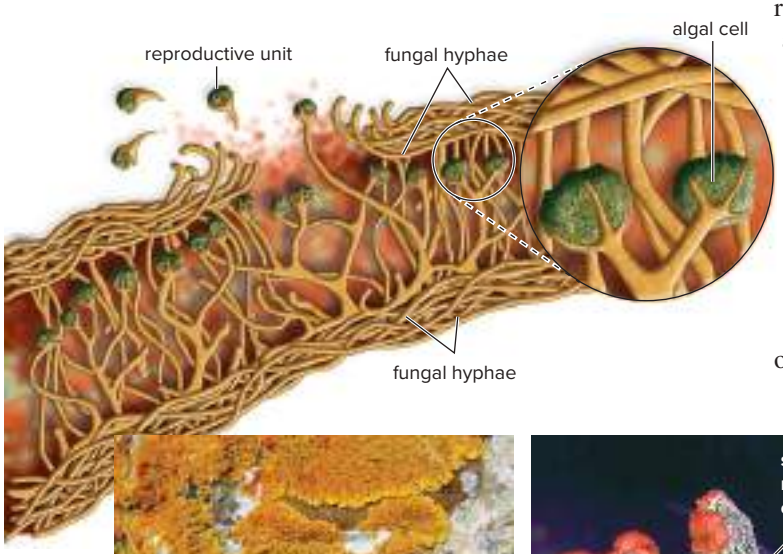
b.

Figure 22.12 Animal parasites. **a.** A dead fly with two fruiting bodies emerging on either side of its head. **b.** White nose syndrome is caused by a parasitic fungus that disrupts hibernating bats and forms the characteristic mold on their muzzles. (a): khlungcenter/Shutterstock; (b): Al Hicks/New York Department of Environmental Conservation/USGS

asexually by releasing fragments that contain hyphae and an algal cell. In fruticose lichens, the sac fungus reproduces sexually (Fig. 22.13*b*).

In the past, lichens were assumed to be mutualistic relationships in which the fungal partner(s) received nutrients from the algal cells, and the algal cells were protected from desiccation by the fungi. Actually, lichens may involve a controlled form of parasitism of the algal cells by the fungi, with the algae not benefiting at all from the association. This idea is supported by experiments in which the fungal and algal components are removed and grown separately. The algae grow faster when they are alone than when they are part of a lichen. In contrast, it is difficult to cultivate the fungus, which does not naturally grow alone. The different lichen species are identified according to their fungal partners.

Three types of lichens are recognized by shape. Compact crustose lichens are often seen on bare rocks or tree bark; fruticose lichens are shrublike; and foliose lichens are leaflike (Fig. 22.13*c*). Lichens are efficient at acquiring nutrients and moisture, and therefore they can survive in areas of low moisture and low temperature, as well as in areas with poor or no soil. They produce and improve the soil, thus making it



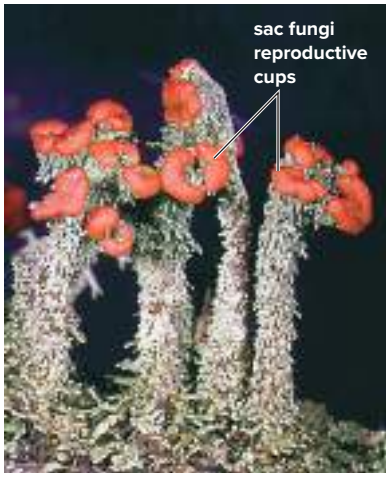


Figure 22.13 Lichen morphology. **a.** A section of a compact crustose lichen shows the placement of the algal cells and the fungal hyphae, which encircle and penetrate the algal cells. **b.** Fruticose lichens are shrublike. **c.** Foliose lichens are leaflike. (a): Dr. Jeremy Burgess/Science Source; (b): Steven P. Lynch; (c): Yogesh More/Alamy Stock Photo

ISTUDY

THEME Nature of Science

Fungi as a Biological Control

Experiment: Fungi that parasitize mosquitoes can be genetically engineered to be more effective killers of mosquitoes.

Observation: Mosquitoes carrying malaria are becoming insecticide resistant in sub-Saharan Africa where malaria is endemic. An ascomycete fungus, called *Metarhizium,* naturally parasitizes mosquitoes.

Hypothesis: Insecticide-resistant mosquitoes can be effectively killed by a *Metarhizium* fungus that has been engineered to deliver an effective spider toxin.

Experimental design: A screened structure with multiple sections containing huts was constructed in a malaria-prone region in the West African country of Burkina Faso. The screens kept the structure impermeable to outside insects, yet remained at the same temperature and humidity as the surrounding environment (Fig. 22C). All experimental huts were stocked with the same number of insecticide-resistant mosquitoes, and were provided with calf blood in the evening. Each hut had a black sheet hung on the wall, which provided a resting area for mosquitoes and a way to transmit fungal spores.

The control hut had a sheet with no fungal spores added. A second hut had a sheet with wild-type *Metarhizium* spores smeared on the sheet. A third hut had the toxin-producing *Metarhizium* spores smeared on the sheet.

In order to verify the fungal infection on dead mosquitoes, wild-type *Metarhizium* were engineered to express a red fluorescent protein, and the transgenic fungi were engineered to express a green fluorescent protein.

Data: For 14 days, mosquitoes that survived were collected, counted, and released back into their hut (Fig. 22D). The number of mosquito survivors in the control hut remained constant, while the survivors in the fungal huts steadily decreased over time. The mosquitoes exposed to toxin-engineered fungi died faster than the wild-type fungi.

Dead mosquitoes were also collected and observed under UV light in order to confirm which strain of *Metarhizium* caused their death (Fig. 22E). The dead mosquitoes exposed to wild-type fungi had spores that fluoresced red, and dead mosquitoes exposed to engineered fungi had spores that fluoresced green. No fluorescence was detected in the control hut.

To test whether the fungus could decrease the population of mosquitoes over time, each hut received a "swarm" of mosquitoes. A swarm includes females and twice as many males for generational reproduction to be successful. Surviving mosquitoes in each hut were allowed to undergo two rounds of the mosquito life cycle. Numbers of surviving adults for the F_1 and F_2 generations were tallied for each of the huts (Fig. 22F).

Results: *Metarhizium* is an ascomycete fungus that parasitizes mosquitoes. Both wild-type and engineered strains are effective in killing mosquitoes, but the engineered

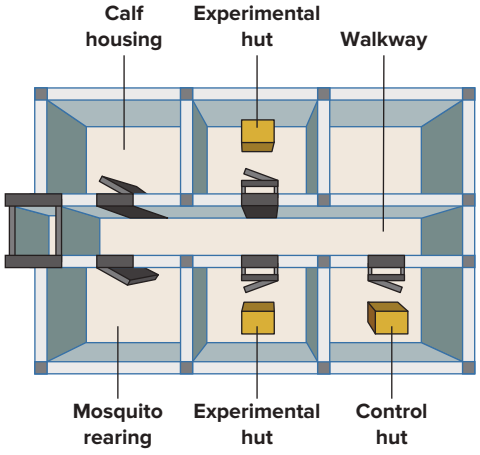


Figure 22C Diagram of the screened building used for the experiment. The mosquito rearing area was used to raise insecticide-resistant mosquitoes. Calves were housed in another section so blood could be provided to the mosquitoes. Three experimental huts are shown, one used as a control, and two others used to test fungal pathogenicity.

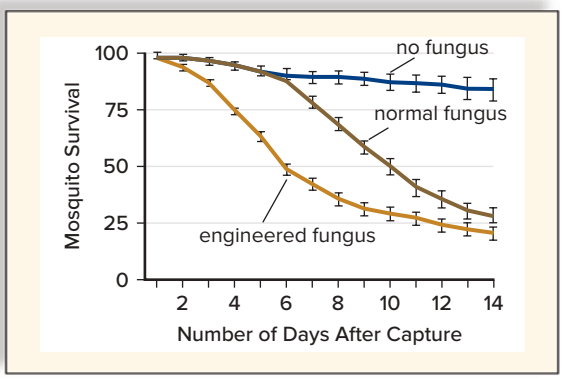


Figure 22D Survival of captured mosquitoes. The genetically engineered *Metarhizium* (orange line) killed mosquitoes faster than the wild-type *Metarhizium* (yellow). The control group of mosquitoes (blue line) were exposed to neither fungus.

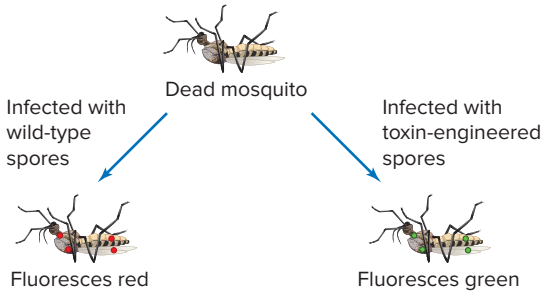


Figure 22E Fluorescent tags were used to track fungal strains. Wild-type fungi fluoresce red, and transgenic fungi fluoresce green.

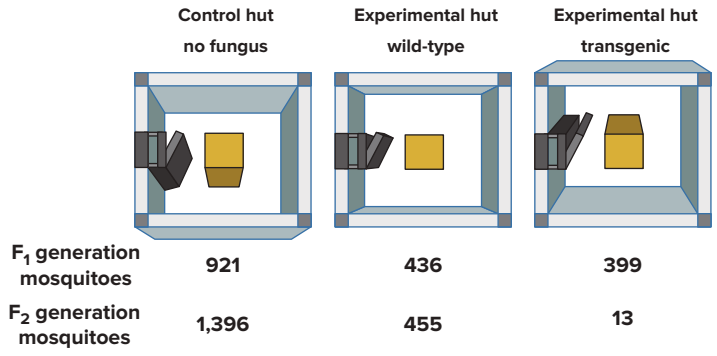


Figure 22F Population counts in the F_1 and F_2 generations. Each hut began with 500 female and 1,000 male mosquitoes. Numbers represent surviving adults after two generations.

mosquitoes kill faster. A faster kill time decreases the ability of mosquitoes to produce future generations because the required number of mosquitoes for a swarm are not achieved, thus decreasing the population overall.

Questions to Consider

1. What other controls not mentioned in the reading do you think were implemented for this experiment?
2. Do you think this could eventually be used as an effective biological control against mosquitoes? Why or why not?

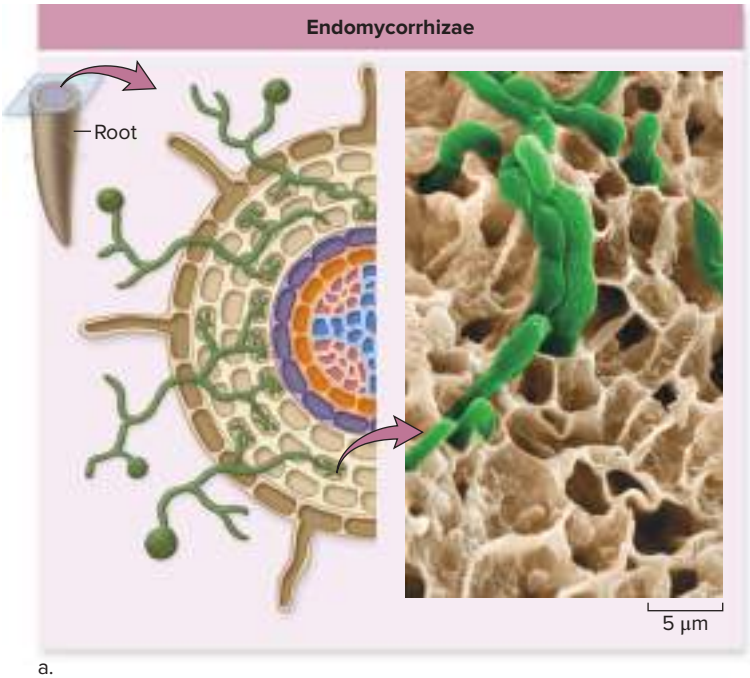
suitable for plants to invade the area. Unfortunately, lichens also take up pollutants, and they cannot survive where the air is polluted. Therefore, lichens can serve as air pollution sensors.

Mycorrhizae: Mutualistic for Both Plant and Fungal Partners

Mycorrhizae (Gk. *mykes,* "fungus"; *rhizion,* "little root") are mutualistic relationships between soil fungi and the roots of most plants. Plants whose roots are invaded by mycorrhizae grow more successfully in poor soils—particularly soils deficient in phosphates—than do plants without mycorrhizae.

The fungal partner can be either a glomeromycete, ascomycete, or basidiomycete. Ectomycorrhizae form a mantle that is exterior to the root, and grow between cell walls. Endomycorrhizae, such as the AM fungi mentioned earlier, penetrate the cell walls (Fig. 22.14). The presence of the fungus gives the plant a greater absorptive surface for the intake of minerals. The fungus also benefits from the association by receiving carbohydrates from the plant. As mentioned, even the earliest fossil plants have mycorrhizae associated with them. It would appear, then, that mycorrhizae helped plants adapt to and flourish on land.

The truffle, an underground fungus that is a gourmet delight and that has an ascocarp somewhat prunelike in appearance, is a



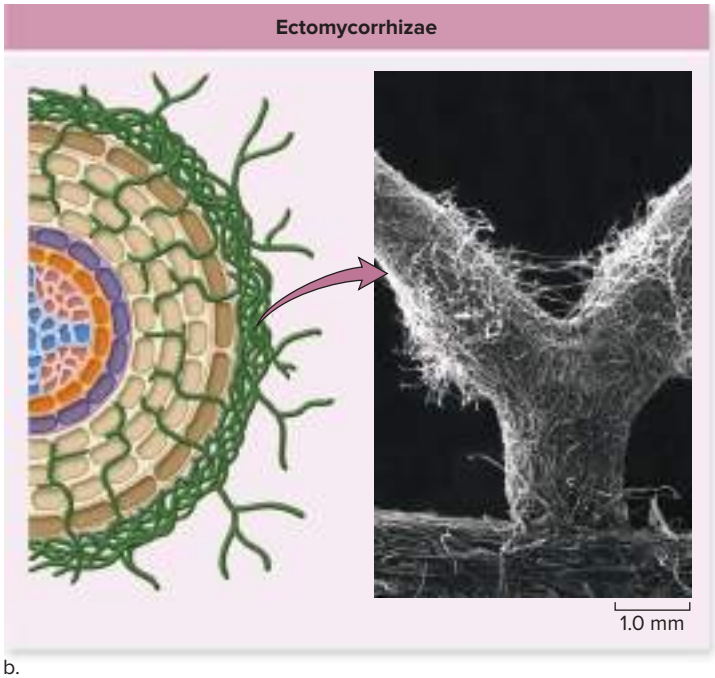


Figure 22.14 Endomycorrhizae and ectomychorrhizae. **a.** In endomycorrhizae, fungal hyphae penetrate the root cell wall. **b.** In ectomycorrhizae, fungal hyphae grow around roots or between cells, but never penetrate the cells. (a): Eye of Science/Science Source; (b): H.B. Massicotte, R.L. Peterson and L.H. Melville

Figure 22.15 A black truffle. The truffle *Tuber melanosporum* (inset) eliminates competition for its mycorrhizal partner, creating a "dead zone" around the tree. (truffles): bonzami emmanuelle/Alamy Stock Photo; (trees): Jean-Louis Le Moigne/Biosphoto/SuperStock

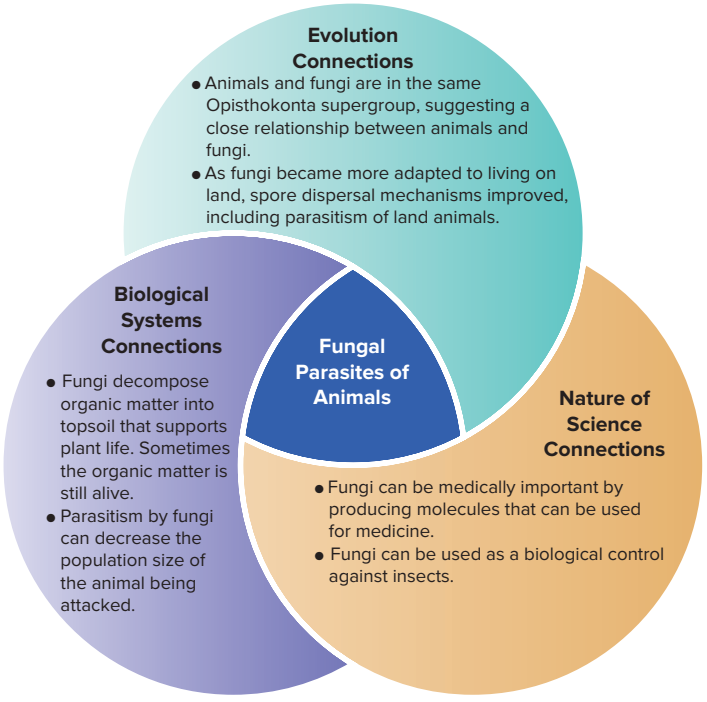
mycorrhizal sac fungus living in association with oak and beech tree roots. In the past, the French used pigs ("truffle-hounds") to sniff out and dig up truffles, but now they have succeeded in cultivating truffles by inoculating the roots of seedlings with the proper mycelium.

In addition to providing water and minerals, some black truffle species, such as *Tuber melanosporum,* engage in chemical warfare against other plants, fungi, and even bacteria. This truffle creates a burnt "dead zone" around its partner tree, so that the tree has no competition (Fig. 22.15).

Check Your Progress 22.3

1. Explain the difference between a mutualistic and parasitic symbiosis.
2. Explain the components of a lichen and how it reproduces.
3. Summarize the symbiotic relationship between mycorrhizae and plants.

Connecting *the* Concepts



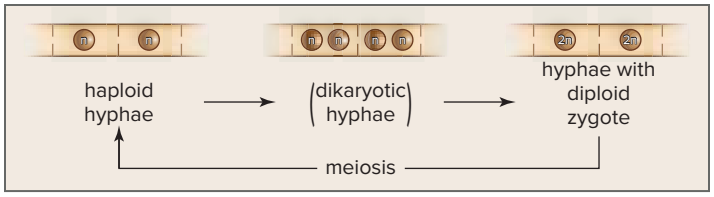
What Did You Learn?

22.1 Evolution and Characteristics of Fungi

Fungi are multicellular, **saprotrophic** eukaryotes. After external digestion, they absorb the resulting nutrient molecules. Saprotrophic fungi aid the cycling of organic molecules in ecosystems by decomposing dead remains. Some fungi are parasitic, especially on plants, and others are mutualistic with plant roots and algae.

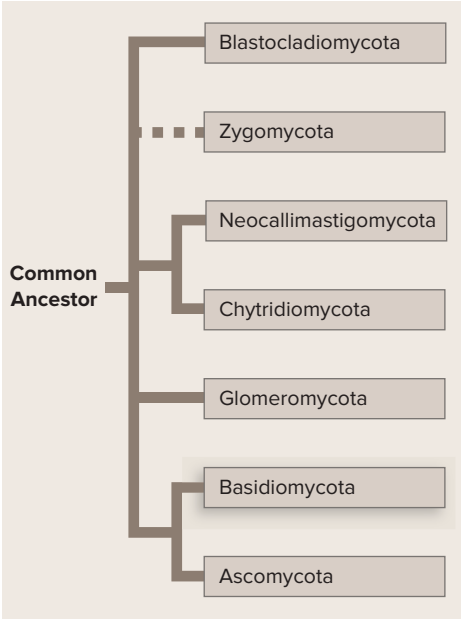
- The body of a fungus is composed of thin filaments called **hyphae,** which collectively are termed a **mycelium.** The cell wall contains **chitin,** and the energy reserve is glycogen. With the notable exception of the chytrids, which have flagellated spores and gametes, fungi do not have flagella at any stage in their life cycle. Aseptate hyphae have no cross-walls; **septate** hyphae have cross-walls called **septa,** but pores allow the cytoplasm and even organelles to pass through.
- Fungi produce **spores** during both asexual and sexual reproduction. Yeasts reproduce asexually by **budding.** During sexual reproduction in some multicellular fungi, hyphae tips fuse, so that **dikaryotic** (n + n) hyphae usually result, depending on the type of fungus. Following nuclear fusion, zygotic meiosis occurs during the production of the sexual spores.

ISTUDY



22.2 Diversity of Fungi

Fungal classification is incomplete and ongoing. Six phyla are recognized: Blastocladiomycota, Neocallimastigomycota, Chytridiomycota (chytrids), Glomeromycota (AM fungi), Ascomycota (sac fungi), and Basidiomycota (club fungi). A seventh phylum, the Zygomycota, is not monophyletic but is included in our discussion.



- **Microsporidia** are single-celled intracellular parasites found in insects and vertebrates, including humans.
- The **zoospore**-producing phyla include the Blastocladiomycota, with members that can exhibit an alternation-of-generation life cycle; the Neocallimastigomycota found living anaerobically in the guts of herbivores; and the **chytrids,** predominately aquatic, single-celled fungi that can be saprotrophic or parasitic.
- Zygomycota, or **zygospore fungi,** are multicellular fungi with aseptate hyphae. During sexual reproduction, they have a dormant stage consisting of a thick-walled **zygospore.** Meiosis occurs in the diploid zygospore. The zygospore germinates a **sporangium** with haploid sporangiospores.
- Glomeromycota, or **AM fungi,** exist in mutualistic associations, called mycorrhizae, with the roots of most land plants.
- The **sac fungi** (ascomycetes) are septate, and during sexual reproduction a saclike **ascus** produces spores. Asci are sometimes located in **fruiting bodies** called ascocarps. Asexual reproduction, which is dependent on the production of **conidiospores,** is more common. Sac fungi include various **yeasts** and **molds,** some of which cause disease in plants and animals, including humans.
- The **club fungi** (basidomycetes) are septate, and during sexual reproduction, club-shaped structures called **basidia** produce spores. Basidia are located in fruiting bodies called basidiocarps. Club fungi have a prolonged dikaryotic stage. A dikaryotic mycelium periodically produces fruiting bodies. Mushrooms and puffballs are examples of club fungi.

22.3 Symbiotic Relationships of Fungi

Symbiotic associations in fungi are generally mutualistic or parasitic. Fungi can parasitize plants like trees and important crops; or animals including invertebrates and vertebrates.

- **Lichens** are an association between one or two fungi and a cyanobacterium or a green alga. Traditionally, this association was considered mutualistic, but experimentation suggests a controlled parasitism by the fungus on the alga. Lichens can live in extreme environments and on bare rocks. They allow other organisms to colonize these harsh environments, and together they eventually form soil.
- The term **mycorrhizae** refers to an association between a fungus and the roots of a plant. The fungus helps the plant absorb minerals, and the plant supplies the fungus with carbohydrates.

Thinking Critically About Biology

1. A fine line seems to exist between symbiosis and parasitism when you examine the relationships between fungi and plants. Under what circumstances might a mutualistic relationship evolve between fungi and plants? Under what circumstances might a parasitic relationship evolve?
2. The importance of fungi in the evolution of terrestrial life is typically understated. Evaluate the importance of fungi in the colonization of land.
3. You are looking through a seed catalog and discover you have a choice between pea seeds that are treated with a fungicide and those that are not. Discuss one reason you might choose the treated seed and one reason you might prefer the untreated seed.

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Nut Fungus

McGraw Hill

Fungi have an important role in the ecosystem as decomposers. How does their unique feeding method allow for this role?

Making It Relevant

1. What are entomopathic fungi and what organic molecule do you suspect these fungi can decompose?
2. Have you ever considered how fungal products have helped you stay healthy? Would you ever drink tea made from the "caterpillar fungus"?
3. In most cases, insects that are parasitized by fungi do not go extinct. Why do you think this is the case?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 5

Plant Evolution and Biology

Life would be impossible without plants. Plants provide nearly all living organisms with the food and oxygen needed for survival. They can be a source of fuel to heat our homes and provide fibers to make our clothes. The frames of our houses and the furniture inside them are often made of wood and other plant products. We even use plants to make the paper for our magazines and textbooks.

Because plants and our everyday lives are intertwined, it's a good idea to know the principles of plant biology: plant structure and function. Plants have the same characteristics of life as animals do, and most reproduce sexually, as we do. Life began in the sea, and plants invaded the terrestrial environment before animals, as you would expect because animals are so dependent on plants. Throughout the course of primate and human evolution, we have been dependent on plants for the majority of our nutrients. Even today, people rely on plants to provide the bulk of their caloric intake. These chapters introduce you to the biology of plants, those incredible organisms that keep the biosphere functioning as it should.

Unit Outline

Chapter 23 Plant Evolution and Diversity

Chapter 24 Flowering Plants: Structure and Organization

Chapter 25 Flowering Plants: Nutrition and Transport

Chapter 26 Flowering Plants: Control of Growth Responses

Chapter 27 Flowering Plants: Reproduction

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Identify the key structural innovations that occurred during the evolution of various plant groups.
Nature of Science	Explain how humans have manipulated plants to better serve our needs.
Biological Systems	Assess how the structure and response of plants to the environment enable them to survive.

ISTUDY

23

Plant Evolution and Diversity

MRaust/iStock/Getty Images

The Uses of Mosses

Angiosperms are the most successful and abundant plants on Earth today because they produce seeds, flowers, and fruit. In contrast, 400 million years ago the predominant plants on Earth were simpler plants like mosses, lycophytes, and ferns. Today, the number of species of these early plants is much smaller, yet humans have still found some interesting uses for them. For example, lycophytes have spores used by the movie industry to create fiery special effects. Ferns can be used for *phytobioremediation,* extracting arsenic out of the soil in heavily polluted areas.

Mosses have been used for over 1,000 years as wound dressing. During World War I, for example, Scottish field doctors were unable to get enough cotton to properly dress wounds. Concerned about bacterial infections, they began to apply *Sphagnum* moss directly to wounds. Many mosses contain dead cells that soak up liquid, and in the case of war-inflicted wounds, this means blood and pus. Mosses also have a unique chemistry where the exchange of ions from the soil results in the exterior of mosses being highly acidic, thus inhibiting bacterial growth in a wound.

As you read through this chapter, think about the following questions:

1. What environmental challenges did plants have to overcome in order to survive on land?
2. What characteristics are unique to each of the major groups of plants?

Chapter Outline

23.1 Ancestry and Features of Land Plants
23.2 Evolution of Bryophytes: Colonization of Land
23.3 Evolution of Lycophytes: Vascular Tissue
23.4 Evolution of Pteridophytes: Megaphylls
23.5 Evolution of Seed Plants: Full Adaptation to Land

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 4.7 What cellular structures are unique to plants?

Section 10.3 What is the end result of meiosis?

Section 16.2 What is the role of natural selection in the evolutionary process?

Following *the* Themes

CHAPTER 23 – PLANT EVOLUTION AND DIVERSITY

Evolution	Key characteristics evolved in each group of plants that enabled them to be more successful than the earlier groups.
Nature of Science	Humans use plants in a variety of ways that have made us dependent upon them for our survival.
Biological Systems	Each group of plants possess key structural adaptations that enable them to perform their functions.

ISTUDY

23.1 Ancestry and Features of Land Plants

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast algae with land plants.
2. List the traits that enabled plants to adapt to life on land.
3. Evaluate the differences in the alternation of generations of land plants.

Plants are multicellular, photosynthetic eukaryotes whose evolution is marked by adaptations to a land existence. Plant systematists use molecular and morphological information to classify plants. Plants are in the supergroup Archaeplastida, along with the red algae, green algae, and charophytes (see Fig. 21.1). In our discussions, the plants will be described as nonvascular (bryophytes), seedless vascular (lycophytes, ferns, and fern allies), and vascular seed plants (gymnosperms and angiosperms). Table 23.1 provides a summary.

Phylogenetic information suggests that the modern plant groups arose in a particular sequence (Fig. 23.1). Liverworts diverged first, mosses diverged next, and hornworts seem to be closely related to vascular plants. Vascular plants are distinguished by tissues that conduct food and water and provide structural support. Lycophytes are then related to all the other modern vascular plants; ferns arose next, then the seed plants.

Plants are thought to have evolved from a freshwater green algae. In this section, we consider similarities between green algae and plants, as well as the different adaptations plants have to facilitate life on land.

Table 23.1 Characteristics of Plants

DOMAIN: Eukarya
SUPERGROUP: Archaeplastida
KINGDOM: Plantae

CHARACTERISTICS

Multicellular; apical tissue produces specialized tissues; alternation-of-generations life cycle; protection of the embryo; gametangia produces gametes; waxy cuticle prevents water loss.

Nonvascular Plants—The Bryophytes

Liverworts, hornworts, and mosses are low-lying and live in moist environments. Dominant gametophyte produces flagellated sperm; unbranched, dependent sporophyte produces windblown spores.

Seedless Vascular Plants—The Lycophytes and Pteridiophytes

Dominant; branched sporophyte has vascular tissue: lignified xylem transports water, and phloem transports organic nutrients; typically has roots, stems, and leaves; sporangia borne on sides of leaves that produce windblown spores; independent and separate gametophyte produces flagellated sperm.

- Lycophyte leaves are microphylls with a single, unbranched vein.
- Pteridiophyte leaves are megaphylls with branched veins.

Seeded Vascular Plants—The Gymnosperms and Angiosperms

Leaves are megaphylls; dominant sporophyte produces heterospores that become dependent male and female gametophytes. Male gametophyte is pollen grain, and female gametophyte occurs within ovule, which becomes a seed.

- Gymnosperms are cone-bearing; sporophyte bears pollen cones, which produce windblown pollen (male gametopyte), and seed cones, which produce seeds.
- Angiosperm sporophytes bear flowers, which produce pollen grains, and bear ovules within an ovary. Following double fertilization, ovules become seeds that enclose a sporophyte embryo and endosperm. Fruit develops from ovary.

The Ancestry of Plants

The evolutionary history of plants begins in the water. Most likely, land plants evolved from a form of freshwater green algae. Green algae are not plants, but like plants they (1) contain chlorophylls *a* and *b* and various accessory pigments, (2) store excess carbohydrates as starch, and (3) have cellulose in their cell walls. One major difference, however, between algae and terrestrial plants is that plants not only protect the zygote but also protect and nourish the resulting embryo—an important feature that separates land plants from green algae (Fig. 23.2).

The commonality with green algae, as well as a comparison of DNA and RNA base sequences, suggests that land plants are most closely related to a particular group of freshwater green algae known as **charophytes.** Molecular data show that charophytes and land plants are in the same clade and form a monophyletic group. Although *Spirogyra* (see Fig. 21.21) is a charophyte, the ancestors of land plants were more closely related to the charophytes *Chara* and *Coleochaete,* shown in Figure 23.3.

Chara are commonly known as stoneworts, because they are encrusted with calcium carbonate deposits. The body consists of a single file of very long cells anchored in mud by thin filaments. Whorls of branches occur at regions called *nodes,* located between the cells of the main axis. Male and female reproductive structures grow at the nodes. *Coleochaete* appears disklike, but the body is actually composed of long, branched filaments of cells. Most important to the evolution of plants, charophytes protect the zygote.

Adaptation to Land

Recall what it is like swimming in a pool. Like algae, you are weightless, surrounded by water and protected from the sun. However, all that changes when you leave the water. When plants moved from water to land, they needed mechanisms to deal with water loss, gravity, and sun exposure. The following features are evolutionary adaptions of most terrestrial plants.

- Unlike algae, plants living on land have a limited amount of water available to them. As an adaptation to living on land, most plants are protected from desiccation (drying out) by a waxy covering called a **cuticle** (Fig. 23.4*a*). The cuticle covers all exposed surfaces and is relatively impermeable. The problem is that this limits gas exchange for photosynthesis and cellular respiration. The solution is tiny openings, found mainly on the underside of leaves, called **stomata** (*sing.,* stoma) (Fig. 23.4*b*). These openings, or pores, allow for gas exchange but can also allow water vapor to escape.

ISTUDY

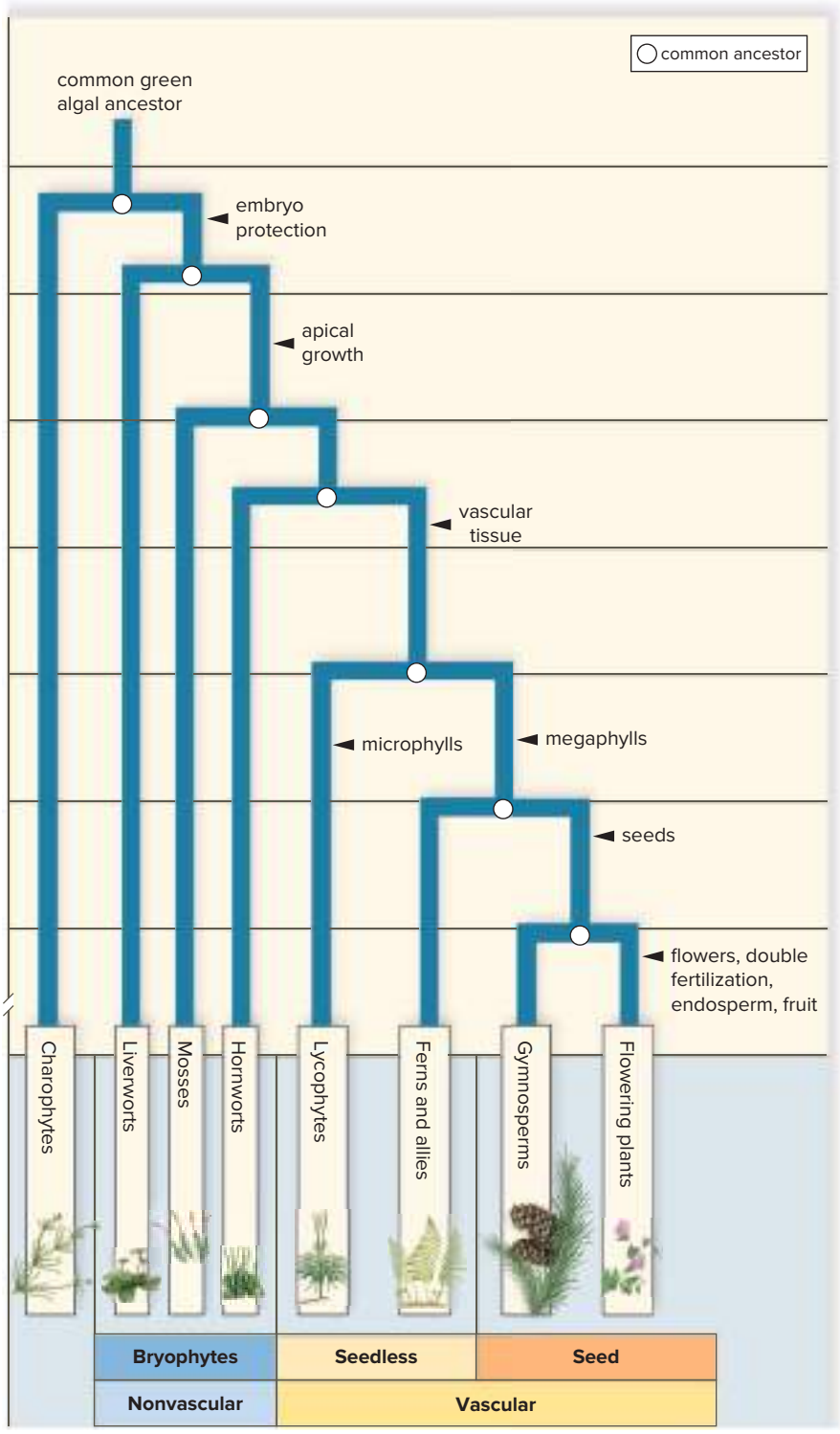


Figure 23.1 Evolutionary history of plants. The evolution of plants involves several significant innovations.

In Section 26.1, we will explore how stomata can be closed at times to limit water loss.

- Moving water within plants is a challenge that increases with plant size. Picture the height of moss compared to the height of a redwood tree. The different members of the land plants can be distinguished based on the presence or absence of **tracheids,** specialized cells with proteins that resist gravity and facilitate the upward transport of water and minerals.
- Compared to green algae, the bodies of land plants are mostly composed of three-dimensional tissues. A *tissue* is an association of many cells of the same type that can lead to specialized structures, such as organs. Tissues provide land plants with an increased ability to avoid water loss at their surfaces because bodies composed of tissues have a lower surface-area-to-volume ratio than do branched filaments, as seen in *Chara* (see Fig. 23.3).
- Plants living on land are exposed to higher intensities of UV rays than are aquatic algae. Exposure to ultraviolet light can increase the chance of mutations, but having a diploid genome can hide the effect of a single, deleterious allele. All terrestrial plants have both a haploid and a diploid generation, and the evolutionary shift toward a more dominant diploid generation allows for greater genetic variability in land plants.

Alternation of Generations

In the life cycle of humans, and other animals, the diploid stage is multicellular, whereas the haploid stage is composed of single-celled gametes (see Fig. 10.9). Plants also have a diploid and a haploid stage, but the significant difference is that both of these stages contain multicellular structures. Therefore, all land plants exhibit an **alternation of generations,** meaning that the plant has two alternating forms in the course of its life cycle. The two types of multicellular bodies are (1) the diploid, spore-producing **sporophyte** generation and (2) the haploid, gamete-producing **gametophyte** generation (Fig. 23.5).

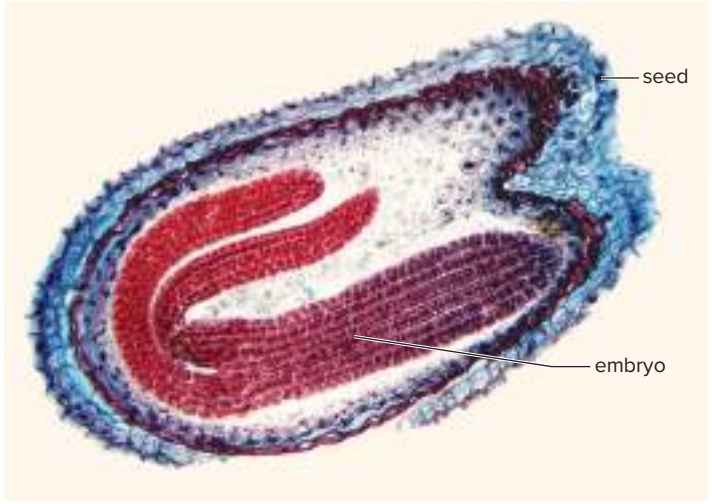
The sporophyte produces a specialized structure called a **sporangium,** where meiosis takes place to produce haploid spores. A **spore** is a reproductive cell that develops into a new organism without the need to fuse with another reproductive cell. The spore then undergoes mitosis and becomes a new multicellular gametophyte.

The gametophyte is a multicellular haploid structure that will develop both male and female gamete-producing regions called *gametangia.* Best seen in the moss life cycle in Figure 23.9, male gametangia are called **antheridia;** they produce sperm. The female gametangia are called **archegonia;** they produce eggs. Sperm and egg will fuse, become a *zygote,* and begin the diploid part of the life cycle.

Dominant Generation

Land plants differ as to which generation is dominant—that is, more conspicuous. In plants such as mosses, the gametophyte is

a. Embryo of the green alga, *Oedogonium* 100×



b. Embryo from the land plant, *Capsella* 60×

Figure 23.2 Embryo difference in green algae and land plants. **a.** The filamentous green alga, *Oedogonium,* with an unprotected embryo. **b.** A seed from the shepherd's purse plant, *Capsella,* showing the embryo protected and nourished. (a): Biology Pics/Science Source; (b): Garry DeLong/Science Source

Chara

Coleochaete

Figure 23.3 Charophytes. The charophytes (represented here by *Chara* and *Coleochaete*) are the green algae most closely related to the land plants. (*Chara*): Natural Visions/Alamy Stock Photo; (*Coleochaete*): Dr. Charles F. Delwiche, University of Maryland

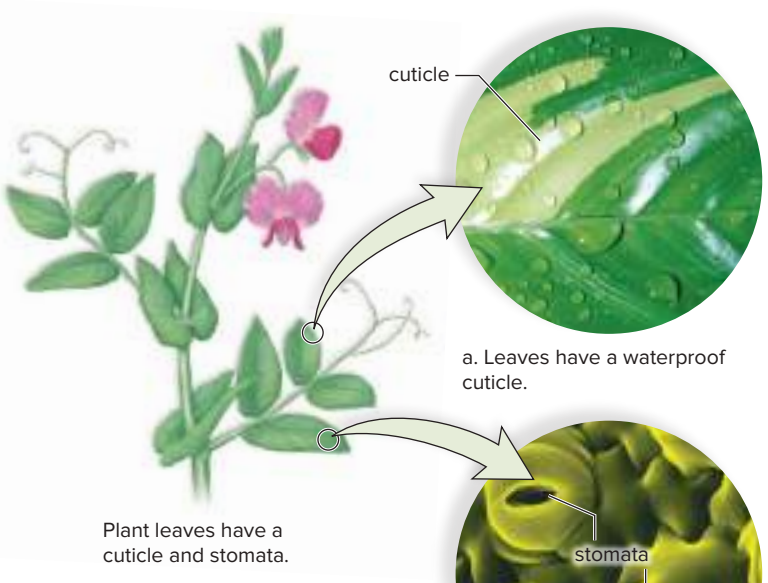


Figure 23.4 Leaf adaptation to prevent desiccation. **a.** A cuticle keeps the underlying cells and tissues from drying out. **b.** The exchange of gases is possible because the cuticle is interrupted by stomata. (a): Ingram Publishing/age fotostock; (b): Dr Jeremy Burgess/Science Source

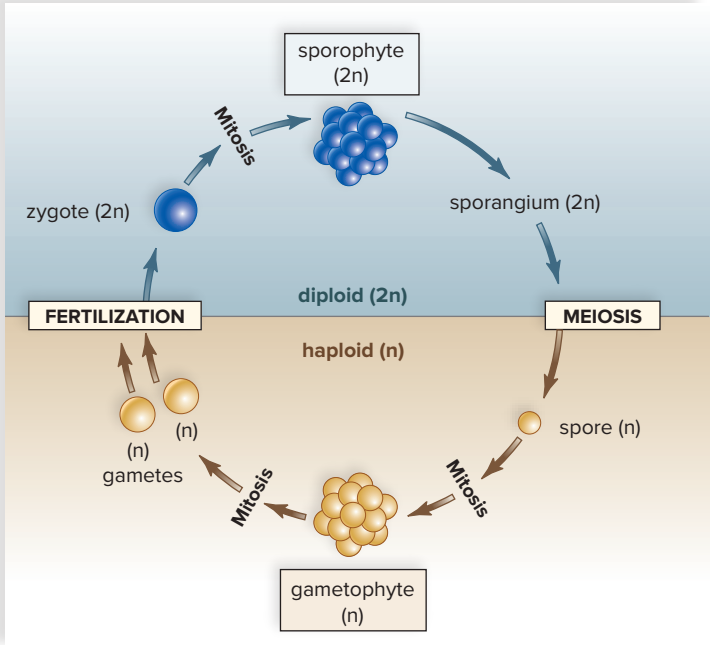


Figure 23.5 Alternation of generations in land plants. The zygote develops into a multicellular 2n generation, and meiosis produces spores in multicellular sporangia. The gametophyte generation produces gametes within multicellular gametangia.

dominant, but in plants such as ferns, pine trees, and peach trees, the sporophyte is dominant (Fig. 23.6). In the history of land plants, a vascular system evolves only in the sporophyte; therefore, the shift to sporophyte dominance is an adaptation to life on land.

Notice that as the sporophyte gains in dominance, the gametophyte becomes microscopic. The gametophyte also becomes dependent on the sporophyte.

ISTUDY

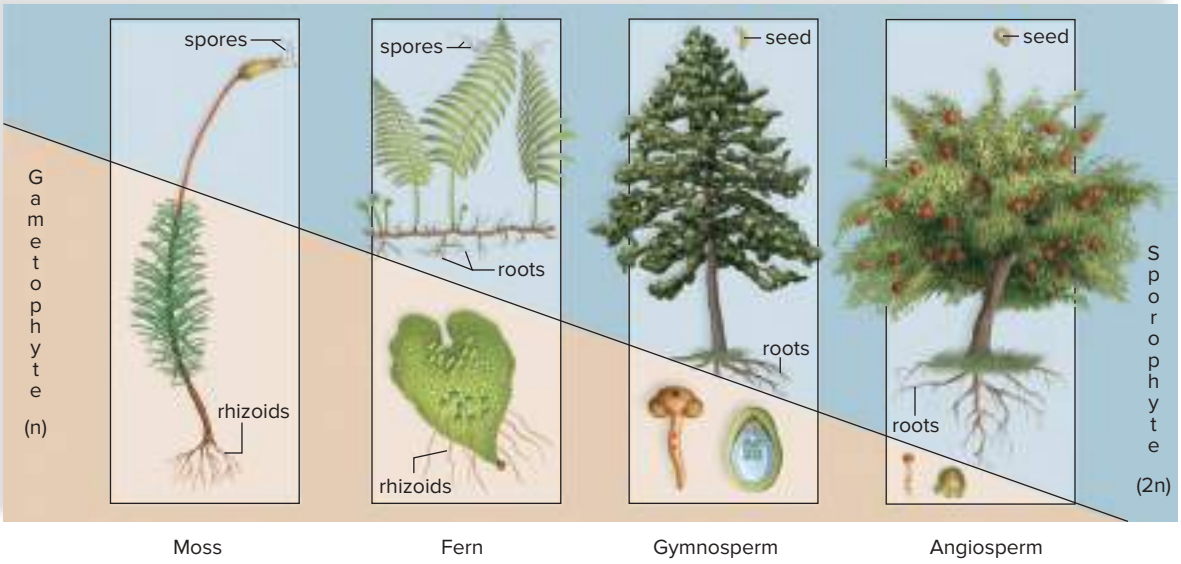


Figure 23.6
Reduction in the size of the gametophyte. Notice the reduction in the size of the gametophyte and the increase in the size of the sporophyte among these representatives of today's land plants. This trend occurred as these plants became adapted for life on land. In the moss and fern, spores disperse the gametophyte. In gymnosperms and angiosperms, seeds disperse the sporophyte.

Check Your Progress 23.1

1. Compare and contrast the traits of charophytes and land plants.
2. List the adaptations that led to a land existence for plants.
3. Identify the role of each generation in the alternation-of-generations life cycle.

23.2 Evolution of Bryophytes: Colonization of Land

Learning Outcomes

Upon completion of this section, you should be able to

1. List the traits that classify a plant as a bryophyte.
2. Compare the three groups of bryophytes.
3. Identify the key structures and stages in the life cycle of a moss.

The **bryophytes**—liverworts, hornworts (Anglo-Saxon *wort,* "herb"), and mosses—were the first plants to colonize land. They only superficially appear to have roots, stems, and leaves, because, by definition, true roots, stems, and leaves must contain vascular tissue. **Vascular tissue** is specialized for the transport of water and organic nutrients throughout the body of a plant. Bryophytes, which lack vascular tissue, are often called **nonvascular plants.** Vascular tissue also provides support to the plant body, so bryophytes typically are low-lying; some mosses reach a maximum height of only about 20 cm.

The fossil record contains some evidence that the various bryophytes evolved during the Ordovician period (about 443.7 MYA, see Table 18.1). An incomplete fossil record makes it difficult to tell how closely related the various bryophytes are. Molecular data, in particular, suggest these plants have individual lines of descent, as shown in Figure 23.1, and that they do not form a monophyletic group. The observation that today's mosses have a rudimentary form of vascular tissue suggests they are more closely related to vascular plants than to hornworts and liverworts.

Bryophyte bodies are covered by a cuticle, which is interrupted in hornworts and mosses by stomata, and they have apical tissue that produces complex tissues. However, bryophytes are the only land plants in which the gametophyte is dominant (see Fig. 23.6). Antheridia produce flagellated sperm, which means they need a film of moisture in order to swim to eggs located inside archegonia. The bryophytes' lack of vascular tissue and the presence of flagellated sperm mean you are apt to find bryophytes in moist locations. Some bryophytes compete well in harsh environments because they can reproduce asexually.

Bryophytes contribute to the lush beauty of forests. They can convert mountain rocks to soil, hold moisture and metals, and resist desiccation. Scientists working in genetic engineering are interested in bryophytes' ability to resist chemical reagents, decay, and herbivory. The idea is that these traits could be transferred to other plants.

Liverworts

Liverworts are divided into two groups—the thallose liverworts with flattened bodies, known as a thallus; and the leafy liverworts, which superficially resemble mosses. The name *liverwort* refers to the lobes of the thallus, which to some resemble the lobes of the liver. The majority of liverwort species are the leafy types.

The liverworts in the genus *Marchantia* have a thin thallus, about 30 cells thick in the center. Each branched lobe of the thallus is approximately a centimeter in length; the upper surface is divided into diamond-shaped segments with a small pore, and the lower surface bears numerous hairlike extensions called **rhizoids** (Gk. *rhizion,* dim. of "root") that project into the soil (Fig. 23.7). Rhizoids serve as anchors and assist in the absorption of water.

Marchantia species reproduce both asexually and sexually. Gemmae cups on the upper surface of the thallus contain gemmae,

ISTUDY

a. Thallus with gemmae cups

b. Male gametophytes bear antheridia

c. Female gametophytes bear archegonia

Figure 23.7 Liverwort, *Marchantia*. **a.** Gemmae can detach and start a new plant. **b.** Antheridia are present in disk-shaped structures. **c.** Archegonia are present in umbrella-shaped structures. (a, c): Ed Reschke/Stone/Getty Images; (b): J. M. Conrader/Science Source

groups of cells that detach from the thallus and can start a new plant. Sexual reproduction depends on disk-headed stalks that bear antheridia, as well as on umbrella-headed stalks that bear archegonia. Following fertilization, tiny sporophytes, composed of a foot, a short stalk, and a capsule, begin growing within archegonia. Wind-blown spores are produced within the capsule.

Hornworts

The **hornwort** gametophyte usually grows as a thin rosette or ribbon-like thallus between 1 and 5 cm in diameter. Although some species of hornworts live on trees, most live in moist, well-shaded areas. They photosynthesize, but also have a symbiotic relationship with cyanobacteria, which, unlike plants, can fix nitrogen from the air.

The small sporophytes of a hornwort resemble tiny green horns rising from a thin gametophyte, usually less than 2 cm in diameter (Fig. 23.8). Like the gametophyte, the sporophyte can photosynthesize, although it has only one chloroplast per cell. Meiosis occurs within the "horn," and spores are released. Alternatively, hornworts can bypass alternation of generations by reproducing asexually through fragmentation.

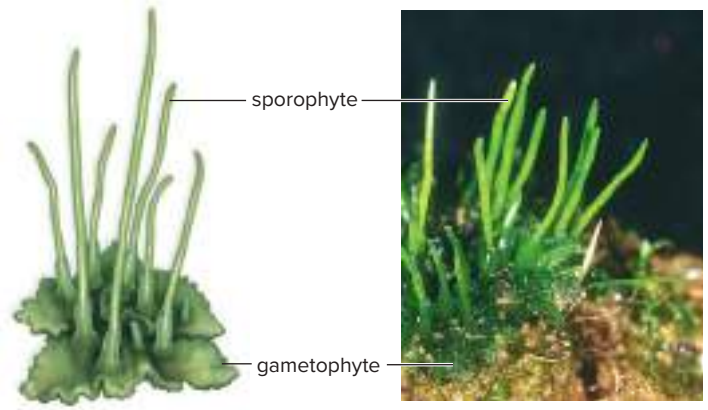


Figure 23.8 Hornwort, *Anthoceros* sp. The "horns" of a hornwort are photosynthetic sporophytes that grow continuously from a base anchored in gametophyte tissue. (photo): Steven P. Lynch

Mosses

Mosses are the largest group of nonvascular plants, with over 12,000 species. There are three distinct groups of mosses: peat mosses, granite mosses, and true mosses. Although most prefer damp, shaded locations in the temperate zone, some survive in deserts, and others inhabit bogs and streams. In forests, they frequently form a mat that covers the ground and rotting logs. In dry environments, they may become shriveled, turn brown, and look completely dead. As soon as it rains, however, the plant becomes green and resumes metabolic activity.

Members of the moss genus *Sphagnum* (peat moss) have great commercial and ecological importance to humans. The cell walls of peat moss have a tremendous ability to absorb water, which is why it is used in gardening to improve the water-holding capacity of the soil, and why it was used historically as wound dressing on battlefields. One percent of Earth's surface is peatlands, where dead *Sphagnum* accumulates but does not decay.

Figure 23.9 describes the life cycle of a typical temperate-zone moss. The gametophyte of mosses begins as an algalike, branching filament of cells, the protonema, which precedes and produces upright, leafy shoots that sprout rhizoids. The shoots bear either antheridia or archegonia. The dependent sporophyte consists of a foot, which is enclosed in female gametophyte tissue; a stalk; and an upper capsule, which contains the sporangium, where spores are produced. A moss sporophyte is always attached to the gametophyte. At first, the sporophyte is green and photosynthetic; at maturity, it is brown and nonphotosynthetic. In some species, the sporangium can produce as many as 50 million spores. The spores disperse the new gametophyte generation.

Check Your Progress 23.2

1. Explain the various methods of bryophyte reproduction.
2. List the characteristics that enabled the bryophytes to successfully colonize land.
3. Explain which portion of the bryophyte life cycle is the most dominant.

ISTUDY

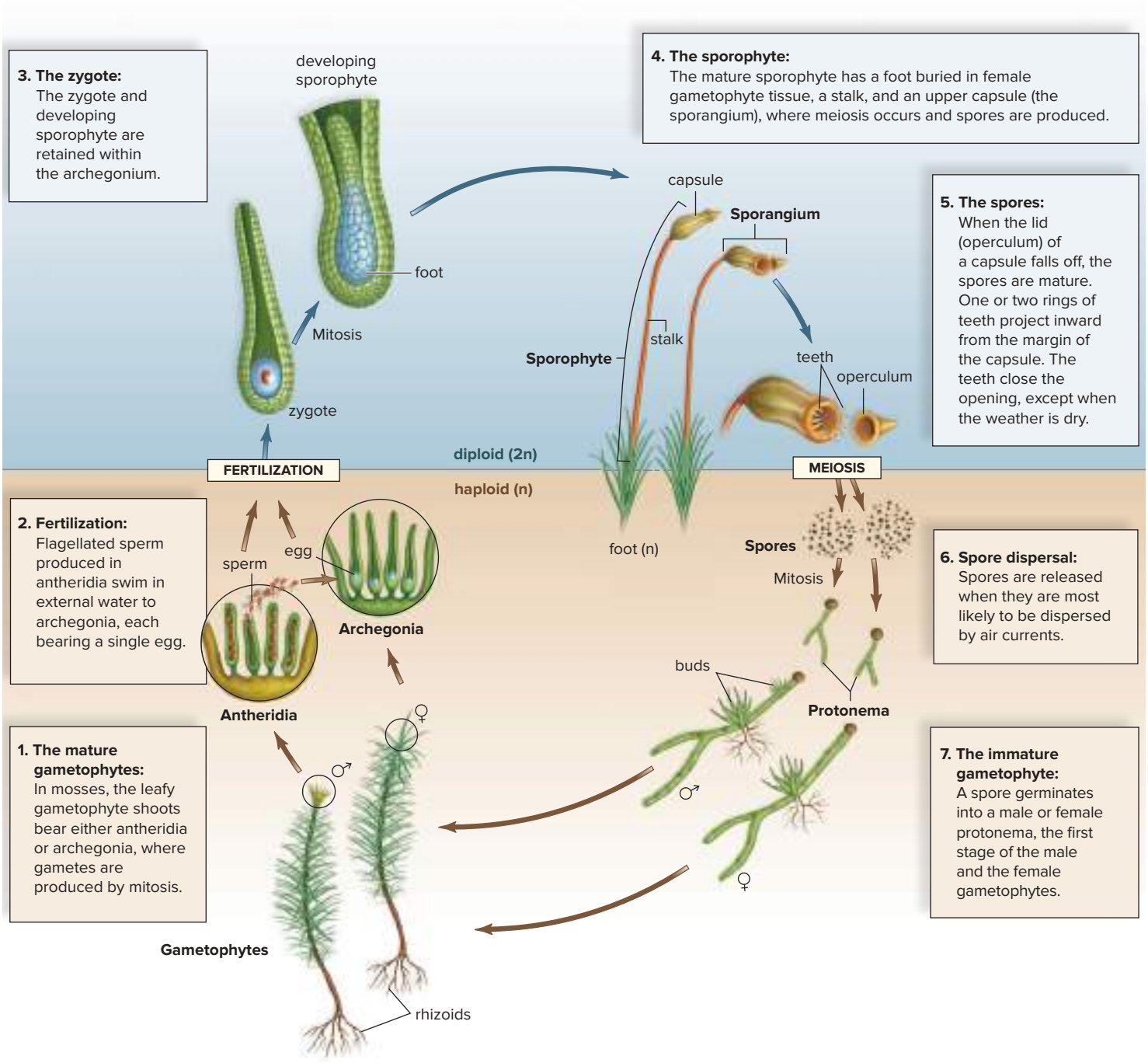


Figure 23.9 Moss life cycle. The majority of a moss life cycle is spent in the haploid (n) state. The sporophyte (2n) is nonphotosynthetic at maturity and produces spores. The leaflike gametophyte (n) is photosynthetic and produces the sperm and egg. The sperm requires water to swim to the egg. This life cycle is from a species of *Polytrichum*.

23.3 Evolution of Lycophytes: Vascular Tissue

Learning Outcomes

Upon completion of this section, you should be able to

1. List the unique structural adaptations found in the lycophytes.
2. Recognize the features present in *Cooksonia* that make it a vascular plant.

Today, **vascular plants,** also known as tracheophytes, dominate the natural landscape in nearly all terrestrial habitats. Vascular plants can achieve great heights because they have roots that absorb water from the soil and a vascular tissue called **xylem** that transports water through the stem to the leaves. Another conducting tissue, called **phloem,** transports nutrients in a plant. Further, the cell walls of the conducting cells in xylem contain **lignin,** a material that strengthens plant cell walls. Thus, the evolution of xylem was essential to the evolution of upright and taller plants.

ISTUDY

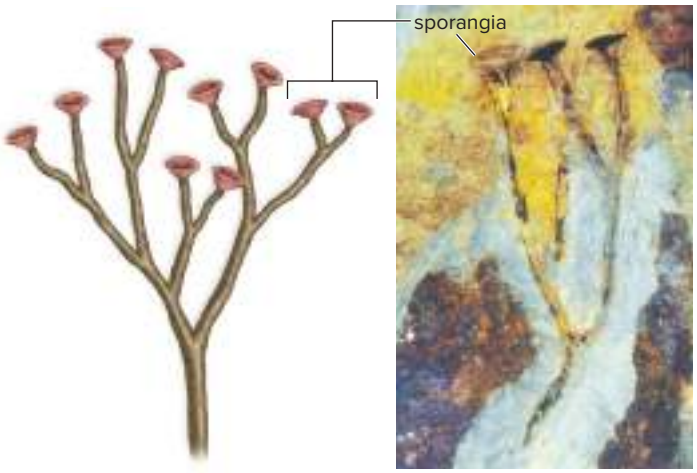


Figure 23.10 A *Cooksonia* fossil. The upright branches of a *Cooksonia* fossil, no more than a few centimeters tall, terminated in sporangia. (photo): Hans Steur, the Netherlands

The Origin of Vascular Plants

The fossil record indicates that the first vascular plants, such as *Cooksonia,* were more likely bushes than trees. *Cooksonia* is a rhyniophyte, a group of vascular plants that flourished during the Silurian period (about 416 MYA, see Table 18.1) but are now extinct. The rhyniophytes were only about 6.5 cm tall and had no roots or leaves. They consisted simply of a stem that forked evenly to produce branches ending in sporangia (Fig. 23.10).

The branching of *Cooksonia* was significant for two reasons. First, instead of the single sporangium in a bryophyte, this plant produced many sporangia, and therefore many more spores. Second, branching is characteristic of plants that have vascular tissue.

The sporangia of *Cooksonia* produced windblown spores, which classifies it as a **seedless vascular plant,** like the rest of the lycophytes and the pteridophytes, discussed in the next section. In addition, the lycophytes and all other vascular plants have a dominant sporophyte generation.

Lycophytes

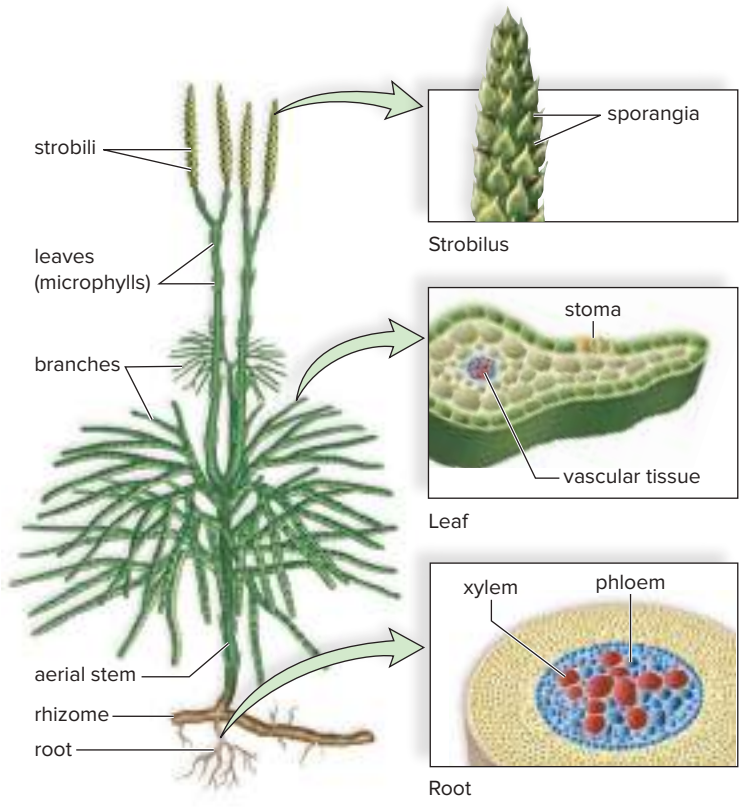
Like the stem of early vascular plants, the first **lycophytes** also had leaves and roots. The leaves are called **microphylls,** because they had only one strand of vascular tissue. Microphylls most likely evolved as simple side extensions of the stem (see Fig. 23.12*a*). Roots evolved simply as lower extensions of the stem; the organization of vascular tissue in the roots of lycophytes today is much as it was in the stems of fossil vascular plants, where the vascular tissue is centrally placed.

Today's lycophytes include around 1,300 species in three main groups: the ground pines or club mosses (*Lycopodium*), spike mosses (*Selaginella,* Fig. 23.11*a*), and quillworts (*Isoetes*). Many lycophytes have "moss" in their name because they look like a moss, yet unlike a moss, these plants have vascular tissue.

Figure 23.11*b* shows the structure of *Lycopodium;* beginning at the top, notice the cone-shaped structure called the **strobili** (*sing.,* strobilus [Gr. *strobilos,* "pinecone"]). The strobili resemble a club or heavy stick, accounting for the common name, club mosses. The sporangia are borne on the strobili, where meiosis takes place and spores are produced. Next, the aerial stem and microphylls contain vascular tissue, as does the underground stem, called the **rhizome.** The roots develop and branch from the rhizome.

Thus far, sporangia have been described as spore-producing structures. Some lycophytes and other vascular plants produce one type of spore that grows into one type of gametophyte. These plant types are called **homosporous** (see Fig. 23.16). Other vascular plants produce two types of spores. They are called **heterosporous.**

a.



b.

Figure 23.11 Lycophytes. **a.** *Selaginella, a* spike moss, is a common landscape or houseplant. **b.** *Lycopodium,* a club moss, has a sporophyte that develops an underground rhizome system. A rhizome is an underground stem. This rhizome produces true roots along its length. (a): Steven P. Lynch

ISTUDY

The two types of spores are microspores and megaspores. **Microspores** grow into a male gametophyte, and **megaspores** grow into a female gametophyte (see Fig. 23.18).

Check Your Progress 23.3

1. Name two features of lycophytes that increase their ability to survive on land.
2. Explain how xylem contributes to an upright body plant.
3. Define the terms *homosporous* and *heterosporous*.

23.4 Evolution of Pteridophytes: Megaphylls

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify three types of pteridophytes.
2. Compare and contrast microphylls and megaphylls.
3. Identify the components of the ferns' life cycle.

Pteridophytes is a broad term used to describe a group of seedless vascular plants that includes ferns and their allies, the horsetails and whisk ferns. Both pteridophytes and the seed plants have megaphylls. **Megaphylls** are broad leaves with several strands of vascular tissue. Figure 23.12*a* shows the difference between microphylls and megaphylls, and Figure 23.12*b* shows how megaphylls could have evolved.

Megaphylls, which evolved independently from the evolution of microphylls, allow plants to efficiently collect solar energy, leading to the production of more food and the possibility of producing more offspring than plants without megaphylls. Therefore, the evolution of megaphylls made plants more fit. Recall that fitness, in an evolutionary sense, is judged by the number of living offspring an organism produces relative to others of its own kind.

The pteridophytes, like the lycophytes, were dominant from the late Devonian period through the Carboniferous period. Today, the lycophytes are quite small, but some of the extinct relatives of today's club mosses were 35 m tall and dominated the Carboniferous swamps. The horsetails (around 18 m tall) and ancient tree ferns (8 m or more in height) also contributed significantly to the great swamp forests of the time (see the following Evolution feature, "Carboniferous Forests").

Horsetails

Today, **horsetails** consist of one genus, *Equisetum,* and approximately 15 species of distinct seedless vascular plants. Most horsetails inhabit wet, marshy environments around the globe. About 299 MYA, horsetails were dominant plants and grew as large as

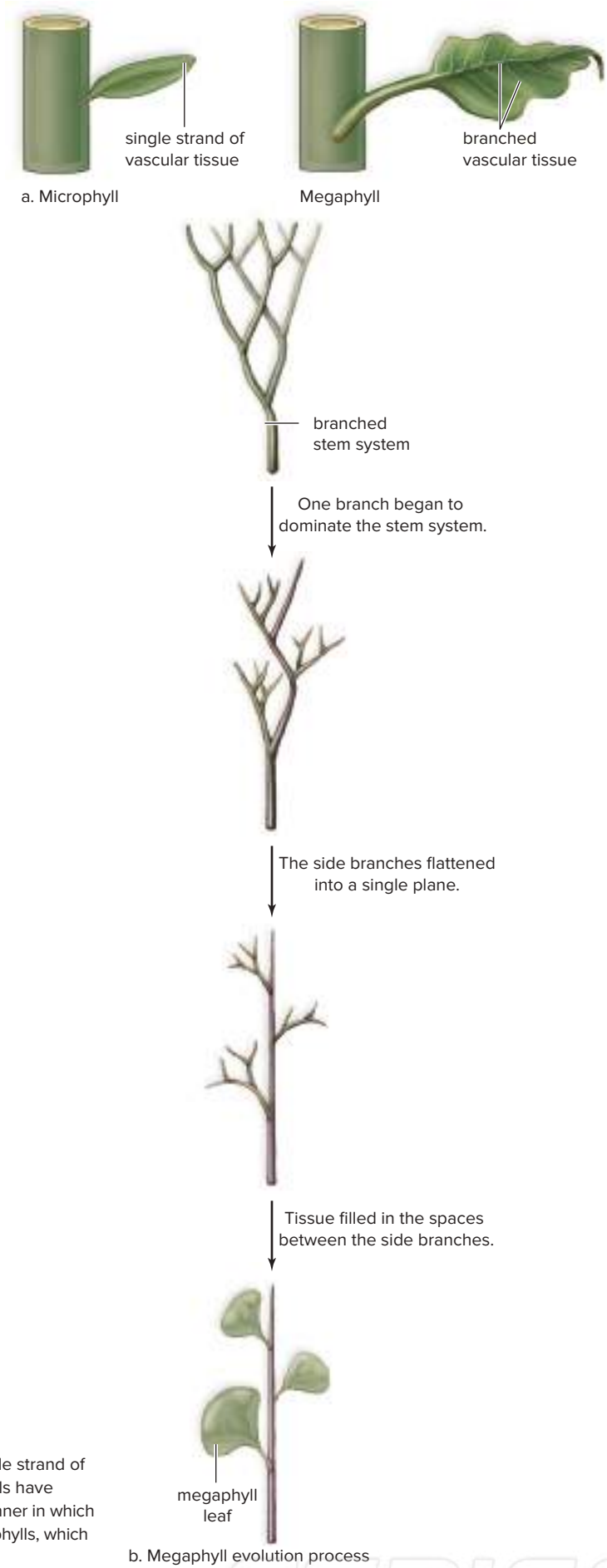


Figure 23.12 Microphylls and megaphylls. **a.** Microphylls have a single strand of vascular tissue, which explains why they are quite narrow. In contrast, megaphylls have several branches of vascular tissue and are broader. **b.** These steps show the manner in which megaphylls may have evolved. All vascular plants, except lycophytes, bear megaphylls, which can gather more sunlight and produce more organic food than microphylls.

ISTUDY

THEME Evolution

Carboniferous Forests

Our industrial society runs on fossil fuels, such as coal. The term *fossil fuel* might seem odd at first until one realizes that it refers to the remains of organic material from ancient times. During the Carboniferous period, around 299 million years ago, a great swamp forest (Fig. 23A) encompassed what is now northern Europe, the Ukraine, and the Appalachian Mountains in the United States. The weather was warm and humid, and the plants grew very tall. These were not plants that would be familiar to us. Instead, they were related to various groups of plants that are less well-known—the lycophytes, horsetails, and ferns.

Lycophytes today may stand as high as 30 cm, but their ancient relatives were 35 m tall and 1 m wide. The strobili were up to 30 cm long, and some had leaves more than 1 m long. Horsetails, too—at 18 m tall—were giants compared to today's specimens. Tree ferns were also taller than the tree ferns found in the tropics today. The progymnosperms, including "seed ferns," so named because their leaves looked like fronds, were significant plants of a Carboniferous swamp.

The amount of biomass in a Carboniferous swamp forest was enormous, and occasionally the swampy water rose, covering the plants that had died. Dead plant material under water does not decompose well. The partially decayed remains became covered by sediment, which changed them into sedimentary rock. Exposed to significant amounts of pressure and over long periods of time, the organic material became coal. This process continued for millions of years, resulting in immense deposits of coal. Geological upheavals raised the deposits to the levels where they can be mined today.

With a change of climate, many of the plants of the Carboniferous period became extinct. Some of their smaller herbaceous relatives have evolved and survived to our time. We owe the industrialization of today's society to these ancient forests.

Questions to Consider

1. How might a geologist use this information to look for new sources of coal?
2. Why is coal not considered to be a renewable resource?

a.

b.

Figure 23A Swamp forest of the Carboniferous period. **a.** Nonvascular plants, early vascular plants, and early gymnosperms (often called "progymnosperms") dominated the swamp forests of the Carboniferous period. **b.** Micrograph of an early gymnosperm called a seed fern. (a): Richard Bizley/Science Source; (b): Sinclair Stammers/Science Source

modern trees. Today, horsetails have a rhizome that produces hollow, ribbed aerial stems and reach a height of 1.3 m (Fig. 23.13).

Equisetum can be branched or unbranched, with leaves that may have been megaphylls at one time but now are reduced and form whorls at the nodes. The skirtlike, slender, green side branches make the plant bear a resemblance to a horse's tail. Many horsetails have strobili at the tips of all stems; others send up buff-colored stems that bear the strobili. The spores germinate into inconspicuous and independent gametophytes. *Equisetum* spores are sensitive to humidity and are "spring-loaded." When conditions are right, the spores have been known to be ejected and travel upward of 2 miles in an hour.

The stems are tough and rigid because of silica deposited in cell walls. Early Americans, in particular, used horsetails for scouring pots and called them "scouring rushes." Today, they are still used as ingredients in a few abrasive powders.

Whisk Ferns

Whisk ferns are represented by the genera *Psilotum* and *Tmesipteris,* which are native to tropical and subtropical regions. The two *Psilotum* species resemble a whisk broom (Fig. 23.14), because they have no leaves. A horizontal rhizome gives rise to aboveground stems that repeatedly fork. The pumpkin-shaped sporangia are borne

ISTUDY

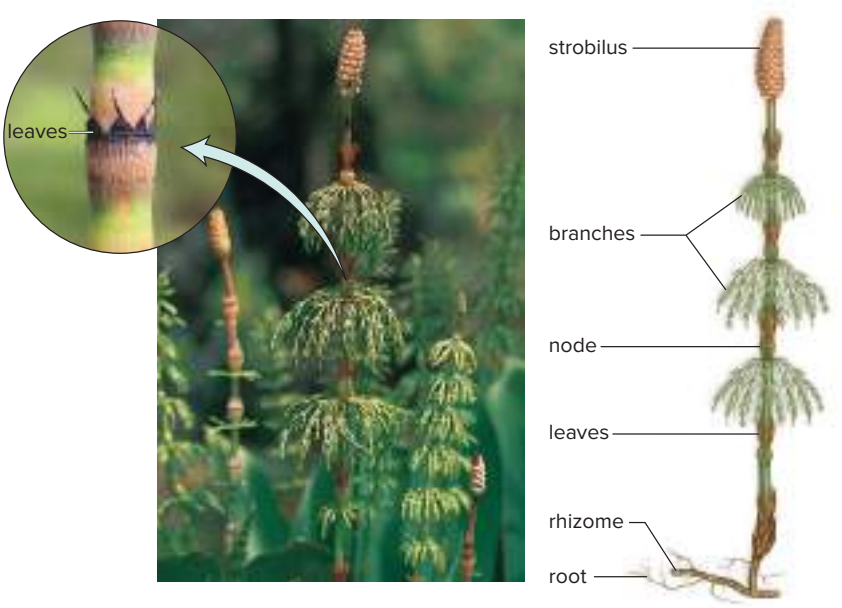


Figure 23.13 Horsetail, *Equisetum*. Whorls of branches and tiny leaves are at the nodes of the stem. Spore-producing sporangia are borne in strobili. (leaves): blickwinkel/Alamy Stock Photo; (plant): Robert Carr/Avalon

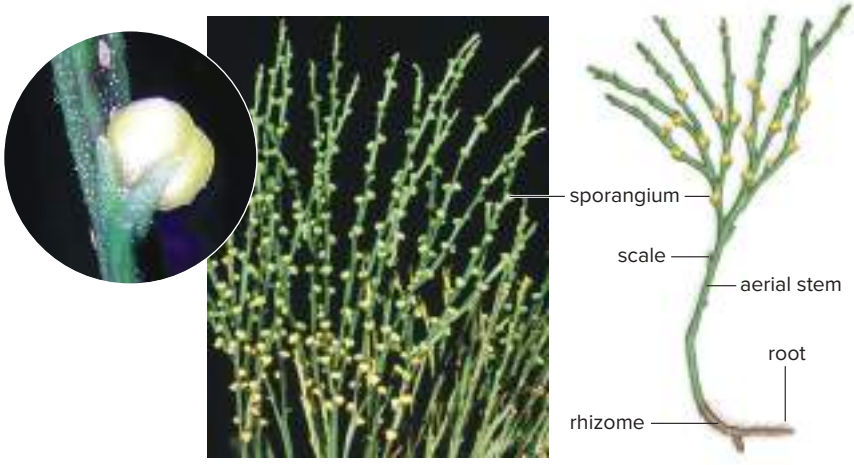


Figure 23.14 Whisk fern, *Psilotum*. *Psilotum* has no leaves—the branches carry on photosynthesis. The sporangia are yellow. (sporangium): Steven P. Lynch; (plant): Biophoto Associates/Science Source

on short side branches. The two or three species of *Tmesipteris* have appendages that some maintain are reduced megaphylls.

Ferns

Ferns are a widespread group of plants (~11,000 species) that are well-known for their attractiveness. Ferns are most abundant in warm, moist, tropical regions, but they can also be found in temperate regions and as far north as the Arctic Circle. Several species live in dry, rocky places, whereas others have adapted to an aquatic life. Ferns range in size from tiny aquatic species less than 1 cm in diameter to modern giant tropical tree ferns that exceed 20 m in height. Recall that unlike lycophytes, ferns have megaphylls, which are more commonly referred to as **fronds.** The leatherleaf fern (used in flower arrangements) has fronds that are broad, with subdivided leaflets; those of a tree fern can be about 1.4 meters long; and those of the hart's tongue fern are straplike and leathery (Fig. 23.15). Sporangia are often located in clusters, called *sori* (*sing., sorus*), on the undersides of the fronds, where they may be shielded by thin, protective structures called *indusia* (*sing., indusium*).

The life cycle of a typical temperate-zone fern, shown in Figure 23.16, applies in general to the other types of vascular seedless plants. The dominant sporophyte produces windblown spores by meiosis within sporangia. The windblown spores disperse, land, and germinate into the gametophyte, the generation that lacks vascular tissue. The separate, heart-shaped gametophyte produces flagellated sperm, which swim in a film of water from the antheridium to the egg within the archegonium, where fertilization occurs. Eventually, the gametophyte disappears and the sporophyte is independent. Both generations of a fern are considered to be independent of one another. Once established, some ferns, such as the bracken fern, can spread into drier areas, because their rhizomes, which grow horizontally in the soil, produce new plants.

Ferns have much economic value and are frequently used by florists in decorative bouquets and as ornamental plants in the home and garden. Wood from tropical tree ferns is often used as a building material, because it resists decay, particularly by termites. Ferns, especially the ostrich fern, are used as food—in the northeastern United States, many restaurants feature fiddleheads (that season's first

Figure 23.15 Diversity of ferns. Structural adaptations found in a variety of ferns. (leatherleaf fern): Gregory Preest/Alamy Stock Photo; (tree fern): ATOM WANG/Shutterstock; (hart's tongue fern): Nic Miller/Organics image library/Alamy Stock Photo

ISTUDY

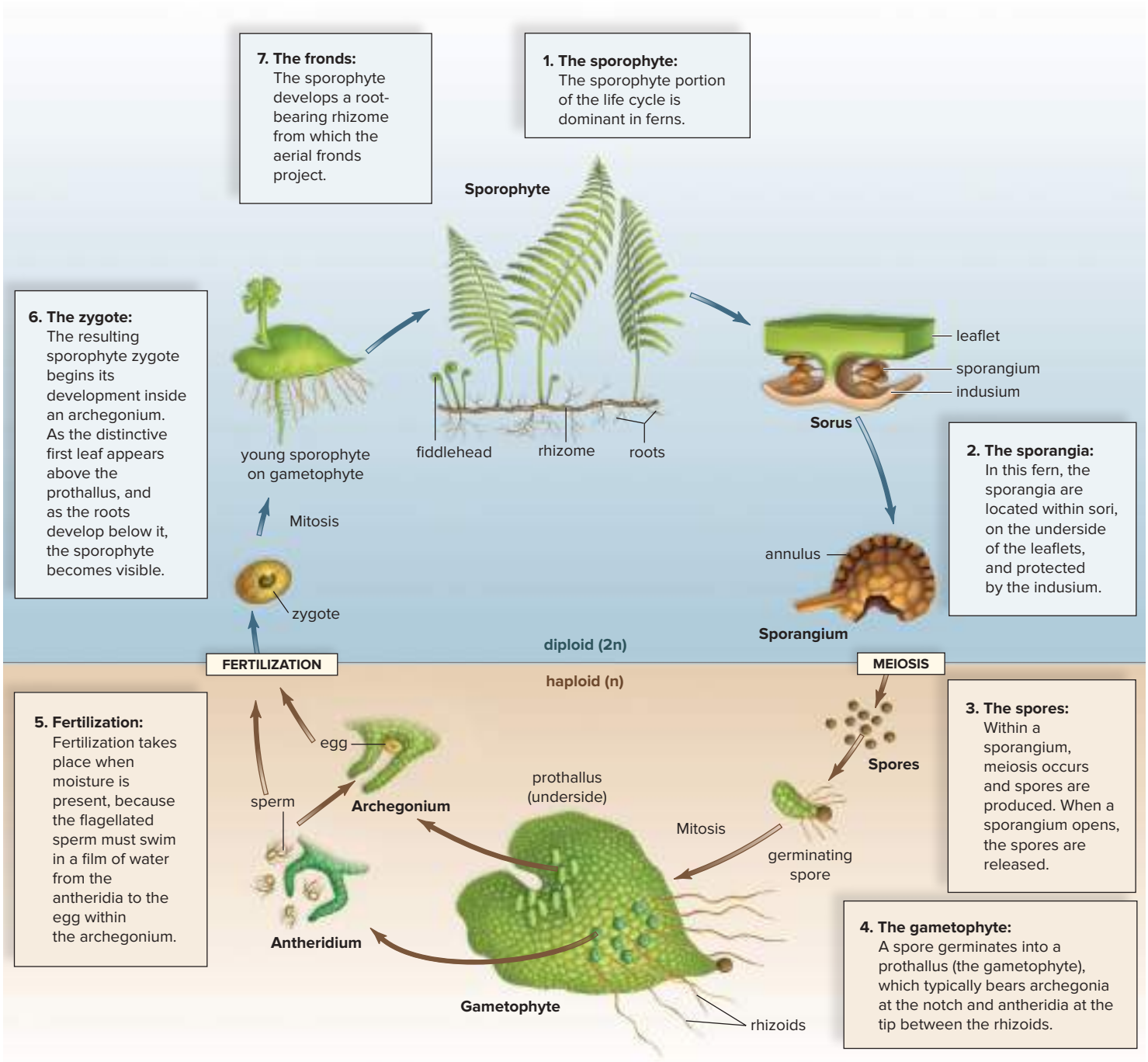


Figure 23.16 Fern life cycle. Stages of a typical fern life cycle.

growth) as a special treat. Ferns also have medicinal value; many Native Americans use them as an astringent during childbirth to stop bleeding, and the maidenhair fern is the source of a cold medicine.

Check Your Progress 23.4

1. Compare the life cycle of a fern to that of a moss.
2. Describe the structures found on a fern megaphyll.
3. Explain the sequence of events in the haploid portion of the fern life cycle.

23.5 Evolution of Seed Plants: Full Adaptation to Land

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast the differences between seed and seedless plants.
2. Describe different groups of gymnosperms.
3. Identify the key components of the gymnosperm and angiosperm life cycles.

ISTUDY

Seed plants are vascular plants that use seeds during the dispersal stage of their life cycle. **Seeds** contain a sporophyte embryo and stored food within a protective seed coat. The seed coat and stored food allow an embryo to survive harsh conditions during long periods of dormancy (arrested state) until environmental conditions become favorable for growth. When a seed germinates, the stored food is a source of nutrients for the growing seedling. The survival of seeds largely accounts for the dominance of seed plants today.

Like a few of the seedless vascular plants, seed plants are heterosporous (have microspores and megaspores), but their innovation was to retain the spores, and not release them into the environment (see Fig. 23.18). As mentioned earlier in this chapter, the microspores become male gametophytes; these are called **pollen grains. Pollination** occurs when a pollen grain is brought into contact with the female gametophyte by wind or a pollinator. Then, sperm move toward the female gametophyte through a growing **pollen tube.** A megaspore develops into a female gametophyte within an **ovule,** which becomes a seed following fertilization.

Note that because the whole male gametophyte (a pollen grain) moves to the female gametophyte, rather than just the sperm (as in seedless plants), no external water is needed to accomplish fertilization. In essence, the less a plant relies on water for reproduction, the farther that plant can radiate onto drier land, take advantage of new resources, and become more abundant.

The two groups of seed plants alive today are gymnosperms and angiosperms. In **gymnosperms** (mostly cone-bearing seed plants), the ovules are not completely enclosed by sporophyte tissue at the time of pollination. In **angiosperms** (flowering plants), the ovules are completely enclosed within diploid sporophyte tissue (ovary), which becomes a fruit.

The first type of seed plant was a woody plant that appeared during the Devonian period (about 359.2 MYA, see Table 18.1); it has been erroneously named a seed fern. The seed ferns of the Devonian were not ferns at all; they were progymnosperms. It's possible these were the type of progymnosperm that gave rise to today's gymnosperms and angiosperms. All gymnosperms are still woody plants, but whereas the first angiosperms were woody, many today are nonwoody. Progymnosperms, including seed ferns, were part of the Carboniferous swamp forests (see the Evolution feature, "Carboniferous Forests," in Section 23.4).

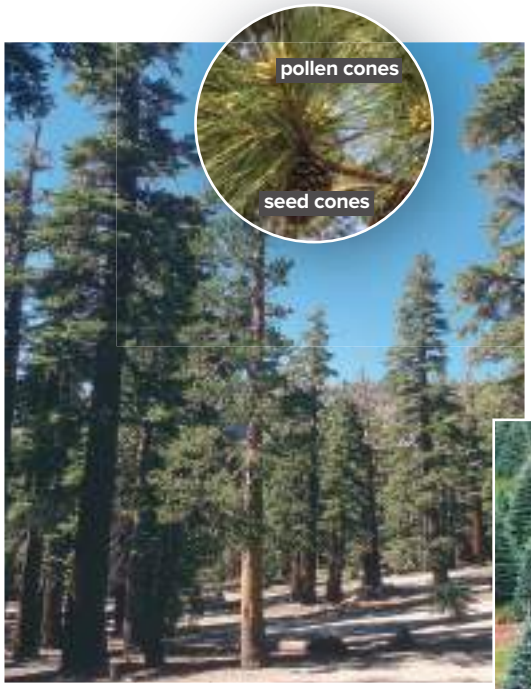
Gymnosperms

The four groups of living gymnosperms (Gk. *gymnos,* "naked"; *sperma,* "seed") are conifers, cycads, ginkgoes, and gnetophytes. Because their seeds are not enclosed by fruit, gymnosperms have "naked seeds." Today, living gymnosperms are classified into more than 1,000 species; the conifers are more plentiful than the other types of gymnosperms.

Conifers

Conifers consist of about 630 species of trees, many evergreen, including pines, spruces, firs, cedars, hemlocks, redwoods, cypresses, yews, and junipers. The name *conifers* signifies plants that bear **cones,** but other gymnosperm phyla are also cone-bearing. The coastal redwood (*Sequoia sempervirens*), a conifer native to northwestern California and southwestern Oregon, is the tallest living vascular plant and may reach nearly 100 m in height. Another conifer, the bristlecone pine (*Pinus longaeva*) of the White Mountains of California, is the oldest living tree; one individual is estimated to be 4,900 years of age.

Vast areas of northern temperate regions are covered in evergreen coniferous forests (Fig. 23.17). The tough, needlelike leaves of pines conserve water, because they have a thick cuticle and



a. Pine

b. Spruce

c. Cypress

Figure 23.17 Conifers. **a.** Pine trees are the most common of the conifers. The pollen cones (male) are smaller than the seed cones (female) and produce pollen. Other conifers include (**b**) the spruces, which make beautiful Christmas trees, and (**c**) the cypress, which can be found in northern temperate regions and is a popular landscape tree. (a): (seed cones): Steven P. Lynch; (pollen cones): Ricochet Creative Productions; (b): Ed Reschke/Stone/Getty Images; (c): Steffen Hauser/botanikfoto/Alamy Stock Photo

ISTUDY

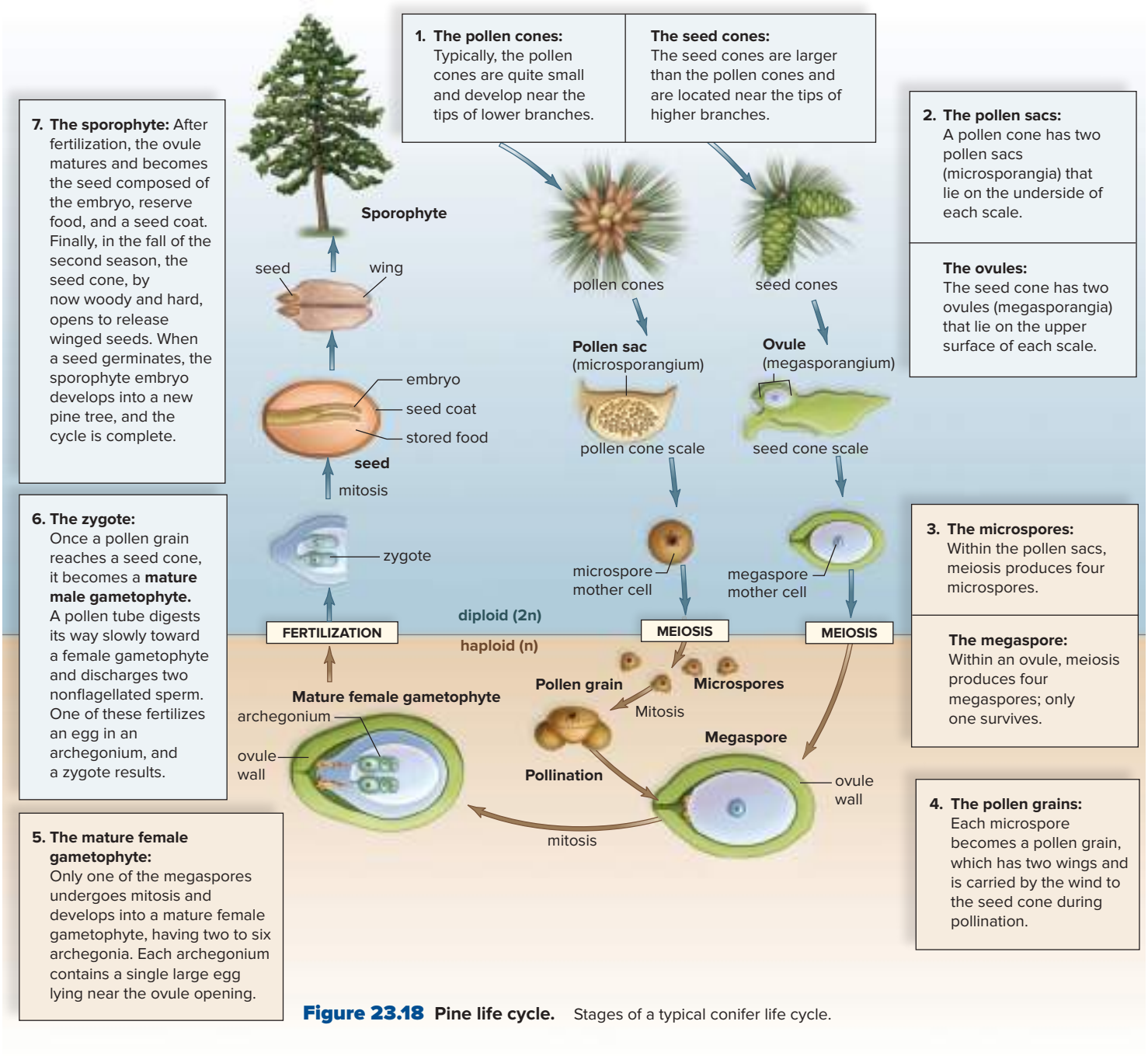


Figure 23.18 Pine life cycle. Stages of a typical conifer life cycle.

recessed stomata. Note that in the life cycle of the pine (Fig. 23.18) the sporophyte is dominant, pollen grains are windblown, and the seed is what is dispersed. Conifers are **monoecious,** which means that a single plant carries both male and female reproductive structures.

Pines, in particular, are well-known for their beauty and pleasant smell; they make attractive additions to parks and gardens. Pines have medicinal value in that the needles and bark are rich in vitamins A and C and can be used to create teas that ease symptoms of a cold or cough. Large pine seeds, called pine nuts, are sometimes harvested for use in cooking, and pine oil is used to scent a number of household and personal care products, such as room sprays and deodorant.

Cycads

Cycads include ten genera and 300 species of distinctive gymnosperms. The cycads are native to tropical and subtropical forests. *Zamia pumila,* found in Florida, is the only species of cycad native to North America. Cycads are commonly used in landscaping. One species, *Cycas revoluta,* referred to as the sago palm, is a common landscaping plant. Their large, finely divided leaves grow in clusters at the top of the stem, and therefore they resemble palms or ferns, depending on their height. The trunk of a cycad is unbranched, even if it reaches a height of 15–18 m, as is possible in some species.

Unlike conifers, cycads have pollen and seed cones on separate plants. The cones, which grow at the top of the stem surrounded by the leaves, can be huge—more than a meter long with a weight of

ISTUDY

a. An African cycad

b. *Ginkgo biloba,* a native of China

Figure 23.19 Three groups of gymnosperms. **a.** Cycads may resemble ferns or palms, but they are cone-producing gymnosperms. **b.** A ginkgo tree has broad leaves and fleshy seeds borne at the end of stalklike megasporophylls. **c.** *Ephedra,* a type of gnetophyte, is a branched shrub. This specimen produces pollen in microsporangia. **d.** *Welwitschia,* another type of gnetophyte, produces two straplike leaves that fray and split. (a): (cycad): Hoberman Collection/Alamy Stock Photo; (cycad cone): PlazaCameraman/iStock/Getty Images; (b): (Ginkgo leaves): blickwinkel/Koenig/Alamy Stock Photo; (ovule): Ken Robertson, University of Illinois/INHS; (c): (plant): Evelyn Jo Johnson/McGraw Hill; (microsporangia): Karl J. Niklas; (d): TravelMuse/Alamy Stock Photo

c. *Ephedra*, a type of gnetophyte

d. *Welwitschia*, a type of gnetophyte

40 kg (Fig. 23.19*a*). Cycads exhibit the life cycle of a gymnosperm, except they are pollinated by insects rather than by wind. Also, the pollen tube bursts in the vicinity of the archegonium, and multiflagellated sperm swim to reach an egg.

Cycads were plentiful in the Mesozoic era at the time of the dinosaurs, and it's likely that dinosaurs fed on cycad seeds. Now, cycads are in danger of extinction, because they grow very slowly.

Ginkgoes

Although **ginkgoes** are plentiful in the fossil record, they are represented today by only one surviving species of tree, the *Ginkgo biloba*. Ginkgoes are **dioecious,** which means that a single plant produces either male or female reproductive structures, but not both (Fig. 23.19*b*). The fleshy seeds, which ripen in the fall, give off such a foul odor that male trees are usually preferred for planting. Ginkgo trees are resistant to pollution and do well along city streets and in city parks. Ginkgo is native to China, and in Asia, ginkgo seeds are considered a delicacy. Extracts from ginkgo trees have been used to improve blood circulation.

Like cycads, the pollen tube of ginkgo bursts to release multiflagellated sperm that swim to the egg produced by the female gametophyte, located within an ovule.

Gnetophytes

Gnetophytes are represented by three living genera and 95 species of plants that are very diverse in appearance. In all gnetophytes, xylem is structured similarly, none have archegonia, and their strobili (cones) have a similar construction. The reproductive structures of some gnetophyte species produce nectar, and insects play a role in the pollination of these species.

Gnetum, which occurs in the tropics, consists of trees or climbing vines with broad, leathery leaves arranged in pairs. *Ephedra,* occurring only in southwestern North America and Southeast Asia, is a shrub with small, scalelike leaves (Fig. 23.19*c*). Ephedrine, a medicine with serious side effects, is extracted from *Ephedra. Welwitschia,* living in the deserts of southwestern Africa, has only two enormous, straplike leaves that fray as it gets older (Fig. 23.19*d*).

Angiosperms

Angiosperms (Gk. *angion,* "vessel"; *sperma,* "seed") are the flowering plants. They are an exceptionally large and successful group, with roughly 370,000 species thought to exist—six times the number of all other plant groups combined. Angiosperms live in all sorts of habitats, from fresh water to desert and from the frigid north to the torrid tropics. They range in size from the tiny, almost microscopic duckweed to *Eucalyptus* trees over 100 m tall.

It would be impossible to exaggerate the importance of angiosperms in our everyday lives. Angiosperms include all the hardwood trees of temperate deciduous forests and all the broadleaved evergreen trees of tropical forests. Also, all herbaceous (nonwoody

ISTUDY

THEME Evolution

The Evolutionary History of Maize

Approximately 12,000 years ago, humans began shifting away from a hunter-gatherer way of life and moved toward an agriculturally based one. As early groups of humans began to spend more time in the same location, and sedentary populations grew, they required larger and more readily available food sources. Over time, these early farmers increasingly selected plants that produced greater yields of food with less of an investment in time and energy.

Maize, which we commonly call corn, was first cultivated in Central America nearly 10,000 years ago in the highlands of Mexico. By the time Europeans were exploring Central America in the 1500s, over 300 varieties were being cultivated (Fig. 23B). All of these varieties can trace their ancestral lineage back to a wild Mexican grass known as teosinte.

Among the earliest structural changes that occurred during corn domestication was the loss of *shattering,* the process by which ears of wild corn break apart and disperse their grains (Fig. 23C). A mutation likely caused the ears of some corn plants to remain intact, a trait that is disadvantageous in nature but beneficial to humans. Nonshattering ears are much easier for humans to harvest.

Figure 23B Varieties of corn. Corn can be cultivated to have a specific color, texture, and sugar content. alonsoleon9/iStock/Getty Images

How did our ancestors artificially select for desirable traits? Hunter-gatherers often followed seasonal migration routes between base camps, returning to the same locations year after year. The disturbance of the natural vegetation at these sites would have provided an ideal site for the colonization of species that were being cultivated by these groups. If next season's seeds were collected from the plants that had the most desirable traits, then over time the frequency of plants with these traits would increase. A modern example, showing how quickly artificial selection can change corn, was conducted in 1896 by agricultural scientists selecting for high oil content in corn kernels. Only the top 20% of oil-producing individuals of each generation were allowed to reproduce, and after 90 generations, the average oil content of the corn kernels had increased 450%.

Harvesting plants with specific traits, however, can decrease the genetic diversity within the population. This decline, in turn, produces a genetic bottleneck, in which various traits are lost, while others increase in frequency, giving rise to a change in the phenotypic frequency within the population (see Section 16.1).

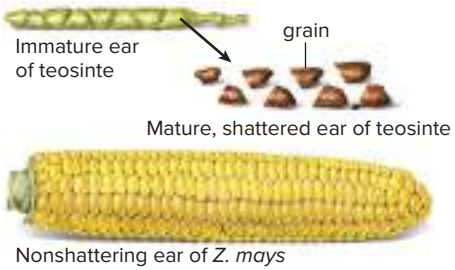


Figure 23C Ears and grain of modern corn and its ancestor, teosinte. Domesticated corn ears are larger than those of the ancestral grass teosinte. In addition, corn fruits are nonshattering, softer, and more edible than are grains of teosinte.

It was the practice of creating corn hybrids in the 1920s that significantly increased yields, as well as the genetic diversity of cultivars. Modern-day corn produces larger grain, generally a more robust plant, an increased amount of apical growth, and fruit with a higher protein and sugar content. In fact, corn has changed so much through artificial selection that domesticated corn can no longer successfully reproduce without human intervention.

Currently, the world's growing population, the loss of farmland, and climate change are straining global food production. The production of corn must now rely on molecular breeding. One method being used to create new strains employs CRISPR/Cas technology that is unique to corn, called the "CRISPR pollen method." This method uses CRISPR-edited genetically modified pollen to carry new versions of a gene to another corn plant's cells. For corn, this method is more effective and versatile than traditional methods of gene modification.

Information gained from genomic studies can also be used to customize corn varieties that increase yields under the growing environmental pressures of drought, pests, and microbial pathogens. Genetic engineering has opened up new windows of evolutionary options for the future of maize and, with that, the future of the humans who rely on this plant.

Questions to Consider

1. What potentially beneficial traits may have been lost during artificial selection in maize?
2. What other crop plants might have undergone a similar form of natural selection?

plants, such as grasses) and most garden plants are flowering plants. This means that all fruits, vegetables, nuts, herbs, and grains, which are the staples of the human diet, are angiosperms. They provide us with clothing, food, medicines, and other commercially valuable products. Over the past 12,000 years, humans have also artificially selected plants to serve us better, resulting in the cultivation of plants as food crops and for other uses. The Evolution feature, "The Evolutionary History of Maize," describes corn (*Zea mays*) as an example.

The flowering plants are called angiosperms because their ovules, unlike those of gymnosperms, are always enclosed within diploid tissues. In the Greek derivation of their name, *angio* ("vessel") refers to the ovary, which develops into a fruit, a unique angiosperm feature.

Origin and Radiation of Angiosperms

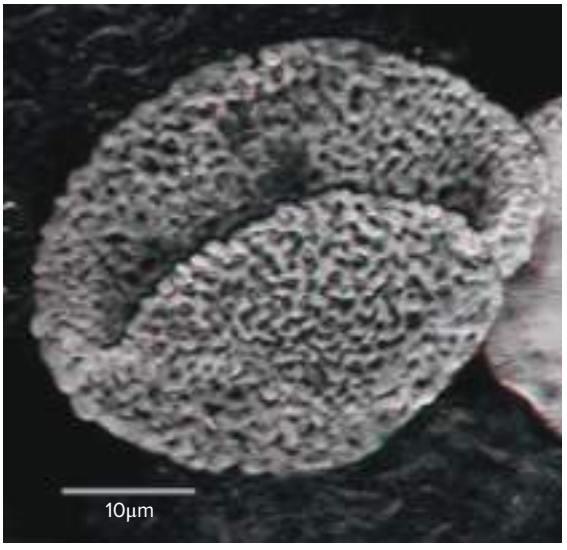
In general, the fossil record of plants is incomplete, because unlike bones and teeth, the soft parts of plants are often eaten or decayed before they can be preserved. This has created a vigorous debate among paleobotanists on the origins of flowering plants. One of the earliest known angiosperm fossils is

ISTUDY

Nanjinganthus dendrostyla, dating back 174 million years to the early Jurassic. *N. dendrostyla* is a stunning fossil with seeds and flowers, giving a glimpse into what some of the early flowering plants may have looked like (see Fig. 19.8). Unlike the soft parts of plants, pollen does preserve well, has characteristic markings, and can be used to date the appearance of a seed plant in an area. Paleobotanists working with drilling cores from Switzerland discovered 240-million-year-old fossilized pollen from the Triassic, pushing the origin of flowering plants back even further (Fig. 23.20*a*).

To find the angiosperm of today that might be most closely related to the first angiosperms, botanists have turned to DNA comparisons. Gene-sequencing data singled out *Amborella trichopoda* (Fig. 23.20*b*) as having ancestral traits. This small, woody shrub, with small, cream-colored flowers, lives only on the island of New Caledonia in the South Pacific. Its flowers are about 4–8 mm wide, and the petals and sepals look the same; therefore, they are called tepals. Plants bear either male or female flowers, with a variable number of stamens or carpels.

Although *A. trichopoda* may not be the original angiosperm species, it is sufficiently close that much may be learned from studying its reproductive biology. Botanists hope that this knowledge will help them understand the early adaptive radiation of angiosperms during the Tertiary period. The gymnosperms were abundant during the Mesozoic era but declined during the mass extinction that occurred at the end of the Cretaceous period (145.5–65.5 MYA). Angiosperms survived and went on to become the dominant plants during modern times.



a. Fossil pollen

b. *Amborella trichopoda*

Figure 23.20 Evolution of flowering plants. **a.** Flowering plant pollen from the Triassic. **b.** Analysis of molecular data suggests that the plant seen here, *Amborella trichopoda,* is the most closely related to the first flowering plants. (a): Susanne Feist-Burkhardt; (b): Doukdouk/Alamy Stock Photo

Monocots and Eudicots

Most flowering plants belong to one of two classes. These classes are the *Monocotyledones,* often shortened to simply the **monocots,** and the *Eudicotyledones,* shortened to **eudicots.** The term *eudicot* (meaning "true dicot") is more specific than the term *dicot.* It was discovered that some of the plants formerly classified as dicots diverged before the evolutionary split that gave rise to the two major classes of angiosperms. These earlier-evolving plants are not included in the designation eudicots.

Cotyledons (Gk. *kotyledon,* "cuplike cavity") are the "seed leaves" containing the nutrients that nourish the plant embryo. If a seed has one cotyledon, it is a monocot, and if a seed has two cotyledons, it is a eudicot (see Fig. 27.8). Several common monocots are corn, tulips, pineapples, and sugarcane; common eudicots include cacti, strawberries, dandelions, poplars, and beans. Table 23.2 lists several fundamental features of monocots and eudicots.

The Flower

Although **flowers** vary widely in appearance (Fig. 23.21), most have certain structures in common (Fig. 23.22).

- The **sepals,** collectively called the calyx, protect the flower bud before it opens. The sepals may drop off or may be colored like the petals. Usually, however, sepals are green and remain attached to the flower stalk.
- The **petals,** collectively called the corolla, are quite diverse in size, shape, and color. The petals are often used to attract a particular pollinator.
- Next are the **stamens.** Each stamen consists of two parts: first, a slender stalk, called a filament, that holds up a second

Table 23.2 Key Features of Monocots and Eudicots

Monocots	Eudicots
One cotyledon	Two cotyledons
Flower parts in threes or multiples of three	Flower parts in fours or fives or multiples of four or five
Pollen grain with one pore	Pollen grain with three pores
Usually herbaceous	Woody or herbaceous
Usually parallel venation	Usually net venation
Scattered bundles in stem	Vascular bundles in a ring
Fibrous root system	Taproot system

ISTUDY

a. Monocot

b. Eudicot

Figure 23.21 Flower diversity. Regardless of size and shape, flowers, such as this (**a**) monocot and (**b**) eudicot, share certain features.
(a): Ed Reschke/Stone/Getty Images; (b): Steven P. Lynch

structure, a saclike container called the anther. Pollen grains develop from microspores produced within the anther.

- At the very center of a flower is the **carpel,** a vaselike structure with three major regions: the stigma, an enlarged, sticky knob; the style, a slender stalk; and the **ovary,** an enlarged base that encloses one or more ovules. The ovule becomes the seed, and the ovary becomes the fruit. Fruit is often instrumental in the distribution of seeds.

Note that not all flowers have all these parts (Table 23.3). A flower is said to be *complete* if it has all four parts; otherwise, it is *incomplete.*

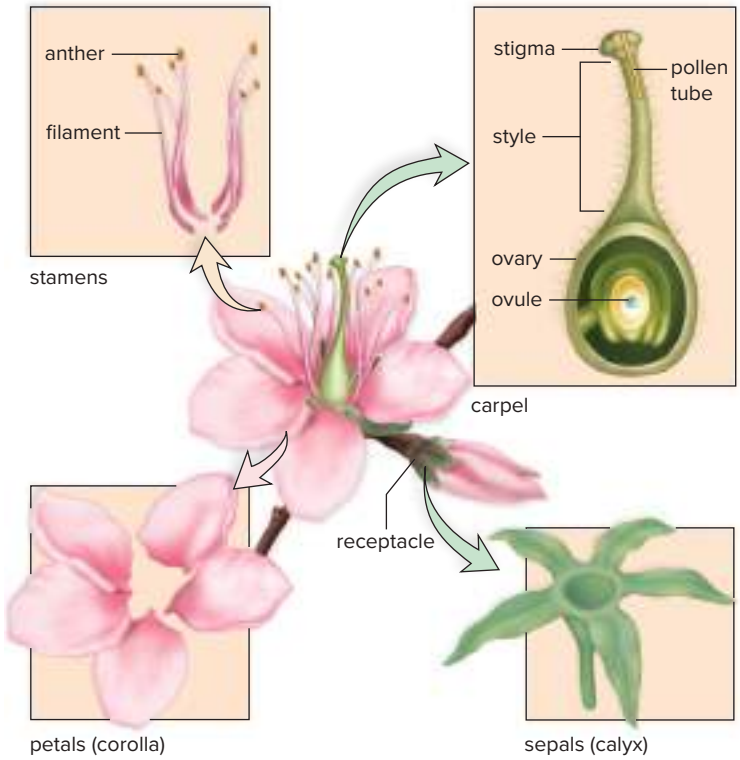


Figure 23.22 Generalized flower. A flower has four main parts: sepals, petals, stamens, and carpels. A stamen has an anther and filament. A carpel has a stigma, style, and ovary. An ovary contains ovules.

Table 23.3 Other Flower Terminology

Term	Description of Flower
Complete	All four parts (sepals, petals, stamens, and carpels) present
Incomplete	Lacks one or more of the four parts
Perfect	Has both stamens and (a) carpel(s)
Imperfect	Has stamens or (a) carpel(s), but not both
Inflorescence	A cluster of flowers
Composite	Appears to be a single flower but consists of a group of tiny flowers

Flowering Plant Life Cycle

Figure 23.23 depicts the life cycle of a typical flowering plant. Like the gymnosperms, flowering plants are heterosporous, producing two types of spores. A megaspore located in an ovule within an ovary of a carpel develops into an egg-bearing female gametophyte, called the embryo sac. In most angiosperms, the embryo sac has seven cells; one of these is an egg, and another contains two polar nuclei. They are called the polar nuclei because they came from opposite ends of the embryo sac.

Microspores, produced within anthers, become pollen grains that, when mature, are male gametophytes with sperm. The mature male gametophyte consists of only three cells: the tube cell and two sperm cells.

During pollination, pollen is transported from the anther to the stigma of a carpel. Here, the tube cell produces a pollen tube that

ISTUDY

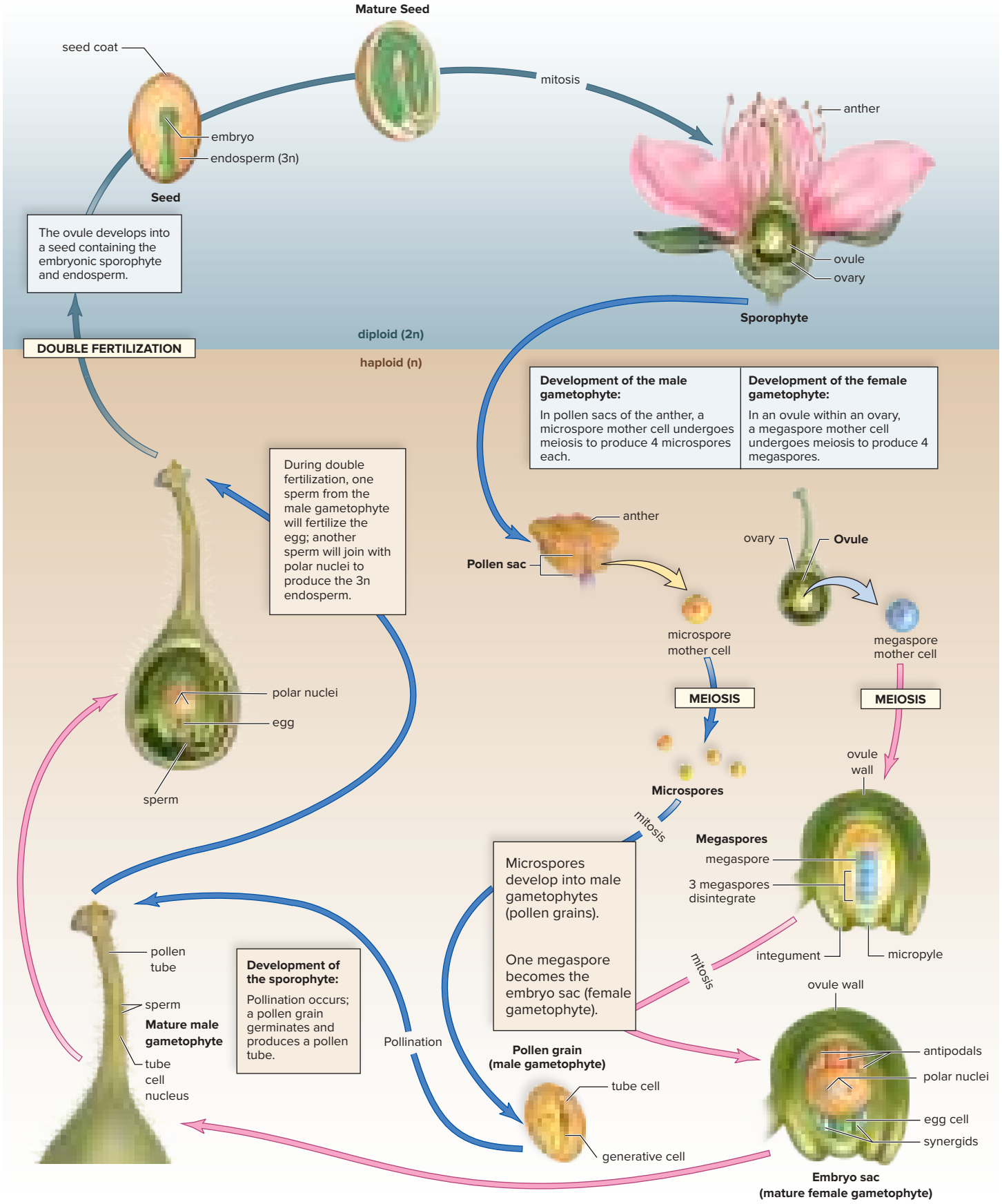


Figure 23.23 Flowering plant life cycle. The parts of the flower involved in reproduction are the stamens and the carpel. Reproduction has been divided into significant stages of female gametophyte development, male gametophyte development, and important stages of sporophyte development.

ISTUDY

carries the two sperm to the micropyle (small opening) of an ovule. Flowering plants undergo **double fertilization:** One sperm unites with an egg, forming a diploid zygote, and the other unites with polar nuclei, forming a triploid endosperm nucleus (see Section 27.1).

Ultimately, the ovule becomes a seed that contains the embryo (the sporophyte of the next generation) and stored food enclosed within a seed coat (see Fig. 23.2*b*). Endosperm in some seeds is absorbed by the cotyledons, whereas, in other seeds, endosperm is digested as the seed matures.

A **fruit** is derived from an ovary, and in some instances it is an accessory part of the flower. Some fruits, such as apples and tomatoes, provide a moist, fleshy covering; other fruits, such as pea pods and acorns, provide a dry covering for seeds.

Flowers and Diversification

Flowers are involved in the production and development of spores, gametophytes, gametes, and embryos enclosed within seeds. Successful completion of sexual reproduction in angiosperms requires the effective dispersal of pollen and then seeds. The various ways pollen and seeds can be dispersed have resulted in many different types of flowers (see the Evolution feature, "Plants and Their Pollinators," in Section 27.1).

Wind-pollinated flowers are usually not showy, whereas many insect- and bird-pollinated flowers are colorful. Night-blooming flowers attract nocturnal mammals or insects; these flowers are usually aromatic and white or cream-colored.

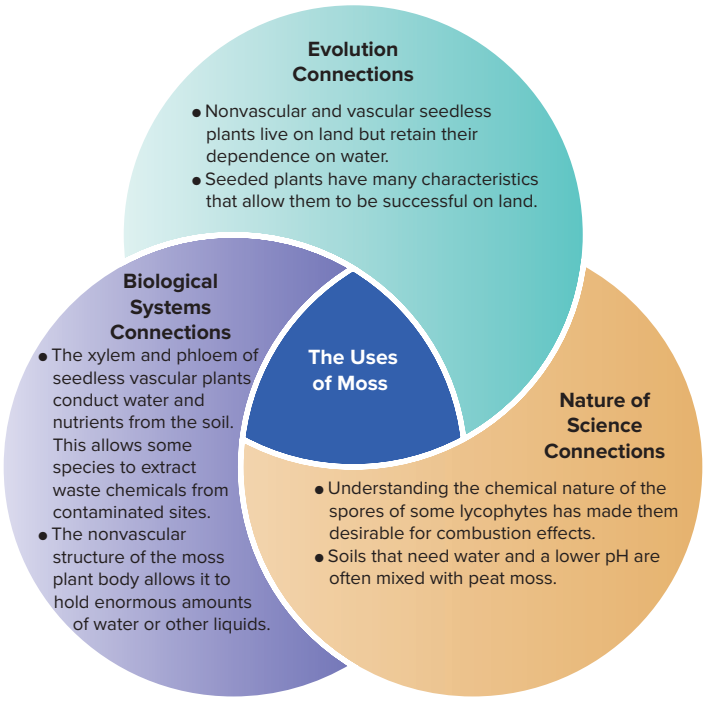
Although some flowers disperse their pollen by wind, many are adapted to attract specific pollinators, such as bees, wasps, flies, butterflies, moths, and even bats, that carry pollen from one flower to another flower of the same type. For example, glands in the region of the ovary produce nectar, a nutrient that is gathered by pollinators as they go from flower to flower. Bee-pollinated flowers are usually blue or yellow and have ultraviolet shadings that lead the pollinator to the location of the nectar. The mouthparts of bees are fused into a long tube, which is able to obtain nectar from the base of the flower. In another example, instead of beautiful colors and tempting nectar, some species of *Ophyrys* orchids use "sexual mimicry" to attract their wasp pollinators. The orchid flowers look like a female wasp and emit pheromones to attract male wasps. The males will engage in "pseudocopulation" with the "female" and end up with pollen attached to its head. Frustrated, the male wasp leaves the flower and attempts to mate with another flower. The male then deposits pollen to the second flower, thereby completing the cross-pollination that the orchid intended.

The fruits of flowers protect and aid in the dispersal of seeds. Dispersal occurs when seeds are transported by wind, gravity, water, and animals to another location. Fleshy fruits may be eaten by animals, which transport the seeds to a new location and then deposit them when they defecate. Because animals live in particular habitats and/or have particular migration patterns, they are apt to deliver the fruit-enclosed seeds to a suitable location for seed germination (when the embryo begins to grow again) and development of the plant.

Check Your Progress 23.5

1. List the life cycle changes that have enabled pines to better adapt to life on land.
2. Compare and contrast the four types of gymnosperms.
3. List the functions of the key structures required for angiosperm reproduction.

Connecting *the* Concepts



ISTUDY

What Did You Learn?

23.1 Ancestry and Features of Land Plants

Land **plants** evolved from a common ancestor with multicellular, freshwater algae about 590 MYA. The **charophytes** *Chara* and *Coleochaete* are green algae related to the ancestors of plants and, like other algae, have characteristics in common with plants, such as photosynthetic pigments, the presence of starch, and cell wall structure.

- Plants evolved modifications for life on land, such as a waxy **cuticle** to combat desiccation and **stomata** for gas exchange.
- Specialized cells called **tracheids** facilitate the movement of water within a plant, and the formation of tissues and organs reduces surface water loss. The **alternation-of-generations** life cycle contains a multicellular diploid stage that can be beneficial if mutations are caused by UV light.
- The **sporophyte** is the diploid portion of the life cycle; here, specialized structures called **sporangia** produce haploid spores. **Spores** divide by mitosis and become the gametophyte.
- The haploid stage is the multicellular **gametophyte,** with male and female gametangia called the **antheridia** and **archegonia.** Sperm and egg are produced, fuse, and form a diploid zygote.

23.2 Evolution of Bryophytes: Colonization of Land

Ancient **bryophytes** were the first plants to colonize land. **Liverworts, hornworts,** and **mosses** are examples of bryophytes, which lack well-developed **vascular tissue** and are therefore called **nonvascular plants.**

- Without true roots, **rhizoids** instead anchor these plants to a substrate. The sporophyte is nutritionally dependent on the gametophyte, because the gametophyte is larger and photosynthetic.
- The life cycle of mosses demonstrates the reproductive strategies of flagellated sperm and dispersal by means of windblown spores.

23.3 Evolution of Lycophytes: Vascular Tissue

Vascular plants, such as rhyniophytes, evolved during the Silurian period.

- The sporophytes contain two types of conducting tissues: **Phloem** is specialized to move organic nutrients, whereas **xylem** is specialized to conduct water and dissolved minerals with the help of **lignin** in xylem cells.
- **Lycophytes** are descended from these first plants. They contain vascular tissue, produce spores, and are called the **seedless vascular plants.**
- Ancient lycophytes were the first plants to have leaves known as **microphylls,** and modern lycophytes have conelike structures called **strobili,** where sporangia form. An underground stem called a **rhizome** has extensions of roots.
- Seedless vascular plants can produce one type of spore, and be **homosporous,** or two types of spores, and be **heterosporous.** The two types of spores made by hetersporous plants are **microspores** and **megaspores.**

23.4 Evolution of Pteridophytes: Megaphylls

In the **pteridophytes** (**ferns** and their allies, **horsetails** and **whisk ferns**) and lycophytes, the sporophyte is the dominant stage of the life cycle and is separate from the tiny gametophyte.

- Windblown spores are the dispersal agents for these plants. The ferns found on Earth today have an obvious **megaphyll;** horsetails and whisk ferns have reduced megaphylls.
- Fern megaphylls are called **fronds,** and there is much diversity in the shape and size of these fronds. Most fronds have sporangia on the underside, found in clusters called **sori.**
- Established ferns can often reproduce asexually by growing new plants from the underground rhizome.

23.5 Evolution of Seed Plants: Full Adaptation to Land

Seed plants also have an alternation-of-generations life cycle, but they are heterosporous, producing both microspores and megaspores.

- Microspores become the windblown or animal-transported male gametophytes—the **pollen grains. Pollination** involves pollen grains carrying and delivering sperm via a **pollen tube** to the female gametophyte (megaspore), which is the **ovule.** Following fertilization, the ovule becomes the **seed,** which contains a sporophyte embryo. Fertilization no longer requires external water, and sexual reproduction is fully adapted to the terrestrial environment.
- The **gymnosperms** (cone-bearing plants) evolved from woody seed ferns during the Devonian period. The **conifers,** represented by the pine tree, exemplify the traits of these plants. Other gymnosperms are the **cycads, ginkgoes,** and **gnetophytes.** Gymnosperms have "naked seeds," because they are not enclosed by fruit, and can be **monoecious** (having both male and female structures on the same plant) or **dioecious** (being either male or female but not both).
- *Nanjinganthus dendrostyla* has been identified as most closely related to the common ancestor for the angiosperms. **Angiosperms** belong to two classes: **Monocotyledones (monocots)** with one **cotyledon,** and **Eudicotyledones (eudicots)** with two cotyledons. The reproductive organs are found in **flowers.** Most flowers have **sepals, petals, stamens,** and **carpels.** After fertilization, the ovules become seeds, which are located in the **ovary.** Seeds are the result of **double fertilization,** with two sperm yielding the embryo, and the endosperm to feed the embryo. This ultimately becomes the **fruit.** Therefore, angiosperms have "covered seeds."
- In many angiosperms, pollen is transported from flower to flower by various pollinators. Both flowers and fruits are found only in angiosperms and may account for the extensive colonization of terrestrial environments by the flowering plants.

Thinking Critically About Biology

1. For each plant listed next, choose all the plant features (a.–j.) that can be used to describe that plant.

Pine tree	a. vascular tissue
Moss	b. no vascular tissue
Apple tree	c. protected embryo
Fern	d. megaphylls
Horsetail	e. gametophyte is the dominant generation
	f. produces seeds
	g. produces fruit
	h. alternation-of-generations life cycle
	i. disperses spores
	j. sporophyte is the dominant generation

ISTUDY

2. You are buying plants at your local garden center and you choose a Boston fern, a ginkgo tree, and a flowering Gerbera daisy. Focusing on evolutionary traits, how are these plants similar? How are these plants different?
3. Compare and contrast meiosis and mitosis in the plant alternation-of-generations life cycle with the human life cycle (see Fig. 10.9).
4. Would you expect the sporophyte generation of a moss or a fern to have more mitotic divisions? Why?
5. What types of environmental stresses are placed on plants living on dry land, and how do plants overcome these stresses?

Making It Relevant to You

1. Flowering plants and other seeded plants are the most common plants found on Earth today. If you could go back in time 400 million years, what types of plants would be the most common?
2. Imagine you and a friend are hiking in a temperate forest and you get lost. You have run out of water but are surrounded by moss. What could you do to stay safely hydrated?
3. In Section 25.1, you will learn that lichens and moss are important in the weathering of rock into soil. What property of moss do you think contributes to the weathering of rock?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

24

Flowering Plants: Structure and Organization

Shutterstock/Candia Baxter

Drought in the Sierra Nevada

The Sierra Nevada in California are home to dozens of conifers and flowering tree species. Trees benefit the planet by cooling the atmosphere and providing niche habitats to many animals. Unfortunately, current rapid climate change is threatening trees because of destructive fires and long droughts.

Historically, fire has been an essential factor in the health and regeneration of forests. Fire releases seeds from cones, recycles nutrients into the soil, and opens spaces for sun exposure and germination. However, in recent years fires have become more severe and longer lasting. This results in severe burns on tree trunks which invites infestation by beetles and other pathogens.

Sierra Nevada trees, such as giant sequoias and black oaks, transport water from the roots to the top of the trees in the outer layers of wood. California's historic decade-long drought is taxing water conduction in woody tissue, while climate change is reducing the amount of snow that would normally melt in spring and summer supplying water to trees.

Forest canopy ecologists monitor the changes in trees by managing networks of weather stations, studying the effects of other organisms in the upper canopies, and researching how the vascular tissues in trees changes in drought, all the while informing the public on the importance of saving these majestic trees.

As you read through this chapter, think about the following questions:

1. How do each of the plant organs contribute to the success of plants?
2. How do modifications of the vegetative organs increase fitness?

Chapter Outline

24.1 Cells and Tissues of Flowering Plants
24.2 Organs of Flowering Plants
24.3 Organization and Diversity of Roots
24.4 Organization and Diversity of Stems
24.5 Organization and Diversity of Leaves

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 4.7 What cellular structures are necessary for plant cells to function?

Section 7.2 What are the reactants, intermediates, and end products of photosynthesis?

Section 23.5 Which structural features helped promote angiosperm success?

Following *the* Themes

CHAPTER 24 – FLOWERING PLANTS: STRUCTURE AND ORGANIZATION

Evolution	A number of structural adaptations were necessary for angiosperm evolution.
Nature of Science	Human life is inextricably linked to plants, and we continue to find a wide variety of uses for plants.
Biological Systems	The structural and physiological systems of angiosperms have enabled them to become the dominant form of plant life on Earth today.

ISTUDY

24.1 Cells and Tissues of Flowering Plants

Learning Outcomes

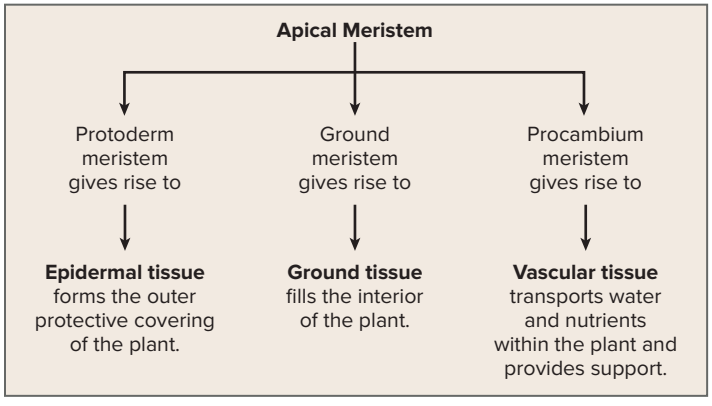
Upon completion of this section, you should be able to

1. Explain how plant cells are different from animal cells.
2. Define *apical meristem* and describe where on a plant it is found.
3. Identify the three types of tissue found in angiosperms.
4. Recognize the differences in the location, structure, and function among various angiosperm tissues.

The body of a plant is organized in a similar fashion to the body of an animal. As in animals, a *cell* is a basic unit of life. A *tissue* is composed of specialized cells that perform a particular function, and an *organ* is a structure made up of multiple tissues. In Section 4.3, you learned about eukaryotic cell features common to both plant and animal cells. Table 24.1 highlights some differences between plant and animal cells.

When a plant embryo begins to develop, the first cells are called **meristem** cells. Like animal stem cells, plant meristem cells are undifferentiated cells that can divide indefinitely and give rise to many types of differentiated cells (Fig. 24.1). As new cells are produced, they are small and boxlike, with a large nucleus and tiny vacuoles. As these cells mature, they assume many different shapes and sizes, each related to the cell's ultimate function.

When a seed germinates and an embryo grows, meristem tissue is present at the tips, or apices, of the young plant and is called **apical meristem.** Apical meristem in turn gives rise to three specialized meristems that create the differing plant tissues (see Fig. 24.14).



Another type of meristem, mainly found in monocots, such as grasses, is the *intercalary meristem,* found at the node (bottom) of the plant instead of the tip. This is why you can cut your lawn, yet the grass continues to grow. In addition, plants not only grow from their apices but also can grow wide. The **vascular cambium** is another type of meristem, which gives rise to new vascular tissue called *secondary growth.* Secondary growth causes a plant to increase in girth.

Table 24.1 Comparison of Plant and Animal Cells

Plant Cell	Animal Cell
Cell wall is present.	No cell wall is present.
Cells connect with plasmodesmata.	Cells connect with various junctions.
Cell division involves a cell plate.	Cell division involves a cleavage furrow.
No centrioles are present during mitosis.	Centrioles are present during mitosis.
Plastids are present.	No plastids are present.
Vacuoles are large.	Vacuoles are small or absent.

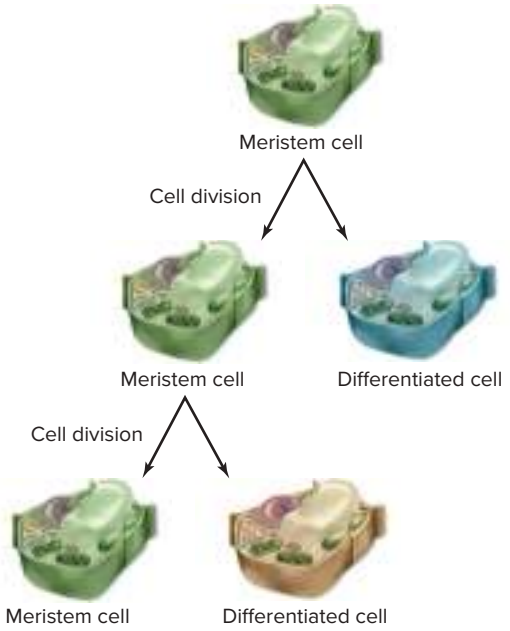


Figure 24.1 Meristem cell division. Meristem cells are located in new and developing parts of a plant. Meristem cells divide to give rise to a differentiating daughter cell and a cell that persists as a meristem cell.

Epidermal Tissue

The entire body of both nonwoody (herbaceous) and young woody plants contains closely packed epidermal cells called the **epidermis** (Gk. *epi,* "over"; *derma,* "skin"). The walls of epidermal cells that are exposed to air are covered with a waxy **cuticle** (L. *cutis,* "skin") to minimize water loss. The cuticle also protects against bacteria and other organisms that might cause disease.

In roots, certain cells of the **epidermal tissue** have long, slender projections called **root hairs** (Fig. 24.2*a*). The hairs increase the surface area of the root for absorption of water and minerals, as well as anchor the plant to various substrates.

On stems, leaves, and reproductive organs, epidermal cells produce hairs, called **trichomes,** that have two important functions: to protect the plant from too much sun and moisture loss and to discourage herbivory (plant eating) (Fig. 24.2*b*). Sometimes trichomes, particularly glandular ones, help protect a plant from herbivores by producing a toxic substance. For example, under the slightest pressure, the stiff trichomes of the stinging nettle lose

ISTUDY

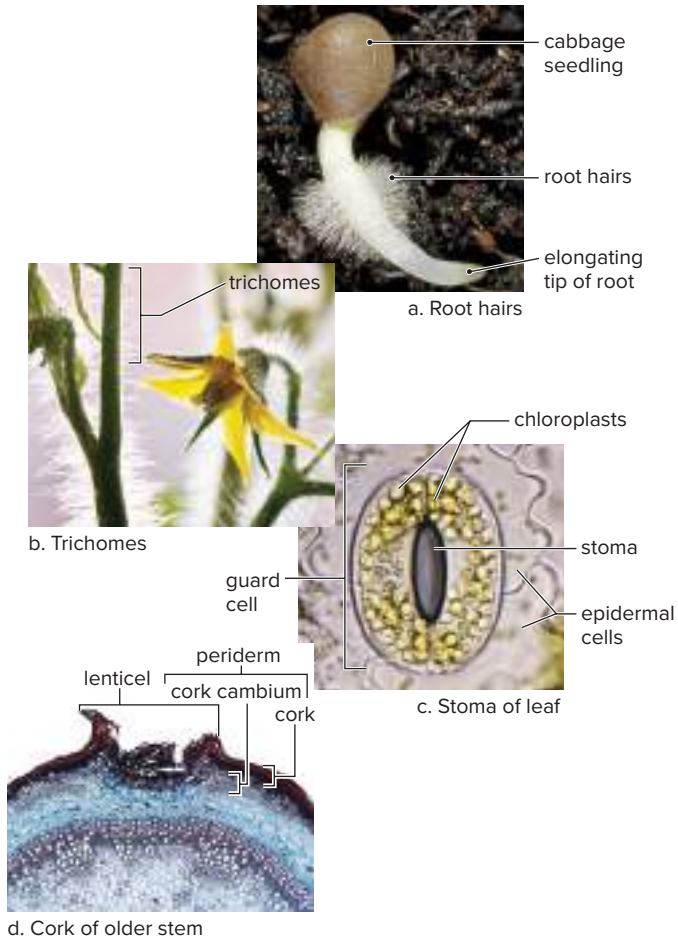


Figure 24.2 Modifications of epidermal tissue. **a.** Root epidermis has root hairs to absorb water. **b.** Trichomes are hairlike epidermal extensions on fruit, leaves, or stems that can protect from herbivory. **c.** Leaf epidermis contains stomata (*sing.,* stoma) for gas exchange. **d.** Periderm includes cork and cork cambium. Lenticels in cork are important in gas exchange. (a): Nigel Cattlin/Alamy Stock Photo; (b): Anthony Pleva/Alamy Stock Photo; (c): Biophoto Associates/Science Source; (d): Kingsley Stern

their tips, forming "hypodermic needles" that inject an intruder with a stinging secretion. See the following Evolution feature, "Survival Mechanisms of Plants," for a discussion of some of the defenses that have evolved in angiosperms.

In leaves, the lower epidermis of eudicots and both surfaces of monocots contain specialized cells called *guard cells* (Fig. 24.2*c*). Guard cells, which are epidermal cells with chloroplasts, surround microscopic pores called **stomata** (*sing.,* stoma). When the stomata are open, gas exchange and water loss occur.

In plants with wood, the epidermis of the stem is replaced by **cork** cells. At maturity, cork cells can be sloughed off (Fig. 24.2*d*). New cork cells are made by a meristem called **cork cambium.** This entire cork area of the plant is called the **periderm** (Gk. *peri,* "around"; *derma,* "skin"). As the new cork cells mature, they increase slightly in volume, and their walls become encrusted with suberin, a lipid material, so they are waterproof and chemically inert. These nonliving cells protect the plant and help it resist fungal, bacterial, and animal attacks. Some cork tissues, notably from the cork oak (*Quercus suber*), are used commercially for bottle corks and other products.

Notice in Figure 24.2*d* how the cork cambium overproduces cork in certain areas of the stem surface, causing ridges and cracks to appear. These features on the surface are called *lenticels.* Lenticels are the site of gas exchange between the interior of a stem and the air.

Ground Tissue

Ground tissue forms the bulk of a flowering plant; it contains parenchyma, collenchyma, and sclerenchyma cells (Fig. 24.3). **Parenchyma** cells are the most abundant and correspond best to the typical plant cell. These are the least specialized of the cell types and are found in all the organs of a plant. They may contain chloroplasts and carry on photosynthesis, or they may contain colorless plastids that store the products of photosynthesis. A juicy bite from an apple yields mostly storage parenchyma cells. Parenchyma cells line the connected air spaces of a water lily and other aquatic plants. Parenchyma cells can divide and give rise to more specialized cells, as when roots develop from stem cuttings placed in water.

Collenchyma cells are like parenchyma cells except they have thicker primary walls. The thickness is uneven and usually involves the corners of the cell. Collenchyma cells often form bundles just beneath the epidermis and give flexible support to immature regions of a plant body. The familiar strands in celery stalks are composed mostly of collenchyma cells.

Figure 24.3 Ground tissue cells. **a.** Parenchyma cells are the least specialized of the plant cells. **b.** Collenchyma cells. Notice how much thicker and irregular the walls are compared to those of parenchyma cells. **c.** Sclerenchyma cells are dead and have very thick walls (stained in red). Their only function is to give strong support. (all): Biophoto Associates/Science Source

ISTUDY

THEME Evolution

Survival Mechanisms of Plants

Plants first made their appearance on land approximately 450 million years ago. Since then, they have evolved a wide variety of mechanisms in order to survive and have established the base of terrestrial ecosystems.

Some groups of plants employ defensive strategies in an attempt to deter predation (Fig. 24A). A defensive strategy is a mechanism that has arisen through a process of natural selection in which the members of a group that possess the strategy compete better than those without it. The more successful competitors usually have a greater opportunity to pass on their genes.

Thorns and spines like those found on black locust trees and cacti are often used to repel large herbivores, but they are generally ineffective against smaller herbivores. If a tree does become injured, the tracheids and vessel elements of xylem immediately plug up with chemicals that block them off above and below the site of the injury. This response prevents the damage from spreading to other locations on the tree.

Other plants produce toxins or sticky secretions in an attempt to deter predation. Anyone who has come in contact with poison ivy or the sap of a pine tree knows the effectiveness of this defense mechanism.

Cellulose is the polysaccharide found in the cell walls of plants. This indigestible substance makes it difficult for many predators to obtain nutrients from eating the leaves, discouraging continued predation.

Dormancy enables plants to survive in environments that have seasonal conditions that do not allow year-round growth. Deciduous trees shed their leaves and transfer their nutrients into their root systems in response to the decrease in light, temperature, and moisture levels that occurs during the fall.

Seed dormancy allows the next generation to wait until growing conditions are optimal before germinating and competing for resources. Many plant life cycles are timed so that the seeds are produced during the summer, sit dormant throughout the winter, and germinate the following spring.

In some plants, the germination of seeds is triggered only after they have undergone some form of physical trauma, or seed scarification. Many species of plants found in chaparral regions, which are hot and dry, germinate only after they have been slightly burned, allowing them to germinate in an environment that has minimal competition for resources.

Evolutionary success is not measured by the survival of a single individual but by the passing of one's genes to the next generation. A number of groups of plants have evolved the ability to reproduce both sexually and asexually. Stolons, rhizomes, and tubers are asexual methods of reproduction. Strawberries, irises, and potatoes are plants that use these methods as well as sexual reproduction.

The wide variety of survival mechanisms has ensured that plants will be present on Earth for a very long time.

Questions to Consider

1. Which plant defense mechanisms would be the most effective against large predators? Small predators?
2. How would the suppression of fires in a chaparral region impact the plant diversity?

a.

b.

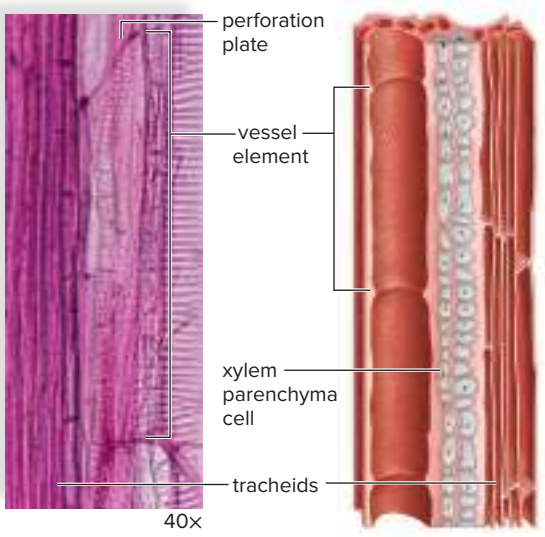
Figure 24A Survival mechanisms used by plants. Plants employ a wide variety of mechanisms to ensure their survival. **a.** Thorns of a locust tree. **b.** Poison ivy contains a toxin within the leaves. (a): Niels Poulsen/Alamy Stock Photo; (b): norcon/iStock/Getty Images

Sclerenchyma cells have thick secondary cell walls impregnated with **lignin,** which is a highly resistant organic substance that makes the walls tough and hard. Most sclerenchyma cells are dead at maturity; their primary function is to support the mature regions of a plant. Two types of sclerenchyma cells are *fibers* and *sclereids.* Although fibers are occasionally found in ground tissue, most are in vascular tissue, which is discussed next. Fibers are long and slender and may be grouped in bundles, which are sometimes commercially important. Hemp fibers can be used to make rope, and flax fibers can be woven into linen. Flax fibers, however, are not lignified, which is why linen is soft. Sclereids, which are shorter than fibers and more varied in shape, are found in seed coats and nutshells. Sclereids, or "stone cells," are responsible for the gritty texture of pears and the hardness of nuts and peach pits.

Vascular Tissue

There are two types of **vascular tissue. Xylem** transports water and minerals from the roots to the leaves, and **phloem** transports sucrose and other organic compounds, usually from the leaves to the roots. Both xylem and phloem are considered complex tissues, because they are composed of two or more kinds of cells. In the roots, the

ISTUDY



a. Xylem photomicrograph (*left*) and drawing (*to side*)

b. Two types of vessels c. Tracheids

Figure 24.4 Xylem structure. **a.** Photomicrograph of xylem vascular tissue and drawing showing general organization of xylem tissue. **b.** Drawing of two types of vessels (composed of vessel elements)—the perforation plates differ. **c.** Drawing of tracheids. (photo): (a): Garry DeLong/ Science Source

vascular tissue is located in the **vascular cylinder;** in the stem, it forms **vascular bundles;** and in the leaves, it is found in **leaf veins.**

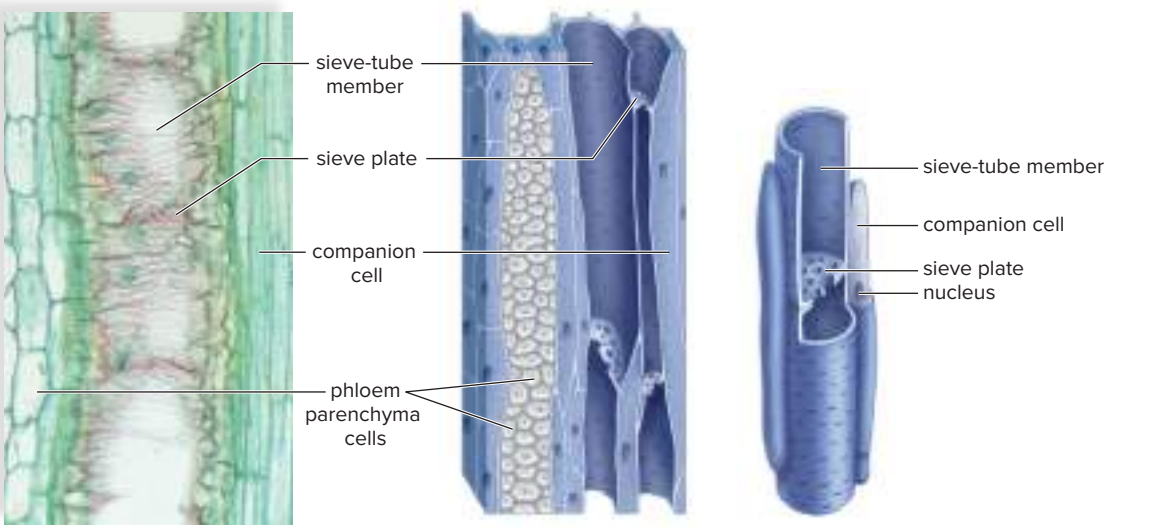
Xylem contains two types of cells that conduct water: tracheids and vessel elements, which are modified sclerenchyma cells (Fig. 24.4). Both types of conducting cells are hollow and dead, but the **vessel elements** are larger, may have perforation plates in their end walls, and are arranged to form a continuous vessel for water and mineral transport.

The elongated **tracheids,** with tapered ends, form a less obvious means of transport, but water can move across the end walls and side walls, because there are *pits,* or depressions, where the secondary wall does not form. In addition to vessel elements and tracheids, xylem can contain sclerenchyma fibers that lend additional support, as well as parenchyma cells that store various substances. Vascular rays, which are flat ribbons or sheets of parenchyma cells located between rows of tracheids, conduct water and minerals across the width of a plant.

The conducting cells of phloem are specialized parenchyma cells called **sieve-tube members,** arranged to form a continuous sieve tube (Fig. 24.5). Sieve-tube members contain cytoplasm but no nuclei. The term *sieve* refers to a cluster of pores in the end walls, which is known as a sieve plate. Each sieve-tube member has a **companion cell,** which contains a nucleus. The two are connected by numerous plasmodesmata, and the nucleus of the companion cell may control and maintain the life of both cells. The companion cells are also believed to be involved in the transport function of phloem. Sclerenchyma fibers also lend support to phloem.

Check Your Progress 24.1

1. List the three specialized meristems that arise from the apical meristem.
2. List the three specialized tissues in angiosperms and the cells that make up these tissues.
3. Compare the transport functions of xylem and phloem.



a. Structure of phloem b. Sieve-tube member and companion cells

Figure 24.5 Phloem structure. **a.** Photomicrograph of phloem vascular tissue and drawing showing general organization of phloem tissue. **b.** Drawing of sieve tube (composed of sieve-tube members) and companion cells. (photo): (a): Robert Knauft/Biology Pics/Science Source

ISTUDY

24.2 Organs of Flowering Plants

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the structure and function of roots, stems, and leaves.
2. List and describe the key features of monocots and eudicots.
3. Explain the difference between annual and perennial plants.

As discussed in Section 23.1, the earliest plants were simple and lacked true stems, roots, and leaves. As plants gained vascular tissue and began moving onto land away from water, organs developed to facilitate living in drier environments. *Vegetative organs* are all the plant parts except the reproductive structures of flowers, fruits, and seeds. Even though all vascular plants have vegetative organs, this chapter focuses on the organs commonly identified with the angiosperms, or flowering plants.

A flowering plant, whether a cactus, a water lily, or an apple tree, has a shoot system and root system (Fig. 24.6). The **shoot system** of a plant is composed of the **stem,** the branches, and the leaves. A stem supports the leaves in a way that exposes each one to as much sunlight as possible. In addition, the stem transports materials between roots and leaves and produces new tissue.

At the end of a stem, a **terminal bud** contains an apical meristem and produces new leaves and other tissues during the initial *primary growth* of a plant (Fig. 24.7). Lateral (side) branches grow from a lateral bud located at the angle where a leaf joins a stem. A **node** occurs where a leaf or leaves are attached to the stem, and an **internode** is the region between nodes. Vascular tissue transports water and minerals from the roots through the stem to the leaves and transports the products of photosynthesis, usually in the opposite direction. The **root system** simply consists of the roots. The root tip also contains an apical meristem and results in primary growth downward.

Ultimately, the three vegetative organs—the root, the stem, and the leaf—perform functions that allow a plant to live and grow. Flowers and fruit are reproductive organs and will be discussed in Section 27.3.

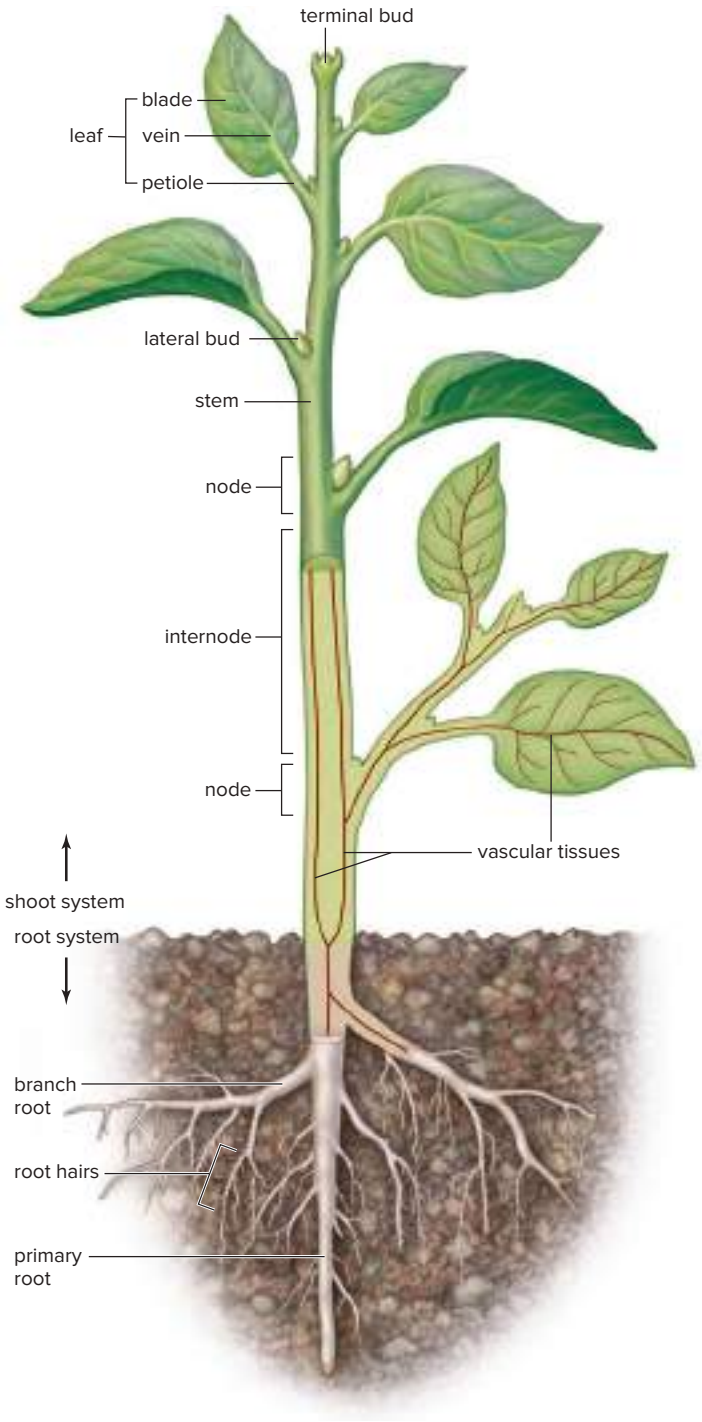


Figure 24.6 Organization of a plant body. The body of a plant consists of a root system and a shoot system. The shoot system contains the stem and leaves: two types of plant vegetative organs. Axillary buds can develop into branches of stems or flowers, the reproductive structures of a plant. The root system is connected to the shoot system by vascular tissue that extends from the roots to the leaves.

Variation of Organs and Organ Systems

As described in Section 23.5, flowering plants are divided into two groups, depending on the number of *cotyledons,* or seed leaves, in the embryonic plant. Plants with a seed containing only one cotyledon are referred to as monocotyledons, or **monocots.** Plants with seeds that contain two cotyledons are known as eudicotyledons, or **eudicots** (Fig. 24.8). Cotyledons of eudicots supply nutrients for seedlings, but the cotyledons of monocots act as transfer tissue, and the nutrients are derived from the endosperm before the true leaves begin photosynthesizing.

The vascular (transport) tissue is organized differently in monocots and eudicots. In the monocot root, vascular tissue occurs in a ring; in the monocot stem, the vascular bundles, which contain vascular tissue surrounded by a bundle sheath, are scattered. In the eudicot root, the xylem, which transports water and minerals, is star-shaped, and the phloem, which transports organic nutrients, is located between the points of the star. In a eudicot stem, the vascular bundles occur in a ring.

Leaf veins are vascular bundles within a leaf. Monocot leaves usually have a smooth margin (edge) and parallel venation, whereas

ISTUDY

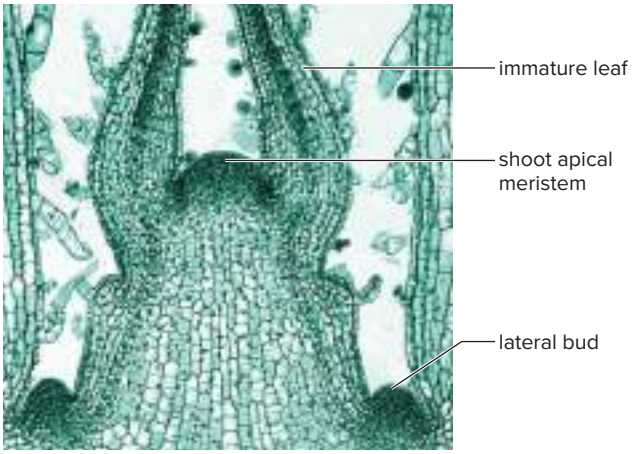


Figure 24.7 A terminal bud. A terminal bud contains an apical meristem and produces new leaves and other tissues during primary growth. Lateral branches grow from the lateral bud. Steven P. Lynch

eudicot leaf margins can be smooth but also commonly seen as lobed or serrated. Additionally, eudicot leaf veins have a netlike pattern.

Adult monocots and eudicots also have structural differences in the number of flower parts. Monocots have their flower parts arranged in multiples of three, whereas eudicots have their flower parts arranged in multiples of four or five. Monocot pollen grains have one pore, and eudicot pollen grains usually have three pores. Recall from Figure 23.20 that it was fossilized pollen characteristic of flowering plants that helped date the appearance of these plants in geologic time.

Although the distinctions between monocots and eudicots may seem of limited importance, they do, in fact, affect many aspects of their structure. The eudicots are the larger group and include some of our most familiar flowering plants—from dandelions to oak trees. The monocots include grasses, lilies, orchids, and palm trees, among others. Some of our most significant food sources are monocots, including rice, wheat, and corn.

Vegetative Organs and Plant Life Cycles

Annual plants live for only one growing season, *biennial* plants for two growing seasons, and **perennial** plants for three or more seasons. Perennials, such as certain roses, irises, and potatoes, expend energy making vegetative structures such as wood, bulbs, underground tubers, and leaf buds. These modified organs help the plant survive year after year. Annual plants, such as sunflowers and peas, produce enough vegetative structures to support flower and seed production. After seeds are produced and dispersed, the entire plant dies.

The difference comes down to two important flower-inducing genes: *LEAFY* and *Apetala 1*. In the shoot system, the shoot apical meristem divides, and vegetative organs, such as leaves, are made. When the *Apetala 1* gene turns on (with the help of the *LEAFY* gene), the apical meristem stops vegetative growth and switches to flower production. The timing of these genetic switches determines whether a plant is an annual or a perennial. Scientists have been able to block flower-inducing genes in annual plants and induce them to grow like perennials by diverting energy into the production of vegetative organs.

Check Your Progress 24.2

1. Describe how the plant body is organized.
2. List the three vegetative organs in a plant, and state their major functions.
3. Compare and contrast the structures of monocots and eudicots.

	Seed	Root	Stem	Leaf	Flower	Pollen
Monocots	One cotyledon in seed	Root xylem and phloem in a ring	Vascular bundles scattered in stem	Leaf veins form a parallel pattern	Flower parts in threes and multiples of three	One pore or slit
Eudicots	Two cotyledons in seed	Root phloem between arms of xylem	Vascular bundles in a distinct ring	Leaf veins form a net pattern	Flower parts in fours or fives and their multiples	Three pores or slits

Figure 24.8 Flowering plants are either monocots or eudicots. Six features, illustrated here, are used to distinguish monocots from eudicots: the number of cotyledons; the arrangement of vascular tissue in roots, stems, and leaves; the number of flower parts; and the differences in pollen structure.

ISTUDY

24.3 Organization and Diversity of Roots

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the tissue types found in each zone of a root.
2. Identify the structural differences between the roots of monocots and eudicots.
3. Describe the various specializations and symbiotic relationships that lead to root diversity.

The root system in the majority of plants is located underground. As a rule of thumb, the root system is at least equivalent in size and extent to the shoot system. An apple tree has a much larger root system than a corn plant, for example. The extensive root system of a plant anchors it in the soil, gives the plant support, and absorbs water and minerals. The cylindrical shape of a root allows it to penetrate the soil as it grows and permits water to be absorbed from all sides. The absorptive capacity of a root is dependent on its many branches, which all bear root hairs in a zone near the tip. Root hairs, which are projections from epidermal root-hair cells, are the structures that absorb water and minerals. In addition, roots produce hormones and can function in the storage of carbohydrates. Carrots and sweet potatoes are examples of storage roots.

Cells and Tissues of a Eudicot Root

The longitudinal section of a eudicot root (Fig. 24.9*a*) reveals zones where cells are in various stages of differentiation. The

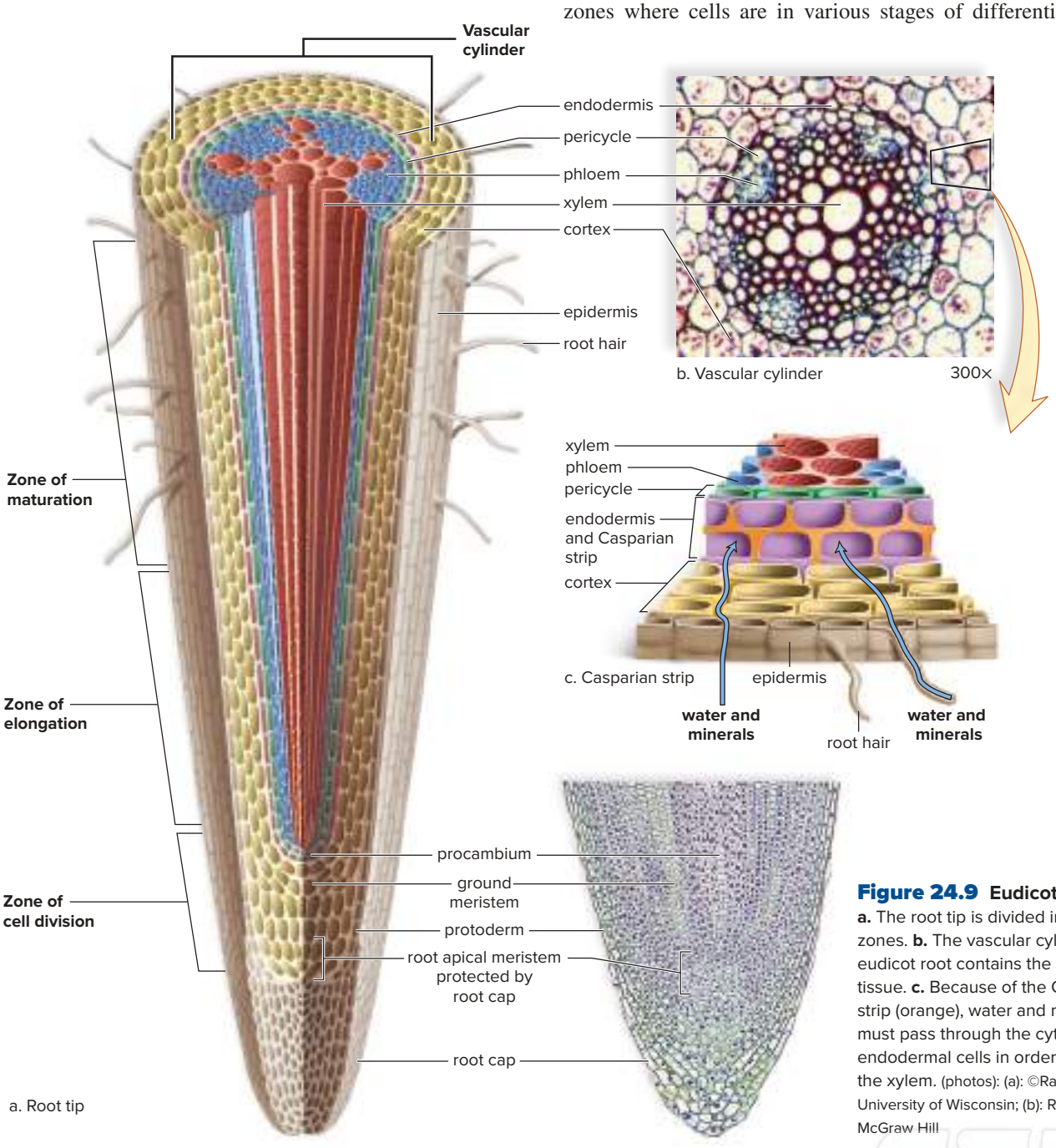


Figure 24.9 Eudicot root tip. **a.** The root tip is divided into three zones. **b.** The vascular cylinder of a eudicot root contains the vascular tissue. **c.** Because of the Casparian strip (orange), water and minerals must pass through the cytoplasm of endodermal cells in order to enter the xylem. (photos): (a): ©Ray F. Evert/ University of Wisconsin; (b): Richard Gross/ McGraw Hill

ISTUDY

root apical meristem is protected by the **root cap.** Root cap cells have to be replaced constantly, because they get ground off by rough soil particles as the root grows. Notice that the root apical meristem is in the *zone of cell division,* which continuously provides new cells to the region above—the zone of elongation. In the *zone of elongation,* the cells grow in length as they become specialized. The *zone of maturation,* which contains fully differentiated cells, is recognizable because here root hairs are found on many of the epidermal cells.

Figure 24.9*a* also shows a cross section of a root at the zone of maturation. These specialized tissues are identifiable.

Epidermis The **epidermis** forms the outer layer of the root and consists of only a single layer of cells. The majority of epidermal cells are thin-walled and rectangular, but in the zone of maturation, many epidermal cells have root hairs. These can project as far as 5–8 mm into the soil particles.

Cortex Moving inward, next to the epidermis are the large, thin-walled parenchyma cells that make up the **cortex** of the root. The cortex mainly functions in food storage. These irregularly shaped cells are loosely packed, making it possible for absorbed water and minerals to weave their way in between the cells or travel through the cells toward the middle of the root.

Endodermis (Gk. *endon,* "within"; *derma,* "skin") The **endodermis** is the next single layer of cells between the cortex and the inner vascular cylinder. The endodermal cells have a very important role in that they regulate the water and ions that can move toward the vascular tissue. Notice the movement of water in Figure 24.9*c*. In one option, water can move in between the cells of the cortex but is suddenly blocked at the endodermis. Like a tight rubber band, the endodermal cells are fitted with a **Casparian strip** made up of the protein lignin and a waterproof lipid substance called *suberin.* This strip prevents the passage of water and mineral ions *between* cell walls; instead, water and ions are forced *through* the endodermal cells, where biological regulation occurs.

Pericycle The **pericycle,** the first layer of cells within the vascular cylinder, can continue to divide and is the starting point where branch, or lateral, roots develop (Fig. 24.10).

Vascular Tissue The main portion of the vascular cylinder contains xylem and phloem. The xylem appears star-shaped in eudicots, because several arms of tissue radiate from a common center (see Fig. 24.9*b*). The phloem is found in separate regions between the arms of the xylem.

Organization of Monocot Roots

Monocot roots have the same growth zones and undergo the same secondary growth as eudicot roots. However, the organization of their tissues is slightly different. The ground tissue of a monocot root's **pith** is centrally located and is surrounded by a vascular ring composed of alternating xylem and phloem bundles (Fig. 24.11). Monocot roots also have pericycle, endodermis, cortex, and epidermis.

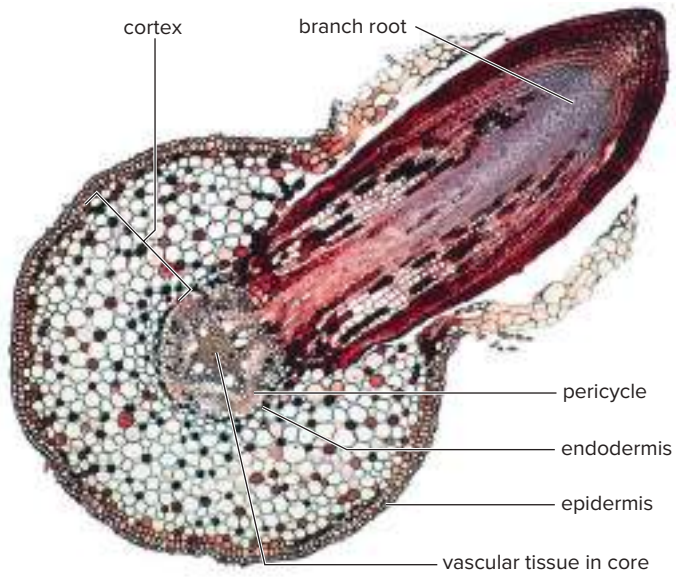


Figure 24.10 Branching of a eudicot root. This cross section of a willow, *Salix,* shows the origination and growth of a branch root from the pericycle. Lee W. Wilcox

Root Diversity

Roots tend to be of two varieties. In eudicots, the first (primary) root grows straight down and is called a **taproot.** Monocots have a **fibrous root system,** which may have large numbers of fine roots

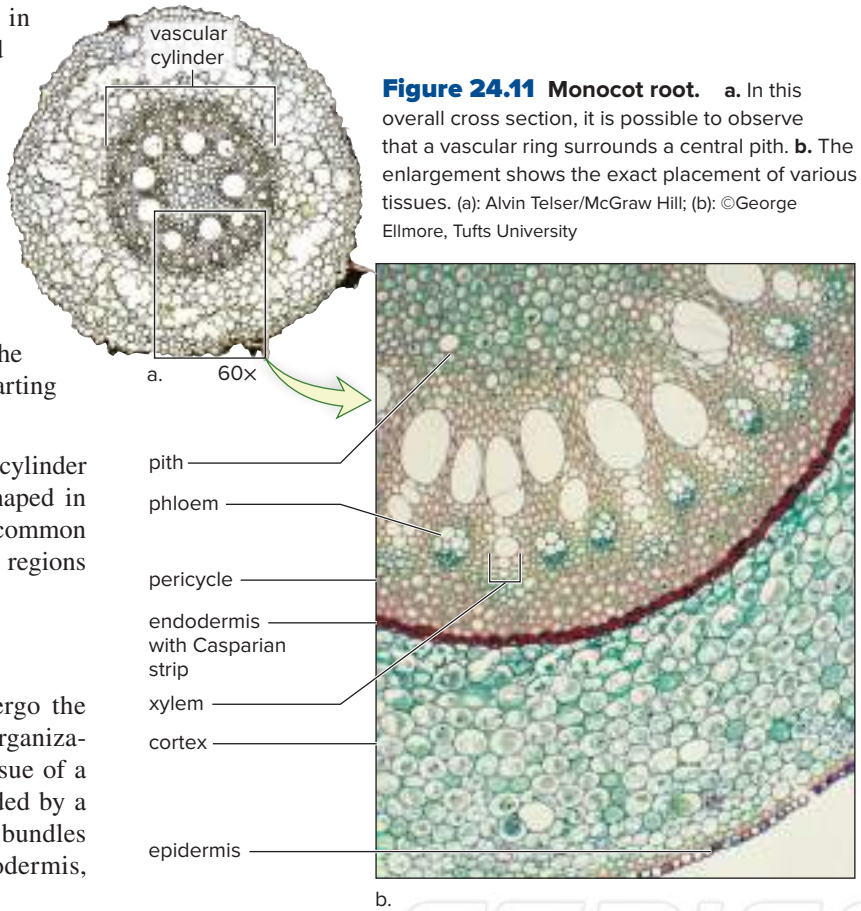


Figure 24.11 Monocot root. **a.** In this overall cross section, it is possible to observe that a vascular ring surrounds a central pith. **b.** The enlargement shows the exact placement of various tissues. (a): Alvin Telser/McGraw Hill; (b): ©George Ellmore, Tufts University

of similar diameter. Many mature plants have a combination of taproot and fibrous root systems.

Taproot Fibrous root

Some plants have roots that possess a variety of adaptations to better perform their functions. Root adaptations can improve anchorage to the ground or the storage of carbohydrates. Other root systems need to increase absorption of water, minerals, oxygen, or sunlight.

Most roots store some food, but in certain plants, such as the sweet potato or carrot, the roots are enlarged and store large quantities of starch. Perennials rely on this stored carbohydrate to grow year after year (Fig. 24.12*a*). Corn plants are straight, top-heavy, and in danger of falling over. Specialized corn prop roots, produced toward the base of stems, support these plants in high winds (Fig. 24.12*b*).

As you will learn in Section 25.1, roots need oxygen to perform cellular respiration. Roots normally get their oxygen from the air pockets in the soil, but oxygen becomes scarce when the plant grows exclusively in water. Large trees growing in water, such as mangrove and bald cypress, have evolved spongy roots called *pneumatophores,* which extend above the water's surface and enhance gas exchange (Fig. 24.12*c*). Many *epiphytes* (plants that live in or on trees) have aerial roots for a variety of reasons. English ivy uses aerial roots to climb the bark of trees. Orchids use their roots to capture moisture in the air and support their weight, and in some instances, green aerial roots perform photosynthesis (Fig. 24.12*d*).

Root Symbiotic Relationships

As described in Section 22.3, *mycorrhizae* are associations between roots and fungi. Plants that have mycorrhizae are able to extract water and minerals from the soil better than those with roots that lack a fungal partner. This relationship is mutualistic, because the fungus receives sugars and amino acids from the plant, while the plant receives increased water and minerals via the fungus.

a. b. c. d.

Figure 24.12 Root specializations. **a.** Sweet potato plants have food storage roots. **b.** Prop roots are specialized for support. **c.** The pneumatophores of this tree allow it to acquire oxygen even though it lives in water. **d.** The aerial roots of orchids offer physical support, water and nutrient uptake, and in some cases even photosynthesis. (a): Madlen/Shutterstock; (b): NokHoOkNoi/iStock/Getty Images; (c): Mark Newman/FLPA/age fotostock; (d): DEA/S Montanari/age fotostock

Peas, beans, and other legumes have **root nodules,** where nitrogen-fixing bacteria live. Plants cannot extract nitrogen from the air, but the bacteria within the nodules can take up and reduce atmospheric nitrogen. The plant gets a source of nitrogen from the bacteria, and the bacteria receive carbohydrates from the plant.

Check Your Progress 24.3

1. Explain the relationship between the root apical meristem and the root cap.
2. List the function of the endodermis and the Casparian strip in a root.
3. Describe the advantages of some root specializations.

ISTUDY

24.4 Organization and Diversity of Stems

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the anatomical structures of a woody twig.
2. Recognize the differences in the arrangement of vascular tissue between a herbaceous dicot and a monocot stem.
3. Describe how secondary growth of a woody stem results in the various tissues within it.
4. Characterize the variations of stem diversity.

A stem is the main axis to a plant's shoot system. It carries leaves and flowers and supports the plant's weight. A stiff stem rising upward against gravity is an ancient adaptation that allowed plants to move into terrestrial habitats.

The anatomy of a woody twig helps us review the organization of a stem (Fig. 24.13). The **terminal bud** contains the shoot tip protected by modified leaves called bud scales. Each spring when growth resumes, bud scales fall off and leave a scar. Each bud-scale scar indicates a year of growth. Leaf scars and bundle scars mark the locations of leaves that have dropped. Dormant axillary buds that will give rise to branches or flowers are also found here.

As seasonal growth resumes, the apical meristem at the shoot tip produces new cells, which increase the height of the stem. The **shoot apical meristem** is protected within the terminal bud, where leaf primordia (immature leaves) envelop it (Fig. 24.14). The leaf primordia mark the locations of nodes; the portion of stem in between nodes is an internode. As a stem grows, the internodes increase in length.

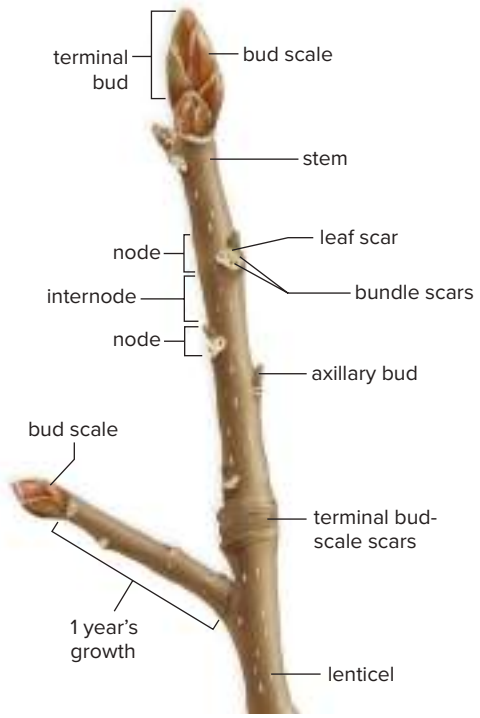


Figure 24.13 Woody twig. The major parts of a stem are illustrated by a woody twig collected in winter.

In addition to leaf primordia, the three specialized types of primary meristem (see Section 24.1) develop from a shoot apical meristem (Fig. 24.14*b*). These primary meristems contribute to the length of a shoot. The *protoderm,* the outermost primary meristem, gives rise to the epidermis. The *ground meristem* produces two tissues composed of parenchyma cells: the pith and the cortex. The *procambium* (see Fig. 24.14*a*) produces the first xylem cells, called primary xylem, and the first phloem cells, called primary phloem.

Differentiation continues as certain cells become the first tracheids or vessel elements of the xylem within a vascular bundle. The first sieve-tube members of a vascular bundle do not have companion cells and are short-lived (some live only a day before being replaced). Mature vascular bundles contain fully differentiated xylem, phloem, and a lateral meristem called **vascular cambium.** Vascular cambium is discussed more fully later in this section.

Herbaceous Stems

Basil, dandelions, and tulips are all examples of plants with **herbaceous stems** (L. *herba,* "vegetation, plant") that exhibit mostly primary growth—no wood or bark. The outermost tissue of herbaceous stems is the epidermis, which is covered by a waxy

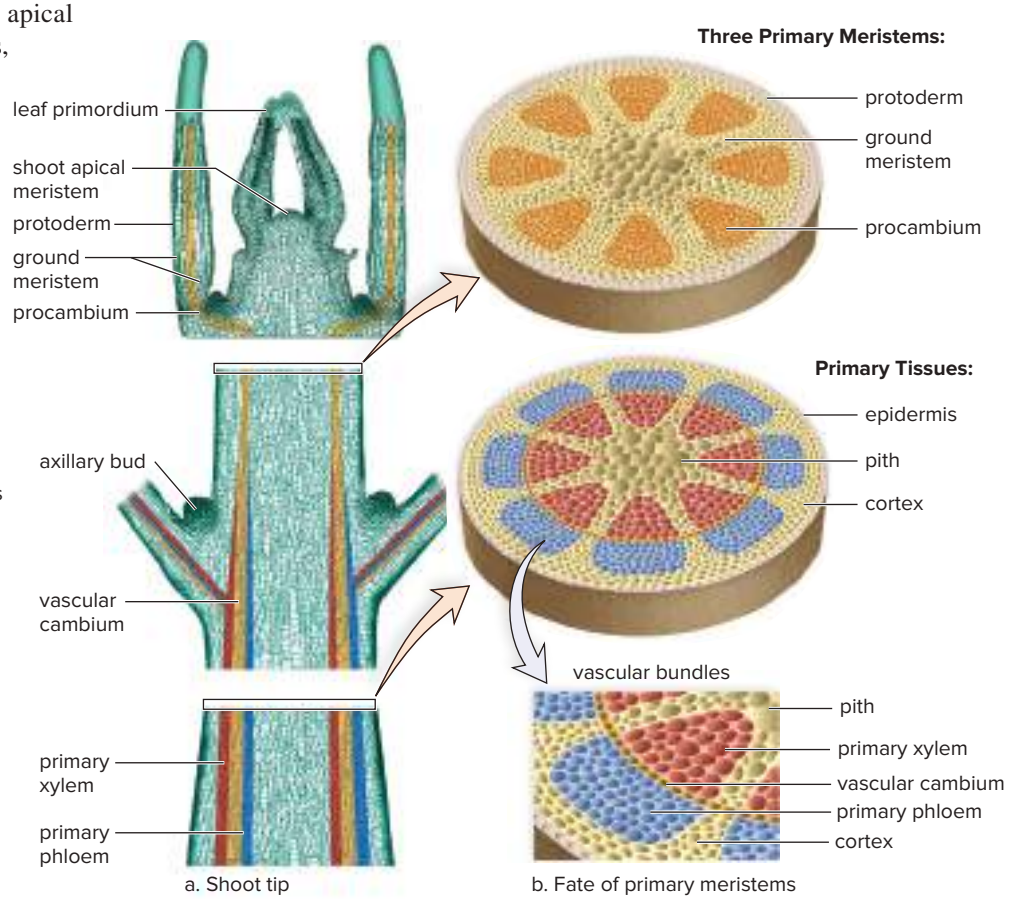


Figure 24.14 Shoot tip and primary meristems. **a.** The shoot apical meristem within a terminal bud is surrounded by leaf primordia. **b.** The shoot apical meristem produces the primary meristems: Protoderm gives rise to epidermis; ground meristem gives rise to pith and cortex; and procambium gives rise to vascular tissue, including primary xylem, primary phloem, and vascular cambium. (a): Steven P. Lynch

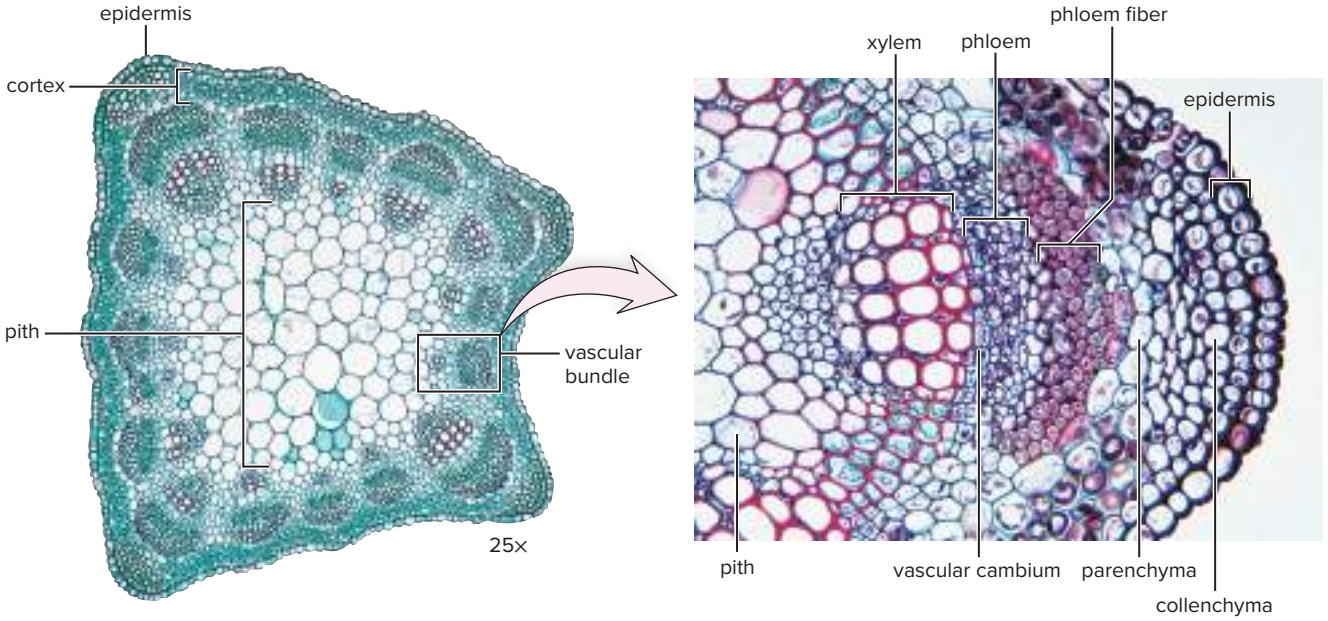


Figure 24.15 Herbaceous eudicot stem. Cross section of a young stem where vascular bundles are arranged toward the periphery of the stem. (top): Ed Reschke; (bottom): Ray F. Evert, University of Wisconsin

cuticle to prevent water loss. These stems have distinctive vascular bundles, where xylem and phloem are found. In each bundle, xylem is typically found toward the inside of the stem, and phloem is found toward the outside.

In the herbaceous eudicot stem, such as a sunflower, the vascular bundles are arranged in a distinct ring where the cortex is separated from the central pith, which stores water and products of photosynthesis (Fig. 24.15). The cortex is sometimes green and carries on photosynthesis.

In a monocot stem, such as a corn stalk, the vascular bundles are scattered throughout the stem, and often the cortex and pith are not clearly distinguishable (Fig. 24.16).

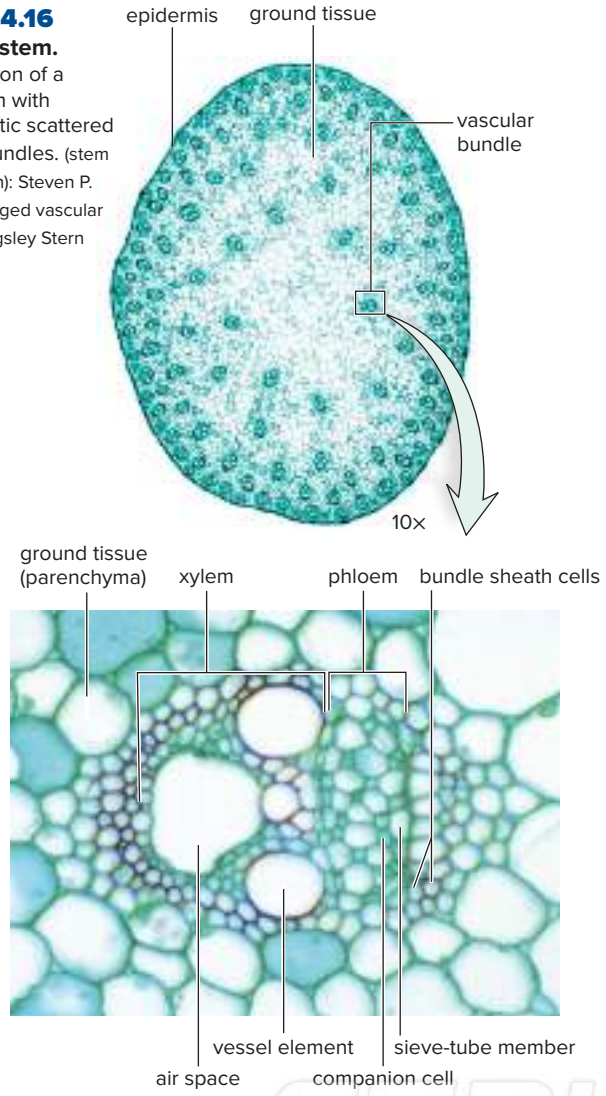
The stems of one monocot in the grass family, bamboo, have been of great benefit in human history, and this plant continues to be useful to us today. The Nature of Science feature, "The Cannabis Plant and Its Products," examines some of the potential benefits of a dicot.

Woody Stems

A woody plant, such as an oak tree, has both primary and secondary tissues. Primary tissues are the new tissues formed each year from the primary meristems. Secondary tissues develop during the first and subsequent years of growth from lateral meristems, forming the vascular cambium and cork cambium. *Primary growth,* which occurs in all plants, increases the length of plant stems and roots; *secondary growth,* which occurs only in conifers and woody eudicots, increases the girth of trunks, stems, branches, and roots.

Trees and shrubs undergo secondary growth because of a change in the location and activity of vascular cambium (Fig. 24.17). In herbaceous plants, vascular cambium is present between the xylem and phloem of each vascular bundle. In woody plants, the vascular cambium develops to form a ring of meristem that divides parallel to the surface of the plant and produces new xylem toward the inside and phloem toward the outside on a yearly basis.

Figure 24.16 Monocot stem. Cross section of a young stem with characteristic scattered vascular bundles. (stem cross section): Steven P. Lynch; (enlarged vascular bundle): Kingsley Stern



THEME Nature of Science

The Cannabis *Plant and Its Products*

Indigenous to Central Asia, *Cannabis* is a dioecious flowering plant with two main species, *Cannabis sativa* and *Cannabis indica*. Like any domesticated plant, many varieties have specific traits selected for by humans over the centuries for a wide range of purposes.

What all varieties of *Cannabis* plants have in common is the production of a group of chemicals called cannabinoids. A total of 113 unique cannabinoids have been characterized from *Cannabis* plants. The two most common are THC (tetrahydrocannabinol) and CBD (cannabidiol).

THC is known to bind receptors in the brain and produce psychotropic or euphoric effects. A few states have legalized THC for recreational use. THC is also one ingredient in what is termed "medical marijuana" and is used for pain management, help with nausea, appetite control, asthma, and other conditions. The U.S. Food and Drug Administration (FDA) has not recognized or approved the *Cannabis* plant as medicine.

CBD products (Fig. 24B) are also considered "medical marijuana" and have recently become very popular. CBD does not have psychotropic effects, is legal, and companies selling CBD-infused products claim to cure muscle pain, insomnia, and mood disorders.

Finally, certain *Cannabis* varieties have very fibrous stems, and seeds with high concentrations of oil. These varieties are what make up the "hemp" industry, which cultivates plants to extract fiber and oils for the cosmetic and textile industry. Figure 24C illustrates the differences between some of these bred varieties of *Cannabis*.

Questions to Consider

1. Why do you believe that most medical remedies obtained from the *Cannabis* plant are not FDA approved?
2. Why is it important to understand the differences between different varieties of the *Cannabis* plant?

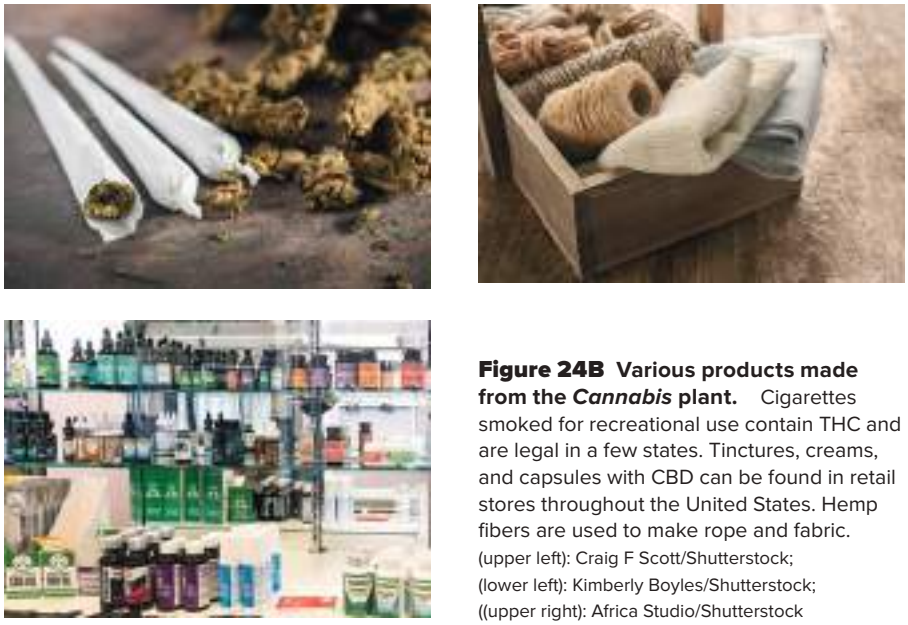


Figure 24B Various products made from the *Cannabis* plant. Cigarettes smoked for recreational use contain THC and are legal in a few states. Tinctures, creams, and capsules with CBD can be found in retail stores throughout the United States. Hemp fibers are used to make rope and fabric.
(upper left): Craig F Scott/Shutterstock;
(lower left): Kimberly Boyles/Shutterstock;
((upper right): Africa Studio/Shutterstock

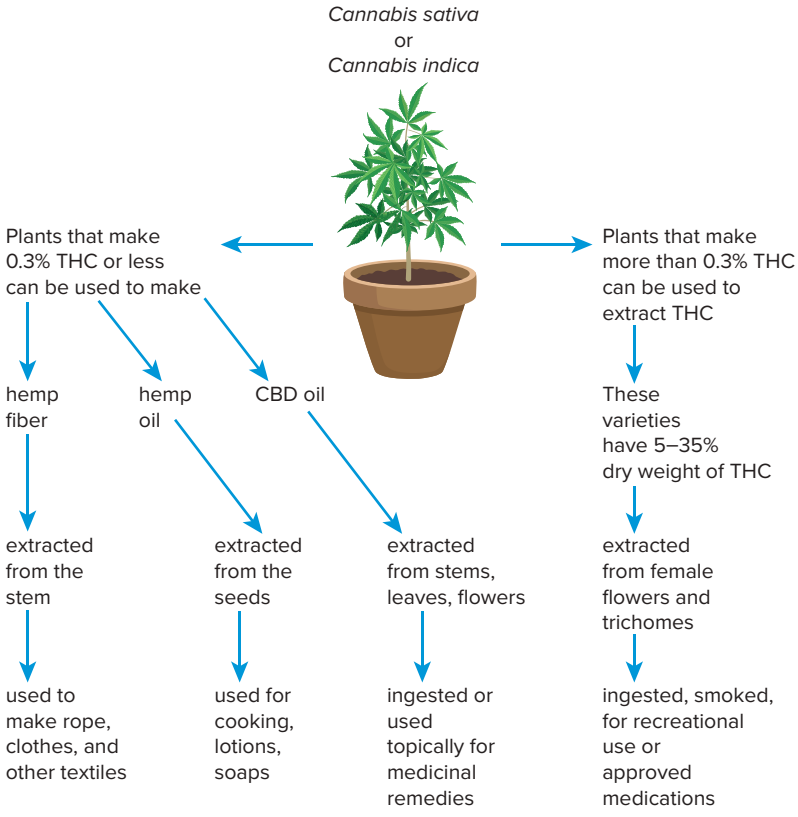


Figure 24C Difference between legal and illegal *Cannabis* plants. *Cannabis* plants legally fall into two categories: Those that produce low amounts of THC can be used to make hemp oil, hemp fiber, and CBD products. Plants that produce high amounts of THC are considered narcotics and are regulated.

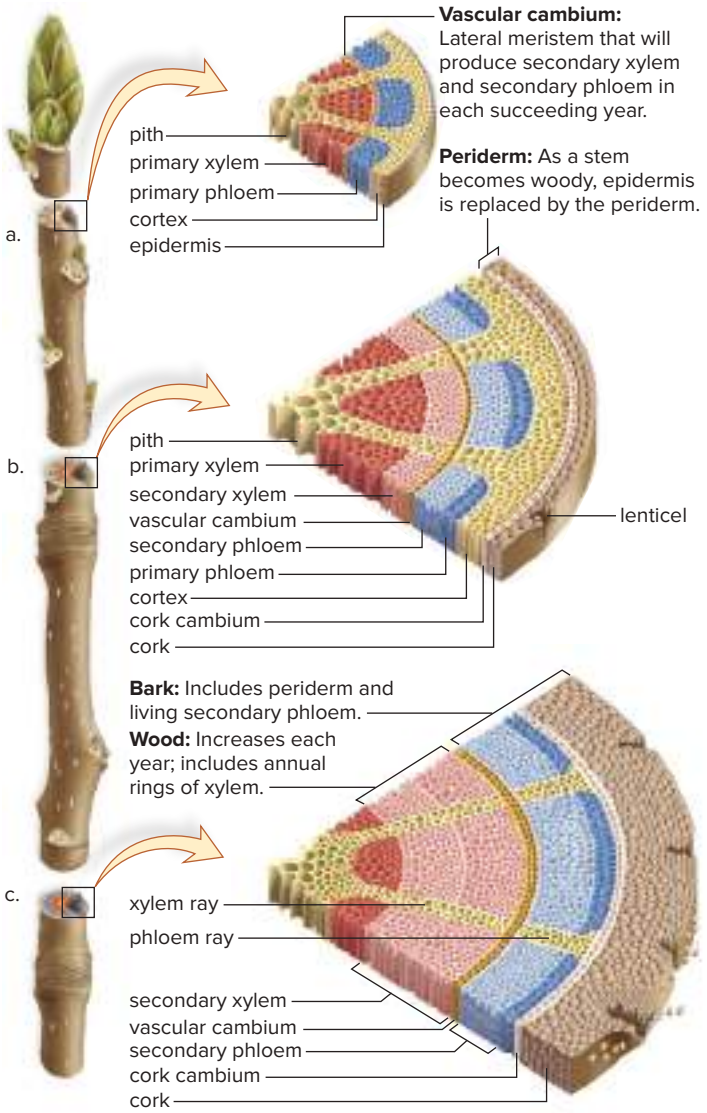


Figure 24.17 Diagrams of secondary growth of stems.
a. Diagram showing eudicot herbaceous stem just before secondary growth begins. **b.** Diagram showing that secondary growth has begun. Periderm has replaced the epidermis. Vascular cambium produces secondary xylem and secondary phloem each year. **c.** Diagram showing a 2-year-old stem. The primary phloem and cortex will eventually disappear, and only the secondary phloem (within the bark) produced by vascular cambium will be active that year. Secondary xylem builds up to become the annual rings of a woody stem.

Also notice in Figure 24.17 the *xylem rays* and *phloem rays* that are visible in the cross section of a woody stem. Rays consist of parenchyma cells that permit lateral conduction of nutrients from the pith to the cortex, as well as some storage of food. A phloem ray can vary in width and is a continuation of a xylem ray.

Bark

The **bark** of a tree contains cork, cork cambium, cortex, and phloem. It is very harmful to remove the bark of a tree, because without phloem, organic nutrients cannot be transported. Although new phloem tissue is produced each year by vascular cambium, it does not build up in the same manner as xylem. In North America, herbivores, such as beavers, elk, and porcupines, eat bark and inadvertently girdle trees. Girdling involves removing bark from around the tree, inevitably leading to the death of the tree. Many forest management agencies wrap the trunks of trees to protect them from animal damage.

In bark, the region of active cell division occurs at the **cork cambium.** When cork cambium first begins to divide, it produces tissue that disrupts the epidermis and replaces it with **cork** cells (see Fig. 24.2*d*). Recall that cork cells are impregnated with suberin, a waxy layer that makes them waterproof but also causes them to die, and that suberin also makes up the Casparian strip in roots. In a woody stem, gas exchange is impeded, except at lenticels, which are pockets of loosely arranged cork cells not impregnated with suberin.

Wood

When a plant first begins growing, the xylem is made by the apical meristem. Later, as the plant matures, xylem is made by the vascular cambium and is called secondary xylem. **Wood** is secondary xylem that builds up year after year, thereby increasing the girth of trees. In trees that have a growing season, vascular cambium is dormant during the winter. In the spring, when moisture is plentiful and leaves require much water for growth, the secondary xylem contains wide vessel elements with thin walls. In this *spring wood,* wide vessels transport sufficient water to the growing leaves. Later in the season, moisture is scarce, and the wood at this time, called *summer wood,* has a lower proportion of vessels (Fig. 24.18). Strength is required,

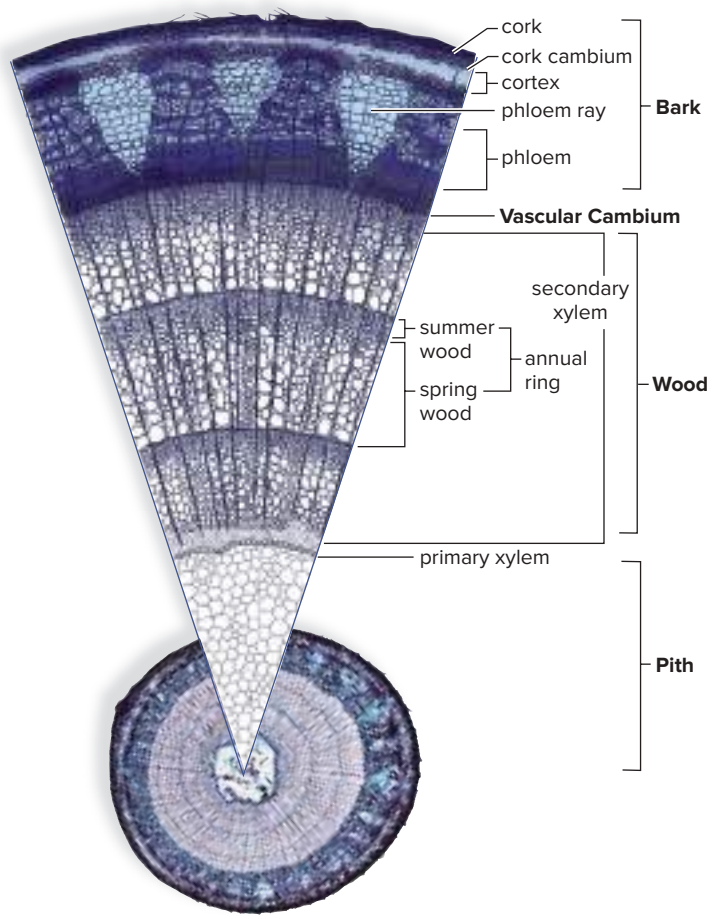


Figure 24.18 Three-year-old woody twig. The buildup of secondary xylem in a woody stem results in annual rings, which tell the age of the stem. The rings can be distinguished because each one begins with spring wood (large vessel elements) and ends with summer wood (smaller and fewer vessel elements). (photo) (circular cross section): Ed Reschke/Stone/Getty Images

ISTUDY

because the tree is growing larger, and summer wood contains numerous, thick-walled tracheids. At the end of the growing season, just before the cambium becomes dormant again, only heavy fibers with especially thick secondary walls may develop.

When the trunk of a tree has spring wood followed by summer wood, the two together make up 1 year's growth, or an **annual ring.** You can tell the age of a tree by counting the annual rings (Fig. 24.19*a*). The outer annual rings, where transport occurs, are called sapwood.

In older trees, the inner annual rings, called heartwood, no longer function in water transport. The cells become plugged with deposits, such as resins, gums, and other substances that inhibit the growth of bacteria and fungi. Heartwood may help support a tree, although some trees stand erect and live for many years after the heartwood has rotted away. Figure 24.19*b* shows the layers of a woody stem in relation to one another.

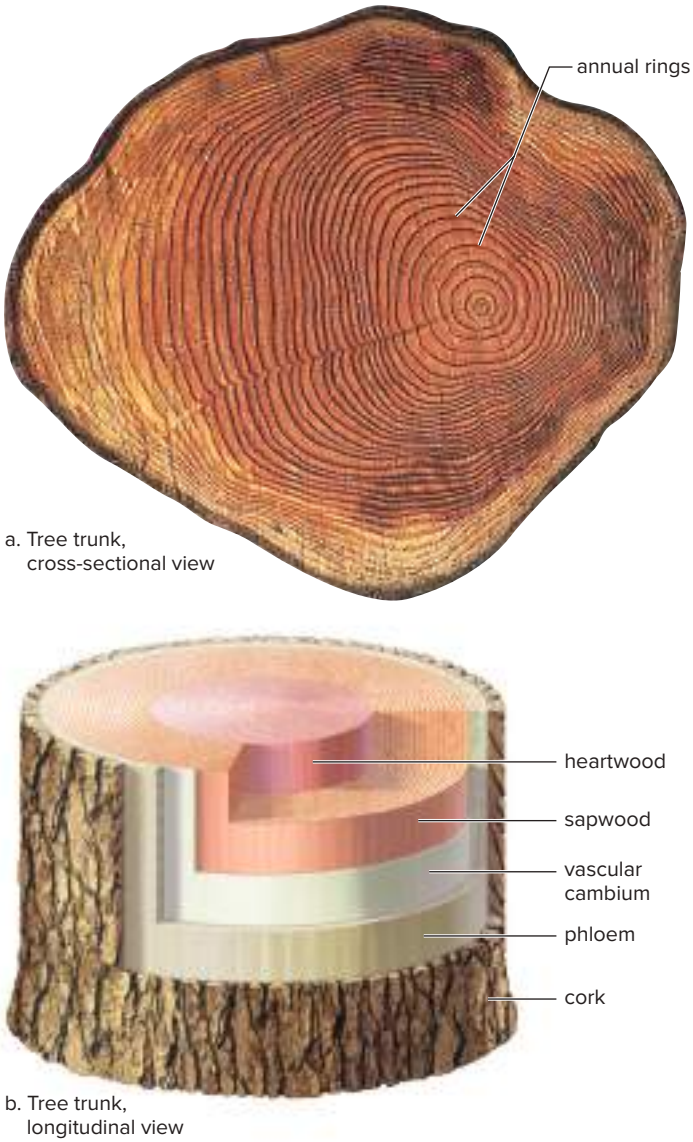


Figure 24.19 Tree trunk. **a.** A cross section of a 39-year-old larch, *Larix decidua.* The xylem within the darker heartwood is inactive; the xylem within the lighter sapwood is active. **b.** The relationship of bark, vascular cambium, and wood is retained in a mature stem. The pith has been buried by the growth of layer after layer of new secondary xylem.
(a): ©Ardea/ardea.com

The annual rings are used to tell the age of a tree, as well as the historical record of tree growth. For example, if rainfall and other conditions have been extremely favorable during a season, the annual ring may be wider than usual. If the tree has been shaded on one side by another tree or a building, the rings may be wider on the sunnier side.

Advantages and Disadvantages of Woody Plants

What are the evolutionary benefits of woody plants? With adequate rainfall, woody plants can grow taller and have more growth, because they have adequate vascular tissue to support and service their leaves. Furthermore, a long life may mean more opportunity to reproduce.

However, it takes energy to produce secondary growth and to prepare the body for winter if the plant lives in the temperate zone. Also, woody plants need more defense mechanisms, because a long-lived plant is likely to be attacked by herbivores and parasites. Trees usually do not reproduce until after they have grown for several seasons, by which time they may have been attacked by predators or been infected with a disease. In certain habitats, it is more advantageous for a plant to put most of its energy into producing a large number of seeds rather than being woody.

Stem Diversity

There are many plants whose ground-level or belowground stems are often mistaken for roots. The diversity of some of these stems is illustrated in Figure 24.20. Horizontal stems, called *stolons* or runners, produce new plants where nodes touch the ground. The strawberry plant is a common example of this type of stem, which functions in vegetative reproduction.

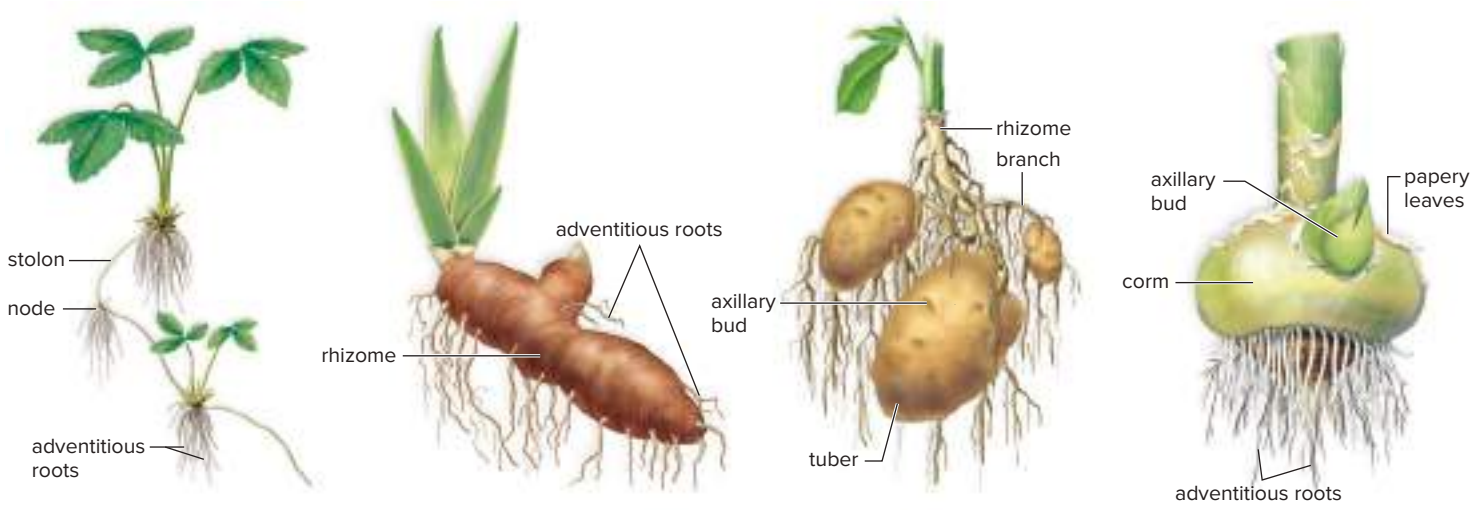
Rhizomes are underground horizontal stems; they may be long and thin, as in sod-forming grasses, or thick and fleshy, as in irises. Rhizomes survive the winter and contribute to asexual reproduction, because each node bears a bud. Some rhizomes have enlarged portions called tubers, which function in food storage. Potatoes are tubers, and the potato "eyes" are buds that mark the nodes.

Corms are bulbous, underground stems that lie dormant during the winter, just as rhizomes do. They also produce new plants the next growing season. Gladiolus corms are referred to as bulbs by laypersons, but botanists reserve the term *bulb* for a structure composed of modified leaves attached to a short, vertical stem. An onion is a bulb.

Aboveground vertical stems can also be modified. For example, cacti have succulent stems specialized for water storage, and the tendrils of grape plants (which are stem branches) allow them to climb. The morning glory and its relatives have stems that twine around support structures. Such tendrils and twining shoots help plants expose their leaves to the sun.

Humans make use of stems in many ways. The stem of the sugarcane plant is a primary source of table sugar; cinnamon and the drug quinine are derived from the bark of *Cinnamomum verum* and various *Cinchona* species, respectively; and wood is necessary for the production of paper and building materials and is used as fuel in many parts of the world.

ISTUDY



a. Stolon b. Rhizome c. Tuber d. Corm

Figure 24.20 Stem diversity. **a.** A strawberry plant has aboveground, horizontal stems called stolons. Every other node produces a new shoot system. **b.** The underground, horizontal stem of an iris is a fleshy rhizome. **c.** The underground stem of a potato plant has enlargements called tubers. We call the tubers potatoes. **d.** The corm of a gladiolus is a stem covered by papery leaves. (photos): (a): Evelyn Jo Johnson/McGraw Hill; (b): jelloyd/Shutterstock; (c): Carlyn Iverson/McGraw Hill; (d): AnkaFed/Shutterstock

Check Your Progress 24.4

1. Describe the transport tissues found in a vascular bundle.
2. Compare the arrangements of the vascular bundles in monocot stems and eudicot stems.
3. Contrast primary growth with secondary growth.
4. List the components of bark.
5. Compare the features in the annual rings of spring wood and summer wood.

24.5 Organization and Diversity of Leaves

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the structures and functions of various leaf tissues.
2. Describe the many forms of leaf diversity.

Leaves are the part of a plant that generally carries on the majority of photosynthesis, a process that requires water, carbon dioxide, and sunlight. Leaves receive water from the root system by way of the stem.

The size, shape, color, and texture of leaves are highly variable. These characteristics are fundamental in plant identification. The leaves of some aquatic duckweeds are less than 1 mm in diameter, while some palms have leaves that exceed 6 m in length. The shape of leaves can vary from cactus spines to deeply lobed white oak leaves. Leaves can exhibit a variety of colors, from various shades of green to deep purple. The texture of leaves varies from smooth and waxy, like a magnolia, to coarse, like a sycamore. Plants that bear leaves the entire year are called **evergreens,** and those that lose all their leaves at the end of their growing season are called **deciduous.**

Broad and thin plant leaves have the maximum surface area for the absorption of carbon dioxide and the collection of solar energy needed for photosynthesis. Unlike stems, leaves are almost never woody. With few exceptions, their cells are living, and the bulk of a leaf contains photosynthetic tissue.

The wide portion of a foliage leaf is called the **blade.** The **petiole** is a stalk that attaches the blade to the stem (Fig. 24.21). The upper acute angle between the petiole and stem is the leaf axil, where the axillary bud is found.

Leaf Morphology

Figure 24.22 shows a cross section of a typical eudicot leaf of a temperate-zone plant. At the top and bottom are layers of epidermal

ISTUDY

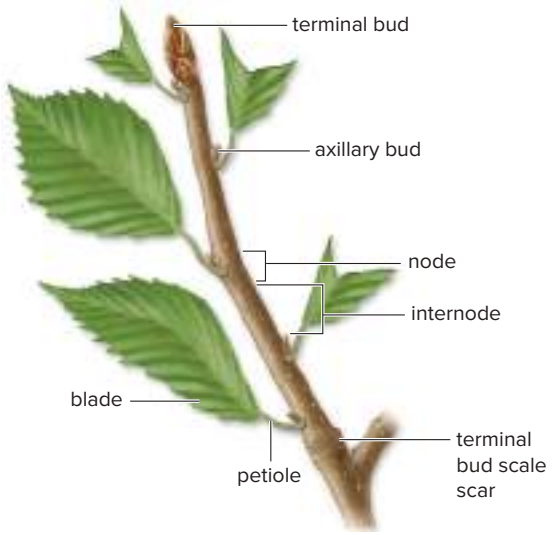


Figure 24.21 Twig with leaves. The major parts of a leaf and attachments are illustrated by a woody twig collected in summer.

tissue that often bear trichomes, protective hairs often modified as glands that secrete irritating substances. These features help deter insects from eating the leaf. The epidermis characteristically has an outer, waxy cuticle that helps keep the leaf from drying out. The cuticle also prevents gas exchange, because it is not gas permeable. However, the lower epidermis of eudicot and both surfaces of monocot leaves contain stomata that allow gases to move into and out of the leaf. Water loss also occurs at stomata, but each stoma has two guard cells that regulate its opening and closing, and stomata close when the weather is hot and dry.

The body of a leaf is composed of **mesophyll** (Gk. *mesos,* "middle"; *phyllon,* "leaf") tissue. Most eudicot leaves have two distinct regions: **palisade mesophyll,** containing tightly packed, elongated cells, and **spongy mesophyll,** containing irregular cells bounded by air spaces. There are important reasons why palisade mesophyll and spongy mesophyll look so different from each other. The long cylindrical cells of the palisade mesophyll are close together, packed with chloroplasts, and are most responsible for photosynthesis. Although close together, the palisade cells are separated from each other by a tiny film of water, which increases the surface area for slow-moving carbon dioxide molecules to diffuse into.

The spongy mesophyll is made up of cells that are loosely packed, irregularly shaped, and have a higher volume of air spaces. This structure increases the chance that carbon dioxide, which does not dissolve easily, will enter the leaf and stay there long enough to diffuse into the palisade layer.

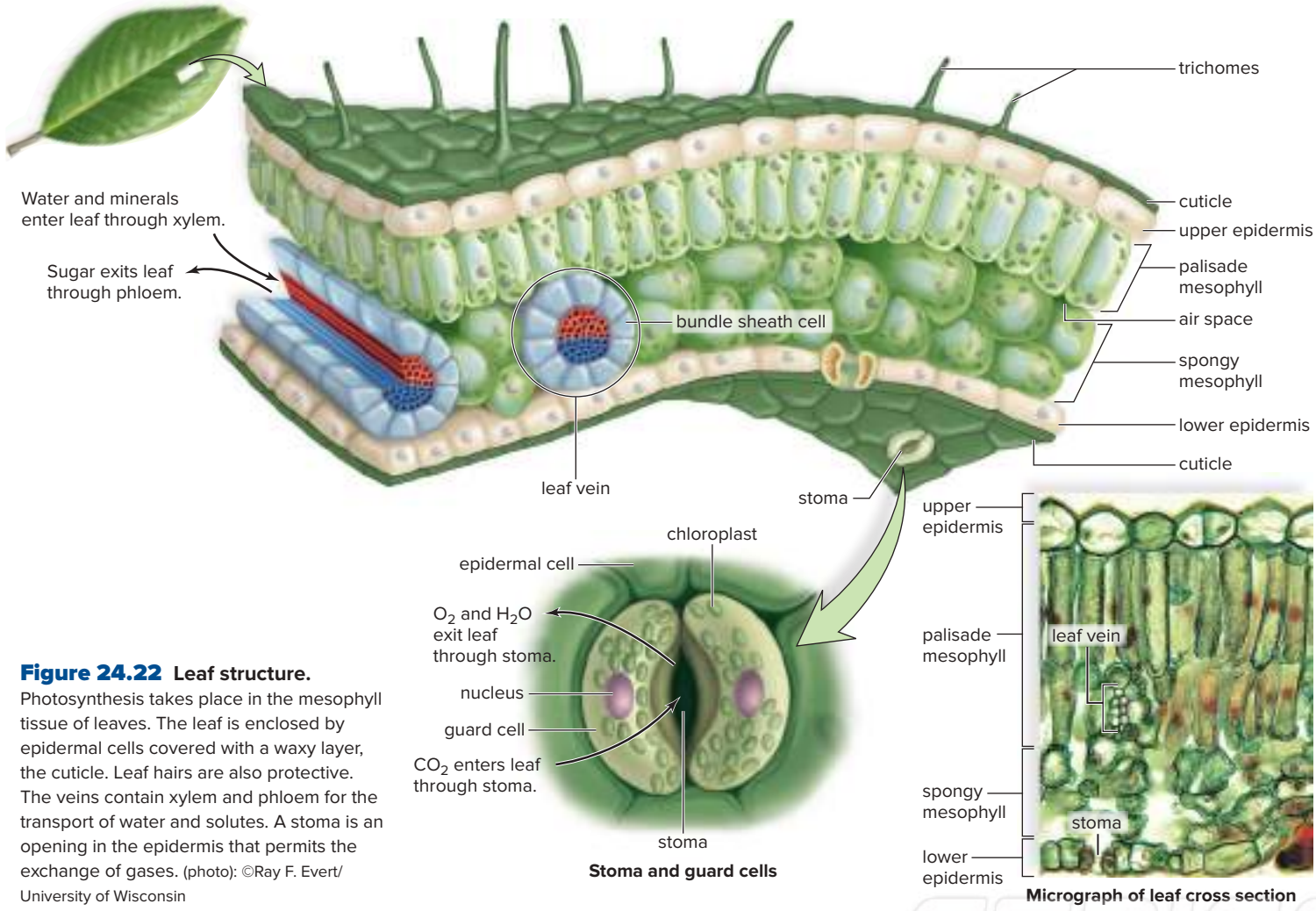


Figure 24.22 Leaf structure.
Photosynthesis takes place in the mesophyll tissue of leaves. The leaf is enclosed by epidermal cells covered with a waxy layer, the cuticle. Leaf hairs are also protective. The veins contain xylem and phloem for the transport of water and solutes. A stoma is an opening in the epidermis that permits the exchange of gases. (photo): ©Ray F. Evert/ University of Wisconsin

Leaf Diversity

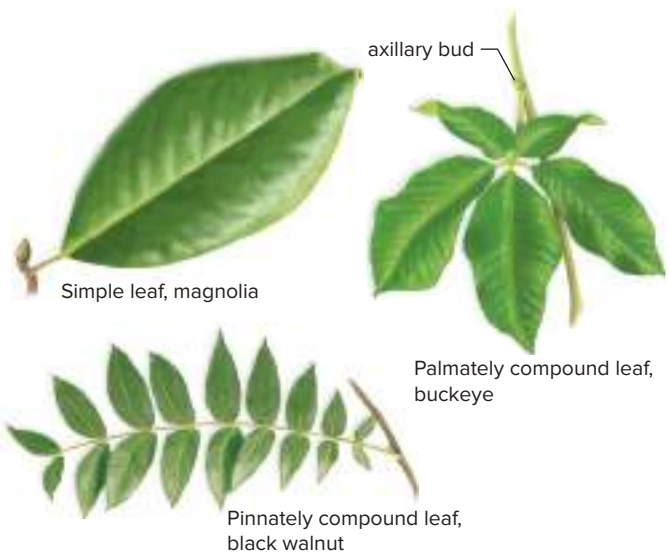
The blade of a leaf can be simple or compound (Fig. 24.23). A simple leaf has a single blade in contrast to a compound leaf, which is divided in various ways into leaflets. An example of a plant with simple leaves is a magnolia, and plants with compound leaves include the buckeye and black walnut. In addition, the black walnut has pinnately compound leaves with leaflets occurring in pairs, while the buckeye has palmately compound leaves with all the leaflets attached to a single point.

Leaves can be arranged on a stem in three ways: alternate, opposite, or whorled. The leaves are alternate in the American beech; in a maple, the leaves are opposite, with two leaves attached to the same node. Bedstraw has a whorled leaf arrangement with several leaves originating from the same node.

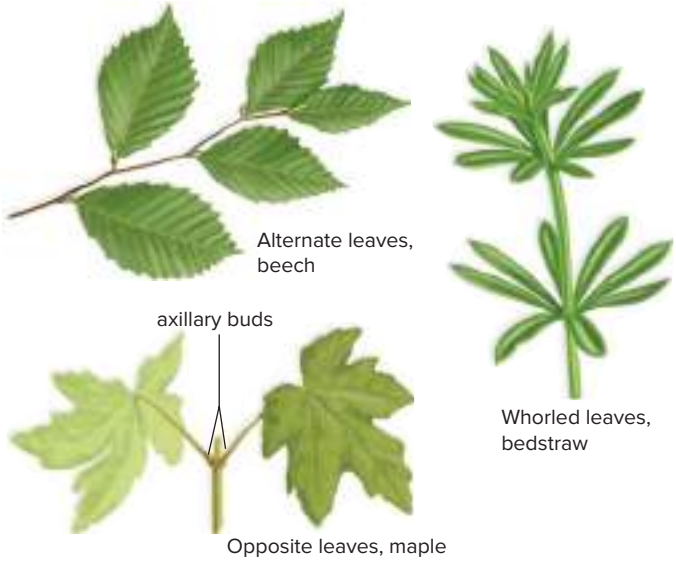
Leaves are adapted to various environmental conditions. Plants that grow in shade tend to have broad, wide leaves, and desert plants tend to have reduced leaves with sunken stomata. The spines of a cactus are actually modified leaves attached to the succulent (water-containing) stem (Fig. 24.24*a*).

An onion bulb is made up of leaves surrounding a short stem. In a head of cabbage, large leaves overlap one another. The petiole of a leaf can be thick and fleshy, as in celery and rhubarb. Climbing leaves, such as those of peas and cucumbers, are modified into tendrils, which can attach to nearby objects (Fig. 24.24*b*).

The leaves of a few plants are specialized for catching insects. A sundew has sticky trichomes that trap insects, as well as other trichomes that secrete digestive enzymes. The Venus flytrap has hinged leaves that snap shut and interlock when an insect triggers



a. Simple versus compound leaves



b. Arrangement of leaves on stem

Figure 24.23 Classification of leaves. **a.** Leaves are either simple or compound, being either pinnately compound or palmately compound. Note the one axillary bud per compound leaf. **b.** Leaf arrangement on a stem can be alternate, opposite, or whorled.

a. Cactus, *Opuntia*

b. Cucumber, *Cucumis*

Figure 24.24 Leaf diversity. **a.** The spines of a cactus plant are modified leaves that protect the fleshy stem from animal predation. **b.** The tendrils of a cucumber are modified leaves that attach the plant to a physical support. **c.** The modified leaves of the Venus flytrap serve as a trap for insect prey. When triggered by an insect, the leaf snaps shut. Once shut, the leaf secretes digestive juices, which break down the soft parts of the insect's body.
(a): NPS Photo by Robb Hannawacker; (b, c): Steven P. Lynch

c. Venus flytrap, *Dionaea*

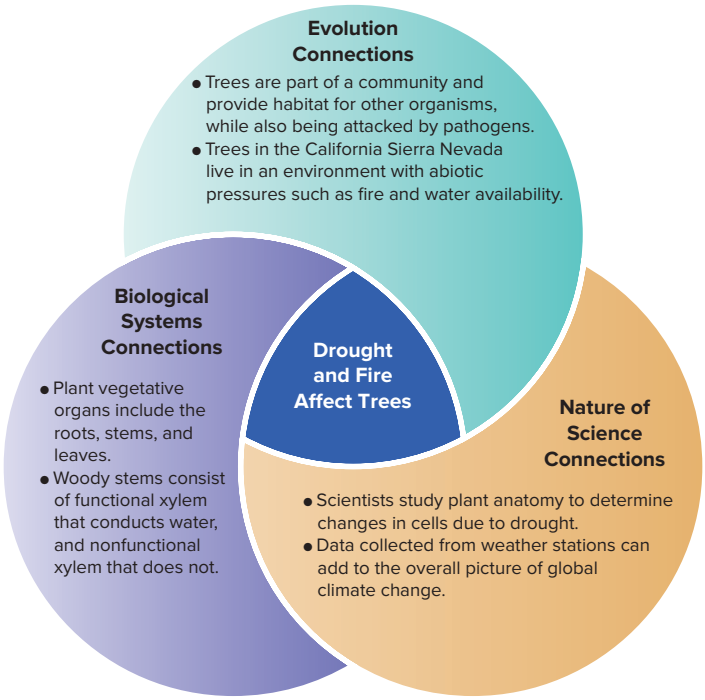
ISTUDY

sensitive trichomes that project from inside the leaves (Fig. 24.24*c*). Certain leaves of a pitcher plant resemble a pitcher and have downward-pointing hairs that lead insects into a pool of digestive enzymes secreted by trichomes. Carnivorous plants commonly grow in marshy regions, where the supply of soil nitrogen is severely limited. The digested insects provide the plants with a source of organic nitrogen.

Check Your Progress 24.5

1. Explain how the cuticle, stomata, and trichomes protect the leaf.
2. Compare the structural and functional differences between the palisade and spongy mesophyll.
3. Give examples of different types of leaves and their functions.

Connecting *the* Concepts



What Did You Learn?

24.1 Cells and Tissues of Flowering Plants

Flowering plants contain **apical meristems,** along with three types of primary **meristems.**

- The protoderm produces the **epidermal tissue.** Within the roots, the epidermal cells can produce a waxy **cuticle, root hairs,** and **trichomes,** and in the leaves the **epidermis** contains the **stomata.** In a woody stem, the epidermis is replaced by the **periderm** layer, made up of the **cork** and **cork cambium.**
- The ground meristem produces **ground tissue,** making up the bulk of a plant. Ground tissue can have living **parenchyma** cells, supportive **collenchyma** cells, and strong, dead **sclerenchyma** cells lined with **lignin.**
- The procambium produces **vascular tissue.** Vascular tissue is located in the **vascular cylinder** of roots, the **vascular bundles** in stems, and **leaf veins** in leaves.

The vascular tissue consists of **xylem** and **phloem.**

- The xylem contains two types of conducting cells: **vessel elements** and **tracheids.** Vessel elements form a continuous pipeline from the roots to the leaves. The elongated tracheids contain tapered ends, which allow water to move through pits in the end walls and sidewalls. Xylem transports water and minerals.
- The phloem contains sieve tubes made up of **sieve-tube members,** each of which is associated with a **companion cell.** The phloem transports sucrose and other organic compounds, including plant hormones.

24.2 Organs of Flowering Plants

Flowering plants have three main vegetative organs.

- The root anchors the plant and absorbs water and minerals, and also stores the products of photosynthesis. **Stems** produce new tissues, support the leaves, transport minerals from the **root system** to the leaves and back, and store plant products. The tip of the stem contains the **terminal bud,** and **nodes** are where leaves attach to stems. Between these regions are the **internodes.** Leaves are specialized for gas exchange and carry on the majority of photosynthesis within the plant. Of the vegetative organs, leaves and stems make up the **shoot system.**

Flowering plants are classified into the **monocots** and the **eudicots.** This distinction is based on the number of cotyledons in the seed; the

arrangement of the vascular tissue in the roots, stems, and leaves; the number of flowering parts; and the structure of pollen.

Plant life cycles depend on gene switches that promote vegetative growth or flower growth. A plant in which flower production is dominant and that lives only one growing season is an **annual;** a plant that produces vegetative structures for multiple growing seasons is a **perennial.**

24.3 Organization and Diversity of Roots

The root tip has three main zones: the zone of cell division (which contains the **root apical meristem**), the zone of elongation, and the zone of maturation. At the very tip of the root is a **root cap,** which protects the growing root from damage.

A cross section of an herbaceous eudicot root reveals the **epidermis,** which functions in protection; the **cortex,** which stores food; the **endodermis,** which along with the **Casparian strip** regulates the movement of water and minerals; and the vascular cylinder, which is composed of a **pericycle,** and then the vascular tissue. Within the vascular cylinder of a eudicot, the xylem appears star-shaped; the phloem is found in separate regions between the points of the star. In contrast, a monocot root has a ring of vascular tissue with alternating bundles of xylem and phloem surrounding the **pith.**

Roots tend to be either **taproots** (one dominant root) or **fibrous roots** with many similarly sized roots. Roots can be highly diverse and include specializations such as storage functions, prop roots, aerial roots, and pneumatophores. Most flowering plant roots have mycorrhizal associations with fungi, which increase water and mineral uptake, and the legume plants house nitrogen-fixing bacteria in **root nodules.**

24.4 Organization and Diversity of Stems

The activity of the **shoot apical meristem** accounts for the primary growth of a stem. The **terminal bud** contains internodes and leaf primordia at the nodes. The lengthening of the internodes allows for stem growth.

A cross section of an **herbaceous** eudicot stem reveals an epidermis, a cortex, vascular bundles in a ring, and an inner pith. Monocot stems have scattered vascular bundles and a cortex and pith that are not well defined.

Secondary growth of a woody stem is due to the **vascular cambium,** which produces new xylem and phloem on an annual basis. The **cork cambium** produces new **cork** cells when needed. Cork is part of the **bark** and replaces the epidermis in woody plants. In a cross section of a woody stem, all the tissue outside the vascular cambium is bark. It consists of secondary phloem, cork cambium, and cork. **Wood** consists of secondary xylem that builds up year after year and forms **annual rings.**

24.5 Organization and Diversity of Leaves

Leaves are the main part of the plant responsible for photosynthesis. The wide portion of a leaf is the **blade,** and the blade attaches to the stem with a **petiole.** Plants that keep their leaves year-round are **evergreens,** while plants that drop their leaves in a growing season are **deciduous.**

The ground tissue of a leaf is made up of **mesophyll** cells. These cells contain chloroplasts and undergo photosynthesis. The **palisade mesophyll** makes up the upper layer; the **spongy mesophyll** is the lower layer. The spongy mesophyll has many air spaces for gas exchange, and its lower epidermal layer contains the stomata. Leaves can be specialized to protect against herbivores, as in a cactus, or catch insects, as found in carnivorous plants.

Thinking Critically About Biology

1. If you look at a cross section of an older woody tree, how would you be able to tell which years experienced a drought, or which years had plentiful rainfall?
2. If you were given an unfamiliar vegetable, how could you tell if it was a root or a stem, based on its external features and a microscopic examination of its cross section?
3. Many plants have roots, stems, and leaves that have become modified in response to specific environmental conditions. For each modification listed next, decide whether it is a root, stem, or leaf.

stolon	Venus flytrap
cactus spine	sweet potato
tendril	baking potato
corm	carrot
rhizome	pneumatophores

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Saltwater Filter

McGraw Hill

Discuss how the vascular tissues xylem and phloem are involved in the experimental process in this video.

Making It Relevant

1. What are some of the benefits of forest fires?
2. Many tourists visit the majestic old growth forests of the Sierra Nevada. How might fire and drought affect the tourism industry?
3. Do you think research conducted in the Sierra Nevada in California would be of interest to scientists in other countries?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

25

Flowering Plants: Nutrition and Transport

Noa/iStock/Getty Images

Using Plants to Cool Buildings

Urban heat islands are extreme inner-city temperatures found in urban areas. Excess heat stems from vehicles, dark road surfaces, and concrete buildings. Metropolitan areas can be up to 10 degrees warmer than the neighboring countryside and have worse air quality. Recent advances in urban planning have begun to use *green facades* to counteract heat and pollution. Greening of facades is being developed in high-rise buildings by attaching plants to vertical spaces and rooftops. Research shows that plants positively affect the mood of city dwellers and decrease heat via shade and transpiration. Transpiration, much like sweating, provides evaporative cooling to the exterior of buildings. As the Earth's climate warms, and urban populations increase, strategically placing plants on buildings may be the norm in the future.

In this chapter, we will explore how plants gather and transport the nutrients required for their growth.

As you read through this chapter, think about the following questions:

1. Which nutrients are essential for plant growth?
2. What structures enable plants to absorb water and minerals from the soil?
3. How do plants affect climate?

Chapter Outline

25.1 Plant Nutrition and Soil
25.2 Water and Mineral Uptake
25.3 Transport Mechanisms in Plants

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 2.3 Which properties of water are essential for the conduction of water from the root system to the leaves?

Section 5.2 How do diffusion and osmosis affect how water, minerals, and nutrients move in a plant?

Section 22.3 Which organisms have evolved symbiotic relationships with flowering plants?

Following *the* Themes

CHAPTER 25 – FLOWERING PLANTS: NUTRITION AND TRANSPORT

Evolution	Vascular plants have evolved symbiotic relationships with a variety of species. These relationships have enabled them to become the dominant form of plant life on Earth today.
Nature of Science	Research has shown that plants that have nutrient deficiencies will not fully develop. Biotechnology helps with understanding how nutrients are conducted through a plant.
Biological Systems	Flowering plants use xylem and phloem to conduct nutrients and water throughout the plant body.

ISTUDY

25.1 Plant Nutrition and Soil

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the macronutrients and micronutrients that plants require.
2. Describe how and why mineral ions enter a plant through the roots.
3. Explain a simplified soil profile.

Plant nutrition is the study of how a plant gains and uses mineral nutrients from the soil. Nutrients are elements such as nitrogen and calcium and are obtained primarily as inorganic ions. In traditional farming, crop plants absorb inorganic nutrients from the soil. Then, humans and other animals consume them. Leftover crop residue and the human and animal manure return the nutrients to the soil. In essence, mineral nutrients continually cycle through all organisms and enter the biosphere predominantly through the root systems of plants. For this reason, many plant scientists refer to plants as the "miners" of Earth's crust.

Water is an essential nutrient for a plant, but much of the water entering a plant evaporates at the leaves. The photosynthetic tissues carry out photosynthesis, a process that uses carbon dioxide and gives off oxygen. Roots, like all plant organs, carry on cellular respiration, a process that uses oxygen and gives off carbon dioxide (Fig. 25.1). Just as roots require water and minerals, it is important that roots always have a continuous supply of oxygen.

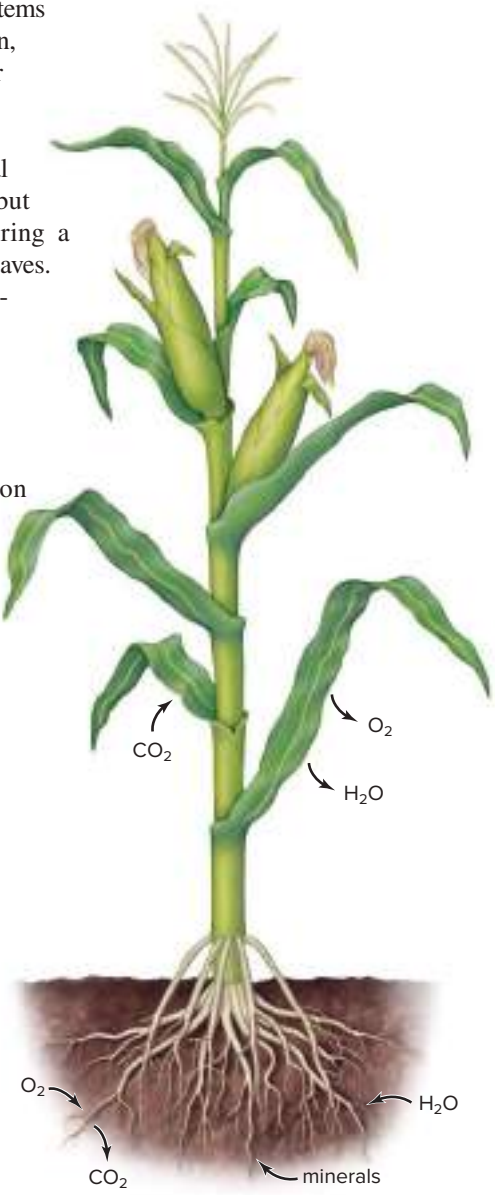


Figure 25.1 Overview of plant nutrition. Carbon dioxide, which enters leaves, and water, which enters roots, are combined during photosynthesis to form carbohydrates, with the release of oxygen from the leaves. Root cells, and all other plant cells, carry on cellular respiration, which uses oxygen and gives off carbon dioxide. Aside from the elements carbon, hydrogen, and oxygen, plants require nutrients, which are absorbed as minerals by the roots.

Essential Inorganic Nutrients

Approximately 95% of a typical plant's dry weight (weight excluding free water) is carbon, hydrogen, and oxygen. Why? Because these are the elements found in most organic compounds, such as carbohydrates. Carbon dioxide (CO_2) supplies the carbon, and water (H_2O) supplies the hydrogen and oxygen found in the organic compounds of a plant.

In addition to carbon, hydrogen, and oxygen, plants require certain other nutrients, which the roots absorb as minerals. A **mineral** is an inorganic substance, usually containing two or more elements. Why do plants need minerals from the soil? In plants, nitrogen is a major component of nucleic acids and proteins, magnesium is a component of chlorophyll, and iron is a building block of cytochrome molecules.

The major **essential nutrients** for plants are listed in Table 25.1. A nutrient is essential if (1) it has an identifiable role, (2) no other nutrient can substitute for it and fulfill the same role, and (3) a deficiency of this nutrient disrupts plant function and metabolism, causing a plant to die without completing its life cycle. Essential nutrients are divided into **macronutrients** and **micronutrients** according to their relative concentrations in plant tissue.

Beneficial nutrients are another category of elements taken up by plants. Beneficial nutrients either are required for growth or enhance the growth of a particular plant. Horsetails require silicon

Table 25.1 Some Essential Inorganic Nutrients in Plants

Elements	Form in Which Element Is Primarily Absorbed
Macronutrients	
Carbon (C)	CO_2
Hydrogen (H)	H_2O
Oxygen (O)	O_2
Phosphorus (P)	$H_2PO_4^-$, HPO_4^{2-}
Potassium (K)	K^+
Nitrogen (N)	NO_3^-, NH_4^+
Sulfur (S)	SO_4^{2-}
Calcium (Ca)	Ca^{2+}
Magnesium (Mg)	Mg^{2+}
Micronutrients	
Iron (Fe)	Fe^{2+}, Fe^{3+}
Boron (B)	BO_3^{3-}, $B_4O_7^{2-}$
Manganese (Mn)	Mn^{2+}
Copper (Cu)	Cu^{2+}
Zinc (Zn)	Zn^{2+}
Chlorine (Cl)	Cl^-
Molybdenum (Mo)	MoO_4^{2-}

ISTUDY

as a mineral nutrient, and sugar beets show enhanced growth in the presence of sodium. Nickel is a beneficial mineral nutrient in soybeans when root nodules are present. Aluminum is used by some ferns, and selenium, which is often fatally poisonous to livestock, is utilized by canola plants.

Determination of Essential Nutrients

Soil is a complex medium, which makes it difficult to figure out what missing nutrient may be affecting a plant's growth. The preferred method for determining the mineral requirements of a plant was developed at the end of the nineteenth century by the German plant physiologists Julius von Sachs and Wilhelm Knop. This method is called **hydroponics** (Gk. *hydrias,* "water"; *ponos,* "hard work"). Hydroponics allows plants to grow well if they are supplied with all the nutrients they need. This method provided proof that plants can fulfill all their needs with simply sunlight, water, and minerals (Fig. 25.2).

In order to test for specific nutrient deficiencies, an investigator omits a particular mineral from the liquid medium and observes the effect on plant growth. If growth suffers, it can be concluded that the omitted mineral is an essential nutrient. Aside from generally stunted growth, there are characteristic symptoms for some of the most common nutrient deficiencies (Fig. 25.3*a*).

Farmers and home gardeners often supplement the soil with fertilizers to avoid nutrient deficiencies. Most mixed fertilizer packaging includes three numbers, referred to as the *NPK ratio* (Fig. 25.3*b*). This ratio describes the percentage by weight of nitrogen (N), phosphorus (P), and potassium (K) contained in the fertilizer mix. For example, if a 100-pound bag of fertilizer has an NPK ratio of 18-24-6 it contains 18 pounds of nitrate, 24 pounds of phosphate (which contains phosphorus), 6 pounds of potash (which contains potassium), and 52 pounds of filler.

Figure 25.2 Hydroponics. Normally, soil provides nutrients and support, but both of these functions can be replaced in a hydroponics system to maximize growth. In this example, plants are suspended and roots are bathed in a nutrient bath.

Figure 25.3 Nutrient deficiencies and fertilizer. **a.** Characteristics of various mineral deficiencies in a tomato plant. **b.** Typical packaging for store-bought fertilizer showing the NPK ratio of 18-24-6. (b): Ricochet Creative Productions LLC

ISTUDY

Soil

Soil is a mixture of mineral particles, decaying organic material, living organisms, air, and water, which together support the growth of plants. All of the essential nutrients, the water, and most of the oxygen the plant requires are absorbed from the soil by the roots. It would not be an exaggeration to say that terrestrial life is dependent on the quality of the soil and its ability to provide plants with the nutrients they need.

Soil Formation

Soil is created when rock is weathered (broken down). Weathering first gradually breaks down rock to rubble and then to smaller particles of sand, silt, and clay. Mechanical weathering includes the forces of wind, rain, the freeze-thaw cycle of ice, and the grinding of rock on rock by the action of glaciers or river flow. Chemical weathering can come in the form of acid rain, the formation of iron oxide, or degradation by lichens and mosses, which can live on bare rock. Lichens are so effective in breaking down rock that they pose a great threat to historic castles and monuments made of stone. Over the weathered rock layer is decaying organic matter called humus. **Humus** (discussed next) supplies nutrients to plants, and the acidity of decomposition releases minerals from rock.

Building soil takes a long time. Under ideal conditions, depending on the type of parent material (the original rock) and the various processes at work, a centimeter of soil may take 15 years to develop.

The Nutritional Function of Soil

In a good agricultural soil, mineral particles, organic matter, and living organisms come together in such a way that there are spaces for air and water. It is best if the soil contains particles of different sizes, because only then can spaces for air be present. It is within these air spaces that roots take up oxygen. Ideally, water clings to particles by capillary action and does not fill the spaces. Flooding or overwatering of plants fills the air spaces with excess water. The plant is therefore deprived of oxygen, cannot undergo cellular respiration, and dies.

Humus Humus, which mixes with the top layer of soil particles, increases the benefits of soil. Humus causes soil to have a loose, crumbly texture that allows water to soak in without doing away with air spaces (Fig. 25.4). After a rain, the presence of humus decreases the chances of runoff. Humus swells when it absorbs water and shrinks as it dries. This action helps aerate soil. Plants do well in soils that contain 10–20% humus.

Soil that contains humus is nutritious for plants. Humus is acidic; therefore, it retains positively charged minerals until plants take them up. When the organic matter in humus is broken down by bacteria and fungi, inorganic nutrients are returned to plants. Although soil particles are the original source of minerals in soil, the recycling of nutrients, as you know, is a major characteristic of ecosystems.

Soil Particles Soil particles vary in size: Sand particles are the largest (0.05–2.0 mm in diameter); silt particles have an intermediate size (0.002–0.05 mm); and clay particles are the smallest (less than 0.002 mm). Soils are a mixture of these three types of particles. Because sandy soils have many large particles, they have large spaces, and the water drains readily between the particles. In contrast to sandy soils, a soil composed mostly of clay particles has small spaces, which fill completely with water.

Figure 25.4 Humus with earthworms. Decomposing organic matter in the soil is accompanied by small burrowing animals such as earthworms. Earthworms ingest organic matter and deposit nutrient-rich casts into the soil. Mikhail Kokhanchikov/Hemera/Getty Images

Most likely, you have experienced the feel of sand and clay: Sand, having no moisture, flows right through your fingers, while clay clumps together in one large mass because of its water content. The ideal soil for agriculture is called *loam* soil. It combines the aeration provided by sand with the mineral- and water-retention capacity of silt and clay.

It is also important that soils have a healthy balance of clay particles and humus. Clay and humus are negatively charged and will bind to positively charged minerals, such as calcium (Ca^{2+}) and potassium (K^{+}), preventing these minerals from being washed away by leaching. Through a process called **cation exchange,** hydrogen ions (and other positive ions) switch places with a positively charged mineral ion, and the root takes up the needed mineral nutrient (Fig. 25.5). The better this exchange, the healthier the soil. In soil science, soils are rated by an index called the cation exchange capacity (CEC), which indicates the availability of negative charge sites able to bind positive cations. As expected, the CEC of sandy soils is less than that of soils with a higher quantity of clay and humus mixed in.

Living Organisms Small plants play a major role in the formation of soil from bare rock. Due to the process of succession (see Section 45.2), larger plants eventually become dominant in certain ecosystems. The roots of larger plants penetrate soil even to the bedrock layer. This action slowly opens up soil layers, allowing water, air, and animals to follow.

A wide variety of animals dwell in the soil, at least part of the time. The largest of them, such as snakes, moles, badgers, and rabbits, disturb and mix soil by burrowing. Smaller animals, such as earthworms, ingest fine soil particles and deposit them on the surface as worm casts (see Fig. 25.4). Earthworms also loosen and aerate the soil. A range of small soil animals, including mites, springtails, and millipedes, help break down leaves and other plant remains by eating them. Soil-dwelling ants construct tremendous colonies with massive chambers and tunnels. These ants also loosen and aerate the soil.

ISTUDY

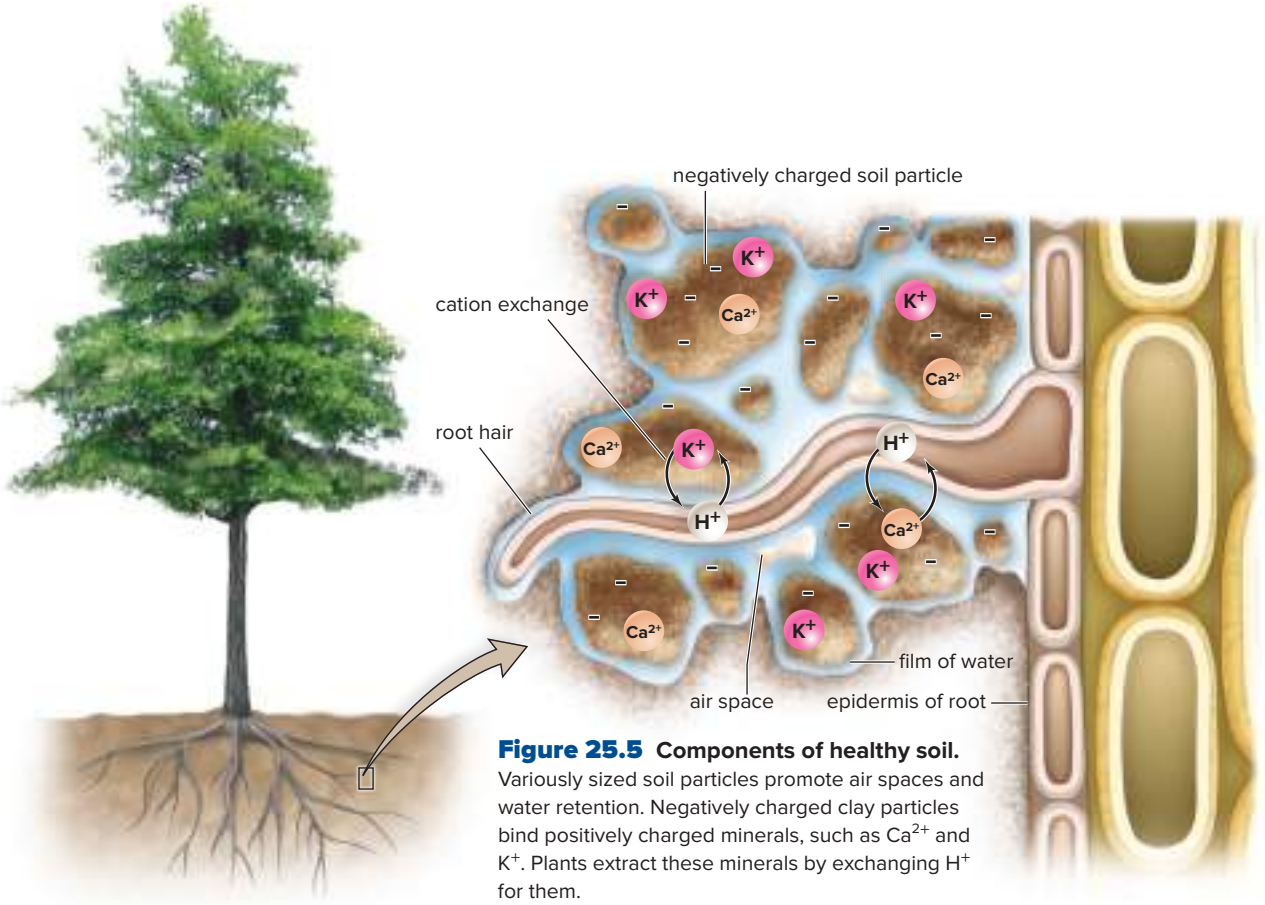


Figure 25.5 Components of healthy soil. Variously sized soil particles promote air spaces and water retention. Negatively charged clay particles bind positively charged minerals, such as Ca^{2+} and K^+. Plants extract these minerals by exchanging H^+ for them.

The microorganisms in soil, such as protozoans, fungi, and bacteria, are responsible for the final decomposition of organic remains in humus to inorganic nutrients. As mentioned, plants are unable to make use of atmospheric nitrogen (N_2), and soil bacteria play an important nutrient role because they make nitrate available to plants.

Soil Profiles

A **soil profile** is a vertical section from the ground surface to the unaltered rock below. Usually, a soil profile has parallel layers known as **soil horizons.** Mature soil generally has three horizons (Fig. 25.6).

Because the parent material (rock) and climate (e.g., temperature and rainfall) differ in various parts of the biosphere, the soil profile varies according to the particular ecosystem. Soils formed in grasslands tend to have a deep A horizon built up from decaying grasses over many years, but because of limited rain, little leaching into the B horizon has occurred. In forest soils, both the A and B horizons have enough inorganic nutrients to allow for root growth. In tropical rain forests, the A horizon is shallower than the generalized profile, and the B horizon is deeper, signifying that leaching is more extensive. Because the topsoil of a rain forest lacks nutrients, it can support crops for only a few years before it is depleted.

Soil Erosion

Soil erosion occurs when water or wind carries soil away to a new location. Soil erosion is the leading cause of water pollution in the United States and is a direct result of overgrazing, the clearing of land for urbanization and roads, and plowing. The uppermost soil layer, topsoil, is difficult to replace, yet erosion is removing it faster than ever before. At least 40% of the world's farmland is seriously degraded, and fertile soil in an area equal to the size of Texas is lost annually. Farmland erosion may be reduced by

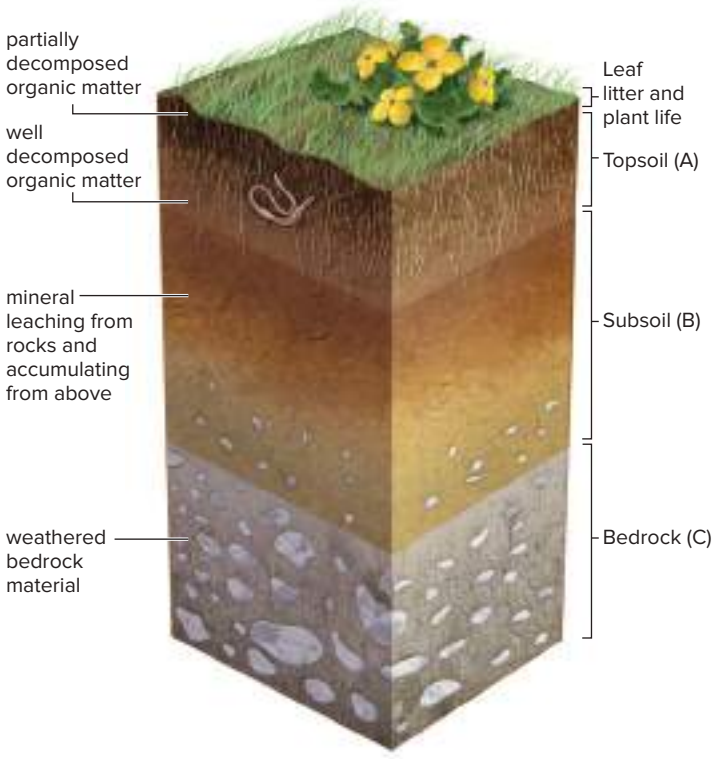


Figure 25.6 Simplified soil profile. The top layer (A horizon) contains most of the humus; the next layer (B horizon) accumulates materials leached from the A horizon; and the lowest layer (C horizon) is composed of weathered parent material. Erosion removes the A horizon, a primary source of humus and minerals in soil.

practices that are aimed at increasing soil organic matter. Effective practices include crop rotation, reduced tillage, and the use of companion crops that hold and nourish soil.

Coastal erosion, often a dramatic result of hurricane or storm surges, can reshape a coastal landscape. One example is in Pacifica, California, where urban development occurred on cliffs that were undercut and collapsed after a major storm. In recent years, there has been a net loss of beach area as a result of an intensification of storms and increases in sea level. Trees are critical for the prevention of coastal erosion. They not only hold soil in place but also break the waves from storm surges and absorb their energy.

Check Your Progress 25.1

1. Name the elements that make up the bulk of commercial fertilizers.
2. List the benefits of humus in the soil.
3. Explain how cation exchange works.
4. Identify the benefits of leaving the remains of the previous year's crops in the field to overwinter.

25.2 Water and Mineral Uptake

Learning Outcomes

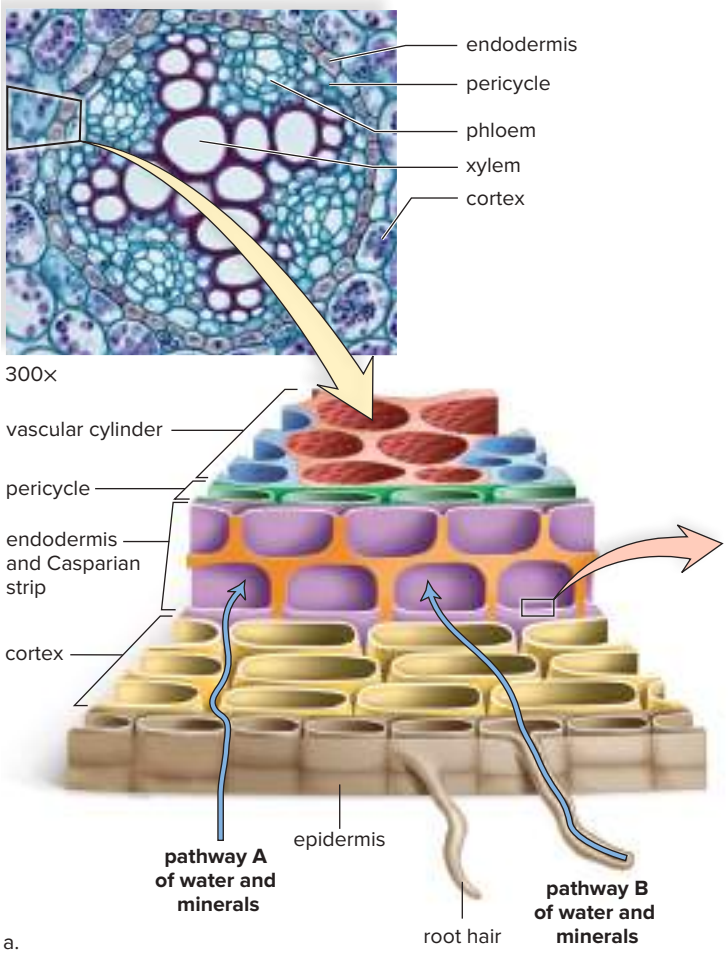
Upon completion of this section, you should be able to

1. Choose the correct order of mineral uptake across the plasma membrane within a plant cell wall.
2. Describe the mutualistic relationships that assist plants in acquiring nutrients from the soil.

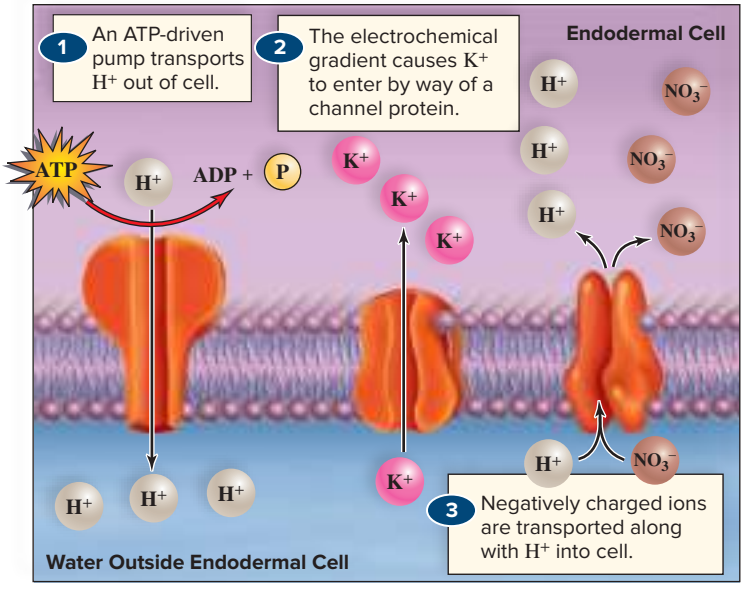
There are two main ways that water and dissolved minerals can enter a root. As seen in Figure 25.7*a,* in pathway A, water weaves its way in between cells, diffusing from the porous cell wall of one cell to the cell wall of adjacent cells. Entry of water past the cell wall is blocked at the *Casparian strip,* a waxy layer that surrounds endodermal cells. Here, the water is forced to enter endodermal cells through the plasma membrane. In pathway B, water travels from the plasma membrane of one cell to the plasma membrane of another cell, connected by openings called plasmodesmata (see Fig. 5.15). Regardless of the pathway, water enters root cells when *osmotic pressure* in the root tissues is lower than that of the soil solution. That is to say, if there is more water outside the root, and less water inside the root, then water moves in by osmosis, causing osmotic pressure.

Mineral Uptake

In contrast to the passive movement of water, minerals can be taken up by passive or active transport. Plants possess an astonishing ability to concentrate minerals until they are many times more concentrated in the plant than in the surrounding medium. The



a.



b.

Figure 25.7 Water and mineral uptake. **a.** Pathways of water and minerals. Water and minerals can travel via porous cell walls but then must enter endodermal cells because of the Casparian strip (pathway A). Alternatively, water and minerals can enter root hairs and move from cell to cell (pathway B). **b.** Transport of minerals across an endodermal plasma membrane. 1 An ATP-driven pump removes hydrogen ions (H^+) from the cell. 2 This establishes an electrochemical gradient that allows potassium (K^+) and other positively charged ions to cross the membrane via a channel protein. 3 Negatively charged mineral ions (e.g., NO_3^-) can cross the membrane by way of a carrier when they co-transport with hydrogen ions (H^+), which are diffusing down their concentration gradient. (photo): Ed Reschke/Stone/Getty Images

ISTUDY

concentration of certain minerals in roots is as much as 10,000 times greater than in the surrounding soil. The presence of the Casparian strip prevents the backflow of minerals and allows the plant to maintain a higher mineral concentration in root xylem than can be found in the surrounding soil.

By what mechanism do minerals cross plasma membranes? As it turns out, the energy of ATP is involved, but only indirectly. Recall that plant cells absorb minerals in the ionic form: Nitrogen is absorbed as nitrate (NO_3^-), phosphorus as phosphate (HPO_4^{2-}), potassium as potassium ions (K^+), and so forth. Ions cannot cross the plasma membrane because they are unable to enter the nonpolar portion of the lipid bilayer. Plant physiologists know that plant cells expend energy to actively take up and concentrate mineral ions. If roots are deprived of oxygen or are poisoned so that cellular respiration cannot occur, mineral ion uptake is diminished.

As shown in Figure 25.7*b,* a plasma-membrane pump, called a proton pump, hydrolyzes ATP and uses the energy released to transport hydrogen ions (H^+) out of the cell. The result is an electrochemical gradient that drives positively charged ions such as K^+ through a channel protein into the cell. Negatively charged mineral ions such as NO_3^- and HPO_4^{2-} are transported, along with H^+, by carrier proteins. Because H^+ is moving down its concentration gradient, no energy is required. Notice that this model of mineral ion transport in plant cells is based on *chemiosmosis,* the establishment of an electrochemical gradient to perform work. The Nature of Science feature, "Plants Can Be Used for Cleaning and Discovery of Minerals," explains how this ability of plants can be exploited for environmental cleanup and mineral exploration.

Following their uptake by root cells, minerals move into xylem and are transported into leaves by the upward movement of water. Along the way, minerals can exit xylem and enter the cells that require them. Some eventually reach leaf cells. In any case, minerals must again cross a selectively permeable plasma membrane when they exit xylem and enter living cells.

Adaptations of Roots for Mineral Uptake

Two mutualistic relationships assist roots in obtaining mineral nutrients. Root nodules are involved in a mutualistic relationship with bacteria, and mycorrhizae have a mutualistic relationship with fungi.

Bacterial nitrogen fixation is responsible for most of the conversion of nitrogen from the air (N≡N) into ammonium (NH_4^+) and is therefore the first step in the introduction of nitrogen into ecological cycles (see Fig. 45.24). Most nitrogen-fixing bacteria live independently in the soil, but some do form symbiotic associations with a host plant. In these associations, the host plant provides food and shelter, while the bacteria provide nitrogen in a form the plant can use.

The most common types of symbiosis occur between various genera of bacteria collectively called **rhizobia** and plants of the legume family, such as beans, clover, and alfalfa. Nitrogen fixation is an energy-intensive process that requires special conditions for the bacteria's enzymes. One of those conditions is an anaerobic environment, as the presence of oxygen disrupts the nitrogen-fixing process. For this reason, nitrogen-fixing bacteria live in organs called **root nodules** (Fig. 25.8), where oxygen levels are maintained high enough for cellular respiration but low enough so as to not inactivate important nitrogen-fixing enzymes. In addition, large-scale farming of legume crops often depletes the native populations of rhizobia, and so farmers must often supplement their crops with pellets containing these bacteria.

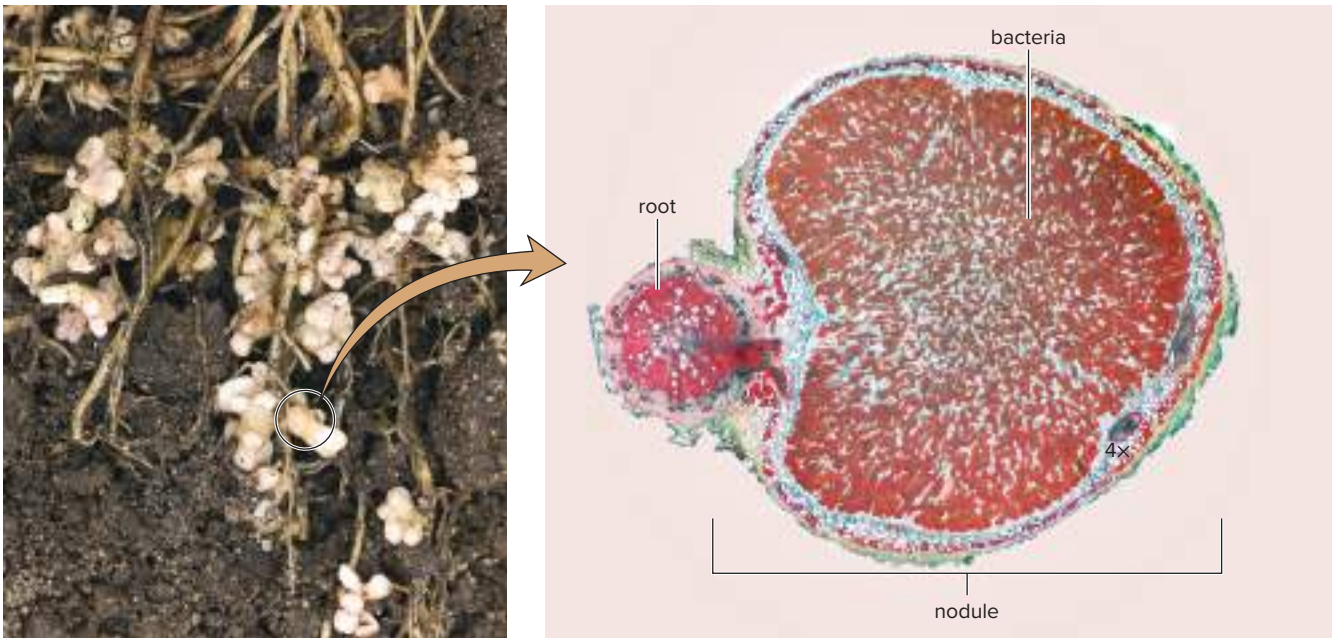


Figure 25.8 Root nodules. Nitrogen-fixing bacteria live in nodules on the roots of plants, particularly legumes. (root nodules): Dr Jeremy Burgess/Science Source; (nodule micrograph): Garry DeLong/Science Source

ISTUDY

THEME Nature of Science

Plants Can Be Used for Cleaning and Discovery of Minerals

Phytoremediation uses plants such as mulberry, poplar, and canola to clean up environmental pollutants. The genetic makeup of these plants allows them to absorb, store, degrade, or transform substances that normally kill or harm other plants and animals. "It's an elegantly simple solution to pollution problems," says Louis Licht, who runs Ecolotree, an Iowa City phytoremediation company.

The idea behind phytoremediation is not new; scientists have long recognized certain plants' abilities to absorb and tolerate toxic substances. But the idea of using these plants on contaminated sites has gained support just in the last 25 years. Different plants work on different contaminants. The mulberry bush, for instance, is effective on industrial sludge; some grasses attack petroleum wastes; and sunflowers remove lead.

The plants clean up sites in different ways depending on the substance involved. If it is an organic contaminant, such as spilled oil, the plants, or the microbes around their roots, break down the substance. The remnants can be either absorbed by the plant or left in the soil or water. For an inorganic contaminant, such as cadmium or zinc, the plants absorb the substance and trap it. The plants must then be harvested and disposed of, or processed to reclaim the trapped contaminant.

Poplars Take Up Excess Nitrates

Most trees planted along the edges of farms are intended to act as a wind break, but another use of poplars is to remove excess minerals from runoff. The poplars act as vacuum cleaners, sucking up nitrate-laden runoff from a fertilized cornfield before this runoff reaches a nearby brook—and perhaps other waters (Fig. 25A). Nitrate runoff into the Mississippi River from Midwest farms is a major cause of the large "dead zone" of oxygen-depleted water that develops each summer in the Gulf of Mexico.

Canola Plants Take Up Selenium

Canola plants (*Brassica rapus* and *B. napa*) are grown in California's San Joaquin Valley to soak up excess selenium in the soil to help prevent an environmental catastrophe like the one that occurred there in the 1980s. Back then, irrigated farming caused naturally occurring selenium to rise to the soil surface. When excess water was pumped onto the fields, some selenium flowed off into drainage ditches, eventually ending up in Kesterson National Wildlife Refuge. The selenium in ponds at the refuge accumulated in plants and fish and subsequently deformed and killed waterfowl.

Eucalyptus Trees Reveal Hidden Gold

The ability of plants to take up minerals, called *biogeochemical absorption,* may not only clean up toxic messes but also serve as a valuable beacon for desirable minerals. Gold, for example, is an element for which worldwide discoveries are down by 45%. Normally, prospectors drill in suspected areas, test soil samples, and disturb the ecosystem in promising areas with no guarantees of success.

In Australia, scientists have found that *Eucalyptus* trees growing over deep deposits of gold have leaves with high concentrations of this sought-after element. Gold is toxic to plants, and when drawn up in the soil through the xylem, gold accumulates in the outermost regions of plants—their leaves and bark. Leaves and bark sampled from various eucalyptus trees were taken to the lab, where X-ray and chemical analyses revealed the levels of gold. The Australian scientists were able to show that the trees growing *directly* over a 35-meter-deep gold deposit were the samples with the unusually high gold readings.

Questions to Consider

1. What happens to the pollutants when the plant dies?
2. Why would one plant be more adapted at absorbing a particular pollutant than another plant?
3. What are the ecological and economic benefits of using plants for gold prospecting?

Figure 25A Poplar trees cleaning up nitrogen. Poplars are able to remove large amounts of nitrates from runoff. USDA Natural Resources Conservation Service/Photograph by Erwin Cole

ISTUDY

The second type of mutualistic relationship, called **mycorrhizae,** involves fungi and almost all plant roots (Fig. 25.9; also see Section 22.3). Only a small minority of plants do not have mycorrhizae, and these plants are most often limited as to the environment in which they can grow. The fungus increases the surface area available for mineral and water uptake and breaks down organic matter in the soil, releasing nutrients the plant can use. In return, the root furnishes the fungus with sugars and amino acids. Plants are extremely dependent on mycorrhizae. Orchid seeds, which are quite small and contain limited nutrients, do not germinate until a mycorrhizal fungus has invaded their cells.

Other means of acquiring nutrients also occur. Parasitic plants, such as dodders, broomrapes, and pinedrops, send out rootlike projections, called haustoria, that tap into the xylem and phloem of the host stem (Fig. 25.10*a*). Carnivorous plants, such as the Venus flytrap and sundew, obtain some nitrogen and minerals when their leaves capture and digest insects (Fig. 25.10*b*).

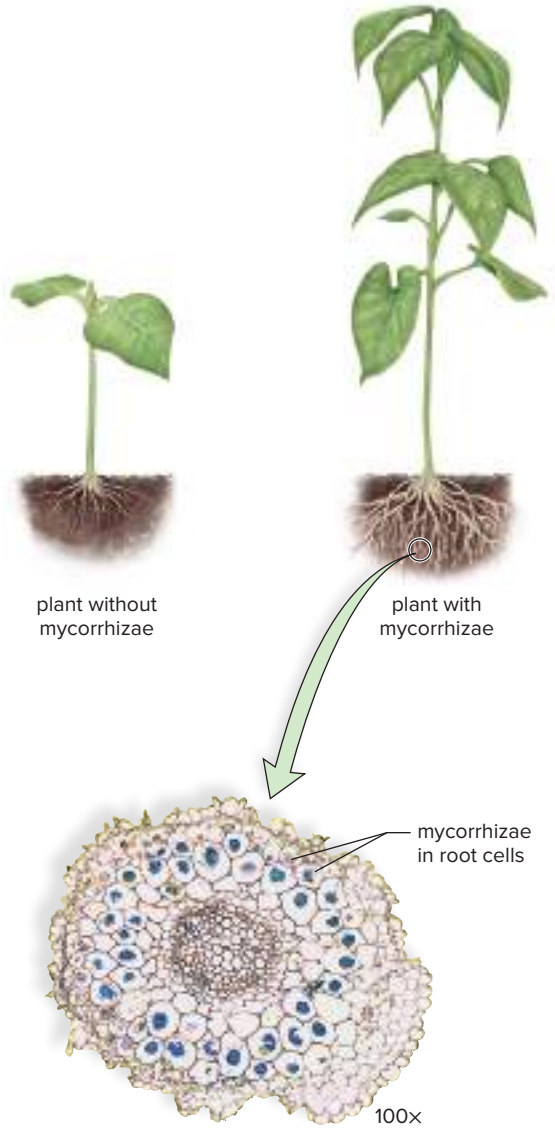


Figure 25.9 Mycorrhizae. Plant growth is better when mycorrhizae are present. (photo): Dr. Keith Wheeler/Science Source

a. Dodder, *Cuscuta* sp.

b. Cape sundew, *Drosera capensis*

Figure 25.10 Other ways plants acquire nutrients.
a. Some plants, such as the dodder, are parasitic. **b.** Other plants, such as the sundew, are carnivorous. (a): Kevin Schafer/The Image Bank/Getty Images; (b): (plant): blickwinkel/Alamy Stock Photo; (leaf inset): Dr. Jeremy Burgess/Science Source

Check Your Progress 25.2

1. Explain the role of the endodermis and Casparian strip in concentrating minerals in a plant.
2. Describe the relationship of nitrogen-fixing bacteria with a host plant.
3. Explain how both partners benefit from a mycorrhizal association.

ISTUDY

25.3 Transport Mechanisms in Plants

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the relationship between water potential and root pressure.
2. Identify the properties of water that influence the upward movement of water in flowering plants.
3. Explain how environmental factors influence the opening and closing of stomata.
4. List the correct sequence of events for the movement of water in xylem and sucrose in phloem.

Flowering plants are well-adapted to living in a terrestrial environment. Their leaves, which carry on photosynthesis, are positioned to catch the rays of the sun because they are held aloft by the stem (see Fig. 25.12). Carbon dioxide enters leaves at the stomata, but water, the other main requirement for photosynthesis, is absorbed by the roots. Water must be transported from the roots through the stem to the leaves.

Reviewing Xylem and Phloem Structure

Vascular plants have a transport tissue, called *xylem,* that moves water and minerals from the roots to the leaves. Xylem, with its strong-walled, nonliving cells, gives trees much-needed internal support. Xylem contains two types of conducting cells: tracheids and vessel elements (Fig. 25.11*a*).

- *Tracheids* are tapered at both ends. The ends overlap with those of adjacent tracheids, and pits allow water to pass from one tracheid to the next.
- *Vessel elements* are long and tubular with perforation plates at each end. Vessel elements placed end to end form a completely hollow pipeline from the roots to the leaves.

The process of photosynthesis results in sugars, which are used as a source of energy and building blocks for other organic molecules throughout a plant. *Phloem* is the type of vascular tissue that transports organic nutrients to all parts of the plant. Roots buried in the soil cannot carry on photosynthesis, but they require a source of energy so they can carry out cellular metabolism. In flowering plants, phloem consists of two types of cells: sieve-tube members and companion cells (Fig. 25.11*b*).

- *Sieve-tube members* are the conducting cells of phloem. The end walls are called sieve plates and have numerous pores; strands of cytoplasm extend from one sieve-tube member to another through the pores. Sieve-tube members lack nuclei.
- *Companion cells,* which do have nuclei, provide proteins to sieve-tube members.

In this way, sieve-tube members form a continuous *sieve tube* for organic nutrient transport throughout the plant.

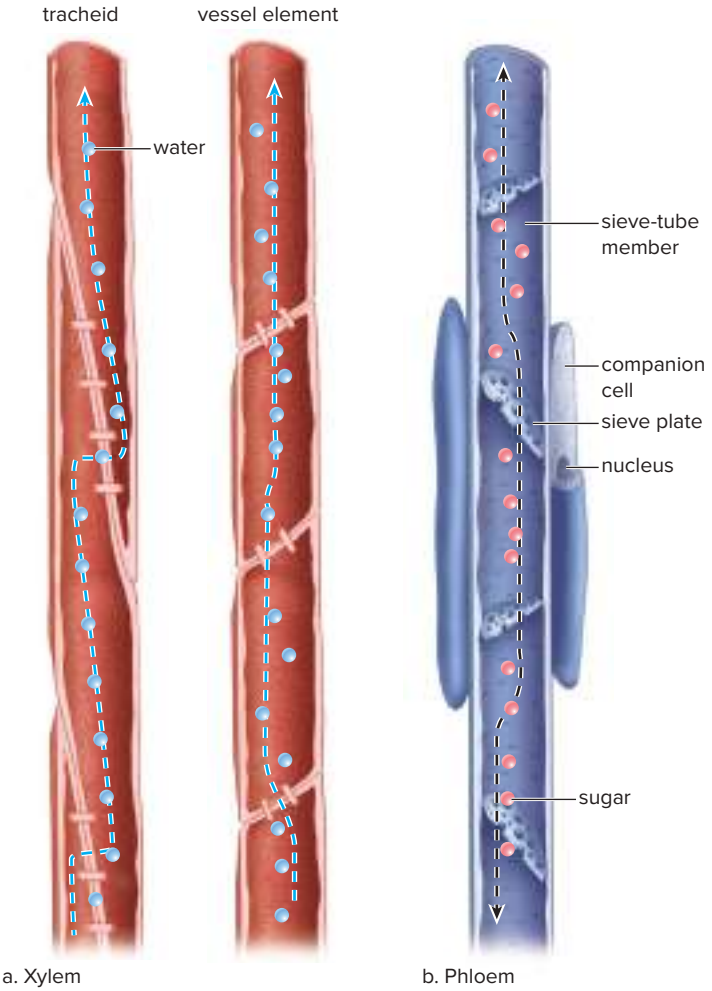


Figure 25.11 Organization of conducting cells in xylem and phloem. The specialized cells making up xylem and phloem form a series of interconnected and parallel cells that form a pathway for the movement of water and carbohydrates in a plant. Xylem cells (**a**) move water from the bottom to the top. Phloem cells (**b**) can move sucrose and other materials in any direction.

The Role of Water Potential

Knowing that vascular plants are structured in a way that allows materials to move from one part to another does not tell us the mechanisms by which these materials move. Plant physiologists have performed numerous experiments to determine how water and minerals rise to the tops of very tall trees in xylem, and how organic nutrients move in the opposite direction in phloem. We might expect that these processes are mechanical and based on the properties of water, because water is a large part of both *xylem sap* and *phloem sap.*

In living systems, water molecules diffuse freely across plasma membranes from the area of higher concentration to the area of lower concentration. Plant scientists prefer describing the movement of water in terms of **water potential:** Water always flows passively from the area of higher water potential to the area of lower water potential (Fig. 25.12). As can be seen in the Biological Systems feature, "The Concept of Water Potential," water potential involves water pressure in addition to osmotic pressure.

ISTUDY

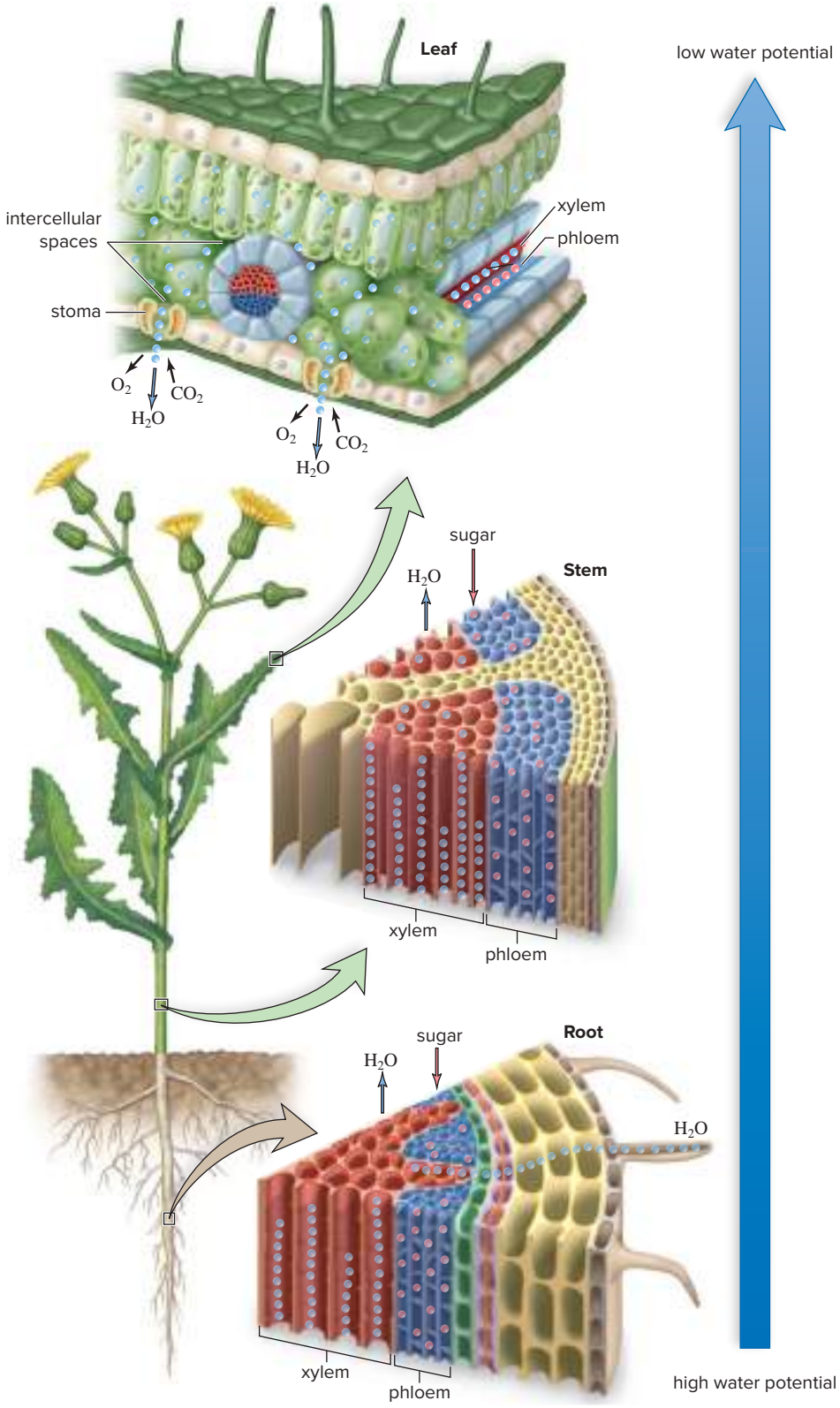


Figure 25.12 Plant transport and water potential. Vascular tissue in plants includes xylem, which transports water and minerals from the roots to the leaves, and phloem, which transports organic nutrients, often in the opposite direction. Notice that xylem and phloem are continuous from the roots through the stem to the leaves, which are the vegetative organs of a plant. Water potential is higher at the roots as water moves in by osmosis. Water potential is lower at the leaves as water escapes through stomata.

ISTUDY

THEME Biological Systems

The Concept of Water Potential

As you learned in Section 6.1, potential energy is stored energy. Potential energy can exist in an object's position, such as that of a boulder at the top of a hill (mechanical energy), or in chemical bonds, such as the bonds between phosphate groups in ATP (chemical energy). Kinetic energy is the energy of motion; it is energy actively engaged in doing work. A boulder rolling down a hillside is exhibiting kinetic energy, as is an enzyme reaction that breaks a bond, converting ATP to ADP and releasing energy in the process.

Water potential is defined as the mechanical energy of water. Just like the boulder, water at the top of a waterfall has a higher water potential than water at the bottom of the waterfall. As illustrated by this example, water moves from a region of higher water potential to a region of lower water potential (Fig. 25B).

In terms of cells, two factors usually determine water potential, which in turn determines the direction in which water moves across a plasma membrane. These factors concern differences in:

- Water pressure across a membrane
- Solute concentration across a membrane

Pressure potential is the effect that pressure has on water potential. Water moves across a membrane from an area of higher pressure to an area of lower pressure. The higher the water pressure, the higher the water potential, and vice versa. Pressure potential is the concept that best explains the movement of sap in xylem and phloem.

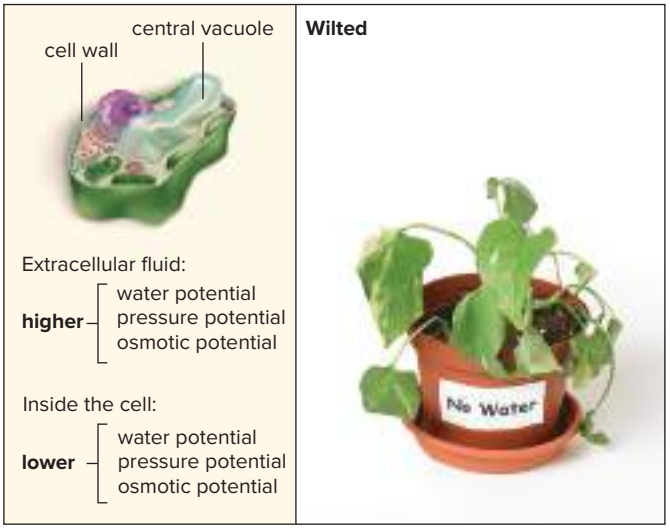
Osmotic potential, in contrast, takes into account the effects of solutes on the movement of water. The presence of solutes restricts the movement of water, because water tends to engage in molecular interactions with solutes, such as hydrogen bonding. Water therefore tends to move across a membrane from an area of lower solute concentration to an area of higher solute concentration. The lower the concentration of solutes, the higher the water potential, and vice versa.

It is not surprising that increasing the water pressure can counter the effects of solute concentration. This situation is common in plant cells. As water enters a plant cell by osmosis because of the higher solute concentration, water pressure increases inside the cell—a plant cell has a strong cell wall that allows water pressure to build up. When the pressure potential inside the cell balances the osmotic potential outside the cell, the flow of water in and out becomes the same.

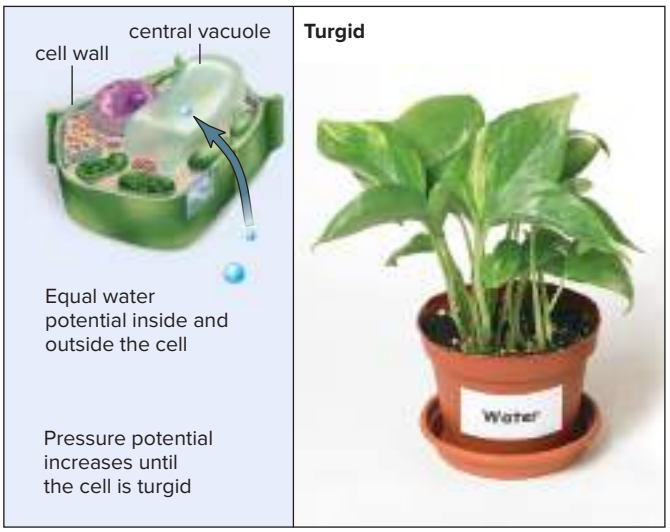
Pressure potential that increases due to osmosis is often called *turgor pressure*. Turgor pressure is critical, because plants depend on it to maintain the turgidity of their bodies (Fig. 25B). The cells of a wilted plant have insufficient turgor pressure, and the plant droops as a result.

Questions to Consider

1. What variables will restrict the movement of water across the plasma membrane?
2. What structures are necessary for a plant to maintain turgidity?
3. What environmental conditions might cause a plant to lose its turgidity?



a. Plant cells need water and are wilted



b. Plant cells have water and are turgid

Figure 25B Water potential and turgor pressure. Water flows from an area of higher water potential to an area of lower water potential. **a.** The cells of a wilted plant have a lower water potential; therefore, water enters the cells. **b.** Equilibrium is achieved when the water potential is equal inside and outside the cell. Cells are now turgid, and the plant is no longer wilted. (both photos): Ken Cavanagh/McGraw Hill

ISTUDY

Chemical properties of water are also important in the movement of xylem sap. The polarity of water molecules and the hydrogen bonding between water molecules allow water to fill xylem cells.

Water Transport

Recall that minerals accumulate at high concentrations beyond the endodermis in the root xylem tissue. This solute concentration difference results in the continuous movement of water into the root, creating **root pressure.** For example, if the stem of a young plant is cut, the cut end will often "leak" xylem sap. If a glass tube is sealed over the cut end, the sap will rise, because root pressure raises the water level in the glass tube (Fig. 25.13*a*). During the day, root pressure is not as obvious, because water is being drawn out from the leaves. At night, water continues to enter the roots, but evaporation slows down at the surface of the leaves. This results in a phenomenon called guttation. **Guttation** occurs when drops of water are forced out of vein endings along the edges of leaves (Fig. 25.13*b*). This morning "dew" effect is the direct result of root pressure.

Cohesion-Tension Model of Xylem Transport

Water that enters xylem must be transported against gravity to all parts of the plant. Transporting water can appear to be a daunting task, especially for plants such as redwood trees, which can exceed 90 m (almost 300 ft) in height.

The **cohesion-tension model** of xylem transport, outlined in Figure 25.14, describes how water and minerals travel upward in xylem, yet requires no expenditure of energy by the plant. To understand how it works, one must start at the bottom of Figure 25.14 in the soil. Recall that there is a higher water potential in the soil and a lower water potential in the plant. Water will move into the root by osmosis. All of the water entering roots creates root pressure, which is helpful for the upward movement of water but is not nearly enough to get it all the way up to the leaves—especially in a tall tree.

Transpiration is the phenomenon that explains how water can completely resist gravity and travel upward. Focusing on the top of the tree (Fig. 25.14), notice the water molecules escaping from the spongy mesophyll and into the air through stomata. The key is that it is not just one water molecule escaping but a chain of water molecules. The movement of water through xylem is like drinking water from a straw. Drinking exerts pressure on the straw, and a chain of water molecules is drawn upward. Water molecules are polar and "stick" together with hydrogen bonds. Water's ability to stay linked in a chain is called **cohesion,** and its ability to stick to the inside of a straw or a xylem vessel is **adhesion.**

In plants, evaporation of water at the leaves exerts *tension,* which pulls on a chain of water molecules. Transpiration is the constant tugging or pulling of the **water column** from the top due to evaporation. Cohesion of water molecules and adhesion to the inside of a xylem vessel facilitate this process. As transpiration occurs, the water column is pulled upward—first within the leaf, then from the stem, and finally from the roots. In addition, unlike other plant cells, xylem vessels offer a simple pipeline, with reinforced lignified walls and low resistance for the movement of water.

a.

b.

Figure 25.13 Root pressure and guttation. **a.** Root pressure, as measured in this experiment, is a positive pressure potential caused by the entrance of water into root cells. **b.** Drops of guttation water on the edges of a leaf. Guttation, which occurs at night, is caused by root pressure. Often, guttation is mistaken for early morning dew. (a): Martyn F. Chillmaid/Science Source; (b): H. Schmidbauer/blickwinkel/age fotostock

The total amount of water a plant loses through transpiration over a long period of time is surprisingly large. At least 90% of the water taken up by roots is eventually lost at the leaves. A single corn plant loses between 135 and 200 liters of water through transpiration during the growing season. A typical tree loses 400 liters of water per day! On a global scale, plant transpiration has enormous effects on climate. For example, an estimated one-half to three-quarters of the rainfall received by the Amazon rain forest originates from water vapor of transpiring plants, often visible as a mist (Fig. 25.15). The evaporation of large amounts of water from plant surfaces dissipates heat and explains how plants cool themselves and their environments.

ISTUDY

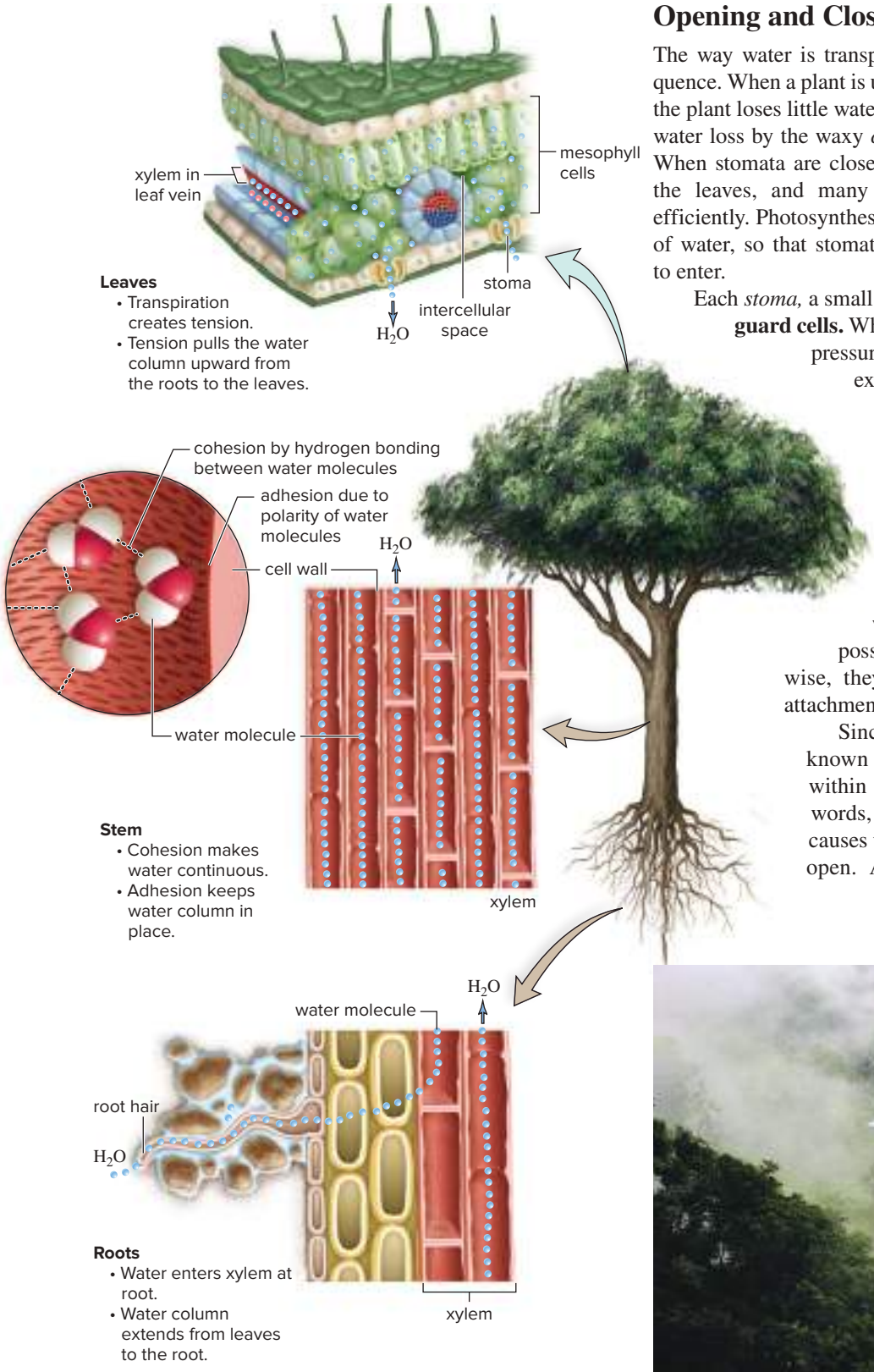


Figure 25.14 Cohesion-tension model of xylem transport. Tension created by evaporation (transpiration) at the leaves pulls water along the length of the xylem—from the roots to the leaves.

Opening and Closing of Stomata

The way water is transported in plants has an important consequence. When a plant is under water stress, the stomata close. Now the plant loses little water, because the leaves are protected against water loss by the waxy *cuticle* of the upper and lower epidermis. When stomata are closed, however, carbon dioxide cannot enter the leaves, and many plants are unable to photosynthesize efficiently. Photosynthesis, therefore, requires an abundant supply of water, so that stomata remain open, allowing carbon dioxide to enter.

Each *stoma,* a small pore in the leaf epidermis, is bordered by **guard cells.** When water enters the guard cells and turgor pressure increases, the stoma opens; when water exits the guard cells and turgor pressure decreases, the stoma closes. Notice in Figure 25.16 that the guard cells are attached to each other at their ends and that the inner walls are thicker than the outer walls. When water enters, a guard cell's radial expansion is restricted because of cellulose microfibrils in the walls, but lengthwise expansion of the outer walls is possible. When the outer walls expand lengthwise, they buckle out from the region of their attachment, and the stoma opens.

Since about 1968, plant physiologists have known that potassium ions (K^+) accumulate within guard cells when stomata open. In other words, active transport of K^+ into guard cells causes water to follow by osmosis and stomata to open. Another interesting observation is that

Figure 25.15 Plant-transpired mist rising from a tropical rain forest. Plants transpire enormous amounts of water, creating water vapor. Water vapor is an important source of rainfall, and the process of evaporative cooling is responsible for cooling the plants and affecting climate. Adalberto Rios Szalay/Sexto Sol/Stockbyte/Getty Images

ISTUDY

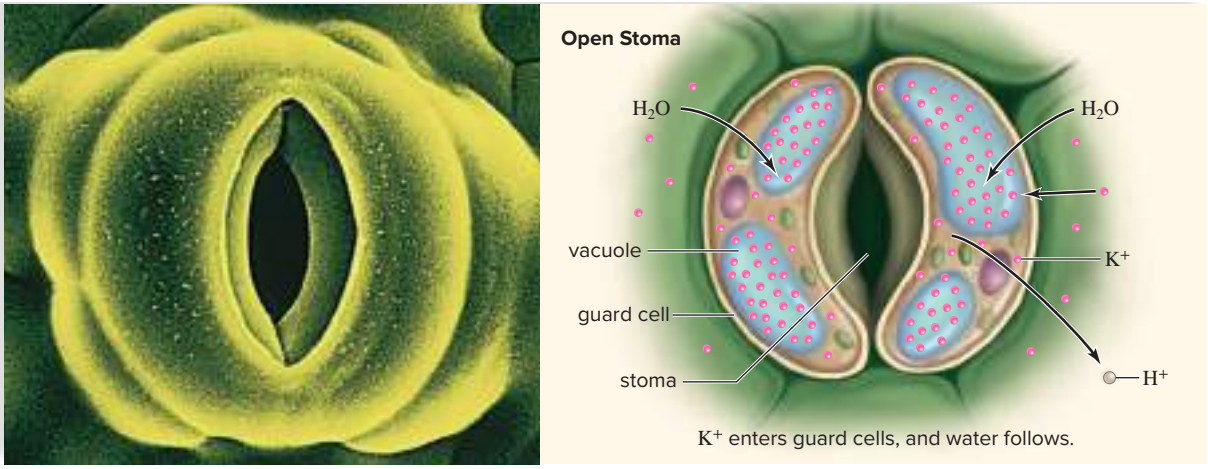


Figure 25.16 Opening and closing of stomata. **a.** A stoma opens when turgor pressure increases in guard cells due to the entrance of K^+, followed by the entrance of water. **b.** A stoma closes when turgor pressure decreases due to the exit of K^+, followed by the exit of water. (both photos): Dr Jeremy Burgess/Science Source

hydrogen ions (H^+) accumulate outside guard cells as K^+ moves into them. A proton pump run by the hydrolysis of ATP transports H^+ to the outside of the cell. This establishes an electrochemical gradient that allows K^+ to enter by way of a channel protein (see Fig. 25.7*b*).

The blue-light component of sunlight has been found to regulate the opening and closing of stomata. Evidence suggests that a flavin pigment absorbs blue light, and then this pigment sets in motion the cytoplasmic response that leads to activation of the proton pump. In a similar way, a receptor in the plasma membrane of guard cells could bring about the inactivation of the pump when carbon dioxide (CO_2) concentration rises, as might happen when photosynthesis ceases. Abscisic acid (ABA), which is produced by cells in wilting leaves, can also cause stomata to close (see Section 26.1). Although photosynthesis cannot occur, water is conserved.

If plants are kept in the dark, stomata open and close about every 24 hours, as though they were responding to the presence of sunlight in the daytime and the absence of sunlight at night. The implication is that some sort of internal biological clock must be keeping time. Circadian rhythms (behaviors that occur nearly every 24 hours) and biological clocks are areas of intense investigation. Other factors that influence the opening and closing of stomata include temperature, humidity, and stress.

Organic Nutrient Transport

Mosses, described in Section 23.2, are short, ancient plants with no vascular tissue. Water, minerals, and the products of photosynthesis all move passively from one cell to the next. As plants evolved and moved onto land, there were many challenges for survival. Plants evolved tissues and organs to acquire water and minerals and to collect sunlight for photosynthesis. As plants grew taller, the shoot system and the root system became increasingly separated, and other systems (xylem and phloem tissue) evolved for long-distance travel. Phloem, specifically, is the tissue that translocates (transfers) the products of photosynthesis. Sugar, produced in mature leaves, moves to areas of development and storage, such as young leaves, fruit, and roots.

ISTUDY

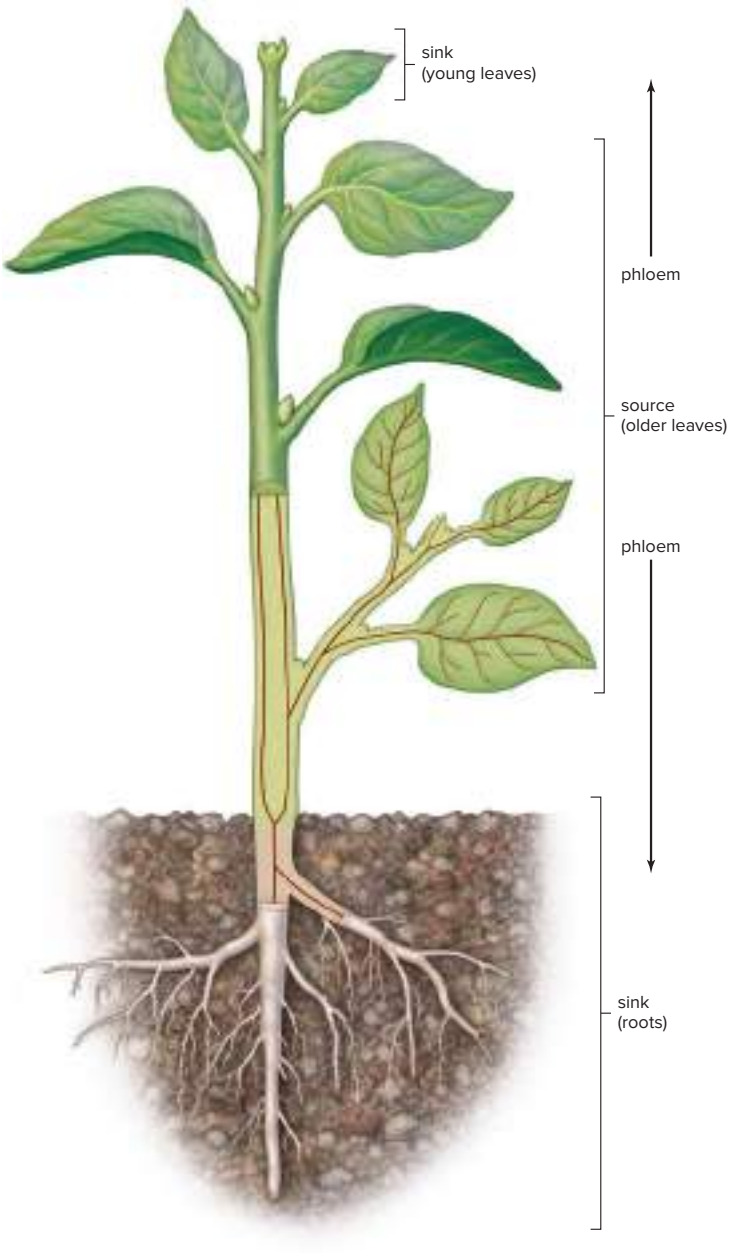


Figure 25.17 The movement of phloem from source to sink. An illustration of an *Arabidopsis* plant, showing the movement of phloem sap from the source leaf to sink leaves and roots.

Role of Phloem

Phloem tissue is typically found external to the xylem in vascular tissues (see Fig. 24.14). In plants with woody stems, phloem makes up the inner bark. It is the location of phloem in woody stems that helps explain why **girdling** a tree will cause the tree to die. If a strip of bark is removed from around the tree, then the lower half of the tree is cut off from its supply of sugar.

Interestingly, phloem sap does not move exclusively upward or downward, and it is not defined by gravity, as water is in xylem. In essence, phloem can travel in any direction, but these directions have a beginning, called the **source** (where sugars originate), and an end, or **sink** (where the sugars are unloaded) (Fig. 25.17).

Radioactive tracer studies with carbon 14(^{14}C) have confirmed that phloem transports organic nutrients from source to sink. When ^{14}C-labeled carbon dioxide (CO_2) is supplied to mature leaves, radioactively labeled sugar is soon found moving down the stem into areas that cannot undergo photosynthesis, such as immature leaves and the roots.

As expected, the liquid traveling in phloem is mostly water and sucrose, but other substances travel in the phloem as well, such as amino acids, hormones, RNA, and proteins involved in defense and protection.

Pressure-Flow Model of Phloem Transport

Figure 25.17 shows one of many experiments that plant scientists have conducted, proving that what starts at a source can end up in a sink. The question now is, how does phloem sap move from the source to the sink? The translocation of sugar can be explained using the **pressure-flow model.** As mentioned earlier, phloem can travel in any direction. For simplicity, Fig. 25.19 will later describe the movement of phloem sap from leaves (the source) to roots (the sink).

Photosynthesizing leaves make sugar, and that sugar is actively transported from cells in the leaf mesophyll into the sieve tubes of phloem. Recall that, like xylem, phloem is a continuous pipeline throughout the plant. Active transport, or *loading,* of sugar into phloem is dependent on an electrochemical gradient established by a proton pump. Sugar is co-transported with hydrogen ions (H^+) that are moving down their concentration gradient (Fig. 25.18).

Next, high concentrations of sugar in the sieve tubes cause water to flow in by osmosis (Fig. 25.19). Like turning the nozzle on a hose, there is an increase in positive pressure as water flows in. The sugar (sucrose) solution, under massive *pressure* at the source, is forced to move by bulk *flow* to areas of lower pressure at the sink, like a root. This step highlights where the pressure-flow model got its name, and indeed, phloem has been measured moving at a velocity of 1 m an hour. The same distance with passive diffusion would take 30 years!

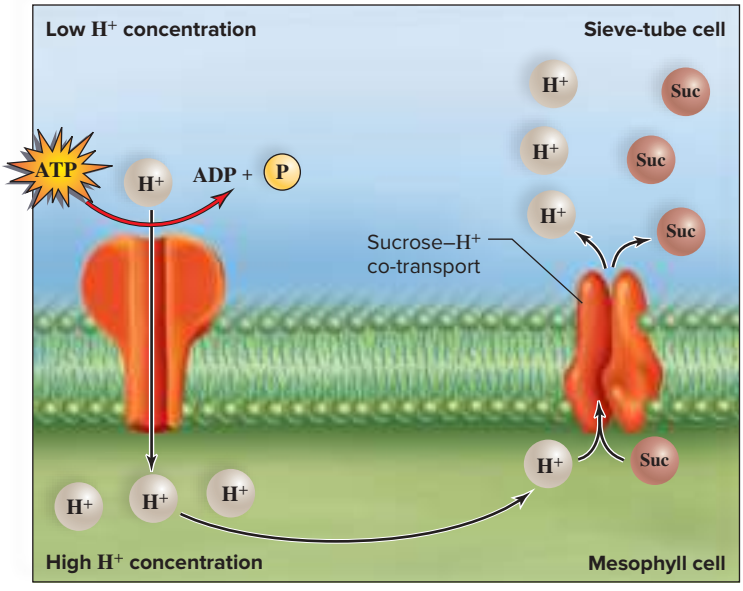


Figure 25.18 Sucrose loading is dependent on a H^+ ion gradient. H^+ ions are actively pumped into the mesophyll cell so they can be co-transported with sucrose into sieve-tube cells.

ISTUDY

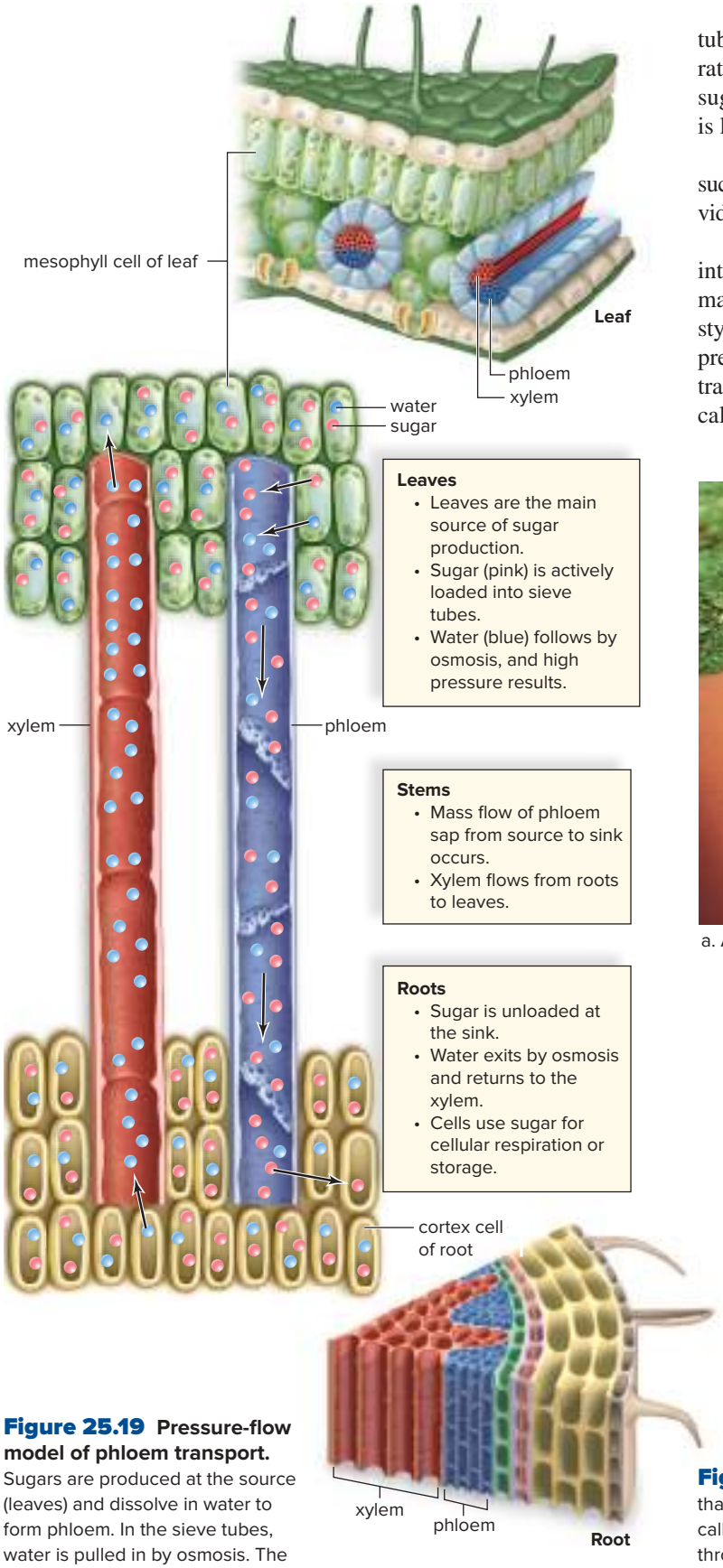


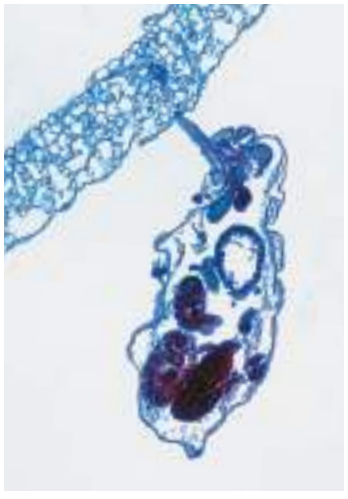
Figure 25.19 Pressure-flow model of phloem transport. Sugars are produced at the source (leaves) and dissolve in water to form phloem. In the sieve tubes, water is pulled in by osmosis. The phloem follows positive pressure and moves toward the sink (root system).

When the sugar arrives at the root, it is transported out of sieve tubes into the root cells. There, the sugar is used for cellular respiration or other metabolic processes. The high concentration of sugar in the root cells causes water to follow by osmosis, where it is later reclaimed by the xylem tissue.

Although leaves are generally the source, storage roots and stems such as carrots, beets, and potatoes are also examples of sources providing much needed sugar during winter or periods of dormancy.

The high pressure of sucrose in phloem has resulted in a very interesting observation of tiny insects called aphids. Aphids, normally a pest in a garden, have a needlelike mouthpart, called a stylet, that can penetrate a stem and tap into a sieve tube. The high-pressure sucrose solution is forced through an aphid's digestive tract very quickly, resulting in a droplet of sucrose on the rear end, called "honeydew" (Fig. 25.20*a*). Many ant species consume this

a. An aphid feeding on a plant stem



b. Aphid stylet in place

Figure 25.20 Acquiring phloem sap. Aphids are small insects that remove nutrients from phloem by means of a needlelike mouthpart called a stylet. **a.** Excess phloem sap appears as a droplet after passing through the aphid's body. **b.** Micrograph of a stylet in plant tissue. When an aphid is cut away from its stylet, phloem sap becomes available for collection and analysis. (a): ©M. H. Zimmermann/Harvard Forest, Harvard University; (b): Steven P. Lynch

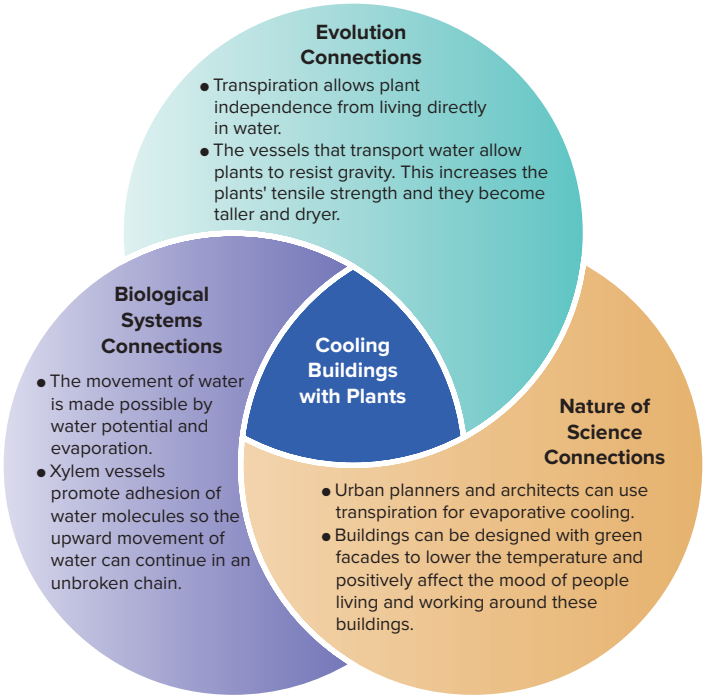
ISTUDY

honeydew and, in turn, protect the aphids. Scientists also take advantage of this natural phloem-tapping system by anesthetizing a drinking aphid, removing its body, and using the inserted stylet to collect phloem for analysis (Fig. 25.20*b*). If the researcher were to insert a needle into the stem, the phloem would clot, but aphids have a unique anticlotting property in their saliva that keeps the initial sieve-tube clot from forming.

Check Your Progress 25.3

1. Describe how water flows upward against gravity.
2. Identify the cohesion and adhesion properties of water that pertain to water transport.
3. Describe the process in which sugars move from source to sink in a plant.

Connecting *the* Concepts



What Did You Learn?

25.1 Plant Nutrition and Soil

Plants need various **essential nutrients** called **minerals.** Essential nutrients needed in large quantities are called **macronutrients;** those needed in smaller quantities are **micronutrients.**

- Carbon, hydrogen, and oxygen make up 95% of a plant's dry weight. The other necessary nutrients are taken up by the roots as mineral ions. **Hydroponics** is a way of growing plants in water, but it is also useful in figuring out mineral deficiencies. Mineral deficiencies can be avoided with the use of fertilizer.
- Plant life is dependent on **soil,** which is formed by the mechanical and chemical weathering of rock. Soil is a mixture of mineral particles, **humus,** living organisms, air, and water. Soil particles are of three types, from the largest to the smallest: sand, silt, and clay. Loam, which contains about equal proportions of all three types, retains water but still has air spaces. Humus contributes to the texture of soil and its ability to provide inorganic nutrients to plants.
- **Cation exchange** is the chemical process by which minerals are absorbed by plant roots.
- Topsoil (a **soil horizon** of a **soil profile**) contains humus, and this is the layer that is lost by **soil erosion.**

25.2 Water and Mineral Uptake

Water, along with minerals, can enter a root by passing between the porous cell walls until it reaches the Casparian strip, after which it passes through an endodermal cell before entering xylem. Water can also enter root hairs and then pass through the cells of the cortex and endodermis to reach xylem.

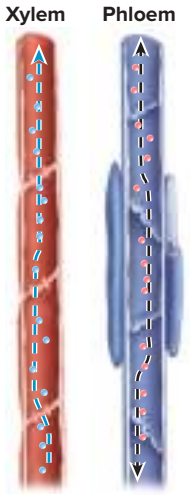
- Mineral ions cross plasma membranes by a chemiosmotic mechanism. A proton pump transports H^+ out of the cell. This establishes an electrochemical gradient that causes positive ions to flow into the cells. Negative ions are carried across the membrane when H^+ moves along its concentration gradient.
- Plants have a number of adaptations that assist them in acquiring nutrients. Legumes have **root nodules** infected with **rhizobia,** which makes nitrogen compounds available to these

plants. Many other plants have **mycorrhizae,** or fungus associated with the root system. The fungus gathers nutrients from the soil, while the root provides the fungus with sugars and amino acids.

25.3 Transport Mechanisms in Plants

As an adaptation to life on land, plants have a vascular system that transports water and minerals from the roots to the leaves. Sugars are then transported in the opposite direction. Vascular tissue includes xylem and phloem.

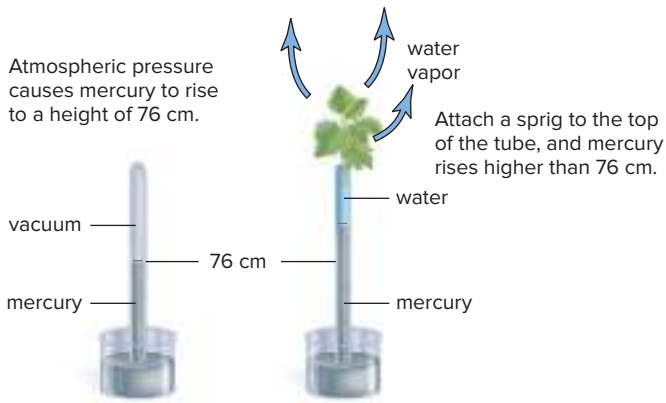
- In xylem, vessels composed of vessel elements aligned end to end form an open pipeline from the roots to the leaves. The difference in **water potential** accounts for the upward movement of water in a plant. At night, **root pressure** can build in the root, causing **guttation.** However, this does not contribute significantly to xylem transport.



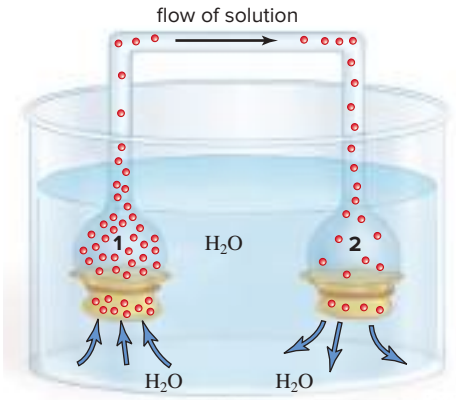
- The **cohesion-tension model** of xylem transport states that **transpiration** creates a tension, which pulls water upward in xylem from the roots to the leaves. This means of transport works only because water molecules are **cohesive** with one another, forming a **water column,** and **adhesive** with xylem walls.
- Most of the water taken in by a plant is lost through stomata by transpiration. Only when there is plenty of water do stomata remain open, allowing carbon dioxide to enter the leaf and photosynthesis to occur.
- Stomata open when **guard cells** take up water. The guard cells stretch lengthwise, because the microfibrils in their walls prevent lateral expansion. Water enters the guard cells after potassium ions (K^+) have entered, causing these cells to buckle outward. Light signals stomata to open, and a high carbon dioxide (CO_2) level may signal stomata to close. Abscisic acid produced by wilting leaves also signals for closure.
- An early observation for the function of phloem came from the effects of girdling a tree. **Girdling** involves the removal of bark, which contains phloem, and once removed would kill a tree.
- In phloem, sieve tubes composed of sieve-tube members aligned end to end form a continuous pipeline from the leaves to the roots. Sieve-tube members have sieve plates through which strands of cytoplasm extend through plasmodesmata from one member to the other.
- The **pressure-flow model** of phloem transport proposes that a positive pressure drives phloem contents in sieve tubes. Sucrose is actively transported into sieve tubes—by a chemiosmotic mechanism—at a **source,** and water follows by osmosis. The resulting increase in pressure creates a flow that moves water and sucrose to a **sink.** A sink can be at the roots or at any other part of the plant that requires organic nutrients.

Thinking Critically About Biology

1. You are a camp counselor and you ask your elementary-school-aged campers to put white carnations into water with blue food coloring. What will happen to the white carnation, and how will you explain this to your campers?
2. Design an experiment to determine whether copper, nitrogen, or phosphate is an essential plant nutrient. State the possible results.
3. Explain why this experiment supports the hypothesis that transpiration can cause water to rise to the tops of tall trees.



4. *Welwitschia* is a genus of plant that lives in the Namib and Mossamedes Deserts of Africa. Annual rainfall averages only 2.5 cm (1 inch) per year. *Welwitschia* plants contain a large number of stomata (22,000 per cm^2), which remain closed most of the time. Can you suggest how a large number of stomata would be beneficial to these desert plants?
5. In the drawing below, explain why the solution flows from the left bulb to the right bulb and how this experiment relates to the pressure-flow model of phloem transport.



ISTUDY

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Saltwater Filter

McGraw Hill

Discuss how the vascular tissue xylem is involved in the experimental process in this video.

Making It Relevant

1. Why are cohesion and adhesion important for the movement of water in a plant?
2. Think about the houseplants you or a family member may own. How does the transport of water affect whether your houseplants are wilted or standing upright?
3. How does transpiration affect the climate of the area you live in?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

26

Flowering Plants: Control of Growth Responses

Bruce Fritz/USDA

Tracking the Sun Across the Sky

The observation that sunflowers track the sun as it moves through the sky is a striking example of a flowering plant's ability to respond to environmental stimuli. Other responses to light can take longer than sun-tracking because they involve hormones and an alteration in growth. For example, flowering plants will exhibit a bend toward the light within a few hours, because a hormone produced by the growing tip has moved from the sunny side to the shady side of the stem. Hormones also help flowering plants respond to stimuli in a coordinated manner. In the spring, seeds germinate and growth begins if the soil is warm enough to contain liquid water. In the fall, when temperatures drop, shoot and root apical growth ceases. Some plants also flower according to the season. The pigment phytochrome is instrumental in detecting the photoperiod and bringing about changes in gene expression, which determine whether a plant flowers or does not flower.

Plant defenses include physical barriers, chemical toxins, and even mutualistic animals. This chapter discusses the variety of ways flowering plants can respond to their environment, including other organisms.

As you read through this chapter, think about the following questions:

1. Which hormones are essential for plant growth?
2. Which environmental stimuli trigger the various plant responses?

Chapter Outline

26.1 Plant Hormones

26.2 Plant Growth and Movement Responses

26.3 Plant Responses to Phytochrome

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 5.3 Which membrane proteins are essential for the control of growth responses?

Figure 5.8 What role does turgor pressure play in the plant response to stimuli?

Figure 7.7 What wavelengths of light are used by plants?

Following *the* Themes

CHAPTER 26 – FLOWERING PLANTS: CONTROL OF GROWTH RESPONSES

Evolution	Plants have evolved responses to specific stimuli that increase their chances of survival.
Nature of Science	Scientific research has found a variety of human applications for a number of plant hormones.
Biological Systems	Plants respond to stimuli by using specific signal transduction pathways.

ISTUDY

26.1 Plant Hormones

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the role of hormones when plant cells utilize signal transduction to respond to stimuli.
2. Compare and contrast the effects of auxins, gibberellins, cytokinins, abscisic acid, and ethylene on plant growth and development.

All organisms are capable of responding to environmental stimuli. Being able to respond to stimuli is a beneficial adaptation, because it leads to organisms' longevity and ultimately to the survival of the species. Flowering plants perceive and react to a variety of environmental stimuli. Some examples include light, gravity, carbon dioxide levels, pathogen infection, drought, and touch. Their responses can be short term, as when stomata open and close in response to light levels, or long term, as when plants respond to gravity with the downward growth of the root and the upward growth of the stem.

Although we think of responses in terms of a plant structure, the mechanism that brings about a response occurs at the cellular level. Research has shown that plant cells respond to stimuli by utilizing **signal transduction,** the binding of a molecular "signal" that initiates and amplifies a cellular response. The concept of signal transducers was first introduced in Section 13.3 in the description of tumor suppressor genes and oncogenes.

Notice in Figure 26.1 that signal transduction involves the following.

- **Receptors**—proteins activated by a specific signal. Receptors can be located in the plasma membrane, the cytoplasm, the nucleus, or even the endoplasmic reticulum. A receptor that responds to light has a pigment component. For example, the phytochrome receptor protein has a region that is sensitive to red light, and the phototropin receptor protein has a region that is sensitive to blue light.
- **Transduction pathway**—a series of relay proteins or enzymes that pass a signal until it reaches the machinery of the cell. In some instances, a bound receptor immediately communicates

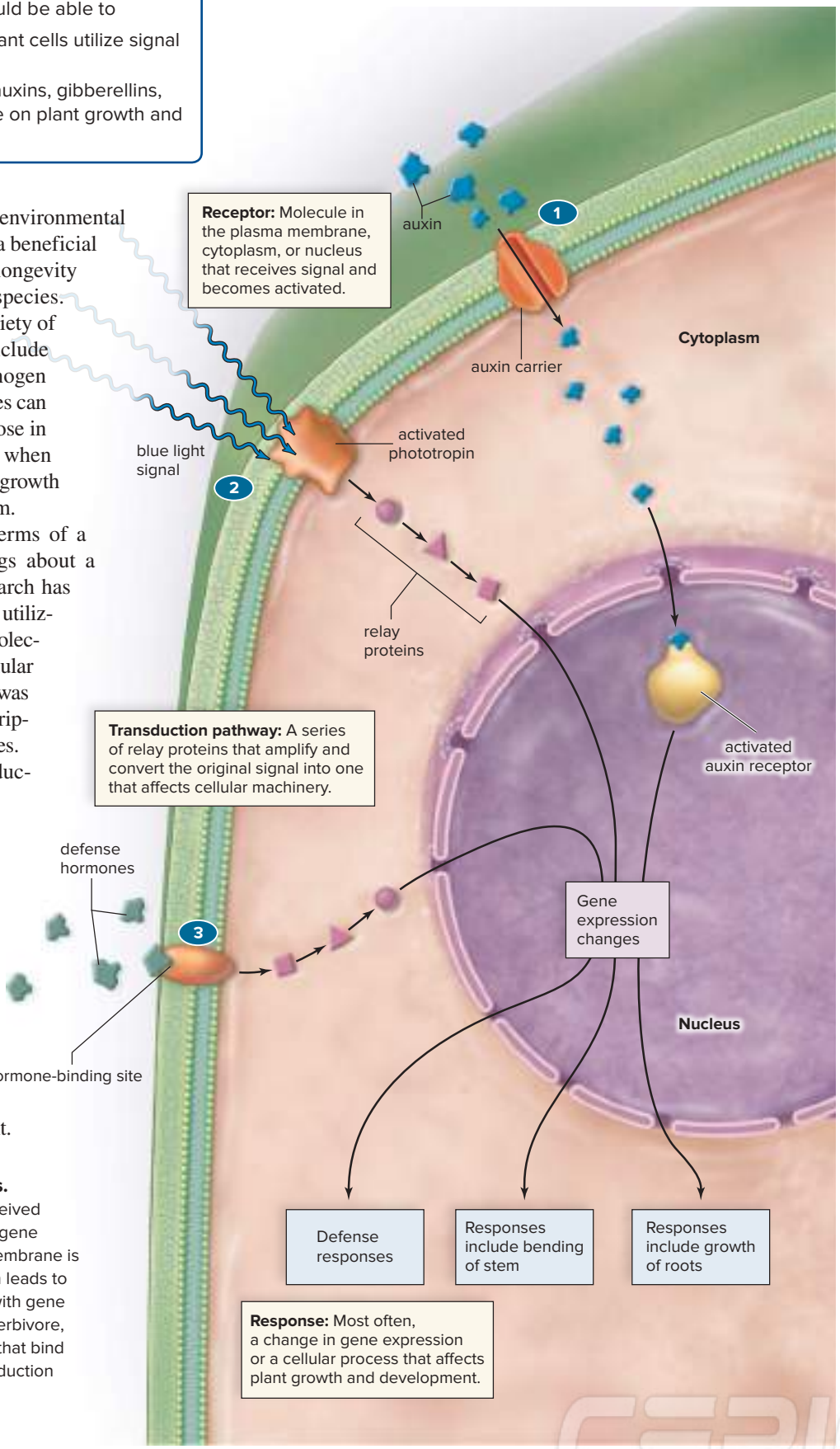


Figure 26.1 Signal transduction in plants.
1 The hormone auxin enters the cell and is received by a receptor in the nucleus. This complex alters gene expression. 2 A light receptor in the plasma membrane is sensitive to and activated by blue light. Activation leads to stimulation of a transduction pathway that ends with gene expression changes. 3 When attacked by an herbivore, the flowering plant produces defense hormones that bind to a plasma membrane receptor. Again, the transduction pathway results in a change in gene expression.

ISTUDY

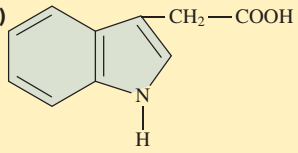
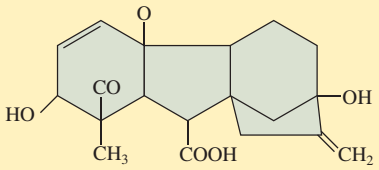
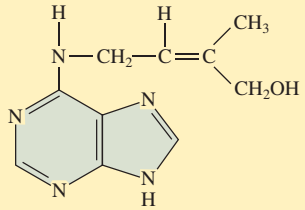
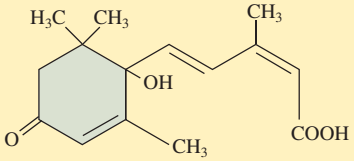
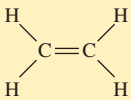
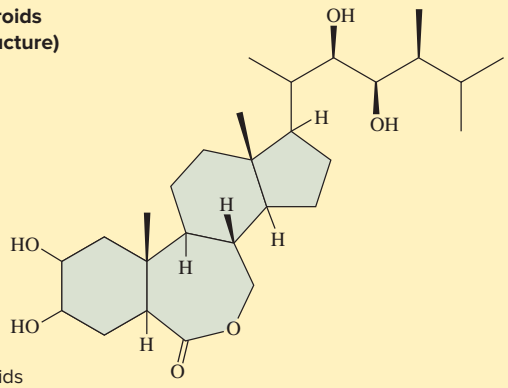
with the transduction pathway, and in other instances, a second messenger, such as Ca^{2+}, initiates the response.

- **Cellular response**—the result of the transduction pathway. Very often, the response is either the transcription of particular genes or the end product of an activated metabolic pathway. The cellular response brings about the overall visible change, such as stomata closing or a stem that turns toward the light.

What role do plant **hormones** play in the ability of flowering plants to respond to a stimulus? The answer is that they serve as chemical signals that coordinate cell responses. These molecules are produced in very low concentrations and are active in another part of the organism. Hormones such as auxins, for example, are synthesized or stored in one part of the plant, but they travel within phloem or from cell to cell to another part of the plant.

In this section, we present descriptions of five major types of plant hormones: auxins, gibberellins, cytokinins, abscisic acid, and ethylene. Each of these affects different aspects of plant responses to stimuli. Table 26.1 summarizes these five hormones, their actions, and their commercial uses.

Table 26.1 Plant Hormones

Hormone	Plant Use	Commercial Use
Auxins (indoleacetic acid; IAA)  Structure of indoleacetic acid (IAA)	Promote growth of roots and fruits; prevent the loss of leaves	Induce fruit production without pollination; used in commercial herbicides (2,4-D)
Gibberellins (gibberellic acid; GA_3)  Structure of gibberellic acid (GA_3)	Cause stem elongation	Increase growth and size of the plants; break the dormancy cycle
Cytokinins (zeatin)  Structure of zeatin	Promote cell division; prevent senescence	Prolong the shelf life of flowers and vegetables
Abscisic acid (abscisic acid; ABA)  Structure of abscisic acid (ABA)	Initiates and maintains seed and bud dormancy	Commercial thinning of fruits promotes growth in the remaining fruit
Ethylene  Structure of ethylene	Causes abscission and ripening of fruit	Causes ripening of fruits and vegetables for market
Brassinosteroids (general structure)  Brassinosteroids	Variety of functions associated with cell growth and differentiation	Reduce stress on plants during cold and drought conditions

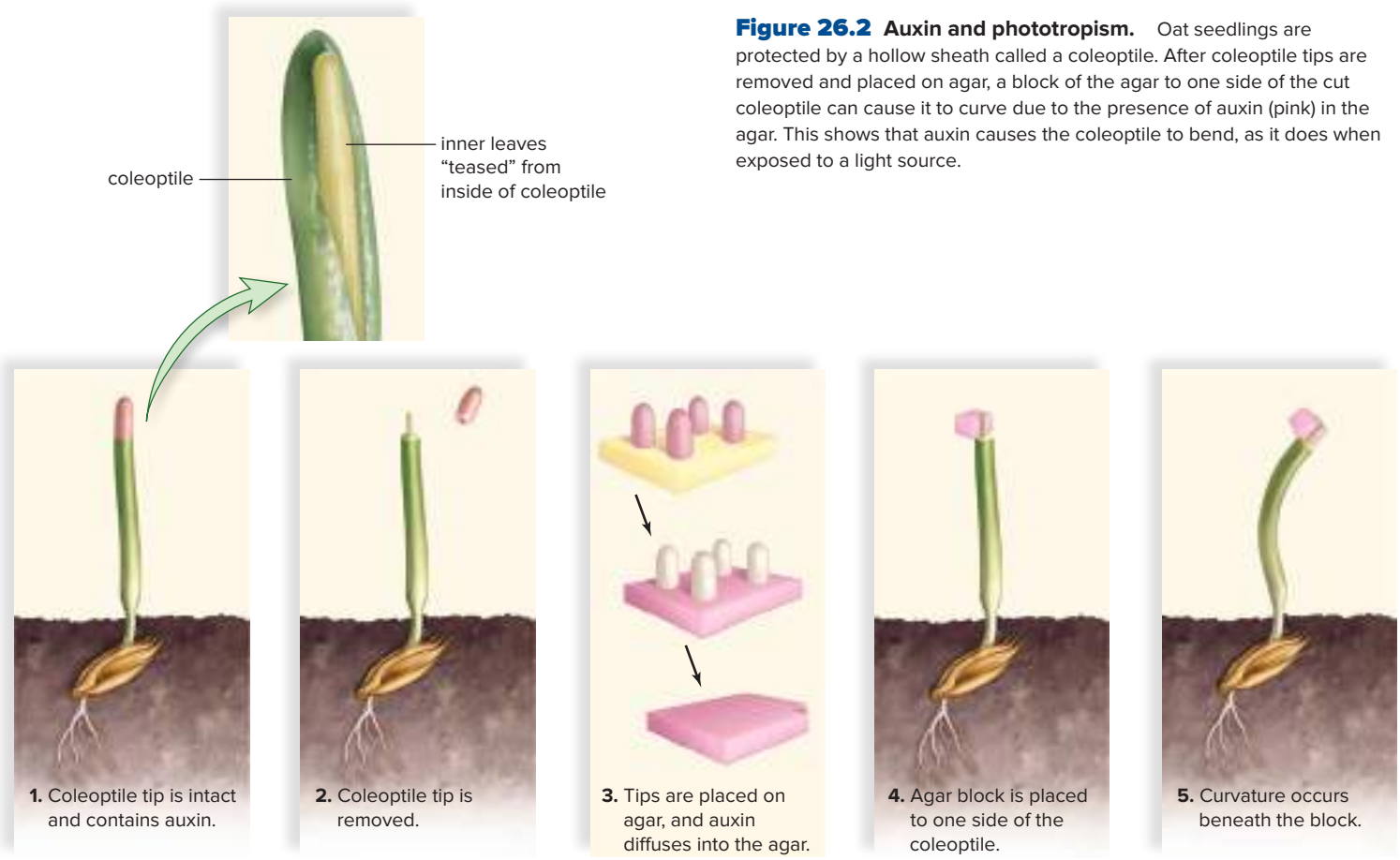


Figure 26.2 Auxin and phototropism. Oat seedlings are protected by a hollow sheath called a coleoptile. After coleoptile tips are removed and placed on agar, a block of the agar to one side of the cut coleoptile can cause it to curve due to the presence of auxin (pink) in the agar. This shows that auxin causes the coleoptile to bend, as it does when exposed to a light source.

Auxins

In 1881, Charles Darwin and his son Francis published a book called *The Power of Movement in Plants.* Here, they described their observation that plants bend toward light. The Darwins used oat seedlings and specifically looked at the young seedlings' coleoptile. A **coleoptile,** much like wearing a rubber glove, is a protective sheath for young leaves. Bending toward light, or *phototropism,* does not occur if the tip of the seedling is cut off or covered by a black cap. They concluded that some influence that causes curvature is transmitted from the coleoptile tip to the rest of the shoot.

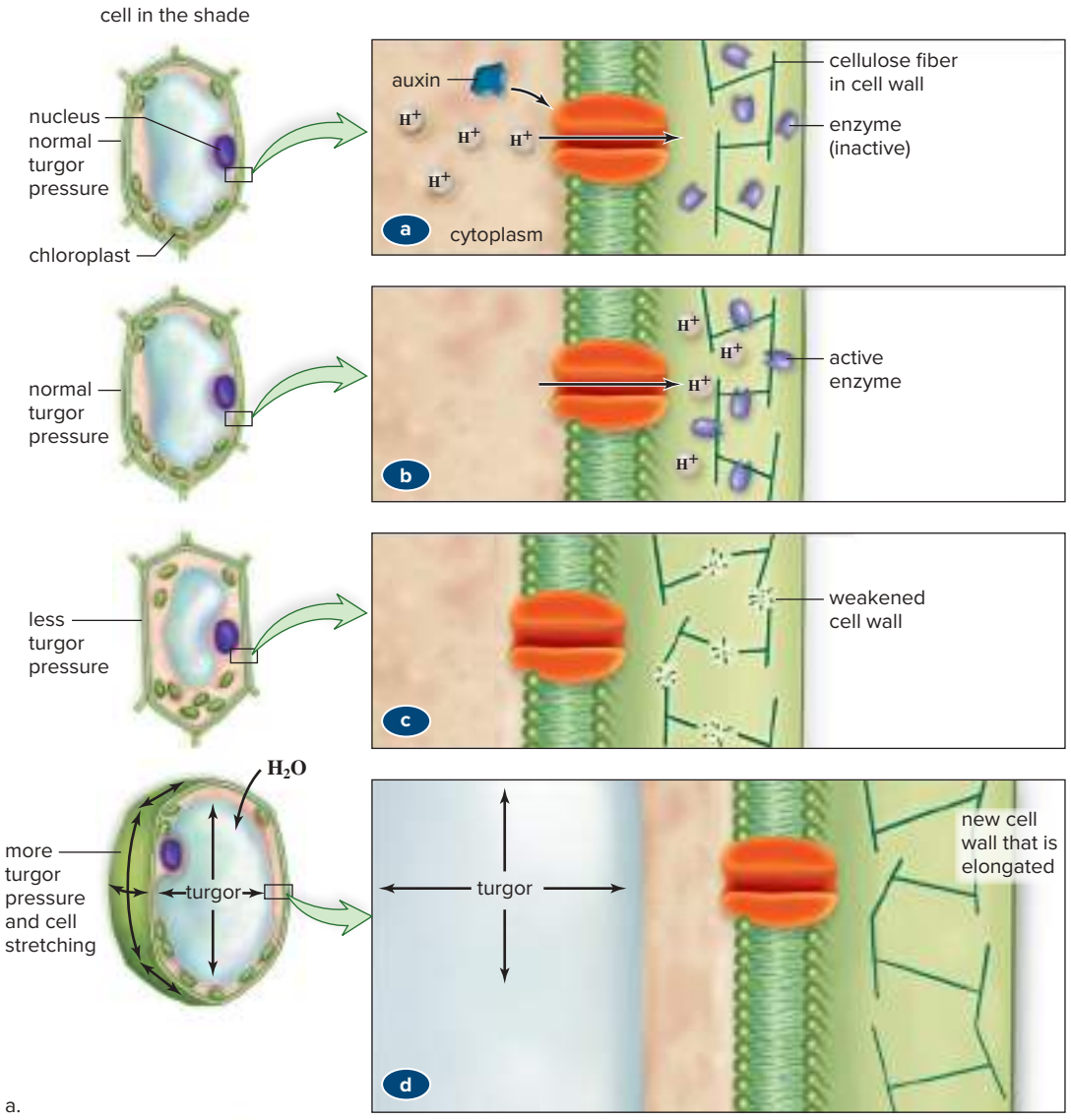
For many years, researchers tried and failed to isolate the chemical involved in phototropism by crushing and analyzing coleoptile tips. In 1926, Frits W. Went cut off the tips of coleoptiles and, rather than crush them, placed them on agar blocks (agar is a gelatin-like material). Then, he placed an agar block to one side of a tipless coleoptile and found that the shoot curved away from that side. The bending occurred even though the seedlings were not exposed to light (Fig. 26.2). Went concluded that the agar block contained a chemical that had been produced by the coleoptile tips. This chemical, he decided, had caused the shoots to bend. He named the chemical substance **auxin** after the Greek word *auximos,* which means "promoting growth." Since then, it has been determined that auxins are also produced in shoot apical meristems (see Fig. 24.7), young leaves, flowers, fruits, and at lower levels in the root apical meristem. There are many variations of auxin, but the most common naturally occurring form is indoleacetic acid (IAA).

How Auxins Cause Stems to Bend

When a stem is exposed to unidirectional light, auxin moves to the shady side, where it enters the nuclei of shady cells and attaches to a receptor (Fig. 26.3*a*). Next, hydrogen ions are pumped into the cell wall, creating an acidic environment. The acid triggers enzymes to dismantle the cellulose fibers in the cell wall, resulting in a weakened wall. To test the hypothesis that acid weakens walls, scientists have neutralized the acid in shady cells and the result was cells that would not grow.

A plant with a cell wall that is weak and loose will result in a decrease in turgor pressure. The water potential inside the cell is now less than outside the cell and, as you learned in Section 25.3, water moves from areas of high potential to low potential. Water flows into the cell from other parts of the plant, attempting to restore turgor pressure once again. The pressure of the water begins to stretch the wall, and the plant cell responds by rebuilding a now longer wall, resulting in cell growth. Without the rebuilding, the cell would burst. Overall, the role of auxin is to cause the wall to weaken, so it can be stretched and rebuilt even longer. The end result of these activities is elongation of the stem, because only the cells on the shady side are getting longer, bending the stem toward the light (Fig. 26.3*b*).

ISTUDY



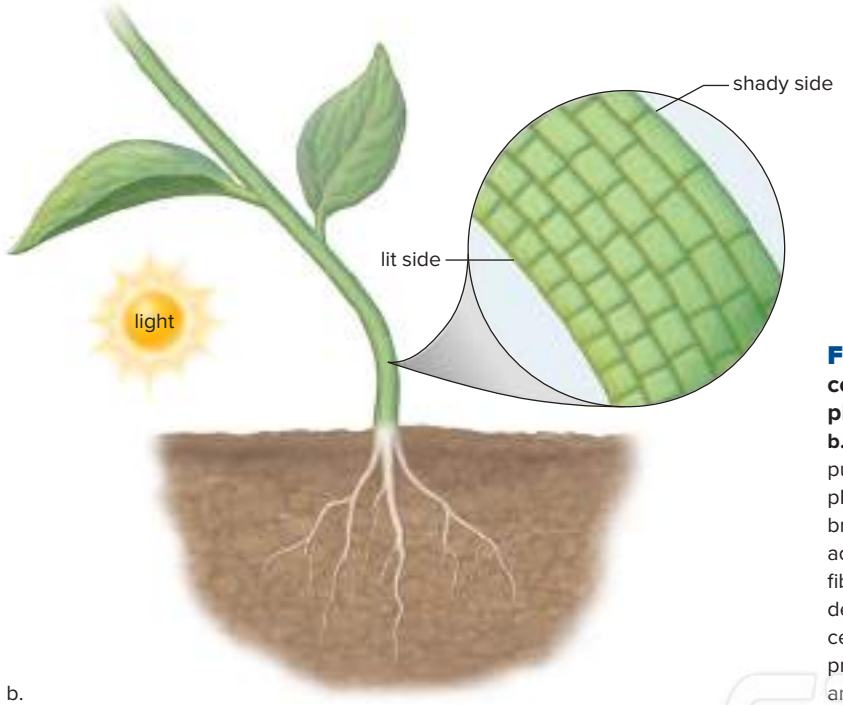


Figure 26.3 Expansion of the cell wall on the shady side of a plant. **a.** Auxin binds to receptors. **b.** Hydrogen ions are actively pumped into the cell wall area. As the pH decreases, enzymes that can break down cellulose fibers are activated in the cell wall. **c.** Cellulose fibers break, loosening the wall and decreasing turgor pressure in the cell. **d.** The cell expands as turgor pressure inside the cell increases, and a new, longer wall is rebuilt.

ISTUDY

Auxins Affect Growth and Development

As mentioned, auxin causes phototropism, maximizing exposure of a plant to the sun. In addition, auxin is responsible for a process called *gravitropism* where, after the direction of gravity has been detected by a flowering plant, auxin moves to the lower surface of roots and stems. Thereafter, roots curve downward and stems curve upward. Phototropism and gravitropism are discussed in greater detail in Section 26.2.

This versatile hormone is also responsible for a phenomenon called **apical dominance.** Experienced gardeners know that, to produce a bushier plant, they must remove the terminal bud (see Fig. 24.6). In a plant that has not been altered, auxin produced in the apical meristem of the terminal bud is transported downward, inhibiting the growth of lateral or axillary buds. Release from apical dominance occurs when pruning (cutting) removes the shoot tip. Then, the axillary buds grow and the plant takes on a fuller appearance. Interestingly, if auxin were to be applied to the broken terminal stem, apical dominance would be restored.

Auxin causes the growth of roots and fruits and prevents the loss of leaves and fruit. The application of an auxin paste to a stem cutting causes adventitious roots to develop more quickly than they would otherwise. Auxin production by seeds promotes the growth of fruit. As long as auxin is concentrated in leaves or fruits rather than in the stem, leaves and fruits do not drop off. Therefore, trees can be sprayed with auxin to keep mature fruit from falling to the ground.

Auxins Have Many Commercial Uses

The most common naturally occurring auxin, IAA, has a relatively simple chemical structure (Table 26.1), allowing it to be easily copied and altered into various synthetic forms. Synthetic auxins are used today in a number of applications. These auxins are sprayed on plants, such as tomatoes, to induce the development of fruit without pollination, creating seedless varieties. Synthetic auxins have been used as herbicides to control broadleaf weeds, such as dandelions and other plants. These substances have little effect on grasses. Agent Orange is a powerful synthetic auxin that was used in extremely high concentrations to defoliate the forests of Vietnam during the Vietnam War. This powerful auxin proved to be carcinogenic and has harmed individuals who were exposed to it.

Gibberellins

Gibberellins were discovered in 1926 when a Japanese scientist was investigating a fungal disease of rice plants called "foolish seedling disease." Rapid stem elongation weakened the plants and caused them to collapse. The fungus infecting the plants produced an excess of a chemical called gibberellin, named after the fungus *Gibberella fujikuroi.* It wasn't until 1956 that a form of gibberellin, now known as gibberellic acid, was isolated from a flowering plant rather than from a fungus. We now know of about 136 gibberellins, and the most common of these is gibberellic acid, GA_3 (the subscript designation distinguishes it from other gibberellins). Young leaves, roots, embryos, seeds, and fruits are places where natural gibberellins can be found.

a. b.

Figure 26.4 Gibberellins cause stem elongation. **a.** The plant on the right was treated with gibberellins; the plant on the left was not treated. **b.** The grapes are larger on the right, because gibberellins caused an increase in the space between the grapes, allowing them to grow larger. (a): Omikron/Science Source; (b): ©Amnon Lichter, The Volcani Center

Gibberellins Have Commercial Uses

When gibberellins are applied externally to plants, the most obvious effect is stem elongation (Fig. 26.4*a*). Gibberellins can cause dwarf plants to grow, cabbage plants to become 2 meters tall, and bush beans to become pole beans.

Gibberellins induce growth in a variety of crops, such as apples, cherries, and sugarcane. A notable example is their use on many of the table grapes grown in the United States. Commercial grapes are a genetically seedless variety that would naturally produce small fruit on very small bunches. Treating with GA_3 substitutes for the presence of seeds, which would normally be the source of endogenous gibberellins for fruit growth. Applications of GA_3 increase both fruit stem length and fruit size (Fig. 26.4*b*).

Dormancy is a period of time when plant growth is suspended. Gibberellins can break the dormancy of buds and seeds. Therefore, application of gibberellins is one way to hasten the development of a flower bud. When gibberellins break the dormancy of barley seeds, a large, starchy endosperm is broken down into sugars to provide energy for the growing seedling. This occurs because amylase, an enzyme that breaks down starch, makes its appearance. In the brewing industry, the production of beer relies on the breakdown of starch. Gibberellins are added to barley seeds, so that they artificially break dormancy and provide sugar for the fermentation process.

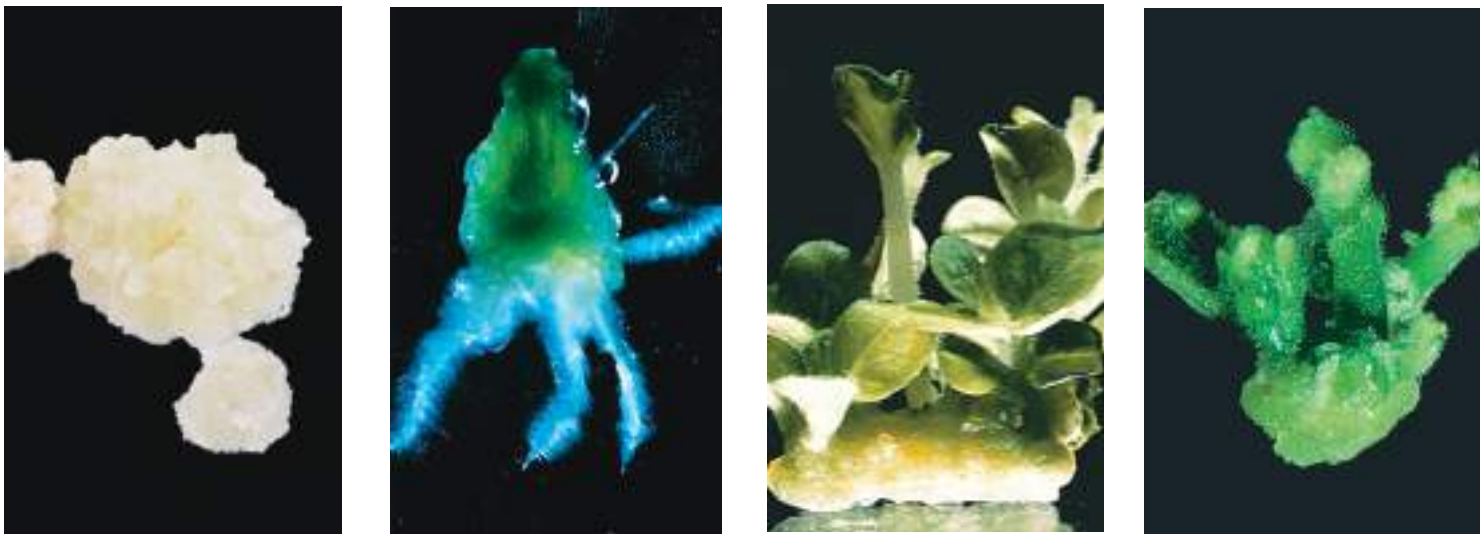
Cytokinins

Cytokinins were discovered as a result of attempts to grow plant tissues and organs in culture vessels in the 1940s. It was found that cell division occurred when coconut milk (a liquid endosperm) and yeast extract were added to the culture medium. Although the specific chemicals responsible could not be isolated at the time, they were collectively called cytokinins, because, as you may recall, *cytokinesis* means "division of the cytoplasm." A naturally occurring cytokinin was not isolated until 1967 and was called zeatin, because it came from corn (*Zea mays*) (Table 26.1).

Cytokinins Promote Cell Division and Organ Formation

Cytokinins influence plant growth by promoting cell division in all tissues of growing plants. In addition, plant organ formation is

ISTUDY



a. b. c. d.

Figure 26.5 The interaction of cytokinins and auxins in organ development. Tissue culture experiments have revealed that auxin and cytokinin interact to affect differentiation during development. **a.** In tissue culture that has the usual amounts of these two hormones, tobacco cells develop into a callus of undifferentiated tissue. **b.** If the ratio of auxin to cytokinin is appropriate, the callus produces roots. **c.** Change the ratio, and vegetative shoots and leaves are produced. **d.** Yet another ratio causes floral shoots. It is now clear that each plant hormone rarely acts alone; it is the relative concentrations of both hormones that produce an effect. (a): Biophoto Associates/Science Source; (b): Alan Darvill and Stefan Eberhard, Complex Carbohydrate Research Center, University of Georgia; (c–d): ©Alan Darvill and Stefan Eberhard, Complex Carbohydrate Research Center, University of Georgia

influenced by cytokinins and its interaction with auxin. Furthermore, cytokinins and auxins are different from all other plant hormones in that they are required for embryo survival. Researchers have tested and are aware that the ratio of auxin and cytokinin and the acidity of the culture medium determine whether plant cells will form an undifferentiated mass of cells, called a *callus,* or a mass of cells with roots, leaves, or flowers (Fig. 26.5). Cytokinins are also responsible for root nodule formation (housing nitrogen-fixing bacteria), as well as gall formation on wounded trees. A gall is a tumorlike growth caused by infections from bacteria, fungi, insects, or nematodes. These organisms can disrupt normal cytokinin function and result in a plant growing uncontrollably in small areas.

Cytokinins Prevent Senescence

The aging of plants is called **senescence.** During senescence, large molecules within the leaf are broken down and transported to other parts of the plant. Senescence does not always affect the entire plant at once; for example, as some plants grow taller, they naturally lose their lower leaves. In the autumn, low levels of cytokinin cause leaves to change color and eventually die (Fig. 26.6). Interestingly, it has been found that senescence of leaves can be prevented by the application of cytokinins. Some varieties of lettuce have been genetically modified to produce cytokinins at the onset of aging. The modified lettuce heads stay fresher longer and avoid brown and wilting leaves.

Abscisic Acid

Abscisic acid (ABA) is a hormone produced in the chloroplast and is derived from carotenoid pigments. Abscisic acid is sometimes called the stress hormone because it initiates and maintains seed and bud dormancy and brings about the closure of stomata. It was once believed that ABA functioned in **abscission,** the dropping of leaves, fruits, and flowers from a plant. But although the external application of ABA promotes abscission, this hormone is no longer believed to function naturally in this process. Instead, the hormone ethylene seems to bring about abscission.

ABA Promotes Dormancy

Recall that dormancy is a period of low metabolic activity and arrested growth. Dormancy occurs when a plant organ readies itself for adverse conditions by ceasing to grow. For example, it is believed that the hormone ABA travels from leaves to vegetative

Figure 26.6 Leaves change colors when cytokinin levels are low. Seasonal changes signal a drop in cytokinin production in deciduous plants, causing the senescence of leaves. Oxford/iStock/360/Getty Images

ISTUDY

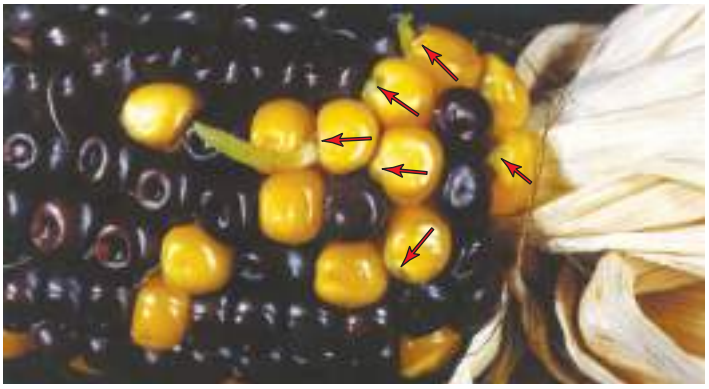


Figure 26.7 Dormancy and germination. Image of viviparous mutant of maize (Indian corn) showing germination on the cob due to reduced sensitivity to abscisic acid. Red arrows indicate emerging seedlings. Dr. Donald R. McCarty, University of Florida

buds in the fall, and thereafter these buds are converted to winter buds. A winter bud is covered by thick, hardened bud scales (see Fig. 24.13). A reduction in the level of ABA and an increase in the level of gibberellins are believed to break seed and bud dormancy. Then, seeds germinate and buds send forth leaves.

Figure 26.7 shows what can happen if a plant becomes insensitive to ABA. In normal corn, a home gardener can leave a cob on the stalk to dry, then collect the dry kernels to plant the following year. ABA will keep the dry kernels from germinating until conditions are right. In the ABA-insensitive mutant corn, the kernels exhibit *vivipary*—an early break in dormancy and germination while still on the cob. The corn seedlings growing from the cob in Figure 26.7 are germinating in the wrong season and will most likely die.

ABA Closes Stomata

The reception of abscisic acid brings about the closing of stomata when a plant is under water stress, as described in Figure 26.8. Investigators have also found that ABA induces rapid depolymerization of actin filaments and formation of a new type of actin that is randomly oriented throughout the cell. This change in actin

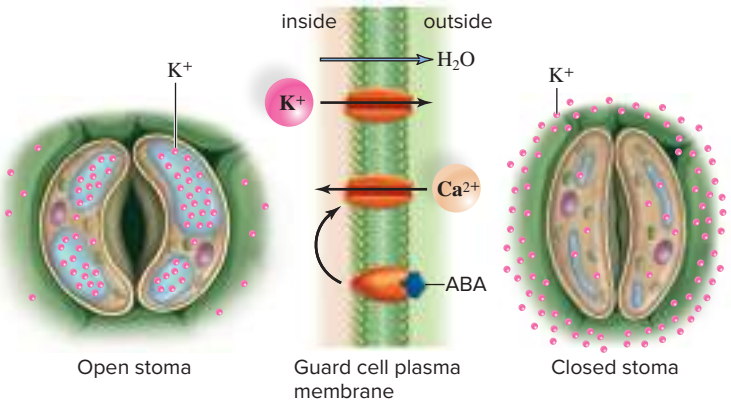


Figure 26.8 Abscisic acid promotes closure of stomata. The stoma is open (*left*). When ABA (the first messenger) binds to its receptor in the guard cell plasma membrane, the second messenger (Ca^{2+}) enters (*middle*). Now, K^+ channels open, and K^+ exits the guard cells. After K^+ exits, so does water. The stoma closes (*right*).

No abscission Abscission

Figure 26.9 Ethylene and abscission. Normally, there is no abscission when a holly twig is placed under a glass jar for a week. When an ethylene-producing ripe apple is also under the jar, abscission of the holly leaves occurs. (both): ©Kingsley Stern

organization may also be part of the transduction pathways involved in stomata closure.

Ethylene

Ethylene ($H_2C = CH_2$) is a gas formed from the amino acid methionine. This hormone is involved in abscission (dropping leaves and fruit) and the ripening of fruits.

Ethylene Causes Abscission

The absence of auxin, and perhaps gibberellin, probably initiates abscission. But once abscission has begun, ethylene stimulates certain enzymes, such as cellulase, which helps cause leaf, fruit, or flower drop. In Figure 26.9, a ripe apple, which gives off ethylene, is under the bell jar on the right, but not under the bell jar on the left. As a result, only the holly plant on the right loses its leaves.

Ethylene Ripens Fruit

In the early 1900s, it was common practice to prepare citrus fruits for market by placing them in a room with a kerosene stove. Only later did researchers realize that an incomplete combustion product of kerosene—ethylene—ripens fruit. It does so by increasing the activity of enzymes that soften fruits. For example, in addition to stimulating the production of cellulase, it promotes the activity of enzymes that produce the flavor and smell of ripened fruits and breaks down chlorophyll, inducing the color changes associated with fruit ripening.

Ethylene moves freely through a plant by diffusion, and because it is a gas, ethylene also moves freely through the air. That is why a basket of ripening apples can induce ripening of a bunch of bananas some distance away. Ethylene is released at the site of a plant wound due to physical damage or infection (which is why one rotten apple spoils the whole barrel).

The use of ethylene in agriculture is extensive. It is used to hasten the ripening of green fruits, such as melons and honeydews, and it is applied to citrus fruits to attain pleasing colors before being put

ISTUDY

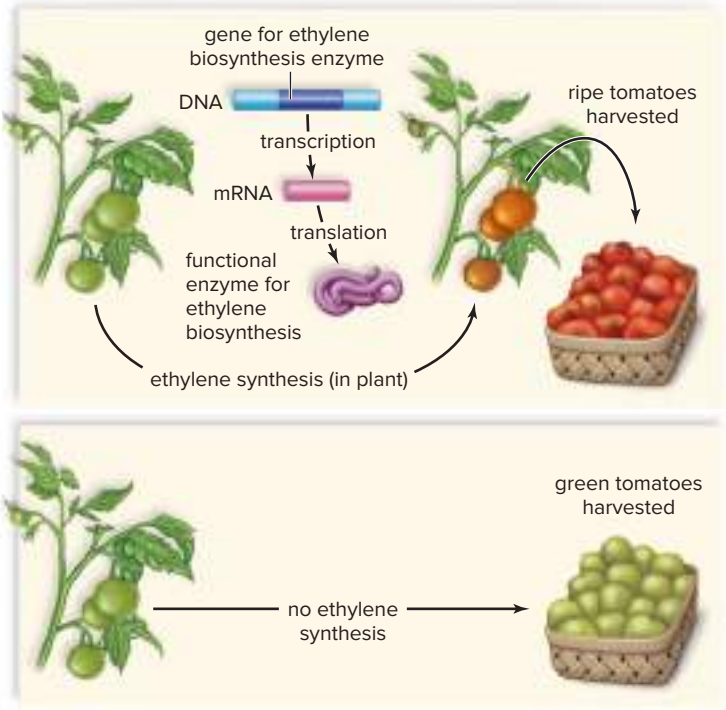


Figure 26.10 Ethylene and fruit ripening. Wild-type tomatoes (*top*) ripen on the vine after producing ethylene. Tomatoes (*bottom*) are genetically modified to produce no ethylene and stay green for shipping.

out for sale. Normally, tomatoes ripen on the vine, because the plants produce ethylene. Today, tomato plants can be genetically modified to not produce ethylene. This facilitates shipping, because green tomatoes are not subject to as much damage (Fig. 26.10). Once the tomatoes have arrived at their destination, they can be exposed to ethylene, so that they ripen. There are many ethylene-absorbing products on the market that consumers can buy. The product usually consists of a tiny, ethylene-permeable pouch filled with potassium permanganate ($KMnO_4$). This chemical absorbs ethylene in a refrigerator and prolongs the life of fruits and vegetables.

Brassinosteroids

Although **brassinosteroids** were first discovered almost four decades ago, they have only recently been recognized as a separate class of plant hormones. These steroid compounds function in the growth and development of the plant, specifically in the specialization of the vascular tissue. They are also recognized in helping with the response of the plant to stress, such as cold or drought conditions.

Responding to the Biotic Environment

The hormones just discussed mostly function for the plant's response to abiotic stimuli, such as light, oxygen, water, pH, and temperature. Plants must also have an arsenal of chemicals to handle biotic stimuli, such as herbivory, parasitism, and competition from other plants.

A plant's epidermis and bark do a good job of discouraging attackers. But, unfortunately, herbivores have ways around a plant's first line of defense. A fungus can invade a leaf via the stomata and set up shop inside the leaf, where it feeds on nutrients meant for the plant. Underground nematodes have sharp mouthparts to break through the epidermis of a root and establish a parasitic relationship. Tiny insects called aphids have piercing mouthparts that allow them to tap into the phloem of a nonwoody stem. These examples illustrate why plants need a variety of defenses that are not dependent on its outer surface. The primary metabolites of plants, such as sugars and amino acids, are necessary to the normal workings of a cell. Plants also produce molecules termed **secondary metabolites** as a defense, or survival, mechanism. Secondary metabolites were once thought to be waste products, but now we know they are part of a plant's arsenal to prevent predation or discourage competition. The Evolution feature, "Plants Can 'Smell' and 'Talk' to Each Other," explains the field of chemical ecology and how plants have evolved various chemical messages to make them more successful in a particular situation.

Check Your Progress 26.1

1. Explain how hormones assist in bringing about responses to stimuli.
2. Describe how auxin causes a plant to bend toward light.
3. Explain why abscisic acid is sometimes referred to as a stress hormone.

26.2 Plant Growth and Movement Responses

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the difference between a tropism and a turgor movement.
2. Describe how a photoreceptor works in phototropism.
3. Explain why a shoot grows upward and a root grows downward.
4. List the internal stimuli involved in closing a Venus flytrap.

All living organisms respond to stimuli and exhibit movement. Plant movements are slow and difficult to notice unless seen in time-lapse video or demonstrated experimentally. Plant movement responses to stimuli can be internal—such as changes in turgor pressure, electrical impulses, or the action of hormones—or they can be external, such as responses to sunlight, water, oxygen, gravity, and barriers such as rocks.

Recall from Figure 26.1 that when there is a stimulus, whether internal or external, the first step is *reception* of the stimulus. The next step is *transduction,* meaning that the stimulus has been changed into a form that is meaningful to the plant. Finally, a *response* is made, usually by the plant's genes. Animals and plants go through the same sequence of events when they respond to a stimulus; however, in the case of plants, no nerves are present. Instead, chemical signals are released, and the binding of these signals brings about transduction and response.

In this section, we consider plant tropisms—responses caused by external stimuli—and turgor movements—responses caused by internal stimuli. Note that tropisms are growth movements, and turgor movements are nongrowth movements.

ISTUDY

THEME Evolution

Plants Can "Smell" and "Talk" to Each Other

Because plants are rooted to the ground, they are unable to escape from herbivores, pathogens, or even competing plants in the area. By producing a variety of chemical defenses, plants have overcome these constraints. With various organic chemicals, plants can attract mycorrhizal partners, pollinators, and the enemies of herbivores. They also repel herbivores, pathogens, and competing plants.

Chemical ecology is the study of the interaction between chemical signals, plants, animals, and the environment in which they live. It brings together scientists from many different fields, such as entomology, chemistry, and plant biology, who work together to study the complex chemical communication systems that occur in nature.

The most common research focus is coevolution of plants and insect herbivores. Studying the constant battle between plants and insects helps us better understand the interactions that have produced the diverse range of species in existence today. Chemical ecology examines how the chemicals within plants are made, how these chemicals contribute to a plant's overall fitness, and how they evolve in response to environmental pressures. In this feature, four examples of chemical interactions are highlighted.

Repelling the Herbivore and Telling the Neighbors

After a predator chews a tomato plant leaf, a small protein called *systemin* is produced in the wounded area in response to the predator's saliva. Systemin is part of a signal transduction pathway that produces other defense compounds, such as *jasmonic acid* and *salicylic acid*. These defense compounds travel in phloem, become widely distributed throughout the plant, and activate the gene expression of proteinase inhibitors. When the next predator begins to eat the same plant, it will be poisoned or repelled by the bad taste of these inhibitors (Fig. 26A). In addition, many of the defense compounds are volatile (evaporate easily) and can stimulate defenses in nearby plants.

Attracting the Enemy's Enemy

The wild tobacco (*Nicotiana attenuate*) plant of the southwestern United States and Mexico produces nicotine to poison herbivores. Unfortunately for the plant, some herbivores, such as the hawkmoth caterpillar, have become resistant to nicotine and decimate the weedy shrubs. Recently, researchers from the Max Planck Institute for Chemical Ecology in Germany found a new way the plant is ridding itself of the hawkmoth caterpillar larva. Dubbed "poison lollipops," the trichomes of some plants produce a sugary substance irresistible to hungry caterpillars (Fig. 26B). The sugar contains a volatile substance, which in turn makes the caterpillar smell good to predatory ants, which grab the caterpillars and carry them off to their nests.

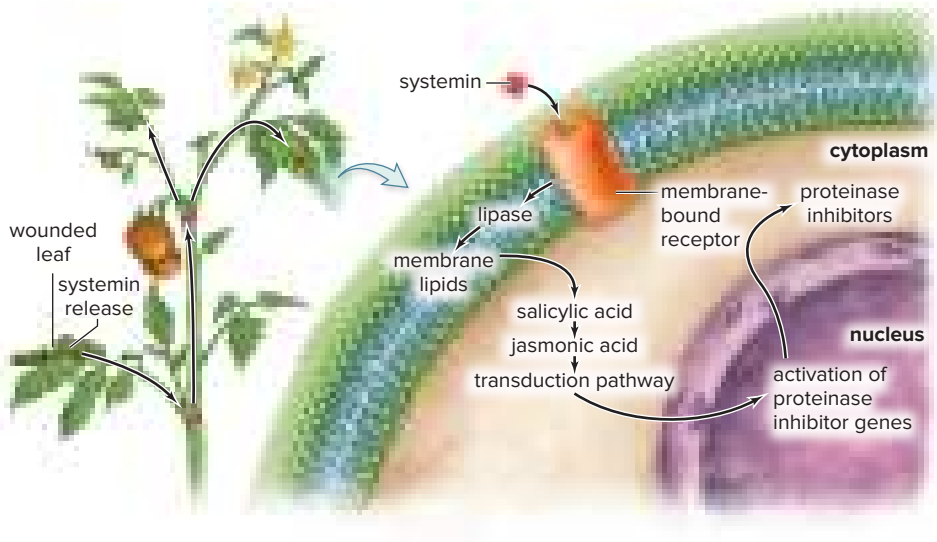


Figure 26A Defense response in tomatoes. Wounded leaves produce systemin, which travels in phloem to all parts of a plant, where it binds to cells that have a systemin receptor. These cells then produce jasmonic acid, a molecule that initiates a transduction pathway, which leads to the production of proteinase inhibitors, which limit insect feeding.

Smelling Your Prey

The dodder vine (*Cuscuta pentagona*) is a parasitic plant that winds itself around a host plant, inserts sharp pegs called haustoria, and feeds on the host's xylem and phloem (see Fig. 25.10). Noticing that dodders seemed to have a preference for certain host plants, scientists hypothesized that dodders could "smell" their preferred meal. To test this, a dodder seedling was placed in a pot between a tomato and a wheat plant (Fig. 26C). After vacillating for several hours, the dodder chose to latch onto the tomato plant. A subsequent experiment confirmed that the attractant was the volatile chemical produced by the tomato. The tomato

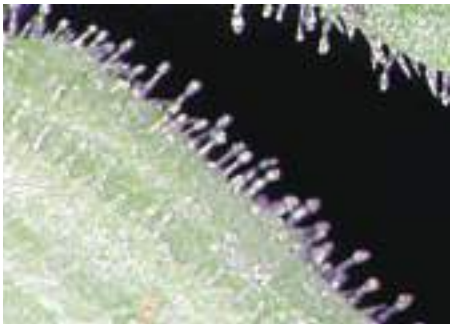


Figure 26B Trichomes. Trichomes, or "poison lollipops," contain sugar to attract caterpillars. Other chemicals increase the predation of the caterpillars. Steven P. Lynch

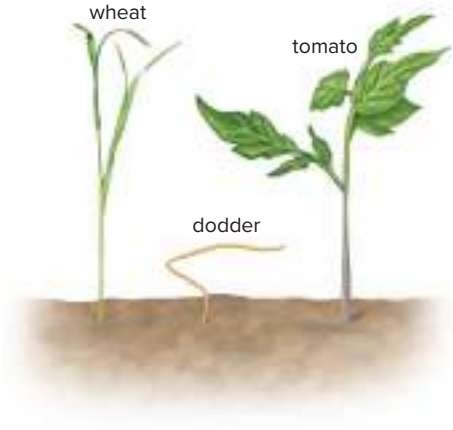


Figure 26C Dodder plant's response to multiple hosts. When placed between two possible hosts, the dodder will choose the host that will sustain the parasite the best.

"perfume" was isolated and placed near the dodder. The dodder "smelled" its preferred prey and moved in that direction.

Keeping Others Away

Some chemical toxins protect plants from other plants. Black walnut trees (*Juglans nigra*) have roots that secrete a chemical toxin that blocks the germination of nearby seeds and inhibits the growth of neighboring plants. This strategy minimizes shading and competition for nutrients while maximizing exposure to the sun.

Questions to Consider

1. How have humans taken advantage of the chemicals that plants produce for their own defenses?
2. How can you test if a defense response is pathogen-specific?
3. During the domestication of crops, humans have intentionally or inadvertently selected for lower levels of toxic compounds. Explain why this type of selection would have occurred.

Movement Caused by External Stimuli

Growth toward or away from a unidirectional stimulus is called a **tropism** (Gk. *tropos,* "turning"). *Unidirectional* means that the stimulus is coming from only one direction instead of multiple directions. Growth toward a stimulus is called a *positive tropism,* and growth away from a stimulus is called a *negative tropism.* Tropisms are due to differential growth—one side of an organ elongates faster than the other, and the result is a curving toward or away from the stimulus.

A number of tropisms have been observed in plants. The three best-known tropisms are phototropism (light), thigmotropism (touch), and gravitropism (gravity).

Phototropism: a movement in response to a light stimulus
Thigmotropism: a movement in response to touch
Gravitropism: a movement in response to gravity

Other tropisms include chemotropism (chemicals), traumotropism (trauma), skototropism (darkness), and aerotropism (oxygen).

Phototropism

A potted plant left in the open with sunlight on all sides will grow and develop vertically. However, if a potted plant is placed on a sunny windowsill with unidirectional light, the stems will begin to bend toward the light (see Fig. 26.12*a*). Positive **phototropism** of stems occurs because the cells on the shady side of the stem elongate due to the presence of auxin. Plant growth that curves away from light is called negative phototropism. Roots, depending on the species examined, are either insensitive to light or exhibit negative phototropism.

Through the study of mutant *Arabidopsis* plants (see the Nature of Science feature, "Why So Many Scientists Work with *Arabidopsis*"), plant scientists know that phototropism occurs because plants have membrane receptors that respond to wavelengths of light. These receptors are called *photoreceptors.* Photoreceptors are proteins embedded with pigment molecules that, in the case of phototropism, respond to blue wavelengths (400–500 nm) of light. Figure 26.11 describes the steps initiating the signal transduction pathway that eventually leads to elongation of cells and the bending of a plant.

When blue light wavelengths are absorbed (Fig. 26.11*a*), the pigment portion of the photoreceptor protein, called *phototropin* (*phot*), changes its shape (Fig. 26.11*b*). This shape change results

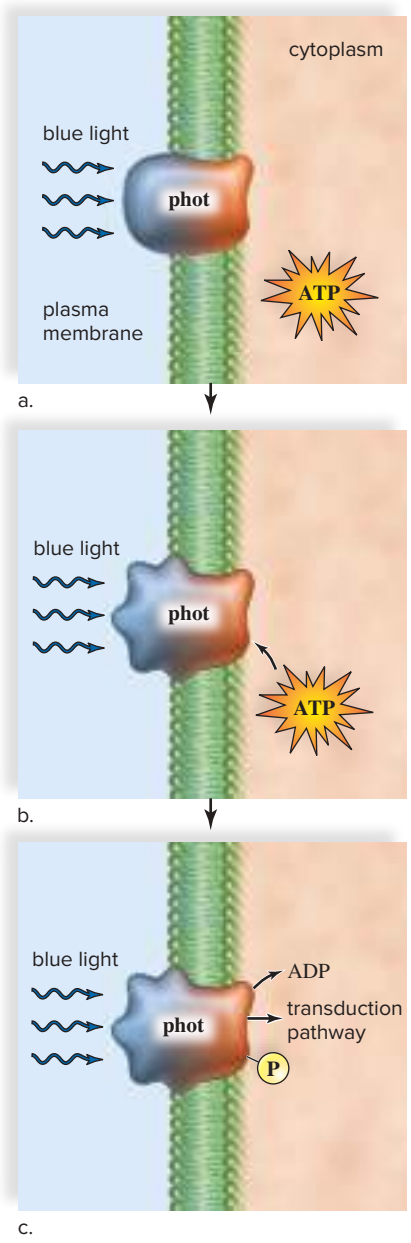


Figure 26.11 Phototropin. In the presence of blue light, (**a**) a photoreceptor called phototropin (phot) is activated (**b**) and becomes phosphorylated (**c**). A transduction pathway begins, leading to the accumulation of auxin.

ISTUDY

THEME Nature of Science

Why So Many Scientists Work with Arabidopsis

Arabidopsis thaliana is a small flowering plant related to cabbage and mustard plants (Fig. 26D). *Arabidopsis* has no commercial value—in fact, it is a weed. However, it has become a model organism for the study of plant molecular genetics, including signal transduction. Unlike the crop plants used formerly, *Arabidopsis* has characteristics that make it an ideal model organism.

- It is small, so many hundreds of plants can be grown in a small amount of space.
- Generation time is short. It takes only 5–6 weeks for plants to mature, and each one produces about 10,000 seeds.
- It can be self-pollinated or cross-pollinated. This feature facilitates the production of strains with multiple mutations.
- It has a relatively small genome: 5 chromosomes, 125 million base pairs of DNA, and 27,400 protein coding genes.

In contrast to *Arabidopsis,* crop plants, such as corn, have generation times of at least several months, and they require a great deal of field space for a large number to grow. Crop plants also have much larger genomes than *Arabidopsis.* For comparison, the genome sizes for rice (*Oryza sativa*), wheat (*Triticum aestivum*), and corn (*Zea mays*) are 420 million, 16 billion, and 2.5 billion base pairs, respectively. However, crop plants have about the same number of functional genes as *Arabidopsis,* and they occur in the same sequence. Therefore, knowledge of the *Arabidopsis* genome can be used to help locate specific genes in the genomes of other plants.

The creation of *Arabidopsis* mutants plays a significant role in discovering what each of its genes does. For example, if a mutant plant lacks stomata, then we know that the affected gene influences the formation of stomata.

The application of *Arabidopsis* genetics to other plants has been demonstrated. For example, one of the mutant genes that alters the development of flowers has been cloned and introduced into tobacco plants, where, as expected, it causes sepals and stamens to appear where petals would normally develop. Knowledge about the development of flowers in *Arabidopsis* can have far-ranging applications. It will someday lead to more productive crops.

A study of the *Arabidopsis* genome will undoubtedly promote plant molecular genetics in general. And because *Arabidopsis* is a model organism, genetic findings from this plant may have applications to humans, just as Mendel's work with pea plants led to the formulation of genetic laws. It's far easier to study signal transduction in *Arabidopsis* cells than in human cells.

Arabidopsis thaliana

Figure 26D Photograph of *Arabidopsis thaliana*. *Arabidopsis* consists of a flat rosette of leaves, from which grows a short flower stalk. Many investigators have turned to this weed as an experimental organism to study the actions of genes, including those that control growth and development. Nigel Cattlin/Alamy Stock Photo

Questions to Consider

1. Why is *Arabidopsis* a better study organism than cabbage or mustard?
2. What are the potential applications of *Arabidopsis* research?

in the transfer of a phosphate group from ATP to the protein portion of the photoreceptor (Fig. 26.11*c*). The phosphorylated photoreceptor triggers a transduction pathway that leads to the entry of auxin into the cell (see Fig. 26.3).

Thigmotropism

Thigmotropism (Gk. *thigma,* "touch"; *tropos,* "turning") is a response to touch from another plant, an animal, rocks, or the wind. An example of this response is the coiling of tendrils or the stems of plants, such as the stems of runner bean and morning glory plants (Fig. 26.12*b*).

A flowering plant grows straight up and down until it touches something. Then, the cells in contact with an object, such as a pole or an underground rock, grow less, while those on the opposite side elongate. Thigmotropism can be quite rapid; tendrils have been observed to encircle an object within 10 minutes. Several minutes of touching can bring about a response that lasts for several days. The response isn't always immediate—tendrils touched in the dark will respond once they are illuminated. ATP rather than light initiates the response. It is possible that auxin and ethylene play a role in the process, because they are capable of inducing the curvature of tendrils even in the absence of touch.

Thigmomorphogenesis is when a plant changes its overall shape due to an environmental touch stimulus, such as a barrier, wind, or rain. For example, when a storm blows across a field, plants in the field respond to these and other mechanical stresses by increasing production of fibers and collenchyma tissue (see Section 24.1). Cell elongation is inhibited, building shorter, sturdier plants. A tree growing in a windy location often has a shorter, thicker trunk than the same type of tree growing in a more protected location. Even simple mechanical stimulation, such as rubbing a plant with a stick, can inhibit cellular elongation and produce a sturdier plant with increased amounts of support tissue.

Gravitropism

Gravitropism is the effect of gravity on plant growth. When a seed germinates, the embryonic shoot exhibits negative gravitropism by growing upward *against* gravity. Increased auxin concentration of the lower side of the young stem causes the cells in that area to grow more than the cells on the upper side,

ISTUDY

a. Phototropism

b. Thigmotropism

Figure 26.12 Phototropism and thigmotropism. **a.** The stem of the plant curves toward the light, exhibiting positive phototropism. **b.** The stem of a runner bean plant, *Phaseolus,* coiling around a pole illustrates thigmotropism. (a): Martin Shields/Alamy Stock Photo; (b): Alison Thompson/Alamy Stock Photo

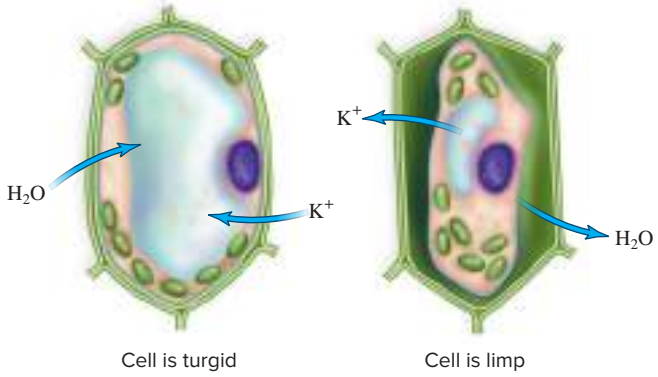
resulting in growth upward. The embryonic root exhibits positive gravitropism by growing *with* gravity downward into the soil (Fig. 26.13*a*).

Charles and Francis Darwin, in addition to studying coleoptiles, studied roots and discovered that if the root cap is removed, roots no longer respond to gravity. Since then, investigators have developed an explanation as to how root cells know which way is down. The root cap contains specialized cells filled with starch granules called **statoliths.** The statoliths are found within organelles called *amyloplasts.* Like marbles in a bag, statoliths settle to the bottom of a cell and put pressure on the other organelles, thus signaling the downward direction (Fig. 26.13*b*). This signal influences the placement of auxin and instructions for growth that follow.

If you place a potted plant on its side, the signals from the statoliths (and light) will change, and auxin will redistribute, causing the roots to bend downward and the shoot to grow upward. If a sideways plant is put in a clinostat, it will grow straight, because gravity will be negated if the plant is in constant motion (Fig. 26.13*c*).

Movement Caused by Internal Stimuli

Internal signals causing plants to exhibit nongrowth movements could be the result of electrical impulses (action potentials), hormonal action, or most commonly, changes in turgor pressure. Recall that a plant cell exhibits turgor when it fills with water:



In general, if water exits many cells of a leaf, the leaf goes limp. Conversely, if water enters a limp leaf, and cells exhibit turgor, the leaf moves as it regains its former position. **Turgor movements** (also called *nastic movements*) are dependent on turgor pressure changes in plant cells. In contrast to tropisms, turgor movements do not involve growth of an organ toward or away from a stimulus.

Turgor Responses to Touch

Turgor movements can result from touch, shaking, or thermal stimulation. Each leaf of the sensitive plant, *Mimosa pudica,* has many smaller leaflets. Touching one leaflet collapses the whole leaf within 1 second, knocking off any munching insect (Fig. 26.14).

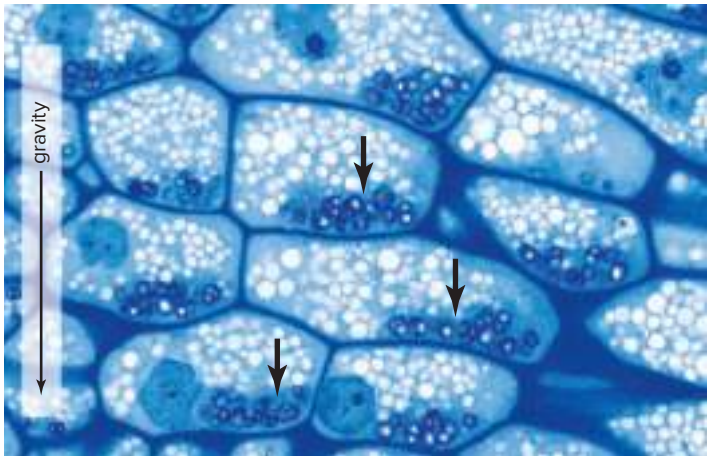
The structure of this plant involved in controlling turgor movement is a thickening called a *pulvinus* at the base of each leaflet. A leaf folds and the petiole drops when the cells in the lower half of the pulvinus, called the motor cells, lose potassium ions (K^+), then water. When the pulvinus cells lose turgor, the leaflets of the leaf collapse. An electrical mechanism may cause the response to move from one leaflet to another. The speed of an electrical charge transmission has been measured at about 1 cm/sec.

A Venus flytrap closes its trap in less than 1 second when three hairs inside the trap, called trigger hairs, are touched by an insect. When the trigger hairs are stimulated by the insect, an electrical charge is transmitted throughout the lobes of a leaf. Exactly what causes this electrical charge is being studied. Two possible explanations have been suggested: (1) Cells located near the outer region of the lobes rapidly secrete hydrogen ions into their cell walls, loosening them and allowing the walls to swell rapidly by osmosis; or (2) cells in the inner portion of the lobes and the midrib rapidly lose ions, leading to a loss of water by osmosis and collapse of these cells. In any case, it appears that turgor movements are involved.

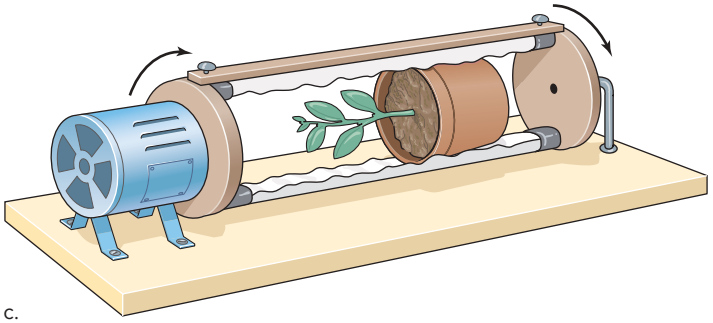
Sleep movements, discussed later, are another example of turgor movement caused by circadian rhythms and the effects of photosensitive pigments.

ISTUDY

a.

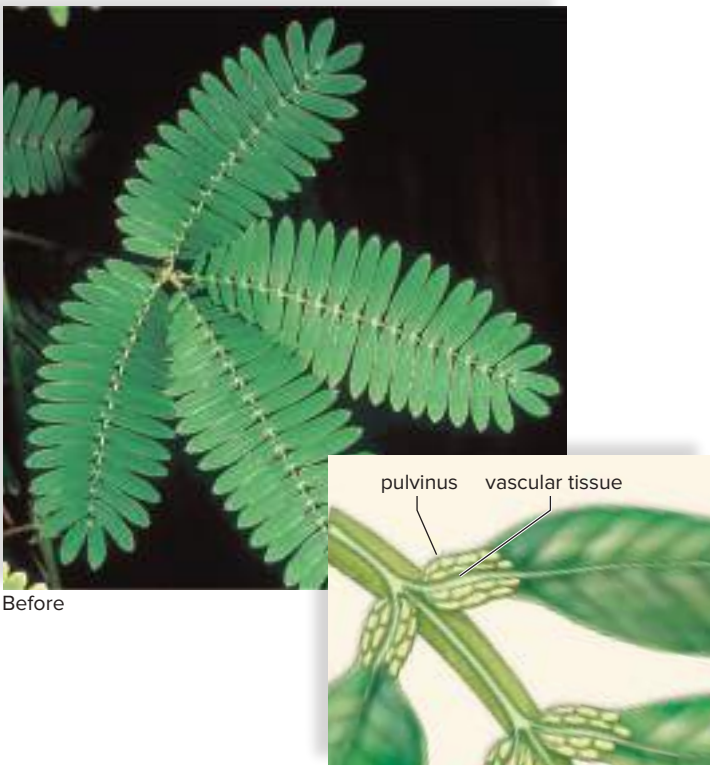


b.



c.

Figure 26.13 Gravitropism. **a.** This corn seed was germinated in a sideways orientation and in the dark. The shoot is growing upward (negative gravitropism) and the root downward (positive gravitropism). **b.** Sedimentation of statoliths (see arrows), which are starch granules, is thought to explain how roots perceive gravity. **c.** A clinostat, a tool used by plant biologists to negate the effects of gravity. Plants are slowly rotated so the statoliths do not settle to the bottom of cells. Thus, the typical bending of shoots and roots in response to gravity does not occur. (a): Martin Shields/Alamy Stock Photo; (b): Randy Moore



Before

After

Figure 26.14 Turgor movement. A leaf of the sensitive plant, *Mimosa pudica,* before and after it is touched. (both photos): John Kaprielian/Science Source

ISTUDY

Check Your Progress 26.2

1. Distinguish between a positive and a negative tropism.
2. Explain how root cells determine the downward direction.
3. List examples of internal and external stimuli.
4. Explain how changes in turgor pressure can cause a leaf to collapse.

26.3 Plant Responses to Phytochrome

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the conversion cycle of phytochrome.
2. Compare flowering in short-day/long-night plants and in long-day/short-night plants as it relates to photoperiodism.
3. Describe phytochrome's role in plant spacing.
4. Describe circadian rhythms and the connection to phytochrome.

Plants sense changes in light in two different ways: (1) They can sense if it is daytime, nighttime, or dawn/dusk and respond by adjusting metabolic processes such as photosynthesis; and (2) they can sense the time of year, affecting seasonal responses for processes such as germination or flowering.

The ability of plants to sense these changes in light lies mostly with the absorption of blue and red wavelengths of the visible light spectrum (see Fig. 7.7) by the light-sensing pigments phototropin and phytochrome. Next, we will focus on phytochrome and its involvement in both day/night cycles and seasonal cycles.

Phytochrome

Phytochrome (Gk. *phyton,* "plant"; *chroma,* "color") is a blue-green leaf pigment that is present in the cytoplasm of plant cells. A phytochrome molecule is composed of two identical proteins (Fig. 26.15).

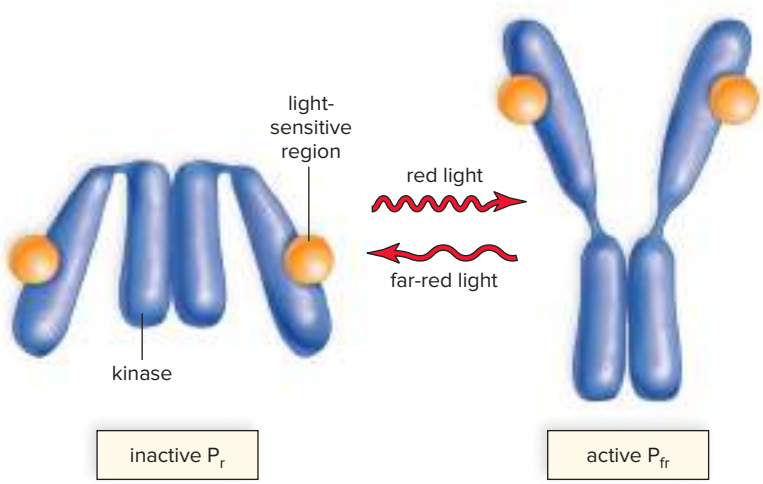


Figure 26.15 Phytochrome conversion cycle. The inactive form of phytochrome (P_r) is converted to the active form P_{fr} in the presence of red light, which is prevalent in daylight. P_{fr} is involved in various plant responses, such as seed germination, shoot elongation, and flowering. P_{fr} is converted to P_r whenever light is limited, such as in the shade or during the night.

Each protein has a larger portion in which a light-sensitive region is located. The smaller portion is a kinase that can link light absorption with a transduction pathway within the cytoplasm. Phytochrome can be said to act as a light switch, because, like a light switch, it can be in the down (inactive) position or in the up (active) position.

Phytochrome can distinguish between red wavelengths (650–680 nm) present during the daytime and far-red wavelengths (710–740 nm) present at dawn or dusk. These two ranges of red wavelengths cause the phytochrome protein to interconvert between two forms:

- P_r (phytochrome red) absorbs red light and is converted into P_{fr} in the daytime.
- P_{fr} (phytochrome far-red) absorbs far-red light and is converted into P_r in the evening.

Functions of Phytochrome

The $P_r \rightarrow P_{fr}$ conversion cycle is known to control various growth functions in plants. P_{fr} promotes seed germination, inhibits shoot elongation, promotes flowering, and affects plant spacing and accumulation of chlorophyll. The effects of phytochrome on flowering, germination, and plant spacing will be discussed in this section.

Flowering and Photoperiodism As just noted, many physiological changes in flowering plants are related to a seasonal change in day length. A physiological response prompted by changes in the length of day or night in a 24-hour daily cycle is called **photoperiodism** (Gk. *photos,* "light"; *periodus,* "completed course"). In some plants, photoperiodism influences flowering; for example, violets and tulips flower in the spring, and asters and goldenrod flower in the fall. Photoperiodism requires the participation of a biological clock (discussed later) and the activity of phytochrome.

Flowering plants can be divided into three groups on the basis of their flowering status.

1. **Short-day plants** flower when the day length is shorter than a *critical length.* (Examples are cocklebur, goldenrod, poinsettia, and chrysanthemum.)
2. **Long-day plants** flower when the day length is longer than a critical length. (Examples are wheat, barley, rose, iris, clover, and spinach.)
3. **Day-neutral plants** are not dependent on day length for flowering. (Examples are tomato and cucumber.)

The criterion for designating plants as short-day or long-day is not an absolute number of hours of light but a critical number that either must be or cannot be exceeded. Spinach is a long-day plant that has a critical day length of 14 hours; ragweed is a short-day plant with the same critical length. Spinach, however, flowers in the summer when the day length increases to 14 hours or more, and ragweed flowers in the fall, when the day length shortens to 14 hours or fewer. In addition, some plants require a specific sequence of day lengths in order to flower.

Soon after the three groups of flowering plants were distinguished, researchers began to experiment with artificial lengths of light and dark that did not necessarily correspond to a normal 24-hour day. These investigators discovered that the cocklebur, a short-day plant, does not flower if a required long dark period is

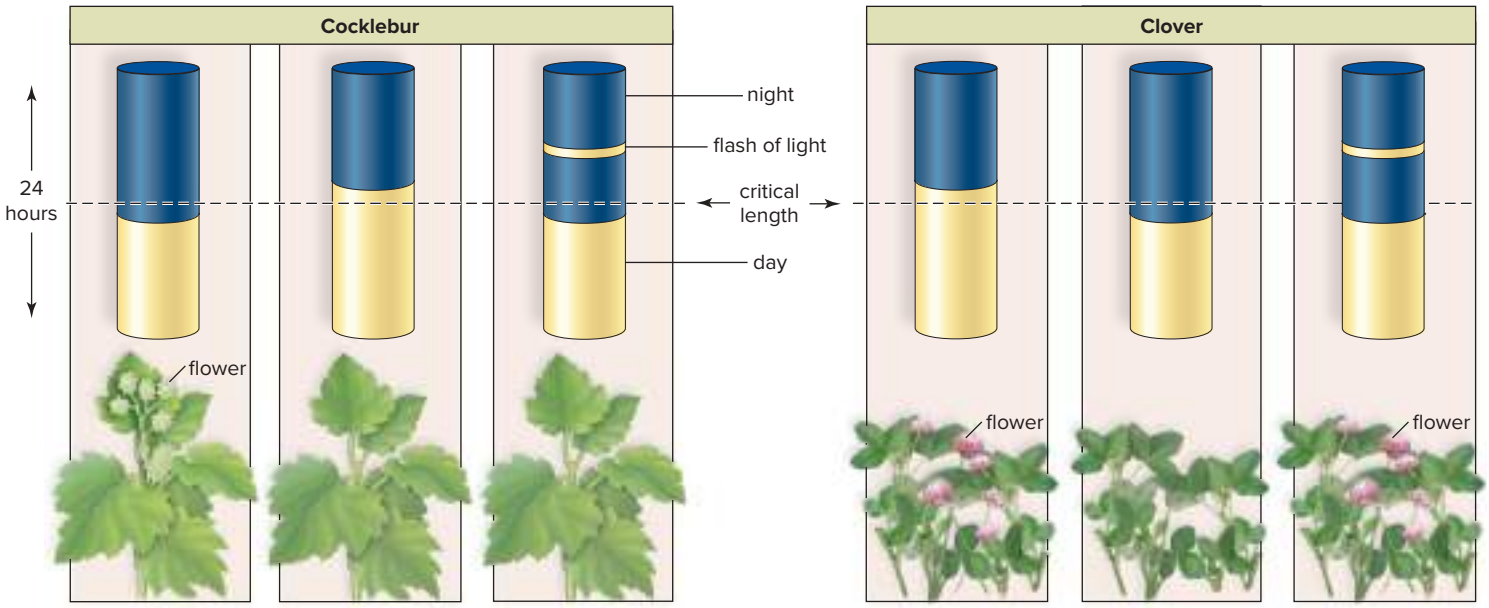


Figure 26.16 Photoperiodism and flowering. **a.** Short-day plant. When the day is shorter than a critical length, this type of plant flowers. The plant does not flower when the day is longer than the critical length. It also does not flower if the longer-than-critical-length night is interrupted by a flash of light. **b.** Long-day plant. The plant flowers when the day is longer than a critical length. When the day is shorter than a critical length, this type of plant does not flower. However, it does flower if the slightly longer-than-critical-length night is interrupted by a flash of light.

interrupted by a brief flash of white light. (Interrupting the light period with darkness has no effect.) In contrast, a long-day plant does flower if an overly long dark period is interrupted by a brief flash of white light. They concluded that the length of the dark period, not the length of the light period, controls flowering. Of course, in nature, short days always go with long nights, and vice versa.

To recap, let's consider Figure 26.16.

- Cocklebur is a short-day plant (Fig. 26.16a). First, when the night is longer than a critical length, cocklebur flowers. Conversely, the plant does not flower when the night is shorter than the critical length. Cocklebur also does not flower if the longer-than-critical-length night is interrupted by a flash of light.
- Clover is a long-day plant (Fig. 26.16*b*). When the night is shorter than a critical length, clover flowers. Conversely, the plant does not flower when the night is longer than a critical length. Finally, unlike the cocklebur, clover does flower when a slightly longer-than-critical-length night is interrupted by a flash of light.

Commercial florists and nursery owners have made extensive use of photoperiods, manipulating with artificial light the flowering times of poinsettias, some lilies, and other plants in order to have them flower at times of biggest demand, such as Christmas and Mother's Day.

Phytochrome and Germination The presence of P_{fr} indicates to some seeds that sunlight is present and conditions are favorable for germination. This explains why some seeds, such as those from snapdragons, poppies, and petunias, must be only partly covered with soil when planted. The seeds of the birch tree need 8 consecutive days of 10 hours of sunlight before they will germinate. Germination of other seeds, such as those of the mustard plant *Arabidopsis* or corn, is inhibited by light, so they must be planted deeper.

When a deeply planted seed germinates, it uses the food reserves stored within that seed to grow roots and elongate the shoot until it can push its way out from the soil. Notice the shoot in Figure 26.13*a*. The stem is elongated and yellow, because this seed has been grown in the dark, simulating growth in soil. The first flash of dim light instantly changes the developmental instructions for seedlings, and a new set of physical changes occur called *photomorphogenesis.* The presence of P_{fr} indicates that sunlight is available, and the seedlings begin to grow normally—the leaves expand and become green and the stem begins branching (see Fig. 27.11).

Next time you are in a produce aisle, notice two vegetables that are grown in the absence of light—white asparagus and alfalfa sprouts. These vegetables are simply elongated stems that are yellow, or **etiolated**—that is, the shoot increases in length, and the leaves remain small and yellow (Fig. 26.17). Plants that are grown in the dark do not receive a different set of instructions from their phytochrome pigments and thus never exhibit photomorphogenesis.

Phytochrome and Competition Plant spacing is another interesting function of phytochrome. A store-bought seed packet will always have instructions on spacing seeds to be planted. In nature, red and far-red light also signal spacing. Leaf shading increases the amount of far-red light relative to red light. Plants measure the amount of far-red light bounced back to them from neighboring plants. The closer together plants are, the more far-red relative to red light they perceive and the more likely they are to grow taller—a strategy for outcompeting others for sunlight.

ISTUDY

a. Normal growth b. Etiolation

Figure 26.17 Phytochrome controls shoot elongation and chlorophyll production. **a.** If red light is prevalent, as it is in bright sunlight, normal growth occurs. **b.** If far-red light is prevalent, as it is in the shade, etiolation occurs. These effects are due to phytochrome. Nigel Cattlin/Alamy Stock Photo

Circadian Rhythms

Many metabolic activities in plants, such as cellular respiration and photosynthesis, cycle through periods of high activity and low activity in a 24-hour period. These cycling changes are referred to as **circadian rhythms.** Jean de Mairan, a French astronomer, first identified circadian rhythms in 1729. He studied the *Mimosa* sensitive plant, which closes its leaves at night. When de Mairan put the plants in total darkness, they continued "sleeping" and "waking" just as they had when exposed to night and day.

Animals, fungi, protists, and plants all experience circadian rhythms in one way or another. In plants, the most visually striking rhythms are the sleep movements first described by de Mairan. Another common example of sleep movements occurs in a plant called *Oxalis,* which is often sold as a "shamrock" before St. Patrick's Day (Fig. 26.18*a*). The leaves (and flowers) open during the day and close at night. This movement is due to changes in the turgor pressure of motor cells in swellings called *pulvini,* located at the base of each leaf.

Morning glory (*Ipomoea leptophylla*) is a plant that opens its flowers in the early part of the day and closes them at night (Fig. 26.18*b*). In most plants, stomata open in the morning and close at night, and some plants secrete nectar at the same time of the day or night.

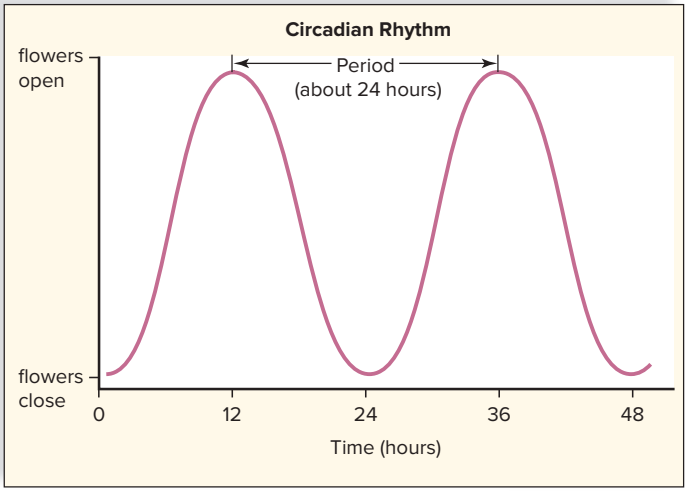
In the natural environment, circadian rhythms are entrained to a daily cycle through the action of phytochrome and blue-light receptors. *Entrainment* means to be synchronized to light at daybreak. Overall, to qualify as a circadian rhythm, the activity must (1) occur every 24 hours, (2) take place whether or not the day/night lighting is present, and (3) be able to be reset if external cues are provided. For example, if you take a transcontinental

Oxalis plant (morning) *Oxalis* plant (night)

a.

Morning glory (morning) Morning glory (night)

b.



c.

Figure 26.18 Circadian rhythms. **a.** The leaves of the *Oxalis* plant fold every 24 hours at night. **b.** The flowers of the morning glory, *Ipomoea leptophylla,* close at night. **c.** Graph of circadian rhythm exhibited by the morning glory. The period is the time between peaks in a repeating cycle. (a) (both): Gretel Guest; (b) (morning): Michael Jenner/Alamy Stock Photo; (night): JupiterImages/360/Photos.com/Getty Images

ISTUDY

flight, you will likely suffer jet lag at the destination, because your body will still be attuned to the day/night pattern of its previous environment. But after several days, you probably will adjust and will be able to go to sleep and wake up according to your new time.

Biological Clock

The internal mechanism by which a circadian rhythm is maintained in the absence of appropriate environmental stimuli is termed a **biological clock.** If organisms are sheltered from environmental stimuli, their biological clock keeps the circadian rhythms going, but the cycle extends. In plants with sleep movements, the sleep cycle changes to 26 hours when the plant is kept in constantly dim light, as opposed to 24 hours when in traditional day/night conditions. Therefore, it is suggested that biological clocks are synchronized by external stimuli to 24-hour rhythms.

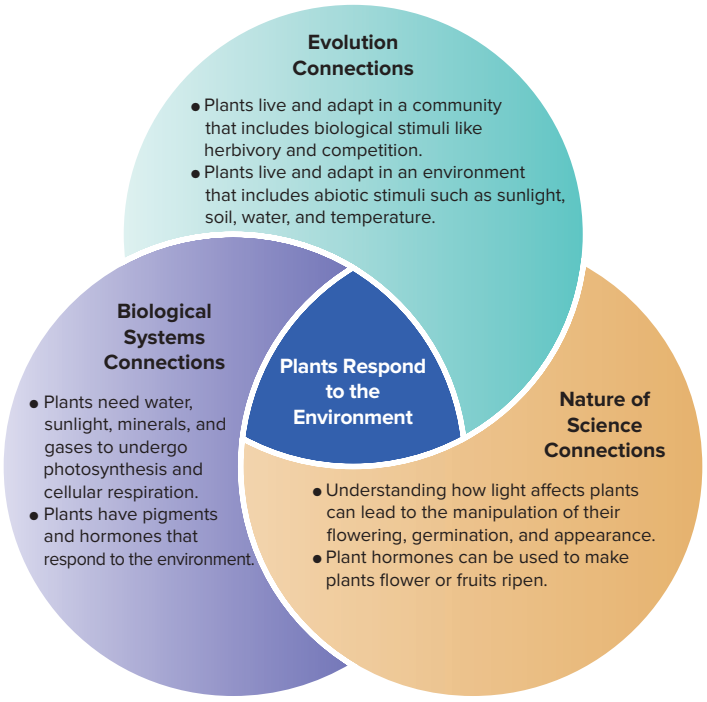
As previously mentioned, the length of daylight compared to the length of darkness, called the photoperiod and influenced by phytochrome, sets the clock. Temperature has little or no effect. This synchronization with light is adaptive, because the photoperiod indicates seasonal changes better than temperature changes. Spring and fall, in particular, can have both warm and cold days.

Work with *Arabidopsis* (see the Nature of Science feature, "Why So Many Scientists Work with *Arabidopsis,*" in Section 26.2) and other organisms suggests that the biological clock involves the transcription of a small number of "clock genes." Although circadian rhythms are outwardly very similar in all species, the clock genes that have been identified are not the same, because biological clocks have evolved several times in different organisms to perform similar tasks.

Check Your Progress 26.3

1. Describe the phytochrome protein and its conversion between two forms.
2. Explain why a long-day plant still flowers if the long day is interrupted by a period of darkness.
3. Describe the various roles of phytochrome.
4. Explain the criteria for circadian rhythms.

Connecting *the* Concepts



What Did You Learn?

26.1 Plant Hormones

Like animals, flowering plants use a **signal transduction pathway** when they respond to a stimulus. The process involves **receptor** activation, **transduction** of the signal by relay proteins, and a **cellular response,** which can consist of the turning on of a gene or an enzymatic pathway.

There are six main plant **hormones** that are often the activation or the response of these pathways.

- Early studies using **coleoptiles** helped discover and define the role of **auxin.** Auxin-controlled cell elongation is involved in phototropism and gravitropism. When a plant is exposed to light, auxin moves laterally from the bright to the shady side of a stem. Auxin is also responsible for the **apical dominance** of a plant.
- **Gibberellin** causes stem elongation between nodes and breaks bud and seed **dormancy.** After this hormone binds to a plasma membrane receptor, a DNA-binding protein activates a gene, leading to the production of amylase.
- **Cytokinins** cause cell division, the effects of which are especially obvious when plant tissues are grown in culture. The absence of cytokinins results in aging, or **senescence.**

ISTUDY

- **Abscisic acid (ABA)** and **ethylene** are two plant growth inhibitors. ABA is well-known for causing stomata to close, and ethylene is known for causing **abscission** and prompting fruits to ripen.
- **Brassinosteroids** are involved in plant cell differentiation and stress responses.

Secondary metabolites are other chemicals produced by plants to combat biotic stimuli, such as herbivory and parasitism.

26.2 Plant Growth and Movement Responses

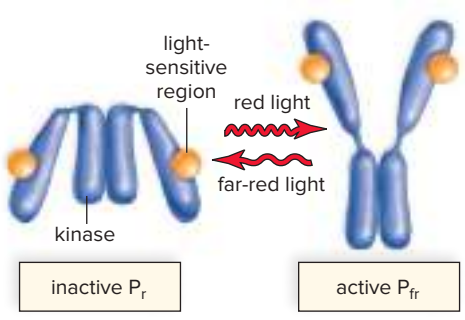
When flowering plants respond to stimuli, this causes growth, movement, or both to occur. **Tropisms** are growth responses toward or away from unidirectional stimuli.

- The positive **phototropism** of stems results in a bending toward light, and the negative **gravitropism** of stems results in a bending away from the direction of gravity. Roots that bend toward the direction of gravity show positive gravitropism. Positive gravitropism of roots is due to **statoliths** settling to the bottom of root cap cells. **Thigmotropism** occurs when a plant part makes contact with an object, as when tendrils coil about a pole.

Turgor movements, or nastic movements, are not directional. Due to turgor pressure changes, some plants respond to touch and some perform sleep movements.

26.3 Plant Responses to Phytochrome

Phytochrome is a pigment that is involved in **photoperiodism,** the ability of plants to sense the length of the day and night during a 24-hour period. This sense leads to seed germination, shoot elongation, and flowering during favorable times of the year. The conformation, and activity, of phytochrome is influenced by daylight. Phytochrome in the P_{fr} form leads to a biological response, such as flowering.



- **Short-day plants** flower only when the days are shorter than a critical length, and **long-day plants** flower only when the days are longer than a critical length. Research has shown that actually it is the length of darkness that is critical. Interrupting the dark period with a flash of white light prevents flowering in a short-day plant and induces flowering in a long-day plant. **Day-neutral plants** are not affected by periods of dark and light. Phytochrome is also involved in plant spacing, germination, and stem elongation. A plant grown in darkness will exhibit **etiolation,** where stems continue to elongate and remain yellow.
- Plants exhibit **circadian rhythms,** which are believed to be controlled by a **biological clock.** The sleep movements of *Oxalis* plants, the closing of stomata, and the daily opening of certain flowers have a 24-hour cycle.

Thinking Critically About Biology

1. You hypothesize that abscisic acid (ABA) is responsible for the turgor pressure changes that permit a plant to track the sun (see the chapter-opening photograph). What observations could you make to support your hypothesis?
2. You formulate the hypothesis that the negative gravitropic response of stems is greater than the positive phototropism of stems. How would you test your hypothesis?
3. Farmers who grow crops that are planted as seedlings may prepare them for their transition from the greenhouse to the field by brushing them gently every day for a few weeks. Why is this beneficial?
4. Because diverse plants exude volatile compounds in response to herbivore or pathogen attack, some experts have written about "talking trees." Explain why someone would use this adage.

Making It Relevant to You

1. Describe examples of environmental stimuli that plants must respond to.
2. Now that you know that plants respond to environmental stimuli, how might you change the way you keep houseplants or a vegetable garden?
3. Much more food is being grown in greenhouses to eliminate the need for farmland and chemicals like herbicides. Why would the study of plant responses be important for maintaining healthy and high-yielding plants in a greenhouse environment?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

27

Flowering Plants: Reproduction

(Orange Canada Lily): Tom Crowning/Shutterstock; (Bacteria): Melissa Brower/CDC

Chapter Outline

27.1 Sexual Reproductive Strategies
27.2 Seed Development
27.3 Fruit Types and Seed Dispersal
27.4 Asexual Reproductive Strategies

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 10.3 What role does meiosis play in sexual reproduction?

Section 14.2 How is biotechnology changing how plants are reproduced?

Table 18.1 How long have angiosperms and modern insects been coevolving?

Microbiomes and Flowers

The term *microbiome* was first introduced in 1988 and is defined as the microbial community occupying a particular habitat. In recent years, the human gut microbiome has been given center stage regarding gut health and overall well-being. Current research has shown that flowering plants have a microbiome too, affecting various reproductive processes, including how long a plant keeps its flowers in bloom.

Previously, flower characteristics were studied based on their interactions with pollinators, and flower longevity was correlated with temperature—the colder the environment, the longer blooms last, and the warmer the environment, the shorter blooms last. Today, scientists have a better understanding of how flowers provide habitats for microbes on glands that secrete nectar and on the surface of the stigma, an area rich with nutrients and humidity for pollen tube growth. Microbe numbers increase over time after a flower opens and can shorten the period that blooms are present. Some bacteria can even impact the quality of the fruit. A plant's microbiome may have more to do with the evolution of angiosperms and their reproductive ecology than previously thought.

As you read through this chapter, think about the following questions:

1. How are flowering plants adapted to a terrestrial lifestyle?
2. What would happen to flowering plants if the surrounding biodiversity increased or decreased?

Following the Themes

CHAPTER 27 – FLOWERING PLANTS: REPRODUCTION

Evolution	The majority of flowering plants owe their success to the animal pollinators with which they have coevolved.
Nature of Science	Plants rely on wind, water, and animals for pollination. Ecologists are studying the potential consequences for flowering plants if the population of specific animal pollinators were to decrease.
Biological Systems	The life cycle of flowering plants is dependent on successful pollination, fertilization, and seed germination.

CEPIEC
ISTUDY

27.1 Sexual Reproductive Strategies

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the key components of the flowering plant life cycle.
2. Recognize the functions of the key flower parts.
3. Compare the male and female gametophytes of flowering plants.
4. Describe the process of double fertilization in flowering plants.

Sexual reproduction in plants is advantageous because it generates variation among the offspring through the process of meiosis and fertilization. In a changing environment, a new variation may be better adapted for survival and reproduction than either parent.

Life Cycle Overview

When plants reproduce sexually, they undergo alternation of generations, in which they alternate between two multicellular stages, one diploid and one haploid.

To review, let's consider Figure 27.1 by beginning with the flowering sporophyte and moving clockwise. In flowering plants, the diploid sporophyte is dominant, and it is the portion of the life cycle that bears flowers. Next, a **flower,** which is the reproductive structure of angiosperms, produces two types of spores by meiosis: microspores and megaspores. The haploid **microspore** (Gk. *mikros,* "small, little") undergoes mitosis and becomes a pollen grain, which is either windblown or carried by an animal to the vicinity of the female gametophyte. A pollen grain is the male gametophyte.

In the meantime, the haploid **megaspore** (Gk. *megas,* "great, large") undergoes mitosis and becomes the female gametophyte. The female gametophyte is an embryo sac within the ovule that is within an ovary. At maturity, a pollen grain contains nonflagellated sperm, which travel by way of a pollen tube to the embryo sac. Once a sperm fertilizes an egg, the zygote becomes an embryo, still within an ovule. The ovule develops into a **seed,** which contains the embryo and stored food surrounded by a seed coat. The ovary becomes a fruit, which aids in dispersing the seeds. When a seed germinates, a new sporophyte emerges and through mitosis and growth becomes a mature organism.

Notice that the sexual life cycle of flowering plants is adapted to a land-based existence. The microscopic female gametophytes develop completely within the sporophyte and are thereby protected from desiccation. Pollen grains (male gametophytes) are not released until they develop a thick wall. Unlike the mosses, no external water is needed to bring about fertilization in flowering plants. Instead, the pollen tube provides passage for a sperm to reach an egg. Following fertilization, the embryo and its stored food are enclosed within a protective seed coat until external conditions are favorable for germination.

Flowers

The flower is unique to angiosperms (Fig. 27.2). Aside from producing the spores and protecting the gametophytes, flowers often attract pollinators, which aid in transporting pollen from plant to plant. Flowers also produce the fruits that enclose the seeds.

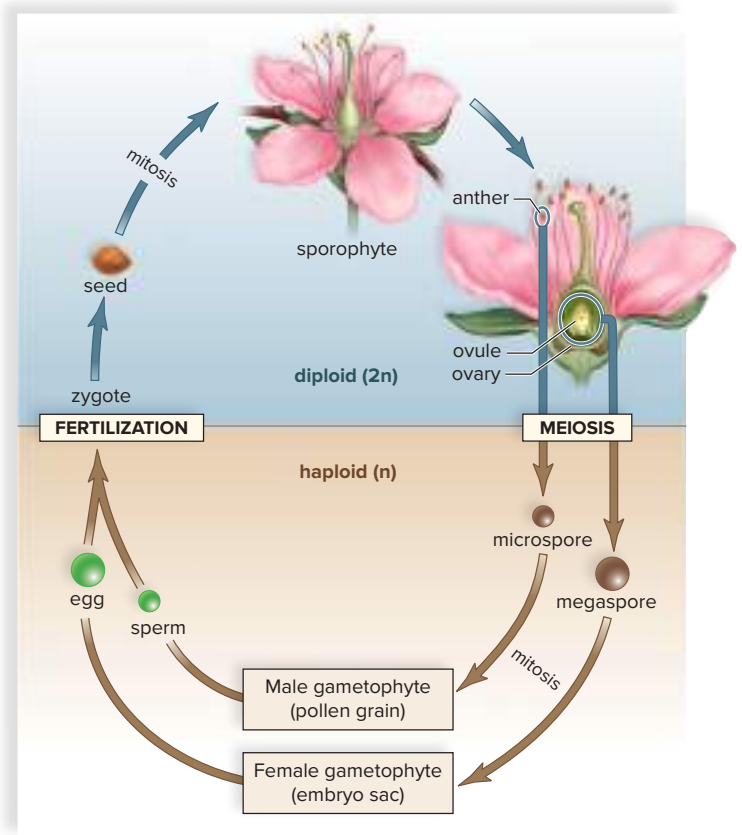


Figure 27.1 Sexual reproduction in flowering plants. The sporophyte bears flowers. The flower produces microspores within anthers and megaspores within ovules by meiosis. A megaspore becomes a female gametophyte, which produces an egg within an embryo sac, and a microspore becomes a male gametophyte (pollen grain), which produces sperm. Fertilization results in a zygote. A seed contains an embryo and stored food within a seed coat. After dispersal, a seed becomes a new sporophyte plant.

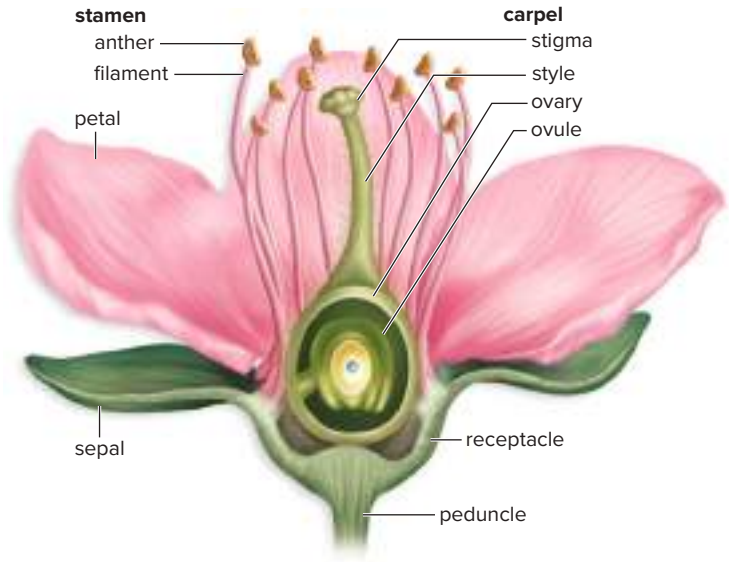
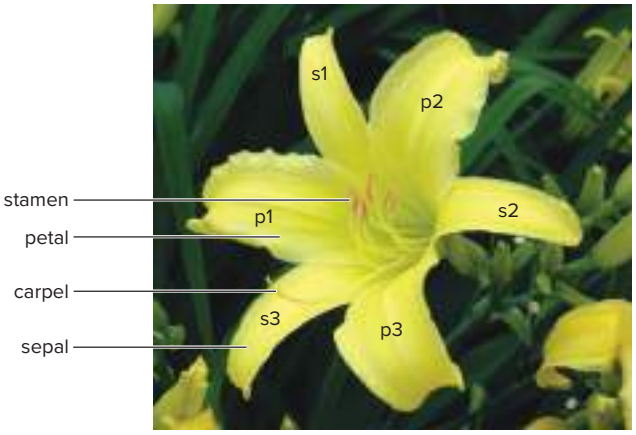
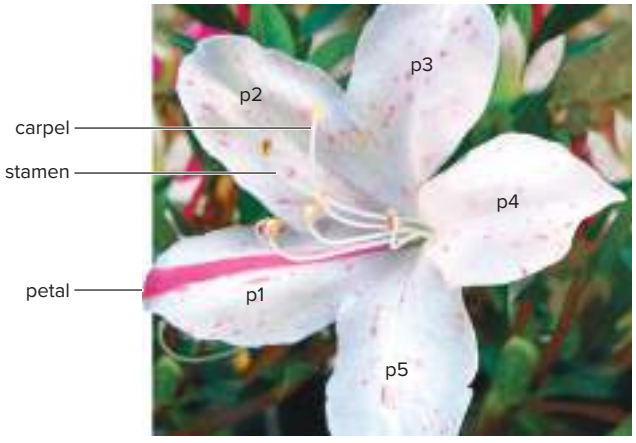


Figure 27.2 Anatomy of a flower. A complete flower has all flower parts: sepals, petals, stamens, and at least one carpel.

The evolution of the flower was a major factor leading to the success of angiosperms, with over 270,000 species. Flowering is often a response to environmental signals, such as the length of the day



a. Daylily, *Hemerocallis* sp.



b. Festive azalea, *Rhododendron* sp.

Figure 27.3 Monocot versus eudicot flowers. **a.** Monocots, such as daylilies, have flower parts in threes. In particular, note the three petals and three sepals, both of which are colored in this flower. **b.** Azaleas are eudicots. They have flower parts in fours or fives; note the five petals of this flower. p = petal, s = sepal (a): Plantography/Alamy Stock Photo; (b): ©Dr. Murray P. Pendarvis

or time of year (see Section 26.3). In many plants, a flower develops when a shoot apical meristem that previously formed leaves suddenly stops producing leaves and starts producing a flower enclosed within a bud. In other plants, axillary buds develop directly into flowers. In monocots, flower parts occur in multiples of three; in eudicots, flower parts occur in multiples of four or five (Fig. 27.3).

Flower Structure

The typical complete eudicot flower has all flower parts: sepals, petals, stamens, and carpels. The four whorls of modified leaves are attached to a receptacle at the end of a flower stalk called a *peduncle.*

- The **sepals** are the most leaflike of all the flower parts and are usually green. Sepals protect the bud as the flower develops within. Collectively, the sepals are called the **calyx.**
- An open flower next has a whorl of **petals,** whose color accounts for the attractiveness of many flowers. The size, shape, and color of petals are attractive to specific pollinators. Some wind-pollinated flowers have no petals at all. Collectively, the petals are called the **corolla.**
- **Stamens** are the "male" portions of flowers. Each stamen contains two parts: the slender stalk called the *filament* (L. *filum,* "thread") and the saclike *anther* supported by the filament. Pollen grains develop from the microspores produced in the anther.
- At the very center of a flower is the **carpel,** a vaselike structure that represents the "female" portion of the flower. A carpel usually has three parts: The **style** is a slender stalk that supports the **stigma,** an enlarged, sticky knob; and the **ovary,** an enlarged base that encloses one or more *ovules* (see Fig. 27.2).

Ovules (L. *ovulum,* "little egg") play a significant role in the production of megaspores and therefore female gametophytes, as described shortly.

A flower can have a single carpel or multiple carpels. Sometimes several carpels are fused into a single structure, in which case the ovary is termed compound; it has several chambers, each of which contains ovules. For example, an orange develops from a compound ovary, and every section of the orange is a chamber.

Variations in Flower Structure

Not all flowers have sepals, petals, stamens, or carpels. Those that do are said to be *complete,* and those that do not are said to be *incomplete.* Flowers that have both stamens and carpels are called *perfect* (bisexual) flowers; flowers with only stamens and those with only carpels are *imperfect* (unisexual) flowers.

If both staminate (male) flowers and carpellate (female) flowers occur on a single plant, the plant is *monoecious* (Gk. *monos,* "one"; *oikos,* "home, house") (Fig. 27.4*a*). If staminate

a.

b.

Figure 27.4 Monoecious and dioecious plants. **a.** The flowers of the mature *Jatropha* plant are monoecious, having both male and female flowers on the same plant. **b.** Holly trees are dioecious; berries are produced only by female plants, and pollen only by male plants. (a): Steven P. Lynch; (b): Martin Hughes-Jones/Garden World Images/age fotostock

ISTUDY

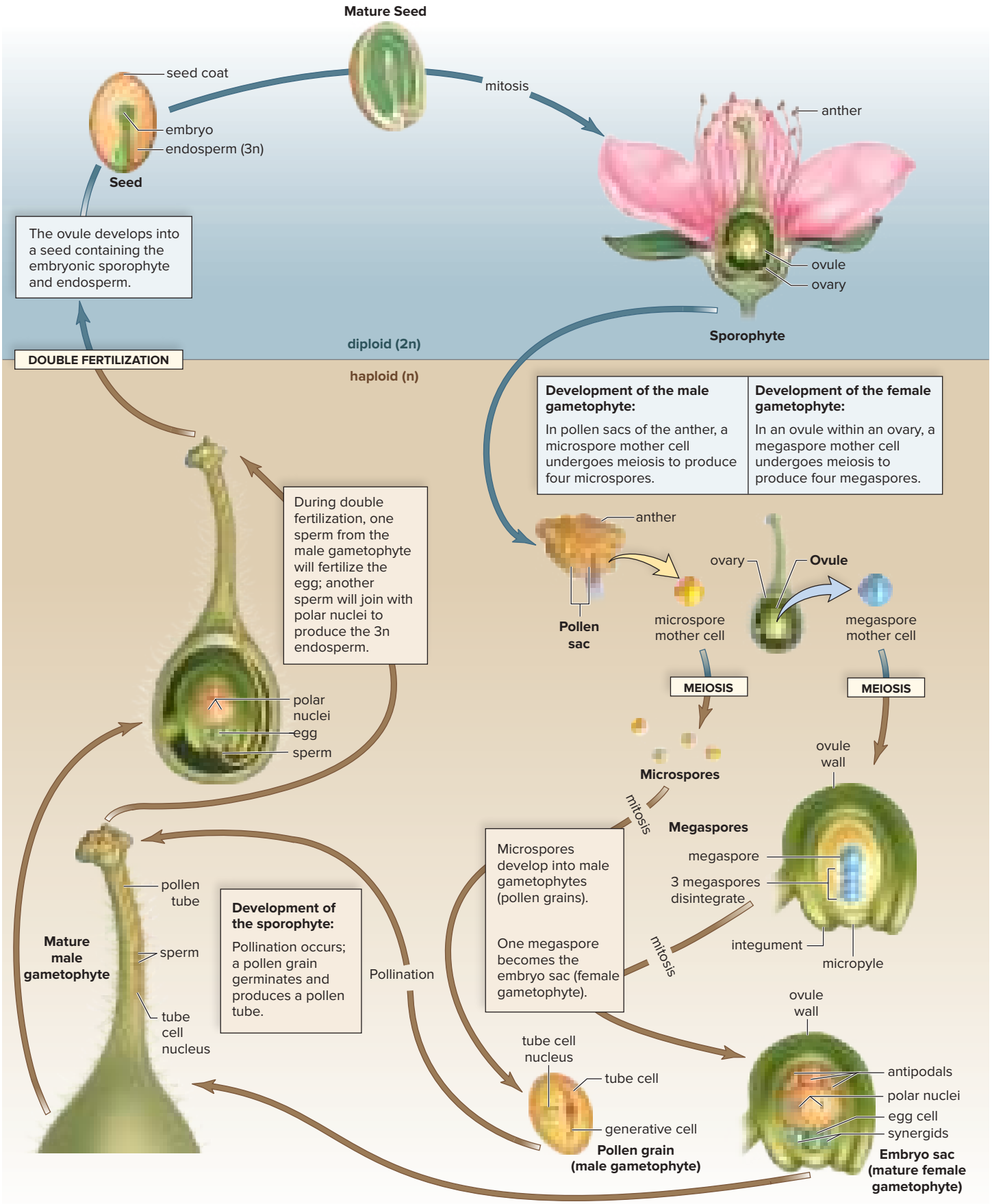


Figure 27.5 **Life cycle of flowering plants.** *Development of gametophyte* (*bottom*): A pollen sac in the anther contains microspore mother cells, which produce microspores by meiosis. A microspore develops into a pollen grain, which germinates and has two sperm. An ovule in an ovary contains a megaspore mother cell, which produces a megaspore by meiosis. A megaspore develops into an embryo sac containing seven cells, one of which is an egg. *Development of sporophyte* (*top*): A pollen grain contains two sperm by the time it germinates and forms a pollen tube. During double fertilization, one sperm fertilizes the egg to form a diploid zygote, and the other fuses with the polar nuclei to form a triploid (3n) endosperm cell. A seed contains the developing sporophyte embryo plus stored food.

ISTUDY

and carpellate flowers occur on separate plants, the plant is *dioecious.* Holly trees are dioecious; if you hope to have holly berries, you need one plant with staminate flowers and another plant with carpellate flowers (Fig. 27.4*b*).

Life Cycle in Detail

In all land plants, the sporophyte produces haploid spores by meiosis. The haploid spores grow and develop into haploid **gametophytes,** which produce gametes by mitotic division. Flowering plants, however, are *heterosporous*—they produce microspores and megaspores. Microspores become mature male gametophytes (sperm-bearing pollen grains), and megaspores become mature female gametophytes (egg-bearing embryo sacs).

Development of Male Gametophyte

Microspores are produced in the anthers of flowers (Fig. 27.5). An anther has four pollen sacs, each containing many microspore mother cells. A microspore mother cell undergoes meiosis to produce four haploid microspores. In each, the haploid nucleus divides mitotically, followed by unequal cytokinesis, and the result is two cells enclosed by a finely sculptured wall. This structure, called the **pollen grain,** is at first an immature *male gametophyte* that consists of a tube cell and a generative cell. The larger tube cell will eventually produce a *pollen tube.* The smaller generative cell divides mitotically either now or later to produce two sperm. Once these events have taken place, the pollen grain has become the mature male gametophyte.

Development of Female Gametophyte

The ovary contains one or more ovules. An ovule has a central mass of parenchyma cells almost completely covered by layers of tissue called integuments, except where there is an opening called a micropyle. One parenchyma cell enlarges to become a megaspore mother cell, which undergoes meiosis, producing four haploid megaspores (Fig. 27.5). Three of these megaspores are nonfunctional; one is functional. In a typical pattern, the nucleus of the functional megaspore divides mitotically until there are eight nuclei in the *female gametophyte.* When cell walls form later, there are seven cells, one of which is binucleate.

The female gametophyte, also called the **embryo sac,** consists of these seven cells:

- One egg cell, associated with two synergid cells
- One central cell, with two polar nuclei
- Three antipodal cells

Development of New Sporophyte

The walls separating the pollen sacs in the anther break down when the pollen grains are ready to be released. **Pollination** is simply the transfer of pollen from an anther to the stigma of a carpel. Self-pollination occurs if the pollen is from the same plant, and cross-pollination occurs if the pollen is from a different plant of the same species.

When a pollen grain lands on the stigma of the same species, it germinates, forming a pollen tube (Fig. 27.5). The germinated pollen grain, containing a tube cell and two sperm, is the mature male gametophyte. As it grows, the pollen tube passes between the cells of the stigma and the style to reach the micropyle, a pore of the ovule. When the pollen tube reaches the micropyle, **double fertilization** occurs: One of the sperm unites with the egg to form a 2n zygote; however, the second sperm unites with the two polar nuclei centrally placed in the embryo sac to form a 3n endosperm nucleus. This latter fertilization is unique to angiosperms. The endosperm nucleus eventually develops into the **endosperm** (Gk. *endon,* "within"; *sperma,* "seed"), a nutritive tissue that the developing embryonic sporophyte will use as an energy source.

Now the ovule begins to develop into a seed. One important aspect of seed development is formation of the seed coat from the ovule wall. A mature seed contains the embryo, stored food, and the seed coat (see Fig. 27.8).

Cross-Pollination

Some species of flowering plants, such as grasses and grains, rely on wind pollination (Fig. 27.6), as do the gymnosperms, the other type of seed plant. Much of the plant's energy goes into making pollen to ensure that some pollen grains actually reach a stigma. The amount of pollen successfully transferred is staggering: A single corn plant may produce 20–50 million pollen grains a season. In corn, the flowers tend to be monoecious, and clusters of tiny male flowers move in the wind, freely releasing pollen into the air.

Most angiosperms rely on animals—insects (e.g., bumblebees, flies, butterflies, and moths), birds (e.g., hummingbirds), or mammals (e.g., bats)—to carry out pollination. The use of animal pollinators is unique to flowering plants, and it helps account for why these plants are so successful on land.

By the time flowering plants appear in the fossil record, around 240 MYA, insects had long been present. For millions of years, plants and their animal pollinators have coevolved. **Coevolution** means that as one species changes, the other

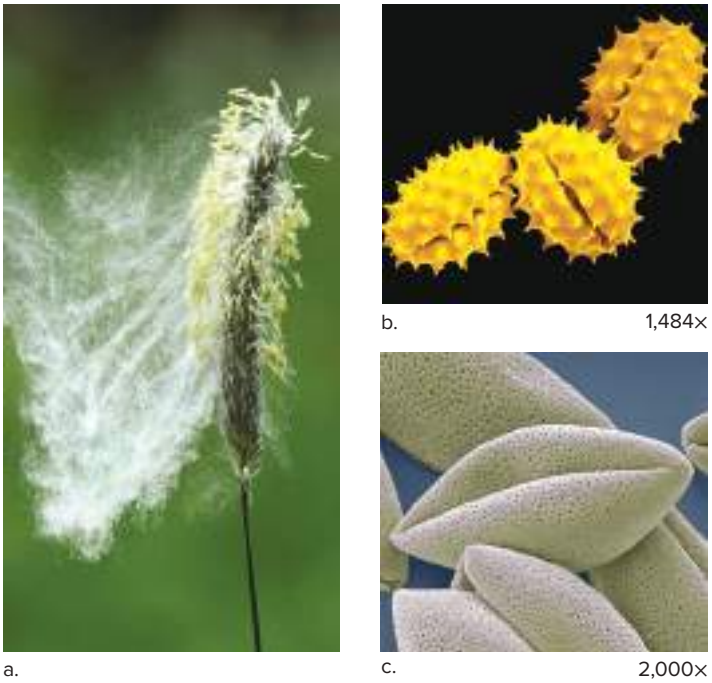


Figure 27.6 Pollination. **a.** Grass releasing pollen. **b.** Pollen grains of Canadian goldenrod, *Solidago canadensis.* **c.** Pollen grains of asparagus, *Asparagus officinalis.* The shape and pattern of pollen grain walls are quite distinctive, and experts can use them to identify the genus, and even sometimes the species, that produced a particular pollen grain. Pollen grains have strong walls resistant to chemical and mechanical damage; therefore, they frequently become fossils. (a): Tim Gainey/Alamy Stock Photo; (b): Mediscan/Alamy Stock Photo; (c): Steve Gschmeissner/Science Photo Library/Getty Images

ISTUDY

THEME Evolution

Plants and Their Pollinators

Plants and their pollinators have adapted to one another. They have a mutualistic relationship in which each benefits—the plant uses its pollinator to ensure that cross-pollination takes place, and the pollinator uses the plant as a source of food. This mutualistic relationship came about through the process of coevolution—that is, the interdependency of the plant and the pollinator is the result of suitable changes in the structure and function of each.

The evidence for coevolution is observational. For example, floral coloring and odor are suited to the sense perceptions of the pollinator; the mouthparts of the pollinator are suited to the structure of the flower; the type of food provided is suited to the nutritional needs of the pollinator; and the pollinator forages at the time of day that specific flowers are open.

Bee- and Wasp-Pollinated Flowers

There are 20,000 known species of bees that pollinate flowers. The best-known pollinators are the honeybees (Fig. 27A*a*). Bee eyes see ultraviolet (UV) wavelengths. Therefore, bee-pollinated flowers are usually brightly colored and are predominantly blue or yellow; they are not entirely red. They may also have ultraviolet shadings called nectar guides, which highlight the portion of the flower that contains the reproductive structures.

The mouthparts of bees are fused into a long tube that contains a tongue. This tube is an adaptation for sucking up nectar provided by the plant, usually at the base of the flower. Bees also collect pollen as a food.

Bee flowers are delicately sweet and fragrant, advertising that nectar is present. The nectar guides often point to a narrow floral tube large enough for the bee's feeding apparatus but too small for other insects to reach the nectar. Bee-pollinated flowers are sturdy and may be irregular in shape because they often have a landing platform where the bee can alight. The flower structure requires the bee to brush up against the anther and stigma as it moves toward the floral tube to feed.

One type of orchid, *Ophrys,* has evolved a unique adaptation. The flower resembles a female wasp, and when the male of that species attempts to copulate with the flower, the flower spring-loads pollen on the wasp's head. When the frustrated wasp attempts to "copulate" with another flower, the pollen is perfectly positioned to come in contact with the stigma of the second flower.

Moth- and Butterfly-Pollinated Flowers

Contrasting moth- and butterfly-pollinated flowers emphasizes the close adaptation between pollinator and flower. Both moths and butterflies have a long, thin, hollow proboscis, but they differ in other characteristics. Moths usually feed at night and have a well-developed sense of smell. The flowers they visit are visible at night, because they are lightly shaded (white, pale yellow, or pink), and they have strong, sweet perfume, which helps attract moths. Moths hover when they feed, and their flowers have deep tubes with open margins that allow the hovering moth to reach the nectar with its long proboscis.

Butterflies, in comparison, are active in the daytime and have good vision but a

a.

b.

Figure 27A Insect pollinators. **a.** A bee-pollinated flower is a color other than red (bees cannot detect this color). The reproductive structures of the flower brush up against the bee's body, ensuring that pollen is transferred. **b.** A butterfly-pollinated flower is often a composite, containing many individual flowers. The broad expanse provides room for the butterfly to land, after which it lowers its proboscis into each flower in turn. (a): Steven P. Lynch; (b): James Urbach/Purestock/Getty Images

(*Continued*)

species undergoes adaptation in response, so that in the end the two species are suited to one another. Plants with flowers that attracted a pollinator had an advantage, because in the end they produced more seeds. Similarly, pollinators that were able to find and remove food from the flower were more successful. Today, we see that the reproductive parts of the flower are positioned so the pollinator picks up pollen from one flower and delivers it to another. And concurrently, the mouthparts of the pollinator are suited to gathering the nectar from these particular plants. A description of the coevolution between plants and their pollinators is given in the Evolution feature, "Plants and Their Pollinators."

As mentioned in the feature above, one well-studied example of coevolution has occurred between bees and the plants they pollinate. Bee-pollinated flowers tend to be yellow, blue, or white because these are the colors bees can see. Bees respond to ultraviolet (UV) markings, called nectar guides, that help them locate nectar. Humans cannot detect light in the ultraviolet range, but bees are sensitive to UV. A bee has a feeding proboscis of the right length to collect nectar from certain flowers and a pollen basket on its hind legs that allows it to carry pollen back to the hive. Many fruits and vegetables are dependent on bee pollination, leading to great concern today because the number of bees is declining due to disease and the use of pesticides.

ISTUDY

weak sense of smell. Their flowers have bright colors—even red, because butterflies can see the color red—but the flowers tend to be odorless. Unable to hover, butterflies need a place to land. Flowers that are visited by butterflies often have flat landing platforms (Fig. 27A*b*). Composite flowers (composed of a compact head of numerous individual flowers) are especially favored by butterflies. Each flower has a long, slender floral tube, accessible to the long, thin butterfly proboscis.

Bird- and Bat-Pollinated Flowers

In North America, the most well-known bird pollinators are the hummingbirds. These small animals have good eyesight but do not have a well-developed sense of smell. Like moths, they hover when they feed. Typical flowers pollinated by hummingbirds are red, with a slender floral tube and margins that are curved back and out of the way. And although they produce copious amounts of nectar, the flowers have little odor. As a hummingbird feeds on nectar with its long, thin beak, its head comes in contact with the stamens and pistil (Fig. 27B*a*).

Bats are adapted to gathering food in various ways, including feeding on the nectar and pollen of plants. Bats are nocturnal and have an acute sense of smell. Those that are pollinators also have keen vision and a long, extensible, bristly tongue. Typically, bat-pollinated flowers open only at night and are light-colored or white. They have a strong, musky smell similar to the odor that bats produce to attract one another. The flowers are generally large and sturdy and are able to hold up when a bat inserts part of its head to reach the nectar. While the bat is at the flower, its head becomes dusted with pollen (Fig. 27B*b*).

Coevolution

How did this coevolution of plants and pollinators come about? Some 250 million years ago, when seed plants were just beginning to evolve and insects were not as diverse as they are today, wind alone carried pollen. Wind pollination, however, is a hit-or-miss affair. Perhaps beetles feeding on vegetative leaves were the first insects to carry pollen directly from plant to plant by chance. Because flowers undergoing direct cross-fertilization would likely produce more fruit, natural selection favored flowers with features that would attract pollinators.

As cross-fertilization continued, more and more flower variations likely developed, and pollinators became increasingly adapted to specific angiosperm species. Today, there are some 270,000 species of flowering plants and over 6 million species of insects. This diversity suggests that the success of angiosperms has contributed to the success of insects, and vice versa.

Questions to Consider

1. What are the potential consequences if honeybees were to go extinct?
2. How does coevolution cause two species to change over time?

a.

b.

Figure 27B Bird and bat pollinators. **a.** Hummingbird-pollinated flowers are curved back, allowing the bird to insert its beak to reach the rich supply of nectar. While doing this, the bird's forehead and other body parts touch the reproductive structures. **b.** Bat-pollinated flowers are large, sturdy flowers that can take rough treatment. Here, the head of the bat is positioned so its bristly tongue can lap up nectar. (a): Daniel Dempster Photography/Alamy Stock Photo; (b): Merlin D. Tuttle/Science Source

Check Your Progress 27.1

1. Compare the development and structure of the male gametophyte with those of the female gametophyte.
2. Describe the products of double fertilization in angiosperms.
3. Explain how a flowering plant may coevolve in response to an increase in the body size of its pollinator.

27.2 Seed Development

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the stages of eudicot development.
2. Compare the main parts of a monocot seed to those of a eudicot.

Development of the embryo within the seed is the next event in the life cycle of the angiosperm. Plant growth and development involves cell division, cell elongation, and differentiation of cells into tissues, which then develop into organs. **Development** is a programmed series of stages from a simple to a more complex form. Cellular **differentiation,** or specialization of structure and function, occurs as development proceeds.

Stages of Eudicot Development

Figure 27.7 shows the stages of development for a eudicot embryo.

Zygote and Proembryo Stages

Immediately after double fertilization, the zygote and the endosperm become visible. The zygote is small with dense cytoplasm (Fig. 27.7*a*). Next, the zygote divides repeatedly in different planes, forming several cells called a proembryo. Also present is

ISTUDY

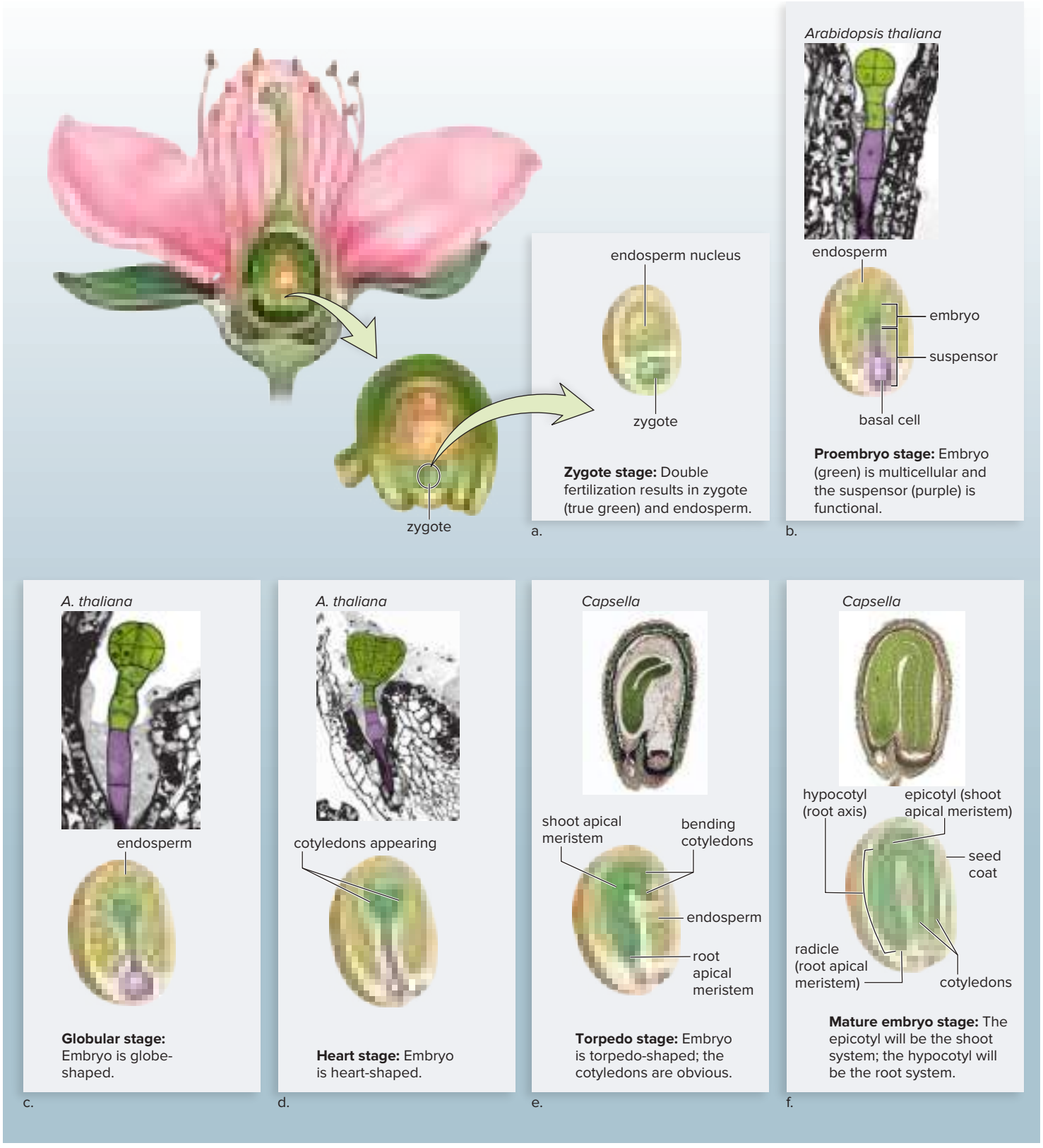


Figure 27.7 Development of a eudicot embryo. After fertilization, the eudicot embryo undergoes multiple stages of differentiation. **a.** In the zygote stage, the nourishing endosperm and zygote are visible. **b.** After multiple cell divisions, a stalklike structure called a suspensor develops in the proembryo. **c.** The axis for shoot and root development is established, and the embryo takes on a globelike appearance. **d.** The heart stage is indicative of the formation of cotyledons. **e.** In the torpedo stage, the shoot, root, and ground meristems are present and actively dividing. **f.** In the mature embryo, a rudimentary shoot and root develop with continued division from the apical meristems. The integuments of the original ovule harden, becoming the seed coat. (photos): (b–d): Dr. Chun-Ming Liu; (e): Biology Pics/Science Source; (f): Steven P. Lynch

ISTUDY

an elongated structure called a suspensor. The suspensor transfers and produces nutrients from the endosperm, which allows the embryo to grow (Fig. 27.7*b*).

Globular Stage

During the globular stage, the proembryo is largely a ball of cells. The root-shoot axis of the embryo is already established at this stage. The embryonic cells near the suspensor will go on to become a root, while those at the other end will ultimately become a shoot (Fig. 27.7*c*).

The outermost cells of the plant embryo will become the dermal tissue. These cells divide with their cell plate perpendicular to the surface; therefore, they produce a single outer layer of cells. Recall that dermal tissue protects the plant from desiccation and includes the stomata, which open and close to facilitate gas exchange and minimize water loss.

The Heart Stage and Torpedo Stage Embryos

The embryo has a heart shape when the **cotyledons,** or seed leaves, appear because of local, rapid cell division (Fig. 27.7*d*). As the embryo continues to enlarge and elongate, it takes on a torpedo shape. Now the root and shoot apical meristems are distinguishable. The shoot apical meristem is responsible for aboveground growth, while the root apical meristem is responsible for underground growth. Ground meristems give rise to the bulk of the embryonic interior, which is now present (Fig. 27.7*e*).

The Mature Embryo

In the mature embryo, the *epicotyl* is the portion between the cotyledons that contributes to shoot development. The *plumule* is found at the tip of the epicotyl and consists of the shoot tip and a pair of small leaves. The *hypocotyl* is the portion below the cotyledon(s). It contributes to stem development and terminates in the radicle or embryonic root.

The cotyledons are quite noticeable in a eudicot embryo and may fold over. Procambium is located at the core of the embryo and is destined to form the future vascular tissue responsible for water and nutrient transport.

As the embryo develops, the integuments of the ovule become the seed coat. The seed coat encloses and protects the embryo and its food supply (Fig. 27.7*f*).

Monocot Versus Eudicot Seeds

Monocots, unlike eudicots, have only one cotyledon. Another important difference between monocots and eudicots is the manner in which nutrient molecules are stored in the seed. In monocots, the cotyledon, in addition to storing certain nutrients, absorbs other nutrient molecules from the endosperm and passes them to the embryo. In eudicots, the cotyledons usually store all the nutrient molecules the embryo uses. Therefore, in Figure 27.7, you can see that the endosperm seemingly disappears. Actually, it has been taken up by the two cotyledons.

Figure 27.8 contrasts the structure of a bean seed (eudicot) and a corn kernel (monocot). The size of seeds may vary from the dust-sized seeds of orchids to the 27-kg seed of double coconuts.

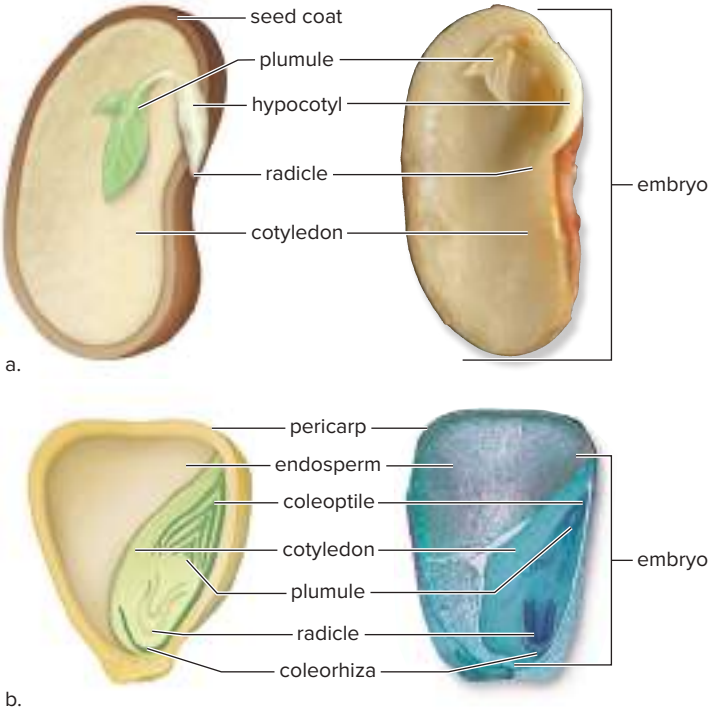


Figure 27.8 Monocot versus eudicot. **a.** In a bean seed (eudicot), the endosperm has disappeared; the bean embryo's cotyledons take over food storage functions. The hypocotyl becomes the shoot system, which will include the plumule (first leaves). The radicle becomes the root system. **b.** The corn kernel (monocot) has endosperm that is still present at maturity. The coleoptile is a protective sheath for the shoot system; the coleorhiza similarly protects the future root system. The pericarp of the fruit develops from the ovary wall. (photos): (bean): Robert and Jean Pollock/Science Source; (corn): ©Ray F. Evert/University of Wisconsin

Check Your Progress **27.2**

1. Identify the origin of each of the three parts of a seed.
2. Explain why the seed coat and the embryo are both 2n.
3. Describe the structure and function of the cotyledon.

27.3 Fruit Types and Seed Dispersal

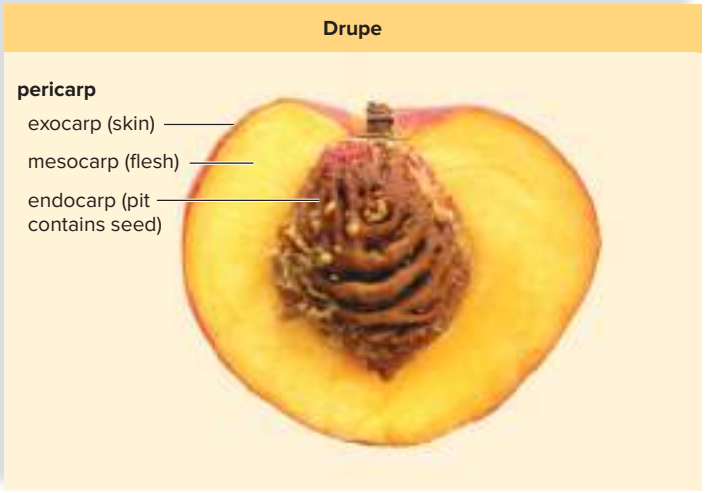
Learning Outcomes

Upon completion of this section, you should be able to

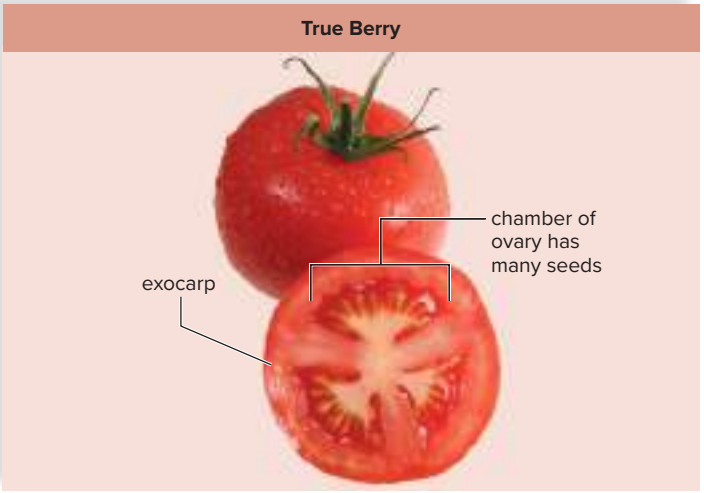
1. Identify examples of fleshy and dry fruits that are simple, compound, aggregate, or accessory fruits.
2. Explain the sequence of events during seed germination.

Most people are unfamiliar with the botanical definition of a fruit. A **fruit** in botany is a mature ovary that can also contain other flower parts, such as the receptacle. This means that pea pods, tomatoes, and what are usually called winged maple seeds are actually fruits (Fig. 27.9). Fruits protect and help disperse seeds. Some fruits are better at one function than the others. The fruit of a peach protects the seed well, but the pit may make it difficult for

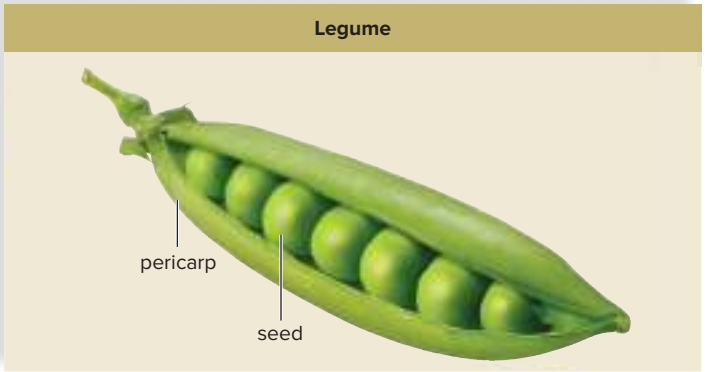
ISTUDY



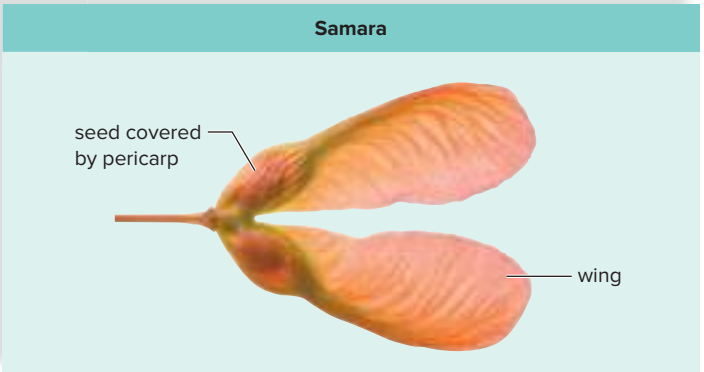
a. A drupe is a fleshy fruit with a pit containing a single seed produced from a simple ovary.



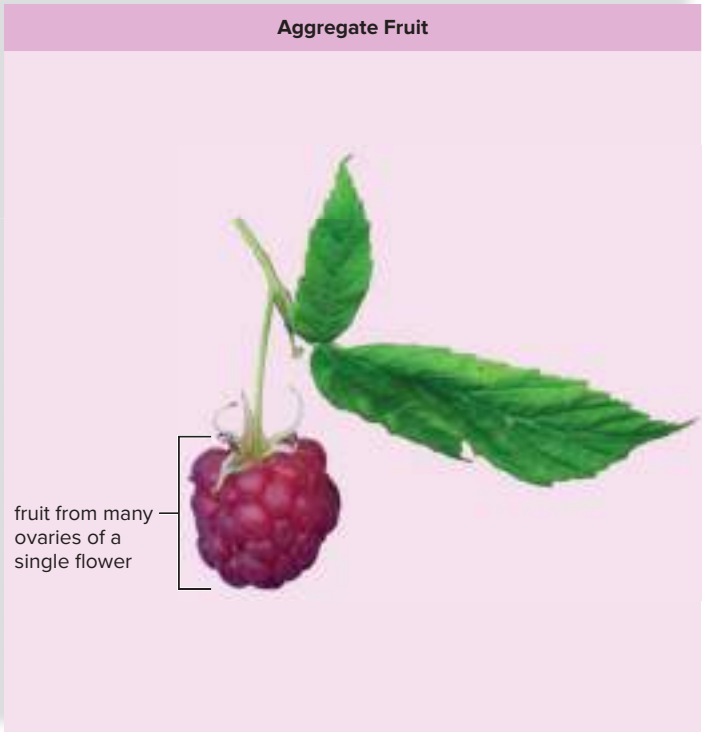
b. A berry is a fleshy fruit having seeds and pulp produced from a compound ovary.



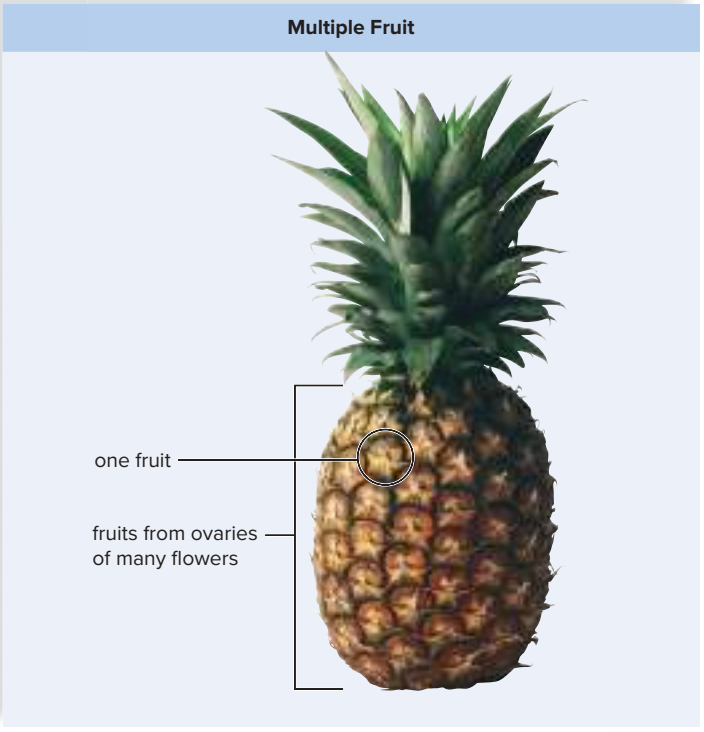
c. A legume is a dry dehiscent fruit produced from a simple ovary.



d. A samara is a dry indehiscent fruit produced from a simple ovary.



e. An aggregate fruit contains many fleshy fruits produced from simple ovaries of the same flower.



f. A multiple fruit contains many fused fruits produced from simple ovaries of individual flowers.

Figure 27.9 Fruits. **a.** A peach is a drupe. **b.** A tomato is a true berry. **c.** A pea is a legume. **d.** The fruit of a maple tree is a samara. **e.** A raspberry is an aggregate fruit. **f.** A pineapple is a multiple fruit. (a): Peter Fakler/Alamy Stock Photo; (b): D. Hurst/Alamy Stock Photo; (c): Danny Smythe/Shutterstock; (d): Robert Llewellyn/Corbis/Getty Images; (e): Ricochet Creative Productions LLC; (f): Ingram Publishing/Alamy Stock Photo

ISTUDY

Table 27.1 Fruit Classification Based on Composition and Texture

Composition (based on type and arrangement of ovaries and flowers)
Simple: develops from a simple ovary or compound ovary
Compound: develops from a group of ovaries
Aggregate: ovaries are from a single flower on one receptacle
Multiple: ovaries are from separate flowers on a common receptacle
Texture (based on mature pericarp)
Fleshy: the entire pericarp or portions of it are soft and fleshy at maturity
Dry: the pericarp is papery, leathery, or woody when the fruit is mature
Dehiscent: the fruit splits open when ripe
Indehiscent: the fruit does not split open when ripe

germination to occur. Peas easily escape from pea pods, but once they are free, they are protected only by the seed coat.

Kinds of Fruits

Fruits can be simple or compound (Table 27.1). A *simple fruit* is derived from a single ovary that can have one or several chambers (Fig. 27.9*a–d*). A *compound fruit* is derived from several groups of ovaries (Fig. 27.9*e–f*). If a single flower has multiple ovaries, as in a blackberry, then it produces an *aggregate fruit* (Fig. 27.9*e*). In contrast, a pineapple comes from many individual ovaries. Because the flowers have only one receptacle, the ovaries fuse to form a large, *multiple fruit* (Fig. 27.9*f*).

As a fruit develops, the ovary wall thickens to become the *pericarp,* which can have as many as three layers: exocarp, mesocarp, and endocarp.

- The *exocarp* forms the outermost skin of a fruit.
- The *mesocarp* is often the fleshy tissue between the exocarp and endocarp of the fruit.
- The *endocarp* serves as the boundary around the seed(s). The endocarp may be hard, as in peach pits, or papery, as in apples.

Some fruits, such as legumes and cereal grains of wheat, rice, and corn, are *dry fruits.* The fruits of grains can be mistaken for seeds, because a dry pericarp adheres to the seed within. Legume fruits such as the pea pod (Fig. 27.9*c*) are dehiscent, which means they split open when ripe. Grains are indehiscent—they do not split open. Humans gather grains before they are released from the plant and then process them to acquire their nutrients.

You are probably more familiar with fleshy fruits, such as the peach and tomato. In these fruits, the mesocarp is well-developed.

Dispersal of Fruits

Generally, it is beneficial for plants to disperse their fruits away from the parent plant, so seedlings do not have to compete with the parent for nutrients. Fruits may drift or be blown to new locations by wind or forced ejection, or may be carried away by animals.

Dispersal by Air

Many dry fruits are dispersed by wind. Woolly hairs, plumes, and wings are all adaptations for this type of dispersal. The somewhat heavier dandelion fruit uses a tiny "parachute" for dispersal (Fig. 27.10*a*). Milkweed pods split open to release seeds that float away on puffy, white threads. The winged fruit of a maple tree has been known to travel up to 10 km from its parent.

Dispersal by Animals

Ripe, fleshy, colorful fruits, such as peaches and cherries, often attract animals and provide them with food (Fig. 27.10*b*). Their hard endocarp protects the seed, so it can pass through the digestive system of an animal and remain unharmed. As the flesh of a tomato is eaten, the small size of the seeds and the slippery seed coat mean that tomato seeds rarely get crushed by the teeth of animals. The seeds swallowed by birds and mammals are defecated (passed out of the digestive tract with the feces) some distance from the parent plant. Squirrels and other animals that gather seeds and fruits bury them some distance away and may even forget where they have been stored. The hooks and spines of clover, bur, and cocklebur attach a dry fruit to the fur of animals and the clothing of humans (Fig. 27.10*c*).

Dispersal by Ejection

Plants that have dehiscent fruit—that is, fruit with slits or openings—tend to disperse their seeds by forced ejection. If you walk by a trumpet vine on a hot, sunny day in late spring, you can hear the explosions of the seedpods as they burst open and send their seeds flying. The side of the seedpod facing the sun dries faster than the shaded side, causing the partially dry pod to buckle and pop open (Fig. 27.10*d*).

Seed Germination

Following dispersal, if conditions are right, seeds may **germinate** to form a seedling. Germination doesn't usually take place until there is sufficient water, warmth, and oxygen to sustain growth. These requirements help ensure that seeds do not germinate until the most favorable growing season has arrived.

Some seeds do not germinate until they have been dormant for a period of time. For seeds, *dormancy* is the time during which no growth occurs, even though conditions may be favorable for growth. In the temperate zone, seeds often have to be exposed to a period of cold weather before dormancy is broken. Fleshy fruits (e.g., apples, pears, oranges, and tomatoes) contain inhibitors, so that germination does not occur while the fruit is still on the plant. For seeds to take up water, bacterial action and even fire may be needed. Once water enters, the seed coat bursts and the seed germinates.

If the two cotyledons of a bean seed are parted, the rudimentary plant with immature leaves is exposed (Fig. 27.11*a*). As the eudicot seedling starts to form, the root emerges first. The shoot is hook-shaped to protect the immature leaves as they emerge from the soil. The cotyledons provide the new seedlings with enough energy for the stem to straighten and the leaves to grow. As the mature leaves of the plant begin photosynthesizing, the cotyledons shrivel up.

ISTUDY

a.

b.

c.

d.

Figure 27.10 Methods of fruit and seed dispersal. **a.** Dandelion fruits float away on air currents. **b.** When birds eat fleshy fruits, seeds pass through their digestive system. **c.** Burdock, a dry fruit, clings to the fur of animals. **d.** Dry pods split open, forcibly ejecting their seeds. (a): Dimitri Vervitsiotis/Digital Vision/Getty Images; (b): Stockbyte/Getty Images; (c): Scott Camazine/Science Source; (d): M. Brodie/Alamy Stock Photo

A kernel of corn is actually a fruit, and therefore its outer covering is the pericarp and seed coat combined (Fig. 27.11*b*). Inside is the single cotyledon. Also, the immature leaves and the radicle are covered, respectively, by a coleoptile and a coleorhiza. These sheaths are discarded as the root grows directly downward into the soil and the shoot of the seedling begins to grow directly upward.

Check Your Progress 27.3

1. Compare the structure and dispersal methods of dry and fleshy fruits.
2. Compare the protective methods of monocot and eudicot seedlings used to protect their first true leaves.

ISTUDY

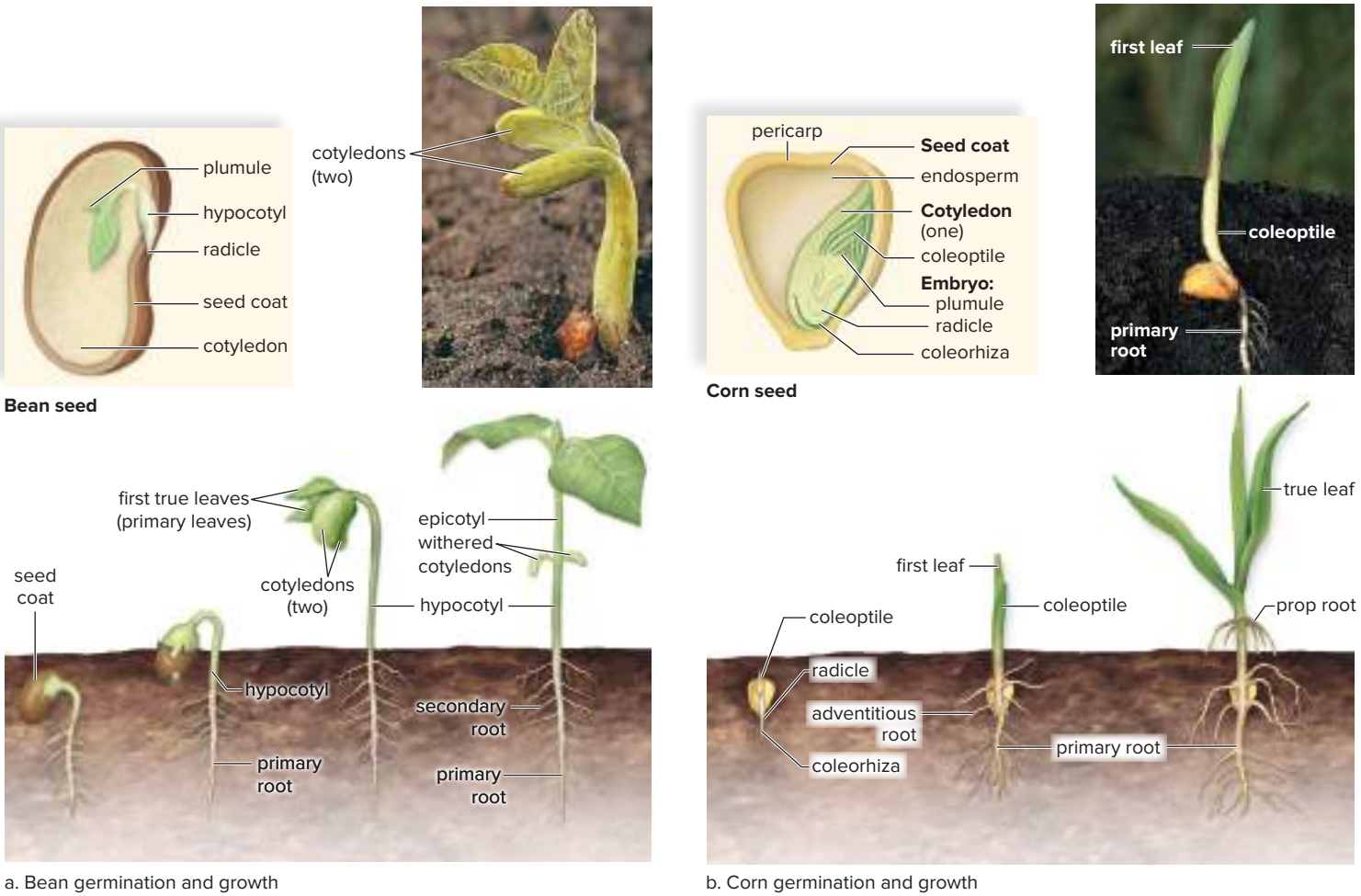


Figure 27.11 Eudicot and monocot seed germination. **a.** Bean (eudicot) seed germinates, revealing two cotyledons. **b.** Corn (monocot) seed germinates, with an emerging coleoptile. (photos): (bean): Ed Reschke; (corn): Ed Reschke/Stone/Getty Images

27.4 Asexual Reproductive Strategies

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the asexual methods of reproduction in plants.
2. Describe how tissue culture can be used to clone plants with desirable traits.

Asexual reproduction is the production of an offspring from a single parent. Because the offspring is genetically identical to the parent, it is called a **clone.** Asexual reproduction is less complicated in plants, because pollination and seed production are not required. Therefore, it can be advantageous when the parent is already well-adapted to a particular environment and the production of genetic variations is not an apparent necessity. In addition, clones are desirable for plant sellers, because they know exactly how the plant will grow and look.

Asexual Reproduction from Stems, Roots, and Cuttings

Figure 27.12 features a strawberry plant that has produced a stolon. *Stolons* are horizontal stems that can be seen because they run aboveground. As you know, the nodes of stems are regions where new growth can occur. In the case of stolons, a new shoot system appears above the node, and a new root system appears below the node. The larger plant on the *left* is the parent plant, and the smaller plant on the *right* is the asexual clone that has arisen at a node. The characteristics of new offspring produced by stolons are identical to those of the parent plant.

Rhizomes are underground stems that produce new plants asexually. Irises are examples of plants that have no aboveground stem because their main stem is a rhizome that grows horizontally underground. As with stolons, new plants arise at the nodes of a rhizome. White potatoes are expanded portions of a rhizome branch called tubers, and each "eye" is a bud that can produce a new potato plant if it is planted with a portion of the swollen tuber. Sweet potatoes, by contrast, are modified roots; they can be propagated by planting sections of the root, which are called slips.

Tissue Culture of Plants

One of the major disadvantages of most asexual propagation techniques is that they also propagate pathogens. Plant pathogens can be viruses, bacteria, or fungi; clones created from an infected

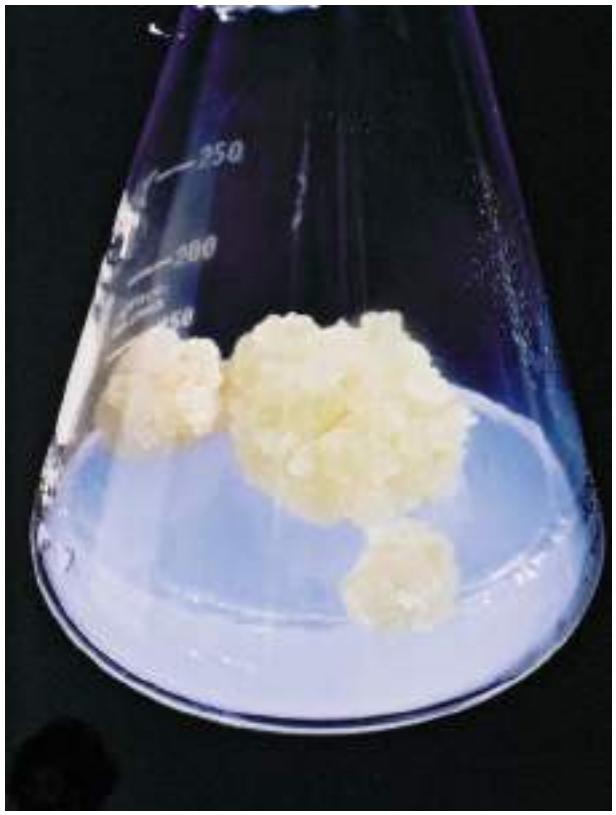
ISTUDY

Figure 27.12 Asexual reproduction in plants. Meristem tissues at nodes can generate new plants, as when the stolons of strawberry plants, *Fragaria,* give rise to new plants. NHPA/SuperStock

parent will also be infected. However, it is possible to maintain plants in a disease-free status if clones from an uninfected parent are made in sterile conditions through tissue culture. Hence, **tissue culture** is simply plant propagation done in a laboratory under sterile conditions.

The key to plant tissue culture, as opposed to animal cell culture, is the totipotency of plant cells. **Totipotency** is the ability of individual plant cells to develop into an entire plant.

Techniques for tissue culture vary, but most begin with cells from the *meristem* of the parent plant. Meristematic cells, which are already actively dividing, are grown on an agar medium in flasks, tubes, or petri dishes. The medium contains auxin, cytokinins, various nutrients, and water. Initially, a mass of undifferentiated (unspecialized) cells, called a callus, forms (Fig. 27.13*a*). As described in Figure 26.5, the addition of different combinations of auxin and cytokinins will signal *organogenesis* (organ formation) and initiate the formation of roots and shoots, until a fully developed plantlet is formed (Fig. 27.13*b, c*). Plantlets can be shipped in their sterile containers to growers for transplantation into a greenhouse or field.



a.

b.

c.

d.

Figure 27.13 Asexual reproduction through tissue culture. **a.** Meristem tissue is placed on sterile media and an undifferentiated mass, called a callus, develops. **b.** From the callus, organogenesis takes place and leaves or roots develop. **c.** The callus develops into a plantlet. **d.** Plantlets can be stored, then shipped in sterile containers and transferred to soil for growth into adult plants. (a): Biophoto Associates/Science Source; (b, c): Courtesy Prof. Dr. Hans-Ulrich Koop from Plant Cell Reports, 17:601–604; (d): Edwin Remsberg/Alamy Stock Photo

ISTUDY

Tissue culture is an important technique for propagating many fruits and vegetables. A notable example is the banana, which is a sterile hybrid that cannot produce seeds. The only way to produce this commercially important fruit is through tissue culture. As another example, asparagus is a dioecious plant, and all commercial stalks are male. The female stalks favor the production of flowers and are undesirable for eating. Tissue culture is a more efficient means of producing disease-free male asparagus for growers.

Many botanical gardens and universities use tissue culture for plant conservation. The Atlanta Botanical Garden has a tissue culture lab that propagates native species of orchids and lilies. The propagated plants are shared with local nurseries, resulting in fewer plants being removed from the wild by collectors. Tissue culture of rare species is also used to help increase the population in the wild, as they are replanted in native habitats (Fig. 27.14).

Cell Suspension Culture

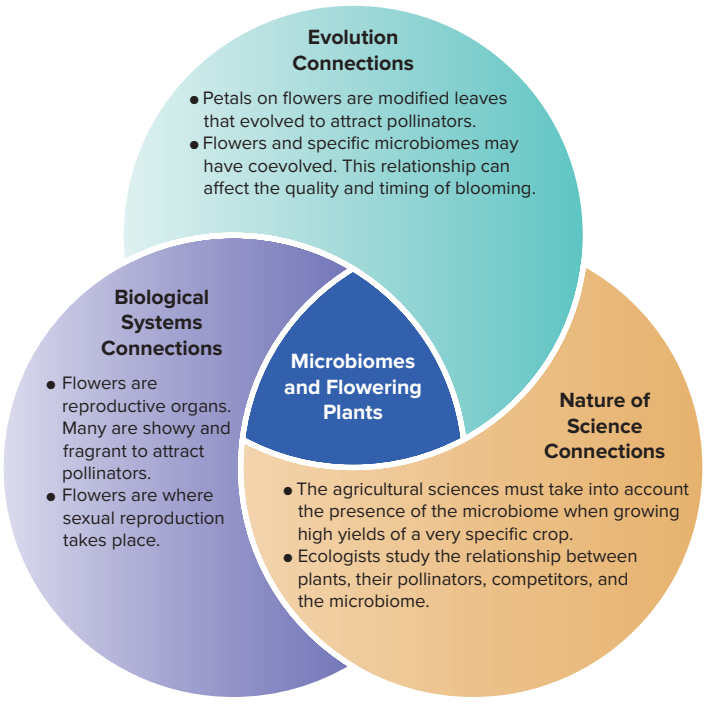
A technique called **cell suspension culture** allows scientists to extract chemicals (secondary metabolites) from plant cells, which may have been genetically modified. This technique allows scientists to avoid overcollection of wild plants from their natural environments. These cells produce the same chemicals that the plant produces. For example, cell suspension cultures of *Cinchona ledgeriana* produce quinine, which is used to treat leg cramping, a major symptom of malaria. Several *Digitalis* species produce digitalis, digitoxin, and digoxin, which are useful in the treatment of heart disease.

Figure 27.14 Tissue culture for plant conservation. This rare Kentucky ladyslipper orchid is grown in a tissue culture lab and will be replanted into native habitats. ©Hillside Nursery, Shelburne Falls, MA

Check Your Progress 27.4

1. Identify the possible benefits of asexual reproduction.
2. Describe methods of asexual reproduction in wild plants.
3. Explain the benefits of plant tissue culture.

Connecting *the* Concepts

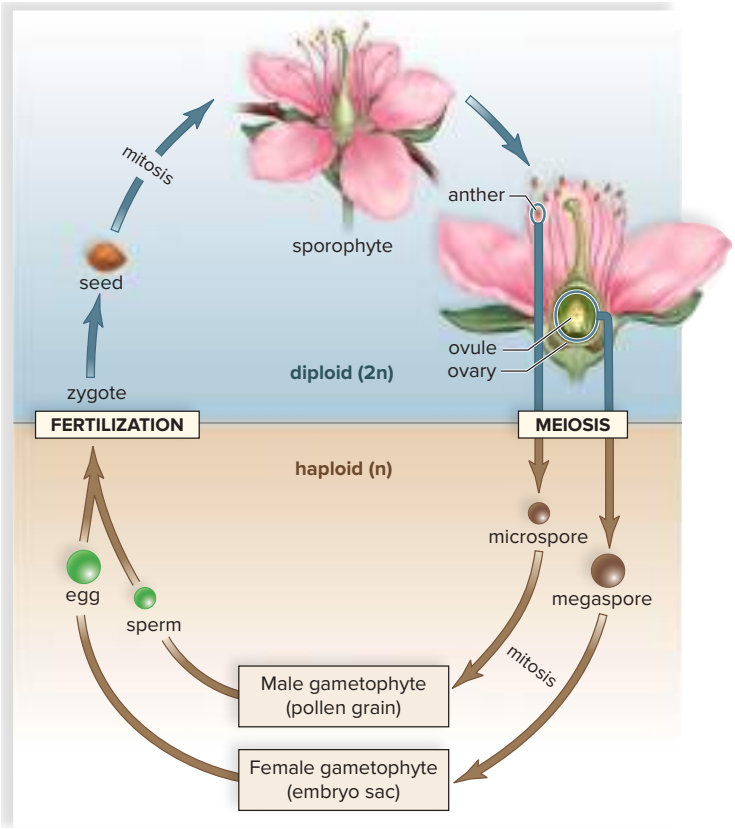


What Did You Learn?

27.1 Sexual Reproductive Strategies

Flowering plants exhibit an alternation-of-generations life cycle. **Flowers** borne by the sporophyte produce **microspores** and **megaspores** by meiosis. Microspores develop into a male **gametophyte,** and megaspores develop into a female gametophyte. Following fertilization, the sporophyte is enclosed within a **seed** covered by the fruit.

- The flowering plant life cycle is adapted to a land existence. The microscopic gametophytes are protected from desiccation by the sporophyte; the pollen grain has a protective wall; and fertilization does not require external water. The seed has a protective seed



coat, and seed germination does not occur until conditions are favorable.

- A typical flower has several parts: **Sepals,** which are usually green, form an outer whorl called a **calyx; petals,** often colored, are the next whorl and are called a **corolla;** and **stamens,** each having a filament and an anther, form a whorl around the base of at least one **carpel.** The carpel, in the center of a flower, consists of a **stigma,** a **style,** and an **ovary.** The ovary contains **ovules.**
- The anthers contain microspore mother cells, which divide meiotically to produce four haploid microspores. These divide into a two-celled **pollen grain,** which is the male gametophyte. One of these cells will divide later to become two sperm cells. After pollination, the pollen grain germinates, and as the pollen tube grows, the sperm cells travel to the embryo sac. **Pollination** is simply the transfer of pollen from anther to stigma.
- Each ovule contains a megaspore mother cell, which divides meiotically to produce an egg cell. The egg cell is in the **embryo sac,** which is the female gametophyte.

Flowering plants undergo **double fertilization.** One sperm nucleus unites with the egg nucleus, forming a 2n zygote, and the other unites with the polar nuclei of the central cell, forming a 3n endosperm cell.

- After fertilization, the **endosperm** cell divides to form multicellular endosperm. The zygote becomes the sporophyte embryo. The ovule matures into the seed (its integuments become the seed coat). The ovary becomes the fruit.

Coevolution occurs when specific traits or behaviors evolve in one species in response to changes that have occurred in another species. The coevolution of flowering plants with animal pollinators may help explain the success of flowering plants on land. Flowering plants would have gained an increased chance of seed production. The pollinators would have gained more feeding opportunities. In the end, both species became dependent on each other for success.

27.2 Seed Development

After fertilization, the cells in the ovule undergo **differentiation** and **development** into a seed.

- As the ovule is becoming a seed, the zygote is becoming an embryo. After the first several divisions, it is possible to discern the embryo and the suspensor. The suspensor attaches the embryo to the ovule and supplies it with nutrients.
- The eudicot embryo becomes first heart-shaped and then torpedo-shaped. Once you can see the two **cotyledons,** it is possible to distinguish the shoot tip and the root tip, which contain the apical meristems. In eudicot seeds, the cotyledons frequently absorb the endosperm.

27.3 Fruit Types and Seed Dispersal

The seeds of flowering plants are enclosed by **fruits.**

- Simple fruits are derived from a single ovary (which can be simple or compound). Some simple fruits are fleshy, such as peaches or apples. Others are dry, such as peas, nuts, and grains.
- Compound fruits consist of aggregate fruits, which develop from a number of ovaries of a single flower, and multiple fruits, which develop from a number of ovaries of separate flowers.

Flowering plants have several ways to disperse seeds. Seeds may be blown by the wind, be attached to animals that carry them away, or be eaten by animals that defecate them some distance away.

Prior to **germination,** you can distinguish the two cotyledons and plumule of a bean seed (eudicot). The plumule is the shoot that bears leaves. Also present are the epicotyl, the hypocotyl, and the radicle. In a corn kernel (monocot), the endosperm, the cotyledon, the plumule, and the radicle are visible.

27.4 Asexual Reproductive Strategies

Many flowering plants undergo **asexual reproduction,** producing **clones,** as when the nodes of stems (either aboveground or underground) give rise to entire plants or when roots produce new shoots.

- The production of clonal plants utilizing **tissue culture** for many common fruits and vegetables is common. Plant cells are **totipotent,** so adult plants can be made from a few parent cells. Leaf, stem, and root culture can result in **cell suspensions** that allow plant chemicals to be produced in large tanks.

Thinking Critically About Biology

1. How might a commercial plant benefit from having just one type of pollinator?
2. The pollinator for a very rare plant has become extinct. What laboratory technique would you use to prevent the plant from also becoming extinct? How might you improve the hardiness of the plant?
3. In most parts of the world, commercial potato crops are produced asexually by planting tubers. However, in some regions, such as Southeast Asia and the Andes, some potatoes are grown from true seeds. Discuss the advantages and disadvantages of growing potatoes from true seeds.

Making It Relevant to You

1. What is the function of a flower?
2. What type of non-plant biodiversity have you observed in a flowering garden?
3. Why do you think angiosperms have different microbiomes on reproductive versus vegetative parts of the plant?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 6

Animal Evolution and Diversity

In his book *On the Origin of Species,* Charles Darwin notes that while the planet Earth has cycled year after year around the sun, "endless forms most beautiful and most wonderful" have appeared and will keep on appearing. This statement can certainly be applied to the evolution of animals, whose variety seems without bounds. The fossil record even reveals myriad animals that are extinct today, of which the dinosaurs are only the most well-known. The search for food, shelter, and mates under a variety of environmental conditions explains why the diversity of animals is so great.

Despite their diversity, animals have an unbroken thread of unity due to their evolution from a common ancestor. At the biosphere level, animals are heterotrophic consumers that require a constant supply of food, ultimately supplied by the autotrophs at the base of the food chain. At the organismal level, their eukaryotic cells usually form tissues and organs with specialized functions. At the molecular level, animals share a common chemistry, including a genetic code that we now know reveals how the many groups of animals are related. In this unit of the text, we concentrate on the characteristics of animals, their origin, and their evolution as revealed by molecular genetics.

Unit Outline

Chapter 28 Invertebrate Evolution

Chapter 29 Vertebrate Evolution

Chapter 30 Human Evolution

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Describe the position of humans in relation to the other animals in the tree of life.
Nature of Science	Explain why, from an evolutionary perspective, animal model organisms can be used in human biomedical research.
Biological Systems	Identify how the environment has shaped the evolution of the animal body plan.

ISTUDY

(jellyfish): PhotoTalk/E+/Getty Images; (bottle): Keith Homan/Shutterstock

28

Invertebrate Evolution

Chapter Outline

28.1 Evolution of Animals
28.2 The Simplest Invertebrates
28.3 Diversity of the Lophotrochozoans
28.4 Diversity of the Ecdysozoans
28.5 Invertebrate Deuterostomes

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 18.9 When did the lineage of animals first appear in the tree of life?

Figure 21.1 Where do animals fit into the evolutionary history of eukaryote supergroups?

Section 21.1 Which single-celled eukaryotes are thought to be the common ancestor of all animals?

Jellyfish—A Cure for Memory Loss?

Jellyfish are a group of invertebrate animals that can trace their evolutionary history back over 500 million years. Their simplistic body plan has enabled them to survive multiple mass extinctions, changes in Earth's oceans, and a host of different predators. Molecular analysis suggests that changes in the developmental genes of their nervous system enabled them to evolve from a stationary lifestyle to an active one. Some scientists feel that they may represent one of the earliest carnivores in the oceans. Many of these same genes are found in animals such as fruit flies and humans, offering evidence of relatedness and common ancestry.

Studying the genome of jellyfish offers insight into the genetics of their nervous system, along with regeneration and the healing of wounds. Recently, a calcium-binding protein was synthesized from the jellyfish *Aequorea victoria* and used to develop a memory-boosting supplement called Prevagen. Aging decreases the number of calcium-binding proteins in our brain, leading to damage to the nerve cells, which result in impaired thought and memory. The company that produces Prevagen has suggested it will help decrease calcium levels in the brain, thus preventing memory loss.

Unfortunately, the scientific evidence that this actually occurs has not supported these claims, thus consumers should be skeptical. However, despite the questionable claims surrounding Prevagen, molecular data support a degree of relatedness between humans and jellyfish. As we learn more about the evolutionary history of jellyfish, we will undoubtedly gain more insight into our own.

As you read through this chapter, think about the following questions:

1. Where do invertebrates fit into the evolutionary history of eukaryotes? Of animals?
2. What evolutionary advantage might there be to invertebrates that have multiple developmental stages, each with a different body form, habitat, and lifestyle?

Following *the* Themes

CHAPTER 28 – INVERTEBRATE EVOLUTION

Evolution	According to the colonial flagellate hypothesis, animals evolved from an invertebrate, single-celled, colonial, choanoflagellate-like protist.
Nature of Science	The phylogeny of invertebrates is a hypothesis of evolutionary history, which like all scientific hypotheses is subject to revision with the addition of new data.
Biological Systems	Invertebrates are widely successful animals that have adapted to an extensive range of habitats and lifestyles, including parasitism, which in some cases has a direct impact on human health.

ISTUDY

28.1 Evolution of Animals

Learning Outcomes

Upon completion of this section, you should be able to

1. List three common characteristics of animals that are not found in other multicellular eukaryotes.
2. Summarize the colonial flagellate hypothesis as it relates to the origin of animals.
3. Distinguish among the different body plans of animals.
4. Compare protostome and deuterostome development.

The traditional five-kingdom classification placed animals in the kingdom Animalia. The modern three-domain system places animals in the domain Eukarya. Within the Eukarya, they are placed in the supergroup Opisthokonta (see Section 21.1), along with fungi and certain protozoans (notably, the choanoflagellates). In this section, we consider what characteristics distinguish animals from other eukaryotes and look at how these traits evolved.

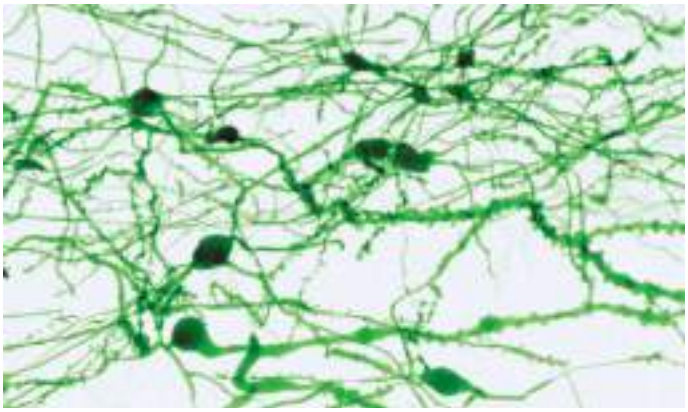
Characteristics of Animals

Animals, fungi, and plants are all multicellular eukaryotes, but unlike plants, which make their food through photosynthesis, fungi and animals are heterotrophs and must acquire nutrients from an external source. Animals differ from fungi, which are saprotrophs, because animals ingest (eat) food and digest it internally, while fungi dissolve food externally and absorb nutrients. Animal cells also lack a cell wall, which is common to both the plants and fungi, although the cell walls in fungi and plants are made of different organic molecules. In general, animals are mobile, have nerves and muscles, and reproduce sexually. However, the tremendous diversity of animals makes assigning specific characteristics to all animals difficult.

Animals have a variety of life cycles. Many animals reproduce sexually, while others reproduce asexually, and some combine both reproductive cycles. Many have a diploid life cycle, while others are haploid or haplodiploid. Animals undergo a series of developmental stages to produce an organism that has specialized tissues that carry on specific functions.

Muscle and nerve tissues, both characteristic of animals, allow motility and a variety of flexible movements. In turn, motility enables animals to search actively for food and to prey on other organisms. Coordinated movements also allow animals to seek mates, shelter, and a suitable climate—behaviors that have allowed animals to live in all habitats and to become vastly diverse.

Animals are descended from a single common ancestor, thus forming a single lineage on the tree of life. Within the animal lineage are two main branches, the vertebrates and the invertebrates. **Vertebrates** are animals that at some stage of their lives have a spinal cord (backbone) running down the center of the back, whereas **invertebrates,** the topic of this chapter, do not have a backbone. Figure 28.1 illustrates some common features of animals, including invertebrates.



a.

b.

c.

d.

Figure 28.1 Common features of animals. **a.** Animals are multicellular, and all animal cells lack a cell wall. **b.** Animals are heterotrophs, meaning they obtain nutrition from external sources. **c.** Animals are typically motile, due to their well-developed nervous and muscular systems. **d.** Most animals reproduce sexually, beginning life as a 2n zygote, which undergoes development to produce a multicellular organism that has specialized tissues.
(a): Salvanegra/iStock/Getty Images; (b): Mark Dierker/McGraw Hill; (c): JonathanC Photography/Shutterstock; (d): ugurhan/E+/Getty Images

ISTUDY

Ancestry of Animals

In Section 23.1, we discussed evidence that plants most likely share a protist (green algae) ancestor with the charophytes. What about animals? The available evidence suggests they most likely evolved from a motile protozoan ancestor that formed early colonies. As we will see, there are several hypotheses that attempt to explain how animals are descended from an ancestor that resembled a hollow, spherical colony of flagellated cells.

Among the protists, the **choanoflagellates** most likely resemble the last single-celled ancestor of animals, and molecular data tell us that they are the closest living relatives of animals. A choanoflagellate is a single cell, 3–10 μm in diameter, with a flagellum surrounded by a collar of 30–40 microvilli. Movement of the flagellum creates water currents, which pull the protist along. As water moves through the microvilli, they engulf bacteria and debris from the water. Choanoflagellates can also exist as a colony of cells (see Fig. 21.25). Several can be found together at the end of a stalk or simply clumped together, like a bunch of grapes.

Figure 28.2 compares two hypotheses on the process of transition from colonial flagellates to multicellular animals. Both begin with an aggregate of a few flagellated cells. In the colonial flagellate hypothesis (Fig. 28.2*a*), this aggregate of cells formed a hollow sphere. Individual cells within the colony would have become specialized for particular functions, such as reproduction. Two tissue layers could have arisen by an infolding of certain cells into a hollow sphere. Tissue layers arise in this manner during the development of animals today. The colonial flagellate hypothesis is also attractive because of its implications regarding animal symmetry, which is discussed shortly. An alternative to the colonial flagellate hypothesis suggests that the first animals evolved ancestral cells called archaeocytes, which were not flagellated (see Fig. 28.2*b*). This hypothesis also presents the idea that cells underwent specialization during the course of their life cycle before multicellularity evolved.

Evolution of Body Plans

All of the various animal body plans we see today were present by the Cambrian period, roughly 500 million years ago (see Table 18.1). How could such diversity have arisen within a relatively short period of geologic time? As an animal develops, there are many possible outcomes regarding the number, position, size, and patterns of its body parts. Different combinations could have led to the great variety of animal forms in the past and present. Scientists now think that slight shifts in the DNA code and the expression of genes called *Hox* (homeotic) genes are responsible for the major differences between animals that arise during development, as described in the Evolution feature, "Evolution of the Animal Body Plan."

The Phylogenetic Tree of Animals

The fossil record of the early evolution of animals is very sparse. Therefore, the best hypothesis of the evolution of animals,

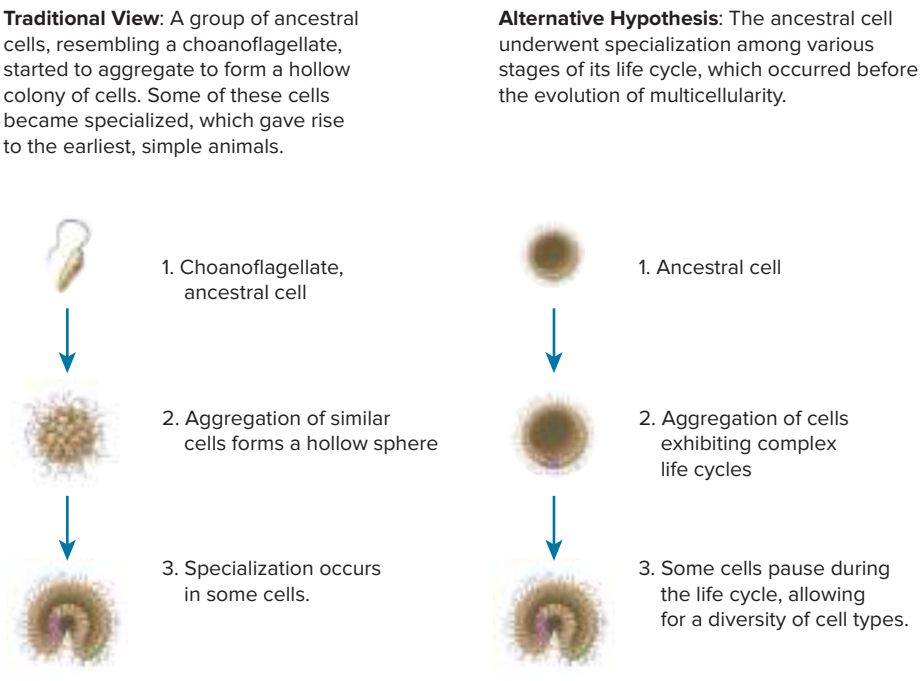


Figure 28.2 Hypotheses on the evolution of early animals. Both models present hypotheses on how the first animals evolved from protists.

ISTUDY

THEME Evolution

Evolution of the Animal Body Plan

The animal body plan can be divided into three categories based on symmetry (see Figs. 28.4 and 28.6). The general trend seems to be for body plans to become increasingly complex, from a lack of symmetry in the sponges, to radial symmetry in the cnidarians, to bilateral symmetry in more recently evolved groups, such as the arthropods and chordates that have multiple tissue types and organ systems.

The body plan of an animal is the result of a carefully orchestrated pattern of genes being expressed (or not expressed) at the right time and in the correct region of the developing embryo. In the first stage of development, the anterior (front) and posterior (rear) ends of the embryo are determined (Fig. 28A). In bilaterally symmetrical animals with segments, such as insects and chordates, the next step in development is to divide the embryo into segments, each of which will become a different part of the body. In fruit flies, genes such as the *gap* and the *pair-rule* genes determine the number of segments. In vertebrates, *FGF8* is one of many genes that determine segmentation pattern.

Once the segmentation pattern is established, *homeotic,* or *HOX,* genes determine the ultimate developmental fate of

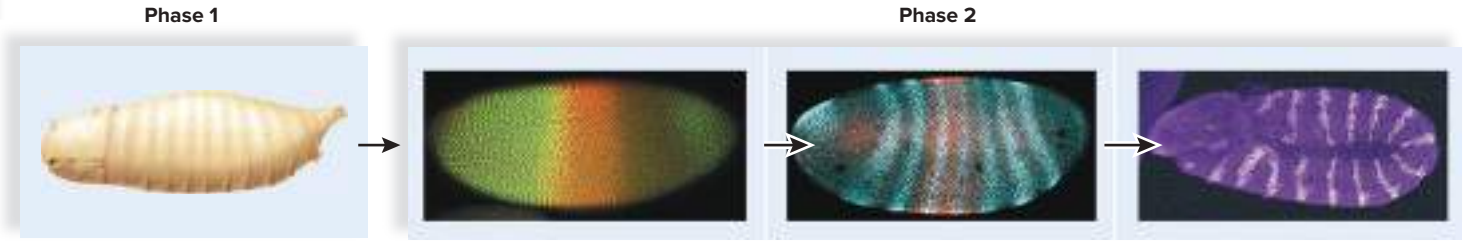


Figure 28A Development stages in the fruit fly embryo. Fruit fly embryos are sectioned and stained during different stages of development. The anterior and posterior regions of the embryo are determined in Phase 1 by genes such as *bicoid.* Genes such as *gap* and *pair-rule* determine the segmentation pattern in Phase 2. (Phase I): Lachina Publishing Services/McGraw Hill; (Phase 2, all): ©Steve Paddock, Howard Hughes Medical Research Institute

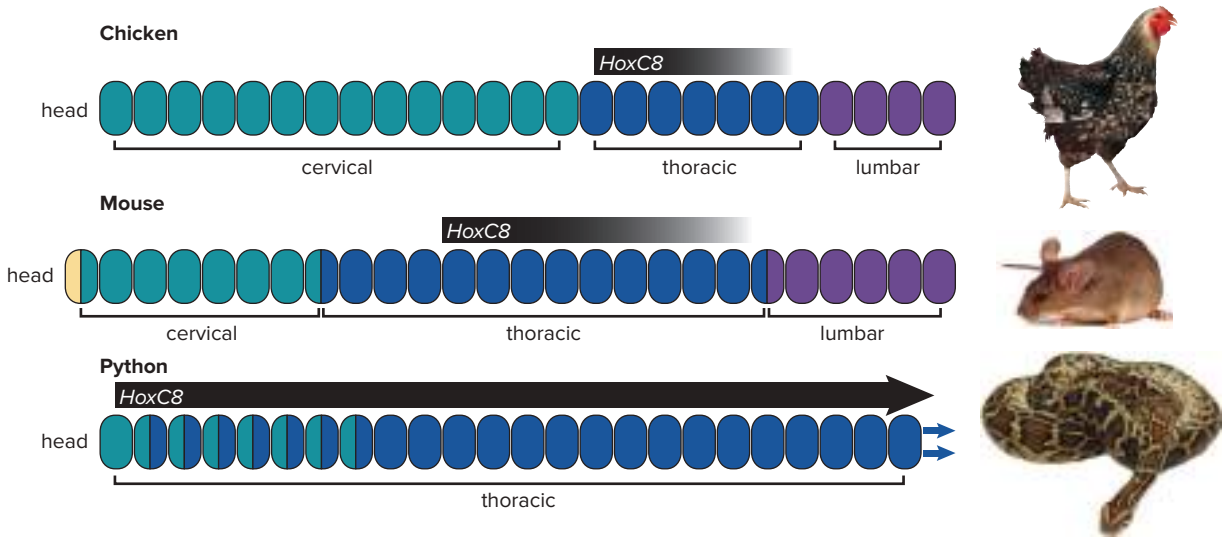


Figure 28B *HOX genes* orchestrate the development of the body plan of animals. In animal development, the number of segments, or vertebrae in vertebrates, is controlled by *HOX* genes. One *HOX* gene, *HOXC8,* determines the number of thoracic vertebrae. Variation in the number of thoracic vertebrae in the spine of a chicken, mouse, and snake is caused by a shift in when, and for how long, *HOXC8* is active during embryonic development. In the chicken, 7 thoracic vertebrae are formed; in the mouse, 12. A large increase in the number of thoracic vertebrae results in the long, thin body plan of a snake. (chicken): GK Hart/Vikki Hart/Photodisc/Getty Images; (mouse): Redmond Durrell/Alamy Stock Photo; (snake): Yuri Arcurs/Design Pics

ISTUDY

each segment. *HOX* genes encode homeotic proteins that bind to the regulatory region of genes that determine the body plan during development. Homeotic proteins act as "switches" that control when, where, and for how long a particular developmental gene is active.

Each *HOX* gene orchestrates the developmental fate of a particular region of the body. In mice, *HOXC8* sets the fate of 12 segments to become thoracic vertebrae, while in snakes, *HOXC8* orchestrates the development of hundreds of thoracic vertebrae (Fig. 28B).

HOX genes have played a role in the development of the body plan since the early stages of animal evolution. We know this because *HOX* genes are found in all animals with multiple types of tissue, and there is a shared similarity of *HOX* genes across animal groups (Fig. 28C). For example, the *HOX* genes that determine the fate of the head region have the same evolutionary origin in flies, worms, and mice. Even cnidarians, with a simple body plan, have some *HOX* genes in common with more complex animals. This implies that all *HOX* genes evolved in animals from a common *HOX* gene ancestor. *HOX* genes are found in linear clusters on the same chromosome. However, not all *HOX* genes are found in all animals. Some groups of animals have more than one cluster of genes, with duplicate copies of genes within each cluster.

Questions to Consider

1. Why are *HOX* genes evidence for a common ancestor of all life?
2. What would you expect to happen if a particular *HOX* gene, such as *HOXC8,* were not functioning properly?
3. What experiments could you plan to test what a particular *HOX* gene controlled at a particular time of development?

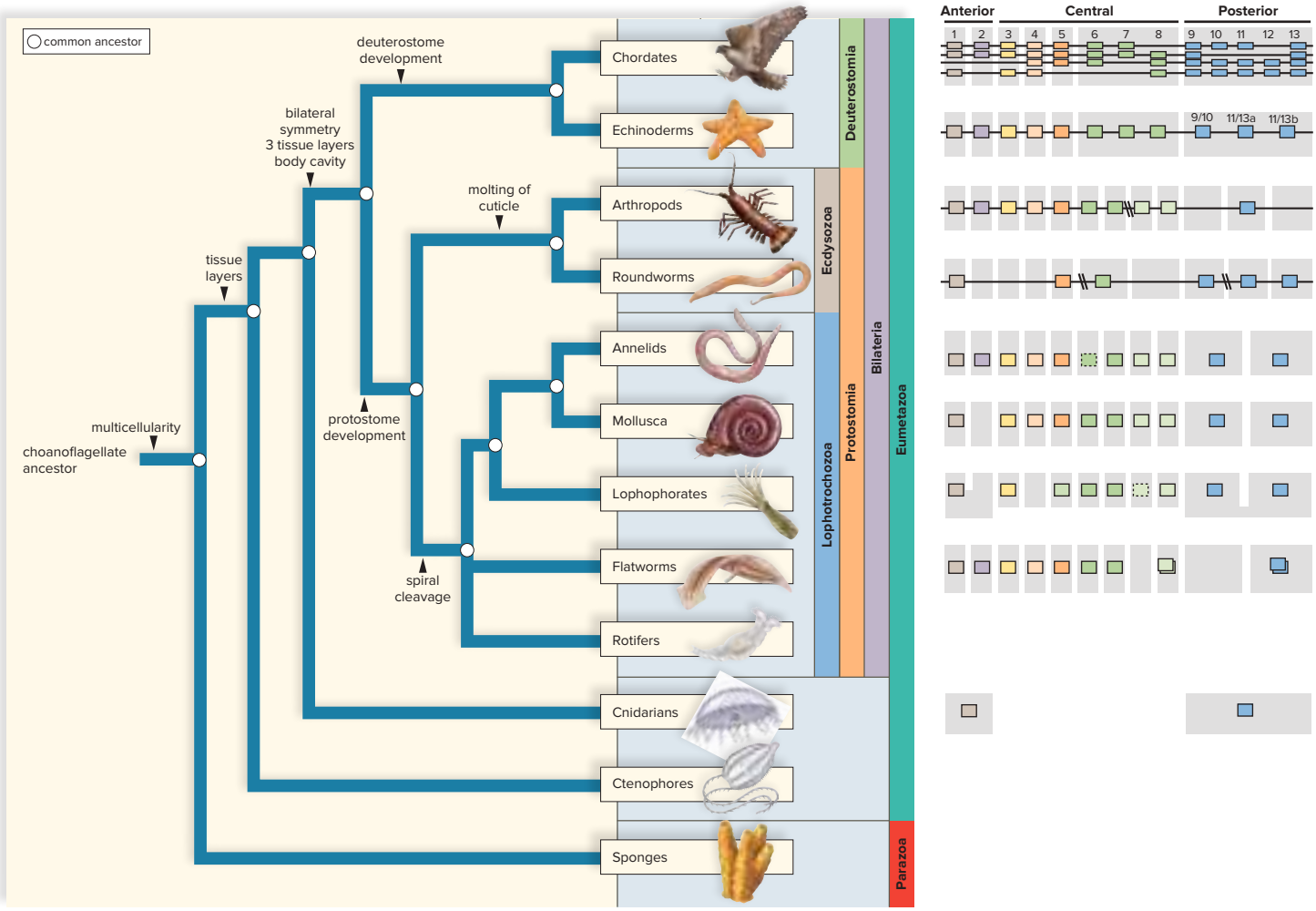


Figure 28C Shared evolutionary history of *HOX* genes in animals. *HOX* genes are found in all groups of animals with multiple tissue layers. A general trend is observed for an increase in the number of *HOX* genes and *HOX* gene clusters and an increase in body plan complexity. Although the number of *HOX* genes varies among animals, there is strong evidence for common ancestry of each *HOX* gene. The color-coding indicates that a particular *HOX* gene is shared between lineages of animals.

ISTUDY

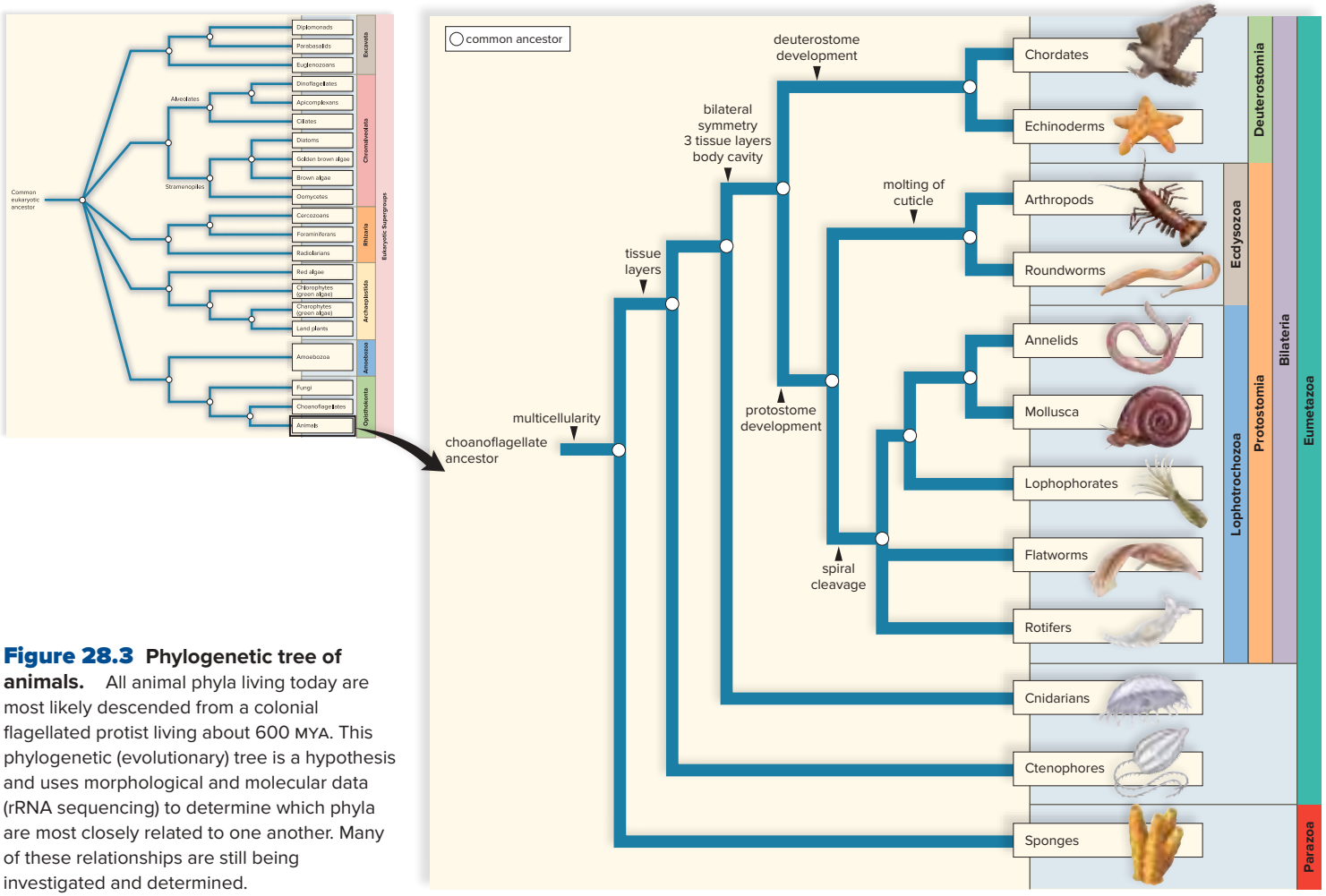


Figure 28.3 Phylogenetic tree of animals. All animal phyla living today are most likely descended from a colonial flagellated protist living about 600 MYA. This phylogenetic (evolutionary) tree is a hypothesis and uses morphological and molecular data (rRNA sequencing) to determine which phyla are most closely related to one another. Many of these relationships are still being investigated and determined.

represented by the phylogenetic tree of animals shown in Figure 28.3, is based on a combination of morphological characters of living and fossil organisms, developmental homologies, and molecular (DNA) characters. The more closely related two organisms are, the more similar their DNA sequences. The addition of molecular data has resulted in updated phylogenetic trees that are quite different from hypotheses that were historically based solely on morphological characteristics.

Table 28.1 provides the general characteristics of each of the phyla shown in Figure 28.3.

Refer to the phylogeny in Figure 28.3 as we discuss the general characteristics shared among animals of each branch of the animal family tree.

Type of Symmetry

Symmetry (Gk. *syn,* "together"; *metron,* "measure") is a pattern of similarity that is observed in objects; three types of symmetry exist in the animal kingdom. **Asymmetry,** or lack of symmetry, is seen in sponges that have no particular pattern to body shape (see Fig. 28.6).

The cnidarians and ctenophores exhibit **radial symmetry**—their bodies are organized circularly, similar to a wheel, such that any longitudinal cut through the central point produces two identical halves (Fig. 28.4*a*). Many adult and immature, or larval, forms of animals are radially symmetrical. Some radially symmetrical animals are mobile and others are immobile, or sessile, when attached to a substrate. Radial symmetry allows an organism to extend out in all directions from one center. Floating animals with radial symmetry, such as jellyfish, also have this feature.

The rest of the animals exhibit **bilateral symmetry** as adults—they have definite left and right halves, and only a single longitudinal cut down the centerline of the animal produces two equal halves (Fig. 28.4*b*). Bilaterally symmetrical animals have defined anterior and posterior ends, and forward movement is guided with the anterior end. The colonial flagellate hypothesis, mentioned earlier, is attractive because it implies that radial symmetry preceded bilateral symmetry in animal history, and a phylogeny supports this hypothesis.

During the evolution of animals, bilateral symmetry was accompanied by **cephalization,** localization of a brain and specialized sensory organs at the anterior end of an animal. This development was critical to an animal's ability to engage in directed movement—toward food or mates and away from danger.

Embryonic Development

The simplest animals are the sponges. Like all animals, sponges are multicellular. However, they do not have truly specialized tissues. True tissues develop in the more complex animals as they

ISTUDY

Table 28.1 Major Phyla of the Animal Kingdom

DOMAIN: Eukarya
SUPERGROUP: Opisthokonta
KINGDOM: Animalia

CHARACTERISTICS Multicellular, usually with specialized tissues; ingest or absorb food; diploid life cycle

Parazoa

Sponges (bony, glass, spongin): Asymmetrical, saclike body perforated by pores; internal cavity lined by choanocytes; spicules serve as internal skeleton. 26,000

Eumetazoa

Ctenophores (comb jellies): Have the appearance of jellyfish; the "combs" are eight visible longitudinal rows of cilia that can assist locomotion; lack the nematocysts of cnidarians but some have two tentacles. 150

Cnidarians (hydra, jellyfish, corals, sea anemones): Radially symmetrical with two tissue layers; sac body plan; tentacles with nematocysts. 10,000

Eumetazoa: Protostomia: Lophotrochozoa

Rotifers (wheel animals): Microscopic animals with a corona (crown of cilia) that looks like a spinning wheel when in motion. 2,500

Flatworms (planarians, tapeworms, flukes): Bilateral symmetry with cephalization; three tissue layers and organ systems; acoelomate with incomplete digestive tract that can be lost in parasites; hermaphroditic. 55,000

Lophophorates (brachiopods, bryozoa): Filter feeders with a circular or horseshoe-shaped ridge around the mouth that bears feeding tentacles. 10,000

Annelids (polychaetes, earthworms, leeches): Segmented with body rings and setae; cephalization in some polychaetes; hydroskeleton; closed circulatory system. 32,000

Mollusks (chitons, clams, snails, squids): Coelom; all have a foot, mantle, and visceral mass; foot is variously modified; in many, the mantle secretes a calcium carbonate shell as an exoskeleton; true coelom and all organ systems. 150,000

Eumetazoa: Protostomia: Ecdysozoa

Roundworms (pinworms, hookworms, filarial worms): Pseudocoelom and hydroskeleton; complete digestive tract; free-living forms in soil and water; parasites common. 61,000

Arthropods (crustaceans, spiders, scorpions, centipedes, millipedes, insects): Chitinous exoskeleton with jointed appendages; undergoes molting; insects—most have wings—are most numerous of all animals. 1,200,000

Eumetazoa: Deuterostomia

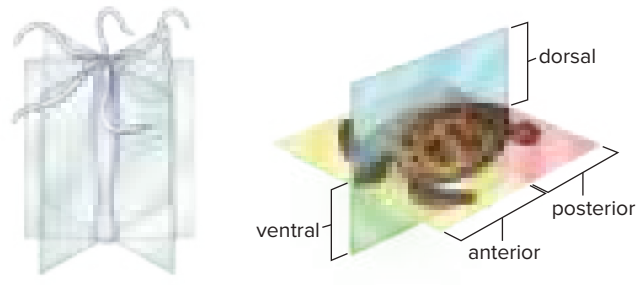
Echinoderms (sea stars, sea urchins, sand dollars, sea cucumbers): Radial symmetry as adults; unique water-vascular system and tube feet; endoskeleton of calcium plates. 12,000

Chordates (tunicates, lancelets, vertebrates): All have notochord, dorsal tubular nerve cord, pharyngeal pouches, and postanal tail at some time; contains mostly vertebrates in which notochord is replaced by vertebral column. 60,000

Not all phyla are listed. Species numbers are estimates of currently identified species.

undergo embryological development. The first tissue layers that appear are called **germ layers,** and they give rise to the organs and organ systems of complex animals.

Animals such as the cnidarians, which as embryos have only two tissue layers—the ectoderm and endoderm—are termed *diploblastic.* Diploblastic animals develop tissues, but no specialized organs. The animals that develop specialized organs are termed *triploblastic* because, as embryos, they have three tissue layers—the ectoderm, mesoderm, and endoderm. All animals are triploblastic except the sponges, ctenophores, cnidarians, and Placozoa (a poorly known phylum with only a single living member discovered).



a. Radial symmetry b. Bilateral symmetry

Figure 28.4 Symmetry in animals. The body plans of animals can be (**a**) radially symmetrical or (**b**) bilaterally symmetrical.

Notice in the phylogenetic tree (see Fig. 28.3) the triploblastic animals are either **protostomes** (Gk. *proto,* "first"; *stoma,* "mouth") or **deuterostomes** (Gk. *deuter,* "second"). These terms have to do with whether the mouth or the anus develops first in the embryo. In protostomes, the mouth develops prior to the anus, whereas in deuterostomes, the anus develops prior to the mouth. Most invertebrates are protostomes, but all vertebrates, including humans, are deuterostomes. Figure 28.5 shows that protostome and deuterostome development are differentiated by three major events: cleavage, blastula formation, and coelom development.

ISTUDY

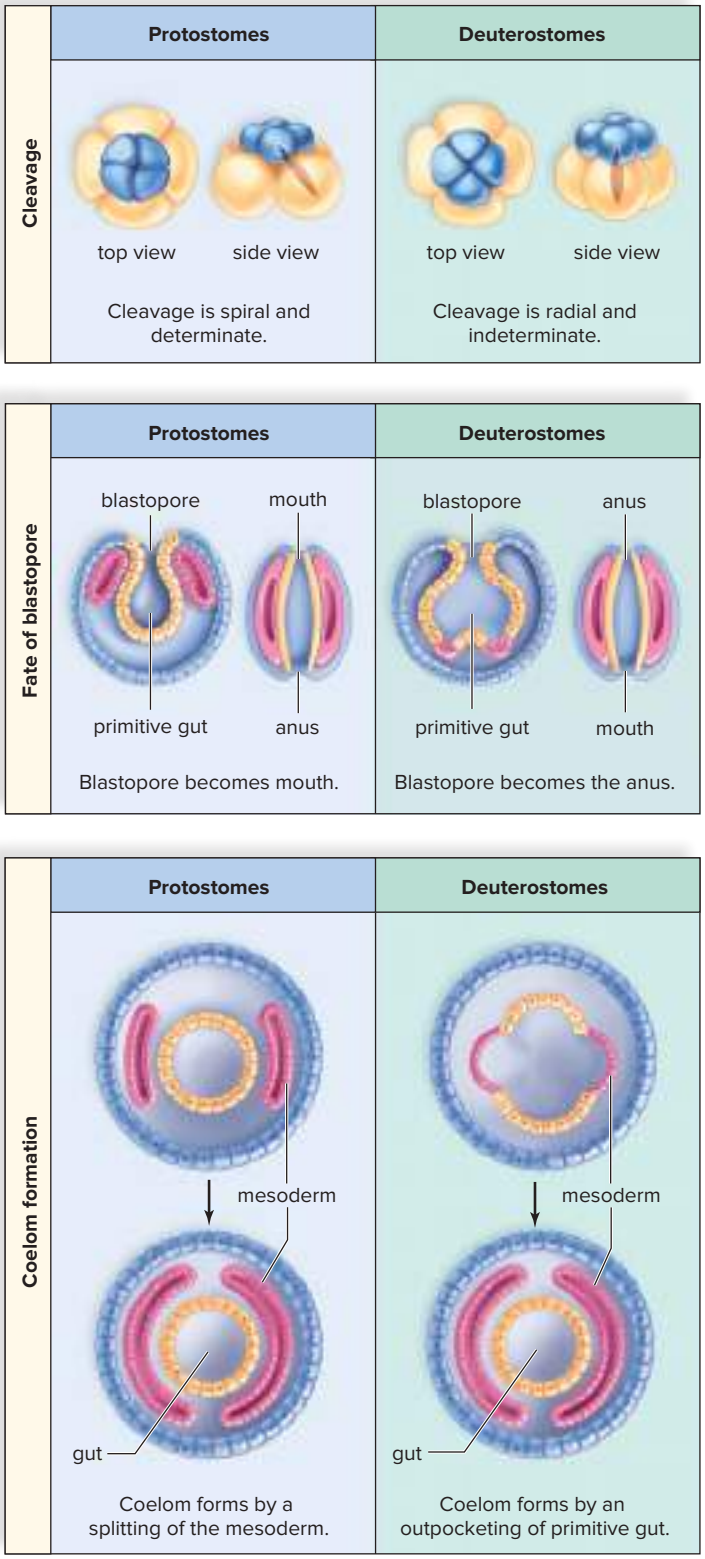


Figure 28.5 Protostomes compared to deuterostomes. *Left:* In the embryo of many protostomes, cleavage is spiral—new cells are at an angle to old cells—and each cell has limited potential and cannot develop into a complete embryo; the blastopore is associated with the mouth; and the coelom, if present, develops by a splitting of the mesoderm. *Right:* In deuterostomes, cleavage is radial—new cells sit on top of old cells—and each one can develop into a complete embryo; the blastopore is associated with the anus; and the coelom, if present, develops by an outpocketing of the primitive gut.

Cleavage The first developmental event after fertilization is **cleavage,** cell division without cell growth. In most protostomes (except ecdysozoans) spiral cleavage occurs, and daughter cells sit in grooves formed by the previous cleavages. The fate of these cells is fixed and determinate in protostomes; each can contribute to development in only one particular way.

In deuterostomes, radial cleavage occurs, and the daughter cells sit right on top of the previous cells. The fate of these cells is indeterminate—that is, if they are separated from one another, each cell can go on to become a complete organism.

Blastula Formation As development proceeds, a hollow sphere of cells, or **blastula,** forms, and the indentation that follows produces an opening called the blastopore. In protostomes, the mouth appears at or near the blastopore.

In deuterostomes, the anus appears at or near the blastopore, and only later does a second opening form the mouth.

Coelom Development Certain protostomes and all deuterostomes have a body cavity lined by mesoderm called a **coelom** (Gk. *koiloma,* "cavity"). More specifically, the coelom in these groups is a **true coelom,** because the mesoderm cells line the cavity completely. However, the coelom develops differently in the two groups. In protostomes, the mesoderm arises from cells located near the embryonic blastopore, and a splitting occurs that produces the coelom.

In deuterostomes, the coelom arises as a pair of mesodermal pouches from the wall of the primitive gut. The pouches enlarge until they meet and fuse. To learn more about animal development, see Chapter 42.

Check Your Progress 28.1

1. State three characteristics that all animals have in common.
2. Explain the colonial flagellate hypothesis about the origin of animals.
3. Compare radial and bilateral symmetry, and provide examples of animals that exemplify each.
4. List two differences between deuterostomes and protostomes.

28.2 The Simplest Invertebrates

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain why sponges are considered to be the simplest animals.
2. Discuss how a sponge respires, feeds, and reproduces.
3. Compare the anatomical features of ctenophores to those of cnidarians, such as hydras.

The Parazoa: Sponges

While all animals are multicellular, **sponges** (phylum Porifera [L. *porus,* "pore"; *ferre,* "to bear"]) are the only animals to lack true tissues and to have only a cellular level of organization. They have only a few cell types, and they lack the nerve and

ISTUDY

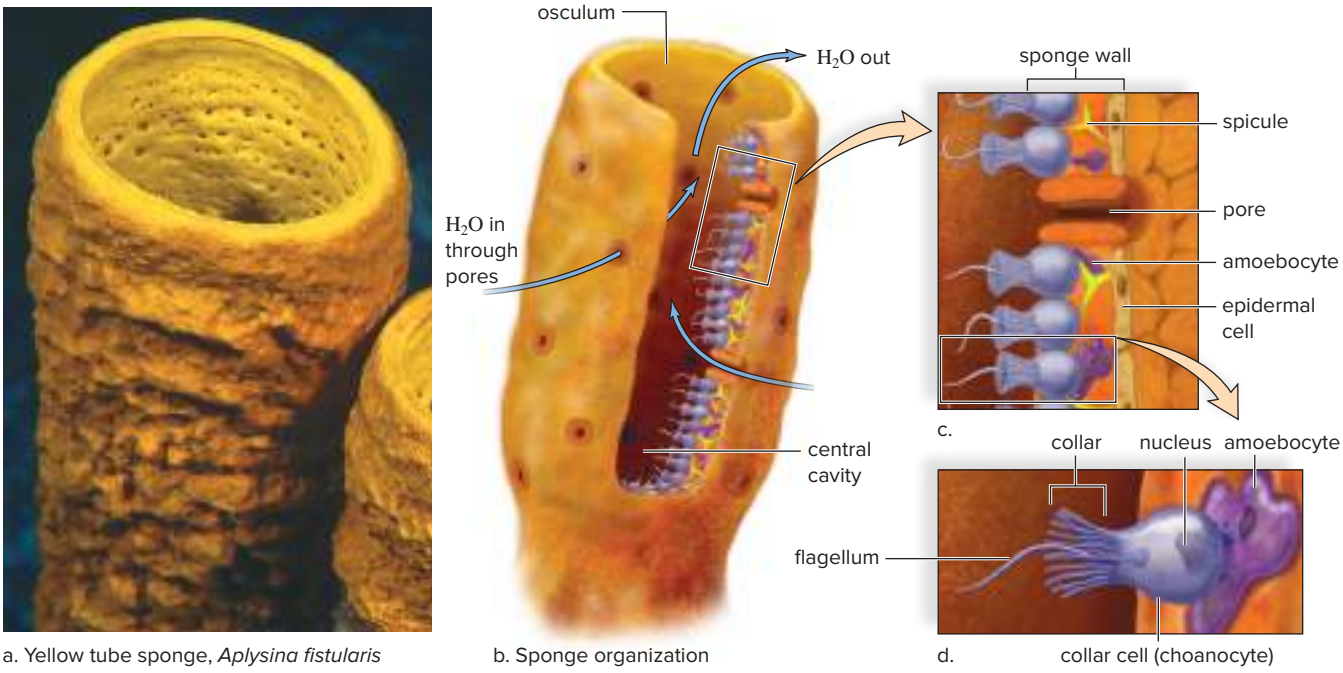


Figure 28.6 Sponge anatomy. **a.** A simple sponge. **b.** The wall of a sponge contains two layers of cells, the outer epidermal cells and the inner collar cells as shown in (**c**). The collar cells (**d,** enlarged) have flagella that beat, moving the water through pores as indicated by the blue arrows in (**b**). Food particles in the water are trapped by the collar cells and digested within their food vacuoles. Amoebocytes transport nutrients from cell to cell. Spicules form an internal skeleton in some sponges. (a): Andrew J. Martinez/Science Source

muscle cells seen in more complex animals. Molecular data place them at the base of the evolutionary tree of animals in a taxon called the **Parazoa** (see Fig. 28.3).

The saclike body of a sponge is perforated by many pores (Fig. 28.6). Sponges are aquatic, largely marine animals that vary greatly in size, shape, and color. However, they all have a canal system of pores of varying complexity that allows water to move through their bodies.

The interior of the canals is lined with flagellated cells that resemble choanoflagellates. In a sponge, these cells are called collar cells, or choanocytes. The beating of the flagella produces water currents, which flow through the pores into the central cavity and out through the osculum, the upper opening of the body. Even a simple sponge only 10 cm tall is estimated to filter as much as 100 liters of water each day. All of a sponge's cells are able to acquire the oxygen they need for cellular respiration by diffusion from this water.

A sponge is a sessile filter feeder, also called a suspension feeder, because it filters suspended particles from the water by means of a straining device—in this case, the pores and the flagella of the collar cells. Microscopic food particles that pass between the flagella and microvilli are engulfed by the collar cells and digested by them in food vacuoles.

The skeleton of a sponge provides structural support for its body. All sponges contain tough fibers made of spongin, a modified form of collagen; a natural bath sponge is the dried spongin skeleton from which all living tissue has been removed. Today, however, commercial "sponges" are usually synthetic.

Typically, the endoskeleton of a sponge also contains small, needle-shaped structures called spicules**.** Traditionally, the type of spicule has been used to classify sponges; there are bony, glass, and spongin sponges. The success of sponges—they have existed longer than any other animal group—is due in part to their spicules. They have few predators, because a mouth full of spicules is an unpleasant experience. Some sponges also produce toxic substances that discourage predators.

Sponges can reproduce both asexually and sexually. They reproduce asexually by fragmentation or by budding. During budding, a small protuberance appears and gradually increases in size until a complete organism forms. Budding produces colonies of sponges that can become quite large. During sexual reproduction, eggs and sperm are released into the central cavity, and the zygote develops into a flagellated larva, which may swim to a new location.

Like many relatively simple organisms, sponges are capable of regeneration, or growth of a whole new sponge from a small part. Moreover, if the cells of a sponge are mechanically separated, they will reassemble into a complete and functioning organism.

The Eumetazoa: Ctenophores and Cnidarians

The **Eumetazoa** are the "true animals," meaning they have true tissues and germ layers during embryonic development. They also all go through similar stages of embryonic development (see Section 42.1).

The first two groups of eumetazoans we will examine all have true tissues, and as embryos, they have two germ layers: ectoderm and endoderm. They are radially symmetrical as adults.

Ctenophora

Ctenophora, or **comb jellies** (Fig. 28.7*a*), are solitary, mostly free-swimming marine invertebrates that are usually found in warm waters. Ctenophores represent the largest of these animals; they are propelled by beating cilia and range in size from a few centimeters to 1.5 m in length. Their body is made up of a transparent, jellylike substance called **mesoglea.** Most ctenophores do not have stinging

ISTUDY

Figure 28.7 Comb jelly compared to cnidarian. **a.** *Pleurobrachia pileus,* a comb jelly. **b.** *Polyorchis penicillatus,* medusan form of a cnidarian. Both animals have similar symmetry, diploblastic organization, and gastrovascular cavities. (a): Jeff Rotman/Alamy Stock Photo; (b): Mark Conlin/Alamy Stock Photo

cells and capture their prey by using sticky, adhesive cells called colloblasts. Some ctenophores are *bioluminescent,* meaning that they are capable of producing their own light.

Traditionally, ctenophores were placed as a group with cnidarians on the animal phylogenetic tree, evolving after sponges. Recent studies, however, suggest that ctenophores contain DNA sequences that are more ancient than those of even sponges. Much debate still exists around their placement because the DNA evidence is uncertain and it contradicts the advanced physiology of ctenophores compared to sponges.

Cnidarians

Cnidarians (phylum Cnidaria) (Fig. 28.7*b*) are tubular or bell-shaped animals that reside mainly in shallow coastal waters; however, some freshwater, brackish, and oceanic forms are known. The term *cnidaria* is derived from the presence of specialized stinging cells called cnidocytes. Each cnidocyte has a fluid-filled capsule called a **nematocyst** (Gk. *nema,* "thread"; *kystis,* "bladder") that contains a long, spirally coiled, hollow thread. When the trigger of the cnidocyte is touched, the nematocyst is discharged. Some threads merely trap a prey; others have spines that penetrate and inject paralyzing toxins.

The body of a cnidarian is a two-layered sac. The outer tissue layer is a protective epidermis derived from ectoderm. The inner tissue layer, which is derived from endoderm, secretes digestive juices into the internal cavity, called the **gastrovascular cavity** (Gk. *gastros,* "stomach") because it functions in the digestion of food and circulation of nutrients. The fluid-filled gastrovascular cavity also serves as a supportive **hydrostatic skeleton,** so called because it offers some resistance to the contraction of muscle but permits flexibility. The two tissue layers are separated by mesoglea.

Two basic body forms are seen among cnidarians. The mouth of a **polyp** is directed upward, while the mouth of a jellyfish, or **medusa,** is directed downward. The bell-shaped medusa has more mesoglea than a polyp, and the tentacles are concentrated on the margin of the bell.

At one time, both body forms may have been a part of the life cycle of all cnidarians. When both are present, the animal is dimorphic: The sessile polyp stage produces medusae by asexual budding, and the motile medusan stage produces egg and sperm. In some cnidarians, one stage is dominant and the other is reduced; in other species, one form is absent altogether.

Cnidarian Diversity Sea anemones (Fig. 28.8*a*) are sessile polyps that live attached to submerged rocks, timbers, or other substrates. Most sea anemones range in size from 0.5 to 20 cm

a. Sea anemone, *Condylactis*

b. Cup coral, *Tubastrea*

c. Portuguese man-of-war, *Physalia*

d. Jellyfish, *Aurelia*

Figure 28.8 Cnidarian diversity. **a.** The sea anemone, which is sometimes called the flower of the sea, is a solitary polyp. **b.** Corals are colonial polyps residing in a calcium carbonate or proteinaceous skeleton. **c.** The Portuguese man-of-war is a colony of modified polyps and medusae. **d.** Jellyfish, *Aurelia*. (a): Comstock Images/PictureQuest/Getty Images; (b): Ron Taylor/Bruce Coleman/Photoshot/Avalon; (c): Image courtesy of Islands in the Sea 2002, NOAA/OER; (d): Vladimir Wrangel/Shutterstock

ISTUDY

in length and 0.5 to 10 cm in diameter and are often colorful. Their upward-turned oral disk contains the mouth and is surrounded by a large number of hollow tentacles containing nematocysts.

Corals (Fig. 28.8*b*) resemble sea anemones that are encased in a calcium carbonate (limestone) house. The coral polyp can extend out of their house to feed on microorganisms and retreat into the house for safety. Some corals are solitary, but the vast majority live in colonies that vary in shape from rounded to branching. Corals are the animals that build coral reefs. They exhibit elaborate geometric designs and stunning colors.

Coral reefs are built from the slow accumulation of limestone produced by corals. Over hundreds of years, this accumulation can result in massive structures, such as the Great Barrier Reef along the eastern coast of Australia and the Mesoamerican Barrier Reef along the coast of Belize. Coral reef ecosystems are very productive, and support an amazing diversity of marine life.

The hydrozoans have a dominant polyp stage. *Hydra* (Fig. 28.9) is a hydrozoan, and so is a Portuguese man-of-war. You might think the Portuguese man-of-war is an odd-shaped medusa, but actually it is a colony of polyps (see Fig. 28.8*c*). The original polyp becomes a gas-filled float that provides buoyancy, keeping the colony afloat. Other polyps, which bud from this one, are specialized for feeding or for reproduction. A long, single tentacle armed with numerous nematocysts arises from the base of each feeding polyp. Swimmers who accidentally come upon a Portuguese man-of-war can receive painful, even serious, injuries from these stinging tentacles.

In true jellyfish, also known as sea jellies (see Fig. 28.8*d*), the medusa is the primary stage and the polyp remains small. Jellyfish are a type of zooplankton which depend on tides and currents for their primary means of movement. They feed on a variety of invertebrates and fishes and are themselves food for various marine animals.

A Typical Cnidarian: Hydra Hydras are often studied as an example of a cnidarian. Hydras are likely to be found attached to underwater plants or rocks in most lakes and ponds. The body of a hydra is a small, tubular polyp about one-quarter of an inch in length. The only opening (which serves as both mouth and anus) is in a raised area surrounded by four to six tentacles that contain a large number of nematocysts.

Figure 28.9 shows the microscopic anatomy of a hydra. The cells of the epidermis are termed epitheliomuscular cells because they contain muscle fibers. Also present in the epidermis are nematocyst-containing cnidocytes and sensory cells that make contact with the nerve cells within a **nerve net.** These interconnected nerve cells allow the transmission of impulses in several directions at once. The body of a hydra can contract or extend, and the tentacles that ring the mouth can reach out and grasp prey and discharge nematocysts.

Hydras reproduce asexually by forming buds, small outgrowths that develop into a complete animal and then detach. Interstitial cells of the epidermis are capable of becoming other types of cells, such as an ovary and/or a testis. When hydras reproduce sexually, sperm from a testis swim to an egg within an ovary. The embryo is encased within a hard, protective shell that allows it to survive until conditions are optimum for it to emerge and develop into a new polyp.

Like the sponges, cnidarians have great regenerative powers, and hydras can grow an entire organism from a small piece.

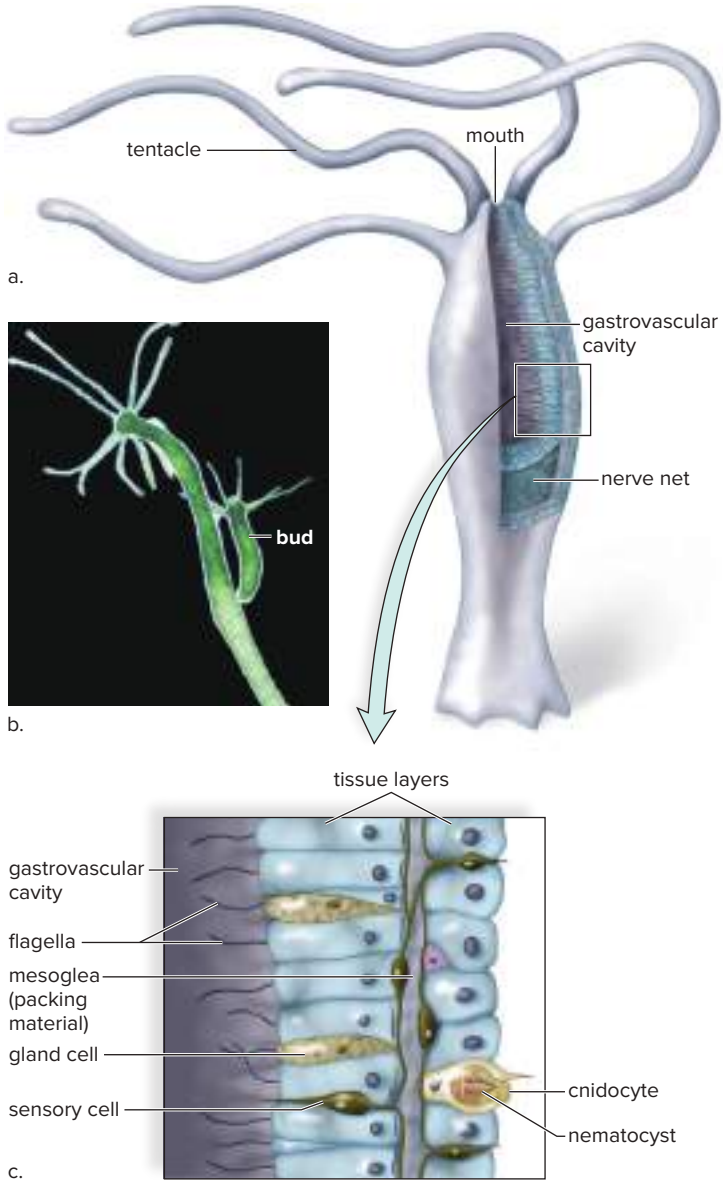


Figure 28.9 Anatomy of a hydra. **a.** The body of a hydra is a small, tubular polyp that reproduces asexually by forming outgrowths called buds. **b.** The buds develop into a complete animal. **c.** The body wall contains two tissue layers separated by mesoglea. Cnidocytes are cells that contain nematocysts. (photo): M. I. Walker/NHPA/Photoshot/AVALON

Check Your Progress 28.2

1. List three ways in which cnidarians are more complex than sponges.
2. Summarize how a sponge obtains nutrients.
3. Describe the medusa and polyp body forms of a cnidarian.
4. Explain how a cnidarian, such as a jellyfish, stings its prey.

ISTUDY

28.3 Diversity of the Lophotrochozoans

Learning Outcomes

Upon completion of this section, you should be able to

1. List the basic features of lophotrochozoans.
2. Summarize the steps in the life cycles of *Schistosoma* and *Taenia*.
3. Identify morphological features of mollusks, bivalves, rotifers, and annelids.
4. Describe the basic anatomy and physiology of a planarian.

The **Lophotrochozoa** are a diverse group of protostomes. The phylogeny of this group is still being investigated. Historically, the group was divided into two subgroups—the lophotrochozoans and platyzoans—under the name Spirilia. However, current trends place these animals in one larger taxon called the Lophotrochozoa. These animals are bilaterally symmetrical during at least one stage of their development. As embryos, they have three germ layers, and as adults they achieve the organ level of organization. Some have a true coelom (see Section 28.1), as exemplified by the annelid worms.

Types of Lophotrochozoa

Lophotrochozoans are protostomes and include the **lophophorans** (bryozoans, phoronids, and brachiopods) and the **trochozoans** (mollusks and annelids). Lophophorans are aquatic and have a feeding apparatus called the lophophore, which is a mouth surrounded by a ciliated tentacle-like structure (Fig. 28.10*a*). Trochozoans all have a trochophore as a larval stage. A trochophore is a free-swimming, marine larva with bands of cilia that control the direction of movement (Fig. 28.10*b*). Flatworms and rotifers are also considered trochozoans because they descended from an ancestor of a trochophore in the past.

Lophophorans

Traditionally, the lophophorans were considered a lineage of protostomes that included the bryozoans and brachiopods, but

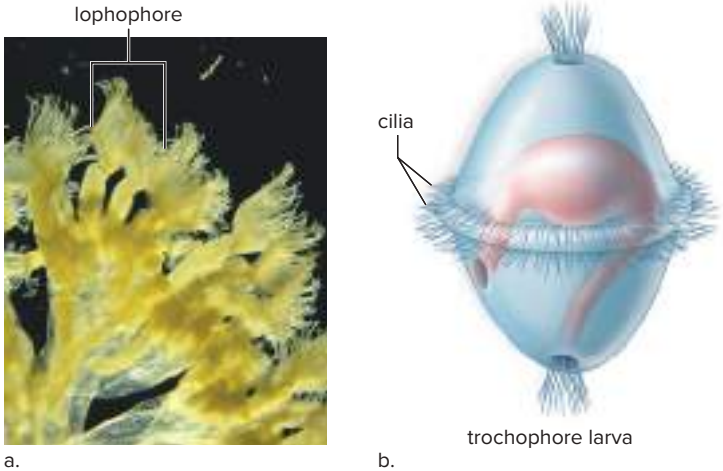


Figure 28.10 Lophotrochozoan characteristics. **a.** The lophophore feeding apparatus. **b.** A trochophore larva. (a): The Natural History Museum/Alamy Stock Photo

Figure 28.11 Lophophorans. A bryozoan (*Canda* sp.) with a horseshoe-shaped tentacle crown. Diane R. Nelson

recent evidence from analysis of the small subunit of ribosomal RNA (rRNA) and other genes suggest that the evolutionary relationships among the lophophorans within the Protostomia are more complex. However, for our purposes, we can represent the lophophorans in the traditional sense as a single lineage (see Fig. 28.3), containing three closely related groups, the bryozoans, the brachiopods, and the phoronids (Fig. 28.11).

Bryozoans (phylum Bryozoa) are aquatic, colonial lophophorans. Colonies are made up of individuals called zooids. Zooids are not independent animals but, rather, individual members within a colony that cooperate as a single organism. Some zooids specialize in feeding and they filter particles from the water with the lophophore; some specialize in reproduction; and some can perform both functions. Zooids coordinate functions within a colony by communicating through chemical signals. Zooids have protective exoskeletons, which they use to attach to substrates, including the bottoms of ships, where they cause a nuisance by increasing drag and impeding maneuverability.

Brachiopods (phylum Brachiopoda) are a small group of lophophorans that have two hinged shells, much as mollusks do—but instead of having a left and right shell, they have a top and a bottom shell. Brachiopods affix themselves to hard surfaces with a muscular pedicle. Like other lophophorans, brachiopods use their lophophore to feed by filtering particles from the water.

Phoronids (phylum Phoronida) live inside a long tube formed from their own chitinous secretions. The tube is buried in the ground and their lophophore extends from it, but it can retract very quickly when needed. Only about 15 species of phoronids exist worldwide.

Trochozoans

Mollusks and annelids are classified as trochozoans, the second major group of lophotrochozoans. They are characterized by a trocophore stage of development.

Mollusks The **mollusks** (phylum Mollusca) are the second most numerous group of animals, with over 150,000 named species. They inhabit marine, freshwater, and terrestrial habitats. This diverse

ISTUDY

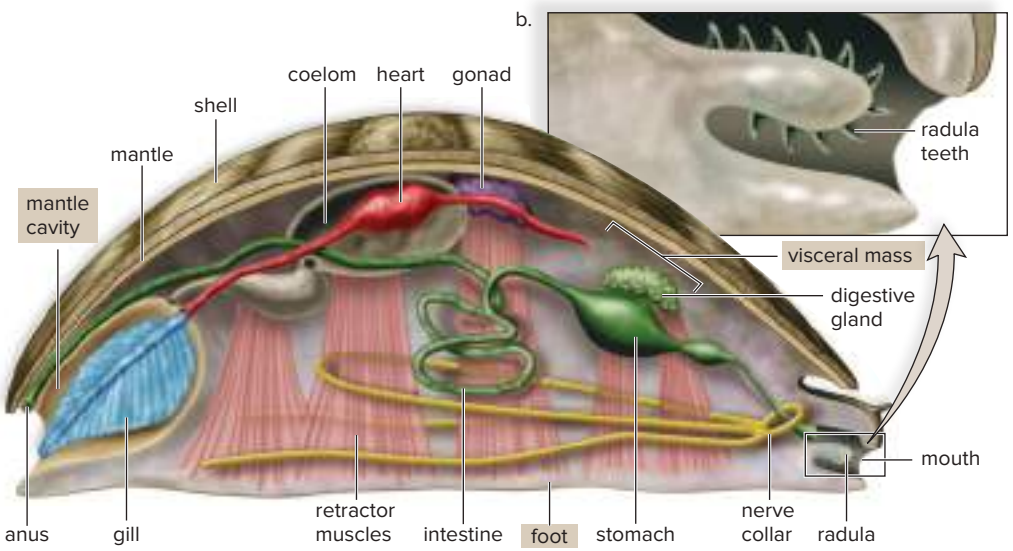


Figure 28.12 Body plan of mollusks. **a.** Mollusks have a three-part body consisting of a ventral, muscular foot that is specialized for various means of locomotion; a visceral mass that includes the internal organs; and a mantle that covers the visceral mass and may secrete a shell. Ciliated gills may lie in the mantle cavity and direct food toward the mouth. **b.** In the mouth of many mollusks, such as snails, the radula is a tonguelike organ that bears rows of tiny teeth that point backward.

phylum includes chitons, limpets, slugs, snails, abalones, conchs, nudibranchs, clams, oysters, scallops, squids, and octopuses. Mollusks vary in size from microscopic to the giant squid, which can attain lengths of over 20 m and weigh over 450 kg. The group includes herbivores, carnivores, filter feeders, and parasites.

Although diverse, mollusks share a three-part body plan consisting of the visceral mass, mantle, and foot (Fig. 28.12*a*). The visceral mass contains the internal organs, including a highly specialized digestive tract, paired kidneys, and reproductive organs. The **mantle** is a covering that lies to either side of, but does not completely enclose, the visceral mass. It may secrete a shell and/or contribute to the development of gills or lungs. The space between the folds of the mantle is called the mantle cavity. The foot is a muscular organ that may be adapted for locomotion, attachment, food capture, or a combination of functions. Another feature often present in mollusks is a rasping, tonguelike *radula,* an organ that bears many rows of teeth and is used to obtain food (Fig. 28.12*b*).

The true coelom is reduced in mollusks and largely limited to the region around the heart. Most mollusks have an open circulatory system. The heart pumps blood, more properly called hemolymph, through vessels into sinuses (cavities) collectively called a **hemocoel** (Gk. *haima,* "blood"; *koiloma,* "cavity"). Blue hemocyanin, rather than red hemoglobin, is the oxygen-carrying pigment. Nutrients and oxygen diffuse into the tissues from these sinuses instead of being carried into the tissues by capillaries, the microscopic blood vessels present in animals with closed circulatory systems.

The nervous system of mollusks consists of several ganglia connected by nerve cords. The amount of cephalization and sensory organs varies from nonexistent in clams to complex in squids and octopuses. The mollusks also exhibit variation in mobility. Oysters are sessile, snails are extremely slow moving, and squids are fast-moving, active predators.

Bivalves Clams, oysters, shipworms, mussels, and scallops are all **bivalves** (class Bivalvia), with a two-part shell that is hinged and closed by one or two powerful adductor muscles (Fig. 28.13). Bivalves have no head, no radula, and very little cephalization. Clams use their hatchet-shaped foot for burrowing in sandy or muddy soil, and mussels use their foot to produce threads that attach them to nearby objects. Scallops both burrow and swim; rapid clapping of the valves releases water in spurts and causes the animal to move forward in a jerky fashion for a few feet.

In freshwater clams such as *Anodonta* (Fig. 28.13*c*), the shell, secreted by the mantle, is composed of protein and calcium carbonate, with an inner layer called mother of pearl. If a foreign body is placed between the mantle and the shell, pearls form as concentric layers of shell are deposited about the particle. The compressed muscular foot of a clam projects ventrally from the shell; by expanding the tip of the foot and pulling the body after it, the clam moves forward.

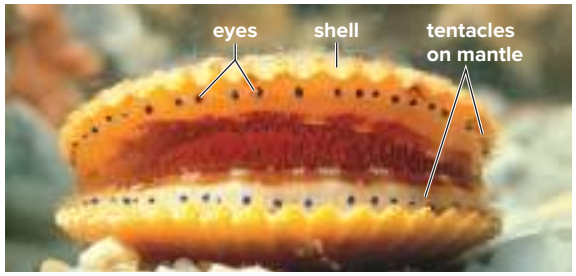
Within the mantle cavity, the ciliated gills hang down on either side of the visceral mass. The beating of the cilia causes water to enter the mantle cavity by way of the incurrent siphon, and to exit by way of the excurrent siphon. The clam is a filter feeder; small particles in this constant stream of water adhere to the gills, and ciliary action sweeps them toward the mouth.

The mouth leads to a stomach and then to an intestine, which coils about in the visceral mass before going right through the heart and ending in an anus. The anus empties at the excurrent siphon. An accessory organ of digestion, called a digestive gland, is also present. The heart lies just below the hump of the shell within the pericardial cavity, the only remains of the coelom. The circulatory system is open; the heart pumps hemolymph into vessels that open into the *hemocoel.* The nervous system is composed of three pairs of ganglia (located anteriorly, posteriorly, and in the foot), which are connected by nerves.

Two excretory kidneys lie just below the heart and remove waste from the pericardial cavity for excretion into the mantle cavity. The clam excretes ammonia (NH_3), a toxic substance that requires the excretion of water at the same time.

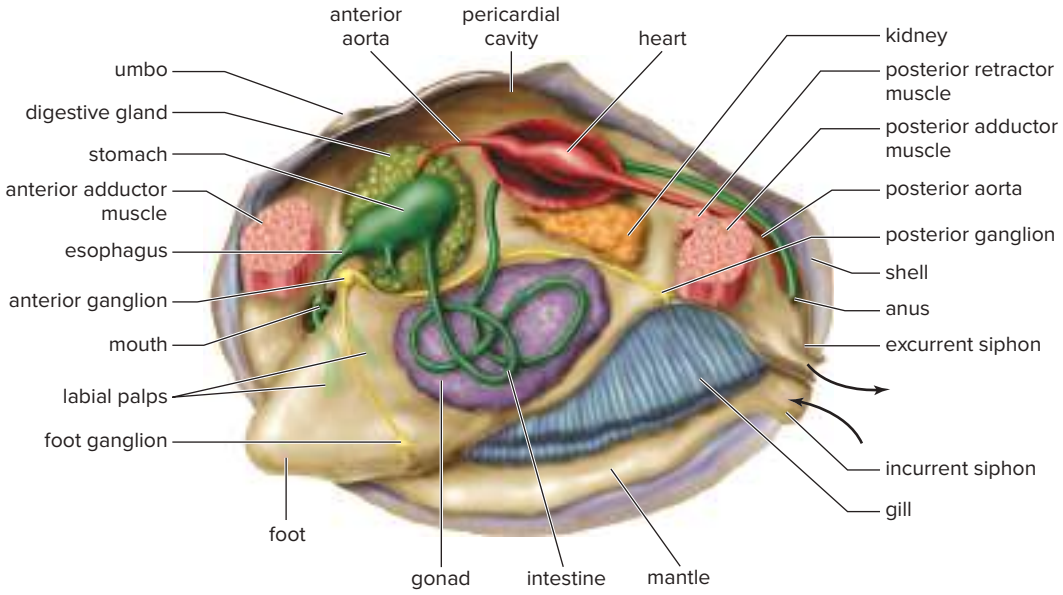
In freshwater clams, the sexes are separate, and fertilization is internal. Fertilized eggs develop into specialized larvae and are released from the clam. Some larvae attach to the gills of a fish and become a parasite before they sink to the bottom and develop into

ISTUDY



a. Scallop, *Pecten* sp.

b. Mussels, *Mytilus edulis*



c. Clam, *Anodonta*

Figure 28.13 Bivalve diversity. Bivalves have a two-part shell. **a.** Scallops clap their valves and swim by jet propulsion. This scallop has sensory organs consisting of blue eyes and tentacles along the mantle edges. **b.** Mussels form dense beds in the intertidal zone of northern shores. **c.** In this drawing of a clam, the mantle has been removed from one side. Follow the path of food from the incurrent siphon to the gills, the mouth, the stomach, the intestine, the anus, and the excurrent siphon. Locate the three ganglia: anterior, foot, and posterior. The heart lies in the reduced coelom. (a): ANT Photo Library/Science Source; (b): Michael Lustbader/Science Source

a clam. Certain clams and annelids have the same type of larva, namely the trochophore larva, and this reinforces the evolutionary relationship between mollusks and annelids.

Other Mollusks The **gastropods** (Gk. *gastros,* "stomach"; *podos,* "foot"), the largest class of mollusks, include slugs, snails, whelks, conchs, limpets, and nudibranchs (Fig. 28.14). Most are marine; however, slugs and garden snails are adapted to terrestrial environments.

Gastropods have an elongated, flattened foot, and most, except for slugs and nudibranchs, have a one-piece, coiled shell that protects the visceral mass. The anterior end bears a well-developed head region with a cerebral ganglion and eyes on the ends of tentacles. Land snails are hermaphroditic; when two snails meet, they shoot calcareous darts into each other's body wall as a part of premating behavior. Then, each inserts a penis into the vagina of the other to provide sperm for the future fertilization of eggs, which are deposited in the soil. Development proceeds directly without the formation of larvae.

Cephalopods (Gk. *kaphale,* "head"; *podos,* "foot") range in length from 2 cm to 20 m, as in the giant squid, *Architeuthis.* *Cephalopod* means head-footed; both squids and octopuses can squeeze their mantle cavity so that water is forced out through a funnel, propelling them by jet propulsion (Fig. 28.15). Also, the tentacles and arms that circle the head capture prey by adhesive secretions or by suckers. A powerful, parrotlike beak is used to tear prey apart. They have well-developed sense organs, including eyes that are similar to those of vertebrates and that focus like a camera. Cephalopods, particularly octopuses, have well-developed brains and show a remarkable capacity for learning.

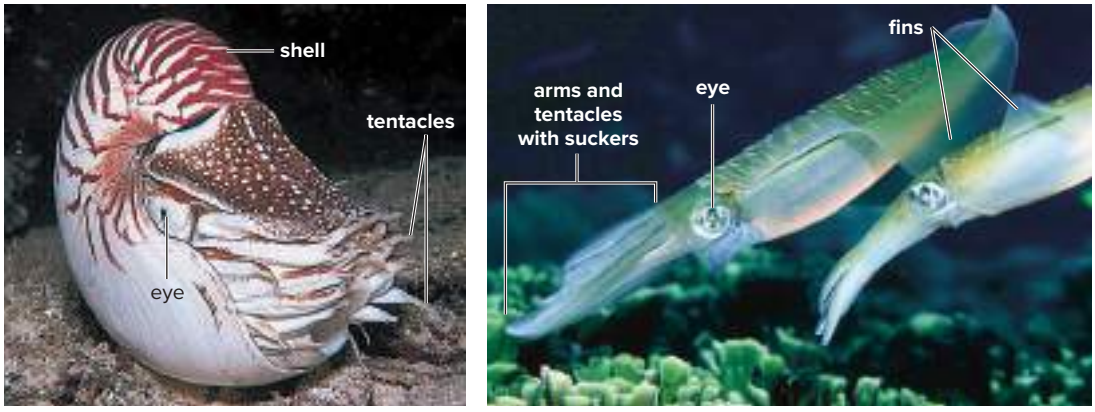
Annelids **Annelids** (phylum Annelida [L. *anellus,* "little ring"]), which are sometimes called the segmented worms, vary in size from microscopic to tropical earthworms that can be over 4 m long. The most familiar members of this group are earthworms, marine worms, and leeches.

Annelids are the only trochozoan with segmentation and a well-developed coelom. **Segmentation** is the repetition of body parts along the length of the body. The well-developed coelom is fluid-filled and serves as a supportive *hydrostatic skeleton.* A hydrostatic skeleton, along with partitioning of the coelom, permits

ISTUDY

a. Flamingo tongue shell, *Cyphoma gibbosum* b. Land snail, *Helix aspersa*

Figure 28.14 Gastropod diversity. Gastropods have the three parts of a mollusk, and the foot is muscular, elongated, and flattened. (a): Wolfgang Polzer/Alamy Stock Photo; (b): ©Farley Bridges



a. Chambered nautilus, *Nautilus belauensis* b. Bigfin reef squid, *Sepioteuthis lessoniana*

Figure 28.15 Cephalopod diversity. Cephalopods have tentacles and/or arms in place of a head. Speed suits their predatory lifestyle, and only the chambered nautilus has a shell. (a): Douglas Faulkner/Science Source; (b): Georgette Douwma/Science Source

independent movement of each body segment. Instead of just burrowing in the mud, an annelid can crawl on a surface.

Setae (L. *seta,* "bristle") are bristles that protrude from the body wall that can anchor the worm and help it move. The *oligochaetes* are annelids with few setae, and the *polychaetes* are annelids with many setae.

Earthworms The common earthworm, *Lumbricus terrestris,* is an oligochaete (Fig. 28.16). Earthworm setae protrude in pairs directly from the surface of the body. Locomotion, which is accomplished section by section, uses muscle contraction and the setae. When longitudinal muscles contract, segments bulge and their setae protrude into the soil; then, when circular muscles contract, the setae are withdrawn, and these segments move forward.

Earthworms reside in soil where there is adequate moisture to keep the body wall moist for gas exchange. They are scavengers that feed on leaves or any other organic matter conveniently taken into the mouth along with dirt. Segmentation and a complete digestive tract have led to increased specialization of digestive system components. Food drawn into the mouth by the action of the muscular pharynx is stored in a crop and ground up in a thick, muscular gizzard (Fig. 28.16). Digestion and absorption occur in a long intestine, whose dorsal surface has an expanded region called a typhlosole, which increases the surface for absorption. Waste is eliminated through the anus.

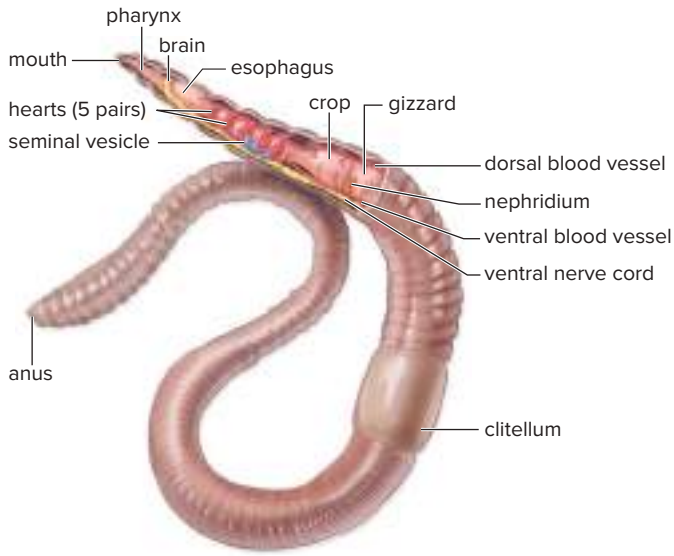


Figure 28.16 Earthworm, *Lumbricus terrestris*. Note the cephalization, specialized parts of the digestive tract, and segmentation.

ISTUDY

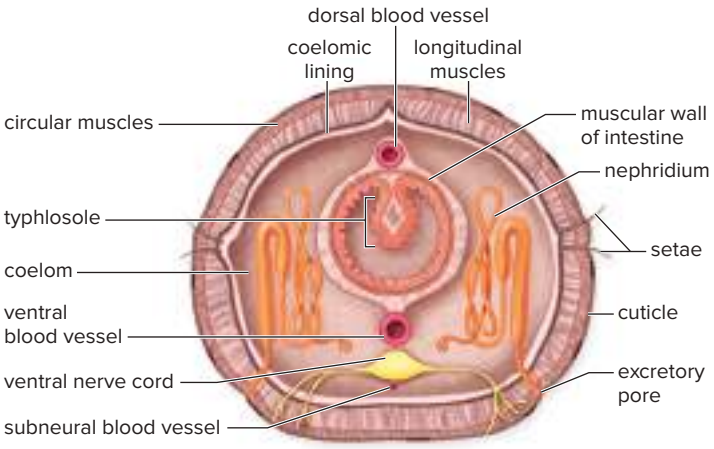


Figure 28.17 Earthworm cross section. In cross section, note the spacious coelom, the paired setae and nephridia, and a ventral nerve cord that has branches in each segment.

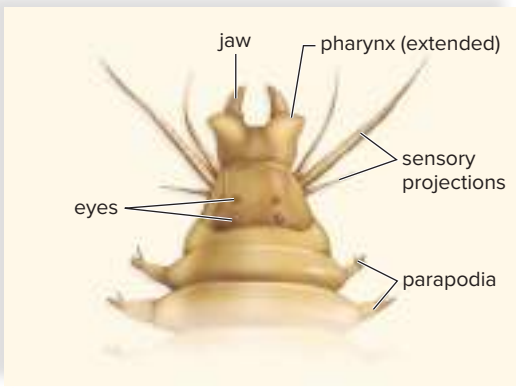


Figure 28.19 Clam worm. Clam worms are predaceous polychaetes that undergo cephalization. Note also the parapodia, which are used for swimming and as respiratory organs.

Earthworm segmentation, which is obvious externally, is also internally evidenced by septa that occur between segments. The long, ventral nerve cord leading from the brain has ganglionic swellings and lateral nerves in each segment. The excretory system consists of paired **nephridia** (Gk. *nephros,* "kidney"), which are coiled tubules in each segment (Fig. 28.17). A nephridium has two openings: One is a ciliated funnel that collects coelomic fluid, and the other is an exit in the body wall. Between the two openings is a convoluted region where waste material is removed from the blood vessels about the tubule.

Annelids have a *closed circulatory system,* which means that the blood is always contained in blood vessels that run the length of the body. Oxygenated blood moves anteriorly in the dorsal blood vessel, which connects to the ventral blood vessel by five pairs of muscular vessels called "hearts." Pulsations of the dorsal blood vessel and the five pairs of hearts are responsible for blood flow. As the ventral vessel takes the blood toward the posterior regions of the worm's body, it gives off branches in every segment.

Earthworms are *hermaphroditic;* the testes, the seminal vesicles, and the sperm ducts are the male organs. The ovaries, the oviducts, and the seminal receptacles are the female organs. During mating, two worms lie parallel to each other, facing in opposite directions (Fig. 28.18). The fused midbody segment, called a *clitellum,* secretes mucus, protecting the sperm from drying out as they pass between the worms. After the worms separate, the clitellum of each produces a slime tube, which is moved along over the anterior end by muscular contractions. As it passes, eggs and the sperm received earlier are deposited, and fertilization occurs. The slime tube then forms a cocoon to protect the hatched worms as they develop. There is no larval stage in earthworms.

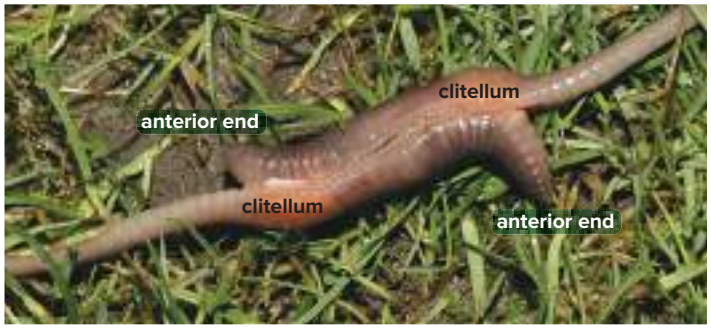


Figure 28.18 Earthworm reproduction. When earthworms mate, they are held in place by a mucus secreted by the clitellum. The worms are hermaphroditic, and when mating, sperm pass from the seminal vesicles of each to the seminal receptacles of the other.

MikeLane45/iStockphoto/Getty Images

Other Annelids Approximately two-thirds of annelids are marine polychaetes. Polychaetes have setae in bundles on parapodia (Gk. *para,* "beside"; *podos,* "foot), which are paddlelike appendages found on most segments. These are used not only in swimming but also as respiratory organs, where the expanded surface area allows for exchange of dissolved gases. Some polychaetes are free-swimming, but the majority live in crevices or burrow into the ocean bottom. Clam worms, such as *Nereis* (Fig. 28.19), are predators. They prey on crustaceans and other small animals, which are captured by a pair of strong, chitinous jaws that extend with a part of the pharynx when the animal is feeding. Associated with its way of life, *Nereis* is cephalized, having a head region with eyes and other sense organs.

Another group of marine polychaetes are sessile tube worms, with *radioles* (ciliated mouth appendages) used to gather food (Fig. 28.20). Christmas tree worms, fan worms, and featherduster worms all have radioles. In featherduster worms, the beautiful radioles cause the animal to look like an old-fashioned feather duster.

Polychaetes have breeding seasons, and only during these times do the worms have sex organs. In *Nereis,* many worms simultaneously shed a portion of their bodies containing either eggs or sperm, and these float to the surface, where fertilization takes place. The zygote rapidly develops into a trochophore larva.

Leeches are annelids that normally live in freshwater habitats. They range in size from less than 2 cm to the medicinal leech, which can be 20 cm in length (Fig. 28.21). They exhibit a variety of patterns and colors but most are brown or olive green. The body of a leech is flattened dorsoventrally. They have the same body plan as other annelids, but they have no setae and each body ring has several transverse grooves.

ISTUDY

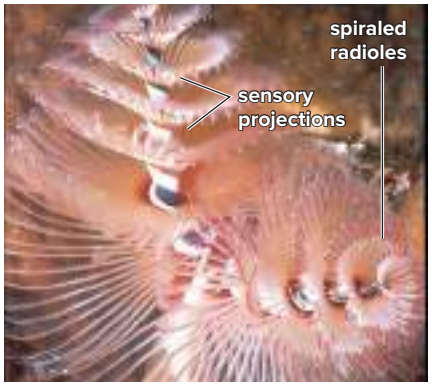


Figure 28.20 Tube worms. Christmas tree worms (a type of tube worm) are sessile feeders whose radioles (ciliated mouth appendages) spiral, as shown here. Diane R. Nelson

Figure 28.21 Leeches. Medical leeches are blood suckers. St. Bartholomew's Hospital/Science Source

While some are free-living, most leeches are fluid feeders that attach themselves to open wounds. Among their modifications are two suckers—a small, oral one around the mouth and a posterior one. Some bloodsuckers, such as the medicinal leech, can cut through tissue. Leeches are able to keep blood flowing by means of hirudin, a powerful anticoagulant in their saliva. Research suggests that leech saliva contains a chemical with local anesthetic properties, which helps explain why the bite of a leech is usually not painful. Medicinal leeches have been used for centuries in blood-letting and other procedures. Today, they are used in reconstructive surgery for severed digits and in plastic surgery.

Flatworms

Flatworms (phylum Platyhelminthes) are trochozoans with an extremely flat body. Like the cnidarians, flatworms have a sac body plan with only one opening, the mouth. Organisms with a single opening are said to have an *incomplete digestive tract,* whereas a *complete digestive tract* has two openings. Flatworms have no body cavity; instead, the third germ layer, mesoderm, fills the space between their organs. Among flatworms, planarians are free-living; flukes and tapeworms are parasitic.

Free-Living Flatworms Planarians are a group of nonparasitic, free-living flatworms. *Dugesia* is a planarian that lives in freshwater lakes, streams, and ponds, where it feeds on small living

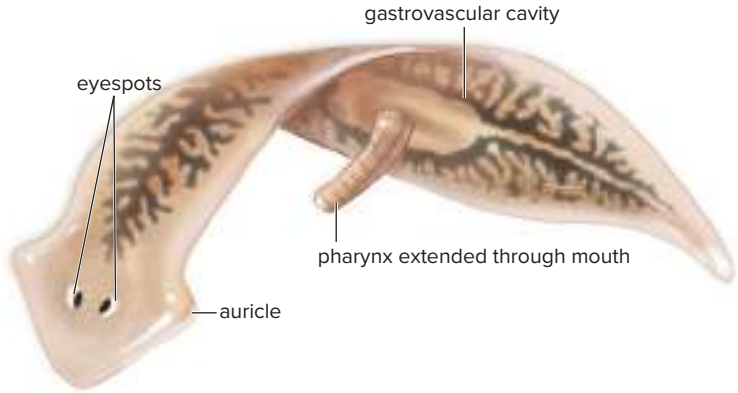


Figure 28.22 Planarian digestive system. When a planarian extends the pharynx, food is sucked up into a gastrovascular cavity, which branches throughout the body.

or dead organisms. A planarian captures food by wrapping itself around the prey, entangling it in slime, and pinning it down. Then, a muscular pharynx is extended through the mouth and a sucking motion takes pieces of the prey into the pharynx. The pharynx leads into a three-branched gastrovascular cavity, in which digestion is both extracellular and intracellular (Fig. 28.22). The digestive system delivers nutrients and oxygen to the cells; the animal has no circulatory system or respiratory system. Waste molecules exit through the mouth.

Planarians have a well-developed excretory system (Fig. 28.23). The excretory organ functions in osmotic regulation, as well as in water excretion. The organ consists of a series of interconnecting canals that run the length of the body on each side. Bulblike structures containing cilia are at the ends of the side branches of the canals. The cilia move back and forth, taking water into the canals that empty at pores. The excretory system often functions as an osmotic-regulating system. The beating of the cilia reminded an early investigator of the flickering of a flame, so the excretory organ of the flatworm is called a flame cell.

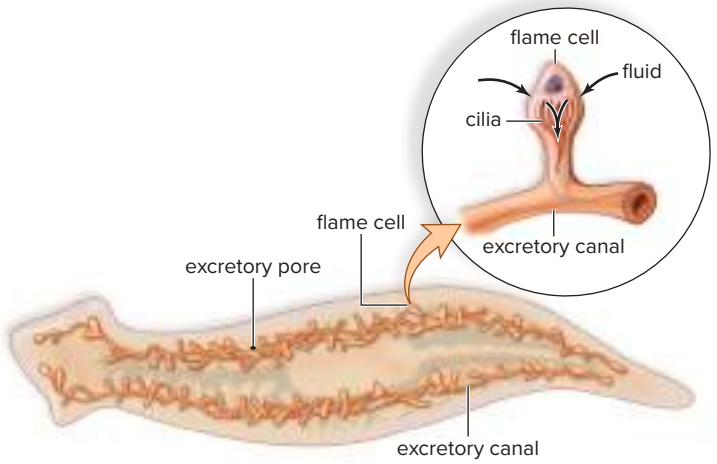


Figure 28.23 Planarian excretory system. The excretory system with flame cells is shown in detail.

ISTUDY

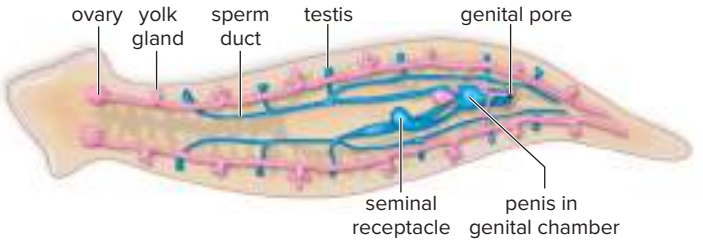


Figure 28.24 Planarian reproductive system. The reproductive system (shown in pink and blue) has both male and female organs.

Planarians usually reproduce sexually. They are monoecious (**hermaphroditic**), which means they possess both male and female sex organs and gametes in a single individual (Fig. 28.24). The worms cross-fertilize when the penis of one is inserted into the genital pore of the other. During this process, each planarian gives and receives sperm. The fertilized eggs are enclosed in a cocoon and hatch in 2 or 3 weeks as tiny worms.

Planarians also can reproduce asexually via regeneration. The tail portion of the planaria breaks off, and each part grows into a new worm. Because planarians have the ability to regenerate, they have been the subject of a field of research called regenerative medicine. Many animals, including humans, do not have the ability to regenerate parts of the body after amputation. The study of how planarians are able to regenerate may lead to advances in regenerative medicine for humans and other animals.

The nervous system of planarians is called *ladder-type* because the two lateral nerve cords plus transverse nerves look like a ladder (Fig. 28.25). Paired **ganglia,** or collections of nerve cells, function as a primitive brain.

Planarians are bilaterally symmetrical and exhibit cephalization. The head of a planarian is bluntly arrow-shaped, with lateral extensions, called auricles, that contain chemosensory cells and tactile cells that they use to detect potential food sources and enemies. They do not have complex eyes but, rather, two pigmented, light-sensitive eyespots on the top of the head (Fig. 28.26).

Planarians have three kinds of muscle layers that allow for quite varied movement: an outer circular layer, an inner longitudinal layer, and a diagonal layer. In larger forms, locomotion is accomplished by the movement of cilia on the ventral and lateral surfaces. Numerous gland cells secrete a mucus, on which the animal glides.

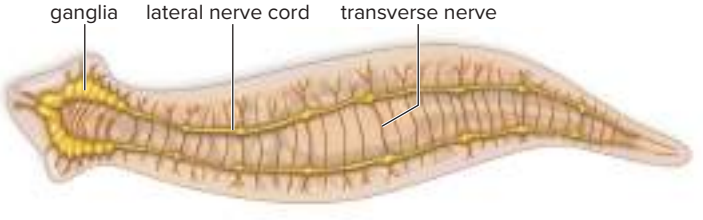


Figure 28.25 Planarian nervous system. The nervous system has a ladderlike appearance.

Figure 28.26 Planarian. The photomicrograph shows that a planarian, *Dugesia,* is bilaterally symmetrical and has a head region with eyespots. M. I. Walker/NHPA/Photoshot/ AVALON

Parasitic Flatworms Flukes (trematodes) and tapeworms (cestodes) are parasitic flatworms. The bodies of both groups are highly evolved for a parasitic mode of life. Flukes and tapeworms feed on nutrients provided by the host and are covered by a protective tegument, which is a specialized body covering that is resistant to host digestive juices.

The parasitic flatworms have lost cephalization, and the head with sensory structures has been replaced by an anterior end with hooks and/or suckers for attachment to the host. They no longer hunt for prey, and the nervous system is not well-developed. In contrast, a well-developed reproductive system helps ensure transmission to a new host.

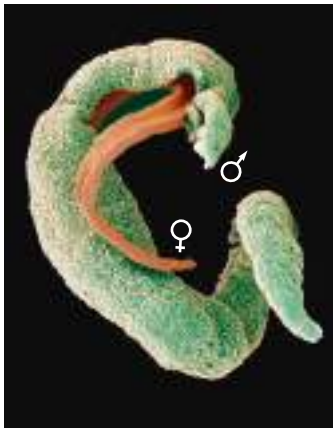
Both flukes and tapeworms utilize a secondary, or intermediate, host to transmit offspring from primary host to primary host. The primary host is infected with the sexually mature adult; the secondary host contains the larval stage or stages. Several human diseases are caused by fluke and tapeworm infections.

Flukes Flukes are named for the organ they inhabit; for example, there are liver, lung, and blood flukes (Fig. 28.27). The almost 11,000 species have an oval to more elongated, flattened body about 2.5 cm long. At the anterior end is an oral sucker surrounded by sensory papillae and at least one other sucker, used for attachment to a host.

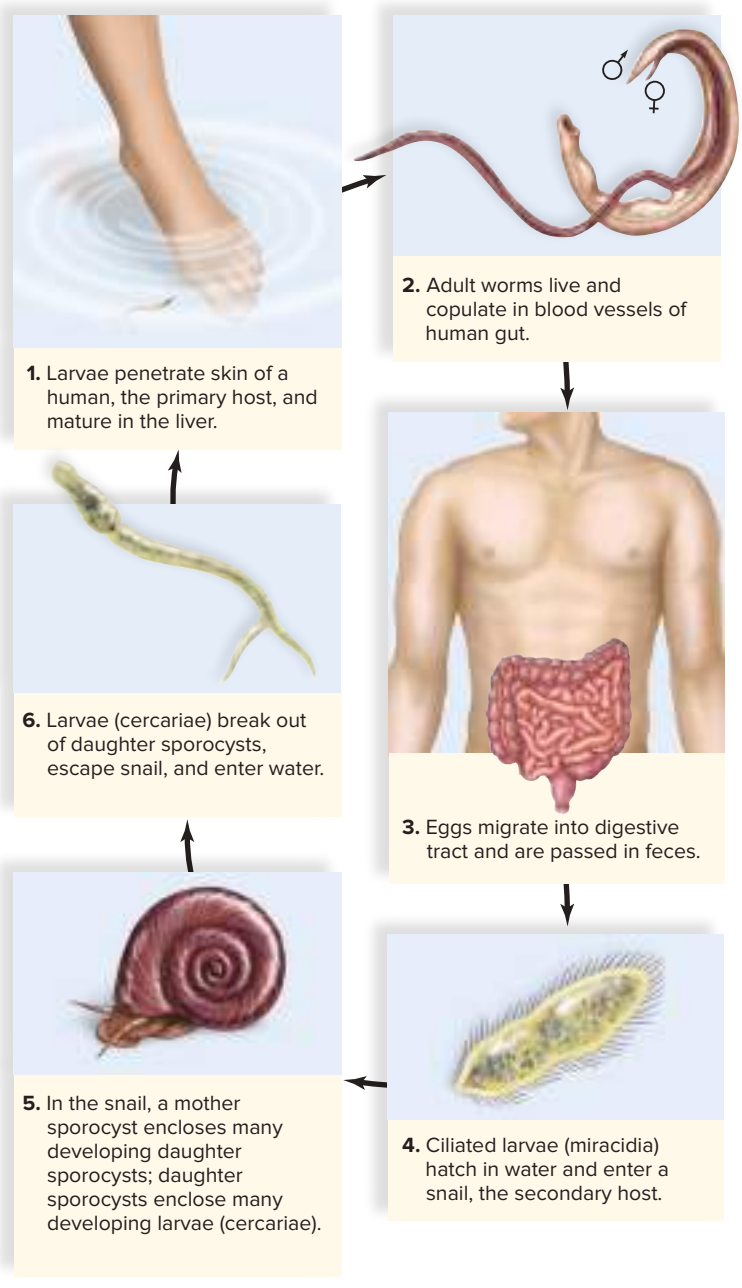
Schistosomiasis is a serious disease caused by a genus of blood fluke, *Schistosoma,* which occurs predominantly in the Middle East, Asia, and Africa. The World Health Organization also lists schistosomiasis as one of several neglected tropical diseases (NTDs) that afflict the people in less-developed nations (see the Biological Systems feature, "African Sleeping Sickness," in Section 21.2). In schistosomiasis, female flukes deposit their eggs in small blood vessels close to the lumen of the human intestine, and the eggs make their way into the digestive tract by a slow migratory process (Fig. 28.27). After the eggs pass out with the feces, they hatch into tiny larvae that swim about in rice paddies and elsewhere until they enter a particular species of snail. Within the snail, asexual reproduction occurs; sporocysts, which are spore-containing sacs, eventually produce new larval forms that leave the snail. If the larvae penetrate the skin of a human, they begin to mature in the liver and implant themselves in the blood vessels of the small intestine.

The flukes and their eggs can cause dysentery, anemia, bladder inflammation, brain damage, and severe liver complications. Infected persons usually die of secondary diseases brought on by

ISTUDY



a.



b.

Figure 28.27 Life cycle of a blood fluke, *Schistosoma*.
a. Micrograph of *Schistosoma*.
b. *Schistosomiasis,* an infection of humans caused by the blood fluke *Schistosoma,* is an extremely prevalent disease in Egypt—especially since the building of the Aswan High Dam. Standing water in irrigation ditches, combined with unsanitary practices, has created the conditions for widespread infection.
(a): JUERGEN BERGER/Science Source

their weakened condition. It is estimated that over 200 million people worldwide are afflicted with this disease.

The Chinese liver fluke, *Clonorchis sinensis,* is a parasite of cats, dogs, pigs, and humans; it requires two secondary hosts: a snail and a fish. The adults reside in the liver and deposit their eggs in bile ducts, which carry them to the intestines for elimination in feces. Nonhuman species generally become infected through the fecal route, but humans usually become infected by eating raw fish. A heavy *Clonorchis* infection can cause severe cirrhosis of the liver and death.

Tapeworms Tapeworms vary in length from a few millimeters to nearly 20 m. They have a highly modified head region called the **scolex,** which contains hooks for attachment to the intestinal wall of the host and suckers for feeding. Tapeworms are hermaphrodites; behind the scolex is a series of reproductive units, called **proglottids,** that contain a full set of female and male sex organs. The number of proglottids may vary depending on the species.

After fertilization, the organs within a proglottid disintegrate and the proglottids become gravid, or full of mature eggs. Gravid proglottids may contain 100,000 eggs. Once mature, the eggs of some species may be released through a pore in the proglottid into the host's intestine, where they exit with feces. In other species, the gravid proglottids break off and are eliminated with the feces. These "segments" can be mistaken for fly maggots—for example, in dog or cat feces—but tapeworm segments are flatter and appear in fresh samples, whereas fly eggs take up to a day to hatch.

Most tapeworms have complicated life cycles, which usually involve several hosts. Figure 28.28 illustrates the life cycle of the pork tapeworm, *Taenia solium,* which has the human as the primary host and the pig as the secondary host. After a pig feeds on feces-contaminated food, the larvae are released. They burrow through the intestinal wall and travel in the bloodstream to finally lodge and encyst in muscle. This **cyst** is a small, hard-walled structure that contains a larva called a bladder worm. Humans become infected with tapeworms when they eat infected meat that has not been thoroughly cooked.

In impoverished nations of the world, more than 50 million people suffer from cysticercosis, an advanced tapeworm infection that interferes with the uptake of nutrients and is therefore particularly serious for growing children. After tainted meat is eaten, the bladder worms break out of the cysts, attach themselves to the intestinal wall, and grow to adulthood. Then the cycle begins again. Generally, tapeworm infections cause diarrhea, weight loss, and fatigue in the primary host.

Rotifers **Rotifers** (phylum Rotifera) are trochozoans related to the flatworms. The seventeenth-century inventor of the microscope, Antonie van Leeuwenhoek, viewed microscopic rotifers and called them the "wheel animalcules." Rotifers have a crown of cilia, known as the corona, on their heads (Fig. 28.29). When in motion, the corona, which looks like a spinning wheel, serves as an organ of locomotion and directs food into the mouth.

ISTUDY

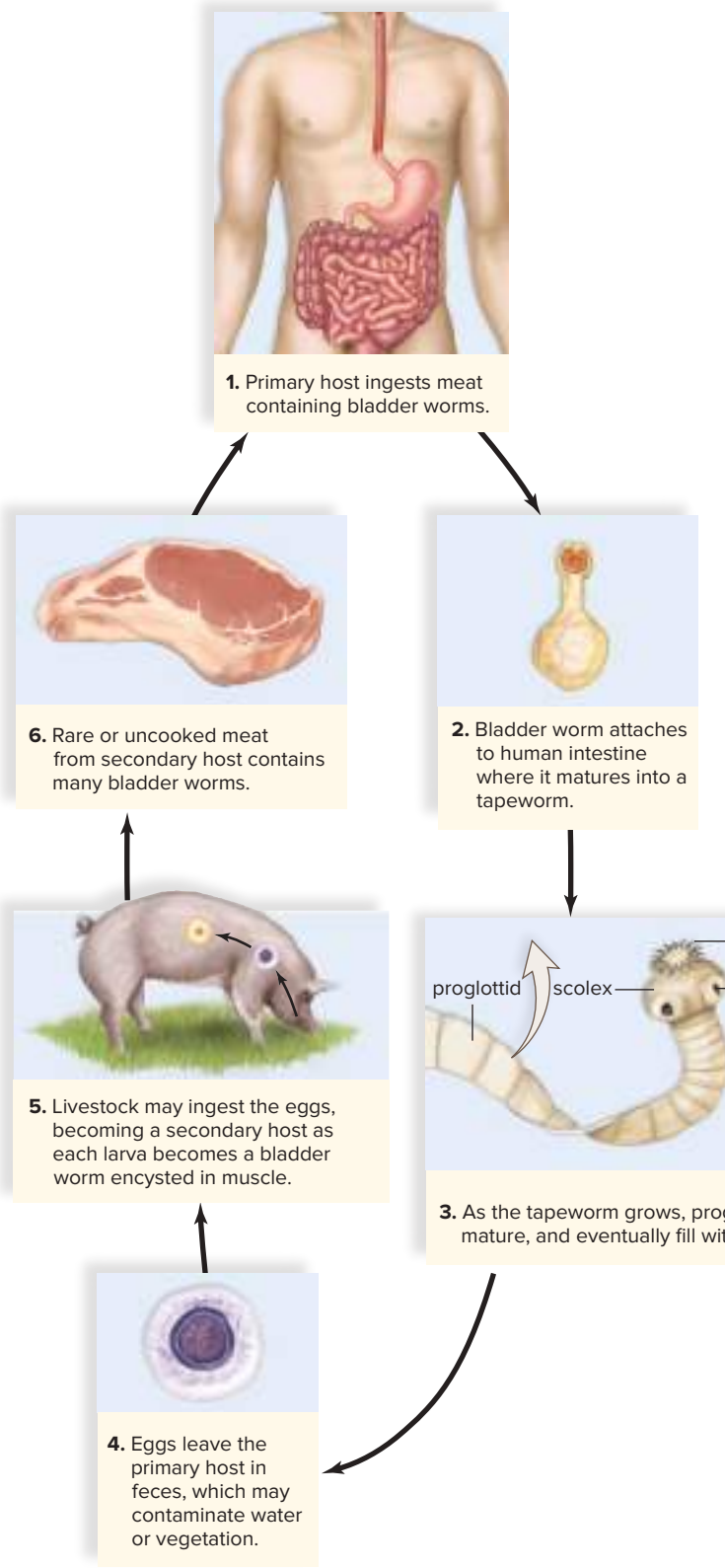


Figure 28.28 Life cycle of a tapeworm, *Taenia*. The life cycle includes a human (primary host) and a pig (secondary host). The adult worm is modified for its parasitic way of life. It consists of a scolex and many proglottids, which become bags of eggs.

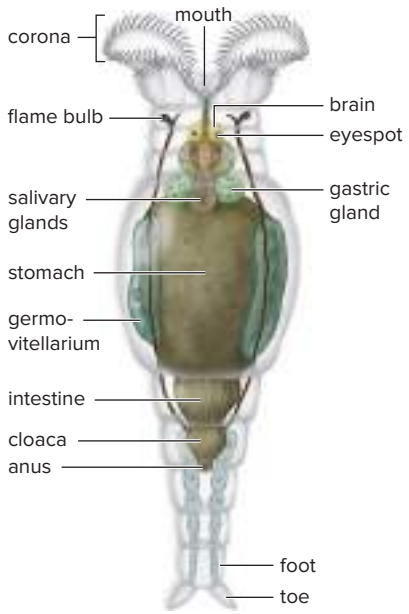


Figure 28.29 Rotifer. Rotifers are microscopic animals only 0.1–3 mm in length. The beating of cilia on two lobes at the anterior end of the animal gives the impression of a pair of spinning wheels.

The approximately 2,000 species primarily live in fresh water; however, some marine and terrestrial forms exist. The majority of rotifers are transparent, but some are very colorful. Many species of rotifers can desiccate during harsh conditions and remain dormant for lengthy periods of time.

Check Your Progress 28.3

1. List the characteristics similar in flatworms, mollusks, and annelids.
2. Compare the features of the flatworm, mollusk, and annelid body cavity, digestive tract, and circulatory system.
3. Describe the life cycle of two trochozoan parasites.
4. List three phyla that feed with lophophores.

28.4 Diversity of the Ecdysozoans

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the characteristics unique to ecdysozoans.
2. Compare the anatomical features of roundworms and arthropods.
3. Describe five characteristics responsible for the success of the arthropods.

The **ecdysozoans** (Gk. *ecdysis,* "stripping off") are protostomes and include roundworms and arthropods. Compared to many of the lophotrochozoans, ecdysozoans do not undergo spiral cleavage. Ecdysozoans construct an outer covering called a cuticle, which protects

ISTUDY

and supports the animal and is periodically shed to allow growth. Unlike many other invertebrate species that reproduce by releasing large amounts of eggs and sperm into their aqueous environment, many ecdysozoans have evolved separate sexes, which come together and copulate to deliver sperm into the female body, or to directly fertilize eggs as they are released. They are one of the major groups of animals, which also contains the largest number of species.

Roundworms

Roundworms (phylum Nematoda) are nonsegmented worms that are prevalent in almost any environment. Around 61,000 species have been identified, but a million species may exist. Nematodes have been characterized as the most numerous multicellular organisms on Earth, mainly because they are so abundant in soil. These worms range in size from microscopic to over 1 m in length. Internal organs, including the tubular reproductive organs, lie within the pseudocoelom. A **pseudocoelom** (Gk. *Pseudes,* "false"; *koiloma,* "cavity") is a body cavity that is incompletely lined by mesoderm. In other words, mesoderm occurs inside the body wall but not around the digestive cavity (gut).

Although many nematodes are free-living, others are important parasites of plants and animals. One free-living species, *Caenorhabditis elegans,* has been used extensively as a model animal to study genetics and developmental biology.

Parasitic Roundworms

The CDC estimates that at any one time, up to a billion people are infected with nematodes of the genus *Ascaris.* Other *Ascaris* species can infect domestic animals and a number of other vertebrates. Though relatively rare in the United States, ascariasis is common in tropical countries, especially in areas with poor sanitation.

As in other nematodes, *Ascaris* males tend to be smaller (15–31 cm long) than females (20–49 cm long) (Fig. 28.30*a*). A typical female *Ascaris* produces over 200,000 eggs daily. The eggs are eliminated with the host's feces and can remain viable in the soil for many months. Eggs enter the human body via uncooked vegetables, soiled fingers, or ingested fecal material and hatch in the intestines. The juveniles make their way into the veins and lymphatic vessels and are carried to the heart and lungs. From the lungs, the larvae travel up the trachea, where they are swallowed and eventually reach the intestines. There, the larvae mature and begin feeding on intestinal contents.

The symptoms of an *Ascaris* infection depend on the stage of the infection. Most infected people have surprisingly few symptoms, but larval *Ascaris* in the lungs can cause coughing or bloody sputum. In the intestines, *Ascaris* can cause malnutrition, blockage of the bile and pancreatic ducts, and poor health.

Trichinosis, caused by the nematode *Trichinella spiralis,* is a serious infection that humans can contract when they eat rare pork containing encysted *Trichinella* larvae. After maturation, the female adult burrows into the wall of the small intestine and produces living offspring, which are carried by the bloodstream to the skeletal muscles, where they encyst (Fig. 28.30*b*). Heavy infections can be painful and lethal.

Filarial worms, another type of roundworm, cause various diseases. For example, mosquitoes can transmit the larvae of a parasitic filarial worm to dogs. Because the worms live in the heart and the arteries that serve the lungs, the infection is called *heartworm disease.* The condition can be fatal; therefore, heartworm medicine is recommended as a preventive measure for all dogs.

Elephantiasis is a disease of humans caused by another filarial worm, *Wuchereria bancrofti.* Restricted to tropical areas of Africa, this parasite uses mosquitoes as a vector to transmit the disease to humans. Because the adult filarial worms reside in lymphatic vessels, the removal of lymph fluid is impeded, and the limbs of an infected person may swell to a monstrous size, resulting in elephantiasis (Fig. 28.30*c*). Elephantiasis is treatable in its early stages but usually not after scar tissue has blocked lymphatic vessels. The World Health Organization lists elephantiasis as one of the 16 neglected tropical diseases (NTDs) that affect impoverished

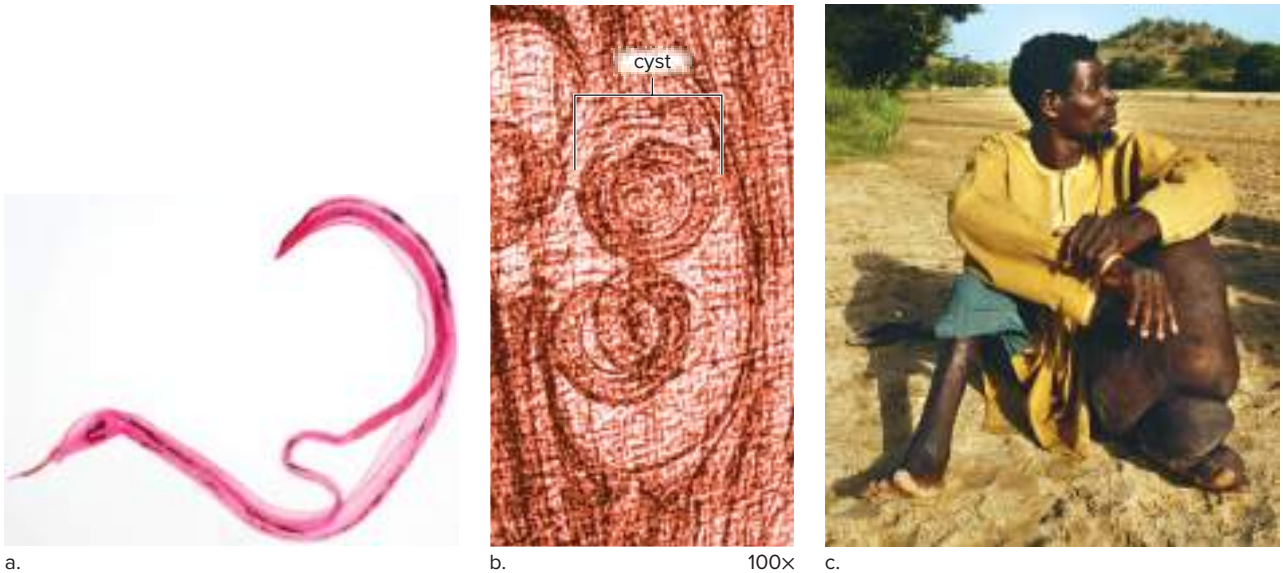
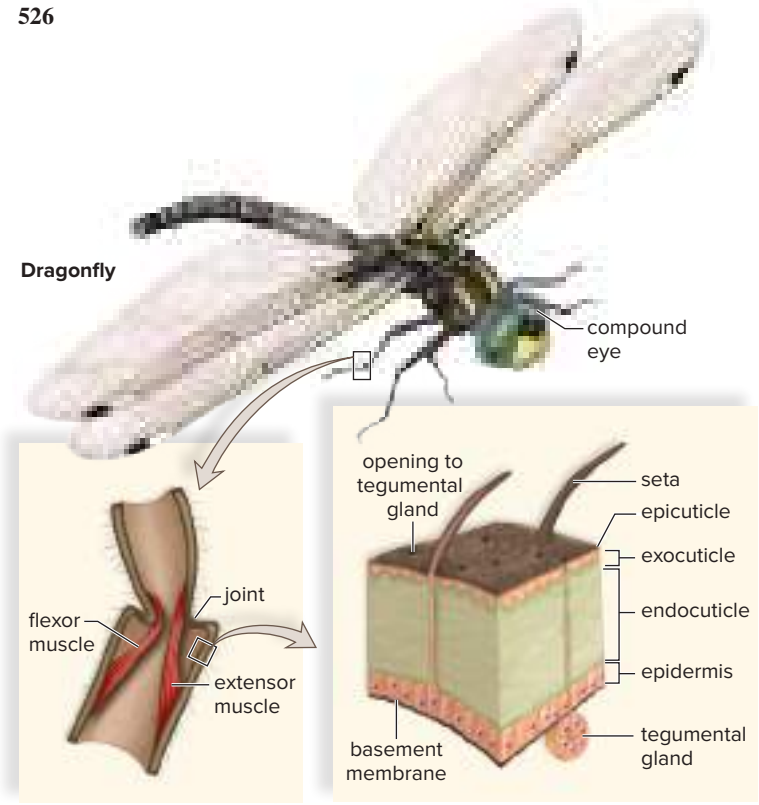


Figure 28.30 Roundworm diversity. **a.** *Ascaris,* a common cause of a roundworm infection in humans. **b.** Encysted *Trichinella* larva in muscle. **c.** A filarial worm infection causes elephantiasis, which is characterized by a swollen body part when the worms block lymphatic vessels. (a): Choksawatdikorn/Shutterstock; (b): Centers for Disease Control and Prevention; (c): Vanessa Vick/The New York Times/Redux Pictures

ISTUDY



a. Joint movement b. Exoskeleton composition

c. Molting

Figure 28.31 Arthropod skeleton and molting. **a.** The joint in an arthropod skeleton is a region where the cuticle is thinner and not as hard as the rest of the cuticle. The direction of movement is toward the flexor muscle or the extensor muscle, whichever one has contracted. **b.** The exoskeleton is secreted by the epidermis and consists of the endocuticle; the exocuticle, hardened by the deposition of calcium carbonate; and the epicuticle, a waxy layer. Chitin makes up the bulk of the exo- and endocuticles. **c.** Because the exoskeleton is nonliving, it must be shed, through a process called molting, for the arthropod to grow. (c): TommyIX/iStockphoto/Getty Images

nations. It is estimated that 120 million people in subtropical and tropical areas of the world have filariasis.

Pinworms are the most common nematode parasite in the United States, occurring most commonly in children. The adult parasites live in the cecum and large intestine. Females migrate to the anal region at night and lay their eggs. Scratching the resultant itch can contaminate hands, clothes, and bedding. The eggs are swallowed, and the life cycle begins again. Fortunately, once diagnosed (usually by microscopic examination of sticky tape used to collect eggs from the skin around the anus), the infection is easily cured.

Arthropods

The **arthropods** (phylum Arthropoda [Gk. *arthron,* "joint"; *podos,* "foot"]) are a very large group of protostomes that have exoskeletons and jointed appendages. The phylum Arthropoda includes insects, crustaceans, millipedes, centipedes, spiders, and scorpions. The parasitic mite measures less than 0.1 mm in length, while the Japanese crab measures up to 4 m in length. Arthropods, which also occupy every type of habitat, are considered the most successful group of all the animals.

Characteristics of Arthropods

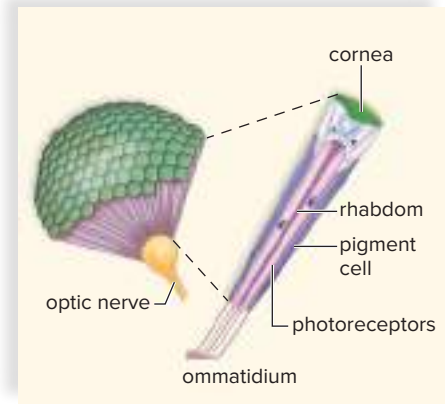
The remarkable success of arthropods is dependent on five characteristics: an exoskeleton, segmentation, a well-developed nervous system, a variety of respiratory systems, and a life cycle that includes metamorphosis.

Exoskeleton Arthropods feature a rigid, jointed **exoskeleton** (Fig. 28.31). The exoskeleton is composed primarily of chitin, a strong, flexible, nitrogenous polysaccharide. The exoskeleton serves many functions, including protection, attachment for muscles, locomotion, and prevention of desiccation. However, because it is hard and nonexpandable, arthropods must molt, or shed, the exoskeleton as they grow larger. Arthropods have this in common with other ecdysozoans.

Before molting, the body secretes a new, larger exoskeleton, which is soft and wrinkled, underneath the old one. After enzymes partially dissolve and weaken the old exoskeleton, the animal breaks it open and wriggles out. The new exoskeleton then quickly expands and hardens (Fig. 28.31c).

Segmentation Segmentation is readily apparent, because each segment has a pair of jointed appendages, even though certain segments are fused into a head, a thorax, and an abdomen. The jointed appendages of arthropods are basically hollow tubes moved by muscles. Typically, the appendages are highly adapted for a particular function, such as food gathering, reproduction, or locomotion. In addition, many appendages are associated with sensory structures and used for tactile purposes.

Nervous System Arthropods have a brain and a ventral nerve cord. The head bears various types of sense organs, including eyes of two types—simple and compound. The compound eye is composed of many complete visual units, each of which operates independently (Fig. 28.32). The lens of each visual unit focuses an image on the light-sensitive membranes of a small number of photoreceptors within that unit. The simple eye, like that of vertebrates, has a single lens that brings the image to focus onto many receptors, each of which receives only a portion of the image.



Compound eye

Figure 28.32 Arthropod eye. Arthropods have a compound eye that contains many individual units, each with its own lens and photoreceptors.

ISTUDY

In addition to sight, many arthropods have well-developed touch, smell, taste, balance, and hearing. Arthropods display many complex behaviors and methods of communication.

Variety of Respiratory Organs Marine forms use gills, which are vascularized, highly convoluted, thin-walled tissue specialized for gas exchange. Terrestrial forms have book lungs (e.g., spiders) or air tubes called tracheae (L. *trachea,* "windpipe"). Tracheae serve as a rapid way to transport oxygen directly to the cells.

Metamorphosis Many arthropods undergo a drastic change in form and physiology that occurs as an immature stage, or larva, becomes an adult. This change is termed **metamorphosis** (Gk. *meta,* "implying change"; *morphe,* "shape, form"). Among arthropods, the larva eats different food and lives in a different environment than does the adult. For example, larval crabs live among and feed on plankton, while adult crabs are bottom dwellers that catch live prey or scavenge dead organic matter. Among insects, such as butterflies, the caterpillar feeds on leafy vegetation, while the adult feeds on nectar. The result is reduced competition for resources, increasing the animal's overall fitness.

While keeping these five arthropod features in mind, let's look at representatives of this large and varied phylum.

Crustaceans

The name **crustacean** is derived from their hard, crusty exoskeleton, which contains calcium carbonate in addition to the typical chitin. Although crustaceans are extremely diverse, the head usually bears a pair of compound eyes and five pairs of appendages. The first two pairs, called antennae and antennules, lie in front of the mouth and have sensory functions. The other three pairs (mandibles and first and second maxillae) lie behind the mouth and are mouthparts used in feeding. Biramous (Gk. *bi*, "two") appendages on the thorax and abdomen are segmentally arranged; one branch is the gill branch, and the other is the leg branch.

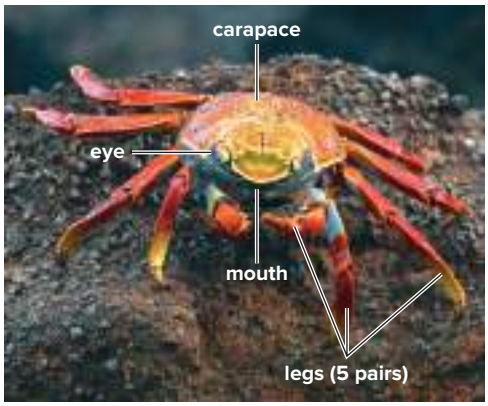
The majority of crustaceans live in marine and aquatic environments (Fig. 28.33). Decapods, which are the most familiar and numerous crustaceans, include lobsters, crabs, crayfish, hermit crabs, and shrimp. These animals have a thorax that bears five pairs of walking appendages. Typically, the gills are positioned above the walking legs. The first pair of walking legs may be modified as claws.

Copepods and krill are small, free-swimming crustaceans that live in the water, where they feed on algae. In the marine environment, they serve as food for fishes, sharks, and whales. They are so numerous that, despite their small size, some believe they are harvestable as food. Barnacles are sessile crustaceans with a thick, heavy, protective shell (Fig. 28.33c). Barnacles can live on wharf pilings, ship hulls, seaside rocks, and even the bodies of whales. They begin life as free-swimming larvae, but they undergo a metamorphosis that transforms their swimming appendages to cirri, feathery structures that are extended and allow them to filter feed when they are submerged.

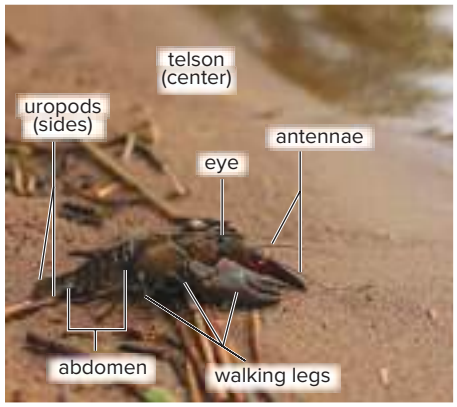
A Typical Crustacean: Crayfish Figure 28.34*a* gives a view of the external anatomy of the crayfish. The head and thorax are fused into a *cephalothorax* (Gk. *kephale,* "head"; *thorax,* "breastplate"), which is covered on the top and sides by a nonsegmented carapace. The abdominal segments are equipped with swimmerets, small, paddlelike structures. The first two pairs of swimmerets in the male, known as claspers, are quite strong and are used to pass sperm to the female. The last two segments bear the uropods and the telson, which make up a fan-shaped tail.

Ordinarily, a crayfish lies in wait for prey. It faces out from an enclosed spot with the claws extended, the antennae moving about. The claws seize any small animal, either dead or living, that happens by and carry it to the mouth. When a crayfish moves about, it generally crawls slowly, but it may swim rapidly by using its heavy abdominal muscles and tail.

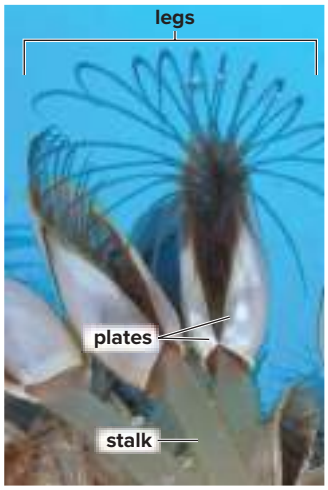
The respiratory system consists of gills that lie above the walking legs, protected by the carapace. As shown in Figure 28.34*b,* the digestive system includes a stomach, which is divided into two main regions: an anterior portion called the gastric mill, equipped with chitinous teeth to grind coarse food; and a posterior region, which acts as a filter to prevent coarse particles from entering the digestive glands, where absorption takes place.



a. Sally lightfoot crab, *Grapsus grapsus*



b. European crayfish, *Actacus fluviatilis*



c. Gooseneck barnacles, *Lepas anatifera*

Figure 28.33 Crustacean diversity. Crabs (**a**) and crayfish (**b**) are decapods—they have five pairs of walking legs. Crabs have a reduced abdomen compared to crayfish. The gooseneck barnacle (**c**) is attached to an object by a long stalk. (a): Michael Lustbader/Science Source; (b): drakuliren/iStock/Getty Images; (c): L. Newman & A. Flowers/Science Source

ISTUDY

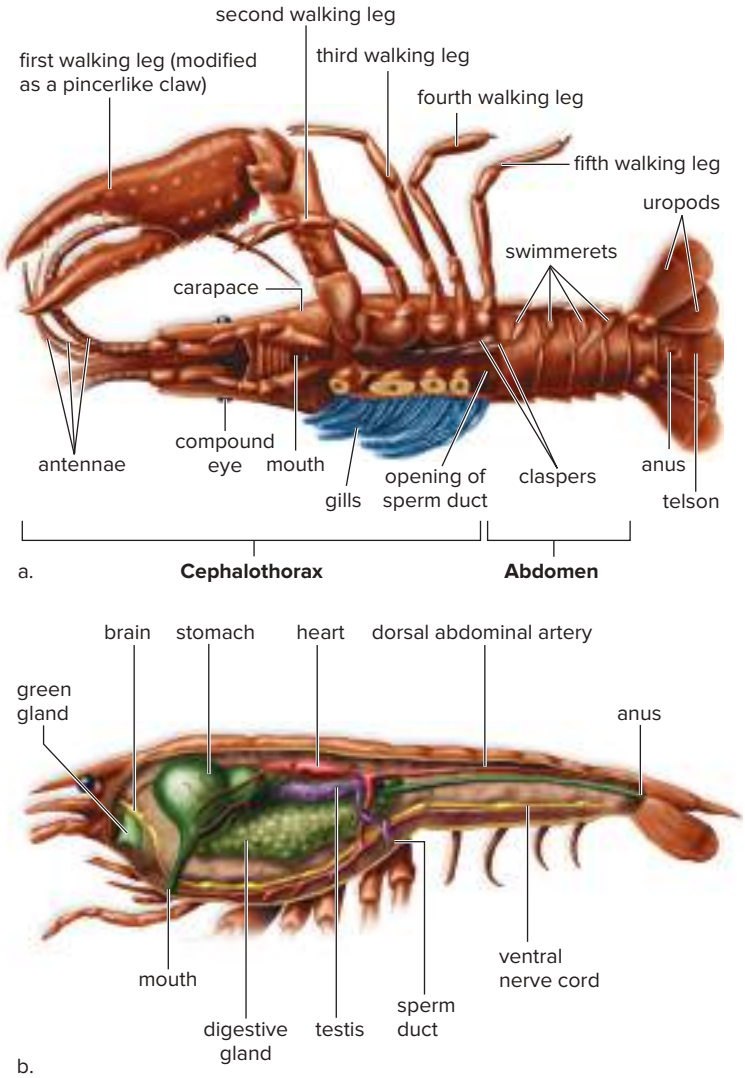


Figure 28.34 Male crayfish, *Cambarus*. **a.** Externally, it is possible to observe the jointed appendages, including the swimmerets, and the walking legs, which include the claws. These appendages, plus a portion of the carapace, have been removed from the right side so the gills are visible. **b.** Internally, the parts of the digestive system are particularly visible. The circulatory system can also be clearly seen. Note the ventral nerve cord.

Green glands lying in the head region, anterior to the esophagus, excrete metabolic wastes through a duct that opens externally at the base of the antennae. The coelom, which is well-developed in the annelids, is reduced in the arthropods and is composed chiefly of the space about the reproductive system. A heart pumps hemolymph containing the blue respiratory pigment hemocyanin into a *hemocoel* consisting of sinuses (open spaces), where the hemolymph flows about the organs. As in the mollusks, this is an open circulatory system because blood is not contained within blood vessels.

The crayfish nervous system is well-developed. Crayfish have a brain and a ventral nerve cord that passes posteriorly. Along the length of the nerve cord, periodic ganglia give off lateral nerves. The crayfish's sensory organs are also well-developed, with compound eyes on the ends of movable eyestalks. These eyes are accurate and can detect motion and respond to polarized light. Other sensory organs include tactile antennae and chemosensitive setae. Crayfish also have statocysts, which serve as organs of equilibrium.

The sexes are separate in the crayfish, and the gonads are located just ventral to the pericardial cavity. In the male, a coiled sperm duct opens to the outside at the base of the fifth walking leg. Sperm transfer is accomplished by the modified first two swimmerets of the abdomen. In the female, the ovaries open at the bases of the third walking legs. A stiff fold between the bases of the fourth and fifth pairs serves as a seminal receptacle. Following fertilization, the eggs are attached to the swimmerets of the female. Young hatchlings are miniature adults, and no metamorphosis occurs.

Centipedes and Millipedes

Centipedes and millipedes are known for their many legs. In **centipedes** ("hundred-leggers"), each of their many body segments has a pair of walking legs (Fig. 28.35*a*). The approximately 3,000 species prefer to live in moist environments, such as under logs, in crevices, and in leaf litter, where they are active predators on worms, small crustaceans, and insects. The head of a centipede includes paired antennae and jawlike mandibles. Appendages on the first trunk segment are clawlike, venomous jaws that kill or immobilize prey, while mandibles chew.

In **millipedes** ("thousand-leggers"), each of four thoracic segments bears one pair of legs (Fig. 28.35*b*), while abdominal segments have two pairs of legs. Millipedes live under stones or burrow in the soil as they feed on leaf litter. Their cylindrical bodies have a tough, chitinous exoskeleton. Some secrete hydrogen cyanide, a poisonous substance.

Insects

Insects are adapted for an active life on land, although some have secondarily invaded aquatic habitats. The body of an insect is divided into a head, a thorax, and an abdomen. The head bears the sense organs and mouthparts (Fig. 28.36). The thorax has three pairs of legs and possibly one or two pairs of wings; and the abdomen contains most of the internal organs. Wings enhance an insect's ability to survive by providing a way of escaping enemies, finding food, facilitating mating, and dispersing offspring.

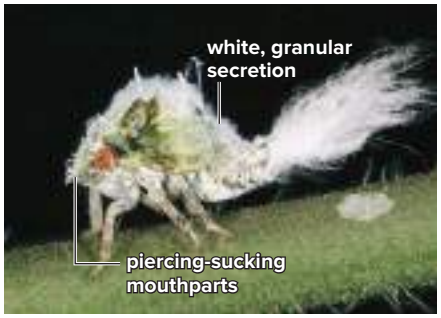
Many insects, such as butterflies, undergo *complete metamorphosis,* involving drastic changes in form. At first, the animal is a wormlike larva (caterpillar) with chewing mouthparts. It then forms

a.

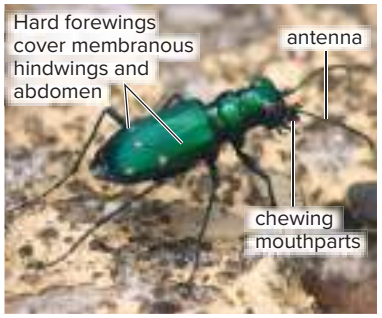
b.

Figure 28.35 Centipede and millipede. **a.** A centipede has a pair of appendages on almost every segment. **b.** A millipede has two pairs of legs on most segments. (a): Larry Miller/Science Source; (b): Tim Zurowski/All Canada Photos/Alamy Stock Photo

ISTUDY

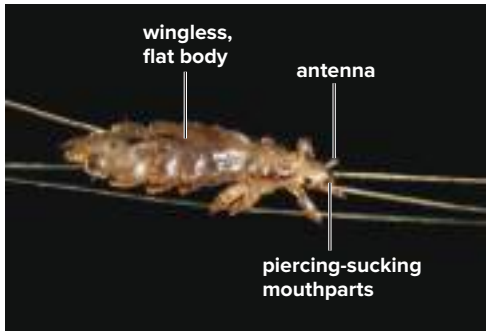


Mealybug, order Homoptera

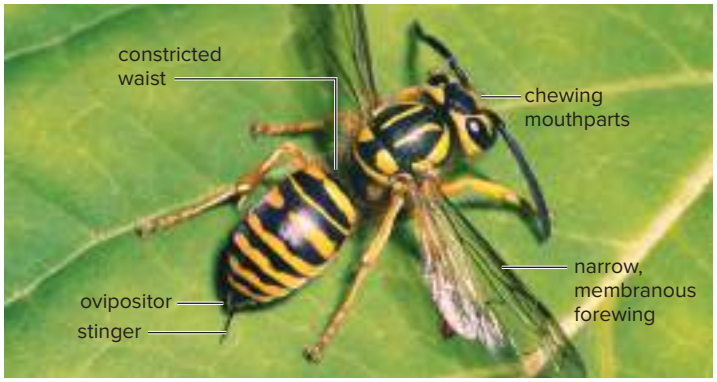


Beetle, order Coleoptera

Leafhopper, order Homoptera



Head louse, order Anoplura



Wasp, order Hymenoptera

Dragonfly, order Odonata

Figure 28.36 Insect diversity. Insects are among the most diverse life-forms on the planet. (mealybug): ©Farley Bridges; (beetle): Tacojim/iStock/Getty Images; (leafhopper): ©Farley Bridges; (louse): blickwinkel/Hecker/Alamy Stock Photo; (wasp): James H. Robinson/Science Source; (dragonfly): ©Farley Bridges

a case, or cocoon, about itself and becomes a pupa. During this stage, the body parts are completely reorganized; the adult then emerges from the cocoon. As mentioned earlier, this life cycle allows the larvae and adults to use different food sources.

Insects show remarkable behavioral adaptations. Bees, wasps, ants, termites, and other colonial insects have complex societies. Insects themselves also serve as an important food source for a variety of other animals (and certain carnivorous plants). The Biological Systems feature, "Would You Eat Insects?," describes entomophagy, or the consumption of insects by humans, which is a common practice in many parts of the world.

A Typical Insect: Grasshopper In the grasshopper (Fig. 28.37), the third pair of legs is suited to jumping. This insect has two pairs of wings. The forewings are tough and leathery, and when folded back at rest, they protect the broad, thin hindwings.

On each lateral surface, the first abdominal segment bears a large tympanum for the reception of sound waves. The posterior region of the exoskeleton in the female has an ovipositor, used to dig a hole in which eggs are laid.

The digestive system is suitable for an herbivorous diet. In the mouth, food is broken down mechanically by mouthparts and enzymatically by salivary secretions. Food is temporarily stored in the crop before passing into the gizzard, where it is finely ground. Digestion is completed in the stomach, and nutrients are absorbed into the hemocoel from outpockets called gastric ceca (*sing.,* cecum). A cecum is a cavity that is open at one end only.

The excretory system consists of *Malpighian tubules,* which extend into a hemocoel and collect nitrogenous wastes, which are concentrated and excreted into the digestive tract. The formation of a solid nitrogenous waste—uric acid—conserves water.

The respiratory system begins with openings in the exoskeleton called spiracles. From here, air enters small tubules called **tracheae** (Fig. 28.37*a*). The tracheae branch and rebranch, finally ending in moist areas where the actual exchange of gases takes place. No individual cell is very far from a site of gas exchange. The movement of air through this complex of tubules is not a passive process; air is pumped through by a series of bladderlike structures (air sacs) attached to the tracheae near the spiracles. Air enters the anterior four spiracles and exits by the posterior six spiracles. Breathing by tracheae may account for the small size of insects (most are less than 6 cm in length), because the tracheae are so tiny and fragile that they would be crushed by any amount of weight.

The circulatory system contains a slender, tubular heart, which lies against the dorsal wall of the abdominal exoskeleton and pumps hemolymph into the hemocoel, where it circulates before returning to the heart again. The hemolymph lacks a respiratory pigment; therefore, it is colorless and transports mainly nutrients and wastes. The highly efficient tracheal system transports respiratory gases.

Grasshoppers undergo *incomplete metamorphosis,* a gradual change in form as the animal matures. The immature grasshopper,

ISTUDY

THEME Biological Systems

Would You Eat Insects?

An urban legend claims that the average person accidentally swallows eight spiders a year while sleeping. No matter how often this myth is repeated, however, there have been no studies performed to confirm it, and, in fact, it is unlikely that most of us ever swallow a spider, whether we're sleeping or awake. However, we do accidentally consume a much larger quantity of insects each year than we might imagine.

Despite our use of pesticides and even genetically resistant crops, insects are commonly present in our food. In its *Defect Levels Handbook,* the U.S. Food and Drug Administration has set some limits on how many insect parts are allowable:

- Broccoli, frozen—60 insects or mites per 100 g
- Chocolate—60 insect fragments per 100 g
- Hops (to make beer)—2,500 aphids per 10 g
- Macaroni—225 insect fragments per 225 grams
- Mushrooms—20 maggots per 100 g
- Peanut butter—30 insect fragments per 100 g

Most of these limits are justified for "aesthetic" reasons, meaning anything more than that might be noticeable enough to gross out the average consumer, but there are no health consequences associated with the consumption of insects. In fact, it's been estimated that each of us probably ingests 1 or 2 pounds of insect parts each year, without ever noticing.

Sometimes, the use of insect products is intentional. Humans have widely accepted the use of honey as a sweetener for thousands of years, even though you might hesitate if someone at the health food store offered to sweeten your tea with a sweet, sticky, yellowish-brown substance made from fluid that bees regurgitate after feeding on nectar (also known as honey)!

As the human population increases to a predicted 9 billion by 2050, global food demand is expected to double, and it is unlikely that conventional agricultural methods will be able to meet this need. In particular, the ways we produce our dietary protein have a huge impact, as more people worldwide increase their consumption of beef, pork, and chicken. The livestock industry currently uses 70% of available agricultural land and accounts for 20% of greenhouse gas production. The United Nations Food and Agriculture Organization recommends considering the potential of insects as a logical food source for the rapidly growing human population.

Figure 28D Using insects for food. Insects being offered for human consumption in Bangkok, Thailand. BSIP/Newscom

Although insects remain decidedly unpopular in the United States and most European countries, over 2 billion people include insects as a part of their regular diet (Fig. 28D). Out of about 1.3 million identified species of insects, approximately 1,900 species are considered edible. Beetles are most commonly consumed, followed by caterpillars, bees, wasps, and ants. Compared to cattle, insects produce nine times as much protein for the same amount of feed. Insects are also lower in fat and a great source of vitamins, minerals, and essential fatty acids. They require less energy to farm and can be cultivated at a much higher density than conventional livestock, and most people don't have the ethical concerns about the treatment of insects that many feel about raising conventional livestock in concentrated animal feeding operations ("factory farms"). Finally, insects are generally resistant to the types of microbial pathogens (e.g., *E. coli, Salmonella*) that seem to be an increasing problem with intensively raised conventional livestock.

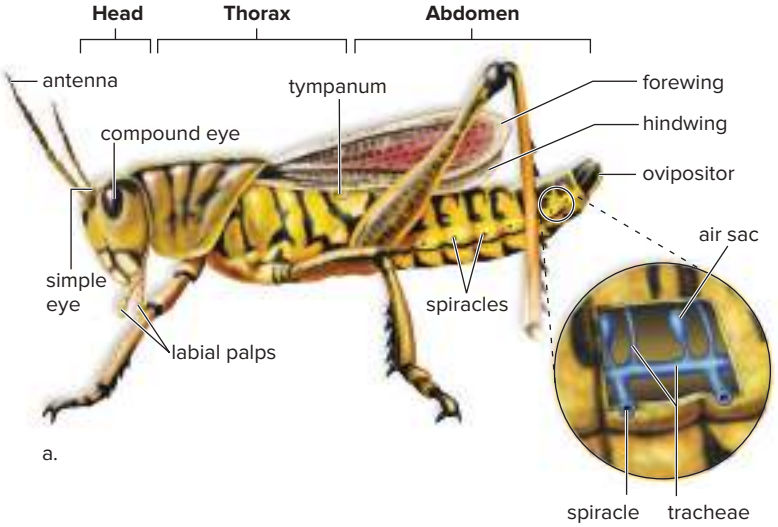
A variety of companies are planning strategies to take advantage of the likely increase in entomophagy, or consumption of insects, in our near future. A London-based company called Entosense has an online marketplace where insect food items and stylish insect-based products are available. Bug Muscle targets bodybuilders and other athletes as the main consumers of its grasshopper- and cricket-based nutritional supplements; Hotlix has produced insect-containing candies for 25 years; and Tiny Farms will sell you kits to begin growing your own edible bugs at home.

You might also be surprised to learn that gourmet dishes featuring insects are already on the menus of some acclaimed restaurants: a grasshopper taco at Oyamel in Washington, DC, or Singapore-style scorpions at Typhoon in Santa Monica. You can also order insects for home cooking and snacking from a variety of online vendors. The next time you are offered an insect-based food (and it is likely you will be), why not give it a try?

Questions to Consider

1. Many people find the idea of eating insects to be disgusting. However, what one person experiences as disgusting may not have the slightest effect on someone else. How can you explain this?
2. Our reaction to certain "disgusting" things seems to be innate—we are born with it. Can you think of why our tendency to avoid certain "disgusting" stimuli might be biologically based?

ISTUDY



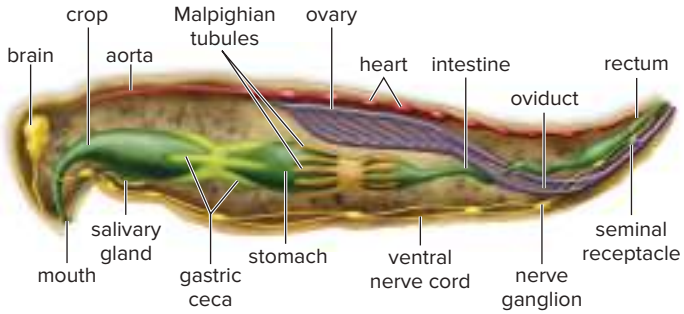


Figure 28.37 Female grasshopper, *Romalea*. **a.** Externally, the body of a grasshopper is divided into three sections and has three pairs of legs. The tympanum receives sound waves, and the jumping legs and the wings are for locomotion. **b.** Internally, the digestive system is specialized. The Malpighian tubules excrete a solid, nitrogenous waste (uric acid). A seminal receptacle receives sperm from the male, which has a penis.

called a nymph, is recognizable as a grasshopper, even though it differs in body proportions from the adult.

Chelicerates

Chelicerates live in terrestrial, aquatic, and marine environments. The first pair of appendages is the pincerlike chelicerae, used in feeding and defense. The second pair is the pedipalps, which can have various functions. A cephalothorax (fused head and thorax) is followed by an abdomen that contains internal organs.

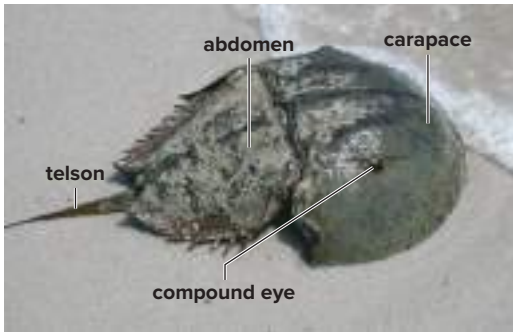
Horseshoe crabs of the genus *Limulus* are familiar along the East Coast of North America (Fig. 28.38*a*). The body is covered by exoskeletal shields. The anterior shield is a horseshoe-shaped carapace, which bears two prominent, compound eyes. Ticks, mites, scorpions, spiders, and harvestmen are all arachnids. Over 25,000 species of mites and ticks have been classified, some of which are parasitic on a variety of other animals. Ticks are ectoparasites of various vertebrates, and they are carriers for such diseases as Rocky Mountain spotted fever and Lyme disease. When not attached to a host, ticks hide on plants and in the soil.

Scorpions can be found on all continents except Antarctica (Fig. 28.38*b*). North America is home to approximately 1,500 species. Scorpions are nocturnal and spend most of the day hidden under a log or a rock. Their pedipalps are large pincers, and their long abdomen ends with a stinger, which contains venom.

Currently, over 35,000 species of spiders have been classified (Fig. 28.38*c*). Spiders, the most familiar chelicerates, have a narrow waist that separates the cephalothorax from the abdomen. Spiders do not have compound eyes; instead, they have numerous simple eyes that perform a similar function. The chelicerae are modified as fangs, with ducts from poison glands, and the pedipalps are used to hold, taste, and chew food. The abdomen often contains silk glands, and spiders spin a web in which to trap their prey.

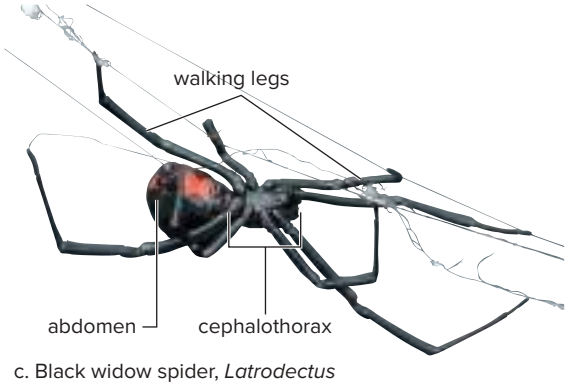
Check Your Progress 28.4

1. Name two ways in which roundworms are anatomically similar to arthropods.
2. List two ways that crustaceans are adapted to an aquatic life and insects are adapted to living on land.
3. Describe the features chelicerates have in common.



a. Horseshoe crab, *Limulus*

b. Kenyan giant scorpion, *Pandinus*



c. Black widow spider, *Latrodectus*

Figure 28.38 Chelicerate diversity. **a.** Horseshoe crabs are common along the East Coast. **b.** Scorpions are more common in tropical and desert areas. **c.** The black widow spider is a venomous spider that spins a web.
(a): Robert Pos/U.S. Fish & Wildlife Service; (b): Tom McHugh/Science Source; (c): Mark Kostich/E+/Getty Images

ISTUDY

28.5 Invertebrate Deuterostomes

Learning Outcomes

Upon completion of this section, you should be able to

1. List the two major groups of animals in the Deuterostomia.
2. Identify the major morphological structures of the sea star, an echinoderm.
3. Describe how sea stars move, feed, and reproduce.

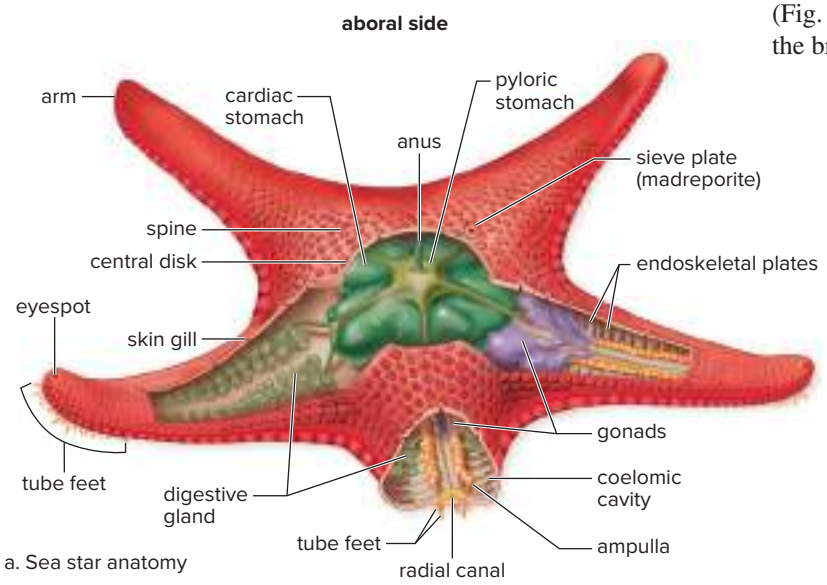
Deuterostomes include invertebrate echinoderms, such as sea stars (also known as starfish) and chordates. Molecular data tell us that echinoderms and chordates are closely related. Morphological data indicate that these two groups share the deuterostome pattern of development (see Fig. 28.5). The echinoderms and a few chordates are invertebrates; most of the chordates are vertebrates (see Section 29.1).

Echinoderms

Echinoderms (phylum Echinodermata [Gk. *echinos,* "spiny"; *derma,* "skin"]) are primarily bottom-dwelling marine animals. They range in size from brittle stars less than 1 cm in length to giant sea cucumbers over 2 m long (Fig. 28.39*c*).

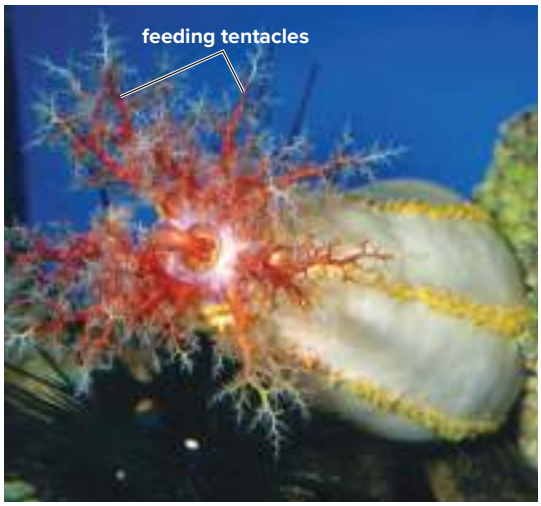
The most striking feature of echinoderms is their five-pointed radial symmetry, as illustrated by a sea star (Fig. 28.39*b*). Although echinoderms are radially symmetrical as adults, their larvae are free-swimming filter feeders with bilateral symmetry. Echinoderms have an endoskeleton of spiny, calcium-rich plates called ossicles (Fig. 28.39*a*). Another innovation of echinoderms is their unique **water vascular system** consisting of canals and appendages that function in locomotion, feeding, gas exchange, and sensory reception.

The more familiar of the echinoderms are the Asteroidea, containing the sea stars (Fig. 28.39*a, b*); the Holothuroidea, including the sea apples, which have long, leathery bodies (Fig. 28.39*c*); and the Echinoidea, including the sea urchin and sand dollar, both of which use their spines for locomotion, defense, and burrowing (Fig. 28.39*d*). Less familiar are the Ophiuroidea, which includes the brittle stars, with a central disk surrounded by radially flexible



a. Sea star anatomy

b. Rainbow star, *Orthasterias koehleri*



c. Sea apple, *Pseudocolochirus*

d. Purple sea urchin, *Strongylocentrotus*

Figure 28.39 Echinoderms. **a.** Sea star (starfish) anatomy. Like other echinoderms, sea stars have a water vascular system that begins with the sieve plate and ends with expandable tube feet. **b.** The rainbow star (*Orthasterias*) uses the suction of its tube feet to open a clam, a primary source of food. **c.** Sea apple (*Pseudocolochirus violaceus*). **d.** Sea urchin (*Strongylocentrotus*). (b): Mark Conlin/Alamy Stock Photo; (c): Roberto Nistri/Alamy Stock Photo; (d): Robert Marien/Fuse/Getty Images

ISTUDY

arms; and the Crinoidea, the oldest group, which includes the stalked feather stars and the motile feather stars.

A Typical Echinoderm: Sea Star

Sea stars number about 1,600 species commonly found along rocky coasts, where they feed on clams, oysters, and other bivalve mollusks. Various structures project through the body wall: (1) Spines from the endoskeletal plates offer some protection; (2) pincerlike structures around the bases of spines keep the surface free of small particles; and (3) skin gills, tiny fingerlike extensions of the skin, are used for respiration. On the oral surface, each arm has a groove lined by little *tube feet* (Fig. 28.39*a*).

To feed, a sea star positions itself over a bivalve and attaches some of its tube feet to each side of the shell. By working its tube feet in alternation, it gradually pulls the shell open. A very small crack is enough for the sea star to evert its cardiac stomach and push it through the crack, so it contacts the soft parts of the bivalve. The stomach secretes enzymes, and digestion begins, even while the bivalve is attempting to close its shell. Later, partly digested food is taken into the sea star's body, where digestion continues in the pyloric stomach using enzymes from the digestive glands found in each arm. A short intestine opens at the anus on the aboral side (side opposite the mouth).

In each arm, the well-developed coelom contains a pair of digestive glands and gonads (either male or female) that open on the aboral surface by very small pores. The nervous system consists of a central nerve ring, which gives off radial nerves into each arm. A light-sensitive eyespot is at the tip of each arm.

Locomotion depends on the water vascular system. Water enters this system through a structure on the aboral side called the *sieve plate,* or *madreporite* (Fig. 28.39*a*). From there, it passes through a stone canal to a ring canal, which circles around the central disc, and then to a radial canal in each arm. From the radial canals, many lateral canals extend into the tube feet, each of which has an ampulla. Contraction of an ampulla forces water into the tube foot, expanding it. When the foot touches a surface, the center is withdrawn, giving it suction so that it can adhere to the surface. By alternating the expansion and contraction of the tube feet, a sea star moves slowly along.

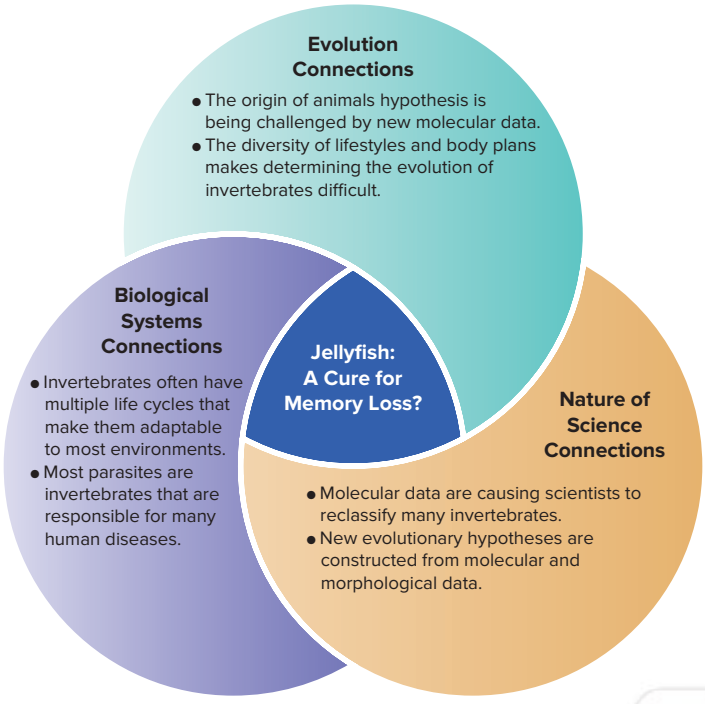
Echinoderms do not have a respiratory, excretory, or circulatory system. Fluids within the coelomic cavity and the water vascular system carry out many of these functions. For example, gas exchange occurs across the skin gills and the tube feet. Nitrogenous wastes diffuse through the coelomic fluid and the body wall. Cilia on the coelom and other structures keep the coelomic fluid moving.

Sea stars reproduce asexually and sexually. If the body is fragmented, each fragment can regenerate a whole animal, as long as a portion of the central disc is present. Sea stars spawn and release either eggs or sperm at the same time. The bilaterally symmetrical larvae undergo metamorphosis to become radially symmetrical adults.

Check Your Progress 28.5

1. Explain why echinoderms and chordates are now considered to be closely related.
2. Delineate the evidence that supports the evolution of echinoderms from bilaterally symmetrical animals.
3. Describe the location and function of skin gills, tube feet, and the stomach.
4. Explain the functions of the water vascular system in sea stars.

Connecting *the* Concepts



ISTUDY

What Did You Learn?

28.1 Evolution of Animals

Animals are multicellular organisms that are heterotrophic and ingest their food. Typically, they are diploid, they move by means of contracting fibers, and they are able to reproduce sexually. Animals can be **vertebrates,** which have a backbone, or **invertebrates,** which do not.

- The colonial flagellate hypothesis proposes that animals evolved from a protist that resembled the **choanoflagellates** of today. The traditional evolutionary tree of invertebrate animals was based only on morphological characters. A modern phylogeny constructed with molecular and morphological data presents an alternative evolutionary history.
- The first developmental event after fertilization is **cleavage,** or cell division without growth. Next, a hollow ball of cells called a **blastula** appears, forming a body cavity that becomes the **coelom.** In animals with a **true coelom,** mesoderm completely lines the coelom cavity.
- Most animals have a particular **symmetry,** or pattern, to their body shape, although a few exhibit **asymmetry.** Animal bodies that are organized in a circular fashion have a **radial symmetry;** those with definite right and left halves have **bilateral symmetry.** Many body plans include **cephalization,** or location of a brain and specialized sensory organs at the anterior end.
- All animals except sponges, cnidarians, and comb jellies are triploblastic, meaning their tissues develop from three embryonic **germ layers.** Triploblastic animals are either **deuterostomes** or **protostomes,** which differ in the development of the coelom and whether the blastopore gives rise to the mouth (protostomes) or the anus (deuterostomes).

28.2 The Simplest Invertebrates

- **Sponges,** are believed to be the oldest lineage of animals. They have a cellular level of organization, lack tissues, and are asymmetrical. Sponges are sessile filter feeders that depend on a flow of water through the body to acquire food. Sponges can be classified by their type of spicules, which are small, needle-shaped structural components of the endoskeleton.
- **Comb jellies** (ctenophores) and cnidarians are diploblastic. They are aquatic and radially symmetrical, with bands of cilia that propel them through the water. The body of a comb jelly is composed largely of jellylike **mesoglea.**
- **Cnidarians** have a sac body plan. They have an internal **gastrovascular cavity** and a **hydrostatic skeleton.** They exist as either a **polyp** or a **medusa,** or they can alternate between the two. Hydras and their relatives—sea anemones and corals—are polyps; in jellyfish, the medusan stage is most common. Hydras and other cnidarians possess tentacles to capture prey and cnidocytes armed with **nematocysts** to stun it. A **nerve net** coordinates movements.

28.3 Diversity of the Lophotrochozoans

There are two major types of **Lophotrochozoa. Lophophorans,** which include the bryozoans, brachiopods, and phoronids, feed with a feathery lophophore that filters microorganisms from water. **Trochozoans** have a trocophore larva, either in living forms or in their ancestors.

- The body of a **mollusk** typically contains a visceral mass, a **mantle,** and a foot. Many also have a head and a radula for feeding. The nervous system consists of several ganglia connected by nerve cords. They have a reduced coelom and an open circulatory system with hemolymph that flows into a **hemocoel.** Clams **(bivalves)** are adapted to a sedentary coastal life, snails **(gastropods)** to life on land, and squids **(cephalopods)** to life in the sea.
- **Annelids** are worms that exhibit **segmentation.** They have a well-developed coelom that acts as a hydrostatic skeleton, a closed circulatory system, a ventral solid nerve cord, **setae** for movement, and paired **nephridia.** Earthworms are oligochaetes ("few bristles") that use the body wall for gas exchange. Polychaetes ("many bristles") are marine worms that may be predators, with a definite head region, or filter feeders, with ciliated tentacles to filter food from the water. Leeches also belong to this phylum.
- Free-living **flatworms** (planarians) have three tissue layers and no coelom. They are **hermaphroditic,** and usually reproduce sexually. Planarians have muscles and a ladder-type nervous system with paired **ganglia** serving as the brain. They take in food through an extended pharynx, leading to a gastrovascular cavity that extends through the body. An osmotic-regulating organ contains flame cells.
- Flukes and tapeworms are common parasites of vertebrate animals. Flukes have two suckers by which they attach to and feed from their hosts. Tapeworms have a **scolex** with hooks and suckers for attaching to the host's intestinal wall. The body of a tapeworm is made up of **proglottids,** which, when mature, contain thousands of eggs. If these eggs are taken up by pigs or cattle, for example, larvae become encysted in their muscles. If humans eat this meat, they, too, may become infested with a tapeworm. Tapeworm infection is a serious concern in impoverished nations.
- **Rotifers** are microscopic and aquatic and have a corona that resembles a spinning wheel when in motion.

28.4 Diversity of the Ecdysozoans

Ecdysozoans are protostomes that produce a cuticle, which is periodically shed.

- **Roundworms,** which have a **pseudocoelom,** are usually small and very diverse; they are present almost everywhere in great numbers. Many have adapted to a parasitic lifestyle; examples in humans include *Ascaris, Trichinella,* and filarial worms. These diseases are more common in developing tropical and subtropical regions where sanitation is poor.
- Many biologists consider **arthropods** to be the most successful and varied group of animals. Their success is largely attributable to a flexible **exoskeleton,** specialized body regions, and jointed appendages. Also important are a high degree of cephalization and a variety of respiratory organs. Many undergo some form of **metamorphosis.** The hard exoskeleton requires a molt at different stages of growth. Crustaceans, insects, and chelicerates are representative groups of arthropods.
- **Crustaceans** (crayfish, lobsters, shrimps, copepods, krill, and barnacles) have a head that bears compound eyes, antennae, antennules, and mouthparts. Crayfish have an open circulatory system, respiration by gills, and a ventral solid nerve cord. **Centipedes** and **millipedes** are known for their many legs.
- **Insects** include butterflies, grasshoppers, bees, and beetles. The anatomy of the grasshopper is adapted to life on land. Like other insects, grasshoppers have wings and three pairs of legs attached to the thorax. They have a tympanum for sound reception, a digestive system specialized for a grass diet, Malpighian tubules for excretion of solid nitrogenous waste, **tracheae** for respiration, internal fertilization, and incomplete metamorphosis.

ISTUDY

- **Chelicerates** (horseshoe crabs, spiders, scorpions, ticks, and mites) have chelicerae, pedipalps, and four pairs of walking legs attached to a cephalothorax.

28.5 Invertebrate Deuterostomes

Echinoderms (sea stars, sea urchins, and sea cucumbers) share the deuterostome pattern of development with the chordates. They are radially symmetrical as adults but not as larvae, and they have internal, calcium-rich plates with spines. Typical of echinoderms, **sea stars** have tiny skin gills, a central nerve ring with branches, and a **water vascular system** for locomotion. Each arm of a sea star contains branches from the nervous, digestive, and reproductive systems.

Thinking Critically About Biology

1. What advantages exist for flowering plants and insect pollinators that coevolved?
2. Multicellular animals are thought to have evolved from single-celled protists resembling the choanoflagellates that exist today. Describe some specific advantages to multicellularity that gave animals an advantage over their single-celled relatives.
3. Cnidarians such as *Hydra* have only one opening, which must serve as both mouth and anus. In contrast, most animal species have a one-way, tubelike digestive system, with separate openings for intake and output. Describe some specific advantages of the latter arrangement.
4. Many species of invertebrates (e.g., flatworms, roundworms, arthropods) are parasites of other animals. List several advantages, as well as challenges, inherent in a parasitic lifestyle.
5. It was recently discovered that the filarial worm *Wuchereria bancrofti,* which causes elephantiasis, itself carries bacterial symbionts of the genus *Wolbachia.* Studies have suggested that the bacteria are essential for reproduction and survival of the worm. What does this suggest regarding potential treatments for elephantiasis?

Making It Relevant to You

1. What other jellyfish features could provide benefits to humans?
2. If your parent began suffering from memory loss, would you feel comfortable using Prevagen despite it not being scientifically tested?
3. What other organisms do you think hold potential cures for human disorders and diseases?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

29

Vertebrate Evolution

Terry Kreeger, Wyoming Game and Fish and Chronic Wasting Disease Alliance (USGS).

Chapter Outline

29.1 The Chordates
29.2 The Vertebrates
29.3 The Fishes
29.4 The Amphibians
29.5 The Reptiles
29.6 The Mammals

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 15.3 How do *Archaeopteryx* and similar transitional fossils support the close relationship between birds and dinosaurs?

Table 18.1 When in the history of life on Earth did the vertebrates first appear?

Figure 21.1 Which eukaryotes are the closest relatives of vertebrate animals?

Zombie Deer

Recently, fall hunters have reported sightings of zombie deer. Zombie symptoms include significant weight loss, stumbling, and drooling. The increase in zombie deer is due to the spreading of a prion-based disease known as chronic wasting disease (CWD). CWD is known to infect white-tailed deer, mule deer, elk, and moose, as well as other members of the deer family. Cows, sheep, goats, and a variety of other mammals are also subject to prion-based diseases. The common thread among all of these mammals is that the misfolded proteins cause rapid degeneration of the neural system leading to death.

Deer are considered one of the most successful groups of mammal vertebrates. Their evolutionary origin can be traced back to the Eocene epoch when most modern mammals appeared and diversified. Significant climate swings during the ice ages led to a rapid evolution and differentiation of deer, resulting in a large number of species and subspecies.

More research needs to be done to gain a greater understanding of the relatedness among members of the deer family (cervids). This will help us know how to best control and possibly prevent the spread of zombie deer.

As you read through this chapter, think about the following questions:

1. When did the first vertebrate ancestor evolve?
2. What characteristics set vertebrates apart from other animals?

Following *the* Themes

CHAPTER 29 – VERTEBRATE EVOLUTION

Evolution	Vertebrates share a common chordate ancestor that had a set of basic characteristics, including a nerve cord.
Nature of Science	The study of vertebrate evolution is key to solving important issues related to Earth's changing environment and other aspects of human health and welfare.
Biological Systems	Vertebrates live in a wide variety of habitats, where they contribute to the structure of the ecosystem.

ISTUDY

29.1 The Chordates

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the four basic characteristics of a chordate.
2. Name the two groups of nonvertebrate chordates.
3. Describe two features of each of the two groups of nonvertebrate chordates.

Chordates (phylum Chordata), like the echinoderms (discussed in Section 28.5), are deuterostomes. In contrast to the exoskeleton of many invertebrates, however, most chordates have an internal skeleton made of bone and/or cartilage, to which the muscles are attached. This arrangement allows the chordates to enjoy a greater freedom of movement and enables many to attain a larger body size than invertebrates.

Most chordates are vertebrates, meaning they have a vertebral column, and most of this chapter is devoted to this very familiar group of animals. First, though, it is important to recognize the nonvertebrate chordates, some of which likely resemble the ancestors of all vertebrate animals, including humans.

Characteristics of Chordates

All chordates have four basic characteristics that are present during some point of their development (Fig. 29.1). Although all of these characteristics were likely present in the adult stage of the common ancestor of chordates, not all of the traits are present in the adults of modern chordates. However, the presence of these characteristics during the development of all chordates indicates their evolutionary ties to a single common ancestor. These four characteristics are:

- *Notochord* (Gk. *notos,* "back"; *chorde,* "string"). A dorsal supporting rod, the **notochord,** is located just below the nerve cord. The majority of vertebrates have an embryonic notochord that is replaced by the vertebral column during embryonic development.
- *Dorsal tubular nerve cord.* In contrast to the arthropods, which have a ventral nerve cord, chordates have a tubular cord situated dorsally. The anterior portion becomes the brain in most chordates. In vertebrates, the nerve cord, often called the spinal cord, is protected by vertebrae.
- *Pharyngeal pouches.* These are seen only during embryonic development in most vertebrates. In the nonvertebrate chordates, the fishes, and amphibian larvae, the pharyngeal pouches become functioning **gills** (respiratory organs). Water passing into the mouth and the pharynx goes through the gill slits, which are supported by gill arches. In terrestrial vertebrates, the pouches are modified for various purposes. For example, in humans the first pair of pouches become the auditory tubes.
- *Postanal tail.* A tail—in the embryo, if not in the adult—extends beyond the anus. In some chordates, including humans, the tail disappears during embryonic development.

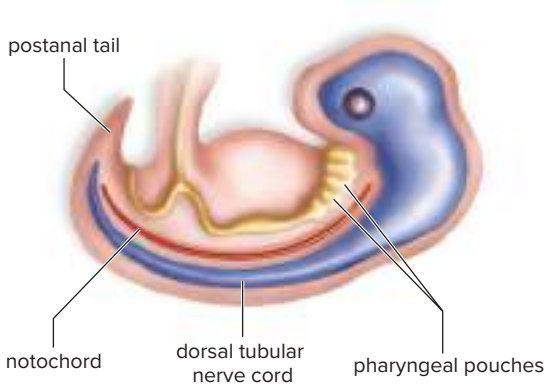
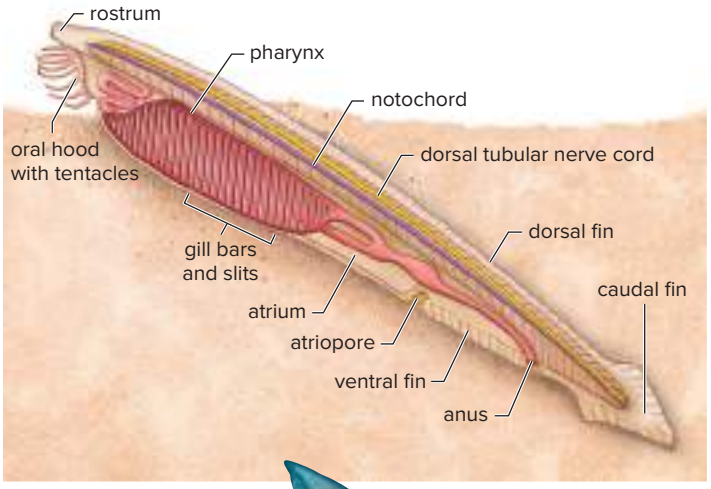


Figure 29.1 Characteristics of the chordates. All chordates have four distinctive traits that are present at some stage of their life cycle.



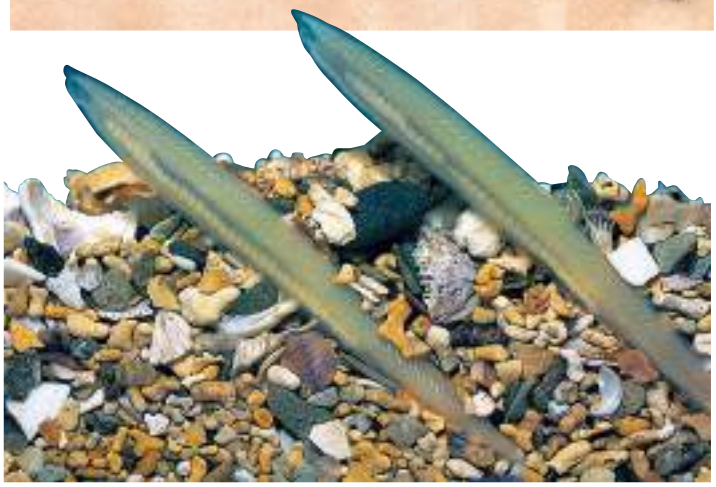


Figure 29.2 Lancelets, *Branchiostoma.* Lancelets are filter feeders. Water enters the mouth and exits at the atriopore after passing through the gill slits. (photo): Heather Angel/Natural Visions

Nonvertebrate Chordates

The nonvertebrate chordates do not have a spine made of bony vertebrae. They are divided into two groups: the cephalochordates and the urochordates.

Lancelets (genus *Branchiostoma,* previously called *Amphioxus*) are **cephalochordates** (Gk. *kephalo,* "head"). These marine chordates, which are only a few centimeters long, are named for their resemblance to a lancet—a small, two-edged surgical knife (Fig. 29.2). Lancelets are found in the shallow water along most coasts, where they usually lie partly buried in the sandy or muddy bottom with only their mouth and gill apparatus exposed. They feed on microscopic particles, which they filter out of a constant stream of water that enters the mouth and passes through the gill slits into a chamber, or *atrium,* before exiting through an opening called an *atriopore.*

ISTUDY

Lancelet adults possess all four general chordate characteristics; therefore, they are important in comparative anatomy and evolutionary studies. In lancelets, the notochord extends from the head to the tail. In addition, segmentation is present, as illustrated by the fact that the muscles are segmentally arranged, and the dorsal tubular nerve cord has periodic branches. Segmentation may not be an important feature in lancelets, but in other vertebrates it leads to specialization of parts, as we observed in the annelids and arthropods in Section 28.4.

Sea squirts, or **urochordates,** live on the ocean floor. They are also called tunicates, because adults have a tunic (outer covering) that makes them look like thick-walled, squat sacs. Sea squirt larvae are bilaterally symmetrical and have all four chordate characteristics. Metamorphosis produces the sessile adult with incurrent and excurrent siphons (Fig. 29.3). When disturbed, they often squirt water from their excurrent siphon, a habit that is reflected in their name. The pharynx is lined by numerous cilia, whose beating creates a current of water that moves into the pharynx and out the numerous gill slits, the only chordate characteristic that remains in the adult.

Many evolutionary biologists hypothesize that the sea squirts are directly related to the vertebrates. It has been suggested that a larva with the four chordate characteristics may have become sexually mature without developing the other adult sea squirt characteristics. Over time, it may have evolved into a fishlike vertebrate. Figure 29.4 shows how the main groups of chordates may have evolved.

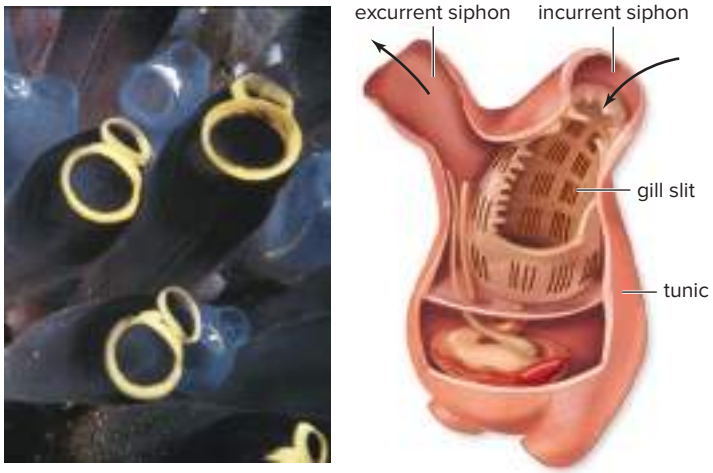


Figure 29.3 Sea squirts, *Clavelina*. Note that the only chordate characteristic remaining in the adult is gill slits. (photo): Diane Nelson

Check Your Progress 29.1

1. Discuss how humans, as chordates, possess all four characteristics either as embryos or adults.
2. Explain why adult sea squirts are classified as chordates, despite their appearance.

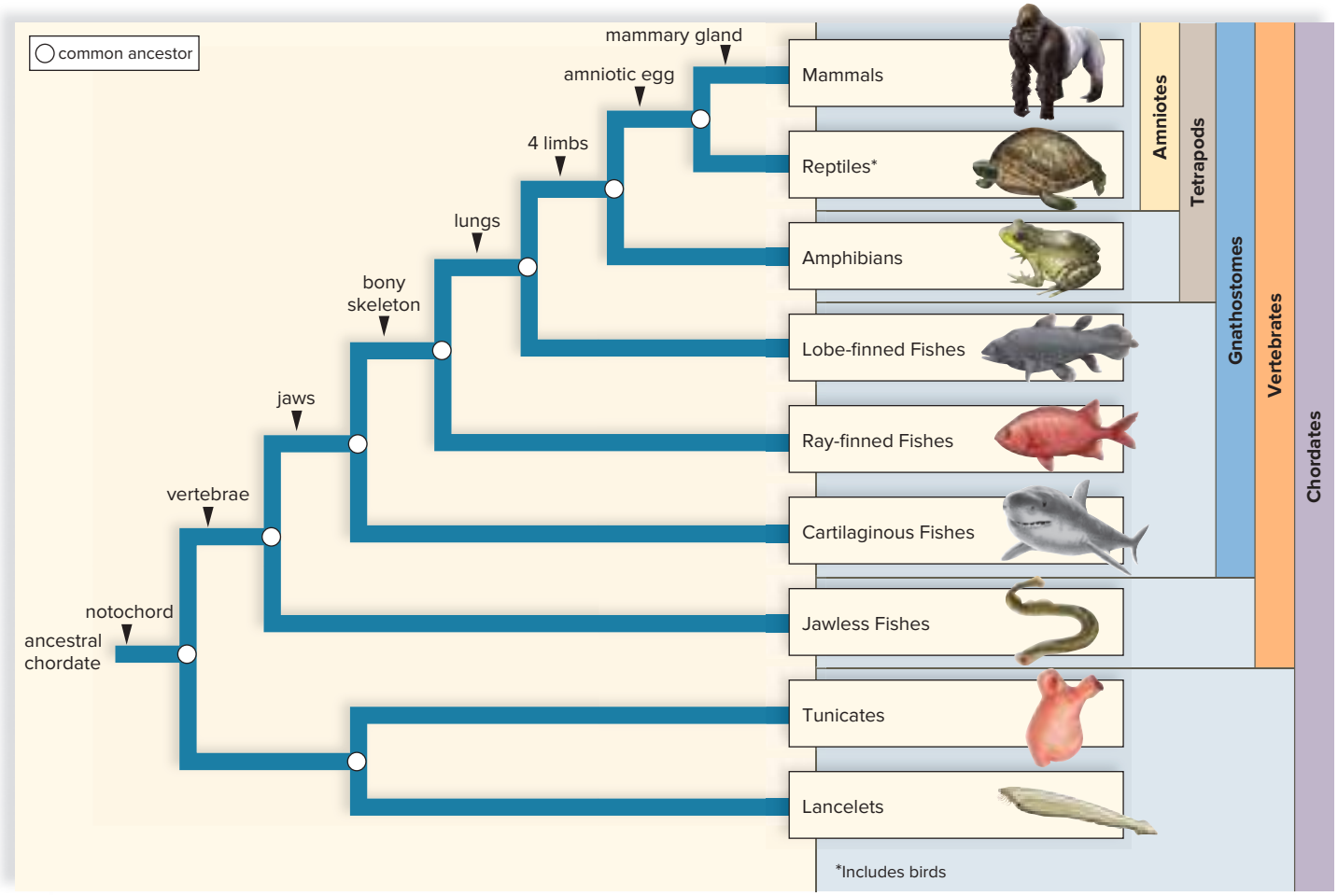


Figure 29.4 Phylogenetic tree of the chordates. For each branch of the tree, new characteristics observed in the common ancestor of each group are noted.

ISTUDY

29.2 The Vertebrates

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the four characteristics unique to vertebrates.
2. Explain how the terms *tetrapod, gnathostome,* and *amniote* relate to vertebrate evolution.
3. Identify the geologic era and periods in which chordates and the first vertebrates appear.

Vertebrates are chordates with vertebral columns and certain other features that distinguish them from the nonvertebrate chordates. There are currently just under 70,000 known species of vertebrates (Table 29.1).

Table 29.1 The Vertebrates

Fishes (jawless, cartilaginous, bony): Endoskeleton, jaws, and paired appendages in most; internal gills; single-loop circulation; usually scales. 32,900
Amphibians (frogs, toads, salamanders): Jointed limbs; lungs; three-chambered heart with double-loop circulation; moist, thin skin. 7,302
Reptiles (snakes, turtles, crocodiles): Amniotic egg; rib cage in addition to lungs; three- or four-chambered heart typical; scaly, dry skin; copulatory organ in males and internal fertilization. 10,711
Birds (songbirds, waterfowl, parrots, ostriches): Endothermy, feathers, and skeletal modifications for flying; lungs with air sacs; four-chambered heart. 10,425
Mammals (monotremes, marsupials, eutherians): Hair and mammary glands. 5,513

Species numbers represent current number of identified species.

Characteristics of Vertebrates

As embryos, vertebrates have the four chordate characteristics (see Fig. 29.1). In addition, vertebrates have the following features.

Vertebral Column

The embryonic notochord is generally replaced by a vertebral column composed of individual vertebrae (Fig. 29.5). Remnants of the notochord give rise to the intervertebral discs, which are compressible, cartilaginous pads between the vertebrae. The vertebral column, which is a part of the flexible but strong endoskeleton, gives evidence that vertebrates are segmented.

Skull

The main axis of the internal skeleton consists of not only the vertebral column but also a skull, which encloses and protects the brain. During vertebrate evolution, the brain has increased in complexity, and specialized regions have developed to carry out specific functions.

The high degree of cephalization is accompanied by complex sense organs. The eyes develop as outgrowths of the brain. The ears are primarily equilibrium devices in aquatic vertebrates, but they also function as sound-wave receivers in land vertebrates. In addition, many vertebrates possess well-developed senses of smell and taste.

Endoskeleton

The vertebrate skeleton (either cartilage or bone) is a living tissue that grows with the animal. It protects internal organs and serves as a place of attachment for muscles. Together, the skeleton and muscles form a system that permits rapid and efficient movement. Two pairs of appendages are characteristic. Fishes typically have pectoral and pelvic fins, while terrestrial tetrapods have four limbs.

Internal Organization

Vertebrates have a large coelom and a complete digestive tract. The blood in vertebrates is contained entirely within blood vessels and is therefore a closed circulatory system. The respiratory system consists of gills or lungs, which obtain oxygen from the environment. The kidneys are important excretory and water-regulating organs that conserve or rid the body of water as necessary. The sexes are generally separate, and reproduction is usually sexual.

Vertebrate Evolution

The Paleozoic era is distinguished by the rise of the chordates and first vertebrates. Chordates appear suddenly on the scene at the start of the Cambrian period, about 485 MYA. By the end of the Ordovician period, which followed, both jawless and jawed fishes had appeared. During this period, too, nonvascular plants moved onto the land.

Even though we do not know the precise origin of vertebrates, we can trace their evolutionary history, as shown in Figure 29.4. (See also Table 18.1 to review evolutionary events on the geologic timescale.) The earliest vertebrates were fishes, organisms that are abundant today in both marine and freshwater habitats. A few of

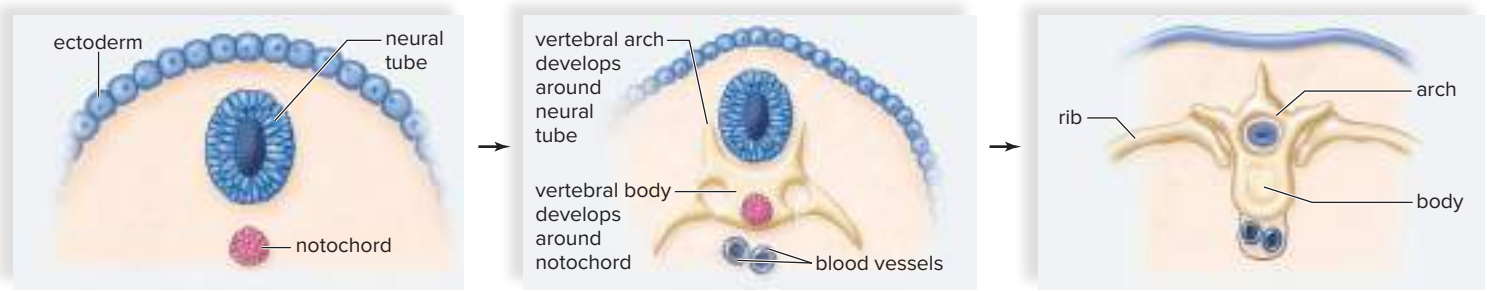


Figure 29.5 Replacement of the notochord by the vertebrae. During vertebrate development, the vertebrae replace the notochord and surround the neural tube. The result is the flexible vertebral column, which protects the nerve cord.

ISTUDY

today's fishes lack jaws and have to suck and otherwise engulf their prey. Most fishes have jaws, which are a more efficient means of grasping and eating prey. The jawed fishes and all the other vertebrates are **gnathostomes**—animals with jaws.

Jawed fishes had dominated the seas by the Silurian period, which followed the Ordovician. Some of these had not only jaws but also a bony skeleton, lungs, and fleshy fins. These characteristics were preadaptive for a land existence, and the amphibians, the first vertebrates to live on land, had evolved from these fishes by the Devonian period, 359.2 MYA. The amphibians were the first vertebrates to have limbs. The terrestrial vertebrates are **tetrapods** (Gk. *tetra,* "four"; *podos,* "foot"), because they have four limbs. Some, such as the snakes, no longer have four limbs, but their evolutionary ancestors did have four limbs.

Many amphibians, such as the frog, reproduce in an aquatic environment. This means that, in general, amphibians are not fully adapted to living on land. Reptiles are fully adapted to life on land, because, among other features, they produce an amniotic egg. The amniotic egg is so named because the embryo is surrounded by an amniotic membrane that encloses the amniotic fluid. Therefore, **amniotes,** animals that exhibit an amniotic membrane, develop within an aquatic environment, but one of their own making. In placental mammals, such as ourselves, the fertilized egg develops inside the female, where it is surrounded by an amniotic membrane.

A watertight skin, which is seen among the reptiles and mammals, is also a good feature to have when living on land.

Check Your Progress 29.2

1. Describe the advantages of an endoskeleton.
2. Explain how four legs would be useful in terrestrial environments.

29.3 The Fishes

Learning Outcomes

Upon completion of this section, you should be able to

1. List several features of jawless fishes.
2. Describe four characteristics shared by all jawed fishes.
3. Compare and contrast cartilaginous fishes and bony fishes.
4. Discuss the evolutionary significance of lobe-finned fishes.

Fishes are the largest group of vertebrates, with nearly 33,000 recognized species. They range in size from a few millimeters in length to the whale shark, which may reach lengths of 12 m. As the oldest and most diverse group of vertebrates, the fossil record of the fishes is extensive.

Jawless Fishes

The earliest fossils of Cambrian origin were the small, filter-feeding, jawless, and finless **ostracoderms.** Several groups of ostracoderms developed heavy dermal armor for protection.

Figure 29.6 Lamprey. Lampreys, such as this one from the genus *Petromyzon,* are agnathans, and have an elongated, rounded body and nonscaly skin. Note the lamprey's toothed oral disk, which is attached to the aquarium glass. ©Heather Angel/Natural Visions

Today's **jawless fishes,** or **agnathans,** have a cartilaginous skeleton and persistent notochord. They are cylindrical, are up to a meter long, and have smooth, nonscaly skin (Fig. 29.6). The hagfishes are exclusively marine scavengers that feed on soft-bodied invertebrates and dead fishes. Many species of lampreys are filter feeders, like their ancestors. Parasitic lampreys have a round, muscular mouth used to attach themselves to another fish and suck nutrients from the host's cardiovascular system. The parasitic sea lamprey gained access to the Great Lakes in 1829, and by 1950 it had almost demolished the resident trout and whitefish populations.

Jawed Fishes

Fishes with jaws have the following characteristics.

Ectothermy

Like all fishes, jawed fishes are **ectotherms** (Gk. *ekto,* "outer"; *therme,* "heat"), which means they depend on the environment to regulate their temperature.

Gills

Like all fishes, jawed fishes breathe with gills and have a single-looped, closed circulatory system with a heart that pumps the blood first to the gills. Then, oxygenated blood passes to the rest of the body.

Cartilaginous or Bony Endoskeleton

The endoskeleton of jawed fishes includes the vertebral column, a skull with jaws, and paired pectoral and pelvic **fins,** projections that are controlled by muscles. The large muscles of the body actually do most of the work of locomotion, but the fins help with balance and turning. Jaws evolved from the first pair of gill arches

ISTUDY

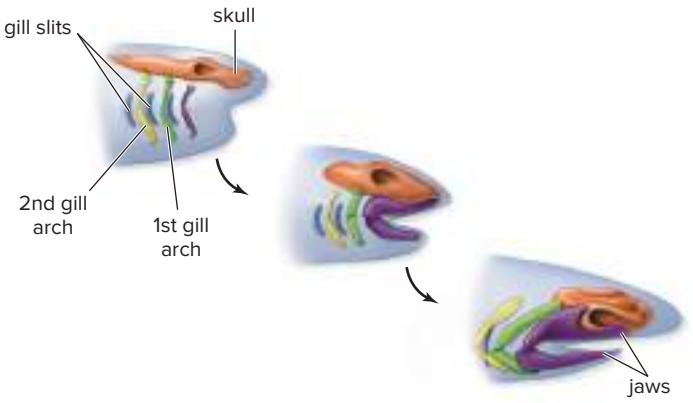


Figure 29.7 Evolution of the jaw. The first jaw evolved from the first and second gill arches of fishes.

present in ancestral agnathans. The second pair of gill arches became support structures for the jaws (Fig. 29.7).

Scales

The skin of the jawed fishes is covered by scales, and therefore, it is not exposed directly to the environment. A scientist can tell the age of a fish from examining the growth of the scales.

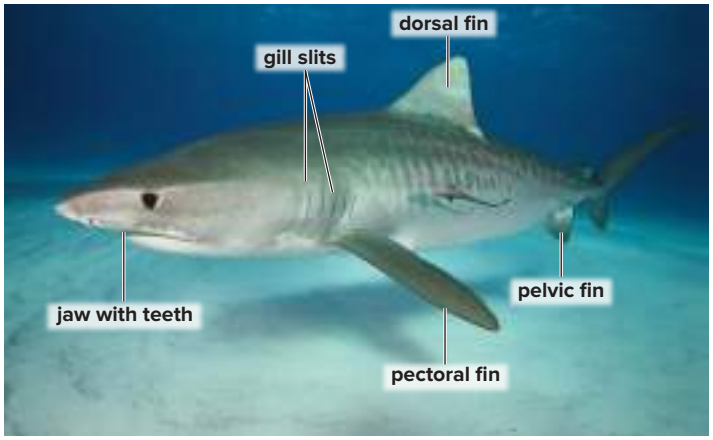
The **placoderms,** extinct jawed fishes of the Devonian period, are probably the ancestors of early sharks and bony fishes. Placoderms were armored with heavy, bony plates and had strong jaws. Like modern-day fishes, they also had paired pectoral and pelvic fins.

Cartilaginous Fishes

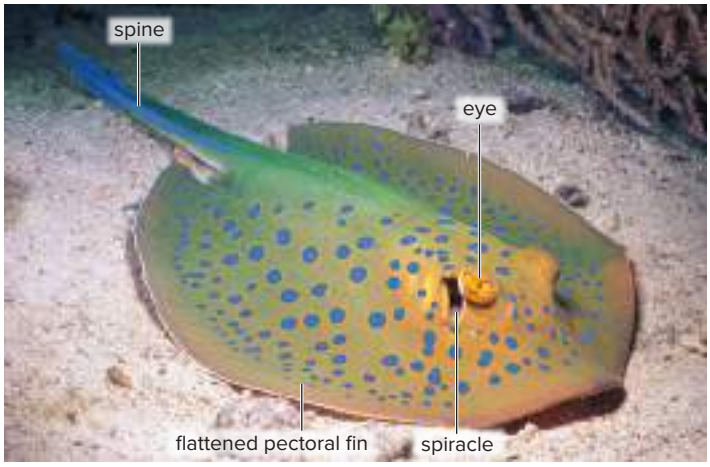
Sharks, rays, skates, and chimaeras are marine **cartilaginous fishes** (Chondrichthyes). The cartilaginous fishes have a skeleton composed of cartilage instead of bone, have five to seven gill slits on both sides of the pharynx, and lack the gill cover of bony fishes. In addition, many have openings to the gill chambers located behind the eyes, called spiracles. Their body is covered with dermal denticles—tiny, teethlike scales that project posteriorly—which is why a shark's skin feels like sandpaper. The menacing teeth of sharks and their relatives are simply larger, specialized versions of these scales. At any one time, a shark such as the great white shark may have up to 3,000 teeth in its mouth, arranged in 6 to 20 rows. Only the first row or two are actively used for feeding; the other rows are replacement teeth.

Three well-developed senses enable sharks and rays (Fig. 29.8) to detect their prey. They have the ability to sense electric currents in water—even those generated by the muscle movements of animals. They have a lateral line system, a series of pressure-sensitive cells that lie within canals along both sides of the body, which can sense pressure caused by a fish or other animal swimming nearby. They also have a keen sense of smell; the part of the brain associated with this sense is very well-developed. Sharks can detect about one drop of blood in 115 liters of water.

The largest sharks are filter feeders, not predators. The basking sharks and whale sharks ingest tons of small crustaceans,



a. Tiger shark, *Galeocerdo cuvier*



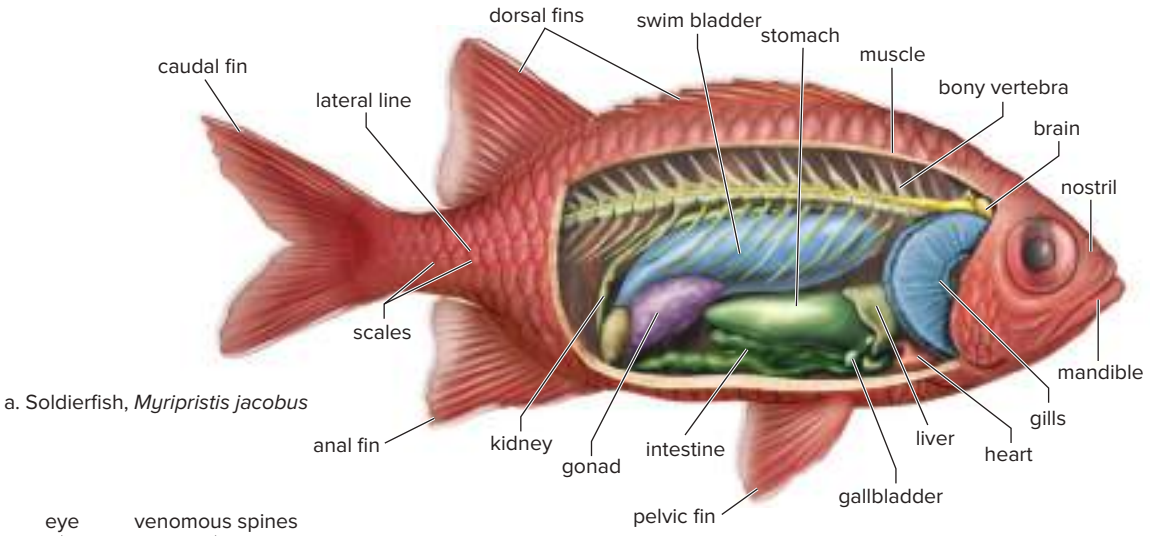
b. Blue-spotted stingray, *Taeniura lymma*

Figure 29.8 Cartilaginous fishes. **a.** Sharks are predators or scavengers that move gracefully through open ocean waters. **b.** Most stingrays grovel in the sand, feeding on bottom-dwelling invertebrates. This blue-spotted stingray is protected by the spine on its whiplike tail. (a): Alastair Pollock Photography/Flickr Open/Getty Images; (b): Comstock Images/PictureQuest/Getty Images

collectively called krill. Many sharks are fast-swimming predators in the open sea (Fig. 29.8*a*). The great white shark, about 7 m in length, feeds regularly on dolphins, sea lions, and seals. Humans are normally not attacked, except when mistaken for sharks' usual prey. Tiger sharks, so named because the young have dark bands, reach 6 m in length and are unquestionably one of the most predaceous sharks. As it swims through the water, a tiger shark will swallow anything. Stomach content analysis has found rolls of tar paper, shoes, gasoline cans, paint cans, and even human body parts. The number of bull shark attacks has increased in the Gulf of Mexico in recent years. This increase is due to more swimmers in the shallow waters and a loss of the sharks' natural food supply.

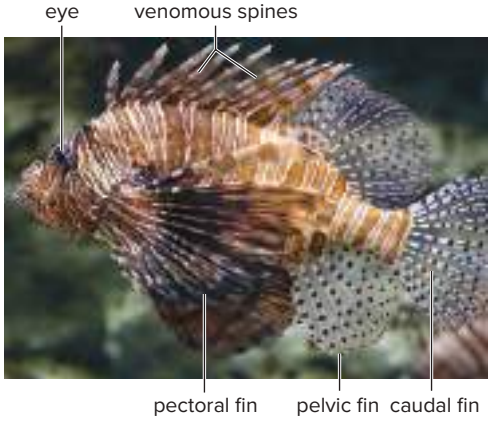
In rays and skates (Fig. 29.8*b*), the pectoral fins are greatly enlarged into a pair of large, winglike fins, and the bodies are dorsoventrally flattened. The spiracles are enlarged and allow them to move water over the gills while resting on the bottom, where they feed on organisms such as crustaceans, small fishes, and mollusks.

ISTUDY

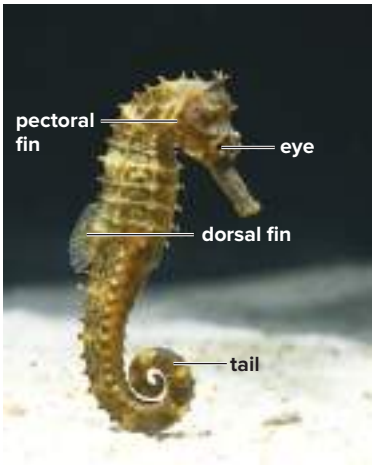


a. Soldierfish, *Myripristis jacobus*

Figure 29.9 Ray-finned fishes. **a.** A soldierfish has the typical appearance and anatomy of a ray-finned fish. A lionfish (**b**), a seahorse (**c**), a flying fish (**d**), and a swordfish (**e**) show how diverse ray-finned fishes can be. (b): powerofforever/iStockphoto/Getty Images; (c): bluehand/Shutterstock; (d): Paulo de Oliveira/NHPA/age fotostock; (e): Deepshine/iStock/Getty Images

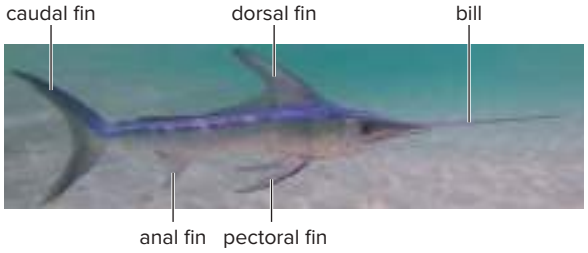


b. Lionfish, *Pterois volitans*



c. Seahorse, *Hippocampus kuda*

d. Flying fish, *Hirundichthys rondeletii*



e. Swordfish, *Xiphias gladius*

Stingrays have a whiplike tail that has serrated spines with venom glands at the base. This tail spike is a defensive weapon, but it can deliver a harmful or even fatal wound. Manta rays are harmless, oceanic, filter-feeding giants with a fin span of up to 6 m and a weight of 2,000 kg. Sawfish rays are named for their large, protruding anterior "saw." Some species of stingrays can deliver an electric shock. Their large electric organs, located at the base of their pectoral fins, can discharge over 300 volts. Skates resemble stingrays but possess two dorsal fins and a caudal fin.

The chimaeras, or ratfishes, are a group of cartilaginous fishes that live in cold marine waters. They are known for their unusual shape and iridescent colors.

Bony Fishes

The majority of living vertebrates, with over 32,000 species, are **bony fishes** (Osteichthyes). The bony fishes range in size from gobies, which are less than 7.5 mm long, to the giant sturgeons, which can obtain a length of 4 m.

The majority of fish species are **ray-finned bony fishes** with fan-shaped fins supported by a thin, bony ray. These fishes are the most successful and diverse of all the vertebrates (Fig. 29.9). Some, such as herrings, are filter feeders; others, such as trout, are opportunists; and still others are predaceous carnivores, such as piranhas and barracudas.

Despite their diversity, bony fishes have many features in common. They lack external gill slits; instead, their gills are covered by an *operculum.* Many bony fishes have a **swim bladder,** a gas-filled sac into which they can secrete gases or from which they can absorb gases, altering its pressure. This results in a change in the fishes' buoyancy and, therefore, their depth in the water. Bony fishes have a single-loop, closed cardiovascular system (see Fig. 29.11*a*).

The nervous system and brain in bony fishes are well-developed, and complex behaviors are common. Bony fishes have separate sexes, and the majority of species undergo external fertilization after females deposit eggs and males deposit sperm into the water.

Lobe-Finned Fishes

Lobe-finned fishes possess fleshy fins supported by bones. Ancestral lobe-finned fishes gave rise to modern-day lobe-finned fishes, lungfishes, and amphibians. **Lungfishes** have lungs as well as gills for gas exchange. The lobe-finned fishes and lungfishes are grouped together as the Sarcopterygii. Today, only two species of lobe-finned fishes (the coelacanths) and six species of lungfishes

ISTUDY

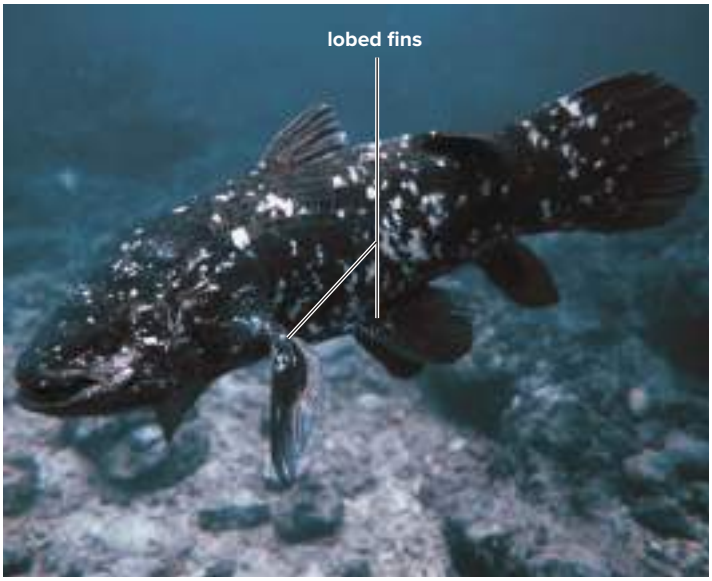


Figure 29.10 Coelacanth, *Latimeria chalumnae*. A coelacanth is a lobe-finned fish once thought to be extinct. Peter Scoones/Science Source

are known. Lungfishes live in Africa, South America, and Australia, either in stagnant fresh water or in ponds that dry up annually.

In 1938, a coelacanth was caught from the deep waters of the Indian Ocean off the eastern coast of South Africa. It took the scientific world by surprise, because these animals were thought to be extinct for 70 million years. Approximately 200 coelacanths have been captured since that time (Fig. 29.10).

Check Your Progress 29.3

1. List and describe the characteristics that fishes have in common.
2. Distinguish between lobe-finned and ray-finned bony fishes.

29.4 The Amphibians

Learning Outcomes

Upon completion of this section, you should be able to

1. List the seven characteristics that define the amphibians.
2. Describe the features of the three groups of living amphibians.
3. Summarize the two hypotheses that explain the evolution of amphibians from lobe-finned fishes.

Amphibians (class Amphibia [Gk. *amphibios,* "living both on land and in water"]) are represented today by frogs, toads, newts, and salamanders. Most members of this group lead a lifestyle that includes a larval stage that lives in the water, and an adult stage that lives on land. Caecilians are also classified as amphibians even though they are fossorial, wormlike amphibians that spend most of their life underground. There are over 7,300 known species of amphibians.

Characteristics of Amphibians

The general characteristics of amphibians are as follows.

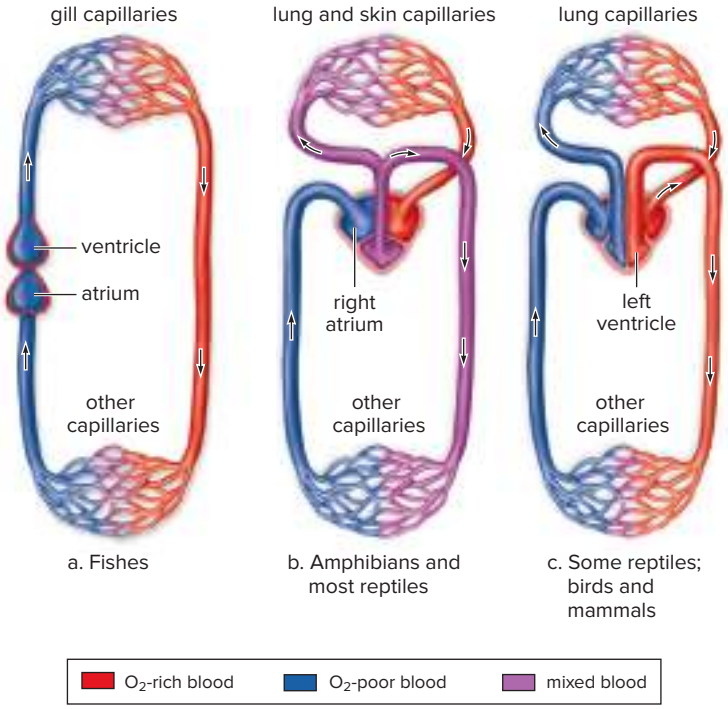


Figure 29.11 Vertebrate circulatory pathways. **a.** The single-loop pathway of fishes has a two-chambered heart. **b.** The double-loop pathway of other vertebrates sends blood to the lungs and to the body. In amphibians and most reptiles, limited mixing of oxygen-rich and oxygen-poor blood takes place in the single ventricle of their three-chambered heart. **c.** The four-chambered heart of some reptiles (crocodilians and birds) and mammals sends only oxygen-poor blood to the lungs and oxygen-rich blood to the body.

Limbs

Typically, amphibians are tetrapods, as mentioned earlier. The skeleton, particularly the pelvic and pectoral girdles, is well-developed to promote locomotion.

Smooth and Nonscaly Skin

The skin, which is kept moist by mucous glands, plays an active role in water balance and respiration, and it can help in temperature regulation when on land through evaporative cooling. A thin, moist skin does mean, however, that most amphibians stay close to water, or else they risk drying out.

Lungs

If lungs are present, they are relatively small, and respiration is supplemented by the exchange of gases across the porous skin (called cutaneous respiration).

Double-Loop Circulatory Pathway

A three-chambered heart, with a single ventricle and two atria, pumps blood to both the lungs and the body (Fig. 29.11*b*, *c*).

Sense Organs

Special senses, such as sight, hearing, and smell, are fine-tuned for life on land. Amphibian brains are larger than those of fish, and the cerebral cortex is more developed. These animals have a specialized tongue for catching prey, eyelids for keeping their eyes moist, and a sound-producing larynx.

ISTUDY

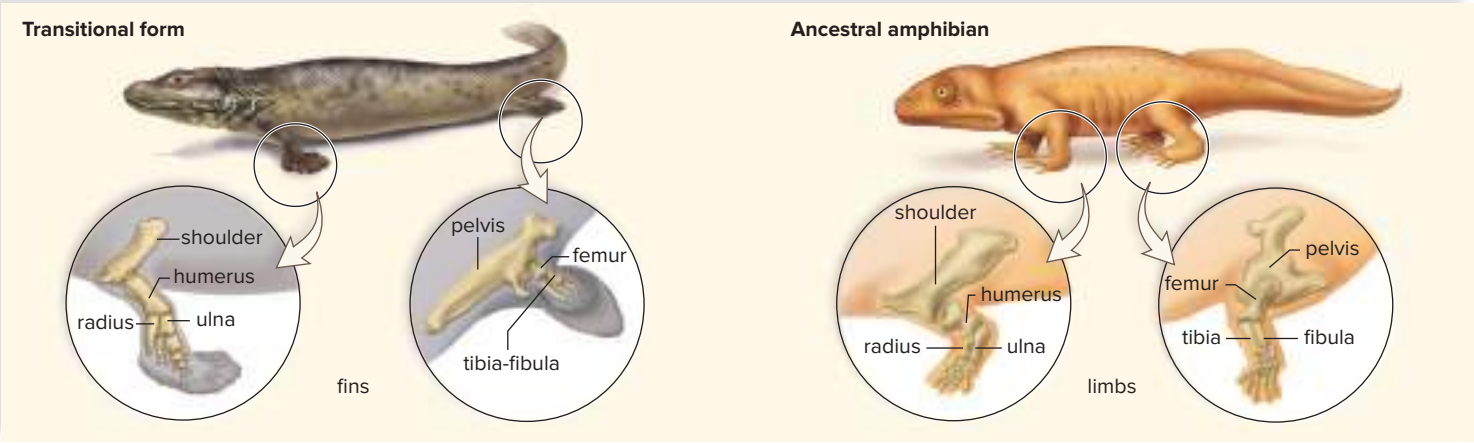


Figure 29.12 Lobe-finned fishes to amphibians. This transitional form links the lobes of lobe-finned fishes to the limbs of ancestral amphibians. Compare the fins of the transitional form (*left*) to the limbs of the ancestral amphibian (*right*).

Ectothermy

Like fishes, amphibians are ectotherms, but they are able to live in environments where the temperature fluctuates greatly. During winters in the temperate zone, they become inactive and enter torpor. The European common frog can survive in temperatures dropping to as low as –6°C.

Aquatic Reproduction

Their name, amphibians, is appropriate because many return to water for the purpose of reproduction. They deposit their eggs and sperm into the water, where external fertilization takes place. Generally, the eggs are protected only by a jelly coat, not by a shell. When the young hatch, they are tadpoles (aquatic larvae with gills) that feed and grow in the water. After amphibians undergo a *metamorphosis* (change in form), they emerge from the water as adults that breathe air. Some amphibians, however, have evolved mechanisms that allow them to bypass this aquatic larval stage and reproduce on land.

Evolution of Amphibians

Amphibians evolved from the lobe-finned fishes with lungs by way of transitional forms. Two hypotheses have been suggested to account for the evolution of amphibians from lobe-finned fishes. Perhaps lobe-finned fishes had an advantage over others because they could use their lobed fins to move from pond to pond. Or perhaps the supply of food on land in the form of plants and insects—and the absence of predators—promoted further adaptations to the land environment. Paleontologists have recently found a well-preserved transitional fossil from the late Devonian period in Arctic Canada that represents an intermediate between lobe-finned fishes and tetrapods with limbs. This fossil, named *Tiktaalik roseae* (see Section 15.3), provides unique insights into how the legs of tetrapods arose (Fig. 29.12).

Diversity of Living Amphibians

The amphibians of today occur in three groups: salamanders and newts; frogs and toads; and caecilians (Fig. 29.13). Salamanders and newts have elongated bodies, long tails, and usually two pairs of limbs. Salamanders and newts range in size from less than 15 cm to the giant Japanese salamander, which exceeds 1.5 m in length. Most have limbs that are set at right angles to the body and resemble the earliest fossil amphibians. They move like a fish, with side-to-side, sinusoidal (S-shaped) movements.

Both salamanders and newts are carnivorous, feeding on small invertebrates, such as insects, slugs, snails, and worms. Salamanders practice internal fertilization; in most, males produce a sperm-containing spermatophore, which females pick up with their *cloaca* (the terminal chamber common to the urinary, digestive, and genital tracts). Then, the fertilized eggs are laid in water or on land, depending on the species. Some amphibians, such as the mudpuppy of eastern North America, remain in the water and retain the gills of the larva.

Frogs and toads, which range in length from less than 1 cm to 30 cm, are common in subtropical to temperate to desert climates around the world. In these animals, which lack tails as adults, the head and trunk are fused, and the long hindlimbs are specialized for jumping (Fig. 29.13*b*). All species are carnivorous and have a tremendous array of specializations, depending on their habitats. Glands in the skin secrete poisons that make the animals distasteful to eat and protect them from microbial infections. Some tropical species with brilliant fluorescent coloration are particularly poisonous, with the bright colors serving as a warning to potential predators. Colombian Indians dip their darts in the deadly secretions of poison-dart frogs, as discussed in the Nature of Science feature, "Vertebrates and Human Medicine." The tree frogs have adhesive toepads that allow them to climb trees, while others, the spadefoots, have hardened spades that act as shovels, enabling them to dig into the soil.

Caecilians are legless, often sightless, worm-shaped amphibians that range in length from about 10 cm to more than 1 m (Fig. 29.13*c*). Most burrow in moist soil, feeding on worms and other soil invertebrates. Some species have folds of skin that make them look like a segmented earthworm.

Check Your Progress 29.4

1. List the characteristics that amphibians have in common.
2. Describe the usual life cycle of amphibians.

ISTUDY

THEME Nature of Science

Vertebrates and Human Medicine

Hundreds of pharmaceutical products come from other vertebrates, and even those that produce poisons and toxins provide medicines that benefit us.

Natural Products with Medical Applications

The black-and-white spitting cobra of Southeast Asia paralyzes its victims with a potent venom, which eventually leads to respiratory arrest. However, that venom is also the source of the drug Immunokine, which can inhibit some harmful effects of an overactive immune system. It is approved in Thailand for use in combating the side effects of cancer therapy, and it is being studied for use in treating AIDS, autoimmune diseases, and other disorders.

Although snakebites can be very painful, certain components found in venom actually relieve pain. The black mamba, found mainly in sub-Saharan Africa, is one of the most lethal snakes on Earth. However, compounds called mambalgins, in its venom, block pain signals by inhibiting the flow of certain ions through nerves that carry pain messages. When tested in mice, these compounds were as effective as morphine, with fewer side effects (Fig. 29A).

Another compound, known as epibatidine, derived from the skin of an endangered Ecuadorian poison-dart frog, is 50–200 times more powerful than morphine in relieving chronic and acute pain, without the addictive properties. Unfortunately, it can also have serious side effects, so companies have synthesized compounds with a similar structure, hoping to improve its safety profile.

Other venoms mainly affect blood clotting. Eptifibatide is derived from the venom of the pigmy rattlesnake, which lives in the southeastern United States. Because it binds to blood platelets and reduces their tendency to clump together, this drug is used to reduce the risk of clot formation in individuals at risk for heart attacks. Alternatively, the venom of several pit vipers, such as the copperhead, contains "clot-busting" (thrombolytic) substances, which can be used to dissolve abnormal clots that have already formed.

Sharks produce a variety of chemicals with potentially medicinal properties. Squalamine is a steroidlike molecule that was first isolated from the liver of dogfish sharks. It has broad antimicrobial properties and can inhibit the abnormal growth of new blood vessels, which is a factor in cancer and a variety of other diseases. Squalamine is also safe enough to be used in the eye and is currently being tested as a potential treatment for macular degeneration, an eye disease that affects almost 20 million adults in the United States.

Animal Pharming

Some of the most powerful applications of genetic engineering can be found in the development of drugs and therapies for human diseases. In fact, this technology has led to a new industry: animal pharming, which uses genetically altered vertebrates, such as mice, sheep, goats, cows, pigs, and chickens, to produce medically useful pharmaceutical products.

To accomplish this, the human gene for some useful product is inserted into the embryo of a vertebrate. That embryo is implanted into a foster mother, which gives birth to the transgenic animal, which contains genes from the two sources. An adult transgenic vertebrate produces large quantities of the pharmed product in its blood, eggs, or milk, from which the product can be easily harvested and purified.

An example of a pharmed product used in human medicine is human antithrombin. This medication is important in the treatment of individuals who have a hereditary deficiency of this protein and so are at high risk for life-threatening blood clots. Approved by the FDA in 2009, the bioengineered drug, known by the brand name ATryn, is purified from the milk of transgenic goats (Fig. 29A*b*).

Xenotransplantation

There is an alarming shortage of human donor organs to fill the need for hearts, kidneys, and livers. One solution is xenotransplantation, the transplantation of nonhuman vertebrate tissues and organs into humans. The first such transplant occurred in 1984 when a team of surgeons implanted a baboon heart into an infant, who, unfortunately, lived only 20 days.

Although apes are more closely related to humans, pigs are considered to be the best source for xenotransplants. Pig organs are similar to human organs in size, anatomy, and physiology, and large numbers of pigs can be produced quickly (Fig. 29A*c*). Most infectious microbes of pigs are unlikely to infect a human recipient. Currently, pig heart valves and skin are routinely used for the treatment of humans. Miniature pigs, whose heart size is similar to that of humans, are being genetically engineered to make their tissues less foreign to the human immune system, to minimize rejection.

Questions to Consider

1. Is it ethical to change the genetic makeup of vertebrates in order to use them as drug or organ factories?
2. What are some health concerns that may arise due to xenotransplantation?

a. Snake

b. Goats

c. Pig heart

Figure 29A Medical uses of animals. **a.** Snake venom may be used to create pain medications. **b.** Mammals, such as these goats, may express pharmaceutical compounds in their milk. **c.** Pigs are now being genetically altered to provide a supply of hearts for heart transplant operations. (a): Marie Holding/iStock/Getty Images; (b): tomophotography/Moment/Getty Images; (c): Victoria Dolidovich/Hemera/iStock/Getty Images

ISTUDY

Figure 29.13 Amphibians. Living amphibians are divided into three orders: **a.** Salamanders and newts. Members of this order have a tail throughout their lives and, if present, unspecialized limbs. **b.** Frogs and toads. Like this frog, members of this order are tailless and have limbs specialized for jumping. **c.** The caecilians are wormlike burrowers. (a): Suzanne L. Collins & Joseph T. Collins/Science Source; (b): T. Kitchin & V. Hurst/NHPA/Photoshot; (c): FLPA/Alamy Stock Photo

29.5 The Reptiles

Learning Outcomes

Upon completion of this section, you should be able to

1. List the seven features that define reptiles.
2. Explain why reptiles are represented by more than one evolutionary lineage.
3. Define the traits of birds that are related to flight.

Reptiles (class Reptilia) are a very successful group of terrestrial animals consisting of more than 21,000 species, including birds. Reptiles have a number of characteristics that enable them to be fully adapted to life on land.

Characteristics of Reptiles

The general characteristics of reptiles include the following.

Paired Limbs

With the exception of snakes, reptiles generally have two pairs of limbs, usually with five toes each. Reptiles are adapted for climbing, running, paddling, or flying.

Skin Adaptations

A thick and dry skin is impermeable to water. Therefore, the skin prevents water loss. In reptiles, the skin is wholly or in part scaly. Many reptiles (e.g., snakes and lizards) molt several times a year.

Efficient Breathing

The lungs (Fig. 29.14) are more developed than in amphibians. Also, in many reptiles, an expandable rib cage assists breathing.

Efficient Circulation

The heart prevents mixing of blood. A septum divides the ventricle either partially or completely. If it partially divides the ventricle,

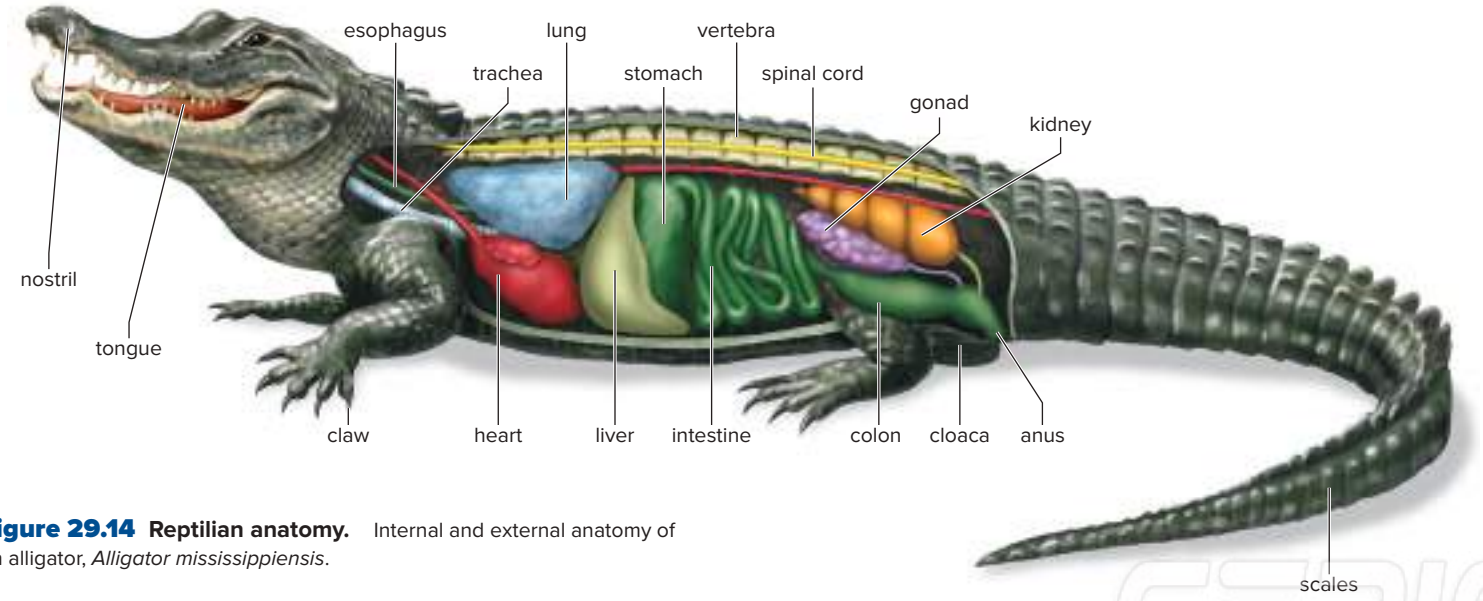


Figure 29.14 Reptilian anatomy. Internal and external anatomy of an alligator, *Alligator mississippiensis*.

ISTUDY

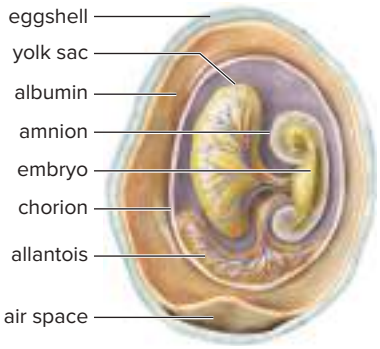


Figure 29.15 Amniotic egg of a reptile. Inside the egg, the embryo is surrounded by three membranes. The chorion aids in gas exchange, the allantois stores waste, and the amnion encloses a fluid that prevents drying out and provides protection. The yolk sac provides nutrients for the embryo.

the mixing of oxygen-poor blood and oxygen-rich blood is reduced. If the septum is complete, oxygen-poor blood is completely separated from oxygen-rich blood (see Fig. 29.11*c*).

Efficient Excretion

The kidneys are well-developed. They excrete uric acid, and therefore less water is required to rid the body of nitrogenous wastes.

Ectothermy

Most reptiles are ectotherms, and this allows them to survive on a fraction of the food per body weight required by birds and mammals. Ectothermic reptiles are adapted behaviorally to maintain a warm body temperature by warming themselves in the sun.

Well-Adapted Reproduction

Sexes are separate and fertilization is internal. Internal fertilization prevents sperm from drying out when copulation occurs. The **amniotic egg** contains extraembryonic membranes, which protect the embryo, remove nitrogenous wastes, and provide the embryo with oxygen, food, and water (Fig. 29.15). These membranes are not part of the embryo itself and are disposed of after development is complete. One of the membranes, the amnion, is a sac that fills with fluid and provides a "private pond," within which the embryo develops.

Evolution of Amniotes

An ancestral amphibian gave rise to the amniotes at some point in the Carboniferous period (see Table 18.1). The amniotes include animals now classified as the reptiles (including birds) and the mammals. The embryo of an amniote has extracellular membranes, including an amnion Fig. 29.15).

Figure 29.16 shows that the amniotes consist of three lineages: (1) the turtles, in which the skull is **anapsid**—that is, it has no openings behind the orbit, or eye socket; (2) all the other reptiles, including the birds, in which the skull is **diapsid,** or has two openings behind the orbit; and (3) the mammals, in which the skull is **synapsid** and has one opening behind the orbit.

Systematists are continuing to explore the evolutionary relationships of reptiles, which also include the birds. For example, the classical view of reptile evolution considers turtles, or anapsids, an independent lineage, separate from that of the rest of the reptiles, which are diapsids. The modern view of turtle evolution places turtles within the archosaurs, along with birds and crocodiles. This would mean that the anapsids are really just highly specialized diapsids. The evidence in support of this modern view is still being debated, and thus the classical view of reptile evolution is presented here.

According to the classical view, all other reptiles, except turtles, are diapsids, because they have a skull with two openings behind the eyes. The thecodonts are diapsids that gave rise to the ichthyosaurs, which returned to the aquatic environment, and the pterosaurs of the Jurassic period, which had a keel for the attachment of large flight muscles, as well as air spaces in their bones to reduce weight. Their wings were membranous and supported by elongated bones of the fourth finger. *Quetzalcoatlus,* the largest flying animal ever to live, had an estimated wingspan of nearly 13.5 m (over 40 ft)!

The thecodonts gave rise to the crocodiles and dinosaurs. A sequence of now known transitional forms occurred between dinosaurs and birds. The crocodilians and birds share derived features, such as skull openings in front of the eyes, and clawed feet. It is customary now to use the designation archosaurs for the crocodilians, dinosaurs, and birds. This means that these animals are more closely related to each other than they are to snakes and lizards.

Dinosaurs varied greatly in size and behavior. The average dinosaur was about the size of a chicken. Some dinosaurs, however, were the largest land animals ever to live. *Brachiosaurus,* a herbivore, was about 23 m long and about 17 m tall. *Tyrannosaurus rex,* a carnivore, was 5 m tall when standing on its hind legs. A bipedal stance freed the forelimbs and allowed them to be used for purposes other than walking, such as manipulating prey. It was also preadaptive for the evolution of wings in the birds. In fact, many fossils of dinosaurs in the family that includes the familiar *T. rex* and velociraptor bear unmistakable impressions of feathers. Prior to the evolution of birds, however, these feathers were not used for flight but, rather, served as insulation against cold weather.

Dinosaurs dominated Earth for about 170 million years before they died out at the end of the Cretaceous period, 65 MYA. One hypothesis for this mass extinction is that a massive meteorite struck Earth near the Yucatán Peninsula (see Section 18.3). The resultant cataclysmic events disrupted existing ecosystems, destroying many living organisms. This hypothesis is supported by the presence of a layer of the mineral iridium, which is rare on Earth but common in meteorites, in the late Cretaceous strata.

Diversity of Living Reptiles

Living reptiles are represented by turtles, lizards, snakes, tuataras, crocodilians, and birds. Figure 29.17 shows representatives of all but the birds.

ISTUDY

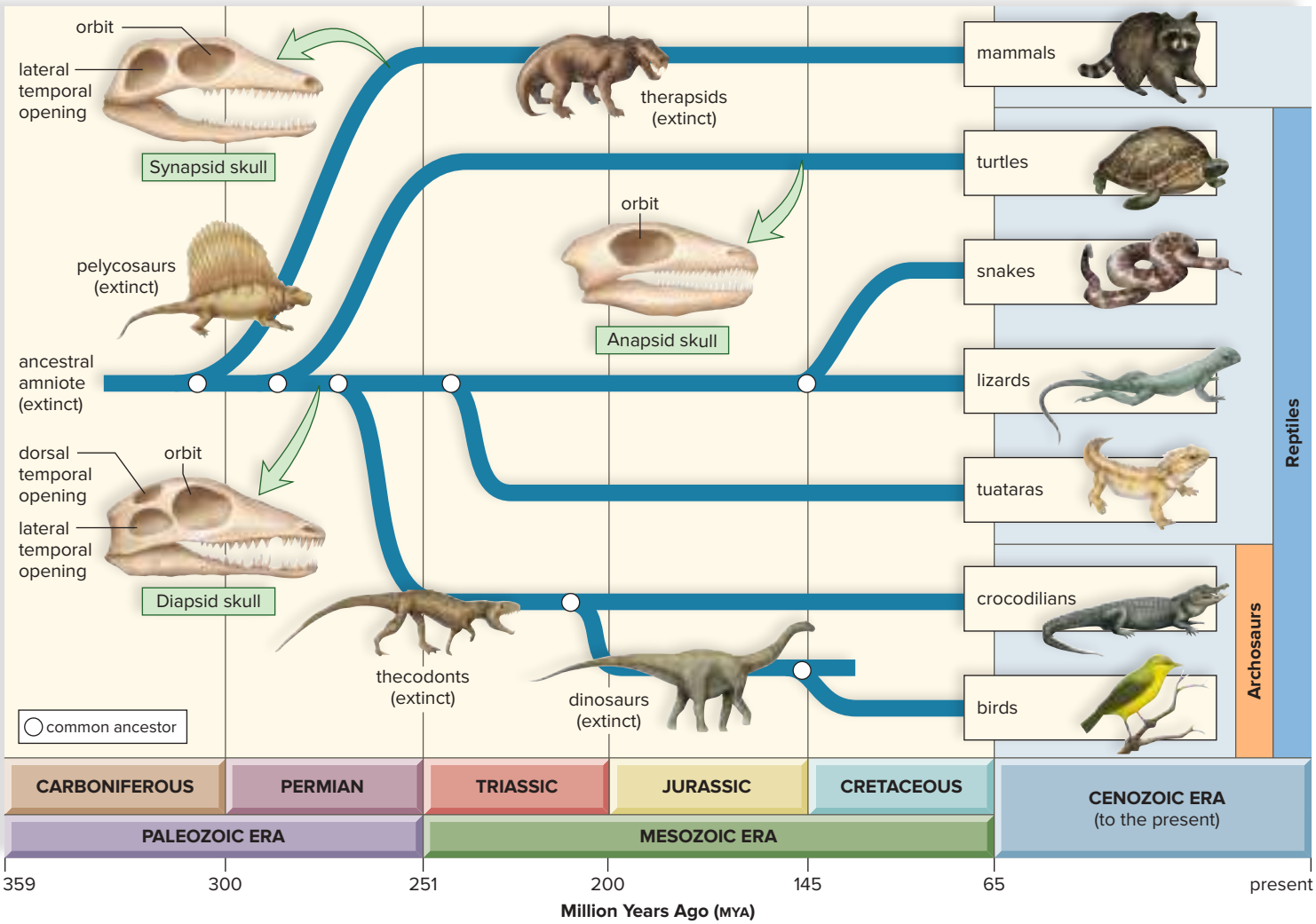


Figure 29.16 Timeline of the evolution of the amniotes. This diagram shows an overview of the presumed evolutionary relationships among amniotes, including the major groups of reptiles. The amniote ancestor evolved in the Paleozoic era. Historically, the openings in the skull were considered evidence that there were two major groups of reptiles, with turtles separate from the other reptile groups. This historical understanding of turtle evolution is presented in this figure. A modern view, not shown, proposes that the turtles are archosaurs, along with crocodiles and birds.

Along with tortoises, turtles can be found in marine, freshwater, and terrestrial environments. Most turtles have ribs and thoracic vertebrae that are fused into a heavy shell. They lack teeth but have a sharp beak. The legs of sea turtles are flattened and paddlelike (Fig. 29.17*a*), while terrestrial tortoises have strong limbs for walking.

Lizards have four clawed feet and resemble their prehistoric ancestors in appearance (Fig. 29.17*b*), although some species have lost their limbs and superficially resemble snakes. Typically, they are carnivorous and feed on insects and small animals, including other lizards. Marine iguanas of the Galápagos Islands are adapted to spending time each day at sea, where they feed on sea lettuce and other algae. Chameleons are adapted to live in trees and have long, sticky tongues for catching insects some distance away. They can change color to blend in with their background. Geckos are primarily nocturnal lizards with adhesive pads on their toes. Skinks are common elongated lizards with reduced limbs and shiny scales. Monitor lizards and Gila monsters, despite their names, are generally not a dangerous threat to humans.

Although most snakes (Fig. 29.17*c*) are harmless, several venomous species, including rattlesnakes, cobras, mambas, and copperheads, have given the whole group a reputation of being dangerous. Snakes evolved from lizards and have lost their limbs as an adaptation to burrowing. A few species such as pythons and boas still possess the vestiges of pelvic girdles. Snakes are carnivorous and have a jaw that is loosely attached to the skull; therefore, they can eat prey that is much larger than their head size. When snakes and lizards flick out their tongues, they are collecting airborne molecules and transferring them to a *Jacobson's organ* at the roof of the mouth and sensory cells on the floor of the mouth. The Jacobson's organ is an olfactory organ for the analysis of airborne chemicals. Snakes possess internal ears that are capable of detecting low-frequency sounds and vibrations. Their ears lack external ear openings.

ISTUDY

a. Green sea turtle, *Chelonia mydas*
b. Gila monster, *Heloderma suspectum*
c. Prairie rattlesnake, *Crotalus viridis*
d. Tuatara, *Sphenodon punctatus*
e. Nile crocodile, *Crocodylus niloticus*

Figure 29.17 Reptilian diversity other than birds. Representative living reptiles include (**a**) the green sea turtle, (**b**) the venomous Gila monster, (**c**) the prairie rattlesnake, and (**d**) the tuatara. **e.** A young crocodile hatches from an egg. The eggshell is leathery and flexible, not brittle, like birds' eggs. (a): Paul Souders/Digital Vision/Getty Images; (b): Design Pics/SuperStock; (c): Steve Hamblin/Alamy Stock Photo; (d): Tim Cuff/Alamy Stock Photo; (e): Catchlight Lens/Shutterstock

Two species of tuataras are found in New Zealand (Fig. 29.17*d*). They are lizardlike animals that can attain a length of 66 cm and can live for nearly 80 years. These animals possess a well-developed "third" eye, known as a pineal eye, which is light-sensitive and buried beneath the skin in the upper part of the head. The tuataras are the only member of an ancient group of reptiles that included the common ancestor of modern lizards and snakes.

The majority of crocodilians (including alligators and crocodiles) live in fresh water, feeding on fishes, turtles, and terrestrial animals that venture too close to the water. They have long, powerful jaws (Fig. 29.17*e*) with numerous teeth and a muscular tail that serves as both a weapon and a paddle. Male crocodiles and alligators bellow to attract mates. In some species, the male protects the eggs and cares for the young.

Birds

As already noted, the fossil record shows that birds evolved from reptiles. This is supported by genetic and anatomical evidence. All **birds** have traits such as the presence of scales (feathers are modified scales), a tail with vertebrae, and clawed feet that show they are indeed reptiles. Interestingly, a close examination of the fossils from several dinosaur specias has revealed that they also had feathers.

To many people, birds are the most conspicuous, melodic, beautiful, and fascinating group of vertebrates. Birds range in size from the tiny "bee" hummingbird at 1.8 g (less than a penny) and 5 cm long to the ostrich at a maximum weight of 160 kg and a height of 2.7 m.

Nearly every anatomical feature of a bird can be related to its ability to fly (Fig. 29.18). These features are involved in the action of flight, providing energy for flight or the reduction of the bird's body weight, making flight less energetically costly.

Feathers

Soft down keeps birds warm, wing feathers allow flight, and tail feathers are used for steering. A feather is a modified reptilian scale with the complex structure shown in Figure 29.18*a*. Nearly all birds molt (lose their feathers) and replace their feathers about once a year.

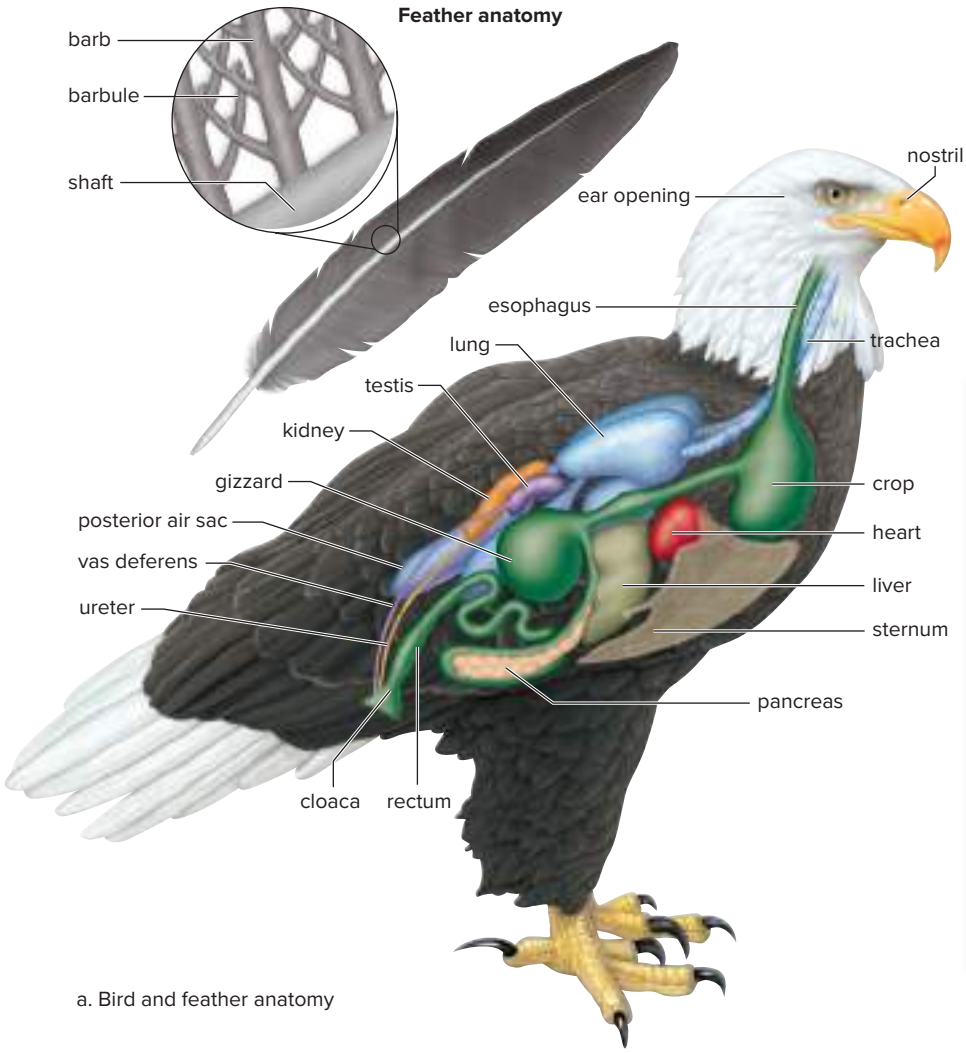
Modified Skeleton

Unique to birds, the collarbone is fused (the wishbone), and the sternum has a keel (Fig. 29.18*b*). Many other bones are fused, making the skeleton more rigid than the reptilian skeleton. The breast muscles are attached to the keel, and their action accounts for a bird's ability to fly. A horny beak has replaced jaws equipped with teeth, and a slender neck connects the head to a rounded, compact torso.

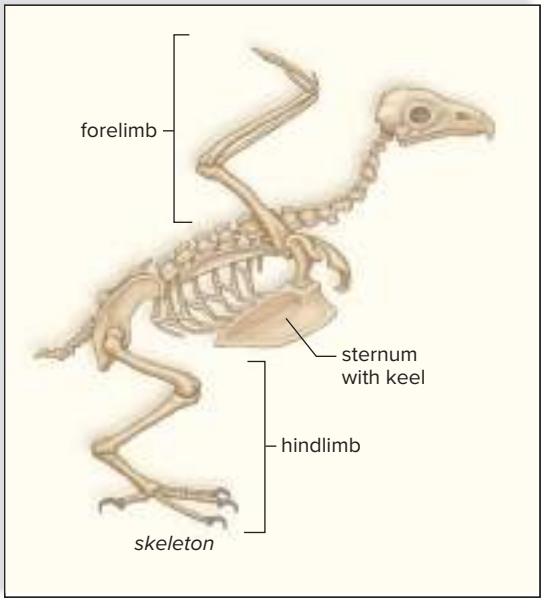
Modified Respiration

In birds, unlike other reptiles, the lobular lungs connect to anterior and posterior air sacs. The presence of these sacs means the air circulates one way through the lungs, and gases are continuously exchanged across respiratory tissues. Another benefit of air sacs is

ISTUDY



a. Bird and feather anatomy



b. Bald eagle, *Haliaetus*

Figure 29.18 Bird anatomy and flight.
a. Bird anatomy. *Top:* In feathers, a hollow central shaft gives off barbs and barbules, which interlock in a latticelike array. *Bottom:* The anatomy of an eagle is representative of bird anatomy. **b.** Bird flight. The skeleton of an eagle shows that birds have a large, keeled sternum to which flight muscles attach. The bones of the forelimb help support the wings.

that they lighten the body and bones for flying. Some of the air sacs are present in cavities within the bones.

Endothermy

Birds, unlike other reptiles, generate internal heat. Many **endotherms** can use metabolic heat to maintain a constant internal temperature. Endothermy may be associated with their efficient nervous, respiratory, and circulatory systems.

Well-Developed Sense Organs and Nervous System

Birds have particularly acute vision and well-developed brains. Their muscle reflexes are excellent. An enlarged portion of the brain seems to be the area responsible for instinctive behavior.

A ritualized courtship often precedes mating. Many newly hatched birds require parental care before they are able to fly away and seek food for themselves.

A remarkable aspect of bird behavior is the seasonal migration of many species over long distances. Birds navigate by day and night, whether it is sunny or cloudy, by using the sun and stars and even Earth's magnetic field to guide them.

Birds are very vocal animals. Their vocalizations are distinctive and so convey an abundance of information.

Diversity of Living Birds

The majority of birds, including eagles, geese, and mockingbirds, have the ability to fly. However, some birds, such as emus, penguins, kiwis, and ostriches, are flightless. Traditionally, birds have been classified according to beak and foot type (Fig. 29.19) and, to some extent, on their habitat and behavior. The birds of prey have notched beaks and sharp talons; shorebirds have long, slender, probing beaks and long, stiltlike legs; woodpeckers have sharp, chisel-like beaks and grasping feet; waterfowl have broad beaks and webbed toes; penguins have wings modified as paddles; songbirds have perching feet; and parrots have short, strong, pliershlike beaks and grasping feet.

Check Your Progress 29.5

1. Contrast the characteristics of crocodilians with those of snakes.
2. Explain what features indicate that birds are reptiles.

ISTUDY

a. Bald eagle, *Haliaetus leucocephalus*

b. Pileated woodpecker, *Dryocopus pileatus*

c. Blue-and-yellow macaw, *Ara ararauna*

d. Cardinal, *Cardinalis cardinalis*

Figure 29.19 Bird beaks. **a.** A bald eagle's beak allows it to tear apart prey. **b.** A woodpecker's beak is used to chisel in wood. **c.** A parrot's beak is modified to pry open nuts. **d.** A cardinal's beak allows it to crack tough seeds. (a): 80s_girl/iStock/Getty Images; (b): Daniel Parent/iStock/Getty Images; (c): Ingram Publishing/SuperStock; (d): Gary W. Carter/Corbis/Getty Images

29.6 The Mammals

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe five features of mammals.
2. Discuss the timeline of the evolution of mammals.
3. Identify several features that define each of the three living lineages of mammals.

The **mammals** (class Mammalia, about 5,500 species) evolved from the reptiles. The first mammals were small, about the size of mice. During all the time the dinosaurs flourished (165–65 MYA), mammals remained small in size and changed little evolutionarily. Some of the earliest mammalian groups are still represented today by the monotremes and marsupials. The placental mammals that evolved later went on to live in many habitats, including air, land, and sea.

Characteristics of Mammals

The following characteristics distinguish mammals.

Hair

One of the most distinguishing characteristics of mammals is the presence of hair. Hair provides insulation against heat loss, and being endothermic allows mammals to be active even in cold weather. The color of hair can camouflage a mammal and help the animal blend into its surroundings. In addition, hair can be ornamental and can serve sensory functions.

Mammary Glands

Mammary glands enable females to feed (nurse) their young without having to leave them to collect food, as birds do. Nursing also creates a bond between mother and offspring that helps ensure parental care while the young are helpless, and it provides antibodies to the young from the mother through the milk.

Skeleton

The mammalian skull accommodates a larger brain relative to body size than does the reptilian skull. Also, mammalian cheek teeth are differentiated as premolars and molars. The vertebrae of mammals are highly differentiated; typically, the middle region of the vertebral column is arched, and the limbs are under the body rather than out to the sides.

Internal Organs

Efficient respiratory and circulatory systems ensure a ready oxygen supply to muscles whose contraction produces body heat. Like birds, mammals have a double-loop circulatory pathway and a four-chambered heart. The kidneys are adapted to conserving water in terrestrial mammals. The nervous system of mammals is highly developed. Special senses in mammals are well-developed, and mammals exhibit complex behavior.

ISTUDY

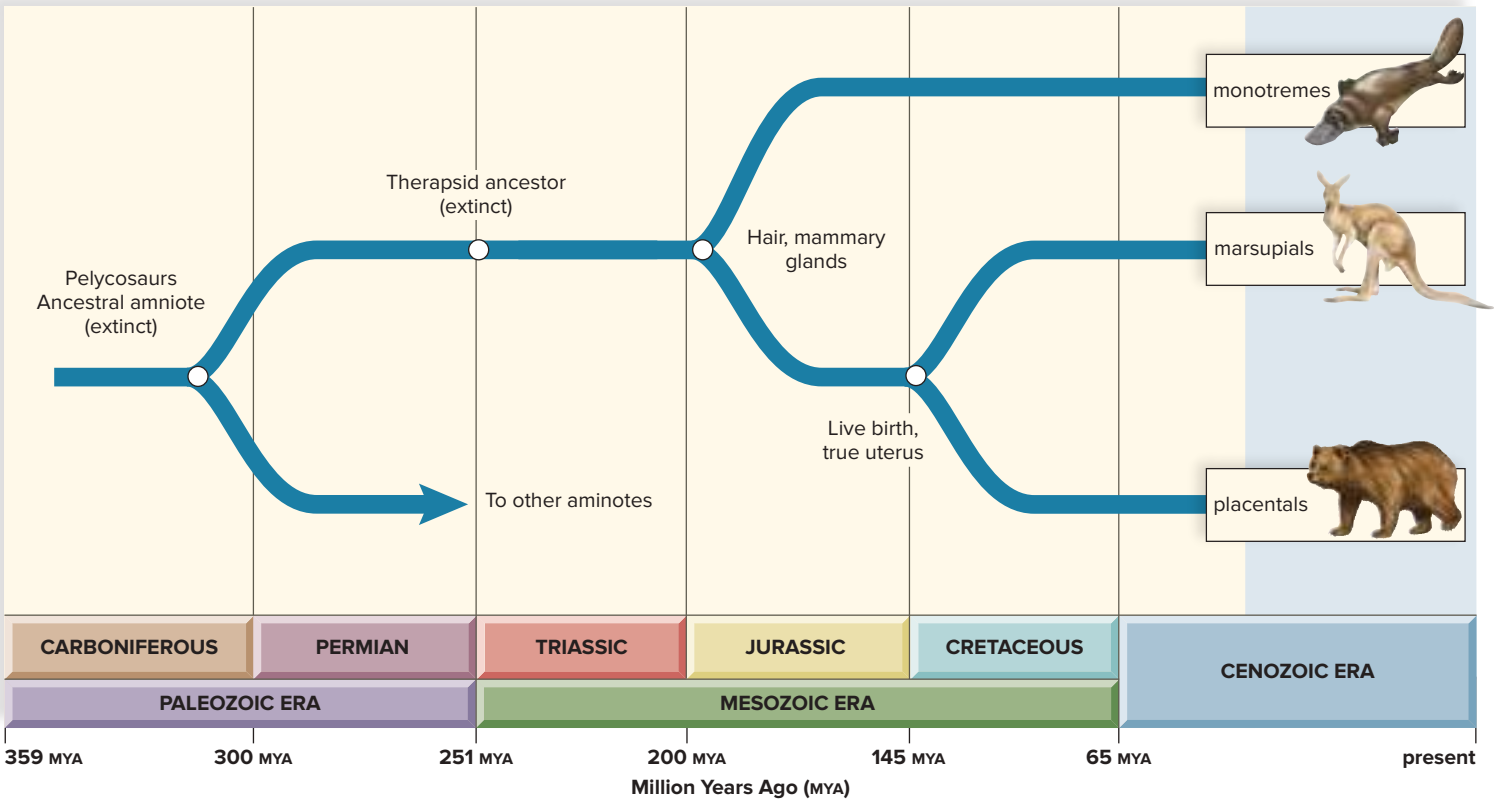


Figure 29.20 Timeline of the evolution of mammal groups. Only three lineages of mammals exist on Earth today, the monotremes, marsupials, and placentals (eutherians). Monotremes are the oldest group of mammals, have hair and mammary glands, but do not give live birth—they lay eggs. The marsupials and placental lineages evolved in the early Cretaceous. Both marsupials and placentals give live birth. In the placental mammals, however, the placenta is more complex and specialized than in the marsupials. Both marsupials and placentals have a true uterus but it is structurally different.

Internal Development

In most mammals, the young are born alive after a period of development in the uterus, a part of the female reproductive tract. Internal development shelters the young and allows the female to move actively about while the young are developing.

The Evolution of Mammals

Mammals share an amniote ancestor with reptiles (see Fig. 29.16). Their more immediate ancestors in the Mesozoic era had a synapsid skull. The first true mammals appeared during the Triassic period, about the same time as the first dinosaurs, and were similar in size to mice. During the reign of the dinosaurs (100 million years), mammals were a minor group that changed little. The common ancestor of all three mammal groups appeared in the late Triassic–early Jurassic period, about 220 MYA (Fig. 29.20). The earliest mammalian group was the monotremes. The marsupials probably originated in the Americas and then spread through South America and Antarctica to Australia before these continents separated. Placental mammals, the third branch of the mammalian lineage (Fig. 29.20), originated in Eurasia and spread to the Americas also by land connections that existed between the continents during the Mesozoic. The placental mammals underwent an adaptive radiation into the habitats previously occupied by the dinosaurs.

Monotremes

Monotremes (Gk. *monos,* "one"; *trema,* "hole") are egg-laying mammals that include only the duckbill platypus (Fig. 29.21*a*) and several species of echidna, or spiny anteaters (Fig. 29.21*b*). The term *monotreme* refers to the presence of a single urogenital opening, the cloaca, which is a shared excretory and reproductive canal. Monotremes, unlike other mammals, lay hard-shelled amniotic eggs (Table 29.2). No embryonic development occurs inside the female's body. The female duckbill platypus lays her eggs in a burrow in the ground. She incubates the eggs, and after hatching, the young lick up milk that seeps from modified sweat glands on the mother's abdomen.

Echidnas, which feed mainly on termites, have pores that seep milk in a shallow belly pouch formed by skin folds on each side. The egg moves from the cloaca to this pouch, where hatching takes place and the young remain for about 53 days. Then, they stay in a burrow, where the mother periodically visits and nurses them.

Marsupials

The **marsupials** (Gk. *marsupium,* "pouch") are also known as the pouched mammals. Marsupials include kangaroos, koalas, Tasmanian devils, wombats, sugar gliders, and opossums. Female marsupials have unique reproductive systems that include two true uteruses and up to three vaginal canals (Table 29.2). The embryos of marsupials are nourished by a yolk-based type of placenta inside the female's body, but they are born in a very immature condition. Newborns are typically hairless and have yet to open their eyes, but they crawl up into a pouch on their mother's abdomen. Inside the pouch, they attach to nipples of mammary glands and continue to develop. Frequently, more are born than can be accommodated by the number of nipples, and it's "first come, first served."

ISTUDY

a. Duckbill platypus, *Ornithorhynchus anatinus*

b. Spiny anteater, *Tachyglossus aculeatu*

Figure 29.21 Monotremes. **a.** The duckbill platypus is a monotreme that inhabits Australian streams. **b.** The short-beaked echidna (anteater) is adapted to a life of burrowing. (a): worldswildlifewonders/Shutterstock; (b): H Lansdown/Alamy Stock Photo

Today, marsupial mammals are most abundant in Australia and New Guinea, filling all the typical roles of placental mammals on other continents. For example, among herbivorous marsupials in Australia today, koalas are tree-climbing browsers (Fig. 29.22*a*), and kangaroos are grazers. A significant number of marsupial species are also found in South and Central America. The opossum is the only North American marsupial (Fig. 29.22*b*).

Table 29.2 Female Reproductive Traits of Mammal Lineages

	Milk	Nipples	Birth	Uterus	Placenta	Birth Canal
Monotremes	Yes	No	Eggs	None	None	Cloaca
Marsupials	Yes	Yes	Live	True	Simple (yolk)	Vagina)
Placentals	Yes	Yes	Live	True	Complex (tissue)	Vagina

a. Koala, *Phascolarctos cinereus*

b. Virginia opossum, *Didelphis virginianus*

Figure 29.22 Marsupials. **a**. The koala is an Australian marsupial that lives in trees. **b.** The opossum is the only marsupial in North America. The Virginia opossum is found in a variety of habitats. (a): John White Photos/Moment/Flickr/Getty Images; (b): mauritius images GmbH/Alamy Stock Photo

Placental Mammals

The **placental mammals,** also known as the eutherians, are the dominant group of mammals on Earth. Developing placental mammals are dependent on the placenta (Table 29.2). The **placenta** in placental mammals is a very specialized organ for the exchange of substances between maternal blood and fetal blood. Nutrients are supplied to the growing offspring, and wastes are passed to the mother for excretion. Although the fetus is clearly parasitic on the female, she has the advantage of being able to freely move about while the fetus develops.

Placental mammals lead an active life. The senses are acute, and the brain is enlarged due to the convolution and expansion of the foremost part—the cerebral hemispheres. The brain is not fully developed for some time after birth, and there is a long period of dependency on the parents, during which the young learn to take care of themselves.

Most mammals live on land, but some (e.g., whales, dolphins, seals, sea lions, and manatees) are secondarily adapted to live in water, and bats are able to fly. Although bats are the only mammal that can actually fly, three types of placentals can glide: the flying squirrels, scaly-tailed squirrels, and flying lemurs. There are 20 different orders of placental mammals (Table 29.3).

Humans are mammals, and in Chapter 30 we consider the evolutionary history of humans, an enormously successful mammalian group.

Check Your Progress 29.6

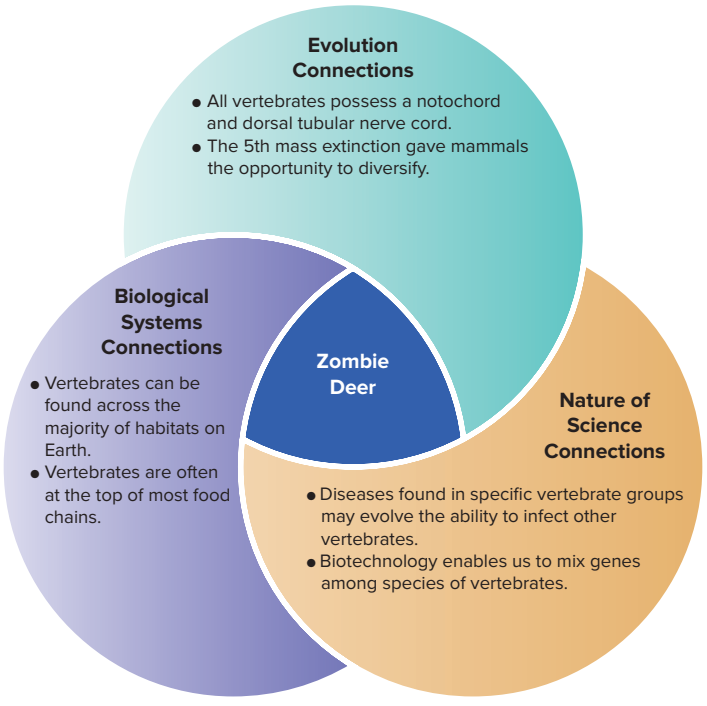
1. Identify two traits that are unique to mammals.
2. Describe features that distinguish the three groups of mammals.

ISTUDY

Table 29.3 Orders of Placental Mammals

Order	Examples	Traits
Cetacea	Whales, dolphins	Marine, no fur, streamlined bodies
Artiodactyla	Cattle, deer, antelope	Even-toed ungulates, horns or antlers, ruminants
Perissodactyla	Horses, rhinoceroses	Odd number toes, nonruminants
Carnivora	Dogs, cats, weasels, minks, stoats	Meat eaters, long canines
Pholidota	Pangolins	Scaly armor, no teeth
Chiroptera	Bats	Fly with membranous wings, some echolocate
Erinaceomorpha	Hedgehogs	Spines, insectivores
Soricomorpha	Shrews, moles	Small, high metabolism, insectivorous
Rodentia	Mice, rats, voles, beavers, squirrels	One pair of specialized incisors that grow continuously
Lagomorpha	Rabbits, hares	Two pairs of incisors that grow continuously, long hindlimbs
Dermoptera	Colugos ("flying lemurs")	Membrane between hands and feet
Scandentia	Tree shrews	Large forward-facing eyes, grasping hand
Primates	Monkeys, apes	Opposable thumb, developed brains, social
Xenarthra	Sloths, armadillos	Special projections on the spine
Afrocoricida	Golden moles, tenrecs	Insectivores
Macroscelidia	Elephant shrews	Small, long snout, long hindlimbs
Tubulidentata	Aardvarks	Insectivore, long snout, coarse fur
Sirenia	Manatees, dugongs	Marine, long whiskers
Hyracoidea	Hyraxes	Small mammals
Proboscidea	Elephants	Largest land mammals, trunk, tusks

CONNECTING *THE* CONCEPTS



ISTUDY

What Did You Learn?

29.1 The Chordates

At some time in their life history, all **chordates** (sea squirts, lancelets, and vertebrates) have a **notochord,** a dorsal tubular nerve cord, pharyngeal pouches, and a postanal tail. Pharyngeal pouches develop into the **gills** of aquatic vertebrates.

- Lancelets (**cephalochordates**) and sea squirts (**urochordates**) are the nonvertebrate chordates. Lancelets are the only chordate to have all four characteristics in the adult stage. Sea squirts lack chordate characteristics (except gill slits) as adults, but they have a larva that could be ancestral to the vertebrates.

29.2 The Vertebrates

As embryos, vertebrates have the four chordate characteristics, plus they develop a vertebral column, a skull, an endoskeleton, and internal organs. The adult vertebral column is derived from the embryonic notochord. Vertebrates have also undergone cephalization, and they have paired appendages and well-developed internal organs.

- Vertebrate evolution is marked by the appearance of vertebrae, jaws, a bony skeleton, lungs, limbs, and the amniotic egg. The **gnathostomes,** or animals with jaws, gave rise to **tetrapods,** or animals with four limbs. **Amniotes** developed an egg surrounded by membranes and amniotic fluid.

29.3 The Fishes

The **fishes** are the largest group of vertebrates. All are ectotherms, meaning their temperature fluctuates with the environment.

- The earliest known fishes, the **ostracoderms,** lacked jaws and paired muscular appendages **(fins).** They are represented today by the **jawless fishes** (also called **agnathans**), such as hagfishes and lampreys.
- Ancestral bony fishes like the **placoderms,** which had jaws and fins, gave rise during the Devonian period to two groups: today's **cartilaginous fishes** (skates, rays, and sharks) and the **bony fishes,** including the **ray-finned bony fishes** and the **lobe-finned fishes.** The ray-finned fishes (Actinopterygii) became the most diverse group among the vertebrates. Many of these have a gas-filled **swim bladder** that affects their buoyancy. Ancient lobe-finned fishes (Sarcopterygii) gave rise to the coelacanths, **lungfishes,** and amphibians.

29.4 The Amphibians

Amphibians are ectothermic tetrapods represented primarily today by frogs and salamanders. Most have thin, moist, nonscaly skin, a three-chambered heart, small lungs supplemented by cutaneous respiration, and finely tuned senses. Most frogs and some salamanders return to the water to reproduce and then metamorphose into terrestrial adults.

29.5 The Reptiles

Reptiles (today's alligators and crocodiles, birds, turtles, tuataras, lizards, and snakes) lay a shelled **amniotic egg,** which allows them to reproduce on land.

- The skull of an amniote may be **anapsid,** with no opening behind the eye socket; **synapsid,** with one opening, or **diapsid,** with two openings. There are two views of reptile evolution. The classical view considers turtles, with an anapsid skull, as having a separate ancestry from other reptiles. The other reptiles, with a diapsid skull, include the crocodilians, dinosaurs, and birds. A more recent view of reptile evolution proposes that turtles are actually very specialized diapsid archosaurs, along with birds and crocodiles.
- **Birds** have reptilian features, including scales (feathers are modified scales), a tail with vertebrae, and clawed feet. The feathers of birds help them maintain a constant body temperature. Birds are adapted for flight: Their bones are hollow, their shape is compact, their breastbone is keeled, and they have well-developed sense organs. Unlike other reptiles, birds are **endotherms** that maintain a constant internal temperature.

29.6 The Mammals

Mammals share an amniote ancestor with reptiles, but they have a synapsid skull. Mammals remained small and insignificant while the dinosaurs existed, but when dinosaurs became extinct at the end of the Cretaceous period, mammals became the dominant land organisms.

Mammals are vertebrates with hair and mammary glands. Hair helps them maintain a constant body temperature, and the mammary glands allow them to feed and establish an immune system in their young.

- **Monotremes** lay eggs, while **marsupials** have a pouch in which the newborn crawls and continues to develop.
- The **placental mammals,** or eutherians, are the most varied and numerous, and they retain offspring inside the female, where they receive nourishment via the **placenta** until birth.

Thinking Critically About Biology

1. What advantages do cartilaginous fishes possess that have allowed for their success over the past 400 million years?
2. *Archaeopteryx* was a birdlike reptile that had a toothed beak. Give an evolutionary explanation for the elimination of teeth in a bird's beak.
3. While amphibians have rudimentary lungs, skin is also a respiratory organ. Why would a thin skin be more sensitive to pollution than lungs?
4. Most paleontologists believe that the major function of feathers in dinosaurs was to insulate against the cold. What are some other possible advantages to feathers, compared to scales?

Making It Relevant to You

1. Which biological systems are impacted the most by a prion-based disease?
2. Would you avoid eating deer, sheep, or beef if you knew that a prion-based disease was present in your area?
3. What are the potential ecological consequences if prion-based diseases evolve the ability to jump the species barrier and infect other mammals?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

30

Human Evolution

(left): Clément Philippe/ArTerra Picture Library/age fotostock; (right): 4x6/iStockphoto/Getty Images

Chapter Outline

30.1 Evolution of Primates
30.2 Evolution of Humanlike Hominins
30.3 Evolution of Early Genus *Homo*
30.4 Evolution of Later Genus *Homo*

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 15.3 What is a transitional fossil?

Section 17.1 What is the difference between the evolutionary species concept and the biological species concept?

Table 29.3 To which order of mammals do the hominins belong?

The Neanderthal in You

Human evolution is a complex topic with a body of knowledge that is constantly growing. With advances in molecular analysis, DNA evidence has been able to provide important information about the relatedness between Neanderthals (*Homo neanderthalensis*) and modern humans (*Homo sapiens*).

Starting around 100,000 years ago, evidence indicates that *Homo sapiens* migrated into the *Homo neanderthalensis* territory. Molecular studies have revealed that as recently as 47,000 years ago, modern humans and Neanderthals were inbreeding. Neanderthals contributed between 1% and 4% of the genes found in modern humans of Eurasian descent. For example, the SPAG17 protein helps move sperm, PCD16 assists in wound healing, TTF1 is involved in the transcription process, and RPTN is a protein found in the hair, skin, and sweat glands.

These discoveries, among others, have renewed interest in discovering the genes that distinguish us from our closest ancestors and, in the process, in developing a greater understanding of the evolutionary history of our species.

As you read through this chapter, think about the following questions:

1. What was the last common ancestor of both Neanderthals and *Homo sapiens*?
2. What is the current thinking on the evolutionary relationship of the Neanderthals to modern humans?

Following *the* Themes

CHAPTER 30 – HUMAN EVOLUTION

Evolution	Like all other organisms, humans have an evolutionary history and are related to all other species through common ancestors.
Nature of Science	By studying the fossil record, and using comparative genomics, scientists are piecing together the story of human evolution.
Biological Systems	The physical and physiological characteristics of the hominins have been shaped by their environment.

ISTUDY

30.1 Evolution of Primates

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the major groups of primates.
2. Discuss the traits common to the primates.
3. Arrange the groups of primates in an evolutionary tree that shows their relationships.

Primates (L. *primus,* "first") are an order of mammals that includes the prosimians, monkeys, apes, and humans (Fig. 30.1). Most primates are adapted for an **arboreal** life—that is, for living in trees. The evolution of primates is characterized by trends toward mobile limbs, grasping hands, a flattened face and stereoscopic vision, a large and complex brain, and a reduced reproductive rate.

Mobile Forelimbs and Hindlimbs

Primates tend to have prehensile hands and feet, meaning they are adapted for grasping and holding. In most primates, flat nails have replaced the claws of ancestral primates, and sensitive pads on the undersides of fingers and toes assist in the grasping of objects. All primates have thumbs, but they are only truly opposable in Old World monkeys, great apes, and humans. Because an **opposable thumb** can touch each of the other fingers, the grip is both powerful and precise (Fig. 30.2). In all but humans, primates with opposable thumbs also have opposable toes.

The evolution of the primate limb was a very important adaptation for their life in trees. Mobile limbs with clawless, opposable digits allow primates to freely grasp and release tree limbs. They also allow primates to easily reach out and bring food, such as fruit, to the mouth.

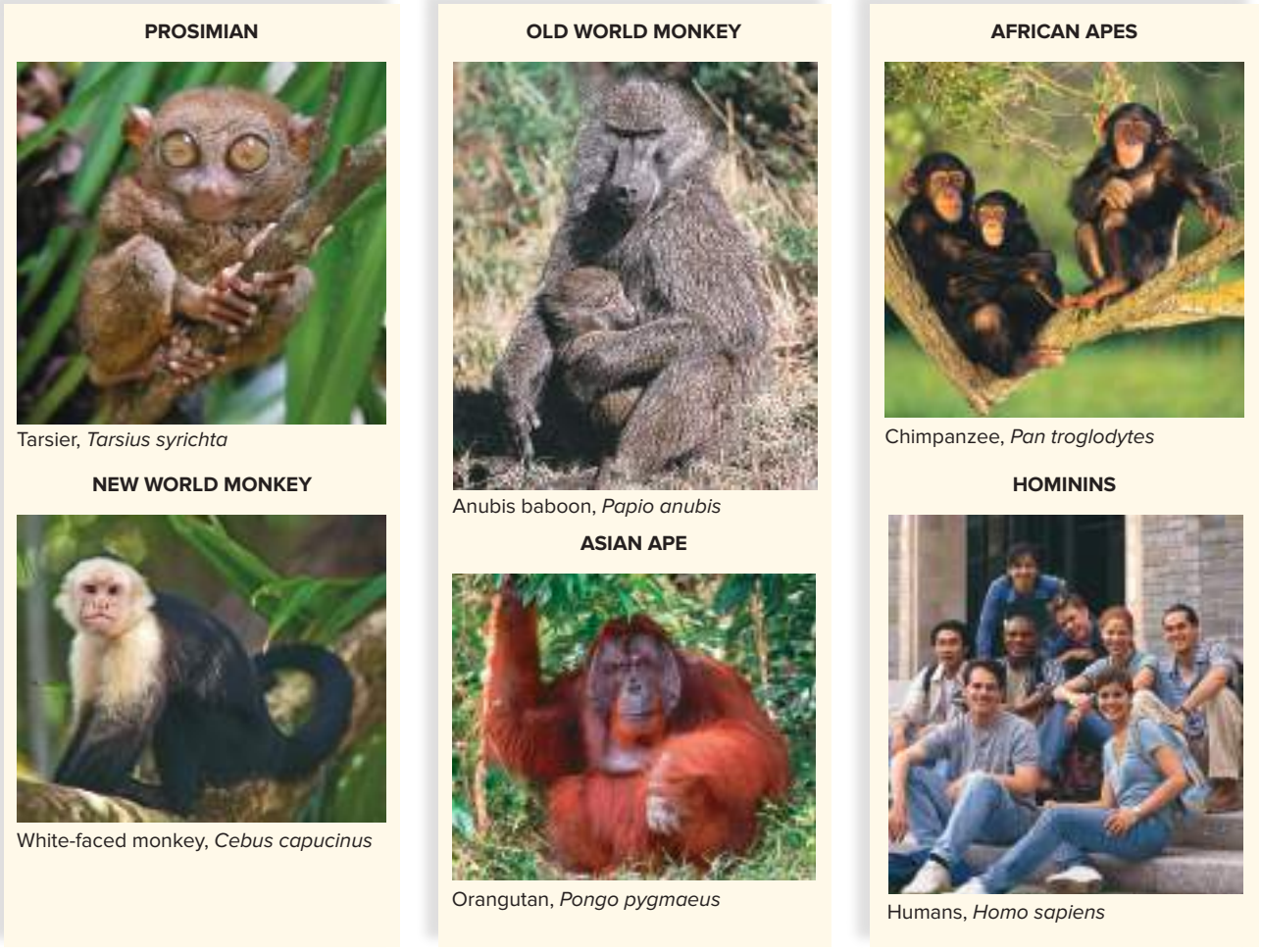


Figure 30.1 Primate diversity. Today's prosimians may resemble the earliest groups of primates. Modern monkeys are divided into the New World monkeys and the Old World monkeys. The apes can be divided into the Asian apes (orangutans and gibbons) and the African apes (chimpanzees and gorillas). Humans (hominins) are also primates. (tarsier): Magdalena Biskup Travel Photography/Flickr/Getty Images; (monkey): Paul Souders/Stone/Getty Images; (baboon): Tierbild Okapia/Science Source; (orangutan): Tim Davis/Science Source; (chimpanzee): Fuse/Getty Images; (humans): Comstock Images/Getty Images

ISTUDY

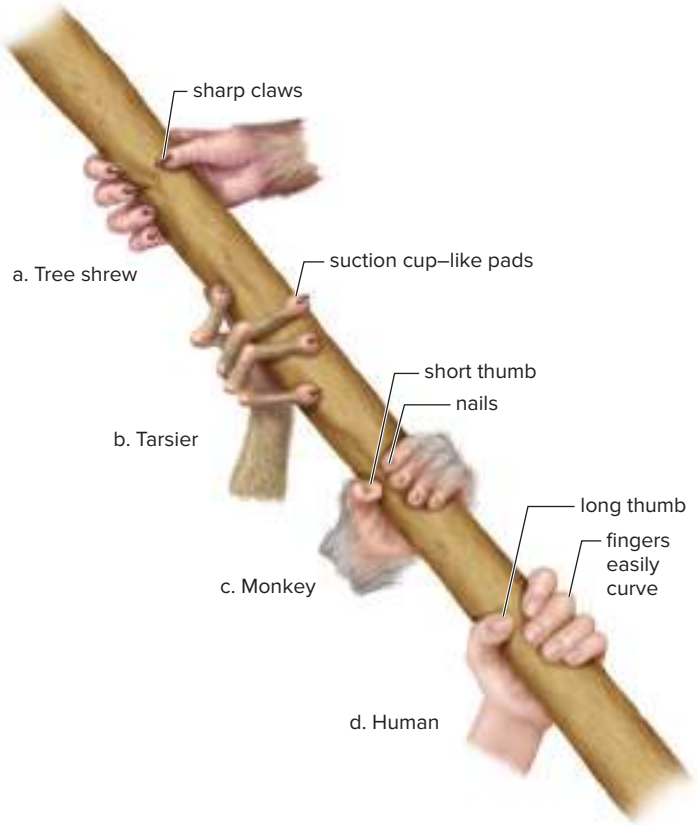


Figure 30.2 Evolution of primate hand. Comparison of primate hands (tarsier, monkey, and human) to that of a tree shrew. The long thumb of a human is opposable.

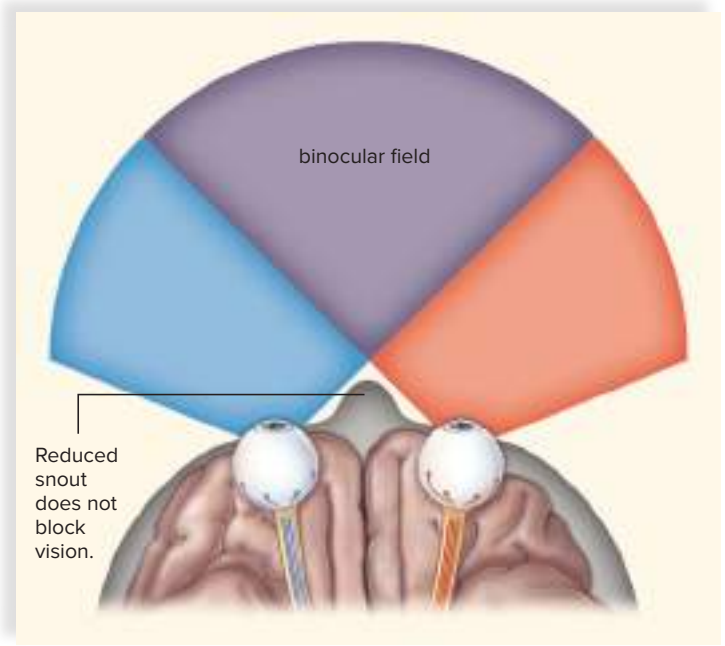


Figure 30.3 Stereoscopic vision. In primates, the snout is reduced, and the eyes are at the front of the head. The result is a binocular field that aids depth perception and provides stereoscopic vision.

Stereoscopic Vision

A foreshortened snout and a relatively flat face are also evolutionary trends in primates. These may be associated with a general decline in the importance of smell and an increased reliance on vision. In most primates, the eyes are located in the front, where they can focus on the same object from slightly different angles (Fig. 30.3). The result is **stereoscopic vision,** a form of vision that combines three-dimensional imaging with good depth perception. This trait permits primates to make accurate judgments about the distance and position of adjacent tree limbs, and allow for effective movement.

Some primates, humans in particular, have color vision and greater visual acuity because the retina contains cone cells in addition to rod cells (see Section 38.3). Rod cells are activated in dim light, but the blurry image is in shades of gray. Cone cells require bright light, but the image is sharp and in color. Humans and the apes have three different kinds of cone cells, which are able to discriminate among greens, blues, and reds.

Large, Complex Brain

Sense organs are only as beneficial as the brain that processes their input. The evolutionary trend among primates is toward a larger and more complex brain. This is evident when comparing the brains of prosimians, such as lemurs and tarsiers, with those of apes and humans. In apes and humans, the portion of the brain devoted to smell is smaller, and the portions devoted to sight have increased in size and complexity. Also, more of the brain is devoted to controlling and processing information received from the hands and the thumbs. The result is good hand-eye coordination. A larger portion of the brain is devoted to communication skills, which supports primates' tendency to live in social groups. Because the brain plays an important role in primate evolution, we will refer back to brain size (in cubic centimeters, or cc) throughout this chapter.

Reduced Reproductive Rate

One other trend in primate evolution is a general reduction in the rate of reproduction, associated with increased age of sexual maturity and extended life spans. Gestation is lengthy, allowing time for forebrain development. One birth at a time is the norm in primates; it is difficult to care for several offspring in the trees while moving from limb to limb. The juvenile period of dependency is extended, and there is an emphasis on learned behavior and complex social interactions.

Sequence of Primate Evolution

Figure 30.4 traces the evolution of primates during the Cenozoic era. The designation **hominins** includes humans and species very closely related to humans. **Hominids** include our closest primate relatives, including chimpanzees, gorillas, and orangutans. Molecular evidence suggests that humans and chimpanzees shared a common ancestor between 5 and 7 MYA. For this reason, the statement that humans descended from chimpanzees (or

ISTUDY

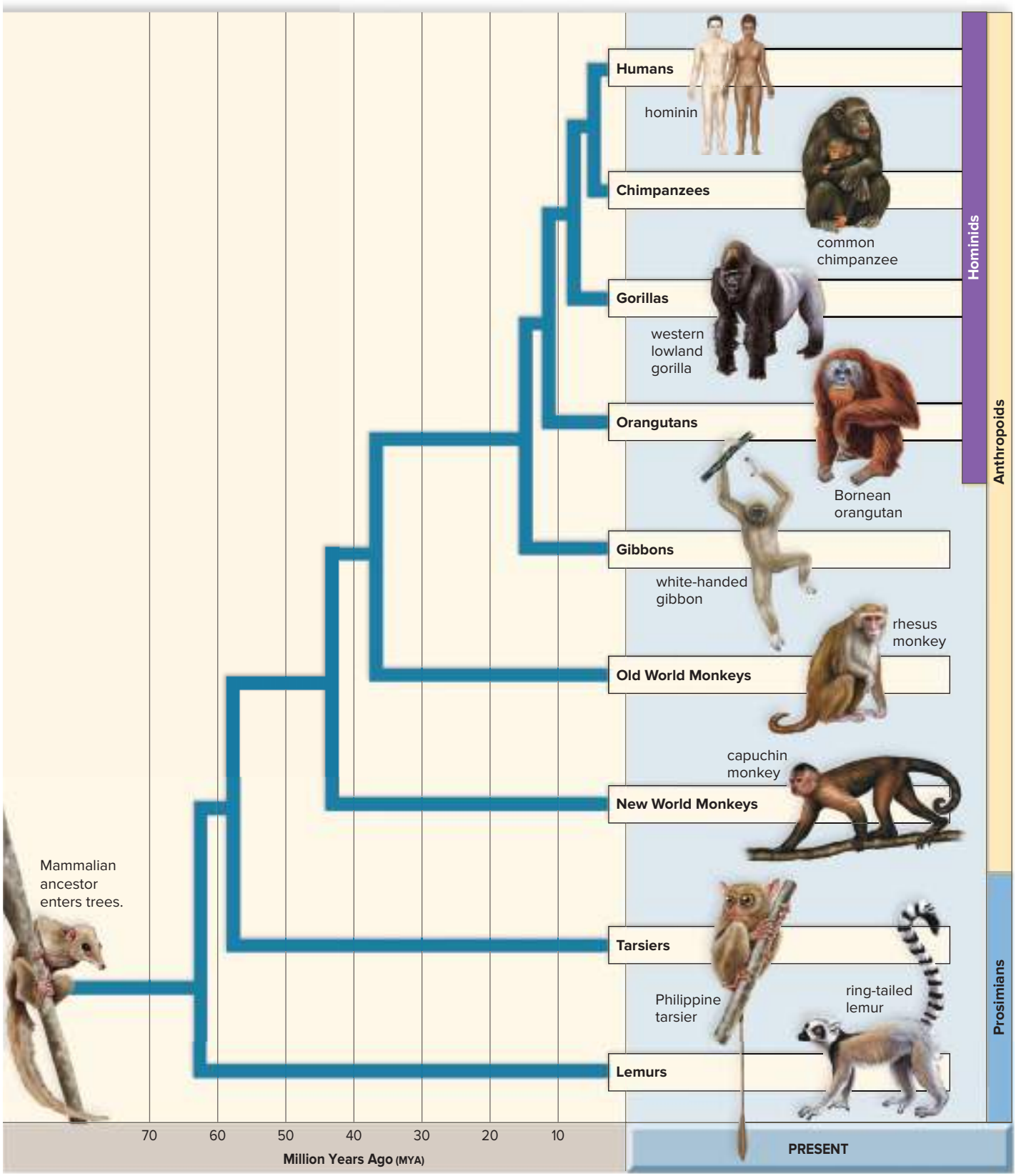


Figure 30.4 Evolution of primates. Primates are descended from an ancestor that may have resembled a tree shrew. The time when each type of primate diverged from the main line of descent is known from the fossil record.

ISTUDY

THEME Nature of Science

A Genomic Comparison of Homo sapiens *and Chimpanzees*

A wealth of genetic evidence suggests that humans and chimpanzees are closely related, despite the fact that chimpanzees have 48 chromosomes and *Homo sapiens* have 46. At first, the difference in chromosome number was considered significant—so significant that for a long time the great apes (chimpanzees, gorillas, and orangutans) and humans were classified into different families. The apes were in the family Pongidae, while humans were in the family Hominidae. However, in 1991, investigators at Yale University showed that human chromosome 2 is actually a fusion of two chimpanzee chromosomes (Fig. 30A).

Additional evidence was provided by the study of transposons. In Section 14.4, you learned that transposons are mobile elements in the genome. Many copies of transposons are present in the human genome, most of which are no longer mobile and instead remain in one location. They are relics of past retrovirus infections. Because these infections are random, any similarity in transposon patterns between two species can be considered evidence of a common ancestor.

Through the course of many studies, investigators have found that humans and chimpanzees have similar patterns of transposons in their genome (see Section 14.4). An example is shown in Fig. 30B of an *Alu* element (a type of transposon) in the vicinity of the hemoglobin gene in both chimpanzees and humans. This similarity suggests that the transposon inserted itself into this location before the chimpanzee–human lineages split.

Pseudogenes are nonfunctional copies of genes that were active in the past. Most pseudogenes are inactive due to mutations that prevent them from coding for a functional protein. The pattern of pseudogenes in humans is most similar to that found in the chimpanzees, but it varies slightly from the patterns found in the other great apes. These and other studies have caused a reclassification of the primates most closely related to us into a group called the hominids.

Modern genomic data show that the base sequence of chimpanzees and humans differs only by 1.5%. Because we now consider that chimpanzees and humans are genetically similar, geneticists have begun to focus on what specific genes make us different—even though our base sequences are quite similar, significant differences do exist. For example, when we compare the two genomes, we find that many DNA stretches (about 5 million in all) are absent from one or the other genomes.

We know that in the evolution of mammals there was an explosion in the amount of noncoding sequences relative to the number of coding genes. So, a common mammalian ancestor may have had a larger quantity of noncoding DNA, and each mammalian ancestor (for example, chimps and humans) may have lost different DNA stretches since their lines of descent separated. These noncoding regions still play a role in gene expression (see Section 13.2), so the stretches of noncoding DNA that were retained by a particular primate may account for the anatomical differences. Evidence suggests that the genes controlling the development of the brain may have been affected the most by the gaps present in the chimpanzee genome compared to the human genome. This may account for the larger size of our brains compared to that of chimpanzees today.

Questions to Consider

1. Which would you consider to have a pattern of transposons and pseudogenes that is closer to that of humans, a prosimian or an Old World monkey?
2. For what additional differences between chimps and humans would you screen the genome for evidence of lost DNA stretches?

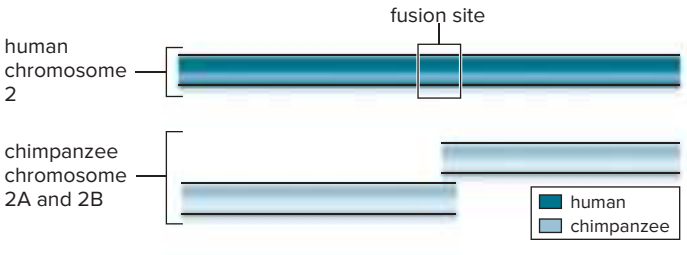


Figure 30A A comparison of human and chimpanzee chromosomes. Human chromosome 2 is a fusion of two chimpanzee chromosomes.

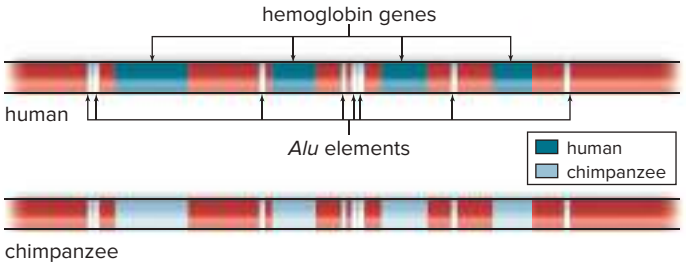


Figure 30B A comparison of transposon patterns. Similarities in transposon patterns, in this case *Alu,* suggest a close common ancestor between humans and chimpanzees.

monkeys and apes) is not accurate. The Nature of Science feature, "A Genomic Comparison of *Homo sapiens* and Chimpanzees," presents DNA comparisons between modern humans and chimpanzees.

The **anthropoids** (Gk. *anthropos,* "man"; *eides,* "like") include all of the hominids, as well as the gibbons, Old World monkeys, and New World monkeys. Old World monkeys, native to Africa and Asia, lack prehensile tails and have protruding noses. Some of the better-known Old World monkeys are the baboon, a ground dweller, and the rhesus monkey, which has been used extensively in medical research. New World monkeys, which often have long, prehensile tails and flat noses, evolved in South America and are now also found in Central America and parts of Mexico. Two of the well-known New World monkeys are the spider monkey and the capuchin, the "organ grinder's monkey."

ISTUDY

Primate fossils similar to monkeys are first found in Africa, dated about 45 MYA. It is hypothesized that ancestors of New World monkeys arrived in South America around 40 MYA. However, it is uncertain how they arrived because the Atlantic crossing was so expansive.

The transitional link between the monkeys, gibbons, and early hominids (around 35 MYA) is best represented by a fossil classified as *Proconsul. Proconsul* was about the size of a baboon, and the size of its brain (165 cc) was also comparable. This fossil species didn't have the tail of a monkey (Fig. 30.5), but its limb proportions suggest it walked as a quadruped on top of tree limbs, as monkeys do. Although primarily a tree dweller, *Proconsul* may have also spent time exploring nearby environs for food.

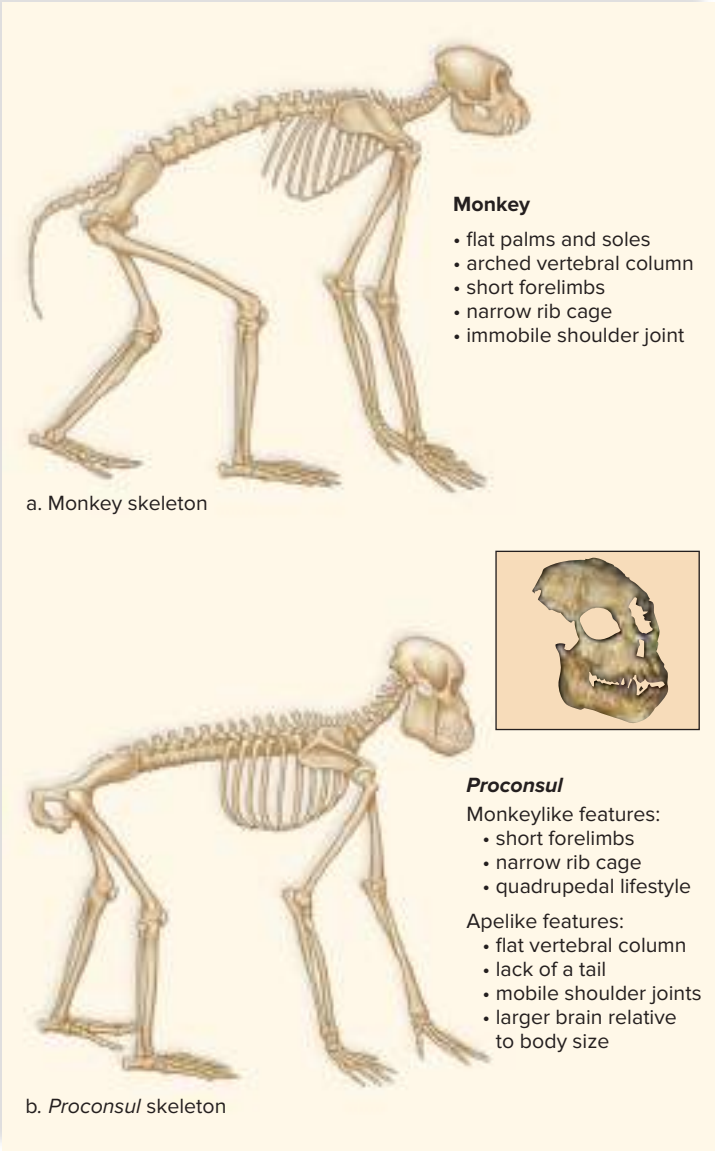


Figure 30.5 Monkey skeleton compared to *Proconsul* skeleton. Comparison of a monkey skeleton (**a**) with that of *Proconsul* (**b**) shows various dissimilarities, indicating that *Proconsul* is more related to today's apes than to today's monkeys. (b): Mary Evans/Natural History Museum/Mary Evans Picture Library Ltd/age fotostock

Proconsul was probably ancestral to the dryopithecines, from which the early hominids arose. About 10 MYA, Africarabia (Africa plus the Arabian Peninsula) joined with Asia, and the apes migrated into Europe and Asia. In 1966, Spanish paleontologists announced the discovery of a specimen of *Dryopithecus* dated at 9.5 MYA near Barcelona. The anatomy of these bones clearly indicates that *Dryopithecus* was a tree dweller and locomoted by swinging from branch to branch, as gibbons do today. They did not walk along the top of tree limbs as *Proconsul* did.

The **prosimians** (L. *pro,* "before"; *simia,* "ape, monkey"), represented by lemurs and tarsiers, were the first type of primate to diverge from the common ancestor for all the primates. All primates share one common mammalian ancestor, which lived about 55 MYA. This ancestor may have resembled today's tree shrews.

Check Your Progress 30.1

1. Identify primate traits that are adaptive for living in trees.
2. Identify the location of hominins, hominids, anthropoids, and prosimians in the evolutionary tree of the primates.
3. Describe the characteristics that humans share with chimpanzees.

30.2 Evolution of Humanlike Hominins

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the significance of bipedalism in hominin evolution.
2. Compare the features of "Ardi" with those of "Lucy."
3. Summarize the importance of ardipithecines and australopithecines in hominin evolution.

There have been many recent advances in the study of the hominins, and recent discoveries of fossils in Africa are challenging our view of how early hominins evolved. Paleontologists use certain anatomical features when they try to determine if a fossil is a hominin. These features include **bipedalism,** or the ability to stand erect and walk on two feet. Humans are bipedal, while apes are quadrupedal (walk on all fours). As we will see, early humanlike hominins are not in the genus *Homo,* but they are considered closely related to humans because they exhibit bipedalism. Although bipedalism places stress on the spinal column, the upright posture frees the hands for tool use.

In addition, characteristics of the skull, such as the shape of the face and brain size are used as identifying criteria. Today's humans have a flatter face and a more pronounced chin than do the apes, because the human jaw is shorter than that of the apes. Then, too, our teeth are generally smaller and less specialized. We don't have the sharp canines of an ape, for example. Chimpanzees have a brain size of about 400 cubic centimeters (cc), and modern humans have a brain size of about 1,360 cc.

ISTUDY

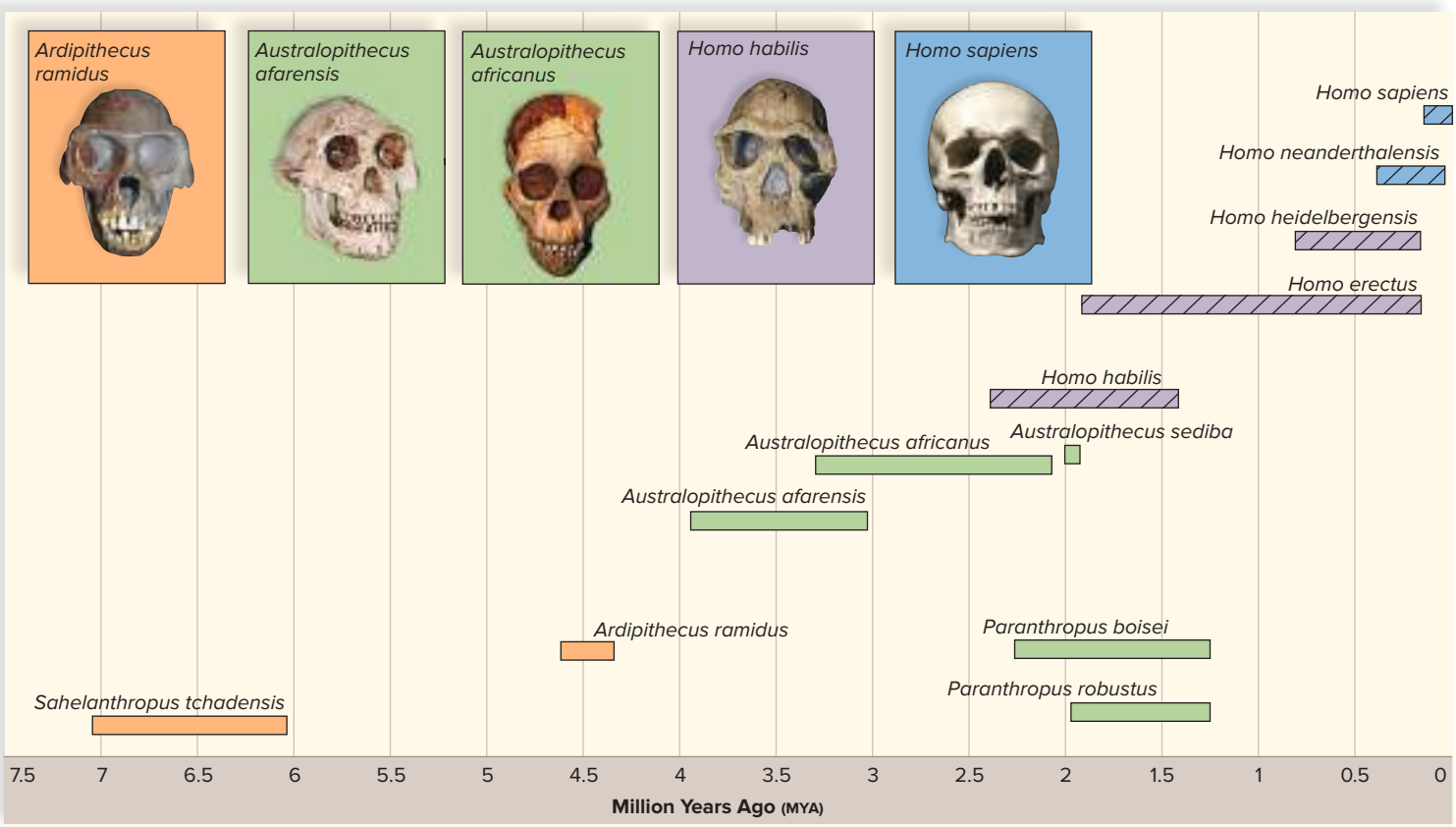


Figure 30.6 Human evolution. Several groups of extinct hominins preceded the evolution of modern humans. These groups have been divided into the early humanlike hominins (orange), later humanlike hominins (green), early species (lavender), and finally the later species (blue). Only modern humans are classified as *Homo sapiens*. (photos): (*Ardipithecus ramidus*): Sabena Jane Blackbird/Alamy Stock Photo; (*Australopithecus afarensis*): Scott Camazine/Alamy Stock Photo; (*Australopithecus africanus*): Philippe Plailly/Science Source; (*Homo habilis*): Sabena Jane Blackbird/Alamy Stock Photo; (*H. sapiens*): JazzIRT/E+/Getty Images

Timeline of Hominin Evolution

Figure 30.6 provides a view of our current understanding on the timeline of hominin evolution. The orange and green bars signify the early humanlike hominins, a lavender bar signifies an early species of the genus *Homo,* while the later members of genus *Homo* are in blue. The length of each bar indicates when evidence of the species first appears in the fossil record until the estimated time of its extinction.

In many cases, it's hard to decide which fossils are hominins, because human features evolved gradually and at different rates. However, paleontologists have identified several fossils dated around the time of the split between the human and ape lineages. One of these is *Sahelanthropus tchadensis,* a species that lived in West Central Africa between 6 and 7 MYA. Initially, only cranial fragments of this species were recovered. However, they suggested that a point on the back of the skull where the neck muscles would have attached indicates bipedalism. This idea that *S. tchadensis* walked upright was further supported by additional fossil discoveries of leg bones from this species. The brain of *S. tchadensis* was similar in size to that of modern chimpanzees, and some scientists suspect it may have been a common ancestor to chimpanzees and humans. The skull of this fossil is very similar to that of the ardipithecines, discussed next.

Ardipithecines

Two species of **ardipithecines** have been uncovered, *Ardipithecus kadabba* and *A. ramidus.* Only teeth and a few bone bits have been found for *A. kadabba,* and these have been dated between 5.2 and 5.8 MYA. A more extensive collection of fossils has been collected for *A. ramidus.* To date, over 100 skeletons, all dated to 4.4 MYA, have been identified from this species; all were collected near a small town in Ethiopia, East Africa. In 2009, scientists announced they had reconstructed these fossils to form a female fossil called Ardi.

Some of Ardi's features are primitive, like that of an ape such as *Dryopithecus* (see Section 30.1), but others are like that of a human. Ardi was about the size of a chimpanzee, standing about 120 cm (4 ft) tall and weighing about 55 kg (110 lb). It appears that males and females were about the same size.

Ardi had a small head compared to the size of her body. The skull has the same features as *Sahelanthropus tchadensis* but is smaller. Ardi's brain size was around 300 to 350 cc, slightly less than that of a chimpanzee brain (around 400 cc), and much smaller than that of a modern human (1,360 cc). The nose and mouth project forward, and the forehead is low with heavy eyebrow ridges, a combination that makes the face more primitive than that of the australopithecines (discussed next). Ardi's teeth were small and

ISTUDY

Figure 30.7 Adaptations for walking erect. A human skeleton compared to an ardipithecine (Ardi). In both skeletons, the spine exits from the center of the skull. A broad pelvis (green) causes the femurs to angle (red arrows) toward the broadened knee joints. However, the skeleton of Ardi has an opposable toe, indicating that this species still lived in trees.

like those of an omnivore. She lacked the strong, sharp canines of a chimpanzee, and her diet probably consisted mostly of soft, rather than tough, plant material.

Ardi could walk erect, but she spent a lot of time in trees. Notice in Figure 30.7 that in both the human and Ardi skeletons, the spine exits from the center of the skull rather than toward the rear. Also, the femurs angle inward toward the knees (red arrows). These skeletal features assist in walking erect by placing the trunk's center of gravity squarely over the feet. Ardi's pelvis and hip joints are broad enough to keep her from swaying from side to side (as chimps do) while walking. The knee joint in both humans and Ardi is modified to support the body's weight, because the bones broaden at this joint.

Ardi's feet have a bone, missing in apes, that would keep her feet squarely on the ground, a sure sign that she was bipedal and not a quadruped like the apes. Nevertheless, like the apes, she has an opposable big toe. Opposable toes allow an animal's feet to grab hold of a tree limb.

The wrists of Ardi's hands were flexible, and most likely she moved along tree limbs on all fours, as ancient apes did. Modern apes *brachiate*—use their arms to swing from limb to limb. Ardi did not do this, but her shoulders were flexible enough to allow her to reach for limbs to the side or over her head. The general conclusion is that Ardi moved carefully in trees. Although the top of her pelvis is like that of a human, and probably served as the attachment of muscles needed for walking, the bottom of the pelvis served as an attachment for the strong muscles needed for climbing trees.

Until recently, it has been suggested that bipedalism evolved when a dramatic change in climate caused the forests of East Africa to be replaced by grassland. However, evidence suggests that Ardi lived in the woods, which questions the advantage that walking erect would have afforded her. Bipedalism does provide an advantage in caring for a helpless infant by allowing it to be carried by hand from one location to another. It is also possible that bipedalism may have benefited the males of the species as they foraged for food on the floor of the forests. More evidence is needed to better understand this mystery, but one thing is clear—the ardipithecines represent a link between our quadruped ancestors and the bipedal hominins.

Australopithecines

The **australopithecines** (called australopiths for short) are a group of hominins that evolved and diversified in Africa from 4 MYA until about 1.5 MYA. In Figure 30.6, the australopiths are represented by green-colored bars. The australopiths had a small brain (an apelike characteristic) and walked erect (a humanlike characteristic). Therefore, it seems that not all human characteristics evolved at the same time. This is an example of **mosaic evolution,** meaning that different body parts change at different rates and, therefore, at different times.

Australopiths stood about 100–115 cm in height and had relatively small brains, averaging about 370–515 cc—slightly larger than that of a chimpanzee. Males were distinctly larger than females. Some australopiths were slight of frame and termed *gracile* (slender). Others were *robust* (powerful) and tended to have massive jaws because of their large grinding teeth. Recent changes in the classification of these groups has separated the gracile types into genus *Australopithecus* and the robust types into genus *Paranthropus.*

There is also evidence that some australopiths consumed meat, as indicated by the recent discovery in Ethiopia of two fossilized bones from large mammals dated 3.4 MYA. The bones bear obvious marks indicating the meat was scraped off with stone tools, providing the earliest evidence of australopiths using stone tools, about 1 million years earlier than scientists had previously thought.

East African Australopiths

The most significant fossil from East Africa is from a species of australopiths called *Australopithecus afarensis.* The female specimen of this species, known as Lucy (Fig. 30.8), had a low forehead and a face that projected forward, with large canine teeth. The body of Lucy was broader than that of an ardipithecine.

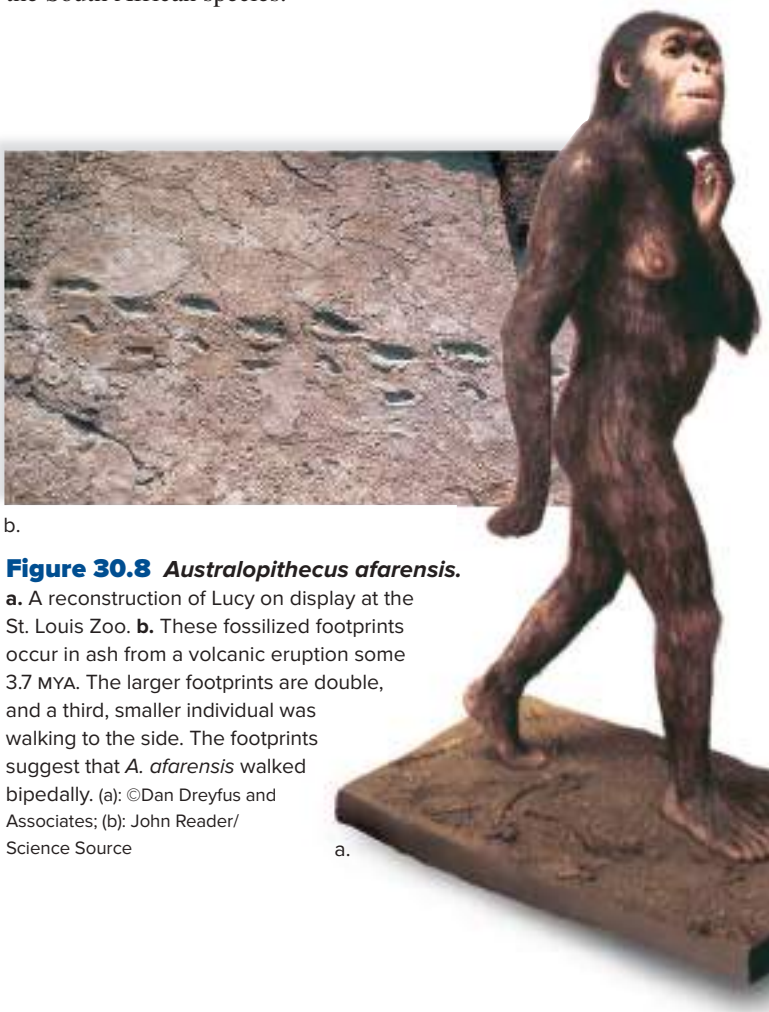
ISTUDY

Although the brain size was small (around 400 cc), Lucy's skeleton indicates she was a biped that stood upright. She stooped a bit like a chimpanzee, and the arms were somewhat proportionally longer than the legs. This suggests brachiation as a possible mode of locomotion in trees. Otherwise, the skeleton was humanlike, even though the pelvis lacked refinements that would have allowed Lucy to walk with a striding gait in a manner similar to modern humans.

Even better evidence of bipedal locomotion comes from a trail of fossilized footprints in Tanzania dated to about 3.7 MYA (Fig. 30.8). The larger footprints are double, as though a smaller being were stepping in the footprints of another, and there are additional footprints off to one side, within hand-holding distance.

Some 30 years after Lucy was discovered, paleontologists discovered a skeleton of a child that has been dated to be 0.1 million year older than Lucy. The face of this skeleton, named Selam but sometimes referred to as "Lucy's baby," looks more like that of an ardipithecine, and the structure of the bones suggests that Selam was not as agile a walker as Lucy.

A. afarensis is a gracile form of australopith and is believed to be ancestral to the robust types found in eastern Africa: *P. boisei* and *P. robustus*. *P. boisei* had a powerful upper body and the largest molars of any hominin. *A. afarensis* is generally considered more directly related to the early members of the genus *Homo* than are the South African species.



b.

a.

Figure 30.8 ***Australopithecus afarensis.*** **a.** A reconstruction of Lucy on display at the St. Louis Zoo. **b.** These fossilized footprints occur in ash from a volcanic eruption some 3.7 MYA. The larger footprints are double, and a third, smaller individual was walking to the side. The footprints suggest that *A. afarensis* walked bipedally. (a): ©Dan Dreyfus and Associates; (b): John Reader/ Science Source

South African Australopiths

The first australopith to be discovered was unearthed in southern Africa in the 1920s. This hominin, named *Australopithecus africanus,* is a gracile type. A second southern African specimen called *P. robustus,* discovered in the 1930s, is a robust type that is believed to have had a brain size of around 530 cc.

In 2008, the American anthropologist Lee Berger discovered the bones of a 2-million-year-old australopithecine he named *A. sediba* ("wellspring"). The small brain (500 cc) and long arms suggest that this species is an australopith that climbed trees. However, it also had a humanlike pelvis, nose, and dentition (teeth). A series of papers published in 2013 by the journal *Science* described *A. sediba*'s mosaic of primitive and modern traits. The discovery has led to hypotheses that this species may have been a direct ancestor of the genus *Homo* or a lineage of australopiths that existed at the same time as early *Homo* species. Research is ongoing to find an answer.

Check Your Progress **30.2**

1. Compare the characteristics of an australopith with those of an ardipithecine.
2. Explain why an understanding of bipedalism and brain size is important in understanding the evolution of the hominins.

30.3 Evolution of Early Genus *Homo*

Learning Outcomes

Upon completion of this section, you should be able to

1. Arrange the early species of *Homo* in evolutionary order.
2. Explain the significance of *Homo habilis* and *H. erectus* in the study of human evolution.

Early *Homo* species (see lavender bars in Fig. 30.6) appear in the fossil record nearly 2.5 MYA. They all have a brain size that is 600 cc or greater, their jaws and teeth resemble those of modern humans, and tool use is evident.

Homo habilis and Homo rudolfensis

Homo habilis and *Homo rudolfensis* are closely related and are considered together here. Recent evidence suggests that these may be a single species, but with so few fossils found, it is uncertain. Some scientists have proposed that *H. rudolfensis* may actually be a previously undiscovered species of *Australopithecus.* For these reasons, only *H. habilis* is shown in Figure 30.6. *Homo habilis* means "handyman," and these two species are credited by some as being the first hominins to use stone tools. Most believe that, although they appear to have been socially organized, they were probably scavengers

rather than hunters. The cheek teeth of these hominins tend to be smaller than even those of the gracile australopiths. This is also evidence that they were omnivorous and ate meat in addition to plant material.

Homo ergaster and Homo erectus

Homo ergaster and *Homo erectus (L. homo, "man"; erectus, "upright")* are traditionally considered separate species, with *H. ergaster* originating in Africa and *H. erectus* evolving from migrants to Europe and Asia. However, many now view them all as members of *H. erectus,* with the variation in fossils resulting from their existence across three continents and nearly two million years—longer than any human ancestor. For our purposes, we will take the latter approach and consider them as a single species of *H. erectus.* Appearing first in Africa around 1.9 million years ago, *H. erectus* evolved either directly from *H. habilis* or from an earlier common ancestor.

Compared to other previous *Homo* species, early *H. erectus* had a larger brain (about 1,000 cc), a rounder jaw, prominent brow ridges, and a projecting nose. This type of nose is adaptive for a hot, dry climate, because it permits water to be removed before air leaves the body. The recovery of an almost complete skeleton of a 10-year-old boy indicates that *H. erectus* was much taller than the hominins discussed thus far (Fig. 30.9). Males were 1.8 m tall, and females were 1.55 m tall. Indeed, these hominins stood erect and most likely had a striding gait like that of modern humans. The robust and probably heavily muscled skeleton still retained some australopithecine features. Even so, the size of the birth canal in female specimens indicates that infants were born in an immature state that required an extended period of care.

As shown in Figure 30.10, *H. erectus* was the first species to migrate into Europe and Asia—sometime between 2.5 MYA and 1 MYA. Once there, *H. erectus* developed variations from the original African population, which is why some scientists consider them separate species. In any case, such an extensive population movement is a first in the history of humankind and a tribute to the intellectual and physical skills of these hominins. They also had a knowledge of fire and may have been the first to cook meat.

There is considerable debate as to when the last of *Homo erectus* died out, with most estimates placing it several hundred thousand years ago. However, in 2019, scientists reported they had determined that *H. erectus* remains on the island of Java in Indonesia were around 100,000 years old. If so, then it is possible that *H. erectus* interacted with other *Homo* species much later than earlier predicted.

Figure 30.9 ***Homo erectus.*** This skeleton of a 10-year-old boy who lived 1.6 MYA in eastern Africa shows femurs that are angled and elongated. SAYYID AZIM/AP Images

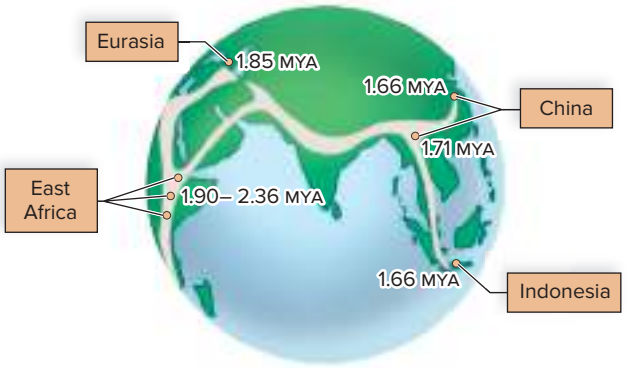


Figure 30.10 **Migration out of Africa.** The dates indicate the migration of early *Homo erectus* from Africa. Source: "Evolution of Early Homo: An Integrated Biological Perspective," S. Anton et al., Science 4 July 2014:345 (6192).

Check Your Progress 30.3

1. Discuss the general evolutionary trends in early *Homo* species.
2. Discuss how the evolution of bipedalism and increased brain size probably contributed to *H. erectus*'s migration from Africa.

ISTUDY

30.4 Evolution of Later Genus *Homo*

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the evidence for the replacement model hypothesis regarding the evolution of later members of the genus *Homo.*
2. Discuss the significance of increased tool use in Cro-Magnons.
3. Summarize how the replacement model explains the major human ethnic groups.

Later *Homo* species are represented by blue-colored bars in Figure 30.6. The evolution of these species from older *Homo* species has been the subject of much debate. While most researchers believe that modern humans (*Homo sapiens*) evolved from an *H. erectus* lineage in Africa, they differ as to the details of when this occurred. Furthermore, recent discoveries of fossil remains, most notably of *H. erectus,* Neanderthals, and Denisovans, but also of previously unidentified species, is challenging much of what we know about this period in human evolution.

Homo heidelbergensis and the Transition to Modern Humans

Populations of *H. erectus* persisted until about 100,000 years ago in Asia, but at some point, several lineages diverged and formed species such as *Homo florensiensis* (see below). Additionally, in Africa, *H. erectus* disappeared around 500,000 years ago, but were replaced by a species called *Homo heidelbergensis.* Whether *H. heidelbergensis* represents a separate lineage or the gradual evolution of *H. erectus* is still under investigation. Either way, it is most likely the lineage that led to modern humans.

A clear dividing line has not been found between fossils from late *H. erectus* and *H. heidelbergensis,* but their fossils begin appearing in Africa around 700,000 years ago. Compared to *H. erectus, H. heidelbergensis* had a larger brain size, and more advanced tool use and social behavior. They appear to be the first species to build shelters.

The most supported scenario of the path to modern humans involves three lineages originating from *H. heidelbergensis.* Between 700,000 and 200,000 years ago, *H. heidelbergensis* split into three geographic populations, each of which generated a different species. Modern *H. sapiens* evolved from the population in Africa. The Asian lineage evolved into recently discovered "Denisovans," and the European lineage evolved into *Homo neanderthalensis* (Neanderthals). Molecular analyses of *Homo sapiens* show DNA evidence from each of these lines may be found in populations of modern humans.

More than any other time period, it is difficult to talk in terms of species beyond *H. erectus* because of fuzzy dividing lines, wide geographic movement, and the presence of multiple species. Keep in mind that between 100,000 and 200,000 years ago, there may have been multiple human lineages existing on the planet, including *H. erectus, H. neanderthalensis, H. sapiens,* and as the Evolution feature, "The Complexity of Our Evolutionary Past," presents, several recently discovered new species.

Evolutionary Hypotheses

The most widely accepted hypothesis for the evolution of modern humans from earlier *Homo* species is referred to as the *replacement model* or *out-of-Africa hypothesis,* which proposes that modern humans evolved from archaic humans only in Africa, and then modern humans migrated to Asia and Europe, where they replaced the early *Homo* species about 100,000 years BP (before the present) (Fig. 30.11).

The replacement model has historically been supported by the fossil record. The earliest remains of modern humans (the Cro-Magnons), dating at least 130,000 years BP, have been found only in Africa. Modern humans are not found in Asia until 100,000 years BP and not in Europe until 60,000 years BP. Recent fossil discoveries in northern Africa are challenging this hypothesis, but more research is needed before scientists can modify the model.

The replacement model is also supported by DNA data. Several years ago, a study showed that the mitochondrial DNA of Africans is more diverse than the DNA of the people in Europe (and the rest of the world). This is significant; if mitochondrial DNA has a constant rate of mutation, Africans should show the greatest diversity, because modern humans have existed the longest in Africa.

A hypothesis opposing the replacement model does exist. This hypothesis, called the *multiregional continuity hypothesis,* proposes that modern humans arose from earlier *Homo* species in essentially the same manner in Africa, Asia, and Europe. The hypothesis is multiregional, because it applies equally to Africa, Asia, and Europe, and it proposes that in these regions genetic continuity will be found between modern populations and early *Homo*. The discovery of new species, such as *H. florensiensis,*

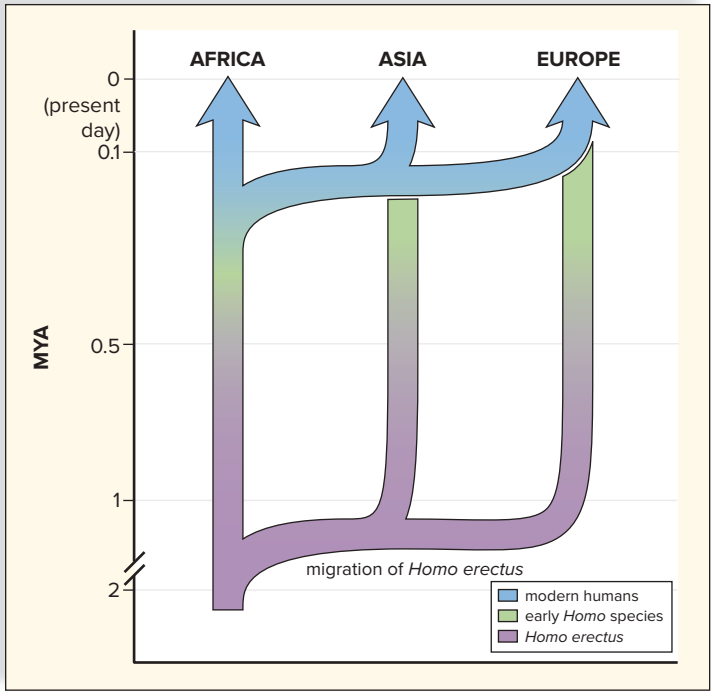


Figure 30.11 Replacement model. Modern humans evolved in Africa and then replaced archaic humans in Asia and Europe.

ISTUDY

THEME Evolution

The Complexity of Our Evolutionary Past

Over the past several years, fossil and genetic evidence has indicated that rather than a simple linear transition between *Homo erectus* and *H. neanderthalensis* and *H. sapiens,* there may have been a period of several hundred thousand years when multiple species of *Homo* occupied what is now Europe and Asia.

Homo floresiensis

In 2004, scientists announced the discovery of the fossil remains of *Homo floresiensis.* The approximately 80,000-year-old fossil of a 1-m-tall, 25-kg adult female was discovered on the island of Flores in the South Pacific. The specimen was the size of a 3-year-old *Homo sapiens* but possessed a braincase only one-third the size of a modern human (Fig. 30C). Due to its small size, this species has been nicknamed "Hobbits," after characters in the book by J. R. R. Tolkien. How *H. floresiensis* evolved is still under investigation. Scientists hypothesize the population split from *H. erectus* after being isolated on the island. Their small size may have evolved on the island or originated in their *H. erectus* ancestors. Interestingly, *H. floresiensis* existed as modern *H. sapiens* evolved and spread across Africa, Europe, and Asia, but is believed to have gone extinct around 12,000 years ago before *Homo sapiens* arrived at the island.

Homo naledi

In 2013, a group of researchers discovered humanlike fossils in a cave outside of Johannesburg, South Africa. The structures of the skull and teeth suggest that these fossils are the remains of a previously unknown species of hominin. The fossils possess characteristics of both australopithecines and the *Homo* genus. Because of a closer similarity to members of the genus *Homo,* the species is called *Homo naledi.* The fossils have been dated to be between 236,000 and 335,000 years old, which places *Homo naledi* in the same time frame as Neanderthals and Denisovans.

Denisovans

Evidence of another member of the human family has been found in a cave in Russia, called Denisova. In 2008, Russian archaeologists working in an area of the cave known to contain deposits between 30,000 and 50,000 years old found a tiny bone fragment, which was later determined to be from the pinkie finger of a 5- to 7-year-old girl. Studies of the bone's DNA revealed that the girl was related to Neanderthals, but she was different enough that scientists consider her to be from a different species, which is now referred to simply as the Denisovans. The discovery of a jawbone (Fig. 30D) in 2019 has provided additional evidence that the Denisovans possessed a hybrid of characteristics between early *Homo* species and what is found in modern humans. Their lineages diverged some 200,000 years ago from *H. heidelbergensis* ancestors in the Middle East.

By comparing Denisovan DNA to that of present-day humans, researchers have discovered that people living in East Asia, and especially the Pacific Islands, share 3–5% of their genome with Denisovans. In addition, Neanderthal DNA has been found when analyzing Denisovans. This indicates that Neanderthals, Denisovans, and early *Homo sapiens* likely coexisted for a period of time and interbred.

Questions to Consider

1. What does the discovery of these species tell you about what we know regarding human evolution?
2. Do you think other species of *Homo* have yet to be discovered? Where would you look?

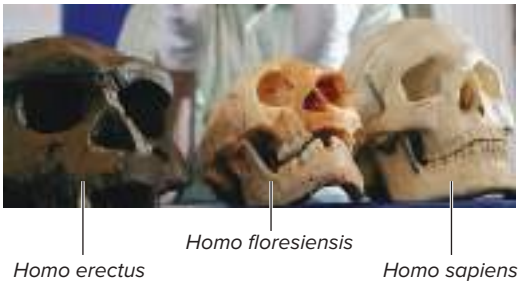


Figure 30C *Homo floresiensis.* The *H. floresiensis* skull is smaller than that of *H. erectus* and *H. sapiens.* Richard Lewis/AP Images

Figure 30D Evidence of Denisovans. This jawbone of a Denisovan was found high in the Tibetan plateaus. Lanzhou University/Dongju Zhang

H. naledi, and the Denisovans, is bringing attention back to this hypothesis, or possibly a combination of the multiregional and replacement models.

Neanderthals

The **Neanderthals,** *Homo neanderthalensis,* are an intriguing species of archaic humans that originated over 400,000 years ago and survived until 30,000 years ago. They appear to have evolved from a lineage of *H. heidelbergensis* migrating out of Africa. Neanderthal fossils are known from the Middle East and throughout Europe. Neanderthals take their name from Germany's Neander Valley, where one of the first Neanderthal skeletons, dated some 200,000 years ago, was discovered. According to the replacement model, the Neanderthals were also supplanted by modern humans.

ISTUDY

Surprisingly, however, the Neanderthal brain was, on average, slightly larger than that of *Homo sapiens* (1,400 cc, compared with 1,360 cc in most modern humans). The Neanderthals had massive brow ridges and wide, flat noses. They also had a forward-sloping forehead and a receding lower jaw. Their nose, jaws, and teeth protruded far forward. Physically, the Neanderthals were powerful and heavily muscled, especially in the shoulders and neck. The bones of Neanderthals were shorter and thicker than those of modern humans. New fossils show that the pubic bone was longer compared to that of modern humans. The Neanderthals lived in Europe and Asia during the last Ice Age, and their sturdy build could have helped conserve heat. As indicated in the chapter opener, inbreeding between Neanderthals and *H. sapiens* may have provided the latter with some genes that increased tolerance to cold weather, as well as other traits.

Fossilized remains of *Homo neanderthalensis* have been found widely throughout what is now Europe and Asia, suggesting that they were the dominant hominin group residing there for hundreds of thousands of years. About 80,000 to 100,000 years ago, ancestors of modern humans entered Europe from the south. The Neanderthals disappeared about 30,000 years ago, but what caused their extinction is still a matter of much scientific debate.

There are many hypotheses as to what may have happened, including conflict between Neanderthals and early *H. sapiens*. New infectious diseases that traveled with *H. sapiens* to Europe could have been difficult for the Neanderthal population to resist. The modern humans may have had some other competitive advantage, such as better hunting and gathering of food, depleting available resources. Or, Neanderthals may have died out for reasons completely unrelated to interactions with their human cousins, such as overhunting of their main food sources or failure to adapt to a changing climate.

Archaeological evidence suggests that Neanderthals were culturally advanced. Some Neanderthals lived in caves; however, others probably constructed shelters. They manufactured a variety of stone tools, including spear points, which could have been used for hunting, and scrapers and knives, which would have helped in food preparation. They most likely successfully hunted bears, woolly mammoths, rhinoceroses, reindeer, and other contemporary animals. They used and could control fire, which probably helped in cooking frozen meat and in keeping warm. They even buried their dead with flowers and tools and may have had a religion.

Cro-Magnons

As mentioned previously, *Homo sapiens* first evolved in Africa from *H. heidelbergensis* ancestors around 200,000 years ago. The **Cro-Magnons** are the oldest fossils to be designated *Homo sapiens.* In keeping with the replacement model, Cro-Magnons then entered Asia from Africa about 100,000 years BP and spread to Europe, interbreeding with Neanderthals and Denisovans. They probably reached western Europe about 40,000 years ago.

Cro-Magnons had a thoroughly modern appearance. They had lighter bones, flat high foreheads, domed skulls housing brains of 1,590 cc, small teeth, and a distinct chin. They were **hunter-gatherers** who collected food from the environment rather than domesticating animals and growing food plants. *H. erectus* were also hunter-gatherers, but Cro-Magnons hunted more efficiently.

Tool Use in Cro-Magnons

Cro-Magnons designed and manipulated tools and weapons of increasing sophistication. They made advanced stone tools, including compound tools, as when stone flakes were fitted to a wooden handle. They may have been the first to make knifelike blades and to throw spears, enabling them to kill animals from a distance. They were such accomplished hunters that some researchers believe they may have been responsible for the extinction of many larger mammals, such as the giant sloth, the mammoth, the saber-toothed tiger, and the giant ox, during the late Pleistocene epoch. This event is known as the Pleistocene overkill.

Language and Cro-Magnons

A more highly developed brain may have also allowed Cro-Magnons to perfect a language composed of patterned sounds. Language greatly enhanced the possibilities for cooperation and a sense of cohesion within the small bands that were the predominant form of human social organization, even for the Cro-Magnons.

The Cro-Magnons were highly creative. They sculpted small figurines and jewelry out of reindeer bones and antlers. These sculptures could have had religious significance or may have been seen as a way to increase fertility. The most impressive artistic achievements of the Cro-Magnons were cave paintings, realistic and colorful depictions of a variety of animals, from woolly mammoths to horses, that have been discovered deep in caverns in southern France and Spain. It has been proposed that these paintings may have been a type of calendar based on the reproductive cycles of the animals that Cro-Magnons hunted. This suggests that Cro-Magnons had the ability to think symbolically, as would be needed in order to speak.

Rise of Agriculture

The Cro-Magnons combined hunting and fishing with gathering fruits, berries, grains, and root crops that grew in the wild. With the rise of agriculture about 10,000 years BP, modern humans are no longer considered Cro-Magnon. However, full dependency on domestic crops and animals did not occur until humans started making tools of bronze (instead of stone), about 4,500 years BP.

Anthropologists previously thought that early humans turned to agriculture because life as a hunter-gatherer had its drawbacks. However, skeletal evidence suggests that early agricultural societies experienced an increase in infectious diseases, malnutrition, and anemia compared to earlier hunter-gatherer groups. Many anthropologists now think that the change in lifestyle was dictated by extinctions of the large game animals, as well as a general warming of the climate. As the glaciers retreated, fertile soil was deposited into rivers and streams full of fish. In suitable locations, such as the fertile crescent of Mesopotamia, fishing villages may have developed, causing populations to settle in one location. Combined with a beneficial climate, the increase in food supplies

ISTUDY

would have resulted in an increase in the population, reducing its ability to migrate easily. An increase in agriculture would also have supported the specialization of tasks in the population and enhanced the process of **biocultural evolution,** in which cultural achievements—not individual phenotypes—are influenced by natural selection.

Human Variation

Humans have been widely distributed about the globe ever since they evolved. As with any other species that has a wide geographic distribution, phenotypic and genotypic variations are noticeable between populations. Today, we say that people have different ethnicities (Fig. 30.12*a*).

a.

b.

c.

Figure 30.12 Ethnic groups. **a.** Some of the differences among the prevalent ethnic groups in the United States may be due to adaptations to their original environments. **b.** The Maasai live in East Africa. **c.** The Inuit live near the Arctic Circle. (a): Anderson Ross/Photodisc/Getty Images; (b): Sylvia S. Mader; (c): Ton Koene/Alamy Stock Photo

Evolutionists have hypothesized that human variations evolved as adaptations to local environmental conditions. One obvious difference among people is skin color. A darker skin is protective against the high UV intensity of bright sunlight. On the other hand, a white skin ensures vitamin D production in the skin when the UV intensity is low. Harvard University geneticist Richard Lewontin points out, however, that this hypothesis concerning the survival value of dark and light skin has never been tested.

Two correlations between body shape and environmental conditions have been noted since the nineteenth century. The first, known as Bergmann's rule, states that animals in colder regions of their range have a bulkier body build. The second, known as Allen's rule, states that animals in colder regions of their range have shorter limbs, digits, and ears. Both of these effects help regulate body temperature by increasing the surface-area-to-volume ratio in hot climates and decreasing the ratio in cold climates. For example, Figure 30.12*b, c* shows that the Maasai of East Africa tend to be slightly built with elongated limbs, while the Inuit, who live in northern regions, tend to have a bulkier build, with shorter limbs.

Other anatomical differences among ethnic groups, such as hair texture, a fold on the upper eyelid (common in Asian peoples), and the shape of lips, cannot be explained as adaptations to the environment. Perhaps these features became fixed in different populations due simply to genetic drift. As far as intelligence is concerned, no significant disparities have been found among different ethnic groups.

The replacement model for the evolution of humans, discussed earlier in this section, pertains to the origin of ethnic groups. This hypothesis proposes that all modern humans have a relatively recent common ancestor—that is, Cro-Magnon—who evolved in Africa and then spread into other regions. Paleontologists tell us that the variation among modern populations is considerably less than existed among archaic human populations some 250,000 years ago. If so, all ethnic groups evolved from the same single ancestral population.

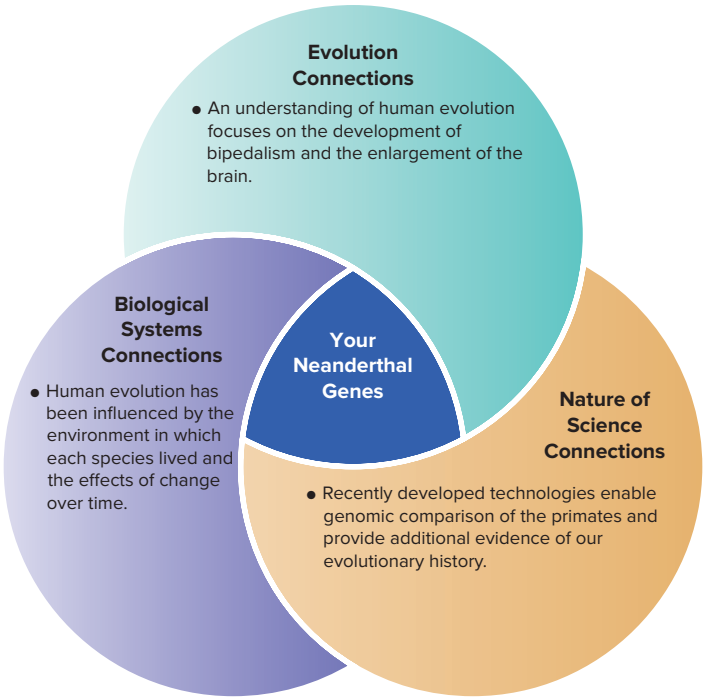
A comparative study of mitochondrial DNA shows that the differences among human populations are consistent with their having a common ancestor no more than a million years ago. Lewontin has also found that the genotypes of different modern populations are extremely similar. He examined variations in 17 genes, including blood groups and various enzymes, among seven major geographic groups that he identified: Europeans (caucasians), Black Africans, mongoloids, South Asian Aborigines, Amerinds, Oceanians, and Australian Aborigines. He found that the great majority of genetic variation—85%—occurs within ethnic groups, not between them. In other words, the amount of genetic variation between individuals of the same ethnic group is greater than the variation between any two ethnic groups.

Check Your Progress 30.4

1. Explain how the replacement model explains both the dominance of Cro-Magnons and the formation of human ethnic groups.
2. Discuss what factors led to the development of biocultural evolution as a factor in human evolution.

ISTUDY

Connecting *the* Concepts



What Did You Learn?

30.1 Evolution of Primates

Most **primates** are adapted for an **arboreal** life.

- The evolution of primates is characterized by trends toward mobile limbs; grasping hands with or without an **opposable thumb;** a flattened face; **stereoscopic vision;** a large, complex brain; and the birth of only one offspring at a time. These traits are particularly useful for living in trees.
- The term **hominin** is used for humans and their closely related, but extinct, relatives. A hominin is a member of a larger group called the **hominids** which includes chimpanzees, gorillas, and orangutans. **Anthropoids** include additional, more distant relatives, such as New and Old World monkeys. The **prosimians** (lemurs, tarsiers) were the first primates to diverge from the common ancestor of all primates.

30.2 Evolution of Humanlike Hominins

Adaptations that allow humans to walk erect have resulted in our anatomy differing from that of the apes. In humans, the spinal cord curves and exits from the center of the skull, rather than from the rear of the skull. The human pelvis is broader and more bowl-shaped to place the weight of the body over the legs. Humans use only the longer, heavier lower limbs for walking upright bipedally; in apes, all four limbs are used for walking, and the upper limbs are longer than the lower limbs.

- Several early humanlike hominin fossils, such as *Sahelanthropus tchadensis,* have been dated around the time of a shared ancestor for apes and humans (7 MYA).
- The **ardipithecines** appeared about 4.5 MYA. Ardi (*Ardipithecus ramidus*) is an example of an ardipithecine. All the early humanlike hominins have a chimp-sized braincase but are believed to have walked erect. They may also have been the first hominins to use tools as an aid in the consumption of meat.
- **Australopithecines** (or australopiths) lived and diversified in Africa from about 4 MYA to 1.5 MYA. They had some apelike and some humanlike characteristics, and thus provide an example of **mosaic evolution.** It is possible that an australopith is a direct ancestor of humans. These hominins walked upright and had a brain size of 370–515 cc. In southern Africa, hominins classified as australopiths include *Australopithecus africanus,* a gracile form, *Paranthropus robustus,* a robust form, and *Australopithecus sediba,* which had both primitive and modern traits. In eastern Africa, hominins classified as australopiths include *A. afarensis* (Lucy), a gracile form, as well as robust forms. Many of the australopiths coexisted, and one of these species is the probable ancestor of the genus *Homo.*

30.3 Evolution of Early Genus *Homo*

Early *Homo,* such as *Homo habilis* and *Homo rudolfensis,* dated around 2 MYA, is characterized by a brain size of at least 600 cc, a jaw with teeth that resemble those of modern humans, and the use of tools.

Homo erectus (2.3–0.3 MYA) had a striding gait, made well-fashioned tools, and could control fire. They migrated into Asia and Europe from Africa between 2 and 1 MYA. *H. erectus* lineages in Asia gave rise to *H. floresiensis.*

30.4 Evolution of Later Genus *Homo*

The replacement model of human evolution says that modern humans originated only in Africa and, after migrating into Europe and Asia, replaced the archaic *Homo* species found there. The multiregional continuity model suggests that modern humans arose in several regions.

- The **Neanderthals,** a group of archaic humans, lived in Europe and Asia. Their sloping chins, squat frames, and heavy muscles are apparently adaptations to the cold. Another group, the Denisovans, may have evolved from a common ancestor that also gave rise to Neanderthals, but the Denisovans migrated more to East Asia and the South Pacific Islands. Genetic evidence suggests that both Neanderthals and Denisovans interbred with early humans.

ISTUDY

- **Cro-Magnon** is a name often given to modern humans. These **hunter-gatherers** used sophisticated tools, and they definitely had a culture, as witnessed by the paintings on the walls of caves. In advanced human populations, specialization of tasks led to **biocultural evolution.**

The human ethnic groups of today differ in ways that can be explained in part by adaptation to the environment. Genetic studies tell us that there are more genetic differences between people of the same ethnic group than between ethnic groups. We are one species.

Thinking Critically About Biology

1. Bipedalism has many selective advantages. However, there is one disadvantage to walking on two feet: Giving birth to an offspring with a large head through a smaller pelvic opening, which is necessitated by upright posture, is very difficult. This situation results in a high percentage of deaths (of both the female parent and offspring) during birth compared to other primates. How do you explain the selection of a trait that is both positive and negative?
2. How might you use biotechnology to show that humans today have Neanderthal genes and, therefore, Cro-Magnons and Neanderthals interbred with one another?
3. Some modern ethnic groups (white Europeans, Asians) have apparently inherited genes from Neanderthals that may influence resistance to cold temperatures. What types of genes might these be?
4. Chimpanzees and humans obviously look and act very differently, but their genomes are remarkably similar. What types of mechanisms could explain how small differences in genotype can result in such large differences in phenotype?

Making It Relevant to You

1. How does genetic analyses of fossils help us understand human evolution better than just a comparison of skulls?
2. Some scientists suggest that the Neanderthals, and possibly the Denisovans, never went extinct, but instead were assimilated into *H. sapiens*. What evidence would you look for to support this hypothesis?
3. Assuming the Denisovans, Neanderthals, and humans were able to interbreed, what does this tell you about their species status according to the biological species concept?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 7

Comparative Animal Biology

In contrast to plants, which are autotrophic and make their own organic food, animals are heterotrophic and feed on organic molecules made by other organisms. An animal's mobility, which is dependent on nerve and muscle fibers, is essential to escaping predators, finding a mate, and acquiring food. Food is then digested, and the nutrients are distributed to the body's cells. Finally, waste products are expelled.

The evolution of animals is marked by an increase in the complexity of the organ systems, which has resulted in a distinct division of labor, with each organ system specialized to carry out specific functions. For example, the cardiovascular system transports materials from one body part to another; the respiratory system carries out gas exchange; and the urinary system filters the blood and removes its wastes. The immune system, along with the lymphatic system, protects the body from infectious diseases. The nervous system and endocrine system coordinate the activities of the other systems.

Our comparative study will show how the different organ systems evolved and how they function to maintain homeostasis, the relative constancy of the internal environment.

Unit Outline

Chapter 31 Animal Organization and Homeostasis
Chapter 32 Circulation and Cardiovascular Systems
Chapter 33 The Lymphatic and Immune Systems
Chapter 34 Digestive Systems and Nutrition
Chapter 35 Respiratory Systems
Chapter 36 Body Fluid Regulation and Excretory Systems
Chapter 37 Neurons and Nervous Systems
Chapter 38 Sense Organs
Chapter 39 Locomotion and Support Systems
Chapter 40 Hormones and Endocrine Systems
Chapter 41 Reproductive Systems
Chapter 42 Animal Development and Aging
Chapter 43 Animal Behavior

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Explain the relationship between the evolution of the animal body plan and the changes to the major organ systems.
Nature of Science	Analyze how specific scientific research studies are advancing our understanding of how animal systems function.
Biological Systems	Review the fundamental structures and functions that are uniquely found in animals.

ISTUDY

31

Animal Organization and Homeostasis

Matthew Horwood/Getty Images

Printing Organs

The first 3D printer was invented in 1984 for the purpose of quickly creating prototype parts in manufacturing. Not long after, scientists imagined another use in regenerative medicine to repair or replace "parts" in human patients. Today, special 3D printers, called bioprinters, are being used to print skin, blood vessels, and even full organs.

Bioprinting begins with making a 3D scan for the printer to follow. For example, scans can replicate a patient's ear, bone structures, or vessels. Bioprinters then create the structure using "bio ink" unique to each patient. The ink contains cells from the patient mixed with an organic, biodegradable gel that solidifies as it is printed, allowing the cells to be held in a 3D shape. When printing is completed, the structure is placed in an incubator where cells multiply and organize naturally into the desired tissue or organ. Once the gel breaks down, the new structure is identical to the patient's original cells and tissues, making it much less likely to be rejected.

What does the future hold for bioprinting? Scientists hope organs can be quickly created for individuals needing transplants, eliminating the need to wait for organs from matching donors.

As you read through this chapter, think about the following questions:

1. How did the evolution of specialized tissues, organs, and organ systems allow animals to better adapt to their environment?
2. What are some of the most important functions of animal skin?
3. How does the disruption of homeostasis lead to disease?

Chapter Outline

31.1 Types of Tissues
31.2 Organs, Organ Systems, and Body Cavities
31.3 The Integumentary System
31.4 Homeostasis

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 1.2 What levels of biological organization are found in animals?

Section 24.1 What types of tissues are found in flowering plants?

Section 28.1 Which types of tissues are most characteristic of animals?

Following *the* Themes

CHAPTER 31 – ANIMAL ORGANIZATION AND HOMEOSTASIS

Evolution	The degree of cell specialization seen in tissues, organs, and organ systems provides evolutionary advantages for animals.
Nature of Science	Scientific research is increasing our understanding of how tissues and organs may be repaired or regenerated.
Biological Systems	Most animals, especially vertebrates, have similar types of tissues, organs, and organ systems.

ISTUDY

31.1 Types of Tissues

Learning Outcomes

Upon completion of this section, you should be able to

1. List and describe the four major types of tissues found in animals.
2. Identify the common locations of the various types of animal tissues.
3. Explain how specialization of cells in tissues enhances tissue function.

Like all living organisms, animals are highly organized. Animals begin life as a single cell—a fertilized egg, or *zygote*. The zygote undergoes cell division, producing cells that will eventually form the variety of tissues that make up organs and organ systems. Although all of these cells carry out a number of common functions—such as obtaining nutrients, synthesizing basic cellular components, and in most cases reproducing themselves—the cells of multicellular organisms further differentiate so they can perform additional, unique functions.

A **tissue** is composed of specialized cells of the same or similar type that perform a common function in the body. The tissues of most complex animals can be categorized into four major types:

- *Epithelial tissue* covers body surfaces, lines body cavities, and forms glands.
- *Connective tissue* binds and supports body parts.
- *Muscular tissue* moves the body and its parts.
- *Nervous tissue* receives stimuli and transmits nerve impulses.

Epithelial Tissue

Epithelial tissue, also called *epithelium* (*pl.,* epithelia), consists of tightly packed cells that form a continuous layer. Epithelial tissue covers surfaces and lines body cavities. Usually, it has a protective function, but it can also be modified to carry out secretion, absorption, excretion, and filtration.

Epithelial cells may be connected to one another by three types of junctions composed of proteins (see Fig. 5.14). Regions where proteins join them together are called tight junctions. In the intestine, the gastric juices stay out of the body, and in the kidneys, the urine stays within kidney tubules because epithelial cells are joined by tight junctions. In the skin, adhesion junctions add strength and allow epithelial cells to stretch and bend, whereas gap junctions are protein channels that permit the passage of molecules between two adjacent cells. These junctions are described in more detail in Section 5.4.

Epithelial tissues are often exposed to the environment on one side, but on the other side they are attached to a *basement membrane* (Fig. 31.1). The basement membrane is simply a thin layer of various types of proteins that anchors the epithelium to the extracellular matrix, which is often a type of connective tissue. The basement membrane should not be confused with the plasma membrane or with the body membranes we will be discussing later in this section.

Simple Epithelia

Epithelial tissue is either simple or complex. **Simple epithelia** have only a single layer of cells and are classified according to cell type. *Squamous epithelium,* which is composed of flattened cells, lines blood vessels and the air sacs of lungs (Fig. 31.1*a*). *Cuboidal epithelium* contains cube-shaped cells and is found lining the

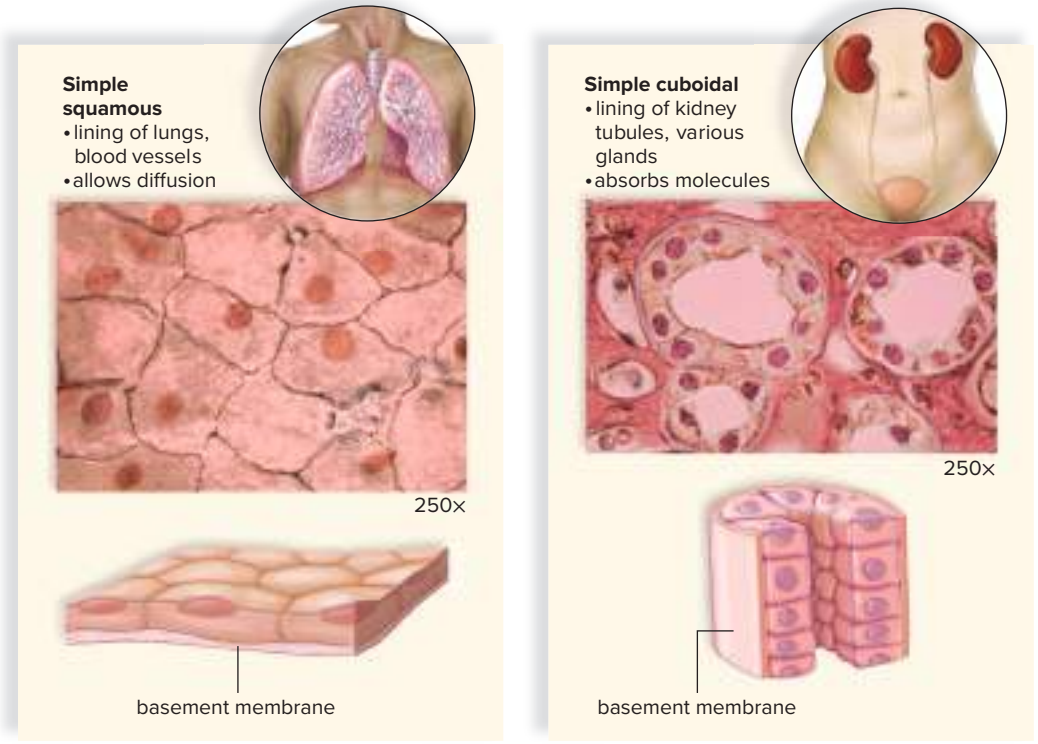


Figure 31.1 Types of epithelial tissues in vertebrates. Basic epithelial tissues found in vertebrates are shown, along with the locations of the tissue and the primary function of the tissue at these locations. (all): ©Ed Reschke

kidney tubules and various glands (Fig. 31.1*b*). *Columnar epithelium* has cells resembling rectangular pillars or columns, with nuclei usually located near the bottom of each cell (Fig. 31.1*c*). This epithelium lines the digestive tract, where it efficiently absorbs nutrients from the small intestine because of minute cellular extensions called microvilli. Ciliated columnar epithelium lines the uterine tubes, where it propels the egg toward the uterus.

Pseudostratified epithelia appears to be layered, but true layers do not exist because each cell touches the basement membrane (Fig. 31.1*d*). The lining of the windpipe, or trachea, is pseudostratified ciliated columnar epithelium. A secreted covering of mucus traps foreign particles, and the upward motion of the cilia carries the mucus to the back of the throat, where it may be either swallowed or expectorated. Smoking can cause a change in mucus secretion and inhibit ciliary action, resulting in a chronic inflammatory condition called bronchitis.

When columnar or pseudostratified epithelium secrete a product, it is said to be glandular. A **gland** can be a single epithelial cell, as in the case of mucus-secreting goblet cells within the columnar epithelium lining the digestive tract, or a gland can contain many cells. Glands that secrete their product into ducts are called **exocrine glands.** Glands that have no duct are known as **endocrine glands.** Endocrine glands (e.g., pituitary gland and thyroid) secrete hormones internally, so they are transported by the bloodstream (see Section 40.1).

Stratified Epithelia

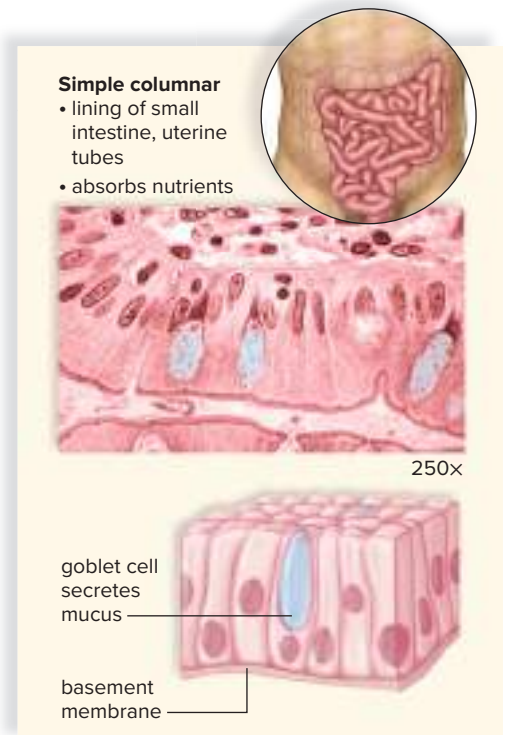
Stratified epithelia have layers of cells piled one on top of the other. Only the bottom layer touches the basement membrane. The nose, mouth, esophagus, anal canal, and vagina are all lined with stratified squamous epithelium (Fig. 31.1*e*). As you'll see, the outer layer of skin is also stratified squamous epithelium, but the cells have been reinforced by keratin, a protein that provides strength. Stratified cuboidal tissue is found in many of the exocrine ducts, and portions of the male and female reproductive system. Stratified columnar epithelia is found in the salivary glands of the mouth.

Transitional epithelium was originally named because it was thought to be an intermediate form of epithelial cell. Now the term is used to imply changeability, because the tissue changes in response to tension. It forms the lining of the urinary bladder, the ureters (the tubes that carry urine from the kidneys to the bladder), and part of the urethra (the single tube that carries urine to the outside). All are organs that may need to stretch. When the bladder is distended, this epithelium stretches and the outer cells take on a squamous appearance.

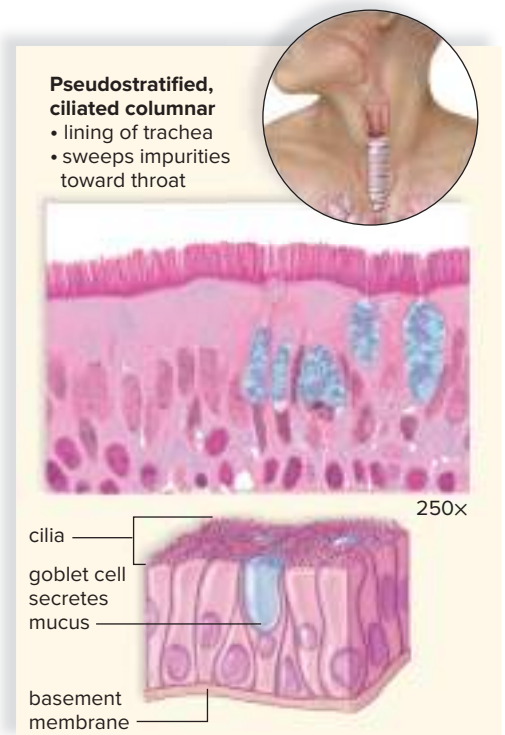
Connective Tissue

Connective tissue is the most abundant and widely distributed tissue in complex animals. It is quite diverse in structure and function; even so, all types have three components: specialized cells, ground substance, and protein fibers (Fig. 31.2).

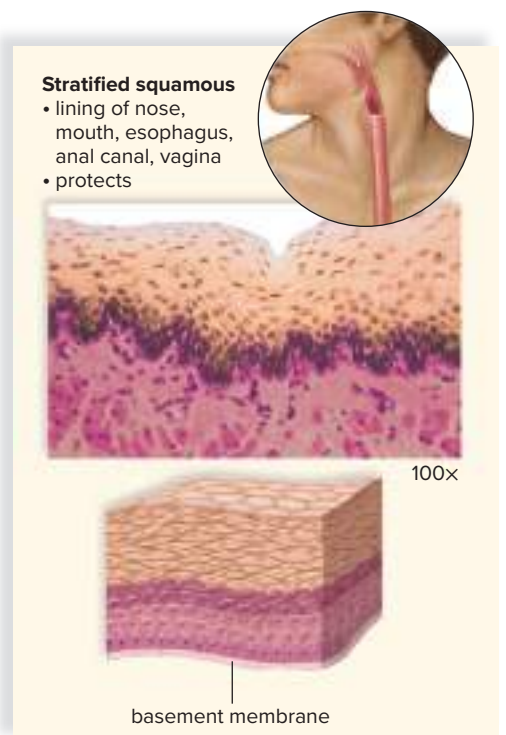
The *ground substance* is a noncellular material that separates the cells and varies in consistency from solid to semifluid to fluid. The term *fiber* is to describe components of multiple tissue types. In connective tissue, there are three possible types of fiber. White collagen fibers contain **collagen,** a protein that gives them flexibility and strength. Reticular fibers are very thin collagen fibers that are highly branched and form delicate supporting networks. Yellow elastic fibers contain **elastin,** a protein that is not as strong as collagen but is more elastic. The combination of the



c.



d.



e.

ISTUDY

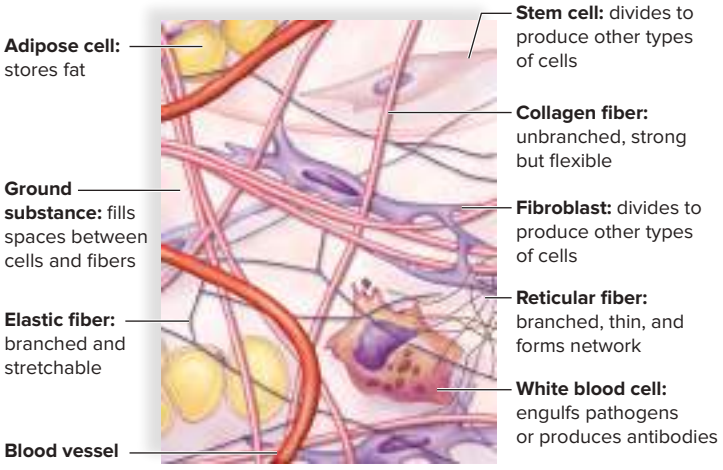


Figure 31.2 Diagram of fibrous connective tissue. Connective tissue is diverse in structure and function in the body.

ground substance plus the fibers are collectively referred to as the connective tissue *matrix.* Connective tissue is classified into three major categories: fibrous, supportive, and fluid.

Fibrous Connective Tissue

Fibrous connective tissues have cells called **fibroblasts** (L. *fibra,* "thread"; Gk. *blastos,* "bud"), located some distance from one another and separated by a jellylike matrix containing white collagen fibers and yellow elastic fibers.

Loose fibrous connective tissue supports epithelium and many internal organs (Fig. 31.3*a*). Its presence in lungs, arteries, and the urinary bladder allows these organs to expand. It forms a protective covering enclosing many internal organs, such as muscles, blood vessels, and nerves.

Adipose tissue serves as the body's primary energy reservoir (Fig. 31.3*b*). It is loose fibrous connective tissue composed mostly of enlarged fibroblasts that store fat. These specialized fibroblasts are called *adipocytes.* Adipose tissue also insulates the body, contributes to body contours, and provides cushioning. In mammals, adipose tissue is found particularly beneath the skin, around the kidneys, and on the surface of the heart.

The number of adipocytes in an individual is fixed. When a person gains weight, the cells become larger, and when weight is lost, the cells shrink. In people who are obese, the individual cells may be up to five times larger than normal. Most adipose tissue is white, but in newborns and hibernating mammals, some is brown due to an increased number of mitochondria that can produce heat.

Dense fibrous connective tissue contains many collagen fibers that are packed together (Fig. 31.3*c*). This type of tissue has more specific functions than does loose connective tissue. For example, dense fibrous connective tissue is found in **tendons** (L. *tendo,* "stretch"), which connect muscles to bones, and in **ligaments** (L. *ligamentum,* "band"), which connect bones to other bones at joints.

Supportive Connective Tissue

Cartilage and bone are the two main supportive connective tissues that provide structure, shape, protection, and leverage for movement. Generally, cartilage is more flexible than bone, because it lacks mineralization of the matrix.

Cartilage In **cartilage,** the cells lie in small chambers called lacunae (*sing.,* lacuna), separated by a matrix that is solid yet flexible. Unfortunately, because this tissue lacks a direct blood supply, it heals very slowly. There are three types of cartilage, distinguished by the type of fiber in the matrix.

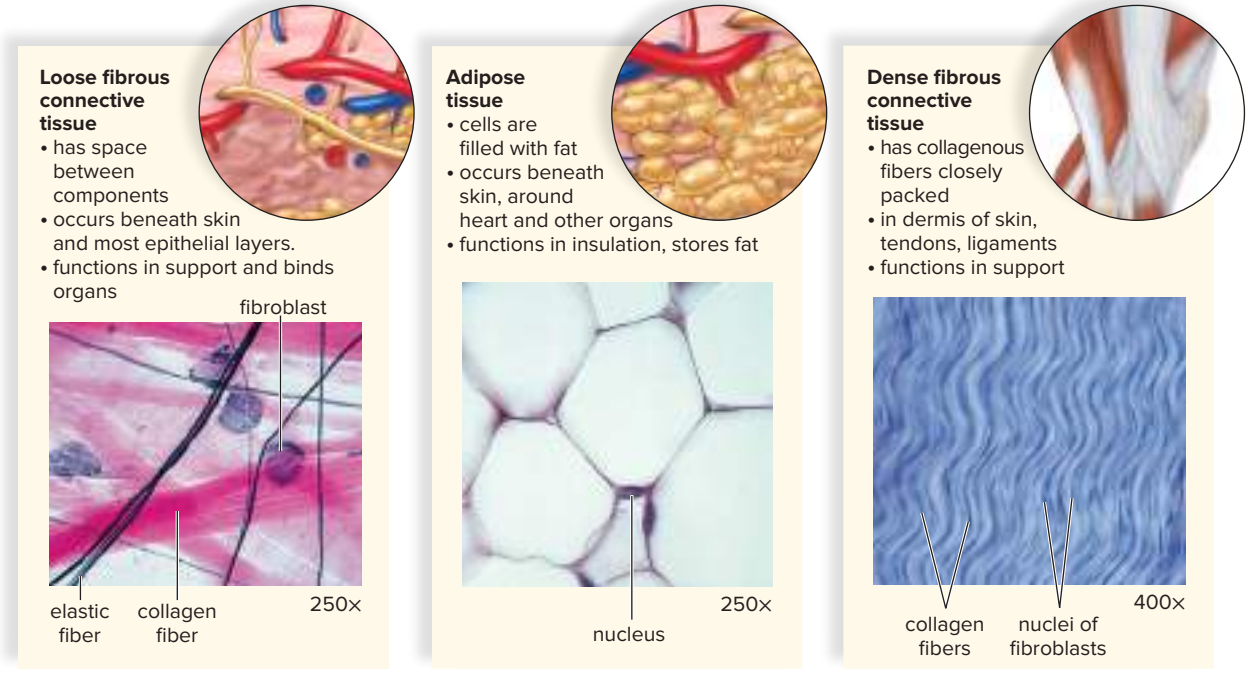


Figure 31.3 Types of connective tissue in vertebrates. While diverse in function, each type of tissue contains specialized cells, ground substance, and protein fibers. (a–b, d–e): ©Ed Reschke; (c): Dennis Strete/McGraw Hill

ISTUDY

Hyaline cartilage (Fig. 31.3*d*), the most common type of cartilage, contains only very fine collagen fibers. The matrix has a white, translucent appearance. Hyaline cartilage is found in the nose and at the ends of the long bones and the ribs, and it forms rings in the walls of respiratory passages. The fetal skeleton also is made of this type of cartilage. Later, the cartilaginous fetal skeleton is replaced by bone.

Elastic cartilage has more elastic fibers than hyaline cartilage. For this reason, it is more flexible and is found, for example, in the framework of the outer ear. **Fibrocartilage** has a matrix containing strong collagen fibers. Fibrocartilage is found in structures that withstand tension and pressure, such as the pads between the vertebrae in the backbone and the wedges in the knee joint.

Bone Of all the connective tissues, **bone** is the most rigid. It consists of an extremely hard matrix of inorganic salts, notably calcium salts, deposited around protein fibers, especially collagen fibers. The inorganic salts give bones rigidity, and the protein fibers provide elasticity and strength, much as steel rods do in reinforced concrete.

Compact bone makes up the shaft of a long bone (Fig. 31.3*e*). It consists of cylindrical structural units called osteons (Haversian systems). The central canal of each osteon is surrounded by rings of hard matrix. Bone cells are located in spaces, called lacunae, between the rings of matrix. Blood vessels in the central canal carry nutrients that allow bone to renew itself. Thin extensions of bone cells within canaliculi (minute canals) connect the cells to each other and to the central canal. The hollow shaft of long bones, such as the femur (thigh bone), is filled with yellow bone marrow (see Section 39.2).

The ends of a long bone contain spongy bone, which has an entirely different structure. **Spongy bone** contains numerous bony bars and plates, separated by irregular spaces. Although lighter than compact bone, spongy bone is still designed for strength. Just as braces are used for support in buildings, the solid portions of spongy bone follow lines of stress. Spongy bone is also the site of red bone marrow, which is critical to the production of blood cells (see Section 32.4).

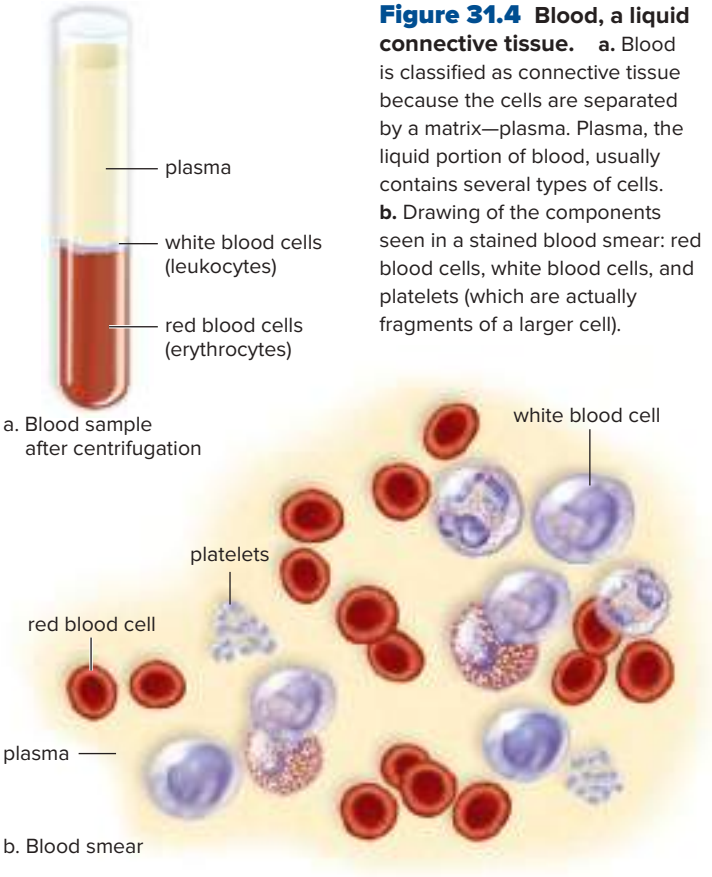
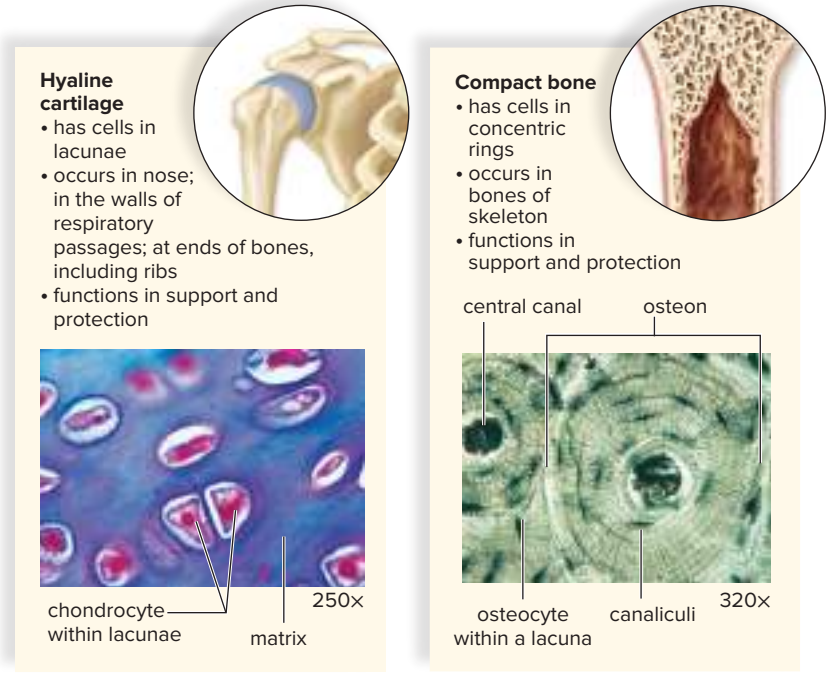


Figure 31.4 Blood, a liquid connective tissue. **a.** Blood is classified as connective tissue because the cells are separated by a matrix—plasma. Plasma, the liquid portion of blood, usually contains several types of cells. **b.** Drawing of the components seen in a stained blood smear: red blood cells, white blood cells, and platelets (which are actually fragments of a larger cell).



Fluid Connective Tissues

Blood, which consists of formed elements and plasma, is a fluid connective tissue located in blood vessels (Fig. 31.4). Formed elements in the blood consist of the many kinds of blood cells and the platelets.

The systems of the body help keep blood composition and chemistry within normal limits, and blood in turn creates **interstitial fluid.** Blood transports nutrients and oxygen to interstitial fluid and removes carbon dioxide and other wastes. It helps distribute heat and plays a role in fluid, ion, and pH balance. The formed elements, discussed next, each have specific functions.

The **red blood cells** are small, disk-shaped cells without nuclei. The absence of a nucleus makes the cells biconcave. The presence of the red pigment hemoglobin makes them red and in turn makes the blood red. Hemoglobin is composed of four units; each unit is composed of the protein globin and a complex, iron-containing structure called heme. The iron forms a loose association with oxygen, and in this way red blood cells transport oxygen and readily give it up in the tissues.

ISTUDY

White blood cells may be distinguished from red blood cells by the fact that they are usually larger, have a nucleus, and without staining would appear translucent. When blood is smeared onto a microscope slide and stained, the nucleus of a white blood cell typically looks blue or purple. White blood cells fight infection, primarily in two ways. Some white blood cells are phagocytic and engulf infectious **pathogens,** while other white blood cells either produce antibodies, molecules that combine with foreign substances to inactivate them, or kill cells outright.

Platelets are not complete cells; rather, they are fragments of large cells present only in bone marrow. When a blood vessel is damaged, platelets help form a plug that seals the vessel, and injured tissues release molecules that help the clotting process.

Lymph is a fluid connective tissue located in lymphatic vessels. Lymphatic vessels absorb excess interstitial fluid and return it to the cardiovascular system. Special lymphatic capillaries, called lacteals, also absorb fat molecules from the small intestine. Lymph nodes, composed of fibrous connective tissue plus specialized white blood cells called lymphocytes, occur along the length of lymphatic vessels. These lymphocytes and other cells remove any foreign material from the lymph as it passes through lymph nodes. Lymph nodes may enlarge when these cells respond to an infection.

Muscular Tissue

Muscular tissue is composed of cells called muscle fibers that are specialized to contract. Muscle fibers contain actin filaments and myosin filaments, whose interaction accounts for movement. The muscles are also important in the generation of body heat. There are three distinct types of muscle tissue: skeletal, smooth, and cardiac. Each type differs in appearance, physiology, and function.

Skeletal muscle, also called voluntary muscle (Fig. 31.5*a*), is attached by tendons to the bones of the skeleton, and when it contracts, body parts move. Contraction of skeletal muscle is under voluntary control and occurs faster than in the other muscle types. Skeletal muscle fibers are cylindrical and quite long—sometimes they run the length of the muscle. They arise during development when several cells fuse, resulting in one fiber with multiple nuclei. The nuclei are located at the periphery of the cell, just inside the plasma membrane. The fibers have alternating light and dark bands, which give them a **striated** appearance, due to the position of actin filaments and myosin filaments in the cell.

Smooth muscle is named because the cells lack striations. The spindle-shaped cells, each with a single nucleus, form layers in which the thick middle portion of one cell is opposite the thin ends of adjacent cells. Consequently, the nuclei form an irregular pattern in the tissue (Fig. 31.5*b*). Smooth muscle is not under voluntary control and therefore is said to be involuntary. Smooth muscle is found in the walls of viscera (intestine, stomach, and other internal organs) and blood vessels. For this reason, it is sometimes called *visceral muscle.* Smooth muscle contracts more slowly than skeletal muscle but can remain contracted for a longer time. When the smooth muscle of the intestine contracts, food

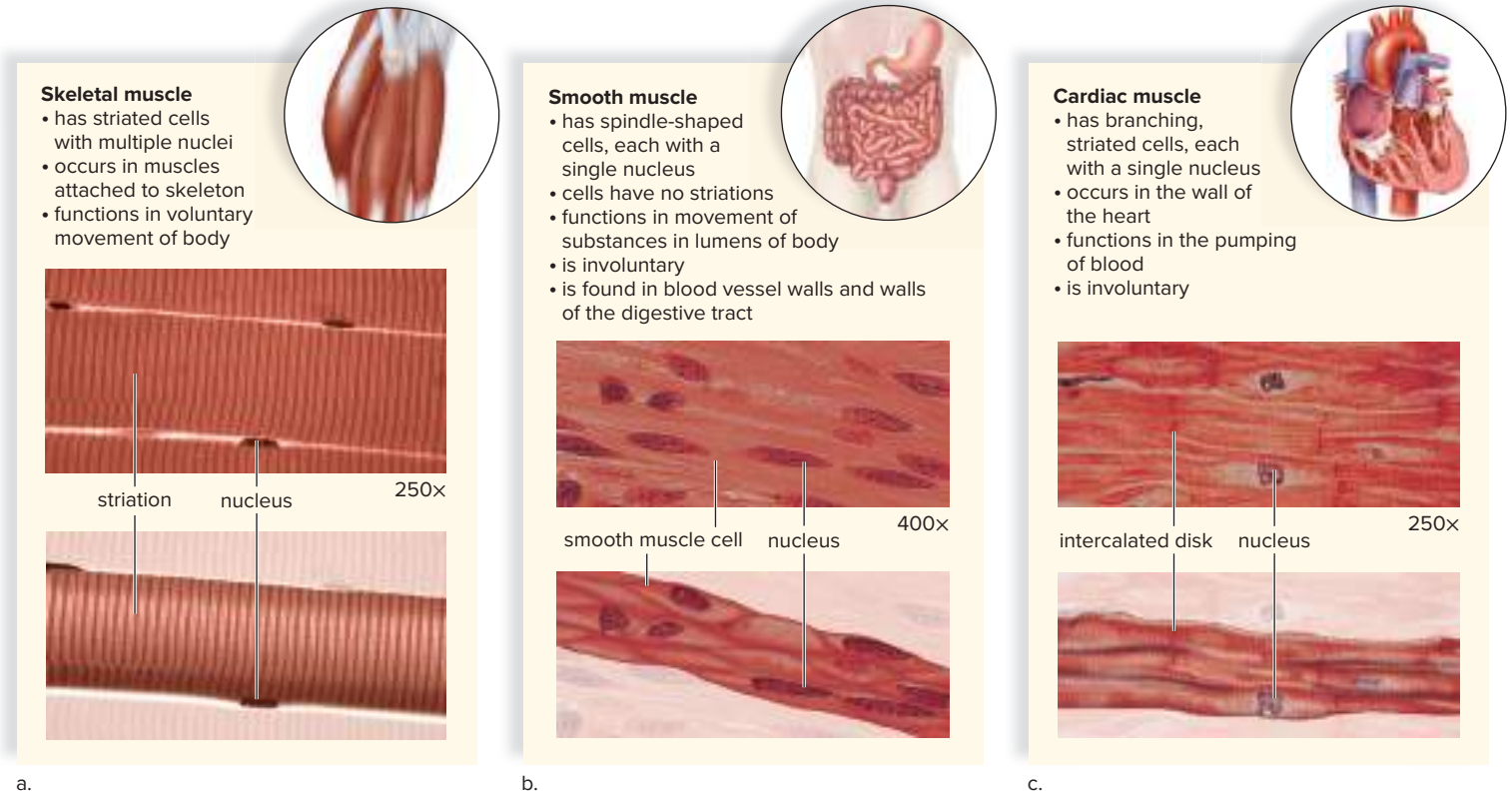


Figure 31.5 Muscular tissue. **a.** Skeletal muscle is voluntary and striated. **b.** Smooth muscle is involuntary and nonstriated. **c.** Cardiac muscle is involuntary and striated. Cardiac muscle cells branch and fit together at intercalated disks. (a, c): ©Ed Reschke; (b): Dennis Strete/McGraw Hill

ISTUDY

moves along its lumen (central cavity). When the smooth muscle of the blood vessels contracts, blood vessels constrict, helping raise blood pressure. Small amounts of smooth muscle are also found in the iris of the eye and in the skin.

Cardiac muscle (Fig. 31.5*c*) is found only in the walls of the heart. Its contraction pumps blood and accounts for the heartbeat. Cardiac muscle combines features of both smooth muscle and skeletal muscle. Like skeletal muscle, it has striations, but the contraction of the heart is involuntary for the most part. Cardiac muscle cells also differ from skeletal muscle cells in that they usually have a single, centrally placed nucleus. The cells are branched and seemingly fused one with the other, and the heart appears to be composed of one large, interconnecting mass of muscle cells. Actually, cardiac muscle cells are separate and individual, but they are bound end to end at *intercalated disks,* areas where folded plasma membranes between two cells contain adhesion junctions and gap junctions.

Nervous Tissue

Nervous tissue contains nerve cells called neurons and supporting cells called neuroglia. An average human body has about 1 trillion neurons. The nervous system conveys signals termed nerve impulses throughout the body.

Neurons

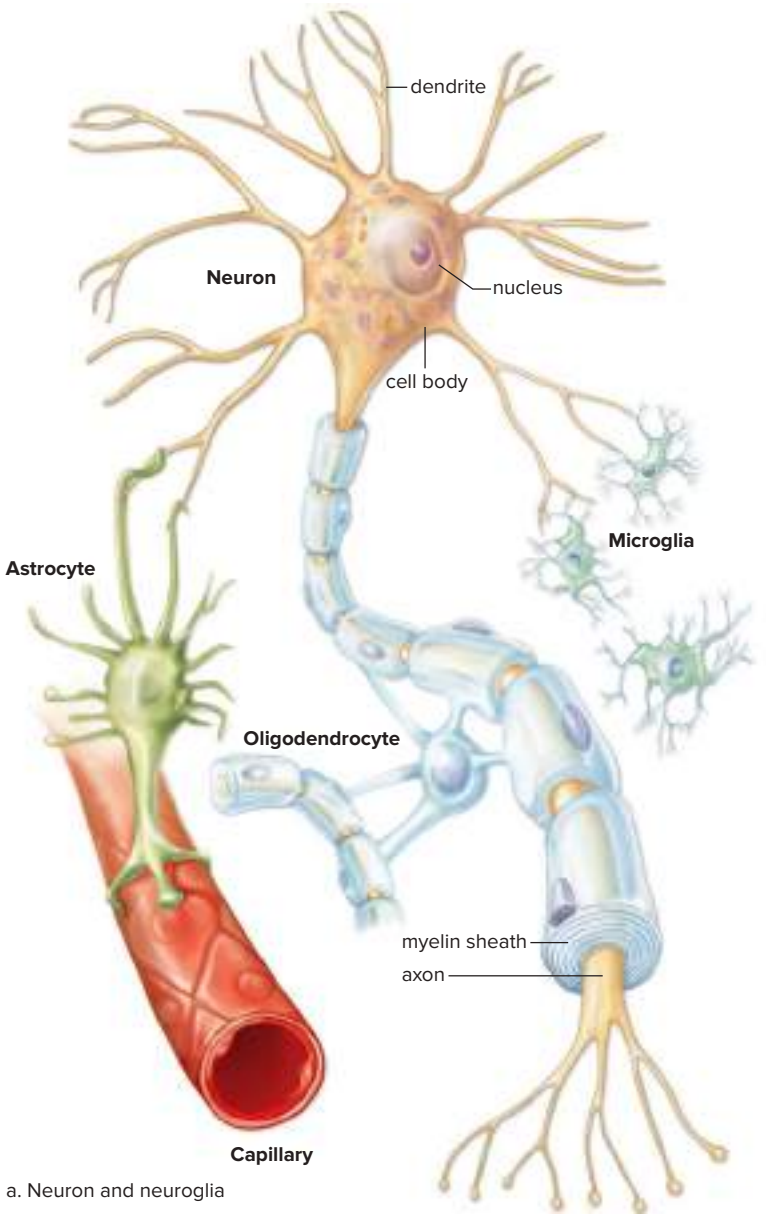
A **neuron** is a specialized cell that has three parts: dendrites, a cell body, and an axon (Fig. 31.6*a*). A dendrite is a process that conducts signals toward the cell body. The cell body contains the major portion of the cytoplasm and the nucleus of the neuron. An axon is a process that typically conducts nerve impulses away from the cell body. Long axons are covered by myelin, a white, fatty substance. The term *fiber* is used here to refer to an axon along with its myelin sheath, if it has one. Outside the brain and spinal cord, fibers bound by connective tissue form **nerves.**

The nervous system has just three functions: sensory input, integration of data, and motor output. Nerves conduct impulses from sensory receptors to the spinal cord and the brain, where integration occurs. The phenomenon called sensation occurs only in the brain, however. Nerves also conduct nerve impulses away from the spinal cord and brain to the muscles and glands, causing them to contract and secrete, respectively. In this way, a coordinated response to the stimulus is achieved.

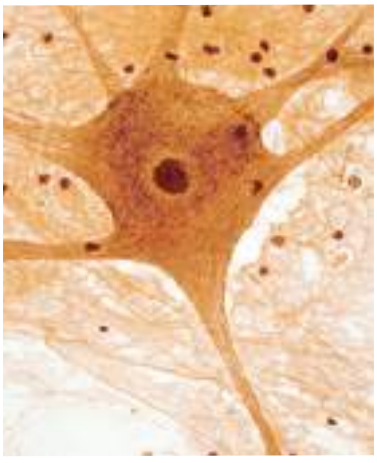
Neuroglia

In addition to neurons, nervous tissue contains cells called **neuroglia.** In the human brain, these cells outnumber neurons as much as ten to one, and they make up approximately half the volume of the organ. Although the primary function of neuroglia is to support and nourish neurons, recent research has shown that some neuroglia directly contribute to brain function.

The four types of neuroglia in the brain are microglia, astrocytes, oligodendrocytes, and ependymal cells (Fig. 31.6*a*). *Microglia,* in addition to supporting neurons, engulf bacterial and cellular debris. *Astrocytes* provide nutrients to neurons and produce a hormone known as glia-derived growth factor, which has



a. Neuron and neuroglia



b. Micrograph of a neuron 200×

Figure 31.6 Neurons and neuroglia. Neurons conduct nerve impulses. Neuroglia consist of cells that support and service neurons and have various functions: Microglia become mobile in response to inflammation and phagocytize debris. Astrocytes lie between neurons and a capillary; therefore, nutrients must first pass through astrocytes before entering neurons. Oligodendrocytes form the myelin sheaths around fibers in the brain and spinal cord. (b): ©Ed Reschke

ISTUDY

potential as a treatment for Parkinson's disease and other diseases caused by neuron degeneration. *Oligodendrocytes* form myelin sheaths. Outside the brain, *Schwann cells* are the type of neuroglia that encircle long nerve fibers and form a myelin sheath. *Ependymal cells* line the fluid-filled spaces of the brain and spinal cord. Neuroglia don't have long processes, but even so, researchers are now beginning to gather evidence that they do communicate among themselves and with neurons.

One fundamental difference between neurons and neuroglia is that the neurons of adult animals usually cannot undergo cell division, but neuroglial cells retain this capacity. As a result, the majority of brain tumors in adults involve actively dividing neuroglial cells. Most of these tumors have to be treated with surgery or radiation therapy, because a large number of tight junctions in the epithelial cells of brain capillaries prevent many substances (including anticancer drugs) from entering the brain. We will take a closer look at the functions of these cell types in Section 37.2.

Check Your Progress 31.1

1. List five types of epithelium and identify an example of where each type can be found.
2. Compare and contrast the three major types of connective tissue.
3. Describe the structure and function of skeletal, smooth, and cardiac muscle.
4. Recall the three parts of a neuron and explain the function of each.

31.2 Organs, Organ Systems, and Body Cavities

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among tissues, organs, and organ systems.
2. List the major life processes carried out by each organ system in vertebrate animals.
3. Describe the two main cavities of the human body and the major organs found in each.

In the biological levels of organization (see Fig. 1.2), the cells of multicellular organisms may be organized into tissues, organs, and organ systems. We explored the major tissue types found in animals in Section 31.1. Here, we will examine some of the common organs and organ systems found in animals. The evolution of organs and organ systems allowed animals to localize biological processes to certain areas of the body and to accomplish them more efficiently.

Organs

An **organ** is composed of two or more types of tissues working together to perform a particular function. For example, a kidney is an organ that contains a variety of epithelial and connective tissues, and these tissues are specialized for the function of eliminating waste products from the blood. While the organs found in animals often have similar functions, in most cases they are adapted to fit the requirements of the species' environment.

Table 31.1 Roles of Animal Body Systems

Life Processes	Organ Systems
Coordinate body activities	Nervous system
	Endocrine system
Acquire food and energy	Muscular system
	Skeletal system
	Nervous system
	Digestive system
Allow movement	Skeletal system
	Muscular system
Exchange gases	Respiratory system
Transport materials	Cardiovascular system
	Lymphatic system
Eliminate wastes	Urinary system
	Digestive system
	Respiratory system
Protect the body from pathogens	Immune system
	Lymphatic system
Produce offspring	Reproductive system

Organ Systems

In most animals, individual organs function as part of an organ system. An **organ system** contains many different organs that cooperate to carry out a general process, such as the digestion of food. Similar types of organ systems are found in most invertebrates, as well as in all vertebrate animals. These organ systems carry out the life processes that all of these animals, including humans, must carry out (Table 31.1).

Body Cavities

Each organ system has a particular distribution within the body. Vertebrates have two main **body cavities:** the smaller dorsal cavity and the larger ventral cavity (Fig. 31.7*a*). The brain and the spinal cord are in the dorsal cavity.

During development, the ventral cavity develops from the coelom. In humans and other mammals, the coelom is divided by a muscular diaphragm that assists breathing. The heart and the lungs are located in the upper (thoracic, or chest) cavity (Fig. 31.7*b*). The major portions of the digestive system, including the accessory organs (e.g., the liver and pancreas), are located in the abdominal cavity, as are the kidneys of the urinary system. The urinary bladder, the female reproductive organs, and certain male reproductive organs are located in the pelvic cavity.

ISTUDY

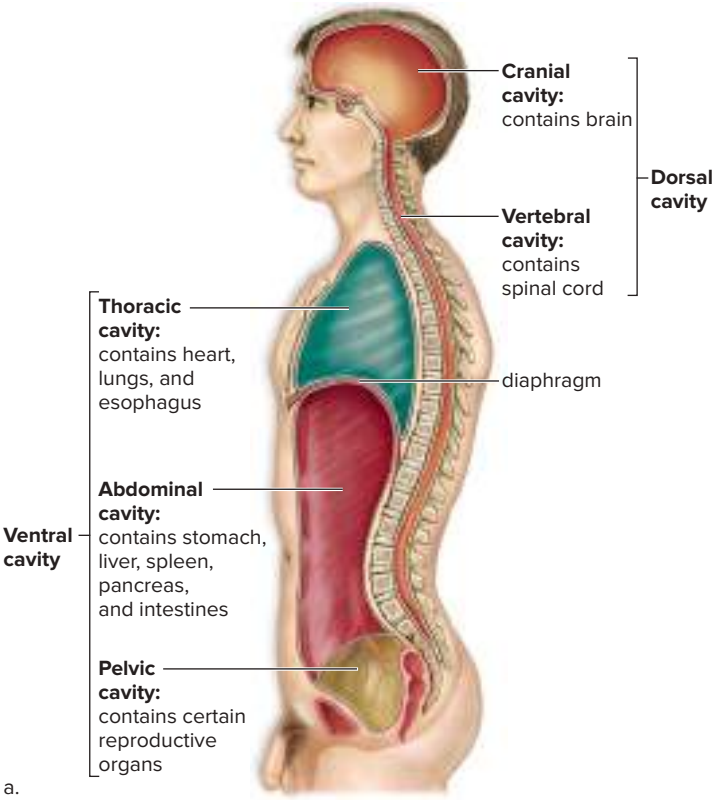


Figure 31.7 Mammalian body cavities. **a.** Side view. The dorsal ("toward the back") cavity contains the cranial cavity and the vertebral canal. The brain is in the cranial cavity, and the spinal cord is in the vertebral canal. The well-developed ventral ("toward the front") cavity is divided by the diaphragm into the thoracic cavity and the abdominopelvic cavity (abdominal cavity and pelvic cavity). The heart and lungs are in the thoracic cavity, and most other internal organs are in the abdominal cavity. **b.** Frontal view of the thoracic cavity.

Check Your Progress 31.2

1. Explain the difference between an organ and an organ system.
2. Identify two organ systems that protect the body from disease.
3. List the body cavities found within the dorsal and ventral cavities of humans.

31.3 The Integumentary System

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the function of the skin in animals.
2. Identify the two main regions of skin and explain their function.
3. Explain the function of melanocytes in the skin and the effects of UV radiation.
4. Describe the makeup and function of the accessory structures of human skin.

The **integumentary system,** consisting of the skin, its derivatives, and its accessory organs, is the largest and most conspicuous organ system in the body. The skin of an average human covers an area of 21 square feet and accounts for nearly 15% of the body weight.

Derivatives of the skin differ throughout the vertebrate world. Most fishes have protective outgrowths of the skin called scales. Amphibian skin is usually covered with mucous glands. Scales are characteristic of reptiles, and they are found on the legs and feet of birds; however, only birds have feathers, which grow from specialized follicles in the skin. Hair is found only on the skin of mammals, and all mammals (even whales) have hair at some stage of their life.

Functions of Skin

Skin covers the body, protecting underlying parts from desiccation, physical trauma, and pathogen invasion. It is also important in regulating body temperature. The skin of small aquatic or semiaquatic animals is often involved in the exchange of gases with the environment (see Section 35.1). In contrast, the dry, scaly skin of reptiles is very poor at gas exchange, but it prevents water loss and thus was probably an important evolutionary adaptation to life on land. Feathers are unique appendages of bird skin that function in insulation, waterproofing, and, of course, flight. Skin is also equipped with a variety of sensory structures that monitor touch, pressure, temperature, and pain. In addition, skin cells manufacture precursor molecules that are converted to vitamin D after exposure to UV light.

Regions of Skin

The **skin** has two main regions: the epidermis and the dermis (Fig. 31.8). A *subcutaneous layer,* also known as the hypodermis, is found between the skin and any underlying structures, such as muscle or bone.

The Epidermis

The **epidermis** (Gk. *epi,* "over"; *derma,* "skin") is made up of stratified squamous epithelium. Human skin can be described as thin skin or thick skin based on the thickness of the epidermis. Thin skin covers most of the body and is associated with hair follicles, sebaceous (oil) glands, and sweat glands. Thick skin appears in regions of wear and tear, such as the palms of the hands and soles of the feet. Thick skin has sweat glands but no sebaceous glands or hair follicles.

In both types of skin, new cells derived from stem (basal) cells become flattened and hardened as they push to the surface

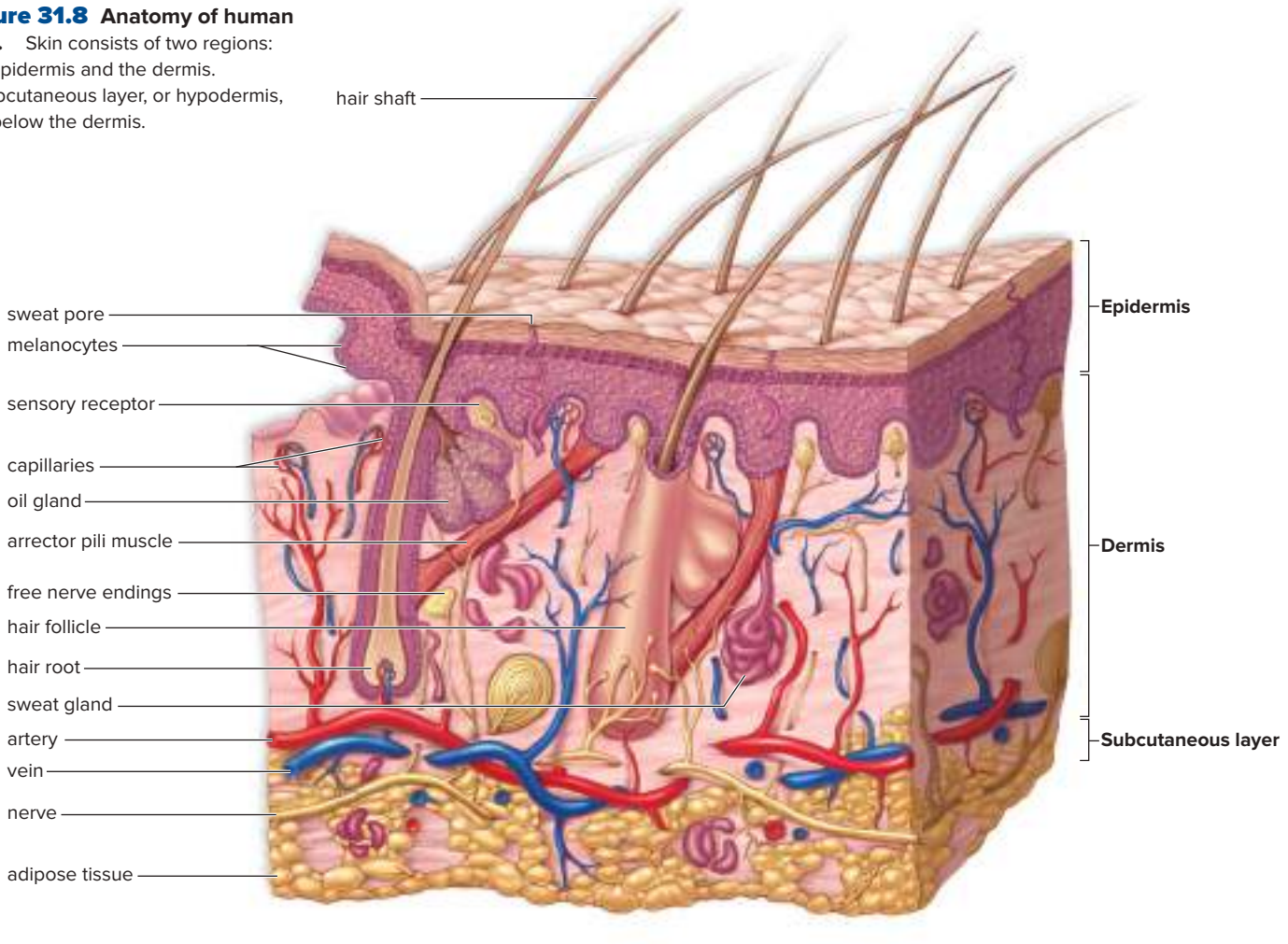


Figure 31.8 Anatomy of human skin. Skin consists of two regions: the epidermis and the dermis. A subcutaneous layer, or hypodermis, lies below the dermis.

(Fig. 31.9). Hardening takes place because the cells produce keratin, a waterproof protein. It is estimated that 1.5 million of these cells are shed from the human body every day. A thick layer of dead, keratinized cells, arranged in spiral and concentric patterns, forms fingerprints (and toe prints), which are thought to increase friction and aid in gripping objects.

Specialized cells in the epidermis called **melanocytes** produce melanin, the pigment responsible for skin color. The amount of melanin varies throughout the body. It is concentrated in freckles and moles. Tanning occurs after a lighter-skinned person is exposed to sunlight because melanocytes produce more melanin, which is distributed to epidermal cells before they rise to the surface. Although some people associate a tan with good health, in reality it signifies that the body is trying to protect itself from the dangerous rays of the sun. Some UV radiation seems to be required for good health, however. The Biological Systems feature, "UV Rays: Too Much Exposure or Too Little?," examines some pros and cons of sun exposure.

Besides skin cancer, the epidermal layer is susceptible to a number of other diseases. Perhaps more than animals with skin covered by tough scales or a thick fur, human skin can be easily traumatized. It is also prone to a variety of bacterial, fungal, and viral infections. Fungi that grow on keratinized tissue are called dermatophytes, and they include the organisms that cause ringworm and athlete's foot. Warts, caused by infection with the

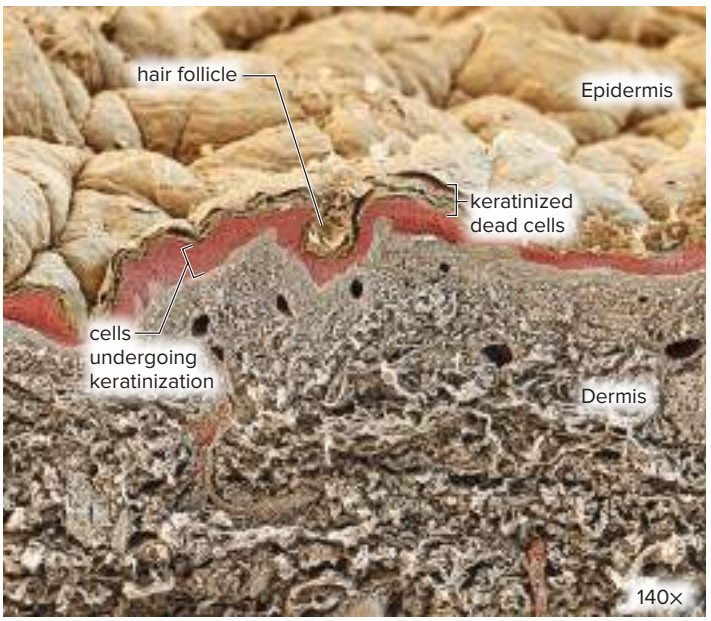


Figure 31.9 Layers of skin. The epidermal surface covers and protects the dermis below. A hair follicle and the dermis are seen in cross section. Cells at the dermal-epidermal border are becoming keratinized and pushed to the surface, from which they are eventually shed. Eye of Science/Science Source

ISTUDY

THEME Biological Systems

UV Rays: Too Much Exposure or Too Little?

The sun is the major source of energy for life on Earth. Without the sun, most organisms would quickly die out. But the sun's energy can also be damaging. In addition to visible light, the sun emits ultraviolet (UV) radiation, which has a shorter wavelength (and thus a higher energy) than visible light. Based on wavelength, UV radiation can be grouped into three types: UVA, UVB, and UVC, but only UVA and UVB reach Earth's surface. Scientists have developed a UV index to determine how powerful the solar rays are in different U.S. cities. In general, the more southern the city, the higher the UV index and the greater the risk for skin cancer. Maps indicating the daily risk levels for various U.S. regions and cities are readily available online from government and commercial websites.

Both UVA and UVB rays can damage skin cells. Tanning occurs when melanin granules increase in keratinized cells at the surface of the skin as a way to prevent further damage by UV rays. Too much exposure to UVB rays can directly damage skin cell DNA, leading to mutations that can cause skin cancer. UVB is also responsible for the pain and redness characteristic of a sunburn. In contrast, UVA rays do not cause sunburn but penetrate more deeply into the skin, damaging collagen fibers and inducing premature aging of the skin, as well as certain types of skin cancer. UVA rays can also induce skin cancer by causing the production of highly reactive chemicals that can indirectly damage DNA.

Skin cancer is the most commonly diagnosed type of cancer in the United States, outnumbering cancers of the lung, breast, prostate, and colon combined. The most common type of skin cancer is basal cell carcinoma (Fig. 31A*a*), which is rarely fatal but can be disfiguring if allowed to grow. Squamous cell carcinoma (Fig. 31A*b*), the second most common type, is fatal in about 1% of cases. The most deadly form is melanoma, which occurs in adolescents and young adults, as well as in older people. If detected early, over 95% of people survive at least 5 years, but if the cancer cells have already spread throughout the body, only 10–20% can expect to live this long.

Melanoma affects pigmented cells and often has the appearance of an unusual mole (Fig. 31Ac). Any moles that become malignant are removed surgically. If the cancer has spread, chemotherapy and a number of other treatments are also available. In March 2007, the USDA approved a vaccine to treat melanoma in dogs, which was the first time a vaccine has been approved as a treatment for any cancer in animals or humans. Clinical trials are under way to test a similar vaccine for use in humans, and initial results have proven to be successful. In March 2011, the FDA approved a drug called ipilimumab, after it was shown to extend the lives of patients with advanced, late-stage melanoma.

According to the Skin Cancer Foundation, about 90% of nonmelanoma skin cancers, and 65% of melanomas, are associated with exposure to UV radiation from the sun. So how can we protect ourselves? First, try to minimize sun exposure between 10 A.M. and 4 P.M. (when the UV rays are most intense) and wear protective clothing, hats, and sunglasses. Second, use sunscreen. Sunscreens generally do a better job of blocking UVB than UVA rays. In fact, the sun protection factor (SPF) printed on sunscreen labels refers only to the degree of protection against UVB. Many sunscreens don't provide as much protection against UVA, and because UVA doesn't cause sunburn, people may have a false sense of security. Some sunscreens do a better job of blocking UVA—look for those that contain zinc oxide, titanium dioxide, avobenzone, or mexoryl. Unfortunately, tanning salons use lamps that emit UVA rays that are two to three times more powerful than the UVA rays emitted by the sun. Because of the potential damage to deeper layers of skin, most medical experts recommend avoiding indoor tanning salons altogether.

Because UV light is potentially damaging, why haven't all humans developed the more protective dark skin that is common to humans living in tropical regions? It turns out that vitamin D is produced in the body only when UVB rays interact with a form of cholesterol found mainly in the skin. This "sunshine vitamin" serves several important functions in the body, including keeping bones strong, boosting the immune system, and reducing blood pressure. Certain foods also contain vitamin D, but it can be difficult to obtain sufficient amounts through diet alone. Therefore, in more temperate areas of the planet, individuals with lighter skin have the advantage of being able to synthesize sufficient vitamin D. Interestingly, dark-skinned people living in such regions may be at increased risk for vitamin D deficiency.

How much sun exposure is enough in temperate regions of the world? During the summer months, an average individual with fair skin will synthesize plenty of vitamin D after exposure to 10–15 minutes of midday sun. During winter months, however, people living north of Atlanta probably receive too few UV rays to stimulate vitamin D synthesis and therefore must fulfill the requirement through their diet.

a. Basal cell carcinoma

b. Squamous cell carcinoma

c. Melanoma

Figure 31A Skin cancer. **a.** Basal cell carcinomas are the most common type of skin cancer. **b.** Squamous cell carcinomas arise from the epidermis, usually in sun-exposed areas. **c.** Malignant melanomas result from proliferation of pigmented cells. Warning signs include changes in size, shape, or color of a normal mole, as well as itching or pain. (a, b): Dr. P. Marazzi/Science Source; (c): James Stevenson/Science Source

Questions to Consider

1. Your friend is considering a visit to an indoor tanning salon and asks your advice. What might you tell your friend about the potential health risks of visiting the salon?
2. If caught early, melanoma is rarely fatal. What are some factors that could delay the detection of a melanoma?
3. There is considerable controversy about the level of dietary vitamin D intake that should be recommended. Why is this the case?

ISTUDY

human papillomavirus, are small areas of epidermal proliferation that are generally harmless. Genital warts (see Section 41.5), however, can be a more serious problem.

The Dermis

The **dermis** is a region of dense fibrous connective tissue beneath the epidermis. As seen in Figure 31.9, the deeper epidermis forms ridges, which interact with projections of the dermis. The dermis contains collagen and elastic fibers. The collagen fibers are flexible but offer great resistance to overstretching; they prevent the skin from being torn. Stretching of the dermis, as occurs in obesity and pregnancy, can produce stretch marks, or striae.

The elastic fibers maintain normal skin tension but also stretch to allow movement of underlying muscles and joints. (The number of collagen and elastic fibers decreases with age and with exposure to the sun, causing the skin to become less supple and more prone to wrinkling.) The dermis also contains blood vessels that nourish the skin. When blood rushes into these vessels, a person blushes, and when blood is minimal in them, a person turns "blue."

Sensory receptors are specialized nerve endings in the dermis that respond to external stimuli. There are sensory receptors for touch, pressure, pain, and temperature. The fingertips contain the most touch receptors, and these add to our ability to use our fingers for delicate tasks.

The Subcutaneous Layer

Technically speaking, the subcutaneous layer (the hypodermis) beneath the dermis is not a part of skin. It is composed of loose connective tissue and adipose tissue. Subcutaneous adipose tissue helps thermally insulate the body from either gaining heat from the outside or losing heat from the inside. Excessive development of the subcutaneous layer occurs with obesity.

Accessory Structures of Human Skin

Nails, hair, and glands are of epidermal origin, even though some parts of hair and glands are largely found in the dermis.

Nails are a protective covering of the distal part of fingers and toes, collectively called digits. Nails grow from epithelial cells at the base of the nail in the portion called the nail root. The cuticle is a fold of skin that hides the nail root. The whitish color of the half-moon-shaped base, or lunula, results from the thick layer of cells in this area. The cells of a nail become keratinized as they grow out over the nail bed. The appearance of nails can reveal clues about a person's health. For example, in clubbing of the nails, the nails turn down instead of lying flat. This condition is associated with a deficiency of oxygen in the blood.

Hair follicles begin in the dermis and continue through the epidermis, where the hair shaft extends beyond the skin. Contraction of the arrector pili muscles attached to hair follicles causes the hairs to "stand on end" and goose bumps to develop. Epidermal cells form the root of hair, and their division causes a hair to grow. The cells become keratinized and die as they are pushed farther from the root.

Hair, except for the root, is formed of dead, hardened epidermal cells; the root is alive and resides at the base of a follicle in the dermis. A person's scalp has about 100,000 hair follicles, on average. The number of follicles varies from one body region to another. The texture of hair is dependent on the shape of the hair shaft. In wavy hair, the shaft is oval, and in straight hair, the shaft is round. Hair color is determined by pigmentation. Dark hair is due to melanin concentration, and blond hair has scanty amounts of melanin. Red hair is caused mainly by the presence of an iron-containing pigment called pheomelanin. Gray or white hair results from a lack of pigment.

A hair on the scalp grows about 1 mm every 3 days. Interestingly, chemical substances in the body, such as illicit drugs and by-products of alcohol metabolism, are incorporated into growing hair shafts, where they can be detected by laboratory tests. A certain amount of hair loss is inevitable. According to the American Academy of Dermatology, most people lose about 50–100 hairs from their scalp every day, but sometimes hair loss is more dramatic. Sudden hair loss is often due to illness, nutritional deficiency, medications, or pregnancy. A gradual loss of hair that becomes more noticeable each year is usually a hereditary condition called *androgenetic alopecia,* which affects about 80 million Americans.

Each hair follicle has one or more **oil glands** (also called *sebaceous glands*), which secrete sebum, an oily substance that lubricates the hair within the follicle and the skin itself. If the oil glands fail to discharge, the secretions collect and form "whiteheads" or "blackheads." The color of blackheads is due to oxidized sebum. Acne is an inflammation of the sebaceous glands that most often occurs during adolescence, when hormonal changes lead to increased oil production.

On average, a person's skin has about 250,000 **sweat glands,** which are present in all regions of skin. A sweat gland is a tubule that begins in the dermis and either opens into a hair follicle or, more often, opens onto the surface of the skin. Sweat glands all over the body play a role in modifying body temperature. When body temperature starts to rise, sweat glands become active. Sweat absorbs body heat as it evaporates. Once the body temperature lowers, sweat glands are no longer active. Other sweat glands occur in the groin and axillary regions and are associated with distinct scents.

Check Your Progress 31.3

1. List one function of skin that is common to all animals and one that is unique to a single group.
2. Compare the structure and function of the epidermal and dermal layers of the skin.
3. Discuss why an individual with darker skin living in northern Canada might develop bone problems.
4. Describe the structure of nails, hair, sweat glands, and oil glands.

31.4 Homeostasis

Learning Outcomes

Upon completion of this section, you should be able to

1. Define *homeostasis* and explain why it is an essential feature of all living organisms.
2. Evaluate the evolutionary benefits of regulating an internal variable, such as body temperature, versus the cost.
3. Differentiate between positive and negative feedback mechanisms and list one specific example of each in animals.

ISTUDY

Overall, the organ systems of animals allow them to maintain a relatively constant internal environment despite fluctuations in the external environment. This process is called **homeostasis.** Although all organisms must carry out some degree of homeostasis, animals vary in the degree to which they regulate their internal environment.

Examples of Homeostatic Regulation

With respect to body temperature, all invertebrates, as well as fish, amphibians, and reptiles, are "cold-blooded," or **poikilothermic,** meaning that their body temperature fluctuates depending on the environmental temperature. This approach saves energy, but it also may restrict the ability of these species to live in extremely cold or hot environments.

Birds and mammals tend to be "warm-blooded," or **homeothermic,** and they have mechanisms for regulating their body temperature toward an optimum. This approach is energetically expensive, but it provides the evolutionary advantage of being able to adapt to many different environments.

Homeostasis does not mean a rigid or unvarying stability but, rather, a dynamic fluctuation around a set point. Besides temperature, animal systems regulate pH, salt balance, and the concentrations of many other body constituents, such as glucose, oxygen, CO_2, and various minerals. All organ systems participate in this regulation:

- The digestive system takes in and digests food, providing nutrient molecules to replace those constantly being consumed by the body cells.
- The respiratory system adds oxygen to the blood and removes carbon dioxide to meet body needs.
- The liver removes glucose from the blood and stores it as glycogen; later, glycogen is broken down to supply the needs of body cells. Blood glucose levels remain fairly constant.
- The pancreas secretes insulin in response to elevated blood glucose; insulin helps regulate glycogen storage.
- Under hormonal control, the kidneys excrete wastes and salts, substances that can affect the pH of the blood.

When homeostasis fails, disease or death often results.

Although homeostasis is, to a degree, controlled by hormones, it is ultimately controlled by the nervous system. In humans, the brain contains regulatory centers that control the function of other organs, maintaining homeostasis. These regulatory centers are often a part of negative feedback systems.

Negative Feedback

Negative feedback is the primary homeostatic mechanism that keeps a variable, such as the blood glucose level, close to a particular value, or set point.

A homeostatic mechanism has at least two components: a sensor and a control center. The sensor detects a change in the internal environment; the control center then initiates an action to bring conditions back to normal again. When normal conditions are reached, the sensor is no longer activated. In other words, a negative feedback mechanism is present when the output of the system dampens the original stimulus. As an example, when blood pressure rises, sensory receptors signal a control center in the brain. The center stops sending nerve impulses to the arterial walls, and they relax. Once the blood pressure drops, signals no longer go to the control center.

A home heating system is often used to illustrate how a more complicated negative feedback mechanism works (Fig. 31.10). You set the thermostat at, say, 68°F. This is the *set point.* The thermostat contains a thermometer, a sensor that detects when the room temperature is above or below the set point. The thermostat also contains a control center; it turns the furnace off when the room is warm and turns it on when the room is cool. When the

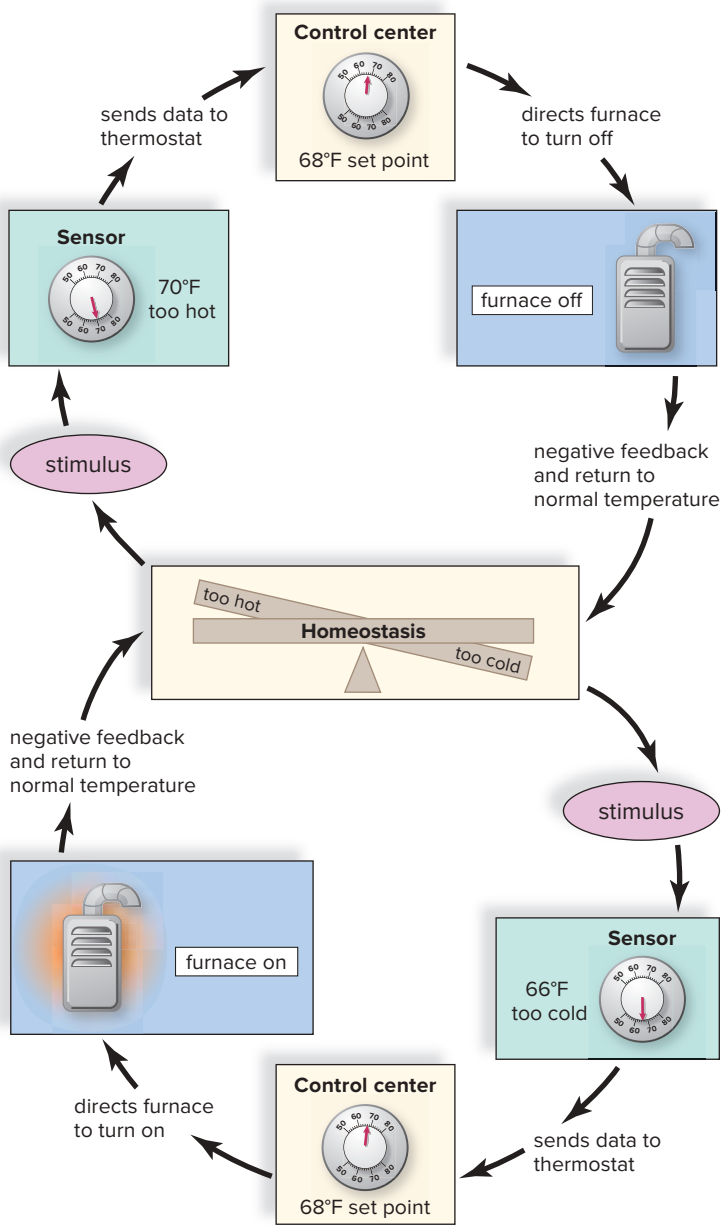


Figure 31.10 Regulation of room temperature using negative feedback. Room temperature is returned to normal when the room becomes too hot (*above*) or too cold (*below*). The thermostat contains both the sensor and the control center. *Above:* The sensor detects that the room is too hot, and the control center turns the furnace off. The room cools, removing the stimulus (negative feedback). *Below:* The sensor detects that the room is too cold, and the control center turns the furnace on. When the temperature returns to normal, the stimulus is no longer present.

ISTUDY

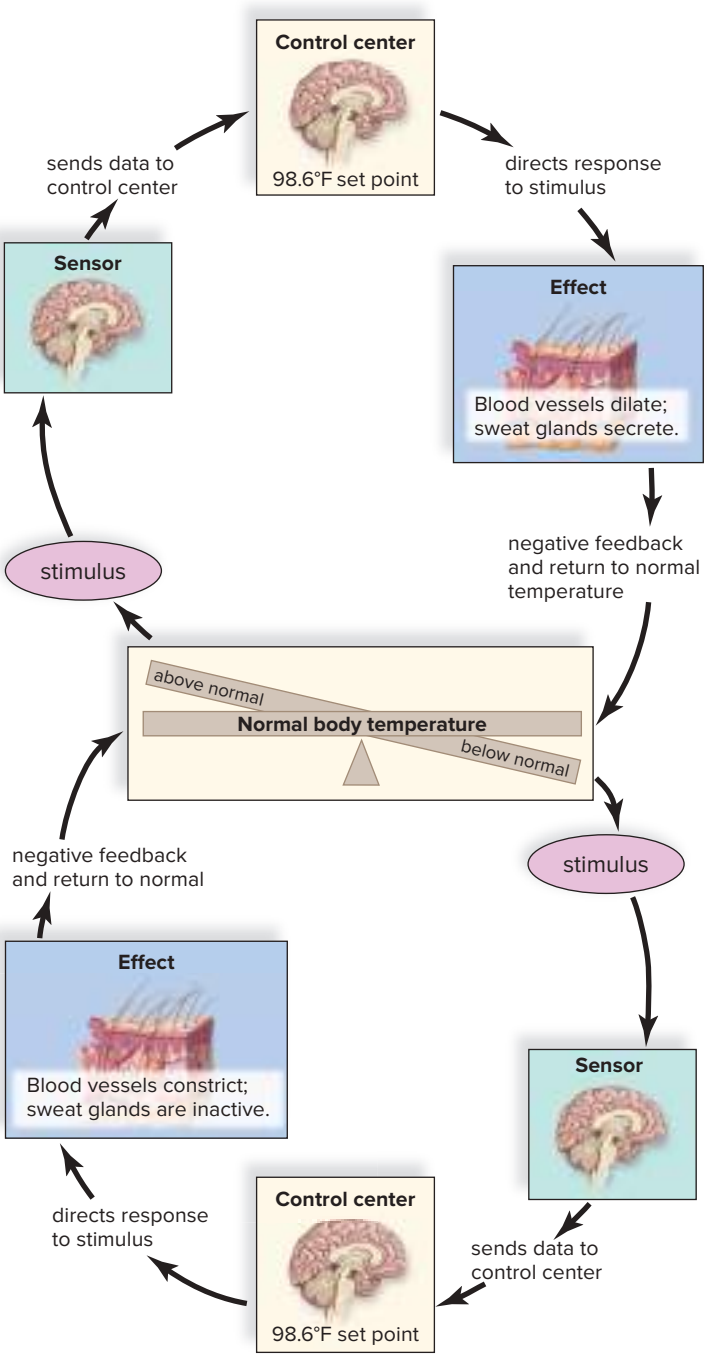


Figure 31.11 Regulation of body temperature by negative feedback. *Above:* When body temperature rises above normal, the hypothalamus senses the change and causes blood vessels to dilate and sweat glands to secrete, so that body temperature returns to normal. *Below:* When body temperature falls below normal, the hypothalamus senses the change and causes blood vessels to constrict. In addition, shivering may occur to bring body temperature back to normal. In this way, the original stimulus is removed (negative feedback).

furnace is off, the room cools a bit, and when the furnace is on, the room warms a bit. In other words, typical of negative feedback mechanisms, there is a fluctuation above and below normal.

Human Example: Regulation of Body Temperature

The sensor and the control center for body temperature are located in a part of the brain called the hypothalamus.

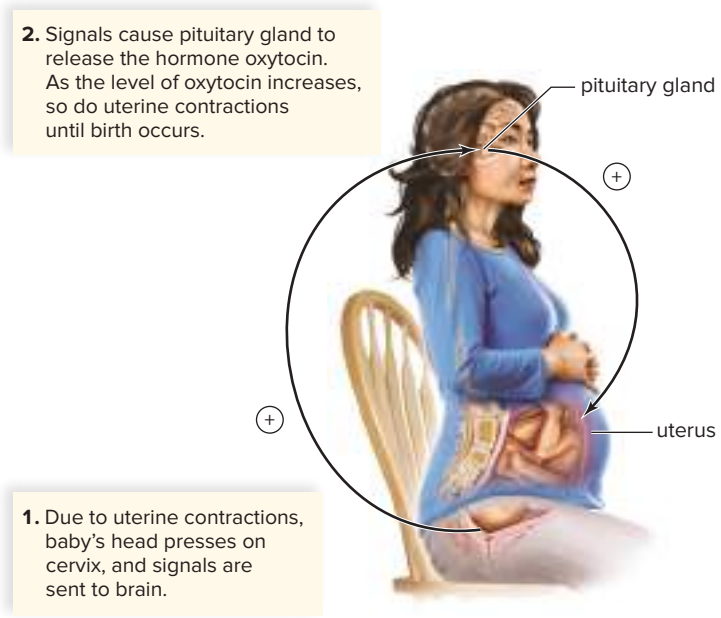


Figure 31.12 Positive feedback. This diagram shows how positive feedback works. The signal causes a change in the same direction until there is a definite cutoff point, such as the birth of a child.

Above Normal Temperature When the body temperature is above normal, the control center directs the blood vessels of the skin to dilate. The result is that more blood flows near the surface of the body, where heat can be lost to the environment. In addition, the nervous system activates the sweat glands, and the evaporation of sweat helps lower body temperature. Gradually, body temperature decreases to 37.0°C (98.6°F). Body temperature does not get colder and colder, because a body temperature below normal brings about a change toward a warmer body temperature.

Below Normal Temperature When the body temperature falls below normal, the control center directs (via nerve impulses) the blood vessels of the skin to constrict (Fig. 31.11). This action conserves heat. If body temperature falls even lower, the control center sends nerve impulses to the skeletal muscles, and shivering occurs. Shivering generates heat, and gradually body temperature rises to 37.0°C. When the temperature rises to normal, the control center is inactivated.

Positive Feedback

Positive feedback is a mechanism that brings about a continually greater change in the same direction. Body processes regulated by positive feedback can produce significant change in a relatively short period of time. For example, when a woman is giving birth, the head of the baby begins to press against the cervix (opening to the birth canal) stimulating sensory receptors there (Fig. 31.12). When nerve impulses reach the brain, the brain causes the pituitary gland to secrete the hormone oxytocin. Oxytocin travels in the blood and causes the uterus to contract. As labor continues, the cervix is increasingly more stimulated, and uterine contractions become stronger until birth occurs.

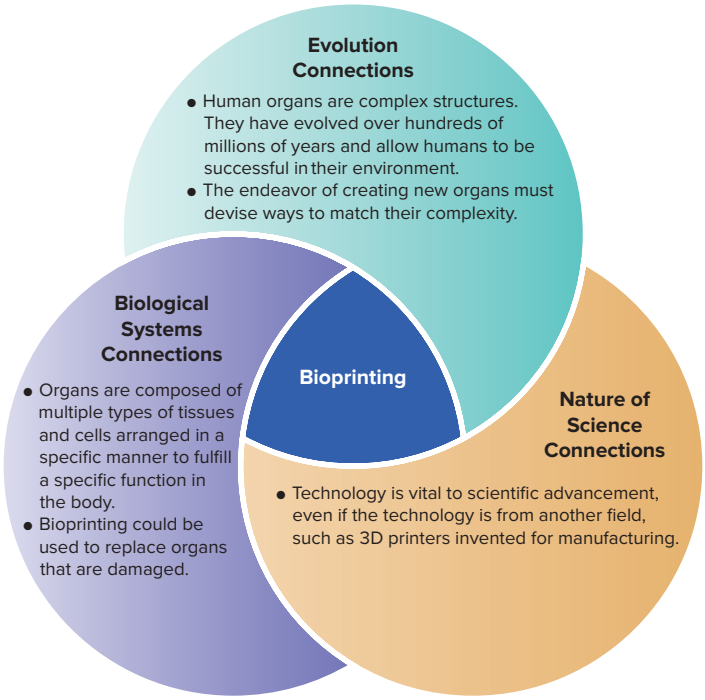
ISTUDY

A positive feedback mechanism can be harmful, as when a fever causes metabolic changes that push the fever still higher. At temperatures higher than about 113°F, many cellular proteins begin to denature, and metabolism stops. Still, positive feedback loops such as those involved in childbirth, blood clotting, and the stomach's digestion of protein assist the body in completing a process that has a definite cutoff point.

Check Your Progress 31.4

1. Define *homeostasis,* and discuss why it is important to body function.
2. Explain how the circulatory, respiratory, and urinary systems specifically contribute to homeostasis.
3. Discuss how negative feedback is similar to the way a thermostat works.

Connecting *the* Concepts



What Did You Learn?

31.1 Types of Tissues

During development, the zygote divides to produce cells that go on to form **tissues** composed of similar cells specialized for a particular function. Tissues make up organs, and organ systems make up the organism. This sequence describes the levels of organization within an organism. Tissues are categorized into four groups:

- **Epithelial tissue** acts to cover the body and line its internal cavities. It is classified by the shape of the cells and the number of layers. Simple epithelial tissue consists of a single layer of cells, all in contact with the basement membrane. Stratified epithelial tissue has multiple cell layers. Epithelial issue may also form **glands,** such as **endocrine glands** and **exocrine glands.**
- **Connective tissue** contains cells and a matrix containing fibers and ground substance. The proteins collagen and elastin give the connective tissue strength and flexibility. **Loose fibrous connective tissue** has both collagen and elastic fibers. **Dense fibrous connective tissue,** like that of **tendons** and **ligaments,** contains closely packed collagen fibers. Both loose fibrous and dense fibrous connective tissues contain cells called **fibroblasts. Adipose tissue** contains cells, called adipocytes, that enlarge and store fat. Supportive connective tissue includes **cartilage** and **bone.** Both have cells within lacunae, but the matrix for cartilage is more flexible than that for bone, which contains calcium salts. Three types of cartilage are **hyaline cartilage, elastic cartilage,** and **fibrocartilage,** while bone can be either **compact bone** or **spongy bone. Blood** is a fluid connective tissue that contains **red blood cells, white blood cells,** and **platelets. Lymph** is also a fluid connective tissue, which is derived from excess **interstitial fluid** that flows through lymphatic vessels.
- **Muscular tissue** can be smooth or striated, as well as involuntary (smooth and cardiac) or voluntary (skeletal). In humans, **skeletal muscle** is attached to bone, **smooth muscle** is in the wall of internal organs, and **cardiac muscle** makes up the heart.
- **Nervous tissue** has one main type of conducting cell, the **neuron,** and several types of **neuroglia.** The majority of neurons have dendrites, a cell body, and an axon. The brain and spinal cord contain complete neurons, while nerves contain only axons. Axons are specialized to conduct nerve impulses.

31.2 Organs, Organ Systems, and Body Cavities

An **organ** contains two or more tissues. An **organ system** is made up of several organs that function together in a particular life process.

ISTUDY

The human body has two main **body cavities.** The dorsal cavity contains the brain and spinal cord. The ventral cavity is divided into the thoracic cavity (containing the heart and lungs) and the abdominal cavity (containing most other internal organs).

31.3 The Integumentary System

The **integumentary system** includes the **skin** and its derivatives, such as scales (in fish, reptiles, and birds), feathers (birds only), and hair (mammals only). This organ system protects underlying tissues against desiccation, trauma, and pathogens, plus functions in thermoregulation, sensory perception, and vitamin D synthesis.

Skin itself has two regions. **Epidermis** (stratified squamous epithelium) overlies the **dermis** (fibrous connective tissue containing sensory receptors, hair follicles, blood vessels, and nerves). The epidermis contains **melanocytes,** cells that produce a pigment, called melanin, that helps protect against the harmful effects of UV light.

A subcutaneous layer is composed of loose connective tissue and adipose tissue. Accessory structures of human skin include **nails, hair follicles, oil glands,** and **sweat glands.**

31.4 Homeostasis

All animals must be capable of maintaining a certain degree of **homeostasis,** or maintenance of a relatively constant internal environment. Animals differ in their approach to regulating certain variables. For example, the body temperature of **poikilothermic** animals varies according to their environment, while **homeothermic** animals maintain an optimal temperature.

All organ systems contribute to homeostasis, but special contributions are made by the liver, which keeps the blood glucose constant, and the kidneys, which regulate the pH. The nervous and hormonal systems regulate the other body systems. Both of these are controlled by **negative feedback** mechanisms, which result in slight fluctuations above and below desired levels. Less commonly, **positive feedback** mechanisms can bring about an increasingly greater change in the same direction.

Thinking Critically About Biology

1. Many cancers develop from epithelial tissue. These include lung, colon, and skin cancers. What are two attributes of this tissue type that make cancer more likely to develop?
2. Bacterial or viral infections can cause a fever. Fevers occur when the hypothalamus changes its temperature set point. Signaling of the hypothalamus can be direct (from the infectious agent itself) or indirect (from the immune system). Which of these would enable the hypothalamus to respond to the greatest variety of infectious agents? Is there any disadvantage to such a signaling system?
3. The risk for human skin cancer, especially melanoma, rises with sun exposure. However, the amount of overall exposure over the years doesn't seem to be as important as the number of brief, intense exposures. A single, blistering sunburn may be even more dangerous than years of moderate tanning. What are some possible explanations for this?
4. What effects can a severe burn have on overall homeostasis of the body? Give a few examples.

Making It Relevant to You

1. Why are structures like blood vessels or ears good candidates for early advancements in bioprinting?
2. Are you designated as an organ donor on your driver's license? What are your reasons for choosing this?
3. Do you think there is a limit to bioprinting? For example, could you imagine an organ as complex as a heart or brain being formed anywhere other than inside a human body during pregnancy?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

32

Circulation and Cardiovascular Systems

Wavebreakmediamicro/Wavebreakmedia/123RF

Synthetic Blood

Every 2 seconds, someone in the United States needs a blood transfusion. The need may have been caused by an injury, or a surgical procedure, or even a disease that requires regular transfusions. To answer these needs, donors provide over 36,000 units of blood every day, or 15 million units of blood every year. Unfortunately, the supply of blood from donors does not always meet demand. Because blood can only be stored for about 42 days, events such as natural disasters place a strain on supplies.

To meet the demand for transfusions, scientists have been developing synthetic, or artificial, blood. The use of an oxygen-carrying blood substitute, called oxygen therapeutics, has the goal of providing a patient's tissues with enough oxygen. In some cases, the blood substitute is completely synthetic and contains chemicals that mimic the oxygen-carrying hemoglobin found in red blood cells. In other cases, scientists are applying biotechnology to manufacture replacement red blood cells. While several biotech companies are currently conducting clinical trials using synthetic blood, most medical professionals believe we are still several years away from being able to produce a synthetic blood supply to meet the needs of society.

As you read through this chapter, think about the following questions:

1. What is the role of blood in an animal?
2. Besides red blood cells, what other elements need to be present in synthetic blood?

Chapter Outline

32.1 Transport in Invertebrates
32.2 Transport in Vertebrates
32.3 The Human Cardiovascular System
32.4 Blood

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 8.1 Animal cells use glucose and oxygen for what specific purpose(s)? Where is carbon dioxide generated?

Figure 29.11 How do the circulatory pathways of fishes, amphibians, reptiles, birds, and mammals resemble each other? How are they different?

Section 31.3 What types of tissues are found in the organs of the cardiovascular system?

Following *the* Themes

CHAPTER 32 – CIRCULATION AND CARDIOVASCULAR SYSTEMS

Evolution	The circulatory systems of animals show a distinct pattern of evolution—from no discrete system to open systems to closed cardiovascular systems.
Nature of Science	Experimentation and observation have increased our understanding of circulatory systems, in some cases resulting in the ability to treat or prevent cardiovascular disease.
Biological Systems	In animals with circulatory systems, the organs and tissues carry out the critical functions of helping provide the cells of the body with oxygen and nutrients, and removing wastes.

ISTUDY

32.1 Transport in Invertebrates

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the purpose of the circulatory system in animals.
2. Explain the differences between blood and hemolymph.
3. Compare and contrast open and closed circulatory systems.

All animal cells require a steady supply of oxygen and nutrients, and their waste products must be removed. In most invertebrates, and all vertebrate animals, these tasks are facilitated by a **circulatory system,** which moves fluid between various parts of the body. However, some invertebrates, such as sponges, cnidarians (e.g., hydras, sea anemones), and flatworms, lack a circulatory system. As we will see, the close proximity of their body tissues to a thin body wall makes a circulatory system unnecessary.

Invertebrates Without a Circulatory System

In hydras (Fig. 32.1*a*), cells either are part of an external layer or line the gastrovascular cavity. Each cell is exposed to water and can independently exchange gases and rid itself of wastes. The cells that line the gastrovascular cavity are specialized to complete the digestive process. They pass nutrient molecules to other cells by diffusion.

In flatworms (Fig. 32.1*b*), the gastrovascular cavity branches throughout the small, flattened body. No cell is very far from one of the three digestive branches, so nutrient molecules can diffuse from cell to cell. Similarly, diffusion meets the respiration and elimination needs of the cells.

In invertebrates with internal body cavities, such as the pseudocoelomate nematodes and the coelomate echinoderms (Fig. 32.1*c*), fluid within the body cavity is used for transporting nutrients and wastes.

Invertebrates with a Circulatory System

Most invertebrates have a circulatory system that transports oxygen and nutrients, such as glucose and amino acids, to their cells. There it picks up wastes, which are later excreted from the body by the lungs or kidneys. There are two types of circulatory fluids: **blood,** which is always contained within blood vessels, and **hemolymph,** a mixture of blood and **interstitial fluid,** which fills the body cavity and surrounds the internal organs.

Open Circulatory Systems

Hemolymph is seen in animals that have an **open circulatory system.** In an open circulatory system, the hemolymph is not always contained within the blood vessels and heart, but enters into the open spaces and cavities of the body. Open circulatory systems were likely the first to evolve, as they are present in simpler and evolutionarily older animals. For example, in most mollusks and arthropods (Fig. 32.2), the heart pumps hemolymph via vessels into tissue spaces that are sometimes enlarged into saclike sinuses. Eventually, hemolymph drains back to the heart. In the grasshopper, an arthropod, the dorsal tubular heart pumps hemolymph into a dorsal aorta, which empties into the hemocoel. When the heart contracts, openings called ostia (*sing.,* ostium) are closed; when the heart relaxes, the hemolymph is sucked back into the heart by way of the ostia.

The hemolymph of a grasshopper is colorless because it does not contain hemoglobin or any other respiratory pigment. It carries

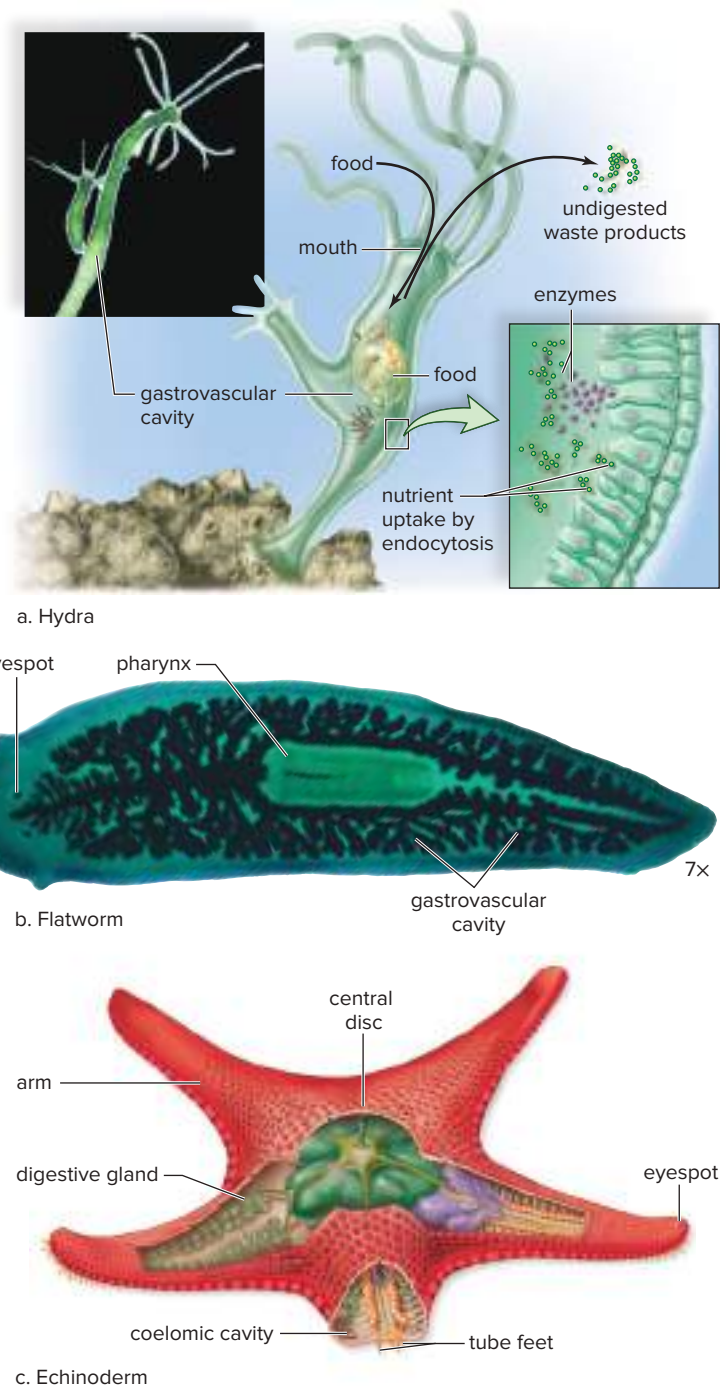


Figure 32.1 Aquatic animals without a circulatory system.
a. In a hydra, a cnidarian, the gastrovascular cavity makes digested material available to the cells that line the cavity. These cells can also acquire oxygen from the watery contents of the cavity and discharge their wastes there. **b.** In a planarian, a flatworm, the gastrovascular cavity branches throughout the body, bringing nutrients to body cells. **c.** In echinoderms, such as this sea star, fluid in the body cavity distributes oxygen and picks up wastes. (photo): (a): M. I. Walker/NHPA/Photoshot/AVALON; (b): Puntasit Choksawatdikorn/123RF

ISTUDY

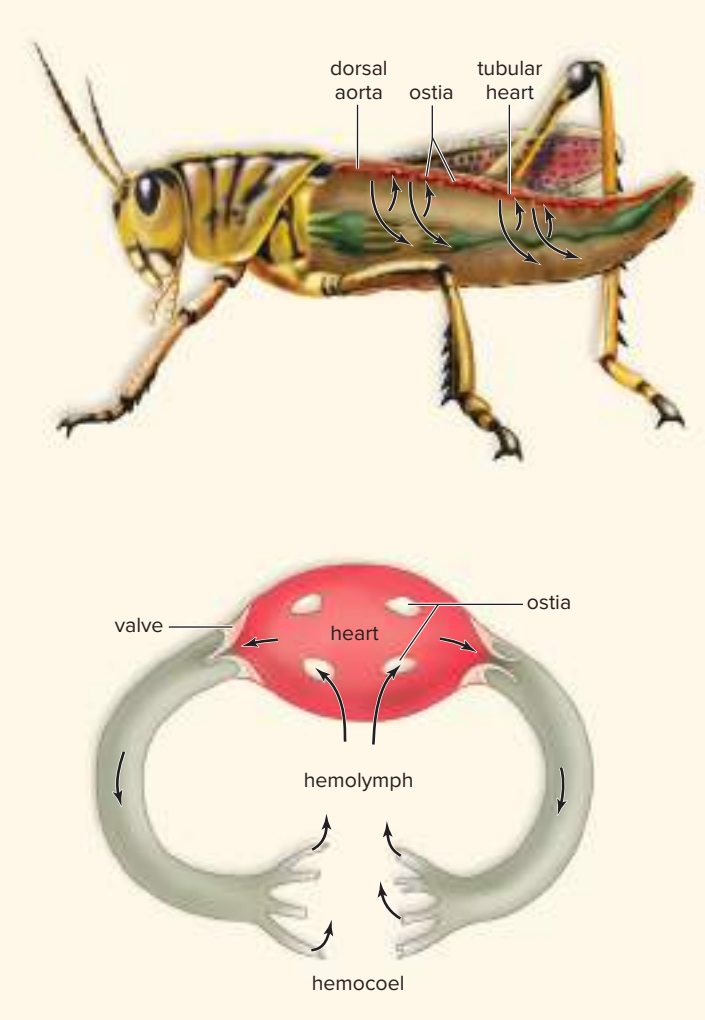


Figure 32.2 Open circulatory systems. *Top:* The grasshopper, an arthropod, has an open circulatory system. *Bottom:* A hemocoel is a body cavity filled with hemolymph, which freely bathes the internal organs. The heart, a pump, sends hemolymph out through vessels and collects it through ostia (openings). The exchange of gases occurs through the use of tracheae (air tubes), which connect directly with the external environment.

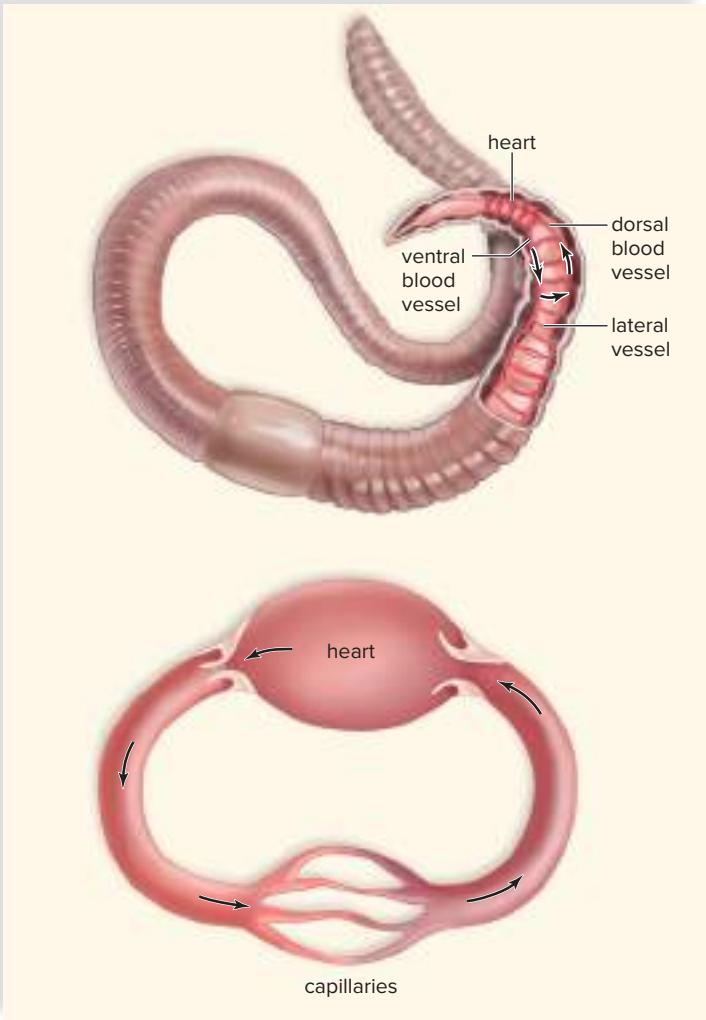


Figure 32.3 Closed circulatory systems. *Top:* The earthworm, an annelid, has a closed circulatory system. The dorsal and ventral blood vessels are joined by five pairs of anterior hearts, which pump blood. *Bottom:* The lateral vessels distribute blood to the rest of the worm.

nutrients but no oxygen. Oxygen is taken to cells, and carbon dioxide is removed from them, by way of air tubes called tracheae, which are found throughout the body. The tracheae provide efficient transport and delivery of respiratory gases, while restricting water loss.

Closed Circulatory Systems

Blood is seen in animals that have a **closed circulatory system,** in which blood does not leave the blood vessels and heart. For example, in annelids, such as earthworms, and in some mollusks, such as squid and octopuses, blood consisting of cells and plasma (a liquid) is pumped by the heart into a system of blood vessels (Fig. 32.3). Valves prevent the backward flow of blood.

In the segmented earthworm, five pairs of anterior hearts (aortic arches) pump blood into the ventral blood vessel (an artery), which has a branch, called a lateral vessel, in every segment of the worm's body. Blood moves through these branches into capillaries, the thinnest of the blood vessels, where exchanges with interstitial fluid take place. Both gas exchange and nutrient-for-waste exchange occur across the capillary walls. Most cells in the body of an animal with a closed circulatory system are not far from a capillary. In an earthworm, after leaving a capillary, blood moves from small veins into the dorsal blood vessel (a vein). This dorsal blood vessel returns blood to the heart for be pumped back out to the tissues.

The earthworm has red blood that contains the respiratory pigment hemoglobin. Hemoglobin is dissolved in the blood and is not contained within blood cells. The earthworm has no specialized organ, such as lungs, for gas exchange with the external environment. Gas exchange takes place across the body wall, which must always remain moist for this purpose.

Check Your Progress 32.1

1. List the general functions of all circulatory systems.
2. Explain how blood differs from hemolymph.
3. Explain why ostia are needed in an open circulatory system, but not a closed circulatory system.

ISTUDY

32.2 Transport in Vertebrates

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among the structure and function of arteries, veins, and capillaries.
2. Compare the path of blood in animals with a one-circuit circulatory pathway versus a two-circuit pathway.
3. Identify the number of atria and ventricles in each type of vertebrate animal.

All vertebrate animals have a closed circulatory system, which is called a **cardiovascular system** (Gk. *kardia,* "heart"; L. *vascular,* "vessel"). It consists of a strong, muscular heart in which the atria (*sing.,* atrium) receive blood and the muscular ventricles pump blood through the blood vessels. There are three kinds of blood vessels (Fig. 32.4): **arteries,** which carry blood away from the heart; **capillaries** (L. *capillus,* "hair"), which exchange materials with interstitial fluid; and **veins** (L. *vena,* "blood vessel"), which return blood to the heart.

Arteries and veins have three distinct layers (Fig. 32.4*a, c*). The outer layer consists of fibrous connective tissue, which is rich in elastic and collagen fibers. The middle layer is composed of smooth muscle and elastic tissue. The innermost layer, called the endothelium, is similar to squamous epithelium.

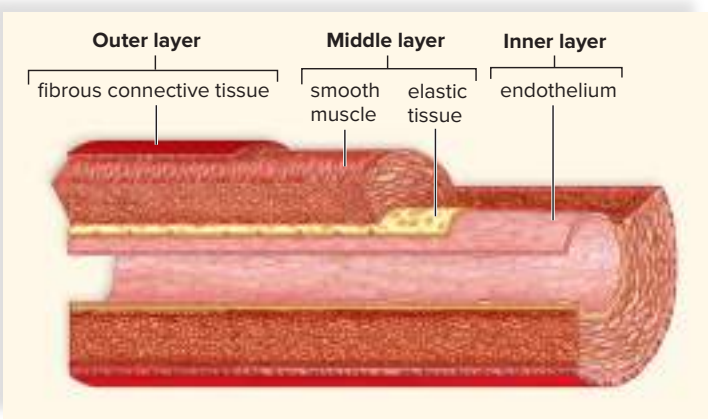
Arteries have thick walls, and those attached to the heart are resilient, meaning they are able to expand and accommodate the sudden increase in blood volume that results after each heartbeat. **Arterioles** are small arteries whose diameter can be regulated by the nervous and endocrine systems. Arteriole constriction and dilation affect blood pressure in general. The greater the number of vessels dilated, the lower the blood pressure.

Arterioles branch into capillaries, which are extremely narrow, microscopic tubes with a wall composed of only one layer of cells (Fig. 32.4*b*). Capillary beds, which consist of many interconnected capillaries (Fig. 32.5), are so prevalent that, in humans, almost all cells are within 60–80 μm of a capillary. But only about 5% of the capillary beds are open at the same time. After an animal has eaten, precapillary sphincters relax, and the capillary beds in the digestive tract are usually open. During muscular exercise, the capillary beds of the muscles are open. Capillaries, which are usually so narrow that red blood cells pass through in single file, allow exchange of nutrient and waste molecules across their thin walls.

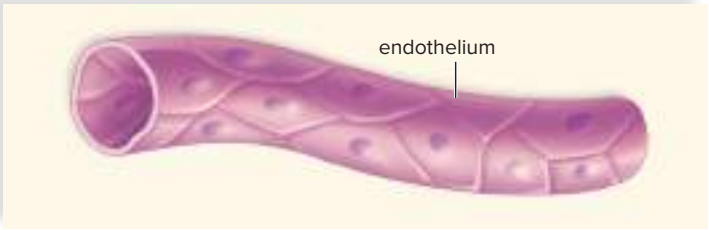
Venules and veins collect blood from the capillary beds and take it to the heart. First, the venules drain the blood from the capillaries, and then they join to form a vein. The wall of a vein is much thinner than that of an artery, and this may be associated with a lower blood pressure in the veins. Valves within the veins point, or open, toward the heart, preventing a backflow of blood when they close (see Fig. 32.4*c*).

Comparison of Circulatory Pathways

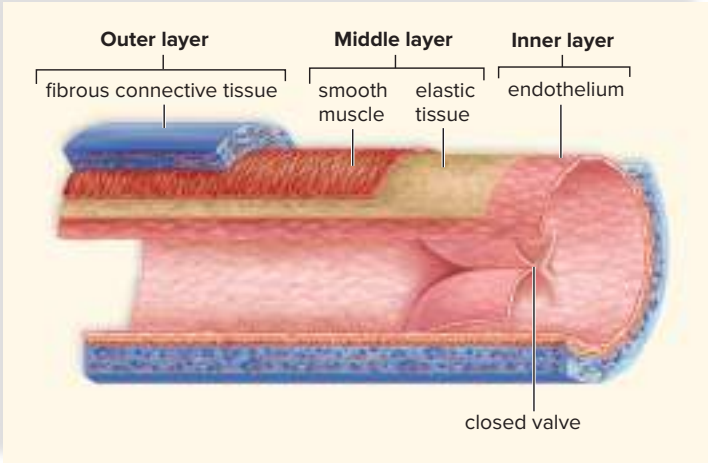
Two different types of circulatory pathways are seen among vertebrate animals. In fishes, blood follows a one-circuit (single-loop) circulatory pathway through the body. The heart has a single atrium and a single ventricle (Fig. 32.6*a*).



a. Artery



b. Capillary



c. Vein

Figure 32.4 Blood vessels of closed circulatory systems.
a. Arteries have well-developed walls with a thick middle layer of elastic tissue and smooth muscle. **b.** Capillary walls are only one cell thick. **c.** Veins have flabby walls, particularly because the middle layer is not as thick as in arteries. Veins have valves, which ensure one-way flow of blood back to the heart.

The pumping action of the ventricle sends blood under pressure to the gills, where gas exchange occurs. After passing through the gills, blood returns to the dorsal aorta, which distributes blood throughout the body. Veins return oxygen-poor blood to an enlarged chamber called the sinus venosus, which leads to the atrium. The atrium pumps blood back to the ventricle. This single circulatory loop has an advantage in that the gill capillaries receive oxygen-poor blood, and the capillaries of the body, called systemic capillaries, receive fully oxygen-rich blood. It is disadvantageous in that after leaving the gills, the blood is under reduced pressure.

As a result of evolutionary changes, other vertebrates have a two-circuit (double-loop) circulatory pathway. The heart pumps blood to the tissues through a **systemic circuit**, as well as pumping blood to the

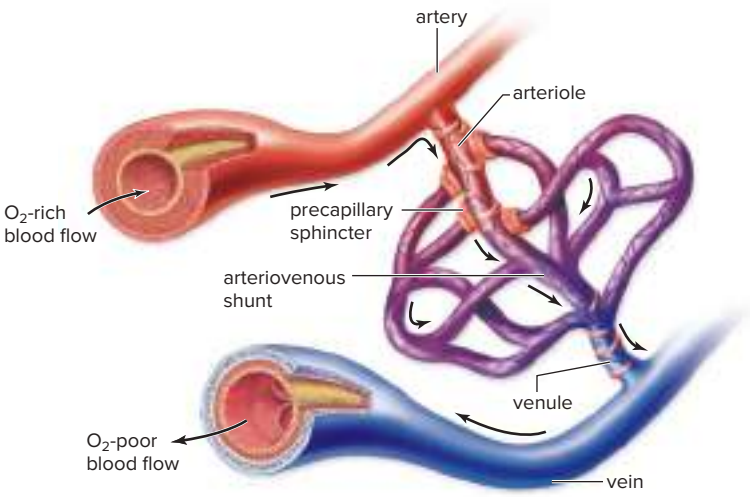


Figure 32.5 Anatomy of a capillary bed. When a capillary bed is open, sphincter muscles are relaxed and blood flows through the capillaries. When precapillary sphincter muscles are contracted, the bed is closed and blood flows through an arteriovenous shunt that carries blood directly from an arteriole to a venule.

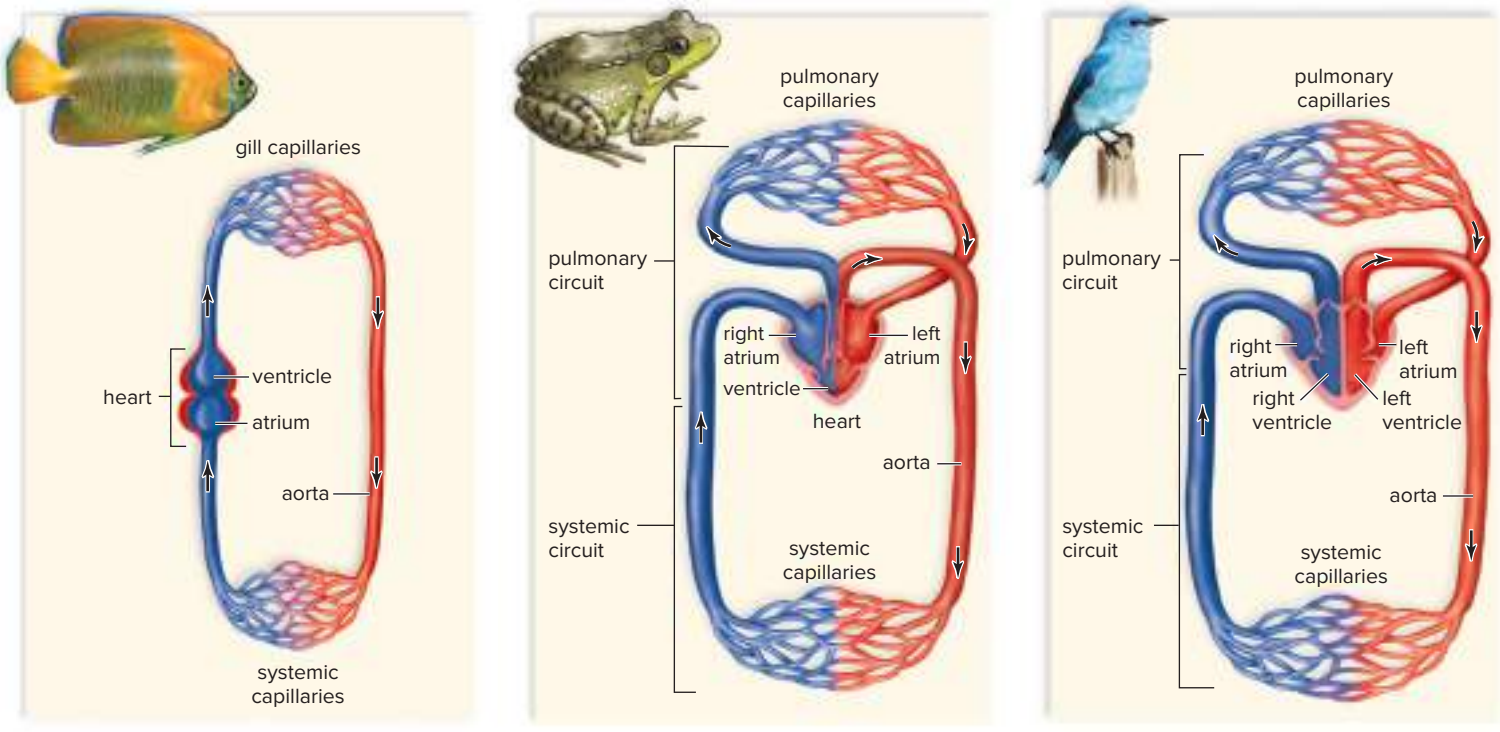
lungs through a **pulmonary circuit** (L. *pulmonarius,* "of the lungs"). This double-pumping action is an adaptation to breathing air on land.

In amphibians, the heart has two *atria* and a single *ventricle* (Fig. 32.6*b*). Oxygen-poor blood from the systemic veins returns to the right atrium. Oxygen-rich blood returning from the lungs passes to the left atrium. Both of the atria empty into the single ventricle. Oxygen-rich and oxygen-poor blood are kept somewhat separate, because oxygen-poor blood is pumped out of the ventricle before the oxygen-rich blood enters. When the ventricle contracts, the division of the main artery also helps keep the blood somewhat separated. More oxygen-rich blood is distributed to the body, and more oxygen-poor blood is delivered to the lungs, and perhaps to the skin, for recharging with oxygen.

In most reptiles, a septum partially divides the ventricle. In these animals, mixing of oxygen-rich and oxygen-poor blood is kept to a minimum. In crocodilians (alligators and crocodiles), the septum completely divides the ventricle into two separate chambers. These reptiles have a four-chambered heart. The heart of birds and mammals is also divided into left and right halves (Fig. 32.6*c*). The right ventricle pumps blood to the lungs, and the larger left ventricle pumps blood to the rest of the body. This arrangement provides adequate blood pressure for both the pulmonary and systemic circuits.

Check Your Progress 32.2

1. List and describe the functions of three types of vessels in a cardiovascular system.
2. Explain why veins are the only blood vessels that contain valves.
3. Discuss the evolutionary benefits of a two-circuit circulatory pathway compared to a one-circuit pathway, especially for animals that breathe air on land.



a. b. c.

Figure 32.6 Comparison of circulatory pathways in vertebrates. **a.** In fishes, the blood moves in a single circuit. Blood pressure created by the pumping of the heart is dissipated after the blood passes through the gill capillaries. This is a disadvantage of the one-circuit system. **b.** Amphibians and most reptiles have a two-circuit system in which the heart pumps blood to both the pulmonary capillaries in the lungs and the systemic capillaries in the body itself. Although there is a single ventricle, little mixing of oxygen-rich and oxygen-poor blood takes place. **c.** The pulmonary and systemic circuits are completely separate in crocodiles (a reptile) and in birds and mammals because the heart is divided by a septum into right and left halves. The right side pumps blood to the lungs, and the left side pumps blood to the rest of the body.

ISTUDY

32.3 The Human Cardiovascular System

Learning Outcomes

Upon completion of this section, you should be able to

1. List the major components of the human heart, including the chambers and major valves.
2. Trace the flow of blood in the systemic and pulmonary circuits.
3. Discuss how the SA and AV nodes control the contractions of the heart muscle, as well as how these electrical changes result in the characteristic patterns seen in an ECG.
4. Describe the major categories of cardiovascular disease that occur in the United States.

In the cardiovascular system of humans, the contractions of the heart keep blood moving into the arteries and then into the pulmonary and systemic circuits. Skeletal muscle contraction against the veins is the main force responsible for the movement of blood in the veins.

The Human Heart

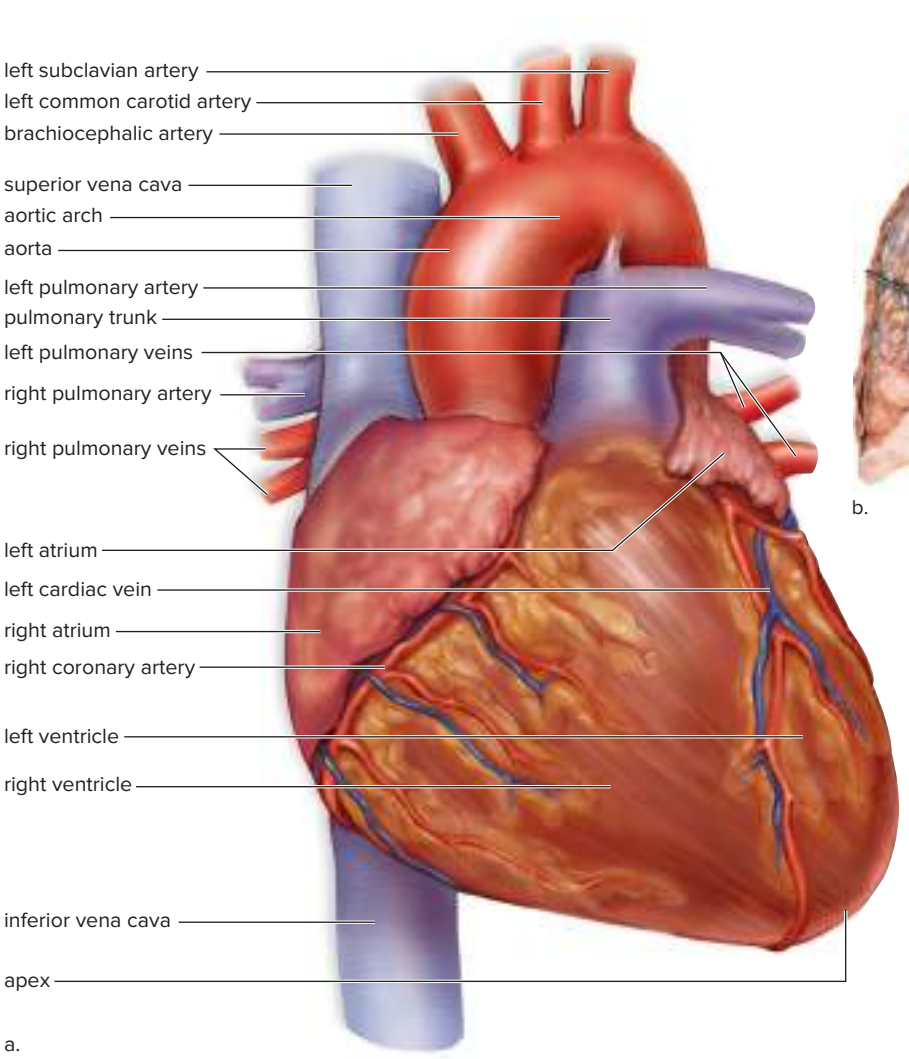
The **heart** is a cone-shaped, muscular organ about the size of a fist (Fig. 32.7). It is located between the lungs directly behind the sternum (breastbone) and is tilted so that the apex (the pointed end) is oriented to the left.

Structure of the Heart

The major portion of the heart, called the *myocardium,* consists largely of cardiac muscle tissue. The myocardium receives oxygen and nutrients from the coronary arteries, not from the blood it pumps. The muscle fibers of the myocardium are branched and tightly joined to one another at intercalated disks.

The heart lies within the *pericardium,* a thick, membranous sac that secretes a small quantity of lubricating liquid. The inner surface of the heart is lined with endocardium, a membrane composed of connective tissue and endothelial tissue. The lining is continuous with the endothelium lining of the blood vessels.

Internally, a wall called the **septum** separates the heart into a right side and a left side (Fig. 32.8). The heart has four chambers. The two upper, thin-walled **atria** (*sing.,* atrium) have wrinkled, protruding appendages called auricles. The two lower chambers



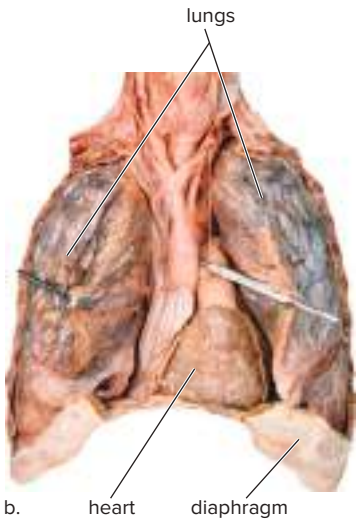
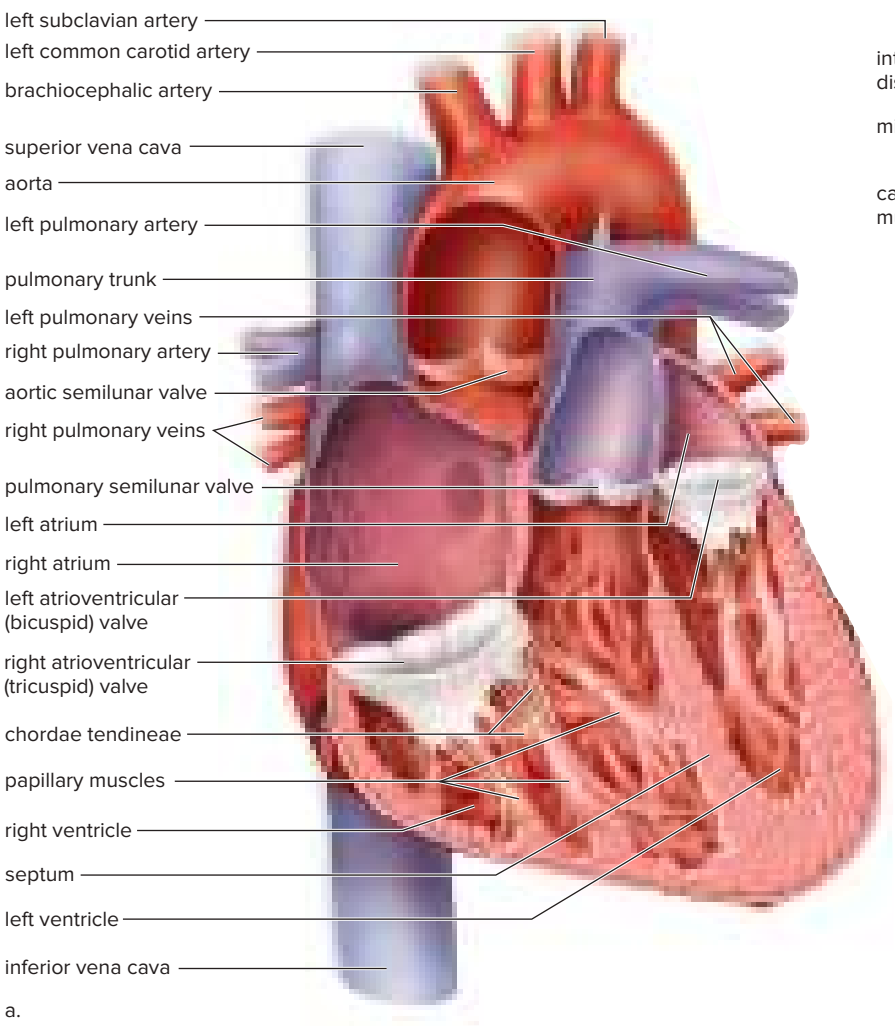


Figure 32.7 External heart anatomy. **a.** The venae cavae and the pulmonary trunk are attached to the right side of the heart. The aorta and the pulmonary veins are attached to the left side of the heart. Blood vessels are colored red if they carry oxygen-rich blood and are colored blue if they carry oxygen-poor blood. **b.** Photograph of a mammalian heart in its natural position in the chest. (b): Christine Eckel/McGraw Hill

ISTUDY



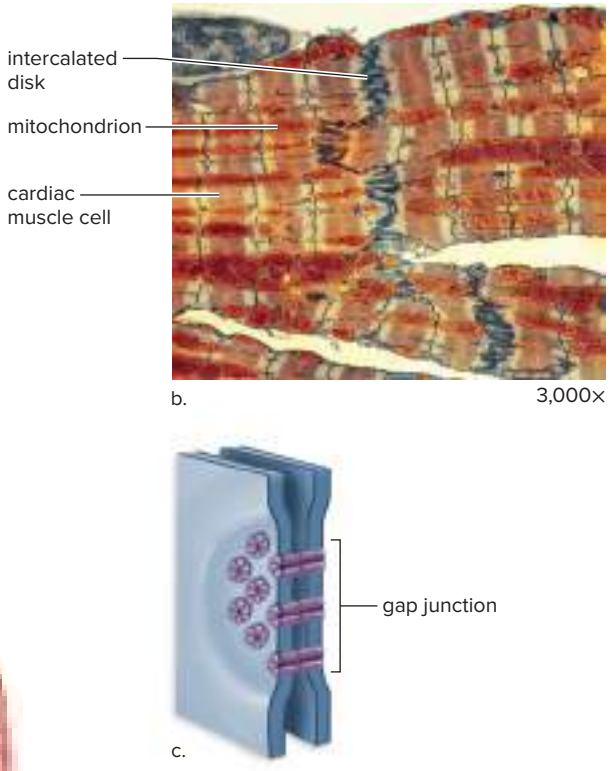


Figure 32.8 Internal anatomy of the heart. **a.** The heart has four chambers; the two chambers on the right are separated from the two chambers on the left by a septum. When the atrioventricular valves open, blood passes from the atria to the ventricles, and when the semilunar valves open, blood passes out of the heart. **b.** Intercalated disks contain (**c**) gap junctions, and these allow muscle cells to contract simultaneously. Desmosomes at the same location allow the cells to bend and stretch. (b): Thomas Deerinck, NCMIR/Science Source

are the thick-walled **ventricles,** which pump the blood away from the heart.

The heart also has four valves, which direct the flow of blood and prevent its backward movement. The two valves that lie between the atria and the ventricles are called the **atrioventricular valves.** These valves are supported by strong, fibrous strings called chordae tendineae. The chordae, which are attached to muscular projections of the ventricular walls, support the valves and prevent them from inverting when the heart contracts. The atrioventricular valve on the right side is called the tricuspid valve, because it has three flaps, or cusps. The valve on the left side is called the bicuspid (or mitral) valve, because it has two flaps.

The remaining two valves are the **semilunar valves,** whose flaps resemble half-moons, between the ventricles and their attached vessels. The pulmonary semilunar valve lies between the right ventricle and the pulmonary trunk. The aortic semilunar valve lies between the left ventricle and the aorta.

Path of Blood Through the Heart

The intercalated disks (Fig. 32.8*b*) join the fibers of cardiac muscle cells, allowing them to communicate with each other. By sending electrical signals between cells, both atria and then both ventricles contract simultaneously. It is possible to trace the flow of blood through the heart and body in the following manner.

1. The superior vena cava and the inferior vena cava, which carry oxygen-poor blood that is relatively high in carbon dioxide, empty into the right atrium.
2. The right atrium sends blood through an atrioventricular valve (the tricuspid valve) to the right ventricle.
3. The right ventricle sends blood through the pulmonary semilunar valve into the pulmonary trunk and the two pulmonary arteries to the lungs.
4. Four pulmonary veins, which carry oxygen-rich blood, empty into the left atrium.
5. The left atrium sends blood through an atrioventricular valve (the bicuspid, or mitral, valve) to the left ventricle.
6. The left ventricle sends blood through the aortic semilunar valve into the aorta and to the rest of the body.

An examination of this pathway should indicate that oxygen-poor blood never mixes with oxygen-rich blood. Furthermore, blood must go through the lungs in order to pass from the right side to the left side of the heart, as is typical in a double-loop circulatory system. Because the left ventricle has the harder job of

pumping blood to the entire body, its walls are thicker than those of the right ventricle, which pumps blood a relatively short distance to the lungs.

It is common to associate oxygen-rich blood with the arteries and oxygen-poor blood with the veins. However, this is true only in the systemic circuit. In the pulmonary circuit, arteries carry deoxygenated blood to the heart, and veins return oxygenated blood to the heart from the lungs. Thus, it is important to recognize that the terms *arteries* and *veins* refer to the direction of blood flow, not the level of oxygenation.

As the ventricles contract, they send out blood under pressure into the arteries. Because the left side of the heart is the stronger pump, blood pressure is greatest in the aorta. Blood pressure then decreases as the cross-sectional area of arteries and then arterioles increases. Therefore, a different mechanism is needed to move blood in the veins, as we will see later in this section.

The Heartbeat

The average human heart contracts, or beats, about 70 times a minute, so each heartbeat lasts about 0.85 second. This adds up to about 100,000 beats per day. Over a 70-year life span, the average human heart will have contracted about 2.5 billion times! The term **systole** (Gk. *systole,* "contraction") refers to contraction of the heart chambers, and the word **diastole** (Gk. *diastole,* "dilation, spreading") refers to relaxation of these chambers. Each heartbeat, or **cardiac cycle,** consists of the phases shown in Fig. 32.9.

Table 32.1 Events of the Cardiac Cycle

Time (sec)	Atria	Ventricles
0.15	Systole	Diastole
0.30	Diastole	Systole
0.40	Diastole	Diastole

Table 32.1 provides the approximate timing of the cardiac cycle, as well as the condition of the atria and ventricles during each part of the cycle.

First, the atria contract (while the ventricles relax), then the ventricles contract (while the atria relax), and then all chambers rest. Note that the heart is in diastole about 50% of the time. The short systole of the atria is appropriate because the atria send blood only into the ventricles. It is the muscular ventricles that actually pump blood out into the cardiovascular system proper.

The volume of blood that the left ventricle pumps per minute into the systemic circuit is called the **cardiac output.** A person with a heartbeat of 70 beats per minute has a cardiac output of 5.25 liters a minute. This is almost equivalent to the amount of blood in the body, and it adds up to about 2,000 gallons a day. During heavy exercise, the cardiac output can increase manyfold.

When the heart beats, the familiar lub-dub sound is heard as the valves of the heart close. The longer and lower-pitched *lub* is caused by vibrations of the heart when the atrioventricular valves

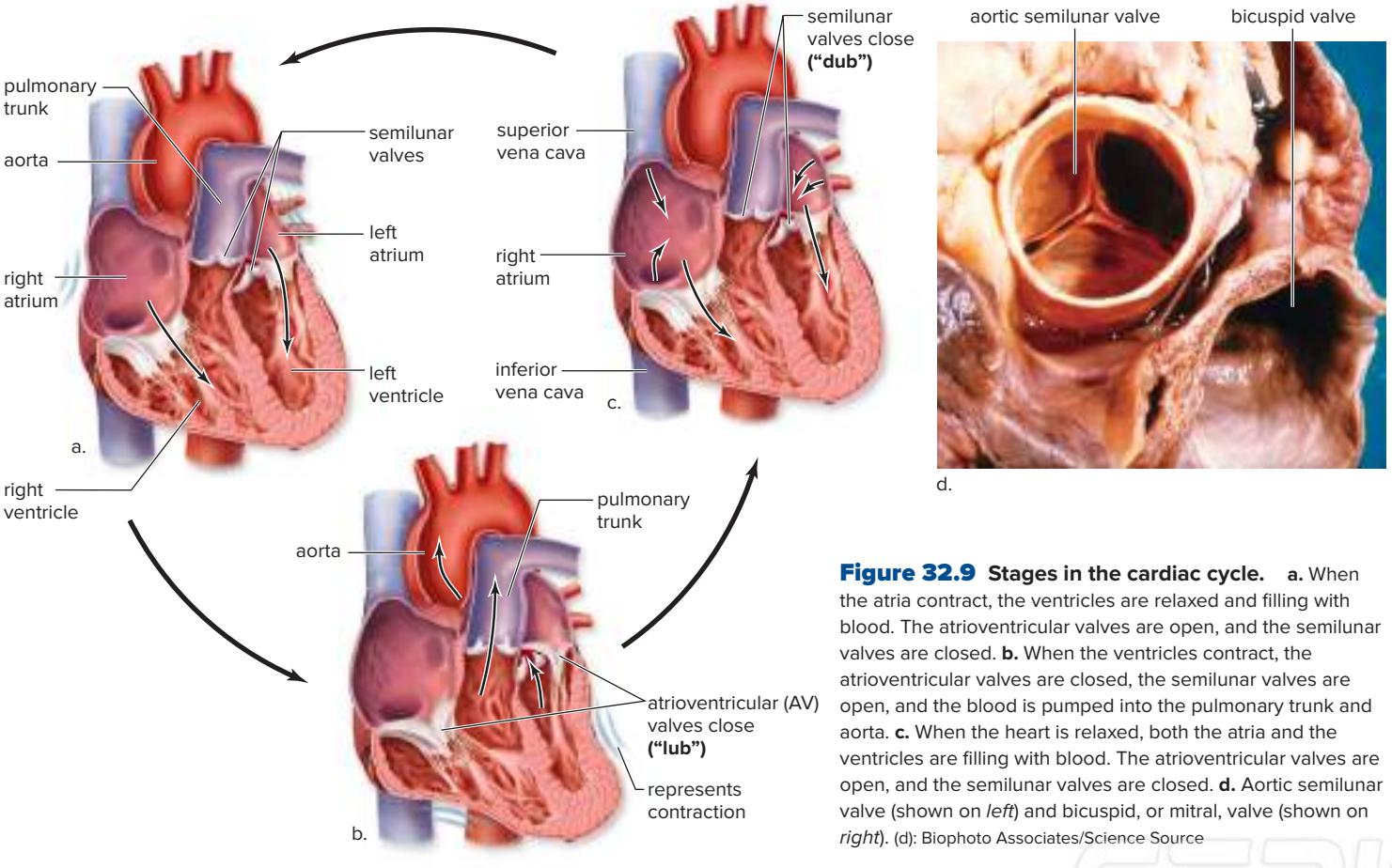


Figure 32.9 Stages in the cardiac cycle. **a.** When the atria contract, the ventricles are relaxed and filling with blood. The atrioventricular valves are open, and the semilunar valves are closed. **b.** When the ventricles contract, the atrioventricular valves are closed, the semilunar valves are open, and the blood is pumped into the pulmonary trunk and aorta. **c.** When the heart is relaxed, both the atria and the ventricles are filling with blood. The atrioventricular valves are open, and the semilunar valves are closed. **d.** Aortic semilunar valve (shown on *left*) and bicuspid, or mitral, valve (shown on *right*). (d): Biophoto Associates/Science Source

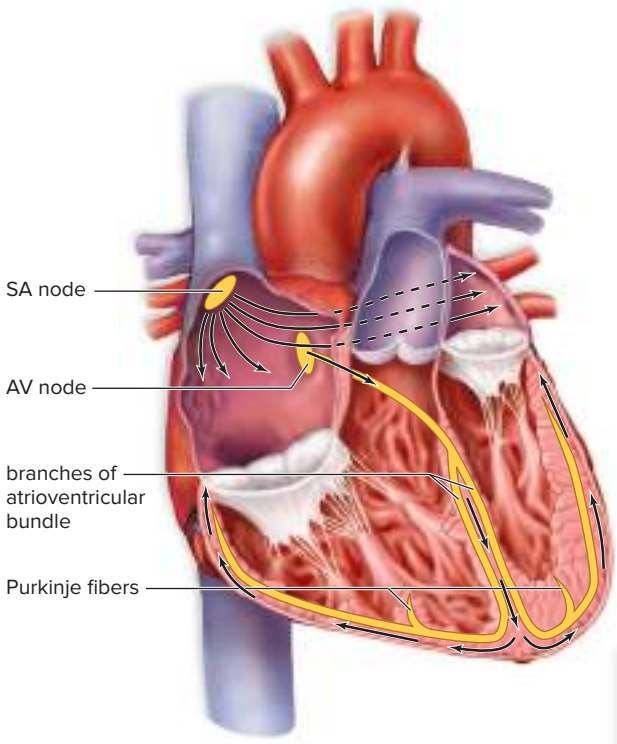


Figure 32.10 Conduction system of the heart. The SA node sends out a stimulus (black arrows), which causes the atria to contract. When this stimulus reaches the AV node, it signals the ventricles to contract. Impulses pass down the two branches of the atrioventricular bundle to the Purkinje fibers, and thereafter the ventricles contract.

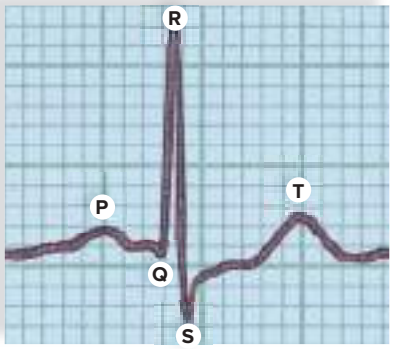
close due to ventricular contraction. The shorter and sharper *dub* is heard when the semilunar valves close due to back pressure of blood in the arteries. A heart murmur, a slight slush sound after the lub, is often due to ineffective valves, which allow blood to pass back into the atria after the atrioventricular valves have closed.

The **pulse** is a wave effect that passes down the walls of the arterial blood vessels when the aorta expands and then recoils following ventricular systole. Because there is one arterial pulse per ventricular systole, the arterial pulse rate can be used to determine the heart rate.

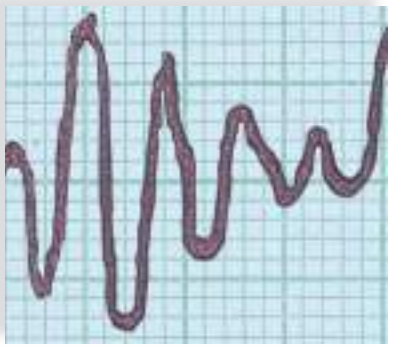
The rhythmic contraction of the atria and ventricles is due to the internal (intrinsic) conduction system of the heart. Nodal tissue, which has both muscular and nervous characteristics, is a unique type of cardiac muscle located in two regions of the heart. The *SA* (*sinoatrial*) *node* is found in the upper dorsal wall of the right atrium; the *AV* (*atrioventricular*) *node* is found in the base of the right atrium, very near the septum (Fig. 32.10). The SA node initiates the heartbeat about every 0.85 second by automatically sending out an excitation impulse, which causes the atria to contract. Therefore, the SA node is called the **pacemaker,** because it usually keeps the heartbeat regular. When the impulse reaches the AV node, the AV node signals the ventricles to contract by way of large fibers terminating in the more numerous and smaller Purkinje fibers.

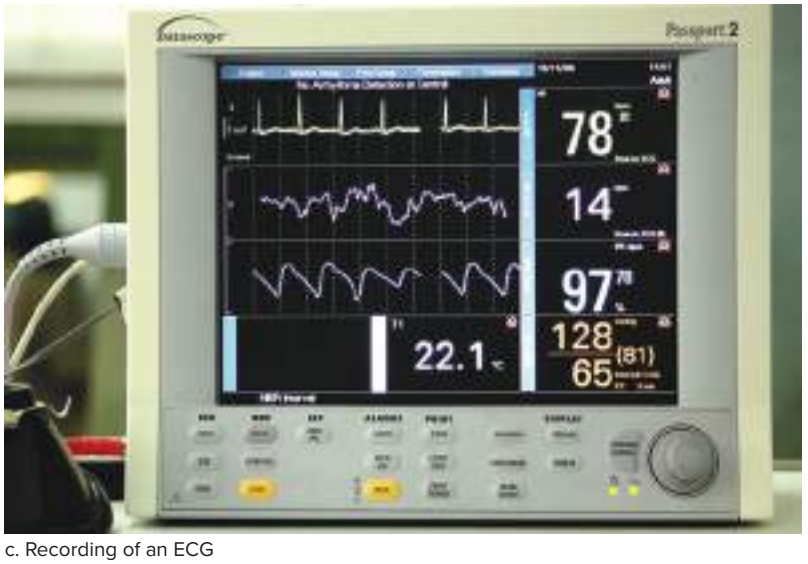
Although the heart muscle will contract without any external nervous stimulation, input from the brain can increase or decrease the rate and strength of heart contractions. In addition, the hormones epinephrine and norepinephrine, secreted into the blood by the adrenal glands, also stimulate the heart. When a person is frightened, for example, the heart pumps faster and stronger due to both nervous and hormonal stimulation.

The Electrocardiogram An **electrocardiogram (ECG)** is a recording of the electrical changes that occur in the myocardium during a cardiac cycle. Body fluids contain ions that conduct electrical currents, and therefore these electrical changes can be detected on the body surface. When an ECG is administered, electrodes placed on the skin are connected by wires to an instrument that detects the myocardium's electrical changes. A normal ECG pattern is shown in Figure 32.11*a*.



a. Normal ECG


b. Ventricular fibrillation



c. Recording of an ECG

Figure 32.11 An electrocardiogram. **a.** A normal ECG indicates that the heart is functioning properly. The P wave occurs just prior to atrial contraction; the QRS complex occurs just prior to ventricular contraction; and the T wave occurs when the ventricles are recovering from contraction. **b.** Ventricular fibrillation produces an irregular ECG due to irregular stimulation of the ventricles. **c.** The recording of an ECG. (a, b): ©Ed Reschke; (c): Medicimage education/Alamy Stock Photo

ISTUDY

When the SA node triggers an impulse, the atrial fibers produce an electrical change called the P wave. The P wave indicates that the atria are about to contract. After that, the QRS complex signals that the ventricles are about to contract and the atria are relaxing. The electrical changes that occur as the ventricular muscle fibers recover produce the T wave.

Various types of abnormalities can be detected by an electrocardiogram. One of these, called ventricular fibrillation, is caused by uncoordinated contraction of the ventricles (Fig. 32.11*b*). Ventricular fibrillation is of special interest, because it can be caused by an injury or a drug overdose. It is the most common cause of sudden cardiac death in a seemingly healthy person. When the ventricles are fibrillating, they can be defibrillated by applying a strong electrical current for a short period of time. Then, the SA node may be able to reestablish a coordinated beat. Many public places, and even private homes, have automatic external defibrillators (AEDs). These are small devices that can be used to determine whether a person is experiencing ventricular fibrillation. If so, the AED administers an appropriate electrical shock to the chest.

Comparison of Circulatory Circuits

As mentioned, the human cardiovascular system includes two major circular pathways: the pulmonary circuit and the systemic circuit (Fig. 32.12).

The Pulmonary Circuit

The pulmonary circuit is responsible for moving blood between the heart and the lungs. In the pulmonary circuit, the path of blood can be traced as follows:

1. Oxygen-poor blood from all regions of the body collects in the right atrium and then passes into the right ventricle, which pumps it into the pulmonary trunk.
2. The pulmonary trunk divides into the right and left pulmonary arteries, which carry blood to the lungs.
3. As blood passes through pulmonary capillaries, carbon dioxide is given off and oxygen is picked up.
4. Oxygen-rich blood returns to the left atrium of the heart, through pulmonary venules that join to form pulmonary veins.

The Systemic Circuit

The majority of the blood vessels in the body belong to the systemic circuit, which is involved in the transportation of blood to and from the organ systems and tissues of the body (with the exception of the lungs). The **aorta** and the **venae cavae** (*sing.,* vena cava) are the major blood vessels in the systemic circuit. In the systemic circuit, the path of blood can be traced as follows:

1. Oxygen-rich blood is pumped from the left ventricle into the major artery of the body, the aorta.
2. Blood then flows from the aorta to smaller arteries and arterioles before arriving at the organ or tissue.
3. From arterioles, blood enters that capillary bed of the organ or tissue. This is the location where gas and nutrient/waste exchange occur.

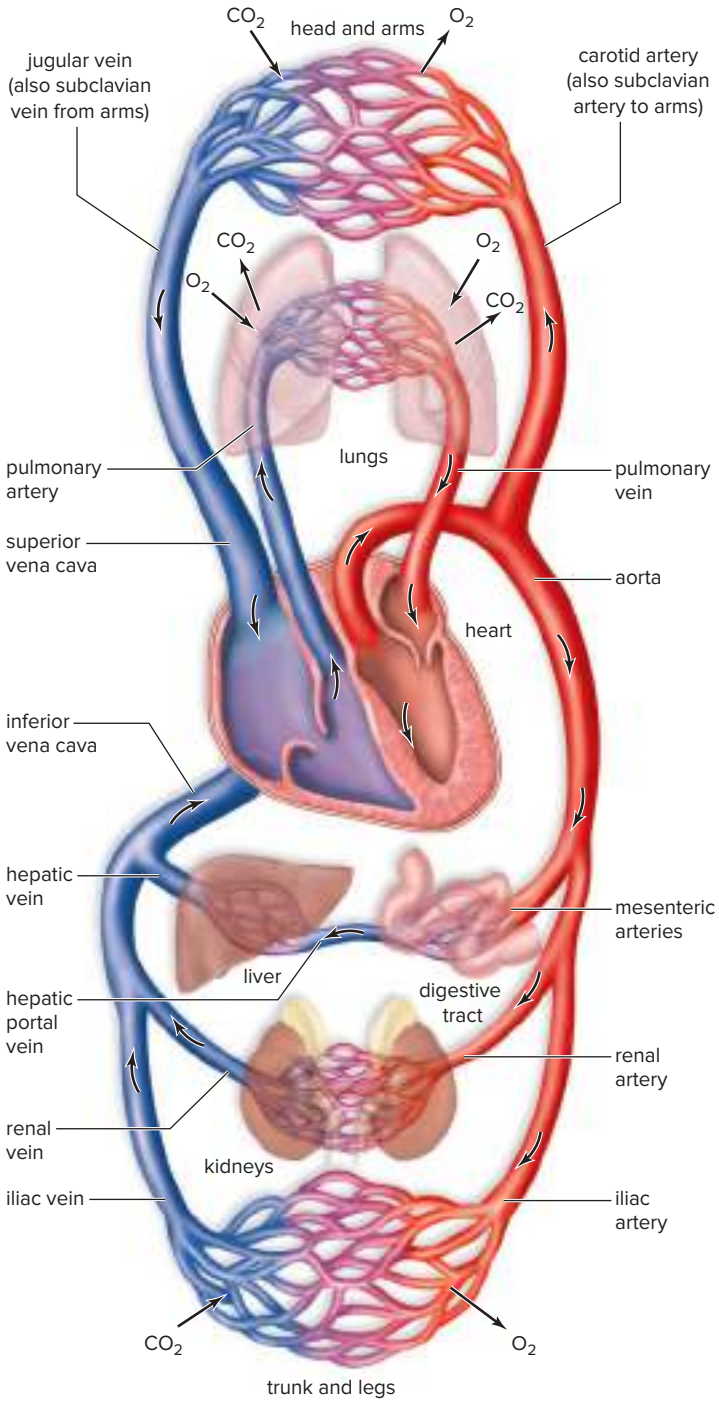


Figure 32.12 Systemic and pulmonary circuits. The pulmonary circuit transports blood between the heart and the lungs. The systemic circuit transports blood between the heart and the remaining tissues and organs of the body. The blue-colored vessels carry blood high in carbon dioxide, and the red-colored vessels carry blood high in oxygen; the arrows indicate the flow of blood.

4. Blood from the capillary bed enters venules, which combine into veins before entering the major vein of the body, the vena cava.
5. The oxygen-poor blood of the vena cava enters the right atrium. It will then be sent to the lungs via the pulmonary circuit.

ISTUDY

In most instances, the artery and the vein that serve the same region are given the same name. For example, iliac and femoral are names applied to both arteries and veins. In the systemic circuit, arteries contain oxygen-rich blood and have a bright red color, but veins contain oxygen-poor blood and appear dull red or, when viewed through the skin, blue.

The coronary arteries are extremely important because they supply oxygen and nutrients to the heart muscle itself (see Fig. 32.7). The coronary arteries arise from the aorta just above the aortic semilunar valve. They lie on the exterior surface of the heart, where they branch into arterioles and then capillaries. In the capillary beds, nutrients, wastes, and gases are exchanged between the blood and the tissues. The capillary beds enter venules, which join to form the cardiac veins, and these empty into the right atrium.

An exception to the path of blood in the systemic circuit occurs between the digestive tract and the liver, where blood must pass through two sets of capillaries because of the hepatic portal system. A **portal system** (L. *porto,* "carry, transport") is a structure in which blood from capillaries travels through veins to reach another set of capillaries, without first traveling through the heart. The hepatic portal system takes blood from the intestines directly to the liver. The liver then performs such functions as metabolizing nutrients and removing toxins (see Section 34.2). Blood leaves the liver by way of the hepatic vein, which enters the inferior vena cava.

Blood Pressure

When the left ventricle contracts, blood is forced into the aorta and then other systemic arteries under pressure. Systolic pressure results from blood being forced into the arteries during ventricular systole, and diastolic pressure is the pressure in the arteries during ventricular diastole. In humans, **blood pressure** can be measured with a *sphygmomanometer,* which has a pressure cuff that determines the amount of pressure required to stop the flow of blood through an artery.

Blood pressure is normally measured on the brachial artery, an artery in the upper arm. Today, digital manometers are often used to take one's blood pressure instead of the older type with a dial. Blood pressure is given in millimeters of mercury (mm Hg). A blood pressure reading consists of two numbers—for example, 120/80—that represent systolic and diastolic pressures, respectively.

As blood flows from the aorta into the various arteries and arterioles, blood pressure falls. Also, the difference between systolic and diastolic pressure gradually diminishes. In the capillaries, there is a slow, fairly even flow of blood. This may be related to the very high total cross-sectional area of the capillaries (Fig. 32.13). It has been calculated that if all the blood vessels in a human body were connected end to end, the total distance would circle around Earth's equator two times! Most of this distance would be due to the large number of capillaries.

Blood pressure in the veins is low and is insufficient for moving blood back to the heart, especially from the limbs of the body. Venous return is dependent on three factors:

- Skeletal muscles near veins put pressure on the collapsible walls of the veins, and therefore on the blood contained in these vessels, when they contract.

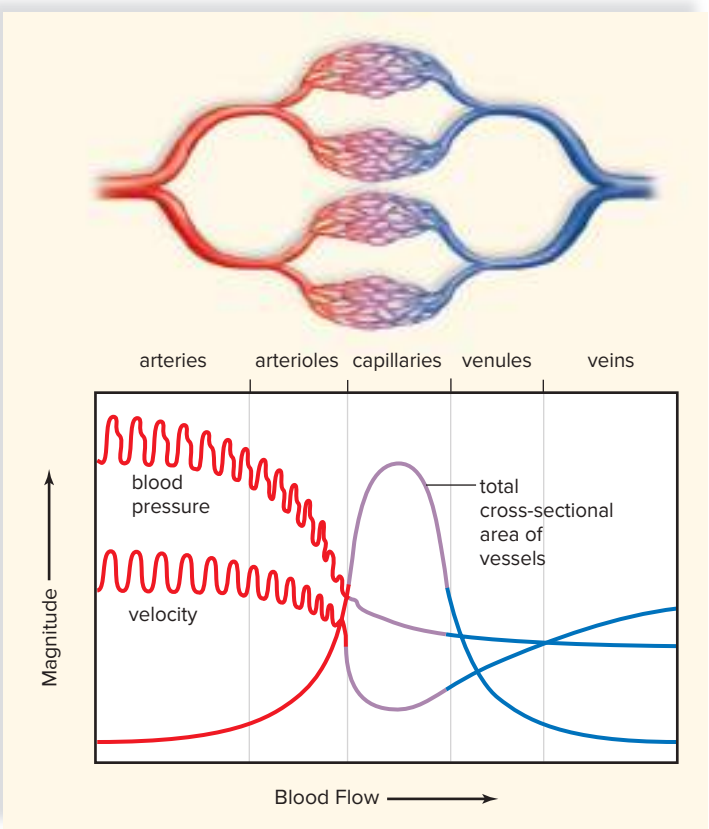


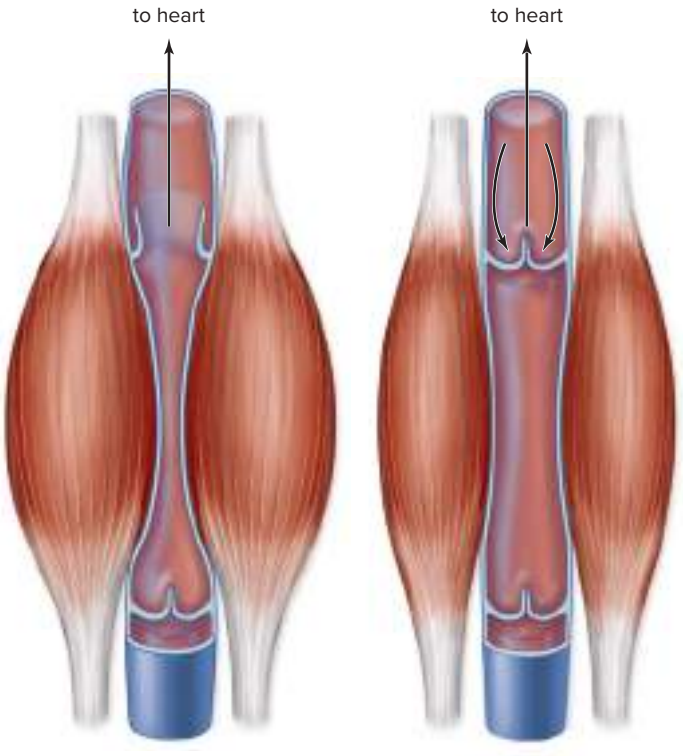
Figure 32.13 Velocity and blood pressure related to vascular cross-sectional area. In capillaries, blood is under minimal pressure and has the least velocity. Blood pressure and velocity drop off, because capillaries have a greater total cross-sectional area than arterioles.

- Variations in pressure in the chest cavity during breathing, also known as the *respiratory pump,* cause blood to flow from areas of higher pressure (such as the abdominal cavity) to lower pressure (in the thoracic cavity) during each inhalation.
- Valves in the veins prevent the backward flow of blood, and therefore pressure from muscle contraction moves blood toward the heart (Fig. 32.14). Varicose veins, abnormal dilations in superficial veins, develop when the valves of the veins become weak and ineffective due to a backward pressure of the blood.

Cardiovascular Disease

Cardiovascular disease (CVD) is the leading cause of death in most Western countries. According to the American Heart Association, CVD has been the most common cause of death in the United States every year since 1900. The only exception to this statistic was 1918, the worst year of a global influenza pandemic. According to statistics from the American Heart Association, about 2,300 Americans die of CVD each day, which is an average of one death every 38 seconds, and about one out of every three deaths overall. The Nature of Science feature, "Recent Findings About Preventing Cardiovascular Disease," examines some possible dietary changes that may reduce CVD risk.

ISTUDY



a. Contracted skeletal muscle pushes blood past open valve.

b. Closed valve prevents backward flow of blood.

Figure 32.14 Valves regulate blood flow in veins. **a.** Pressure on the walls of a vein, exerted by skeletal muscles, increases blood pressure within the vein and forces a valve open. **b.** When external pressure is no longer applied to the vein, blood pressure decreases, and back pressure forces the valve closed. Closure of the valves prevents the blood from flowing in the opposite direction.

Hypertension

It is estimated that approximately 47% of Americans have **hypertension,** which is high blood pressure. Table 32.2 outlines the blood pressure levels associated with hypertension.

Table 32.2 Hypertension Levels

Blood Pressure Category	Systolic Pressure (mm Hg)		Diastolic Pressure (mm Hg)
Normal	below 120	and	below 80
Elevated	120–129	and	below 80
Stage 1 Hypertension	130–139	or	80–89
Stage 2 Hypertension	140 or higher	or	90 or higher
Hypertension Crisis (immediate care needed)	180 or higher	and/or	120 or higher

Source: American Heart Association.

Hypertension is most often caused by a narrowing of arteries due to atherosclerosis (described next). This narrowing causes the heart to work harder to supply the required amount of blood. The resulting increase in blood pressure can damage the heart, arteries, and other organs. Other risk factors that can contribute to hypertension include obesity, smoking, chronic stress, genetics, and a high dietary sodium intake (which causes retention of fluid). Only about two-thirds of people with hypertension seek medical help for their condition, and it is likely that many people with high blood pressure are unaware of it.

Atherosclerosis

Atherosclerosis is an accumulation of soft masses of fatty materials, particularly cholesterol, beneath the inner linings of arteries (Fig. 32.15). Such deposits are called plaque. As deposits occur, plaque tends to protrude into the lumen of the vessel, interfering with the flow of blood. Plaque can also cause a clot to form on the irregular arterial wall. As long as the clot remains stationary, it is called a *thrombus,* but if and when it dislodges and moves along with the blood, it is called an *embolus.* If thromboembolism is not treated, complications such as a stroke or heart attack can arise.

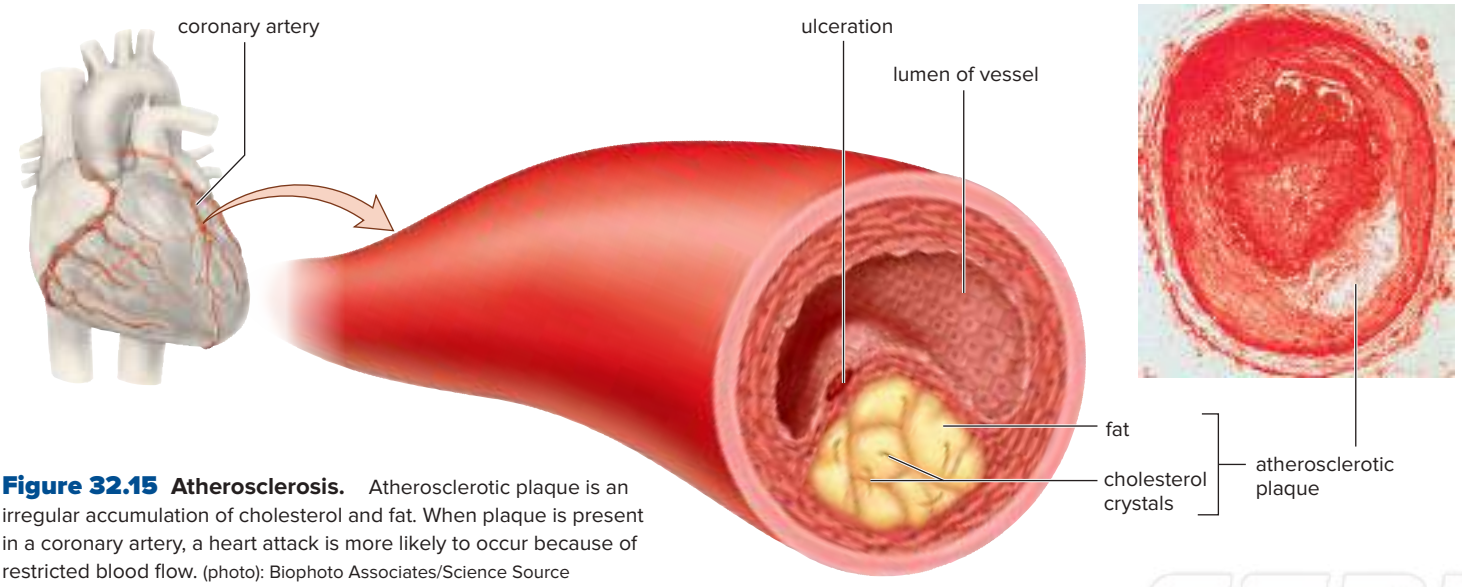


Figure 32.15 Atherosclerosis. Atherosclerotic plaque is an irregular accumulation of cholesterol and fat. When plaque is present in a coronary artery, a heart attack is more likely to occur because of restricted blood flow. (photo): Biophoto Associates/Science Source

ISTUDY

THEME Nature of Science

Recent Findings About Preventing Cardiovascular Disease

For decades, several factors have been associated with an increased risk of cardiovascular disease (CVD), especially atherosclerosis. Some of these cannot be avoided, such as increasing age, male sex, family history of heart disease, and belonging to certain races, including African American, Mexican American, and Native American. Other risk factors—smoking, obesity, high cholesterol, hypertension, diabetes, physical inactivity—can be avoided or at least affected by changing one's behavior or taking medications. In recent years, however, other factors have been under consideration, such as the following.

- **Alcohol.** Excessive alcohol use can destroy just about every organ in the body, the heart included. Based on several large research studies, however, the AHA notes that people who consume one or two drinks per day have a 30–50% reduced risk of CVD compared to people who do not drink. However, the maximum protective effect is achieved with only one or two drinks per day—consuming more than that increases the risk of many alcohol-related problems. The American Heart Association does *not* recommend that people who do not drink start using alcohol, or that people who drink increase their consumption, based on these findings.
- **Resveratrol.** The "red wine effect," or "French paradox," refers to the observation that levels of CVD in France are relatively low, despite the consumption of a high-fat diet. One possible explanation is that wine is frequently consumed with meals. In addition to its alcohol content, red wine contains especially high levels of antioxidants, including resveratrol. Resveratrol is mainly produced in the skin of grapes, so it is also found in grape juice. Resveratrol supplements are also available at health food stores. The benefits of resveratrol alone are questionable, however, and most controlled studies to date have demonstrated no beneficial effects. The lower incidence of CVD in the French may be due to multiple factors, including lifestyle and genetic differences.
- **Omega-3 fatty acids.** The influence that diet has on blood cholesterol levels has been well studied. It is generally beneficial to minimize our intake of foods high in saturated fat (red meat, cream, and butter) and trans fats (most margarines, commercially baked goods, and deep-fried foods). Replacing these harmful fats with healthier ones, such as monounsaturated fats (olive and canola oils) and polyunsaturated fats (corn, safflower, and soybean oils), is beneficial. In addition, the American Heart Association now recommends eating at least two servings of fish a week, especially salmon, mackerel, herring, lake trout, sardines, and albacore tuna, which are high in omega-3 fatty acids. These essential fatty acids can decrease triglyceride levels, slow the growth rate of atherosclerotic plaque, and lower blood pressure. However, children and pregnant women are advised to limit their fish consumption because of the high levels of mercury contamination in some fishes. For middle-aged and older men and postmenopausal women, the benefits of fish consumption far outweigh the potential risks.

Questions to Consider

1. Would you be helping your health if you decided to eat mackerel every day and drink two glasses of red wine with it? Why or why not?
2. What would be some difficulties in trying to determine the true cause of the "French paradox"?

Atherosclerosis often begins in early adulthood and develops progressively through middle age, but symptoms may not appear until an individual is 50 or older.

Stroke and Heart Attack

Strokes and heart attacks are associated with hypertension and atherosclerosis. A **stroke,** or disruption of blood supply to the brain, often results when a small cranial arteriole bursts or is blocked by an embolus. A lack of oxygen causes a portion of the brain to die, and paralysis or death can result. A person is sometimes forewarned of a stroke by a feeling of numbness in the hands or the face, difficulty speaking, or temporary blindness in one eye.

If a coronary artery becomes partially blocked, the individual may experience **angina pectoris,** characterized as a squeezing sensation or a flash of burning. If a coronary artery is completely blocked, perhaps by a thromboembolism, a portion of the heart muscle dies due to a lack of oxygen. This is a myocardial infarction, also called a **heart attack.** It may be necessary to place a stent, or self-expanding wire mesh tube, inside a blocked artery to keep it open (Fig. 32.16). If this approach is unsuccessful, a coronary bypass may be required, in which a surgeon replaces one or more blocked coronary arteries with an artery taken from elsewhere in the patient's body.

Check Your Progress 32.3

1. Name each blood vessel and heart chamber that blood passes through on its way from the venae cavae to the aorta, and identify which artery carries oxygen-poor blood.
2. Explain what specifically causes the sounds of the heartbeat.
3. Discuss why systolic blood pressure is higher than diastolic.
4. Predict what type of conditions might occur as a result of chronic hypertension and plaque.

ISTUDY

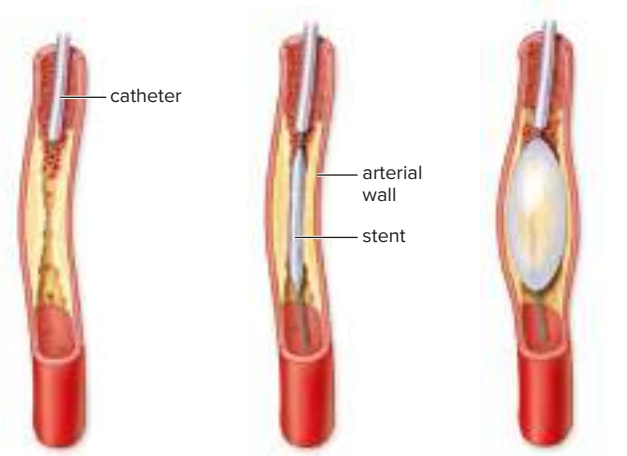


Figure 32.16 Angioplasty with stent placement. **a.** A plastic tube (catheter) is inserted into the coronary artery until it reaches the clogged area. **b.** A metal stent with a balloon inside it is pushed out the end of the plastic tube into the clogged area. **c.** When the balloon is inflated, the vessel opens, and the stent is left in place to keep the vessel open.

32.4 Blood

Learning Outcomes

Upon completion of this section, you should be able to

1. List the major types of blood cells and their functions.
2. Identify the processes that result in the formation of a blood clot.
3. Compare and contrast the ABO and Rh blood classification systems.
4. Define *capillary exchange* and describe the two major forces involved.

Blood is classified as a form of connective tissue that contains cells within a fluid matrix (see Section 31.1). In contrast to the hemolymph found in open circulatory systems, blood is normally contained within the blood vessels of a closed circulatory system. In mammals, blood has a number of functions that help maintain homeostasis:

- Transporting gases, nutrients, waste products, and hormones throughout the body
- Combating pathogenic microorganisms
- Helping maintain water balance and pH
- Regulating body temperature
- Carrying platelets and factors that ensure clotting to prevent blood loss

Blood has two main portions: a liquid portion, called plasma, and the formed elements, consisting of cells and platelets (Fig. 32.17).

Plasma

Plasma contains many types of molecules, including nutrients, wastes, salts, and hundreds of different types of proteins. Some of these proteins are involved in buffering the blood, effectively keeping the pH near 7.4. They also maintain the blood's osmotic pressure, so that water has an automatic tendency to enter blood capillaries. Several plasma proteins are involved in blood clotting, and others transport large organic molecules in the blood.

Albumin, the most plentiful of the plasma proteins, transports bilirubin, a breakdown product of hemoglobin, and various types of lipoproteins transport cholesterol. Another very significant group of plasma proteins are the antibodies, which are proteins produced by the immune system in response to specific pathogens and other foreign materials (see Section 33.4).

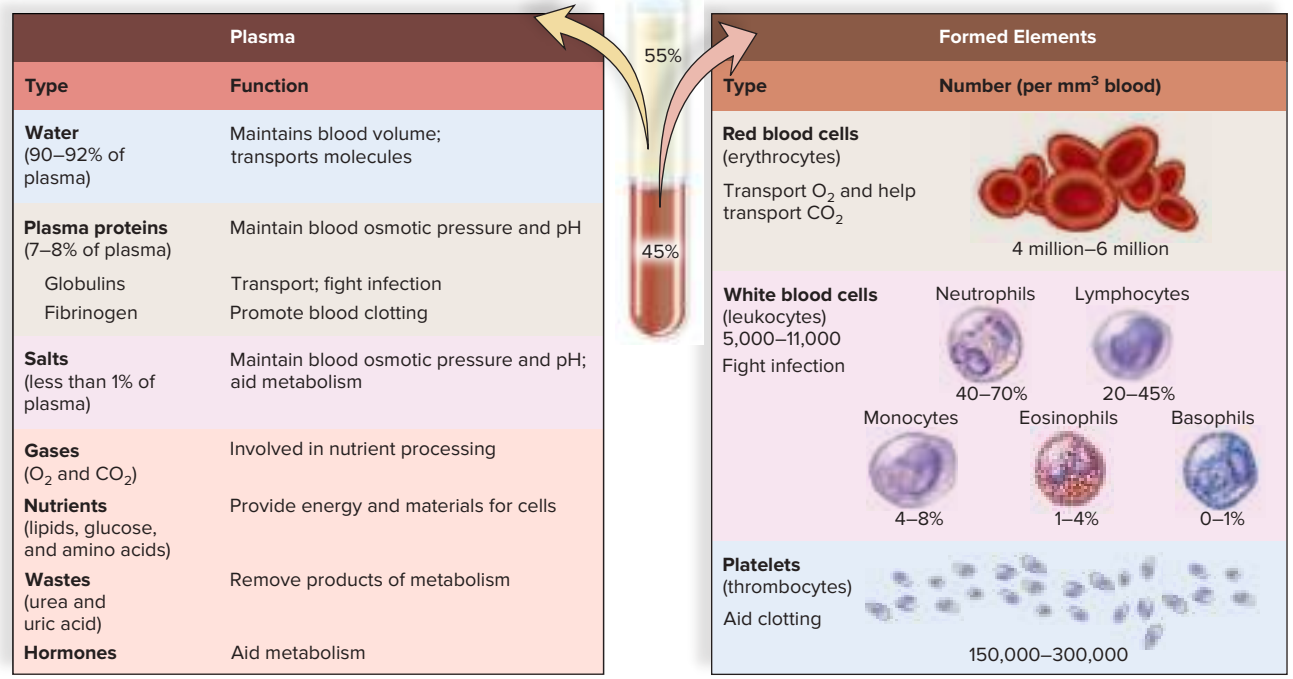


Figure 32.17 Composition of blood. Blood contains formed elements (cells and platelets) within a liquid connective matrix (plasma).

ISTUDY

Formed Elements

The **formed elements** are of three types: red blood cells, or erythrocytes (Gk. *erythros,* "red"; *kytos,* "cell"); white blood cells, or leukocytes (Gk. *leukos,* "white"); and platelets, or thrombocytes (Gk. *thrombos,* "blood clot").

Red Blood Cells

Red blood cells (RBCs) are small, biconcave disks that at maturity lack a nucleus and contain the respiratory pigment hemoglobin. The average adult human has 5 to 6 million RBCs per cubic millimeter (mm^3) of whole blood, and each one of these cells contains about 250 million hemoglobin molecules. **Hemoglobin** (Gk. *haima,* "blood"; L. *globus,* "ball") contains four globin protein chains, each associated with heme, an iron-containing group. Iron combines loosely with oxygen, and in this way oxygen is carried in the blood. If the number of RBCs is insufficient, or if the cells do not have enough hemoglobin, the individual has *anemia* and has a tired, run-down feeling.

In adults, RBCs are manufactured in the red bone marrow of the skull, the ribs, the vertebrae, and the ends of the long bones. The hormone erythropoietin, produced by the kidneys, stimulates RBC production. Now available as a drug, erythropoietin is helpful to persons with anemia and is sometimes misused by athletes who want to enhance the oxygen-carrying capacity of their blood.

Before they are released from the bone marrow into blood, RBCs synthesize hemoglobin and lose their nuclei. After living about 120 days, they are destroyed chiefly in the liver and the spleen, where they are engulfed by large phagocytic cells. When RBCs are destroyed, hemoglobin is released. The iron is recovered and returned to the red bone marrow for reuse. The heme portions of the molecules undergo chemical degradation and are excreted by the liver as bile pigments in the bile. The bile pigments are primarily responsible for the color of feces.

Blood Types The earliest attempts at blood transfusions resulted in illness and even the death of some recipients. Eventually, it was discovered that only certain transfusion donors and recipients are compatible, because red blood cell membranes carry specific proteins or carbohydrates that are antigens to blood recipients. An antigen (Gk. *anti,* "against"; L. *genitus,* "forming, causing") is a molecule, usually a protein or carbohydrate, that can trigger a specific immune response. Several groups of RBC antigens exist, the most significant being the ABO and Rh systems. Clinically, it is very important that the blood groups be properly cross-matched to avoid a potentially deadly transfusion reaction.

ABO System In the ABO system, the presence or absence of type A and type B antigens on RBCs determines a person's blood type. For example, if a person has type A blood, the A antigen is on their RBCs. Because it is considered by the immune system to be "self," this molecule is not recognized as an antigen by this individual, although it can be an antigen to a recipient who does not have type A blood.

In the ABO system, there are four blood types: A, B, AB, and O. Because the A and B antigens are also commonly found on microorganisms present in and on our bodies, a person's plasma usually contains antibodies to the A or B antigens not present on their RBCs. These antibodies are called anti-A and anti-B. The following chart explains which antibodies are present in the plasma of each blood type:

Blood Type	Antigen on Red Blood Cells	Antibody in Plasma
A	A	Anti-B
B	B	Anti-A
AB	A, B	None
O	None	Anti-A and anti-B

Because type A blood has anti-B but not anti-A antibodies in the plasma, a donor with type A blood can give blood to a recipient with type A blood (Fig. 32.18). However, if type A blood is given to a type B recipient, **agglutination** (Fig. 32.19), the clumping of RBCs, can cause blood to stop circulating in small blood vessels, leading to organ damage.

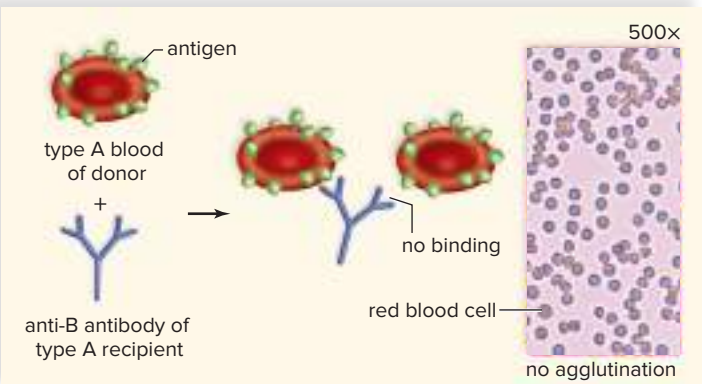


Figure 32.18 Matched blood transfusion. No agglutination occurs when the donor and recipient have the same type blood.
(photo): toeytoey2530/iStock/Getty Images

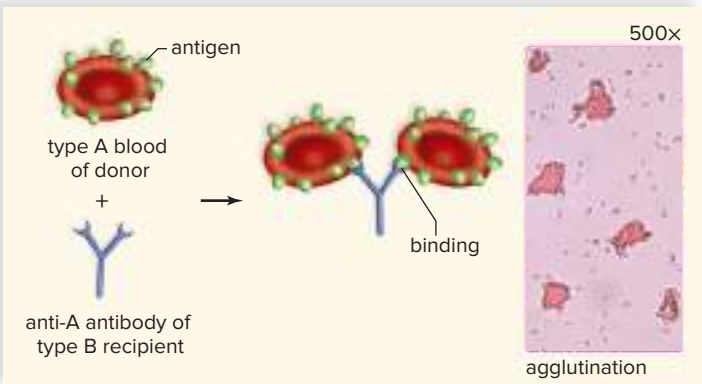


Figure 32.19 Mismatched blood transfusion. Agglutination occurs, because blood type B has anti-A antibodies in the plasma.
(photo): toeytoey2530/iStock/Getty Images

ISTUDY

Theoretically, a person with type O blood may donate to all types of recipients because type O RBCs have no A or B antigens. Therefore, individuals with type O blood are sometimes referred to as universal donors. A person with which blood type can receive blood from any donor? Type AB blood has no anti-A or anti-B antibodies, and thus it is sometimes called the universal recipient. In practice, however, it is not safe to rely solely on the ABO system when matching blood. Instead, samples of the two types of blood are physically mixed, and the result is microscopically examined for agglutination before blood transfusions are done.

An equally important concern when transfusing blood is to make sure that the donor is free from transmissible infectious agents, such as the microbes that cause AIDS, hepatitis, and syphilis.

Rh System Another important antigen on RBCs is the Rh factor. Eighty-five percent of the U.S. population have this particular antigen on their RBCs and are Rh-positive. Fifteen percent do not have the antigen and are Rh-negative. The designation of blood type usually includes whether the individuals have or do not have the Rh factor on their RBCs—for example, type A-positive (A+). Unlike the case with the A and B antigens, Rh-negative individuals normally do not have antibodies to the Rh factor, but they may make them when exposed to the Rh factor.

During pregnancy, if the female parent is Rh-negative and the male parent is Rh-positive, the child may be Rh-positive. If the Rh-positive fetal RBCs leak across the placenta, the female parent may produce anti-Rh antibodies. In this or a subsequent pregnancy with another Rh-positive baby, these antibodies may cross the placenta and destroy the child's RBCs. This condition, called hemolytic disease of the newborn (HDN), can be fatal without an immediate blood transfusion after birth.

The problem of Rh incompatibility can be prevented by giving Rh-negative females an Rh immunoglobulin injection toward the end of pregnancy and within 72 hours of giving birth to an Rh-positive child. This treatment contains a relatively low level of anti-Rh antibodies that help destroy any Rh-positive blood cells in the female parent's blood before her immune system produces high levels of anti-Rh antibodies.

White Blood Cells

Because they are a critical component of the immune system, the functions of white blood cells are discussed in detail in Section 33.4 and only briefly here. **White blood cells (WBCs),** also called leukocytes, differ from RBCs in that they are usually larger, have a nucleus, lack hemoglobin, and without staining, appear translucent. With staining, WBCs appear light blue unless they have granules that bind with certain stains (see Fig. 32.17). There are far fewer WBCs than RBCs in the blood, with approximately 5,000–11,000 WBCs per mm^3 in humans. On the basis of their structure, WBCs can be divided into granular and agranular leukocytes.

Granular Leukocytes The cytoplasm of **granular leukocytes** (neutrophils, eosinophils, and basophils) contains spherical vesicles, or granules, filled with enzymes and proteins, which these cells use to help defend the body against invading microbes and other parasites.

Neutrophils have a multilobed nucleus, resulting in their sometimes being called polymorphonuclear cells. They are the most abundant of the WBCs and are able to squeeze through capillary walls and enter the tissues, where they phagocytize and digest bacteria. The thick, yellowish fluid called *pus* that develops in some bacterial infections contains mainly dead neutrophils that have fought the infection. **Basophil** granules stain a deep blue and contain inflammatory chemicals, such as histamine. The prominent granules of **eosinophils** stain a deep red, and these WBCs are involved in fighting parasitic worms, among other actions.

Agranular Leukocytes The **agranular leukocytes,** which are also called mononuclear cells, lack obvious granules and include the monocytes and the lymphocytes.

Monocytes are the largest of the WBCs, and they tend to migrate into tissues in response to chronic, ongoing infections, where they differentiate into large phagocytic **macrophages** (Gk. *makros,* "long"; *phagein,* "to eat"). These long-lived cells not only fight infections directly but also release growth factors that increase the production of different types of WBCs by the bone marrow. Some of these factors are available for medicinal use and may be helpful to people with low immunity, such as people with AIDS or people on chemotherapy for cancer. A third function of macrophages is to interact with lymphocytes to help initiate the adaptive immune response (see Section 33.4).

Lymphocytes are the second most common type of WBC in the blood. The two major types of lymphocytes, T cells and B cells, each play a distinct role in adaptive immune responses to specific antigens. One type of T cell, the helper T cell, initiates and influences most of the other cell types involved in adaptive immunity. The other type, the cytotoxic T cell, attacks infected cells that contain viruses. In contrast, the main function of B cells is to produce antibodies. Each B cell produces just one type of antibody, which is specific for one type of antigen. Antigens are molecules that cause a specific immune response because the immune system recognizes it as "foreign." When antibodies combine with antigens, the complex is often phagocytized by a macrophage. The activities of lymphocytes, along with other aspects of animal immune systems, are discussed in more detail in Section 33.4.

Platelets and Blood Clotting

Platelets (thrombocytes) result from fragmentation of large cells, called *megakaryocytes,* in the red bone marrow. Platelets are produced at a rate of 200 billion a day, and the blood contains 150,000–300,000 per mm^3. These formed elements are involved in blood **clotting,** or coagulation (Fig. 32.20).

When a blood vessel in the body is damaged, platelets clump at the site of the puncture and partially seal the leak. Platelets and the injured tissues release a clotting factor called prothrombin activator, which converts prothrombin in the plasma to thrombin. This reaction requires calcium ions (Ca^{2+}). *Thrombin,* in turn, acts as an enzyme that severs two short amino acid chains from a *fibrinogen* molecule, one of the proteins in plasma. These activated fragments then join end to end, forming long threads of *fibrin.*

Fibrin threads wind around the platelet plug in the damaged area of the blood vessel and provide the framework for the clot. Red blood cells also are trapped within the fibrin threads; these cells

ISTUDY

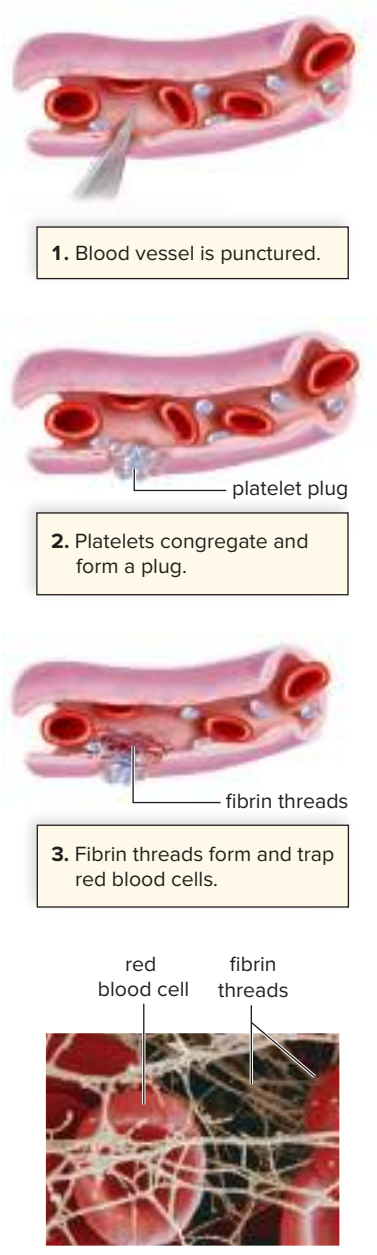


Figure 32.20 Blood clotting. A number of plasma proteins participate in a series of enzymatic reactions that lead to the formation of fibrin threads. (photo): Eye of Science/Science Source

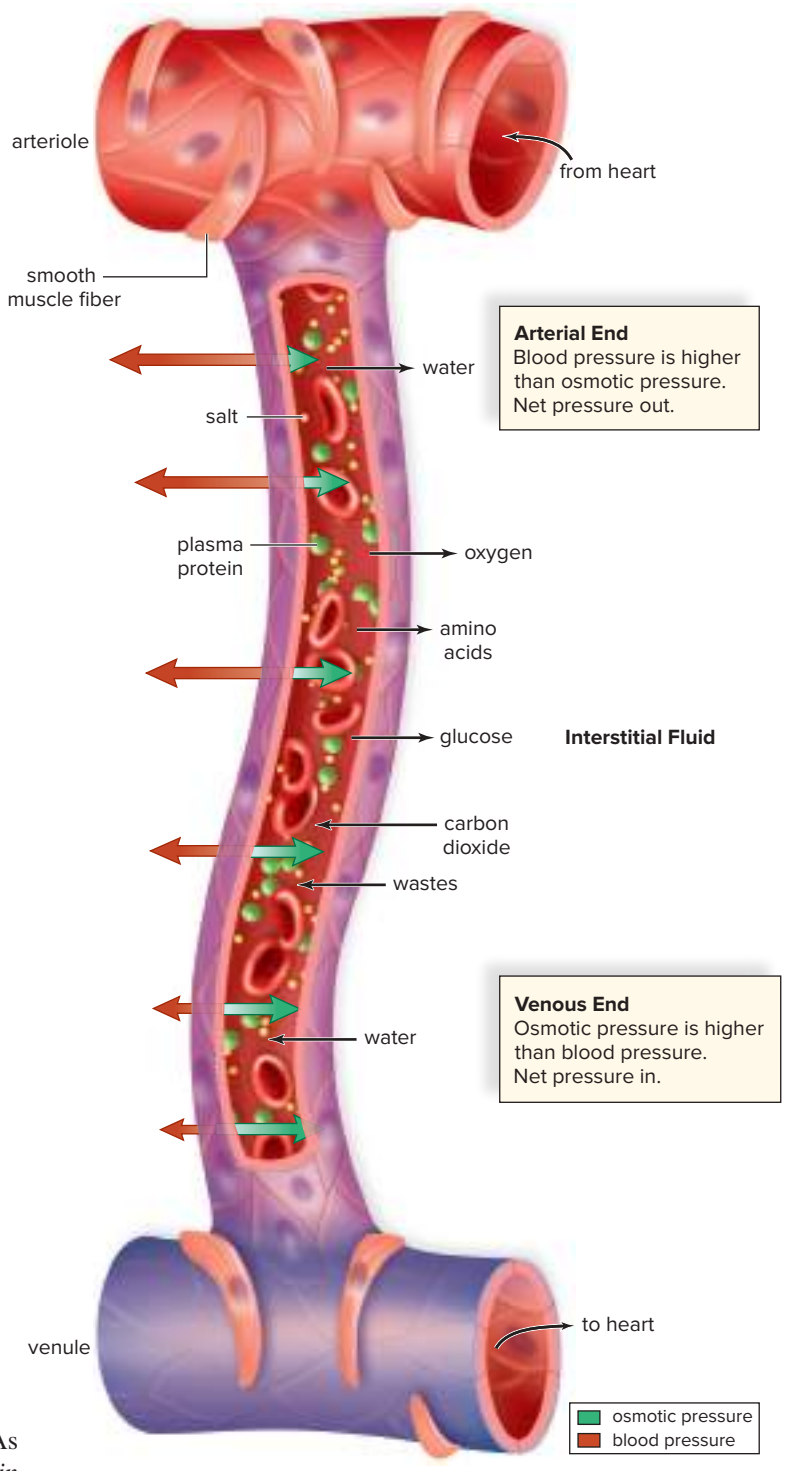


Figure 32.21 Capillary exchange. A capillary, illustrating the exchanges that take place and the forces that aid the process. At the arterial end of a capillary, the blood pressure is higher than the osmotic pressure; therefore, water (H_2O) tends to leave the bloodstream. In the midsection, molecules, including oxygen (O_2) and carbon dioxide (CO_2), follow their concentration gradients. At the venous end of a capillary, the osmotic pressure is higher than the blood pressure; therefore, water tends to enter the bloodstream. Notice that the red blood cells and the plasma proteins are too large to exit a capillary.

make a clot appear red. A fibrin clot is present only temporarily. As soon as blood vessel repair is initiated, an enzyme called *plasmin* destroys the fibrin network and restores the fluidity of plasma.

The Nature of Science feature, "Medicinal Leeches," describes how the use of leeches can be used to prevent clotting during medical procedures and increase blood flow.

Capillary Exchange

Figure 32.21 illustrates capillary exchange between a systemic capillary and interstitial fluid. Blood that enters a capillary at the arterial end is rich in oxygen and nutrients, and it is under pressure created by the pumping of the heart. Two forces primarily control

ISTUDY

THEME Nature of Science

Medicinal Leeches

Although it may seem more like an episode of a popular TV show than a real-life medical treatment, the U.S. Food and Drug Administration has approved the use of leeches as medical "devices" for treating conditions involving poor blood supply to various tissues.

Leeches are bloodsucking, aquatic creatures whose closest living relatives are earthworms (Fig. 32A). Prior to modern times, medical practitioners frequently applied leeches to patients, mainly in an attempt to remove the bad "humors" that they thought were responsible for many diseases. This practice was abandoned, thankfully, in the nineteenth century when it was realized that the "treatment" often harmed the patient.

Due to their tenacious nature, however, leeches are making a comeback in twenty-first-century medicine. By applying leeches to tissues that have been injured by trauma or disease, blood supply can be improved. When reattaching a finger, for example, it is easier to suture together the thicker-walled arteries than the veins. Poorly draining blood from the veins can pool in the appendage and threaten its survival. It turns out that leech saliva contains chemicals that dilate blood vessels and prevent blood from clotting by blocking the activity of thrombin. These effects can improve the circulation to the body part. Another substance in leech saliva actually anesthetizes the bite wound. In a natural setting, this allows the leech to feast on the blood supply of its victim undetected, but in a medical setting, it makes the whole experience more tolerable, at least physically. Mentally, however, the application of leeches can still be a rather unsettling experience, and patient acceptance is a major factor limiting their more widespread use.

Figure 32A Medicinal leeches. Leeches can attach to human skin and help prevent clotting and increase blood flow. Astrid & Hanns-Frieder Michler/Science Source

Questions to Consider

1. Leeches were historically used for bloodletting, one of the oldest medical practices. Can you think of several reasons why, prior to the mid-nineteenth century, early physicians might have gotten the idea that removing blood could help cure various diseases?
2. The use of actual living organisms in medical treatments is sometimes called "biotherapy." Can you think of other situations in which living creatures might be used for human health benefits, either directly or indirectly?

the movement of fluid through the capillary wall: (1) osmotic pressure, which tends to cause water to move from interstitial fluid into blood, and (2) blood pressure, which tends to cause water to move in the opposite direction. At the arterial end of a capillary, the osmotic pressure of blood (21 mm Hg) is lower than the blood pressure (30 mm Hg). Osmotic pressure is created by the presence of salts and the plasma proteins. Because osmotic pressure is lower than blood pressure at the arterial end of a capillary, water exits a capillary at this end.

Midway along the capillary, where blood pressure is lower, the two forces essentially cancel each other, and there is no net movement of water. Solutes now diffuse according to their concentration gradient: Oxygen and nutrients (glucose and amino acids) diffuse out of the capillary; carbon dioxide and wastes diffuse into the capillary. Red blood cells and almost all plasma proteins remain in the capillaries.

The substances that leave a capillary contribute to interstitial fluid. Because plasma proteins are too large to readily pass out of the capillary, interstitial fluid tends to contain all components of plasma but has much lower amounts of protein.

At the venule end of a capillary, where blood pressure has fallen even more, osmotic pressure is greater than blood pressure, and water tends to move into the capillary. Almost the same amount of fluid that left the capillary returns to it, although some excess interstitial fluid is always collected by the lymphatic capillaries (Fig. 32.22). Interstitial fluid contained within lymphatic vessels is called **lymph.** Lymph is returned to the systemic venous blood when the major lymphatic vessels enter the subclavian veins in the shoulder region. See Section 33.2 for more information about the lymphatic system.

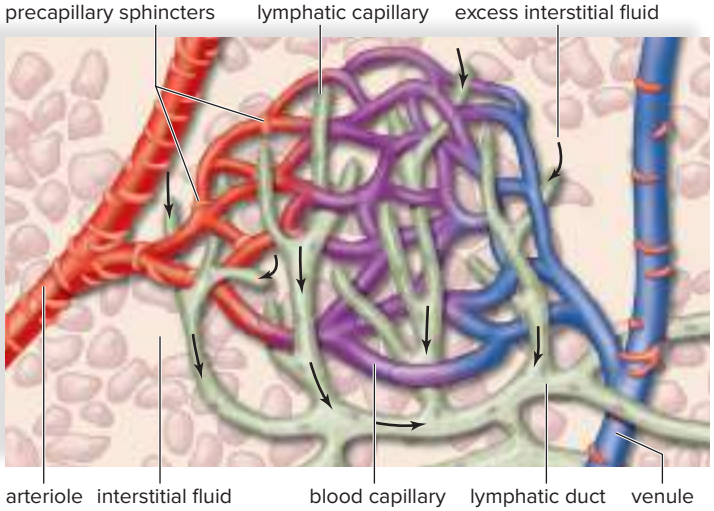


Figure 32.22 Capillary bed. A lymphatic capillary bed lies near a blood capillary bed. When lymphatic capillaries take up excess interstitial fluid, it becomes lymph. Precapillary sphincters can shut down a blood capillary, and blood then flows through the shunt.

ISTUDY

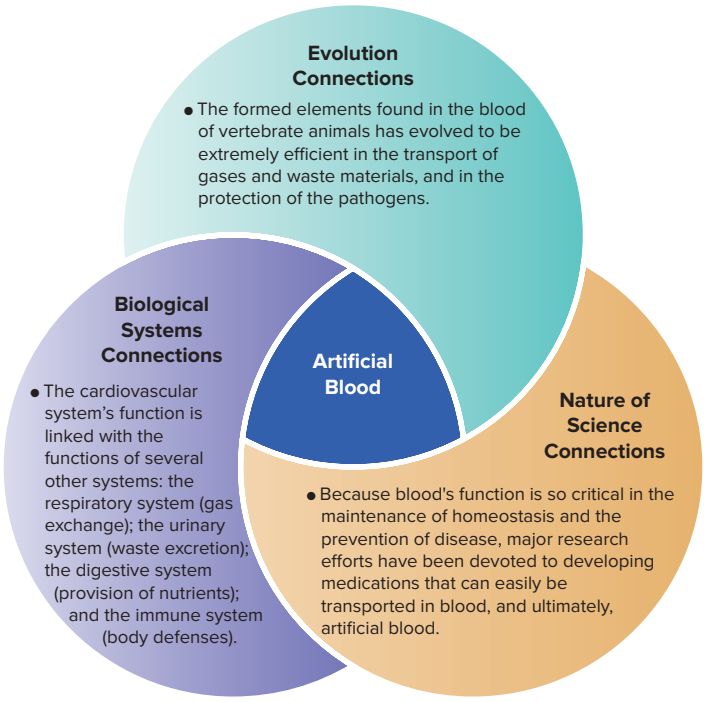
Not all capillary beds are open at the same time. When the precapillary sphincters (circular muscles) shown in Figure 32.5 are relaxed, the capillary bed is open and blood flows through the capillaries. When precapillary sphincters are contracted, blood flows through a shunt that carries blood directly from an arteriole to a venule.

In addition to nutrients and wastes, the blood distributes heat to body parts. When you are warm, many capillaries that serve the skin are open, and your face is flushed. This helps rid the body of excess heat. When you are cold, skin capillaries close, conserving heat, and your skin takes on a bluish tinge.

Check Your Progress 32.4

1. List the major components of blood and the functions of each.
2. Name the major events, in chronological order, that result in a blood clot.
3. Explain why Rh incompatibility is a problem only when a fetus is Rh-positive and the female parent is Rh-negative, but not vice versa.
4. Describe the major factors that affect the rate of capillary exchange.

Connecting *the* Concepts



What Did You Learn?

32.1 Transport in Invertebrates

Most animals have a **circulatory system** that provides their tissues with oxygen and nutrients, and removes wastes. However, some invertebrates lack such a transport system. The presence of a gastrovascular cavity allows diffusion alone to supply the needs of cells in cnidarians and flatworms. Roundworms make use of their pseudocoelom in the same way that echinoderms use their coelom to circulate materials.

Other invertebrates do have a transport system. Insects have an **open circulatory system** that transports **hemolymph,** and earthworms have a **closed circulatory system** that transports **blood** and excess **interstitial fluid.**

32.2 Transport in Vertebrates

Vertebrates have a closed **cardiovascular system** in which **arteries** carry blood away from the heart, branch into smaller **arterioles,** and then into **capillaries,** where exchange takes place. **Venules** collect blood from the capillaries, and merge into **veins** that carry blood to the heart.

Fishes have a one-circuit circulatory pathway, because the heart, with the single atrium and ventricle, pumps blood to the gills and then to the body, without a second pass through the heart. The other vertebrates have both pulmonary and systemic circuits. Amphibians have two atria but a single ventricle. Crocodilians, birds, and mammals, including humans, have a heart with two atria and two ventricles, in which oxygen-rich blood is always separate from oxygen-poor blood. Animals with such a two-circuit circulatory pathway have a **systemic circuit** and a **pulmonary circuit.**

32.3 The Human Cardiovascular System

The human **heart** is largely made of cardiac muscle (myocardium), and it is surrounded by pericardium. The internal chambers of the heart—the left and right **atria** and **ventricles**—are separated by a **septum.**

- During **systole,** the chambers contract to pump blood, and during **diastole** they relax, allowing filling. The **cardiac cycle** (heartbeat) in humans begins when the SA (sinoatrial) node **(pacemaker)** causes the two atria to contract, and blood moves through the **atrioventricular valves** to the two ventricles. The SA node also stimulates the AV (atrioventricular) node, which in turn causes the

two ventricles to contract. This electrical activity can be measured with an **electrocardiogram (ECG)**. Ventricular contraction sends blood through the **semilunar valves** to the pulmonary trunk and the aorta. Then, all chambers rest. The heart sounds, lub-dub, are caused by the closing of the valves, and the wave effect of blood being pumped through arteries can be felt as the **pulse.** The amount of blood pumped by the ventricles each minute is the **cardiac output.**

- In the pulmonary circuit, blood travels to and from the lungs. In the systemic circuit, the **aorta** divides into blood vessels that serve the body's cells. The **venae cavae** return oxygen-poor blood to the heart. In a **portal system,** blood from one set of capillaries (e.g., in the intestine) travels through veins directly to another set of capillaries (e.g., in the liver).
- **Blood pressure** created by the pumping of the heart accounts for the flow of blood in the arteries, but skeletal muscle contraction is largely responsible for the flow of blood in the veins, which have valves preventing a backward flow.

Hypertension and **atherosclerosis** are two circulatory disorders that can lead to **angina pectoris,** a **heart attack,** or a **stroke.** Following a heart-healthy diet, getting regular exercise, maintaining a healthy weight, and not smoking cigarettes can help protect against the development of these conditions.

32.4 Blood

Blood has two main parts: **plasma** and the formed elements. Plasma is mostly water (90–92%), but it also contains 7–8% proteins (such as albumin and antibodies), nutrients, and wastes.

- The formed elements include **red blood cells (RBCs),** white blood cells, and platelets. Red blood cells contain hemoglobin, which functions in oxygen transport.
- **White blood cells (WBCs),** also called leukocytes, which include **granular leukocytes** and **agranular leukocytes,** defend the body against infections. Three types of granular leukocytes are the **neutrophils,** which are phagocytes; **basophils,** which are involved in inflammation; and **eosinophils,** which are important in parasitic infections.
- The two types of agranular leukocytes are the **monocytes,** which enter tissues to become phagocytic macrophages, and **lymphocytes,** which carry out adaptive (specific) immunity to infection.
- The **platelets** and two plasma proteins, prothrombin and fibrinogen, function in blood **clotting,** an enzymatic process that results in fibrin threads. Blood clotting includes three major events: (1) Platelets and injured tissue release prothrombin activator, which (2) enzymatically changes prothrombin to thrombin, which is an enzyme that (3) causes fibrinogen to be converted to fibrin threads.
- The ABO blood-typing system is based on the presence or absence of A and B antigens on the red blood cells (A and B). If a mismatched transfusion is given, antibodies in the recipient's blood may react to these antigens, causing **agglutination** of red blood cells. A second type of red blood cell antigen is the Rh factor. If an Rh-negative woman becomes pregnant with an Rh-positive fetus, they may produce anti-Rh antibodies that could damage any Rh-positive fetus she carries.

When blood reaches a capillary, water moves out at the arterial end due to blood pressure. At the venous end, water moves in due to osmotic pressure. In between, nutrients diffuse out of, and wastes diffuse into, the capillary according to concentration gradients. Any excess interstitial fluid is absorbed into lymphatic vessels, where it is known as **lymph.**

Thinking Critically About Biology

1. A few specialized human tissues do not contain any blood vessels, including capillaries. Can you think of two or three? How might these tissues survive without a direct blood supply?
2. Assume your heart rate is 70 beats per minute (bpm), and each minute your heart pumps 5.25 liters of blood to your body. Based on your age to the nearest day, about how many times has your heart beat so far, and what volume of blood has it pumped?
3. There are not enough living hearts available to meet the need for heart transplants. What are some of the major problems manufacturers have had to overcome when attempting to design an artificial heart that will last for years inside an individual's body?

Making It Relevant to You

1. In addition to being used for transfusions, what other benefits do you think that artificial blood may have in the delivery of drugs and the prevention of disease?
2. How might artificial blood be engineered to prevent accidental poisoning from carbon monoxide?
3. Human blood groups have been used as the basis for paternity tests for decades. How will that change once artificial blood becomes more widely used?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

33

The Lymphatic and Immune Systems

Oz Digital/Alamy Stock Photo

The COVID-19 Vaccine

For most people, infection with the SARS-CoV-2 virus produces mild symptoms, such as a fever, cough, or loss of smell and taste. Some individuals are asymptomatic and do not experience any symptoms and thus may not know they have been infected. However, many groups, including older individuals and people who are immunocompromised, are highly susceptible to the virus. As we began to understand the nature of this virus, the scientific community realized that the only way to combat COVID-19 was to develop a vaccine.

To create the vaccine, researchers first identified the parts of the virus that cause our immune system to respond to the infection and build up antibodies to fight the virus. These antibodies then later provide us with immunity when exposed to SARS-CoV-2.

As you read through this chapter, think about the following questions:

1. How does your body respond to its first exposure to a new pathogen?
2. How do vaccines provide immunity?

Chapter Outline

33.1 Evolution of Immune Systems
33.2 The Lymphatic System
33.3 Innate Immune Defenses
33.4 Adaptive Immune Defenses
33.5 Immune System Disorders and Hypersensitivity Reactions

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 5.3 Which of the major protein functions are most important in the immune system?

Section 21.6 In what ways do the amoeboid protists resemble macrophages of the mammalian immune system?

Section 31.2 What life processes are carried out by the lymphatic and immune systems?

Following *the* Themes

CHAPTER 33 – THE LYMPHATIC AND IMMUNE SYSTEMS

Evolution	Even the simplest of multicellular organisms have developed cells that specialize in immune defenses. This provides clues about how increasingly complex immune system responses have evolved.
Nature of Science	Because the immune system is essential to our health, studies of patients with immune system disorders, such as AIDS, have improved our basic understanding of how the immune system functions.
Biological Systems	The lymphatic and immune systems are both essential to homeostasis, with the lymphatic system serving some additional functions unrelated to immunity.

ISTUDY

33.1 Evolution of Immune Systems

Learning Outcomes

Upon completion of this section, you should be able to

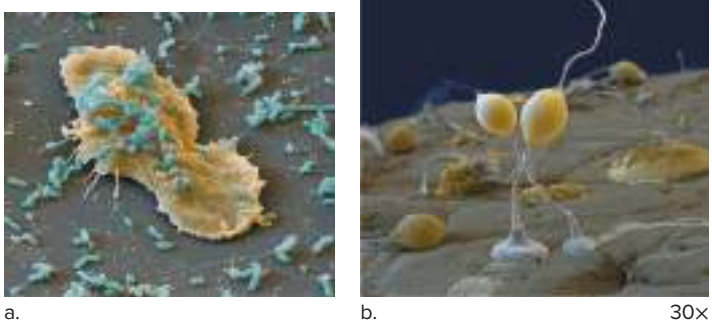
1. Summarize the forms of immune systems in invertebrate animals.
2. Describe the difference between innate and adaptive immune responses.

Our **immune system** protects us from all sorts of harmful invaders, including bacterial and viral pathogens, various toxins, and perhaps even the cancerous cells that occasionally arise. It is a very intricate system made up of many components that work together, rivaling the nervous system in its complexity. Researchers are now beginning to understand not only the evolution of the immune system in animals but also how to apply this information to help humans combat disease.

Immunity in Cellular Slime Molds

Researchers at Baylor College of Medicine in Houston, Texas, have discovered that very simple creatures called cellular slime molds can exhibit signs of a rudimentary immune system. These organisms are commonly known as "social amoebas," because of their unique life cycle (see Section 21.6). When food is plentiful, these protists live as separate amoeboid cells (Fig. 33.1*a*), ingesting bacteria through phagocytosis and reproducing by binary fission (dividing into identical copies). As the food supply dwindles, many thousands of these individuals can join together to form a slug 2–4 millimeters long (Fig. 33.1*b*). The slug migrates toward light as a single, multicellular body, then differentiates into a reproductive structure that releases spores, many of which disperse to become new organisms.

For several years, biologists have known that individual cells within the slug can become specialized to perform various functions, but the Baylor group discovered a new type of cell within the slug that they named *sentinel cells.* These cells circulate throughout the slug, engulfing bacteria and toxins. Eventually, the sentinel cells are sloughed from the body of the slug, thereby "sacrificing themselves" for the good of the organism. Some scientists believe that phagocytic white blood cells in the human body, such as neutrophils and macrophages, may have evolved from these types of cells.



a. b. 30×

Figure 33.1 Early immune responses. **a.** The single-celled form of the cellular slime mold, *Dictyostelium discoideum.* **b.** Under certain conditions, thousands of individuals can form a multicellular slug, in which some cells develop protective functions. (a–b): Eye of Science/Science Source

Immunity in *Drosophila*

Although social amoebas may provide an example of how phagocytic cells first became specialized to protect multicellular organisms, the immune system is far more developed in invertebrates, such as the fruit fly, *Drosophila melanogaster.* It was in this well-studied insect that scientists discovered the existence of a group of cellular receptors that could recognize common components found in many pathogenic microbes, but not in the insect's own cells. When these receptors bind to these *pathogen-associated molecular patterns,* or *PAMPs,* they trigger an immune reaction, increasing the odds that the pathogen can be eliminated from the fly.

Examples of PAMPs found on pathogenic microbes include the double-stranded RNA that is produced during the replication cycle of many viruses and certain arrangements of carbohydrates, lipids, or proteins found only on bacterial or fungal cell walls. Receptors for PAMPs have been found in organisms as diverse as fruit flies, plants, and humans, suggesting they were one of the earliest, and most successful, types of cellular receptors that evolved for the recognition of pathogens. As you will see in Section 33.3, these two examples illustrate a type of host defense known as **innate immunity,** which can recognize common microbial invaders very quickly but shows no signs of an increased response on repeated exposure to the same invader.

The Rise of Adaptive Immunity

In addition to innate immunity, vertebrate animals also exhibit **adaptive immunity,** characterized by the production of a very large number of diverse receptors on the surface of specialized white blood cells (such as B and T lymphocytes in humans). These receptors bind very specifically to molecules called **antigens,** much as a key fits a lock. This binding stimulates lymphocytes to divide and become much more numerous, resulting in the characteristic features of adaptive immunity, such as greatly increased responses to specific antigens and immunological memory after an initial exposure to an antigen.

The generation of such a diverse array of antigen receptors depends on a rearrangement of the DNA that codes for these receptors, somewhat like choosing different combinations of cards from a deck. Scientists have now discovered that this process developed quite suddenly in an ancestor that gave rise to the jawed vertebrates—including the cartilaginous fishes (sharks and rays), bony fishes, amphibians, reptiles, birds, and mammals.

The precise mechanism by which this "explosion" of adaptive immunity occurred is still incompletely understood, although it now seems quite likely that it involved the insertion of a small piece of DNA (a transposon, or "jumping gene"; see Section 14.4) into a gene coding for a more primitive, less variable antigen receptor—perhaps similar to the receptors for PAMPs, mentioned earlier. In contrast to the relatively "fixed" receptors for PAMPs recognized by the innate immune system, the generation of antigen receptors by gene rearrangement allows the adaptive immune system to be able to respond to new antigens that evolve, for example, in emerging infectious agents. In other words, the vertebrate immune system has evolved an ability to respond to the continuing evolution of pathogenic microbes and other threats to our health.

ISTUDY

Check Your Progress 33.1

1. Describe the function of sentinel cells in cellular slime molds.
2. List three specific types of PAMPs found on microbes.
3. Describe three ways the evolution of receptors for specific antigens increased the effectiveness of the immune system.

33.2 The Lymphatic System

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe three major functions of the lymphatic system.
2. Distinguish between the roles of primary and secondary lymphoid tissues and list examples of each.

The **lymphatic system,** which works closely with both the cardiovascular and immune systems, includes the lymphatic vessels and the lymphoid organs. It has three main functions that contribute to homeostasis:

- Lymphatic capillaries absorb excess interstitial fluid and return it to the bloodstream.
- In the small intestine, lymphatic capillaries called lacteals absorb fats in the form of lipoproteins and transport them to the bloodstream.
- The lymphoid organs and lymphatic vessels are sites of production and distribution of lymphocytes, which help defend the body against pathogens.

Lymphatic Vessels

Lymphatic vessels form a one-way system, beginning with **lymphatic capillaries**—tiny, closed-ended vessels that are found throughout the body (Fig. 33.2). Lymphatic capillaries take

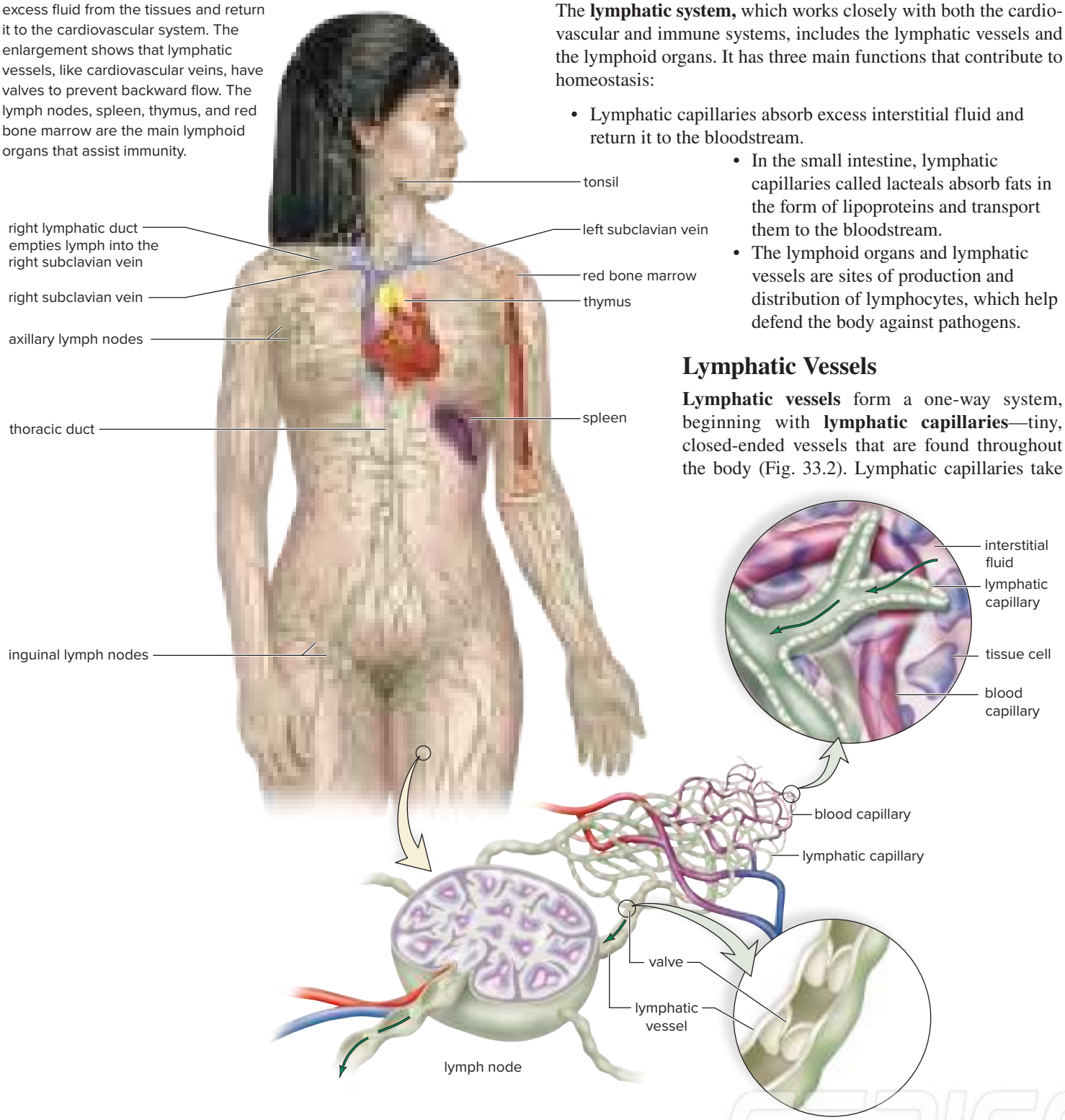
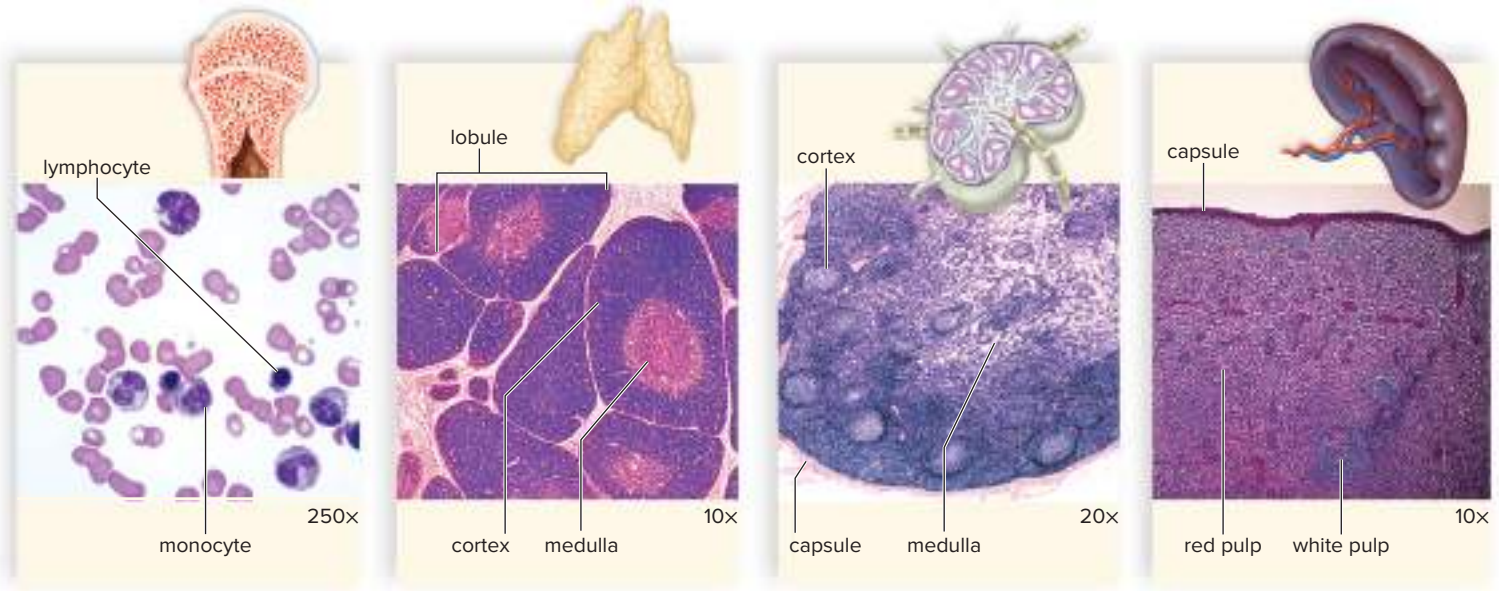


Figure 33.2 Lymphatic system. Lymphatic vessels drain excess fluid from the tissues and return it to the cardiovascular system. The enlargement shows that lymphatic vessels, like cardiovascular veins, have valves to prevent backward flow. The lymph nodes, spleen, thymus, and red bone marrow are the main lymphoid organs that assist immunity.

ISTUDY



a. Red bone marrow b. Thymus c. Lymph node d. Spleen

Figure 33.3 The lymphoid organs. **a.** Blood cells, including lymphocytes, are produced in red bone marrow. B cells mature in the bone marrow, but (**b**) T cells mature in the thymus. **c.** Lymph is cleansed in lymph nodes, while (**d**) blood is cleansed in the spleen. (a–b): (photo): Ed Reschke/Stone/Getty Images; (c): (photo): Alvin Telser/McGraw Hill; (d): (photo): ©Ed Reschke

up excess interstitial fluid. The fluid inside lymphatic capillaries is called **lymph.**

The lymphatic capillaries join to form lymphatic vessels, which merge before entering either the thoracic duct or the right lymphatic duct. The larger thoracic duct returns lymph to the left subclavian vein. The right lymphatic duct returns lymph to the right subclavian vein.

The construction of the larger lymphatic vessels is similar to that of cardiovascular veins. Skeletal muscle contraction forces lymph through lymphatic vessels, and it is prevented from flowing backward by one-way valves.

A number of diseases may result in an increased amount of fluid leaving the blood capillaries, or an insufficient return of fluid to the blood via the lymphatic vessels. In either case, a localized accumulation of interstitial fluid called *edema* may result, illustrating the importance of this aspect of lymphatic system function.

Lymphoid Organs

The **lymphoid (lymphatic) organs** are reviewed in Figures 33.2 and 33.3. Lymphocytes develop and mature in **primary lymphoid organs,** such as bone marrow and the thymus. The **secondary lymphoid organs** are sites where some lymphocytes are activated by antigens.

A major primary lymphoid organ is the **red bone marrow,** a spongy, semisolid, red tissue where hematopoietic stem cells divide and produce all the types of blood cells, including lymphocytes (Fig. 33.3*a*). In a child, most of the bones have red bone marrow, but in an adult, it is present only in the bones of the skull, the sternum (breastbone), the ribs, the clavicle (collarbone), the pelvic bones, the vertebral column, and the proximal heads of the femur and humerus.

There are two main types of lymphocytes: B lymphocytes (B cells) and T lymphocytes (T cells). Although both types begin their development in the red bone marrow, **B cells** remain there until they are mature. In contrast, immature **T cells** migrate from the bone marrow via the bloodstream to the thymus, where they mature.

The soft, bilobed **thymus** is a primary lymphoid organ located in the thoracic cavity between the trachea and the sternum ventral to the heart (see Fig. 33.2). It is in the thymus that T cells learn to recognize the combinations of self-molecules and foreign molecules; this recognition characterizes mature T-cell responses (see Section 33.4). The thymus varies in size, but it is largest in children and shrinks as we get older. When well-developed, it contains many lobules (Fig. 33.3*b*).

Once lymphocytes are mature, they enter the bloodstream. From there, they frequently migrate into secondary lymphoid organs, such as the lymph nodes and spleen. Here, lymphocytes may encounter foreign molecules or cells; in response, they proliferate and become activated. Activated lymphocytes then reenter the bloodstream, searching for signs of infection or inflammation, like a squadron of highly trained military personnel seeking to destroy a specific enemy.

Lymph nodes are small (about 1–25 mm in diameter), ovoid structures occurring along lymphatic vessels. They are a major type of secondary lymphoid organ. As lymph percolates through the cortex and medulla of a lymph node (Fig. 33.3*c*), resident phagocytic cells engulf any foreign debris and pathogens. These phagocytes can then "present" these foreign materials to T cells in the lymph node (see Section 33.4).

Sometimes incorrectly called "lymph glands," lymph nodes are named for their location. For example, inguinal lymph nodes are in the groin, and axillary lymph nodes are in the armpits. Physicians often feel for the presence of swollen, tender lymph nodes as evidence that the body is fighting an infection. Unfortunately, cancer cells sometimes enter lymphatic vessels and congregate in lymph nodes. Therefore, when a person undergoes surgery for cancer, it is a common procedure to remove some lymph nodes and examine them to determine whether the cancer has spread to other regions of the body.

ISTUDY

The **spleen,** an oval secondary lymphoid organ with a dull purplish color, is located in the upper left side of the abdominal cavity posterior to the stomach. Most of the spleen contains red pulp, which filters and cleanses the blood. Red pulp consists of blood vessels and sinuses, where macrophages remove old and defective blood cells. The spleen also has white pulp, consisting of small areas of secondary lymphoid tissue (Fig. 33.3*d*). Much as the lymph nodes serve as sites for lymphocytes to respond to foreign material from the tissues, the spleen serves a similar role for the blood.

The spleen's outer capsule is relatively thin, and an infection or a trauma can cause the spleen to burst, necessitating surgical removal. Although some of the spleen's functions can be largely replaced by other organs, an individual with no spleen is more susceptible to certain types of infections and may require antibiotic therapy indefinitely.

Patches of lymphoid tissue in the body include the *tonsils,* located in the pharynx; *Peyer patches,* located in the intestinal wall; and the *appendix,* attached to the cecum (large intestine). These structures encounter pathogens and antigens that enter the body by way of the mouth.

Check Your Progress 33.2

1. Distinguish between the lymphatic and circulatory systems.
2. Summarize the functions of the lymphatic system.
3. Describe the general location and function of the lymphoid organs.

33.3 Innate Immune Defenses

Learning Outcomes

Upon completion of this section, you should be able to

1. Define *innate immunity.*
2. Describe four mechanisms of innate immunity and the major tissues, molecules, and/or cells involved.
3. Explain some specific ways that the innate immune system interacts with and influences the adaptive immune system.

We are constantly exposed to microbes, such as viruses, bacteria, and fungi, in our environment. **Immunity** is the capability of removing or killing foreign substances, pathogens, and cancer cells in the body. Mechanisms of *innate immunity* are fully functional without previous exposure to these invaders, while *adaptive immunity* (see Section 33.4) is initiated and amplified by exposure.

As summarized in Figure 33.4, innate immune defenses include the following:

- Physical and chemical barriers
- The inflammatory response
- Phagocytes and natural killer cells
- Protective proteins, such as complement and interferons

Innate defenses occur immediately or very shortly after an infection occurs. With innate immunity, there is no recognition that a pathogen has attacked before, and therefore no immunological "memory" is present for the attacker.

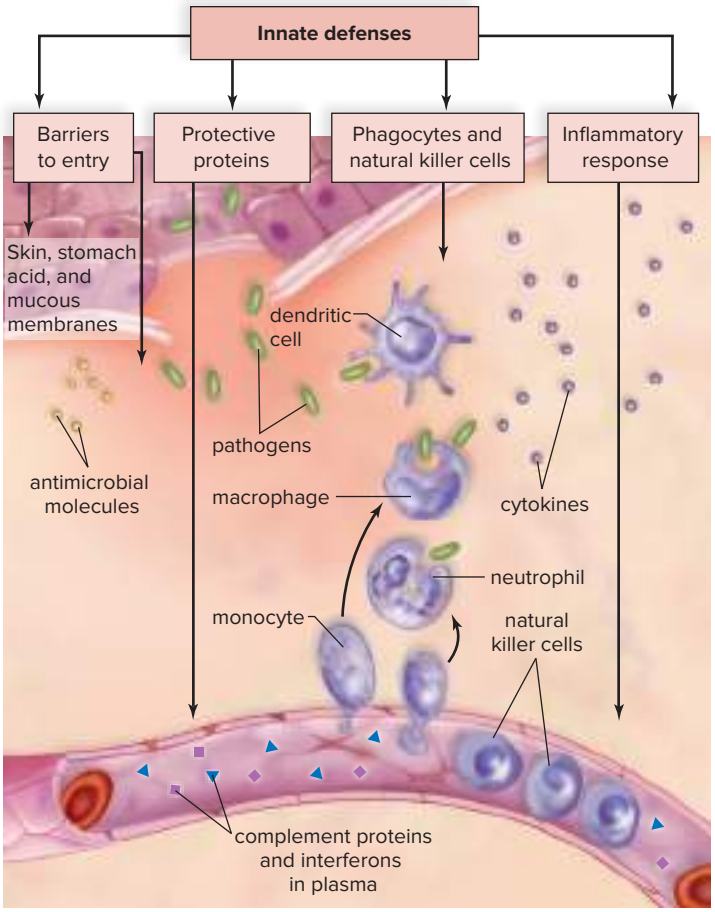


Figure 33.4 Overview of innate immune defenses. Most innate defenses act rapidly to detect and respond to various highly conserved molecules expressed by pathogens.

Physical and Chemical Barriers

The body has built-in barriers, both physical and chemical, that serve as the first line of defense against an infection by pathogens.

The intact skin is generally an effective physical barrier that prevents infection. As you saw in Section 31.3, the outer layers of our skin are composed of dead, keratinized cells that form a relatively impermeable barrier. The skin also has the benefit of a chemical barrier in the form of secretions of sebaceous (oil) glands of the skin. This acidic mixture contains chemicals that weaken or kill certain bacteria on the skin. But when the skin has been injured, one of the first concerns is the possibility of an infection.

Mucous membranes lining the respiratory, digestive, reproductive, and urinary tracts are also physical barriers to entry by pathogens. For example, the ciliated cells that line the upper respiratory tract sweep mucus and trapped particles up into the throat, where they can be coughed or spit out or swallowed.

Perspiration, saliva, and tears contain an antibacterial enzyme called lysozyme. Saliva also helps wash microbes off the teeth and tongue, and tears wash the eyes. Similarly, as urine is voided from the body, it flushes bacteria from the urinary tract.

The acidic pH of the stomach inhibits growth or kills many types of bacteria. At one time, it was thought that no bacterium could survive the acidity of the stomach. But now we know that ulcers are

caused by the bacterium *Helicobacter pylori*. Similarly, the acidity of the vagina and its thick walls discourage the presence of pathogens.

Finally, a significant chemical barrier to infection is created by the normal flora, microbes that usually reside in the mouth, large intestine, and other areas. This is commonly referred to as the **microbiome,** and research is showing that these bacteria play a significant role in our health. By using available nutrients and releasing their own waste, these resident bacteria prevent potential pathogens from taking up residence. For this reason, chronic use of antibiotics can make a person susceptible to pathogenic infection by killing off the normal flora.

Inflammatory Response

When tissues are damaged by a variety of causes, including pathogens, a series of events known as the **inflammatory response** occurs. An inflamed area has at least four common signs: redness, heat, swelling, and pain. Most of these signs are due to capillary changes in the damaged area, as illustrated in Figure 33.5. Chemical mediators released by damaged cells, including **histamine** that is mainly secreted by tissue-dwelling cells of the innate immune system called **mast cells,** cause capillaries to dilate and become more permeable. Increased blood flow to the area causes the skin to redden and become warm. Increased permeability of the capillaries allows proteins and fluids to escape into the tissues, resulting in swelling. Various chemicals released by damaged cells stimulate free nerve endings, causing the sensation of pain.

Inflammation also causes various types of white blood cells to migrate from the bloodstream into damaged tissues. Once in the tissues, monocytes can differentiate into dendritic cells and macrophages, both of which are able to devour many pathogens and still survive (Fig. 33.6). Macrophages also release colony-stimulating factors, namely cytokines, which pass by way of the blood to the red bone marrow where they stimulate the production and release of white blood cells.

Sometimes an inflammation persists, and the result is chronic inflammation that can itself become damaging to tissues. Examples include the chronic responses to the bacterium that causes tuberculosis or to asbestos fibers, which once inhaled into the lungs cannot be removed. Some cases of chronic inflammation are treated by administering anti-inflammatory drugs, such as aspirin, ibuprofen, or cortisone. These medications inhibit the responses to inflammatory chemicals being released in the damaged area.

From the site of their production in damaged tissues, various inflammatory mediators are absorbed into the bloodstream, where they can affect several other organs. Although the liver is not normally thought of as a part of the immune system, the liver responds to these chemicals by increasing production of various *acute phase proteins,* some of which can coat microbial invaders, making them easier for phagocytes to engulf. One type of acute phase protein, called C-reactive protein, is frequently measured to assess levels of inflammation in patients with certain diseases.

Inflammatory chemicals in the blood may also act on the brain to initiate an elevated body temperature, or *fever.* The body's fever response has two primary functions: first, to create an environment that makes it difficult for bacteria and viruses to reproduce; and second, to elevate the body temperature slightly to make the cells

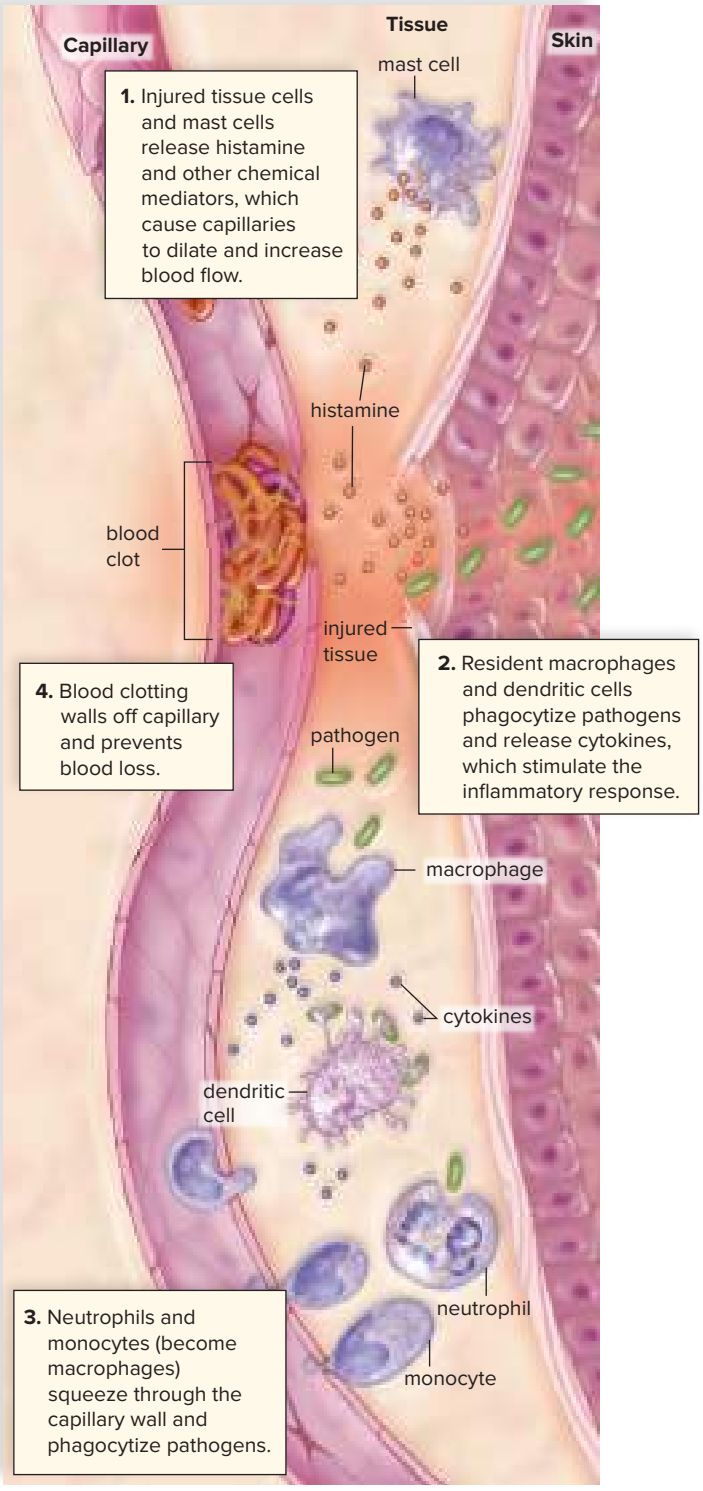
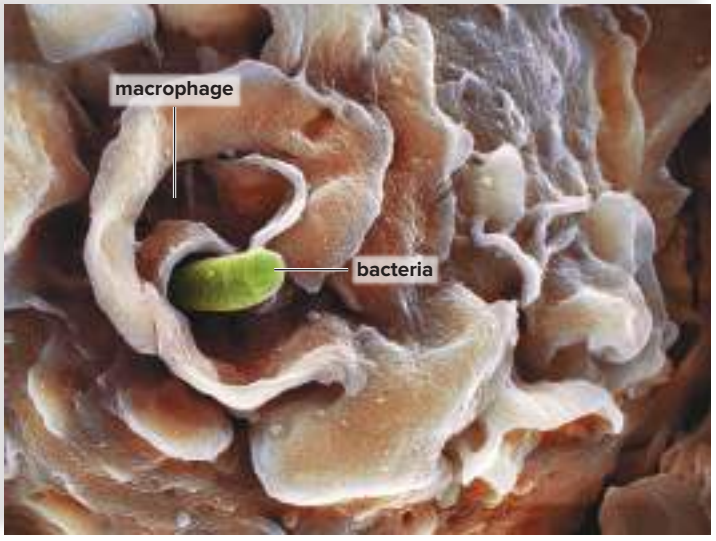


Figure 33.5 Inflammatory response. Due to capillary changes in a damaged area and the release of chemical mediators, such as histamine by mast cells, an inflamed area exhibits redness, heat, swelling, and pain. The inflammatory response can be accompanied by other reactions to the injury. Macrophages and dendritic cells, present in the tissues, phagocytize pathogens, as do neutrophils, which squeeze through capillary walls from the blood. Macrophages and dendritic cells release cytokines, which stimulate the inflammatory and other immune responses. A blood clot can form to seal a break in a blood vessel.

ISTUDY



SEM 1,075×

Figure 33.6 Macrophage engulfing bacteria. Monocyte-derived macrophages are the body's scavengers. They engulf microbes and debris in the body's fluids and tissues, as illustrated in this colorized scanning electron micrograph. SPL/Science Source

of the immune system more effective in fighting pathogens. Since mild to moderate fever appears to help the body fight off invaders more effectively, the wisdom of using drugs to treat mild fevers is being questioned. However, a body temperature higher than about 107°F can be fatal, especially in children, so obviously this situation must be treated as an emergency.

Inflammatory responses are accompanied by other responses to the injury. The clotting system can be activated to seal a break in a blood vessel. Antigens, chemical mediators, dendritic cells, and macrophages move from the damaged tissue via the lymph to the lymph nodes. There, these phagocytes interact with T cells and B cells to activate a specific adaptive response to the infection (see Section 33.4). Inflammation also initiates the healing response, in which macrophages play an essential role.

Phagocytes and Natural Killer Cells

Several types of white blood cells are phagocytic, meaning that they can engulf and digest relatively large particles, such as viruses and bacteria. **Neutrophils** are able to leave the bloodstream and phagocytize bacteria in tissues. They have multiple ways of killing bacteria. The cytoplasm of a neutrophil is packed with granules that contain antimicrobial peptides, as well as enzymes that can digest bacteria. Other enzymes inside neutrophil granules generate highly reactive free radicals such as superoxide and hydrogen peroxide, all of which participate in killing engulfed bacteria.

As the infection is being overcome, some neutrophils die. These—along with dead tissue cells, dead bacteria, and living white blood cells—may form pus, a whitish material. The presence of pus usually indicates that the body is trying to overcome a bacterial infection.

Eosinophils can be phagocytic, but they are better known for mounting an attack against animal parasites, such as tapeworms, that are too large to be phagocytized.

As mentioned, the two longer-lived types of phagocytic white blood cells are **macrophages** (Fig. 33.6) and **dendritic cells.** Macrophages are found in all sorts of tissues, whereas dendritic cells are especially prevalent in the skin. Both cell types engulf pathogens, which are then digested and broken down into smaller molecular components. They then travel to lymph nodes, where they stimulate T cells, which are responsible for initiating adaptive immune responses.

Natural killer (NK) cells are large, granular lymphocytes that kill virus-infected cells and cancer cells by cell-to-cell contact. NK cells do their work while adaptive defenses are still mobilizing, and they produce cytokines that promote adaptive immunity.

What makes NK cells attack and kill a cell? NK cells seek out cells that lack a particular type of "self" molecule, called MHC-I (major histocompatibility complex I), on their surface. Because some virus-infected and cancer cells may lack these MHC-I molecules, they may be recognized by NK cells, which kill these cells by inducing them to undergo cellular suicide (apoptosis). Because NK cells do not recognize specific viral or tumor antigens, and do not proliferate when exposed to a particular antigen, their numbers do not increase after stimulation.

Protective Proteins

Complement is composed of a number of blood plasma proteins, produced mainly by the liver, that "complement" certain immune responses. These proteins are continually present in the blood plasma but must be activated by pathogens to exert their effects. The complement system helps destroy pathogens in three ways:

1. *Enhanced inflammation.* Complement proteins are involved in and amplify the inflammatory response because certain ones can bind to mast cells and trigger histamine release, and others can attract phagocytes to the scene.
2. *Increased phagocytosis.* Some complement proteins bind to the surface of pathogens, increasing the odds that the pathogens will be phagocytized by a neutrophil or macrophage.
3. *Membrane attack complexes.* Certain other complement proteins join to form a membrane attack complex, which produces holes in the surface of some bacteria and viruses. Fluids and salts then enter the bacterial cell or virus to the point that it bursts (Fig. 33.7).

Interferons come in several different types, but all are **cytokines,** soluble proteins that affect the behavior of other cells. Most interferons are made by virus-infected cells. They bind to the receptors of noninfected cells, causing them to produce substances that slow cellular metabolism and interfere with viral replication. Interferons are used to treat certain cancers and viral infections, such as hepatitis C.

Check Your Progress 33.3

1. List three physical and three chemical barriers.
2. Describe the inflammatory response and explain how this response is beneficial.
3. Name five cell types involved in innate immunity and the major functions of each.
4. Summarize three specific functions of the complement system.

ISTUDY

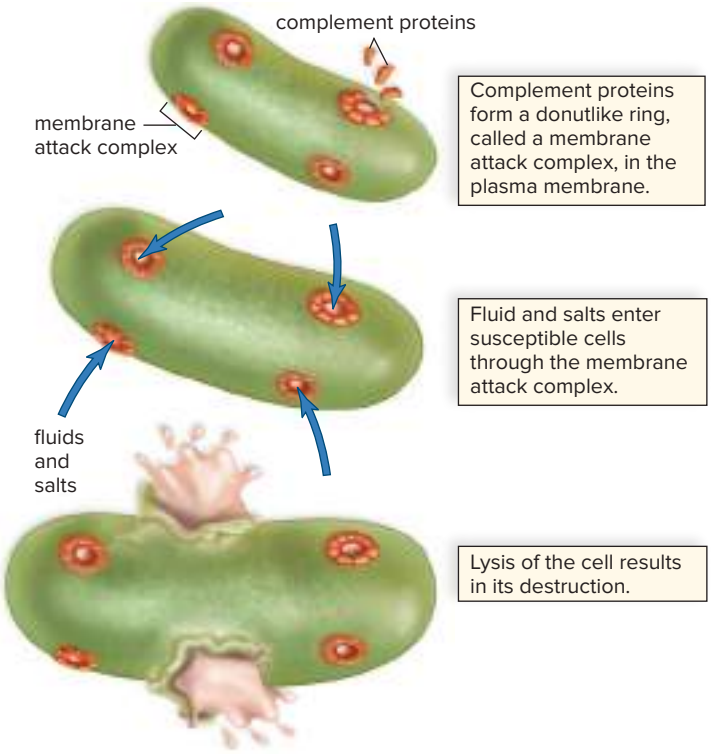


Figure 33.7 Action of the complement system against a bacterium. When complement proteins in the blood plasma are activated by an immune response, they form a membrane attack complex, which makes holes in bacterial cell walls and plasma membranes, allowing fluids and salts to enter until the cell eventually bursts.

33.4 Adaptive Immune Defenses

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast the activities of B cells and T cells.
2. Describe the basic structure of an antibody molecule and explain the different functions of IgG, IgA, IgM, and IgE.
3. Define *monoclonal antibodies* and list some specific applications of this technology.
4. Discuss active and passive immune responses, giving specific examples of each.

Even while innate defenses are trying to fight an infection, adaptive defenses also begin to respond. Because these defenses do not normally react to our own cells or molecules, it is said that the adaptive immune system can distinguish "self" from "nonself." Adaptive defenses usually take 5–7 days to become activated, but they may last for years. This explains why once we recover from some infectious diseases, we usually do not get the same disease a second time. Because we are not born with these defenses, some prefer the term *acquired immunity* to describe this type of immunity.

Adaptive defenses depend primarily on the activities of B cells and T cells (Fig. 33.8). Both B cells and T cells are manufactured in the red bone marrow. As mentioned earlier, B cells mature there, but T cells mature in the thymus. Both cell types are capable of binding to and thus "recognizing" specific antigens because they have **antigen receptors** on their plasma membrane. Pathogens,

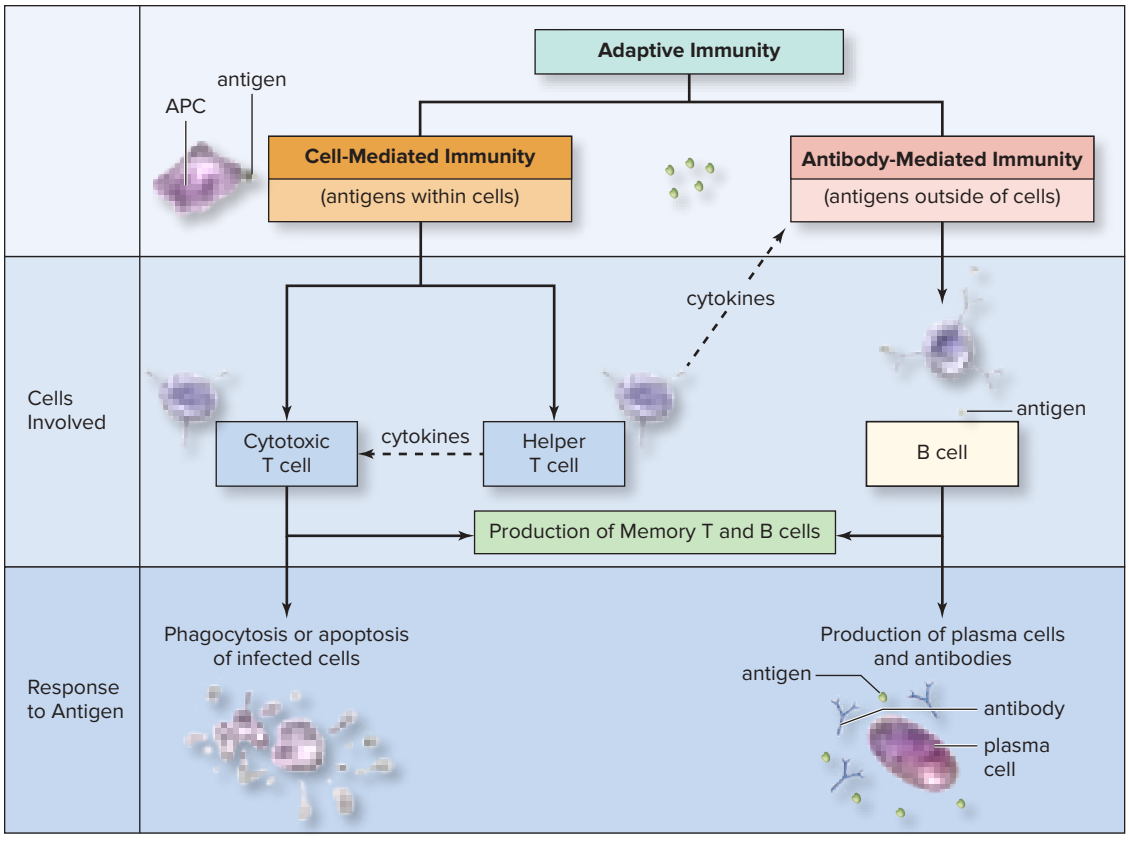


Figure 33.8 Overview of adaptive immune defenses. The adaptive responses involve two branches. The cell-mediated immunity branch targets cells that are presenting a specific antigen. The antibody-mediated immunity branch produces antibodies that are located outside of the cells, such as in the interstitial fluid.

ISTUDY

cancer cells, and transplanted tissues and organs bear antigens that the immune system usually recognizes as nonself.

During our lifetime, we need a diversity of B cells and T cells to recognize these antigens and protect us against them. Remarkably, diversification occurs during the lymphocyte maturation process to so great an extent that there are specific B cells and/or T cells for almost every possible antigen. Because B and T cells defend us from disease by specifically reacting to antigens, they can be likened to special forces that can attack selected targets without harming nearby residents (uninfected cells).

B Cells and Antibody-Mediated Immunity

The receptor for an antigen on the surface of a B cell is called a **B-cell receptor (BCR).** B cells are usually activated in a lymph node or the spleen, after their BCRs bind to a specific antigen. Subsequently, the B cell divides by mitosis many times, making many copies (clones) of itself. The **clonal selection theory** states that the antigen receptor of each B cell or T cell binds to only a single type of antigen.

As illustrated in Figure 33.9, many B cells are present, but only those that have BCRs that can combine with the specific antigen(s) present go on to divide and produce many new cells. Therefore, the antigen is said to "select" the B cells that will begin dividing. At the same time, cytokines secreted by helper T cells stimulate B cells to differentiate. Many of these B cells become **plasma cells,** which are specialized for secretion of antibodies. Plasma cells are larger than regular B cells because they have extensive rough endoplasmic reticulum for the mass production and secretion of **antibodies** that bind to a specific antigen. Antibodies are the secreted form of the BCR of an activated B cell, and these antibodies react to the same antigen as the original B cell. As the antigen that stimulated the B-cell response is removed from the body, the development of new plasma cells ceases, and those present undergo apoptosis. Other progeny of the dividing B cells become **memory B cells,** so named because these cells always "remember" a particular antigen and make us immune to a particular illness, but not to any other illness.

Defense of the body by B cells is known as **antibody-mediated immunity** (see Fig. 33.8). It is also called humoral immunity, because these antibodies are present in blood and lymph. Historically, the term *humor* referred to any fluid normally occurring in the body.

The Structure of Antibodies

The basic unit of antibody structure is a Y-shaped protein molecule with two arms. Each arm has a "heavy" (long) polypeptide chain and a "light" (short) polypeptide chain (Fig. 33.10). These chains have constant (*C*) regions, located at the trunk of the Y, where the sequence of amino acids is set. The variable (*V*) regions at the tips of the Y form two antigen-binding sites, and their shape is specific to a particular antigen. The antigen combines with the antibody at the antigen-binding site in a lock-and-key manner.

The binding of antibodies to an antigen can have several outcomes. Often, the reaction produces a clump of antigens combined with antibodies, termed an *immune complex.* The antibodies in an immune complex are like a beacon, attracting white blood cells that move in for the kill. For example, immune complexes may be engulfed by neutrophils or macrophages, or they may activate NK cells to destroy a cell coated with antibodies. Immune complexes

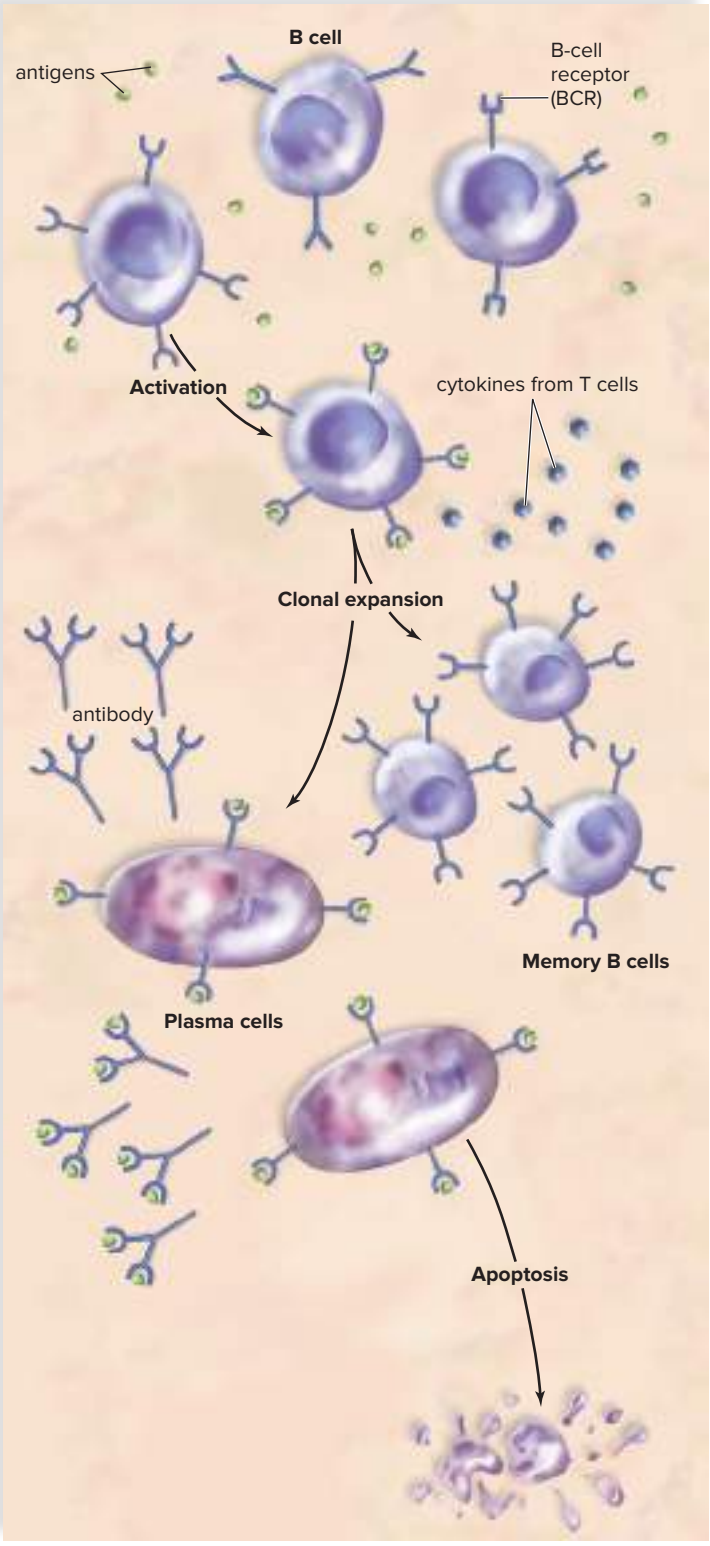


Figure 33.9 Clonal selection theory as it applies to B cells. Each B cell has a B-cell receptor (BCR) designated by shape that will combine with a specific antigen. Activation of a B cell occurs when its BCR can combine with an antigen (colored green). In the presence of cytokines, the B cell undergoes clonal expansion, producing many plasma cells and memory B cells. These plasma cells secrete antibodies specific to the antigen, and memory B cells immediately recognize the antigen in the future. After the infection passes, plasma cells undergo apoptosis, also called programmed cell death.

ISTUDY

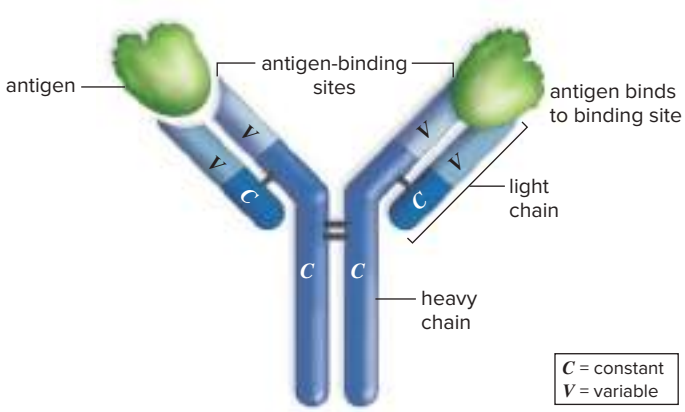


Figure 33.10 Structure of antibodies. An antibody contains two heavy (long) polypeptide chains and two light (short) chains arranged so there are two variable regions, where a particular antigen is capable of binding with the antibody. The shape of the antigen fits the shape of the binding site.

may also activate the complement system. Antibodies may also "neutralize" viruses or toxins by preventing them from binding to specific receptors on cells.

There are several types, or classes, of antibodies, also called **immunoglobulins (Igs).** The class of antibody is determined by the structure of the antibody's constant region. The major class of antibody found in the blood, called *IgG,* is a single Y-shaped molecule (Fig. 33.10). IgG antibodies are the major type that can cross the placenta from the female parent to their fetus, to provide some temporary protection to the newborn. IgG is also found in breast milk, along with a type called IgA. *IgA* is also the main class secreted in milk, tears, and saliva, and at mucous membranes. Antibodies of the *IgM* class are pentamers—that is, clusters of five Y-shaped molecules linked together. IgM antibodies are the first antibodies produced during most B-cell responses. As such, their presence is often interpreted as indicating a recent infection. The other major type of antibody is *IgE,* which is mainly bound to receptors on eosinophils and on mast cells in the tissues. A summary of the major classes of immunoglobulins in humans is provided in Table 33.1.

Table 33.1 Human Immunoglobins

Class	Distribution	Function
IgG	Main antibody type in circulation	Binds to pathogens, activates complement, and enhances phagocytosis
IgM	Antibody type found in circulation; largest antibody	Activates complement; clumps cells
IgA	Main antibody type in secretions such as saliva and milk	Prevents pathogens from attaching to epithelial cells in digestive and respiratory tract
IgD	Antibody type found on surface of immature B cells	Presence signifies readiness of B cell to respond to antigens
IgE	Antibody type found as antigen receptors on eosinophils in blood and on mast cells in tissues	Responsible for immediate allergic response and protection against certain parasitic worms

Monoclonal Antibodies

As has been noted, every plasma cell derived from a single B cell secretes antibodies that bind to a single antigen. This discovery, along with the development of techniques for growing cells in the laboratory, led to the production of **monoclonal antibodies.** Niels Jerne, George Köhler, and César Milstein were awarded the Nobel Prize in 1984 for their development of this technology.

Monoclonal antibodies are produced by cells derived from a single plasma cell; thus, all these antibodies have identical specificity for one antigen. Monoclonal antibodies are typically produced by first immunizing an animal (usually a mouse) with the antigen of interest (Fig. 33.11). The animal is then killed, its spleen is removed, and the spleen cells (including a large number of B cells) are fused with mouse myeloma cells (malignant plasma cells that live and divide indefinitely). The fused cells are called *hybridomas: hybrid* because they result from the fusion of two different cells, and *oma* because one of the cells is a cancer cell. The hybridomas are then isolated as individual cells and screened to select only those that are producing the desired monoclonal antibody.

Research Uses for Monoclonal Antibodies

The ability to quickly produce monoclonal antibodies in the laboratory has made them an important tool for academic research. Monoclonal antibodies are very useful because of their extreme specificity for only a particular molecule. A monoclonal antibody can be used to select out a specific molecule among many others, much like finding needles in a haystack. In this way, monoclonal antibodies have simplified formerly tedious laboratory tasks.

Medical Uses for Monoclonal Antibodies

Monoclonal antibodies also have many applications in medicine. In one application, they can be used to make quick and certain diagnoses of infections and other conditions. For example, a particular hormone called hCG is present in the urine of a female only if they are pregnant. An anti-hCG monoclonal antibody can be used to detect this hormone. Thanks to this technology, pregnancy tests that once required a visit to a doctor's office and the use of expensive laboratory equipment can now be performed at home, at minimal expense.

Monoclonal antibodies can be used not only to diagnose infections and illnesses but also to fight them. RSV, a common virus that causes serious respiratory tract infections in very young children, is now being successfully treated with a monoclonal antibody drug. The antibody recognizes a protein on the viral surface, and when it binds very tightly to the surface of the virus, the patient's own immune system can more easily recognize the virus and destroy it before it has a chance to cause serious illness.

Because monoclonal antibodies can distinguish some cancer cells from normal tissue cells, they may also be used to identify cancers at very early stages, when treatment can be most effective. Trastuzumab (Herceptin) is a monoclonal antibody used to treat breast cancer. Given intravenously, it binds to a protein receptor found on some breast cancer cells and prevents them

ISTUDY

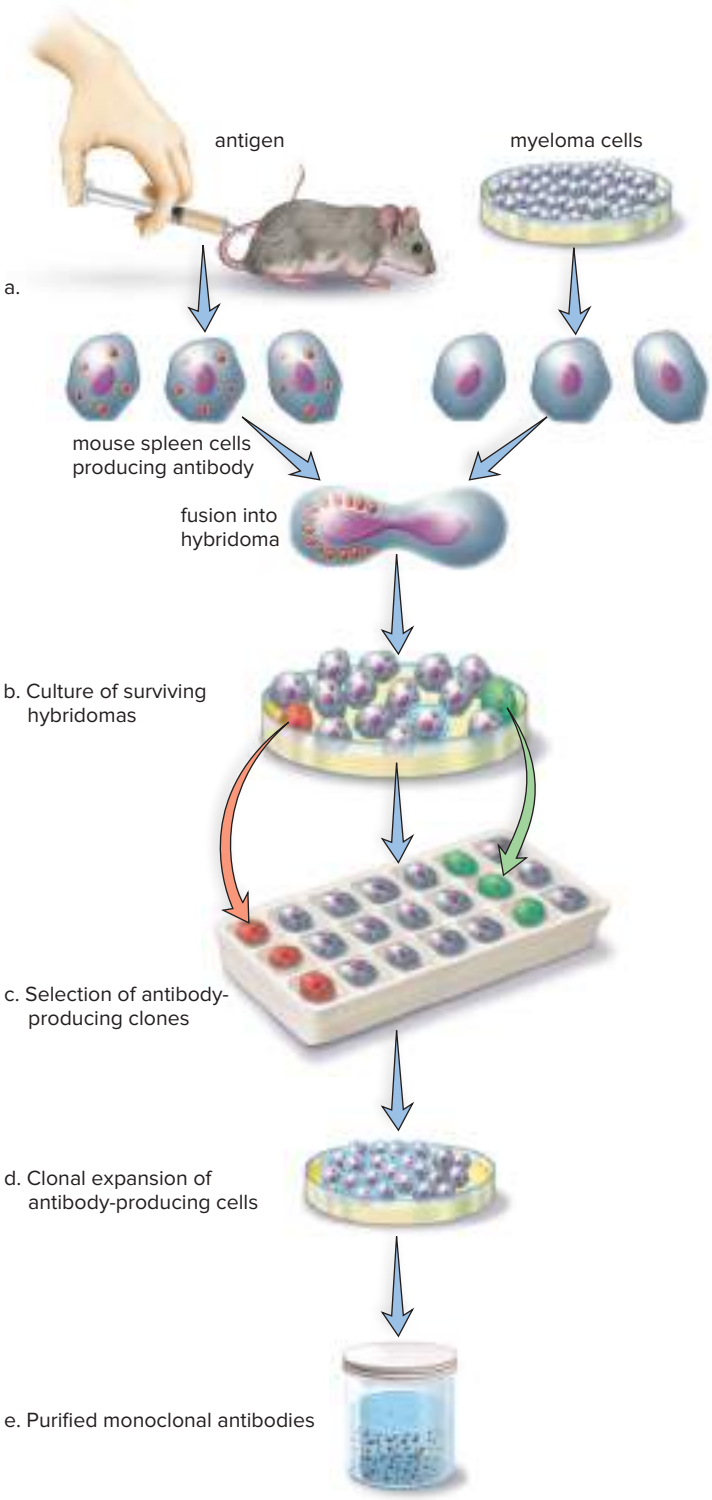


Figure 33.11 Production of monoclonal antibodies. **a.** Spleen cells derived from mice immunized with a specific antigen are fused with myeloma (cancerous) cells. **b.** This produces hybridoma cells that are "immortal." **c.** Hybridoma cells are cultured and clones producing the desired antibody are selected and expanded (**d**), producing monoclonal antibodies (**e**).

from dividing so quickly. Other cells of the immune system may also kill tumor cells that have the monoclonal antibody attached to their surface.

Since the first therapeutic monoclonal antibody was approved by the FDA in 1986, many are now available, and hundreds more are currently being tested. Adalimumab (Humira) is a monoclonal antibody that binds to and inhibits tumor necrosis factor, a cytokine associated with the exaggerated inflammatory reactions that characterize several autoimmune diseases, and it is used to treat rheumatoid arthritis, psoriatic arthritis, and Crohn's disease, among others. Monoclonal antibodies against the SARS-CoV-2 virus have been developed to assist high-risk individuals in their fight against COVID-19.

T Cells and Cell-Mediated Immunity

After a T cell completes its development in the thymus, it has a unique **T-cell receptor (TCR)** similar to the BCR on B cells. Unlike B cells, however, T cells are unable to recognize an antigen without help. The antigen must be displayed, or "presented," to the TCR by an **MHC (major histocompatibility complex) protein** on the surface of another cell.

There are two major types of T cells: **helper T cells** (T_H cells), which regulate adaptive immunity, and **cytotoxic T cells** (T_C cells), which attack and kill virus-infected cells and cancer cells. Each type of T cell has a TCR that can recognize an antigen fragment in combination with an MHC molecule. A major difference, however, is that the T_H cells recognize and respond only to antigens presented by specialized **antigen-presenting cells (APCs)** with MHC class II proteins on their surface, while T_C cells recognize and respond only to antigens presented by various types of cells with MHC class I proteins on their surface.

Figure 33.12 illustrates the important role that APCs such as macrophages and dendritic cells play in stimulating T_H cells. After phagocytizing a pathogen, APCs travel to a lymph node or the spleen, where T_H cells, T_C cells, and B cells congregate. In the meantime, the APC has broken the pathogen apart in a lysosome. A fragment of the pathogen is then displayed in association with an MHC class II protein on the cell's surface, which can bind to and select any T_H cell that has a TCR capable of combining with a particular antigen/MHC combination. This interaction stimulates the T_H cell to divide, thereby cloning itself. Some of these proliferating T_H cells secrete various cytokines that influence B cells, T_C cells, and other cell types. Many other cloned T_H cells become **memory T cells.** Like memory B cells, memory T cells are long-lived, and their number is far greater than the original number of T cells that could recognize a specific antigen. Therefore, when the same antigen enters the body later on, the immune response may occur so rapidly that no detectable illness occurs. As the antigen is cleared, activated T cells become susceptible to apoptosis, just as plasma cells do during a B-cell response.

By releasing a variety of cytokines, activated helper T cells are largely responsible for **cell-mediated immunity** (see Fig. 33.8), which is the destruction or elimination of pathogens and other threats by T_C cells, macrophages, natural killer cells, or other cells. For example, certain cytokines cause T_C cells to

ISTUDY

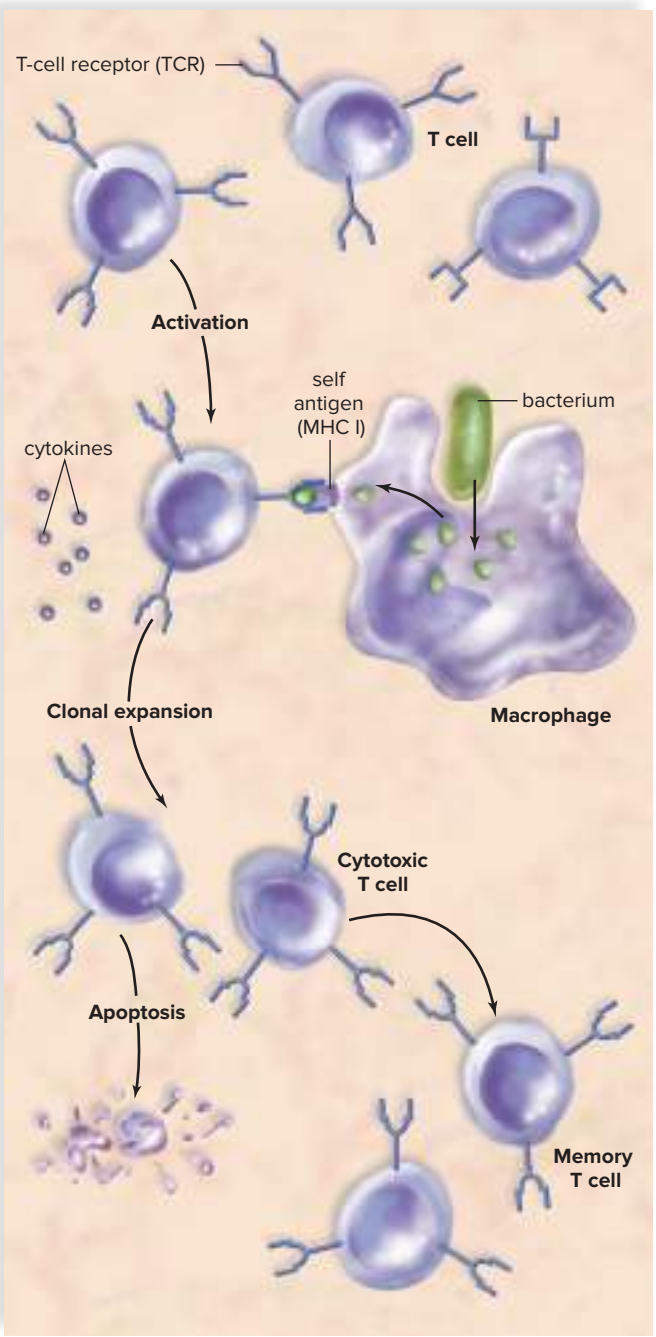


Figure 33.12 Clonal selection theory as it applies to T cells. Activation of a T cell occurs when its T-cell receptor (TCR) can combine with an antigen presented by a macrophage. In this example, cytotoxic T cells are produced. When the immune response is finished, they undergo apoptosis. A small number of memory T cells remain.

proliferate (see Fig. 33.8), while others activate macrophages to more actively seek, engulf, and destroy pathogens. Other cytokines released by T_H cells influence B-cell activities, even though B cells are responsible for antibody-mediated immunity. Cytokines are also used for immunotherapy purposes, as we will see later in this section.

Functions of Cytotoxic T Cells

A major difference in recognition of an antigen by helper T cells and cytotoxic T cells is that helper T cells recognize an antigen only in combination with MHC class II proteins, while T_C cells recognize an antigen only in combination with MHC class I proteins, which are found on almost all types of cells. The outcome of this recognition is also very different—activated T_H cells secrete a variety of cytokines that "help" other cells, but activated T_C cells specialize in killing other cells.

The cytoplasm of a T_C cell contains storage vacuoles that are filled with a chemical called *perforin,* as well as enzymes called *granzymes* (Fig. 33.13). After a T_C cell is activated, it travels via the bloodstream to areas of inflammation in the body. On migrating into the tissues, if the T_C cell recognizes its unique target combination of antigen plus MHC-I protein on a virus-infected cell or a cancer cell, it releases perforin molecules, which perforate the plasma membrane, forming a pore. Granzymes then use the pore to enter the abnormal cell, causing it to undergo apoptosis and die. Once T_C cells have released the perforins and granzymes, they move on to the next target cell. Because of this ability, T_C cells have been referred to as the "serial killers" of the immune system. As with B cells and T_H cells, most activated T_C cells are short-lived, while others become long-lived memory T_C cells, ever ready to defend against the same virus or kill the same type of cancer cell again.

HIV Infection

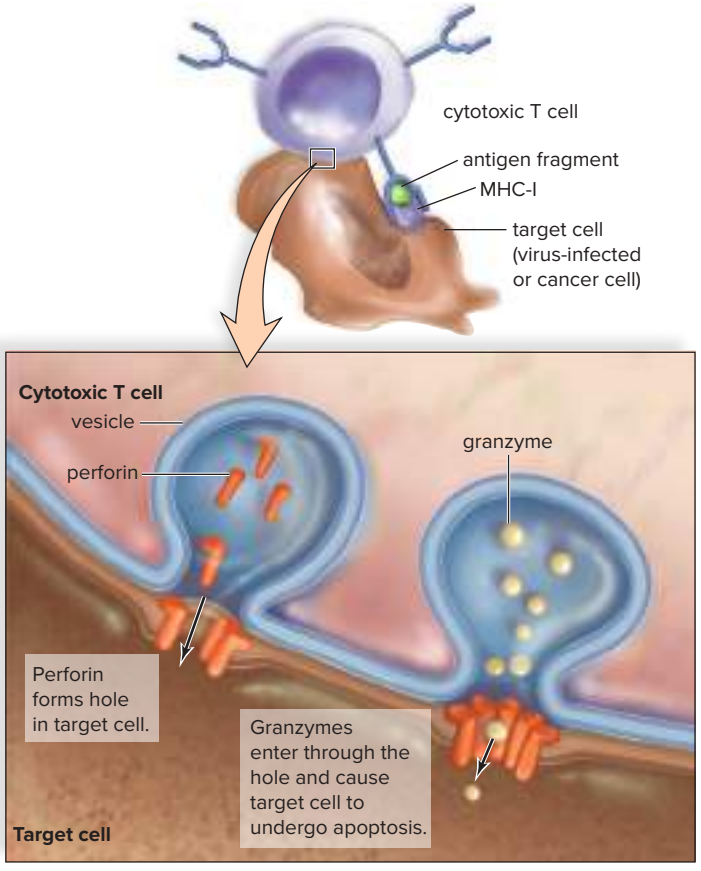
It is estimated that over 38 million people worldwide are infected with **human immunodeficiency virus (HIV),** and more than 40 million people have died of AIDS since 1981. Over 4,000 people become infected with HIV every day.

Helper T cells are the major host cell for HIV, but macrophages and dendritic cells can also be infected. After HIV enters a host cell, it reproduces (see Figure 20.3), inserting itself into the host cell DNA, where it will remain for the life of the cell. Perhaps years later, this hidden viral DNA will begin producing more HIV particles, which go on to infect and destroy more and more helper T cells. The Biological Systems feature, "AIDS and Opportunistic Infections," details some key features of the progression of an HIV infection over time.

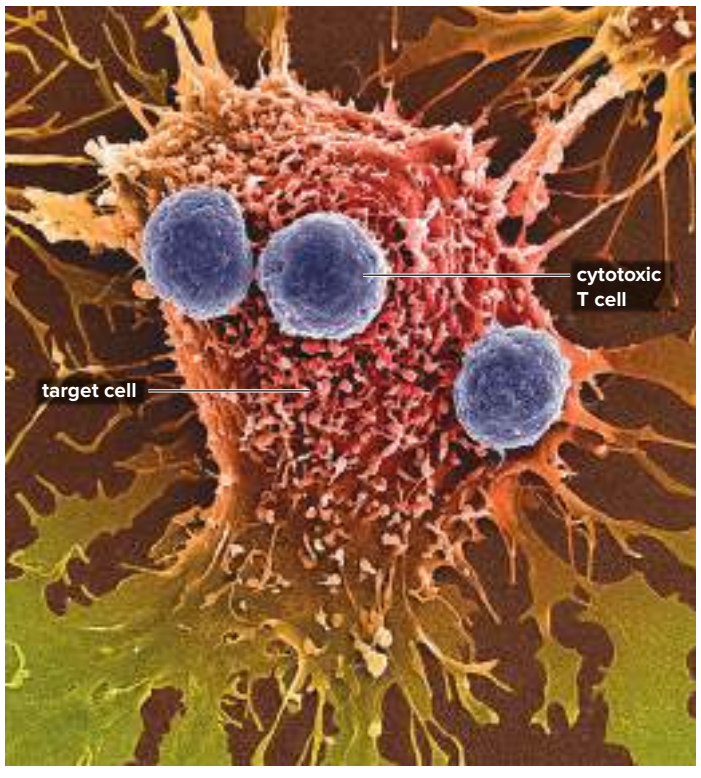
At first, the immune system is able to control the virus, probably through a combination of antibodies and cell-mediated immunity. But gradually the virus mutates, so that the immune system no longer recognizes it, and as the HIV count rises, the helper T-cell count drops to way below normal. As noted in the Biological Systems feature, "AIDS and Opportunistic Infections," the person with an HIV infection begins to develop opportunistic infections—infections that would be unable to take hold in a person with a healthy immune system. Now the individual has **acquired immunodeficiency syndrome (AIDS).**

Fortunately, antiretroviral drugs have made HIV a manageable infection in most individuals who receive treatment, although the drugs have many side effects. Unfortunately, there is still a shortage of these expensive drugs in many less developed countries.

ISTUDY



a.



b. SEM 1,250×

Figure 33.13 Cell-mediated immunity. **a.** How a T cell destroys a virus-infected cell or cancer cell. **b.** The scanning electron micrograph shows cytotoxic T cells attacking and destroying a cancer cell (target cell).
(b): Steve Gschmeissner/Science Source

Cytokines as Therapeutic Agents

Cytokines are produced by T cells, macrophages, and many other cells. Because they affect white blood cell formation and/or function, some cytokines are approved for use as therapy for cancer and certain other conditions. As mentioned, interferon is used to treat hepatitis C and certain cancers, and it may slow the progression of multiple sclerosis. The T-cell activating cytokine interleukin-2 is used to treat some forms of melanoma and kidney cancer.

In some cases, it may be desirable to inhibit the activity of certain cytokines, especially those involved in promoting chronic inflammatory diseases. Tumor necrosis factor (TNF), mentioned earlier, is a major cytokine produced by macrophages that has the ability to promote the inflammatory response. Several treatments that inhibit the TNF response, including anti-TNF monoclonal antibodies, are being developed as potential treatments for inflammatory diseases such as rheumatoid arthritis, Crohn's disease, and asthma. The drug etanercept (Enbrel) uses a different approach to block TNF; it is a protein that incorporates part of the receptor for TNF that is normally found on cells. However, the protein lacks the other parts of the receptor that normally trigger inflammatory responses, so instead it acts as a "decoy," binding to TNF and blocking its normal functions. As would be expected, these medications may also inhibit some of the beneficial effects of inflammation, and patients taking them may be more prone to infections.

Active Versus Passive Immunity

In general, adaptive immune responses can be induced either actively or passively. **Active immunity** occurs when an individual produces their own immune response against an antigen. For example, when you catch a cold, you recover because your body produces the T-cell and B-cell responses that eventually clear the offending viruses from your body. Active immunity can also be induced artificially when a person is well, to prevent infection in the future.

Immunization involves the use of vaccines, substances that contain an antigen to which the immune system responds. Traditionally, vaccines are the pathogens themselves, or their products, that have been treated, so that they are no longer virulent (able to cause disease). The use of vaccines has been effective in reducing the rates of bacterial diseases such as diphtheria, tetanus, and whooping cough, as well as viral diseases such as measles, mumps, and rubella. In fact, vaccination against smallpox was so successful that the disease was eradicated from the planet in 1977.

Today, it is possible to genetically engineer bacteria to mass-produce a protein from pathogens, and this protein can be used as a vaccine (see Section 14.2). This method was used to produce a vaccine against hepatitis B, a viral disease, and recently has been used to produced a vaccine against malaria, called Mosquirix, that is designed to help prevent malaria infections of children in less developed countries.

ISTUDY

THEME Biological Systems

AIDS and Opportunistic Infections

AIDS (acquired immunodeficiency syndrome) is caused by HIV, the human immunodeficiency virus. An HIV infection leads to the eventual destruction of helper T (T_H) cells. HIV kills T_H cells by directly infecting them, and it causes many uninfected T_H cells to die by a variety of mechanisms, including rare cancers such as Kaposi sarcoma (Fig. 33A). Many HIV-infected T_H cells are also killed by the person's own immune system.

Figure 33A Kaposi sarcoma. Karposi sarcoma is a form of cancer that occurs in the skin or gastrointestinal tract. National Cancer Institute (NCI)

A healthy individual typically has 800–1,000 T_H cells per mm^3 of blood (Fig. 33B). After an initial HIV infection, it may take several years for an individual's T_H-cell numbers to drop below 500 cells per mm^3 of blood, at which point the individual infected with HIV usually begins to contract many unusual types of infections that would not cause disease in a person with a healthy immune system.

Such infections are known as opportunistic infections (OIs). The U.S. Centers for Disease Control and Prevention (CDC) has defined three categories of HIV infection, based mainly on the types of OIs seen in the person. These OIs also tend to be associated with a decreasing number of T_H cells in the blood:

CDC category A (> 500 T_H cells/mm^3 blood)

- Typically, no OIs seen; may have persistently enlarged lymph nodes

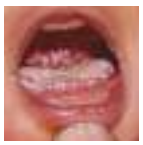
CDC category B (200–499 T_H cells/mm^3 blood)

- Shingles, a painful infection with *Varicella zoster* (chickenpox) virus
- Candidiasis, or thrush, a fungal infection of the mouth, throat, or vagina

Shingles
Centers for Disease Control and Prevention

CDC category C (< 200 T_H cells/mm^3 blood)

- Pneumocystis pneumonia, a fungal infection, causing the lungs to become useless as they fill with fluid and debris
- Kaposi sarcoma, a cancer of blood vessels due to human herpesvirus 8, giving rise to reddish-purple, coin-sized spots and lesions on the skin (Fig. 33A)



Candidiasis
Dr. M.A. Ansary/ Science Source

- Toxoplasmic encephalitis, a protozoan infection characterized by severe headaches, fever, seizures, and coma
- *Mycobacterium avium* complex (MAC), a bacterial infection resulting in persistent fever, night sweats, fatigue, weight loss, and anemia
- Cytomegalovirus, a viral infection that leads to blindness, inflammation of the brain, and throat ulcerations

Pneumocystis pneumonia
Dr. Edwin P. Ewing, Jr./CDC

Thanks to the development of powerful drug therapies that inhibit the life cycle of HIV, people infected with HIV in the United States have a lower incidence of opportunistic infections than occurred in the 1980s and 1990s.

Questions to Consider

1. Why has it been so difficult to develop an effective vaccine for HIV?
2. What are some possible differences between the types of OIs typically seen in category B and those seen in category C (i.e., why do shingles and candidiasis occur in B, but others more commonly in C)?
3. What, if any, obligation do relatively wealthy countries, such as the United States, have in providing anti-HIV drugs to less developed countries?

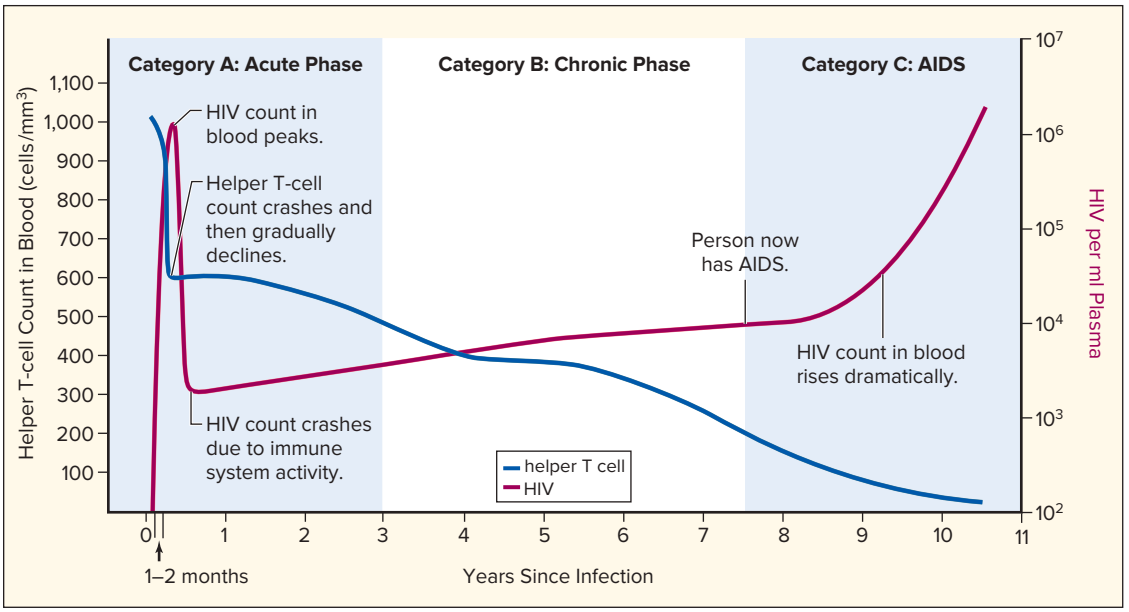


Figure 33B Progression of HIV infection during its three stages, called categories A, B, and C. In category A, the individual may have no symptoms or very mild symptoms associated with the infection. By category B, opportunistic infections—such as candidiasis, shingles, and diarrhea—have begun to occur. Category C is characterized by more severe opportunistic infections and is clinically described as AIDS.

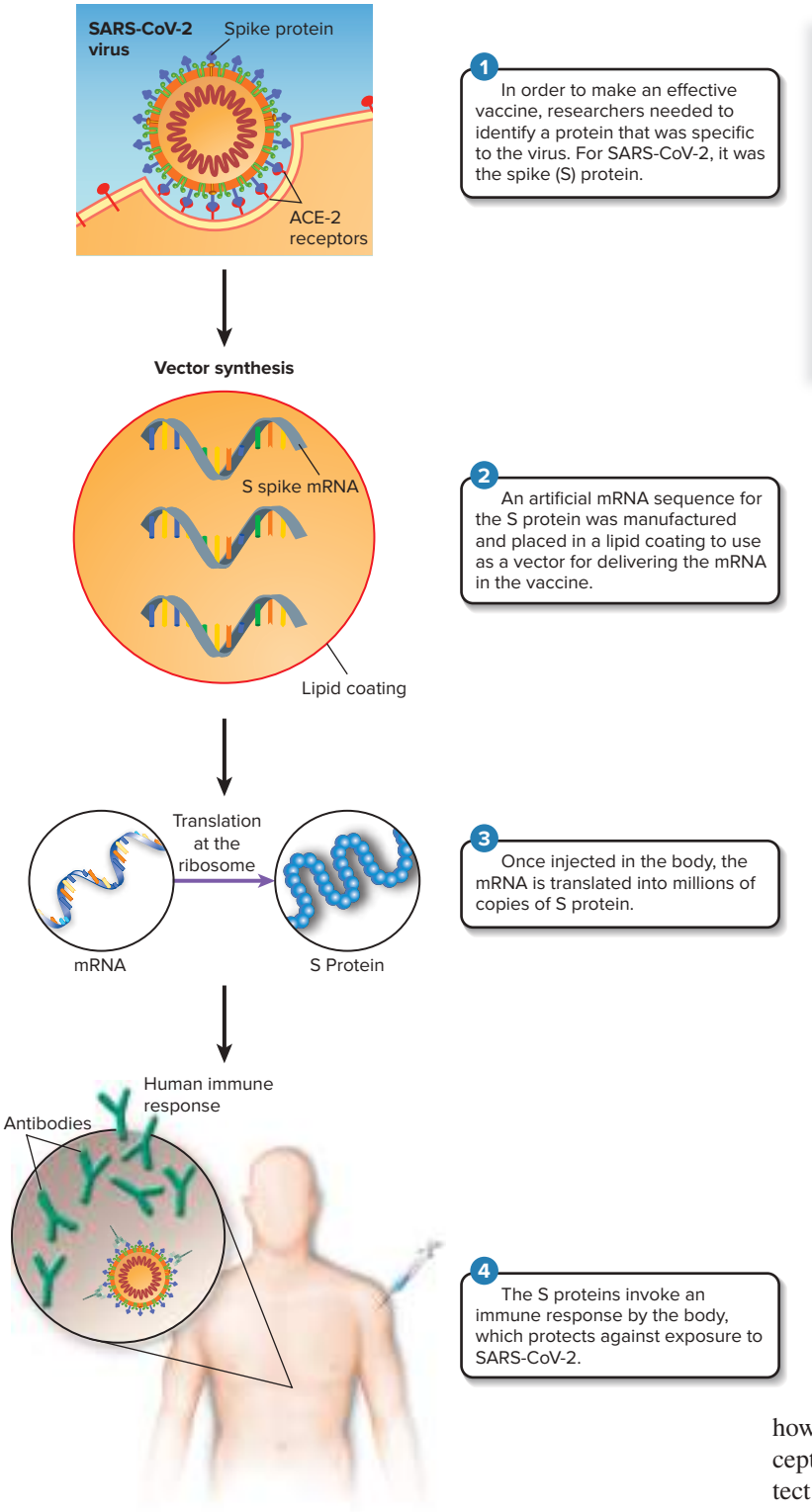


Figure 33.14 The development of mRNA vaccines. Using the mRNA of the S protein of SARS-CoV-2, researchers were able to develop a new method of producing vaccines in response to the pandemic.

Newer technologies, such as those being used against the SARS-CoV-2 virus, are employing mRNA molecules to produce specific parts of the virus to act as antigens (Fig. 33.14).

This approach is now being used to develop vaccines against a variety of other pathogens and diseases, including HIV, hepatitis, influenza, and even cancer.

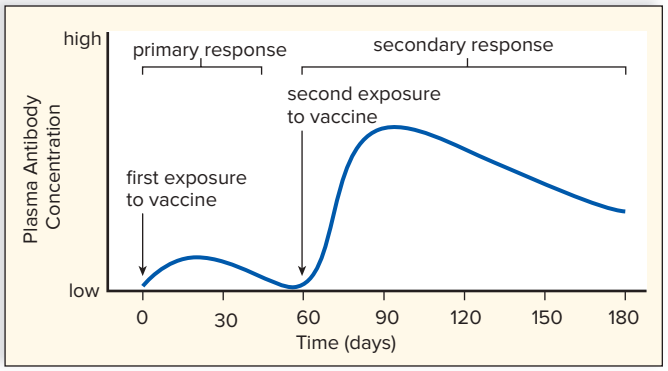


Figure 33.15 Antibody titers. During immunization, the primary response after the first injection of a vaccine is minimal, but the secondary response, which occurs after the second injection, shows a dramatic rise in the amount of antibody present in plasma.

The Nature of Science feature, "Adult Vaccinations," lists the current recommended vaccination schedule for adults over the age of 18.

After most vaccines are given, it is possible to determine the antigen-specific antibody titer (the amount of antibody present in a sample of plasma). After the first vaccination, a primary response occurs. For several days, no antibodies are present; then the titer rises slowly, followed by first a plateau and then a gradual decline as the antibodies bind to the antigen or simply break down (Fig. 33.15). After a second exposure, a secondary response occurs. The second exposure is called a "booster," because it boosts the immune response to a high level. The high levels of antigen-specific T cells and antibodies are expected to prevent disease symptoms if the individual is later exposed to the disease-causing agent. Even years later, if the antigen enters the body, memory B cells can quickly give rise to more plasma cells capable of producing the correct type of antibody.

Boosters were used successfully during the COVID-19 pandemic to raise immunity in an individual against variants of the rapid-evolving SARS-CoV-2 virus.

Passive immunity occurs when an individual receives another person's antibodies or immune cells. The passive transfer of antibodies is a common natural process. For example, newborn infants are passively immune to some diseases, because antibodies have crossed the placenta from the female parent's blood. These antibodies soon disappear, however, so that within a few months, infants become more susceptible to infections as their own immune system must now protect them. Breast-feeding may prolong the natural passive immunity an infant receives from their female parent, because antibodies are present in the female parent's milk.

Even though passively administered antibodies last only a few weeks, they can sometimes be used to prevent illness in a patient who has been unexpectedly exposed to certain infectious agents or toxins. Examples of diseases that may be prevented or treated in this manner include rabies, tetanus, botulism, and snakebites. Individuals with certain types of genetic immunodeficiencies may also benefit greatly from regular intravenous injections of human IgG, extracted from a large, diverse adult population.

ISTUDY

THEME Nature of Science

Adult Vaccinations

Many people mistakenly believe that individuals receive their full complement of vaccinations by the time they leave high school. In reality, being vaccinated is a lifelong activity. The Centers for Disease Control and Prevention (CDC) has identified a series of vaccinations that are recommended as of the age of 19 (Table 33A).

In many cases, vaccinations are recommended when an individual is determined to be at risk for a specific disease or condition. For example, while the vaccination for hepatitis B (HepB) may not be required, it is recommended for individuals who have had more than one sexual partner during a 6-month period, have been diagnosed with a sexually transmitted disease (see Section 41.5), use injection drugs, or may have been exposed to blood or infected body fluid. In most cases, vaccinations are recommended as a protective measure even if you are not in an at-risk category.

As always, if you have questions regarding any of these diseases, or about your personal need for vaccinations, consult with your health-care provider. For more information on vaccination schedules from birth through adulthood, visit the CDC's website at www.cdc.gov/vaccines or the Immunization Action Coalition (www.immunize.org).

Questions to Consider

1. Why would some vaccines require multiple doses over an adult's lifetime?
2. Why would people over the age of 60 or 65 require different vaccinations?

Table 33A List of Recommended Adult Vaccines by Age

Vaccine	19–26 years	27–49 years	50–64 years	≥ 65 years
COVID-19	2- or 3- dose primary series and booster (see physician)			
Influenza inactivated (IIV4) or **Influenza recombinant** (RIV4)	1 dose annually			
or	or			
Influenza live, attenuated (LAIV4)	1 dose annually			
Tetanus, diphtheria, pertussis (Tdap or Td)	1 dose Tdap each pregnancy; 1 dose Td/Tdap for wound management (see physician)			
	1 dose Tdap, then Td or Tdap booster every 10 years			
Measles, mumps, rubella (MMR)	1 or 2 doses depending on indication (if born in 1957 or later)			For healthcare personnel, see physician
Varicella (VAR)	2 doses (if born in 1980 or later)		2 doses	
Zoster recombinant (RZV)	2 doses for immunocompromising conditions (see physician)		2 doses	
Human papillomavirus (HPV)	2 or 3 doses depending on age at initial vaccination or condition	27 through 45 years		
Pneumococcal (PCV15, PCV20, PPSV23)	1 dose PCV15 followed by PPSV23 or 1 dose PCV20 (see physician)			See physician
				See physician
Hepatitis A (HepA)	2, 3, or 4 doses depending on vaccine			
Hepatitis B (HepB)	2, 3, or 4 doses depending on vaccine or condition			
Meningococcal A, C, W, Y (MenACWY)	1 or 2 doses depending on indication, see physician for booster recommendations			
Meningococcal B (MenB)	2 or 3 doses depending on vaccine and indication, see physician for booster recommendations			
	19 through 23 years			
***Haemophilus influenzae* type b** (Hib)	1 or 3 doses depending on indication			

Recommended vaccination for adults who meet age requirement, lack documentation of vaccination, or lack evidence of past infection

Recommended vaccination for adults with an additional risk factor or another indication

Recommended vaccination based on shared clinical decision-making

No recommendation/Not applicable

CDC

ISTUDY

Instead of antibodies, cells of the immune system can be transferred into a patient. The best example is a bone marrow transplant in patients with cancer, in which stem cells that produce blood cells are replenished after the person's own marrow has been intentionally destroyed by radiation or chemotherapy.

Check Your Progress 33.4

1. Distinguish between antibody-mediated immunity and cell-mediated immunity, and list the types of cells involved in each.
2. Explain the diversity of antibodies.
3. Explain the difference between active and passive immunity, and list three examples of each.

33.5 Immune System Disorders and Hypersensitivity Reactions

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the two main types of immunodeficiency disorders and provide examples of each.
2. Discuss the most common immunological mechanisms responsible for allergies and how these may be treated.
3. Define *autoimmune disease* and list several specific examples of these diseases.
4. Explain the types of precautions that must be taken when transplanting organs.

In most cases, the immune system is essential for our health and survival, as demonstrated by the diseases, some of them fatal, that occur in people who are immunodeficient. In other instances, the immune system may work against the best interests of the body, as occurs in allergies, autoimmune disorders, and rejection of transplanted organs.

Immunodeficiencies

A number of immunodeficiency disorders are known, but all result in some degree of increased susceptibility to infections. Primary immunodeficiencies are genetic, meaning they are passed from parents to offspring. For example, in severe combined immunodeficiency (SCID), both T cells and B cells are either lacking completely or not functioning well enough to protect the body from a variety of infections that are not a problem for most people. SCID occurs in only about 1 in 100,000 births.

A variety of faulty genes can cause SCID. In most cases, however, by about 3 months of age, when most of the antibodies that infants have obtained from their female parent have been degraded, infants with untreated SCID usually die. Possible treatments include a bone marrow transplant to replace the stem cells that form all of our white blood cells, and gene therapy to replace the faulty DNA. If these treatments are unsuccessful, the outcome is usually poor.

Another primary immunodeficiency is X-linked agammaglobulinemia (XLA), which is due to a mutated gene on the X chromosome that is needed for proper development of B cells. XLA affects only males because their cells have only one X chromosome. A female with a normal gene on at least one of the two X chromosomes does not develop the disease. Because their T cells are unaffected, boys with XLA usually experience few health complications as long as they receive regular injections of human IgG. About 1 in 190,000 males has XLA.

We have already seen that HIV infection causes AIDS, which is an example of a secondary immunodeficiency. These disorders are not genetic but, instead, are acquired after birth. Besides infections, other potential causes of secondary immunodeficiencies include malnutrition, irradiation, certain drugs and toxins, and certain cancers. Some of these can be cured by addressing their cause.

Allergies

Allergies are hypersensitivities to substances, such as pollen, food, or animal hair, that ordinarily would do no harm to the body. The response to these antigens, called allergens, usually includes some degree of tissue damage.

An **immediate allergic response** can occur within seconds of contact with an allergen. The response is caused by antibodies of the IgE class (Table 33.2). IgE antibodies are attached to receptors on the plasma membrane of mast cells in the tissues, as well as to basophils and eosinophils in the blood. When an allergen attaches to these IgE antibodies, the cells release histamine and other substances that bring about the symptoms of an allergy (Fig. 33.16). When an allergen such as pollen is inhaled, histamine stimulates the inflammatory response in the mucous membranes of the nose and eyes typical of *hay fever.* If a person has **asthma,** the airways leading to the lungs constrict, resulting in difficult breathing accompanied by wheezing. An allergen in food typically causes nausea, vomiting, and diarrhea.

Anaphylactic shock is an immediate allergic response that occurs after an allergen has entered the bloodstream. Bee stings, foods, various medications, and latex rubber are all known to cause this reaction in some individuals. Anaphylactic shock is characterized by a sudden and life-threatening drop in blood pressure due to an

Table 33.2 Comparison of Immediate and Delayed Allergic Responses

	Immediate Response	Delayed Response
Onset of Symptoms	Takes several minutes	Takes 2 to 3 days
Lymphocytes Involved	B cells	T cells
Immune Reaction	IgE antibodies	Cell-mediated immunity
Type of Symptoms	Hay fever, asthma, and many other allergic responses	Contact dermatitis (e.g., poison ivy)
Therapy	Antihistamine and epinephrine	Cortisone

ISTUDY

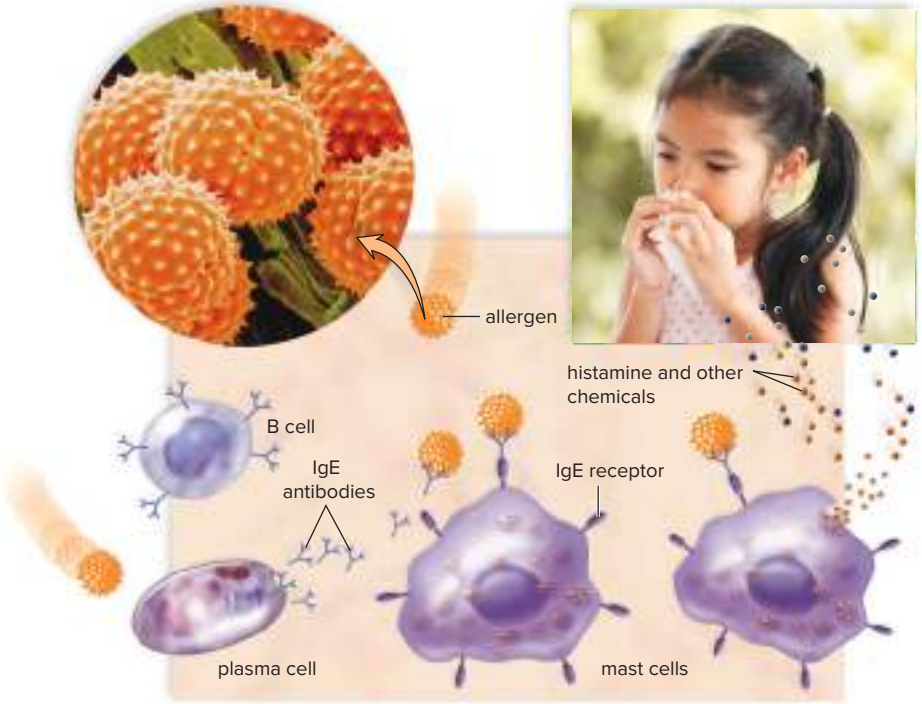


Figure 33.16 An allergic reaction. An allergen attaches to IgE antibodies, which then cause mast cells to release histamine and other chemicals that are responsible for the allergic reaction. (photos): (allergens): David Scharf/Science Source; (girl): A3pfamily/Shutterstock

increased dilation of the capillaries throughout the body due to a release of histamine. The smooth muscle lining the bronchi may also be strongly stimulated to constrict, resulting in an inability to breathe. Injecting epinephrine can counteract this reaction until medical help is available, and some people carry an epinephrine-containing, spring-loaded syringe (sometimes called an EpiPen) for this purpose.

Mild to moderate allergies are usually treated with antihistamines, which compete with histamine for binding to histamine receptors. In more serious cases, injections of the allergen can be given in an effort to stimulate the immune system to produce high quantities of IgG against the allergen. The hope is that these IgG antibodies will combine with the allergen molecules before they have a chance to reach the IgE antibodies. A monoclonal antibody called Xolair is also available; it blocks the binding of IgE to its receptor on inflammatory cells.

A **delayed allergic response** is initiated by memory T cells at the site of allergen contact in the body. The allergic response is regulated by the cytokines secreted by these "sensitized" T cells at the site. A classic example of a delayed allergic response is the skin test for tuberculosis (TB). When the test result is positive, the tissue where the antigen was injected becomes red and hardened. This indicates prior exposure to *Mycobacterium tuberculosis,* the bacterium that causes TB. Contact dermatitis, which occurs when a person's skin reacts to poison ivy, jewelry, cosmetics, or many other substances that touch the skin, is another example of a delayed allergic response.

Autoimmune Diseases

When a person has an **autoimmune disease,** the immune system mistakenly attacks the body's own cells or molecules. Exactly what causes autoimmune diseases is not known. In some cases, there appears to be a genetic tendency to develop autoimmune diseases. Many autoimmune diseases also seem to occur after an individual has recovered from an infection. It is also known that certain antigens of microbial pathogens can resemble antigens found in their host, a phenomenon known as molecular mimicry. A good example of this is rheumatic fever, which sometimes follows an infection with bacteria of the genus *Streptococcus.* Certain proteins in the cell wall of these bacteria are known to resemble proteins in the heart; as a result, antibodies formed against the bacterial proteins may cause inflammation of the heart, which can continue even after the bacteria have been cleared from the body.

Some autoimmune diseases affect only specific tissues. Rheumatoid arthritis is a common autoimmune disorder that causes recurring inflammation in synovial joints (Fig. 33.17). Complement proteins, T cells, and B cells all participate in the destruction of the joints, which eventually become immobile. In myasthenia gravis, antibodies interfere with the functioning of neuromuscular junctions, causing muscular weakness. In multiple sclerosis, T cells attack the myelin sheath of nerve fibers, causing a variety of symptoms related to the defective transmission of messages by nerves.

Systemic lupus erythematosus (lupus) is a chronic autoimmune disorder that affects multiple tissues and organs. It is characterized by the production of antibodies that react with the DNA contained in almost every cell of the body. The symptoms vary somewhat, but most people experience a characteristic skin rash (Fig. 33.18), joint pain, and kidney damage, which may be life-threatening. About 1.5 million people in the United States have lupus, 90% of whom are women of childbearing age. Because little is known about the origin of autoimmune disorders, no cures are available. The symptoms can sometimes be controlled using immunosuppressive drugs, such as cortisone, but these drugs can have serious side effects.

Transplant Rejection

Certain organs, such as the skin, heart, and kidneys, could be transplanted relatively easily from one person to another if the body did

Figure 33.17 Rheumatoid arthritis. Rheumatoid arthritis is caused by recurring inflammation in synovial joints, due to immune system attack. chaowalit407/iStock/Getty Images

ISTUDY

Figure 33.18
Systemic lupus.
The characteristic rash of systemic lupus. Scott Camazine/Alamy Stock Photo

not attempt to reject them. Unfortunately, like the immune system itself, MHC proteins are a double-edged sword. In addition to their beneficial role in presenting antigens to T cells, MHC proteins are major targets of the immune response during the rejection of a transplanted organ.

The MHC proteins of different individuals differ by the sequence of their amino acids. The immune system will attack any foreign tissue that bears MHC antigens that are different from those of the individual. Ideally, a transplant donor would have exactly the same type of MHC proteins as those of the recipient; however, while it is difficult to find a perfect MHC match, the odds of organ rejection can be reduced by administering immunosuppressive drugs. Two commonly used drugs, cyclosporine and tacrolimus, inhibit the production of certain cytokines by T cells.

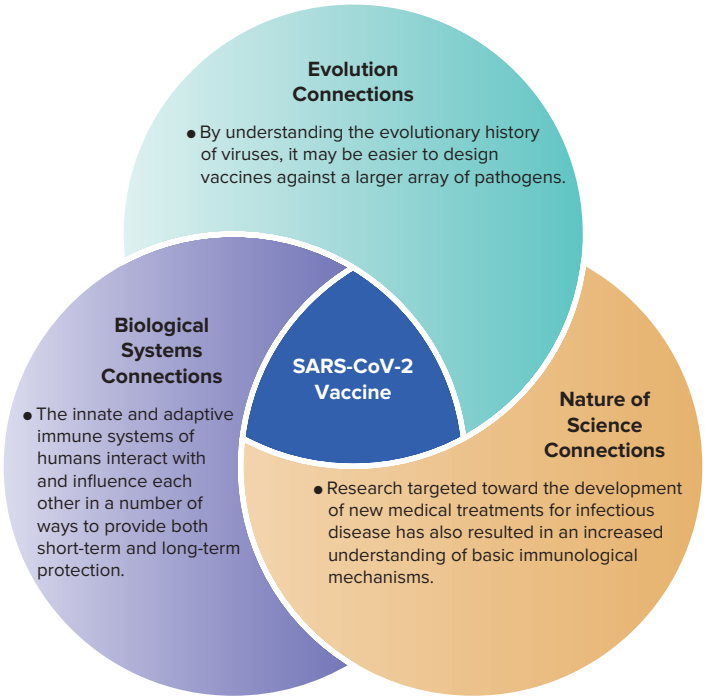
Xenotransplantation, the transplantation of animal tissues and organs into humans, is a potential way to solve the shortage of organs from human donors. The pig is the most popular choice as an organ source, because pig organs are generally the right size, and pigs are a common meat source and thus readily available. Genetic engineering can make pig organs less antigenic by removing the MHC antigens. The ultimate goal is to make pig organs as widely accepted as type O blood cells.

Other researchers hope that tissue engineering, including the production of human organs from stem cells, will someday do away with the problem of rejection. Scientists have recently grown new heart valves in the laboratory using stem cells gathered from amniotic fluid following amniocentesis, and surgeons have successfully used lab-grown urinary bladder tissue to rebuild defective bladders in human patients.

Check Your Progress 33.5

1. Explain the general type of abnormality that causes most primary immunodeficiencies.
2. Describe the treatments for autoimmune diseases.
3. Define *xenotransplantation.*

Connecting *the* Concepts



What Did You Learn?

33.1 Evolution of Immune Systems

The **immune system** protects an organism from a variety of threats. From the available evidence, it appears that **innate immunity** is much older than adaptive immunity because aspects of innate immunity are present in relatively simple multicellular animals. Innate immunity is quite well-developed in insects and often involves the recognition of pathogen-associated molecular patterns (PAMPs). Available evidence indicates that **adaptive immunity,** or the ability to generate a diverse array of antigen receptors that can recognize very specific **antigens,** first appeared in an ancestor of the jawed vertebrates.

ISTUDY

33.2 The Lymphatic System

The **lymphatic system** includes the **lymphatic vessels** and **lymphoid (lymphatic) organs.** The lymphatic system:

- Removes excess interstitial fluid (called **lymph**) collected by **lymphatic capillaries**
- Absorbs fats from the small intestine
- Produces and distributes lymphocytes

Lymphocytes are produced in the **primary lymphoid organs** (**red bone marrow** and **thymus**), and lymphocytes respond to antigens in **secondary lymphoid organs** (e.g., **lymph nodes** and **spleen**). **T cells** (T lymphocytes) mature in the thymus, while **B cells** (B lymphocytes) mature in red bone marrow.

33.3 Innate Immune Defenses

Two types of **immunity** protect the body from infections and other threats. Innate defenses are always present, or they occur very soon after exposure to an infection. These include:

- Barriers to entry
- The **microbiome**
- The **inflammatory response** (which includes **histamine** released from tissue-dwelling **mast cells**)
- Phagocytes (including **neutrophils, eosinophils, macrophages,** and **dendritic cells**) and **natural killer (NK) cells**
- Protective proteins (including **complement** and **interferons**)

33.4 Adaptive Immune Defenses

Adaptive (also called acquired) immunity involves B cells and T cells. These cells have unique **antigen receptors** on their surface. According to the **clonal selection theory,** upon binding of its antigen receptor with a specific antigen, B cells and T cells divide and differentiate.

- The antigen receptor on a B cell is called a **B-cell receptor (BCR),** and clonal selection of B cells forms **plasma cells** and **memory B cells.** Plasma cells secrete **antibodies** and are responsible for **antibody-mediated immunity.** Antibodies, or **immunoglobulins (Igs),** are Y-shaped molecules that have at least two binding sites for a specific antigen. **Monoclonal antibodies** are produced by cells derived from a single plasma cell fused to a myeloma cancer cell, and have a variety of applications in science and medicine.
- The two main types of T cells are **helper T cells** (T_H cells) and **cytotoxic T cells** (T_C cells, or CTLs). For a T_H cell to recognize an antigen, its **T-cell receptor (TCR)** must bind to an antigen presented by class II **MHC (major histocompatibility complex) proteins** on the surface of **antigen-presenting cells (APCs).** Types of APCs include dendritic cells and macrophages. Thereafter, the activated T cell undergoes clonal expansion, forming activated T cells and **memory T cells** (T_H cells, in this case). Activated T_H cells produce cytokines that affect many other immune cells, resulting in **cell-mediated immunity. HIV (human immunodeficiency virus)** infects and destroys T_H cells, resulting in **acquired immunodeficiency syndrome (AIDS).**
- T_C cells recognize antigens presented by MHC class I proteins on the surface of virus-infected or cancer cells. They then kill these cells by releasing perforin and granzymes, inducing apoptosis.
- **Active immunity** occurs as a response to an illness or to **immunization** with a vaccine. **Passive immunity** is needed when an individual is in immediate danger of succumbing to an infectious disease. Passive immunity can occur naturally (as in the transfer of antibodies from female parent to infant) or may be used as a medical treatment.

33.5 Immune System Disorders and Hypersensitivity Reactions

Immunodeficiencies can be primary (genetic) or secondary (due to some other cause).

- **Allergies,** as seen in hay fever or in **asthma,** occur when the immune system reacts to substances not normally recognized as dangerous. An **immediate allergic response** is mediated by IgE antibodies, and in its most severe form can result in **anaphylactic shock.**
- A **delayed allergic response,** such as occurs in the tuberculosis skin test, is mediated by T cells.
- **Autoimmune disease** occurs when the immune system attacks the body's own cells or tissues. In a transplant rejection, the immune system is usually responding to unmatched MHC proteins on the cells of a donated organ.

Thinking Critically About Biology

1. Some primitive organisms, such as invertebrates, have no lymphocytes and thus lack an adaptive immune system, but they have some components of an innate immune system, including phagocytes and certain protective proteins. What are some general features of innate immunity that make it very valuable to organisms lacking more specific antibody- and cell-mediated responses? What are some disadvantages to having only an innate immune system?
2. A person bitten by a venomous snake should be given antivenom (antibodies) to prevent death. If bitten by the same type of snake 3 years after the initial bite, will this person have immunity to the venom, or should the person get another shot of antivenom? Justify your response with an explanation of the type of immunity someone gains from a shot of antibodies.
3. Why is it that Rh incompatibility can be a serious problem when an Rh-negative female parent is carrying an Rh-positive fetus, but ABO incompatibility between female parent and fetus is usually no problem? That is, a type A female parent can usually safely carry a type B fetus. (*Hint:* The antibodies produced by an Rh-negative female parent against the Rh antigen are usually IgG, whereas the antibodies produced against the A or B antigen are IgM.) Because the Rh antigen obviously serves no vital function (most humans lack it), why do you think it hasn't been completely eliminated during human evolution?

Making It Relevant to You

1. Scientists are working to develop vaccines that would work against multiple mosquito-borne diseases at a single time. Based on what you learned about adaptive immunity, what should they be focusing on in their research?
2. Have you ever skipped getting a flu vaccine because you either got it the year before, or you thought it gave you the flu? Why do you now know that this information is incorrect?
3. There has been an increase in the number of people who do not believe in vaccinations. What dangers might this belief pose to the population?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

34

Digestive Systems and Nutrition

Image Source/Getty Images

Lactose Intolerance

Lactose intolerance is usually thought of as a digestive disorder. It occurs when the digestive system slows the production of lactase—the enzyme required to digest the sugar lactose. If lactase is absent, lactose passes into the large intestine and is fermented by bacteria that produce gas.

Lactose is present in milk for the nourishment of newborn offspring. Other species do not drink milk into maturity and neither did humans until the domestication of animals. Before that time, all human ancestors were lactose intolerant. What changed?

Geneticists have found several independent mutations that allowed for lactase persistence, or continued lactase production in adulthood. The trait became more common in human populations over the past 10,000 years, especially in those that raised livestock.

Today, only ancestors of those populations are lactase persistent. If you are not one of those people, know that lactose intolerance isn't a disorder where something went wrong. You, along with most of the world's current population (65%), simply did not inherit a recently evolved trait.

As you read through this chapter, think about the following questions:

1. In what ways are digestive systems of other animals adapted to their environments?
2. What role do enzymes play in digestion?
3. How can food choices and digestive disorders impact your health?

Chapter Outline

34.1 Digestive Tracts
34.2 The Human Digestive System
34.3 Digestive Enzymes
34.4 Nutrition and Human Health

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Chapter 3 What are some structural differences among carbohydrates, lipids, proteins, and nucleic acids?

Figure 6.1 How does energy flow from the sun, into chemical energy, to be ultimately dissipated as heat?

Figure 8.12 How do components of the human diet enter common metabolic pathways?

Following *the* Themes

CHAPTER 34 – DIGESTIVE SYSTEMS AND NUTRITION

Evolution	Animals have evolved a wide variety of strategies to acquire nutrients from their environment.
Nature of Science	Scientific research on the relationships between nutrients and disease provides the foundation for healthy diet recommendations.
Biological Systems	The digestive system provides excellent examples of organ specialization and the relationships between structure and function.

ISTUDY

34.1 Digestive Tracts

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the structural features of incomplete versus complete digestive tracts.
2. Explain the difference between a continuous and discontinuous feeder.
3. Discuss some specific adaptations that are seen in omnivores, herbivores, and carnivores.

A **digestive system** includes all the organs, tissues, and cells involved in ingesting food and breaking it down into smaller components. Digestion contributes to homeostasis by providing the body with the nutrients needed to sustain the life of cells. A digestive system:

- Ingests food
- Breaks down food into small molecules that can cross plasma membranes
- Absorbs nutrient molecules
- Eliminates undigestible remains

A digestive tract, or *gut,* is typically defined as a long tube through which food passes as it is being digested. The majority of animals have some sort of digestive tract, but some (e.g., sponges) have no digestive tract at all. Instead, as water from the aqueous environment flows through the sponge (see Fig. 28.6), food particles are removed by cells that make up the inner lining of the organism. Cells in the sponge, called *archaeocytes,* may also ingest and distribute food to the rest of the organism.

Incomplete Versus Complete Tracts

An **incomplete digestive tract** has a single opening, usually called a mouth; however, the single opening is used as both an entrance for food and an exit for wastes. Planarians, which are flatworms, have an incomplete tract (Fig. 34.1). It begins with a mouth and muscular pharynx, and then the tract, a gastrovascular cavity, branches throughout the body.

Planarians are primarily carnivorous and feed largely on smaller, aquatic animals, as well as bits of organic debris. When a planarian is feeding, the pharynx actually extends beyond the mouth. The body is wrapped about the prey and the pharynx sucks up small quantities at a time. Digestive enzymes in the tract allow some extracellular digestion to occur. Digestion is finished intracellularly by the cells that line the tract. No cell in the body is far from the digestive tract; therefore, diffusion alone is sufficient to distribute nutrient molecules.

The digestive tract of a planarian is notable for its lack of specialized parts. It is saclike, because the pharynx serves not only as an entrance for food but also as an exit for undigestible material. This use of the same body parts for more than one function tends to minimize the evolution of more specialized parts, such as those seen in complete tracts.

Planarians have some modified parasitic relatives. Tapeworms, which are parasitic flatworms, lack a digestive system. Nutrient molecules are absorbed by the tapeworm from the intestinal juices of the host, which surround the tapeworm's body. The integument and body wall of the tapeworm are highly modified for this purpose. They have millions of microscopic, fingerlike projections that increase the surface area for absorption.

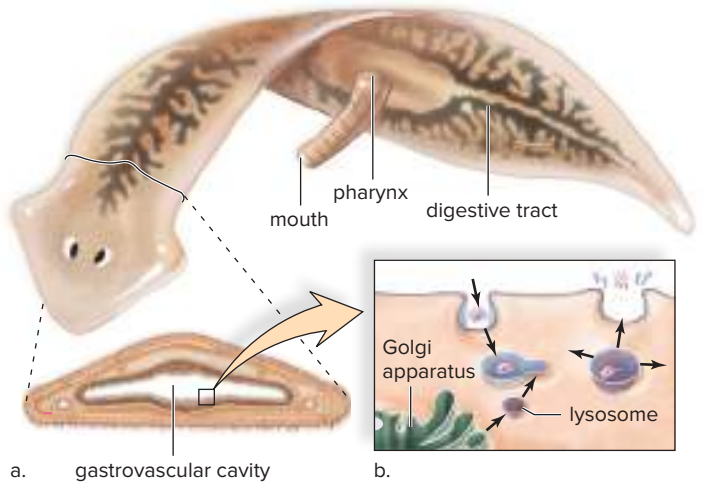


Figure 34.1 Incomplete digestive tract of a planarian.
a. Planarians, which are flatworms, have a gastrovascular cavity with a single opening that acts as both an entrance and an exit. Like hydras, planarians rely on intracellular digestion to complete the digestive process. **b.** Phagocytosis produces a vacuole, which joins with an enzyme-containing lysosome. The digested products pass from the vacuole into the cytoplasm before any undigestible material is eliminated at the plasma membrane.

In contrast to planarians, earthworms, which are annelids, have a **complete digestive tract,** meaning that the tract has a mouth and an anus (Fig. 34.2). Earthworms feed mainly on the decayed organic matter found in soil. The muscular pharynx draws in a large amount of soil with a sucking action. Soil then enters the crop, which is a storage area with thin, expansive walls. From there, it goes to the gizzard, where thick, muscular walls crush the food and ingested sand grinds it. Digestion is extracellular within

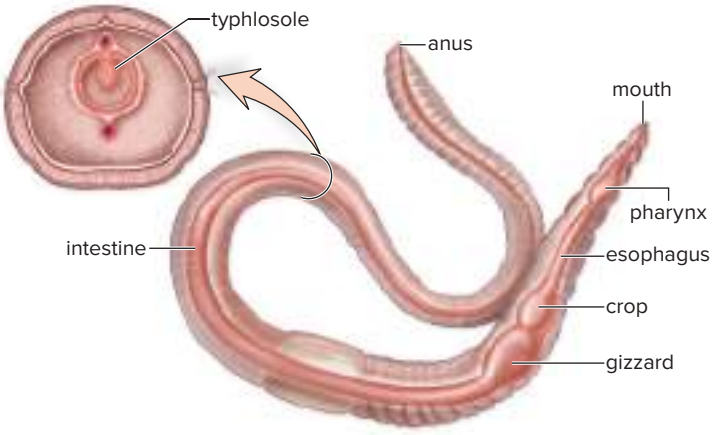
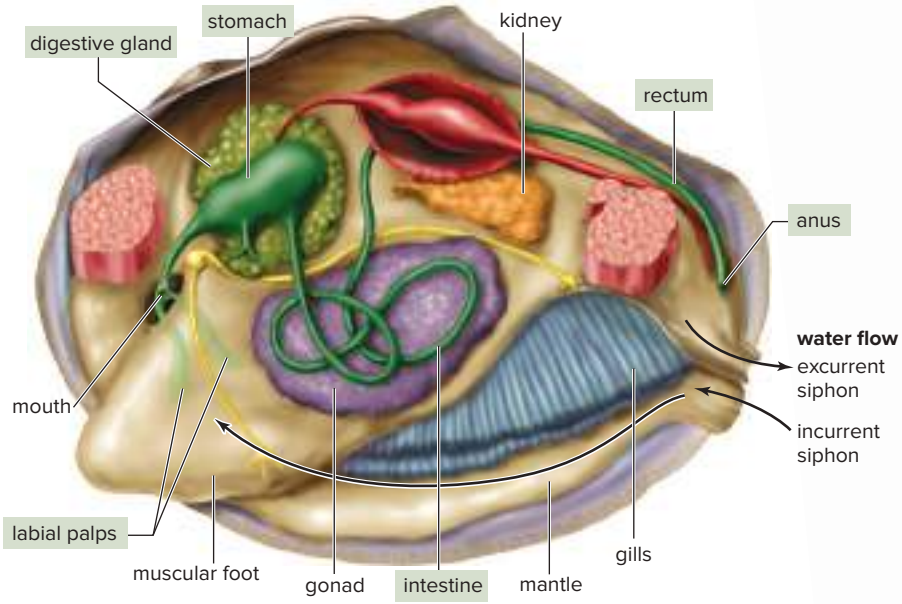


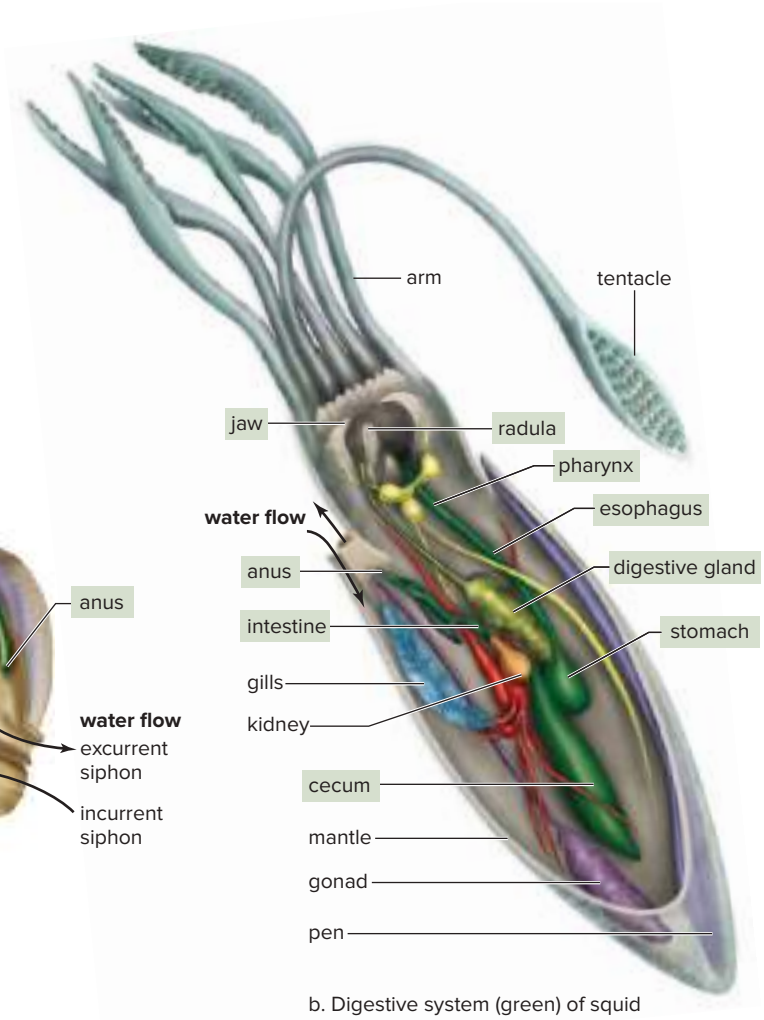
Figure 34.2 Complete digestive tract of an earthworm.
Complete digestive tracts have both a mouth and an anus and can have many specialized parts, such as those labeled in this drawing. Also in earthworms, which are annelids, the absorptive surface of the intestine is increased by an internal fold called the typhlosole.

ISTUDY

Figure 34.3 Nutritional mode of a clam compared to a squid. Clams and squids are mollusks. A clam burrows in the sand or mud, where it filter feeds, whereas a squid swims freely in open waters and captures prey. In keeping with their lifestyles, a clam (**a**) is a continuous feeder, and a squid (**b**) is a discontinuous feeder. Digestive system labels are shaded green.



a. Digestive system (green) of clam



b. Digestive system (green) of squid

the intestine. The surface area of digestive tracts is often increased for absorption of nutrient molecules, and in earthworms, this is accomplished by an intestinal fold called the *typhlosole*. Undigested remains pass out of the body at the anus. Specialization of parts is obvious in the earthworm, because the pharynx, crop, gizzard, and intestine have particular functions as food passes through the digestive tract.

Continuous Versus Discontinuous Feeders

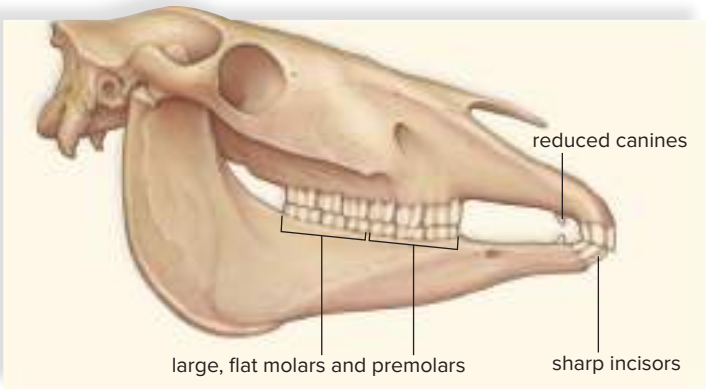
Some aquatic animals acquire their nutrients by continuously passing water through some type of apparatus that captures food. Clams, which are mollusks, are filter feeders (Fig. 34.3*a*). Water is always moving into the mantle cavity by way of the incurrent siphon (slitlike opening) and depositing particles, including algae, protozoans, and minute invertebrates, on the gills. The size of the incurrent siphon permits the entrance of only small particles, which adhere to the gills. Ciliary action moves suitably sized particles to the labial palps, which force them through the mouth into the stomach. Digestive enzymes are secreted by a large digestive gland, but amoeboid cells throughout the tract are believed to complete the digestive process by intracellular digestion.

Not all filter feeders are small invertebrates. A baleen whale, such as the blue whale, is an active filter feeder. Baleen—a keratinized, curtainlike fringe—hangs from the roof of the mouth and filters small shrimp, called krill, from the water. A baleen whale filters up to a ton of krill every few minutes.

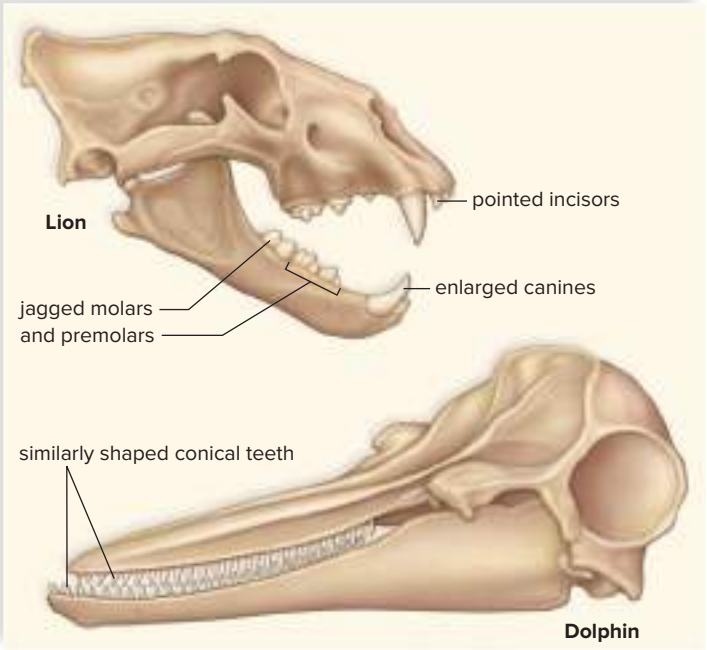
Discontinuous feeders have evolved the ability to store food temporarily while it is being digested, enabling them to spend less time feeding and more time engaging in other activities. Discontinuous feeding requires a storage area for food, which can be a crop, where no digestion occurs, or a stomach, where digestion begins.

Squids, which are mollusks, are discontinuous feeders (Fig. 34.3*b*). The body of a squid is streamlined, and the animal moves rapidly through the water using jet propulsion (forceful expulsion of water from a tubular funnel). The head of a squid is surrounded by ten arms, two of which have developed into long, slender tentacles whose suckers have toothed, horny rings. These tentacles seize prey (fishes, shrimps, and worms) and bring it to the squid's beaklike jaws, which bite off pieces pulled into the mouth by the action of a *radula,* a tonguelike structure. An esophagus leads to a stomach and a cecum (blind sac), where digestion occurs. The stomach, supplemented by the cecum, retains food until digestion is complete.

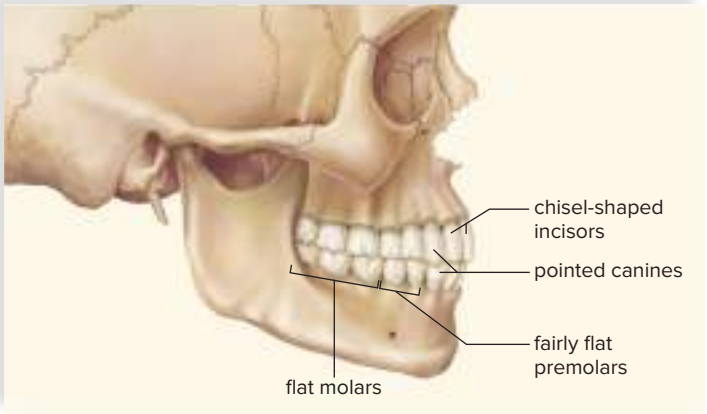
ISTUDY



a. Horses are herbivores.



b. Lions and dolphins are carnivores.



c. Humans are omnivores.

Figure 34.4 Dentition among mammals. **a.** Horses are herbivores and have teeth suitable to clipping and chewing grass. **b.** Lions and dolphins are carnivores. Dentition in a lion is suitable for killing large animals, such as zebras and wildebeests, and tearing apart their flesh. Dentition in a dolphin is suitable for grasping small animals, such as fish, which are swallowed whole. **c.** Humans are omnivores and have teeth suitable for a mixed diet of vegetables and meat.

Adaptations to Diet

Beyond the general categories of continuous versus discontinuous feeders, some animals have further adapted to more specialized diets. Some animals are **omnivores;** they eat both plants and animals. Others are strict **herbivores;** they feed only on plants. Still others are strict **carnivores;** they eat only other animals. Among invertebrates, filter feeders such as clams and tube worms are omnivores. Land snails, which are terrestrial mollusks, and some insects, such as grasshoppers and locusts, are herbivores. Spiders (arthropods) are carnivores, as are sea stars (echinoderms), which feed on clams. Some invertebrates are cannibalistic. A female praying mantis (an insect), if starved, will feed on her mate as the reproductive act is taking place!

Mammals have also adapted to consume a variety of food sources. Among herbivores, the koala of Australia is famous for its diet of only eucalyptus leaves, and likewise many other mammals are browsers, feeding on bushes and trees. Grazers, such as the horse, feed off grasses. The horse has sharp, even incisors for neatly clipping off blades of grass and large, flat premolars and molars for grinding and crushing the grass (Fig. 34.4*a*). Extensive grinding and crushing disrupts plant cell walls, allowing bacteria located in the part of the digestive tract called the cecum to digest cellulose.

Ruminants such as cattle, sheep, and goats have a large, four-chambered stomach. In contrast to horses, they graze quickly and swallow partially chewed grasses into the **rumen,** which is the first chamber. The rumen serves as a fermentation vat, where microorganisms break down material, such as cellulose, that the animal could not otherwise digest. Later on, when the ruminant is no longer feeding, undigested, solid material called cud is regurgitated and chewed again to facilitate more complete digestion.

Many mammals, including dogs, lions, toothed whales, and dolphins, are carnivores. Lions use pointed canine teeth for killing, short incisors for scraping bones, and pointed molars for slicing flesh (Fig. 34.4*b, top*). Dolphins and toothed whales swallow food whole without chewing it first; they are equipped with many identical, conical teeth that are used to catch and grasp their slippery prey before swallowing (Fig. 34.4*b, bottom*). Meat is rich in protein and fat and is easier to digest than plant material. The intestine of a rabbit, a herbivore, is much longer than that of a similarly sized cat, a carnivore.

Humans, like pigs, raccoons, mice, and most bears, are omnivores. Therefore, the dentition has a variety of specializations to accommodate both a vegetable diet and a meat diet. An adult human has 32 teeth. One-half of each jaw has teeth of four types: two chisel-shaped incisors for shearing; one pointed canine (cuspid) for tearing; two fairly flat premolars (bicuspids) for grinding; and three molars, well flattened for crushing (Fig. 34.4*c*). Omnivores are generally better able to adapt to different food sources, which can vary by location and season.

Check Your Progress 34.1

1. Compare the digestive tract of a planarian with that of an earthworm.
2. Describe some of the limitations of an incomplete digestive tract.
3. Compare the teeth of carnivores to those of herbivores.

ISTUDY

34.2 The Human Digestive System

Learning Outcomes

Upon completion of this section, you should be able to

1. List all the major components of the human digestive tract, from the mouth to the anus.
2. Compare and contrast the structural features of the small intestine and the large intestine.
3. Discuss the major functions of the pancreas, liver, and gallbladder.

Humans have a complete digestive tract, which begins with a mouth and ends in an anus. The major structures of the human digestive tract are illustrated in Figure 34.5. The salivary glands, pancreas, liver, and gallbladder are accessory organs that aid digestion.

The digestion of food in humans is an extracellular event and requires a cooperative effort between different parts of the body. Digestion consists of two major stages: mechanical digestion and chemical digestion.

Mechanical digestion involves the physical breakdown of food into smaller particles. This task is accomplished through the chewing of food in the mouth and the physical churning and mixing of food in the stomach and small intestine. Chemical digestion requires enzymes that are secreted by the digestive tract or by accessory glands that lie nearby. Specific enzymes break down particular macromolecules into smaller molecules that can be absorbed (see Section 34.3).

Mouth

The **mouth,** or oral cavity, serves as the beginning of the digestive tract. The *palate,* or roof of the mouth, separates the oral cavity

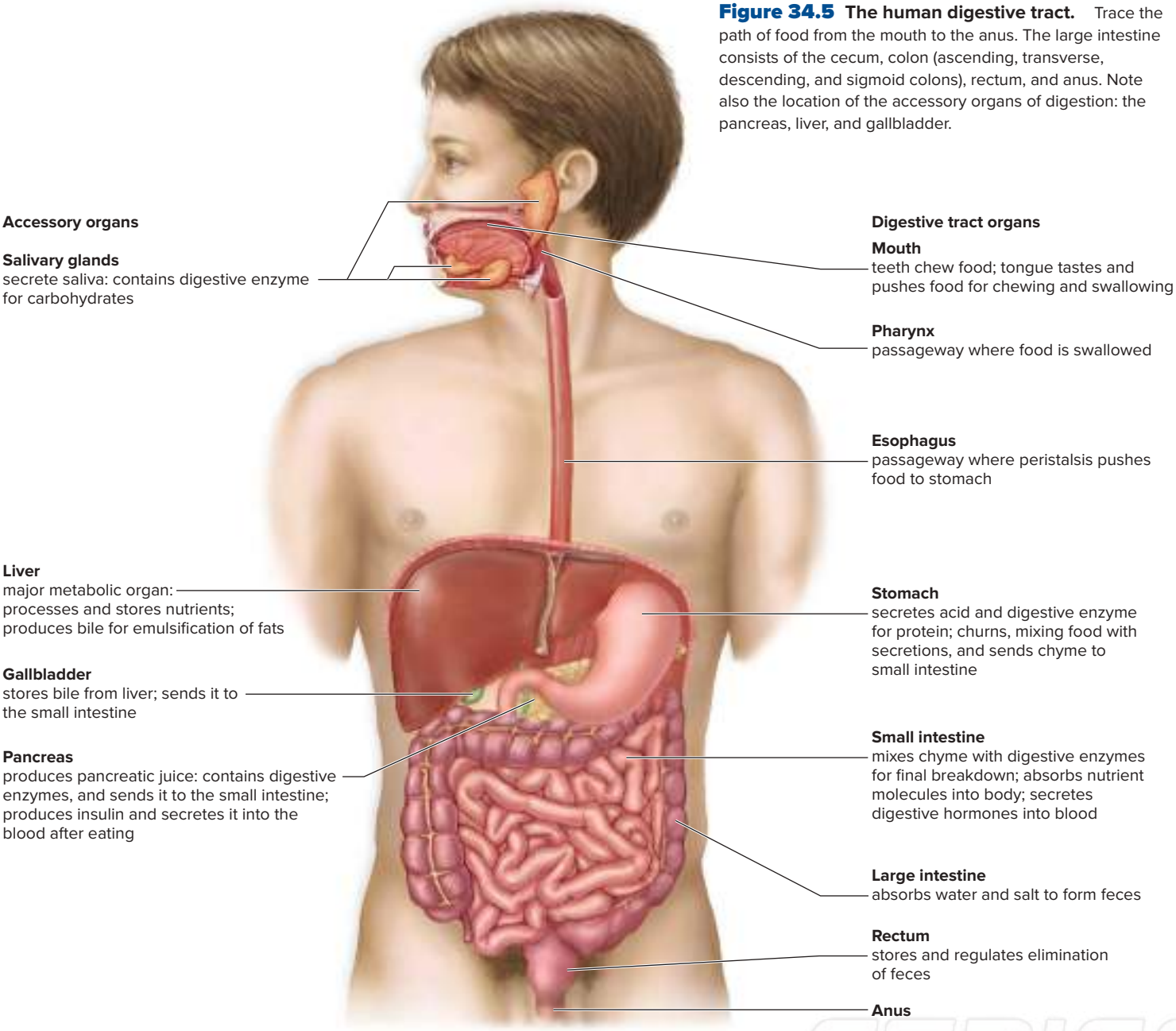


Figure 34.5 The human digestive tract. Trace the path of food from the mouth to the anus. The large intestine consists of the cecum, colon (ascending, transverse, descending, and sigmoid colons), rectum, and anus. Note also the location of the accessory organs of digestion: the pancreas, liver, and gallbladder.

ISTUDY

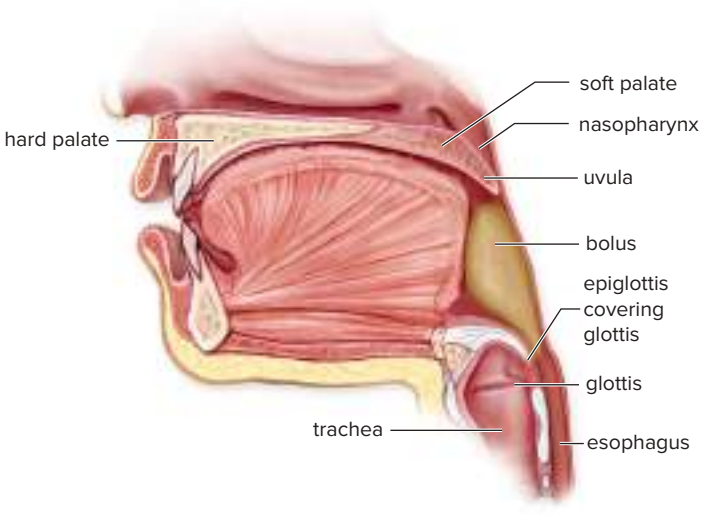


Figure 34.6 Swallowing. Respiratory and digestive passages converge and diverge in the pharynx. When food is swallowed, the soft palate closes off the nasopharynx, and the epiglottis covers the glottis, forcing the bolus to pass down the esophagus. Therefore, a person does not breathe when swallowing.

from the nasal cavity. It consists of the anterior hard palate and the posterior soft palate. The fleshy *uvula* is the posterior extension of the soft palate (Fig. 34.6). The cheeks and lips retain food while it is chewed by the teeth and mixed with saliva.

Three major pairs of **salivary glands** send their excretions by way of ducts into the mouth. The main enzyme in saliva is **salivary amylase,** which begins to digest the starch that is present in many foods of plant origin (see Section 34.3). Other components include salivary lipase, white blood cells and antimicrobial proteins and antibodies.

While in the mouth, food is manipulated by a muscular tongue, which has touch and pressure receptors similar to those in the skin. Taste buds, sensory receptors that are stimulated by the chemical composition of food, are also found primarily on the tongue as well as on the surface of the mouth (see Section 38.2). The tongue, which is composed of striated muscle and an outer layer of mucous membrane, mixes the chewed food with saliva. It then forms this mixture into a mass called a *bolus* in preparation for swallowing.

Pharynx and Esophagus

The digestive and respiratory passages come together in the **pharynx** and then separate. The **esophagus** is a tubular structure, about 25 cm in length, that takes food to the stomach. *Sphincters* are muscles that encircle tubes and act as valves; tubes close when sphincters contract, and they open when sphincters relax. The lower gastroesophageal sphincter is located where the esophagus enters the stomach. When food enters the stomach, the sphincter relaxes for a few seconds and then closes again. Heartburn occurs due to acid reflux, when some of the stomach's contents escape into the esophagus. When vomiting occurs, the abdominal muscles and the diaphragm, a muscle that separates the thoracic and abdominal cavities, contract.

When food is swallowed, the soft palate, the rear portion of the mouth's roof, moves back to close off the nasopharynx. A flap of tissue called the epiglottis covers the glottis, or opening into the trachea. Now the bolus must move through the pharynx into the esophagus, because the air passages are blocked (Fig. 34.6).

The central space of the digestive tract, through which food passes as it is digested, is called the **lumen** (Fig. 34.7). From the esophagus to the large intestine, the wall of the digestive tract is composed of four layers. The innermost layer next to the lumen is called the **mucosa.** The mucosa is a type of mucous membrane, and therefore it produces mucus, which protects the wall from the digestive enzymes inside the lumen.

The second layer in the digestive wall is called the **submucosa.** The submucosal layer is a broad band of loose connective tissue that contains blood vessels, lymphatic vessels, and nerves. Lymph nodules, called Peyer patches, are also in the submucosa. Like other secondary lymphoid tissues, they are sites of lymphocyte responses to antigens (see Section 33.2).

The third layer is termed the **muscularis,** and it contains two layers of smooth muscle. The inner, circular layer encircles the tract; the outer, longitudinal layer lies in the same direction as the tract. The contraction of these muscles, which are under involuntary nervous control, accounts for the movement of the gut contents from the esophagus to the rectum by **peristalsis** (Gk. *peri,* "around"; *stalsis,* "compression"), a rhythmic contraction that moves the contents along in various tubular organs (Fig. 34.8).

The fourth layer of the wall is the **serosa,** which secretes a watery fluid that lubricates the outer surfaces of the digestive tract

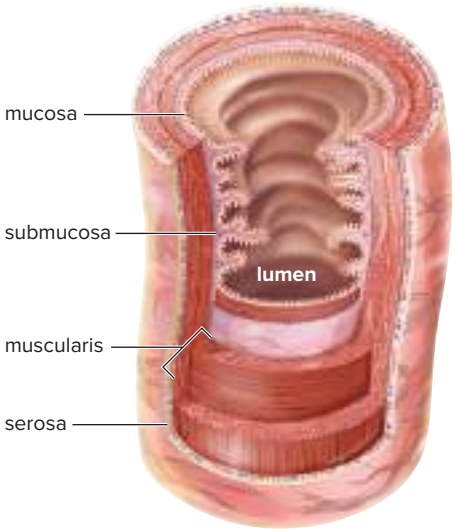


Figure 34.7 Wall of the digestive tract. The esophagus, stomach, small intestine, and large intestine all have a lumen and walls composed of similar layers.

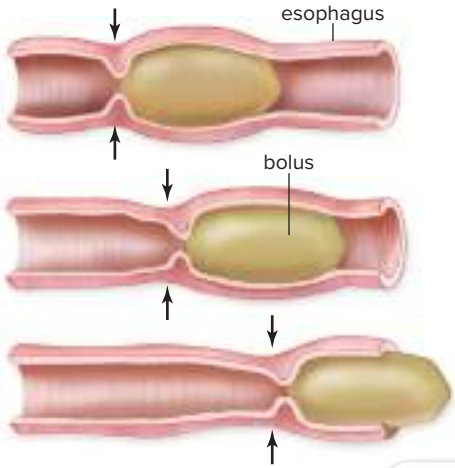


Figure 34.8 Peristalsis in the digestive tract. These three drawings show how a peristaltic wave moves through a single section of the esophagus over time. The arrows point to areas of contraction.

ISTUDY

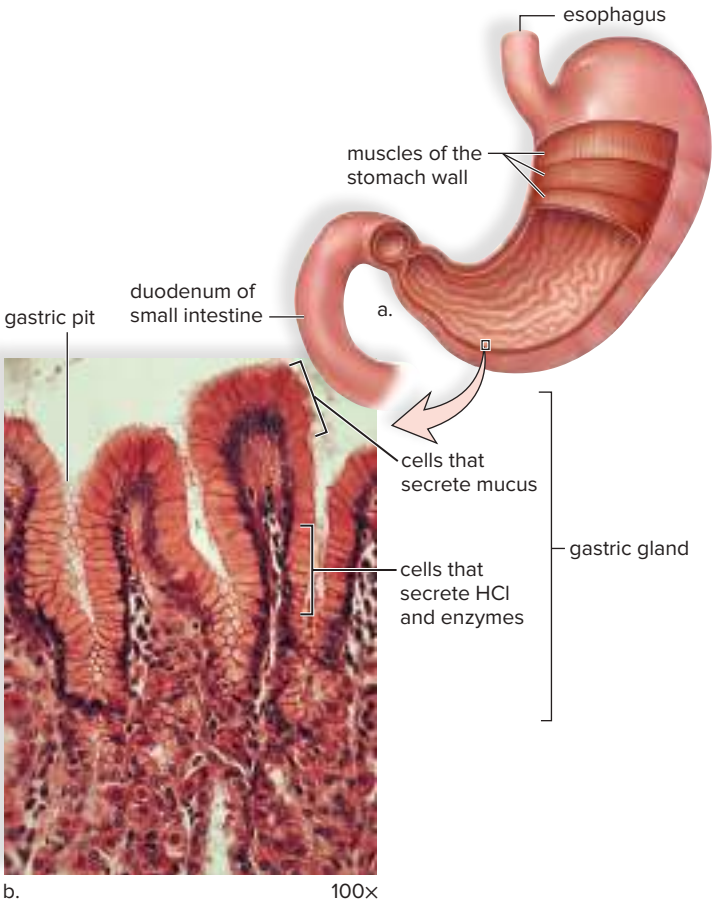


Figure 34.9 Anatomy of the stomach. **a.** The stomach, which has thick walls, expands as it fills with food. **b.** The mucous membrane layer of its walls secretes mucus and contains gastric glands, which secrete a gastric juice active in the digestion of protein. (b): (photo): Ed Reschke/ Stone/Getty Images

and reduces friction as various parts rub against each other and other organs. The serosa is actually a part of the *peritoneum,* the internal lining of the abdominal cavity.

Stomach

The **stomach** (Fig. 34.9) is a thick-walled, J-shaped organ that lies on the left side of the body beneath the diaphragm. The wall of the stomach has deep folds (rugae), which disappear as the stomach fills to its capacity (approximately 1 liter in humans). Therefore, many animals can periodically eat relatively large meals and spend the rest of their time at other activities.

The stomach is more than a food storage organ, as was discovered by William Beaumont in the mid-nineteenth century. Beaumont, an American doctor, had a patient who had been shot in the stomach, and when the wound healed, he was left with a fistula, or opening, that allowed Beaumont to look inside the stomach and collect the juices produced by gastric glands. Beaumont was able to determine that the muscular walls of the stomach contract vigorously and mix food with juices that are secreted whenever food enters the stomach. He found that gastric juice contains hydrochloric acid (HCl) and a substance, now called pepsin, that is active in digestion. Beaumont's work pioneered the study of digestive physiology. For similar reasons, modern animal scientists can surgically create fistulas into the rumen of cattle in order to study ruminant nutrition.

The epithelial lining of the stomach has millions of *gastric pits,* which lead into *gastric glands* (Fig. 34.9). The gastric glands produce gastric juice. So much hydrochloric acid is secreted by the gastric glands that the stomach routinely has a pH of about 2.0. Such a high acidity usually is sufficient to kill bacteria and other microorganisms that might be in food. This low pH also stops the activity of salivary amylase, which functions optimally at the near-neutral pH of saliva.

A thick layer of mucus protects the wall of the stomach from enzymatic action. Sometimes, however, gastric acid can leak upward, through the lower esophageal sphincter, where its acidic pH can irritate the mucosal lining of the esophagus. This gastroesophageal reflux disease (GERD) can cause heartburn and a number of other symptoms. In people with chronic GERD, the epithelial cells of the esophagus may change from stratified squamous (see Fig. 31.1) to a more columnar shape characteristic of the small intestine. This condition, known as "Barrett's esophagus," can lead to esophageal cancer.

Other individuals may develop gastric ulcers, which are areas where the protective epithelial layer of the stomach has been damaged. For many years, these stomach ulcers were attributed mainly to stress, but through the work of Australian scientists Barry Marshall and Robin Warren, we now know that they can be caused by an acid-resistant bacterium, *Helicobacter pylori.* Wherever the bacterium attaches to the epithelial lining, the lining stops producing mucus, and the area becomes damaged by acid and digestive enzymes. If the condition is promptly diagnosed, antibiotic treatment is usually curative. Marshall and Warren were awarded the Nobel Prize in Medicine in 2005.

Eventually, food mixing with gastric juice in the stomach contents becomes **chyme,** which has a thick, creamy consistency. At the base of the stomach is a narrow opening controlled by a sphincter. Whenever the sphincter relaxes, a small quantity of chyme passes through the opening into the small intestine. When chyme enters the small intestine, it sets off a neural reflex, which causes the muscles of the sphincter to contract vigorously and close the opening temporarily. Then, the sphincter relaxes again and allows more chyme to enter. The slow manner in which chyme enters the small intestine allows for thorough digestion.

Small Intestine

The **small intestine** is named for its small diameter (compared to that of the large intestine), but perhaps it should be called the long intestine. The small intestine averages about 6 m in length, compared to the large intestine, which is about 1.5 m in length.

The first 25 cm of the small intestine is called the **duodenum.** A duct brings bile from the liver and gallbladder, and pancreatic juice from the pancreas, into the small intestine (see Fig. 34.12*a*). **Bile** emulsifies fat—emulsification causes fat droplets to disperse in water. The intestine has a slightly basic pH, because pancreatic juice contains sodium bicarbonate ($NaHCO_3$), which neutralizes chyme. The enzymes in pancreatic juice and enzymes produced by the intestinal wall complete the process of food digestion.

It has been estimated that the surface area of the small intestine is approximately that of a tennis court. What factors contribute

ISTUDY

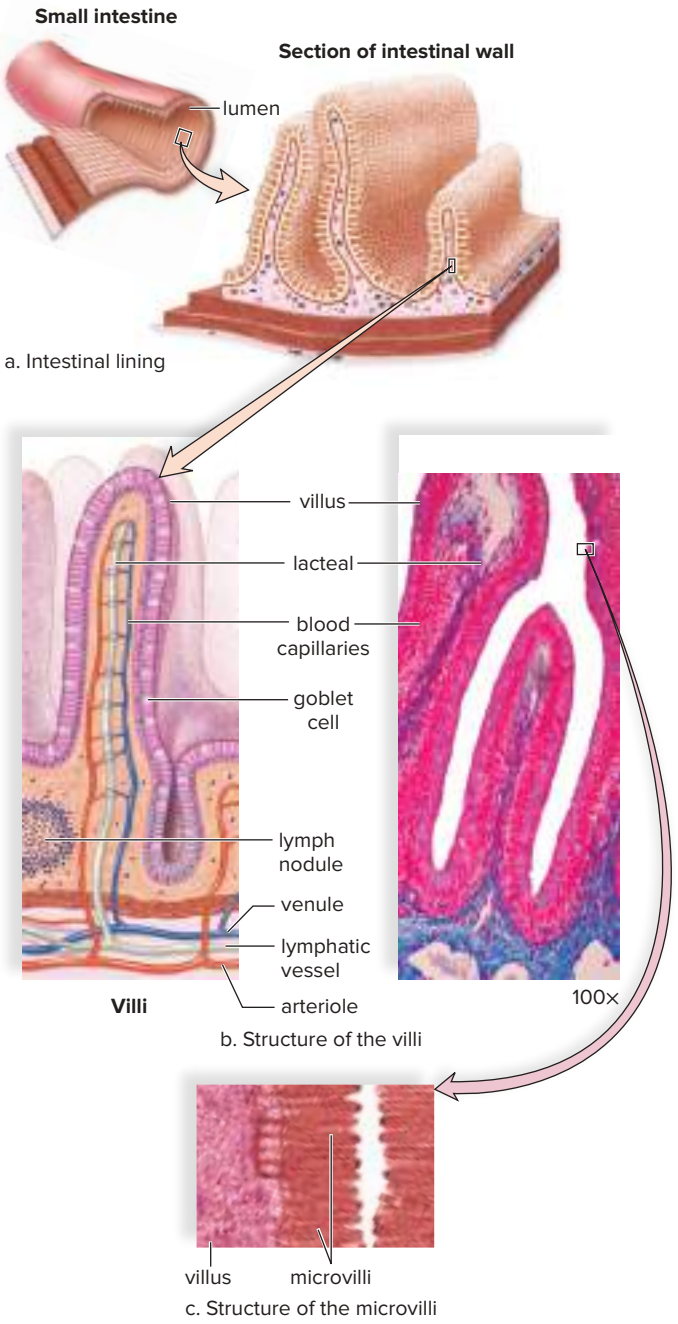


Figure 34.10 Anatomy of the small intestine. **a.** The wall of the small intestine has folds that bear fingerlike projections called villi. **b.** The products of digestion are absorbed by microvilli into the blood capillaries and the lacteals of the villi. **c.** A micrograph of the microvilli. (b): (photo): Choksawatdikorn/Science Photo Library/Getty Images; (c): Steve Gschmeissner/SPL/Science Source

to increasing its surface area? First, the wall of the small intestine contains fingerlike projections called villi (*sing.,* **villus**), which give the intestinal wall a soft, velvety appearance (Fig. 34.10). Second, a villus has an outer layer of columnar epithelial cells, and each of these cells has thousands of microscopic extensions called microvilli. Collectively, in electron micrographs, microvilli give the villi a fuzzy border, known as a "brush border." Because the microvilli bear the intestinal enzymes, these enzymes are called brush-border enzymes. The microvilli greatly increase the surface area of the villus for the absorption of nutrients.

Nutrients are absorbed into the vessels of a villus, which contains blood capillaries and a lymphatic capillary, called a **lacteal.** Sugars (digested from carbohydrates) and amino acids (digested from proteins) enter the blood capillaries of a villus. Glycerol and fatty acids (digested from fats) enter the epithelial cells of the villi, and within these cells they are joined and packaged as lipoprotein droplets, which enter a lacteal. After nutrients are absorbed, they are eventually carried to all the cells of the body by the bloodstream.

Celiac disease is the result of an autoimmune response against a protein called gluten. Gluten is naturally found in grains such as wheat, barley, and rye. In a person with celiac disease, the presence of gluten in the small intestine results in an inflammatory response, which damages the villi and microvilli of the small intestine. This can lead to a number of medical conditions, including abdominal pain, diarrhea, and malnutrition.

There is a difference between celiac disease and gluten sensitivity. Individuals with gluten sensitivity have some of the same symptoms, but there is no damage to the intestines. Individuals who have celiac disease are placed on a strict gluten-free diet. Many gluten-free foods can be found in grocery stores, but people who have not been diagnosed with celiac disease or gluten sensitivity will typically not benefit from a gluten-free diet.

Large Intestine

The **large intestine,** which includes the cecum, colon, rectum, and anus, is larger in diameter (6.5 cm) but shorter in length (1.5 m) than the small intestine. The large intestine absorbs water, salts, and some vitamins. It also stores undigestible material until it is eliminated as feces.

The cecum, which lies below the junction with the small intestine, is the blind end of the large intestine. The cecum has a small projection called the vermiform **appendix** (L. *verm,* "worm"; *form,* "shape"; *append,* "an addition") (Fig. 34.11). The function of the

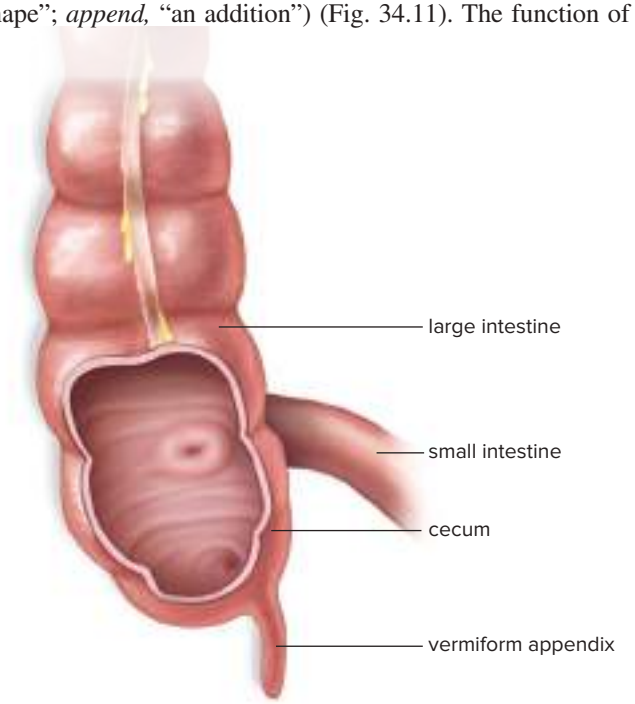


Figure 34.11 The appendix. The appendix is found at the junction of the small intestine and the large intestine.

ISTUDY

human appendix is unclear, although many experts suggest it may serve as a reservoir for the "good bacteria" that help maintain our intestinal health. In the case of appendicitis, the appendix becomes infected and so filled with fluid that it may burst. If an infected appendix bursts before it can be removed, it can lead to a serious, generalized infection of the abdominal lining called peritonitis.

The colon joins the rectum, the last 20 cm of the large intestine. About 1.5 liters of water enter the digestive tract daily as a result of eating and drinking. An additional 8.5 liters enter the digestive tract each day carrying the various substances secreted by the digestive glands. About 95% of this water is absorbed by the small intestine, and much of the remaining portion is absorbed by the colon. If this water is not reabsorbed, **diarrhea,** the passing of watery feces, can lead to serious dehydration and ion loss, especially in children.

The large intestine has a large population of bacteria, including *Escherichia coli* and perhaps 400 other species. By taking up space and nutrients, these bacteria provide protection against more pathogenic species. They also produce some vitamins—such as vitamin K, which is necessary to blood clotting. Digestive wastes, or feces, eventually leave the body through the **anus,** the opening of the anal canal.

Feces are normally about 75% water and 25% solid matter. Almost one-third of this solid matter is made up of intestinal bacteria. In fact, there are about 100 billion bacteria per gram of feces! The rest of the solids are undigested plant material, fats, waste products (such as bile pigments), inorganic material, mucus, and dead cells from the intestinal lining. The color of feces is the result of bilirubin breakdown and the presence of oxidized iron. The foul odor is the result of bacterial action.

The colon is subject to the development of polyps, which are small growths arising from the mucosa. Polyps, whether they are benign or cancerous, can be removed surgically. Research suggests that dietary fat increases the likelihood of colon cancer. Dietary fat causes an increase in bile secretion, and it could be that intestinal bacteria convert bile salts to substances that promote the development of colon cancer. Dietary fiber absorbs water and adds bulk, thereby diluting the concentration of bile salts and facilitating the movement of substances through the intestine. Regular elimination reduces the time that the colon wall is exposed to any cancer-promoting agents in feces.

The Accessory Organs of the Small Intestine

The pancreas, liver, and gallbladder are all accessory digestive organs of the small intestine. Figure 34.12*a* shows how the pancreatic duct from the pancreas and the common bile duct from the liver and gallbladder enter the duodenum.

Pancreas

The **pancreas** lies deep in the abdominal cavity, resting on the posterior abdominal wall. It is an elongated and somewhat flattened organ that has both an endocrine and an exocrine function. As an endocrine gland, it secretes insulin and glucagon, hormones that help keep the blood glucose level within normal limits (see Section 40.3). In this chapter, however, we are interested in its exocrine function. Most pancreatic cells produce pancreatic juice, which contains sodium bicarbonate ($NaHCO_3$) and digestive enzymes for all types of food. Sodium bicarbonate neutralizes acid chyme from the stomach. Pancreatic amylase digests starch, trypsin digests protein, and lipase digests fat.

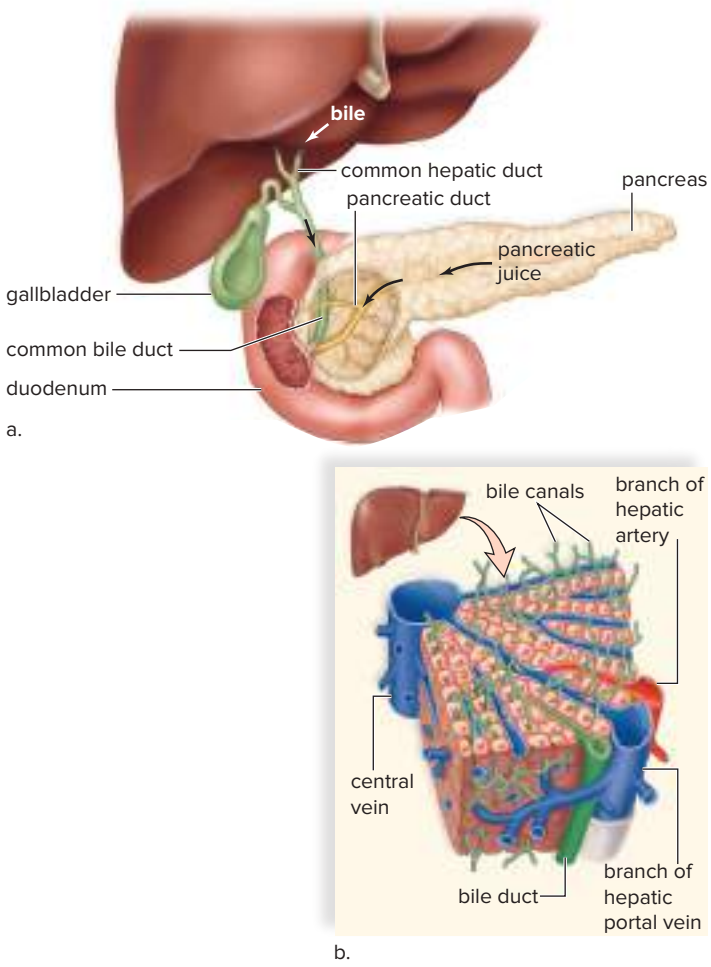


Figure 34.12 Liver, gallbladder, and pancreas. **a.** The liver makes bile, which is stored in the gallbladder and sent (black arrow) to the small intestine by way of the common bile duct. The pancreas produces digestive enzymes that are sent (black arrows) to the small intestine by way of the pancreatic duct. **b.** The liver contains over 100,000 lobules. Each lobule contains many cells that perform the various functions of the liver. They remove and add materials to the blood and deposit bile in a duct.

Liver

The **liver,** which is the largest gland in the body, lies mainly in the upper right section of the abdominal cavity, under the diaphragm (see Fig. 34.5). The liver contains approximately 100,000 lobules, which serve as its structural and functional units (Fig. 34.12*b*). Triads, located between the lobules, consist of a bile duct, which takes bile away from the liver; a branch of the hepatic artery, which brings oxygen-rich blood to the liver; and a branch of the hepatic portal vein, which transports nutrients to the liver from the intestines (Fig. 34.12*a*). The central veins of lobules enter a hepatic vein. Blood moves from the intestines to the liver via the hepatic portal vein and from the liver to the inferior vena cava via the hepatic veins.

In some ways, the liver acts as the gatekeeper to the blood. As blood in the hepatic portal vein passes through the liver, it removes many toxic substances and metabolizes them. The liver also removes and stores iron and the vitamins A, B_{12}, D, E, and K. The

ISTUDY

liver makes many of the proteins found in blood plasma and helps regulate the quantity of cholesterol in the blood.

The liver maintains the blood glucose level at about 100 mg/100 mL (0.1%), even though a person eats intermittently. When insulin is present, any excess glucose present in blood is removed and stored by the liver as glycogen. Between meals, glycogen is broken down to glucose, which enters the hepatic veins. In this way, the blood glucose level remains constant.

If the supply of glycogen is depleted, the liver converts glycerol (from fats) and amino acids to glucose molecules. The conversion of amino acids to glucose necessitates deamination, the removal of amino groups. By a complex metabolic pathway, the liver then combines ammonia with carbon dioxide to form urea. Urea is the usual nitrogenous waste product from amino acid breakdown in humans.

The liver produces bile, which is stored in the gallbladder. Bile has a yellowish green color, because it contains the bile pigment *bilirubin,* derived from the breakdown of hemoglobin, the red pigment of red blood cells. Bile also contains bile salts. Bile salts are derived from cholesterol, and they emulsify fat in the small intestine. When fat is emulsified, it breaks up into droplets, providing a much larger surface area, which can be acted upon by a digestive enzyme from the pancreas.

Liver Disorders Because the liver performs so many vital functions, serious disorders of the liver can be life-threatening. When a person has a liver ailment, a yellowing of the skin and the sclera of the eyes called **jaundice** may occur. Jaundice results when the liver is not helping the body excrete excess bilirubin, which is then deposited in the tissues.

Regardless of the cause, inflammation of the liver is called **hepatitis.** The most common causes of hepatitis are viruses. Hepatitis A virus is usually acquired from food or water that has been contaminated with feces. Hepatitis B, which is usually spread by sexual contact, can also be spread by blood transfusions or contaminated needles. The hepatitis B virus is more contagious than the AIDS virus, which is spread in the same way. A vaccine is now available for hepatitis B, however. Hepatitis C, which is usually acquired by contact with infected blood and for which no vaccine is available, can lead to chronic hepatitis, liver cancer, and death.

Cirrhosis is another chronic disease of the liver. First, the organ becomes fatty, and then liver tissue is replaced by inactive, fibrous scar tissue. Cirrhosis of the liver is often seen in people with alcohol use disorder, due to malnutrition and to the excessive amounts of alcohol (a toxin) the liver is forced to break down.

The liver has amazing regenerative powers and can recover if the rate of regeneration exceeds the rate of damage. During liver failure, however, there may not be enough time to let the liver heal itself. Liver transplantation is usually the preferred treatment for liver failure, but the wait list for liver transplants is extensive.

Because the liver serves so many functions, artificial livers have been difficult to develop. One type is a cartridge that contains cultured liver cells (either human or pig). Like kidney dialysis, the patient's blood is passed outside of the body and through an apparatus containing the liver cells, which perform their normal functions, and the blood is returned to the patient. Progress is also being made in the area of transplanting a smaller number of liver cells, as opposed to the entire organ. These cells can be either grown from stem cells or derived from the livers of donors who have died, but whose livers as entire organs are not suitable for transplantation.

Gallbladder

The **gallbladder** is a pear-shaped, muscular sac attached to the surface of the liver (see Fig. 34.5). About 1,000 ml of bile are produced by the liver each day, and any excess is stored in the gallbladder. Water is reabsorbed by the gallbladder, so that bile becomes a thick, mucuslike material. When bile is needed, the gallbladder contracts, releasing bile into the duodenum via the common bile duct (Fig. 34.12).

The cholesterol content of bile can come out of solution and form crystals called gallstones. These stones can be as small as a grain of sand or as large as a golf ball. The passage of the stones from the gallbladder may block the common bile duct, causing pain as well as possible damage to the liver or pancreas. In such cases, the gallbladder must be removed.

Check Your Progress 34.2

1. Trace the path of food from the mouth to the large intestine.
2. Describe the likely selective pressures that resulted in the evolution of taste buds.
3. Explain how the stomach, small intestine, and large intestine are each adapted to perform their particular functions.
4. Discuss how each accessory organ contributes to the digestion of food.

34.3 Digestive Enzymes

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the overall characteristics and functions of digestive enzymes.
2. Compare the specific types of nutrients that are digested in the mouth, stomach, and small intestine.

Digestive enzymes, like other enzymes, are proteins that speed up specific chemical reactions. Table 34.1 lists the major enzymes produced by the digestive tract. These enzymes help break down carbohydrates, proteins, nucleic acids, and fats, which are the major nutrient components of food.

Starch is a polysaccharide, and its digestion begins in the mouth. Saliva from the salivary glands has a neutral pH and contains salivary amylase, the first enzyme to act on starch.

$$\text{starch} + H_2O \xrightarrow{\text{salivary amylase}} \text{maltose}$$

In this equation, salivary amylase is written above the arrow to indicate it is neither a reactant nor a product in the reaction. It

ISTUDY

Table 34.1 Enzymes of the Human Digestive System

Enzyme	Target	Produced By	Site of Action	Function
Salivary amylase	Carbohydrates	Salivary glands	Mouth	Begins the breakdown of starch to maltose
Salivary lipase	Lipids	Salivary glands	Mouth	Begins the breakdown of triglycerides to fatty acids
Pepsin	Proteins	Stomach	Stomach	Breakdown of proteins to peptides and amino acids
Pancreatic lipase	Lipids	Pancreas	Small intestine	Breakdown of lipids to glycerol and fatty acids
Nuclease	Nucleic acids	Pancreas	Small intestine	Breakdown of nucleic acids to nucleotides
Pancreatic amylase	Carbohydrates	Pancreas	Small intestine	Breakdown of starch to maltose
Trypsin	Proteins	Pancreas	Small intestine	Breakdown of proteins to peptides and amino acids
Maltase	Carbohydrates	Small intestine	Small intestine	Breakdown of maltose to glucose
Nucleosidases	Nucleic acids	Small intestine	Small intestine	Breakdown of nucleotides to phosphates, bases, and sugars
Peptidases	Proteins	Small intestine	Small intestine	Breakdown of peptides to amino acids

merely speeds the reaction in which its substrate, starch, is digested to many molecules of maltose, a disaccharide. Maltose molecules cannot be absorbed by the intestine. Additional digestive action in the small intestine converts maltose to glucose, a monosaccharide, which then can be absorbed.

Protein digestion begins in the stomach. Gastric juice secreted by gastric glands has a very low pH—about 2.0—because it contains hydrochloric acid (HCl). Pepsinogen, a precursor that is converted to pepsin when exposed to HCl, is also present in gastric juice. Pepsin acts on protein to produce peptides.

$$\text{protein} + H_2O \xrightarrow{\text{pepsin}} \text{peptides}$$

Peptides vary in length, but they always consist of a number of linked amino acids. Peptides from the stomach are usually too large to be absorbed by the intestinal lining, but later in the small intestine, they are broken down to amino acids, the monomers of a protein molecule.

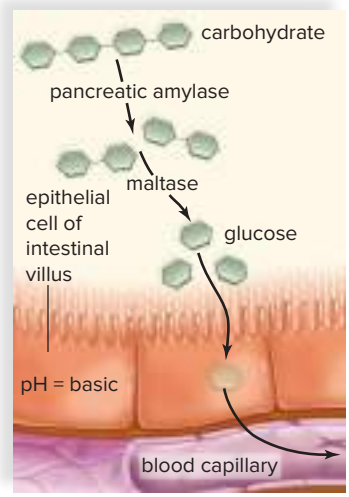
Pancreatic juice, which enters the duodenum, has a basic pH because it contains sodium bicarbonate ($NaHCO_3$). Sodium bicarbonate neutralizes the acid in chyme, producing the slightly basic pH that is optimum for pancreatic enzymes. One of these is pancreatic amylase, which digests starch (Fig. 34.13*a*).

$$\text{starch} + H_2O \xrightarrow{\text{pancreatic amylase}} \text{maltose}$$

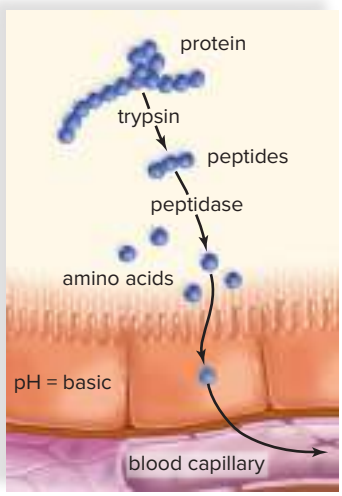
Another pancreatic enzyme, **trypsin,** digests protein (Fig. 34.13*b*). Trypsin is secreted as trypsinogen, which is converted to trypsin in the small intestine.

$$\text{protein} + H_2O \xrightarrow{\text{trypsin}} \text{peptides}$$

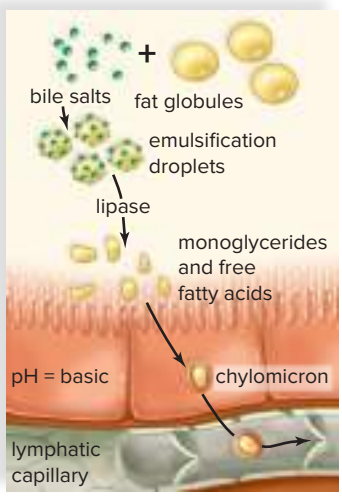
Maltase and peptidases, enzymes produced by the small intestine, complete the digestion of starch to glucose and protein to



a. Carbohydrate digestion



b. Protein digestion



c. Fat digestion

Figure 34.13 Digestion and absorption of nutrients. **a.** Starch is digested to glucose, which is actively transported into the epithelial cells of intestinal villi. From there, glucose moves into the bloodstream. **b.** Proteins are digested to amino acids, which are actively transported into the epithelial cells of intestinal villi. From there, amino acids move into the bloodstream. **c.** Fats are emulsified by bile and digested to monoglycerides and fatty acids. These diffuse into epithelial cells, where they recombine and join with proteins to form lipoproteins, called chylomicrons. Chylomicrons enter a lacteal.

ISTUDY

amino acids, respectively. Glucose and amino acids are small molecules that cross into the cells of the villi and enter the blood (Fig. 34.13*a*, *b*).

Another enzyme, pancreatic lipase, digests fat molecules in fat droplets after they have been emulsified by bile salts (Fig. 34.13*c*).

$$\text{fat} \xrightarrow{\text{bile salts}} \text{fat droplets}$$

$$\text{fat droplets} + H_2O \xrightarrow{\text{lipase}} \text{glycerol} + 3 \text{ fatty acids}$$

Fats are triglycerides, which means each one is composed of glycerol and three fatty acids. The end products of lipase digestion—glycerol and fatty acid molecules—are small enough to cross the cells of the intestinal villi, where absorption takes place. As mentioned in Section 34.1, glycerol and fatty acids enter the cells of the villi, and within these cells, they are rejoined and packaged as lipoprotein droplets before entering the lacteals (see Fig. 34.10*b*).

Peptidases and maltase, enzymes secreted by the surface cells of the small intestinal villi, complete the digestion of protein to amino acids and starch to glucose, respectively. These small molecules can then be absorbed by the cells of the villi. Peptides, which result from the first step in protein digestion, are digested to amino acids by peptidases:

$$\text{peptides} + H_2O \xrightarrow{\text{peptidases}} \text{amino acids}$$

Maltose, a disaccharide that results from the first step in starch digestion, is digested to glucose by maltase.

$$\text{maltose} + H_2O \xrightarrow{\text{maltase}} \text{glucose} + \text{glucose}$$

Other disaccharides, each of which has its own enzyme, are digested in the small intestine. The absence of any one of these enzymes can cause illness. The most common situation involves a deficiency of lactase, the enzyme that digests lactose, the sugar found in milk. Interestingly, the rate of this enzyme deficiency varies between human races. For example, only 25% of white adults, but 75–90% of adult African Americans, are lactase-deficient. Consuming dairy products often gives these individuals the symptoms of *lactose intolerance* (diarrhea, gas, cramps), caused by the fermentation of nondigested lactose by intestinal bacteria. However, lactose-reduced products are available for most kinds of dairy products, including milk, cheese, and ice cream.

As mentioned in Section 6.3, enzymes function best at an optimum temperature and pH, which helps them maintain the proper shape to fit their substrate. The compartmentalization of the digestive system assists in the regulation of enzymatic activity. Because the entire digestive system of humans is maintained at a constant temperature, usually 37°C (98.6°F), enzymatic activity is largely controlled by pH. As discussed in Section 34.1, the pH of the stomach is between 1 and 2, allowing the activation of pepsinogen enzyme. However, shortly after chyme proceeds into the duodenum (small intestine), the sodium bicarbonate released from the pancreas returns the pH to around 7.4 to 7.8. This "switch" serves to activate digestive enzymes released by the pancreas, thus allowing the digestion of carbohydrates, fats, and proteins.

Check Your Progress 34.3

1. Describe the location(s) in the digestive tract where each of the major types of nutrients is broken down.
2. Explain what final molecule (monomer) results from the digestion of carbohydrates, proteins, and fats.

34.4 Nutrition and Human Health

Learning Outcomes

Upon completion of this section, you should be able to

1. List the major types of nutrients and provide examples of foods that are a good source of each.
2. Describe the connection between a person's diet and the likely development of obesity, type 2 diabetes, and cardiovascular disease.
3. Distinguish among vitamins, coenzymes, and minerals.

This section of the chapter discusses the components of the human diet, as well as some problems that may arise from consuming a diet that does not provide nutrients in the correct balance. The Biological Systems feature, "Dietary Guidelines for Human Health," summarizes the current recommendations surrounding diet and human health.

Carbohydrates

Carbohydrates are present in food in the form of sugars, starch, and fiber. Fruits, vegetables, milk, and honey are natural sources of sugars. Glucose and fructose are monosaccharide sugars, and lactose (milk sugar) and sucrose (table sugar) are disaccharides. Disaccharides are broken down in the small intestine, and monosaccharides are absorbed into the bloodstream and delivered to cells. Once inside animal cells, monosaccharides are converted to glucose, much of which is used for the production of ATP by cellular respiration (see Section 8.1).

Plants store glucose as starch, and animals store glucose as glycogen. Good sources of starch are beans, peas, cereal grains, and potatoes. Starch is digested to glucose in the digestive tract, and excess glucose is stored as glycogen. The human liver and muscles can only store a total of about 600 g of glucose in the form of glycogen; excess glucose is converted into fat and stored in adipose tissues.

Although other animals likewise store glucose as glycogen in liver or muscle tissue (meat), little is left by the time an animal is eaten for food. Except for honey and milk, which contain sugars, animal foods do not contain high levels of carbohydrates.

Fiber includes various undigestible carbohydrates derived from plants. Food sources rich in fiber include beans, peas, nuts,

ISTUDY

THEME Biological Systems

Dietary Guidelines for Human Health

Dietary guidelines are typically revised by the U.S. government every 5 years to reflect changes in nutrition science. The latest guidelines were released in 2020 by the Departments of Agriculture and Health and Human Services. The overall purposes of these guidelines are to:

- Promote health
- Prevent chronic (long-term) disease
- Assist people in reaching and maintaining a healthy weight

The new guidelines focus less on prescribing quantitative levels of nutrients, and more on establishing healthy eating patterns. These patterns include the following foods:

- A variety of vegetables, including leafy vegetables, beans, red and yellow vegetables, and starches
- Fruits
- Grains (At least half of all grains should be whole grains.)
- Fat-free or low-fat dairy products (including soy)
- Proteins in the form of seafood, lean meats, poultry, eggs, legumes, nuts, and soy products
- Oils

To establish these healthy eating patterns, the following specific recommendations were made to limit certain nutrients recognized as raising health concerns:

- Consume less than 10% of calories per day from added sugar.
- Consume less than 10% of calories per day from saturated fats.
- Consume less than 2,300 milligrams (mg) per day of sodium.
- Consume alcohol only in moderation: a maximum of one drink per day for women and up to two drinks per day for men (and only by adults of legal drinking age).

For more information on these guidelines, including the science of how they were determined, visit https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials.

Questions to Consider

1. Why do you think quantitative values were removed for many of the nutrients?
2. Reflect on your diet over the past 24 hours. How does your diet relate to these general guidelines?

fruits, and vegetables. Whole-grain products are also a good source of fiber and are therefore more nutritious than food products made from refined grains. During refinement, fiber as well as vitamins and minerals are removed from grains, so that primarily starch remains. For example, a slice of bread made from whole-wheat flour contains 3 g of fiber; a slice of bread made from refined wheat flour contains less than 1 g of fiber.

Technically, fiber is not a nutrient for humans, because it cannot be digested to small molecules that enter the bloodstream. Insoluble fiber, however, adds bulk to fecal material, which stimulates movement in the large intestine, preventing constipation. Soluble fiber combines with bile acids and cholesterol in the small intestine and prevents them from being absorbed. In this way, high-fiber diets may protect against heart disease. The typical American consumes only about 15 g of fiber each day; the recommended daily intake is between 25 and 30 grams per day. To increase your fiber intake, eat whole-grain foods, snack on fresh fruits and raw vegetables, and include nuts and beans in your diet.

A meta-analysis of over 250 published papers indicated that diets that are high in fiber (at the 25–29 grams per day level) reduced the risk of dying from diseases such as cardiovascular disease, stroke, type 2 diabetes, and colon cancer by 16–24%.

A current controversy in human nutrition is the relative risk of consuming high levels of high-fructose corn syrup (HFCS), compared to other sweeteners. HFCS, or corn sugar, is now the most commonly used sweetening agent, found in soft drinks and a huge variety of foods that end up on our plates. Many websites and a few research studies have suggested that HFCS is a major factor in the rising epidemic of obesity and related diseases, but many nutritionists contend that the *type* of sugar consumed is not as important as the *amount.* As an example, the typical American obtains about one-sixth of their daily caloric intake from HFCS and other sugars. It is recognized that consuming such a high percentage of "empty calories" in the form of simple sugars is contributing to the increasing incidence of obesity in the United States.

Lipids

Like carbohydrates, *triglycerides* (fats and oils) supply energy for cells, but *fat* is also stored for the long term in the body. Dietary experts generally recommend that people include unsaturated, rather than saturated, fats in their diets (see Fig. 3.9 to review these nutrients). Two unsaturated fatty acids, alpha-linolenic and linoleic acids (also called omega-3 fatty acids), are *essential* in the diet, meaning that we cannot synthesize them. Delayed growth and skin problems can develop in people who consume an insufficient amount of these essential unsaturated fatty acids, which are found in high amounts in certain fish and in plant oils such as canola and soybean oils.

Another type of lipid, *cholesterol,* is a necessary component of the plasma membrane of all animal cells. It is also a precursor for the synthesis of various compounds, including bile, steroid hormones, and vitamin D. Plant foods do not contain cholesterol, but animal foods such as cheese, egg yolks, liver, and certain shellfish (shrimp and lobster) are rich in cholesterol. Elevated blood cholesterol levels are associated with an increased risk of cardiovascular disease, the number one cause of disease-related death in the United States (described later in this section).

Animal-derived foods, such as butter, red meat, whole milk, and cheeses, contain saturated fats, which are also associated with cardiovascular disease. Statistical studies suggest that trans fatty acids (trans fats) are even more harmful than saturated fatty acids. Trans fatty

ISTUDY

THEME Nature of Science

The Rise and Fall of Artificial Trans Fatty Acids

The chemical process for creating artificial trans fats was patented in the early 1900s. At the height of their use in the 1990s, nearly all processed foods, such as cookies, crackers, donuts, frozen breakfasts, and frozen pizzas, contained trans fats. In addition, they were used for frying fast food and cooking at home in the form of Crisco and other vegetable shortenings. It all ended in 2018 when trans fats were banned by the FDA. Both the rise of artificial trans fats and their subsequent fall from favor were the result of continuous scientific research.

Trans Fats in Food Products

Oils are relatively cheap to include in food products compared to solid fats like butter and lard. Unfortunately, they don't last long on the shelf before going rancid, nor do they provide the texture of solid fats. The scientific discovery of hydrogenation solved these problems by turning the unsaturated fatty acids in oils into trans fatty acids that have long shelf life and are semisolid.

The use of artificial trans fats in food drastically increased as scientific research began establishing a link between diets high in saturated fat and heart disease. Artificial trans fats were thought to be a more healthy alternative and were even promoted for a period of time.

Trans Fat Research

As their use increased, so did the research into trans fats. In the 1980s and 1990s, several studies discovered that even small amounts of artificial trans fats increased the risk of heart disease. Because the link was apparent, scientists and health officials started to push for bans and give more information to the public.

Some parts of the food industry tried to push back and funded a scientific study with the hope of finding trans fats were not harmful to health. However, it found similar results. Despite the industry objections, the FDA required food manufacturers to list trans fats on nutrition facts labels. This led to consumers wanting products without trans fats and the industry scientists adapting by creating alternatives to lower the amount of trans fats in new food products (Fig. 34A).

Trans Fat Ban

Bans on artificial trans fats began appearing in the early 2000s. For example, Denmark established a ban in 2003, New York City in 2006, and the state of California in 2011.

Having a full ban was important because food manufacturers could claim a food had 0 grams of trans fat as long as a serving had less than 0.5 gram of trans fat. Because even small amounts impact health, consumers could unknowingly be increasing their risk of heart disease.

Nearly 115 years after the first trans fats were artificially created in a lab, they were banned from foods in the United States. The FDA established the ban in 2015 and companies had until 2018 to fully comply. It's estimated the elimination of artificial trans fats will prevent 50,000 heart attacks per year.

Average trans fat content of new food products

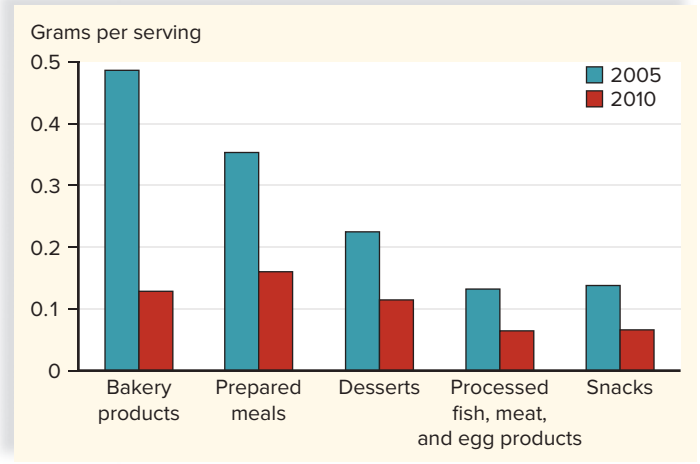


Figure 34A Reduction of artificial trans fats. The amount of trans fats used in food products significantly decreased between 2005 and 2010 after food manufacturers were required to list trans fats on food labels. USDA, Economic Research Service calculations using Mintel Global New Products Database

Questions to Consider

1. Concerns about trans fats began in the 1950s. Why did it take so many decades for studies to provide results of their effects on heart disease?
2. Why do you think some parts of the food industry were opposed to listing trans fats on nutrition labels?
3. How does the history of trans fats reflect the objective nature of science?

acids arise when unsaturated oils are hydrogenated to produce a solid fat, as in shortening and some margarines. Trans fats may reduce the function of the plasma membrane receptors that clear cholesterol from the bloodstream. Trans fats were commonly found in commercially packaged foods, such as cookies and crackers; in commercially fried foods, such as french fries; and in packaged snacks. However, as discussed in the Nature of Science feature, "The Rise and Fall of Artificial Trans Fatty Acids," their use was banned several years ago by the U.S. Food and Drug Administration (FDA).

Proteins

Dietary *proteins* are digested to amino acids, which cells use to synthesize thousands of different cellular proteins. Of the 20 different amino acids, 9 are *essential amino acids* that adult humans cannot synthesize and thus must be present in their diet. Animal products such as beef, pork, poultry, eggs, and dairy products contain all these essential amino acids and are considered "complete" or "high-quality" protein sources.

ISTUDY

Most foods derived from plants do not have as much protein per serving as those derived from animals, and some types of plant foods lack one or more of the essential amino acids. For example, the proteins in corn have a low content of the essential amino acid lysine (although high-lysine corn has been produced through genetic engineering technology). Approximately 6% of Americans (and millions of people in other countries) are either vegetarians, who avoid eating animal flesh, or vegans, who avoid consuming any products derived from animals.

Vegetarians, vegans, and in fact, all adults can meet their protein needs by consuming grains, beans, and nuts in various combinations. Also, tofu, soymilk, and other foods made from processed soybeans are complete protein sources. Medical organizations, such as the American Dietetic Association, recognize that vegetarian diets, when well-planned, are suitable for individuals at all stages of life.

Although a severe deficiency in dietary protein intake can be life-threatening, some Americans may consume too much protein. Health food stores are actually full of protein supplements, aimed mainly at athletes who are trying to build muscle mass. While athletes are recommended to consume only 1–1.5 grams of protein per kilogram of body weight per day, this is just slightly higher than the 0.8 gram per kilogram recommended for people who are inactive. This means a 150-pound person who is inactive would need to consume only about 54 grams of protein per day, which is about the amount contained in two cheeseburgers.

When amino acids are broken down, the liver removes the nitrogen portion (*deamination*) and uses it to form urea, which is excreted in urine. The water needed for the excretion of urea can cause dehydration when a person is exercising and losing water by sweating. High-protein diets can also increase calcium loss in the urine and encourage the formation of kidney stones. Furthermore, high-protein foods derived from animals often contain a high amount of fat, and some plant proteins may cause problems for those who have immune reactions to gluten (see Section 34.2).

Vitamins and Minerals

Vitamins are organic compounds other than carbohydrates, fats, and proteins that regulate various metabolic activities and must be present in the diet. Many vitamins are part of *coenzymes;* for example, niacin is the name for a portion of the coenzyme NAD^+, and riboflavin is a part of FAD. Coenzymes are needed in small amounts, because they are used over and over again in cells. Not all vitamins are coenzymes, however; vitamin A, for example, is a precursor for the pigment that prevents night blindness.

It has been known for some time that the absence of a vitamin can be associated with a particular disorder. Vitamins are especially abundant in fruits and vegetables, so it is suggested that we eat about 4½ cups of fruits and vegetables per day. Although many foods are now enriched or fortified with vitamins, some individuals are still at risk for vitamin deficiencies because of poverty, food availability, and diet choices.

The body also needs about 20 elements called *minerals* for various physiological functions, including the regulation of biochemical reactions, the maintenance of fluid balance, and as components of certain structures, such as bone. Some individuals (especially females) do not get enough iron, calcium, magnesium, or zinc in their diets. Adult females, if they are menstruating, need more iron in the diet each month than males (18 mg compared to 10 mg). Many people take calcium and vitamin D supplements, as directed by a physician, to enhance bone health and counteract osteoporosis, a degenerative bone disease. Most Americans consume too much sodium, even double the amount needed. Excess sodium can cause water retention and contribute to hypertension.

Obesity

As mentioned, the consumption of an excess amount of calories (relative to calories expended) from any source causes storage of these calories in the form of body fat. Obesity can be defined in several ways: (1) a condition in which excess body fat has an adverse effect on normal activity and health; (2) weight over 20% more than the ideal for your height and body build; and (3) a body mass index (BMI) over 30. A person's BMI can be calculated by dividing weight in kilograms by height in meters squared, or by using an online BMI calculator. Most estimates indicate that about 30% of Americans meet the criteria for obesity. Obesity raises the risk of many medical conditions, including type 2 diabetes and cardiovascular disease. The seriousness of obesity as a health-care problem is evidenced by the increasing popularity of surgical procedures designed to reduce food consumption.

Type 2 Diabetes

Diabetes mellitus occurs when the hormone insulin is not functioning properly, resulting in abnormally high levels of glucose in the blood. This may occur due to a deficiency of insulin secretion by the pancreas, as in type 1 diabetes, or to an inability of cells to respond to insulin (also called insulin resistance), defined as type 2 diabetes. In both types, the excess blood glucose spills over into the urine, leading to increased urination, thirst, and weight loss. Over time, the high levels of blood glucose, and lack of other insulin functions, can lead to damage to blood vessels, nerves, eyes, and kidneys, and even to death.

Type 1 diabetes is an autoimmune disorder where your body attacks insulin-producing cells in the pancreas. It typically occurs at a young age and can usually be successfully managed with insulin injections. There is no cure for type 1 diabetes, nor is the exact cause known.

Type 2 diabetes is linked to obesity, lack of exercise, high blood pressure, high blood cholesterol, and a family history of diabetes. People are usually diagnosed at older ages, but it is becoming more common in youth. Healthy eating patterns outlined by the Dietary Guidelines can reduce the risk for developing type 2 diabetes.

Cardiovascular Disease

Cardiovascular disease is the leading cause of death in the United States. Heart attacks and strokes often occur when arteries become blocked by plaque, which contains saturated fats and cholesterol. Cholesterol is carried in the blood by two types of lipoproteins: low-density lipoprotein (LDL) and high-density lipoprotein (HDL). LDL molecules are considered "bad," because they are like delivery trucks that carry cholesterol from the liver to the cells and to the arterial walls. HDL molecules are considered "good," because they are like garbage trucks that dispose of cholesterol.

ISTUDY

HDL transports cholesterol from the cells to the liver, which converts it to bile salts that enter the small intestine.

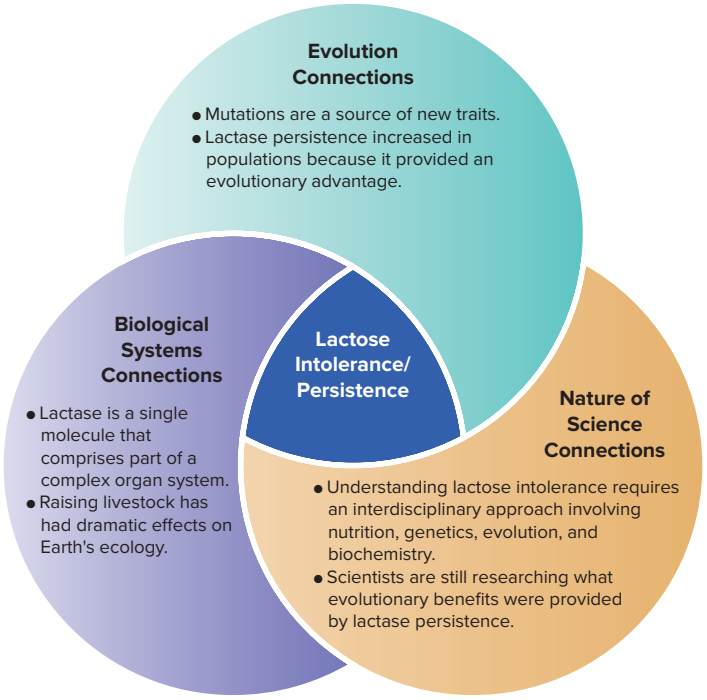
According to the American Heart Association, diets high in saturated fats, trans fats, and/or cholesterol tend to raise LDL cholesterol levels, while eating unsaturated fats may actually lower LDL cholesterol levels. Furthermore, coldwater fish (e.g., herring, sardines, tuna, and salmon) contain polyunsaturated fatty acids and especially omega-3 fatty acids, which are believed to reduce the risk of cardiovascular disease. However, taking fish oil supplements to obtain omega-3s is not recommended without a physician's approval, because too much of these fatty acids can interfere with normal blood clotting.

A physician can determine whether blood lipid levels are normal. If a person's cholesterol and triglyceride levels are elevated, modifying the fat content of their diet, losing excess body fat, and exercising regularly can reduce them. If lifestyle changes do not lower blood lipid levels enough to reduce the risk of cardiovascular disease, a physician may prescribe cholesterol-lowering medications.

Check Your Progress 34.4

1. State several reasons why a diet that includes plenty of vegetables is generally better for you than a diet that includes excess protein.
2. Describe the relationship among blood cholesterol, saturated fat intake, and cardiovascular disease.
3. Define *vitamin*.

Connecting *the* Concepts



What Did You Learn?

34.1 Digestive Tracts

Digestive systems include all organs, tissues, and cells involved in ingesting food and breaking it down into smaller components.

- A few animals (e.g., sponges) lack a digestive tract; others, such as planarians, have an incomplete digestive tract that has only one opening. An **incomplete digestive tract** has little specialization. Many other animals, such as earthworms, have a complete digestive tract that has both a mouth and an anus. A **complete digestive tract** tends to have specialized regions.
- Some animals are continuous feeders (e.g., clams, which are filter feeders); others are discontinuous feeders (e.g., squids). Discontinuous feeders need a storage organ for food.
- Most mammals have teeth. **Herbivores** need teeth that can clip off plant material and grind it up. Also, many herbivores have a **rumen** inhabited by bacteria that can digest cellulose. **Carnivores** need teeth that can tear and rip animal flesh into pieces. Meat is easier to digest than plant material, so the digestive system of carnivores has fewer specialized regions and the intestine is shorter than that of herbivores. **Omnivores** tend to have a combination of teeth types.

34.2 The Human Digestive System

In the human digestive system, both mechanical and chemical digestion begin in the **mouth,** where food is chewed and mixed with saliva produced by **salivary glands.** Saliva contains **salivary amylase,** which begins carbohydrate digestion.

Food then passes to the **pharynx** and down the **esophagus** by **peristalsis** to the **stomach.** The stomach stores and mixes food with mucus and gastric juices to produce chyme. Pepsin begins protein digestion in the stomach. Beginning with the esophagus, the wall of the digestive tract has a central space or **lumen,** an inner **mucosa,** then a **submucosa, muscularis,** and outer **serosa.**

ISTUDY

Partially digested food **(chyme)** passes into the **small intestine.** The **duodenum** of the small intestine receives bile from the liver and pancreatic juice from the pancreas. **Bile** emulsifies fat and readies it for digestion by pancreatic lipase. The pancreas also produces amylase and proteases. These and other intestinal enzymes finish the process of chemical digestion.

- The walls of the small intestine have fingerlike projections called villi (*sing.,* **villus**), where small nutrient molecules are absorbed. Amino acids and glucose enter the blood vessels of a villus. Glycerol and fatty acids are packaged as lipoproteins before entering lymphatic vessels, called **lacteals,** in the villi.

The **large intestine** consists of the cecum, colon, and rectum, which ends at the **anus.** The cecum, a blind pouch at the junction of the small and large intestines, has a small projection called the **appendix,** which sometimes becomes infected and inflamed, necessitating its removal. The large intestine does not produce digestive enzymes; it does absorb water, salts, and some vitamins. Reduced water absorption results in **diarrhea.** The intake of water and fiber helps prevent constipation. **Polyps** in the colon are small growths that can be benign or cancerous.

Three accessory organs of digestion—the **pancreas, liver,** and **gallbladder**—send secretions to the duodenum via ducts. The pancreas produces pancreatic juice, which contains digestive enzymes for carbohydrates, protein, and fat.

The liver produces bile, which is stored in the **gallbladder.** The liver receives blood from the small intestine by way of the hepatic portal vein. **Hepatitis** and **cirrhosis** are common disorders of the liver, and either can result in **jaundice** due to accumulation of bilirubin.

34.3 Digestive Enzymes

Digestive enzymes are present in digestive juices and break down food into the nutrient molecules glucose, amino acids, fatty acids, and glycerol.

- Salivary amylase and pancreatic amylase begin the digestion of starch, and maltase completes the digestion of starch to glucose.
- Pepsin and trypsin digest protein to peptides, then intestinal peptidases digest these to amino acids.
- Following emulsification by bile, lipase digests fat to glycerol and fatty acids.
- Nucleases break down nucleic acids into nucleotides.

Each digestive enzyme is present in a particular part of the digestive tract. Salivary amylase functions in the mouth; pepsin functions in the stomach; trypsin, lipase, and pancreatic amylase occur in the intestine along with the various enzymes that digest disaccharides and peptides.

34.4 Nutrition and Human Health

The nutrients released by the digestive process should provide us with an adequate amount of major nutrients, essential amino acids and fatty acids, and all necessary vitamins and minerals.

Carbohydrates are necessary in the diet, but simple sugars and refined starches are not as healthy, because they provide calories but little or no fiber, vitamins, or minerals. Proteins supply us with essential amino acids, but many Americans consume more protein than is healthy. It is also wise to restrict one's intake of meats that are fatty, because animal fats are saturated fats. Unsaturated fatty acids, particularly the omega-3 fatty acids, are protective against cardiovascular disease, whereas saturated fatty acids may lead to plaque formation, which blocks blood vessels.

Obesity is becoming an increasingly serious problem, especially because it is associated with the development of type 2 diabetes and cardiovascular disease.

Thinking Critically About Biology

1. Imagine you could track a single molecule of carbon that is part of a glucose molecule that, in turn, is part of a starch molecule in an apple. Describe what happens to that molecule of carbon, in a chemical sense, after you bite into the apple. What enzymes are involved, and where, as the starch molecule is broken down into simpler carbohydrates, then into glucose, which is absorbed into your bloodstream and eventually into cells, which use it to produce ATP?
2. Snakes often swallow whole animals, a process that takes a long time. Then, snakes spend some time digesting their food. What structural modifications to the digestive tract would allow the slow swallowing and storage of a whole animal to occur? What chemical modifications would be necessary to digest a whole animal?
3. A drug for leukemia is not damaged in the stomach and is well-absorbed by the small intestine. However, the molecular form of the drug collected from the blood is not the same as the form that was swallowed by the patient. What explanation is most likely?
4. Suppose you are taking large doses of creatine, an amino acid supplement advertised for its ability to enhance muscle growth. Because your muscles can grow only at a limited rate, what do you suppose happens to the excess creatine that is not used for the synthesis of new muscle?

Making It Relevant to You

1. What types of foods would you have to avoid if you had lactose intolerance? Can the nutrients, such as vitamins and minerals, in those foods be gained through other sources?
2. Being lactase persistent provided an evolutionary advantage when the trait arose. If you are lactase persistent, do you think it provides any evolutionary advantage today? Conversely, if you are lactose intolerant, do you think you are at an evolutionary disadvantage?
3. Congenital lactose intolerance is a disorder that occurs when an infant cannot digest lactose. It is extremely rare compared to the form of lactose intolerance experienced by 65% of the world's adult population. Why would the infant form of the trait not be as common as the adult form?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

35

Respiratory Systems

Wavebreakmedia/iStock/Getty Images

Chapter Outline

35.1 Gas-Exchange Surfaces
35.2 Breathing and Transport of Gases
35.3 Respiration and Human Health

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 7.6 During which specific parts of photosynthesis do plants produce oxygen and use carbon dioxide?

Section 8.4 How are carbon dioxide and oxygen related to the process of cellular respiration?

Figure 32.6 What path does blood travel from the heart to the site of gas exchange in fish, amphibians, and birds?

Asthma

Asthma is a disease in which the airways become constricted and inflamed, resulting in difficulty breathing. The symptoms often include wheezing and shortness of breath, a frequent cough (often at night), and a feeling of being very tired or weak. Over 235 million children and adults worldwide have asthma, and the incidence seems to be increasing. Experts offer various explanations for this. One hypothesis is that we may be "too clean," in the sense that we are not exposed to enough common bacteria, viruses, and parasites as children. As a result, our immune systems may react to harmless material we inhale. There are many harmful forms of air pollution, but of increasing concern are tiny, "ultrafine" particles, those less than 0.1 micrometer (μm) across, which are produced at high levels by diesel engines. These particles can bypass the normal defenses of the upper respiratory tract and end up lodging deep in the lungs, with damaging effects.

Recently, breakthroughs have occurred in asthma research. There appear to be a number of genes that increase susceptibility to asthma. For example, one set of genes on chromosome 5 is associated with the body's inflammatory response. Variations in a gene on chromosome 17 increase susceptibility in individuals who have been exposed to respiratory viruses.

As you read through this chapter, think about the following questions:

1. What structures in the lungs would be most susceptible to asthma?
2. How would asthma influence the ability of the body to maintain homeostasis?

Following *the* Themes

CHAPTER 35 – RESPIRATORY SYSTEMS

Evolution	Strategies have evolved in virtually all multicellular animals that allow them to efficiently extract oxygen from their environment and to eliminate carbon dioxide.
Nature of Science	Research studies into respiration are providing the basic biological framework needed for treating respiratory disease and building functioning lung tissues in the laboratory.
Biological Systems	Animal respiratory systems help maintain homeostasis by being responsive to changing demands for oxygen within the body of the organism.

ISTUDY

35.1 Gas-Exchange Surfaces

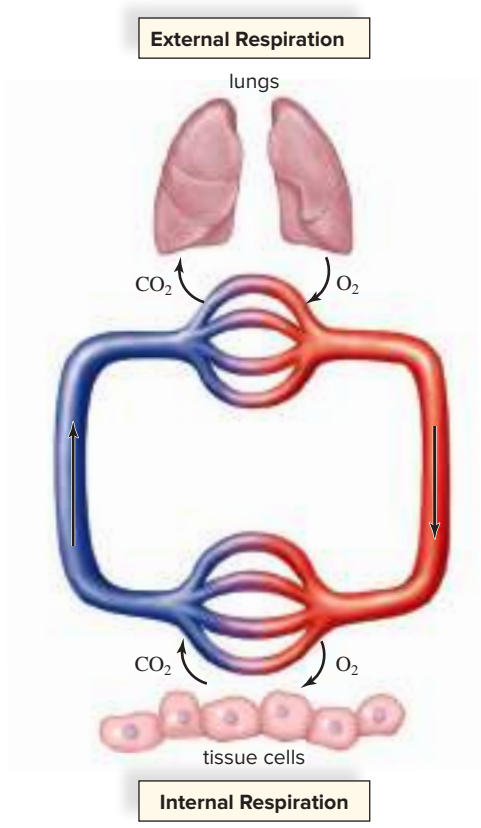
Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish among ventilation, external respiration, and internal respiration.
2. Compare and contrast the gas-exchange mechanisms of invertebrates, aquatic vertebrates, and terrestrial vertebrates.
3. Trace the path of a molecule of oxygen as it passes from the human nose to the alveoli of the lungs.

Respiration is the sequence of events that results in gas exchange between the body's cells and the environment. In terrestrial vertebrates, respiration includes these steps:

- **Ventilation** (breathing) includes inspiration (the entrance of air into the lungs) and expiration (the exit of air from the lungs).
- **External respiration** is gas exchange between the air and the blood within the lungs. Blood then transports oxygen from the lungs to the tissues.
- **Internal respiration** is gas exchange between the blood and the interstitial fluid. (The body's cells exchange gases with the interstitial fluid.) The blood then transports carbon dioxide to the lungs.



Gas exchange takes place by the physical process of diffusion (see Section 5.2). For external respiration to be effective, the gas-exchange region must be moist, thin, and large in relation to the size of the body.

Some animals are small and shaped in a way that allows the surface of the animal to be the gas-exchange surface. Most complex animals have evolved specialized tissues for external respiration, such as gills in aquatic animals and lungs in terrestrial animals. The effectiveness of diffusion is enhanced by vascularization (the presence of many capillaries), and delivery of oxygen to the cells is promoted when the blood contains a respiratory pigment, such as hemoglobin.

Regardless of the particular external respiration surface and the manner in which gases are delivered to the cells, in the end, oxygen enters mitochondria, where cellular respiration takes place (see Section 8.1). A rare exception to this was discovered in 2010, when a team of Italian and Danish deep-sea divers discovered a new species of tiny, jellyfish-like animals called loriciferans living in sediment more than 10,000 feet below the surface of the Mediterranean Sea, a depth that contains almost no oxygen. This discovery represents the first known multicellular animals that do not appear to require oxygen! Subsequent studies have indicated that the cells of these animals may lack mitochondria, but instead contain structures that resemble those used by anaerobic bacteria to undergo cellular respiration in the absence of oxygen. For most animals, however, if internal respiration does not occur, ATP production declines dramatically, and life ceases.

Overview of Gas-Exchange Surfaces

It is more difficult for animals to obtain oxygen from water than from air. Water fully saturated with air contains only a fraction of the amount of oxygen that is present in the same volume of air. Also, water is more dense than air. Therefore, aquatic animals expend more energy carrying out gas exchange than do terrestrial animals. Fish use as much as 25% of their energy output to respire, while terrestrial mammals use only 1–2% of their energy output for that purpose.

Hydras, which are cnidarians, and planarians, which are flatworms, have a large surface area in comparison to their size. This makes it possible for most of their cells to exchange gases directly with the environment. In hydras, the outer layer of cells is in contact with the external environment, and the inner layer can exchange gases with the water in the gastrovascular cavity (Fig. 35.1).

The earthworm is an example of a terrestrial invertebrate that is able to use its body surface for respiration because the capillaries come close to the surface (Fig. 35.2). An earthworm keeps its body surface moist by secreting mucus and by releasing fluids from excretory pores. Further, the worm is behaviorally adapted to remain in damp soil during the day, when the air is driest.

Aquatic invertebrates (e.g., clams and crayfish) and aquatic vertebrates (e.g., fish and tadpoles) have gills that extract oxygen from a watery environment. **Gills** are finely divided, vascularized outgrowths of the body surface or the pharynx (Fig. 35.3*a*). Various mechanisms are used to pump water across the gills, depending on the organism.

Insects have a system of air tubes called **tracheae** through which oxygen is delivered directly to the cells without entering the blood (Fig. 35.3*b*). Air sacs located near the wings, legs, and abdomen act as bellows to help move the air into the tubes through external openings.

ISTUDY

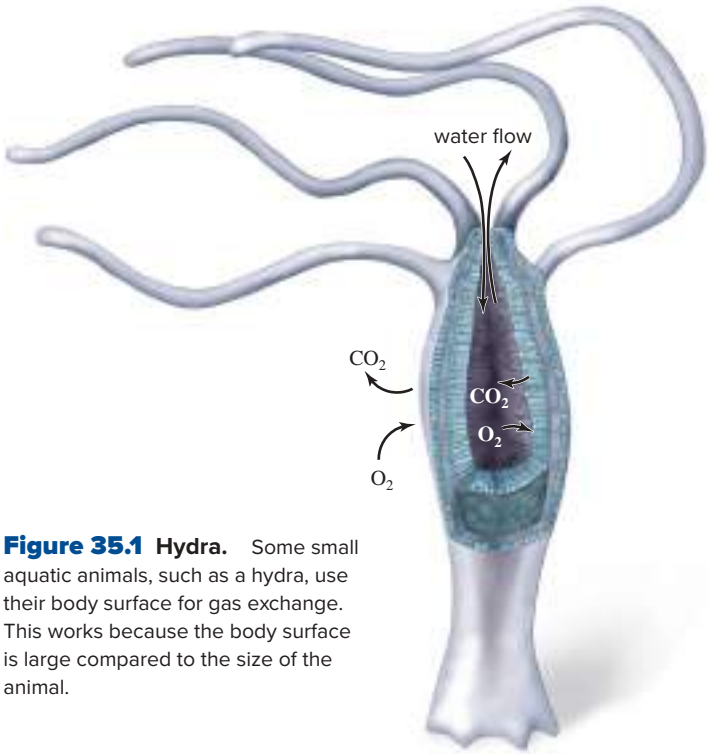


Figure 35.1 Hydra. Some small aquatic animals, such as a hydra, use their body surface for gas exchange. This works because the body surface is large compared to the size of the animal.

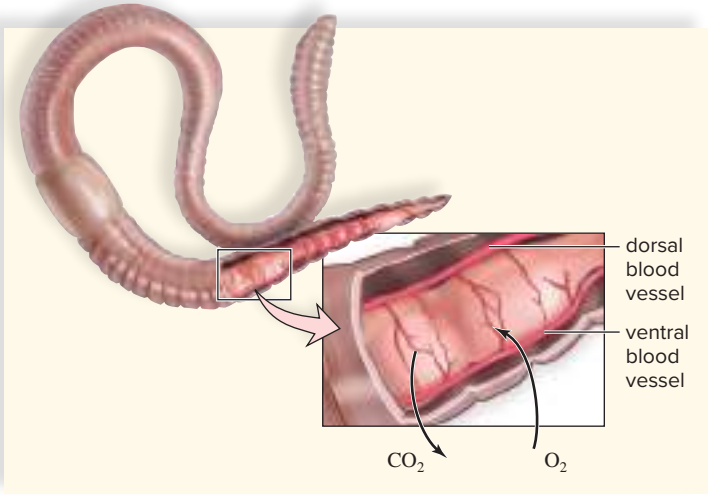
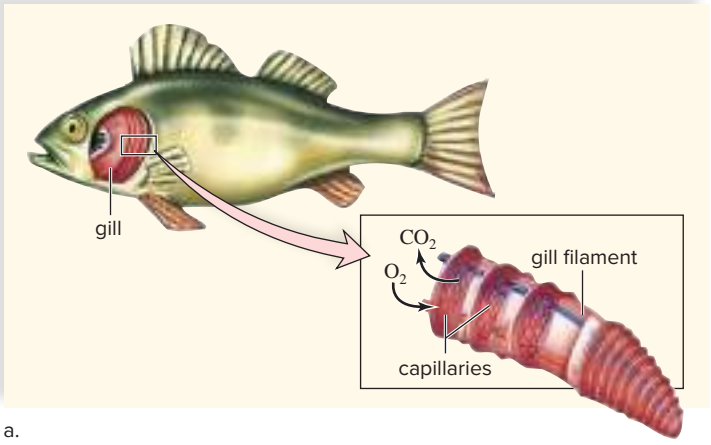
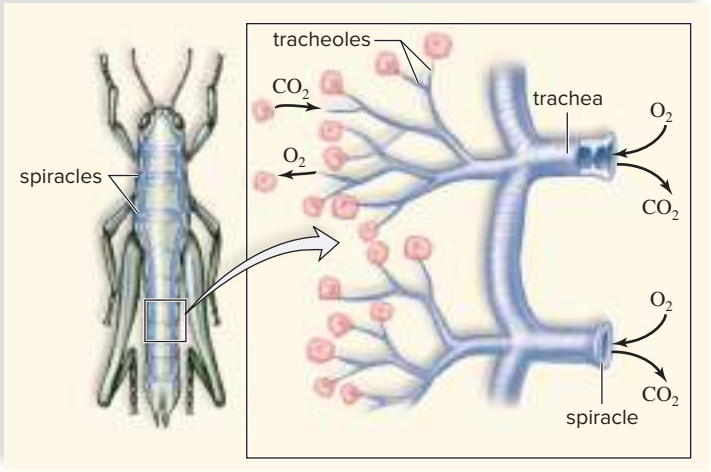


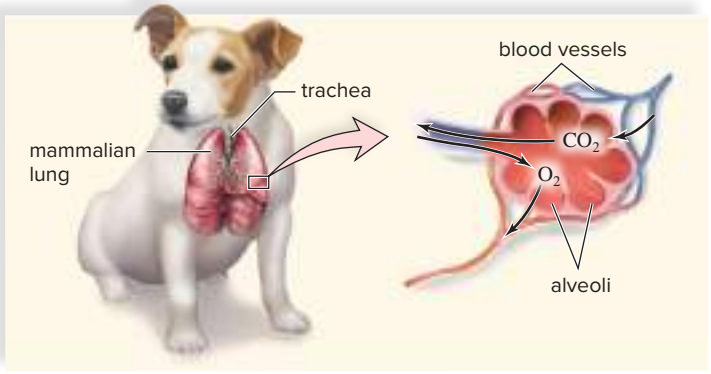
Figure 35.2 Earthworm. An earthworm's entire external surface functions in external respiration.



a.



b.



c.

Figure 35.3 Respiratory organs. **a.** Fish have gills to assist with external respiration. **b.** Insects have a tracheal system that delivers oxygen directly to their cells. **c.** Terrestrial vertebrates have lungs with a large total external respiration surface.

Terrestrial vertebrates usually have **lungs,** which are vascularized outgrowths from the lower pharyngeal region. The tadpoles of frogs live in the water and have gills as external respiratory organs, but adult amphibians possess simple, saclike lungs. Most amphibians respire to some extent through the skin, and some salamanders depend entirely on the skin, which is kept moist by mucus produced by numerous glands on the surface of the body.

The lungs of birds and mammals are elaborately subdivided into small passageways and spaces (Fig. 35.3*c*). It has been estimated that human lungs have a total surface area of about 70 square meters, which is about 50 times the skin's surface area. Air is a rich source of oxygen compared to water; however, it does have a drying effect on external respiratory surfaces. A human loses about 350 ml of water per day through respiration when the air has a relative humidity of only 50%. To keep the lungs from drying out, air is moistened as it moves in through the passageways leading to the lungs.

The Gills of a Fish

Animals with gills use various means of ventilation. Among mollusks, such as clams or squids, water is drawn into the mantle

ISTUDY

cavity, where it passes through the gills. In crustaceans such as crabs and shrimp, the gills are located in thoracic chambers covered by the exoskeleton. The action of specialized appendages located near the mouth keeps the water moving. In fish, ventilation is brought about by the combined action of the mouth and gill covers, or opercula (*sing.,* operculum; L. *operculum,* "small lid"). When the mouth is open, the opercula are closed and water is drawn in. Then the mouth closes, and the opercula open, drawing the water from the pharynx through the gill slits located between the gill arches.

As mentioned, the gills of bony fishes are outward extensions of the pharynx (Fig. 35.4). On the outside of the gill arches, the gills are composed of filaments that are folded into platelike lamellae. Fish use **countercurrent exchange** to transfer oxygen from the surrounding water into their blood. *Con*current flow would mean that oxygen-rich water passing over the gills would flow in the same direction as oxygen-poor blood in the blood vessels. This arrangement results in an equilibrium point, at which only half the oxygen in the water is captured. *Counter*current flow, in contrast, means that the two fluids flow in opposite directions. With countercurrent flow, as blood gains oxygen, it always encounters water having an even higher oxygen content. A countercurrent mechanism prevents an equilibrium point from being reached, and about 80–90% of the initial dissolved oxygen in water is extracted.

The Tracheal System of Insects

Arthropods are coelomate animals, but the coelom is reduced and the internal organs lie within a cavity called the hemocoel, because it contains hemolymph, a mixture of blood and lymph (see Section 32.1). Hemolymph flows freely through the hemocoel, making circulation in arthropods inefficient. Many insects are adapted for flight, and their flight muscles require a steady supply of oxygen.

Insects overcome the inefficiency of their blood flow by having a respiratory system that consists of tracheae that take oxygen directly to the cells (Fig. 35.5). The tracheae branch into even smaller tubules called tracheoles, which also branch and rebranch until finally the air tubes are only about 0.1 μm in diameter. There are so many fine tracheoles that almost every cell is near one. Also,

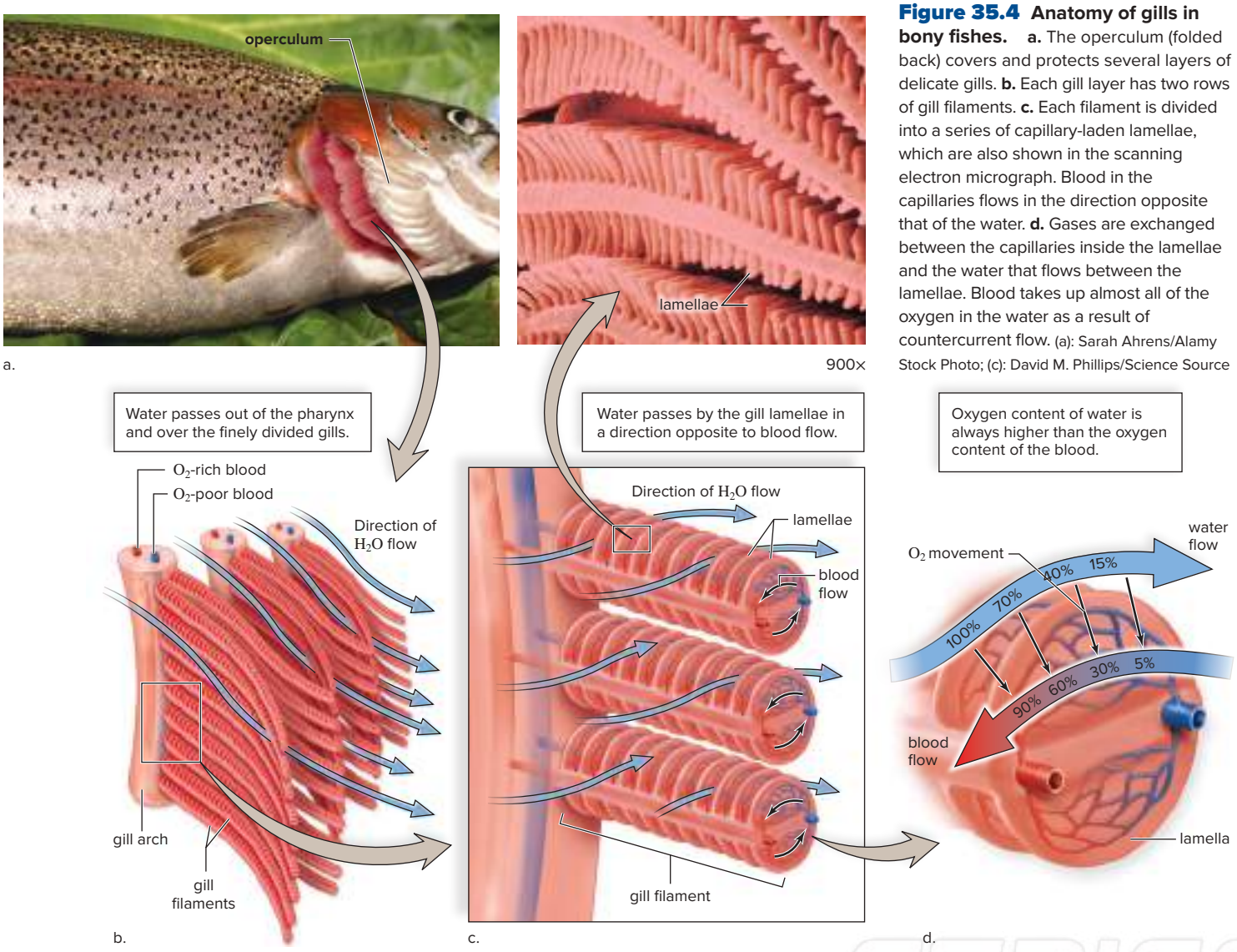


Figure 35.4 Anatomy of gills in bony fishes. **a.** The operculum (folded back) covers and protects several layers of delicate gills. **b.** Each gill layer has two rows of gill filaments. **c.** Each filament is divided into a series of capillary-laden lamellae, which are also shown in the scanning electron micrograph. Blood in the capillaries flows in the direction opposite that of the water. **d.** Gases are exchanged between the capillaries inside the lamellae and the water that flows between the lamellae. Blood takes up almost all of the oxygen in the water as a result of countercurrent flow. (a): Sarah Ahrens/Alamy Stock Photo; (c): David M. Phillips/Science Source

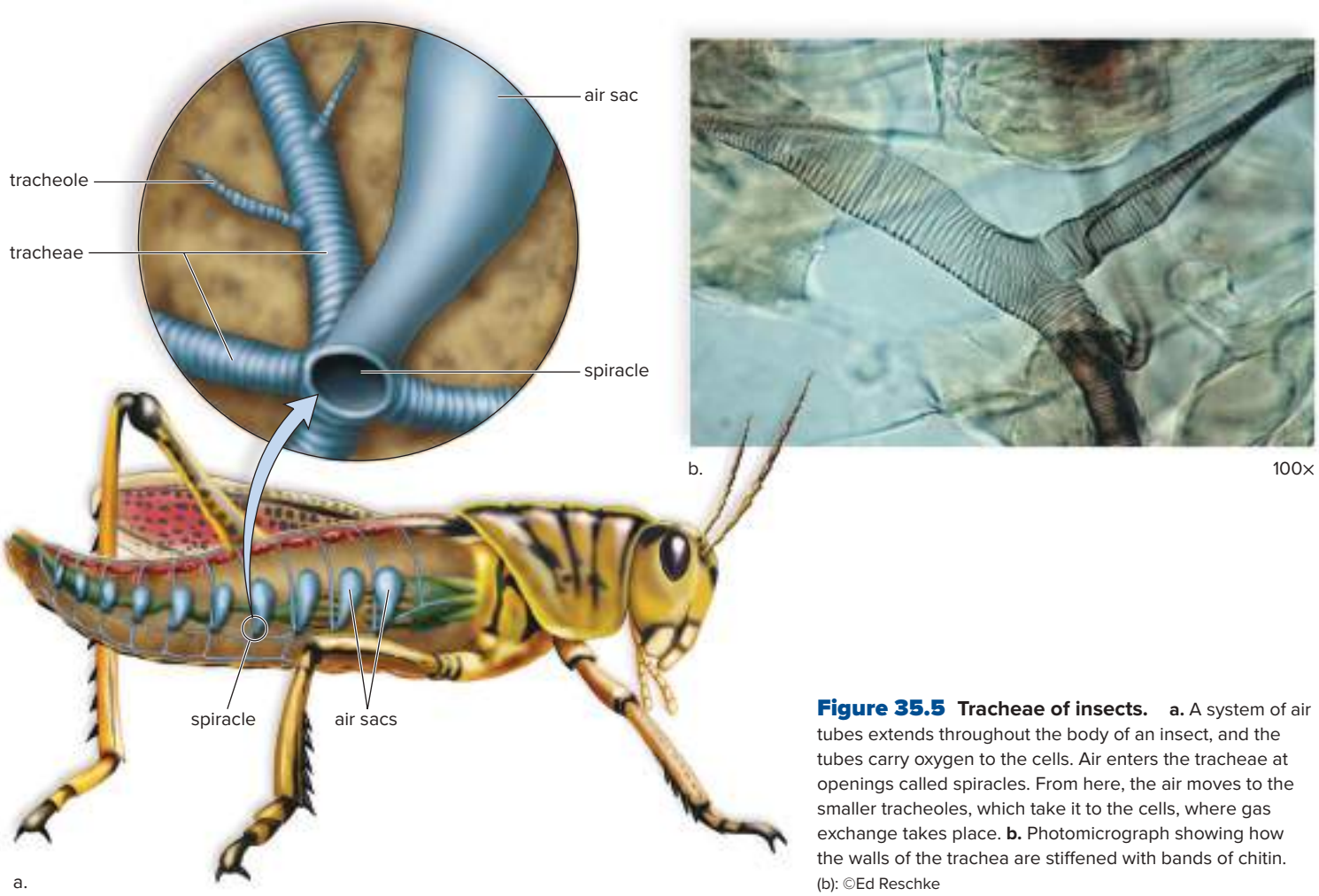


Figure 35.5 Tracheae of insects. **a.** A system of air tubes extends throughout the body of an insect, and the tubes carry oxygen to the cells. Air enters the tracheae at openings called spiracles. From here, the air moves to the smaller tracheoles, which take it to the cells, where gas exchange takes place. **b.** Photomicrograph showing how the walls of the trachea are stiffened with bands of chitin. (b): ©Ed Reschke

the tracheoles indent the plasma membrane, so that they terminate close to mitochondria. Therefore, O_2 can flow more directly from a tracheole to mitochondria, where cellular respiration occurs. The tracheae also dispose of CO_2.

The tracheoles are fluid-filled, but the larger tracheae contain air and open to the outside by way of spiracles (Fig. 35.5). Usually, the spiracle has some sort of closing device that reduces water loss, and this may be why insects have no trouble inhabiting drier climates. Scientists have determined that the tracheae can actually expand and contract, thereby drawing air into and out of the system. To improve the efficiency of the tracheal system, many larger insects also have air sacs, which are thin-walled and flexible, located near major muscles. Contraction of these muscles causes the air sacs to empty, and relaxation causes the air sacs to expand and draw in air. This method is comparable to the way that human lungs expand to draw air into them.

Even with all these adaptations, insects still lack the efficient circulatory system of birds and mammals, which is able to pump oxygen-rich blood through arteries to all the cells of the body. This may be why insects have remained relatively small (despite the attempts of science-fiction movies to make us think otherwise).

A tracheal system is an adaptation to breathing air, yet some insect larval stages, and even some adult insects, live in the water. In these instances, the tracheae do not receive air by way of spiracles. Instead, diffusion of oxygen across the body wall supplies the tracheae with oxygen. Mayfly and stonefly nymphs have thin extensions of the body wall called tracheal gills—the tracheae are particularly numerous in this area. This is an interesting adaptation, because it dramatizes that tracheae function to deliver oxygen in the same manner as vertebrate blood vessels.

Some aquatic insects have developed a different strategy. Like most insects, water beetles breathe through spiracles. Because they live in water, however, they capture a bubble of air from the surface and carry it with them, exchanging the oxygen inside for CO_2. Water spiders even spin an underwater web, which they fill with air bubbles, forming what some scientists have called an "external lung."

The Lungs of Humans

The human respiratory system includes all of the structures that conduct air in a continuous pathway to and from the lungs (Fig. 35.6*a*). The lungs lie deep in the body, within the thoracic cavity, where they are protected from drying out. As air moves through the nose, pharynx, trachea, and bronchi to the lungs, it is filtered, so that it is free of debris, warmed, and humidified. By

ISTUDY

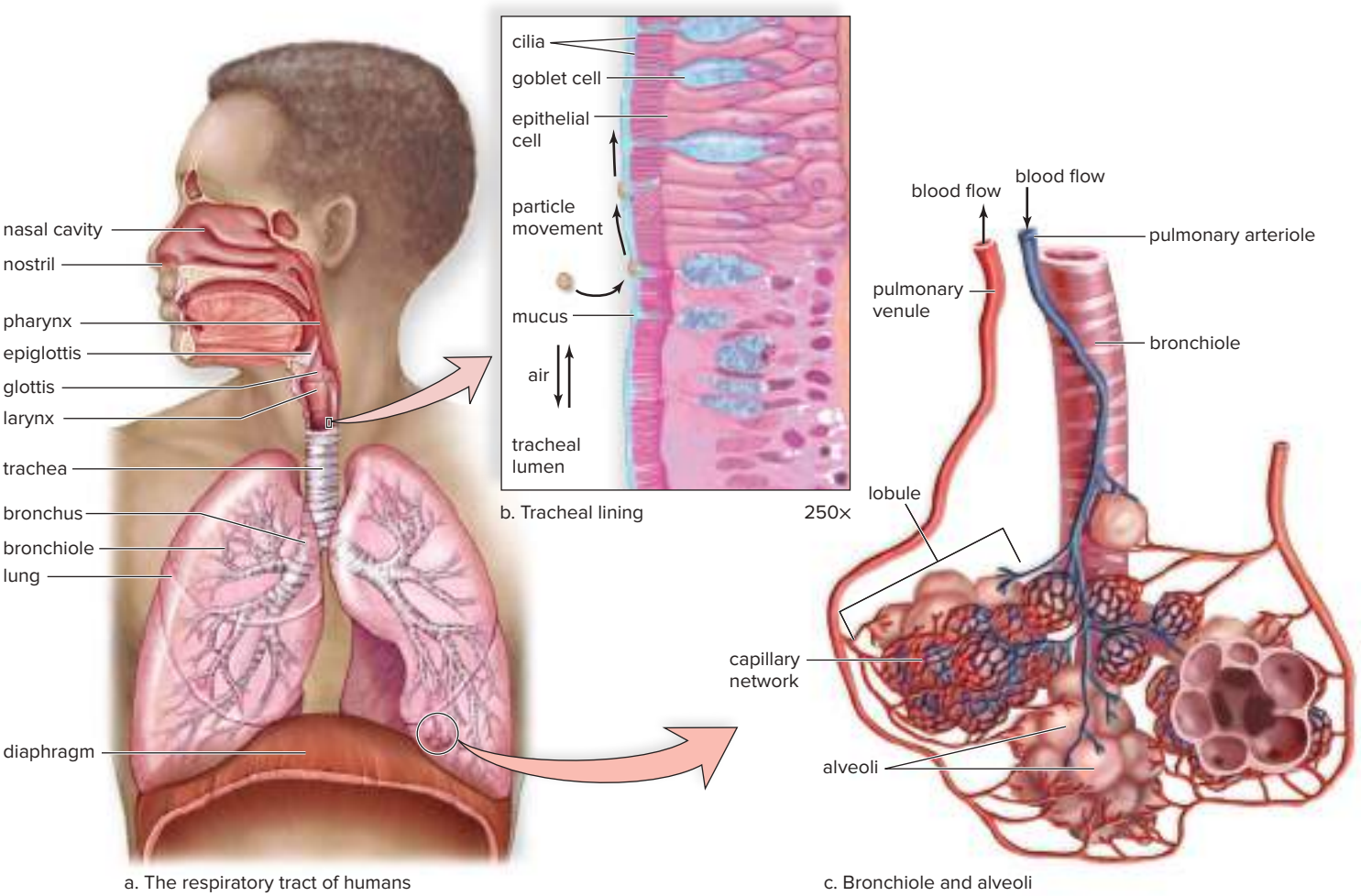


Figure 35.6 The human respiratory tract. **a.** The respiratory tract extends from the nose to the lungs, which are composed of air sacs called alveoli. **b.** The lining of the trachea is a ciliated epithelium with mucus-producing goblet cells. The lining prevents inhaled particles from reaching the lungs: The mucus traps the particles, and the cilia help move the mucus toward the throat, where it can be swallowed or expectorated. **c.** Gas exchange occurs between air in the alveoli and blood within a capillary network that surrounds the alveoli. (photo): ©Ed Reschke

the time the air reaches the lungs, it is at body temperature and saturated with water.

In the nose, hairs and cilia act as a screening device. In the trachea and the bronchi, cilia beat upward, carrying mucus, dust, and occasionally small bits of food that "went down the wrong way" back into the throat, where the accumulation may be swallowed or expectorated (Fig. 35.6*b*).

The hard and soft palates separate the nasal cavities from the oral cavity, but the air and food passages cross in the **pharynx.** This arrangement may seem inefficient, and there is a danger of choking if food accidentally enters the trachea; however, it has the advantage of letting you breathe through your mouth in case your nose is plugged up. In addition, it permits greater intake of air during heavy exercise, when greater gas exchange is required.

Air passes from the pharynx through the **glottis,** which is an opening into the **larynx,** or voice box. At the edges of the glottis are two folds of connective tissue covered by mucous membrane called the **vocal cords.** These flexible and pliable bands vibrate against each other, producing sound when air is expelled past them through the glottis from the larynx.

The trachea divides into two primary **bronchi,** which enter the right and left lungs. Branching continues, eventually forming a great number of smaller passages called **bronchioles.** The two bronchi resemble the trachea in structure, but as the bronchial tubes divide and subdivide, their walls become thinner, and rings of cartilage are absent. Each bronchiole terminates in an elongated space enclosed by a multitude of air pockets, or sacs, called **alveoli,** which fill the internal region of the lungs (Fig. 35.6*c*). Internal gas exchange occurs between the air in the alveoli and the blood in the capillaries.

Check Your Progress 35.1

1. List some features common to all animals that are able to exchange gases directly with their environment.
2. Explain why the countercurrent flow that occurs in the gills of fish is much more efficient than concurrent flow would be.
3. For humans, list the respiratory structures that air passes through during inspiration.

ISTUDY

35.2 Breathing and Transport of Gases

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the mechanisms used by amphibians, mammals, and birds to inflate their lungs.
2. Explain how the breathing rate in humans is influenced by both physical and chemical factors.
3. Describe how carbon dioxide (CO_2) is carried in the blood and the effect that blood carbon dioxide concentration has on blood pH.

During breathing, the lungs are ventilated. Oxygen (O_2) moves into the blood, and carbon dioxide (CO_2) moves out of the blood into the lungs. Blood transports O_2 to the body's cells and CO_2 from the cells to the lungs.

Breathing

Terrestrial vertebrates ventilate their lungs by moving air into and out of the respiratory tract. Amphibians use positive pressure to force air into the respiratory tract. With the mouth and nostrils firmly shut, the floor of the mouth rises and pushes the air into the lungs. Reptiles, birds, and mammals use negative pressure to move air into the lungs and positive pressure to move it out. **Inspiration** (inhalation) is the act of moving air into the lungs, and **expiration** (exhalation) is the act of moving air out of the lungs.

Ventilation is governed by Boyle's law, which states that, at a constant temperature, the pressure of a given quantity of gas is inversely proportional to its volume (Fig. 35.7). When the sides of the container move outward, air pressure decreases inside the container and air moves in, just as air automatically enters the lungs because the rib cage moves up and out during inspiration. Conversely, if the sides of the container are pressed inward, air flows out because of increased air pressure inside the container. Similarly, air automatically exits the lungs when the rib cage moves down and in during expiration. The analogy is not exact, however, because no force is required for the rib cage to move down, and inspiration is the only active phase of breathing. Forced expiration can occur if we so desire, however.

Reptiles have jointed ribs that can be raised to expand the lungs, but mammals have both a rib cage and a diaphragm. The **diaphragm** is a horizontal muscle that divides the thoracic cavity (above) from the abdominal cavity (below). During inspiration in mammals, the rib cage moves up and out, and the diaphragm contracts and moves down (Fig. 35.8*a*). As the thoracic (chest) cavity expands and lung volume increases, air flows into the lungs due to decreased air pressure in the thoracic cavity and lungs. Inspiration is the active phase of breathing in reptiles and mammals.

During expiration in mammals, the rib cage moves down, and the diaphragm relaxes and moves up to its former position (Fig. 35.8*b*). No muscle contraction is required; thus, expiration is the passive phase of breathing in reptiles and mammals. During expiration, air flows out as a result of increased pressure in the thoracic cavity and lungs.

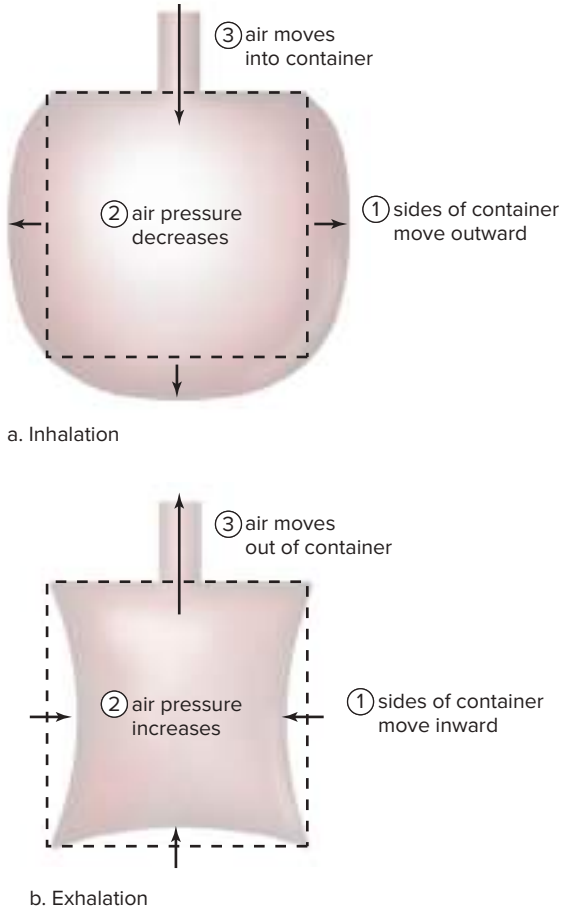


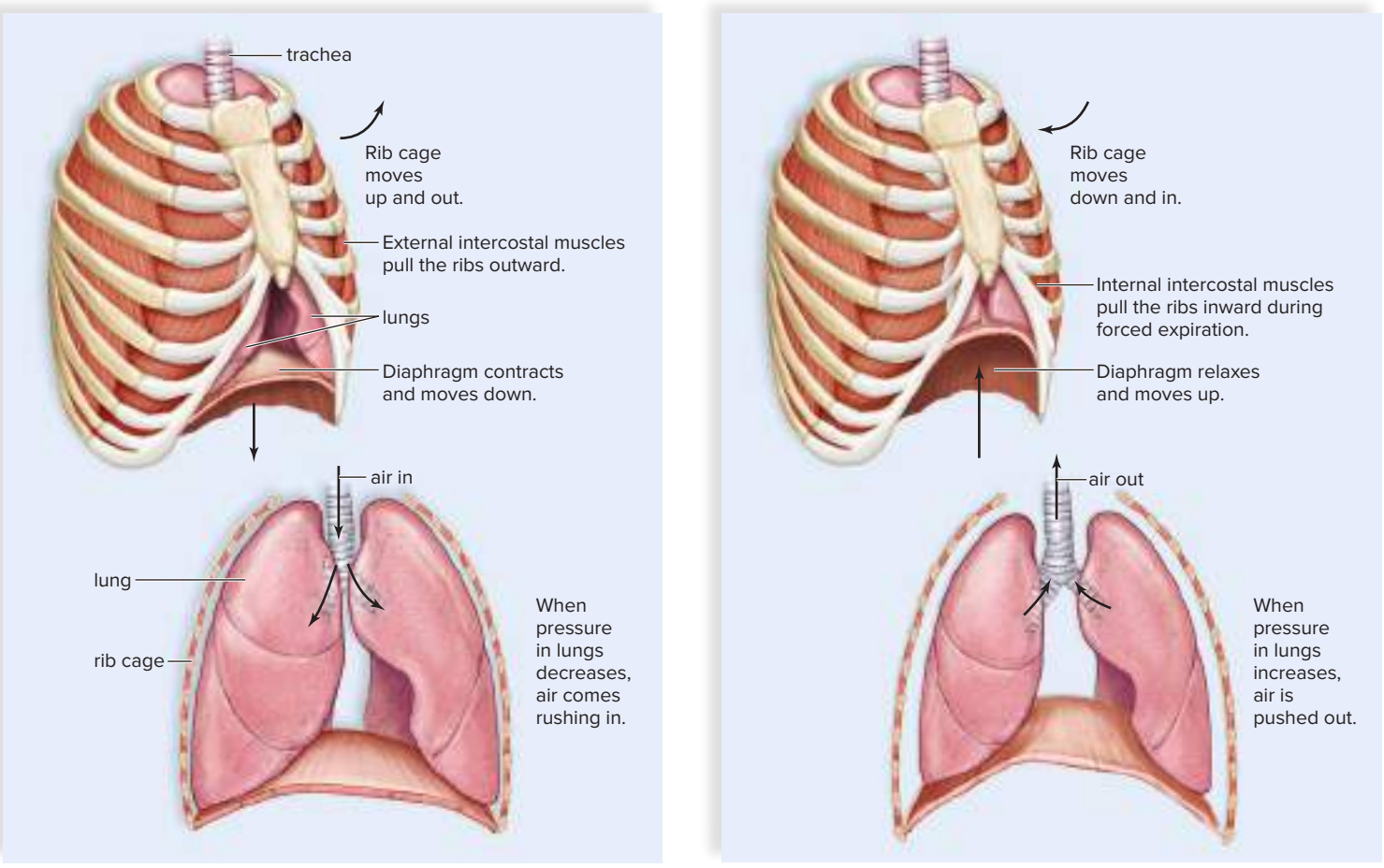
Figure 35.7 The relationship between air pressure and volume. **a.** Similar to what happens during inhalation, when the sides of a flexible container move outward, the volume of the container increases and the air pressure decreases. **b.** During exhalation, increased air pressure in the lungs causes air to flow out, similar to the effects of moving the sides of the container inward.

All terrestrial vertebrates, except birds, use a *tidal ventilation mechanism,* so named because the air moves in and out by the same route. This means that the lungs of amphibians, reptiles, and mammals are not completely emptied and refilled during each breathing cycle. Because of this, the air entering mixes with used air remaining in the lungs. Although this does help conserve water, it also decreases gas-exchange efficiency. In contrast, birds use a *one-way ventilation mechanism* (Fig. 35.9). Incoming air is carried past the lungs by a trachea, which takes it to a set of posterior air sacs. The air then passes forward through the lungs into a set of anterior air sacs. From here, it is finally expelled. Notice that fresh air never mixes with used air in the lungs of birds, thereby greatly improving gas-exchange efficiency.

Modifications for Breathing in Humans

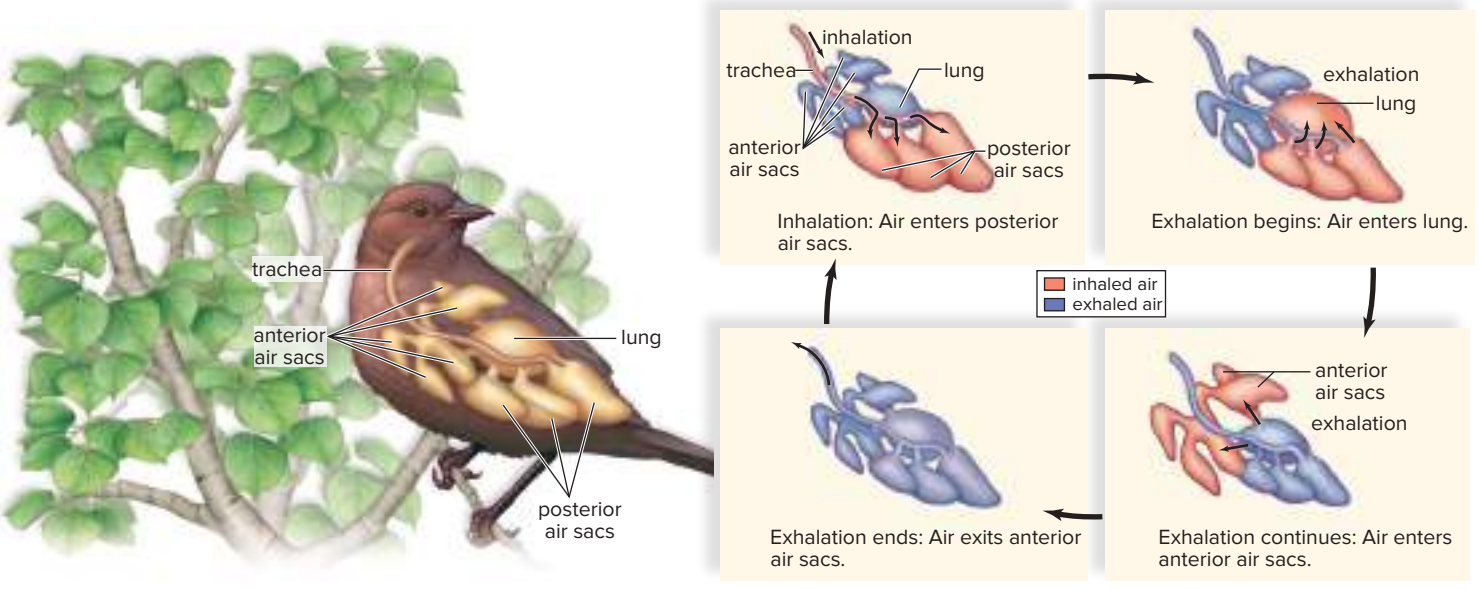
Normally, adults have a breathing rate of 12 to 20 ventilations per minute. The rhythm of ventilation is controlled by a **respiratory center** in the medulla oblongata of the brain. The respiratory center automatically sends out impulses by way of a spinal nerve to the diaphragm (phrenic nerve) and intercostal nerves to the

ISTUDY



a. Inspiration b. Expiration

Figure 35.8 The thoracic cavity during inspiration and expiration. **a.** During inspiration, the thoracic cavity and lungs expand, so that air is drawn in. **b.** During expiration, the thoracic cavity and lungs resume their original positions and pressures. Then, air is forced out.



a. b.

Figure 35.9 Respiratory system in birds. **a.** Air sacs are attached to the lungs of birds. **b.** These allow birds to have a one-way mechanism of ventilating their lungs.

ISTUDY

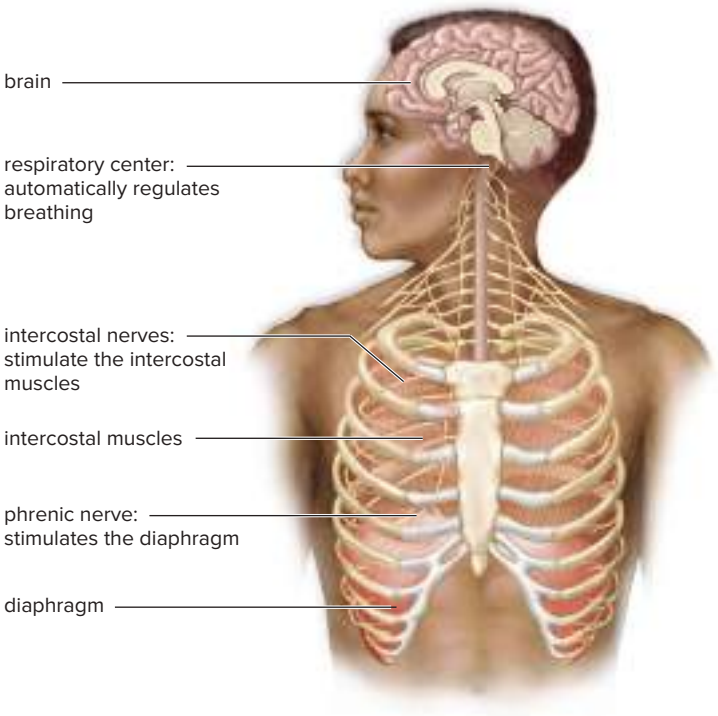


Figure 35.10 Nervous control of breathing. The breathing rate can be modified by nervous stimulation of the intercostal muscles and diaphragm.

intercostal muscles of the rib cage (Fig. 35.10). Now inspiration occurs. Then, when the respiratory center stops sending neuronal signals to the diaphragm and the rib cage, expiration occurs.

Although the respiratory center automatically controls the rate and depth of breathing, its activity can also be influenced by nervous input and chemical input. Following forced inhalation, stretch receptors in the alveolar walls initiate inhibitory nerve impulses that travel from the inflated lungs to the respiratory center. This stops the respiratory center from sending out nerve impulses.

The respiratory center is directly sensitive to the levels of hydrogen ions (H^+). However, when carbon dioxide enters the blood, it reacts with water and releases hydrogen ions. In this way, CO_2 participates in regulating the breathing rate. When hydrogen ions rise in the blood and the pH decreases, the respiratory center increases the rate and depth of breathing. The chemoreceptors in the **carotid bodies,** located in the carotid arteries, and in the **aortic bodies,** located in the aorta, stimulate the respiratory center during intense exercise due to a reduction in pH, or if arterial oxygen decreases to 50% of normal.

Gas Exchange and Transport

Respiration includes the exchange of gases in our lungs, called external respiration, as well as the exchange of gases in the tissues, called internal respiration (Fig. 35.11). The principles of diffusion largely

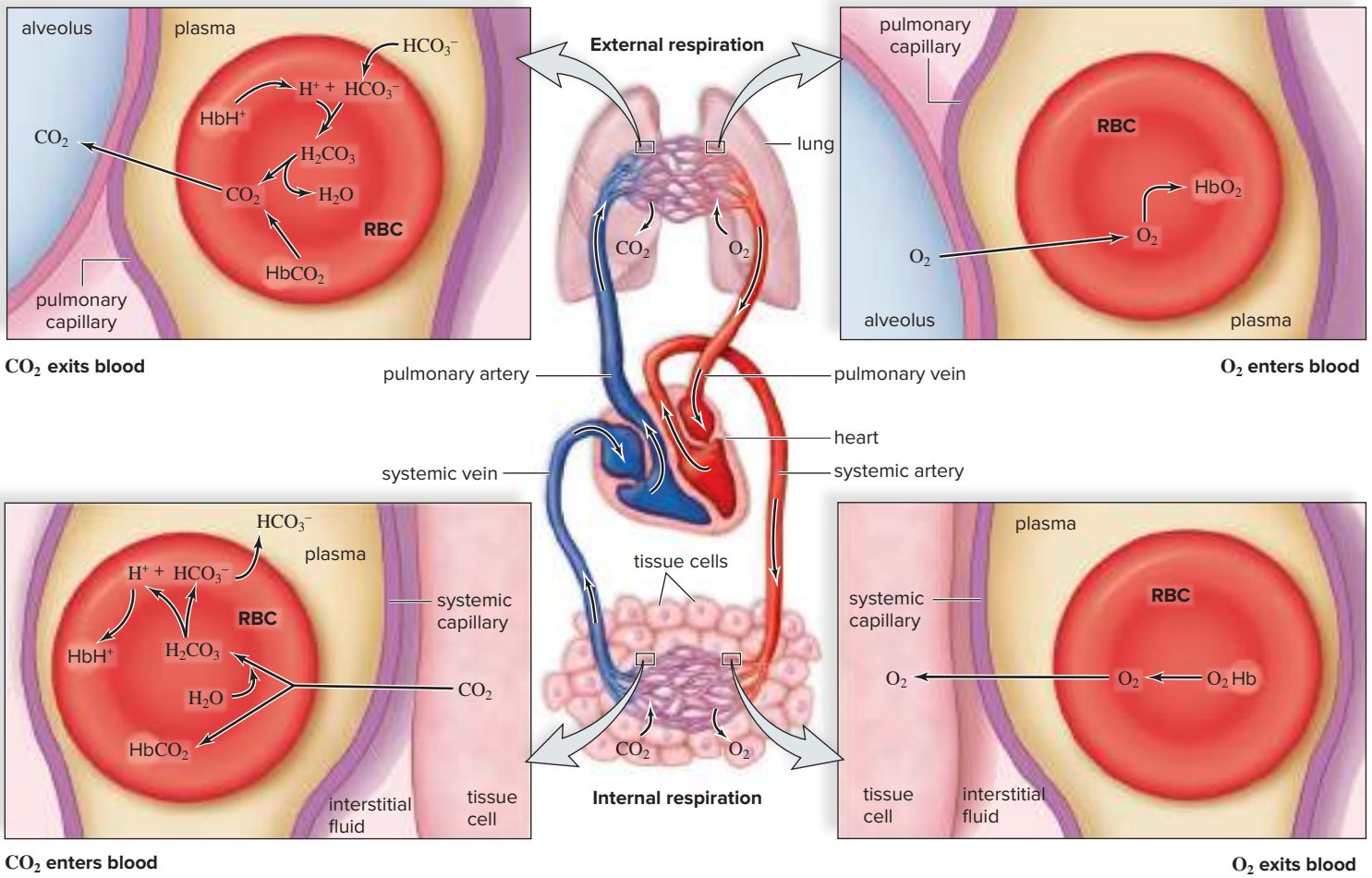


Figure 35.11 External and internal respiration. During external respiration (*top*) in the lungs, carbon dioxide (CO_2) leaves blood, and oxygen (O_2) enters blood. During internal respiration (*bottom*) in the tissues, oxygen leaves blood, and carbon dioxide enters blood.

ISTUDY

govern the movement of gases into and out of blood vessels in the lungs and in the tissues. Gases exert pressure, and the amount of pressure each gas exerts is called the **partial pressure,** symbolized as P_{O_2} and P_{CO_2}. If the partial pressure of oxygen differs across a membrane, oxygen will diffuse from the higher to the lower pressure. Similarly, carbon dioxide diffuses from the higher to the lower partial pressure.

Ventilation causes the alveoli of the lungs to have a higher P_{O_2} and a lower P_{CO_2} than the blood in pulmonary capillaries, and this accounts for the exchange of gases in the lungs. When blood reaches the tissues, cellular respiration in cells causes the interstitial fluid to have a lower P_{O_2} and a higher P_{CO_2} than the blood in the systemic capillaries, and this accounts for the exchange of gases in the tissues.

Transport of Oxygen and Carbon Dioxide

The transport of O_2 and CO_2 is somewhat different in external respiration than in internal inspiration, although the driving forces of diffusion are the same.

External Respiration As blood enters the lungs, a small amount of CO_2 is being carried by hemoglobin with the formula $HbCO_2$. Also, some hemoglobin is carrying hydrogen ions with the formula HbH^+. Most of the CO_2 in the pulmonary capillaries is carried as bicarbonate ions (HCO_3^-) in the plasma. As the free CO_2 from the following equation begins to diffuse out, this reaction is driven to the right:

$$\underset{\text{hydrogen ion}}{H^+} + \underset{\text{bicarbonate ion}}{HCO_3^-} \longrightarrow \underset{\text{carbonic acid}}{H_2CO_3} \longrightarrow \underset{\text{water}}{H_2O} + \underset{\text{carbon dioxide}}{CO_2}$$

The reaction occurs in red blood cells, where the enzyme **carbonic anhydrase** speeds the breakdown of carbonic acid (Fig. 35.11, *top left*). Pushing this equation to the far right by breathing fast can cause you to stop breathing for a time; pushing this equation to the left by not breathing is even more temporary, because breathing will soon resume due to the rise in H^+.

Most oxygen entering the pulmonary capillaries from the alveoli of the lungs combines with **hemoglobin** in red blood cells (RBCs) to form **oxyhemoglobin** (Fig. 35.11, *top right*):

$$\underset{\text{deoxyhemoglobin}}{Hb} + \underset{\text{oxygen}}{O_2} \longrightarrow \underset{\text{oxyhemoglobin}}{HbO_2}$$

At the normal P_{O2} in the lungs, hemoglobin is practically saturated with oxygen. Each hemoglobin molecule contains four polypeptide chains, and each chain is folded around an iron-containing group called **heme** (Fig. 35.12). The iron forms a loose association with oxygen. Because there are about 250 million hemoglobin molecules in each red blood cell, each red blood cell is capable of carrying at least 1 billion molecules of oxygen.

Carbon monoxide (CO) is an air pollutant that is produced by the incomplete combustion of natural gas, gasoline, kerosene, and even wood and charcoal. Because CO is a colorless, odorless gas, people can be unaware that they are breathing it. But once CO is in the bloodstream, it combines with the iron of hemoglobin 200 times more tightly than oxygen, and the result can be death. This is the reason that homes are equipped with CO detectors.

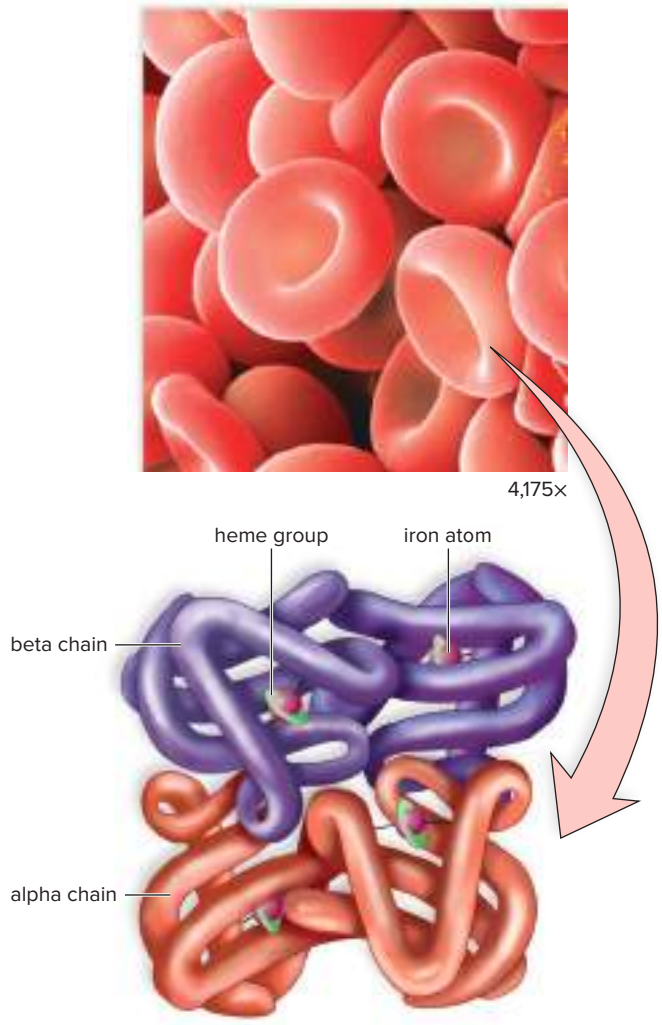


Figure 35.12 Hemoglobin. Hemoglobin consists of four polypeptide chains, two alpha (*red*) and two beta (*purple*), each associated with a heme group. Each heme group contains an iron atom, which can bind to O_2. (photo): Andrew Syred/Science Source

Internal Respiration Blood entering the systemic capillaries is a bright red color because RBCs contain oxyhemoglobin. Because the temperature in the tissues is higher and the pH is lower than in the lungs, oxyhemoglobin has a tendency to give up oxygen:

$$HbO_2 \longrightarrow Hb + O_2$$

Oxygen diffuses out of the blood into the tissues because the P_{O_2} of interstitial fluid is lower than that of blood (Fig. 35.11, *bottom right*). The lower P_{O_2} is due to cells continuously using up oxygen in cellular respiration. After oxyhemoglobin gives up O_2, this oxygen leaves the blood and enters interstitial fluid, where it is taken up by cells.

Carbon dioxide, in contrast, enters blood from the tissues because the P_{CO_2} of interstitial fluid is higher than that of blood. Carbon dioxide, produced continuously by cells, collects in interstitial fluid. After CO_2 diffuses into the blood, it enters the red blood cells, where a small amount combines with the protein portion of hemoglobin to form **carbaminohemoglobin** ($HbCO_2$). Most of the CO_2, however, is transported in the form of the **bicarbonate ion**

ISTUDY

(HCO_3^-). First, CO_2 combines with water, forming carbonic acid, and then this dissociates to a hydrogen ion (H^+) and HCO_3^-:

$$\underset{\text{carbon dioxide}}{CO_2} + \underset{\text{water}}{H_2O} \longrightarrow \underset{\text{carbonic acid}}{H_2CO_3} \longrightarrow \underset{\text{hydrogen ion}}{H^+} + \underset{\text{bicarbonate ion}}{HCO_3^-}$$

Carbonic anhydrase also speeds this reaction. The HCO_3^- diffuses out of the red blood cells to be carried in the plasma (see Fig. 35.11, *bottom left*).

The release of H^+ from this reaction could drastically change the pH of the blood, which is highly undesirable, because cells require a normal pH in order to remain healthy. However, the H^+ is absorbed by the globin portions of hemoglobin. Hemoglobin that has combined with H^+ is called reduced hemoglobin and has the formula HbH^+. HbH^+ plays a vital role in maintaining the normal pH of the blood. Blood that leaves the systemic capillaries is a dark maroon color because red blood cells contain reduced hemoglobin.

Check Your Progress 35.2

1. Describe how the mechanism of ventilation in reptiles and mammals relates to Boyle's law.
2. Explain how the carotid bodies and aortic bodies affect the rate of respiration.
3. Define the role of oxyhemoglobin, reduced hemoglobin, and carbaminohemoglobin in homeostasis.

35.3 Respiration and Human Health

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe several common disorders that mainly affect the upper respiratory tract, as well as several that affect the lower respiratory tract.
2. Classify several common respiratory disorders according to whether they are mainly caused by allergies, infections, a genetic defect, or toxin exposure.

The human respiratory tract is constantly exposed to environmental air, which may contain infectious agents, allergens, tobacco smoke, or other toxins. This results in the respiratory tract being susceptible to a number of diseases. Some of the most important of these are summarized here.

Disorders of the Upper Respiratory Tract

The upper respiratory tract consists of the nasal cavities, sinuses, pharynx, and larynx. Because the upper part of the respiratory tract filters out many pathogens and other materials that may be present in the air, it is commonly affected by a variety of infections, which may also spread to the middle ear or the sinuses.

The Common Cold

Most "colds" are relatively mild viral infections of the upper respiratory tract characterized by sneezing, rhinitis (runny nose), and perhaps a mild fever. Most colds last a few days, after which the immune response is able to eliminate the inciting virus. Because colds are caused by several different viruses (such as coronaviruses), often each with hundreds of different strains, making vaccines may be difficult, especially because it is difficult to predict the type or occurrence of a new strain. As with all viral infections, antibiotics, such as penicillin, have no effect in treating colds.

Figure 35.13 Strep throat. Pharyngitis caused by the bacterium *Streptococcus pyogenes* can cause swollen tonsils, as shown here. The whitish patches are areas of pus formation, indicating that white blood cells are fighting the infection. Dr. P. Marazzi/Science Source

Strep Throat

Most cases of **pharyngitis,** or inflammation of the pharynx, are caused by viruses, but strep throat is an acute pharyngitis caused by the bacterium *Streptococcus pyogenes*. Typical symptoms include severe sore throat, high fever, and white patches in the tonsillar area (Fig. 35.13). Adults experience about half as many sore throats as do children, who average about five upper respiratory infections per year and about one strep throat infection every 4 years. Many untreated strep infections probably resolve on their own, but some can lead to more serious conditions, such as scarlet fever or rheumatic fever. Fortunately, infection with *S. pyogenes* can be easily and quickly diagnosed with specific laboratory tests, and it is usually curable with antibiotics.

Disorders of the Lower Respiratory Tract

Several common disorders affecting the lower respiratory tract are summarized in Figure 35.14.

Disorders Affecting the Tracheae and Bronchi

One of the most obvious and life-threatening disorders that can affect the trachea is choking. The best way for a person without extensive medical training to help someone who is choking is to perform the Heimlich maneuver, which involves grabbing the choking person around the waist from behind and forcefully pulling both hands into the person's upper abdomen to expel whatever is lodged. If this fails, trained medical personnel may be able to quickly insert a breathing tube through an incision made in the

ISTUDY

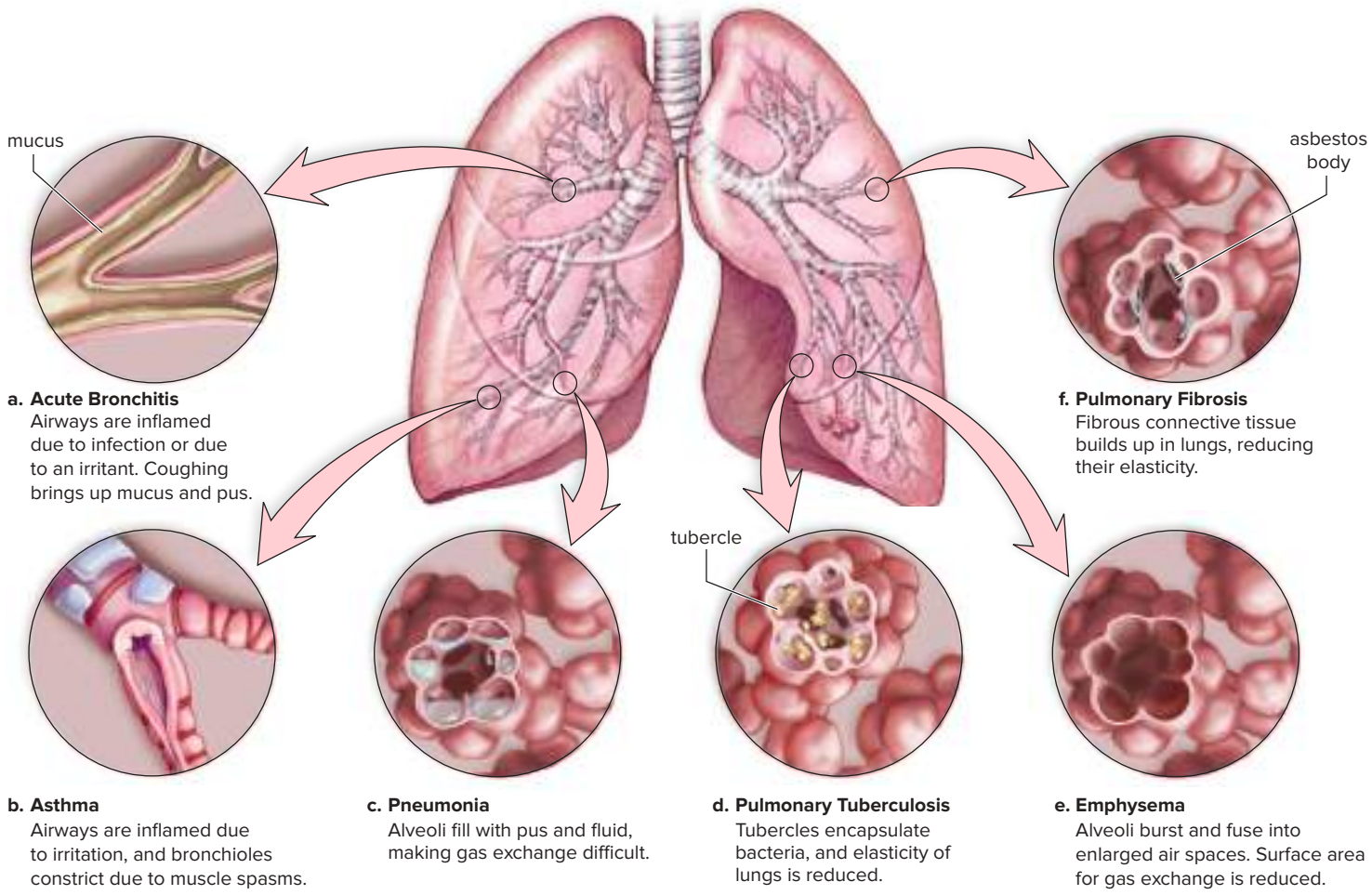


Figure 35.14 Common bronchial and pulmonary diseases. Exposure to infectious pathogens and/or polluted air, including tobacco smoke, causes the diseases and disorders shown here.

trachea. This procedure is called a tracheotomy, and the opening is a *tracheostomy*.

If infections of the upper respiratory tract spread into the lower respiratory tract, *acute bronchitis,* or inflammation of the bronchi, often results (Fig. 35.14*a*). Other causes of acute bronchitis include allergic reactions and damage from environmental toxins, such as those present in cigarette smoke. It is estimated that approximately 5% of the U.S. population experiences a bout of acute bronchitis in any given year. Symptoms include fever, a cough that produces phlegm or pus, and chest pain. Depending on the cause, acute bronchitis may be treatable with antibiotics, or it may resolve with time or progress to more serious conditions.

If the inciting cause (such as smoking) persists, acute bronchitis can develop into *chronic bronchitis,* in which the airways are inflamed and filled with mucus. Over time, the bronchi undergo degenerative changes, including the loss of cilia and their normal cleansing action. Under these conditions, infections are more likely to occur. Smoking and exposure to other airborne toxins are the most frequent causes of chronic bronchitis. Along with emphysema, chronic bronchitis is a major component of *chronic obstructive pulmonary disease* (*COPD*), the fourth leading cause of death in the United States.

Asthma is a disease of the bronchi and bronchioles marked by coughing, wheezing, and breathlessness. The airways are unusually sensitive to various irritants, which include allergens such as pollen, animal dander, dust, and cigarette smoke. Even cold air or exercise can be an irritant.

An asthmatic attack results from inflammation in the airways and the contraction of smooth muscle lining their walls, resulting in a narrowing of the diameter of the airways (Fig. 35.14*b*). All estimates indicate that the incidence of asthma in American children has been increasing steadily since the early 1980s. Possible explanations for this include a more sedentary lifestyle, with more exposure to indoor toxins, and less frequent exposure to beneficial microbes. Asthma is not currently curable, but it is treatable. Drugs administered by inhalers can help prevent the inflammation and dilate the bronchi.

Disorders Affecting the Lungs

Altogether, the various diseases of the lung cause over 500,000 deaths per year in the United States, and they affect hundreds of millions of people worldwide. Depending on the cause, treatments for lung disease include antibiotics, supplemental oxygen, and anti-inflammatory drugs. For individuals with serious lung conditions who have exhausted all other treatment options, a lung transplant

ISTUDY

THEME Nature of Science

SARS-CoV-2 and the Respiratory System

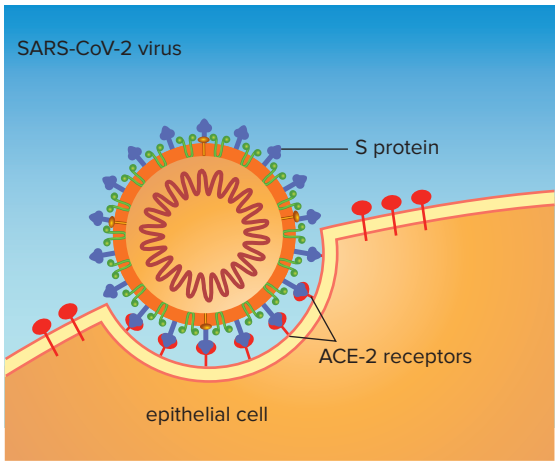


Figure 35A SARS-CoV-2 and the lungs. The S (spike) protein on the surface of the SARS-CoV-2 virus targets the ACE-2 receptors found on the surface of the epithelial cells that line the respiratory system.

SARS-CoV-2, the virus that causes COVID-19, enters the body through the respiratory system, usually attached to aerosols and small droplets of fluid expelled from the lungs of a person who is infected. These travel down the respiratory tract into small air sacs called alveoli. There, the S (spike) protein in the surface of the virus target receptors, called ACE-2, that are present on the epithelial cells that line the interior surfaces of the lungs (Fig. 35A).

These receptors are involved in the regulation of a hormone (angiotensin) that aids in controlling blood pressure. In the lungs, this hormone helps the body regulate the amount of blood flowing into the alveoli, and thus the exchange of gases.

The viral infection not only reduces the body's ability to exchange gases but also causes the activation of the inflammatory response, which moves additional fluids into the lungs (see Section 33.3).

ACE-2 receptors do not exist only in the tissues of the lungs. A number of body tissues contain ACE-2, including blood vessels, the organs of the gastrointestinal tract, and the kidneys. The ability of the virus to infect these cells can interfere with the operation of these organ systems and produce additional COVID-19 complications.

Questions to Consider

1. How might this information help develop treatments for COVID-19?
2. Because the number of ACE-2 receptors is not the same in all individuals, how might this explain the severity of COVID-19 in some people?

may be the best option, but often there aren't enough donor lungs to meet the need.

While we have all experienced respiratory ailments of some form, the COVID-19 pandemic made it clear just how dangerous these infections can be. The Nature of Science feature, "SARS-CoV-2 and the Respiratory System," explores how this virus interacts with the respiratory system.

Pneumonia is a viral, bacterial, or fungal infection of the lungs in which bronchi and alveoli fill with a discharge, such as pus and fluid (Fig. 35.14*c*). Along with coughing and difficulty breathing, people with pneumonia often have a high fever, sharp chest pain, and a cough that produces thick phlegm or even pus. Several bacteria can cause pneumonia, as can the influenza virus, especially in very young people, very old people, and people with a suppressed immune system. People with AIDS are subject to a particularly rare form of pneumonia caused by a fungus of the genus *Pneumocystis,* but they may contract many other types of pneumonia as well.

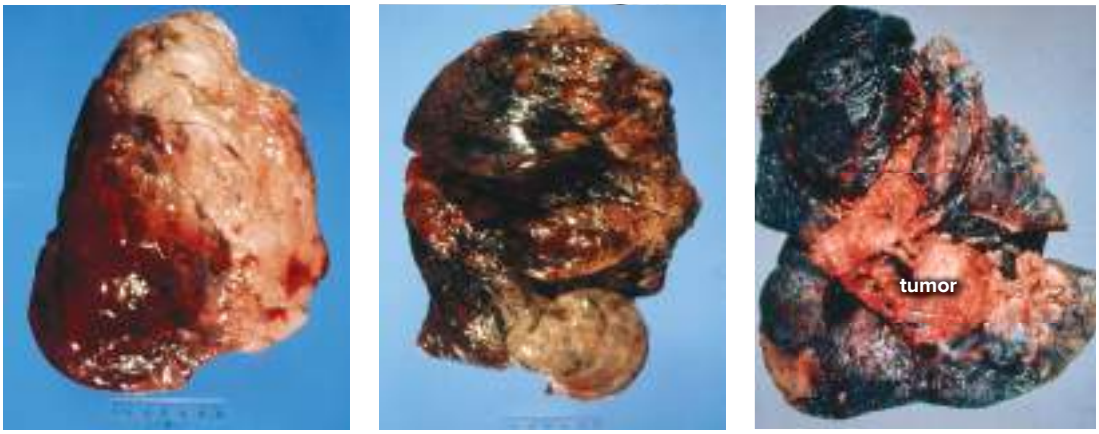
Pulmonary tuberculosis is caused by the bacterium *Mycobacterium tuberculosis.* Tuberculosis (TB) was a major killer in the United States before the middle of the twentieth century, after which antibiotic therapy brought it largely under control. However, the incidence of TB is rising in certain areas of the world, especially where HIV infection (which reduces immunity to *M. tuberculosis*) is common and treatments are not widely available. According to the Centers for Disease Control and Prevention, approximately one-third of the world's population is infected with *M. tuberculosis,* and TB is the cause of death for as many as half of all persons with AIDS. In 2013, the United States had fewer than 10,000 cases, the lowest number ever recorded, but infection rates remain high in certain ethnic groups and among immigrants from areas where TB is more prevalent.

When tubercle bacilli invade the lung tissue, the cells build a protective capsule around the organisms, isolating them from the rest of the body. This tiny capsule is called a tubercle (Fig. 35.14*d*). If the resistance of the body is high, the imprisoned organisms die, but if the resistance is low, the organisms can escape and spread.

It is possible to tell if a person has ever been exposed to *M. tuberculosis* with a TB skin test, in which a highly diluted extract of the bacterium is injected into the skin of the individual. A person who has never been exposed to the bacterium shows no reaction, but one who has previously been infected develops an area of inflammation that peaks in about 48 hours.

Emphysema is a chronic and incurable lung disorder in which the alveoli are distended and their walls damaged, so that the surface area available for gas exchange is reduced (Fig. 35.14*e*). As mentioned, emphysema often contributes to COPD in people who smoke. Air trapped in the lungs leads to alveolar damage and a noticeable ballooning of the chest. The elastic recoil of the lungs is reduced, so not only are the airways narrowed but the driving force behind expiration is also reduced. The person is breathless and may have a cough. Because the

ISTUDY



a. Normal lung b. Emphysema c. Lung cancer

Figure 35.15 Smoking and lung disorders. Smoking causes 90% of all lung cancers and is a major cause of emphysema. **a.** Normal lung. **b.** The lung of a person who died from emphysema, shrunken and blackened from trapped smoke. **c.** The lung of a person who died from lung cancer, blackened from smoke except for the presence of the tumor, which is a mass of malformed soft tissue. (a–b): Southern Illinois University/Science Source; (c): Clinical Photography, Central Manchester University Hospitals NHS Foundation Trust, UK/Science Source

surface area for gas exchange is reduced, less oxygen reaches the heart and brain, leaving the person feeling depressed, sluggish, and irritable. Exercise, drug therapy, and supplemental oxygen, along with cessation of smoking, may relieve the symptoms and slow the progression of emphysema and COPD (Fig. 35.15*b*).

Inhaling particles such as silica (sand), coal dust, or asbestos can lead to **pulmonary fibrosis,** a condition in which fibrous connective tissue builds up in the lungs. The lungs cannot inflate properly and are always tending toward deflation (see Fig. 35.14*f*). Breathing asbestos is also associated with the development of cancer, including a type called mesothelioma. In the United States, the use of asbestos as a fireproofing and insulating agent has been limited since the 1970s; however, many thousands of lawsuits are filed each year by people with asbestos-related illnesses.

Lung cancer is the leading cause of cancer-related death worldwide. More people die from lung cancer each year than from cancer of the colon, breast, and prostate combined. Lung cancer is more common in males than females, but rates in females have increased in recent years due to an increasing number of females who smoke. Lung cancer rates remain low until about age 40, when they gradually start to rise, peaking at around age 70. Symptoms may include coughing, shortness of breath, blood in the sputum, and chest pain. Many other symptoms can occur if the cancer spreads to other parts of the body, which is common.

Lung cancer may be treated with a combination of surgery, chemotherapy, and radiation. Even with treatment, lung cancer is highly lethal—5-year survival rates range from 18% in the United States to less than 8% in less-developed countries. About 150,000 people in the United States die of lung cancer each year (Fig. 35.15*c*). The American Cancer Society links about 90% of these deaths to smoking. Smoking is also associated with bronchitis, emphysema, heart disease, and other types of cancer. Considering that the nicotine in cigarette smoke is addictive, it is better never to start smoking than to try quitting later on. This advice applies to e-cigarettes as well (see the Nature of Science feature, "Are E-cigs Safe?").

Cystic fibrosis (CF) is an example of a lung disease that is genetic rather than infectious, although infections also play a role in the disease. One in 31 Americans carries the defective gene, but a child must inherit two copies of the gene to have the disease. Still, CF is the most common genetic disease in the U.S. white population.

The gene that is defective in CF codes for cystic fibrosis transmembrane regulator (CFTR), a protein needed for proper transport of chloride (Cl^-) ions out of the epithelial cells of the lung. Because this also reduces the amount of water transported out of lung cells, the mucus secretions become very sticky and can form plugs that interfere with breathing.

Symptoms of CF include coughing and shortness of breath; part of the treatment involves clearing mucus from the airways by vigorously slapping the patient on the back, as well as by administering mucus-thinning drugs. None of these treatments is curative, however, and because the lungs can be severely affected, the median survival age for people with CF is only 30 years. Researchers are attempting to develop gene therapy strategies to replace the faulty *CFTR* gene.

Check Your Progress 35.3

1. Explain why antibiotic drugs such as penicillin are ineffective at treating the common cold.
2. Name two disorders of the lower respiratory tract that mainly cause a narrowing of the airways and two that restrict the lungs' ability to expand.
3. List six illnesses associated with smoking cigarettes.

ISTUDY

THEME Nature of Science

Are E-cigs Safe?

For at least 50 years, the many health risks of smoking have been clear. Despite this, over 30 million adult Americans smoke. Once a person starts smoking, the addictive power of nicotine is strong. But why do young people start smoking? Some may want to look mature or "cool," to be accepted by friends, or to rebel against authority. Some people who smoke believe that smoking helps them control their weight; others state they simply enjoy the "buzz" that nicotine can provide.

Because of the unhealthy side effects of smoking, people often look for alternatives, which explains the growing popularity of electronic cigarettes (e-cigarettes). E-cigarettes are often designed to look like real cigarettes (Fig. 35B), but instead of tobacco, they contain a cartridge filled with an "e-liquid" that consists mainly of nicotine plus propylene glycol or vegetable glycerin. When the device is used, a battery heats the liquid, turning it into a vapor that can be inhaled (explaining the popular term *vaping* for this practice). Often, an LED light at the tip glows, mimicking a lit cigarette. There is no cigarette smell, though, because no tobacco is burning.

Manufacturers claim that the vapor from an e-cigarette is much safer than cigarette smoke. However, a number of studies have indicated that this is not the case. In fact, researchers are just beginning to understand the influence of higher levels of nicotine in the body. Nicotine is known to damage the mitochondria of cells, and may have a negative effect on the nervous system over time. It is also known to increase hardening of the arteries, which can affect the overall health of the cardiovascular system over time. These facts are further compounded by the fact that some e-cigs may have 20 times the nicotine levels of traditional cigarettes.

In addition, a variety of contaminants—including metals, aldehydes, and trace levels of certain carcinogens—have been detected. Some of these chemicals have been directly linked to oral and respiratory system cancers. While some people who smoke say using e-cigarettes helped them quit smoking cigarettes, some studies have indicated that people who first become hooked on vaping might "graduate" to smoking.

Another concern is that, although companies claim they aren't marketing to children, the evidence suggests otherwise. Research has indicated that there has been a 78% increase in the use of e-cigs in high school students in just the past few years, and an alarming 48% increase in middle school use. The CDC estimates that 38% of high school students, and 13% of middle school students, have tried vaping.

So, is vaping safer than smoking? While it may have been initially considered a less dangerous alternative, evidence suggests there are considerable health risks associated with the use of e-cigarettes. According to the American Lung Association, the best way to prevent these health risks is to never start smoking, or if you currently smoke, cease all forms of smoking completely.

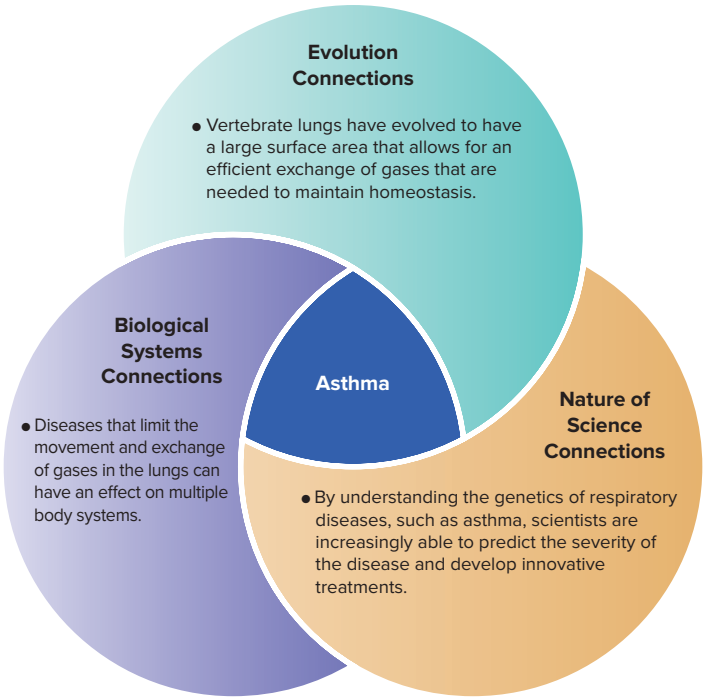
Questions to Consider

1. Compared to most drugs, nicotine is unusual, because at low doses it is mainly a stimulant, but at high doses it has more sedative effects. How might these properties contribute to nicotine's addictive potential?
2. Suppose you've never "vaped" before, and someone offers you an e-cigarette at a party. Would you be tempted to try it? Why or why not?

Figure 35B Electronic cigarettes. Recent studies have indicated that the use of e-cigs presents significant health risks. Martina Paraninfi/Moment/Getty Images

ISTUDY

Connecting *the* Concepts



What Did You Learn?

35.1 Gas-Exchange Surfaces

During **respiration,** gas exchange occurs between an animal's cells and its environment. In terrestrial vertebrates, **ventilation** includes inspiration and expiration. **External respiration** refers to gas exchange between the air and the blood, and **internal respiration** is gas exchange between the blood and the interstitial fluid.

- Some aquatic animals, such as hydras, earthworms, and some amphibians, use their entire body surface for gas exchange. However, most animals have a specialized gas-exchange area. Large aquatic animals usually pass water through **gills.** In bony fishes, blood in the capillaries flows in the direction opposite that of the water. Blood takes up almost all of the oxygen in the water as a result of this **countercurrent exchange.**
- For gas exchange on land, insects have tracheae**,** and vertebrates have **lungs.** In insects, air enters the tracheae at openings called spiracles. From there, the air moves to ever smaller tracheoles until gas exchange takes place at the cells themselves. Lungs are found inside the body, where water loss is reduced. To ventilate the lungs, some vertebrates use positive pressure, but most inhale, using muscular contraction to produce a negative pressure that causes air to rush into the lungs. When the breathing muscles relax, air is exhaled.
- Birds have a series of air sacs attached to the lungs. When a bird inhales, air enters the posterior air sacs, and when a bird exhales, air moves through the lungs to the anterior air sacs before exiting the respiratory tract. The one-way flow of air through the lungs allows more fresh air to be present in the lungs with each breath, and this leads to greater uptake of oxygen from one breath of air. Mammals have two-way or tidal airflow in and out of the lungs, and as a result some mixing is always occurring between fresh air and previously inhaled air, with less uptake of oxygen than in birds.
- In humans, inhaled air passes from the mouth or nasal cavities to the **pharynx,** through the **glottis, larynx, trachea, bronchi,** and **bronchioles,** finally reaching the **alveoli** of the lungs. The glottis contains **vocal cords** that produce sound. Because both air and food pass through the pharynx, the glottis is covered by the epiglottis when food is being swallowed.

35.2 Breathing and Transport of Gases

During **inspiration,** air moves into the lungs, where exchange occurs; during **expiration,** air passes in the opposite direction.

- Humans breathe by negative pressure, as do other mammals. During inspiration, the rib cage goes up and out, and the **diaphragm** lowers. The lungs expand and air rushes in. During expiration, the rib cage goes down and in, and the diaphragm rises. Therefore, air rushes out.
- A **respiratory center** in the brain controls the rate and depth of ventilation. The rate of breathing increases when the amount of H^+ and carbon dioxide in the blood rises, as detected by chemoreceptors such as the **aortic bodies** and **carotid bodies.**
- Gas exchange in the lungs and tissues is brought about by diffusion, according to the **partial pressure** of each gas. In the lungs, **hemoglobin** takes up oxygen, forming **oxyhemoglobin,** which transports oxygen in the blood. Within Hb, the oxygen molecules are actually bound to iron, which is a component of **heme.** Carbon dioxide is mainly transported in plasma as the **bicarbonate ion,** but some combines with Hb, forming **carbaminohemoglobin.** Excess hydrogen ions are transported by hemoglobin. The enzyme carbonic anhydrase found in red blood cells speeds the formation of the bicarbonate ion.

ISTUDY

35.3 Respiration and Human Health

The respiratory tract is susceptible to a wide variety of infections and other disease conditions.

- Infections of the upper respiratory tract include the common cold, caused by several different viruses, and strep throat, which is a form of **pharyngitis** caused by a bacterium.
- In the lower respiratory tract, bronchitis is an inflammation of the bronchi, usually due to infections, whereas in **asthma** the inflammation and smooth muscle contraction in the bronchi and bronchioles are due to sensitivity to various irritants. Many important diseases affect the lungs, including **pneumonia, pulmonary tuberculosis, pulmonary fibrosis, emphysema, lung cancer,** and **cystic fibrosis (CF)**.

Thinking Critically About Biology

1. The respiratory system of birds is quite different from that of mammals. Unlike mammalian lungs, the lungs of birds are relatively rigid and do not expand much during ventilation. Most birds also have thin-walled air sacs that fill most of the body cavity not occupied by other organs. Inspired air passes through the bird's lungs, into these air sacs, and back through the lungs again on expiration. In what ways might this system be more efficient than the mammalian lung?
2. Children may also have obstructive sleep apnea (OSA). Symptoms experienced during the daytime include breathing through the mouth and difficulty focusing. Children don't often have the excessive sleepiness during the day that adults may have. Children with OSA often have enlarged tonsils or adenoids. Removal of the tonsils and adenoids often alleviates the problem. Continuous positive airway pressure (CPAP) treatment might be necessary if the OSA continues after surgery. Why would enlarged tonsils or adenoids cause OSA?
3. Carbon monoxide (CO) binds to hemoglobin over 200 times more strongly than oxygen does, which means less O_2 is being delivered to tissues. What would be the specific cause of death if too much hemoglobin became bound to CO instead of to O_2?
4. As mentioned in the chapter, sometimes it is necessary to install a permanent tracheostomy—for example, if people who smoke develop laryngeal cancer. Besides interfering with the ability to speak, what sorts of health problems would you expect to see in an individual with a permanent tracheostomy, and why?

Making It Relevant to You

1. How do you think exercise benefits individuals with asthma?
2. How might an understanding of the genetic basis of respiratory diseases, such as asthma and cystic fibrosis, help researchers develop personalized treatments?
3. Asthma can cause side effects such as anxiety, irritability, sleep problems, and restlessness, all of which are associated with the nervous system. Why do you think a respiratory disease would cause problems with the nervous system?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

36

Body Fluid Regulation and Excretory Systems

McGraw Hill

Is Your Urine Sterile?

There is a common misconception, dating from studies in the 1950s, that because your urine is produced within your body, it is sterile. As a result, any bacteria that are found during a urinalysis must be the result of either a urinary tract infection or some other disease of the urinary system.

But recent studies contradict that idea. Based on studies of the natural bacterial populations of other body systems, such as the digestive system, researchers have determined that there are natural populations of microbiota that consist of a number of different types of bacteria that normally inhabit the urinary system. Now research is focusing on determining what the normal environment is, and what role these bacteria are performing in the urinary tract. By determining this, it may be possible to develop better diagnostics on how to treat disorders and infections of the urinary system.

As you read through this chapter, think about the following questions:

1. Besides salt glands, what other types of strategies have animals developed to either conserve or excrete excess salt?
2. By what mechanisms is the human kidney able to regulate the salt concentration of urine it produces?

Chapter Outline

36.1 Animal Excretory Systems

36.2 The Human Urinary System

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 5.8 What occurs when a cell is surrounded by a solution having a higher or lower solute concentration than that inside the cell?

Section 8.5 What happens when proteins are used as an energy source?

Section 34.4 What happens to excess nutrients and minerals that cannot be stored?

Following *the* Themes

CHAPTER 36 – BODY FLUID REGULATION AND EXCRETORY SYSTEMS

Evolution	Many different mechanisms for maintaining water-salt balance and excreting metabolic wastes have evolved, molded by the environments in which animals live.
Nature of Science	By studying many different animals in a variety of environments, scientists have learned how animal excretory systems regulate water-salt balance, excrete metabolic wastes, and maintain a healthy pH.
Biological Systems	The urinary system plays an important role in maintaining pH and ion concentration in vertebrates.

ISTUDY

36.1 Animal Excretory Systems

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the overall, specific functions of animal excretion systems.
2. List the costs and benefits of the excretion of ammonia, urea, or uric acid as nitrogenous waste products.
3. Compare and contrast the excretory organs of earthworms, arthropods, aquatic vertebrates, and terrestrial vertebrates.

An important part of maintaining homeostasis in animals involves **osmoregulation,** or balancing the levels of water and salts in the body. Often, the osmoregulatory system of an animal also removes metabolic wastes from the body, a process called **excretion.**

Nitrogenous Waste Products

The breakdown of nitrogen-containing molecules, such as amino acids and nucleic acids, results in excess nitrogen, which must be excreted. When the body breaks down amino acids to generate energy, or converts them to fats or carbohydrates, the amino ($—NH_2$) groups must be removed because they are not needed, and they may be toxic at high levels. Depending on the species, this excess nitrogen may be excreted in the form of ammonia, urea, or uric acid. The removal of amino groups from amino acids requires a fairly constant amount of energy; however, the amount of energy required to convert amino groups to ammonia, urea, or uric acid differs, as indicated in Figure 36.1.

Ammonia

Amino groups removed from amino acids immediately form **ammonia** (NH_3) by the addition of a third hydrogen ion (H^+). This reaction requires little or no energy. Ammonia is quite toxic, but it can be a nitrogenous excretory product if sufficient water is available to wash it from the body. Ammonia is excreted by most fishes and other aquatic animals whose gills and/or body surfaces are in direct contact with the water of the environment.

Urea

Sharks, adult amphibians, and mammals usually excrete **urea** as their main nitrogenous waste. Urea is much less toxic than ammonia and can be excreted in a moderately concentrated solution. This elimination strategy allows body water to be conserved, an important advantage for terrestrial animals with limited access to water. The production of urea requires the expenditure of energy, however, because it is produced in the liver by a set of energy-requiring enzymatic reactions, known as the urea cycle. In this cycle, carrier molecules take up carbon dioxide and two molecules of ammonia, finally releasing urea.

Uric Acid

Uric acid is synthesized by a long, complex series of enzymatic reactions that requires the expenditure of even more energy than does urea synthesis. Uric acid is not very toxic, and it is poorly soluble in water. Poor solubility is an advantage if water conservation is needed, because uric acid can be concentrated even more readily than can urea.

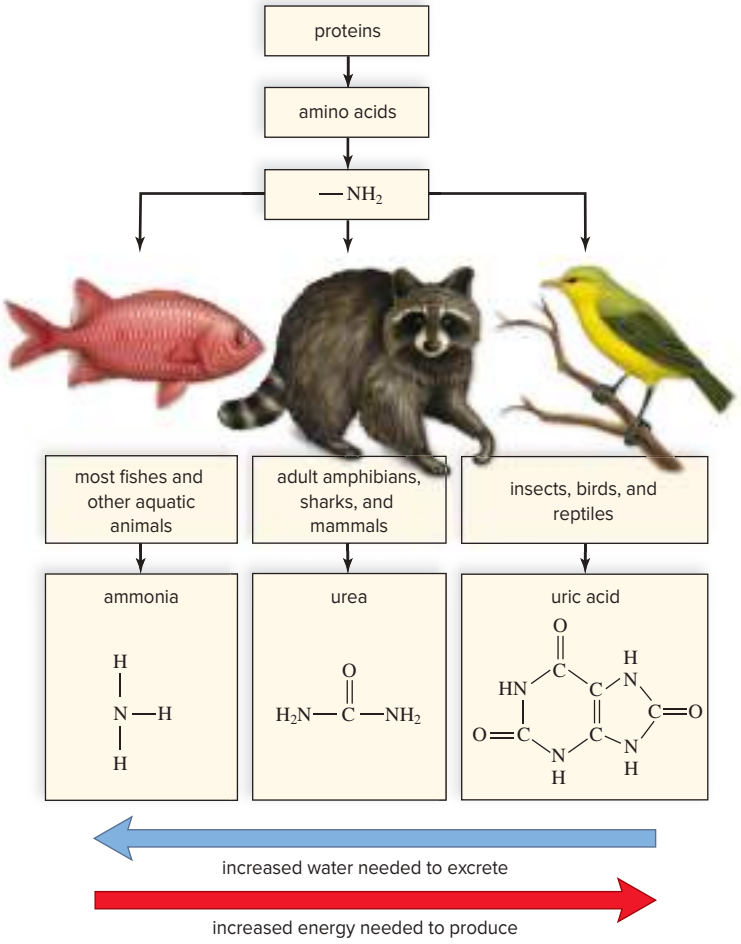


Figure 36.1 Nitrogenous wastes. Proteins are hydrolyzed to amino acids, whose breakdown results in carbon chains and amino groups ($—NH_2$). The carbon chains can be used as an energy source, but the amino groups must be excreted as ammonia, urea, or uric acid.

Uric acid is routinely excreted by insects, reptiles, and birds. In reptiles and birds, a dilute solution of uric acid passes from the kidneys to the *cloaca,* a common reservoir for the products of the digestive, urinary, and reproductive systems. The cloacal contents are refluxed into the large intestine, where water is reabsorbed. The white substance in bird feces is uric acid.

Embryos of reptiles and birds develop inside completely enclosed shelled eggs. The production of insoluble, relatively nontoxic uric acid is advantageous for shelled embryos, because all nitrogenous wastes are stored inside the shell until hatching takes place. For all these reasons, the evolutionary advantages of uric acid production have outweighed the disadvantage of energy expenditure needed for its synthesis.

Humans have retained the ability to produce uric acid, mainly from the breakdown of excess purine and pyrimidine nucleic acids in the diet. Although the exact causes are unknown, in some individuals uric acid builds up in the blood and can precipitate in and around the joints, producing a painful ailment called *gout.*

Excretory Organs Among Invertebrates

Most invertebrates have tubular excretory organs that regulate the water-salt balance of the body and excrete metabolic wastes into the environment.

ISTUDY

The planarians, flatworms that live in fresh water, have two strands of branching excretory tubules that open to the outside of the body through excretory pores (Fig. 36.2*a*). Located along the tubules are bulblike *flame cells,* each of which contains a cluster of beating cilia that looks like a flickering flame under the microscope. The beating of flame-cell cilia propels fluid through the excretory tubules and out of the body. The system is believed to function in ridding the body of excess water and in excreting wastes.

The body of an earthworm is divided into segments, and nearly every body segment has a pair of excretory structures called *nephridia.* Each nephridium is a tubule with a ciliated opening and an excretory pore (Fig. 36.2*b*). As fluid from the coelom is propelled through the tubule by beating cilia, its composition is modified. For example, nutrient substances are reabsorbed and carried away by a network of capillaries surrounding the tubule. **Urine** is a liquid that contains metabolic wastes, excreted salts, and water; the urine of an earthworm is passed out of the body via the excretory pore. Although the earthworm is considered a terrestrial animal, it excretes a very dilute urine. Each day, an earthworm may produce a volume of urine equal to 60% of its body weight.

Insects have a unique excretory system consisting of long, thin *Malpighian tubules* attached to the gut. Uric acid is actively transported from the surrounding hemolymph into these tubules, and water follows a salt gradient established by active transport of K^+. Water and other useful substances are reabsorbed at the rectum, but the uric acid leaves the body through the anus. Insects that live in water, or eat large quantities of moist food, reabsorb little water. But insects in dry environments reabsorb most of the water and excrete a dry, semisolid mass of uric acid.

The excretory organs of other arthropods are given different names, although they function similarly. In aquatic crustaceans (e.g., crabs, crayfish), nitrogenous wastes are generally removed by diffusion across the gills. Some crustaceans also possess excretory organs, called *green glands,* in the ventral portion of the head region. Fluid collects within the tubules from the surrounding blood of the hemocoel, but this fluid is modified before it leaves the tubules. The secretion of salts into the tubules regulates the amount of urine excreted.

In shrimp and pillbugs, the excretory organs are located in the maxillary segments and are called *maxillary glands.* Spiders, scorpions, and other arachnids possess *coxal glands,* which are located near one or more appendages and are used for excretion. Coxal glands are spherical sacs resembling annelid nephridia. Wastes are collected from the surrounding blood of the hemocoel and discharged through pores at one to several pairs of appendages.

Osmoregulation by Aquatic Vertebrates

In most vertebrates, the kidneys are the most important organs involved in osmoregulation. As described later in this chapter, the kidneys perform several functions critical to homeostasis, including maintaining the balance between water and several types of salts. This is a necessity, because ions such as Na^+, Ca^{2+}, K^+, and PO_4^- greatly affect the workings of the body systems, such as the skeletal, nervous, and muscular systems.

The kidneys produce urine, a liquid that contains a number of different metabolic wastes. The concentration of the urine produced by an animal varies depending on its environment, as well as on factors such as water and salt intake.

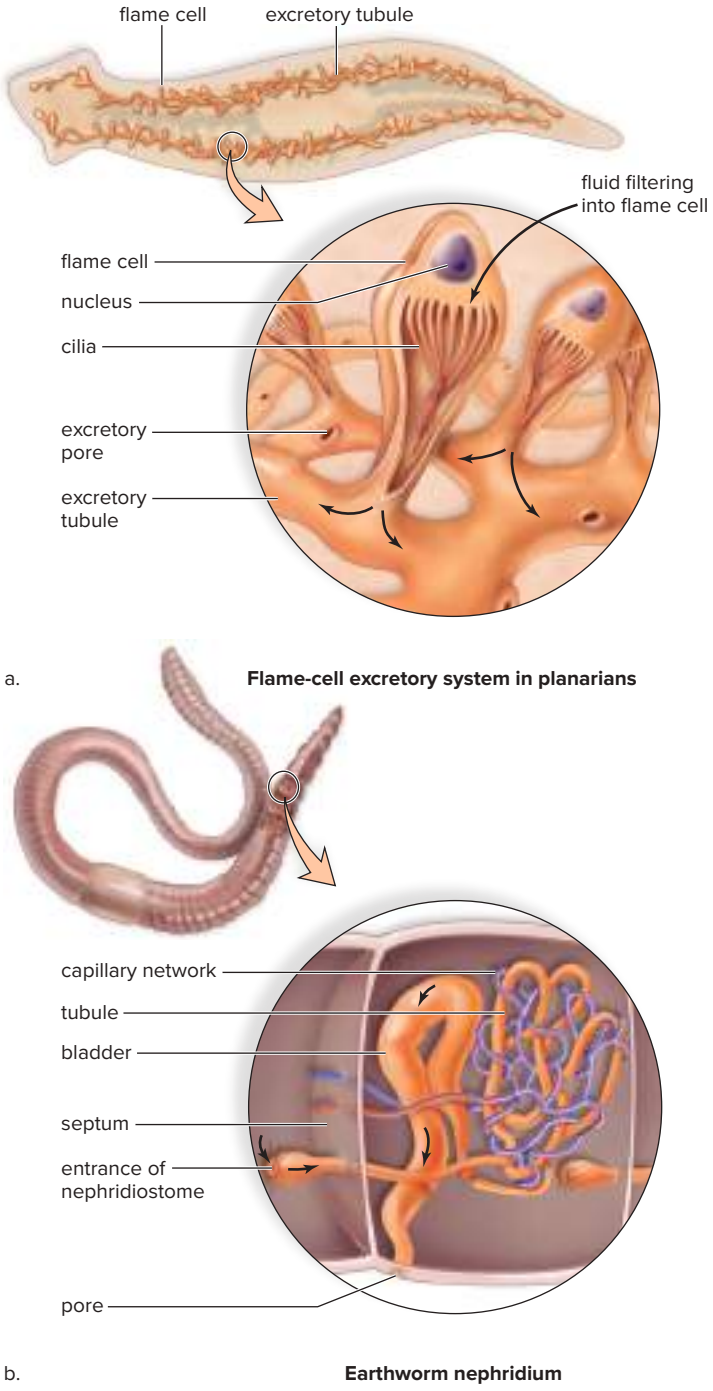
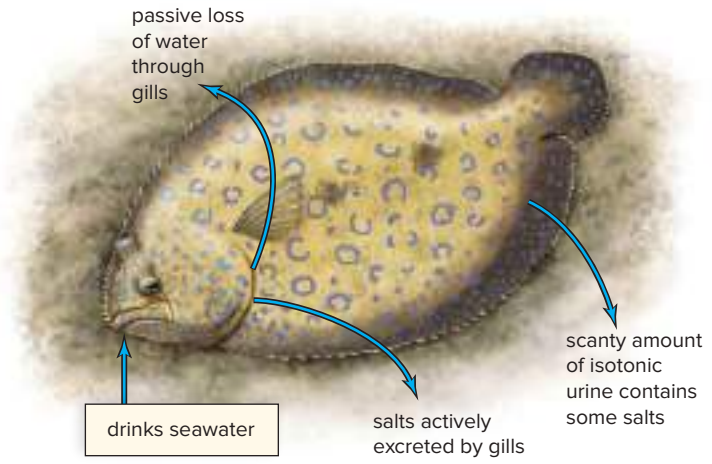


Figure 36.2 Excretory organs in animals. **a.** Two or more tracts of branching tubules run the length of the body and open to the outside by pores. At the ends of side branches are small, bulblike cells called flame cells. **b.** The nephridium has a ciliated opening, the nephridiostome, which leads to a coiled tubule surrounded by a capillary network. Urine can be temporarily stored in the bladder before being released to the outside via a pore called a nephridiopore.

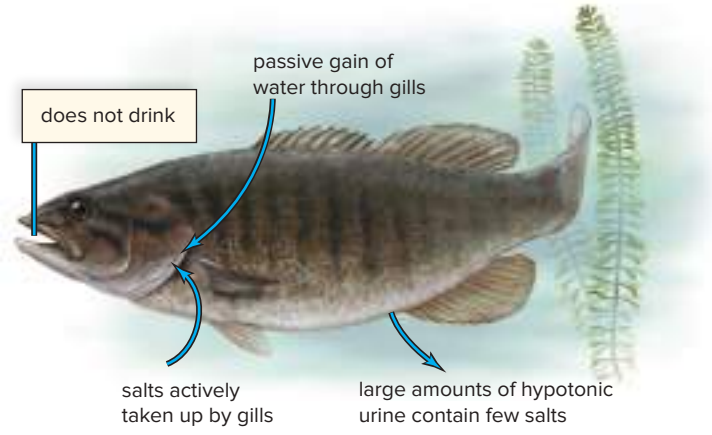
Cartilaginous Fishes

The total concentration of the various ions in the blood of sharks, rays, and skates is less than that in seawater. Their blood plasma is nearly isotonic to seawater, because they pump it full of urea, and this molecule gives their blood the same tonicity as seawater. Excess salts are secreted by the kidneys and by an excretory organ, the rectal gland.

ISTUDY



a. Marine bony fish (a flounder)



b. Freshwater bony fish (a trout)

Figure 36.3 Body fluid regulation in bony fishes. Marine bony fishes (**a**) use different mechanisms than do freshwater bony fishes (**b**) to osmoregulate their body fluids.

Marine Bony Fishes

The marine environment, which is high in salts, is hypertonic to the blood plasma of bony fishes. The common ancestor of marine fishes evolved in fresh water, and only later did some groups invade the sea. Therefore, marine bony fishes must avoid the tendency to become dehydrated (Fig. 36.3*a*).

As the sea washes over their gills, marine bony fishes lose water by osmosis. To counteract this, they drink seawater almost constantly. On average, marine bony fishes swallow an amount of water equal to 1% of their body weight every hour. This is equivalent to a human drinking about 700 ml of water every hour around the clock. But while they get water by drinking, this habit also causes these fishes to acquire salt. To rid their body of excess salt, they actively transport it into the surrounding seawater at the gills. The kidneys conserve water, and marine bony fishes produce a scant amount of isotonic urine.

Freshwater Bony Fishes

The osmotic problems of freshwater bony fishes and the response to their environment are exactly opposite those of marine bony fishes (Fig. 36.3*b*). Freshwater fishes tend to gain water by osmosis across the gills and the body surface. As a consequence, these fishes never drink water. They actively transport salts into the blood across the membranes of their gills. They eliminate excess water by producing large quantities of dilute (hypotonic) urine. They discharge a quantity of urine equal to one-third their body weight each day.

Osmoregulation by Terrestrial Vertebrates

An important evolutionary adaptation that allowed animals to survive on land was the development of a kidney that could produce a concentrated (hypertonic) urine. The need for water conservation is particularly well illustrated in desert mammals, such as the kangaroo rat, as well as in animals that drink seawater.

Kangaroo Rat

Dehydration threatens all terrestrial animals, especially those that live in a desert, as does the kangaroo rat. During daylight hours, kangaroo rats remain in a cool burrow, a behavioral adaptation to conserve water. In addition, the kangaroo rat's nasal passages have a highly convoluted mucous membrane surface that captures condensed water from exhaled air. Exhaled air is usually full of moisture, which is why you can see it on cold winter mornings—the moisture in exhaled air is condensing.

A major adaptation that allows the kangaroo rat to conserve water is the ability to form a very hypertonic urine—20 times more concentrated than its blood plasma. The kidneys of a kangaroo rat are able to accomplish this feat because the structure in their kidneys that is largely responsible for producing concentrated urine, called the loop of the nephron (see Fig. 36.8), is much longer and more efficient than that in most other animals. Also, kangaroo rats produce fecal material that is almost completely dry.

Most terrestrial animals need to drink water at least occasionally to make up for the water lost from the skin and respiratory passages and through urination. However, the kangaroo rat is so adapted to conserving water that it can survive by using metabolic water derived from cellular respiration, and it never drinks water (Fig. 36.4).

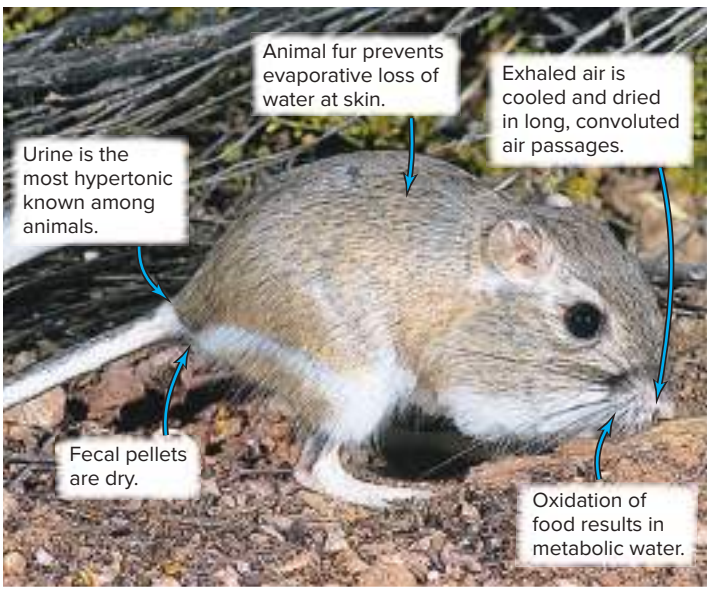


Figure 36.4 Adaptations of a kangaroo rat to a dry environment. The kangaroo rat minimizes water loss through a variety of ways. Bruce Coleman/Photoshot/AVALON

ISTUDY

Figure 36.5 Adaptations of marine birds to a high-salt environment. Many marine birds and reptiles have glands that pump salt out of the body. David Hosking/Science Source

Marine Mammals and Sea Birds

Reptiles, birds, and mammals evolved on land, and their kidneys are especially good at conserving water. However, some of these vertebrates have become secondarily adapted to living in or near the sea. They can drink seawater and still manage to survive. If humans drink seawater, we lose more water than we take in just ridding the body of all that salt!

Little is known about how whales manage to get rid of extra salt, but we know that their kidneys are enormous. In some marine animals, however, the kidneys are not efficient enough to secrete all the excess salt. As mentioned in the chapter-opening story, some animals living in high-salt environments have developed specialized glands for excreting these salts. These glands work by actively transporting salt from the blood into the gland, where it can be excreted as a concentrated solution.

In sea birds, salt-excreting glands are located near the eyes. The glands produce a salty solution that is excreted through the nostrils and moves down grooves on their beaks until it drips off (Fig. 36.5). In marine turtles, the salt gland is a modified tear (lacrimal) gland, and in sea snakes, a salivary sublingual gland beneath the tongue gets rid of excess salt. The work of these glands is regulated by the nervous system. Osmoreceptors, perhaps located near the heart, are thought to stimulate the brain, which then directs the gland to excrete salt until the salt concentration in the blood decreases to a tolerable level.

Check Your Progress 36.1

1. Distinguish between osmoregulation and excretion.
2. Describe two advantages of excreting urea instead of ammonia or uric acid.
3. Summarize the strategies used by kangaroo rats to conserve water.

36.2 The Human Urinary System

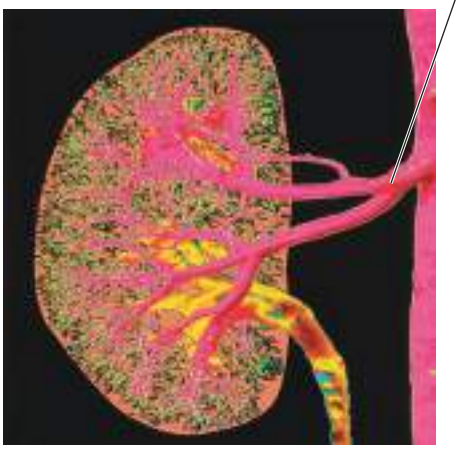
Learning Outcomes

Upon completion of this section, you should be able to

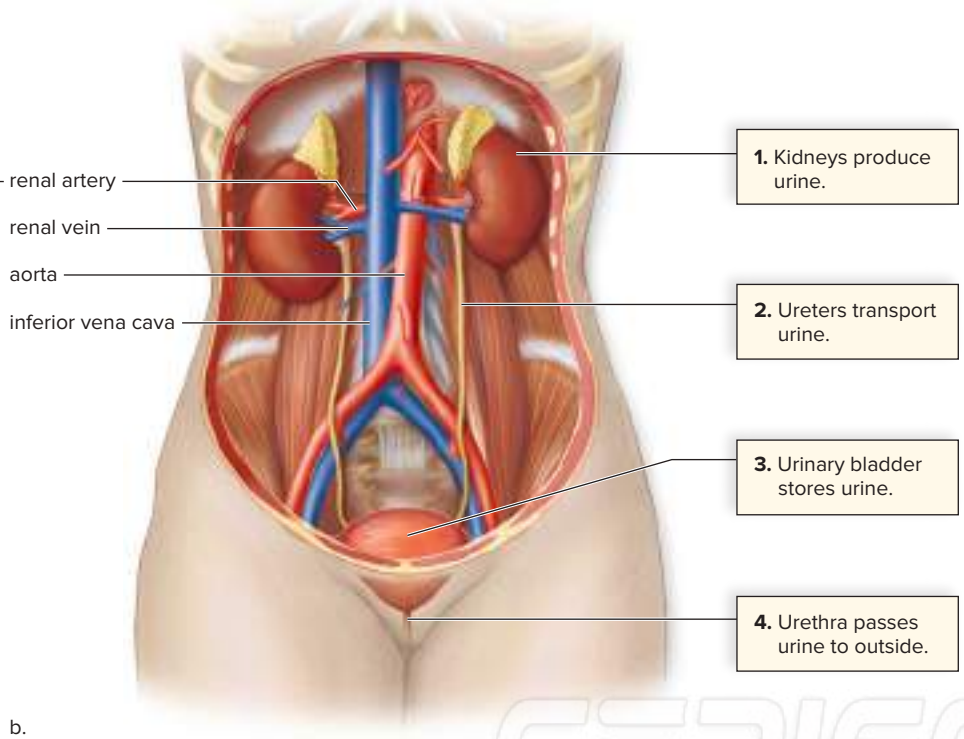
1. Trace the anatomical path that urine takes from the glomeruli to its exit from the body.
2. Discuss the contributions of glomerular filtration, tubular reabsorption, and tubular secretion to the formation of urine.
3. Summarize the four major functions of human kidneys in maintaining homeostasis.

The major excretory organs of humans, as with most other vertebrates, are the kidneys (Fig. 36.6). The kidneys are the ultimate regulators of blood composition because they can remove various unwanted products from the body.

Figure 36.6 The human urinary system. **a.** The kidneys are well supplied with blood, as shown in the angiogram. **b.** Urine is found only in the kidneys, ureters, urinary bladder, and urethra. (a): James Cavallini/Science Source



a.



b.

ISTUDY

Human **kidneys** are bean-shaped, reddish-brown organs, each about the size of a fist. They are located on each side of the vertebral column just below the diaphragm, in the lower back, where they are partially protected by the lower rib cage. The right kidney is slightly lower than the left kidney.

Urine made by the kidneys is conducted from the body by the other organs in the urinary system. Each kidney is connected to a **ureter,** a duct that takes urine from the kidney to the **urinary bladder,** where it is stored until it is voided from the body through the single **urethra.** In males, the urethra passes through the penis; in females, the opening of the urethra is ventral to that of the vagina. No connection exists between the genital (reproductive) and urinary systems in females, but in males the urethra also carries sperm during ejaculation.

Kidneys

If a kidney is sectioned longitudinally, three major parts can be distinguished (Fig. 36.7). The *renal cortex,* which is the outer region of a kidney, has a somewhat granular appearance. The *renal medulla* consists of six to ten cone-shaped renal pyramids that lie on the inner side of the renal cortex. The innermost part of the kidney is a hollow chamber called the *renal pelvis.* Urine collects in the renal pelvis and then is carried to the bladder by a ureter.

Nephrons

Microscopically, each kidney is composed of over 1 million tiny tubules called **nephrons** (Gk. *nephros,* "kidney"). The nephrons of a kidney produce urine. Some nephrons are located primarily in the renal cortex, but others dip down into the renal medulla, as shown in Figure 36.7*b*. Each nephron is made of several parts

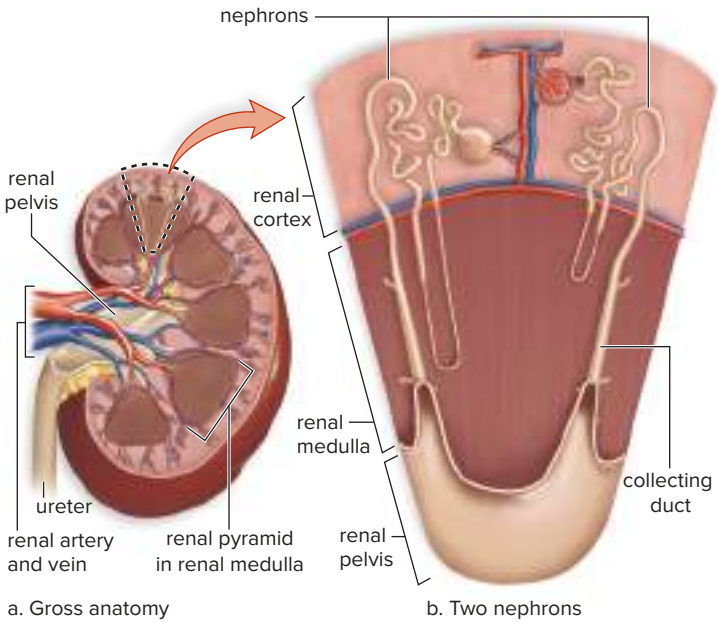


Figure 36.7 Macroscopic and microscopic anatomy of the kidney. **a.** Longitudinal section of a kidney, showing the location of the renal cortex, the renal medulla, and the renal pelvis. **b.** An enlargement of one renal lobe, showing the placement of nephrons.

(Fig. 36.8). The blind end of a nephron is pushed in on itself to form a cuplike structure called the **glomerular capsule** (L. *glomeris,* "ball"), also known as Bowman's capsule. The outer layer of the glomerular capsule is composed of squamous epithelial cells; the inner layer is composed of specialized cells that allow easy passage of molecules.

Leading from the glomerular capsule is a portion of the nephron known as the **proximal convoluted tubule** (L. *proximus,* "nearest"), which is lined by cells with many mitochondria and tightly packed microvilli. Then, simple squamous epithelium appears in the **loop of the nephron** (loop of Henle), which has a descending limb and an ascending limb. This is followed by the **distal convoluted tubule** (L. *distantia,* "far"). Several distal convoluted tubules enter one **collecting duct.** The collecting duct transports urine down through the renal medulla and delivers it to the renal pelvis.

Each nephron has its own blood supply (Fig. 36.8). The renal artery branches into numerous small arteries, which branch into arterioles, one for each nephron. Each arteriole, called an afferent arteriole, divides to form a capillary bed, the **glomerulus** (L. *glomeris,* "ball"), which is surrounded by the glomerular capsule. The glomerulus drains into an efferent arteriole, which subsequently branches into a second capillary bed around the tubular parts of the nephron. These capillaries, called peritubular capillaries, lead to venules that join to form veins leading to the renal vein, a vessel that enters the inferior vena cava.

Urine Formation

An average human produces between 1 and 2 liters of urine daily. The fundamental process of urine formation involves initially filtering a large amount of water and a collection of solutes out of the blood, then reabsorbing much of the water, along with other material the body needs to conserve.

Urine production requires three distinct processes (Fig. 36.9*a*):

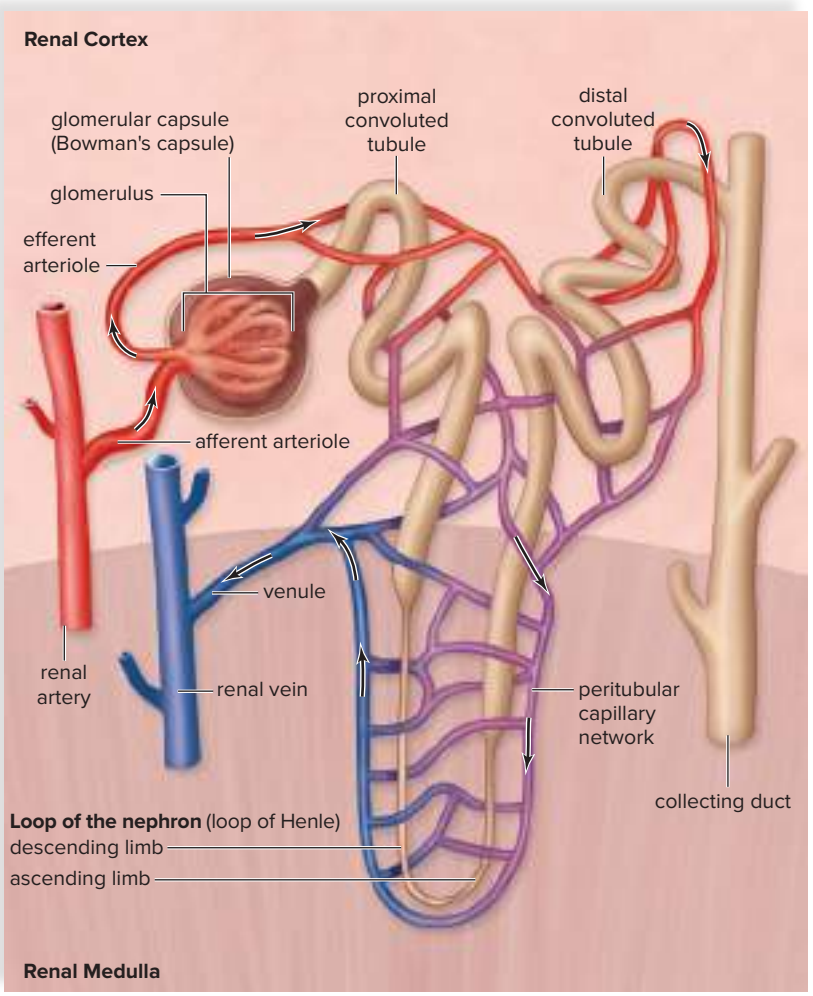
- Glomerular filtration at the glomerular capsule
- Tubular reabsorption at the convoluted tubules
- Tubular secretion at the convoluted tubules

Glomerular Filtration

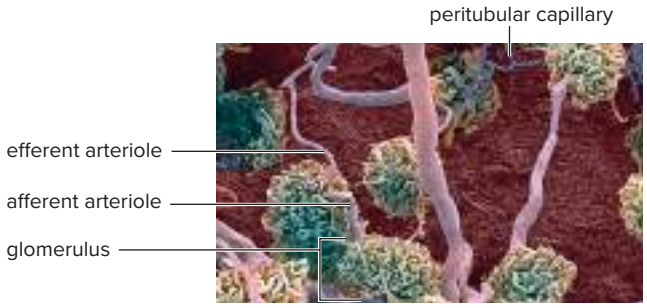
Glomerular filtration (Fig. 36.9*a*) is the movement of small molecules across the glomerular wall into the glomerular capsule as a result of blood pressure. When blood enters the glomerulus, blood pressure is sufficient to cause small molecules, such as water, nutrients, salts, and wastes, to move from the glomerulus to the inside of the glomerular capsule, especially because the glomerular walls are 100 times more permeable than the walls of most capillaries elsewhere in the body. The molecules that leave the blood and enter the glomerular capsule are called the *glomerular filtrate.* Plasma proteins and blood cells are too large to be part of this filtrate, so they remain in the blood as it flows into the efferent arteriole.

Glomerular filtrate is essentially protein-free, but otherwise it has the same composition as blood plasma. If this composition were not altered in other parts of the nephron, death from starvation (loss of nutrients) and dehydration (loss of water) would

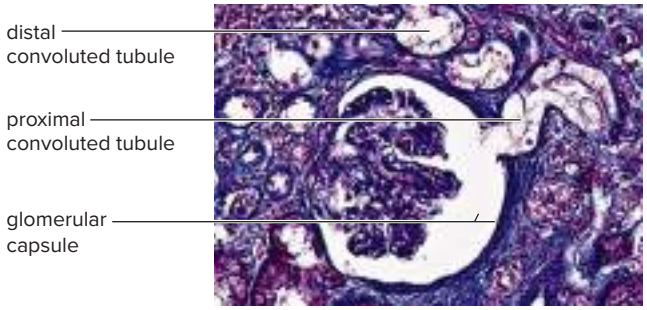
ISTUDY



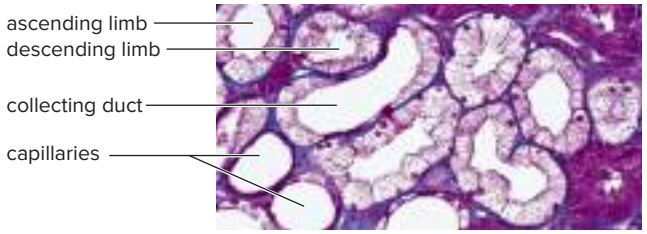
a. A nephron and its blood supply



b. Surface view of glomerulus and its blood supply



c. Cross section of glomerulus and convoluted tubules 50×



d. Cross sections of loop of the nephron limbs and collecting duct 100×

Figure 36.8 Nephron anatomy. **a.** You can trace the path of blood about a nephron by following the arrows. A nephron is made up of a glomerular capsule, the proximal convoluted tubule, the loop of the nephron, the distal convoluted tubule, and the collecting duct. The micrographs in (**b**), (**c**), and (**d**) show these structures. (b): Steve Gschmeissner/Science Photo Library/Brand X Pictures/Getty Images; (c–d): Ed Reschke/Stone/Getty Images

quickly follow. The total blood volume averages about 5 liters, and this amount of fluid is filtered every 40 minutes. Thus, 180 liters of filtrate are produced daily, some 60 times the amount of blood plasma in the body. Most of the filtered water is obviously quickly returned to the blood, or a person would die from urination. Tubular reabsorption prevents this from happening.

Tubular Reabsorption

Tubular reabsorption (Fig. 36.9*a*) takes place when substances move across the walls of the tubules into the associated peritubular capillary network (Fig. 36.9*a, b*). Here, osmosis comes into play. You may remember that *osmosis* is the diffusion of water down its concentration gradient across a membrane (see Section 5.2). *Osmolarity* is a measure of the potential for osmosis; water tends to move from a solution with low osmolarity into a solution with high osmolarity.

The osmolarity of the blood is essentially the same as that of the filtrate within the glomerular capsule, and therefore osmosis of water from the filtrate into the blood cannot yet occur. However, sodium ions (Na^+) are actively pumped into the peritubular capillary, and then chloride ions (Cl^-) follow passively. The osmolarity of the blood then is such that water moves passively from the tubule into the blood. About 60–70% of salt and water are reabsorbed at the proximal convoluted tubule, and 20–25% at the loop of the nephron.

Nutrients such as glucose and amino acids also return to the blood, mostly at the proximal convoluted tubule. This is a selective process, because only molecules recognized by carrier proteins in plasma membranes are actively reabsorbed. The cells of the proximal convoluted tubule have numerous microvilli, which increase the surface area, and numerous mitochondria, which supply the energy needed for active transport (Fig. 36.9*b*).

Glucose is an example of a molecule that ordinarily is reabsorbed completely because the supply of carrier molecules for it is plentiful. However, if the filtrate contains more glucose than there are carriers to handle it, glucose exceeds its renal threshold, or transport maximum. When this happens, the excess glucose in the filtrate appears in the urine. In diabetes mellitus, an abnormally large amount of glucose is present in the filtrate, because the liver cannot store all the excess glucose as glycogen. The presence of glucose in

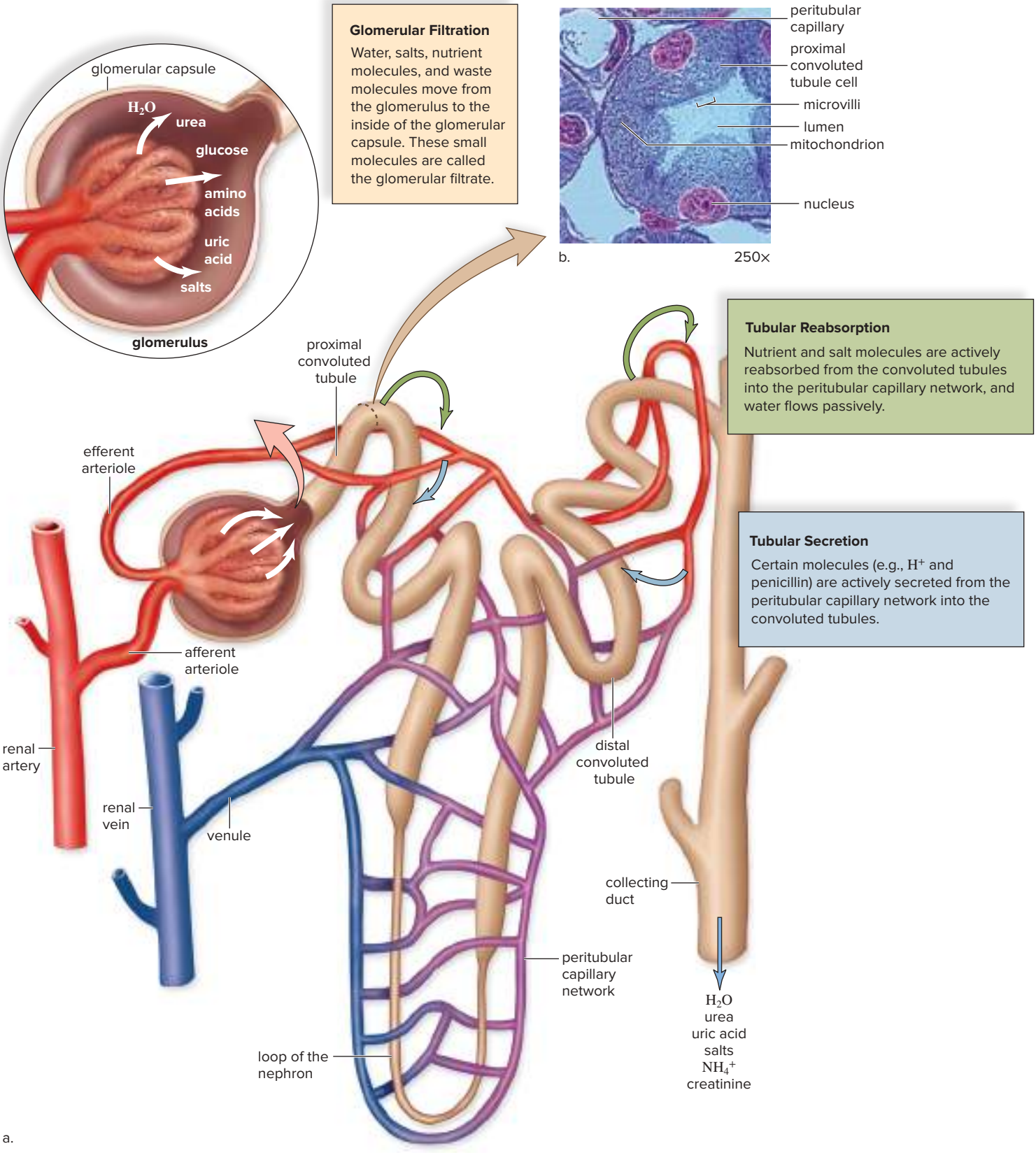


Figure 36.9 Processes in urine formation. **a.** The three main processes in urine formation are described in boxes and color-coded to arrows that show the movement of molecules into or out of the nephron at specific locations. In the end, urine is composed of the substances within the collecting duct (blue arrow). **b.** This photomicrograph shows that the cells lining the proximal convoluted tubule have a brush border composed of microvilli, which greatly increases the surface area exposed to the lumen. The peritubular capillary adjoins the cells. (b): Joseph F. Gennaro Jr./Science Source

ISTUDY

the filtrate results in less water being absorbed; the increased thirst and frequent urination in people with untreated diabetes are a result of less water being reabsorbed into the peritubular capillary network.

Urea is an example of a substance that is passively reabsorbed from the filtrate. At first, the concentration of urea within the filtrate is the same as that in blood plasma. But after water is reabsorbed, the urea concentration is greater than that of peritubular plasma. In the end, about 50% of the filtered urea is reabsorbed.

Tubular Secretion

Tubular secretion is the second way substances are removed from blood and added to tubular fluid (Fig. 36.9*a*). Substances such as hydrogen ions, uric acid, salts, ammonia, creatinine, and penicillin are eliminated by tubular secretion. The process of tubular secretion may be viewed as helping rid the body of potentially harmful compounds that were not filtered into the glomerulus.

The Kidneys and Homeostasis

The kidneys are organs of homeostasis for four main reasons:

- The kidneys *excrete metabolic wastes,* such as urea, which is the primary nitrogenous waste of humans.
- They *maintain the water-salt balance,* which in turn affects blood volume and blood pressure.
- Kidneys *maintain the acid-base balance* and therefore the pH balance.
- They *secrete hormones.*

One hormone secreted by the kidneys, **erythropoietin (EPO),** stimulates the stem cells in bone marrow to produce more red blood cells. Another substance produced by the kidneys, called renin, is discussed later in this section. The Biological Systems feature, "Urinalysis," examines how these, and other compounds, are detected during a routine urine test.

Maintaining the Water-Salt Balance

Most of the water and salt (NaCl) present in filtrate is reabsorbed across the wall of the proximal convoluted tubule. The excretion of a hypertonic urine (one that is more concentrated than blood) is dependent on the reabsorption of water from the loop of the nephron and the collecting duct. During the process of reabsorption, water passes through water channels called **aquaporins,** which were first discovered in 1992.

Loop of the Nephron A long loop of the nephron, which typically penetrates deep into the renal medulla, is made up of a descending (downward) limb and an ascending (upward) limb. Salt (NaCl) passively diffuses out of the lower portion of the ascending limb, but the upper, thick portion of the limb actively extrudes salt out into the tissue of the outer renal medulla (Fig. 36.10). Less and less salt is available for transport as fluid moves up the thick portion of the ascending limb. Because of these circumstances, an osmotic gradient is created within the tissues of the renal medulla: The concentration of salt is greater in the direction of the inner medulla. Note that water cannot leave the ascending limb, because this limb is impermeable to water.

The innermost portion of the inner medulla has the highest concentration of solutes. This cannot be due to salt, because active transport of salt does not start until fluid reaches the thick portion

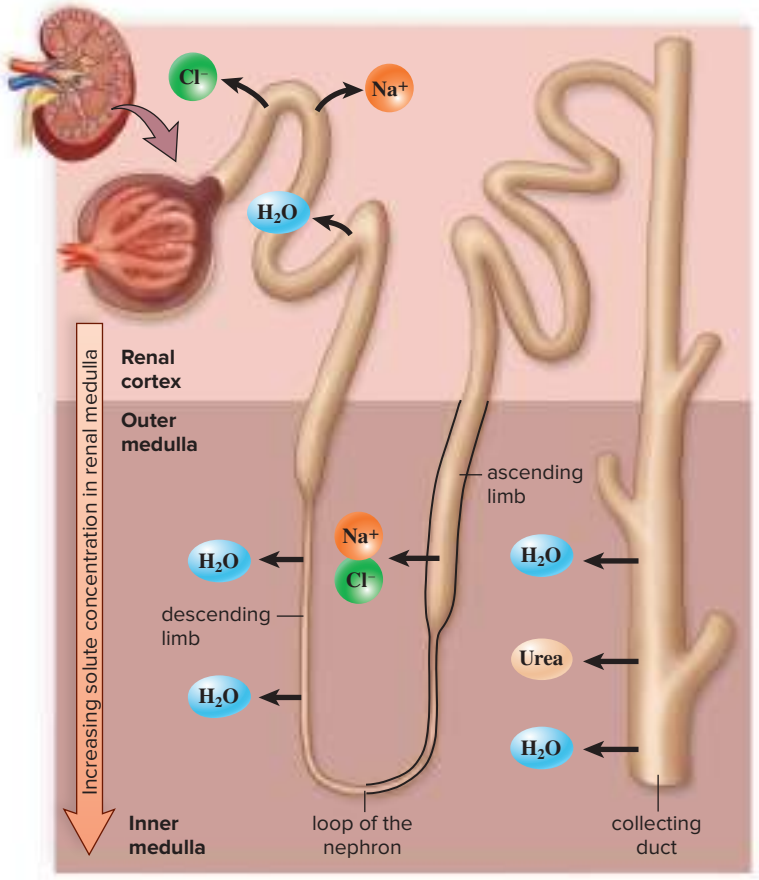


Figure 36.10 Reabsorption of salt and water. Salt (NaCl) diffuses and is actively transported out of the ascending limb of the loop of the nephron into the renal medulla; also, urea leaks from the collecting duct and enters the tissues of the renal medulla. These actions create a hypertonic environment, which draws water out of the descending limb and the collecting duct. This water is returned to the cardiovascular system.

of the ascending limb. Urea is believed to leak from the lower portion of the collecting duct, and it is this molecule that contributes to the high solute concentration of the inner medulla.

Because of the osmotic gradient within the renal medulla, water leaves the descending limb along its entire length. This is a countercurrent mechanism: As water diffuses out of the descending limb, the remaining fluid within the limb encounters an even greater osmotic concentration of solute; therefore, water continues to leave the descending limb from the top to the bottom. Filtrate within the collecting duct also encounters the same osmotic gradient mentioned earlier (Fig. 36.10). Therefore, water diffuses out of the collecting duct into the renal medulla, and the urine within the collecting duct becomes hypertonic to blood plasma.

Antidiuretic hormone (ADH) is released by the posterior lobe of the pituitary in response to an increased concentration of salts in the blood. To understand the action of this hormone, consider its name. *Diuresis* means increased amount of urine, and *antidiuresis* means decreased amount of urine. When ADH is present, more water is reabsorbed (blood volume and pressure rise), and a decreased amount of more concentrated urine is produced. One way by which ADH accomplishes this change is by causing the insertion of additional aquaporin water channels into the epithelial cells of the distal convoluted tubule and collecting duct, allowing more water to be reabsorbed.

ISTUDY

THEME Biological Systems

Urinalysis

A routine urinalysis can detect abnormalities in urine color, concentration, and content. Significant information about the general state of one's health, particularly in the diagnosis of renal and metabolic diseases, is provided by this analysis also. A complete urinalysis consists of three phases of examination: physical, chemical, and microscopic.

Physical Examination

The physical examination describes urine color, clarity, and odor. The color of normal, fresh urine is usually pale yellow. However, the color may vary from almost colorless (dilute) to dark yellow (concentrated). Pink, red, or smoky brown urine is usually a sign of bleeding that may be due to a kidney, bladder, or urinary tract infection. Liver disorders can also produce dark brown urine. The clarity of normal urine may be clear or cloudy. However, a cloudy urine sample is also characteristic of abnormal levels of bacteria. Finally, urine odor is usually "nutty" or aromatic, but a foul-smelling odor is characteristic of a urinary tract infection. A sweet, fruity odor is characteristic of glucose in the urine or diabetes mellitus. Urine's odor is also affected by the consumption of garlic, curry, asparagus, and vitamin C.

Chemical Examination

The chemical examination is often done with a dipstick, a thin strip of plastic impregnated with chemicals that change color upon reaction with certain substances present in urine (Fig. 36A). The color change on each segment of the dipstick is compared with a standardized color chart. Dipsticks can be used to determine urine's specific gravity; pH; and the content of glucose, bilirubin, urobilinogen, ketone, protein, nitrite, blood, and white blood cells (WBCs) in the urine.

- *Specific gravity* is an indicator of how well the kidneys are able to adjust tonicity in urine. Normal values for urine specific gravity range from 1.002 to 1.035. A high value (concentrated urine) may be a result of dehydration or diabetes mellitus.
- Normal *urine pH* can be as low as 4.5 and as high as 8.0. In people with kidney stone disease, urine pH has a direct effect on the type of stones formed.
- *Glucose* is normally not present in urine. If present, diabetes mellitus is suspected.
- *Bilirubin* (a by-product of hemoglobin degradation) is not normally present in the urine. *Urobilinogen* (a by-product of bilirubin degradation) is normally present in very small amounts. High levels of bilirubin or urobilinogen may indicate liver disease.
- *Ketones* are not normally found in urine. Urine ketones are a by-product of fat metabolism. The presence of ketones in the urine may indicate diabetes mellitus or use of a low-carbohydrate diet, such as the Atkins diet.
- Plasma *proteins* should not be present in urine. A significant amount of urine protein (proteinuria) is usually a sign of kidney damage.
- Urine typically does not contain *nitrates* or *nitrites.* The presence of these nitrogen compounds is a sign of urinary tract infection.
- It is not abnormal for blood to show up in the urine when a female is menstruating. Blood present in the urine at other times may indicate a bacterial infection or kidney damage.
- The chemical test for *WBCs* is normally negative. A high urine WBC count usually indicates a bacterial infection somewhere in the urinary tract.

Figure 36A Urinalysis test. A urinalysis test can detect problems with a number of the body's systems. The brown results on this test indicate that the individual may have a form of diabetes. Alexander Raths/Shutterstock

Microscopic Examination

For the microscopic examination, the urine is centrifuged, and the sediment (solid material) is examined under a microscope. When renal disease is present, the urine often contains an abnormal amount of cellular material. Urinary casts are sediments formed by the abnormal coagulation of protein material in the distal convoluted tubule or the collecting duct. The presence of crystals in the urine is characteristic of kidney stones, kidney damage, or problems with metabolism.

Forensic Analysis

Urinalysis is also used by the federal government and some businesses to screen potential employees for the use of numerous illegal drugs. Drugs associated with date rape (such as flunitrazepam [Rohypnol], known as "roofies," and GHB, or gamma-hydroxybutyrate) can be detected in the urine of people who were sexually assaulted to determine if they were drugged.

Questions to Consider

1. What might be the difference between the levels of water-soluble and fat-soluble chemicals in the urine?
2. Why does diabetes result in high levels of glucose in the urine?

ISTUDY

In practical terms, if an individual does not drink much water on a certain day, the posterior lobe of the pituitary releases ADH, causing more water to be reabsorbed and less urine to form. On the other hand, if an individual drinks a large amount of water and does not perspire much, ADH is not released. More water is excreted, and more urine forms. Diuretics, such as caffeine and alcohol, increase the flow of urine by interfering with the action of ADH. ADH production also increases at night, an adaptation that allows longer periods of sleep without the need to wake up to urinate.

Hormones Control the Reabsorption of Salt Usually, more than 99% of the Na^+ filtered at the glomerulus is returned to the blood. Most sodium (67%) is reabsorbed at the proximal convoluted tubule, and a sizable amount (25%) is extruded by the ascending limb of the loop of the nephron. The rest is reabsorbed from the distal convoluted tubule and collecting duct.

Blood volume and pressure are, in part, regulated by salt reabsorption. When blood volume, and therefore blood pressure, is not sufficient to promote glomerular filtration, a cluster of cells near the glomerulus called the *juxtaglomerular apparatus* secretes renin. **Renin** is an enzyme that changes angiotensinogen (a large plasma protein produced by the liver) into angiotensin I. Later, angiotensin I is converted to **angiotensin II,** a powerful vasoconstrictor that also stimulates the adrenal glands, which lie on top of the kidneys, to release aldosterone (Fig. 36.11). **Aldosterone** is a hormone that promotes the excretion of potassium ions (K^+) and the reabsorption of sodium ions (Na^+) at the distal convoluted tubule. The reabsorption of sodium ions is followed by the reabsorption of water. Therefore, blood volume and blood pressure increase.

Atrial natriuretic hormone (ANH) is a hormone secreted by the atria of the heart when cardiac cells are stretched due to increased blood volume. ANH inhibits the secretion of renin by the juxtaglomerular apparatus and the secretion of aldosterone by the adrenal cortex. Its effect, therefore, is to promote the excretion of Na^+—that is, natriuresis. When Na^+ is excreted, so is water, and therefore blood volume and blood pressure decrease.

These examples show that the kidneys regulate the water balance in blood by controlling the excretion and reabsorption of ions. Sodium is an important ion in plasma that must be regulated, but the kidneys also excrete or reabsorb other ions, such as potassium ions (K^+), bicarbonate ions (HCO_3^-), and magnesium ions (Mg^{2+}), as needed.

Maintaining the Acid-Base Balance

The functions of cells are influenced by pH. Therefore, the regulation of pH is extremely important to good health. The *bicarbonate (HCO_3^-) buffer system* and breathing work together to help maintain the pH of the blood. Central to the mechanism is the following reaction, which we first explored in our discussion of the respiratory system (see Section 35.2):

$$\underset{\text{hydrogen ion}}{H^+} + \underset{\text{bicarbonate ion}}{HCO_3^-} \longrightarrow \underset{\text{carbonic acid}}{H_2CO_3} \longrightarrow \underset{\text{water}}{H_2O} + \underset{\text{carbon dioxide}}{CO_2}$$

The excretion of carbon dioxide (CO_2) by the lungs helps keep the pH within normal limits because when carbon dioxide is exhaled, this reaction is pushed to the right and hydrogen ions are tied up in water. When blood pH decreases, chemoreceptors in the carotid bodies (in the carotid arteries) and in aortic bodies (in the aorta) stimulate the respiratory control center, and the rate and depth of breathing increase. And when blood pH begins to rise, the respiratory control center is depressed, and the amount of bicarbonate ion increases in the blood (see Section 35.2).

As powerful as this system is, only the kidneys can rid the body of a wide range of acidic and basic substances. The

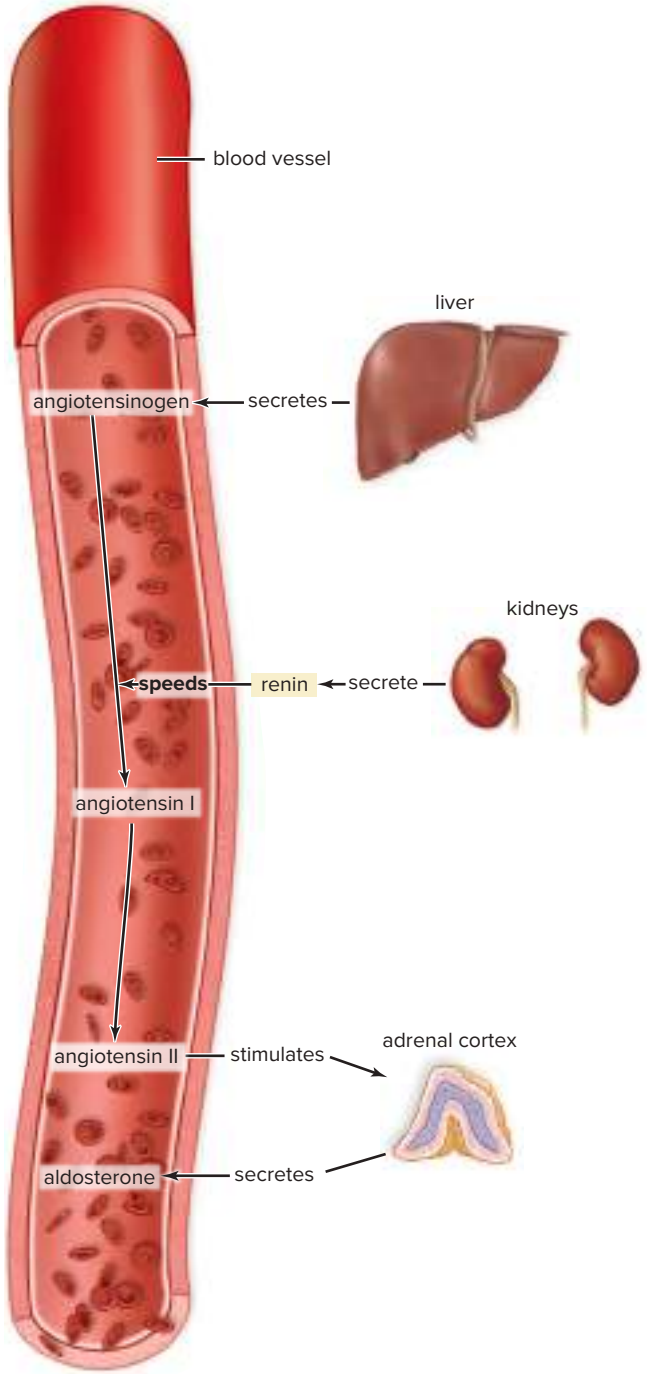
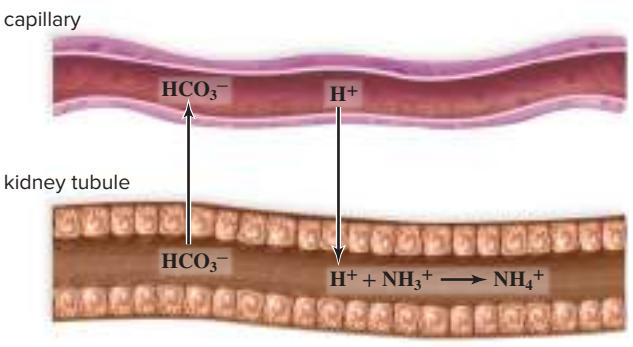


Figure 36.11 The renin-angiotensin-aldosterone system. The liver secretes angiotensinogen into the bloodstream. Renin from the kidneys initiates the chain of events that results in angiotensin II. Angiotensin II acts on the adrenal cortex to secrete aldosterone, which causes reabsorption of sodium ions by the kidneys and a subsequent rise in blood pressure.

ISTUDY

kidneys are slower-acting than the buffer/breathing mechanism, but they have a more powerful effect on pH. For the sake of simplicity, we can think of the kidneys as reabsorbing bicarbonate ions and excreting hydrogen ions as needed to maintain the normal pH of the blood:



If the blood is acidic, hydrogen ions are excreted and bicarbonate ions are reabsorbed. If the blood is basic, hydrogen ions are not excreted and bicarbonate ions are not reabsorbed. The fact that urine is typically acidic (pH about 6) shows that usually an excess of hydrogen ions are excreted. Ammonia (NH_3) provides a means for buffering these hydrogen ions in urine:

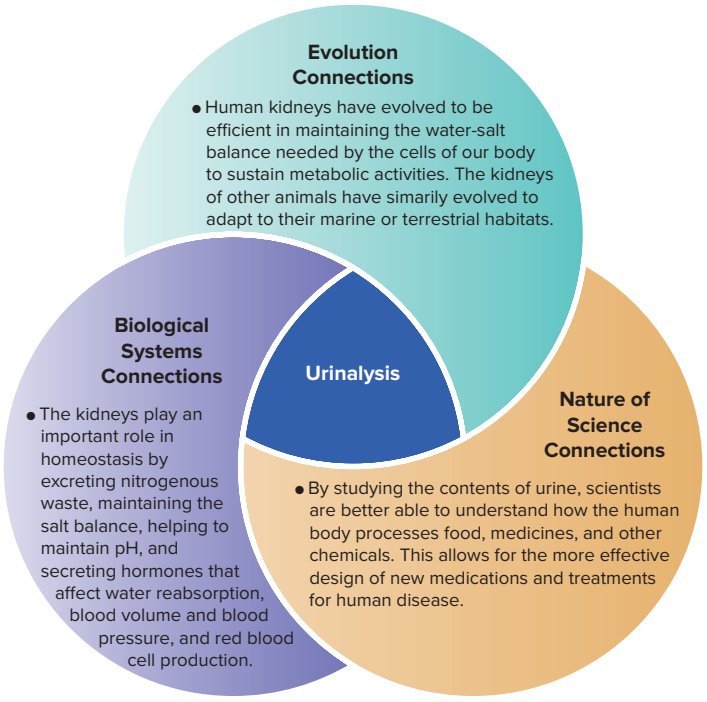
$$\underset{\text{ammonia}}{NH_3} + \underset{\text{hydrogen ions}}{H^+} \longrightarrow \underset{\text{ammonium}}{NH_4^+}$$

Ammonia is produced in tubule cells by the deamination of amino acids. Phosphate provides another means of buffering hydrogen ions in urine.

Check Your Progress 36.2

1. Describe which of the four major functions of the human urinary system are accomplished solely by the kidneys, and which are shared with other body systems.
2. Explain the effect of the renin-angiotensin-aldosterone system on water-salt balance.
3. Describe how the kidneys contribute to the maintenance of normal blood pH.

Connecting *the* Concepts



What Did You Learn?

36.1 Animal Excretory Systems

Osmoregulation, or balancing the body's water and salt levels, includes **excretion** of metabolic wastes, including nitrogenous wastes.

- Aquatic animals usually excrete **ammonia** (which requires little energy but much water to excrete), and land animals excrete either **urea** or **uric acid** (which requires much energy to produce, but can conserve water).
- Most animals have specialized excretory organs. The flame cells of planarians rid the body of excess water. Earthworm nephridia exchange molecules with the blood in a manner similar to that of vertebrate kidneys. Malpighian tubules in insects take up metabolic wastes and water from the hemolymph. Later, the water is absorbed by the gut.
- Osmotic regulation is important to animals. Most must balance their water and salt intake and excretion to maintain normal solute and water concentration in body fluids. Marine fishes constantly

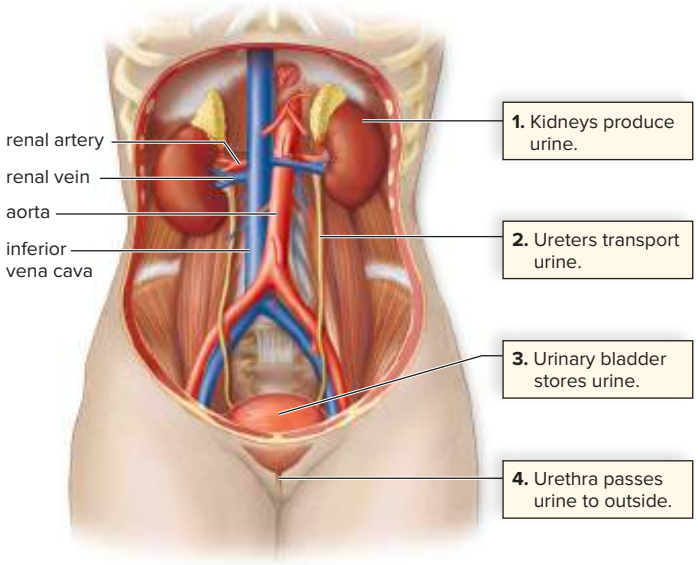
ISTUDY

drink water, excrete salts at the gills, and pass an isotonic **urine.** Freshwater fishes never drink water; they take in salts at the gills and excrete a hypotonic urine.

- Some terrestrial animals have adapted to extreme environments. For example, the desert kangaroo rat can survive on metabolic water; marine birds and reptiles have glands that extrude salt.

36.2 The Human Urinary System

The human urinary system includes the **kidneys, ureters, urinary bladder,** and **urethra.** The kidneys serve four basic homeostatic functions: excretion of metabolic waste; maintenance of water-salt balance; maintenance of pH balance; and production of hormones, such as erythropoietin.



- Kidneys are made up of **nephrons,** each of which has several parts. The renal artery branches into many smaller arteries, each of which forms numerous arterioles. Each afferent arteriole forms a tuft of capillaries, called the **glomerulus.** From the **glomerular capsule,** a **proximal convoluted tubule** leads to the **loop of the nephron,** which is followed by the **distal convoluted tubule.** Several nephrons enter one **collecting duct,** which transports urine through the renal medulla into the renal pelvis.
- Urine formation by a nephron requires three steps: **glomerular filtration,** during which nutrients, water, and wastes enter the nephron's glomerular capsule; **tubular reabsorption,** when nutrients and most water are reabsorbed into the peritubular capillary network; and **tubular secretion,** during which additional wastes are added to the convoluted tubules.
- In order to excrete a hypertonic urine, the ascending limb of the loop of the nephron extrudes salt, so that the renal medulla is increasingly hypertonic. Because urea leaks from the lower end of the collecting duct, the inner renal medulla has the highest concentration of solute. Therefore, a countercurrent mechanism ensures that water diffuses out of the descending limb and the collecting duct. During this process, water passes through membrane channels called **aquaporins.**
- Three hormones are involved in maintaining the water-salt balance of the blood. **Antidiuretic hormone (ADH)**, which makes the collecting duct more permeable to water, is secreted by the posterior pituitary in response to an increase in the osmotic pressure of the blood. **Aldosterone** is secreted by the adrenal cortex after low blood pressure has caused the kidneys to release **renin.** The presence of renin leads to the formation of **angiotensin II,** which causes the adrenal cortex to release aldosterone. Aldosterone causes the kidneys to retain Na^+; therefore, water is reabsorbed and blood pressure rises. **Atrial natriuretic hormone,** in contrast, prevents the secretion of renin and aldosterone.
- The kidneys also keep blood pH within normal limits. They reabsorb (HCO_3^-) and excrete H^+ as needed to maintain the pH at about 7.4. Finally, the kidneys help regulate red blood cell production by secreting **erythropoietin (EPO).**

Thinking Critically About Biology

1. High blood pressure often is accompanied by kidney damage. In some people, the kidney damage is subsequent to the high blood pressure, but in others the kidney damage is the cause of the high blood pressure. Explain how a low-salt diet would enable you to determine whether the high blood pressure or the kidney damage came first.
2. The renin-angiotensin-aldosterone system can be inhibited in order to reduce high blood pressure. Usually, the angiotensin-converting enzyme, which converts angiotensin I to angiotensin II, is inhibited by drug therapy. Why would this enzyme be an effective point at which to disrupt the system?
3. A form of diabetes that may be unfamiliar to most people is diabetes insipidus. When the word *diabetes* is used, everyone tends to think of the type related to insulin. That form of diabetes is diabetes mellitus, which can be type 1 or type 2. In diabetes insipidus, the kidneys are unable to perform their function of water and salt homeostasis due to a lack of ADH in the body. In both types of diabetes, kidney function is affected. Why would frequent urination be a symptom of both diabetes mellitus and diabetes insipidus?

Making It Relevant to You

1. Why might extended use of antibiotics for urinary tract infections produce problems for an individual over time?
2. How might an understanding of the bacteria that inhabit our urinary tracts help us in preventing diseases and disorders of this body system?
3. Given what you have learned in this chapter, why do you think physicians regularly order a urinalysis? What do these results tell them about the overall health of the individual?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

37

Neurons and Nervous Systems

PA Images/Alamy Stock Photo

Chapter Outline

37.1 Evolution of the Nervous System
37.2 Nervous Tissue
37.3 The Central Nervous System
37.4 The Peripheral Nervous System

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 5.10 How does the sodium-potassium pump work in a cell?

Section 29.1 From what embryonic structures are the brain, spinal cord, and (in vertebrates) vertebral column derived?

Section 30.1 How has the evolution of a large, complex brain allowed humans to become the most dominant species on the planet?

Multiple Sclerosis

Christina Applegate is an actor and producer who has appeared in multiple movies. In 2021, she revealed she had multiple sclerosis, or MS. MS is a disease that affects the nervous system. The first symptoms of MS tend to be weakness or tingling in the arms and legs, fatigue, a loss of coordination, and blurred vision. As the disease progresses, the individual may experience problems with speech, vision, movement, and numbness in the extremities. MS is an inflammatory disease that affects the myelin sheaths, which wrap parts of some nerve cells like insulation around an electrical cord. As these sheaths deteriorate, the nerves no longer conduct impulses normally. Most researchers think MS results from a misdirected attack on myelin by the body's immune system. Recent research indicates that, in some cases, viruses such as Epstein–Barr virus may be involved.

Almost one million people in the United States have MS, and it is the most common disease of the nervous system in young adults. Typically, the first symptoms occur between the ages of 20 and 40. The disease is not contagious, is rarely fatal, and does not appear to be inherited, although some studies suggest a genetic component associated with susceptibility to MS. Like many diseases that affect the nervous system, there is no cure. The good news is that the disease is not severe in almost 45% of cases.

As you read through this chapter, think about the following questions:

1. How did the evolution of the nervous system provide advantages to animals?
2. What specific types of processes occur uniquely in nervous tissues?

Following *the* Themes

CHAPTER 37 – NEURONS AND NERVOUS SYSTEMS

Evolution	Nervous systems have evolved in all types of animals except the simplest multicellular animals and range from relatively simple nerve nets to the highly complex human brain.
Nature of Science	Scientific studies have revealed many new insights into how the nervous system functions, leading to new treatments for some neurological diseases. However, much is yet to be discovered about this body system.
Biological Systems	Although animal nervous systems vary greatly, all are based on the use of neurons to respond to stimuli and control and coordinate the activity of the other body systems.

ISTUDY

37.1 Evolution of the Nervous System

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the nervous systems of cnidarians, planarians, and annelids.
2. Describe the essential features of a typical vertebrate nervous system.
3. Explain the major adaptations that evolved in the brains of mammals.

The nervous system is vital in complex animals, enabling them to seek food and mates and to avoid danger. It ceaselessly monitors internal and external conditions and makes appropriate changes to maintain homeostasis. A comparative study of animal nervous systems shows the evolutionary trends that led to the nervous system of mammals.

Invertebrate Nervous System Organization

The simplest multicellular animals, such as sponges, lack neurons (nerve cells) and therefore have no nervous system. However, their cells can respond to their environment and can communicate with each other, perhaps by releasing calcium or other ions; the most common example is closure of the osculum (central opening) in response to various stimuli.

Hydras, which are cnidarians with the tissue level of organization and radial symmetry, have a **nerve net** composed of neurons in contact with one another and with contractile cells in the body wall (Fig. 37.1*a*). They can contract and extend their bodies, move their tentacles to capture prey, and even turn somersaults. Sea anemones and jellyfish, which are also cnidarians, seem to have two nerve nets: A fast-acting one allows major responses, particularly in times of danger; the slower one coordinates slower and more delicate movements.

Planarians (flatworms) possess bilateral symmetry, and the organization of their nervous system reflect this form of symmetry. The system resembles a ladder, with two ventrally located lateral or longitudinal nerve cords (bundles of nerves) that extend from the cerebral ganglia to the posterior end of their body. Transverse nerves connect the nerve cords, as well as the cerebral ganglia, to the eyespots. **Cephalization,** or concentration of nervous tissue in the anterior or head region, has occurred. A cluster of neuron cell bodies is called a **ganglion** (*pl.,* ganglia), and the anterior cerebral ganglia of flatworms receive sensory information from photoreceptors in the eyespots and sensory cells in the auricles (Fig. 37.1*b*). The two lateral nerve cords allow a rapid transfer of information from the cerebral ganglia to the posterior end, and the transverse nerves between the nerve cords keep the movement of the two sides coordinated. Bilateral symmetry plus cephalization are two significant organizational trends in the development of the nervous system that provide adaptations for an active way of life.

Annelids (such as earthworms) (Fig. 37.1*c*) and arthropods (such as crabs) (Fig. 37.1*d*) are complex animals with the typical invertebrate nervous system. A brain is present, and a ventral nerve cord has a ganglion in each segment. The brain, which normally receives sensory information, controls the activity of the ganglia and assorted nerves, so that the muscle activity of the entire animal is coordinated.

A group of mollusks called cephalopods (such as squids) (Fig. 37.1*e*) show marked cephalization—the anterior end has a well-defined brain and well-developed sense organs, such as eyes. The cephalopods are widely regarded as the most intelligent invertebrates. Many are highly social creatures, and some, such as the octopus, have been observed to collect, transport, and assemble coconut shells for later use as a shelter, as well as demonstrating the ability to learn over time.

Vertebrate Nervous System Organization

Vertebrates have many more neurons than do invertebrates. For example, an insect's entire nervous system contains a total of about 1 million neurons, while a cat's nervous system may contain many thousand times that number (Fig. 37.1*f*). The human brain contains an estimated 86 billion neurons, and it is believed that some species of whales may have even more.

All vertebrates have a brain that controls the nervous system. It is customary to divide the vertebrate brain into the hindbrain, midbrain, and forebrain (Fig. 37.2), although the relative sizes of the parts vary greatly among species. The hindbrain, the most ancient part of the brain, regulates motor activity below the level of consciousness. For example, the lungs and heart function even when an animal is sleeping. The medulla oblongata contains control centers for breathing and heart rate. Coordination of motor activity associated with limb movement, posture, and balance eventually became centered in the cerebellum.

Several types of paired sensory receptors, including the eyes, ears, and olfactory structures, allow the animal to gather information from the environment (see Section 38.1). These sense organs are generally located at the anterior end of the animal, because this end is usually the first to enter new environments. The optic lobes are part of the midbrain, which was originally a center for coordinating reflexes involving the eyes and ears. In early vertebrate evolution, the forebrain was concerned mainly with the sense of smell. Beginning with the amphibians and continuing in the other vertebrates, the forebrain processes sensory information. Later, the thalamus evolved to receive sensory input from the midbrain and the hindbrain and pass it on to the cerebrum, the anterior part of the forebrain in vertebrates. In the forebrain, the hypothalamus is particularly concerned with homeostasis, and in this capacity, the hypothalamus communicates with the medulla oblongata and the pituitary gland.

The **central nervous system (CNS)** consists of the brain and spinal cord (Fig. 37.3). The **peripheral nervous system (PNS)** (Gk. *periphereia,* "circumference") consists of all the nerves and ganglia that lie outside the central nervous system. The CNS and PNS are considered in more detail in Sections 37.3 and 37.4.

The Mammalian Nervous System

The hindbrain and midbrain of mammals are similar to those of other vertebrates. However, the forebrain of mammals is greatly enlarged, due to the addition of an outermost layer called the *neocortex,* which is seen only in mammals. It functions in higher

ISTUDY

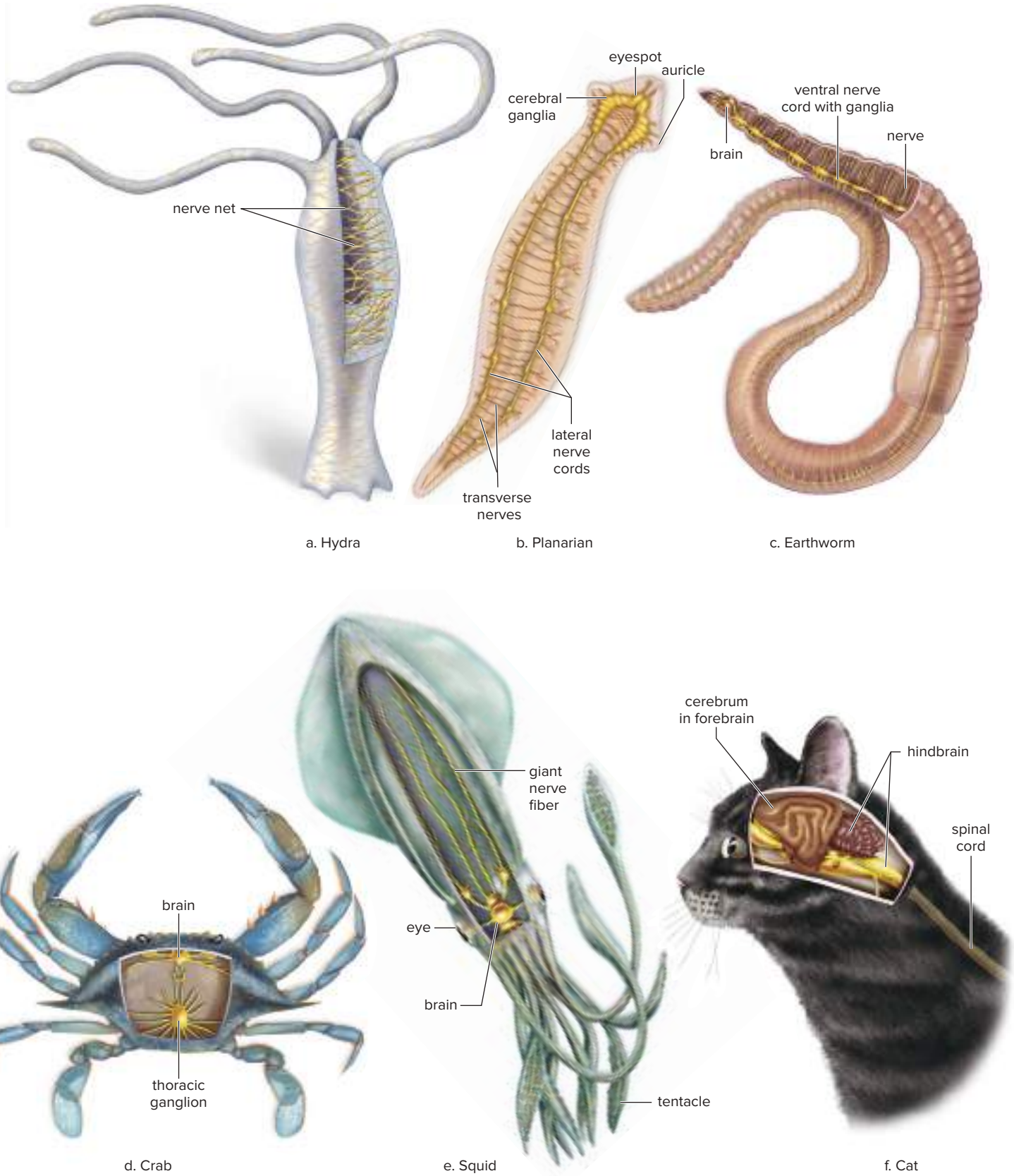


Figure 37.1 Evolution of the nervous system. **a.** The nerve net of a hydra, a cnidarian. **b.** In a planarian, a flatworm, the paired nerve cords with transverse nerves have the appearance of a ladder. **c.** The earthworm, an annelid, has a central nervous system consisting of a brain and a ventral solid nerve cord. It also has a peripheral nervous system consisting of nerves. **d.** The crab, an arthropod, has a nervous system that resembles that of annelids, but the ganglia are larger. **e.** The squid, a mollusk, has a definite brain with well-developed, giant nerve fibers that produce rapid muscle contraction, so the squid can move quickly. **f.** A cat, like other vertebrates, has a spinal cord (a dorsal tubular nerve cord) in the central nervous system.

ISTUDY

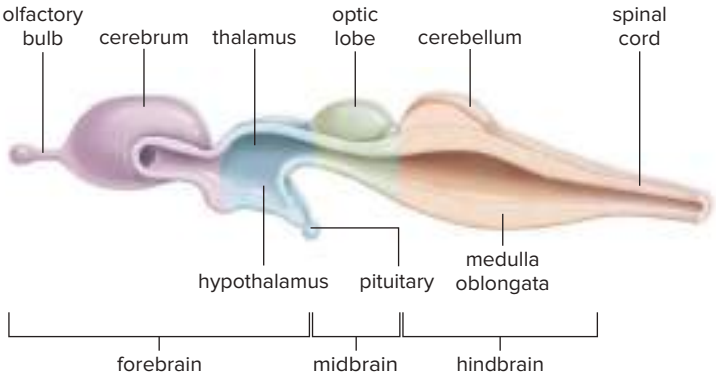


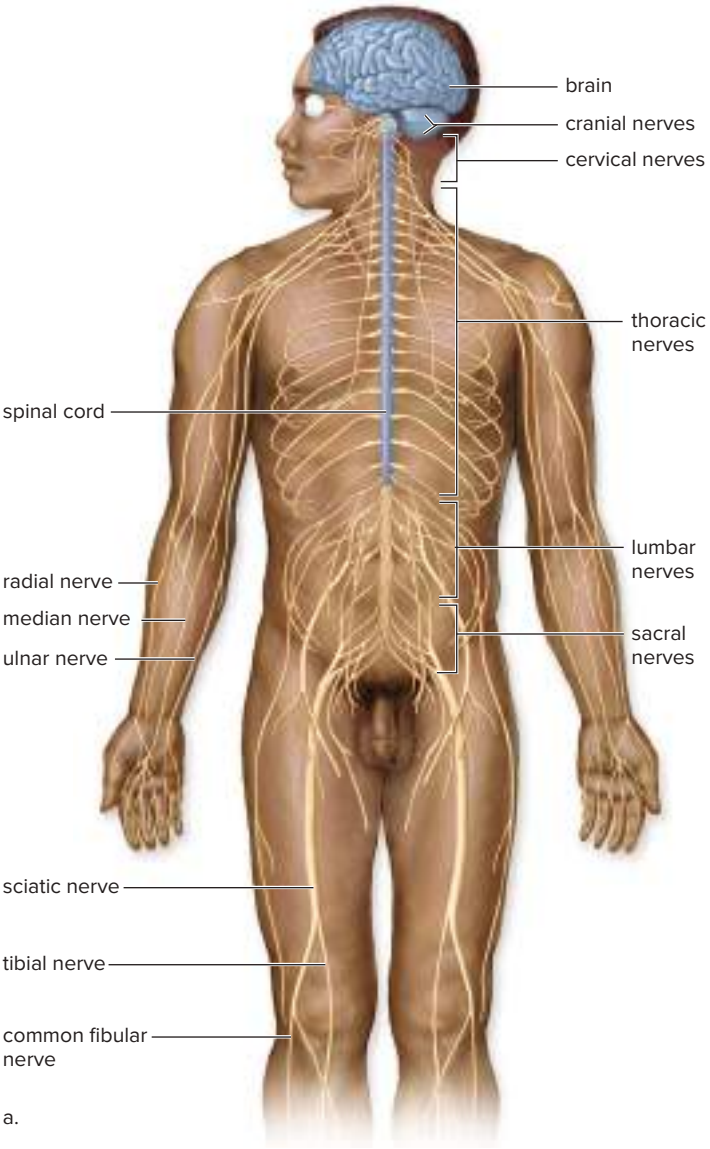
Figure 37.2 Organization of the vertebrate brain. The vertebrate brain is divided into three regions: the forebrain, the midbrain, and the hindbrain.

mental processes, such as spatial reasoning, conscious thought, and language.

Although all mammals have a neocortex, all neocortexes are not the same. Large variation has been observed in the number of crevices and folds, which can greatly increase the surface area and numbers of connections between regions. The frontal lobes (see Section 37.3) are especially large and complex in primates, and in humans other parts of the cortex are also enlarged and form very complex connections with other parts of the brain. It is likely that this greatly increased brain capacity allowed mammals, and especially humans, to become increasingly adept at higher mental activities, such as manipulating the environment, complex learning, and anticipating the future, all of which have provided tremendous evolutionary advantages.

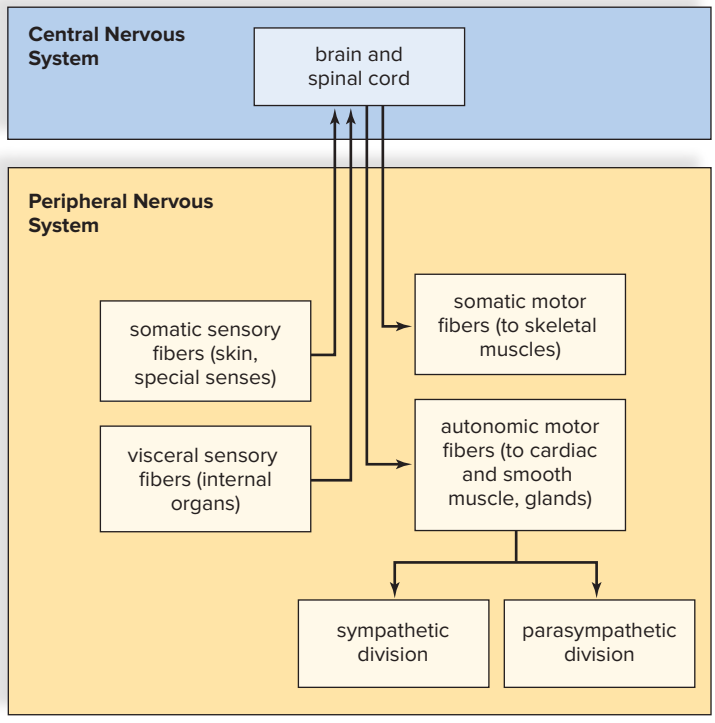
Check Your Progress 37.1

1. Define the terms *nerve net, ganglion,* and *brain.*
2. Describe the major functions of the hindbrain, midbrain, and forebrain.
3. Identify the specific location of the more recently evolved parts of the brain.



a.

Figure 37.3 Organization of the nervous system in humans. **a.** The central nervous system (CNS) is composed of the brain and spinal cord; the peripheral nervous system (PNS) consists of nerves. **b.** In the somatic system of the PNS, nerves conduct impulses from sensory receptors in the skin and internal organs to the CNS, and motor impulses from the CNS to the skeletal muscles. In the autonomic system, consisting of the sympathetic and parasympathetic divisions, motor impulses travel to smooth muscle, cardiac muscle, and glands.



b.

ISTUDY

37.2 Nervous Tissue

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the basic structure of a neuron and compare the functions of the three types of neurons.
2. Discuss the changes in ion concentrations inside and outside a neuron that result in an action potential.
3. Summarize the role of various neurotransmitters in propagating nerve impulses.

Although complex, nervous tissue is composed of just two principal types of cells. **Neurons,** also known as nerve cells, are the functional units of the nervous system. They receive sensory information, convey the information to an integration center such as the brain, and conduct signals from the integration center to effector structures, such as the glands and muscles. **Neuroglia** (sometimes just referred to as glial cells) serve as supporting cells, providing support and nourishment to the neurons.

Neurons and Neuroglia

Neurons vary in appearance depending on their function and location. They consist of three major parts: a cell body, dendrites, and an axon (Fig. 37.4). The **cell body** contains a nucleus and a variety of organelles. The **dendrites** (Gk. *dendron,* "tree") are short, highly branched processes that receive signals from the sensory receptors or other neurons and transmit them to the cell body. The **axon** (Gk. *axon,* "axis") is the portion of the neuron that conveys information to another neuron or to other cells. Axons can be bundled together to form nerves. For this reason, axons are often called **nerve fibers.** Many axons are covered by a white insulating layer called the **myelin sheath** (Gk. *myelos,* "spinal cord").

Neuroglia greatly outnumber neurons in the brain. Named for the Greek word for "glue," neuroglia cells were once thought to simply provide structural and nutritional support for neurons. However, some researchers now characterize neuroglia cells as the "supervisors" of the neurons, because some neuroglia cells play an important role in synapse formation and help neurons process information.

There are several types of neuroglia in the CNS, each with some specific known functions. The most numerous type of cell in the brain is the **astrocyte,** which serves many roles in maintaining neuron health and function. **Microglia** are phagocytic cells that help remove bacteria and debris. The myelin sheath is formed from the membranes of tightly spiraled neuroglia. In the CNS, neuroglial cells called **oligodendrocytes** form the myelin sheath. In the PNS, **Schwann cells** (also known as neurolemmocytes) perform this function, leaving gaps called **nodes of Ranvier.** Other forms of neuroglia include *ependymal cells* which line the ventricles of the brain, where they produce the cerebrospinal fluid, and *satellite cells* that surround neuron cell bodies in ganglia (ganglia are discussed in Section 37.4) and insulate the neurons and assist in the regulation of materials in and out of the neurons.

Types of Neurons

Neurons can be described in terms of their function and shape. **Motor neurons** take nerve impulses from the CNS to muscles or glands. Motor neurons are said to have a *multipolar* shape, because they have many dendrites and a single axon (Fig. 37.4*a*). Motor neurons cause muscle fibers to contract or glands to secrete, and therefore they are said to innervate these structures.

Sensory neurons take nerve impulses from sensory receptors to the CNS. The sensory receptor, which is the distal end of the

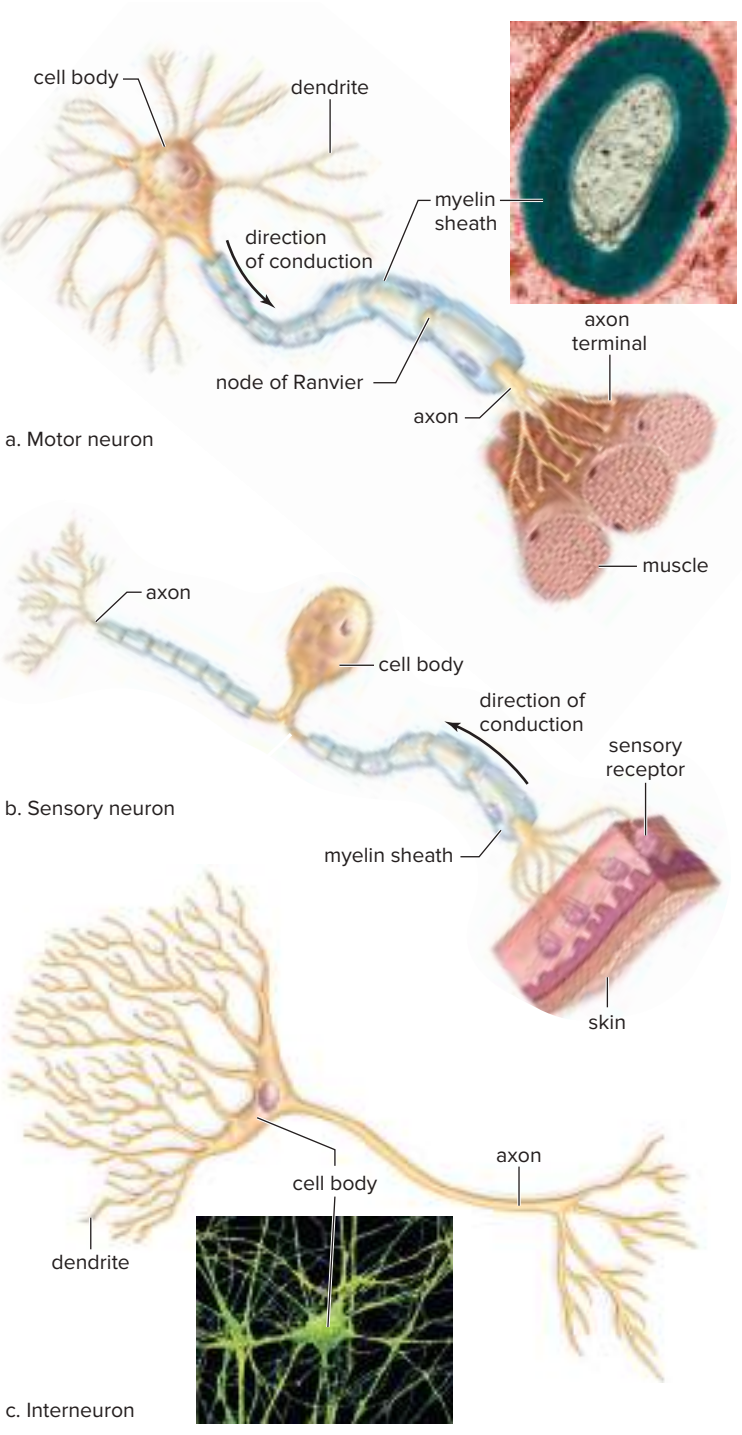


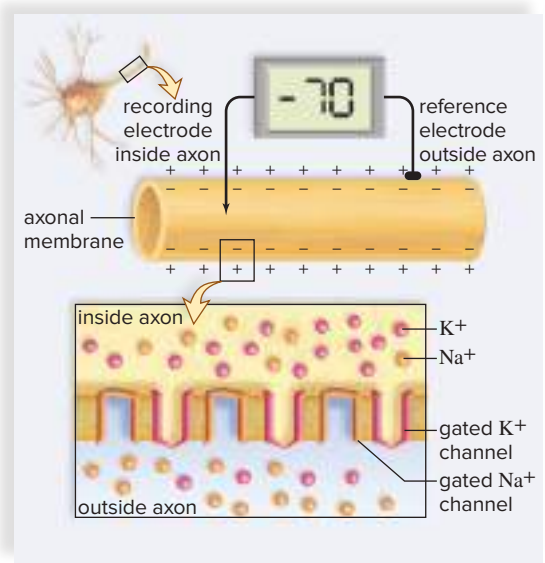
Figure 37.4 Neuron anatomy. **a.** Motor neuron. Note the branched dendrites and the single, long axon, which branches only near its tip. **b.** Sensory neuron with dendritelike structures projecting from the peripheral end of the axon. **c.** Interneuron (from the cortex of the cerebellum) with very highly branched dendrites. (a): (photo): Biophoto Associates/Science Source; (c): (photo): Steve Gschmeissner/Science Source

long axon of a sensory neuron, may be as simple as a naked nerve ending (a pain receptor), or it may be built into a highly complex organ, such as the eye or ear. Almost all sensory neurons have a structure that is termed *unipolar* (Fig. 37.4*b*). In unipolar neurons, the process that extends from the cell body divides into a branch that extends to the periphery and another that extends to the CNS.

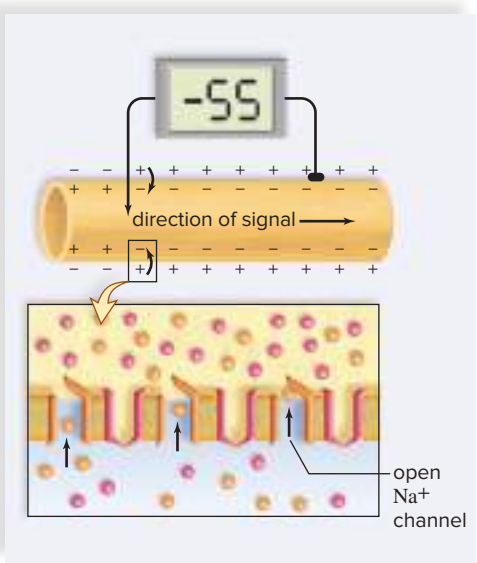
Interneurons (L. *inter,* "between") occur entirely within the CNS. Interneurons, which are typically multipolar (Fig. 37.4*c*), convey nerve impulses between various parts of the CNS. Some lie between sensory neurons and motor neurons; some take messages from one side of the spinal cord to the other or from the brain to the cord, and vice versa. They also form complex pathways in the brain, leading to higher mental functions, such as thinking, memory, and language.

Transmission of Nerve Impulses

In the early 1900s, scientists first hypothesized that the nerve impulse is an electrochemical phenomenon involving the movement of unequally distributed ions on either side of an axonal membrane, the plasma membrane of an axon. It was not until the 1960s, however, that experimental techniques were developed to test this hypothesis. Investigators were able to insert a tiny electrode into the giant axon of the squid *Loligo.* This internal electrode was then connected to a voltmeter, an instrument with a screen that shows voltage differences over time (Fig. 37.5). Voltage is a measure of the electrical potential difference between two points, which in this case is the difference between the electrode placed

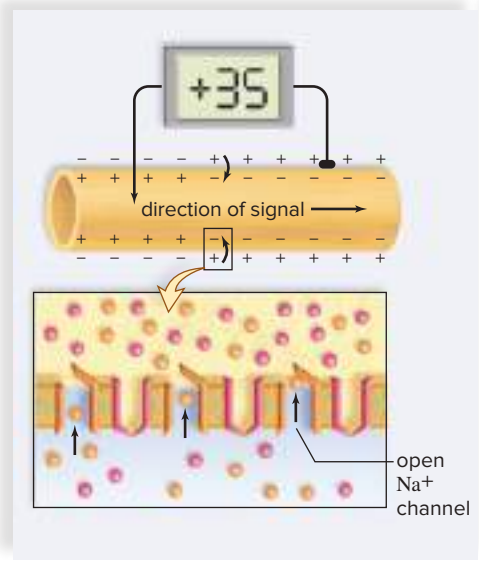


a. Resting potential: Na^+ outside the axon, K^+ and large anions inside the axon. Separation of charges polarizes the cell and causes the resting potential.

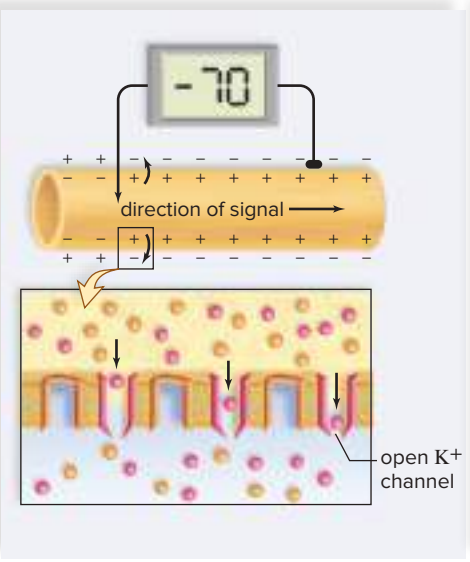


b. Stimulus causes the axon to reach its threshold; the axon potential increases from –70 to –55. The action potential has begun.

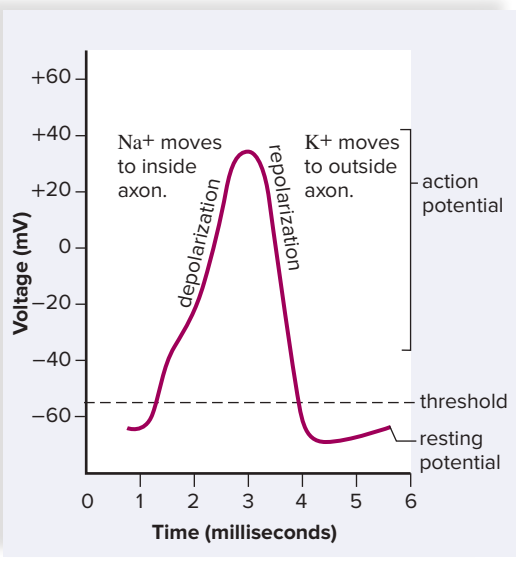
Figure 37.5 Resting and action potential of the axonal membrane.
a. Resting potential. A voltmeter indicates that the axonal membrane has a resting potential of –70 mV. There is a preponderance of Na^+ outside the axon and a preponderance of K^+ inside the axon. The permeability of the membrane to K^+ compared to Na^+ causes the inside to be negative compared to the outside. **b.** During an action potential, depolarization occurs when Na^+ gates open and Na^+ begins to move inside the axon. **c.** Depolarization continues until a potential of +35 mV is reached. **d.** Repolarization occurs when K^+ gates open and K^+ moves outside the axon. **e.** Graph of an action potential.



c. Depolarization continues as Na^+ gates open and Na^+ moves inside the axon.



d. Action potential ends: Repolarization occurs when K^+ gates open and K^+ moves to outside the axon. The sodium-potassium pump returns the ions to their resting positions.



e. An action potential can be visualized if voltage changes are graphed over time.

inside and another placed outside the axon. An electrical potential difference across a membrane is called a *membrane potential.*

Resting Potential

When the axon is not conducting an impulse, the voltmeter records a membrane potential equal to about −70 mV (millivolts), indicating that the inside of the neuron is more negative than the outside (Fig. 37.5*a*). This is called the **resting potential,** because the axon is not conducting an impulse.

The existence of this polarity can be correlated with a difference in ion distribution on either side of the axonal membrane. As Figure 37.5*a* shows, there is a higher concentration of sodium ions (Na^+) outside the axon, and a higher concentration of potassium ions (K^+) inside the axon.

The unequal distribution of these ions is due in part to the activity of the *sodium-potassium pump* (described in Section 5.3; see Fig. 5.10). This pump is an active transport system in the plasma membrane that pumps three sodium ions out of the axon and two potassium ions into the axon. The pump is always working, because the membrane is somewhat permeable to these ions, and they tend to diffuse toward areas of lesser concentration. Because the membrane is more permeable to potassium than to sodium, there are always more positive ions outside the membrane than inside; this accounts for some of the membrane potential recorded by the voltmeter. The axon cytoplasm also contains large, negatively charged proteins. Altogether, then, the voltmeter records that the resting potential is –70 mV inside the cell.

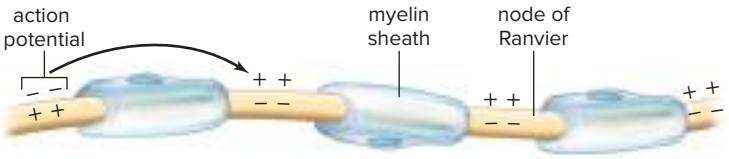
Action Potential

An **action potential** is a rapid change in polarity across a portion of an axonal membrane as the nerve impulse occurs. An action potential involves two types of gated ion channels in the axonal membrane, one that allows the passage of Na^+ and one that allows the passage of K^+. In contrast to ungated ion channels, which constantly allow ions across the membrane, gated ion channels open and close in response to a stimulus, such as a signal from another neuron.

The *threshold* is the minimum change in polarity across the axonal membrane that is required to generate an action potential. Therefore, the action potential is an all-or-none event. During *depolarization,* the inside of a neuron becomes positive because of the sudden entrance of sodium ions. If threshold is reached, many more sodium channels open, and the action potential begins. As sodium ions rapidly move across the membrane to the inside of the axon, the action potential swings up from –70 mV to +35 mV (Fig. 37.5*c*). This reversal in polarity causes the sodium channels to close and the potassium channels to open. As potassium ions leave the axon, the membrane potential swings down from +35 mV to –70 mV. In other words, a *repolarization* occurs (Fig. 37.5*d*). An action potential takes only 2 msec (milliseconds). To visualize such rapid fluctuations in voltage across the axonal membrane, researchers generally find it useful to plot the voltage changes over time (Fig. 37.5*e*).

Propagation of Action Potentials

In nonmyelinated axons (such as sensory receptors in the skin), the action potential travels down an axon one small section at a time, at a speed of about 1 m/sec (meter per second). In myelinated axons, the gated ion channels that produce an action potential are concentrated at the nodes of Ranvier. *Saltar* in Spanish means "to jump," so this mode of conduction, called **saltatory conduction,** means that the action potential "jumps" from node to node:



Speeds of 200 m/sec (about 450 miles per hour) have been recorded. As you can see, this speed is considerably greater than the rate of travel in nonmyelinated axons and allows what seems to be an instantaneous response.

As soon as an action potential has moved on, the previous section undergoes a **refractory period,** during which the Na^+ gates are unable to open. Notice, therefore, that the action potential cannot move backward and instead always moves down an axon toward its terminals. The intensity of a signal traveling down a nerve fiber is determined by how many nerve impulses are generated within a given time span.

Transmission Across a Synapse

Every axon branches into many fine endings, each tipped by a small swelling, called an axon terminal (Fig. 37.6). Each terminal lies very close to the dendrite (or the cell body) of another neuron. This region of close proximity is called a **synapse.** At a synapse, the membrane of the first neuron is called the *pre*synaptic membrane, and the membrane of the next neuron is called the *post*synaptic membrane. The small gap between the neurons is called the **synaptic cleft.**

A nerve impulse cannot cross a synaptic cleft. Transmission across a synapse is carried out by molecules called **neurotransmitters,** which are stored in synaptic vesicles. When nerve impulses traveling along an axon reach an axon terminal, gated channels for calcium ions (Ca^{2+}) open, and calcium enters the terminal. This sudden rise in Ca^{2+} stimulates synaptic vesicles to merge with the presynaptic membrane, and neurotransmitter molecules are released into the synaptic cleft. They diffuse across the cleft to the postsynaptic membrane, where they bind with specific receptor proteins.

Depending on the type of neurotransmitter and/or the type of receptor, the response of the postsynaptic neuron can be toward excitation or toward inhibition. Excitatory neurotransmitters that use gated ion channels are fast-acting. Other neurotransmitters affect the metabolism of the postsynaptic cell and therefore are slower-acting.

Neurotransmitters

More than 100 substances are known or suspected to be neurotransmitters in both the CNS and the PNS. Many of these can have opposing effects on different tissues. *Acetylcholine* (*ACh*) excites skeletal muscle but inhibits cardiac muscle. It has either an excitatory or an inhibitory effect on smooth muscle or glands, depending on their location. In the CNS, *norepinephrine* is important to dreaming, waking, and mood. *Dopamine* is involved in emotions, learning, and attention, and *serotonin* is involved in thermoregulation, sleeping, emotions, and perception. *Endorphins* are neurotransmitters that bind to natural opioid receptors in the brain. They

ISTUDY

are associated with the "runner's high" of exercisers, because they also produce a feeling of tranquility. Endorphins are produced by the brain not only when there is physical stress but also when emotional stress is present.

After a neurotransmitter has been released into a synaptic cleft and has initiated a response, it is removed from the cleft. In some synapses, the postsynaptic membrane contains enzymes that rapidly inactivate the neurotransmitter. For example, the enzyme *acetylcholinesterase* (*AChE*) breaks down acetylcholine. In other synapses, the presynaptic cell is responsible for reuptake, a process in which it rapidly reabsorbs the neurotransmitter, possibly for repackaging in synaptic vesicles or for molecular breakdown. The short existence of neurotransmitters at a synapse prevents continuous stimulation (or inhibition) of postsynaptic membranes.

Many drugs affecting the nervous system act by interfering with or potentiating the action of neurotransmitters. Such drugs can enhance or block the release of a neurotransmitter, mimic the action of a neurotransmitter or block the receptor, or interfere with the removal of a neurotransmitter from a synaptic cleft. Depression, a common mood disorder, appears to involve imbalances in norepinephrine and serotonin. Some antidepressant drugs, such as fluoxetine (Prozac), prevent the reuptake of serotonin, and others, including bupropion hydrochloride (Wellbutrin), prevent the reuptake of both serotonin and norepinephrine. Blocking reuptake prolongs the effects of these two neurotransmitters in networks of neurons in the brain that are involved in the emotional state.

Substance abuse disorders occur when a person takes a drug at a dose level and under circumstances that increase the potential for a harmful effect. In many cases, the individual often takes more of the drug than was intended. Individuals who abuse drugs are apt to display a psychological and/or physical dependence on the drug. With physical dependence, formerly called "addiction," more of the drug is needed to get the same effect, and withdrawal symptoms occur when the individual stops taking the drug. The Biological Systems feature, "Drug Use and Substance Abuse Disorders," describes the more common types of these drugs.

Synaptic Integration

A single neuron has many dendrites, plus the cell body, and both can have synapses with many other neurons. One thousand to 10,000 synapses per single neuron is not uncommon. Therefore, a neuron is on the receiving end of many excitatory and inhibitory signals. An excitatory signal produces a potential change that causes the neuron to become less polarized, or closer to triggering an action potential. An inhibitory signal causes the neuron to become hyperpolarized, or farther from an action potential.

Neurons integrate these incoming signals, and they do so specifically at the area of the neuron cell body where the axon emerges, called the *axon hillock*. **Integration** is the summing up of

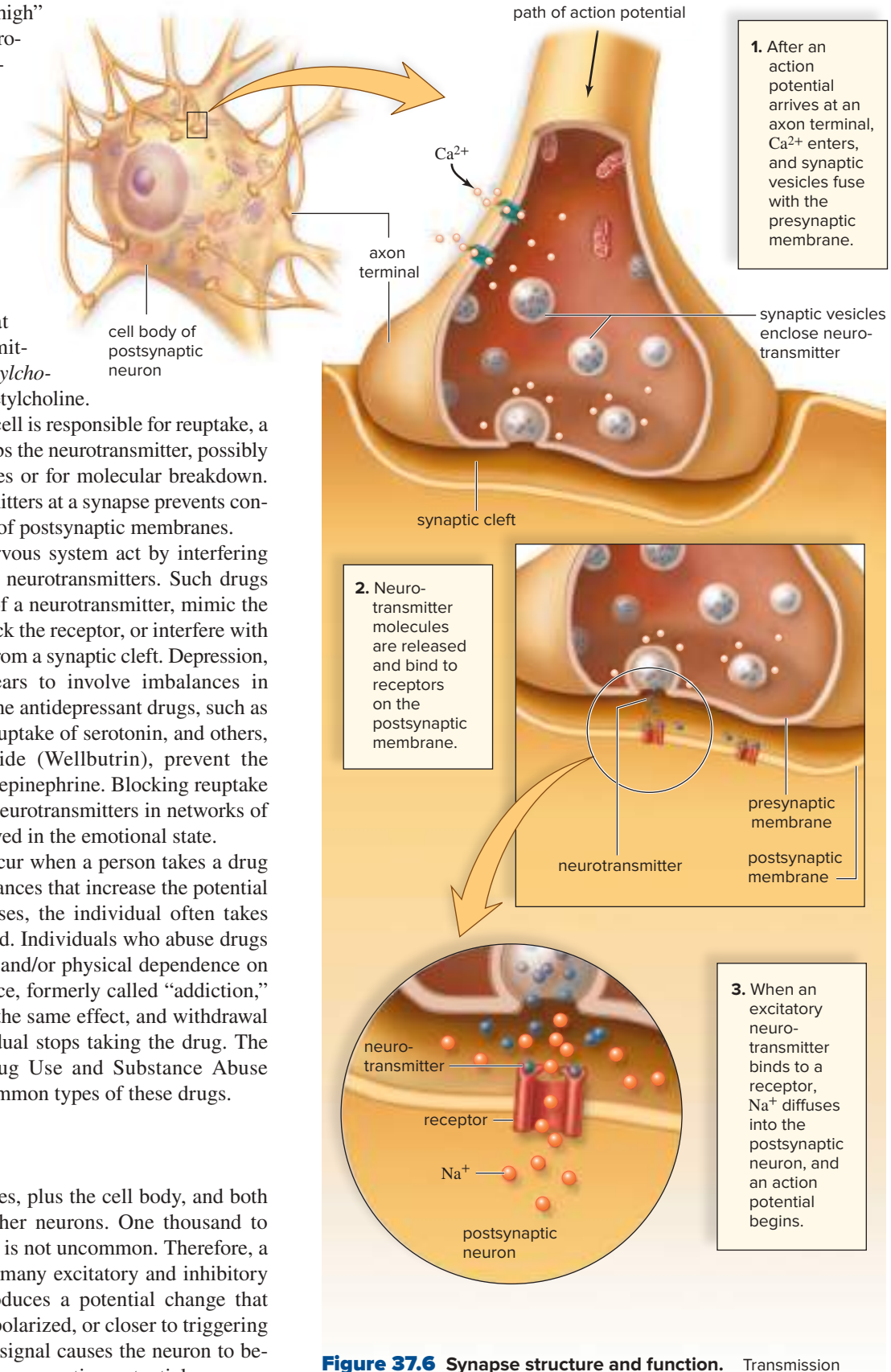


Figure 37.6 Synapse structure and function. Transmission across a synapse from one neuron to another occurs when a neurotransmitter is released at the presynaptic membrane, diffuses across a synaptic cleft, and binds to a receptor in the postsynaptic membrane. An action potential may begin.

ISTUDY

THEME Biological Systems

Drug Use and Substance Abuse Disorders

Substance use problems are apparent when a person takes a drug at a dose level and under circumstances that increase the potential for a harmful outcome. Addiction is present when more and more of the drug is needed to get the same effect, and withdrawal symptoms occur when the person stops taking the drug. This is true not only for teenagers and adults but also for newborn babies of females who are addicted to drugs.

Alcohol

With the exception of caffeine, alcohol (ethanol) consumption is the most socially accepted form of drug use in the United States, although consuming alcohol is illegal for those under 21. According to recent national surveys, 30% of high school students reported drinking some alcohol (down from 37.4% in 2014), and 14% binge drank (five-plus drinks in one setting) during the 30 days preceding the survey. Among adults, 86.4% reported they had consumed alcohol during their lifetime, with 56% stating they had used alcohol in the past month. Notably, 80% of college-age young adults drink. According to the CDC, underage drinking in college contributes to over 3,900 deaths per year and over 120,000 emergency room visits.

Alcohol acts as a *depressant* on many parts of the brain by increasing the action of GABA, an inhibitory neurotransmitter. Depending on the amount consumed, the effects of alcohol on the brain can lead to a feeling of relaxation, lowered inhibitions, impaired concentration and coordination, slurred speech, and vomiting. If the blood level of alcohol becomes too high, coma or death can occur.

Alcohol is a carbohydrate, and as such, it can be converted into energy for the body to use. However, alcoholic beverages lack the vitamins, minerals, essential amino acids, and fatty acids the body needs. For this reason, many people with alcohol use disorder are vitamin-deficient, undernourished, and prone to illness.

Nicotine

According to the CDC, the numbers have been decreasing since 2005. Still, statistics indicate that 12.5% of adults and 7.4% of 18- to 24-year-olds in the United States smoked cigarettes. While this has traditionally involved using tobacco cigarettes, increasingly this is now associated with "vaping," the use of e-cigarettes (see the Nature of Science feature, "Are E-cigs Safe?," in Section 35.3).

When tobacco is smoked or chewed, nicotine is rapidly delivered throughout the body. It causes a release of epinephrine from the adrenal glands, increasing blood sugar levels and initially causing a feeling of stimulation. As blood sugar falls, depression and fatigue set in, causing the individual to seek more nicotine. In the CNS, nicotine stimulates neurons to release dopamine, a neurotransmitter that promotes a temporary sense of pleasure, reinforcing dependence on the drug. About 70% of people who try smoking become addicted.

As mentioned in earlier chapters, smoking is strongly associated with serious diseases of the cardiovascular and respiratory systems. Once addicted, however, only 10–20% of people who smoke are able to quit. Most medical approaches to quitting involve the administration of nicotine in safer forms, such as skin patches, gum, or a nicotine inhaler, so that withdrawal symptoms can be minimized while dependence is gradually reduced.

"Club" Drugs

Ecstasy (MDMA [3,4-methylenedioxymethamphetamine]), Rohypnol (flunitrazepam), and ketamine are examples of drugs used by many teens and young adults who attend night-long dances called raves or trances. Of course, these drugs may be used by others as well. MDMA (or "XTC") is chemically similar to methamphetamine, and many people who use the drug say it increases their feelings of well-being and friendliness, even love, for other people. However, it can cause many of the same side effects as other stimulants, such as an increase in heart rate and blood pressure, muscle tension, and blurred vision. In high doses, MDMA can interfere with temperature regulation, leading to hyperthermia, followed by damage to the liver, kidneys, and heart. Chronic MDMA use can also damage memory and may lead to depression. Perhaps because of these side effects, several recent surveys have reported that MDMA use has been decreasing among youths aged 12 to 17.

Rohypnol is in the same class of drugs as diazepam (Valium), a popular sedative. When mixed with alcohol, Rohypnol (or "roofies") can render a person incapable of resisting and has been used by the perpetrators of sexual assault. It can also produce anterograde amnesia, which means survivors of an assault may not remember events they experienced while under the influence of the drug. Ketamine, or "Special K," is an anesthetic that veterinarians use to perform surgery on animals. When administered to humans, it typically induces a dreamlike state, sometimes with hallucinations. Because it can render the person unable to move, ketamine has also been used as a date rape drug. At higher doses, ketamine can cause dangerous reductions in heart and respiratory functions.

Methamphetamine

Methamphetamine, also called "meth" or "crank," is a powerful CNS stimulant. Meth is often synthesized or "cooked" in makeshift home laboratories, usually starting with either ephedrine or pseudoephedrine, common ingredients in many cold and asthma medicines. Many states have passed laws making these medications more difficult to purchase. Meth is usually produced as a powder (speed) that can be snorted, or as crystals (crystal meth) that can be smoked. The most immediate effect is an initial "rush" of euphoria, due to high amounts of dopamine released in the brain. The person may also experience an increased energy level, alertness, and elevated mood. After the initial rush, a state of high agitation typically occurs that, in some individuals, leads to violent behavior. Chronic use can lead to paranoia, irritability, hallucinations, insomnia, tremors, hyperthermia, cardiovascular collapse, and death. Similar to cocaine, long-term meth use renders the brain less able to produce normal levels of dopamine. The individuals may have strong cravings for more meth, without being able to reach a satisfactory high.

Cathinones or "Bath Salts"

A type of newer illicit drug that's become available is the cathinones, commonly called "bath salts." These synthetic powders are sold in colorful containers under innocent-sounding names like "Blue Wave" and "Vanilla Sky." However, unlike the innocuous sodium salts and fragrances that have been added to bathwater for centuries, these new products often contain synthetic amphetamine-like chemicals such as methylenedioxypyrovalerone and mephedrone. Although they are often labeled "Not For Human Consumption," when ingested, injected, or snorted, these chemicals inhibit the reuptake of several neurotransmitters, producing a euphoric

ISTUDY

sensation. Unfortunately, a long list of side effects includes severe chest pain, hallucinations, extreme paranoia, and violent episodes often leading to suicide. In response to dramatic increases in bath salts–related emergency room visits and deaths, in 2012 a law was passed that makes it illegal to possess or sell many of the chemicals required to make bath salts. Unfortunately, illicit drugmakers are presumably working to develop new "designer drugs" to circumvent these bans.

Cocaine and Crack

Cocaine is an alkaloid derived from the shrub *Erythroxylon coca*. Approximately 35 million Americans have used cocaine by sniffing/snorting, injecting, or smoking it. Cocaine is a powerful *stimulant* in the CNS that interferes with the reuptake of dopamine at synapses, increasing overall brain activity. The result is a rush of well-being that lasts 5–30 minutes. This is followed by a crash period, characterized by fatigue, depression, irritability, and lack of interest in sex. In fact, men who use cocaine often develop erectile disfunction.

Crack is the name given to cocaine that is processed to a free base form for smoking. The term *crack* refers to the crackling sound heard when the drug is smoked. Smoking allows high doses of the drug to reach the brain rapidly, providing an intense and immediate high, or "rush." Approximately 8 million Americans use crack.

Cocaine is highly addictive; with continued use, the brain makes less dopamine to compensate for a seemingly endless supply. The individual experiences withdrawal symptoms and an intense craving for the drug. Over time, the brain of a person who uses cocaine becomes less active (Fig. 37A).

Heroin

Heroin is derived from the resin or sap of the opium poppy plant, which is widely grown—from Turkey to Southeast Asia and parts of Latin America. Drugs derived from opium are called opiates, a class that also includes morphine and oxycodone. The number of people using heroin in the United States has nearly doubled in the last few years, partly due to an increased supply.

As with other illicit drugs, addiction is common. Heroin binds to receptors in the brain that normally bind to endorphins, naturally occurring neurotransmitters that kill pain and produce feelings of tranquility. After heroin is injected, snorted, or smoked, a feeling of euphoria, along with the relief of any pain, occurs within a few minutes. With repeated heroin use, the body's production of endorphins decreases. Tolerance develops, so that the individual needs to take more of the drug just to prevent withdrawal symptoms (tremors, restlessness, cramps, vomiting), and the original euphoria is no longer felt.

People who use heroin long-term are at increased risk for acquiring hepatitis, HIV/AIDS, and various bacterial infections from the use of shared needles, and people who use heroin heavily may experience convulsions and death by respiratory arrest.

Heroin addiction can be treated with synthetic opiate compounds, such as methadone or suboxone, that decrease withdrawal symptoms and block heroin's effects. However, methadone itself can be addictive, and methadone-related deaths are on the rise.

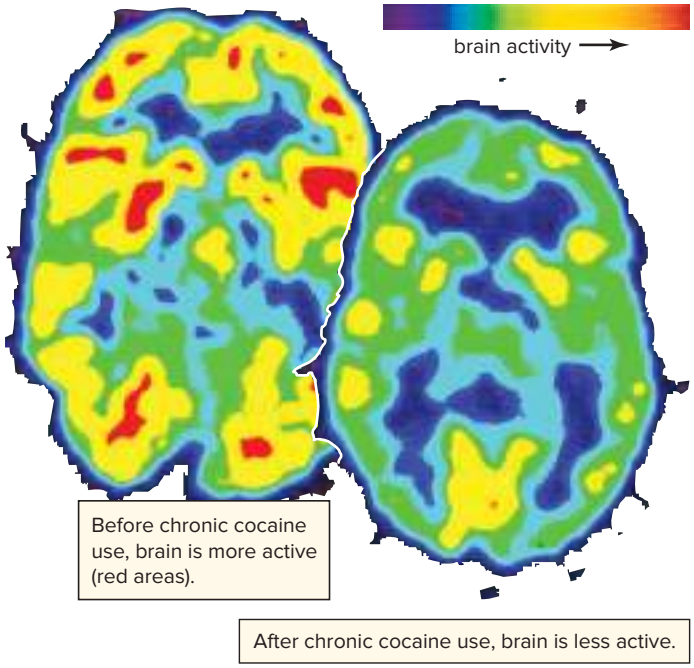


Figure 37A Cocaine use and the brain. Brain activity before and after the use of cocaine. Brookhaven National Laboratory/Science Source

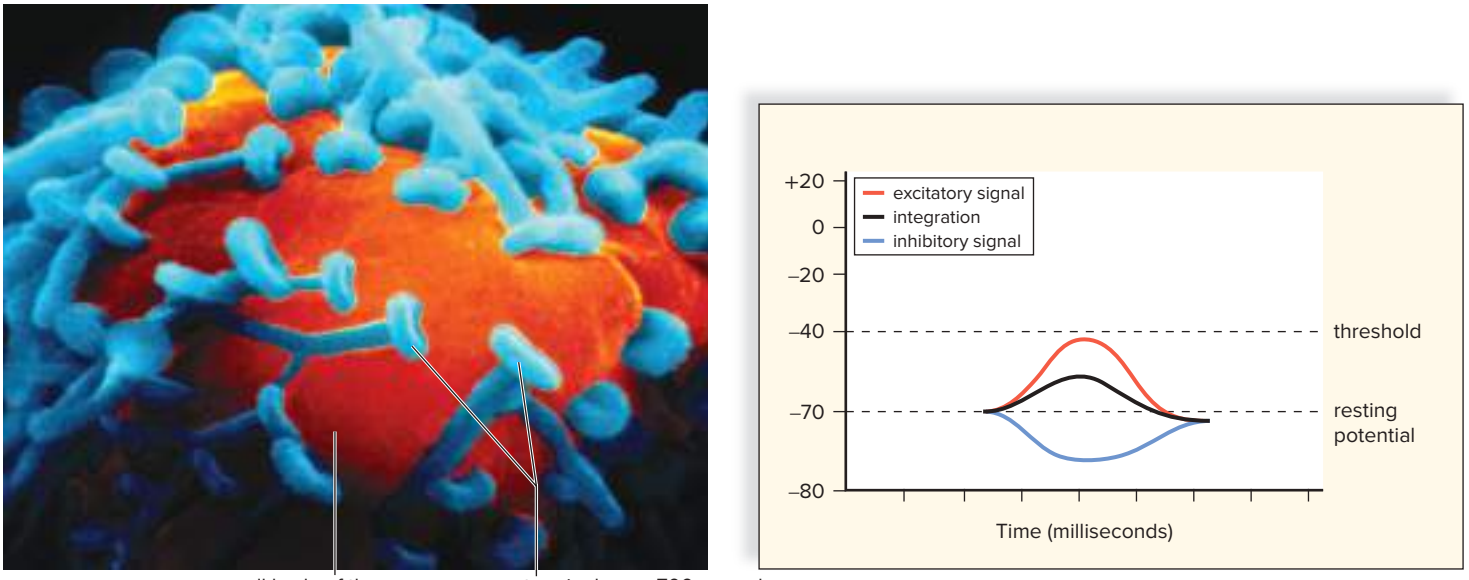


Figure 37.7 Synaptic integration. **a.** Many neurons synapse with a cell body. **b.** Both inhibitory signals (blue) and excitatory signals (red) are summed up in the dendrite and cell body of the postsynaptic neuron. Only if the combined signals cause the membrane potential to rise above threshold does an action potential occur. In this example, threshold was not reached. (a): Omikron/Science Source

excitatory and inhibitory signals (Fig. 37.7). If a neuron receives many excitatory signals (either from different synapses or at a rapid rate from one synapse), chances are the axon will transmit a nerve impulse. In Figure 37.7*b,* the inhibitory signals (shown in blue) are canceling out the excitatory signals, resulting in no nerve impulse.

Check Your Progress 37.2

1. Explain why a nerve impulse travels more quickly down a myelinated axon than down an unmyelinated axon.
2. Describe the movement of specific ions during the generation of a nerve impulse.
3. Describe how the bite of a black widow spider, which contains a powerful AChE inhibitor, might cause the common symptoms of muscle cramps, salivation, fast heart rate, and high blood pressure.

37.3 The Central Nervous System

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the anatomy of the spinal cord and spinal nerves.
2. List the major regions of the human brain and describe some major functions of each.
3. Compare the causes and types of symptoms seen in some common CNS disorders.

The central nervous system (CNS) consists of the spinal cord and brain. It has three specific functions:

- *Receives sensory input*—sensory receptors in the skin and other organs respond to external and internal stimuli by generating nerve impulses that travel to the CNS.
- *Performs integration*—the CNS sums up the input it receives from all over the body.
- *Generates motor output*—nerve impulses from the CNS go to the muscles and glands; muscle contractions and gland secretions are responses to stimuli received by sensory receptors.

As an example of the operation of the CNS, consider the events that occur as a person raises a glass to their lips. Continuous sensory input to the CNS from the eyes and hand informs the CNS of the position of the glass, and the CNS continually sums up the incoming data before commanding the hand to proceed. At any time, integration with other sensory data might cause the CNS to command a different motion instead. The lips detect the arrival of the glass, passing this information to the CNS, which then directs the actions of drinking.

The spinal cord and the brain are both protected by bone; the spinal cord is surrounded by vertebrae, and the brain is enclosed by the skull. Both the spinal cord and the brain are wrapped in three protective membranes known as **meninges** (Fig. 37.8). The spaces between the meninges are filled with **cerebrospinal fluid,** which cushions and protects the CNS. Cerebrospinal fluid, produced by a type of glial cell, is contained in the central canal of the spinal cord and within the **ventricles** of the brain, which are interconnecting spaces that produce cerebrospinal fluid and serve as reservoirs for it. **Meningitis** (inflammation of the meninges) is a serious disorder caused by a number of bacteria or viruses that can invade the meninges.

The Spinal Cord

The **spinal cord** is a bundle of nervous tissue enclosed in the vertebral column (see Fig. 37.13); it extends from the base of the brain to the vertebrae just below the rib cage. The spinal cord has two

ISTUDY

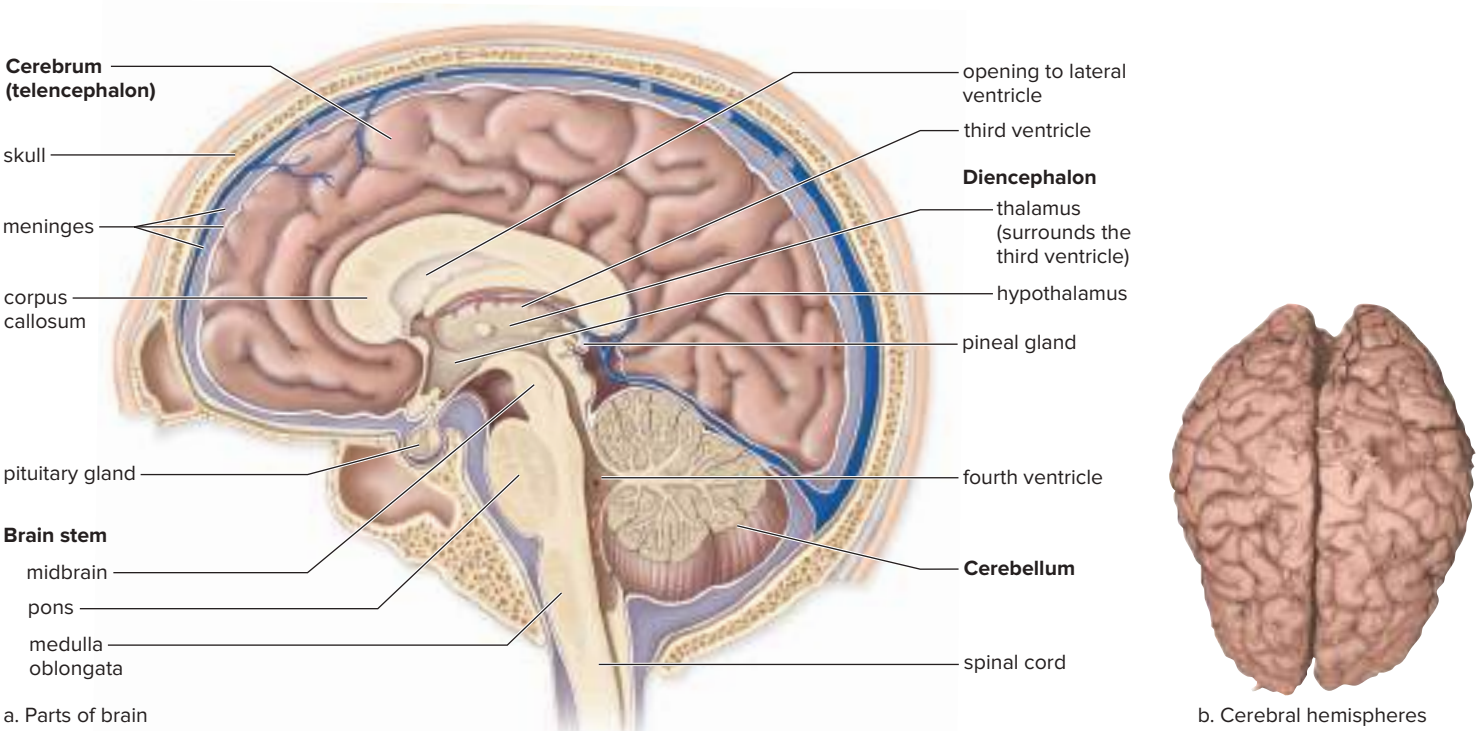


Figure 37.8 The human brain. **a.** The right cerebral hemisphere is shown, along with other, closely associated structures. The hemispheres are connected by the corpus callosum. **b.** The cerebrum is divided into the right and left cerebral hemispheres. (b): Dr. Colin Chumbley/Science Source

main functions: (1) it is the center for many **reflex actions,** which are automatic responses to external stimuli, and (2) it provides a means of communication between the brain and the spinal nerves, which leave the spinal cord.

A cross section of the spinal cord reveals that it is composed of a central portion of gray matter and a peripheral region of white matter. The **gray matter** consists of cell bodies and unmyelinated fibers. In cross section, it is shaped like a butterfly, or the letter *H,* with two dorsal (posterior) horns and two ventral (anterior) horns surrounding a central canal. The gray matter contains portions of sensory neurons and motor neurons, as well as short interneurons that connect sensory and motor neurons.

Myelinated long fibers of interneurons that run together in bundles called **tracts** give **white matter** its color. These tracts connect the spinal cord to the brain. They are like a busy superhighway, by which information continuously passes between the brain and the rest of the body. In the dorsal part of the cord, the tracts are primarily ascending, taking information *to* the brain. Ventrally, the tracts are primarily descending, carrying information *from* the brain. Because the tracts cross over at one point, the left side of the brain controls the right side of the body and the right side of the brain controls the left side of the body. Researchers estimate there are about 100,000 miles of myelinated nerve fibers in the adult human brain.

If the spinal cord is damaged as a result of an injury, paralysis may result. If the injury occurs in the cervical (neck) region, all four limbs are usually paralyzed, a condition known as quadriplegia. If the injury occurs in the thoracic region, the lower body may be paralyzed, a condition called paraplegia.

Other disease processes can cause paralysis. In **amyotrophic lateral sclerosis (ALS)**, or Lou Gehrig disease, motor neurons in the brain and spinal cord degenerate and die, leaving people weakened, then paralyzed, then unable to breathe properly. Although there is no cure, some drugs slow the disease; others are currently in clinical trials.

The Brain

Nerve impulses are the same in all neurons, so how is it that the stimulation of our eyes causes us to see and the stimulation of our ears causes us to hear? Essentially, the central nervous system carries out the function of integrating incoming data. The **brain** allows us to perceive our environment, to reason, and to remember. The exact processes by which the brain generates these higher functions remain largely mysterious.

The Cerebrum

The **cerebrum** is the largest, outermost portion of the brain in humans (Fig. 37.8*a*). The cerebrum is the last center to receive sensory input and carry out integration before commanding voluntary motor responses. It communicates with and coordinates the activities of the other parts of the brain. The cerebrum also contains the two lateral ventricles; the third ventricle is surrounded by the diencephalon, and the fourth ventricle lies between the cerebellum and the pons.

Cerebral Hemispheres The cerebrum is divided into two halves, called **cerebral hemispheres** (Fig. 37.8*b*). A deep groove known as the *longitudinal fissure* divides the cerebrum into the right and left hemispheres. Each hemisphere receives information from and controls the opposite side of the body. Although the hemispheres appear the same, the right hemisphere is associated with artistic and musical ability, emotion, spatial relationships, and

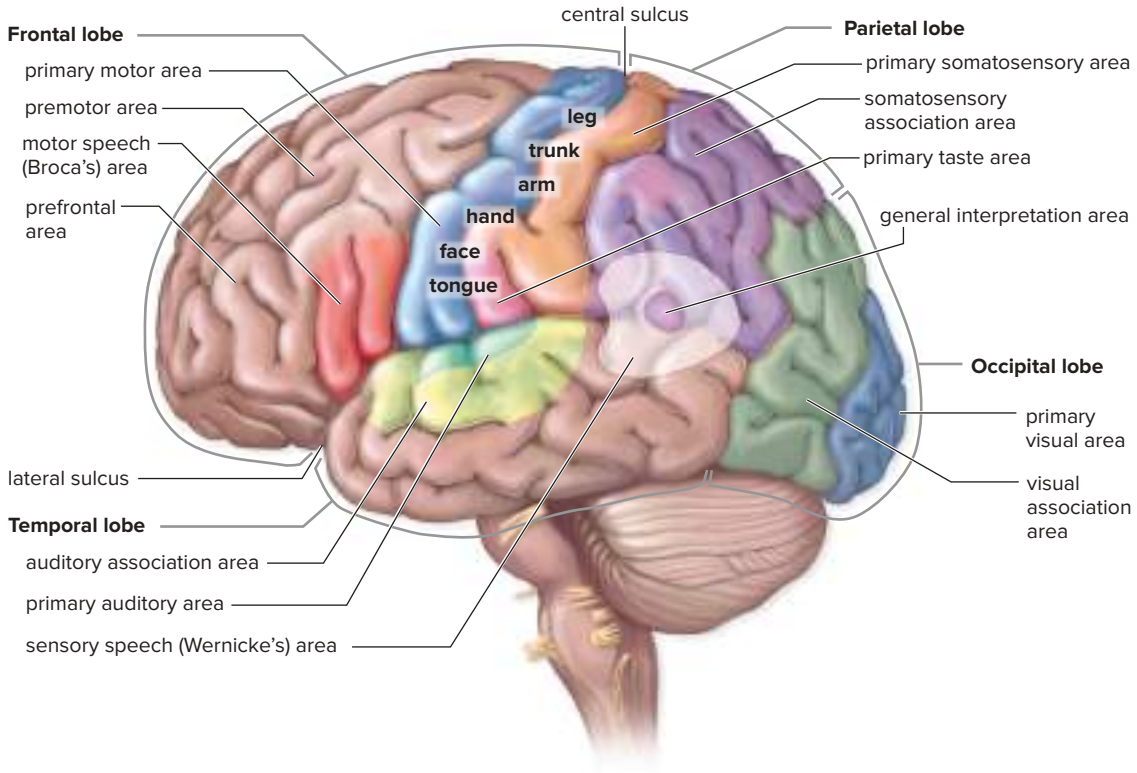


Figure 37.9 The lobes of a cerebral hemisphere. Each cerebral hemisphere is divided into four lobes: frontal, parietal, temporal, and occipital. These lobes contain centers for reasoning and movement, somatic sensing, hearing, and vision, respectively.

pattern recognition. The left hemisphere is more adept at mathematics, language, and analytical reasoning. The two cerebral hemispheres are connected by a bridge of tracts within the *corpus callosum.*

Shallow grooves called *sulci* (*sing., sulcus*) divide each hemisphere into paired lobes (Fig. 37.9). The *frontal lobes* lie toward the front of the hemispheres and are associated with motor control, memory, reasoning, and judgment. For example, if a fire occurs, the frontal lobes enable you to decide whether to exit via the stairs or the window, or if it is winter, how to dress if the temperature plummets to subzero. The left frontal lobe contains *Broca's area,* which organizes motor commands to produce speech.

The *parietal lobes* lie posterior to the frontal lobe and are concerned with sensory reception and integration, as well as taste. A *primary taste area* in the parietal lobe accounts for taste sensations.

The *temporal lobes* are located laterally. A primary auditory area in each temporal lobe receives information from the ears. The *occipital lobes* are the most posterior lobes. A *primary visual area* in each occipital lobe receives information from the eyes.

The Cerebral Cortex The **cerebral cortex** is a thin (less than 5 mm thick) but highly convoluted outer layer of gray matter that covers the cerebral hemispheres. The convolutions increase the surface area of the cerebral cortex. It contains tens of billions of neurons and is the region of the brain that accounts for sensation, voluntary movement, and all the thought processes required for learning, memory, language, and speech.

Two regions of the cerebral cortex are of particular interest. The *primary motor area* is in the frontal lobe just ventral to (before) the central sulcus. Voluntary commands to skeletal muscles begin in the primary motor area, and each part of the body is controlled by a certain section. The size of the section indicates the precision of motor control. For example, controlling the muscles of the face and hands takes up a much larger portion of the primary motor area than controlling the entire trunk. The *primary somatosensory area* is just dorsal to the central sulcus in the parietal lobe. Sensory information from the skin and skeletal muscles arrives here.

When the blood supply to any area of the brain is disrupted, a **stroke** results. Stroke is the third leading cause of death in the United States. The most common type is *ischemic* stroke, in which there is a sudden loss of blood supply to an area of the brain, usually due to arterial blockage or clot formation. The area(s) of the brain affected by a stroke will determine what type of symptoms arise. For example, a stroke that affects only the motor areas of the cerebral cortex might paralyze one side of the body, while a stroke involving Broca's area might render the person unable to speak.

Although strokes are most common in older people, recent studies have indicated up to a 51% increase in strokes in males ages 15 through 34, as well as a 17% increase in females of the same age. Because many of the risk factors for stroke are similar to those for cardiovascular disease, see the Nature of Science feature, "Recent Findings About Preventing Cardiovascular Disease," in Section 32.3 to learn how to reduce your risk.

Basal Nuclei Although the bulk of the cerebrum is composed of white matter (tracts), masses of gray matter are located deep within the white matter. These are called basal nuclei (or basal ganglia) and they integrate motor commands, ensuring that proper muscle groups are activated or inhibited. **Parkinson's disease (PD)** is a brain disorder characterized by tremors, speech difficulties, and difficulty standing and walking. PD results from a loss of the cells in the basal nuclei that normally produce the neurotransmitter dopamine.

ISTUDY

THEME Nature of Science

Understanding Parkinson's Disease

Parkinson's disease (PD) was first described as the "shaking palsy" in 1817 by the English surgeon James Parkinson, for whom the disease was later named. It affects about 1 million people in the United States and over 6 million worldwide. PD is most common in people over 60 and rarely affects those under 40 (actor Michael J. Fox is one exception). The symptoms of PD include bradykinesis (slow movements), tremors, rigidity of the limbs and trunk, and impaired speech, balance, and coordination. PD is caused by injury to, or the death of, dopamine-producing neurons in the basal ganglia (specifically, the substantia nigra) deep in the forebrain, which normally help control voluntary movement. The initial events that cause the neuron damage are not well understood, and there is no cure.

Early Studies of PD

In 1976, Barry Kidston was a graduate student in chemistry at the University of Maryland. He wanted to experiment with hard drugs and decided to synthesize a narcotic he had read about in a 1947 scientific paper. The drug, called MPPP, was said to be less addictive than morphine and was technically not illegal (although it is now). Kidston was successful initially and was apparently able to achieve a satisfactory "high" by intravenously injecting himself with the compound, which he had synthesized in a makeshift lab in his parent's basement. His luck ran out, however, when he accidentally produced a related compound, called MPTP, instead of MPPP. Soon after injecting the MPTP, Kidston's speech became slurred, he had trouble walking, and within 3 days he could hardly move.

Kidston's doctors were baffled, but after a neurologist noted that his symptoms resembled Parkinson's disease, he was treated with anti-PD drugs, and he dramatically improved. For 2 years, he was able to function well on medication, but then he died, somewhat ironically, from a cocaine overdose. His autopsy revealed a substantial loss of dopamine-producing cells in the substantia nigra, which is a hallmark of PD (Fig. 37B).

Kidston's case was published in a medical journal, but it was barely noticed until 1981, when six people who used IV drugs turned up at various emergency rooms in the San Francisco area, all showing very similar symptoms, as if they had "turned to stone." Some quick investigative work turned up the fact that all had tried some new heroin that had become available on the street. When some of this material was analyzed, one of the toxicologists remembered seeing the article about Kidston a few years earlier and quickly determined that the drug all the new patients had injected contained MPTP.

Developing Treatments for PD

As news that PD could be induced by MPTP in humans reached the biomedical research community, scientists were quickly able to demonstrate that MPTP could also induce PD in animals such as monkeys and mice. Since the mid-1980s, animals with MPTP-induced PD have been used in hundreds of studies that have advanced our understanding of how PD develops, and they have been instrumental in the development of therapies, such as the following:

- *Drugs that increase dopamine.* These vary from L-dopa, which is converted into dopamine in the brain, to newer drugs, such as entacapone, which inhibit the normal breakdown of dopamine.
- *Surgical treatments.* The most common of these is deep brain stimulation, in which an electrode is inserted into an area of the brain called the subthalamic nucleus, which seems to be overactive in PD. Coincidentally, one of the original patients who developed Parkinson's symptoms after using IV drugs had this procedure done and has made very significant improvements in their condition.
- *Stem cell transplants.* The use of fetal stem cells to replace diseased neurons in PD remains controversial. However, results from animal models are encouraging, and groups such as the California Institute for Regenerative Medicine and the Michael J. Fox Foundation for Parkinson Disease Research continue to support stem cell research as a potential cure for PD.
- *Gene therapy.* A 2014 study showed significant reduction in Parkinson's symptoms in 15 patients who were injected with ProSavin, which delivers genes for dopamine synthesis directly into the brains of people with the disease.

Questions to Consider

1. What are some other instances in which accidental findings have resulted in scientific discoveries?
2. Should the families of people like Kidston, who accidentally benefit medicine, be compensated financially?
3. Why is it so difficult to determine the specific causes of PD and other brain diseases?

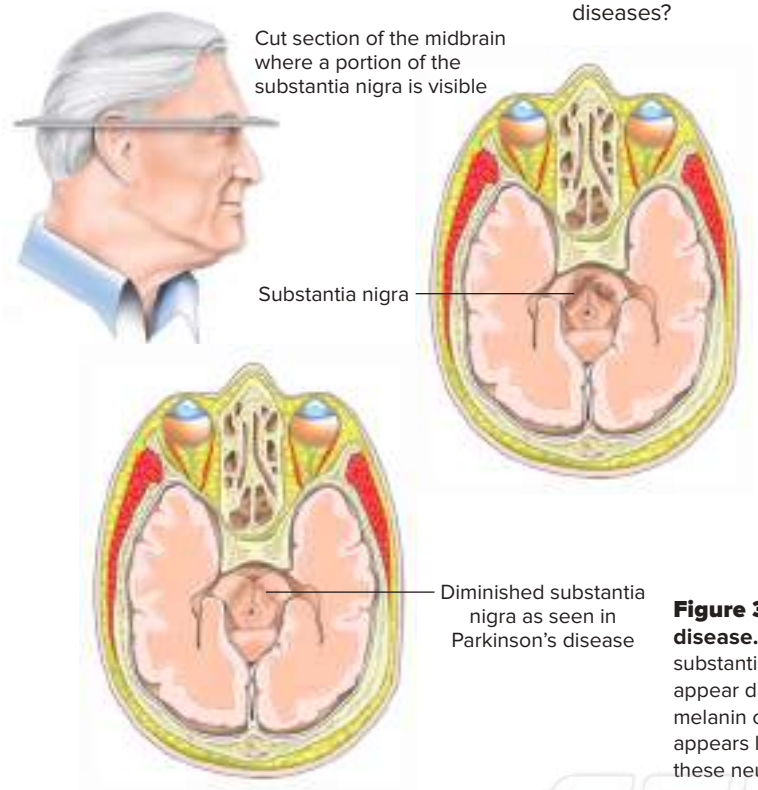


Figure 37B Parkinson's disease. Neurons in the substantia nigra of the brain appear dark because of a high melanin content. In PD, this area appears lighter due to a loss of these neurons.

Other Parts of the Brain

The hypothalamus and the thalamus are in the *diencephalon,* a region that encircles the third ventricle. The **hypothalamus** forms the floor of the third ventricle. It is an integrating center that helps maintain homeostasis by regulating hunger, sleep, thirst, body temperature, and water balance. The hypothalamus controls the pituitary gland and, thereby, serves as a link between the nervous and endocrine systems (see Section 40.2).

The **thalamus** consists of two masses of gray matter in the sides and roof of the third ventricle. It receives all sensory input except smell. The thalamus integrates this information and sends it on to the appropriate portions of the cerebrum. For this reason, the thalamus is often referred to as the gatekeeper for sensory information en route to the cerebral cortex. The thalamus also participates in higher mental functions, such as memory and emotions.

The **pineal gland,** which secretes the hormone melatonin, is also located in the diencephalon. *Melatonin* is a hormone involved in maintaining a normal sleep-wake cycle. It is sometimes recommended for people with insomnia, but many side effects can occur. Relatively high levels of melatonin are a key ingredient in the "relaxation brownies" sold at convenience stores and online; many states are banning their sale, however.

The **cerebellum** lies under the occipital lobe of the cerebrum and is separated from the brain stem by the fourth ventricle. It is the largest part of the hindbrain. The cerebellum receives sensory input from the eyes, ears, joints, and muscles about the present position of body parts, and it receives motor output from the cerebral cortex about where those parts should be located. After integrating this information, the cerebellum sends motor impulses by way of the brain stem to the skeletal muscles. In this way, the cerebellum maintains posture and balance. It also ensures that all the muscles work together to produce smooth, coordinated voluntary movements, such as those in playing the piano or hitting a baseball.

The **brain stem** contains the midbrain, the pons, and the medulla oblongata (see Fig. 37.8). The **midbrain** acts as a relay station for tracts passing between the cerebrum and the spinal cord or cerebellum. The tracts cross in the brain stem, so the right side of the body is controlled by the left portion of the brain, and the left side of the body is controlled by the right portion of the brain.

The **pons** (L. *pons,* "bridge") contains bundles of axons that form a bridge between the cerebellum and the rest of the CNS. The pons also works with the medulla oblongata to regulate many basic body functions.

The **medulla oblongata** lies just superior to the spinal cord, and it contains tracts that ascend or descend between the spinal cord and higher brain centers. It regulates the heartbeat, breathing, swallowing, and blood pressure. It also contains reflex centers for vomiting, coughing, sneezing, hiccuping, and swallowing.

The most common neurological disease of young adults is **multiple sclerosis (MS)**. It typically affects myelinated nerves in the cerebellum, brain stem, basal ganglia, and optic nerve. MS is considered an autoimmune disease, in which the person's own white blood cells attack the myelin, oligodendrocytes, and eventually neurons in the CNS. The word *sclerosis* refers to the multiple scars, or plaques, that can be seen through various types of scans. The myelin damage affects the transmission of nerve impulses, resulting in the most common symptoms: fatigue, vision problems, weakness, numbness, and tingling. Nearly 400,000 people in the United States have MS, and about 10,000 new cases are diagnosed each year, mainly in young adults.

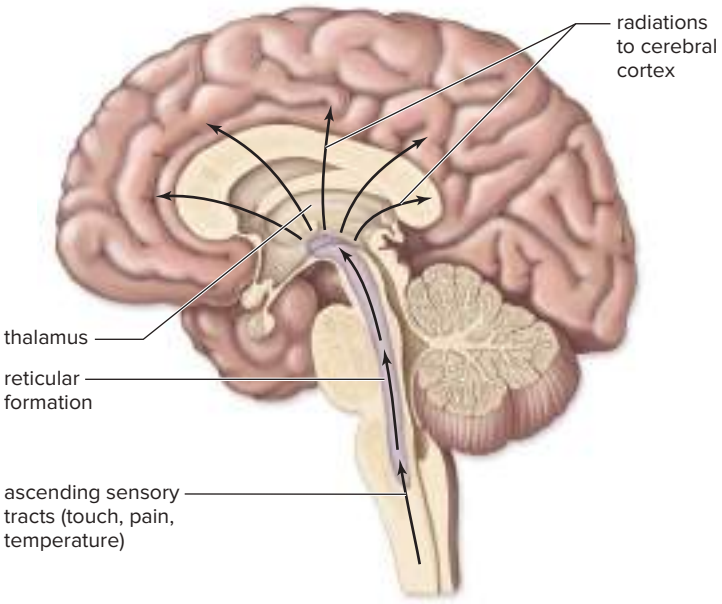


Figure 37.10 The reticular activating system. The reticular formation receives and sends on motor and sensory information to various parts of the CNS. One portion, the reticular activating system (RAS) (arrows), arouses the cerebrum and, in this way, controls alertness versus sleep.

The Reticular Activating System The **reticular activating system (RAS)** contains the reticular formation, a complex network of nuclei and nerve fibers extending the length of the brain stem (Fig. 37.10). The reticular formation receives sensory signals, which it sends up to higher centers, and motor signals, which it sends to the spinal cord.

The RAS arouses the cerebrum via the thalamus and causes a person to be alert. Apparently, the RAS can filter out unnecessary sensory stimuli, which explains why some individuals can study with the TV on. If you want to awaken the RAS, surprise it with a sudden stimulus, such as splashing your face with cold water; if you want to deactivate it, remove visual and auditory stimuli. A severe injury to the RAS can cause a person to become comatose, from which recovery may be impossible.

The Limbic System

The **limbic system** is a complex group of brain structures that lie just under the cortex, near the thalamus. Although definitions vary somewhat, the limbic system includes the hypothalamus, hippocampus, amygdala, olfactory bulb, and other nearby structures (Fig. 37.11). The limbic system blends higher mental functions and primitive emotions into a united whole. It accounts for why activities such as sexual behavior and eating seem pleasurable and why, for instance, mental stress can cause high blood pressure.

Two significant components of the limbic system, the hippocampus and the amygdala, are essential for learning and memory. The **hippocampus,** a seahorse-shaped structure deep in the temporal lobe, is well situated in the brain to make the prefrontal area

ISTUDY

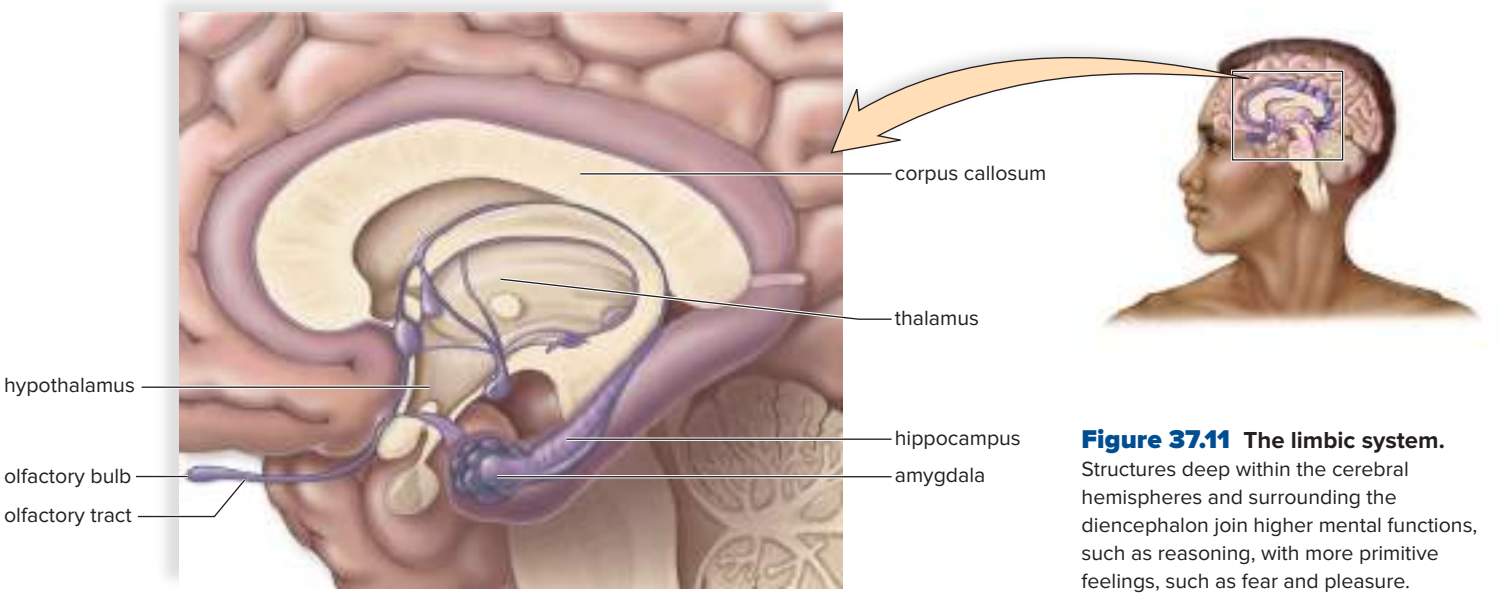


Figure 37.11 The limbic system. Structures deep within the cerebral hemispheres and surrounding the diencephalon join higher mental functions, such as reasoning, with more primitive feelings, such as fear and pleasure.

aware of past experiences stored in sensory association areas. The **amygdala,** in particular, can cause these experiences to have emotional overtones. For example, the smell of smoke may serve as an alarm to search for fire in the house. The inclusion of the frontal lobe in the limbic system gives us the ability to restrain ourselves from acting out on strong feelings by using reason.

Learning and Memory **Memory** is the ability to hold a thought in mind or recall events from the past, ranging from a word we learned only yesterday to an early emotional experience that has shaped our lives. Learning takes place when we retain and use memories.

The prefrontal area in the frontal lobe is active during short-term memory, as when we temporarily recall a phone number. However, some phone numbers go into long-term memory. Think of a phone number you know by heart, and see if you can bring it to mind without also thinking about the place or person associated with the number. Most likely you cannot, because typically long-term memory is a mixture of what is called semantic memory (numbers, words, etc.) and episodic memory (persons, events, etc.). Skill memory is a type of memory that can exist independently of episodic memory. Skill memory is being able to perform motor activities, such as riding a bike or playing ice hockey.

What parts of the brain are functioning when you remember something from long ago? The hippocampus gathers long-term memories, which are stored in bits and pieces throughout the sensory association areas, and makes them available to the frontal lobe. Why are some memories so emotionally charged? The amygdala is responsible for fear conditioning and associating danger with sensory information received from the thalamus and the cortical sensory areas.

Several diseases of the brain can affect memory. **Alzheimer's disease** is the most common cause of dementia, or a loss of reasoning, memory, and other higher brain functions, especially in people over age 65. Individuals with Alzheimer's disease have abnormal neurons throughout the brain, but especially in the hippocampus and amygdala. These neurons have two abnormalities: (1) plaques, containing a protein called beta amyloid, which accumulate around the axons; and (2) neurofibrillary tangles (bundles of fibrous protein) surrounding the nucleus. The cause of these protein abnormalities is unknown, although several genes that predispose a person to develop Alzheimer's disease have been identified. Although no cure is available, most of the drugs currently approved to treat symptoms of Alzheimer's disease are cholinesterase inhibitors, which effectively increase the levels of acetylcholine in the person's brain. This in turn can improve learning and memory, at least temporarily.

Check Your Progress 37.3

1. Trace the path of a nerve impulse from a stimulus in an internal organ (such as food in the large intestine stimulating peristalsis) to the brain and back.
2. Name the four major lobes of the human brain.
3. List at least two common diseases of the CNS, and describe their symptoms and causes.

37.4 The Peripheral Nervous System

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the overall anatomy of the PNS, including the cranial nerves and spinal nerves.
2. Explain how the somatic system differs from the autonomic system.
3. Contrast the functions of the sympathetic and parasympathetic divisions of the autonomic nervous system.

The peripheral nervous system (PNS) lies outside the central nervous system and contains **nerves,** which are bundles of axons. Axons that occur in nerves are also called *nerve fibers* (Fig. 37.12).

ISTUDY

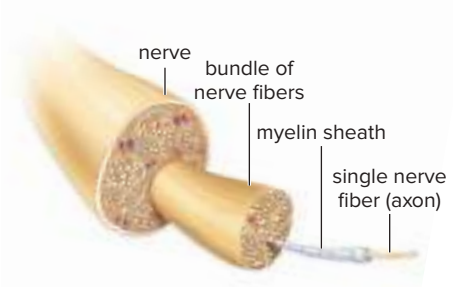


Figure 37.12 Structure of a nerve. Nerves are composed of bundles of nerve fibers.

The cell bodies of neurons are found in the CNS and in ganglia, collections of cell bodies outside the CNS.

The paired cranial and spinal nerves are part of the PNS. In the PNS, the somatic nervous system has sensory and motor functions that control the skeletal muscles. Ascending tracts carry sensory information to the brain, and descending tracts carry motor commands to the neurons in the spinal cord that control the muscles. The autonomic nervous system controls smooth muscle, cardiac muscle, and the glands. It is further divided into the sympathetic and parasympathetic divisions.

Humans have 12 pairs of **cranial nerves** attached to the brain (Fig. 37.13*a*). Some of these are sensory nerves; they contain only sensory nerve fibers. Some are motor nerves, containing only motor fibers; others are mixed nerves, with both sensory and motor fibers.

Cranial nerves are largely concerned with the head, neck, and facial regions of the body. However, the vagus nerve has branches not only to the pharynx and larynx but also to most of the internal organs.

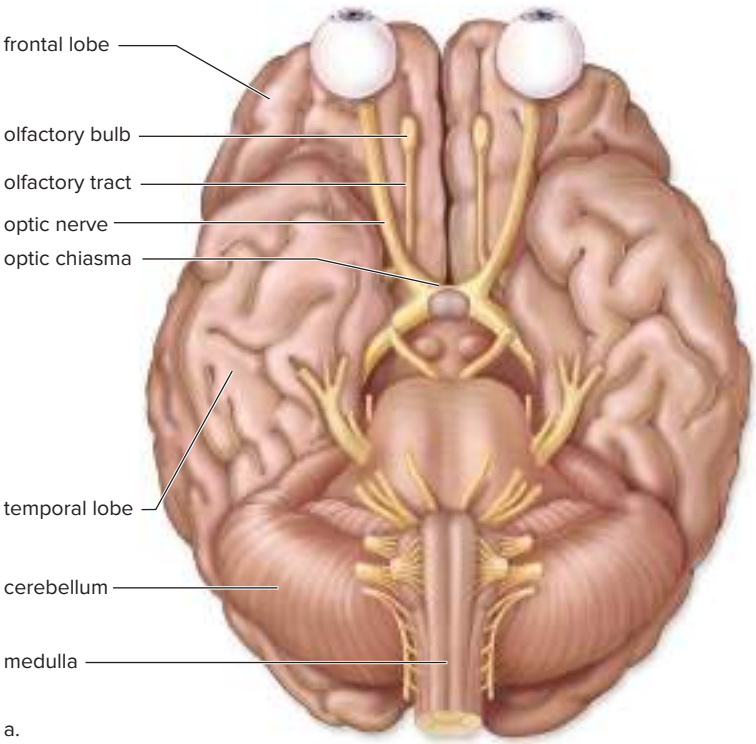
Humans also have 31 pairs of **spinal nerves** (Figs. 37.13*b* and 37.14), which emerge from the spinal cord via two short branches, or roots. The dorsal roots contain axons of sensory neurons, which conduct impulses to the spinal cord from sensory receptors. The cell body of a sensory neuron is in the **dorsal root ganglion.** The ventral roots contain axons of motor neurons, which conduct impulses away from the spinal cord to effectors. These two roots join to form a spinal nerve. All spinal nerves are mixed nerves, containing many sensory and motor fibers.

Somatic System

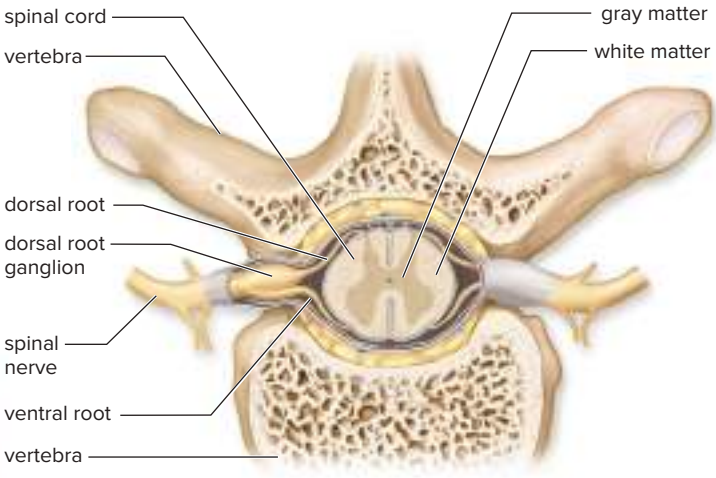
The PNS has two divisions—somatic and autonomic (Table 37.1). The nerves in the **somatic system** serve the skin, joints, and skeletal muscles. Therefore, the somatic system includes nerves with the following functions:

- Take sensory information from external sensory receptors in the skin and joints to the CNS
- Carry motor commands away from the CNS to the skeletal muscles

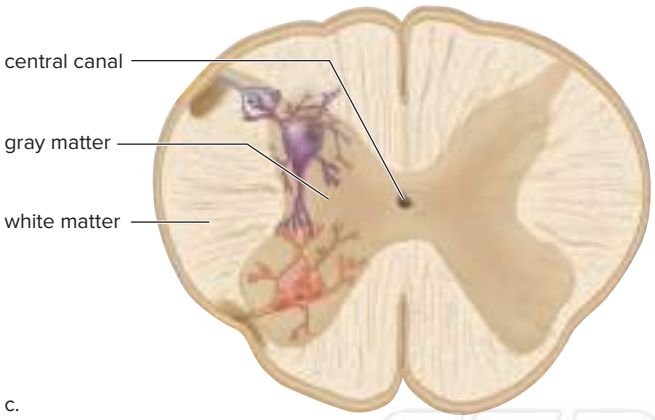
The neurotransmitter acetylcholine (ACh) is active in the somatic system.



a.



b.



c.

Figure 37.13 Cranial and spinal nerves. **a.** Ventral surface of the brain, showing the attachment of the cranial nerves. **b.** Cross section of the vertebral column and spinal cord, showing a spinal nerve. Each spinal nerve has a dorsal root and a ventral root attached to the spinal cord. **c.** Cross section of the spinal cord.

ISTUDY

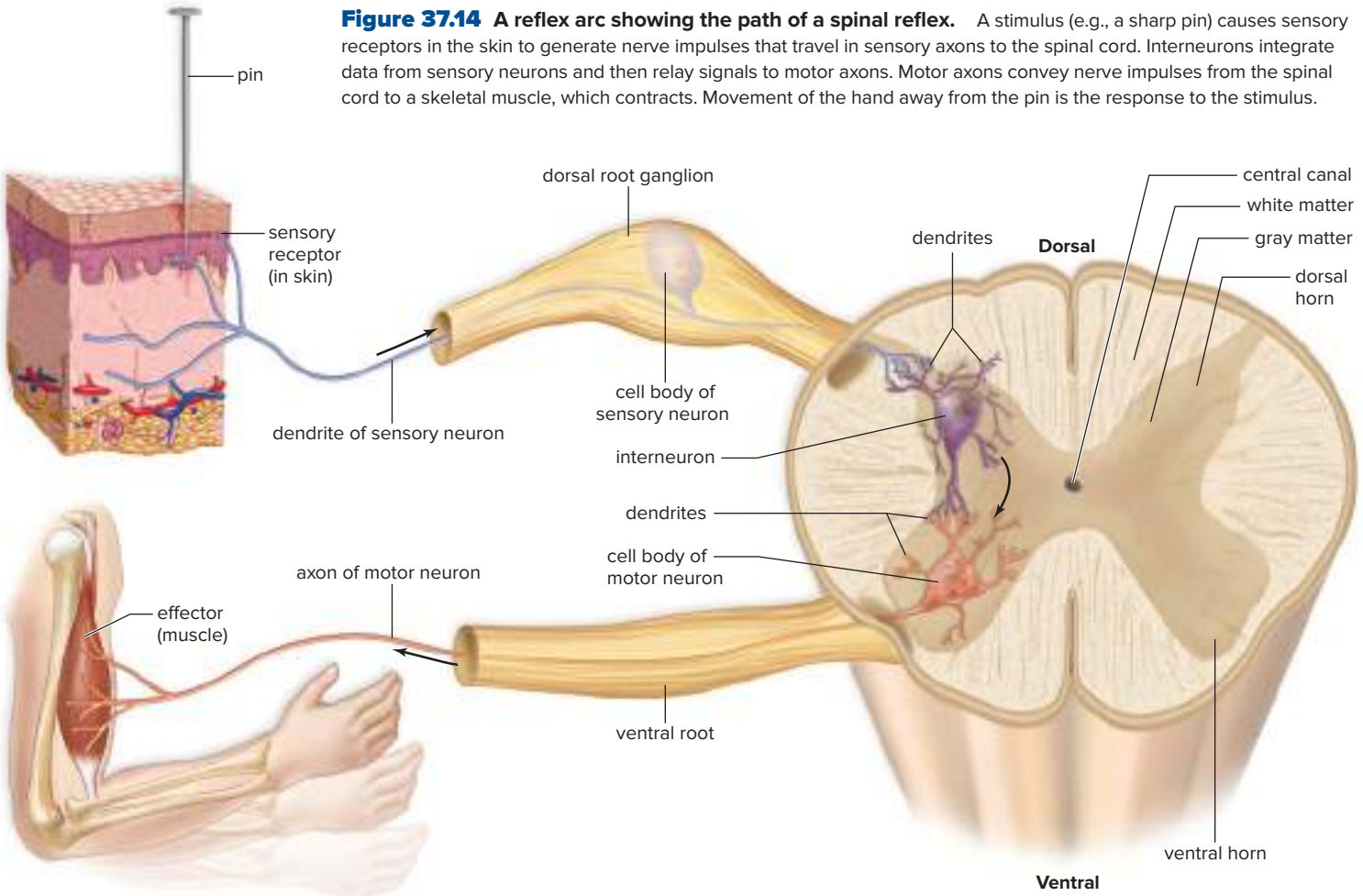


Figure 37.14 A reflex arc showing the path of a spinal reflex. A stimulus (e.g., a sharp pin) causes sensory receptors in the skin to generate nerve impulses that travel in sensory axons to the spinal cord. Interneurons integrate data from sensory neurons and then relay signals to motor axons. Motor axons convey nerve impulses from the spinal cord to a skeletal muscle, which contracts. Movement of the hand away from the pin is the response to the stimulus.

Voluntary control of skeletal muscles always originates in the brain. Involuntary responses to stimuli, called reflex actions, can involve only the spinal cord. Reflexes enable the body to react swiftly to stimuli that could disrupt homeostasis. For example, flying objects cause our eyes to blink, and sharp pins cause our hands to jerk away, even without our having to think about it.

The Reflex Arc Figure 37.14 illustrates the path of a reflex that involves only the spinal cord. For instance, if your hand touches a sharp pin, **sensory receptors** generate nerve impulses that move along sensory axons through a dorsal root ganglion toward the spinal cord. Sensory neurons that enter the cord dorsally pass signals on to many interneurons in the gray matter of the

Table 37.1 Comparison of Somatic Motor and Autonomic Motor Pathways

	Somatic Motor Pathway	Autonomic Motor Pathways	
		Sympathetic	Parasympathetic
Type of control	Voluntary/involuntary	Involuntary	Involuntary
Number of neurons per message	One	Two (preganglionic shorter than postganglionic)	Two (preganglionic longer than postganglionic)
Location of motor fiber	Most cranial nerves and all spinal nerves	Thoracolumbar spinal nerves	Cranial (e.g., vagus) and sacral spinal nerves
Neurotransmitter	Acetylcholine	Norepinephrine	Acetylcholine
Effectors	Skeletal muscles	Smooth and cardiac muscle, glands	Smooth and cardiac muscle, glands

ISTUDY

spinal cord. Some of these interneurons synapse with motor neurons. The short dendrites and the cell bodies of motor neurons are also in the spinal cord, but their axons leave the cord ventrally. Nerve impulses travel along motor axons to an **effector,** which brings about a response to the stimulus. In this case, a muscle contracts, so you withdraw your hand from the pin. However, sometimes the effector may be a gland.

Various other reactions are possible—you will most likely look at the pin, wince, and cry out in pain. This series of responses is explained by the fact that some of the interneurons in the white matter of the cord carry nerve impulses in tracts to the brain. The brain makes you aware of the stimulus and directs subsequent reactions to the situation. You don't feel pain until the brain receives the information and interprets it. Visual information received directly by way of a cranial nerve may make you aware that your finger is bleeding. Then, you might decide to look for a bandage.

Autonomic System

The **autonomic system** of the PNS regulates the activity of cardiac and smooth muscle and glands. It carries out its duties without our awareness or intent. The system is divided into the sympathetic and parasympathetic divisions (Table 37.1). Both of these divisions function automatically and usually in an involuntary manner, innervate all internal organs, and use two neurons and one ganglion for each impulse. The first neuron has a cell body within the CNS and a preganglionic fiber. The second neuron has a cell body within the ganglion and a postganglionic fiber (Fig. 37.15).

Reflex actions, such as those that regulate blood pressure and breathing rate, are especially important in the maintenance of homeostasis. These reflexes begin when the sensory neurons in contact with internal organs send information to the CNS. They are completed by motor neurons in the autonomic system.

Sympathetic Division

Most preganglionic fibers of the **sympathetic division** arise from the middle, or thoracolumbar, portion of the spinal cord and almost immediately terminate in ganglia lying near the cord. Therefore, in this division, the preganglionic fiber is short, but the postganglionic fiber that makes contact with an organ is long.

The sympathetic division is especially important during emergency situations and is associated with "fight or flight" responses (Fig. 37.16). In order to fend off a foe or flee from danger, active muscles require a ready supply of glucose and oxygen. The sympathetic division accelerates the heartbeat and dilates the bronchi. At the same time, the sympathetic division inhibits the digestive tract, because digestion is not an immediate necessity if you are under attack. The neurotransmitter released by the postganglionic axon is primarily norepinephrine (NE). Structurally, NE resembles epinephrine (adrenaline), an adrenal medulla hormone that usually increases heart rate and contraction (see Section 40.3).

Parasympathetic Division

The **parasympathetic division** includes a few cranial nerves (e.g., the vagus nerve) and fibers that arise from the sacral (bottom) portion of the spinal cord. Therefore, this division is often referred to as the craniosacral portion of the autonomic system. In the parasympathetic division, the preganglionic fiber is long and the postganglionic fiber is short, because the ganglia lie near or within the organ.

The parasympathetic division, sometimes called the "housekeeper" or "rest and digest division," promotes all the internal responses we associate with a relaxed state; for example, it causes the pupil of the eye to contract, promotes the digestion of food, and slows the heartbeat. The neurotransmitter used by the parasympathetic division is acetylcholine (ACh).

Several disorders can affect the peripheral nerves. Guillain-Barré syndrome (GBS) results from an abnormal immune reaction to one of several types of infectious agents. Antibodies formed against these microbes cross-react with the myelin coating of peripheral nerves, causing demyelination of peripheral nerve axons. The first symptom of GBS is usually weakness in the legs 2 to 4 weeks after an infection or immunization. Soon the arms are affected, and in some cases the respiratory muscles may be weakened to the point that mechanical ventilation is required. Fortunately, the inflammation usually subsides in a few weeks, and most people fully recover within 6 to 12 months. In myasthenia gravis (MG), abnormal antibodies react with the acetylcholine (ACh) receptor at the neuromuscular junction of the skeletal muscles. When an action potential arrives at the synaptic cleft of a peripheral nerve, these antibodies block the normal action of ACh, resulting in muscle weakness. Although there is no cure, people with MG often respond well to drugs that inhibit acetylcholinesterase, an enzyme that normally destroys ACh after it is released into synapses.

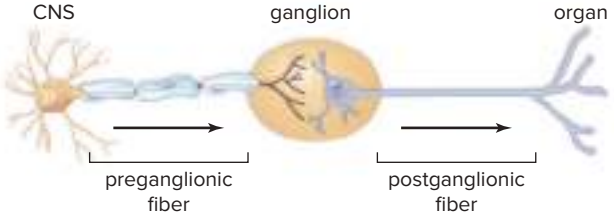


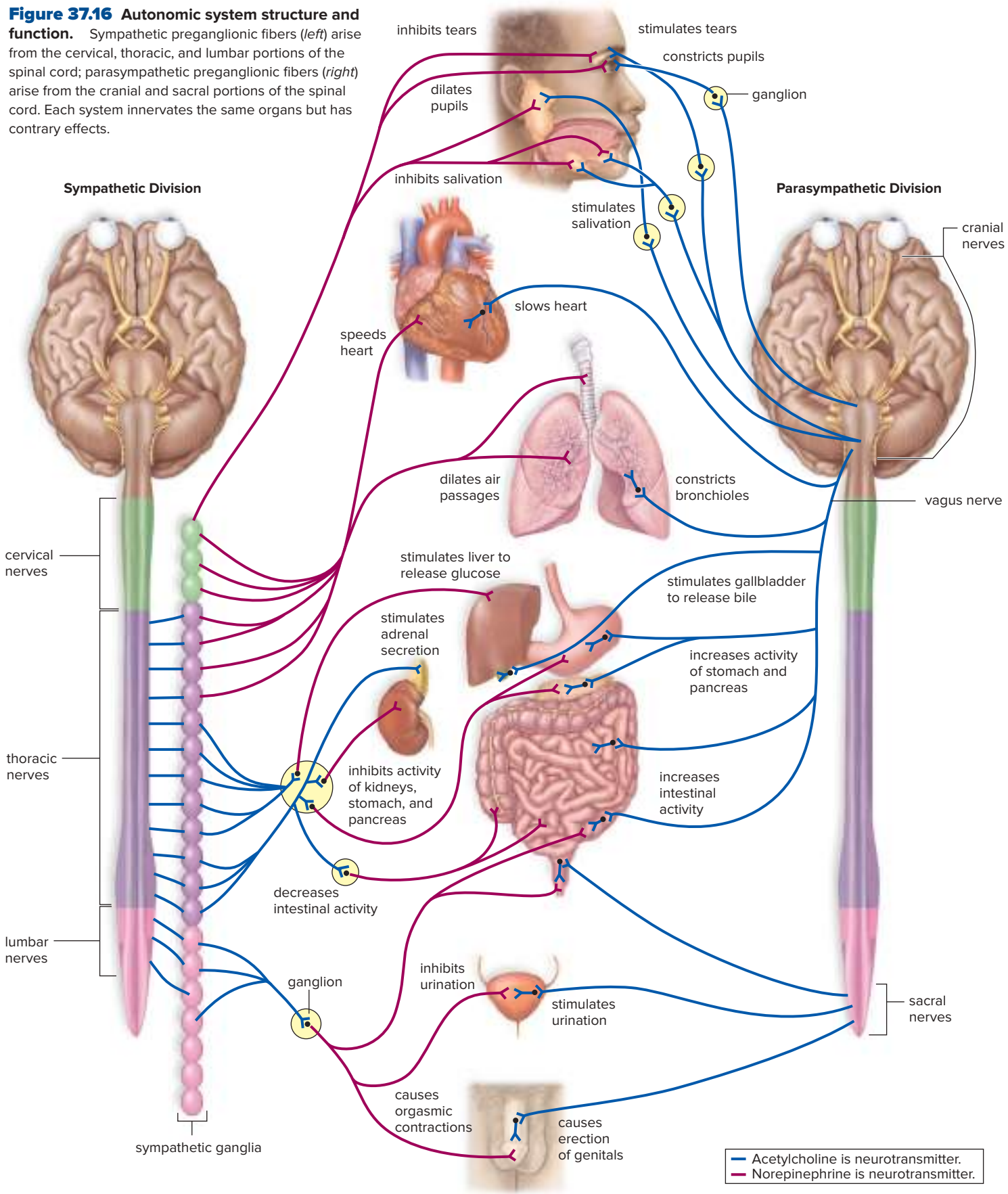
Figure 37.15 Structure of a ganglion. The relationship of preganglionic and postganglionic nerve fibers in a ganglion.

Check Your Progress 37.4

1. Review the neurological explanation for the observation that, after you touch a hot stove, you withdraw your hand before you feel any pain.
2. Apply your knowledge of the autonomic nervous system to explain why your stomach may ache if you exercise after a meal.
3. Describe the shift in autonomic system activity that occurs when you are startled from your sleep.

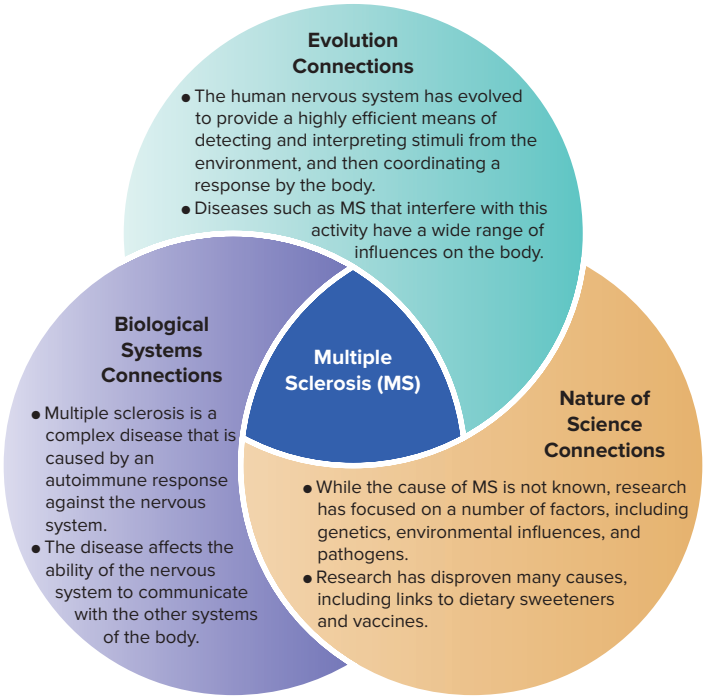
ISTUDY

Figure 37.16 Autonomic system structure and function. Sympathetic preganglionic fibers (*left*) arise from the cervical, thoracic, and lumbar portions of the spinal cord; parasympathetic preganglionic fibers (*right*) arise from the cranial and sacral portions of the spinal cord. Each system innervates the same organs but has contrary effects.



ISTUDY

Connecting *the* Concepts

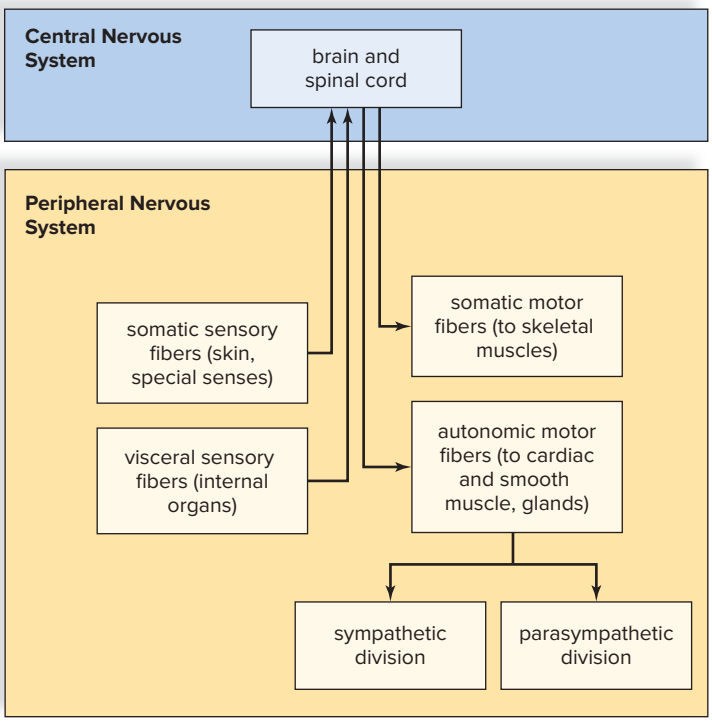


What Did You Learn?

37.1 Evolution of the Nervous System

A comparative study of the invertebrates shows a gradual increase in the complexity of the nervous system.

- Cnidarians have a **nerve net** made up of neurons that communicate with each other and with contractile cells in the body wall.



- Flatworms have a ladderlike nervous system, with **cephalization** and a cluster of nerve cell bodies, or **ganglion,** at the anterior end.
- Annelids have a brain, plus a ganglion present in each body segment.
- Vertebrate nervous systems are much more complex, with a brain and spinal cord making up the **central nervous system (CNS)**, and an additional **peripheral nervous system (PNS)**. In mammals, the forebrain is greatly enlarged and has an additional outer layer called the neocortex.

37.2 Nervous Tissue

The anatomical unit of the nervous system is the **neuron,** of which there are three types: sensory neuron, motor neuron, and interneuron. Each of these is made up of a **cell body,** an **axon,** and **dendrites.** Axons are often bundled together as **nerve fibers,** forming nerves, and many axons are covered by a **myelin sheath.**

- **Motor (efferent) neurons** take nerve impulses from the CNS to muscles or glands, and **sensory (afferent) neurons** deliver impulses to the CNS from sensory receptors. **Interneurons** convey messages between different areas of the CNS.
- **Neuroglia** (glial cells) serve important roles in supporting neuron health and function. The types of glial cells include **astrocytes, microglia, oligodendrocytes, Schwann cells,** ependymal cells, and satellite cells.
- When an axon is not conducting an **action potential** (nerve impulse), the **resting potential** indicates that the inside of the fiber is negative compared to the outside. The sodium-potassium pump helps maintain this resting potential. When the axon is conducting a nerve impulse, an action potential (a change in membrane potential) travels along the fiber. Depolarization occurs (inside becomes positive) due to the movement of Na^+ to the inside, and then repolarization occurs (the inside becomes negative again) due to the movement of K^+ to the outside of the fiber. In

myelinated axons, the nerve impulse "jumps" from one nonmyelinated area (node of Ranvier) to the next, a mode known as **saltatory conduction.** Once an action potential occurs, that section of the axon undergoes a brief **refractory period,** during which the sodium gates are unable to open.

- Transmission of the nerve impulse from one neuron to another takes place across a **synapse.** Synaptic vesicles usually release a chemical, known as a **neurotransmitter,** into the **synaptic cleft.** The binding of neurotransmitters to receptors in the postsynaptic membrane can either increase the chance of an action potential (stimulation) or decrease the chance of an action potential (inhibition) in the next neuron. A neuron can have as many as 10,000 synapses with other neurons; **integration** is the summing up of these signals.

37.3 The Central Nervous System

The CNS consists of the brain and **spinal cord,** which are both protected by bone and covered by **meninges.**

- **Cerebrospinal fluid** fills the spaces between these meninges, as well as the **ventricles** of the brain. When infections invade the meninges, **meningitis** results. The CNS receives and integrates sensory input and formulates motor output. The gray matter of the spinal cord contains neuron cell bodies; the white matter consists of myelinated axons that occur in bundles called tracts. The spinal cord sends sensory information to the brain, receives motor output from the brain, and carries out **reflex actions.**
- The CNS is composed of **gray matter,** containing unmyelinated fibers, and **white matter,** with myelinated fibers that run in **tracts.** In the brain, the outer **cerebrum** has two **cerebral hemispheres,** connected by the corpus callosum. Sensation, reasoning, learning and memory, and language and speech take place in the cerebrum. The **cerebral cortex** is a thin layer of gray matter covering the cerebrum.
- The cerebral cortex of each cerebral hemisphere has four lobes: a frontal, a parietal, an occipital, and a temporal lobe. The primary motor area in the frontal lobe sends out motor commands to lower brain centers, which pass them on to motor neurons. The primary somatosensory area in the parietal lobe receives sensory information from lower brain centers in communication with sensory neurons. Association areas for vision are in the occipital lobe, and those for hearing are in the temporal lobe. Several areas of the cortex are involved in **memory.**

The brain has a number of other regions.

- The **hypothalamus** controls homeostasis, and the **thalamus** specializes in sending sensory input on to the cerebrum.
- The **pineal gland** secretes melatonin, and the **cerebellum** primarily coordinates skeletal muscle contractions.
- In the **brainstem,** the **midbrain** is the area where nerve tracts controlling one side of the body cross over to the other side and the **medulla oblongata** and **pons** have centers for vital functions such as breathing and the heartbeat.
- The **reticular activating system (RAS)** is involved in alertness and filtering out irrelevant information.
- The **limbic system** connects higher reasoning and **memory** with emotions; the **hippocampus** and **amygdala** also play a role.

A number of diseases affect the human nervous system. In **amyotrophic lateral sclerosis (ALS)**, motor neurons in the brain and spinal cord degenerate and die. When a **stroke** occurs, the blood supply to the brain is disrupted. **Parkinson's disease (PD)** and **Alzheimer's disease** are brain disorders that mainly affect older individuals. **Multiple sclerosis (MS)** is an autoimmune disease that affects the myelin sheaths, disrupting nerve transmission.

37.4 The Peripheral Nervous System

The **nerves** of the peripheral nervous system (PNS) lie outside the CNS, and their cell bodies make up ganglia. In humans, these nerves include 12 paired **cranial nerves,** and 31 paired **spinal nerves.** The cell body of a sensory nerve lies in a **dorsal root ganglion.**

- The PNS contains the **somatic system** and the **autonomic system.** Reflexes are automatic, and some do not require the involvement of the brain. A simple reflex uses neurons that make up a reflex arc. In the somatic system, a sensory neuron conducts nerve impulses from a **sensory receptor** to an interneuron, which in turn transmits impulses to a motor neuron, which stimulates an **effector** to react.
- The motor portion of the somatic system of the PNS controls skeletal muscle; in contrast, the motor portion of the autonomic system controls smooth muscle of the internal organs and glands. The **sympathetic division,** which is often associated with reactions that occur during times of stress, and the **parasympathetic division,** which is often associated with activities that occur during times of relaxation, are both parts of the autonomic system.

Thinking Critically About Biology

1. In individuals with panic disorder, the fight-or-flight response is activated without a clear trigger or danger. How might it be possible to directly control this response in order to treat panic disorder? Why is such control often impractical?
2. A person who lost a leg several years ago continues to experience pain as though it were coming from the missing limb. What hypothesis could explain the neurological basis of this pain?
3. Dopamine is a key neurotransmitter in the brain. People with Parkinson's disease are lacking in this chemical, whereas people who use drugs like nicotine, cocaine, and methamphetamine experience an enhancement of their dopamine activity when taking these substances. Do you think these drugs might be a possible treatment, or even a cure, for Parkinson's disease? Why or why not?
4. What are some factors that make brain diseases such as Alzheimer's disease, Parkinson's disease, and MS so difficult to treat?

Making It Relevant to You

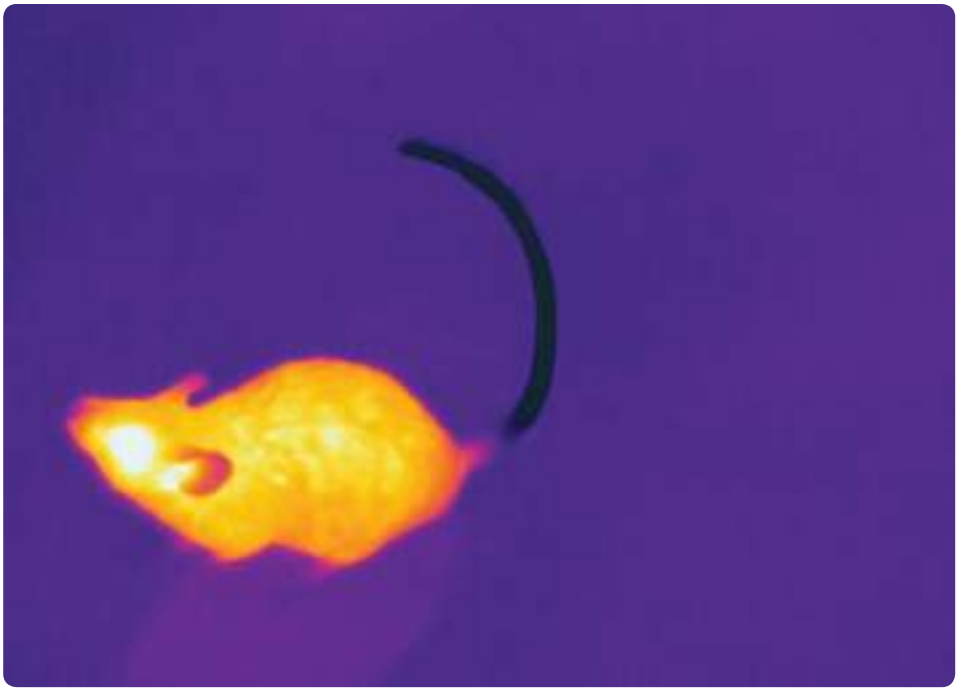
1. While MS is not considered to be a hereditary disease, some people may carry genes that make them more susceptible. Scientists are actively looking for the genes that may increase susceptibility. Would you want to know that you carried a gene that gave you an increased likelihood of diseases such as MS or Parkinson's?
2. Recent research suggests that MS susceptibility may have a connection between the microbes (microbiome) found in the digestive tract and the immune system. How might this relationship occur?
3. If diseases such as Alzheimer's and Parkinson's have a detrimental effect on humans, why have we not evolved mechanisms to protect ourselves from these illnesses?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

38

Sense Organs



Ted Kinsman/Science Source

Chapter Outline

38.1 Sensory Receptors
38.2 Chemical Senses
38.3 Sense of Vision
38.4 Senses of Hearing and Equilibrium
38.5 Somatic Senses

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Sections 37.3 and 37.4 What are the roles of the central and peripheral nervous systems in an animal's responses to its environment?

Figure 37.9 How is the cerebral cortex involved in the processing of sensory information?

Figure 37.14 How do sensory receptors in the skin stimulate a spinal reflex?

Seeing in the Dark

Visible light is made up of waves of electromagnetic energy with different wavelengths. Sensory receptors in the human eye contain pigments that change when exposed to electromagnetic energy at various wavelengths. This is then perceived by the brain as a visual image.

A few types of animals have evolved the ability to detect electromagnetic energy at longer wavelengths, known as the infrared spectrum. Just about any source of heat—the sun, a fire, or a warm body—emits energy in the infrared spectrum. Certain kinds of snakes, such as the pit vipers, have evolved specialized infrared sensory organs. Located in a pit below each eye, these organs are very sensitive to infrared waves emitted by their warm-blooded prey; the photo above, taken using an infrared-sensitive camera, shows how a mouse might "appear" to a pit viper. Even when placed in total darkness, snakes with this ability can track and find such prey quickly.

As you read through this chapter, think about the following questions:

1. How does the ability to detect infrared energy provide certain snakes with a competitive advantage over predators lacking this ability?
2. Of the types of sensory receptors described in this chapter—chemoreceptors, photoreceptors, mechanoreceptors, or thermoreceptors—which is essential for the survival of the animal?

Following *the* Themes

CHAPTER 38 – SENSE ORGANS

Evolution	The sense organs that have evolved in animals are essential to their ability to maintain homeostasis, avoid danger, find food, and locate mates.
Nature of Science	Biomedical scientists are using the knowledge of the transmission of sensory information to restore the senses of people who are deaf or blind.
Biological Systems	Four types of sensory receptors—chemoreceptors, photoreceptors, mechanoreceptors, and thermoreceptors—provide animals with information about their internal and external environments.

ISTUDY

38.1 Sensory Receptors

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the differences among sensory receptors, sensory transduction, and perception.
2. Describe four types of sensory receptors and list examples of each.

In order to survive, animals must be able to maintain homeostasis, as well as locate required nutrients, avoid dangers, and learn from previous experiences. These kinds of selective pressures have resulted in the evolution of the many different types of **sensory receptors,** which are specialized cells capable of detecting changes in internal or external conditions, and of communicating that information to the central nervous system.

A sensory receptor is able to convert some type of event, or *stimulus,* into a nerve impulse. This process is known as **sensory transduction.** Some sensory receptors are modified neurons, and others are specialized cells closely associated with neurons.

The plasma membrane of a sensory receptor contains proteins that react to a stimulus. For example, these membrane proteins might be sensitive to temperature or react with a certain chemical. When this happens, ion channels open allowing ions to flow through the plasma membrane. If the stimulus is sufficient, nerve impulses begin and are carried by a sensory nerve fiber within the PNS to the CNS.

Note that there is no difference between the nerve impulses generated by different types of sensory receptors. All these impulses are simply the action potentials discussed in Section 37.2, regardless of whether they arise in the eyes, ears, nose, mouth, skin, or internal organs. The interpretation of these nerve impulses by appropriate areas of the brain brings about a response that is appropriate for the particular type of stimulus. That is why artificial stimulation of the nerves that normally carry impulses generated in the ear or the eye are interpreted by the brain as sound or light, respectively (see the Nature of Science feature, "Artificial Retinas Come into Focus," in Section 38.3).

What's more, not all of these sensory impulses are received at the conscious levels of the brain—for example, we are not aware of the constant adjustments that are occurring in response to various internal stimuli. Any sensory stimuli we become conscious of fall under the category of **perception.**

Based on the source of the stimulus, sensory receptors can be classified as *interoceptors* or *exteroceptors.* Interoceptors receive stimuli from inside the body, such as changes in blood pressure, blood volume, and the pH of the blood. Interoceptors located within internal organs are sometimes called viscero ceptors; those that help maintain muscle tone and posture are proprioceptors (discussed in Section 38.5).

A few types of exteroceptors enable an animal to detect stimulus from the environment. **Chemoreceptors** can respond to a diverse range of chemical substances, from oxygen levels in the blood to molecules of food in the mouth or nasal passages. Electromagnetic receptors respond to heat or light energy. The infrared sensors of snakes are an example, as are the **photoreceptors** (Gk. *photos,* "light") of the eyes that detect light. **Mechanoreceptors** are stimulated by mechanical forces, usually pressure of some sort. Mechanoreceptors are responsible for detecting changes that are perceived as sound or touch, as well as for maintaining equilibrium (balance). **Thermoreceptors,** located in the hypothalamus and skin, are stimulated by changes in temperature.

Although the extent to which nonhuman animals have perceptions is largely unknown, it is likely that some of them perceive their world in very different ways (Fig. 38.1). As noted in the

a.

b.

c.

Figure 38.1 Perception in the animal kingdom. Animals have a variety of mechanisms by which they perceive their environment. **a.** Some snakes detect thermal energy. **b.** Whales use echolocation. **c.** Dogs detect chemicals in the environment. (a): Robert Marien/Corbis/Getty Images; (b): James R.D. Scott/Moment/Getty Images; (c): Jim Parkin/iStock/360/Getty Images

ISTUDY

chapter opener, some snakes can detect infrared energy that is completely invisible to humans. Bats, dolphins, and whales are capable of *echolocation,* meaning they can produce very-high-frequency sounds, and then learn about objects in their environment by listening for echoes. Some whales can also hear very-low-frequency sounds emitted by other whales hundreds of miles away. Dogs have a sense of smell that is more sensitive than that of humans, and they can be trained to detect drugs, human remains, explosives, and even bedbugs. Several scientific studies have confirmed that dogs can detect some types of human cancer just by sniffing the appropriate samples. Clearly, the sensory systems are a subject of fascination to biologists, because through them we experience our world.

Check Your Progress 38.1

1. Define *sensory transduction.*
2. List three examples of sensory capabilities found in animals that are lacking in humans.

38.2 Chemical Senses

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the role of chemoreceptors in animals.
2. Compare and contrast the senses of taste and smell in humans.
3. Describe how the human brain processes information about taste and smell.

Chemoreception is found almost universally in animals and is therefore believed to be the most primitive sense. Possessing chemoreceptors that are sensitive to certain chemical substances can be important in locating food, finding a mate, and detecting potentially dangerous chemicals in the environment.

The location and sensitivity of chemoreceptors vary throughout the animal kingdom. Although chemoreceptors are present throughout the body of planarians, they are concentrated in the auricles located on the sides of the head (see Fig. 28.22). Many insects have taste receptors on their mouthparts, but in the housefly, chemoreceptors are located primarily on the feet. Many insects are also able to detect airborne pheromones, which are chemical messages passed between individuals.

In crustaceans such as lobsters and crabs, chemoreceptors are widely distributed on their appendages and antennae. Many fish have chemoreceptors scattered over the surface of their skin. Snakes possess *Jacobson's organs,* a pair of pitlike sensory organs in the roof of the mouth. When a snake flicks its forked tongue, scent molecules are carried to the Jacobson's organs, and sensory information is transmitted to the brain for interpretation.

Sense of Taste in Humans

In humans, the sense of taste is associated with approximately 10,000 **taste buds** that are located primarily on the tongue (Fig. 38.2). Many taste buds lie along the walls of the papillae, the small elevations on

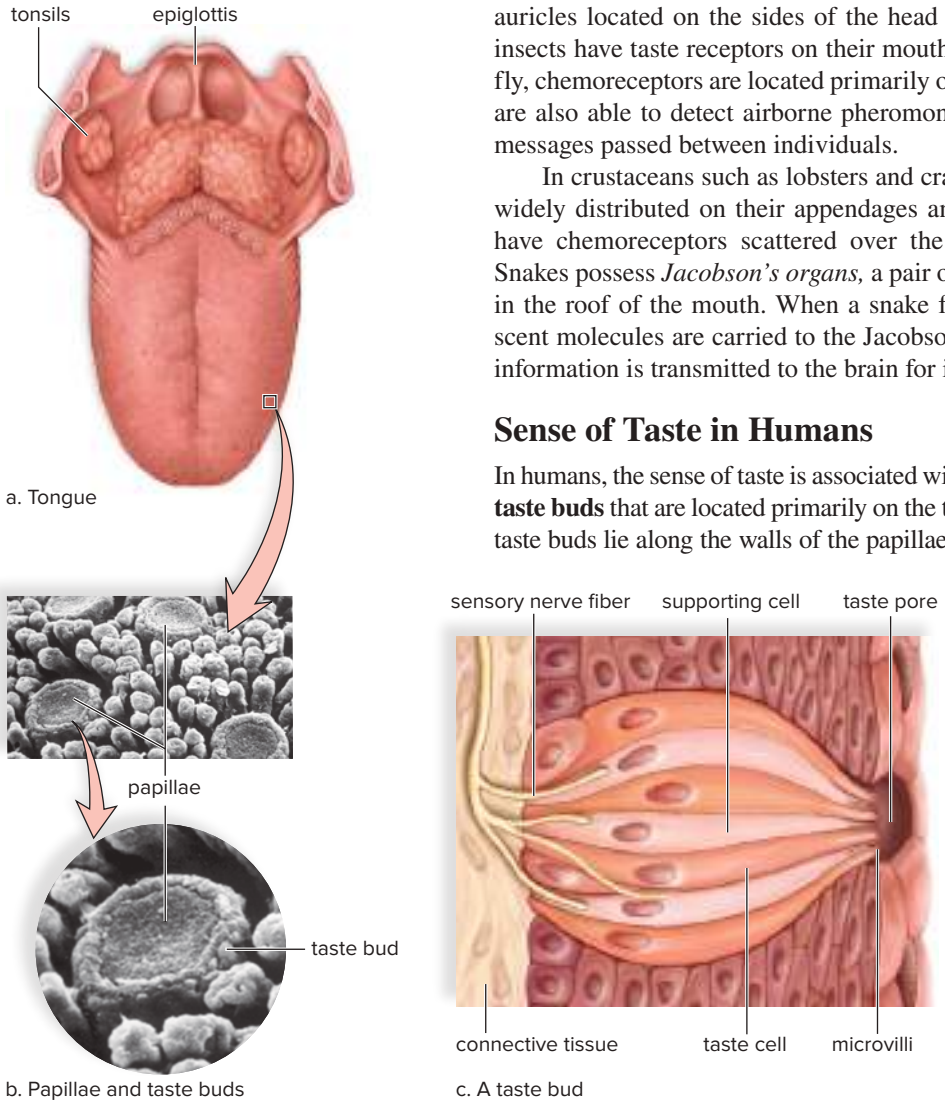


Figure 38.2 Taste buds in humans. **a.** Papillae on the tongue contain taste buds that are sensitive to sweet, sour, salty, bitter, and umami. **b.** Photomicrograph and enlargement of papillae. **c.** Taste cells end in microvilli that bear receptor proteins for certain molecules. When molecules bind to the receptor proteins, nerve impulses are generated and go to the brain, where the sensation of taste occurs. (b): (both): Omikron/Science Source

ISTUDY

the tongue that are visible to the unaided eye. Isolated taste buds are also present on the hard palate, the pharynx, and the epiglottis.

Taste buds on the tongue open at a taste pore. Taste buds have supporting cells and a number of elongated taste cells that end in microvilli. The microvilli, which project into the taste pore, bear receptor proteins for certain molecules. When molecules bind to receptor proteins, nerve impulses are generated in associated sensory nerve fibers. These nerve impulses travel to the brain, where they are interpreted as tastes.

Humans have five main types of taste receptors: sweet, sour, salty, bitter, and umami (Japanese, "savory, delicious"). Foods rich in certain amino acids, such as the common seasoning monosodium glutamate (MSG), as well as certain flavors of cheese, beef broth, and some seafood, produce the taste of umami. Taste buds for each of these tastes are located throughout the tongue, although certain regions may be slightly more sensitive to particular tastes. Various foods can stimulate a variety of different types of taste buds at the same time. The brain appears to survey the overall pattern of incoming sensory impulses and take a "weighted average" of their taste messages as the perceived taste.

Researchers have found chemoreceptors in the human lung that are sensitive only to chemicals that normally taste bitter. These receptors are not clustered in buds, and they do not send taste signals to the brain. Stimulation of these receptors causes the airways to dilate, leading scientists to speculate about implications for new medications to treat diseases such as asthma.

Sense of Smell in Humans

In humans, the sense of smell, or olfaction, is dependent on between 10 and 20 million **olfactory cells.** These structures are located within olfactory epithelium high in the roof of the nasal cavity (Fig. 38.3). Olfactory cells are modified neurons. Each cell ends in a tuft of about five olfactory cilia, which bear receptor proteins for odor molecules. Each olfactory cell has only 1 out of 1,000 different types of receptor proteins. Nerve fibers from similar olfactory cells lead to the same neuron in the olfactory bulb, an extension of the brain.

An odor contains many odor molecules that activate a characteristic combination of receptor proteins. A rose might stimulate certain olfactory cells, designated by blue and green in Figure 38.3, whereas a gardenia might stimulate a different combination. When the neurons communicate this information via the olfactory tract to the olfactory areas of the cerebral cortex, we perceive that we have smelled a rose or a gardenia.

For decades, scientists have estimated that humans can discern only about 10,000 different odors. However, according to a study in which human volunteers were asked to distinguish between very similar odorant molecules, it is likely that the average human can actually perceive over 1 trillion different smells!

You may have noticed that a certain aroma vividly brings to mind a certain person or place. For example, a whiff of perfume may remind you of someone you knew, or the smell of boxwood may remind you of your grandfather's farm. The olfactory bulbs have direct connections with the limbic system and its centers for emotions and memory. One study found that participants with previous negative experiences of visiting the dentist rated the smell of a chemical often encountered in dentists' offices as unpleasant, while those lacking such negative experiences rated it as pleasant.

The number of olfactory cells declines with age, and the remaining population of receptors becomes less sensitive. Thus,

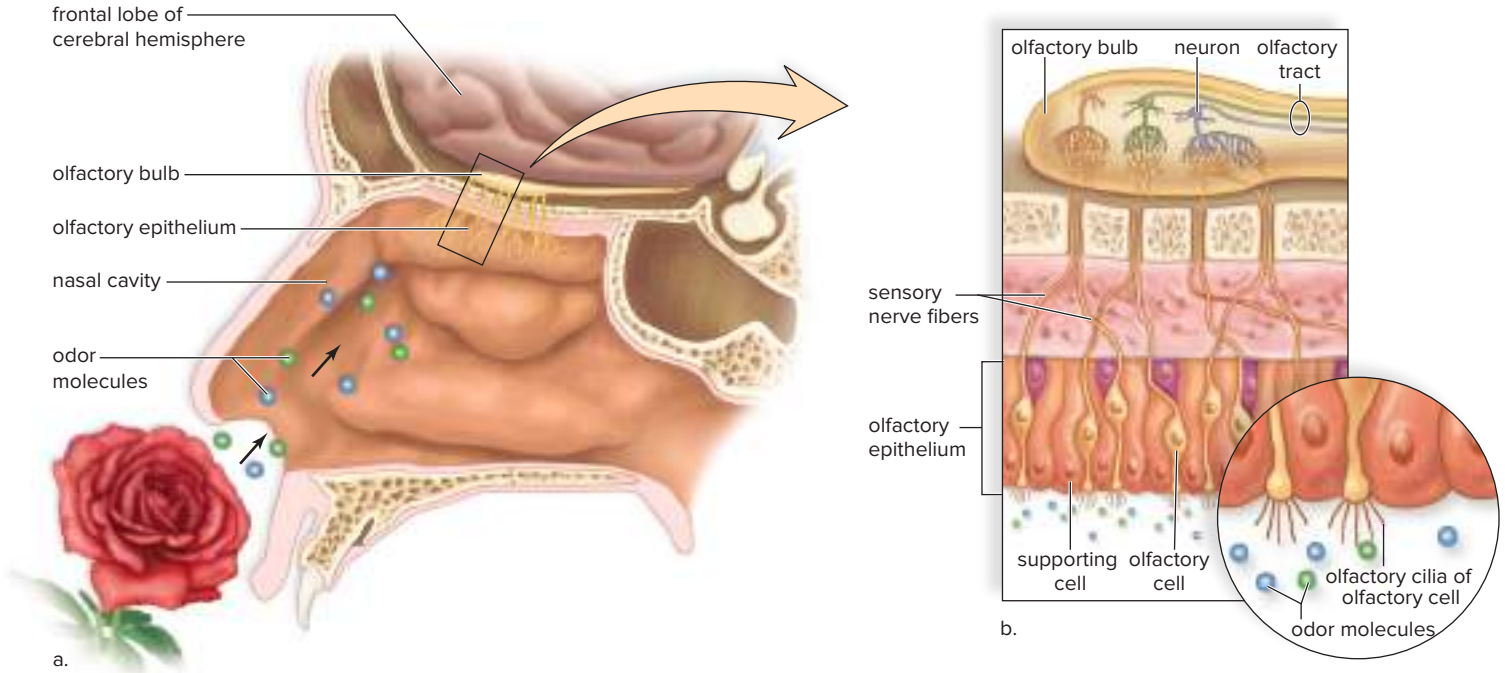


Figure 38.3 Olfactory cell location and anatomy. **a.** The olfactory epithelium in humans is located high in the nasal cavity. **b.** Olfactory cells end in cilia that bear receptor proteins for specific odor molecules. The cilia of each olfactory cell can bind to only one type of odor molecule (signified here by color). For example, if a rose causes olfactory cells sensitive to "blue" and "green" odor molecules to be stimulated, then neurons designated by blue and green in the olfactory bulb are activated. The primary olfactory area of the cerebral cortex interprets the pattern of stimulation as the scent of a rose.

ISTUDY

older people may tend to apply excessive amounts of perfume or aftershave. The ability to smell can also be lost as a result of head trauma, respiratory infection, or brain disease. This condition can become dangerous if these individuals cannot smell spoiled food, a gas leak, or smoke.

Usually, the senses of taste and smell work together to create a combined effect when interpreted by the cerebral cortex. For example, when you have a cold, you may think food has lost its taste, but most likely you have lost the ability to detect its smell. This method works in reverse also. When you smell something, some of the molecules move from the nose down into the mouth region and stimulate the taste buds there. Therefore, part of what we refer to as smell may, in fact, be taste.

Check Your Progress 38.2

1. Compare and contrast the senses of smell and taste.
2. List the five types of taste receptors in humans.
3. Discuss what could account for how a nerve impulse would be interpreted by the different sense organs.

38.3 Sense of Vision

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the structure of the eyes of arthropods with the eyes of vertebrates.
2. List all tissues or cell layers through which light passes from when it enters the eye until it is converted to a nerve impulse.
3. Discuss the distinct roles of rod cells, cone cells, and rhodopsin in converting a light stimulus into a nerve impulse.
4. Describe several common disorders affecting vision.

The ability of organisms to detect wavelengths of light, what is called vision, is an important capability for many, but not all, animals. Like the senses of smell and hearing, vision allows us to perceive the environment at a distance, and as a result this can have survival value. In this section, we review how animals detect light and how the human eye accomplishes vision.

How Animals Detect Light

As mentioned previously, photoreceptors are sensory receptors that are sensitive to light. Some animals lack photoreceptors and instead depend on senses such as smell and hearing; other animals have photoreceptors but live in environments that do not require them. For example, moles live underground and use their senses of smell and touch rather than eyesight.

Not all photoreceptors form images. The "eyespots" of planarians allow these animals to sense and move away from light. Image-forming eyes are found among four invertebrate groups: cnidarians, annelids, mollusks, and arthropods. Arthropods have

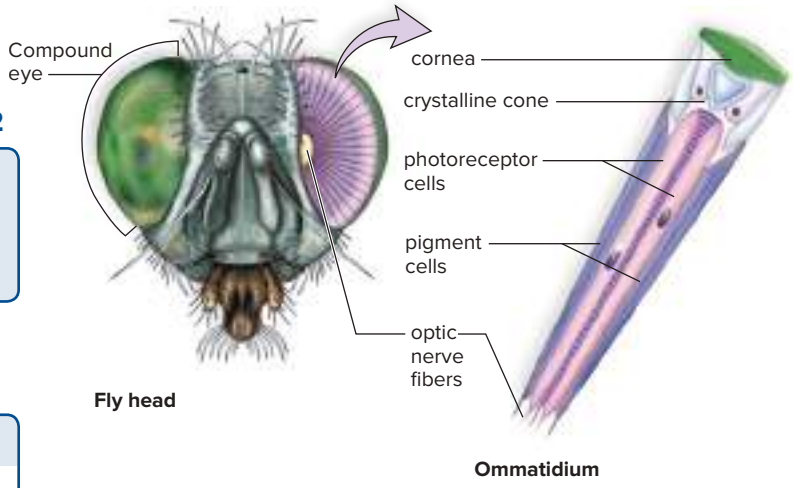


Figure 38.4 Compound eye. Each visual unit of a compound eye has a cornea and a lens, which focus light onto photoreceptors. The photoreceptors generate nerve impulses, which are transmitted to the brain, where interpretation produces a mosaic image. (photo): ©Farley Bridges

compound eyes composed of many independent visual units called ommatidia (Gk. *ommation,* dim. of *omma,* "eye"), each possessing all the elements needed for light reception (Fig. 38.4). Both the cornea and the crystalline cone function as lenses to direct light rays toward the photoreceptors. The photoreceptors generate nerve impulses, which pass to the brain by way of optic nerve fibers. The outer pigment cells absorb stray light rays, so that the rays do not pass from one visual unit to the other.

Flies and mosquitoes can see only a few millimeters in front of them, but dragonflies can see small prey insects several meters away. Research has shown that foraging bees use their sense of vision as a sort of "odometer" to estimate how far they have flown from their hive.

Most insects have color vision, but they see a limited number of colors compared to humans. However, many insects can also see some ultraviolet rays, and this enables them to locate the particular parts of flowers, such as nectar guides, that have ultraviolet patterns (Fig. 38.5).

Some fishes, all reptiles, and most birds are believed to have color vision, but most mammals lack full color vision and can only see part of the visible spectrum. Among the mammals, only the primates have color vision. It would seem, then, that the color vision trait was adaptive for being diurnal (active during the day), which accounts for its retention in only a few groups of mammals.

ISTUDY

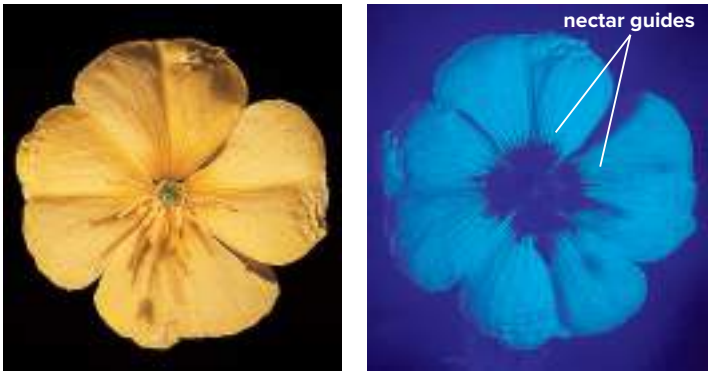


Figure 38.5 Nectar guides. Evening primrose, *Oenothera,* as seen by humans (*left*) and insects (*right*). Humans see no markings, but insects see distinct lines and central blotches, because their eyes respond to ultraviolet rays. These types of markings, known as nectar guides, often highlight the reproductive parts of flowers, where insects feed on nectar and pick up pollen at the same time. (both): Heather Angel/Natural Visions

Vertebrates (including humans) and certain mollusks, such as the squid and the octopus, have a **camera-type eye.** Because mollusks and vertebrates are not closely related, this similarity is an example of convergent evolution. A single lens focuses an image of the visual field on photoreceptors, which are closely packed together. In vertebrates, the lens changes shape to aid focusing, but in mollusks the lens moves back and forth. All of the photoreceptors taken together can be compared to a piece of film in a camera. The human eye is more complex than a camera, however, as you will see.

Animals with two eyes facing forward have three-dimensional vision, or **stereoscopic vision.** The visual fields overlap, and each eye is able to view an object from a different angle. Humans and most predatory animals tend to have stereoscopic vision. Animals with eyes facing sideways, such as rabbits, don't have stereoscopic vision, but they do have **panoramic vision,** meaning their visual field is very wide. Panoramic vision is useful to prey animals because it makes it more difficult for a predator to sneak up on them.

Many vertebrates have a membrane in the back of their eye called a *tapetum lucidum,* which reflects light back into the photoreceptor cells of the retina to increase sensitivity to light. This explains the eerie glowing appearance of some animals' eyes at night.

The Human Eye

The human eye, which is an elongated sphere about 2.5 cm in diameter, has three layers: the sclera, the choroid, and the retina (Fig. 38.6). The outer **sclera** is an opaque, white, fibrous layer that covers most of the eye; in front of the eye, the sclera becomes the transparent **cornea,** the window of the eye. A thin layer of epithelial cells forms a mucous membrane called the **conjunctiva,** which covers the surface of the sclera and keeps the eyes moist.

The middle, thin, dark-brown **choroid** layer contains many blood vessels and a brown pigment that absorbs stray light rays. Toward the front of the eye, the choroid thickens and forms the ring-shaped ciliary body and a thin, circular, muscular diaphragm: the iris. The **iris,** the colored portion of the eye, regulates the size of an opening called the pupil. The **pupil,** like the aperture of a camera lens, regulates light entering the eye. The **lens,** which is attached to the ciliary body by ligaments, divides the cavity of the eye into two portions and helps form images. A basic, watery solution called aqueous humor fills the anterior compartment between the cornea and the lens. The aqueous humor provides a fluid cushion, as well as nutrient and waste transport, for the eye.

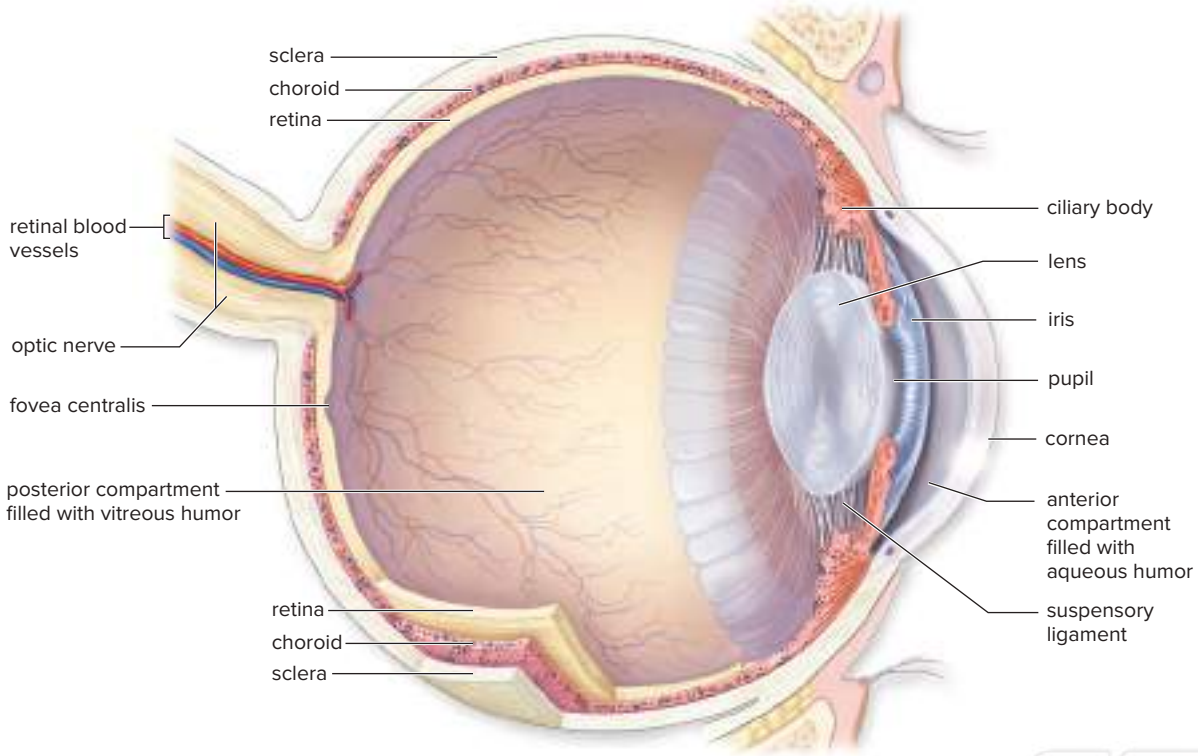


Figure 38.6 Anatomy of the human eye. Notice that the sclera, the outer layer of the eye, becomes the cornea and that the choroid, the middle layer, is continuous with the ciliary body and the iris. The retina, the inner layer, contains the photoreceptors for vision. The fovea centralis is the region where vision is most acute.

ISTUDY

The inner layer of the eye, the **retina,** is located in the posterior compartment. The retina contains photoreceptors called **rod cells** and **cone cells.** The rods are very sensitive to light, but they do not respond to colors; therefore, at night or in a darkened room, we see only shades of gray. Rods are distributed in the peripheral regions of the retina. The cones, which require bright light, are sensitive to different wavelengths of light, and provide humans with the ability to distinguish colors. The retina has a central region called the **fovea centralis,** where cone cells are densely packed. Light is normally focused on the fovea when we look directly at an object. This is helpful because vision is most acute in the fovea centralis.

Sensory fibers form the optic nerve, which takes nerve impulses to the brain. No rods or cones are present where the optic nerve exits the retina (see Fig. 38.10). Therefore, no vision is possible in this area, and it is termed the **blind spot.** You can detect your own blind spot by putting a dot to the right of center on a piece of paper. Use your right hand to move the paper slowly toward your right eye while you look straight ahead. The dot will disappear at one point—this is your blind spot.

How the Eye Focuses

When we look directly at something, such as the printed letters on this page, light rays pass through the pupil and are focused on the retina. The image produced is much smaller than the object, because light rays are bent (refracted) when they are brought into focus. The image on the retina is also upside down and reversed from left to right. When information from the retina reaches the brain, it is processed so that we perceive our surroundings in the correct orientation.

Focusing starts at the cornea and continues as the rays pass through the lens. The lens provides additional focusing power as **visual accommodation** occurs for close vision. The shape of the lens is controlled by the **ciliary muscle** within the ciliary body. When we view a distant object, the ciliary muscle is relaxed, causing the suspensory ligaments attached to the ciliary body to be taut; therefore, the lens remains relatively flat (Fig. 38.7*a*). When we view a near object, the ciliary muscle contracts, releasing the tension on the suspensory ligaments, and the lens becomes more round due to its natural elasticity (Fig. 38.7*b*). Because close work requires contraction of the ciliary muscle, it very often causes muscle fatigue known as eyestrain. With normal aging, the lens loses its ability to accommodate for near objects; thus, many people need reading glasses once they reach middle age.

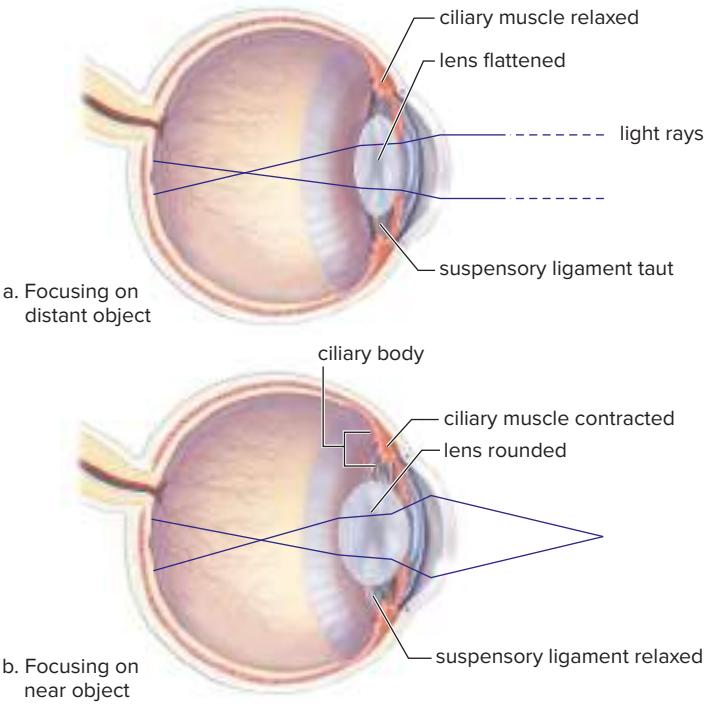


Figure 38.7 Focusing of the human eye. Light rays from each point on an object are bent by the cornea and the lens in such a way that an inverted and reversed image of the object forms on the retina. **a.** When focusing on a distant object, the lens is flat, because the ciliary muscle is relaxed and the suspensory ligament is taut. **b.** When focusing on a near object, the lens accommodates; that is, it becomes rounded, because the ciliary muscle contracts, causing the suspensory ligament to relax.

Photoreceptors of the Eye

Sensory transduction occurs once light has been focused on the photoreceptors in the retina. Figure 38.8 illustrates the structure of these rod cells and cone cells. Both rods and cones have an outer segment joined to an inner segment by a stalk. Pigment molecules that react to light are embedded in the membrane of the many disks present in the outer segment. Synaptic vesicles are located at the synaptic endings of the inner segment.

The visual pigment in rods is called rhodopsin. **Rhodopsin** is a complex molecule made up of the protein opsin and a light-absorbing molecule called *retinal,* which is a derivative of vitamin A. When a rod absorbs light, rhodopsin splits into opsin and retinal, leading to a cascade of reactions and the closure of ion channels in the rod cell's plasma membrane. The release of inhibitory transmitter molecules from the rod's synaptic vesicles ceases. Thereafter, nerve impulses go to the visual areas of the cerebral cortex. Rods are very sensitive to light and therefore are suited to night vision. Carrots and other brightly colored vegetables are rich in *carotenoids,* some of which can be easily converted to vitamin A in the body, so eating these foods may improve night vision. Rod cells are plentiful in the peripheral region of the retina; therefore, they also provide us with peripheral vision and perception of motion.

The cones, by contrast, are located primarily in the fovea centralis and are activated by bright light. They allow us to detect the fine detail and the color of an object. Color vision depends on three different kinds of cones, which contain B (blue), G (green), and R (red) pigments. Each pigment is made up of retinal and opsin, but a slight difference is present in the opsin structure of each, which accounts for the difference in their absorption patterns. Various combinations of cones are believed to be stimulated by in-between shades of color. For example, the color yellow is perceived when green cones are highly stimulated, red cones are partially stimulated, and blue cones are not stimulated. In color blindness, an individual lacks certain visual pigments. As indicated in Section 11.4, some forms of color blindness are a sex-linked hereditary disorder.

ISTUDY

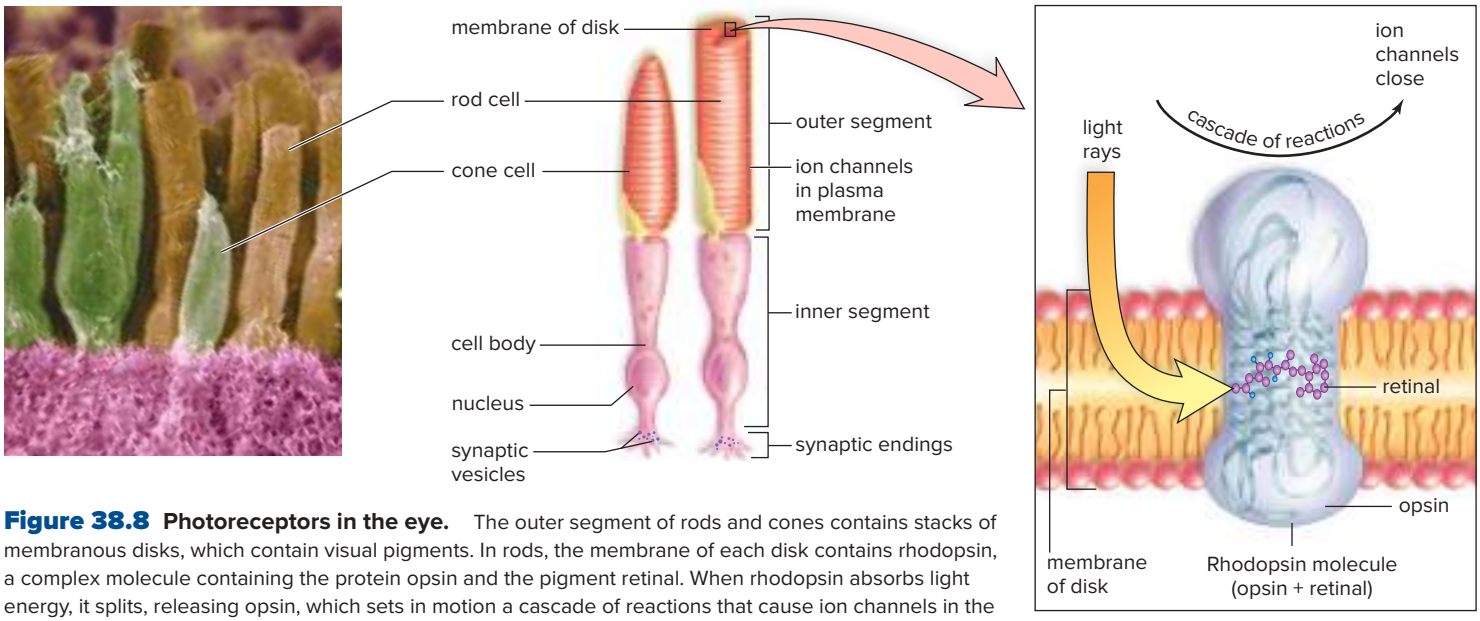


Figure 38.8 Photoreceptors in the eye. The outer segment of rods and cones contains stacks of membranous disks, which contain visual pigments. In rods, the membrane of each disk contains rhodopsin, a complex molecule containing the protein opsin and the pigment retinal. When rhodopsin absorbs light energy, it splits, releasing opsin, which sets in motion a cascade of reactions that cause ion channels in the plasma membrane to close. Thereafter, nerve impulses go to the brain. (photo): Omikron/Science Source

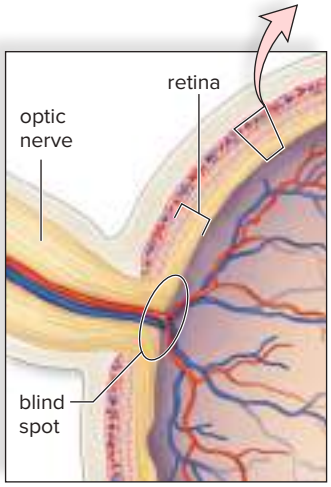
Integration of Visual Signals in the Retina

The retina has three layers of neurons (Fig. 38.9). The layer closest to the choroid contains the rod cells and cone cells; the middle layer contains bipolar cells; and the innermost layer contains ganglion cells, whose sensory fibers become the optic nerve. Only the rod cells and cone cells are sensitive to light; therefore, before these photoreceptors can be stimulated, light must penetrate through the other cell layers.

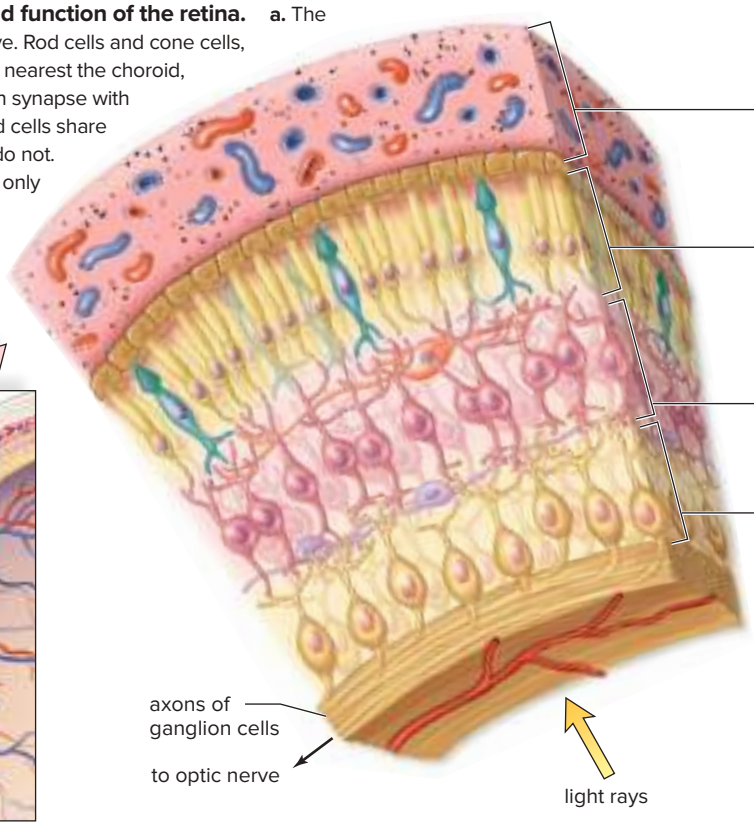
The rod cells and cone cells synapse with the bipolar cells, which in turn synapse with ganglion cells that initiate nerve impulses. Notice in Figure 38.9 that there are many more rod cells and cone cells than ganglion cells. In fact, the human retina has about 150 million rod cells, 6 million cone cells, but only 1 million ganglion cells.

The sensitivity of cones versus rods is mirrored by how directly they connect to ganglion cells. As many as 150 rods may excite the same ganglion cell, meaning that each ganglion cell receives signals

Figure 38.9 Structure and function of the retina. **a.** The retina is the inner layer of the eye. Rod cells and cone cells, located at the back of the retina nearest the choroid, synapse with bipolar cells, which synapse with ganglion cells. Further, many rod cells share one bipolar cell, but cone cells do not. Certain cone cells synapse with only one ganglion cell. Cone cells, in general, distinguish more detail than do rod cells. **b.** Micrograph of the retina. (b): Biophoto Associates/Science Source



a. Location of retina



b. Micrograph of retina

ISTUDY

from rod cells covering about 1 mm^2 of retina (about the size of a thumbtack hole). Therefore, the stimulation of rods results in vision that is blurred and indistinct. In contrast, some cone cells in the fovea centralis excite only one ganglion cell. This explains why cones provide us with a sharper, more detailed image of an object.

From the Retina to the Visual Cortex

The axons of ganglion cells in the retina assemble to form the optic nerves. The optic nerves carry nerve impulses from the eyes to the optic chiasma. The optic chiasma has an X shape, formed by a crossing-over of optic nerve fibers (Fig. 38.10). Fibers from the right half of each retina converge and continue on together in the right optic tract, and fibers from the left half of each retina converge and continue on together in the left optic tract.

The optic tracts sweep around the hypothalamus, and most fibers synapse with neurons in nuclei (masses of neuron cell bodies) within the thalamus. Axons from the thalamic nuclei form optic radiations that take nerve impulses to the visual area within the occipital lobe. Notice in Figure 38.10 that the image arriving at the thalamus, and therefore the visual area, has been split, because the left optic tract carries information about the right portion of the visual field (shown in green) and the right optic tract carries information about the left portion of the visual field (shown in red). Therefore, the right and left visual areas must communicate with each other for us to see the entire visual field. Also, because the image is inverted and reversed, it must be righted in the brain for us to correctly perceive the visual field.

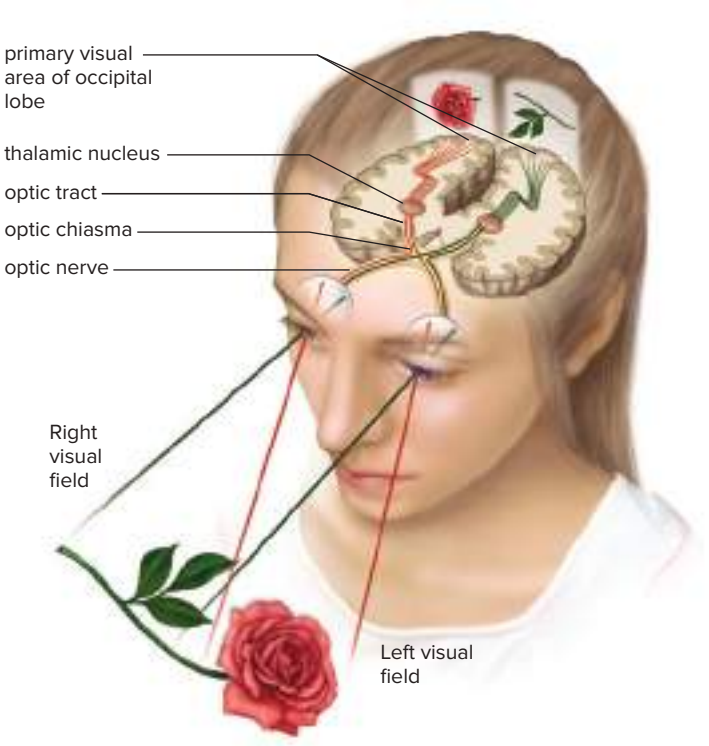


Figure 38.10 Optic chiasma. Both eyes "see" the entire visual field. Because of the optic chiasma, data from the right half of each retina (red lines) go to the right visual cortex, and data from the left half of each retina (green lines) go to the left visual cortex. These data are then combined to allow us to see the entire visual field. Note that the visual pathway to the brain includes the thalamus, which has the ability to filter sensory stimuli.

The most surprising finding has been that the primary visual area acts as a post office, parceling out information regarding color, form, motion, and possibly other attributes to different portions of the adjoining visual association area. Therefore, the brain has taken the visual field apart, even though we see a unified visual field. The visual association areas are believed to rebuild the field and give us an understanding of it at the same time.

Disorders of Vision

Common disorders of vision include diseases of the retina, glaucoma and cataracts, and problems with visual focus.

Diseases of the Retina

Diseases of the retina are the most common cause of blindness in adults. One of these is **diabetic retinopathy,** in which the capillaries to the retina become damaged secondary to diabetes. In **macular degeneration,** the leading cause of blindness for people in the United States under the age of 65, capillaries supplying the retinas become damaged, and hemorrhages and blocked vessels can occur. People who smoke are 20 times more likely to acquire this disorder than people who don't smoke. If diagnosed early, both of these retinal diseases can be treated, using drugs that are injected directly into the vitreous humor. With **retinal detachment,** the retina peels away from the supportive choroid layer, due to eye trauma or other diseases. A detached retina can often be reattached using a laser. As noted in the Nature of Science feature, "Artificial Retinas Come into Focus," artificial retina technology is becoming an option for some people who are blind whose visual pathways remain intact.

Glaucoma and Cataracts

Glaucoma is the second most common cause of blindness in the United States. Glaucoma occurs when the drainage system of the eyes fails, so that aqueous humor builds up and increases intraocular pressure. In the early stages, this pressure tends to destroy the nerve fibers responsible for peripheral vision, but untreated glaucoma can result in total blindness. Eye doctors always check intraocular pressure, but the disorder can come on quickly, and any nerve damage is permanent. Treatment usually involves drugs that increase the outflow of aqueous humor, but surgery may be required.

With aging, the eye is increasingly subject to **cataracts,** in which the lens can become opaque and therefore incapable of transmitting light rays. Cataracts occur in 50% of people between the ages of 65 and 74 and in 70% of those age 75 or older. In most cases, vision can be restored by surgical removal of the unhealthy lens and replacement with a clear plastic artificial lens. Cataract surgery is one of the most frequently performed surgeries in the United States, and the number is increasing as the average age of the population increases. Surgeons may be able to replace cataracts with multifocal lenses that allow the eye to focus, eliminating the need for glasses in some individuals.

Visual Focus Disorders

People who can read what are designated as size 20 letters on an optometrist's chart 20 feet away are said to have 20/20 vision. Those who cannot read these letters but can focus on close objects are said to be **nearsighted (myopic).** These individuals often have an elongated eye, and when they attempt to look at a distant object,

ISTUDY

THEME Nature of Science

Artificial Retinas Come into Focus

Over 25 million people worldwide are blind due to diseases of the retinas. In most cases, there are no cures and few effective treatments. However, after years of work, medical researchers have now developed artificial retina technology that is restoring some vision for people who were completely blind.

Retinal diseases such as retinitis pigmentosa (a genetic disease) cause the rods and cones to die but leave the ganglion cells and neurons intact. In 1988, researchers first demonstrated that a blind person could see light if the nerve cells behind the retina were stimulated with an electrical current. Since then, scientists have been working on developing technological approaches that could take the place of the diseased photoreceptors. In March 2011, a California company received approval in Europe to sell the Argus II, a retinal prosthesis designed for implantation in people who are blind who still have intact connections between their retinas and brain. Clinical trials began in 2015, and there are hopes that the product will become available soon.

The device includes a tiny digital camera embedded in a pair of glasses worn by the individual (Fig. 38A). Images from the camera are translated into electrical signals representing patterns of light and dark. These signals are transmitted wirelessly to a receiver implanted above the ear or near the eye. This receiver in turn sends signals via a tiny wire attached to a 1 mm × 1 mm microchip that has been surgically implanted under the retina (Fig. 38A). The chip contains 60 electrodes, which can stimulate the ganglion cells, producing electrical impulses that travel via the optic nerve to the brain, where they are interpreted as vision.

Although it takes time for people to learn to interpret the patterns of light and dark transmitted by the device, some individuals who are blind have been able to recognize simple objects, see people in front of them follow their movement, and even read large print slowly. Moreover, researchers are already working on upgraded models that incorporate up to 1,500 electrodes, to increase the resolution of the image produced. This is anticipated to be the threshold of information required for people who were formerly blind to recognize faces. Other groups are working on artificial retinas that can be implanted inside the eye itself.

Although the amount of information that is transmitted by an artificial retina device is important, other scientists are focusing on the type of information that is transmitted. A 2010 study conducted at Weill Cornell Medical College in New York by investigators Shiela Nirenberg and Chethan Pandarinath demonstrated that a different approach could restore normal vision in blind mice. First, these scientists focused on deciphering the patterns by which the ganglion cells normally transmit information from the photoreceptors to the brain. Next, they tested these codes in blind mice that had been genetically altered so that their ganglion cells expressed a protein called channelrhodopsin, which initiates a nerve impulse when it is exposed to light. That way, the investigators didn't have to implant artificial retinas in the mice but instead exposed the mice to different patterns of flashing light. The results of these experiments showed that the mice receiving light information using the proper code appeared to be able to see at nearly normal levels.

While eliminating the need for surgery is a major advantage of this approach, it is impossible (or at least unethical) to genetically engineer humans to express certain genes. The solution to this problem may lie in gene therapy, in which the ganglion cells of individuals who are blind would be treated so that they express the *channelrhodopsin* gene. If that approach were successful, people would wear a pair of glasses with an embedded camera that would emit light that had been translated into patterns the brain can understand.

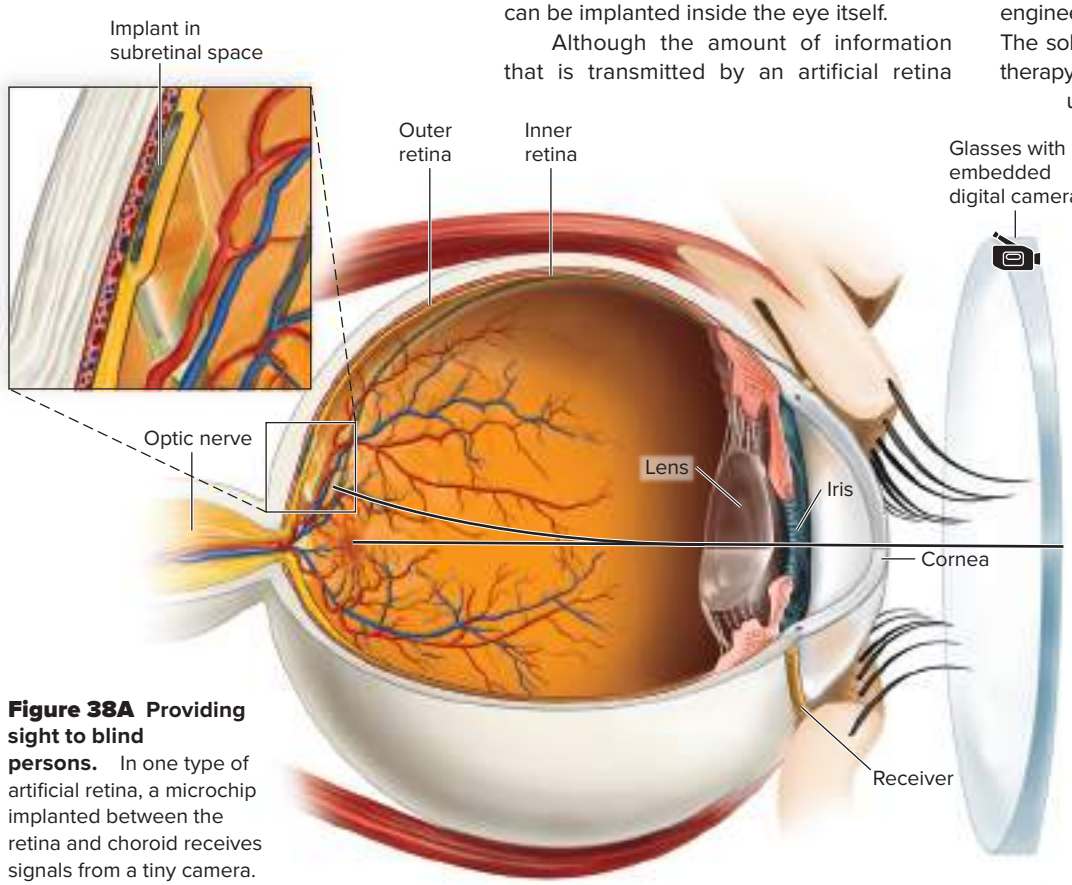
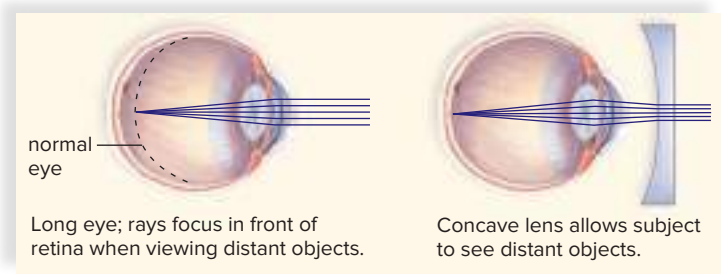


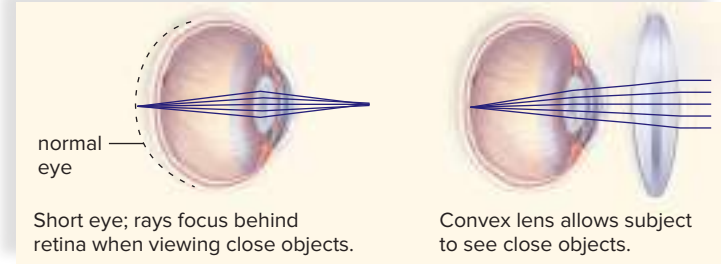
Figure 38A Providing sight to blind persons. In one type of artificial retina, a microchip implanted between the retina and choroid receives signals from a tiny camera.

Questions to Consider

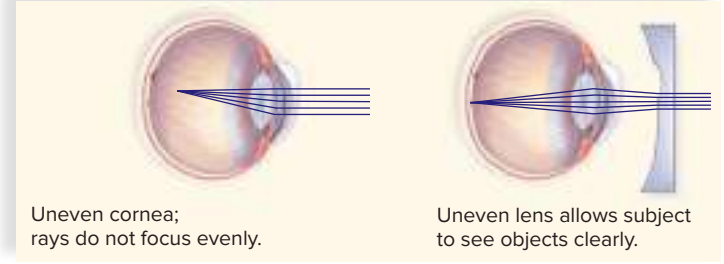
1. Although this technology is very exciting, it is also expensive. Do you believe health insurance companies should pay the large cost, over $100,000, of an artificial retina?
2. Suppose this technology advances to the point where vision can not only be restored but also greatly enhanced. Should it then be legally available to anyone who can afford it—for example, highly paid professional baseball players? What about for children who want to be professional athletes?



a. Nearsightedness



b. Farsightedness



c. Astigmatism

Figure 38.11 Common abnormalities of the eye, with possible corrective lenses. **a.** A concave lens in people who are nearsighted focuses light rays on the retina. **b.** A convex lens in people who are farsighted focuses light rays on the retina. **c.** An uneven lens in persons with astigmatism focuses light rays on the retina.

the image is brought to focus in front of the retina. Concave lenses, which diverge the light rays so the image can be focused on the retina, usually correct this problem (Fig. 38.11*a*).

Those who can easily read the optometrist's chart but cannot easily focus on near objects are said to be **farsighted (hyperopic).** They often have a shortened eye, and when they try to view near objects, the image is focused behind the retina. Convex lenses that increase the bending of light rays allow the image to be focused on the retina (Fig. 38.11*b*).

When the cornea or lens is uneven, the image appears fuzzy. This condition, called **astigmatism,** can sometimes be corrected by an unevenly ground lens to compensate for the uneven cornea (Fig. 38.11*c*).

Rather than wear glasses or contact lenses, many people with visual focus problems are now choosing to undergo laser-assisted in situ keratomileusis, or LASIK, surgery. First, specialists determine how much the cornea needs to be flattened to achieve optimum vision. Controlled by a computer, the laser then removes this amount of the cornea. Each year, around 600,000 LASIK procedures are performed in the United States. Various surveys indicate that about 95% of people who have the LASIK procedure are satisfied with the results.

Check Your Progress 38.3

1. Compare rods and cones in terms of their main functions, their light sensitivity, and the excitation of ganglion cells.
2. List the three layers of cells in the human retina.
3. Summarize how the shape of the eye can result in nearsightedness or farsightedness, and describe the cause of astigmatism.

38.4 Senses of Hearing and Equilibrium

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare several strategies that different types of animals use to hear sounds.
2. Distinguish among the parts of the human ear that make up the outer, middle, and inner ear.
3. Review the differences between rotational and gravitational equilibrium and how each is accomplished.
4. Describe several common disorders affecting hearing and/or equilibrium.

Mechanoreception is sensing physical contact on the surface of the skin or movement of the surrounding environment (such as sound waves in air or water). In animals, mechanoreception is associated with the sense of hearing.

The evolutionary advantage of hearing is that it allows animals to receive information from a distance, as well as from any direction. Hearing plays an important role in avoiding danger, detecting prey, finding mates, and communication. In the most basic sense, hearing is caused by the vibration in a surrounding medium which resonates some part of an animal's body. This resonance is converted into electrical signals that can then be interpreted by the animal's brain. The simplest form of hearing is that associated with mechanoreceptors that exist as free nerve endings on the surface of the animal. In vertebrates, hearing is most often associated with specialized organs, such as ears.

How Animals Detect Sound Waves

Many insects can detect sounds. A common structure involved in insect hearing is a thin membrane, or tympanum, that stretches across an air space, such as the tracheae, which also function in insect respiration (see Section 35.1). Tympanal organs are also located on the thorax (chest) of grasshoppers and on the front legs of crickets. Similar to a mammalian eardrum, the membrane is stimulated to vibrate by sound waves, but in insects this directly activates nerve impulses in attached receptor cells.

The **lateral line** system of fishes (Fig. 38.12) guides them in their movements and in locating other fishes, including predators, prey, and mates. Usually running along both sides of a fish from the gills to the tail, the system detects water currents and pressure waves from nearby objects in a manner similar to the sensory receptors in the human ear. Water from the environment enters tiny

ISTUDY

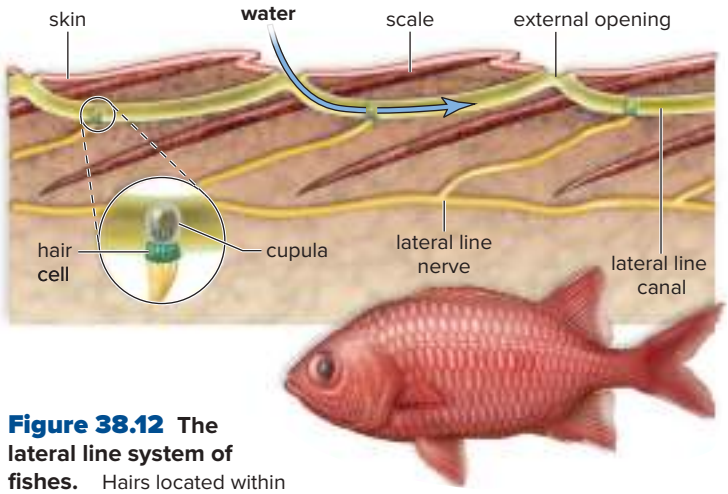


Figure 38.12 The lateral line system of fishes. Hairs located within cupulas near the skin surface detect wave vibrations and currents, helping guide fish movements in order to locate predators and prey, as well as mates.

canals containing hair cells with cilia embedded in a gelatinous cupula. When the cupula bends due to pressure waves, the hair cells initiate nerve impulses.

Most terrestrial vertebrates can hear sound traveling in air, but some, such as amphibians and snakes, are also sensitive to vibrations from the ground, which travel to their inner ear via various parts of their skeleton. A middle ear with an eardrum that transmits sound waves to the inner ear via three small bones is unique to mammals.

The Human Ear

The ear has two sensory functions: hearing and balance (equilibrium). The mechanoreceptors for both of these are located in the inner ear, and each consists of *hair cells* with stereocilia (long microvilli) that are sensitive to mechanical stimulation.

The ear has three distinct divisions: the outer, inner, and middle ear (Fig. 38.13). The **outer ear** consists of the pinna (external "ear") and the auditory canal. The opening of the auditory canal is lined with fine hairs and glands. Glands that secrete earwax, or cerumen, are located in the upper wall of the auditory canal. Earwax helps guard the ear against the entrance of foreign materials, such as air pollutants and microorganisms.

The **middle ear** begins at the **tympanic membrane** (eardrum) and ends at a bony wall containing two small openings covered by membranes. These openings are called the *oval window* and the *round window.* Three small bones are found between the tympanic membrane and the oval window. Collectively called the **ossicles,** individually they are the *malleus* (hammer), the *incus* (anvil), and the *stapes* (stirrup), which are named because their shapes resemble these objects. The malleus adheres to the tympanic membrane, and the stapes touches the oval window.

An **auditory tube** (eustachian tube), which extends from each middle ear to the nasopharynx, permits the equalization of air pressure. When changing elevation, such as in an airplane, the act of chewing gum, yawning, or swallowing opens the auditory tubes wider. As this occurs, we often feel the ears "pop."

Whereas the outer ear and the middle ear contain air, the inner ear is filled with fluids. Anatomically speaking, the **inner ear** has three areas: the **semicircular canals** and the **vestibule** are both concerned with equilibrium; the **cochlea** is concerned with hearing. The cochlea resembles the shell of a snail, because it spirals.

The Auditory Canal and Middle Ear

The process of hearing begins when sound waves enter the auditory canal. Just as ripples travel across the surface of a pond, sound waves travel by the successive vibrations of molecules. Ordinarily, sound waves do not carry much energy, but when a large number of waves strike the tympanic membrane, it moves back and forth (vibrates) ever so slightly. The malleus then takes the pressure from the inner surface of the tympanic membrane and passes it by means of the incus to the stapes in such a way that the pressure is multiplied about 20 times as it moves. The stapes strikes the membrane of the oval window, causing it to vibrate, and in this way, the pressure is passed to the fluid within the cochlea.

Inner Ear

When the snail-shaped cochlea is examined in cross section (Fig. 38.14), the vestibular canal, the cochlear canal, and the tympanic canal become apparent. The cochlear canal contains endolymph, which is similar in composition to interstitial fluid. The vestibular and tympanic canals are filled with perilymph, which is continuous

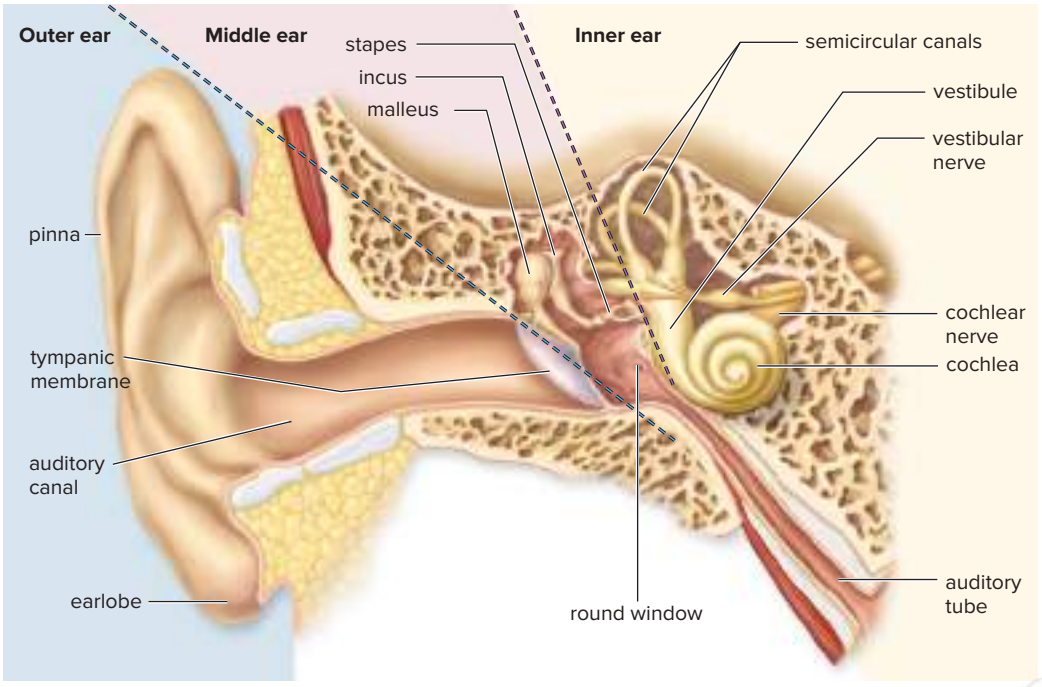
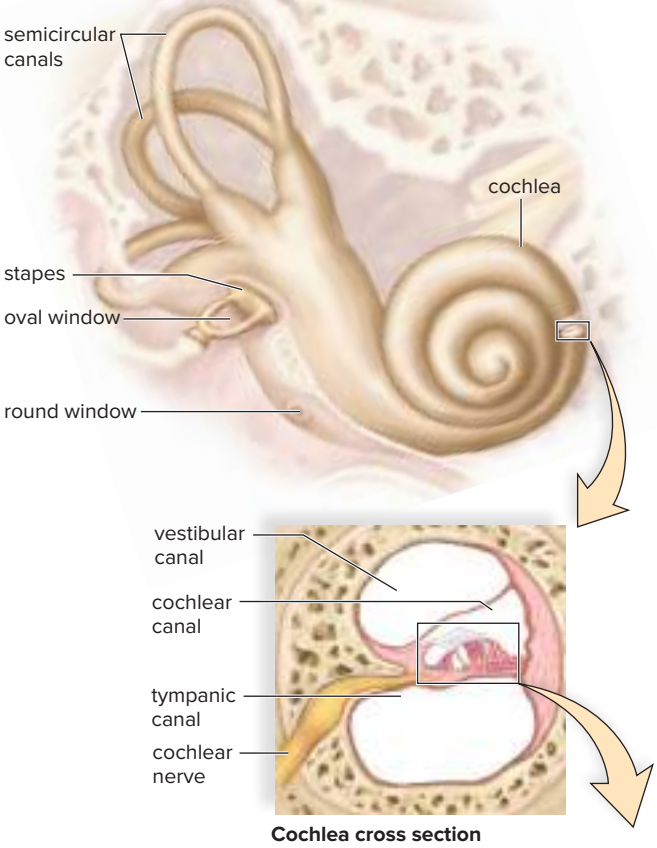
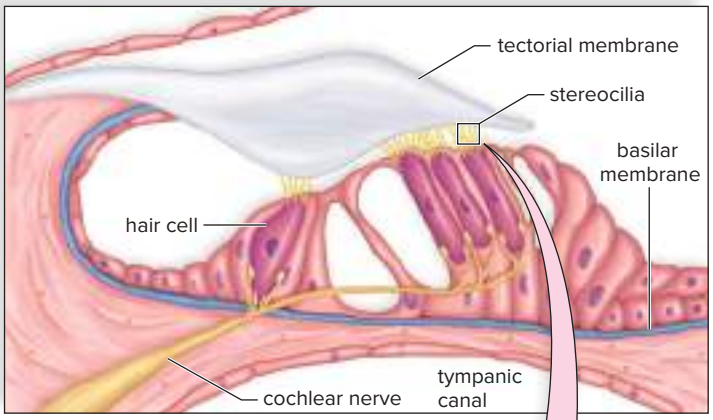


Figure 38.13 Anatomy of the human ear. In the middle ear, the malleus (hammer), the incus (anvil), and the stapes (stirrup) amplify sound waves. In the inner ear, the mechanoreceptors for equilibrium are in the semicircular canals and the vestibule, and the mechanoreceptors for hearing are in the cochlea.

ISTUDY



Cochlea cross section



Organ of Corti

Stereocilia 3,900×

Figure 38.14 Mechanoreceptors for hearing. The organ of Corti is located within the cochlea. In the uncoiled cochlea, the organ consists of hair cells resting on the basilar membrane, with the tectorial membrane above. Pressure waves move from the vestibular canal to the tympanic canal, causing the basilar membrane to vibrate. This causes the stereocilia (or at least a portion of the more than 20,000 hair cells) embedded in the tectorial membrane to bend. Nerve impulses traveling in the cochlear nerve result in hearing. (photo): Prof. P.M. Motta/Univ. "La Sapienza", Rome/Science Source

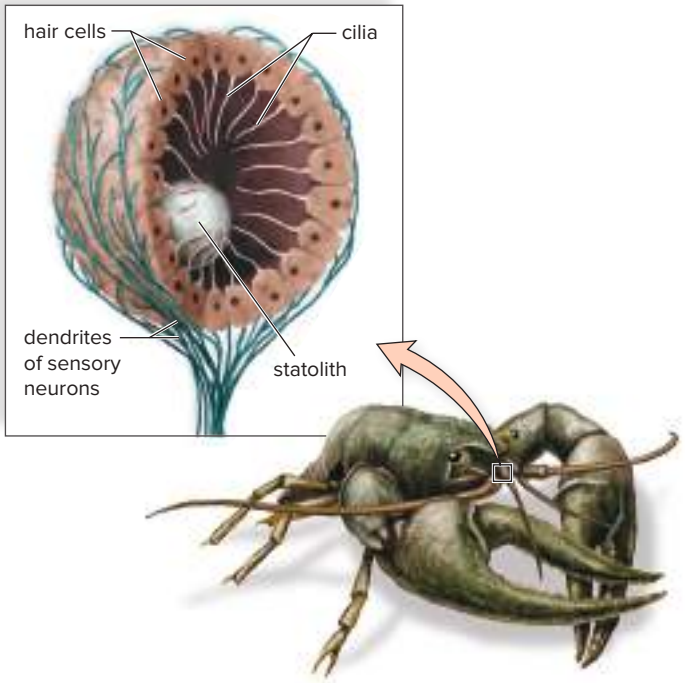


Figure 38.15 Sense of equilibrium in an invertebrate. Within a statocyst, a small particle (the statolith) comes to rest on hair cells and allows a crustacean to sense the position of its head.

with the cerebrospinal fluid. Along the length of the basilar membrane, which forms the lower wall of the *cochlear canal,* are little hair cells whose stereocilia are embedded within a gelatinous material called the *tectorial membrane.* The hair cells of the cochlear canal, called the **organ of Corti,** or spiral organ, synapse with nerve fibers of the *cochlear nerve* (auditory nerve).

When the stapes strikes the membrane of the oval window, pressure waves move from the vestibular canal to the tympanic canal across the basilar membrane, and the round window membrane bulges. The basilar membrane moves up and down, and the stereocilia of the hair cells embedded in the tectorial membrane bend. Then, nerve impulses begin in the cochlear nerve and travel to the brain stem. When they reach the auditory areas of the cerebral cortex, they are interpreted as a sound.

Each part of the organ of Corti is sensitive to a different wave frequency, or pitch of sound. Near the tip, the organ of Corti responds to low pitches, such as the sound of a tuba, and near the base, it responds to higher pitches, such as that of a bell or whistle. The nerve fibers from each region along the length of the organ of Corti lead to slightly different areas in the brain. The pitch sensation we experience depends on which region of the basilar membrane vibrates and which area of the brain is stimulated.

Volume is a function of the amplitude of sound waves. Loud noises cause the fluid in the vestibular canal to exert more pressure and the basilar membrane to vibrate to a greater extent. The resulting increased stimulation is interpreted by the brain as volume. It is believed that the brain interprets the tone of a sound based on the distribution of the hair cells that are stimulated.

Sense of Equilibrium

Gravitational equilibrium organs, called statocysts (Fig. 38.15), are found in cnidarians, mollusks, and crustaceans, which are

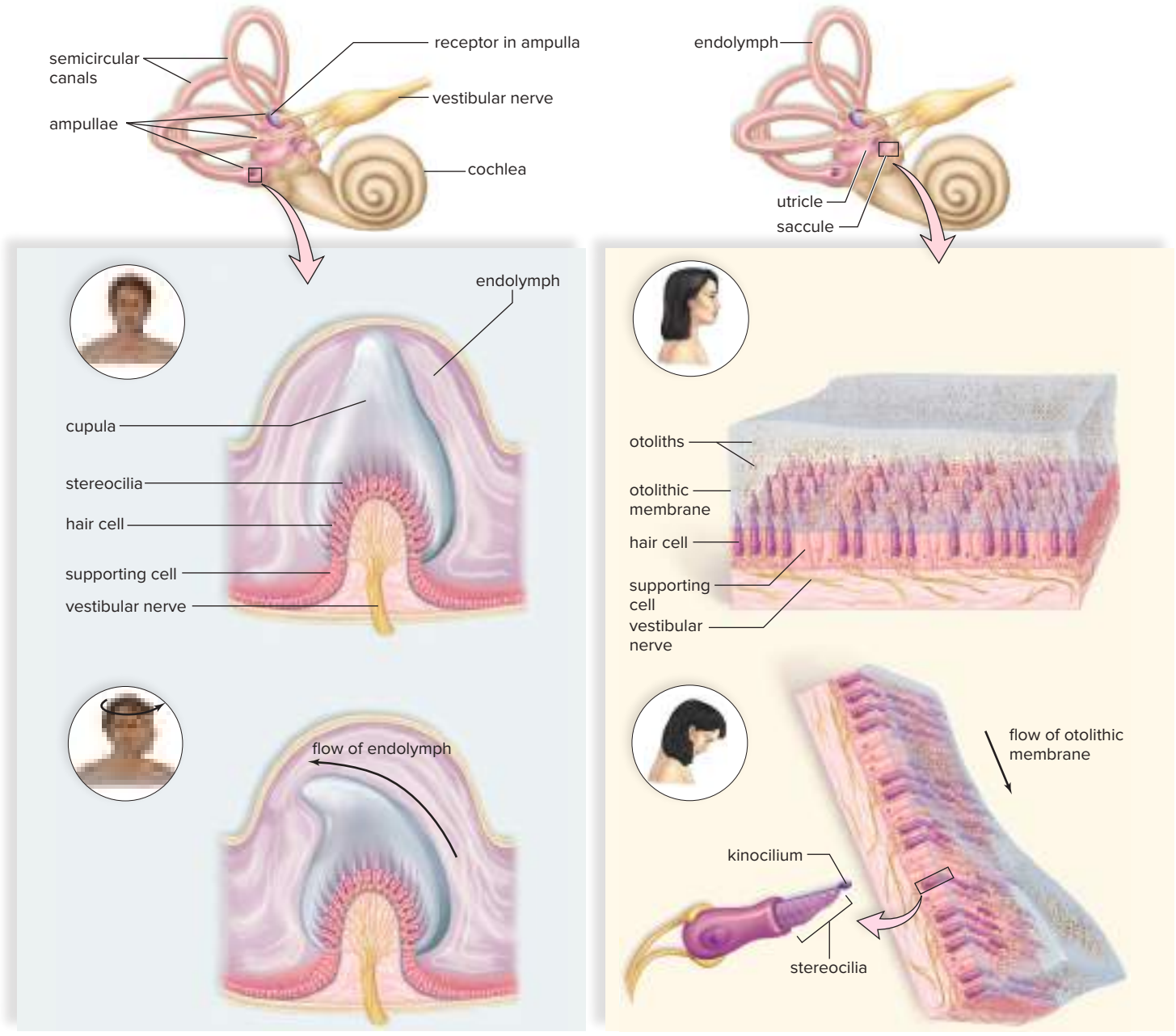
ISTUDY

arthropods. When the head stops moving, a small particle called a statolith stimulates the cilia of the closest hair cells, and these cilia generate impulses that are interpreted as the position of the head.

In the human ear, mechanoreceptors in the semicircular canals detect rotational and/or angular movement of the head (rotational equilibrium), while mechanoreceptors in the utricle and saccule detect straight-line movement of the head in any direction (gravitational equilibrium).

Rotational Equilibrium

Rotational equilibrium (Fig. 38.16*a*) involves the semicircular canals, which are arranged so that there is one in each dimension of space. The base of each of the three canals, called the ampulla, is slightly enlarged. Little hair cells, whose stereocilia are embedded within a gelatinous material called a cupula, are found within the ampullae. Because there are three semicircular canals, each



a. Rotational equilibrium: Receptors in ampullae of semicircular canal

b. Gravitational equilibrium: Receptors in utricle and saccule of vestibule

Figure 38.16 Mechanoreceptors for equilibrium. **a.** Rotational equilibrium. The ampullae of the semicircular canals contain hair cells with stereocilia embedded in a cupula. When the head rotates, the cupula is displaced, bending the stereocilia. Thereafter, nerve impulses travel in the vestibular nerve to the brain. **b.** Gravitational equilibrium. The utricle and the saccule contain hair cells with stereocilia embedded in an otolithic membrane. When the head bends, otoliths are displaced, causing the membrane to sag and the stereocilia to bend. If the stereocilia bend toward the kinocilium, the longest of the stereocilia, nerve impulses increase in the vestibular nerve. If the stereocilia bend away from the kinocilium, nerve impulses decrease in the vestibular nerve. This difference tells the brain in which direction the head moved.

ISTUDY

ampulla responds to head movement in a different plane of space. As fluid (endolymph) within a semicircular canal flows over and displaces a cupula, the stereocilia of the hair cells bend, and the pattern of impulses carried by the vestibular nerve to the brain changes. The brain uses information from the semicircular canals to maintain equilibrium through appropriate motor output to various skeletal muscles that can right our position in space as need be.

Gravitational Equilibrium

Gravitational equilibrium (Fig. 38.16*b*) depends on the **utricle** and **saccule,** two membranous sacs located in the vestibule. Both of these sacs contain little hair cells, whose stereocilia are embedded within a gelatinous material called an otolithic membrane. Calcium carbonate ($CaCO_3$) granules, or **otoliths,** rest on this membrane. The utricle is especially sensitive to horizontal (back and forth) movements of the head, while the saccule responds best to vertical (up and down) movements.

When the head is still, the otoliths in the utricle and the saccule rest on the otolithic membrane above the hair cells. When the head moves in a straight line, the otoliths are displaced and the otolithic membrane sags, bending the stereocilia of the hair cells beneath. If the stereocilia move toward the largest stereocilium, called the kinocilium, nerve impulses increase in the vestibular nerve. If the stereocilia move away from the kinocilium, nerve impulses decrease in the vestibular nerve. If you are upside down, nerve impulses in the vestibular nerve cease. These data tell the brain the direction of the movement of the head.

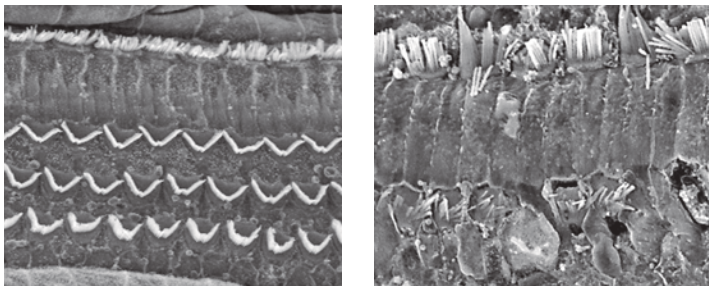
Disorders of Hearing and Equilibrium

Hearing loss can develop gradually or suddenly and has many potential causes. Especially in children, the middle ear is subject to infections that, in severe cases, can lead to hearing loss. Age-associated hearing loss usually develops gradually, beginning at around age 20. By age 60, about one in three people report significant hearing loss.

Most cases of hearing loss can be attributed to the effect of years of frequent (and preventable) exposure to loud noise, which can damage the stereocilia of the spiral organ. Prolonged exposure to noise above a level of 80 decibels can damage the hair cells of the organ of Corti (Fig. 38.17). The first signs of noise-induced hearing loss are usually muffled hearing, pain or a "full" feeling in the ears, or tinnitus (ringing in the ears). If you have any of these symptoms, take steps immediately to prevent further damage. If exposure to loud noise is unavoidable, noise-reducing earmuffs or earplugs are available.

Some types of deafness can be present at birth. Several genetic disorders can interfere with the ability to hear, as can infections with German measles (rubella) or mumps virus when contracted by a female during pregnancy. For this reason, all females should be vaccinated against these viruses before they reach childbearing age.

People with certain types of deafness may be able to hear again with cochlear implants. A cochlear implant consists of an external device that sits behind the ear and an internal device surgically implanted under the skin. The external part picks up sounds from the environment and converts them into electrical impulses, which are sent directly to different regions of the auditory nerve and then to the brain.



a. b.

Figure 38.17 Studies of hearing loss. **a.** Microscopic view of normal hair cells in the organ of Corti of a guinea pig. **b.** Damage to these hair cells occurred after 24 hours of exposure to a noise level typical of a rock concert. (both photos): Dr. Yehoash Raphael/University of Michigan

Disorders of equilibrium often manifest as *vertigo,* the feeling that a person or the environment is moving when no motion is occurring. It is possible to simulate a feeling of vertigo by spinning your body rapidly and then stopping suddenly. Vertigo can be caused by problems in the brain, as well as the inner ear. An estimated 20% of those who experience these symptoms have *benign positional vertigo (BPV),* which may result from the formation of abnormal particles in the semicircular canals. When individuals with BPV move their head suddenly, especially when lying down, these particles shift like pebbles inside a tire. When the movement stops, the particles tumble down with gravity, stimulating the stereocilia and resulting in the sensation of movement.

Because the senses of hearing and equilibrium are anatomically linked, certain disorders can affect both. An example of this is *Ménière's disease,* which is usually characterized by vertigo, a feeling of fullness in the affected ear(s), tinnitus, and hearing loss. The disease usually occurs between the ages of 20 and 50, and only one ear is affected in about 80% of cases. The exact cause of this disease is unknown, but it seems related to an increased volume of fluid in the semicircular canals, vestibule, and/or cochlea. For this reason, it has been called "glaucoma of the ear." No cure is known for Ménière's disease, but the vertigo and feeling of pressure can often be managed by adhering to a low-salt diet, which is thought to decrease the amount of fluid present. Any hearing loss that occurs, however, is usually permanent.

Check Your Progress 38.4

1. Determine whether each of the following belongs to the outer, middle, or inner ear: a. ossicles, b. pinna, c. semicircular canals, d. cochlea, e. vestibule, f. auditory canal.
2. List, in order, the structures that must conduct a sound wave from the time it enters the auditory canal until it reaches the cochlea.
3. Identify which structures of the inner ear are responsible for gravitational equilibrium and for rotational equilibrium.

ISTUDY

38.5 Somatic Senses

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast the functions of proprioceptors, cutaneous receptors, and pain receptors.
2. List the specific types of cutaneous receptors that are sensitive to fine touch, pressure, pain, and temperature.

Senses whose receptors are associated with the skin, muscles, joints, and viscera are termed the *somatic senses.* These receptors can be categorized into three types: proprioceptors, cutaneous receptors, and pain receptors. All of these send nerve impulses via the spinal cord to the primary somatosensory areas of the cerebral cortex (see Fig. 37.10).

Proprioceptors

Proprioceptors are mechanoreceptors involved in reflex actions that maintain muscle tone, and thereby the body's equilibrium and posture. For example, proprioceptors called *muscle spindles* are embedded in muscle fibers (Fig. 38.18). If a muscle relaxes too much, the muscle spindle stretches, generating nerve impulses that cause the muscle to contract slightly. Conversely, when muscles are stretched too much, proprioceptors called *Golgi tendon organs,* buried in the tendons that attach muscles to bones, generate nerve impulses that cause the muscles to relax. Both types of receptors act together to maintain a functional degree of muscle tone.

Cutaneous Receptors

The skin is composed of several layers, including the epidermis and dermis (see Fig. 31.8). The dermis contains many **cutaneous receptors,** which make the skin sensitive to touch, pressure, pain, and temperature.

Four types of cutaneous receptors are sensitive to fine touch. *Meissner corpuscles* and *Krause end bulbs* are concentrated in the fingertips, palms, lips, tongue, nipples, penis, and clitoris. *Merkel disks* are found where the epidermis meets the dermis. A free nerve ending called a *root hair plexus* winds around the base of a hair follicle and fires if a hair is touched.

Two types of cutaneous receptors are sensitive to pressure. *Pacinian corpuscles* are onion-shaped sensory receptors deep inside the dermis. *Ruffini endings* are encapsulated by sheaths of connective tissue and contain lacy networks of nerve fibers.

At least two types of free nerve endings in the epidermis are thermoreceptors. Both cold and warm receptors contain ion channels with activities that are affected by temperature. Cold receptors generate nerve impulses at an increased frequency as the temperature drops; warm receptors increase activity as the temperature rises. Some chemicals (e.g., menthol) can stimulate cold receptors.

Pain Receptors

The skin and many internal organs and tissues have pain receptors, also called free nerve endings or *nociceptors.* Regardless of the cause, damaged cells release chemicals that cause nociceptors to generate nerve impulses, which the brain interprets as pain. Other types of nociceptors are sensitive to extreme temperatures or excessive pressure.

Pain receptors have arisen in evolution because they alert us to potential danger. If you accidentally reach too close to a fire, for example, the reflex action of withdrawing your hand will help protect you from further tissue damage, while the unpleasant sensation your brain perceives will help you remember not to reach too close to a fire again.

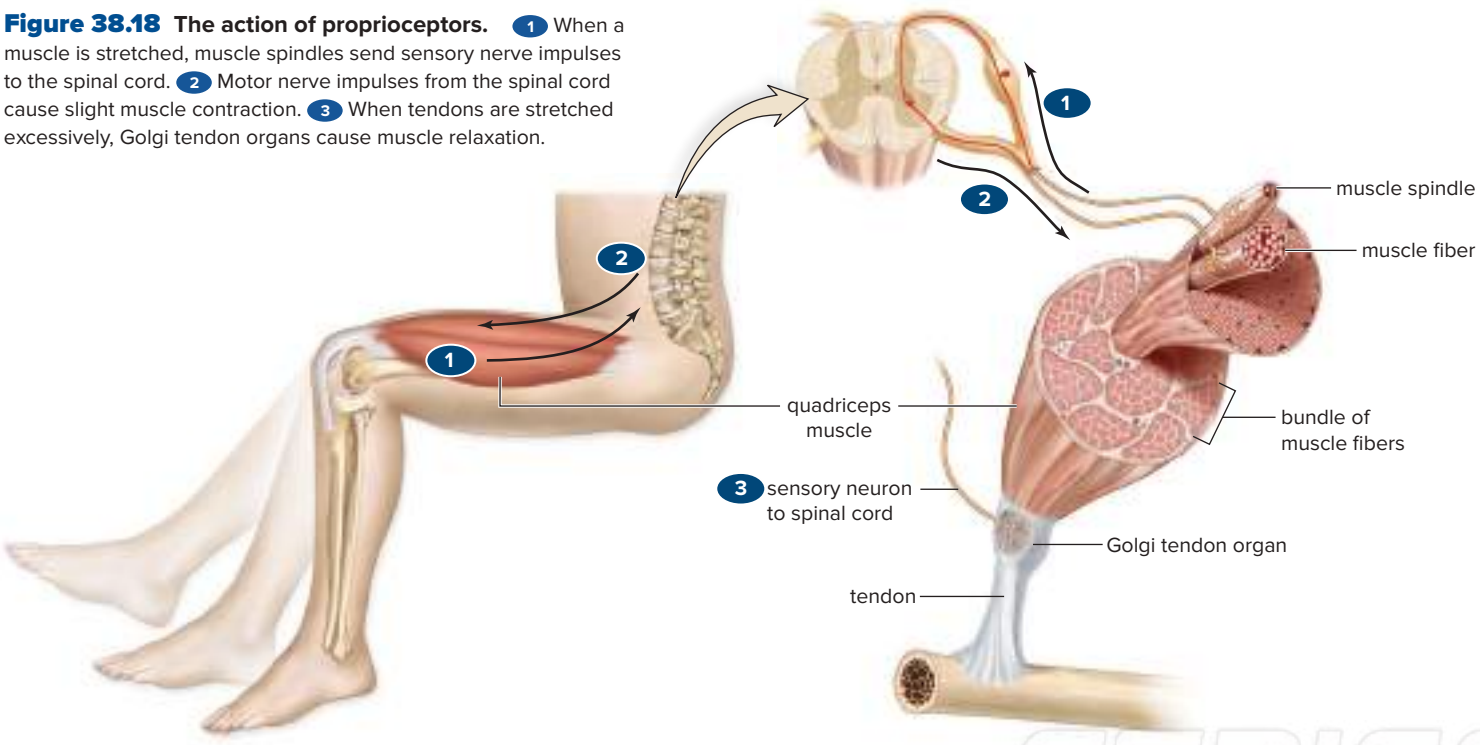


Figure 38.18 The action of proprioceptors. 1 When a muscle is stretched, muscle spindles send sensory nerve impulses to the spinal cord. 2 Motor nerve impulses from the spinal cord cause slight muscle contraction. 3 When tendons are stretched excessively, Golgi tendon organs cause muscle relaxation.

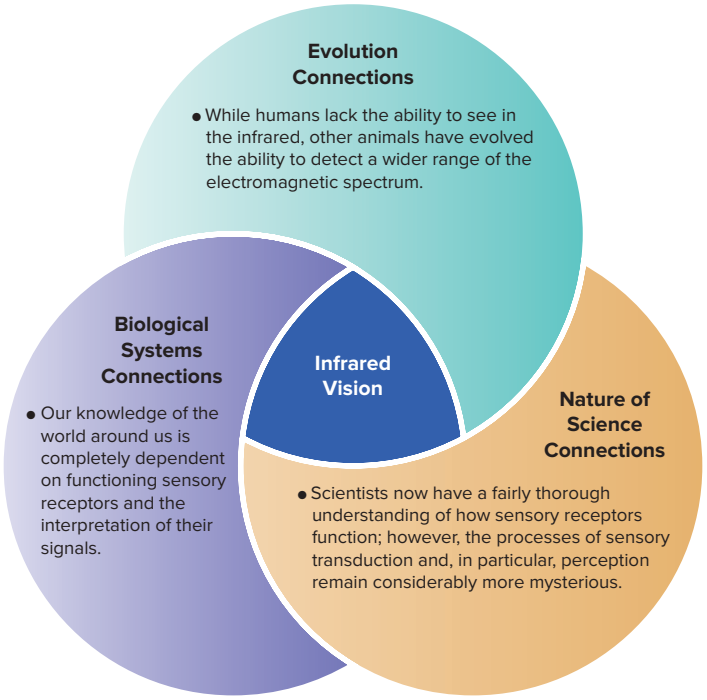
ISTUDY

Unfortunately, despite the evolutionary benefit of acute pain, chronic pain often serves no such purpose. To relieve such nonadaptive pain, a variety of painkilling medications, or *analgesics,* have been developed. If the source of the pain is inflammation, anti-inflammatory medications can be used. These include natural anti-inflammatory compounds like corticosteroids, or nonsteroidal anti-inflammatory drugs (NSAIDs) like aspirin or ibuprofen. Each day, millions of individuals use NSAIDs, which are generally available over-the-counter, and work by inhibiting enzymes that generate inflammatory chemicals called prostaglandins. For more intense pain, opioid medications such as morphine or oxycodone can be prescribed. These stimulate receptors in the brain for another class of naturally occurring analgesics, the *endorphins.*

Check Your Progress 38.5

1. Identify the problems that would likely occur if a person lacked muscle spindles or nociceptors.
2. In evolutionary terms, assess why cutaneous receptors quickly become adapted to stimuli (e.g., why we don't continue to feel a chair once we settle in), whereas the sense of pain seems to be much less adaptable (e.g., many people have chronic pain).

Connecting *the* Concepts



What Did You Learn?

38.1 Sensory Receptors

Sensory receptors evolved to enable animals to receive and respond to information about their internal and external environments. Sensory receptors are specialized cells capable of detecting various stimuli, then performing **sensory transduction,** or conversion of those events into nerve impulses. **Perception** is the awareness of these stimuli.

- Four types of sensory receptors are **chemoreceptors, electromagnetic receptors** (including **photoreceptors**), **mechanoreceptors,** and **thermoreceptors.**

38.2 Chemical Senses

Chemoreception is found universally in animals and is therefore believed to be the most primitive sense.

- Chemoreceptors are present in a wide variety of locations in animals. Examples of chemoreceptors in humans include **taste buds** and **olfactory cells.**

38.3 Sense of Vision

Most animals have **photoreceptors** that are sensitive to light.

- Arthropods have **compound eyes;** vertebrates and some mollusks have a **camera-type eye.**
- Animals with eyes that face sideways have **panoramic vision,** while those with eyes facing forward have **stereoscopic vision.**
- In humans, vision is dependent on the eye, the optic nerves, and the visual areas of the cerebral cortex. The human eye has three layers. The **sclera** can be seen as the white of the eye; it also becomes the transparent bulge in the front of the eye called the **cornea.** The surface of the sclera is covered by the **conjunctiva.** The middle **choroid** layer contains blood vessels and pigment that absorbs stray light rays. It also forms the **iris,** which regulates the size of the **pupil.** The **lens** helps form images. The **retina,** or inner layer, contains **rod cells** (sensory receptors for dim light) and **cone cells** (sensory receptors for bright light and color). Many cones are present in the **fovea centralis,** but no rods or cones are in the **blind spot.**

ISTUDY

- The cornea begins focusing light, and the lens provides further **visual accommodation,** controlled by the **ciliary muscle.** The retina is composed of three layers of cells: the rod and cone layer, the bipolar cell layer, and the ganglion cell layer. When light strikes **rhodopsin** in the membranous disks of rod cells, rhodopsin splits into opsin and retinal, generating nerve impulses that are transmitted from the ganglion cells, via the optic nerve, to the brain. On their way, the optic nerves cross at the optic chiasma and pass through the thalamus before images are interpreted by the visual cortex.

Common causes of blindness include retinal disorders like **diabetic retinopathy, macular degeneration,** and **retinal detachment. Glaucoma** is an abnormal increase in intraocular pressure. **Cataracts,** or opaque lenses, affect transmission of light. People with visual focus disorders may be **nearsighted (myopic), farsighted (hyperopic),** or have **astigmatism.**

38.4 Senses of Hearing and Equilibrium

Animals use a variety of strategies to detect sounds or vibrations in their environment.

- Insects have tympanal organs to detect sounds; fishes have a **lateral line** system to detect vibrations in water.
- Hearing in humans is dependent on the ear, the cochlear nerve, and the auditory areas of the cerebral cortex. The ear is divided into three parts. The **outer ear** consists of the pinna and the auditory canal, which direct sound waves to the middle ear. The **middle ear** begins with the **tympanic membrane** and contains the **ossicles** (malleus, incus, and stapes). The malleus is attached to the tympanic membrane, and the stapes is attached to the oval window, which is covered by membrane. An **auditory tube** connects each middle ear to the nasopharynx. The **inner ear** contains the **cochlea** for hearing, and the **semicircular canals** and **vestibule** for equilibrium.
- Hearing begins when the outer and middle portions of the ear convey and amplify the sound waves that strike the oval window. Its vibrations set up pressure waves within the cochlea, which contains the **organ of Corti,** consisting of hair cells whose stereocilia are embedded in the tectorial membrane. When the stereocilia of the hair cells bend, nerve impulses begin in the cochlear nerve and are carried to the brain.

The ear also contains receptors for our sense of equilibrium.

- **Rotational equilibrium** is dependent on the stimulation of hair cells in the ampullae of the semicircular canals.
- **Gravitational equilibrium** relies on the stimulation of hair cells by **otoliths** in the **utricle** and **saccule,** inside the vestibule.
- Disorders affecting hearing and balance include age-associated hearing loss, **vertigo,** and Ménière's disease.

38.5 Somatic Senses

The somatic sensory receptors include proprioceptors, sensory receptors, and pain receptors.

- **Proprioceptors,** such as muscle spindles and Golgi tendon organs, help maintain equilibrium and posture.
- **Cutaneous receptors** in the skin detect touch, pressure, pain, and temperature.
- Pain receptors (nociceptors) are free nerve endings that respond to chemicals released by damaged cells.

Thinking Critically About Biology

1. Some sensory receptors, such as those for taste, smell, and pressure, readily undergo the process of sensory adaptation, or decreased response to a stimulus. In contrast, receptors for pain are less prone to adaptation. Why does this make good biological sense? What do you think happens to children who are born without the ability to feel pain normally?
2. The density of taste buds on the tongue can vary. Some individuals with obesity have a lower density of taste buds than usual. Assume that taste perception is related to taste bud density. If so, what hypothesis would you test to see if there is a relationship between taste bud density and obesity?
3. Imagine you are in a room with very little light, and you are having trouble finding the light switch. You notice you can see better with your peripheral vision than in the center of your visual field. Why is this the case?

Making It Relevant to You

1. Artificial retina systems, such as those being developed for certain types of blindness, may be altered in the future to allow detection of other wavelengths of light, such as infrared. Why might perception of these wavelengths in humans be a problem? What other body systems would potentially have to be altered?
2. Why would animals such as snakes have evolved the ability to see infrared, but not humans or primates?
3. How would you determine that a disease affecting vision or hearing has a genetic component?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

39

Locomotion and Support Systems

Soeren Stache/picture alliance/Getty Images

Chapter Outline

39.1 Diversity of Skeletons
39.2 The Human Skeletal System
39.3 The Muscular System

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 37.6 What is the function of acetylcholine in the transmission of a nerve impulse to skeletal muscle?

Sections 37.3 and **37.4** How does the primary motor area of the cerebral cortex generate commands to skeletal muscle, and how does the somatic division of the PNS control the muscles?

Section 38.1 How do the various types of sensory receptors provide information and feedback to the brain?

A World Record Marathon

At the 2018 Berlin Marathon, Eliud Kipchoge of Kenya set a new world record of 2 hours 1 minute and 39 seconds on the 26.2-mile course, for an average pace of 4 minutes and 38.4 seconds per mile. However, this was not his personal best. A year earlier, Kipchoge ran a similar (but unratified) marathon in 2 hours and 25 seconds. Any sort of movement, including marathons, requires a close coordination between the muscular and skeletal systems. The same is true when eagles fly, fish swim, or animals feed, escape prey, reproduce, or simply play. Although some animals lack muscles and bones, they all use contractile fibers to move about at some stage of their lives. In many invertebrates, muscles push against body fluids located inside a body cavity.

Only in vertebrates are muscles attached to a bony endoskeleton. Aside from giving the body shape and protecting internal organs, the skeleton serves as a storage area for inorganic calcium and produces blood cells. While contributing to body movement, the skeletal muscles give off heat, which warms the body.

As you read through this chapter, think about the following questions:

1. How does the nervous system specifically control the skeletal system?
2. How do the sensory systems exert influence over the nervous system and therefore over the muscular and skeletal systems?

Following *the* Themes

CHAPTER 39 – LOCOMOTION AND SUPPORT SYSTEMS

Evolution	The need for protection, support, and mobility in different environments has led to the evolution of a number of different skeletal systems.
Nature of Science	Investigations into bone and muscle functions have provided an understanding of how animals are adapted to their lifestyles, and have generated many applications to human health.
Biological Systems	In humans, the skeletal system protects the internal organs, stores ions and contributes to homeostasis, produces blood cells for the circulatory system, and works together with the muscular system.

ISTUDY

39.1 Diversity of Skeletons

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the structure of a hydrostatic skeleton and list some examples of animals that possess one.
2. Discuss some advantages of having an endoskeleton versus an exoskeleton.
3. Provide several examples of how mammalian skeletons are adapted to particular forms of locomotion.

Skeletons serve as support systems for animals, providing rigidity, protection, and surfaces for muscle attachment. Several different kinds of skeletons occur in the animal kingdom. As we will see, these range from the use of fluid-filled body cavities to provide support, to tough external skeletons, to the hard skeletons that consist of the bones and cartilage you are familiar with in our internal skeletons.

Hydrostatic Skeleton

In animals that lack a hard skeleton, a fluid-filled gastrovascular cavity or a fluid-filled coelom can act as a hydrostatic skeleton. A **hydrostatic skeleton** utilizes fluid pressure to offer support and resistance to the contraction of muscles, so that mobility results. As analogies, consider that a garden hose stiffens when filled with water, and that a water-filled balloon changes shape when squeezed at one end. Similarly, an animal with a hydrostatic skeleton can change shape and perform a variety of movements.

Hydras and planarians use their fluid-filled gastrovascular cavity as a hydrostatic skeleton. The tentacles of a hydra also have hydrostatic skeletons, allowing them to be extended to capture food. Roundworms have a fluid-filled pseudocoelom and move in a whiplike manner when their longitudinal muscles contract.

The coelom of annelids, such as earthworms, is segmented and has septa that divide it into compartments (Fig. 39.1). Each segment has its own set of longitudinal and circular muscles and its own nerve supply, so each segment or group of segments may function independently. When circular muscles contract, the segments become thinner and elongate. When longitudinal muscles contract, the segments become thicker and shorten. By alternating circular muscle contraction and longitudinal muscle contraction and by using its setae to hold its position during contractions, the animal moves forward.

Use of Muscular Hydrostats

Even animals that have an exoskeleton or an endoskeleton move selected body parts by means of *muscular hydrostats,* meaning that fluid contained within the individual cells that constitute a muscle assists with movement of that part. Muscular hydrostats are used by clams to extend their muscular foot and by sea stars to extend their tube feet. Spiders depend on them to move their legs, and moths rely on them to extend their proboscis. In vertebrates, movement of an elephant's trunk involves a muscular hydrostat that allows the animal to reach high into trees, pick up a morsel of food off the ground, or manipulate other objects.

Exoskeleton and Endoskeleton

Mollusks and arthropods have a rigid **exoskeleton,** an external covering composed of a stiff material. The strength of an exoskeleton can be improved by increasing its thickness and weight, but this leaves less room for internal organs.

In mollusks, such as snails and clams, a thick and nonmobile calcium carbonate shell is primarily used for protection against the environment and predators. A mollusk's shell can grow as the animal grows.

The exoskeleton of arthropods, such as insects and crustaceans, is composed of chitin, a strong, flexible, nitrogenous polysaccharide. Their exoskeleton protects them against wear and tear, predators, and desiccation (drying out)—an important feature for arthropods that live on land. Working together with muscles, the jointed and movable appendages of arthropods allow them to crawl, fly, and/or swim. Because their exoskeleton is of a fixed

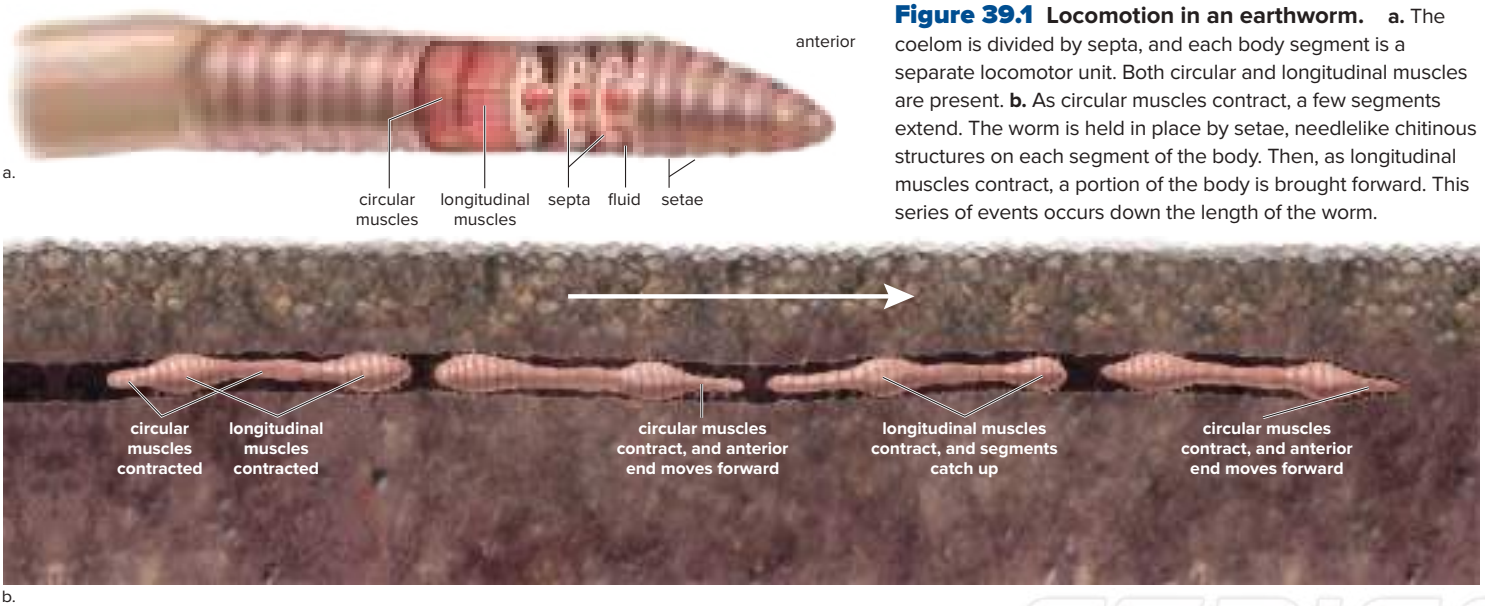


Figure 39.1 Locomotion in an earthworm. **a.** The coelom is divided by septa, and each body segment is a separate locomotor unit. Both circular and longitudinal muscles are present. **b.** As circular muscles contract, a few segments extend. The worm is held in place by setae, needlelike chitinous structures on each segment of the body. Then, as longitudinal muscles contract, a portion of the body is brought forward. This series of events occurs down the length of the worm.

ISTUDY

Figure 39.2 Exoskeleton. Exoskeletons support muscle contraction and prevent drying out. The chitinous exoskeleton of an arthropod is shed as the animal molts; until the new skeleton dries and hardens, the animal is more vulnerable to predators, and muscle contractions may not translate into body movements. In this photo, a cicada has just finished molting. Martin Shields/Alamy Stock Photo

size, however, arthropods must *molt,* or shed their skeleton, in order to grow (Fig. 39.2).

Both echinoderms and vertebrates have an **endoskeleton,** which is made up of rigid internal structures. The skeleton of echinoderms consists of spicules and plates of calcium carbonate embedded in the living tissue of the body wall. In contrast, the vertebrate endoskeleton is living tissue. Sharks and rays have skeletons composed only of cartilage. Other vertebrates, such as bony fishes, amphibians, reptiles, birds, and mammals, have endoskeletons composed of bone and cartilage.

The advantages of the jointed vertebrate endoskeleton are listed in Figure 39.3. An endoskeleton grows with the animal, so molting is not required. It supports the weight of a large animal without limiting the space for internal organs. An endoskeleton also offers protection to vital internal organs, but it is protected by the soft tissues around it. Injuries to soft tissue are usually easier to repair than injuries to a hard skeleton. Compared to the relatively limited mobility of arthropod appendages, vertebrate limbs are generally more flexible and have different types of joints, allowing for even more complex movements.

The skeletons of mammals come in many sizes and shapes, which are often adapted to a particular mode of locomotion. Aquatic animals such as seals, sea lions, whales, and dolphins have a streamlined, torpedo-shaped skeleton that facilitates movement through water. Many animals that jump, such as kangaroos and rabbits, have a compact skeleton with elongated hindlimbs that propel them forward. Carnivores, such as members of the cat family, walk on their toes, which is an adaptation to running and chasing prey. (Note that when humans run, we push off with our toes to move faster.) Hoofed mammals, such as horses and deer, have evolved long legs and run on the tips of elongated phalanges. The lowest part of each limb of a horse consists entirely of a modified third digit.

Humans are bipedal and walk on the soles of the feet formed by the tarsal and metatarsal bones. This form of locomotion allows the hands to be free and may have evolved from monkeys' and apes' habit of using only forelimbs as they swing through the

a.

Advantages of Jointed Endoskeleton
Can grow with the animal
Supports the weight of large animals
Protects vital internal organs
Is protected by outer tissues
Allows flexible movements

b.

Figure 39.3 The vertebrate endoskeleton. **a.** The jointed vertebrate endoskeleton has (**b**) a number of advantages. (a): revers/Shutterstock

branches of trees. Dexterity of hands and feet is actually the ancestral mammalian condition. In humans and apes, the bones of the hands and feet are not fused, and the wrist and ankle can rotate in three dimensions.

Check Your Progress 39.1

1. List the types of skeletons found in animals.
2. Describe the type of support system that makes it possible to stick out your tongue.
3. Explain why an earthworm loses its cylindrical shape when it dies.

39.2 The Human Skeletal System

Learning Outcomes

Upon completion of this section, you should be able to

1. List the five major functions of the skeletal system.
2. Describe the macroscopic and microscopic structure of bone.
3. Identify the major bones that constitute the human axial and appendicular skeletons.

The skeletal system has many functions that contribute to homeostasis:

- *Support of the body.* The rigid skeleton provides an internal framework that largely determines the body's shape.

ISTUDY

- *Protection of vital internal organs,* such as the brain, heart, and lungs. The bones of the skull protect the brain; the rib cage protects the heart and lungs. The vertebrae protect the spinal cord.
- *Sites for muscle attachment.* The pull of muscles on the bones makes movement possible. Articulations (joints) occur between all the bones, but we associate body movement particularly with jointed appendages.
- *Storage reservoir for ions.* All bones have a matrix that contains calcium phosphate, a source of calcium ions and phosphate ions in the blood.
- *Production of blood cells.* Blood cells and other blood elements are produced in the red bone marrow of the skull, ribs, sternum, pelvis, and long bones.

In this section, we describe the characteristics of human bones, the components of various regions of the skeleton, and the different types and functions of joints.

Bone Growth and Renewal

During prenatal development, the structures that will form the bones of the human skeleton are composed of cartilage. Because these cartilaginous structures are shaped like the future bones, they provide "models" of these bones. The models are converted to bones as calcium salts are deposited in the matrix (nonliving material), first by the cartilage cells and later by bone-forming cells called **osteoblasts** (Gk. *osteon,* "bone"; *blastos,* "bud"). The conversion of cartilaginous models to bones is called endochondral ossification.

In some cases, ossification occurs without any previous cartilaginous model. This type of ossification occurs in the dermis and forms bones called dermal bones. Examples include the mandible (lower jaw), certain bones of the skull, and the clavicle (collarbone). During intramembranous ossification, fibrous connective tissue membranes give support as ossification begins.

Endochondral ossification of a long bone begins in a region called a primary ossification center, located in the middle of the cartilaginous model. In the primary ossification center, the cartilage is broken down and invaded by blood vessels, and cells in the area mature into bone-forming osteoblasts. Later, secondary ossification centers form at the ends of the model. A cartilaginous *growth plate* remains between the primary ossification center and each secondary center. As long as these plates remain, growth is possible. The rate of growth is controlled by hormones, particularly growth hormone (GH) and the sex hormones. Eventually, the plates become ossified, causing the primary and secondary centers of ossification to fuse, and the bone stops growing.

In the adult, bone is continually being broken down and built up again. Bone-absorbing cells called **osteoclasts** (Gk. *osteon,* "bone"; *klastos,* "broken in pieces") break down bone, remove worn cells, and deposit calcium in the blood. In this way, osteoclasts help maintain the blood calcium level and contribute to homeostasis.

Among many other functions, calcium ions play a major role in muscle contraction and nerve conduction. The blood calcium level is closely regulated by the antagonistic hormones parathyroid hormone (PTH) and calcitonin. PTH promotes the activity of osteoclasts, and calcitonin inhibits their activity to keep the blood calcium level within normal limits.

Assuming that the blood calcium level is normal, bone destruction caused by the work of osteoclasts is repaired by osteoblasts. As they form bone, some of these osteoblasts get caught in the matrix (nonliving material) they secrete and are converted to **osteocytes** (Gk. *osteon,* "bone"; *kytos,* "cell"). These cells live within the lacunae of osteons, where they continue to affect the timing and location of bone remodeling.

While a child is growing, the rate of bone formation is greater than the rate of bone breakdown. The skeletal mass continues to increase until ages 20 to 30. After that, the rate of formation and rate of breakdown of bone mass are equal, until ages 40 to 50. Then, resorption begins to exceed formation, and the total bone mass slowly decreases.

As people age, an abnormal thinning of the bones called **osteoporosis** can lead to an increased risk of fractures, especially of the wrist, vertebrae, and pelvis. About 12 million people in the United States have osteoporosis, which results in 1.5 million fractures each year.

Females are twice as likely as males to have an osteoporosis-related fracture in their lifetime, and about one in four females will experience such a fracture. This is partly due to the fact that females have about 30% less bone mass than males to begin with. Also, because sex hormones play an important role in maintaining bone strength, females typically lose about 2% of their bone mass each year after menopause. Estrogen replacement therapy has been shown to increase bone mass and reduce fractures, but it can also increase the risk of cardiovascular disease and certain types of cancer. Other strategies for decreasing osteoporosis risk include consuming 1,000–1,200 mg of calcium per day (depending on age), obtaining sufficient vitamin D, and engaging in regular physical exercise.

Anatomy of a Long Bone

A long bone, such as the humerus, illustrates principles of bone anatomy. When the bone is split open, as in Figure 39.4, the longitudinal section shows it is not solid but has a cavity called the medullary cavity bounded at the sides by compact bone and at the ends by spongy bone. Beyond the spongy bone, there is a thin shell of compact bone and finally a layer of hyaline cartilage. The cavity of an adult long bone usually contains yellow bone marrow, which is a fat-storage tissue.

Compact bone contains many osteons (also called Haversian systems), where osteocytes lie in tiny chambers called lacunae. The lacunae are arranged in concentric circles around central canals that contain blood vessels and nerves. The lacunae are separated by a matrix of collagen fibers and mineral deposits, primarily calcium and phosphorus salts.

Spongy bone has numerous bony bars and plates separated by irregular spaces. Although lighter than compact bone, spongy bone is still designed for strength. Just as braces are used for support in buildings, the solid portions of spongy bone follow lines of stress. The spaces in spongy bone are often filled with **red bone marrow,** a specialized tissue that produces blood cells. This is an additional way the skeletal system assists homeostasis since red blood cells transport oxygen, and white blood cells are a part of the immune system, which fights infection.

ISTUDY

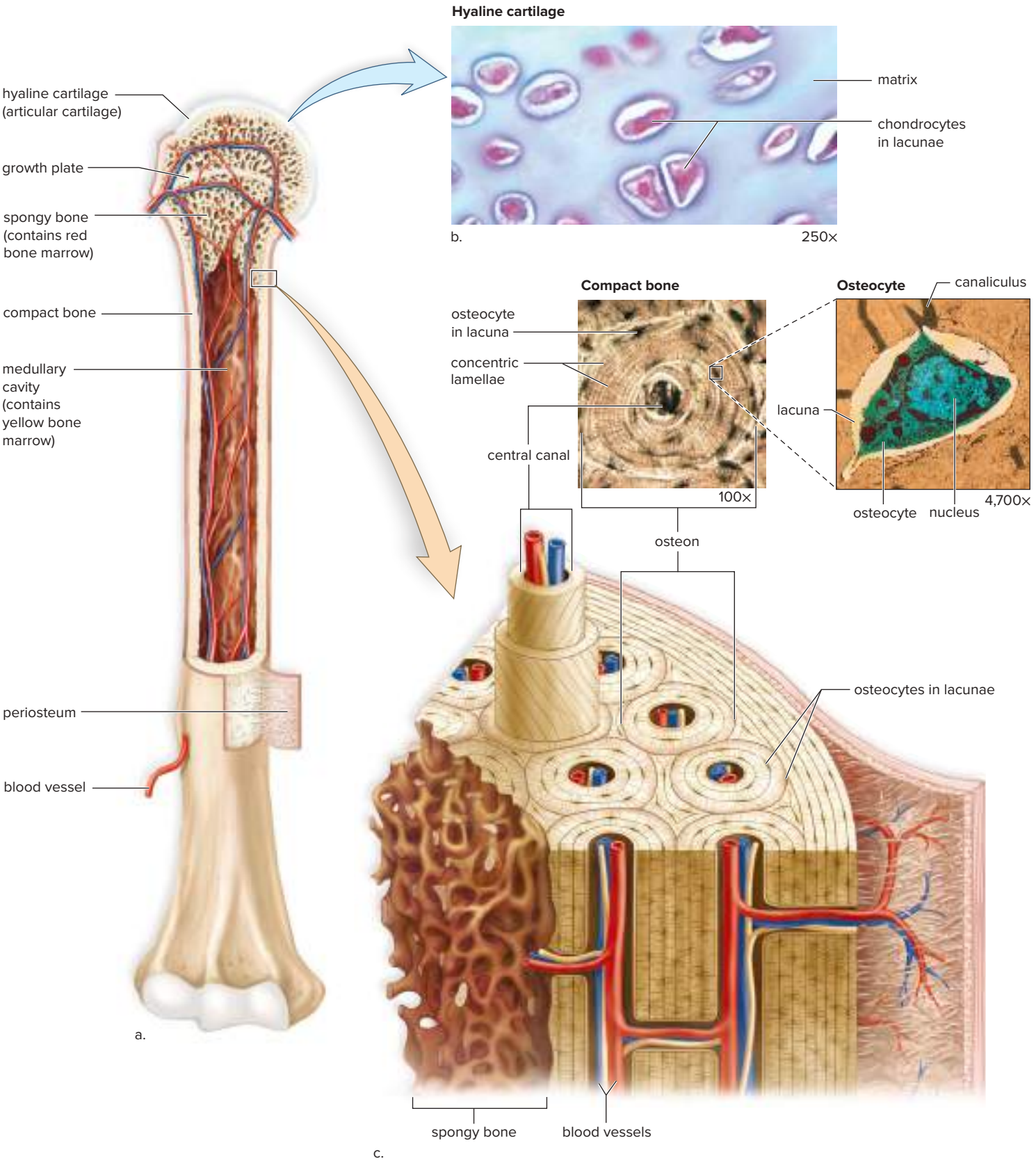


Figure 39.4 Anatomy of a long bone. **a.** A long bone is encased by fibrous membrane (periosteum), except where it is covered at the ends by (**b**) hyaline cartilage. **c.** Spongy bone located beneath the cartilage may contain red bone marrow. The central shaft contains yellow bone marrow and is bordered by compact bone, which is shown in the enlargement and micrograph. (photos): (hyaline cartilage, compact bone): ©Ed Reschke; (osteocyte): Biophoto Associates/Science Source

ISTUDY

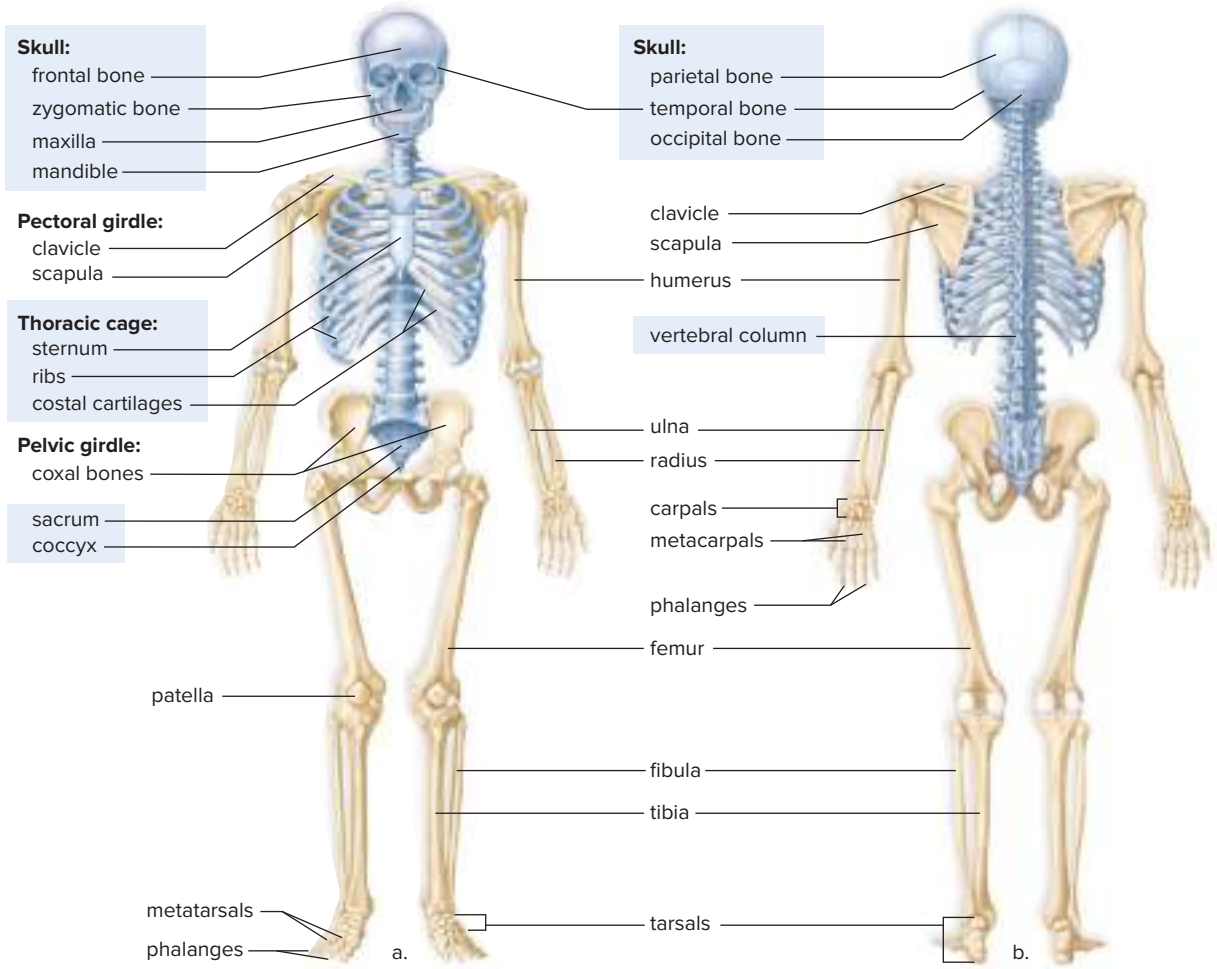


Figure 39.5 The human skeleton. **a.** Anterior view. **b.** Posterior view. The bones of the axial skeleton are in blue, and the rest is the appendicular skeleton. Not all bones are shown.

The Axial Skeleton

Approximately 206 bones make up a human skeleton. A total of 80 bones make up the **axial skeleton** (L. *axis,* "axis, hinge"; Gk. *skeleton,* "dried body"), which lies in the midline of the body and consists of the skull, the vertebral column, the thoracic cage, the sacrum, and the coccyx (blue labels in Fig. 39.5).

The Skull

The skull, which protects the brain, is formed by the cranium and the facial bones (Fig. 39.6). In newborns, certain bones of the cranium are joined by membranous regions called *fontanels* ("soft spots"), all of which usually close and become **sutures** by the age of 2 years. The bones of the cranium contain the sinuses (L. *sinus,* "hollow"), air spaces lined by mucous membrane that reduce the weight of the skull and give a resonant sound to the voice. Two sinuses, called the mastoid sinuses, drain into the middle ear. Mastoiditis, a condition that can lead to deafness, is an inflammation of these sinuses.

The major bones of the cranium have the same names as the lobes of the brain. On the top of the cranium, the frontal bone forms the forehead, and the parietal bones extend to the sides. Below the much larger parietal bones, each temporal bone has an opening that leads to the middle ear. In the rear of the skull, the occipital bone curves to form the base of the skull. At the base of the skull, the spinal cord passes upward through a large opening, called the *foramen magnum,* and becomes the brain stem.

The temporal and frontal bones are cranial bones that contribute to the face. The sphenoid bones account for the flattened areas on each side of the forehead, which we call the temples. The frontal bone not only forms the forehead but also has supraorbital ridges, where the eyebrows are located. Glasses sit where the frontal bone joins the nasal bones.

The most prominent of the facial bones are the mandible, the maxillae, the zygomatic bones, and the nasal bones. The mandible, or lower jaw, is the only freely movable portion of the skull (Fig. 39.6), and its action permits us to chew our food. It also forms the "chin." Tooth sockets are located in the mandible and on the maxillae, which form the upper jaw and a portion of the hard palate. The zygomatic bones are the cheekbone prominences, and the nasal bones form the bridge of the nose. Other bones make up the nasal septum, which divides the nose cavity into two regions.

Whereas the ears are formed only by cartilage and not by bone, the nose is a mixture of bones, cartilage, and connective tissues. The lips and cheeks have a core of skeletal muscle.

ISTUDY

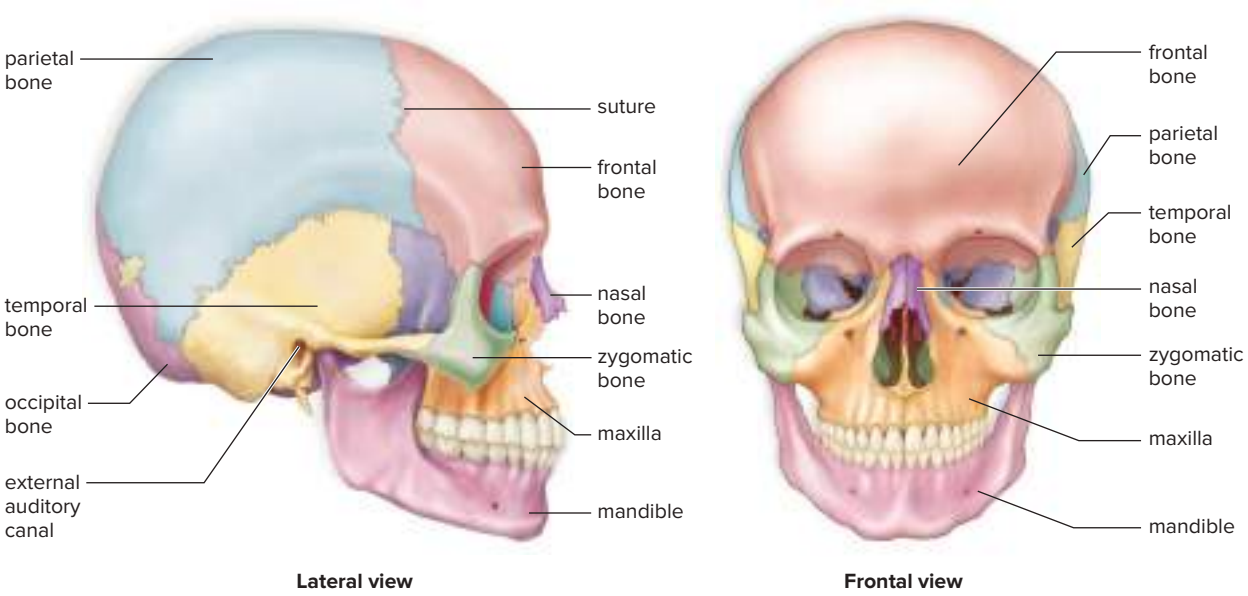


Figure 39.6 The skull. The skull consists of the cranium and the facial bones. The frontal bone is the forehead, the zygomatic arches form the cheekbones, and the maxillae form the upper jaw. The mandible has a projection we call the chin.

The Vertebral Column and Rib Cage

The **vertebral column** (L. *vertebra,* "bones of backbone") supports the head and trunk and protects the spinal cord and the roots of the spinal nerves. It is a longitudinal axis that serves either directly or indirectly as an anchor for all the other bones of the skeleton.

Thirty-three vertebrae make up the vertebral column. Seven cervical vertebrae are located in the neck; 12 thoracic vertebrae are in the thorax; 5 lumbar vertebrae are in the lower back; 5 fused sacral vertebrae form a single sacrum; and 4 fused vertebrae are in the coccyx, or tailbone. Normally, the vertebral column has four curvatures, which provide more resilience and strength for an upright posture than could a straight column.

Intervertebral discs, composed of fibrocartilage, between the vertebrae provide padding. They prevent the vertebrae from grinding against one another and absorb shock caused by movements such as running, jumping, and even walking. The presence of the discs allows the vertebrae to move as we bend forward, backward, and from side to side. Unfortunately, these discs become weakened with age and can herniate and rupture. Pain results if a disc presses against the spinal cord and/or spinal nerves. The body may heal itself, or the disc can be removed surgically. If the latter, the vertebrae can be fused together, but this limits the flexibility of the body.

The thoracic vertebrae are a part of the *rib cage,* sometimes called the thoracic cage. The rib cage also contains the ribs, the costal cartilages, and the sternum, or breastbone (Fig. 39.7).

There are 12 pairs of ribs. The upper 7 pairs are "true ribs," because they attach directly to the sternum. The lower 5 pairs do not connect directly to the sternum and are called the "false ribs." Three pairs of false ribs attach by means of a common cartilage, and 2 pairs are "floating ribs," because they do not attach to the sternum at all.

The rib cage demonstrates how the skeleton is protective yet flexible. The rib cage protects the heart and lungs, yet it swings outward and upward on inspiration and then downward and inward on expiration.

The Appendicular Skeleton

The **appendicular skeleton** (L. *appendicula,* dim. of *appendix,* "appendage") consists of the bones within the pectoral and pelvic girdles and the attached limbs (see Fig. 39.5). The pectoral (shoulder) girdle and upper limbs are specialized for flexibility, but the pelvic girdle (hipbones) and lower limbs are specialized for strength. A total of 126 bones make up the appendicular skeleton.

The Pectoral Girdle and Upper Limb

The components of the **pectoral girdle** are only loosely linked together by ligaments (Fig. 39.8). Each clavicle (collarbone) connects with the sternum and the scapula (shoulder blade), but the

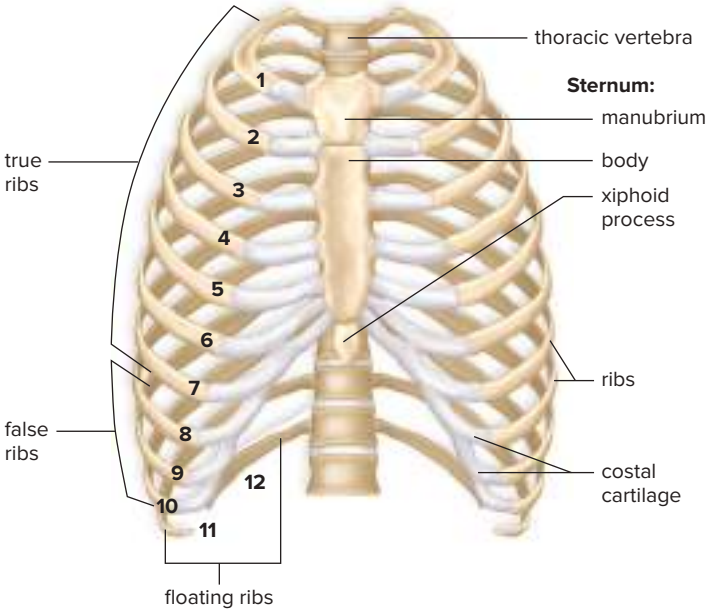


Figure 39.7 The rib cage. The rib cage consists of the thoracic vertebrae, the 12 pairs of ribs, the costal cartilages, and the sternum, or breastbone.

ISTUDY

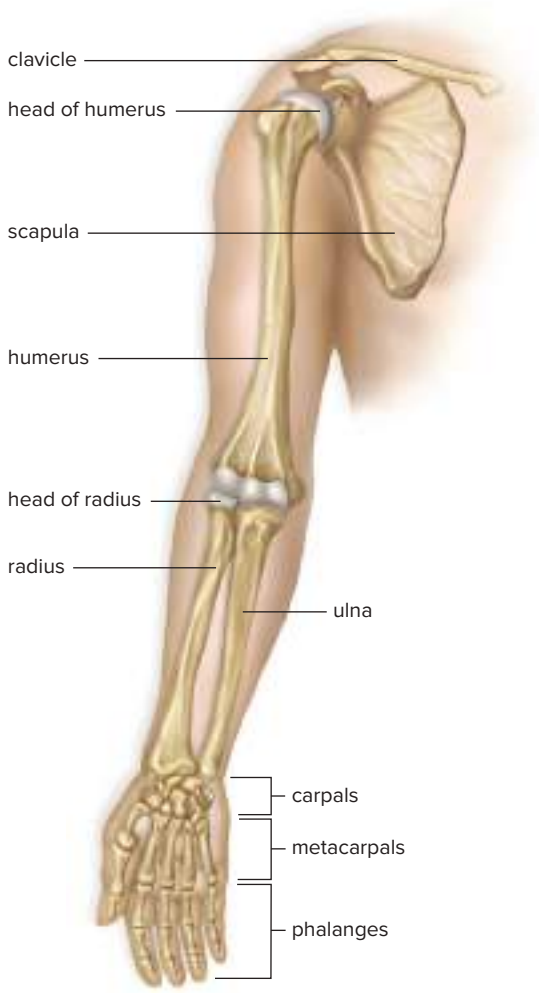


Figure 39.8 Bones of the pectoral girdle and upper limb. The humerus is known as the "funny bone" of the elbow. The sensation on bumping it is due to the activation of a nerve that passes across its end.

Figure 39.9 Bones of the pelvic girdle and lower limb. The femur is our strongest bone—it withstands a pressure of 540 kg per 2.5 cm^3 when we walk.

scapula is held in place only by muscles. This allows it to glide and rotate on the clavicle.

The single long bone in the arm, the humerus, has a smoothly rounded head that fits into a socket of the scapula. The socket, however, is very shallow and much smaller than the head. Although this means that the arm can move in almost any direction, the joint lacks stability. Therefore, this is the joint that is most apt to dislocate. The opposite end of the humerus meets the two bones of the lower arm, the ulna and the radius, at the elbow. The prominent bone in the elbow is the topmost part of the ulna. When the upper limb is held so that the palm is turned frontward, the radius and ulna are about parallel to one another. When the upper limb is turned so that the palm is next to the body, the radius crosses in front of the ulna, a feature that contributes to the easy twisting motion of the forearm.

The many bones of the hand increase its flexibility. The wrist has eight carpal bones, which look like small pebbles. From these, five metacarpal bones fan out to form a framework for the palm. The metacarpal bone that leads to the thumb is placed in such a way that the thumb can reach out and touch the other digits (*digit* is a term that refers to either fingers or toes). Beyond the metacarpals are the phalanges, the bones of the fingers and the thumb. The phalanges of the hand are long, slender, and lightweight.

The Pelvic Girdle and Lower Limb

The **pelvic girdle** (L. *pelvis,* "basin") (Fig. 39.9) consists of two heavy, large coxal bones (hip bones). The coxal bones are anchored to the sacrum, and together these bones form a hollow cavity called the pelvic cavity. The wider pelvic cavity in females compared to that of males accommodates pregnancy and childbirth. The weight of the body is transmitted through the pelvis to the lower limbs and then onto the ground. The largest bone in the body is the femur, or thighbone.

In the leg, the larger of the two bones, the tibia, has a ridge we call the shin. Both of the bones of the leg have a prominence that contributes to the ankle—the tibia on the inside of the ankle and the fibula on the outside of the ankle.

Although there are seven tarsal bones in the ankle, only one receives the weight and passes it on to the heel and the ball of the foot. If you wear high-heeled shoes, the weight is thrown toward the front of your foot. The metatarsal bones participate in forming the arches of the foot. There is a longitudinal arch from the heel to the toes and a transverse arch across the foot. These provide a stable, springy base for the body. If the tissues that bind the metatarsals together become weakened, "flat feet" are apt to result. The bones of the toes are called phalanges, just as are those of the fingers, but in the foot the phalanges are stout and extremely sturdy.

ISTUDY

Classification of Joints

Bones are connected at the **joints,** which are classified as fibrous, cartilaginous, or synovial. Most fibrous joints, such as the sutures between the cranial bones, are immovable. Cartilaginous joints, such as those between the vertebrae, are slightly movable. The vertebrae are also separated by discs, which increase their flexibility. The two hip bones are slightly movable, because they are ventrally joined by cartilage at the pubic symphysis. Owing to hormonal changes, this joint becomes more flexible during late pregnancy, allowing the pelvis to expand during childbirth.

In freely movable **synovial joints,** two bones are separated by a cavity. **Ligaments,** composed of fibrous connective tissue, bind the two bones to each other, holding them in place as they form a capsule. In a "double-jointed" individual, the ligaments are unusually loose. The joint capsule is lined by synovial membrane, which produces synovial fluid, a lubricant for the joint.

The shoulders, elbows, hips, and knees are examples of synovial joints (Fig. 39.10). In the knee, as in other freely movable joints, the bones are capped by a layer of articular cartilage. In addition, crescent-shaped pieces of cartilage called menisci (Gk. *meniscus,* "crescent") lie between the bones. These give added stability, helping support the weight placed on the knee joint. Unfortunately, athletes often experience injury of the meniscus, known as torn cartilage. Thirteen fluid-filled sacs called bursae (*sing.,* bursa) (L. *bursa,* "purse") occur around the knee joint. Bursae ease the friction between tendons and ligaments and between tendons and bones. Inflammation of the bursae is called bursitis. Tennis elbow is a form of bursitis.

Different types of synovial joints can be distinguished. The knee and elbow joints are *hinge joints* because, like a hinged door, they largely permit movement in one direction only. The joint between the first two cervical vertebrae, which permits side-to-side movement of the head, is an example of a *pivot joint,* which allows only rotation. More movable are the *ball-and-socket joints;* for example, the ball of the femur fits into a socket on the hip bone. Ball-and-socket joints allow movement in all planes and even rotational movement.

All types of joints are subject to **arthritis,** or inflammation of the joints. The Arthritis Foundation estimates that nearly one in three adult Americans has some degree of chronic joint pain, and arthritis is secondary only to heart disease as a cause of work disability in the United States. The most common type of arthritis is osteoarthritis, which results from the deterioration of the cartilage in one or more synovial joints. The hands, hips, knees, lower back, and neck are most commonly affected. Rheumatoid arthritis is considered to be an autoimmune disease, in which the immune system attacks the joints for reasons that are not well understood.

Regardless of the cause, arthritis is most often treated by over-the-counter anti-inflammatory drugs, such as aspirin and ibuprofen, or more powerful prescription drugs, such as the corticosteroids or gel-like substances called hyaluronates. In severe cases, certain joints, usually the knees and hips, can be replaced with artificial versions made of ceramic, metal, and/or plastic. These joint replacements generally last over 20 years, after which they may need to be replaced, although research is increasing the durability of artificial joints.

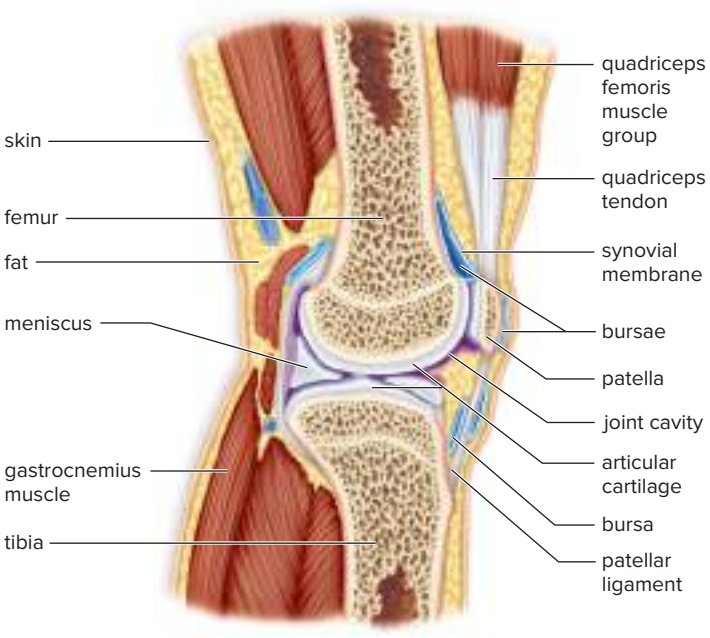


Figure 39.10 Knee joint. The knee is an example of a synovial joint. The cavity between the bones is encased by ligaments and lined by synovial membrane. The patella (kneecap) guides the quadriceps tendon over the joint when flexion or extension occurs.

Check Your Progress 39.2

1. Describe the functions of osteoblasts, osteoclasts, and osteocytes.
2. Distinguish between the structure and function of spongy bone and those of compact bone.
3. Determine whether each of the following bones belongs to the axial or appendicular skeleton: sacrum, frontal bone, humerus, tibia, vertebra, coxal bone, temporal bone, scapula, and sternum.

39.3 The Muscular System

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the macroscopic and microscopic structure of a muscle fiber.
2. Explain the molecular mechanism of muscle contraction.
3. Indicate three ways that muscle cells can generate ATP.
4. Explain the specific role of acetylcholine in stimulating a muscle fiber to contract.

Muscles are composed of contractile tissue that is capable of changing its length by contracting and relaxing. Most animals rely on muscle tissue to produce movement—to swim, crawl, walk, run, jump, or fly. Clearly, there are strong evolutionary advantages

ISTUDY

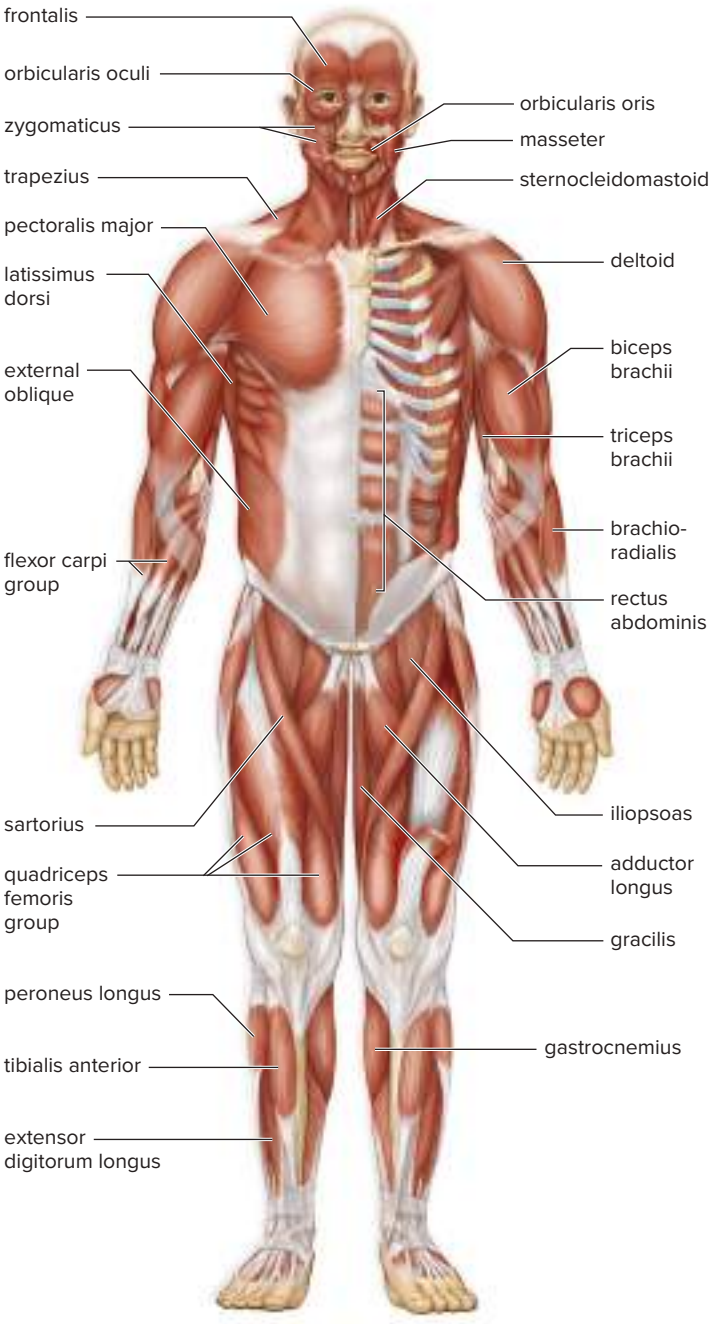


Figure 39.11 Human musculature. Anterior view of some of the major superficial skeletal muscles.

to being able to move into new environments, to flee from danger, to seek and/or capture food, and to find new mates. Only a few animals are nonmotile (also called *sessile*), and most of these live in water, where currents can bring a supply of food to them.

As discussed in Section 31.1, humans and other vertebrates have three distinct types of muscle tissue: smooth, cardiac, and skeletal. Most of the focus in this section will be on skeletal muscle, or striated voluntary muscle, which is important in maintaining posture, providing support, and allowing for movement. The processes responsible for skeletal muscle contraction also release heat, which is distributed throughout the body, helping maintain a constant body temperature.

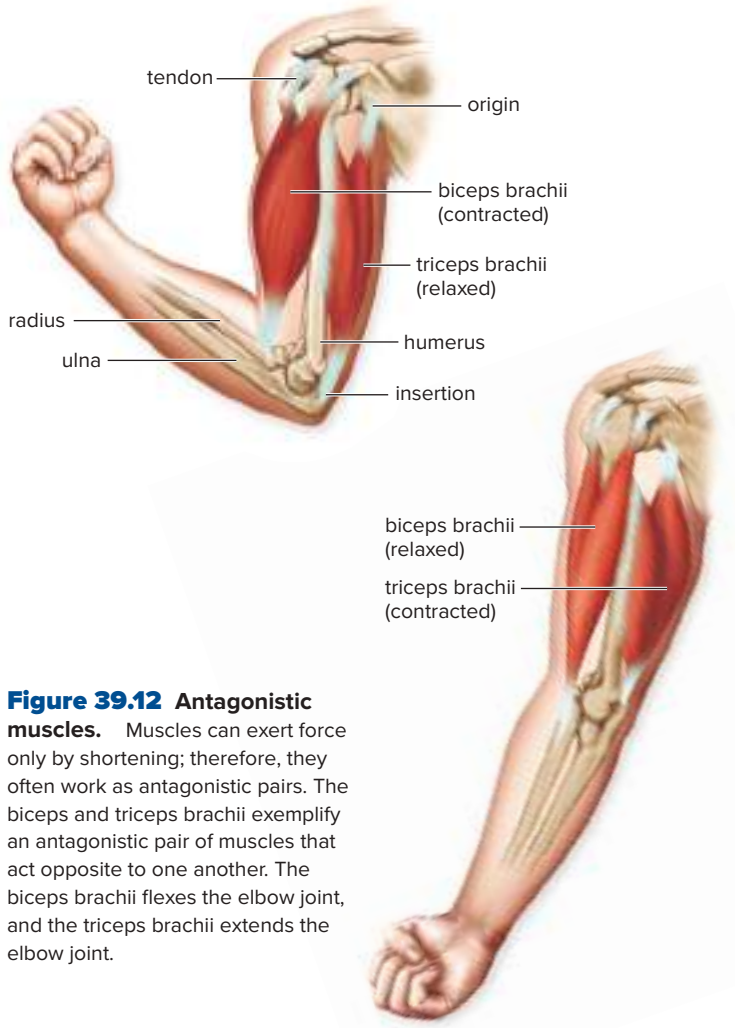


Figure 39.12 Antagonistic muscles. Muscles can exert force only by shortening; therefore, they often work as antagonistic pairs. The biceps and triceps brachii exemplify an antagonistic pair of muscles that act opposite to one another. The biceps brachii flexes the elbow joint, and the triceps brachii extends the elbow joint.

Macroscopic Anatomy and Physiology

The over 650 skeletal muscles and their associated tissues make up approximately 40% of the weight of an average human. Muscle tissue is approximately 15% more dense than fat tissue, so a pound of muscle takes up less space than does a pound of fat. However, even at rest, muscle tissue consumes about three times more energy than adipose tissue. Several of the major human superficial muscles are illustrated in Figure 39.11.

Skeletal muscles are attached to the skeleton by bands of fibrous connective tissue called **tendons** (L. *tendo,* "stretch"). When muscles contract, they shorten. Therefore, muscles can only pull; they cannot push. Because of this, skeletal muscles must work in antagonistic pairs. One muscle of an antagonistic pair flexes the joint and bends the limb; the other one extends the joint and straightens the limb. Figure 39.12 illustrates this principle.

In the laboratory, if a muscle is given a rapid series of threshold stimuli—that is, stimuli strong enough to bring about action potentials, as described in Section 37.2.—it can respond to the next stimulus without relaxing completely. In this way, muscle contraction builds, or summates, until maximal sustained contraction, called **tetany,** is achieved. Tetanic contractions ordinarily occur in the body's muscles whenever skeletal muscles are actively used.

Even when muscles appear to be at rest, they exhibit **tone,** in which some of their fibers are contracting. As you saw in

THEME Nature of Science

The Accidental Discovery of Botox

Several of the most important bacterial pathogens that cause human diseases—including cholera, diphtheria, tetanus, and botulism—do so by secreting potent toxins capable of sickening or killing their hosts. The botulinum toxin, produced by the bacterium *Clostridium botulinum,* is one of the most lethal substances known. Less than a microgram of the purified toxin can kill an average-size person, and 4 kilograms (8.8 pounds) would be enough to kill all the humans on Earth! Given this scary fact, it seems that the scientists who discovered the lethal activity of this bacterial toxin nearly 200 years ago could never have anticipated that the intentional injection of a very dilute form of botulinum toxin (now known as Botox) would become the most common nonsurgical cosmetic procedure performed by many physicians.

As with many breakthroughs in science and medicine, the pathway from thinking about botulism as a deadly disease to using botulinum toxin as a beneficial treatment involved the hard work of many scientists, mixed with a considerable amount of luck.

In the 1820s, a German scientist, Justinus Kerner, was able to prove that the deaths of several people had been caused by their consumption of spoiled sausage (in fact, botulism is named for the Latin word for "sausage," *botulus*). A few decades later, a Belgian researcher named Emile Pierre van Ermengem identified the specific bacterium responsible for producing the botulinum toxin, which could cause symptoms ranging from droopy eyelids to paralysis and respiratory failure.

By the 1920s, medical scientists at the University of California had obtained the toxin in pure form, which allowed them to determine that it acted by preventing nerves from communicating with muscles, specifically by interfering with the release of acetylcholine from the axon terminals of motor nerves (see Section 39.3).

Scientists soon began testing very dilute concentrations of the toxin as a treatment for conditions in which the muscles contract too much, such as crossed eyes or spasms of the facial muscles or vocal cords, and in 1989 the FDA first approved diluted botulinum toxin (Botox) for treating specific eye conditions called blepharospasm (eyelid spasms) and strabismus (crossing of the eyes).

Right around this time, a lucky break occurred that eventually would open the medical community's eyes to the greater potential of the diluted toxin. A Canadian ophthalmologist, Jean Carruthers, had been using it to treat her patients' eye conditions, when she noticed that some of their wrinkles had also subsided. One night at a family dinner, Dr. Carruthers shared this information with her husband, a dermatologist, who decided to investigate whether he could reduce the deep wrinkles of some of his patients by injecting the dilute toxin into their skin. The treatment worked well, and after trying it on several more patients (as well as on themselves!), the Canadian doctors spent several years presenting their findings at scientific meetings and in research journals. Although their ideas were initially met with skepticism, the Carrutherses eventually were able to convince the scientific community that diluted botulinum toxin is effective in treating wrinkles; however, they never patented it for that use, so they missed out on much of the $1.3 billion in annual sales the drug now earns for the company that did patent it.

Figure 39A Treating wrinkles with diluted botulinum toxin. Thinkstock/Comstock Images/Getty Images

The uses of diluted botulinum toxin have been growing since it was first FDA-approved for the treatment of frown lines in 2002 (Fig. 39A). It has now been approved for the treatment of chronic migraine headaches, excessive underarm sweating, and facial wrinkles known as "crow's-feet." The company is seeking approval for many other uses of diluted botulinum toxin.

Perhaps all of this would have eventually happened even without the observations of an alert eye doctor, but progress would have very likely been slower. As the French microbiologist Louis Pasteur observed in 1854, "Chance favors the prepared mind," meaning that many scientific discoveries involve many investigators, and years of work, mixed with a flash of inspiration.

Questions to Consider

1. Considering that botulism is caused by a preformed toxin, how do you suppose it can be treated?
2. Do you think companies should be allowed to patent a naturally occurring molecule such as botulinum toxin? Why or why not?

Section 38.5, sensory receptors called muscle spindles and Golgi tendon organs are partly responsible for maintaining tone. Muscle tone is particularly important in maintaining posture. If all the fibers in the muscles of the neck, trunk, and legs were to suddenly relax, the body would collapse.

Muscle tone has also been implicated in the formation of facial wrinkles. As described in the Nature of Science feature, "The Accidental Discovery of Botox," medical injections of Botox interfere with muscle contraction, smoothing wrinkles.

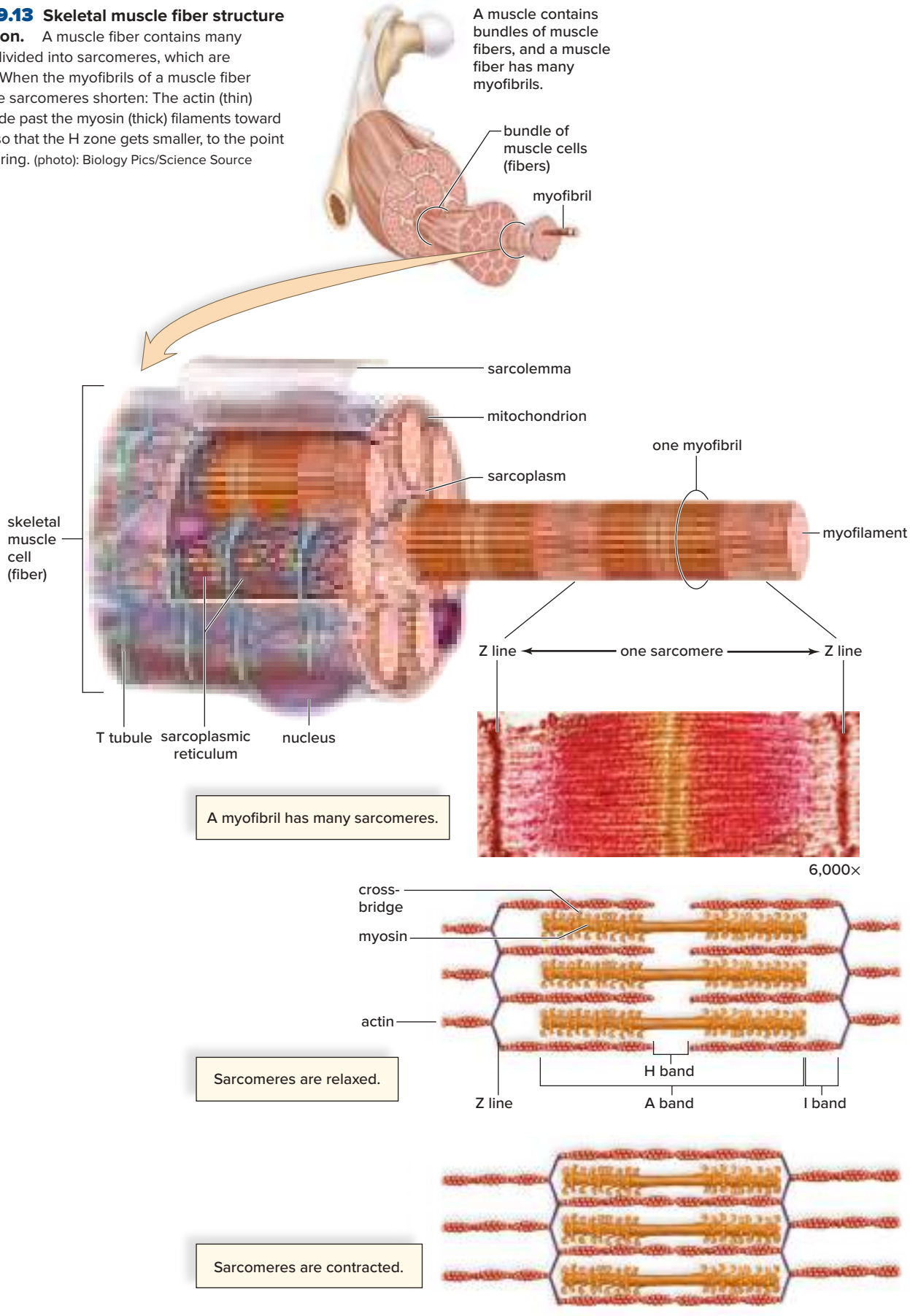
Microscopic Anatomy and Physiology

A vertebrate skeletal muscle is composed of a number of muscle fibers in bundles. Each muscle fiber is a cell containing the usual cellular components, but some components have special features (Fig. 39.13).

The **sarcolemma,** or plasma membrane, forms a transverse system, or T system. The T tubules penetrate, or dip down, into the cell, so that they come in contact—but do not fuse—with the

ISTUDY

Figure 39.13 Skeletal muscle fiber structure and function. A muscle fiber contains many myofibrils, divided into sarcomeres, which are contractile. When the myofibrils of a muscle fiber contract, the sarcomeres shorten: The actin (thin) filaments slide past the myosin (thick) filaments toward the center, so that the H zone gets smaller, to the point of disappearing. (photo): Biology Pics/Science Source



ISTUDY

sarcoplasmic reticulum, which consists of expanded portions of modified endoplasmic reticulum. These expanded portions serve as storage sites for calcium ions (Ca^{2+}), which are essential for muscle contraction. The sarcoplasmic reticulum encases hundreds and sometimes even thousands of **myofibrils** (Gk. *myos,* "muscle"; L. *fibra,* "thread"), which are the contractile portions of a muscle fiber.

Myofibrils are cylindrical and run the length of the muscle fiber. The light microscope shows that a myofibril has light and dark bands, termed striations. These bands are responsible for skeletal muscle's striated appearance. The electron microscope reveals that the striations of myofibrils are formed by the placement of protein filaments within contractile units called **sarcomeres.**

Examining sarcomeres when they are relaxed shows that a sarcomere extends between two dark lines called Z lines (Fig. 39.13). There are two types of protein filaments: thick filaments, made up of **myosin,** and thin filaments, made up of **actin.** The I band is light-colored, because it contains only actin filaments attached to a Z line. The dark regions of the A band contain overlapping actin and myosin filaments, and its H zone has only myosin filaments.

Sliding Filament Model

Examining muscle fibers when they are contracted reveals that the sarcomeres within the myofibrils have shortened. When a sarcomere shortens, the actin (thin) filaments slide past the myosin (thick) filaments and approach one another. This causes the I band to shorten and the H zone to nearly or completely disappear.

The movement of actin filaments in relation to myosin filaments is called the **sliding filament model** of muscle contraction. During the sliding process, the sarcomere shortens, even though the filaments themselves remain the same length. When you play "tug of war," your hands grasp the rope, pull, let go, attach farther down the rope, and pull again. The myosin heads are like your hands—grasping, pulling, letting go, and then repeating the process.

The participants in muscle contraction have the functions listed in Table 39.1. ATP supplies the energy for muscle contraction. Although the actin filaments slide past the myosin filaments, it is the myosin filaments that do the work. Myosin filaments break down ATP and form cross-bridges that attach to and pull the actin filaments toward the center of the sarcomere.

Use of ATP in Contraction

ATP provides the energy for muscle contraction. Although muscle cells contain *myoglobin,* a molecule that stores oxygen, cellular respiration does not immediately supply all the ATP that is needed. In the meantime, muscle fibers rely on *creatine phosphate* (phosphocreatine), a storage form of high-energy phosphate. Creatine phosphate cannot directly participate in muscle contraction. Instead, it anaerobically regenerates ATP by the following reaction:

$$\text{creatine—Ⓟ} + \text{ADP} \longrightarrow \text{ATP} + \text{creatine}$$

This reaction occurs in the midst of sliding filaments, and therefore this method of supplying ATP is the speediest energy source available to muscles.

When all of the creatine phosphate is depleted, mitochondria may by then be producing enough ATP for muscle contraction to continue. If not, fermentation is a second way for muscles to supply ATP without consuming oxygen. Fermentation, which is apt to occur when strenuous exercise first begins, supplies ATP for only a short time, and lactate builds up. Whether lactate causes muscle aches and fatigue on exercising is now being questioned.

We all have had the experience of needing to continue deep breathing following strenuous exercise. This continued intake of oxygen, which is required to complete the metabolism of lactate and restore cells to their original energy state, offsets what is known as **oxygen debt.** The lactate is transported to the liver, where 20% of it is completely broken down to carbon dioxide (CO_2) and water (H_2O). The ATP gained by this respiration is then used to reconvert 80% of the lactate to glucose.

In persons who regularly exercise, such as athletes in training, the number of mitochondria increases, and muscles rely on them rather than on fermentation to produce ATP. Less lactate is produced, and there is less oxygen debt.

Muscle Innervation

Muscles are stimulated to contract by motor nerve fibers. Nerve fibers have several branches, each of which ends at an axon terminal in close proximity to the sarcolemma of a muscle fiber. A small gap, called a synaptic cleft, separates the axon terminal from the sarcolemma. This entire region is called a **neuromuscular junction** (Fig. 39.14).

Axon terminals contain synaptic vesicles filled with the neurotransmitter acetylcholine (ACh). When nerve impulses traveling down a motor neuron arrive at an axon terminal, the synaptic vesicles release ACh into the synaptic cleft. ACh quickly diffuses across the cleft and binds to receptors in the sarcolemma. Now, the sarcolemma generates impulses that spread over the sarcolemma and down T tubules to the sarcoplasmic reticulum. The release of calcium from the sarcoplasmic reticulum causes the filaments in sarcomeres to slide past one another. Sarcomere contraction results in myofibril contraction, which in turn results in muscle fiber and finally muscle contraction.

Once a neurotransmitter has been released into a neuromuscular junction and has initiated a response, it is removed from the junction. When the enzyme acetylcholinesterase (AChE) breaks down acetylcholine, muscle contraction ceases due to reasons we will discuss next.

Table 39.1 Components of Muscle Contraction

Name	Function
Actin filaments	Slide past myosin, causing contraction
Ca^{2+}	Needed for myosin to bind to actin
Myosin filaments	Pull actin filaments by means of cross-bridges; are enzymatic and split ATP
ATP	Supplies energy for muscle contraction

ISTUDY

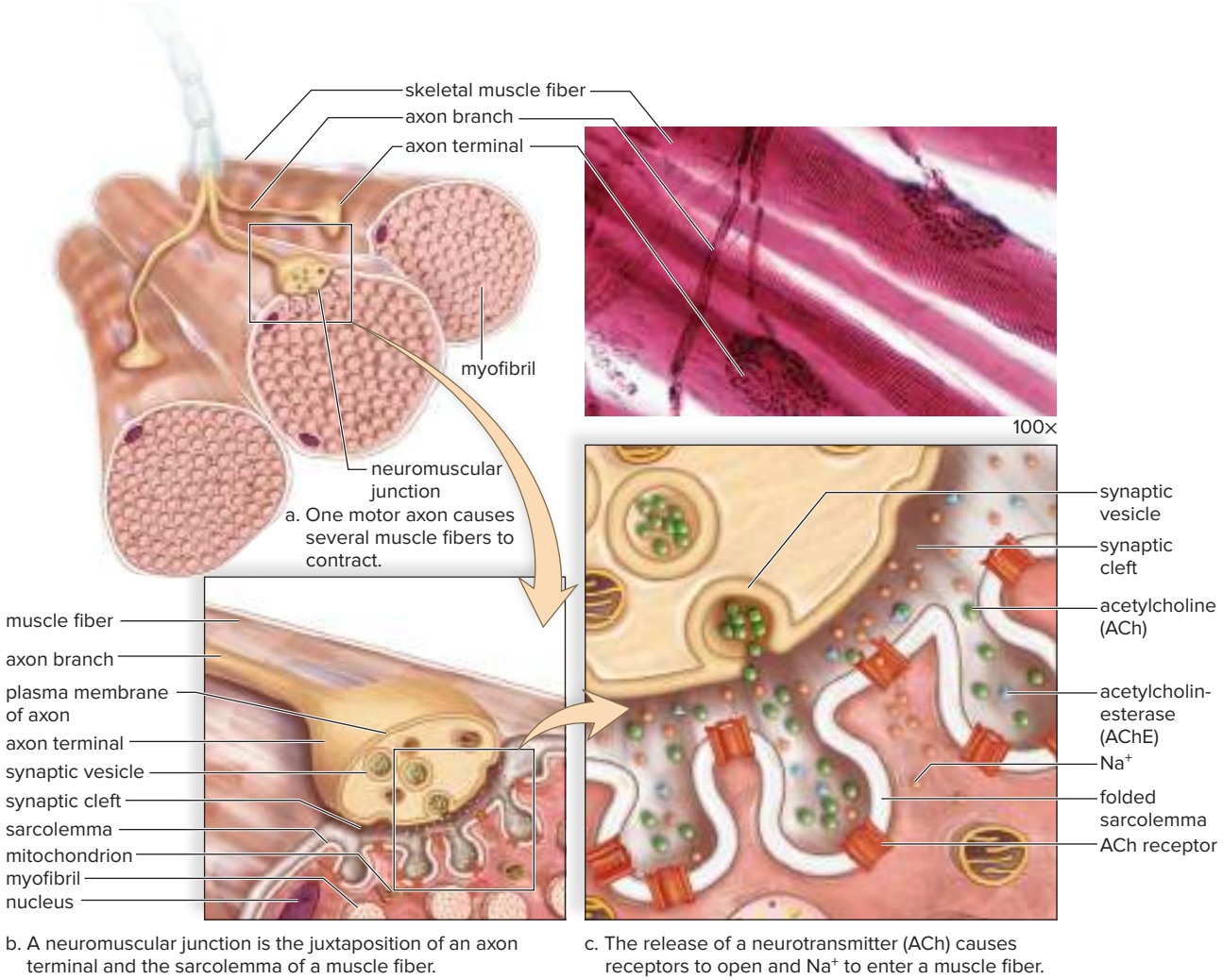


Figure 39.14 Neuromuscular junction. The branch of a motor nerve fiber (**a**) ends in an axon terminal (**b**) that meets but does not touch a muscle fiber. A synaptic cleft separates the axon terminal from the sarcolemma of the muscle fiber. Nerve impulses traveling down a motor fiber cause synaptic vesicles (**c**) to discharge a neurotransmitter that diffuses across the synaptic cleft. When the neurotransmitter is received by the sarcolemma of a muscle fiber, impulses begin, leading to muscle fiber contraction. (a): (right): Ed Reschke/Stone/Getty Images

Role of Calcium in Muscle Contraction

Figure 39.15 illustrates the placement of two other proteins associated with a thin filament, which is composed of a double row of twisted actin molecules. Threads of tropomyosin wind about an actin filament, and troponin occurs at intervals along the threads. Calcium ions (Ca^{2+}) that have been released from the sarcoplasmic reticulum combine with troponin. After binding occurs, the tropomyosin threads shift their position, and myosin binding sites are exposed.

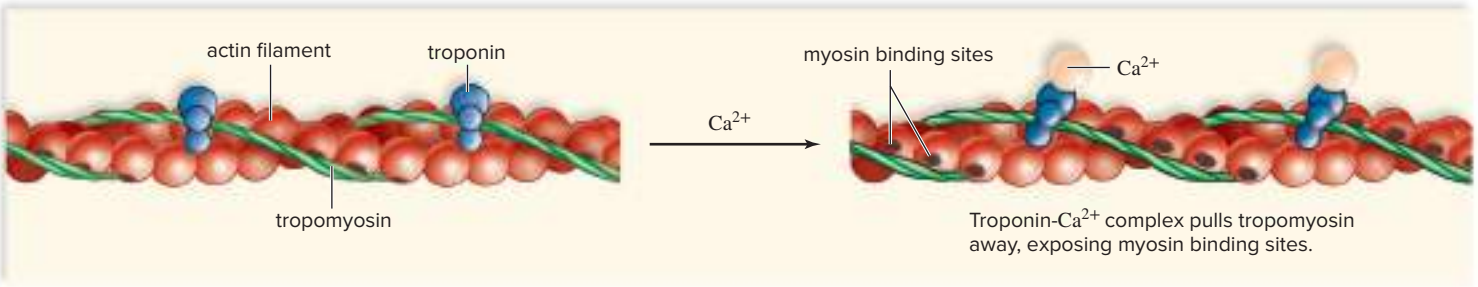
Thick filaments are bundles of myosin molecules with double globular heads. Myosin heads function as ATPase enzymes, splitting ATP into ADP and Ⓟ. This reaction activates the heads, so that they can bind to actin. The ADP and Ⓟ remain on the myosin heads until the heads attach to actin, forming cross-bridges. Now, ADP and Ⓟ are released, and this causes the cross-bridges to change their positions. This is the power stroke that pulls the thin filaments toward the middle of the sarcomere. When more ATP molecules bind to myosin heads, the cross-bridges are broken as the heads detach from actin. The cycle begins again; the actin filaments move nearer the center of the sarcomere each time the cycle is repeated.

Contraction continues until nerve impulses cease and calcium ions are returned to their storage sites. The membranes of the sarcoplasmic reticulum contain active transport proteins that pump calcium ions back into the calcium storage sites, and muscle relaxation occurs. When a person or an animal dies, ATP production ceases. Without ATP, the myosin heads cannot detach from actin, nor can calcium be pumped back into the sarcoplasmic reticulum. As a result, the muscles remain contracted, a phenomenon called *rigor mortis.*

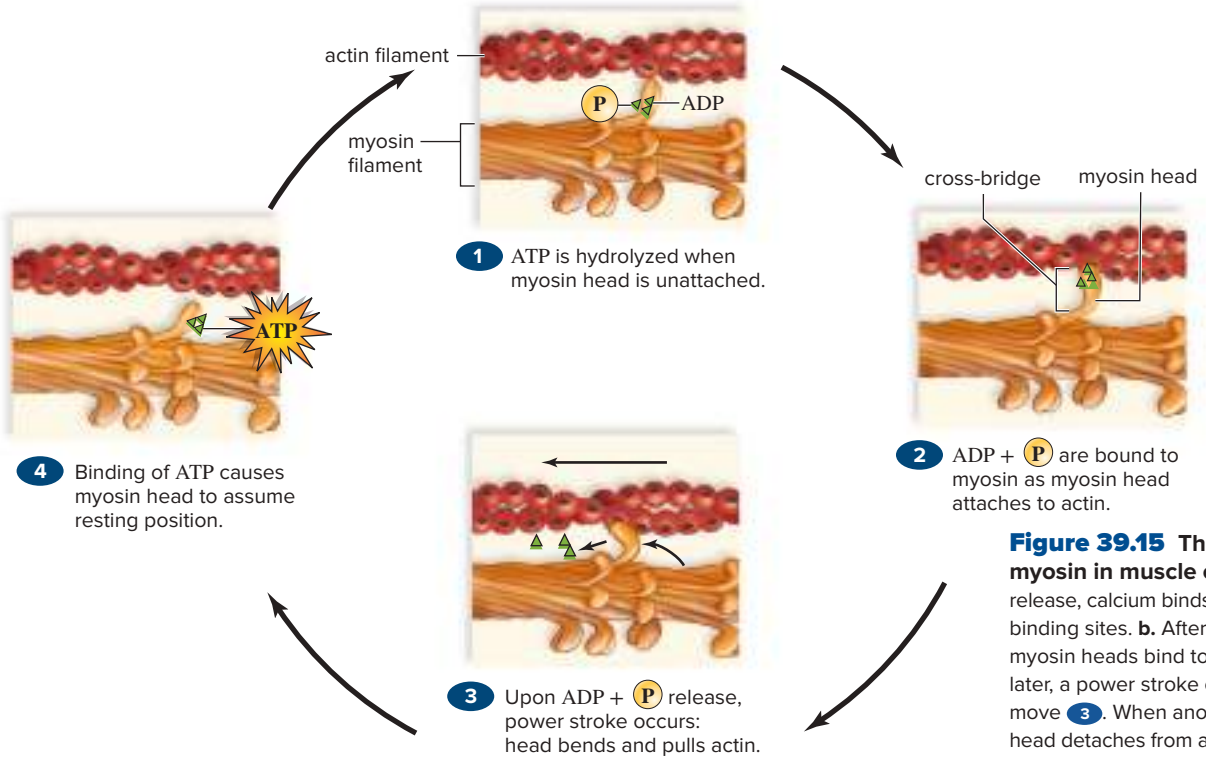
Check Your Progress 39.3

1. Define an *antagonistic pair* of muscles.
2. Describe the microscopic levels of structure in a skeletal muscle.
3. Discuss the specific role of ATP in muscle contraction.

ISTUDY



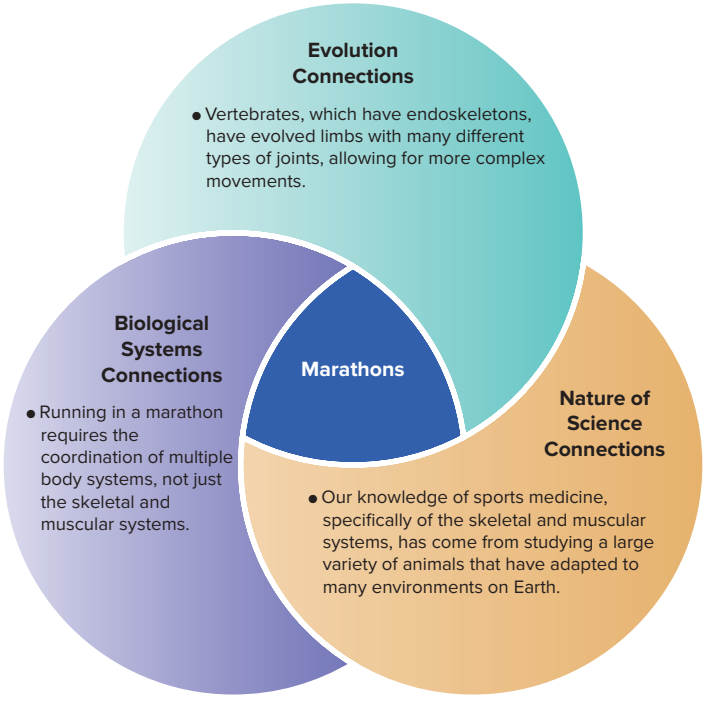
a. Function of Ca^{2+}



b. Function of myosin

Figure 39.15 The role of calcium and myosin in muscle contraction. **a.** Upon release, calcium binds to troponin, exposing myosin binding sites. **b.** After breaking down ATP 1, myosin heads bind to an actin filament 2, and later, a power stroke causes the actin filament to move 3. When another ATP binds to myosin, the head detaches from actin 4, and the cycle begins again. Although only one myosin head is featured, many heads are active at the same time.

CONNECTING *THE* CONCEPTS



ISTUDY

What Did You Learn?

39.1 Diversity of Skeletons

Three types of skeletons are found in the animal kingdom: a **hydrostatic skeleton** (cnidarians, flatworms, and segmented worms); an **exoskeleton** (certain mollusks and arthropods); and an **endoskeleton** (sponges, echinoderms, and vertebrates). The rigid but jointed skeleton of arthropods and vertebrates helped them colonize the terrestrial environment. The overall shape of an animal's skeleton is adapted to its environment and the type(s) of locomotion it uses.

39.2 The Human Skeletal System

The human skeleton gives support to the body, helps protect internal organs, provides sites for muscle attachment, and is a storage area for calcium and phosphorus salts, as well as a site for blood cell formation.

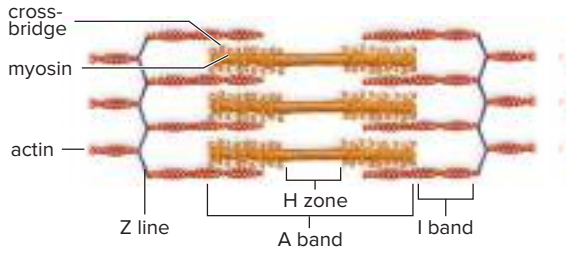
Most bones are cartilaginous in the fetus but are converted to bone during development. A long bone undergoes endochondral ossification, in which a cartilaginous growth plate remains between the primary ossification center in the middle and the secondary centers at the ends of the bones. Growth of the bone is possible as long as the growth plates are present, but eventually they, too, are converted to bone.

- Bone is constantly being renewed; **osteoclasts** break down bone, and **osteoblasts** build new bone. **Osteocytes** are in the lacunae of osteons; a long bone has a shaft of **compact bone** and two ends that contain **spongy bone.** The shaft contains a medullary cavity with yellow marrow, and the ends contain **red bone marrow,** which produces blood cells. **Osteoporosis,** or loss of bone density, is a common disease in older adults.
- The human skeleton is divided into two parts: (1) the **axial skeleton,** which is made up of the skull, the **vertebral column,** the sternum, and the ribs; and (2) the **appendicular skeleton,** which is composed of the **pectoral girdle,** the **pelvic girdle,** and their appendages. The skull of a newborn has membranous fontanels which usually close and become **sutures** by 2 years of age.
- **Joints** are classified as immovable, as are those of the cranium; slightly movable, as are those between the vertebrae; and freely movable (synovial joints), as are those in the knee and hip. In **synovial joints, ligaments** bind the two bones together, forming a capsule containing synovial fluid. **Arthritis** is an inflammation of a joint.

39.3 The Muscular System

Muscle tissue can change its length by contracting and relaxing. Skeletal muscles attach to bones via **tendons.** Whole skeletal muscles can shorten only when they contract; therefore, they work in antagonistic pairs. For example, if one muscle flexes the joint and brings the limb toward the body, the other muscle of the antagonistic pair extends the joint and straightens the limb. A muscle at rest exhibits **tone,** meaning some fibers are contracting. **Tetany** refers to maximum sustained muscle contractions.

- A whole skeletal muscle is composed of muscle fibers. Each muscle fiber is a cell that contains **myofibrils** in addition to the usual cellular components. The plasma membrane **(sarcolemma)** forms a transverse (T) system. T tubules penetrate the cell and contact the **sarcoplasmic reticulum,** which stores calcium ions. Longitudinally, myofibrils are divided into **sarcomeres,** which display the arrangement of actin and myosin filaments.
- The **sliding filament model** of muscle contraction states that **myosin** filaments have cross-bridges, which attach to and detach from **actin** filaments, causing actin filaments to slide and the sarcomere to shorten. (The H zone disappears as actin filaments approach one another.) Myosin breaks down ATP, and this supplies the energy for muscle contraction. During **oxygen debt,** anaerobic creatine phosphate breakdown and fermentation can quickly generate ATP. Sustained exercise requires cellular respiration for the generation of ATP.



- Nerves innervate muscles. A nerve impulse travels down a motor neuron to a **neuromuscular junction,** causing the release of ACh, which binds to receptors on the sarcolemma. Impulses begin and move down T tubules that approach the sarcoplasmic reticulum, where calcium is stored. Thereafter, calcium ions are released and bind to troponin. The troponin-Ca^{2+} complex causes tropomyosin threads winding around actin filaments to shift their position, revealing myosin binding sites.
- Myosin filaments are composed of many myosin molecules with double globular heads. When myosin heads break down ATP, they are ready to attach to actin. The release of ADP + Ⓟ causes myosin heads to change their position. This is the power stroke that causes the actin filament to slide toward the center of a sarcomere. When more ATP molecules bind to myosin, the heads detach from actin, and the cycle begins again.

Thinking Critically About Biology

1. The dystrophin protein is located between the sarcolemma and the outer myofilaments of the muscle. It is responsible for conducting the force of the muscle contraction from the myofilaments to the connective tissue of the muscle. In many cases, a mutation in this protein causes the symptoms of muscular dystrophy.
   a. Why would a loss of dystrophin protein cause weakness and a loss of coordination?
   b. Muscular dystrophy is frequently referred to as a muscle-wasting disease, in which the muscles lose mass over time. How would a defect in dystrophin contribute to the wasting of muscle tissue?
2. Similar to the activity in our muscle cells, certain bacteria can utilize sugars by aerobic cellular respiration (which requires oxygen) or by fermentation (which does not). However, like muscle cells, these bacteria use fermentation only when oxygen is not available. Why is aerobic respiration preferable? What is the specific role of oxygen in this process?
3. Most authorities emphasize the importance of calcium consumption for the prevention of osteoporosis. However, surveys show that osteoporosis is rare in some countries, such as China, where calcium intake is lower than in the United States. These studies often point to the higher level of animal protein consumed in the United States as having a detrimental effect on bone mass. If this is true, what physiological mechanism might explain it? Can you think of any other differences between the lifestyles in China and the United States that might be factors?

Making It Relevant to You

1. Why might participation in high-impact sports, such as marathons, lead to problems such as osteoarthritis later in life?
2. Artificial knees and hips are increasingly becoming common as we live longer. Why do physicians suggest that patients undergo strength training and exercise regimes before getting these surgeries?
3. From an evolutionary perspective, why might the body view the bones as nothing more than calcium storage sites, and how might this explain the increased risk of osteoporosis later in life?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

40

Hormones and Endocrine Systems

David McNew/Getty Images

Chapter Outline

40.1 Animal Hormones
40.2 Hypothalamus and Pituitary Gland
40.3 Other Endocrine Glands and Hormones

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 3.3 What are the general structure and function of steroids?

Chapter 5 What are the general classifications of chemical signaling molecules?

Section 37.3 What are the location and function of the hypothalamus?

Hormone Disrupters

You may have heard of concerns regarding hormone disruptor chemicals, such as BPA (bisphenol A) in plastics and food items, and the possible impacts on human health. However, you may not know that hormone disruptors play an important role in regulating pest species. Hormones regulate the metamorphosis of many insects, from their worm-like larval stages to their adult forms. One of these is called juvenile hormone, which plays an important role in insect development. Methoprene, a common compound found in flea sprays, is an insect-growth regulator that interferes with the juvenile hormone of insects such as fleas and mosquitoes. While methoprene does not kill the insects directly, it prevents their ability to develop from eggs to larvae and then adults.

Because humans do not have the same developmental stages as insects, these compounds are safe for us. However, in some cases, chemicals such as BPA mimic hormones found in humans, and have been associated with an increased risk of some diseases.

As you read through this chapter, think about the following questions:

1. Why would vertebrates use some of the same hormones as more primitive invertebrates?
2. How does the nervous system work with the endocrine system to control body functions?

Following *the* Themes

CHAPTER 40 – HORMONES AND ENDOCRINE SYSTEMS

Evolution	The nervous and endocrine systems in animals have evolved to respond to changes in their internal and external environments.
Nature of Science	Research into how hormones function has led to effective treatments for many common disorders.
Biological Systems	A relatively limited number of hormones control a wide range of homeostatic processes in animals.

ISTUDY

40.1 Animal Hormones

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between the mode of action of a neurotransmitter and that of a hormone.
2. Identify the major endocrine glands of the human body.
3. Compare the mechanisms of action of peptide and steroid hormones.

The nervous and endocrine systems work together to regulate the activities of the other organs. Both systems use chemical signals when they respond to changes that might threaten homeostasis. However, they have different means of delivering these signals (Fig. 40.1).

Sensory receptors detect changes in the internal and external environments and transmit that information to the central nervous system (CNS), which responds by stimulating muscles and glands. Communication depends on nerve signals, conducted in axons, and neurotransmitters, which cross synapses. Axon conduction occurs rapidly, as does the diffusion of a neurotransmitter across the short distance of a synapse. In other words, the nervous system is organized to respond rapidly to stimuli. This is particularly useful if the stimulus is an external event that endangers our safety—helping us move quickly to avoid being hurt.

The **endocrine system** functions differently. The endocrine system is largely composed of glands (Fig. 40.2). These glands secrete **hormones,** such as insulin, which are carried by the bloodstream to target cells throughout the body. It takes time to deliver hormones, and it takes time for cells to respond, but the effect is longer-lasting. In other words, the endocrine system is organized for a slower but prolonged response.

There is a difference between the glands of the endocrine glands and exocrine glands that are found elsewhere in the body. **Exocrine glands** secrete their products into ducts, which take them to the lumens of other organs or outside the body. For example, the salivary glands of the digestive system send saliva into the mouth by way of the salivary ducts (see Section 34.2).

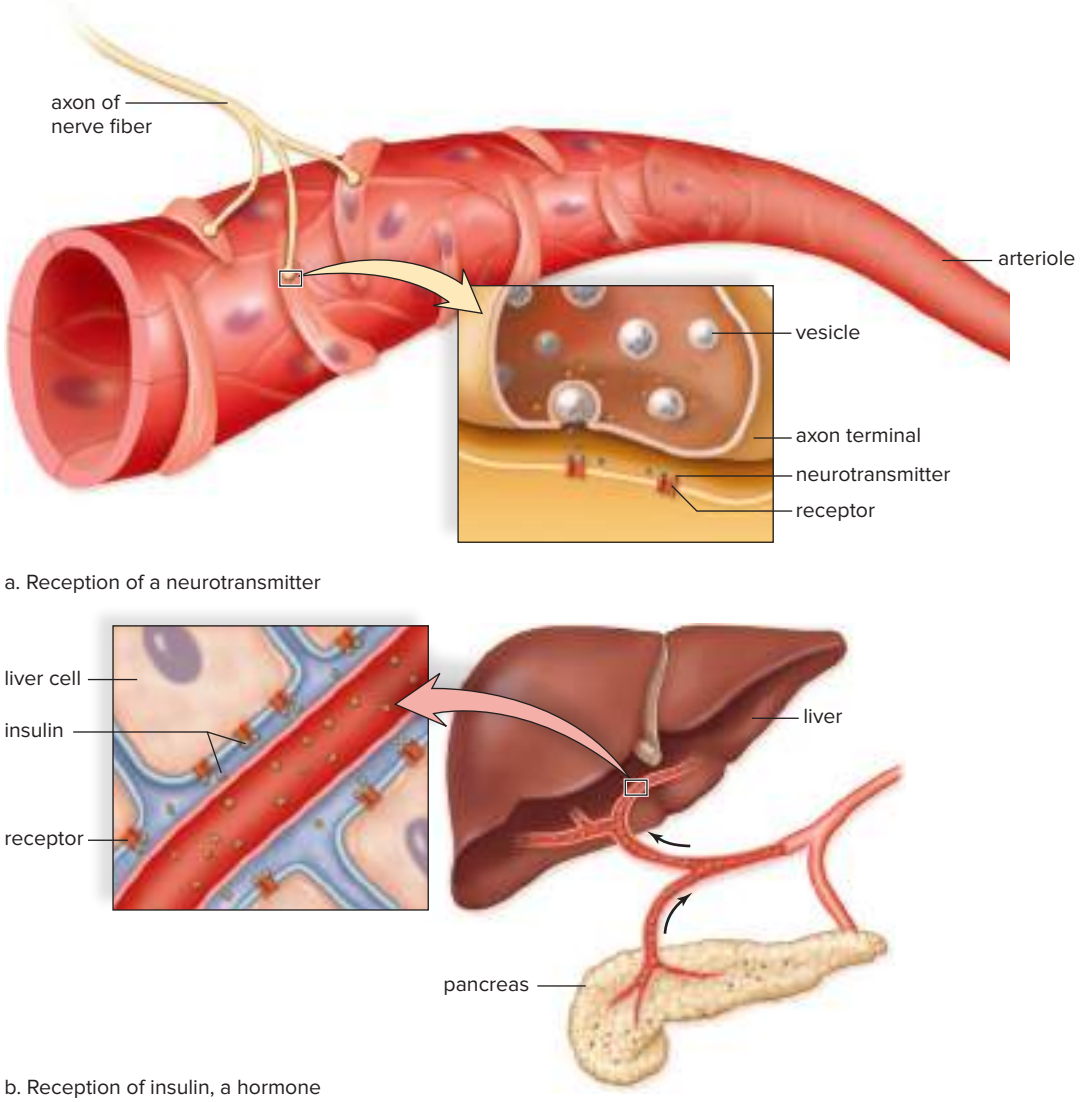


Figure 40.1 Modes of action of the nervous and endocrine systems. **a.** Nerve impulses passing along an axon cause the release of a neurotransmitter. The neurotransmitter, a chemical signal, binds to a receptor and causes the wall of an arteriole to constrict. **b.** The hormone insulin, a chemical signal, travels in the cardiovascular system from the pancreas to the liver, where it binds to a receptor and causes the liver cells to store glucose as glycogen.

ISTUDY

HYPOTHALAMUS

Releasing and inhibiting hormones: regulate the anterior pituitary

PITUITARY GLAND

Posterior Pituitary

Antidiuretic hormone (ADH): controls reabsorption of water by kidneys

Oxytocin: stimulates uterine contraction and milk letdown

Anterior Pituitary

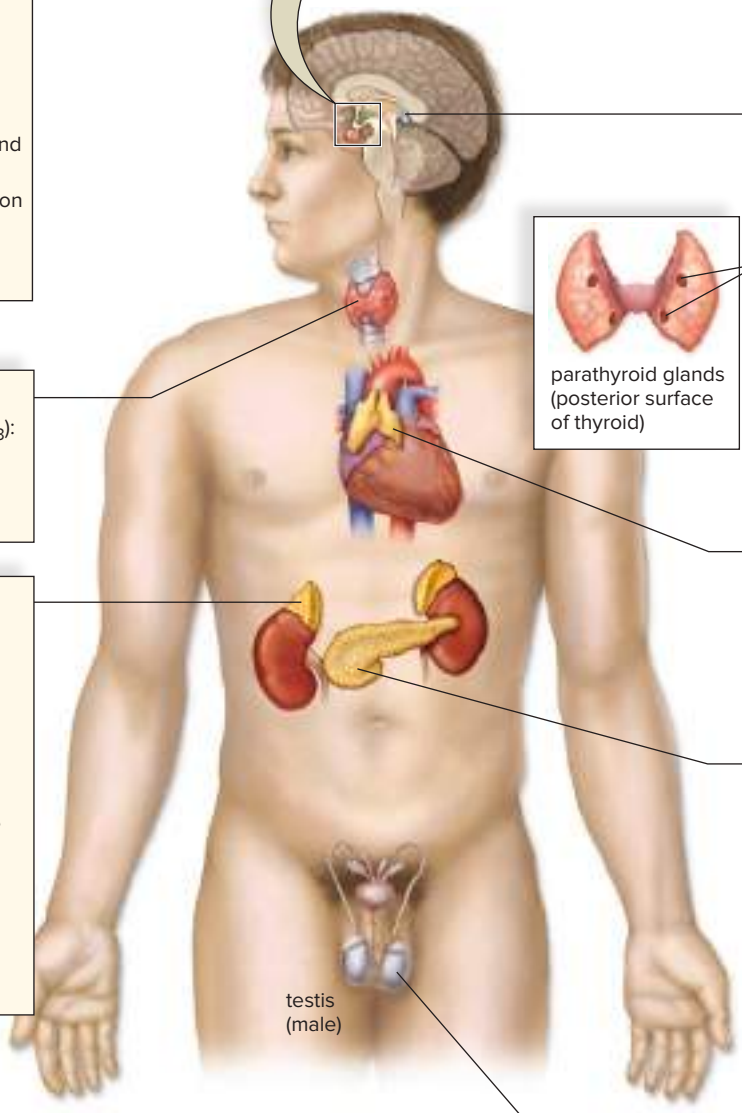
Thyroid-stimulating hormone (TSH): stimulates thyroid

Adrenocorticotropic hormone (ACTH): stimulates adrenal cortex

Gonadotropic hormones (FSH, LH): regulate production of eggs, sperm, and sex hormones

Prolactin (PL): stimulates milk production

Growth hormone (GH): controls bone growth, protein synthesis, and cell division



PINEAL GLAND

Melatonin: controls circadian and circannual rhythms

parathyroid glands (posterior surface of thyroid)

PARATHYROIDS

Parathyroid hormone (PTH): raises blood calcium level

THYROID

Thyroxine (T_4) and triiodothyronine (T_3): increase metabolic rate; regulate growth and development

Calcitonin: lowers blood calcium level

THYMUS

Thymosins: regulate production and maturation of T lymphocytes

ADRENAL GLAND

Adrenal Cortex

Glucocorticoids (cortisol): raise blood glucose level; stimulate breakdown of protein

Mineralocorticoids (aldosterone): regulate reabsorption of sodium and excretion of potassium

Sex hormones: stimulate reproductive functions and bring about sex characteristics

Adrenal Medulla

Epinephrine and norepinephrine: are active in emergency situations; raise blood glucose level

PANCREAS

Insulin: lowers blood glucose level and promotes glycogen buildup

Glucagon: raises blood glucose level and promotes glycogen breakdown

testis (male)

ovary (female)

GONADS

Testes

Androgens (testosterone): produce male sex characteristics

Ovaries

Estrogens and progesterone: produce female sex characteristics

Figure 40.2 Major glands of the human endocrine system. Major glands and the hormones they produce are depicted. Also, the endocrine system includes other organs, such as the kidneys, the gastrointestinal tract, and the heart, which also produce hormones but not as a primary function of these organs.

ISTUDY

Endocrine glands secrete their products into the bloodstream, which delivers them throughout the body.

The Role of Hormones

Hormones influence almost every basic homeostatic function of an organism, including metabolism, growth, reproduction, osmoregulation, and digestion. Therefore, it is not surprising that hormones are produced by invertebrates as well as vertebrates. For example, the hormone insulin is a key regulator of metabolism in vertebrates, and insulin-related peptides have been identified in insects and mollusks, suggesting an early evolutionary origin of this hormone.

Hormones also control some processes that are unique to invertebrates. As mentioned in the chapter-opening story, hormones control insect *metamorphosis,* the dramatic transformation that some insects undergo while hatching from an egg as a wormlike larva, going through several molts in which the exoskeleton is shed, and maturing into adults. Several hormones control this process.

Sometimes evolution produces new uses for the same hormones. In some species of freshwater snails a peptide related to insulin is involved in body and shell growth, as well as in energy metabolism. Variable hormone functions are seen in vertebrates as well. All vertebrates synthesize thyroid hormones, which generally increase metabolism, as you'll see later in this chapter. In amphibians, a surge of thyroid hormones also seems to promote metamorphosis from a tadpole into an adult. In contrast, the hormone prolactin inhibits metamorphosis in amphibians, stimulates skin pigmentation in reptiles, initiates incubation of eggs in birds, and stimulates milk production in mammals.

Hormones Are Chemical Signals

Like other chemical signals, hormones are a means of communication between cells, between body parts, and even between individuals. However, only certain cells, called target cells, can respond to a specific hormone. A target cell for a particular hormone carries a receptor protein for that hormone (Fig. 40.3). The hormone and receptor protein bind together the way a key fits a lock. The target cell then responds to that hormone. For example, in a condition called *androgen insensitivity,* an individual has X and Y sex chromosomes, and the testes, which remain in the abdominal cavity, produce the sex hormone testosterone. However, the body cells lack receptors that are able to combine with testosterone, and the individual appears to be biologically female.

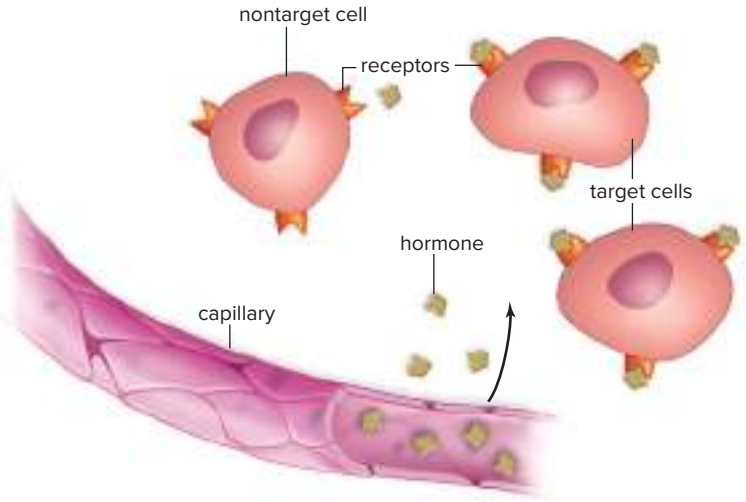


Figure 40.3 Target cell concept. Most hormones are distributed by the bloodstream to target cells. Target cells have receptors for the hormone, and the hormone combines with the receptor, as a key fits a lock.

Chemical signals that influence the behavior of other individuals of the species are called **pheromones.** Pheromones have been well documented in a wide variety of animal species, including those that established the mating rituals of insects and aquatic invertebrates, and those involving the social hierarchy of bees. While pheromones are found in other mammals (such as mice), their influence has been more difficult to prove in humans. To date, no specific human pheromones have been identified, but it is recognized that females who live in the same household tend to have synchronized menstrual cycles, perhaps because pheromones released by a female who is menstruating affect the menstrual cycle of other females in the household, although other factors may also play a role.

The Action of Hormones

Hormones exert a wide range of effects on cells. Some hormones induce target cells to increase their uptake of particular molecules, such as glucose, or ions, such as calcium. Others bring about an alteration of the target cell's structure in some way.

Most endocrine glands secrete **peptide hormones.** These hormones are peptides, proteins, glycoproteins, and modified amino acids. **Steroid hormones,** in contrast, all have the same molecular complex of four carbon rings, because they are all derived from cholesterol (see Fig. 3.12).

The Action of Peptide Hormones The actions of peptide hormones can vary, and as an example in this section, we concentrate on what happens in muscle cells after the hormone epinephrine binds to a receptor in the plasma membrane (Fig. 40.4). In muscle cells, the reception of epinephrine leads to the breakdown of glycogen to glucose, which provides energy for ATP production.

The immediate result of epinephrine binding is the formation of **cyclic adenosine monophosphate (cAMP),** which contains one phosphate group attached to adenosine at two locations. Therefore, the molecule is cyclic. cAMP activates a protein kinase enzyme in the cell, and this enzyme in turn activates another enzyme, and so forth. The series of enzymatic reactions that follows cAMP formation is called an enzyme cascade or signaling cascade. Because each enzyme can be used over and over again, more enzymes become involved at every step of the cascade. Finally, many molecules of glycogen are broken down to glucose, which enters the bloodstream.

Typical of a peptide hormone, epinephrine never enters the cell. Therefore, the hormone is called the first messenger**,** whereas cAMP, which sets the metabolic machinery in motion, is called the **second messenger.** For example, imagine that the adrenal medulla, which produces epinephrine, is like a company's home office, which sends out a courier (the hormone epinephrine—the first messenger) to its factory (the cell). The courier doesn't have a

ISTUDY

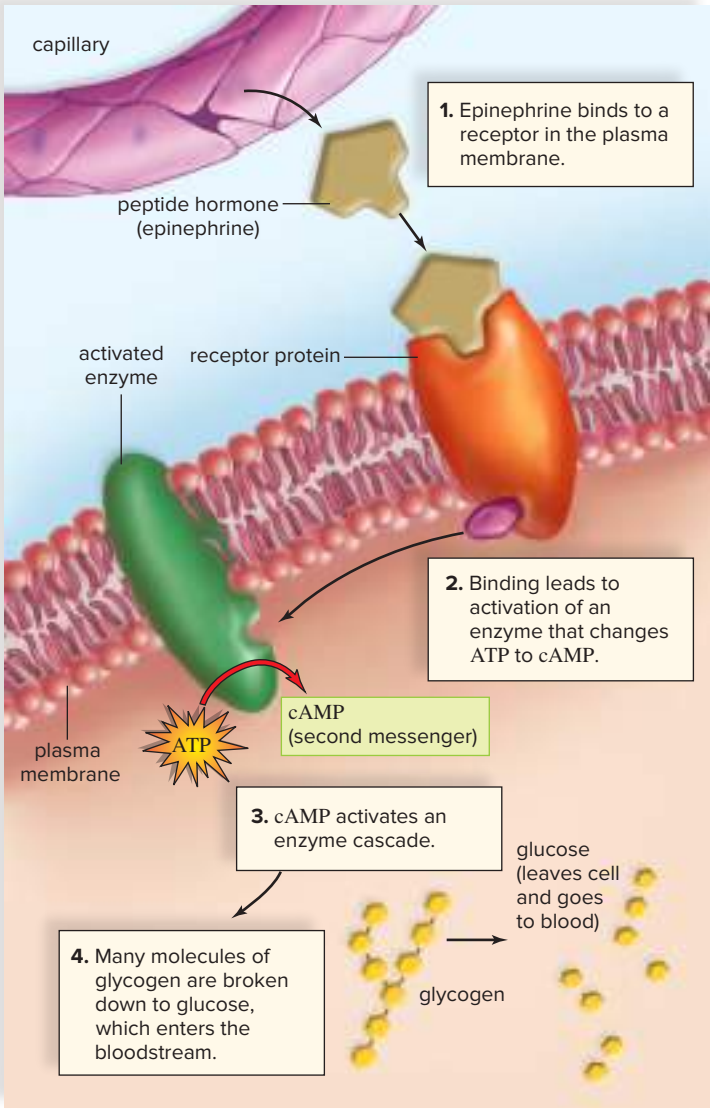


Figure 40.4 Action of a peptide hormone. Peptide hormones (epinephrine, in this example) act as first messengers, binding to specific receptors in the plasma membrane. First messengers activate second messengers (cAMP, in this case), which influence various cellular processes.

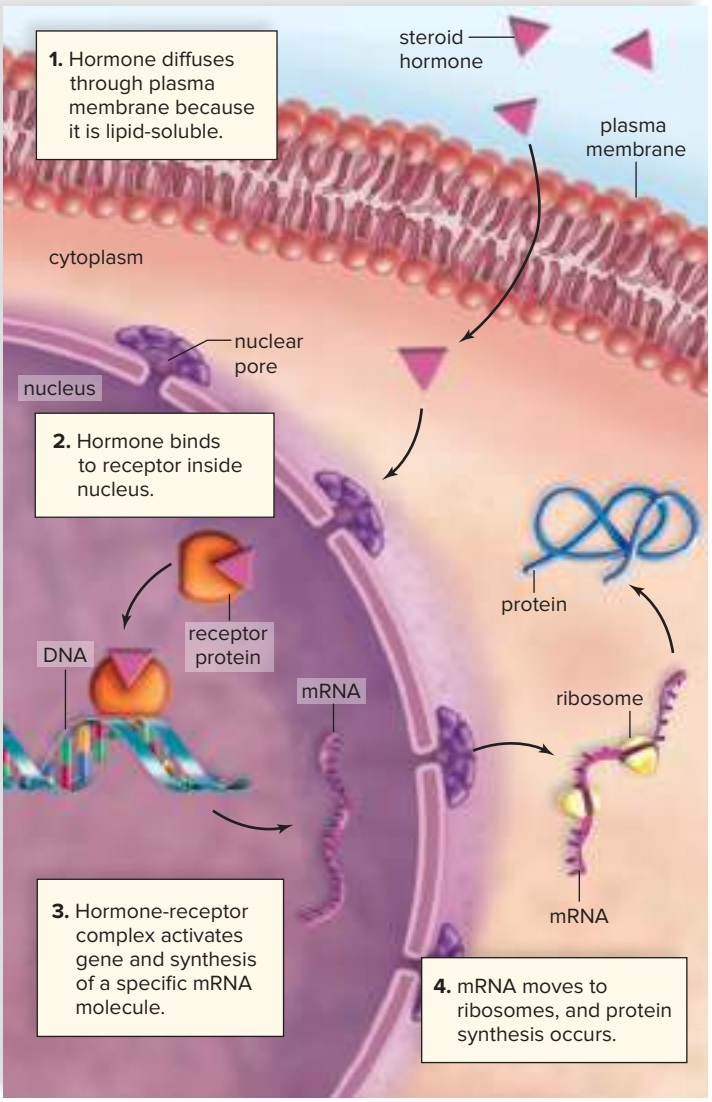


Figure 40.5 Action of a steroid hormone. A steroid hormone passes directly through the target cell's plasma membrane before binding to a receptor in the nucleus or cytoplasm. The hormone-receptor complex binds to DNA, and gene expression follows.

pass to enter the factory but tells a supervisor through a screen door that the home office wants the factory to produce a particular product. The supervisor (cAMP—the second messenger) walks over and flips a switch that starts the machinery (the enzymatic pathway), and a product is made.

The Action of Steroid Hormones Only the adrenal cortex, the ovaries, and the testes produce steroid hormones. Steroid hormones do not bind to plasma membrane receptors; instead, they are able to enter the cell because they are lipids (Fig. 40.5). Once inside, a steroid hormone binds to an internal receptor, usually in the nucleus but sometimes in the cytoplasm. Inside the nucleus, the hormone-receptor complex binds with DNA and activates certain genes. Messenger RNA (mRNA) moves to the ribosomes in the cytoplasm, and protein synthesis (e.g., an enzyme) follows. To continue the analogy, a steroid hormone is like a courier who has a pass to enter the factory (the cell). Once inside, the courier makes contact with the plant manager (DNA), who sees to it that the factory (cell) is ready to produce a product.

Steroids act more slowly than peptides because it takes more time to synthesize new proteins than to activate enzymes already present in cells. Their action lasts longer, however.

Check Your Progress 40.1

1. Compare and contrast the nervous and endocrine systems with regard to their function and the types of signals they use.
2. Compare the location of the receptors for peptide and steroid hormones.
3. Explain why second messengers are needed for most peptide hormones.

ISTUDY

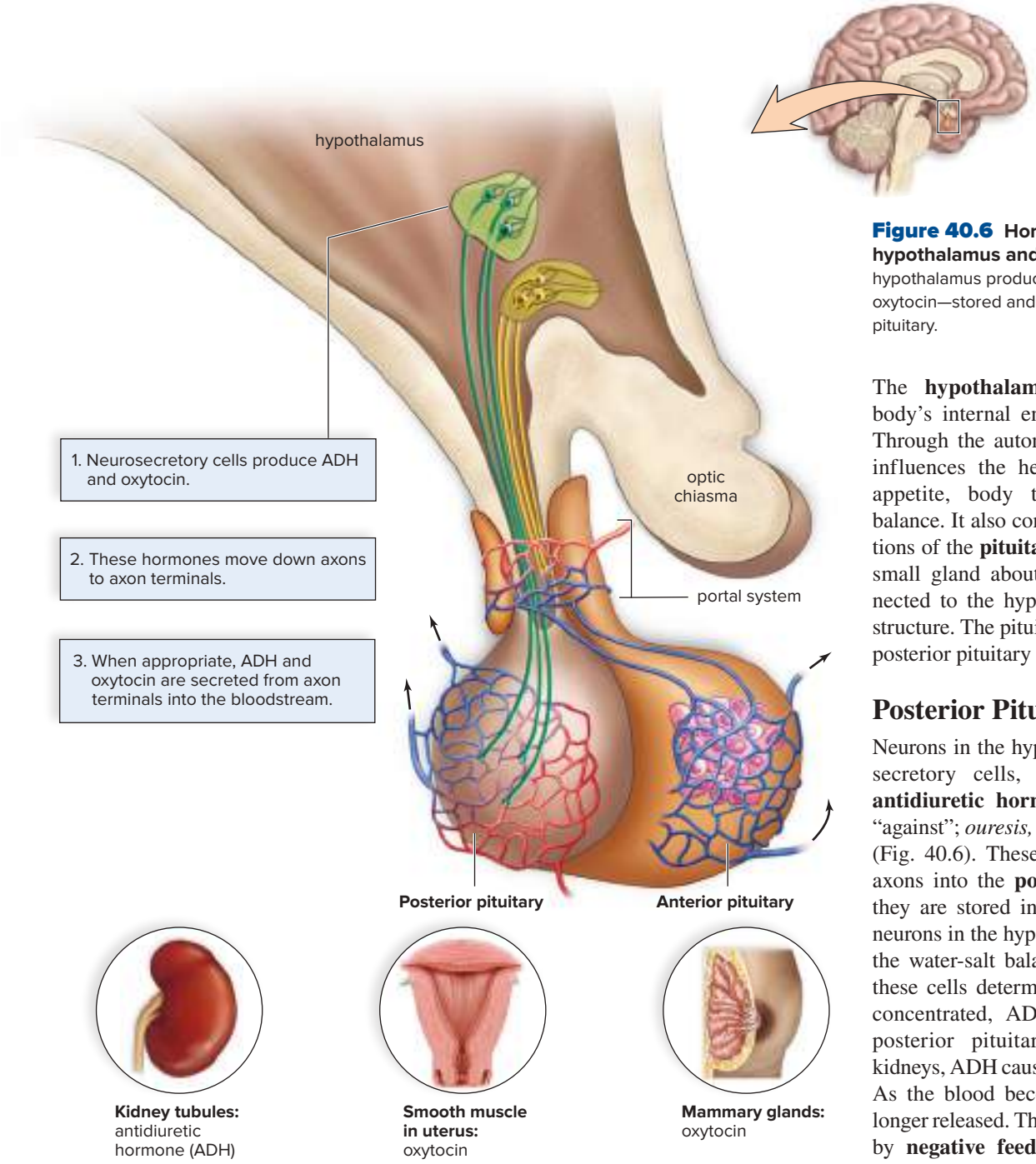


Figure 40.6 Hormones produced by the hypothalamus and posterior pituitary. The hypothalamus produces two hormones—ADH and oxytocin—stored and secreted by the posterior pituitary.

40.2 Hypothalamus and Pituitary Gland

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the relationship between the hypothalamus and the pituitary gland.
2. List and describe the functions of the hormones released by the anterior and posterior pituitary gland.
3. Explain how some hormones are regulated by negative feedback, and some by positive feedback, and give an example of each.

The **hypothalamus** helps regulate the body's internal environment in two ways. Through the autonomic nervous system, it influences the heartbeat, blood pressure, appetite, body temperature, and water balance. It also controls the glandular secretions of the **pituitary gland** (hypophysis), a small gland about 1 cm in diameter connected to the hypothalamus by a stalklike structure. The pituitary has two portions: the posterior pituitary and the anterior pituitary.

Posterior Pituitary

Neurons in the hypothalamus, called neurosecretory cells, produce the hormones **antidiuretic hormone (ADH)** (Gk. *anti,* "against"; *ouresis,* "urination") and oxytocin (Fig. 40.6). These hormones pass through axons into the **posterior pituitary,** where they are stored in axon terminals. Certain neurons in the hypothalamus are sensitive to the water-salt balance of the blood. When these cells determine that the blood is too concentrated, ADH is released from the posterior pituitary. Upon reaching the kidneys, ADH causes water to be reabsorbed. As the blood becomes dilute, ADH is no longer released. This is an example of control by **negative feedback**—the *effect* of the hormone (to dilute blood) shuts down the *release* of the hormone. Negative feedback maintains stable conditions and homeostasis.

If too little ADH is secreted, or if the kidneys become unresponsive to ADH, a condition known as *diabetes insipidus* results. Individuals with this condition are usually very thirsty; they produce copious amounts of urine and can become severely dehydrated if the condition is untreated.

The consumption of alcohol inhibits ADH release. This effect helps explain the frequent urination associated with drinking alcohol.

Oxytocin (Gk. *oxys,* "quick"; *tokos,* "birth"), the other hormone made in the hypothalamus, causes uterine contractions during childbirth and milk letdown when a baby is nursing. The more the uterus contracts during labor, the more nerve impulses reach the

hypothalamus, causing oxytocin to be released. Similarly, the more a baby suckles, the more oxytocin is released. In both instances, the release of oxytocin from the posterior pituitary is controlled by **positive feedback**—that is, the stimulus continues to bring about an effect that ever increases in intensity. Oxytocin may also play a role in the propulsion of semen through the male reproductive tract and may affect feelings of sexual satisfaction and emotional bonding.

Anterior Pituitary

A portal system, which consists of two capillary networks connected by a vein, lies between the hypothalamus and the anterior pituitary (Fig. 40.7). The hypothalamus controls the anterior pituitary by producing **hypothalamic-releasing hormones** and in some instances **hypothalamic-inhibiting hormones.** For example, one hypothalamic-releasing hormone stimulates the anterior pituitary to secrete a thyroid-stimulating hormone, and a particular hypothalamic-inhibiting hormone prevents the anterior pituitary from secreting prolactin.

Anterior Pituitary Hormones Affecting Other Glands

Some of the hormones produced by the **anterior pituitary** affect other glands. **Gonadotropic hormones** stimulate the gonads—the

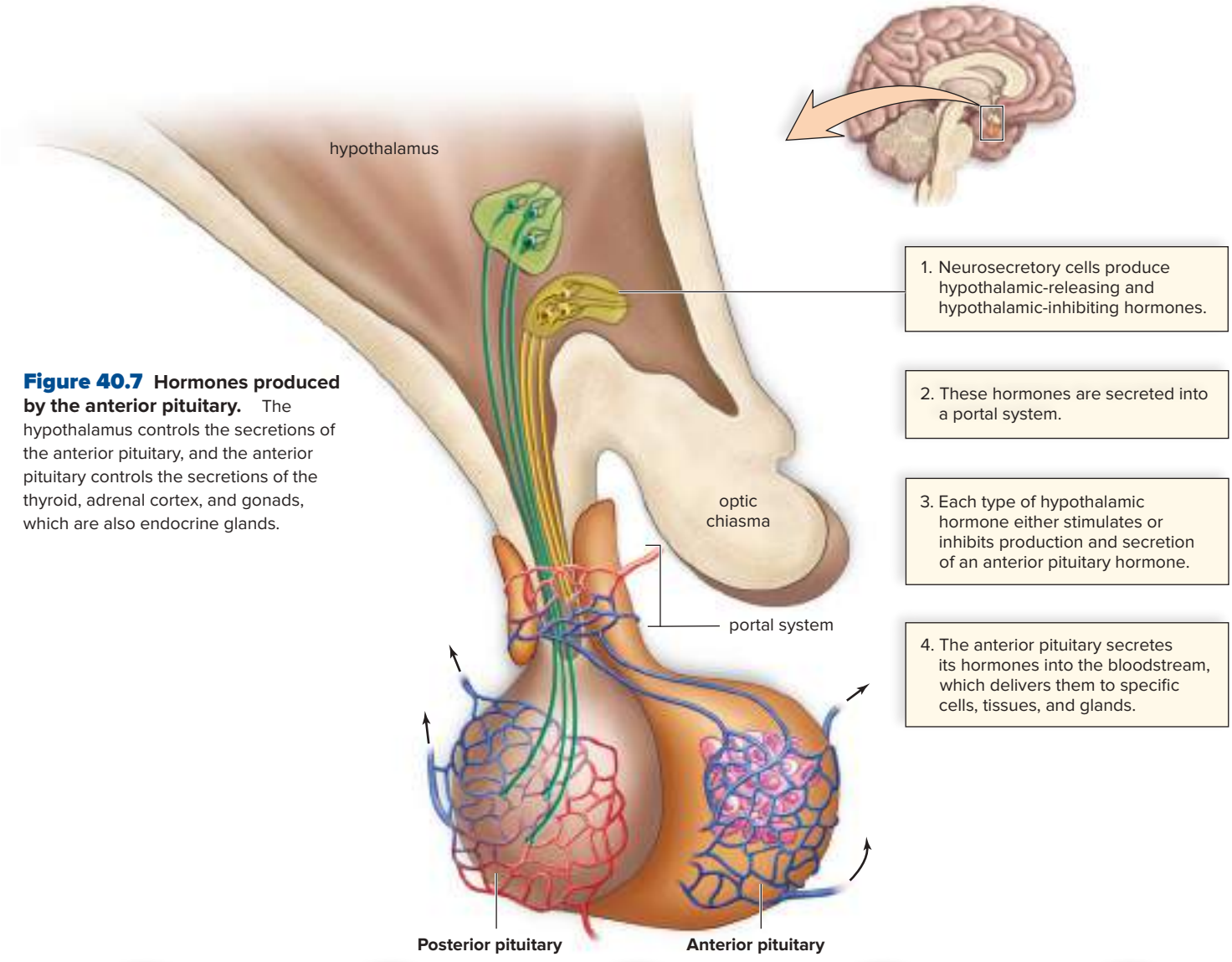


Figure 40.7 Hormones produced by the anterior pituitary. The hypothalamus controls the secretions of the anterior pituitary, and the anterior pituitary controls the secretions of the thyroid, adrenal cortex, and gonads, which are also endocrine glands.

ISTUDY

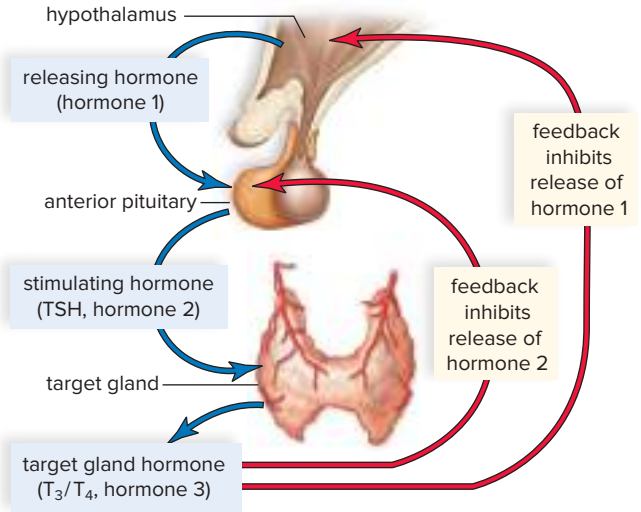


Figure 40.8 Negative feedback mechanisms in the endocrine system. Feedback mechanisms (red arrows) provide a means of controlling the amount of hormones produced (blue arrows) by the hypothalamus and pituitary glands.

testes in males and the ovaries in females—to produce gametes and sex hormones. **Adrenocorticotropic hormone (ACTH)** stimulates the adrenal cortex to produce cortisol. **Thyroid-stimulating hormone (TSH)** stimulates the thyroid to produce thyroxine (T_4) and triiodothyronine (T_3). In each instance, the blood level of the last hormone in the sequence exerts negative feedback control over the secretion of the first two hormones. This is how it works for TSH (Fig. 40.8):

Anterior Pituitary Hormones Not Affecting Other Glands

Three hormones produced by the anterior pituitary do not affect other endocrine glands. **Prolactin (PRL)** (L. *pro,* "before"; *lactis,* "milk") is produced in quantity only after childbirth. It causes the mammary glands in the breasts to develop and produce milk. It also plays a role in carbohydrate and fat metabolism.

Melanocyte-stimulating hormone (MSH) (Gk. *melanos,* "black"; *kytos,* "cell") causes skin-color changes in many fishes, amphibians, and reptiles having melanophores, skin cells that produce color variations. The concentration of this hormone in humans is very low.

Growth hormone (GH), or somatotropic hormone, promotes skeletal and muscular growth. It increases the rate at which amino acids enter cells and protein synthesis occurs. It also promotes fat metabolism, as opposed to glucose metabolism. The amount of GH produced is greatest during childhood and adolescence.

If too little GH is produced during childhood, the individual has *pituitary dwarfism,* characterized by normal proportions but small stature. Such children also have problems with low blood sugar (hypoglycemia) because GH normally helps oppose the effect of insulin on glucose uptake. Through the administration of GH, growth patterns can be restored and blood sugar problems alleviated. If too much GH is secreted during childhood, the person may develop gigantism. Individuals with gigantism usually have health problems, primarily because elevated GH cancels out the effects of insulin, promoting diabetes mellitus (see Section 40.3).

Normal hand Acromegaly hand

Figure 40.9 Acromegaly. Acromegaly is caused by overproduction of GH in the adult. It is characterized by enlargement of the bones in the face, fingers, and toes as a person ages. (both): Wysocka Malgorzata/Shutterstock

On occasion, GH is overproduced in the adult, and a condition called *acromegaly* results. Because long bone growth is no longer possible in adults, only the feet, hands, and face (particularly the chin, nose, and eyebrow ridges) can respond, and these portions of the body become overly large (Fig. 40.9).

A quick Internet search reveals that many websites offer human growth hormone for sale as a "fountain of youth" that can help adults lose weight, add muscle, and reduce the effects of aging. Indeed, several professional actors and athletes have admitted to using human GH to build muscle and reduce body fat. Note that GH is not a *steroid,* which refers to testosterone or related steroid hormones. However, using human GH in this manner can have many undesired side effects, such as joint and muscle pain, high blood pressure, and diabetes mellitus.

Check Your Progress 40.2

1. Explain how the hypothalamus communicates with the endocrine system.
2. List the hormones produced by the posterior pituitary and describe a function of each.
3. List the hormones produced by the anterior pituitary and describe a function of each.

40.3 Other Endocrine Glands and Hormones

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the functions of hormones of the thyroid and parathyroid glands.
2. Compare and contrast the mineralocorticoids and glucocorticoids.
3. Identify the causes and major symptoms of the major medical conditions associated with the endocrine system.

ISTUDY

In this section, we discuss the thyroid and parathyroid glands, adrenal glands, pancreas, pineal gland, thymus, and other tissues that produce hormones secondarily. All the hormone products of these glands and tissues play a role in health and homeostasis.

Thyroid and Parathyroid Glands

The **thyroid gland** is attached to the trachea just below the larynx (see Fig. 40.2). Weighing approximately 20 grams, the thyroid gland is composed of a large number of follicles, each a small, spherical structure made of thyroid cells that produce the hormones triiodothyronine (T_3), which contains three iodine atoms, and **thyroxine** (T_4), which contains four iodine atoms. Cells that reside outside the follicles of the thyroid gland produce the hormone calcitonin. The parathyroid glands, which produce parathyroid hormone, are embedded in the posterior surface of the thyroid gland.

Effects of T_3 and T_4

Both of these thyroid hormones increase the overall metabolic rate. They do not have a single target organ; instead, they stimulate most of the cells of the body to metabolize at a faster rate. More glucose is broken down, and more energy is used. Interestingly, even though T_3 and T_4 are peptide hormones because they are derived from the amino acid tyrosine, their receptor is actually located inside cells, more like a steroid hormone receptor.

To produce T_3 and T_4, the thyroid gland actively acquires iodine. The concentration of iodine in the thyroid gland is approximately 25 times that found in the blood. If a person consumes an insufficient amount of iodine, the thyroid gland is unable to produce the required amount of T_3 and T_4. This results in constant stimulation of the thyroid by the TSH released by the anterior pituitary. The thyroid gland enlarges, resulting in a condition called *endemic goiter* (Fig. 40.10*a*). In the 1920s, scientists discovered that the use of iodized salt allows the thyroid to produce thyroid hormones and therefore helps prevent endemic goiter. However, iodine deficiency is still extremely common in some parts of the world, with an estimated 2 billion people (one-third of the world's population) still having some degree of iodine deficiency.

An insufficiency of T_3 and T_4 in the newborn is called *congenital hypothyroidism* (Fig. 40.10*b*). Babies with this condition are short and stocky, and many develop an intellectual disability. The causes vary from iodine deficiency in a female who is pregnant to genetic defects affecting the production of TSH, T_3, T_4, or the receptor for any of these hormones. Once detected, iodine deficiency is easily treated by ensuring appropriate levels of iodine consumption. However, according to the American Thyroid Association, congenital hypothyroidism due to iodine deficiency remains the most common preventable cause of intellectual disability in the world.

Even mild iodine deficiency during pregnancy, which may be present in some females in the United States, may be associated with intellectual disability in children. In cases due to problems with the thyroid gland itself, thyroxine therapy is curative, but it must begin as early as possible to avoid permanently stunted growth and intellectual disability.

Hypothyroidism can also occur in adults, most often when the immune system produces antibodies that destroy the thyroid gland. Untreated hypothyroidism in adults results in *myxedema,* which is characterized by lethargy, weight gain, loss of hair, slower heart rate, lowered body temperature, and thickness and puffiness of the skin. The administration of thyroxine usually restores normal body functions and appearance.

Hyperthyroidism results from the oversecretion of T_3 or T_4. In Graves' disease, antibodies are produced and react with the TSH receptor on thyroid follicular cells, mimicking the effect of TSH and causing the overproduction of T_3 and T_4. One typical sign of Graves' disease is **exophthalmos** (exophthalmia), or excessive protrusion of the eyes due to edema in eye socket tissues and

a. Endemic goiter b. Congenital hypothyroidism c. Exophthalmic goiter

Figure 40.10 Abnormalities of the thyroid. **a.** An enlarged thyroid gland is often caused by a lack of iodine in the diet. Without iodine, the thyroid is unable to produce its hormones, and continued anterior pituitary stimulation causes the gland to enlarge. **b.** Individuals who develop hypothyroidism during infancy or childhood do not grow and develop as others do. Unless medical treatment is begun, the body is short and stocky; intellectual disability is also likely. **c.** In exophthalmic goiter, a goiter is due to an overactive thyroid, and the eye protrudes because of edema in eye socket tissue. (a): Bruce Coleman Inc/Avalon/Alamy Stock Photo; (b): Mediscan/Alamy Stock Photo; (c): Dr. P. Marazzi/Science Source

ISTUDY

inflammation of the muscles that move the eyes (Fig. 40.10*c*). The individual usually becomes hyperactive, nervous, and irritable and may have insomnia. Graves' disease is the most common cause of hyperthyroidism in children and adolescents, and it is five to ten times more common in females. Available treatments include drugs that block iodine uptake by the thyroid gland, surgical removal of part or all of the gland, and the administration of radioactive iodine to destroy the overactive tissue.

Hyperthyroidism can also be caused by thyroid cancer, the most common cancer of the endocrine system. Thyroid cancer is usually detected as a lump during physical examination. Again, the treatment is surgery in combination with the administration of radioactive iodine. The prognosis for most patients is excellent.

Effects of Calcitonin

Calcium (Ca^{2+}) plays a significant role in both nervous conduction and muscle contraction. It is also necessary for blood clotting and the maintenance of healthy bones and teeth. The blood calcium level is regulated in part by **calcitonin,** a hormone secreted by the thyroid gland when the blood calcium level rises. The primary effect of calcitonin is to bring about the deposit of calcium in the bones (Fig. 40.11, *top*). It does this by temporarily reducing the activity and number of osteoclasts. When the blood calcium lowers to normal, the thyroid's release of calcitonin is inhibited.

Although calcitonin appears to play a very important role in regulating calcium homeostasis in fish and a few other animals, it appears to be less significant in humans. As evidence for this, a deficiency of calcitonin (as occurs when the thyroid glands are removed) is not linked with any specific disorder. However, because of its bone-building effects, calcitonin is an FDA-approved drug for reducing bone loss in osteoporosis.

Parathyroid Glands

Parathyroid hormone (PTH), produced by the **parathyroid glands,** causes the blood calcium level to increase and the blood phosphate (HPO_4^{2-}) level to decrease. Low blood calcium stimulates the release of PTH, which promotes the activity of osteoclasts, releasing calcium from the bones. PTH also promotes the kidneys' reabsorption of calcium, lessening its excretion. In the kidneys, PTH also brings about activation of vitamin D. Vitamin D, in turn, stimulates the absorption of calcium from the small intestine (Fig. 40.11, *bottom*). These effects bring the blood calcium level back to the normal range, and PTH secretion stops. Calcitonin and PTH are therefore considered to be *antagonistic hormones,* because their action is opposite one another and both hormones work together to regulate the blood calcium level.

Many years ago, the four parathyroid glands were sometimes mistakenly removed during thyroid surgery because of their small size and hidden location. Gland removal caused insufficient parathyroid hormone production, or *hypoparathyroidism.* This condition leads to a dramatic drop in the blood calcium level, followed by excessive nerve excitability. Nerve signals happen spontaneously and without rest, causing a phenomenon called tetany. In *tetany,* the body shakes from continuous muscle contraction. Without treatment (usually with intravenous calcium), severe hypoparathyroidism causes seizures, heart failure, and death.

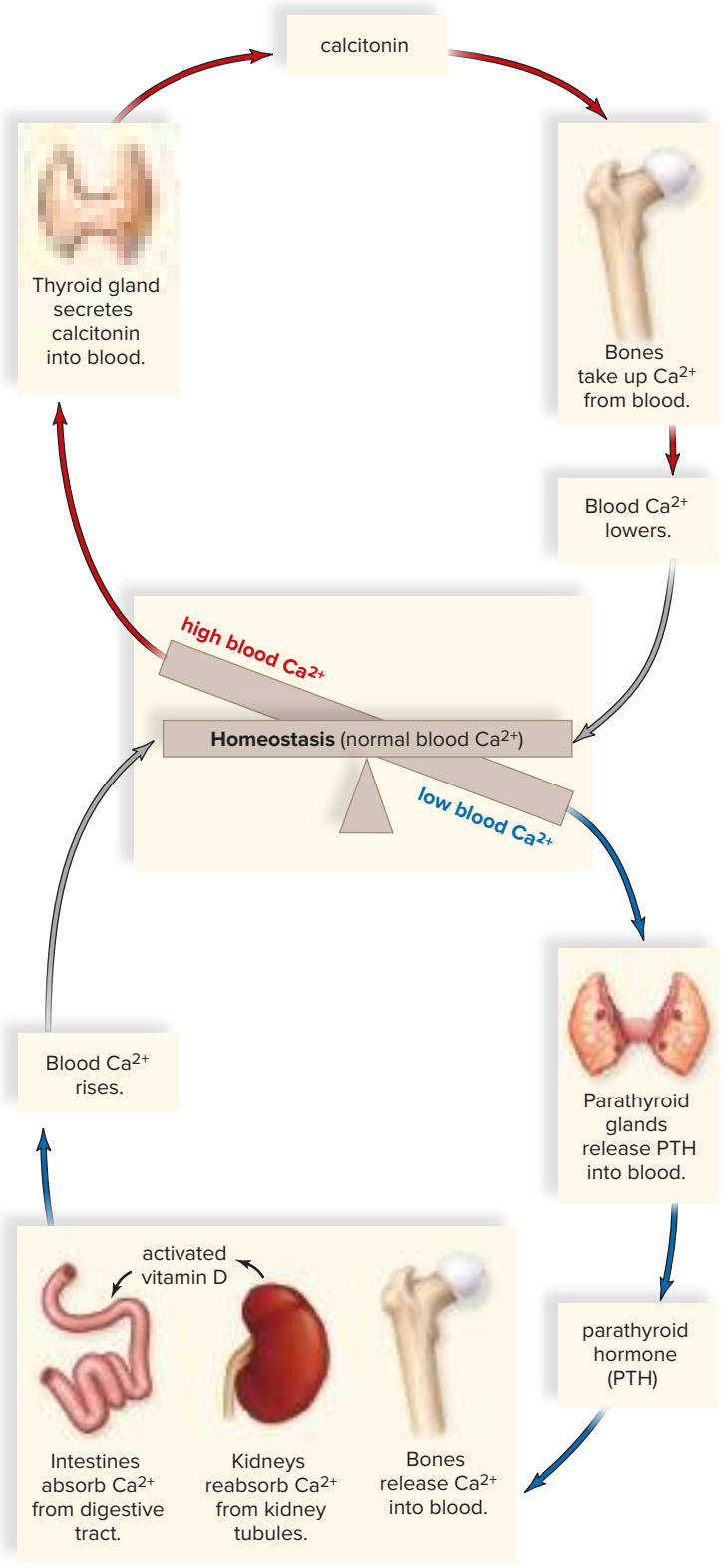


Figure 40.11 Regulation of blood calcium level. *Top:* When the blood calcium (Ca^{2+}) level is high, the thyroid gland secretes calcitonin. Calcitonin promotes the uptake of Ca^{2+} by the bones, and therefore the blood Ca^{2+} level returns to normal. *Bottom:* When the blood Ca^{2+} level is low, the parathyroid glands release parathyroid hormone (PTH). PTH causes the bones to release Ca^{2+} and the kidneys to reabsorb Ca^{2+} and activate vitamin D; thereafter, the intestines absorb Ca^{2+}. Therefore, the blood Ca^{2+} level returns to normal.

ISTUDY

Untreated *hyperparathyroidism* (oversecretion of PTH) can result in osteoporosis because of continuous calcium release from the bones. Hyperparathyroidism can also cause the formation of calcium kidney stones.

Adrenal Glands

The **adrenal glands** sit atop the kidneys (see Fig. 40.2). Each adrenal gland is about 5 cm long and 3 cm wide, weighs about 5 g, and consists of an inner portion called the **adrenal medulla** and an outer portion called the **adrenal cortex.** These portions, like the anterior and posterior pituitary, are two functionally distinct endocrine glands. Stress of all types, including both emotional and physical trauma, prompts the hypothalamus to stimulate both portions of the adrenal glands. The adrenal cortex is also involved in regulating the water-salt balance, as well as secreting a small amount of male and female sex hormones.

Adrenal Medulla

During emergency situations that call for a "fight-or-flight" reaction, the hypothalamus sends nerve impulses by way of sympathetic nerve fibers to many organs, including the adrenal medulla (see Fig. 37.16). This neurological response to danger quickly dilates the pupils, speeds the heart, dilates the air passages, and inhibits many nonessential bodily functions. Meanwhile, the adrenal medulla secretes **epinephrine** (adrenaline) and **norepinephrine** (noradrenaline) into the bloodstream (Fig. 40.12). These hormones continue the response to stress throughout the body—for example, by accelerating the breakdown of glucose to form ATP, triggering the mobilization of glycogen reserves in skeletal muscle, and increasing the cardiac rate and force of contraction. These effects are usually short-lived however, as these two hormones are rapidly metabolized by the liver.

Adrenal Cortex

In contrast to the rapid response of the sympathetic nervous system and adrenal medulla, the hypothalamus produces a longer-term response to stress by stimulating the anterior pituitary to produce ACTH, which in turn causes the adrenal cortex to secrete glucocorticoids.

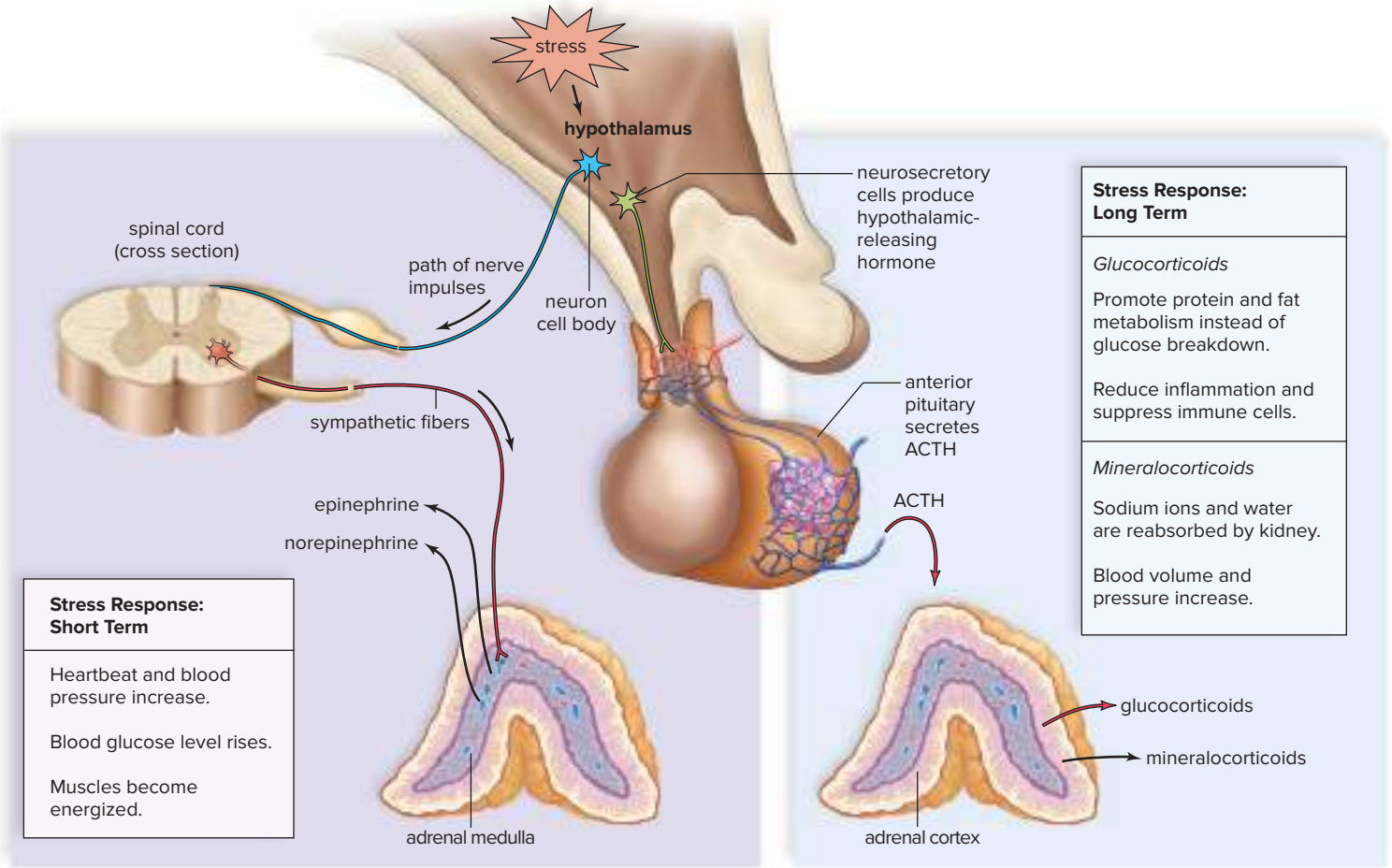


Figure 40.12 Adrenal glands. Both the adrenal cortex and the adrenal medulla are under the control of the hypothalamus when they help us respond to stress. *Left:* Nervous stimulation causes the adrenal medulla to provide a rapid, but short-term, stress response. *Right:* ACTH from the anterior pituitary causes the adrenal cortex to release glucocorticoids. Independently, the adrenal cortex releases mineralocorticoids. The adrenal cortex provides a slower, but long-term, stress response.

ISTUDY

Glucocorticoids **Cortisol** is the most important **glucocorticoid** produced by the human adrenal cortex. Cortisol raises the blood glucose level in at least two ways:

- It promotes the breakdown of muscle proteins to amino acids, which are taken up by the liver from the bloodstream and converted into glucose.
- It promotes the catabolism of fatty acids rather than carbohydrates, and this spares glucose. The rise in blood glucose is beneficial to an animal under stress, because glucose is the preferred energy source for neurons.

Glucocorticoids also counteract the inflammatory response, including the type of reaction that leads to the pain and swelling of joints in arthritis and bursitis. Cortisone, a glucocorticoid, is often used to treat these conditions, because it reduces inflammation. However, very high levels of glucocorticoids in the blood can suppress the body's defense system, rendering an individual more susceptible to injury and infection.

Mineralocorticoids **Mineralocorticoids** produced by the adrenal cortex regulate salt and water balance, leading to increases in blood volume and blood pressure. **Aldosterone** is the most important of the mineralocorticoids. Aldosterone primarily targets the kidneys, where it promotes the renal absorption of sodium (Na^+) and renal excretion of potassium (K^+), thereby helping regulate blood volume and blood pressure.

The secretion of mineralocorticoids is not controlled by the anterior pituitary. In Section 36.2, we explored that when the atria of the heart are stretched due to increased blood volume, cardiac cells release a hormone called **atrial natriuretic hormone (ANH),** also called *atrial natriuretic peptide,* which inhibits the secretion of aldosterone from the adrenal cortex. (Note that the heart is one of several organs in the body that release a hormone but are not considered among the major endocrine organs.) The effect of ANH is to cause *natriuresis,* the excretion of sodium ions (Na^+). When sodium is excreted, so is water, and therefore blood pressure lowers to normal (Fig. 40.13, *top*). ANH can further lower blood pressure by dilating smooth muscle in blood vessels.

We also discovered in Section 36.2 that when the blood sodium (Na^+) level, and therefore blood pressure, is low, the kidneys secrete **renin** (Fig. 40.13, *bottom*). Renin is an enzyme that converts the plasma protein angiotensinogen to angiotensin I, which is changed to angiotensin II by an enzyme in lung capillaries. Angiotensin II stimulates the adrenal cortex to release aldosterone. The effect of this renin-angiotensin-aldosterone system is to raise blood pressure in two ways: (1) Angiotensin II constricts the arterioles, and (2) aldosterone causes the kidneys to reabsorb sodium. When the blood sodium level rises, water is reabsorbed, in part because the hypothalamus secretes ADH (see Section 40.2). Then, blood pressure rises to normal.

Diseases and Conditions of the Adrenal Cortex Insufficient secretion of hormones by the adrenal cortex, also known as *Addison's disease,* is relatively rare. The most common cause is an inappropriate attack on the adrenal cortex by the immune system.

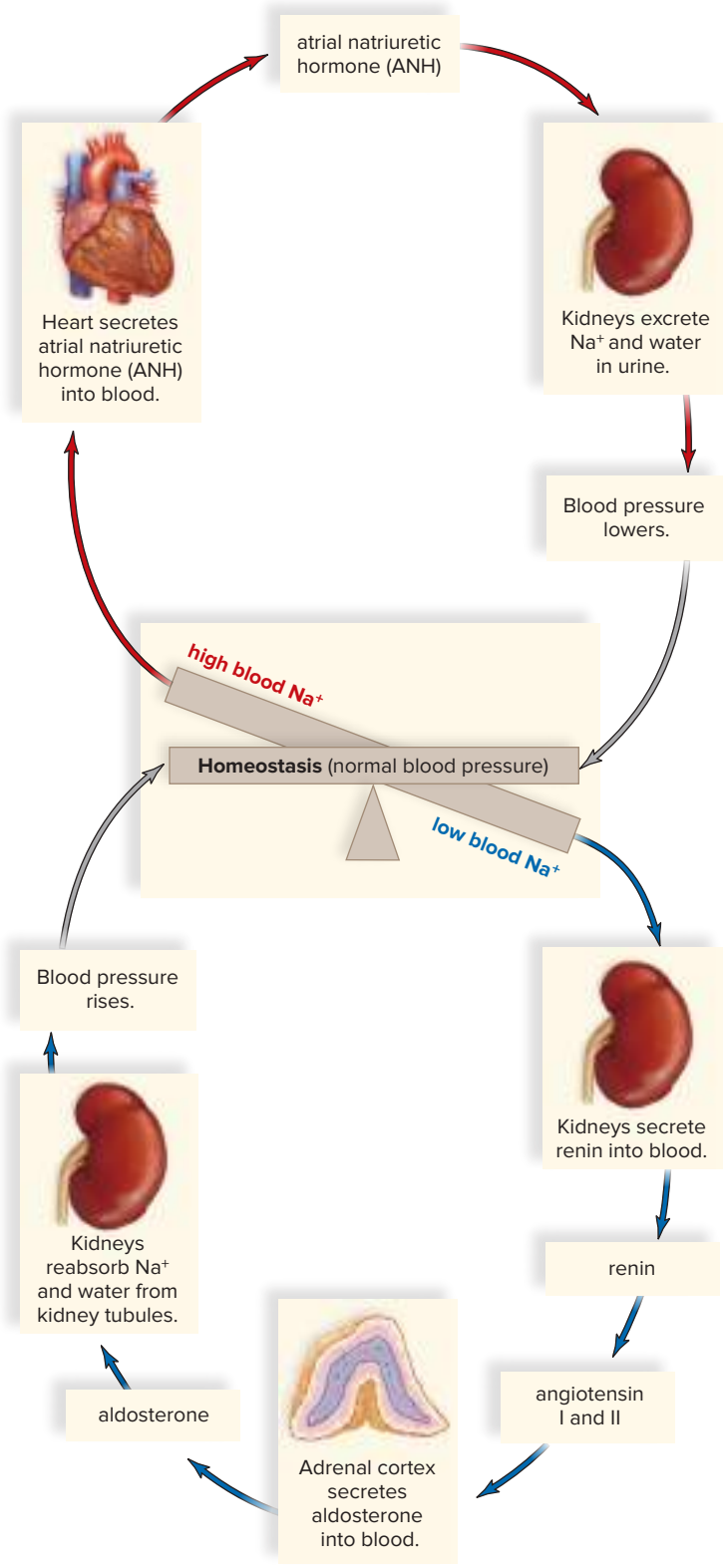
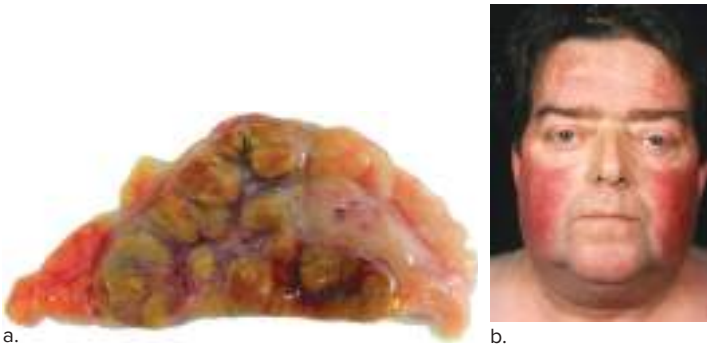


Figure 40.13 Regulation of blood pressure and volume. *Top:* When the blood Na^+ is high, a high blood volume causes the heart to secrete atrial natriuretic hormone (ANH). ANH causes the kidneys to excrete Na^+, and water follows. The blood volume and pressure return to normal. *Bottom:* When the blood sodium (Na^+) level is low, a low blood pressure causes the kidneys to secrete renin. Renin leads to the secretion of aldosterone from the adrenal cortex. Aldosterone causes the kidneys to reabsorb Na^+, and water follows, so that blood volume and pressure return to normal.

ISTUDY

Figure 40.14 Addison's disease. Addison's disease is characterized by an atypical bronzing of the skin, particularly noticeable in individuals with light-colored skin. Note the color of the hands of someone with Addison's disease (*left*) compared to the hand of an individual without the disease (*right*). BSIP/Science Source



a. b.

Figure 40.15 Cushing's syndrome. **a.** The tumors associated with adrenal hyperplasia often contribute to Cushing's syndrome. **b.** A man showing some of the common characteristics of Cushing's syndrome. (gland): Dennis K. Burns, MD/Travis G. Brown, MD/Walter L. Kemp, MD/McGraw Hill; (man): Biophoto Associates/Science Source

Because the disease affects the secretion of both glucocorticoids and mineralocorticoids, a variety of symptoms may occur, such as dehydration, weakness, weight loss, and hypotension (low blood pressure). The presence of excessive but ineffective ACTH often causes increased pigmentation of the skin, because ACTH, like MSH, can stimulate melanocytes to produce melanin (Fig. 40.14). Treatment involves the replacement of the missing hormones. Left untreated, Addison's disease can be fatal.

Excessive levels of glucocorticoids result in *Cushing's syndrome*. This disorder can be caused by tumors that affect either the pituitary gland, resulting in excess ACTH secretion, or the adrenal cortex itself. The most common cause, however, is the administration of glucocorticoids to treat other conditions (e.g., to suppress chronic inflammation). Regardless of the source, excess glucocorticoids cause muscle protein to be metabolized and subcutaneous fat to be deposited in the midsection (Fig. 40.15). Excess production of adrenal male sex hormones in females may result in masculinization, including an increase in body hair, deepening of the voice, and beard growth. Depending on the cause, the treatment of Cushing's syndrome may involve a careful reduction in the amount of cortisone being taken, the use of cortisol-inhibiting drugs, or surgery to remove any existing pituitary or adrenal tumor.

Pancreas

The **pancreas** is a slender, fish-shaped organ that stretches across the abdomen behind the stomach and near the duodenum of the small intestine (see Fig. 40.2). Approximately 6 inches long and weighing about 80 grams, the pancreas is composed of two types of tissue: Exocrine tissue produces and secretes digestive juices that go by way of ducts to the small intestine; endocrine tissue, called the **pancreatic islets** (islets of Langerhans), produces and secretes the hormones insulin and glucagon directly into the blood. The Nature of Science feature, "Identifying Insulin as a Chemical Messenger," describes the scientific process surrounding the discovery of insulin.

The pancreas is not under pituitary control. **Insulin** is secreted when there is a high blood glucose level, which usually occurs just after eating (Fig. 40.16, *top*). Insulin stimulates the uptake of glucose by cells, especially liver cells, muscle cells, and adipose tissue cells. In liver and muscle cells, glucose is then stored as glycogen, and in fat cells the breakdown of glucose supplies glycerol for the formation of fat. In these ways, insulin lowers the blood glucose level.

Glucagon is secreted from the pancreas, usually between meals, when blood glucose is low (Fig. 40.16, *bottom*). The major target tissues of glucagon are the liver and adipose tissue. Glucagon stimulates the liver to break down glycogen to glucose and to use fat and protein in preference to glucose as energy sources. Adipose tissue cells break down fat to glycerol and fatty acids. The liver takes these up and uses them as substrates for glucose formation. In these ways, glucagon raises the blood glucose level. Insulin and glucagon are another example of antagonistic hormones, which work together to maintain the blood glucose level.

Diabetes Mellitus

An estimated 37.3 million Americans (11.3% of the population) have diabetes mellitus, often referred to simply as diabetes. Of these, an estimated 8.1 million are undiagnosed. Diabetes is characterized by an inability of the body's cells, especially liver and muscle cells, to take up glucose as they should. This causes blood glucose to be higher than normal, and cells rely on other fuels, such as fatty acids, for energy. Therefore, cellular famine exists in the midst of plenty. As the blood glucose level rises, glucose, along with water, is excreted in the urine (Gr. *mellitus,* "honey, sweetness"). This results in frequent urination and causes the person with diabetes to be extremely thirsty.

Other symptoms of diabetes include fatigue, constant hunger, and weight loss. People with diabetes often experience vision problems due to diabetic retinopathy (see Section 38.2) and swelling in the lens of the eye due to the high blood sugar levels. If untreated, these individuals often develop serious and even fatal complications. Sores that don't heal develop into severe infections. Blood vessel damage causes kidney failure, nerve destruction, heart attack, or stroke.

ISTUDY

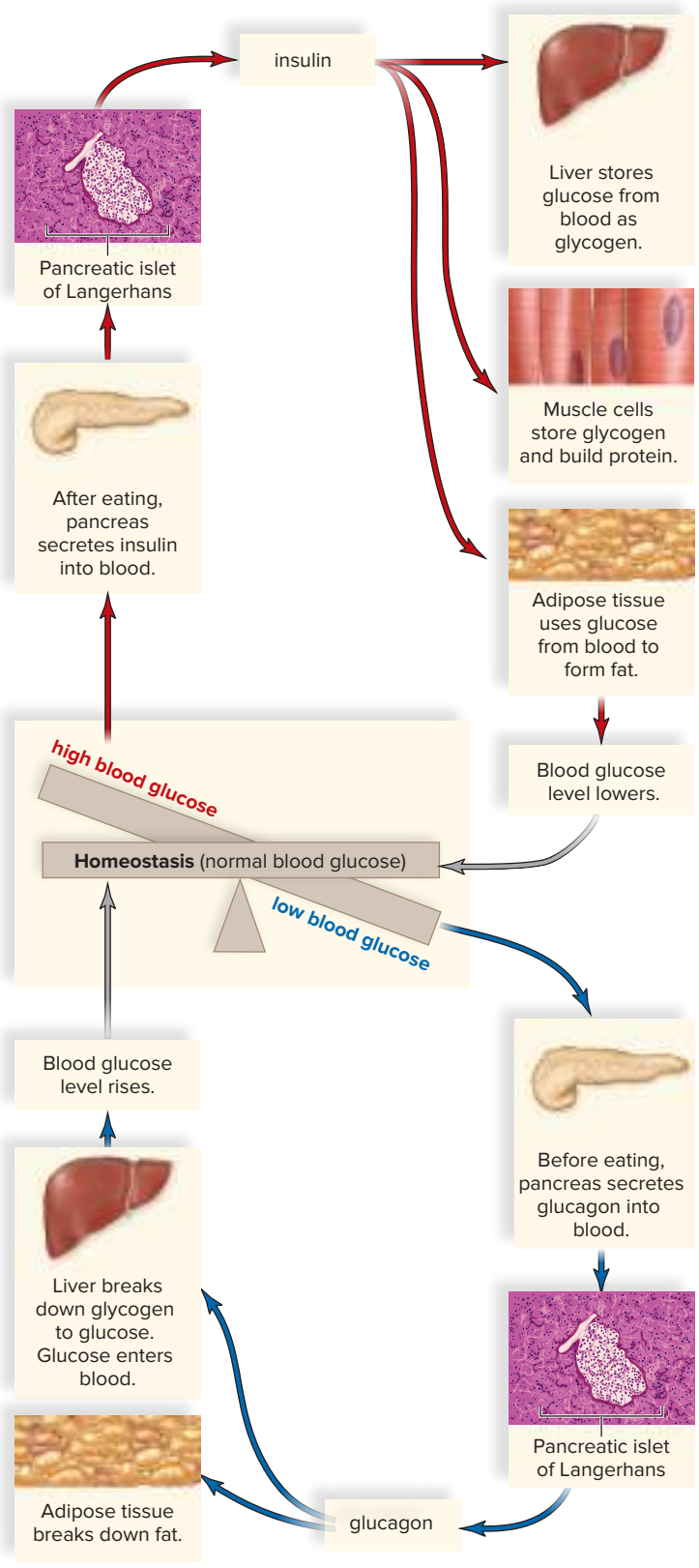


Figure 40.16 Regulation of blood glucose level. *Top:* When the blood glucose level is high, the pancreas secretes insulin. Insulin promotes the storage of glucose as glycogen and the synthesis of proteins and fats (as opposed to their use as energy sources). Therefore, insulin lowers the blood glucose level to normal. *Bottom:* When the blood glucose level is low, the pancreas secretes glucagon. Glucagon acts opposite to insulin; therefore, glucagon raises the blood glucose level to normal.

Two types of diabetes have been identified: type 1 diabetes, sometimes called juvenile diabetes, and type 2 diabetes, or adult-onset diabetes. Despite their names, however, both diseases may occur in either children or adults.

Type 1 Diabetes In *type 1 diabetes,* the pancreas is not producing enough insulin. This condition is believed to be brought on by exposure to an environmental agent, most likely a virus, whose presence causes cytotoxic T cells to destroy the pancreatic islets. The body turns to the metabolism of fat, which leads to the buildup of ketones in the blood, called ketoacidosis, increasing the acidity of the blood and potentially leading to coma and death.

Individuals with type 1 diabetes must have daily insulin injections. These injections control the diabetic symptoms but can cause unpredictable changes, because the blood sugar level may swing between hypoglycemia (low blood glucose) and hyperglycemia (high blood glucose). Without testing the blood glucose level, it is difficult to be certain which of these is present, because the symptoms can be similar. These symptoms include perspiration, pale skin, shallow breathing, and anxiety. Whenever these symptoms appear, immediate attention is required to bring the blood glucose back within the normal range. If the problem is hypoglycemia, the treatment is one or two glucose tablets, hard candy, or orange juice. If the problem is hyperglycemia, the treatment is insulin. Better control of blood glucose levels can often be achieved with an insulin pump, a small device worn outside the body that is connected to a plastic catheter inserted under the skin.

Because diabetes is such a common problem, many researchers are working to develop more effective methods for treating it. The most desirable would be an artificial pancreas, defined as an automated system that would provide insulin based on real-time changes in blood sugar levels. It is possible to transplant a working pancreas, or even fetal pancreatic islet cells, into patients with type 1 diabetes. Another possibility is *xenotransplantation,* in which insulin-producing islet cells of another species, such as pigs, are placed inside a capsule that allows insulin to exit but prevents the immune system from attacking the foreign cells. Finally, researchers are now testing a "reverse vaccine," which, instead of stimulating an immune response, seems to block the immune system's attack on the islet cells, perhaps by inducing T cells capable of suppressing these responses.

Type 2 Diabetes Most adults with diabetes have *type 2 diabetes.* Often, the individual is overweight or obese, and adipose tissue produces a substance that impairs insulin receptor function. However, complex genetic factors can be involved, as shown by the tendency for type 2 diabetes to occur more often in certain families, or even ethnic groups. For example, the condition is 77% more common in African Americans than in non-Hispanic whites.

Normally, the binding of insulin to its plasma membrane receptor causes the number of protein carriers for glucose to increase, causing more glucose to enter the cell. In a person with

ISTUDY

THEME Nature of Science

Identifying Insulin as a Chemical Messenger

The pancreas is both an exocrine gland and an endocrine gland. It sends digestive juices to the duodenum by way of the pancreatic duct, and it secretes the hormones insulin and glucagon into the bloodstream.

In 1920, physician Frederick Banting decided to try to isolate insulin in order to identify it as a chemical messenger. Previous investigators had been unable to do this, because the enzymes in the digestive juices destroyed insulin (a protein) during the isolation procedure. Banting hit upon the idea of ligating (tying off) the pancreatic duct, which he knew from previous research would lead to the degeneration only of the cells that produce digestive juices and not of the pancreatic islets (of Langerhans), where insulin is made. His professor, J. J. Macleod, made a laboratory available to him at the University of Toronto and assigned a graduate student, Charles Best, to assist him.

Figure 40A Early insulin experiments. Charles H. Best and Sir Frederick Banting in 1921 with the first dog to be kept alive by insulin. Bettmann/Getty Images

Banting and Best (Fig. 40A) had limited funds and spent that summer working, sleeping, and eating in the lab. By the end of the summer, they had obtained pancreatic extracts that lowered the blood glucose level in diabetic dogs. Macleod then brought in biochemists, who purified the extract. Insulin therapy for the first human patient began in 1922, and large-scale production of purified insulin from pigs and cattle followed. Banting and Macleod received a Nobel Prize for their work in 1923.

The amino acid sequence of insulin was determined in 1953. Insulin is now synthesized using recombinant DNA technology, using the bacterium *E. coli* to produce the hormone. Banting and Best performed the required steps (given in the accompanying table) to identify a chemical messenger.

Questions to Consider

1. What type of disease or symptoms would you expect to occur after ligating the pancreatic ducts of dogs?
2. What are some advantages, and potential disadvantages, of producing a medicine destined to be injected into humans (such as insulin) in a bacterium such as *E. coli*?
3. Some people oppose the use of animals for medical research. Do you think that insulin would have eventually been discovered without animal experimentation? Why or why not?

Steps to Identify a Chemical Messenger as Used by Banting and Best

Experimental Procedure	Results from Banting and Best Experiment
1. Identify the source of the chemical	Pancreatic islets are source
2. Identify the effect to be studied	Presence of pancreas in body lowers blood glucose
3. Isolate the chemical	Insulin isolated from pancreatic secretions
4. Show that the chemical has the desired effect	Insulin lowers blood glucose

type 2 diabetes, insulin still binds to its receptor, but the number of glucose carriers does not increase. Therefore, the cell is said to be insulin-resistant.

It is possible to prevent or at least control type 2 diabetes by adhering to a low-fat, low-sugar diet and exercising regularly. If this fails, oral drugs are available that stimulate the pancreas to secrete more insulin and enhance the metabolism of glucose in the liver and muscle cells. Millions of Americans may have type 2 diabetes without being aware of it; however, the effects of untreated type 2 diabetes are as serious as those of type 1 diabetes.

Testes and Ovaries

The activity of the testes and ovaries is controlled by the hypothalamus and pituitary. The **testes** are located in the scrotum, and the **ovaries** are located in the pelvic cavity. The testes produce **androgens** (e.g., **testosterone**), the male sex hormones. The ovaries produce **estrogens** and **progesterone,** the female sex hormones. These hormones provide feedback that controls the hypothalamic secretion of gonadotropin-releasing hormone (GnRH). The pituitary gland secretion of follicle-stimulating hormone (FSH) and luteinizing hormone (LH), the gonadotropic hormones, is controlled by feedback from the sex hormones, too. The activities of FSH and LH are discussed in Section 41.3.

Under the influence of the gonadotropic hormones, the testes release an increased amount of testosterone at the time of puberty, which stimulates growth of the penis and testes. Testosterone also brings about and maintains the male secondary sex characteristics that develop during puberty. These include the growth of facial, axillary (underarm), and pubic hair. It

ISTUDY

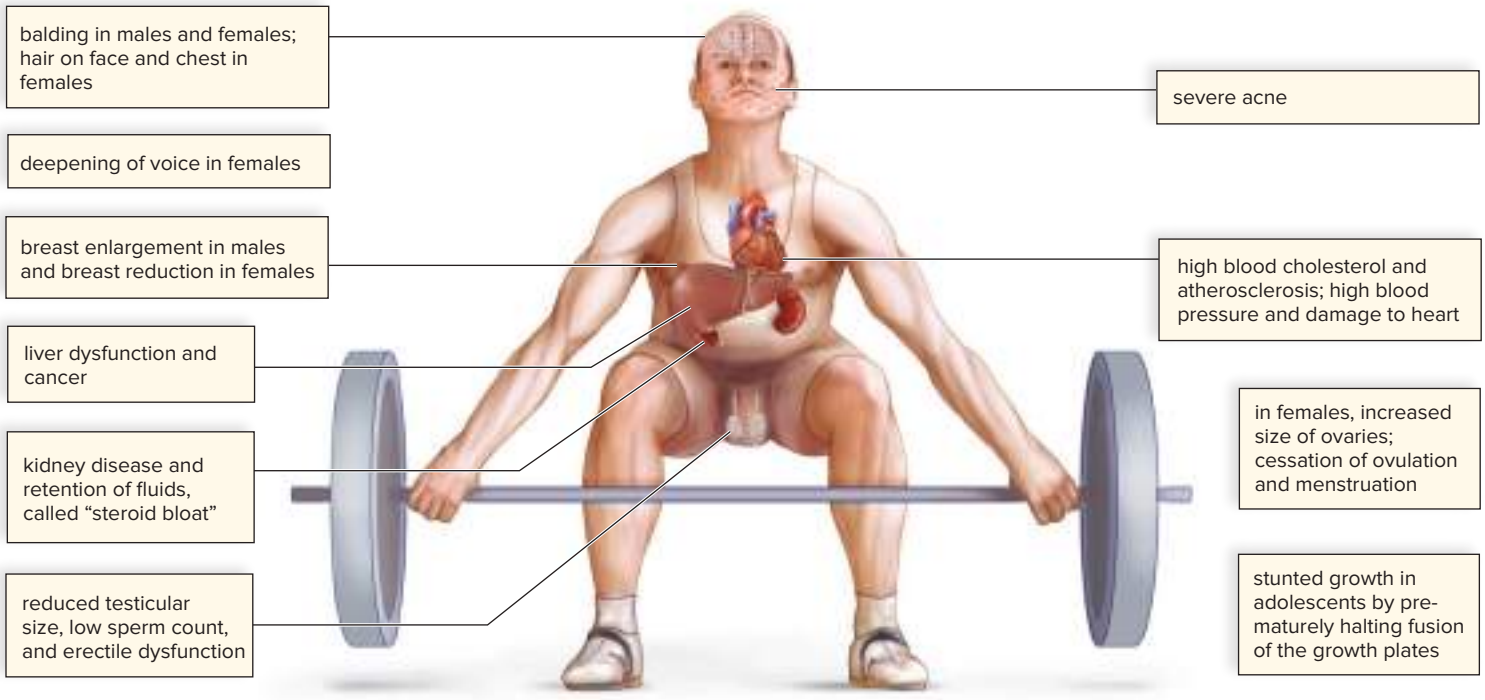


Figure 40.17 The effects of anabolic steroid use. Improper use of anabolic steroids may interfere with a number of the body systems.

prompts the larynx and vocal cords to enlarge, causing the voice to lower. Testosterone also stimulates the activity of oil and sweat glands in the skin. Another side effect of testosterone is baldness. Genes for baldness are inherited by both sexes, but baldness is seen more often in males because of the presence of testosterone.

Testosterone is partially responsible for the muscular strength of males, and this is the reason some athletes take supplemental amounts of **anabolic steroids,** which are either testosterone or related chemicals. The dangerous side effects of taking anabolic steroids are listed in Figure 40.17.

The female sex hormones, estrogens (often referred to in the singular) and progesterone, have many effects on the body. In particular, estrogen secreted at the time of puberty stimulates the growth of the uterus and vagina. Estrogen is necessary for egg maturation and is largely responsible for the secondary sex characteristics in females, including female body hair and fat distribution. In general, females have a more rounded appearance than males because of a greater accumulation of fat beneath the skin. Also, the pelvic girdle is wider in females, resulting in a larger pelvic cavity. Both estrogen and progesterone are required for breast development and regulation of the uterine cycle. This includes monthly menstruation (discharge of blood and mucosal tissues from the uterus). The ovaries and adrenal glands of females also normally produce a small amount of testosterone, which plays a role in the development of muscle and bone strength, overall energy level, sex drive (libido), and sexual pleasure.

Pineal Gland

The **pineal gland,** located deep in the human brain (see Fig. 40.2), produces the hormone **melatonin,** primarily at night. Melatonin is involved in our daily sleep-wake cycle; normally, we grow sleepy at night when melatonin levels increase and awaken once daylight returns and melatonin levels are low (Fig. 40.18). Daily 24-hour cycles such as this are called **circadian rhythms** (L. *circum,* "around"; *dies,* "day"), and circadian rhythms are controlled by an internal timing mechanism called a biological clock.

Instead of being buried deep in the brain, the pineal gland of some vertebrates is on top of the brain, and in certain fossilized reptiles and even some primitive extant reptiles and amphibians, an additional opening in the skull is present, covered only by a thin layer of skin. This, along with the presence of light-sensing cells in the pineal gland, has led some investigators to conclude that this gland functioned as a "third eye" at some point in evolution. The exact functions of this structure are not completely understood, although it may have aided in determining the position of the sun or in establishing circadian rhythms.

Animal research suggests that melatonin also regulates sexual development. In keeping with these findings, it has been noted that children whose pineal glands have been destroyed due to a brain tumor experience early puberty.

Thymus

The lobular **thymus** lies just beneath the sternum (see Fig. 40.2). This organ reaches its largest size and is most active during

ISTUDY

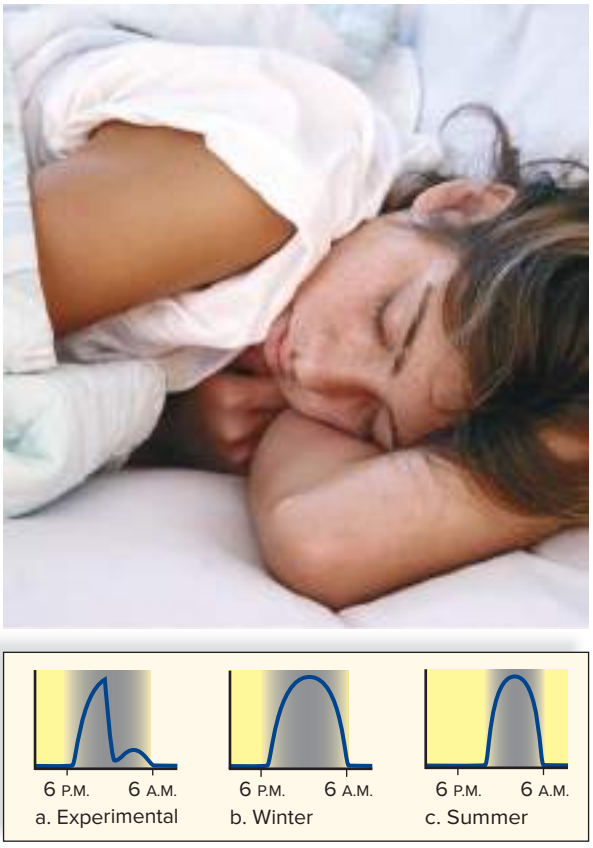


Figure 40.18 Melatonin production. Melatonin production is greatest at night when we are sleeping. Light suppresses melatonin production (**a**), so its duration is longer in the winter (**b**) than in the summer (**c**). (photo): ©Evelyn Jo Johnson

childhood. With aging, the thymus gets smaller and becomes fatty. Lymphocytes that originate in the bone marrow and then pass through the thymus become T lymphocytes. The lobules of the thymus are lined by epithelial cells that secrete hormones called thymosins. These hormones aid in the differentiation of T lymphocytes packed inside the lobules.

Hormones from Other Organs or Tissues

Some organs not usually considered endocrine glands do secrete hormones. Two examples already mentioned are renin excreted by the kidneys and atrial natriuretic hormone excreted by the heart. A number of other tissues produce hormones.

Leptin

Leptin is a peptide hormone produced by adipose tissue throughout the body. Leptin acts on the hypothalamus, where it signals satiety, or fullness. After leptin was discovered in the 1990s, researchers hoped that it could be used to control obesity in humans. Unfortunately, the trials have not yielded satisfactory results. In fact, the blood of individuals with obesity may be rich in leptin. It is possible that the leptin they produce is ineffective because of a genetic mutation or because their hypothalamic cells lack a suitable number of receptors for leptin.

Erythropoietin

As mentioned in Section 36.2, the kidneys secrete **erythropoietin (EPO)** in response to a low blood oxygen level. EPO stimulates the production of red blood cells in the red bone marrow. People with anemia, which is common in kidney disease, cancer, and AIDS, may be effectively treated with injections of recombinant EPO. In recent years, some athletes have practiced blood doping, in which EPO is used to improve performance by increasing the oxygen-carrying capacity of the blood. The potential dangers of blood doping far outweigh the temporary advantages, however. Because EPO increases the number of red blood cells, the blood becomes thicker, blood pressure can become elevated, and the athlete is at increased risk of heart attack or stroke.

Prostaglandins

Prostaglandins are potent chemical signals produced in cells from arachidonate, a fatty acid. Prostaglandins are not distributed in the blood. They act locally, quite close to where they were produced. In the uterus, prostaglandins cause muscles to contract; therefore, they are implicated in the pain and discomfort of menstruation in some females. Also, prostaglandins mediate the effects of *pyrogens,* chemicals believed to reset the temperature regulatory center in the brain. Aspirin reduces body temperature and controls pain because it prevents the synthesis of prostaglandins.

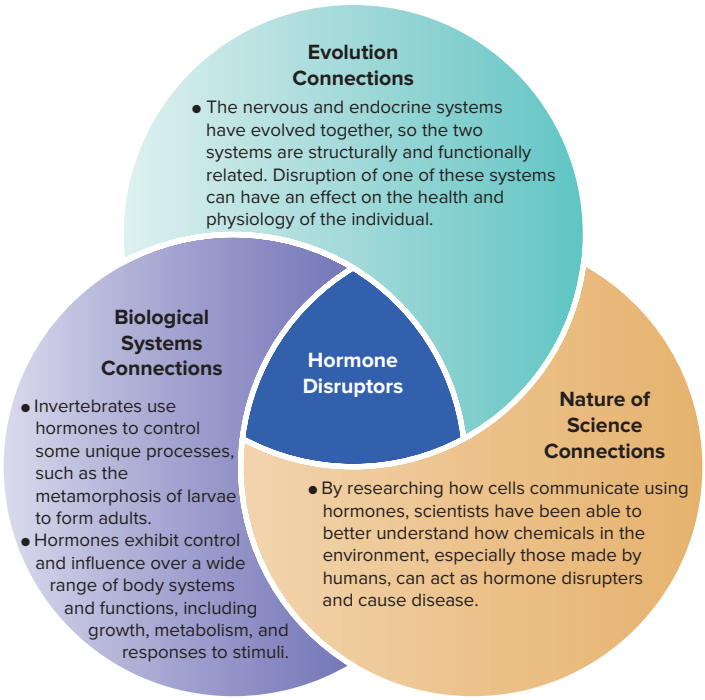
Certain prostaglandins increase the secretion of protective mucus in the stomach and thus are used to prevent gastric ulcers. Others lower blood pressure and have been used to treat hypertension. Still others inhibit platelet aggregation and have been used to prevent thrombosis. Because prostaglandins can affect various tissues, however, unwanted side effects can be a problem. For example, Misoprostol, a prostaglandin commonly used to prevent stomach ulcers, should not be taken by females who are pregnant, as it may cause uterine contractions, resulting in miscarriage or premature labor.

Check Your Progress 40.3

1. Explain how the angiotensin-aldosterone system raises blood pressure.
2. List the endocrine gland that secretes each of the following hormones: aldosterone, melatonin, epinephrine, EPO, leptin, glucagon, ANH, cortisol, and calcitonin.
3. Name one hormone that stimulates the activity of osteoclasts and one that inhibits them.

ISTUDY

Connecting *the* Concepts

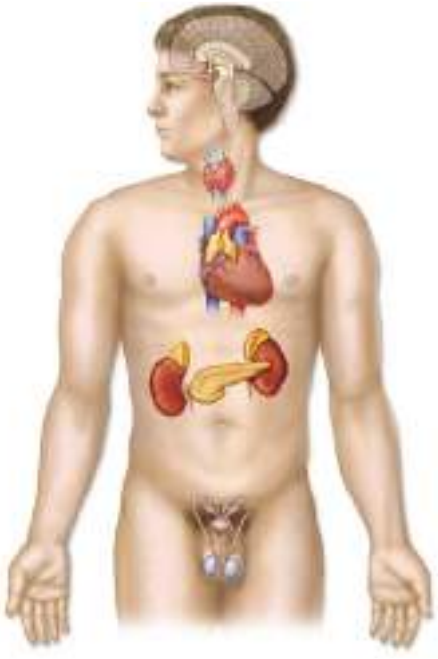


What Did You Learn?

40.1 Animal Hormones

The nervous system and **endocrine system** both use chemical signals to maintain homeostasis.

Endocrine glands secrete **hormones** into the bloodstream, and from there they are distributed to target organs or tissues. In contrast, **exocrine glands** secrete their products via ducts.



Hormones are chemical signals that usually act at a distance between body parts. **Pheromones** are chemical signals that influence the behavior of other individuals. Hormones are either peptides or steroids.

- Reception of **peptide hormones** at the plasma membrane activates an enzyme cascade inside the cell. The hormone is the first messenger**,** and key intermediates that are formed inside the cell, such as **cyclic adenosine monophosphate (cAMP),** are the **second messengers.**
- **Steroid hormones** combine with a receptor inside the cell, and the complex attaches to and activates DNA. Protein synthesis follows.

40.2 Hypothalamus and Pituitary Gland

The **hypothalamus** and **pituitary gland** exert control over the endocrine system.

- Neurosecretory cells in the hypothalamus produce **antidiuretic hormone (ADH)** and **oxytocin,** which are stored in axon endings in the **posterior pituitary** until they are released. As with most hormones, secretion of ADH is regulated by **negative feedback;** the effect of the hormone shuts down its release. In contrast, during childbirth and milk letdown, oxytocin secretion is regulated by **positive feedback;** a stimulus brings about ever-increasing hormone levels.
- The hypothalamus produces **hypothalamic-releasing hormones** and **hypothalamic-inhibiting hormones,** which pass to the anterior pituitary by way of a portal system.
- The **anterior pituitary** produces several types of hormones. Some of these stimulate other endocrine glands: **gonadotropic hormones** influence the gonads; **adrenocorticotropic hormone (ACTH)** stimulates the adrenal glands; and **thyroid-stimulating hormone (TSH)** acts on the thyroid gland.

ISTUDY

- Three anterior pituitary hormones act directly on tissues: **prolactin (PRL)** causes mammary gland development and milk production; **melanocyte-stimulating hormone (MSH)** affects skin color; and **growth hormone (GH)** causes skeletal and muscular growth.

40.3 Other Endocrine Glands and Hormones

- The **thyroid gland,** controlled by TSH, requires iodine to produce **thyroxine** (T_4) and triiodothyronine (T_3), which increase the metabolic rate. Depending on the age of an individual, a deficiency of T_3 and T_4 may result in congenital **hypothyroidism,** simple goiter, or myxedema. **Hyperthyroidism** may be the result of Graves' disease (**exophthalmos** is a common sign) or thyroid cancer. The thyroid gland also produces **calcitonin,** which lowers blood calcium levels by increasing calcium deposition in bones.
- The **parathyroid glands** secrete **parathyroid hormone (PTH),** which raises the blood calcium and decreases the blood phosphate levels.
- The **adrenal glands** respond to stress: Immediately, the **adrenal medulla** secretes **epinephrine** and **norepinephrine,** which bring about responses we associate with emergency situations. On a long-term basis, the **adrenal cortex,** controlled by ACTH, produces the **glucocorticoids** (e.g., cortisol) and the **mineralocorticoids** (e.g., aldosterone). **Cortisol** stimulates hydrolysis of proteins to amino acids that are converted to glucose; in this way, it raises the blood glucose level. **Aldosterone** causes the kidneys to reabsorb sodium ions (Na^+) and to excrete potassium ions (K^+). Addison's disease develops when the adrenal cortex is underactive, and Cushing's syndrome occurs when the adrenal cortex is overactive.
- **Atrial natriuretic hormone (ANH)** is secreted by the heart when its atria are stretched due to high blood pressure. ANH increases renal excretion of sodium ions and water. When blood pressure is too low, the kidneys release renin, resulting in the formation of angiotensin II, which causes arterioles to constrict, and aldosterone to be released.
- The **pancreas** has **pancreatic islets** that secrete **insulin,** which lowers the blood glucose level, and **glucagon,** which has the opposite effect. The most common illness caused by hormonal imbalance is diabetes mellitus, which is due to the failure of the pancreas to produce insulin or the failure of the cells to take it up.
- The testes and ovaries, controlled by gonadotropic hormones, produce the sex hormones. The major male sex hormone **(androgen)** is **testosterone,** and the major female sex hormones are **estrogen** and **progesterone.** Some athletes misuse **anabolic steroids** to increase strength and athletic performance.
- Tissue and organs having other functions also produce hormones. The **pineal gland** produces **melatonin,** which may be involved in **circadian rhythms** and the development of the reproductive organs. The thymus produces hormones that aid in T-lymphocyte development, **leptin** from adipose tissue regulates appetite, and **erythropoietin (EPO)** from the kidneys stimulates the production of red blood cells. **Prostaglandins** are produced and act locally, with a variety of effects on different tissues.

Thinking Critically About Biology

1. Even though some of their functions overlap, why is it advantageous for animals to have both a nervous system and an endocrine system?
2. Caffeine inhibits the breakdown of cAMP in the cell. According to Figure 40.4, how would this affect blood glucose levels?
3. Certain endocrine disorders, such as Cushing's syndrome, can be caused by excessive secretion of a hormone (in this case, ACTH) by the pituitary gland or by a problem with the endocrine gland itself (in this case, the adrenals). If you determined the ACTH levels of a person with Cushing's syndrome, how could you tell the difference between a pituitary problem and a primary adrenal problem?
4. In animals, pheromones can influence many different behaviors. Because humans produce a number of airborne hormones, what human behaviors might be influenced by these? How would these hormones be received by others, and would we necessarily be aware of their effects on us?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Quail Hormones

McGraw Hill

From your understanding of the endocrine system in humans, what endocrine glands and hormones are most likely involved in this response of the quails to the changes in their environment?

Making It Relevant

1. Why do you think pheromones have not been identified in humans? What could explain this from an evolutionary perspective?
2. BPA is a chemical used in the manufacture of plastics that acts as a hormone disruptor in humans. It is suspected in causing fertility problems. Suppose you were to initiate a research study on BPA, what human hormones would you include in that study?
3. Why do you think it is harder to understand the role of hormones than the chemical signals of the nervous system?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

41

Reproductive Systems

MedicalRF.com

Three-Parent Babies

In vitro fertilization (IVF) technology has been used for many years to help couples conceive a child. The process involves taking an egg from the female parent and fertilizing it with the male parent's sperm to create embryos that are then implanted into the female parent (or a surrogate) to carry the child to term. A new technique allows for a donor egg to be used in cases where the female parent's cells contain defective genes in the mitochondria.

Mitochondria are organelles that contain genetic material (DNA) for 37 genes, 14 of which are related to specific proteins related to ATP production. Any defect in these genes leads to serious conditions such as Alpers' disease and various mitochondrial myopathies (muscle disorders).

A new form of in vitro fertilization creates what is sometimes called a "three-parent baby," because a third party donates an egg to be fertilized with the nucleus from the female parent and the sperm from the male parent. The donor's nuclear genes are removed, but the genes inside the mitochondria remain behind. The resulting baby has all of the nuclear genes of the female and male parents, but the 37 genes found in the mitochondria belong to the donor of the original egg cell.

As you read through this chapter, think about the following questions:

1. What is the role of male and female reproductive systems in producing gametes?
2. Which sex normally contributes the mitochondrial genes to the offspring?

Chapter Outline

41.1 How Animals Reproduce
41.2 Human Male Reproductive System
41.3 Human Female Reproductive System
41.4 Control of Human Reproduction
41.5 Sexually Transmitted Diseases

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 10.3 How does meiosis halve the chromosome number to produce haploid cells?

Section 16.2 What influences does sexual selection have on mating patterns in animals?

Section 40.2 What is the relationship between the hypothalamus/pituitary and the reproductive systems of males and females?

Following *the* Themes

CHAPTER 41 – REPRODUCTIVE SYSTEMS

Evolution	Animals have evolved a large number of strategies, both sexual and asexual, for producing offspring.
Nature of Science	Knowledge about human reproduction has led to the development of efficient methods of birth control, solutions for infertility, and better prevention and treatment of STDs.
Biological Systems	Animals reproducing asexually produce identical copies of themselves, whereas animals reproducing sexually produce gametes, and accessory organs for the storage and/or passage of these cells into or out of the body.

ISTUDY

41.1 How Animals Reproduce

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast asexual and sexual reproduction and list several animals that undergo each.
2. Describe some advantages of each of the following life histories: oviparous, ovoviviparous, and viviparous.

For a species to survive, the individuals of that species must reproduce. In **asexual reproduction,** a single parent gives rise to offspring that are identical to the parent, unless mutations have occurred. The adaptive advantage of asexual reproduction is that organisms can reproduce rapidly and colonize favorable environments quickly.

Sexual reproduction involves a shuffling of the genetic material during meiosis (see Section 10.1). In animals, this process forms the sex cells, or **gametes,** which subsequently unite to form genetically unique offspring. The tremendous amount of genetic variation that is created during meiosis can be advantageous to the survival of a species, especially when the environment is changing. The majority of animal species are capable of sexual reproduction, but before we take a look at that process we will explore some of the asexually reproducing animals.

Asexual Reproduction

Several types of invertebrates, such as sponges, cnidarians, flatworms, annelids, and echinoderms, can reproduce asexually. In cnidarians, such as hydras, new individuals may arise asexually as an outgrowth (bud) of the parent (Fig. 41.1). Some species of hydras can also reproduce sexually.

Many flatworms can reproduce asexually by splitting in half, generating two identical individuals. In the laboratory, a planarian can be cut into as many as ten pieces, each of which will grow into a new planarian. To varying degrees, many other animals, including sponges, annelids, and echinoderms, also have the ability to regenerate from fragments. Scientists are studying the ability of some animals to regenerate, with the hope of learning how to regenerate certain human tissues.

Sexual Reproduction

The majority of animals are *dioecious,* which means having separate sexes. Usually during sexual reproduction, the egg of one parent is fertilized by the sperm of another. *Monoecious,* or **hermaphroditic,** animals have both male and female sex organs in a single body. Some hermaphroditic organisms, such as tapeworms, are capable of self-fertilization, but the majority, such as earthworms, practice cross-fertilization with other individuals. Sequential hermaphroditism, or sex reversal, also occurs. In coral reef fishes called wrasses, a male has a harem of several females. If the male dies, the largest female becomes a male.

Animals usually produce gametes in specialized organs called **gonads.** Sponges are an exception to this rule, because the collar cells lining the central cavity of a sponge give rise to sperm and eggs. Hydras and other cnidarians produce only temporary gonads in the fall, when sexual reproduction occurs. Animals in other phyla have permanent reproductive organs.

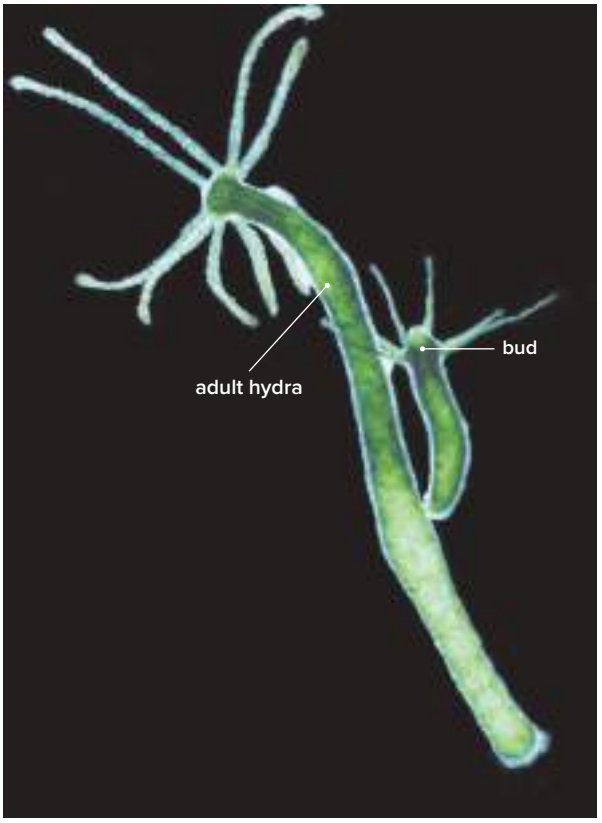


Figure 41.1 Reproduction in hydras. Hydras reproduce asexually and sexually. During asexual reproduction, a new polyp buds from the parental polyp. During sexual reproduction, temporary gonads develop in the body wall. M. I. Walker/NHPA/Photoshot/AVALON

The gonads are **testes,** which produce sperm, and **ovaries,** which produce eggs. Eggs or sperm are derived from **germ cells,** which become specialized for this purpose during early development. Other cells in a gonad support and nourish the developing gametes or produce hormones necessary to the reproductive process. The reproductive system also usually has a number of accessory structures, such as ducts and storage areas, that aid in bringing the gametes together.

Several types of flatworms, roundworms, crustaceans, annelids, insects, fishes, lizards, and even some turkeys have the ability to reproduce parthenogenetically. **Parthenogenesis** is a modification of sexual reproduction in which an unfertilized egg develops into a complete individual. In honeybees, the queen bee makes and stores sperm, which she uses to selectively fertilize eggs. Any unfertilized eggs become haploid males.

Many aquatic animals practice *external fertilization,* in which the gametes are released and unite outside the bodies of the reproducing animals. Among fishes, methods of fertilization and care of the developing embryos vary greatly. Some species simply scatter large numbers of eggs and sperm in the environment, and they may even eat some of their eggs or offspring. Mouthbrooders scoop up their eggs or newly hatched young into their mouths, protecting them from predators. Other fish species build elaborate nests for their eggs, made from plant debris and saliva-coated bubbles.

An invertebrate aquatic species called palolo worms, which inhabit coral reefs in the South Pacific, exhibit a unique fertilization strategy. Always at the same time of year and during a particular phase of the moon, the worms break in half. During a period of just

ISTUDY

Figure 41.2 Damselflies mating on land. On land, animals have various means to ensure that the gametes do not dry out. Here, a female damselfly receives sperm from the male. Herbert Kehrer/The Image Bank/Getty Images

a few hours, hundreds of thousands of tail sections, containing the gonads, swim to the surface, releasing eggs and sperm. Meanwhile, local human populations take to the sea, using nets to scoop up large numbers of the half-worms, which are eaten raw or saved in buckets to be baked or fried and consumed at an annual feast.

Copulation is sexual union to facilitate the reception of sperm, resulting in *internal fertilization.* In terrestrial vertebrates, males typically have a penis for depositing sperm into the vagina of females. But not so in birds, which lack a penis and vagina. They have a cloaca, a chamber that receives products from the digestive, urinary, and reproductive tracts. A male transfers sperm to a female after placing his cloacal opening against hers. In damselflies, the female curls her abdomen forward to receive sperm previously deposited in a pouch by the male, but copulation in the usual sense doesn't occur (Fig. 41.2).

Reproductive Strategies

Any animal that deposits an egg in the external environment is **oviparous.** Many oviparous animals have a larval stage, an immature form capable of feeding. Because the larva has a different lifestyle, it is able to use a different food source than the adult. Other animals, such as crayfish, do not have a larval stage; the egg hatches into a tiny juvenile with the same form as the adult.

Some invertebrates, fish, amphibians, and reptiles are **ovoviviparous,** meaning they retain their eggs in some way and release young that are able to fend for themselves. Notice that while developing, the young are still receiving nourishment from a yolk, as opposed to directly from the parent's body. Examples include certain mollusks, such as oysters, which retain their eggs in the mantle cavity, and male seahorses, which have a brood pouch for development of the fertilized eggs from which live offspring are released.

Most reptiles (e.g., turtles and crocodiles) lay a leathery-shelled egg that contains **extraembryonic membranes** to serve the needs of the embryo and prevent it from drying out (Fig. 41.3*a*). One membrane surrounds an abundant supply of **yolk,** which is a nutrient-rich material. The shelled egg with its internal extraembryonic membranes is a significant adaptation to the terrestrial environment. Birds (Fig. 41.3*b*) lay and care for hard-shelled eggs; the newly hatched birds usually have to be fed before they are able to fly away and seek food for themselves. Complex hormonal and neural regulation is involved in the reproductive behavior of parental birds.

Among mammals, the duckbill platypus and spiny anteater lay shelled eggs. Marsupials and placental mammals do not lay eggs. In marsupials, the yolk sac membrane briefly supplies the unborn embryo with nutrients acquired internally from the female parent. Immature young finish their development in a pouch, where they are nourished on milk.

The placental mammals are termed **viviparous,** because they do not lay eggs and development occurs inside the female's body until offspring can live independently. Their **placenta** is a complex structure derived, in part, from the chorion, another of the reptilian extraembryonic membranes. The evolution of this type of placenta allowed the developing young to internally exchange materials with the female until they are able to function on their own. Viviparity represents the ultimate in caring for the unborn, and in placental mammals, the female continues to supply milk to nourish the offspring after birth. It should be noted that certain fish, amphibians, and reptiles are viviparous, though they lack a placenta.

In the sections that follow, we will take an in-depth look at the reproductive systems of humans and the issues of birth control and health.

a. b.

Figure 41.3 Examples of eggs. **a.** Reptiles, such as these crocodiles, were the first vertebrates to evolve amniotic eggs. **b.** Birds lay hard-shelled eggs and are well-known for incubating their eggs and caring for their offspring after they hatch. (a): Catchlight Lens/Shutterstock; (b): Chokchai Silarug/Moment/Getty Images

ISTUDY

Check Your Progress 41.1

1. List one advantage of asexual reproduction and one advantage of sexual reproduction.
2. Distinguish among oviparous, ovoviviparous, and viviparous.
3. Explain how a shelled egg allows reproduction on land.

41.2 Human Male Reproductive System

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the structures of the human male reproductive system and provide a function for each.
2. Describe the location and stages of spermatogenesis.
3. Summarize how hormones regulate the male reproductive system.

The human male reproductive system includes the organs pictured in Figure 41.4. The primary sex organs, or gonads, of males are the paired **testes** (*sing.,* testis), which are suspended within the sacs of the scrotum.

In this section, we will explore the structure and function of the major organs and structures of the male reproductive system (Table 41.1).

Table 41.1 Male Reproductive System

Organ/Structure	Function
Testes	Produce sperm and sex hormones
Epididymides	Sites of maturation and some storage of sperm
Vasa deferentia	Conduct and store sperm
Seminal vesicles	Contribute fluid to semen
Prostate gland	Contributes fluid to semen
Urethra	Conducts sperm (and urine)
Bulbourethral glands	Contribute fluid to semen
Penis	Involved in copulation

The Testes

The testes produce **sperm** and the male sex hormones. They begin their development inside the abdominal cavity but descend into the scrotal sacs during the last 2 months of fetal development. If the testes fail to descend—and the male does not receive hormone therapy or undergo surgery to place the testes in the scrotum—male infertility may result. The reason is that undescended testes developing in the body cavity are subject to higher body temperatures than testes that have descended into the scrotum. The scrotum helps regulate testicular temperature by holding them closer to or farther away from the body. In fact, any activity that increases testicular temperature, such as taking hot baths, can decrease sperm production. However, despite the frequently held belief that

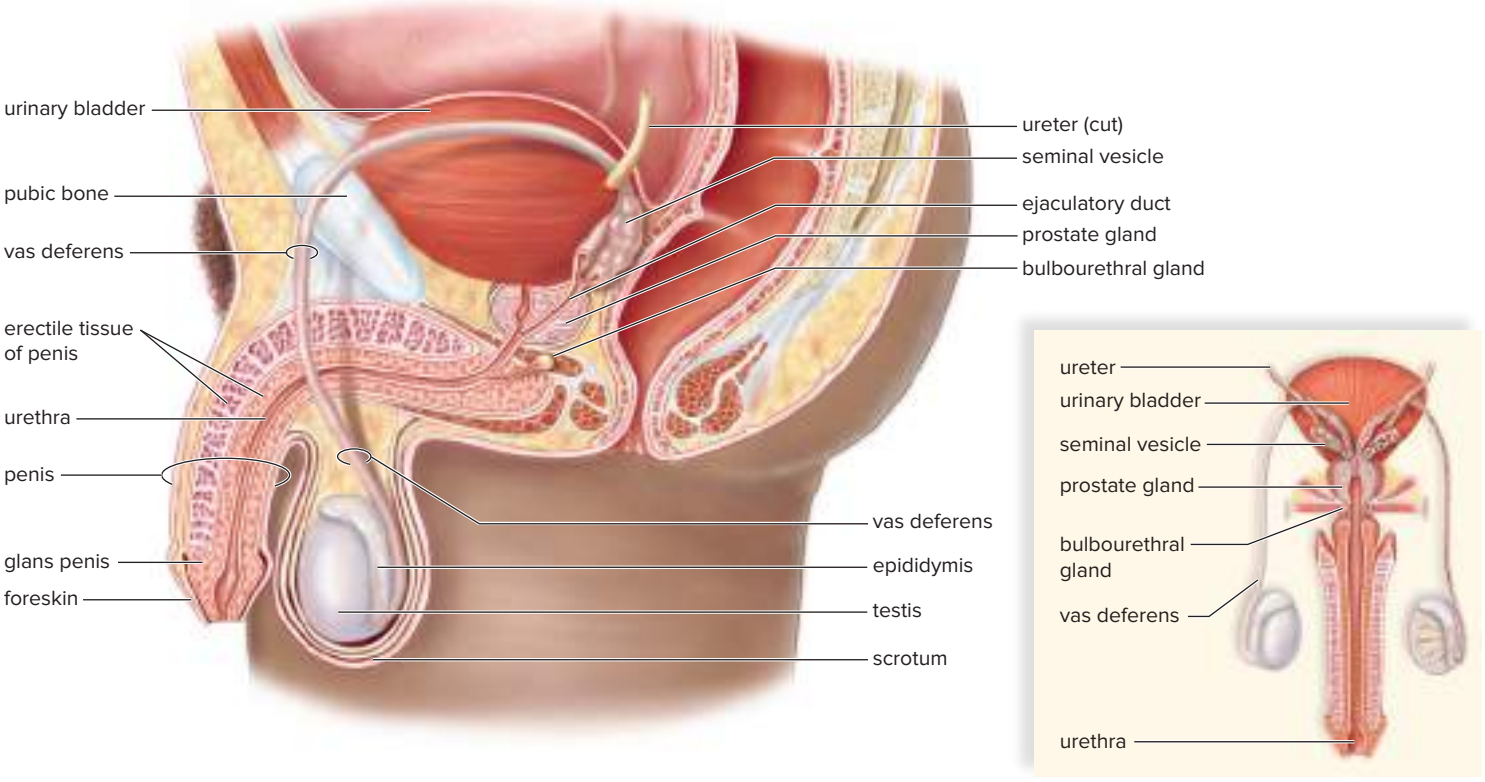


Figure 41.4 The male reproductive system. The testes produce sperm. The seminal vesicles, the prostate gland, and the bulbourethral glands provide a fluid medium for the sperm, which move from the vas deferens through the ejaculatory duct to the urethra in the penis. The foreskin (prepuce) is removed when a penis is circumcised.

ISTUDY

wearing boxer-style underwear favors higher sperm production compared to briefs, this assumption is unsupported by research.

Sperm produced in the testes mature within the **epididymis** (*pl.,* epididymides), a tightly coiled duct just outside each testis. Maturation seems to be required for sperm to swim to the egg. Once the sperm have matured, they enter the **vas deferens** (*pl.,* vasa deferentia), also called the ductus deferens. Sperm may be stored for a time in the vasa deferentia. Each vas deferens passes into the abdominal cavity, where it curves around the bladder and empties into an ejaculatory duct. The vasa deferentia are severed or blocked in a surgical form of birth control called a *vasectomy.* Sperm are then unable to complete their journey through the male reproductive tract; they are phagocytized and destroyed by macrophages.

Testicular cancer is the most common type of cancer in young males between the ages of 15 and 34. The cure rate is high if early treatment is received. All males should perform regular self-exams to detect unusual lumps or swellings of the testes and report any changes to their doctor.

Production of Sperm

A longitudinal section of a testis shows it is composed of compartments called lobules, each of which contains one to three tightly coiled **seminiferous tubules** (Fig. 41.5*a*). Altogether, these tubules have a combined length of approximately 250 meters (over two and a half football fields). This provides a large amount of surface area for sperm development. A microscopic cross section

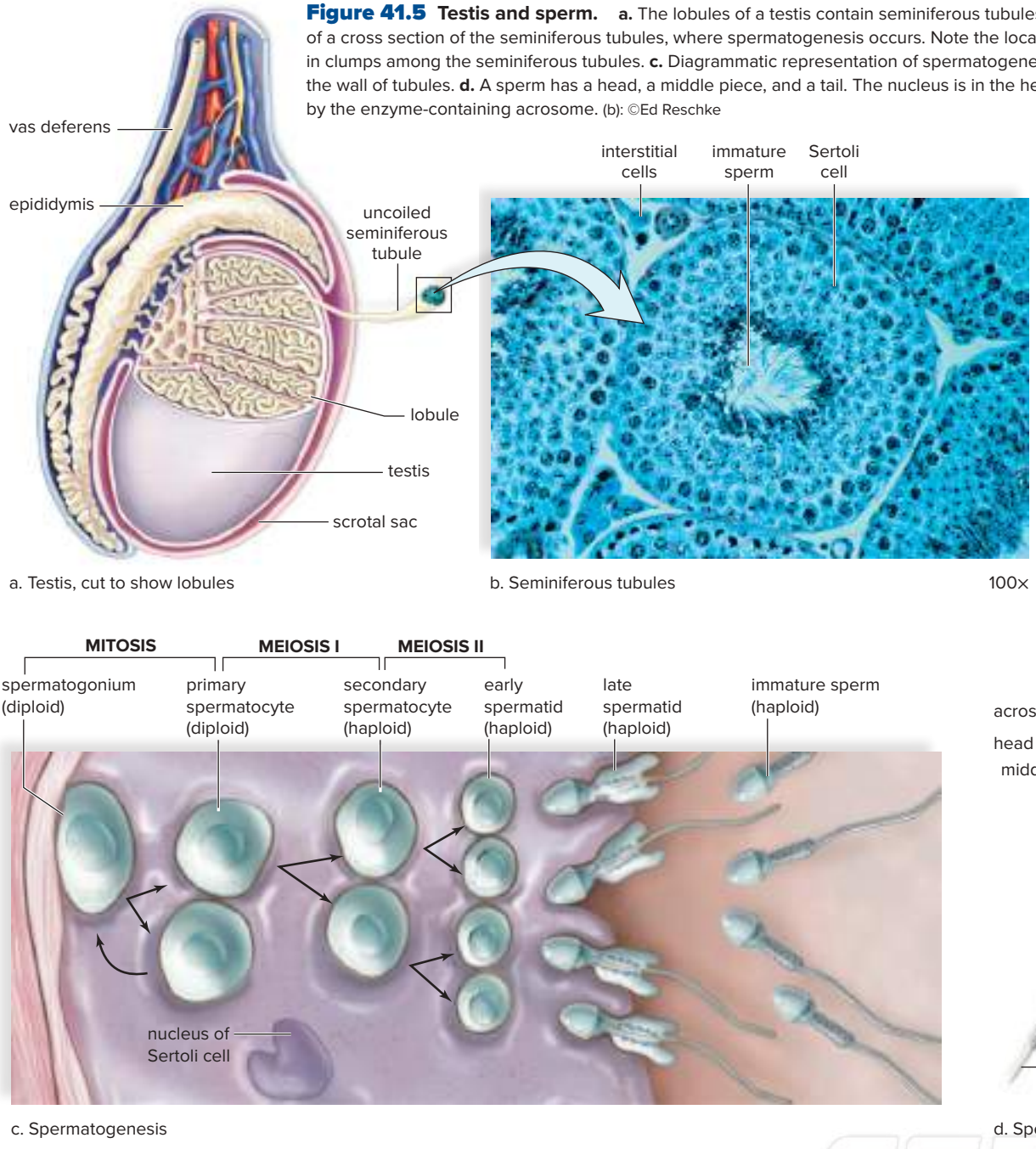


Figure 41.5 Testis and sperm. **a.** The lobules of a testis contain seminiferous tubules. **b.** Light micrograph of a cross section of the seminiferous tubules, where spermatogenesis occurs. Note the location of interstitial cells in clumps among the seminiferous tubules. **c.** Diagrammatic representation of spermatogenesis, which occurs in the wall of tubules. **d.** A sperm has a head, a middle piece, and a tail. The nucleus is in the head, which is capped by the enzyme-containing acrosome. (b): ©Ed Reschke

of a seminiferous tubule shows that it is packed with cells undergoing *spermatogenesis* (Fig. 41.5*b*), the production of sperm.

During spermatogenesis, spermatogonia divide to produce primary spermatocytes (which have the 2n number of chromosomes). Primary spermatocytes move away from the outer wall, increase in size, and undergo meiosis I to produce secondary spermatocytes. Each secondary spermatocyte has only 23 chromosomes (Fig. 41.5*c*). Secondary spermatocytes (n) undergo meiosis II to produce four spermatids, each of which also has 23 chromosomes. Spermatids then differentiate into sperm (Fig. 41.5*d*). Note the presence of *Sertoli cells* (purple in Fig. 41.5*c*), which support and nourish the developing sperm and regulate spermatogenesis. It takes approximately 74 days for a spermatogonium to develop into sperm.

Mature sperm, or *spermatozoa,* have three distinct parts: a head, a middle piece, and a tail (Fig. 41.5*d*). Mitochondria in the middle piece provide energy for the movement of the tail, which is a flagellum. Like other eukaryotic flagella and cilia, microtubules in the sperm tail form the characteristic 9 + 2 pattern (see Fig. 4.21). The head contains a nucleus covered by a cap called the *acrosome* (Gk. *akros,* "at the tip"; *soma,* "body"), which stores enzymes needed to penetrate the thick membrane surrounding the egg.

The Penis and Male Orgasm

The **penis** (Fig. 41.6) is the male organ of sexual intercourse. The penis has a long shaft and an enlarged tip called the glans penis. The glans penis is normally covered by a layer of skin called the *foreskin.* Male *circumcision,* the surgical removal of the foreskin, is performed on more than 50% of newborn males in the United States, often for cultural, religious, or health reasons. Research shows that males who are circumcised are 25–35% less likely to acquire certain STDs, including an HIV infection.

Three cylindrical columns of spongy, erectile tissue containing distensible blood spaces extend through the shaft of the penis. During sexual arousal, autonomic nerves release nitric oxide (NO). This stimulus leads to the production of cGMP (cycline guanosine monophosphate), a type of second messenger similar to cAMP (see Section 40.1). The cGMP causes the smooth muscle of the incoming arterial walls to relax and the erectile tissue to fill with blood. The veins that normally take blood away from the penis are compressed, and the penis becomes erect.

Semen (seminal fluid) is a thick, whitish fluid that contains sperm and secretions from three glands (see Fig. 41.4 and Table 41.1). The paired **seminal vesicles** lie at the base of the bladder, and each has a duct that joins with a vas deferens. As sperm pass from the vasa deferentia into the ejaculatory ducts, the seminal vesicles secrete a thick, viscous fluid containing nutrients (fructose) for possible use by the sperm. The fluid also contains prostaglandins that stimulate smooth muscle contraction along the male and female reproductive tracts.

The **prostate gland** is a single, doughnut-shaped gland that surrounds the upper portion of the urethra just below the bladder. The prostate gland secretes a milky, alkaline fluid believed to activate or increase the motility of sperm. By age 50, over half of all males experience a noncancerous enlargement of the prostate gland, or benign prostatic hypertrophy (BPH). Because of its location, BPH can result in an increased frequency of urination and a weak urinary stream. Also, prostate cancer is the second most common type of cancer among males in the United States. Age, ethnicity, and a family history of prostate cancer are a few of the risk factors associated with this disease.

Bulbourethral glands are pea-sized organs that lie posterior to the prostate on each side of the urethra. They produce a clear, viscous secretion known as *pre-ejaculate.* Secretions from the bulbourethral glands are the first to enter the urethra, where they may cleanse the urethra of acidic residue from urine. Because this fluid can also pick up sperm that remain in the urethra from previous ejaculations, it is possible for sperm to enter the genital tract of a female during sexual intercourse even when ejaculation has not yet occurred.

As sexual stimulation intensifies, *ejaculation* may occur. During the first phase of ejaculation, called emission, the spinal cord sends sympathetic nerve impulses to the epididymides and vasa deferentia. Their muscular walls contract, causing sperm to enter the ejaculatory duct, whereupon the seminal vesicles, prostate gland, and bulbourethral glands release their secretions.

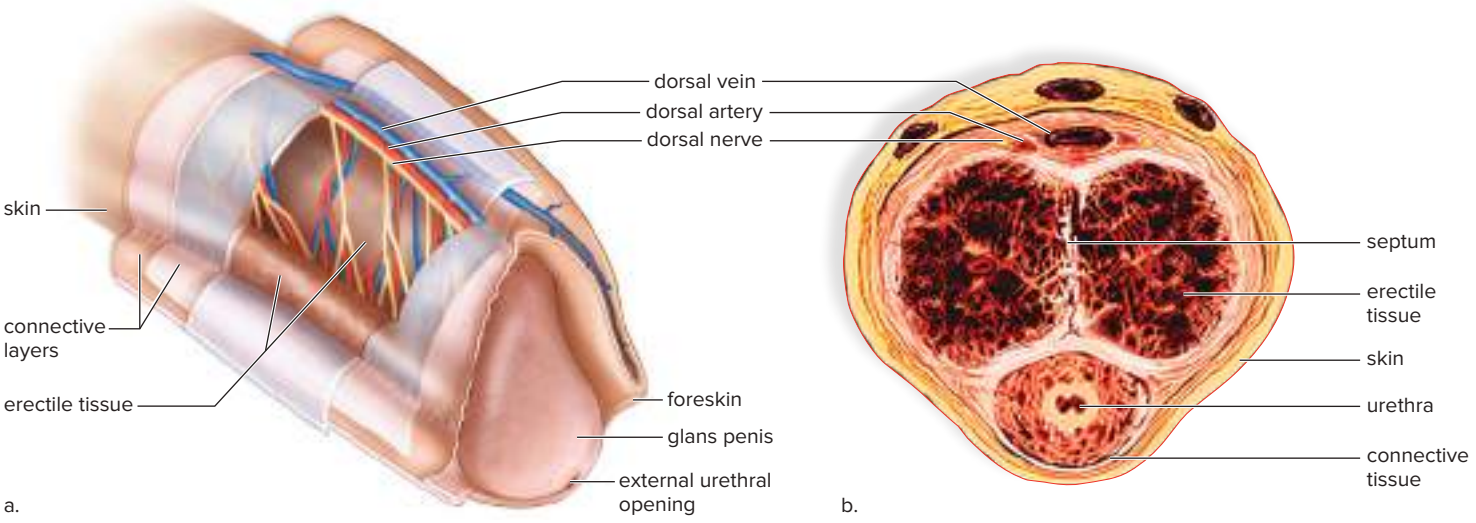


Figure 41.6 Penis anatomy. **a.** Beneath the skin and the connective tissue lies the urethra, surrounded by erectile tissue. This tissue expands to form the glans penis, which is covered by the foreskin (prepuce) in males who are uncircumcised. **b.** Micrograph of shaft in cross section showing the location of erectile tissue. One column surrounds the urethra. (b): Anatomical Travelogue/Science Source

ISTUDY

During the second phase of ejaculation, called expulsion, rhythmical contractions of muscles at the base of the penis and within the urethral wall expel semen in spurts from the opening of the urethra. During ejaculation, the urinary bladder sphincter normally contracts, so that no semen can enter the bladder. (The urethra carries either urine or semen at different times.)

The contractions that expel semen from the penis are a part of male *orgasm,* the physiological and psychological sensations that occur at the climax of sexual stimulation. The physiological reactions of both male and female orgasm involve the genital (reproductive) organs and associated muscles, as well as the entire body. Marked muscular tension is followed by contraction and relaxation. In the male, following ejaculation and/or loss of sexual arousal, the penis returns to its flaccid state. After ejaculating, males typically experience a period of time, called the *refractory period,* during which stimulation does not bring about an erection. The length of the refractory period increases with age.

An average male releases 2–5 ml of semen during orgasm. The ejaculated semen of a typical human male contains between 50 and 150 million sperm per ml, meaning there may be more than 400 million sperm released in one ejaculation. According to the World Health Organization, a low sperm concentration, or *oligozoospermia,* is defined as less than 15 million sperm per ml of ejaculate. Because only one sperm typically enters an egg, however, males with low sperm counts can still be fertile.

Erectile dysfunction (*ED*), formerly called impotency, is the inability to achieve or maintain an erection suitable for sexual intercourse. ED may have a number of causes, including any condition that affects blood flow, certain medications or illicit drugs, and psychological factors. Since 1998, medications have been available for the treatment of ED. All these drugs—such as Viagra, Levitra, and Cialis—work by inhibiting an enzyme called PDE-5, found mainly in the penis, which normally breaks down cGMP. This extends the activity of cGMP, increasing the likelihood of achieving a full erection. Because the PDE-5 enzyme is also found in the retinas and certain other tissues, ED medications can cause vision problems and other undesirable side effects.

Hormonal Regulation in Males

The hypothalamus has ultimate control of the testes' sexual function, because it secretes a hormone called **gonadotropin-releasing hormone (GnRH).** GnRH stimulates the anterior pituitary to produce the **gonadotropic hormones, follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH),** which are present in both males and females.

In males, FSH promotes spermatogenesis in the seminiferous tubules. LH in males controls the production of the androgen testosterone by the interstitial cells (Leydig cells), which are scattered in the spaces between the seminiferous tubules (see Fig. 41.5*b*). All these hormones, including *inhibin,* a hormone released by the seminiferous tubules, are involved in a negative feedback relationship that maintains the fairly constant production of sperm and testosterone (Fig. 41.7).

Testosterone is the main sex hormone in males. It is essential for the normal development and functioning of the organs listed in Table 41.1. Testosterone also brings about and maintains the male *secondary sex characteristics* that develop at the time of puberty. Males are generally

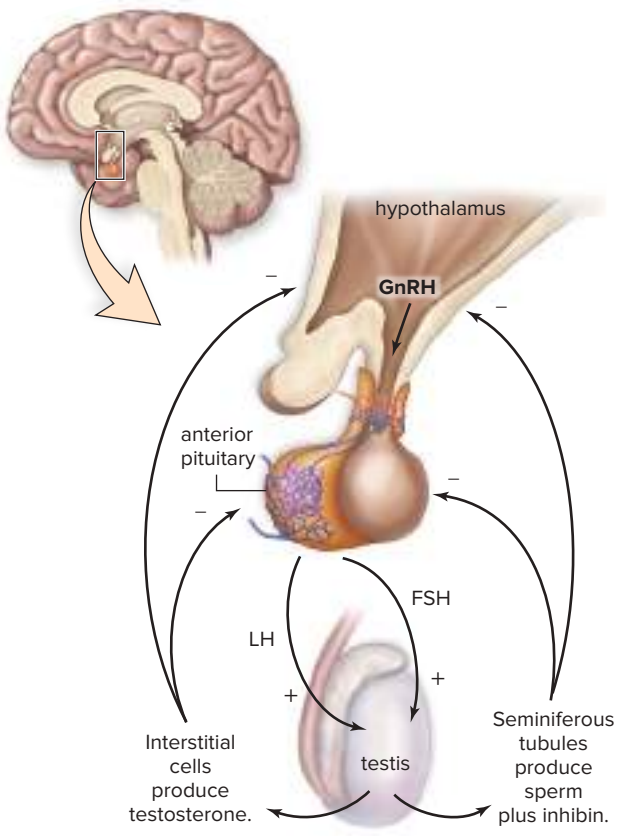


Figure 41.7 Hormonal control of the testes. GnRH (gonadotropin-releasing hormone) stimulates the anterior pituitary to produce FSH and LH. FSH stimulates the testes to produce sperm, and LH stimulates the testes to produce testosterone. Testosterone from interstitial cells and inhibin from the seminiferous tubules exert negative feedback control over the hypothalamus and the anterior pituitary, and this ultimately regulates the level of testosterone in the blood.

taller than females and have broader shoulders and longer legs relative to trunk length. The deeper voice of males compared to females is due to the development in males of a larger larynx with longer vocal cords. The "Adam's apple," which is a protrusion at the front of the larynx, is usually more prominent in males than in females. Testosterone causes males to develop noticeable hair on the face, chest, and occasionally other regions of the body, such as the back. A chemical called dihydrotestosterone (DHT), which is synthesized from testosterone in the hair follicles, leads to the receding hairline and male-pattern baldness in genetically susceptible individuals.

Testosterone is responsible for the greater muscular development in males. Knowing this, both males and females sometimes take anabolic steroids, either testosterone or related steroid hormones resembling testosterone, to build up their muscles. For more information about the risks of anabolic steroids, review Figure 40.17.

Check Your Progress 41.2

1. Trace the pathway a sperm must follow from its origin to its exit from the male reproductive tract.
2. List the glands that contribute fluids to the semen.
3. Describe the effects of FSH and LH in males.

ISTUDY

41.3 Human Female Reproductive System

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the structures of the human female reproductive system and list a function for each.
2. Describe the process of oogenesis.
3. List the stages of the ovarian and uterine cycles and explain what is occurring in each stage.
4. Discuss the changes that occur in the uterus during menstruation, pregnancy, and menopause.

The human female reproductive system includes the organs depicted in Figure 41.8. The female gonads, or primary sex organs, are paired **ovaries** (*sing.,* ovary), located on each side of the upper pelvic cavity.

In this section, we will explore the structure and function of the major organs and structures of the female reproductive system (Table 41.2).

The Ovaries, Uterus, and Vagina

The ovaries produce the female sex hormones and a secondary **oocyte** each month. The uterine tubes, also called fallopian tubes, extend from the ovaries to the uterus; however, these tubes are not attached to the ovaries. Instead, they have fingerlike projections called *fimbriae* (*sing.,* fimbria) that sweep over the ovaries. When an oocyte bursts from an ovary during ovulation, it usually is swept into a uterine tube by the combined action of the fimbriae and the beating of cilia that line the uterine tubes. Fertilization, if it occurs, normally takes place in a uterine tube. The developing embryo is propelled slowly by ciliary movement and tubular muscle contraction to the uterus. Some females opt to permanently prevent pregnancy by having a surgery called *tubal ligation,* in which the uterine tubes are closed off, so that sperm cannot reach the oocyte.

Table 41.2 Female Reproductive System

Organ/Structure	Function
Ovaries	Produce egg and sex hormones
Uterine tubes (fallopian tubes)	Conduct egg to the uterus and are the location of fertilization
Uterus (womb)	Houses developing embryo and fetus
Vagina	Receives penis during copulation and serves as birth canal

Normally, the ovaries alternate in producing one oocyte each month. For the sake of convenience, we will refer to the released oocyte as an **egg.** The ovaries also produce the female sex hormones—the **estrogens,** collectively called estrogen, and **progesterone**—during the ovarian cycle.

The **uterus** is a thick-walled, muscular organ about the size and shape of an inverted pear. The narrow end of the uterus is called the **cervix.** The embryo completes its development after embedding itself in the uterine lining, called the **endometrium** (Gk. *endon,* "within"; *metra,* "womb"). If, by chance, the embryo should embed itself in another location, such as a uterine tube, an *ectopic pregnancy* results.

A small opening in the cervix leads to the vaginal canal. The **vagina** is a tube at a 45-degree angle to the small of the back. The mucosal lining of the vagina lies in folds, and the vagina can be distended. This is especially important when the vagina serves as the birth canal, and it can facilitate intercourse, when the vagina receives the penis during copulation. Several different types of bacteria normally reside in the vagina and create an acidic environment. This environment is protective against the possible growth of pathogenic bacteria, but sperm prefer the basic environment provided by semen.

The external genital organs of a female are known collectively as the *vulva* (Fig. 41.9). The *mons pubis* and two folds of skin called *labia minora* and *labia majora* are on each side of the urethral and vaginal openings. Beneath the labia majora, pea-sized greater *vestibular glands* (Bartholin glands) open on each side of the vagina. They keep the vulva moist and lubricated during intercourse.

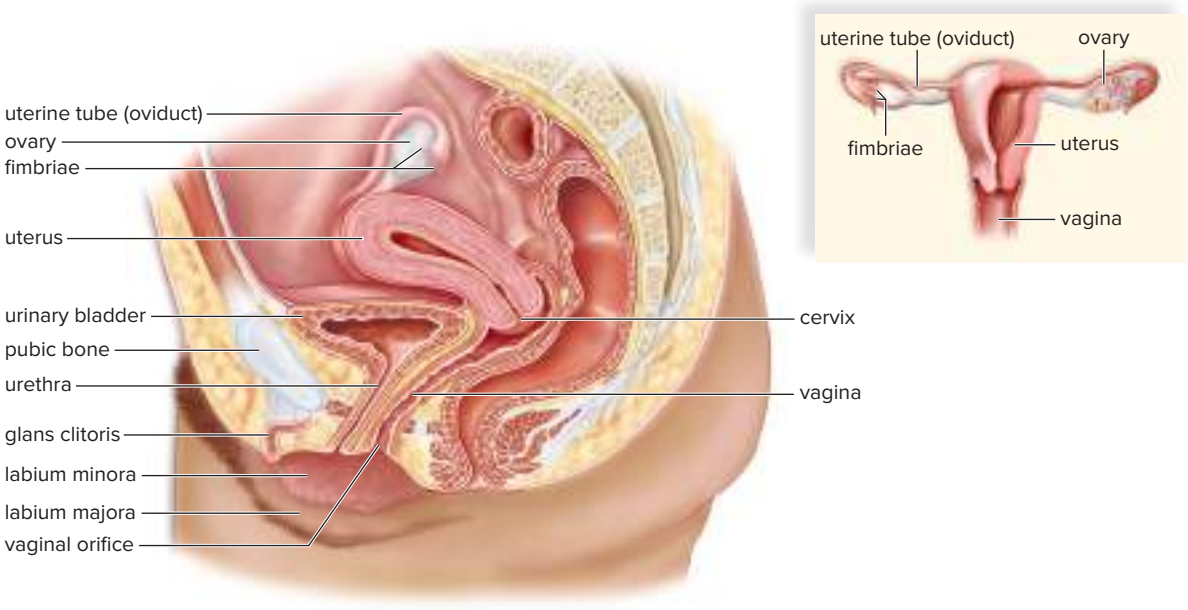


Figure 41.8 Female reproductive system. The ovaries produce one oocyte per month. Fertilization occurs in the uterine tube, and development occurs in the uterus. The vagina is the birth canal and organ of sexual intercourse.

ISTUDY

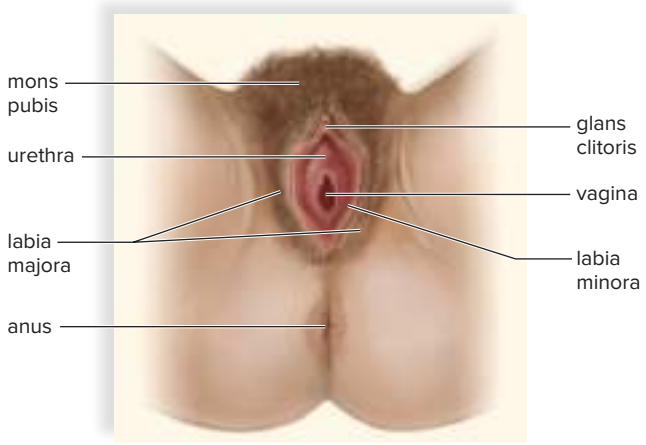


Figure 41.9 External genitals of a female. The external genitals of the female include the labia majora, labia minora, and the glans clitoris.

At the juncture of the labia minora is the *clitoris,* which is homologous to the penis in males. The clitoris has a shaft of erectile tissue and is capped by a pea-shaped glans. The many sensory receptors of the clitoris allow it to function as a sexually sensitive organ. The clitoris has twice as many nerve endings as the penis. As in males, female orgasm involves both psychological and physiological reactions. In the female, there is also a release of neuromuscular tension in the muscles of the genital area, vagina, and uterus.

The Female Orgasm

Upon sexual stimulation, the labia minora, the vaginal wall, and the clitoris become engorged with blood. The breasts also swell, and the nipples become erect. The labia majora enlarge, redden, and spread away from the vaginal opening.

The vagina expands and elongates. Blood vessels in the vaginal wall release small droplets of fluid that seep into the vagina and lubricate it. Mucus-secreting glands beneath the labia minora on either side of the vagina also provide lubrication for entry of the penis into the vagina. Although the vagina is the organ of sexual intercourse in females, the extremely sensitive clitoris plays a significant role in the female sexual response.

Orgasm occurs at the height of the sexual response. Blood pressure and pulse rate rise, breathing quickens, and the walls of the vagina, uterus, and uterine tubes contract rhythmically. A sensation of intense pleasure is followed by relaxation when organs return to their normal size. Females have little or no refractory period between orgasms, and multiple orgasms can occur during a single sexual experience.

The Ovarian Cycle

The **ovarian cycle** occurs as a **follicle** (L. *folliculus,* "little bag") changes from a primary to a secondary and finally to a vesicular (Graafian) follicle (Fig. 41.10*a*). Epithelial cells of a primary

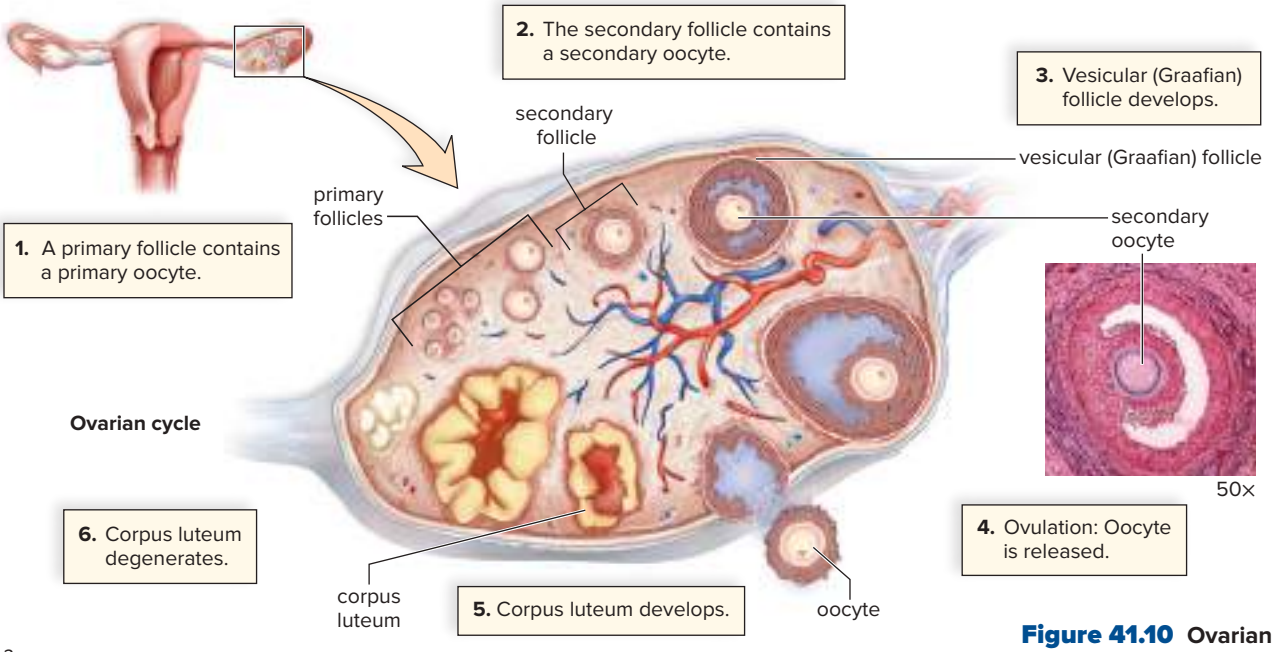


Figure 41.10 Ovarian cycle. **a.** A single follicle actually goes through all stages (1–6) in one place within the ovary. As a follicle matures, layers of follicle cells surround a secondary oocyte. Eventually, the mature follicle ruptures, and the secondary oocyte is released. The follicle then becomes the corpus luteum, which eventually disintegrates. **b.** During oogenesis, the chromosome number is reduced from 46 to 23. Fertilization restores the full number of chromosomes. (photo): Ed Reschke/Stone/Getty Images

ISTUDY

follicle surround a primary oocyte. Pools of follicular fluid surround the oocyte in a secondary follicle. In a vesicular follicle, a fluid-filled cavity increases to the point that the follicle wall balloons out on the surface of the ovary.

Oogenesis and Ovulation

As a follicle matures, *oogenesis,* a form of meiosis depicted in Figure 41.10*b,* is initiated and continues. The primary oocyte divides, producing two haploid cells. One cell is a secondary oocyte, and the other is a polar body. The vesicular follicle bursts, releasing the secondary oocyte (egg) surrounded by a clear membrane. This process is referred to as **ovulation.** Once a vesicular follicle has lost the secondary oocyte, it develops into a **corpus luteum** (L. *corpus,* "body"; *luteus,* "yellow"), a glandlike structure.

The secondary oocyte enters a uterine tube. If fertilization occurs, a sperm enters the secondary oocyte; then, the oocyte completes meiosis and becomes an egg. An egg with 23 chromosomes and a second polar body results. When the sperm nucleus unites with the egg nucleus, a zygote with 46 chromosomes is produced. If zygote formation and pregnancy do not occur, the corpus luteum begins to degenerate after about 10 days.

Control and Phases of the Ovarian Cycle

This cycle is regulated by both follicle-stimulating hormone (FSH) and luteinizing hormone (LH) from the anterior pituitary (Fig. 41.11).

The gonadotropic hormones FSH and LH are not present in constant amounts and instead are secreted at different rates during the cycle (Fig. 41.12). For simplicity's sake, it is convenient to emphasize that during the first half, or **follicular phase,** of the cycle, FSH promotes the development of a follicle that primarily

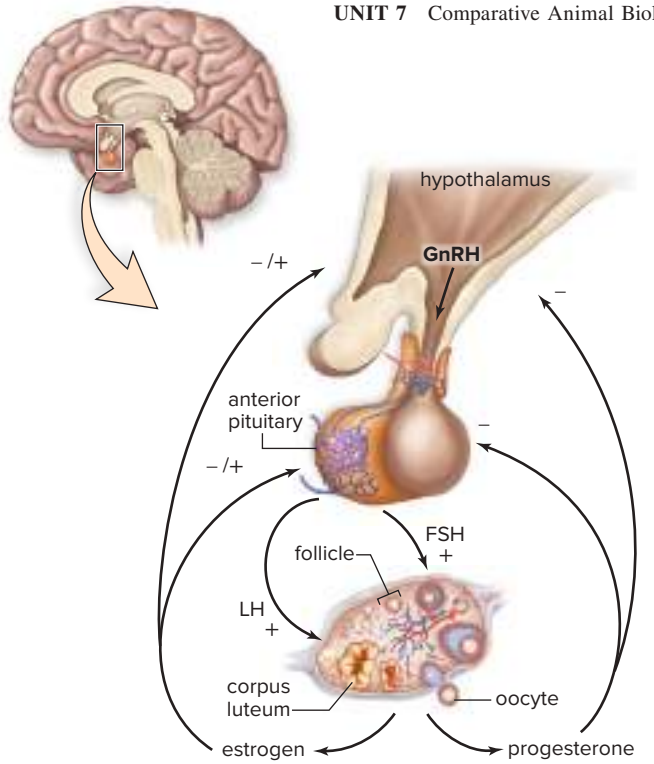


Figure 41.11 Hormonal control of ovaries. The hypothalamus produces GnRH (gonadotropic-releasing hormone). GnRH stimulates the anterior pituitary to produce FSH (follicle-stimulating hormone) and LH (luteinizing hormone). FSH stimulates the follicle to produce primarily estrogen, and LH stimulates the corpus luteum to produce primarily progesterone. Estrogen and progesterone maintain the sex organs (e.g., uterus) and the secondary sex characteristics, and they exert feedback control over the hypothalamus and the anterior pituitary. Feedback control regulates the relative amounts of estrogen and progesterone in the blood.

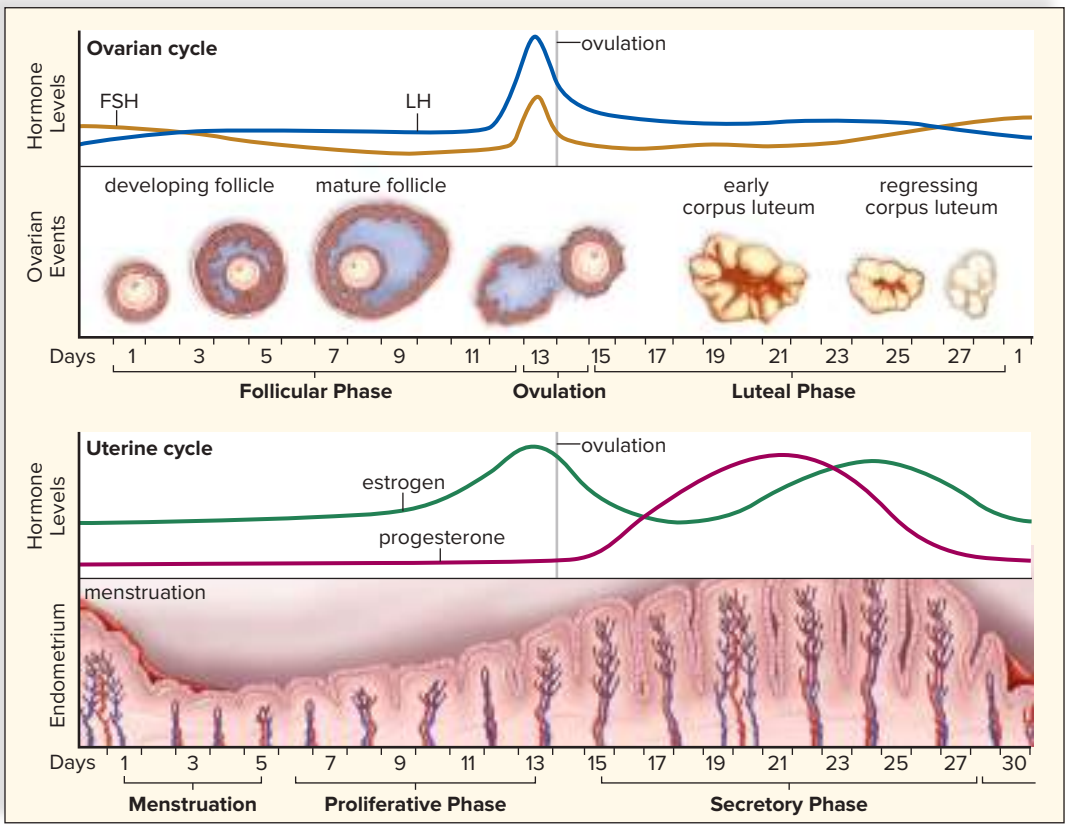


Figure 41.12 Female hormone levels during the ovarian and uterine cycles. During the follicular phase of the ovarian cycle (*top*), FSH released by the anterior pituitary promotes the maturation of a follicle in the ovary. The ovarian follicle produces increasing levels of estrogen, which causes the endometrium to thicken during the proliferative phase of the uterine cycle (*bottom*). After ovulation and during the luteal phase of the ovarian cycle, LH promotes the development of the corpus luteum. This structure produces increasing levels of progesterone, which causes the endometrium to become secretory. Menstruation and the proliferative phase begin when progesterone production declines to a low level.

ISTUDY

secretes estrogen. As the estrogen level in the blood rises, it exerts negative feedback control over the anterior pituitary secretion of FSH, so that the follicular phase comes to an end. At the same time that FSH along with LH release is being dampened by moderate amounts of estrogen, the synthesis of the gonadotropic hormones continues, and they build up in the anterior pituitary.

When the level of estrogen in the blood becomes very high, it exerts positive feedback on the hypothalamus and anterior pituitary. The hypothalamus is stimulated to suddenly secrete a large amount of GnRH. This leads to a surge of LH (and to a lesser degree, FSH) by the anterior pituitary and to ovulation at about the fourteenth day of a 28-day cycle (Fig. 41.12, *top*).

During the second half, or **luteal phase,** of the ovarian cycle, LH promotes the development of the corpus luteum, which primarily secretes progesterone. As the blood level of progesterone rises, it exerts negative feedback control over anterior pituitary secretion of LH, so that the corpus luteum begins to degenerate. As the luteal phase comes to an end, the low levels of progesterone and estrogen in the body cause menstruation to begin, as discussed next.

The Uterine Cycle

The female sex hormones produced in the ovarian cycle (estrogen and progesterone) affect the endometrium of the uterus, causing the cyclical series of events known as the **uterine cycle** (Fig. 41.12, *bottom*). The 28-day cycle is divided into the three phases, described next.

Phases of the Uterine Cycle

During *days 1–5,* there is a low level of female sex hormones in the body, causing the endometrium to disintegrate and its blood vessels to rupture. A flow of blood passes out of the vagina during **menstruation** (L. *menstrualis,* "happening monthly"), also known as the menstrual period. This is the *menstrual phase* of the uterine cycle.

During *days 6–13,* increased production of estrogen by an ovarian follicle causes the endometrium to thicken and to become vascular and glandular. This is called the *proliferative phase* of the uterine cycle.

Ovulation usually occurs on *day 14* of the 28-day cycle. During *days 15–28,* increased production of progesterone by the corpus luteum causes the endometrium to double in thickness and the uterine glands to mature, producing a thick, mucoid secretion. This is called the *secretory phase* of the uterine cycle. The endometrium now is prepared to receive the developing embryo. If pregnancy does not occur, the corpus luteum degenerates, and the low level of sex hormones in the female body causes the endometrium to break down as menstruation occurs. Table 41.3 compares the stages of the uterine cycle with those of the ovarian cycle.

Menstruation

Seven to ten days before the start of menstruation, some females experience premenstrual syndrome (PMS). During this time, a female may exhibit breast enlargement and tenderness, achiness, headache, and irritability. The exact cause for PMS has yet to be discovered.

During menstruation, arteries that supply the lining constrict and the capillaries weaken. Blood spilling from the damaged vessels detaches layers of the lining, not all at once but in random patches. Mucus, blood, and degenerating endometrium descend from the uterus, through the vagina, creating menstrual flow. *Fibrinolysin,* an enzyme released by dying cells, prevents the blood from clotting. Menstruation lasts from 3 to 5 days, as the uterus sloughs off the thick lining that was 3 weeks in the making.

The first menstrual period, called *menarche,* typically occurs between the ages of 11 and 13. Menarche signifies that the ovarian and uterine cycles have begun. If menarche does not occur by age 16, or if normal uterine cycles are interrupted for 6 months or more without pregnancy being the cause, the condition is termed amenorrhea. Primary amenorrhea is usually caused by nonfunctional ovaries or developmental abnormalities. Secondary amenorrhea may be caused by weight loss and/or excessive exercise.

Menopause, which usually occurs between ages 45 and 55, is the time in a female's life when menstruation ceases because the ovaries are no longer functioning. Menopause is not complete until menstruation is absent for a year.

Fertilization and Pregnancy

If fertilization does occur, an embryo begins development even as it travels down the uterine tube to the uterus. The endometrium is now prepared to receive the developing embryo, which becomes embedded in the lining several days following fertilization.

The placenta originates from both the female parent's and embryonic tissues. It is shaped like a large, thick pancake and is the site of the exchange of gases and nutrients between fetal and the female parent's blood, although the two rarely mix. At first, the placenta produces **human chorionic gonadotropin (HCG),** which maintains the corpus luteum until the placenta begins its own production of progesterone and estrogen. HCG is the

Table 41.3 Ovarian and Uterine Cycles (Simplified)

Ovarian Cycle	Events	Uterine Cycle	Events
Follicular phase (Days 1–13)	FSH	Menstrual phase (Days 1–5))	Endometrium breaks down
	Follicle maturation	Proliferative phase (Days 6–13	Endometrium rebuilds
	Estrogen		
Ovulation (Day 14*)	LH spike		
Luteal phase (Days 15–28)	LH	Secretory phase (Days 15–28)	Endometrium thickens and glands are secretory
	Corpus luteum		
	Progesterone		

*Assuming a 28-day cycle

ISTUDY

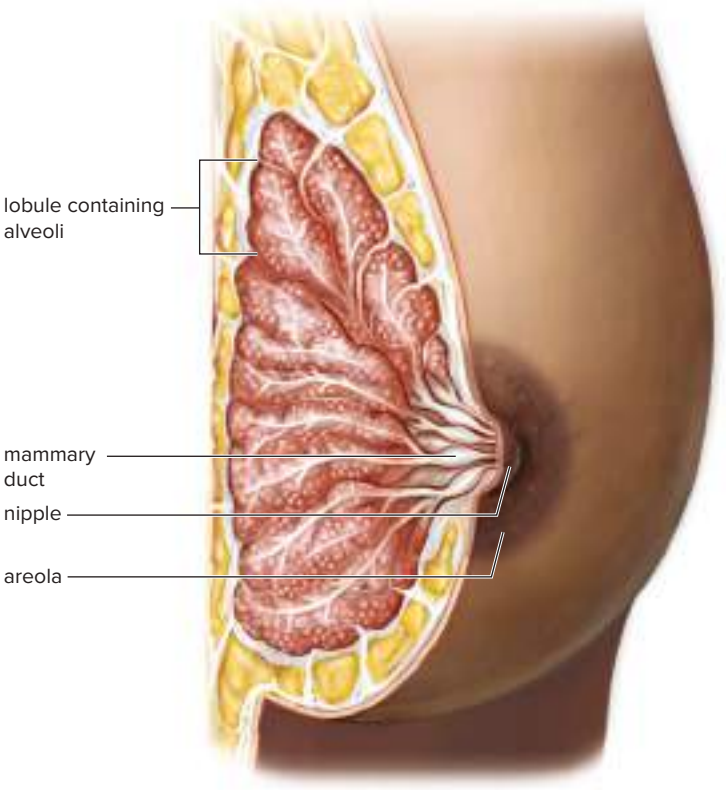


Figure 41.13 Anatomy of the breast. The female breast contains lobules consisting of ducts and alveoli. In the lactating breast, cells lining the alveoli have been stimulated to produce milk by the hormone prolactin.

hormone detected in pregnancy tests; it is present in the female parent's blood and urine as early as 10 days after conception.

Progesterone and estrogen have two effects. They shut down the anterior pituitary, so no new follicles mature, and they maintain the lining of the uterus, so the corpus luteum is not needed. No menstruation occurs during pregnancy.

Female Secondary Sex Characteristics

Estrogen, in particular, is essential for the normal development and functioning of the female reproductive organs listed in Table 41.2. Estrogen is also largely responsible for the secondary sex characteristics in females, including body hair and fat distribution. In general, females have a more rounded appearance than males because of a greater accumulation of fat beneath their skin. Also, the pelvic girdle enlarges, so that females have wider hips than males, and the thighs converge at a greater angle toward the knees. Both estrogen and progesterone are required for breast development.

The Female Breast

A female breast contains between 15 and 24 lobules, each with its own mammary duct (Fig. 41.13). A duct begins at the nipple and divides into numerous other ducts that end in blind sacs called alveoli.

Lactation, the production of milk by the cells of the alveoli, is stimulated by the hormone prolactin. Milk is not produced during pregnancy, because the production of prolactin is suppressed by the feedback inhibition effect of estrogen and progesterone on the anterior pituitary. A couple of days after delivery of a baby, milk production begins. In the meantime, the breasts produce a watery, yellowish-white fluid called colostrum, which is similar in composition to milk but contains more protein and less fat. Colostrum is rich in antibodies, which may provide some degree of immunity to the newborn. Milk contains water, proteins, amino acids, sugars, and lysozymes (enzymes with antibiotic properties). Human milk contains about 750 calories per liter, which is considerably higher than even whole cow's milk (about 500 calories per liter).

Breast cancer is the second most common type of cancer among females in the United States; skin cancer is the most common. Based on current rates, about 12.5% of females born today can expect to be diagnosed with cancer of the breast during their lifetime. Females should regularly check their breasts for lumps and other irregularities and have mammograms as recommended by their physician. Although breast cancer genes have been described, most forms of breast cancer are nonhereditary. Breast cancer treatment options have expanded in recent years, resulting in increased survival rates. Currently, depending on the stage of the cancer, about 91% of females survive at least 5 years after their initial diagnosis.

Besides more traditional surgery, radiation, and chemotherapy, two newer types of breast cancer treatments are hormonal therapy and targeted therapy. Some breast cancers have receptors for estrogen, and they grow in response to increased estrogen levels. To treat these tumors, various drugs are available to either decrease estrogen synthesis or block the interaction of estrogen with its receptor. Targeted therapies usually involve the administration of a monoclonal antibody (see Section 33.4) that can react very specifically with antigens on tumor cells. The best-known example, called Herceptin, blocks a protein called HER2, expressed on 20–30% of breast cancer cells, where it is involved in stimulating uncontrolled cell growth.

Check Your Progress 41.3

1. Name the structures of the female reproductive system that (a) produce the egg, (b) transport the egg, (c) house a developing embryo, and (d) serve as the birth canal.
2. Explain the effects of FSH and LH on the ovarian cycle.
3. Summarize the roles of estrogen and progesterone in the ovarian and uterine cycles.

41.4 Control of Human Reproduction

Learning Outcomes

Upon completion of this section, you should be able to

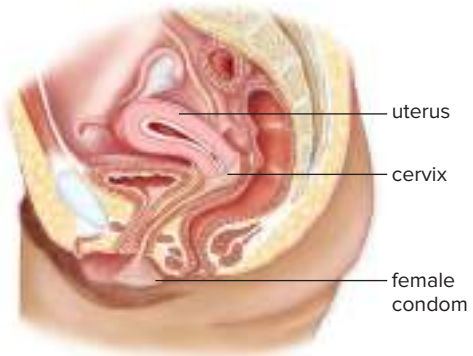
1. Recognize the commonly available methods of birth control that are designed to prevent conception.
2. Describe which birth control methods are effective at preventing the spread of sexually transmitted diseases.
3. Describe several reproductive technologies that can help couples experiencing infertility have children.

Several means are available to reduce or enhance the human reproductive potential. Contraceptives are medications and devices that reduce the chance of pregnancy.

ISTUDY

a. Intrauterine device placement

Intrauterine devices



b. Female condom placement

Female condom

c. Male condom placement

Male condom

Figure 41.14 Various birth control methods. **a.** Intrauterine devices mechanically prevent implantation and can contain progesterone to prevent ovulation, prevent implantation, and thicken cervical mucus. **b.** A female condom fitted inside the vagina prevents sperm entry and protects against STDs. **c.** A male condom fits over the penis, preventing sperm from entering the vagina and protecting against STDs. (a): (photo): Saturn Stills/Science Source; (b): (photo): Keith Brofsky/Photodisc/Getty Images; (c): (photo): Lars A. Niki

Traditional Birth Control Methods

The most reliable method of birth control is abstinence—that is, not engaging in sexual intercourse. This form of birth control has the added advantage of preventing the transmission of sexually transmitted diseases. Male and female condoms also offer some protection against sexually transmitted diseases in addition to helping prevent pregnancy. Some of the common means of birth control used in the United States are shown in Figure 41.14.

Contraceptive Injections and Vaccines

Contraceptive vaccines are in development. For example, a vaccine intended to immunize females against HCG, the hormone so necessary to maintaining the implantation of the embryo, was successful in a limited clinical trial. Because HCG is not normally present in the body, no autoimmune reaction is expected, but the immunization does wear off with time. Others believe it would also be possible to develop a safe antisperm vaccine that could be used in females.

Contraceptive implants use a synthetic progesterone to prevent ovulation by disrupting the ovarian cycle. Most versions consist of a single capsule that remains effective for about 3 years. Contraceptive injections are available as progesterone only or a combination of estrogen and progesterone. The length of time between injections can vary from 1 to several months.

Emergency Contraception

Emergency contraception, or "morning-after pills," consists of medications that can prevent pregnancy after unprotected intercourse. The expression "morning-after" is a misnomer, in that some treatments can be started up to 5 days after unprotected intercourse.

The first FDA-approved medication produced for emergency contraception was a kit called Preven. Preven includes four synthetic progesterone pills; two are taken up to 72 hours after unprotected intercourse, and two more are taken 12 hours later. The hormone upsets the normal uterine cycle, making it difficult for an embryo to implant in the endometrium. One study estimated that Preven was 85% effective in preventing unintended pregnancies. The Preven kit also includes a pregnancy test; females are instructed to take the test first before using the hormone, because the medication is not effective on an established pregnancy.

In 2006, the FDA approved another drug, called Plan B One-Step, which is up to 89% effective in preventing pregnancy if taken within 72 hours after unprotected sex. It is available without a prescription to females age 17 and older. In August 2010, ulipristal acetate (also known as ella) was also approved for emergency contraception. It can be taken up to 5 days after unprotected sex, and studies indicate it is somewhat more effective than Plan B One-Step. Unlike Plan B One-Step, however, a prescription is required.

Mifepristone, also known as RU-486 or the "abortion pill," can cause the loss of an implanted embryo by blocking the progesterone receptors of endometrial cells. This causes the endometrium to slough off, carrying the embryo with it. When taken in conjunction with a prostaglandin to induce uterine contractions, RU-486 is 95% effective at inducing an abortion up to the 49th day of gestation. Because of its mechanism of action, the use of

RU-486 is more controversial compared to other medications, and while it is currently available in the United States for early medical abortion, it is not approved for emergency contraception.

Male Birth Control

As noted in Section 41.2, a vasectomy is a surgical form of birth control for males. Over 500,000 American males have a vasectomy each year. Because this procedure is not easily reversed, a great deal of research is being devoted to developing safe and effective hormonal birth control for males. The use of condoms represents the most effective form of birth control currently available to males.

Currently, there are no vaccines or hormonal injections that have been approved for male contraceptions. However, research has been focusing on several different approaches, some of which are showing promise in animal models and could be available in the next few years. Implants, pills, patches, gels, and injections are being explored as ways to deliver testosterone and/or progesterone at adequate levels to suppress sperm production. Another approach uses injections of a gel into the vas deferens to reduce sperm mobility.

Reproductive Technologies

Infertility is the inability of a couple to achieve pregnancy after 1 year of regular, unprotected intercourse. The American Medical Association estimates that 15% of all couples experience infertility. The cause of infertility can be evenly attributed to the male and female partners.

Sometimes, the causes of infertility can be corrected by medical intervention, so that couples can have children. If no obstruction is apparent and body weight is normal, it is possible for females to take fertility drugs, which are gonadotropic hormones that stimulate the ovaries and bring about ovulation. Such hormone treatments may cause multiple ovulations and multiple births.

When reproduction does not occur in the usual manner, many couples adopt a child. Others sometimes try one of the assisted reproductive technologies (ARTs) developed to increase the chances of pregnancy. In these cases, sperm and/or oocytes are often retrieved from the testes and ovaries, and fertilization takes place in a clinical or laboratory setting.

Artificial Insemination by Donor (AID)

During artificial insemination, sperm from a donor are introduced into the female's vagina by a physician. This technique is especially helpful if the male partner has a low sperm count, because the sperm can be collected over a period of time and concentrated, so that the sperm count is sufficient to result in fertilization. Often, however, a female is inseminated by sperm acquired from a different donor. At times, a combination of partner and donor sperm is used.

A variation of AID is *intrauterine insemination* (*IUI*). In IUI, fertility drugs are given to stimulate the ovaries, and then the donor's sperm is placed into the uterus, rather than into the vagina.

If the prospective parents wish, sperm can be sorted into those believed to be X-bearing or Y-bearing, to increase the chances of having a child of the desired sex. First, the sperm are treated with a DNA-staining chemical. Because the X chromosome has slightly more DNA than the Y chromosome, it takes up more dye. When a laser beam shines on the sperm, the X-bearing sperm shine a little more brightly than the Y-bearing sperm. A machine sorts the sperm into two groups on this basis. Parents can expect about a 65% success rate when selecting for males and about 85% when selecting for females.

In Vitro Fertilization (IVF)

During IVF, fertilization occurs in laboratory glassware. Ultrasound machines can spot follicles in the ovaries that hold immature oocytes; therefore, the latest method is to forgo the administration of fertility drugs and retrieve immature oocytes by using a needle. The immature oocytes are then brought to maturity in glassware before concentrated sperm are added. After about 2–4 days, the embryos are ready to be transferred to the uterus of the female, who is now in the secretory phase of the uterine cycle. If desired, the embryos can be tested for a genetic disease, as discussed in the Nature of Science feature, "Preimplantation Genetic Diagnosis." If implantation is successful, development continues to term.

Gamete Intrafallopian Transfer (GIFT)

Recall that the term **gamete** refers to a sex cell, either a sperm or an oocyte. Gamete intrafallopian transfer was devised to overcome the low success rate (15–20%) of in vitro fertilization. The method is the same as for in vitro fertilization, except the oocytes and sperm are placed in the uterine tubes immediately after they have been brought together. GIFT has the advantage of being a one-step procedure for a female since the oocytes are removed and reintroduced at the same time. A variation on this procedure is to fertilize the eggs in the laboratory and then place the zygotes into the uterine tube.

Surrogate Parents

In some instances, females are contracted and paid to carry babies as *surrogate parents.* The sperm and even the oocyte can be contributed by the contracting parents.

Intracytoplasmic Sperm Injection (ICSI)

In this highly sophisticated procedure, a single sperm is injected into an oocyte. It is used effectively when a man has severe infertility problems.

If all the assisted reproductive technologies discussed were employed simultaneously, it would be possible for a baby to have five parents: a sperm donor, an oocyte donor, a surrogate parent, the contracting female parent, and the contracting male parent. The legal definitions of "parent" can become a matter of contention in some cases.

Check Your Progress 41.4

1. Identify which birth control methods discussed in this section physically block sperm from entering the uterus.
2. Rank these birth control methods regarding their effectiveness at preventing pregnancy: abstinence, female birth control pill, condoms, and vasectomy.
3. Briefly describe the following processes: artificial insemination by donor (AID), in vitro fertilization (IVF), gamete intrafallopian transfer (GIFT), and intracytoplasmic sperm injection (ICSI).

ISTUDY

THEME Nature of Science

Preimplantation Genetic Diagnosis

If prospective parents are heterozygous for a genetic disorder, they may want the assurance that their offspring will be free of the disorder. Determining the genotype of an embryo can provide this assurance. For example, if both parents are *Aa* for a recessive disorder, the embryo will develop normally if it has the genotype *AA* or *Aa*. In contrast, if one of the parents is *Aa* for a dominant disorder, the embryo will develop normally only if it has the genotype *aa*.

Following in vitro fertilization (IVF), the zygote (fertilized egg) divides. When the embryo has six to eight cells (Fig. 41A*a*), the removal of one of these cells for testing purposes has no effect on normal development. Only embryos with a cell that tests negative for the genetic disorder of interest are placed into the uterus to continue developing.

So far, several thousand children with normal genotypes for genetic disorders that run in their families have been born worldwide following embryo testing. It is possible that embryos that test positive for a disorder can be treated by gene therapy or CRISPR technology (see Section 14.1), although currently these procedures are not considered ethical in many countries.

Testing the oocyte is possible if the condition of concern is recessive. Recall that meiosis in females results in a single egg and at least two polar bodies. Polar bodies later disintegrate, and they receive very little cytoplasm, but they do receive a haploid number of chromosomes. When a female is heterozygous for a recessive genetic disorder, about half the first polar bodies have received the genetic defect, and in these instances, the egg has received the normal allele. Therefore, if a polar body tests positive for a recessive defect, then the egg has received the normal dominant allele. Only normal eggs are then used for IVF. Even if the sperm should happen to carry the same mutant allele, the zygote will, at worst, be heterozygous. But the phenotype will appear normal (Fig. 41A*b*).

If, in the future, gene therapy becomes routine, an egg could be given genes that control traits desired by the parents, such as musical or athletic ability, prior to IVF.

Questions to Consider

1. Some biomedical ethicists are concerned that embryo testing might lead to a form of eugenics, in which wealthier parents are able to afford to improve their children's physical traits. Others argue that this technology should be developed as much as possible to reduce the occurrence of debilitating diseases. Which argument is more compelling to you, and why?
2. Some experts feel that preimplantation screening should be required for all embryos that are conceived by in vitro fertilization, to reduce the occurrence of disease. Do you agree or disagree?

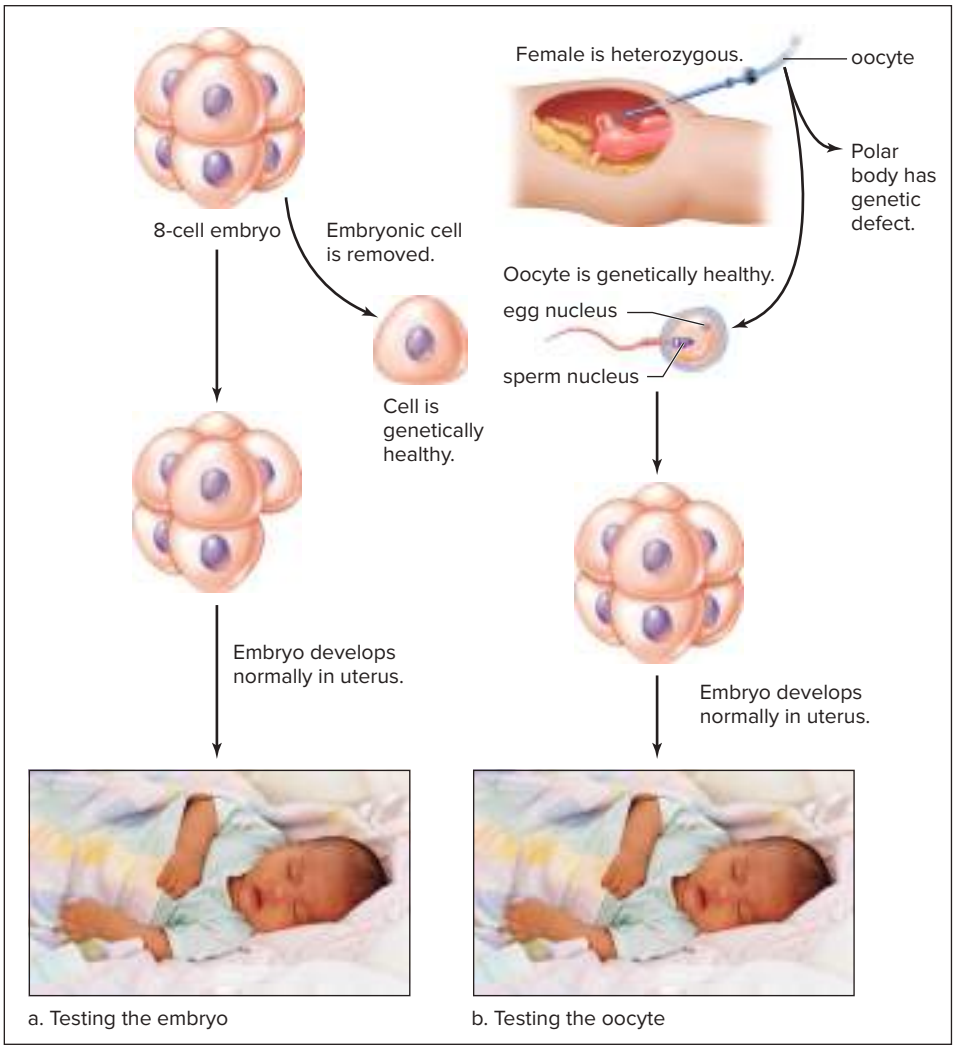


Figure 41A Preimplantation genetic diagnosis. **a.** Following IVF and cleavage, genetic analysis is performed on one cell removed from an eight-celled embryo. If it is found to be free of the genetic defect of concern, the seven-celled embryo is implanted in the uterus and develops into a fetus. **b.** Chromosome and genetic analysis is performed on a polar body attached to an oocyte. If the oocyte is free of a genetic defect, the egg is used for IVF, and the embryo is implanted in the uterus for further development. (both): (photo): Elyse Lewin/Exactostock-1555/SuperStock

ISTUDY

41.5 Sexually Transmitted Diseases

Learning Outcomes

Upon completion of this section, you should be able to

1. Categorize whether an STD is caused by viruses or bacteria.
2. Discuss how HIV affects the immune system and how this is related to the three categories of HIV infection.
3. Describe three STDs that are preventable by vaccination.

The CDC estimates that over 26 million new cases of sexually transmitted diseases (STDs) occur each year in the United States. About half of these new cases occur in young people, ages 15–24. At any given time, about 110 million Americans, including about one in four college students, are infected with one or more of the 30 different STD-causing organisms ranging from viruses to arthropods; however, this discussion centers on the most prevalent and serious ones. The Biological Systems feature, "Preventing Transmission of STDs," describes some methods to avoid the transmission of STDs.

Viral STDs

A number of STDS, including AIDS, genital herpes, genital warts, and hepatitis A and B, are caused by viruses. Therefore, they do not respond to treatment with traditional antibiotics. Antiviral drugs have been developed to treat some of these viral infections, but in many cases they persist for life.

HIV Infections and AIDS

Acquired immunodeficiency syndrome (AIDS) is caused by the **human immunodeficiency virus (HIV)**. An introduction to HIV and AIDS was provided in Section 33.4. Currently, an estimated 38.4 million people worldwide are infected with HIV, and an estimated 650,000 people die from AIDS each year. The great majority of these people live in less-developed countries, and an estimated 71% of individuals infected with HIV in the world live in sub-Saharan Africa.

In the United States, approximately 1.2 million people are infected with HIV. Notably, about one in eight Americans with HIV are unaware of their infection. Gay men make up the largest proportion of people living with HIV in the United States, but the fastest rate of increase is now seen in people who are heterosexual. Over half of new HIV infections are now occurring in people under the age of 25.

Transmission HIV is transmitted by sexual contact with a person who has the virus, including vaginal or anal intercourse and oral-genital contact. Also, needle-sharing among people who use intravenous drugs can increase the risk of HIV transmission. Some of the earliest cases of AIDS resulted from the transfusion of HIV-infected blood, but this is now extremely rare, thanks to effective screening methods. Babies born to females with an HIV infection may become infected before or during birth or through breast-feeding. Transmission through tears, saliva, or sweat is thought to be highly unlikely.

Cultural factors may also play an important role in the transmission of HIV. For example, polygamy is common in many African countries, as are misconceptions about the effectiveness of condoms in preventing HIV transmission, and even about the fact that HIV causes AIDS. Such practices and beliefs may significantly increase the risk of HIV infection.

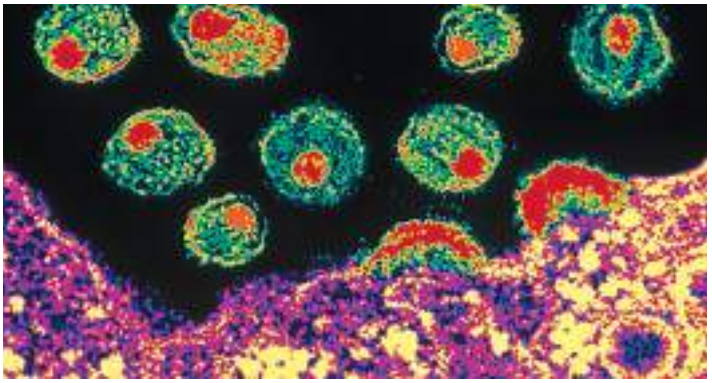


Figure 41.15 HIV, the AIDS virus. False-colored micrograph showing HIV particles budding from an infected helper T cell. These viruses can infect other helper T cells and macrophages, which work with helper T cells to stem infection. Scott Camazine/Science Source

Stages and Symptoms of an HIV Infection HIV infects helper T cells, macrophages, and other cells that have a molecule called CD4 on their surface (Fig. 41.15). Helper T cells (see Section 33.4) control the activities of many other cells in the immune system. After HIV enters a host cell, it converts its viral RNA to DNA, which is then integrated into the host cell DNA.

In the early stages after an HIV infection, the person's blood contains a high amount of virus, even though the individual may experience only mild, flulike symptoms, which soon disappear. Even if a person with HIV undergoes an HIV test at this time, it would probably be negative, because this tests for the presence of anti-HIV antibodies, which are usually not detectable for an average of 25 days after infection. However, the person can still be highly infectious at this stage.

After the initial surge of HIV replication, the immune system temporarily reduces the virus levels in the blood. Over an average period of 8–10 years, the HIV-infected cells gradually produce increasing amounts of the virus, which infect and destroy more and more helper T cells. As the virus reproduces, it also mutates, so it is no longer recognized by the immune system.

Three categories of HIV infection are distinguished, based mainly on the type of disease seen in the person. Within each category, the number of $CD4^+$ T cells is also monitored. People in category A usually have greater than 500 $CD4^+$ T cells per microliter of blood, and they either lack symptoms completely or have persistently enlarged lymph nodes. People in category B have a $CD4^+$ T-cell count between 200 and 499 per microliter, and they begin to develop opportunistic infections (OIs)—infections that would be unable to take hold in a person with a healthy immune system. Once the $CD4^+$ T-cell count falls below 200 per microliter, the individual is in category C, or true AIDS. The deficiency of helper T-cell functions in these people results in their contracting one or more AIDS-defining OIs (see Section 33.4).

Treating or Preventing HIV Infection Before the development of antiretroviral drugs, most individuals with HIV eventually progressed to category C AIDS, which was almost invariably fatal.

ISTUDY

THEME Biological Systems

Preventing Transmission of STDs

- Abstain from sexual intercourse or develop a long-term monogamous (always the same partner) sexual relationship with a partner who is free of STDs (Fig. 41B).
- Refrain from having multiple sex partners or having relations with someone who has multiple sex partners. If you have sex with two people and each of these persons has sex with two people, and so forth, the number of individuals involved becomes quite large.
- Be aware that behaviors associated with intravenous drug use (e.g., needle sharing) may increase the risk of AIDS and hepatitis B, which can then be transmitted sexually to other partners. Be aware that anyone who already has another sexually transmitted disease is more susceptible to an HIV infection.
- Be aware that certain types of sexual activities are associated with an increased risk of HIV infection. Anal intercourse (in which the penis is inserted into the rectum) may increase the risk of HIV. The lining of the rectum is thin, and infected CD4 T cells can easily enter the body there. Also, the rectum is supplied with many blood vessels, and insertion of the penis into the rectum is likely to cause tearing and bleeding that facilitate the entrance of HIV. The vaginal lining is thick and difficult to penetrate, but the lining of the uterus is only one cell thick at certain times of the month and does allow CD4 T cells to enter.
- Males who are uncircumcised are more likely to become infected than males who are circumcised. This is because vaginal secretions, viruses, and bacteria may remain under the foreskin for a longer time.

Practice Safer Sex

- Always use a latex condom during sexual intercourse if you are not in a monogamous relationship. Be sure to follow the directions supplied by the manufacturer for the use of a condom. At one time, people who used condoms were advised to use nonoxynol-9 in conjunction with a condom, but testing shows that this spermicide has no effect on viruses, including HIV.
- Be aware of the risks associated with fellatio (kissing and insertion of the penis into a partner's mouth) and cunnilingus (kissing and insertion of the tongue into the vagina), because they may be a means of transmission. The mouth and gums often have cuts and sores that facilitate catching an STD.
- Practice penile, vaginal, oral, and hand cleanliness. Be aware that hormonal contraceptives make the female genital tract receptive to the transmission of sexually transmitted diseases, including HIV.
- Be cautious about using alcohol or any drug that may prevent you from being able to control your behavior.

Drug Use Increases the Risk of Hepatitis and HIV Transmission

- Stop, if necessary, or do not start the habit of injecting drugs into your veins. Be aware that HIV and hepatitis B can be spread by blood-to-blood contact.
- If you are struggling with a substance use disorder and are unable to stop using drugs, always use a new sterile needle for injection or one that has been cleaned in bleach (Fig. 41C).

Questions to Consider

1. How can you be certain that a potential sexual partner doesn't have any STDs?
2. What does it mean if a person just had an HIV test and it was negative?
3. Which methods of birth control are most effective at preventing the transmission of STDs? Which are least effective or provide no protection?

Figure 41C Sharing needles increases the risk of STDs. Intravenous drug use with unclean, shared needles can transmit viral STDs. Don Mason/Stockbyte/Getty Images

ISTUDY

Although there is still no cure for HIV infection, total AIDS-related deaths have been decreasing worldwide since about 2004, mainly due to increased availability of antiretroviral therapy (ART).

According to the National Institute of Allergy and Infectious Diseases (NIAID), six major types of drugs have been developed for ART:

- Entry inhibitors stop HIV from entering a cell by, for example, preventing the virus from binding to CD4.
- Fusion inhibitors prevent the virus from fusing with a cellular membrane, which also interferes with viral entry.
- Reverse transcriptase inhibitors, such as AZT, interfere with the conversion of viral RNA into DNA.
- Integrase inhibitors prevent HIV from inserting its DNA into the DNA of the host cells.
- Protease inhibitors prevent a viral enzyme from processing newly created polypeptides.
- Multiclass combination products combine more than one of these five types of ART into a single product.

Treating individuals with a combination of drug types means that the virus is less likely to become resistant to therapy, because several viral mutations would need to occur at once. This approach is sometimes called highly active antiretroviral therapy (HAART).

When the first antiretroviral drug, known as AZT, became available in 1995, it was initially very effective at reducing the viral load in many individuals, but the virus tended to become resistant to the drug very quickly. This is because the HIV reverse transcriptase enzyme is very error-prone, which means the virus mutates at a very high rate, resulting in drug-resistant mutants. However, today, individuals infected with HIV are typically treated with three or four antiretroviral drugs at the same time, a combination called *highly active antiretroviral therapy (HAART).* Because it is unlikely that a particular mutant virus will be resistant to several drugs all at once, HAART is usually able to inhibit HIV replication to the point that no virus is detectable in the blood. It is important to note, however, that these people presumably still have large numbers of HIV-infected cells, and thus they can still transmit the virus. Other potential problems with HAART include various side effects, which can be severe, the high cost, and the complexity of dosing schedules for the different medications.

Investigators have found that when HAART is discontinued, the virus rebounds. Therefore, therapy is usually continued indefinitely. A female who is pregnant and is HIV-positive who takes reverse transcriptase inhibitors during their pregnancy reduces the chances of HIV transmission to the newborn. If possible, drug therapy should be delayed until the 10th to 12th weeks of pregnancy to minimize any adverse effects of antiretroviral drugs on fetal development.

In recent years, claims that people have been "cured" of HIV/AIDS have surfaced in the media. In one example, a male from Berlin, with both leukemia and AIDS, received a bone marrow transplant that apparently cured both diseases. More recently, several cases have been reported where individuals have been cured of the disease following stem cell transplants. However, this form of treatment is currently under investigation since it does not work with all individuals with AIDS.

Another approach is the development of an AIDS vaccine. Although there are many difficulties in vaccine development, AIDS vaccine trials are underway. The process can take many years. After a vaccine has been tested in animals, it must pass through three phases of clinical trials before it is marketed or administered to the public. In Phases I and II, the vaccine is tested for 1 to 2 years in a small number of volunteers who are not HIV-infected. The most effective vaccines move into Phase III. In Phase III, the vaccine is tested for 3 to 4 years in thousands of people who are not HIV-infected. A Phase III trial of the RV144 HIV vaccine was conducted from 2003 to 2009 in Thailand and involved over 16,000 volunteers. The vaccine was reported to reduce the rate of HIV infection by 32%. In February 2015, a second trial commenced in South Africa.

Although the results from the RV144 trial yielded a lower percentage than will be needed for a vaccine to be considered effective, most scientists have many reasons to be optimistic that an AIDS vaccine can and will be developed. The RV144 results, and other successes with vaccinating monkeys, suggest that the research is beginning to yield results. But the most compelling reason for optimism is the human body's ability to suppress the infection. The immune system is able to successfully and effectively decrease the HIV viral load in the body, helping delay the onset of AIDS an average of 10 years in 60% of people who are HIV-infected in the United States. Studies have shown that a small number of people remain HIV uninfected after repeated exposure to the virus, and a few individuals who are HIV-infected maintain a healthy immune system for over 15 years. It is these stories of the human body's ability to fight HIV infection that keep scientists hopeful that there is a way to help the body fight HIV infection. Most scientists agree that—despite the obstacles or the setbacks—an AIDS vaccine is possible in our future.

Human Papillomavirus

An estimated 42 million Americans are infected with the *human papillomavirus* (*HPV*). Most people will be infected with HPV during their lifetime, although about 90% will recover from the infection without any consequences, or even knowledge that they were infected.

There are over 100 types of HPV. Many of these cause warts, and about 30–40 types specifically cause genital warts, which are an STD. Genital warts may appear as flat or raised lesions on the penis and foreskin of males (Fig. 41.16*a*) and on the vulva, vagina, and cervix in females. Note that if warts are present only on the cervix, there may be no outward signs of the disease. Newborns can also be infected with HPV during passage through the birth canal.

Warts can usually be treated effectively by surgery, freezing, application of an acid, or laser burning. Even after treatment, however, the virus can sometimes be transmitted. Therefore, once someone has been diagnosed with genital warts, abstinence or the use of a condom is recommended.

About ten types of HPV are associated with cancer, which may affect the cervix, vulva, vagina, anus, penis, or oral cavity. Decades ago, cervical cancer was the leading cause of cancer death for females in the United States, but those rates have decreased significantly due to improved methods for diagnosis and treatment. However, cervical cancer is still a significant cause of death in less-developed countries. It can often be detected in its early stages by a *Pap smear,* in which a

ISTUDY

a. b.

Figure 41.16 Two STDs caused by viruses. **a.** Genital warts, caused by human papillomavirus and seen here on the penis, can also affect the anus, vulva, vagina, or cervix. **b.** Genital herpes, caused by the herpes simplex virus, can cause blisters on the labia (seen here), vagina, rectum, or penis. (a): Dr. M. F. Rein/Centers for Disease Control and Prevention; (b): Dr P. Marazzi/Science Source

few cells are removed from the cervix for microscopic examination. If the cells are cancerous, a hysterectomy (removal of the uterus) may be recommended. Additionally, HPV now causes more cancers of the oropharynx than does tobacco, perhaps due to an increase in oral sex in combination with a decline in tobacco use.

In June 2006, the U.S. Food and Drug Administration licensed Gardasil, an HPV vaccine that is effective against the four most common types of HPV found in the United States, including the two types that cause about 70% of cervical cancers. Because the vaccine doesn't protect those who are already infected, ideally children should be vaccinated before they become sexually active. In 2009, the U.S. Food and Drug Administration approved Gardasil for use in males. The Centers for Disease Control and Prevention recommends that 11- to 12-year-old females and males receive three doses of the vaccine. Nonpregnant females between ages 13 and 26, and males from age 13 to 21, can also be vaccinated if they did not receive any or all of the three recommended doses when they were younger. Older individuals should speak with their doctor to find out if getting vaccinated is right for them.

Genital Herpes

Genital herpes is caused by the herpes simplex virus, of which there are two types. Type 1 usually causes cold sores and fever blisters, while type 2 more often causes genital herpes. In the United States, about one out of every six people ages 14–49 currently have genital herpes, but most people with the infection don't have symptoms. Even without symptoms, however, the virus can still be transmitted.

After becoming infected, most people experience a tingling or itching sensation before blisters appear on the genitals within 2–20 days (Fig. 41.16*b*). Once the blisters rupture, they leave painful ulcers, which take from 5 days to 3 weeks to heal. These symptoms may be accompanied by fever, pain on urination, swollen lymph nodes in the groin, and in females a watery vaginal discharge. At this time, the individual also has an increased risk of acquiring an HIV infection. Exposure to herpes during a vaginal delivery can cause an infection in the newborn, which can lead to neurological disorders and even death. Birth by cesarean section prevents this possibility.

After the blisters heal, the virus becomes latent (dormant). Blisters can recur repeatedly, at variable intervals and with milder symptoms. Sunlight, sexual intercourse, menstruation, and stress are associated with these recurrences. While the virus is latent, it resides in the ganglia of the sensory nerves associated with the infected skin. The ability of the virus to "hide" in this manner has made it difficult to develop an effective vaccine for herpes simplex viruses.

Viral Hepatitis

Several different viruses can cause *hepatitis,* or inflammation of the liver. Of these, hepatitis A and B viruses are the most important. Hepatitis A is usually acquired from sewage-contaminated drinking water, but this infection can also be sexually transmitted through oral-anal contact. Hepatitis B is spread through sexual contact and by blood-borne transmission (e.g., accidental needle-stick on the job, a contaminated blood transfusion, needles shared while injecting drugs, or transfer from the female parent to the fetus). Simultaneous infection with hepatitis B virus and HIV is common, because both share the same routes of transmission. Fortunately, a combined vaccine is available for hepatitis A and B. It is recommended that all children receive the vaccine to prevent these infections.

Bacterial STDs

Chlamydia, gonorrhea, and syphilis are major bacterial STDs that are usually curable with antibiotic therapy, if diagnosed early enough. If properly used, latex condoms can help prevent the spread of most STDs.

Chlamydia

Infection with the tiny bacterium *Chlamydia trachomatis,* simply known as *chlamydia,* is the most commonly reported bacterial STD in the United States, with the highest rates of infection occurring in 15- to 19-year-olds.

ISTUDY

Figure 41.17 Chlamydial eye infection. This newborn's eyes were infected after passing through the birth canal of a female parent infected with chlamydia. Western Ophthalmic Hospital/Science Source

Chlamydial infections of the lower reproductive tract are usually mild or asymptomatic, especially in females. About 8–21 days after infection, males may experience a mild burning sensation on urination and a mucoid discharge. Females may have a vaginal discharge along with symptoms resembling a urinary tract infection. Chlamydia also causes cervical ulcerations, which increase the risk of acquiring HIV. If the infection is misdiagnosed or if a female does not seek medical help, the infection may spread to the uterine tubes, and pelvic inflammatory disease (PID) results.

PID is characterized by inflamed uterine tubes that become partially or completely blocked by scar tissue. As a result, a female can become infertile or at increased risk of ectopic pregnancy, which can be a medical emergency. If a baby comes in contact with chlamydia during delivery, pneumonia or eye infections can result (Fig. 41.17).

Gonorrhea

Gonorrhea is caused by the bacterium *Neisseria gonorrhoeae* (Fig. 41.18). Most cases occur in people between 15 and 29 years of age, and rates among African Americans are about 20 times greater than among whites. Although as many as 20% of males with gonorrhea are asymptomatic, usually they experience pain on urination and a thick, greenish-yellow urethral discharge.

Untreated gonorrhea can cause infertility in both males and females by causing inflammation and scarring in either the vas deferens or the uterine tubes, respectively. If a baby is exposed during birth, an eye infection can lead to blindness. All vaginally delivered newborns are treated with antibiotic eyedrops to prevent this possibility.

Oral-genital contact can also cause *N. gonorrhoeae* infection of the mouth, throat, and tonsils. Anal intercourse can lead to gonorrhea proctitis, an inflammation of the anus characterized by anal pain and blood or pus in the feces. Gonorrhea can spread to internal parts of the body, causing heart damage or arthritis. If, by chance, the person touches infected genitals and then touches the eyes, a severe eye infection can result.

Until recently, gonorrhea was considered to be curable by antibiotics. However, antibiotic-resistant strains of *N. gonorrhoeae* are becoming an increasingly urgent problem.

Syphilis

Syphilis is an STD that was first described hundreds of years ago. It is caused by the bacterium *Treponema pallidum* (Fig. 41.19). Untreated syphilis progresses through three stages. In the primary stage, a hard chancre (ulcerated sore with hard edges) indicates the site of infection. During the secondary stage, the individual breaks out in a rash that does not itch, coinciding with the replication and spread of the bacteria all over the body.

Not all cases of syphilis progress to the tertiary stage, during which syphilis may affect the cardiovascular or nervous system. People at this stage may become blind or develop an intellectual disability, walk with a shuffle, or show signs of serious mental illness. Gummas, which are large, destructive ulcers, may develop on the skin or in the internal organs.

T. pallidum bacteria can cross the placenta, causing birth defects or a stillbirth. If diagnosed and treated in its early stages, syphilis is easy to cure with antibiotics.

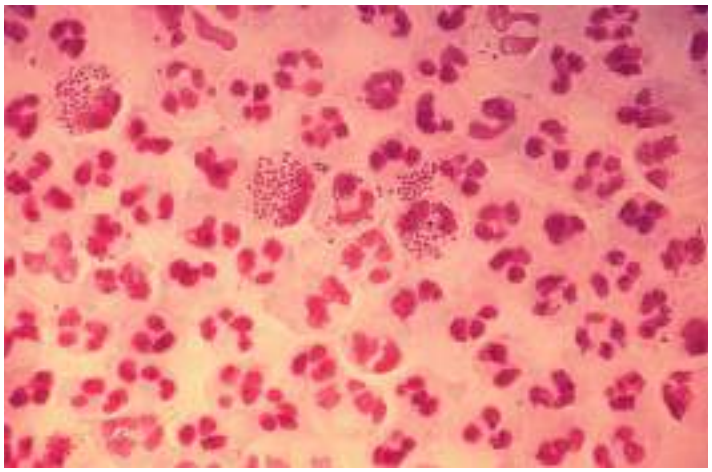


Figure 41.18 Gonorrhea. Photomicrograph of a urethral discharge from a male infected with gonorrhea. In this image, the gonorrheal bacteria (*Neisseria gonorrhoeae*) have been phagocytosed by some of the neutrophils present in the discharge. Joe Miller/Centers for Disease Control and Prevention

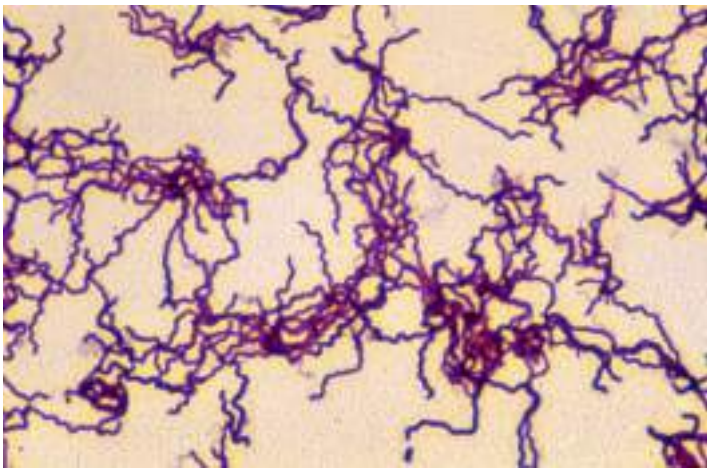


Figure 41.19 Syphilis. *Treponema pallidum,* the species of bacteria that cause syphilis, are spirochetes, as indicated by their corkscrewlike shape. Melba Photo Agency/Alamy Stock Photo

ISTUDY

Vaginal Infections

Vaginitis, meaning vaginal inflammation from any cause, is the most commonly diagnosed gynecologic condition. Bacterial vaginosis (BV), or overgrowth of certain bacteria inhabiting the vagina, accounts for 40–50% of vaginitis cases in female Americans. A common culprit is the bacterium *Gardnerella vaginalis.* How females acquire this infection is not well understood, but it is more common in females who are sexually active.

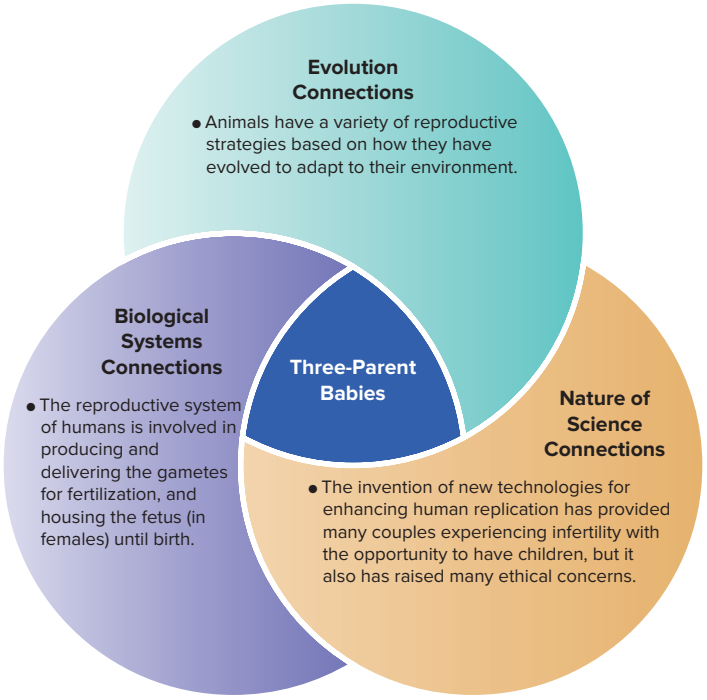
In addition to BV, the yeast *Candida albicans* can also cause vaginitis. Yeast is normally found living in the vagina; under certain circumstances, its growth increases above normal, causing vaginitis. For example, females who take birth control pills or antibiotics may be prone to yeast infections. Both can alter the normal balance of vaginal organisms, causing a yeast infection. Typical signs of a yeast infection include a thickish, white vaginal discharge and itching of the vulva and/or vagina. Antifungal medications inserted into the vagina are used to treat yeast infections.

The protozoan *Trichomonas vaginalis* can also infect the urethra of males and the vagina of females. Males with this infection are usually asymptomatic, but they can pass the parasite to their partner through sexual intercourse. Symptoms of trichomoniases in females are a foul-smelling, yellow-green exudate and itching of the vulva/vagina. Effective antiprotozoal drugs are available by prescription, but if one partner remains infected, reinfection will occur. For this reason, both people should be treated simultaneously.

Check Your Progress 41.5

1. Describe four classes of antiretroviral drugs and the specific stage of the HIV reproduction cycle targeted by each class.
2. Describe the medical complications in females that are associated with HPV infection.
3. Identify a serious condition that can occur in females due to both chlamydia and gonorrhea.

Connecting *the* Concepts



What Did You Learn?

41.1 How Animals Reproduce

Ordinarily, **asexual reproduction** may quickly produce a large number of offspring genetically identical to the parent. Sexual reproduction involves **gametes** and produces offspring that are genetically slightly different from the parents.

- The **gonads** are the primary sex organs, which produce **germ cells.** The **testes** produce sperm, and the **ovaries** produce eggs. The accessory sex organs include storage areas for sperm and ducts that conduct the gametes. Fertilization can be external or internal. Internal fertilization is facilitated by **copulation,** or sexual union.
- Exceptions to the usual male-female patterns exist. In **hermaphroditic** animals, single individuals have both male and female gonads. In **parthenogenesis,** an unfertilized egg develops into an adult.
- The egg of **oviparous** animals contains **yolk,** and in terrestrial animals a shelled egg prevents it from drying out. **Ovoviviparous** animals retain their eggs until the offspring have hatched; **viviparous** animals retain the embryo and give birth. Eggs of reptiles have **extraembryonic membranes** that allow them to develop on land; the same membranes are modified into a **placenta** for internal development in mammals. Placental mammals exemplify viviparous animals.

ISTUDY

41.2 Human Male Reproductive System

In human males, **sperm** are produced in the **testes,** mature in each **epididymis,** and may be stored in each **vas deferens** before entering the urethra, along with **semen (seminal fluid)** produced by the **seminal vesicles, prostate gland,** and **bulbourethral glands.** Sperm and semen is ejaculated during male orgasm, when the **penis** is erect.

Spermatogenesis occurs in the **seminiferous tubules** of the testes, which also produce testosterone in interstitial cells. **Testosterone** brings about the maturation of the primary sex organs during puberty and promotes the secondary sex characteristics of males, such as low voice, facial hair, and increased muscle strength.

- Follicle-stimulating hormone (FSH) from the anterior pituitary stimulates spermatogenesis, and luteinizing hormone (LH) stimulates testosterone production.
- **Gonadotropin-releasing hormone (GnRH)** released from the hypothalamus controls anterior pituitary production of two **gonadotropic hormones, follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH).** FSH stimulates spermatogenesis, and LH stimulates testosterone production. In males, the level of testosterone in the blood controls the secretion of GnRH and the gonadotropic hormones by a negative feedback system.

41.3 Human Female Reproductive System

In females, an **oocyte** (also called an **egg**) produced by an **ovary** enters a uterine tube, which leads to the **uterus.** At the lower end of the uterus, the **cervix** connects with the **vagina.** The external genital area of human females includes the vaginal opening, the clitoris, the labia minora, and the labia majora.

- During the **ovarian cycle,** one **follicle** a month (in either ovary) matures and produces first a primary, then a secondary oocyte. After **ovulation** (release of a secondary oocyte), the vesicular follicle develops into a **corpus luteum.** The follicle and the corpus luteum produce **estrogen** and **progesterone,** the female sex hormones. Ovulation marks the end of the **follicular phase** of the ovarian cycle, and the beginning of the **luteal phase.**
- The **uterine cycle** occurs concurrently with the ovarian cycle. In the first half of these cycles (days 1–13, before ovulation), the anterior pituitary produces FSH and the follicle produces estrogen. Estrogen causes the **endometrium** to increase in thickness. In the second half of these cycles (days 15–28, after ovulation), the anterior pituitary produces LH and the follicle produces progesterone. Progesterone causes the endometrium to become secretory. Feedback control of the hypothalamus and anterior pituitary causes the levels of estrogen and progesterone to fluctuate. When they are at a low level, **menstruation** begins.
- If fertilization occurs, a zygote is formed, and development begins. The resulting embryo travels down the oviduct and implants itself in the endometrium. A placenta, which is the region of exchange between the fetal blood and the female parent's blood, forms. At first, the placenta produces **human chorionic gonadotropin** (**HCG**), which maintains the corpus luteum; later, it produces progesterone and estrogen.
- The female sex hormones, estrogen and progesterone, also affect other traits of the body. Primarily, estrogen brings about the maturation of the primary sex organs during puberty and promotes the secondary sex characteristics of females, including body hair (usually less than in males), a wider pelvic girdle, a more rounded appearance, and the development of breasts. **Lactation** is the production of milk. Usually between the ages of 45 to 55, **menopause** occurs as a female's ovarian function ceases.

41.4 Control of Human Reproduction

Numerous birth control methods and devices, such as the birth control pill, diaphragm, and condom, are available for those who wish to prevent pregnancy. A "morning-after pill," RU-486, is now available. Contraceptive vaccines are being developed that could prevent pregnancy.

If couples have **infertility,** or trouble conceiving a child, they may use assisted reproductive technologies. Artificial insemination and in vitro fertilization have been followed by more sophisticated techniques, such as intracytoplasmic sperm injection.

41.5 Sexually Transmitted Diseases

Important sexually transmitted diseases (STDs) caused by viruses include **acquired immunodeficiency syndrome (AIDS),** caused by the **human immunodeficiency virus (HIV).** Other viral STDs include human papillomavirus (HPV), which causes genital warts and several types of cancer, as well as genital herpes and hepatitis A and B. Major bacterial STDs include chlamydia and gonorrhea, both of which may cause pelvic inflammatory disease (PID), and syphilis, which has cardiovascular and neurological complications if untreated. Vaginal infections have many causes; the most common are bacterial vaginosis (BV), yeast infection, and *Trichomonas* infection.

Thinking Critically About Biology

1. The average sperm count in males is now lower than it was several decades ago. The reasons for the lower sperm count usually seen today are not known. What data might be helpful in order to formulate a testable hypothesis?
2. Female athletes who train intensively often stop menstruating. The important factor appears to be the reduction of body fat below a certain level. Give a possible evolutionary explanation for a relationship between body fat in females and reproductive cycles.
3. In the animal kingdom, only primates menstruate. Other mammals come into season, or "heat," during certain times of the year, while still others ovulate only after having sex. What would be some possible advantages of a monthly menstrual cycle?
4. Human females undergo menopause typically between the ages of 45 and 55. In most other animal species, both males and females maintain their reproductive capacity throughout their lives. Why do you suppose human females lose that ability in midlife?

Making It Relevant to You

1. New molecular tools, such as CRISPR (see Section 14.3), are allowing for direct genetic modifications to embryos. If you were a potential parent, under what circumstances would you want to modify the genetics of your offspring?
2. Why would some people state that the use of reproductive technologies represents an "unnatural" process? What are your thoughts?
3. In the near future, it may be possible for human reproduction to occur completely outside of the female parent. How might this be beneficial for the developing fetus, and what may be the drawbacks?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

42

Animal Development and Aging

Image Source/Getty Images

The Genetics of Longevity

Due to advances in medicine and nutrition, our life span expectancy has increased from about 45 years of age in 1900, to just under 79 years of age in the United States today. Increasingly, more individuals are living past the age of 100 (called centenarians). In the United States, there are over 80,000 individuals, or around 0.02% of the population, who are centenarians. The rate is almost triple this in Japan and Okinawa.

Over the past several years, researchers have begun to explore the genetic basis of living longer. A variation of one gene, *FOX03A,* has been identified to be more prevalent in centenarians than in the general population. This gene regulates the insulin pathways of the body and appears to control the genetic mechanisms that protect cells against free radicals. Both insulin regulation (including caloric restriction diets) and protection against free radicals have been known for some time to enhance longevity in a variety of organisms. Another gene, called *SIRT6,* also known as the "longevity gene," is believed to protect our DNA against damage, thus prolonging cell life. While these are not the only genes contributing to longevity, they provide a starting point for larger genomic studies on human aging.

As you read through this chapter, think about the following questions:

1. What genetic factors control the process of development in humans?
2. What are the major causes of aging at the cellular and physiological levels?

Chapter Outline

42.1 Early Developmental Stages
42.2 Developmental Processes
42.3 Human Embryonic and Fetal Development
42.4 The Aging Process

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 10.5 How is meiosis involved in the production of gametes in animals?

Section 10.5 What are the differences between oogenesis and spermatogenesis in humans?

Section 41.3 Where does fertilization occur in human females?

Following *the* Themes

CHAPTER 42 – ANIMAL DEVELOPMENT AND AGING

Evolution	Similarities in the developmental stages of embryos among fish, amphibians, reptiles, and mammals provide evidence for the evolutionary relatedness of all animals.
Nature of Science	Research into the development of model organisms has advanced the science of developmental biology and our knowledge of the mechanisms of aging.
Biological Systems	Animal development begins when egg and sperm nuclei unite, and it progresses through complex stages controlled by the differential expression of a common set of genes found in all cells of an animal's body.

ISTUDY

42.1 Early Developmental Stages

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the structures of an egg and a sperm that are directly involved in fertilization.
2. Compare and contrast the cellular, tissue, and organ stages of embryonic development.
3. Describe the major steps in the development of the central nervous system of vertebrates.

As we explored in Section 41.1, animals that reproduce sexually use a wide variety of strategies to achieve **fertilization,** the union of a sperm and an egg to form a zygote. Figure 42.1 illustrates how fertilization occurs in mammals, including humans.

Details of Fertilization in Humans

Human sperm have three distinct parts: a tail, a middle piece, and a head (see Fig. 41.5*d*). The head contains the sperm nucleus and is capped by a membrane-bound *acrosome.* The egg of a mammal (actually, the secondary oocyte) is surrounded by a few layers of adhering follicular cells, collectively called the *corona radiata.* These cells nourished the oocyte when it was in a follicle of the ovary. The oocyte also has an extracellular matrix termed the *zona pellucida* just outside the plasma membrane, but beneath the corona radiata.

Fertilization requires a series of events that result in the diploid zygote. Although only one sperm actually fertilizes the oocyte, millions of sperm begin this journey, and perhaps a few hundred succeed in reaching the oocyte. The sperm cover the surface of the oocyte and secrete enzymes that help weaken the corona radiata. They squeeze through the corona radiata and bind to a molecule called Juno (after the Roman goddess of fertility) under the zona pellucida. After a sperm head binds tightly to the zona pellucida, the acrosome releases digestive enzymes that forge a pathway for the sperm through the zona pellucida to the oocyte plasma membrane.

When the first sperm binds to the oocyte plasma membrane, the next few events prevent *polyspermy* (entrance of more than one sperm). As soon as a sperm touches the plasma membrane of an oocyte, the oocyte's plasma membrane depolarizes, and this change in charge, known as the "fast block," repels sperm only for a few seconds. Then, vesicles in the oocyte called *cortical granules* secrete enzymes that turn the zona pellucida into an impenetrable *fertilization membrane.* This longer-lasting cortical reaction is known as the "slow block." During this phase, Juno molecules are also released from the oocyte, preventing further specific interactions.

The last events of fertilization include the formation of a diploid zygote. Microvilli extending from the plasma membrane of the oocyte (Fig. 42.1) bring the entire sperm into the oocyte. The

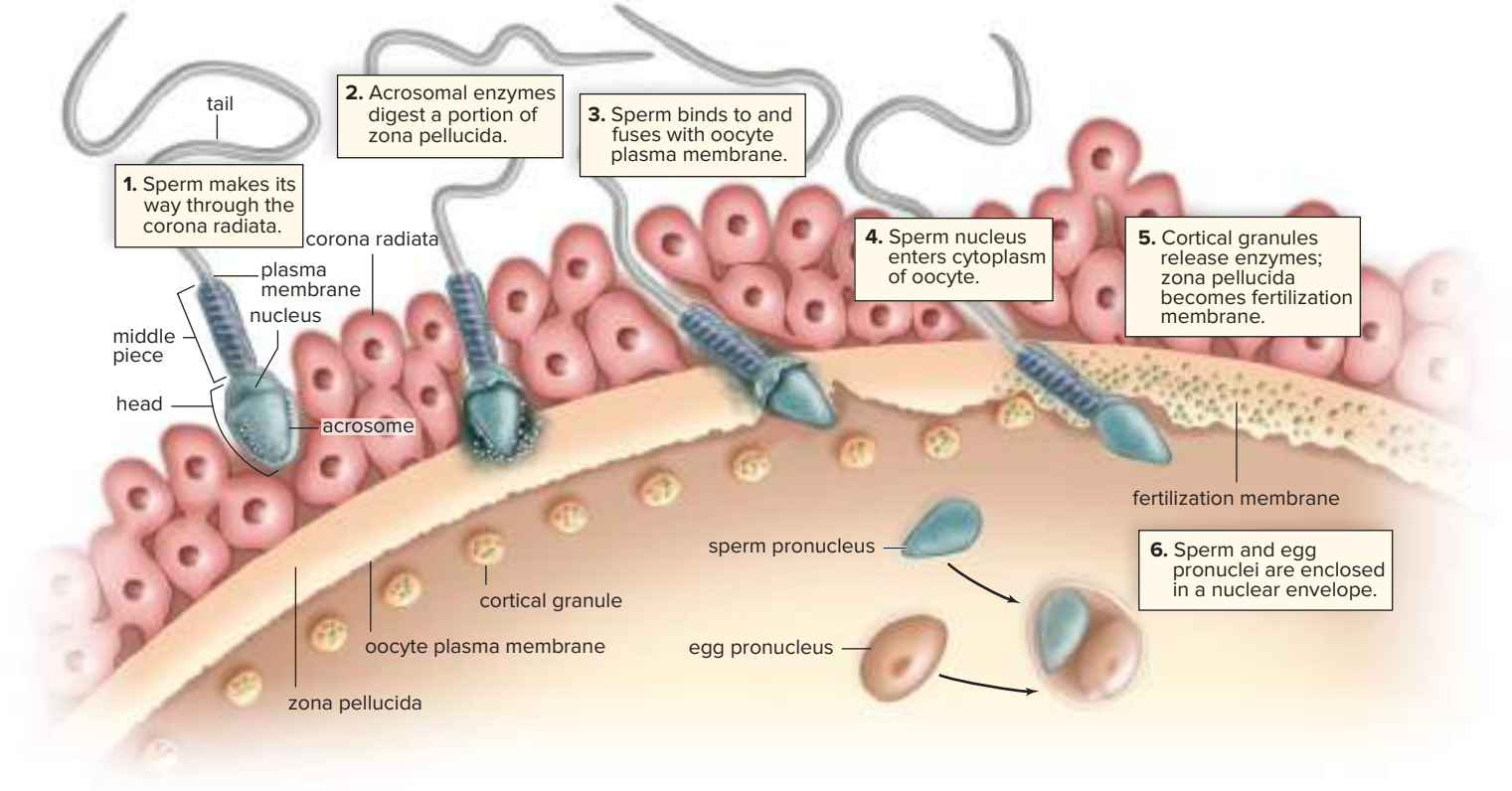


Figure 42.1 Fertilization. During fertilization, a single sperm is drawn into the oocyte by microvilli of its plasma membrane (micrograph). The head of a sperm has a membrane-bound acrosome filled with enzymes. When released, these enzymes digest a pathway for the sperm through the zona pellucida. After a sperm binds to the plasma membrane of the oocyte, changes occur that prevent other sperm from entering the oocyte. The oocyte finishes the second meiotic division and is now an egg. Fertilization is complete when the sperm pronucleus and the egg pronucleus contribute chromosomes to the zygote.

ISTUDY

sperm nucleus releases its chromatin, which re-forms into chromosomes enclosed within the sperm *pronucleus*. Meanwhile, the secondary oocyte completes meiosis, becoming an egg whose chromosomes are also enclosed in a pronucleus. A single nuclear envelope soon surrounds both sperm and egg pronuclei. Cell division is imminent, and the centrosomes that give rise to a spindle apparatus are derived from the basal body of the sperm's flagellum. The two haploid sets of chromosomes share the first spindle apparatus of the newly formed zygote.

Embryonic Development

Development includes all the changes that occur during the life cycle of an organism. Although development is a continuous, complex process, it can be divided into three major stages: (1) cellular stages, (2) tissue stages, and (3) organ stages.

Cellular Stages of Development

During the early stages of development, an organism is called an **embryo.** The cellular stages of development are:

- Cleavage, resulting in a multicellular embryo;
- Formation of the blastula.

Cleavage is cell division without growth; DNA replication and mitotic cell division occur repeatedly, and the cells get smaller with each division.

As shown in Figure 42.2, cleavage in a primitive animal called a lancelet results in uniform cells that form a **morula** (L. *morula,* "little mulberry"), which is a solid ball of cells. The morula continues to divide, forming a **blastula** (Gk. *blastos,* "bud"; L. *ula,* "small"), a hollow ball of cells having a fluid-filled cavity called a **blastocoel.** The blastocoel forms when the cells of the morula extrude Na^+ into extracellular spaces, and water follows by osmosis, collecting in the center.

All vertebrates have a blastula stage, but the appearance of the blastula can be different from that of a lancelet. Chickens lay a hard-shelled egg containing plentiful **yolk,** a dense nutrient material. Because yolk-filled cells do not participate in cleavage, the blastula is a layer of cells that spreads out over the yolk. The blastocoel then forms as a space that separates these cells from the yolk:

(chick): small_frog/iStock/E+/Getty Images

The zygotes of vertebrates, such as frogs, chickens, and humans, also undergo cleavage and form a morula. In frogs, cleavage is not equal because of the presence of the yolk. When yolk is present, the zygote and embryo exhibit polarity, and the embryo has an *animal pole* and a *vegetal pole.* The animal pole contains faster-growing, smaller cells that will eventually develop into the ectoderm and endoderm layers; the vegetal pole contains slower-growing, larger cells that develop into endoderm (Fig. 42.3).

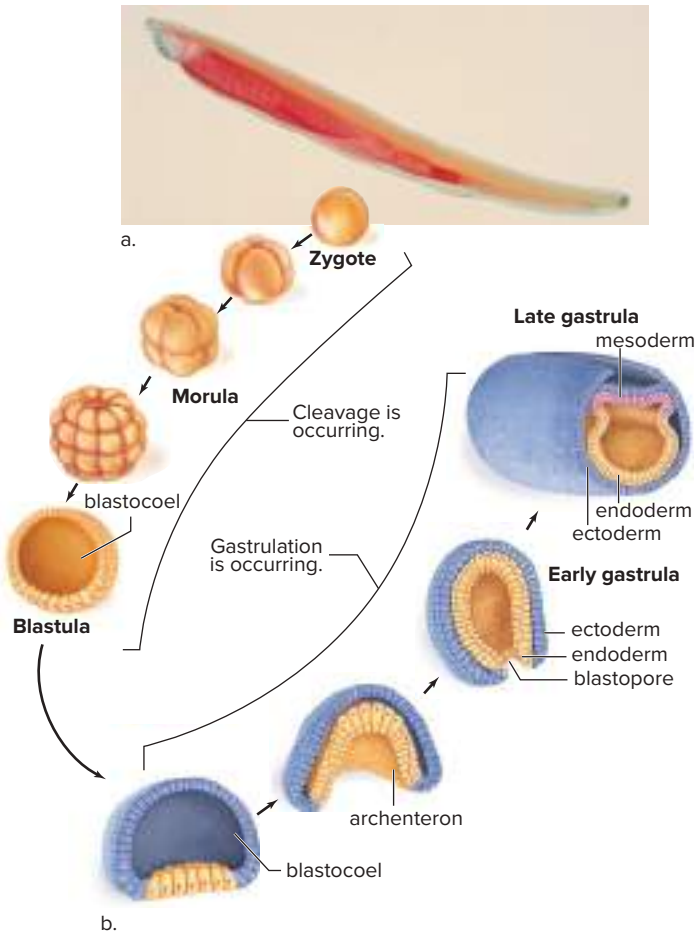


Figure 42.2 Embryonic stages of development. **a.** The early stages of development are exemplified in the lancelet. **b.** Cleavage produces a number of cells that form a cavity, the blastocoel. Invagination during gastrulation produces the germ layers ectoderm and endoderm. Then, the mesoderm arises. (a): (photo): Patrick J. Lynch/Science Source

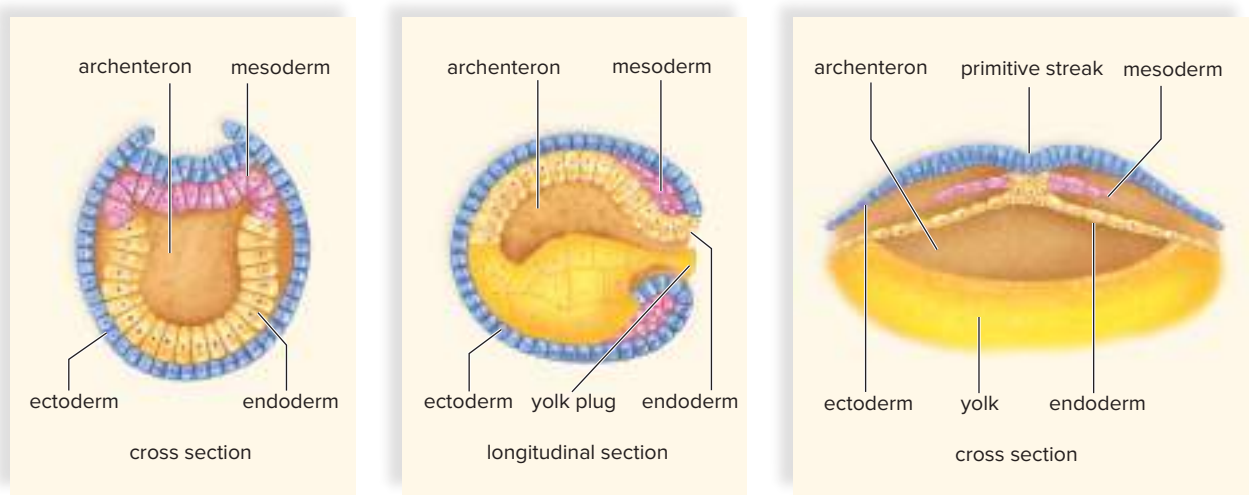
Tissue Stages of Development

The tissue stages of development involve **gastrulation,** the formation of a **gastrula** (Gk. *gaster,* "belly"; L. *ula,* "small") from the blastula. The two tissue stages are the early and late gastrula.

The early gastrula stage begins when certain cells begin to push, or invaginate, into the blastocoel, creating a double layer of cells (see Fig. 42.2). Cells migrate during this and other stages of development, sometimes traveling quite a distance before reaching a destination, where they continue developing. As cells migrate, they "feel their way" by changing their pattern of adhering to extracellular proteins.

An early gastrula has two layers of cells. The outer layer of cells is called the **ectoderm,** and the inner layer is called the **endoderm.** The endoderm borders the gut, but at this point, it is termed either the archenteron or the primitive gut. The pore, or hole, created by invagination (inward folding) is the

ISTUDY



a. Lancelet late gastrula b. Frog late gastrula c. Chick late gastrula

Figure 42.3 Comparative development of mesoderm. **a.** In the lancelet, mesoderm forms by an outpocketing of the archenteron. **b.** In the frog, mesoderm forms by migration of cells between the ectoderm and endoderm. **c.** In the chick, mesoderm also forms by invagination of cells.

blastopore, and in a lancelet, the blastopore eventually becomes the anus.

Gastrulation is not complete until three layers of cells that will develop into adult organs are produced. In addition to ectoderm and endoderm, the late gastrula has a middle layer of cells called the **mesoderm.**

Figure 42.2 illustrates gastrulation in a lancelet, and Figure 42.3 compares the lancelet, frog, and chicken late gastrula stages. In the lancelet, mesoderm formation begins as outpockets from the primitive gut (Fig. 42.3). These outpockets grow in size until they meet and fuse, forming two layers of mesoderm. The space between them is the coelom (see Section 28.1). The coelom is a body cavity lined by mesoderm that contains internal organs. In humans, the coelom becomes the thoracic and abdominal cavities of the body.

In the frog, cells containing yolk do not participate in gastrulation and, therefore, they do not invaginate. Instead, a slitlike blastopore is formed when the animal pole cells begin to invaginate from above, forming endoderm. Animal pole cells also move down over the yolk, to invaginate from below. Some yolk cells, which remain temporarily in the region of the blastopore, are called the yolk plug. Mesoderm forms when cells migrate between the ectoderm and endoderm. Later, the mesoderm splits, creating the coelom.

A chicken egg contains so much yolk that endoderm formation does not occur by invagination. Instead, an upper layer of cells becomes ectoderm, and a lower layer becomes endoderm. Mesoderm arises by an invagination of cells along the edges of a longitudinal furrow in the midline of the embryo. Because of its appearance, this furrow is called the *primitive streak*. Later, the newly formed mesoderm splits to produce a coelomic cavity.

Ectoderm, mesoderm, and endoderm are called the embryonic **germ layers.** No matter how gastrulation takes place, the result is the same: Three germ layers are formed. It is possible to relate the development of future organs to these germ layers (Table 42.1).

Table 42.1 Embryonic Germ Layers

Embryonic Germ Layer	Vertebrate Adult Structures
Ectoderm (outer layer)	Nervous system; epidermis of skin and derivatives of the epidermis (hair, nails, glands); tooth enamel, dentin, and pulp; epithelial lining of oral cavity and rectum
Mesoderm (middle layer)	Musculoskeletal system; dermis of skin; cardiovascular system; urinary system; lymphatic system; reproductive system—including most epithelial linings; outer layers of respiratory and digestive systems
Endoderm (inner layer)	Epithelial lining of digestive tract and respiratory tract, associated glands of these systems; epithelial lining of urinary bladder; thyroid and parathyroid glands

Organ Stages of Development

The organs of an animal's body develop from the three embryonic germ layers. Much study has been devoted to how the nervous system develops in chordates, so we summarize that process here.

The newly formed mesoderm cells lie along the main longitudinal axis of the animal and coalesce to form a dorsal supporting rod called the **notochord.** The notochord persists in lancelets, but in frogs, chickens, and humans, it is replaced later in development by the vertebral column, giving these animals the name vertebrates.

The nervous system develops from midline ectoderm located just above the notochord. At first, a thickening of cells, called the **neural plate,** is seen along the dorsal surface of the embryo. Then, neural folds develop on each side of a neural groove, which becomes the **neural tube** when these folds fuse. Figure 42.4 shows cross sections of frog development to illustrate the formation of the neural tube. At this point, the embryo is called a *neurula.* Later, the anterior end of the neural tube develops into the brain, and the rest becomes the spinal cord.

The *neural crest* is a band of cells that develops where the neural tube pinches off from the ectoderm. These cells

ISTUDY

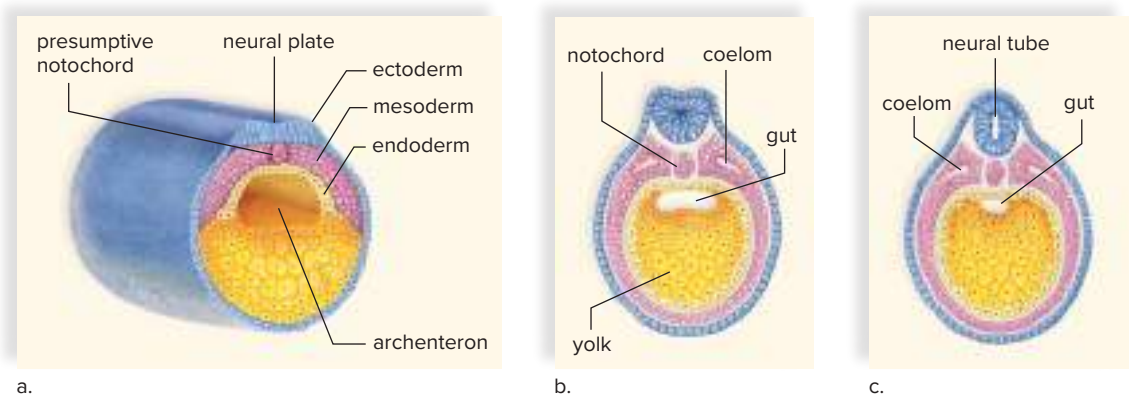


Figure 42.4 Development of neural tube and coelom in a frog embryo. **a.** Ectodermal cells that lie above the future notochord (called the presumptive notochord) thicken to form a neural plate. **b.** A splitting of the mesoderm produces a coelom, which is completely lined by mesoderm. **c.** A neural tube and a coelom have now developed.

migrate to various locations, where they help form skin, muscles, the adrenal medulla, and the ganglia of the peripheral nervous system.

Midline mesoderm cells that did not contribute to the formation of the notochord now become two longitudinal masses of tissue. These two masses become blocked off into *somites,* which are serially arranged along both sides along the length of the notochord. Somites give rise to muscles associated with the axial skeleton, and to the vertebrae. The serial origin of axial muscles and the vertebrae illustrates that vertebrates (including humans) are segmented animals. Lateral to the somites, the mesoderm splits, forming the mesodermal lining of the coelom.

In the development of other organs, a primitive gut tube is formed by endoderm as the body itself folds into a tube. The heart, too, begins as a simple tubular pump. Organ formation continues until the germ layers have given rise to the specific organs listed in Table 42.1. Figure 42.5 helps you relate the formation of vertebrate structures and organs to the three embryonic layers of cells: the ectoderm, the mesoderm, and the endoderm.

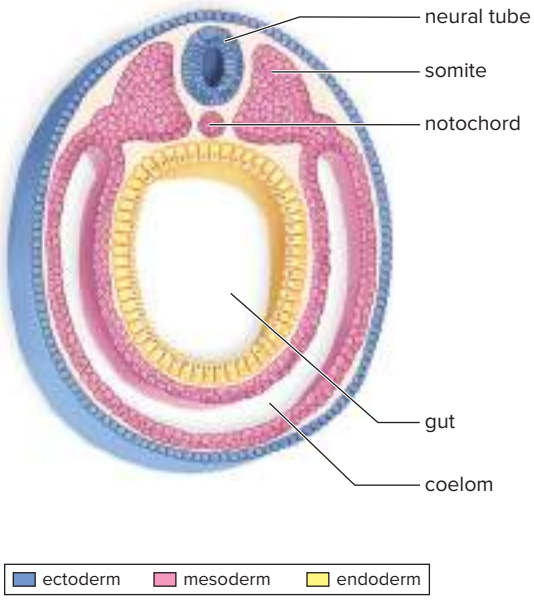


Figure 42.5 Vertebrate embryo, cross section. At the neurula stage, each of the germ layers, indicated by color (see key), can be associated with the later development of particular parts. The somites give rise to the muscles of each segment and to the vertebrae, which replace the notochord in vertebrates.

Check Your Progress 42.1

1. Describe the fast and slow blocks to polyspermy.
2. Name the germ layer of the gastrula that gives rise to the notochord, thyroid and parathyroid glands, nervous system, epidermis, skeletal muscles, kidneys, bones, and pancreas.
3. Identify the stage(s) of embryonic development in which cross sections of all chordate embryos closely resemble one another.

42.2 Developmental Processes

Learning Outcomes

Upon completion of this section, you should be able to

1. Distinguish between cellular differentiation and morphogenesis.
2. Define *cytoplasmic segregation* and *induction* and explain how these phenomena contribute to cellular differentiation.
3. Describe the major steps of morphogenesis in *Drosophila melanogaster.*

Development requires three interconnected processes: growth, cellular differentiation, and morphogenesis. **Cellular differentiation** occurs when cells become specialized in structure and function; that is, a muscle cell looks different and acts differently than a nerve cell. **Morphogenesis** produces the shape and form of the body. One of the earliest indications of morphogenesis is cell movement. Later, morphogenesis includes **pattern formation,** which means how tissues and organs are arranged in the body. **Apoptosis,** or programmed cell death (see Fig. 9.3), is an important part of pattern formation.

Developmental genetics has benefited from research using the roundworm *Caenorhabditis elegans* and the fruit fly *Drosophila melanogaster.* These organisms are referred to as model organisms, because the study of their development produced concepts that help us understand development in general.

ISTUDY

Cellular Differentiation

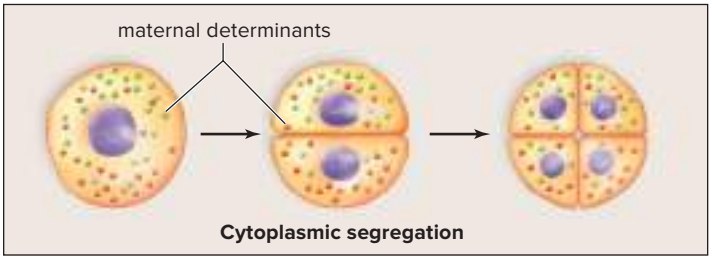
At one time, investigators mistakenly believed that different cell types, such as human liver cells and brain cells, must inherit different DNA sequences from the original single-celled zygote. Perhaps, they speculated, the genes are parceled out as development occurs, and that is why cells of the body have a different structure and function. We now know that is not the case; rather, every cell in the body (except for the sperm and unfertilized egg) has a full complement of genes.

The zygote is **totipotent;** it has the ability to generate the entire organism and, therefore, must contain all the instructions needed by any other specialized cell in the body. For the first few days of cell division, all the embryonic cells are totipotent. When the embryonic cells begin to specialize and lose their totipotency, they do not lose genetic information. In fact, our ability to clone mammals such as sheep, mice, and cats from specialized adult cells shows that every cell in an organism's body has the same collection of genes.

The answer to this puzzle becomes clear when we consider that only muscle cells produce the proteins myosin and actin; only red blood cells produce hemoglobin; and only skin cells produce keratin. In other words, we now know that specialization is not due to a parceling out of genes; rather, it is due to differential gene expression. Certain genes, but not others, are turned on (transcribed) in differentiated cells. In recent years, investigators have turned their attention to discovering the mechanisms that lead to differential gene expression. Two mechanisms—cytoplasmic segregation and induction—seem to be especially important.

Cytoplasmic Segregation

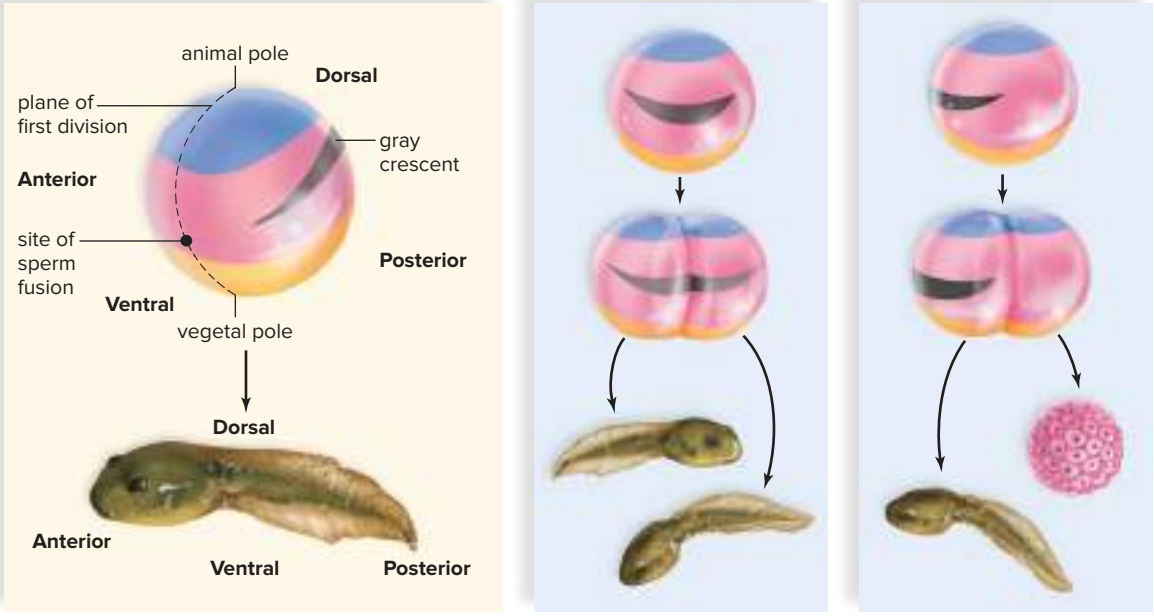
Differentiation must begin long before we can recognize specialized types of cells. Ectodermal, endodermal, and mesodermal cells in the gastrula look quite similar, but they must be different, because they develop into different organs. The egg is now known to contain substances in the cytoplasm called **maternal determinants,** which influence the course of development. **Cytoplasmic segregation** is the parceling out of maternal determinants as mitosis occurs:



An experiment conducted in 1935 showed that the cytoplasm of a frog's egg is not uniform. It is polar, having both an anterior/posterior axis and a dorsal/ventral axis, which can be correlated with the *gray crescent,* a gray area that appears after the sperm fertilizes the egg (Fig. 42.6*a*). If the gray crescent is divided equally by the first cleavage, each experimentally separated daughter cell develops into a complete embryo (Fig. 42.6*b*). However, if the zygote divides so that only one daughter cell receives the gray crescent, only that cell becomes a complete embryo (Fig. 42.6*c*). This experiment allowed scientists to speculate that the gray crescent must contain particular chemical signals that are needed for development to proceed normally.

Induction and Frog Experiments

As development proceeds, the specialization of cells and formation of organs are influenced not only by maternal determinants but also



a. Zygote of a frog is polar and has axes.

b. Each cell receives a part of the gray crescent.

c. Only the cell on the left receives the gray crescent.

Figure 42.6
Cytoplasmic influence on development. **a.** The zygote of a frog has anterior/posterior and dorsal/ventral axes that correlate with the position of the gray crescent. **b.** The first cleavage normally divides the gray crescent in half, and each daughter cell is capable of developing into a complete tadpole. **c.** But if only one daughter cell receives the gray crescent, then only that cell can become a complete embryo. This shows that maternal determinants are present in the cytoplasm of a frog's egg.

ISTUDY

by signals given off by neighboring cells. **Induction** is the ability of one embryonic tissue to influence the development of another tissue.

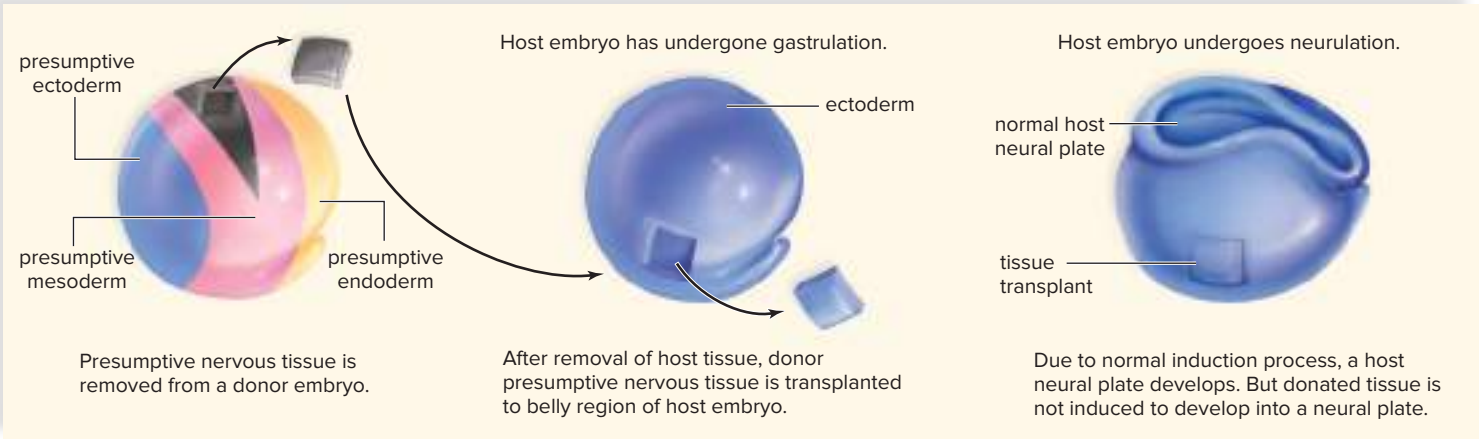
A frog embryo's gray crescent becomes the dorsal lip of the blastopore, where gastrulation begins. Because this region is necessary for complete development, the dorsal lip of the blastopore is called the *primary organizer.* The cells closest to the primary organizer become endoderm, those farther away become mesoderm, and those farthest away become ectoderm. This suggests that a molecular concentration gradient may act as a chemical signal to induce germ layer differentiation.

The gray crescent in the zygote of a frog marks the dorsal side of the embryo where the mesoderm becomes notochord and ectoderm becomes nervous system. In a classic experiment, researchers showed that presumptive (potential) notochord tissue induces the formation of the nervous system (Fig. 42.7). If presumptive nervous system tissue, located just above the presumptive notochord, is cut out and transplanted to the belly region of the embryo, it does not form a neural tube. But in contrast, if presumptive notochord tissue is cut out and transplanted beneath what would be belly ectoderm, this ectoderm does differentiate into neural tissue. Many other examples of induction are now known.

Induction in Caenorhabditis elegans

The tiny nematode *Caenorhabditis elegans* is only 1 mm long, and vast numbers can be raised in the laboratory either in petri dishes or a liquid medium. The worm is hermaphroditic, and self-fertilization is the rule. Therefore, even though induced mutations may be recessive, the next generation will yield individuals that are homozygous recessive and show the mutation. Many modern genetic studies have been performed on *C. elegans,* and the entire genome has been sequenced. Individual genes have been altered and cloned and their products injected into cells or extracellular fluid.

As a result of genetic studies, much has been learned about *C. elegans.* Development of *C. elegans* takes only 3 days, and the adult worm contains only 959 cells. Investigators have been able to watch the process from beginning to end, because the worm is



a.

b.

Figure 42.7 Control of nervous system development. **a.** In this experiment, the presumptive nervous system (*blue*) does not develop into the neural plate if moved from its normal location. **b.** In this experiment, the presumptive notochord (*pink*) can cause even belly ectoderm to develop into the neural plate (*blue*). This shows that the notochord induces ectoderm to become a neural plate, most likely by sending out chemical signals.

ISTUDY

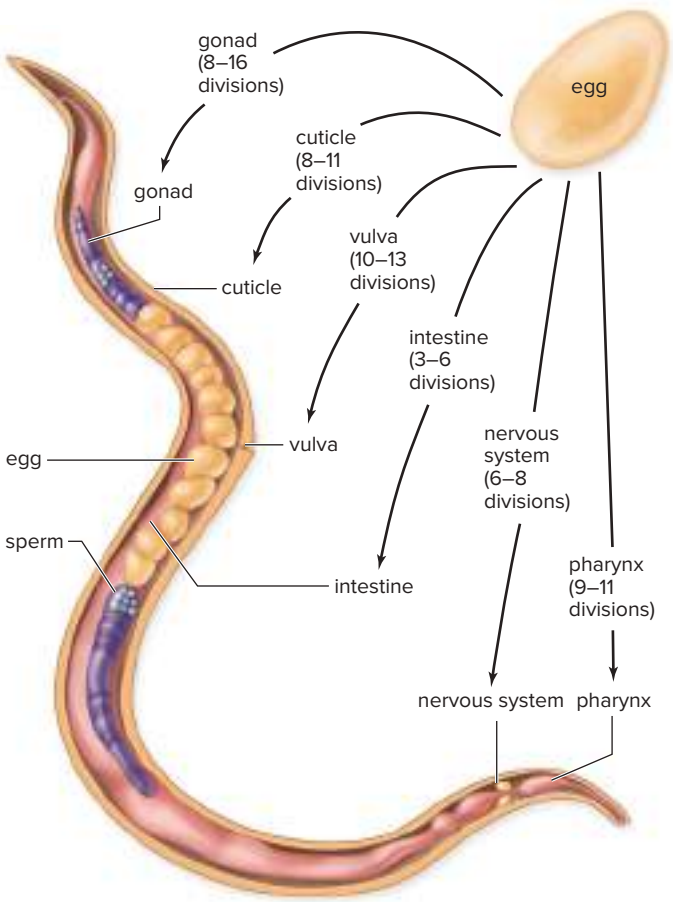


Figure 42.8 Development of *C. elegans*, a nematode. A fate map of the worm showing that as cells arise by cell division, they are destined to become particular structures.

transparent. *Fate maps* have been developed that show the destiny of each cell as it arises following successive cell divisions (Fig. 42.8).

Some investigators have studied in detail the development of the worm's vulva, a pore through which eggs are laid. A cell called the anchor cell induces the vulva to form. The cell closest to the anchor cell receives the most inducer and becomes the inner vulva. This cell, in turn, produces another inducer, which acts on its two neighboring cells, and they become the outer vulva. The inducers are growthlike factors that alter the metabolism of the receiving cell and activate particular genes. Work with *C. elegans* has shown that induction requires the transcriptional regulation of genes in a particular sequence.

Morphogenesis

An animal achieves its ordered and complex body form through morphogenesis, which requires that cells associate to form tissues, and tissues give rise to organs. Pattern formation is the process that enables morphogenesis. In pattern formation, cells of the embryo divide and differentiate, taking up orderly positions in tissues and organs.

Although animals display an amazingly diverse array of morphologies, or body forms, most share common sets of genes that direct pattern formation. When pattern formation has ensured that key cells are properly arranged, then morphogenesis, the construction of the ultimate body form, can take place.

Morphogenesis in Drosophila melanogaster

Much of the study of pattern formation and morphogenesis has been done using relatively simple models, such as the fruit fly *Drosophila melanogaster.* A *Drosophila* egg is only about 0.5 mm long, and it develops into an adult in about 2 weeks. The fly's genome is considerably smaller than that of humans or even mice (see Table 14.1). However, the genes that direct pattern formation appear to be highly conserved among animals with segmented body morphologies, including humans.

As pattern formation occurs in *Drosophila,* the embryonic cells begin to express genes differently in graded, periodic, and eventually striped arrangements. Boundaries between large body regions are established first, before the refinement of smaller, subdivided parts can take place.

The Anterior/Posterior Axes The first step in *Drosophila* pattern formation and morphogenesis begins with the establishment of anteroposterior polarity, meaning that the anterior (head) and posterior (abdomen) ends are different from one another. Such polarity is present in the egg before it is fertilized by a sperm.

Egg polarity results from maternal determinants, mRNAs that are deposited in specific positions within the egg while it is still in the ovary. The protein products of these genes diffuse away from the areas of their highest concentration in the embryo, forming gradients that influence patterns of tissue development. These proteins are also known as *morphogens* due to their crucial influence in morphogenesis. For example, the Bicoid protein is most concentrated anteriorly, where it prevents the formation of the posterior region. The Nanos protein is most concentrated posteriorly, and it is required for abdomen formation.

The Segmentation Pattern Once the anterior and posterior ends of the embryo have been established, a group of zygotic genes called *gap* genes come into play. The task of gap genes is to divide the anteroposterior axis into broad regions (Fig. 42.9*a*). They are called gap genes because mutations in these genes result in gaps in the embryo, where large blocks of segments are missing. Gap genes are temporarily activated by the gradients of anterior and posterior morphogens and in turn activate the *pair-rule* genes.

The pair-rule genes are expressed periodically, in alternating stripes (Fig. 42.9*b*). They "rough out" a preliminary segmentation pattern along the anteroposterior axis. The products of pair-rule genes may stimulate or suppress the expression of other genes, particularly the *segment-polarity* genes. These genes ensure that each segment has boundaries, with distinct anterior and posterior halves. The segment-polarity genes are also expressed in a striped fashion, but with twice as many stripes as the pair-rule genes (Fig. 42.9*c*). Mutations in segment-polarity genes result in the loss of one part of each segment, as well as the duplication of another portion of the same segment.

ISTUDY

a. Protein products of gap genes

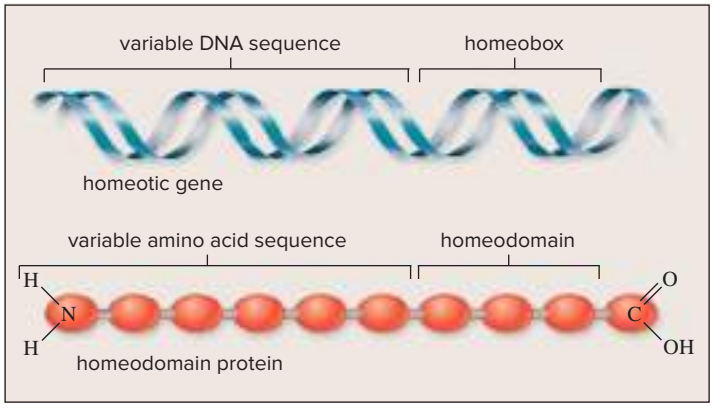
b. Protein products of pair-rule genes

c. Protein products of segment-polarity genes

Figure 42.9 Development in *Drosophila*, a fruit fly. **a.** The different colors show that two different gap gene proteins are present from the anterior to the posterior end of an embryo. **b.** The green stripes show that a pair-rule gene is being expressed as segmentation of the fly occurs. **c.** Now segment-polarity genes help bring about the division of each segment into an anterior end and a posterior end. (a–c): ©Steve Paddock, Howard Hughes Medical Institute

Homeotic Genes The **homeotic genes** are often referred to as *selector genes,* because they select for segmental identity—in other words, they dictate which body parts arise from the segments. Mutations in homeotic genes may result in the development of body parts in inappropriate areas, such as legs instead of antennae, or wings instead of tiny, balancing organs called halteres (Fig. 42.10*a*). Such alterations in morphology are known as *homeotic transformations.*

Interestingly, homeotic genes in *Drosophila* and other organisms have all been found to share a structural feature called a **homeobox.** (*HOX,* the term used for mammalian homeotic genes, is a shortened form of *homeobox.*) A homeobox is a sequence of nucleotides that encodes a 60-amino-acid sequence called a **homeodomain:**



a. SEM 50×

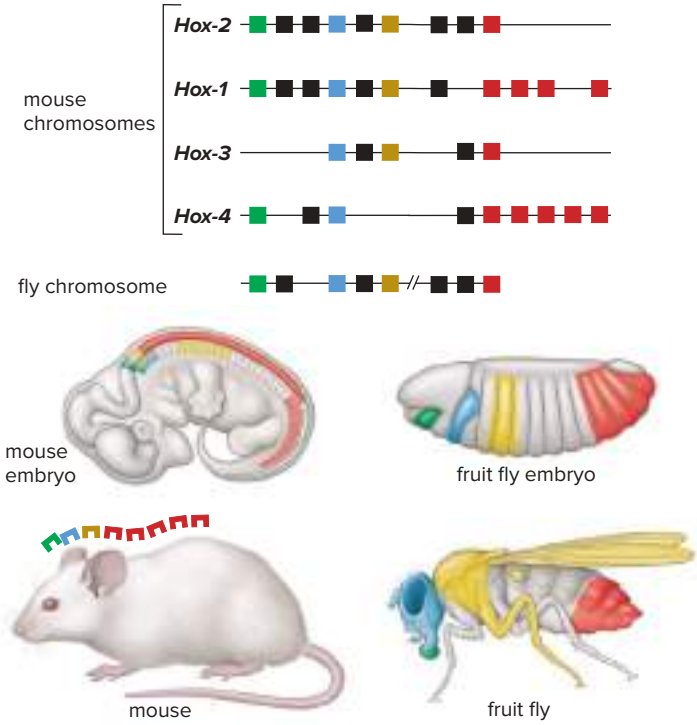


Figure 42.10 Pattern formation in *Drosophila*. Homeotic genes control pattern formation, an aspect of morphogenesis. **a.** If homeotic genes are activated at inappropriate times, abnormalities such as a fly with four wings occur. **b.** The green, blue, yellow, and red colors show that homologous homeotic genes occur on four mouse chromosomes and on a fly chromosome in the same order. These genes are color coded to the region of the embryo, and therefore the adult, where they regulate pattern formation. The black boxes are homeotic genes that are not identical between the two animals. In mammals, homeotic genes are called *HOX* genes. (a): David Scharf/ Science Source

ISTUDY

The homeodomain is a functionally important part of the protein encoded by a homeotic gene. Homeotic genes code for transcription factors, proteins that bind to regulatory regions of DNA and determine whether or not specific target genes are turned on. The homeodomain is the DNA-binding portion of the transcription factor, although the other, more variable sequences of a transcription factor determine which target genes are turned on. It is thought that the homeodomain proteins encoded by homeotic genes direct the activities of various target genes involved in morphogenesis, such as those involved in cell-to-cell adhesion. In the end, this orderly process determines the morphology of particular segments.

The importance of homeotic genes is underscored by the finding that the homeotic genes are highly conserved, being present in the genomes of many organisms, including mammals such as mice and even humans. Most homeotic genes have their loci on the same chromosome in *Drosophila,* while in mammals there are four clusters that reside on different chromosomes.

Notice that, in both flies and mammals, the position of the homeotic genes on the chromosome matches their anterior-to-posterior expression pattern in the body (Fig. 42.10*b*). The first gene clusters determine the final development of anterior segments, whereas those later in the sequence determine the final development of posterior segments of the animal's body.

Mutations in homeotic genes have effects in the mammalian body similar to those of the homeotic transformations observed in *Drosophila.* For instance, mutations in two adjacent *HOX* genes in the mouse result in shortened forelimbs that are missing the radius and ulna bones. In humans, mutations in a different *HOX* gene cause synpolydactyly, a rare condition in which there are extra digits (fingers and toes), some of which are fused to their neighbors.

Apoptosis We have already discussed the importance of apoptosis (programmed cell death) in the normal, day-to-day operation of the immune system (for example, see Fig. 33.8) and in the regulation of the cell cycle (see Fig. 9.3). Apoptosis is also an important part of morphogenesis. In tadpoles, for example, apoptosis is largely responsible for the disappearance of the tail. During human development, apoptosis is needed to remove the webbing between fetal fingers and toes.

Fate maps of *C. elegans* (see Fig. 42.8) indicate that apoptosis occurs in 131 cells as development takes place. When a cell-death signal is received, an inhibiting protein becomes inactive, triggering an apoptotic cascade that ends in enzymes destroying the cell.

Check Your Progress 42.2

1. Name and define two mechanisms of cellular differentiation.
2. Define the term *morphogen.*
3. Describe the function of the homeobox sequence in a homeotic gene.

42.3 Human Embryonic and Fetal Development

Learning Outcomes

Upon completion of this section, you should be able to

1. Name the membranes surrounding the human embryo and list their functions.
2. Chronologically list the major events that occur during embryonic and fetal development.
3. Describe the structure and functions of the placenta.

In humans, the length of time from conception (fertilization followed by implantation in the endometrium) to birth (parturition) is approximately 9 months. It is customary to calculate the time of birth by adding 280 days to the start of the last menstruation because this date is usually known, whereas the day of fertilization is usually unknown. Because the time of birth is influenced by so many variables, only about 5% of babies actually arrive on the forecasted date.

Human development is often divided into embryonic development (months 1 and 2) and fetal development (months 3 through 9). During **embryonic development,** the major organs are formed, and during fetal development, these structures become larger and are refined.

Development can also be divided into *trimesters.* Each trimester can be characterized by specific developmental accomplishments. During the first trimester, embryonic and early fetal development occur. The second trimester is characterized by the development of organs and organ systems. By the end of the second trimester, the fetus appears distinctly human. In the third trimester, the fetus grows rapidly and the major organ systems become functional.

Extraembryonic Membranes

Before we consider human development chronologically, we must understand the placement of **extraembryonic membranes** (L. *extra,* "on the outside"). Extraembryonic membranes are best understood by considering their function in reptiles and birds. In reptiles, these membranes made development on land first possible. If an embryo develops in the water, the water supplies oxygen for the embryo and takes away waste products. The surrounding water prevents desiccation, or drying out, and provides a protective cushion. For an embryo that develops on land, all these functions are performed by the extraembryonic membranes.

In the chick, the extraembryonic membranes develop from extensions of the germ layers, which spread out over the yolk. Figure 42.11 shows the chick surrounded by the membranes. The **chorion** (Gk. *chorion,* "membrane") lies next to the shell and carries on gas exchange. The **amnion** (Gk. *amnion,* "membrane around fetus") contains the protective amniotic fluid, which bathes the developing embryo. The **allantois** (Gk. *allantos,* "sausage") collects nitrogenous wastes, and the **yolk sac** surrounds the remaining yolk, which provides nourishment.

ISTUDY

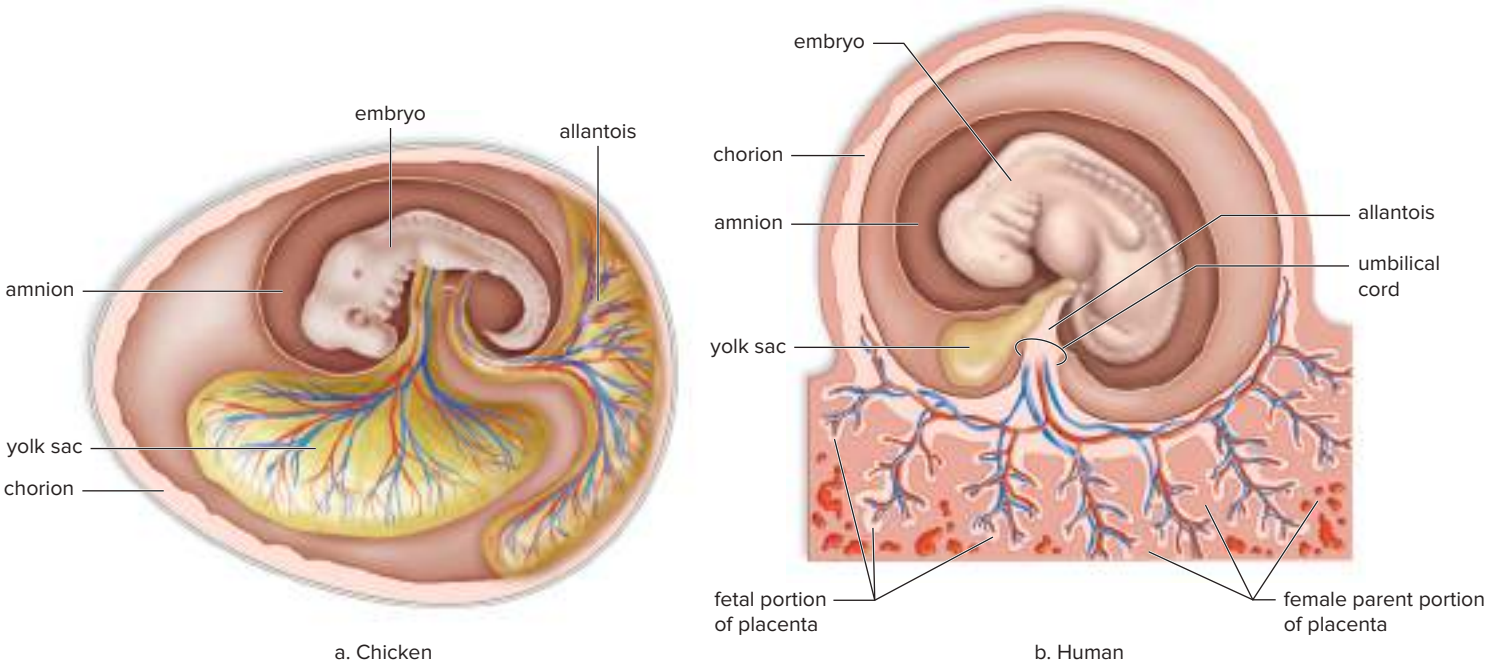


Figure 42.11 Extraembryonic membranes. Extraembryonic membranes, which are not part of the embryo, are found during the development of (**a**) chickens and (**b**) humans.

The function of the extraembryonic membranes in humans has been modified to suit internal development. Their presence, however, shows that we are related to the reptiles.

The chorion develops into the fetal half of the placenta, the organ that provides the embryo/fetus with nourishment and oxygen and takes away its wastes. Blood vessels within the chorionic villi are continuous with the umbilical blood vessels. The blood vessels of the allantois become the umbilical blood vessels, and the allantois accumulates the small amount of urine produced by the fetal kidneys and later gives rise to the urinary bladder. The yolk sac, which lacks yolk, is the first site of blood cell formation. The amnion contains fluid to cushion and protect the embryo, which develops into a fetus.

It is interesting to note that all chordate animals develop in water—either in bodies of water or surrounded by amniotic fluid within a shell or uterus.

Embryonic Development

Embryonic development comprises the first 2 months of development.

The First Week

Fertilization usually occurs in the upper third of the uterine tube (Fig. 42.12). Cleavage begins 30 hours after fertilization and continues as the embryo passes through the uterine tube to the uterus. By the time the embryo reaches the uterus on the third day, it is a morula. By about the fifth day, the morula has transformed into a blastula, and in mammals, a blastocyst then forms.

The **blastocyst** has a fluid-filled cavity, a single layer of outer cells called the **trophoblast** (Gk. *trophe,* "food"; *blastos,* "bud"), and an inner cell mass. The early function of the trophoblast is to provide nourishment for the embryo. Later, the trophoblast, reinforced by a layer of mesoderm, gives rise to the chorion, one of the extraembryonic membranes (see Fig. 42.11). The inner cell mass eventually becomes the embryo, which develops into a fetus.

The Second Week

At the end of the first week, the embryo begins the process of **implantation** in the wall of the uterus. The trophoblast secretes enzymes to digest away some of the tissue and blood vessels of the endometrium of the uterus (Fig. 42.12). The embryo is now about the size of the period at the end of this sentence. The trophoblast begins to secrete **human chorionic gonadotropin (HCG),** the hormone that is the basis for the pregnancy test and that maintains the corpus luteum past the time it normally disintegrates (see Fig. 41.9). Because of this, the endometrium is maintained, and menstruation does not occur.

As the week progresses, the inner cell mass detaches itself from the trophoblast, and two more extraembryonic membranes form (Fig. 42.13*a*). The yolk sac, which forms below the embryonic disk, has no nutritive function as it does in chickens, but it is the first site of blood cell formation. However, the amnion and its cavity are where the embryo (and then the fetus) develop. In humans, amniotic fluid acts as an insulator against cold and heat and absorbs shock, such as that caused when the female parent exercises.

Gastrulation occurs during the second week. The inner cell mass now has flattened into the **embryonic disk,** composed of two

ISTUDY

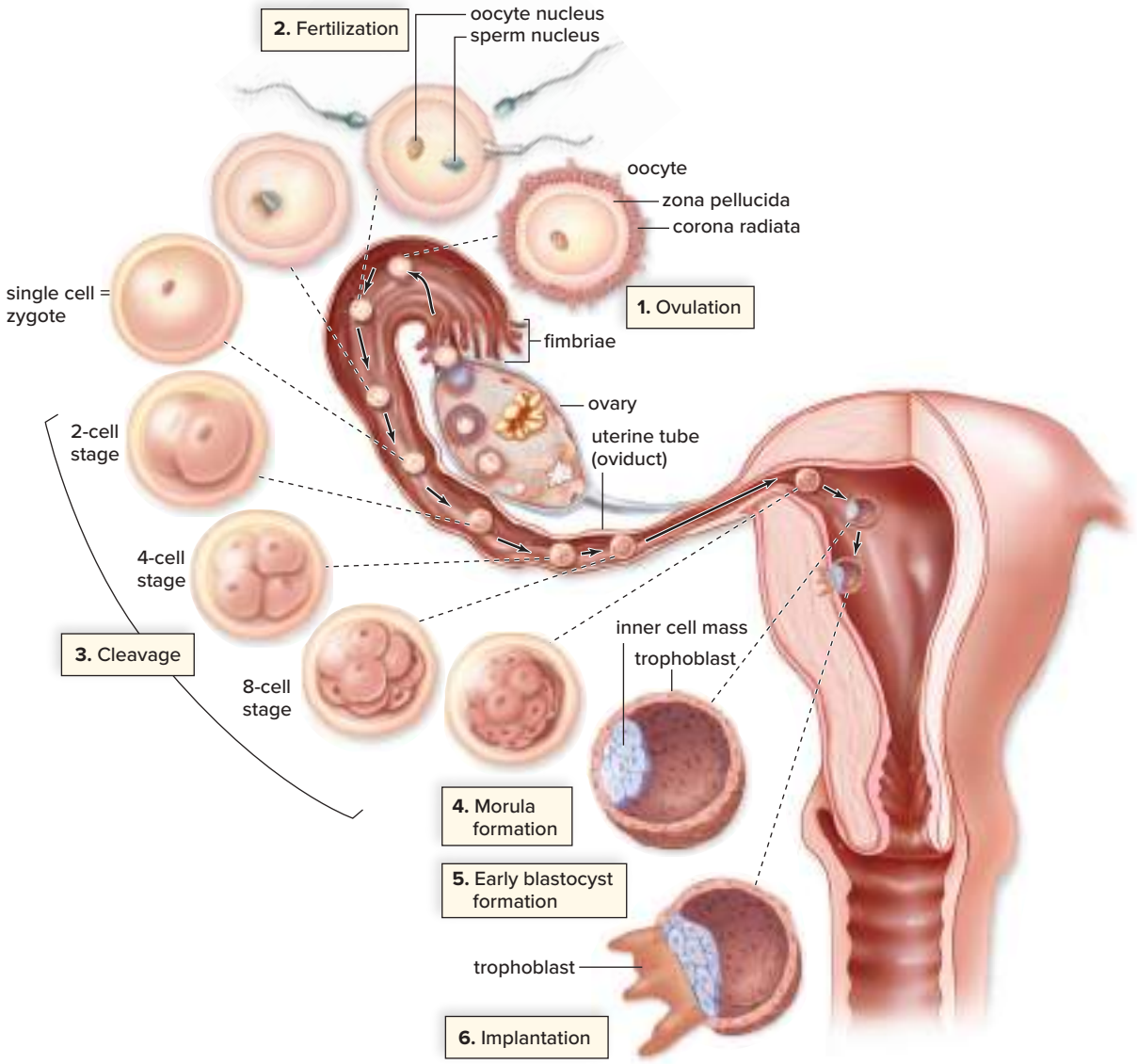


Figure 42.12 Human development before implantation. **(1)** At ovulation, the secondary oocyte leaves the ovary. A single sperm nucleus enters the oocyte, and **(2)** fertilization of the egg occurs in the uterine tube. As the zygote moves along the uterine tube, it undergoes **(3)** cleavage to produce **(4)** a morula. **(5)** The blastocyst forms and **(6)** implants itself in the uterine lining.

layers of cells: ectoderm above and endoderm below. Once the embryonic disk elongates to form the primitive streak, the third germ layer, mesoderm, forms by invagination of cells along the streak. The trophoblast is reinforced by mesoderm and becomes the chorion (Fig. 42.13*b*). It is possible to relate the development of future organs to these germ layers (see Table 42.1).

The Third Week

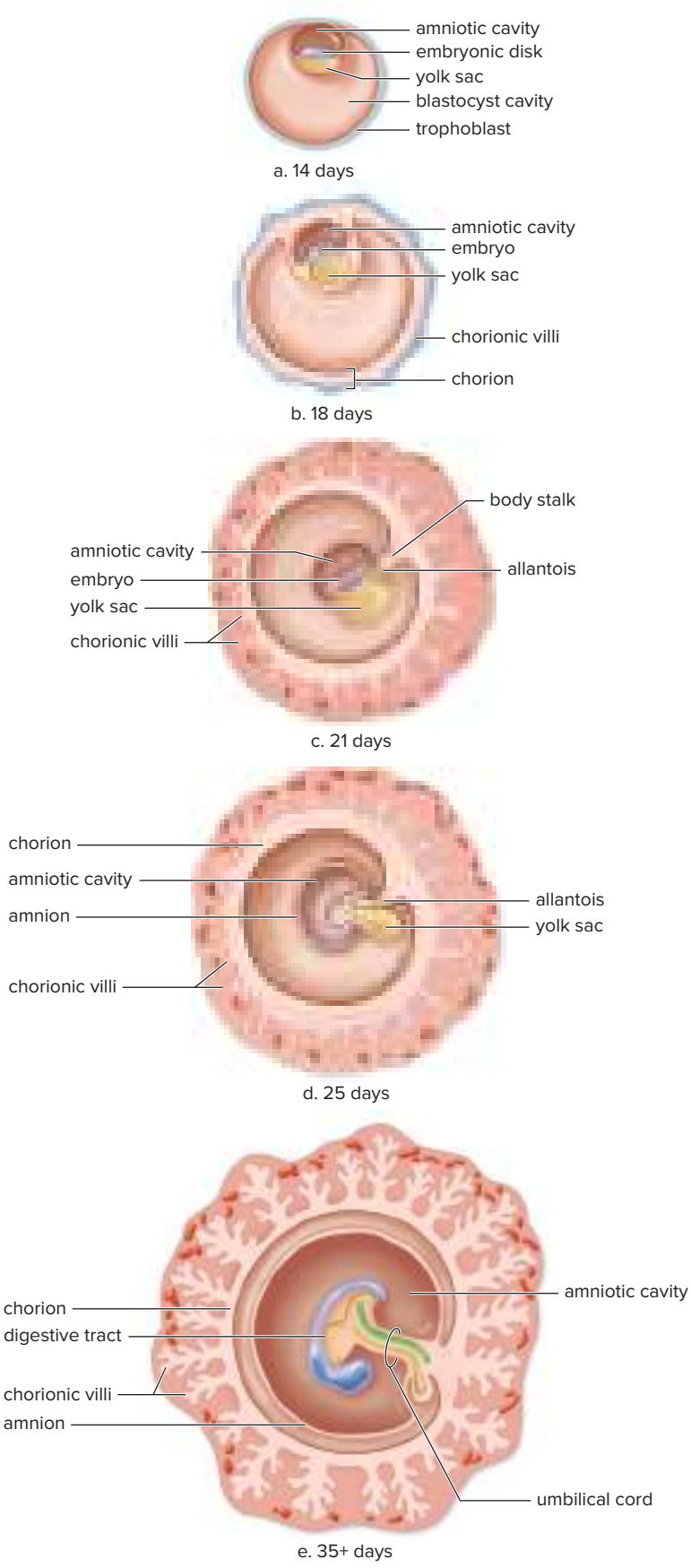
Two important organ systems make their appearance during the third week. The nervous system is the first organ system to be visually evident. At first, a thickening appears along the entire dorsal length of the embryo; then the neural folds appear. When the neural folds meet at the midline, the neural tube, which later develops into the brain and the nerve cord, is formed (see Fig. 42.4). After the notochord is replaced by the vertebral column, the nerve cord is called the spinal cord.

Development of the heart begins in the third week and continues into the fourth week. At first, there are right and left heart tubes; when these fuse, the heart begins pumping blood, even though the chambers of the heart are not fully formed. The veins enter posteriorly, and the arteries exit anteriorly from this largely tubular heart, but later the heart twists, so that all major blood vessels are located anteriorly.

The Fourth and Fifth Weeks

At 4 weeks, the embryo is barely larger than the height of this print. A bridge of mesoderm called the body stalk connects the caudal (tail) end of the embryo with the chorion, which has tree-like projections called **chorionic villi** (Gk. *chorion,* "membrane";

ISTUDY



L. *villus,* "shaggy hair") that increase surface area for contact with the placenta in the female parent (Fig. 42.13*c, d, e*) . The fourth extraembryonic membrane, the allantois, is contained within this stalk, and its blood vessels become the umbilical blood vessels. These structures form the **umbilical cord** (L. *umbilicus,* "navel"), which connects the developing embryo to the placenta (Fig. 42.13*e*).

Little protrusions called *limb buds* appear (Fig. 42.14); later, the arms and legs develop from the limb buds, and even the hands and feet become apparent. At the same time—during the fifth week—the head enlarges, and the developing eyes, ears, and nose are discernable.

The Sixth Through Eighth Weeks

During the sixth through eighth weeks of development, the embryo becomes easily recognizable as human. Concurrent with brain development, the head achieves its normal relationship with the body as a neck region develops. The nervous system is developed well enough to permit reflex actions, such as a startle response to touch. At the end of this period, the embryo is about 38 mm (1.5 in) long and weighs less than 1 g, even though all organ systems are established.

The Structure and Function of the Placenta

The **placenta** is a mammalian structure that functions in gas, nutrient, and waste exchange between the embryonic (and later, fetal) and female parent's cardiovascular systems. The placenta begins formation once the embryo is fully implanted. At first, the entire chorion has chorionic villi that project into endometrium. Later, these disappear in all areas except where the placenta develops. By the tenth week, the placenta (Fig. 42.15) is fully formed and is producing progesterone and estrogen. These hormones have two effects:

- They prevent any new follicles from maturing because of negative feedback control of the hypothalamus and anterior pituitary.
- They maintain the lining of the uterus, so now the corpus luteum is not needed. No menstruation occurs during pregnancy.

The placenta has a fetal side contributed by the chorion and a side consisting of uterine tissues contributed by the female parent. Notice in Figure 42.15 how the chorionic villi are surrounded by the female parent's blood, yet the female parent's and fetal blood

Figure 42.13 Human embryonic development. **a.** At first, the embryo contains no organs, only tissues. The amniotic cavity is above the embryo, and the yolk sac is below. **b.** The chorion develops villi, the structures so important to the exchange between the female parent and child. **c, d.** The allantois and yolk sac, two more extraembryonic membranes, are positioned inside the body stalk as it becomes the umbilical cord. **e.** At 35+ days, the embryo has a head region and a tail region. The umbilical cord takes blood vessels between the embryo and the chorion (placenta).

ISTUDY

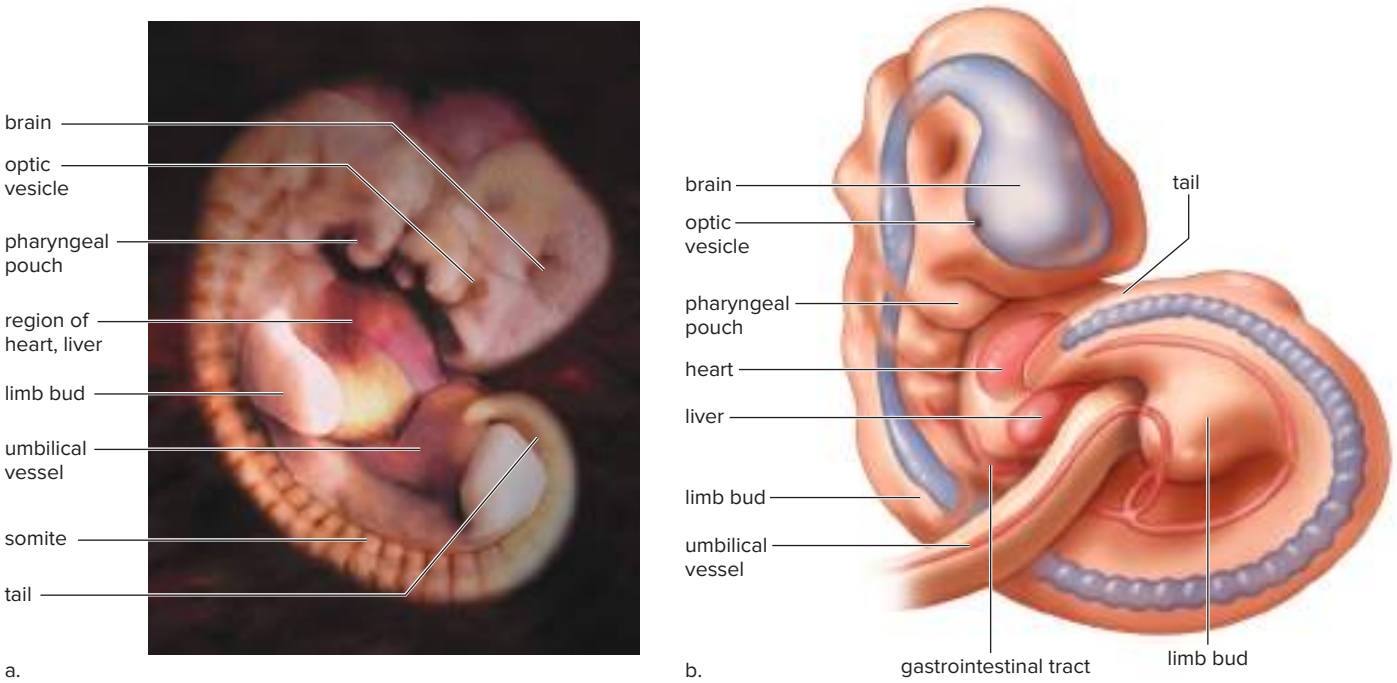


Figure 42.14 Human embryo at beginning of fifth week. **a.** Scanning electron micrograph. **b.** The embryo is curled, so that the head touches the heart, the two organs whose development is farther along than the rest of the body. The organs of the gastrointestinal tract are forming, and the arms and legs develop from the bulges called limb buds. The tail is an evolutionary remnant; its bones regress and become those of the coccyx (tailbone). The pharyngeal arches become functioning gills only in fishes and amphibian larvae; in humans, the first pair of pharyngeal pouches becomes the auditory tubes. The second pair becomes the tonsils, while the third and fourth become the thymus and the parathyroid glands, respectively. (a): Anatomical Travelogue/Science Source

do not mix under normal conditions, because exchange always takes place across plasma membranes.

The umbilical cord is the lifeline of the fetus because it contains the umbilical arteries and vein, which transport waste molecules (carbon dioxide and urea) to the placenta for disposal into the female parent's blood and take oxygen and nutrient molecules from the placenta to the rest of the fetal circulatory system. If the placenta prematurely tears from the uterine wall, the lives of the fetus and the female parent are endangered.

Late in fetal development, and peaking just before birth, the female parent's antibodies are transported across the placenta, into the fetal circulation. This helps ensure that the newborn will be protected against common pathogens and other microbes to which the female parent has been exposed, until the newborn's immune system can mature and respond on its own. Harmful chemicals can also cross the placenta, as discussed in the Biological Systems feature, "Preventing and Testing for Birth Defects."

Fetal Development and Birth

Fetal development (months 3–9) is marked by an extreme increase in size. Weight multiplies 600 times, going from less than 28 g to 3 kg. During this time, too, the fetus grows to about 50 cm in length. The genitalia appear in the third month, so it is possible to tell if the fetus is male or female.

Soon, hair, eyebrows, and eyelashes add finishing touches to the face and head. In the same way, fingernails and toenails complete the hands and feet. A fine, downy hair (lanugo) covers the limbs and trunk, only to disappear later. The fetus looks very old, because the skin is growing so fast that it wrinkles. A waxy, almost cheeselike substance (vernix caseosa) (L. *vernix,* "varnish"; *caseus,* "cheese") protects the wrinkly skin from the watery amniotic fluid.

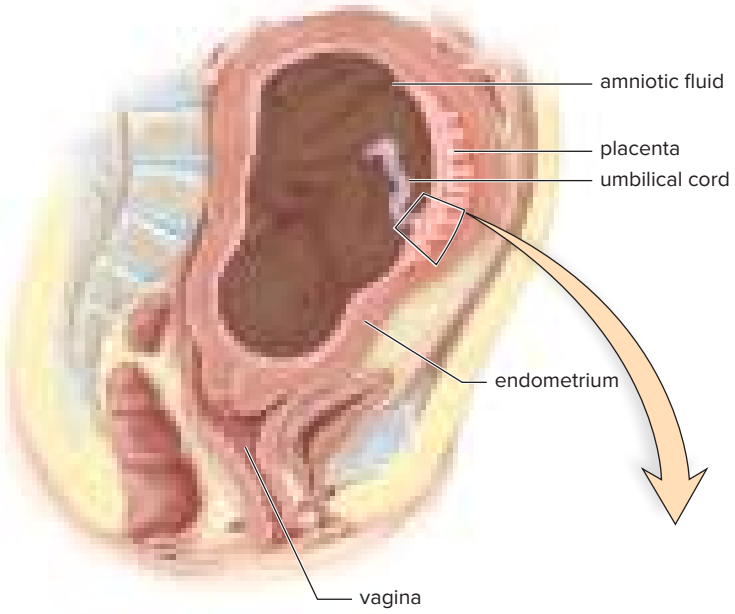
From about the fourth month on, the female parent can feel movements of the fetal limbs. At about the same time, the fetal heartbeat can be heard through a stethoscope. Conversely, the fetus is able to hear and respond to sounds by about 18 weeks. A fetus born at 24 weeks has a chance of surviving, although the lungs are still immature and often cannot capture oxygen adequately. As a fetus rapidly grows during the third trimester, its chances of surviving being born a month or two prematurely increase dramatically.

The Stages of Birth

When the fetal brain is sufficiently mature, the fetal hypothalamus stimulates the pituitary to release ACTH, causing the adrenal cortex to release androgens into the bloodstream. These diffuse into the placenta, which uses androgens as a precursor for estrogens, hormones that stimulate the production of prostaglandins and oxytocin. All three of these molecules cause the uterus to contract and expel the fetus.

The process of birth (parturition) includes three stages. During the first stage, the cervix dilates to allow passage of the baby's head and body. The amnion usually bursts about this time, an event known as the female parent's "water breaking." During the second stage, the baby is born and the umbilical cord is cut. During the third stage, the placenta is delivered (Fig. 42.16).

ISTUDY



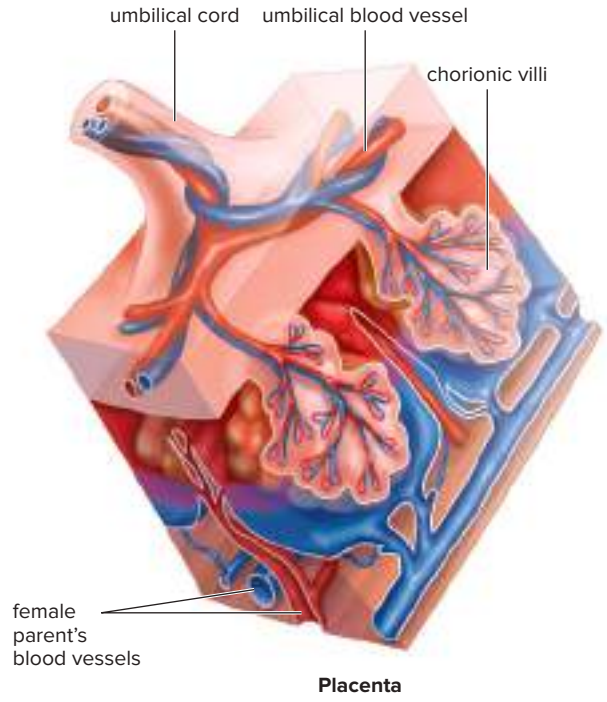
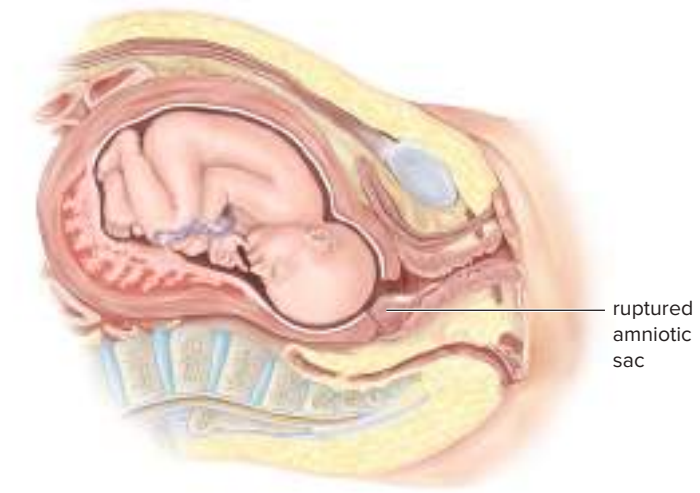


Figure 42.15 Anatomy of the placenta in a fetus at 6–7 months. The placenta is composed of both the fetal and female parent's tissues. Chorionic villi penetrate the uterine lining and are surrounded by the female parent's blood. The exchange of molecules between the fetal and female parent's blood takes place across the walls of the chorionic villi.

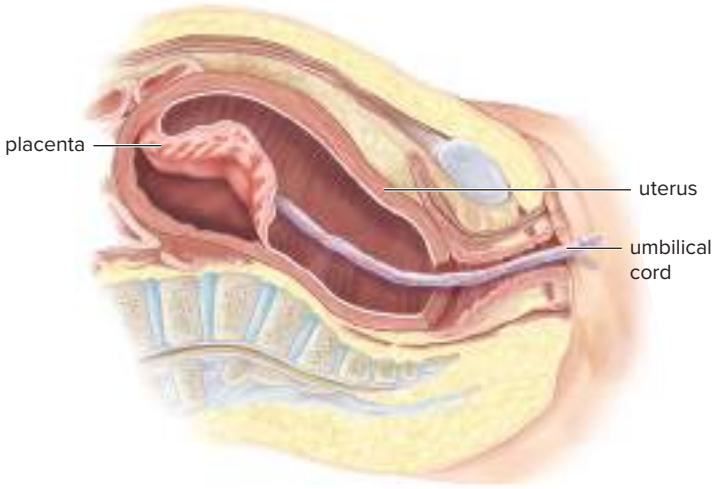
Check Your Progress 42.3

1. Identify the location where fertilization usually occurs in the human female.
2. Name the extraembryonic membrane that gives rise to each of the following: umbilical blood vessels, the first blood cells, and the fetal half of the placenta.
3. Describe the major changes that occur in the human embryo between the third and fifth weeks of development.



a. First stage of birth: Cervix dilates

b. Second stage of birth: Baby emerges



c. Third stage of birth: Afterbirth is expelled

Figure 42.16 Three stages of parturition. **a.** Position of fetus just before birth begins. **b.** Dilation of cervix and birth of baby. **c.** Expulsion of afterbirth.

ISTUDY

THEME Biological Systems

Preventing and Testing for Birth Defects

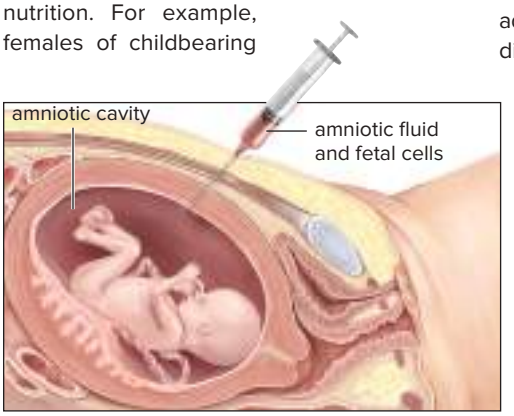
Birth defects, or congenital disorders, are abnormal conditions that are present at birth. According to the Centers for Disease Control and Prevention (CDC), about 1 in 33 babies born in the United States has a birth defect. Those due to genetics can sometimes be detected before birth by one of the methods shown in Figure 42A; however, these methods are not without risk.

Some birth defects are not serious, and not all can be prevented. But females can take steps to increase their chances of delivering a healthy baby.

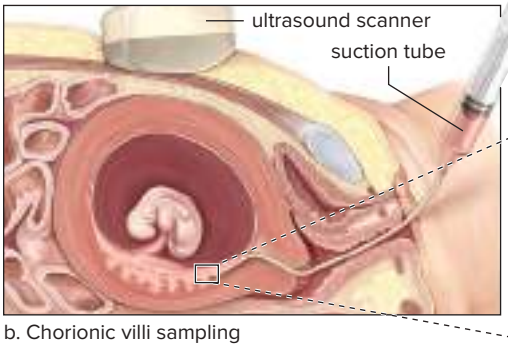
Eat a Healthy Diet

Certain birth defects occur because the developing embryo does not receive sufficient nutrition. For example, females of childbearing age are urged to make sure they consume adequate amounts of the vitamin folic acid in order to prevent neural tube defects, such as spina bifida and anencephaly. In spina bifida, part of the vertebral column fails to develop properly and cannot adequately protect the delicate spinal cord. With anencephaly, most of the fetal brain fails to develop. Infants with anencephaly are stillborn or survive for only a few days after birth.

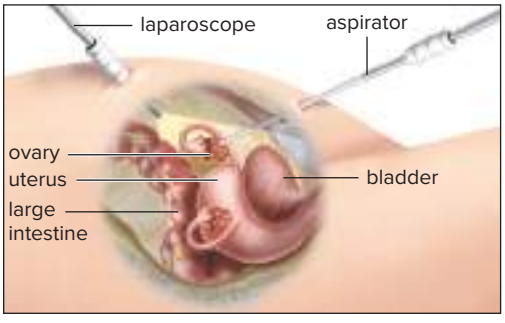
Fortunately, folic acid is plentiful in leafy green vegetables, nuts, and citrus fruits. It is also present in multivitamins, and many breads and cereals are fortified with it. The CDC recommends that all females of childbearing age get at least 400 micrograms of folic acid every day through supplements, in addition to eating a healthy, folic acid–rich diet. Unfortunately, neural tube birth defects can occur just a few weeks after conception, when many females are still unaware that they are pregnant—especially if the pregnancy is unplanned.



a. Amniocentesis



b. Chorionic villi sampling



c. Preimplantation genetic diagnosis

Figure 42A Three methods for genetic defect testing before birth. About 20% of all birth defects are due to genetic or chromosomal abnormalities, which may be detected before birth. **a.** Amniocentesis is usually performed from the fifteenth to the seventeenth weeks of pregnancy. **b.** Chorionic villi sampling is usually performed from the eighth to the twelfth weeks of pregnancy. **c.** Preimplantation genetic diagnosis can be performed prior to in vitro fertilization, either on oocytes that have been collected from the female or on the early embryo.

Avoid Alcohol, Smoking, and Illicit Drug Use

Alcohol consumption during pregnancy is a leading cause of birth defects. In severe instances, the baby is born with fetal alcohol syndrome (FAS), estimated to occur in 0.5 to 3.0 of every 1,000 live births in the United States. Children with FAS are frequently underweight, have an abnormally small head, abnormal facial development, and intellectual disability. As they mature, children with FAS often exhibit short attention span, impulsiveness, and poor judgment, as well as serious difficulties with learning and memory. Heavy alcohol use also reduces a female's folic acid level, increasing the risk of the neural tube defects just described.

Cigarette smoking causes many birth defects. Babies born to females who smoke typically have low birth weight, and are more likely to have defects of the face, heart, and brain than those born to females who do not smoke.

Illegal drugs should also be avoided. For example, cocaine causes blood pressure fluctuations that deprive the fetus of oxygen. Babies who are exposed to cocaine may have problems with vision and coordination, and they may have intellectual disability.

Alert Medical Personnel If You Are or May Be Pregnant

Several medications that are safe for healthy adults may pose a risk to a developing fetus. If females who are pregnant need to be immunized, they are usually given killed or inactivated forms of the vaccine, because live forms often present a danger to the fetus.

Because the rapidly dividing cells of a developing embryo or fetus are very susceptible to damage from radiation, females who are pregnant should avoid unnecessary X-rays. If a medical X-ray is unavoidable, the female should notify the X-ray technician, so the fetus can be protected as much as possible—for example, by draping a lead apron over the female's abdomen if this area is not targeted for X-ray examination.

Avoid Infections that Cause Birth Defects

Certain pathogens, such as the virus that causes rubella (German measles), can result in birth defects. In the past, this virus caused many birth defects, specifically intellectual disability, deafness, blindness, and heart defects. Rubella is much less of a danger in more-developed countries today, because most females have been vaccinated against rubella as children. Other infections that can cause birth defects include toxoplasmosis, herpes simplex, and cytomegalovirus.

Questions to Consider

1. Why is it likely that the actual rate of birth defects is higher than what is reported?
2. Besides tobacco, alcohol, and illegal drugs, what other potential risks should a female who is pregnant avoid? Why?

ISTUDY

42.4 The Aging Process

Learning Outcomes

Upon completion of this section, you should be able to

1. Discuss the effects of aging on various body systems.
2. Compare and contrast the preprogrammed theories with the damage accumulation theories of aging.

Aging and death are as much a part of biology as are development and birth. If animals—and indeed, all forms of life—did not age and eventually die, their exponentially increasing numbers would presumably overwhelm the ability of the Earth to support life. Moreover, aging and death are a critical part of the evolutionary process, because animals that age and die are continuously replaced by individuals with new genetic combinations that can adapt to environmental changes that occur.

Why Do We Age?

Aging is a complex process affected by multiple factors. Most scientists who study gerontology believe that aging is partly genetically preprogrammed. This idea is supported by the observation that longevity runs in families—that is, the children of long-lived parents tend to live longer than those of short-lived parents. As would also be expected, studies show that identical twins have a more similar life span than nonidentical twins. In this section, we will focus primarily on the hypotheses of cellular aging and damage accumulation.

Cellular Aging

Hormones Many laboratory studies of aging have been performed in the nematode *Caenorhabditis elegans,* in which single-gene mutations have been shown to influence the life span. For example, mutations that decrease the activity of a hormone receptor similar to the insulin receptor more than double the life span of the worms, which also behave and look like much younger worms throughout their prolonged lives. Interestingly, small-breed dogs, such as poodles and terriers, which may live 15 to 20 years, have lower levels of an analogous receptor compared to large-breed dogs that live 6 to 8 years.

Telomeres Studies of the behavior of cells grown in the lab also suggest a genetic influence on aging. Most types of differentiated cells can divide only a limited number of times. One factor that may control the number of cell divisions is the length of the *telomeres,* sequences of DNA, at the ends of chromosomes. Telomeres protect the ends of chromosomes from deteriorating or fusing with other chromosomes. Each time a cell divides, the telomeres normally shorten, and cells with shorter telomeres tend to undergo fewer divisions. Some cells, such as stem cells, possess an enzyme called telomerase, which replenishes the length of the telomeres, effectively making stem cells immortal. Cancer cells, which behave in a similar manner to stem cells, frequently possess an active telomerase enzyme, which allows them to replicate continuously (see Section 9.4). Studies using stem cells and cancer cells have begun to close in on the genetic factors that cause cellular aging. For example, in 2012 researchers used gene therapy (see Section 14.3) to introduce an active telomerase enzyme into mice, thus slowing the aging process.

Mitochondria and Diet The mitochondria are the powerhouses of the cell (see Section 4.7). As the mitochondria harvest the energy contained in carbohydrates, fats, and proteins, they generate free radicals. Free radicals are unstable molecules that carry an extra electron. To become stable, free radicals donate an electron to another molecule, such as DNA, proteins (e.g., enzymes), or lipids found in plasma membranes. Eventually, these molecules become unable to function, and the cell loses internal functions. This may lead to cell death. Scientists have determined that high-calorie diets increase the levels of free radicals, thus accelerating cellular aging. Multiple studies on model organisms, including mice, have supported that a low-calorie diet can extend the life span. It is also possible to reduce the negative effects of free radicals by increasing one's consumption of natural antioxidants, such as those present in brightly colored and dark-green vegetables and fruits. Chemicals in nuts, fish, shellfish, and red wine have also been shown to reduce our exposure to free radicals and slow the aging process.

However, if aging were mainly a function of genes, such as the *FOX03A* gene mentioned at the start of this chapter, we would expect to see much less variation in life span among individuals of a given species than is actually seen. For this reason, experts have estimated that, in most cases, genes account only for about 25% of what determines the length of life.

Damage Accumulation

Another set of hypotheses proposes that aging involves the accumulation of damage over time. In 1900, the average human life expectancy was around 45 years. A baby born in the United States this year has a life expectancy of just under 79 years. Because human genes have presumably not evolved significantly in such a short time, most of this gain in life span is due to better medical care, along with the application of scientific knowledge about how to prolong our lives.

Two basic types of cellular damage can accumulate over time. The first type can be thought of as agents that are unavoidable—for example, the accumulation of harmful DNA mutations or the buildup of harmful metabolites. A second form involves the fact that proteins—such as the collagen fibers present in many support tissues—may become increasingly cross-linked as people age. This cross-linking may account for the inability of such organs as the blood vessels, heart, and lungs to function as they once did. Some researchers have now found that glucose has the tendency to attach to any type of protein, which is the first step in a cross-linking process. They are currently experimenting with drugs that can prevent cross-linking. However, other sources of cellular damage may be avoidable, such as an unhealthy diet or exposure to the sun.

ISTUDY

The Effects of Aging on Human Organ Systems

Today's human population is older on average than ever before. In the next half-century, the number of people over age 65 will increase by 147%. Before discussing some possible mechanisms behind the aging process, it seems appropriate to first consider the effects of aging on the various body systems.

Integumentary System

As aging occurs, the skin becomes thinner and less elastic, because the number of elastic fibers decreases and the collagen fibers become increasingly cross-linked to each other, reducing their flexibility. There is also less adipose tissue in the subcutaneous layer; therefore, older people are more likely to feel cold. Together, these changes typically result in sagging and wrinkling of the skin.

As people age, the sweat glands also become less active, resulting in decreased tolerance to high temperatures. There are fewer hair follicles, so the hair on the scalp and the limbs thins out. Older people also experience a decrease in the number of melanocytes, making their hair gray and their skin pale. In contrast, some of the remaining pigment cells are larger, and pigmented blotches ("age spots") appear on the skin.

Cardiovascular System

Common problems with cardiovascular function are usually related to diseases, especially atherosclerosis (see Section 32.3). However, even with normal aging, the heart muscle does weaken somewhat, and it may increase slightly in size as it compensates for its decreasing strength. The maximum heart rate decreases even among the most fit older athlete, and it takes longer for the heart rate and blood pressure to return to normal resting levels following stress. Some part of this decrease in heart function may also be due to blood leaking back through heart valves that have become less flexible.

Aging also affects the blood vessels. The middle layer of arteries contains elastic fibers, which, like collagen fibers in the skin, become more cross-linked and rigid with time. These changes, plus a frequent decrease in the internal diameter of arteries due to atherosclerosis, contribute to a gradual increase in blood pressure with age. Indeed, nearly 50% of older adults have chronic hypertension. Such changes are common in individuals living in more-developed countries but not in agricultural societies. This indicates that a diet low in cholesterol and saturated fatty acids, along with a sensible exercise program, may help prevent age-related cardiovascular disease.

Immune System

As people age, many of their immune system functions become compromised. Because a healthy immune system normally protects the entire body from infections, toxins, and at least some types of cancer, some investigators believe that losses in immune function can play a major role in the aging process.

The thymus is an important site for T-cell maturation (see Section 33.4). Beginning in adolescence, the thymus begins to *involute,* gradually decreasing in size and eventually becoming replaced by fat and connective tissue. The thymus of a 60-year-old adult is about 5% of the size of the thymus of a newborn, resulting in a decrease in the ability of older people to generate T-cell responses to new antigens.

The evolutionary rationale for this may be that the thymus is energetically expensive for an organism to maintain, and that compared to younger animals that must respond to a high number of new infections and other antigens, older animals have already responded to most of the antigens to which they will be exposed in their life.

Aging also affects other immune functions. Because most B-cell responses are dependent on T cells, antibody responses also begin to decline. This, in turn, may explain why older adults do not respond as well to vaccinations as young people do. This presents challenges in protecting older people against diseases such as influenza and pneumonia, which can otherwise be prevented by annual vaccination.

Not all immune functions decrease in older animals, though. The activity of natural killer cells, which are a part of the innate immune system, seems to change very little with age. Perhaps by investigating how these cells remain active throughout a normal human life span, researchers can learn to preserve other aspects of immunity in older individuals.

Digestive System

The digestive system is perhaps less affected by the aging process than other systems. Because the secretion of saliva decreases, more bacteria tend to adhere to the teeth, causing more decay and periodontal disease. Blood flow to the liver is reduced, resulting in less efficient metabolism of drugs or toxins. This means that, as a person gets older, less medication is needed to maintain the same level in the bloodstream.

Respiratory System

Cardiovascular problems are often accompanied by respiratory disorders, and vice versa. Decreasing elasticity of lung tissues means that ventilation is reduced. Because we rarely use the entire vital capacity, these effects may not be noticed unless the demand for oxygen increases, such as during exercise.

Excretory System

Blood supply to the kidneys is also reduced. The kidneys become smaller and less efficient at filtering wastes. Salt and water balance are difficult to maintain, and older people dehydrate faster than young people. *Urinary incontinence* (lack of bladder control) increases with age, especially in females. In males, an enlarged prostate gland may reduce the diameter of the urethra, causing frequent or difficult urination.

Nervous System

Between the ages of 20 and 90, the brain loses about 20% of its weight and volume. Neurons are extremely sensitive to oxygen deficiency, and neuron death may be due not to aging itself but to reduced blood flow in narrowed blood vessels. However, contrary

ISTUDY

to previous scientific opinion, recent studies using advanced imaging techniques show that most age-related loss in brain function is not due to whatever loss of neurons is occurring. Instead, decreased function may occur due to alterations in complex chemical reactions, or increased inflammation in the brain. For example, an age-associated decline in levels of dopamine can affect the brain regions involved in complex thinking.

Perhaps more important than the molecular details, recent studies have confirmed that lifestyle factors can affect the aging brain. For example, animals on restricted-calorie diets developed fewer Alzheimer's-like changes in their brains. Other positive factors that may help maintain a healthy brain include lifelong learning (the "use it or lose it" idea), regularly engaging in exercise, and getting sufficient sleep.

Sensory Systems

In general, with aging, more stimulation is needed for taste, smell, and hearing receptors to function as before. A majority of people over age 80 experience a significant decline in their sense of smell, and about 15% develop *anosmia,* a total inability to smell. The latter condition can be a serious health hazard, due to the inability to detect smoke, gas leaks, or spoiled food. After age 50, most people gradually begin to lose the ability to hear tones at higher frequencies, and this can make it difficult to identify individual voices and understand conversation in a group.

Starting at about age 40, the lens of the eye does not accommodate as well, resulting in *presbyopia,* or difficulty focusing on near objects, which causes many people to require reading glasses as they reach middle age. Finally, cataracts and other eye disorders become much more common in older people.

Musculoskeletal System

For the average person, muscle mass peaks between ages 16 and 19 for females and 18 and 24 for males. Beginning in the twenties or thirties, but accelerating with increasing age, muscle mass generally decreases, due to decreases in both the size and the number of muscle fibers. Most people who reach age 90 have 50% less muscle mass than when they were 20. Although some of this loss may be inevitable, regular exercise can slow this decline.

Like muscles, bones tend to shrink in size and density with age. Due to compression of the vertebrae, along with changes in posture, most of us lose height as we age. Those who reach age 80 will be about 2 inches shorter than they were in their twenties. Due to osteoporosis (see Section 39.2), females lose bone mass more rapidly than males do, especially after menopause.

Although some decline in bone mass is a normal result of aging, certain extrinsic factors are also important. A healthy diet and a moderate exercise program have been found to slow the progressive loss of bone mass.

Endocrine System

As with the immune system, aging of the hormonal system can affect many different organs of the body. These changes are complex, however, with some hormone levels tending to decrease with age and others increasing. The activity of the thyroid gland generally declines, resulting in a lower basal metabolic rate. The production of insulin by the pancreas may remain stable, but cells become less sensitive to its effects, resulting in a rise in fasting glucose levels of about 10 mg/dl each decade after age 50.

Human growth hormone (HGH) levels also decline with age, but it is very unlikely that taking HGH injections will "cure" aging, despite Internet claims. In fact, one study found that people with lower levels of HGH actually lived longer than those with higher levels.

Reproductive System

Testosterone levels are highest in males in their twenties. After age 30, testosterone levels decrease by about 1% per year. Extremely low testosterone levels have been linked to a decreased sex drive, excessive weight gain, a loss of muscle mass, osteoporosis, general fatigue, and depression. However, the levels below which testosterone treatment should be initiated remain controversial. Testosterone replacement therapy, whether through injection, patches, or gels, is associated with side effects such as enlargement of the prostate, acne or other skin reactions, and the production of too many red blood cells.

Menopause, the period in a female's life during which the ovarian and uterine cycles cease, usually occurs between ages 45 and 55. The ovaries become unresponsive to the gonadotropic hormones produced by the anterior pituitary, and they no longer secrete estrogen or progesterone. At the onset of menopause, the uterine cycle becomes irregular, but as long as menstruation occurs, it is still possible for a female to conceive. Therefore, a female is usually not considered to have completed menopause (and thus be infertile) until menstruation has been absent for a year.

The hormonal changes during menopause often produce physical symptoms such as "hot flashes" (caused by circulatory irregularities), dizziness, headaches, insomnia, sleepiness, and depression. To ease these symptoms, female hormone replacement therapy (HRT) was routinely prescribed until 2002, when a large clinical study showed that in the long term, HRT caused more health problems than it prevented. For this reason, most doctors no longer recommend long-term HRT for the prevention of postmenopausal conditions.

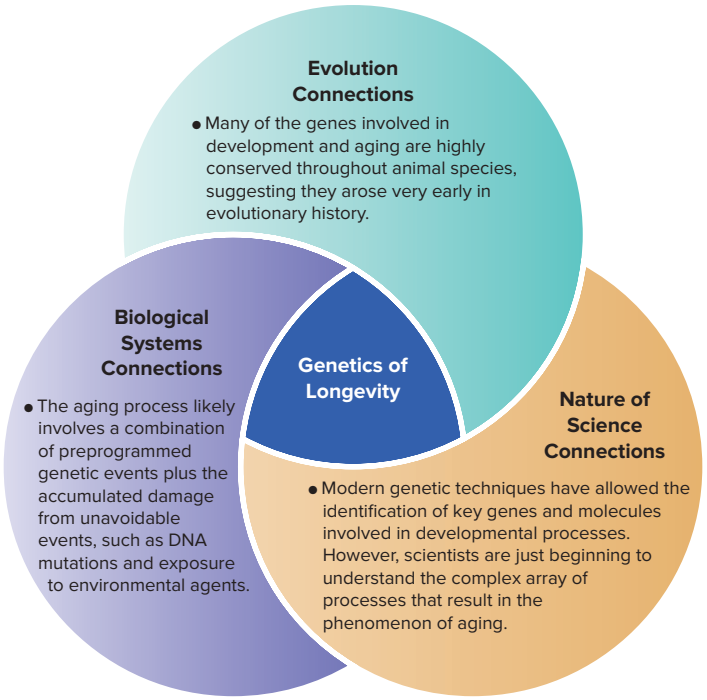
It is also of interest that, as a group, females live longer than males. It is likely that estrogen offers females some protection against cardiovascular disorders when they are younger. Males experience a marked increase in heart disease in their forties, but an increase is not noted in females until after menopause, when females lead males in the incidence of stroke. Males remain more likely than females to have a heart attack at any age, however.

Check Your Progress 42.4

1. Describe how involution of the thymus can lead to decreased responses to vaccines in older people.
2. Define *menopause,* and explain why it occurs.
3. Distinguish between the two hypotheses regarding aging, and give examples of each.

ISTUDY

Connecting *the* Concepts



What Did You Learn?

42.1 Early Developmental Stages

Development begins at **fertilization.** Only one sperm actually enters the oocyte. Both sperm and egg contribute chromosomes to the diploid zygote. **Development** includes all changes that occur during the life cycle of an organism.

- The early developmental stages in animals proceed from cellular stages to tissue stages to organ stages. During the cellular stage, **cleavage** (cell division) occurs, but there is no overall growth. The result is a **morula,** which becomes the **blastula** when an internal cavity (the **blastocoel**) appears. During these early stages, the organism is also called an **embryo.**
- During the tissue stage, a **gastrula** forms from the blastula. **Gastrulation** (invagination of cells into the blastocoel) results in formation of the **germ layers: ectoderm, mesoderm,** and **endoderm.** Inward folding of the early gastrula creates the **blastopore.** Both the cellular and tissue stages can be affected by the amount of **yolk.**
- Organ formation can be related to germ layers. For example, the nervous system develops from midline ectoderm, just above the **notochord.** During this process, a layer of cells called the **neural plate** contributes to formation of a **neural tube,** which gives rise to the brain and spinal cord.

42.2 Developmental Processes

The single-celled zygote is **totipotent,** meaning it can produce all types of specialized cells in an organism. **Cellular differentiation** begins with **cytoplasmic segregation** in the egg. After the first cleavage of a frog embryo, only a daughter cell that receives certain **maternal determinants** (i.e., the gray crescent) is able to develop into a complete embryo.

- **Induction** is also part of cellular differentiation. For example, the notochord induces the formation of the neural tube in frog embryos. In *C. elegans,* investigators have shown that induction is an ongoing process in which one type of tissue sequentially regulates the development of other types, through chemical signals coded for by particular genes.
- **Morphogenesis** involves cell movement, which defines body shape and form. Some morphogen genes determine the axes of the body, and others regulate the development of segments. An important concept has emerged: During development, sequential sets of master genes code for morphogen gradients that activate the next set of master genes in turn. **Pattern formation,** the arrangement of tissues and organs, often involves **apoptosis** (programmed cell death).
- **Homeotic genes** control pattern formation, such as the presence of antennae, wings, and limbs on the segments of *Drosophila.* Homeotic genes all contain a **homeobox,** a shared nucleotide sequence that codes for a sequence of 60 amino acids called a **homeodomain.** These proteins are also transcription factors, and the homeodomain is the portion of the protein that binds to DNA. Homologous homeotic genes have been found in a wide variety of organisms, and therefore they must have arisen early in the history of life and been conserved.

ISTUDY

42.3 Human Embryonic and Fetal Development

Human development can be divided into **embryonic development** (months 1 and 2) and fetal development (months 3 through 9).

- Fertilization occurs in a uterine tube, and cleavage occurs as the embryo moves toward the uterus. The morula becomes the blastocyst before implanting in the endometrium of the uterus. During the second week, the **embryonic disk** forms from ectoderm and endoderm, and becomes the primitive streak that gives rise to the mesoderm layer of cells.
- The **extraembryonic membranes** appear early in human development. The **trophoblast** of the **blastocyst** is the first sign of the **chorion,** which goes on to become the fetal part of the **placenta.** After **implantation,** the trophoblast secretes **human chorionic gonadotropin (HCG),** which maintains the corpus luteum and is also the basis of pregnancy testing.
- Exchange occurs between the fetal and female parent's blood at the placenta. The chorion has **chorionic villi** that increase contact with the female parent's blood vessels in the placenta. The **amnion** contains amniotic fluid, which cushions and protects the embryo. The **yolk sac** and **allantois** form the **umbilical cord,** which is a lifeline between the female parent and the developing embryo and fetus.
- Organ development begins with neural tube and heart formation. There follows a steady progression of organ formation during embryonic development. During fetal development, refinement of organ systems occurs, and the fetus adds weight.

42.4 The Aging Process

The aging process is an essential part of the biology and evolution of animals. Aging affects every organ system of the human body, in mostly predictable ways. One example is **menopause,** during which a female's ovarian and uterine cycles cease. Various hypotheses have been advanced to explain the aging process, which can be grouped into those suggesting that aging is genetically preprogrammed and those emphasizing the accumulation of various types of cellular damage over time.

Thinking Critically About Biology

1. Mitochondria contain their own genetic material. Mitochondrial mutations are inherited from the female parent to the child, without any contributions from the male parent. Why?
2. A variety of fertility tests are available for females to purchase over the counter—for example, at drugstores. Especially as more females delay their first pregnancy until they are older, these tests are becoming more popular. Many of these tests incorporate antibodies that specifically react with hormones found in a female's urine. What specific hormones could be measured in these tests, and how might these tests actually work?
3. The Biological Systems feature, "Preventing and Testing for Birth Defects" (see Section 42.3), suggests that females of childbearing age begin taking precautions to prevent birth defects (good nutrition, rubella vaccination, etc.). Why should these precautions begin before a female becomes pregnant?

Making It Relevant to You

1. Suppose it was possible to identify specific alleles of aging genes for a greatly shortened, or lengthened, life span. Would you want to know this information about your genes?
2. How might an identification of aging genes also allow us to better understand the environmental influences of aging?
3. Why do you think animal species evolved an aging process? Why not make all individuals immortal?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

43

Animal Behavior

florintt/E+/Getty Images

Chapter Outline

43.1 Inheritance Influences Behavior
43.2 The Environment Influences Behavior
43.3 Animal Communication
43.4 Behavioral Ecology

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 13.3 How can the environment influence gene expression?

Section 16.2 What roles do sexual selection and male competition play in the evolution of the species?

Section 40.2 What role does the endocrine system play in controlling behavior?

Behavior of Bees

Honeybees (*Apis mellifera*) live in hives with a complex social structure that is controlled by the queen bee. A typical hive will contain a single queen who can live for up to five years. Her job is to lay eggs that become the worker bees. Unfertilized eggs become drones. During the queen's mating flight, she releases a pheromone to attract drones for mating. Another pheromone, known as queen substance, suppresses the sexual development of the worker bees. If the level of the queen substance drops, it can trigger the worker bees to "rear" a new queen from the young larvae, ensuring the colony always has a healthy, viable queen.

In the spring, colonies swarm if they are overcrowded or have multiple queens. During swarming, the queen and a large number of bees leave to establish a new colony. The old colony will then attempt to rebuild itself with a new queen. Because the queen lays all of the eggs, every bee in the hive is genetically related. Members of the hive see to the propagation of their own genes when they help the queen reproduce. Animal behavior, as discussed in this chapter, is dedicated to the principle that natural selection shapes behavior, just as it does the anatomy and physiology of an animal.

As you read through this chapter, think about the following questions:

1. What are the benefits and drawbacks of living in a social group?
2. How do genetics and the environment work together to influence behavior?

Following *the* Themes

CHAPTER 43 – ANIMAL BEHAVIOR

Evolution	Natural selection has helped shape behavior by selecting for individuals with the most well-adapted traits.
Nature of Science	Scientists have conducted experiments to show that both genetics and the environment help shape behavior.
Biological Systems	Animal communication requires a sender and a receiver who can interpret chemical, auditory, visual, and tactile signals.

ISTUDY

43.1 Inheritance Influences Behavior

Learning Outcomes

Upon completion of this section, you should be able to

1. Explain the key aspects of studies that suggest behavior has a genetic basis.
2. Describe the body systems that play a role in influencing behavior.

Behavior encompasses any action that can be observed and described. For example, in the observation step in the scientific method (see Section 1.3), investigators may observe the behavior of a chimpanzee through visual means. The disposition of a skunk can be observed through olfactory means, and the warning of a rattlesnake can provide auditory information of imminent danger. In the same manner, scientists pose the question of whether genetics can determine the behavior an animal is capable of performing.

The "nature versus nurture" question asks to what extent nature (our genes) and nurture (the environment) influence behavior. We would expect that genes, which control the development of neural and hormonal mechanisms, will help influence the behavior of an animal. The results of experiments done to discover the degree to which genetics controls behavior support the hypothesis that most behaviors have, at least in part, a genetic basis.

Experiments that Suggest Behavior Has a Genetic Basis

An animal behavior study involving lovebirds and another involving garter snakes suggest that behavior has a genetic basis, and one type of study in humans attempts to evaluate nature versus nurture.

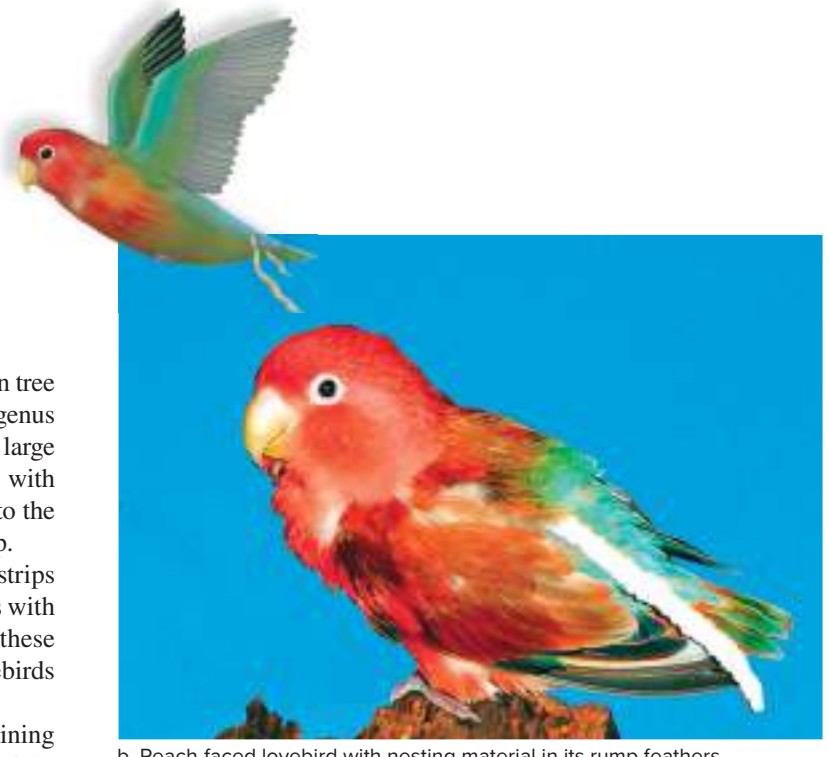
Behavior with a Genetic Basis: Nest Building in Lovebirds

Lovebirds are small, green-and-pink African parrots that nest in tree hollows. Several closely related species of lovebirds in the genus *Agapornis* build their nests differently. Fischer lovebirds cut large leaves (or in the laboratory, pieces of paper) into long strips with their bills. They use their bills to carry the strips (Fig. 43.1*a*) to the nest, where they weave them in with others to make a deep cup.

In contrast, peach-faced lovebirds cut somewhat shorter strips and carry them to the nest by inserting them into their feathers with their bill (Fig. 43.1*b*). In this way, they can carry several of these short strips with each trip to the nest, whereas Fischer lovebirds can carry only one of the longer strips at a time.

Researchers hypothesized that if the behavior for obtaining and carrying nesting material is inherited, then hybrids might show intermediate behavior. When the two species of birds were mated, their offspring were hybrid birds that were observed to have difficulty carrying nesting materials. They cut strips of intermediate length and then attempted to tuck the strips into their rump feathers. They did not push the strips far enough into the feathers, however, and when they walked or flew, the strips always came out. Hybrid birds eventually learned, after about 3 years, to carry

a. Fischer lovebird with nesting material in its beak



b. Peach-faced lovebird with nesting material in its rump feathers

Figure 43.1 Nest-building behavior in lovebirds. **a.** Fischer lovebirds, *Agapornis fischeri,* carry strips of nesting material in their bills, as do most other birds. **b.** Peach-faced lovebirds, *Agapornis roseicollis,* tuck strips of nesting material into their rump feathers before flying back to the nest. (a): Juniors Bildarchiv/age fotostock; (b): ©Refuge for Saving the Wildlife, Inc.

ISTUDY

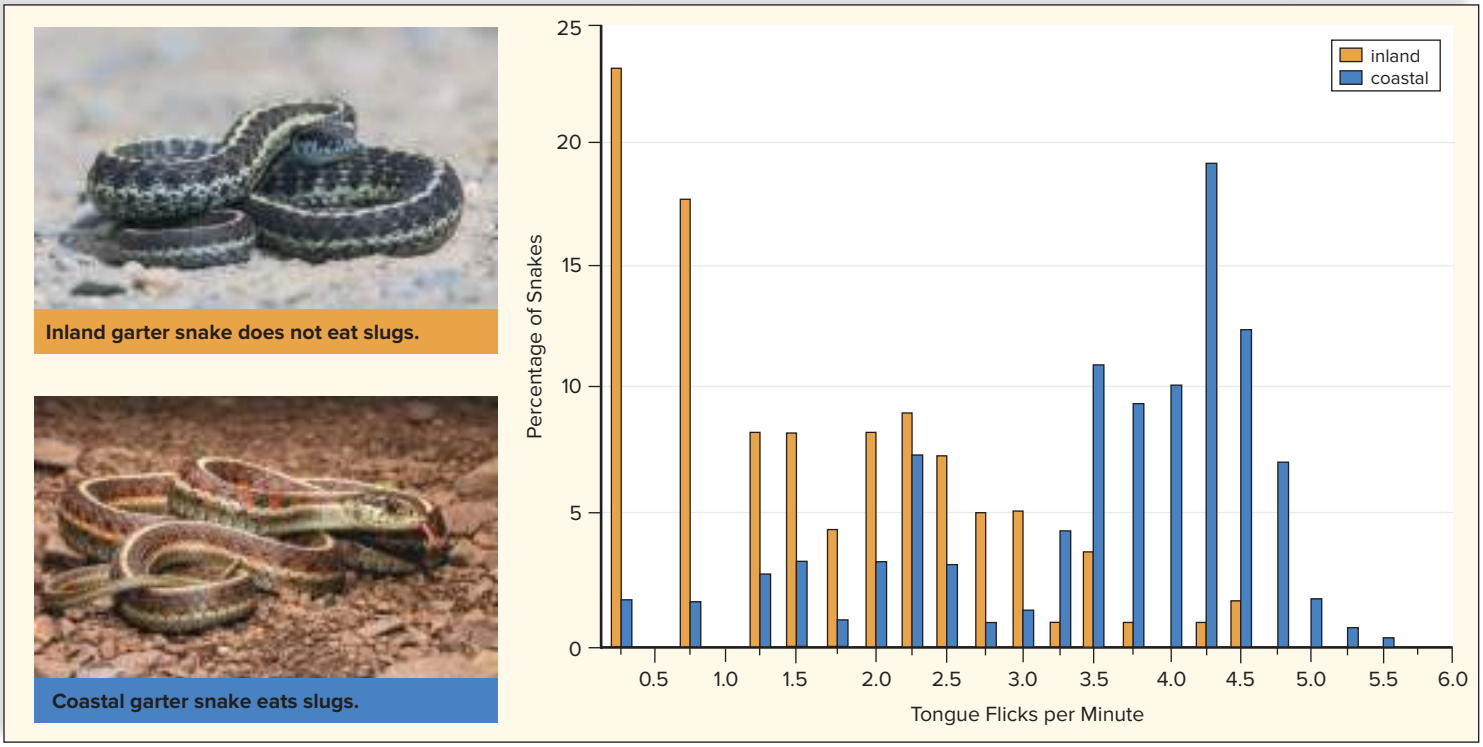


Figure 43.2 Feeding behavior in garter snakes. The number of tongue flicks by inland and coastal garter snakes, *Thamnophis elegans,* is measured in terms of their response to slug extract on cotton swabs. Coastal snakes tongue-flicked more than inland snakes. (photo): (inland): Carl Corbidge/Alamy Stock Photo; (photo): (coastal): Anton Sorokin/Alamy Stock Photo

the cut strips in their beaks, but they still briefly turned their heads toward their rump before flying off. These studies support the hypothesis that behavior has a genetic basis.

Behavior with a Genetic Basis: Food Choice in Garter Snakes

A variety of experiments have been conducted to determine if food preference in garter snakes has a genetic basis. There are two types of garter snake populations in California. Inland populations are aquatic and commonly feed under water on frogs and fish. Coastal populations are terrestrial and feed mainly on slugs. In one study, inland adult snakes refused to eat slugs in the laboratory, whereas coastal snakes readily did so. The experimental results of mating between snakes from the two populations (inland and coastal) show that their offspring have a partial preference for slugs when feeding.

Differences between slug acceptors and slug rejecters appear to be inherited. Experiments were able to determine the physiological difference between the two populations. When snakes eat, their tongues carry chemicals to an odor receptor in the roof of the mouth. They use tongue flicks to recognize their prey. Even newborns will flick their tongues at cotton swabs dipped in fluids of their prey. Swabs were dipped in slug extract, and the number of tongue flicks were counted for newborn inland snakes and newborn coastal snakes. Coastal snakes had a higher number of tongue flicks than did inland snakes (Fig. 43.2).

Apparently, inland snakes do not eat slugs because they cannot detect their smell as easily as coastal snakes can. A genetic difference between the two populations of snakes results in a physiological difference in their nervous systems. Although hybrids showed a great deal of variation in the number of tongue flicks, they tended to average between the number performed by coastal and inland snakes, as predicted by the genetic hypothesis.

Twin Studies in Humans

On occasion, human identical twins have been separated at birth and raised under different environmental conditions. Studies of these twins have shown that they have similar food preferences and activity patterns, and they even select mates with similar characteristics. This type of study lends support to the hypothesis that at least certain behaviors are strongly influenced by nature (genes).

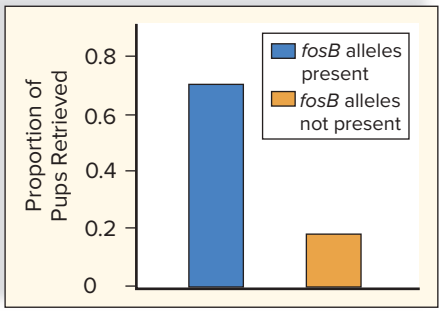
Examples of the Genetic Basis of Behavior

A number of studies have isolated genes that contribute to behavior. Generally, these genes have been associated with the nervous and endocrine systems, because both are responsible for the overall coordination of body systems.

Egg-Laying Behavior in Marine Snails

The egg-laying behavior in the marine snail *Aplysia* involves a specific sequence of events. Following copulation, the animal

ISTUDY



a.

b. *fosB* alleles present

c. *fosB* alleles not present

Figure 43.3 Female parent care in mice. **a.** A female mouse with *fosB* alleles spends time retrieving and crouching over their young, whereas mice that lack these alleles do not display these behaviors. **b.** This typical female mouse retrieves their young and crouches over them. **c.** This mouse, lacking *fosB*, does not retrieve their young and does not crouch over them. (b): Tom McHugh/Science Source; (c): Jean-Louis Klein/Marie-Luce Hubert/Science Source

extrudes long strings of more than a million egg cases. The animal takes the egg case string in its mouth, covers it with mucus, waves its head back and forth to wind the string into an irregular mass, and attaches the mass to a solid object, such as a rock.

Several years ago, scientists isolated and analyzed an egg-laying hormone (ELH) that causes the snail to lay eggs, even if it has not mated. ELH was found to be a small protein of 36 amino acids that diffuses into the circulatory system and excites the smooth muscle cells of the reproductive duct, causing them to contract and expel the egg string. Using recombinant DNA techniques, the investigators isolated the gene for ELH. The gene's product turned out to be a protein with 271 amino acids. The protein can be cleaved into as many as 11 products, and ELH is one of these. ELH alone, or in conjunction with the other products, is thought to control all the components of egg-laying behavior in *Aplysia*.

Nurturing Behavior in Mice

In another study, investigators found that maternal behavior in mice was dependent on the presence of a gene called *fosB*. Normally, when female parents first inspect their newborns, various sensory information from their eyes, ears, nose, and touch receptors travels to the hypothalamus. This incoming information causes *fosB* alleles to be activated, resulting in the production of a particular protein. The protein begins a process that activates cellular enzymes and additional genes. The end result is a change in the neural circuitry within the hypothalamus, which manifests itself in maternal nurturing behavior toward the newborn mice. Mice that do not engage in nurturing behavior were found to lack *fosB* alleles, and the hypothalamus failed to make any of the products or to activate any of the enzymes and other genes that lead to maternal nurturing behavior. Female mice with *fosB* alleles tended to retrieve their young if they become separated and crouch over them once they bring them back (Fig. 43.3).

Check Your Progress **43.1**

1. Compare the studies that show how behavior has a genetic basis.
2. Identify the body systems that influence behavior.

43.2 The Environment Influences Behavior

Learning Outcomes

Upon completion of this section, you should be able to

1. Choose the correct sequence of events required for classical and operant conditioning.
2. Identify examples of how the environment influences behavior.

Even though genetic inheritance serves as a basis for behavior, it is possible that environmental influences (nurture) also affect behavior. For example, behaviorists originally believed that

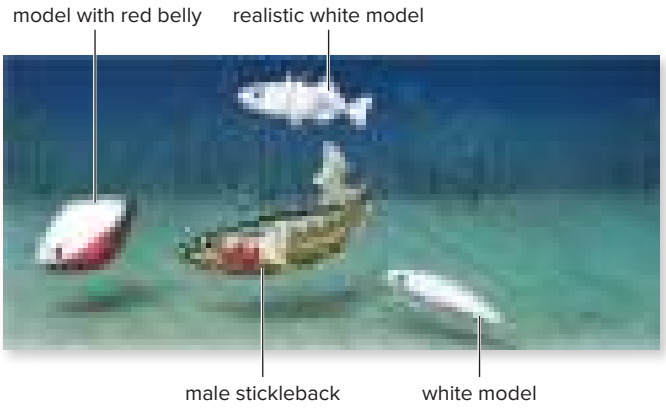


Figure 43.4 Fixed action pattern in a male stickleback. When confronted with a model with a red belly, a male stickleback will elicit a fixed action pattern and attack the model. He will ignore models that lack the red belly. Source: Raven, Peter et al. Biology, 11 ed. New York, NY: McGraw Hill Education, 2017.

some behaviors were unchanging behavioral responses known as **fixed action patterns (FAPs).** An FAP is elicited by a **sign stimulus**—a particular trigger in the environment. For example, male stickleback fish aggressively defend a territory against other males. In laboratory studies, the male reacts more aggressively to any model that has a red belly like he has (Fig. 43.4), rather than to a model that looks like a female stickleback fish. In this instance, the color red is a sign stimulus that triggers the aggressive behavior.

Investigators discovered that many behaviors that were originally thought to be FAPs could improve with practice due to the organism's learning. In this context, **learning** is defined as a durable change in behavior brought about by experience. Learning can be brought about by a wide variety of experiences. Habituation is a form of learning in which an animal no longer responds to a particular stimulus due to experience. Deer grazing on the side of a busy highway, ignoring traffic, is an example of habituation.

Instinct and Learning

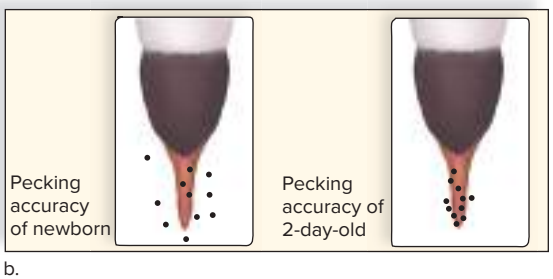
Laughing gull chicks' begging behavior appears to be an FAP, because it is always performed the same way in response to the parent's red bill (the sign stimulus). A chick directs a pecking motion toward the parent's bill, grasps it, and strokes it downward (Fig. 43.5). Parents can bring about the begging behavior by swinging their bill gently from side to side. After the chick responds, the parent regurgitates food onto the floor of the nest. If need be, the parent then encourages the chick to eat. This interaction between the chicks and their parents suggests that the begging behavior involves learning.

To test this hypothesis, diagrammatic pictures of gull heads were painted on small cards, and then eggs were collected from nests in the field. The eggs were hatched in a dark incubator to eliminate visual stimuli before the test. On the day of hatching, each chick was allowed to make about a dozen pecks at the model. The chicks were returned to the nest, and then each was retested.

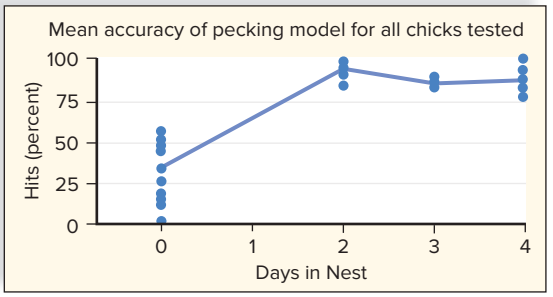
The tests showed that, on average, only one-third of the pecks by a newly hatched chick strike the model. But 1 day after hatching, more than half the pecks strike the model, and 2 days after hatching, the accuracy reaches a level of more than 75%. Investigators concluded that improvement in motor skills and learning can lead to an improvement in the performance of an animal's instinctive behaviors.

Figure 43.5 Pecking behavior in laughing gulls. **a.** At about 3 days, a laughing gull chick grasps the red bill of a parent, stroking it downward, and the parent then regurgitates food. **b.** The accuracy of a chick when striking a test probe, painted red. **c.** The chick-pecking accuracy graphically. Note from these diagrams that a chick markedly improves its ability (within only 2 days) to peck a bill, a behavior that normally causes a parent to regurgitate food.

a. Laughing gull adult and chick, *Leucophaeus atricilla*



b.



c.

ISTUDY

Figure 43.6 Konrad Lorenz and imprinting. The goslings are following Konrad Lorenz because they imprinted on him at a young age and think he is their female parent. Thomas D. Mcavoy/The LIFE Picture Collection/Shutterstock

Imprinting

Imprinting, a form of learning, occurs when a young animal forms an association with the first moving object it sees. Konrad Lorenz was one of the first individuals to study imprinting in birds. He observed chicks, ducklings, and goslings following the first moving object they saw after hatching, which was typically their female parent (Fig. 43.6). Imprinting in the wild has survival value that can lead to increased reproductive success. This behavior enables an individual to recognize its own species and thus eventually to find an appropriate mate.

In the laboratory, however, investigators found that birds can seemingly be imprinted on any object—even a human or a red ball—if it is the first moving object they see during a sensitive period of 2 to 3 days after hatching. The term *sensitive period* means the period of time during which a particular behavior develops. A chick imprinted on a red ball follows it around and chirps whenever the ball is moved out of sight.

Social interactions between parent and offspring during the sensitive period seem to be a key to normal imprinting. For example, female mallards cluck the entire time that imprinting is occurring. In some species, vocalization before and after hatching is necessary for normal imprinting.

Social Interactions and Learning

White-crowned sparrows sing a species-specific song, but males of a particular region have their own dialect. Birds were caged in order to test the hypothesis that young white-crowned sparrows learn how to sing from older members of their species.

Three groups of birds were tested. Birds in the first group heard no songs at all. When grown, these birds sang a song that had a slight resemblance to the adult song. Birds in the second group heard tapes of white-crowns singing. When grown, they sang in the dialect of the tapes that had been played during a sensitive period, about age 10–50 days. White-crowned sparrows' dialects (or other species' songs) played before or after this sensitive period had no effect on the birds. Birds in a third group did not hear tapes and instead were given an adult tutor. No matter when the tutoring began, these birds sang the song of the tutor species. Results like these show that social interactions with the appropriate species assist learning in birds.

Associative Learning

A change in behavior that involves an association between two events is termed **associative learning.** For example, birds that get sick after eating a monarch butterfly no longer prey on monarch butterflies, even though they may be readily available. The smell of freshly baked bread may entice you to have a piece, even though you may have just eaten. If so, perhaps you associate the taste of bread with a pleasant memory, such as being at home. Two types of associative learning are classical conditioning and operant conditioning.

Classical Conditioning

Classical conditioning is a method of modifying behavior by pairing two different types of stimuli (at the same time), causing an animal to form an association between them. The best-known laboratory example of classical conditioning is that of an experiment by the Russian psychologist Ivan Pavlov. First, Pavlov observed that dogs salivate when presented with food, so he began to ring a bell whenever the dogs were fed. Eventually, the dogs salivated whenever the bell was rung, regardless of whether food was present or not (Fig. 43.7). The dogs had come to associate the ringing of the bell with being fed.

Classical conditioning suggests that an organism can be trained—that is, conditioned—to associate a response with a specific stimulus. Unconditioned responses occur naturally, as when salivation follows the presentation of food. Conditioned responses are learned, as when a dog learns to salivate when it hears a bell.

Advertisements attempt to use classical conditioning to sell products. Commercials often pair attractive people with a product being advertised in the hope that viewers will associate attractiveness with the product. This pleasant association may cause them to buy the product.

Some types of classical conditioning can be helpful when trying to increase beneficial behaviors. For example, it has been suggested that you hold a child on your lap when reading to them in the hope that the child will associate a pleasant feeling with reading.

ISTUDY

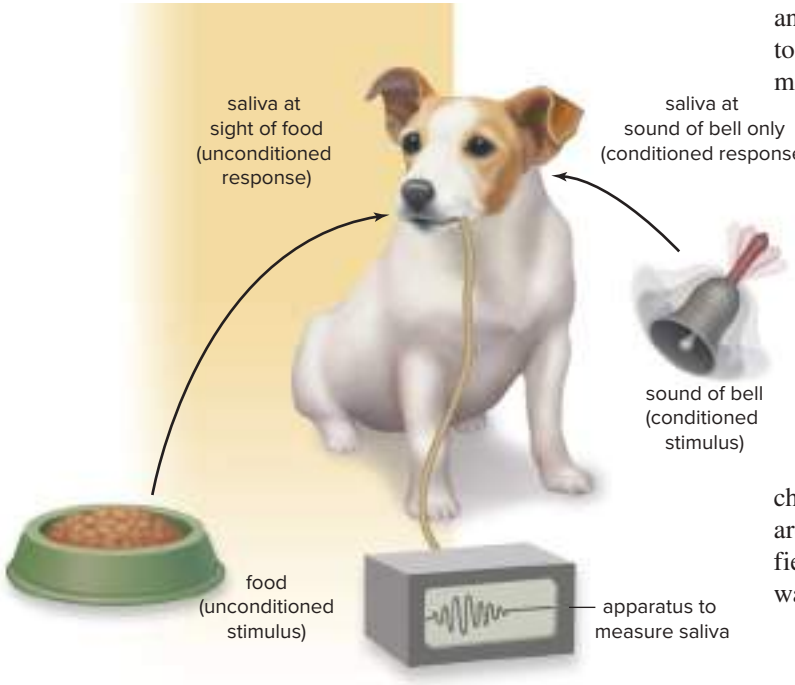


Figure 43.7 Classical conditioning. Ivan Pavlov discovered classical conditioning by performing this experiment with dogs. A bell is rung when a dog is given food. Salivation is measured. Eventually, the dog salivates when the bell is rung, even though no food is presented. Food is an unconditioned stimulus, and the sound of the bell is a conditioned stimulus that brings about the response of salivation.

Operant Conditioning

Operant conditioning is a method of modifying behavior in which a stimulus-response connection is strengthened. Most people know it is helpful to give an animal a reward, such as food or affection, when teaching it to perform a specific behavior. Animal trainers use operant conditioning to get many animals to perform a specific behavior. They present a stimulus to the animal—say, a hoop—and then give a reward (food) for the proper response (jumping through the hoop). Sometimes, the reward does not need to be immediate—in latent operant conditioning, an animal makes an association without the immediate reward, as when squirrels make a mental map of where they have hidden nuts.

B. F. Skinner was well-known for studying this type of learning in the laboratory. In the simplest type of experiment he performed, a caged rat inadvertently pressed a lever and was rewarded with sugar pellets, which it avidly consumed. Thereafter, the rat learned to press the lever whenever it wanted a sugar pellet. In more sophisticated experiments, Skinner even taught pigeons to play Ping-Pong by reinforcing the desired responses to stimuli.

In child rearing, it has been suggested that parents who give a positive reinforcement for good behavior will be more successful than parents who punish behaviors they believe are undesirable.

Orientation and Migratory Behavior

Migration is long-distance travel from one location to another. Loggerhead sea turtles hatch on a Florida beach and then travel across the Atlantic Ocean to the Mediterranean Sea, which offers an abundance of food. After several years, pregnant females return to the same beaches to lay their eggs. Every year, populations of monarch butterflies fly from North America to Mexico, where they overwinter, and then return in the spring to breed.

At the very least, migration requires **orientation,** the ability to travel in a particular direction, such as south in the winter and north in the spring. Most of the research studying orientation has been done in birds. Many birds can use the sun during the day or the stars at night to orient themselves. The sun moves across the sky during the day, but the birds are able to compensate for this, because they have a sense of time. They are presumed to have a biological clock that allows them to know where the sun will be in relation to the direction they should be going any time of the day.

Experienced birds can also **navigate,** which is the ability to change direction in response to environmental clues. These clues are apt to come from Earth's magnetic field. Figure 43.8 shows a study that was done with starlings, which

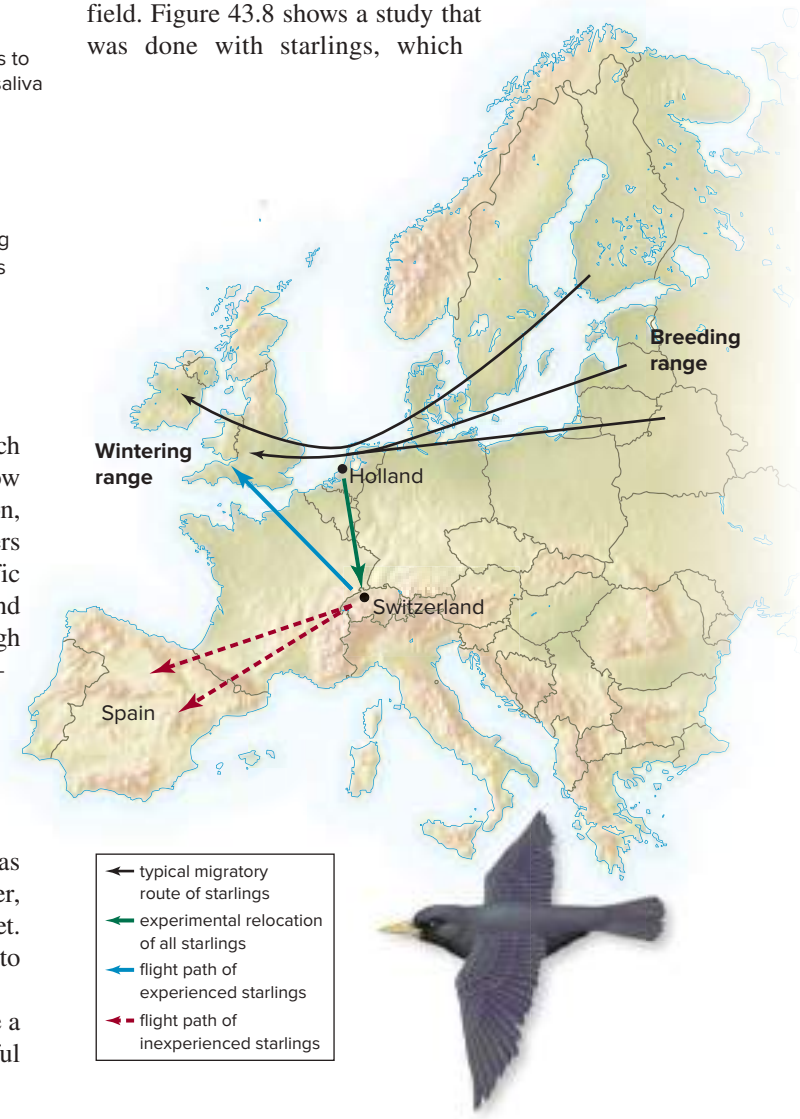


Figure 43.8 Starling migratory experiment. Starlings, *Sturnus vulgaris,* on their way from the Baltics to Great Britain were captured and released in Switzerland. Inexperienced birds kept flying in the same direction and ended up in Spain. Experienced birds had learned to navigate, as witnessed by the fact that they still arrived in Great Britain.

ISTUDY

typically migrate from the Baltics to Great Britain and return. Test starlings were captured in Holland and transported to Switzerland. Experienced birds corrected their flight pattern and still navigated their way to Great Britain. Young, inexperienced birds ended up in Spain instead of Great Britain.

Migratory behavior has a proximate cause and an ultimate cause. The proximate cause consists of environmental stimuli that tell the birds it is time to travel. The ultimate cause is the possibility of reaching an environment more favorable for survival and reproduction. In order for the behavior to persist, the benefit must outweigh the cost.

Cognitive Learning

In addition to the modes of learning already discussed, animals may learn through observation, imitation, and insight. For example, Japanese macaques learn to wash sweet potatoes before eating them by imitating others.

Insight learning occurs when an animal solves a problem without having any prior experience with the situation. The animal appears to call upon prior experience with other circumstances to solve the problem. For example, chimpanzees have been observed stacking boxes to reach bananas in laboratory settings.

Other animals, too, aside from primates, seem to be able to reason things out. In one experiment, ravens were offered meat that was attached to string hanging from a branch in a confined aviary. The ravens were accustomed to eating meat, but they had no knowledge of how strings work. It took several hours, but eventually one raven flew to the branch, reached down, grabbed the string with its beak, and pulled the string up over and over again, each time securing the string with its foot (Fig. 43.9). Eventually, the meat was within reach, and the raven was able to grab the meat with its beak. Ravens that observed this behavior were then able to obtain the meat by pulling up the string with the meat on it as well.

Ravens learn to retrieve food.

Figure 43.9 Insight learning. The raven determined that it could reach the food if it pulled up on the string. ©Dr. Bernd Heinrich

It seems that animals are capable of planning ahead. A sea otter will save a particular rock to act as a hard surface against which to bash open clams. A chimpanzee strips leaves off a twig, which it then uses to secure termites from a termite nest.

If animals can think, many people wonder if they have emotions. This, too, is an unexplored area that is now of interest. The Nature of Science feature, "Jaguar Behavior," explores how understanding animal behavior can help in conservation efforts.

Check Your Progress 43.2

1. Name the type of learning that occurs when an animal no longer tries to eat bumblebees after being stung by one.
2. Give an example that shows how instinct and learning may interact as behavior develops.
3. Discuss evidence that shows how the environment influences behavior.

43.3 Animal Communication

Learning Outcomes

Upon completion of this section, you should be able to

1. Recognize the various ways animals communicate.
2. Describe the advantages and disadvantages of chemical, auditory, visual, and tactile communication.

Animals exhibit a wide diversity of social behaviors. Some animals are largely solitary and join with a member of the opposite sex only for the purpose of reproduction. Others find mates and cooperate in raising offspring. Some form **societies** in which members of a species are organized in a cooperative manner, extending beyond reproductive and parental behavior. We have already explored how the social groups of honeybees and red deer relate to natural selection (see Section 16.2). These social behaviors in animals require that they communicate with one another.

Communicative Behavior

Communication is an action by a sender that is intended to influence the behavior of a receiver. The communication can be purposeful, but it does not have to be. Bats send out a series of sound pulses and listen for the corresponding echoes to find their way through dark caves and locate food at night. Some moths have an ability to hear these sound pulses, causing them to begin evasive tactics when they sense a bat is near. Bats are not purposefully communicating with the moths. The bat sounds are simply a cue to the moths that danger is near.

ISTUDY

THEME Nature of Science

Jaguar Behavior

Jaguars (*Panthera onca*) are a top-level carnivore and the largest cat in the Americas, and they play a critical role in maintaining the biodiversity and stability of the rain forest ecosystem. Deer, peccary, tapirs, armadillo, and agoutis are just a few of the species that jaguars prey upon. They are known to be opportunistic hunters who adjust their diet based upon the availability of suitable prey. The hunting tactics of the jaguar have been finely honed over the course of its evolution to make it an incredibly efficient predator. To capture prey, the jaguar will either stalk the unsuspecting animal or wait motionless and ambush it. When the prey comes into range, the jaguar will leap upon its back and often kill it with a single crushing bite to the back of the skull. Once the prey is dead, the jaguar will drag the animal to a more secluded spot to feed. Jaguars are extremely elusive creatures, rarely coming into contact with humans in their native environment.

When jaguars become old, sick, or injured, their ability to hunt their natural prey often diminishes. A decrease in their native prey base, often due to overhunting or human-caused habitat loss, can prompt jaguars to modify their hunting behavior. This modification is what can turn a normal jaguar into a "problem cat" that begins hunting livestock or domestic animals. This often brings the jaguar into direct conflict with humans. In many less-developed countries, the local response to a problem jaguar is to kill the cat in order to protect the rancher's or farmer's livelihood. With jaguar populations decreasing across their entire range, various groups have stepped up and developed different methods of dealing with the problem cats.

Since 2003, the Belize Zoo along with the Belize government and the U.S. Fish and Wildlife Service have been working together to help minimize jaguar killings by the locals in response to predation upon their livestock and domestic animals. Jaguars identified as problem cats are now trapped and transferred to the zoo's offsite facility. Here, the jaguars are introduced to a daily behavior modification routine (Fig. 43A). Typically, jaguars who enter the program exhibit a high level of aggression and stress toward humans. During the first phase of the program, positive reinforcement training is used to help reduce stress and make the cats more accepting of human interaction. Once they exhibit behaviors that show positive responses to humans, they are moved to phase two of the program. At this stage, they are given time in a larger enclosure that contains climbing logs, a pool, and other enrichment activities that mimic environments where the jaguars can express more of their natural behaviors.

When cats have been fully trained and are considered "graduates" of the program, they are then used for public education at the Belize Zoo. Many of their instincts and natural behaviors have been modified so they can live successful lives within a captive environment. While undergoing the training process, the animals are given a complete medical examination. The results of the examinations show that the majority of the jaguars who became problem cats were at one time shot and are often found to have shotgun pellets lodged inside them. In many cases, it is believed that their weakened physical condition, due to these injuries, is what led them to become problem cats.

Figure 43A Captive environments attempt to mimic the natural environment of jaguars so they will exhibit their natural behaviors.
Ricochet Creative Productions LLC

Questions to Consider

1. Even with training to modify the behavior of "problem cats," do you think they can truly be acclimated to living in captivity?
2. What other strategies could be used to help minimize human–jaguar conflict?

Chemical Communication

Chemical signals have the communicative advantage of being effective both night and day. A **pheromone** (Gk. *phero,* "bear, carry"; *monos,* "alone") is a chemical signal in low concentration that is passed among members of the same species. Some animals are capable of secreting a variety of different pheromones, each with a different meaning. Female moths secrete chemicals from abdominal glands, which are detected downwind by receptors on male antennae. The antennae are especially sensitive, and this ensures that only male moths of the correct species (and not predators) will be able to detect them.

ISTUDY

Figure 43.10 Use of a pheromone. This male cheetah is spraying urine onto a tree to mark its territory. Gregory G. Dimijian/Science Source

Ants and termites mark their trails with pheromones. Cheetahs and other cats mark their territories by depositing urine, feces, and anal gland secretions at the boundaries (Fig. 43.10). Klipspringers (small antelope) use secretions from a gland below the eye to mark the twigs and grasses of their territory. Pheromones are also known to control the behavior of social insects, as when worker bees suppress their own reproduction in order to care for the offspring produced by the queen.

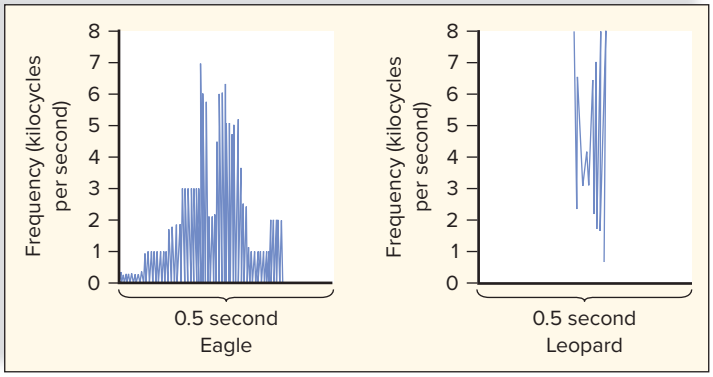
Researchers are trying to determine to what degree pheromones, along with hormones, affect the behavior of animals. For example, they are studying whether pheromones are responsible for animals carrying out parental care, becoming aggressive, or engaging in courtship behavior. Some researchers also maintain that human behavior is influenced by pheromones wafting through the air—pheromones that are not perceived consciously. They have discovered that, like mice, humans have an organ in the nose, called the vomeronasal organ (VNO), that can detect not only odors but also pheromones. The neurons from this organ lead to the hypothalamus, the part of the brain that controls the release of many hormones in the body.

Auditory Communication

Auditory communication is communication through sound. In certain environments, this has advantages over other kinds of communication. It is faster than chemical communication, and unlike visual communication, it is effective both night and day. Further, auditory communication can be modified not only by loudness but also by pattern, duration, and repetition. In an experiment with rats, a researcher discovered that an intruder can avoid attack by increasing the frequency with which it makes an appeasement sound.

Male crickets have calls, and male birds have songs for a variety of occasions. For example, birds may have one song for distress, another for courting, and still another for marking

a.



b.

Figure 43.11 Auditory communication. **a.** Vervet monkeys, *Cercopithecus aethiops,* are responding to an alarm call. Vervet monkeys can give different alarm calls according to whether a troop member sights an eagle or a leopard, for example. **b.** The frequency per second of the sound differs for each type of call. (monkeys running): Arco Images/Alamy Stock Photo; (monkey calling): Flavio Vallenari/E+/Getty Images

territories. Sailors have long heard the songs of humpback whales transmitted through the hulls of ships. But only recently has it been shown that the song has six basic themes. Each theme is composed of its own phrases that can vary in length and be interspersed with sundry cries and chirps. The purpose of the song is probably sexual, advertising the singer's availability. Bottlenose dolphins have one of the most complex languages in the animal kingdom.

Language is the ultimate auditory communication. Humans can produce a large number of different sounds and put them together in many ways. Nonhuman primates have different vocalizations, each having a definite meaning, as when vervet monkeys give alarm calls (Fig. 43.11). Although chimpanzees can be taught to use an artificial language, they do not progress beyond the capability level of a 2-year-old human child. It has also been difficult to prove that chimps understand the concept of grammar or can use their language to reason. It still seems as though humans possess a communication ability greater than that of other animals.

ISTUDY

Figure 43.12 Male baboon displaying full threat. The dominant male will "yawn" to show off his large canine teeth in an effort to intimidate his rivals. Raul Arboleda/AFP/Getty Images

Visual Communication

Visual signals are most often used by species that are active during the day. Contests between males often make use of threat postures that possibly prevent outright fighting. A male baboon displaying full threat is an awesome sight that establishes his dominance and keeps peace within the baboon troop (Fig. 43.12). Hippopotamuses perform territorial displays that include mouth opening.

Many animals use complex courtship behaviors and displays. The plumage of a male Raggiana Bird of Paradise allows him to put on a spectacular courtship dance to attract a female, giving her a basis on which to select a mate. Defense and courtship displays are exaggerated and always performed in the same way, so their meaning is clear. Fireflies use a flash pattern to signal females of the same species (Fig. 43.13).

Visual communication allows animals to signal others of their intentions without the need to provide any auditory or chemical messages. The body language of students during an exam provides an example. Students who have a hand in their lap, and are looking down at their lap, are often using their cell phone to look up answers during the exam. Students who have both arms and hands on the desk and are actively filling out their exam are less likely to be cheating. Instructors can use students' body language to determine whether cheating is occurring and take appropriate actions.

Psychologists have long tried to understand how visual cues can be used to better understand human emotions and behavior. Similarly, researchers today evaluate body language in animals to determine whether they also have emotions.

Tactile Communication

Tactile communication occurs when information is passed from one animal to another by touch. For example, recall that laughing gull chicks peck at the parent's bill to induce the parent to feed them (see Fig. 43.5). A male leopard nuzzles the female's neck to calm her and to stimulate her willingness to mate. In primates, grooming—one animal cleaning the coat and skin of another—helps cement social bonds within a group.

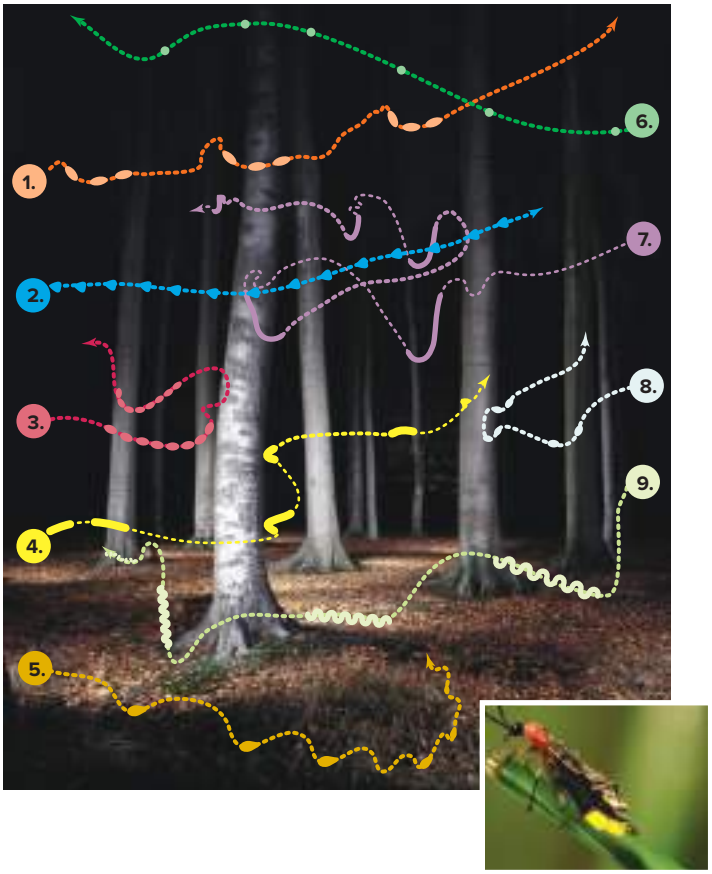


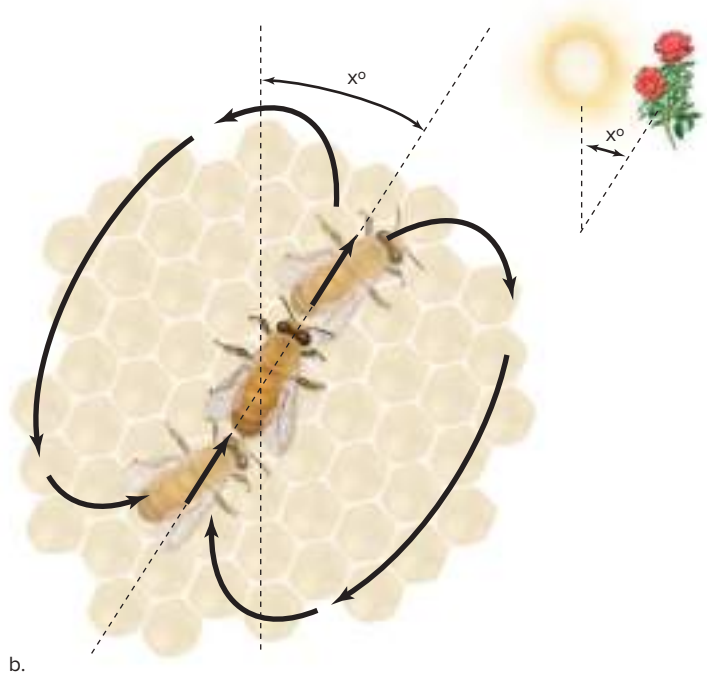
Figure 43.13 Fireflies use visual communication. Each number represents the male flash pattern of a different species. The patterns are a behavioral reproductive isolation mechanism. (trees): Ghislain & Marie David de Lossy/Cultura/Getty Images; (firefly): James Jordan Photography/Moment/Getty Images

Honeybees use a combination of communication methods, but especially tactile ones, to impart information about the environment. When a foraging bee returns to the hive, it performs a "waggle dance," which indicates the distance and direction of a food source (Fig. 43.14). As the bee moves between the two loops of a figure 8, it buzzes noisily and shakes its entire body in so-called waggles. Outside the hive, the dance is done on a horizontal surface, and the straight run indicates the direction of the food. Inside the hive, the angle of the straight run to that of the direction of gravity is the same as the angle of the food source to the sun. In other words, a 40° angle to the left of vertical means that food is 40° to the left of the sun.

Bees can use the sun as a compass to locate food because they have a biological clock, an internal means of telling time, which allows them to compensate for the sun's movement in the sky. Today, we know that the timing of the clock, in both insects and mammals (including humans), requires alterations in the expression of a gene called *period*.

ISTUDY

a.



b.

Figure 43.14 Communication among bees. **a.** Honeybees do a waggle dance to indicate the direction of food. **b.** If the dance is done outside the hive on a horizontal surface, the straight run of the dance will point to the food source. If the dance is done inside the hive on a vertical surface, the angle of the straightaway to that of the direction of gravity is the same as the angle of the food source to the sun. (a): Scott Camazine/Alamy Stock Photo

Check Your Progress 43.3

1. Describe examples of how communication is meant to affect the behavior of the receiver.
2. Give an advantage and disadvantage of each type of communication.
3. Identify the human receptor for each type of communication.

43.4 Behavioral Ecology

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how territoriality and the foraging techniques associated with it increase fitness.
2. Recognize various strategies that increase an individual's fitness.

Behavioral ecology is the study of how natural selection shapes the behavior of a species. It has been shown that behavior has a genetic basis, and investigators believe that certain behaviors lead to increased survival and production of offspring. Therefore, the behavior we can observe today in a species is still present because it has an adaptive value, or at minimum, does not have a negative value. The following types of behaviors, in particular, have been studied for their adaptive value: territoriality, reproductive strategies, social behaviors, and altruistic behaviors.

Territoriality and Fitness

In order to gather food, many animals have a home range where they carry out daily activities. The portion of the animal's range that it defends for its exclusive use, in which competing members of its species are not welcome, is called its **territory.** The behavior of defending one's territory is called **territoriality.** An animal's territory may have a good food source, and it may be the area in which the animal will reproduce.

As an example, gibbons live in the tropical rain forest of South and Southeast Asia. Normally, they can travel their home range in about 3–4 days. Gibbons are also monogamous and territorial. Territories are maintained by loud singing (Fig. 43.15). Males sing just before sunrise, and mated pairs sing duets during the morning. Males, but not females, show evidence of fighting to defend their territory, in the form of broken teeth and scars. Defense of a territory has a certain cost; it takes energy to sing and fight off others.

In order for territoriality to continue, it must have an adaptive value. Chief among the benefits of territoriality are access to a source of food, breeding opportunities, and a place to rear young. The territory has to be the right size for the animal: Too large a territory cannot be defended, and too small a territory may not contain enough resources.

Cheetahs require a large territory in order to hunt for their prey. As a result, they need to mark their territory in a fashion that will last for a while. As shown in Figure 43.10, cheetahs, like many dogs, use urine to mark their territory. Hummingbirds are known to defend a very small territory because they depend on only a small patch of flowers as their food source.

The size of a territory will often change from one season to the next. Territoriality is more likely to occur during reproductive periods. Seabirds have very large home ranges consisting of hundreds of kilometers of open ocean, but when they reproduce they become fiercely territorial. Each bird has a very small territory, consisting of only a small patch of beach where they place their nest.

ISTUDY

Figure 43.15 Male and female gibbons. Siamang gibbons, *Hylobates syndactylus,* are monogamous, and they share the task of raising offspring. They also share the task of marking their territory by singing. As is often the case in monogamous relationships, the sexes are similar in appearance. Male is above and female is below. Nicole Duplaix/Corbis Documentary/Getty Images

Foraging for Food

Foraging for food is when animals are actively looking for nutrition. An animal needs to acquire a food source that will provide more energy than is expended in the effort of acquiring the food. In one study, it was shown that shore crabs prefer to eat intermediate-sized mussels, if given equal numbers of a variety of sizes. The net energy gained from medium mussels is greater than if the crabs eat larger mussels. The large mussels take too much energy to open per the amount of energy they provide (Fig. 43.16). The **optimal foraging model** states that it is adaptive for foraging behavior to be as energetically efficient as possible.

Even though it can be demonstrated that animals that take in more energy during foraging are more likely to produce a greater number of offspring, other factors also come into play in survival. If an animal forages in a way that puts it in danger and it is killed and eaten, it has no chance of producing offspring. Animals often face trade-offs that lead to modification of their behavior toward maximizing their success.

Reproductive Strategies and Fitness

Many primate species are **polygamous,** meaning that a single male mates with multiple females. Because of gestation and lactation, females tend to invest more energy in their offspring than most males. Under these circumstances, it is adaptive for females to be

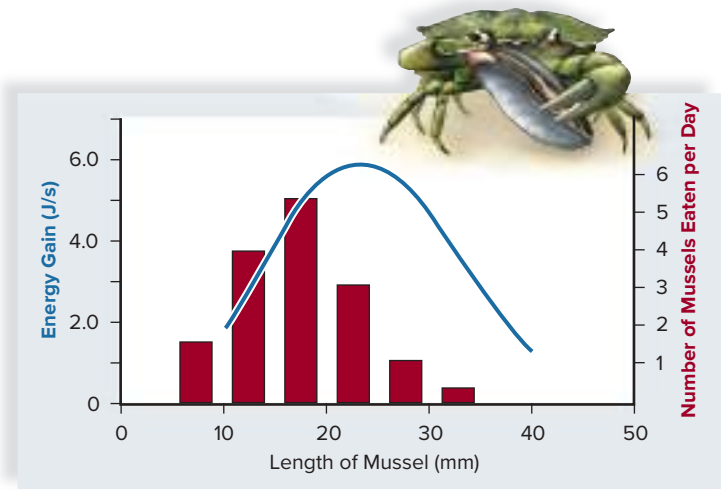


Figure 43.16 Foraging for food. When offered a choice of an equal number of each size of mussel, the shore crab, *Carcinus maenas,* prefers the intermediate size. This size provides the highest rate of net energy return measured in Joules per second (J/s). Net energy is determined by the energetic yield of the mussel minus the energetic costs of breaking open the shell and digesting the mussel.

concerned about a good food source. When food sources are clumped, females often congregate in small groups around the food. Because only a few females are expected to be receptive at a time, males will attempt to defend these few from other males. Males often compete with other males for the limited number of receptive females that are available (Fig. 43.17).

A limited number of primates are **polyandrous,** meaning that a female mates with multiple males. Tamarins are squirrel-sized New World monkeys that live in Central and South America. They live together in groups of one or more families in which one female mates with more than one male. The female tamarin normally gives birth to twins of such a large size that the male parent, not the female parent, carries them about. This may be the reason these animals are polyandrous. Polyandry also occurs when the environment does not have sufficient resources to support several young at a time.

Figure 43.17 Hamadryas baboons. Among Hamadryas baboons, *Papio hamadryas,* a male, which is silver-white and twice the size of a female, keeps and guards a harem of females with which he mates exclusively. Thomas Dobner/Alamy Stock Photo

ISTUDY

We have already mentioned the reproductive strategy of gibbons. They are **monogamous,** which means that they pair bond, and both males and females help with the rearing of the young. Males are active parents, frequently grooming and handling infants. Monogamy is relatively rare in primates, which includes prosimians, monkeys, and apes (only about 18% are monogamous). In primates, monogamy occurs when males have limited mating opportunities, territoriality exists, and the male is fairly certain the offspring are his. In gibbons, females are evenly distributed in the environment, which can lead to an increase in aggression between females. Investigators note that females will attack an electronic speaker when it plays female sounds in their territory.

Sexual Selection

Sexual selection is a form of natural selection that favors physical or behavioral features that increase an animal's chances of mating. In other words, these features are adaptive in the sense that they lead to increased fitness.

Sexual selection most often occurs due to female choice, which leads to male competition. Because most females produce a limited number of eggs relative to the number of sperm produced by the male, it is more adaptive for females to be choosier about their mate than males. If they choose a mate that passes on features to a male offspring that will cause him to be chosen by other females, the parents' fitness has increased.

Whether females actually choose features that are adaptive to the environment is in question. For example, peahens are likely to choose peacocks that have the most elaborate tails. Such a fancy tail could otherwise be detrimental to the male and make him more likely to be captured by a predator. In one study, an extra ornament was attached to the male parent of a zebra finch, and the female offspring of this bird underwent the process of imprinting (see Section 43.2). Then, those females were more likely to choose a mate that also had the same artificial ornament. The Evolution feature, "Sexual Selection in Male Bowerbirds," explores how courtship influences selection in a species of bird.

While females can always be sure an offspring is theirs, males do not have this certainty. However, males often produce a large number of sperm. The best strategy for males to increase their fitness is to mate with as many females as possible to increase their potential number of offspring. Competition may be required for them to gain access to females, and ornaments such as antlers can enhance a male's ability to fight (Fig. 43.18). When bull elk compete, they issue a number of loud screams that give way to a series of grunts. If a clear winner hasn't emerged, the two bulls walk in parallel to show each other their physique. If this doesn't convince one of them to back off, the pair resorts to ramming each other with their antlers. The intent of the fight is for one male to gain dominance over the other, not to kill his rival—although these encounters may have fatal consequences due to injury or mishap.

Societies and Fitness

The principles of evolutionary biology can be applied to the study of social behavior in animals. Sociobiologists hypothesize that living in a society has a greater reproductive benefit than reproductive cost. A cost-benefit analysis can help determine whether this hypothesis is supported.

Group living has advantages. It can help an animal avoid predators, rear offspring, and find food. For instance, a group of impalas is more likely than a single impala to hear an approaching predator. Many fish moving rapidly in multiple directions might distract a would-be predator. Weaver birds form giant colonies, which help protect them from predators, and the birds may share information about food sources. Members of a baboon troop signal to one another when they have found an especially bountiful fruit tree. Lions working together are able to capture large prey, such as zebra and buffalo.

Figure 43.18 Competition. During the mating season, bull elk, *Cervus elaphus,* males may find it necessary to engage in antler wrestling in order to have sole access to females in a territory. imageBROKER/Alamy Stock Photo

Group living also has disadvantages. When animals are crowded together into a small area, disputes can arise over access to the best feeding places and sleeping sites. Dominance hierarchies are one way to apportion resources, but this often puts subordinates at a disadvantage. However, in many mammal species, males are preferred due to their dominant status in a family group, but since they are larger than females, they need to be fed more frequently. Subordinate females do not have access to enough food resources to adequately nurse male offspring, and therefore they tend to rear only female offspring. Still, like the subordinate males in a baboon troop, subordinate females in a red deer harem may be better off in terms of fitness if they stay with a group, despite the cost involved.

Living in close quarters exposes individuals to illness and parasites that can easily pass from one animal to another. Social behavior helps offset some of the proximity disadvantages. For example, baboons and other types of social primates invest a significant amount of time grooming one another. This most likely decreases their chances of contracting parasites. Humans use extensive medical care to help offset the health problems that arise from living in the densely populated cities of the world.

Altruism Versus Self-Interest

Altruism (L. *alter,* "the other") is a behavior that has the potential to decrease the lifetime reproductive success of the altruist, while benefiting the reproductive success of another member of the society. In some insect societies, reproduction is limited to only one pair: the queen and her mate. For example, among army ants, the queen is inseminated only during her nuptial flight. The rest of her life is spent producing her offspring (Fig. 43.19).

The army-ant society has three sizes of sterile female workers. The smallest workers (3 mm), called the nurses, take care of the

ISTUDY

Figure 43.19 Queen ant. A queen ant (*Philidris* sp.) has a large abdomen for egg production and is cared for by small ants, called nurses. The idea of inclusive fitness suggests that relatives, in addition to offspring, increase an individual's reproductive success. Therefore, sterile nurses are being altruistic when they help the queen produce offspring to which they are closely related. Patrick Landmann/Science Source

queen and larvae, feeding them and keeping them clean. The intermediate-sized workers, constituting most of the population, go out on raids to collect food. The soldiers (14 mm), with huge heads and powerful jaws, run along the sides and rear of raiding parties and protect the column of ants from attack by intruders.

The altruistic behavior of sterile workers can be explained in terms of fitness, which is judged by reproductive success. Genes are passed from one generation to the next in two, quite different ways. The first way is direct: A parent can pass a gene directly to an offspring. The second way is indirect: A relative that reproduces can pass a shared gene to the next generation. *Direct selection* is adaptation to the environment due to the reproductive success of an individual. *Indirect selection,* called **kin selection,** is adaptation to the environment due to the reproductive success of the individual's relatives. The **inclusive fitness** of an individual includes personal reproductive success, as well as the reproductive success of relatives.

Examples of Inclusive Fitness

Among social bees, social wasps, and ants, the queen is diploid (2n), but her mate is haploid (n). If the queen has had only one mate, sister workers will be closely related to each other. They will share, on average, 75% of their genes, because they inherit the same allele from their male parent. Their potential offspring would share, on average, only 50% of their genes with the queen. Therefore, a worker can achieve a greater inclusive fitness by helping her female parent (the queen) produce additional sisters than by trying to reproduce herself. Under these circumstances, a behavior that appears altruistic is more likely to evolve.

Indirect selection can also occur among animals whose offspring receive only a half set of genes from both parents. Consider that your brother or sister shares 50% of your genes, your niece or nephew shares 25% of your genes, and so on. Therefore, the survival of two nieces or nephews is worth the survival of one sibling, assuming they both go on to reproduce.

Among chimpanzees in Africa, a female in estrus frequently copulates with several members of the same group, and the males make no attempt to interfere with each other's matings. Genetic relatedness appears to underlie their apparent altruism. Members of a group share more than 50% of their genes, because members never leave the territory in which they are born.

Reciprocal Altruism

In some bird species, offspring from a previous clutch of eggs may stay at the nest to help parents rear the next batch of offspring. In a study of Florida scrub jays, the number of fledglings produced by an adult pair doubled when they had helpers. In certain mammalian groups, like meerkats, the offspring have been observed to help their parents (Fig. 43.20).

Figure 43.20 Inclusive fitness. A meerkat is acting as a babysitter for their younger siblings while their female parent is away. Researchers point out that the helpful behavior of the older meerkat can lead to increased inclusive fitness. Thiebaud Gontard/Biosphoto/SuperStock

ISTUDY

THEME Evolution

Sexual Selection in Male Bowerbirds

At the start of the breeding season, male bowerbirds use small sticks and twigs to build elaborate display areas called bowers. They clear the space around the bower and decorate the area with items used to attract females. After the bower is complete, the male spends most of his time near his bower, calling to females, renewing his decorations, and guarding his work against possible raids by other males.

When a female approaches, the male begins his display. He faces her, fluffs up his feathers, and flaps his wings to the beat of a call. If the female enters the bower, the two will mate. Some males mate with up to 25 females per year. Biologists discovered that females often chose males that had well-built bowers with well-decorated platforms.

Male bowerbirds are not gaudy in appearance, but their displays are highly intense and aggressive. Their courtship displays are similar to those used by males during aggressive encounters with other males. Males must display intensely to be attractive, but males that are too intense too soon can startle females. Females may benefit from mating with the most intensely displaying males, but if they are startled, they may not be able to efficiently assess male traits.

Communication between the sexes might maximize the potential benefits of intense male courtship displays, while minimizing the potential costs. Females make crouching motions, and the degree of crouching reflects the level of display intensity she will tolerate without being startled. By giving higher-intensity displays when females increase their crouching, males can increase their courtship success by displaying intensely enough to be attractive without threatening the females.

The hypothesis tested was that males respond to female crouching signals by adjusting their intensity, and that a particular male's ability to respond to female signals is related to his success in courtship. A male's ability to modify his courtship display was difficult to measure in natural courtships, because it was not clear whether males were responding to females, or vice versa. To solve this problem, a robotic female bowerbird was used.

Using these "fembots," researchers were able to control female signals and measure male responses in experimental courtships. In general, male satin bowerbirds modulated their displays in response to robotic female crouching (Fig. 43B*a*). Video cameras monitored their behaviors at bowers, allowing each male's courtship/mating success to be measured. It was found that males who modulated their displays more effectively in response to robotic female signals startled real females less often in natural courtships and were thus more successful in courting females (Fig. 43B*b*).

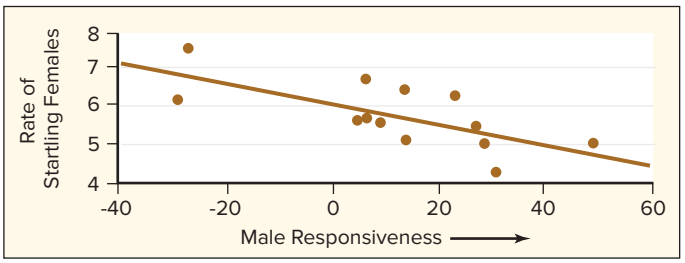
The results suggest that females prefer intensely displaying males as mates and that successful males modulate their intensity in response to female signals, thus producing displays attractive to females without threatening them.

Male responsiveness to female signals may be an important part of successful courtship in many species. Sexual selection may favor the ability of males to read female signals and adjust courtship displays accordingly.

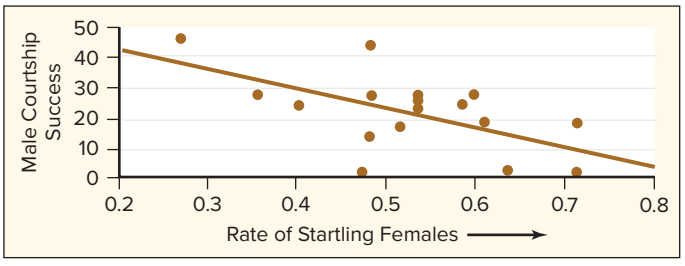
Questions to Consider

1. Why would it be more adaptable for a male bowerbird to be aggressive and have an intense courtship display instead of being gaudy?
2. How would an extremely intense male bowerbird mate?

a.



b.



c.

Figure 43B Courtship success of males. **a.** Among satin bowerbirds, *Ptilonorhynchus violaceus,* the male bowerbird prepares the bower and typically decorates its platform with blue objects. **b.** Some males are better able to vary the intensity of their courtship display depending on the crouch rate of a robotic female, and these males startle real females less often. **c.** Experimenters found that males that respond best under experimental conditions startle live females less and have better courtship success. (a): Dave Watts/Alamy Stock Photo

ISTUDY

Among jackals in Africa, solitary pairs managed to rear an average of 1.4 pups, whereas pairs with helpers reared 3.6 pups.

There are several benefits of staying behind to help raise young. First, a helper is contributing to the survival of its own kin. Therefore, the helper actually gains a fitness benefit. Second, a helper is more likely than a nonhelper to inherit a parental territory, which may include other helpers. Helping, then, involves making a short-term reproductive sacrifice in order to increase future reproductive potential. Therefore, helpers at the nest are also practicing a behavior called **reciprocal altruism.**

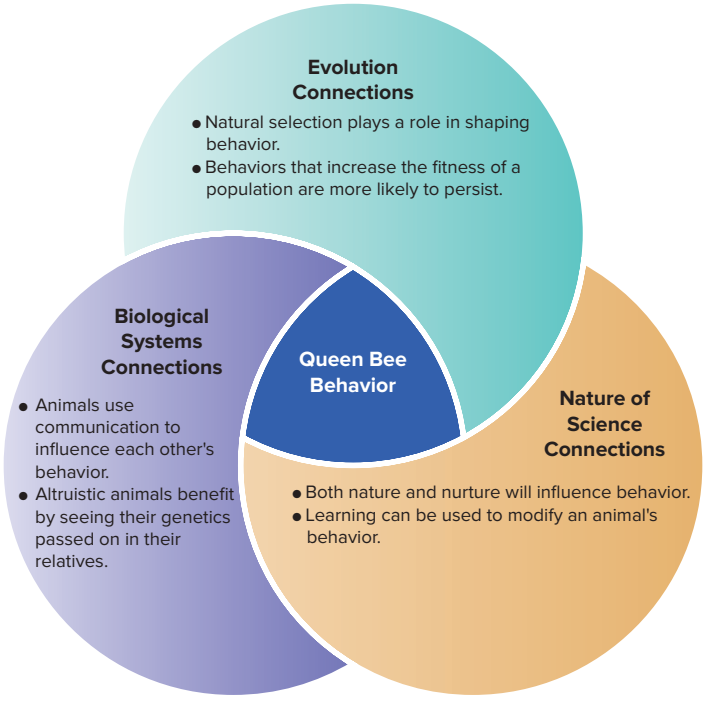
Reciprocal altruism also occurs in animals that are not closely related. In this event, an animal helps or cooperates with another animal, gaining no immediate benefit. However, the animal that was helped repays the debt at some later time. Reciprocal altruism usually occurs in groups of animals that are mutually dependent. Cheaters in reciprocal altruism are recognized and are not reciprocated in future events.

An example of this type of reciprocal altruism occurs in vampire bats that live in the tropics. Bats returning to the roost after a feeding activity share their blood meal with other bats in the roost. If a bat fails to share blood with one that had previously shared blood with it, the cheater bat will be excluded from future blood sharing.

Check Your Progress 43.4

1. Explain how territoriality is related to foraging for food.
2. Compare and contrast reproductive strategies and forms of sexual selection.
3. Describe examples of how altruistic behavior increases an individual's fitness.

Connecting *the* Concepts



What Did You Learn?

43.1 Inheritance Influences Behavior

Investigators have long been interested in the degree to which nature (genetics) or nurture (environment) influences **behavior.** Hybrid studies with lovebirds produce results consistent with the hypothesis that behavior has a genetic basis. Garter snake experiments indicate that the nervous system controls behavior. Studies of human identical twins have shown that certain types of behavior are apparently inherited. DNA studies of marine snails indicate that the endocrine system also controls behavior.

43.2 The Environment Influences Behavior

- **Fixed action patterns (FAPs)** are behavioral responses elicited by a **sign stimulus.** Research has shown that FAPs can be modified by **learning.** The red bill of laughing gulls initiates chick begging behavior. However, with experience, the chick's begging behavior improves as the chick learns. It demonstrates an increased ability to recognize its parents.
- **Imprinting** is a form of learning in which an animal develops an association with the first moving object it sees during a sensitive period. Song learning in birds involves various elements—including

ISTUDY

the existence of a sensitive period, during which an animal is primed to learn—and the positive benefit of social interactions.

- **Associative learning** is a change in behavior that involves an association between two events. **Classical conditioning** and **operant conditioning** are two forms of associative learning. In classical conditioning, the pairing of two different types of stimuli causes an animal to form an association between them. In this way, dogs can be conditioned to salivate at the sound of a bell. In operant conditioning, animals learn behaviors because they are rewarded when they perform them.
- **Migration** occurs when an animal travels from one location to another. In order to do this, **orientation,** or the ability to travel in a particular direction, is required. **Navigation** is the ability to change direction in response to environmental cues. Animals often use the sun, the stars, and Earth's magnetic field during migration.
- **Insight learning** occurs when an animal solves a problem without having prior experience with the situation.

43.3 Animal Communication

Members of various species of animals are organized in a cooperative manner, forming **societies.** Social behavior requires the passing of information from one individual to another. **Communication** is an action by a sender that affects the behavior of a receiver. Chemical, auditory, visual, and tactile signals are forms of communication that foster the exchange of information between the sender and the receiver.

- **Pheromones** are chemical signals passed among members of the same species.
- **Auditory communication** is the passing of information through sound, which can include language.
- **Visual communication** is the passing of information through the use of visual cues. This allows animals to signal others without the need for auditory or chemical messages.
- **Tactile communication** is the passing of information through the use of touch; it is often associated with sexual behavior.

43.4 Behavioral Ecology

Behavioral ecology is the study of how natural selection shapes behavior. Traits that promote reproductive success are expected to provide an overall advantage to the individual. Animals often have a **territory** in which they can obtain food, acquire living space, and reproduce. The act of defending one's territory is called **territoriality.**

- When animals choose foods that return the most net energy, they have more energy left over for reproduction. The **optimal foraging model** is an adaptive behavior in which foraging is as energetically efficient as possible.
- Depending on the environment and social structure, animals use a specific type of reproductive strategy. Many primates are **polygamous,** meaning one male will mate with multiple females. **Polyandrous** means one female mating with multiple males. Gibbons are **monogamous,** meaning that a male and a female form a pair bond and mate only with each other.
- **Sexual selection** is a form of natural selection for traits that increase an animal's fitness. Males produce many sperm and are expected to compete to inseminate females. Females produce few eggs and are expected to be selective about their mates.
- Living in a social group can have its advantages (e.g., ability to avoid predators, raise young, and find food). It also has disadvantages (e.g., tension between members, spread of illness and parasites, and reduced reproductive potential). When animals live in groups, the benefits must outweigh the costs, or the behavior would not exist.
- In some animal societies, **altruism** is present. This is a behavior in which an individual benefits from the reproductive success of another member of society while limiting its own. **Kin selection** occurs when individuals increase the reproductive success of their relatives. In this context, it is necessary to consider **inclusive fitness,** which includes personal reproductive success, as well as the reproductive success of relatives. Sometimes, individuals help their parents rear siblings. Social insects help increase their female parent's chance of reproductive success, but this behavior seems reasonable when we consider that siblings share 75% of their genes. Among mammals, a parental helper may be likely to inherit the parent's territory. In **reciprocal altruism,** animals aid one another for future benefits.

Thinking Critically About Biology

1. Meerkats are said to exhibit altruistic behavior because certain members of a population act as sentries. How would you test the hypothesis that sentries are engaged in altruistic behavior?
2. Different chemicals used in our environment have been shown to disrupt the production of pheromones in animals and humans alike. Indicate the potential consequences that could occur if a colony of bees were exposed to a pheromone-disrupting chemical.
3. The 2013 film *Blackfish* attempts to expose the darker side of keeping killer whales in captivity. It chronicles the 2010 death of a SeaWorld trainer, along with multiple other incidents in which 5-ton orca whales appeared to intentionally attack their trainers. The movie has many people questioning the ethical nature of keeping animals in captivity.

   If we keep animals in captivity, are we depriving them of their freedom? Some point out that freedom is never absolute. Even an animal in the wild is restricted in various ways by its abiotic and biotic environments. Many modern zoos and aquariums keep animals in habitats that nearly match their natural ones, so that they have some freedom to roam and behave naturally—but is this really possible with an animal such as an orca?

   Today, reputable zoos and aquariums rarely go out and capture animals in the wild—they usually get their animals from other facilities. Most people feel it is not a good idea to take animals from the wild. Certainly, zoos and aquariums should not be involved in the commercial and often illegal trade of wild animals that still goes on today. In the twenty-first century, many zoos and aquariums are instead actively engaged in the conservation of animals, rather than just their exploitation for exhibits. These facilities provide the best home possible for animals that are recovering from injury or increasing their numbers, until they can be released back into the wild. In your opinion, which animals are suited to a life in captivity, and which are not?

Making It Relevant to You

1. What is the advantage to a beehive if the queen decides to swarm?
2. Which pheromones do humans produce and how do they influence behavior?
3. From an evolutionary perspective, what are the pros and cons of living in a group in which all members share a genetic relatedness?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

UNIT 8

Ecology

Ecology is the science that describes the interaction between the living and nonliving components of the environment, the role that species play within the environment, and the impact that humans have upon the Earth.

Ecologists have two statements that summarize these interactions: "Everything is affected by everything else" and "There is no free lunch." For example, when tall smokestacks were constructed in the Midwest, the pollutants didn't go away: They caused acid rain in the Northeast. As is so often the case, the only way to solve an ecological problem is to stop the behavior or activity that caused the problem. It must be dealt with head-on.

The word *ecology* initially described the biosphere (the portions of the sea, land, and air that contain living organisms), but today ecology is also an experimental science. Ecologists use the scientific process to determine how ecosystems function and how they will likely respond to human impact. Ecologists also use complex models to make predictions as to how future ecological changes will impact human society. One of the main goals of conservation biology is to help preserve species and manage ecosystems for sustainable human welfare.

Unit Outline

Chapter 44 Population Ecology

Chapter 45 Community and Ecosystem Ecology

Chapter 46 Major Ecosystems of the Biosphere

Chapter 47 Conservation of Biodiversity

Unit Learning Outcomes

The learning outcomes for this unit focus on three major themes in the life sciences.

Evolution	Discuss how natural selection and coevolution have played a significant role in shaping the diversity within ecosystems.
Nature of Science	Identify the aspects of biology that conservation biologists study in order to determine how best to manage natural resources.
Biological Systems	Explain the complex web of interactions among individuals, species, communities, and their environment.

ISTUDY

44

Population Ecology

Christopher Wiederspahn/Alamy Stock Photo

The Invasive Spotted Lanternfly

The spotted lanternfly (*Lycorma delicatula*) is naturally found in China where it is preyed upon by various insects, birds, and even some snakes. These predators have evolved with the spotted lanternfly and recognize it as a prey species, thus helping to keep its population in check. Unfortunately, the spotted lanternfly was accidentally introduced to the United States in 2014 where it doesn't have natural predators. The earliest sighting was September 2014 in Berks County, Pennsylvania. Since then, it has spread across 45 counties in Pennsylvania and into eight neighboring states.

The adults are about one inch long and half an inch wide, with large colorful wings that have spots on them. They go through complete metamorphosis, starting life as an egg, progressing to the nymph stage, and then maturing into an adult. Each stage of the life cycle will use different resources, making these invasive pests very destructive. They use their piercing mouthparts to draw sap from over 70 different species of plants. Many of these plants, such as grapevines, fruit trees, and hardwoods, are economically important to the economy of Pennsylvania and many of the surrounding states.

Spotted lanternflies are not strong fliers and instead rely on hitchhiking for the dispersal of their population. They will lay a mudlike egg mass on trees, cars, bikes, and other objects. Humans then accidentally move the egg masses to new locations, enabling their population to spread. People need to be vigilant and monitor their environment for spotted lantern flies. If discovered, they should contact their local Department of Agriculture for assistance. If allowed to spread, this invasive insect could cause serious damage and destruction to the U.S. grape, orchard, and logging industries.

As you read through this chapter, think about the following questions:

1. What ecological consequences may occur as a species expands its range?
2. How do population growth models apply to the study of ecology?

Chapter Outline

44.1 Scope of Ecology
44.2 Demographics of Populations
44.3 Population Growth Models
44.4 Regulation of Population Size
44.5 Life History Patterns
44.6 Human Population Growth
44.7 A Sustainable Future

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 6.1 How does energy flow from one organism to another impact population growth?

Figure 10.9 Explain what role the human life cycle plays in determining population growth.

Section 16.1 Describe how microevolution is measured within a population.

Following *the* Themes

CHAPTER 44 – POPULATION ECOLOGY

Evolution	A change in one population, or in the environment, will have an impact on other populations in the community.
Nature of Science	Ecologists study the interactions among a variety of living organisms and their environment.
Biological Systems	Population growth is based on birthrates, mortality rates, the age structure of the population, and a variety of other factors.

ISTUDY

44.1 Scope of Ecology

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify what aspects of biology the study of ecology encompasses.
2. Identify the ecological levels that exist within the field of ecology.

In 1866, the German zoologist Ernst Haeckel coined the word **ecology** (Gk. *oikos,* "home"; *logy,* "study of"), which he defined as the study of the interactions among all organisms and with their physical environment. Haeckel also pointed out that ecology and evolution are intertwined. Ecological interactions act as selection pressures that result in evolutionary change, which in turn affects ecological interactions.

Ecology, like so many biological disciplines, is wide-ranging. At one of its simplest levels, ecologists study how an individual organism is adapted to its environment. For example, they study how a fish is adapted to and survives in its **habitat** (the place where the organism lives). Most organisms do not exist singly; rather, they are part of a population, the functional unit that interacts with the environment and on which natural selection operates (Fig. 44.1). A **population** is defined as all the organisms belonging to the same species within an area at the same time. At this level of study, ecologists are interested in factors that affect the growth and regulation of population size. The population is part of the larger collection of individuals that make up a particular **species,** all of the populations of similar organisms that are capable of interbreeding and producing viable offspring.

A **community** consists of all the populations of different species interacting in a specific area. In a coral reef, there are numerous populations of algae, corals, crustaceans, fishes, and so forth. At this level, ecologists want to know how different populations interact with each other. An **ecosystem** consists of all of the populations within the community and the associated abiotic variables (e.g., the availability of sunlight for plants). Energy flow and chemical cycling are significant aspects of how an ecosystem functions. Ecosystems rarely have rigid boundaries. Usually, a transition zone called an ecotone, which has a mixture of organisms from adjacent ecosystems, exists between ecosystems. The **biosphere** encompasses the zones of the Earth's soil, water, and air where living organisms are found.

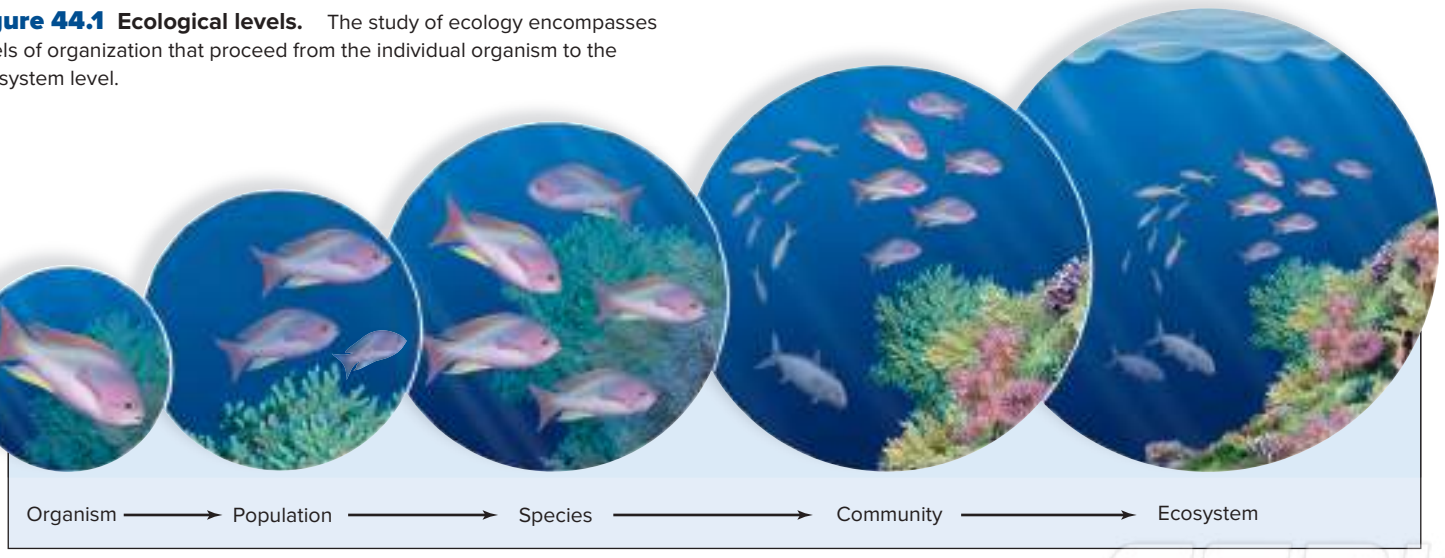
Modern ecology is not just descriptive; it is predictive. It analyzes levels of organization and develops models and hypotheses that can be tested. A central goal of modern ecology is to develop models that explain and predict patterns of distribution and the potential abundance of organisms. Ultimately, ecology considers not one particular area, but the distribution and abundance of populations all across the planet. For instance, ecologists try to determine which factors have selected for the mix of plants and animals in a tropical rain forest and how climate change may alter this in the near future.

Although modern ecology is useful in and of itself, it also has unlimited application possibilities, including the proper management of plants and wildlife, the identification of and efficient use of renewable and nonrenewable resources, the preservation of habitats and natural cycles, the maintenance of food resources, and the ability to predict the impact and course of diseases, such as malaria or coronaviruses.

Check Your Progress 44.1

1. Distinguish between a population and a community.
2. Explain what is meant by *abiotic variables.*
3. Describe the central goal of modern ecological studies.

Figure 44.1 Ecological levels. The study of ecology encompasses levels of organization that proceed from the individual organism to the ecosystem level.



ISTUDY

44.2 Demographics of Populations

Learning Outcomes

Upon completion of this section, you should be able to

1. Recognize how environmental conditions affect the density and distribution patterns of a population.
2. Interpret survivorship curves and life tables.
3. Recognize how the proportion of individuals at varying reproductive stages determines a population's age distribution.

Demography is the statistical study of a population that includes information about its density, distribution, and rate of growth (which are dependent on the population's mortality pattern and age distribution).

Density and Distribution

Population density is the number of individuals per unit area; for example, there are about 94 people per square mile in the United States. Population density figures make it seem that individuals are uniformly distributed, but this often is not the case. For example, most people in the United States live in cities, where the number of people per unit area is dramatically higher than in the country. And even within a city, more people live in particular neighborhoods than in others, and such distributions can change over time. Therefore, basing ecological models solely on population density can lead to inaccurate results.

Population distribution is the pattern of dispersal of individuals across a given area. The availability of resources will influence where populations live. **Resources** are nonliving (abiotic) and living (biotic) components of an environment that are used to support living organisms. The availability of adequate levels of light, water, space, enough mates, and food are some important resources required by all populations. **Limiting factors** are environmental aspects that determine where an organism lives and how large their population can become. For example, trout live only in cool mountain streams, where the oxygen content is high, but carp and catfish are found in rivers near the coast because they can tolerate warm waters with a lower concentration of oxygen. The timberline is the limit of tree growth in mountainous regions or in high latitudes. Trees cannot grow above the high timberline because of low temperatures and the fact that water remains frozen most of the year. Biotic factors can also play a role in the distribution of organisms. In Australia, the red kangaroo does not live outside arid inland areas because it is adapted to feeding on the grasses that grow there.

Three descriptions—*clumped, random,* and *uniform*—are often used to characterize patterns of distribution. Suppose you considered the distribution of a species across its full range. A range is that portion of the globe where the species can be found; for example, red kangaroos live in Australia. On that scale, you would expect to find a clumped distribution. However, organisms are located in areas suitable to their adaptations; as mentioned, red kangaroos live in grasslands, and catfish live in warm river water near the coast.

Within a smaller area, such as a single body of water or a single forest, the availability of resources influences which pattern of distribution is exhibited for a particular population. For example, a study of the distribution of hard clams in a bay on the south shore of Long Island, New York, showed that clam abundance is associated with sediment levels. Investigators hope to use this information to increase the density and number of clams in areas that have low clam populations.

Distribution patterns can vary from one species to the next. Patterns can also change based upon the availability of resources. Social animals, such as elephants, live and travel in groups, thus exhibiting a clumped type of distribution. In plants, such as dandelions, that have windblown seeds, a random distribution often occurs. A different situation occurs when some coastal birds gather for the breeding season, a time when nesting sites are in high demand. This can lead to a uniform distribution as each pair of birds builds and defends its nesting territory (Fig. 44.2).

A variety of factors influence the distribution patterns of organisms. These patterns may shift as a population increases or decreases in size. They can also shift as individuals within the population move through different stages of their life cycles. For example, creosote bushes in the desert shift from clumped to random to uniform as they go from immature to mature shrubs.

Figure 44.2 Distribution patterns in nature. **a.** Photograph of an elephant family shows a clumped distribution. **b.** Windblown dispersal of daisy seeds leads to a random distribution. **c.** Competition for limited nesting sites can result in a uniform distribution. (a): Guenterguni/E+/Getty Images; (b): Omersukrugoksu/iStock/Getty Images; (c): Laszlo Podor/Moment Open/Getty Images

a.

b.

Figure 44.3 Biotic potential. A population's maximum growth rate under ideal conditions—that is, its biotic potential—is greatly influenced by the number of offspring produced in each reproductive event. **a.** Mice, which produce many offspring that quickly mature to produce more offspring, have a much higher biotic potential than the rhinoceros (**b**), which produces only one or two offspring per infrequent reproductive event. (a): Tom McHugh/Science Source; (b): Tracey Thompson/Corbis/Getty Images

Population Growth

The **rate of natural increase (*r*),** or growth rate, is determined by the number of individuals born each year minus the number of individuals that die each year. It is assumed that immigration and emigration are equal and need not be considered in the calculation of the growth rate. Populations grow when the number of births exceeds the number of deaths. If the number of births is 30 per year and the number of deaths is 10 per year per 1,000 individuals, the growth rate is

$$(30 - 10) / 1{,}000 = 0.02 = 2.0\%$$

The highest possible rate of natural increase for a population is called its **biotic potential** (Fig. 44.3). For a population to reach its biotic potential, the number of limiting factors needs to be minimized. Some of these limiting factors can include the following:

- Number of offspring per reproductive event that survive until they are old enough to reproduce themselves
- Amount of competition within the population
- Age distribution of the members of the population, especially the number of reproductive females present
- Presence of disease and predators

Mortality Patterns

Population growth patterns assume that populations are made up of identical individuals. Actually, the individuals of a population are in different stages of their life span. A **cohort** consists of the members of a population born at the same time. Some investigators study population dynamics and construct life tables that show how many members of a cohort are still alive after certain intervals of time.

For example, Table 44.1 is a life table for a bluegrass cohort. The cohort started with 843 individuals. The table tells us that after 3 months, 121 individuals have died, and therefore the mortality rate is 0.143 per capita. Another way to express the same statistic, however, is to consider that 722 individuals are still alive—have survived—after 3 months. **Survivorship** is the probability of newborn individuals of a cohort surviving to particular ages. If we plot the number surviving at each age, a survivorship curve is produced.

Table 44.1 Life Table for a Bluegrass Cohort

Age (months)	Number Observed Alive	Number Dying	Mortality Rate per Capita	Avg. Number of Seeds per Individual
0–3	843	121	0.143	0
3–6	722	195	0.271	300
6–9	527	211	0.400	620
9–12	316	172	0.544	430
12–15	144	95	0.660	210
15–18	49	39	0.796	60
18–21	10	7	0.700	30
21–24	3	3	1.000	10
24	0	—	—	—

The results of investigations like this have revealed that each species generally fits one of three typical survivorship curves: I, II, and III (Fig. 44.4*a*). The type I curve is characteristic of a population in which most individuals survive well past the midpoint of the life span, and death does not come until near the end of the life span. Animals that have this type of survivorship curve include large mammals and humans in more-developed countries. In contrast, the type III curve is typical of a population in which most individuals die very young. This type of survivorship curve occurs in many invertebrates, fishes, and most plants. In the type II curve, survivorship decreases at a constant rate throughout the life span. For many songbirds and small mammals, death is usually unrelated to age; thus, they represent a type II survivorship curve.

The survivorship curves of natural populations do not always fit these three idealized curves. In a bluegrass cohort, as shown in

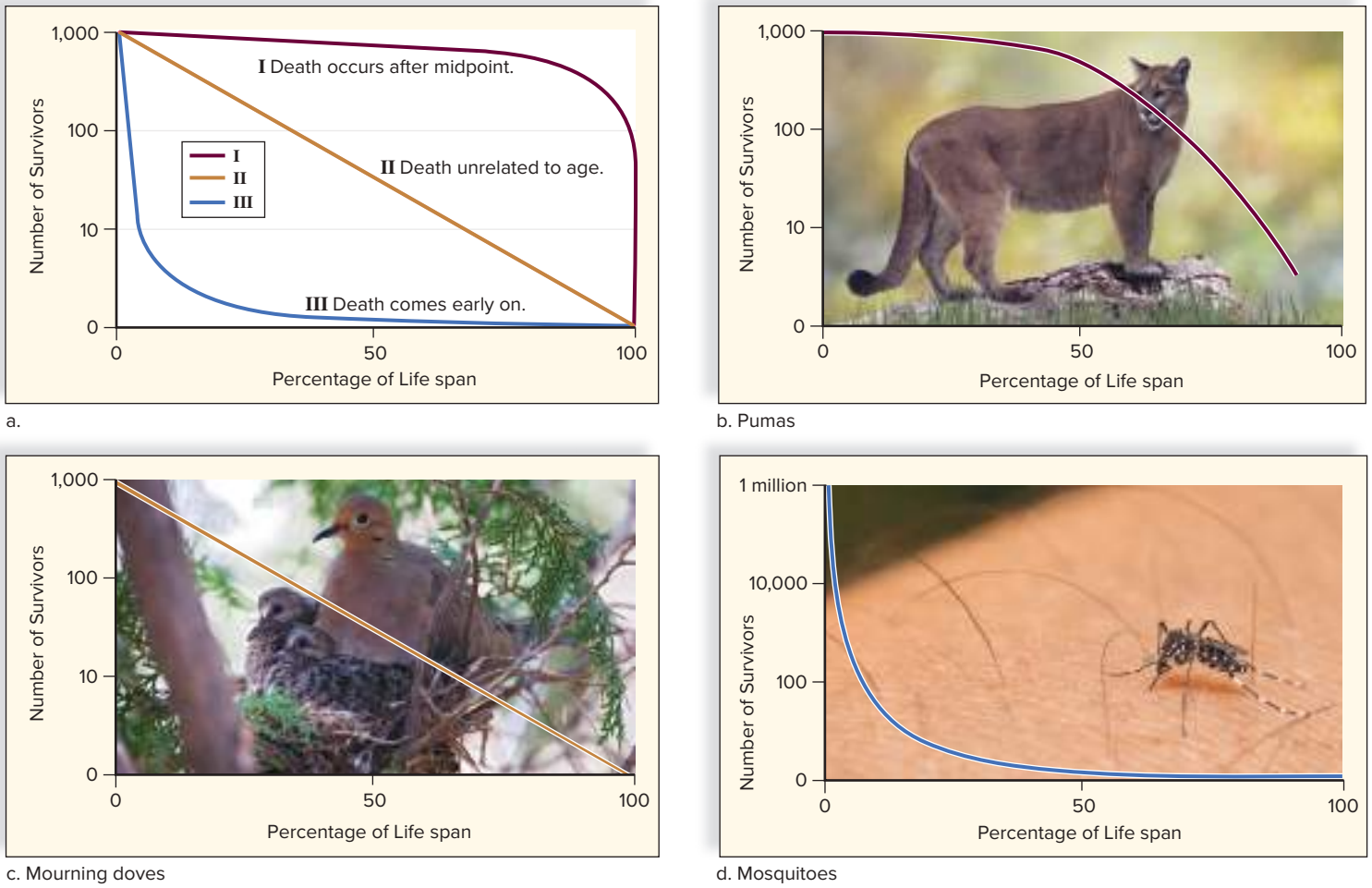


Figure 44.4 Survivorship curves. Survivorship curves show the number of individuals of a cohort that are still living over time. **a.** Three generalized survivorship curves. **b.** The survivorship curve for a puma represents a combination of the type I curve. **c.** The survivorship curve for a mourning dove generally fits a type II curve. **d.** The survivorship curve for mosquitoes is a type III curve. (photos): (b): Svetlana Foote/Shutterstock; (c): Klaus Fischel/Natural Selection/Design Pics; (d): Chik_77/Shutterstock

Table 44.1, for example, most individuals survive until 6–9 months, and then the chances of survivorship diminish at an increasing rate. Pumas have a similar type I survivorship curve (Fig. 44.4*b*), where most individuals survive the first few years of their 8- to 13-year life span. Statistics for a mourning dove cohort are close enough to classify the survivorship curve in the type II category (Fig. 44.4*c*), while a mosquito cohort has a type III curve (Fig. 44.4*d*).

Much can be learned about the life history of a species by studying its life table and the survivorship curve that can be constructed based on that table. In a population with a type III survivorship curve, it would be predicted that natural selection would favor those individuals that had more offspring. Because death comes early for most members, only a few would live long enough to reproduce, and therefore natural selection would favor those that produced more offspring.

Other types of information can be obtained from studying life tables. Looking again at Table 44.1, we can see that per-capita seed production increases as plants mature, and then seed production drops off.

Age Distribution

When the individuals in a population reproduce repeatedly, several generations may be alive at the same time. From the perspective of population growth, a population contains three major age groups: prereproductive, reproductive, and postreproductive. Populations differ according to what proportion of the population falls into each age group. At least three **age structure diagrams** are possible (Fig. 44.5).

Having a birthrate that is higher than the death rate results in the prereproductive group being the largest of the three groups. This produces a pyramid-shaped diagram. Under such conditions, even if the growth for that year were matched by the deaths for that year, the population would continue to grow in the following years. More individuals would be entering than leaving the reproductive group. Eventually, as the size of the reproductive group equals the size of the prereproductive group, a bell-shaped diagram results. The postreproductive group is still the smallest, however, because of mortality. If the birthrate falls below the death rate, the prereproductive group becomes smaller than the reproductive group. The age structure diagram is urn-shaped, because the postreproductive group is now the largest.

The age distribution reflects the past and future history of a population. Because a postwar baby boom occurred in the United States between 1946 and 1964, the postreproductive group will soon be the largest group.

ISTUDY

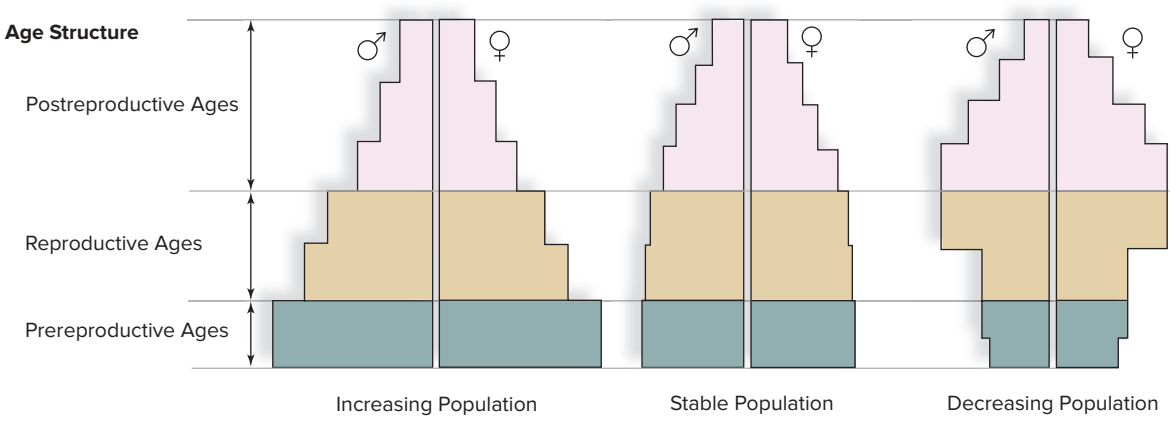


Figure 44.5 Age structure diagrams. Typical age structure diagrams for hypothetical populations that are increasing, stable, or decreasing. Different numbers of individuals in each age class create these distinctive shapes. In each diagram, the left half represents males, whereas the right half represents females.

Check Your Progress 44.2

1. Distinguish between population density and population distribution.
2. Describe the differences among type I, II, and III survivorship curves.
3. Explain why a pyramid-shaped age diagram indicates a growing population.

44.3 Population Growth Models

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe exponential population growth and the circumstances that encourage it.
2. Identify the features of logistic growth and the carrying capacity of a population.

Based on observation and natural selection principles, ecologists have developed two working models for population growth. In the pattern called **semelparity** (Gk. *seme,* "once"; L. *parous,* "to bear or bring forth"), the members of the population have only a single reproductive event in their lifetime. When the time for reproduction draws near, the mature adults cease to grow and expend all their energy in reproduction, and then die. Many insects, such as winter moths, and annual plants, such as zinnias, follow this pattern of reproduction growth. They produce a resting stage of development, such as eggs or seeds that can survive unfavorable conditions and resume growth the next favorable season. In other words, semelparity is an adaptation to an unstable environment.

In the pattern called **iteroparity** (Gk. *itero,* "repeat"), members of the population experience many reproductive events throughout their lifetime. They continue to invest energy in their future survival, which increases their chances of reproducing again. Iteroparity is an adaptation to a stable environment in which chances of survival for offspring are relatively high. Most vertebrates, shrubs, and trees have this pattern of reproduction.

Reproduction does not always fit these two patterns, though (Fig. 44.6). Even so, ecologists have found it useful to develop mathematical models of population growth based on these two very different patterns of reproduction. Although the mathematical models described in this section are simplifications, they still may

a.

b.

Figure 44.6 Patterns of reproduction. In general, organisms tend to reproduce continuously or have a single reproductive event. **a.** Aphids use asexual reproduction during the summer months, and then shift to sexual reproduction right before the onset of winter. **b.** Annual plants tend to produce a large number of seeds per reproductive event. The number that germinate often depends on environmental factors. (a): JasonOndreicka/iStock/Getty Images; (b): Lyn Holly Coorg/Photographer's Choice/Getty Images

ISTUDY

be used to predict the distribution and abundance of organisms, or the responses of populations when their environment changes in some way. Testing predictions permits the development of new hypotheses, which can then be evaluated.

Exponential Growth

As an example of semelparous reproduction, consider a population of insects in which females reproduce only once a year and then the adult population dies. Each female produces on the average 4.8 eggs per generation, half of which will develop into female offspring, which reproduce the following year. In the next generation, the females produce an average of 4.8 eggs. In this case of discrete breeding, R = net reproductive rate.[1] Net reproductive rate is used because it is the observed rate of natural increase after deaths have occurred.

Figure 44.7*a* shows how the population would grow year after year for 10 years, assuming that R stayed constant from generation to generation. This growth is equal to the size of the population, because all members of the previous generation have died. Mayflies, featured in Figure 44.7, have one reproductive event, which occurs in the spring. The development of the next generation requires as many as 50 molts during the winter. Figure 44.7*b* shows the growth curve for such a population. This growth curve, which is roughly J-shaped, depicts exponential growth. With **exponential growth,** the number of individuals added each generation increases rapidly due to the total number of reproductive females increasing in the population.

Notice that the growth curve in Figure 44.7*b* has two phases:

Lag phase: During this phase, growth is slow, because the population is small.
Exponential growth phase: During this phase, growth is accelerating.

Figure 44.7*c* gives the mathematical equation that allows you to calculate growth and size for any population that has discrete (nonoverlapping) generations. In other words, all members of the previous generation die off before the new generation appears. To use this equation to determine future population size, it is necessary to know R, which is the net reproductive rate determined after gathering mathematical data regarding past population increases. Notice that even though R remains constant, growth is exponential, because the number of individuals added each year is increasing. Therefore, the growth of the population is accelerating.

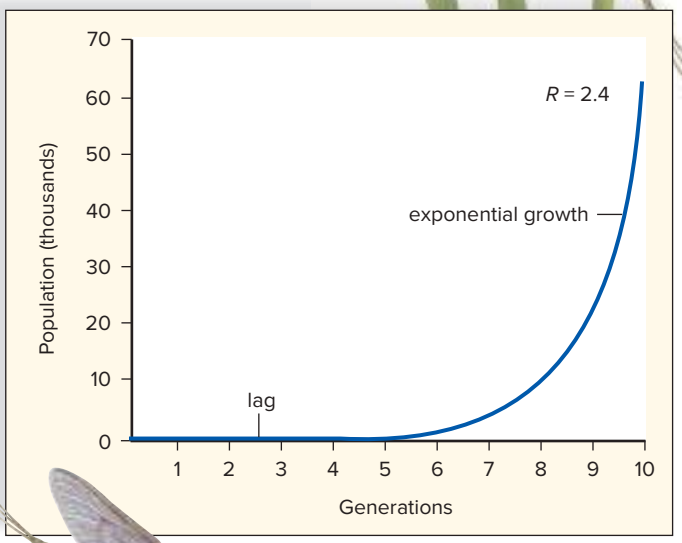
For exponential growth to continue unchecked, plenty of habitat, food, shelter, and any other requirements to sustain growth must be available. But in reality, environmental conditions prevent exponential growth. Eventually, any further growth is impossible because of limiting factors—the food supply runs out, and competition for limited resources begins to increase. Also, as the population increases in size, so do limiting factors like predation, parasites, disease, and competition among members.

Logistic Growth

A second type of growth curve results when limiting environmental factors that oppose growth come into play. In 1930, Raymond Pearl developed a method for estimating the number of yeast cells accruing every 2 hours in a laboratory culture vessel. His data are shown in Figure 44.8*a*. When the data are plotted, the growth curve has the appearance shown in Figure 44.8*b*. This type of growth curve is a sigmoidal (S) or S-shaped curve.

Generation	Population Size	Number of Females
0	10.0	5
1	24.0	12
2	57.6	28.8
3	138.2	69.1
4	331.7	165.9
5	796.1	398.1
6	1,910.6	955.3
7	4,585.4	2,292.7
8	11,005.0	5,502.5
9	26,412.0	13,206.1
10	63,388.8	31,694.5

a.



b.

To calculate population size from year to year, use this formula:

$$N_{t+1} = RN_t$$

N_t = number of females already present
R = net reproductive rate
N_{t+1} = population size the following year

c.

Figure 44.7 Model for exponential growth. When the data for discrete reproduction in (**a**) are plotted, the exponential growth curve in (**b**) results. **c.** This formula produces the same results as (**a**) and generates the same graph.

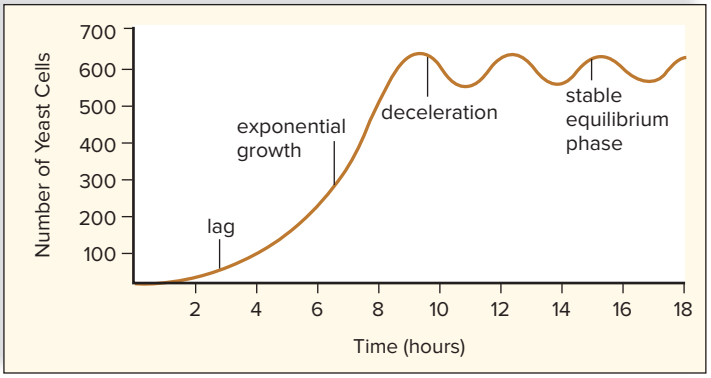
[1]Changing r to R is simply customary in discrete (nonoverlapping) breeding calculations; both coefficients deal with the same thing (birth minus death).

ISTUDY

Growth of Yeast Cells in Laboratory Culture

Time (t) (hours)	Number of individuals (N)	Number of individuals added per 2-hour period $\left(\frac{\Delta N}{\Delta t}\right)$
0	9.6	0
2	29.0	19.4
4	71.1	42.1
6	174.6	103.5
8	350.7	176.1
10	513.3	162.6
12	594.4	81.1
14	640.8	46.4
16	655.9	15.1
18	661.8	5.9

a.



b.

To calculate population growth as time passes, use this formula:

$$\frac{N}{t} = rN\left(\frac{K-N}{K}\right)$$

N = population size

N/t = change in population size

r = rate of natural increase

K = carrying capacity

$\frac{K-N}{K}$ = effect of carrying capacity on population growth

c.

Figure 44.8 Model for logistic growth. When the data for repeated reproduction (**a**) are plotted, the logistic growth curve in (**b**) results. **c.** This formula produces the same results as (**a**) and generates the same graph.

Notice that this **logistic growth** has four phases:

Lag phase: During this phase, growth is slow, because the population is small.

Exponential growth phase: During this phase, growth is accelerating.

Deceleration phase: During this phase, growth slows down.

Stable equilibrium phase: During this phase, there is little if any growth, because births and deaths are about equal. The population size tends to oscillate around the environmental carrying capacity.

Figure 44.8*c* gives the mathematical equation that allows us to calculate logistic growth. (The curve is termed *logistic* because the exponential portion of the curve produces a straight line when the log of N is plotted.) The entire equation for logistic growth is below but let's consider each portion of the equation separately:

$$\frac{N}{t} = rN\frac{(K-N)}{K}$$

Because the population has repeated reproductive events, we need to consider growth as a function of change in time (Δ):

$$\frac{\Delta N}{\Delta t} = rN$$

If the change in time is very small, then we can use differential calculus, and the instantaneous population growth (d) is given by:

$$\frac{dN}{dt} = rN$$

This portion of the equation applies to the first two phases of growth—the lag phase and the exponential growth phase. Here, also, we do not expect exponential growth to continue. Charles Darwin calculated that a single pair of elephants could have over 19 million live descendants after 750 years. Others have calculated that a single female housefly could produce over 5 trillion flies in 1 year! Such explosive growth does not occur, however, because environmental conditions, both abiotic and biotic, cause population growth to slow down. The yeast population mentioned earlier was grown in a vessel in which food would run short and waste products would accumulate. Such environmental conditions prevent exponential growth from continuing.

Look again at Figure 44.8. Following exponential growth, a population is expected to enter a deceleration phase and then a stable equilibrium phase of the logistic growth curve. Now the population is at the carrying capacity of the environment.

Carrying Capacity

The environmental **carrying capacity (*K*)** is the maximum number of individuals of a given species the community can support. As the population size approaches the carrying capacity of the community, the more likely it is that resources will become scarce and that biotic effects, such as competition and predation, will put pressure on the population. The birthrate is expected to decline, and the death rate is expected to increase. This results in a decrease in population growth; eventually, the population stops growing and its size remains stable. Carrying capacity in any community can vary throughout time, depending on fluctuating conditions—for example, the amount of rainfall from one year to the next.

ISTUDY

How does the mathematical model for logistic growth take this process into account? To our equation for growth under conditions of exponential growth, we add the following term:

$$\frac{(K - N)}{K}$$

In this expression, K is the carrying capacity of the environment. The easiest way to understand the effects of this term is to consider two extreme possibilities. First, consider a time at which the population size is well below carrying capacity. Resources are relatively unlimited, and we expect rapid, nearly exponential growth to take place. The model predicts this. When N is very small relative to K, the term $(K - N)/K$ is very nearly $(K - 0)/K$, or approximately 1. Therefore, $(dN)/(dt)$ is approximately equal to rN.

Similarly, consider what happens when the population reaches carrying capacity. Here, we predict that growth will stop and the population will stabilize. When N is equal to K, the term $(K - N)/K$ declines from nearly 1 to 0, and the population growth slows to zero.

As mentioned, the model predicts that exponential growth occurs only when population size is much lower than the carrying capacity. So as a practical matter, if humans are using a fish population as a continuous food source, it is best to maintain the size of the fish population in the exponential phase of the logistic growth curve. Biotic potential can have its full effect, and the birthrate is the highest it can be during this phase. If we overfish, the population will sink into the lag phase, and it will be years before exponential growth recurs.

In contrast, if we are trying to limit the growth of a pest, it is best, if possible, to reduce the carrying capacity rather than reduce the population size. Reducing the population size only encourages exponential growth to begin once again. Farmers can reduce the carrying capacity for a pest by alternating rows of different crops, rather than growing one type of crop throughout an entire field.

Check Your Progress 44.3

1. Explain how carrying capacity (K) limits exponential growth.
2. Explain the conditions that would cause a population to undergo logistic growth.

44.4 Regulation of Population Size

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the density-independent and density-dependent factors that affect population size.
2. Describe the intrinsic factors that can impact population size and growth.

In a study of winter moth population dynamics, researchers discovered that a large proportion of eggs did not survive the winter and exponential growth never occurred. Perhaps a low number of individuals at the start of each season helps prevent the occurrence of exponential growth.

It is possible that exponential growth causes population size to rise above the carrying capacity of the environment, and as a consequence a population crash may occur. As one example, in 1911, 4 male and 21 female reindeer were released on St. Paul Island in the Bering Sea off Alaska. St. Paul Island had an undisturbed environment, and there was little hunting pressure and no predators. The herd grew exponentially to about 2,000 reindeer by 1938, overgrazed the habitat, and then abruptly declined to only 8 animals by 1950 (Fig. 44.9).

This pattern of population explosion, eventually followed by a population crash, is called irruptive, or Malthusian, growth. It is named in honor of the eighteenth-century economist Thomas Robert Malthus, who had a great influence on Charles Darwin. Populations do not ordinarily undergo Malthusian growth because limiting factors regulate population growth.

Ecologists have long recognized that both biotic and abiotic conditions play an important role in regulating population size in natural environments.

Density-Independent Factors

Abiotic factors include events such as droughts, freezes, hurricanes, floods, and forest fires. Any one of these natural disasters can kill individuals and lead to a sudden and catastrophic reduction in population size. However, the density of the population does not

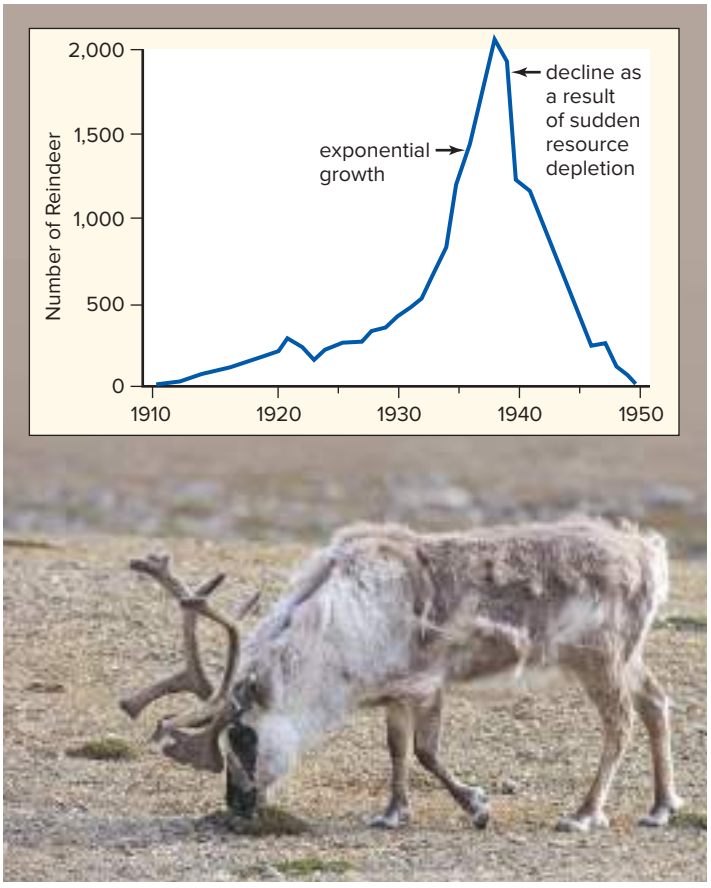


Figure 44.9 Density-dependent effect. On St. Paul Island, Alaska, reindeer, *Rangifer,* grew exponentially for several seasons and then underwent a sharp decline as a result of overgrazing the available range. (photo): SuperStock/Alamy Stock Photo

ISTUDY

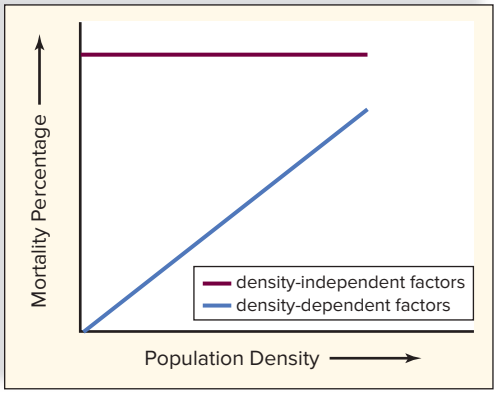


Figure 44.10 Effects of density-dependent versus density-independent factors. This shows the difference in the percentage of individuals that die compared to the density of population.

increase or decrease the frequency of these events. Therefore, an abiotic factor is usually a **density-independent factor,** meaning that the intensity of the effect does not increase or decrease in response to the population density (Fig. 44.10).

For example, the percentage of individuals killed in a flash flood event is independent of density—floods do not necessarily kill a larger percentage of a dense population than of a less dense population. Nevertheless, the larger the population, the greater the number of individuals probably affected. In Figure 44.11, the impact of a flash flood on a low-density population of mice living in a field was 3 out of 5 (60% mortality), whereas the impact on a high-density population was 12 out of 20 (also 60% mortality).

Density-Dependent Factors

Biotic factors are considered **density-dependent factors,** because the percentage of the population affected increases as the density of the population increases. The intensity of competition, predation, disease, and parasitism within a population will increase in intensity as the density of the population increases.

Competition can occur when members of a species attempt to use the same resources (such as light, food, space) that are in limited supply. As a result, not all members of the population will have adequate access to the resources necessary to ensure their survival or reproduction. As an example, let's consider a western bluebird (*Sialia mexicana*) population in which members have to compete for nesting sites. Each pair of birds requires a tree hole to raise offspring. If there are more holes than breeding pairs, each pair can have a hole in which to lay eggs and rear young birds. But if there are fewer holes than there are breeding pairs, then each pair must compete to acquire a nesting site. Pairs that fail to gain access to holes will be unable to reproduce and raise young.

Competition for food also controls population growth. However, resource partitioning among age groups is a way to reduce competition for food. As mentioned earlier in this text, the life cycle of butterflies includes caterpillars, which require a different food from the adults. The caterpillars graze on leaves, while the adults feed on nectar produced by flowers. This prevents the parents from competing with their offspring for food.

Predation occurs when one living organism, the predator, eats another, the prey. In the broadest sense, predation impacts every species within a community at some level. The effect of predation on a prey population generally increases as the prey population grows larger, because prey are easier to find. Consider a field inhabited by a population of mice (Fig. 44.12). Each mouse must have a hole in which to hide to avoid being eaten by a hawk. If there are 100 holes, and a low density of 102 mice, then only 2 mice will be left out in the open. It might be hard for the hawk to find only 2 mice in the field. If neither mouse is caught, then the predation rate is 0/2 = 0%. However, if there are 100 holes, and a high density of 200 mice, then the chance is greater that the hawk will be able to find some of these 100 mice without holes. If half of the exposed mice are caught, the predation rate is 50/100 = 50%. Therefore, increasing the density of the available prey has increased the proportion of the population preyed upon.

Parasites, such as blood-sucking ticks, are generally much smaller than their hosts. Parasites do not attempt to kill their hosts, but they can weaken them over time. A highly parasitized individual is less apt to produce as many offspring than it would if it were healthy. In this way, parasitism also plays a role in regulating population size.

Disease is much more likely to spread in a dense population due to the amount of contact between individuals who are healthy versus those who are infected. Consider the spread of the *Yersinia pestis* plague, known as the Black Death, in Europe during the fourteenth century. This outbreak of bacterial disease reduced the population of Europe by 30–60%.

Other Considerations

Density-independent and density-dependent factors are extrinsic to the organism. Intrinsic factors can also affect the size and growth of a

a. Low density of mice

b. High density of mice

Figure 44.11 Density-independent effects. The impact of a density-independent factor, such as weather or natural disasters, is not influenced by population density. The impact of a flash flood on (**a**) a low-density population (mortality rate of 3/5, or 60%) is similar to the impact on (**b**) a high-density population (mortality rate of 12/20, also 60%).

ISTUDY

a. Low density of mice

b. High density of mice

Figure 44.12 Density-dependent effects—predation. The impact of predation on a population is directly proportional to the density of the population. In a low-density population (**a**), the chances of a predator finding the prey are low, resulting in little predation. But in the higher-density population (**b**), there is a greater likelihood of the predator locating potential prey, resulting in a greater predation rate.

population. Intrinsic factors are those based on the anatomy, physiology, or behavior of the organism. Territoriality and dominance hierarchies are behaviors that affect population size and growth rates. Recruitment and migration are other intrinsic social means by which the population sizes of more complex organisms are regulated.

Outside of any regulating factors, it could be that some populations have an innate instability. Ecologists have developed models that predict complex, erratic changes in even simple systems. For example, a computer model of Dungeness crab populations assumed that adults produce many larvae and then die. Most of the larvae do not survive, and those that do, stay close to home. Under these circumstances, the model predicted wild fluctuations in population size without a recurring pattern. This type of complex, nonrandom generation is called *deterministic chaos,* or simply *chaos.*

Population growth–regulating factors can serve as selective agents. Some members of a population may possess traits that make it more likely that they, rather than other members of the population, will survive and reproduce when these particular density-independent or density-dependent factors are present in the environment. Therefore, these traits will be more prevalent in the next generation whenever these factors are a part of the environment.

Check Your Progress 44.4

1. Describe the effect that population density can have on competition and predation.
2. Provide examples that show how a density-independent factor can act as a selective agent.
3. Identify various intrinsic factors that influence population size and growth.

44.5 Life History Patterns

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the two life history patterns of species.
2. List organisms that exemplify *r*-selection and those that exhibit *K*-selection.

Populations vary on such particulars as the number of births per reproductive event, the age at reproduction, the life span, and the probability of living the entire life span. These particulars are part of a species' life history. Life histories contain characteristics that can be thought of as trade-offs. Each population is able to capture only so much of the available energy, and how this energy is distributed among its life span (short versus long), reproduction events (few versus many), care of offspring (little versus much), and so forth has evolved over time. Natural selection shapes the life history of individual species. Related species, such as frogs and toads, may have different life history patterns if they occupy different environmental roles (Fig. 44.13).

The logistic population growth model has been used to suggest that members of some populations are subject to *r*-selection and members of other populations are subject to *K*-selection.

r-Selected Populations

In fluctuating or unpredictable environments, density-independent factors keep populations in the lag or exponential phase of population growth. Population size is low relative to *K,* and ***r*-selection** favors *r*-strategists, which tend to be small individuals that mature early and have a short life span. Most energy typically goes into producing a large number of relatively small offspring, with minimal energy going into parental care. The more offspring, the more likely it is that some of them will survive to reproductive age.

When there are low population densities, density-dependent mechanisms, such as predation and intraspecific competition, are unlikely to play a major role in regulating population size and growth rates most of the time. Such organisms are often very good dispersers and colonizers of new habitats. Classic examples of such *opportunistic species* are bacteria, some fungi, many insects, rodents, and annual plants (Fig. 44.14).

K-Selected Populations

Some environments are relatively stable and predictable, and in these environments populations tend to be near *K,* with minimal fluctuations in size. Resources such as food and shelter are relatively scarce for these individuals, and those that are best able to compete have the greatest reproductive success. ***K*-selection** favors *K*-strategists,

ISTUDY

a. Mouth-brooding frog, *Rhinoderma darwinii*

b. Strawberry poison arrow frog, *Dendrobates pumilio*

c. Midwife toad, *Alyces obstetricans*

Figure 44.13 Parental care among frogs and toads. **a.** In mouth-brooding frogs of South America, the male carries the larvae in a vocal pouch (directly under the mouth), which elongates the full length of his body, before the froglets are released. **b.** In poison arrow frogs of Costa Rica, after the eggs hatch, the tadpoles wiggle onto the parent's back and are then carried to water. **c.** The midwife toad of Europe carries strings of eggs entwined around his hind legs and takes them to water when they are ready to hatch. (a): Danita Delimont/Alamy Stock Photo; (b): Riccardo Oggioni/Alamy Stock Photo; (c): Tom McHugh/Science Source

species that allocate the majority of their energy to their own growth and survival, as well as to the growth and survival of their offspring. Therefore, they tend to be fairly large, are slow to mature, and have relatively few offspring (fecundity). They also have a limited number of reproductive events (parity) and have a fairly long life span.

Because these organisms, termed *equilibrium species,* are strong competitors, once they become established they often outcompete opportunistic species. They are specialists rather than generalizers and tend to become extinct when their normal way of life is destroyed. The best possible examples of *K*-strategists include long-lived plants (saguaro cacti, oaks, cypresses, and pines), birds of prey (hawks and eagles), and large mammals (whales, elephants, bears, and humans) (Fig. 44.14). Another example of a *K*-strategist is the Florida panther, the largest mammal in the Florida Everglades. It requires a very large range and produces few offspring, which require parental care. Currently, the Florida panther is unable to compensate for a reduction in its range and is therefore on the verge of extinction.

The Two Strategies in Nature

Nature is actually more complex than the two possible life history patterns suggest. It now appears that *r*-strategist and *K*-strategist populations are at the ends of a spectrum, and most populations lie somewhere between these two extremes. For example, recall that plants have an alternation-of-generations life cycle—the sporophyte generation and the gametophyte generation. Ferns, which could be classified as *r*-strategists, distribute many spores and leave the gametophyte to fend for itself. In contrast, gymnosperms (e.g., pine trees) and angiosperms (e.g., oak trees), which could be classified as *K*-strategists, retain and protect the gametophyte. They produce seeds that contain the next sporophyte generation plus stored food. The added investment is significant, but these plants still release large numbers of seeds.

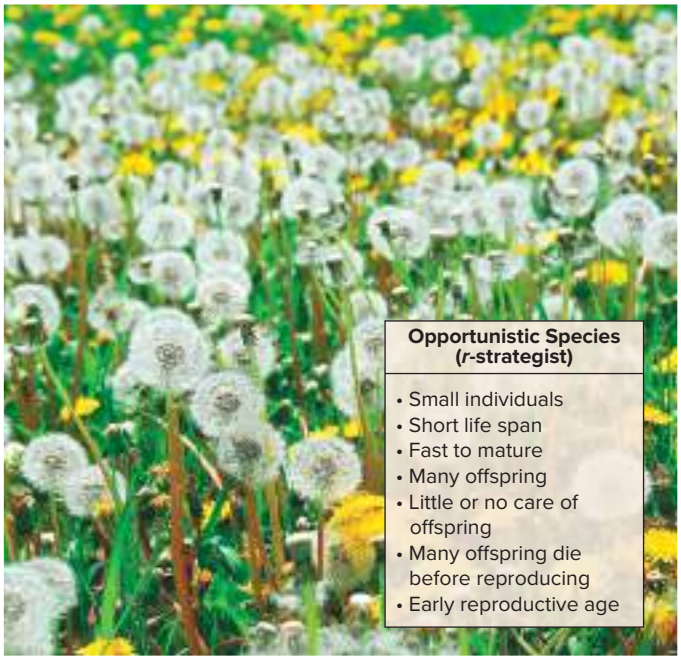


Figure 44.14 Life history strategies. Are dandelions *r*-strategists with the characteristics noted, and are bears *K*-strategists with the characteristics noted? Most often, the distinctions between these two possible life strategies are not clear-cut. (dandelions): Elena Elisseeva/Alamy Stock Photo; (bears): Barrett Hedges/National Geographic/Getty Images

ISTUDY

THEME Biological Systems

Sustainability in the Lobster Industry

Conservation and sustainability are often viewed to be in opposition to economic growth, but for the lobster industry, as well as many others, the exact opposite is true. Maine is home to the largest lobster industry in the United States. During the early days of lobster fishing, there were relatively few regulations and quotas on the number or sizes of lobsters that could be harvested. In the 1870s, several laws were implemented that prevented the harvesting of egg-bearing females and also established the minimum size of a lobster that could be legally harvested. The minimum legal size for a harvested lobster is 3¼ inches from the eye socket to the start of the tail (carapace length), resulting in only 1 lb out of every 14 lb of lobsters being harvested. The science behind the decision was based on the fact that when lobsters were harvested at smaller sizes, over 90% of them had not reached their breeding size. Lobsters go through a variety of different life stages before they reach adulthood. During many of these early life stages, they can become a quick meal to a large number of ocean predators. If they are fortunate enough to reach maturity, a big female can produce up to 100,000 eggs, while a smaller female may produce only 10,000 eggs (Fig. 44A). Unfortunately, the vast majority of their offspring never make it to adulthood. Depending on water temperatures, it may take a lobster 5 to 8 years to reach adulthood.

Harvesting smaller immature lobsters meant that there would only be a 10% breeding population available to sustain the entire population. Increasing the minimum harvest size meant that approximately 50% of the breeding population would still remain in the water.

Unfortunately, some lobster fishers ignored the sustainability laws and harvested lobsters that were undersized, as well as egg-bearing females. They would scrub the eggs off the females and hide the undersized lobsters to avoid getting caught. This contributed to the decline of the lobster harvest, causing it to drop to less than half its historical numbers by the early 1930s.

At this point, many lobster fishers became desperate to try anything to save their industry. In 1933, a sustainability law was enacted that placed a maximum size on lobsters that could be harvested. The additional requirement prohibits the collection of lobsters that have a carapace length over 5 inches. This was implemented to allow the largest breeders, lobsters who have the potential to produce 100,000 eggs, to remain in the population. Many of these large females will have their tails notched with a "V" to signify they are part of the breeding pool. Since the implementation of these new policies, the numbers of lobsters taken in during harvests has been on the rise.

A conservation ethic has taken hold of the Maine lobster industry, which resulted in a shift to their culture. While many of the rules that limit which and how many lobsters can be harvested during a particular season may seem harsh, it was the lobster fishers who came up with most of the rules. Innovations in lobster trap design that minimize the number of illegal lobsters harvested, as well as trap limits, zoning laws, and additional licensing and management laws, have been implemented.

Many Maine lobster fishers have a "conservation ethic" that dates back over a century and has enabled the industry to not only be sustainable but thrive. In fact, the Maine lobster industry is one of the world's most sustainable fisheries.

Figure 44A Female lobster with eggs. A mature female lobster has the potential to produce 100,000 eggs during a single reproductive cycle. Owen Humphreys/PA Images/Getty Images

Questions to Consider

1. If the lobster population continues to increase, should the minimum and/or maximum harvestable size of a lobster be changed?
2. Who should ultimately determine the harvest rules for lobsters?
3. What would be the ecological consequence to the ocean if lobsters were to go extinct?

Also, adult size is not always a determining factor for the life history pattern. For example, a cod is a rather large fish weighing up to 12 kg and measuring nearly 2 m in length—but cod release gametes in vast numbers, the zygotes form in the sea, and the parents make no further investment in the developing offspring. Of the 6–7 million eggs released by a single female cod, only a few will become adult fish. Cod are generally considered *r*-strategists.

The Biological Systems feature, "Sustainability in the Lobster Industry," describes how biologists used the life history of lobsters to help rebuild and maintain their population. This helped revitalize the dying lobster industry in Maine.

ISTUDY

Check Your Progress 44.5

1. Compare and contrast the general characteristics of a *K*-strategist and an *r*-strategist.
2. Explain why a population may vary between *K* and *r* strategies.

44.6 Human Population Growth

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the past and present growth patterns of the human population.
2. Identify the pressures the human population places on the Earth's resources.

The human population has risen steadily to its present size of over 8.0 billion people (Fig. 44.15). Prior to 1750, the growth of the human population was relatively slow, but as more reproducing individuals were added, growth increased, until the curve began to angle steeply upward, indicating that the population was undergoing exponential growth. The number of people added annually to the world population peaked at about 87 million around 1990, and currently around 83 million people are added per year. This is roughly equal to the current populations of Argentina, Ecuador, and Peru combined.

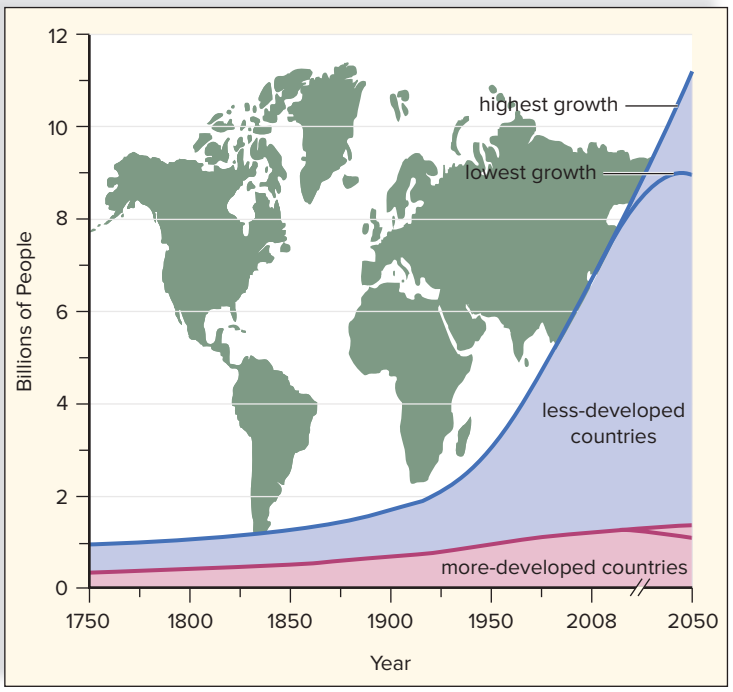


Figure 44.15 World population growth. **a.** The world's population size is now slightly over 8.0 billion. It is predicted that it may level off to between 9 and 11 billion by 2050, depending on the speed with which the growth rate declines.

The potential for future population growth can be appreciated by considering the **doubling time,** the length of time it takes for the population size to double. This was considered to be in the range of 35–60 years for humans, but most population experts recognize that it will probably take longer to reach 9–11 billion. Such an increase in population size would put extreme demands on our ability to produce and distribute resources. In 50 to 60 years, the world would need double the amount of food, jobs, water, energy, and so on just to maintain everyone at their present standard of living.

Many people are gravely concerned about the increased amount of resources that will be needed to support the additional growth in the human population. The first billion didn't occur until 1800; the second billion was attained in 1930; the fourth billion in 1974; and today there are over 8.0 billion. The world's population may level off to between 9 and 11 billion, depending on the speed at which the growth rate declines. Zero population growth cannot be achieved until the birthrate is equal to the death rate.

More-Developed vs. Less-Developed Countries

The countries of the world can be divided into two general groups. In the **more-developed countries (MDCs),** typified by the United States, Germany, Japan, and Australia, population growth is low, and the people enjoy a high standard of living. In the **less-developed countries (LDCs),** typified by Afghanistan, Ethiopia, Rwanda, and Haiti, population growth is high, and the majority of people live in poverty.

The MDCs doubled their populations between 1850 and 1950. This change was largely due to a decline in the death rate, the development of modern medicine, and improved socioeconomic conditions. The decline in the death rate was followed shortly thereafter by a decline in the birthrate, so populations in the MDCs experienced only modest growth between 1950 and 1975. This sequence of events (decreased death rate followed by decreased birthrate) is termed a **demographic transition.** Yearly growth of the MDCs has now stabilized at about 0.3%.

In contrast to most other MDCs, there is no end in sight for U.S. population growth. Although yearly growth of the U.S. population is only 0.72%, many people immigrate to the United States each year. The baby boomers, individuals who were born between 1946 and 1964, have all now left the reproductive age group, which has contributed to a decline in birthrates. In the future, immigration will be one of the biggest drivers of population growth rates in most MDCs.

Most of the world's population (approximately 80%) now lives in LDCs. Although the death rate began to decline steeply in the LDCs following World War II because of the importation of modern medicine from the MDCs, the birthrate remains fairly high. The yearly growth of the LDCs peaked at 2.5% between 1960 and 1965. Since that time, a demographic transition has been occurring, and the death rate and birthrate have fallen. The collective growth rate for the LDCs is now around 1.5%, but many countries in sub-Saharan Africa have not experienced this decline. In some of these, females average more than five children each.

The collective population of the LDCs and countries like India and China, who are transitioning out of being an LDC, may

ISTUDY

explode from 5.9 billion today to over 8 billion by 2050. Some of this increase will occur in Africa, but most will occur in Asia, because the Asian population is now about four times the size of the African population. Asia already has 60% of the world's population, living on 31% of the world's arable (farmable) land. Therefore, Asia is expected to experience acute water scarcity, a significant loss of biodiversity, and increased urban pollution. Twelve of the world's most polluted cities are in Asia.

The following are suggestions for reducing the expected population increase in the LDCs:

- Establish or strengthen family planning programs. A decline in growth is seen in countries with good family planning programs supported by community leaders.
- Use social progress to reduce the desire for large families. Providing education, raising the status of females, and reducing child mortality are social improvements that could reduce this desire.
- Delay the onset of childbearing. This, along with wider spacing of births, could help birthrate decline.

Age Distributions

The populations of the MDCs and LDCs can be divided into the three age groups introduced earlier: prereproductive, reproductive, and postreproductive (Fig. 44.16). Currently, the LDCs are experiencing a population momentum, because they have more young females entering the reproductive years than older females leaving them.

Most people are under the impression that if each couple has two children, **zero population growth** (no increase in population size) will take place immediately. However, most countries will continue growing due to the age structure of the population and the human life span. If there are more young females entering the reproductive years than there are females who are older leaving them, **replacement reproduction** will still result in population growth.

Many MDCs have a stable age structure, but most LDCs have a youthful profile—a large proportion of the population is younger than 15 years old. For example, in Nigeria, 41.7% of the population is under age 15. On average, Nigerian females have a fertility rate of approximately 4.62 per female. As a result of these numbers, the population of Nigeria is expected to increase from 214 million to over 392 million by 2050. The population of the continent of Africa is projected to increase from slightly over 1.3 billion to around 2.5 billion by 2050. This means that the LDC populations will still expand, even after replacement reproduction is attained. The more quickly replacement reproduction is achieved, however, the sooner zero population growth can also be achieved.

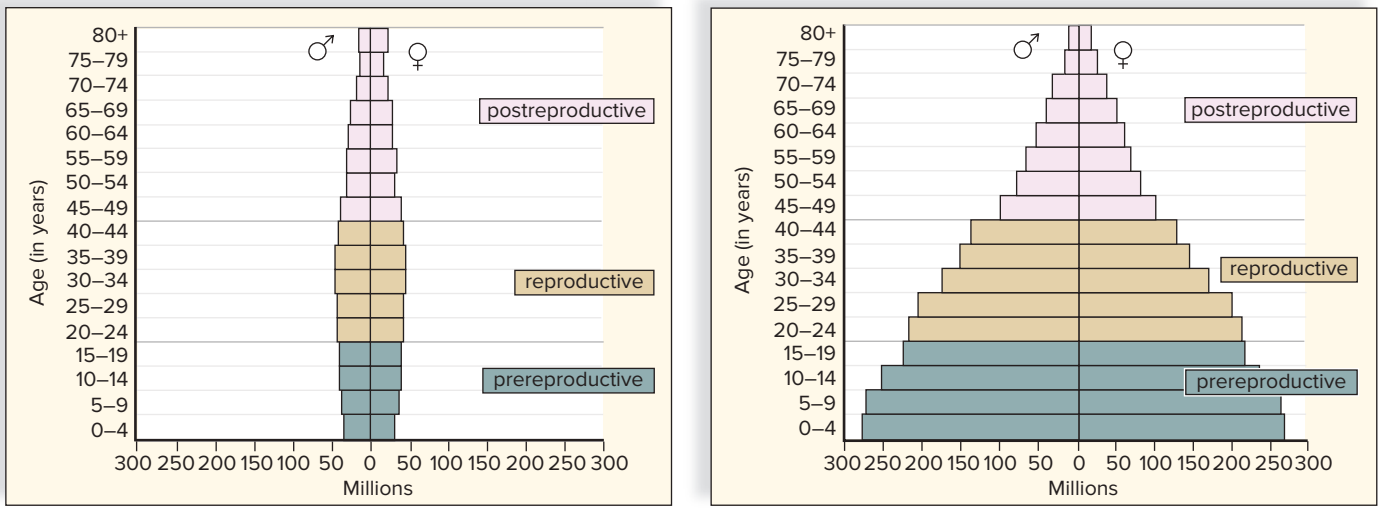
Population Growth and Environmental Impact

Population growth is putting extreme pressure on each country's social organization, the Earth's resources, and the biosphere. Because the population of the LDCs is still growing at a significant rate, it might seem that their population increase will be the greater cause of future environmental degradation. But this is not necessarily the case, because the MDCs consume a much larger proportion of the Earth's resources than do the LDCs. This consumption leads to environmental degradation, which is a major concern.

Environmental Impact

The environmental impact (EI) of a population is measured not only in terms of population size but also in terms of resource consumption per capita and the pollution produced by this consumption. In other words,

EI = population size × resource consumption per capita
= pollution per unit of resource used



a. More-developed countries (MDCs)

b. Less-developed countries (LDCs)

Figure 44.16 Age-structure diagrams. The shape of these age-structure diagrams allows us to predict that (**a**) the populations of MDCs are approaching stabilization and (**b**) the populations of LDCs will continue to increase for some time.

ISTUDY

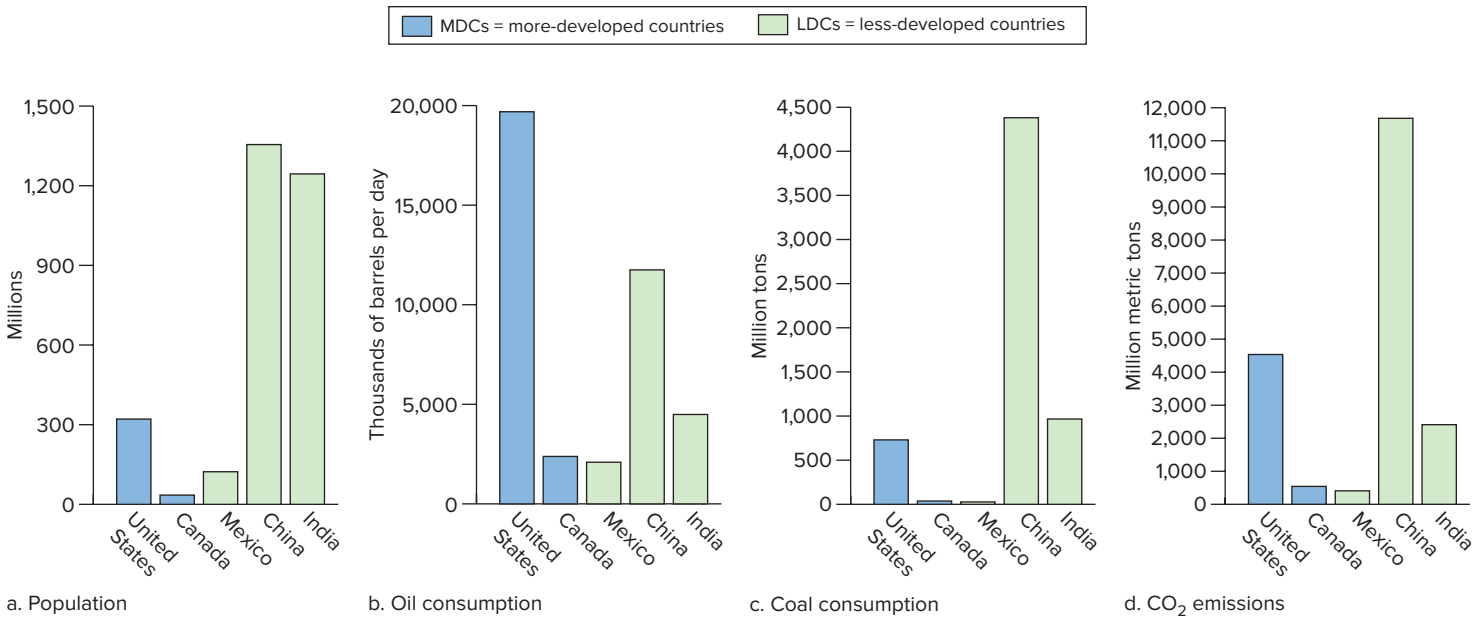


Figure 44.17 Environmental impact caused by MDCs and LDCs. **a.** MDCs have significantly smaller populations than LDCs. **b.** In the United States, per-person oil consumption is 7.25 times greater than that in China. **c.** In the United States, per-person coal consumption is 3.24 times greater than that in India. **d.** In the United States, per-person CO_2 production is 4.2 times greater than that in Mexico.

Therefore, there are two possible types of overpopulation: The first is simply due to population growth, and the second is due to increased resource consumption caused by population growth. The first type of overpopulation is more obvious in LDCs, and the second type is more obvious in MDCs, because the per-capita consumption is so much higher in the MDCs. For example, an average family in the United States, in terms of per-capita resource consumption and waste production, is the equivalent of approximately 30 people in India. We need to realize, therefore, that only a limited number of people can be sustained at anywhere near the standard of living in the MDCs.

The current environmental impact caused by MDCs and LDCs is shown in Figure 44.17. The MDCs account for only about one-fourth of the world population (Fig. 44.17*a*). But the MDCs have a significantly greater per-person consumption of oil (Fig. 44.17*b*) and coal (Fig. 44.17*c*). While China leads the world in CO_2 emissions (Fig. 44.17*d*), it also has the largest percentage of the human population, making its per-person impact smaller than that of MDCs.

As the LDCs become more industrialized, their per-capita consumption will also rise and, in some LDCs, it may nearly equal that of a more-developed country. For example, China's economy has been growing rapidly and, because China has such a large population (1.44 billion), it is competing with the United States for oil and metals in world markets. It could be that as less-developed countries consume more, people in the United States will have no choice but to consume less.

Consumption of resources has a negative effect on the environment. In Section 45.4, you'll learn how resource consumption leads to various pollution problems, and in Section 47.3, you'll see how our increasing environmental impact may cause a mass extinction of wildlife that will significantly impact our entire planet.

Check Your Progress 44.6

1. Compare the population growth of the LDCs with that of the MDCs.
2. Describe why replacement reproduction still leads to continued population growth.
3. Explain how an increase in the consumption of resources by LDCs would affect consumption by MDCs.

44.7 A Sustainable Future

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare renewable versus nonrenewable resources.
2. List different methods of transitioning a society toward renewable energy resources.
3. Explain how agricultural practices can be adjusted to minimize their environmental impacts.

Human consumption of natural resources, resulting in the loss or alteration of natural habitats (Fig. 44.18), is the main cause for the current loss of biodiversity. Some resources are nonrenewable, and some are renewable. **Nonrenewable resources** are those that are limited in supply. For example, the amount of land, fossil fuels, and minerals is finite and is being exhausted. Even though better extraction methods can make more fossil fuels and minerals available, and efficient use and recycling can make the supply last longer, eventually these resources will run out. **Renewable resources** are not limited in supply. For example, certain forms of energy, such as solar and wind energy, can be replenished.

ISTUDY

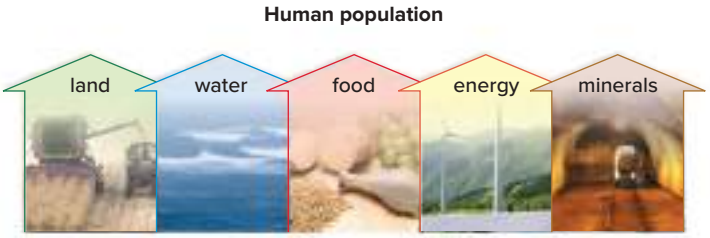


Figure 44.18 Resources. Humans use land, water, food, energy, and minerals to meet their basic needs. This provides us with a place to live, food to eat, and products to support our lifestyles. (land): Budimir Jevtic/Shutterstock; (water): Evelyn Jo Johnson; (food): John Thoeming/McGraw Hill; (energy): Geniusksy/Shutterstock; (minerals): TTstudio/Shutterstock

When a society exhibits the following characteristics, it is an indication that it is not sustainable in its current form (Fig. 44.19*a*):

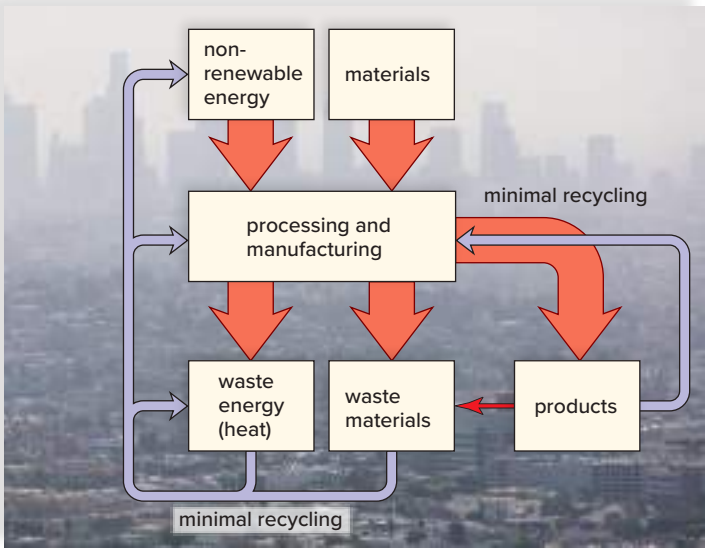
- A large percentage of the natural ecosystems have been altered for human purposes.
- Nonrenewable fossil fuels provide the majority of energy needed, which results in global warming, acid precipitation, and smog.
- Fresh water is being used at a rate faster than it can be replenished in aquifers and other sources.
- Agriculture requires large inputs of nonrenewable fossil fuel energy, fertilizer, and pesticides, which create much pollution.
- At least half of the agricultural yield in the United States goes toward feeding animals; it takes 10 lb of grain to produce 1 lb of meat.
- Minerals are nonrenewable, and the mining, manufacture, and use of mineral products is responsible for a significant amount of environmental pollution.

To move toward a more sustainable society, we need to move away from the use of nonrenewable resources and use more renewable ones. For a society to be considered sustainable, it needs to be able to provide the same goods and services for future generations that it provides for the current one. At the same time, biodiversity is conserved.

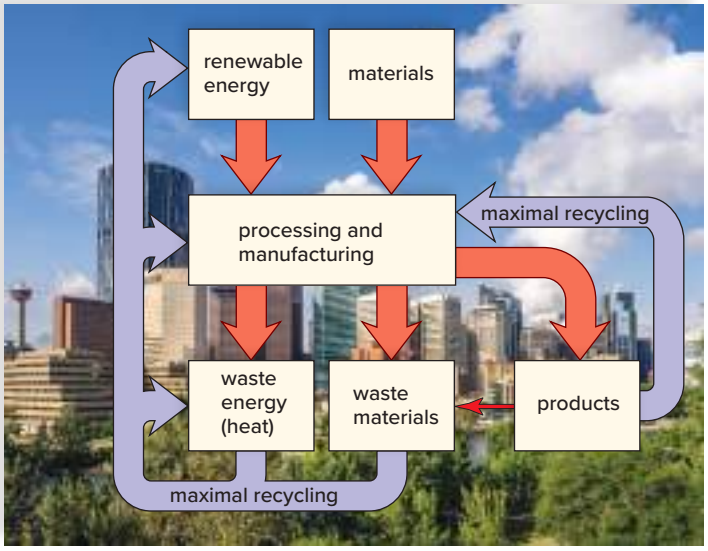
A natural ecosystem can offer clues as to what a sustainable human society would be like. A natural ecosystem uses only renewable solar energy, and its materials cycle through producers, herbivores, carnivores, and detritivores, and back to producers once again. It is clear that if we want to develop a sustainable society (Fig 44.19*b*), we too should use renewable energy sources and recycle materials. In addition to recycling and reuse of many of Earth's minerals (e.g., aluminum, copper, iron, and gold), which are nonrenewable, we should also consider alternative energy use, water conservation, and modifications to the way we conduct modern agriculture.

Energy

Approximately 10% of the world's energy supply comes from nuclear power, while 63% comes from fossil fuels. Both of these are finite, nonrenewable sources. Fossil fuels (oil, natural gas, and coal) are so named because they are derived from the compressed



a. Human society at present



b. Sustainable society

Figure 44.19 Society's current consumption of resources versus a sustainable one. **a.** At present, most societies in more developed countries live with a "throwaway" mentality. There is a high input of energy and materials, which results in a large output of waste materials and loss of energy in the form of heat (red arrows) and minimal recycling (purple arrows). **b.** A sustainable society would use only renewable energy sources, reuse heat and waste materials, and maximize the recycling of products (purple arrows). (a): imantsu/Shutterstock; (b): Jeff Whyte/Shutterstock

remains of plants and animals that died millions of years ago. Even with the discovery of new sources of natural gas, oil, and coal, these resources remain limited in supply. The consumption of these types of resources can also damage the environment badly. They are some of the main contributing factors to climate change. Comparatively speaking, each person in a more-developed country (MDC) uses approximately as much energy in one day as a person in a less-developed country (LDC) uses in one year. Increasing our reliance on renewable energy resources is a major step toward becoming a sustainable society.

ISTUDY

Figure 44.20 Hydropower. Hydropower dams provide a clean form of energy, but they can be ecologically disastrous in other ways. Maxim Burkovskiy/Shutterstock

Renewable Energy Sources

Renewable energy sources include hydropower, geothermal energy, wind power, and solar energy. Hydroelectric plants convert the energy of falling water into electricity (Fig. 44.20). Worldwide, hydropower presently generates approximately 16% of all electricity utilized. In the United States, hydropower accounts for approximately 6.3% of the total U.S. electricity generation. Brazil, New Zealand, and Switzerland produce between 50–66% of their electricity using hydropower, but Norway is the world's leading hydropower producer, accounting for 90% of its total power production. One way to start reducing our reliance on nonrenewable resources such as fossil fuels is to increase our reliance on hydropower. However, much of the recent hydropower development has been due to construction of enormous dams that have detrimental environmental effects. Small-scale dams that generate less power per dam, but do not have the same environmental impact, are believed to be the more environmentally responsible choice.

Geothermal energy is produced because Earth has an internal source of heat. Elements such as uranium, thorium, radium, and plutonium undergo radioactive decay underground and then heat the surrounding rocks to hundreds of degrees Celsius. When the rocks are in contact with underground aquifers, streams, or lakes, huge amounts of steam and hot water are produced. This steam can be piped up to the surface to supply hot water for home heating or to run steam-driven turbogenerators. The California Geysers Project, for example, is one of the world's largest geothermal electricity-generating complexes.

In some regions of the United States, wind power is capable of producing a significant percentage of our renewable energy needs (Fig. 44.21*a*). Despite the common belief that a huge amount of land is required for "wind farms" that produce commercial electricity, the actual amount of space for a wind farm compares favorably to the amount of land required by a coal-fired power plant or a solar thermal energy system. A community that generates its own electricity by using wind power can solve the problem of uneven energy production by selling electricity to a local public utility when an excess is available and buying electricity from the same facility when wind power is in short supply. The environmental impact of wind power is significantly lower than that of nonrenewable fossil fuels.

Solar energy is diffuse energy that must be first collected, then converted to another energy form for transport, and finally stored if it is to compete with other available forms of energy. Passive solar heating of a house is successful when the windows of the house face the sun, the building is well insulated, and heat can be stored in water tanks, rocks, bricks, or some other suitable material.

a. Wind

b. Solar

c. Hydrogen

Figure 44.21 Other renewable energy sources. **a.** Wind power requires land on which to place enough windmills to generate energy. **b.** Photovoltaic cells on rooftops and (**c**) the use of hydrogen cars will reduce air pollution and dependence on fossil fuels. (a): Glen Allison/Photodisc/Getty Images; (b): Danita Delimont/Getty Images; (c): Itsuo Inouye/AP Images

ISTUDY

In a photovoltaic (solar) cell, a wafer of an electron-emitting metal is in contact with another metal that collects the electrons and passes them along into wires in a steady stream. Spurred by the oil shocks of the 1970s, in 2008 the U.S. government began supporting the development of photovoltaic cells. As a result, the price of buying one has dropped from about $100 per watt (in the early 1970s) to around $1 to $1.50 (in 2022). The photovoltaic cells placed on roofs, for example, generate electricity that can be used inside a building and/or sold back to a power company (Fig. 44.21*b*).

Traditional cars have internal combustion engines that run on gasoline, while hybrid cars run on both gasoline and electricity, which significantly increases their miles driven per gallon of gas (Fig. 44.21*c*). Now that we have better ways to capture solar energy, scientists are investigating the possibility of using solar energy to extract hydrogen from water via electrolysis. The hydrogen can then be used as a clean-burning fuel that produces water as a waste product.

Increasingly, vehicles are powered by fuel cells that use hydrogen to produce electricity. Fuel cells are now powering buses in Vancouver and Chicago. Hydrogen cars are already in limited production. Hydrogen fuel can be produced locally or in central locations, using energy from photovoltaic cells. If in central locations, hydrogen can be piped to filling stations using the natural gas pipes already in place in the United States. The advantages of such a solar–hydrogen revolution would be at least twofold: First, the world would no longer be dependent on nonrenewable fossil fuels, and second, environmental problems, such as global warming, acid rain, and smog, would begin to lessen.

All electric vehicles or battery electric vehicles (BEVs) completely rely on an electric motor instead of an internal combustion engine. These vehicles are charged via special charging stations located throughout many communities. Since they do not burn fossil fuels, they do not produce the same emissions that are contributing to pollution and global climate change. The majority of auto manufacturers currently have or will be releasing an all electric vehicle in the near future.

Water

In many areas of the less-developed world, people do not have ready access to clean and safe drinking water. It's considered a human right for people to have clean drinking water, but actually the majority of fresh water is utilized by industry and agriculture. Worldwide, approximately 60% of all fresh water is used to irrigate crops! Domestically, in MDCs, more water is usually used for bathing, flushing toilets, and watering lawns than for drinking and cooking.

Although the needs of the human population overall do not exceed the renewable supply of water, this is not the case in certain regions of the United States and the rest of the world. In some areas, humans increase the supply of fresh water by damming rivers and withdrawing water from aquifers. Dams have drawbacks: (1) Reservoirs behind the dam will lose water due to evaporation and seepage into underlying rock beds. The amount of water lost sometimes equals the amount that dams make available! (2) The salt left behind by evaporation and agricultural runoff increases salinity and can make a river's water unusable farther downstream. (3) Over time, dams hold back less water because of sediment buildup. Sometimes, a reservoir becomes so full of silt that it is no longer useful for storing water. (4) The alteration of a river or stream has a negative impact on the native wildlife both upstream and downstream from the dam.

To meet their freshwater needs, people are pumping vast amounts of water from underground reservoirs called **aquifers.** Aquifers hold about 1,000 times the amount of water that falls on land as precipitation each year. In the past 50 years, groundwater depletion has become a problem in many areas of the world. Removal of water is causing land **subsidence,** settling of the soil as it dries out. In California's San Joaquin valley, an area of more than 13,000 km^2 has subsided at least 30 cm due to groundwater depletion, and in the worst spot, the surface of the ground has dropped more than 9 m! Subsidence damages canals, buildings, and underground pipes.

The American west has been experiencing one of the most significant droughts in history. In 2022, Lake Mead was reduced to its lowest level since the lake began filling in 1937, impacting over 25 million people, threatening hydropower production, and closing most of the recreational activities of the lake.

Conservation of Water

By 2025, two-thirds of the world's population may be facing serious water shortages. Some solutions for expanding water supplies are being explored. Planting drought- and salt-tolerant crops will help decrease the amount of water needed for irrigation. Development of many such crops is already underway through new genetic engineering techniques. Using drip irrigation is a more efficient method of delivering water to crops, which can save about 50% more water than traditional irrigation methods while also increasing crop yields. Although the first drip systems were developed in 1960, they are currently used on less than 1% of irrigated land. Most governments subsidize irrigation so heavily that farmers have little incentive to invest in drip systems or other water-saving methods.

Reusing water and adopting conservation measures could help the world's industries cut their water demands by more than half. For example, recycling washing machine water, shower water, or water used to wash cars through a filter before it is discarded as sewage could significantly reduce domestic water usage. Home yard irrigation should occur during dusk and dawn hours, as opposed to the middle of the day when evaporation is at its highest. Purchasing and using dual-flush toilets can also save millions of gallons of water per year.

Agriculture

In 1950, the global human population numbered 2.5 billion, and at that time, there was only enough food to provide less than 2,000 calories per person per day. Today, current agricultural practices offer enough food to provide everyone on Earth with a healthy diet consisting of 2,500 calories per day. However, one-sixth of the world's population (over 1 billion people) is currently considered malnourished due to lack of proper distribution and the redirection of large amounts of grain to feed livestock. In addition, modern farming methods are environmentally destructive in the following

ISTUDY

ways. First, planting only a few genetic varieties, or monocultures (a genetically identical crop), means that a single destructive parasite or pathogen can cause huge crop losses. Second, modern farming relies on heavy use of fertilizers, pesticides, and herbicides. Pesticides reduce soil fertility because they kill off beneficial soil organisms, as well as pests. Use of pesticides leads to resistance in insects, increasing eradication costs via increased continued pesticide use. In addition, fertilizers, pesticides, and herbicides all contribute to pollution. Third, modern agriculture uses significant amounts of fresh water through irrigation. Fourth, modern agriculture uses large amounts of fossil fuels. Fertilizer production is energy-intensive; irrigation pumps require energy to remove water from aquifers; large tractors and even airplanes and helicopters are used to spread fertilizers, pesticides, and herbicides; and huge machines are often used to harvest crops.

Several alternatives exist to employing modern agricultural methods (Fig. 44.22). Polyculture, or the planting of several varieties of crops in a field, can simultaneously reduce the susceptibility of crops to pests or diseases. Polyculture also reduces the amount of herbicides necessary to kill competing weeds and can be used to replenish nutrients to topsoil. Crop rotation—where, for example, nitrogen-fixing crops, such as legumes, are alternated across harvest years with crops that utilize soil nitrogen, such as corn—can help reduce the use of nitrogen-containing fertilizers. Such multiuse farming techniques generally help increase the amount of organic matter and nutrients in the soil.

Organic farms are also increasing in number. Organic farms, as mandated by the U.S. Department of Agriculture, are those in which synthetic pesticides and herbicides are avoided or largely excluded from use. Organic farming has become increasingly profitable in recent years due to consumer market demands that produce be grown in healthier ways. Despite the increased cost relative to nonorganic produce, many states are experiencing a rapid increase in the number of registered organic farms. Health concerns surrounding pesticide content in nonorganic produce, as well as better tasting food, have encouraged this trend.

One way that organic farmers have eliminated the need for pesticides is by using integrated pest management. This technique encourages the growth of competitive beneficial insects, and so uses biological pest control methods (also called "biocontrol"). Biocontrol has also helped reduce pesticide use in traditional farms as well. Such methods include using natural predators, such as spiders and ladybird beetles, to reduce the numbers of pests. Use of natural or engineered pathogens has also been suggested for biocontrol of pests, but there is concern that such diseases will escape crop areas and affect natural plants or wildlife.

Several types of farming now exist that reduce erosion and help minimize topsoil loss, such as contour farming. Terrace farming involves converting steep slopes into steplike hills to minimize erosion, and some farmers plant "natural fences," such as rows of trees, around crops to prevent topsoil loss due to wind or other factors. These trees can also be used as other products; mature rubber trees provide us with rubber, and tagua nuts are an excellent substitute for ivory, for example. Cover crops, which are often a mixture of legumes and grasses, also help stabilize soil between rows of cash crops. Finally, avoiding farming on steep slopes also helps reduce erosion. Soil nutrients can be increased through composting, organic farming techniques, or other self-renewable methods. In general, we should consider using precision farming (PF) techniques that rely on accumulated knowledge to reduce habitat destruction, while improving crop yields.

Urban Growth

More and more people are leaving rural communities and moving to cities. Growth of cities increases pollution via many sources, including automobile exhaust, runoff of pollutants on impervious surfaces (e.g., roads) into waterways, noise pollution, and industrial and domestic wastes. Thus, the future growth of cities should

a. Polyculture

b. Contour farming

c. Biological pest control

Figure 44.22 Methods that make farming more friendly to the environment. **a.** Polyculture reduces the ability of one parasite to wipe out an entire crop and also reduces the need to use an herbicide to kill weeds. This farmer has planted alfalfa in between strips of corn, which also replenishes the nitrogen content of the soil (instead of adding fertilizers). Alfalfa, a legume, has root nodules that contain nitrogen-fixing bacteria. **b.** Contour farming with no-till conserves topsoil, because water has less tendency to run off. **c.** Instead of pesticides, it is possible to use natural predators. Here, ladybugs are feeding on aphids, an insect pest species. (a): Tim McCabe, USDA Natural Resources Conservation Service/U.S. Department of Agriculture (USDA); (b): Karin de Mamiel/iStock/Getty Images; (c): Perennou Nuridsany/Science Source

ISTUDY

involve careful planning to serve the needs of new arrivals, without an overexpansion of the city.

Energy use in cities can be decreased by providing an energy-efficient public transportation system. Many U.S. cities are now encouraging carpooling by having HOV (high-occupancy vehicle) lanes on highways, as well as reduced toll costs for cars occupied by more than one person. Portland, Oregon, even has short-term electric car rental stations around the city, where people can rent cars for an evening and easily drop them off when finished. Maintaining a network of safe bicycle lanes also encourages people to ride their bikes to work.

A way to curtail urban sprawl—the ongoing expansion of cities outward—is to build them upward. Several cities, such as Vancouver, Canada, are building many high-rise apartment buildings to accommodate more people in a smaller area. As new buildings are built, they should be as "green" as possible—by using solar or geothermal energy for heating—and be constructed out of sustainable or recycled materials. Space on the tops of buildings could be used to make "green roofs," where a garden of grasses, herbs, and vegetables is planted to assist temperature control, supply food, reduce rainwater runoff, and be visually appealing. As more people move into cities, city officials can focus on renovating older parts of the city, as opposed to spreading out further.

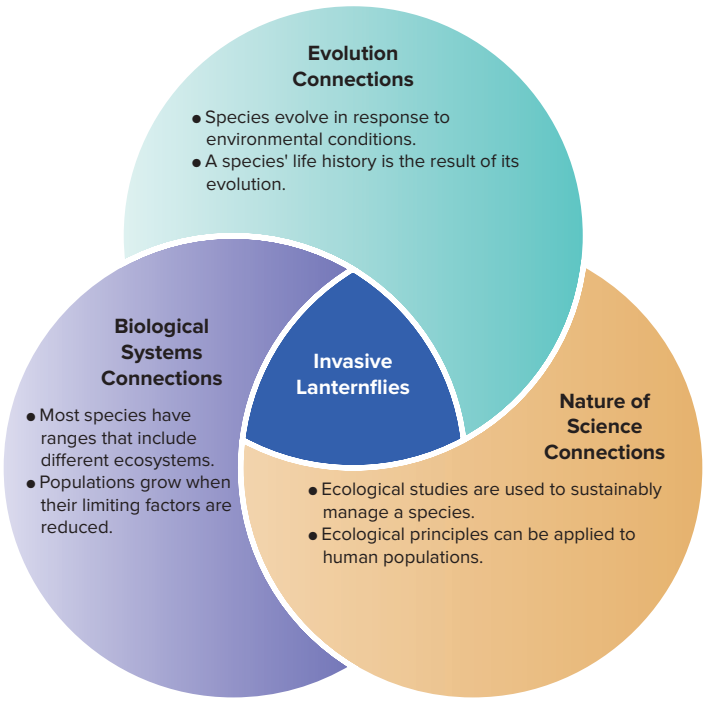
Modern cities can also have more planned green spaces, including plentiful walking and bicycle paths. In city parks, native species that attract bees and butterflies, and which also require less water and fertilizers, could be planted, as opposed to nonnative grasses.

Sustainable cities can also improve storm-water management by using sediment traps for storm drains, artificial wetlands, and holding ponds. As new development occurs, cities can increase the use of porous surfaces for walking paths, parking lots, and roads. These surfaces reflect less heat, while soaking up rainwater runoff.

Check Your Progress 44.7

1. Identify which nonrenewable energy sources are environmentally harmful.
2. List ways in which we can conserve water to minimize negative environmental impacts.
3. Discuss two alternative agricultural practices that are environmentally friendly.

Connecting *the* Concepts



What Did You Learn?

44.1 Scope of Ecology

Ecology is the study of the interactions of organisms with other organisms and with the physical environment. Some ecologists study how an organism is adapted to its individual **habitat,** whereas others look at larger levels of ecology. The additional levels of study include the organism, **population, species, community, ecosystem,** and **biosphere.** Ecologists are particularly interested in how interactions affect the distribution, abundance, and life history strategies of organisms.

44.2 Demographics of Populations

Demography studies provide statistical data about a population.

- For example, **population density** is the number of individuals per unit area or volume. Distribution of individuals can be uniform, random, or clumped. A **population's distribution** is often

determined by abiotic factors, which include **resources** such as water and sunlight, temperature, and availability of nutrients. **Limiting factors** determine where an organism will live.

- Population growth is dependent on number of births and number of deaths, as well as immigration and emigration. The number of births minus the number of deaths results in the **rate of natural increase (*r*)** (growth rate). **Biotic potential** is the highest possible rate of natural increase for a population. Population growth is measured by tracking a **cohort,** or members of a population that were born at the same time.
- Mortality (deaths per capita) within a population can be recorded in a life table and illustrated by a **survivorship** curve.
- The pattern of population growth is reflected in **age-structure diagrams,** which are a reflection of the age distribution of a population, which consists of prereproductive, reproductive, and postreproductive segments.

44.3 Population Growth Models

Two working models are used to describe population growth. **Semelparity** describes organisms that have one reproductive event before they die. **Iteroparity** describes organisms that have multiple reproductive events throughout their lifetime. If the environment offers unlimited resources and the number of reproductive females in the population reaches an optimum level, **exponential growth** results and produces a J-shaped curve.

- Most environments restrict growth, and exponential growth cannot continue indefinitely. Under these circumstances, an S-shaped, or **logistic, growth** curve results. The growth of the population is given by the equation $N/t = rN(K - N)/K$ for populations in which individuals have repeated reproductive events. The term $(K - N)/K$ represents the unused portion of the **carrying capacity (*K*).** When the population reaches carrying capacity, the population stops growing, because environmental conditions oppose biotic potential.

44.4 Regulation of Population Size

Population growth can be limited by **density-independent factors** (e.g., abiotic factors, such as weather events) and **density-dependent factors** (biotic factors such as **competition** and **predation**). Other means of regulating population growth exist in some populations. For example, territoriality is possibly a means of population regulation. Other populations do not seem to be regulated, and their population size fluctuates widely.

44.5 Life History Patterns

The logistic growth model has been used to suggest that the environment promotes either ***r*-selection** or ***K*-selection.** In unpredictable environments, *r*-selection occurs due to density-independent factors. Energy is allocated to producing as many small offspring as possible. Adults remain small and do not invest in parental care of offspring. In relatively stable environments, *K*-selection occurs and density-dependent factors affect population size. Energy is allocated to survival and repeated reproductive events. The adults are large and invest in parental care of offspring. Actual life histories contain trade-offs between these two patterns, and such trade-offs are subject to natural selection.

44.6 Human Population Growth

The potential for human population growth can be predicted by determining the **doubling time.** Currently, the human population is still expanding, but deceleration has begun. **More-developed countries (MDCs)** experience a high standard of living and low birthrates. **Less-developed countries (LDCs),** on the other hand, experience a low standard of living and high birthrates.

- A **demographic transition** occurs when a country experiences a decrease in the death rate followed by a decrease in the birthrate. This helps slow the growth rate of a country. Encouraging people to follow a **zero population growth** pattern or **replacement reproduction** will also help slow population growth. It is unknown when the world population size will level off, but it may occur by 2050. The majority of population increase will come from the LDCs, but we also need to consider the significant environmental impact that MDCs have on the planet as we look to address future concerns.

44.7 A Sustainable Future

The current biodiversity crisis is primarily caused by human actions. Our society is considered unsustainable due to our dependence on **nonrenewable resources** of energy production. A shift toward more dependence on **renewable resources** (wind, solar, geothermal, and hydropower) of energy production, along with **sustainable** use of water, will help decrease human impact.

Globally, many people lack access to clean drinking water. In the United States, we are rapidly depleting **aquifers,** which is leading to many social and environmental problems. As aquifers are drained, the land experiences settling or **subsidence.** Water conservation methods such as drip irrigation, planting drought- and salt-tolerant crops, and decreasing household and industrial water consumption is necessary to reduce humanity's impact on the environment. Agricultural practices can also be adjusted to help reduce topsoil loss and decrease their environmental impact.

Thinking Critically About Biology

1. In the winter moth life cycle, parasites are a less important cause of mortality than cold winter weather and predators. Give an evolutionary explanation for the inefficiency of parasites in controlling population size.
2. When populations grow exponentially, they often overshoot the carrying capacity of their environment and eventually crash. Why should an ecologist be concerned about the Asian carp population if it is predicted it will eventually overshoot its carrying capacity and crash?
3. The answer to how to curb the expected increase in the world's population lies in discovering how to curb the rapid population growth of the less-developed countries. In these countries, population experts have discovered what they call the "virtuous cycle." Family planning leads to healthier women, healthier women have healthier children, and the cycle continues. Women no longer need to have many babies for only a few to survive. More education is also helpful, because it teaches people about the benefits of postponing childbearing and promoting women's rights. Are you in favor of foreign aid to help countries develop family planning programs? Why or why not?

Making It Relevant to You

1. What are the main factors that limit the ability of the spotted lanternfly to expand its range?
2. Do you think humans should work to prevent the spotted lanternfly from becoming established in new environments?
3. Explain the larger ecological consequences of the spotted lanternfly expanding their range as a result of human actions.

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

Chimperil59/iStock/Getty Images

45

Community and Ecosystem Ecology

Restoring Wetland Communities

Producers are the foundation of most ecosystems. Communities with a diverse producer base are often more stable and resilient to change. Historically, the Illinois River was one of the most productive inland commercial fisheries in the United States due to seasonal flooding that would bring nutrient-laden sediment to the floodplain. This supported a diverse producer base which in turn supported multiple trophic levels.

Continued human alteration of the river contributed to a decline in the productivity of the floodplain. In 2007, The Nature Conservancy began restoring the Emiquon Preserve by reviving seasonal flooding. This flooding provided more nutrients for wetland plants, as shown in the increase of over 100 documented species since the restoration began. However, the uncontrolled hydrology of the floodplain caused over 75% of the dense cattail stands to disappear and prompted a decline in the muskrat population.

In 2016, a water control structure was implemented to help regulate seasonal water levels, leading to an increase in the cattail population. Other populations responded to the increased producer productivity as well.

However, 2022 saw a decrease in submerged aquatic vegetation that left The Nature Conservancy's management staff puzzled. They soon realized that the diversity and stability of the floodplain ecosystem is highly connected to the natural ebb and flow of the river. The next step is to identify the abiotic features of the natural hydrological cycle that made Emiquon such a diverse community. The challenge then becomes trying to artificially mimic these.

As you read through this chapter, think about the following questions:

1. In a community, are some species interactions more common than others?
2. What would happen to a community's structure if the producer base were changed?

Chapter Outline

45.1 Ecology of Communities
45.2 Community Development
45.3 Dynamics of an Ecosystem
45.4 Ecological Consequences of Climate Change

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 6.1 What role does thermodynamics play in the flow of energy through an ecosystem?

Section 43.2 Does a species' behavior determine the type of relationship it has with other species?

Section 44.4 How do density-dependent and density-independent factors influence community dynamics?

Following *the* Themes

CHAPTER 45 – COMMUNITY AND ECOSYSTEM ECOLOGY

Evolution	Changes to a species' niche can cause the niche of other species to change as well. This is the basis of coevolution.
Nature of Science	By studying the biogeochemical cycles of ecosystems, humans can better understand how to maintain ecosystem health.
Biological Systems	Populations interact to form communities, which then interact with the abiotic environment as part of the greater ecosystem.

ISTUDY

45.1 Ecology of Communities

Learning Outcomes

Upon completion of this section, you should be able to

1. Recognize the features of a biological community and species richness.
2. Describe the factors that define an organism's ecological niche within its community.
3. Identify how the interactions among species organize a community.

Populations do not occur as single entities; they are part of a community. A **community** is a group of populations of different species interacting with one another in the same environment. Communities come in different sizes and compositions, and it is sometimes difficult to decide where one community ends and another begins.

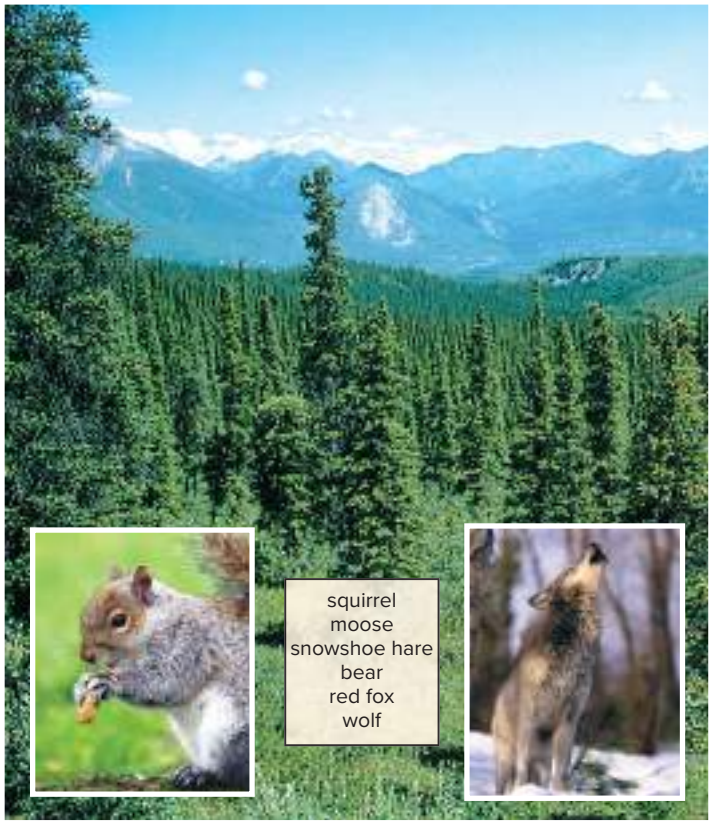
For example, a fallen log can be considered a community, because the many different populations living on and in it interact with one another. The fungi break down the log and provide food for the earthworms and insects living in and on the log. Those insects may feed on one another, too. If birds flying throughout the forest feed on the insects and worms living in and on the log, then they are also part of the larger forest community. In this section, we consider the characteristics of communities and the interactions that take place in them.

Community Structure

Two characteristics—species composition and diversity—allow us to compare communities. The species composition of a community, also known as **species richness,** is simply a listing of the different species found in that community. **Species diversity** is the measure of species richness and the number of individuals within each species within the community.

Species Composition

It is apparent, by comparing the photographs in Figure 45.1, that a coniferous forest has a different composition than that of a tropical rain forest. The narrow-leaved evergreen trees are prominent in a coniferous forest, and broad-leaved evergreen trees are numerous in a tropical rain forest. As the list of mammals demonstrates, a coniferous community and a tropical rain forest community contain different types of mammals. Ecologists comparing these two communities would also find differences in the other groups of plants and animals as well. In the end, ecologists would conclude that not only are the species compositions of these two communities different but the tropical rain forest has more species, and therefore, higher species richness.



a.

b.

Figure 45.1 Community structure. Communities differ in their composition, as witnessed by their predominant plants and animals. The diversity of communities is described by the richness of species and their relative abundance. **a.** Examples of trees and mammals commonly found in a coniferous forest. **b.** Examples of trees and mammals commonly found in a tropical rain forest. (a-1): (forest): Charlie Ott/Science Source; (a-2): (squirrel): Keven Law of London, England/Moment/Getty Images; (a-3): (wolf): Art Wolfe/The Image Bank/Stone/Getty Images; (b-1): (rain forest): TravelStrategy/Shutterstock; (b-2): (sloth): McGraw Hill; (b-3): (kinkajou): Alan & Sandy Carey/Science Source

ISTUDY

Species Diversity

The *diversity* of a community goes beyond composition, because it includes not only a listing of all the species in a community but also the relative abundance of each species. To take an extreme example, a forest sampled in West Virginia has 76 yellow poplar trees but only one American elm (among other species). If we were simply walking through this forest, we could miss seeing the American elm. If, instead, the forest had 36 poplar trees and 41 American elms, the forest would seem more diverse to us, and indeed it would be more diverse. The greater the species richness and the more even the distribution of the species in a community, the greater the species diversity.

The **island biogeography model** postulates that species diversity on an island is influenced by the distance of the island from the mainland (closer islands have more diversity than islands farther away) and the total area of the island (large islands have more diversity than small islands). Island biogeography studies provide information that is used in determining the ideal size for conservation areas, among other questions. Islands of habitat come in many different forms. Central Park in New York City is an 840-acre island of woods and lakes located in Manhattan. Marine protected areas (MPAs) are islands of protected areas within the ocean that serve to increase the populations of many marine species. The Nature of Science feature, "Marine Protected Areas and Ocean Biodiversity," explores the nature of MPAs in greater detail.

Community Interactions

This chapter examines the various types of community interactions and their importance to the structure of a community. Such interactions illustrate some of the most important evolutionary selection pressures acting on individuals. They also help us develop an understanding of how biodiversity can be preserved.

Habitat and Ecological Niche

Each species occupies a particular position in the community, both in a spatial sense (where it lives) and in a functional sense (what role it plays). A particular place where a species lives and reproduces is its **habitat.** The habitat might be the forest floor, a swift stream, or the seashore. The **ecological niche** is the role a species plays in its community. The niche includes the methods the species uses to acquire the resources it needs to meet energy, nutrient, and survival demands.

For a dragonfly larva, its habitat is a pond or lake, where it eats other insects. The pond must contain vegetation where the dragonfly larva can hide from its predators, such as fish and birds. In addition, the water must be clear enough for it to see its prey and warm enough for it to be in active pursuit.

Because it is difficult to study all aspects of a niche, some observations focus only on one aspect, such as study of the Galápagos finches (see Fig. 15.9). Each of these bird species is a specialist species, because each has a limited diet, tolerates only small changes in environmental conditions, and lives in a specific habitat. Some species, such as raccoons, roaches, and house sparrows, are generalists, with a broad range of niches. These organisms have a diversified diet, tolerate a wide range of environmental conditions, and can live in a variety of places.

Because a species' niche is determined by both abiotic factors (such as temperature and precipitation) and biotic factors (such as competitors, parasites, and predators), ecologists like to distinguish between the fundamental and the realized niche. The *fundamental niche* comprises the range of abiotic conditions under which a species can survive when adverse biotic conditions are absent. The *realized niche* comprises those conditions under which a species actually survives when adverse biotic interactions, such as competition and predation, are present. Therefore, a species' fundamental niche tends to be larger than its realized niche.

Competition Between Populations

Competition occurs when members of different species try to use the same resource (such as light, space, or nutrients) that is in limited supply. If the resource is not in limited supply, there is no competition. In the 1930s, Russian ecologist G. F. Gause grew two species of *Paramecium* in one test tube containing a fixed amount of food. Although each population survived when grown separately, only one survived when they were grown together (Fig. 45.2). The successful *Paramecium* population had a higher biotic potential than the unsuccessful population. After observing the outcome of other, similar experiments in the laboratory, ecologists formulated the **competitive exclusion principle,** which

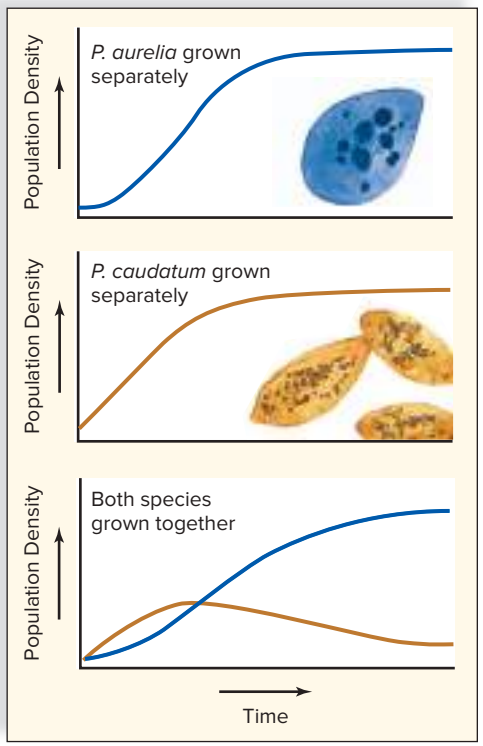


Figure 45.2 Competition between two laboratory populations of *Paramecium*. When grown alone in pure culture, *Paramecium aurelia* and *Paramecium caudatum* exhibit sigmoidal growth. When the two species are grown together in mixed culture, *P. aurelia* is the better competitor, and *P. caudatum* dies out. Both attempted to exploit the same resources, which led to competitive exclusion. Source: Gause, G. F., *The Struggle for Existence*. Baltimore, MD: Williams & Wilkins Company, 1934.

ISTUDY

THEME Nature of Science

Marine Protected Areas and Ocean Biodiversity

The ocean is home to a wide diversity of marine ecosystems, many of which play critical roles in the health, well-being, and economies of human populations. Mangrove forests are composed of trees and shrubs found in the coastal intertidal zone. They are one of a few plants adapted to life in and near salt water. Their tangle of prop roots arch into the water, allowing sediment to accumulate, and provide shelter for hundreds of marine species. Many of the juvenile fish and invertebrates found in the mangroves will move out to the coral reefs once they mature. Worldwide, more than one in six species of mangroves are threatened with extinction. Coastal development, climate change, and agriculture are the primary factors behind the current status of mangrove forests. Over 35% of the world's mangroves have already been cleared. In some countries, such as India and the Philippines, over 50% have been cleared.

Coral reefs contain the greatest levels of diversity within the marine environment. It is estimated that 25% of all marine species will use coral reefs for food and shelter at some point in their life cycle. Over 500 million people depend on coral reefs for their food, protection, and livelihoods. Worldwide, billions of dollars a year are generated from activities associated with coral reefs.

Coral reefs are found primarily in tropical waters around the world, covering less than 2% of the ocean bottom. Their biodiversity is parallel to that of the tropical rain forest on land. Sadly, 75% of the reefs around the world are facing human-driven threats in the form of climate change, ocean acidification, and invasive species, just to name a few. If these threats continue unabated, scientists estimate that 90% of the world's corals will be pushed into extinction by 2050.

In an effort to preserve and establish a source of biodiversity within the ocean, marine protected areas (MPAs) have been established. An MPA is defined as an area of the marine environment that has been protected by federal, state, or local laws to provide lasting protection of the natural resources within them. Fishing grounds in central California, coastal areas of Cape Cod, and most of the Florida Keys are just a few examples of marine protected areas. Globally, only about 4% of the world's oceans are protected (Fig. 45A).

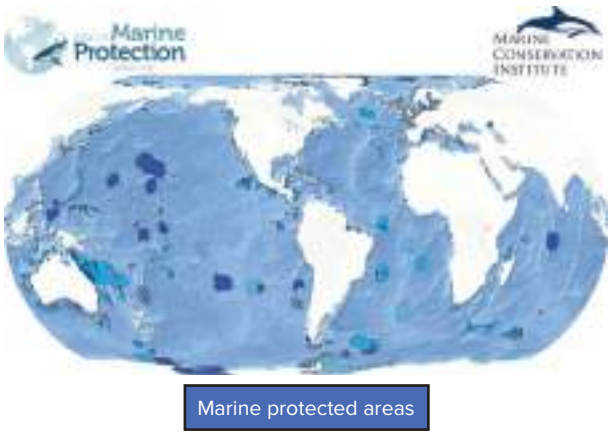


Figure 45A Marine protected areas. MPAs are areas of protection in the oceans that are designed to protect as much marine biodiversity as possible. Marine Conservation Institute. http://www.mpatlas.org/media/filer_public/49/30/49309c5a-738b-4de3-b1c2-fb92c87c2dfa/globalmpas_logo_2020.jpg

MPAs provide a wide range of benefits for the marine environment and the communities that rely on them for their livelihoods. They can be used to maintain biodiversity and provide habitat for endangered and commercially valuable species. Fish and other marine species are given a place where they can safely reproduce and grow without direct human interference. When MPAs are established and managed correctly, they act as a source of new individuals that will leave the MPA and populate the surrounding communities, rebuilding the size of the adjacent populations. Perhaps, one of the most important things an MPA can do is help maintain the economy and the cultural and historical aspects of communities that are linked to the marine environment. Without the continued management and development of new MPAs, it will be difficult for humans to sustainably manage marine ecosystems.

Questions to Consider

1. Should individual countries around the world be required to establish MPAs within their coastal waters?
2. Should select groups be granted permission to use the resources in an MPA?

states that no two species can indefinitely occupy the same niche at the same time.

It is possible for similar species to have different ecological niches, so that competition between the species is minimized or avoided altogether. In another laboratory experiment, Gause found that two species of *Paramecium* could occupy the same tube when one species fed on food at the bottom of the tube and the other fed on food suspended in solution. Individuals of each species that can avoid competition have a reproductive advantage. **Resource partitioning** is the division of resources in order to decrease competition between two species. This often leads to increased niche specialization and less niche overlap. An example of resource partitioning involves owl and hawk populations. Owls and hawks feed on similar prey (small rodents), but owls are nocturnal hunters and hawks are diurnal hunters. What could have been one niche became two, more specialized niches because of a divergence of behavior.

It is possible to observe the process of niche specialization in nature. When three species of ground finches of the Galápagos Islands occur on three separate islands, their beaks tend to be the same intermediate size, enabling each to feed on a wider range of seeds (Fig. 45.3). When the three species live on the same island, selection has favored divergence in beak size, because the size of the beak affects the kinds of seeds that can be eaten. In other words, competition has led to resource partitioning and, therefore, niche specialization.

Character displacement is the tendency for characteristics to be more divergent when populations belong to the same community than when they are isolated. Character displacement is often

ISTUDY

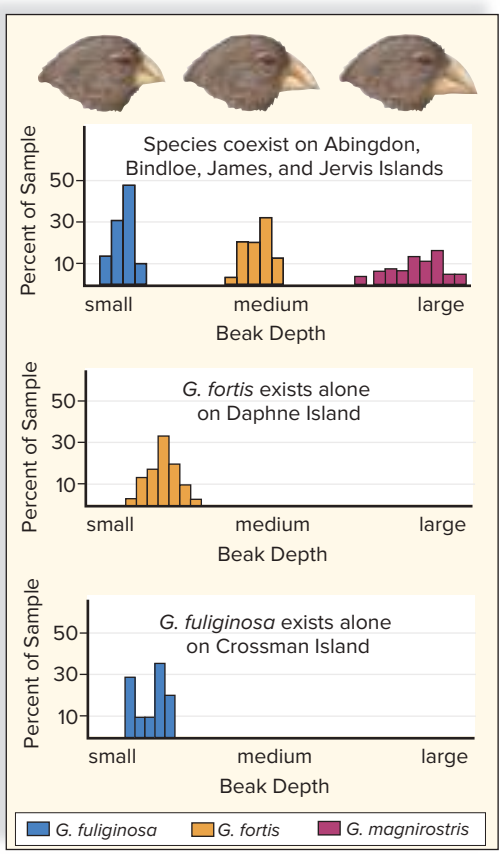


Figure 45.3 Character displacement in finches on the Galápagos Islands. When *Geospiza fuliginosa, G. fortis,* and *G. magnirostris* are on the same island, their beak sizes are appropriate to eating small-, medium-, and large-sized seeds, respectively. When *G. fortis* and *G. fuliginosa* are on separate islands, their beaks have the same intermediate size, which allows them to eat seeds that vary in size.

used as evidence that competition and resource partitioning have taken place.

Niche specialization can be subtle. Five different species of warblers that occur in North American forests are all nearly the same size, and all feed on a type of spruce tree caterpillar. For all these species to exist in the same community, they must be avoiding direct competition. In a famous study, ecologist Robert MacArthur recorded the amount of time each of five warbler species spent in different regions of spruce trees to determine where each species did most of its feeding (Fig. 45.4). He discovered that each species primarily used different parts of the tree and, in that way, had a more specialized niche.

As another example, consider that swallows, swifts, and martins all eat flying insects and parachuting spiders. These birds even frequently fly together in mixed flocks. But each type of bird has a different nesting site and migrates at a slightly different time of year. In doing so, they are not competing for the same food source at the same time.

In all these cases of niche partitioning, we have merely assumed that what we observe today is due to competition in the past. Joseph Connell has studied the distribution of barnacles on the Scottish coast. Small barnacles live on the high part of the intertidal zone, and large barnacles live on the lower part (Fig. 45.5). Free-swimming larvae of both species attach themselves to rocks randomly throughout the intertidal zone; however, in the lower zone, the large *Balanus* barnacles seem to prevent the smaller *Chthamalus* individuals from attaching to the rocks, or instead, *Balanus* will simply grow over them.

To test his observation, Connell removed the larger barnacles and found that the smaller barnacles grew equally well on all parts of the rock. The entire intertidal zone is the fundamental niche for *Chthamalus,* but competition is restricting the range of *Chthamalus* on the rocks. *Chthamalus* is more resistant to drying out than is *Balanus;* therefore, it has an advantage that permits it to grow in the upper intertidal zone, where it is exposed more often to the air. In other words, the upper intertidal zone becomes the realized niche for *Chthamalus*.

Predator–Prey Interactions

Predation occurs when one living organism, called a **predator,** feeds on another, called **prey.** In its broadest sense, predaceous consumers include not only animals such as lions, which kill zebras, but also filter-feeding blue whales, which strain krill from ocean waters, and herbivorous deer, which browse on trees and shrubs.

Parasitism can be considered a type of predation, because one individual obtains nutrients from another. Parasitoids are organisms that lay their eggs inside a host. The developing larvae feed on the host and sometimes cause the host to die.

Predation and parasitism tend to increase the abundance of the predator and parasite populations at the expense of the prey or host population.

Predator–Prey Population Dynamics

Predators play a role in determining the population density of prey. In another classic experiment, G. F. Gause reared the ciliated protozoans *Paramecium caudatum* (prey) and *Didinium nasutum* (predator) together in a culture medium. He observed that *Didinium* ate all the *Paramecia* and then died of starvation.

In nature, we can find a similar example. When a gardener took prickly-pear cacti to Australia from South America, the cacti spread out of control until millions of acres were covered with nothing but cacti. The cacti were brought under control when a moth from South America, whose caterpillar feeds only on the cacti, was introduced. The caterpillar was a voracious predator on the cacti, efficiently reducing the cacti population. Now both cacti and moth are found at greatly reduced densities in Australia.

This raises an interesting point: The population density of the predator can be affected by the population density of the prey. In other words, the predator–prey relationship is actually a two-way street. In that context, consider that at first the biotic potential (maximum reproductive rate) of the prickly-pear cacti was maximized, but factors that oppose biotic potential came into play after the moth was introduced. And the biotic potential of the moth was maximized when it was first introduced, but the carrying capacity decreased after its food supply was diminished.

ISTUDY

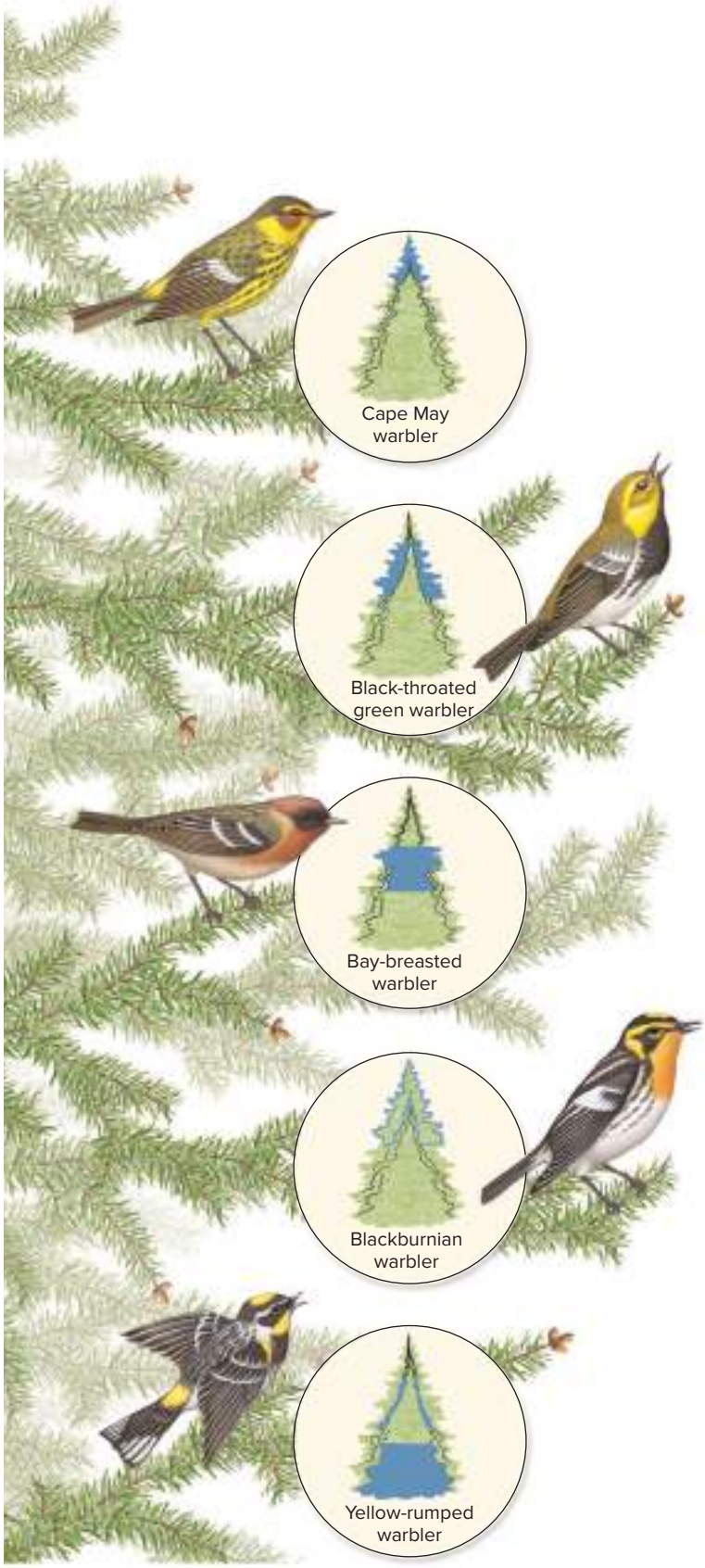


Figure 45.4 Niche specialization among five species of coexisting warblers. The diagrams represent spruce trees. The time each species spent in various portions of the trees was determined; each species spent more than half its time in the blue regions.

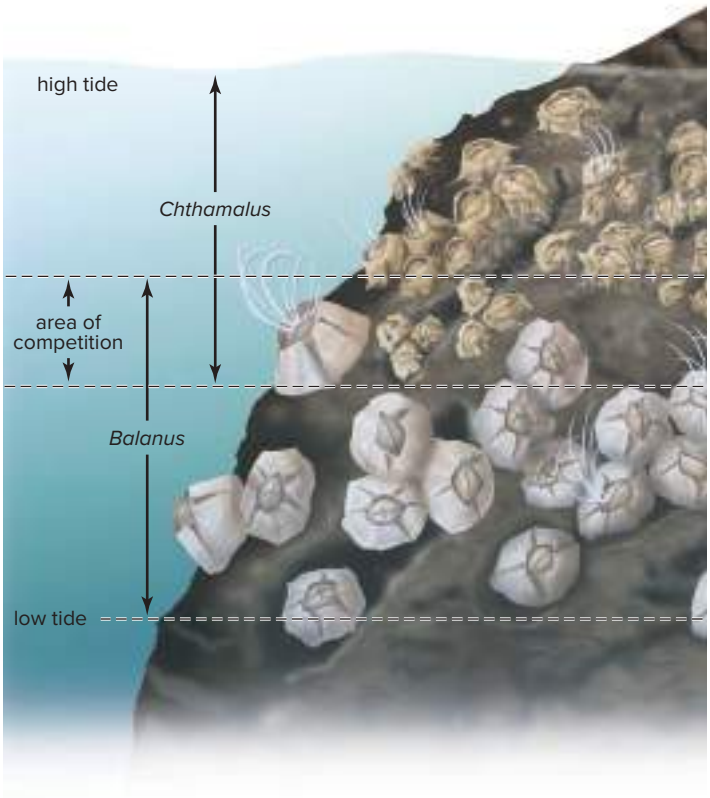


Figure 45.5 Competition between two species of barnacles. Competition prevents two species of barnacles from occupying as much of the intertidal zone as possible. Both exist in the area of competition between *Chthamalus* and *Balanus.* Above this area, only *Chthamalus,* a small barnacle, survives, and below it only *Balanus,* a large barnacle, survives.

Under most environmental conditions, predator and prey populations will fluctuate in size. We can appreciate that an increase in the size of the predator population is dependent on an increase in the size of the prey population. A decrease in population size instead of the establishment of a steady population size can also occur. At least two possibilities account for the reduction:

- Perhaps the biotic potential (reproductive rate) of the predator is so great that its increased numbers overconsume the prey, and then as the prey population declines, so does the predator population.
- Perhaps the biotic potential of the predator is unable to keep pace with the prey, and the prey population overshoots the carrying capacity and suffers a crash. Now the predator population follows suit because of a lack of food.

In either case, the result is a series of peaks and valleys, with the predator population size lagging slightly behind and below that of the prey population.

A famous example of predator–prey cycles occurs between the snowshoe hare (*Lepus americanus*) and the Canada lynx (*Lynx canadensis*), a type of small, predatory cat (Fig. 45.6). The snowshoe hare is a common herbivore in the coniferous forests of North America, where it feeds on terminal twigs of various shrubs and small trees. The Canada lynx feeds on snowshoe hares but also on ruffed and spruce grouse, two types of birds. Studies have

ISTUDY

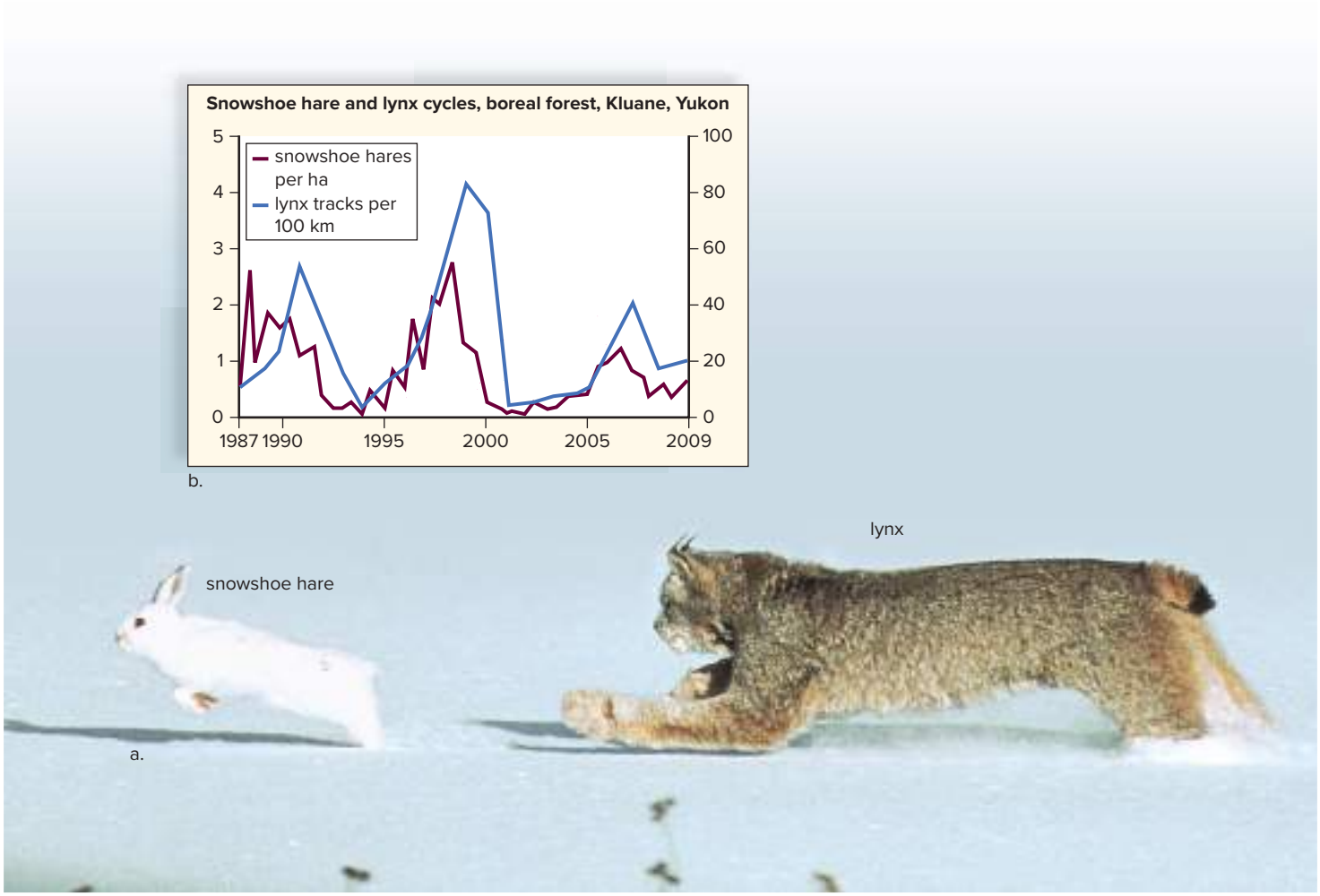


Figure 45.6 Predator–prey interaction between a lynx and a snowshoe hare. **a.** A Canada lynx, *Lynx canadensis,* is a solitary predator. A long, strong forelimb with sharp claws grabs its main prey, the snowshoe hare, *Lepus americanus*. **b.** Research indicates that the snowshoe hare population reaches a peak abundance before that of the lynx by a year or more. A study conducted from 1987 to 2009 shows a cycling of the lynx and snowshoe hare populations. (a): Alan & Sandy Carey/Science Source; (b): Source: Sheriff, Michael J., Charles J. Krebs, and Rudy Boonstra. "The Ghosts of Predators Past: Population Cycles and the Role of Maternal Programming Under Fluctuating Predation Risk." Ecology 91, no. 10 (2010): 2983–2994.

revealed that the hare and lynx populations cycle regularly, as graphed in Figure 45.6*b*.

Investigators at first assumed that the lynx was the only factor that would bring about a decline in the hare population, and that this accounts for the cycling. But others have noted that the decline in snowshoe hare abundance is accompanied by low growth and reproductive rates, which could be signs of a food shortage. Experiments were done to test whether predation or lack of food causes the decline in the hare population. The results indicate that the hare cycle is produced by interactions among three trophic levels: the predators, the hares, and the producer base. Its biological impacts produce a ripple effect across multiple species of predator and prey species within their community.

Prey Defenses

Prey defenses are mechanisms that decrease the possibility of being eaten by a predator. Prey species have evolved a variety of mechanisms that enable them to avoid predators, including heightened senses, speed, protective armor, protective spines or thorns, tails and appendages that break off, and chemical defenses.

One common strategy to avoid capture by a predator is **camouflage,** or the ability to blend into the background, which helps the prey avoid being detected by the predator. Some animals have cryptic coloration that blends them into their surroundings. For example, flounders can take on the same coloration as their background (Fig. 45.7*a*). Many examples of protective camouflage are known: Walking sticks look like twigs; katydids look like sprouting green leaves; some caterpillars resemble bird droppings; and some insects, moths, and lizards blend into the bark of trees.

Another common antipredator defense among animals is *warning coloration,* which tells the predator that the prey is potentially dangerous. As a warning to possible predators, poison arrow frogs are brightly colored (Fig. 45.7*b*). Also, many animals, including caterpillars, moths, and fishes, possess false eyespots that are meant to confuse or startle the predator. Other animals have elaborate structures that cause the *startle response*. The South

ISTUDY

a. Camouflage b. Warning colorization c. Fright

Figure 45.7 Antipredator defenses. **a.** Flounder is a fish that blends in with its background. **b.** The poison arrow frog is brightly colored to warn predators it is dangerous to the touch. **c.** The large false head of the South American lanternfly may startle a potential predator. (a): Richard Thompson/Alamy Stock Photo; (b): Philip Coblentz/MedioImages/SuperStock; (c): Pablo Reinsch Photography/Moment Open/Getty Images

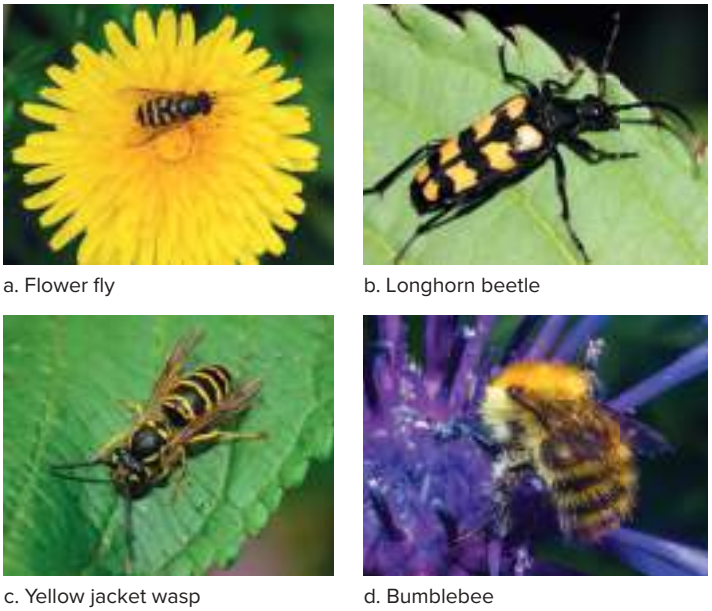
American lanternfly has a large false head with false eyes, making it resemble the head of an alligator (Fig. 45.7*c*). However, antipredator defenses are not always false. A porcupine certainly looks formidable, and for good reason. Its arrowlike quills have barbs that dig into the predator's flesh and penetrate even deeper as the predator struggles after being impaled. In the meantime, the porcupine runs away.

Association with other prey species is another common strategy that may help avoid capture. Flocks of birds, schools of fish, and herds of mammals stick together as protection against predators. Baboons detect predators visually, whereas antelopes detect predators by smell. Sometimes they will forage together, gaining double protection against stealthy predators. The gazellelike springboks of southern Africa jump stiff-legged 2–4 m into the air when alarmed. Such a jumble of shapes and motions alert other members of the herd, along with distracting an attacking lion, allowing the herd to escape.

Mimicry

Mimicry is the resemblance of one species to another that possesses an overt antipredator defense. A mimic that lacks the defense of the organism it resembles is called a Batesian mimic (named for Henry Bates, who described the phenomenon). Once an animal experiences the defense of the model organism, it remembers the defense mechanism and avoids most animals that look similar. Figure 45.8*a, b* shows two insects (flower fly and longhorn beetle) that resemble the yellow jacket wasp shown in Figure 45.8*c* but lack the wasp's ability to sting. Classic examples of Batesian mimicry include the scarlet kingsnake mimicking the venomous coral snake, and the *Papilio memnon* butterfly mimicking foul-tasting butterflies.

Some species that actually have the same defense resemble each other. Many stinging insects—bees, wasps, hornets, and bumblebees—have the familiar alternating black and yellow bands. Once a predator has been stung by a black and yellow insect, it is wary of that color pattern in the future. Mimics that share the same protective defense are called Müllerian mimics, after Fritz Müller, who discovered this type of mimicry. The bumblebee in Figure 45.8*d* and the yellow jacket wasp are Müllerian mimics, because both of them can sting.



a. Flower fly b. Longhorn beetle c. Yellow jacket wasp d. Bumblebee

Figure 45.8 Mimicry among insects. A flower fly (**a**) and a longhorn beetle (**b**) are Batesian mimics, because they are incapable of stinging another animal, yet they have the same appearance as the yellow jacket wasp (**c**). A bumblebee (**d**) and a yellow jacket wasp are Müllerian mimics, because they have a similar appearance and both use stinging as a defense. (a): ErikAgar/iStock/Getty Images; (b): Ger Bosma/Moment Open/Getty Images; (c): Paul Reeves Photography/Shutterstock; (d): Jon666/iStock/Getty Images

Symbiotic Relationships

Symbiosis is the close association between two different species over long periods of time. There are three types of symbiotic relationships, as shown in Table 45.1. Some biologists argue that the amount of harm or good two species do to one another is dependent on which variable the investigator is measuring.

Parasitism is similar to predation in that an organism, called a **parasite,** derives nourishment from another, called a **host.** Parasitism is a type of symbiosis in which one of the species causes some harm to the other but does not attempt to kill it (Table 45.1). Viruses (such as HIV, which reproduces inside

ISTUDY

Table 45.1 Symbiotic Relationships

Interaction	Species 1	Species 2
Parasitism*	Benefited	Harmed
Commensalism	Benefited	No effect
Mutualism	Benefited	Benefited

*Can be considered a type of predation.

human lymphocytes) are always parasitic, and parasites occur in all the kingdoms of life. Bacteria (e.g., strep infection), protists (e.g., malaria), fungi (e.g., rusts and smuts), plants (e.g., Indian pipe), and animals (e.g., tapeworms and fleas) all have parasitic members. Although small parasites can be endoparasites (e.g., heartworms) (Fig. 45.9), larger ones are more likely to be ectoparasites (e.g., leeches).

The effects of parasites on the health of the host can range from slightly weakening them to actually killing them over time. When host populations are at a high density, parasites readily spread from one host to the next, causing intense infestations and a subsequent decline in host density. Parasites that do not kill their host can still play a role in reducing the host's population density, because an infected host is less fertile and becomes more susceptible to other causes of death.

In addition to nourishment, host organisms also provide their parasites with a place to live and reproduce, as well as a mechanism for dispersing offspring to new hosts. Many parasites have both a primary and a secondary host. The secondary host may be a vector that transmits the parasite to the next primary host. Usually, both hosts are required in order to complete the life cycle, as in malaria (see Fig. 21.8). The association between parasite and host is so intimate that parasites will often have a narrow range of host species.

Commensalism

Commensalism is a symbiotic relationship between two species in which one species is benefited and the other is neither benefited nor harmed.

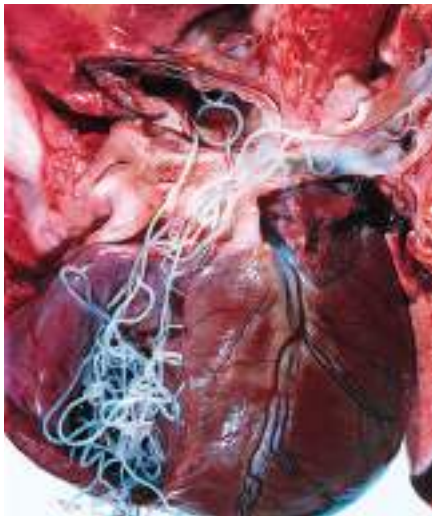


Figure 45.9 Heartworm. *Dirofilaria immitis* is a parasitic nematode spread by mosquitoes. The worms, which live in the heart and pulmonary blood artery, can cause the death of the host. ©Sharon Patton/University of Tennessee College of Veterinary Medicine

Instances are known in which one species provides a home and/or transportation for the other species. Barnacles that attach themselves to the backs of whales or the shells of horseshoe crabs are provided with both a home and transportation. It is possible, though, that the movement of the host is impeded by the presence of the attached animals, and therefore some are reluctant to use these as instances of commensalism.

Epiphytes, such as Spanish moss and some species of orchids and ferns, grow in the branches of trees, where they receive light, but they take no nourishment from the trees. Instead, their roots obtain nutrients and water from the air. Clownfishes live within the waving mass of tentacles of sea anemones. Because most fishes avoid the stinging tentacles of the anemones, clownfishes are provided with some protection from predators. Perhaps this relationship borders on mutualism, because the clownfishes actually may attract other fishes on which the anemones can feed, or they may provide some cleaning services for the anemones.

Commensalism often turns out, on closer examination, to be an instance of either mutualism or parasitism. Cattle egrets are so named because these birds stay near cattle, which flush out their prey—insects and other animals—from vegetation. The relationship becomes mutualistic when egrets remove ectoparasites from the cattle (Fig. 45.10). Remoras are fishes that attach themselves to the bellies of sharks by means of a modified dorsal fin acting as a suction cup. Remoras benefit by getting a free ride and feeding on a shark's leftovers. However, the shark benefits when remoras remove its ectoparasites.

Mutualism

Mutualism is a symbiotic relationship in which both members benefit. As with other symbiotic relationships, it is possible to find numerous examples among all organisms. Bacteria that reside in the human intestinal tract acquire food, but they also provide us with vitamins that we cannot synthesize for ourselves. Termites would not be able to digest wood if not for the protozoans that inhabit their intestinal tracts and digest cellulose. Mycorrhizae are mutualistic associations between the roots of plants and fungal hyphae. The hyphae improve the uptake of nutrients for the plant, protect the plant's roots against pathogens, and produce plant growth hormones. In return, the plant provides the fungus with carbohydrates.

Figure 45.10 Egret symbiosis. Cattle egrets eat insects off and around various animals, such as this African cape buffalo. slowmotiongli/iStock/Getty Images

ISTUDY

Some sea anemones make their homes on the backs of crabs. The crab uses the stinging tentacles of the sea anemone to gather food and to protect itself, while the sea anemone gets a free ride that allows it greater access to food. Lichens can grow on rocks, because their fungal member conserves water and leaches minerals, which are provided to the algal partner. The algae photosynthesize and provide organic food for both organisms. However, it's been suggested that the fungus is parasitic, at least to a degree, on the algae.

In Mexico and Central America, the bullhorn acacia tree is adapted to provide a home for ants of the species *Pseudomyrmex ferruginea.* Unlike other acacias, this species has swollen thorns with a hollow interior, where ant larvae can grow and develop. In addition to housing the ants, acacias provide them with food. The ants feed from nectaries at the base of the leaves and eat fat- and protein-containing nodules called Beltian bodies, found at the tips of the leaves. The ants constantly protect the plant from herbivores and other plants that might shade it.

The relationship between plants and their pollinators, mentioned previously, is a good example of mutualism. Perhaps the relationship began when certain herbivores feasted on certain plants. The plant's provision of nectar may have spared the pollen while allowing the animal to become an agent of pollination. Eventually, pollinator mouthparts became very specific to gathering the nectar of a particular plant species. The plant species, in turn, often becomes dependent on the pollinator for dispersing pollen. The mutualistic relationships between plants and their pollinators are examples of **coevolution.** The Evolution feature, "Interactions and Coevolution," discusses some examples of coevolution.

The outcome of mutualism is an intricate web of species interactions that are critical to the foundation of every community. For example, in areas of the western United States, the branches and cones of whitebark pine are turned upward, meaning that the seeds do not fall to the ground when the cones open. Birds called Clark's nutcrackers eat the seeds of the whitebark pine trees and store them in the ground (Fig. 45.11), thus acting as critical seed dispersers for the trees. Also, grizzly bears find the stored seeds and consume them.

Figure 45.11 Clark's nutcrackers. Mutualism can take many forms, as when birds such as Clark's nutcrackers, *Nucifraga columbiana,* feed on but also disperse the seeds of whitebark pine trees. Frank Fichtmüller/iStock/Getty Images

Figure 45.12 Cleaning symbiosis. A cleaner wrasse, *Labroides dimidiatus,* in the mouth of a spotted sweetlip, *Plectorhinchus chaetodonoides,* is feeding off parasites. Does this association improve the health of the sweetlip, or is the sweetlip being exploited? Investigation is underway. Bill Wood/Bruce Coleman/Photoshot

Whitebark pine seeds do not germinate unless their seed coats are exposed to fire. When natural forest fires in the area are suppressed, whitebark pine trees decline in number, and so do Clark's nutcrackers and grizzly bears. When lightning-ignited fires are allowed to burn, or prescribed burning is used in the area, the whitebark pine populations increase, as do the populations of Clark's nutcrackers and grizzly bears.

Cleaning symbiosis is a symbiotic relationship in which crustaceans, fish, and birds act as cleaners for a variety of vertebrates. Large fish in coral reefs line up at cleaning stations and wait their turn to be cleaned by small fish and various crustaceans, which will often enter the mouths of the large fish (Fig. 45.12) in order to clean their mouth and gills. Whether cleaning symbiosis is an example of mutualism is still being questioned.

Check Your Progress 45.1

1. Explain the difference between species richness and diversity.
2. Identify the difference between an organism's habitat and its niche.
3. Describe the two factors that can cause predator and prey populations to cycle in a predictable manner.

45.2 Community Development

Learning Outcomes

Upon completion of this section, you should be able to

1. Choose the correct sequence of events that occurs during ecological succession.
2. Compare the two types of ecological succession.

Each community has a history that can be surveyed over time. We know that the distribution of life has been influenced by dynamic changes that have occurred during the history of Earth. We have

ISTUDY

THEME Evolution

Interactions and Coevolution

Coevolution is present when two species adapt in response to selective pressure imposed by the other. Symbiosis (close association between two species), which includes parasitism, commensalism, and mutualism, is especially prone to the process of coevolution.

Flowering plants often use brightly colored petals or enticing scents to attract animal pollinators (see the Evolution feature, "Plants and Their Pollinators," in Section 27.1). Butterfly-pollinated flowers often contain a platform for the butterfly to land on, and the butterfly has a proboscis that allows it to feed on the flower.

Coevolution also occurs between predators and prey. For example, a cheetah sprints forward to catch its prey, which selects for the gazelles that are fast enough to avoid capture. Over generations, the adaptation of the prey may put selective pressure on the predator for an adaptation to the prey's defense mechanism. In this way, an evolutionary "arms race" can develop.

The process of coevolution has been studied in the brown-headed cowbird, a social parasite that reproduces at the expense of other birds by laying its eggs in their nests. It is a strange sight to see a small bird feeding a cowbird nestling several times its size. Investigators discovered that in order to "trick" a host bird, the female cowbird has to quickly lay an egg that mimics the host's egg while the host is away from the nest. The cowbird will leave most of the host's eggs in the nest to prevent the host from deserting a nest with only one egg. (The cowbird chick hatches first and is behaviorally adapted to growing faster and larger than its nestmates [Fig. 45B*b*].)

At this stage in the "arms race," the cowbird appears to have the upper hand; however, selection may favor host birds that are able to distinguish the cowbird eggs from their own. In the case of the yellow warbler, the adults have evolved the mechanism of building a new nest on top of the cowbird eggs in order to avoid being brood parasitized.

Coevolution can take many forms. The sexual portion of the life cycle of *Plasmodium,* the cause of malaria, occurs within mosquitoes (the vector), while the asexual portion occurs in humans. The human immune system uses surface proteins to detect pathogens. *Plasmodium* stays one step ahead of the host's immune system by changing the surface proteins of its cells to avoid detection. A similar capability of HIV has added to the difficulty of producing an AIDS vaccine.

The relationship between parasite and host can even include the ability of parasites to seemingly manipulate the behavior of their hosts in self-serving ways. Ants infected with the lance fluke mysteriously cling to blades of grass with their mouthparts. There, the infected ants are eaten by grazing sheep, transmitting the flukes to the next host in their life cycle. Similarly, snails of the genus *Succinea* are parasitized by worms of the genus *Leucochloridium*. As the worms mature, they invade the snail's eyestalks, making them resemble edible caterpillars. Birds eat the infected snails, allowing the parasites to complete their development inside the urinary tracts of birds.

The traditional view was that as host and parasite coevolved, each would become more tolerant of the other. Eventually, parasites could become commensal, or harmless to the host. Then over time, the parasite and host might even become mutualists. In fact, the evolution of the eukaryotic cell by endosymbiosis is based on the supposition that some early bacteria took up residence inside a larger cell, and then the parasite and cell became mutualists.

However, this argument is too teleological for some; after all, no organism is capable of "looking ahead" at its evolutionary fate. Rather, if an aggressive parasite could transmit more of itself in less time than a benign one, aggressiveness would be favored by natural selection, because the most aggressive would reproduce. Other factors, such as the life cycle of the host, can determine whether aggressiveness is beneficial or not. For example, a benign parasite of newts would do better than an aggressive one. Why? Because newts take up solitary residence in the forest, outside of ponds, for 6 years, and parasites have to wait that long before they are likely to meet up with another potential host. If a parasite kills its host before it can reach another, it loses not only its food source but also its home.

Questions to Consider

1. What happens to a species if it cannot coevolve along with the species it is interacting with?
2. Why is it necessary for a parasite to avoid killing its host?

a. b.

Figure 45B Social parasitism. The brown-headed cowbird, *Molothrus ater,* is a social parasite of more than 220 species of birds. **a.** The blue eggs of the eastern bluebird and the speckled egg of the cowbird in a nest. **b.** A cowbird chick that is outcompeting its nestmates for food. (a): Daniel Dempster Photography/Alamy Stock Photo; (b): Marie Read/Science Source

ISTUDY

previously discussed how continental drift contributed to various mass extinctions during the past. For example, as the continents slowly came together, forming the supercontinent Pangaea, many forms of marine life became extinct. And when the continents drifted toward the poles, immense glaciers drew water from the oceans and even chilled once-tropical lands. During ice ages, glaciers moved southward; then between ice ages, glaciers retreated, changing the environment and allowing life to colonize the land once again. Over time, complex communities evolved. Many ecologists, however, try to observe changes as they occur during much shorter timescales.

Ecological Succession

Communities are subject to disturbances that can range in severity from a storm blowing down a patch of trees, to a beaver damming a pond, to a volcanic eruption. We know from observation that after disturbances, changes occur in the community over time.

Ecological succession is a change within a community involving a series of species replacements. *Primary succession* is the formation of soil from exposed rock due to wind, water, and other abiotic factors. This type of succession occurs in areas where there is no base soil, such as following a volcanic eruption or a glacial retreat (Fig. 45.13). *Secondary succession* is a disturbance-based succession in which there is a progressive change from grasses to shrubs to a mixture of shrubs and trees. This type of succession occurs in areas where soil is already present, as when a cultivated field returns to a natural state.

Pioneer species are the first producers to inhabit a community after a disturbance. These are the first species that begin establishing the producer base of the community. Secondary succession begins and the area progresses through a series of stages over time (Fig. 45.14). The initial stage begins with small, short-lived species and proceeds through stages of species of mixed sizes and life spans until, finally, large, long-lived species of trees predominate. Ecologists have tried to determine the processes and mechanisms by which the changes described in Figures 45.13 and 45.14 take place—and whether these processes always have the same end point of community composition and diversity.

Models About Succession

In 1916, F. E. Clements proposed that succession in a particular area will always lead to a mature and stable community, which he called a **climax community.** He hypothesized that climate, in

a. Rock/lichen stage

b. Shrub stage

c. Low tree stage

d. High tree stage

Figure 45.13 Primary succession. This is an example of primary succession as glaciers retreated from an area called Glacier Bay, Alaska. **a.** Lichens and mosses invade the area left vacant by the retreating glacier. **b.** Small bushes grow in the newly abundant soil. **c.** Fast-growing alder trees help build the soil nutrients and water content. **d.** Improved soil conditions allow the white spruce–Western hemlock community to develop, eventually moving toward a more mature and stable community. (a-1): (rock/lichen stage): Kevin Smith/Design Pics/Getty Images; (a-2): (lichens): Background Abstracts/Photodisc/Getty Images; (b-1): (shrub stage): Don Paulson/Alamy Stock Photo; (b-2): (*Dryas*): Martin Fowler/Shutterstock; (c): (low tree stage): Bruce Heinemann/Photodisc/Getty Images; (d): (high tree stage): Don Paulson/Alamy Stock Photo

ISTUDY

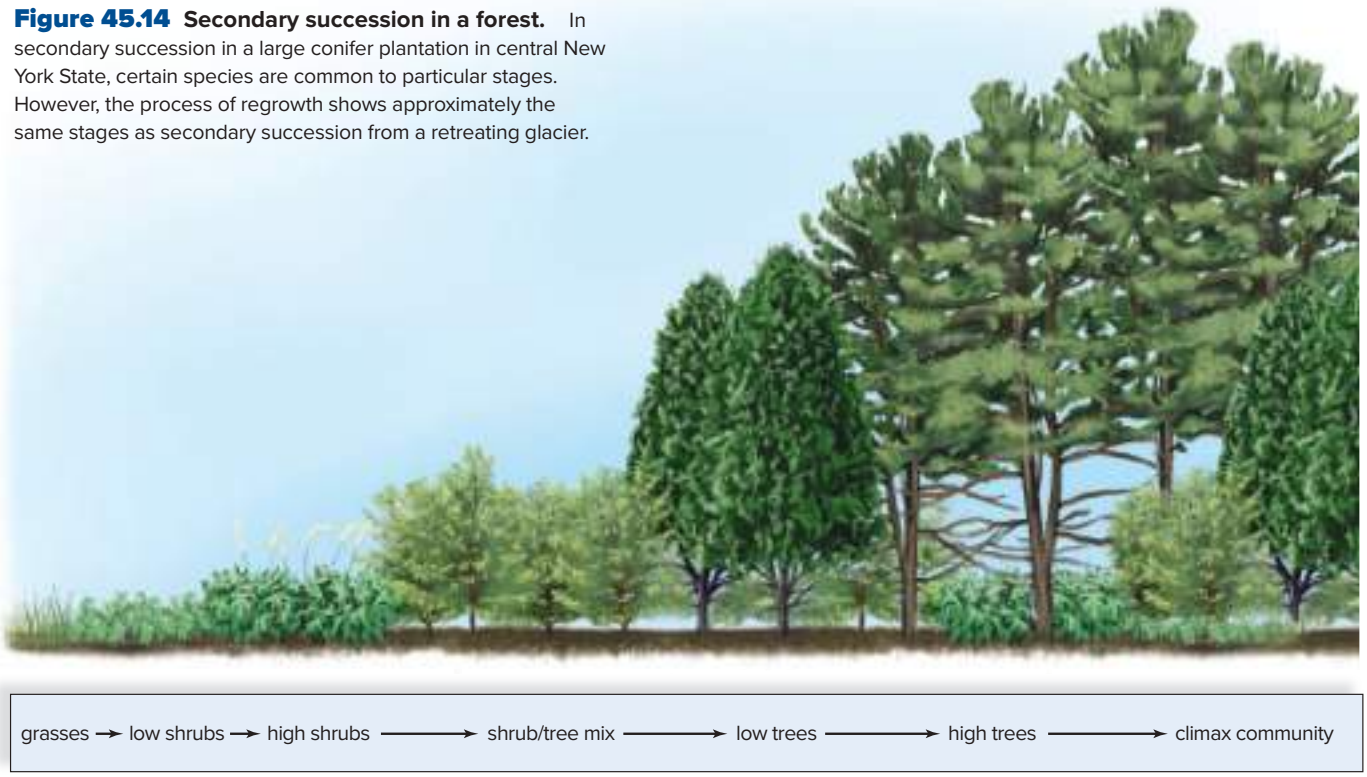


Figure 45.14 Secondary succession in a forest. In secondary succession in a large conifer plantation in central New York State, certain species are common to particular stages. However, the process of regrowth shows approximately the same stages as secondary succession from a retreating glacier.

particular, determined whether succession resulted in a desert, a type of grassland, or a particular type of forest. This is the reason, he said, that coniferous forests occur in northern latitudes, deciduous forests in temperate zones, and tropical rain forests in the tropics. Secondarily, he hypothesized that soil conditions might also affect the results. Shallow, dry soil might produce a grassland where otherwise a forest might be expected, or the rich soil of a riverbank might produce a woodland where a prairie would be expected.

Further, Clements hypothesized that each stage facilitated the invasion and replacement by organisms for the next stage. Shrubs can't grow on dunes until dune grass has caused soil to develop. Similarly, in the example in Figure 45.14, shrubs can't arrive until grasses have made the soil suitable for them. Each group of species prepares the way for the next, so that grass–shrub–forest development occurs in a sequential way. Therefore, in what is sometimes called "climax theory," this is known as the *facilitation model* of succession.

Aside from this facilitation model, an *inhibition model* also has been proposed to help explain succession. That model predicts that colonists hold on to their space and inhibit the growth of other plants until the colonists die or are unable to successfully compete against other species. Still another model, called the *tolerance model,* predicts that different types of plants can colonize an area at the same time. Sheer chance determines which seeds arrive first, and successional stages may simply reflect the length of time it takes species to mature. This alone could account for the grass–shrub–forest development that is often seen (Fig. 45.14).

The length of time it takes for trees to develop might give the impression that there is a recognizable series of plant communities, from the simple to the complex. In reality, succession occurs gradually, and the models mentioned here are probably not mutually exclusive but a mixture of multiple, complex processes.

Although the dynamic nature of natural communities may not have been apparent to early ecologists, we now recognize it as a community's most outstanding characteristic. Also, it seems obvious now that the most complex communities are ones that contain various stages of succession. Each successional stage has its own mix of plants and animals, and if a sample of all stages is present, community diversity is greatest. Furthermore, we do not know whether succession continues to specific end points, because the process may not be complete anywhere on Earth.

Check Your Progress 45.2

1. Identify the events that occur during succession.
2. Compare and contrast the various models used to explain succession.

45.3 Dynamics of an Ecosystem

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the interactions of organisms with their environment that comprise an ecosystem.
2. Identify the ways autotrophs, photoautotrophs, and heterotrophs obtain nutrients.
3. Interpret the energy flow and biogeochemical cycling within and among ecosystems.
4. Explain the energy flow among populations through food webs and ecological pyramids.

ISTUDY

An **ecosystem** is composed of the interactions between populations as well as their physical environment. The abiotic (nonliving) components of an ecosystem include the atmosphere, water, and soil. The biotic (living) components can be categorized according to their food source as autotrophs or heterotrophs (Fig. 45.15).

Autotrophs

Autotrophs are organisms that require only inorganic nutrients and an outside energy source to produce organic nutrients for their own use and for all the other members of a community. **Producers** are a type of autotroph, because they produce food from inorganic nutrients. Autotrophs include photosynthetic organisms, such as land plants and algae (photoautotrophs). They possess photosynthetic pigments such as chlorophyll and carry on photosynthesis in freshwater and marine habitats. Algae are aquatic autotrophs that make up the phytoplankton load in most bodies of water. Green plants are the dominant photosynthesizers on land.

Some autotrophic bacteria are chemosynthetic. They obtain energy by oxidizing inorganic compounds, such as ammonia, nitrites, and sulfides, and they use this energy to synthesize organic compounds. Chemoautotrophs have been found to support communities in some caves as well as at hydrothermal vents along deep-sea oceanic ridges where sunlight is unavailable.

Heterotrophs

Heterotrophs are organisms that need preformed organic nutrients they can use as an energy source. They are called **consumers** because they consume food that was generated by a producer.

Herbivores are animals that graze directly on plants or algae. In terrestrial habitats, various insects are small herbivores; antelopes and bison are large herbivores. In aquatic habitats, zooplankton are small herbivores; various fishes and manatees are large herbivores. **Carnivores** feed on other animals; various birds that feed on insects are carnivores, and so are hawks that feed on other birds and small mammals. **Omnivores** are animals that feed on both plants and animals. Chickens, raccoons, and humans are examples of omnivores. Some animals are scavengers, such as vultures and jackals, which eat the carcasses of dead animals.

Detritivores are organisms that feed on detritus, which consists of decomposing particles of organic matter. Marine fan worms filter detritus from the water, whereas burrowing clams take it from the substratum. Earthworms, some beetles, termites, and ants are all terrestrial detritivores.

Bacteria and fungi, including mushrooms, are **decomposers;** they acquire nutrients by breaking down dead organic matter, including animal wastes. Decomposers perform a valuable service

a. Producers

c. Carnivores

b. Herbivores

d. Decomposers

Figure 45.15 Biotic components. **a.** Diatoms and green plants are producers (photoautotrophs). **b.** Caterpillars and rabbits are herbivores. **c.** Snakes and hawks are carnivores. **d.** Bacteria and mushrooms are decomposers. (a-1): (diatoms): Ed Reschke; (a-2): (tree): Mats Andersson/Maskot/Getty Images; (b-1): (caterpillar): Ken Cavanagh/McGraw Hill; (b-2): (rabbit): Avirut S/Shutterstock; (c-1): (snake): Mark Kostich/iStock/Getty Images; (c-2): (hawk): Tze-hsin Woo/Flickr/Getty Images; (d-1): (bacteria): Image Source/Getty Images; (d-2): (mushroom): Denise McCullough

ISTUDY

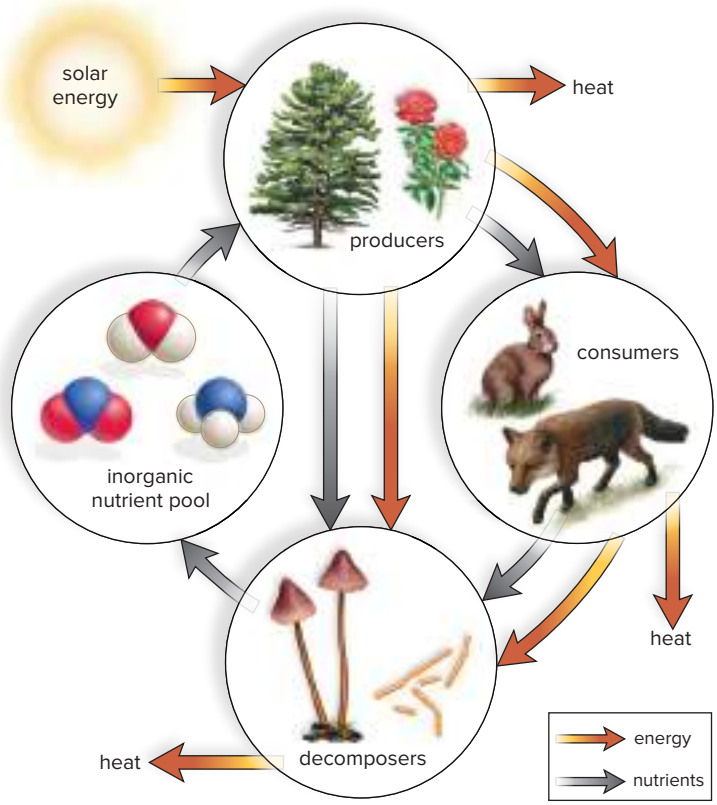


Figure 45.16 Nature of an ecosystem. Chemicals cycle, but energy flows through an ecosystem. As energy transformations repeatedly occur, all the energy derived from the sun eventually dissipates as heat.

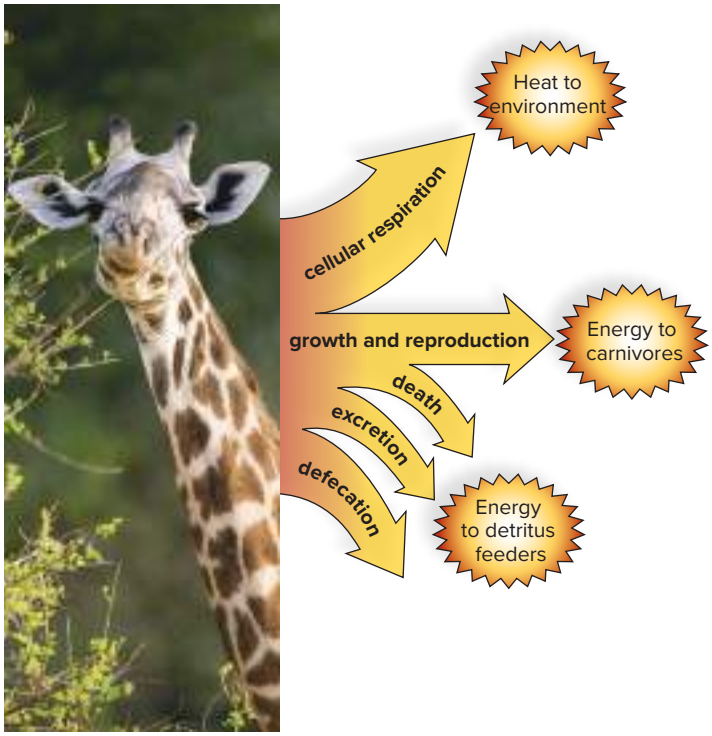


Figure 45.17 Energy balances. Only about 10% of the food energy taken in by a herbivore is passed on to carnivores. A large portion goes to detritus feeders via defecation, excretion, and death, and another large portion is used for cellular respiration. (photo): fStop Images GmbH/ Shutterstock

because they release inorganic substances, which are taken up by plants once more. Otherwise, plants would be completely dependent only on physical processes, such as the release of minerals from rocks, to supply them with inorganic nutrients.

A diagram of all the biotic components of an ecosystem illustrates that every ecosystem is characterized by two fundamental phenomena: energy flow and chemical cycling (Fig. 45.16). Energy flow begins when producers absorb solar energy, and chemical cycling begins when producers take in inorganic nutrients from the physical environment. Thereafter, via photosynthesis, producers make organic nutrients (food) directly for themselves and indirectly for the other populations within the ecosystem.

Energy Flow

Energy flows through an ecosystem following various pathways. As organic nutrients pass from one component of the ecosystem to another, as when a herbivore eats a plant or a carnivore eats a herbivore, only a small portion of the original amount of energy is transferred. Eventually, the majority of the energy dissipates into the environment as heat. This is why the vast majority of ecosystems cannot exist without a continual input of solar energy.

Only a portion of the organic nutrients made by producers is passed on to consumers because plants use organic molecules to fuel their own cellular respiration. Similarly, only a small percentage of nutrients consumed by lower-level consumers, such as herbivores, is available to higher-level consumers, or carnivores. As Figure 45.17 demonstrates, a certain amount of the food eaten by a herbivore is not digested and is eliminated as feces. Metabolic nitrogenous wastes are excreted as urine. Of the assimilated energy, a large portion is used during cellular respiration for the production of ATP and thereafter becomes heat. Only the remaining energy, which is converted into increased body weight or additional offspring, becomes available for carnivores.

The elimination of feces and urine by a heterotroph, and the death of an organism, does not mean that organic nutrients are lost to an ecosystem. Instead, they represent the organic nutrients made available to decomposers. Decomposers convert the organic nutrients, such as glucose, back into inorganic chemicals, such as carbon dioxide and water, and release them to the soil or atmosphere. When producers absorb these inorganic chemicals, they have completed their cycle within an ecosystem.

The laws of thermodynamics, which were first described in Section 6.1, support the concept that energy flows through an ecosystem. The first law states that energy can be neither created nor destroyed. This explains why ecosystems are dependent on a continual outside source of energy, usually solar energy, which photosynthesizers use to produce organic nutrients. The second law states that, with every transformation, some energy is converted into a less available form, such as heat. Because plants carry on cellular respiration, for example, only about 55% of the original energy absorbed by plants is available to an ecosystem.

ISTUDY

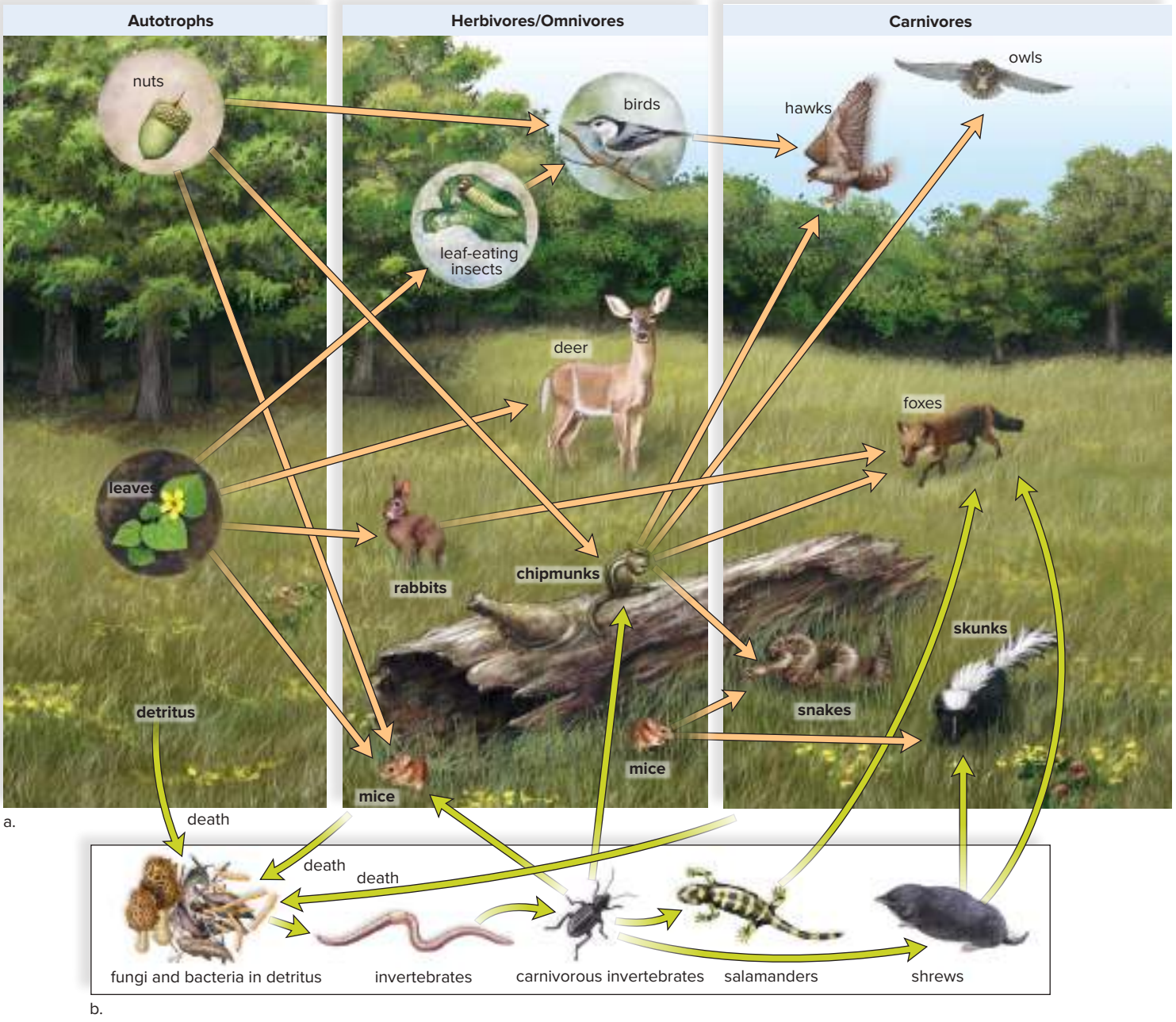


Figure 45.18 Grazing and detrital food web. Food webs are descriptions of who eats whom. **a.** Tan arrows illustrate the possible transfer of energy within a grazing food web. For example, nuts provide energy to birds, which are then fed upon by a hawk. Autotrophs such as the tree are producers (first trophic, or feeding, level), the first series of animals are primary consumers (second trophic level), and the next group of animals are secondary consumers (third trophic level). **b.** Green arrows illustrate the possible transfer of energy within a detrital food web, which begins with detritus—the bacteria and fungi of decay and the remains of dead organisms. A large portion of these remains are from the grazing food web illustrated in (**a**). The organisms in the detrital food web are sometimes fed on by animals in the grazing food web, as when robins feed on earthworms. Thus, the grazing food web and the detrital food web are connected to one another.

An Example of Energy Flow

Let's apply the principles presented so far to an actual ecosystem—a forest of 132,000 m^2 in New Hampshire. The various interconnecting paths of energy flow are represented by a **food web,** a diagram that describes trophic (feeding) relationships. Figure 45.18*a* is a grazing food web that begins with a producer and moves through successive trophic levels. A portion of the energy available in the oak tree will be transferred to the organisms that feed on the acorns and leaf tissue. A portion of the energy available in the form of herbivores such as caterpillars, mice, rabbits, and deer will be transferred to the carnivores, such as the hawks and owls. The organisms that feed on both the producers and the herbivores are classified as omnivores.

Figure 45.18*b* is a detrital food web, which begins with detritus. Detritus provides energy to soil organisms such as earthworms. Earthworms in turn provide energy to carnivorous invertebrates, which then provide energy to carnivores, such as shrews or

ISTUDY

salamanders. Because the members of a detrital food web may become food for aboveground carnivores, the detrital and grazing food webs are connected.

We naturally tend to think that aboveground plants, such as trees, are the largest storage form of organic matter and energy, but this is not necessarily the case. In a deciduous forest, the organic matter lying on the forest floor and mixed into the soil contains over twice as much energy as the leaf matter of living trees. Therefore, more energy in a forest may be funneling through the detrital food web than through the grazing food web.

Trophic Levels

The arrangement of the species in Figure 45.18 suggests that organisms are linked to one another in a straight line, according to feeding relationships, or who eats whom. Diagrams that show a single path of energy flow in an ecosystem are called **food chains.** For example, in the grazing food web, we might find this grazing food chain:

leaves→caterpillars→warblers→hawks

And in the detrital food web, we might find this detrital food chain:

detritus→earthworms→salamanders

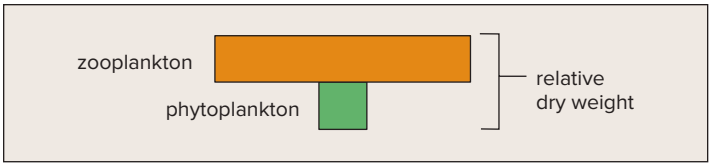
A **trophic level** is a level of nutrients within a food web or chain. In the grazing food web in Figure 45.18*a,* the green plants are the producers in the first trophic level, the animals in the center are the primary consumers in the second trophic level, and the last group of animals are the secondary consumers in the third trophic level.

Ecological Pyramids

The shortness of food chains can be attributed to the loss of energy between trophic levels. In general, only about 10% of the energy of one trophic level is available to the next trophic level. Therefore, if a herbivore population consumes 1,000 kg of plant material, only about 100 kg is converted into the body tissue of a herbivore, 10 kg is converted into the body tissue of the first-level carnivores, and 1 kg into the body tissue of the second-level carnivores. This "10% rule" explains why few top-level carnivores can be supported in a food web and require large communities to sustain their populations. The flow of energy with large losses between successive trophic levels is sometimes depicted as an **ecological pyramid.**

Energy flow from one trophic level to the next produces a pyramid based on the number of organisms or the amount of biomass at each trophic level (Fig. 45.19). When constructing such pyramids, problems arise, however. For example, in Figure 45.18, each tree would contain numerous caterpillars; therefore, there would be more herbivores than autotrophs. The explanation, of course, has to do with size. An autotroph can be as tiny as a microscopic alga or as big as a beech tree; similarly, a herbivore can be as small as a caterpillar or as large as an elephant.

Pyramids of biomass eliminate size as a factor because **biomass** is the number of organisms multiplied by the dry weight of the organic matter within one organism. The biomass of the producers is expected to be greater than the biomass of the herbivores, and that of the herbivores is expected to be greater than that of the carnivores. In aquatic ecosystems, such as some lakes and open seas where algae are the only producers, the herbivores may have a greater biomass than the producers, because the algae are consumed at such a high rate. Such pyramids, which have more herbivores than producers, are called inverted pyramids:



These kinds of drawbacks are making some ecologists hesitant about using pyramids to describe ecological relationships. One more problem is what to do with the decomposers, which are rarely included in pyramids, even though a large portion of energy becomes detritus in many ecosystems.

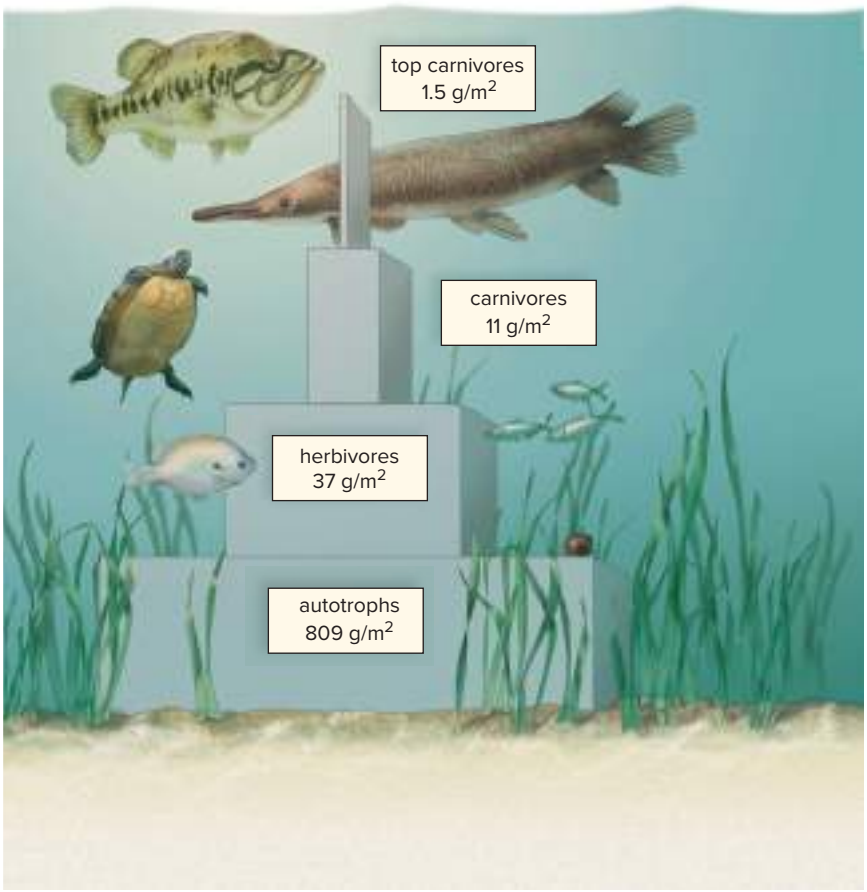


Figure 45.19 Biomass within an ecological pyramid. The biomass, or dry weight (g/m^2), for trophic levels in a grazing food web in a bog at Silver Springs, Florida. There is a sharp drop in biomass between the producer level and herbivore level, which is consistent with the common knowledge that the detrital food web plays a significant role in bogs.

ISTUDY

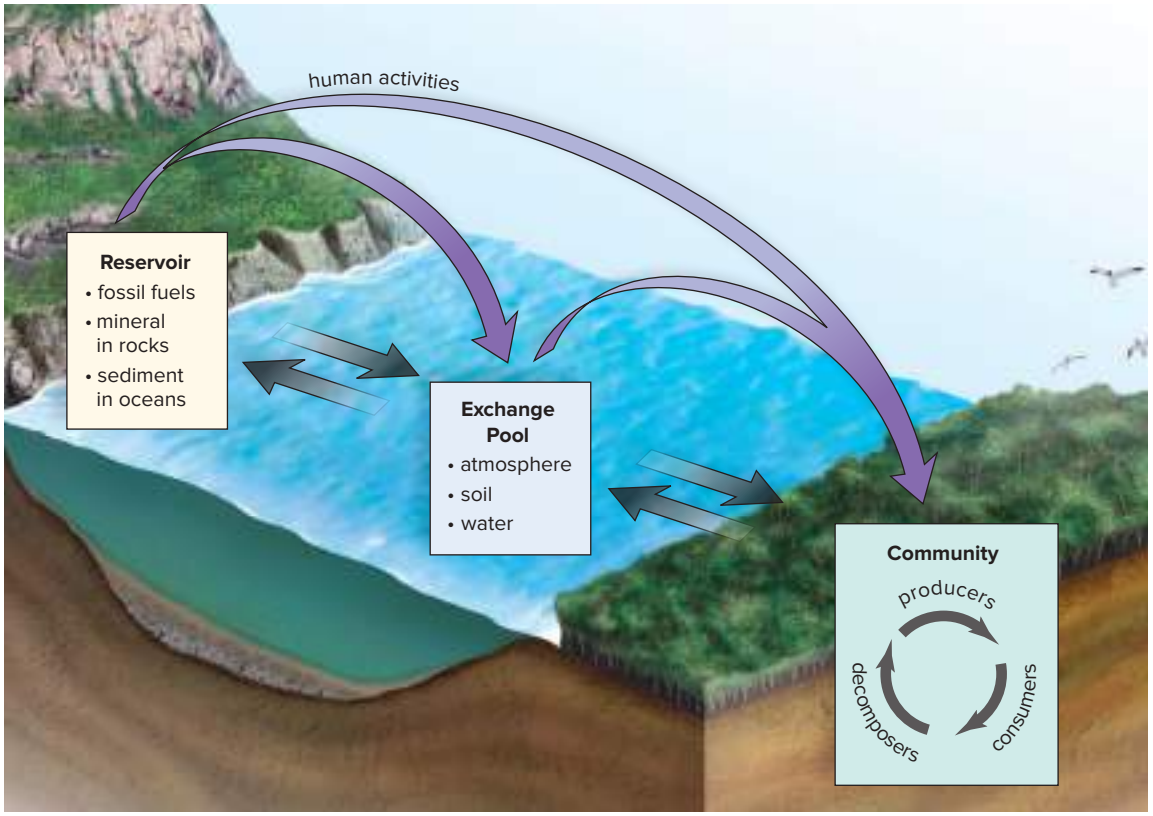


Figure 45.20 Model for chemical cycling. Chemical nutrients cycle between these components of ecosystems. Reservoirs, such as fossil fuels, minerals in rocks, and sediments in oceans, are normally relatively unavailable sources, but exchange pools, such as those in the atmosphere, soil, and water, are available sources of chemicals for the biotic community. When human activities (purple arrows) remove chemicals from reservoirs and pools and make them available to the biotic community, pollution can result.

Chemical Cycling

The pathways by which chemicals circulate through ecosystems involve both living (biotic) and nonliving (geologic) components known as the **biogeochemical cycles.** Here we focus on the four main biogeochemical cycles: water, carbon, phosphorus, and nitrogen.

A biogeochemical cycle may be sedimentary or gaseous. The phosphorus cycle is a sedimentary cycle; the chemical is absorbed from the soil by plant roots, passed to heterotrophs, and eventually returned to the soil by decomposers. The carbon and nitrogen cycles are gaseous, meaning that the chemical returns to and is withdrawn from the atmosphere as a gas.

Chemical cycling involves the components of ecosystems shown in Figure 45.20. A *reservoir* is a source normally unavailable to producers, such as the carbon present in calcium carbonate shells on ocean bottoms. An *exchange pool* is a source from which organisms can obtain chemicals, such as the atmosphere or soil. Certain chemicals can move along food chains in a *biotic community* and never enter an exchange pool.

Human activities (purple arrows in Fig. 45.20) remove chemicals from reservoirs and exchange pools and make them available to the biotic community. In this way, human activities result in pollution, because they upset the normal balance of nutrients for producers in the environment.

The Water Cycle

The **water cycle,** also called the *hydrologic cycle,* is described in Figure 45.21. A **transfer rate** is defined by the amount of a substance that moves from one component of the environment to another within a specified period of time. The width of the arrows in Figure 45.21 indicates the relative transfer rate of water.

During the water cycle, fresh water is first distilled from salt water through evaporation. During evaporation, a liquid, in this case water, changes to a gaseous state. The sun's rays cause fresh water to evaporate from the seawater, and the salts are left behind. Next, condensation occurs. During condensation, a gas is converted into a liquid. For example, vaporized fresh water rises into the atmosphere, collects in the form of a cloud, cools, and falls as rain over the oceans and the land.

Water evaporates from land and from plants (evaporation from plants is called transpiration) and from bodies of fresh water. Because land lies above sea level, gravity eventually returns all fresh water to the sea. In the meantime, water is contained within standing bodies (lakes and ponds), flowing bodies (streams and rivers), and groundwater.

Some of the water from precipitation (e.g., rain, snow, sleet, hail, and fog) makes its way into the ground and saturates the earth to a certain level. The top of the saturation zone is called the groundwater table, or simply, the water table. Because water

ISTUDY

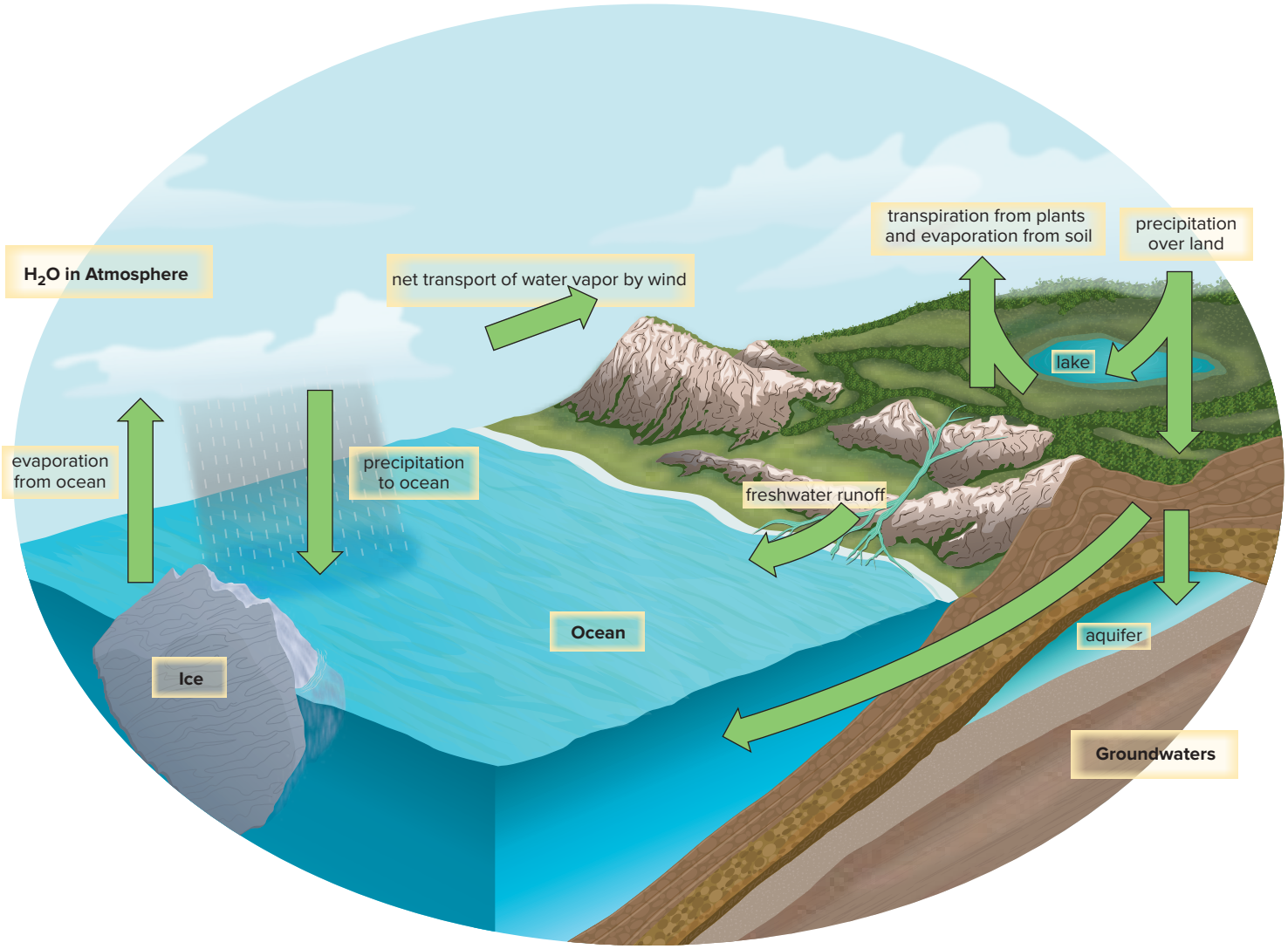


Figure 45.21 The water cycle. Evaporation from the ocean exceeds precipitation, so there is a net movement of water vapor onto land, where precipitation results in surface water and groundwater, which flow back to the sea. On land, transpiration by plants contributes to evaporation.

infiltrates through the soil and rock layers, groundwater can be located in rock layers called aquifers. Water is usually released in appreciable quantities to wells or springs. Aquifers are recharged when rainfall and melted snow percolate into the soil.

Human Activities and the Water Cycle In some parts of the United States, especially the arid West and southern Florida, withdrawals from aquifers are exceeding the level of recharge. This is called "groundwater mining." In these locations, the groundwater level is dropping, and residents may run out of groundwater, at least for irrigation purposes, by 2030.

Fresh water makes up about 3% of the world's supply of water. Water is considered a renewable resource, because a new supply is always being produced as a result of the water cycle. It is possible to run out of fresh water, however, when the rate of consumption exceeds the rate of production or when the water has become so polluted that it is not usable.

The Carbon Cycle

In the carbon cycle, organisms in both terrestrial and aquatic ecosystems exchange carbon dioxide (CO_2) with the atmosphere (Fig. 45.22). Therefore, the CO_2 in the atmosphere is the exchange pool for the carbon cycle. On land, plants take up CO_2 from the air through photosynthesis. The CO_2 is incorporated into nutrients, which are used by both autotrophs and heterotrophs alike. When organisms, including plants, conduct cellular respiration, carbon is returned to the atmosphere as CO_2. Carbon dioxide is then incorporated back into plants when they carry out photosynthesis.

In aquatic ecosystems, the exchange of CO_2 with the atmosphere is indirect. Carbon dioxide from the air combines with water to produce bicarbonate ion (HCO_3^-). This is the main source of carbon for algae. Similarly, when aquatic organisms engage in cellular respiration, the CO_2 they give off becomes HCO_3^-. The amount of bicarbonate in the water is in equilibrium with the amount of CO_2 in the air.

ISTUDY

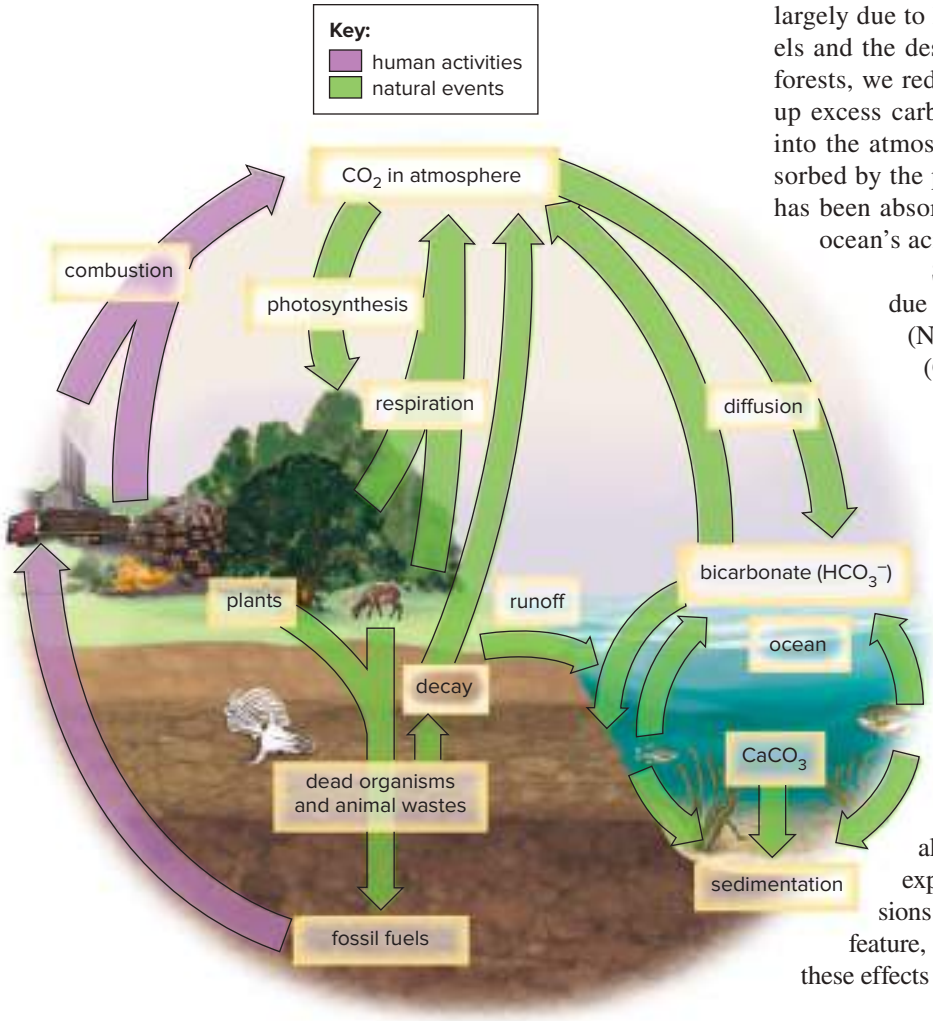


Figure 45.22 The carbon cycle. The transfer rate of carbon into the atmosphere due to respiration approximately matches the rate of withdrawal by plants for photosynthesis. Carbon dioxide from the air combines with water to form bicarbonate ion. Decay, as well as the burning of fossil fuels and the destruction of vegetation by human activities (purple arrows), is placing more carbon dioxide in the atmosphere than can be withdrawn.

Reservoirs Hold Carbon Living and dead organisms contain organic carbon and serve as a reservoir for the carbon cycle. The world's biotic components, particularly trees, contain 800 billion tons of organic carbon. An additional 1,000–3,000 billion tons are estimated to be held in the remains of plants and animals in the soil. Ordinarily, decomposition of organisms returns CO_2 to the atmosphere.

Some 300 mya, plant and animal remains were transformed into coal, oil, and natural gas, the materials we call *fossil fuels*. Fossil fuels contain significant quantities of carbon, which is released during combustion of the fuel. Another reservoir for carbon is the inorganic carbonate that accumulates in limestone and in calcium carbonate shells. Many marine organisms have calcium carbonate shells that remain in sediments long after the organisms have died. Geological forces change these sediments into limestone.

Human Activities and the Carbon Cycle More CO_2 is being released into the atmosphere than is being removed, largely due to human activities such as the burning of fossil fuels and the destruction of forests. When humans do away with forests, we reduce a reservoir and the very organisms that take up excess carbon dioxide. Today, the amount of CO_2 released into the atmosphere is about twice the amount that can be absorbed by the producers on the planet. Much of the excess CO_2 has been absorbed by the ocean, contributing to an increase in ocean's acidity.

Several other gases are emitted into the atmosphere due to human activities. These include nitrous oxide (N_2O) from fertilizers and animal wastes and methane (CH_4) from bacterial decomposition that occurs in anaerobic environments. These gases are known as **greenhouse gases,** because, just like the panes of a greenhouse, they allow solar radiation to pass through but hinder the escape of infrared rays (heat) back into space. This phenomenon has come to be known as the **greenhouse effect.**

The greenhouse gases are contributing significantly to an overall rise in the Earth's ambient temperature, a trend called **global warming,** which is leading to climate change. The expression **climate change** refers to the recent changes in the Earth's climate. The global climate has already warmed about 0.6°C since the Industrial Revolution. Computer models are unable to consider all possible variables, but the Earth's temperature is expected to rise 1.5–4.5°C by 2100 if greenhouse emissions continue at the current rates. The Nature of Science feature, "Global Climate Change," in Section 45.4 discusses these effects further.

The Phosphorus Cycle

Figure 45.23 depicts the phosphorus cycle. Phosphorus from oceanic sediments moves onto land due to a geological uplift. On land, the very slow weathering of rocks places phosphate ions (PO_3^- and HPO_4^+) into the soil. Some of these become available to plants, which use phosphate in a variety of molecules, including phospholipids, ATP, and the nucleotides that become a part of DNA and RNA. Animals eat producers and incorporate some of the phosphate into their teeth, bones, and shells. However, eventually the death and decay of all organisms and the decomposition of animal wastes make phosphate ions available to producers once again. Because the available amount of phosphate is already being used within food chains, phosphate is usually a limiting inorganic nutrient for plants. Phosphate levels influence the size of populations within an ecosystem.

Some phosphate naturally runs off into aquatic ecosystems, where algae acquire it from the water before it can become trapped in sediments. Phosphate in marine sediments does not become available to producers on land until a geological upheaval exposes sedimentary rocks on land. The cycle then begins again.

Human Activities and the Phosphorus Cycle Humans boost the supply of phosphate by mining phosphate ores that are used in the production of fertilizer and detergents. Runoff of phosphate and nitrogen from agricultural and residential fertilizer use, animal

ISTUDY

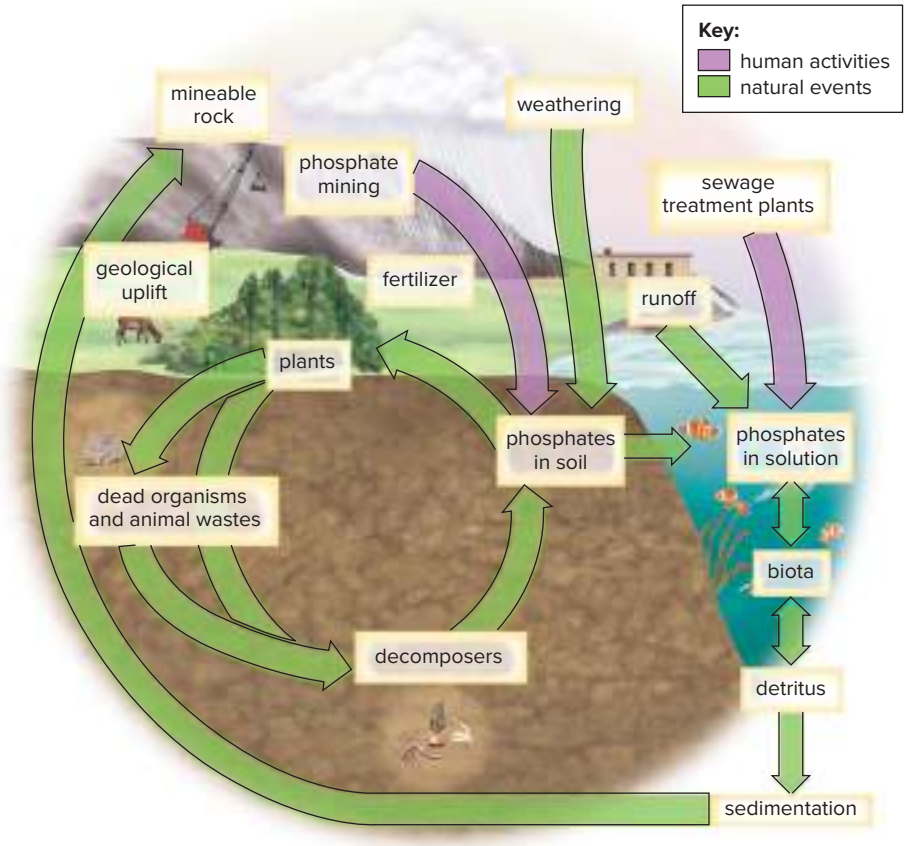


Figure 45.23 The phosphorus cycle. Both the weathering of rocks and human activities produce phosphorus for terrestrial and aquatic biota. When phosphorus becomes a part of oceanic sediments, it is lost to biotic communities until geological uplift makes it available again.

wastes from livestock feedlots, and discharge from sewage treatment plants result in **eutrophication** (over-enrichment) of waterways.

The Nitrogen Cycle

Nitrogen gas (N_2) makes up about 78% of the gases in the atmosphere, but plants cannot make use of nitrogen in its gaseous form. The availability of nitrogen in an ecosystem can limit the size of the producer populations. First, let's consider that **nitrogen fixation** occurs when nitrogen gas (N_2) is converted to ammonium (NH_4^+), a form plants can use (Fig. 45.24). Some cyanobacteria in aquatic ecosystems and some free-living bacteria in soil are able to fix atmospheric nitrogen in this way. Other nitrogen-fixing bacteria live in nodules on the roots of legumes, such as beans, peas, and clover. They make organic compounds containing nitrogen available to the host plants, so that the plant can form proteins and nucleic acids.

Plants can also use nitrates (NO_3^-) as a source of nitrogen. The production of nitrates during the nitrogen cycle is called **nitrification.** Nitrification can occur in two ways:

- Nitrogen gas (N_2) is converted to NO_3^- in the atmosphere when cosmic radiation, meteor trails, and lightning provide the energy needed for nitrogen to react with oxygen.
- Ammonium (NH_4^+) in the soil from various sources, including the decomposition of organisms and animal wastes, is converted to NO_3^- by nitrifying bacteria in soil. Specifically, NH_4^+ (ammonium) is converted to NO_2^- (nitrite), and then NO_2^- is converted to NO_3^- (nitrate).

During the process of **assimilation,** plants take up NH_4^+ and NO_3^- from the soil and use these ions to produce proteins and

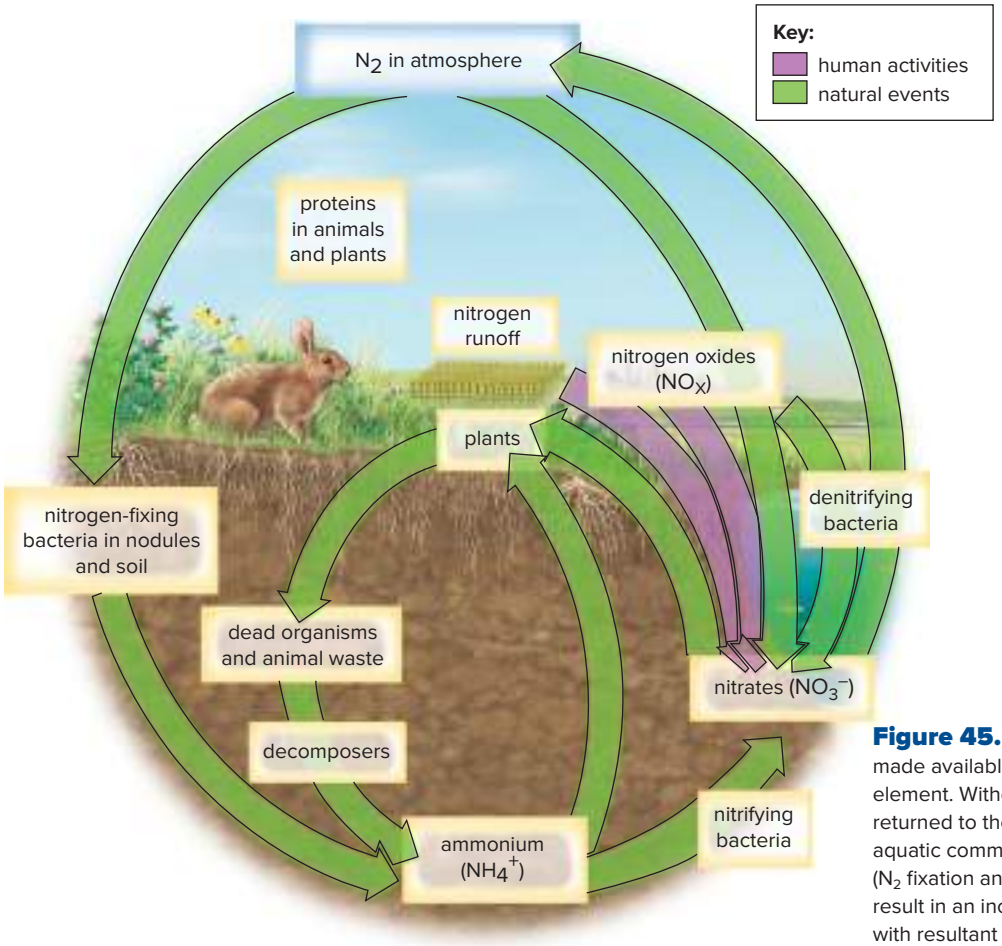


Figure 45.24 The nitrogen cycle. Nitrogen is primarily made available to biotic communities by internal cycling of the element. Without human activities, the amount of nitrogen returned to the atmosphere (denitrification in terrestrial and aquatic communities) exceeds withdrawal from the atmosphere (N_2 fixation and nitrification). Human activities (purple arrows) result in an increased amount of NO_3^- in terrestrial communities, with resultant runoff to aquatic biotic communities.

ISTUDY

nucleic acids. Notice in Figure 45.24 that the subcycle involving the biotic community, which occurs on land and in the ocean, need not depend on the presence of nitrogen gas at all.

Finally, **denitrification** is the conversion of nitrate back to nitrogen gas, which then enters the atmosphere. Denitrifying bacteria living in the anaerobic mud found at the bottom of lakes, bogs, and estuaries carry out this process as a part of their own metabolism. In the nitrogen cycle, denitrification would counterbalance nitrogen fixation if not for human activities.

Human Activities and the Nitrogen Cycle Humans significantly alter the transfer rates in the nitrogen cycle by producing fertilizers from N_2. Fertilizer runs off into rivers and lakes and this results in an overgrowth of algae and rooted aquatic plants. When the algae die off, enlarged populations of decomposers use up all the oxygen in the water, and the result is a massive fish kill.

Acid deposition occurs because nitrogen oxides (NO_x) and sulfur dioxide (SO_2) enter the atmosphere from the burning of fossil fuels. Both of these gases combine with water vapor to form acids, which eventually return to the earth. Acid deposition has drastically affected forests and lakes in northern Europe, Canada, and the northeastern United States by altering the pH of their soils and surface waters. Acid deposition also reduces agricultural yields and corrodes marble, metal, and stonework.

Check Your Progress 45.3

1. Identify the various types of populations that are at the base of an ecological pyramid and the start of a food chain.
2. Compare the flow of energy to the flow of chemicals through an ecosystem.
3. Provide examples of how human activities can alter the biogeochemical cycles.

45.4 Ecological Consequences of Climate Change

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the abiotic and biotic factors that are contributing to climate change.
2. Describe the consequences of climate change upon different biomes.

Throughout the history of the Earth, the climate has changed. Many of these changes were the result of small variations in the Earth's orbit around the sun. This caused the amount of solar energy received by various regions to shift, thus changing their climate. The scientific community overwhelmingly agrees that the current changes to the Earth's climate are the result of human activities, and these changes are occurring at a rate faster than anything experienced in the past 1,300 years.

Contributors to Climate Change

Various abiotic factors are contributing to climate change, with the leading factor being the "greenhouse effect." This is a warming trend that occurs when the Earth's atmosphere traps the heat that should be radiated back into space. Various gases, including water vapor, carbon dioxide, nitrous oxide, chlorofluorocarbons (CFCs), and methane, are the chief contributing factors to the greenhouse effect.

As the Earth's atmosphere warms, it produces a larger amount of water vapor. The water vapor in turn retains additional heat that should have radiated back into the atmosphere, thus forming an important feedback mechanism associated with the greenhouse effect. Consider how warm a summer night is when there is abundant cloud cover, versus a clear night that lacks cloud cover.

Figure 45.22 shows how carbon dioxide is released naturally through processes such as cellular respiration and decomposition, and then absorbed through photosynthesis and often stored in the bodies of photosynthetic organisms. Significant amounts of carbon are also stored in the form of fossil fuels. Human actions have created an imbalance in the carbon cycle by the burning of fossil fuels, which releases more carbon dioxide than can be absorbed naturally through photosynthesis. However, carbon dioxide is not the only gas that contributes to the greenhouse effect:

- Nitrous oxide is produced through many modern agricultural practices, including the use of fertilizers, burning of fossil fuels, and nitric acid production.
- Chlorofluorocarbons (CFCs) are human-produced synthetic compounds that have been used across a wide variety of commercial and residential applications. The destructive nature of CFCs have played an important role in the destruction of the ozone layer, which has led to international agreements regulating their usage.
- Methane is a hydrocarbon gas that is produced through natural and human-related activities. Landfill decomposition, various forms of agriculture, and domestic livestock production all contribute to the production and release of methane into the atmosphere.

As humans alter landscapes, we are undoubtedly contributing to global climate change. Deforestation is occurring at an alarming rate on a global scale. As large regions of forest are cleared, it deprives the forest of its canopy (Fig. 45.25). In 2019, nearly 3,000 square km of the Amazon rain forest were cleared and burned to make way for agriculture and ranching. This causes the soil to dry out, often preventing the growth of new trees. The lack of mature trees leads to large swings in temperature, which can cause the regions to become more desertlike. During deforestation, the trees are often burned, releasing millions of tons of carbon dioxide into the atmosphere, while reducing the forest's ability to store carbon.

Consequences of Climate Change

Over 97% of the scientific community supports the belief that human activities are the primary reason behind the climate-changing events currently being experienced by the Earth. Industrial activities that are necessary to support modern society

ISTUDY

Figure 45.25 Deforestation of rain forests. Rain forest destruction is a significant contributor to global climate change. If current rates continue, the world's rain forests could be gone within the next hundred years. guentermanaus/Shutterstock

have caused atmospheric levels of carbon dioxide to increase from 280 parts per million to over 410 parts per million over the past 150 years. All of the different factors contributing to climate change are expected to produce a wide range of consequences.

The Earth's temperature, as predicted by the Intergovernmental Panel on Climate Change (IPCC), is expected to increase between 2.5° and 10°F within the next century. The IPCC is composed of over 1,300 scientists from the United States and other countries. They are predicting that the rise in temperature may actually benefit various regions, while harming others.

In some regions, the growing season will begin earlier than in the past. This may mean that agricultural crops will be able to be planted earlier, with a potential of increased harvests. The potential for longer growing seasons and increased harvests may be countered by changes in precipitation patterns. In early 2021, Kentucky experienced a period of abnormally dry conditions, followed by historic rainfall. This led to flash flooding that washed away homes and roadways, leaving more than 25 people dead as a result of this extreme weather event. Around the same time, over 26.8% of the United States experienced extreme to exceptional drought conditions. The United States was not alone in witnessing these types of extreme weather events. Sixty-three percent of the European Union and United Kingdom were under drought warnings or alerts during this time, while nearly a third of Pakistan encountered flooding due to excessive monsoon rains. These droughts and floods caused significant environmental and property damage.

Hurricanes and other severe precipitation events have become stronger and more intense as a result of climate change. Category 4 and 5 hurricanes have increased in frequency and duration in the North Atlantic region since the early 1980s. The storm intensity associated with hurricanes, as well as the rates of rainfall, are also projected to increase. Hurricane Ida blasted ashore as a Category 4 storm that inundated portions of Louisiana and Mississippi, destroying businesses and neighborhoods, and killing at least 28 people. Ida then moved northeast to saturate portions of New York, New Jersey, and the East Coast, causing the deaths of 54 people. (Fig. 45.26).

As the Earth's temperatures continue to increase, it is predicted that there will be a significant decrease in the amount of ice found in the Arctic during the summer. The decrease in the ice pack, as well as the warming of the ocean, has led to a global increase in sea level of about 8 inches since the 1880s. Projections indicate that sea levels will rise another 1 to 4 feet by the year 2100. This increase, along with stronger storm surges and higher tides, is expected to produce significant flooding of major coastal cites like Miami, New York, and New Orleans. The Nature of Science feature, "Global Climate Change," examines some of the consequences of a changing climate, as well as some of the efforts that have been developed to reduce the severity of this crisis.

Every individual on the planet is responsible for contributing to the issue of climate change. Individuals in more-developed countries such as the United States, Germany, and Great Britain play a much larger role in the production of greenhouse gases,

a. Before

b. After

Figure 45.26 Devastation to the Bahama Islands. Hurricane Dorian devastated the Bahama Islands in 2019 when it passed through as a historically slow-moving and powerful hurricane. (a): Daniel Carlson/Shutterstock; (b): Reuters/Alamy Stock Photo

ISTUDY

THEME Nature of Science

Global Climate Change

Scientists around the world continue to collect and interpret environmental data to help us understand how and why the Earth's climate is changing. Changes in the average temperature of a region, precipitation patterns, sea levels, and greenhouse gas concentrations are all indicators of this.

Since the late nineteenth century, the average temperature across the Earth has risen by about 1.8 degrees Fahrenheit. Temperatures have increased globally over the past two decades, and the eight hottest years on record have all occurred since 2015.

Average precipitation rates have also increased by 6% over the past century. Since 1990, the United States has experienced 8 of the top 10 years of extreme precipitation events. Increases in sea surface temperatures produce a more active hurricane season. Since the mid-1990s, the Atlantic Ocean, the Caribbean, and the Gulf of Mexico have seen six of the ten most active hurricane seasons.

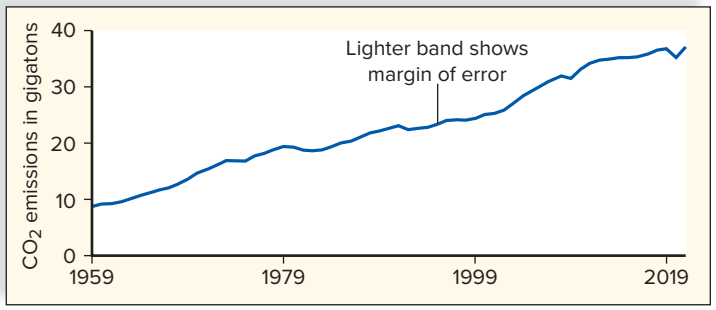
Sea levels worldwide have risen an average of 1 inch per decade due to the overall increase in the surface temperature of the world's oceans. Over half of the human population lives within 60 miles of the coast. Climate models suggest we will see a rise in sea levels of 3 to 4 feet over the next century. New York City ranges from 5 to 16 feet above sea level, while the Florida Keys are an average of 3 to 4 feet above sea level. Even if the rising waters don't produce flooding, many coastal areas will be exposed to increasingly severe storms and storm surges that could lead to significant economic losses.

Global carbon dioxide emissions are increasing annually (Fig. 45C*a*). The United States is the largest producer of carbon dioxide emissions per person, followed by China and the EU. We are facing a worldwide problem that will require global cooperation in order to be solved.

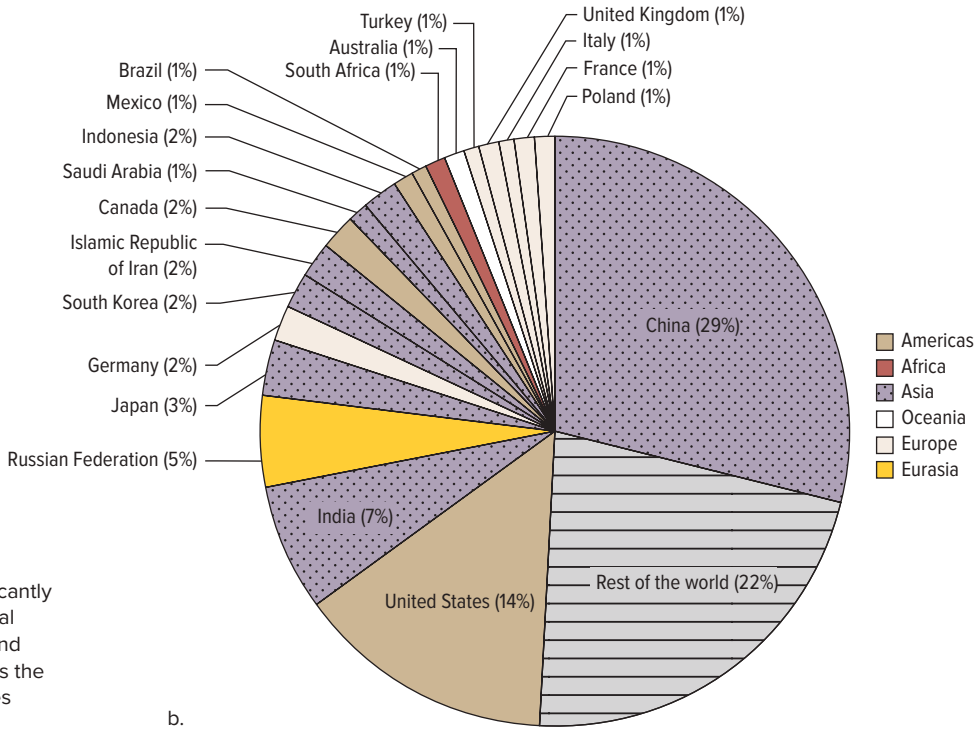
The Kyoto Protocol was initially adopted on December 11, 1997, and enforced on February 16, 2005. The goal of the protocol was to lower the overall emissions of the six greenhouse gases. By November 2009, 187 countries had ratified the protocol, with the goal of reducing emissions by an average of 5.2% by the year 2012. The United States is responsible for approximately 14% of the global CO_2 emissions from fossil fuel combustion and is one of the largest per-capita emission producers in the world (Fig. 45C*b*). Unfortunately, at that time the United States had not ratified the agreement, which raised concerns that other countries may not uphold their commitment to combating climate change as well.

The Copenhagen conference in 2009 ended without any type of binding agreement for long-term action against climate change. It did produce a collective commitment by many developed nations to raise $30 billion to be used to help poor nations combat climate change and cope with its effects.

The 2010 Cancun summit on climate change helped solidify this agreement. Because deforestation produces about 15% of the global carbon emissions, many developing



a.



b.

Figure 45C Global warming and climate change. **a.** CO_2 emissions have increased significantly in the past hundred years, causing changes in global climate patterns, including more severe droughts and erratic patterns of precipitation. **b.** Although China is the largest producer of CO_2, the United States produces more CO_2 per person (or capita).

ISTUDY

countries can now receive incentives to prevent the destruction of their rain forests.

The 2013 UN Climate Change Conference in Warsaw was successful in establishing ways to help less-developed nations reduce greenhouse emissions as a result of deforestation, as well as establishing finance commitments to assist less-developed nations. The conference also helped keep governments on track toward a universal climate agreement to be implemented in 2020.

The 2015 Paris Climate Change conference is considered the most successful of the meetings to date. The Paris agreement strengthens the global response to the threat of climate change. Global efforts would limit the increase in global average temperature to less than 1.5°C, as well as strengthen the ability of countries to deal with climate change in the future.

Sadly, in 2019 the United States became the first and only country in the world to withdraw from the Paris Climate Agreement, and at the time was no longer seen as a global leader on climate change. A few years later, under a new presidential administration, the United States rejoined the agreement and made science a central focus of combatting climate change. Many countries that signed the Paris agreement, however, are concerned that the United States' inconsistent mindset may give other countries an excuse to slow their actions or withdraw from the agreement altogether, which will cause even more dire consequences for a planet that is already changing.

Questions to Consider

1. Should the United States and other more-developed nations pay less-developed nations to preserve their forests?
2. Should individual U.S. states be allowed to remain in global climate agreements if the federal government decides not to ratify a specific treaty?

carbon dioxide in particular, that are produced per person. It is up to each of us, individually, to make small changes that then add up to large-scale benefits in decreasing the impact of climate change.

Small changes, at home, that can help reduce your personal impact and contribution to climate change include things like replacing light fixtures and light bulbs with ENERGY STAR products. This can save up to $70 per year on electric bills. Caulking and insulating drafty areas of your home can help reduce heating and cooling costs by 20%. In addition, all appliances are now required to provide an Energy Guide (Fig. 45.27) so consumers can easily compare energy usage between brands.

At the office, small steps can be taken that will save money while also reducing your contribution to climate change. Activating power management features on computers and monitors decreases the amount of energy being used while they are in idle modes. Reducing, reusing, and recycling in the office can save a company thousands of dollars and significantly decrease its environmental impact.

If everyone on Earth were to make a handful of small, environmentally friendly changes to their lifestyle, it would add up to a significant global impact. Domestic and international efforts, in regard to production and usage of CFCs, are enabling the ozone layer to repair itself. It is predicted that by 2065 the layer will be fully restored. Correcting the problems that are causing climate change are not things that can be solved individually or quickly, but without making some form of concerted efforts, we can be guaranteed that the problems we face will only continue to worsen.

Check Your Progress 45.4

1. List three different factors contributing to climate change.
2. Explain the consequences of climate change to cities such as New York, Miami, and New Orleans.
3. Describe a simple plan that can be implemented in your own home to help decrease your impact on climate change.

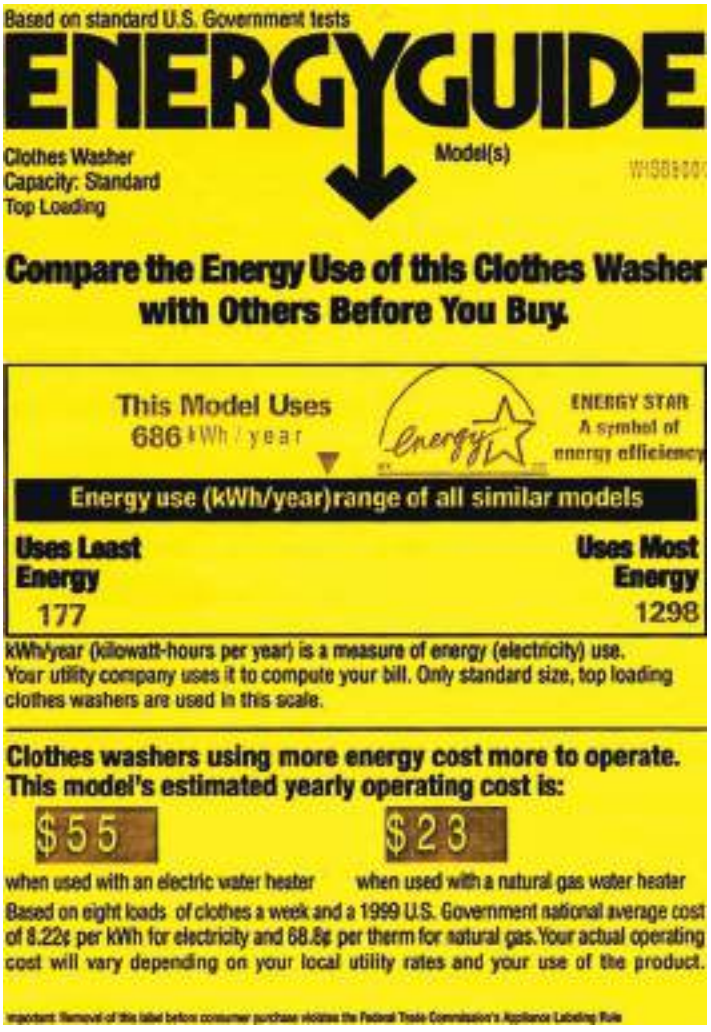
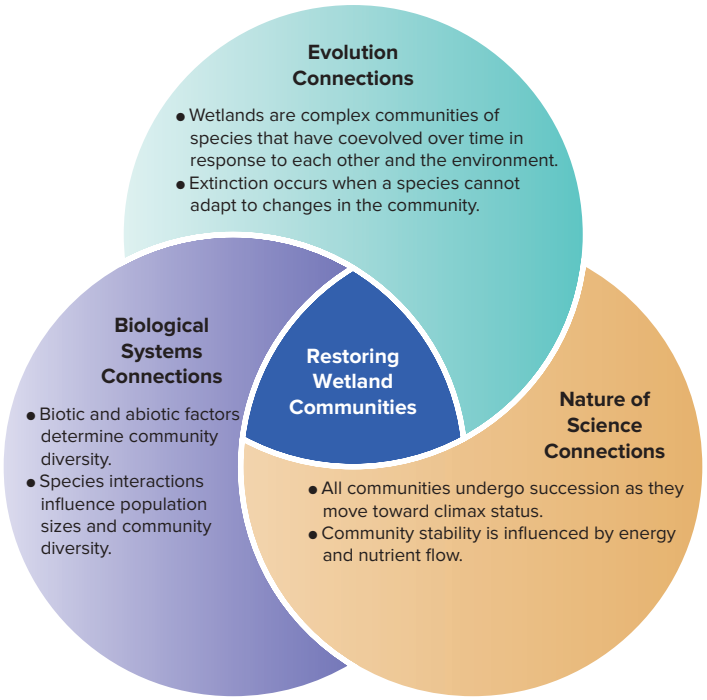


Figure 45.27 Energy Guides. Energy Guide labels help consumers compare the energy use of similar brands of major appliances. Roger Loewenberg/McGraw Hill

Connecting *the* Concepts



What Did You Learn?

45.1 Ecology of Communities

A **community** is an assemblage of populations interacting with one another in the same environment.

- Communities differ in their composition, also known as **species richness.** The **species diversity** includes both species richness and relative abundance. The **island biogeography model** of diversity helps us determine the ideal size of conservation areas.
- An organism's **habitat** is where it lives in the community. An **ecological niche** is defined by the role an organism plays in its community, including its habitat and how it interacts with other species in the community. The **competitive exclusion principle** states that no two species can indefinitely occupy the same niche at the same time.
- **Resource partitioning** decreases competition between species, which can lead to **character displacement.** When resources are partitioned between two or more species, increased niche specialization occurs. But the difference between species can be more subtle, as when warblers feed at different parts of the tree canopy. Barnacles competing on the Scottish coast may be an example of ongoing competition.
- **Predation** occurs when a **predator** feeds upon **prey.** This type of interaction is especially influenced by the size of the predator population and the amount of resources available for the prey. A cycling of population densities may occur. Prey defenses take many forms: **Camouflage,** warning coloration, and **mimicry** are a few mechanisms. Batesian mimicry occurs when one species has the warning coloration but lacks the defense. Müllerian mimicry occurs when two species with the same warning coloration have the same defense.
- **Symbiosis,** the close association of two species over a long period of time, comes in several forms. **Parasitism** occurs when a **parasite** takes nourishment from its **host.** Whether parasites are aggressive (kill their host) or benign probably depends on which results in the highest fitness. **Commensalism** occurs when one species gains a benefit from the relationship and there is no impact on the other. Mutualism occurs when both species involved in a relationship receive a positive benefit. The continuous interaction between two species results in **coevolution,** a change in one species in response to a change in the other.

45.2 Community Development

Ecological succession involves a series of species replacements in a community. Primary succession occurs where there is no soil present. Secondary succession occurs where soil is present and certain plant species can begin to grow. **Pioneer species** are the first producers to inhabit an area after a disturbance. A **climax community** forms when stages of succession lead to a particular type of community.

45.3 Dynamics of an Ecosystem

Populations interact with each other and the abiotic environment to form an **ecosystem.**

- The biotic components of an ecosystem consist of **autotrophs,** which are the producers of the community, and **heterotrophs,** which are the consumers of the community. Heterotrophs can be classified as **herbivores, carnivores, omnivores, detritivores,** and **decomposers** based on how they obtain their nutrients.
- Abiotic components are resources such as nutrients and conditions such as type of soil and temperature.

Ecosystems are characterized by energy flow and chemical cycling. Energy flows in one direction, because as food passes from one population to the next, each population makes energy conversions that result in a loss of usable energy. Chemicals cycle because they pass from one population to the next until decomposers return them once more to the producers.

ISTUDY

- Ecosystems contain **food webs** in which the various organisms are connected by trophic relationships. In grazing food webs, **food chains** begin with a producer. In a detrital food web, food chains begin with detritus. **Trophic levels** depict the levels of nutrients available in a food web or chain. As energy flows from one level to the next, it is depicted as an **ecological pyramid.** Ecological pyramids are graphic representations of the number of organisms multiplied by their **biomass,** representing the energy content of trophic levels.

Biogeochemical cycles are the pathways in which chemicals circulate through ecosystems. Chemical cycling involves a reservoir, an exchange pool, and a biotic community.

- In the **water cycle** (hydrologic cycle), evaporation over the ocean is not compensated for by precipitation. The **transfer rate** is the amount of substance that moves from one component of the environment to another. Precipitation over land results in bodies of fresh water plus groundwater, including aquifers. Eventually, all water returns to the oceans. In the carbon cycle, carbon dioxide in the atmosphere is an exchange pool; both terrestrial and aquatic plants and animals exchange carbon dioxide with the atmosphere. Living and dead organisms serve as reservoirs for the carbon cycle, because they contain organic carbon. Human activities increase the level of CO_2 and other **greenhouse gases,** which help cause a phenomenon known as the **greenhouse effect. Global warming** is leading to **climate change** as the ambient temperature of the Earth continues to increase.
- In the phosphorus cycle, geological upheavals move phosphorus from the ocean to land. Slow weathering of rocks returns phosphorus to the soil. Most phosphorus is recycled within a community. Excessive accumulation of phosphorus in an aquatic community can lead to **eutrophication.**
- In the nitrogen cycle, plants cannot use nitrogen gas from the atmosphere. During **nitrogen fixation,** N_2 converts to ammonium, making nitrogen available to plants. **Nitrification** is the production of nitrates, whereas **denitrification** is the conversion of nitrate back to N_2, which enters the atmosphere. Human activities increase transfer rates in the nitrogen cycle. **Acid deposition** occurs when nitrogen oxides enter the atmosphere, combine with water vapor, and return to Earth in precipitation.

45.4 Ecological Consequences of Climate Change

The trapping of heat in the Earth's atmosphere, called the greenhouse effect, is due to the influence of gases such as water vapor, carbon dioxide, nitrous oxide, chlorofluorocarbons (CFCs), and methane. Many of these greenhouse gases are produced as the result of our lifestyle as a modern society.

- Carbon dioxide is released when we burn fossil fuels.
- Large amounts of nitrous oxide are produced through modern agricultural practices.
- Chlorofluorocarbons (CFCs) are the result of our production of synthetic compounds.
- Methane comes from domestic livestock production, along with landfill decomposition.

Increases in the concentrations of these gases in the atmosphere is leading to climate change across the globe. The consequences of climate change will vary depending on how much the Earth's temperature increases. The Intergovernmental Panel on Climate Change (IPCC) expects the temperature to increase between 2.5° and 10°F within the next century. An increase in the severity of hurricanes, flooding, and drought are all predicted to occur as a result of this increase in temperature. A decrease in the amount of ice in the Arctic has already caused sea levels to rise by about 8 inches.

All of these events are due to the collective actions of humans, but the collective actions of humans can *also* help minimize the potential consequences of climate change. Reducing, reusing, and recycling are all ways we can make a difference.

Thinking Critically About Biology

1. Residents that live 25 miles downstream from a new coal-burning power plant are complaining about a strange taste that has started to appear in their drinking water. How would you explain the connection between the new power plant and the odd taste in the drinking water?
2. If communities naturally move through the stages of succession, why was it so important for wolves to be reintroduced into Yellowstone National Park?
3. If you observe three species of frogs in the same general area, how would you test a hypothesis that they occupy different niches?
4. In order to improve species richness, you decide to add phosphate to a pond. How might you determine how much phosphate to add in order to avoid eutrophication?

Making It Relevant to You

BioNOW

Want to know how this science is relevant to your life? Check out the BioNOW video:

- Biodiversity

McGraw Hill

What was the effect of the exotic (alien) species on the biodiversity? Why do you think this was the case?

Making It Relevant

1. Should federal or local tax dollars be spent on restoration projects like Emiquon?
2. Can private land be claimed by the government if it is necessary for large-scale restoration projects?
3. How does a restoration project like Emiquon benefit citizens who live in other states?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

46

Major Ecosystems of the Biosphere

(inset): mj795/iStock/Getty Images; (art): Panthera Corporation.

Chapter Outline

46.1 Climate and the Biosphere

46.2 Terrestrial Ecosystems

46.3 Aquatic Ecosystems

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 7.5 How does the climate determine the producer base of a biome?

Section 45.1 How do symbiotic relationships enable species to survive in harsh environments?

Figure 45.21 How does the water cycle connect terrestrial and aquatic habitats?

The Central American Jaguar Corridor

Species such as jaguars sit on the top of the food chain and require a large diversity of habitats to support their species. The jaguars' range extends from Mexico in the north to northern Argentina in the south. Some of their prey base includes animals such as armadillos, peccary, turtles, and deer. Currently, jaguars are listed as "near threatened" on the International Union for Conservation of Nature (IUCN) Red List of Threatened Species.

The Jaguar Corridor Initiative was established in an effort to preserve the genetic continuity of jaguars across their entire range. Countries in Central and South America have come together and agreed to place tracts of land into a protected status. The goal is to create a continuous corridor of crucial habitats and ecosystems to support jaguars and other species.

Unfortunately, the Jaguar Corridor Initiative, in its entirety, has not come to fruition. It has proven challenging to establish coordination among the various countries. Panthera, a global conservation organization, is working with 11 of the 18 countries that contain jaguar populations. If enough effort is made at the local level, it may provide the support needed to maintain the initiative. If not, we will likely see the extinction of many of these species within our lifetimes.

As you read through this chapter, think about the following questions:

1. Should some level of protection be extended to the prey species of the jaguar?
2. How would changes in one country impact the jaguar population in another country?

Following *the* Themes

CHAPTER 46 – MAJOR ECOSYSTEMS OF THE BIOSPHERE

Evolution	The biodiversity of Earth's biomes has evolved over billions of years.
Nature of Science	New species are being discovered as scientists study previously unexplored biomes.
Biological Systems	Over time, changes in one biome influence changes in other biomes.

ISTUDY

46.1 Climate and the Biosphere

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe how solar radiation produces variations in Earth's climate.
2. Explain how global air circulation patterns and physical geographic features are associated with the Earth's temperature and rainfall patterns.

Climate is the prevailing weather conditions in a particular region that occur over the course of a long period of time. Weather, on the other hand, consists of the environmental conditions that occur over a short period of time. Climate is dictated by temperature and rainfall, which are influenced by the following factors:

- Variations in solar radiation distribution due to the tilt of the Earth as it orbits the sun
- Other effects caused by topography (surface features) and whether a body of water is nearby

Effect of Solar Radiation

Because the Earth is a sphere, it receives direct sunlight at the equator but indirect sunlight at the poles (Fig. 46.1*a*). The region between latitudes approximately 26.5° north and south of the equator is considered the tropics. The tropics are warmer than the areas north of 23.5°N and south of 23.5°S, known as the temperate regions.

The tilt of the Earth as it orbits around the sun causes one pole or the other to be angled toward the sun (except at the spring and fall equinoxes, when sunlight aims directly at the equator). This accounts for the seasons that occur in all parts of the Earth except at the equator (Fig. 46.1*b*). When the Northern Hemisphere is having winter, the Southern Hemisphere is having summer, and vice versa.

If the Earth were standing still and were a solid, uniform ball, all air movements would occur in two directions. Warm air at the equator would rise and move toward the poles where it would cool and sink. Rising air creates zones of lower air pressure. However, because the Earth rotates on its axis daily and its surface consists of continents and oceans, the flow of warm and cold air is modified into three large circulation cells in each hemisphere (Fig. 46.2).

At the equator, the sun heats the air and evaporates water. Warm, moist air rises, and as it cools it loses most of its moisture as rain. The greatest amounts of rainfall on Earth are near the equator. The rising air flows toward the poles, but at about 30° north and south latitude, it cools and sinks toward the Earth's surface before reheating. As the dry air descends and warms, areas of high pressure are generated. High-pressure regions are zones of low rainfall. The great deserts of Africa, Australia, and the Americas occur at these latitudes.

At the Earth's surface, the air flows toward the poles and the equator. As dry air moves across the Earth, moisture from both land and water gets absorbed. At about 60° north and south latitude, the warmed air rises and cools, producing another low-pressure area with high rainfall. This moisture supports the great forests of the temperate zone. Part of this rising air flows toward the equator, and part continues toward the poles, where it will descend. The poles are high-pressure areas that have low amounts of precipitation.

Besides affecting precipitation, the spinning of the Earth also affects the winds (Fig. 46.2). In the Northern Hemisphere, large-scale winds generally bend clockwise, and in the Southern Hemisphere, they bend counterclockwise. The curving pattern of the winds, ocean currents, and cyclones is the result of the Earth rotating in an eastward direction. At about 30° north latitude and 30° south latitude, the winds blow from the east-southeast in the Southern Hemisphere and from the east-northeast in the Northern Hemisphere (the east coasts of continents at these latitudes are wet). The doldrums, regions of calm, occur at the equator. The winds blowing

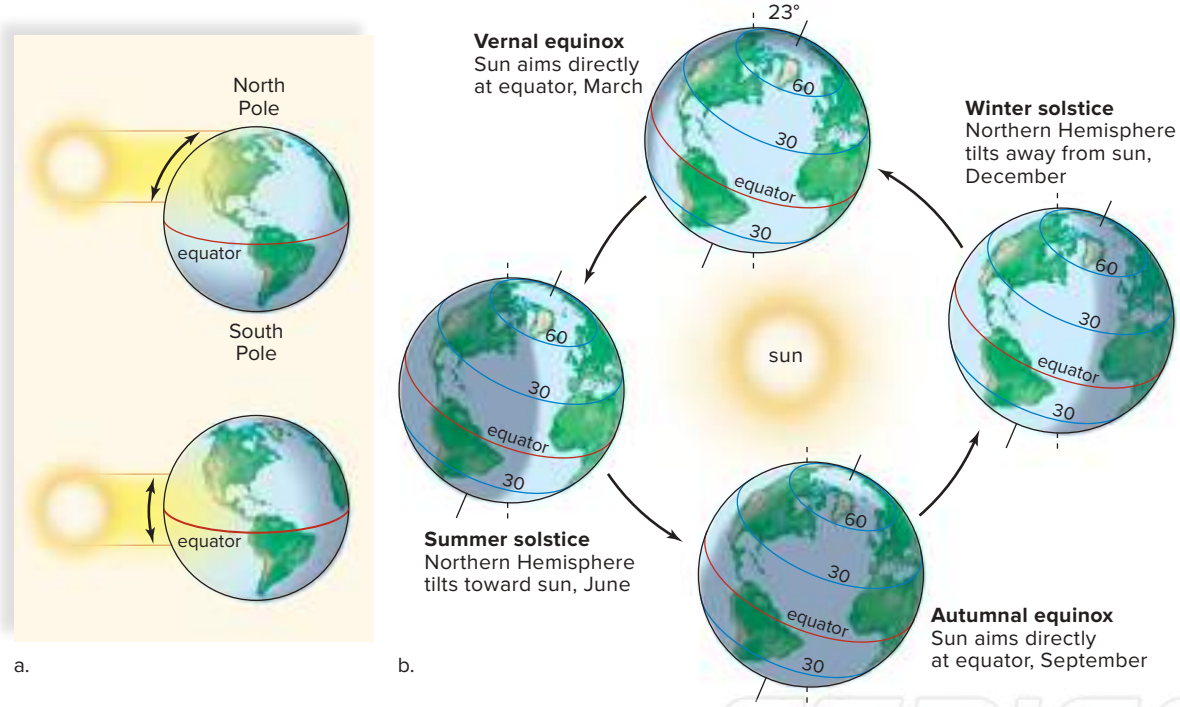


Figure 46.1
Distribution of solar energy. **a.** Because the Earth is a sphere, beams of solar energy striking the Earth near one of the poles are spread over a wider area than similar beams striking the Earth at the equator. **b.** The seasons of the Northern and Southern Hemispheres are due to the tilt of the Earth on its axis as it rotates about the sun.

ISTUDY

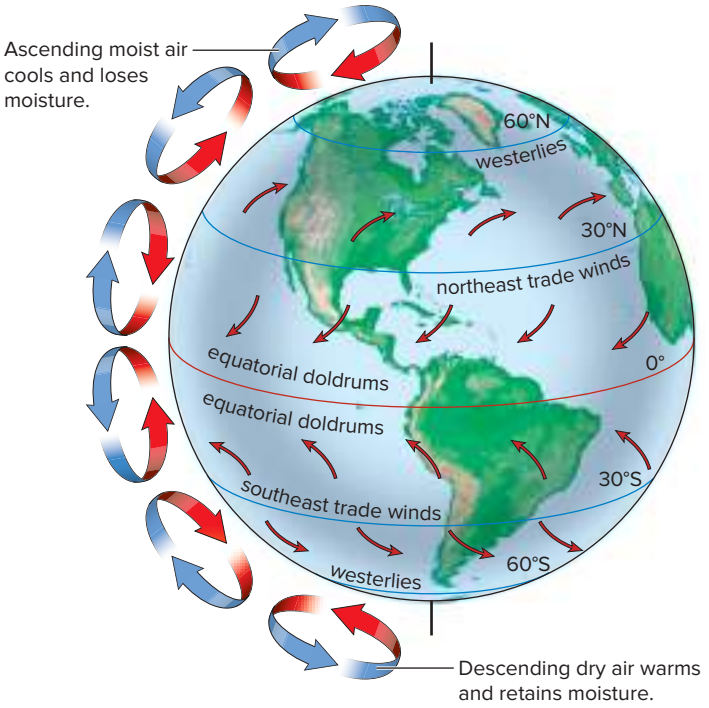


Figure 46.2 Global wind circulation. Air ascends and descends as shown because the Earth rotates on its axis. Also, the trade winds move from the northeast to the west in the Northern Hemisphere, and from the southeast to the west in the Southern Hemisphere. The westerlies move toward the east.

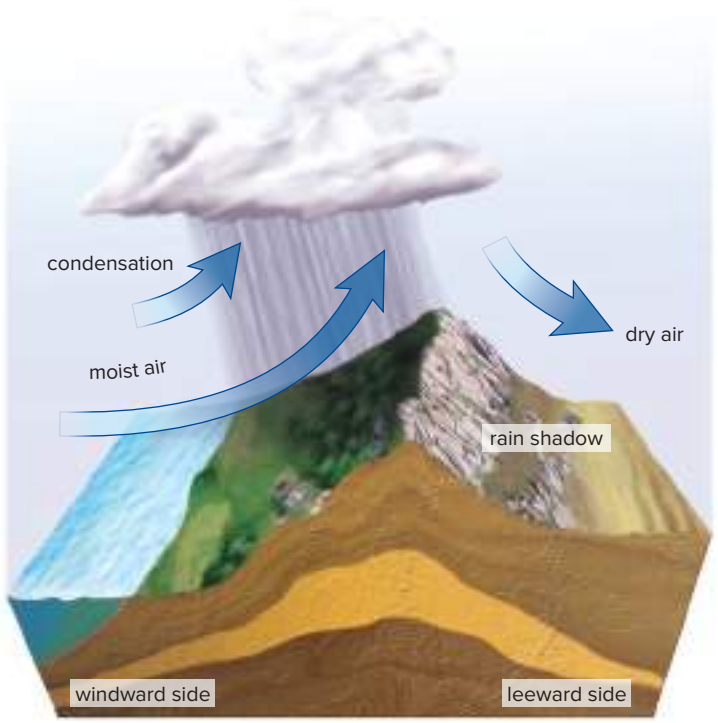


Figure 46.3 Formation of a rain shadow. When winds from the sea cross a coastal mountain range, they rise and release their moisture as they cool this side of a mountain, called the windward side. The leeward side of a mountain receives relatively little rain and is therefore said to lie in a "rain shadow."

from the doldrums toward the poles are called trade winds because sailors depended on them to fill the sails of their trading ships.

Between 30° and 60° north and south latitudes, strong winds called the prevailing westerlies blow from west to east. The west coasts of the continents at these latitudes are wet, as is the Pacific Northwest, where a massive evergreen forest is located. Weaker winds, called the polar easterlies, blow from east to west at still higher latitudes of their respective hemispheres.

Other Effects

Topography means the physical features of the land. One physical feature that affects climate is the presence of mountains. As air blows up and over a coastal mountain range, it cools as it rises. The coastal side of the mountain, called the windward side, receives more rainfall than the other side. On the interior side of the mountain or leeward side, the air descends, absorbs moisture from the ground, and produces dry air and clear weather (Fig. 46.3). The leeward side of the mountain or **rain shadow,** tends to be much drier than the windward side.

In the Hawaiian Islands, for example, the windward side of the mountains receives more than 750 cm of rain a year, while the leeward side gets, on average, only 50 cm of rain and is generally sunny. In the United States, the western side of the Sierra Nevada is lush, while the eastern side is a semidesert.

The temperature of the oceans is more stable than that of landmasses. Oceanic water gains or loses heat more slowly than do terrestrial environments. This difference causes coasts to have a unique weather pattern that is not observed inland. During the day, the land warms more quickly than the ocean, and the air above the land rises, pulling a cool sea breeze in from the ocean. At night, the reverse happens: The breeze blows from the land toward the sea.

India and some other countries in southern Asia have a **monsoon** climate, in which wet ocean winds blow onshore for almost half the year. The land heats more rapidly than the waters of the Indian Ocean during spring. The difference in temperature between the land and the ocean causes an enormous circulation of air: Warm air rises over the land, and cooler air comes in off the ocean to replace it. As the warm air rises, it loses its moisture, and the monsoon season begins. As just discussed, rainfall is particularly heavy on the windward side of hills. Cherrapunji, a city in northern India, receives an annual average of 1,090 cm of rain a year because of its high altitude. This weather pattern has reversed by November. The land is now cooler than the ocean; therefore, dry winds blow from the Asian continent across the Indian Ocean. In the winter, the air over the land is dry, the skies cloudless, and temperatures pleasant. The chief crop of India is rice, which starts to grow when the monsoon rains begin.

In the United States, people often speak of the "lake effect," meaning that, in the winter, arctic winds blowing over the Great Lakes become warm and moisture-laden. When these winds rise and lose their moisture, snow begins to fall. Places such as Buffalo, New York, receive heavy snowfalls due to the lake effect, and snow is on the ground there for an average of 90–140 days every year.

Check Your Progress 46.1

1. Identify the conditions that account for a warm climate at the equator.
2. Name two physical features that can affect rainfall.

ISTUDY

46.2 Terrestrial Ecosystems

Learning Outcomes

Upon completion of this section, you should be able to

1. Recognize the geographic distribution of the major terrestrial biomes.
2. Identify the key characteristics of the major terrestrial biomes.

Major terrestrial ecosystems, called biomes, are characterized by their climate and geography. A **biome** has a particular mix of plants and animals that are adapted to living within a specific range of environmental conditions in a given region.

When terrestrial biomes are plotted according to their mean annual temperature and mean annual precipitation (Fig. 46.4*a*), particular patterns result across the globe. The distribution of biomes is shown in Figure 46.4*b*. Even though this figure shows definite dividing lines, keep in mind that the change from one biome to another is gradual. Each biome has a connection to all the other terrestrial and aquatic ecosystems of the biosphere.

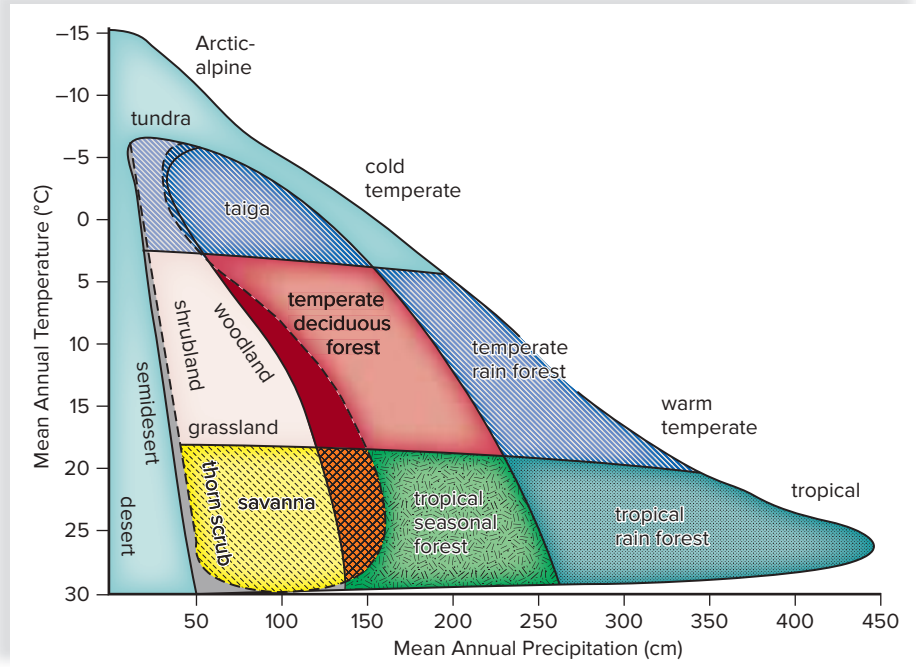
The distribution of the biomes and their corresponding populations are determined principally by differences in climate. Latitude and altitude both play a role in determining temperature gradients. If you travel north from the equator to the North Pole, you can observe tropical rain forests, followed by a temperate deciduous forest, a coniferous forest, and tundra. A similar sequence is seen when ascending a mountain (Fig. 46.5). The coniferous forest of a mountain is called a **montane coniferous forest,** and the tundra near the peak of a mountain is called an **alpine tundra.** When going south from the equator to the South Pole, you would not reach a region corresponding to a coniferous forest or tundra because of the lack of landmasses in the Southern Hemisphere.

Each biome supports characteristic types of plants and animals; however, as described in the opening essay, the range of jaguars encompasses a wide variety of habitats. Individual cats establish large home ranges that shift seasonally and include different biomes. The Nature of Science feature, "Wildlife Conservation and DNA," describes how DNA analysis is helping to clarify not only species relationships but also migration patterns and sources of human diseases.

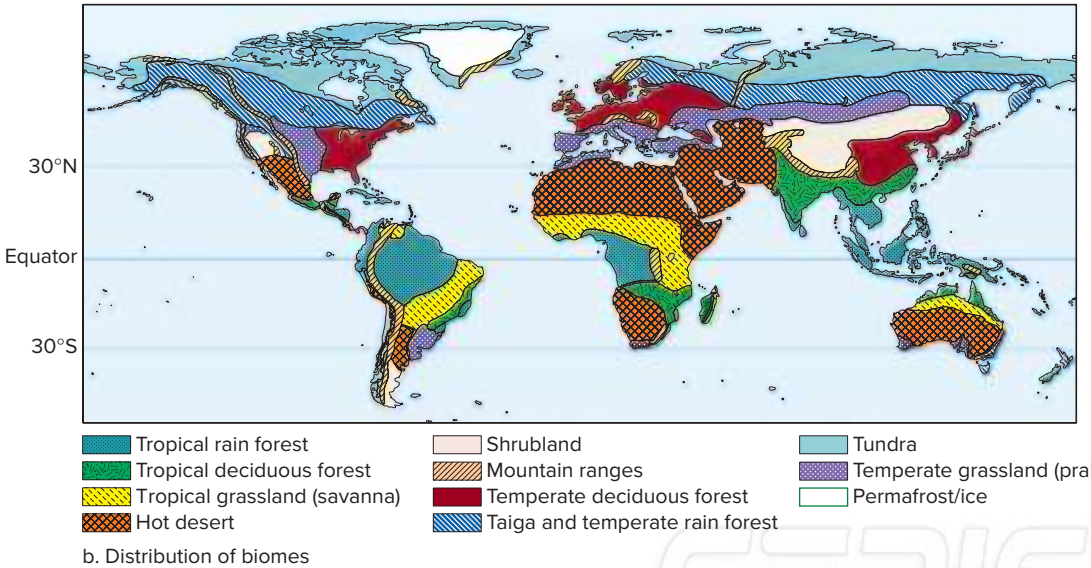
We now examine the features of the major biomes, beginning with the tundra.

Tundra

The **tundra** biome is characterized as being cold and dark much of the year. Found in the Arctic regions, it has extremely long, cold, harsh winters and short summers (6–8 weeks). Because rainfall amounts to only about 20 cm a year, the tundra could be considered a desert, but melting snow creates a landscape of pools and bogs in



a. Biome pattern of temperature and precipitation



b. Distribution of biomes

Figure 46.4 Pattern of biome distribution. **a.** Pattern of world biomes in relation to temperature and moisture. The dashed line encloses a wide range of environments in which either grasses or woody plants can dominate the area, depending on the soil type. **b.** The same type of biome can occur in different regions of the world, as shown on this global map.

ISTUDY

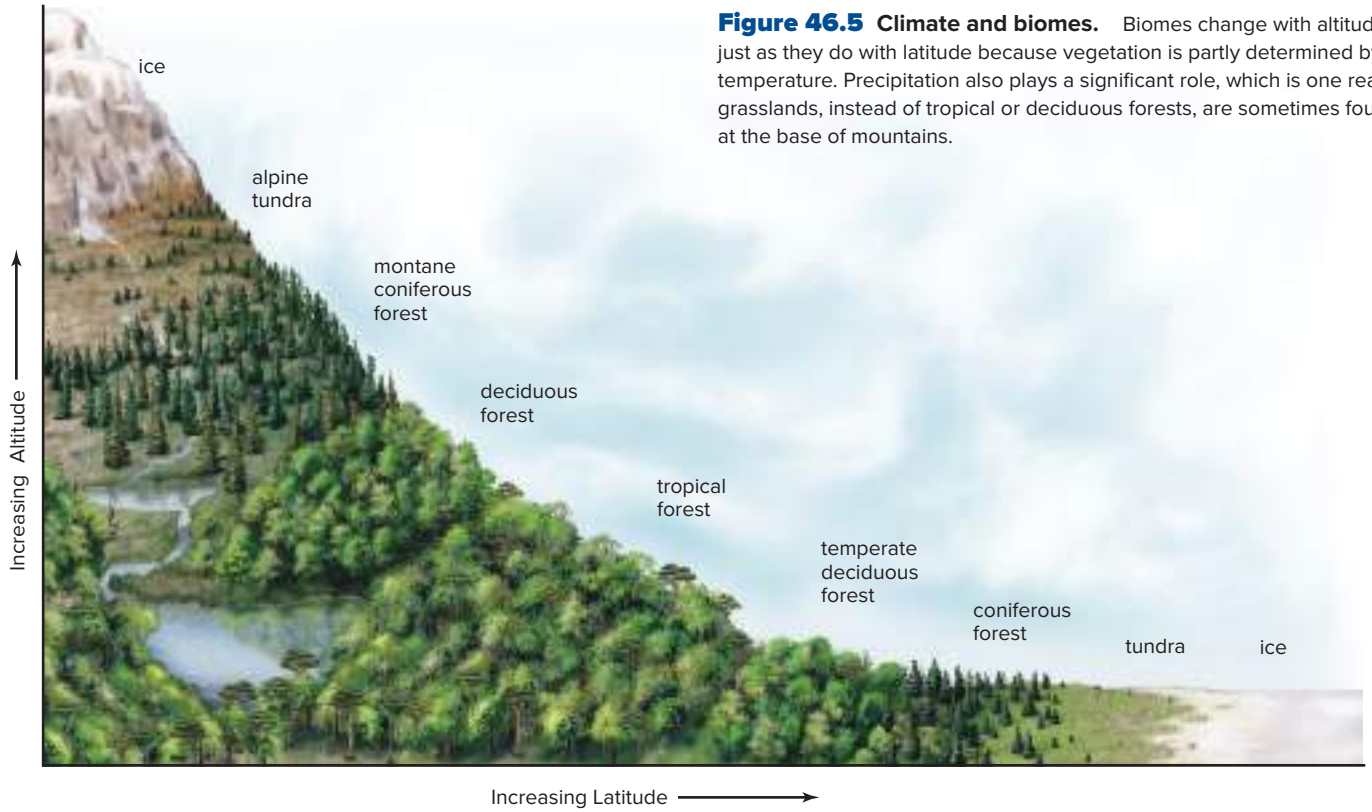


Figure 46.5 Climate and biomes. Biomes change with altitude just as they do with latitude because vegetation is partly determined by temperature. Precipitation also plays a significant role, which is one reason grasslands, instead of tropical or deciduous forests, are sometimes found at the base of mountains.

the summer. Only the topmost layer of soil thaws; **permafrost** is the layer that remains permanently frozen, resulting in minimal drainage. The available soil in the tundra is nutrient-poor. Arctic tundra, which encircles the Earth just south of ice-covered polar seas in the Northern Hemisphere, covers about 20% of the Earth's land surface (Fig. 46.6). (As mentioned, a similar ecosystem, the alpine tundra, occurs above the timberline on mountain ranges.)

Trees are not found in the tundra because the growing season is too short. The roots cannot penetrate the permafrost and they cannot become anchored in the shallow boggy soil during the summer. In the summer, the ground is covered with short grasses and sedges, as well as numerous patches of lichens and mosses. Dwarf woody shrubs, such as dwarf birch, flower and produce seeds during the short growing season.

A few animals live in the tundra year-round. For example, the mouselike lemming stays beneath the snow; the ptarmigan, a

a. Distribution of tundras

b. Tundra

c. Bull caribou

Figure 46.6 The tundra. **a.** Distribution of tundras. In this biome, which is nearest the polar regions, the vegetation consists principally of lichens, mosses, grasses, and low-growing shrubs. **b.** The melting snow forms pools of water on the permanently frozen ground. **c.** Caribou will feed on lichens, grasses, and shrubs during the summer, then migrate south during the winter. (b): U.S. Fish & Wildlife Service; (c): Jeff McGraw/Shutterstock

THEME Nature of Science

Wildlife Conservation and DNA

After DNA analysis, scientists were amazed to find that some 60% of loggerhead turtles drowning in the nets and hooks of fisheries in the Mediterranean Sea were from beaches in the southeastern United States. Because the unlucky creatures were a good representative sample of the turtles in the area, that meant more than half of the young turtles living in the Mediterranean Sea had hatched from nests on beaches in Florida, Georgia, and South Carolina (Fig. 46A*a*). Some 20,000–50,000 loggerheads die each year due to the Mediterranean fisheries, which may partly explain the decline in loggerheads nesting on southeastern U.S. beaches for the last 25 years.

Jaguars (*Panthera onca*) (Fig. 46A*b*) are the third-largest cat in the world, after lions and tigers, and the largest cat in the Western Hemisphere. Their natural range is greater than that of any other mammal in North and Central America. In order to conserve a top-level predator that has an extensive range, it requires support from all of the various countries that are home to jaguars.

Detailed genetic analysis of jaguar DNA has indicated that, whether they live in Mexico, Argentina, or anywhere in between, they are all the same species. They are the only wide-ranging carnivore in the world that shows genetic continuity across their entire range. This genetic information led to the formation of the Jaguar Corridor Initiative (JCI), which was featured at the start of the chapter. The goal of the initiative is to create a genetic corridor that links jaguar populations in all of the 18 countries in Latin America, from Mexico to Argentina, hopefully ensuring the survival of this species.

In a classic example of how DNA analysis might be used to protect endangered species from future ruin, scientists from the United States and New Zealand carried out discreet experiments in a Japanese hotel room on whale sushi bought in local markets. Sushi, a staple of the Japanese diet, is a rice and fish mixture wrapped in seaweed. Armed with a miniature DNA sampling machine, the scientists found that, of the 16 pieces of whale sushi they examined, many were from whales that are endangered or protected under an international moratorium on whaling. "Their findings demonstrated the true power of DNA studies," says David Woodruff, a conservation biologist at the University of California, San Diego.

One sample was from an endangered humpback, four were from fin whales, one was from a northern minke, and another from a beaked whale. Stephen Palumbi of the University of Hawaii says the technique could be used for monitoring and verifying catches. Until then, he says, "no species of whale can be considered safe."

a.

b.

Figure 46A DNA studies. **a.** Many loggerhead turtles found in the Mediterranean Sea are from the southeastern United States. **b.** No matter where they are found in their range, all jaguars belong to the same species. (a): Thomas Barwick/DigitalVision/Getty Images; (b): Linda More/iStock/Getty Images

a.

b.

Figure 46B Potential coronavirus sources. **a.** Bats are known to be carriers of a wide variety of viruses. **b.** Pangolin are believed to be a possible intermediate animal that transmitted SARS-CoV-2 from bats to humans. (a): Charoenchai Tothaisong/123RF; (b): Imagevixen/RooM/Getty Images

Genetic testing showed that the coronavirus shares 96% of its genes with those found in bats (Fig. 46B*a*). Its genetic profile shows that the virus can invade and hijack cells in the same manner as the SARS virus. During the early stages of the viral outbreak, bats appeared to be the most likely source of the virus.

Determining how the virus jumped from bats to humans is important. Early speculation pointed the blame at snakes. This idea arose

(*Continued*)

ISTUDY

when the viral genetic profile was compared to that of animal species found in the seafood market. This hypothesis, however, was quickly dismissed after further genetic testing.

The pangolin (Fig. 46B*b*), a small, scaled mammal that was sold in the seafood market, is also suspected to be a bat-to-human link. While this seems plausible, further genetic analysis is necessary to determine the species that enabled transmission of the virus to humans. Until a confirmed source of transmission is determined, it will be difficult to control and prevent further outbreaks of the novel coronavirus.

Questions to Consider

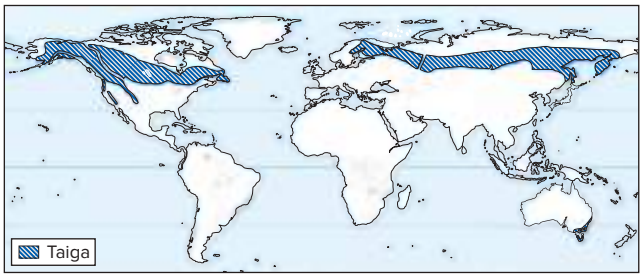
1. If new DNA analysis of jaguars indicated that there are two species of jaguars in Central America, how could this impact their conservation status?
2. If genetic analysis proves that COVID-19 jumped from bats to pangolins to humans, how could future viral spreads be prevented?

grouse, burrows in the snow during storms; and the musk ox conserves heat because of its thick coat and short, squat body. Other animals that live in the tundra include snowy owls, lynxes, voles, Arctic foxes, and snowshoe hares. In the summer, the tundra is alive with numerous insects and birds, particularly shorebirds and waterfowl that migrate inland. Caribou in North America and reindeer in Asia and Europe also migrate to and from the tundra, as do the wolves that prey upon them. Polar bears are common near the coastal regions. All species have adaptations for living in extreme cold with a short growing season.

Coniferous Forests

Coniferous forests have long, cold, snowy winters with warm and humid summers. They tend to be dominated by cone-bearing trees and can be found in three primary locations: in the **taiga,** which extends around the world in the northern part of North America and Eurasia; near mountaintops (where it is called a montane coniferous forest); and along the Pacific coast of North America, as far south as Northern California.

The taiga typifies the coniferous forest with its cone-bearing trees, such as spruce, fir, and pine. These trees are well-adapted to the cold because both the leaves and bark have thick coverings. Also, the needlelike leaves can withstand the weight of heavy snow. There is a limited understory of plants, but the forest floor is covered by low-lying mosses and lichens beneath a layer of needles. Birds harvest the seeds of the conifers, while bears, deer, moose, beavers, and muskrats live around the lakes and streams. Wolves prey on these larger mammals. A montane coniferous forest also harbors the wolverine and the mountain lion. The taiga, or boreal forest, exists south of the tundra and covers approximately 11% of the Earth's landmasses (Fig. 46.7). There are no comparable biomes in the Southern Hemisphere because large landmasses do not exist at that latitude.



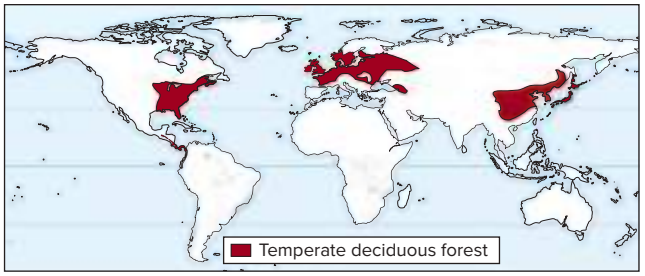
a. Distribution of taiga biomes

b. Spruce trees in the taiga biome

c. Bull moose

Figure 46.7 The taiga. **a.** Distribution of taiga biomes. The taiga, which means "swampland," spans northern Europe, Asia, and North America. This biome has a dominant presence of (**b**) spruce trees and (**c**) moose, which frequent the ponds. (b): Perry Mastrovito/Creatas Images/Jupiterimages; (c): Jupiterimages/360/liquidlibrary/Getty Images

ISTUDY



a. Distribution of temperate deciduous forests

b. Temperate deciduous forest

c. Bobcat

Figure 46.8 Temperate deciduous forests. **a.** Distribution of temperate deciduous forests. **b.** Trees such as maples, beech, and oak, and mammals such as this (**c**) bobcat are found in this biome. (b): Bereczki Barna/Alamy Stock Photo; (c): Kevin Schafer/Stockbyte/Getty Images

The coniferous forest that runs along the west coast of Canada and the United States is sometimes called a **temperate rain forest** due to the plentiful rainfall and rich soil. As the prevailing winds move in off the Pacific Ocean, they drop their moisture when they meet the coastal mountain range. This environment has produced some of the tallest conifer trees ever in existence, including the coastal redwoods. This forest is also called an old-growth forest because some trees are as old as 800 years. It truly is an evergreen forest because of the abundance of green plants year-round.

Squirrels, lynxes, and numerous species of amphibians, reptiles, and birds inhabit the temperate rain forest. The northern spotted owl, *Strix occidentalis caurina,* which has been the focus of conservation efforts, is an endangered species found in this ecosystem.

Temperate Deciduous Forests

Temperate deciduous forests have a moderate climate with relatively high rainfall (75–150 cm per year). The seasons are well defined, and the growing season ranges between 140 and 300 days. The trees, such as oak, beech, sycamore, and maple, have broad leaves. These are termed *deciduous trees* because they lose their leaves in the fall and regrow them in the spring. Deciduous forests can be found south of the taiga in eastern North America, eastern Asia, and much of Europe (Fig. 46.8).

The tallest trees form a canopy, an upper layer of leaves that are the first to receive sunlight. Enough sunlight penetrates to provide energy for another layer of trees, called understory trees. Beneath these trees are shrubs and herbaceous plants that may flower in the spring before the trees have put forth their leaves. Still another layer of plant growth—mosses, lichens, and ferns—resides beneath the shrub layer. This stratification provides a variety of habitats for insects and birds. Ground animals are also plentiful. Squirrels, rabbits, woodchucks, and chipmunks are small herbivores. These and ground birds such as turkeys, pheasants, and grouse are preyed on by various carnivores. In contrast to the taiga, amphibians and reptiles occur in this biome because the winters are not as cold. Frogs and turtles prefer an aquatic existence, as do beavers and muskrats. Autumn fruits, nuts, and berries provide a supply of food for the winter, and the leaves, after turning brilliant colors and falling to the ground, contribute to the rich layer of humus. The minerals within the rich soil are washed deep into the ground by spring rains, but the tree roots capture most of them and take it back up into the forest system again.

Tropical Rain Forests

Tropical rain forests are characterized by a climate that is always warm (between 20° and 25°C), rainfall that is plentiful (with a minimum of 190 cm per year), and approximately 12 hours of sunlight every day across the entire year. These are the richest land biomes on Earth. It is estimated that approximately 50% of the Earth's biodiversity can be found in its tropical rain forests. South America, Africa, and the Indo-Malayan region near the equator all contain tropical rain forests.

A tropical rain forest has a complex structure, with many levels of life, including the forest floor, understory, and canopy. The vegetation of the forest floor tends to be very sparse because most of the sunlight is absorbed by the canopy layer. The understory consists of smaller plants that are specialized for life in the shade. The canopy, topped by the crowns of tall trees, is the most productive level of the tropical rain forest (Fig. 46.9).

Some of the broadleaf evergreen trees can grow from 15 to 50 m or more in height. These tall trees often have trunks buttressed at ground level to prevent their toppling over. Lianas, or woody vines, encircle the tree as it grows and help strengthen the trunk. The diversity of species is enormous—a 10-km^2 area of tropical rain forest may contain 750 species of trees and 1,500 species of flowering plants.

ISTUDY

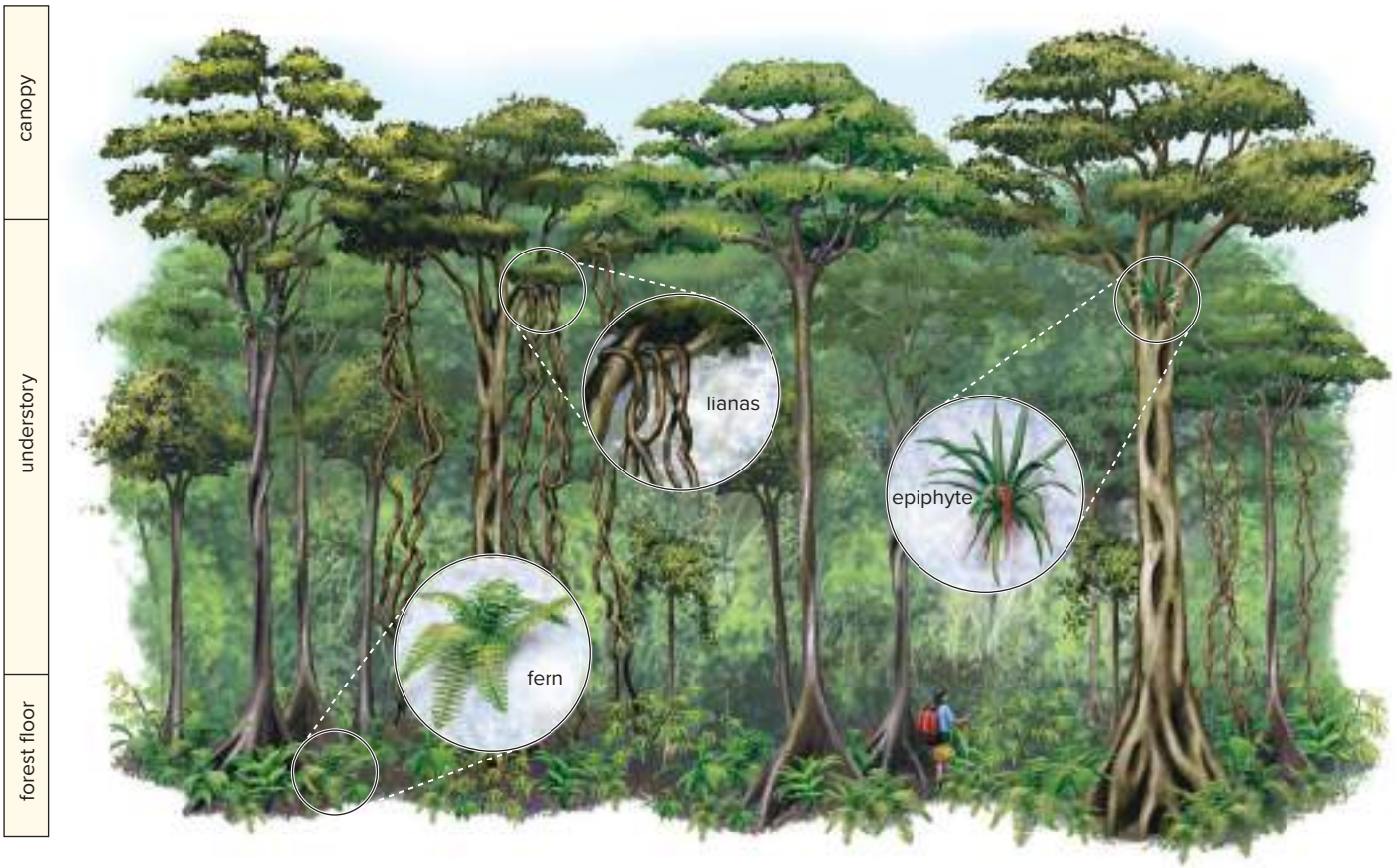


Figure 46.9 Levels of life in a tropical rain forest. The primary levels in a tropical rain forest are the canopy, the understory, and the forest floor. But the canopy (solid layer of leaves) contains levels as well, and some organisms spend their entire lives in one particular level. Long lianas (hanging vines) climb into the canopy, where they produce leaves. Epiphytes are air plants that grow on the trees but do not parasitize them.

Although some animals live on the forest floor (e.g., pacas, agoutis, peccaries, and armadillos), most live in the trees (Fig. 46.10). Insect life is so abundant that the majority of species have not been identified yet. Termites play a vital role in the decomposition of woody plant material, and ants are found everywhere, particularly in the trees. The various birds, such as hummingbirds, parakeets, parrots, and toucans, are often beautifully colored. Amphibians and reptiles are well represented by many types of frogs, snakes, and lizards. Lemurs, orangutan, and monkeys are well-known primates that feed on the fruits of the trees. The largest carnivores are the big cats—the jaguars in Central and South America, and the leopards in Africa and Asia.

Many animals spend their entire lives in the canopy, as do some plants. **Epiphytes** are plants that grow on other plants but usually have roots of their own, which absorb moisture and minerals from the air. Epiphytes such as bromeliads catch rain and debris by forming vases of overlapping leaves. Examples of the most common epiphytes include pineapples and orchids.

We usually think of tropical forests as being nonseasonal rain forests, but tropical forests that have wet and dry seasons are found in India, Southeast Asia, West Africa, South and Central America, the West Indies, and northern Australia. Here, there are deciduous trees, with many layers of growth beneath them.

Contrary to popular belief, the soil of a tropical rain forest biome is nutrient-poor. Due to the high amount of herbivory, only a small amount of leaf litter makes it to the forest floor. This remaining leaf litter is quickly decomposed and its nutrients taken up by the producers. Productivity is high because of high temperatures, a year-long growing season, and nearly 12 hours of sunlight year-round. It is possible to conduct slash-and-burn agriculture on a small scale, but it becomes destructive and unsustainable on a large scale. The ash produced from cutting down and burning trees provides nutrients for several years of growing crops. Thereafter, the forest must be allowed to regrow, and a new section must be cut and burned. In addition, in humid tropical forests, iron and aluminum oxides occur at the surface, causing a reddish residue known as laterite. When the trees are cleared, laterite bakes in the hot sunlight to a bricklike consistency, which will not support crops.

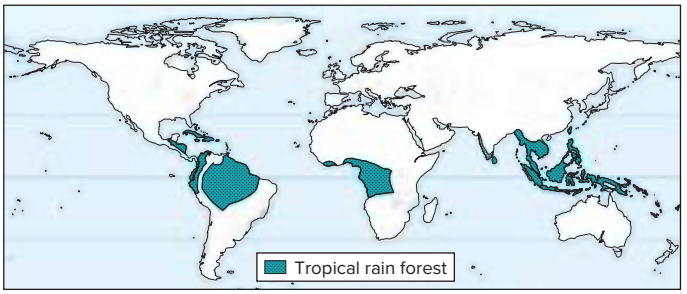
It is estimated that 2.4 acres of rain forest are destroyed per second. This rate equates to nearly 78 million acres annually. Unless conservation strategies are employed soon, rain forests will be destroyed beyond recovery, taking with them unique and important life-forms. Ecologists estimate that approximately 137 species are driven to extinction every day in rain forests, primarily due to deforestation. For more on the impact of deforestation, see the Biological Systems feature, "Destruction of the Amazon Rain Forest."

Shrublands

Shrublands are regions that tend to have dry summers and receive most of their rainfall in the winter. In general, shrubs (4.5–6 m) are shorter than trees and possess woody, persistent stems but no large, central trunk. Shrubs have small but thick evergreen leaves, which

ISTUDY

b. An epiphyte growing on a broadleaf evergreen



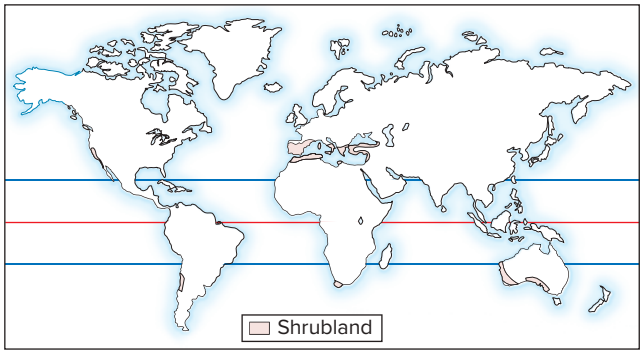
a. Distribution of tropical rain forests

c. Jaguar

Figure 46.10 Tropical rain forests. **a.** Distribution of tropical rain forests. Broadleaf evergreen trees, often with (**b**) epiphyte attached, and mammals such as this (**c**) Jaguar, are found in tropical rain forests. (b): ©Ricochet Creative Productions LLC; (c): AlexanderDavid/iStockphoto/Getty Images

are often coated with a waxy material that prevents loss of moisture. Their thick underground roots can survive dry summers and frequent fires and take moisture from the deep soil. Shrubs are adapted to withstand arid conditions and can quickly sprout new growth after a fire. You may recall from Figure 45.19 that a shrub stage is part of the process of both primary and secondary succession.

Shrublands occur along the cape of South Africa, the western coast of North America, and the southwestern and southern shores of Australia, as well as around the Mediterranean Sea and in central Chile. **Chaparral** (Fig. 46.11) is a type of shrubland that lacks an understory and ground litter and is highly flammable. The seeds of many species require the heat and scarring action of fire to induce germination. Other shrubs sprout from the roots after a fire. Typical animals of the chaparral include mule deer, rodents, lizards, and greater roadrunners. Chaparral regions occur in California.



a. Distribution of shrublands

b. Shrubland

c. Roadrunner

Figure 46.11 Shrubland. **a.** Distribution of shrublands. **b.** Shrublands, such as this chaparral in California, are subject to raging fires, but the shrubs are adapted to quickly regrow. **c.** Greater roadrunners find a home in the chaparral. (b): Steven P. Lynch; (c): Gary Kramer/U.S. Fish & Wildlife Service

ISTUDY

THEME Biological Systems

Destruction of the Amazon Rain Forest

The Amazon rain forest is one of the most biologically diverse ecosystems on Earth. It is home to approximately 10% of the species found on our planet, with over 40,000 species of plants, 3,000 species of freshwater fish, and 370 known species of reptiles living there. This ecosystem serves as one of Earth's last refuges for harpy eagles, jaguars, and pink dolphins (Fig. 46C*a*). Approximately 350 indigenous and ethnic groups, comprised of more than 30 million people, call the Amazon home. The producers of the rain forest help stabilize the Earth's climate by sequestering between 90 and 140 billion metric tons of carbon.

Sadly, numerous factors threaten Amazon's rain forest with destruction. Poorly planned roads provide access to illegal and unregulated logging activities. The development of hydropower dams in regions of high conservation value have led to a decline in the water quality in some of the Amazon's freshwater rivers. Numerous species have been pushed to the edge of extinction due to the unsustainable and illegal harvesting of the Amazon's natural resources (Fig. 46C*b*).

The largest contributor to the destruction of the Amazon rain forest is deforestation. In 2021, over 4.8 million acres were burned to clear more land for agricultural development, most of which is used for grazing cattle. To make way for the planting of soy, the slash-and-burn method is practiced. This is when the natural vegetation is cut down, allowed to dry, and then burned, releasing billions of tons of greenhouse gases into the atmosphere. Farming depletes most of the soil nutrients within a few years, and the land is no longer capable of supporting crops without applying larger amounts of fertilizers. All of these practices combine to destroy the ecological balance that has been part of the Amazon rain forest for thousands of years.

These actions cause the Earth to lose a vast reservoir of carbon dioxide and other greenhouse gases. We are also losing untold amounts of biodiversity and potentially new sources of medicines. Thus, the Amazon has a global importance and it will take a global effort to halt its destruction and restore the stability of its ecosystem.

Questions to Consider

1. Should countries outside of South America help pay to protect the Amazon?
2. What type of global consequences will occur if the Amazon continues to be destroyed?
3. How much of the Amazon needs to be protected to ensure that the ecosystem will remain stable?

a.

b.

Figure 46C Amazon rain forest. **a.** The Amazon rain forest covers 1.4 billion acres. **b.** Jaguars, harpy eagles, and pink dolphins are just a few of the many species threatened with extinction in the Amazon. (b-1): Hans Wagemaker/Shutterstock; (b-2): Barry B. Doyle/Moment/Getty Images; (b-3): Aniroot/iStock/Getty Images

A northern shrub area that lies to the west of the Rocky Mountains is sometimes classified as a cold desert. This region is dominated by sagebrush, which provides the resources to support various species of birds.

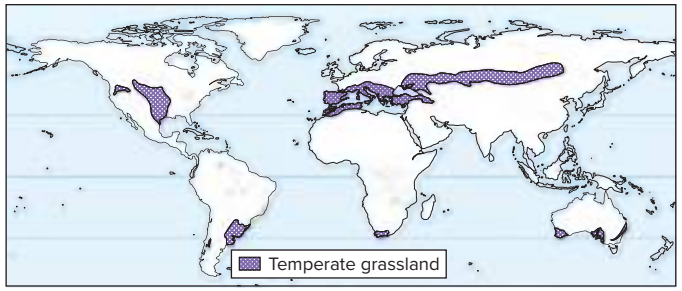
Grasslands

Grasslands occur where annual rainfall is greater than 25 cm but is generally insufficient to support trees, despite the fertile soil. For example, in temperate areas, where rainfall is between 25 and 75 cm, it is too dry for forests and too wet for deserts to form.

Grasses are well-adapted to a changing environment and can tolerate a high degree of grazing, flooding, drought, and sometimes fire. Where rainfall is high, tall grasses that reach more than 2 m in height (e.g., pampas grass) can flourish. In drier areas, shorter grasses (between 5 and 10 cm) are dominant. Low-growing bunch grasses (e.g., grama grass) grow in the United States near deserts.

The growth of grasses is seasonal. As a result, grassland animals (such as bison) migrate and others (such as ground squirrels) hibernate when there is little grass for them to eat.

ISTUDY

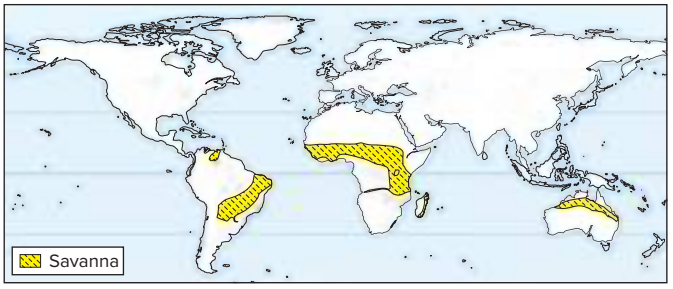


a. Distribution of temperate grasslands

b. Tall-grass prairie

c. American bison

Figure 46.12 Temperate grassland. **a.** Distribution of temperate grasslands. **b.** Tall-grass prairies are seas of grasses dotted by pines and junipers. **c.** Bison, once abundant, are now being reintroduced into certain areas. (b): KenCanning/E+/iStock/Getty Images; (c): Fuse/Getty Images



a. Distribution of savannas

b. Acacia trees of the savanna

c. Mammals of the savanna

Figure 46.13 The savanna. **a.** Distribution of savannas. **b.** The African savanna varies from grassland to widely spaced shrubs and Acacia trees. **c.** This biome supports a large assemblage of animal species. (b): TomazKunst/iStock/Getty Images; (c): Jim Tampin/Alamy Stock Photo

Temperate Grasslands

The **temperate grasslands** are characterized as having winters that are bitterly cold and summers that are hot and dry. When traveling across the United States from east to west, the line between the temperate deciduous forest and a tall-grass prairie is roughly along the border between Illinois and Indiana. The tall-grass prairie receives more rainfall than does the short-grass prairie, which occurs near deserts. Temperate grasslands include the Russian steppes, the South American pampas, and the North American prairies (Fig. 46.12).

Large herds of bison—estimated at hundreds of thousands—once roamed the prairies, as did herds of pronghorn antelope. Now, small mammals, such as mice, prairie dogs, and rabbits, typically live below ground, but usually feed above ground. Hawks, snakes, badgers, coyotes, and foxes feed on these mammals. Virtually all of these grasslands, however, have been converted to agricultural lands because of their fertile soils.

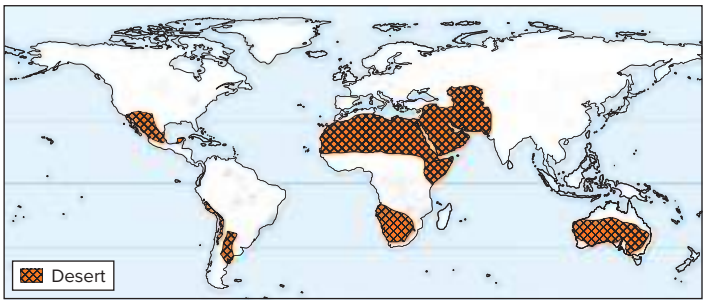
Savannas

Savannas occur in regions where a relatively cool dry season is followed by a hot rainy season (Fig. 46.13). The savanna is characterized by large expanses of grasses with relatively few trees. The plants of the savanna have extensive and deep root systems, which enable them to survive drought and fire. One tree that can survive the severe dry season is the thorny flat-topped *Acacia,* which sheds its leaves during a

ISTUDY

b. Plants of the desert

Figure 46.14 The desert. **a.** Distribution of deserts. **b.** Plants and animals that live in a desert are adapted to arid conditions. The plants are either succulents, which retain moisture, or shrubs with woody stems and small leaves, which lose little moisture. **c.** Among the animal life, the kit fox is an example of a desert carnivore. (b): Kevin Burke/Corbis/Getty Images; (c): Kevin Schafer/The Image Bank/Getty Images



a. Distribution of deserts

c. Kit fox

drought. The largest savannas are in central and southern Africa. Other savannas exist in Australia, Southeast Asia, and South America.

The African savanna supports the greatest variety and number of large herbivores of all the biomes. Elephants and giraffes are browsers that feed on tree vegetation. Antelopes, zebras, wildebeests, water buffalo, and rhinoceroses are grazers that feed on the grasses. Any plant litter that is not consumed by grazers is attacked by a variety of small organisms, among them termites. Termites build towering nests in which they tend fungal gardens, their source of food. The herbivores support a large population of carnivores. Lions, hyenas, cheetahs, and leopards all prey upon the abundant herbivore populations found in the savanna.

Deserts

Deserts are characterized by days that are hot because a lack of cloud cover allows the sun's rays to penetrate easily, but nights are cold because heat escapes easily into the atmosphere. The winds that descend in these regions lack moisture, resulting in an annual rainfall of less than 25 cm. Deserts are usually found at latitudes of about 30°, in both the Northern and Southern Hemispheres, and cover nearly 30% of the Earth's land surface.

The Sahara, which stretches all the way from the Atlantic coast of Africa to the Arabian peninsula, along with a few other deserts, has little or no vegetation. However, most deserts contain a variety of plants that are highly adapted to survive long droughts, extreme heat, and extreme cold (Fig. 46.14*b*). Adaptations to these conditions include thick epidermal layers, water-storing succulent stems and leaves, and the ability to set seeds quickly in the spring. The best-known desert perennials in North America are the spiny cacti, which have stems that store water and have the ability to run photosynthesis. Also common are nonsucculent shrubs, such as the many-branched sagebrush with silvery gray leaves and the spiny-branched ocotillo, which produces leaves during wet periods and sheds them during dry periods.

Some animals are adapted to the desert environment. To conserve water, many desert animals, such as reptiles and insects, are nocturnal or burrowing and have a protective outer body covering. The desert is home to numerous insects, which pass through the stages of development in synchrony with the periods of rain. Reptiles, especially lizards and snakes, are perhaps the most characteristic group of vertebrates found in deserts, but burrowing birds (e.g., burrowing owls) and rodents (e.g., the kangaroo rat) are also well-known. Larger mammals, such as the kit fox (Fig. 46.14*c*), prey on the rodents, as do hawks.

Check Your Progress 46.2

1. Identify the progression of biomes, starting at the equator and moving toward the North Pole.
2. Contrast the vegetation of the tropical rain forest with that of a temperate deciduous forest.

ISTUDY

46.3 Aquatic Ecosystems

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the characteristics of freshwater and saltwater ecosystems.
2. Interpret the manner in which ocean currents affect the climate and weather over the continents.

Aquatic ecosystems are classified into two broad categories: freshwater (inland) systems or saltwater systems. Figure 46.15 shows how these ecosystems are interconnected.

In the water cycle, the sun's rays cause seawater to evaporate, and the salts are left behind. The evaporated fresh water then rises into the atmosphere, cools, and falls as rain (see Fig. 45.21). When rain falls, some of the water sinks, or percolates, into the ground and saturates it. The layer of water in the ground is called the groundwater table, or simply the water table.

Because land has a higher elevation than the oceans, gravity will eventually return all fresh water to the sea. Along this route to the sea, it passes through streams and rivers and can be contained in lakes and ponds. Sometimes groundwater is also located in underground rivers called aquifers. Whenever the Earth contains basins or channels, water will rise to the level of the water table.

Mountain Streams and Rivers

When frozen precipitation at high elevations melts, it begins its long journey toward the sea as freshwater in a mountain stream or river. These bodies of water tend to be found in high elevations, are cold, and tend to be low in nutrients and biodiversity. Mayflies and trout are species that can be found in clean mountain streams. As the water makes its way to lower elevations, it accumulates nutrients and becomes warmer.

Lakes

Lakes are bodies of fresh water that can be classified by their nutrient status. Oligotrophic (nutrient-poor) lakes are characterized by a small amount of organic matter and low productivity (Fig. 46.16*a*). Eutrophic (nutrient-rich) lakes are characterized by plentiful organic matter and high productivity (Fig. 46.16*b*). These latter lakes are usually situated in naturally nutrient-rich regions or are enriched by agricultural or urban and suburban runoff. **Eutrophication** is the process in which a body of water receives a large input of nutrients in a relatively short period of time.

Lake Overturn

In the temperate zone, deep lakes are stratified in the summer and winter and have distinct vertical zones. In summer, lakes in the temperate zone have three layers of water that differ in temperature

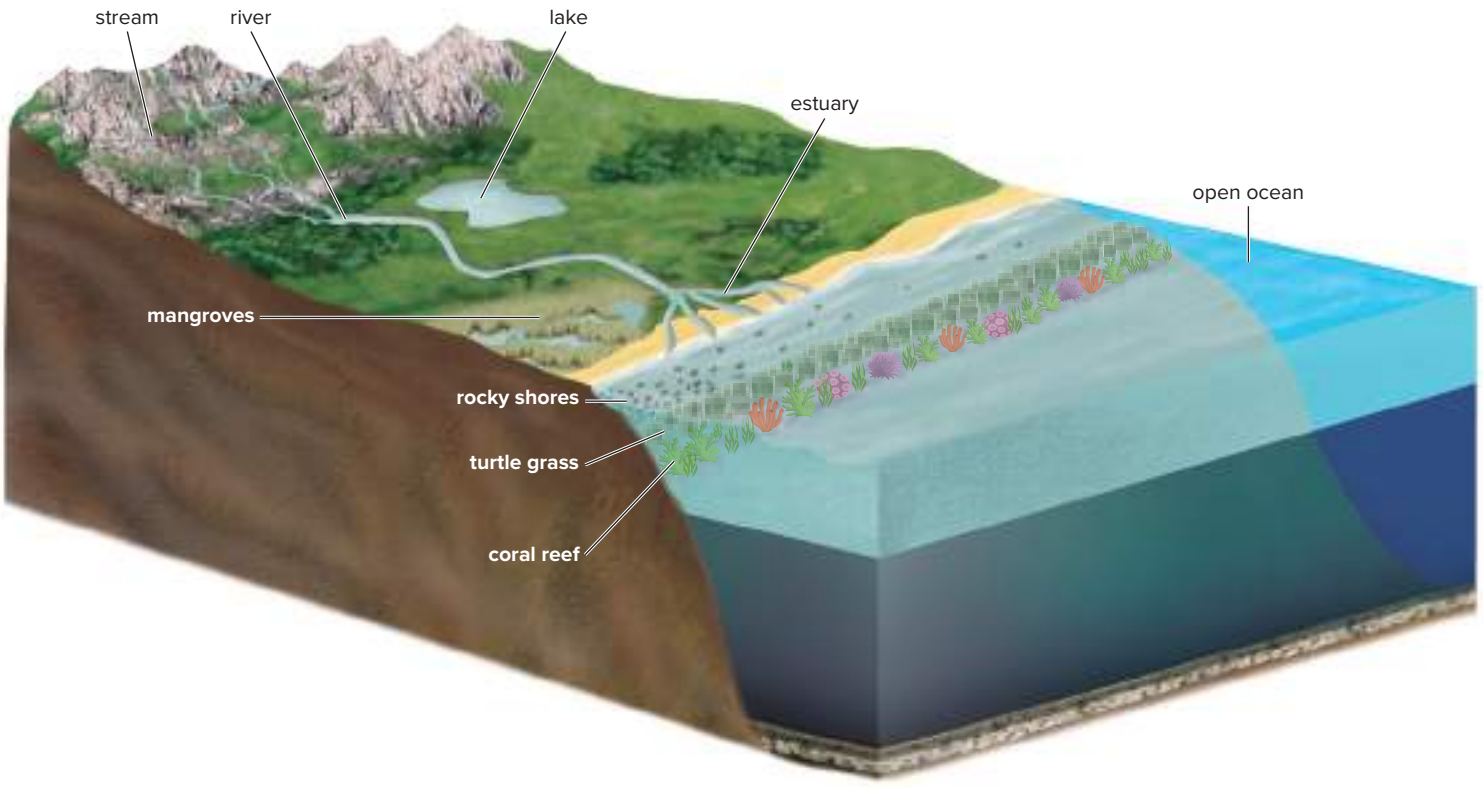


Figure 46.15 Freshwater and saltwater ecosystems. Precipitation enters the aquatic ecosystem in the mountains in the form of snow and rain. As it makes its way downstream toward the ocean, the water collects nutrients and pollutants. When it reaches the ocean, it deposits this material into the estuary, which may damage coral reefs.

ISTUDY

a. Oligotrophic lake

b. Eutrophic lake

Figure 46.16 Types of lakes. Lakes can be classified according to whether they are (**a**) oligotrophic (nutrient-poor) or (**b**) eutrophic (nutrient-rich). Eutrophic lakes tend to have large populations of algae and rooted plants, resulting in a large population of decomposers, which use up much of the oxygen and leave little oxygen for fishes. (a): Arutthaphon Poolsawasd/Moment/Getty Images; (b): Pat Watson/McGraw Hill

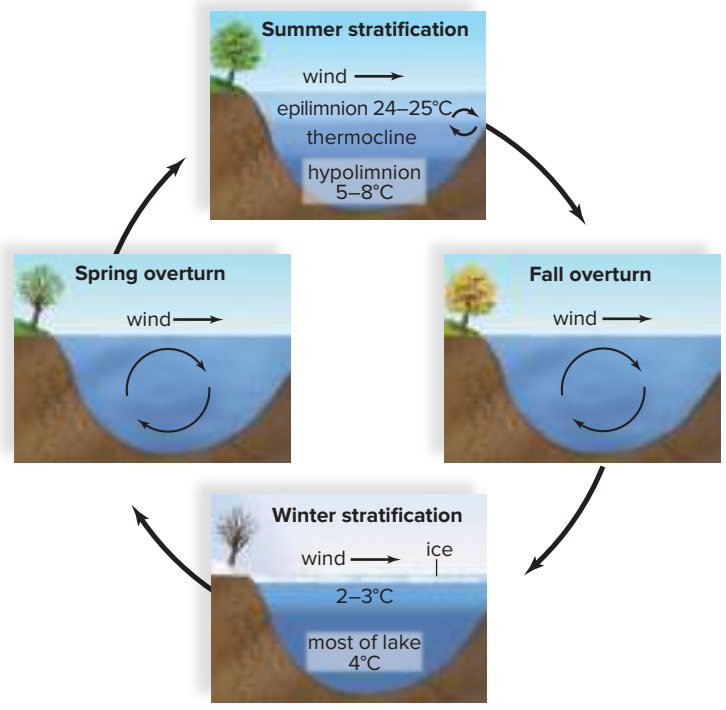


Figure 46.17 Lake stratification in a temperate region. Temperature profiles of a large oligotrophic lake in a temperate region vary with the season. During the spring and fall overturns, the deep waters receive oxygen from surface waters, and surface waters receive inorganic nutrients from deep waters.

(Fig. 46.17). The surface layer, the epilimnion, is warmed from solar radiation; the middle layer, the thermocline, experiences an abrupt drop in temperature; and the lowest layer, the hypolimnion, is cold. These differences in temperature prevent mixing. The warmer, less dense water of the epilimnion "floats" on top of the colder, more dense water of the thermocline, which floats on top of the hypolimnion.

Phytoplankton found in the sunlit epilimnion use up various nutrients in order to photosynthesize. Photosynthesis gives this layer a ready supply of oxygen. Detritus accumulates at the bottom of the lake where oxygen is used up as decomposition occurs. Decomposition releases nutrients, however. As the season progresses, the epilimnion becomes nutrient-poor, while the hypolimnion begins to be depleted of oxygen.

Overturns occur in the fall, as the epilimnion cools, and in the spring, as it warms. In the fall, the upper epilimnion waters become cooler than the hypolimnion waters. This causes the surface water to sink and the deep water to rise. The **fall overturn** is the mixing of the layers until the temperature is uniform throughout the lake. At this point, wind aids in the circulation of water so that mixing occurs. Eventually, oxygen and nutrients become evenly distributed.

As winter approaches, the water cools and ice forms. Ice has an insulating effect, preventing further cooling of the water below. This permits aquatic organisms to survive the winter in the water beneath the ice.

In the spring, as the ice melts, the cooler water on top sinks below the warmer water on the bottom. The **spring overturn** is the mixing of the layers until the temperature is uniform throughout the lake. At this point, wind again aids in the circulation of the water. As the surface waters absorb solar radiation, thermal stratification occurs once more.

The vertical stratification and seasonal change of temperatures in a lake influence the seasonal distribution of fish and other aquatic life. For example, coldwater fish move to the deeper water in summer and inhabit the upper water in winter. In the fall and spring just after mixing occurs, phytoplankton growth at the surface is most abundant.

Life Zones

In both fresh and salt water, free-drifting microscopic organisms called *plankton* (Gk. *planktos,* wandering) are important components of the ecosystem. **Phytoplankton** (Gk. *phyton,* plant; *planktos,* wandering) are photosynthesizing algae that act as the producer base of the lake ecosystem. They become more noticeable when a green scum

ISTUDY

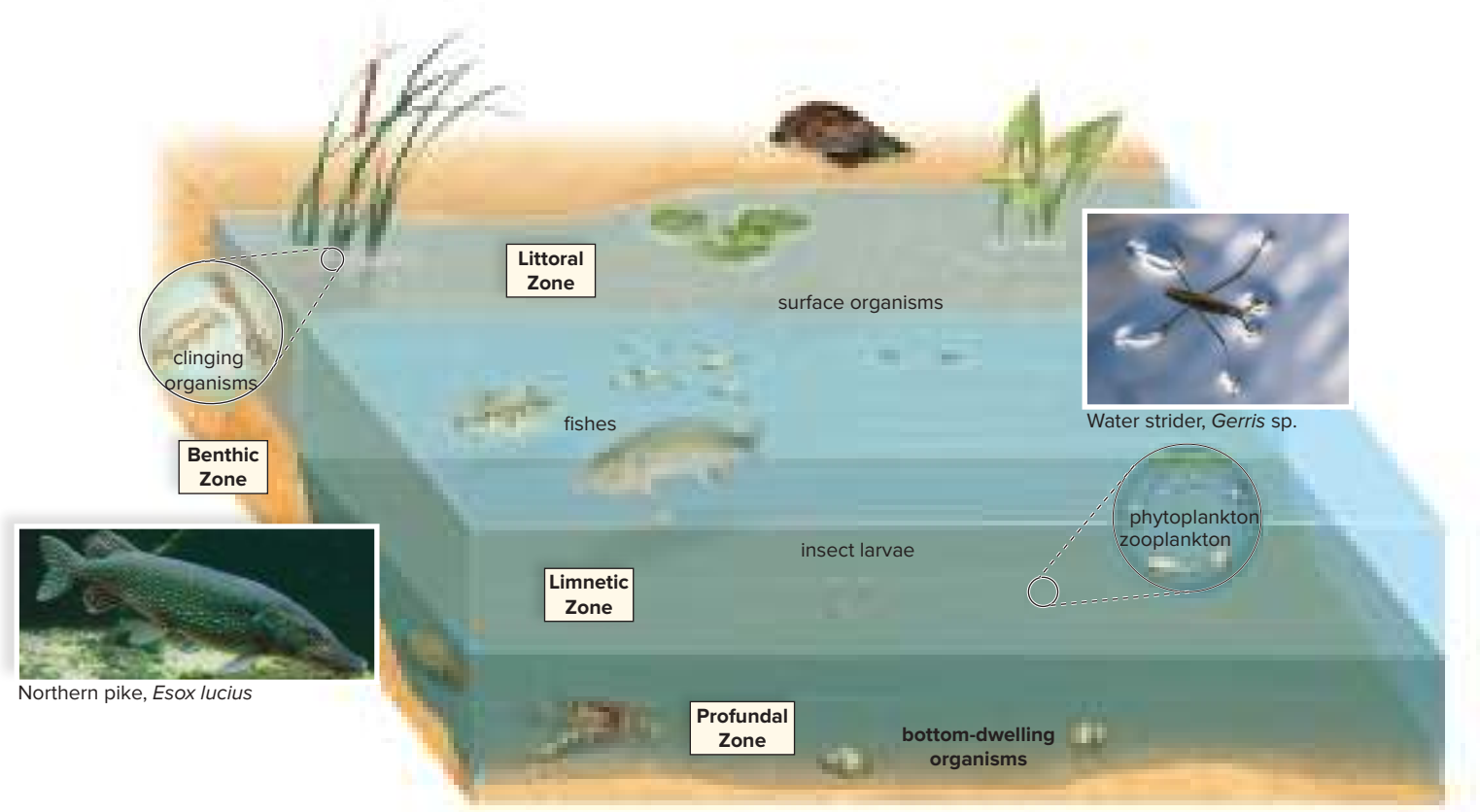


Figure 46.18 Zones of a lake. Rooted plants and clinging organisms live in the littoral zone. Phytoplankton, zooplankton, and fishes are in the sunlit limnetic zone. Water striders (photo) stand on the surface film of water with water-repellent feet. Crayfishes and mollusks (art) are in the profundal zone, as well as the littoral zone. Pike (photo and art) are top carnivores prized by anglers. (photos): (pike): abadonian/360/iStock/Getty Images; (water strider): Jan Miko/Shutterstock

or red tide appears on the water. **Zooplankton** (Gk. *zoon,* animal; *planktos,* wandering) are tiny animals that feed on the phytoplankton.

Lakes and ponds can be divided into several life zones (Fig. 46.18). The *littoral zone* is closest to the shore, the *limnetic zone* forms the sunlit body of the lake, and the *profundal zone* is below the level of light penetration. The *benthic zone* includes the sediment at the soil-water interface.

Aquatic plants are rooted in the shallow littoral zone of a lake, providing habitat for numerous protozoans, invertebrates, fishes, amphibians, and some reptiles. Largemouth bass are a type of ambush predator that waits among vegetation around the margins of lakes and surges out to capture passing prey. Wading birds are commonly seen feeding in the littoral zone. Some organisms, such as the water strider, live at the water-air interface and can literally walk on water. In the limnetic zone, small fishes, such as minnows and killifish, feed on plankton and serve as food for larger fish, such as bass. In the profundal zone, zooplankton, invertebrates, and fishes such as catfish and whitefish feed on debris that falls from higher zones.

The bottom of the lake, the benthic zone, is composed of mostly silt, sand, inorganic sediment, and dead organic material (detritus). Bottom-dwelling organisms are known as benthic species and include worms, snails, clams, crayfishes, and some insect larvae. Decomposers, such as bacteria, are also found in the benthic zone and break down wastes and dead organisms into nutrients that are eventually used by the producers.

Wetlands

Wetlands are areas that hold some amount of water during part of the year. Generally, wetlands are classified by their vegetation. **Marshes** are wetlands that are frequently or continually inundated by water. They are characterized by the presence of rushes, reeds, and other grasses, which provide excellent habitat for waterfowl and small mammals. Marshes are one of the most productive ecosystems on Earth. **Swamps** are wetlands that are dominated by either woody plants or shrubs. Common swamp trees include cypress, red maple, and tupelo. The American alligator is a top predator in many swamp ecosystems. **Bogs** are wetlands characterized by acidic waters, peat deposits, and sphagnum moss. Bogs receive most of their water from precipitation and are nutrient-poor. Several species of plants thrive in bogs, including cranberries, orchids, and insectivorous plants such as Venus flytraps and pitcher plants. Moose and a number of other animals are inhabitants of bogs in the northern United States and Canada.

Humans have historically channeled and diverted rivers and filled in wetlands with the idea that "useless land" was being improved. However, these activities degrade ecosystems, can cause seasonal flooding, and eliminate food and habitats for many unique fishes, waterfowl, and other wildlife. Wetlands also purify waters by filtering them and by diluting and breaking down toxic wastes and excess nutrients. Wetlands directly absorb storm waters and overflow from lakes and rivers. In this way, they protect farms, cities, and towns from the devastating effects of floods. Federal and local laws have been enacted for the protection of wetlands as more people recognize their value.

ISTUDY

a. Mudflat

b. Mangrove swamp

Figure 46.19 Coastal ecosystems. **a.** Mudflats are frequented by migrant birds. **b.** Mangrove swamps skirt the coastlines of many tropical and subtropical lands. (a): MikeLane45/iStock/Getty Images; (b): shakzu/360/iStock/Getty Images

Estuaries

An **estuary** is a portion of the ocean where fresh water and salt water meet and mix. Mudflats, mangrove swamps, and rocky shores, featured in Figure 46.19, are examples of estuaries. Mangrove swamps can be found in subtropical and tropical zones, while marshes and mudflats occur in temperate zones. Coastal bays, fjords (inlets of water between high cliffs), and some lagoons (bodies of water separated from the sea by a narrow strip of land) are also classified as estuaries. Therefore, the term *estuary* has a very broad definition.

Organisms living in an estuary must be able to withstand constant mixing of waters and rapid changes in salinity. Organisms adapted to the estuarine environment benefit from its abundance of nutrients. An estuary acts as a nutrient trap because the sea prevents the rapid escape of nutrients brought by a river. Estuaries are biologically diverse and highly productive communities.

Phytoplankton and shore plants thrive in the nutrient-rich estuaries, providing an abundance of food and habitat for animals. It is estimated that nearly two-thirds of marine fishes and shellfish spawn and develop in the protective and rich environment of estuaries, making the estuarine environment the nursery of the sea. An abundance of larval, juvenile, and mature fish, as well as shellfish, attract a wide variety of predators.

Rocky Shores

Rocky shores (Fig.46.20) and sandy shores are constantly bombarded by the sea as the tides roll in and out. The **intertidal zone** is the region of shoreline that lies between the high- and low-tide marks (see Fig.46.15). In the upper portion of the intertidal zone, barnacles are glued so tightly to the stone by their own secretions that their calcareous outer plates remain in place even after the animal dies. In the midportion of the intertidal zone, brown algae, known as rockweed, may overlie the barnacles. Below the intertidal zone, macroscopic seaweeds, which are the main photosynthesizers, anchor themselves to the rocks by holdfasts.

Figure 46.20 Organisms of rocky shores. Organisms, such as these sea stars and mollusks, live in the tidal pools of rocky shores and coasts. NOAA National Estuarine Research Reserve Collection

Organisms cannot attach themselves to shifting, unstable sands on a sandy beach; therefore, nearly all the permanent residents dwell underground. Either they burrow during the day and surface to feed at night or they remain permanently within their burrows and tubes. Ghost crabs and sandhoppers (amphipods) burrow themselves above the high-tide mark and feed at night when the tide is out. Sandworms and sand (ghost) shrimp remain within their burrows in the intertidal zone and feed on detritus whenever possible. Still lower in the sand, clams, cockles, and sand dollars are found. A variety of shorebirds visit the beaches and feed on various invertebrates and fishes.

Seagrass Beds

Seagrass beds can be found in relatively shallow salt and brackish waters, extending out from coastal areas. There are 72 known species of seagrasses that form dense underwater meadows. Crabs, shrimp, juvenile fish, and hundreds of species of invertebrates make the blades of grass their home. The sheer number and diversity of smaller organisms in this area attracts larger predators, such as sharks, turtles, manatees, and herbivorous fish. It is estimated that one acre of seagrass can support over 40,000 fish and 50 million small invertebrates.

Seagrasses help remove nutrients, pollutants, and carbon dioxide deposited into the ocean by human actions. Research estimates that one acre of seagrass meadow can capture 740 pounds of carbon every year. The grasses also slow wave actions as waves move toward shore, making the grasses a buffer against storm damage.

ISTUDY

Coral Reefs

Coral reefs are areas of biological abundance that are primarily found in shallow, warm, tropical waters. Their chief constituents are stony corals, animals that have a calcium carbonate (limestone) exoskeleton, and calcareous red and green algae. Corals provide a home for microscopic algae called *zooxanthellae.* The corals feed at night, and the algae photosynthesize during the day, forming a mutualistic relationship. The algae need sunlight for photosynthesis, which is why coral reefs typically develop in shallow, sunlit waters.

A reef is densely populated with life. The large number of crevices and caves provide shelter for filter feeders (sponges, sea squirts, and fanworms) and for scavengers (crabs and sea urchins). The barracuda, moray eel, and shark are top predators in coral reefs. Many types of small, beautifully colored fishes live here. These become food for larger fishes, including snappers that are caught for human consumption.

Oceans

Shallow ocean waters (called the *euphotic zone*) contain the greatest concentration of organisms in the sea (see Fig. 46.15). Here, phytoplankton is food not only for zooplankton but also for small fishes. These attract a number of predatory and commercially valuable fishes. Seaweed can be found growing on the continental shelf, and even on outcroppings as the water gets deeper. Clams, worms, and sea urchins are preyed upon by sea stars, lobsters, crabs, and brittle stars.

Most of the ocean's volume lies within the **pelagic zones,** as noted in Figure 46.21. The *epipelagic zone* lacks the inorganic nutrients of shallow waters, which means it does not have as high a concentration of phytoplankton as the shallow areas. Still, the photosynthesizers are food for a large assembly of zooplankton, which then become food for schools of various fishes. A number of porpoise and dolphin species visit and feed in the epipelagic zone. A variety of whales can be found in this zone. Baleen whales strain krill (small crustaceans) from the water, and toothed sperm whales feed primarily on the common squid.

Animals in the deeper waters of the *mesopelagic zone* are carnivores and are adapted to the absence of light. Many of these organisms tend to be translucent, red-colored, or even luminescent. Among these are some species of shrimps, squids, and fishes, including lantern and hatchet fishes. Various species of zooplankton, invertebrates, and fishes migrate from the mesopelagic zone to the surface to feed at night.

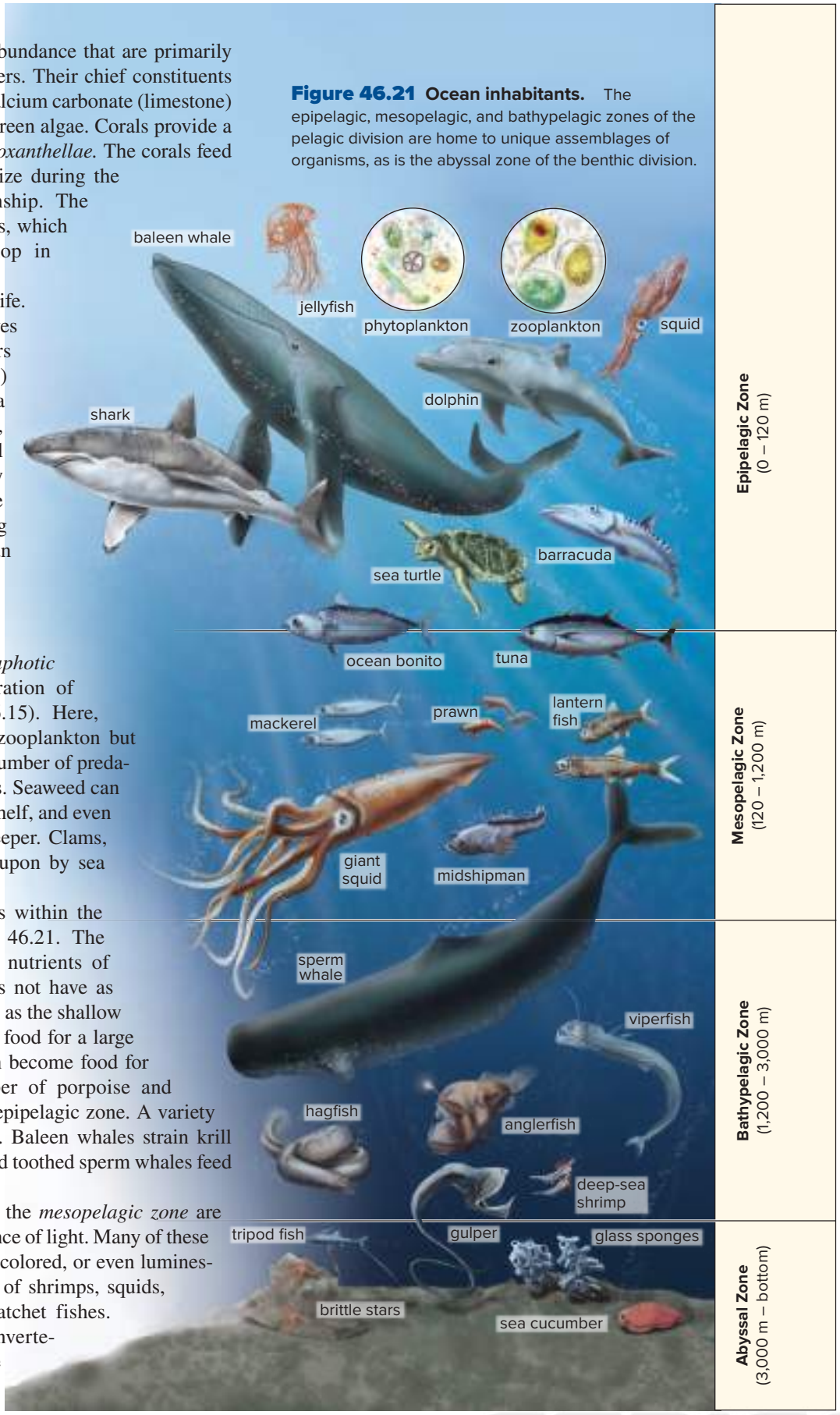


Figure 46.21 Ocean inhabitants. The epipelagic, mesopelagic, and bathypelagic zones of the pelagic division are home to unique assemblages of organisms, as is the abyssal zone of the benthic division.

ISTUDY

The deepest waters of the *bathypelagic zone* are in complete darkness except for an occasional flash of bioluminescent light. Carnivores and scavengers are found in this zone. Strange-looking fishes with distensible mouths and abdomens, as well as small tubular eyes, feed on infrequent prey.

It once was thought that minimal life existed in the *abyssal zone* beneath the bathypelagic zone due to the intense pressure and extreme cold. However, many invertebrates survive there by feeding on debris floating down from the mesopelagic zone. Sea lilies (crinoids) rise above the seafloor; sea cucumbers and sea urchins crawl around on the sea bottom; and tube worms burrow in the mud.

The flat abyssal zone is interrupted by enormous underwater mountain chains called oceanic ridges. Along the axes of the ridges, crustal plates spread apart, and molten magma rises to fill the gap. At **hydrothermal vents,** seawater percolates through cracks and is heated to about 350°C, causing sulfate to react with water and form hydrogen sulfide (H_2S). Chemoautotrophic bacteria that obtain energy from oxidizing hydrogen sulfide exist freely or live mutualistically with organisms at the vents. They are the start of food chains for an ecosystem that includes huge tube worms, clams, crustaceans, echinoderms, and fishes. This ecosystem can exist where light never penetrates because, unlike photosynthesis, chemosynthesis does not require light energy.

In Section 45.3, we discussed food webs and energy flow through ecosystems. Although energy is lost with transfer to each level, from producers to primary and then secondary consumers, pollutants can become concentrated as they move upto higher trophic levels—an effect termed **biomagnification.**

Ocean Currents

Climate is driven by the sun, but the oceans play a major role in redistributing heat in the biosphere. Water tends to be warmer at the equator and much cooler at the poles because of the distribution of the sun's rays (see Fig. 46.1*a*). Air takes on the temperature of the water below, and warm air moves from the equator to the poles. In other words, the oceans make the winds blow. Landmasses also play a role, but the oceans hold heat longer and remain cool longer during periods of changing temperature than do continents.

When wind blows strongly and steadily across a great expanse of ocean for a long time, it causes friction on the water and sets it in motion. The momentum of the water, aided by the wind, keeps it moving in a steady flow called a current. Because the ocean currents eventually strike land, they move in a circular path—clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere (Fig. 46.22). As the currents flow, they take warm water from the equator to the poles. One such current, called the Gulf Stream, brings tropical Caribbean water to the east coast of North America and the higher latitudes of western Europe. Without the Gulf Stream, Great Britain, which has a relatively warm temperature, would be as cold as Greenland. In the Southern Hemisphere, another major ocean current warms the eastern coast of South America.

Also in the Southern Hemisphere, a current called the Humboldt Current flows toward the equator. The Humboldt Current carries phosphorus-rich cold water northward along the west coast of South America. During a process called **upwelling,** cold offshore winds cause cold nutrient-rich waters to rise and take the place of

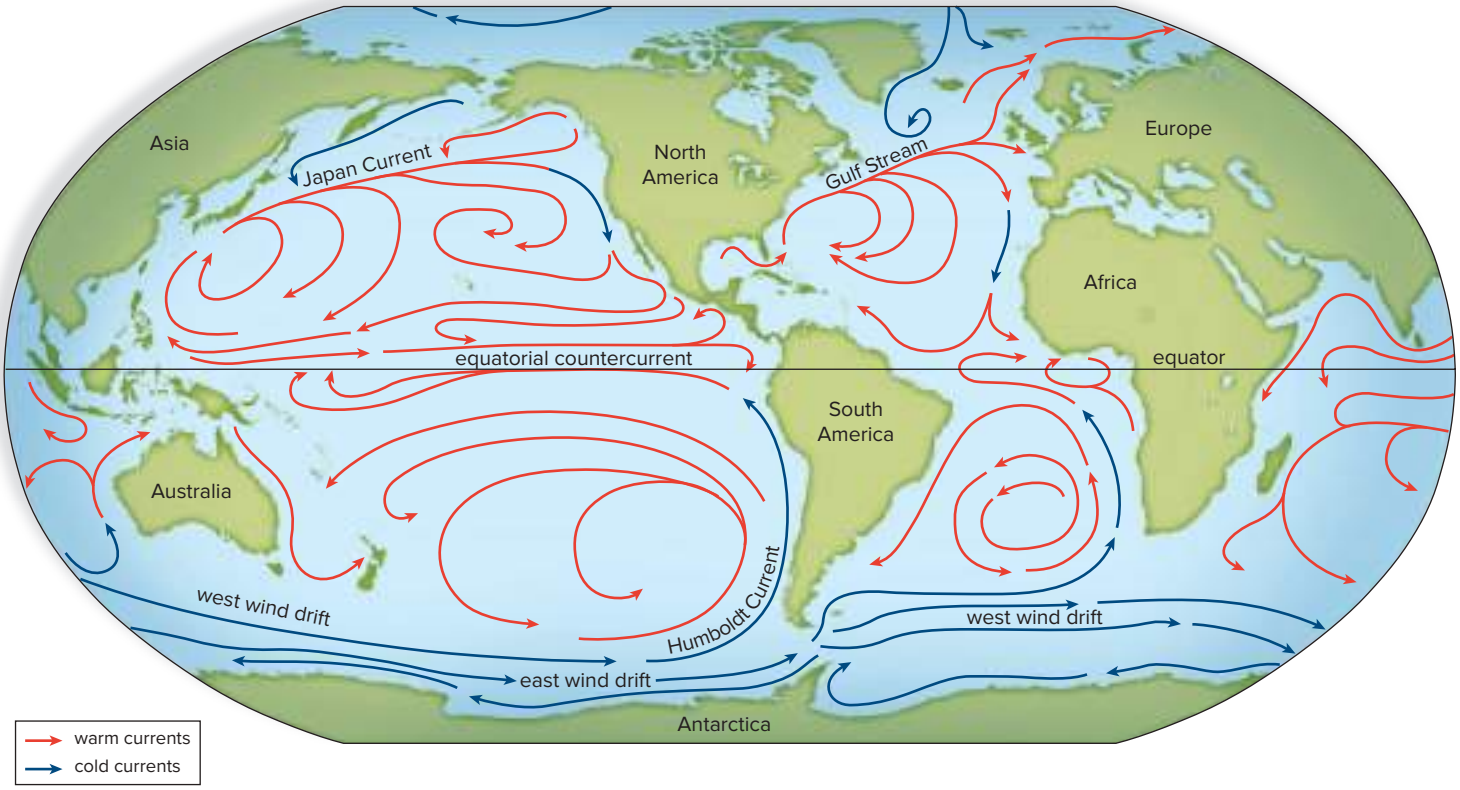


Figure 46.22 Ocean currents. The arrows on this map indicate the locations and directions of the major ocean currents set in motion by the global wind circulation. By carrying warm water to cool latitudes (e.g., the Gulf Stream) and cool water to warm latitudes (e.g., the Humboldt Current), these currents have a major effect on the world's climates.

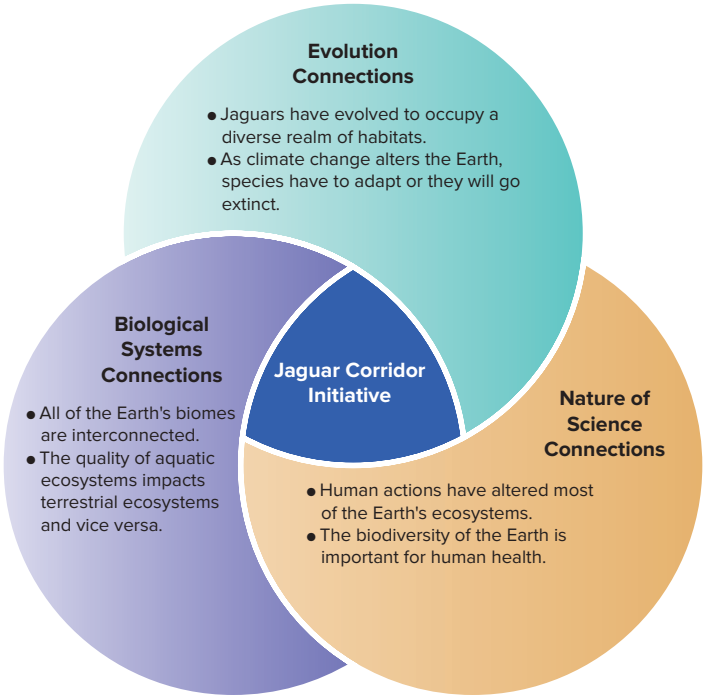
ISTUDY

warm nutrient-poor waters. In South America, the enriched waters cause an abundance of marine life, which supports the fisheries of Peru and northern Chile. Birds feeding on these organisms deposit their droppings on land, where they are mined as guano, a commercial source of phosphorus. When the Humboldt Current is not as cool as usual, upwelling does not occur, stagnation results, the fisheries decline, and climate patterns change globally. This phenomenon is called an **El Niño–Southern Oscillation (ENSO).**

Check Your Progress 46.3

1. Identify the abiotic features of freshwater and saltwater ecosystems.
2. Describe what occurs during an upwelling.

Connecting *the* Concepts



What Did You Learn?

46.1 Climate and the Biosphere

Climate refers to the prevailing weather conditions in a particular region. The position of a region on the Earth, along with the amount of solar radiation, plays a role in determining the region's weather patterns.

- Warm air rises near the equator, loses its moisture, and then descends at about 30° north and south latitude to the poles. When the air descends, it absorbs moisture from the land, and therefore the great deserts of the world are formed at 30° latitudes. Because the Earth rotates on its axis daily, the winds blow in opposite directions above and below the equator.
- **Topography** also plays a role in the distribution of moisture. Air rising over coastal ranges loses its moisture on the windward side. The **rain shadow** is on the leeward side, which tends to be arid. A **monsoon** climate occurs in southern Asia when wet ocean winds blow onshore, carrying heavy rains.

46.2 Terrestrial Ecosystems

A **biome** is a major type of terrestrial community. Biomes are distributed according to climate—that is, temperature and rainfall influence the pattern of biomes about the world. The effect of temperature causes the same sequence of biomes when traveling to northern latitudes as when traveling up a mountain.

- A **montane coniferous forest** is located on the side of a mountain, while the tundra near the peak is the **alpine tundra.**
- The **tundra** is the northernmost biome and consists largely of short grasses, sedges, and dwarf woody plants. Because of cold winters and short summers, most of the water in the soil is frozen year-round. This is called **permafrost.**
- The **taiga,** a coniferous forest, has less rainfall than other types of forests.
- The **temperate rain forest** is a coniferous forest on the west coast of Canada and the United States.
- The **temperate deciduous forest** has trees that gain and lose their leaves because of the alternating seasons of summer and winter.
- **Tropical rain forests** are continually warm and wet. These are the most diverse and productive of all biomes. **Epiphytes** are plants found in the rain forest that grow on other plants.
- **Shrublands** usually occur along coasts that have dry summers and receive most of their rainfall in the winter. **Chaparral** regions are a type of shrubland that is highly flammable.

ISTUDY

- **Grasslands** receive more rainfall than deserts but not enough to be a forest. **Temperate grasslands,** such as that found in the central United States, have a limited variety of vegetation and animal life.
- **Savanna,** a tropical grassland, supports the greatest number of different types of large herbivores.
- **Deserts** are characterized by a lack of water—they are usually found in places with less than 25 cm of precipitation per year. Some desert plants, such as cacti, are succulents with thick stems and leaves, and others are shrubs that are deciduous during dry periods.

46.3 Aquatic Ecosystems

Aquatic ecosystems are divided into freshwater and saltwater systems.

- **Wetlands** are areas that contain water for part of the year. **Marshes, swamps,** and **bogs** are different types of wetlands.
- **Lakes** are classified by their level of nutrients. **Eutrophication** is a rapid increase in the nutrient load of an aquatic ecosystem. In deep lakes of the temperate zone, the temperature and concentration of nutrients and gases in the water vary with depth. The **fall overturn** and **spring overturn** cycle the nutrients and oxygen throughout the lake. **Phytoplankton** and **zooplankton** are the microscopic organisms that live in aquatic ecosystems. Lakes and ponds have three life zones. Rooted plants and clinging organisms live in the littoral zone, plankton and fishes live in the sunlit limnetic zone, and bottom-dwelling organisms such as crayfishes and mollusks live in the profundal zone.
- Marine ecosystems are divided into coastal ecosystems and the oceans. The **estuary** is where fresh water meets salt water. Estuaries (and associated salt marshes, mudflats, and mangrove forests) are near the mouth of a river and are considered the nurseries of the sea. The **intertidal zone** is the region between the high- and low-tide marks.
- An ocean is divided into the pelagic zone and the ocean floor. **Coral reefs** are areas of high biological abundance in shallow, warm, tropical waters. The **pelagic zone** (open waters) has three zones. The epipelagic zone receives adequate sunlight and supports the most life. The mesopelagic zone contains organisms adapted to minimal or no light. The bathypelagic zone is in complete darkness. The ocean floor includes the continental shelf, the continental slope, and the abyssal zone. **Hydrothermal vents** form where seawater percolates through cracks in the floor of the ocean. Chemoautotrophic bacteria form the base of these communities.
- As pollutants move through aquatic ecosystems, they undergo an increase in concentration called **biomagnification.**
- Ocean currents move in a circular pattern. During an **upwelling,** cold, nutrient-rich waters rise to the surface. The **El Niño–Southern Oscillation (ENSO)** is a reversal of the normal circulation pattern that prevents upwelling from occurring.

Thinking Critically About Biology

1. Pharmaceutical companies are "bioprospecting" the tropical rain forests. These companies are looking for naturally occurring compounds in plants or animals that can be used as drugs for a variety of diseases. The most promising compounds act as antibacterial or antifungal agents. Even discounting the fact that the higher density of species in tropical rain forests would produce a wider array of compounds than another biome, why would antibacterial and antifungal compounds be more likely to evolve in this biome?
2. Explain the consequences that climate change will have upon the annual migration of Canada geese.
3. Builders in Traverse County, Minnesota, are required to control soil erosion with filter fences, to steer rainwater away from exposed soil, to build sediment basins, and to plant protective buffers. Presently, homeowners must have a 25-foot setback from wetlands and a 50-foot setback from lakes and creeks. They are also encouraged to pump out their septic systems every 2 years. Do you approve of legislation that requires farmers and homeowners to protect freshwater supplies? Why or why not?

Making It Relevant to You

1. What are the potential consequences to the jaguars in Brazil if Mexico were to lose 20% of its native habitat?
2. Identify two actions we can take as individuals that would benefit the jaguar populations in Central and South America.
3. Should there be "global laws" that prevent habitat destruction and the endangerment of species that have ranges spanning multiple countries?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

47

Conservation of Biodiversity

Dirk-Jan Mattaar/iStock/360/Getty Images

Invasion of the Lionfish

Lionfish are native to the coral reefs around Southeast Asia and Indonesia. In their natural environment, lionfish play an important role as one of the top predators, relying on their rapid reflexes and camouflage to capture small fish. While they are a venomous fish, the chemicals they produce are designed mostly for defense. Their warning coloration acts as a hazard sign to potential predators. However, their beautiful coloration also made them attractive for aquarium owners. Around 1985, some of these fish were dumped into the Atlantic Ocean, probably off the coast of Miami. DNA analyses suggest that the initial population of lionfish may have been as few as six individuals.

Lionfish lack natural predators in the Atlantic, and by the year 2000 their populations had increased to the point that they had become one of the most predominant predators in the Caribbean. Lionfish are known to eat over 50 species of Caribbean fish. Some, like grouper and snapper, are commercially important. A few lionfish can remove up to 80% of the small feeder fish on a reef. This upsets the natural ecology of the reef and adversely affects the populations of larger reef predators.

As you read through this chapter, think about the following questions:

1. In what ways do scientists measure biodiversity?
2. What are some threats to biodiversity?

Chapter Outline

47.1 Conservation Biology and Biodiversity
47.2 Value of Biodiversity
47.3 Causes of Extinction
47.4 Conservation Techniques

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Section 18.3 Why are the current extinctions of such great concern to scientists?

Section 44.6 How is human population growth contributing to extinction rates?

Section 45.3 What role does biodiversity play in ecosystem health?

Following *the* Themes

CHAPTER 47 – CONSERVATION OF BIODIVERSITY

Evolution	For most species, evolutionary change does not occur rapidly enough to allow for adaptations to the human-related changes in the Earth's environment. The result is an increase in the rate of extinction.
Nature of Science	Conservation biology is supported by multiple fields of science, ranging from ecology and systematics to evolution and genetics.
Biological Systems	The identification of both biodiversity hotspots and keystone species is critical for successful conservation.

ISTUDY

47.1 Conservation Biology and Biodiversity

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify the role of conservation biology with regard to biodiversity.
2. Describe how conservation biology is an applied, goal-oriented, multidisciplinary field.
3. Recognize the spectrum of biodiversity.

Conservation biology (L. *conservatio,* "keep, save") is a field of biology that studies biodiversity with the goal of conserving natural resources for this generation and all future generations. Conservation biology is unique in that it is concerned with both the development of scientific concepts and the application of these concepts to the everyday world. The primary goal of conservation biology is the management of **biodiversity,** the diversity of life on Earth. To achieve this goal, many subfields of biology have been brought together to form the field of conservation biology (Fig. 47.1).

Conservation biologists must be aware of the latest theoretical and practical findings. They use this knowledge to identify the source of problems and develop courses of action to correct the problems. Often, conservation biologists work with government officials from the local to the federal level. Public education is an essential component of conservation biology.

For conservation biology to be an effective field of study, scientists must look at the larger connections within the biosphere. Overall, a high level of biodiversity is beneficial, because it positively influences the stability of ecosystems. In turn, stable ecosystems contribute to the overall health of the human population. The opposite tends to be true—if biodiversity decreases, then ecosystem stability decreases, leading to negative consequences for the human population.

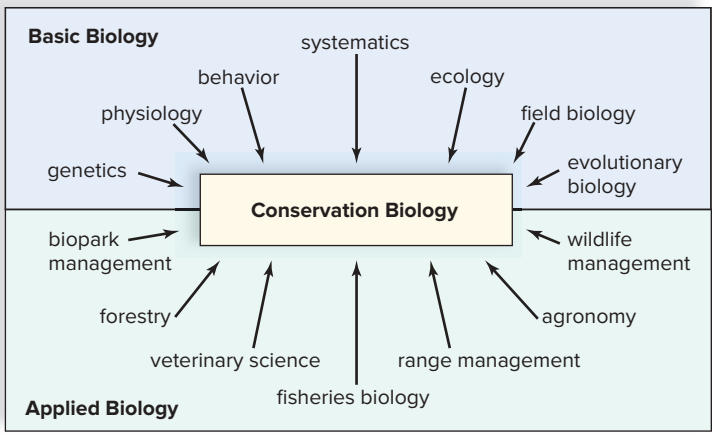


Figure 47.1 Subdisciplines that support conservation biology. The field of conservation biology is supported by a large diversity of biological fields. Each field contributes components of information that are synthesized into the large picture.

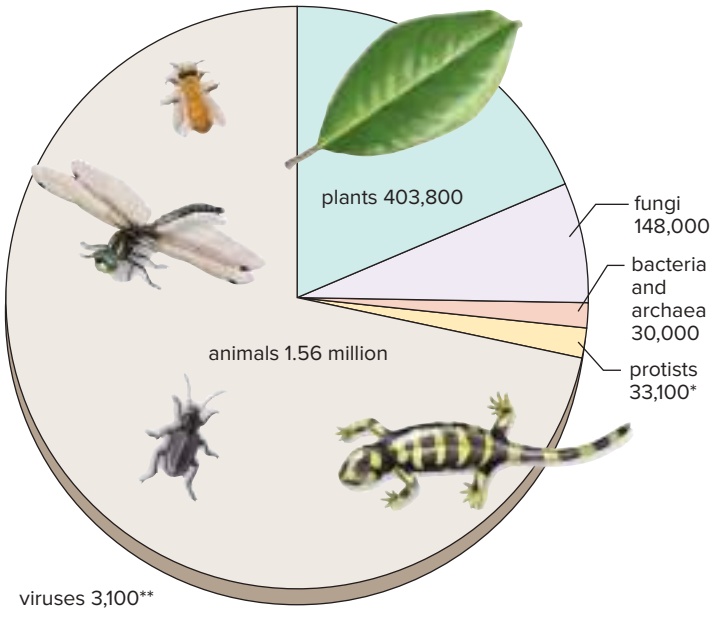
Conservation biology has emerged in response to the extinction crisis the Earth is experiencing. Estimates vary, but at least 10–20% of all species now living most likely will become extinct in the next 20 to 50 years unless planned, coordinated actions are taken. It is urgent that all citizens understand the concept and importance of biodiversity, the causes of present-day extinctions, how to prevent future extinctions from occurring, and the potential consequences of decreased biodiversity.

To protect biodiversity, scientists can apply the science of **bioinformatics:** the collection and analysis of biological information and making it readily available using modern computer technology. Throughout the world, molecular, descriptive, and biogeographical information on organisms is being collected for study purposes.

Biodiversity—Three Levels of Organization

It is common practice to describe biodiversity in terms of the number of species present among various groups of organisms. To date, over 2.1 million species have been described and cataloged across the globe (Fig. 47.2). This may be only a small fraction of Earth's species, however. Recent scientific studies of biodiversity suggest that there are probably around 8.7 million species (not counting bacteria or viruses) on the planet, but some think this number is much higher. Obviously, most species have yet to be discovered and described.

According to the U.S. Fish and Wildlife Service (FWS), as of 2022, there were over 733 animal species and 940 plant species in the United States in danger of extinction. Worldwide, there are more than 41,000 species in danger of extinction. An **endangered species** is one that is in peril of immediate extinction throughout all or most



*Includes protozoans and chromists
**Not considered by some to be living

Figure 47.2 Number of described species. There are over 2.1 million described species, with insects making up over half of them. Estimates place the total species diversity on Earth at around 8.7 million species.

ISTUDY

of its natural range. Examples of endangered species include the monarch butterfly, whooping crane, and black-footed ferret. **Threatened species** are organisms that are likely to become endangered in the near future. Examples of threatened species include the jaguar, orange-breasted falcon, and rainbow parrotfish.

To develop a meaningful understanding of life on Earth, we need to know more about species than just their total number. Ecologists describe biodiversity as a combination of three levels of biological organization:

- Genetic diversity
- Community diversity
- Landscape diversity

Genetic Diversity

Genetic diversity refers to genetic variations among the members of a population. Populations with high genetic diversity are more likely to have some individuals that can survive a change in the structure of their ecosystem. These populations tend to be more adaptable than those with limited biodiversity. The 1846 potato blight in Ireland, the 1922 wheat failure in the Soviet Union, and the 1984 outbreak of citrus canker in Florida were all made worse by limited genetic variation among these crops. If a species' population is small and isolated, it is more likely to exhibit a limited genetic diversity. Small populations, with limited genetic diversity, are more likely to go extinct when ecosystems change.

Community Diversity

Community diversity is dependent on the interactions between species in a community. The species composition of one community can be completely different from that of another community. Diverse community compositions increase the levels of biodiversity in the biosphere.

Past conservation efforts frequently concentrated on saving a single species with human appeal, termed a charismatic species—such as the black-footed ferret or spotted owl. This approach, however, is short-sighted. A more effective approach is to conserve species that play a critical role in an ecosystem. Saving an entire community can save many species, whereas disrupting a community threatens the existence of all species.

One example of a short-sighted approach was the introduction of opossum shrimp, *Mysis relicta,* into Flathead Lake in Montana and its tributaries in the early 1980s as food for salmon. The shrimp ate so much zooplankton that, in the end, far less food was available for the fish. This then resulted in a decrease in the available food for the grizzly bears and bald eagles (Fig. 47.3).

Landscape Diversity

Landscape diversity involves studying **landscape** interactions. A landscape is a group of interacting ecosystems, such as mountains, rivers, and grasslands. Sometimes, ecosystems are so fragmented that they are connected only by small patches or corridors of undeveloped land that allow organisms to move from one ecosystem to the next. Fragmentation of the landscape reduces reproductive capacity and food availability and can disrupt seasonal behaviors.

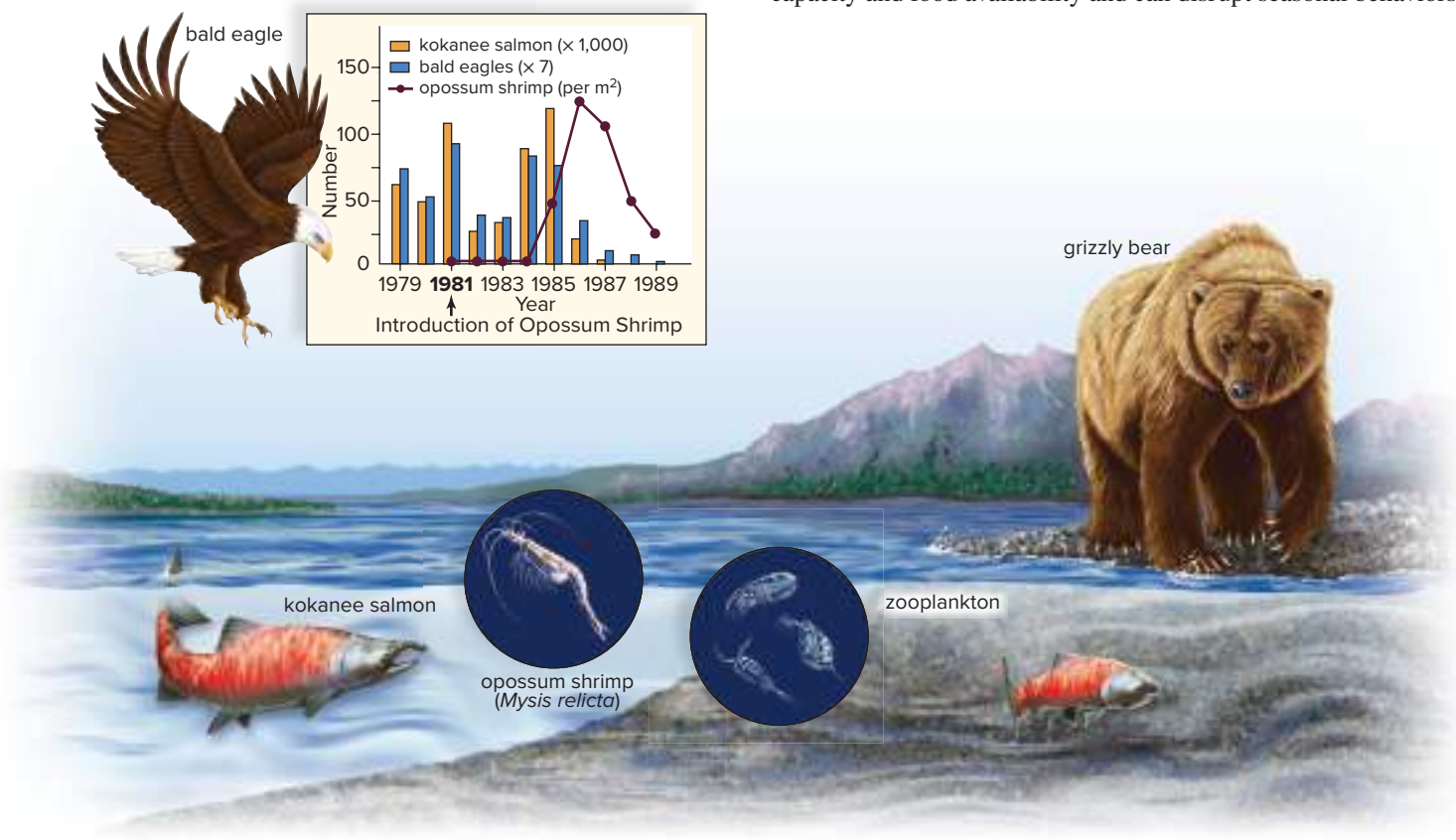


Figure 47.3 Example of community diversity. Humans introduced the opossum shrimp as prey for salmon. Instead, the shrimp competed with salmon for zooplankton as a food source. The salmon, eagle, and bear populations subsequently declined.

ISTUDY

Distribution of Biodiversity

Biodiversity is not evenly distributed throughout the biosphere. If we are forced to protect a limited number of areas, we should focus our efforts on regions that contain the greatest biodiversity. Biodiversity is typically higher in the tropics, and declines toward the poles. The coral reefs of the Indonesian archipelago contain the greatest biodiversity of all aquatic ecosystems.

Some regions of the world are called **biodiversity hotspots,** because they contain a large concentration of species. The biodiversity found in the hotspots accounts for about 44% of all known higher plant species and 35% of all terrestrial vertebrate species. Hotspots cover only about 1.4% of the Earth's land area. The island of Madagascar, the Cape region of South Africa, Indonesia, the coast of California, and the Great Barrier Reef of Australia are locations of biodiversity hotspots.

Check Your Progress 47.1

1. Explain the role of conservation biology.
2. Describe how conservation biology is supported by a variety of disciplines.
3. Define *biodiversity,* and explain what is meant by a biodiversity hotspot.

47.2 Value of Biodiversity

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare the direct and indirect values of biodiversity.
2. Describe the role biodiversity plays in a natural ecosystem.

Conservation biology strives to reverse the trend of species extinction. Educating people about the value of biodiversity is a key step in making conservation possible.

Direct Value

Species that perform services that provide an economic value to humans are considered to have a direct value. Here, we discuss a few of the key direct values provided by various species.

Medicinal Value

Humans often associate the worth of a resource with its value as expressed in monetary terms. Medicines derived from living organisms are a good example (Fig. 47.4).

Most of the prescription drugs used in the United States, which are valued at over $350 billion, were originally derived from living organisms. Quinine, which is extracted from the bark of the cinchona tree (Fig. 47.4*a*) of South America, is an excellent example of a tropical plant that has provided us with useful medicines. Potent chemicals from this plant are now used to treat malaria. Researchers estimate that hundreds of new drugs are yet to be found in tropical rain forests, and they may have an economic value of over $200 billion.

a.

b.

Figure 47.4 Direct value of wildlife. Many wild species have a significant monetary value to humans. **a.** The cinchona tree (*Cinchona calisaya*) is a source of medicine. **b.** *Hirudo medicinalis* is a leech approved by the U.S. Food and Drug Administration for medicinal use. (a): flowerphotos/Alamy Stock Photo; (b): Photocrea/iStock/Getty Images

The antibiotic penicillin is derived from a fungus, and certain species of bacteria produce the antibiotics tetracycline and streptomycin. These drugs have proved to be indispensable in the treatment of bacterial infections.

Leeches have been used in medicinal practices for centuries. Today, medicinal leeches are used to help mend skin after plastic surgery and speed healing by preventing blood clotting when a person has had a digit reattached (Fig. 47.4*b*). The blood of horseshoe crabs, *Limulus,* contains a substance called limulus amoebocyte lysate, which is used to ensure that medical devices, such as pacemakers, surgical implants, and prosthetic devices, are free of bacteria.

Agricultural Value

Crops such as wheat, corn, and rice have been derived from wild plants that were modified to increase their yields. The same high-yield, genetically similar strains tend to be grown worldwide. When rice crops in Africa were being devastated by a virus, researchers grew wild rice plants from thousands of seed samples until they found one that contained a gene for resistance to the

ISTUDY

virus. These wild plants were then used in a breeding program to transfer the gene into high-yield rice plants. If this variety of wild rice had become extinct before it could be discovered, rice cultivation in Africa could have collapsed.

Animals are the main pollinators for flowering plants. The domesticated honeybee, *Apis mellifera,* pollinates billions of dollars' worth of food crops annually in the United States (Fig. 47.5*a*). This dependency on a single species has inherent dangers: Colony collapse disorder has wiped out more than 30% of the commercial honeybee population in the United States. The USDA estimates that honeybee pollination supports $15 billion worth of agricultural production, including more than 130 different types of fruits and vegetables, per year. In order to find honeybees resistant to the mites, researchers must look to the wild populations.

Using natural pest controls is preferable to using chemical pesticides. When a rice pest called the brown planthopper became resistant to pesticides through natural selection, farmers began to use the natural predator of the brown planthopper. The economic savings were calculated at well over a billion dollars. Similarly, cotton growers in Cañete Valley, Peru, found that pesticides were no longer working against the cotton aphid because resistance had evolved in the aphid population. Researchers have identified the aphids' natural predators and are using them with great success (Fig. 47.5*b*).

a.

b.

Figure 47.5 Examples of the agricultural value of biodiversity. **a.** *Apis mellifera* provides over $3 billion worth of agricultural value in pollination services. **b.** Ladybugs can be used as a biological control for aphids. (a): Daniel Prudek/Shutterstock; (b): D. Hurst/Alamy Stock Photo

Figure 47.6 The consumptive value of biodiversity. Many wild species, such as these fish, are used as sources of food. Monty Rakusen/Cultura/Getty Images

Consumptive Use Value

Humans have had much success cultivating crops, domesticating animals, growing trees in plantations, and so forth. But so far, aquaculture, the growing of fish and shellfish for human consumption, has provided only a limited amount of resources. Most freshwater and marine harvests depend on catching wild animals, such as fishes (e.g., trout, cod, tuna, and flounder), crustaceans (e.g., lobsters, shrimps, and crabs), and mammals (e.g., whales). These aquatic organisms are an invaluable source of biodiversity (Fig. 47.6).

The environment provides a variety of other products that are sold commercially worldwide, including wild fruits and vegetables, skins, fibers, beeswax, and seaweed. Hunting and fishing are the primary methods of obtaining meat for many human populations. In one study, researchers calculated that the economic value of wild pigs in the diet of indigenous hunters in Sarawak, East Malaysia, was approximately $40 million per year.

Similarly, many trees are still felled in the natural environment for their wood. Researchers have calculated, however, that a species-rich forest in the Peruvian Amazon is worth far more if the forest is used for fruit and rubber production than for timber production. Fruit and the latex needed to produce rubber can be brought to market for an unlimited number of years. Unfortunately, once the trees are cut down, none of these products, including timber, can be sustainably harvested.

Indirect Value

The wild species we have been discussing all play a role in their respective ecosystems. If we want to preserve them, it is more beneficial to save large portions of their ecosystems. The indirect value of these ecosystems is based on services they provide that do not have an immediate measurable economic value. Biogeochemical

cycles, waste recycling, and provision of fresh water all have indirect values to human society. In addition, these natural regions have intangible value; a walk in the woods or time spent at the local park can have a calming and restorative effect on many people. Reconnecting with nature has proven physiological and psychological benefits to human health. Our very survival depends on the functions that ecosystems perform for us.

Biogeochemical Cycles

Ecosystems are characterized by energy flow and chemical cycling (see Fig. 45.16). The biodiversity within ecosystems contributes to the workings of the water, carbon, nitrogen, phosphorus, and other biogeochemical cycles. We are dependent on these cycles for fresh water, the removal of carbon dioxide from the atmosphere, the uptake of excess soil nitrogen, and the provision of phosphate. When human activities upset the natural workings of biogeochemical cycles, the environmental consequences often result in negative consequences for humans. Our current technology is limited in its ability to replicate the biogeochemical cycles.

Waste Recycling

Decomposers break down dead organic matter and other types of wastes into inorganic nutrients, which are used by the producers in ecosystems. This function aids humans immensely, because we dump millions of tons of waste material into natural ecosystems each year. If it were not for decomposition, waste would soon cover the entire surface of our planet. We can build sewage treatment plants, but they are expensive, and few of them break down solid wastes completely to inorganic nutrients. It is less expensive and more efficient to provide plants and trees with partially treated wastewater and let soil bacteria cleanse it completely.

Biological communities are also capable of breaking down and immobilizing pollutants, such as heavy metals and pesticides. A review of wetland functions in Canada assigned a value of $50,000 per hectare (100 acres, or 10,000 m^2) per year to the ability of natural areas to purify water and take up pollutants.

Provision of Fresh Water

Few terrestrial organisms are adapted to living in a salty environment—they need fresh water. The water cycle continually supplies fresh water to terrestrial ecosystems. Humans use fresh water in innumerable ways, from drinking water to irrigating crops. Freshwater ecosystems, such as rivers and lakes, provide us with a large diversity of species we can use as a source of food.

Unlike other commodities, there is no substitute for fresh water. We can remove salt from seawater to obtain fresh water, but the cost of desalination is about four to eight times the average cost of fresh water acquired via the water cycle.

Forests and other natural ecosystems exert a "sponge effect." They soak up water and then release it at a steady rate. When rain falls in a natural area, plant foliage and dead leaves lessen its impact, and the soil slowly absorbs it, especially if it has been aerated by soil organisms. The water-holding capacity of forests reduces the possibility of flooding. The value of a marshland outside Boston, Massachusetts, has been estimated at $72,000 per hectare per year, solely for its ability to reduce floods. Forests release water slowly for days or weeks after the rains have ceased. Rivers flowing through forests in West Africa release twice as much water halfway through the dry season, and between three and five times as much at the end of the dry season, as do rivers from agricultural lands.

Prevention of Soil Erosion

Intact ecosystems naturally retain soil and prevent soil erosion. The importance of this ecosystem attribute is especially observed following deforestation. In Pakistan, the world's largest dam, the Tarbela Dam, is losing its storage capacity of 12 billion m^3 many years sooner than expected, because sediment is building up behind the dam due to deforestation. At one time, the Philippines exported $100 million worth of oysters, mussels, clams, and cockles each year. Now, silt carried down rivers following deforestation is smothering the mangrove ecosystem that serves as a nursery for the sea. Most coastal ecosystems are not as bountiful as they once were because of deforestation and myriad other problems.

Regulation of Climate

At the local level, trees provide shade and reduce the need for fans and air conditioners during the summer. Proper placement of shade trees near a home can reduce energy bills 10–20%.

Globally, forests restore the climate because they take up carbon dioxide. The leaves of trees use carbon dioxide when they photosynthesize, the bodies of the trees store carbon, and oxygen is released as a by-product. When trees are cut and burned, carbon dioxide is released into the atmosphere. The destruction of forests reduces the amount of carbon dioxide removed from the atmosphere and the amount of oxygen produced. This change in the atmospheric gases, especially greenhouse gases such as CO_2, affects the amount of solar radiation retained on the Earth's surface. Large-scale deforestation is affecting the global atmosphere, and in turn changing the Earth's climate.

Ecotourism

In the United States, millions of people enjoy vacationing in a natural setting. To do so, they spend billions of dollars each year on fees, travel, lodging, and food. Many tourists want to go sport fishing, whale watching, boating, hiking, birdwatching, and the like. With 1,200 miles of sand beaches and over 86 million tourists a year, tourism generates $67 billion a year for Florida's economy. Others simply want to immerse themselves in the beauty and serenity of a natural environment. Many less-developed countries in tropical regions are taking advantage of this by offering "ecotours" of the local biodiversity. Providing guided tours of forests is more profitable than destroying them.

Biodiversity and Natural Ecosystems

Massive changes in biodiversity, such as deforestation, have a significant impact on ecosystems. Researchers are interested in determining whether a high degree of biodiversity also helps ecosystems function more efficiently. To test the benefits of biodiversity in a Minnesota grassland habitat, researchers sowed plots with seven levels of plant diversity. Their study found that ecosystem performance improves with increasing diversity. A similar study in California also showed greater overall resource use in more diverse plots because of resource partitioning among the plants.

ISTUDY

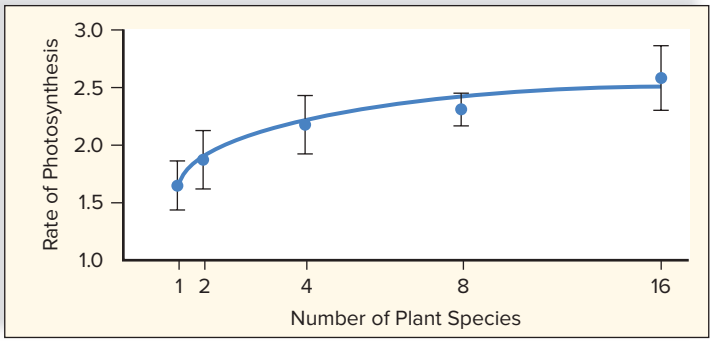


Figure 47.7 The influence of biodiversity on community productivity. Research results show that the higher the biodiversity (measured by the number of plant species), the greater the rate of photosynthesis in an experimental community.

Another group of experimenters tested the effects of an increase in diversity among four groups: producers, herbivores, parasites, and decomposers. They found that the rate of photosynthesis increased as diversity increased (Fig. 47.7). A computer simulation has shown that the response of a deciduous forest to elevated carbon dioxide is a function of species diversity. A more complex community, composed of nine tree species, exhibited a 30% greater amount of photosynthesis than a community composed of a single species.

More studies are needed to test whether biodiversity maximizes resource acquisition and retention in an ecosystem. Diverse ecosystems are better able to withstand environmental changes and invasions by other species. Fragmentation affects the functioning of and distribution of organisms in an ecosystem. These topics and many more are the focus of current conservation biology research.

Check Your Progress 47.2

1. Explain the difference between a direct value and an indirect value of biodiversity.
2. Describe the benefits of biodiversity to natural ecosystems.

47.3 Causes of Extinction

Learning Outcomes

Upon completion of this section, you should be able to

1. Classify the causes of extinction.
2. Compare natural and human-influenced causes of extinction.

To stem the tide of extinction due to human activities, it is first necessary to identify its causes. Researchers examined the records of 1,880 threatened and endangered wild species in the United States and found that habitat loss was involved in 85% of the cases (Fig. 47.8*a*). Nonnative species had a hand in nearly 50%, pollution was a factor in 24%, overexploitation in 17%, and disease in 3%. The percentages add up to more than 100%, being that most of these species are threatened or endangered as a result of more than one reason. The Nature of Science feature, "The Sixth Mass Extinction Event," explores the extent of this species loss in more detail.

Macaws are a good example of how a combination of factors can lead to a species decline (Fig. 47.8*b*). Not only has their habitat been reduced by encroaching timber and mining companies, but macaws are also hunted for food and collected for the pet trade.

Habitat Loss

Habitat loss is occurring in all ecosystems, but concern has now centered on tropical rain forests and coral reefs because they are particularly rich in species.

A sequence of events in Brazil offers a fairly typical example of the manner in which rain forest is converted to land uninhabitable for wildlife. The construction of a major highway into the forest first provided a way to reach the interior of the forest (Fig. 47.8*c*). Small towns and industries sprang up along the highway, and roads branching off the main highway gave rise to even more roads. The result was fragmentation of the once immense forest.

The government offered subsidies to anyone willing to take up residence in the forest, and the people who came cut and burned trees in patches (Fig. 47.8*c*). Tropical soils contain limited nutrients, but when the trees are burned, nutrients are released that support a lush growth for the grazing of cattle for about 3 years. However, once the land was degraded (Fig. 47.8*c*), the farmers moved on to another portion of the forest to start over again.

Loss of habitat also affects freshwater and marine biodiversity. Coastal degradation is mainly due to the large concentration of people living on or near the coast. Already, 60% of coral reefs have been destroyed or are on the verge of destruction; it is possible that all coral reefs will disappear during the next 40 years unless our behaviors drastically change. Mangrove forest destruction is also a problem. Indonesia, with the most mangrove acreage, has lost over 45% of its mangroves, and the percentage is even higher for other tropical countries. Wetland areas, estuaries, and seagrass beds are also being rapidly destroyed by human actions.

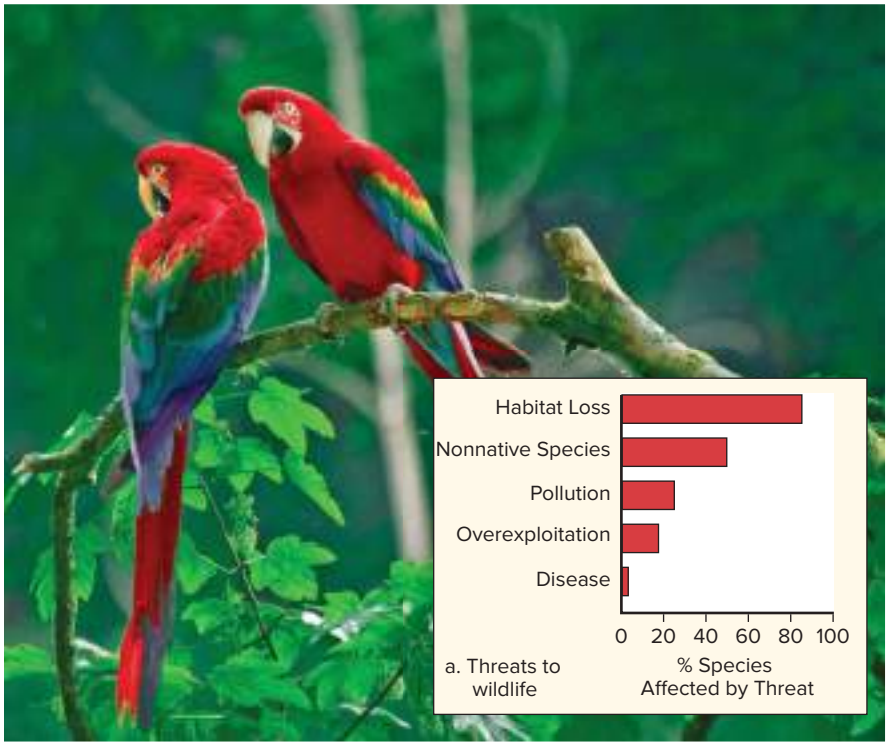
Nonnative Species

Nonnative species, sometimes called *invasive species,* are nonnative members of an ecosystem. Ecosystems around the globe are characterized by unique assemblages of organisms that have evolved in response to each other and changes in their environment. Migration of some species to a new location is not usually possible because of barriers such as oceans, deserts, mountains, and rivers. Humans, however, have introduced nonnative species into new ecosystems in a variety of ways.

Colonization

Europeans, in particular, took a number of familiar species with them when they colonized new places. For example, the pilgrims brought the dandelion to the United States as a familiar salad green. In addition, they introduced pigs to North America that have since become feral, reverting to their wild state. In some parts of the United States, feral pigs have become quite destructive.

ISTUDY



b. Macaws

Roads cut through forest

Patches of forest

Destroyed areas

c. Wildlife habitat is reduced

Figure 47.8 Habitat loss. **a.** In a study that examined records of imperiled U.S. plants and animals, habitat loss emerged as the greatest threat to wildlife. **b.** Macaws that reside in South American tropical rain forests are endangered for many of the same reasons listed in the graph in (**a**). **c.** Habitat loss due to road construction in Brazil. *Top:* Road construction opened up the rain forest and subjected it to fragmentation. *Middle:* The result was patches of forest and degraded land. *Bottom:* Wildlife could not live in destroyed portions of the forest.
(b): (photo): Charan Reddy/EyeEm/Getty Images; (photo): (c): (roads): Universal Images Group North America LLC/Alamy Stock Photo; (photo): (c): (forest patches): Ronnachai Limpakdeesavasd/Alamy Stock Photo; (photo): (c): (destroyed areas): Spaces Images/Blend Images LLC

Horticulture and Agriculture

Some nonnative species were originally introduced with good intentions. Japanese honeysuckle (*Lonicera japonica*) was introduced in 1806 as an ornamental plant and to help control erosion. Since then, it has taken over forests and crowded out many native plant species (Fig. 47.9*a*). The water hyacinth was introduced to the United States from South America because of its beautiful flowers. Today, it clogs up waterways and diminishes natural diversity.

Accidental Transport

Global trade and travel accidentally bring new species from one region to another. Researchers found that the ballast water released from ships into Coos Bay, Oregon, contained 367 marine species from Japan. The zebra mussel from the Caspian Sea was accidentally introduced into the Great Lakes in 1988. It now forms dense beds that squeeze out native mussels. Other organisms accidentally introduced into the United States include the Formosan termite, the Argentine fire ant, and the nutria, a type of large rodent.

Nonnative species can disrupt food webs. As mentioned earlier, opossum shrimp introduced into a lake in Montana added a trophic level that, in the end, meant less food for bald eagles and grizzly bears (see Fig. 47.3).

The Impact of Nonnative Species on Islands

Islands are particularly susceptible to environmental discord caused by the introduction of nonnative species. Islands have unique assemblages of native species that are closely adapted to one another and cannot compete well against nonnatives. Myrtle trees, *Myrica faya,* introduced into the Hawaiian Islands from the Canary Islands, are symbiotic with a type of bacterium that is capable of nitrogen fixation. This feature allows the species to establish itself on nutrient-poor volcanic soil, a distinct advantage in Hawaii. However, once established, myrtle trees disrupt the normal succession of native plants on volcanic soil.

The brown tree snake has been introduced onto a number of islands in the Pacific Ocean. The snake eats adult birds, their eggs, and nestlings. On Guam, it has reduced ten native bird species to the point of extinction. On the Galápagos Islands, introduced animals, such as goats and feral pigs, have changed the vegetation from highland forest to pampas-like grasslands, which has also destroyed stands of cacti. In Australia, mice and rabbits have stressed native marsupial populations. Mongooses introduced into the Hawaiian Islands to control rats also prey on native birds (see Fig. 47.9*b*).

ISTUDY

THEME Evolution

The Sixth Mass Extinction Event

Extinction occurs when all the members of a species die off without having been able to reproduce and replace themselves. Throughout the history of the Earth, species have naturally gone extinct. Extinction opens up ecological niches for new life-forms to evolve into. When the rate of extinction outpaces the rate of replacement, however, it can lead to a mass extinction. A mass extinction is usually defined as 75% or more of the species on Earth going extinct in a relatively short span of time. In geological terms, a short span of time is less than 3 million years. Geological and fossil records show that the Earth has undergone five mass extinction events (Table 47A). While asteroids hitting the Earth is the most identifiable reason for a mass extinction, the biggest driver behind mass extinctions is major changes to the Earth's carbon cycle. Historically, this has been the result of large-scale volcanic eruptions, causing an accumulation of heat-trapping gases in the atmosphere that leads to global warming, ocean acidification, and decreased oxygen concentrations in the oceans.

Historical rates of extinction are calculated at 0.1 to 2.0 species going extinct per million years. The current rate of extinction is 10 to 10,000 times higher than historical rates. Because of this, most scientists believe we are in the beginning stages of a sixth mass extinction, largely due to human exploitation of the planet. Our direct and indirect activities—such as habitat loss and destruction, pollution, climate change, the introduction of nonnative species, and overexploitation of the Earth's natural resources—are pushing a record number of species to the brink of, and into, extinction.

The International Union for Conservation of Nature's Red List of Threatened Species is one of the most important indicators of the health of the world's biodiversity. The IUCN has assessed more than 147,500 species and determined that over 41,000 species are threatened or endangered. Currently, 27% of mammals (Fig. 47A), 41% of amphibians, and 13% of bird species are threatened with extinction.

Questions to Consider

1. How many species need to go extinct before it becomes problematic for humans?
2. Is the extinction of certain species more important than others?
3. Identify the major causes of the current mass extinction.

Table 47A Mass Extinction Events and Their Causes

Time Frame	Extinction Rate	Causes
Cretaceous-Tertiary (65 MYA)	80%	• Asteroid/Comet impact • Volcanic activity • Climate change
Triassic (200 MYA)	76%	• Increases in methane/CO_2 • Rapid climate change
Permian (250 MYA)	95%	• Volcanic activity • Increases in methane/CO_2 • Rapid climate change
Devonian (340 MYA)	70%	• Asteroid impact • Rapid global cooling
Ordovician (445 MYA)	85%	• Rapid global cooling • Falling sea levels

Figure 47A In 2019, the Bramble Cay melomys (*Melomys rubicola*) was officially declared extinct as the result of a rise in sea level due to climate change. Polaris/Newscom

a.

b.

Figure 47.9 Nonnative species. **a.** *Lonicera japonica,* Japanese honeysuckle, has taken over forests and pushed native species into extinction. **b.** Mongooses were introduced into Hawaii to control rats, but they also prey on native birds. (a): Tomsmith585/iStock/Getty Images; (b): Malcolm Schuyl/Alamy Stock Photo

Pollution

In the present context, **pollution** can be defined as any substance introduced into the environment that adversely affects the lives and health of living organisms. Pollution has been identified as the third main cause of extinction. Pollution can also weaken organisms and lead to disease. Biodiversity is particularly threatened by acid deposition, eutrophication, ozone depletion, and synthetic organic chemicals.

Acid Deposition

Both sulfur dioxide from power plants and nitrogen oxides in automobile exhaust are converted to acids

when they combine with water vapor in the atmosphere. These acids return to Earth as either wet deposition (acid rain or snow) or dry deposition (sulfate and nitrate salts). Sulfur dioxide and nitrogen oxides are not always deposited in the same location where they are emitted. They may instead be carried far downwind. Acid deposition weakens trees, which increases their susceptibility to disease and insects. It also kills small invertebrates and decomposers, so that entire ecosystems are disrupted. Many lakes in the northern United States are now lifeless because of the effects of acid deposition.

Eutrophication

Lakes are also under stress due to over-enrichment. When lakes receive excess nutrients due to runoff from agricultural fields and fertilized lawns, as well as wastewater from sewage treatment, algae begin to grow in abundance. An algal bloom is apparent as a thick, green layer of algae on the surface of the water. Upon death, the decomposers break down the algae, but in so doing, they use up oxygen. A decreased amount of oxygen is available to fish, leading sometimes to a massive fish kill.

Ozone Depletion

The ozone shield is a layer of ozone (O_3) in the stratosphere, some 50 km above the Earth. The ozone shield absorbs most of the wavelengths of harmful ultraviolet (UV) radiation, so that they do not strike the Earth. The cause of ozone depletion can be traced to chlorine atoms (Cl^-) that come from the breakdown of chlorofluorocarbons (CFCs). The best-known CFC is Freon, a heat-transfer agent still found in older refrigerators and air conditioners. Severe ozone shield depletion can impair crop and tree growth and kill plankton (microscopic plant and animal life) that sustain oceanic life. The immune system and the ability of all organisms to resist infectious diseases can also become weakened.

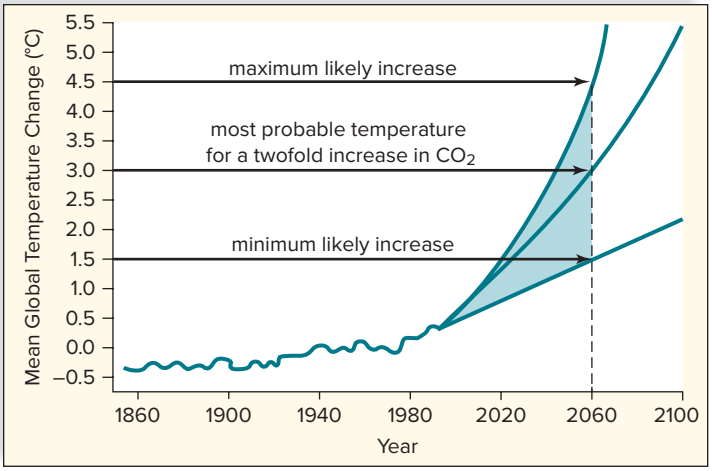
Organic Chemicals

Our modern society uses synthetic organic chemicals in all sorts of ways. Organic chemicals called nonylphenols are used in products ranging from pesticides to dishwashing detergents, cosmetics, plastics, and spermicides. These chemicals mimic the effects of hormones, which can harm wildlife.

As one example, salmon are born in fresh water but mature in salt water. After investigators exposed young fish to nonylphenol, they found that 20–30% were unable to make the transition between fresh and salt water. Nonylphenols cause the pituitary to produce prolactin, a hormone that may prevent saltwater adaptation in these fish.

Climate Change

As mentioned in Section 45.4, **climate change** refers to recent changes in the Earth's climate. Our planet is experiencing erratic temperature patterns, more severe storms, and melting glaciers as a result of an increase in the Earth's temperature. You may also recall from Figure 45.22 that the majority of carbon dioxide and methane produced today is the result of human-based activities. These gases are known as greenhouse gases, because they allow solar radiation to reach the Earth but hinder the escape of its heat back into space. Data collected around the world show a steady



a.

b.

Figure 47.10 Climate change. **a.** Mean global temperature is expected to rise due to the introduction of greenhouse gases into the atmosphere. **b.** The resulting changes in climate have the potential to significantly affect the world's biodiversity. A temperature rise of only a few degrees causes coral reefs to "bleach" and become lifeless. If, in the meantime, migration occurs, coral reefs could move northward. (b): buttchi 3 Sha Life/Shutterstock

rise in CO_2 concentration. These data are used to generate computer models that predict how the Earth will continue to warm in the near future (Fig. 47.10*a*). An upward shift in temperatures could influence everything from growing seasons in plants to migratory patterns of animals.

As temperatures rise, regions of suitable climate for human needs may shift toward the poles and higher elevations. Extinctions are expected to increase as species attempt to migrate to more suitable climates. (Plants migrate when seeds disperse and growth occurs in a new locale.) For example, for beech trees to remain in a favorable habitat, it's been calculated that the rate of beech migration would have to be 40 times faster than has ever been observed. It seems unlikely that beech or any other type of tree would be able to meet the pace required.

Many species are confined to relatively small patches of habitat surrounded by agricultural or urban areas that prevent natural migrations. Even if species have the capacity to disperse to new sites, suitable habitats may not be available. If the global climate changes faster than organisms can migrate, extinction of

ISTUDY

many species is likely to occur, while some species may experience population increases. For example, parasites and pests that are usually killed by cold winters would be able to survive in greater numbers. The tropics may expand, and whether present-day temperate-zone agriculture will survive is questionable.

Overexploitation

Overexploitation is the process of taking more individuals from a wild population than can be naturally replaced, resulting in a decrease in the population. A positive feedback cycle explains overexploitation: The smaller the population, the more valuable its members, and the greater the incentive to capture the few remaining organisms. Poachers are very active in the collecting and sale of endangered and threatened species, because it has become so lucrative. The overall international value of trading wildlife species is approximately $20 billion, of which $8 billion is attributed to the illegal sale of rare species.

Markets for rare plants and exotic pets support both legal and illegal trade in wild species. Rustlers dig up rare cacti, such as the crested saguaros, and sell them to gardeners for as much as $15,000 each. Parrots are among the birds taken from the wild for sale to pet owners. For every bird delivered alive, many more have died in the collection process. The same holds true for tropical fish, which often come from the coral reefs of Indonesia and the Philippines. Divers dynamite reefs or use plastic squeeze-bottles of cyanide to stun the fish; in the process, many fish and valuable corals are killed.

The Convention of International Trade of Endangered Species (CITES) was an agreement established in 1973 to ensure that international trade of species does not threaten their survival. Today, over 36,000 species of plants and animals receive some level of protection from over 182 countries worldwide.

Poachers still hunt for hides, claws, tusks, horns, and bones of many endangered mammals. Because of its rarity, a single Siberian tiger is now worth more than $500,000—its bones are pulverized and used as a medicinal powder. The horns of rhinoceroses become ornate carved daggers, and their bones are ground up to sell as a medicine. The ivory of an elephant's tusk is used to make art objects, jewelry, and piano keys. The fur of a Bengal tiger sells for as much as $100,000 in Tokyo.

The UN Food and Agricultural organization tells us that we have now overexploited 11 of 15 major oceanic fishing areas. Fish are a renewable resource if harvesting does not exceed the ability of the fish to reproduce. Modern society uses larger and more efficient fishing fleets to decimate fishing stocks. Pelagic species, such as tuna, are captured by purse-seine fishing, in which a very large net surrounds a school of fish. The net is then closed in the same manner as a drawstring purse. Thousands of dolphins that swim above schools of tuna are often captured and then die in this type of net. Many tuna suppliers now advertise their product as "dolphin safe"; however, this label's meaning varies, depending on the agency or organization that licenses the label.

Other fishing boats drag huge trawling nets, large enough to accommodate 12 jumbo jets, along the seafloor to capture bottom-dwelling fish (Fig. 47.11*a*). Only large fish are kept, while the undesirable small fish and sea turtles are discarded back into the ocean to die. Trawling has been called the marine equivalent of clear-cutting forest, because after the net goes by, the sea bottom is devastated (Fig. 47.11*b*).

Today's fishing practices don't allow fisheries to recover. Cod and haddock, once the most abundant bottom-dwelling fish along the northeastern coast of the United States, are now often outnumbered by dogfish and skate.

A marine ecosystem can be disrupted by overfishing, as exemplified on the U.S. West Coast. When sea otters began to decline in numbers, investigators found they were being eaten by orcas (killer whales). Usually, orcas prefer seals and sea lions over sea otters. The orcas began eating sea otters when the seal and sea lion populations decreased due to the overfishing of the perch and herring.

a. Fishing by use of a drag net

b. Result of drag net fishing

Figure 47.11 Effects of trawling. **a.** These Alaskan pollock were caught by dragging a net along the seafloor. **b.** Appearance of the seafloor after the net has passed. (a): StrahilDimitrov/360/iStock/Getty Images; (b): ©Peter Auster/University of Connecticut

ISTUDY

Ordinarily, sea otters keep the population of sea urchins under control. With fewer sea otters around, the sea urchin population exploded and decimated the kelp beds. Overfishing set in motion a chain of events that detrimentally altered the food web of this ecosystem.

Check Your Progress 47.3

1. Identify the five main causes of extinction.
2. Explain why the introduction of exotic species can be detrimental to biodiversity.

47.4 Conservation Techniques

Learning Outcomes

Upon completion of this section, you should be able to

1. Describe the value of preserving biodiversity hotspots.
2. Distinguish between keystone species and flagship species.
3. Identify and explain the most useful procedures for habitat restoration.

Despite the value of biodiversity to our very survival, human activities are causing the extinction of thousands of species a year. Clearly, we need to reverse this trend and preserve as many species as possible. Habitat preservation and restoration are important in preserving biodiversity.

Habitat Preservation

Preservation of a species' habitat is of primary concern, but first we must prioritize which species to preserve. As mentioned previously, the biosphere contains biodiversity hotspots, relatively small areas having a concentration of endemic (native) species not found anyplace else. In the tropical rain forests of Madagascar, 93% of the primate species, 99% of the frog species, and over 80% of the plant species are endemic, found only in Madagascar. Preserving these forests and other hotspots would save a wide variety of organisms.

Keystone Species

Keystone species are species that influence the stability and viability of a community, although their population may not be very large. The extinction of a keystone species can lead to other extinctions and a loss of biodiversity. For example, bats are designated a keystone species in tropical forests of Africa and Asia. They are pollinators that also disperse the seeds of trees. When bats are killed off and their roosts destroyed, the trees fail to reproduce. The grizzly bear is a keystone species in the northwestern United States and Canada (Fig. 47.12*a*). Bears disperse the seeds of berries; as many as 7,000 seeds may be in one dung pile. Grizzly bears kill and eat the young of many hoofed animals and thereby keep their populations under control. Grizzly bears are also a principal mover of soil when they dig up roots and prey upon hibernating ground squirrels and marmots. Other examples of keystone species are beavers in wetlands, bison in grasslands, alligators in swamps, and elephants in grasslands and forests.

a. Grizzly bears, *Ursus arctos horribilis*

b. Old-growth forest; northern spotted owl, *Strix occidentalis caurina* (inset)

Figure 47.12 Habitat preservation. When particular species are protected, other wildlife benefits. **a.** The Greater Yellowstone Ecosystem has been delineated in an effort to save grizzly bears, which need a very large habitat. **b.** Currently, the remaining portions of old-growth forests in the Pacific Northwest are not being logged in order to save the northern spotted owl (*inset*). (a): Deb Garside/Design Pics; (b): (forest): Timothy Epp/Shutterstock; (b): (owl): All Canada Photos/Alamy Stock Photo

Keystone species should not be confused with **flagship species,** which evoke a strong emotional response in humans. Flagship species are considered charismatic and are treasured for their beauty, "cuteness," strength, and/or regal nature. These species can help motivate the public to preserve biodiversity. Flagship species include Monarch butterflies, lions, tigers, dolphins, and giant pandas. Some species can be both keystone and flagship species—for example, the grizzly bear—but a keystone species is valued because of its role in an ecosystem.

ISTUDY

Metapopulations

The grizzly bear population is actually a **metapopulation** (Gk. *meta,* "higher-order"), a large population that has been subdivided into several small, isolated populations due to habitat fragmentation. Originally, there were an estimated 50,000–100,000 grizzly bears south of Canada. This number has been reduced as the result of human housing communities encroaching on their home range. Bears are often killed when they come in contact with people. Currently, there are six virtually isolated subpopulations totaling about 1,000 individuals. The Yellowstone National Park population numbers 200, but the others are even smaller.

Saving metapopulations sometimes requires determining which of the populations is a source and which are sinks. A **source population** is one that is stable or growing in size, producing an excess of individuals. Often, its birthrate is higher than its death rate. The excess individuals from source populations will move into **sink populations,** where the environment is not as favorable and where the birthrate equals or is often lower than the death rate. When trying to save the northern spotted owl, conservationists determined that it was best to avoid having owls move into sink habitats, because they could not sustain the owls. The northern spotted owl reproduces successfully in old-growth rain forests of the Pacific Northwest (Fig. 47.12*b*), but not in nearby immature forests that are in the process of recovering from logging. Scientists have also recognized that competition from the barred owl is also a major threat to the recovery of the northern spotted owl.

Landscape Preservation

Grizzly bears inhabit a number of different types of ecosystems, including plains, mountains, and rivers. Saving only one of these ecosystems would not be sufficient enough to preserve the grizzly bears, however. Instead, it is necessary to save as many of the connected ecosystems that form the entire range of the grizzly. As mentioned earlier, a landscape encompasses different types of ecosystems. An area called the Greater Yellowstone Ecosystem, where bears are free to roam, has been defined. It contains millions of acres in Yellowstone National Park; state lands in Montana, Idaho, and Wyoming; five national forests; various wildlife refuges; and even private lands.

Landscape protection for one species is beneficial for the rest of the wildlife that shares the same space. The last of the contiguous 48 states' harlequin ducks, bull trout, westslope cutthroat trout, lynx, pine martens, wolverines, mountain caribou, and great gray owls are found in areas occupied by grizzly bears. Gray wolves have also returned to this territory. Grizzly bear range also overlaps with 40% of Montana's vascular plants of special conservation status.

The Edge Effect An *edge* is a transition from one habitat to another. In the case of land development, an edge may be very defined, as when a forest abuts farm fields. The edge reduces the amount of habitat typical of an ecosystem because the edge habitat is slightly different from that of the interior of the patch—this is termed the **edge effect.** For example, forest edges are brighter, warmer, drier, and windier, with more vines, shrubs, and weeds than the forest interior. Also, Figure 47.13*a* shows that a small and a large patch of habitat both have the same depth of edge; therefore, the effective habitat shrinks as a patch gets smaller.

Many popular game animals, such as turkeys and white-tailed deer, are more plentiful in the edge region of a particular area. However, today it is known that creating edges can be detrimental to wildlife because of habitat fragmentation.

The edge effect can also have a serious impact on population size. Songbird populations west of the Mississippi have been declining, and ornithologists have noticed that the nesting success of songbirds is quite low at the edge of a forest. The cause turns out to be the brown-headed cowbird, a social parasite of songbirds. Adult cowbirds prefer to feed in open agricultural areas, and they only briefly enter the forest when searching for a host nest in which to lay their eggs (Fig. 47.13*b*). Cowbirds are therefore benefited, while songbirds are harmed, by the edge effect.

Habitat Restoration

Restoration ecology is a subdiscipline of conservation biology that looks for scientific ways to return ecosystems to their natural state. Restoration ecology has three key principles. First, it should start as quickly as possible to avoid losing the remaining fragments of the original habitat.

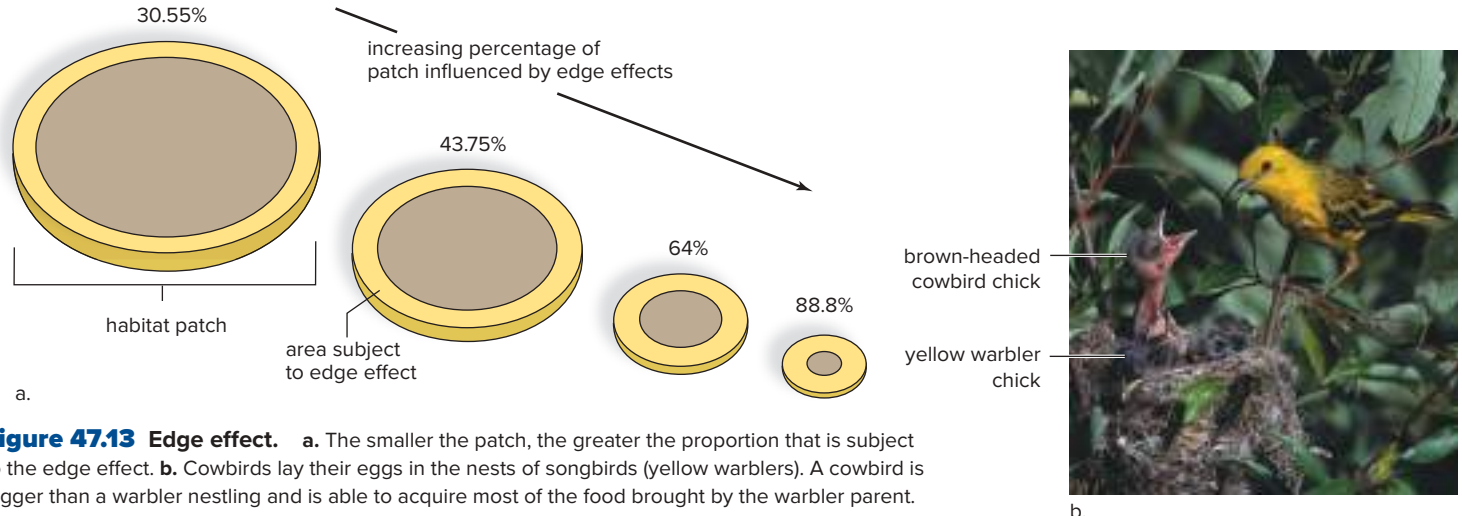


Figure 47.13 Edge effect. **a.** The smaller the patch, the greater the proportion that is subject to the edge effect. **b.** Cowbirds lay their eggs in the nests of songbirds (yellow warblers). A cowbird is bigger than a warbler nestling and is able to acquire most of the food brought by the warbler parent.
(b): Avalon.red/Alamy Stock Photo

ISTUDY

Second, once the natural histories of the species in the habitat are understood, it is best to mimic natural processes to bring about restoration. Natural processes can include using controlled burns to bring back grassland habitats, biological pest controls to rid the area of nonnative species, and bioremediation techniques to clean up pollutants.

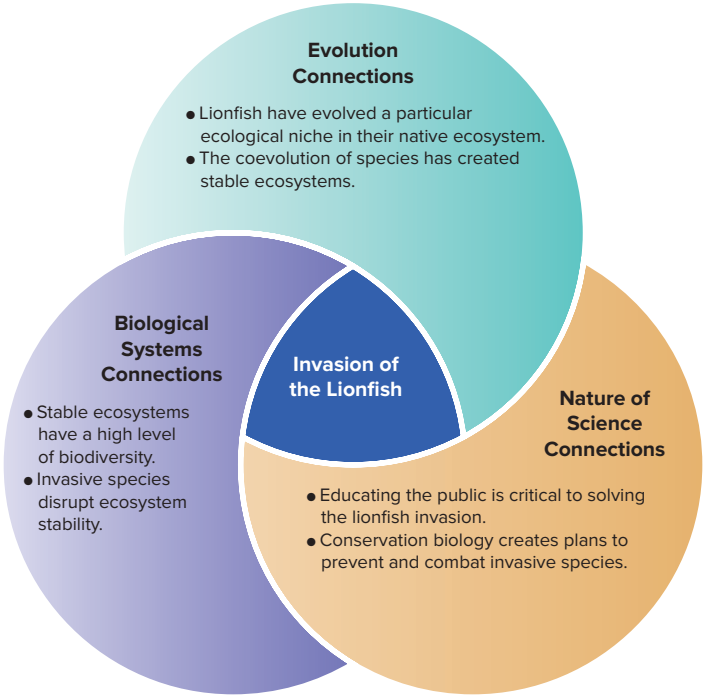
Third, the goal is sustainable development, the ability of an ecosystem to be self-sustaining while still providing services to humans.

If all of the species listed on the IUCN Red List were to go extinct within the next century, scientists believe we could approach the 75% threshold level required for a mass extinction event within the next 240 to 540 years. Because of the intricate web of ecological connections, when one species goes extinct, it creates a ripple effect in which stress is placed upon multiple other species in the ecosystem. This may in turn cause several of them to go extinct, which places stress upon multiple other species, and so on. Unfortunately, humans have created a perfect storm of conditions that have pushed the biodiversity of our planet into the beginning stages of a sixth mass extinction. Even though we may never reach the 75% threshold, the damage to biodiversity will have a ripple effect that will disrupt every ecosystem on Earth. While life has recovered after each of the previous five mass extinctions, it is unclear how devastating this will be for us humans.

Check Your Progress 47.4

1. Explain why it is more important to preserve hotspots than other areas.
2. Identify examples of keystone species.
3. List and explain the three principles of habitat restoration.

Connecting *the* Concepts



What Did You Learn?

47.1 Conservation Biology and Biodiversity

Conservation biology is the scientific study of **biodiversity** and its management for sustainable human welfare. The unequaled present rate of extinctions has drawn together scientists and environmentalists in basic and applied fields to address the problem by using the science of **bioinformatics.**

- Biodiversity is described as the number of species in a given habitat. Conservation biology looks to identify which species are **endangered** or **threatened,** making them the primary focus of conservation. Biodiversity is a combination of three levels of organization: **genetic diversity, community diversity,** and **landscape diversity,** in which the interacting ecosystems form a **landscape.**
- Conservationists have discovered that biodiversity is not evenly distributed in the biosphere, and therefore saving particular areas may protect more species than saving other areas. Areas with exceptionally high concentrations of biodiversity are termed **biodiversity hotspots.**

47.2 Value of Biodiversity

The direct value of biodiversity is seen in the economic value that a species provides to humans. Wild species are our best source of new medicines to treat human ills, and they help meet other medical needs as well.

- Wild species have agricultural value. Domesticated plants and animals are derived from wild species, and wild species can be used as a source of genes for the improvement of their genetics. Instead of synthetic pesticides, wild species can be used as biological controls, and most flowering plants benefit from animal

pollinators. Much of our food, particularly fish and shellfish, is still caught in the wild. Hardwood trees from natural forests supply us with lumber for many purposes.

- The indirect services provided by ecosystems are largely unseen and difficult to quantify, but they are absolutely necessary to our well-being. These services include the workings of biogeochemical cycles, waste recycling, the provision of fresh water, the prevention of soil erosion, and the regulation of climate. Many people enjoy living and vacationing in natural settings. Studies show that more diverse ecosystems function better than less diverse systems.

47.3 Causes of Extinction

Researchers have identified the major causes of extinction. Habitat loss is the most frequent cause, followed by the introduction of **nonnative species, pollution** and resulting diseases, **climate change,** and **overexploitation.**

- Habitat loss has occurred in all parts of the biosphere, but concern has now centered on tropical rain forests and coral reefs, where biodiversity is especially high.
- Nonnative species have been introduced into foreign ecosystems through colonization, horticulture and agriculture, and accidental transport.
- Various causes of pollution include fertilizer runoff, industrial emissions, and improper disposal of wastes.
- Climate change is contributing to species extinction, but at this point, the full impact is still being explored.
- Overexploitation is exemplified by commercial fishing, which is so efficient that the fisheries of the world are collapsing.

47.4 Conservation Techniques

To preserve species, it is necessary to preserve key habitats that contain the most species.

- **Keystone species** are ones that influence the viability of the community and are of significant importance. **Flagship species** are those that evoke an emotional response by humans and can be used to help promote conservation. Some emphasize the need to preserve biodiversity hotspots because of their richness.
- Today, it is often necessary to save **metapopulations** because of past habitat fragmentation. If this happens, it is best to determine the **source populations** and prioritize those over the **sink populations.**

When preserving landscapes, it is best to preserve large tracts of land due to the decreased biodiversity caused by habitat fragmentation and the **edge effect.**

Because many ecosystems have been degraded, habitat restoration may be necessary before sustainable development is possible. Three principles of restoration are (1) start before sources of wildlife and seeds are lost; (2) use simple biological techniques that mimic natural processes; and (3) aim for sustainable development, so that the ecosystem can persist while fulfilling the needs of humans.

Thinking Critically About Biology

1. The scale at which conservation biologists work often makes direct experimentation difficult. But computer models can assist in predicting the future of populations or ecosystems. Some scientists feel that these models are inadequate because they cannot reproduce all the variables found in the real world. If you were trying to predict the impact on songbirds of clear-cutting a portion of a forest, what information would you need to develop a good model?
2. Bioprospecting is the search for medically useful molecules derived from living organisms. The desire for monetary and medicinal gains from such discoveries can lead to the overexploitation of endangered species. However, bioprospecting can protect ecosystems, and in this way it saves many species rather than individual species. What types of living organisms would bioprospectors be most interested in?
3. Describe the biotic potential of the lionfish and explain how it is contributing to the destruction of coral reefs in the Caribbean. Propose potential solutions to minimize the impact that lionfish have upon the coral reef ecosystem in the Caribbean.

Making It Relevant to You

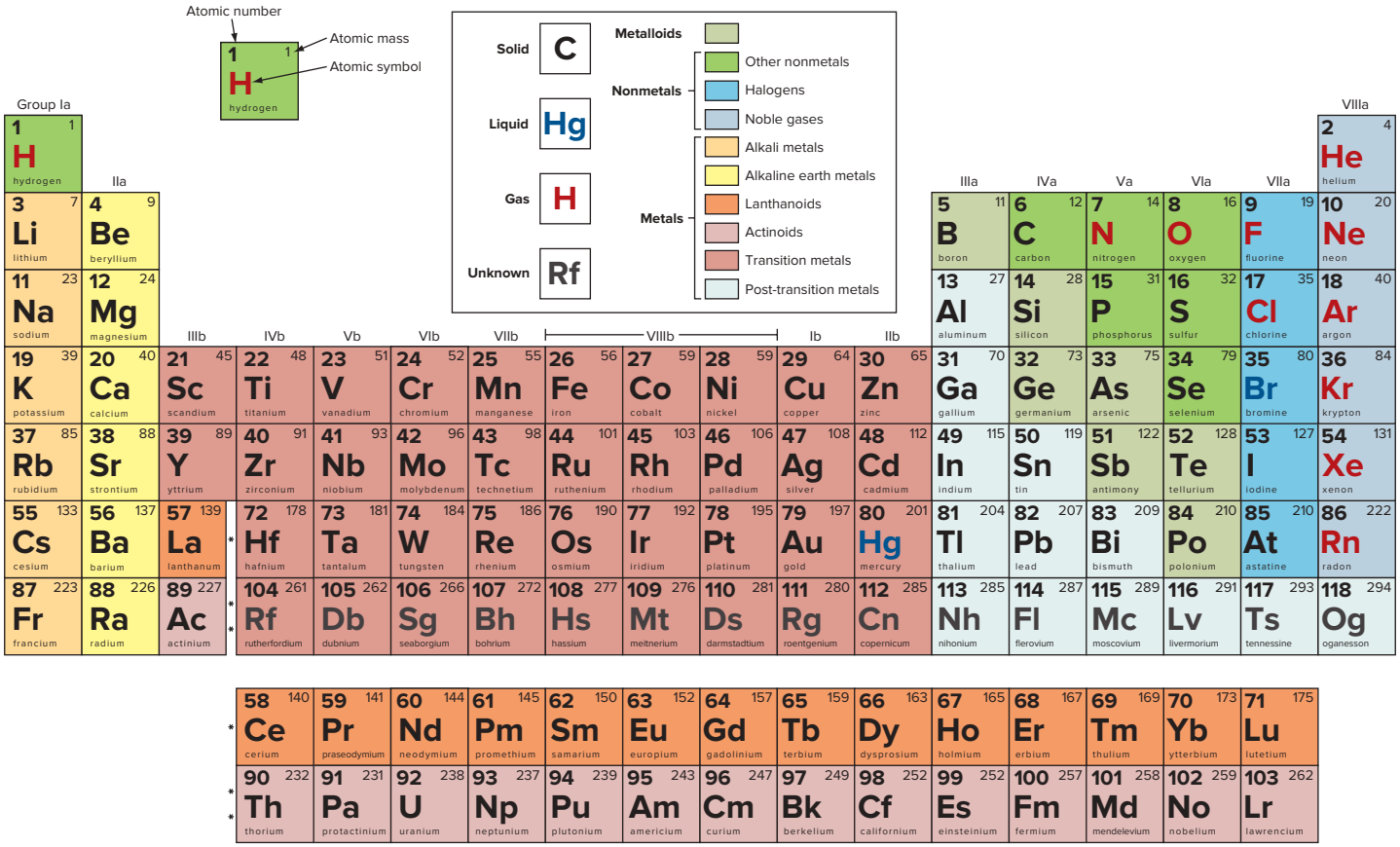
1. What is the ecological consequence of lionfish being introduced into the Caribbean?
2. If your economic livelihood was destroyed by the invasion of the lionfish, would you want to hold the people who introduced them into the Atlantic responsible?
3. Is it possible for an invasive species to play a beneficial role in the ecosystem it was introduced into?

Find answers to all of the chapter questions at connect.mheducation.com

ISTUDY

Appendix

Periodic Table of Elements



ISTUDY

Table 1: The Elements

Atomic Number	Symbol	Name	Atomic Mass	Form	Class
1	H	hydrogen	1	gas	other nonmetal
2	He	helium	4	gas	noble gases
3	Li	lithium	7	solid	alkali metals
4	Be	beryllium	9	solid	alkaline earth metals
5	B	boron	11	solid	metalloid
6	C	carbon	12	solid	other nonmetal
7	N	nitrogen	14	gas	other nonmetal
8	O	oxygen	16	gas	other nonmetal
9	F	fluorine	19	gas	halogens
10	Ne	neon	20	gas	noble gases
11	Na	sodium	23	solid	alkali metals
12	Mg	magnesium	24	solid	alkaline earth metals
13	Al	aluminum	27	solid	post-transition metals
14	Si	silicon	28	solid	metalloid
15	P	phosphorus	31	solid	other nonmetal
16	S	sulfur	32	solid	other nonmetal
17	Cl	chlorine	35	gas	halogens
18	Ar	argon	40	gas	noble gases
19	K	potassium	39	solid	alkali metals
20	Ca	calcium	40	solid	alkaline earth metals
21	Sc	scandium	45	solid	transition metal
22	Ti	titanium	48	solid	transition metal
23	V	vanadium	51	solid	transition metal
24	Cr	chromium	52	solid	transition metal
25	Mn	manganese	55	solid	transition metal
26	Fe	iron	56	solid	transition metal
27	Co	cobalt	59	solid	transition metal
28	Ni	nickel	59	solid	transition metal
29	Cu	copper	64	solid	transition metal
30	Zn	zinc	65	solid	transition metal
31	Ga	gallium	70	solid	post-transition metals
32	Ge	germanium	73	solid	metalloid
33	As	arsenic	75	solid	metalloid
34	Se	selenium	79	solid	other nonmetal
35	Br	bromine	80	liquid	halogens
36	Kr	krypton	84	gas	noble gases
37	Rb	rubidium	85	solid	alkali metals
38	Sr	strontium	88	solid	alkaline earth metals
39	Y	yttrium	89	solid	transition metal
40	Zr	zirconium	91	solid	transition metal
41	Nb	niobium	93	solid	transition metal
42	Mo	molybdenum	96	solid	transition metal
43	Tc	technetium	98	solid	transition metal
44	Ru	ruthenium	101	solid	transition metal
45	Rh	rhodium	103	solid	transition metal
46	Pd	palladium	106	solid	transition metal
47	Ag	silver	108	solid	transition metal
48	Cd	cadmium	112	solid	transition metal
49	In	indium	115	solid	post-transition metals
50	Sn	tin	119	solid	post-transition metals
51	Sb	antimony	122	solid	metalloid
52	Te	tellurium	128	solid	metalloid
53	I	iodine	127	solid	halogens
54	Xe	xenon	131	gas	noble gases
55	Cs	cesium	133	solid	alkali metals
56	Ba	barium	137	solid	alkaline earth metals
57	La	lanthanum	139	solid	lanthanoids

(*Continued*)

ISTUDY

Atomic Number	Symbol	Name	Atomic Mass	Form	Class
58	Ce	cerium	140	solid	lanthanoids
59	Pr	praseodymium	141	solid	lanthanoids
60	Nd	neodymium	144	solid	lanthanoids
61	Pm	promethium	145	solid	lanthanoids
62	Sm	samarium	150	solid	lanthanoids
63	Eu	europium	152	solid	lanthanoids
64	Gd	gadolinium	157	solid	lanthanoids
65	Tb	terbium	159	solid	lanthanoids
66	Dy	dysprosium	163	solid	lanthanoids
67	Ho	holmium	165	solid	lanthanoids
68	Er	erbium	167	solid	lanthanoids
69	Tm	thulium	169	solid	lanthanoids
70	Yb	ytterbium	173	solid	lanthanoids
71	Lu	lutetium	175	solid	lanthanoids
72	Hf	hafnium	178	solid	transition metal
73	Ta	tantalum	181	solid	transition metal
74	W	tungsten	184	solid	transition metal
75	Re	rhenium	186	solid	transition metal
76	Os	osmium	190	solid	transition metal
77	Ir	iridium	192	solid	transition metal
78	Pt	platinum	195	solid	transition metal
79	Au	gold	197	solid	transition metal
80	Hg	mercury	201	liquid	transition metal
81	Tl	thallium	204	solid	post-transition metals
82	Pb	lead	207	solid	post-transition metals
83	Bi	bismuth	209	solid	post-transition metals
84	Po	polonium	210	solid	metalloid
85	At	astatine	210	solid	halogens
86	Rn	radon	222	gas	noble gases
87	Fr	francium	223	solid	alkali metals
88	Ra	radium	226	solid	alkaline earth metals
89	Ac	actinium	227	solid	actinoids
90	Th	thorium	232	solid	actinoids
91	Pa	protactinium	231	solid	actinoids
92	U	uranium	238	solid	actinoids
93	Np	neptunium	237	solid	actinoids
94	Pu	plutonium	239	solid	actinoids
95	Am	americium	243	solid	actinoids
96	Cm	curium	247	solid	actinoids
97	Bk	berkelium	249	solid	actinoids
98	Cf	californium	252	solid	actinoids
99	Es	einsteinium	252	solid	actinoids
100	Fm	fermium	257	solid	actinoids
101	Md	mendelevium	258	solid	actinoids
102	No	nobelium	259	solid	actinoids
103	Lr	lawrencium	262	solid	actinoids
104	Rf	rutherfordium	261	unknown	transition metal
105	Db	dubnium	262	unknown	transition metal
106	Sg	seaborgium	266	unknown	transition metal
107	Bh	bohrium	272	unknown	transition metal
108	Hs	hassium	277	unknown	transition metal
109	Mt	meitnerium	276	unknown	transition metal
110	Ds	darmstadtium	281	unknown	transition metal
111	Rg	roentgenium	280	unknown	transition metal
112	Cn	copernicium	285	unknown	transition metal
113	Nh	nihonium	285	unknown	post-transition metals
114	Fl	flerovium	287	unknown	post-transition metals
115	Mc	moscovium	289	unknown	post-transition metals
116	Lv	livermorium	291	unknown	post-transition metals
117	Ts	tennessine	293	unknown	post-transition metals
118	Og	oganesson	294	unknown	post-transition metals

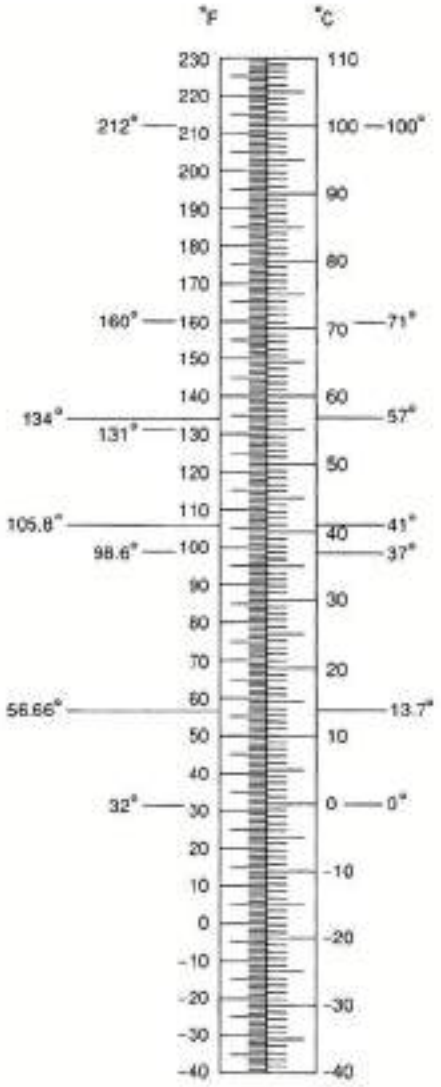
ISTUDY

The Metric System

Unit and Abbreviation	Metric Equivalent	Approximate English-to-Metric Equivalents
Length		
nanometer (nm)	= 10^{-9} m (10^{-3} μm)	
micrometer (μm)	= 10^{-6} m (10^{-3} mm)	
millimeter (mm)	= 0.001 (10^{-3}) m	
centimeter (cm)	= 0.01 (10^{-2}) m	1 inch = 2.54 cm 1 foot = 30.5 cm
meter (m)	= 100 (10^{2}) cm = 1,000 mm	1 foot = 0.30 m 1 yard = 0.91 m
kilometer (km)	= 1,000 (10^{3}) m	1 mi = 1.6 km
Weight (mass)		
nanogram (ng)	= 10^{-9} g	
microgram (μg)	= 10^{-6} g	
milligram (mg)	= 10^{-3} g	
gram (g)	= 1,000 mg	1 ounce = 28.3 g 1 pound = 454 g
kilogram (kg)	= 1,000 (10^{3}) g	= 0.45 kg
metric ton (t)	= 1,000 kg	1 ton = 0.91 t
Volume		
microliter (μl)	= 10^{-6} l (10^{-3} ml)	
milliliter (ml)	= 10^{-3} l = 1 cm^3 (cc) = 1,000 mm^3	1 tsp = 5 ml 1 fl oz = 30 ml
liter (l)	= 1,000 ml	1 pint = 0.47 l 1 quart = 0.95 l 1 gallon = 3.79 l
kiloliter (kl)	= 1,000 l	

°C	°F	
100	212	Water boils at standard temperature and pressure.
71	160	Flash pasteurization of milk
54	130	Highest recorded temperature in the world, Death Valley, July 2021
41	105.8	Average body temperature of a marathon runner in hot weather
37	98.6	Human body temperature
13.7	56.66	Human survival is still possible at this temperature.
0	32.0	Water freezes at standard temperature and pressure.

Units of Temperature



To convert temperature scales:

$$°C = \frac{(°F - 32)}{1.8}$$

$$°F = 1.8(°C) + 32$$

ISTUDY

Glossary

A

abiogenesis Origin of life from nonliving matter, such as occurred on the early Earth.

abiotic synthesis Process of chemical evolution that resulted in the formation of organic molecules (amino acids, monosaccharides, etc.) from inorganic material.

abscisic acid (ABA) Plant hormone that causes stomata to close and initiates and maintains dormancy.

abscission Dropping of leaves, fruits, or flowers from a land plant.

absolute dating Determining the age of a fossil by direct measurement, usually involving radioisotope decay.

absorption spectrum For photosynthetic pigments, a graph of how much solar radiation is absorbed versus the wavelength of light.

accessory pigment Protein that assists in the photosynthetic process by transferring energy from the photons to the central chlorophyll molecules.

acid Molecules tending to raise the hydrogen ion concentration in a solution and thus to lower its pH numerically.

acid deposition The return to Earth in rain or snow of sulfate or nitrate salts of acids produced by commercial and industrial activities.

acquired immunodeficiency syndrome (AIDS) Disease caused by the HIV virus that destroys helper T cells and macrophages of the immune system, thus preventing an immune response to pathogens; caused by unprotected sexual contact with an infected person, intravenous drug use, or transfusions of contaminated blood (rare).

actin One of two major proteins of muscle, along with myosin, that make up the thin filaments in myofibrils of muscle fibers.

actin filament Component of the cytoskeleton; plays a role in the movement of the cell and its organelles; a protein filament in a sarcomere of a muscle, its movement shortens the sarcomere, yielding muscle contraction.

action potential Electrochemical changes that take place across the axon membrane; the nerve impulse.

active immunity Ability to produce antibodies due to the immune system's response to a microorganism or a vaccine.

active site Region of an enzyme where the substrate binds and where the chemical reaction occurs.

active transport Use of a plasma membrane carrier protein to move a molecule or an ion from a region of lower concentration to one of higher concentration; it opposes equilibrium and requires energy.

adaptation Species modification in structure, function, or behavior that makes a species more suitable to its environment.

adaptive immunity Type of immunity characterized by the reaction of lymphocytes to specific antigens.

adaptive radiation Rapid evolution of several species from a common ancestor into new ecological or geographic zones.

adenine (A) One of four nitrogen-containing bases in the nucleotides composing the structure of DNA and RNA. Pairs with uracil (U) and thymine (T).

adhesion Tendency of water molecules to form hydrogen bonds with polar surfaces, such as the inside of capillaries and transport vessels.

adhesion junction Junction between cells in which the adjacent plasma membranes do not touch but are held together by intercellular filaments attached to buttonlike thickenings.

adipose tissue Connective tissue in which fat is stored.

ADP (adenosine diphosphate) Nucleotide with two phosphate groups that can accept another phosphate group and become ATP.

adrenal cortex Outer portion of the adrenal gland that secretes mineralocorticoids, such as aldosterone, and glucocorticoids, such as cortisol.

adrenal gland Gland that lies atop a kidney; the *adrenal medulla* produces the hormones epinephrine and norepinephrine, and the *adrenal cortex* produces the glucocorticoid and mineralocorticoid hormones.

adrenal medulla Inner portion of the adrenal gland that secretes the hormones epinephrine and norepinephrine.

adrenocorticotropic hormone (ACTH) Hormone secreted by the anterior lobe of the pituitary gland that stimulates activity in the adrenal cortex.

aerobic Chemical process that requires air (oxygen); phase of cellular respiration that requires oxygen.

age structure diagram In demographics, a display of the age groups of a population; a growing population has a pyramid-shaped diagram.

agglutination Clumping of red blood cells due to a reaction between antigens on red blood cell plasma membranes and antibodies in the plasma.

agnathan Group of fishes that lack jaws; namely, the lampreys and hagfishes.

agranular leukocyte Form of white blood cell that lacks spherical vesicles (granules) in the cytoplasm.

aldosterone Hormone secreted by the adrenal cortex that regulates the sodium and potassium ion balance of the blood.

allantois Extraembryonic membrane that accumulates nitrogenous wastes in the eggs of reptiles, including birds; contributes to the formation of umbilical blood vessels in mammals.

allele Alternative form of a gene; alleles occur at the same locus on homologous chromosomes.

allele frequency Relative proportion of each allele for a gene in the gene pool of a population.

allergy Immune response to substances that usually are not recognized as foreign.

allopatric speciation Model that proposes that new species arise due to an interruption of gene flow between populations that are separated geographically.

alloploidy Polyploid organism that contains the genomes of two or more different species.

allosteric site Site on an allosteric enzyme that binds an effector molecule; binding alters the activity of the enzyme.

alpine tundra Tundra near the peak of a mountain.

alternation of generations Life cycle, typical of land plants, in which a diploid sporophyte alternates with a haploid gametophyte.

altruism Social interaction that has the potential to decrease the lifetime reproductive success of the member exhibiting the behavior, but to benefit the reproductive success of another member of the society.

alveolate Group of protists that includes single-celled dinoflagellates, apicomplexans, and ciliates; alveoli support plasma membrane.

alveolus (*pl.*, alveoli) In humans, terminal, microscopic, grapelike air sac found in lungs.

Alzheimer's disease Disease of the central nervous system (brain) that is characterized by an accumulation of beta amyloid protein and neurofibrillary tangles in the hippocampus and amygdala.

AM fungi (Glomeromycota) Fungi with branching invaginations (arbuscular mycorrhizae, or AM) used to invade plant roots.

amino acid Organic molecule composed of an amino group and an acid group; covalently bonds to produce peptide molecules.

ammonia Nitrogenous end product that takes a limited amount of energy to produce but requires much water to excrete because it is toxic.

amnion Extraembryonic membrane of birds, reptiles, and mammals that forms an enclosing, fluid-filled sac.

amniote Vertebrate that produces an egg surrounded by four membranes, one of which is the amnion; amniote groups are the reptiles (including birds) and mammals.

amniotic egg Egg that has an amnion, as seen during the development of reptiles (including birds) and mammals.

amoeboid Cell that moves and engulfs debris with pseudopods.

amoebozoan Supergroup of eukaryotes that includes amoebas and slime molds and is characterized by lobe-shaped pseudopodia.

ISTUDY

amphibian Member of vertebrate class Amphibia that includes frogs, toads, and salamanders; they are tied to a watery environment for reproduction.

amygdala Part of the limbic system of the brain; associated with emotional experiences.

amyotrophic lateral sclerosis (ALS) Neurodegenerative disease affecting the motor neurons in the brain and spinal cord, resulting in paralysis and death; also known as Lou Gehrig disease.

anabolic steroid Synthetic steroid that mimics the effect of testosterone.

anabolism Chemical reaction in which smaller molecules (monomers) are combined to form larger molecules (polymers); anabolic metabolism.

anaerobic Chemical reaction that occurs in the absence of oxygen; an example is the fermentation reactions.

analogous structure Structure that has a similar function in separate lineages but differs in anatomy and ancestry.

analogy Similarity of function but not of origin.

anaphase Fourth phase of mitosis; chromosomes move toward the poles of the spindle.

anaphylactic shock Severe systemic form of anaphylaxis involving bronchiolar constriction, impaired breathing, vasodilation, and a rapid drop in blood pressure with a threat of circulatory failure.

anapsid Characteristic of a vertebrate skull in which there is no opening in the skull behind the eye socket (orbit).

ancestral traits Traits that are found in a common ancestor and its descendants.

androgen Male sex hormone (e.g., testosterone).

aneuploidy Condition in which a cell does not contain the correct number, or combinations, of chromosomes.

angina pectoris Condition characterized by thoracic pain resulting from occluded coronary arteries; may precede a heart attack.

angiogenesis Formation of new blood vessels; rapid angiogenesis is a characteristic of cancer cells.

angiosperm Flowering land plant; the seeds are borne within a fruit.

angiotensin II Hormone produced from angiotensinogen (a plasma protein) by the kidneys and lungs; raises blood pressure.

animal Multicellular, heterotrophic eukaryote that undergoes development to achieve its final form. In general, animals are mobile organisms, characterized by the presence of muscular and nervous tissue.

annelid Segmented worm, such as the earthworm and the clam worm.

annual A plant living only one year or season.

annual ring Layer of wood (secondary xylem) usually produced during one growing season.

anterior pituitary Portion of the pituitary gland that is controlled by the hypothalamus and produces six types of hormones, some of which control other endocrine glands.

antheridium (*pl.*, antheridia) Sperm-producing structure, as in the moss life cycle.

anthropoid Group of primates that includes monkeys, apes, and humans.

antibody Protein produced in response to the presence of an antigen; each antibody combines with a specific antigen.

antibody-mediated immunity Specific mechanism of defense in which plasma cells derived from B cells produce antibodies that combine with antigens.

anticodon Three-base sequence in a transfer RNA molecule base that pairs with a complementary codon in mRNA.

antidiuretic hormone (ADH) Hormone secreted by the posterior pituitary that increases the permeability of the collecting ducts in a kidney.

antigen Foreign substance, usually a protein or a polysaccharide, that stimulates the immune system to react, such as to produce antibodies.

antigen receptor Receptor protein in the plasma membrane of immune system cells whose shape allows them to combine with a specific antigen.

antigen-presenting cell (APC) Cell that displays an antigen to certain cells of the immune system so they can defend the body against that particular antigen.

anus Outlet of the digestive tube.

aorta In humans, the major systemic artery that takes blood from the heart to the tissues.

aortic body Sensory receptor in the aortic arch sensitive to the O_2, CO_2, and H^+ content of the blood.

apical dominance Influence of a terminal bud in suppressing the growth of axillary buds.

apical meristem In vascular land plants, masses of cells in the root and shoot that reproduce and elongate as primary growth occurs.

apicomplexan Group of parasitic protozoans, formerly called sporozoans, that lack mobility and form spores; now named for a unique collection of organelles.

apoptosis Programmed cell death; involves a cascade of specific cellular events leading to death and destruction of the cell.

appendicular skeleton Part of the vertebrate skeleton comprising the appendages, shoulder girdle, and hip girdle.

appendix In humans, small, tubular appendage that extends outward from the cecum of the large intestine.

aquaporin Channel protein through which water can diffuse across a membrane.

aquifer Rock layers that contain water and release it in appreciable quantities to wells or springs.

arboreal Living in trees.

archaeans (archaea) Prokaryotic organisms that are members of the domain Archaea.

archaeplastid Supergroup of eukaryotes that includes land plants and red and green algae. Developed from endosymbiotic cyanobacteria.

archegonium (*pl.*, archegonia) Egg-producing structures, as in the moss life cycle.

ardipithecine Extinct hominines of the genus *Ardipithecus* that existed between 4.5 and 5.5 MYA.

arteriole Vessel that takes blood from an artery to capillaries.

artery Blood vessel that transports blood away from the heart.

arthritis Condition characterized by an inflammation of the joints; two common forms are osteoarthritis and rheumatoid arthritis.

arthropod Group of invertebrates, with an exoskeleton and jointed appendages, such as crustaceans and insects.

artificial selection Intentional breeding of certain traits, or combinations of traits, over others to produce a desirable outcome.

ascus (*pl.*, asci) Fingerlike sac in which nuclear fusion, meiosis, and ascospore production occur during sexual reproduction of sac fungi.

asexual reproduction Reproduction that produces genetically identical offspring, requires only one parent, and does not involve gametes.

assimilation Process in which plants take up ammonia and nitrate from the soil and use it to produce proteins and nucleic acids.

associative learning Acquired ability to associate two stimuli or between a stimulus and a response.

assortative mating Mating of individuals with similar phenotypes.

asthma Condition in which bronchioles constrict and cause difficulty in breathing.

astigmatism Uneven shape of the cornea or lens of the eye; causes a distortion in the light reaching the retina of the eye.

astrocyte Nervous system cell that provides metabolic and structural support to the neurons.

asymmetry Lack of any symmetrical relationship in the morphology of an organism.

atherosclerosis Form of cardiovascular disease characterized by the accumulation of fatty materials (usually cholesterol) in the arteries.

atom Smallest particle of an element that displays the properties of the element.

atomic mass Average of atom mass units for all the isotopes of an atom.

atomic number Number of protons within the nucleus of an atom.

atomic symbol One or two letters that represent the name of an element (e.g., H stands for a hydrogen atom, and Na stands for a sodium atom).

ATP (adenosine triphosphate) Nucleotide with three phosphate groups. The breakdown of ATP into ADP + Ⓟ makes energy available for energy-requiring processes in cells.

ATP synthase complex Complex of proteins in the cristae of mitochondria and thylakoid membrane of chloroplasts that produces ATP from the diffusion of hydrogen ions across a membrane.

atria (*sing.*, atrium) Chambers; particularly upper chambers of a vertebrate heart, lying above a ventricle.

atrial natriuretic hormone (ANH) Hormone secreted by the heart that increases sodium excretion.

ISTUDY

atrioventricular valve Heart valve located between an atrium and a ventricle.
auditory communication Sound that an animal makes to send a message to another individual.
auditory tube Also called the eustachian tube; connects the middle ear to the nasopharynx for the equalization of pressure.
australopithecine (australopith) One of several species of *Australopithecus,* a genus that contains the first generally recognized humanlike hominins.
autoimmune disease Disease that results when the immune system mistakenly attacks the body's own tissues.
autonomic system Portion of the peripheral nervous system that regulates internal organs.
autophagy Breakdown of the internal structures of a eukaryotic cell by the lysosomes.
autoploidy Polyploid organism with multiple chromosome sets all from the same species.
autosome Chromosome pairs that are the same between the sexes; in humans, all but the X and Y chromosomes.
autotroph Organism that can capture energy and synthesize organic molecules from inorganic nutrients.
auxin Plant hormone regulating growth, particularly cell elongation; also called indoleacetic acid (IAA).
axial skeleton Part of the vertebrate skeleton forming the vertical support or axis, including the skull, the rib cage, and the vertebral column.
axon Elongated portion of a neuron that conducts nerve impulses, typically from the cell body to the synapse.

B

B cell Lymphocyte that matures in the bone marrow and, when stimulated by the presence of a specific antigen, gives rise to antibody-producing plasma cells.
B-cell receptor (BCR) Molecule on the surface of a B cell that binds to a specific antigen.
bacillus A rod-shaped bacterium; also a genus of bacteria, *Bacillus*.
bacteriophage Virus that infects bacteria.
bacterium (*pl.,* bacteria) Member of the domain Bacteria.
bark External part of a tree, containing cork, cork cambium, cortex, and phloem.
Barr body Dark-staining body in the cell nuclei of female mammals that contains a condensed, inactive X chromosome; named after its discoverer, Murray Barr.
base Molecules tending to lower the hydrogen ion concentration in a solution and thus raise the pH numerically.
basidium Clublike structure in which nuclear fusion, meiosis, and basidiospore production occur during sexual reproduction of club fungi.
basophil White blood cell with a granular cytoplasm; able to be stained with a basic dye.
behavior Observable, coordinated responses to environmental stimuli.
behavioral ecology Study of how natural selection shapes behavior.
benign Mass of cells derived from a single mutated cell that has repeatedly undergone cell division but has remained at the site of origin.
bicarbonate ion Ion that participates in buffering the blood, and the form in which carbon dioxide is transported in the bloodstream; chemical formula is HCO_3^-.
bilateral symmetry Body plan having two corresponding or complementary halves.
bile Secretion of the liver that is temporarily stored and concentrated in the gallbladder before being released into the small intestine, where it emulsifies fat.
binary fission Splitting of a parent cell into two daughter cells; serves as an asexual form of reproduction in bacteria.
binomial nomenclature Scientific name of an organism, the first part of which designates the genus and the second part of which designates the specific epithet.
biocultural evolution Phase of human evolution in which cultural events affect natural selection.
biodiversity Total number of species, the variability of their genes, and the communities in which they live.
biodiversity hotspot Region of the world that contains unusually large concentrations of species.
biogeochemical cycle Circulating pathway of elements such as carbon and nitrogen, involving exchange pools, storage areas, and biotic communities.
biogeography Study of the geographic distribution of organisms.
bioinformatics Area of scientific study that utilizes computer technologies to analyze large sets of data, typically in the study of genomics and proteomics.
biological clock Internal mechanism that maintains a biological rhythm in the absence of environmental stimuli.
biological species concept The concept that defines species as groups of populations that have the potential to interbreed and that are reproductively isolated from other groups.
biology The branch of science that is concerned with the study of life and living organisms.
biomagnification The accumulation of pollutants as they move up the food web.
biomass The number of organisms multiplied by their weight.
biome One of the biosphere's major communities, characterized in particular by certain climatic conditions and particular types of plants and animals.
biomolecule Organic molecule such as a protein, nucleic acid, carbohydrate, or fat.
biosphere Zone of air, land, and water at the surface of the Earth in which living organisms are found.
biotechnology Use of DNA technology and genetic engineering to alter an organism to produce a product that benefits an ecosystem or human activity.
biotic potential Maximum population growth rate under ideal conditions.
bipedalism The ability to walk on two feet, a characteristic that is unique in the animal kingdom to the hominins.
bird Endothermic reptile that has feathers and wings, is often adapted for flight, and lays hard-shelled eggs.
bivalent Homologous chromosomes, each having sister chromatids that are joined by a nucleoprotein lattice during meiosis; also called a tetrad.
bivalve Type of mollusk with a shell composed of two valves; includes clams, oysters, and scallops.
blade Broad, expanded portion of a land plant leaf that may be single or compound leaflets.
blastocoel Fluid-filled cavity of a blastula.
blastocyst Early stage of human embryonic development that consists of a hollow, fluid-filled ball of cells.
blastopore Opening into the primitive gut formed at gastrulation.
blastula Hollow, fluid-filled ball of cells occurring during animal development prior to gastrula formation.
blind spot Region of the retina, lacking rods or cones, where the optic nerve leaves the eye.
blood Fluid circulated by the heart through a closed system of vessels; type of connective tissue.
blood pressure Force of blood pushing against the inside wall of blood vessels.
body cavity In vertebrates, defined region of the body in which organs reside.
bog Wet, spongy ground in a low-lying area, usually acidic and low in organic nutrients. They often contain peat deposits and sphagnum moss.
bone Connective tissue having protein fibers and a hard matrix of inorganic salts, notably calcium salts.
bony fishes Vertebrates belonging to the class of fish called Osteichthyes that have a bony, rather than cartilaginous, skeleton.
bottleneck effect Type of genetic drift; occurs when a majority of genotypes are prevented from participating in the production of the next generation as a result of a natural disaster or human interference.
brain Ganglionic mass at the anterior end of the nerve cord; in vertebrates, the brain is located in the cranial cavity of the skull.
brain stem In mammals, the portion of the brain consisting of the medulla oblongata, pons, and midbrain.
brassinosteroids A class of plant hormone that has a wide variety of functions in the plant, including cell differentiation and protecting of the plant against cold and drought stress.
bronchiole In terrestrial vertebrates, small tube that conducts air from a bronchus to the alveoli.
bronchus (*pl.,* bronchi) In terrestrial vertebrates, branch of the trachea that leads to the lungs.
brown algae Marine photosynthetic protists with a notable abundance of xanthophyll pigments; this group includes well-known seaweeds of northern rocky shores.

ISTUDY

bryophyte A nonvascular land plant—including the mosses, liverworts, and hornworts—in which the gametophyte is dominant.
budding Asexual form of reproduction whereby a new organism develops as an outgrowth of the body of the parent.
buffer Substance or group of substances that tends to resist pH changes of a solution, thus stabilizing its relative acidity and basicity.
bulbourethral glands Male sex glands that produce pre-ejaculate fluid that neutralizes acid in the urethra.
bulk transport Movement of substances, usually large particles, across the plasma membrane using vesicles.

C

C_3 plant Plant that fixes carbon dioxide via the Calvin cycle; the first stable product of C_3 photosynthesis is a 3-carbon compound.
C_4 plant Plant that fixes carbon dioxide to produce a C_4 molecule that releases carbon dioxide to the Calvin cycle.
calcitonin Hormone secreted by the thyroid gland that decreases blood calcium level and increases the deposit of calcium in the bones.
calorie Amount of heat energy required to raise the temperature of 1 gram of water 1°C.
Calvin cycle reaction Portion of photosynthesis that takes place in the stroma of chloroplasts and can occur in the dark; it uses the products of the light reactions to reduce CO_2 to a carbohydrate.
calyx The sepals collectively; the outermost flower whorl.
CAM Crassulacean-acid metabolism; a form of photosynthesis in succulent plants that separates the light-dependent and Calvin reactions by time.
camera-type eye Type of eye found in vertebrates and certain mollusks; a single lens focuses an image on closely packed photoreceptors.
camouflage Process of hiding from predators in which an organism's behavior, form, and pattern of coloration allow it to blend into the background and prevent detection.
cancer Malignant tumor whose nondifferentiated cells exhibit loss of contact inhibition, uncontrolled growth, and the ability to invade tissue and metastasize.
capillary Microscopic blood vessel; gases and other substances are exchanged across the walls of a capillary between blood and interstitial fluid.
capsid Protective protein containing the genetic material of a virus.
capsule A form of glycocalyx that consists of a gelatinous layer; found in blue-green algae and certain bacteria.
carbaminohemoglobin Hemoglobin carrying carbon dioxide; sometimes expressed as $HbCO_2$.
carbohydrate Class of organic compounds that typically contain carbon, hydrogen, and oxygen in a 1:2:1 ratio; includes the monosaccharides, disaccharides, and polysaccharides.
carbon dioxide fixation Process by which carbon dioxide gas is attached to an organic compound; in photosynthesis, this occurs in the Calvin cycle reactions.
carbonic anhydrase Enzyme in red blood cells that speeds the formation of carbonic acid from water and carbon dioxide.
carcinogen Environmental agent that causes mutations leading to the development of cancer.
cardiac cycle One complete cycle of systole and diastole for all heart chambers.
cardiac muscle Striated, involuntary muscle tissue found only in the heart.
cardiac output Blood volume pumped by each ventricle per minute (not total output pumped by both ventricles).
cardiovascular system Organ system in animals that is involved with the circulation of the blood or hemolymph.
carnivore Consumer in a food chain that eats other animals.
carotenoid An accessory photosynthetic pigment of plants and algae that is often yellow or orange in color; consists of two classes—the xanthophylls and the carotenes.
carotid body Structure located at the branching of the carotid arteries; contains chemoreceptors sensitive to the O_2, CO_2, and H^+ content in blood.
carpel Ovule-bearing unit that is a part of the pistil of a flower.
carrier Heterozygous individual who has no apparent abnormality but can pass on an allele for a recessively inherited genetic disorder.
carrier protein Protein in the plasma membrane that combines with and transports a molecule or an ion across the plasma membrane.
carrying capacity (*K*) Largest number of organisms of a particular species that can be maintained indefinitely by a given environment.
cartilage Connective tissue in which the cells lie within lacunae embedded in a flexible, proteinaceous matrix.
cartilaginous fishes Vertebrates that belong to the class of fish called the Chondrichthyes: possess a cartilaginous, rather than bony, skeleton; include sharks, rays, and skates.
Casparian strip Layer of impermeable lignin and suberin bordering four sides of root endodermal cells; prevents water and solute transport between adjacent cells.
catabolism Metabolic process that breaks down large molecules into smaller ones; catabolic metabolism.
cataracts Condition where the lens of the eye becomes opaque, preventing the transmission of light to the retina.
catastrophism Belief, proposed by Georges Cuvier, that periods of catastrophic extinctions occurred, after which repopulation of surviving species took place, giving the appearance of change through time.
cation exchange Mechanism by which plant roots obtain minerals by exchanging hydrogen ions for other positively charged mineral ions.
cell The smallest unit of life that displays all the properties of life; composed of cytoplasm surrounded by a plasma membrane.
cell body Portion of a neuron that contains a nucleus and from which dendrites and an axon extend.
cell cycle An ordered sequence of events in eukaryotes that involves cell growth and nuclear division; consists of the stages G_1, S, G_2, and M.
cell envelope In a prokaryotic cell, the portion composed of the plasma membrane, the cell wall, and the glycocalyx.
cell plate Structure across a dividing plant cell that signals the location of new plasma membranes and cell walls.
cell recognition protein A glycoprotein in the plasma membrane that identifies self and helps the body defend itself against pathogens.
cell suspension culture Small clumps of naked plant cells grown in tissue culture that produce drugs, cosmetics, or agricultural chemicals.
cell theory One of the major theories of biology, which states that all organisms are made up of cells, cells are capable of self-reproduction, and come only from preexisting cells.
cell wall Cellular structure that surrounds a plant, protistan, fungal, or bacterial cell and maintains the cell's shape and rigidity; composed of polysaccharides.
cell-mediated immunity Specific mechanism of defense in which T cells destroy antigen-presenting cells.
cellular differentiation Process and developmental stages by which a cell becomes specialized for a particular function.
cellular respiration Metabolic reactions that use the energy from carbohydrate, fatty acid, or amino acid breakdown to produce ATP molecules.
cellular response Response to the transduction pathway in which proteins or enzymes change a signal to a format that the cell can understand, resulting in the appropriate response.
cellular slime mold Free-living amoeboid cells that feed on bacteria and yeasts by phagocytosis and aggregate to form a plasmodium that produces spores.
cellulose Polysaccharide that is the major complex carbohydrate in plant cell walls.
centipede Elongated arthropod characterized by having one pair of legs to each body segment; they may have 15 to 173 pairs of legs.
central dogma Processes that dictate the flow of information from the DNA to RNA to protein in a cell.
central nervous system (CNS) Portion of the nervous system consisting of the brain and spinal cord.
central vacuole In a plant cell, a large, fluid-filled sac that stores metabolites. During growth, it enlarges, forcing the primary cell wall to expand and the cell surface-area-to-volume ratio to increase.
centriole Cell structure, existing in pairs, that occurs in the centrosome and may help organize a mitotic spindle for chromosome movement during animal cell division.

ISTUDY

centromere Constriction where sister chromatids of a chromosome are held together.
centrosome Central microtubule-organizing center of cells. In animal cells, it contains two centrioles.
cephalization Having a well-recognized anterior head with a brain and sensory receptors.
cephalochordate Small, fishlike invertebrate that is a member of the phylum Chordata. Commonly called lancelets, they are probably the closest living relative to vertebrates.
cephalopod Type of mollusk in which the head is prominent and the foot is modified to form two arms and several tentacles; includes squids, cuttlefish, octopuses, and nautiluses.
cerebellum In terrestrial vertebrates, portion of the brain that coordinates skeletal muscles to produce smooth, graceful motions.
cerebral cortex Outer layer of cerebral hemispheres; receives sensory information and controls motor activities.
cerebral hemisphere Either of the two lobes of the cerebrum in vertebrates.
cerebrospinal fluid Fluid found in the ventricles of the brain, in the central canal of the spinal cord, and in association with the meninges.
cerebrum Largest part of the brain in mammals.
cervix Narrow end of the uterus, which leads into the vagina.
channel protein Protein that forms a channel to allow a particular molecule or ion to cross the plasma membrane.
chaparral Biome characterized by broad-leafed evergreen shrubs forming dense thickets that are adapted to being exposed to fire.
character displacement Tendency for characteristics to be more divergent when similar species belong to the same community than when they are isolated from one another.
charophyte Type of living green algae that, on the basis of nucleotide sequencing and cellular features, is most closely related to land plants.
chelicerates Arthropods (e.g., horseshoe crabs, sea spiders, arachnids) that exhibit a pair of pointed appendages used to manipulate food.
chemical energy Energy associated with the interaction of atoms in a molecule.
chemical messengers Any molecule that is used by a cell for communication.
chemiosmosis Process by which mitochondria and chloroplasts use the energy of an electron transport chain to create a hydrogen ion gradient that drives ATP formation.
chemoautotroph Organism able to synthesize organic molecules by using carbon dioxide as the carbon source and the oxidation of an inorganic substance (such as hydrogen sulfide) as the energy source.
chemoheterotroph Organism that is unable to produce its own organic molecules and therefore requires organic nutrients in its diet.
chemoreceptor Sensory receptor that is sensitive to chemical stimulation—for example, receptors for taste and smell.
chitin Strong but flexible nitrogenous polysaccharide found in the exoskeleton of arthropods and in the cell walls of fungi.
chlorophyll Green photosynthetic pigment of algae and plants that absorbs solar energy; occurs as chlorophyll *a* and chlorophyll *b*.
chlorophyte Most abundant and diverse group of green algae, including freshwater, marine, and terrestrial forms that synthesize. Chlorophytes share chemical and anatomical characteristics with land plants.
chloroplast Membrane-bound organelle in algae and plants with chlorophyll-containing membranous thylakoids; where photosynthesis takes place.
choanoflagellate The classification of protists that are most closely related to animals.
chordate Animal that has a dorsal tubular nerve cord, a notochord, pharyngeal gill pouches, and a postanal tail at some point in its life cycle; includes a few types of invertebrates (e.g., sea squirts and lancelets) and the vertebrates.
chorion Extraembryonic membrane functioning for respiratory exchange in birds and reptiles; contributes to placenta formation in mammals.
chorionic villus In placental mammals, treelike extension of the chorion, projecting into the maternal tissues at the placenta.
choroid Vascular, pigmented middle layer of the eyeball.
chromatid Following replication, a chromosome consists of a pair of sister chromatids, held together at the centromere; each chromatid is comprised of a single DNA helix.
chromatin Network of DNA strands and associated proteins observed within a nucleus of a cell.
chromosome The structure that transmits the genetic material from one generation to the next; composed of condensed chromatin; each species has a particular number of chromosomes that is passed on to the next generation.
chyme Thick, semiliquid food material that passes from the stomach to the small intestine.
chytrids (Chytridiomycota) Mostly aquatic fungi with flagellated spores that may represent the most ancestral fungal lineage.
cilia (*sing.*, cilium) Short, hairlike projections from the plasma membrane, occurring usually in large numbers.
ciliary muscle Within the ciliary body of the vertebrate eye, the muscle that controls the shape of the lens.
ciliate Complex single-celled protist that moves by means of cilia and digests food in food vacuoles.
circadian rhythm Biological rhythm with a 24-hour cycle.
circulatory system In animals, an organ system that moves substances to and from cells, usually via a heart, blood, and blood vessels.
cirrhosis Chronic, irreversible injury to liver tissue; commonly caused by frequent alcohol consumption.
citric acid cycle Cycle of reactions in mitochondria that begins with citric acid. This cycle breaks down an acetyl group and produces CO_2, ATP, NADH, and $FADH_2$; also called the Krebs cycle.
clade Evolutionary lineage consisting of an ancestral species and all of its descendants, forming a distinct branch on a cladogram.
cladistics Method of systematics that uses derived characters to determine monophyletic groups and construct cladograms.
cladogram In cladistics, a branching diagram that shows the relationship among species in regard to their shared derived characters.
class One of the categories, or taxa, used by taxonomists to group species; the taxon above the order level.
classical conditioning Type of learning whereby an unconditioned stimulus that elicits a specific response is paired with a neutral stimulus, so that the response becomes conditioned.
classification Process of naming organisms and assigning them to taxonomic groups (taxa).
cleavage Cell division without cytoplasmic addition or enlargement; occurs during the first stage of animal development.
cleavage furrow Indentation in the plasma membrane of animal cells during cell division; formation marks the start of cytokinesis.
climate Generalized weather patterns of an area, primarily determined by temperature and average rainfall.
climate change Recent changes in the Earth's climate; evidence suggests that this is primarily due to human influence, including the increased release of greenhouse gases.
climax community In ecology, a mature and stable community that results when succession has come to an end.
clonal selection theory States that the antigen selects which lymphocyte will undergo clonal expansion and produce more lymphocytes bearing the same type of receptor.
clone Offspring that is genetically identical to its parent organism.
cloning Production of identical copies. In organisms, the production of organisms with the same genes; in genetic engineering, the production of many identical copies of a gene.
closed circulatory system A type of circulatory system where blood is confined to vessels and is kept separate from the interstitial fluid.
clotting Also called coagulation, the response of the body to an injury in the vessels of the circulatory system; involves platelets and clotting proteins.
club fungi (Basidiomycota) Fungi that produce spores in club-shaped basidia within a fruiting body; includes mushrooms, shelf fungi, and puffballs.
cnidarian Invertebrate existing as either a polyp or medusa, with two tissue layers and radial symmetry.
coccus A spherical-shaped bacterium.
cochlea Spiral-shaped structure of the vertebrate inner ear containing the sensory receptors for hearing.
codominance Inheritance pattern in which both alleles of a gene are equally expressed in a heterozygote.

ISTUDY

codon Three-base sequence in messenger RNA that during translation directs the addition of a particular amino acid into a protein or directs termination of the process.

coelom Body cavity of an animal; the method by which the coelom is formed (or lack of formation) is an identifying characteristic in animal classification.

coenzyme Nonprotein organic molecule that aids the action of the enzyme to which it is loosely bound.

coevolution Mutual evolution in which two species exert selective pressures on the other species.

cofactor Nonprotein assistant required by an enzyme in order to function; many cofactors are metal ions, others are coenzymes.

cohesion Tendency of water molecules to cling to each other or to form hydrogen bonds with other water molecules.

cohesion-tension model Explanation for upward transport of water in xylem based upon transpiration-created tension and the cohesive properties of water molecules.

cohort Group of individuals having a statistical factor in common, such as year of birth, in a population study.

coleoptile Protective sheath that covers the young leaves of a seedling.

collagen Protein found in the collagen fibers of connective tissue that gives the tissue flexibility and strength.

collecting duct Duct within the kidney that receives fluid from several nephrons; the reabsorption of water occurs here.

collenchyma Plant tissue composed of cells with unevenly thickened walls; supports growth of stems and petioles.

colony Loose association of cells, each remaining independent for most functions.

comb jelly Invertebrate that resembles a jelly fish and is the largest animal to be propelled by beating cilia.

commensalism Symbiotic relationship in which one species is benefited, and the other is neither harmed nor benefited.

common ancestor Ancestor common to at least two lines of descent.

communication Signal by a sender that influences the behavior of a receiver.

community Assemblage of species interacting with one another within the same environment.

community diversity Variety of species in a particular locale, dependent on the species interactions.

compact bone Type of bone that contains osteons, consisting of concentric layers of matrix, and osteocytes in lacunae.

companion cell Plant cell in the vascular tissue of plants that metabolically supports the conducting cells of the phloem.

comparative genomics Study of genomes through the direct comparison of their genes and DNA sequences from multiple species.

competition Results when members of a species attempt to use a resource that is in limited supply.

competitive exclusion principle Theory that two species cannot occupy the same niche in the same place and at the same time.

competitive inhibition Form of enzyme inhibition where the substrate and inhibitor are both able to bind to the enzyme's active site. Only when the substrate is at the active site will product form.

complement Collective name for a series of enzymes and activators in the blood, some of which may bind to an antibody and may lead to rupture of a foreign cell.

complementary base pairing Hydrogen bonding between particular purines and pyrimidines; responsible for the structure of DNA, and some RNA, molecules.

complete digestive tract Digestive tract that has both a mouth and an anus.

compound Substance having atoms of two or more different elements in a fixed ratio.

compound eye Type of eye found in arthropods; it is composed of many independent visual units.

concentration gradient Gradual change in chemical concentration between two areas of differing concentrations.

cone Reproductive structure in conifers made up of scales bearing sporangia; pollen cones bear microsporangia, and seed cones bear megasporangia.

cone cell Photoreceptor in vertebrate eyes that responds to bright light and makes color vision possible.

conidiospore Spore produced by sac and club fungi during asexual reproduction.

conifer Member of a group of cone-bearing gymnosperm land plants that includes pine, cedar, and spruce trees.

conjugation Transfer of genetic material from one cell to another.

conjugation pilus (*pl.*, conjugation pili) In a bacterium, elongated, hollow appendage used to transfer DNA to other cells.

conjunctiva Delicate membrane that lines the eyelid, protecting the sclera.

connective tissue Type of animal tissue that binds structures together, provides support and protection, fills spaces, stores fat, and forms blood cells; adipose tissue, cartilage, bone, and blood are types of connective tissue; living cells in a nonliving matrix.

conservation biology Discipline that seeks to understand the effects of human activities on species, communities, and ecosystems and to develop practical approaches to preventing the extinction of species and the destruction of ecosystems.

consumer Organism that feeds on another organism in a food chain generally; primary consumers eat plants, and secondary consumers eat animals.

continental drift The movement of the Earth's crust by plate tectonics, resulting in the movement of continents with respect to one another.

control Sample that goes through all the steps of an experiment but does not contain the variable being tested; a standard against which the results of an experiment are checked.

convergent evolution Similarity in structure in distantly related groups generally due to similiar selective pressures in like environments.

copulation Sexual union between a male and a female.

coral reef Coral formation in shallow tropical waters that supports an abundance of diversity.

corepressor Molecule that binds to a repressor, allowing the repressor to bind to an operator in a repressible operon.

cork Outer covering of the bark of trees; made of dead cells that may be sloughed off.

cork cambium Lateral meristem that produces cork.

cornea Transparent, anterior portion of the outer layer of the eyeball.

corolla The petals, collectively; usually, the conspicuously colored flower whorl.

corpus luteum Follicle that has released an egg and increases its secretion of progesterone.

cortex In plants, ground tissue bounded by the epidermis and vascular tissue in stems and roots; in animals, outer layer of an organ, such as the cortex of the kidney or adrenal gland.

cortisol Glucocorticoid secreted by the adrenal cortex that responds to stress on a long-term basis; reduces inflammation and promotes protein and fat metabolism.

cost–benefit analysis A weighing-out of the costs and benefits (in terms of contributions to reproductive success) of a particular strategy or behavior.

cotyledon Seed leaf for embryo of a flowering plant; provides nutrient molecules for the developing plant before photosynthesis begins.

countercurrent exchange Fluids flow side-by-side in opposite directions, as in the exchange of fluids in the kidneys.

covalent bond Chemical bond in which atoms share one or more pairs of electrons.

cranial nerve Nerve that arises from the brain.

crenation In animal cells, shriveling of the cell due to water leaving the cell when the environment is hypertonic.

cristae (*sing.*, crista) Short, fingerlike projections formed by the folding of the inner membrane of mitochondria.

Cro-Magnon Common name for the first fossils to be designated *Homo sapiens*.

crossing-over Exchange of segments between nonsister chromatids of a bivalent during meiosis.

crustacean Member of a group of aquatic arthropods that contains, among others, shrimps, crabs, crayfish, and lobsters.

cryptic species Species that are very similar in appearance but are considered separate species based on other characteristics, such as behavior or genetics.

cutaneous receptor Sensory receptor of the dermis that is activated by touch, pain, pressure, and temperature.

cuticle Waxy layer covering the epidermis of plants that protects the plant against water loss and disease-causing organisms.

cyanobacterium (*pl.*, cyanobacteria) Photosynthetic bacterium that contains chlorophyll and releases oxygen; formerly called a blue-green alga.

ISTUDY

cycad Type of gymnosperm with palmate leaves and massive cones; cycads are most often found in the tropics and subtropics.
cyclic adenosine monophosphate (cAMP) ATP-related compound that acts as the second messenger in peptide hormone transduction; it initiates activity of the metabolic machinery.
cyclin Protein that cycles in quantity as the cell cycle progresses; combines with and activates the kinases that promote the events of the cycle.
cyst In protists and invertebrates, resting structure that contains reproductive bodies or embryos.
cystic fibrosis (CF) Genetic disease caused by a defect in the *CFTR* gene, which is responsible for the formation of a transmembrane chloride ion transporter; causes the mucus of the body to be viscous.
cytochrome Any of several iron-containing protein molecules that are members of the electron transport chain in photosynthesis and cellular respiration.
cytokine Type of protein secreted by a T lymphocyte that attacks viruses, virally infected cells, and cancer cells.
cytokinesis Division of the cytoplasm following mitosis or meiosis.
cytokinin Plant hormone that promotes cell division; often works in combination with auxin during organ development in plant embryos.
cytoplasm Region of a cell between the nucleus, or the nucleoid region of a bacterium, and the plasma membrane. It is composed of a semifluid solution composed of water and organic and inorganic materials.
cytoplasmic segregation Process that parcels out the maternal determinants, which play a role in development, during mitosis.
cytosine (C) One of four nitrogen-containing bases in the nucleotides composing the structure of DNA and RNA; pairs with guanine.
cytoskeleton Internal framework of the cell, consisting of microtubules, actin filaments, and intermediate filaments.
cytotoxic T cell T lymphocyte that attacks and kills antigen-bearing cells.

D

data (*sing.*, datum) Facts or information collected through observation and/or experimentation.
day-neutral plant Plant whose flowering is not dependent on day length (e.g., tomato and cucumber).
deamination Removal of an amino group (—NH_2) from an amino acid or other organic compound.
deciduous Land plant that sheds its leaves annually.
decomposer Organism, usually a bacterium or fungus, that breaks down organic matter into inorganic nutrients that can be recycled in the environment.
deductive reasoning The use of general principles to predict specific outcomes. Often uses "if . . . then" statements.
dehydration reaction Chemical reaction in which a water molecule is released during the formation of a covalent bond.
delayed allergic response Allergic response initiated at the site of the allergen by sensitized T cells, involving macrophages and regulated by cytokines.
deletion Change in chromosome structure in which the end of a chromosome breaks off or two simultaneous breaks lead to the loss of an internal segment; often causes abnormalities (e.g., cri du chat syndrome).
demographic transition Due to industrialization, a decline in the birthrate following a reduction in the death rate, so that the population growth rate is lowered.
demography Properties of the rate of growth and the age structure of populations.
denaturation Loss of a protein's or an enzyme's normal shape, so that it no longer functions; usually caused by a less than optimal pH and temperature.
dendrite Part of a neuron that sends signals toward the cell body.
dendritic cell Antigen-presenting cell of the epidermis and mucous membranes.
denitrification Conversion of nitrate or nitrite to nitrogen gas by bacteria in soil.
dense fibrous connective tissue Type of connective tissue containing many collagen fibers packed together; found in tendons and ligaments, for example.
density-dependent factor Biotic factor, such as disease or competition, that affects population size in a direct relationship to the population's density.
density-independent factor Abiotic factor, such as fire or flood, that affects population size independent of the population's density.
deoxyribose Pentose sugar found in DNA.
derived trait Structural, physiological, or behavioral trait that is present in a specific lineage and is not present in the common ancestor for several related lineages.
dermis In mammals, thick layer of the skin underlying the epidermis.
desert Ecological biome characterized by a limited amount of rainfall; deserts have hot days and cool nights.
detritivore Any organism that obtains most of its nutrients from the detritus in an ecosystem.
deuterostome Group of coelomate animals in which the second embryonic opening is associated with the mouth; the first embryonic opening, the blastopore, is associated with the anus.
development Process of regulated growth and differentiation of cells and tissues.
diabetic retinopathy Complication of diabetes that causes the capillaries in the retina to become damaged, potentially causing blindness.
diagnostic traits Characteristics that distinguish species from one another.
diaphragm In mammals, dome-shaped, muscularized sheet separating the thoracic cavity from the abdominal cavity; contraceptive device that prevents sperm from reaching the egg.
diapsid Characteristic of a vertebrate skull in which there are two openings in the skull behind the eye socket (orbit).
diarrhea Excessively frequent and watery bowel movements.
diastole Relaxation period of a heart chamber during the cardiac cycle.
diatom Golden brown alga with a cell wall in two parts, or valves; significant component of phytoplankton.
differentiation Specialization in the structure or function of a cell typically caused by the activation of specific genes.
diffusion Movement of molecules or ions from a region of higher to lower concentration; it requires no energy and tends to lead to an equal distribution (equilibrium).
digestive system Organ system of animals that is involved in the processing of nutrients and elimination of solid waste material.
dihybrid cross Cross between parents that differ in two traits.
dikaryotic Having two haploid nuclei that stem from different parent cells; during sexual reproduction, sac and club fungi have dikaryotic cells.
dinoflagellate Photosynthetic single-celled protist with two flagella, one whiplash and the other located within a groove between protective cellulose plates; significant part of phytoplankton.
dinosaur General term used to describe the large reptiles that existed prior to the start of the Cretaceous period.
dioecious Having unisexual flowers or cones, with the male flowers or cones confined to certain land plants and the female flowers or cones of the same species confined to different plants.
diploid (2n) Cell condition in which two of each type of chromosome are present.
diplomonad Protist that has modified mitochondria, two equal-sized nuclei, and multiple flagella.
directional selection Outcome of natural selection in which an extreme phenotype is favored, usually in a changing environment.
disaccharide Sugar that contains two monosaccharide units (e.g., maltose).
disruptive selection Outcome of natural selection in which the two extreme phenotypes are favored over the average phenotype, leading to more than one distinct form.
distal convoluted tubule Final portion of a nephron that joins with a collecting duct; associated with tubular secretion.
diverge Process by which a new evolutionary path begins; on a phylogenetic tree, this is indicated by branching lines.
DNA (deoxyribonucleic acid) Nucleic acid polymer produced from covalent bonding of nucleotide monomers that contain the sugar deoxyribose; the genetic material of nearly all organisms.

ISTUDY

DNA fingerprinting Use of variations in the number of small segments of DNA to identify an individual or species.
DNA helicase Enzyme involved in DNA replication that is responsible for separating the DNA strands by breaking the hydrogen bonds between the strands.
DNA ligase Enzyme that links DNA fragments; used during production of recombinant DNA to join foreign DNA to vector DNA.
DNA microarray Glass or plastic slide containing thousands of single-stranded DNA fragments arranged in an array (grid); used to detect and measure gene expression; also called gene chips.
DNA polymerase During replication, an enzyme that joins the nucleotides complementary to a DNA template.
DNA primase Enzyme involved in DNA replication that adds short sequences of RNA (a primer) to start the replication process.
DNA repair enzyme One of several enzymes that restore the original base sequence in an altered DNA strand.
DNA replication Synthesis of a new DNA double helix prior to mitosis and meiosis in eukaryotic cells and during prokaryotic fission in prokaryotic cells.
domain Largest of the categories, or taxa, used by taxonomists to group species; the three domains are Archaea, Bacteria, and Eukarya.
domain Archaea One of the three domains of life; contains prokaryotic cells that often live in extreme habitats and have unique genetic, biochemical, and physiological characteristics; its members are sometimes referred to as *archaea*.
domain Bacteria One of the three domains of life; contains prokaryotic cells that differ from archaea because they have their own unique genetic, biochemical, and physiological characteristics.
domain Eukarya One of the three domains of life, consisting of organisms with eukaryotic cells; includes protists, fungi, plants, and animals.
dominance hierarchy Organization of animals in a group that determines the order in which the animals have access to resources.
dominant allele Allele that exerts its phenotypic effect in the heterozygote; it masks the expression of the recessive allele.
dormancy In plants, a cessation of growth under conditions that seem appropriate for growth.
dorsal root ganglion Mass of sensory neuron cell bodies located in the dorsal root of a spinal nerve.
double fertilization In flowering plants, one sperm nucleus unites with the egg nucleus, and a second sperm nucleus unites with the polar nuclei of an embryo sac.
double helix Double spiral; describes the three-dimensional shape of DNA.
doubling time Number of years it takes for a population to double in size.
duodenum First part of the small intestine, where chyme enters from the stomach.
duplication Change in chromosome structure in which a particular segment is present more than once in the same chromosome.

E

ecdysozoan Protostome characterized by periodic molting of its exoskeleton. Includes the roundworms and arthropods.
echinoderm Group of invertebrates such as sea stars, sea urchins, and sand dollars; characterized by radial symmetry and a water vascular system.
ecological niche Role an organism plays in its community, including its habitat and its interactions with other organisms.
ecological pyramid Visual depiction of the biomass, number of organisms, or energy content of various trophic levels in a food web—from the producer to the final consumer populations.
ecological release In an ecosystem, the freedom of a species to expand its use of available resources due to an elimination of competition.
ecological succession The gradual change in the makeup of a community following a disturbance (secondary succession) or beginning with the creation of new soil (primary succession).
ecology Study of the interactions of organisms with other organisms and with the physical and chemical environment.
ecosystem Biological community together with the associated abiotic environment; characterized by a flow of energy and a cycling of inorganic nutrients.
ectoderm Outermost primary tissue layer of an animal embryo; gives rise to the nervous system and the outer layer of the integument.
ectotherm Organism having a body temperature that varies according to the environmental temperature.
edge effect Phenomenon in which the edges around a landscape patch provide a slightly different habitat than the favorable habitat in the interior of the patch.
effector Muscle or gland that receives signals from motor fibers and thereby allows an organism to respond to environmental stimuli.
egg Also called an ovum; haploid cell that is usually fertilized by a sperm to form a diploid zygote.
El Niño–Southern Oscillation (ENSO) Warming of water in the eastern Pacific equatorial region such that the Humboldt Current is displaced, with possible negative results, such as reduction in marine life.
elastic cartilage Type of cartilage composed of elastic fibers, allowing greater flexibility.
elastin Protein found in elastic fibers of connective tissue that gives the connective tissue flexibility.
electrocardiogram (ECG) Recording of the electrical activity associated with the heartbeat.
electron Negative subatomic particle, moving about in an energy level around the nucleus of an atom that is considered to have no atomic mass units.
electron shell The average location, or energy level, of an electron in an atom. Often drawn as concentric circles around the nucleus.
electron transport chain (ETC) Process in a cell that involves the passage of electrons along a series of membrane-bound electron carrier molecules from a higher to lower energy level; the energy released is used for the synthesis of ATP.
electronegativity The ability of an atom to attract electrons toward itself in a chemical bond.
element Substance that cannot be broken down into substances with different properties; composed of only one type of atom.
elongation Middle stage of translation in which additional amino acids specified by the mRNA are added to the growing polypeptide.
embryo Stage of a multicellular organism that develops from a zygote before it becomes free-living; in seed plants, the embryo is part of the seed.
embryo sac Female gametophyte (megagametophyte) of flowering plants.
embryonic development In humans, the first 2 months of development following fertilization, during which the major organs are formed.
embryonic disk During human development, flattened area during gastrulation from which the embryo arises.
emerging diseases An infectious disease, usually caused by a virus or bacterium, that has not previously been detected in humans.
emerging viruses Newly identified viruses that are becoming more prominent, usually because they cause serious disease.
emphysema Disorder of the respiratory system, specifically the lungs, that is characterized by damage to the alveoli, thus reducing the ability to exchange gases with the external environment.
endangered species Species that is in peril of immediate extinction throughout all or most of its range (e.g., monarch butterfly, whooping crane).
endergonic reaction Chemical reaction that requires an input of energy; opposite of exergonic reaction.
endocrine gland Ductless organ that secretes hormone(s) into the bloodstream.
endocrine system Organ system involved in the coordination of body activities; uses hormones as chemical signals secreted into the bloodstream.
endocytosis Process by which substances are moved into the cell from the environment; includes phagocytosis, pinocytosis, and receptor-mediated endocytosis.
endoderm Innermost primary tissue layer of an animal embryo that gives rise to the linings of the digestive tract and associated structures.
endodermis Internal plant root tissue forming a boundary between the cortex and the vascular cylinder.
endomembrane system Cellular system that consists of the nuclear envelope, endoplasmic reticulum, Golgi apparatus, and vesicles.

ISTUDY

endometrium Mucous membrane lining the interior surface of the uterus.
endoplasmic reticulum (ER) System of membranous saccules and channels in the cytoplasm, often with attached ribosomes.
endoskeleton Protective internal skeleton that is found in echinoderms and vertebrates.
endosperm In flowering plants, nutritive storage tissue that is derived from the union of a sperm nucleus and polar nuclei in the embryo sac.
endospore Spore formed within a cell; certain bacteria form endospores.
endosymbiotic theory Explanation of the evolution of eukaryotic organelles by phagocytosis of prokaryotes.
endotherm Organism in which maintenance of a constant body temperature is independent of the environmental temperature.
energy Capacity to do work and bring about change; occurs in a variety of forms.
energy of activation (E_a) Energy that must be added in order for molecules to react with one another.
enhancer DNA sequence that acts as a regulatory element to increase the level of transcription when regulatory proteins, such as transcription activators, bind to it.
entropy Measure of disorder or randomness in a system.
enzymatic protein Protein that catalyzes a specific reaction; may be found in the plasma membrane or the cytoplasm of the cell.
enzyme Organic catalyst, usually a protein, that speeds a reaction in cells due to its particular shape.
enzyme inhibition Means by which cells regulate enzyme activity; may be competitive or noncompetitive inhibition.
eosinophil White blood cell containing cytoplasmic granules that stain with acidic dye.
epidermal tissue Exterior tissue, usually one cell thick, of leaves, young stems, roots, and other parts of plants.
epidermis In mammals, the outer, protective layer of the skin; in plants, tissue that covers roots, leaves, and stems of nonwoody or young woody plants.
epididymis Location of sperm maturation in an adult human; located just outside of the testis.
epigenetic inheritance An inheritance pattern in which a nuclear gene has been modified but the changed expression of the gene is not permanent over many generations; the transmission of genetic information by means that are not based on the coding sequences of a gene.
epinephrine Hormone secreted by the adrenal medulla in times of stress; adrenaline.
epiphyte Plant that takes its nourishment from the air because its placement in other plants gives it an aerial position.
epistatic interaction Pattern of genetic inheritance where two or more genes interact to control a phenotype. Often associated with metabolic pathways.
epithelial tissue Tissue that lines organs and body cavities.
erythropoietin (EPO) Hormone produced by the kidneys that speeds red blood cell formation.
esophagus Muscular tube for moving swallowed food from the pharynx to the stomach.
essential nutrient In plants, substance required for normal growth, development, or reproduction; in humans, a nutrient that cannot be produced in sufficient quantities by the body and must be obtained from the diet.
estrogen Female sex hormone that helps maintain sexual organs and secondary sex characteristics.
estuary Portion of the ocean located where a river enters and fresh water mixes with salt water. This biome is considered the nursery of the marine ecosystem.
ethylene Plant hormone that causes ripening of fruit and is involved in abscission.
etiolate The yellowing of leaves and increase in shoot length that occur when a plant is grown in the dark.
euchromatin Chromatin with a lower level of compaction and therefore accessible for transcription.
eudicot (Eudicotyledone) Flowering plant group; members have two embryonic leaves (cotyledons), net-veined leaves, vascular bundles in a ring, flower parts in fours or fives and their multiples, and other characteristics.
euglenid Flagellated and flexible freshwater single-celled protist that usually contains chloroplasts and has a semirigid cell wall.
eukaryotic cell (eukaryote) Type of cell that has a membrane-bound nucleus and membranous organelles; found in organisms within the domain Eukarya.
Eumetazoa Taxon of animals that possess true tissues, germ layers during embryonic development, and an increase in cellular specialization. All animals except the Porifera (sponges) are eumetazoans.
euploidy Condition in which a cell contains the correct number, and combinations, of chromosomes.
eutrophication Enrichment of water by inorganic nutrients used by phytoplankton. Often, over-enrichment caused by human activities leads to excessive bacterial growth and oxygen depletion.
evergreen Land plant that sheds leaves over a long period, so some leaves are always present.
evolution Genetic change in a species over time, resulting in the development of genetic and phenotypic differences that are the basis of natural selection; descent of organisms from a common ancestor.
evolutionary species concept Every species has its own evolutionary history, which is partly documented in the fossil record.
ex vivo gene therapy Gene therapy in which cells are removed from an organism, and DNA is injected to correct a genetic defect; the cells are returned to the organism to treat a disease or disorder.
excavate Supergroup of eukaryotes that includes euglenids, kinetoplastids, parabasalids, and diplomonads.
excretion Elimination of metabolic wastes by an organism at exchange boundaries such as the plasma membrane of single-celled organisms and excretory tubules of multicellular animals.
exergonic reaction Chemical reaction that releases energy; opposite of endergonic reaction.
exocrine gland Gland that secretes its product to an epithelial surface directly or through ducts.
exocytosis Process in which an intracellular vesicle fuses with the plasma membrane, so that the vesicle's contents are released outside the cell.
exon Segment of mRNA containing the protein-coding portion of a gene that remains within the mRNA after splicing has occurred.
exophthalmos Condition associated with an enlargement of the thyroid gland accompanied by an abnormal protrusion of the eyes.
exoskeleton Protective external skeleton, as in arthropods.
experiment A test of a hypothesis that examines the influence of a single variable. Often involves both control and test groups.
experimental design Methodology by which an experiment will seek to support the hypothesis.
experimental variable Factor of the experiment being tested.
expiration Act of expelling air from the lungs; exhalation.
exponential growth Growth of a population in which there is a rapid increase over a short period of time due to an increase in the number of reproductive females in the population. This produces a J-shaped growth curve.
extant Species, or other levels of taxa, that are still living.
external respiration Exchange of oxygen and carbon dioxide between alveoli of the lungs and blood.
extinction Total disappearance of a species or higher group.
extracellular matrix (ECM) Nonliving substance secreted by some animal cells; is composed of protein and polysaccharides.
extraembryonic membrane Membrane that is not a part of the embryo but is necessary to the continued existence and health of the embryo.

F

facilitated transport Passive transfer of a substance into or out of a cell along a concentration gradient by a process that requires a protein carrier.
facultative anaerobe Prokaryote that is able to grow in either the presence or the absence of gaseous oxygen.

ISTUDY

FAD Flavin adenine dinucleotide; a coenzyme of oxidation-reduction that becomes $FADH_2$ as oxidation of substrates occurs in the mitochondria during cellular respiration; FAD then delivers electrons to the electron transport chain.

fall overturn Mixing process that occurs in fall in stratified lakes, whereby oxygen-rich top waters mix with nutrient-rich bottom waters.

family One of the categories, or taxa, used by taxonomists to group species; the taxon located above the genus level.

farsighted (hyperopic) Condition in which an individual cannot focus on objects closely; caused by the focusing of the image behind the retina of the eye.

fat Organic molecule that contains glycerol and three fatty acids; energy storage molecule.

fatty acid Molecule that contains a hydrocarbon chain and ends with an acid group.

fermentation Anaerobic breakdown of glucose that results in a gain of two ATP and end products, such as alcohol and lactate; occurs in the cytoplasm of cells.

fern Member of a group of land plants that have large fronds; in the sexual life cycle, the independent gametophyte produces flagellated sperm, and the vascular sporophyte produces windblown spores.

fertilization Fusion of sperm and egg nuclei, producing a zygote that develops into a new individual.

fibroblast Cell found in loose connective tissue that synthesizes collagen and elastic fibers in the matrix.

fibrocartilage Cartilage with a matrix of strong collagenous fibers.

fibrous root system In most monocots, a mass of similarly sized roots that cling to the soil.

filament End-to-end chains of cells that form as cell division occurs in only one plane; in plants, the elongated stalk of a stamen.

fimbria (*pl.,* fimbriae) Small, bristlelike fiber on the surface of a bacterial cell, which attaches bacteria to a surface; also fingerlike extension from the oviduct near the ovary.

fin In fishes and other aquatic animals, membranous, winglike, or paddlelike process used to propel, balance, or guide the body.

fishes Aquatic, gill-breathing vertebrates that usually have fins and skin covered with scales; fishes were among the earliest vertebrates that evolved.

fitness Ability of an organism to reproduce and pass its genes to the next fertile generation; measured against the ability of other organisms to reproduce in the same environment.

five-kingdom system System of classification that contains the kingdoms Monera, Protista, Plantae, Animalia, and Fungi.

fixed action pattern (FAP) Innate behavior pattern that is stereotyped, spontaneous, independent of immediate control, genetically encoded, and independent of individual learning.

flagellum (*pl.,* flagella) Long, slender extension used for locomotion by some bacteria, protozoans, and sperm.

flagship species Species that evoke a strong emotional response in humans; charismatic, cute, regal (e.g., lions, tigers, dolphins, pandas).

flatworms Invertebrates such as planarians and tapeworms with extremely thin bodies, a three-branched gastrovascular cavity, and a ladder-type nervous system.

flower Reproductive organ of a flowering plant, consisting of several kinds of modified leaves arranged in concentric rings and attached to a modified stem called the receptacle.

fluid-mosaic model Model for the plasma membrane based on the changing location and pattern of protein molecules in a fluid phospholipid bilayer.

follicle Structure in the ovary of animals that contains an oocyte; site of oocyte production.

follicle-stimulating hormone (FSH) Hormone released by the anterior pituitary; in males it promotes the production of sperm; in females it promotes the development of the follicle in the ovary.

follicular phase First half of the ovarian cycle, during which the follicle matures and much estrogen (and some progesterone) is produced.

food chain The order in which one population feeds on another in an ecosystem, thereby showing the flow of energy from a detritivore (detrital food chain) or a producer (grazing food chain) to the final consumer.

food web In ecosystems, a complex pattern of interconnected food chains that represent the feeding relationships between the organisms in the community.

foraminiferan A protist bearing a calcium carbonate test with many openings through which pseudopods extend.

formed elements Portion of the blood that consists of erythrocytes, leukocytes, and platelets (thrombocytes).

formula A group of symbols and numbers used to express the composition of a compound.

fossil Any past evidence of an organism that has been preserved in the Earth's crust.

founder effect Cause of genetic drift due to colonization by a limited number of individuals who, by chance, have different genotype and allele frequencies than the parent population.

fovea centralis Region of the retina consisting of densely packed cones; responsible for the greatest visual acuity.

frameshift mutation Insertion or deletion of at least one base, so that the reading frame of the corresponding mRNA changes.

free energy Energy in a system that is capable of performing work.

frond Leaf of a fern, palm, or cycad.

fruit Flowering plant structure consisting of one or more ripened ovaries that usually contain seeds.

fruiting body Spore-producing and spore-disseminating structure found in sac and club fungi.

functional genomics Study of gene function at the genome level. It involves the study of many genes simultaneously and the use of DNA microarrays.

functional group Specific cluster of atoms attached to the carbon skeleton of organic molecules that enters into reactions and behaves in a predictable way.

fungus (*pl.,* fungi) Eukaryotic, saprotrophic decomposer; the body is made up of filaments called hyphae that form a mass called a mycelium.

G

G protein A protein that binds GTP molecules and acts as an intermediary between the receptor and a protein in the cell membrane that activates the signal transduction pathway.

gallbladder Organ attached to the liver that stores and concentrates bile.

gamete Haploid sex cell (e.g., egg or sperm).

gametogenesis Development of the male and female sex gametes.

gametophyte Haploid generation of the alternation-of-generations life cycle of a plant; produces gametes that unite to form a diploid zygote.

ganglion (*pl.,* ganglia) Collection or bundle of neuron cell bodies, usually outside the central nervous system.

gap junction Junction between cells formed by the joining of two adjacent plasma membranes; it lends strength and allows ions, sugars, and small molecules to pass between cells.

gastropod Mollusk with a broad, flat foot for crawling (e.g., snails and slugs).

gastrovascular cavity Blind digestive cavity in animals that have a sac body plan.

gastrula Stage of animal development during which the germ layers form, at least in part, by invagination.

gastrulation Formation of a gastrula from a blastula; characterized by an invagination to form cell layers of a caplike structure.

gene Unit of heredity existing as alleles on the chromosomes; in diploid organisms, typically two alleles are inherited—one from each parent.

gene cloning DNA cloning to produce many identical copies of the same gene.

gene expression Transfer of information from the genetic material (DNA) to RNA, and then onto a protein.

gene flow Sharing of genes between two populations through interbreeding.

gene mutation Altered gene whose sequence of bases differs from the original sequence.

gene pharming Production of pharmaceuticals using transgenic organisms, usually agricultural animals.

gene pool Total of the alleles of all the individuals in a population.

gene therapy Correction of a detrimental mutation by the insertion of DNA sequences into the genome of a cell.

ISTUDY

genetic code Universal code that has existed for eons and allows for conversion of DNA and RNA's chemical code to a sequence of amino acids in a protein. Each codon consists of three bases that stand for one of the 20 amino acids found in proteins or directs the termination of translation.

genetic diversity Genetic variation that exists among members of a population.

genetic drift Mechanism of evolution due to random changes in the allelic frequencies of a population; more likely to occur in small populations or when only a few individuals of a large population reproduce.

genetic engineering Alterations to the genome of an organism, usually for the purpose of producing an enhancement of a trait or a product beneficial to humans.

genetic profile An individual's genome, including any possible mutations.

genetic recombination Process in which chromosomes are broken and rejoined to form novel combinations; in this way, offspring receive alleles in combinations different from their parents.

genetically modified organism (GMO) Organism whose genetic material has been altered or enhanced using DNA technology.

genome editing Form of DNA targeting that utilizes endoncleases to target specific sequences of DNA for the addition or removal of nucleotides.

genomics Area of study that examines the genome of a species or group of species.

genotype Genes of an organism for a particular trait or traits; often designated by letters—for example, *BB* or *Aa*.

genus One of the categories, or taxa, used by taxonomists to group species; contains those species that are most closely related through evolution.

geologic timescale History of the Earth based on the fossil record and divided into eras, periods, and epochs.

germ cells During zygote development, cells that are set aside from the somatic cells and that will eventually undergo meiosis to produce gametes.

germ layer Primary tissue layer of a vertebrate embryo—namely, ectoderm, mesoderm, or endoderm.

germinate Beginning of growth of a seed, spore, or zygote, especially after a period of dormancy.

gibberellin Plant hormone promoting increased stem growth; also involved in flowering and seed germination.

gills Respiratory organs of most aquatic animals; in fish, an outward extension of the pharynx.

ginkgo Member of phylum Ginkgophyta; maidenhair tree.

girdling Removing a strip of bark from around a tree.

gland Epithelial cell or group of epithelial cells that are specialized to secrete a substance.

glaucoma Condition in which the fluid in the eye (aqueous humor) accumulates, increasing pressure in the eye and damaging nerve fibers.

global warming Predicted increase in the Earth's temperature due to human activities that promote the greenhouse effect.

glomerular capsule Cuplike structure that is the initial portion of a nephron.

glomerular filtration Movement of small molecules from the glomerulus into the glomerular capsule due to the action of blood pressure.

glomerulus Capillary network within the glomerular capsule of a nephron.

glottis Opening for airflow in the larynx.

glucagon Hormone produced by the pancreas that stimulates the liver to break down glycogen, thus raising blood glucose levels.

glucocorticoid Type of hormone secreted by the adrenal cortex that influences carbohydrate, fat, and protein metabolism; *see also* cortisol.

glucose Six-carbon monosaccharide; used as an energy source during cellular respiration and as a monomer of the structural polysaccharides.

glycerol Three-carbon carbohydrate with three hydroxyl groups attached; a component of fats and oils.

glycocalyx Gel-like coating outside the cell wall of a bacterium. If compact, it is called a capsule; if diffuse, it is called a slime layer. This coating helps protect the bacterium against drying out and evade the host's immune system.

glycogen Storage polysaccharide found in animals; composed of glucose molecules joined in a linear fashion but having numerous branches.

glycolipid Lipid in plasma membranes that contains an attached carbohydrate chain; assembled in the Golgi apparatus.

glycolysis Anaerobic breakdown of glucose that results in a gain of two ATP and the production of pyruvate; occurs in the cytoplasm of cells.

glycoprotein Protein in plasma membranes that has an attached carbohydrate chain; assembled in the Golgi apparatus.

gnathostome Term used to describe any vertebrate with jaws.

gnetophyte Member of one of the four phyla of gymnosperms; Gnetophyta has only three living genera, which differ greatly from one another (e.g., *Welwitschia* and *Ephedra*).

golden brown algae Single-celled organism that contains pigments, including chlorophyll *a* and *c* and carotenoids, that produce its color.

Golgi apparatus Organelle consisting of sacs and vesicles that processes, packages, and distributes molecules about or from the cell.

gonad Organ that produces gametes; the ovary produces eggs, and the testis produces sperm.

gonadotropic hormone Substance secreted by the anterior pituitary that regulates the activity of the ovaries and testes; principally, follicle-stimulating hormone (FSH) and luteinizing hormone (LH).

Gonadotropin-releasing hormone (GnRH) Hormone released by the hypothalamus that influences the secretion of hormones by the anterior pituitary, thus regulating the activity of the testes.

G protein A protein that binds GTP molecules and acts as an intermediary between the receptor and a protein in the cell membrane that activates the signal transduction pathway.

granular leukocyte Form of white blood cell that contains spherical vesicles (granules) in its cytoplasm. Examples are neutrophils, eosinophils, and basophils.

granum (*pl.*, grana) Stack of chlorophyll-containing thylakoids in a chloroplast.

grassland Biome characterized by rainfall greater than 25 cm/yr, grazing animals, and warm summers; includes the prairie of the U.S. Midwest and the African savanna.

gravitational equilibrium Maintenance of balance when the head and body are motionless.

gravitropism Growth response of roots and stems of plants to the Earth's gravity; roots demonstrate positive gravitropism, and stems demonstrate negative gravitropism.

gray matter Nonmyelinated axons and cell bodies in the central nervous system.

green algae Members of a diverse group of photosynthetic protists; contain chlorophylls *a* and *b* and have other biochemical characteristics like those of plants.

greenhouse effect Reradiation of solar heat toward the Earth, caused by an atmosphere that allows the sun's rays to pass through but traps the heat in the same manner as the glass of a greenhouse.

greenhouse gases Gases in the atmosphere, such as carbon dioxide, methane, water vapor, ozone, and nitrous oxide, that are involved in the greenhouse effect.

ground tissue Tissue that constitutes most of the body of a plant; consists of parenchyma, collenchyma, and sclerenchyma cells that function in storage, basic metabolism, and support.

growth factor A hormone or chemical, secreted by one cell, that may stimulate or inhibit growth of another cell or cells.

growth hormone (GH) Substance secreted by the anterior pituitary; controls size of an individual by promoting cell division, protein synthesis, and bone growth.

guanine (G) One of four nitrogen-containing bases in the nucleotides composing the structure of DNA and RNA; pairs with cytosine.

guard cell One of two cells that surround a leaf stoma; changes in the turgor pressure of these cells cause the stoma to open or close.

guttation Liberation of water droplets from the edges and tips of leaves, resulting from root pressure.

gymnosperm Type of woody seed plant in which the seeds are not enclosed by fruit and are usually borne in cones, such as those of the conifers.

H

habitat Place where an organism lives and is able to survive and reproduce.

hair follicle Tubelike depression in the skin in which a hair develops.

halophile Type of archaean that lives in extremely salty habitats.

ISTUDY

haploid (n) Cell condition in which only one of each type of chromosome is present.

Hardy-Weinberg equilibrium (HWE) Genotypic and allele frequencies will remain constant (no evolution will occur) in a population from generation to generation when there are no other evolutionary influences. Influences include nonrandom mating, selection, migration, and genetic drift.

Hardy-Weinberg principle A mathematical model that proposes that the genotypic frequencies of a nonevolving population can be described by the expression $p^2 + 2pq + q^2$.

heart Muscular organ whose contraction causes blood to circulate in the body of an animal.

heart attack Damage to the myocardium due to blocked circulation in the coronary arteries; myocardial infarction.

heat Type of kinetic energy associated with the random motion of molecules.

helper T cell Secretes lymphokines, which stimulate all kinds of immune cells.

heme Iron-containing group found in hemoglobin.

hemizygous Possessing only one allele for a gene in a diploid organism; males are hemizygous for genes on the X chromosome.

hemocoel Residual coelom found in arthropods, which is filled with hemolymph.

hemoglobin Iron-containing respiratory pigment occurring in vertebrate red blood cells and in the blood plasma of some invertebrates.

hemolymph Circulatory fluid that is a mixture of blood and interstitial fluid; seen in animals that have an open circulatory system, such as mollusks and arthropods.

hemolysis The rupturing or bursting of a red blood cell as the result of being placed in a hypotonic solution, which causes water to enter the cell.

hepatitis Inflammation of the liver. Viral hepatitis occurs in several forms.

herbaceous stem Nonwoody stem; herbaceous plants tend to die back to ground level at the end of the growing season.

herbivore Primary consumer in a grazing food chain; a plant eater.

hermaphroditic Type of animal that has both male and female sex organs.

heterochromatin Highly compacted chromatin that is not accessible for transcription.

heterosporous Seed plant that produces two types of spores—microspores and megaspores. A plant that produces only one type of spore is *homosporous*.

heterotroph Organism that cannot synthesize needed organic compounds from inorganic substances and therefore must take in organic food.

heterozygote advantage Situation in which individuals heterozygous for a trait have a selective advantage over those who are homozygous dominant or recessive; an example is sickle-cell disease.

heterozygous Possessing unlike alleles for a particular trait.

hexose Any monosaccharide that contains six carbons; examples are glucose and galactose.

hippocampus Region of the central nervous system associated with learning and memory; part of the limbic system.

histamine Substance, produced by basophils in blood and mast cells in connective tissue, that causes capillaries to dilate.

homeobox Developmental genes, shared by all animals, that orchestrate the development of the body plan.

homeodomain Conserved DNA-binding region of transcription factors encoded by the homeobox of homeotic genes.

homeostasis Maintenance of normal internal conditions in a cell or an organism by means of self-regulating mechanisms.

homeothermic Ability to regulate the internal body temperature around an optimum value.

homeotic genes Genes that control the overall body plan by controlling the fate of groups of cells during development.

hominid Taxon that includes humans, chimpanzees, gorillas, and orangutans.

hominin Taxon that includes humans and species very closely related to humans, plus chimpanzees.

homologous chromosome (homologue) Member of a pair of chromosomes that are alike and come together in synapsis during prophase of the first meiotic division.

homologous gene Gene that codes for the same protein, even if the base sequence may be different.

homologous structure A structure that is similar in different types of organisms because these organisms descended from a common ancestor.

homology Similarity of parts or organs of different organisms caused by evolutionary derivation from a corresponding part or organ in a remote ancestor, usually having a similar embryonic origin.

homosporous A plant that produces only one type of asexual spore.

homozygous Possessing two identical alleles for a particular trait.

hormone Chemical messenger produced in one part of an organism's body that controls the activity of other parts.

hornwort A bryophyte of the phylum Anthocerophyta, with a thin gametophyte and a tiny sporophyte that resembles a broom handle.

horsetail A seedless vascular plant having only one genus (*Equisetum*) in existence today; characterized by rhizomes, scalelike leaves, strobili, and tough, rigid stems.

host Organism that provides nourishment and/or shelter for a parasite.

human chorionic gonadotropin (HCG) Gonadotropic hormone produced by the chorion that maintains the corpus luteum.

Human Genome Project (HGP) Initiative to determine the complete sequence of the human genome and to analyze this information.

human immunodeficiency virus (HIV) Virus responsible for the disease AIDS.

humus Decomposing organic matter in the soil.

hunter-gatherer A hominin that hunted animals and gathered plants for food.

hyaline cartilage Cartilage whose cells lie in lacunae separated by a white, translucent matrix containing very fine collagen fibers.

hybridization Interbreeding between two different species; typically prevented by prezygotic isolation mechanisms.

hydrogen bond Weak bond that arises between a slightly positive hydrogen atom of one molecule and a slightly negative atom of another molecule or between parts of the same molecule.

hydrogen ion (H^+) Hydrogen atom that has lost its electron and therefore bears a positive charge.

hydrolysis reaction Splitting of a chemical bond by the addition of water, with the H^+ going to one molecule and the OH^- going to the other.

hydroponics Technique for growing plants by suspending them with their roots in a nutrient solution.

hydrostatic skeleton Fluid-filled body compartment that provides support for muscle contraction, resulting in movement; seen in cnidarians, flatworms, roundworms, and segmented worms.

hydrothermal vent Hot springs in the seafloor along ocean ridges where heated seawater and sulfate react to produce hydrogen sulfide; here, chemosynthetic bacteria support a community of varied organisms.

hydroxide ion (OH^-) One of two ions that result when a water molecule dissociates; it has gained an electron and therefore bears a negative charge.

hypertension Form of cardiovascular disease characterized by elevated blood pressure.

hyperthyroidism Caused by the oversecretion of hormones from the thyroid gland; symptoms include hyperactivity, nervousness, and insomnia.

hypertonic solution Higher-solute concentration (less water) than the cytoplasm of a cell; causes cell to lose water by osmosis.

hypha Filament of the vegetative body of a fungus.

hypothalamic-inhibiting hormone One of many hormones produced by the hypothalamus that inhibits the secretion of an anterior pituitary hormone.

hypothalamic-releasing hormone One of many hormones produced by the hypothalamus that stimulates the secretion of an anterior pituitary hormone.

hypothalamus In vertebrates, part of the brain that helps regulate the internal environment of the body, for example, heart rate, body temperature, and water balance, in addition to the function of the pituitary gland.

hypothesis Supposition established by reasoning after consideration of available evidence; it can be tested by obtaining more data, often by experimentation.

hypothyroidism Caused by the undersecretion of hormones from the thyroid gland; symptoms include weight gain, lethargic behavior, and depression.

hypotonic solution Solution that contains a lower-solute (more water) concentration than the cytoplasm of a cell; causes cell to gain water by osmosis.

ISTUDY

I

immediate allergic response Allergic response that occurs within seconds of contact with an allergen; caused by the attachment of the allergen to IgE antibodies.
immune system System associated with protection against pathogens, toxins, and some cancerous cells; in humans, this is an organ system.
immunity Ability of the body to protect itself from foreign substances and cells, including disease-causing agents.
immunization Strategy that utilizes vaccines for achieving immunity to the effects of specific disease-causing agents.
immunoglobulin (Ig) Globular plasma protein that functions as an antibody.
implantation In placental mammals, the embedding of an embryo at the blastocyst stage into the endometrium of the uterus.
imprinting Learning that occurs when a young animal forms an association with the first moving object it is exposed to.
in vivo gene therapy Form of gene therapy where vectors containing the genes of interest are delivered directly to the cells of the organism.
inbreeding Mating between closely related individuals; influences the genotype ratios of the gene pool.
inclusive fitness Fitness that results from personal reproduction and from helping nondescendant relatives reproduce.
incomplete digestive tract Digestive tract that has a single opening, usually called a mouth.
incomplete dominance Inheritance pattern in which an offspring has an intermediate phenotype, as when a red-flowered plant and a white-flowered plant produce pink-flowered offspring.
incomplete penetrance Dominant alleles that are either not always or partially expressed.
independent assortment Alleles of unlinked genes segregate independently of each other during meiosis, so that the gametes can contain all possible combinations of alleles.
index fossil Deposit found in certain layers of strata; similar fossils can be found in the same strata around the world.
induced fit model Change in the shape of an enzyme's active site that enhances the fit between the active site and its substrate(s).
induced mutation Mutation that is caused by an outside influence, such as organic chemicals or ionizing radiation.
inducer Molecule that brings about activity of an operon by joining with a repressor and preventing it from binding to the operator.
induction Ability of a chemical or a tissue to influence the development of another tissue.
inductive reasoning Using specific observations and the process of logic and reasoning to arrive at general scientific principles.
infertility Inability of a couple to achieve pregnancy after 1 year of regular, unprotected intercourse.
inflammatory response Tissue response to injury that is characterized by redness, swelling, pain, and heat.
ingroup In a cladistic study of evolutionary relationships among organisms, the group that is being analyzed.
inheritance of acquired characteristics Lamarckian belief that characteristics acquired during the lifetime of an organism can be passed on to offspring.
initiation First stage of translation in which the translational machinery binds an mRNA and assembles.
innate immunity An immune response that does not require a previous exposure to the pathogen.
inner ear Portion of the ear consisting of a vestibule, semicircular canals, and the cochlea; where equilibrium is maintained and sound is transmitted.
inorganic chemistry Branch of science that studies the chemical reactions and properties of all of the elements except hydrogen and carbon.
insect Type of arthropod. The head has mouthparts, antennae, compound eyes, and simple eyes; the thorax has three pairs of legs and often wings; and the abdomen has internal organs.
insight learning Ability to apply prior learning to a new situation without trial-and-error activity.
inspiration Act of taking air into the lungs; inhalation.
insulin Hormone released by the pancreas that lowers blood glucose levels by stimulating the uptake of glucose by cells, especially muscle and liver cells.
integration Summing up of excitatory and inhibitory signals by a neuron or by some part of the brain.
integumentary system Body system that includes the skin and its accessory organs. Derivatives of the skin include scales (in fish, reptiles, and birds), feathers (birds only), and hair (mammals only)
interferon Antiviral agent produced by an infected cell that blocks the infection of another cell.
intergenic sequence Region of DNA that lies between genes on a chromosome.
interkinesis Period of time between meiosis I and meiosis II during which no DNA replication takes place.
intermediate filament Ropelike assembly of fibrous polypeptides in the cytoskeleton that provides support and strength to cells; so called because it is intermediate in size between actin filaments and microtubules.
internal respiration Exchange of oxygen and carbon dioxide between blood and interstitial fluid.
interneuron Neuron located within the central nervous system that conveys messages between parts of the central nervous system.
internode In vascular plants, the region of a stem between two successive nodes.
interphase Stages of the cell cycle (G_1, S, G_2) during which growth and DNA synthesis occur when the nucleus is not actively dividing.
interspersed repeat Repeated DNA sequence that is spread across several regions of a chromosome or across multiple chromosomes.
interstitial fluid Fluid that surrounds the body's cells; consists of dissolved substances that leave the blood capillaries by filtration and diffusion.
intertidal zone Region along a coastline where the tide recedes and returns.
intron Intervening sequence found between exons in mRNA; removed by RNA processing before translation.
inversion Change in chromosome structure in which a segment of a chromosome is turned around 180°; this reversed sequence of genes can lead to altered gene activity and abnormalities.
invertebrate Animal without a vertebral column or backbone.
ion Charged particle that carries a negative or positive charge.
ionic bond Chemical bond in which ions are attracted to one another by opposite charges.
iris Muscular ring that surrounds the pupil and regulates the passage of light through this opening.
iron–sulfur world hypothesis Hypothesis that ocean thermal vents provided all of the materials needed for abiotic synthesis of the first molecules.
island biogeography model Proposes that the biodiversity on an island is dependent on its distance from the mainland, with islands located at a greater distance having a lower level of diversity.
isomers Molecules with the same molecular formula but a different structure, and therefore a different shape.
isotonic solution Solution that is equal in solute concentration to that of the cytoplasm of a cell; causes cell to neither lose nor gain water by osmosis.
isotopes Atoms of the same element having the same atomic number but a different mass number due to a variation in the number of neutrons.
iteroparity Repeated production of offspring at intervals throughout the life cycle of an organism.

J

jaundice Yellowish tint to the skin caused by an abnormal amount of bilirubin (bile pigment) in the blood, indicating liver malfunction.
jawless fishes (agnathans) Type of fishes that lack jaws; includes hagfishes and lampreys.
joint Articulation between two bones of a skeleton.
junction protein Protein in the cell membrane that assists in cell-to-cell communication.

K

***K*-selection** Favorable life history strategy under stable environmental conditions characterized by the production of a few offspring with much attention given to offspring survival.

ISTUDY

karyotype A pictorial display of human chromosomes arranged by pairs according to their size, shape, and general appearance in mitotic metaphase.
keystone species Species whose activities significantly affect community structure.
kidneys Paired organs of the vertebrate urinary system that regulate the chemical composition of the blood and produce a waste product called urine.
kin selection Indirect selection; adaptation to the environment due to the reproductive success of an individual's relatives.
kinetic energy Energy associated with motion.
kinetochore An assembly of proteins that attaches to the centromere of a chromosome during mitosis.
kinetoplastid Single-celled, flagellate protist characterized by the presence in their single mitochondrion of a kinetoplast (a structure containing a large mass of DNA).
kingdom One of the categories, or taxa, used by taxonomists to group species; the taxon above phylum.

L

lactation Secretion of milk by mammary glands, usually for the nourishment of an infant.
lacteal Lymphatic vessel in an intestinal villus; aids in the absorption of fats.
lake Body of fresh water, often classified by nutrient status, such as oligotrophic (nutrient-poor) or eutrophic (nutrient-rich).
landscape A number of interacting ecosystems.
landscape diversity Variety of habitat elements within an ecosystem (e.g., plains, mountains, and rivers).
large intestine In vertebrates, portion of the digestive tract that follows the small intestine; in humans, consists of the cecum, colon, rectum, and anal canal.
larynx Cartilaginous organ located between the pharynx and the trachea; in humans, contains the vocal cords; sometimes called the voice box.
last universal common ancestor (LUCA) The first living organism on the planet, from which all life evolved.
lateral line Canal system containing sensory receptors that allow fishes and amphibians to detect water currents and pressure waves from nearby objects.
law Universal principle that describes the basic functions of the natural world.
law of independent assortment Mendelian principle that explains how combinations of traits appear in gametes; *see also* independent assortment.
law of segregation Mendelian principle that explains how, in a diploid organism, alleles separate during the formation of the gametes.
laws of thermodynamics Two laws explaining energy and its relationships and exchanges. The first, also called the "law of conservation," says that energy cannot be created or destroyed but can only be changed from one form to another; the second says that energy cannot be changed from one form to another without a loss of usable energy.
leaf vein Vascular tissue within a leaf.
learning Relatively permanent change in an animal's behavior that results from practice and experience.
lens Transparent, disclike structure found in the vertebrate eye, behind the iris; brings objects into focus on the retina.
leptin Hormone produced by adipose tissue that acts on the hypothalamus to signal satiety (fullness).
less-developed country (LDC) Country that is becoming industrialized; typically, population growth is expanding rapidly, and the majority of people live in poverty.
lichen Symbiotic relationship between one or two certain fungi and either cyanobacteria or algae, in which the fungi possibly provide inorganic food or water and the algae or cyanobacteria provide organic food.
life cycle Recurring pattern of genetically programmed events by which individuals grow, develop, maintain themselves, and reproduce.
ligament Tough cord or band of dense, fibrous tissue that binds bone to bone at a joint.
light reaction Portion of photosynthesis that captures solar energy and takes place in thylakoid membranes of chloroplasts; it produces ATP and NADPH.
lignin Chemical that hardens the cell walls of land plants.
limbic system In humans, functional association of various brain centers, including the amygdala and hippocampus; governs learning and memory and various emotions, such as pleasure, fear, and happiness.
limiting factor Resource or environmental condition that restricts the abundance and distribution of an organism.
lineage Line of descent represented by a branch in a phylogenetic tree.
lipid Class of organic compounds that tend to be soluble in nonpolar solvents; includes fats and oils.
liposome Droplet of phospholipid molecules formed in a liquid environment.
liver Large, dark red internal organ that produces urea and bile, detoxifies the blood, stores glycogen, and produces the plasma proteins, among other functions.
liverwort Bryophyte with a lobed or leafy gametophyte and a sporophyte composed of a stalk and capsule.
lobe-finned fishes Fishes with limblike fins; also called the sarcopterygians.
locus Physical location of a trait (or gene) on a chromosome.
logistic growth Population increase that results in an S-shaped curve; growth is slow at first, steepens, and then levels off due to environmental resistance.
long-day plant Plant that flowers when day length is longer than a critical length (e.g., wheat, barley, clover, and spinach).
loop of the nephron Portion of a nephron between the proximal and distal convoluted tubules; functions in water reabsorption; also called the loop of Henle.
loose fibrous connective tissue Tissue composed mainly of fibroblasts widely separated by a matrix containing collagen and elastic fibers.
lophophoran A general term to describe several groups of lophotrochozoans that have a feeding structure called a lophophore.
lophotrochozoa Main group of protostomes; widely diverse. Includes the lophophorans (bryozoans, phoronids, and brachiopods) and trochozoans (mollusks and annelids).
lumen Cavity inside any tubular structure, such as the lumen of the digestive tract.
lung cancer Uncontrolled cell growth that affects any component of the respiratory system.
lungfishes Type of lobe-finned fishes that utilize lungs in addition to gills for gas exchange.
lungs Internal respiratory organ containing moist surfaces for gas exchange.
luteal phase Second half of the ovarian cycle, during which the corpus luteum develops and much progesterone (and some estrogen) is produced.
luteinizing hormone (LH) Hormone released by the anterior pituitary; in males it regulates the production of testosterone by the interstitial cells; in females it promotes the development of the corpus luteum in the ovary.
lycophytes The first vascular plants to evolve and to have leaves. The leaves are microphylls (e.g., club mosses).
lymph Fluid, derived from interstitial fluid, that is carried in lymphatic vessels.
lymph node Mass of lymphatic tissue located along the course of a lymphatic vessel.
lymphatic capillary Smallest vessel of the lymphatic system; closed-ended; responsible for the uptake of fluids from the surrounding tissues.
lymphatic system Organ system consisting of lymphatic vessels and lymphatic organs; transports lymph and lipids, and aids the immune system.
lymphatic vessel Vessel that carries lymph.
lymphocyte Specialized white blood cell that functions in specific defense; occurs in two forms—T lymphocytes and B lymphocytes.
lymphoid (lymphatic) organ Organ other than a lymphatic vessel that is part of the lymphatic system; the lymphatic organs are the lymph nodes, spleen, thymus, and bone marrow.
lysogenic cell Cell that contains a prophage (virus incorporated into DNA), which is replicated when the cell divides.
lysogenic cycle Bacteriophage life cycle in which the virus incorporates its DNA into that of a bacterium; occurs preliminary to the lytic cycle.
lysosome Membrane-bound vesicle that contains hydrolytic enzymes for digesting macromolecules and bacteria; used to recycle worn-out cellular organelles.
lytic cycle Bacteriophage life cycle in which the virus takes over the operation of the bacterium immediately upon entering it and subsequently destroys the bacterium.

ISTUDY

M

macroevolution Large-scale evolutionary change, such as the formation of new species.

macronutrient Essential element needed in large amounts for plant growth, such as nitrogen, calcium, or sulfur.

macrophage In vertebrates, large phagocytic cell derived from a monocyte that ingests microbes and debris.

macular degeneration Condition in which the capillaries supplying the retina of the eye become damaged, resulting in reduced vision and blindness.

malignant The power to threaten life; cancerous.

maltase Enzyme produced in the small intestine that breaks down maltose to two glucose molecules.

mammal Endothermic vertebrate characterized especially by the presence of hair and mammary glands.

mantle In mollusks, an extension of the body wall that covers the visceral mass and may secrete a shell.

marsh Soft wetland that is treeless and continuously covered in water that often contains rushes, reeds, or grasses.

marsupial Member of a group of mammals bearing immature young nursed in a marsupium, or pouch—for example, kangaroo and opossum.

mass extinction Episode of large-scale extinction in which large numbers of species disappear in a few million years or less.

mass number Mass of an atom equal to the number of protons plus the number of neutrons within the nucleus.

mast cell Connective tissue cell that releases histamine in allergic reactions.

maternal determinant One of many substances present in the egg that influences the course of development.

matrix Unstructured, semifluid substance that fills the space between cells in connective tissues or inside organelles.

matter Anything that takes up space and has mass.

mechanical energy Energy possessed by an object as the result of its motion or position.

mechanoreceptor Sensory receptor that responds to mechanical stimuli, such as pressure, sound waves, or gravity.

medulla oblongata In vertebrates, part of the brain stem that is continuous with the spinal cord; controls heartbeat, blood pressure, breathing, and other vital functions.

medusa Among cnidarians, bell-shaped body form that is directed downward and contains much mesoglea.

megafauna Large animals, such as humans, bears, and deer. Often defined as those over 100 pounds (44 kg) in adult size.

megaphyll Term used to classify the types of leaves on a plant. Megaphylls are broad leafs with multiple veins. They are characteristic of vascular plants.

megaspore One of the two types of spores produced by seed plants; develops into a female gametophyte (embryo sac).

meiosis Type of nuclear division that reduces the chromosome number from 2n to n; daughter cells receive the haploid number of chromosomes in varied combinations.

melanocyte Specialized cell in the epidermis that produces melanin, the pigment responsible for skin color.

melanocyte-stimulating hormone (MSH) Substance that causes melanocytes to secrete melanin in most vertebrates.

melatonin Hormone, secreted by the pineal gland, that is involved in biorhythms.

membrane-first hypothesis Proposes that the plasma membrane was the first component of the early cells to evolve.

memory Capacity of the brain to store and retrieve information about past sensations and perceptions; essential to learning.

memory B cell Forms during a primary immune response but enters a resting phase until a secondary immune response occurs.

memory T cell T cell that differentiated during an initial infection and responds rapidly during subsequent exposure to the same antigen.

meninges Protective membranous coverings around the central nervous system.

meningitis Inflammation of the brain or spinal cord meninges (membranes).

menopause Termination of the ovarian and uterine cycles in females who are older.

menstruation Periodic shedding of tissue and blood from the inner lining of the uterus in primates.

meristem Undifferentiated embryonic tissue in the active growth regions of plants.

mesoderm Middle primary tissue layer of an animal embryo that gives rise to muscle, several internal organs, and connective tissue layers.

mesoglea Transparent, jellylike substance located between the endoderm and ectoderm of some sponges and cnidarians.

mesophyll Inner, thickest layer of a leaf consisting of palisade and spongy mesophyll; the site of most of photosynthesis.

messenger RNA (mRNA) Type of RNA formed from a DNA template and bearing coded information for the amino acid sequence of a polypeptide.

metabolic pathway Series of linked reactions, beginning with a particular reactant and terminating with an end product.

metabolic pool Metabolites that are the products of and/or the substrates for key reactions in cells, allowing one type of molecule to be changed into another type, such as carbohydrates converted to fats.

metabolism The sum of the chemical reactions that occur in a cell.

metamorphosis Change in shape and form that some animals, such as insects, undergo during development.

metaphase Third phase of mitosis; chromosomes are aligned at the metaphase plate.

metapopulation Population subdivided into several small and isolated populations due to habitat fragmentation.

metastasis Spread of cancer from the place of origin throughout the body; caused by the ability of cancer cells to migrate and invade tissues.

methanogen Type of archaean that lives in oxygen-free habitats, such as swamps, and releases methane gas.

MHC (major histocompatibility complex) protein Protein marker that is a part of cell-surface markers anchored in the plasma membrane, which the immune system uses to identify "self."

micelle Single layer of fatty acids (or phospholipids) that orient themselves in an aqueous environment.

microbiome The term used to describe all of the bacteria (and also fungi and viruses) that naturally inhabit the human body.

microbodies Membrane-bound vesicles that contain enzymes used for metabolic functions.

microevolution Change in gene frequencies between populations of a species over time.

microglia Supportive cells of the nervous system that help remove bacteria and debris, thus supporting the activity of the neurons.

micronutrient Essential element needed in small amounts for plant growth, such as boron, copper, and zinc.

microphyll Small leaf with one vein.

microsphere Formed from proteinoids exposed to water; has properties similar to those of today's cells.

microspore One of the two types of spores produced by seed plants; develops into a male gametophyte (pollen grain).

microsporidia A fungal lineage consisting of single-celled obligate, intracellular parasites.

microtubule Small, cylindrical organelle composed of tubulin protein around an empty central core; present in the cytoplasm, centrioles, cilia, and flagella.

midbrain In mammals, the part of the brain located below the thalamus and above the pons which serves as a relay station for tracts passing between the cerebrum and the spinal cord or cerebellum.

middle ear Portion of the ear consisting of the tympanic membrane, the oval and round windows, and the ossicles; where sound is amplified.

migration Regular back-and-forth movement of animals between two geographic areas at particular times of the year.

millipede More-or-less cylindrical arthropod characterized by having two pairs of short legs on most of its body segments; may have 13 to almost 200 pairs of legs.

mimicry Superficial resemblance of two or more species; a survival mechanism that avoids predation by appearing to be noxious.

mineral Naturally occurring inorganic substance containing two or more elements; certain minerals are needed in the diet.

mineralocorticoids Hormones secreted by the adrenal cortex that regulate salt and water balance, leading to increases in blood volume and blood pressure.

ISTUDY

mitochondria (*sing.*, mitochondrion) Membrane-bound organelles in which ATP molecules are produced during the process of cellular respiration.
mitosis The stage of cellular reproduction in which nuclear division occurs; process in which a parent nucleus produces two daughter nuclei, each having the same number and kinds of chromosomes as the parent nucleus.
mitotic spindle A complex of microtubules and associated proteins that assist in separating the chromatids during cell division.
mixotrophic Organism that can use autotrophic and heterotrophic means of gaining nutrients.
model Simulation of a process that aids conceptual understanding until the process can be studied firsthand; a hypothesis that describes how a particular process could be carried out.
molds Various fungi whose body consists of a mass of hyphae (filaments) that grow on and receive nourishment from organic matter, such as human food and clothing.
mole The molecular weight of a molecule expressed in grams; contains 6.023×10^{23} molecules.
molecular clock Idea that the rate at which mutational changes accumulate in certain genes is constant over time and is not involved in adaptation to the environment.
molecule Union of two or more atoms of the same element; also, the smallest part of a compound that retains the properties of the compound.
mollusk Member of a group of invertebrates including squids, clams, snails, and chitons; characterized by a visceral mass, a mantle, and a foot.
monoclonal antibody One of many antibodies produced by a clone of hybridoma cells that all bind to the same antigen.
monocot (Monocotyledone) Flowering plant group; members have one embryonic leaf (cotyledon), parallel-veined leaves, scattered vascular bundles, flower parts in threes or multiples of three, and other characteristics.
monocyte Type of agranular leukocyte that functions as a phagocyte, particularly after it becomes a macrophage, which is also an antigen-presenting cell.
monoecious Having male flowers or cones and female flowers or cones on a single plant.
monogamous Breeding pair of organisms that reproduce only with each other through their lifetime.
monohybrid cross Cross between parents that differ in only one trait.
monomer Small molecule that is a subunit of a polymer (e.g., glucose is a monomer of starch).
monophyletic Group of species including the most recent common ancestor and all its descendants.
monosaccharide Simple sugar; a carbohydrate that cannot be broken down by hydrolysis (e.g., glucose); also, any monomer of the polysaccharides.
monosomy Chromosome condition in which a diploid cell has one less chromosome than normal; designated as $2n - 1$.
monotreme Egg-laying mammal (e.g., duckbill platypus or spiny anteater).
monsoon Climate in India and southern Asia caused by wet ocean winds that blow onshore for almost half the year.
montane coniferous forest Coniferous forest located on a mountain.
more-developed country (MDC) Country that is industrialized; typically, population growth is low, and the people enjoy a good standard of living overall.
morphogenesis Emergence of shape in tissues, organs, or entire embryo during development.
morphological species concept Definition of a species that defines species by specific diagnostic traits.
morphology Physical characteristics that contribute to the appearance of an organism.
morula Spherical mass of cells resulting from cleavage during animal development prior to the blastula stage.
mosaic evolution Concept that human characteristics did not evolve at the same rate; for example, some body parts are more humanlike than others in early hominins.
moss Bryophyte that is typically found in moist habitats.
motor neuron Nerve cell that conducts nerve impulses away from the central nervous system and innervates effectors (muscle and glands). Also called efferent neurons.
mouth In humans, organ of the digestive tract where food is chewed and mixed with saliva.
mRNA transcript mRNA molecule formed during transcription that has a sequence of bases complementary to a gene.
mucosa Epithelial membrane containing cells that secrete mucus; found in the inner cell layers of the digestive (first layer) and respiratory tracts.
multicellular Organism composed of more than one cell; usually has organized tissues, organs, and organ systems.
multifactorial traits Genetic traits that are under the influence of both the environment and multiple genes (polygenes).
multiple alleles Inheritance pattern in which there are more than two alleles for a particular trait; each individual has only two of all possible alleles.
multiple sclerosis (MS) Disease of the central nervous system characterized by the breakdown of myelin in the neurons; considered to be an autoimmune disease.
muscular tissue Type of animal tissue composed of fibers that shorten and lengthen to produce movements.
muscularis Smooth muscle layer found in the digestive tract.
mutagen Chemical or physical agent that increases the chance of mutation.
mutation Any change made in the nucleotide sequence of DNA; source of new variation for a species.
mutualism Symbiotic relationship in which both species benefit in terms of growth and reproduction.
mutualistic Form of symbiotic relationship where both species benefit.
mycelium Tangled mass of hyphal filaments composing the vegetative body of a fungus.
mycorrhizae (*sing.*, mycorrhiza) Mutualistic relationship between fungal hyphae and roots of vascular plants.
myelin sheath White, fatty material—derived from the membrane of neurolemmocytes—that forms a covering for nerve fibers.
myofibril Specific muscle cell organelle containing a linear arrangement of sarcomeres, which shorten to produce muscle contraction.
myosin Muscle protein making up the thick filaments in a sarcomere; it pulls actin to shorten the sarcomere, yielding muscle contraction.

N

NAD^+ (nicotinamide adenine dinucleotide) Coenzyme in oxidation-reduction reactions that accepts electrons and hydrogen ions to become $NADH + H^+$ as oxidation of substrates occurs. During cellular respiration, NADH carries electrons to the electron transport chain in mitochondria.
nail Flattened epithelial tissue from the stratum lucidum of the skin; located on the tips of fingers and toes.
natural group In systematics, a group of organisms that possess a shared evolutionary history.
natural killer (NK) cell Lymphocyte that causes an infected or cancerous cell to burst.
natural selection Mechanism of evolutionary change caused by environmental selection of organisms that have heritable variation and compete for resources, and are thus most fit to reproduce; results in adaptation to the environment.
navigate To steer or manage a course by adjusting one's bearings and following the result of the adjustment.
Neanderthal Hominin with a sturdy build that lived during the last Ice Age in Europe and the Middle East; hunted large game and left evidence of being culturally advanced.
nearsighted (myopic) Condition in which an individual cannot focus on objects at a distance; caused by the focusing of the image in front of the retina of the eye.
negative feedback Mechanism of homeostatic response by which the output of a system suppresses or inhibits activity of the system.
nematocyst In cnidarians, a capsule that contains a threadlike fiber, the release of which aids in the capture of prey.
nephridia (*sing.*, nephridium) Segmentally arranged, paired excretory tubules of many invertebrates, as in the earthworm.
nephron Microscopic kidney unit that regulates blood composition by glomerular filtration, tubular reabsorption, and tubular secretion.
nerve Bundle of long axons outside the central nervous system.
nerve fiber Axon; conducts nerve impulses away from the cell. Axons are classified as either myelinated or unmyelinated based on the presence or absence of a myelin sheath.

ISTUDY

nerve net Diffuse, noncentralized arrangement of nerve cells in cnidarians.
nervous tissue Tissue that contains nerve cells (neurons), which conduct impulses, and neuroglia, which support, protect, and provide nutrients to neurons.
neural plate Region of the dorsal surface of the chordate embryo that marks the future location of the neural tube.
neural tube Tube formed by closure of the neural groove during development. In vertebrates, the neural tube develops into the spinal cord and brain.
neurodegenerative disease Disease, usually caused by a prion, virus, or bacterium, that damages or impairs the function of nervous tissue.
neuroglia Nonconducting nerve cells that are intimately associated with neurons and function in a supportive capacity; also called glial cells.
neuromuscular junction Region where an axon bulb approaches a muscle fiber; contains a presynaptic membrane, a synaptic cleft, and a postsynaptic membrane.
neuron Nerve cell that characteristically has three parts: dendrites, cell body, and an axon.
neurotransmitter Chemical stored at the ends of axons that is responsible for transmission across a synapse.
neutron Neutral subatomic particle, located in the nucleus and assigned one atomic mass unit.
neutrophil Granular leukocyte that is the most abundant of the white blood cells; first to respond to infection.
nitrification Process by which nitrogen in ammonia and organic compounds is oxidized to nitrites and nitrates by soil bacteria.
nitrogen fixation Process whereby free atmospheric nitrogen is converted into compounds, such as ammonium and nitrates, usually by bacteria. This makes the nitrogen available for plants to use.
node In plants, the place where one or more leaves attach to a stem.
nodes of Ranvier Gaps in the myelin sheath around a nerve fiber.
nomenclature In systematics, the process of assigning names to taxonomic groups; usually determined by governing organizations.
noncompetitive inhibition Form of enzyme inhibition where the inhibitor binds to an enzyme at a location other than the active site; while at this site, the enzyme shape changes, the inhibitor is unable to bind to its substrate, and no product forms.
noncyclic pathway Light-dependent photosynthetic pathway that is used to generate ATP and NADPH; because the pathway is noncyclic, the electrons must be replaced by the splitting of water (photolysis).
nondisjunction Failure of the homologous chromosomes or sister chromatids to separate during either mitosis or meiosis; produces cells with abnormal chromosome numbers.
nonnative species Species that migrate or are introduced by humans into a new ecosystem; also called alien, invasive, or exotic species.
nonpolar covalent bond Bond in which the sharing of electrons between atoms is fairly equal.
nonrandom mating Mating among individuals on the basis of their phenotypic similarities or differences, rather than mating on a random basis.
nonrenewable resources Minerals, fossil fuels, and other materials present in essentially fixed amounts (within the human timescale) in our environment.
nonvascular plants Bryophytes, such as mosses and liverworts, that have no vascular tissue and either occur in moist locations or have special adaptations for living in dry locations.
norepinephrine Neurotransmitter of the postganglionic fibers in the sympathetic division of the autonomic system that is produced by the adrenal medulla; also called noradrenaline.
notochord Cartilage-like supportive dorsal rod in all chordates at some time in their life cycle; replaced by vertebrae in vertebrates.
nuclear envelope Double membrane that surrounds the nucleus in eukaryotic cells and is connected to the endoplasmic reticulum; has pores that allow substances to pass between the nucleus and the cytoplasm.
nuclear pore Opening in the nuclear envelope that permits the passage of proteins into the nucleus and ribosomal subunits out of the nucleus.
nucleic acid Polymer of nucleotides; both DNA and RNA are nucleic acids.
nucleoid Region of prokaryotic cells where DNA is located; it is not bound by a nuclear envelope.
nucleolus Dark-staining, spherical body in the nucleus that produces ribosomal subunits.
nucleoplasm Semifluid medium of the nucleus containing chromatin.
nucleotide Monomer of DNA and RNA consisting of a 5-carbon sugar bonded to a nitrogenous base and a phosphate group.
nucleus Membrane-bound organelle within a eukaryotic cell that contains chromosomes and controls the structure and function of the cell.

O

obligate anaerobe Prokaryote unable to grow in the presence of free oxygen.
observation Initial step in the scientific method that often involves the recording of data from an experiment or natural event.
octet rule The observation that an atom is most stable when its outer shell is complete and contains eight electrons; an exception is hydrogen, which requires only two electrons in its outer shell to have a completed shell.
oil Triglyceride, usually of plant origin, that is composed of glycerol and three fatty acids and is liquid in consistency due to many unsaturated bonds in the hydrocarbon chains of the fatty acids.
oil gland Gland of the skin, associated with a hair follicle, that secretes sebum; sebaceous gland.
olfactory cell Modified neuron that is a sensory receptor for the sense of smell.
oligodendrocyte Type of glial cell that forms myelin sheaths around neurons in the CNS.
omnivore Organism in a food chain that feeds on both plants and animals.
oncogene Cancer-causing gene formed by a mutation in a proto-oncogene; codes for proteins that stimulate the cell cycle and inhibit apoptosis.
oocyte Immature egg that is undergoing meiosis; upon completion of meiosis, the oocyte becomes an egg.
oogenesis Production of eggs in females by the process of meiosis and maturation.
Oomycetes Filamentous organisms having cell walls made of cellulose; typically decomposers of dead freshwater organisms, but some are parasites of aquatic or terrestrial organisms. Also called water molds.
open circulatory system Arrangement of internal transport in which the hemolymph bathes the organs directly, and there is no distinction between the hemolymph and interstitial fluid.
operant conditioning Learning that results from rewarding or reinforcing a particular behavior.
operator In an operon, the sequence of DNA that serves as a binding site for a repressor, and thereby regulates the expression of structural genes.
operon Group of structural and regulating genes that function as a single unit.
Opisthokonta Supergroup of eukaryotes that includes choanoflagellates, animals, and fungi.
opposable thumb Fingers arranged in such a way that the thumb can touch the ventral surface of the fingertips of all four fingers.
optimal foraging model Analysis of behavior as a compromise of feeding costs versus feeding benefits.
order One of the categories, or taxa, used by taxonomists to group species; the taxon located above the family level.
organ Combination of two or more different tissues performing a common function.
organ of Corti Structure in the vertebrate inner ear that contains auditory receptors (also called spiral organ).
organ system Group of related organs working together; examples are the digestive and endocrine systems.
organelle Small, membranous structure in the cytoplasm having a specific structure and function.
organic chemistry Branch of science that deals with organic molecules, including those that are unique to living organisms.
organism An individual multicelled or single-celled entity.
orientation In birds, the ability to know present location by tracking stimuli in the environment.
osmoregulation Regulation of the water-salt balance to maintain a normal balance within internal fluids.
osmosis Diffusion of water through a selectively permeable membrane from an area of higher concentration to one of lower concentration.

ISTUDY

osmotic pressure Measure of the tendency of water to move across a selectively permeable membrane; visible as an increase in liquid on the side of the membrane with higher solute concentration.
ossicle One of the small bones of the vertebrate middle ear—malleus, incus, and stapes.
osteoblast Bone-forming cell.
osteoclast Cell that is responsible for bone resorption.
osteocyte Mature bone cell located within the lacunae of bone.
osteoporosis Condition characterized by a loss of bone density; associated with levels of sex hormones and diet.
ostracoderm Member of a group of earliest vertebrate fossils of the Cambrian and Devonian periods; these fishes were small, jawless, and finless.
otolith Calcium carbonate granule associated with sensory receptors for detecting movement of the head; in vertebrates, located in the utricle and saccule.
outer ear Portion of the ear consisting of the pinna and the auditory canal.
outgroup In a cladistic study of evolutionary relationships among organisms, a group that has a known relationship to, but is not a member of, the taxa being analyzed.
ovarian cycle Monthly changes occurring in the ovary that determine the level of sex hormones in the blood.
ovary In flowering plants, the enlarged, ovule-bearing portion of the carpel that develops into a fruit; female gonad in animals, which produces an egg and female sex hormones.
overexploitation When the number of individuals taken from a wild population is so great that the population becomes severely reduced in numbers.
oviparous Type of reproduction in which development occurs in an egg, laid by the female, in reptiles.
ovoviviparous Animals that produce eggs that develop internally and hatch at around the same time as they are released to the environment; mostly aquatic.
ovulation Bursting of a follicle when a secondary oocyte is released from the ovary; if fertilization occurs, the secondary oocyte becomes an egg.
ovule In seed plants, a structure that contains the female gametophyte and has the potential to develop into a seed.
oxygen debt Amount of oxygen required to oxidize lactic acid produced anaerobically during strenuous muscle activity.
oxyhemoglobin Compound formed when oxygen combines with hemoglobin.
oxytocin Hormone released by the posterior pituitary that causes contraction of the uterus and milk letdown.
ozone shield Accumulation of O_3, formed from oxygen in the upper atmosphere; a filtering layer that protects the Earth from ultraviolet radiation.

P

pacemaker The sinoatrial node of the heart; electrical device designed to mimic the normal electrical patterns of the heart.
paleontology Study of fossils that results in knowledge about the history of life.
palisade mesophyll Layer of tissue in a plant leaf containing elongated cells with many chloroplasts.
pancreas Internal organ that produces digestive enzymes and the hormones insulin and glucagon.
pancreatic islet Masses of cells that constitute the endocrine portion of the pancreas.
panoramic vision Vision characterized by having a wide field of vision; found in animals with eyes to the side.
parabasalid Single-celled protist that lacks mitochondria; possesses flagella in clusters near the anterior of the cell.
parasite Species that is dependent on a host species for survival, usually to the detriment of the host species.
parasitic Form of symbiotic relationship in which one species benefits while the second is harmed.
parasitism Symbiotic relationship in which one species (the *parasite*) benefits in terms of growth and reproduction to the detriment of the other species (the *host*).
parasympathetic division Division of the autonomic system that is active under normal conditions; uses acetylcholine as a neurotransmitter.
parathyroid gland Gland embedded in the posterior surface of the thyroid gland; produces parathyroid hormone.
parathyroid hormone (PTH) Hormone secreted by the four parathyroid glands; increases the blood calcium level and decreases the phosphate level.
Parazoa Taxon of animals that lack true tissues and have only a cellular level of organization. The sponges (Porifera) are the only phylum in this group.
parenchyma Plant tissue composed of the least specialized of all plant cells; found in all organs of a plant.
Parkinson's disease (PD) Progressive deterioration of the central nervous system due to a deficiency in the neurotransmitter dopamine.
parsimony In systematics, the simplest solution in the analysis of evolutionary relationships.
parthenogenesis Development of an egg cell into a whole organism without fertilization.
partial pressure Pressure exerted by each gas in a mixture of gases.
passive immunity Protection against infection acquired by transfer of antibodies to a susceptible individual.
pathogens Disease-causing agents, such as viruses, parasitic bacteria, fungi, and animals.
pattern formation Positioning of cells during development that determines the final shape of an organism.
pectoral girdle Portion of the vertebrate skeleton that provides support and attachment for the upper (fore) limbs; consists of the scapula and clavicle on each side of the body.
pedigree Graphical representation of how a trait is being inherited across generations.
pelagic zone Open portion of the ocean that contains the majority of the ocean volume.
pelvic girdle Portion of the vertebrate skeleton to which the lower (hind) limbs are attached; consists of the coxal bones.
penis Male copulatory organ; in humans, the male organ of sexual intercourse.
pentose Five-carbon monosaccharide. Examples are deoxyribose found in DNA and ribose found in RNA.
pepsin Enzyme secreted by gastric glands that digests proteins to peptides.
peptide Two or more amino acids joined together by covalent bonding.
peptide bond Type of covalent bond that joins two amino acids.
peptide hormone Type of hormone that is a protein, a peptide, or derived from an amino acid.
peptidoglycan Polysaccharide that contains short chains of amino acids; found in bacterial cell walls.
perception Sensory stimulation that we become aware of, such as a smell or sound.
perennial Flowering plant that lives more than one growing season because the underground parts regrow each season.
pericycle Layer of cells surrounding the vascular tissue of roots; produces branch roots.
periderm Protective tissue that replaces epidermis; includes cork, cork cambium.
peripheral nervous system (PNS) Nerves and ganglia that lie outside the central nervous system.
peristalsis Wavelike contractions that propel substances along a tubular structure, such as the esophagus.
permafrost Permanently frozen ground, usually occurring in the tundra, a biome of Arctic regions.
peroxisome Enzyme-filled vesicle in which fatty acids and amino acids are metabolized to hydrogen peroxide that is broken down to harmless products.
petal A flower part that occurs just inside the sepals; often conspicuously colored to attract pollinators.
petiole The part of a plant leaf that connects the blade to the stem.
pH scale Measurement scale for hydrogen ion concentration. Based on the formula $-\log[H^+]$.
phagocytosis Process by which cells engulf large substances, forming an intracellular vacuole.
pharyngitis Inflammation of the pharynx; often caused by viruses or bacteria.
pharynx In vertebrates, common passageway for both food intake and air movement; located between the mouth and the esophagus.
phenotype Visible expression of a genotype (e.g., brown eyes or attached earlobes).

ISTUDY

pheromone Chemical messenger that works at a distance and alters the behavior of another member of the same species.

phloem Vascular tissue that conducts organic solutes in plants; contains sieve-tube members and companion cells.

phospholipid Molecule that forms the bilayer of the cell's membranes; has a polar, hydrophilic head bonded to two nonpolar, hydrophobic tails.

photoautotroph Organism able to synthesize organic molecules by using carbon dioxide as the carbon source and sunlight as the energy source.

photoperiod (photoperiodism) Relative lengths of daylight and darkness that affect the physiology and behavior of an organism.

photoreceptor Sensory receptor that responds to light stimuli.

photorespiration Series of reactions that occurs in plants when carbon dioxide levels are depleted but oxygen continues to accumulate, and the enzyme RuBP carboxylase fixes oxygen instead of carbon dioxide.

photosynthesis Process, usually occurring within chloroplasts, that uses solar energy to reduce carbon dioxide to carbohydrate.

photosystem Photosynthetic unit where solar energy is absorbed and high-energy electrons are generated; contains a pigment complex and an electron acceptor; occurs as PS (photosystem) I and PS II.

phototropism Growth response of plant stems to light; stems demonstrate positive phototropism.

phylogenetic species concept Definition of a species that is determined by analysis of a phylogenetic tree to determine a common ancestor.

phylogeny Evolutionary history of a group of organisms.

phylum One of the categories, or taxa, used by taxonomists to group species; the taxon located above the class level.

phytochrome Photoreversible plant pigment that is involved in photoperiodism and other responses of plants, such as etiolation.

phytoplankton Part of plankton containing organisms that photosynthesize, releasing oxygen to the atmosphere and serving as the main producers in aquatic ecosystems.

pineal gland Gland—either at the skin surface (fish, amphibians) or in the third ventricle of the brain (mammals)—that produces melatonin.

pinocytosis Process by which vesicle formation brings macromolecules into the cell.

pioneer species Early colonizer of barren or disturbed habitats that usually has rapid growth and a high dispersal rate.

pith Parenchyma tissue in the center of some stems and roots.

pituitary gland Small gland that lies just inferior to the hypothalamus; consists of the anterior and posterior pituitary, both of which produce hormones.

placenta Organ formed during the development of placental mammals from the chorion and the uterine wall; allows the embryo, and then the fetus, to acquire nutrients and rid itself of wastes; produces hormones that regulate pregnancy.

placental mammal Also called eutherian; species that rely on internal development whereby the fetus exchanges nutrients and wastes with its female parent via a placenta.

placoderms First jawed vertebrates; heavily armored fishes of the Devonian period.

plankton Freshwater and marine organisms that are suspended on or near the surface of the water; include phytoplankton and zooplankton.

plants Multicellular, photosynthetic eukaryotes that increasingly became adapted to live on land.

plasma In vertebrates, the liquid portion of blood; contains nutrients, wastes, salts, and proteins.

plasma cell Mature B cell that mass-produces antibodies.

plasma membrane Membrane surrounding the cytoplasm that consists of a phospholipid bilayer with embedded proteins; functions to regulate the entrance and exit of molecules from cell.

plasmid Extrachromosomal ring of accessory DNA in the cytoplasm of prokaryotes.

plasmodesmata In plants, cytoplasmic connections in the cell wall that connect two adjacent cells.

plasmodial slime mold Free-living mass of cytoplasm that moves by pseudopods on a forest floor or in a field, feeding on decaying plant material by phagocytosis; reproduces by spore formation.

plasmolysis Contraction of the cytoplasm in the plant cell due to the loss of water.

plastid Organelle of plants and algae that is bound by a double membrane and contains internal membranes and/or vesicles (i.e., chloroplasts, chromoplasts, leucoplasts).

plate tectonics Concept that the Earth's crust is divided into a number of fairly rigid plates whose movements account for continental drift.

platelet Component of blood that is necessary to blood clotting; also called a thrombocyte.

pleiotropy Inheritance pattern in which one gene affects many phenotypic characteristics of the individual.

pneumonia Condition of the respiratory system characterized by the filling of the bronchi and alveoli with fluid; caused by a viral, fungal, or bacterial pathogen.

poikilothermic Body temperature that varies depending on environmental conditions; informally termed "cold-blooded."

point mutation Change of only one base in the sequence of bases in a gene.

polar body Nonfunctional product of oogenesis produced by the unequal division of cytoplasm in females during meiosis; in humans three of the four cells produced by meiosis are polar bodies.

polar covalent bond Bond in which the sharing of electrons between atoms is unequal.

pollen grain In seed plants, structure that is derived from a microspore and develops into a male gametophyte.

pollen tube In seed plants, a tube that forms when a pollen grain lands on the stigma and germinates. The tube grows, passing between the cells of the stigma and the style to reach the egg inside an ovule, where fertilization occurs.

pollination In gymnosperms, the transfer of pollen from pollen cone to seed cone; in angiosperms, the transfer of pollen from anther to stigma.

pollution Any contaminant introduced into the environment that adversely affects the lives and health of living organisms.

polyandrous Practice of female animals having several male mates; found in the New World monkeys where the males help in rearing the offspring.

polygamous Practice of males having several female mates.

polygenic inheritance Pattern of inheritance in which a trait is controlled by several allelic pairs.

polygenic trait Trait that is under the control of multiple genes as opposed to monogenic (single-gene) trait.

polymer Macromolecule consisting of covalently bonded monomers; for example, a polypeptide is a polymer of monomers called amino acids.

polymerase chain reaction (PCR) Technique that uses the enzyme DNA polymerase to produce millions of copies of a particular piece of DNA.

polyp Among cnidarians, body form that is directed upward and contains much mesoglea; in anatomy, small, abnormal growth that arises from the epithelial lining.

polypeptide Polymer of many amino acids linked by peptide bonds.

polyploidy Having a chromosome number that is a multiple greater than twice that of the monoploid number.

polyribosome String of ribosomes simultaneously translating regions of the same mRNA strand during protein synthesis.

polysaccharide Polymer made from carbohydrate monomers; the polysaccharides starch and glycogen are polymers of glucose monomers.

pons Portion of the brain stem above the medulla oblongata and below the midbrain; assists the medulla oblongata in regulating the breathing rate.

population Group of organisms of the same species occupying a certain area and sharing a common gene pool.

population density The number of individuals per unit area or volume living in a particular habitat.

population distribution The pattern of dispersal of individuals living within a certain area.

population genetics The study of gene frequencies and their changes within a population.

portal system Pathway of blood flow that begins and ends in capillaries, such as the portal system located between the small intestine and liver.

ISTUDY

positive feedback Mechanism of homeostatic response in which the output of the system intensifies and increases the activity of the system.
posterior pituitary Portion of the pituitary gland that stores and secretes oxytocin and antidiuretic hormone produced by the hypothalamus.
posttranscriptional control Gene expression following transcription that regulates the way mRNA transcripts are processed.
posttranslational control Alteration of gene expression by changing a protein's activity after it is translated.
postzygotic isolating mechanism Anatomical or physiological difference between two species that prevents successful reproduction after mating has taken place.
potential energy Stored energy in a potentially usable form, as a result of location or spatial arrangement.
predation Interaction in which one organism (the predator) uses another (the prey) as a food source.
predator Organism that practices predation.
prediction Step of the scientific process that follows the formulation of a hypothesis and assists in creating the experimental design.
preparatory (prep) reaction Reaction that oxidizes pyruvate with the release of carbon dioxide; results in acetyl CoA and connects glycolysis to the citric acid cycle.
pressure-flow model Explanation for phloem transport; osmotic pressure following active transport of sugar into phloem produces a flow of sap from a source to a sink.
prey Organism that provides nourishment for a predator.
prezygotic isolating mechanism Anatomical, physiological, or behavioral difference between two species that prevents the possibility of mating.
primary lymphoid organs Location in the lymphatic system where lymphocytes develop and mature (e.g., bone marrow and thymus).
primate Mammalian order that includes prosimians, monkeys, apes, and hominins, all of whom have adaptations for living in trees.
primordial soup hypothesis Another name for the hypothesis of abiogenesis proposed by Alexander Oparin and J. B. S. Haldane that the first cells evolved in the oceans from chemicals present in the early atmosphere.
principle Theory that is generally accepted by an overwhelming number of scientists; also called a law.
prion Infectious particle consisting of protein only and no nucleic acid. Prions can cause normal proteins to misfold.
producer Photosynthetic organism at the start of a grazing food chain that makes its own food (e.g., green plants on land and algae in water).
product Substance that forms as a result of a reaction. These can act as a substrate for a following reaction.
progesterone Female sex hormone that helps maintain sexual organs and secondary sex characteristics.
proglottid Segment of a tapeworm that contains both male and female sex organs and becomes a bag of eggs.
prokaryote Organism that lacks the membrane-bound nucleus and the membranous organelles typical of eukaryotes.
prokaryotic cell (prokaryote) Cell that generally lacks a membrane-bound nucleus and organelles; the cell type within the domains Bacteria and Archaea.
prolactin (PRL) Hormone secreted by the anterior pituitary that stimulates the production of milk from the mammary glands.
prometaphase Second phase of mitosis; chromosomes are condensed but not fully aligned at the metaphase plate.
promoter In an operon, a sequence of DNA where RNA polymerase binds prior to transcription.
prophase First phase of mitosis; characterized by the condensation of the chromatin; chromosomes are visible, but scattered in the nucleus.
proprioceptor Class of mechanoreceptors responsible for maintaining the body's equilibrium and posture; involved in reflex actions.
prosimian Group of primates that includes lemurs and tarsiers, and may resemble the first primates to have evolved.
prostaglandin Hormone that has various and powerful local effects.
prostate gland Gland in male humans that secretes an alkaline, cloudy, fluid that increases the motility of sperm.
protease Enzyme that breaks the peptide bonds between amino acids in proteins, polypeptides, and peptides.
proteasome Cellular structure containing proteases that is involved in the destruction of tagged proteins; used by cells for posttranslational control of gene expression.
protein Polymer of amino acids; often consisting of one or more polypeptides and having a complex three-dimensional shape.
protein-first hypothesis In chemical evolution, the proposal that protein originated before other macromolecules and made possible the formation of protocells.
proteinoids Abiotically polymerized amino acids that, when exposed to water, become a microsphere having cellular characteristics.
proteome Sum of the expressed proteins in a cell.
proteomics Study of the structure, function, and interaction of the complete collection of proteins in a cell.
protists The group of eukaryotic organisms that are not a plant, fungus, or animal. Protists are generally a microscopic complex single cell; they evolved before other types of eukaryotes in the history of Earth.
proto-oncogene Gene that promotes the cell cycle and prevents apoptosis; may become an oncogene through mutation.
protocell (protobiont) In biological evolution, a possible cell forerunner with a spherical grouping of lipids that became a cell once it acquired genes.
proton Positive subatomic particle located in the nucleus and assigned one atomic mass unit.
protostome Group of coelomate animals in which the first embryonic opening (the blastopore) is associated with the mouth.
proximal convoluted tubule Portion of a nephron, following the glomerular capsule, where tubular reabsorption of filtrate occurs.
pseudocoelom Body cavity, lying between the digestive tract and body wall, that is incompletely lined by mesoderm.
pseudopod Cytoplasmic extension of some protists; used for locomotion and engulfing food.
pteridophyte A group of seedless vascular plants that includes ferns and their allies (horsetail and whisk fern).
pulmonary circuit Circulatory pathway between the lungs and the heart.
pulmonary fibrosis Respiratory condition characterized by the buildup of connective tissue in the lungs; typically caused by inhalation of coal dust, silica, or asbestos.
pulmonary tuberculosis Respiratory infection caused by the bacterium *Mycobacterium tuberculosis*.
pulse Vibration felt in arterial walls due to expansion of the aorta following ventricle contraction.
Punnett square Visual representation developed by Reginald Punnett that is used to calculate the expected results of simple genetic crosses.
pupil Opening in the center of the iris of the vertebrate eye that regulates the amount of light entering the eye.

R

***r*-selection** Favorable life history strategy under certain environmental conditions; characterized by a high reproductive rate with little or no attention given to offspring survival.
radial symmetry Body plan in which similar parts are arranged around a central axis, like spokes of a wheel.
radiocarbon dating Process of radiometric dating that measures the decay of ^{14}C to ^{14}N.
radiolarian Protist that has a glassy silicon test, usually with a radial arrangement of spines; pseudopods are external to the test.
rain shadow Leeward side (side sheltered from the wind) of a mountainous barrier, which receives much less precipitation than the windward side.
rate of natural increase (*r*) Growth rate that is determined by calculating the number of individuals that are born each year and subtracting the number of individuals that die each year.
ray-finned bony fishes Group of bony fishes with fins supported by parallel bony rays connected by webs of thin tissue.
reactant Substance that participates in a reaction.
receptor Type of membrane protein that binds to specific molecules in the environment, providing a mechanism for the cell to sense and adjust to its surroundings.

ISTUDY

receptor protein Protein located in the plasma membrane or within the cell; binds to a substance that alters some metabolic aspect of the cell.
receptor-mediated endocytosis Selective uptake of molecules into a cell by vacuole formation after they bind to specific receptor proteins in the plasma membrane.
recessive allele Allele that exerts its phenotypic effect only in the homozygote; its expression is masked by a dominant allele.
reciprocal altruism The trading of helpful or cooperative acts, such as helping at the nest, by individuals—the animal that was helped will repay the debt at some later time.
recombinant DNA (rDNA) DNA that contains genes from more than one source.
red algae Marine photosynthetic protists with a notable abundance of phycobilin pigments; includes coralline algae of coral reefs.
red blood cell (RBC) Erythrocyte; contains hemoglobin and carries oxygen from the lungs or gills to the tissues in vertebrates.
red bone marrow Vascularized, modified connective tissue that is sometimes found in the cavities of spongy bone; site of blood cell formation.
red tide A population bloom of dinoflagellates that causes coastal waters to turn red. Releases a toxin that can lead to paralytic shellfish poisoning.
redox reaction A paired set of chemical reactions in which one molecule gives up electrons (oxidized) while another molecule accepts electrons (reduced); also called an oxidation-reduction reaction.
reflex action Automatic, involuntary response of an organism to a stimulus.
refractory period Time following an action potential when a neuron is unable to conduct another nerve impulse.
regulator gene Gene that controls the expression of another gene or genes; in an operon, regulator genes code for repressor proteins.
reinforcement Connection between natural selection and reproductive isolation that occurs when two closely related species come back into contact after a period of isolation.
relative dating Determining the age of fossils by noting their sequential relationships in strata; *absolute dating* relies on radioactive dating techniques to assign an actual date.
renewable resource Resource normally replaced or replenished by natural processes and not depleted by moderate use.
renin Enzyme released by the kidneys that leads to the secretion of aldosterone and a rise in blood pressure.
repetitive DNA element Sequence of DNA on a chromosome that is repeated several times.
replacement reproduction Population in which each person is replaced by only one child.
replication fork In eukaryotic DNA replication, the location where the two parental DNA strands separate.
repressor In an operon, protein molecule that binds to an operator, preventing transcription of structural genes.
reproduce To produce a new individual of the same kind.
reproductive cloning Used to create an organism that is genetically identical to the original individual.
reproductive isolation Model by which new species arise when gene flow is disrupted between two populations, genetic changes accumulate, and the populations are subsequently unable to mate and produce viable offspring.
reptile Terrestrial vertebrate with internal fertilization, scaly skin, and an egg with a leathery shell; includes snakes, lizards, turtles, crocodiles, and birds.
resource partitioning Mechanism that increases the number of niches by dividing the resource, such as food or living space, among species.
resources Abiotic and biotic components of an environment that support or are needed by living organisms.
respiration Sequence of events that results in gas exchange between the cells of the body and the environment.
respiratory center Group of nerve cells in the medulla oblongata that sends out nerve impulses on a rhythmic basis, resulting in involuntary inspiration on an ongoing basis.
responding variable Result or change that occurs when an experimental variable is utilized in an experiment.
resting potential Membrane potential of an inactive neuron.
restriction enzyme Bacterial enzyme that stops viral reproduction by cleaving viral DNA; used to cut DNA at specific points during production of recombinant DNA; also referred to as a restriction endonuclease.
reticular activating system (RAS) Area of the brain that contains the reticular formation; acts as a relay for information to and from the peripheral nervous system and higher processing centers of the brain.
retina Innermost layer of the vertebrate eyeball, containing the photoreceptors—rod cells and cone cells.
retinal detachment Condition characterized by the separation of the retina from the choroid layer of the eye.
retrovirus RNA virus containing the enzyme reverse transcriptase that carries out RNA/DNA transcription.
reverse transcriptase Viral enzyme found in retroviruses that is capable of converting their RNA genome into a DNA copy.
rhizarian Group of eukaryotes that includes foraminiferans and radiolarians.
rhizobia A general term used for bacteria that undergo nitrogen fixation in a symbiotic relationship with plants of the legume family.
rhizoid Hairlike structure found on the roots of some plants that assists in anchoring the plant and also provides for some absorption of water.
rhizome Rootlike underground stem.
rhodopsin Light-absorbing molecule in rod cells and cone cells that contains a pigment and the protein opsin.
ribose Pentose sugar found in RNA.
ribosomal RNA (rRNA) Structural form of RNA found in the ribosomes.
ribosome Site of protein synthesis in a cell; composed of proteins and ribosomal RNA (rRNA).
ribozyme RNA molecule that functions as an enzyme that can catalyze chemical reactions.
RNA (ribonucleic acid) Nucleic acid produced from covalent bonding of nucleotide monomers that contain the sugar ribose; occurs in many forms, including messenger RNA, ribosomal RNA, and transfer RNA.
RNA interference Cellular process that utilizes miRNA and siRNA molecules to reduce or inhibit the expression of specific genes.
RNA polymerase During transcription, an enzyme that creates an mRNA transcript by joining nucleotides complementary to a DNA template.
RNA-first hypothesis In chemical evolution, the proposal that RNA originated before other macromolecules and allowed the formation of the first cell(s).
rod cell Photoreceptor in vertebrate eyes that responds to dim light.
root apical meristem Meristem tissue located under the root cap; the site of the majority of cell division in the root.
root cap Protective cover of the root tip, whose cells are constantly replaced as they are ground off when the root pushes through rough soil particles.
root hair Extension of a root epidermal cell that collectively increases the surface area for the absorption of water and minerals.
root nodule Structure on plant root that contains nitrogen-fixing bacteria.
root pressure Osmotic pressure caused by active movement of minerals into root cells; elevates water in xylem for a short distance.
root system Includes the main root and any and all of its lateral (side) branches.
rotational equilibrium Maintenance of balance when the head and body are suddenly moved or rotated.
rotifer Microscopic invertebrate characterized by a ciliated corona that when beating looks like a rotating wheel.
rough ER (endoplasmic reticulum) Membranous system of tubules, vesicles, and sacs in cells; has attached ribosomes.
roundworm Invertebrate with a nonsegmented cylindrical body covered by a cuticle that molts; some forms are free-living in water and soil, and many are parasitic.
RuBP carboxylase An enzyme that starts the Calvin cycle reactions by catalyzing attachment of the carbon atom from CO_2 to RuBP.
rumen The first chamber in the digestive tract of a ruminant mammal; microorganisms here break down complex carbohydrates.

S

sac fungi (Ascomycota) Fungi that produce spores in fingerlike sacs called asci within a fruiting body; includes morels, truffles, yeasts, and molds.

ISTUDY

saccule Saclike cavity in the vestibule of the vertebrate inner ear; contains sensory receptors for gravitational equilibrium.
salivary amylase In humans, enzyme in saliva that digests starch to maltose.
salivary gland In humans, gland associated with the mouth that secretes saliva.
saltatory conduction Movement of nerve impulses from one node to another along a myelinated axon.
saprotroph Organism that secretes digestive enzymes and absorbs the resulting nutrients back across the plasma membrane.
sarcolemma Plasma membrane of a muscle fiber; also forms the tubules of the T system involved in muscular contraction.
sarcomere One of many units, arranged linearly within a myofibril, whose contraction produces muscle contraction.
sarcoplasmic reticulum Smooth endoplasmic reticulum of skeletal muscle cells; surrounds the myofibrils and stores calcium ions.
saturated fatty acid Fatty acid molecule that lacks double bonds between the carbons of its hydrocarbon chain. The chain bears the maximum number of hydrogens possible.
savanna Terrestrial biome that is a grassland in Africa, characterized by few trees and a severe dry season.
Schwann cell Cell that surrounds a fiber of a peripheral nerve and forms the myelin sheath; also known as a neurolemmocyte.
scientific method Process by which scientists formulate a hypothesis, gather data by observation and experimentation, and come to a conclusion.
sclera White, fibrous, outer layer of the eyeball.
sclerenchyma Plant tissue composed of cells with heavily lignified cell walls; functions in support.
scolex Tapeworm head region; contains hooks and suckers for attachment to host and feeding.
sea star An echinoderm with noticeable 5-pointed radial symmetry; found along rocky coasts where they feed on bivalves.
second messenger Chemical signal such as cyclic AMP that causes the cell to respond to the first messenger—a hormone bound to plasma membrane receptor protein.
secondary lymphoid organ Location in the lymphatic system where lymphocytes are activated by antigens; example is the lymph nodes.
secondary metabolite Molecule not directly involved in growth, development, or reproduction of an organism; in plants, these molecules, which include nicotine, caffeine, tannins, and menthols, can discourage herbivores.
secondary oocyte In oogenesis, the functional product of meiosis I; becomes the egg.
sediment Particulate material (sand, clay, etc.) that is carried by streams and rivers and deposited in areas of slow water movement.
sedimentation Process by which particulate material accumulates and forms a stratum.
seed Mature ovule that contains an embryo, with stored food enclosed in a protective coat.
seed plant Vascular plant that disperses seeds; the gymnosperms and angiosperms.
seedless vascular plant Collective name for club mosses (lycophytes) and ferns (pteridophytes). Characterized by windblown spores.
segmentation Repetition of body units, as seen in the earthworm.
selectively permeable Property of the plasma membrane that allows some substances to pass but prohibits the movement of others.
semelparity Condition of having a single reproductive effort in a lifetime.
semen (seminal fluid) Thick, whitish fluid consisting of sperm and secretions from several glands of the male reproductive tract.
semicircular canal One of three half-circle-shaped canals of the vertebrate inner ear; contains sensory receptors for rotational equilibrium.
semiconservative replication Process of DNA replication that results in two double-helix molecules, each having one parental and one new strand.
semilunar valve Valve resembling a half-moon, located between the ventricles and their attached vessels.
seminal vesicles Glands in male humans that secrete a viscous fluid that provides nutrition to the sperm cells.
seminiferous tubule Long, coiled structure contained within chambers of the testis where sperm are produced.
senescence Sum of the processes involving aging, decline, and eventual death of a plant or plant part.
sensory neuron Nerve cell that transmits nerve impulses to the central nervous system after a sensory receptor has been stimulated. Also called afferent neurons.
sensory receptor Structure that receives either external or internal environmental stimuli and is a part of a sensory neuron or transmits signals to a sensory neuron.
sensory transduction Process by which a sensory receptor converts an input to a nerve impulse.
sepal Outermost, leaflike covering of the flower; usually green in color.
septate Having cell walls; some fungal species have hyphae that are septate.
septum Partition or wall that divides two areas; the septum in the heart separates the right half from the left half.
serosa A thin, fluid secreting membrane that surrounds internal organs. Alternatively, the outer embryonic membrane of birds and reptiles; chorion.
seta (*pl.*, setae) A needlelike, chitinous bristle in annelids, arthropods, and others.
sex chromosomes Chromosomes that differ between the sexes; in humans, these represent the X and Y chromosomes.
sexual dimorphism Species that have distinct differences between the sexes, resulting in male and female forms.
sexual reproduction Reproduction involving meiosis, gamete formation, and fertilization; produces offspring with chromosomes inherited from each parent with a unique combination of genes.
sexual selection Changes in males and females, often due to male competition and female selectivity, leading to increased fitness.
shoot apical meristem Group of actively dividing embryonic cells at the tips of plant shoots.
shoot system Aboveground portion of a plant consisting of the stem, leaves, and flowers.
short tandem repeat (STR) profiling Procedure of analyzing DNA in which PCR and gel electrophoresis are used to create a banding pattern; these are usually unique for each individual; process used in DNA barcoding.
short-day plant Plant that flowers when day length is shorter than a critical length (e.g., cocklebur, poinsettia, and chrysanthemum).
shrubland Arid terrestrial biome characterized by shrubs and tending to occur along coasts that have dry summers and receive most of their rainfall in the winter.
sieve-tube member Member that joins with others in the phloem tissue of plants as a means of transport for nutrient sap.
sign stimulus The environmental trigger that causes a fixed action pattern or unchanging behavioral response.
signal Molecule that stimulates or inhibits an event in the cell.
signal transduction Process that occurs within a cell when a molecular signal (protein, hormone, etc.) initiates a response within the interior of the cell.
simple epithelia Epithelial tissue that possesses only a single layer of cells. Simple epithelia are classified by the shape of the cells.
single-stranded binding proteins Proteins that are involved in the process of DNA replication. Their purpose is to prevent the newly separated DNA strands from re-forming a double helix.
sink In the pressure-flow model of phloem transport, the location (roots) from which sugar is constantly being removed. Sugar will flow to the roots from the source.
sink population Population that is found in an unfavorable area where at best the birthrate equals the death rate; sink populations receive new members from source populations.
sister chromatid One of two genetically identical chromosomal units that are the result of DNA replication and are attached to each other at the centromere.
skeletal muscle Striated, voluntary muscle tissue that comprises skeletal muscles; also called striated muscle.
skin Outer covering of the body; can be called the integumentary system because it contains organs such as sense organs.
sliding filament model An explanation for muscle contraction based on the movement of actin filaments in relation to myosin filaments.
small intestine In vertebrates, the portion of the digestive tract that precedes the large intestine; in humans, consists of the duodenum, jejunum, and ileum.

ISTUDY

smooth ER (endoplasmic reticulum) Membranous system of tubules, vesicles, and sacs in eukaryotic cells; site of lipid synthesis; lacks attached ribosomes.

smooth muscle Nonstriated, involuntary muscles found in the walls of internal organs; sometimes also called visceral muscle.

society Group in which members of a species are organized in a cooperative manner, extending beyond sexual and parental behavior.

sodium-potassium pump Carrier protein in the plasma membrane that moves sodium ions out of cells, and potassium ions into cells. It is important in the function of nerve and muscle cells in animals.

soil Accumulation of inorganic rock material and organic matter that is capable of supporting the growth of vegetation.

soil erosion Movement of topsoil to a new location due to the action of wind or running water.

soil horizon Major layer of soil visible in vertical profile; for example, topsoil is the A horizon.

soil profile Vertical section of soil from the ground surface to the unaltered rock below.

solute Substance that is dissolved in a solvent, forming a solution.

solution Fluid (the solvent) that contains a dissolved solid (the solute).

solvent Liquid portion of a solution that dissolves a solute.

somatic cell Body cell; excludes cells that undergo meiosis and become sperm or eggs.

somatic system Portion of the peripheral nervous system containing motor neurons that serve the skin, joints, and skeletal muscles.

source In the pressure-flow model of phloem transport, the location (leaves) of sugar production. Sugar will flow from the leaves to the sink.

source population Population that can provide members to other populations of the species because it lives in a favorable area, and the birthrate is most likely higher than the death rate.

speciation Origin of new species due to the evolutionary process of descent with modification.

species Group of similarly constructed organisms capable of interbreeding and producing fertile offspring; organisms that share a common gene pool.

species diversity Variety of species that make up a community.

species richness Number of species in a community.

specific epithet In the binomial system of taxonomy, the second part of an organism's name; it may be descriptive.

sperm Male gamete having a haploid number of chromosomes and the ability to fertilize an egg, the female gamete.

spermatogenesis Production of sperm in males by the process of meiosis and maturation.

spicule Skeletal structure of sponges composed of calcium carbonate or silicate.

spinal cord In vertebrates, the nerve cord that is continuous with the base of the brain and housed within the vertebral column.

spinal nerve Nerve that arises from the spinal cord.

spirillum (*pl.*, spirilla) Long, rod-shaped bacterium that is twisted into a rigid spiral; if the spiral is flexible rather than rigid, it is called a spirochete.

spirochete Long, rod-shaped bacterium that is twisted into a flexible spiral; if the spiral is rigid rather than flexible, it is called a spirillum.

spleen Large, glandular organ located in the upper left region of the abdomen; stores and filters blood.

sponge Invertebrate that is a pore-bearing filter feeder whose inner body wall is lined by collar cells that resemble a single-celled choanoflagellate.

spongy bone Type of bone that has an irregular, meshlike arrangement of thin plates of bone.

spongy mesophyll Layer of tissue in a plant leaf containing loosely packed cells, increasing the amount of surface area for gas exchange.

spontaneous mutation Mutation that arises as a result of anomalies in normal biological processes, such as mistakes made during DNA replication.

sporangium (*pl.*, sporangia) Structure that produces spores.

spore Asexual reproductive or resting cell capable of developing into a new organism without fusion with another cell, in contrast to a gamete.

sporophyte Diploid generation of the alternation-of-generations life cycle of a plant; produces haploid spores that develop into the haploid generation.

spring overturn Mixing process that occurs in spring in stratified lakes whereby oxygen-rich top waters mix with nutrient-rich bottom waters.

stabilizing selection Outcome of natural selection in which extreme phenotypes are eliminated and the average phenotype is conserved.

stamen In flowering plants, the portion of the flower that consists of a filament and an anther containing pollen sacs where pollen is produced.

starch Storage polysaccharide found in plants that is composed of glucose molecules joined in a linear fashion with few side chains.

statolith Sensor found in root cap cells that causes a plant to demonstrate gravitropism.

stem Usually the upright, vertical portion of a plant that transports substances to and from the leaves.

stereoscopic vision Vision characterized by depth perception and three-dimensionality.

steroid Type of lipid molecule having a complex of four carbon rings (e.g., cholesterol, estrogen, progesterone, and testosterone).

steroid hormone Type of hormone that has the same complex of four carbon rings, but each one has different side chains.

stigma In flowering plants, portion of the carpel where pollen grains adhere and germinate before fertilization can occur.

stomach In vertebrates, muscular sac that mixes food with gastric juices to form chyme, which enters the small intestine.

stomata (*sing.*, stoma) Small openings between two guard cells on the underside of leaf epidermis, through which gases pass.

stramenopile Group of protists that includes brown algae, water molds, diatoms, and golden brown algae and is characterized by a "hairy" flagellum.

strata (*sing.*, stratum) Ancient layers of sedimentary rock; result from slow deposition of silt, volcanic ash, and other materials.

stratified epithelia Epithelial tissue that consists of multiple layers of epithelial cells, with the bottom layer in contact with the basement membrane.

striated Having bands; in cardiac and skeletal muscle, alternating light and dark bands produced by the distribution of contractile proteins.

strobilus (*pl.*, strobili) In club mosses, terminal cluster of leaves that bears sporangia.

stroke Condition resulting when an arteriole in the brain bursts or becomes blocked by an embolism; cerebrovascular accident.

stroma Region within a chloroplast that surrounds the grana; contains enzymes involved in the synthesis of carbohydrates during the Calvin cycle of photosynthesis.

stromatolite Domed structure found in shallow seas consisting of cyanobacteria bound to calcium carbonate.

structural gene Gene that codes for the amino acid sequence of a peptide or protein.

structural genomics Study of the sequence of DNA bases and the amount of genes in organisms.

style Elongated, central portion of the carpel between the ovary and stigma.

submucosa Tissue layer just under the epithelial lining of the lumen of the digestive tract (second layer).

subsidence When a portion of the Earth's surface gradually settles downward.

substrate Reactant in an enzyme-controlled reaction.

substrate-level ATP synthesis Process in which ATP is formed by transferring a phosphate from a metabolic substrate to ADP.

supergroup Systematic term that refers to the major groups of eukaryotes.

surface tension Force that holds moist membranes together due to the attraction of water molecules through hydrogen bonds.

surface-area-to-volume ratio Ratio of a cell's outside area to its internal volume; the relationship limits the maximum size of a cell.

survivorship Probability of newborn individuals of a cohort surviving to particular ages.

suture Line of union between two nonarticulating bones, as in the skull.

swamp Wet, spongy land that is saturated and sometimes partially or intermittently covered with water. They are dominated by woody plants or shrubs.

sweat gland Skin gland that secretes a fluid substance for evaporative cooling; sudoriferous gland.

ISTUDY

swim bladder In fishes, a gas-filled sac whose pressure can be altered to change buoyancy.
symbiosis Relationship that occurs when two different species interact together in a unique way; it may be beneficial, neutral, or detrimental to one or both species.
symmetry Pattern of similarity in an object.
sympathetic division Division of the autonomic system that is active when an organism is under stress; uses norepinephrine as a neurotransmitter.
sympatric speciation Origin of new species in populations that overlap geographically.
synapse Junction between neurons consisting of the presynaptic (axon) membrane, the synaptic cleft, and the postsynaptic (usually dendrite) membrane.
synapsid Characteristic of a vertebrate skull in which there is a single opening in the skull behind the eye socket (orbit).
synapsis Pairing of homologous chromosomes during meiosis I.
synaptic cleft In a synapse, the small gap between the membranes of the presynaptic and postsynaptic cells.
synaptonemal complex Protein structure that forms between the homologous chromosomes of prophase I of meiosis; promotes the process of crossing-over.
synovial joint Freely moving joint in which two bones are separated by a cavity.
systematics Study of the diversity of life for the purpose of understanding the evolutionary relationships between species.
systemic circuit Circulatory pathway of blood flow between the tissues and the heart.
systole Contraction period of the heart during the cardiac cycle.

T

T cell Lymphocyte that matures in the thymus and exists in four varieties, one of which kills antigen-bearing cells outright.
T-cell receptor (TCR) Molecule on the surface of a T cell that can bind to a specific antigen fragment in combination with an MHC molecule.
tactile communication Communication through touch; for example, when a chick pecks its female parent for food, chimpanzees groom each other, and honeybees "dance."
taiga Terrestrial biome that is a coniferous forest extending in a broad belt across northern Eurasia and North America.
tandem repeat Repetitive DNA sequence in which the repeats occur one after another in the same region of a chromosome.
taproot Main axis of a root that penetrates deeply and is used by certain plants (such as carrots) for food storage.
taste bud Structure in the vertebrate mouth containing sensory receptors for taste; in humans, most taste buds are on the tongue.
taxon (*pl.*, taxa) Group of organisms that fills a particular classification category.
taxonomist Scientist that investigates the identification and naming of new organisms.
taxonomy Branch of science associated with the identification and classification of organisms.
technology Application of scientific knowledge for a practical purpose.
telomere Tip of the end of a chromosome that shortens with each cell division and may thereby regulate the number of times a cell can divide.
telophase Final phase of mitosis; daughter cells are located at each pole.
temperate deciduous forest Forest found south of the taiga; characterized by deciduous trees such as oak, beech, and maple, moderate climate, relatively high rainfall, stratified plant growth, and plentiful ground life.
temperate grassland Grassland characterized by very cold winters and dry, hot summers; examples are the North American prairies and Russian steppes.
temperate rain forest Coniferous forest for example, the forest that runs along the west coast of Canada and the United States—characterized by plentiful rainfall and rich soil.
tendon Strap of fibrous connective tissue that connects skeletal muscle to bone.
terminal bud Bud that develops at the apex of a shoot.
termination End of translation that occurs when a ribosome reaches a stop codon on the mRNA that it is translating, causing release of the completed protein.
territoriality Marking and/or defending a particular area against invasion by another species member. The area is often used for the purpose of feeding, mating, and caring for young.
territory Area occupied and defended exclusively by an animal or a group of animals.
testcross Cross between an individual with a dominant phenotype and an individual with a recessive phenotype to determine whether the dominant individual is homozygous or heterozygous.
testes Male gonads that produce sperm and the male sex hormones.
testosterone Male sex hormone that helps maintain sexual organs and secondary sex characteristics.
tetany Severe spasm caused by involuntary contraction of the skeletal muscles due to a calcium imbalance.
tetrapod Four-footed vertebrate; includes amphibians, reptiles, birds, and mammals.
thalamus In vertebrates, the portion of the diencephalon that passes on selected sensory information to the cerebrum.
theory An explanation of a specific aspect of the natural world that has been widely supported by scientific studies and experimentation.
therapeutic cloning Used to create mature cells of various cell types. Facilitates study of specialization of cells and provides cells and tissue to treat human illnesses.
thermoacidophile Type of archaean that lives in hot, acidic, aquatic habitats, such as hot springs or near hydrothermal vents.
thermoreceptor Sensory receptor that detects heat.
thigmotropism In plants, unequal growth due to contact with solid objects, as the coiling of tendrils around a pole.
threatened species Species that is likely to become an endangered species in the foreseeable future (e.g., jaguar, rainbow parrotfish).
thylakoid Flattened sac within a granum of a chloroplast; membrane contains chlorophyll; location where the light reactions of photosynthesis occur.
thymine (T) One of four nitrogen-containing bases in the nucleotides composing the structure of DNA; pairs with adenine.
thymus Lymphoid organ involved in the development and functioning of the immune system; T lymphocytes mature in the thymus.
thyroid gland Large gland in the neck that produces several important hormones, including thyroxine, triiodothyronine, and calcitonin.
thyroid-stimulating hormone (TSH) Substance produced by the anterior pituitary that causes the thyroid to secrete thyroxine (T_4) and triiodothyronine(T_3).
thyroxine Hormone secreted from the thyroid gland that promotes growth and development; in general, it increases the metabolic rate in cells.
tight junction Junction between cells when adjacent plasma membrane proteins join to form an impermeable barrier.
tissue Group of similar cells combined to perform a common function.
tissue culture Process of growing tissue artificially, usually in a liquid medium in laboratory glassware.
tone Continuous, partial contraction of muscle.
tonicity The solute concentration (osmolarity) of a solution compared to that of a cell. If the solution is isotonic to the cell, there is no net movement of water; if the solution is hypotonic, the cell gains water; and if the solution is hypertonic, the cell loses water.
topography Surface features of the Earth.
totipotent Cell that has the full genetic potential of the organism, including the potential to develop into a complete organism.
toxin Poisonous substance produced by living cells or organisms. Toxins are often proteins that are capable of causing disease on contact with or absorption by body tissues.
trachea (*pl.*, tracheae) In insects, air tube located between the spiracles and the tracheoles. In tetrapod vertebrates, air tube (windpipe) that runs between the larynx and the bronchi.
tracheid In vascular plants, type of cell in xylem that has tapered ends and pits through which water and minerals flow.
tract Bundle of myelinated axons in the central nervous system.

ISTUDY

trait A characteristic of an organism; may be based on the physiology, morphology, or the genetics of the organism.
trans fat Unsaturated fatty acid chain in which the configuration of the carbon-carbon double bonds is such that the hydrogen atoms are across from each other, as opposed to being on the same side (cis).
transcription First stage of gene expression; process whereby a DNA strand serves as a template for the formation of mRNA.
transcription activator Protein that participates in the initiation of transcription by binding to the enhancer regulatory regions.
transcription factor In eukaryotes, protein required for the initiation of transcription by RNA polymerase.
transcriptional control Control of gene expression by the use of transcription factors, and other proteins, that regulate either the initiation of transcription or the rate at which it occurs.
transduction Exchange of DNA between bacteria by means of a bacteriophage.
transduction pathway Series of proteins or enzymes that change a signal to one understood by the cell.
transfer rate Amount of a substance that moves from one component of the environment to another within a specified period of time.
transfer RNA (tRNA) Type of RNA that transfers a particular amino acid to a ribosome during protein synthesis; at one end it binds to the amino acid, and at the other end it has an anticodon that binds to an mRNA codon.
transformation Taking up of extraneous genetic material from the environment by bacteria.
transgenic organism An organism whose genome has been altered by the insertion of genes from another species.
transitional fossil Fossil that bears a resemblance to two groups that in the present day are classified separately.
translation During gene expression, the process whereby ribosomes use the sequence of codons in mRNA to produce a polypeptide with a particular sequence of amino acids.
translational control Gene expression regulated by influencing the interaction of the mRNA transcripts with the ribosome.
translocation Movement of a chromosomal segment from one chromosome to another nonhomologous chromosome, leading to abnormalities (e.g., Down syndrome).
transpiration Plant's loss of water to the atmosphere, mainly through evaporation at leaf stomata.
transposon DNA sequence capable of randomly moving from one site to another in the genome.
trichocyst Found in ciliates; contains long, barbed threads useful for defense and capturing prey.
trichomes In plants, specialized outgrowths of the epidermis (e.g., root hairs).
triglyceride Neutral fat composed of glycerol and three fatty acids; typically involved in energy storage.
trisomy Chromosome condition in which a diploid cell has one more chromosome than normal; designated as $2n + 1$.
trochozoan Type of protostome that produces a trochophore larva; also has two bands of cilia around its middle.
trophic level Feeding level of one or more populations in a food web.
trophoblast Outer membrane surrounding the embryo in mammals; when thickened by a layer of mesoderm, it becomes the chorion, an extraembryonic membrane.
tropical rain forest Biome near the equator in South America, Africa, and the Indo-Malayan regions; characterized by warm weather and plentiful rainfall, and contains the greatest diversity of species of any terrestrial biome.
tropism In plants, a growth response toward or away from a directional stimulus.
true coelom Body cavity completely lined with mesoderm; found in certain protostomes and all deuterostomes.
trypsin Protein-digesting enzyme secreted by the pancreas.
tubular reabsorption Movement of primarily nutrient molecules and water from the contents of the nephron into blood at the proximal convoluted tubule.
tubular secretion Movement of certain molecules from blood into the distal convoluted tubule of a nephron, so that they are added to urine.
tumor Cells derived from a single mutated cell that has repeatedly undergone cell division; benign tumors remain at the site of origin, while malignant tumors metastasize.
tumor suppressor gene Gene that codes for a protein that ordinarily suppresses the cell cycle; inactivity due to a mutation can lead to a tumor.
tundra Biome that encircles the Earth just south of ice-covered polar seas in the Northern Hemisphere. It is characterized as being cold and dark most of the year and containing permafrost.
turgor movement In plant cells, pressure of the cell contents against the cell wall when the central vacuole is full.
turgor pressure Pressure of the cell contents against the cell wall; in plant cells, determined by the water content of the vacuole; provides internal support.
tympanic membrane Membranous region that receives air vibrations in an auditory organ; in humans, the eardrum.

U

umbilical cord Cord connecting the fetus to the placenta, through which blood vessels pass.
uniformitarianism Belief, supported by James Hutton, that geological forces act at a continuous, uniform rate.
unsaturated fatty acid Fatty acid molecule that contains double bonds between some carbons of its hydrocarbon chain; contains fewer hydrogens than a saturated hydrocarbon chain.
upwelling Upward movement of deep, nutrient-rich water along coasts; it replaces surface waters that move away from shore when the direction of prevailing wind shifts.
uracil (U) Pyrimidine base that occurs in RNA, replacing thymine.
urea Main nitrogenous waste of terrestrial amphibians and most mammals.
ureter Tubular structure conducting urine from the kidney to the urinary bladder.
urethra Tubular structure that receives urine from the bladder and carries it to the outside of the body.
uric acid Main nitrogenous waste of insects, reptiles, and birds.
urinary bladder Organ where urine is stored.
urine Liquid waste product made by the nephrons of the vertebrate kidney through the processes of glomerular filtration, tubular reabsorption, and tubular secretion.
urochordate Group of aquatic invertebrate chordates that consists of the tunicates (sea squirts).
uterine cycle Cycle that runs concurrently with the ovarian cycle; it prepares the uterus to receive a developing zygote.
uterus In mammals, expanded portion of the female reproductive tract through which eggs pass to the environment or in which an embryo develops and is nourished before birth.
utricle Cavity in the vestibule of the vertebrate inner ear; contains sensory receptors for gravitational equilibrium.

V

vacuole Membrane-bound sac, larger than a vesicle; usually functions in storage and can contain a variety of substances. In plants, the central vacuole fills much of the interior of the cell.
vagina Component of the female reproductive system that serves as the birth canal; receives the penis during copulation.
valence shell The outer electron shell of an atom. Contains the valence electrons, which determine the chemical reactivity of the atom.
vas deferens Also called the ductus deferens; storage location for mature sperm before they pass into the ejaculatory duct.
vascular bundle In plants, primary phloem and primary xylem enclosed by a bundle sheath.
vascular cambium In plants, lateral meristem that produces secondary phloem and secondary xylem.
vascular cylinder In eudicots, the tissues in the middle of a root, consisting of the pericycle and vascular tissues.
vascular plant Plant that has xylem and phloem.
vascular tissue Transport tissue in plants, consisting of xylem and phloem.
vector In genetic engineering, a means to transfer foreign genetic material into a cell (e.g., a plasmid). Alternatively, an organism, such as a biting insect, that transmits a disease or parasite between other organisms.
vein Blood vessel that arises from venules and transports blood toward the heart.
vena cava Large systemic vein that returns blood to the right atrium of the heart in tetrapods; either the superior or inferior vena cava.
ventilation Process of moving air into and out of the lungs; breathing.
ventricle Cavity in an organ, such as a lower chamber of the heart or the ventricles of the brain.
venule Vessel that takes blood from capillaries to a vein.

ISTUDY

vertebral column Portion of the vertebrate endoskeleton that houses the spinal cord; consists of many vertebrae separated by intervertebral discs.
vertebrate Chordate in which the notochord is replaced by a vertebral column.
vertigo Equilibrium disorder that is often associated with problems in the receptors of the semicircular canals in the ear.
vesicle Small, membrane-bound sac that stores substances within a cell.
vessel element Cell that joins with others to form a major conducting tube found in xylem.
vestibule Space or cavity at the entrance to a canal, such as the cavity that lies between the semicircular canals and the cochlea.
vestigial structure Remnant of a structure that was functional in some ancestor but is no longer functional in the organism in question.
villus Small, fingerlike projection of the inner small intestinal wall.
viroid Infectious strand of RNA devoid of a capsid and much smaller than a virus.
virus Noncellular parasitic agent consisting of an outer capsid and an inner core of nucleic acid.
visual accommodation Ability of the eye to focus at different distances by changing the curvature of the lens.
visual communication Form of communication between animals using their bodies, including various forms of display.
vitamin Organic nutrient that is required in small amounts for metabolic functions. Vitamins are often part of coenzymes.
viviparous Animal that gives birth after partial development of offspring within the female parent.
vocal cord In humans, fold of tissue within the larynx; creates vocal sounds when it vibrates.

W

water column In plants, water molecules joined together in xylem from the leaves to the roots.
water cycle The continuous movement of water through the environment.
water potential Potential energy of water; a measure of the capability to release or take up water relative to another substance.
water vascular system Series of canals that takes water to the tube feet of an echinoderm, allowing them to expand.
wax Sticky, solid, water-repellent lipid consisting of many long-chain fatty acids usually linked to long-chain alcohols.
wetland Area that is covered by water at some point in the year. (*See also* bog, marsh, swamp.)
whisk fern Common name for seedless vascular plant that consists only of stems and has no leaves or roots.
white blood cell (WBC) Leukocyte, of which there are several types, each having a specific function in protecting the body from invasion by foreign substances and organisms.
white matter Myelinated axons in the central nervous system.
wood Secondary xylem that builds up year after year in woody plants and becomes the annual rings.

X

X-linked Allele that is located on an X chromosome; not all X-linked genes code for sexual characteristics.
xylem Vascular tissue that transports water and mineral solutes upward through the plant body; it contains vessel elements and tracheids.

Y

yeast Single-celled fungus that has a single nucleus and reproduces asexually by budding or fission, or sexually through spore formation.
yolk Dense nutrient material in the egg of a bird or reptile.
yolk sac One of the extraembryonic membranes that, in shelled vertebrates, contains yolk for the nourishment of the embryo, and in placental mammals is the first site for blood cell formation.

Z

zero population growth No growth in population size.
zooplankton Part of plankton containing protozoans and other types of microscopic animals that act as the primary consumers in the aquatic ecosystem.
zoospore Spore that is motile by means of one or more flagella.
zygospore Thick-walled resting cell formed during sexual reproduction of zygospore fungi.
zygospore fungi Fungi, such as black bread mold, that reproduce by forming windblown spores in sporangia; sexual reproduction involves a thick-walled zygospore.
zygote Diploid cell formed by the union of two gametes; the product of fertilization.

ISTUDY

Index

Note: Page numbers followed by *f* and *t* represent figures and tables, respectively.

A

ABA. *See* Abscisic acid
Abiogenesis, 314, 331
Abiotic resource and factor
  ecosystem dynamics, 814, 837, 848
  population density and distribution, 815, 821–822
Abiotic synthesis, 314, 331
ABO blood group, 192–193
  inheritance of, 193
ABO system, 603–604
Abscisic acid (ABA), 463
  dormancy, 475–476, 476f
  stomata, closing of, 462–463, 476, 476f
  stomata opening, 462–463
Abscission, 475
Absolute dating, 319, 331
  of fossils, 319–320
Absorption spectrum, 116
Abyssal plain, 879, 880*f*
Acacia, bullhorn, 844
Acacia trees, 873*f*
Accessory organ, 637–638
  gallbladder, 633f, 638
  liver, 633f, 637–638
  pancreas, 633f, 635, 637, 734f, 744–746
Accessory pigments, 381
Accidental transport of exotic species, 890
Acer rubrum, 335
Acetobacter aceti, 132
Acetylcholine (ACh), 682
  botulinum toxin, 726
  defined, 682
  muscle contraction, 729
  parasympathetic system, 694, 695f
  somatic system, 692–694
Acetylcholinesterase (AChE), 683, 728
ACh (acetylcholine). *See* Acetylcholine (ACh)
AChE (acetylcholinesterase), 683, 728
Achromatopsia, 283–284
Acid-base balance, 671, 673–674
Acid deposition, 33, 856, 891–892
Acid rain, 33, 33*f*
Acid reflux, 634
Acids, 32
Acne, 584
Acquired immunity. *See* Adaptive immune defense
Acquired immunodeficiency syndrome (AIDS). *See* AIDS (acquired immunodeficiency syndrome)
Acromegaly, 739, 739*f*
Acrosome, 755*f*, 756, 774, 774*f*
Acrylamide, 233
ACTH (adrenocorticotropic hormone), 738*f*, 739, 742, 742*f*, 743
Actin and myosin, in muscle cells, 47
Actin filaments, 74, 75*f*
  muscle fibers and contraction, 728, 728t, 729
Action potential, 681*f*, 682
Activation of enzymes, 106
Active immunity, 621
Active site (enzyme), 103, 104*f*
Active transport, 88–93
  endocytosis, 90–91, 91f
  energy use for, 90
  exocytosis, 90, 90f
  movement against concentration gradient, 90
  sodium-potassium pump, 90, 91f
Acute bronchitis, 657, 657*f*
Acute phase protein, 614
Acyclovir, 355
AD (Alzheimer disease), 691
ADA (adenosine deaminase), 249
Adalimumab, 620
Adam's apple, 757
Adaptations, 5–6, 5*f*, 267
  adaptive radiation, 303–304, 303f
  definition, 5
  desert ecosystem, 874
  diet
    carnivores, 632
    herbivores, 632
  and evolutionary change, 6
  human, to life at high elevations, 7
  land, 408–409
  natural selection and, 266–268
  to prevent desiccation, 409f
  seed plants, evolution of, 418–426
    angiosperms, 421–424
    gymnosperms, 419–424
Adaptive immune defense, 616–625, 616*f*
  active vs. passive immunity, 621
  B cells and antibody-mediated immunity, 617–619
  evolution, 610
  T cells and cell-mediated immunity, 620
Adaptive radiation, 303–304, 303*f*
Addison disease, 743–744
Adenine (A), 205, 205*f*. *See also* DNA (deoxyribonucleic acid)
Adenosine diphosphate (ADP), 729
Adenosine triphosphate (ATP), 317
  mineral uptake in plants, 454–457
  muscle contraction, 728, 728t, 729
Adenovirus, 351*f*, 353
ADH (antidiuretic hormone), 671, 671, 737, 737*f*
Adhesion, 30, 461
Adhesion junctions, 93, 93*f*
Adipocytes, 576
Adipose tissue, 576
  glucagon, 744
  glucose, 744
  leptin, 748
ADP (adenosine diphosphate),53, 102, 729
Adrenal cortex, 742–744
  anatomy, 734f, 742
  anterior pituitary, 737f
  cortisol, 743
  functions, 742, 743–744
  malfunctions, 743–744
    Addison disease, 743, 744
    Cushing syndrome, 744
  steroid hormones, 735
Adrenal gland, 734*f*, 742, 742*f*
  adrenal cortex, 742–744
  adrenal medulla, 734f, 742
  anatomy, 734f, 742, 742f
    anterior pituitary, 737*f*
    cortisol, 743
    functions, 742–744
    malfunctions, 743–744
    steroid hormones, 735
Adrenaline. *See* Epinephrine
Adrenal medulla, 734*f*, 742
Adrenocorticotropic hormone (ACTH), 738*f*, 739, 742, 742*f*, 744
Adrenoleukodystrophy (ALD), 71, 200
Adult-onset diabetes. *See* Type 2 diabetes
Adult stem cells, 152, 153
Adult vaccinations, 624, 624*t*
Aedes mosquitoes, 357
Aequorea victoria, 505
Aeration, soil, 452
Aerial root, 438
Aerobic chemoheterotrophs, 366
Aerobic metabolism, athletes, 126
Aerobic reaction, 128
AF-2 (mutagen), 233
Afferent arteriole, 668
Aflatoxin, 232
African cape buffalo, 843*f*
African human trypanosomiasis, 374
African sleeping sickness, 374
Afterbirth. *See* Placenta
Agapornis, 795, 795*f*
Agar, 382*f*
Agarose, 382*f*
Age distribution of population, 827, 827*f*
Age of Amphibians, 325. *See* Carboniferous period
Age of Cycads, 325, 331
Age of Fishes, 325
Age spot, 790
Age structure diagram, 817, 818*f*
Aggregate fruit, 497*f*, 498
Aging process, 789–791
  cardiovascular system, 790
  digestive system, 790
  endocrine system, 791
  excretory system, 790
  hypotheses of aging, 789
    damage accumulation theories, 789
    preprogrammed theories, 789
  immune system, 790
  integumentary system, 790
  musculoskeletal system, 791
  nervous system, 790–791
  reproductive system, 791
  respiratory system, 790
  sensory system, 791
Agnathans, 540

ISTUDY

Agranular leukocyte. *See also* Lymphocyte
biodiversity direct value, 886–887
Agranular leukocytes, 604
Agrobacterium-mediated plant transformation, 361
AID (artificial insemination by donor), 764
AIDS (acquired immunodeficiency syndrome), 766–768. *See also* HIV (human immunodeficiency virus)
opportunistic infections, 622, 766
treatment and prevention, 766–768
AIDS patients, 377, 397
Air sac
birds, 652
insects, 649, 650
Alagille syndrome, 178
Alarm call, 803
Albinism, 187
Albumin, 602
Alcohol
abuse, 684
birth defects, 788
urination, 737
Alcoholic fermentation, 131
Aldosterone, 673, 673*f*, 743
Alexandrium catanella, 375
Algae. *See also* Protist evolution and diversity
algal bloom, 375, 892
archaeplastida supergroup, 381–386
brown algae, 379f, 389
charophytes, 383, 408, 409f
chlorophytes, 381–383
coral reef, 381, 879
ecology dynamics, 848
golden brown algae, 378, 378f, 379
green algae, 381–391, 408
charophytes, 383–384
chlorophytes, 381–382
plants, evolutionary history of, 409f
red algae, 381, 381f
Stramenopiles, 378–380
symbiotic relationships, 375, 381, 844
Algal bloom, 892
Allantois, 782, 784
Allele frequency, 277
Alleles, 163, 185, 277
dominant, 185
at gene locus, 185f
recessive, 185
Allen's rule, 569
Allergy
anaphylactic shock, 625
delayed allergic response, 626
immediate allergic response, 625
Alligator mississippiensis, 335
Allopatric speciation, 300–302, 300*f*, 301*f*
Alloploidy, 302, 302*f*
Allosteric site, 106
Alpha-linolenic fatty acid, 641
Alpine tundra, 865, 866*f*
ALS (amyotrophic lateral sclerosis), 687
Alternation of generation, 409
brown algae, 379
land plants, 409, 410f
Ulva, 383
Alternation of generations (plants), 169
Alternative mRNA splicing, 214, 232
Alternative pre-mRNA splicing, 232
Altman, Sidney, 315
Altruism *vs.* self-interest, 808–809
Alu sequence, 251
Alvarez, Luis, 329
Alvarez, Walter, 329
Alveolates, 388
Alveoli, lungs, 651, 651*f*
Alyces obstetricans, 824*f*
Alzheimer disease (AD), 48, 233, 356, 691
Amanita mushrooms, 213
Amanita phalloides, 398, 398*f*
Amanitas, 398
Amborella trichopoda, 423, 423*f*
Ambulocetus natans, 269, 270*f*
Amenorrhea, 761
American alligator, 335
American bison, 873, 874*f*
Ames test for mutagenicity, 234, 234*f*
AM fungi, 395, 403
Amia calva, 335
Amino acids, 47, 48*f*, 138, 202
conversion to glucose, 638
essential, 138, 641
nonessential, 138
peptide, 48, 49f
polypeptide, 48
sequence, 48
structure of, 46
Aminoacyl-tRNA synthetases, 215
Ammonia, 32, 138, 664, 674
Ammonium, 855
Amniocentesis, 788*f*
Amnion, 782
Amniotes, 540
evolution, timeline of, 548f
Amniotic egg, 547
Amoeba proteus, 385, 385*f*
Amoeba, social, 610
Amoeboids, 389
Amoeba proteus, 385
cellular slime molds, 386
Entamoeba histolytica, 385
organelles, 391
Amoebozoans, 385
Amoebozoa supergroup, 385
amoeboids, 385
slime molds, 385–386
Amphibians, 539, 544*f*
Age of Amphibians, 325
aquatic reproduction, 543
characteristics lungs, 648
double-loop circulatory pathway, 543
ectothermy, 543
evolution of, 543–544
hormones and endocrine system, 735
limbs, 543
lungs, 543
metamorphosis, 735
respiration, 648, 652
sense organs, 543
smooth/nonscaly skin, 543
Amphipathic molecules. *See* Phospholipids
Amygdala, 691
Amylase
pancreatic, 639
salivary, 634, 638
Amyloplasts, 481
Amyotrophic lateral sclerosis (ALS), 687
Anabolic steroid, 747, 747*f*, 757
Anabolism, 138
Anaerobes, obligate, 362
Anaerobic reaction, 129
Anaerobic respiration, in cytoplasm, 126
Analgesic, 714
Analogous structures, 271, 344, 347
Analogous trait, 306
Analogy, 344, 347
Anaphase, 149, 150*f*
Anaphylactic shock, 625
Anapsid, 547
Anatomical evidence, 271–272, 271*f*
Ancestral angiosperm, 344*f*
Ancestral traits, 342, 347
Ancestry. *See also* Common ancestor; Common ancestor
land plants, 408–411
LUCA (last universal common ancestor), 330f, 331
phylogenetic species concept, 294
Anchor cell, 780
Andreas, San, 329
Androgen, 746
Androgen insensitivity, 236, 735
Anencephaly, 788
Aneuploidy, 174–177
Angina pectoris, 601
Angiogenesis, 156
Angioplasty, with stent placement, 602*f*
Angiosperm evolution and full adaption to land, 418–426. *See also* Flowering plant
flowering plant life cycle, 424–426
flowers and diversification, 426
life history strategy, 824
origin and radiation of, 422–424
ovules, 422
Angiosperm fossil, 422
Angiosperms, 419, 421, 422
Angiotensin I, 673, 673*f*, 743
Angiotensin II, 673, 673*f*, 743
ANH (atrial natriuretic hormone), 673, 743, 743*f*
Animal, 9. *See also* Human; Invertebrate; Vertebrate
behavior, 794–811
animal communication, 801–805
behaviors that increase fitness, 805–809
environment influences behavior, 797–801
inheritance influences behavior, 795–797
common features of, 504f
communication, 801–805
auditory communication, 802
chemical communication, 801–802
tactile communication, 803
visual communication, 802–803
development and aging, 773–793
aging process, 789–791
early developmental stages, 772–776
human embryonic and fetal development, 758, 744–776, 762–767
embryonic development
blastula formation, 775
cleavage, 775, 783
developmental processes, 776–782
organ stages of development, 776–776
excretory systems, 664–667, 665
invertebrate excretory organs, 664–665
nitrogenous waste products, 664, 671
osmoregulation by aquatic vertebrates, 665–666
osmoregulation by terrestrial vertebrates, 666–667

ISTUDY

Animal—(*Cont.*)
hormone and endocrine system, 733–736
action, 735–736
chemical signals and pheromones, 735
HOX genes, evolutionary history of, 509f
hypotheses, 505, 505f
medical uses of, 545f
pollination, 844, 887
phylogenetic tree of, 506f
reproductive systems, 752–753
asexual reproduction, 752
life history strategies, 753
sexual reproduction, 752–753
Animal body plan, evolution of, 508–509
Animal body systems, roles of, 580*t*
Animalia, 339
Animal kingdom, 509*t*
Animal parasites, 401*f*
Animal pharming, 545
Animal phyla, 506
Animal pole, 775
Ankle, 723
Ankyrin-1 gene, 192
Annelid. *See also* Earthworm
hydrostatic skeleton, 717
Annelids, 516
Annual plants, 435, 818
Annual rings, 442
Anodonta, 517. *See also* Clam
Anolis, 301
Anopheles mosquitoes, 377
Anosmia, 791
Anoxygenic photosynthesis, 363
Ant, 465
altruism vs. self-interest, 807–810
army, 807, 808f
coevolution, 844
mutualism, 844
pheromones, 802
Antagonistic hormone, 741
Antagonistic pairing of muscles, 725, 725*f*
Antenna, chemical communication, 802
Anterior pituitary, 737, 738*f*, 738–739
hormones affecting other glands
adrenocorticotropic hormone (ACTH), 739, 742*f*, 744
gonadotropic hormones, 738, 746, 757, 760
thyroid-stimulating hormone (TSH), 740
hormones not affecting other glands, 737–739
growth hormone (GH), 739
melanocyte-stimulating hormone (MSH), 739
hypothalamic-inhibiting hormones, 738
hypothalamic-releasing hormones, 738
major endocrine glands, 734f
Anteroposterior polarity, 780
Antheridia, 409
Antheridium, 417, 418*f*
Anthoceros sp.,412*f*
Anthropoids, 560
Antibiotics, 364
resistance, 276, 361
use, 364
Antibodies, 47
Antibody
aging process, 790
development, 617
monoclonal antibodies, 618, 762
plasma, 617
structure, 617–618, 618f
Antibody-mediated immunity, 617, 620
Antibody titer, 623, 623*f*
Anticodon, 214
Antidepressant drug, 683
Antidiuretic hormone (ADH), 671, 673, 737, 737*f*
Antigen
molecular mimicry, 626
receptors, 610, 616
transplant rejection, 626
Anti-inflammatory medication, 714, 724
Antiretroviral drug (ART), 767–768
Anus
complete digestive tract, 631
earthworms, 630f, 631
human, 633f, 638, 771
Anvil. *See* Incus
Aorta, 598
Aortic body, 654
Apatosaurus, 326
APCs (antigen-presenting cells), 619
Apetala gene, 435
Apetala 1 gene, 435
Aphids, 465–466, 465*f*, 478, 818*f*
Apical dominance, 472
Apical meristem, 430
Apical meristem differentiation
plant life cycle, 435
root system, 436, 438
shoot system, 439
Apicomlexans, 388
Apis mellifera, 794, 887. *See also* Bee
Apoptosis, 145, 145*f*
and cell division, 145
morphogenesis, 776, 782
NK cells, 615
Appendicitis, 636
Appendicular skeleton, 722–723
pectoral girdle and upper limb, 722–723
pelvic girdle and lower limb, 723
Appendix, 613, 636, 636*f*
Applegate, Christina, 676
AquAdvantage salmon, 248–249, 249*f*
Aquaporins, 84, 671
Aquatic ecosystem
coastal ecosystems, 875f, 878f
lakes, 875–876
oceans, 879–880
wetlands, 877
Aquifer, 853
Arabidopsis genome, 480, 486
Arabidopsis plants, 479
Arabidopsis thaliana, 480
photograph of, 480f
Arboreal life, 557
Arbuscular mycorrhizal fungi, 395
Archaea, 8–9, 8*f*, 61, 365–367
domain classification, 365
in moderate habitats, 367
structure of, 366
flagella, 361
S-layer, 361
symbiotic relationships
commensalism, 363
mutualism, 363
parasitism, 363
types of, 366–367
halophiles, 366
methanogens, 367
moderate habitats, 367
thermoacidophiles, 367
Archaeocyte, 630
Archaeopteryx, 269, 326
Archaeplastida supergroup, 389, 408
green algae, 381–385
charophytes, 383–385
chlorophytes, 381–383
red algae, 381, 381f
Archegonium, 409, 417, 421
Archenteron, 776
Architeuthis, 518
Arctic Apple, 246, 246*f*
Ardi, 562
Ardipithecus kadabba, 562
Ardipithecus ramidus, 562, 562*f*, 563*f*
Ardi's feet, 562
Argus II artificial retina, 707
Aristotle, 260, 334, 339
Arm
bones, 723
muscles, 725f
Army ant, 807, 808*f*
ART (antiretroviral drug), 767–768
Arteries, 592
Arterioles, 592
Artery
aging process, 790
hepatic, 637
Arthritis, 626, 724
rheumatoid, 626, 626f
Arthropods, 526
characteristics of, 526–527
exoskeleton, 526
chelicerates
horseshoe crab, 886
crustacean
barnacles, 839, 840*f*, 843, 878
chemical senses, 700
krill, 631
respiratory system, 649
excretory organs, 665
exoskeleton, 717–718
eye, 526f
insects
chemical senses, 700
excretory organs, 665
omnivorous diet, 630
respiratory system, 648*f*, 649–650
vision, 702
nervous system
crab, 678*f*
respiratory system
crustacean, 649
insects, 647, 649–650, 650*f*
sensory organs
chemical senses, 700
compound eye, 702
equilibrium, 710–711
skeleton, 526f
Artificial insemination by donor (AID), 764
Artificial joint, 724
Artificial pancreas, 745

ISTUDY

Artificial selection, 267, 422
dogs and, 267, 267f
of plants, 267, 267f
Artiodactyla, 342
Ascaris genus, 525
Ascaris infection, 525
symptoms of, 525
Ascocarps, 396
Ascomycota, 393*t*, 397
evolution and diversity of fungi, 393
lichen, 400–403
mycorrhizae, 403–404
symbiotic relationships, 400
Ascospores, 397
Ascus, 397
Asexual reproduction, 168, 378, 380, 500
defined, 752
flowering plant, 501f
cell suspension culture, 502
stems, roots, and cuttings, 500
tissue culture, 500–502
invertebrates, cnidarians, 752
parthenogenesis, 752
protists, 387
apicomplexans, 376–377
characteristics of, 371
ciliates, 377–378
diatoms, 378
dinoflagellates, 375
euglenids, 373–374
evolution and diversity of, 371
green algae, 381–388
water molds, 379, 379*f*, 380
strategies, 500–503
from stems, roots, and cuttings, 500
tissue culture, of plants, 500–503, 503*f*
through tissue culture, 501f
Asparagus officinalis, 492*f*
Asparagus, tissue culture propagation, 500–503
Aspergillus genus, 397
Aspergillus nidulans, 397
Assimilation (nitrogen cycle), 855
Associative learning
classical conditioning, 799
defined, 799
operant conditioning, 800
Assortative mating, 283
Asteroid, 329
Asthma, 628, 657, 657*f*
Astigmatism, 708, 708*f*
Astrocytes, 579, 680
Asymmetry, 510
Atherosclerosis, 790
Atomic mass, 24
Atomic mass unit (AMU), 21
Atomic number, 21
Atomic symbol, 21
Atomic theory, 20
Atoms, 2, 3*f*, 21–22
Bohr model of, 25, 25f
structure of, 22f
subatomic particles, 21, 21f
ATP (adenosine triphosphate), 51, 72, 103, 114, 272. *See also* Adenosine triphosphate (ATP)
ATP cycle, 102, 102f
coupled reactions, 102
as energy currency of cells, 53, 102
functions of, 103
hydrolysis of, 53, 103
production, 136–137, 136f, 138
structure of, 52–53, 53f, 103
use of, during muscle contraction, 103–104, 103f
ATP synthase complex, 117, 136
Atria, 594
Atrial natriuretic hormone (ANH), 673, 743, 743*f*
Atrioventricular valves, 595
Attachment, viruses, 353
Auditory canal, 709
Auditory communication, 803, 803*f*
Auditory tube, 709
Augustus, Edward, 199
Australopithecines, 563
Australopithecus afarensis, 563, 564*f*
Australopithecus africanus, 564
Australopithecus sediba, 564
Australopiths, 563
Autoimmune disease
defined, 626
monoclonal antibodies, 618
Automatic external defibrillators (AEDs), 598
Autonomic system, 693*t*
functions, 694
parasympathetic division, 694, 695f
sympathetic division, 694, 695f
Autophagy, 69
Autoploidy, 302
Autosomal dominant disorders, 191–192, 191*f*
hereditary spherocytosis, 192
Huntington disease, 192
osteogenesis imperfecta, 192
Autosomal recessive disorders, 190–191
cystic fibrosis, 191, 191f
methemoglobinemia, 190–191, 190f
phenylketonuria, 191
Autosomal recessive pedigree, 190, 190*f*
Autosome, 189
Autotrophic, 363, 387
chemoautotrophs bacteria, 363
photoautotrophic bacteria, 363
Autotrophs, 112, 112*f*. *See also* Photosynthesis
autotrophic bacteria, chemoautotrophic bacteria, 880
ecosystem dynamics, 848, 850f
and heterotrophs, 112, 114f
Auxin, 472, 472*f*
chemical substance, 472
growth and development, 474
stems to bend, 472
tissue culture experiments, 475f
Avery, Oswald, 203, 204
Avian influenza, 15, 356
Axial skeleton, 721–722
skull, 721
vertebral column and rib cage, 722
Axillary bud, leaves, 445
Axillary hair, 747
Axillary lymph node, 611*f*
Axon
neuron component, 680
terminal and muscle innervation, 728
Axonal membrane
action potential, 682
resting potential, 682
Axon hillock, 683
AZT, 355

B

Baboon, 288*f*
Hamadryas, 806f
visual communication, 804, 804f
Baby boom, 826
Bacillus, 61
Bacillus anthracis, 339
Bacteria, 8, 9, 61. *See also* Prokaryotic cells
bacillus, 61
coccus, 61
DNA replication in, 209f, 210domain, 8, 9, 9f
generalized structure, 61–62, 62f
genetically modified, 240, 242–243
spirilla, 61
spirochetes, 61
structure of, 61–62, 62f
transformation of, 203–205
Bacterial diseases, in humans, 364*t*
Bacterial fermentation, 132, 133
Bacterial nitrogen fixation, 455
Bacteriophages, 353, 368
Bacterium (bacteria). *See also* Cyanobacterium; *specific bacteria name antibiotics*
antibiotics, 276, 364
characteristics of, 361–362
Gram-negative bacteria, 362, 362*f*
Gram-positive bacteria, 361–362, 362*f*
peptidoglycan (PG), 361, 362*f*
cyanobacteria, 364–365
diversity of, 362f
ecosystem dynamics, 848
intestinal, 636
metabolism, 362–363
autotrophic bacteria, 363
heterotrophic bacteria, 363
oxygen, 362–363
nitrogen fixation, 855
nitrogen fixation, 363, 363f, 365, 455
reproduction, 361
structure of, 366
flagella, 360
glycocalyx, 360
S-layer, 360
summary, 368–369
symbiotic relationships, 363–364
antibiotics, 364
commensalism, 363, 843
mutualism, 363, 843–844
parasitic bacteria, 363
parasitism, 363, 842–843
transformation, prokaryote reproduction, 361
Balanus, 839, 840*f*
Bald eagle, 885, 885*f*, 890
Baldness, 746, 757
Baleen whale, 631
Ball-and-socket joint, 724
Bangham, Alec, 316
Banting, Frederick, 746
Barcoding concept
illustration of, 337f
Barnacle, 839, 840*f*, 843, 878
Barnes maze test, 229, 229*f*
Barr body, 1, 76, 226
Barrett's esophagus, 635
Barr, Murray, 176

ISTUDY

Bartholin gland. *See* Vestibular gland
Basal body, 76
Basal ganglia. *See* Basal nuclei
Basal nuclei, 688
Base, acid-base balance, 671, 673–674
Basement membrane, 574
Bases, 31
Base substitution, 233
Basidiocarps, 397, 400
Basidiomycete yeast, 400
Basidiomycota, 393
  evolution and diversity of fungi, 397
  smuts and rusts, 397, 400
Basidium, 397
Basilar membrane, 710
Basophil, 604
Bat
  communication, 801
  echolocation, 700
  reciprocal altruism, 810
  Vampire, 810
Bates, Henry, 842
Batesian mimic, 842
Bath salts, 684–685
Bathypelagic zone, 879, 879*f*
Batrachochytrium dendrobatidis, 394
Bats, 221
Bay-breasted warbler, 840*f*
B cell
  aging process, 790
  antibody-mediated immunity, 617–619
  autoimmune disorders, 626
B-cell receptor (BCR), 617–619
  development, 612, 616
  memory B cell, 617
BCR (B-cell receptor), 617–619
Beadle, George, 210
Beak, morphology of finches, 304–305, 838
Bean seed, structure of, 496
Bear
  community diversity, 885, 890
  Grizzly, 885, 890, 895
  keystone species, 894
  K-selected populations, 824f
  landscape preservation, 895
  metapopulation, 895
Beaumont, William, 635
Becquerel, Antoine-Henri, 21
Becquerel, Henri, 23
Bee
  animal behavior, 794
  bumblebee, 842
  inclusive fitness, 808
  mimicry, 842
  parthenogenesis, 752
  pheromones, 802–803
  pollination, 887
  pollination, 419
  tactile communication, 804
  vision, 702
  wasp-pollinated flowers, 493
Beeswax, 45, 46*f*
Beetle
  Longhorn, 842
  mimicry, 842
Begging behavior, 798
Behavioral isolation, 298*f*, 299
Behavioral traits, 344
Behavior, animal, 794–811
  animal communication, 801–804
    auditory communication, 802
    chemical communication, 801–802
    tactile communication, 804
    visual communication, 803–804
  behaviors that increase fitness, 805–810
    altruism *vs.* self-interest, 807–810
    reproductive strategies and fitness, 806–807
    societies and fitness, 807
    territoriality and fitness, 805–806
  defined, 795
  environment influences behavior, 797–801
    associative learning, 799–800
    cognitive learning, 801
    imprinting, 799
    instinct and learning, 798–799
    orientation and migratory behavior, 800–801
    social interactions and learning, 799
  inheritance influences behavior, 795–797
    garter snakes, 796
    lovebirds, 795–797
    marine snails, 796–797
    mice, 797
    twin studies, 796
Behavior of organism, 5
Belize Zoo, 801
Beltian body, 844
Beneficial nutrients, 450
Benign growth, 154
Benign positional vertigo (BPV), 712
Benign prostatic hypertrophy (BPH), 756
Benthic zone, 876
Berger, Lee, 564
Best, Charles, 746
Bicarbonate ion
  buffer system, 673
  respiration, 655
Bicep, 725*f*
Biennial plants, 435
Bilaterally symmetrical animals, 510
Bilateral symmetry, 510
  animal evolution, 677
  nervous system, 677
Bile, 635, 638
Bile duct, 637
Bile salt, 638
Bilirubin, 638, 672
Binary fission, 361, 361*f*, 368
Binomial nomenclature, 9, 260, 334, 346
Biochemical evidence, 272, 272*f*, 273
Biodiversity, 14. *See also* Diversity; Systematic biology
  causes of extinction, 889–894
    climate change, 892–93
    disease, 889, 892
    habitat loss, 889, 890*f*
    nonnative species, 883, 889–890
    overexploitation, 889, 893–894
    pollution, 889, 891–92
  conservation, 883–97
    causes of extinction, 889–894
    conservation biology and biodiversity, 884–886
    techniques, 894–96
  conservation biology and biodiversity, 884–886
    biodiversity organization, 884–85
    defined, 884
    distribution of biodiversity, 886
    defined, 884
    described species, 884f
    distribution, 886
    hotspot, 886
    organization, 884–85
      community diversity, 885, 885*f*
      landscape diversity, 885
    value, 886–889
      biodiversity and natural ecosystems, 888–889
      direct value, 886–87
      indirect value, 887–888
Biogeochemical absorption, 456
Biogeochemical cycle, 852–856
  carbon cycle, 853–854
  indirect value of biodiversity, 888
  nitrogen cycle, 855–856
  phosphorus cycle, 854
  water cycle, 852–853, 888
Biogeography, 263, 270–271, 271*f*, 837
Bioinformatics, 254, 884
Bio ink unique, 573
Biological clock, 486
  circadian rhythms, 747
  orientation and migration, 800–801
Biological organization, levels of, 2, 3*f*
Biological species concept, 297
Biology, 1, 2
Bioluminescence, 375
Bioluminescence, bathypelagic zone, 880
Bioluminescent, 513
Biomagnification, 880
Biomass, 851, 851*f*
Biome
  climate, 865
  coniferous forest, 865, 867–868
  defined, 865
  deserts, 873–874
  global distribution, 865f
  grassland, 872–873
  shrublands, 870–872
  temperate deciduous forest, 868–869
  tropical forest, 866f, 869–871
  tundra, 865, 867, 866f
Biomolecules, 37, 313, 331
  carbon as ideal building block for, 37–38
  carbon skeleton, 38
  of cells, 39, 39t
  functional groups, 38, 38t
  macromolecules, 39
  synthesis and degradation of, 39, 39f
Bioprinters, 573
Bioremediation, 245–246, 245*f*
Biosphere, 2, 3*f*, 815. *See also* Ecosystem
Biotechnology, 238
  biotechnology products, 245, 245–249
    genetically modified animals, 246–248
    genetically modified bacteria, 245
    genetically modified plants, 246, 246*f*
  DNA technology, 238–245
    cloning, 238, 238*f*
    DNA analysis, 241–242
    DNA sequencing, 239, 239*f*
    genome editing, 241
    polymerase chain reaction, 239–240, 240*f*
    recombinant DNA technology, 238–239, 239*f*
  gene therapy, 249–250
    blindness, 707
    ex vivo gene therapy, 249, 249*f*

ISTUDY

preimplantation genetic diagnosis, 765, 788*f*
RNA interference, 250
in vivo gene therapy, 249–250
genomics
bioinformatics, 884
and genomics, 237–257
bioinformatics, 254
definition of gene, 252
eukaryotic genome, structure of, 250, 251*f*
functional and comparative genomics, 252–253
intergenic sequences, 251–252, 251*f*
proteomics, 253
sequencing genome, 250
summary, 255–256
Biotic potential, 816, 816*f*, 839
Biotic resource and factor
ecosystem dynamics, 837, 848
population density and distribution, 815, 822–824
Bipedalism, 561
Bird flu virus, 350
spikes of, 352f
Birds
anatomy and flight, 550f
bat-pollinated flowers, 494, 494f
beaks, 550f
communication, 803
endothermy, 549–550
feathers, 549
hormones and endocrine system, 735
learning, 799
lung, 649
migration, 800–801
modified respiration, 549
modified skeleton, 549
nervous system, 550
orientation, 800–801
osmoregulation, 666–667
pollination, 426
reciprocal altruism, 808
reproduction, 753
reptilian diversity, 549f
respiratory system, 648, 652, 653f
well-developed sense organs, 550
Bird's-nest fungi, 397, 399*f*
Birth control, 762–765
abstinence, 763
condoms, 763f
contraceptive vaccines, 763–764
intrauterine devices, 763f
male, 755
morning-after pills, 763
population control, 827
traditional methods, 763, 763f
tubal ligation, 758
vasectomy, 755, 764
Birth defect, 788
Birthrate
age distribution, 817, 826, 827
population growth
human, 827
models, 820
Birth, stages, 787*f*
Birth weight, 285*f*
Bison, American, 873, 873*f*
Biston betularia, 268
Bitrogen fixation, bacteria, 455
Bitter taste receptor, 701
Bivalent, 164
Bivalve diversity, 518*f*
Bivalves, 517
Black bread mold, 395*f*
Blackburnian warbler, 840*f*
Black-throated green warbler, 840*f*
Black truffle, 404*f*
Black walnut trees (*Juglans nigra*), 479
Blade, leaf, 444. 445*f*
BLAST (*b*asic *l*ocal *a*lignment *s*earch *t*ool), 254
Blastocad, 393
Blastocladiomycota, 392–393, 393*t*
Blastocoel, 775
Blastocyst, 783
Blastopore, 776
Blastula, 775
Blastula formation, 512
Bleeder's disease. *See* Hemophilia
Blending concept of inheritance, 182. *See also* Mendelian patterns of inheritance
Blindness
color blindness, 283–284, 704
diabetic retinopathy, 706
eye disorders, 706–708
Blind spot, 704
Blood, 590
calcium, 719
clotting, 604–605, 605f
composition of, 602f
liver functions, 638
spleen, 613
synthetic, 589
types, 603
Blood fluke, life cycle of, 523*f*
Blood glucose level
liver, 638, 744, 745f
pancreas, 744
Blood pressure
aging process, 745
anaphylactic shock, 625
hypertension, 745
velocity, 599f
water-salt balance in humans, 743–744
Blood vessel
arteries
aging process, 745
hepatic, 637
capillaries
inflammatory response, 615
small intestines, 636
veins
hepatic portal vein, 637
hepatic vein, 637
Bloom, algal, 892
Bluebird, western, 822
Blue Fugates, 190
Bluegrass, 816, 817*f*
Blue-green algae. *See* Cyanobacterium
Blue-green bacteria. *See* Cyanobacteria
BMI (body mass index), 643
BMP4 gene (bone morphogenetic protein 4), 305*f*
Bobcat, 869*f*
Body cavities, 580–581, 5851*f*
Body fluid regulation, 663–665
animal excretory systems, 664–667
invertebrate excretory organs, 664–665
nitrogenous waste products, 664, 671
osmoregulation by aquatic vertebrates, 665–666
osmoregulation by terrestrial vertebrates, 666–667
human urinary system, 667–674
aging process, 790
homeostasis and kidneys, 671–674
kidneys, 668
urine formation, 668–671
Body language, 804
Body mass index (BMI), 643
Bogs, 877
Bohr model, of atoms, 22, 22*f*
Bohr, Niels, 22
Boletus
fruiting bodies of, 399f
pore mushroom, 399f
Bolus, 634
Bone
calcium deposits, 741
compact bone, 724
growth and renewal, 724
human
compact bone, 724
long bone anatomy, 724
osteoporosis, 724
spongy bone, 724
Bone, connective tissues, 577
Bone marrow
red, 611f, 612, 724
spongy bone, 719
transplant, 625
Bone mass, 791
Bone morphogenetic protein 4 (*BMP4* gene), 305*f*
Bony fish
gills, 649f
osmoregulation, 666, 666f
Bony fishes, 542
Booster vaccine, 625
Boreal forest, 868
Borgia, G., 809
Botox, 726
Bottleneck effect, 282
Bottlenose dolphin, 803
Botulinum toxin, 726
Botulism, 726
Bovine growth hormone (bGH), 346
Bovine spongiform encephalopathy (BSE), 355
Bowerbird, 809
Bowman's capsule. *See* Glomerular capsule
BPA (bisphenol A), 732
BPH (benign prostatic hypertrophy), 756
BPV (benign positional vertigo), 712
Brachiate, 563
Brachiopods, 516
Brachiosaurus, 547
Brain
aging process, 790–791
anatomy, 687f
brain stem, 690
central nervous system, 686–691
brain stem, 690
cerebellum, 677, 690
cerebral cortex, 688
cerebrum, 677, 688
hypothalamus, 677, 690
limbic system, 690–691
medulla oblongata, 677, 690
reticular activating system (RAS), 690, 690*f*

ISTUDY

Brain—(*Cont.*)
cerebellum, 677, 690
cerebrospinal fluid, 686
cerebrum, 687–688
basal nuclei, 688
cerebral cortex, 688
cerebral hemispheres, 687–688
nervous system organization, 677
cranial nerve, 692
embryonic development, 785
hypothalamus, 677, 690
learning and memory, 691
limbic system, 690–691
medulla oblongata, 677, 690
meninge, 686
pineal gland, 690, 734f, 747
reticular activating system (RAS), 690, 690f
thalamus, 677, 690
ventricles, 686
vertebrate nervous system organization, 677
Brain eating amoeba, 385
Brain stem, 690
Bramble Cay melomys, 891*f*
Branching, of *Cooksonia*, 414
Branch, phylogeny, 295
Brassica napa, 456
Brassica oleracea, 267, 267*f*
Brassica rapus, 456
BRCA2 gene, mutation in, 244
BRCA1 oncogene, 157
Bread
Neurospora crassa, 397
Rhizopus stolonifer, 394, 395f
Breast
feeding, 739
Breathing and transport of gases, 652–656
breathing, 647, 652–654
amphibians, 652
fish, 649
mammals, 652
modifications in human, 652–654
reptiles, 652
gas exchange and transport
partial pressure, 655
surfaces, 647–651
transport of oxygen and carbon dioxide, 655–656
oxygen debt, 728
Broca's area, 688, 688*f*
Bronchiole, 651, 651*f*
Bronchitis, 656–657
Bronchus
defined, 651
respiratory tract disorders, 656–657
asthma, 625, 657
bronchitis, 656–657
choking, 656
Brown algae, 379*f*, 380. *See also* Chromalveolata supergroup
Brown-headed cowbird, 845, 845*f*, 895, 895*f*
Brown planthopper, 887
Brown, Robert, 57
Brown tree snake, 890
Brush-border enzyme, 636
Bryophytes, 411
evolution and colonization of land
hornworts, 412
liverworts, 411–412
mosses, 412, 413*f*
Bryozoans (phylum Bryozoa), 516
Bud
leaf, 445f
shoot system, 435, 439
Budding, 392
asexual, 396
hydras, 515, 752
sponges, 513
viral, 354
Buffers, 32
and pH, 32, 34
Buffer system, 673
Buffon, Count, 260
Bulbourethral gland, 757, 754*t*, 756
Bulk transport, 84
Bullhorn acacia, 844
Bumblebee, 842, 842*f*
Burdock, 499
Bursa, 724
Bursitis, 724
Butter, 43
Butterfly
competition for food, 822
migration, 800–801
mimicry, 842
Monarch, 800
Butterfly-pollinated flowers, 493–494

C

Cactus, 874
Cactus, prickly-pear, 839
Caecilians, 543
Caenorhabditis elegans, 779–780, 780*f*, 782, 789
Caenorhabditis elegans, 225
Caffeine, 684
Cairns-Smith, 318
Calcitonin, 719, 740, 741
Calcium
bones, 719
calcitonin, 719, 740, 741
parathyroid, 741–742, 741f
Calcium ion
binding of myosin and actin, 728, 728t
blood, 719
role in muscle contraction, 7, 730f
storage in muscles, 728
Callus, 745, 501*f*, 501
Calmodulin (*CaM*) gene, 305*f*
Calorie, 29
Calvin cycle, 119, 121*f*, 139
importance of, 121
steps of
carbon dioxide fixation, 119
carbon dioxide reduction, 119
regeneration of RuBP, 119
Calvin cycle reaction, 115, 115*f*. *See also* Photosynthesis
Calvin, Melvin, 22, 115, 115*f*
Calyx, 490
CAM (crassulacean-acid metabolism), 123
Cambarus, 528*f*. *See also* Crayfish
Cambrian animals, 323
Camera-type eye, 703
CaM (calmodulin) gene, 305*f*
cAMP (cyclic adenosine monophosphate), 95, 95*f*, 735
CAM photosynthesis, 123
Canada lynx, 840, 841*f*
Cancer, 143, 154, 209, 244
benign growth, 154
breast, 762
cancer cells, 154
characteristics of, 154–156
versus normal cells, 154*t*
causes of, 157, 156f
cell cycle and, 154–157
esophagus, 635
gene mutation and, 234
lung, 659, 659f
lymph nodes, 612
malignant growth, 154
monoclonal antibodies, 618, 762
origin of, 156
progression of, 154, 154f
prostate, 756
teste, 755
testicular, 755
thyroid, 741
translocations and, 178
Candida albicans, 397, 771
Candidiasis, 622
Canine tooth, 632
Canis familiaris, 334
Canis lupus, 334
Cannabidiol oil (CBD oil), 36
Cannabis. *See* Marijuana
Cannabis indica, 441
Cannabis plant
legal and illegal, 441f
products, 441f
Cannabis sativa, 36, 441, 685
Cannabis varieties, 441
Cannibalism, 632
Canola plants, 456
selenium, 456
Canopy, forest, 869
Cape buffalo, African, 843*f*
Cape May warbler, 840*f*
Capillaries, 592
Capillary
inflammatory response, 615
small intestines, 635
Capillary bed, 606*f*
anatomy of, 593f
Capillary exchange, 605, 605*f*
Capsid, viral, 353, 368
Capsule, 61, 62*f*
Carbaminohemoglobin, 655–656
Carbohydrate
fiber, 640
human digestion and nutrition, 641
Carbohydrates, 39–43, 99, 112
disaccharides, 40, 40f
monosaccharides, 40, 40f
polysaccharides, 41–43, 41f
storage in roots, 438
as structural materials, 39, 40f
Carbon, 37
as building block for biomolecules, 37–38
and life, 37, 37f
skeleton and functional groups, 38, 38t
Carbon cycle, 853, 854*f*
Carbon dioxide, 119
carbon cycle, 853
climate change, 856

ISTUDY

and climate change, 120
concentration in atmosphere, 14, 14f
deforestation, 888
emissions, 888
human respiratory system
acid-base balance, 673–674
breathing rate, 654
transport, 655–656
Carbon dioxide fixation, 119
in CAM plants, 123, 123f
in C_3 plants, 122f
in C_4 plants, 123, 123f
Carbonic acid, 34
Carbonic anhydrase, 655
Carboniferous forests, 416
Carboniferous period, 416
dinosaurs of, 326f
swamp forest of, 325f, 416f
Carbon monoxide (CO), 655
detector, 107, 107f
and enzyme inhibition, 107
poisoning, 107
Carboxyl groups, 38
Carcinogenesis, 154, 234
Carcinogens, 232
Cardiac cycle, 596
stages, 596f
Cardiac muscle, 579
Cardiac output, 596
Cardiovascular disease (CVD), 599–602
aging process, 790
alcohol, 601
atherosclerosis, 600–601, 600f, 790
cholesterol, 643
hypertension, 600, 600t
obesity, 643
omega-3 fatty acids, 601
resveratrol, 601
stroke and heart attack, 643
stroke/heart attack, 601–602
valves regulate blood flow, in veins, 600f
Cardiovascular system, 592
aging process, 790
blood pressure, 743–744, 790
cardiovascular disease, 641–642, 643, 790
blood vessels
arteries, 790
capillaries, 614, 636
veins, 637
Caribou, 868
Carnivore
diet adaptation, 632
ecosystem dynamics, 848, 849
Carotenoid accessory pigment, 375, 378
Carotenoids, 116, 704
Carotid body, 654
Carpal bone, 723
Carpel, 424, 490
Carrageenan, 381
chocolate milk, 382f
Carrier proteins, 83, 83*f*, 84, 88
Carriers, 189
Carroll, Sean, 268
Carruthers, Jean, 726
Carrying capacity, 820–821
K-selected population, 823–824
predator-prey interactions, 839, 840
Cartilage, 576
endochondral ossification, 719
knee, 724
osteoarthritis, 724
Cartilaginous fishes, 541, 541*f*, 665
Cartilaginous joint, 724
Cas9 (endonuclease enzyme), 242
Casparian strip, 436*f*, 437, 454
Caspases, 145
Cat
chemical communication, 802
nervous system, 677, 678f
Catabolism, 138
Catabolite activator protein (CAP), 224, 225*f*
Catalase, 71
Cataract, 706
Catastrophism, 262
Caterpillar, 848*f*
Cation exchange capacity (CEC), 452
Cattle
African sleeping sickness, 347
bovine spongiform encephalopathy, 335
commensalism, 843
digestive system, 632
Cattle egret, 843
Cech, Thomas, 315
Cecum, 631, 636
Celiac disease, 636
Cell, 2, 3*f*, 6, 56. *See also* Cellular respiration
as basic units of life, 57
definition of, 57
and entropy, 100, 100f
and flow of energy, 99–101
organisms and, 57, 57f
size, 58, 58f
structure and function, 56–79
cellular level of organization, 57–60
cytoskeleton, 74–77
endomembrane system, 68–70
energy-related organelles, 72–74
eukaryotic cells, 63–66
microbodies and vacuoles, 71–72
nucleus and ribosomes, 66–68
prokaryotic cells, 61–62
summary, 78–79
study of, use of microscopes for, 59–60, 59f
Cell body (neuron component), 680
Cell cycle, 144–145, 144*f*
cancer and, 154–157
and cellular reproduction, 143–161
cell cycle and cancer, 154–157
eukaryotic chromosome, 147–148, 147*f*
mitosis and cytokinesis, 148–153
prokaryotic cell division, 158–160
summary, 160–161
checkpoints, 145
control of, 145
G_0 stage, 144
interphase, 144, 144f
G_1 stage, 144
G_2 stage, 144
S stage, 144
M (mitotic) stage, 144
Cell division, 144–145
in eukaryotes, 147–153 (see also Cytokinesis; Mitosis)
functions of, 159t
prokaryotic, 158–159
Cell envelope, 360
bacteria, 61
Cell-mediated immunity, 617, 621*f*
Cell recognition proteins, 83*f*
Cells, 30
Cell signaling, 92, 94–95, 94*f*, 94*t*
Cell structure and function, 56–79
cellular level of organization, 57–60
cytoskeleton, 74–77, 75f
actin filaments, 74
centrioles, 76, 76*f*
cilia, 76
flagella, 76, 77*f*
intermediate filaments, 74
microtubules, 74–76
protein components of, 74
endomembrane system, 68–70
endoplasmic reticulum, 68, 68*f*
Golgi apparatus, 68–69, 69*f*
lysosomes, 69, 69*f*
summary, 70, 70*f*
energy-related organelles, 72–74
chloroplasts, 72, 72*f*, 73*f*
mitochondria, 72–74, 73*f*
eukaryotic cells, 63–66
animal cell, 64*f*
definition of, 61
origin of, 63, 63*f*
plant cell, 65*f*
structure of, 64–66, 64*f*, 65*f*
microbodies and vacuoles, 71–72
peroxisomes, 71, 71*f*
vacuoles, 71–72, 71*f*
nucleus, 66–67, 66f
prokaryotic cells, 61–62
definition of, 61
size and shape, 61
structure of, 61–62, 62*f*
ribosomes, 67–68
summary, 78–79
Cell surfaces, in animals, 91–93
extracellular matrix, 92, 92f
junctions between cells, 92–92, 93f
adhesion junctions, 92, 93*f*
gap junctions, 92, 93*f*
tight junction, 92, 93*f*
Cell suspension culture, 502
Cell theory, 57
Cell-to-cell communication
cell signaling, 94–95, 94f, 94t
second messenger systems, 95, 95f
Cellular damage theory of aging, 789
Cellular differentiation, 494
animal development, 777–782
cytoplasmic segregation, 777
induction, 777–782
Cellular organization, 57–60
Cellular respiration, 72, 73, 126–141
definition, 127
efficiency of, 137
fermentation, 131–133, 131f
advantages and disadvantages of, 131, 132
as anaerobic process, 131
efficiency of, 133
NAD^+ regeneration, 131

ISTUDY

Cellular respiration—(*Cont.*)
glycolysis, 129–131, 130f
energy-harvesting steps, 129, 130*f*
energy-investment step, 129, 130*f*
inputs and outputs of, 131, 133
metabolism, 137–139
anabolism, 138
catabolism, 138
energy organelles, 138–139
flow of energy, 139
mitochondria, 133–137
citric acid cycle, 134–135, 134*f*
electron transport chain, 135–136
preparatory reaction, 134, 134*f*
structure of, 133, 133*f*
mitochondria and, 108–109, 108f
overview, 127–129, 127f
breakdown of glucose, 127, 127*f*
NAD^+ and FAD, 128
oxidation-reduction reaction, 128
phases of, 128–128, 128f
aerobic phase, 128
anaerobic phase, 128
citric acid cycle, 128
electron transport chain, 128
glycolysis, 128
preparatory reaction, 128
and photosynthesis, 139f
summary, 140–141
Cellular response and signal transduction, 471
Cellular slime molds, 386, 389, 610
Cellulose, 42, 42*f*, 121
Cell wall
antibiotic, 364
archaea, 366
bacterium, 61
plant, 93
Cenozoic era, 331
mammalian diversification, 326–327
primates, evolution of, 327
Centipedes, 528, 528*f*
and millipede, 528f
Central American Jaguar Corridor, 826
Central dogma of molecular biology, 68, 211
Central nervous system, 677, 686–691. *See also* Brain; Spinal cord
Central vacuole, 71–72, 71*f*
Centrioles, 76, 76*f*, 149
Centromere, 148, 149*f*, 164
Centrosome, 75, 76, 148, 174
division of, 148
Cephalization, 510, 522
animal evolution, 677
invertebrate nervous systems, 677
Cephalochordates, 537
Cephalopods, 518
diversity, 519f
Cephalothorax, 527
Cercozoa, 389
Cercozoans, 380, 380*f*
Cerebellum, 677, 690
Cerebral cortex, 688
Cerebral hemisphere, 687–688, 688*f*
Cerebrospinal fluid, 686
Cerebrum, 687–688
basal nuclei, 688
cerebral cortex, 688
cerebral hemispheres, 687–688
nervous system organization, 677
Cerumen, 702
Cervical vertebra, 722
Cervix
anatomy, 758
birth, 787f
Cervus elaphus, 807*f*
Cesium-134, 24
Cesium-137, 24
Cestode. *See* Tapeworm
CF (cystic fibrosis), 290, 659
CFC (chlorofluorocarbon), 892
CFTR gene, 659
cGMP (cycline guanosine monophosphate), 756, 757
Chagas disease, 374
Chambered nautilus, 519*f*
Chameleons, 548
Channel proteins, 83, 83*f*
Channelrhodopsin, 707
Chaparral, 871
Chaperone molecules, 216
Chaperone proteins, 50
Chara, 384, 385*f*, 408
Character displacement, 838–839
Chargaff, Erwin, 205
Chargaff's rules, 205, 207
Charpentier, Emmanuelle, 242
Charismatic species, 885
Charophytes, 372, 381, 389, 408, 410*f*
Chase, Martha, 204
Cheetah, 803, 803*f*, 805
Chelicerate diversity, 531*f*
Chelicerate, horseshoe crab, 886
Chemical-bond energy, 26
Chemical communication, 733, 802–803
Chemical cycling
ecosystem dynamics, 852–856, 852f
carbon cycle, 853–854
nitrogen cycle, 855–856
phosphorus cycle, 854
water cycle, 847–848, 884
Chemical element. *See* Element
Chemical energy, 99
Chemical messengers, 94
Chemical mutagens, 232
Chemical sense organ, 700–702
sense of smell in humans, 701–702
sense of taste in humans, 700–701, 700f, 702
vomeronasal organ (VNO), 803
Chemiosmosis, 119, 136, 455
Chemistry, 19–35. *See also* Organic chemistry
acids and bases, 31–34
acidic solutions, 32
basic solutions, 32
buffers and pH, 32, 32
pH scale, 32, 32*f*
chemical elements, 20–25
atomic number and mass number, 21
atoms, 20–21, 21*f*
electrons and energy, 22
elements, 20, 20*f*
periodic table, 21, 22*f*
radioactive isotopes, 22–25
molecules and compounds, 25–27
covalent bonding, 26–27, 27*f*
ionic bonding, 26–, 26*f*
summary, 34–35
water chemistry, 27–31
hydrogen bonding, 28*f*, 28
models for structure of water, 27, 28*f*
properties of water, 28–31, 29*f*, 30*f*
Chemoautotroph, 848, 880
bacteria, 880
Chemoautotrophs, 363, 368
bacteria, 315, 363
Chemoheterotroph, 363, 366
Chemoheterotrophic prokaryotes, 368
Chemoheterotrophs, 362
Chemoreception, 700–702
sense of smell in humans, 701–702
sense of taste in humans, 700–701, 700f, 702
Chemoreceptor, 699, 700–702
Chemosynthesis, 363
Chemotropism, 479
Chiasmata, 166*f*, 167
Chicken
development
extraembryonic membrane, 782
gastrula, 775–776
yolk, 775, 776, 782
Chimpanzees, 558
auditory communication, 803
human, comparison of, 560f
inclusive fitness, 808
Chinese liver fluke, 523
Chitin, 42–43, 392
arthropod exoskeleton, 717
Chlamydia, 769, 770*f*
Chlamydomonas, 381
haploid life cycle of, 383f
reproduces sexually, 383
vegetative cells, 383
electron micrograph of, 383*f*
Chlorine, 892
Chlorofluorocarbon (CFC), 892
Chlorophyll, 112
Chlorophyll production, 485*f*
Chlorophytes, 381–383, 389
Chloroplasts, 72, 72*f*, 73*f*
endosymbiotic theory, 72
origin of life, 322
and photosynthesis, 108, 108f
photosynthesis in, 112
structure of, 72, 73f
Choanoflagellates, 386, 389, 507
Chocolate milk
without carrageenan, 382f
Choking, 656
Cholera, 191
Cholesterol, 45, 46*f*
accumulation in walls of arterial blood vessels, 91
in animal plasma membrane, 81
cardiovascular disease, 614, 643
and cardiovascular disorders, 45
functions, 642
sources, 642
Chordates (phylum Chordata), 342, 347, 537–538
characteristics of, 537, 537f, 539
dorsal tubular nerve cord, 537
endoskeleton, 539
internal organization, 539
notochord, 539
pharyngeal pouches, 537
postanal tail, 537

ISTUDY

skull, 539
vertebral column, 539
evolution of, 539–540
nonvertebrate chordates, 537–538
phylogenetic tree of, 538f
Chorion, 782, 783
Chorionic villi sampling, 778*f*
Chorionic villus, 784–785
Choroid, 703
Christmas tree worms, 520
Chromalveolata supergroup, 388
alveolates, 375–378
apicomplexans, 376–377
ciliates, 377–378, 377*f*
dinoflagellates, 375, 376*f*
stramenopiles
brown algae, 378–379
diatoms, 378
golden brown algae, 378
oomycetes, 379–380
Chromatid, 144, 148
Chromatin, 66*f*, 67, 225
packing, 225
structure, 225–228, 227f
Chromatin remodeling complex, 226
Chromoplasts, 72
Chromosome, 67
duplication, 148, 149f
of eukaryotes, 210
eukaryotic, 147–148, 147f
homologous, 163–164, 163f
karyotype, 163
nondisjunction, 174, 175f
number, changes in, 174–177
Jacobs syndrome, 177
Klinefelter syndrome, 177, 177*f*
monosomy, 174
poly-X females, 177
trisomy 21, 175, 176*f*
Turner syndrome, 176, 177, 176*f*
polyploidy speciation, 302
structure, changes in, 177–179, 178f
deletion, 176*f*
duplication, 176, 178*f*
inversion, 177, 178*f*
translocation, 177–178, 178*f*
in zygote, 174
Chronic bronchitis, 657
Chronic inflammation, 614
Chronic obstructive pulmonary disease (COPD), 657
Chronic wasting disease (CWD), 536
Chthamalus, 839, 840*f*
Chyme, 635
Chytridiomycota, 393–394, 393*t*
Chytrids, 394
Cicada, 718*f*
Cichlid, 302
Cigarette smoking, 660, 660*f*, 788
Cilia, 76
Ciliary muscle, 704
Ciliates, 377–378, 377*f*, 388
Cilium
human lungs, 648
Cinchona ledgeriana, 502
Cinchona species, 443
Cinnamomum verum, 443
Circadian rhythms, 463, 485, 486, 485*f*
biological clock, 747
Circulation and cardiovascular system. *See also* Blood
blood, 602–607
blood clotting, 604–605
capillary exchange, 605–607
plasma, 602
platelets, 604–605
red blood cells, 603–604
white blood cells, 604
blood vessels
arteries, 643
capillaries, 614, 636
veins, 637
cardiovascular disease (CVD), 599–601
atherosclerosis, 600–601
hypertension, 600
stroke/heart attack, 601–602
circulatory circuits, comparison of
pulmonary circuit, 598
systemic circuit, 598–599
human cardiovascular system, 594–601
blood, path of, 595–596
blood pressure, 599, 743–744, 790
cardiovascular disease, 641–642, 644, 790
electrocardiogram, 597–598
heartbeat, 596–597
heart, structure of, 594–595
invertebrates, transport, 590–591
with circulatory system, 590–591
without circulatory system, 590
vertebrates, transport, 592–593
blood vessels, of closed circulatory systems, 592*f*
circulatory pathways, comparison of, 592–593, 593*f*
pulmonary circuit, 593
Circulatory pathways
comparison of, 593f
Circulatory systems, 529, 590
aquatic animals, 590f
blood vessels, 650
closed, 591, 591f, 592f
open, 590–591, 591f
Circumcision, 756
Cirrhosis, 638
Citric acid cycle, 128, 134–135, 134*f*
inputs and outputs of, 135
production of CO_2, 135
substrate-level ATP synthesis in, 132
Citrus canker, 885
Clades, 342, 347
Cladistics, 342, 347
Cladogram, 342, 347
constructing, 342f
phylogenetic tree, 343f
species and traits, 342f
Clam
filter feeders, 631f
foot, 717
population distribution, 815
Clam worms, 520, 520*f*
Clarkia, 302
Clark's nutcracker, 844, 844*f*
Class, 8, 335, 335*f*, 336. *See also* Taxonomy
Classical conditioning, 799, 800*f*
Classification of life, 334, 336*f*, 336
Clavelina, 538*f*
Claviceps purpurea, 398, 398*f*
Clavicle, 722–723
Cleaning symbiosis, 844
Cleavage, 512
defined, 775
human embryo, 783
Cleavage furrow, 150
Clements, F. E., 846
Climate
biomes, 865
coniferous forests, 865
desert, 874
grassland, 872–874
shrubland, 870–872
temperate deciduous forests, 866*f*, 869
tropical rain forests, 866*f*, 869–870
tundra, 865–868, 866*f*
defined, 863
ecosystem, 863–864
solar radiation, 863–864
ocean currents, 880
regulation and biodiversity, 888
Climate change, 14, 14*f*, 120, 120*f*
carbon cycle, 853
defined, 854, 892
and extinction, 14, 14f
extinction threat, 892
global, 858
greenhouse gases, 854, 857, 892
tropical rain forest destruction and, 120
Climax community, 846
Clitellum, 520
Clitoris, 759
Cloaca
birds, 753
uric acid, 664
Clock genes, 486
Clonal selection theory, 617, 617*f*, 620*f*
Clone, 500
Cloning, 238, 238*f*
reproductive, 152, 153, 153f
therapeutic, 152, 153, 153f
Clonorchis infection, 523
Clonorchis sinensis, 523
Closed circulatory system, 520. *See also* Cardiovascular system
Clostridium botulinum, 726
Clostridium tetani, 339, 364
endospore of, 364f
Clover, 483
Clownfish, 843
"Club" drugs, 684
Club fungi, 397, 399*f*
Clumped distribution, 815
Cnidarian diversity, 514
Cnidarian, reproduction, 752
Coacervate droplets, 316, 331
Coastal ecosystem, 875*f*, 878*f*
Cocaine and crack, 685, 788
and brain activity, 685f
Coccus, 61
Cochlea, 709
Cochlear canal, 709, 710*f*
Cochlear implant, 712
Cochlear nerve, 709*f*
Cocklebur, 483
Cod, 825
Codominance, 193
Codon, 211

ISTUDY

Codonosiga, 387
Coelacanth, 543*f*
Coelom, 511
  embryonic development in animals, 776, 777f
Coelom development, 511
Coenzymes, 51, 105, 643
Coevolution, 493, 494, 844
  of plants, 494
Cofactors, 104, 105, 106*f*
Cognitive learning
  imitation, 801
  insight, 801
  observation, 801
Cohesion, 30, 461
Cohort, 816
Colchicine, 76
Cold, common, 656
Coleochaete, 408
Coleoptile, 472
Collagen, 47, 92, 192
Collarbone, human, 722
Collecting duct, 668, 669*f*
Collenchyma cell, 431, 431*f*
Colon, 637
Colonial choanoflagellates, 387*f*
Colonial flagellate hypothesis, 507
Colonization and exotic species, 889
Colony, 383
  protists
    choanoflagellates, 386
    ciliates, 378
Coloration
  camouflage, 841, 842f
  mimicry, 842
  warning, 841
Color blindness, 198, 198*f*, 283–284, 704
Color vision, 702, 704
Columnar epithelium, 575
Comb jellies, 513
Commensal, 367
Commensalism, 363
  bacteria and archaea, 363
  defined, 843
  ecology of communities, 843
Common ancestor, 341, 347
Common cold, 656
Communication
  animal, 801–805
    auditory communication, 803
    chemical communication, 802–803, 804
    visual communication, 804
  defined, 801
Community, 2, 3*f*. *See also* Community and ecosystem ecology
  climax, 846
  defined, 814, 836, 836f
  diversity, 885, 885f
Community and ecosystem ecology, 835–861
  climate change, ecological consequences of, 856–859
    consequences of, 856–859
    contributors to, 856
  community development, 844–847
  dynamics of ecosystem, 847–856
    autotrophs, 848
    chemical cycling, 852–856
    energy flow, 849–851
    heterotrophs, 847–849
  ecology of communities, 836–844
    community interactions, 837–839
    community structure, 836–837
    predator-prey interactions, 839–842
    symbiotic relationships, 842–844
Compact bone, 577, 719
Companion cells, 433, 458
Comparative genomics, 253, 253*t*
Competition
  male competition, 287–288
  between populations, 837–839, 837f
  regulation of population size, 821
  resource, 837–839
  sexual selection, 807
Competitive exclusion principle, 837
Competitive inhibition of enzyme, 106
Complementary base pairing, 52, 52*f*, 207, 208–209, 209*f*
Complement protein, 615, 616*f*
Complete digestive tract, 519
  earthworms, 630–631
Complete metamorphosis, 528
Compound, definition, 25
Compound eye, 702, 702*f*
Compound light microscope, 59, 59*f*
Concentration gradient, 84
Concurrent flow, 649
Condensation, 852
Condom, 763*f*
Cone cell
  compound eye, 702
  human eye, 704, 705
Cones, 419
  conifers, 419–420
  cycads, 420–421
Confocal microscopy, 60
Confuciusornis, 269
Congenital hypothyroidism, 740, 740*f*
Conidiophores, 396
Conidiospores, 396
Coniferous forest, 865, 866*f*
Conifers, 419–420, 419*f*
Conjugation, 361, 368, 383, 389
  ciliates, 378
  prokaryotes, 361
  Spirogyra, 383
Conjugation pili, 62
Conjugation pilus, 361, 368
Conjunctiva, 703
Connective tissue, types of, 576*f*
Connell, Joseph, 839
Conservation biologist, 884
Conservation biology, 884–886
Conservation of biodiversity, 883–897
  causes of extinction, 889–894
    climate change, 892–893
    diseases, 889, 892
    nonnative species, 889–891
    habitat loss, 889, 890*f*
    overexploitation, 889, 893–894
    pollution, 889, 891–892
  conservation biology and biodiversity, 884–886
    biodiversity organization, 884–885
    defined, 884
    distribution of biodiversity, 886
  conservation techniques, 894–896
    habitat preservation, 894–895
    habitat restoration, 895–896
  and DNA, 867
  value of biodiversity, 886–889
    biodiversity and natural ecosystems, 888–889
    direct value, 886–887
    indirect value, 887–888
Consortium for the Barcode of Life (CBOL), 337
Consumer, 848
Consumptive use value of biodiversity, 887
Contact dermatitis, 626
Continental drift, 328–329, 328*f*
  plate tectonics, 329
Continental shelf, 879
Continuous feeder, 631
Contraction, muscular
  adenosine triphosphate (ATP), 728, 729
  role of calcium, 729
  sliding filament model, 728
Control center
  negative feedback, 585–586, 585f–586f
  positive feedback, 586–587, 586f
Control group, 11
Convergent evolution, 306, 306*f*, 344, 347
Cooksonia, 414
  fossil, 414f
  sporangia of, 414
COPD (chronic obstructive pulmonary disease), 657
Copulation, 753
Copy number variations, 254
Coral, reef community, 814, 879
Coral reefs, 15, 515, 838, 879
  algae, 879
  bleaching, 892f
  community, 814
  habitat loss, 889
  ocean ecosystems, 878
Coral snake, 842
Cordyceps fungi, 390
Corepressor, 222
Corey, Robert, 49
Cork cambium, 431, 442
  bark of, 442
Cork cambium, 440, 442
Cork cells, 431
Corn (*Zea mays*), 480
  ears and grain of, 422f
  varieties of, 422f
Cornea, 702, 703
Corn kernel, structure of, 496
Corn oil, 43
Corn smut, 400*f*
Corn stalk, 440
Corolla, 490
Corona, 523
Corona radiata, 774, 774*f*
Coronary arteries, 599
β-Coronavirus, 349
Coronavirus (COVID-19), 15, 276, 349, 356
  emerging infectious diseases, 356, 358, 358f
  origin of, 867
  respiratory system, 658, 658f
  severity of, 867
  sources of, 867, 867f
  vaccine, 609
Corpus callosum, 687*f*, 688
Corpus luteum, 759*f*, 760, 783

ISTUDY

Cortex, 437
bark, 442
roots, 437
stem system, 440
Cortical granule, 774
Cortisol, 743
Cortisone, 743
Cost-benefit analysis, 287
Cotyledons, 423, 496
monocots and eudicots, 434
Countercurrent exchange, 649
Countercurrent flow, 649*f*
Coupled reactions, 102
Courtship
birds, 803, 809, 809f
displays, 809
fireflies, 804, 804f
Covalent bond, 26–27, 27*f*
double, 27, 27f
nonpolar, 27
polar, 27
COVID-19. *see* Coronavirus (COVID-19)
Cowbird, brown-headed, 845, 845*f*, 895*f*
Coxal bone, 723
Coxal gland, 665
C_3 plants, 122, 122*f*
C_4 plants, 123, 123*f*
Crabs, 878
mutualism, 843
nervous system, 678f
optimal foraging model, 806
Crack cocaine, 685
Cranial nerve, 692, 692*f*
Cranium, 721, 722*f*
Crank. *See* Methamphetamine
Crayfish, 527
Crayfish nervous system, 528
Crayfish's sensory organs, 528
C-reactive protein, 614
Creatine phosphate, 728
Crenation, 87
Creosote bush, 815
Cretinism. *See* Congenital hypothyroidism
Cricket, 803
Crick, Francis, 206, 208, 215
Cri du chat (cat's cry) syndrome, 178
CRISPR (*c*lustered *r*egularly *i*nterspaced *s*hort palindromic *r*epeats), 237, 242
and genome editing, 242, 243f
pioneers, 242
use in humans, 244, 244f
CRISPR-Cas9 system, 244
CRISPR pollen method, 422
Cristae, mitochondria, 73, 73*f*
Crop, 630, 631
Crossing-over, 166–167, 166*f*, 174
errors in, 174
Cross-pollination, in garden pea, 183, 183*f*. *See also* Mendelian patterns of inheritance
Crustacean. *See also* Crab
barnacles, 839, 840f, 843, 878
chemical senses, 700
excretory organs, 665
krill, 631
respiratory system, 649
Crustacean diversity, 527
Crustaceans, 527
Cryptic species, 294, 296
Ctenophora, 513
Cuboidal epithelium, 574
Cucumber, 446*f*
Curie, Marie, 22, 23
Current, ocean, 880
Cushing syndrome, 744
Cutaneous receptor
Krause end bulbs, 713
Meissner corpuscles, 713
Merkel disks, 713
Pacinian corpuscles, 713
root hair plexus, 713
Ruffini endings, 713
Cuticle, 408, 430
leaf, 445f
Cuvier, Baron Georges, 260–261, 274
CVD. *See* Cardiovascular disease (CVD)
Cyanobacteria, 62, 364–365, 365
cyanobacterial cells, 365
diversity, 365f
lichen, 365
Cyanobacterium, nitrogen fixation, 855
Cycads, 420–421
Cycas revoluta, 420
Cyclic adenosine monophosphate (cAMP), 95, 95*f*, 735
Cyclic AMP (cAMP), 222, 224
Cyclic pathway, of photosynthesis, 117, 118*f*
Cyclin-dependent-kinase (CDK), 146
Cycline guanosine monophosphate (cGMP), 756, 757
Cyclins, 145
Cyclospora cayetanensis, 377
Cystic fibrosis (CF), 90, 188, 191, 191*f*, 249, 290, 659
Cystic fibrosis transmembrane regulator (CFTR), 659
Cysts, 371, 523
Cytochrome *c*, 272, 345
Cytochrome c oxidase, 107
Cytochrome oxidase complex, 136
Cytochrome reductase complex, 136
Cytokine
inflammatory response, 614
interferons, 615
therapeutic agents, 621, 623
Cytokinesis, 144, 150
in animal cells, 150–151, 152f
in plant cells, 151–152, 152f
Cytokinins
cell division and organ formation, 474
leaves, color changes, 475, 475f
in organ development, 470f
senescence, 475
tissue culture experiments, 475f
Cytolysis, 87
Cytomegalovirus, 622, 788
Cytoplasm, 364
Cytoplasmic segregation, 778, 778*f*
Cytoplasm, prokaryotes, 60–61
Cytoplasm, prokaryotic cell, 360
Cytosine (C), 205, 205*f*. *See also* DNA (deoxyribonucleic acid)
Cytoskeleton, 64, 74–76, 75*f*
actin filaments, 74
centrioles, 76, 76f
cilia, 76
flagella, 76, 77f
intermediate filaments, 74
microtubules, 74–76
protein components of, 74
Cytotoxic T cell, 619

D

Dalton, John, 20
Damage accumulation theory of aging, 789
Damselfly, 753, 753*f*
Dandelion, 824*f*
Darwin and evolution, 258–275. *See also* Evolution
Darwin's theory of natural selection, 260–263
biogeographical observations, 264, 264*f*
natural selection and adaptation, 265–266
observations of change over time, 263
selection at work, 267–269
evidence for evolution, 269–273
anatomical evidence, 271–272
biochemical evidence, 272–273
biogeographical evidence, 270–271
criticism of evolution, 273
from developmental biology, 273
fossil evidence, 269–270, 270*f*
finches, 303, 304–305
history of evolutionary thought, 260–262
late eighteenth/early nineteenth century influences, 260–262
mid-eighteenth century influences, 260
summary, 274–275
Darwin, Charles, 182, 260, 338, 472, 481, 504. *See also* Darwin and evolution
Darwin, Erasmus, 260
Darwin, Francis, 481
Darwin's theory of natural selection, 182, 263–269
biogeographical observations, 263–265, 264f
natural selection and adaptation, 265–266
observations of change over time, 263
selection at work, 267–269
Data, 11–12
presentation of, 11, 11f
statistical, 12
Daughter cells, 148
Deadly Fungi, 397, 398
Deafness, 712
Deamer, David, 316
Deamination, 138, 643
Death rate
defined, 817
human population growth, 872
Deceleration phase, 820
Deciduous forest
temperate, 866f, 869
tropical, 870
Deciduous plant, 432, 444, 869–870
Decomposition
ecosystem dynamics, 849
indirect value of biodiversity, 888
Deductive reasoning, 10
Deep brain stimulation, 689
Deer, Red, 808
Defect Levels, 530
Defense strategy
plant, 432, 443
prey defenses, 841–842
camouflage, 841, 842*f*
mimicry, 842
warning coloration, 841
territoriality, 805–806
trichomes, 431f
Deforestation, 856, 857*f*
biodiversity, 888
Dehydration reactions, 39, 51

ISTUDY

De la Chapelle syndrome, 176
de Lamarck, Jean-Baptiste, 262, 274
Delayed allergic response, 625*t*, 626
Deletion, chromosome, 177, 177*f*
Deletion syndromes, 178, 178*f*
Dementia, 691
Demographics of populations, 815–818.
    See also Population ecology
  density and distribution, 815
    clumped distribution, 815
    random distribution, 815
    regulation of population size, 821–823
    uniform distribution, 815
  population growth, 816–817
    age distribution, 817, 827, 827*f*
    mortality patterns, 816–817
Demographic transition, 826
Denaturation, 104
Denatured protein, 50
Dendrite, 680
Dendritic cell, 615
Dendrobates pumilio, 824*f*
Denitrification, 856
Density-dependent factor, 822, 822*f*
Density-independent factor,
    822–823, 822*f*
Dentition. *See* Tooth
Deoxyribose, 40, 51
Depolarization, 682
Depression, 683
Derived traits, 341, 347
Dermis, 584, 713
Described species, 884*f*
Desert
  cold, 872
  ecosystem, 874
Designer drug
  bath salts, 684–685
  K2, 685
  MDMA (methylenedioxymethamphetamine), 684
  methamphetamine, 684
Desmodus rotundus, 98
Desmosomes, 92
Detritivores, 848
Deuterostomes, 512
  protostomes compared, 512f
Development, 5
Developmental genes and macroevolution, 307–309
  eye development, 308
  human evolution, 309
  limb development, 306
  pelvic fin genes, 308
  shape development, 308
Developmental similarities, in vertebrates, 272, 272*f*
Development and aging, 773–793
  aging process, 789–791
    cardiovascular system, 790
    digestive system, 790
    endocrine system, 791
    excretory system, 790
    hypotheses of aging, 789
    immune system, 790
    integumentary system, 790
    musculoskeletal system, 791
    nervous system, 790–791
    reproductive system, 791
    respiratory system, 790
    sensory system, 791
  cellular stages of development, 775
  defined, 775
  developmental processes
    morphogenesis, 77, 780–782
  early developmental stages, 774–777
    embryonic development, 775–777, 783–785
    fertilization, 774–775, 783, 784*f*
  human
    aging process, 789–791
    embryonic and fetal development, 758,
      774–777, 782–788, 786*f*
  organ stages of development, 776–777
  tissue stages of development, 775–776
DHT (dihydrotestosterone), 757
Diabetes, 153
Diabetes insipidus, 737
Diabetes mellitus, 744–745
  diabetic retinopathy, 706
  obesity, 643
  type 1 diabetes, 643, 745
  type 2 diabetes, 643, 745–746
Diabetic retinopathy, 706
Diagnostic traits
  evolutionary species concept, 296, 297f
  morphological species concept, 294
Diamondback rattlesnakes (*Crotalus atrox*), 267
Diaphragm
  breathing, 652
  vomiting, 634
Diapsid, 547
Diarrhea, 637
  tapeworm infections, 523
Diastole, 596
Diatom, 388
Diatoms, 378, 378*f*, 389
  asexual reproduction, 378
  sexual reproduction, 378
Dictyostelium discoideum, 610*f*
Didinium nasutum, 839
Diencephalon, 690
Diet
  adaptation
    carnivores, 632
    herbivores, 632
    omnivores, 632
  genetic basis for behavior, 796
  human
    birth defect prevention, 788
    obesity, 643
    vitamins and minerals, 643
Dietary Guidelines (2020), 641
Dietary Guidelines, human health, 641
Differential reproductive success, 266
Diffusion, 85, 85*f*
  gas-exchange surfaces and respiration, 647
Digestive system, 629–645
  digestive enzymes, 638–640
    lipase, 640
    maltase, 640
    pancreatic amylase, 640
    pepsin, 635, 639
    peptidase, 639
    salivary amylase, 633, 638
    trypsin, 639
  digestive tracts, 630–632
    adaptations to diet, 632
    complete digestive tract, 630–631
    continuous feeders, 631
    discontinuous feeders, 631
    human, 633–638
    incomplete, 630–631
  functions, 630
  human digestive system, 633–638
    accessory organs, 633*f*, 637–638
    aging process, 790
    large intestine, 633*f*, 636, 637
    mouth, 633–634
    pharynx and esophagus, 633*f*, 634–635
    small intestine, 633*f*, 633–636
    stomach, 633*f*, 633
  human nutrition and health, 640–644
    carbohydrates, 640
    diet and obesity, 643–644
    lipids, 641–642
    proteins, 642–643
    vitamins and minerals, 643
Digestive tract, 630–632
  adaptations to diet, 632
    carnivores, 632
    herbivores, 632
    omnivores, 632
  complete digestive tract, 630–632
  continuous feeders, 631
  discontinuous feeders, 631
  filter feeders, 631, 632
  human, 633–638, 633f
    accessory organs, 633*f*, 637–638
    large intestine, 633*f*, 634, 636, 637
    mouth, 633–634
    pharynx and esophagus, 633*f*, 634–635
    small intestine, 633*f*, 635–636, 637
    stomach, 633*f*, 635
  incomplete digestive tract, 630–631
  organ stages of development, 776–777
Digitalis species, 502
Dihybrid cross, 186, 186*f*
Dihydrotestosterone (DHT), 757
Dikaryotic hypha, 392
Dimorphism, 287*f*
Dinoflagellate bloom, and fish kill, 376*f*
Dinoflagellates, 388
  typical, 375f
Dinosaurs, 547
Dioecious, 421
Dioecious animal, 752
Diploblastic animals, 511
Diploid (2n) number, 148, 163
Diplomonads, 388
Directional selection, 285–286, 285*f*, 286*f*
Direct selection, 808
Direct value of biodiversity, 886–887
  agricultural value, 886–887
  consumptive use value, 887
  medicinal value, 886
Disaccharides, 40, 40*f*
Discontinuous feeder, 631
Disease
  extinction cause, 889, 891, 892
  regulation of population size, 822
  sexually transmitted disease (STD), 766–771
Disruptive selection, 825, 825*f*, 287*f*
Distal convoluted tubule, 668, 669*f*
Distribution patterns, 815, 815*f*
Distribution, population, 815
Diuresis, 671
Diverge, 341, 347

ISTUDY

Diversity
community diversity, 885, 885f
landscape, 885
maintenance, 288–290
heterozygote advantage, 290
natural selection, 289–290
species, 836, 837, 869, 889
Diversity of life, 2*f*, 5
Division of labor, cell, 80
DNA (deoxyribonucleic acid), 551, 202, 202, 272
barcoding, 338
Chargaff's rules, 205
complementary base pairing, 52, 52f
composition of various species, 205, 205t
conservation of wildlife, 867
eukaryotic chromosome, 147–148, 147f
as genetic material, 203–207, 204f
junk DNA, 251, 252
nucleotide composition of, 205, 205f
origin of life, 322
phylogenetic species concept, 296
replication, 208–210, 209f
accuracy of replication, 210
complementary base pairing, 208–209, 207*f*
eukaryotic DNA replication, 209*f*, 210
joining of new DNA strands, 209
lagging strand, 209
leading strand, 209
meiosis and, 166
Okazaki fragments, 209
prokaryotic DNA replication, 209*f*, 210
semiconservative replication, 208, 208*f*
unwinding of DNA molecule, 208, 209*f*
sequence alignment, 344f, 510
steroid hormones, 736
structure of, 51–52, 51f, 52f, 52t, 205–206
Watson and Crick model, 206–207, 207f
wildlife conservation, 867
X-ray diffraction pattern of, 206, 206f
DNA Barcoding of Life, 337
DNA fingerprinting, 193, 241, 241*f*
DNA helicase, 208
DNA ligase, 209, 238, 239
DNA methylation, 228, 228*f*
DNA microarrays, 252–253, 253*f*
DNA polymerase, 209, 210, 239
DNA primase, 208
DNA profiling. *See* DNA fingerprinting
DNA repair enzyme, 233
DNA sequencing, 239, 239*f*
DNA splicing, 239
DNA technology, 238–242
cloning, 238, 238
DNA analysis, 240–241
DNA sequencing, 239, 239f
genome editing, 240
polymerase chain reaction, 239, 240
recombinant DNA technology, 238–239, 238f
Dodder, 457, 457*f*
Dodder plant, multiple hosts, 478*f*
Dodder vine (*Cuscuta pentagona*), 478
Dog
chemoreceptors, 699f, 699
classical conditioning, 799, 800f
Dolphin
auditory communication, 803
bottlenose, 803
convergent evolution, 306
echolocation, 700
teeth, 632
Domain, 8–9, 335, 335*f*, 346
Domain Archaea, 338, 338, 339*t*, 346
Domain bacteria, 338, 339*f*, 339*t*, 346
Domain Eukarya, 338, 339*f*, 339*t*, 346
Domain, tree of life, 339*f*
Dominance hierarchy, 287
Dominant allele, 185
Dominant generation, 409–410
Dopamine
cocaine, 685
functions, 684
tobacco use, 684
Dormancy, 498
Dorsal root, 692, 692*f*
Dorsal root ganglion, 695, 695*f*
Double fertilization, 426, 492
Double helix model, DNA, 205, 06*f*
Doubling time, 826
Doudna, Jennifer A., 242
Down syndrome, 178. *See* Trisomy 21
Dragonfly
habitat, 837
vision, 702
Drone, 794
Drosera capensis, 457*f*
Drosophila biarmipes, 268
wing spot in, 268, 269f
Drosophila melanogaster, 197, 253
immune system evolution, 610
morphogenesis, 780–782
segmentation, 780
Drug abuse, 684–685
alcohol, 684
birth defects, 788
cocaine and crack, 685
designer drugs, 685
growth hormone, 739
heroin, 685
marijuana, 685
nicotine, 684
sexually transmitted diseases, 767
Dry fruits, 498
Dryopithecus, 561, 562
Duchenne muscular dystrophy, 198, 200
Duckbill platypus, 753
Due date, 782
Dugesia, 521
Duodenum, 635
Duplication, chromosome, 177, 178*f*
Dwarfism, pituitary, 739
Dynein, 76
Dysentery, 364
Dystrophin, 198, 200

E

Eagle, bald, 885, 885*f*, 890
Ear
disorders
aging process, 790
hearing loss, 712, 791
vertigo, 712
human, 709–712, 709f
functions, 709
inner ear, 709–710
middle ear, 709
outer ear, 709
mammal characteristic, 709
Eardrum. *See* Tympanic membrane
Earth, temperature, 854, 856, 865, 865*f*
Earthworms, 519–520
complete digestive tract, 630–631, 630f
cross section, 520f
excretory organs, 665, 665f
hydrostatic skeleton, 717, 717f
nervous system, 677
reproduction, 520f
respiration, 647, 648f
Earwax, 45, 709
Ebola outbreaks, 15
Ebola virus, 354
Ecdysozoans, 524, 525
Echinoderms, 532, 532*f*, 533
sea stars, 632
Echolocation, 699*f*, 700
E-cigarette, 660, 660*f*, 684
Ecological niche
character displacement, 838–839
competitive exclusion principle, 837
defined, 837
niche specialization, 839, 840f
resource partitioning, 838
Ecological pyramid, 851, 851*f*
Ecological release, 303
Ecological succession, 846
Ecology, 812
community and ecosystem ecology, 835–861
climate change, ecological consequences of, 856–859
community development, 844–847
dynamics of ecosystem, 847–856
ecology of communities, 836–844
defined, 814
ecological succession, 846–847
population ecology, 813–834
demographics of populations, 815–818
human population growth, 826–828
life history patterns, 823–826
population growth models, 816–821
regulation of population size, 821–823
scope of ecology, 814, 814*f*
restoration, 895
scope, 814
Ecosystem, 2, 3*f*
aquatic, 875–881
coastal ecosystems, 878*f*
lakes, 875–876, 876*f*
oceans, 879–880
biodiversity and performance, 888–889
biologically diverse, 4
biosphere, 862–882
aquatic ecosystems, 875–881
climate and the biosphere, 863–864
terrestrial ecosystems, 865–874
chemical cycling and energy flow in, 4, 4f
community and ecosystem ecology, 835–859
climate change, ecological consequences of, 856–859
community development, 844–847
dynamics of ecosystem, 847–856
ecology of communities, 836–844

ISTUDY

Ecosystem—(*Cont.*)
defined, 814, 848
dynamics of ecosystem, 842–851
autotrophs, 848
chemical cycling, 852–856
energy flow, 849–851
heterotrophs, 847–849
ecology of communities, 836–844
community interactions, 837–839
community structure, 831–837
predator-prey interactions, 839–842
symbiotic relationships, 842–844
landscape diversity, 885
terrestrial, 864–874
biome distribution, 865*f*
coniferous forest, 865, 866*f*, 868–869
deserts, 874
grasslands, 872–874
shrublands, 870–873
temperate deciduous forest, 866*f*, 869
tropical rain forest, 866*f*, 869–870
tundra, 865–868, 865*f*
Ecotone, 814
Ecotourism, 888
Ecstasy. *See* MDMA (methylenedioxymethamphetamine)
Ectoderm, 775, 775*f*, 776
Ectomychorrhizae, 403*f*
Ectoparasite, 843
Ectopic pregnancy, 758
ED (erectile dysfunction), 757
Edge effect, 895, 895*f*
Ediacara Hills of Australia, 322
Ediacaran fossils, 324
Edmonstone, John, 265
Effector, 694
Efferent neuron. *See* Motor neuron
E2*F* protein, 146
Egg fungi, 379
Egg-laying hormone (ELH), 797
Eggs, 162
birds, 753
extraembryonic membrane, 753, 782
human oocyte, 758, 760, 774–775
mammal, 753
oviparous animals, 753
ovoviviparous animals, 753
reptiles, 753, 782
yolk, 753, 775, 782
Egret, cattle, 843
Ejaculation, 756–757
Ejaculatory duct, 754*f*, 755, 756
Elastic cartilage, 577
Elastin, 91, 575
Elbow, 723, 724
Electrocardiogram, 597, 597*f*
Electromagnetic receptor, 699
Electromagnetic spectrum, 116, 116*f*
Electronegativity, 27
Electrons, 21, 21*f*
and energy, 22
in valence shells, 25
Electron shells, 25
Electron transport chain, 128
Electron transport chain (ETC), 117, 135–137, 139
and chemiosmosis, 136
cycling of carriers, 136
hydrogen ions in intermembrane space, 136
members of chain, 136
organization and function of, 135f
Elements, 20, 20*f*
atomic number and mass number, 21
atoms, 20–21, 21f
electrons and energy, 22
periodic table, 21, 22f
radioactive isotopes, 21–22
Elephantiasis, 525
ELH (egg-laying hormone), 797
Elk, 807
El Niño-Southern Oscillation (ENSO), 881
Elongation, translation, 217, 218*f*
Embolus, 600
Embryo
defined, 775
development in animals
blastula formation, 775
cellular stages of development, 775
cleavage, 775, 783
organ stages of development, 776–777
human
development, 758, 775–777, 783–785
fertilization, 761, 774–775, 783
preimplantation genetic diagnosis, 765
Embryonic development, 510
Embryonic disk, 783
Embryonic stem cells (*ESCs*), 152, 153
Embryo sac, 492. *See also* Gametophyte
Emergency contraception, 763–764
Emerging diseases, 15, 356, 356*f*
Emerging viruses, 356, 368
SARS-CoV-2, 356–358
Zika virus, 356
Emission, 756
Emphysema, 657*f*, 658–659, 659*f*
Encephalitis, 662
Encephalitozoon cuniculi, 393, 393*f*
Endangered species, 884–885
Endemic goiter, 740, 740*f*
Endergonic reactions, 101
Endochondral ossification, 719
Endocrine glands, 575
defined, 735
pancreas, 637, 774, 746
Endocrine system, 732–750
aging process, 791
animal hormones, 733–736
action, 735–736
chemical signals and pheromones, 735
antagonistic hormones, 741
erythropoietin (EPO), 671
functions, 732, 733
hypothalamus and pituitary gland, 737–739
anterior pituitary, 737, 738–739
posterior pituitary, 737–738, 737*f*
kidney secretion, 671
major glands, 734f
mode of action, 733f, 735–736
peptide hormones, 735–736, 736*f*
steroid hormones, 736, 736*f*
other endocrine glands and hormones, 739–748
adrenal glands, 734*f*, 742–744
erythropoietin (EPO), 748
leptin, 748
pancreas, 734*f*, 744–747
parathyroid, 734*f*, 741–742
pineal gland, 734*f*, 746
prostaglandins, 748, 756
testes and ovaries, 734*f*, 736, 746–747, 754–755, 758, 791
thymus, 734*f*, 747–748
thyroid, 734*f*, 739, 740–742
pancreas, 633f, 635, 637
pineal gland, 690
thymus, 611f, 612
Endocytosis, 84, 90–91, 91*f*
phagocytosis, 90, 91f
pinocytosis, 90, 91f
receptor-mediated, 90–91, 91f
Endoderm, 775, 775*f*, 776
Endodermis, 437Endomembrane system, 68–70
endoplasmic reticulum, 68, 68f
Golgi apparatus, 68–69, 69f
lysosomes, 69, 69f
summary, 70, 70f
Endometrium, 758, 760*f*, 761
Endomycorrhizae, 403*f*
Endoplasmic reticulum (ER), 67, 68, 68*f*
rough ER, 68
smooth ER, 68
Endorphin, 682, 714
Endoskeleton, 539
advantages, 718
composition, 718, 718
defined, 718, 718f
echinoderms, 718
vertebrate characteristic, 718
Endosperm, 426, 492, 496
Endospores, 364, 368
Endosymbiosis, 371, 845
aerobic bacterium, 371
cercozoans, 380
Endosymbiotic theory, 63, 322, 331, 371
Energy, 4, 71. *See also* ATP (adenosine triphosphate)
of activation, 104, 104f
definition of, 99
energy flow and ecosystem dynamics, 849–851
ecological pyramids, 851
food web, 850, 850*f*
thermodynamic laws, 849
flow of, 97–98, 97f, 139
forms, 97
from glucose metabolism, 136, 137f
laws of thermodynamics, 97–98
for life processes, 97
transformation of, 100
metabolic reactions and, 101–102
Energy flow
ecosystem dynamics, 849–851, 849f
ecological pyramids, 851
food web, 850, 845*f*
thermodynamic laws, 849
trophic levels, 851
Energy-related organelles, 72–74, 72*f*, 138–139
chloroplasts, 72, 72f, 73f
mitochondria, 72–74, 73f
Enhancers, 228
Ensatina salamander, 300–301
ENSO (El Niño-Southern Oscillation), 881
Entamoeba histolytica, 385
Entomopathogenic, 390
Entosense, 530
Entrainment, 485
Entropy, 100
Enveloped viruses, 353

ISTUDY

Environment
influence on behavior, 797–801
associative learning, 799–800
cognitive learning, 801
imprinting, 799
instinct and learning, 798
orientation and migratory behavior, 800–801
social interactions and learning, 799
Environmental influences, on phenotype, 196
Enzymatic proteins, 82, 82*f*
Enzyme, 103, 107
active site, 101, 104f, 107
brush-border enzymes, 636
carbonic anhydrase, 655
as catalysts of chemical reactions, 104
definition of, 103
digestive, 638–640
lipase, 640
maltase, 639
pancreatic amylase, 639
pepsin, 635, 639
peptidase, 639
salivary amylase, 634, 638
trypsin, 639
induced fit model, 104
inhibition, 106, 106f, 105
metabolic pathways and, 101–102
activation energy, 104, 104*f*
enzymatic rate, factors affecting, 104–106
enzyme activation, 106
enzyme inhibition, 106, 106*f*
enzyme-substrate complex, 103
Enzyme cascade, 735
Enzymes, 39, 46
Eosinophil, 615
Eosinophils stain, 604
Ependymal cell, 680
Ependymal cells, 580
Ephedra, 421
Epicotyl, 496
Epidermal tissue, 430–431, 431*f*
Epidermis, 581–584
plants, 430, 437
Epididymis, 754*f*, 754*t*, 755, 755*f*, 756
Epidiolex, 36
Epigenetic inheritance, 225, 226, 228
Epigenetics, 228
Epiglottis, 634
Epilimnion, 876
Epinephrine, 626, 735, 735
Epipelagic zone, 879
Epiphysis. *See* Pineal gland
Epiphyte, 843, 870
Epiphytes, 438
Episodic memory, 691
Epistatic interactions, 195
Epithelial tissue, animal, 574*f*
simple epithelia, 574–575
stratified epithelia, 575
EPO (erythropoietin), 671, 748
E protein Equilibrium (balance)
gravitational, 710–710, 712
rotational, 711–712
Equilibrium and hearing, 708–712
disorders, 712
hearing loss, 712
vertigo, 712
human ear, 709–712
anatomy, 709–710
pitch, 710
volume, 710
sense of equilibrium, 710–712
gravitational equilibrium, 710–711, 711
invertebrates, 710*f*
rotational equilibrium, 711–712
sound wave detection, 708–709
Equilibrium (population), species, 824
Equisetum genus, 415, 417*f*
Equisetum spores, 416
Equus, 309–310, 309*f*
Erectile dysfunction (ED), 757
Ergot infection, of rye, 398*f*
Ergotism, 398
Erithacus rubecula, 335
Erosion, soil, 888
Erythrocyte. *See* Red blood cells (RBCs)
Erythromycin, 364
Erythropoietin (EPO), 671, 748
Erythroxylon coca, 685
Escherichia coli, 205, 205*t*, 222, 238, 339, 362*f*, 363
Esophagus
cancer, 635
gastroesophageal reflux disease (GERD), 635
human digestion, 633f, 633–638
An Essay on the Principle of Population (Thomas Malthus), 262
Essential nutrient
human, 642
Estrogen, 45
breast cancer, 762
function, 745–747, 761–762
ovarian cycle, 759–760
production, 746, 758
uterine cycle, 761
Estuary, 878
Ethanol, 112
Ethylene
abscission, 476, 476f
fruit, ripens, 476–477, 477f
tomato plants, 477
Ethylene dibromide (EDB), 233
Etiolated, 484
Eubacteria, 61
Eucalyptus trees, 421, 456
gold, 456
reveal hidden gold, 456
Euchromatin, 148, 225
Eudicot embryo, development of, 495*f*
Eudicot root, branching of, 437*f*
Eudicot root tip, 436*f*
Eudicots, 423, 434
flowering plants, 435f
herbaceous stems, 439–440
key features of, 423t
organs and organ systems, 434–435
root system, 434, 436, 444
seed development, 494–496
seed germination, 500f
Eudicotyledons, 434
Euglena deses, 371, 373, 373*f*
Euglenids, 373–374, 388
Euglenozoans, 373, 388
Eukarya, 8, 9*f*, 61
Eukaryote, reproduction, 361
Eukaryotic cells, 8, 61, 63–66
animal cell, 63f
definition of, 61
DNA replication in, 209f, 210
origin of, 61, 61f
plant cell, 65f
and prokaryotic cells, difference between, 63, 78t
regulation of gene expression in, 225–232
structure of, 64–65, 64f, 65f
Eukaryotic cells arise, 322
Eukaryotic chromosome, 147–148, 147*f*
Eukaryotic genome, structure of, 250, 251*f*
Eukaryotic ribosomes, 364
Eukaryotic supergroups, 371, 371*t*
Eumetazoa, 513–516
Euphotic zone, 879
Euploidy, 174
Eutrophication, 855, 875, 892
Eutrophic lake, 875, 876*f*
Evaporation (water cycle), 852–853, 875
Evening primrose, 703*f*
Evergreens, 444
Evolution, 6, 260. *See also* Darwin and evolution
cellular slime molds, 610
and classification of life, 6–10
evolutionary tree of life, 7, 8*f*
natural selection, 6–7, 6*f*
taxonomic classification, 8–10, 8*t*
classification systems, 340
connecting, 330
criticism, 273
Darwin's theory of, 313
finches, 303, 304–305
Darwin's theory of natural selection and, 263–269
biogeographical observations, 263–265, 263*f*
natural selection and adaptation, 265–266
observations of change over time, 263
selection at work, 267–269
Drosophila, 610
evidence for, 269–273
anatomical evidence, 271–272
biochemical evidence, 272, 273
biogeographical evidence, 270–271
from developmental biology, 273
fossil evidence, 269–270, 270*f*
flowering plants, 423f
fungi
diversity of, 393–399
evolution of, 391–392
reproduction of, 392
structure of, 392
symbiotic relationships, 400–404
geologic timescale, 320, 321t, 331
time period, evolutionary events, 321*t*
history of life, 319f
absolute dating fossils, 319
Cenozoic era, 326–327
eukaryotic cells arise, 322
fossils tell story, 318–320, 320*f*
Mesozoic era, 325–326
monomers, 314–315
multicellularity arises, 322–323
Paleozoic era, 323–325
polymers, 315
Precambrian time, 320–323
protocells, 315–317
relative dating fossils, 319

ISTUDY

Evolution—(*Cont.*)
self-replication system, 318
stages of, 313*f*, 314–318
immune systems, 610–611
invertebrate
nervous system, 677
macroevolution and speciation, 293–311
anatomy of speciation, 295, 295*f*
defined, 294
how new species evolve, 294–300
modes of speciation, 300–306
phylogenetic species concept, 295
principles of macroevolution, 307–310
maize, evolutionary history of, 422
mammal
nervous system, 677–679
placental mammals, 753
microevolution, 277–282, 295
natural selection, 6–7, 6f
nervous system
invertebrates, 677
mammals, 677–680
vertebrates, 677, 678*f*, 679*f*
plants
alternation of generations, 409–411
ancestry and features of, 408
ancestry of, 408
bryophytes and colonization of land, 411–412
characteristics of, 408*t*
evolutionary history of, 409*f*
land adaptation, 408–409
population, 276–292
genes, populations, and evolution, 277–284
maintenance of diversity, 288–290
natural selection, 284–288
protists, 370–389
Amoebozoa, Opisthokonta, and Rhizaria supergroups, 371*t*
Archaeplastida supergroup, 381–385
biology of protists, 371
Excavata supergroup, 371–374
rise of adaptive immunity, 610
tree of life, 8f
vertebrate nervous system, 677
Evolutionary species concept, 294–297, 297*f*
Excavata supergroup, 371–375, 388
diplomonad, 372–373
euglenozoans, 373–374
kinetoplastids, 374
parabasalids, 373
Excavates, 371
Exchange pool, 852
Excretion, 664
Excretory system, 663–675
animal excretory systems, 664–667
invertebrate excretory organs, 664–665
nitrogenous waste products, 664, 671
osmoregulation by aquatic vertebrates, 665–666
osmoregulation by terrestrial vertebrates, 666–667
defined, 664
human urinary system, 667–674
aging process, 790
homeostasis and kidneys, 671–674
kidneys, 668
urine formation, 668–671
Exercise and gene expression, 229
Exergonic reactions, 101
Exhalation. *See* Expiration
Exocrine gland
defined, 733
pancreas, 744, 746
Exocrine glands, 575
Exocytosis, 84, 90, 90*f*
Exon shuffling, 214
Exons of pre-mRNA, 213
Exophthalmia. *See* Exophthalmos
Exophthalmic goiter, 740*f*
Exophthalmos, 740
Exoskeleton, 526, 717–712
arthropods, 717
mollusks, 717
Experiment, 10–11
Experimental design, 11
Experimental variable, 11
Expiration, 647, 652, 653*f*
Exponential growth, 819
Exponential growth phase, 819, 820
Expulsion, 757
Extant species, 260
External fertilization, 752
External respiration, 647, 647*f*, 654–656, 654
Exteroceptor, 699
Extinction, 14, 323, 331, 891
causes, 889–894
climate change, 892–893
habitat loss, 889, 890*f*
nonnative species, 889–890
overexploitation, 893–894
pollution, 891–892
endangered species, 884–885
fossil record, 294
habitat preservation, 894–895
habitat restoration, 895–896
keystone species, 894
landscape preservation, 895
mass events, 891, 891t
phylogeny, 295
threatened species, 885
Extracellular matrix (ECM), 82, 92, 92*f*
Extraembryonic membrane, 753, 782, 783*f*
Extraterrestrial origins hypothesis, 314–315
Eye
anatomy, 703–704, 703f
artificial retinas, 70, 707
camera-type, 703
compound, 702, 702f
development of, 308, 308f
disorders, 706–708
glaucoma and cataracts, 706
retinal diseases, 706
visual focus disorders, 706–708
focusing, 704, 706–708
human, 703–708
aging process, 791
anatomy, 703–704, 703*f*
disorders, 706–708, 791
focusing, 704, 706–708
integration of visual signals, 705–706
photoreceptors, 703, 704
from the retina to the visual cortex, 706
primary visual area of occipital lobes, 688
Eye color, 181
gene interactions and, 195, 195f
genetics of, 181

F

Facchini, Roberto, 335
Face, facial bones, 721
Facial hair, 747, 757
Facilitated transport, 88, 88*f*
Facilitation model of succession, 847
Facultative anaerobes, 362, 368
FAD (coenzyme of oxidation-reduction), 128
Fall overturn, 876
False rib, 722
Familial hypercholesterolemia (FH), 90, 193, 249
Family, 8, 335, 335*f*, 346
FAP (fixed action patterns), 797–798
Farsightedness, 708, 708*f*
FAS (fetal alcohol syndrome), 788
Fate map, 779
Fatigue, tapeworm infections, 523
Fats, 43, 43*t*. *See also* Triglycerides
human digestion and nutrition, 641, 643
Fatty acid, 43
human digestion and nutrition, 641, 643
saturated, 43, 44f
unsaturated, 43, 44f
Feces
energy flow, 849
human, 636, 637
Federal Bureau of Investigation (FBI), 241
Female choice, 286–287
Female gametophyte, 492
Female grasshopper, 531*f*
Female reproductive system, human, 758–762
aging process, 789
anatomy, 758–759, 758f, 760f
estrogen and progesterone
breast cancer, 762
function, 746
ovarian cycle, 759–761
production, 746, 754
uterine cycle, 761
female breast, 762
female orgasm, 759
fertilization, 758, 760, 761–762, 774–775, 774f, 783
ovaries, 734f, 746–747, 758, 791
gonadotropic hormones, 738, 760
ovarian cycle, 759–761, 761*t*
steroid hormones, 736
uterus, 761
vagina, 758–759, 758t
Femur, 723, 723*f*
Fermentation, 129, 131–133, 131*f*, 728
advantages and disadvantages of, 131, 133
as anaerobic process, 131
bacterial, 131, 133
efficiency of, 131, 133
and food production, 130
NAD^+ regeneration, 131
products from, 133f
yeast, 131, 133
Ferns
diversity of, 417f
evolution of, 417–418
Hart's tongue, 417
leatherleaf, 417
life cycle, 417, 418f
life history strategy, 824
whisk, 416–417

ISTUDY

Fertilization, 168, 173, 512
double, 426
external, 752
human, 758, 760, 761–762, 774–775, 774f, 783
internal, 753
IVF (in vitro fertilization), 764, 765
Fertilization membrane, 774
Fertilizer, 451*f*
greenhouse gases, 854
nitrogen cycle, 856
Fetal alcohol syndrome (FAS), 788
Fetus, human
embryonic development, 782–787
early, 744–777
extraembryonic membranes, 782
fetal development, 786–787
placenta structure and function, 785–786
preimplantation genetic diagnosis, 765
Fever, 614
FGF8 genes, 508
Fiber (carbohydrate), 640
Fibers, 432
muscle, 727f, 728, 728f
sclerenchyma cells, 431–433
Fibrillin, 194
Fibrin, 98, 604
Fibrinolysin, 761
Fibrocartilage, 577
Fibronectin, 92
Fibrous connective tissue
dense, 576
dense fibrous connective tissues, 724
diagram of, 576f
loose, 576
Fibrous joint, 724
Fibrous proteins, 49, 50*f*
Fibrous root system, 437, 438
Fibula, 723, 723*f*
"Fight or flight" response, 694, 742
Filament model, sliding, 728
Filaments, 383, 389
flowers, 424f
Filarial worms, 525
Filter feeder
baleen whale, 631
clam, 631–632
Fimbria
human ovaries, 758
Fimbriae, 62, 62*f*, 360
Neisseria gonorrhoeae, 360
Shigella dysenteriae, 364
Fin
convergent evolution, 306
pelvic, 308
Finch
ecological niche, 837, 838
Galápagos, 303, 837, 838, 839f
sexual selection, 807
speciation, 303, 307–305
Zebra, 807
Fire, Andrew, 232
Firefly, 804, 804*f*
First messenger, 735
Fischer lovebird, 795, 795*f*
Fishes
amphibians, 544–546
characteristics of, 544
evolution of, 544
animals, medical uses of, 545f
bony fishes, 542–543
lobe-finned fishes, 542, 544*f*
nervous system and brain, 543
ray-finned bony fishes, 542
breathing and transport of gases, 648–649
cartilaginous fishes, 542
chemical senses, 700
cleaning symbiosis, 844
convergent evolution, 306
fertilization and reproduction, 753
jawed fish, 540–542
cartilaginous/bony endoskeleton, 540
ectothermy, 540
gills, 540
scales, 541–542
jawless fishes, 540
lateral line system, 708–709
limiting factors, 816
living amphibians, diversity of, 544
mammals
characteristics of, 551–552
evolution of, 552–554
osmoregulation, 665–666
reptiles
amniotes, evolution of, 547
birds, 550–551
characteristics of, 546
living birds, diversity of, 551
living reptiles, diversity of, 547–548
skin color, 739
vertebrates/human medicine, 545f
Fishing industry, 887, 893–894, 893*f*
Fitness, 267
behaviors that increase fitness, 805–810
altruism *vs.* self-interest, 807–810
inclusive fitness, 808, 810*f*
reproductive strategies and fitness, 806–807
societies and fitness, 8078
territoriality and fitness, 805–806
sexual selection, 286
Five-kingdom system, 338, 346
Fixed action patterns (FAP), 798
Flagella, 76, 77*f*, 360
prokaryotic, 62
Flagship species, 894
Flame cell, 665
Flatworm
asexual reproduction, 752
incomplete digestive tract, 630
parasitic, 630
Flatworms, 521
Flemming, Alexander, 10
Fletcher, Neil, 335
Florida panther, 824
Florida scrub jay, 808
Flow, ecosystem dynamics, 849–851
Flower fly, 842, 842*f*
Flowering plants
cells and tissues of, 430–433
epidermal tissue, 430–431
ground tissue, 431–432
vascular tissue, 432–433
commercial florists, 484
day-neutral plants, 484
evolution and full adaption to land, 418, 423f
diversification, 426
monocots and eudicots, 423
origination and radiation, 422–424
ovules, 416
features, 423t
life cycle, 424–426, 425f
life history strategy, 818
long-day plants, 483
monocots/eudicots, 435f
organs of, 434–435
plant life cycles, 435
variation of, 434–435
vegetative, 435
short-day plants, 483
structure and organization, 429
cells and tissues, 430–432
leaf organization and diversity, 444–447
roots organization and diversity, 436–438
stem organization and diversity, 439–444
summary, 447–448
Flowers, 484. *See also* Flowering plant
diversification, 424f
generalized, 424f
imperfect, 490
life cycle of flowering plants, 424–426, 489, 491f
perfect, 490
petals, 423
sepals, 423
stamens, 423
structure, 423–424, 489–492
calyx, 423, 424*f*, 490
carpels, 423, 424, 490
monocots and eudicots, 423, 423*t*, 434–435, 496
petals, 423, 424*f*, 490
sepals, 423, 424*f*, 490
stamen, 423, 424*f*, 490
terminology, 424*t*
terminology, 424t
Fluid connective tissues, 577
Fluid-mosaic model, of plasma membrane, 81*f*, 82
Fluid regulation and excretory system, 663–675
animal excretory systems, 664–667
invertebrate excretory organs, 664–665
nitrogenous waste products, 664, 671
osmoregulation by aquatic vertebrates, 665–666
osmoregulation by terrestrial vertebrates, 666–667
human urinary system, 667–674
homeostasis and kidneys, 671–674
kidneys, 668
urine formation, 668–671
Flukes, 522
coevolution, 845
lance, 845
Flu viruses, 352
seasonal versus pandemic flu, 352
Flu Viruses: Seasonal Versus Pandemic Flu, 350
Fly
compound eye, 702f
flower, 842, 842f
South American lantern, 841–842
Focusing the human eye, 704, 704*f*
disorders
astigmatism, 708
farsightedness, 708
nearsightedness, 706–708
presbyopia, 791

ISTUDY

Folic acid, 788
Foliose lichen, 4012, 401*f*
Follicle (ovarian cycle), 759
Follicle, hair, 790
Follicle-stimulating hormone (FSH), 757, 759–760
Follicular phase, 760
Fontanel, 721
Food
  biodiversity direct value, 886–887
  foraging, 805–806, 806f
Food chain, 851
Food vacuole, 885
Food web, 850, 850*f*
Foot
  human
    bones, 7234, 723*f*
    dexterity, 718
  muscular hydrostats, 717
Foraging for food, 805–806, 806*f*
Foramen magnum, 721
Foraminiferans, 380, 380, 389
Forebrain, 677
Foreskin, 754*f*, 756, 756*f*
Forest
  climate regulation, 888
  coniferous forest, 865, 866f
  deforestation, 856, 870, 888
  ecological succession, 846, 846f
  temperate deciduous forest, 867f, 868–869
  tropical rain, 869–870, 870f
  water-holding capacity, 889
Forest floor
  temperate deciduous forests, 869
  tropical rain forest, 869
Formula, 26
FosB gene, 797
Fossil fuels, 112
Fossil fungi, 391
Fossils, 269–270, 270*f*, 320*f*, 331
  absolute dating methods, 319
  evolutionary species concept, 294
  horse, 309
  morphological species concept, 294
  prokaryotic, 360
  relative dating of, 319
Founder effect, 282
Fovea centralis, 703*f*, 704
FOX03A gene, 789
Fox, George, 365
Fox, Michael J., 689
Fox, Sidney, 315
Fracture from osteoporosis, 719
Fragile-X syndrome, 244
Fragmentation
  biodiversity, 885, 888–889
  habitat, 888, 890f
Frameshift mutations, 233, 233*f*
Franklin, Rosalind, 206, 206*f*
Free energy, 100
Free-living flatworms, 521
Freeze-fracture method, 82
French paradox, 601
Frog
  coloration, 841
  development
    cleavage, 775
    cytoplasmic segregation, 778
    gastrula, 775, 776
    induction, 779
    neural tube, 776
  leopard, 294, 296f
  parental care, 824f
  Poison arrow, 824f, 841
  speciation, 294, 296f
  temporal isolation, 297–299
Fronds, 417
Frontal bone, 721, 722*f*
Frontal lobe, 679, 688, 688*f*, 691
Fruit flies, 197
Fruit fly
  embryo development stages, 508f
  immune system evolution, 610
  morphogenesis, 780, 781f
  segmentation, 780
Fruiting body, 396
Fruits, 426
  aggregate, 496
  composition/texture, classification, 496t
  compound, 496
  dry, 496
  kinds of, 496
  maple tree, 497f
  multiple, 496
  pea, 497f
  peach, 497f
  pineapple, 497f
  raspberry, 497f
  seed dispersal, 496–500
    by air, 496
    by animals, 498
    by ejection, 498
    methods of, 499*f*
  simple, 496
  tomato, 497f
FSH (follicle-stimulating hormone), 746, 757, 759–760
Fucus, 379
Fukushima disaster, 24
Functional genomics, 252–253
Functional groups, 38, 38*t*
Fundamental niche, 837
Fungal hyphae, 403*f*
Fungal infections, 397*f*
Fungal mycelium, 386*f*
Fungal roots, 390
Fungal strains, fluorescent tags, 402*f*
Fungus (fungi), 9, 338, 340, 391. *See also* Water mold; Water mold; Water mold
  diversity of, 393–399
    Basidiomycota, 397
    Glomeromycota, 395
    microsporidia, 393
    sac fungi, 395–397
    zoospore, 393–394
    zygospore fungi, 394–395
  ecosystem dynamics, 848
  evolution of, 391–392
  features of, 393f
  lichen, 365
  mycorrhizae, 438
  reproduction of, 392
  structure of, 392
  symbiotic relationships, 400–404
    fungal parasitism, 400
    lichens, 400–403
    mycorrhizae, 403–404
  with zoospores, 394f

G

Galactose, 222
β-galactosidase, 222
Galápagos finches, 264, 268, 268*f*, 303, 837, 838, 839*f*
Galápagos tortoises, 268, 268*f*
Gallbladder, 633*f*, 637*f*, 638
Gallstones, 638
Gametangia, 409
Gametangium, 395
Gamete intrafallopian transfer (GIFT), 764
Gametes, 162, 162, 752, 764
  formation of, 170
  isolation, 299
Gametogenesis, 173
Gametophytes, 173, 409, 410, 492
  alternation of generations, 409
  angiosperms, 421–422
  bryophytes, 411–413
  gymnosperms, 421f, 422
  size reduction, 411f
Ganglia, 522
Ganglion, 677
Gap gene, 780
Gap junctions, 91, 93*f*
Garrod, Archibald, 210
Garter snake, 796, 796*f*
Gart gene, 175
Gaseous biogeochemical cycle, 852
Gas exchange, animal
  breathing and transport of gases, 654–656
    amphibians, 648, 652
    birds, 652
    earthworms, 647, 648*f*
    fish, 649
    mammals, 652
    modifications in human, 652–654
    reptiles, 652
  gas exchange and transport
    partial pressure, 655
    surfaces, 647–651
    transport of oxygen and carbon dioxide, 655–656
  surfaces, 674–651
    gills of fish, 648–649
    human lungs, 650–651
    overview, 647–648
    respiration steps, 647
    trachea system of insects, 649–650
Gas exchange
  in lungs, 85, 86f
  plant, 431, 438, 442, 445
Gastric gland, 635
Gastric juice, 635
Gastric pit, 635
Gastric ulcer, 635
Gastroesophageal reflux disease (GERD), 635
Gastroesophageal sphincter, 634
Gastropods, 518
  diversity, 519f
Gastrovascular cavity, 514
Gastrula, 775, 775*f*, 776, 776*f*
Gastrulation, 775–776, 775*f*
Gause, G. F., 837, 839
GBS (Guillain-Barré syndrome), 694
G_1 checkpoint, 145
  CDK regulation of, 145f
  importance of, 145
  p53 regulation of, 145f

ISTUDY

Gehring, Walter, 308
Gel electrophoresis, 240
Gelidium, 381
Gene, 5, 182, 250
  Ankyrin-1, 192
  BMP4, 305f
  BRCA2, 244
  CaM, 305f
  CFTR, 659
  fosB, 797
  FOX03A, 789
  gap, 780
  Gart, 175
  Gfap, 229
  HERC2, 181, 195
  Hox, 308
  molecular biology of, 202–220
    DNA replication, 208–210, 209*f*
    genetic material, 203–207
    RNA and genetic code, 210–212
    transcription, 212–214
    translation, 216–218
  OCA2, 195
  p53, 143, 157, 234, 250
  pair-rule, 780
  Pax6, 307–308, 308f
  Pitx1, 308
  regulator, 222
  structural, 222
  Tbx5, 308
Gene cloning, 238, 238*f*
Gene expression, 210, 212
  exercise and, 229
  overview of, 211, 211f
  regulation of (see Gene expression regulation)
  RNA and genetic code, 210–212
  summary of, 218, 219f
  transcription, 212–214
  translation, 214–218
Gene expression regulation, 221–236
  eukaryotic regulation, 225–232
    chromatin structure, 225–228
    posttranscriptional control, 225, 230–231
    posttranslational control, 225, 231
    transcriptional control, 225, 228
    translational control, 225, 231
  gene mutation, 232–234
    and cancer, 234
    causes, 232–233
    effect on protein activity, 233–234
  prokaryotic regulation, 222–224
    lac operon, 222, 224, 224*f*
    trp operon, 222, 223*f*
  summary, 235
Gene flow, 280, 280*f*
Gene locus, 185, 185*f*
Gene mutation, 232–234
  and cancer, 234
  causes, 232–233
  effect on protein activity, 233–234
  evolution, 309
  induced mutations, 232–233
  spontaneous mutations, 232
Gene pharming, 246
Gene pool, 277
Gene therapy, 249, 689
  blindness, 707
  ex vivo gene therapy, 249, 249f
  preimplantation genetic diagnosis, 765, 788f
  RNA interference, 250
  in vivo gene therapy, 249–250
Genetically modified animals, 246–248
  cloning transgenic animals, 246–247
  gene pharming, 246
Genetically modified bacteria, 245, 244–245
Genetically modified organism (GMO), 238, 245. *See also* Transgenic organism
Genetically modified plants, 245–246, 246*f*
Genetic code, 211–212, 212*f*, 272
Genetic diagnosis, 765, 788*f*
Genetic diversity, 885
Genetic drift, 282, 282*f*
Genetic engineering, 238
  in animals, 246–249
  in plants, 245–246, 246f
Genetics, longevity of, 773
Genetic profile, 252–253
Genetic recombination, 166–167
Genetic testing, 867, 868
Genetic variation, 266
Genital herpes, 768*f*, 769, 788
Genital wart, 769, 769*f*
Genomic imprinting, 228
Genomics, 250–255. *See also* Biotechnology
  bioinformatics, 254, 254, 884
  definition of gene, 252
  eukaryotic genome, structure of, 250, 251f
  functional and comparative genomics, 252–253
  intergenic sequences, 251–252, 251f
    repetitive elements, 251
    transposons, 251, 252*f*
    unique noncoding DNA, 252
  proteomics, 253
  sequencing genome, 250
Genotype, 185
Genus, 8, 335, 335*f*, 337
Geographic isolation, 300–301, 303
Geologic timescale, 320, 320*t*, 331
  eukaryotic cells evolution, 322
  multicellularity, 322–323
  prokaryotes evolution, 320
  time period, evolutionary events, 321t
Geometric ratio of increase, 266
Geospiza fortis, 268, 839*f*
German measles, 788
Germ cell, 752
Germinate, 498
Germination, of seeds, 483
Germ layers, 510, 775*f*, 776, 776*t*
Germ-line mutations, 232
Gerontology, 789
Gfap gene, 229
GFP (Green fluorescent protein), 402
GH (growth hormone), 719, 739, 791
Giardia lamblia, 372, 373*f*
Gibberellafujikuroi, 474
Gibberellins, 474
  commerical uses, 474
  dormancy, 474
  stem elongation, 474f
Gibbon, 805
GIFT (gamete intrafallopian transfer), 764
Gill
  defined, 647
  fish, 648–649
    bony fishes, 649*f*
Gill (mushroom), 399*f*
Ginkgo biloba, 421
Ginkgoes, 421
Giraffa camelopardalis, 296, 298*f*
Girdling, 464
Gizzard, earthworm, 630, 631
Gladiolus corms, 443
Gland, 575
  major human glands, 732f
Glans penis, 756, 756*f*
Glaucoma, 706
Global warming, 14, 854
Globular proteins, 49
Gloeocapsa, 365*f*
Glomeromycota, 393*t*, 395
Glomerular capsule, 668, 669*f*
Glomerular filtrate, 668
Glomerular filtration, 668–669, 670*f*
Glomerulus, 668
G protein, 95
Glossopteris supports, biogeography of, 328*f*
Glottis, 634, 651
Glucagon, 744
Glucocorticoid, 742*f*, 743, 744
Glucose, 40, 40*f*, 672
  breakdown of, during cellular respiration, 127, 127f
  cortisol, 743
  diabetes, 643, 744, 745
  liver, 733f, 744
  metabolism, 136, 137f
  tubular reabsorption, 669–671
Glucose phosphate, 121
Gluten, 636
Gluten-free diet, 636
Gluten sensitivity, 636
Glycerol, 43, 638
Glycocalyx, 61, 83, 360
Glycogen, 40*f*, 42, 43
  human liver, 638, 733f, 744
Glycolipids, 82–83
Glycolysis, 128, 128*f*, 129–131, 130*f*, 317
  in cytoplasm, 129
  definition of, 129
  energy-harvesting steps, 129, 130f
  energy-investment step, 129, 130f
  inputs and outputs of, 129, 131
  substrate-level ATP synthesis in, 129, 129f
Glycoproteins, 82–83, 349, 353, 354
Glyptodon, 263, 263*f*
Gnathostomes, 540
Gnetophytes, 421
Gnetum, 421
GnRH (gonadotropin-releasing hormone), 746, 757, 757*f*, 760*f*, 761, 772
Goiter, 740
Golden brown algae, 378*f*, 389
Golden Rice, 246
Golgi apparatus, 68–69, 69*f*
Golgi, Camillo, 68
Golgi tendon organ, 713, 726

ISTUDY

Gonad
hormone and endocrine system, 734f
sexual reproduction, 752, 754, 757
Gonadotropic hormone
follicle-stimulating hormone (FSH), 757, 760, 772
function, 738
gonadotropin-releasing hormone (GnRH), 746, 757, 757f, 760f, 761, 772
luteinizing hormone (LH), 746, 757, 760–761, 761f
Gonadotropin-releasing hormone (GnRH), 746, 757, 757*f*, 760*f*, 761, 772
Gondwana and Laurasia, 328
Gonorrhea, 276, 769–770, 770*f*
Good genes hypothesis, 286
Gould, John, 264
Gout, 664
G3P (glyceraldehyde-3-phosphate), 119
fate of, 121, 123f
during glycolysis, 129
oxidation of, 129
Gracilaria, 381
Gradualistic model of macroevolution, 307, 307*f*
Grain refinement, 640
Gram staining, 361*f*
Grana, 112
Grant, Peter, 268
Grant, Rosemary, 268
Granular leukocyte
eosinophils, 615
neutrophils, 615
Granular leukocytes, 604
Granum, chloroplast, 72, 73*f*
Granzyme, 620, 621, 621*f*
Grass
bluegrass life table, 816, 816t
ecological succession, 846
Grasshoppers, 529
hemolymph of, 590
incomplete metamorphosis, 529
Grassland
ecosystem, 873–874
savannas, 872–873
Graves' disease, 741
Gravitational equilibrium, 710–711, 712
Gravitropism, 474, 480
Gray crescent, 778, 778*f*
Gray matter
basal nuclei, 688
cerebral cortex, 688
spinal cord, 687
Grazer, 632
Grazing food web, 850, 851*f*
Greater roadrunner, 872*f*
Greater Yellowstone Ecosystem, 895
Green algae, 381, 382, 389, 408
biofuel, 382
DNA sequencing, 385
embryo difference, 410f
Green fluorescent protein (GFP), 402
Green gland, 665
Greenhouse effect, 14, 854
Greenhouse gases, 120, 854, 857, 892
Griffith, Frederick, 203
transformation experiment, 203, 203f
Grizzly bear, 885, 885*f*, 890, 894, 894*f*, 8951
Ground meristem, 439
Ground substance, 575
Ground tissue, 432, 432*f*
Groundwater mining, 852
Groundwater table, 852–853
Growth factors, 145
Growth hormone (GH), 47, 719, 739, 843
Growth plate, 719
Growth response control of flowering plants
growth and movement responses
circadian rhythms, 485–486
external stimuli, 477–480
internal stimuli, 480–482
phytochrome, 482–484
plant hormones, 470–477, 471t
abscisic acid, 475–476
auxins, 471–474
biotic environment, 477
cellular response, 471
cytokinin, 474–475
ethylene, 476–477
gibberellins, 474
receptors, 470
transduction pathway, 471
Growth signals, 146
Guanine (G), 205, 205*f*. *See also* DNA (deoxyribonucleic acid)
Guard cells, 431, 462
Guillain-Barré syndrome (GBS), 694
Gulf Stream, 880
Gull, 798
Guttation, 461
Gymnosperms, 419, 423
conifers, 419–420
cycads, 420–421
ginkgoes, 421
gnetophytes, 421
groups of, 421f

H

HAART (highly active antiretroviral therapy), 768
HAB (Harmful algal bloom), 375
Habitat
community interactions, 837
defined, 814, 837
extreme, 366f
isolation, 297, 298f
loss, 889, 890f
preservation, 894–895
keystone species, 894
landscape preservation, 895
metapopulations, 895
restoration, 895–896
Habituation, 798
Haeckel, Ernst, 814
Hair, human
axillary and pubic, 747
baldness, 747, 757
facial hair, 747, 757
Hair cell, inner ear, 709
Hair follicles, human, 785, 790
Haldane, J. B. S., 314
Haldane's hypothesis, 314
Halophiles, 366, 368
Halophilic archaea, 366*f*
Hamadryas baboon, 806*f*
Hand
bones, 723
dexterity, 718
Haploid (n) number, 148, 163
Hardy, Godfrey H., 277
Hardy-Weinberg equilibrium (HWE), 277–280, 278*f*
migration, 280
mutation, 280–280
natural selection, 284
non-random mating, 282
small population size, 280–282
Hardy-Weinberg principle, 278
Hare, snowshoe, 840–841
Harmful algal bloom (HAB), 375
Hart's tongue fern, 417
Haversian systems, 577
Hawaiian honeycreeper, 303
Hawk, 848
Hay fever, 625
HCG (human chorionic gonadotropin), 761, 783
HCO3-buffer system, 673–674
HDL (high-density lipoprotein), 643
Hearing and equilibrium, 708–712
disorders, 712
aging process, 790
hearing loss, 712, 791
vertigo, 712
human ear, 709–712
aging process, 791
anatomy, 709–710
pitch, 710
volume, 710
sense of equilibrium, 710–712
gravitational equilibrium, 710–711, 712
invertebrates, 710*f*
rotational equilibrium, 711–712
sound wave detection, 708–709
Hearing loss, 712, 712*f*, 791
Heart
attack, 601
blood path, 595–596
cardiac cycle, events of, 590t
conduction system of, 597f
external heart anatomy, 594f
human aging process, 790
internal anatomy of, 595f
structure of, 594–595
Heart attack, 643
Heartburn, 634
Heartworm, 843*f*
Heartworm disease, 525
Heat, 98
Heimlich maneuver, 656
HeLa cells, 155, 155*f*
Helicobacter pylori, 635
Helper T cell, 619
Hemagglutinin spike, 352
Heme, 655
Hemidesmosome, 92
Hemizygous, for X-linked traits, 198
Hemocoel, 517
Hemoglobin, 46, 49, 190, 603
oxygen transport, 655, 655f
red blood cells, 655
Hemolymph, 690
arthropods respiration, 649
Hemolysis, 88

ISTUDY

Hemolytic disease of the newborn (HDN), 604
Hemophilia
hemophilia A, 199
hemophilia B, 199
royal families of Europe and, 199-
Hemp fibers, 432
Henslow, John, 263
Hepatic artery, 637
Hepatic portal vein, 637
Hepatic vein, 637
Hepatitis, 638, 767, 769
Hepatitis B vaccine, 245
Herbaceous eudicot stem, 440*f*
Herbaceous stems, 439
Herbivore
diet adaptation, 632
ecosystem dynamics, 848, 850f, 874
Herceptin, 617
HERC2 gene, 195
mutation in, 181
Hereditary spherocytosis, 192
Hermaphroditic, 520, 522
Hermaphroditism
animals, 752
sequential, 752
Heroin, 685
Herpes, genital, 768*f*, 770, 788
Herpes simplex virus, 768*f*, 7878
Hershey, Alfred, 204
Heterochromatin, 147*f*, 148, 225, 226
Heterosporous, 414
flowering plants, 492
Heterotroph
animals, 848–849
ecosystem dynamics, 848–849
Heterotrophic, 387
Heterotrophs, 112, 114*f*
Heterozygosity, 290–291
Heterozygote advantage, 191
Heterozygous individual, 185
Hexoses, 40
HFCS (high-fructose corn syrup), 641
High blood pressure (hypertension), 790
High-density lipoprotein (HDL), 644
High-fructose corn syrup (HFCS), 641
Highly active antiretroviral therapy (HAART), 768
Himalayan rabbits, 196
Hindbrain, 677
Hindlimb reduction, 308
Hinge joint, 724
Hip
artificial, 724
joint, 724, 724f
Hippocampus, 229, 229*f*, 690, 691
Hippopotamus, 804
Histamine, 614, 625
Histone modification, 226
Histones, 147, 225, 226
History of life, 319*f*
Cenozoic era, 326–328
fossils tell story, 318–320, 320f
absolute dating fossils, 319
relative dating fossils, 319
Geologic timescale, 320–323
prokaryotes evolution, 320
eukaryotic cells evolution, 322
multicellularity, 322–323
Mesozoic era, 325–326
origin of life
evolution of monomers, 314–315
evolution of polymers, 315
evolution of protocells, 315–318
evolution of self-replication system, 318
stages of, 313*f*, 314–318
Paleozoic era, 323–325
Precambrian time, 320–323
eukaryotic cells arise, 322
multicellularity arises, 322–323
HIV (human immunodeficiency virus), 766–768, 766*f*
categories, 622, 767
opportunistic infections, 620, 622
pregnancy, 768
T cells, 620, 622, 766
treatment and prevention, 766–768
HMS *Beagle* (British naval vessel), 260, 261*f*, 263
H1N1 virus, 350, 356
H5N1 virus, 350, 352, 356
H7N9 virus, 352, 356
Hodgkin disease, 886
Holdfast structure, 379
Homeobox genes, 273, 780–781
Homeodomain, 781–782
Homeostasis, 4–5, 584–587, 585
animal organization
negative feedback, 737
positive feedback, 738
homeostatic regulation, examples of, 585
human kidneys, 671–674
acid-base balance, 671, 674–675
water-salt balance, 671–674, 742, 743
hypothalamus, 677
negative feedback, 586
positive feedback, 586–587
Homeothermic, 585
Homeotic transformation, 781
Hominins, 558
Homo erectus, 565, 565*f*
lineage, 565
Homo florensiensis, 565, 567
Homo genus, 570
early evolution of
Africa, migration out of, 565*f*
Homo erectus, 565, 565*f*
Homo ergaster, 565
Homo habilis, 564
Homo rudolfensis, 565
Homo genus, later evolution of
complexity of, 567
Denisovans, 567, 567*f*
Homo floresiensis, 567, 567*f*
Homo naledi, 567
Cro-Magnons, 567–568
language, 568
rise of agriculture, 568
tool use, 568
ethnic groups, 568f
evolutionary hypotheses, 566
Homo heidelbergensis, 565–566
human variation, 568–566
Neanderthals, 566–567
Homo heidelbergensis, 565, 566
Homologous chromosomes, 163–164, 163*f*, 185*f*
independent assortment of, 167, 167f
Homologous genes, 254
Homologous structures, 271, 271*f*, 343, 347
Homologous trait, 306
Homologues. *See* Homologous chromosomes
Homology, 347
Homo naledi, 566
Homo neanderthalensis, 556, 566
Homo sapiens, 556, 560, 565, 566, 567
vs. chimpanzees, genomic comparison of, 560
Homosporous, 414
Homozygous individual, 185
Honeybee, 794, 804, 805*f*, 887
Hook, 360, 360*f*
Horizontal gene transfer, 394
Hormonal therapy
hormone replacement therapy (HRT), 791
Hormone, 732–750
animal hormones, 733–736
action, 735–736
chemical signals and pheromones, 735
antagonistic hormones, 741
functions, 732, 733
hypothalamus and pituitary gland, 737–739
anterior pituitary, 737, 738–739
male hormone regulation, 757
posterior pituitary, 737–738, 737*f*
kidney secretion, 671
major glands, 734f
mode of action, 733f, 735–736
peptide hormones, 735–736, 736*f*
steroid hormones, 736, 736*f*
other endocrine glands and hormones, 739–748
adrenal glands, 734*f*, 742–743
erythropoietin (EPO), 671, 748
leptin, 748
pancreas, 633*f*, 635, 637, 734*f*, 743–747
parathyroid, 734*f*, 741–742
pineal gland, 734*f*, 747
prostaglandins, 748, 756
testes and ovaries, 734*f*, 736, 746–746, 754–755, 758, 791
thymus, 734*f*, 747–748
thyroid, 734*f*, 739, 740–742
pineal gland, 690
regulation in human males, 757, 757f
thymus, 611f, 611
Hormone replacement therapy (HRT), 791
Hornet, 842
Hornworts, 412, 412*f*
Horse
evolution, 309–310
herbivore diet, 632
Horseshoe crab, 886
Horseshoe crabs, 531
Horsetails, 415, 417*f*
Horticulture and exotic species, 890
Host (parasitism)
coevolution, 844
defined, 842
secondary, 843
Host specificity, 353
Hotspot, biodiversity, 886
Hox, 780–781
HOXC8 genes, 508
Hox gene, 273, 308, 505, 508, 509*f*, 780–781
shared evolutionary history of, 509f
HPV (human papillomavirus), 768–769
HRT (hormone replacement therapy), 791

ISTUDY

16 H-type spike, 350
Huber, Claudia, 315
Human
cardiovascular system
blood pressure, 743–744
cardiovascular disease, 640–641, 643
digestive system and nutrition, 633–638
accessory organs, 633*f*, 637–638
large intestine, 633*f*, 636, 637
mouth, 633–634
pharynx and esophagus, 633*f*, 634–635
small intestine, 633*f*, 635–636
stomach, 633*f*, 635
ear, 709–710
anatomy, 709–710
functions, 709
inner ear, 709–710
middle ear, 709
outer ear, 709
eye, 703–708
anatomy, 703–704
disorders, 706–708
focusing, 704, 706–708
integration of visual signals, 705–706
photoreceptors, 703, 704
from the retina to the visual cortex, 706
hormone and endocrine system, 732–750
hypothalamus and pituitary gland, 737–739
other endocrine glands and hormones, 739–748
immune and lymphatic system, 609–628
adaptive immune defenses, 616–621
innate immune defenses, 613–615
lymphatic system, 611–613
male reproductive system, 754–757
muscular system, 724–729
macroscopic anatomy and physiology, 725–726
microscopic anatomy and physiology, 726–729
neuron and nervous system
central nervous system, 677, 686–691
embryonic development, 729*f*
nervous tissue, 680–686
peripheral nervous system, 677, 691–694
population growth, 826–829
age distributions, 827–828, 827*f*
more-developed *vs.* less-developed countries, 826*f*, 827
reproductive system
aging process, 791
control of reproduction, 762–764
female reproductive system, 758–762, 791
sexually transmitted diseases, 766–770
respiratory system
aging process, 790
lungs, 648, 650–651
modifications in breathing, 652–654
oxygen debt, 728
skeletal system, 718–724, 721f
aging process, 791
appendicular skeleton, 722–723
axial skeleton, 721–722
bone growth and renewal, 719
joint classification, 724
long bone anatomy, 719
somatic senses, 713–714
teeth, 632
urinary system, 667–674, 667f
homeostasis and kidneys, 671–674
kidneys, 668
urine formation, 668–671
vision, 703–708
anatomy, 703–704
disorders, 706–708
focusing, 704, 706–708
integration of visual signals, 705–706
photoreceptors, 703, 704
from the retina to the visual cortex, 706
Human chorionic gonadotropin (HCG), 761, 783
Human evolution, 556, 562*f*
hindlimbs, 557
mobile forelimbs, 557
primate evolution, sequence of, 558–561
primates, evolution of, 557–561
reduced reproductive rate, 558
stereoscopic vision, 557–558
Human genome, 253
Human Genome Project (HGP), 250
Human growth hormone, 245
Human immunodeficiency virus (HIV), 353
host specificity, 353
lytic and lysogenic cycles, 353–355
reproduction, 353–355
Human influenza virus, 352
Humanlike hominins, evolution of
ardipithecines, 561–563
australopithecines, 536–537
timeline of, 561
Human papillomavirus (HPV), 768–769
Humans, bacterial diseases, 364
Human skin, anatomy of, 581
Humboldt Current, 880
Humerus, 723
Hummingbird-pollinated flowers, 493*f*
Hummingbird, territoriality, 805
Humor, 617
Humpback whale, 803
Humus, 452
with earthworms, 452f
Hunting, subsistence, 887
Huntington disease, 192
Huntington protein, 192
Hurricane, 857
Hutton, James, 262
Hyaline cartilage, 577
Hybridization
alloploidy, 302
postzygotic isolating mechanisms, 298f, 299–300
prezygotic isolating mechanisms, 297–299
Hybridoma, 618
Hydra, 515
anatomy of, 515f
asexual reproduction, 752
nerve net, 677
reproduce asexually, 515
respiration, 647, 648f
sexual reproduction, 752
skeleton, 717
Hydrocarbons, 37
Hydrochloric acid (HCl), 32, 635
Hydrogen bonding, 28*f*, 29
Hydrogen ions (H^+), 32
human respiratory system, 652
mineral uptake in plants, 454f, 456
opening and closing of stomata, 462
pressure-flow model of phloem transport, 464
Hydrologic cycle, 852
Hydrolysis reaction, 39
Hydrolyzes ATP, 455
Hydroponics, 451, 451*f*
Hydrostatic skeleton, 514, 518
annelids, 717
Hydrothermal, chemical evolution, 315*f*
Hydrothermal vents
origin of life, 315f
Hydrothermal vents, ecology and ecosystems, 848, 879
Hydroxide ions (OH^-), 32
Hylobates syndactylus, 806*f*
Hyperglycemia, 745
Hyperopic vision, 708
Hyperparathyroidism, 742
Hypertension, 600, 790
Hyperthyroidism, 741
Hypertonic solution, 88–89
Hyphae, 386, 386*f*
fungi, 392
lichen, 400
Hyphal filaments, 386*f*
Hypocotyl, 495
Hypoglycemia, 745
Hypolimnion, 875
Hypoparathyroidism, 741
Hypothalamus and pituitary gland, 737–739
anterior pituitary, 737, 738–739
hormones affecting other glands, 738–739
hormones not affecting other glands, 737–739
hypothalamic-inhibiting hormones, 738
hypothalamic-releasing hormones, 738
major endocrine glands, 734*f*
hypothalamus
anatomy, 890
vertebrate organization, 677
ovarian cycle, 761
posterior pituitary, 734f, 737–738, 737f
Hypotheses, 350
Hypothesis, 11
Hypothyroidism, 740
Hypotonic solution, 88
Hyracotherium, 309

I

Iberomesornis, 269
Ice, lake ecosystems, 876
Ichthyophthirius, 378
Ick, 378
ICSI (intracytoplasmic sperm injection), 764
ICTV (International Committee on Taxonomy of Viruses), 350
Identical twins, 228
IgA, 618, 618*t*
IgD, 618*t*
IgE, 618, 618*t*, 626
IgG, 618, 618*t*, 626
IgM, 618, 618*t*
Immediate allergic response, 626

ISTUDY

Immune and lymphatic system, 609–628
adaptive immune defenses, 616–625
active *vs.* passive immunity, 621
B cells and antibody-mediated immunity, 617–619
evolution, 610–611
T cells and cell-mediated immunity, 617–625
aging process, 790
evolution of immune systems, 610–611
cellular slime molds, 610
Drosophila, 610
rise of adaptive immunity, 610
immune system disorders and hypersensitivity reactions
allergies, 625–626
autoimmune diseases, 626
immunodeficiencies, 625
transplant rejection, 626
innate immune defenses, 610, 613–615
inflammatory response, 614–615
phagocytes and natural killer cells, 615
physical and chemical barriers, 613–614
protective proteins, 615
lymphatic system, 611–613
lymphatic vessels, 611–612
lymphoid organs, 612–613
Immunization and active immunity, 621
Immunodeficiency disorder, 625
Immunoglobulins (Igs)
development, 618
plasma, 602, 603, 617
structure, 6198
Implantation, 783
Imprinting, 799
Inborn errors of metabolism, 210
Inbreeding, 282, 283–284, 284*f*
Incisor, 632
Inclusive fitness, 808, 808*f*
Incomplete digestive tract, 521, 630, 630*f*
Incomplete dominance, 193, 191*f*
Incomplete penetrance, 191
Incontinence, urinary, 790
Incus, 709
Independent assortment, 168, 168*f*
Index fossils, 319, 331
Indirect value of biodiversity, 887–886
biogeochemical cycles, 888
climate regulation, 888
ecotourism, 888
fresh water provision, 888
soil erosion protection, 888
waste recycling, 888
Induced fit model, 104
Induced mutations, 232–233
Induced pluripotent stem (iPS) cells, 153
Inducer, 222
Inducible operon, 222
Induction, 779–780
Inductive reasoning, 10
Indusia, 417
Industrial melanism, 268
Infection, opportunistic, 620, 622, 658
Infertility, 764
Inflammatory response, 614–615
analgesics, 714
anti-inflammatory medication, 714, 724
anti-TNF treatments, 621
capillaries, 614
chronic inflammation, 614
complement, 615
fever, 614
glucocorticoids, 742
white blood cells, 614
Influenza A, 350
Influenza virus
classification, 350
emerging viruses, 358
human, 352
Infrared spectrum, 698, 700
Ingroup (cladistics), 342, 347
Inguinal lymph nodes, 611*f*
Inheritance, 182
animal behavior, 795–796
garter snakes, 796
lovebirds, 795
marine snails, 796–797
mice, 797
twin studies, 796
autosomal patterns of, 189–190
beyond Mendelian inheritance, 192–200
epistatic interactions, 195
incomplete dominance and incomplete penetrance, 193
multiple alleles and codominance, 192–193
pleiotropic effects, 194–195
polygenic inheritance, 195–196
X-linked inheritance, 197–200
blending concept of, 182
Mendel's laws of inheritance, 183–189
law of independent assortment, 186, 187*f*
law of segregation, 183–184
Mendel and laws of probability, 187–188
Mendel's cross, 185–186
testcrosses, 188–189, 189*f*
Mendel's patterns of inheritance and human disease, 189–192
autosomal dominant disorder, 191–192
autosomal patterns of inheritance, 189–190
autosomal recessive disorder, 190–191
particulate theory of, 182
polygenic, 195–196, 196f
X-linked, 197–200
Inheritance of acquired characteristics, 262, 262*f*
Inhibin, 757
Inhibition model of succession, 847
Initiation, translation, 217, 217*f*
Innate (genetically modified potato), 246
Innate immune defense, 613–615, 613*f*, 614*f*
cellular slime molds and fruit flies, 610
inflammatory response, 614–615
capillaries, 614
chronic inflammation, 614
complement, 615
fever, 614
white blood cells, 614
phagocytes and natural killer cells, 615
eosinophils, 615
macrophages, 615
natural killer (NK) cells, 615
neutrophils, 615
physical and chemical barriers, 613–614
protective proteins, 615
complement, 615
interferons, 615
Innate immunity, 610
Inner ear, 709–710
Innervation, muscle, 728
Inorganic chemistry, 37
Insect/bird-pollinated flowers, 426
Insect diversity, 529*f*
Insects, 528
chemical senses, 700
excretory organs, 665
metamorphosis, 735
omnivorous diet, 632
respiratory system, 648f, 649–650
sound wave detection, 708
using for food, 530f
vision, 702
Insight learning, 801, 801*f*
Insoluble fiber, 641
Inspiration, 647, 652, 653*f*
Instinct and learning, 798
Insulin, 47, 245, 643, 733*f*, 744, 745
Integral proteins, plasma membrane, 82
Integration, 353
central nervous system, 683, 686
synaptic, 683, 686f
visual signals, 705–706, 705f
Integrins, 92
Integumentary system, 581–584
aging process, 790
human skin, accessory structures of, 584
skin, functions of, 581
skin, regions of, 581–584
Intercalary meristem, 430
Intercalated disks, 579
Intercostal muscle, 652, 654
Interferon, 615
Intergenic sequences, in human genome, 251–252, 251*f*
repetitive elements, 251
transposons, 251, 252f
unique noncoding DNA, 252
Interkinesis, 169
Intermediate filaments, 74, 75*f*
Internal fertilization, 753
Internal organization, 539
Internal respiration, 647, 647*f*, 654, 654*f*, 655–656
International Code of Zoological Nomenclature (ICZN), 338
International Committee on Taxonomy of Viruses (ICTV), 350
Interneuron, 680*f*, 681, 693–694
Internode
leaf/leaves, 434
shoot system, 434, 439
Interoceptor, 699
Interspersed repeats, 251
Interstitial cell, 755*f*, 757
Interstitial fluid, 577, 590
lymphatic vessels, 611–612
Intertidal zone, 878
Intervertebral disk, 722
Intestine
earthworms, 631
human
large, 633*f*, 634, 636, 637
small, 633*f*, 635–636
Intracytoplasmic sperm injection (ICSI), 764
Intrauterine device (IUD), 763*f*
Intrauterine insemination (IUI), 764
Introns, 213, 250
function of, 214

ISTUDY

Invagination, 775*f*
Inversion, chromosome, 177, 178*f*
Invertebrates, 505
animals evolution, 506–512
ancestry of, 507
body plans, 507
characteristics, 506
embryonic development, 510–512
phylogenetic tree, 507
symmetry, type of, 510
cannibalism, 632
deuterostomes
echinoderms, 532–533
sea stars, 533
water vascular system, 532
ecdysozoans, diversity, 524–531
arthropods, 526–531
roundworms, 525–526
equilibrium, 710f
evolution
nervous system, 677
excretory organs
coxal glands, 665
flame cells, 665
green glands, 665
Malpighian tubules, 665
maxillary glands, 665
nephridia, 655
hormones and endocrine system, 735
nervous system, 677
cephalization, 677
nerve net, 677
reproduction
cnidarians, 752
sponges, 752
simplest invertebrates
animal kingdom, 511*f*
cnidarians, 514–515
ctenophora, 513–514
sponge anatomy, 513*f*
sponges, 512–513
spiralians, diversity
lophotrochozoa, 516–518
In vitro fertilization (IVF), 751, 764, 765
Involution of thymus, 790
Iodine, 740
Iodine-131, 24
Ionic bonding, 26, 26*f*
Ionizing radiation, 233
Ions, 27
Iris, 703
Iris versicolor, 335
Iron-sulfur world hypothesis, 314, 315, 331
Irruptive growth, 821
Ischemic stroke, 688
Island
nonnative species, 890
island biogeography model, 837
Isomers, 38–38, 39*f*
Isoniazid, 233
Isotonic solution, 86–87
Isotopes, 21
Iteroparity, 818
IUD (intrauterine device), 763*f*
IUI (intrauterine insemination), 764
Ivanowsky, Dimitri, 350
IVF (in vitro fertilization), 751, 764, 765

J

Jackal, 810
Jacob, François, 222
Jacobson's organ, 548, 700
Jacobs syndrome, 177
Jaguar Corridor Initiative, 862
Jaguars, 862, 867
behavior of, 802
Japanese crab, 526
Jasmonic acid, 478
Jaundice, 638
Jawed fish
cartilaginous/bony endoskeleton, 540, 541
ectothermy, 540
evolution of, 541f
fins, 540
gills, 540
scales, 541
Jawless fishes, 540
Jeholodens, 326
Jellyfish, 505, 514, 515
nerve net, 677
Jerne, Niels, 618
Joint, human, 724
Jumping genes, 252. *See also* Transposons
Junction proteins, 83, 83*f*
Juno molecule, 774
Jurassic fossil, 343
Juxtaglomerular apparatus, 673

K

K2, 685
Kangaroo rat, 666, 666*f*, 874
Kangaroo, red, 815
Kaposi sarcoma, 622
Karenia brevis, 370
karyotype, 163
Keratin, 47, 49
Kerner, Justinus, 726
Ketamine, 684
Ketoacidosis, 745
Ketones, 672
Keystone species, 894
Kidney
human urinary system, 667–674
homeostasis and kidneys, 671–674
kidneys, 668, 668*f*
urine formation, 668–671
osmoregulation, 665, 671
parathyroid hormone (PTH), 741–742
Kidney stone, 742
Kidston, Barry, 689
Kinase, 106
Kinesin, 75
Kinetic energy, 99
Kinetochores, 149, 149*f*
Kinetochore spindle fibers, 149
Kinetoplastids, 374, 388
Kingdom, 8, 9, 335, 336*f*, 337
Kingsnake, scarlet, 842
Kinky hair syndrome. *See* Menkes syndrome
Kinocilium, 808
Kin selection, 808
Kipchoge, Eliud, 716
Klinefelter syndrome, 177, 177*f*
Klipspringer, 803
Knee, human, 724, 724*f*
Koala, 632
Köhler, George, 618
Kokanee salmon, 885*f*
Krause end bulb, 713
Krebs cycle. *See* Citric acid cycle
Krill, 631
K-selection, 823
Kuru, 355
Kyoto Protocol, 858

L

Labia majora, 759
Labia minora, 759
Labroidesdimidiatus, 844*f*
lac operon, 22, 224, 224*f*
Lacks, Henrietta, 155
Lactate, 728
Lactation
defined, 762
passive immunity, 621
prolactin (PRL), 739
Lacteal, 636
Lactic acid fermentation, 129
Lactobacillus bulgaricus, 132
Lactose, 40, 629
Lactose intolerance, 40, 629
Lacunae, 576
Ladder-type, 522
Ladybug, 887*f*
Ladybugs, 887*f*
Lag phase, 819, 820
Lake
acid deposition, 891
ecosystem, 875–876, 875f
lake effect, 864
life zones, 876
benthic zone, 877
limnetic zone, 877
littoral zone, 877
profundal zone, 877
overturn, 876, 877f
LAL (limulus amoebocyte lysate), 886
Lamarck, Jean-Baptiste, 262
Laminaria, 379, 379*f*
Laminarin, 378
Lampreys, 540, 540*f*
Lance fluke, 845
Lancelet
cleavage, 775
gastrula, 776
Lancelets, 537, 538
Land
landscape diversity and preservation, 885, 895
Land colonization
bryophytes, evolution of, 411–412
Land, invasion of, 323
invertebrates, 324–325
plants, 323–324
vertebrates, 325
Land plants
embryo difference, 410f
generations, alternation of, 410f
Landscape
diversity, 885
preservation, 895

ISTUDY

Land snail, 518
omnivorous diet, 632
Language use
chimpanzees, 803
dolphins, 803
humans, 803
Large intestine, 633*f*, 636, 637
Larval stage, 753
Larynx, human, 651, 757
LASIK (laser-assisted in situ keratomileusis), 708
Last universal common ancestor (LUCA), 313, 331, 338
Latent, 353
Lateral line system, 708–709, 709*f*
Laterite, 870
Latimeria chalumnae, 543*f*
Laughing gull, 798, 798*f*
Law, 12
Law of independent assortment, 186, 186*f*, 187*f*
Law of segregation, 183–185, 183*f*, 184*f*
Mendel's experimental design and results, 183–184, 183f
F_1 generation, 183
F_2 generation, 184
monohybrid cross, 184, 184*f*
P generation, 183
Punnett square, 183, 183*f*
statement of, 185
Laws of thermodynamics, 99–100
first law, 99–100
second law, 100
LDC (less-developed country), 826*f*, 826
LDL (low-density lipoprotein), 643
L-dopa, 689
Leaf hairs, 445*f*
Leaf/leaves
adaptation to prevent desiccation, 410f
classification of, 446f
color changes, 476, 477f
diversity of, 446f
internode, 434
node, 434
onion bulb, 446
shoot system, 434
structure and organization, 445, 445f
structure and organization, 444–447
classification, 446*f*
diversity, 446–447
morphology, 444–445
veins, 433, 434, 435
and vascular tissue, 432–433
twig, 442f
LEAFY gene, 435
Learning
associative learning, 799–800
classical conditioning, 799
operant conditioning, 800
cognitive learning, 801
defined, 798
habituation, 798
imprinting, 799
instinct and learning, 798
memory in human brain, 691
orientation and migratory behavior, 800–801
social interactions and learning, 799
Leclerc, Georges-Louis, 260
Leeches, 521, 521*f*
medicinal, 886, 886f
Leg bones, human, 723
Legionnaires' disease, 15
Legume
nitrogen fixation, 855
nodules of, 363f
Lens
anatomy, 703–704
cataracts, 706
focusing disorders, 706–708
Lenticels, 431
Lentiviruses, 250
Leonardo DiCaprio Foundation, 120
Leopard frog, 294, 296*f*
Leprosy, 364*t*
Leptin, 741
Lepus americanus, 840, 841*f*
Less-developed country (LDC), 826, 826*f*
Leucochloridium, 845
Leucoplasts, 72
Leukemia, 768
Lewontin, Richard, 569
LH (luteinizing hormone), 746, 757, 760–761, 760*f*
Liana, 869
Libby, Willard, 319
Lichens, 365, 368, 400, 401
cyanobacterium, 365
morphology, 401f
symbiotic relationships, 844
symbiotic relationships, 365
Lidy Hot Springs, 367
Life, 1–17
challenges facing science and society, 13–15
biodiversity and habitat loss, 14–15, 14*f*
climate change, 14, 14*f*
emerging diseases, 15
characteristics, 2–6
adaptations, 5–6, 5*f*
homeostasis, 4–5, 4*f*
material and energy requirements, 4, 4*f*
organization, 2, 3*f*
reproduction and development, 5
response, 5
diverse forms, 2, 2f
evolution and classification, 6–10
evolutionary tree of life, 7, 8*f*
natural selection, 6–7, 6*f*
taxonomic classification, 8–10, 8*t*
process of science (scientific method), 10–13, 11f
hypothesis, 10
observation, 10
predictions and experiments, 11
presenting and analyzing data, 11, 11*f*
scientific theory, 12–13
summary, 16–17
tree of, 323f
Life cycle, 172–174
of humans, 173f
oogenesis, 173–174, 174*f*
spermatogenesis, 173, 173*f*
of plants, 173
Life expectancy, 789
Life history patterns, 823–825
K-selected populations, 823–824
r-selected populations, 823
Life history strategy, 753
Life table, 816–817, 816*t*
Life zone
lake, 876
benthic zone, 877
limnetic zone, 877
littoral zone, 872
profundal zone, 872
ocean
abyssal plain, 880, 879*f*
euphotic zone, 879
pelagic zone, 879
Ligament
dense fibrous connective tissue, 724
synovial joints, 724
Ligament, dense fibrous connective tissue, 576
Light
animal detection and vision, 702–703
blue light, 463
human eye, 703–704
sense of vision, 698
ultraviolet
nectar guides, 702, 703*f*
wavelengths, 698
Light reactions, 114–115
cyclic pathway, 116, 117f
electron flow in, 116–117, 117f, 118f
noncyclic pathway, 116–117, 117f
Lignin, 93, 413, 432
Lilium bulbiferum, 335, 335*f*
Lilium canadense, 335, 335*f*
Limb development, 308, 784
Limb bud, 785
Limbic system, 690–691, 691*f*
amygdala, 691
hippocampus, 690, 691
Limiting factor, 815
Limulus, 886
Limulus amoebocyte lysate (LAL), 889
Limulus genus, horseshoe crabs of, 531
Lineages, 295, 340, 347
Linnaeus, Carolus, 269, 294, 334, 335*f*
Linnaeus, hierarchical classification system of, 341
Lion, 632
Lionfish, 883
Lipase, 637
Lipid-protein microspheres, 316
Lipids, 43–45, 43*t*
human diet and nutrition, 640–642
phospholipids, 45
steroids, 45, 46f
triglycerides, 43–45, 44f
waxes, 45, 46f
Liposomes, 249, 316, 331
Liquid connective tissue, 577
Liver
anatomy, 633f, 637–638, 637f
deamination, 643
disorders, 637–638
cirrhosis, 638
hepatitis, 638
glycogen storage, 733f, 744
inflammatory response, 613
insulin, 733f, 744
Liverworts, 411–412, 412*f*
Living amphibians, diversity of, 544
Living cells, 313
Living organisms, 452–453
Lizards, 547
Lizard, survivorship, 817

ISTUDY

Loam, 452
Lobaria, 401*f*
Lobe-finned fishes, 542
Lobe-finned fishes, to amphibians, 546*f*
Lobster industry, sustainability in, 825
Lobule, 637
Locomotion and support system, 716–731
  diversity of skeletons, 717–718
    endoskeletons, 717, 718
    exoskeletons, 717–718
    hydrostatic skeleton, 717
  human skeletal system, 718–724
    appendicular skeleton, 722–723
    axial skeleton, 721–722
    bone growth and renewal, 719
    joint classification, 724
    long bone anatomy, 719
  muscular system, 717–722
    macroscopic anatomy and physiology, 725–726
    microscopic anatomy and physiology, 726–729
Locus, gene, 185, 185*f*
Loggerhead sea turtle, 867
Logistic growth, 819–821, 820*f*
Long bone
  anatomy, 719, 720f
    compact bone, 719
    spongy bone, 719
  endochondral ossification, 719
Longevity, 773, 789
Longhorn beetle, 842, 842*f*
Longitudinal fissure, 687
Long-term memory, 691
Long-term stress response, 742*f*
Loop of the nephron, 668, 669*f*, 671–673, 671*f*
Lophophorans, 516, 516*f*
Lophotrochozoans, 516
  characteristics, 516f
Lorica, 378
Loriciferan, 647
Lovebird, 795–797, 795*f*
Low-density lipoprotein (LDL), 90, 643
Lower respiratory tract disorder, 656–659
  lungs, 657–658
    cystic fibrosis (CF), 659
    emphysema, 657*f*, 658–659
    lung cancer, 659
    pneumonia, 657, 658
    pulmonary fibrosis, 657*f*, 659
    pulmonary tuberculosis, 657*f*, 657
  pneumonia, 622
  trachea and bronchi, 656–657
    asthma, 625, 657
    bronchitis, 656–657
    choking, 656
    chronic obstructive pulmonary disease (COPD), 657
Lucy, 563
Lumbar vertebra, 722
Lumbricusterrestris, 519
Lumen, 634
Lung
  amphibians, 648
  bird, 648
  chemoreceptors, 701
  human, 648, 650–651, 657–659
  lower respiratory tract disorders
    cystic fibrosis (CF), 659
    pneumonia, 622, 658
  respiratory tract disorders, 656–660
    cystic fibrosis (CF), 659
    emphysema, 657*f*, 658–659
    lung cancer, 659
    pneumonia, 658
    pulmonary fibrosis, 657*f*, 659
    pulmonary tuberculosis, 657*f*, 658
Lung cancer, 232, 659, 659*f*
Lungfishes, 542
Luteal phase, 761
Luteinizing hormone (LH), 746, 757, 760–761, 760*f*
Lycophytes, 408, 414, 414*f*, 416
  evolution of, 415–418
Lycopodium, 414, 414*f*
  structure of, 414
Lycorma delicatula, 813
Lyell, Charles, 262
Lymph, 606, 606
  fluid connective tissue, 578
Lymphatic and immune system, 609–628
  adaptive immune defenses, 616–625
    active *vs.* passive immunity, 621
    B cells and antibody-mediated immunity, 617–620
    evolution, 610
    T cells and cell-mediated immunity, 617–625
  evolution of immune systems, 610–611
    cellular slime molds, 610
    Drosophila, 610
    rise of adaptive immunity, 610
  immune system disorders and hypersensitivity reactions
    allergies, 625–626
    autoimmune diseases, 626
    immunodeficiencies, 625
    transplant rejection, 626
  innate immune defenses, 610, 613–615
    inflammatory response, 614–615
    phagocytes and natural killer cells, 615
    physical and chemical barriers, 613–614
    protective proteins, 615
  lymphatic system, 611–613
    lymphatic vessels, 611–612
    lymphoid organs, 612–613
Lymphatic capillary
  function, 606f
  small intestines, 636
Lymphatic duct, 611*f*, 612
Lymphatic vessel, 611–612
Lymph node, 611*f*, 612
Lymphocyte, 244
  development, 612
  natural killer (NK) cells, 615
  primary lymphoid organs, 612
Lymphocytes, 604
Lymphoid organ
  primary
    red bone marrow, 611*f*, 612, 719
    thymus, 611*f*, 612, 734*f*, 747–748
  secondary, 611–612
    lymph nodes, 611*f*, 612
    Peyer patches, 613, 634
    spleen, 611*f*, 612
    tonsils, 611*f*, 613
Lynch, Steven P., 495
Lynx, Canada, 840–841, 841*f*
Lysogenic bacterial cells, 353
Lysogenic cells, 353, 368
Lysogenic cycle, 353, 368
Lysosomal storage diseases, 69
Lysosomes, 69, 69*f*
  by Golgi apparatus, 69
  hydrolytic-digestive enzymes in, 69, 70
  Tay-Sachs disease, 69
Lytic cycle, 353
  of viral replication, 368

M

MAC (*Mycobacterium avium* complex), 622
Macaw, 889, 890*f*
MacLeod, Colin, 204
Macleod, J. J., 746
Macrocystis, 379, 379*f*
Macroevolution and speciation, 293–311
  anatomy of speciation, 295, 295f
  defined, 294, 300
  how new species evolve, 294–300
    reproductive isolating mechanisms, 297–300
    species, 294–297
  modes of speciation, 300–306
    adaptive radiation, 303–306
    allopatric speciation, 300–302
    convergent evolution, 306
    sympatric speciation, 302
  phylogenetic species concept, 295
  principles of macroevolution, 307–309
    developmental genes and macroevolution, 307–309
Macromolecules, 39, 39*t*
Macronutrients, 450
Macrophage
  functions, 615, 615f
  inflammatory response, 615
Macrophages, 604
Macular degeneration, 56, 706
Mad cow disease, 50, 231, 355
Madreporite, 533
Maine lobster industry, 825
Maize, evolutionary history of, 422
Malaria, 376
Male competition, 287–288
Male crayfish, 528*f*
Male gametophyte, 492
Male reproductive system, human, 754–757
  aging process, 791
  hormonal regulation, 757
  penis and male orgasm, 754f, 756–757
  production of sperm, 755–756
  teste
    cancer, 755
    functions, 746–745, 752, 754–755
    gonadotropic hormones, 738
    major glands, 734*f*
    steroid hormones, 736
Malignant growth, 154. *See also* Cancer
Malleus, 709
Malpighian tubule, 529, 665
Maltase, 639
Malthusian growth, 821
Malthus, Thomas R., 262, 821
Maltose, 40, 41*f*, 639
Mammalian body cavities, 581*f*
Mammal lineages, female reproductive traits of, 553*t*
Mammals, 551
  characteristics of
    ear, 709
    hair, 551

ISTUDY

internal development, 552
internal organs, 551
mammary glands, 552
skeleton, 551
diet adaptations, 632
evolution
female reproductive traits, 553*t*
marsupials, 553, 553*f*
monotremes, 552, 552*f*
nervous system, 677
placental mammals, 553, 554*t*
placental mammals, 653
timeline of, 552*f*
hormones and endocrine system, 735
middle ear, 709
nervous system organization, 677
origin of, 270
osmoregulation in marine mammals, 667
respiration, 652
Mammary gland, prolactin (PRL), 739
Mandible, human, 721
Mangrove swamp, 878, 878*f*
Mantle, 517
Marchantia genus, 411, 412*f*
Marfan syndrome, 194, 194*f*
Marijuana, 685
Marine mammal osmoregulation, 667
Marine protected areas (MPAs), 838, 838*f*
Marine snail, 796–797
Marshall, Barry, 635
Marshes, 877–878
Marsupials, 271, 271*f*
Mass extinctions, 323, 329–330, 329*f*, 331
Mass number, 21
Massospondylus, 312
Mast cell, 614
Mastoiditis, 721
Mastoid sinus, 721
Maternal care behavior, 797*f*
Maternal determinant, 778
Mating and genetic diversity, 282
Matrix, mitochondria, 73, 73*f*
Matter, 20
Matthei, J. Heinrich, 211
Mature embryo, 490
Mawr, Bryn, 165
Maxilla, human, 721
Maxillary gland, 665
Mayr, Ernst, 300
McCarty, Maclyn, 204
McClintock, Barbara, 251
MDC (more-developed country), 826*f*, 827
MDMA (methylenedioxymethamphetamine), 684
Measles, German, 788
Mechanical energy, 98
Mechanical isolation, 299
Mechanical weathering, 452
Mechanoreception, 708
Mechanoreceptor, 699, 709, 710*f*, 711*f*
Medicine, medicinal value of biodiversity, 886
Medulla oblongata, 677, 690
Medullary cavity, 719
Medusa, 514
Meerkat, 810, 810*f*
Megafauna, 331
Megaphylls, 415, 415*f*
evolution process, 415f
Megaspores, 415, 489
angiosperms, 422–423
gymnosperms, 421f
Meiosis, 162, 173, 185, 410
genetic variation, 166–168
crossing-over, 166–167, 166*f*
independent assortment, 168, 168*f*
significance of, 168
and mitosis, comparison of, 170–172, 171f, 172t
overview of, 163–166, 164f
fate of daughter cells, 166
homologous pairs of chromosomes, 163–164, 163*f*
meiosis I, 164–166, 164*f*, 169, 169*f*
meiosis II, 164–166, 164*f*, 170, 170*f*
reduction division, 164–166, 164*f*
and parthenogenic lizards, 167
phases of, 168–169
anaphase I, 169, 169*f*
interkinesis, 169
meiosis II and gamete formation, 170, 170*f*
metaphase I, 169, 169*f*
prophase I, 169, 169*f*
telophase I, 169, 169*f*
and sexual reproduction, 160–178
chromosome number and structure, changes in, 171–176
life cycle, 172–174
parthenogenesis, 752
spermatogenesis and oogenesis, 755*f*, 756, 760
Meiosis I, 755*f*, 756
Meiosis II, 755*f*, 756
Meissner corpuscle, 713
Melanin, 195
Melanocytes, 582
Melanocyte-stimulating hormone (MSH), 739
Melanoma, 583
Melanophore, 739
Melatonin, 690, 747, 748*f*
Mello, Craig, 231
Melomysrubicola, 891*f*
Membrane attack complex, 615
Membrane-first hypothesis, 317, 331
Membrane structure and function, 8096
active transport across membrane, 89–92
endocytosis, 90–92, 92*f*
energy use for, 90
exocytosis, 90, 91*f*
movement against concentration gradient, 90
sodium-potassium pump, 90, 91*f*
cell-to-cell communication
cell signaling, 94–95, 94*f*, 94*t*
second messenger systems, 95, 95*f*
cell surfaces modification, 93–94
animal cell, 93–94
plant cell, 92
passive transport across membrane, 85–87
diffusion, 85, 85*f*
facilitated transport, 88, 88*f*
osmosis, 85–88, 86*f*
plasma membrane
components of, 81–82, 81*f*, 82*f*
fluid-mosaic model, 81*f*, 82
glycoproteins and glycolipids, 82–83
permeability of, 84, 84*f*
proteins and functions, 83
structure and function, 81–85
summary, 95–96
Memory
brain, 691
sense of smell, 691
Memory B cell, 617
Memory T cell, 625
Menarche, 761
Mendeleev, Dmitri, 21
Mendel, Gregor, 165, 182, 182*f*, 480
Mendelian patterns of inheritance, 181–200
blending concept of inheritance, 182
garden pea plants as true-breeding, 182
cross-pollination experiments, 182, 182*f*
and human disease, 189–200
autosomal dominant disorder, 187–200
autosomal patterns of inheritance, 189–190
autosomal recessive disorder, 190–191
Mendel's laws of inheritance, 183–189
law of independent assortment, 186, 186*f*
law of segregation, 183–185
Mendel and laws of probability, 187–188
Mendel's cross, 185–186
testcrosses, 188–189, 189*f*
other patterns of inheritance since, 192–200
epistatic interactions, 195
incomplete dominance and incomplete penetrance, 193
multiple alleles and codominance, 192–193
pleiotropic effects, 194–195
polygenic inheritance, 195–196
X-linked inheritance, 197–200
particulate theory of inheritance, 182
summary, 200–201
Meninge, 686
Meningitis, 686
Meniscus, 724
Menkes syndrome, 198
Menopause, 761, 791
Menstrual phase, 761
Menstruation, 735, 761
Meristem, 430
Meristematic cells, 501
Meristematic tissue, 152
Meristem cell division, 430
Merkel disk, 713
Merychippus, 310
Meselson, Matthew, 208
Mesoderm
coelom development, 775
embryonic development, 775f, 775, 778f
organ development, 775–777, 777f
Mesoglea, 513
Mesopelagic zone, 879, 879*f*
Mesophyll, 445
Mesosomes, 61
Mesozoic era, 331
Messenger RNA (mRNA), 51, 67, 210
steroid hormones, 736
Messenger RNA codons, 211, 212*f*
Metabolic pathway, 103
activation energy, 103–104, 104f
definition, 103
enzymatic rate, factors affecting, 104–105
cofactors and coenzymes, 105, 106*f*
optimal pH, 104, 105, 105*f*
substrate concentration, 104
temperature, 105105*f*, 106*f*
enzyme activation, 106
enzyme inhibition, 106, 106f

ISTUDY

Metabolic pathway—(*Cont.*)
enzymes in, 102
enzyme-substrate complex, 102
Metabolic pool concept, 137, 138*f*
Metabolic waste, 671
Metabolism, 4, 137–139
anabolism, 138
bacteria, 362–363
autotrophic bacteria, 363
heterotrophic bacteria, 363
oxygen, 362–363
catabolism, 137–138
definition of, 100
energy and enzymes, 96–110
cells and flow of energy, 99–100
metabolic pathways and enzymes, 103–106
metabolic reactions and energy transformations, 101–103
oxidation-reduction reactions and metabolism, 107–109
summary, 109–110
energy organelles, 138–139
flow of energy, 139
Metacarpal bone, 723
Metagenomes, 255
Metagenomics, 255
Metamorphosis, 527
amphibians, 735
arthropods, 735
Metaphase, 149, 150*f*
Metaphase chromosomes, 147*f*, 148
Metaphase plate, 149
Metapopulations, 895
Metarhizium spores, 402
Metastasis, 157
Metatarsal bone, 723
Meteor, with schreibersite, 316*f*
Methamphetamine, 684
Methane, 27, 27*f*
greenhouse effect, 854
Methanogenic archaea, 367
Methanogens, 367, 368
Methemoglobinemia, 190–191, 190*f*
Methicillin-resistant *S. aureus* (MRSA), 364
Methoprene, 732
Methylene blue, 191
Methylenedioxymethamphetamine (MDMA), 684
MG (myasthenia gravis), 626, 694
MHC protein
natural killer (NK) cells, 615
pheromones, 735
T cells, 619, 620f
transplant rejection, 626
Micelles, 316, 331
Microbiome, 488
Microcephaly, 357
Microevolution, 277–282, 279*f*, 295
Microglia, 579
Micronutrients, 4450
Microorganism, 296, 359
Microphylls, 414, 415*f*
MicroRNAs (miRNAs), 214
Microscopy, 59–60
contrast, 60, 60f
illumination and viewing, 60
magnification, 60
recording, 60
resolution, 60
types of microscopes, 59, 59f
Microspheres, 315, 331
Microspores, 415, 424, 489
Microsporidia, 393
Microtubule-organizing center (MTOC), 75
Microtubules, 74–76, 75*f*, 149
Microvillus, 636
Midbrain, 677, 690
Middle ear, 709
Middle East respiratory syndrome (MERS), 14, 354
Midwife toad, 824*f*
Migration
birds, 800–802
defined, 800
gene flow, 281
orientation and migratory behavior, 800–802
Milk of magnesia, 32
Miller, Stanley, 314
Miller-Urey experiment, 314
Millipedes, 528, 528*f*
Milstein, César, 618
Mimicry, molecular mimicry and autoimmune diseases, 626
Mimosa pudica, 480
Mimosa sensitive plant, 484
Mineral, 450
Mineral
cohesion-tension model of xylem transport, 461–462
discovery with plants, 455
plant nutrition and uptake, 454–457
Mineralocorticoid, 742*f*, 743, 744
Mineral uptake, 454–457, 454*f*
roots adaptations, 455
Misfolded proteins, 49
Mislabeled fish, 337*f*
Mitochondria, 72–74, 73*f*, 126, 133, 371, 751
and cellular respiration, 108–109, 133–137
chloroplasts and, 108–109, 108f
citric acid cycle, 134–135, 134f
diseases, 74
electron transport chain, 135–137
number of, 72
as powerhouse of cell, 73, 133
preparatory reaction, 133–134, 134f
structure of, 73, 73f, 133, 133f
cristae, 73
matrix, 73
Mitochondrial DNA (mtDNA), mutation, 343
Mitosis, 144, 148
chromosome number, 148, 148t
functions of, 152
human sperm production, 755f
and meiosis, comparison of, 170–172, 171f, 172t
phases of, 149–150, 150f
anaphase, 149
metaphase, 149
prometaphase, 149
prophase, 149
telophase, 149
preparation for, 148–149
chromosome duplication, 148, 149*f*
division of centrosome, 148–149
Mitotic spindle, 144
Mixotrophic, 371, 387
Model, 11
Mode of speciation, 300–306
adaptive radiation, 303–306
allopatric speciation, 300–302
convergent evolution, 306
sympatric speciation, 302
Modern humans (*Homo sapiens*), 556
Molar, 632
Mold
bread
Neurospora crassa, 397
Rhizopus stolonifer, 394, 395*f*
food industry, 396–397
sac fungi, 395, 397
slime (See Slime mold)
Mole, 102
Molecular biology of gene, 202–220
DNA replication, 208–210
overview of, 208–209, 208*f*, 209*f*
prokaryotic *versus* eukaryotic replication, 209–210, 209*f*
gene expression, 210
RNA and genetic code, 210–212
summary of, 218, 219*f*
transcription, 212–214
translation, 214–218
genetic material, 203–207
DNA structure, 205–207
transformation of bacteria, 203–205
summary of, 220
Molecular clock, 345, 347
Molecular mimicry, 626
Molecular traits, 344
Molecule, 2, 3*f*, 26
Mollusks
body plan of, 517f
digestive systems, 631
exoskeleton, 717
nervous system of, 517
omnivorous diet, 632
Molothrusater, 845*f*
Molting, 526*f*
arthropods, 717, 718f, 735
Monarch butterfly, 800
Monera, 338
Mongoose, 890
Monkey, auditory communication, 803
Monkey skeleton, *Proconsul* skeleton, comparison of, 561
Monoclonal antibodies, 618, 762
Monocot pollen grains, 435
Monocot root, 437*f*
Monocots, 423, 434
features, 435f
flowering plants, 435f
herbaceous stems, 439–450
key features of, 423t
organs and organ systems, 343–435
root system, 437–438, 437f
woody plants, 450, 443
Monocot seed germination, 500*f*
Monocot stem, 450, 450*f*
Monocot *vs.* eudicot, 496*f*
Monocotyledones, 423
Monocytes, 604
Monod, Jacques, 222
Monoecious, 420
Monoecious animal, 752
Monogamy, 807

ISTUDY

Monohybrid cross, 184
Monomer, 39
Monophyletic, 371, 391, 394
Monophyly, phylogenetic species concept, 295, 296
Monosaccharides, 40, 40*f*
Monosomy, 174
Monotremes, 552
Monsoon climate, 864
Mons pubis, 759, 759*f*
Montane coniferous forest, 865, 866*f*, 867
Moose, 868, 869*f*
More-developed country (MDC), 826*f*, 827
Morgan, Thomas Hunt, 197
Morning-after pill, 763
Morning glory (*Ipomoea leptophylla*), 484
Morphine, 80
Morphogen, 780
Morphogenesis, 777, 780–782
Morphological species concept, 294
Morphological traits, 344
Morphology
  defined, 249
  morphological species concept, 249
Mortality patterns, 816–817
Morula, 775, 783
Mosquitoes
  captured, survival of, 402f
  insecticide-resistant, 402f
  population counts, 403f
  survivorship, 817
Mosses, 407, 412
  gametophytes, 412
  reproduction, 412
  traits, 411
Moss life cycle, 413*f*
Moth
  communication, 801
  genetic basis of body color, 277–284
  Peppered, 277–280
Moth/butterfly-pollinated flowers, 493–494
Motor molecules, 74
Motor neuron, 680–681, 680*f*
Motor output and control, 686, 688
Mountain
  alpine tundra, 865, 866, 866f
  climate, 864
  montane coniferous forest, 865, 866f, 867
Mouse
  biotic potential, 816
  nurturing behavior, 797
  pattern formation, 781f
  predation and population size, 822
Mouth
  digestive tracts, 630
    earthworms, 630, 631
    human, 633–634
    incomplete digestive tracts, 630
  human, 633–634
    palate, 633–634
    salivary glands, 633*f*, 634
    uvula, 634
Mouthbrooder, 752, 824*f*
M protein, 358
MPPP drug, 689
MPTP drug, 689
mRNA transcript, 212, 213*f*
MS (Multiple Sclerosis), 626, 676, 690
MSH (melanocyte-stimulating hormone), 739
mtDNA nucleotide changes, 345
Mucosa, 634
Mucus
  digestive tract, 634
  innate immune defense, 613
  respiratory system, 651
Mudflat, 878, 878*f*
Müllerian mimic, 842
Mullis, Kary, 240
Multicellular brown algae, 378
Multicellularity arises, 322–323
Multicellular molds, 395
Multicellular organisms, 2, 57
Multifactorial traits, 196
Multiple alleles, 192–193
Multiple Sclerosis (MS), 626, 676, 690
Multiregional continuity hypothesis, 566
Muscle contraction
  adenosine triphosphate (ATP), 728, 729
  sliding filament model, 728
Muscle fiber, 727*f*, 728, 729*f*
Muscle mass, 791
Muscle relaxation, 729
Muscle spindle, 713, 726
Muscle tone, 713, 716
Muscular dystrophy, 198
Muscular hydrostat, 717
Muscularis, 634
Muscular system, human, 724–729
  aging process, 791
  antagonistic pairing of muscles, 725, 725f
  macroscopic anatomy and physiology, 725–726
  microscopic anatomy and physiology, 726–729
    ATP use in contraction, 728, 729
    muscle innervation, 728
    role of calcium in muscle contraction, 729
    sliding filament model, 728
  muscular tissue
    skeletal muscle, 724–729
  superficial musculature, 725f
Muscular tissue, 578*f*
  skeletal muscle, 724–729
Mushrooms, 397
  Amanita, 398
  Boletus, 399f
  ecosystem dynamics, 848
  life cycle, 399f
  morel, 396f
  poisonous species, 398f
  puffball, 399f
Mus musculus
  nurturing behavior, 797
  predation and population size, 822
Mutagens, 232
Mutation
  evolution, 308
  genetic diversity, 280–281
Mutation rate, 345
Mutations, 5
Mutualism, 363, 368, 400
  bacteria, 363, 844
  defined, 843
  ecology of communities, 843t
  fungi, 456, 458, 845
  lichen, 845
  lichen, 368
  mycorrhizae, 458, 458f, 845
Myasthenia gravis (MG), 626, 694
Mycelium, 387, 392, 395*f*, 397, 399*f*, 400
Mycobacterium avium complex (MAC), 622
Mycobacterium tuberculosis, 626, 658
Mycorrhizae, 387, 395, 403–404, 438, 457, 457*f*, 845
Myelin
  axons, 680
  Guillain-Barré syndrome (GBS), 6943
  multiple sclerosis, 262
  oligodendrocytes, 680
Mylodon darwinii, 263
Myocardium, 594
Myofibril, 727*f*, 728
Myoglobin, 728
Myopic vision, 706
Myosin, 74
Myosin filament, 728, 728*t*, 730*f*
Myrica faya, 890
Myrtle tree, 890
Mysis relicta, 885, 885*f*
Myxedema, 780

N

NAD^+, 128
NADH-Q reductase complex, 136
Naegleria fowleri, 385
Nails, human, 584
Nanjinganthus dendrostyla, 423
Nasal bone, 721, 722*f*
Nasal septum, 721
Nasopharynx, 634
Nastic movements, 481
Natriuresis, 743
Natural groups, 334, 346
Natural killer (NK) cell, 615
Natural selection, 6–7, 6*f*, 260, 265, 273. *See also* Darwin and evolution
  and adaptation, 265–266
  and observations
    adaptation to **e**nvironment, 267
    competition for resources, 266
    difference in reproductive success, 266–267
    variation in population, 266, 266*f*
  populations, 284, 285–286
  selection at work, 267–268
"Nature *vs.* nurture" question, 795
Nautilus belauensis, 519*f*
Navigation, 800
NE (norepinephrine), 682, 694, 695*f*, 742
Neanderthals (*Homo neanderthalensis*), 556
Nearsightedness, 706–708, 708*f*
Nectar guide, 702, 703*f*
Nectar guides, 493
Negative feedback
  blood concentration, 737
  homeostasis, 737
  production of sperm and testosterone, 757
Neglected tropical diseases (NTDs), 374, 522
Neisseria gonorrhoeae, 360, 362, 770, 770*f*
Nematodes, 400
Neocallimastigomycota, 393–394, 393*t*, 394
Neocallimastix genus, 394
Neocortex, 677
Nephridiopore, 665*f*
Nephridiostome, 665*f*
Nephridium, 665, 665*f*
Nephron, 668, 668*f*

ISTUDY

Nereis, 520
Nereocystis, 379
Nerve, 691–692, 692*f*
Nerve impulse transmission, 681–682
  action potential, 682
  propagation of action potentials, 682
  resting potential, 682
  transmission across a synapse, 682–683
Nerve net, 515, 677
Nerves, connective tissue, 579
Nervous system, 526, 676–697
  breathing, 652–656, 654f
  central nervous system, 686–691
  embryonic development, 779f, 783–785
  evolution of nervous system, 677–679
    invertebrates, 677
    vertebrates, 677
  mode of action, 682, 733, 733f
  nervous tissue, 680–686
    anatomy, 680–681
    neuroglia, 680
    neurons, 680–681
    synaptic integration, 683, 686
    transmission of nerve impulses, 681–682
  neurons
    anatomy, 680, 680*f*
    defined, 680
    types, 680–682
  peripheral nervous system, 677, 691–694
    autonomic system, 694
    somatic system, 692–694, 693*t*
Nervous tissue, 680–686
  anatomy, 680–681
  neuroglia, 579–580, 680
    astrocytes, 680
    microglia, 680
    oligodendrocytes, 680
  neuron, 579
  neurons, 680–681
    axon, 680
    cell body, 680
    defined, 680
    dendrites, 680
  synaptic integration, 683, 686
  transmission of nerve impulses, 681–682
    action potential, 682
    propagation of action potentials, 682
    resting potential, 682
    transmission across a synapse, 682–683
Nest
  fish, 753
  lovebirds, 795–796, 795f
Neural crest, 776
Neural plate, 776, 779*f*
Neural tube
  development, 778–777, 777f
  human, 783–784
Neuraminidase spike, 352
Neurodegenerative diseases, 355, 368
Neuroglia, 579*f*
  astrocytes, 680
  ependymal cell, 680
  functions, 680
  microglia, 680
  node of Ranvier, 680
  oligodendrocytes, 680
  satellite cell, 680
  Schwann cell, 680
Neuromuscular junction, 728, 729*f*
Neuron, 579*f*, 676–694
  anatomy, 680, 680f
  central nervous system, 686–691
  defined, 680
  evolution of nervous system, 677–679
  mode of action, 682
  nervous tissue, 680–686
    anatomy, 680–681
    neuroglia, 680
    neurons, 680–681
    synaptic integration, 683, 686
    transmission of nerve impulses, 681–682
  types, 680–681
Neurospora crassa, 397
Neurotransmitter, 682–683
Neurula, 776
Neurulation, 779*f*
Neutrons, 21, 21*f*
Neutrophils, 604, 615
Nevada, Sierra, 301
Niacin deficiency, 106
Niche, ecological
  character displacement, 838–839
  competitive exclusion principle, 837
  defined, 837
  niche specialization, 839, 840f
  realized niche, 837
Niche specialization, 839, 840*f*
Nicotiana attenuate, 478
Nicotine
  abuse, 684
  vaping, 660
Nirenberg, Marshall, 211
Nitrate, 855–856
Nitrification, 855
Nitrite, 855
Nitrogen cycle, 855–856, 855*f*
Nitrogen fixation, 363
  bacteria, 363, 363f, 365, 855
  cyanobacteria, 855
  cyanobacterial cells, 365
  legume, 855
Nitrogenous waste product
  ammonia, 664, 674
  urea, 664, 671
  uric acid, 664, 665
Nitrogen oxide, 891–892
Nitrogen, poplar trees cleaning up, 455*f*, 455
Nitrous oxide, 854
NK (natural killer) cell, 615
Noble gases, 21, 25
Nociceptors, 713
Noctiluca, 375
Node (phylogeny), 295
Node (plant), 408
  leaf, 445f
  shoot system, 434, 439
Node of Ranvier, 680, 682
Nomenclature, 346
  standardized rules of, 338
Noncompetitive inhibition of enzyme, 106, 106*f*
Noncyclic pathway, of photosynthesis, 116–117, 117*f*
Nondisjunction, 174, 175*f*. *See also* Aneuploidy
  primary, 173, 175f
  secondary, 172, 175f
Nonfunctional proteins, 233–234
Nonnative species, extinction, 889–891
  accidental transport, 890
  colonization, 889
  horticulture and agriculture, 890
  impact on islands, 890
Nonpolar covalent bond, 27
Nonrandom mating, 284
Nonrenewable resources, 829
Nonsteroidal anti-inflammatory drugs (NSAIDS), 714
Nonvascular plants, 411, 412
Nonylphenol, 892
Norepinephrine (NE), 682, 694, 695*f*, 742
Northern pike, 877*f*
Northern spotted owl, 869, 895
Notochord
  development of, 776, 779f
  replacement of, 539f
NPK ratio, 451
NSAIDS (nonsteroidal anti-inflammatory drugs), 714
9 N-type spike, 350
Nuclear envelope, 67
Nuclear pore complex, 67
Nuclear pores, 67
Nucleic acids, 50–53
  ATP, 52–53, 53f
  DNA and RNA structure, 51–52, 51f, 52f, 51t
  nucleotides, 50, 51f
Nucleoid, 61, 62*f*, 360
Nucleolus, 67
Nucleoplasm, 66, 66*f*
Nucleosomes, 226
Nucleotides, 51, 51*f*
Nummulites, 381
Nurse (army ant), 807
Nurturing behavior, 797
NutraSweet®, 245
Nutrient deficiencies, 451*f*
Nutrition, 317–318
  flowering plant
    plant nutrition and soil, 450–454
    summary, 466–468
    transport mechanisms, 449–454, 458–466
    water and mineral uptake, 454–457
  human health and, 640–644
    carbohydrates, 640
    diet and obesity, 643–644
    lipids, 641–642
    proteins, 642–643
    vitamins and minerals, 643
  origin of life, 318–319

O

Oat seedlings, 472*f*
Obesity, 643
  carbohydrates, 640
  high-fructose corn syrup (HFCS), 641
Obligate anaerobes, 363, 368
  bacteria, 363
  methanogens, 367
Obligate intracellular parasites, 350, 353
Observation, 10
OCA2 gene, 195
Occipital bone, 721, 722*f*
Occipital lobe, 688, 688*f*

ISTUDY

Ocean
biodiversity, 838
currents, 880–881
ecosystems, 879–880
pelagic zone, 879
temperature, 864
water cycle, 852–853, 874
Ochromonas, 378, 378*f*
golden brown algae, 378f
Octet rule, 25
Oenothera, 703*f*
Oil glands, 613
human hair follicle, 584
Oils, 43, 43*t*. *See also* Triglycerides
Okazaki fragments, 209
Okazaki, Reiji, 209
Olfactory bulb, 701–702
Olfactory cell, 701–702
Olfactory organ
vertebrate nervous system organization, 677
Oligocene epoch
mammals of, 327f
Oligochaetes, 519
Oligodendrocyte, 680
Oligodendrocytes, form myelin sheaths, 580
Oligotrophic lake, 875. 875*f*, 876*f*
Oligozoospermia, 757
Omega–3 fatty acid, 641, 644
Omnivore
diet adaptation, 632
ecosystem dynamics, 848, 850f
Oncogenes, 157, 157*f*
One gene, one enzyme hypothesis, 397
One-way ventilation mechanism, 652
Oocytes, 173, 758, 759*f*, 760, 774*f*, 775
Oogenesis, 162, 172, 760
in mammals, 173–174, 174f
Oomycetes, 379, 389
Oparian-Haldane hypothesis, 331
Oparin, Alexander, 314, 316
Oparin-Haldane hypothesis, 314
Open circulatory system, 590
Operant conditioning, 800
Operator, 222
Operculum, 549, 642
Operon, 222
inducible, 222
repressible, 222
Ophiostoma, 400
Ophrys, 493
Ophyrys orchids, 426
Opiate, 685
Opioids
abuse and addiction of, 80
crisis, 80
medications, 714
Opisthokonta supergroup, 386–387, 389
Opossum shrimp, 890
Opportunistic infection, 620, 622, 766
Opportunistic species, 823, 824, 824*f*
Opposable thumb, 557
human, 723
Optic chiasma, 706, 706*f*
Optic lobe, 677
Optimal foraging model, 806
Opuntia, 446*f*
Oral sex and sexually transmitted disease, 769
Oral thrush, 397*f*
Orchid seeds, 457
Order, 8, 335, 335*f*, 346
Ordovician period, 411
Organ, 430
aging processes, 790–791
animal
xenotransplantation, 627, 745
formation
embryonic development, 783–785
stages of development, 777
plants
definition, 430
flowering plant life cycle, 435
flowering plant structure, 434–435
printing, 573
Organelles, 63. *See also* Eukaryotic cells
origin of, 63, 63f
Organic chemistry, 36–52
carbohydrates, 39–43, 40f
disaccharides, 40–41
monosaccharides, 40
polysaccharides, 41–42
definition, 37
lipids, 43–46, 43t
phospholipids, 45, 45*f*
steroids, 45, 46*f*
triglycerides, 43–45, 44*f*
waxes, 45, 46*f*
nucleic acids, 50–53
ATP, 50–53, 53*f*
DNA and RNA structure, 51–52, 51*f*, 52*f*, 51*t*
organic molecules, 37–39
biomolecules of cells, 39, 39*t*
carbon atom, 37–38
carbon skeleton and functional group, 38–39, 38*t*
proteins, 46–50
amino acids, 47, 48*f*
folding, 50
structural organization, 47–50, 49*f*
summary, 54–55
Organic molecules, 37–39
biomolecules of cells, 39, 39t
carbon atom, 37–38
carbon skeleton and functional group, 38–39, 38t
Organic monomers, 313
Organic polymers, 313
Organism, 2, 3*f*
Organization of life, 2, 3*f*
Organ of Corti, 710, 710*f*, 712
Organogenesis, 501
Organs, 2, 3*f*, 580, 581
Organ systems, 2, 3*f*, 580, 581*f*
Orgasm
female, 759
male, 757
Orientation and migratory behavior, 800–801
Origin and history of life
geological factors that influence evolution
continental drift, 328–329
mass extinctions, 329–330
history of life
Cenozoic era, 326–328
fossils tell story, 318–320, 320*f*
Geologic time, 320–323
Mesozoic era, 325–326
Paleozoic era, 323–325
origin of life
evolution of monomers, 314–315
evolution of polymers, 315
evolution of protocells, 315–317
evolution of self-replication system, 318
stages of, 313*f*, 314–318
Origin of life
evolution of monomers, 314–315
evolution of polymers, 315
evolution of protocells, 315–317
evolution of self-replication system, 318
stages of, 313f, 314–318
On the *Origin of Species* (Charles Darwin), 261*f*, 504
Ornament and sexual selection, 807
Oscillatoria, 365*f*
Osmolarity, 669
Osmoregulation
aquatic vertebrates, 665–666
cartilaginous fishes, 665
freshwater bony fishes, 666
kidneys, 665
marine bony fishes, 666
defined, 664
human kidneys, 671–673
terrestrial vertebrates, 666–667
kangaroo rat, 666, 666*f*
marine mammals and sea birds, 667, 667*f*
Osmosis, 85–86, 86*f*
in animal cell, 87f
defined, 669
human urinary system, 669
in plant cell, 87f
Osmotic gradient, 671
Osmotic potential, 460
Osmotic pressure, 86, 454
water potential, 458–461
water uptake, 454–458
Osmotic-regulating system, 521
Ossicle (middle ear), 709
Ossification, 719
Osteoarthritis, 724
Osteoblast, 719
Osteoclast, 719
Osteocyte, 719
Osteogenesis imperfecta, 192
Osteons, 577
Osteoporosis, 719, 742, 791
Ostracoderms, 540
Otolith, 711*f*, 712
Otolithic membrane, 8711*f*, 712
Outer ear, 709
Outgroup (cladistics), 342, 347
Oval window, 709
Ovarian cycle, 760–761, 761*t*
Ovary, 424, 490
functions, 752
human, 734f, 746–747, 758
aging process, 791
gonadotropic hormones, 738, 760–761
ovarian cycle, 760–761, 761*t*
steroid hormones, 736
Overexploitation and extinction, 889, 893–894
Overturn, lake, 875–876, 876*f*
Oviparous animal, 753
Ovoviviparous animal, 753

ISTUDY

Ovulation, 760, 784*f*
Ovules, 419, 490
angiosperm, 422
flowering plant life cycle, 424–426
pine life cycle, 420f
seed plants, 418–419
Owl, Northern Spotted, 869, 895
Oxalis, 485
Oxidation, 107–108
Oxidation-reduction reactions, 107–108
chloroplasts and photosynthesis, 108, 108f, 113
and metabolism, 107–109
mitochondria and cellular respiration, 108–109
Oxygen debt, 728
Oxygen, transport in human respiratory system, 655
Oxyhemoglobin, 655
Oxytocin, 737–738, 737*f*
Oyster, 753
Ozone shield, 320, 331
Ozone shield depletion, 892

P

Pacemaker, 597
Pacinian corpuscle, 713
Pain medication, 713
Pain receptor, 713–714
Pair-rule gene, 780
Pakicetus, 270, 293
Palate, 633–634
Paleontology, 261, 318, 331
Palisade mesophyll, 445
Palmately compound leaf, 446*f*
Palolo worm, 752
Palumbi, Stephen, 867
Pancreas, 633*f*, 635, 637, 637*f*, 734*f*, 744–746
Pancreatic amylase, 637
Pancreatic islet, 744, 745
Pancreatic juice, 635, 637
Pangaea, 328
Pangolin, 867, 867*f*
Panoramic vision, 703
Panspermia, 315
Panther
Florida, 824
K-selection, 823, 824
Panthera onca, 333, 802, 867. *see also* Jaguars
Papilio memnon, 842
Papillae, 700*f*
Papio hamadryas, 806*f*
Pap smear, 768
Papua New Guinea, 355
Parabasalids, 388
Paralysis, 687
Paramecium, 377
competition, 837, 837f, 838
predator-prey interactions, 839
structure of, 377f
Paramylon, 373
Paranthropus genus, 563
Paranthropus robustus, 563
Parasaurolophus walkeri, 326, 326*f*
Parasite and parasitism
bacteria, 843
coevolution, 844
defined, 842
ecology of communities, 842–843
ectoparasite, 843
flatworm
tapeworms, 630
predator-prey interactions, 839
regulation of population size, 822
secondary host, 843
viruses, 842
Parasitic, 368, 400
Parasitic flatworm
tapeworms, 630
Parasitic flatworms, 522*f*
Parasitic kinetoplastids, 373
Parasitic lampreys, 540
Parasitic plants, 457
Parasitic roundworms, 525
Parasitism, 363
Parasympathetic division, 694, 695*f*
Parathyroid, 734*f*, 741–742
Parathyroid hormone (PTH), 741–742
Parazoz, 512–513
Parenchyma cell
ground tissue, 431–432
root cortex, 437
vascular tissue, 432–433
Parent cell, 148
Parietal bone, 721, 722*f*
Parietal lobe, 688, 688*f*
Parkinson disease (PD), 232, 580, 688, 689, 689*f*
treatments for, 688
Parkinson, James, 688
Parsimony, 342, 347
Parthenogenesis, 167, 752
Partial pressure, 655
Particulate theory of inheritance, 182. *See also* Mendelian patterns of inheritance
Parturition, 787*f*
Passive immunity, active *vs.*, 621
Passive transport, 85–88
diffusion, 85, 85f
facilitated transport, 88, 88f
osmosis, 85–88
Pasteurization, 132
Pasteur, Louis, 132, 350, 359, 726
Pasteur's experiments, 359*f*
Patella, 724*f*
Pathogen-associated molecular pattern (PAMP), 610
Pathogens, 363, 369, 578
Patricelli, G. L., 809
Pattern formation, 777, 780, 781*f*
Patterson, David J., 380
Pauling, Linus, 49
Pavlov, Ivan, 799
Pax6 gene, 307–308, 308*f*
P. boisei, 564
PD (Parkinson disease), 688, 689, 689*f*
PD–1 (programmed cell death protein 1), 244
PDE–5 enzyme, 757
Peach-faced lovebird, 795, 795*f*
Peahen, 807
Pearl, Raymond, 819
Pectins, 92
Pectoral girdle, 722–723, 723*f*
Pedigree, 190
autosomal dominant, 191f
autosomal recessive, 190, 190f
X-linked recessive, 198f
Pelagic zone, 879
Pellagra, 106
Pelvic cavity, 723
Pelvic fin, 308
Pelvic girdle, 723, 723*f*
Pelvic inflammatory disease (PID), 770
Penicillin, fungal derivation, 886
Penicillium genus, 397
Penis, human, 754*f*, 754*t*, 756–757, 756*f*
Pentoses, 40
PEP (phosphoenolpyruvate), 123
PEP carboxylase (PEPCase), 123
Peppered moth, 277–280, 277*f*
Pepsin, 635, 639
Peptidase, 639
Peptide, 47
synthesis and degradation of, 48f
Peptide (human digestion), 640
Peptide bond, 47
Peptide hormone, 735, 736*f*
Peptidoglycan, 42, 61, 361, 368
Perception, 699
Perennial plants, 435
Perforin, 620, 621*f*
Pericardium, 594
Pericarp, 498
Pericycle, 437
Periderm, 431
Perilymph, 709–710
Periodic table, 21, 22*f*
groups, 21
periods, 21
Peripheral nervous system, 691–695
autonomic system
defined, 694
parasympathetic division, 694, 695*f*
sympathetic division, 694, 695*f*
defined, 677, 691
somatic system, 692–694, 693t
Peripheral proteins, plasma membrane, 82, 82*f*
Peristalsis, 634, 634*f*
Peritoneum, 635
Peritubular capillary network, 669
Periwinkle, rosy, 886, 886*f*
Permafrost, 866
Permeability of plasma membrane, 84, 84*f*, 84*t*
Peroxisomes, 71, 71*f*
functions of, 71
in liver, 71
in plant cells, 71
Pest control, 887
Petals, 423, 424*f*, 490
Peter, Rosemary, 268
Petiole, 444
Peyer patch, 613, 634
3PG (3-phosphoglycerate), 119
p53 gene, 143, 157, 234, 250
pH
buffer, 673–674
human acid-base balance, 671, 673–674
thermoacidophiles, 367
Phagocyte and natural killer cell
dendritic cell, 615
eosinophils, 615
macrophages
functions, 615
inflammatory response, 615

ISTUDY

natural killer (NK) cells, 615
neutrophils, 615
Phagocytosis, 90, 91*f*
complement, 615
evolution of immune systems, 610
Phalange, 723
Pharyngeal pouches, 272
Pharyngitis, 656
Pharynx
earthworms, 631
fish gills, 649
human
digestion, 633*f*, 634–635
respiration, 650–651
planarians, 630
Phenotype, 185–186
Phenylalanine, 245
Phenylketonuria (PKU), 191, 233
Pheromone
chemical communication, 802–803
chemoreception, 700
humans, 735, 803, 803f
Phloem, 413, 432, 458
acquiring phloem sap, 465f
bark of, 442
conducting cells, organization of, 458f
liquid traveling in, 464
movement from source to sink, 464f
plant transport systems
organic nutrient transport, 463–466
structure, 458
pressure-flow model, 464
procambium, 439
role of, 464
root system, 437
sap
nutrient transport, 463–466
water transport, 461
sink, 461
source, 461
structure, 432, 433, 434
woody plants, 440
Phloem rays, 440
Phloem sap, 458
nutrient transport, 463–466
water transport, 461
Phloem structure, 433, 458
Phloem transports, 464, 465*f*
pressure-flow model of, 465f
tapping system, 466
Phoronids, 516
Phosphate
NPK ratio of fertilizer, 451
phosphorus cycle, 854
Phospholipids, 45, 46*f*
amphipathic nature of, 81–82
in plasma membrane, 81–82, 82f
Phosphorus
cycle, 854, 855f
guano, 881
NPK ratio of fertilizer, 451
Photoautotrophs, 363, 368
bacteria, 363
ecosystem dynamics, 848
Photomorphogenesis, 484
Photoperiodism, 484
and flowering, 485f
Photoreceptors, 479
animals, 699, 702
human eye, 703, 704, 705f
Photorespiration, 122, 123
Photosynthesis, 4, 72, 99, 108, 111–125, 360, 363, 366
and adaptation to environment, 123
CAM photosynthesis, 123
carbon dioxide fixation, 119–122
versus cellular respiration, 139f
C_4 photosynthesis, 122–123
equation for, 113
in flowering plants, 112, 113f
improvement, 111
leaves and, 112, 113f
photosynthetic organisms, 112, 112f, 114
process of, 114–116
Calvin cycle reaction, 115, 115*f*
light reactions, 114–115
$NADP^+$/NADPH, role of, 114, 114*f*
summary of, 115, 115*f*
raw materials for, 112
solar energy in, use of, 116–119, 116f
ATP production, 119
electron flow in light reactions, 116–117, 117*f*, 118*f*
organization of thylakoid membrane, 117, 118*f*
pigments and photosystems, 116, 116*f*
summary, 124
Photosynthetic organisms. *See* Autotrophs
Photosystem, 116
photosystem I (PS I), 116, 117
photosystem II (PS II), 116–117
Phototropin, 479
Phototropism, 474, 472*f*, 479, 479*f*
pH scale, 32, 32*f*
Phylogenetic species concept, 296, 298*f*, 295
Phylogenetic tree, 343
Phylogeny, 334, 340–345, 346
anatomy of speciation, 295
cladistics, 342
interpreting, 340–342
from molecular data, 345f
monophyly, 296
phylogenetic species concept, 296, 298f, 295, 295f
tracing, 342–346
behavioral traits, 344
fossil traits, 343
molecular traits, 344–345
morphological traits, 343–344
traits, classification, 341f
Phylum, 8, 335, 335*f*, 346
Phylum Basidiomycota, 397
Physoderma maydis, 393, 400
Phytobioremediation, 407
Phytochrome, 483
Phytochrome controls, shoot elongation, 485*f*
Phytoplankton
lake ecosystems, 875
ocean ecosystems, 879
Phytoremediation, 456
PID (pelvic inflammatory disease), 770
Pigments
and photosystems, 116, 116f
visual, 704, 705f
Pig, *xenotransplantation*, 627
Pike, Northern, 877*f*
Pine
mutualism, 844
whitebark, 844
Pineal gland, 690, 734, 747
Pine life cycle, 420*f*
Pingelap, 283–284
Pinna, 709
Pinnately compound leaf, 446*f*
Pinocytosis, 90, 91*f*
Pinus longaeva, 419
Pinworms, 526
Pioneer species, 846
Pisum sativum, 182
Pitch (hearing), 710
Pith, 437
root system, 437
stem system, 439–440
Pituitary dwarfism, 739
Pituitary gland and hypothalamus, 737–739
anterior pituitary, 737–739
hormones affecting other glands, 738–739
hormones not affecting other glands, 737–739
hypothalamic-inhibiting hormones, 738
hypothalamic-releasing hormones, 738
major endocrine glands, 734*f*
defined, 737
posterior pituitary, 734f, 737–738, 737f
Pit viper, 698
Pitx1 gene, 308
Pivot joint, 724
Placenta
human, 785–786, 787f
placental mammals, 753
Placental mammals, 553
evolution, 753
life history strategy, 753
Placoderms, 541
Planarian
asexual reproduction, 752
chemical senses, 700
excretory organs, 665
incomplete digestive tract, 630, 630f
light detection, 702
nerve net, 677
respiration, 647
skeleton, 717
Planarian digestive system, 521*f*
Planarian excretory system, 521*f*
Planarian nervous system, 522*f*
Plankton, 371, 387
lake ecosystems, 876
Plantae, 338
Plant cell central vacuole, 71–72, 71*f*
Plant cell walls, 92
Plants, 9. *See also* Flowering plant; *Photosynthesis*
cell wall, expansion of, 473f
cool buildings, 449
commerical uses, 474
defenses, 469
dioecious, 492
evolution and diversity
alternation of generations, 409–411
ancestry and features of, 408
ancestry of, 408
characteristics of, 408
evolutionary history of, 409*f*
land adaptation, 408–409

ISTUDY

Plants—(*Cont.*)
evolution and full adaption to land, 418, 423f
diversification, 426
monocots and eudicots, 423
origination and radiation, 422–424
ovules, 419
growth and movement responses, 477–483
hormones, 470–477, 471, 471t
abscisic acid, 475–476
auxins, 472–474
biotic environment, 477
cellular response, 471
cytokinins, 474–475
ethylene, 476–477
gibberellins, 474
receptors, 470
transduction pathway, 471
minerals, cleaning/discovery of, 456
monoecious, 490
nitrogen, poplar trees cleaning up, 455, 455t
nutrition and transport
acquirement, 457*f*
essential inorganic nutrients, 450–451, 450*t*
essential nutrients, determination of, 451
overview of, 450*f*
soil, 450–454
water and mineral uptake, 454–466
xylem and phloem of, 457*f*
physiologists, 455
phytoremediation, 455
reproduction, 488–502
asexual reproductive strategies, 500–502
fruit types and seed dispersal, 496–500
life cycle overview, 489
seed development, 494–496
sexual reproductive strategies, 489–494
response, 477
roots, growth of, 474
secondary metabolites, 477
smell and talk, 477
structure and organization, 429–449
body, organization of, 434*f*
cells and tissues, 430–433
leaf organization and diversity, 436–439
primary growth of, 434, 435*f*
shoot system, 434
stem organization and diversity, 439–444
survival mechanisms of, 432, 432*f*
terminal bud, 435*f*
transport mechanisms, 458–466
Plants sense, 483
Plant systematists, 408
Plant transport, 458, 459*f*
Plaque, 600
Plasma, 617
Plasma cell, 617
Plasma membrane, 45, 45*f*, 315–317
animal, 81, 81f
animal viruses, 354
components of, 81–82, 81f, 82f
integral proteins, 82
peripheral proteins, 82
phospholipids, 81–82, 82*f*
fluid-mosaic model, 81f, 82
function of, 61
glycoproteins and glycolipids, 82–83
origin of life, 315–317
permeability of, 84, 84f, 84t
carrier proteins, 84
concentration gradient, 84
molecules passage into and out of cell, 84, 84*t*
selectively permeable, 84
in prokaryotes, 61, 62f
prokaryotic cell, 360
proteins and functions, 83
carrier proteins, 83, 83*f*, 85
cell recognition proteins, 83, 83*f*
channel proteins, 83, 83*f*
enzymatic proteins, 83, 83*f*
junction proteins, 83, 83*f*
receptor proteins, 83, 83*f*
protocell, comparison of, 317f
pump, 455
sensory organs, 699
structure and function, 81–85
structure and growth of, 317f
Plasmids, 61–62, 238, 360
prokaryotic cell, 360
Plasmin, 605
Plasminogen, 98
Plasmodesmata, 93, 93*f*
Plasmodial, 389
slime molds, 386f
Plasmodium, 845
Plasmodium vivax, 376, 377
life cycle of, 376f
Plasmolysis, 87
Plasmopara viticola, 379
Plastids, 72
Platelets, 578, 604–605
Plato, 260
Platypus, duckbill, 753
Platyzoans, 516
Pleiotropic effects, 194–195
Pleiotropy, 194
Pleistocene epoch, woolly mammoths of, 327*f*
Plumule, 495
PMS (premenstrual syndrome), 761
P muscle contraction, 729
Pneumatophores, 438
Pneumocystis pneumonia, 622, 658
Pneumonia, 657*f*, 658
HIV opportunistic infection, 622, 658
Poikilothermic, 585
Pointed molar, 632
Point mutations, 233, 234*f*
Poison arrow frog, 824*f*, 841, 842*f*
Poison lollipops, 478, 478*f*
Polar body, 174, 764, 760
Polar covalent bond, 27
Polarity development, 775, 780
Polar spindle fibers, 149
Pollen grains, 419, 492. *See also* Pollination
angiosperms, 421–422
conifers, 419–420
fossil, 422–423
Pollen tube, 419, 420*f*, 421, 424, 492
Pollination, 419, 492, 492*f*
animal, 844, 887
animal, 493
bee, 419, 493, 887
bird, 426, 494
coevolution, 844
coevolution, 494
cross-pollination, 492–494
flower diversity, 424f
self-pollination, 492
wind, 426, 492, 494
Pollinators, 493–494
bird and bat, 494f
coevolution of plants, 494
insect, 493f
Pollution
acid deposition, 891–892
carbon dioxide emissions, 853, 888
defined, 891
extinction, 889
acid deposition, 891–892
eutrophication, 892
organic chemicals, 892
ozone depletion, 892
waste recycling, 888
water, 888
Polyandry, 806–807
Polychaetes, 519, 520
Polydactyly, 193
Polygamy, 806
Polygenic inheritance, 195–196, 196*f*
multifactorial traits, 195
skin color, 194, 195f
Polygenic trait, 284
Polymerase chain reaction (PCR), 239, 240*f*
Polymers, 39
Polymers, evolution of, 315
Polyp, 514
colon, 637
Hydra asexual reproduction, 752f
Polypeptide, 47
Polyploidy, 303
Polyribosomes, 67, 216*f*, 217, 218
Polysaccharides, 41–43, 41*f*
glycogen, 41f, 42
starch, 40, 41f
as storage molecules, 39, 4140, 41
structural, 42–43
cellulose, 42, 42*f*
chitin, 42–43
peptidoglycan, 42
Polyspermy, 774
Polytrichum, 413*f*
Poly-X females, 177
Pons, 690
Poplars take up excess nitrates, 456*f*, 456
Population, 2, 3*f*
defined, 277, 814
Population density, 815
Population distribution, 815
Population ecology, 812–834
demographics of populations, 815–818
density and distribution, 815
population growth, 816–817
human population growth, 826–828
age distributions, 827, 827*f*
doubling time, 826
more-developed *vs.* less-developed countries, 826*f*, 827
life history patterns, 823–826
K-selected populations, 823–824
r-selected populations, 823
population growth models, 818–821
exponential growth, 819

ISTUDY

iteroparity, 818
logistic growth, 819–821
semelparity, 818
regulation of population size, 821–823
density-dependent factors, 822
density-independent factors, 821–822
predator-prey interactions, 822, 823*f*, 839
scope of ecology, 814, 814f
Population evolution, 276–292
genes, populations, and evolution, 277–285
migration, 280–281
mutation, 280
natural selection, 284
nonrandom mating, 284
peppered moth, microevolution in, 277–280
small population size, 281–282
maintenance of diversity, 288–291
heterozygote advantage, 290
natural selection, 289–290
natural selection, 284–288
sexual selection, 286–288
types of, 285–286, 286*f*
Population growth, 816–817
age distribution, 817, 827, 827f
biotic potential, 816
human, 826–828
age distributions, 827, 827*f*
more-developed *vs.* less-developed countries, 826*f*, 827
models, 818, 821
exponential growth, 819
iteroparity, 818
logistic growth, 819–821
semelparity, 818
mortality patterns, 816–817
Porcupine, 842
Porphyra blades, 381
Porphyria, 194
Portal system, 599
Positive feedback, 737
Positron-emission tomography (PET), 22
Posterior/anterior axis, 780
Posterior pituitary, 734*f*, 737–738, 737*f*
Postganglionic fiber, 694
Postreproductive age group, 817
Posttranscriptional control, of gene expression, 225, 230, 231
Posttranslational control, 225, 231
Postzygotic isolating mechanism, 298*f*, 299–300
Potassium
NPK ratio of fertilizer, 451
opening and closing of stomata, 452
Potassium-argon (K-Ar) dating, 319
Potassium permanganate ($KMnO_4$), 477
Potato, 500
Potential energy, 99
Potential energy
water potential, 458–461
Potts, D. M., 199
Power stroke, 729, 730*f*
p53 protein, 145, 146, 157
Prairie, 873*f*
Praying mantis, 632
Precambrian, prokaryote fossils of, 322*f*
Precipitation
biome
coniferous forests, 868–869
desert, 874
distribution, 865, 865*f*
grassland, 872–874
shrubland, 870–872
temperate deciduous forests, 869
tropical rain forests, 869–870
tundra, 865–868
lake effect, 864
monsoon, 864
rain shadow, 864
water cycle, 875
Predation
defined, 822, 839
predator-prey interactions, 839–842
parasitism, 839
population dynamics, 822, 822*f*, 839–841
prey defenses, 841–842
Prediction, 11
Pre-ejaculate, 756
Preganglionic fiber, 694
Pregnancy
development, 782–787
due date, 789
early, 774–777
extraembryonic membranes, 782–7832
fetal, 786–787
placenta structure and function, 785–786
stages of birth, 786
trimester, 782
ectopic, 758
fertilization, 758, 760, 761–762, 764, 774–775, 783
HIV/AIDS, 768
implantation, 783
placenta structure and function, 785–786
seahorses, 753
tests, 618
toxoplasmosis, 786
Premenstrual syndrome (PMS), 761
Pre-mRNA, 230
Preparatory (prep) reaction, 134, 134*f*
Prereproductive age group, 817
Presbyopia, 791
Pressure, partial, 655
Pressure potential, 460
Preven, 763
Prezygotic isolating mechanisms, 297–299
behavioral isolation, 298f, 299
gamete isolation, 299
habitat isolation, 297, 298
mechanical isolation, 299
reinforcement, 302
temporal isolation, 297–299
Prickly-pear cactus, 839
Primary follicle, 759–760
Primary growth
plants, 434, 435f
woody plants, 440–443
Primary host, 843
Primary lymphoid organ
defined, 615
red bone marrow, 614f, 615, 719
thymus, 614f, 615, 734f, 747–748
Primary motor area, 688, 688*f*
Primary olfactory area, 701*f*
Primary oocyte, 759*f*, 760
Primary organizer, 779
Primary somatosensory area, 688, 688*f*, 713
Primary spermatocyte, 756
Primary succession, 846, 846*f*
Primary tissue (woody plants), 440
Primary visual area, 688, 688*f*
Primate diversity, 557*f*
Primate fossils, 561
Primate hand, evolution of, 558*f*
Primate order, monogamy, 806–807
Primates (L. *primus*), 341, 557
evolution of, 559f
Primitive streak, 776
Primordial soup hypothesis314, 331
Primroses, 196
Principle, 12
Principles of Geology (Charles Lyell), 262
Prions, 50, 355, 368
PRL (prolactin), 739
Problem solving, 801
Procambium, 439
Proconsul, 561
Producer, 848
Production, 746–747
Product rule of probability, 187
Products, 101, 103
Profundal zone, 877
Progesterone, 746–747, 758, 761–762
Proglottids, 523
Progymnosperms, 419
Prokaryotes, 320, 359–361, 360, 368
reproduction, 361
structure of, 360–361
Prokaryotic cell division, 158–159
asexual reproduction, 158
binary fission, 158, 158f
and eukaryotic division, 158–159, 159t
prokaryotic chromosome, 158
Prokaryotic cells, 8, 61
definition of, 61
DNA replication in, 209f, 210
features of, 360f
regulation of gene expression in, 222–224
size and shape, 61
structure of, 61–62, 62f
cell envelope, 61
cytoplasm, 61–62
external structures, 62
Prokaryotic fossils, 360
Prolactin (PRL), 739
Proliferative phase, 760*f*, 761
Prometaphase, 149, 150*f*
Promoter, 212, 213*f*, 222
Promoting growth, 472
Pronucleus, 774
Propagation of action potentials, 682
Prophage, 353
Prophase, 149, 150*f*
Proprioceptor, 713, 713*f*
Prosimians, 561
Prostaglandin, 748, 756
Prostate cancer, 756
Prostate gland, 754*f*, 754*t*, 756, 790
Proteases, 232
Proteasomes, 232
Protective protein, 615
complement, 615
interferons, 615
Protein
origin of life, 318
protective proteins, 615

ISTUDY

Protein (dietary)
human diet and nutrition, 642–643
human digestion, 642
Proteinaceous infectious, 355
Protein-first hypothesis, 315, 331
Protein kinases, 74
Proteinoids, 315, 331
Protein phosphatases, 74
Proteins, 46–50, 672
amino acids, 46–47, 47f
amino acid sequence in, 47
denatured, 48
as energy source, 138
folding, 48
function of, 46
importance of, 46
membrane, 82, 83
structural organization, 48
primary structure, 48, 49*f*
quaternary structure, 48, 49*f*
secondary structure, 48, 49*f*
tertiary structure, 48, 459*f*
synthesis, 361
Proteoglycans, 92
Proteome, 253
Proteomics, 217–218, 253–254
Proterospongia, 386
Protista, 338
Protist evolution and diversity
Amoebozoa, Opisthokonta, and Rhizaria supergroups, 371t
Archaeplastida supergroup, 381–385
excavata supergroup, 371–375
diplomonad, 372–373
euglenozoans, 373–375
kinetoplastids, 374
parabasalids, 373
protists, evolution, 370–390
overview of, 371
Excavata supergroup, 371–375
Protists, 9, 387, 391
characteristics of, 371
evolution and diversity of, 371
Protobiont, 315
Protocells (protobiont), 313, 315, 331
evolution of, 315–318
Protoderm, 439
Proton pump, 455
Protons, 21, 21*f*
Proto-oncogenes, 156, 234
mutation in, 156, 158f
Protoplasts, 246
Protostomes, 511
Proxima b, 19
Proximal convoluted tubule, 668, 669
Pseudocoelom, 525
Pseudogymnoascus destructans, 400
Pseudomyrmex ferruginea, 844
Pseudopods, 380, 389
Pseudostratified epithelia, 575
Psilocybe mexicana, 398
Psilotum, 416, 417*f*
Psilotum species, 416, 417*f*
Pteridophytes, 415
evolution of
ferns, 417–418
horsetails, 415–416
megaphylls, 415–418
whisk ferns, 416–417
PTH (parathyroid hormone), 741–742
Puberty, 757
Pubic hair, 746
Pubic symphysis, 724
Puffballs, 399*f*
spores, 397
Pulinella chromatophora, 380*f*
Pulmonary circuits, 593, 598*f*
Pulmonary fibrosis, 657*f*, 659
Pulmonary tuberculosis, 657*f*, 658
Pulse, 597
Pulvinus, 481
Punctuated equilibrium model of macro-evolution, 307, 307*f*
Punnett, Reginald, 183
Punnett square, 183, 187–188, 188*f*
Pupil, 703
Purkinje fibers, 597
Pus, 703
P wave, 598
Pyramid, ecological, 851, 851*f*
Pyrenoid, 373, 381
Pyruvate, 128–129, 137

Q

QRS complex, 598
Quadriceps tendon, 724*f*
Queen substance, 794
Queen Victoria of England, 199
Queen Victoria's Gene (D. M. Potts), 199
Quercus suber, 431
Quetzalcoatlus, 547
Quillwort, 414

R

Rabbit, 849*f*
Radial symmetry, 510
Radioactive decay, 24
Radioactive isotopes, 21–22
high levels of radiation, 22, 22, 22f
low levels of radiation, 22, 22f
Radioactivity, 22
Radiocarbon dating, 319, 331
Radiolarians, 380, 380*f*, 389
Radioles, 520
Radiometric techniques, 329
Radius, 723
Radula, 517, 631
Raggiana Bird of Paradise, 804
Rain forest
species richness, 836
temperate deciduous forest, 869–870
tropical rain forest
biome, 866*f*, 870–872
Rain shadow, 864, 864*f*
Random distribution, 815, 815*f*
Rangifer, 821*f*
RAS (reticular activating system), 690, 690*f*
Ras proteins, 234
Rat kangaroo, 666, 666*f*
Rate of natural increase
calculation, 816
r-selected population, 823
Raven, 801
Ray-finned fishes, 542, 542*f*
bony fishes, 541
RB protein, 145, 148, 155
Reactants, 101
Realized niche, 837
Realizing Increased Photosynthetic Efficiency (RIPE), 111
Receptor
antigen, 610, 616
B-cell receptor (BCR), 617–619
chemoreceptors, 699, 700, 701
cutaneous receptors, 713
pain receptors, 713–714
sensory receptors, 699–700
target cells, 735, 735f
Receptor-mediated endocytosis, 90–91, 91*f*
Receptor proteins, 83, 83*f*
Receptors, plasma membrane, 470
Recessive allele, 185
Reciprocal altruism, 808–809
Recombinant DNA (rDNA), 238–239, 238*f*, 239*f*
Rectum, 633*f*, 637
Red algae, 381, 381*f*, 389. *See also* Archaeplastida supergroup
agar and agarose, 383f
in Asian cuisine, 382f
in foods, 382
in laboratory research, 383
for skin care, 383
Red blood cells (RBCs), 577, 603
Red deer, 287, 288*f*, 810
Red kangaroo, 815
Redox reaction, 107
Red pulp, 613
Red seaweed, 382
Red tides, 375, 388
Reduction, 107–107
Red wine effect, 601
Reemerging diseases, 15
Reflex action, 687, 693
Reflex arc, 693–694, 693*f*
Refractory period, 682, 757
Regeneration sponges, 752
Regenerative medicine, 522
Regulated secretion, 90
Regulation of population size, 821–823
density-dependent factors, 822, 822f
density-independent factors, 821–822, 822f
Regulator gene, 222
Reindeer, 821, 821*f*
Reinforcement of reproductive isolation, 301–303
Relative dating, 319, 331
Remora, 843
Renal cortex, 668, 668*f*, 669*f*
Renal medulla, 668, 668*f*, 669*f*
Renal pelvis, 668
Renewable resources, 828
Renin, 673, 673*f*, 743
Repetitive DNA elements, 251
interspersed repeats, 251
tandem repeats, 251
Replacement model of human evolution, 567, 567*f*, 569
Replacement reproduction, 826–827
Replication fork, 209*f*, 210
Repolarization, 682
Repressible operon, 222
Repressor, 222

ISTUDY

Reproduction, 6. *See also* Asexual reproduction; Reproductive system; Sexual reproduction
Reproductive age group, 817
Reproductive cloning, 152, 153, 153*f*
Reproductive isolating mechanisms, 297–300
  defined, 297
  postzygotic isolating mechanisms, 298f, 299–300
  prezygotic isolating mechanisms, 297–299
    behavioral isolation, 298*f*, 299
    gamete isolation, 299
    habitat isolation, 298*f*
    mechanical isolation, 299
    temporal isolation, 297–299
  reinforcement, 301–303
Reproductive strategies and fitness, 806–807
  monogamy, 807
  polyandry, 806–807
  polygamy, 806
  sexual selection, 807
Reproductive system, 751–772. *See also* Asexual reproduction; Sexual reproduction
  animals, 752–754
    asexual reproduction, 752
    life history strategies, 755
    sexual reproduction, 752–755
  in animal viruses, 354
  bacteria, 360–361
  eukaryote, 361
    conjugation, 361
    transduction, 361
    transformation, 361
  flowering plant, 488–503
    asexual reproductive strategies, 500–502
    fruit types and seed dispersal, 498–500
    life cycle overview, 489
    seed development, 494–495
    sexual reproductive strategies, 489–494
  human
    aging process, 791
    control of reproduction, 755, 762–764
    female reproductive system, 758–762
    male reproductive system, 754–757
    reproductive technology, 764
    sexually transmitted diseases, 768–771
  iteroparity, 818
  of retrovirus HIV, 355f
  viruses, viroids, and prions, 354
    animal viruses, 354–355
    bacteriophages, 353–354
    emerging viruses, 355–356
    retroviruses, 355
Reproductive technology, 764
  artificial insemination by donor (AID), 764
  gamete intrafallopian transfer (GIFT), 764
  intracytoplasmic sperm injection (ICSI), 764
  in vitro fertilization (IVF), 764
Reptiles
  characteristics, amniotic egg, 782
  ectothermy, 547
  efficient breathing, 547
  efficient circulation, 547
  efficient excretion, 547
  hormones and endocrine system, 735
  paired limbs, 547
  reproduction and life history strategy, 751
  respiration, 652
  skin, 547
  skin color, 739
  well-adapted reproduction, 547
Reptilian anatomy, 547*f*
Reservoir (chemical cycling), 852, 854
Resistance, 269
Resource and population, 815
Resource partitioning, 838
Respiration
  birds, 652, 653f
  cellular, 647
  external respiration, 647, 647f, 654–655, 654f
  internal respiration, 647, 647f, 654, 654f, 655–656
  oxygen debt, 728
  reptiles, 652
  steps, 647
    external respiration, 647, 647*f*, 654–655, 654*f*
    internal respiration, 647, 647*f*, 654, 654*f*, 655–656
    ventilation, 647
  ventilation, 647
Respiratory center, 652–654
Respiratory organs, variety of, 527
Respiratory pigment
  hemocyanin, 528
  hemolymph lacks, 529
Respiratory syncytial virus (RSV),618
Respiratory system, 646–662
  aging process, 790
  breathing and transport of gases, 652–656
    breathing, 652–654
    gas exchange and transport, 654–656
  disorders, 656–659
    lower, 656–659
    upper, 656
  gas-exchange surfaces, 647–651
    gills of fish, 648–649
    human lungs, 650–651
    overview, 647–648
    respiration steps, 647
    trachael system of insects, 649–650
  respiration and human health, 656–659
    lower respiratory tract disorders, 656–659
    upper respiratory tract disorders, 656
  respiration steps, 647
    external respiration, 647, 647*f*, 654–655, 654*f*
    internal respiration, 647, 647*f*, 654, 654*f*, 655–656
    ventilation, 647
Respiratory tract disorder, 656–659
  lower, 656–659
    lungs, 657–659
    trachea and bronchi, 656–657
  upper, 656
    common cold, 656
    strep throat, 656
Responding variable, 11
Resting potential, 681*f*, 682
Restoration ecology, 895
Restriction enzyme, 238–239, 239*f*
Reticular activating system (RAS), 690, 690*f*
Reticular fibers, 575
Retina
  artificial, 706, 707, 707f
  diseases, 706
  eye anatomy, 703, 704
  focusing eye, 704, 706–708
  integration of visual signals, 705–706
  from retina to visual cortex, 706
  structure and function, 705f
Retinal, 704, 705*f*
Retinal detachment, 706
Retinoblastoma protein (RB), 234
Retroviruses, 355, 368
  HIV, reproduction, 355f
Reverse transcriptase, 355, 368
Rheumatic fever, 626
Rheumatoid arthritis, 626, 626*f*, 724
Rhinoceros, biotic potential, 816*f*
Rhinodermadarwinii, 824*f*
Rhizaria supergroup, 380–381, 389
  cercozoa, 380
  foraminiferans, 380, 380f
  radiolarians, 380–381, 380f
Rhizobia, 455
Rhizoids, 394, 411, 412
  liverworts, 411–412
  mosses, 412
Rhizomes, 414, 443, 500
Rhizopus stolonifer, 394, 395*f*
Rhodopsin, 704, 705*f*
Rh system, 604
Rhyniophyte, 414
Rib, 722
Rib cage, 722, 722*f*
Riboflavin deficiency, 105
Ribonucleic acid (RNA). *See* RNA (ribonucleic acid)
Ribose, 40, 51, 316
Ribosomal RNA (rRNA), 51, 67, 210, 338, 516*f*
  nucleotide sequences of, 338
  role in translation, 216–217, 216f
Ribosomes, 6668
  binding to endoplasmic reticulum, 66
  in eukaryotic cells, 67
  function of, 216–217, 218f
  number of, 67
  prokaryotic, 62
  protein synthesis in, 66, 66
  structure of, 216, 216f
Ribozyme, 103, 23
Rice (*Oryza sativa*), 480, 886–887
Rigor mortis, 729
Ringworm tinea, 397*f*
River ecosystem, 878
RNA (ribonucleic acid),51, 210, 772. *See also* Messenger RNA (mRNA); Ribosomal RNA (rRNA); Transcription; Transfer RNA (tRNA)
  classes of, 210
  messenger RNA (mRNA), steroid hormones, 736
  nucleotides in, 210
  in nucleus, 67
  origin of life, 322
  polymerase, 398
  possible extraterrestrial components of, 316
  structure of, 51–52, 51f, 52f, 52t, 210, 211f
    and DNA structure, 210*t*
RNA-first hypothesis, 315, 331
RNA interference (RNAi), 231, 250
RNA polymerase, 212, 213*f*, 216, 222
RNase, 204
RNA splicing, 213
Roadrunner, greater, 873*f*

ISTUDY

Rock
ecological succession, 844
lichen, 844
phosphorus cycle, 854
Rocky shore ecosystem, 877–878, 878*f*
Rod cell, human eye, 704, 705–706
Rodhocetus, 270
Rohypnol, 684
Romalea, 531*f*
Roofie, 684
Root (phylogeny), 295
Root (plant)
Casparian strip, 442
mycorrhizae, 438
organization/diversity of, 436–438
diversity, 435–438
eudicot root, cells/tissues of, 436–435
monocot roots, organization of, 435
symbiotic relationships, 438
vascular cylinder, 432
Root adaptations, 438
Root apical meristem, 435
Root cap, 436
Root diversity, 435–438
Root hair plexus, 713
Root hairs, 430
function, 436
plant epidermal cells, 430–431
zone of maturation, 435
Root nodules, 438, 455, 456*f*
nitrogen fixation, 455
nitrogen fixation, 438
Root pressure, 461
guttation, 461
Root specializations, 438*f*
Root system, 434
Rosy periwinkle, 886, 886*f*
Rotational equilibrium, 711–712
Rotifers, 523524, 524*f*
Round window, 709
Roundworm diversity, 525*f*
Roundworms, 525
hydrostatic skeleton, 717
Royal disease, 199. *See also* Hemophilia
rRNA sequencing, 510
R-selection, 823
RSV (respiratory syncytial virus), 617
Rubber tree, 887
Rubella, 788
RuBP (ribulose-1,5-bisphosphate), 119, 122
RuBP carboxylase, 119, 122
Ruffini ending, 713
Rumen, 632
Runaway hypothesis, 286

S

Saccharomyces cerevisiae, 132, 397
Saccule, 712
Sac fungi, 395, 401, 403, 404
asexual reproduction, 396f
Sacral vertebra, 722
Sacrum, 722
Safrole, 232
Sago palm, 420
Sahelanthropus tchadensis, 562, 562*f*
Salamander, speciation, 300–301
Salicylic acid, 478
Saliva, 634
Salivary amylase, 634, 638
Salivary gland, 633*f*
Salix, 437*f*
Salmon
community diversity, 885
Kokanee, 885f
organic chemical extinction threat, 892
Sockeye, 301, 301f
speciation, 300f
Salmonella, 364
Saltatory conduction, 682
Salt gland, 667
Salt, water-salt balance, human, 671–673, 742
Salty taste receptor, 701
Sandworms, 878
Sandy shore ecosystem, 878
Saprolegnia, 379
Saprotrophs, 363, 368, 391
Sarcolemma, 726, 727*f*, 728
Sarcomere, 727*f*, 728
Sarcoplasmic reticulum, 727*f*, 728
SARS-CoV-2 virus, 15, 349
in-coronavirus genus and family Coronaviridae, 350
emerging infectious diseases, 356, 358–359, 358f
lifecycle of, 358, 358f
respiratory system, 658, 658f
Satellite cell, 680
Satin bowerbird, 809
Saturated fat, 641, 644
Saturated fatty acid, 43, 44*f*, 51
Savanna, 873–874, 873*f*
Scanning electron microscope (SEM), 59, 59*f*
Scapula, 722–723, 723*f*
Scarlet kingsnake, 842
Schistosoma, 523*f*
Schistosomiasis, 522
Schleiden, Matthias, 57
Schwann cells, 580, 680
Schwann, Theodor, 57
SCID (severe combined immunodeficiency), 249, 625
Science, 1
Scientific method, 10–14, 11*f*
Scientific name, 10
Scientific publications, 12
Scientific theory, 12
Sclereids, 432
Sclerenchyma cell, 432
Sclerenchyma fibers, 433
Sclerosis, 690
Scolex, 523
Scorpions, 531
Scrotum, 754, 754*f*
Scrub jay, Florida, 808
Sea anemone
mutualism, 844
nerve net, 677
Seabird
osmoregulation, 667
territoriality, 805
Seagrass beds, 878
Seahorse, 753
Sea jellies, 515
Seasonal flu, spikes of, 352*f*
Sea squirts, 538, 538*f*
Sea stars, 533
carnivorous diet, 632
Sea turtle
Loggerhead, 867
Seaweed
brown algae, 378
intertidal zone, 878
multicellular green algae, 381, 381f
red, 382
Sebaceous glands, 581, 584
Secondary follicle, 759*f*, 760
Secondary growth, 430
herbaceous stems, 442f
vascular cambium, 430
woody plants, 440
Secondary host, 843
Secondary immunodeficiency disorder, 625
Secondary lymphoid organ, 612–613
lymph nodes, 612f, 612
Peyer patches, 613, 634
spleen, 611f, 612
tonsils, 611f, 613
Secondary oocyte, 174, 759*f*, 760, 775, 784*f*
Secondary sex characteristic, 746, 747, 762
Secondary spermatocyte, 756
Secondary succession, 846, 87*f*
Second messenger, 95, 95*f*, 735
Secretory phase, 760*f*, 761
Sedgewick, Adam, 263
Sediment, 318, 331
Sedimentary biogeochemical cycle, 852
Sedimentation, 318, 331
Seed, 489
development
eudicot, stages of, 494–496
globular stage, 496
heart stage and torpedo stage embryos, 496
mature embryo, 496, 495*f*
monocot *vs.* eudicot seeds, 496, 496*f*
zygote and proembryo stages, 494
dispersal, 498
dormancy, 498
flowering plant life cycle, 489
germination, 498–499
Seed coat, 496, 495*f*
Seed dispersal, 496–499
methods of, 499f
Seed dormancy, 432
Seed ferns, 416
Seed germination, 498–499
Seed leaves, 423
Seedless vascular plant, 414
pteridophytes, 415
ferns, 417–418
horsetails, 415–416
megaphylls, 415–418
whisk ferns, 416–417
Seed plants, 419
evolution of
adaptation to land, 418–426
angiosperms, 421–422
angiosperms, origin/radiation of, 422–424
diversification, 426
flowering plant life cycle, 424–426
flowers, 426

ISTUDY

gymnosperms, 418
conifers, 419–420
cycads, 420–421
ginkgoes, 421
gnetophytes, 421
Segmentation, 518, 526
arthropods, 780–782
Segment-polarity gene, 780
Selaginella, 414
Selectively permeable, plasma membrane, 84
Selector genes, 781
Self-interest *vs.* altruism, 807–808
Self-splicing, 213
Semantic memory, 691
Semelparity, 818
Semen, 756
Semicircular canal, 709
Semiconservative replication, 208, 208*f*. *See also* DNA (deoxyribonucleic acid)
Semilunar valves, 759
Seminal fluid, 756
Seminal vesicle, 754*f*, 754*t*, 756
Seminiferous tubule, 755–756, 755*f*
Senescence, 475
Sense organs, 698–715
aging process, 791
chemical senses, 700–702
crustaceans, 700
fish, 700
insects, 700
planarians, 700
sense of smell in humans, 701–702
sense of taste in humans, 700–701, 700*f*, 702
snakes, 700
hearing and equilibrium, 708–712
disorders, 712
human ear, 709–710
sense of equilibrium, 710–712
sound wave detection, 708–709
sensory receptors, 699–700
chemoreceptors, 699, 700, 701
mechanoreceptors, 699, 709
photoreceptors, 699, 702, 703, 704
thermoreceptors, 698, 699
somatic senses, 713–714
cutaneous receptors, 713
pain receptors, 713–714
proprioceptors, 713
vision, 702–708
animal detection of light, 702–703
human eye, 703–704
vomeronasal organ (VNO), 803
Sensitive period, 799
Sensory input, 686
Sensory neuron, 680–681, 680*f*
Sensory receptors, 584, 698, 699–700
exteroceptors
chemoreceptors, 699, 700, 701
electromagnetic receptors, 698
mechanoreceptors, 699, 709
photoreceptors, 699, 702, 703, 704
thermoreceptors, 698, 699
interoceptors, 699
reflex action, 693
Sensory transduction, 699
Sentinel cell, 610
Sepals, 423, 490
Sepioteuthis lessoniana, 519*f*
Septa, 392
Septate fungus, 392*f*, 392
Septum, 594
nasal, 721
Sequential hermaphroditism, 752
Sequoia sempervirens, 419
Serosa, 634–635
Serotonin, 682
Sertoli cell, 756
Setae, 519
Severe acute respiratory syndrome (SARS), 15
Severe combined immunodeficiency (SCID), 625
Sex chromosome number, changes in, 176–177
Jacobs syndrome, 177
Klinefelter syndrome, 177, 177f
poly-X females, 177
Turner syndrome, 176, 176f
Sex hormone, 746–747, 757, 758
discovery of, 165
Sex reversal, 752
Sex selection, 764
Sexual development
pineal gland, 747
testes and ovaries, 746–747, 752, 754–755, 758
Sexual dimorphism, 286
Sexually transmitted disease (STD), 276, 766–771
bacterial, 769–770
chlamydia, 769
gonorrhea, 770
syphilis, 770
trichomoniasis, 771
viral, 766–771
genital herpes, 769*f*, 769, 788
hepatitis, 767, 769
HIV/AIDS, 766–768
human papillomavirus (HPV), 768–769
Sexual mimicry, 426
Sexual reproduction, 162, 163, 378
animal, 752–753
conjugation, 377f, 378
defined, 752
human
aging process, 791
control of reproduction, 754, 762–764
female reproductive system, 758–762, 791
male reproductive system, 754–757, 791
sexually transmitted diseases, 766–771
invertebrates
cnidarians, 752
sponges, 752
meiosis
parthenogenesis, 752
spermatogenesis and oogenesis, 755*f*, 756, 760
parthenogenesis, 752
protists, 387
apicomplexans, 376–377
characteristics of, 731
ciliates, 377–378
diatoms, 378
dinoflagellates, 375
euglenids, 373–374
evolution and diversity of, 371
green algae, 381–385
water molds, 379, 379*f*, 380
in sac fungi, 396f
Spirogyra, 383
trichomoniasis, 373
Sexual reproduction, in flowering plants, 489*f*
dioecious plants, 490f
flowers, 489–492
anatomy of, 489*f*
structure, 490
variations, 490–492
life cycle, 489, 491f, 492–494
cross-pollination, 492–494, 492*f*
female gametophyte, development of, 492
male gametophyte, development of, 492
sporophyte, development of, 492
monocot vs. eudicot flowers, 490f
monoecious plants, 490f
Sexual selection, 286–288, 807
Shamrock, 485
Shape
animal development, 308
bacteria, 362, 362f
viruses, 350
Shelf fungus, 399*f*
Shiga toxin, 364
Shigella dysenteriae, 364
Shingles, 622
Shoot apical meristem, 435, 439, 439*f*
Shoot system, 434
Shoot tip, and primary meristems, 439*f*
Short tandem repeats (STRs), 251
sequences, 241
Short-term memory, 691
Short-term stress response, 742*f*
Shotgun sequencing method, 255, 255*f*
Shrimp, 665
Shrub
chaparral, 871
desert ecosystems, 874
ecological succession, 846
shrubland ecosystem, 870–872
Shrubland ecosystem, 870–872, 871*f*
Sialia mexicana, 822
Siamang gibbon, 806*f*
Sickle-cell disease, 194–195, 233, 244, 290, 290*f*
Sieve plate, 533
Sieve tube, 458, 464, 465
Sieve-tube member, 433, 458
Signal, 145
Signaling cascade, 735
Signal peptide, 67
Signal recognition particle (SRP), 67
Signal transduction, 470
pathway, 471
in plants, 470f
Signal transduction pathway, 94, 95*f*
Sign stimulus, 798
Silversword alliance, 306
Single-celled organisms, 57
Single nucleotide polymorphisms (SNPs), 250
Single-stranded binding proteins (SSB), 208, 209*f*
Sink (phloem), 464, 464*f*
cancer, 582, 583, 583f
layers of, 582f
Sink population, 895
Sinornis, 269
Sinus, 721
Siphon, 631
SIRT6 gene, 773
Sister chromatids, 144, 145*f*, 148, 149*f*, 163–164, 169

ISTUDY

Skeletal muscle, 578, 724–730
anatomy, 725–726
macroscopic anatomy and physiology, 725–726
microscopic anatomy and physiology, 726–729
ATP use in contraction, 728, 729
muscle innervation, 728
role of calcium in muscle contraction, 729
sliding filament model, 728
Skeleton
carbon, 38
diversity, 717–718
endoskeletons, 717, 718
exoskeletons, 717–718
hydrostatic skeleton, 717
endoskeleton, 718
advantages, 718
composition, 717, 718
echinoderms, 718
vertebrate characteristic, 718
exoskeleton, 717–718
arthropods, 717
mollusks, 717
human, 718–724, 721f
aging process, 791
appendicular skeleton, 722–723
axial skeleton, 721–722
bone growth and renewal, 719
joint classification, 724
long bone anatomy, 719
hydrostatic skeleton
annelids, 717
vertebrate characteristic, 718
Skill memory, 691
Skin
amphibians, 739
cutaneous receptor, 713
fish, 739
human
aging process, 790
hair follicles, 790
oil glands, 613
sweat glands, 790
wrinkles, 726
innate immune defense, 613
reptiles, 739
Skin color, 195
melanocyte-stimulating hormone (MSH), 739
polygenic basis of, 195, 196f
Skinner, B. F., 800
Skull, 539
human, 721, 725f
S-layer, 360
Sleep
melatonin, 690, 747
Sleeping sickness
areas affected by, 374f
Sleep movements, 481
Sliding filament model, 728
Slime molds, 389
amoeboid, 386
cellular slime molds, 386
immune system evolution, 610
plasmodial slime molds, 385f
Slug (social amoeba), 610
Small-interfering RNAs (siRNAs), 225, 231
Small intestine
anatomy, 633f, 635–636, 636f
lumen, 634
Small nuclear RNAs (snRNAs), 213, 230
Small RNA (sRNA) molecules, and gene expression regulation, 230–231, 231*f*
Smell
anosmia, 791
human sense of smell, 701–702, 791
Smoking
birth defects, 788
cigarettes, 660, 684
nicotine abuse, 684
"vaping" vs., 660
Smooth muscle, 578
Snail
coevolution, 845
egg-laying, 796–797
land, 632
marine, 796–797
omnivorous diet, 632
Snakes, 548
chemical senses, 699f, 700
coloration, 841
food choice in garter snakes, 796
garter, 796
infrared sensory organ, 698, 699–700
Snowshoe hare, 840–841
Social amoeba, 610
Social behavior
animal communication, 801–804
auditory communication, 803
chemical communication, 802–803
tactile communication, 804
visual communication, 804
learning and social interactions, 799
Social interactions and learning, 799
Social parasite, 845
Society
defined, 801
fitness and societies, 807
Sockeye salmon, 301, 301*f*
Sodium bicarbonate, 637
Sodium chloride, formation of, 26–27, 26*f*
Sodium hydroxide (NaOH), 32
Sodium-potassium pump, 89, 89*f*, 682
Soft palate, 634
Soil
erosion, 453–454, 888
coastal erosion, 454
healthy soil, components of, 453f
loam, 452
nutritional function of, 452
permafrost, 866
tropical rain forest, 869–870
Soil-dwelling ants, 452
Soil formation, 452
Soil horizons, 453
Soil particles, 452
Soil profiles, 453
simplified, 453f
Solar energy, 99, 108. *See also* Photosynthesis
in light reaction, 114–115
Solar radiation effect on climate,–864
Solar system, 19
Sollfors, Stefan, 337
Soluble fiber, 641
Solute, 85
Solute concentration, 450, 460
Solutes, 29
Solution, 29, 85
Solvent, 85
Somatic cells, 145
Somatic mutations, 232
Somatic sense organ, 713–714
cutaneous receptors, 713
pain receptors, 713–714
proprioceptors, 713
Somatic system, 692–694, 693*t*
pathways comparison, 693t
somatic sense organ, 713–714
cutaneous receptors, 713
pain receptors, 713–714
proprioceptors, 713
Somatosensory area, primary, 688, 688*f*, 713
Somite, 777
Songbird
auditory communication, 803
social interactions and learning, 799
Sorus, 417
Sound wave detection, 708–709
Source, 461
Source population, 895
Sour taste receptor, 701
South American lantern fly, 841–842
Soybeans, mineral nutrient, 451
Soy sauce production, 132j
S protein, 358
SPAG17 protein, 556
Speciation and macroevolution, 293–311
anatomy of speciation, 295, 295f
defined, 294, 300
how new species evolve, 294–300
reproductive isolating mechanisms, 297–300
species, 294–297
modes of speciation, 300–306
adaptive radiation, 303–306
allopatric speciation, 300–302
convergent evolution, 306
sympatric speciation, 302
phylogenetic species concept, 295
principles of macroevolution, 307–310
developmental genes and macroevolution, 307–309
Species, 2, 3*f*, 8, 335, 335*f*, 346
community structure, 836–837
cryptic species, 294, 296
defined, 296
definition, 8
described, 884f
keystone, 894
Species composition, 836
Species diversity, 836, 837, 869, 889
Species richness, 836
Specific epithet, 335, 346
Specific gravity, of urine, 672
Sperm, 162, 173
Spermatid, 756
Spermatocyte, 756
Spermatogenesis, 162, 173, 755*f*, 756
in mammals, 173, 173f
Spermatogonia, 173
Spermatozoa, 756
Sperm DNA, 250
Sperm editing, 244
Sperm, human
fertilization, 774–777, 783
production, 755–756
spermatogenesis, 755f, 756

ISTUDY

Sphagnum genus, 412
Sphagnum moss, 407
Sphenoid bone, 721
Sphincter, gastroesophageal, 634
Sphygmomanometer, 599
Spice. *See* K2
Spicules, 513
Spider, carnivorous diet, 632
Spike, 352, 352*f*
Spinal cord
  functions, 686–686
  meninge, 686
  parasympathetic division, 694, 695f
  reflex arc, 693–694
  spinal nerve, 691–692
  sympathetic division, 694, 695f
  vertebrate nervous system organization, 678f, 679f
Spinal nerve, 691–692, 692*f*
Spindle, 76
Spindle-shaped cells, 578
Spiracles, 529
  insects, 650
Spirilla, 61
Spirillum volutans, 361
Spirochetes, 61
Spirogyra, 371, 383, 384*f*, 389, 408
Spleen, 611*f*, 612
Spliceosomes, 213
Sponge
  anatomy, 513f
  asexually and sexually, 513
  digestion, 630
  regeneration, 752
  skeleton of, 513
Spongy bone, 571
Spongy mesophyll, 445
Spontaneous generation, 359
Spontaneous mutations, 232
Sporangia, 380, 414
Sporangiophores, 394
Sporangiospores, 394
Sporangium, 385, 389, 394, 409
  bryophytes, 411
  lycophytes, 415
  plants, 411
  pteridophytes
    fern, 417
    whisk ferns, 416–417
  rhyniophytes, 414
Spores, 170, 392, 409
  from black bread mold, 395
  bryophytes, 411
    hornworts, 412
    liverworts, 411–412
    mosses, 412, 413*f*
  fungi, 393
    sac fungi, 395, 396*f*
    smuts and rusts, 400, 400*f*
    zygospore fungi, 394
  lycophytes, 414–415
  pteridophytes, 415–418
  vascular plants, 414
Sporophyte, 173, 409
  alternation of generation, 409–411
  bryophytes
    hornworts, 412
    liverworts, 411–412
    mosses, 412
  dominant generation, 409–411
  gymnosperms, 421
  lycophytes, 414
Sporozoan. *See* Apicomplexans
Sporozoans, 376
Spotted lanternfly (*Lycorma delicatula*), 813
Spotted owl, 869, 895
Springbok, 842
Spring overturn, 876
Spruce
  conifer diversit, 419–420
  coniferous forest, 868, 868f
Squamous epithelium, 574
Squid
  discontinuous feeding, 631, 631f
  nervous system, 678f
S (spike) protein, 658
Stabilizing selection, 285, 285*f*
Stable equilibrium phase, 820
Stahl, Frank, 208
Stamens, 423, 490
  flowering plant life cycle, 424–426
  flowers structure, 423, 424f
Stanley Miller's experiment, 314*f*
St. Anthony's Fire, 398
Stapes, 709, 710
Staph arrangement describes, 362
Staphylococcus aureus strains, 364
Starch, 41, 41*f*, 121
Starfish, 532. *See* Sea star
Starling, 800–801, 800*f*
Start codon, 217
Statistical data, 11
Statistical significance, 12
Statocyst, 710–711, 710*f*
Statolith, 710*f*, 711
Statoliths, 481
STD. *See* Sexually transmitted disease (STD)
Stegosaurus, 326
Stem cell research, 56
Stem cells, 56, 152, 153
Stem cell transplants, 689
Stem diversity, 444*f*
Stems
  organization/diversity of, 439
    herbaceous stems, 439–440
    leaf diversity, 446–447
    leaf morphology, 444–445
    stem diversity, 443–444
    woody stems, 440–443
  secondary growth, diagrams of, 442f
  shoot system, 434, 439
  underground, 443
  vascular bundle, 434
Stentor, 377
Stereoscopic vision, 557, 558*f*, 703
Sterility
  postzygotic isolating mechanisms, 298f, 300
  sympatric speciation, 302
Steroid hormone
  actions, 736, 736f
  anabolic, 747, 747f, 757
Steroids, 45, 46*f*
Stevens, Nettie, 165, 165*f*
Stickleback fish, 798, 798*f*
Sticky ends, DNA, 239, 239*f*
Stigma, 490
Stimulatory pathway, 157, 156*f*
Stimulus
  classical conditioning, 799
  fixed action patterns (FAP), 797–798
  operant conditioning, 800
  reception of, 477
  sign stimulus, 798
  transduction, 477
  vision, 702
Stingrays, 542
Stinkhorn, 399*f*
Stirrup. *See* Stapes
Stoeckle, Kate, 337
Stolons, 394, 500
Stomach
  discontinuous feeding, 631
  human
    anatomy, 633*f*, 635*f*
    gastroesophageal sphincter, 634
    lumen, 634
  squid, 631
Stomata, 112, 122, 408, 431
  guard cells, 462
  opening and closing, 463f
    abscisic acid (ABA), 463
  water transport, 461–462
Stone cell. *See* Sclereids
Stramenopiles, 378, 388
  algae, 378–380
  brown algae, 378–379
  diatoms, 378
  golden brown algae, 378
  oomycetes, 379–380
Strata, 261, 331
Stratification
  lake overturn, 875
  temperate deciduous forest, 869
Stratified epithelia, 575
Stratum, 318
Strauss, Louisa, 337
Strawberry, 443, 444*f*
Strep throat, 656, 656*f*
Streptococcus pneumoniae, 203
Streptococcus pyogenes, 362, 626, 656, 656*f*
Streptococcus thermophilus, 132, 362
Stress and adrenal glands, 742–743
Striation, skeletal muscle, 578, 728
Strigophilusgarylarsonii, 335
Strix occidentalis caurina, 869
Strobili, 414
Stroke, 601, 643, 688
Stroma, 72, 73*f*, 112
Stromatolites, 320, 331
Structural genes, 222
Structural genomics, 250
Sturnus vulgaris, 800*f*
Style, 490
Subcutaneous layer, 584
  of skin, 581
Subduction zones, 329
Suberin, 431, 437, 442
Submucosa, 634
Subspecies, 289*f*
Substrate-level ATP synthesis, 129, 129*f*, 136
Substrate-level phosphorylation. *See* Substrate-level ATP synthesis
Substrates, 103
Succession, ecological, 846–847

ISTUDY

Succinea, 845
Sucrose, 41
  H^+ ion gradient, 64f
Suctoria, 377
Sulfur dioxide, 892–892
Summer wood, 442
Sun and solar radiation, 863–864, 863*f*
Sundew, Cape, 457*f*
"Superbug," 276. *See also* Antibiotics
Superfemale. *See* Poly-X females
Supergroups, 8, 9, 9*t*, 340, 346, 371, 387
Supportive connective tissue, 574
  cartilage knee, 724
Support systems. *See* Locomotion and support system; Skeleton
Supraorbital ridge, 721
Surface-area-to-volume ratio, 58, 58*f*
Surface tension, 30
Surrogate mother, 764
Survivorship, 816–817, 817*f*
Sushi, 867
Suture (skull), 721, 722*f*
Swallowing, 634, 634*f*
Swamp
  ecosystem, 877
  mangrove swamp, 87
Swamps, 877
Sweat glands
  aging process, 790
  human skin, 584
Sweet taste receptor, 701
Swim bladder, 542
Swine flu, 15, 350
Swyer syndrome, 176
Symbiosis, cleaning, 844
Symbiotic relationships, 363, 368, 400–404
  algae, 844
  archaea, 363–364
  bacteria, 843, 844
  bacteria, 363–364
  commensalism, 363
    ecology of communities, 843, 843*t*
  ecology of communities, 842–844, 843t
    commensalism, 843, 843*t*
    mutualism, 843–844, 843*t*
    parasitism, 842, 843*t*
  fungal parasitism, 400
  lichens, 400–403, 844
  mutualism, 363
    bacteria, 843
    ecology of communities, 843*t*
    fungi, 844
    lichen, 844
    lichens, 400
    mycorrhizae, 843
  mycorrhizae, 403–303, 843
  parasitism, 363
    bacteria, 843
    coevolution, 844
    defined, 842
    ecology of communities, 842–843, 843*t*
    ectoparasite, 843
    secondary host, 843
    viruses, 842
Symmetry
  animal evolution, 677
  bilateral symmetry, 677
Sympathetic division, 694, 695*f*
Sympatric speciation, 302
  alloploidy, 302
  autoploidy, 302
  polyploidy, 302
Synapse
  defined, 682
  integration, 683, 686, 686f
  transmission across, 682–683
Synapsid, 547
Synapsis, 164
Synaptic cleft, 682, 782
Synaptic vesicle, 728
Synaptonemal complex, 164
Synovial joint, 724
Synovial membrane, 724, 724*f*
Synthetic auxins, 474
Synthetic blood, 589
Synthorx (biotechnology company), 202
Syphilis, 770, 770*f*
Systematic biology, 334–338, 346
  field of taxonomy, 334
  Linnaean classification hierarchy, 335–338
  Linnaean taxonomy, 334–333
Systematics, 7, 334
Systemic circuits, 592, 598*f*
Systemic lupus erythematosus, 626, 627*f*
Systemin, 478
Systole, 596

T

T_3. *See* Triiodothyronine
T_4. *See* Thyroxine
Tactile communication, 804
Taenia, 524*f*
Taenia solium, 523
Taiga, 867–868, 869*f*
Tall-grass prairie, 873*f*
Tamarin, 806–807
Tandem repeats, 251
Tapetum lucidum, 703
Tapeworms, 523, 630
  infections, 523
  life cycle of, 524f
Taproot, 437
Target cell, 735, 735*f*
Tarsal bone, 723
Taste
  parietal lobes, 688, 688f
  sense of taste in humans, 700–701, 700f, 702
  taste bud, 634, 700, 700f
Taste bud, 634, 700, 700*f*
Taste pore, 701
Tatum, Edward, 210
Taxon, 334
Taxonomists, 294, 334, 337
Taxonomy, 8, 260, 334, 337
  classification of organisms, 334f
Tay-Sachs disease, 69
TB (tuberculosis), 626, 657*f*, 658
Tbx5 gene, 308
T cell, 244
  aging process, 790
  allergic responses, 626
  autoimmune disorders, 626
  cell-mediated immunity, 619
  delayed allergic response, 619
  development, 612, 616
  helper T cell, 619
  HIV/AIDS, 620, 621, 658, 766
  thymus, 748
Technology, 15
Tectorial membrane, 710, 710*f*
Telomerase, 210
Telomere, 789
Telomeres, 154, 210
Telophase, 149, 150*f*
Temperate deciduous forest, 867–868
Temperate grassland, 872, 873*f*
Temperate rain forest, 865*f*, 868
Temperate-zone fern, typical, 436
Temperature
  body temperature, 586
  control center, negative feedback, 585–586, 585f–586f
  control center, positive feedback, 586–587, 586f
  Earth
    biome distribution, 865, 865*f*
    climate change, 854, 857, 858
  fever, 615
  lake, 876
Temperature regulation, prostaglandins, 748
Temporal bone, 721, 722*f*
Temporal isolation, 297–299, 299*f*
Temporal lobe, 688, 688*f*
Tendons
  dense fibrous connective tissue, 576
  skeletal muscle, 725
Tension, 461
Terminal bud, 434, 441
  plants, 435f
  shoot system, 435, 439
Termination of polypeptide synthesis, 217–218, 218*f*
Termite
  mutualism, 843
  pheromones, 801
Terrestrial ecosystem, 865–874
  biome distribution, 865f
  coniferous forest, 865, 866f
  deserts, 873–874
  grasslands, 872–873
  shrublands, 871–872
  temperate deciduous forest, 866f, 868–869
  tropical rain forest, 866f, 869–872
  tundra, 865–867, 868f
Territoriality, 287, 805–806
Territory
  defined, 287, 805
  territoriality, 805–806
TESS telescope, 19
Testcrosses, 188–189
  one-trait testcross, 188, 189f
  two-trait testcross, 188–189
Test group, 11
Testis
  cancer, 755
  functions, 746–747, 752, 754–755, 754t
  gonadotropic hormones, 738
  major glands, 734f
  steroid hormones, 736
Testosterone, 45
  aging process, 791
  production, 745–747, 757
Tetany, 725, 741
Tetracyclines, 364
Tetrahydrocannabinol (THC), 685

ISTUDY

Tetrapods, 256, 539
Thalamus, 677, 690
Thallus, 411
Thamnophis elegans, 796*f*
THC (tetrahydrocannabinol), 441, 685
Thecodonts, 548
The Evolution of Classification Systems, 338
Theory of evolution, 12–13
Theory, scientific, 12–13
The Power of Movement in Plants, 471
Therapeutic cloning, 152, 153, 153*f*
Thermoacidophiles, 367, 368
Thermocline, 876
Thermodynamic law
  ecosystem dynamics, 850
Thermoreceptor, 699, 713
Thermus aquaticus, 239
Thigmomorphogenesis, 480
Thigmotropism, 479–480
Thoracic cage. *See* Rib cage
Thoracic cavity, 653*f*
Thoracic duct, 611*f*
Thoracic vertebra, 722
Threatened species, 885
Three-domain system, 337–341
  definition of, 338
  domain archaea, 339
  domain bacteria, 339
  domain eukarya, 339–340
Threshold, 682
Thrombocyte. *See* Platelets
Thrush, 622
Thylakoids, 62, 72, 73*f*, 112
Thymine (T), 205, 205*f*. *See also* DNA (deoxyribonucleic acid)
Thymine dimers, 234
Thymus
  aging process, 790
  endocrine system, 734f
  functions, 612, 747–748
  lymphatic system organization, 611f
Thyroid, 734*f*, 739, 740–742
Thyroid-stimulating hormone (TSH), 739, 740
Thyroxine, 739, 740
Tibia, 723
Tidal ventilation mechanism, 652
Tight junction, 92, 93*f*
Tiktaalik roseae, 259, 269, 270*f*
Tineas, 397
Tinsel flagellum, 373
Tissue, animal, 574
  connective tissue, 575–579
  epithelial tissue, 574–575, 574f
  nervous tissue, 579–578
  tissue engineering, 626, 658
Tissue culture, 500, 503
  for plant conservation, 503f
Tissue engineering of organs, 626, 658
Tissue, plant, 430–433
  definition, 430
  flowering plant cells and tissues, 430–433
    epidermal tissue, 430–431
    ground tissue, 431–432
    vascular tissue, 432–433
Tissues, 2, 3*f*
Tissue stages of development, 775–776
Titer, antibody, 621, 621*f*
Tmesipteris species, 416, 417
TNF (tumor necrosis factor), 619, 621
Toad
  Midwife, 824f
  parental care, 824f
Tobacco
  birth defects, 788
  nicotine abuse, 684
  vaping, 660
Tobacco mosaic virus, 353
Tobacco plant, 478
Tobacco smoke, 233
Tolerance model of succession, 847
Tomatoes, defense response, 477*f*
Tomato plants, 476
Tone, muscle, 713, 725–726
Tongue
  human, 634
  sense of taste, 700–701, 700f, 702
Tonicity, 87
Tonsil, 611*f*
Tooth
  adaptations to diet, 632, 632f
  canines, 632
  human, 632
  incisors, 632
  molars, 632
  premolars, 632
Topography and climate, 864
Topsoil, 453
Totipotency, 501
Toxins, 364, 368
  botulinum, 726
  Shiga toxin, 364
  tetanus, 363–364, 364f
Toxoplasma gondii, 377
Toxoplasmic encephalitis, 622
Toxoplasmosis, 788
t-PA (tissue plasminogen activator), 241
Trachea, 529
  arthropods, 648f, 649–650, 650f
  mammals
    dog, 648*f*
    human, 650
  respiratory tract disorders, 656–657
    asthma, 625, 657
    bronchitis, 657
    choking, 656
Tracheids, 410, 433, 458
  plant survival mechanisms, 432
  structure, 433
Tracheostomy, 656
Tracheotomy, 656
Tract (interneuron bundle), 687
Trait evolution, 341*f*
Trait, polygenic trait, 283
Traits, 334, 337
Transacetylase, 222
Transamination, 138
Transcription, 211, 212–214, 213*f*
  function of introns, 214
  mRNA transcript, 212, 213f
  pre-mRNA, 213
  production of RNA, 212
  promoter, 212, 213f
  RNA molecules processing, 213–214, 214f
  RNA polymerase, 212, 213f
  stages of, 212–213, 213f
  template strand, 212
Transcription activator, 228
Transcriptional control, eukaryotic, 225, 229, 229*f*
Transcription factors, 229
Transduction, 361, 368
  prokaryotes, 361
Trans fat, 43, 45
  nutrition and health, 641–642, 644
Transfer rate of carbon cycle, 852, 854*f*
Transfer RNA (tRNA),51, 67, 210
  amino acid-tRNA complex, 215
  charged tRNA, 215
  role in translation, 214–215
  structure of, 214–215, 215f
Transformation, 361, 368
  prokaryotes, 361
Transform boundary, 329
Transgenic animals, 248–250
  applications, 249250
  cloning of, 248–249
Transgenic bacteria, 242, 244–245
Transgenic organism, 239, 242
Transgenic salmon, 249–250, 249*f*
Transitional epithelium, 575
Transitional fossils, 269, 270*f*
Translation, 211, 214–218
  elongation, 217, 218f
  initiation, 217, 217f
  ribosomal RNA, role of, 216–217, 216f
  steps of, 217–218
  termination, 217–218, 218f
  transfer RNA, role of, 214–215, 215f
Translational control, 225, 233
Translocation, chromosome, 176–178, 178*f*
  balanced, 176
  unbalanced, 176–178
Translocation syndromes, 179
Transmissible spongiform encephalopathies (TSEs), 49, 355
Transmission across a synapse, 682–683
Transmission electron microscope (TEM), 59, 59*f*
Transmission of nerve impulses, 681–682
  action potential, 682
  propagation of action potentials, 682
  resting potential, 682
  transmission across a synapse, 682–683
Transpiration, 461
  water cycle, 852
Transplant
  artificial lungs, 658
  artificial retina, 706, 707
  bone marrow, 621
  pancreas, 745
  rejection, 621
  xenotransplantation, 627, 745
Transport, animal
  breathing and transport of gases, 654–656
    amphibians, 648, 652
    birds, 652
    fish, 652
    human, 652–656
    mammals, 652
    reptiles, 652
  gas exchange and transport
    human respiratory system, 655–656
    partial pressure, 655
    surfaces, 647–651
    transport of oxygen and carbon dioxide, 655–656

ISTUDY

Transport, plant, 458, 459*f*
organic nutrient transport, 463–466
reviewing xylem and phloem structure, 458
stomata open and close, 462–463
water potential, 458, 461
water transport
cohesion-tension model of xylem, 461, 462*f*
transpiration, 461
Transposon, 251, 252*f*
Transposon patterns, 560*f*
Tree
annual rings, 443
ecological succession, 846, 847
trunk, 443, 443f
Tricep, 725*f*
Triceratops, 326
Trichinella larvae, 525
Trichinella spiralis, 525
Trichocysts, 377, 388
Trichomes, 430, 478*f*
Trichomonas vaginalis, 373, 373*f*, 770
Trichomoniasis, 771
Triglycerides, 43–45, 44*f*, 640
energy storage in fats, 43, 45
fatty acids with saturated bonds, 43, 44f
fatty acids with unsaturated bonds, 43, 44f
glycerol subunit, 43
oils and fat, 43
synthesis and degradation of, 43, 44f
trans fat, 43
Triiodothyronine, 739
Trimester, 782
Triploid plant, 302
Trisomy, 174
Trisomy 21, 175, 176*f*
tRNA (Transfer RNA). *See* Transfer RNA (tRNA)
Trochozoan, 516
flatworms, parasitic, 630
Trophic level, 849, 850*f*
Trophoblast, 783
Tropical forest, 869–871
Tropical rain forests, 16, 120
biome, 869–871, 870f, 871f
destruction and climate change, 120
plant-transpired mist, 462f
role of, 120
species richness, 836
Tropism, 478
negative, 478
positive, 478
Tropomyosin, 729
Troponin, 729
trp operon, 222, 223*f*
True coelom, 512
True rib, 722
Truffle, 404*f*
Trumpet vine, 496
Trypanosoma brucei, 374, 374*f*, 375*f*
transmission pattern of, 374f
Trypanosoma cruzi, 374
Trypsin, 637, 639
Tryptophan, 222, 223*f*
TSH (thyroid-stimulating hormone), 739, 740
T system, 726
T tubule, 726, 727*f*, 728
Tubal ligation, 758
Tube feet, 533
Tube foot, 632, 717
Tuber, 443, 444*f*
Tuberculosis (TB), 626, 657*f*, 658
Tuber melanosporum, 404, 404*f*
Tube worm, 632
Tube worms, 520*f*
Tubular reabsorption, 669–671, 670*f*
Tubular secretion, 670*f*, 671
Tubulin, 74
Tumor, 154
Tumor necrosis factor (TNF), 621, 623
Tumor suppressor genes, 143, 154
mutation in, 154, 158f
Tuna, 893
Tundra, 865–867, 866*f*
Turdus migratorius, 335
Turgor movements, 480, 481*f*
Turgor pressure, 88, 460*f*
opening and closing of stomata, 462, 463
wilting, 463
Turner syndrome, 176, 177*f*
T wave, 598
Twin, studies, 796
Tympanic canal, 709
Tympanic membrane, 709
Tympanum, 708
Type 1 diabetes, 644, 745
Type 2 diabetes, 644, 745–746
Typhlosole, 630*f*, 631
Tyrannosaurus rex, 326, 547

U

Ulcer, 635
Ulna, 723
Ultraviolet (UV) light, nectar guides, 702, 703*f*
Ultraviolet (UV) radiation, 583
Ulva lactuca, 383, 389*f*
Umami taste receptor, 701
Umbilical cord, 785, 785*f*, 786
Understory, 868
Undescended testis, 754
Uniform distribution, 815
Uniformitarianism, 262
Unique noncoding DNA, 252
Unsaturated fat, 641
Unsaturated fatty acid, 43, 44*f*
Upper respiratory tract disorder
common cold, 656
strep throat, 656, 656f
Upwelling, 876
Uracil (U), 210. *See also* RNA (ribonucleic acid)
Urban heat islands, 449
Urea
defined, 664
production, 638
tubular reabsorption, 671
Ureter, 668, 668*f*
Urethra, 754*f*, 757*t*, 756–757
defined, 668
Urey, Harold, 314
Uric acid
defined, 664
gout, 664
insects, 665
Urinalysis
chemical examination, 672
forensic analysis, 672
microscopic examination, 672
physical examination, 672
Urinary bladder, 668, 757
Urinary incontinence, 790
Urinary system, human, 667–674, 667*f*
antidiuretic hormone (ADH), 737, 737f
homeostasis and kidneys, 671–674
acid-base balance, 673–674
water-salt balance, 671–673, 742, 743
kidneys
nephrons, 668, 669*f*
urine formation, 668–671
glomerular filtration, 668–669, 670*f*
tubular reabsorption, 669–671, 670*f*
tubular secretion, 670*f*, 671
Urine, 672
defined, 665
earthworms, 677
energy flow, 849
fish, 666
human, 668–671
formation, 668–671
glomerular filtration, 668–669, 670*f*
tubular reabsorption, 669–671, 670*f*
tubular secretion, 670*f*, 671
kangaroo rat, 666
Urobilinogen, 672
Urochordates, 538
Uterine cycle, 761, 761*t*
estrogen and progesterone, 761
menstrual phase, 761
proliferative phase, 760f, 761
prostaglandins, 748
secretory phase, 760f, 761
Uterine tube, 758, 758*f*, 758*t*
Uterus
human
defined, 758
uterine cycle, 761
uterine cycle, 761
Utricle, 712
Uvula, 634
Uy, J. A. C., 809

V

Vaccine
active immunity vs., 621
by age, 624, 624t
booster, 623
contraceptive vaccines, 763–764
HIV, 768
Vacuole amoeboids, planarians, 630
Vacuoles, 71–72, 71*f*
in animal cells, 71
central vacuole, plant cell, 71–72, 71f
plant, 71
Vagina, 758–759
Valence shell, 25
Vampire bat, 98
Vampire bats, 810
van Beneden, Pierre-Joseph, 163
Van Ermengem, Emile Pierre, 726
van Leeuwenhoek, Antonie, 359, 523
Van Niel, C. B., 114
"Vaping," 660, 684
Variation in population, 266, 266*f*

ISTUDY

Vascular bundles, 433
  definition, 433
  stem structure, 434
Vascular cambium, 430, 439
  definition, 430
  secondary growth, 430
Vascular cylinder, 433
Vascular plants, 408, 413–417, 413, 414
Vascular ray, 433
Vascular tissue, plant, 411, 432
  lycophytes, 414–415
  lycophytes, evolution of, 413–415
  monocots and eudicots, 434
  origin of, 414
  root system, 437
Vas deferens, 754*f*, 754*t*, 755, 756
Vasectomy, 755, 764
Vector, 238, 376
  transmitting, 388
Vegetal pole, 775
Vegetative cells, 383
Vegetative organs, 434
Vein, human
    hepatic portal vein, 637
    hepatic vein, 637
Veins, 592
Velociraptor, 326
Venae cavae, 598
Venation, 434
Ventilation. *See also* Breathing and transport of gases
  defined, 647
  fish, 649
  methods, 652
Ventricle (brain), 680
Ventricles, 595
Venules, 592
Venus flytrap, 267, 446, 481
Vermiform appendix. *See* Appendix
Vertebra, 722, 777, 777*f*
Vertebral column, 539
  human, 722, 776
Vertebrate circulatory pathways, 543*f*
Vertebrate forelimbs, 271, 271*f*
Vertebrates, 505, 506. *See also* Reproductive system
  evolution
    nervous system, 677, 678–679*f*
  human medicine, 545
  nervous system, 677, 678–679f
  osmoregulation, 665–666
    kangaroo rat, 666, 666*f*
    marine mammals and sea birds, 667
    terrestrial vertebrates, 666–667
Vertigo, 712
Vesicles, 65, 316, 331
Vesicular follicle, 760
Vessel elements, 433, 458
Vestibular canal, 710
Vestibular gland, 758
Vestibule, 709
Vestigial structures, 260, 271
Vibrio cholerae, 339
Video-enhanced contrast microscopy, 60
Villus, 636
Viral diseases, in humans, 350*t*
Virchow, Rudolph, 57
Viroids, 355, 368
Viruses, 350, 351*f*. *See also* Archaea; Bacterium (bacteria)
  classification system of, 350
  discovery of, 350
  host specific, 353
  parasitic nature of, 353
  reproduction of, 353–356, 354f
    animal viruses, reproduction of, 354–355
    attachment, 353
    biosynthesis, 353
    emerging infection diseases, 356–359
    lysogenic cycle, 353–354
    lytic cycle, 353
    maturation, 353
    penetration, 353
    release, 353
  structure of, 350–353
Visceral muscle, 578. *See also* Smooth muscle
Visceroceptor, 699
Vision, 702–708
  animal detection of light, 702–703
  artificial retinas, 706, 707
  camera-type eye, 703
  compound eye, 702
  human eye, 703–704
    aging process, 791
    anatomy, 703–704, 703*f*
    disorders, 706–708, 791
    focusing, 704, 706–708
    integration of visual signals, 705–706
    photoreceptors, 703, 704
    from the retina to the visual cortex, 706
  panoramic, 703
  stereoscopic, 703
Visual accommodation, 704
Visual communication, 804–804
Vitamin A, 420, 704
Vitamins, 106
  defined, 643
  folic acid, 788
Vitamins C, 672
Vitamins K, 363
Viviparity, 753
VNO (vomeronasal organ), 803
Vocal cord, 651, 757
Vocal pouch, 824*f*
Voltage, 681, 682
Volume, 710
Volume and breathing, 652, 652*f*
Voluntary muscle. *See* Skeletal muscle
Volvox, 371, 383, 383, 384*f*
Vomeronasal organ (VNO), 803
Vomiting, 634
von Linne, Karl, 335
Vulva, 758, 780

W

Wächtershäuser, Günter, 314, 315
Waggle dance, 804, 805*f*
Walking erect, adaptations for, 563*f*
Wallace, Alfred Russel, 265
Walsh, G., 809
Warbler
  Bay-breasted, 840f
  Blackburnian, 840f
  Black-throated green, 840f
  Cape May, 840f
  speciation, 840f
  Yellow-rumped, 840f
Warning coloration, 841, 842*f*
Warren, Robin, 635
Wart, genital, 768
Wasp
  inclusive fitness, 808
  mimicry, 842
  yellow jacket, 842
Water, 27. *See also* Aquatic ecosystem
  chemistry, 27–31
  column, 461
  cycle, 852–853, 875
  groundwater mining, 852
  hydrogen bonding in, 28f, 28
  ice, lake ecosystems, 875–876
  mineral uptake, 454–457, 454f
  models for structure of, 27
    ball-and-stick model, 27, 28*f*
    electron model, 27, 28*f*
    space-filling model, 27, 27*f*
  molds, 380f, 380
  properties of, 28–31, 29f, 30f
    cohesion and adhesion, 30–31, 30*f*
    high heat capacity, 29, 29*f*
    high heat of evaporation, 29, 29*f*
    ice less dense than liquid water, 30, 31*f*
    water as solvent, 29
  water-salt balance, human, 671–673, 742
Water cycle, 852–853, 853*f*, 875
Water potential, 458, 459*f*, 460, 460*f*
  concept of, 460
Water-salt balance. *See also* Osmoregulation
  adrenal glands, 742
  human kidneys, 671–673
Water strider, 877*f*
Water table, 852, 875
Water transport
  cohesion-tension model of xylem, 461, 462f
  transpiration, 461
Water vascular system, 532
Watson and Crick model, of DNA, 206–207, 207*f*
  complementary base pairing, 207
  double helix model, 206, 207f
Watson, James, 206, 208
Waxes, 45, 46*f*
Weather
  biomes
    coniferous forests, 868–869
    desert, 874
    grassland, 872–874
    shrubland, 870–872
    temperate deciduous forests, 869
    tropical rain forests, 869–870
    tundra, 865–868
  bodies of water, 880
  ocean currents, 880
  solar radiation, 863–864
  topography, 864
Webb telescope, 19
Wegener, Alfred, 328
Weight, human birth, 285*f*
Weight loss, tapeworm infections, 523

Weinberg, Wilhelm, 278
Welwitschia, 421
Went, Frits W., 472
Western bluebird, 822
Wetland
  ecosystem, 876–877
  habitat loss, 889
  waste recycling, 888
Wetland community restoration, 835
Whale, 293
  auditory communication, 803
  Baleen, 631
  echolocation, 699f, 700
  filter feeding, 631
  Humpback, 803
  sushi, 867
  teeth, 632
Wheat (*Triticum aestivum*), 480
Wheat rust, micrograph of, 400*f*
Wheel animalcules, 523
Whiptail lizard, 167
Whisk ferns, 416, 417*f*
Whitebark pine, 844
White blood cells (WBCs), 578, 604, 672. *See also* B cell; T cell
  antigen binding, 610
  granular leukocytes
    eosinophils, 615
    neutrophils, 615
  inflammatory response, 614
  natural killer (NK) cells, 615
White matter, 687
Whittaker, R. H., 338, 393
Wildlife conservation. *See* Conservation of biodiversity
Wilkins, Maurice H. F., 206
Williams syndrome, 178
Willow, 437*f*
Wilson, Edmund Beecher, 165
Wind
  circulation, 863–864
  fruit dispersal, 498
  lake overturn, 875
  ocean currents, 880
  pollination, 492, 494
Windblown spores, 417
Windpipe. *See* Trachea
Wind-pollinated flowers, 426
Wine, 397
Wobble hypothesis, 215
Woese, Carl, 365
Woodruff, David, 867
Wolf, 267, 271*f*
Womb. *See* Uterus
Wood, 442–443
  spring, 442
  summer, 442
Woody plants, advantages and disadvantages of, 443
Woody stem, 440–443
  bark, 442
  organization, 436
  stem diversity, 443–444
  trees and shrubs, 440
  wood, 442–443
Woody twig, 439*f*
  three-year-old, 442f
Worker (army ant), 807
World Health Organization (WHO), 374, 525
Wrasse, 844*f*
Wrinkle, 726
Wrist, 723
Wuchereria bancrofti, 525

X

Xanthoria, 401*f*
Xenon-133, 24
Xenotransplantation, 545, 627, 745
Xeroderma pigmentosum, 233
X-inactivation, in mammalian females, 226, 227*f*
X-linked agammaglobulinemia (XLA), 625
X-linked inheritance, 197–200
  disorders in humans, 198
    adrenoleukodystrophy, 200
    color blindness, 198, 198*f*
    hemophilia, 198
    Menkes syndrome, 198
    muscular dystrophy, 198, 200
  Morgan's experiment, 197, 197f
  solving X-linked genetics problems, 197–198
X-ray, birth defects, 788
Xylem, 413, 432, 433
  conducting cells, organization of, 458f
  features, 432
  leaves, 445f
  plant survival mechanisms, 432
  procambium, 439
  structure, 433f
  woody plants, 443
Xylem rays, 458
Xylem sap, 458
  leak, 461
Xylem structure, 458
Xylem transport, cohesion-tension model of, 461, 462*f*

Y

Y chromosome, 197
Yeast fermentation, 131, 132
Yeast, logistic growth, 819–820
Yeasts, 392, 395
  Candida albicans, 397
  importance for humans, 397
  sac fungi, 396f
Yellow jacket wasp, 842, 842*f*
Yellow-rumped warbler, 840*f*
Yellowstone National Park, 894*f*, 895
Yersinia pestis, 822
Yolk, 753, 775, 782
Yolk sac, 782, 783

Z

Zamia pumila, 420
Zea mays, 205, 205*t*, 422
Zeatin. *See* Cytokinin
Zebra finch, 807
Zebra mussel, 890
Zero population growth, 826, 827
Zika virus, 356, 357
Z lines, 727*f*, 728
Zombie deer, 536
Zombie symptoms, 536
Zona pellucida, 774
Zone of cell division, 437
Zone of elongation, 437
Zone of maturation, 437
Zooids, 516
Zooplankton, 887, 879
Zoosporangia, 380
Zoospores, 380, 383, 383*f*, 389, 393
  chytrids, 394
Zooxanthellae, 375, 879
Zygomatic bone, 721, 722*f*
Zygomycota, 393*t*, 394–395
Zygospore fungi, 383, 384, 394–395
Zygospore fungus. *See* Zygomycota
Zygote, 163, 173, 174, 409, 574
  angiosperms, 489
  animal development, 775
  conifer life cycle, 420f
  defined, 297
  fern, 418f
  human fertilization, 760, 774–775, 783
  land plants, 408
  mosses, 413f
  seed development stage, 491, 495f
  totipotency, 778

ISTUDY